Mathematics

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Notation List

Collision Index and Scope Rules

bare $A, B, \ldots$; $\mathcal{A}, \mathcal{B}, \ldots$	unadorned Latin and calligraphic letters denote local objects unless a decorated form is listed below; local declarations in a theorem, proof, or subsection take precedence
B, $B_r(x)$, $\overline{B}_r(x)$, $\mathrm{B}(x, y)$, $\mathcal{B}(\mathbb{R}^n)$	matrix or set variable / open ball / closed ball / beta function / Borel σ-algebra; font, arguments, and ambient subject determine the referent
$I_{\mathcal{V}}$, I_n, I^k, I_R	identity operator, identity matrix, k-cell, and power-series interval
P, $\mathcal{P}(X)$, $\mathscr{P}$, $P_{\mathcal{U}}$, P_j	locally declared projection or map / power set / umbral polynomial algebra / orthogonal projection / spectral projection
$\mathcal{F}$, $\mathscr{F}$, $\mathcal{F}(K)$, $\mathcal{F}(\mathcal{V})$	local family of functions or operators / umbral formal-series algebra / finite subsets of K / finite-rank operators
$\mathcal{L}(\mathcal{V}, \mathcal{W})$, $\mathcal{L}^n$, $\mathscr{L}^1$, $\mathrm{LS}(A, \mathbf{y})$, $\mathcal{L}(\theta, \alpha)$	linear maps / Lebesgue measurable sets / integrable functions / least-squares solution set / local Lagrangian or loss notation
$R(f, \Pi, \{c_i\})$, R, $R_k(A)$, $A = QR$	Riemann sum, locally declared radius or scalar, row-sum quantity, and QR factorization; arity and argument type disambiguate the notation
Δx_i, ΔA, $\Delta(A)$, $\Delta(\mathcal{V})$, Δ_{k-1}	increment, matrix perturbation, defect from normality, metric discriminant, and coefficient simplex
Ω, Ω_T, Ω_n, $\mathrm{vol}(\Omega)$	local domain or set / finite spectrum of T / primitive roots of unity or DFT diagonal, as specified by context / volume domain
Q^k, Q_n, Φ_n	standard k-simplex / normalized Fourier matrix / cyclotomic polynomial; cyclotomic polynomials use Φ_n
f^{-1}, $f^{-1}(B)$	inverse function when f is bijective / preimage of B under any function f; bijectivity, argument type, and local declaration determine the meaning
$\wedge$, $\vee$, $x \wedge y$, $x \vee y$, $dx_I \wedge dx_J$	logical conjunction and disjunction / lattice meet and join / exterior product; subject and operand types determine the meaning
$\mathrm{supp}(f)$	nonzero set $\{x : f(x) \neq 0\}$ when the codomain has a distinguished zero / its closure when the domain is topological; the local definition determines the meaning
$\overline{z}$, $\overline{E}$, $\overline{\mathcal{A}}$, A^*, $T^*\omega$, f', $\mathcal{V}'$	conjugation, closure, uniform closure, adjoint or conjugate transpose, pullback, derivative, and dual space; the decorated object determines the operation

Global Conventions and Basic Notation

Convention. Notation is scoped by subject, argument type, font, decoration, and any local definition in force. Unadorned letters denote local objects by default; persistent meanings are carried by subscripts, superscripts, operators, fonts, or declared argument patterns. When a glyph is reused, the distinguishing feature is recorded below.

Set, Function, and Logical Notation

$x \in E$, $x \notin E$	membership / non-membership
$E \subseteq F$, $\emptyset$	subset inclusion, not necessarily proper / empty set
$X \times Y$, $f : X \to Y$	Cartesian product / function with domain X and codomain Y
$f(A)$, $f^{-1}(B)$, f^{-1}	image of A / preimage of B / inverse function when f is bijective; argument type and local context distinguish preimages from inverse-function values
$\mathrm{range}(f)$, $f\|_A$	range of f / restriction of f to A
$g \circ f$, id_X	composition of functions with compatible domain and codomain / identity function on X
$\mathbf{1}_E$	indicator of E, equal to 1 on E and 0 outside E
$[x]_\sim$, $X/\sim$	equivalence class of x / quotient set for the relation $\sim$
$\{A_i\}_{i \in I}$, $\bigcup_{i \in I} A_i$, $\bigcap_{i \in I} A_i$	indexed family of sets / set of points lying in some A_i / set of points lying in every A_i
$\sup E$, $\inf E$	least upper bound / greatest lower bound in the ambient ordered set; extended values may be used when explicitly allowed
$x \wedge y$, $x \vee y$, $\bot$, $\top$	meet / join / bottom / top in a lattice
$\neg$, $\wedge$, $\vee$	logical negation / conjunction / inclusive disjunction
$\Rightarrow$, $\Longrightarrow$, $\Longleftrightarrow$	logical implication (either arrow length) / logical equivalence
$\forall$, $\exists$, $\exists!$	universal / existential / unique-existence quantifier
$:=$	definition by equality

Number, Size, and Scalar Notation

$\mathbb{N}$, $\mathbb{N}_0$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$	positive integers / nonnegative integers / integers / rationals / reals / complexes
$n!$, $(n)_r$, $\binom{n}{r}$	factorial / falling factorial / binomial coefficient
$\mathfrak{S}_n$, $\mathrm{sign}(\sigma)$	permutations of $\{1, \ldots, n\}$ / sign of a permutation
$\mathbb{F}$, $\mathbb{F}_0$	ambient scalar field unless fixed as $\mathbb{R}$ or $\mathbb{C}$ / prime subfield of $\mathbb{F}$; probabilities use explicit Pr or locally declared probability spaces
$\|x\|$, $\|E\|$, $\|\|x\|\|$	absolute value or complex modulus for scalars; interval length or cardinality according to argument type; norm of a vector, function, or operator
a, b, c	locally declared scalars or interval endpoints

Linear and Operator Structures Notation

Vector Spaces and Lists

$\mathbb{F},\ \mathbb{F}_0$	ambient scalar field, typically $\mathbb{R}$ or $\mathbb{C}$ / prime subfield of $\mathbb{F}$
$V(n,q),\ \mathbb{F}_q^n$	n-dimensional coordinate space over the finite field with q elements
$\mathbb{F}^{(\mathbb{N})}$	vector space of finitely supported sequences over $\mathbb{F}$
$\mathbf{u},\mathbf{v},\mathbf{w},\mathbf{0}$	vectors / zero vector
$\mathcal{U},\mathcal{V},\mathcal{W}$	vector spaces or subspaces; other calligraphic letters such as $\mathcal{A},\mathcal{B},\mathcal{F},\mathcal{M}$ are local unless listed below; umbral algebras use script $\mathscr{P},\mathscr{F}$
$I_{\mathcal{V}},\ I_n,\ 0$	identity operator on $\mathcal{V}$ / identity matrix / zero map or zero matrix; distinct from the cell I^k and interval I_R
$\beta=(\mathbf{v}_1,\dots,\mathbf{v}_m)$	ordered vector list; a basis iff it is linearly independent and spans
$(),\ \beta_{\widehat{j}}$	empty list / list β with the jth entry omitted
$(\beta,\mathbf{v}),\ \beta_{j\leftarrow\mathbf{v}}$	list with $\mathbf{v}$ appended / list with the jth entry replaced by $\mathbf{v}$
$\mathcal{B}$	locally declared basis set, such as a Hamel, Hilbert, module, or affine basis; $\mathcal{B}(\mathbb{R}^n)$ denotes the Borel σ-algebra in analysis
$\operatorname{span}(E),\ \dim(\mathcal{V})$	linear span of E / dimension of $\mathcal{V}$
$\operatorname{codim}_{\mathcal{V}}(\mathcal{U})$	codimension of $\mathcal{U}$ in $\mathcal{V}$; equal to $\dim(\mathcal{V}/\mathcal{U})$ in finite dimension
$\bigwedge_{i\in I}\mathcal{U}_i,\ \bigvee_{i\in I}\mathcal{U}_i$	meet / join of a family of subspaces
$\sum_{i\in I}\mathcal{U}_i$	sum of an indexed family of subspaces; equivalently the join
$\mathcal{U}+\mathcal{W},\ \mathcal{U}\oplus\mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$\mathcal{V}_1\times\cdots\times\mathcal{V}_m$	product vector space with componentwise operations
$\prod_{i\in I}\mathcal{V}_i,\ \bigoplus_{i\in I}^{\text{ext}}\mathcal{V}_i$	external direct product / external direct sum of finite-support families
$\mathcal{V}\cong\mathcal{W},\ \mathbf{e}_{j,k}$	isomorphic vector spaces / basis vector supported in one product component
$\mathbf{v}+\mathcal{U},\ \pi(\mathbf{v})$	coset modulo $\mathcal{U}$ / locally declared quotient map $\mathcal{V}\to\mathcal{V}/\mathcal{U}$

Linear Maps and Duality

$T:\mathcal{V}\to\mathcal{W},\ \mathcal{L}(\mathcal{V},\mathcal{W})$	linear map / vector space of linear maps; distinct from measure-theoretic $\mathcal{L}^n$, least-squares $\mathrm{LS}(A,\mathbf{y})$, and local loss notation $\mathcal{L}(\theta)$
$T_A:\mathbb{F}^n\to\mathbb{F}^m$	linear map $\mathbf{x}\mapsto A\mathbf{x}$ induced by $A\in\mathrm{Mat}(m,n,\mathbb{F})$
$ST,\ T\vert_{\mathcal{U}},\ T/\mathcal{U}$	composition $S\circ T$ / restriction to $\mathcal{U}$ / induced quotient operator
$T^{-1},\ T^m,\ p(T)$	inverse of an invertible map or operator / operator power / polynomial functional calculus
$\ker(T),\ \operatorname{range}(T)$	kernel and range of T
$\operatorname{rank}(T),\ \operatorname{nullity}(T)$	dimensions of the range and kernel of a finite-domain linear map T
$\rho_{\mathcal{U},\mathcal{W}},\ \rho\perp\sigma$	projection onto $\mathcal{U}$ along $\mathcal{W}$ / algebraically orthogonal idempotents satisfying $\rho\sigma=\sigma\rho=0$; geometric orthogonality is listed separately below
$\mathcal{V}_{\mathbb{C}},\ \mathrm{cpx},\ T_{\mathbb{C}}$	complexification of a real vector space / canonical embedding $\mathbf{v}\mapsto\mathbf{v}+i\mathbf{0}$ / complexification of a real operator
$\operatorname{graph}(T),\ [S,T]$	graph of a map / commutator $ST-TS$
$\mathrm{Comm}(\mathcal{F}),\ \mathrm{Comm}(A)$	commutant of a family / commutant of a single matrix or operator
$A\sim_{\text{row}}B,\ S\sim T$	row-equivalent matrices / operator similarity when declared; distinct from cardinal, a.e., and coefficient equivalence
$\mathcal{V}',\ \beta',\ T'$	dual space, dual basis, and dual map; prime denotes duality in this context
$\varphi,\psi,\ \mathcal{U}^0$	linear functionals / annihilator
$\mathcal{V}'',\ \Lambda_{\mathcal{V}}$	double dual / natural evaluation map $\mathcal{V}\to\mathcal{V}''$
$[B]_{\beta,\gamma},\ \Phi_B,\ \Psi_B$	matrix of a bilinear pairing in ordered bases / induced maps $\mathcal{V}\to\mathcal{W}'$ and $\mathcal{W}\to\mathcal{V}'$ in a pairing context; module coordinate maps redeclare Φ_B locally
$[\mathbf{v}]_\beta,\ [T]_{\gamma\leftarrow\beta}$	coordinate vector in basis β / matrix of T from β-coordinates to γ-coordinates
$C_{\gamma\leftarrow\beta}$	change-of-basis matrix sending β-coordinate columns to γ-coordinate columns

Matrices and Entrywise Structure

$A,B,C,\ \mathrm{Mat}(m,n,\mathbb{F})$	matrix variables / space of $m\times n$ matrices over $\mathbb{F}$; decorated forms such as C_f and C_{ij} have separate meanings
$A_{ij},\ A_{j,\cdot},\ A_{\cdot,k}$	entry in row i and column j / jth row / kth column
$[a_{ij}],\ E_{j,k},\ A\mathbf{x}$	matrix with entries a_{ij} / matrix unit / matrix-vector product
$A^{-1},\ \mathrm{Mat}(n,n,\mathbb{F})^\times$	inverse matrix when it exists / set of invertible $n\times n$ matrices
$A^t,\ A^T,\ A^*$	transpose / alternate transpose notation / conjugate transpose or adjoint
$\operatorname{diag}(\lambda_1,\dots,\lambda_n)$	diagonal matrix with diagonal entries $\lambda_1,\dots,\lambda_n$
$\overline{z},\ \operatorname{Re}z,\ \operatorname{Im}z$	complex conjugate / real part / imaginary part; for sets, overline denotes closure
$\operatorname{tr}(A),\det(A)$	trace and determinant
$\operatorname{adj}(A)$	adjugate matrix
$M_{ij},\ C_{ij},\ A_j(\mathbf{b})$	minor matrix / cofactor / matrix obtained from A by replacing column j with $\mathbf{b}$
$\operatorname{col}(A),\ \operatorname{row}(A)$	column space and row space of A
$\operatorname{rank}(A),\ \operatorname{nullity}(A)$	rank and nullity of A
$A\oplus B,\ A\otimes B$	matrix direct sum / Kronecker product; vector-space direct sums use subspace notation
$A\otimes I_q+I_n\otimes B$	Kronecker sum of square matrices A and B in compatible dimensions
$A\circ B,\ \operatorname{vec}(A)$	Hadamard product / column-stacking vectorisation
$M/A,\ M/D$	Schur complement of the block A or D in a compatible block matrix M; distinct from quotient-space notation
$B,\ L=B^tB$	directed incidence matrix / associated incidence Laplacian in a graph context

Inner Product Geometry

$\|\mathbf{v}\|$, $\|T\|_{\mathrm{op}}$	vector norm / operator norm
$\|\cdot\|_1$, $\|\cdot\|_2$, $\|\cdot\|_\infty$	coordinate one-norm / Euclidean or spectral two-norm according to argument type / sup norm
$\|\cdot\|_F$	Frobenius norm
$\langle \mathbf{u}, \mathbf{v}\rangle$, $\langle A, B\rangle_F$	inner product / Frobenius inner product
$\mathbf{u} \perp \mathbf{v}$, $\mathcal{U}^\perp$, δ_{ij}	orthogonal vectors, orthogonal complement, Kronecker delta
$\mathrm{proj}_{\mathbf{v}}(\mathbf{u})$, $P_{\mathcal{U}}$	orthogonal projection of $\mathbf{u}$ onto $\mathrm{span}(\mathbf{v})$ / orthogonal projection onto $\mathcal{U}$
$\mathrm{LS}(A, \mathbf{y})$	least-squares solution set $\arg\min_{\mathbf{x}\in\mathbb{F}^n} \|A\mathbf{x} - \mathbf{y}\|_2$ for $A \in \mathrm{Mat}(m, n, \mathbb{F})$; roman notation avoids collision with $\mathcal{L}$ and $\mathcal{L}^n$
$d(\mathbf{v}, \mathcal{U})$	distance $\inf_{\mathbf{u}\in\mathcal{U}} \|\mathbf{v} - \mathbf{u}\|$ from $\mathbf{v}$ to $\mathcal{U}$
$\mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram matrix of $\mathbf{u}_1, \ldots, \mathbf{u}_n$
$\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram determinant
$P_{\mathbf{u}}$, $U_{\mathbf{w}}$	rank-one projection matrix / Householder matrix, with defining vector indicated by the subscript
$\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$	ordered basis; orthonormal when declared in an inner-product context
$\mathbf{e} = (1, \ldots, 1)^t$	all-ones vector
$\mathrm{Ray}_A(\mathbf{x})$, $\mathbf{x}^* A\mathbf{x}$	Rayleigh quotient notation; for unit $\mathbf{x}$, $\mathrm{Ray}_A(\mathbf{x}) = \mathbf{x}^* A\mathbf{x}$ / Hermitian quadratic form
$\mathrm{Sym}(B)$, $\mathrm{Skew}(B)$, $\mathrm{Alt}(B)$	predicates that the bilinear form B is symmetric, skew-symmetric, or alternating
$\mathrm{rad}_L(B)$, $\mathrm{rad}_R(B)$	left / right radical of a declared bilinear pairing B
$S^{\perp_R}$, ${}^{\perp_L}T$	right / left orthogonal of a subset under a declared bilinear pairing
M_β, $\Delta(\mathcal{V})$, $\mathrm{rad}(\mathcal{V})$	matrix of a square bilinear form in basis β / discriminant set or square class / radical of a metric vector space
$\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, $\mathcal{H}$, $w(\mathcal{V})$	orthogonal direct sum / hyperbolic space or plane in metric-vector-space contexts / Witt index; affine hyperplanes use $\mathcal{H}(\mathbf{N}, b)$
$\mathcal{O}(\mathcal{V})$, $\mathrm{Sp}(\mathcal{V})$, $\tau_{\mathbf{v},a}$, $\sigma_{\mathbf{u}}$	orthogonal group / symplectic group / symplectic transvection / orthogonal symmetry

Affine, Projective, and Convex Geometry

$\mathcal{A}(\mathcal{V})$, $\mathcal{A}_0(\mathcal{V})$, $\mathrm{Sub}(\mathcal{V})$	flats in $\mathcal{V}$ / $\mathcal{A}(\mathcal{V}) \cup \{\emptyset\}$ / subspaces of $\mathcal{V}$; $\mathcal{A}$ without arguments remains local
$\mathrm{affspan}(E)$, $\mathrm{affhull}(E)$	affine span / least flat containing E
$X \parallel Y$, $X \vee Y$	parallel flats / join of flats or projective subspaces; meet is intersection in the relevant lattice
$\mathbb{R}^n_+$, $\mathbb{R}^n_{++}$, $\mathbf{x} \geq \mathbf{0}$, $\mathbf{x} > \mathbf{0}$, $\mathbf{x} \gg \mathbf{0}$	nonzero nonnegative orthant / coordinatewise positive orthant / coordinatewise nonnegative, nonzero nonnegative, and coordinatewise positive inequalities
$\mathrm{conv}(S)$, Δ_{k-1}, C_A	convex hull / coefficient simplex $\{(t_i) : t_i \geq 0, \sum_i t_i = 1\}$ / nonnegative cone generated by the columns of A; distinct from cofactor and companion notation
$\mathcal{H}(\mathbf{N}, b)$, $\mathcal{H}_\pm(\mathbf{N}, b)$, $\langle \mathbf{N}, X\rangle < b$	affine hyperplane / closed half-spaces / setwise inequality, meaning $\langle \mathbf{N}, \mathbf{x}\rangle < b$ for all $\mathbf{x} \in X$

Spectral and Polynomial Structure

$E_\lambda(T)$, $\mathrm{spec}(T)$	eigenspace $\ker(T - \lambda I_{\mathcal{V}})$ / spectrum
$\mathrm{Inv}(T)$	lattice or set of T-invariant subspaces
$\lambda_i(A)$, $m_A(\lambda)$	eigenvalues in the locally declared order / algebraic multiplicity of λ as a root of p_A; distinct from measure notation $m(E)$
$\mathcal{G}_\lambda(T)$, $J_k(\lambda)$	generalized eigenspace for λ / Jordan block of size k and eigenvalue λ
p_A, q_T, χ_T	matrix characteristic polynomial / operator characteristic polynomial / characteristic polynomial determined by elementary divisors
μ_A, μ_T, $w_p(\lambda, A)$	minimal polynomial of a matrix or operator / Weyr count of λ-Jordan blocks of size at least p; distinct from measure notation $\mu(E)$
P_j, $T = \sum_j \mu_j P_j$	spectral projections / spectral decomposition with distinct eigenvalues μ_j
$\sqrt{T}$, $A^{1/2}$, $\rho(A)$	positive square root of a positive operator or matrix / spectral radius of A; distinct from the interval parameter in K_ρ
$\mathbb{F}[z]$	polynomials in z over $\mathbb{F}$
$\mathscr{P} = \mathbb{C}[x]$, $\mathscr{P}'$, $\mathscr{F} = \mathbb{C}[[t]]$	umbral polynomial algebra / algebraic dual / formal power-series algebra; script notation distinguishes these from $\mathcal{P}(X)$ and local families $\mathcal{F}$
$\mathcal{V}_T$, $\mathcal{C}_T(\mathbf{v})$, $\mathrm{ann}_T(\mathbf{v})$	$\mathbb{F}[z]$-module induced by T / T-cyclic subspace generated by $\mathbf{v}$ / annihilator ideal of $\mathbf{v}$
$m_{\mathbf{v}}$, C_f, $\mathrm{InvFact}(T)$, $\mathrm{ElemDiv}(T)$	order polynomial of $\mathbf{v}$ under T / companion matrix of f / invariant factors / elementary divisors
e^{tA}	matrix exponential $\sum_{k\geq 0} (tA)^k/k!$

Matrix Analysis and Numerical Structure

$A \succeq 0,\ A \succ 0$	Hermitian positive semidefinite / Hermitian positive definite
$\mathrm{in}(A),\ n_+(A), n_-(A), n_0(A)$	inertia of a Hermitian matrix / positive, negative, and zero eigenvalue counts
$A = QR,\ A = LL^*,\ A = LDL^t$	QR factorization / Cholesky factorization / symmetric or Hermitian LDL^t factorization
$A = V\Sigma W^*,\ A = X\Sigma_r Y^*$	singular value decomposition / compact SVD with rank r
$\sigma_j(A),\ \sigma_{\max}(A),\ \sigma_{\min}(A)$	singular values and extreme singular values
$\lvert T\rvert,\ \lvert A\rvert,\ \mathrm{vol}(\Omega)$	operator or matrix modulus / volume of a set; scalar absolute value remains $\lvert x\rvert$
$A^\dagger,\ A_k$	Moore–Penrose pseudoinverse / rank-k truncated SVD approximation in an SVD context
$\kappa_2(A),\ \Delta A,\ \Delta\mathbf{x}$	spectral condition number of an invertible matrix / matrix and solution perturbations
$\Delta(A)$	defect from normality; distinct from perturbation notation ΔA
$R_k(A),\ R_k^{\mathrm{off}}(A),\ R_{\max}(A)$	absolute row sum $\sum_j \lvert a_{kj}\rvert$ / deleted row sum $\sum_{j\neq k} \lvert a_{kj}\rvert$ / maximum absolute row sum
$\mathrm{Ger}_k(A),\ \mathrm{Ger}(A)$	kth Gersgorin disk / union of all Gersgorin disks
$\mathrm{tridiag}(a, b, c)$	tridiagonal matrix with subdiagonal a, diagonal b, and superdiagonal c
$w,\ F_n,\ Q_n,\ \omega_n$	half-bandwidth / unnormalised Fourier matrix / normalised Fourier matrix / primitive nth root used for the DFT
$\mathcal{K}_m(A, \mathbf{b})$	Krylov subspace $\mathrm{span}(\mathbf{b}, A\mathbf{b}, \ldots, A^{m-1}\mathbf{b})$
$Q_m,\ H_m$	Krylov basis matrix / projected Hessenberg or tridiagonal matrix
$A = S - T,\ B = S^{-1}T,\ \mathbf{e}_k$	matrix splitting / stationary iteration matrix / kth error vector
$A = D + L + U,\ B_J,\ B_{\mathrm{GS}}$	diagonal, lower, and upper splitting / Jacobi and Gauss–Seidel iteration matrices
$\omega,\ S_\omega,\ T_\omega,\ B_\omega$	SOR parameter / SOR splitting matrices / SOR iteration matrix; distinct from the differential form ω
$M,\ M^{-1}A,\ AM^{-1}$	preconditioner / left and right preconditioned matrices

Algebraic Structures Notation

Algebraic Language and Categories

$\mathcal{C},\ A \xrightarrow{f} B,\ \mathrm{id}_A$	ambient category / morphism $f : A \to B$ / identity morphism of A; $\mathcal{C}$ remains local unless declared
$\mathrm{Hom}_{\mathcal{C}}(A, B),\ g \circ f$	set of morphisms $A \to B$ in $\mathcal{C}$ / composition for compatible morphisms
$A \simeq B,\ F : \mathcal{C} \to \mathcal{D}$	isomorphic objects / functor with domain $\mathcal{C}$ and codomain $\mathcal{D}$

Groups, Actions, Rings, and Ideals

$G, H, K,\ e_G,\ \langle S\rangle$	groups or subgroups / identity element of G / substructure generated by S in the ambient algebraic category
$H \leq G,\ H \trianglelefteq G,\ G/H,\ [G : H]$	subgroup / normal subgroup / quotient group / subgroup index
$\mathrm{ord}(g),\ \mathrm{Aut}(G)$	order of an element / automorphism group
$R, S,\ 0_R,\ 1_R,\ R^\times$	rings / zero and identity elements of R / unit group of R
$Z(R),\ \mathrm{char}(R)$	center of R / characteristic of R
$I, J,\ \mathfrak{a}, \mathfrak{p},\ R/I$	ideals; fraktur letters denote distinguished ideals, such as prime ideals, when declared / quotient ring
$R[x],\ \mathbb{F}[z],\ \deg f$	polynomial rings in the displayed indeterminate / degree of a polynomial
$U_n,\ \Omega_n,\ \Phi_n(x),\ \varphi(n)$	complex nth roots of unity / primitive nth roots / nth cyclotomic polynomial / Euler totient; Ω_n is algebraic root notation, distinct from domain notation

Modules and Associative Algebras

$M, N,\ S \leq M,\ M/N$	modules / submodule S of M / quotient module
$\langle\langle S\rangle\rangle_R$	submodule generated by S over R; double brackets distinguish module generation from group and ideal generation
$\mathrm{Hom}_R(M, N),\ \mathrm{End}_R(M)$	set of R-module homomorphisms $M \to N$ / endomorphism ring of M
$\ker\tau,\ \mathrm{im}\,\tau,\ \overline{\tau}$	kernel / image / quotient-induced map $M/\ker\tau \to \mathrm{im}\,\tau$ for a homomorphism $\tau : M \to N$
$\bigoplus_i M_i,\ M = N \oplus H$	external direct sum of modules / internal direct-sum decomposition
$\mathcal{C}_S(N),\ \mathcal{L}_\sigma,\ \mathcal{R}_\sigma$	set $\{H \leq N : N = S \oplus H\}$ of complements / left inverses of σ / right inverses of σ; subscript distinguishes these from $\mathcal{L}$ and $\mathcal{L}^n$
$\mathcal{F}(\mathcal{V})$	finite-rank operators $\{T \in \mathcal{L}(\mathcal{V}) : \dim\mathrm{im}\,T < \infty\}$ in associative-algebra contexts; distinct from local function-family notation

Algebraic Geometry and Commutative Algebra Notation

Status. This part of the notes is still mostly scaffolded. The entries below reserve the notation promised by the section headings and keep it distinct from already-active algebra and analysis notation.

Commutative Algebra

$\mathrm{Spec}\,R,\ \mathfrak{p} \in \mathrm{Spec}\,R$	set of prime ideals of a commutative ring R / prime ideal $\mathfrak{p} \subseteq R$
$D(f),\ V(I)$	standard open set $\{\mathfrak{p} \in \mathrm{Spec}\,R : f \notin \mathfrak{p}\}$ / Zariski closed set $\{\mathfrak{p} : I \subseteq \mathfrak{p}\}$
$S^{-1}R,\ R_{\mathfrak{p}},\ k(\mathfrak{p})$	localization at a multiplicative set S / localization $(R\setminus\mathfrak{p})^{-1}R$ / residue field $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$
$\mathrm{rad}(I),\ \sqrt{I}$	radical ideal $\{r \in R : r^n \in I \text{ for some } n \geq 1\}$; use $\mathrm{rad}(\mathcal{V})$ for metric-vector-space radicals

Affine and Projective Geometry

$\mathbb{A}^n_k,\ \mathbb{P}^n_k$	affine n-space over a field k / projective n-space over k; geometric notation distinct from probability notation
$I(X),\ Z(I),\ k[X]$	vanishing ideal $\{f \in k[x_1,\dots,x_n] : f\vert_X = 0\}$ / zero locus of an ideal / coordinate ring, typically $k[x_1,\dots,x_n]/I(X)$
$f^{\#},\ \varphi^*$	coordinate-ring pullback, e.g. $f^{\#} : k[Y] \to k[X]$ for $f : X \to Y$; distinct from adjoints and differential-form pullbacks by context

Analysis and Continuous Structure Notation

Sets and Functions in Analysis

$\mathcal{P}(X)$	power set $\{A : A \subseteq X\}$; projection symbols use $P_{\mathcal{U}}$ or P_j
$J_n,\ \#E,\ A \sim_{\#} B$	finite initial segment $J_0 = \emptyset$, $J_n = \{1,\dots,n\}$ / cardinality of a finite set / existence of a bijection $A \to B$

Order and Completeness

$\vert x\vert,\ \vert a-b\vert$	real absolute value or complex modulus / usual real-line distance $d(a,b) = \vert a-b\vert$
$\overline{\mathbb{R}}$	extended real line $\mathbb{R} \cup \{-\infty,+\infty\}$ ordered by $-\infty < x < +\infty$ for $x \in \mathbb{R}$; the bar is conventional, not topological closure
$x < y,\ x \le y$	strict order / non-strict order in the ambient ordered set

Sequences and Series

$(a_n),\ a_n \to a$	sequence indexed by $n \in \mathbb{N}$ / convergence of the sequence to a
$(a_{n_k}),\ n_1 < n_2 < \cdots$	subsequence indexed by a strictly increasing sequence (n_k) of positive integers
$s_n = \sum_{k=1}^{n} a_k$	partial-sum sequence associated to $\sum a_k$
$\limsup a_n = s^*,\ \liminf a_n = s_*$	upper and lower limits in $\overline{\mathbb{R}}$, equivalently $\inf_N \sup_{n\ge N} a_n$ and $\sup_N \inf_{n\ge N} a_n$ for real sequences
$\sum_{n=1}^{\infty} a_n,\ \prod_{n=1}^{\infty} a_n$	limit of partial sums / limit of finite partial products when defined

Metric and Topological Notation

$d(x,y),\ B_r(x)$	metric distance / open ball $\{y : d(x,y) < r\}$
τ_d	metric topology induced by the metric d
$B_\varepsilon(a) = V_\varepsilon(a)$	real-line open ball $(a-\varepsilon, a+\varepsilon)$; $V_\varepsilon(a)$ is Abbott's notation
$\overline{B}_r(x),\ \{G_\alpha\}_{\alpha\in A}$	closed ball $\{y : d(x,y) \le r\}$ / indexed open cover
$d_E,\ d_{\max},\ d_{\text{sum}}$	Euclidean, maximum, and sum product metrics on a product space
$E^\circ,\ \overline{E},\ E^{\text{acc}}$	interior / topological closure / set of limit points of E
$\text{bd}(E),\ E^c$	boundary $\overline{E} \setminus E^\circ$ / complement in the ambient space
$\text{diam}\, E$	diameter $\sup\{d(p,q) : p,q \in E\}$ of a nonempty metric set
$X \simeq Y$	homeomorphic topological spaces, i.e. related by a homeomorphism
$A \cap \overline{B} = \emptyset,\ \overline{A} \cap B = \emptyset$	definition of separated subsets A and B
$F_\sigma,\ G_\delta,$ meager/thin/first category	countable union of closed sets / countable intersection of open sets / subset of a countable union of nowhere-dense sets
residual/thick, generic	set containing a countable intersection of dense open sets / property whose truth set is residual

Functions, Differentiation, and Integration

$\lim_{x\to p} f(x),\ f(x+),\ f(x-)$	limit of f at p / right and left limits at x
$C(X),\ C([a,b]),\ C_b(X),\ \Vert f\Vert_\infty$	continuous functions on X and $[a,b]$ / bounded continuous functions / supremum norm
$f_n \to f,\ f_n \rightrightarrows f$	pointwise convergence / uniform convergence on the ambient domain
$f'(x),\ Df(a),\ D_j f_i,\ \partial_i f,\ J_f(x)$	one-variable derivative, total derivative at a, component partial $D_j f_i$, scalar partial, and Jacobian determinant; J_f is distinct from J_n and $J_k(\lambda)$
$\nabla f,\ D_{\mathbf{u}} f,\ C^r,\ C^\infty,\ C^\omega$	gradient, directional derivative, and classes C^r, C^∞, C^ω
$\Pi = \{x_0,\dots,x_n\},\ c_i,\ \Delta x_i,\ \Delta\alpha_i$	partition of $[a,b]$, tags $c_i \in [x_{i-1},x_i]$, and increments $\Delta x_i = x_i - x_{i-1}$, $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$
$R(f,\Pi,\{c_i\}),\ \delta,\ \delta(x)$	$R(f,\Pi,\{c_i\}) = \sum_{i=1}^n f(c_i)\Delta x_i$ / constant mesh bound / gauge; uppercase $R(\cdots)$ is local to Riemann sums
$U(\Pi,f,\alpha),\ L(\Pi,f,\alpha)$	Riemann–Stieltjes upper and lower sums with integrator α
$f \in \mathcal{R},\ f \in \mathcal{R}(\alpha)$	ordinary Riemann integrability / Riemann–Stieltjes integrability with respect to α
$\int_a^b f,\ \int_a^b f\,d\alpha,\ \int_E f\,d\mu$	ordinary Riemann integral / Riemann–Stieltjes integral / measure integral
$\int_a^\infty f,\ \int_{-\infty}^\infty f$	one-sided and two-sided improper integrals as limits over finite truncations
$\ell(\Pi,\gamma),\ \ell(\gamma)$	polygonal length of γ along Π / curve length
$I^k,\ Q^k,\ \text{supp}(f)$	k-cell, standard k-simplex, and topological support $\overline{\{x : f(x) \ne 0\}}$; I^k is not an identity operator
$dx_I,\ dx_I \wedge dx_J$	basic form indexed by I / wedge product of basic forms
$\omega,\ d\omega,\ T^*\omega,\ \partial_{\text{ch}}\Psi$	differential form, exterior derivative, pullback by T, and chain boundary
$\int_\Psi \omega,\ \partial_{\text{ch}}[\mathbf{p}_0,\dots,\mathbf{p}_k]$	integral of ω over the chain Ψ / oriented boundary $\sum_{j=0}^k (-1)^j[\mathbf{p}_0,\dots,\widehat{\mathbf{p}_j},\dots,\mathbf{p}_k]$
$dV,\ dA,\ ds$	volume, area, and arc-length elements

Power Series, Fourier, and Special Functions

$\sum_{n\geq 0} c_n(x-a)^n,\ R$	power series centered at a / radius of convergence in this context, distinct from $R(\cdots)$ and the R in QR
$I_R,\ K_\rho$	centered interval $I_R = (-R, R)$ / compact subinterval $K_\rho = [-\rho, \rho] \subset I_R$ for $0 < \rho < R$
$\widehat{f}(m),\ s_N(f;x),\ s_N(f)$	$\widehat{f}(m) = (2\pi)^{-1}\int_{-\pi}^{\pi} f(x)e^{-imx}\,dx$ / $s_N(f;x) = \sum_{\lvert m\rvert\leq N}\widehat{f}(m)e^{imx}$, viewed pointwise or as a function
$f \sim_{\text{coef}} \sum \widehat{f}(n)e^{inx},\ f \sim_{\text{coef}} \sum c_n\phi_n$	formal Fourier or orthonormal coefficient expansion; no convergence assertion is implied
$D_N(t)$	Dirichlet kernel $D_N(t) = \sum_{m=-N}^{N} e^{imt}$; uppercase D_N is not derivative notation
$\mathsf{E},\ \mathsf{L},\ \mathsf{C},\ \mathsf{S}$	power-series exponential, inverse logarithm, cosine, and sine
$\Gamma(x),\ \mathrm{B}(x,y)$	gamma function / beta function in roman type, distinct from balls $B_r(x)$ and variables B

Measure and Function Spaces

$E \in \mathcal{M},\ E \in \mathcal{M}(\mu)$	E measurable in the ambient σ-algebra / Rudin measurable set generated by μ
$\mathcal{B}(\mathbb{R}^n),\ \mathcal{L}^n$	Borel σ-algebra on $\mathbb{R}^n$ / Lebesgue measurable subsets of $\mathbb{R}^n$; not a basis $\mathcal{B}$, a linear-map space $\mathcal{L}$, or a least-squares set LS
$\mathcal{E},\ \mathcal{M}_F(\mu),\ A\triangle B,\ d_\mu(A,B)$	elementary sets, finitely μ-measurable sets, symmetric difference, and $d_\mu(A,B) = \mu^*(A\triangle B)$; $\mathcal{M}$ is measure-theoretic here
$\mu^*(E),\ m^*(E)$	μ-outer measure / Lebesgue outer measure
$\mathrm{vol}_n(B),\ m_n^*(\Omega)$	n-dimensional box volume / box-cover outer measure $\inf\{\sum_{j\in J}\mathrm{vol}_n(B_j) : (B_j)_{j\in J} \text{ covers } \Omega,\ J \text{ at most countable}\}$
$m(E),\ \mu(E)$	Lebesgue or elementary volume of E / measure of E; not algebraic multiplicity $m_A(\lambda)$ or minimal polynomial μ_A
$\lambda^+,\ \lambda^-,\ \lvert\lambda\rvert,\ \lambda \ll \mu,\ \lambda \perp \mu,\ \frac{d\lambda}{d\mu}$	positive variation, negative variation, total variation, absolute continuity, mutual singularity, and Radon–Nikodym derivative
$\bowtie,\ \{f \bowtie a\}$	relation in $\{>, \geq, <, \leq\}$ / level set $\{x : f(x) \bowtie a\}$
$f^+,\ f^-,\ \mathcal{S}^+(f),\ \mathcal{I}_E(s)$	$f^+ = \max(f, 0)$, $f^- = -\min(f, 0)$ / simple minorants of f / simple-function integral on E
$f \sim_{\text{a.e. }} g \text{ on } E,$ a.e.	$\mu(\{x \in E : f(x) \neq g(x)\}) = 0$ / outside a null set
$E_x,\ E^y,\ f_x,\ f^y$	product slices $E_x = \{y : (x,y) \in E\}$, $E^y = \{x : (x,y) \in E\}$, and $f_x(y) = f^y(x) = f(x,y)$
$\mathcal{U}f,\ F_a$	undergraph $\{(x,t) : 0 < t < f(x)\}$ for $f \geq 0$ / superlevel set $F_a = \{x : f(x) > a\}$; $\mathcal{U}f$ is not a vector subspace
$[f : A],\ [B : Q],\ \mathrm{dp}(E)$	$[f : A] = m(A)^{-1}\int_A f$ / $[B : Q] = m(B \cap Q)/m(Q)$ / density-point set of E
$\mathscr{L}^1(\mu),\ \mathscr{L}^1_{\text{loc}}(\mathbb{R}^n)$	measurable f with $\int \lvert f\rvert\,d\mu < +\infty$ / locally integrable functions; script $\mathscr{L}$ is distinct from $\mathcal{L}^n$ and $\mathcal{L}$
$L^p(\mu),\ L^p(E),\ \lVert f\rVert_p$	Lebesgue space modulo a.e. equality with respect to μ or the ambient measure on E / $\lVert f\rVert_p = (\int \lvert f\rvert^p\,d\mu)^{1/p}$

Probability and Stochastic Structure Notation

Probability Spaces and Distributions

$(\Omega, \mathcal{F}, \mathbb{P}),\ \Pr(A)$	probability space / event probability $\Pr(A) = \mathbb{P}(A)$ for $A \in \mathcal{F}$; explicit probability notation keeps $\mathbb{P}_k^n$ and $\mathbb{F}_0$ unambiguous
$X, Y,\ X_*\mathbb{P},\ \mathcal{L}(X)$	random variables / pushforward law $X_*\mathbb{P}(B) = \mathbb{P}(X^{-1}(B))$ / law of X when locally declared; not Lebesgue $\mathcal{L}^n$
$X \sim F,\ X \overset{\text{dist}}{=} Y,\ X_1, \ldots, X_n$ iid	X has distribution F / X and Y have the same distribution / independent and identically distributed sample
$F(t),\ g(t),\ \Phi(t)$	distribution function $F(t) = \Pr(X \leq t)$ / density g / standard normal distribution function in probabilistic contexts
$\mathbb{E}[X],\ \mathrm{Var}(X),\ \mathrm{Cov}(X,Y)$	expectation / $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$ / $\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, when finite
$\mathrm{Corr}(X,Y),\ \rho,\ \sigma_{XY}$	correlation coefficient / local correlation parameter / covariance shorthand $\sigma_{XY} = \mathrm{Cov}(X,Y)$ when declared
$\psi_X(t)$	moment-generating function $\psi_X(t) = \mathbb{E}e^{tX}$ on its domain of finiteness
$X \perp\!\!\!\perp Y$	independence of random variables, equivalently independence of $\sigma(X)$ and $\sigma(Y)$ when defined
$X \perp\!\!\!\perp Y \mid Z$	conditional independence of X and Y given Z
$X_n \xrightarrow{\mathbb{P}} X,\ X_n \rightsquigarrow X,\ X_n \xrightarrow{L^p} X,\ X_n \xrightarrow{\text{a.s.}} X,\ X_n \xrightarrow{qm} X$	convergence in probability / convergence in distribution / convergence in L^p / almost-sure convergence / convergence in quadratic mean

Stochastic Processes and Markov Structure

$(X_t)_{t\in T},\ \mathcal{F}_t$	stochastic process indexed by T / filtration $(\mathcal{F}_t)$ when locally declared; this $\mathcal{F}_t$ is not the umbral algebra $\mathscr{F}$
$\mathbf{x},\ \mathbf{e},\ A\mathbf{e} = \mathbf{e}$	probability vector $\mathbf{x} \geq 0$, $\sum_i x_i = 1$ / all-ones vector $\mathbf{e} = (1, \ldots, 1)^t$ / row-stochastic Markov convention $A = [a_{ij}] \geq 0$, $A\mathbf{e} = \mathbf{e}$
$\pi,\ \pi A = \pi,\ A^p$	stationary distribution written as a row probability vector / stationarity equation / p-step Markov matrix
$P_t,\ p_t(x,y)$	transition semigroup / transition probability, kernel, or density from x to y, when introduced

Numerical and Computational Mathematics Notation

Status. Apart from numerical linear algebra, already covered under Linear and Operator Structures, the numerical and computational chapters remain placeholders. The entries below reserve notation only; they do not introduce PDE or machine-learning conventions.

Floating Point, Approximation, and Error

$\mathrm{fl}(x),\ u,\ \varepsilon_{\mathrm{mach}}$	floating-point rounded value, unit roundoff, and machine epsilon
$x_h,\ e_h,\ \mathcal{O}(h^p)$	approximation at discretization scale h, corresponding error, and Landau order as $h \to 0$
$\widehat{x},\ \Delta x,\ r = b - A\widehat{x}$	computed approximation, perturbation or error in x, and residual for $Ax = b$

Monte Carlo and Stochastic Computation

$X_1, \ldots, X_N,\ \widehat{\mu}_N$	samples / sample-mean estimator $\widehat{\mu}_N = N^{-1} \sum_{i=1}^{N} X_i$ when estimating $\mu = \mathbb{E}X$
$\mathrm{Var}(\widehat{\mu}_N),\ \mathrm{SE}(\widehat{\mu}_N)$	estimator variance / standard error $\mathrm{SE}(\widehat{\mu}_N) = \sqrt{\mathrm{Var}(\widehat{\mu}_N)}$
$w_i,\ \widehat{I}_N$	importance weights / Monte Carlo estimate of an integral or expectation, when locally introduced

Part I
Set Theory, Logic, and Foundations

Foundational prerequisites collect the logical, set-theoretic, functional, order-theoretic, cardinal, and research-practice language used throughout the notes.

1 Sets

Sets provide the membership language used throughout the notes: elements, subsets, products, power sets, operations, and indexed families.

1.1 Introduction to Sets

Definition 1.1.1. *Sets and Membership*
Sets and the membership relation are taken as primitive notions.

(i) *Membership*: Given an object x and a set E, the assertion $x \in E$ means that x is an element of E; its negation is written $x \notin E$.
(ii) *Empty set*: The symbol $\emptyset$ denotes a set satisfying $x \notin \emptyset$ for every object x.

Definition 1.1.2. *Set-Builder Notation*
For a set X and a condition $P(x)$ on its elements, $\{x \in X : P(x)\}$ denotes the set of all $x \in X$ satisfying $P(x)$.

1.2 Cartesian Products

Definition 1.2.1. *Ordered Pairs and Cartesian Products*
Coordinate order is part of both constructions.

(i) *Ordered pair*: For objects a, b, (a, b) has first coordinate a and second coordinate b.
(ii) *Equality*: For objects a, b, a', b', $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$.
(iii) *Cartesian product*: For sets X and Y, $X \times Y := \{(x, y) : x \in X,\ y \in Y\}$.

For intervals A and B, the Cartesian product $A \times B$ has a rectangular geometric representation.

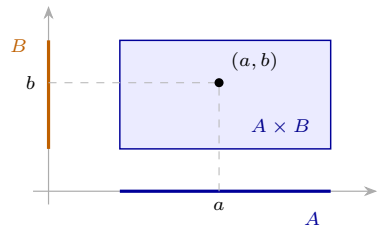

Figure 1: The product region $A \times B$ for horizontal interval A and vertical interval B.

Figure 1 illustrates Definition 1.2.1: $(a, b) \in A \times B$ precisely when $a \in A$ and $b \in B$, in that order.

1.3 Subsets and Power Sets

Definition 1.3.1. *Basic Subset Terminology*
Let A, B be sets.

(i) *Subset*: $A \subseteq B$ if and only if $x \in A$ implies $x \in B$ for every object x.
(ii) *Proper subset*: $A \subsetneq B$ iff $A \subseteq B$ and $A \neq B$.
(iii) *Singleton*: $A = \{x\}$ for some object x.
(iv) *Nonempty*: $A \neq \emptyset$.

Assumption 1.3.2. *Set Extensionality*
Throughout these notes, set equality is governed by extensionality.
For sets A and B, $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proof Method 1.3.3. *Proof by Two Containments*
To prove equality of sets A and B, discharge the following obligations.

(i) Fix an arbitrary $x \in A$ and prove $x \in B$; this establishes $A \subseteq B$.
(ii) Fix an arbitrary $x \in B$ and prove $x \in A$; this establishes $B \subseteq A$.
(iii) Invoke Assumption 1.3.2 to conclude $A = B$.

If both sets have explicit membership criteria, the two inclusions may be proved together.
Fix an arbitrary object x and prove $x \in A \iff x \in B$.

Definition 1.3.4. *Power Set*
For a set X, its *power set* is $\mathcal{P}(X) := \{S : S \subseteq X\}$. The two smallest cases are as follows.

(i) The empty set has only itself as a subset, so $\mathcal{P}(\emptyset) = \{\emptyset\}$.
(ii) If $X = \{a, b\}$ with $a \neq b$, then $\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, X\}$.

1.4 Union, Intersection, Difference, and Complement

Definition 1.4.1. *Elementary Set Operations*
Let X be a set and let $A, B \subseteq X$.

(i) *Union*: $A \cup B := \{x \in X : x \in A \text{ or } x \in B\}$.
(ii) *Intersection*: $A \cap B := \{x \in X : x \in A \text{ and } x \in B\}$.
(iii) *Difference*: $A \setminus B := \{x \in X : x \in A \text{ and } x \notin B\}$.
(iv) *Complement*: $A^c := X \setminus A$, relative to the ambient set X.

Definition 1.4.2. *Symmetric Difference*
Let X be a set and let $A, B \subseteq X$. The *symmetric difference* is $A \triangle B := (A \setminus B) \cup (B \setminus A)$.
It consists exactly of the elements belonging to one of A and B, but not both.

Proposition 1.4.3. *Basic Set Algebra*
Let X be a set and let $A, B, C \subseteq X$. Then

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
(ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
(iii) $(A \cup B)^c = A^c \cap B^c$;
(iv) $(A \cap B)^c = A^c \cup B^c$;
(v) $A \triangle B = (A \cup B) \setminus (A \cap B)$.

Proof. Fix an arbitrary $x \in X$. By set extensionality, each identity follows once its two membership conditions are equivalent.

(i) Membership in $A \cap (B \cup C)$ means that $x \in A$ and that either $x \in B$ or $x \in C$. This holds exactly when either $x \in A \cap B$ or $x \in A \cap C$. Hence it is equivalent to membership in $(A \cap B) \cup (A \cap C)$.
(ii) Membership in $A \cup (B \cap C)$ means that $x \in A$, or that both $x \in B$ and $x \in C$. This holds exactly when two conditions are satisfied. First, $x \in A$ or $x \in B$; second, $x \in A$ or $x \in C$. Hence it is equivalent to membership in $(A \cup B) \cap (A \cup C)$.
(iii) The condition $x \in (A \cup B)^c$ is equivalent to $x \notin A \cup B$. This means that $x \notin A$ and $x \notin B$, or equivalently $x \in A^c \cap B^c$.
(iv) The condition $x \in (A \cap B)^c$ is equivalent to $x \notin A \cap B$. This means that $x \notin A$ or $x \notin B$, or equivalently $x \in A^c \cup B^c$.
(v) The condition $x \in A \triangle B$ is equivalent to $x \in A \setminus B$ or $x \in B \setminus A$. Thus x belongs to one of A and B, but not both. Equivalently, $x \in A \cup B$ and $x \notin A \cap B$. By the definition of difference, this is equivalent to $x \in (A \cup B) \setminus (A \cap B)$.

Structural Commentary: Each identity reduces to an equivalence of membership conditions.
Key Insight: Membership turns set operations into logical connectives. □

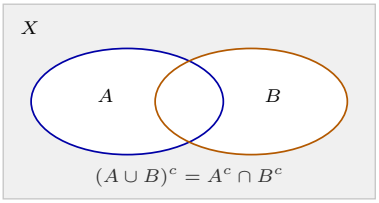

Figure 2: The shaded region represents $(A \cup B)^c = A^c \cap B^c$ inside the ambient set X.

Figure 2 illustrates part (iii). The shaded region lies outside both A and B, so it is outside $A \cup B$.

1.5 Indexed Families and Products

Definition 1.5.1. *Families, Indexed Families, Unions, and Intersections*
Let X be a set and let I be an index set.

(i) *Family*: A set whose members are sets, also called a *collection* of sets.
(ii) *Indexed family*: An assignment $i \mapsto A_i$ over I, written $(A_i)_{i \in I}$ or $\{A_i\}_{i \in I}$. It is an indexed family in X iff $(\forall i \in I)\, A_i \subseteq X$.
(iii) *Indexed union*: For an indexed family $\{A_i\}_{i \in I}$ in X, $\bigcup_{i \in I} A_i := \{x \in X : (\exists i \in I)\, x \in A_i\}$.
(iv) *Indexed intersection*: For an indexed family $\{A_i\}_{i \in I}$ in X, $\bigcap_{i \in I} A_i := \{x \in X : (\forall i \in I)\, x \in A_i\}$.

Theorem 1.5.2. *Arbitrary De Morgan Laws*
Let X and I be sets, and let $\{E_i\}_{i \in I}$ be an indexed family in X. Take all complements relative to X.

(i) $\left(\bigcup_{i \in I} E_i\right)^c = \bigcap_{i \in I} E_i^c$.
(ii) $\left(\bigcap_{i \in I} E_i\right)^c = \bigcup_{i \in I} E_i^c$.

Proof. Fix an arbitrary $x \in X$. Extensionality reduces both identities to pointwise membership equivalences.

(i) $x \in \left(\bigcup_{i \in I} E_i\right)^c \iff (\forall i \in I)\, x \notin E_i \iff (\forall i \in I)\, x \in E_i^c \iff x \in \bigcap_{i \in I} E_i^c$.
(ii) $x \in \left(\bigcap_{i \in I} E_i\right)^c \iff (\exists i \in I)\, x \notin E_i \iff (\exists i \in I)\, x \in E_i^c \iff x \in \bigcup_{i \in I} E_i^c$.

Structural Commentary: The proof checks both identities pointwise and applies quantifier negation.
Key Insight: Complement exchanges existential and universal membership conditions. □

Definition 1.5.3. *Disjoint Families and Parametrizations*
Let I be an index set and let $\mathcal{F}$ be a family of sets. Let $\{A_i\}_{i\in I}$ be an indexed family of sets.

(i) *Pairwise disjoint*: $(\forall E, G \in \mathcal{F})\,(E \neq G \implies E \cap G = \emptyset)$.
(ii) *Parametrization*: $\{A_i\}_{i\in I}$ parametrizes $\mathcal{F}$ iff $\mathcal{F} = \{A_i : i \in I\}$.

Definition 1.5.4. *Cartesian Product of an Indexed Family*
Let I be an index set and let $\{E_i\}_{i\in I}$ be an indexed family of sets.

(i) *Product*: $\prod_{i\in I} E_i := \{(x_i)_{i\in I} : (\forall i \in I)\, x_i \in E_i\}$.
(ii) *Binary case*: If $I = \{1, 2\}$, identify $(x_i)_{i\in I}$ with the ordered pair (x_1, x_2). Under this identification, $\prod_{i\in I} E_i = E_1 \times E_2$.
(iii) *Constant family*: If E is a set and $(\forall i \in I)\, E_i = E$, write $E^I := \prod_{i\in I} E_i$. Thus E^I is the set of I-indexed families in E.
(iv) *Dependence*: Different parametrizations of the same family can yield different products.

1.6 Russell's Paradox and Axioms

Theorem 1.6.1. *Russell's Paradox*
For a condition $P(x)$ on sets, unrestricted comprehension asserts $(\exists R)(\forall x)\,[x \in R \iff P(x)]$.

(i) No set R satisfies $(\forall x)\,[x \in R \iff x \notin x]$.
(ii) The unrestricted comprehension schema is inconsistent.

Proof.

(i) Suppose that such a set R exists. Setting $x = R$ gives $R \in R \iff R \notin R$. If $R \in R$, then $R \notin R$; if $R \notin R$, then $R \in R$. Both cases are contradictory, so no such set exists.
(ii) Take $P(x) := x \notin x$. The corresponding instance asserts $(\exists R)(\forall x)\,[x \in R \iff x \notin x]$. This contradicts (i), proving (ii).

Structural Commentary: Specialize at $x = R$, then apply the matching comprehension instance.
Key Insight: Self-membership makes the defining condition assert its own negation. □

Perspective 1.6.2. *From Comprehension to Axioms*
The obstruction in Theorem 1.6.1 explains the need for controlled set formation.
Separation forms $\{x \in B : P(x)\}$ only after a bounding set B is available. The standard ZFC axioms have the following informal roles.

(i) *Extensionality* makes membership determine set equality.
(ii) *Separation* extracts a condition-defined subset from an existing bounding set.
(iii) *Pairing* supplies $\{a, b\}$ from given sets a and b, including $\{a\}$ when $a = b$.
(iv) *Union* collects into one set the elements of the members of a given set.
(v) *Power set* supplies $\mathcal{P}(A)$, the set of all subsets of a given set A.
(vi) *Infinity* supplies an inductive set from which the natural numbers are constructed.
(vii) *Replacement* forms the image of a set under a definable functional relation.
(viii) *Foundation* requires nonempty sets to contain an $\in$-minimal member.
(ix) *Choice* supplies a choice function on a set-indexed family of nonempty sets.

This perspective is a functional map, not a formal axiom list or a source of proof authority.
Formal statements and dependencies belong to the advanced set-theory section.

2 Logic and Proofs

Logic makes explicit the statement grammar already used informally in the elementary set development and fixes it for the definitions, theorems, and proofs that follow.

2.1 Statements and Well-Formed Language

Definition 2.1.1. *Mathematical Statements and Truth Values*
The *truth values* used here are *true* and *false*.
A *mathematical statement* is a declarative sentence having exactly one truth value.

Principle 2.1.2. *Well-Formed Statements*
A mathematical expression may serve as a statement only when all the following conditions hold.

(i) *Symbols*: All symbols have declared syntactic roles.
(ii) *Meaning*: All constituent expressions have defined meanings.
(iii) *Scope*: All variable occurrences have unambiguous scope.

Failure of any condition makes the expression ill formed rather than false; no truth value is assigned.

2.2 Connectives and Negation

Definition 2.2.1. *Truth-Functional Connectives*
A connective is *truth functional* when its output depends only on input truth values. Let a and b be mathematical statements.

(i) *Negation*: $\neg a$ is true iff a is false.
(ii) *Conjunction*: $a \wedge b$ is true iff both a and b are true.
(iii) *Disjunction*: $a \vee b$ is true iff at least one of a and b is true; both may be true.
(iv) *Implication*: $a \Rightarrow b$ is false iff a is true and b is false. It is true in the remaining cases.
(v) *Equivalence*: $a \iff b$ is true iff a and b have the same truth value.

Proposition 2.2.2. *Negating Truth-Functional Connectives*
For mathematical statements a and b, the following equivalences hold.

(i) $\neg(a \wedge b) \iff (\neg a \vee \neg b)$.

(ii) $\neg(a \vee b) \iff (\neg a \wedge \neg b)$.
(iii) $\neg(a \Rightarrow b) \iff (a \wedge \neg b)$.

Proof. By Definition 2.2.1, the three equivalences reduce to the following truth conditions.

(i) The statement $\neg(a \wedge b)$ is true iff $a \wedge b$ is false. This occurs iff at least one of a and b is false, precisely when $\neg a \vee \neg b$ is true.
(ii) The statement $\neg(a \vee b)$ is true iff $a \vee b$ is false. This occurs iff both a and b are false, precisely when $\neg a \wedge \neg b$ is true.
(iii) The statement $\neg(a \Rightarrow b)$ is true iff $a \Rightarrow b$ is false. This occurs iff a is true and b is false, precisely when $a \wedge \neg b$ is true.

Thus each pair has the same truth value, proving all three equivalences.
Structural Commentary: Each item reduces both sides to one constituent truth-value condition.
Key Insight: Negation selects the cases excluded by the original connective. □

2.3 Quantifiers and Bound Variables

Definition 2.3.1. *Quantifiers, Bound Variables, and Scope*
Let X be a set and let $P(x)$ be a condition on its elements.

(i) *Universal quantifier*: $(\forall x \in X)\, P(x)$ means $\{x \in X : \neg P(x)\} = \emptyset$.
(ii) *Existential quantifier*: $(\exists x \in X)\, P(x)$ means $\{x \in X : P(x)\} \neq \emptyset$.
(iii) *Binding and scope*: A binding declaration governs a region, called the variable's *scope*.
(iv) *Bound occurrence*: An occurrence governed by a binding declaration is *bound*.
(v) *Free occurrence*: An occurrence outside every binding declaration is *free*.
(vi) *Standard binders*: Quantifiers, summation signs, integrals, and set-builder notation bind variables.
(vii) *Renaming*: A bound variable may be renamed consistently within its scope to a fresh symbol.
The statement remains unchanged provided no free occurrence is captured.

Proposition 2.3.2. *Quantifier Order Can Change Truth Value*
There exist sets X, Y and a condition $P(x, y)$ on $X \times Y$ such that

(i) $(\forall x \in X)(\exists y \in Y)\, P(x, y)$ *is true;*
(ii) $(\exists y \in Y)(\forall x \in X)\, P(x, y)$ *is false.*

Consequently, the two quantifier orders are not logically equivalent.

Proof. Take $X = Y = \{0, 1\}$ and let $P(x, y)$ be the condition $x = y$.

(i) The choice $y = x$ verifies $(\forall x \in X)(\exists y \in Y)\, P(x, y)$.
(ii) If $(\exists y \in Y)(\forall x \in X)\, P(x, y)$ were true, choose its witness $y \in Y$. Then $P(0, y) \wedge P(1, y)$, so $y = 0$ and $y = 1$, contradicting $0 \neq 1$.

Thus (i) and (ii) hold, so the quantifier orders are not logically equivalent.
Structural Commentary: The proof separates the orders using equality on a two-point set.
Key Insight: Pointwise witnesses exist, but no single witness works uniformly. □

Remark 2.3.3. *Index Variables*
Let X and I be sets, and let $\{A_i\}_{i \in I}$ be an indexed family in X. For $x \in X$, $x \in \bigcup_{i \in I} A_i \iff (\exists i \in I)\, x \in A_i$.
Likewise, $x \in \bigcap_{i \in I} A_i \iff (\forall i \in I)\, x \in A_i$. The indexed operator binds i in both formulas.
The index may be renamed consistently to any fresh symbol without changing either formula.

2.4 Proof Grammar

Proof Method 2.4.1. *Proof Grammar*
Given a mathematical claim, expose its proof obligations before selecting a proof rule.

(i) *Objects*: Identify the ambient sets or structures and declare the role of each symbol.
(ii) *Logical form*: Separate the hypotheses from the exact conclusion.
(iii) *Quantifiers*: Record the quantifiers in order. Note the permitted dependence of witnesses.
By Proposition 2.3.2, quantifier order can change a statement's truth value.
(iv) *Well-formedness*: Verify that every expression has a defined meaning.
Check that every variable occurrence has unambiguous scope. Principle 2.1.2 records this requirement.

A completed parse specifies the data, hypotheses, quantifier dependencies, and exact conclusion.
An undefined expression is ill formed rather than false and must be repaired before a proof begins.

2.5 Proof by Contradiction

Definition 2.5.1. *Contradiction*
A *contradiction* is a mathematical statement whose truth conditions cannot be satisfied.
For a mathematical statement c, the expression $c \wedge \neg c$ is the canonical contradiction form.
By Definition 2.2.1, its two conjuncts cannot be true simultaneously.

Proof Method 2.5.2. *Proof by Contradiction*
Let a and b be mathematical statements. To prove $a \Rightarrow b$ by classical contradiction, discharge the following obligations.

(i) *Temporary assumption*: Assume $\neg(a \Rightarrow b)$. By Proposition 2.2.2, this is equivalent to assuming $a \wedge \neg b$.
(ii) *Contradiction*: Identify a mathematical statement c. Derive both c and $\neg c$ from the hypotheses and temporary assumption.
(iii) *Conclusion*: Since the temporary assumption yields a contradiction, reject it and conclude $a \Rightarrow b$.

Justify each inference by the hypotheses, definitions, earlier results, or temporary assumption.

2.6 Mathematical Induction and Well-Ordering

Assumption 2.6.1. *Well-Ordering of* $\mathbb{N}$
Throughout these notes, $(\forall S \subseteq \mathbb{N})\,[S \neq \emptyset \Rightarrow (\exists m \in S)(\forall n \in S)\, m \leq n]$.

Theorem 2.6.2. *Mathematical Induction*
Let $(P_n)_{n\in\mathbb{N}}$ be mathematical statements. Suppose the following conditions hold.

(i) Basis: P_1 is true.
(ii) Induction step: $(\forall n \in \mathbb{N})\,(P_n \Rightarrow P_{n+1})$.

Then $(\forall n \in \mathbb{N})\, P_n$.

Proof. Let $F := \{n \in \mathbb{N} : \neg P_n\}$ and suppose $F \neq \emptyset$. By the standing well-ordering assumption, choose the least $m \in F$.
By (i), $1 \notin F$, so $m \neq 1$. Hence $m-1 \in \mathbb{N}$ and $m-1 < m$. Minimality gives $m-1 \notin F$, so P_{m-1} is true.
Applying (ii) at $n = m-1$ gives P_m, contradicting $m \in F$. Thus $F = \emptyset$, so $(\forall n \in \mathbb{N})\, P_n$.
Structural Commentary: Well-ordering supports a least-counterexample contradiction.
Key Insight: A least failure has a smaller successful predecessor. □

3 Functions

Functions assign unique outputs; their images, preimages, restrictions, compositions, and inverses organize nearly every later construction.

3.1 Functions, Images, and Preimages

Definition 3.1.1. *Functions, Images, and Preimages*
Let X, Y be sets. A *function* $f : X \to Y$ assigns a unique value $f(x) \in Y$ to each $x \in X$.

(i) $X = \mathrm{dom}(f)$ is the *domain*, and Y is the *codomain*.
(ii) $\mathrm{range}(f) := f(X) := \{f(x) : x \in X\}$ is the *range* of f.
(iii) If $A \subseteq X$, then $f(A) := \{f(x) : x \in A\}$ is the *image* of A.
(iv) If $B \subseteq Y$, then $f^{-1}(B) := \{x \in X : f(x) \in B\}$ is the *preimage* of B; no inverse function is required.

Assumption 3.1.2. *Function Extensionality*
Let X, Y be sets and let $f, g : X \to Y$. If $(\forall x \in X)\, f(x) = g(x)$, then $f = g$.

Proposition 3.1.3. *Image and Preimage Calculus*
Let X, Y, I be sets with $I \neq \emptyset$, and let $f : X \to Y$. Let $\{A_i\}_{i\in I}$ and $\{B_i\}_{i\in I}$ be indexed families in X and Y, respectively.

(i) $f\left(\bigcup_{i\in I} A_i\right) = \bigcup_{i\in I} f(A_i)$.
(ii) $f\left(\bigcap_{i\in I} A_i\right) \subseteq \bigcap_{i\in I} f(A_i)$.
(iii) $f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i)$.
(iv) $f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i)$.
(v) $(\forall B \subseteq Y)\, f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$.
(vi) $(\forall B, C \subseteq Y)\, f^{-1}(B \setminus C) = f^{-1}(B) \setminus f^{-1}(C)$.
(vii) $(\forall B \subseteq Y)\, f(f^{-1}(B)) = B \cap \mathrm{range}(f)$.
(viii) $(\forall A \subseteq X)(\forall B \subseteq Y)\, [f(A) \subseteq B \iff A \subseteq f^{-1}(B)]$.

Proof.

(i) Fix $y \in Y$. Then $y \in f\left(\bigcup_{i\in I} A_i\right) \iff (\exists i \in I)(\exists x \in A_i)\, y = f(x)$. This is equivalent to $y \in \bigcup_{i\in I} f(A_i)$.

(ii) If $y \in f\left(\bigcap_{i\in I} A_i\right)$, choose $x \in \bigcap_{i\in I} A_i$ with $y = f(x)$. Then $(\forall i \in I)\, x \in A_i$, so $(\forall i \in I)\, y \in f(A_i)$ and $y \in \bigcap_{i\in I} f(A_i)$.

(iii) Fix $x \in X$. Then $x \in f^{-1}\left(\bigcup_{i\in I} B_i\right) \iff (\exists i \in I)\, f(x) \in B_i$. This is equivalent to $(\exists i \in I)\, x \in f^{-1}(B_i)$, hence to $x \in \bigcup_{i\in I} f^{-1}(B_i)$.

(iv) Fix $x \in X$. Then $x \in f^{-1}\left(\bigcap_{i\in I} B_i\right) \iff (\forall i \in I)\, f(x) \in B_i$. This is equivalent to $(\forall i \in I)\, x \in f^{-1}(B_i)$, hence to $x \in \bigcap_{i\in I} f^{-1}(B_i)$.

(v) Fix $B \subseteq Y$ and $x \in X$. Then $x \in f^{-1}(Y \setminus B) \iff f(x) \notin B \iff x \notin f^{-1}(B) \iff x \in X \setminus f^{-1}(B)$.

(vi) Fix $B, C \subseteq Y$ and $x \in X$. Then $x \in f^{-1}(B \setminus C) \iff [f(x) \in B \wedge f(x) \notin C]$. This is equivalent to $x \in f^{-1}(B) \setminus f^{-1}(C)$.

(vii) Fix $B \subseteq Y$ and $y \in Y$. Then $y \in f(f^{-1}(B)) \iff (\exists x \in X)\, [f(x) \in B \wedge y = f(x)]$. This is equivalent to $y \in B \cap \mathrm{range}(f)$.

(viii) ($\Rightarrow$) Fix $A \subseteq X$ and $B \subseteq Y$. Suppose $f(A) \subseteq B$ and fix $x \in A$. Then $f(x) \in B$, so $x \in f^{-1}(B)$; hence $A \subseteq f^{-1}(B)$.

(viii) ($\Leftarrow$) Fix $A \subseteq X$ and $B \subseteq Y$. Suppose $A \subseteq f^{-1}(B)$ and fix $y \in f(A)$. Choose $x \in A$ with $y = f(x)$.
Then $x \in f^{-1}(B)$, so $y = f(x) \in B$ and $f(A) \subseteq B$.

Structural Commentary: Membership chains prove the identities; direct containments prove the adjunction.
Key Insight: Preimages translate codomain membership into domain membership. □

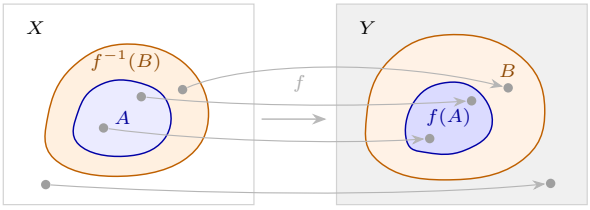

Figure 3: Image–preimage adjunction: $f(A) \subseteq B \iff A \subseteq f^{-1}(B)$.

Figure 3 illustrates Proposition 3.1.3(viii). The labeled regions show that $f(A) \subseteq B$ exactly when $A \subseteq f^{-1}(B)$.

Definition 3.1.4. *Algebraic Support of a Function*
Let $f : X \to Y$ and let $0 \in Y$ be distinguished. The *algebraic support* of f is $\operatorname{supp}(f) := \{x \in X : f(x) \neq 0\}$.

3.2 Injective, Surjective, and Bijective Maps

Definition 3.2.1. *Injective, Surjective, and Bijective Maps*
Let X, Y be sets and let $f : X \to Y$.

(i) *Injective*: $(\forall x_1, x_2 \in X)\,[f(x_1) = f(x_2) \implies x_1 = x_2]$.
(ii) *Surjective*: $\operatorname{range}(f) = Y$.
(iii) *Bijective*: f is both injective and surjective.

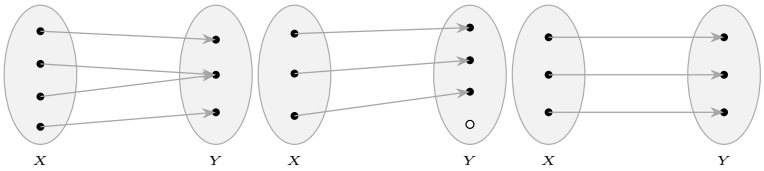

Figure 4: Left: surjective with a collision; center: injective but not surjective; right: bijective.

Figure 4 contrasts the three conditions in Definition 3.2.1. The left map is surjective but not injective, whereas the center map is injective but not surjective. The right map is bijective.

Example 3.2.2. *Image and Preimage for the Square Map*
Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := x^2$. If $x \in [-2, 1]$, then $0 \leq f(x) \leq 4$. Conversely, if $y \in [0, 4]$, then $-\sqrt{y} \in [-2, 1]$ and $f(-\sqrt{y}) = y$.
Hence $f([-2, 1]) = [0, 4]$. For $x \in \mathbb{R}$, $x \in f^{-1}([1, 4]) \iff 1 \leq x^2 \leq 4 \iff 1 \leq |x| \leq 2 \iff x \in [-2, -1] \cup [1, 2]$.
Since $f(-1) = f(1)$ and $-1 \neq 1$, the map is not injective. Since $(\forall x \in \mathbb{R})\, f(x) \geq 0$, one has $-1 \notin \operatorname{range}(f)$, so the map is not surjective.
Key Insight: Repeated values obstruct injectivity; omitted codomain values obstruct surjectivity.

3.3 Composition, Restriction, and Inverse Functions

Definition 3.3.1. *Composition, Restriction, Extension, and Identity*
Let X, Y be sets and let $f : X \to Y$.

(i) *Restriction*: If $A \subseteq X$, then $f|_A : A \to Y$ satisfies $(\forall x \in A)\,(f|_A)(x) = f(x)$.
(ii) *Extension*: If U is a set with $X \subseteq U$, an extension of f is a map $\widetilde{f} : U \to Y$ satisfying $\widetilde{f}|_X = f$.
(iii) *Composition*: If C, Z are sets, $g : C \to Z$, and $f(X) \subseteq C$, then $g \circ f : X \to Z$ satisfies $(\forall x \in X)\,(g \circ f)(x) = g(f(x))$.
(iv) *Identity*: The map $\mathrm{id}_X : X \to X$ satisfies $(\forall x \in X)\,\mathrm{id}_X(x) = x$.

Theorem 3.3.2. *Bijective Means Invertible*
Let X, Y be sets and let $f : X \to Y$.

(i) The map f is bijective iff $(\exists g : Y \to X)\,[g \circ f = \mathrm{id}_X \wedge f \circ g = \mathrm{id}_Y]$.
(ii) If $g, h : Y \to X$ both satisfy the two identities in (i), then $g = h$. When it exists, this map is denoted $f^{-1} : Y \to X$.

Proof.

(i) ($\Rightarrow$) Suppose f is bijective. For $y \in Y$, choose $x_y \in X$ with $f(x_y) = y$ by surjectivity; it is unique by injectivity.
Define $g : Y \to X$ by $g(y) := x_y$. For $x \in X$, $g(f(x)) = x$, and for $y \in Y$, $f(g(y)) = y$.
Thus $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$ by function extensionality.
(i) ($\Leftarrow$) Suppose $g : Y \to X$ satisfies the two identities. If $x_1, x_2 \in X$ and $f(x_1) = f(x_2)$, then $x_1 = g(f(x_1)) = g(f(x_2)) = x_2$.
Hence f is injective. For $y \in Y$, $y = f(g(y)) \in \operatorname{range}(f)$, so f is surjective and hence bijective.
(ii) Suppose $g, h : Y \to X$ satisfy the two identities. For $y \in Y$, $g(y) = g(f(h(y))) = h(y)$. Thus $g = h$ by function extensionality.

Structural Commentary: Construction proves existence; cancellation recovers bijectivity and proves uniqueness.
Key Insight: A bijection pairs each output with exactly one input. □

Proposition 3.3.3. *Composition and Restriction Calculus*
Let X, Y, Z, W be sets, and let $f : X \to Y$, $g : Y \to Z$, and $h : Z \to W$.

(i) $h \circ (g \circ f) = (h \circ g) \circ f$.
(ii) $\mathrm{id}_Y \circ f = f$ and $f \circ \mathrm{id}_X = f$.
(iii) $(\forall A \subseteq X)\,(g \circ f)(A) = g(f(A))$.
(iv) $(\forall C \subseteq Z)\,(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$.
(v) $(\forall A \subseteq X)\,(g \circ f)|_A = g \circ (f|_A)$.
(vi) If f and g are bijective, then $g \circ f$ is bijective.

(vii) If f and g are bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof.

(i) For $x \in X$, both sides evaluate to $h(g(f(x)))$. Hence the two maps are equal by function extensionality.
(ii) For $x \in X$, $(\mathrm{id}_Y \circ f)(x) = f(x) = (f \circ \mathrm{id}_X)(x)$. Hence both identities hold by function extensionality.
(iii) Fix $A \subseteq X$ and $z \in Z$. Then $z \in (g \circ f)(A) \iff (\exists x \in A)\, z = g(f(x)) \iff (\exists y \in f(A))\, z = g(y) \iff z \in g(f(A))$.
(iv) Fix $C \subseteq Z$ and $x \in X$. Then $x \in (g \circ f)^{-1}(C) \iff g(f(x)) \in C \iff f(x) \in g^{-1}(C) \iff x \in f^{-1}(g^{-1}(C))$.
(v) Fix $A \subseteq X$ and $x \in A$. Then $((g \circ f)|_A)(x) = g(f(x)) = (g \circ (f|_A))(x)$, so the restrictions are equal by function extensionality.
(vi) Suppose f and g are bijective. Fix $x_1, x_2 \in X$ and suppose $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then $f(x_1) = f(x_2)$ by the injectivity of g. Hence $x_1 = x_2$ by the injectivity of f, so $g \circ f$ is injective.
For $z \in Z$, choose $y \in Y$ with $g(y) = z$ by the surjectivity of g, then choose $x \in X$ with $f(x) = y$ by the surjectivity of f. Thus $(g \circ f)(x) = z$, so $g \circ f$ is surjective and therefore bijective.
(vii) Suppose f and g are bijective. The inverse functions $f^{-1} : Y \to X$ and $g^{-1} : Z \to Y$ exist by Theorem 3.3.2(i).
For $x \in X$, $(f^{-1} \circ g^{-1})((g \circ f)(x)) = f^{-1}(g^{-1}(g(f(x)))) = f^{-1}(f(x)) = x$.
For $z \in Z$, $(g \circ f)((f^{-1} \circ g^{-1})(z)) = g(f(f^{-1}(g^{-1}(z)))) = g(g^{-1}(z)) = z$. Thus $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ by Theorem 3.3.2(ii).

Structural Commentary: Pointwise evaluation proves the calculus; injectivity and surjectivity pass through the factors.
Key Insight: Inverse maps cancel compositions in reverse order. □

4 Counting and Finite Sums

Counting records finite constructions before cardinality and infinite sets are treated abstractly.

4.1 Finite Sums

Definition 4.1.1. *Finite Sums Indexed by Integers*
Let $\mathbb{F}$ be a field and let $(a_i)_{i\in\mathbb{Z}}$ be a family in $\mathbb{F}$. Fix $m, n \in \mathbb{Z}$. If $m \le n$, define $\sum_{i=m}^{n} a_i := a_m + a_{m+1} + \cdots + a_n$.

If $m > n$, set $\sum_{i=m}^{n} a_i := 0$; this is the empty sum.

Proposition 4.1.2. *Basic Laws for Finite Sums*
Let $\mathbb{F}$ be a field. Let $(a_i)_{i\in\mathbb{Z}}$ and $(b_i)_{i\in\mathbb{Z}}$ be families in $\mathbb{F}$. Fix $m, n \in \mathbb{Z}$ with $m \le n$.

(i) Splitting: Fix $r \in \mathbb{Z}$ with $m \le r < n$. Then $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) Additivity: $\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i$.

(iii) Homogeneity: For $c \in \mathbb{F}$, $\sum_{i=m}^{n} ca_i = c \sum_{i=m}^{n} a_i$.

(iv) Triangle inequality: If $\mathbb{F} = \mathbb{C}$, then $\left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n} |a_i|$.

(v) Monotonicity: Suppose $\mathbb{F} = \mathbb{R}$ and $(\forall i \in \mathbb{Z})\,[m \le i \le n \Rightarrow a_i \le b_i]$. Then $\sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i$.

Proof. For (ii)–(v), fix m and set $\ell := n - m + 1$; by Theorem 2.6.2 with induction parameter ℓ, it suffices in each item to verify the case $n = m$ and the step from n to $n + 1$.

(i) By Definition 4.1.1 and associativity of addition in $\mathbb{F}$, $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) If $n = m$, both sides equal $a_m + b_m$. Assume additivity through n. By (i), $\sum_{i=m}^{n+1} (a_i + b_i) = \sum_{i=m}^{n} (a_i + b_i) + a_{n+1} + b_{n+1}$.
Substitute the induction hypothesis and regroup to obtain $\sum_{i=m}^{n+1} a_i + \sum_{i=m}^{n+1} b_i$.

(iii) If $n = m$, both sides equal ca_m. Assume homogeneity through n. By (i), $\sum_{i=m}^{n+1} ca_i = \sum_{i=m}^{n} ca_i + ca_{n+1}$.
The induction hypothesis and distributivity turn the right side into $c(\sum_{i=m}^{n} a_i + a_{n+1})$. By (i), this equals $c \sum_{i=m}^{n+1} a_i$.

(iv) If $n = m$, equality holds. Assume the inequality through n. By (i) and the triangle inequality, $\left|\sum_{i=m}^{n+1} a_i\right| \le \left|\sum_{i=m}^{n} a_i\right| + |a_{n+1}|$.
The induction hypothesis and (i) bound the right side by $\sum_{i=m}^{n+1} |a_i|$.

(v) If $n = m$, the conclusion is the hypothesis $a_m \le b_m$. Assume monotonicity through n. By (i), the sums through $n + 1$ split at n.
The induction hypothesis, order compatibility, and $a_{n+1} \le b_{n+1}$ give $\sum_{i=m}^{n+1} a_i \le \sum_{i=m}^{n+1} b_i$.

Structural Commentary: Splitting reduces each law to one new summand and the scalar axioms.
Key Insight: Finite-sum laws propagate one term at a time. □

4.2 Lists and Counting Principles

Counting represents finite constructions by explicit lists and converts disjoint alternatives or successive choices into cardinality identities.

Definition 4.2.1. *Finite Sets and Standard Index Sets*
Set $J_0 := \emptyset$ and $J_n := \{1, \ldots, n\}$ for $n \in \mathbb{N}$. A set A is *finite* if it admits a bijection $e : J_n \to A$ with $n \in \mathbb{N}_0$.

Proposition 4.2.2. *Finite Cardinality Is Well Defined*
Let A be a set and let $m, n \in \mathbb{N}_0$. If $e : J_m \to A$ and $g : J_n \to A$ are bijections, then $m = n$.
Thus each finite set A admits a unique $n \in \mathbb{N}_0$ such that a bijection $J_n \to A$ exists. Define the finite cardinality of A by $\#A := n$.

Proof. The composite $g^{-1} \circ e : J_m \to J_n$ is bijective.
It therefore suffices to prove that a bijection between two standard index sets forces their indices to agree.
For $r \in \mathbb{N}$, let P_r assert that a bijection $J_{r-1} \to J_k$ with $k \in \mathbb{N}_0$ forces $k = r - 1$. It suffices to prove P_1 and $(\forall r \in \mathbb{N})\,(P_r \Rightarrow P_{r+1})$.
A bijection $J_0 \to J_k$ forces $J_k = \emptyset$, hence $k = 0$; thus P_1 holds. Assume P_r and let $\varphi : J_r \to J_k$ be bijective. Since $J_r \neq \emptyset$, $k \in \mathbb{N}$.
Put $q := \varphi(r)$. The restriction $\varphi|_{J_{r-1}}$ is a bijection $J_{r-1} \to J_k \setminus \{q\}$.
Define $\rho : J_{k-1} \to J_k \setminus \{q\}$ by $\rho(i) := i$ when $i < q$ and $\rho(i) := i + 1$ when $i \geq q$.
Its branch images J_{q-1} and $J_k \setminus J_q$ are disjoint and cover $J_k \setminus \{q\}$. Both branches are injective, so ρ is bijective.
The composite $\rho^{-1} \circ \varphi|_{J_{r-1}} : J_{r-1} \to J_{k-1}$ is therefore bijective. Thus $k - 1 = r - 1$ by P_r. Hence $k = r$, and P_{r+1} holds.
Consequently, $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2. Finally, $n = m$ by P_{m+1} with the bijection $g^{-1} \circ e$.
Structural Commentary: Induction deletes one matched point and canonically relists the remaining index set.
Key Insight: Deletion reduces both finite indices while preserving bijectivity. □

Definition 4.2.3. *Lists, Words, and Product Sets*
Let $p \in \mathbb{N}$ and let $S_1, \ldots, S_p$ be sets.

(i) A *list* of length p in $S_1 \times \cdots \times S_p$ is a tuple $s = (s_1, \ldots, s_p)$ in that product.
(ii) If $S_1 = \cdots = S_p = S$, a list in S^p is a *word* of length p over the alphabet S.
(iii) A counting model is *ordered* when its outcomes are lists and coordinate positions are retained.
Thus lists $s = (s_1, \ldots, s_p)$ and $t = (t_1, \ldots, t_p)$ are distinct iff $(\exists i \in J_p)\, s_i \neq t_i$.

Proposition 4.2.4. *Addition, Subtraction, and Division Principles*

(i) Addition: Let $r \in \mathbb{N}$ and let $A_1, \ldots, A_r$ be pairwise disjoint finite sets. Then $\bigcup_{i=1}^{r} A_i$ is finite and $\#\big(\bigcup_{i=1}^{r} A_i\big) = \sum_{i=1}^{r} \#A_i$.
(ii) Subtraction: If A is finite and $B \subseteq A$, then B and $A \setminus B$ are finite and $\#(A \setminus B) = \#A - \#B$.
(iii) Division: Let $f : Y \to X$ be a surjection between finite sets. Its fibers are finite.
If $d \in \mathbb{N}$ satisfies $(\forall x \in X)\, \#f^{-1}(\{x\}) = d$, then $\#Y = d\#X$ and $\#X = \frac{\#Y}{d}$.

Proof.

(i) For $k \in \mathbb{N}$, let P_k denote claim (i) for k sets. If $k = 1$, then $\bigcup_{i=1}^{1} A_i = A_1$ and $\sum_{i=1}^{1} \#A_i = \#A_1$, so P_1 holds.
Assume P_k and let $A_1, \ldots, A_{k+1}$ be pairwise disjoint finite sets. Put $U := \bigcup_{i=1}^{k} A_i$. By P_k, U is finite and $\#U = \sum_{i=1}^{k} \#A_i$.
Put $q := \#U$ and $n := \#A_{k+1}$, and choose bijections $e : J_q \to U$ and $g : J_n \to A_{k+1}$.
Define $h : J_{q+n} \to U \cup A_{k+1}$ as follows. Set $h(t) := e(t)$ when $t \in J_q$ and $h(t) := g(t - q)$ when $t \in J_{q+n} \setminus J_q$.
The two domain blocks partition J_{q+n}, and their images are the disjoint sets U and A_{k+1}.
Both restrictions of h are bijective, so h is bijective.
Therefore $U \cup A_{k+1}$ is finite with cardinality $q + n = \sum_{i=1}^{k+1} \#A_i$, so P_{k+1} holds. Hence $(\forall k \in \mathbb{N})\, P_k$ by Theorem 2.6.2.

(ii) For $r \in \mathbb{N}$, let P_r assert claim (ii) whenever $\#A = r - 1$ and $B \subseteq A$. If $r = 1$, then $A = B = \emptyset$, so P_1 holds.
Assume P_r, let $\#A = r$, and choose a bijection $e : J_r \to A$. Put $a := e(r)$ and $A' := A \setminus \{a\}$.
The restriction $e|_{J_{r-1}} : J_{r-1} \to A'$ is bijective, so $\#A' = r - 1$. If $a \in B$, put $B' := B \setminus \{a\}$.
By P_r, the sets B' and $A' \setminus B'$ are finite and $\#(A' \setminus B') = \#A' - \#B'$.
By part (i) for $B = B' \cup \{a\}$, B is finite with $\#B = \#B' + 1$.
Since $A \setminus B = A' \setminus B'$, the set $A \setminus B$ is finite and $\#(A \setminus B) = (r - 1) - (\#B - 1) = \#A - \#B$.
If $a \notin B$, then B and $A' \setminus B$ are finite with $\#(A' \setminus B) = \#A' - \#B$ by P_r.
By part (i) for $A \setminus B = (A' \setminus B) \cup \{a\}$, $\#(A \setminus B) = ((r - 1) - \#B) + 1 = \#A - \#B$.
Thus P_{r+1} holds. Consequently, $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2.
An arbitrary finite A and subset $B \subseteq A$ satisfy $P_{\#A+1}$, proving claim (ii).

(iii) Put $m := \#X$. If $m = 0$, then $X = \emptyset$ and hence $Y = \emptyset$ because $f : Y \to X$; both identities follow.
Suppose $m \in \mathbb{N}$, choose a bijection $e : J_m \to X$, and set $F_i := f^{-1}(\{e(i)\})$ for $i \in J_m$.
Each F_i is finite by part (ii). The sets $F_1, \ldots, F_m$ are pairwise disjoint with union Y.
The equality $(\forall i \in J_m)\, \#F_i = d$ holds by the uniform-fiber hypothesis.
The identity $\#Y = \sum_{i=1}^{m} \#F_i$ holds by part (i). Hence $\#Y = \sum_{i=1}^{m} d = md = d\#X$. Since $d \neq 0$, $\#X = \frac{\#Y}{d}$.

Structural Commentary: Induction constructs union and complement listings; fiber decomposition supplies division.
Key Insight: Disjoint blocks convert set decompositions into arithmetic. □

Theorem 4.2.5. *Multiplication Principle*
Let $p \in \mathbb{N}$ and let $n_1, \ldots, n_p \in \mathbb{N}_0$.

(i) Successive choices: Set $H_0 := \{\emptyset\}$.
(ii) At stage $i \in J_p$, let $(C_i(h))_{h \in H_{i-1}}$ be a family of finite sets satisfying $(\forall h \in H_{i-1})\, \#C_i(h) = n_i$.
Set $H_i := \{(h, c) : h \in H_{i-1},\ c \in C_i(h)\}$. Then H_p is finite and $\#H_p = \prod_{i=1}^{p} n_i$.
(iii) Finite products: Let $S_1, \ldots, S_p$ be finite sets satisfying $(\forall i \in J_p)\, \#S_i = n_i$.
Then $S_1 \times \cdots \times S_p$ is finite and $\#(S_1 \times \cdots \times S_p) = \prod_{i=1}^{p} n_i$.

Proof.

(i) For $r \in \mathbb{N}$, let P_r denote claim (i) with $p = r$. The map $(\emptyset, c) \mapsto c$ is a bijection $H_1 \to C_1(\emptyset)$.
Thus H_1 is finite with $\#H_1 = n_1$, so P_1 holds. Assume P_r and consider an $(r + 1)$-stage model.

By P_r for its first r stages, H_r is finite with $\#H_r = \prod_{i=1}^{r} n_i$.

If $H_r = \emptyset$, then $H_{r+1} = \emptyset$ and $\#H_{r+1} = 0 = n_{r+1}\#H_r = \prod_{i=1}^{r+1} n_i$, so P_{r+1} holds.

Suppose $H_r \neq \emptyset$. Put $m := \#H_r$; then $m \in \mathbb{N}$. Choose a bijection $e : J_m \to H_r$.
Define the family $(A_j)_{j\in J_m}$ by $A_j := \{e(j)\} \times C_{r+1}(e(j))$. The sets $A_1, \dots, A_m$ are pairwise disjoint and have union H_{r+1}.
The map $c \mapsto (e(j), c)$ is a bijection $C_{r+1}(e(j)) \to A_j$. Thus $A_1, \dots, A_m$ are finite and $(\forall j \in J_m)\, \#A_j = n_{r+1}$.

Therefore $\#H_{r+1} = \sum_{j=1}^{m} \#A_j = mn_{r+1} = \prod_{i=1}^{r+1} n_i$ by Proposition 4.2.4(i). Thus P_{r+1} holds.

Consequently, $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2; in particular, P_p holds.

(ii) In (i), take the constant families $C_i(h) := S_i$. Then H_p is finite with $\#H_p = \prod_{i=1}^{p} n_i$.

Given $s = (s_1, \dots, s_p) \in S_1 \times \cdots \times S_p$, set $h_0 := \emptyset$. At stage $i \in J_p$, set $h_i := (h_{i-1}, s_i)$, and define $\Psi(s) := h_p$.
The map $\Psi : S_1 \times \cdots \times S_p \to H_p$ is bijective. Indeed, successive first and second coordinates recover the unique tuple s.

Hence $S_1 \times \cdots \times S_p$ is finite. Its cardinality is $\#(S_1 \times \cdots \times S_p) = \#H_p = \prod_{i=1}^{p} n_i$.

Structural Commentary: Induction counts recursive histories; a coordinate bijection transfers the count to products.
Key Insight: Uniform extension fibers multiply the number of histories. □

Example 4.2.6. *Finite Words by the Multiplication Principle*
The following slot counts multiply by Theorem 4.2.5.

(i) A card in a standard deck is a pair consisting of one of 4 suits and one of 13 ranks. Thus the deck has $4 \cdot 13 = 52$ cards.
(ii) For $B := \{0, 1\}$, $\#(B^8) = 2^8 = 256$, so the binary alphabet has 256 words of length 8.
(iii) For $D := \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$, $(\forall N \in \mathbb{N})\, \#(D^N) = 4^N$. At $N = 3$, the DNA alphabet therefore has 64 codons.

Key Insight: Fixed-size coordinate choices multiply, so constant alphabets produce exponential word counts.

4.3 Permutations, Subsets, and Binomial Coefficients

Definition 4.3.1. *Permutations and Sign*
Let $n \in \mathbb{N}_0$. A *permutation* of J_n is a bijection $\sigma : J_n \to J_n$, and $\mathfrak{S}_n$ denotes the set of all such permutations.
Given $\sigma \in \mathfrak{S}_n$, set $\mathrm{Inv}(\sigma) := \{(j, k) \in J_n^2 : j < k,\ \sigma(j) > \sigma(k)\}$ and define its *sign* by $\mathrm{sign}(\sigma) := (-1)^{\#\,\mathrm{Inv}(\sigma)}$.

Definition 4.3.2. *Factorials and Binomial Coefficients*

(i) *Factorial*: Set $0! := 1$ and $n! := \prod_{j=1}^{n} j$ for $n \in \mathbb{N}$.

(ii) *Falling factorial*: Set $(n)_0 := 1$ for $n \in \mathbb{N}_0$. If $n, r \in \mathbb{N}$ and $r \leq n$, set $(n)_r := \prod_{j=0}^{r-1} (n - j)$.

(iii) *Binomial coefficient*: If $n, r \in \mathbb{N}_0$ and $r \leq n$, set $\binom{n}{r} := \frac{n!}{r!(n-r)!}$; this is read *n choose r*.

Proposition 4.3.3. *Ordered Sampling With and Without Replacement*
Let $n, r \in \mathbb{N}_0$ and let S be finite with $\#S = n$. Set $S^0 := \{()\}$ and use $x^0 := 1$ for $x \in \mathbb{N}_0$.
Define $D_r(S) := \{s = (s_1, \dots, s_r) \in S^r : (\forall i, j \in J_r)\, [i < j \Rightarrow s_i \neq s_j]\}$.

(i) With replacement: The ordered samples form S^r, and $\#(S^r) = n^r$.
(ii) Without replacement: If $r \leq n$, the ordered samples with distinct entries form $D_r(S)$, and $\#D_r(S) = (n)_r$.
(iii) Full orderings: $\#D_n(S) = n!$ and $\#\mathfrak{S}_n = n!$.

Proof.

(i) If $r = 0$, then $\#S^0 = 1 = n^0$. If $r \in \mathbb{N}$, then $\#S^r = \prod_{k=1}^{r} n = n^r$ by Theorem 4.2.5(ii).

(ii) Suppose $r \leq n$. If $r = 0$, then $D_0(S) = S^0$ and $\#D_0(S) = 1 = (n)_0$.
Suppose $r \in \mathbb{N}$. At stage $k \in J_r$, an injective prefix $(s_1, \dots, s_{k-1})$ uses a set of cardinality $k - 1$.
The available set $S \setminus \{s_1, \dots, s_{k-1}\}$ is finite with cardinality $n - k + 1$ by Proposition 4.2.4(ii).

Successive-choice histories flatten bijectively to $D_r(S)$, so $\#D_r(S) = \prod_{k=1}^{r} (n - k + 1) = (n)_r$ by Theorem 4.2.5(i).

(iii) The identity $\#D_n(S) = n!$ holds by (ii). Define $\Phi : \mathfrak{S}_n \to D_n(J_n)$ by $\Phi(\sigma) := (\sigma(i))_{i\in J_n}$.
Given $s \in D_n(J_n)$, the map $i \mapsto s_i$ is injective and its image has cardinality n. Its complement in J_n has cardinality 0 by Proposition 4.2.4(ii), so the map is surjective.
Thus $s \mapsto (i \mapsto s_i)$ is the inverse of Φ. Hence Φ is bijective and $\#\mathfrak{S}_n = \#D_n(J_n) = n!$.

Structural Commentary: Product counting handles all lists; available-set histories handle injective lists.
Key Insight: Removing used entries decreases each successive choice count by one. □

Proposition 4.3.4. *Unordered Samples Without Replacement*
Let $n, r \in \mathbb{N}_0$ with $r \leq n$, let S be finite with $\#S = n$, and set $\mathcal{P}_r(S) := \{A \subseteq S : \#A = r\}$.
Then $\mathcal{P}_r(S)$ is finite and $\#\mathcal{P}_r(S) = \binom{n}{r} = \frac{(n)_r}{r!} = \frac{n!}{(n-r)!\, r!}$.

Proof. Choose a bijection $e : J_n \to S$. Define $\chi : \mathcal{P}_r(S) \to \{0, 1\}^n$ by $\chi(A)_i := 1$ when $e(i) \in A$ and $\chi(A)_i := 0$ otherwise.
If $A \neq B$, choose $x \in (A \setminus B) \cup (B \setminus A)$ and then choose $i \in J_n$ with $e(i) = x$. Thus $\chi(A)_i \neq \chi(B)_i$, so χ is injective.
The set $\{0, 1\}^n$ is finite by Proposition 4.3.3(i). Its subset $\chi(\mathcal{P}_r(S))$ is finite by Proposition 4.2.4(ii).
Hence $\mathcal{P}_r(S)$ is finite because χ is a bijection onto its image. Define $\pi : D_r(S) \to \mathcal{P}_r(S)$ by $\pi(s_1, \dots, s_r) := \{s_1, \dots, s_r\}$.
Given $A \in \mathcal{P}_r(S)$, choose a bijection $f : J_r \to A$. Then $(f(i))_{i\in J_r} \in \pi^{-1}(\{A\})$, so π is surjective.
The fiber $\pi^{-1}(\{A\})$ is bijective with $D_r(A)$, so $\#\pi^{-1}(\{A\}) = r!$ by Proposition 4.3.3(iii).
The identity $\#D_r(S) = (n)_r$ holds by Proposition 4.3.3(ii). The equality $\#D_r(S) = r!\,\#\mathcal{P}_r(S)$ holds by Proposition 4.2.4(iii).
Therefore $\#\mathcal{P}_r(S) = \frac{(n)_r}{r!} = \frac{n!}{(n-r)!\, r!} = \binom{n}{r}$.

Structural Commentary: Binary encoding proves finiteness; uniform ordering fibers convert ordered samples to subsets.
Key Insight: Each subset has exactly $r!$ ordered listings. □

Example 4.3.5. *Committees*
Let C be a class of 20 people. The 3-person committees form $\mathcal{P}_3(C)$. Thus $\#\mathcal{P}_3(C) = \binom{20}{3} = \frac{20\cdot 19\cdot 18}{3\cdot 2\cdot 1} = 1140$ by Proposition 4.3.4.
Key Insight: A committee is an unordered subset, so permutations of its members are not recounted.

Proposition 4.3.6. *Edge Values and Symmetry*
Let $n, k \in \mathbb{N}_0$ with $k \le n$.

(i) Edge values: $\binom{n}{0} = \binom{n}{n} = 1$.
(ii) Symmetry: $\binom{n}{k} = \binom{n}{n-k}$.

Proof.

(i) Since $0! := 1$, $\binom{n}{0} = \frac{n!}{0!\,n!} = 1$ and $\binom{n}{n} = \frac{n!}{n!\,0!} = 1$.
(ii) Since $0 \le n-k \le n$, $\binom{n}{n-k} = \frac{n!}{(n-k)!\,k!} = \frac{n!}{k!\,(n-k)!} = \binom{n}{k}$.

Structural Commentary: Direct factorial simplification proves the edge values and symmetry.
Key Insight: The denominator is invariant under $k \leftrightarrow n-k$. □

Proposition 4.3.7. *Binomial Theorem and Subset Count*
Let $n \in \mathbb{N}$ and let $a, b \in \mathbb{C}$.

(i) Binomial theorem: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.

(ii) Subset count: If S is finite with $\#S = n$, then $\mathcal{P}(S)$ is finite and $\#\mathcal{P}(S) = 2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Proof.

(i) Write $(a+b)^n = \prod_{j=1}^{n} (a+b)$. A subset $I \subseteq J_n$ records the factors contributing a; the corresponding term is $a^{\#I} b^{n-\#I}$.
For $k \in \{0, \dots, n\}$, $\#\mathcal{P}_k(J_n) = \binom{n}{k}$ by Proposition 4.3.4. Grouping terms by $k = \#I$ proves the identity.
(ii) Fix a finite set S with $\#S = n$ and choose a bijection $e : J_n \to S$.
Define $\chi : \mathcal{P}(S) \to \{0,1\}^n$ by $\chi(A)_i := 1$ when $e(i) \in A$ and $\chi(A)_i := 0$ otherwise.
The inverse sends $v \in \{0,1\}^n$ to $\{e(i) : i \in J_n,\ v_i = 1\}$, so χ is bijective.
Hence $\mathcal{P}(S)$ is finite with $\#\mathcal{P}(S) = 2^n$ by Proposition 4.3.3(i).
The sets $\mathcal{P}_0(S), \dots, \mathcal{P}_n(S)$ are pairwise disjoint with union $\mathcal{P}(S)$. Hence $\#\mathcal{P}(S) = \sum_{k=0}^{n} \#\mathcal{P}_k(S)$ by Proposition 4.2.4(i).
For $k \in \{0, \dots, n\}$, $\#\mathcal{P}_k(S) = \binom{n}{k}$ by Proposition 4.3.4. Therefore $\#\mathcal{P}(S) = \sum_{k=0}^{n} \binom{n}{k}$.

Structural Commentary: Expansion and subset partitioning classify the same binary choices by cardinality.
Key Insight: Choosing the a-positions is equivalent to choosing a subset of J_n. □

Definition 4.3.8. *Multisets*
Let S be a set. A *multiset on S* is a pair (S_0, m) such that

(i) $S_0 \subseteq S$ is the *underlying set*;
(ii) $m : S_0 \to \mathbb{N}$ is the *multiplicity function*, and $m(s)$ is the *multiplicity* of $s \in S_0$;
(iii) if S_0 is finite, set $E_m := \{(s, j) : s \in S_0,\ j \in J_{m(s)}\}$ and define the *size* by $|m| := \#E_m$.

The graph $\{(s, m(s)) : s \in S_0\}$ is an equivalent record of (S_0, m).

Example 4.3.9. *Multiset Bookkeeping*
The list (a, a, b, b, b, c) determines the multiset (S_0, m) with $S_0 = \{a, b, c\}$, $m(a) = 2$, $m(b) = 3$, and $m(c) = 1$.
The set E_m has $2 + 3 + 1 = 6$ elements, so $|m| = 6$. The same bookkeeping records eigenvalues counted with algebraic multiplicity.
Key Insight: A multiset forgets order while retaining the number of occurrences of each element.

Definition 4.3.10. *Multinomial Coefficients*
Let $r \in \mathbb{N}$ and let $\mathbf{n} = (n_1, \dots, n_r) \in \mathbb{N}_0^r$. Set $n := \sum_{i=1}^{r} n_i$ and define $\binom{n}{n_1, \dots, n_r} := \frac{n!}{n_1! \cdots n_r!}$.

Proposition 4.3.11. *Multinomial Count and Expansion*
Let $r \in \mathbb{N}$, let $\mathbf{n} = (n_1, \dots, n_r) \in \mathbb{N}_0^r$, and let $x_1, \dots, x_r \in \mathbb{C}$. Set $n := \sum_{i=1}^{r} n_i$.

Let $\mathcal{M}_{\mathbf{n}}$ be the set of tuples $(A_1, \dots, A_r)$ of pairwise disjoint subsets of J_n satisfying $\bigcup_{i=1}^{r} A_i = J_n$ and $(\forall i \in J_r)\, \#A_i = n_i$.

Set $K_{r,n} := \{\mathbf{m} = (m_1, \dots, m_r) \in \mathbb{N}_0^r : \sum_{i=1}^{r} m_i = n\}$.

(i) Multinomial count: $\#\mathcal{M}_{\mathbf{n}} = \binom{n}{n_1, \dots, n_r} = \frac{n!}{n_1! \cdots n_r!}$.

(ii) Multinomial expansion: $(x_1 + \cdots + x_r)^n = \sum_{\mathbf{m} \in K_{r,n}} \binom{n}{m_1, \dots, m_r} \prod_{i=1}^{r} x_i^{m_i}$.

Proof.

(i) Set $N_0 := n$ and $N_j := n - \sum_{i=1}^{j} n_i$ for $j \in J_r$; thus $N_r = 0$. After $A_1, \dots, A_{j-1}$ are chosen, put $R_j := J_n \setminus \bigcup_{i=1}^{j-1} A_i$.
The equality $\#R_1 = N_0$ holds directly. If $j \ge 2$, then $\#R_j = N_{j-1}$ by Proposition 4.2.4(i)–(ii).
Since $N_{j-1} = \sum_{i=j}^{r} n_i \ge n_j$, the number of choices for $A_j \in \mathcal{P}_{n_j}(R_j)$ is $\binom{N_{j-1}}{n_j}$ by Proposition 4.3.4.

Since $N_r = 0$, the successive-choice histories are exactly the elements of $\mathcal{M}_{\mathbf{n}}$. Hence $\#\mathcal{M}_{\mathbf{n}} = \prod_{j=1}^{r} \binom{N_{j-1}}{n_j}$ by Theorem 4.2.5(i).

The factorial factors telescope, so $\#\mathcal{M}_{\mathbf{n}} = \prod_{j=1}^{r} \frac{N_{j-1}!}{n_j!\,N_j!} = \frac{n!}{n_1!\cdots n_r!} = \binom{n}{n_1,\ldots,n_r}$.

(ii) Expand $(x_1 + \cdots + x_r)^n = \prod_{k=1}^{n} (x_1 + \cdots + x_r)$. Each term is determined by a function $\phi : J_n \to J_r$ and equals $\prod_{k=1}^{n} x_{\phi(k)}$.
Set $m_i := \#\phi^{-1}(\{i\})$ for $i \in J_r$. Then $\mathbf{m} = (m_1, \ldots, m_r) \in K_{r,n}$. The fibers $(\phi^{-1}(\{1\}), \ldots, \phi^{-1}(\{r\}))$ form an element of $\mathcal{M}_{\mathbf{m}}$. Conversely, each element of $\mathcal{M}_{\mathbf{m}}$ determines a unique function with those fibers.
Thus the number of terms equal to $\prod_{i=1}^{r} x_i^{m_i}$ is $\binom{n}{m_1,\ldots,m_r}$ by (i). Grouping the terms over $\mathbf{m} \in K_{r,n}$ proves the expansion.

Structural Commentary: Successive subset choices count labeled partitions; fiber profiles classify product terms.
Key Insight: Multinomial coefficients count functions with prescribed fiber sizes. □

Example 4.3.12. *Repeated Letters, Committees, and Pairings*

(i) The position sets of the letters in (A, A, G, G, G, G, C, C, C) form a labeled partition of sizes $2, 4, 3$.
Thus the number of arrangements is $\binom{9}{2,4,3} = \frac{9!}{2!\,4!\,3!} = 1260$ by Proposition 4.3.11(i).

(ii) Splitting seven people into labeled subcommittees of sizes $3, 2, 2$ gives $\binom{7}{3,2,2} = \frac{7!}{3!\,2!\,2!} = 210$ assignments by Proposition 4.3.11(i).

(iii) Let $m \in \mathbb{N}$ and let S be finite with $\#S = 2m$.
Let $\mathcal{Q}_m(S)$ be the set of pairwise disjoint m-element families $\mathcal{A} \subseteq \mathcal{P}_2(S)$ satisfying $\bigcup \mathcal{A} = S$; its elements are the pairings of S.
The sets $\mathcal{P}(S)$ and $\mathcal{P}(\mathcal{P}(S))$ are finite by two applications of Proposition 4.3.7(ii). Hence $\mathcal{Q}_m(S)$ is finite by Proposition 4.2.4(ii).
Define $q : D_{2m}(S) \to \mathcal{Q}_m(S)$ by $q(s) := \{\{s_{2j-1}, s_{2j}\} : j \in J_m\}$. Each $s \in D_{2m}(S)$ lists S exactly once, so $q(s) \in \mathcal{Q}_m(S)$.
Given $\mathcal{A} \in \mathcal{Q}_m(S)$, choose a bijection $p : J_m \to \mathcal{A}$. Choose labels $((a_j, b_j))_{j \in J_m}$ satisfying $(\forall j \in J_m)\, p(j) = \{a_j, b_j\}$.
Then $(a_1, b_1, \ldots, a_m, b_m) \in q^{-1}(\{\mathcal{A}\})$, so q is surjective.
Each pairing has $m!\,2^m$ preimages: the m pairs may be ordered arbitrarily, and the two members of each pair may be interchanged.
The identity $\#D_{2m}(S) = (2m)!$ holds by Proposition 4.3.3(iii). Therefore the number of pairings is $\frac{(2m)!}{m!\,2^m}$ by Proposition 4.2.4(iii).

Key Insight: Multinomial counts labeled blocks; quotienting symmetries handles unlabeled blocks.

4.4 Inclusion-Exclusion and Combinatorial Proof

Proposition 4.4.1. *Finite Inclusion-Exclusion for Two and Three Sets*
Let A, B, C be finite sets.

(i) Two sets: The set $A \cup B$ is finite. Its cardinality is $\#(A \cup B) = \#A + \#B - \#(A \cap B)$.

(ii) Three sets: The set $A \cup B \cup C$ is finite.
Its cardinality is $\#(A \cup B \cup C) = \#A + \#B + \#C - \#(A \cap B) - \#(A \cap C) - \#(B \cap C) + \#(A \cap B \cap C)$.

Proof.

(i) Set $D := A \cap B$ and $E := B \setminus D$. The sets D and E are finite with $\#E = \#B - \#D$ by Proposition 4.2.4(ii).
The sets A and E are disjoint with union $A \cup B$. Hence $A \cup B$ is finite and $\#(A \cup B) = \#A + \#E$ by Proposition 4.2.4(i).
Substituting $D = A \cap B$ and $\#E = \#B - \#D$ proves (i).

(ii) The set $A \cup B$ is finite by (i). Set $F := (A \cup B) \cap C$.
The set $A \cup B \cup C$ is finite and $\#(A \cup B \cup C) = \#(A \cup B) + \#C - \#F$ by (i) with $A \cup B$ and C.
The sets $A \cap C$ and $B \cap C$ are finite by two applications of Proposition 4.2.4(ii).
Since $F = (A \cap C) \cup (B \cap C)$, the identity $\#F = \#(A \cap C) + \#(B \cap C) - \#(A \cap B \cap C)$ holds by (i) with $A \cap C$ and $B \cap C$.
The identity $\#(A \cup B) = \#A + \#B - \#(A \cap B)$ holds by (i). Substitution of the three identities proves (ii).

Structural Commentary: The two-set decomposition supplies the three-set formula through distributivity.
Key Insight: Correct each overlap after adding the individual set sizes. □

Method 4.4.2. *Combinatorial Proof by Classification*
Use When: Both sides of a finite identity can be expressed by weighting the same classified outcomes.
Fix target counts $N_L, N_R \in \mathbb{N}_0$.

(i) Choose finite sets Ω and P, a pattern map $p : \Omega \to P$, and weights $w_L, w_R : P \to \mathbb{Z}$.

(ii) Establish the exact fiber decompositions $(\forall \nu \in \{L, R\})\, N_\nu = \sum_{\alpha \in P} w_\nu(\alpha) \#p^{-1}(\{\alpha\})$.

(iii) Prove $(\forall \alpha \in P)\, w_L(\alpha) = w_R(\alpha)$, then substitute into the decompositions to obtain $N_L = N_R$.

Endpoint Check: Confirm that the fibers cover Ω disjointly and that both weight sums record every intended contribution.
Key Idea: Common classification reduces global equality to equality on each pattern.

5 Relations, Orders, and Lattices

Relations package comparisons, identifications, order structures, and quotient constructions.

5.1 Relations and Equivalence

Definition 5.1.1. *Relations*
Let X, Y be sets.

(i) *From X to Y*: A subset $R \subseteq X \times Y$ is a relation from X to Y.

(ii) *Relation notation*: Given a relation R from X to Y, write $xRy \iff (x, y) \in R$ for $(x, y) \in X \times Y$.

(iii) *On X*: A subset $R \subseteq X^2$ is a relation on X.

Definition 5.1.2. *Equivalence Relations*
Let X be a set and let $\sim$ be a relation on X.

(i) *Reflexive*: $\sim$ is reflexive iff $(\forall x \in X)\, x \sim x$.

(ii) *Symmetric*: $\sim$ is symmetric iff $(\forall x, y \in X)\, [x \sim y \Rightarrow y \sim x]$.

(iii) *Transitive*: $\sim$ is transitive iff $(\forall x, y, z \in X)\,[(x \sim y \wedge y \sim z) \Rightarrow x \sim z]$.
(iv) *Equivalence relation*: $\sim$ is an equivalence relation iff conditions (i)–(iii) hold.

Definition 5.1.3. *Equivalence Classes, Invariants, and Canonical Forms*
Let X, Y be sets and let $\sim$ be an equivalence relation on X.

(i) *Equivalence class*: For $x \in X$, set $[x]_\sim := \{y \in X : y \sim x\}$.
(ii) *Invariant*: A map $f : X \to Y$ is an invariant of $\sim$ iff $(\forall x, y \in X)\,[x \sim y \Rightarrow f(x) = f(y)]$.
(iii) *Complete invariant*: A map $f : X \to Y$ is a complete invariant of $\sim$ iff $(\forall x, y \in X)\,[x \sim y \iff f(x) = f(y)]$.
(iv) *Canonical forms*: A subset $\mathcal{C} \subseteq X$ is a set of canonical forms for $\sim$ iff $(\forall x \in X)(\exists! c \in \mathcal{C})\, c \sim x$.

5.2 Partitions and Quotient Sets

Definition 5.2.1. *Partition of a Set*
Let X be a set. A *partition* of X is a set $\mathcal{F} \subseteq \mathcal{P}(X) \setminus \{\emptyset\}$ satisfying $(\forall x \in X)(\exists! B \in \mathcal{F})\, x \in B$.
The members of $\mathcal{F}$ are its *blocks*.

Proposition 5.2.2. *Equivalence Classes Partition the Set*
Let X be a set, let $\sim$ be an equivalence relation on X, and set $\mathcal{F}_\sim := \{[x]_\sim : x \in X\}$. Then $\mathcal{F}_\sim$ is a partition of X.

Proof. For $x \in X$, reflexivity gives $x \in [x]_\sim$; hence the classes are nonempty and cover X.
If $u \in [x]_\sim \cap [y]_\sim$, then $u \sim x$ and $u \sim y$, whence $x \sim y$ and $y \sim x$ by symmetry and transitivity.
If $v \in [x]_\sim$, then $v \sim x \sim y$, so $v \in [y]_\sim$. If $v \in [y]_\sim$, then $v \sim y \sim x$, so $v \in [x]_\sim$.
Thus intersecting classes are equal, proving unique class membership; hence $\mathcal{F}_\sim$ partitions X.
Structural Commentary: Reflexivity supplies classes; symmetry and transitivity collapse intersections.
Key Insight: Two classes sharing an element have the same members. □

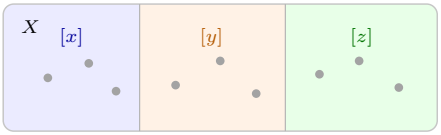

Figure 5: Equivalence classes form disjoint blocks that cover the underlying set.

Figure 5 depicts the partition in Proposition 5.2.2. Every element of X lies in one equivalence class, and distinct classes are disjoint.

Proposition 5.2.3. *Partitions Determine Equivalence Relations*
Let X be a set and let $\mathcal{F}$ be a partition of X. For $x \in X$, let $B_x \in \mathcal{F}$ be the unique block containing x.
For $(x, y) \in X^2$, define $x \sim_{\mathcal{F}} y \iff B_x = B_y$.

(i) Equivalence relation: $\sim_{\mathcal{F}}$ is an equivalence relation on X.
(ii) Classes: $(\forall x \in X)\,[x]_{\sim_{\mathcal{F}}} = B_x$, and $\{[x]_{\sim_{\mathcal{F}}} : x \in X\} = \mathcal{F}$.

Proof.

(i) For $x, y, z \in X$, $B_x = B_x$, so $x \sim_{\mathcal{F}} x$. If $x \sim_{\mathcal{F}} y$, then $B_x = B_y$, hence $B_y = B_x$ and $y \sim_{\mathcal{F}} x$.
Moreover, $(x \sim_{\mathcal{F}} y \wedge y \sim_{\mathcal{F}} z) \Rightarrow B_x = B_y = B_z \Rightarrow x \sim_{\mathcal{F}} z$. Thus $\sim_{\mathcal{F}}$ is reflexive, symmetric, and transitive.
(ii) Fix $x \in X$. If $y \in [x]_{\sim_{\mathcal{F}}}$, then $B_y = B_x$, so $y \in B_y = B_x$. If $y \in B_x$, then B_x is a block containing y.
Uniqueness gives $B_y = B_x$, so $y \sim_{\mathcal{F}} x$ and $y \in [x]_{\sim_{\mathcal{F}}}$. Hence $[x]_{\sim_{\mathcal{F}}} = B_x$. Therefore $\{[x]_{\sim_{\mathcal{F}}} : x \in X\} \subseteq \mathcal{F}$.
If $B \in \mathcal{F}$, then $B \neq \emptyset$; choose $x \in B$. The uniqueness of B_x gives $B = B_x = [x]_{\sim_{\mathcal{F}}}$, so the reverse containment holds.

Structural Commentary: Block equality proves the axioms; uniqueness identifies classes with blocks.
Key Insight: Equivalent elements are assigned to the same block. □

Example 5.2.4. *Rational Equivalence on $\mathbb{R}$*
For $(x, y) \in \mathbb{R}^2$, define $x \sim_{\mathbb{Q}} y \iff x - y \in \mathbb{Q}$. Fix $x, y, z \in \mathbb{R}$.

(i) *Reflexive*: $x - x = 0 \in \mathbb{Q}$.
(ii) *Symmetric*: $x - y \in \mathbb{Q} \Rightarrow y - x = -(x - y) \in \mathbb{Q}$.
(iii) *Transitive*: $(x - y \in \mathbb{Q} \wedge y - z \in \mathbb{Q}) \Rightarrow x - z = (x - y) + (y - z) \in \mathbb{Q}$.

Thus $\sim_{\mathbb{Q}}$ is an equivalence relation, and $[x]_{\sim_{\mathbb{Q}}} = x + \mathbb{Q} = \{x + q : q \in \mathbb{Q}\}$.
A subset $\mathcal{R} \subseteq \mathbb{R}$ is a representative set iff $(\forall x \in \mathbb{R})(\exists! r \in \mathcal{R})\, r \in x + \mathbb{Q}$.
Key Insight: Rational translation orbits are exactly the equivalence classes.

Definition 5.2.5. *Quotient Set*
Let X be a set and let $\sim$ be an equivalence relation on X. The *quotient set* of X by $\sim$ is $X/\!\sim := \{[x]_\sim : x \in X\}$.

Proposition 5.2.6. *Maps Defined on Equivalence Classes*
Let X, Y be sets, let $\sim$ be an equivalence relation on X, and let $f : X \to Y$.
The assignment $g([x]_\sim) := f(x)$ defines $g : X/\!\sim \to Y$ iff $(\forall x, y \in X)\,[x \sim y \Rightarrow f(x) = f(y)]$.

Proof.

($\Rightarrow$) Suppose g is well-defined, $x \sim y$. Then $x \in [x]_\sim \cap [y]_\sim$, so $[x]_\sim = [y]_\sim$ by Proposition 5.2.2. Thus $f(x) = g([x]_\sim) = g([y]_\sim) = f(y)$.
($\Leftarrow$) Suppose $(\forall x, y \in X)\,[x \sim y \Rightarrow f(x) = f(y)]$. If $[x]_\sim = [y]_\sim$, then $x \in [y]_\sim$, whence $x \sim y$ and $f(x) = f(y)$.
Thus the assignment is independent of the representative. Hence it defines a map $g : X/\!\sim \to Y$.

Structural Commentary: Equal classes prove necessity; representative independence proves sufficiency.
Key Insight: A class function cannot distinguish equivalent representatives. □

5.3 Ordered Sets, Well-Orders, and Lattices

Definition 5.3.1. *Ordered Set Vocabulary*
Let S be a set, let $E, C \subseteq S$, and let $\leq_S \subseteq S^2$. Set $x <_S y \iff (x \leq_S y \wedge x \neq y)$.
The relation $\leq_S$ is *antisymmetric* iff $(\forall x, y \in S)\,[(x \leq_S y \wedge y \leq_S x) \Rightarrow x = y]$. Elements $x, y \in S$ are *comparable* iff $x \leq_S y \vee y \leq_S x$.

(i) *Partial order*: The relation $\leq_S$ is a partial order iff it is reflexive, antisymmetric, and transitive. Then $(S, \leq_S)$ is a *poset*.
(ii) *Total order*: A partial order $\leq_S$ is total iff $(\forall x, y \in S)\,(x \leq_S y \vee y \leq_S x)$. Then $(S, \leq_S)$ is an *ordered set*.
(iii) *Chain*: If $\leq_S$ is a partial order, then C is a chain iff $(\forall x, y \in C)\,(x \leq_S y \vee y \leq_S x)$.
(iv) *Well-order*: The order $\leq_S$ well-orders S iff it is total and the following condition holds. $(\forall Y \in \mathcal{P}(S)\setminus\{\emptyset\})(\exists \ell \in Y)(\forall y \in Y)\,\ell \leq_S y$.
(v) *Bounds*: An element $\beta \in S$ is an upper bound of E iff $(\forall x \in E)\,x \leq_S \beta$. It is a lower bound iff $(\forall x \in E)\,\beta \leq_S x$.
(vi) *Boundedness*: The set E is bounded above iff $(\exists \beta \in S)(\forall x \in E)\,x \leq_S \beta$. It is bounded below iff $(\exists \beta \in S)(\forall x \in E)\,\beta \leq_S x$.
(vii) *Supremum*: For $\alpha \in S$, $\alpha = \sup E$ iff the following two conditions hold.
  1. *Upper bound*: $(\forall x \in E)\,x \leq_S \alpha$.
  2. *Upper bound*: $(\forall x \in E)\,x \leq_S \alpha$.
  3. *Least*: $(\forall \beta \in S)\,[((\forall x \in E)\,x \leq_S \beta) \Rightarrow \alpha \leq_S \beta]$.

(viii) *Infimum*: For $\gamma \in S$, $\gamma = \inf E$ iff the following two conditions hold.
  1. *Lower bound*: $(\forall x \in E)\,\gamma \leq_S x$.
  2. *Greatest*: $(\forall \beta \in S)\,[((\forall x \in E)\,\beta \leq_S x) \Rightarrow \beta \leq_S \gamma]$.

(ix) *Maximal and minimal elements*: An element $m \in E$ is maximal iff $(\nexists x \in E)\,m <_S x$. It is minimal iff $(\nexists x \in E)\,x <_S m$.
(x) *Least-upper-bound property*: This property is characterized by the following condition.
$(\forall A \in \mathcal{P}(S) \setminus \{\emptyset\})\,[((\exists \beta \in S)(\forall x \in A)\,x \leq_S \beta) \Rightarrow (\exists \alpha \in S)\,\alpha = \sup A]$.

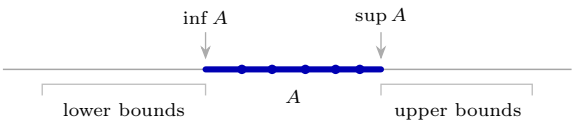

Figure 6: The supremum is the least upper bound, while the infimum is the greatest lower bound.

Figure 6 illustrates the bounds in Definition 5.3.1 for a set A.
It marks $\inf A$ and $\sup A$ as extremal bounds; Example 5.3.2 shows that neither need belong to the set.

Example 5.3.2. *Suprema Need Not Be Elements*
In $\mathbb{Q}$, set $E_1 := \{r \in \mathbb{Q} : r < 0\}$ and $E_2 := \{r \in \mathbb{Q} : r \leq 0\}$. Both sets have supremum 0, but $0 \notin E_1$ and $0 \in E_2$.
For $E := \{\frac{1}{n} : n \in \mathbb{N}\}$, $\sup E = 1 \in E$ and $\inf E = 0 \notin E$.
Key Insight: An extremal bound need not be an element of the set.

Theorem 5.3.3. *Least Upper Bounds Give Greatest Lower Bounds*
Let $(S, \leq)$ be an ordered set with the least-upper-bound property. Let $B \subseteq S$ be nonempty and bounded below. Then $\inf B$ exists in S.

Proof. Set $L := \{\ell \in S : (\forall b \in B)\,\ell \leq b\}$. $L \neq \emptyset$ by the boundedness of B below. Choose $b_0 \in B$.
Then $(\forall \ell \in L)\,\ell \leq b_0$, so L is bounded above. Set $\alpha := \sup L$; it exists in S by the least-upper-bound property.
For $b \in B$, the element b is an upper bound of L, so $\alpha \leq b$. Hence $\alpha \in L$. Moreover, $(\forall \ell \in L)\,\ell \leq \alpha$ by $\alpha = \sup L$.
Thus α is the greatest lower bound of B, and $\alpha = \inf B$.
Structural Commentary: The proof takes the supremum of the lower-bound set.
Key Insight: A nonempty set bounds its lower bounds above. □

Definition 5.3.4. *Lattice Vocabulary*
Let $(L, \leq)$ be a partially ordered set.

(i) *Meet*: For $x, y \in L$, the meet is $x \wedge y := \inf\{x, y\}$ when this infimum exists.
(ii) *Join*: For $x, y \in L$, the join is $x \vee y := \sup\{x, y\}$ when this supremum exists.
(iii) *Lattice*: The partially ordered set L is a lattice iff the following condition holds. $(\forall x, y \in L)\,[x \wedge y \in L \wedge x \vee y \in L]$.
(iv) *Complete lattice*: A lattice L is complete iff $(\forall E \subseteq L)\,[\inf E \in L \wedge \sup E \in L]$.
(v) *Bottom*: An element $\bot \in L$ is a bottom element iff $(\forall x \in L)\,\bot \leq x$.
(vi) *Top*: An element $\top \in L$ is a top element iff $(\forall x \in L)\,x \leq \top$.
(vii) *Modular lattice*: A lattice L is modular iff the following law holds. $(\forall x, y, z \in L)\,[x \leq z \Rightarrow z \wedge (y \vee x) = (z \wedge y) \vee x]$.

5.4 Relations Between Sets and Closures

Definition 5.4.1. *Inverse and Composition of Relations*
Let X, Y, Z be sets, let $R \subseteq X \times Y$, and let $S \subseteq Y \times Z$.

(i) *Inverse*: $R^{-1} := \{(y, x) \in Y \times X : (x, y) \in R\}$.
(ii) *Composition*: $S \circ R := \{(x, z) \in X \times Z : (\exists y \in Y)\,[(x, y) \in R \wedge (y, z) \in S]\}$.
(iii) *Diagonal*: $\Delta_X := \{(x, x) : x \in X\}$.

Proposition 5.4.2. *Inverse and Composition Laws*
Let W, X, Y, Z be sets. Let $R \subseteq X \times Y$, let $S \subseteq Y \times Z$, and let $T \subseteq Z \times W$.

(i) Identity: $\Delta_Y \circ R = R = R \circ \Delta_X$.
(ii) Associativity: $T \circ (S \circ R) = (T \circ S) \circ R$.
(iii) Involution: $(R^{-1})^{-1} = R$.
(iv) Reversal: $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

Proof.

(*i*) For $(x,y) \in X \times Y$, $(x,y) \in \Delta_Y \circ R \iff (\exists u \in Y)\,[(x,u) \in R \wedge u = y] \iff (x,y) \in R$.
For the right identity, $(x,y) \in R \circ \Delta_X \iff (\exists v \in X)\,[v = x \wedge (v,y) \in R] \iff (x,y) \in R$.
(*ii*) ($\Rightarrow$) Suppose $(x,w) \in T \circ (S \circ R)$. Choose $z \in Z$ with $(x,z) \in S \circ R$ and $(z,w) \in T$.
Choose $y \in Y$ with $(x,y) \in R$ and $(y,z) \in S$. Thus $(y,w) \in T \circ S$ and $(x,w) \in (T \circ S) \circ R$.
(*ii*) ($\Leftarrow$) Suppose $(x,w) \in (T \circ S) \circ R$. Choose $y \in Y$ with $(x,y) \in R$ and $(y,w) \in T \circ S$.
Choose $z \in Z$ with $(y,z) \in S$ and $(z,w) \in T$. Thus $(x,z) \in S \circ R$ and $(x,w) \in T \circ (S \circ R)$.
(*iii*) For $(x,y) \in X \times Y$, $(x,y) \in (R^{-1})^{-1} \iff (y,x) \in R^{-1} \iff (x,y) \in R$.
(*iv*) ($\Rightarrow$) Suppose $(z,x) \in (S \circ R)^{-1}$. Then $(x,z) \in S \circ R$.
Choose $y \in Y$ with $(x,y) \in R$ and $(y,z) \in S$. Hence $(z,y) \in S^{-1}$ and $(y,x) \in R^{-1}$, so $(z,x) \in R^{-1} \circ S^{-1}$.
(*iv*) ($\Leftarrow$) Suppose $(z,x) \in R^{-1} \circ S^{-1}$. Choose $y \in Y$ with $(z,y) \in S^{-1}$ and $(y,x) \in R^{-1}$.
Then $(y,z) \in S$ and $(x,y) \in R$, so $(x,z) \in S \circ R$ and $(z,x) \in (S \circ R)^{-1}$.

Structural Commentary: Membership expansion and witness comparison prove the four laws.
Key Insight: Composition records witnesses; inversion reverses their order. □

Definition 5.4.3. *Relation Powers and Closures*
Let X be a set and let $R \subseteq X^2$. Set $R^0 := \Delta_X$ and $R^{n+1} := R \circ R^n$ for $n \in \mathbb{N}_0$.

(i) *Reflexive closure*: $\mathrm{Ref}(R) := R \cup \Delta_X$.
(ii) *Transitive closure*: $R^+ := \bigcup_{n \in \mathbb{N}} R^n$.
(iii) *Reflexive-transitive closure*: $R^* := \bigcup_{n \in \mathbb{N}_0} R^n = \Delta_X \cup R^+$.

Lemma 5.4.4. *Addition Law for Relation Powers*
Let X be a set and let $R \subseteq X^2$. Then $(\forall m,n \in \mathbb{N}_0)\, R^n \circ R^m = R^{m+n}$.

Proof. Fix $m \in \mathbb{N}_0$. For $r \in \mathbb{N}$, let P_r assert $R^{r-1} \circ R^m = R^{m+r-1}$.
The identity $R^0 \circ R^m = \Delta_X \circ R^m = R^m$ holds by Proposition 5.4.2(i), so P_1 holds.
Assume P_r. Then $R^r \circ R^m = R \circ (R^{r-1} \circ R^m)$ by Proposition 5.4.2(ii) and the recursion.
Hence $R^r \circ R^m = R \circ R^{m+r-1} = R^{m+r}$ by P_r and the recursion, so P_{r+1} holds.
Consequently, $(\forall r \in \mathbb{N})\, P_r$ by induction. For $n \in \mathbb{N}_0$, take $r = n+1$ to obtain $R^n \circ R^m = R^{m+n}$.
Structural Commentary: Reindexed induction combines the recursive power law with associativity.
Key Insight: Relation composition adds power indices. □

Theorem 5.4.5. *Minimality of Reflexive and Transitive Closures*
Let X be a set and let $R \subseteq X^2$.

(i) Reflexive: $\mathrm{Ref}(R)$ is the smallest reflexive relation on X containing R.
(ii) Transitive: R^+ is the smallest transitive relation on X containing R.
(iii) Reflexive and transitive: R^* is the smallest reflexive and transitive relation on X containing R.

Proof.

(i) The relation $\mathrm{Ref}(R)$ contains R and Δ_X, so it is reflexive.
If $Q \subseteq X^2$ is reflexive and $R \subseteq Q$, then $\Delta_X \subseteq Q$, whence $\mathrm{Ref}(R) = R \cup \Delta_X \subseteq Q$.
(ii) Since $R = R^1$, $R \subseteq R^+$. Suppose $(x,y) \in R^+$ and $(y,z) \in R^+$. Choose $m,n \in \mathbb{N}$ with $(x,y) \in R^m$ and $(y,z) \in R^n$.
Then $(x,z) \in R^{m+n} \subseteq R^+$ by Lemma 5.4.4, so R^+ is transitive. Let $Q \subseteq X^2$ be transitive with $R \subseteq Q$.
For $n \in \mathbb{N}$, let P_n assert $R^n \subseteq Q$. P_1 holds by $R^1 = R \subseteq Q$. Assume P_n. Then $R^{n+1} = R \circ R^n \subseteq Q \circ Q$ by $R \subseteq Q$ and P_n.
Also $Q \circ Q \subseteq Q$ by the transitivity of Q, so P_{n+1} holds. Thus $(\forall n \in \mathbb{N})\, P_n$ by induction, and $R^+ \subseteq Q$.
(iii) The relation R^* contains R and is reflexive by $R^* = \Delta_X \cup R^+$. Suppose $(x,y) \in R^*$ and $(y,z) \in R^*$.
Choose $m,n \in \mathbb{N}_0$ with $(x,y) \in R^m$ and $(y,z) \in R^n$. Then $(x,z) \in R^{m+n} \subseteq R^*$ by Lemma 5.4.4, so R^* is transitive.
If $Q \subseteq X^2$ is reflexive and transitive with $R \subseteq Q$, then $\Delta_X \subseteq Q$ and $R^+ \subseteq Q$ by (ii). Thus $R^* = \Delta_X \cup R^+ \subseteq Q$.

Structural Commentary: Containment, closure, and minimality are verified in parallel.
Key Insight: Closure is generated by diagonal pairs together with finite composites. □

Example 5.4.6. *Closures of the Successor Relation*
Define $R \subseteq \mathbb{Z}^2$ by $m\,R\,n \iff n = m+1$. For $k \in \mathbb{N}$, let P_k assert $(\forall m,n \in \mathbb{Z})\,[m\,R^k\,n \iff n = m+k]$. P_1 holds by the definition of R.
Assume P_k. For $(m,n) \in \mathbb{Z}^2$, $m\,R^{k+1}\,n \iff (\exists u \in \mathbb{Z})\,[u = m+k \wedge n = u+1] \iff n = m+k+1$.
Thus P_{k+1} holds, and $(\forall k \in \mathbb{N})\, P_k$ by induction.

(i) $\mathrm{Ref}(R) = \{(m,n) \in \mathbb{Z}^2 : n = m \vee n = m+1\}$.
(ii) $R^+ = \{(m,n) \in \mathbb{Z}^2 : m < n\}$.
(iii) $R^* = \{(m,n) \in \mathbb{Z}^2 : m \le n\}$.

Key Insight: Finite successor compositions generate the strict and nonstrict integer orders.

Definition 5.4.7. *Order Isomorphism*
Let $(S, \le_S)$ and $(T, \le_T)$ be partially ordered sets.
A bijection $f : S \to T$ is an *order isomorphism* when the following condition holds. $(\forall x,y \in S)\,[x \le_S y \iff f(x) \le_T f(y)]$.
The two partially ordered sets are *order-isomorphic* when such a map exists.

Example 5.4.8. *Basic Order Isomorphisms*

(i) Define $f : \mathbb{Z} \to \{2n : n \in \mathbb{Z}\}$ by $f(n) := 2n$. Its inverse is $2n \mapsto n$, so f is bijective.
Moreover, $m \le n \iff 2m \le 2n$; hence f is an order isomorphism for the usual orders.
(ii) Let X be a set and define $c : \mathcal{P}(X) \to \mathcal{P}(X)$ by $c(A) := X \setminus A$. Since $c(c(A)) = A$, the map is bijective.
Moreover, $A \subseteq B \iff c(A) \supseteq c(B)$. Hence c is an order isomorphism from $(\mathcal{P}(X), \subseteq)$ to $(\mathcal{P}(X), \supseteq)$.
(iii) Let $(S, \le)$ be a poset and set $\downarrow s := \{x \in S : x \le s\}$. Define $\eta : S \to \{\downarrow s : s \in S\}$ by $\eta(s) :=\downarrow s$.
If $s \le t$, then $\downarrow s \subseteq \downarrow t$ by transitivity. If $\downarrow s \subseteq \downarrow t$, then $s \in \downarrow s \subseteq \downarrow t$, so $s \le t$. Thus $s \le t \iff \eta(s) \subseteq \eta(t)$.
If $\eta(s) = \eta(t)$, then $s \le t$ and $t \le s$, so $s = t$ by antisymmetry. Thus η is injective.
It is surjective by the definition of its codomain, so it is an order isomorphism.

Key Insight: Order isomorphisms preserve comparisons while changing their representation.

6 Cardinality, Choice, and Infinite Sets

Choice and maximality principles come first as foundational tools; cardinality then compares sets by functions rather than by informal size.

6.1 Choice and Zorn's Lemma

Definition 6.1.1. *Choice Function*
Let $\mathcal{F}$ be a family of nonempty sets, let I be a set, and let $(X_i)_{i\in I}$ be an indexed family of nonempty sets.

(i) A *choice function* on $\mathcal{F}$ is a map $c : \mathcal{F} \to \bigcup_{F\in\mathcal{F}} F$ satisfying $(\forall F \in \mathcal{F})\, c(F) \in F$.

(ii) An *indexed choice function* for $(X_i)_{i\in I}$ is a map $c : I \to \bigcup_{i\in I} X_i$ satisfying $(\forall i \in I)\, c(i) \in X_i$.

Principle 6.1.2. *Choice Principle*
Let I be a set and let $(X_i)_{i\in I}$ be an indexed family of nonempty sets. The axiom of choice asserts $(\exists c : I \to \bigcup_{i\in I} X_i)(\forall i \in I)\, c(i) \in X_i$. Equivalently, $\prod_{i\in I} X_i \neq \varnothing$. An explicit c proves existence directly; the principle supplies simultaneous choices without an explicit rule.

Principle 6.1.3. *Zorn's Lemma*
Let $(P, \leq)$ be a poset with $P \neq \varnothing$. Assume the following chain-upper-bound condition.
$(\forall C \subseteq P)\,[((\forall x, y \in C)\,(x \leq y \vee y \leq x)) \Rightarrow (\exists u \in P)(\forall c \in C)\, c \leq u]$. Then $(\exists m \in P)(\forall x \in P)\,(m \leq x \Rightarrow x = m)$.

Definition 6.1.4. *Inclusion and Containment Orders*
Let $\mathcal{F}$ be a family of sets and let $\mathcal{A} \subseteq \mathcal{F}$.

(i) *Inclusion*: for $A, B \in \mathcal{F}$, define $A \leq B$ iff $A \subseteq B$. An upper bound of $\mathcal{A}$ is $U \in \mathcal{F}$ satisfying $(\forall A \in \mathcal{A})\, A \subseteq U$.
An element $M \in \mathcal{F}$ is maximal iff $(\forall B \in \mathcal{F})\,(M \subseteq B \Rightarrow B = M)$.

(ii) *Reverse inclusion*: for $A, B \in \mathcal{F}$, define $A \leq B$ iff $B \subseteq A$. An upper bound of $\mathcal{A}$ is $U \in \mathcal{F}$ satisfying $(\forall A \in \mathcal{A})\, U \subseteq A$.
An element $M \in \mathcal{F}$ is maximal iff $(\forall B \in \mathcal{F})\,(B \subseteq M \Rightarrow B = M)$.

Proposition 6.1.5. *Inclusion Orders Are Partial Orders*
Let $\mathcal{F}$ be a family of sets. Inclusion and reverse inclusion are partial orders on $\mathcal{F}$.

Proof. Fix $A, B, C \in \mathcal{F}$. Since $A \subseteq A$, inclusion is reflexive. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$, so inclusion is transitive. If $A \subseteq B$ and $B \subseteq A$, then $A = B$ by Assumption 1.3.2, so inclusion is antisymmetric. Hence inclusion is a partial order.
For reverse inclusion, $A \supseteq A$ gives reflexivity, $A \supseteq B \supseteq C$ gives $A \supseteq C$, and $A \supseteq B \supseteq A$ gives $A = B$ by Assumption 1.3.2. Thus reverse inclusion is also a partial order.
Structural Commentary: Each order law follows from the corresponding containment law and set extensionality.
Key Insight: Reversing containment preserves reflexivity, transitivity, and antisymmetry. □

Proof Method 6.1.6. *Zorn Maximality Template*
Use this method when the target follows from a candidate admitting no proper admissible extension.
Let P be the set of candidates, and let $\leq$ record admissible extension.

(i) Verify that $P \neq \varnothing$ and that $\leq$ is a partial order on P.
(ii) Fix a chain $C \subseteq P$, construct $u \in P$, and prove $(\forall p \in C)\, p \leq u$.
(iii) Obtain a maximal $m \in P$ by Principle 6.1.3, and prove that maximality forces the target property.

For $C \neq \varnothing$, common candidates are $\bigcup_{A\in C} A$ under inclusion and $\bigcap_{A\in C} A$ under reverse inclusion.
Under reverse inclusion, another object contained in every member of C may be necessary.
If $C = \varnothing$, any element of P is an upper bound. In every case, prove $u \in P$; this is the indispensable admissibility check.
Key Idea: A chain bound must preserve every admissibility condition defining the candidate set.

6.2 Same Cardinality, Finite, Countable, and Uncountable

Definition 6.2.1. *Same Cardinality*
For sets A, B, write $A \sim_{\#} B$ iff a bijection $f : A \to B$ exists. Then A and B have *the same cardinality*, or are *equivalent*.

Proposition 6.2.2. *Same Cardinality Is an Equivalence Relation*
Let A, B, C be sets.

(i) $A \sim_{\#} A$.
(ii) $A \sim_{\#} B \Rightarrow B \sim_{\#} A$.
(iii) $(A \sim_{\#} B \wedge B \sim_{\#} C) \Rightarrow A \sim_{\#} C$.

Consequently, $\sim_{\#}$ is an equivalence relation among sets.

Proof.

(i) The identity $\mathrm{id}_A : A \to A$ is bijective, so $A \sim_{\#} A$.
(ii) Suppose $A \sim_{\#} B$ and choose a bijection $f : A \to B$. The inverse $f^{-1} : B \to A$ exists, and f is its two-sided inverse. Hence f^{-1} is bijective by Theorem 3.3.2(i), so $B \sim_{\#} A$.
(iii) Suppose $A \sim_{\#} B$ and $B \sim_{\#} C$, and choose bijections $f : A \to B$ and $g : B \to C$. The composite $g \circ f : A \to C$ is bijective by Proposition 3.3.3(vi), so $A \sim_{\#} C$.

Structural Commentary: Identity, inversion, and composition verify the three equivalence-relation laws.
Key Insight: Bijections are closed under reversal and composition. □

Theorem 6.2.3. *Schroeder–Bernstein Theorem*
Let A, B be sets, and let $f : A \to B$ and $g : B \to A$ be injections. Then $A \sim_{\#} B$.

Proof. Set $A_0 := A \setminus g(B)$ and recursively set $B_n := f(A_n)$ and $A_{n+1} := g(B_n)$ for $n \in \mathbb{N}_0$.
Put $U_A := \bigcup_{n\in\mathbb{N}_0} A_n$, $U_B := \bigcup_{n\in\mathbb{N}_0} B_n$, $A_* := A \setminus U_A$, and $B_* := B \setminus U_B$.
The restriction $f_* := f|_{U_A} : U_A \to U_B$ is bijective because f is injective and $U_B = f(U_A)$. For $b \in B_*$, $g(b) \notin A_0$.
Suppose $g(b) \in A_{n+1} = g(B_n)$ with $n \in \mathbb{N}_0$. Then injectivity of g gives $b \in B_n \subseteq U_B$, a contradiction. Thus $g(b) \in A_*$.
Conversely, if $a \in A_*$, then $a \notin A_0$, so $a \in g(B)$. Choose $b \in B$ satisfying $a = g(b)$. Suppose $b \in B_n$ with $n \in \mathbb{N}_0$.
Then $a \in A_{n+1} \subseteq U_A$, a contradiction. Hence $b \in B_*$. Therefore $g_* := g|_{B_*} : B_* \to A_*$ is bijective.
Define $h : A \to B$ by $h(x) := f_*(x)$ for $x \in U_A$ and $h(x) := g_*^{-1}(x)$ for $x \in A_*$. The domain parts U_A, A_* and image parts U_B, B_* are disjoint partitions, and both branches are bijective. Hence h is bijective, so $A \sim_{\#} B$.
Structural Commentary: Forward layers are matched by f; the complementary remainders are matched by g^{-1}.
Key Insight: Alternate the injections until the unmatched remainder can be reversed. □

Definition 6.2.4. *Finite, Countable, and Uncountable*
Let A be a set.

(i) A is *finite* iff $(\exists n \in \mathbb{N}_0)\, A \sim_{\#} J_n$.
(ii) A is *infinite* iff it is not finite.
(iii) A is *countable* iff $A \sim_{\#} \mathbb{N}$.
(iv) A is *uncountable* iff it is neither finite nor countable.
(v) A is *at most countable* iff it is finite or countable.

Countable sets are also called denumerable. Here *enumerable* more generally describes the image of a sequence.

Proposition 6.2.5. *The Natural Numbers Are Infinite*

(i) $(\forall m \in \mathbb{N}_0)$, *no injection* $\mathbb{N} \to J_m$ *exists.*
(ii) The set $\mathbb{N}$ *is infinite.*
(iii) If A *is countable, then* A *is infinite.*
(iv) If F *is finite and* $A \subseteq F$, *then* A *is not countable.*

Proof.

(i) For $r \in \mathbb{N}$, let Q_r assert that no injection $\mathbb{N} \to J_{r-1}$ exists. Since $J_0 = \emptyset$, no map $\mathbb{N} \to J_0$ exists, so Q_1 holds.
Assume Q_r and suppose $u : \mathbb{N} \to J_r$ is injective. If $r \notin u(\mathbb{N})$, then $u : \mathbb{N} \to J_{r-1}$ contradicts Q_r.
Otherwise let $s \in \mathbb{N}$ be the unique index satisfying $u(s) = r$.
Define $\tau : \mathbb{N} \to \mathbb{N} \setminus \{s\}$ by $\tau(k) := k$ for $k < s$ and $\tau(k) := k + 1$ for $k \geq s$.
The map τ is injective and avoids s, so $u \circ \tau : \mathbb{N} \to J_{r-1}$ is injective, again contradicting Q_r. Thus Q_{r+1} holds.
Hence $(\forall r \in \mathbb{N})\, Q_r$ by Theorem 2.6.2, proving (i).
(ii) If $\mathbb{N}$ were finite, a bijection $J_m \to \mathbb{N}$ would have an inverse injection $\mathbb{N} \to J_m$, contradicting (i).
(iii) Let A be countable. If A were finite, composing a bijection $\mathbb{N} \to A$ with a bijection $A \to J_m$ would contradict (i).
(iv) If a countable A were contained in a finite F, then $\mathbb{N} \to A \hookrightarrow F \to J_m$ would be injective, contradicting (i).

Structural Commentary: Induction deletes the unique preimage of the final point of a finite target.
Key Insight: Removing one domain point leaves $\mathbb{N}$ injectively equivalent to itself. □

Proposition 6.2.6. *Finite Sums Are Invariant under Reindexing*
Let $n \in \mathbb{N}_0$, *let* $c : J_n \to \mathbb{C}$, *and let* $\sigma : J_n \to J_n$ *be a bijection. Then* $\sum_{i=1}^{n} c(\sigma(i)) = \sum_{i=1}^{n} c(i)$.

Proof. For $r \in \mathbb{N}$, let P_r be the assertion of the proposition with $n = r - 1$. Then P_1 is the empty-sum identity.
Assume P_r, put $m := r - 1$, and fix $c : J_{m+1} \to \mathbb{C}$ and a permutation σ of J_{m+1}.
Put $j := \sigma^{-1}(m + 1)$ and define $\theta : J_m \to J_{m+1} \setminus \{j\}$ by $\theta(i) := i$ for $i < j$ and $\theta(i) := i + 1$ for $i \geq j$.
Then θ and $\sigma \circ \theta : J_m \to J_m$ are bijective. Since $\sigma(j) = m + 1$, commutativity gives $\sum_{i=1}^{m+1} c(\sigma(i)) = c(m + 1) + \sum_{i=1}^{m} c(\sigma(\theta(i)))$.
Latter expression equals $c(m + 1) + \sum_{i=1}^{m} c(i) = \sum_{i=1}^{m+1} c(i)$ by P_r. Thus P_{r+1} holds. Hence $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2, proving P_{n+1}.
Structural Commentary: Induction removes the position mapped to the final index and reindexes the remainder.
Key Insight: A permutation changes positions without changing the multiset of summands. □

Definition 6.2.7. *Summation over a Finite Set*
Let X be finite, put $n := \#X$, let $f : X \to \mathbb{C}$, and choose a bijection $g : J_n \to X$. Define $\sum_{x\in X} f(x) := \sum_{i=1}^{n} f(g(i))$.
If $X = \emptyset$, then $n = 0$ and this value is the empty sum 0. If $k : J_n \to X$ is another bijection, put $\sigma := g^{-1} \circ k$.
Then σ is a permutation of J_n, $k = g \circ \sigma$, and the two listings give equal sums by Proposition 6.2.6.
Key Insight: Reindexing changes the listing, not the sum.

Proposition 6.2.8. *Finite Set Summation Laws*
Let X, Y *be finite sets.*

(i) Well-definedness: If $f : X \to \mathbb{C}$, *then* $\sum_{x\in X} f(x)$ *is independent of the bijection* $J_{\#X} \to X$.
(ii) Substitution: If $h : Y \to X$ *is a bijection and* $f : X \to \mathbb{C}$, *then* $\sum_{x\in X} f(x) = \sum_{y\in Y} f(h(y))$.
(iii) Linearity: If $f, g : X \to \mathbb{C}$, *then* $\sum_{x\in X} (f(x) + g(x)) = \sum_{x\in X} f(x) + \sum_{x\in X} g(x)$. *If* $c \in \mathbb{C}$, *then* $\sum_{x\in X} cf(x) = c \sum_{x\in X} f(x)$.
(iv) Disjoint unions: If $X \cap Y = \emptyset$ *and* $f : X \cup Y \to \mathbb{C}$, *then* $X \cup Y$ *is finite. The identity* $\sum_{z\in X\cup Y} f(z) = \sum_{x\in X} f(x) + \sum_{y\in Y} f(y)$ *holds.*
(v) Finite Fubini: The set $X \times Y$ *is finite. If* $f : X \times Y \to \mathbb{C}$, *then the following identities hold.*
$\sum_{x\in X} \sum_{y\in Y} f(x, y) = \sum_{(x,y)\in X\times Y} f(x, y)$, *and* $\sum_{y\in Y} \sum_{x\in X} f(x, y) = \sum_{(x,y)\in X\times Y} f(x, y)$.

Proof.

(i) Let $g, k : J_{\#X} \to X$ be bijections and put $\sigma := g^{-1} \circ k$. Then $k = g \circ \sigma$, so the two sums agree by Proposition 6.2.6.

(ii) Choose a bijection $g : J_{\#Y} \to Y$. The composite $h \circ g : J_{\#Y} \to X$ is bijective, so $\#Y = \#X$ by Proposition 4.2.2.
After expanding the finite-set sums, $\sum_{x \in X} f(x) = \sum_{i=1}^{\#Y} f(h(g(i))) = \sum_{y \in Y} f(h(y))$ by part (i).

(iii) Put $m := \#X$, choose a bijection $\gamma : J_m \to X$, and set $a_i := f(\gamma(i))$ and $b_i := g(\gamma(i))$ for $i \in J_m$, with $a_i = b_i = 0$ otherwise.
After expanding the three finite-set sums, the first identity holds by Proposition 4.1.2(ii).
After the analogous expansions, the scalar-multiple identity holds by Proposition 4.1.2(iii).

(iv) Put $m := \#X$ and $n := \#Y$. If $m = 0$ or $n = 0$, then one set is empty and both conclusions follow directly.
Suppose $m, n \in \mathbb{N}$, and choose bijections $\alpha : J_m \to X$ and $\beta : J_n \to Y$.
Define $\delta : J_{m+n} \to X \cup Y$ by $\delta(i) := \alpha(i)$ when $i \leq m$ and $\delta(i) := \beta(i - m)$ when $i > m$.
The two branches have disjoint images and make δ bijective because $X \cap Y = \emptyset$. Thus $X \cup Y$ is finite.
The identity in (iv) holds by Proposition 4.1.2(i) after expansion along δ.

(v) Put $m := \#X$ and $n := \#Y$. The set $X \times Y$ is finite with $\#(X \times Y) = mn$ by Theorem 4.2.5(iii).
If $m = 0$ or $n = 0$, all three sums equal 0. Otherwise choose bijections $\alpha : J_m \to X$ and $\beta : J_n \to Y$.
Define bijections $\ell, r : J_{mn} \to X \times Y$ by $\ell((i-1)n + j) := (\alpha(i), \beta(j))$ and $r((j-1)m + i) := (\alpha(i), \beta(j))$ for $i \in J_m$ and $j \in J_n$.
The first identity in (v) holds by Proposition 4.1.2(i) and part (i) after expansion along ℓ.
The second identity in (v) holds by Proposition 4.1.2(i) and part (i) after expansion along r.

Structural Commentary: Reindexing, concatenation, and lexicographic listings transfer ordinary finite-sum laws to sets.
Key Insight: A finite-set sum depends on values, not listing order. □

Definition 6.2.9. *Sequences as Enumerations*
Let A be a set.

(i) A *sequence* in A is a map $x : \mathbb{N} \to A$, written $(x_n)_{n \in \mathbb{N}}$ with $x_n := x(n)$.
(ii) A sequence $x : \mathbb{N} \to A$ *enumerates* A iff $x(\mathbb{N}) = A$.
(iii) A sequence $x : \mathbb{N} \to A$ is a *distinct enumeration* of A iff $x : \mathbb{N} \to A$ is bijective.

Thus A is countable iff it has a distinct enumeration.

Proposition 6.2.10. *The Integers Are Countable*
The set $\mathbb{Z}$ is countable.

Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ by $f(2k) := k$ and $f(2k-1) := 1 - k$ for $k \in \mathbb{N}$.
Define $e : \mathbb{Z} \to \mathbb{N}$ by $e(z) := 2z$ for $z \geq 1$ and $e(z) := 1 - 2z$ for $z \leq 0$.
The even and odd branches partition $\mathbb{N}$, while the positive and nonpositive branches partition $\mathbb{Z}$.
Direct substitution gives $e \circ f = \mathrm{id}_{\mathbb{N}}$ and $f \circ e = \mathrm{id}_{\mathbb{Z}}$. Thus f is bijective by Theorem 3.3.2(i), so $\mathbb{Z}$ is countable.
Structural Commentary: Parity separates the positive and nonpositive integer branches.
Key Insight: Interleave the two one-sided integer lists. □

Theorem 6.2.11. *Infinite Subsets of Countable Sets*
Let A be countable and let $E \subseteq A$ be infinite. Then E is countable.

Proof. Choose a bijection $x : \mathbb{N} \to A$, write $x_n := x(n)$, and set $I_E := \{n \in \mathbb{N} : x_n \in E\}$. The restriction $x|_{I_E} : I_E \to E$ is bijective.
Hence I_E is infinite, since otherwise this bijection would make E finite. Each intersection $I_E \cap J_n$ is finite by Proposition 4.2.4(ii).
For $k \in \mathbb{N}$, set $M_k := \{n \in I_E : \#(I_E \cap J_n) = k\}$. We prove $(\forall k \in \mathbb{N})\, M_k \neq \emptyset$ by induction.
Let $m := \min I_E$ by well-ordering. Then $I_E \cap J_m = \{m\}$, so $m \in M_1$. Suppose $M_k \neq \emptyset$ and choose $m \in M_k$.
The set $I_E \setminus J_m$ is nonempty; otherwise $I_E \subseteq J_m$ would be finite by Proposition 4.2.4(ii). Let $r := \min(I_E \setminus J_m)$ by well-ordering.
Then $I_E \cap J_r = (I_E \cap J_m) \dot\cup \{r\}$, so $\#(I_E \cap J_r) = k + 1$ by Proposition 4.2.4(i). Thus $r \in M_{k+1}$; $(\forall k \in \mathbb{N})\, M_k \neq \emptyset$ by Theorem 2.6.2.
If $n < m$ belong to I_E, then $I_E \cap J_m$ is the disjoint union of $I_E \cap J_n$ and the nonempty set $I_E \cap (J_m \setminus J_n)$.
Hence $\#(I_E \cap J_n) < \#(I_E \cap J_m)$ by Proposition 4.2.4(i), so each M_k is a singleton. Let n_k be its unique element.
If $n \in I_E$ and $k := \#(I_E \cap J_n)$, then $n \in M_k$, so $n = n_k$. Thus $k \mapsto n_k$ is a bijection $\mathbb{N} \to I_E$.
The composite $k \mapsto x_{n_k}$ is a bijection $\mathbb{N} \to E$, so E is countable.
Structural Commentary: Finite initial-segment ranks canonically enumerate the indices that hit the subset.
Key Insight: Rank each selected index by the number of earlier selected indices. □

Proposition 6.2.12. *Subsets and Images of At Most Countable Sets*
Let A be at most countable.

(i) If $B \subseteq A$, then B is at most countable.
(ii) If $f : A \to Y$ is a function, then $f(A)$ is at most countable.

Proof.

(i) Suppose $B \subseteq A$. If A is finite, then B is finite by Proposition 4.2.4(ii).
If A is countable, then B is finite or, when infinite, countable by Theorem 6.2.11. Thus B is at most countable.

(ii) Suppose $f : A \to Y$ and first assume A is finite. Put $m := \#A$, choose a bijection $e : J_m \to A$.
Set $I := \{i \in J_m : (\forall j \in J_m)\, (j < i \Rightarrow f(e(j)) \neq f(e(i)))\}$. The set I is finite by Proposition 4.2.4(ii).
The map $i \mapsto f(e(i))$ on I is injective by the definition of I. Fix $y \in f(A)$. The nonempty set $\{i \in J_m : f(e(i)) = y\}$ has a least element by well-ordering, and that element belongs to I. Thus the map is also surjective onto $f(A)$, so $f(A)$ is finite.
Now suppose A is countable. Choose a bijection $e : \mathbb{N} \to A$ and set $I := \{i \in \mathbb{N} : (\forall j \in \mathbb{N})\, (j < i \Rightarrow f(e(j)) \neq f(e(i)))\}$.
The set I is at most countable by (i). The map $i \mapsto f(e(i))$ on I is injective by the definition of I.
Fix $y \in f(A)$. The nonempty set $\{i \in \mathbb{N} : f(e(i)) = y\}$ has a least element by well-ordering, and that element belongs to I.
Thus the map is also surjective onto $f(A)$, so $f(A)$ is at most countable.

Structural Commentary: Subsets restrict a listing; images retain the first index of each attained value.
Key Insight: Deleting entries or repetitions cannot increase cardinality. □

Proposition 6.2.13. *At Most Countable Sets Are Exactly Enumerable Images*
For a set B, the following are equivalent:

(i) B is at most countable.
(ii) $B = \emptyset$ or a surjection $f : \mathbb{N} \to B$ exists.

Thus a nonempty set is at most countable iff it is the image of a sequence.

Proof.

(*i*) $\Rightarrow$ (*ii*) If $B = \emptyset$, then (ii) holds. If B is countable, any bijection $\mathbb{N} \to B$ is surjective.
If B is finite and nonempty, put $m := \#B$, choose a bijection $e : J_m \to B$.
Define $f : \mathbb{N} \to B$ by $f(n) := e(n)$ for $n \leq m$ and $f(n) := e(m)$ for $n > m$. Then f is surjective.

(*ii*) $\Rightarrow$ (*i*) If $B = \emptyset$, then B is finite. If $f : \mathbb{N} \to B$ is surjective, then $B = f(\mathbb{N})$ is at most countable by Proposition 6.2.12(ii).

Structural Commentary: A finite list can repeat its final term; a countable list already supplies a surjection.
Key Insight: At most countable sets are sequence images or the empty set. □

Proposition 6.2.14. *Pairs of Positive Integers Are Countable*
The set $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. For $r \in \mathbb{N}_0$, put $t_r := \frac{r(r+1)}{2}$. Define $p : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by $p(i,j) := t_{i+j-2} + i$.
Fix $(i,j) \in \mathbb{N} \times \mathbb{N}$ and put $r := i + j - 1$; then $t_{r-1} < p(i,j) \leq t_r$. The intervals $(t_{r-1}, t_r]$ are disjoint for $r \in \mathbb{N}$.
Hence equality of two values of p forces equality of the coordinate sums and then equality of the first, and therefore also the second, coordinates. Thus p is injective. Fix $q \in \mathbb{N}$. The set $R_q := \{r \in \mathbb{N} : q \leq t_r\}$ is nonempty because $t_q \geq q$.
Let $r := \min R_q$ by well-ordering. Minimality gives $t_{r-1} < q \leq t_r$. Set $i := q - t_{r-1}$ and $j := r + 1 - i$.
Then $i, j \in \mathbb{N}$ and $p(i,j) = q$, so p is surjective. Hence p is a bijection and $\mathbb{N} \times \mathbb{N}$ is countable.
Structural Commentary: Triangular intervals separate the diagonals and recover both coordinates from each index.
Key Insight: Coordinate-sum diagonals occupy disjoint finite intervals. □

Theorem 6.2.15. *Countable Union of Countable Sets*
Let $(E_n)_{n\in\mathbb{N}}$ be countable sets and set $S := \bigcup_{n\in\mathbb{N}} E_n$. Then S is countable.

Proof. For $n \in \mathbb{N}$, let $\mathcal{B}_n$ be the set of bijections $\mathbb{N} \to E_n$. Then $(\forall n \in \mathbb{N})\, \mathcal{B}_n \neq \emptyset$.
Choose $(x_n)_{n\in\mathbb{N}}$ satisfying $(\forall n \in \mathbb{N})\, x_n \in \mathcal{B}_n$ by the axiom of choice, and write $x_{n,k} := x_n(k)$.
Choose a bijection $d : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ by Proposition 6.2.14. For $t \in \mathbb{N}$, write $d(t) = (n_t, k_t)$ and define $a : \mathbb{N} \to S$ by $a(t) := x_{n_t,k_t}$.
Then $a(\mathbb{N}) = S$, so S is at most countable by Proposition 6.2.13.
The countable set E_1 is contained in S, so S is not finite by Proposition 6.2.5(iv). Hence S is countable.
Structural Commentary: Choice selects row enumerations, and a pairing bijection flattens their two indices.
Key Insight: A countable array becomes one sequence after pairing its indices. □

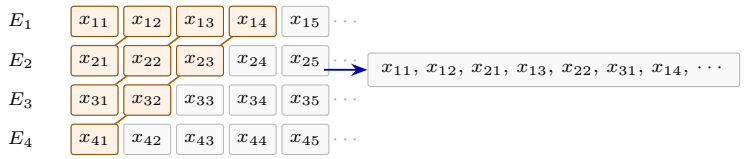

Figure 7: Diagonal scan of countable rows into one surjective sequence.

Figure 7 illustrates the scan used in Theorem 6.2.15.
Every indexed entry occurs on a finite diagonal; repeated values do not affect surjectivity.

Corollary 6.2.16. *At Most Countable Unions*
Let A be at most countable and let $(B_\alpha)_{\alpha\in A}$ be an indexed family of at most countable sets. Then $\bigcup_{\alpha\in A} B_\alpha$ is at most countable.

Proof. Set $T := \bigcup_{\alpha\in A} B_\alpha$. If $T = \emptyset$, then the conclusion holds. Suppose $T \neq \emptyset$ and set $I := \{\alpha \in A : B_\alpha \neq \emptyset\}$.
The set I is nonempty and at most countable by Proposition 6.2.12(i).
Choose a surjection $e : \mathbb{N} \to I$ by Proposition 6.2.13, and put $\alpha_j := e(j)$. For $j \in \mathbb{N}$, let $\mathcal{S}_j$ be the set of surjections $\mathbb{N} \to B_{\alpha_j}$.
Then $(\forall j \in \mathbb{N})\, \mathcal{S}_j \neq \emptyset$ by Proposition 6.2.13. Choose $(b_j)_{j\in\mathbb{N}}$ satisfying $(\forall j \in \mathbb{N})\, b_j \in \mathcal{S}_j$ by the axiom of choice, and write $b_{j,k} := b_j(k)$.
Choose a bijection $d : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ by Proposition 6.2.14. For $t \in \mathbb{N}$, write $d(t) = (j_t, k_t)$ and define $a : \mathbb{N} \to T$ by $a(t) := b_{j_t,k_t}$.
Then $a(\mathbb{N}) = T$, so T is at most countable by Proposition 6.2.13.
Structural Commentary: Surjections index the nonempty pieces and their elements before a pairing map flattens both indices.
Key Insight: Empty pieces can be discarded; all remaining rows admit sequence images. □

Corollary 6.2.17. *Countable Cartesian Products*
Let A, B be sets.

(i) If A and B are at most countable, then $A \times B$ is at most countable.
(ii) If A and B are countable, then $A \times B$ is countable.

Proof.

(i) For $a \in A$, set $S_a := \{a\} \times B$. The map $b \mapsto (a,b)$ is a bijection $B \to S_a$, so $(S_a)_{a\in A}$ is an indexed family of at-most-countable sets. Since $A \times B = \bigcup_{a\in A} S_a$, the product is at most countable by Corollary 6.2.16.

(ii) By (i), $A \times B$ is at most countable. Choose $b_0 \in B$. The map $a \mapsto (a, b_0)$ is a bijection $A \to A \times \{b_0\}$, so $A \times B$ contains the countable subset $A \times \{b_0\}$. Thus $A \times B$ is not finite by Proposition 6.2.5(iv), and it is therefore countable.

Structural Commentary: Slicing proves at-most-countability; one fixed-coordinate slice rules out finiteness.
Key Insight: A product is a union of copies of either factor. □

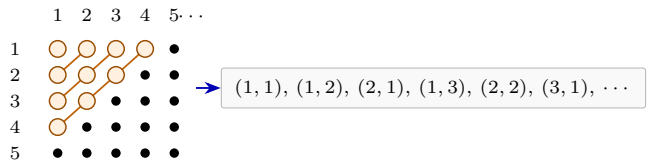

Figure 8: Diagonal enumeration of $\mathbb{N} \times \mathbb{N}$ by finite $i + j$ diagonals.

Figure 8 depicts the $\mathbb{N} \times \mathbb{N}$ pairing in Proposition 6.2.14.
Each pair lies on one finite coordinate-sum diagonal, which determines its interval of indices.

Theorem 6.2.18. *Finite Tuples over a Countable Set*
Let A be countable. For $n \in \mathbb{N}$, set $T_n := \{(a_1, \dots, a_n) : (\forall k \in J_n)\, a_k \in A\}$. Then T_n is countable.

Proof. For $r \in \mathbb{N}$, let P_r assert that T_r is countable. The map $a \mapsto (a)$ is a bijection $A \to T_1$, so P_1 holds. Assume P_r.
The product $T_r \times A$ is countable by Corollary 6.2.17(ii), and concatenation bijects $T_r \times A$ with T_{r+1}.
Thus P_{r+1} holds, so $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2.
Structural Commentary: Induction appends one coordinate and transfers countability across a binary product bijection.
Key Insight: Every fixed finite product is built one coordinate at a time. □

Corollary 6.2.19. *The Rationals Are Countable*
The set $\mathbb{Q}$ is countable.

Proof. Set $D := \{(p, q) \in \mathbb{Z} \times \mathbb{Z} : q \neq 0\}$ and define $\pi : D \to \mathbb{Q}$ by $\pi(p, q) := \frac{p}{q}$.
The set $\mathbb{Z} \times \mathbb{Z}$ is countable by Proposition 6.2.10 and Corollary 6.2.17(ii), so D is at most countable by Proposition 6.2.12(i).
Since $\mathbb{Q} = \pi(D)$, the set $\mathbb{Q}$ is at most countable by Proposition 6.2.12(ii).
The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ identifies a countable subset, so $\mathbb{Q}$ is not finite by Proposition 6.2.5(iv), and therefore $\mathbb{Q}$ is countable.
Structural Commentary: Integer pairs form an at-most-countable parameter set whose quotient map covers $\mathbb{Q}$.
Key Insight: Fractions are images of nonzero-denominator integer pairs. □

Proof Method 6.2.20. *Cardinality Proof Patterns*
Let A, B be sets. Choose the route matching the target cardinality claim.

(i) *Same cardinality*: Construct $f : A \to B$, then verify that f is injective and surjective.
(ii) *Countable*: Construct a bijection $\mathbb{N} \to A$. Alternatively, prove that A is at most countable and not finite.
(iii) *Uncountable*: Prove $A \neq \emptyset$. Fix an arbitrary map $s : \mathbb{N} \to A$ and construct $a_s \in A \setminus s(\mathbb{N})$.
Then no sequence surjects onto A, so A is not at most countable by Proposition 6.2.13.

Key Idea: Bijections certify equal cardinality; diagonal witnesses obstruct every proposed enumeration.

6.3 Ordinals, Transfinite Recursion, and Well-Foundedness

Definition 6.3.1. *Ordinals and Order Types*

(i) A set T is *transitive* iff $(\forall x \in T)\, x \subseteq T$.
(ii) An *ordinal* is a transitive set α on which $\in$ is a strict well-order. The relation $\in$ is irreflexive and transitive on α, distinct elements are $\in$-comparable, and each nonempty $A \subseteq \alpha$ contains m satisfying $(\forall a \in A)\, a \notin m$.
(iii) For ordinals α, β, write $\alpha < \beta$ iff $\alpha \in \beta$ and $\alpha \leq \beta$ iff $\alpha \subseteq \beta$.
(iv) Set $0 := \emptyset$. The *successor* of α is $\alpha + 1 := \alpha \cup \{\alpha\}$. The *finite ordinals* are 0 and its finite iterated successors.
An ordinal is *infinite* iff it is not finite. It is a *limit ordinal* iff it is nonzero and is not a successor.
(v) The least infinite ordinal is ω; its elements are the finite ordinals $0, 1, 2, \dots$.
If a well-ordered set $(S, \leq_S)$ is order-isomorphic to an ordinal δ in the sense of Definition 5.4.7, then δ is the *order type* of $(S, \leq_S)$.

Principle 6.3.2. *Ordinal Foundation Principles*
The following standard properties of von Neumann ordinals form the ordinal framework used below.

(i) *Ordinal comparison*: two ordinals are equal or one belongs to the other, and each nonempty set of ordinals has a least member.
(ii) *Strong transfinite induction*: if P is a property of ordinals satisfying $(\forall \alpha)\, [((\forall \beta < \alpha)\, P(\beta)) \Rightarrow P(\alpha)]$, then $(\forall \alpha)\, P(\alpha)$.
(iii) *Order types*: a well-ordered set is order-isomorphic to a unique ordinal.
(iv) *Well-ordering theorem*: under Principle 6.1.2, each set admits a well-order and is therefore bijective with an ordinal by (iii).

These principles concern ordinal comparison and recursive construction; do not prescribe a particular well-order on an arbitrary set.

Definition 6.3.3. *Well-Founded Relation*
Let $\prec$ be a relation on a set X.

(i) The *predecessor set* of $x \in X$ is $\mathrm{Pred}_{\prec}(x) := \{y \in X : y \prec x\}$.
(ii) For $A \subseteq X$, an element $m \in A$ is *$\prec$-minimal in A* iff $\mathrm{Pred}_{\prec}(m) \cap A = \emptyset$.
(iii) The relation $\prec$ is *well-founded* iff $(\forall A \subseteq X)\, [A \neq \emptyset \Rightarrow (\exists m \in A)\ \mathrm{Pred}_{\prec}(m) \cap A = \emptyset]$.

Neither totality nor transitivity is required.

Theorem 6.3.4. *Well-Founded Induction*
Let $\prec$ be a well-founded relation on X, and let P be a property of elements of X.
If $(\forall x \in X)\, [((\forall y \in \mathrm{Pred}_{\prec}(x))\, P(y)) \Rightarrow P(x)]$, then $(\forall x \in X)\, P(x)$.

Proof. Set $B := \{x \in X : \neg P(x)\}$. Suppose $B \neq \emptyset$, and choose a $\prec$-minimal $b \in B$. If $y \in \mathrm{Pred}_{\prec}(b)$, then $y \notin B$ by minimality.
Thus $(\forall y \in \mathrm{Pred}_{\prec}(b))\, P(y)$. The induction premise gives $P(b)$, contradicting $b \in B$. Hence $B = \emptyset$ and $(\forall x \in X)\, P(x)$.
Structural Commentary: A least-counterexample contradiction applies the induction premise at one minimal failure.
Key Insight: A minimal failure has only successful predecessors. □

Principle 6.3.5. *Transfinite and Well-Founded Recursion*
Recursive values may depend on the full restriction of the desired function to earlier arguments.

(i) *Ordinal recursion*: let θ be an ordinal, and suppose $G(\alpha, g)$ is assigned whenever $\alpha < \theta$ and $\operatorname{dom} g = \alpha$.
Then $(\exists! F)\,[\operatorname{dom} F = \theta \wedge (\forall \alpha < \theta)\, F(\alpha) = G(\alpha, F|_\alpha)]$.

(ii) *Well-founded recursion*: let $\prec$ be well-founded on X, and suppose $H(x, g)$ is assigned whenever $x \in X$ and $\operatorname{dom} g = \operatorname{Pred}_\prec(x)$.
Then $(\exists! F)\,[\operatorname{dom} F = X \wedge (\forall x \in X)\, F(x) = H(x, F|_{\operatorname{Pred}_\prec(x)})]$.

In (i), the least disagreement proves uniqueness; in (ii), uniqueness follows by Theorem 6.3.4 for the property that two candidates agree. The rules apply only when the value at an argument depends on values at its predecessors.

Definition 6.3.6. *Ordinal Arithmetic*
For a set A of ordinals, set $\sup A := \bigcup A$. Fix an ordinal α.
Transfinite recursion in the second argument defines the following operations.
Each limit clause assumes that λ is a nonzero limit ordinal.

(i) *Addition*: $\alpha + 0 := \alpha$ and $\alpha + (\beta + 1) := (\alpha + \beta) + 1$. For a limit λ, $\alpha + \lambda := \sup\{\alpha + \beta : \beta < \lambda\}$.
(ii) *Multiplication*: $\alpha \cdot 0 := 0$ and $\alpha \cdot (\beta + 1) := \alpha \cdot \beta + \alpha$. For a limit λ, $\alpha \cdot \lambda := \sup\{\alpha \cdot \beta : \beta < \lambda\}$.
(iii) *Exponentiation*: $\alpha^0 := 1$ and $\alpha^{\beta+1} := \alpha^\beta \cdot \alpha$. For a limit λ, $\alpha^\lambda := \sup\{\alpha^\beta : \beta < \lambda\}$.

The successor clause for addition agrees with Definition 6.3.1.

Example 6.3.7. *Ordinal Arithmetic Remembers Order*
The limit clauses give $1 + \omega = \sup\{1 + n : n < \omega\} = \omega$ and $2 \cdot \omega = \sup\{2 \cdot n : n < \omega\} = \omega$.
The successor clauses give $\omega + 1 > \omega$ and $\omega \cdot 2 = \omega + \omega > \omega$. Thus $1 + \omega < \omega + 1$ and $2 \cdot \omega < \omega \cdot 2$.
Ordinal addition and multiplication retain the order of concatenation and are not commutative.

Definition 6.3.8. *Cardinals as Initial Ordinals*

(i) An ordinal κ is a *cardinal*, or *initial ordinal*, iff $(\forall \beta < \kappa)\, \beta \not\sim_\# \kappa$.
(ii) If a set X is bijective with an ordinal, its *cardinality* $|X|$ is the least such ordinal.
The minimum exists by Principle 6.3.2(i), applied to the nonempty set of candidates no larger than a fixed witness.
Under Principle 6.1.2, every set has a cardinality by Principle 6.3.2(iv).
(iii) A cardinal is *infinite* iff it is not a finite ordinal, and $\aleph_0 := |\omega|$.

Definition 6.3.9. *Cofinal Subsets and Cofinality*
Let α be an ordinal.

(i) A subset $C \subseteq \alpha$ is *cofinal in* α, or *unbounded in* α, iff $(\forall \beta < \alpha)(\exists \gamma \in C)\, \beta \le \gamma$.
(ii) A function $f : \delta \to \alpha$ is *cofinal* iff $f(\delta)$ is cofinal in α.
(iii) The *cofinality* of α is $\operatorname{cf}(\alpha) := \min\{\delta : (\exists f : \delta \to \alpha)(\forall \beta < \alpha)(\exists \xi < \delta)\, \beta \le f(\xi)\}$.

The minimum exists because id_α is cofinal and nonempty sets of ordinals have least elements.
If α is a limit ordinal, then C is cofinal iff $\sup C = \alpha$. Indeed, if C is not cofinal, then some member of α strictly bounds C from above. Conversely, cofinality above each successor below α forces $\bigcup C = \alpha$. The empty function is cofinal in 0, while $0 \mapsto \alpha$ is cofinal from 1 to $\alpha + 1$. Since an empty range is not cofinal in a successor, $\operatorname{cf}(0) = 0$ and $\operatorname{cf}(\alpha + 1) = 1$.

Lemma 6.3.10. *Nondecreasing Cofinal Witness*
Let λ be a nonzero limit ordinal and set $\delta := \operatorname{cf}(\lambda)$.

(i) The ordinal δ is nonzero and is a limit ordinal.
(ii) There is a cofinal function $f : \delta \to \lambda$ satisfying $(\forall \xi \le \eta < \delta)\, f(\xi) \le f(\eta)$.

Proof.

(i) Choose a cofinal function $h : \delta \to \lambda$. Since an empty range is not cofinal in λ, $\delta \neq 0$.
Suppose $\delta = \varepsilon + 1$. The restriction $h|_\varepsilon$ is not cofinal by the minimality of δ, so $(\exists \beta < \lambda)(\forall \eta < \varepsilon)\, h(\eta) < \beta$.
Set $\rho := \max\{\beta, h(\varepsilon)\} + 1$. Since λ is a limit ordinal, $\rho < \lambda$ and $(\forall \eta < \delta)\, h(\eta) < \rho$, contradicting the cofinality of h. QED.
(ii) For $\xi < \delta$, set $f(\xi) := \sup\{h(\eta) : \eta \le \xi\}$. Since $\xi + 1 < \delta$, range of $h|_{\xi+1}$ is not cofinal and is bounded below λ; hence $f(\xi) < \lambda$.
If $\xi \le \eta < \delta$, then the set defining $f(\xi)$ is contained in the set defining $f(\eta)$, so $f(\xi) \le f(\eta)$.
Finally, $h(\xi) \le f(\xi)$ for $\xi < \delta$, and therefore f is cofinal.

Structural Commentary: Minimality first excludes successor domains, then successive suprema monotone-ize a cofinal witness.
Key Insight: No proper initial segment of a minimal cofinal domain is cofinal. □

Definition 6.3.11. *Regular and Singular Cardinals*
An infinite cardinal κ is *regular* iff $\operatorname{cf}(\kappa) = \kappa$, and it is *singular* iff $\operatorname{cf}(\kappa) < \kappa$.

Theorem 6.3.12. *Idempotence and Cardinality of Cofinality*
Let α be an ordinal and set $\delta := \operatorname{cf}(\alpha)$.

(i) $\operatorname{cf}(\delta) = \delta$.
(ii) The ordinal δ is a cardinal.
(iii) If α is a nonzero limit ordinal, then δ is an infinite regular cardinal.

Proof.

(i) If $\alpha = 0$, then $\delta = 0$ and $\operatorname{cf}(\delta) = 0$. If α is a successor, then $\delta = 1$ and $\operatorname{cf}(\delta) = 1$.
Suppose α is a nonzero limit ordinal. A nondecreasing cofinal function $f : \delta \to \alpha$ exists by Lemma 6.3.10(ii); fix one.
The ordinal δ is a nonzero limit ordinal by Lemma 6.3.10(i).
A nondecreasing cofinal function $g : \operatorname{cf}(\delta) \to \delta$ therefore exists by Lemma 6.3.10(ii).
For $\beta < \alpha$, choose $\xi < \delta$ with $\beta \le f(\xi)$ and then $\eta < \operatorname{cf}(\delta)$ with $\xi \le g(\eta)$. Monotonicity gives $\beta \le f(g(\eta))$, so $f \circ g$ is cofinal in α.
The minimality of δ gives $\delta \le \operatorname{cf}(\delta)$, while id_δ gives $\operatorname{cf}(\delta) \le \delta$. Hence $\operatorname{cf}(\delta) = \delta$.
(ii) If δ were not a cardinal, choose $\beta < \delta$ and a bijection $b : \beta \to \delta$. Its range is cofinal, so $\operatorname{cf}(\delta) \le \beta < \delta$, contradicting (i). QED.
(iii) Let α be a nonzero limit ordinal. The ordinal δ is nonzero and limit by Lemma 6.3.10(i), hence it is infinite.
It is a cardinal by (ii), and $\operatorname{cf}(\delta) = \delta$ by (i), so it is regular.

Structural Commentary: Composition forbids cofinality from shrinking twice; initiality and regularity then follow.
Key Insight: Minimal cofinal domains are fixed points of cofinality. □

Example 6.3.13. *Basic Ordinal Cofinalities*
An empty range is bounded. For a nonempty finite range in a limit ordinal, recursion produces a maximum; its successor still lies below the limit and bounds the range. Hence no finite ordinal is a cofinal domain for a nonzero limit ordinal.
The maps $n \mapsto n$, $n \mapsto \omega + n$, and $n \mapsto \omega^n$ from ω are cofinal in ω, $\omega \cdot 2$, and ω^ω, respectively.
Each target therefore has cofinality at most ω and no finite cofinality. Thus $\mathrm{cf}(\omega) = \mathrm{cf}(\omega \cdot 2) = \mathrm{cf}(\omega^\omega) = \omega$.
Cofinality measures the shortest unbounded approach, not the order type or cardinality of its target.

Proposition 6.3.14. *Cardinality Is Canonical and Bijection-Invariant*
Assume Principle 6.1.2. For sets X, Y:

(i) $X \sim_\# |X|$, *and* $|X|$ *is a cardinal.*
(ii) $|X| = |Y| \iff X \sim_\# Y$.

Proof.

(*i*) The relation $X \sim_\# |X|$ holds by the definition of $|X|$. If $\beta < |X|$ and $\beta \sim_\# |X|$, then $\beta \sim_\# X$ by Proposition 6.2.2, contradicting the minimality of $|X|$. Thus $|X|$ is a cardinal.
(*ii*) ($\Rightarrow$) If $|X| = |Y|$, then $X \sim_\# |X| = |Y| \sim_\# Y$, so $X \sim_\# Y$ by Proposition 6.2.2.
(*ii*) ($\Leftarrow$) If $X \sim_\# Y$, then $(\forall \gamma)\,[\gamma \sim_\# X \iff \gamma \sim_\# Y]$ by Proposition 6.2.2.
The least ordinals satisfying these equivalent conditions are equal, so $|X| = |Y|$.

Structural Commentary: Minimal ordinal representatives identify exactly the equivalence classes of bijections.
Key Insight: Cardinality retains only bijection type. □

Proposition 6.3.15. *Finite Cardinality and $\aleph_0$ Agree with Earlier Notation*
Assume Principle 6.1.2.

(i) If X is finite and $\#X = n$, then $|X| = n$, where n on the right is the corresponding finite ordinal.
(ii) $\aleph_0 = \omega$.
(iii) A set X is countable iff $|X| = \aleph_0$.
(iv) The cardinal $\aleph_0$ is the least infinite cardinal.

Proof.

(*i*) The unique map $J_0 \to 0$ is bijective, and for $n \in \mathbb{N}$ the map $q_n : J_n \to n$ defined by $q_n(k) := k - 1$ is bijective.
Since $X \sim_\# J_n$ and $J_n \sim_\# n$, $X \sim_\# n$ by Proposition 6.2.2.
If $\beta < n$ and $\beta \sim_\# X$, choose $m < n$ with $\beta = m$. Then $J_m \sim_\# J_n$ by Proposition 6.2.2.
Hence $m = n$ by Proposition 4.2.2, contradicting $m < n$. Thus n is the least ordinal bijective with X, so $|X| = n$.
(*ii*) The map $q : \mathbb{N} \to \omega$ defined by $q(k) := k - 1$ is bijective. If $\beta < \omega$, choose $m \in \mathbb{N}_0$ with $\beta = m$.
If $\beta \sim_\# \omega$, then a witnessing bijection, q, and the bijection $J_m \to m$ from (i) compose to give an injection $\mathbb{N} \to J_m$, which is impossible by Proposition 6.2.5(i). Hence ω is the least ordinal bijective with itself, so $\aleph_0 = |\omega| = \omega$.
(*iii*) ($\Rightarrow$) If X is countable, then $X \sim_\# \mathbb{N}$ by Definition 6.2.4. Since $\mathbb{N} \sim_\# \omega$ by (ii), $X \sim_\# \omega$ by Proposition 6.2.2.
Hence $|X| = \aleph_0$ by Proposition 6.3.14(ii) and (ii).
(*iii*) ($\Leftarrow$) If $|X| = \aleph_0$, then $X \sim_\# \omega$ by Proposition 6.3.14(ii) and (ii). Since $\omega \sim_\# \mathbb{N}$ by (ii), $X \sim_\# \mathbb{N}$ by Proposition 6.2.2.
Thus X is countable by Definition 6.2.4.
(*iv*) An infinite cardinal is an infinite ordinal and is therefore at least the least infinite ordinal $\omega = \aleph_0$.

Structural Commentary: Explicit finite and countable bijections identify the initial ordinals selected by cardinality.
Key Insight: General cardinality extends both finite size and countability. □

Proposition 6.3.16. *Cardinality Comparison by Injections*
Assume Principle 6.1.2. For sets X, Y, $|X| \le |Y|$ iff an injection $X \to Y$ exists.

Proof. Set $\kappa := |X|$ and $\lambda := |Y|$. The ordinals κ, λ are cardinals and satisfy $X \sim_\# \kappa$ and $Y \sim_\# \lambda$ by Proposition 6.3.14(i).
Choose bijections $u : X \to \kappa$ and $v : Y \to \lambda$.
($\Rightarrow$) If $\kappa \le \lambda$, then $v^{-1} \circ \iota \circ u : X \to Y$ is injective, where $\iota : \kappa \hookrightarrow \lambda$ is inclusion.
($\Leftarrow$) Suppose $e : X \to Y$ is injective. Then $v \circ e \circ u^{-1} : \kappa \to \lambda$ is injective.
If $\lambda < \kappa$, inclusion injects λ into κ, so $\kappa \sim_\# \lambda$ by Theorem 6.2.3.
This contradicts the initiality of κ. Hence ordinal comparison gives $\kappa \le \lambda$.

Structural Commentary: Conjugate an injection to initial ordinals and exclude the reverse strict comparison.
Key Insight: Schroeder–Bernstein prevents a cardinal from embedding into a smaller cardinal. □

Lemma 6.3.17. *Infinite Cardinals Are Limit Ordinals*
Every infinite cardinal is a limit ordinal.

Proof. Suppose an infinite cardinal has the form $\kappa = \alpha + 1$. The ordinal α is infinite, otherwise its successor would be finite, so $\omega \le \alpha$.
Define $f : \alpha + 1 \to \alpha$ by $f(\alpha) := 0$, by $f(\xi) := \xi + 1$ when $\xi < \omega$, and by $f(\xi) := \xi$ when $\omega \le \xi < \alpha$.
The three domain pieces are disjoint, and their images are $\{0\}$, $\omega \setminus \{0\}$, and $\{\xi : \omega \le \xi < \alpha\}$.
These images are disjoint and cover α, so f is bijective. Thus κ is bijective with the smaller ordinal α, contradicting that κ is a cardinal.
Hence κ is neither zero nor a successor ordinal, and therefore it is a limit ordinal.
Structural Commentary: A direct folding bijection contradicts initiality for any alleged infinite successor cardinal.
Key Insight: An infinite successor ordinal absorbs its final endpoint. □

6.4 Diagonalization and Power Sets

Theorem 6.4.1. *Cantor's Theorem*
Let X be a set. No function $f : X \to \mathcal{P}(X)$ is surjective; hence $X \not\sim_{\#} \mathcal{P}(X)$.

Proof. Fix $f : X \to \mathcal{P}(X)$ and set $D_f := \{x \in X : x \notin f(x)\}$. Then $D_f \subseteq X$, so $D_f \in \mathcal{P}(X)$.
If $a \in X$ and $D_f = f(a)$, then $a \in D_f \iff a \in f(a)$, contrary to $a \in D_f \iff a \notin f(a)$.
Hence $(\forall a \in X)\, D_f \neq f(a)$, so $D_f \notin f(X)$ and f is not surjective. Since a bijection must be surjective, $X \not\sim_{\#} \mathcal{P}(X)$.
Structural Commentary: The diagonal subset differs from each image set at its own indexing element.
Key Insight: Diagonal self-membership forces $(\forall a \in X)\, D_f \neq f(a)$. □

Proposition 6.4.2. *Binary Sequences Model a Power Set*
Let $A := \{0,1\}^{\mathbb{N}}$. Define $\Phi : A \to \mathcal{P}(\mathbb{N})$ by $\Phi(s) := \{n \in \mathbb{N} : s(n) = 1\}$. Then Φ is bijective.

Proof. For $E \subseteq \mathbb{N}$, define $\chi_E : \mathbb{N} \to \{0,1\}$ by $\chi_E(n) := 1$ when $n \in E$ and $\chi_E(n) := 0$ otherwise. Define $\Psi : \mathcal{P}(\mathbb{N}) \to A$ by $\Psi(E) := \chi_E$.
If $E \subseteq \mathbb{N}$ and $n \in \mathbb{N}$, then $n \in \Phi(\Psi(E)) \iff \chi_E(n) = 1 \iff n \in E$. Hence $\Phi(\Psi(E)) = E$ by Assumption 1.3.2.
If $s \in A$ and $n \in \mathbb{N}$, then $\chi_{\Phi(s)}(n) = 1 \iff s(n) = 1$. Both values lie in $\{0,1\}$, so $\chi_{\Phi(s)}(n) = s(n)$. Thus $(\forall n \in \mathbb{N})\, \chi_{\Phi(s)}(n) = s(n)$.
Hence $\Psi(\Phi(s)) = s$ by Assumption 3.1.2. The maps Φ and Ψ are inverse, so Φ is bijective by Theorem 3.3.2.
Structural Commentary: Characteristic functions construct a two-sided inverse to the subset-extraction map.
Key Insight: A membership bit records exactly one subset decision. □

Theorem 6.4.3. *Binary Sequences Are Uncountable*
Let $A := \{0,1\}^{\mathbb{N}}$. Then A is uncountable.

Proof. Define $j : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ by $j(n) := \{n\}$, and define $\Phi : A \to \mathcal{P}(\mathbb{N})$ by $\Phi(s) := \{n \in \mathbb{N} : s(n) = 1\}$.
The map j is injective because $j(m) = j(n)$ implies $m = n$. The map Φ is bijective by Proposition 6.4.2, so $\Phi^{-1} \circ j : \mathbb{N} \to A$ is injective.
If A were finite, choose $m \in \mathbb{N}_0$ and a bijection $q : A \to J_m$.
Then $q \circ \Phi^{-1} \circ j : \mathbb{N} \to J_m$ would be injective. No such injection exists by Proposition 6.2.5(i), a contradiction.
Thus A is not finite. Suppose A were countable. Choose a bijection $e : \mathbb{N} \to A$, and write $s_n := e(n)$.
Define $s : \mathbb{N} \to \{0,1\}$ by $s(k) := 1 - s_k(k)$. Then $s \in A$ and $(\forall n \in \mathbb{N})\, s(n) \neq s_n(n)$, so $(\forall n \in \mathbb{N})\, s \neq e(n)$.
This contradicts the surjectivity of e. Hence A is neither finite nor countable, so it is uncountable.
Structural Commentary: An embedded copy of $\mathbb{N}$ excludes finiteness; a diagonal flip excludes enumerability.
Key Insight: The nth flipped bit separates the new sequence from row n. □

$$
\begin{array}{lcl}
s_1 & = & \boxed{a_{11}}\, a_{12}\ a_{13}\ a_{14}\ a_{15} \\
s_2 & = & a_{21}\ \boxed{a_{22}}\, a_{23}\ a_{24}\ a_{25} \\
s_3 & = & a_{31}\ a_{32}\ \boxed{a_{33}}\, a_{34}\ a_{35} \quad \cdots \\
s_4 & = & a_{41}\ a_{42}\ a_{43}\ \boxed{a_{44}}\, a_{45} \\
s_5 & = & a_{51}\ a_{52}\ a_{53}\ a_{54}\ \boxed{a_{55}}
\end{array}
$$

Figure 9: Boxed entries $a_{nn} = s_n(n)$ whose bit flips define a sequence distinct from every listed row.

Figure 9 marks the diagonal entries used in Theorem 6.4.3.
Flipping the nth boxed bit forces $s(n) \neq s_n(n)$, so the constructed sequence cannot equal row n.

6.5 Cardinal Comparison and the Continuum

Throughout this subsection, assume Principle 6.1.2 and use the initial-ordinal cardinality $|X|$ from Definition 6.3.8.

Definition 6.5.1. *Cardinal Comparison and Arithmetic*
For sets A, B, define the *function set* by $A^B := \{f \mid f : B \to A\}$. Let κ, λ be cardinals.

(i) *Comparison*: the relations $\kappa \leq \lambda$ and $\kappa < \lambda$ are the corresponding ordinal comparisons.
(ii) The *cardinal sum* is $\kappa + \lambda := |(\{0\} \times \kappa) \cup (\{1\} \times \lambda)|$.
(iii) The *cardinal product* is $\kappa\lambda := |\kappa \times \lambda|$.
(iv) The *cardinal power* is $\kappa^\lambda := |\{f \mid f : \lambda \to \kappa\}|$.

These are cardinal operations. Cardinal arguments distinguish them from the ordinal operations of Definition 6.3.6.

Proposition 6.5.2. *Cardinal Comparison and Operations on Representatives*
Let A, B be sets, and set $\kappa := |A|$ and $\lambda := |B|$.

(i) $\kappa \leq \lambda$ iff an injection $A \to B$ exists.
(ii) $\kappa < \lambda$ iff an injection $A \to B$ exists and $A \not\sim_{\#} B$.
(iii) $\kappa + \lambda = |(\{0\} \times A) \cup (\{1\} \times B)|$. If $A \cap B = \emptyset$, then $\kappa + \lambda = |A \cup B|$.
(iv) $\kappa\lambda = |A \times B|$.
(v) $\kappa^\lambda = |A^B|$.

Proof. Choose bijections $u : A \to \kappa$ and $v : B \to \lambda$ by Proposition 6.3.14(i).

(i) ($\Rightarrow$) If $\kappa \leq \lambda$, then an injection $A \to B$ exists by Proposition 6.3.16.
(i) ($\Leftarrow$) If an injection $A \to B$ exists, then $\kappa \leq \lambda$ by Proposition 6.3.16.
(ii) ($\Rightarrow$) Suppose $\kappa < \lambda$. Then $\kappa \leq \lambda$ and $\kappa \neq \lambda$. An injection $A \to B$ exists by (i), and $A \not\sim_{\#} B$ by Proposition 6.3.14(ii).
(ii) ($\Leftarrow$) Suppose an injection $A \to B$ exists and $A \not\sim_{\#} B$. Then $\kappa \leq \lambda$ by (i) and $\kappa \neq \lambda$ by Proposition 6.3.14(ii), so $\kappa < \lambda$.
(iii) Set $U := (\{0\} \times A) \cup (\{1\} \times B)$ and $V := (\{0\} \times \kappa) \cup (\{1\} \times \lambda)$. Define $F : U \to V$ by $F(0,a) := (0, u(a))$ and $F(1,b) := (1, v(b))$.
The inverse sends $(0,\xi)$ to $(0, u^{-1}(\xi))$ and $(1,\eta)$ to $(1, v^{-1}(\eta))$, so F is bijective by Theorem 3.3.2.
Hence $|U| = |V|$ by Proposition 6.3.14(ii), proving the first equality.
If $A \cap B = \emptyset$, define $G : U \to A \cup B$ by $G(0,a) := a$ and $G(1,b) := b$.
Its inverse sends $x \in A$ to $(0,x)$ and $x \in B$ to $(1,x)$, so G is bijective. Thus $\kappa + \lambda = |A \cup B|$ by Proposition 6.3.14(ii).

(iv) Define $P : A \times B \to \kappa \times \lambda$ by $P(a,b) := (u(a), v(b))$. Its inverse is $(\xi,\eta) \mapsto (u^{-1}(\xi), v^{-1}(\eta))$, so P is bijective by Theorem 3.3.2. Therefore $\kappa\lambda = |A \times B|$ by Proposition 6.3.14(ii).
(v) Define $T : A^B \to \{g \mid g : \lambda \to \kappa\}$ by $T(f) := u \circ f \circ v^{-1}$. Its inverse sends $g : \lambda \to \kappa$ to $u^{-1} \circ g \circ v$, so T is bijective by Theorem 3.3.2.
Hence $\kappa^\lambda = \left|A^B\right|$ by Proposition 6.3.14(ii).

Structural Commentary: Chosen bijections transport comparison and all three cardinal operations.
Key Insight: Bijections act coordinatewise on tags, pairs, and functions. □

Definition 6.5.3. *Continuum Cardinal*
The *continuum cardinal* is $\mathfrak{c} := 2^{\aleph_0}$.

Proposition 6.5.4. *Models of the Continuum Cardinal*
$\mathfrak{c} = \left|\{0,1\}^{\mathbb{N}}\right| = |\mathcal{P}(\mathbb{N})|$.

Proof. The map $q : J_2 \to \{0,1\}$ defined by $q(k) := k - 1$ is bijective. Thus $|\{0,1\}| = 2$ and $|\mathbb{N}| = \aleph_0$ by Proposition 6.3.15(i),(iii).
Hence $\mathfrak{c} = \left|\{0,1\}^{\mathbb{N}}\right|$ by Proposition 6.5.2(v). The relation $\{0,1\}^{\mathbb{N}} \sim_{\#} \mathcal{P}(\mathbb{N})$ holds by Proposition 6.4.2.
Hence $\left|\{0,1\}^{\mathbb{N}}\right| = |\mathcal{P}(\mathbb{N})|$ by Proposition 6.3.14(ii).
Structural Commentary: Finite and countable representatives reduce cardinal exponentiation to binary functions and subsets.
Key Insight: A binary function is the characteristic function of one subset. □

Corollary 6.5.5. *The Continuum Strictly Exceeds $\aleph_0$*
$\aleph_0 < \mathfrak{c}$.

Proof. The map $j : \mathbb{N} \to \mathcal{P}(\mathbb{N})$, $j(n) := \{n\}$, is injective since $j(m) = j(n) \Rightarrow m = n$.
No bijection $\mathbb{N} \to \mathcal{P}(\mathbb{N})$ exists by Theorem 6.4.1. Therefore $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$ by Proposition 6.5.2(ii).
The identities $|\mathbb{N}| = \aleph_0$ and $|\mathcal{P}(\mathbb{N})| = \mathfrak{c}$ hold by Proposition 6.3.15(iii) and Proposition 6.5.4, respectively. Hence $\aleph_0 < \mathfrak{c}$.
Structural Commentary: Singleton embedding gives weak comparison, while Cantor's theorem rules out equality.
Key Insight: The diagonal obstruction makes the continuum strictly larger. □

6.6 Cardinal Arithmetic

Throughout this subsection, assume Principle 6.1.2 and use the cardinal operations from Definition 6.5.1.

Proposition 6.6.1. *Cardinal Arithmetic Reference Laws*
Let κ, λ, μ be cardinals.
(i) $\kappa + \lambda = \lambda + \kappa$ *and* $\kappa\lambda = \lambda\kappa$.
(ii) $\kappa + (\lambda + \mu) = (\kappa + \lambda) + \mu$ *and* $\kappa(\lambda\mu) = (\kappa\lambda)\mu$.
(iii) $\kappa(\lambda + \mu) = \kappa\lambda + \kappa\mu$.
(iv) $\kappa^{\lambda+\mu} = \kappa^\lambda \kappa^\mu$, $(\kappa^\lambda)^\mu = \kappa^{\lambda\mu}$, *and* $(\kappa\lambda)^\mu = \kappa^\mu \lambda^\mu$.

Proof. Set $A := \{0\} \times \kappa$, $B := \{1\} \times \lambda$, and $C := \{2\} \times \mu$.
The sets are pairwise disjoint, and their coordinate projections are bijections onto κ, λ, μ, respectively.
Thus $|A| = \kappa$, $|B| = \lambda$, and $|C| = \mu$ by Proposition 6.3.14(ii) and Definition 6.3.8.
(i) The identity is a bijection $A \cup B \to B \cup A$. The map $(a,b) \mapsto (b,a)$ is a bijection $A \times B \to B \times A$ with itself as inverse.
(ii) The equality $A \cup (B \cup C) = (A \cup B) \cup C$ holds.
The map $(a,(b,c)) \mapsto ((a,b),c)$ is a bijection $A \times (B \times C) \to (A \times B) \times C$ with inverse $((a,b),c) \mapsto (a,(b,c))$.
(iii) The equality $A \times (B \cup C) = (A \times B) \cup (A \times C)$ holds. The union on the right is disjoint because $B \cap C = \emptyset$.
(iv) Define $R : A^{B \cup C} \to A^B \times A^C$ by $R(f) := (f|_B, f|_C)$. Its inverse is $R^{-1}(g,h) := g \cup h$; disjointness makes this union a function.
Define $Q : (A^B)^C \to A^{B \times C}$ by $Q(F)(b,c) := F(c)(b)$. Its inverse sends h to $c \mapsto (b \mapsto h(b,c))$. Let $\pi_A(a,b) := a$ and $\pi_B(a,b) := b$.
The map $D : (A \times B)^C \to A^C \times B^C$ defined by $D(h) := (\pi_A \circ h, \pi_B \circ h)$ has inverse $D^{-1}(f,g)(c) := (f(c), g(c))$.
Claims (i)–(iv) follow from these set equalities and bijections by Proposition 6.5.2(iii)–(v).
Structural Commentary: Set identities and explicit bijections transport representative arithmetic to cardinal laws.
Key Insight: Restriction, currying, and projections control the exponent laws. □

Theorem 6.6.2. *Infinite Cardinal Square Law*
If κ is an infinite cardinal, then $|\kappa \times \kappa| = \kappa$.

Proof.
(i) *Induction setup*: For an ordinal α, let $P(\alpha)$ mean that, if α is an infinite cardinal, then $|\alpha \times \alpha| = \alpha$.
Fix an ordinal κ and assume $(\forall \beta < \kappa)\, P(\beta)$. It suffices by Principle 6.3.2(ii) to prove $P(\kappa)$.
The assertion $P(\kappa)$ is vacuous unless κ is an infinite cardinal, so assume that it is.
(ii) *Well-order construction*: Define $r : \kappa \times \kappa \to \kappa$ by $r(\xi,\eta) := \max\{\xi,\eta\}$.
For $p = (\beta,\beta')$ and $q = (\gamma,\gamma')$ in $\kappa \times \kappa$, declare $p \prec q$ when the first applicable condition holds:
  (a) $r(p) < r(q)$;
  (b) $r(p) = r(q)$ and $\beta < \gamma$;
  (c) $r(p) = r(q)$, $\beta = \gamma$, and $\beta' < \gamma'$.

Put $p \preceq q$ iff $p = q$ or $p \prec q$. Ordinal comparison makes these clauses exhaustive for distinct pairs.
Comparison by the first differing entry of $(r(p), \beta, \beta')$ is transitive and antisymmetric; hence $\preceq$ is a total order.
If $\emptyset \neq S \subseteq \kappa \times \kappa$, set $R_S := \{r(p) : p \in S\}$. The nonempty set R_S has a least element ρ by Principle 6.3.2(i).
Set $I_S := \{\xi < \kappa : (\exists \eta < \kappa)\,[(\xi,\eta) \in S \wedge r(\xi,\eta) = \rho]\}$; it is nonempty by the definition of ρ.
The minimum $\xi_0 := \min I_S$ exists by Principle 6.3.2(i).
Set $J_S := \{\eta < \kappa : (\xi_0,\eta) \in S \wedge r(\xi_0,\eta) = \rho\}$; it is nonempty by the definition of I_S.
The minimum $\eta_0 := \min J_S$ exists by Principle 6.3.2(i). Then (ξ_0,η_0) is least in S, so $\preceq$ is a well-order.
(iii) *Ordinal comparison*: Choose an ordinal δ and an order isomorphism $f : (\delta, \leq) \to (\kappa \times \kappa, \preceq)$ by Principle 6.3.2(iii).
The map $\xi \mapsto (\xi, 0)$ injects κ into $\kappa \times \kappa$. Since $|\kappa| = \kappa$ by Definition 6.3.8, $\kappa \leq |\kappa \times \kappa|$ by Proposition 6.3.16.
The equality $|\kappa \times \kappa| = |\delta|$ holds by Proposition 6.3.14(ii), and $|\delta| \leq \delta$ by Definition 6.3.8. Thus $\kappa \leq |\kappa \times \kappa| = |\delta| \leq \delta$.

(iv) *Initial-segment contradiction*: Suppose $\kappa < \delta$. Write $f(\kappa) = (\beta, \beta')$ and set $X := \{q \in \kappa \times \kappa : q \prec (\beta, \beta')\}$.
The restriction $f|_\kappa : \kappa \to X$ is bijective because f preserves and reflects order.
Hence $|X| = \kappa$ by Proposition 6.3.14(ii) and Definition 6.3.8.
Set $\alpha := \max\{\beta, \beta'\} + 1$. Since $\beta, \beta' < \kappa$ and κ is a limit ordinal by Lemma 6.3.17, $\alpha < \kappa$.
Set $\lambda := |\alpha|$. Then λ is a cardinal, and a bijection $u : \alpha \to \lambda$ exists by Proposition 6.3.14(i).
Since $\lambda \leq \alpha$ by Definition 6.3.8, $\lambda < \kappa$. If $q \in X$, then $r(q) \leq \max\{\beta, \beta'\} < \alpha$, so $q \in \alpha \times \alpha$.
The coordinate map $(\xi, \eta) \mapsto (u(\xi), u(\eta))$ is a bijection $\alpha \times \alpha \to \lambda \times \lambda$, whose restriction to X is injective.
Thus $\kappa = |X| \leq |\lambda \times \lambda|$ by Proposition 6.3.16. If λ is finite, then $|\lambda \times \lambda| < \aleph_0 \leq \kappa$ by Theorem 4.2.5 and Proposition 6.3.15.
If λ is infinite, then $|\lambda \times \lambda| = \lambda < \kappa$ by induction hypothesis. In either case, $|\lambda \times \lambda| < \kappa$, contradicting $\kappa \leq |\lambda \times \lambda|$; hence $\kappa \not< \delta$.
(v) *Conclusion*: Since $\kappa \leq \delta$, $\delta = \kappa$ by Principle 6.3.2(i). Thus f is a bijection $\kappa \to \kappa \times \kappa$, so $|\kappa \times \kappa| = \kappa$ by Proposition 6.3.14(ii) and Definition 6.3.8. Hence $P(\kappa)$ holds, completing the induction.

Structural Commentary: Strong induction bounds max-ranked initial segments by smaller cardinal squares.
Key Insight: Every predecessor lies in a square of smaller cardinality. □

Theorem 6.6.3. *Infinite Cardinal Absorption*
Let κ, λ be cardinals, and let $\mu := \max\{\kappa, \lambda\}$ be infinite.

(i) $\kappa + \lambda = \mu$.
(ii) If $\kappa, \lambda > 0$, *then* $\kappa\lambda = \mu$.
(iii) $\aleph_0 + \aleph_0 = \aleph_0\aleph_0 = \aleph_0$.

Proof. The equality $|\mu \times \mu| = \mu$ holds by Theorem 6.6.2. Set $U := (\{0\} \times \mu) \cup (\{1\} \times \mu)$.
Since $0, 1 \in \mu$, the maps $\xi \mapsto (0, \xi)$ and $(i, \xi) \mapsto (\xi, i)$ inject μ into U and U into $\mu \times \mu$, respectively.
The identities $|U| = \mu + \mu$ and $|\mu| = \mu$ hold by Definition 6.5.1 and Definition 6.3.8, respectively.
Thus $\mu \leq \mu + \mu \leq |\mu \times \mu| = \mu$ by Proposition 6.3.16, so $\mu + \mu = \mu$.

(i) Set $V := (\{0\} \times \kappa) \cup (\{1\} \times \lambda)$. If $\mu = \kappa$, then $\xi \mapsto (0, \xi)$ injects μ into V; if $\mu = \lambda$, then $\xi \mapsto (1, \xi)$ does so.
The inclusions $\kappa \hookrightarrow \mu$ and $\lambda \hookrightarrow \mu$ induce an injection $V \to U$, and $|V| = \kappa + \lambda$ by Definition 6.5.1.
Thus $\mu \leq \kappa + \lambda \leq \mu + \mu = \mu$ by Proposition 6.3.16, so $\kappa + \lambda = \mu$.
(ii) Assume $\kappa, \lambda > 0$, so $0 \in \kappa \cap \lambda$. If $\mu = \kappa$, then $\xi \mapsto (\xi, 0)$ injects μ into $\kappa \times \lambda$; if $\mu = \lambda$, then $\xi \mapsto (0, \xi)$ does so.
The coordinate inclusions induce an injection $\kappa \times \lambda \to \mu \times \mu$, and $\kappa\lambda = |\kappa \times \lambda|$ by Definition 6.5.1.
Thus $\mu \leq \kappa\lambda \leq |\mu \times \mu| = \mu$ by Proposition 6.3.16, so $\kappa\lambda = \mu$.
(iii) The cardinal $\aleph_0$ is infinite by Proposition 6.3.15(iv). The asserted equalities follow by (i),(ii) with $\kappa = \lambda = \aleph_0$.

Structural Commentary: The square law sandwiches each sum and nonzero product between μ and itself.
Key Insight: An infinite maximum absorbs every bounded factor. □

7 Axiomatic and Advanced Set Theory

This section reserves space for deeper axiomatic and metatheoretic material: formal set axioms, models, forcing, large cardinals, and sets of reals.

7.1 Extensionality, Specification, and Replacement

Variables range over sets; formulas use first-order logic with equality and the sole nonlogical symbol $\in$.
For y' fresh in $\varphi(y)$, use $(\exists! y)\,\varphi(y) := (\exists y)[\varphi(y) \wedge (\forall y')(\varphi(y') \Rightarrow y' = y)]$.

Assumption 7.1.1. *Axiom of Extensionality*
Assume $(\forall A)(\forall B)[(\forall x)(x \in A \iff x \in B) \Rightarrow A = B]$.
The converse is equality substitution, so this axiom is the formal source of Assumption 1.3.2.

Assumption 7.1.2. *Separation/Specification Schema*
Fix $n \in \mathbb{N}_0$, put $\mathbf{z} = (z_1, \ldots, z_n)$, and let $\varphi(x, \mathbf{z})$ have no other free variables.
Require B not free in φ; the associated axiom is the universal closure in $A, \mathbf{z}$ of the next formula. $(\exists B)(\forall x)[x \in B \iff (x \in A \wedge \varphi(x, \mathbf{z}))]$.
In set-builder notation, its witness is $B = \{x \in A : \varphi(x, \mathbf{z})\} \subseteq A$.

Assumption 7.1.3. *Replacement Schema*
Fix $n \in \mathbb{N}_0$, put $\mathbf{z} = (z_1, \ldots, z_n)$, and let $\varphi(x, y, \mathbf{z})$ have no other free variables.
Require B not free in φ; the associated axiom is the universal closure in $A, \mathbf{z}$ of the next implication.
$[(\forall x \in A)(\exists! y)\,\varphi(x, y, \mathbf{z})] \Rightarrow (\exists B)(\forall y)[y \in B \iff (\exists x \in A)\,\varphi(x, y, \mathbf{z})]$.
The antecedent makes φ functional on A; the consequent collects its full range into a set.

Perspective 7.1.4. *Selection versus Functional Image*
Assumption 7.1.2 selects elements of a bounding set A, so its witness satisfies $B \subseteq A$.
Assumption 7.1.3 collects the range of a definable function on A, whose values may lie outside A.
An axiom schema comprises one first-order axiom per permitted formula and finite parameter list.

7.2 Models, Constructibility, and Independence

Fix a classical proof calculus with assumption discharge and double-negation elimination.
Use a ZFC metatheory formalizing first-order syntax, satisfaction, and the required set constructions.
Write Ord for the class of ordinals. The theories ZF, ZFC, and their extensions are *object theories* interpreted in models.
No claim below assumes that the metatheory proves its own consistency.

Definition 7.2.1. *Models and Inner Models of Set Theory*
Fix an object theory T in the first-order language of membership.
Let $\mathcal{M} = (M, E)$ satisfy $\emptyset \neq M$ and $E \subseteq M \times M$.

(i) The structure $\mathcal{M}$ is a *set model* of T iff $(\forall \sigma \in T)\,\mathcal{M} \models \sigma$.
(ii) It is a *transitive set model* of T iff it is a set model, $(\forall x \in M)\,x \subseteq M$, and $E = \{(x, y) \in M \times M : x \in y\}$.
(iii) A class N is an *inner model* of T iff $(\forall x \in N)\,x \subseteq N$, $\mathrm{Ord} \subseteq N$, and $(N, \in) \models T$.

Countability of a set model is an external metatheoretic property.
Thus it can regard one of its sets as uncountable when the relevant bijection is external.

Definition 7.2.2. *Relative Consistency, Consistency Strength, and Independence*
Let T, U be object theories in the fixed proof calculus; for a sentence φ, set $T + \varphi := T \cup \{\varphi\}$.

(i) $\mathrm{Con}(T)$ abbreviates $T \nvdash \bot$.
(ii) A *relative-consistency result from T to U* is a metatheoretic proof of $\mathrm{Con}(T) \Longrightarrow \mathrm{Con}(U)$.
(iii) The theories T and U are *equiconsistent* iff $[\mathrm{Con}(T) \Longrightarrow \mathrm{Con}(U)] \wedge [\mathrm{Con}(U) \Longrightarrow \mathrm{Con}(T)]$ is metatheoretically provable.
(iv) The theory U has at least the *consistency strength* of T iff the metatheory proves $\mathrm{Con}(U) \Longrightarrow \mathrm{Con}(T)$.
(v) A sentence φ is *independent of T* iff $T \nvdash \varphi$ and $T \nvdash \neg\varphi$.

Relative consistency does not assert its antecedent; independence selects no ambient truth.
The sentence CH asserts $2^{\aleph_0} = \aleph_1$. For a cardinal κ, let κ^+ be the least cardinal strictly greater than κ.
The sentence GCH asserts $(\forall \kappa \geq \aleph_0)\, 2^\kappa = \kappa^+$, with the quantifier restricted to cardinals.

Proposition 7.2.3. *Independence via Two Consistent Extensions*
Let T be a classical object theory and let φ be a sentence.
Then φ is independent of T iff $\mathrm{Con}(T + \varphi) \wedge \mathrm{Con}(T + \neg\varphi)$.

Proof.

($\Rightarrow$) Assume $T \nvdash \varphi$ and $T \nvdash \neg\varphi$. By assumption discharge, $T + \varphi \vdash \bot \Longrightarrow T \vdash \neg\varphi$.
By assumption discharge and double-negation elimination, $T + \neg\varphi \vdash \bot \Longrightarrow T \vdash \neg\neg\varphi \Longrightarrow T \vdash \varphi$.
Both consequents contradict the hypothesis, so $\mathrm{Con}(T + \varphi)$ and $\mathrm{Con}(T + \neg\varphi)$.
($\Leftarrow$) Assume $\mathrm{Con}(T + \varphi)$ and $\mathrm{Con}(T + \neg\varphi)$. Then $T \vdash \varphi \Longrightarrow T + \neg\varphi \vdash \bot$ and $T \vdash \neg\varphi \Longrightarrow T + \varphi \vdash \bot$.
Thus $T \nvdash \varphi$ and $T \nvdash \neg\varphi$, so φ is independent of T.

Structural Commentary: The two directions translate extension consistency into the two nonprovability obligations.
Key Insight: Opposite consistent extensions encode both nonprovability conditions. □

Definition 7.2.4. *Constructible Hierarchy*
For a set M, fix $n \in \mathbb{N}_0$ and $\mathbf{p} = (p_1, \ldots, p_n) \in M^n$. Let $\varphi(x, z_1, \ldots, z_n)$ be a formula with no other free variables.
For these data, set $A := \{x \in M : (M, \in) \models \varphi(x, p_1, \ldots, p_n)\}$. The family $\mathrm{Def}(M)$ consists exactly of these sets as the data vary.
Transfinite recursion sets $L_0 := \emptyset$ and $L_{\alpha+1} := \mathrm{Def}(L_\alpha)$. At a nonzero limit ordinal λ, set $L_\lambda := \bigcup_{\beta<\lambda} L_\beta$.

The *constructible universe* is $L := \bigcup_{\alpha \in \mathrm{Ord}} L_\alpha$; a set is *constructible* iff it belongs to L.

Proposition 7.2.5. *Basic Properties of the Constructible Hierarchy*
For ordinals $\alpha \leq \beta$:

(i) L_α is transitive;
(ii) $L_\alpha \subseteq L_\beta$;
(iii) L is transitive;
(iv) $L_\alpha \cap \mathrm{Ord} = \alpha$;
(v) $\mathrm{Ord} \subseteq L$.

Proof.

(i) Set $\mathrm{Trans}(M) := (\forall x \in M)\, x \subseteq M$ and $P(\gamma) := [\mathrm{Trans}(L_\gamma) \wedge L_\gamma \subseteq L_{\gamma+1}]$. To prove $(\forall \gamma \in \mathrm{Ord})\, P(\gamma)$ by transfinite induction.
Since $L_0 = \emptyset$ is transitive and $L_0 \subseteq L_1$, the assertion $P(0)$ holds. Assume $P(\gamma)$.
If $x \in y \in L_{\gamma+1}$, then $y \subseteq L_\gamma$ and $x \in L_\gamma \subseteq L_{\gamma+1}$; hence $L_{\gamma+1}$ is transitive.
For $x \in L_{\gamma+1}$, the parameter x and formula $z \in x$ define x over $L_{\gamma+1}$. Thus $x \in L_{\gamma+2}$ and $P(\gamma+1)$ holds.
At a nonzero limit ordinal λ, assume $(\forall \gamma < \lambda)\, P(\gamma)$. If $x \in y \in L_\lambda$, choose $\gamma < \lambda$ with $y \in L_\gamma$.
Then $x \in L_\gamma \subseteq L_\lambda$ by $P(\gamma)$, so L_λ is transitive. For $x \in L_\lambda$, the parameter x and formula $z \in x$ define x over L_λ.
Thus $L_\lambda \subseteq L_{\lambda+1}$ and $P(\lambda)$ holds. Thus $(\forall \alpha \in \mathrm{Ord})\ \mathrm{Trans}(L_\alpha)$.
(ii) Fix α and prove $(\forall \beta \in \mathrm{Ord})\, [\alpha \leq \beta \Rightarrow L_\alpha \subseteq L_\beta]$ by transfinite induction.
At $\beta = 0$, the premise $\alpha \leq 0$ gives $\alpha = 0$, so the inclusion is equality. If $\beta = \gamma + 1$ and $\alpha \leq \beta$, then either $\alpha = \beta$ or $\alpha \leq \gamma$.
In the latter case, $L_\alpha \subseteq L_\gamma \subseteq L_{\gamma+1}$ by the induction hypothesis and the simultaneous induction in (i).
At a nonzero limit β, the case $\alpha = \beta$ is trivial.
If $\alpha < \beta$, then $\alpha + 1 < \beta$ and $L_\alpha \subseteq L_{\alpha+1} \subseteq L_\beta$ by the simultaneous induction in (i) and the definition of L_β.
(iii) If $x \in y \in L$, choose $\gamma \in \mathrm{Ord}$ with $y \in L_\gamma$. Then $x \in L_\gamma \subseteq L$ by (i), so L is transitive.
(iv) Set $Q(\gamma) := [L_\gamma \cap \mathrm{Ord} = \gamma]$ and proceed by transfinite induction. At 0, $L_0 \cap \mathrm{Ord} = 0$.
At a nonzero limit ordinal λ, $L_\lambda \cap \mathrm{Ord} = \bigcup_{\xi<\lambda} (L_\xi \cap \mathrm{Ord}) = \bigcup_{\xi<\lambda} \xi = \lambda$ by the induction hypothesis.
Assume $Q(\gamma)$. If $\delta \in L_{\gamma+1} \cap \mathrm{Ord}$, then $\delta \subseteq L_\gamma \cap \mathrm{Ord} = \gamma$, so $\delta \leq \gamma$. Conversely, $\delta < \gamma \Longrightarrow \delta \in \gamma = L_\gamma \cap \mathrm{Ord} \subseteq L_{\gamma+1}$.
Ordinalhood is bounded and defines $\gamma = L_\gamma \cap \mathrm{Ord}$ over the transitive set L_γ; hence $\gamma \in L_{\gamma+1}$.
Therefore $L_{\gamma+1} \cap \mathrm{Ord} = \gamma + 1$, which proves $Q(\gamma + 1)$.
(v) $(\forall \alpha \in \mathrm{Ord})\, \alpha \in L_{\alpha+1}$ by (iv); hence $\mathrm{Ord} \subseteq L$.

Structural Commentary: Simultaneous induction gives transitivity and nesting; a second induction identifies the stage ordinals.
Key Insight: Definability carries each stage and its index forward. □

Principle 7.2.6. *Constructible Inner Model Result*

(i) The theory ZF proves that L is an inner model of ZFC + GCH.
(ii) The metatheory proves $\mathrm{Con}(\mathrm{ZF}) \Longrightarrow \mathrm{Con}(\mathrm{ZFC} + \mathrm{GCH})$.
(iii) The reverse consistency implication holds because ZFC + GCH contains ZF; hence the theories are equiconsistent.

Reflection and condensation are not developed here, so this principle records framework and supplies no proof authority.

Definition 7.2.7. *Forcing Posets, Generic Filters, and Extensions*
Let M be a transitive set. Let $\mathbb{P} = (P, \leq, 1_\mathbb{P}) \in M$ be a partial order whose greatest element is $1_\mathbb{P}$.
The element $1_\mathbb{P}$ is the weakest condition, and $p \leq q$ means that p is stronger than q.

(i) A set D is *dense in P*, written $\mathrm{Dense}_P(D)$, iff $D \subseteq P$ and $(\forall p \in P)(\exists q \in D)\, q \leq p$.

(ii) A *filter on P* is a set $\emptyset \neq G \subseteq P$ satisfying $(\forall p \in G)(\forall q \in P)\,[p \leq q \Rightarrow q \in G]$.
Its directedness condition is $(\forall p, q \in G)(\exists r \in G)\,[r \leq p \wedge r \leq q]$.
(iii) The filter G is *M-generic* iff $(\forall D \in M)\,[\mathrm{Dense}_P(D) \Rightarrow G \cap D \neq \emptyset]$.
(iv) Transfinite recursion defines $\mathcal{N}_0^P := \emptyset$ and $\mathcal{N}_{\alpha+1}^P := \mathcal{P}(\mathcal{N}_\alpha^P \times P)$.
At a nonzero limit ordinal λ, set $\mathcal{N}_\lambda^P := \bigcup_{\beta<\lambda} \mathcal{N}_\beta^P$. Set $\mathcal{N}^P := \bigcup_{\alpha \in \mathrm{Ord}} \mathcal{N}_\alpha^P$; its elements are the *$P$-names*.
(v) For a filter G and $\tau \in \mathcal{N}^P$, recursion on name rank defines $\tau^G := \{\sigma^G : (\exists p \in G)\,[(\sigma, p) \in \tau]\}$.
(vi) For an M-generic filter G, the *forcing extension* is $M[G] := \{\tau^G : \tau \in M \cap \mathcal{N}^P\}$.

Theorem 7.2.8. *Existence of Generic Filters over Countable Transitive Sets*
In the metatheory, let M be a countable transitive set. Let $\mathbb{P} = (P, \leq, 1_{\mathbb{P}}) \in M$ be a partial order, and fix $p \in P$. Then an M-generic filter $G \subseteq P$ satisfying $p \in G$ exists.

Proof.
(i) *Enumeration*: External countability gives $(D_n)_{n<\omega}$ enumerating $\{D \in M : \mathrm{Dense}_P(D)\}$. Repetitions are allowed; set $p_0 := p$.
(ii) *Descending construction*: Recursively choose $p_{n+1} \in D_n$ with $p_{n+1} \leq p_n$ by the density of D_n.
(iii) *Filter verification*: Set $G := \{q \in P : (\exists n < \omega)\, p_n \leq q\}$. Then $p = p_0 \in G$.
If $q \in G$, $q \leq r$, and $r \in P$, choose $n < \omega$ with $p_n \leq q$; transitivity gives $p_n \leq r$, so $r \in G$.
If $q_1, q_2 \in G$, choose $i, j < \omega$ with $p_i \leq q_1$ and $p_j \leq q_2$.
Set $k := \max\{i, j\}$. Then $p_k \in G$, $p_k \leq q_1$, and $p_k \leq q_2$ by the descending construction. Thus G is a filter on P.
(iv) *Genericity*: The construction gives $(\forall n < \omega)\, p_{n+1} \in G \cap D_n$.
If $D \in M$ and $\mathrm{Dense}_P(D)$, then $(\exists n < \omega)\, D = D_n$; hence $G \cap D \neq \emptyset$. Therefore G is M-generic and contains p.

Structural Commentary: A descending chain turns enumerated dense sets into one filter.
Key Insight: Later conditions strengthen all earlier choices. □

Principle 7.2.9. *Forcing Extension Framework*
Let M be a countable transitive model of ZFC. Let $\mathbb{P} = (P, \leq, 1_{\mathbb{P}}) \in M$, and let $G \subseteq P$ be M-generic.
The forcing theorem has the following conclusions.

(i) $M \subseteq M[G]$.
(ii) $M \cap \mathrm{Ord} = M[G] \cap \mathrm{Ord}$.
(iii) $M[G] \models \mathrm{ZFC}$.

Its proof recursively defines the forcing relation on formulas and names before proving the Truth Lemma.
That machinery is not developed here, so this principle supplies no proof authority.

Principle 7.2.10. *Relative Consistency and Independence of the Continuum Hypothesis*
The constructible-inner-model and forcing constructions establish the following relative-consistency implications.

(i) $\mathrm{Con}(\mathrm{ZFC}) \Longrightarrow \mathrm{Con}(\mathrm{ZFC} + \mathrm{CH})$.
(ii) $\mathrm{Con}(\mathrm{ZFC}) \Longrightarrow \mathrm{Con}(\mathrm{ZFC} + \neg\mathrm{CH})$.

Under Con(ZFC), the sentence CH is independent of ZFC by Proposition 7.2.3.
Each extension contains ZFC, so both are equiconsistent with ZFC. The constructible universe supplies (i).
For (ii), conditions are finite partial functions from $\omega \times \omega_2$ to 2.
Reverse inclusion orders them, and the resulting poset has the countable chain condition.
It preserves ω_1 and ω_2 and adds ω_2 distinct subsets of ω.
The relative-consistency proof is syntactic, or equivalently Boolean-valued. It does not assume a transitive model of all ZFC.
The required constructibility and forcing theorems are not proved here, so this principle supplies no proof authority.

7.3 Baire and Cantor Spaces

Definition 7.3.1. *Sequence Spaces and Cylinders*
Let $A \in \{2, \omega\}$, where $2 = \{0, 1\}$.

(i) For $n < \omega$, let A^n be the set of functions $n \to A$, and let A^ω be the set of functions $\omega \to A$. Set $A^{<\omega} := \bigcup_{n<\omega} A^n$.
(ii) For $s \in A^{<\omega}$, its *length* is $|s| := \mathrm{dom}\, s$. For $u \in A^{<\omega} \cup A^\omega$, write $s \sqsubseteq u$ iff $s = u|_{|s|}$; then s is an *initial segment of u*.
(iii) Strings $s, t \in A^{<\omega}$ are *compatible* iff $s \sqsubseteq t$ or $t \sqsubseteq s$; otherwise they are *incompatible*.
The *basic cylinder* determined by s is $N_s^A := \{x \in A^\omega : s \sqsubseteq x\}$.
(iv) A set $U \subseteq A^\omega$ is *open* iff it is a union of basic cylinders. It is *closed* iff $A^\omega \setminus U$ is open and *clopen* iff it is both open and closed.
(v) The spaces 2^ω and ω^ω are *Cantor space* and *Baire space*, respectively. Write N_s for N_s^A when A is fixed.

Proposition 7.3.2. *Basic Cylinders Form a Clopen Basis*
Let $A \in \{2, \omega\}$, let $s, t \in A^{<\omega}$, and let $U \subseteq A^\omega$ be open.

(i) If $s \sqsubseteq t$, then $N_s^A \cap N_t^A = N_t^A$. If $t \sqsubseteq s$, then $N_s^A \cap N_t^A = N_s^A$. If s, t are incompatible, then $N_s^A \cap N_t^A = \emptyset$.
(ii) The cylinder N_s^A is clopen.
(iii) $(\forall x \in U)(\exists n < \omega)\, x \in N_{x|_n}^A \subseteq U$.

Proof.
(i) If $s \sqsubseteq t$, then $(\forall x \in N_t^A)\, s = t|_{|s|} = x|_{|s|}$; hence $N_t^A \subseteq N_s^A$ and $N_s^A \cap N_t^A = N_t^A$.
If $t \sqsubseteq s$, then $(\forall x \in N_s^A)\, t = s|_{|t|} = x|_{|t|}$; hence $N_s^A \subseteq N_t^A$ and $N_s^A \cap N_t^A = N_s^A$.
If $x \in N_s^A \cap N_t^A$, then s, t are finite initial segments of x and are comparable by length.
Thus incompatible s, t have disjoint cylinders.
(ii) The cylinder N_s^A is open because it is a union of itself. Fix $x \notin N_s^A$. Since $s \neq x|_{|s|}$, choose $j < |s|$ with $x(j) \neq s(j)$.
If $y \in N_{x|_{j+1}}^A$, then $y(j) = x(j) \neq s(j)$. Therefore $N_{x|_{j+1}}^A \subseteq A^\omega \setminus N_s^A$.
The complement is a union of cylinders and is open, so N_s^A is clopen.
(iii) Fix $x \in U$. Since U is a union of cylinders, choose $r \in A^{<\omega}$ with $x \in N_r^A \subseteq U$. Then $r = x|_{|r|}$, so $n := |r|$ has required property.

Structural Commentary: Comparability controls intersections; first disagreements construct complementary cylinders.
Key Insight: Finite prefixes determine all basic neighborhoods. □

Definition 7.3.3. *Prefix Metric and Polish Space*
Let $A \in \{2, \omega\}$.

(i) For distinct $x, y \in A^\omega$, set $\Delta_A(x,y) := \min\{n < \omega : x(n) \neq y(n)\}$. Define $d_A(x,x) := 0$ and $d_A(x,y) := 2^{-(\Delta_A(x,y)+1)}$ for $x \neq y$.
(ii) A metric d on X is an *ultrametric* iff $(\forall x, y, z \in X)\, d(x,z) \leq \max\{d(x,y), d(y,z)\}$.
(iii) A sequence $(x_k)_{k<\omega} \in X^\omega$ is *d-Cauchy* iff $(\forall \varepsilon > 0)(\exists K < \omega)(\forall k, \ell \geq K)\, d(x_k, x_\ell) < \varepsilon$.
It *converges to $x \in X$* iff $(\forall \varepsilon > 0)(\exists K < \omega)(\forall k \geq K)\, d(x_k, x) < \varepsilon$.
The metric d is *complete* iff each d-Cauchy sequence in X converges to a point of X.
(iv) A set $D \subseteq A^\omega$ is *dense* iff $(\forall s \in A^{<\omega})\, D \cap N_s^A \neq \emptyset$.
(v) A topological space is *Polish* iff its topology is induced by some complete metric and it has a countable dense subset.

Theorem 7.3.4. *Baire and Cantor Spaces Are Polish*
For $A \in \{2, \omega\}$, set $E_A := \{x \in A^\omega : (\exists n < \omega)(\forall k \in \omega \setminus n)\, x(k) = 0\}$.

(i) The prefix metric d_A is an ultrametric.
(ii) The metric d_A is complete.
(iii) The metric d_A induces the cylinder topology on A^ω.
(iv) The eventually-zero set E_A is countable.
(v) The set E_A is dense in A^ω.
(vi) The space A^ω is Polish.

Proof.

(i) Symmetry of the disagreement set makes d_A symmetric. By definition, $(\forall x, y \in A^\omega)\, [d_A(x,y) \geq 0 \wedge (d_A(x,y) = 0 \iff x = y)]$.
Fix $x, y, z \in A^\omega$ and set $r := \max\{d_A(x,y), d_A(y,z)\}$. If $r = 0$, then $x = y = z$. If $r > 0$, choose $m < \omega$ with $r = 2^{-(m+1)}$.
Both pairs x, y and y, z agree below m, so $d_A(x,z) \leq r$. Thus the strong triangle inequality holds.
The ordinary triangle inequality follows from $r \leq d_A(x,y) + d_A(y,z)$, so d_A is an ultrametric.
(ii) Let $(x_k)_{k<\omega}$ be d_A-Cauchy. For $m < \omega$, choose $K_m < \omega$ such that $(\forall k, \ell \geq K_m)\, d_A(x_k, x_\ell) < 2^{-m}$.
Set $s_m := x_{K_m}|_m \in A^m$. If $k \geq K_m$, then $x_k|_m = s_m$. If $m \leq n$, comparison at an index $k \geq \max\{K_m, K_n\}$ gives $s_m = s_n|_m$.
Hence $x := \bigcup_{m<\omega} s_m$ is a function $\omega \to A$. Given $\varepsilon > 0$, choose $m < \omega$ with $2^{-m} < \varepsilon$.
For $k \geq K_m$, the sequences x_k, x agree below m, so $d_A(x_k, x) < 2^{-m} < \varepsilon$. Thus $x_k \to x$, and d_A is complete.
(iii) For $x \in A^\omega$ and $m < \omega$, $N^A_{x|_m} = \{y \in A^\omega : d_A(x,y) < 2^{-m}\}$.
Both sides require agreement below m, so each cylinder containing x is a metric ball.
Conversely, given $\varepsilon > 0$, choose $m < \omega$ with $2^{-m} < \varepsilon$.
Then $N^A_{x|_m} \subseteq \{y \in A^\omega : d_A(x,y) < \varepsilon\}$, so the metric and cylinder topologies coincide.
(iv) The map $k \mapsto k - 1$ is a bijection $\mathbb{N} \to \omega$, so ω is countable.
For $n \in \mathbb{N}$, the coordinate map $f \mapsto (f(0), \ldots, f(n-1))$ bijects ω^n with the n-tuples over ω.
Hence ω^n is countable by Theorem 6.2.18. Since $A^n \subseteq \omega^n$, the set A^n is at most countable by Proposition 6.2.12(i).
The set $A^0 = \{\emptyset\}$ is finite, so $A^{<\omega}$ is at most countable by Corollary 6.2.16.
For $s \in A^{<\omega}$, let $\widehat{s} \in A^\omega$ extend s and equal 0 on $\omega \setminus |s|$. Then $E_A = \{\widehat{s} : s \in A^{<\omega}\}$.
Thus E_A is at most countable by Proposition 6.2.12(ii), applied to the image of $s \mapsto \widehat{s}$.
For $j < \omega$, let $e_j \in A^\omega$ equal 1 at j and 0 elsewhere.
Distinct indices give distinct $e_j \in E_A$, so E_A is infinite and therefore countable.
(v) The relation $(\forall s \in A^{<\omega})\, \widehat{s} \in E_A \cap N_s^A$ holds, so E_A is dense.
(vi) The complete metric d_A induces the cylinder topology by (ii) and (iii).
The set E_A is a countable dense subset by (iv) and (v). Hence A^ω is Polish.

Structural Commentary: Prefix stabilization proves completeness; finite supports supply a countable dense set.
Key Insight: Finite information controls both distance and density. □

Definition 7.3.5. *Products, Projections, and Continuity*
Let $A, B \in \{2, \omega\}$.

(i) The *product topology* on $A^\omega \times B^\omega$ has basis $\{N_s^A \times N_t^B : s \in A^{<\omega},\ t \in B^{<\omega}\}$. Its open sets are the unions of these rectangles.
(ii) For $E \subseteq A^\omega \times B^\omega$, its first-coordinate projection is $\pi_A(E) := \{x \in A^\omega : (\exists y \in B^\omega)\, (x,y) \in E\}$.
(iii) A function $f : A^\omega \to B^\omega$ is *continuous* iff $(\forall t \in B^{<\omega})\, f^{-1}(N_t^B)$ is open.

Proposition 7.3.6. *Continuity by Finite Prefixes*
Let $A, B \in \{2, \omega\}$ and let $f : A^\omega \to B^\omega$. Then f is continuous iff the following finite-prefix condition holds.
$(\forall x \in A^\omega)(\forall m < \omega)(\exists n < \omega)(\forall y \in A^\omega)\, [y|_n = x|_n \implies f(y)|_m = f(x)|_m]$.

Proof.

($\Rightarrow$) Assume f is continuous, and fix $x \in A^\omega$ and $m < \omega$. Set $V := f^{-1}(N^B_{f(x)|_m})$. Then V is open and $x \in V$.
Choose $n < \omega$ with $x \in N^A_{x|_n} \subseteq V$ by Proposition 7.3.2(iii). If $y|_n = x|_n$, then $y \in V$ and $f(y)|_m = f(x)|_m$.
($\Leftarrow$) Assume the prefix condition, fix $t \in B^{<\omega}$, and set $m := |t|$.
For $x \in f^{-1}(N_t^B)$, choose $n < \omega$ such that $(\forall y \in A^\omega)\, [y|_n = x|_n \Rightarrow f(y)|_m = f(x)|_m]$ by the prefix condition.
If $y \in N^A_{x|_n}$, then $f(y)|_m = f(x)|_m = t$, so $N^A_{x|_n} \subseteq f^{-1}(N_t^B)$.
Thus $f^{-1}(N_t^B)$ is a union of cylinders and is open; hence f is continuous.

Structural Commentary: Each direction translates target prefixes into source-cylinder neighborhoods.
Key Insight: Finite output information depends on finite input information. □

Definition 7.3.7. *Trees and Infinite Branches*
Let $A \in \{2, \omega\}$, let T be a set, and let $C \subseteq A^\omega$.

(i) A *tree on A* is a possibly empty set $T \subseteq A^{<\omega}$ satisfying the next condition; write $\mathrm{Tree}_A(T)$. $(\forall t \in T)(\forall s \in A^{<\omega})\, [s \sqsubseteq t \implies s \in T]$.
(ii) The *body* of T is $[T] := \{x \in A^\omega : (\forall n < \omega)\, x|_n \in T\}$.
(iii) The *tree of initial segments* of C is $T_C := \{x|_n : x \in C,\ n < \omega\}$.

Theorem 7.3.8. *Closed Sets Are Exactly Tree Bodies*
Let $A \in \{2, \omega\}$, let $T \subseteq A^{<\omega}$, and let $C \subseteq A^{\omega}$.

(i) If $\mathrm{Tree}_A(T)$, then $[T]$ is closed.
(ii) $\mathrm{Tree}_A(T_C)$.
(iii) If C is closed, then $C = [T_C]$.
(iv) The set C is closed iff $(\exists S \subseteq A^{<\omega})\,[\mathrm{Tree}_A(S) \wedge C = [S]]$.

Proof.

(i) Fix $x \notin [T]$ and choose $n < \omega$ with $x|_n \notin T$. The cylinder $N^A_{x|_n}$ lies in $A^{\omega} \setminus [T]$ because each $y \in N^A_{x|_n}$ satisfies $y|_n = x|_n \notin T$. Thus the complement of $[T]$ is open.
(ii) By definition, $T_C \subseteq A^{<\omega}$. If $t = x|_n \in T_C$ and $s \in A^{<\omega}$ satisfies $s \sqsubseteq t$, then $s = x|_{|s|} \in T_C$; hence $\mathrm{Tree}_A(T_C)$.
(iii) By definition, $C \subseteq [T_C]$. Fix $x \notin C$ and choose $n < \omega$ with $x \in N^A_{x|_n} \subseteq A^{\omega} \setminus C$ by Proposition 7.3.2(iii).
If $x|_n \in T_C$, then some $y \in C$ extends $x|_n$, contradicting $N^A_{x|_n} \cap C = \emptyset$. Hence $x \notin [T_C]$, so $[T_C] \subseteq C$.
(iv) ($\Rightarrow$) If C is closed, then $\mathrm{Tree}_A(T_C)$ by (ii) and $C = [T_C]$ by (iii).
(iv) ($\Leftarrow$) If $\mathrm{Tree}_A(S)$ and $C = [S]$, then C is closed by (i) with $T = S$.

Structural Commentary: Complement cylinders prove closure; realized prefixes construct the representing tree.
Key Insight: Closed membership is determined by all finite approximations. □

7.4 Filters, Ultrafilters, and Sets of Reals

Definition 7.4.1. *Filters, Ideals, and Duality*
Let X be a set, let $\mathcal{F}, \mathcal{I} \subseteq \mathcal{P}(X)$, and let $A \subseteq X$.

(i) The family $\mathcal{F}$ is a *filter* on X iff $X \in \mathcal{F}$, $\emptyset \notin \mathcal{F}$, $(\forall B, C \in \mathcal{F})\, B \cap C \in \mathcal{F}$, and $(\forall B \in \mathcal{F})(\forall C \subseteq X)\,[B \subseteq C \Rightarrow C \in \mathcal{F}]$.
(ii) The family $\mathcal{I}$ is an *ideal* on X iff $\emptyset \in \mathcal{I}$, $X \notin \mathcal{I}$, $(\forall B, C \in \mathcal{I})\, B \cup C \in \mathcal{I}$, and $(\forall B \in \mathcal{I})(\forall C \subseteq B)\, C \in \mathcal{I}$.
(iii) If $\mathcal{F}$ is a filter, its *dual ideal* is $\mathcal{I}_{\mathcal{F}} := \{X \setminus B : B \in \mathcal{F}\}$. If $\mathcal{I}$ is an ideal, its *dual filter* is $\mathcal{F}_{\mathcal{I}} := \{X \setminus B : B \in \mathcal{I}\}$.
(iv) If $\mathcal{I}$ is an ideal, then A is *$\mathcal{I}$-positive* iff $A \notin \mathcal{I}$.

Definition 7.4.2. *Generated, Principal, and Fréchet Filters*
Let X be a set and let $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfy $\mathcal{B} \neq \emptyset$, $\emptyset \notin \mathcal{B}$, and $(\forall B_0, B_1 \in \mathcal{B})\, B_0 \cap B_1 \in \mathcal{B}$.

(i) The *filter generated by $\mathcal{B}$* is $\langle \mathcal{B} \rangle := \{A \subseteq X : (\exists B \in \mathcal{B})\, B \subseteq A\}$.
(ii) For $\emptyset \neq D \subseteq X$, the *principal filter generated by D* is $\mathcal{F}_D := \{A \subseteq X : D \subseteq A\}$.
(iii) For infinite X, the *Fréchet filter*, or *cofinite filter*, is $\mathrm{Fr}(X) := \{A \subseteq X : X \setminus A \text{ is finite}\}$.

Definition 7.4.3. *κ-Complete Filter*
Let X be a set, let κ be an infinite cardinal, and let $\mathcal{F}$ be a filter on X.
The filter $\mathcal{F}$ is *κ-complete* iff $(\forall \lambda < \kappa)(\forall A \in \mathcal{F}^{\lambda})\, \bigcap_{\xi < \lambda} A(\xi) \in \mathcal{F}$.

Proposition 7.4.4. *Basic Filter and Ideal Constructions*
Let X be a set, let $\mathcal{F}$ be a filter on X, let $\mathcal{I}$ be an ideal on X, and let $\emptyset \neq D \subseteq X$.
Let $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfy $\mathcal{B} \neq \emptyset$, $\emptyset \notin \mathcal{B}$, and $(\forall B_0, B_1 \in \mathcal{B})\, B_0 \cap B_1 \in \mathcal{B}$.

(i) $\mathcal{I}_{\mathcal{F}}$ is an ideal and $\mathcal{F}_{\mathcal{I}}$ is a filter on X.
(ii) $\langle \mathcal{B} \rangle$ is the least filter containing $\mathcal{B}$.
(iii) $\mathcal{F}_D$ is a filter; if X is infinite, then $\mathrm{Fr}(X)$ is a filter whose dual ideal is the family of finite subsets of X.
(iv) The filter $\mathcal{F}$ is $\aleph_0$-complete.

Proof.

(i) The equalities $X \setminus X = \emptyset$ and $X \setminus \emptyset = X$ show that $\mathcal{I}_{\mathcal{F}}$ contains $\emptyset$ but not X. If $A, B \in \mathcal{F}$, then $(X \setminus A) \cup (X \setminus B) = X \setminus (A \cap B) \in \mathcal{I}_{\mathcal{F}}$.
Let $A \in \mathcal{F}$ and $E \subseteq X \setminus A$. Then $A \subseteq X \setminus E$, so $X \setminus E \in \mathcal{F}$ and $E \in \mathcal{I}_{\mathcal{F}}$. Thus $\mathcal{I}_{\mathcal{F}}$ is an ideal.
The same two complement equalities give $X \in \mathcal{F}_{\mathcal{I}}$ and $\emptyset \notin \mathcal{F}_{\mathcal{I}}$. If $A, B \in \mathcal{I}$, then $(X \setminus A) \cap (X \setminus B) = X \setminus (A \cup B) \in \mathcal{F}_{\mathcal{I}}$.
Let $A \in \mathcal{I}$ and $X \setminus A \subseteq E \subseteq X$. Then $X \setminus E \subseteq A$, so $X \setminus E \in \mathcal{I}$ and $E \in \mathcal{F}_{\mathcal{I}}$. Hence $\mathcal{F}_{\mathcal{I}}$ is a filter.
(ii) Choose $B_0 \in \mathcal{B}$. Then $B_0 \subseteq X$ gives $X \in \langle \mathcal{B} \rangle$. Since no member of $\mathcal{B}$ is contained in $\emptyset$, one has $\emptyset \notin \langle \mathcal{B} \rangle$.
If $A, A' \in \langle \mathcal{B} \rangle$, choose $B, B' \in \mathcal{B}$ with $B \subseteq A$ and $B' \subseteq A'$. Then $B \cap B' \in \mathcal{B}$ and $B \cap B' \subseteq A \cap A'$.
Thus $\langle \mathcal{B} \rangle$ is closed under intersections, and upward closure follows from transitivity of inclusion. The inclusion $\mathcal{B} \subseteq \langle \mathcal{B} \rangle$ holds.
If a filter $\mathcal{G}$ contains $\mathcal{B}$ and $A \in \langle \mathcal{B} \rangle$, choose $B \in \mathcal{B}$ with $B \subseteq A$. Then $A \in \mathcal{G}$, so $\langle \mathcal{B} \rangle \subseteq \mathcal{G}$.
(iii) Since $\mathcal{F}_D = \langle \{D\} \rangle$, the family $\mathcal{F}_D$ is a filter by (ii). Suppose X is infinite. The set X is cofinite, whereas $\emptyset$ is not. The intersection of two cofinite sets is cofinite because its complement is the union of two finite sets, and every superset of a cofinite set is cofinite. Thus $\mathrm{Fr}(X)$ is a filter. Its complement transform is $\{A \subseteq X : A \text{ is finite}\}$.
(iv) Let $\lambda < \aleph_0$ and $A \in \mathcal{F}^{\lambda}$. The ordinal λ is finite because $\aleph_0$ is the least infinite cardinal. The empty intersection is $X \in \mathcal{F}$, and adjoining one $A(\xi)$ preserves membership by binary closure. Hence the required intersection belongs to $\mathcal{F}$ by induction on λ.

Structural Commentary: Complementation proves duality; base closure and finite induction handle the filter constructions.
Key Insight: Upward closure turns finite-intersection bases into filters. □

Definition 7.4.5. *Ultrafilters and Principal Ultrafilters*
Let X be a set and let $\mathcal{U}$ be a filter on X.

(i) The filter $\mathcal{U}$ is an *ultrafilter* iff $(\forall A \subseteq X)\, \#(\mathcal{U} \cap \{A, X \setminus A\}) = 1$.
(ii) The ultrafilter $\mathcal{U}$ is *principal* iff $(\exists x \in X)\, \mathcal{U} = \{A \subseteq X : x \in A\}$; otherwise it is *nonprincipal*.

Theorem 7.4.6. *Ultrafilters Are Exactly the Maximal Filters*
Let X be a set and let $\mathcal{F}$ be a filter on X. Then $\mathcal{F}$ is an ultrafilter iff $(\forall \mathcal{G} \subseteq \mathcal{P}(X))\,[\mathcal{G}$ is a filter on $X \wedge \mathcal{F} \subseteq \mathcal{G} \Rightarrow \mathcal{G} = \mathcal{F}]$.

Proof.

($\Rightarrow$) Suppose $\mathcal{F}$ is an ultrafilter and let $\mathcal{G}$ be a filter with $\mathcal{F} \subseteq \mathcal{G}$
If $A \in \mathcal{G} \setminus \mathcal{F}$, then $X \setminus A \in \mathcal{F} \subseteq \mathcal{G}$, so $\emptyset = A \cap (X \setminus A) \in \mathcal{G}$, contradicting properness. Thus $\mathcal{G} = \mathcal{F}$.

($\Leftarrow$) Suppose the displayed maximality condition holds, and fix $A \subseteq X$. Assume $A, X \setminus A \notin \mathcal{F}$, and set $\mathcal{B}_A := \{B \cap A : B \in \mathcal{F}\}$.
The relation $A = X \cap A$ gives $\mathcal{B}_A \neq \emptyset$, and $(B \cap A) \cap (C \cap A) = (B \cap C) \cap A \in \mathcal{B}_A$ for $B, C \in \mathcal{F}$.
If $\emptyset \in \mathcal{B}_A$, choose $B \in \mathcal{F}$ with $B \cap A = \emptyset$. Then $B \subseteq X \setminus A$, so $X \setminus A \in \mathcal{F}$, a contradiction.
Hence $\langle \mathcal{B}_A \rangle$ is a filter by Proposition 7.4.4(ii). For $B \in \mathcal{F}$, the relation $B \cap A \subseteq B$ gives $B \in \langle \mathcal{B}_A \rangle$, so $\mathcal{F} \subseteq \langle \mathcal{B}_A \rangle$.
Since $A \in \mathcal{B}_A \subseteq \langle \mathcal{B}_A \rangle$ but $A \notin \mathcal{F}$, the containment is proper, contradicting maximality.
Thus $A \in \mathcal{F}$ or $X \setminus A \in \mathcal{F}$; properness prevents both, so $\mathcal{F}$ is an ultrafilter.

Structural Commentary: One direction excludes proper extensions; the other adjoins any undecided set through a filter base.
Key Insight: Maximality is equivalent to deciding every complementary pair. □

Principle 7.4.7. *Ultrafilter Lemma*
Let X be a set and let $\mathcal{F}$ be a filter on X. The Ultrafilter Lemma asserts $(\exists \mathcal{U} \subseteq \mathcal{P}(X))\,[\mathcal{U}$ is an ultrafilter on $X \wedge \mathcal{F} \subseteq \mathcal{U}]$.

Theorem 7.4.8. *Ultrafilter Extension under Zorn's Lemma*
Principle 6.1.3 implies Principle 7.4.7.

Proof.
(i) *Extension poset*: Assume Zorn's Lemma, let X be a set, and let $\mathcal{F}$ be a filter on X.
Let $\mathscr{P}$ be the set of filters on X containing $\mathcal{F}$, ordered by inclusion.
Then $\mathcal{F} \in \mathscr{P}$, and inclusion is a partial order on $\mathscr{P}$ by Proposition 6.1.5.
(ii) *Chain bounds*: The filter $\mathcal{F}$ bounds the empty chain. Let $\mathcal{C} \subseteq \mathscr{P}$ be a nonempty chain and set $\mathcal{G} := \bigcup_{\mathcal{H} \in \mathcal{C}} \mathcal{H}$.
Then $X \in \mathcal{G}$, $\emptyset \notin \mathcal{G}$, and $\mathcal{F} \subseteq \mathcal{G}$. If $A, B \in \mathcal{G}$, choose $\mathcal{H}_0, \mathcal{H}_1 \in \mathcal{C}$ with $A \in \mathcal{H}_0$ and $B \in \mathcal{H}_1$.
Since $\mathcal{C}$ is a chain, one of these filters contains both A and B, so $A \cap B \in \mathcal{G}$.
Upward closure passes from any member of $\mathcal{C}$ containing A. Thus $\mathcal{G} \in \mathscr{P}$ and it bounds $\mathcal{C}$.
(iii) *Maximal extension*: Every chain in $\mathscr{P}$ has an upper bound in $\mathscr{P}$, so a maximal $\mathcal{U} \in \mathscr{P}$ exists by Zorn's Lemma.
The filter $\mathcal{U}$ is an ultrafilter by Theorem 7.4.6, and $\mathcal{F} \subseteq \mathcal{U}$ by the definition of $\mathscr{P}$.

Structural Commentary: Unions preserve filters along chains, so Zorn produces a maximal extension.
Key Insight: Chain compatibility makes binary filter closure survive union. □

Remark 7.4.9. *Choice Strength of Ultrafilter Extension*
Over ZF, Principle 7.4.7 is equivalent to the Boolean Prime Ideal Theorem and is strictly weaker than the full Axiom of Choice. This metamathematical comparison is not used below as proof authority. Theorem 7.4.8 establishes only the implication from the stronger Zorn principle available earlier. No choice principle enters Theorem 7.4.6 or the definition of principal ultrafilters.

Corollary 7.4.10. *Nonprincipal Ultrafilters on Infinite Sets*
Assume Principle 7.4.7, and let X be infinite. Then $(\exists \mathcal{U} \subseteq \mathcal{P}(X))\,[\mathcal{U}$ is a nonprincipal ultrafilter on $X]$.

Proof. The family $\mathrm{Fr}(X)$ is a filter by Proposition 7.4.4(iii). Choose an ultrafilter $\mathcal{U} \supseteq \mathrm{Fr}(X)$ by the hypothesis of the corollary.
If $\mathcal{U}$ were principal at $x \in X$, then $\{x\} \in \mathcal{U}$. Since $X \setminus \{x\} \in \mathrm{Fr}(X) \subseteq \mathcal{U}$, closure under intersections would give $\emptyset \in \mathcal{U}$, contradicting properness. Hence $\mathcal{U}$ is nonprincipal.
Structural Commentary: Extend the cofinite filter and exclude principality by intersecting complementary singleton data.
Key Insight: Cofinite membership prevents concentration at one point. □

Definition 7.4.11. *Measurable Cardinal*
Let κ be an uncountable cardinal. In ZFC, the cardinal κ is *measurable* iff $(\exists \mathcal{U} \subseteq \mathcal{P}(\kappa))\,[\mathcal{U}$ is a nonprincipal κ-complete ultrafilter on $\kappa]$.

Remark 7.4.12. *Large-Cardinal Assumption Boundary*
The assertion that a measurable cardinal exists is an additional large-cardinal hypothesis; it is not assumed elsewhere in this part. The ordinary Ultrafilter Lemma does not supply such a cardinal, as extending a filter to an ultrafilter need not preserve κ-completeness. Thus the large-cardinal connection enters through the extra completeness requirement, not through ultrafilter maximality alone.

Assumption 7.4.13. *Club and Stationary Setting*
For the club-and-stationary results below, assume Principle 6.1.2 and fix a regular cardinal $\kappa > \omega$.

Definition 7.4.14. *Closed Unbounded Subset*
A set $C \subseteq \kappa$ is *closed in κ* iff $(\forall \nu < \kappa)\,[\nu$ is a nonzero limit ordinal $\wedge\, C \cap \nu$ is cofinal in $\nu \Rightarrow \nu \in C]$.
It is *unbounded in κ* iff it is cofinal in κ, and it is a *club subset of κ* iff it is both closed and unbounded.

Theorem 7.4.15. *Intersection of Fewer Than κ Clubs*
If $0 < \lambda < \kappa$ and $(C_i)_{i<\lambda}$ is a family of club subsets of κ, then $\bigcap_{i<\lambda} C_i$ is club in κ.

Proof. Set $C := \bigcap_{i<\lambda} C_i$. If $\nu < \kappa$ is a nonzero limit ordinal and $C \cap \nu$ is cofinal in ν, then $(\forall i < \lambda)\, C_i \cap \nu$ is cofinal in ν.
Hence $(\forall i < \lambda)\, \nu \in C_i$, so $\nu \in C$ and C is closed.
If $\theta < \kappa$ and $g : \theta \to \kappa$, then $\sup g(\theta) < \kappa$; otherwise g would be cofinal and $\mathrm{cf}(\kappa) \leq \theta < \kappa$, contradicting regularity.
Fix $\beta < \kappa$ and set $\beta_0 := \beta + 1 < \kappa$ by Lemma 6.3.17.
Recursively, after β_n is defined, choose $(\gamma_{n,i})_{i<\lambda}$ with $(\forall i < \lambda)\, \gamma_{n,i} \in C_i \setminus (\beta_n + 1)$ by the standing Choice Principle.
Set $\beta_{n+1} := \sup_{i<\lambda}(\gamma_{n,i} + 1)$. The limit-ordinal property of κ gives $(\forall i < \lambda)\, \gamma_{n,i} + 1 < \kappa$, so the preceding bound gives $\beta_{n+1} < \kappa$.
Set $\delta := \sup_{n<\omega} \beta_n$. The same observation with $\theta = \omega$ gives $\delta < \kappa$.
Fix $i < \lambda$. Since $(\forall n < \omega)\, \beta_n < \gamma_{n,i} < \beta_{n+1} \leq \delta$, the sequence $(\gamma_{n,i})_{n<\omega}$ is cofinal in the nonzero limit ordinal δ.
Closedness of C_i gives $\delta \in C_i$. As $i < \lambda$ was arbitrary, $\delta \in C$; moreover, $\beta < \delta$, so C is unbounded.
Structural Commentary: Regularity bounds a countable recursion whose stages visit every club.
Key Insight: Each club contributes a cofinal subsequence to one common limit. □

Definition 7.4.16. *Club Filter, Stationary Sets, and the Nonstationary Ideal*
(i) The *club filter* on κ is $\mathcal{F}_{\mathrm{club}}(\kappa) := \{A \subseteq \kappa : (\exists C \subseteq A)\, C$ is club in $\kappa\}$.
(ii) A set $S \subseteq \kappa$ is *stationary* iff $(\forall C \subseteq \kappa)\,[C$ is club in $\kappa \Rightarrow S \cap C \neq \emptyset]$.
(iii) The *nonstationary ideal* is $\mathrm{NS}_\kappa := \{A \subseteq \kappa : (\exists C \subseteq \kappa)\,[C$ is club in $\kappa \wedge A \cap C = \emptyset]\}$. Thus S is stationary iff $S \notin \mathrm{NS}_\kappa$.

Proposition 7.4.17. *Club-Filter Completeness and Stationary-Set Basics*
Let $C, S \subseteq \kappa$.

(i) The family $\mathcal{F}_{\mathrm{club}}(\kappa)$ is a κ-complete filter.
(ii) The ideal NS_κ is dual to $\mathcal{F}_{\mathrm{club}}(\kappa)$.
(iii) If C is club and S is stationary, then $C \cap S$ is stationary.
(iv) If S is stationary, then S is unbounded.
(v) If S is stationary, then $\{\nu \in S : \nu$ is a nonzero limit ordinal$\}$ is stationary.

Proof.

(i) The set κ is club in itself, so $\kappa \in \mathcal{F}_{\mathrm{club}}(\kappa)$. No club is empty, so $\emptyset \notin \mathcal{F}_{\mathrm{club}}(\kappa)$, and upward closure follows from the definition.
Let $\lambda < \kappa$ and $(A_i)_{i<\lambda} \in \mathcal{F}_{\mathrm{club}}(\kappa)^\lambda$. If $\lambda = 0$, then $\bigcap_{i<\lambda} A_i = \kappa$. If $\lambda > 0$, choose clubs $C_i \subseteq A_i$ by the standing Choice Principle.
The intersection $\bigcap_{i<\lambda} C_i$ is club by Theorem 7.4.15, and it is contained in $\bigcap_{i<\lambda} A_i$. Thus the club filter is κ-complete.

(ii) For $A \subseteq \kappa$, $\kappa \setminus A \in \mathcal{F}_{\mathrm{club}}(\kappa)$ iff $(\exists C \subseteq \kappa \setminus A)\,[C$ is club$]$, equivalently iff $A \in \mathrm{NS}_\kappa$.
Hence $\mathrm{NS}_\kappa = \mathcal{I}_{\mathcal{F}_{\mathrm{club}}(\kappa)}$, which is an ideal by (i) and Proposition 7.4.4(i).

(iii) Fix a club $D \subseteq \kappa$. The intersection $C \cap D$ is club by Theorem 7.4.15 with $\lambda = 2$, so $(C \cap S) \cap D \neq \emptyset$.
As D was arbitrary, $C \cap S$ is stationary.

(iv) Suppose $S \subseteq \kappa$ is bounded, and choose $\beta < \kappa$ with $S \subseteq \beta + 1$. Set $D_\beta := \{\xi < \kappa : \beta < \xi\}$.
Since κ is a limit ordinal, $(\forall \eta < \kappa)\ \max\{\eta, \beta\} + 1 \in D_\beta$, so D_β is unbounded.
If $D_\beta \cap \nu$ is cofinal in a nonzero limit ordinal $\nu < \kappa$, then $D_\beta \cap \nu \neq \emptyset$, so $\beta < \nu$ and $\nu \in D_\beta$.
Hence D_β is closed and therefore club. Since $S \cap D_\beta = \emptyset$, the set S is not stationary.

(v) Set $L := \{\nu < \kappa : \nu$ is a nonzero limit ordinal$\}$. If $L \cap \nu$ is cofinal in a nonzero limit ordinal $\nu < \kappa$, then $\nu \in L$, so L is closed.
For $\beta < \kappa$, the sequence $(\beta + n)_{n<\omega}$ is bounded below κ by regularity, so $\beta + \omega < \kappa$.
Since $\beta + \omega \in L$ lies above β, the set L is unbounded and therefore club. The set $S \cap L$ is stationary by (iii).

Structural Commentary: Club intersections prove completeness; closed tails and limit ordinals test stationarity.
Key Insight: Stationary sets survive every additional club restriction. □

Definition 7.4.18. *Diagonal Intersection and Normality*

(i) For $(X_\xi)_{\xi<\kappa} \in \mathcal{P}(\kappa)^\kappa$, its *diagonal intersection* is $\triangle_{\xi<\kappa} X_\xi := \{\beta < \kappa : (\forall \xi < \beta)\ \beta \in X_\xi\}$.

(ii) A filter $\mathcal{F}$ on κ is *normal* iff $(\forall (X_\xi)_{\xi<\kappa} \in \mathcal{F}^\kappa)\ \triangle_{\xi<\kappa} X_\xi \in \mathcal{F}$.

Theorem 7.4.19. *The Club Filter Is Normal*

(i) If $(C_\xi)_{\xi<\kappa}$ is a sequence of clubs in κ, then $\triangle_{\xi<\kappa} C_\xi$ is club.
(ii) The club filter on κ is normal.

Proof.

(i) We prove clubness through decreasing refinement, closure, and recursive unboundedness.

(a) *Decreasing refinement*: Define $(C'_\xi)_{\xi<\kappa}$ by $C'_\xi := \bigcap_{\eta\le\xi} C_\eta$.
Each C'_ξ is club by Theorem 7.4.15 with $\lambda = \xi + 1$, since $0 < \xi + 1 < \kappa$; moreover, $(C'_\xi)_{\xi<\kappa}$ is decreasing.
Fix $\rho < \kappa$. Then $(\forall \xi < \rho)\ \rho \in C'_\xi$ iff $(\forall \xi < \rho)(\forall \eta \le \xi)\ \rho \in C_\eta$ iff $(\forall \eta < \rho)\ \rho \in C_\eta$.
As $\rho < \kappa$ was arbitrary, the diagonal intersections agree. Hence assume $(C_\xi)_{\xi<\kappa}$ is decreasing and set $D := \triangle_{\xi<\kappa} C_\xi$.

(b) *Closedness*: Suppose $\nu < \kappa$ is a nonzero limit ordinal and $D \cap \nu$ is cofinal in ν. Fix $\xi < \nu$.
Then $\{\rho \in D \cap \nu : \xi < \rho\} \subseteq C_\xi$ is cofinal in ν, so $\nu \in C_\xi$ by closedness.
As $\xi < \nu$ was arbitrary, $\nu \in D$; thus D is closed.

(c) *Unboundedness*: Fix $\eta < \kappa$ and set $\beta_0 := \eta + 1 < \kappa$.
Recursively set $\beta_{n+1} := \min(C_{\beta_n} \setminus (\beta_n + 1))$; unboundedness of C_{β_n} makes the defining set nonempty.
Set $\beta := \sup_{n<\omega} \beta_n < \kappa$; otherwise $n \mapsto \beta_n$ would be cofinal in κ, contrary to $\mathrm{cf}(\kappa) = \kappa > \omega$.
The strict increase makes β a nonzero limit ordinal. Fix $\xi < \beta$ and choose $n < \omega$ with $\xi < \beta_n$.
Decreasingness gives $(\forall m \ge n)\ \beta_{m+1} \in C_{\beta_m} \subseteq C_\xi$. This tail is cofinal in β, so $\beta \in C_\xi$ by closedness.
As $\xi < \beta$ was arbitrary, $\beta \in D$; since $\eta < \beta$, the set D is unbounded and therefore club.

(ii) Let $(A_\xi)_{\xi<\kappa} \in \mathcal{F}_{\mathrm{club}}(\kappa)^\kappa$, and choose clubs $C_\xi \subseteq A_\xi$ by the standing Choice Principle.
The set $\triangle_{\xi<\kappa} C_\xi$ is club by (i), and $\triangle_{\xi<\kappa} C_\xi \subseteq \triangle_{\xi<\kappa} A_\xi$. Hence $\triangle_{\xi<\kappa} A_\xi \in \mathcal{F}_{\mathrm{club}}(\kappa)$.

Structural Commentary: Decreasing refinement separates closure from recursive unboundedness; contained clubs establish normality.
Key Insight: Each point satisfies only club conditions indexed below it. □

Theorem 7.4.20. *Fodor's Pressing-Down Lemma*
Let $S \subseteq \kappa$ be stationary and let $f : S \to \kappa$ be regressive, meaning $(\forall \alpha \in S)\ f(\alpha) < \alpha$. Then $(\exists \gamma < \kappa)\,[\{\alpha \in S : f(\alpha) = \gamma\}$ is stationary$]$.

Proof. For $\gamma < \kappa$, set $S_\gamma := \{\alpha \in S : f(\alpha) = \gamma\}$. Suppose $(\forall \gamma < \kappa)\,[S_\gamma$ is nonstationary$]$.
Choose $(C_\gamma)_{\gamma<\kappa}$ satisfying $(\forall \gamma < \kappa)\,[C_\gamma$ is club $\wedge\ C_\gamma \cap S_\gamma = \emptyset]$ by the standing Choice Principle.
The set $D := \triangle_{\gamma<\kappa} C_\gamma$ is club by Theorem 7.4.19(i). Choose $\alpha \in S \cap D$ and set $\gamma := f(\alpha)$. Then $\gamma < \alpha$ and $\alpha \in D$, so $\alpha \in C_\gamma$.
Since $\alpha \in S_\gamma$, this contradicts $C_\gamma \cap S_\gamma = \emptyset$. Hence $(\exists \gamma < \kappa)\,[S_\gamma$ is stationary$]$.
Structural Commentary: Avoid every fiber by a club, combine the clubs diagonally, and use regressiveness.
Key Insight: The regressive value indexes a club already tested at its argument. □

Assumption 7.4.21. *Descriptive-Set-Theory Setting*
The cardinal fixed in Assumption 7.4.13 is no longer in force. Work in ZFC, and let X be Baire space, Cantor space, or a finite product of these spaces with its product topology.

Definition 7.4.22. *Borel Sets and the Borel Pointclasses*
The *Borel σ-algebra* $\mathcal{B}(X)$ is the least family of subsets of X containing all open sets and closed under complements and countable unions. Its members are the *Borel sets*.
Let ω_1 be the least uncountable ordinal. The *Borel pointclasses* are defined recursively for $0 < \alpha < \omega_1$.

(i) $\mathbf{\Sigma}^0_1(X) := \{A \subseteq X : A \text{ is open}\}$ and $\mathbf{\Pi}^0_1(X) := \{X \setminus A : A \in \mathbf{\Sigma}^0_1(X)\}$.
(ii) Fix $1 < \alpha < \omega_1$ after all lower levels have been defined.
Set $\mathbf{\Sigma}^0_\alpha(X) := \{ \bigcup_{n<\omega} A(n) : A \in (\bigcup_{0<\beta<\alpha} \mathbf{\Pi}^0_\beta(X))^\omega\}$. Set $\mathbf{\Pi}^0_\alpha(X) := \{X \setminus A : A \in \mathbf{\Sigma}^0_\alpha(X)\}$.
(iii) $\mathbf{\Delta}^0_\alpha(X) := \mathbf{\Sigma}^0_\alpha(X) \cap \mathbf{\Pi}^0_\alpha(X)$.

Thus $\mathbf{\Sigma}^0_1(X)$, $\mathbf{\Pi}^0_1(X)$, and $\mathbf{\Delta}^0_1(X)$ are the open, closed, and clopen subsets of X, respectively.

Theorem 7.4.23. *The Borel Hierarchy Exhausts the Borel Sets*
A set $A \subseteq X$ is Borel iff $(\exists \alpha < \omega_1)\,[0 < \alpha \wedge A \in \mathbf{\Sigma}^0_\alpha(X)]$.

Proof.
($\Rightarrow$) Set $\mathcal{H} := \bigcup_{0<\alpha<\omega_1} \mathbf{\Sigma}^0_\alpha(X)$. Then $\mathbf{\Sigma}^0_1(X) \subseteq \mathcal{H}$.
The relation $(\forall \alpha < \omega_1)\,[0 < \alpha \Rightarrow \mathbf{\Pi}^0_\alpha(X) \subseteq \mathbf{\Sigma}^0_{\alpha+1}(X)]$ holds by constant unions, so $\mathcal{H}$ is complement-closed.
Fix $B \in \mathcal{H}^\omega$ and choose $\alpha \in (\omega_1 \setminus \{0\})^\omega$ satisfying $(\forall n < \omega)\, B(n) \in \mathbf{\Sigma}^0_{\alpha(n)}(X)$ by Choice.
One has $(\forall n < \omega)\, B(n) \in \mathbf{\Pi}^0_{\alpha(n)+1}(X)$ by constant unions and complementation.
Set $\gamma := \sup_{n<\omega}(\alpha(n)+1) + 1 < \omega_1$ by Corollary 6.2.16. Since $(\forall n < \omega)\,\alpha(n) + 1 < \gamma$, one has $\bigcup_{n<\omega} B(n) \in \mathbf{\Sigma}^0_\gamma(X) \subseteq \mathcal{H}$.
Thus $\mathcal{H}$ is a σ-algebra containing $\mathbf{\Sigma}^0_1(X)$, so $\mathcal{B}(X) \subseteq \mathcal{H}$ by minimality.
($\Leftarrow$) Set $P(\alpha) := \mathbf{\Sigma}^0_\alpha(X) \cup \mathbf{\Pi}^0_\alpha(X) \subseteq \mathcal{B}(X)$. We prove $(\forall \alpha < \omega_1)\,[0 < \alpha \Rightarrow P(\alpha)]$ by strong transfinite induction.
The property $P(1)$ holds by the definition of $\mathcal{B}(X)$. Fix $1 < \alpha < \omega_1$ and assume $(\forall \beta < \alpha)\,[0 < \beta \Rightarrow P(\beta)]$.
The inclusion $\mathbf{\Sigma}^0_\alpha(X) \subseteq \mathcal{B}(X)$ holds by the recursive definition, induction hypothesis, and countable-union closure.
Thus $\mathbf{\Pi}^0_\alpha(X) \subseteq \mathcal{B}(X)$ by complementation. Hence $P(\alpha)$.

Structural Commentary: Forward closure identifies the generated σ-algebra; transfinite induction proves the reverse inclusion.
Key Insight: Countable Borel operations have ranks below ω_1. □

Definition 7.4.24. *The Projective Pointclasses*
For $E \subseteq X \times \omega^\omega$, set $\pi_X(E) := \{x \in X : (\exists y \in \omega^\omega)\,(x, y) \in E\}$. Define the projective pointclasses recursively.

(i) $\mathbf{\Sigma}^1_1(X) := \{\pi_X(E) : E \in \mathcal{B}(X \times \omega^\omega)\}$; its members are the *analytic sets*.
(ii) For $n \in \mathbb{N}$, $\mathbf{\Pi}^1_n(X) := \{X \setminus A : A \in \mathbf{\Sigma}^1_n(X)\}$.
(iii) For $n \in \mathbb{N}$, $\mathbf{\Sigma}^1_{n+1}(X) := \{\pi_X(E) : E \in \mathbf{\Pi}^1_n(X \times \omega^\omega)\}$.
(iv) For $n \in \mathbb{N}$, $\mathbf{\Delta}^1_n(X) := \mathbf{\Sigma}^1_n(X) \cap \mathbf{\Pi}^1_n(X)$.

The members of $\mathbf{\Pi}^1_1(X)$ are the *coanalytic sets*. A set $A \subseteq X$ is *projective* iff $(\exists n \in \mathbb{N})\,[A \in \mathbf{\Sigma}^1_n(X) \vee A \in \mathbf{\Pi}^1_n(X)]$.

Proposition 7.4.25. *Borel Sets Are Analytic and Coanalytic*
(i) $\mathcal{B}(X) \subseteq \mathbf{\Delta}^1_1(X)$.
(ii) The class $\bigcup_{n\in\mathbb{N}} (\mathbf{\Sigma}^1_n(X) \cup \mathbf{\Pi}^1_n(X))$ *is closed under complements.*

Proof.
(i) Set $Y := \omega^\omega$ and $\mathscr{S} := \{C \subseteq X : C \times Y \in \mathcal{B}(X \times Y)\}$.
Then $(\forall C \in \mathscr{S})\,[(X \setminus C) \times Y = (X \times Y) \setminus (C \times Y) \in \mathcal{B}(X \times Y)]$, so $\mathscr{S}$ is complement-closed.
One has $(\forall C \in \mathscr{S}^\omega)\,[(\bigcup_{n<\omega} C(n)) \times Y = \bigcup_{n<\omega} (C(n) \times Y) \in \mathcal{B}(X \times Y)]$. Thus $\mathscr{S}$ is countable-union-closed.
The inclusion $\mathbf{\Sigma}^0_1(X) \subseteq \mathscr{S}$ holds because products of open sets are open. Thus $\mathscr{S}$ is a σ-algebra and $\mathcal{B}(X) \subseteq \mathscr{S}$ by minimality.
Since $Y \neq \emptyset$, $(\forall C \in \mathcal{B}(X))\, C = \pi_X(C \times Y) \in \mathbf{\Sigma}^1_1(X)$ by the definition of $\mathbf{\Sigma}^1_1(X)$.
Consequently, $(\forall A \in \mathcal{B}(X))\, A \in \mathbf{\Sigma}^1_1(X) \cap \mathbf{\Pi}^1_1(X) = \mathbf{\Delta}^1_1(X)$ by the pointclass definitions.
(ii) Set $\mathscr{P} := (\mathbf{\Sigma}^1_n(X) \cup \mathbf{\Pi}^1_n(X))_{n\in\mathbb{N}}$.
Then $(\forall n \in \mathbb{N})(\forall A \in \mathscr{P}(n))\, X \setminus A \in \mathscr{P}(n)$ by the definition of $\mathbf{\Pi}^1_n(X)$. Thus $\bigcup_{n\in\mathbb{N}} \mathscr{P}(n)$ is complement-closed.

Structural Commentary: Product closure proves the level-one inclusion; pointclass duality proves complement closure.
Key Insight: A constant witness realizes every Borel set as a projection. □

Remark 7.4.26. *Determinacy and Large-Cardinal Boundary*
The material from Definition 7.4.22 through Proposition 7.4.25 is developed in ZFC alone, without measurable-cardinal or determinacy hypotheses. No projective-regularity theorem is asserted; any later theorem of that kind must state its additional hypothesis explicitly.

8 Mathematical Writing and Research Practice

This section fixes the communication, evidence, integrity, and reading practices used to create and evaluate mathematical and machine-learning research.

8.1 Theorem-Style Results and Formal Statements

Formal environments identify a block's function, evidential status, and permissible use as authority.

Convention 8.1.1. *Theorem-Style Result Hierarchy*
In these notes, the theorem-style result environments are exactly the following; each contains a proved claim.

(i) *Theorem*: a major result that advances a principal objective of the surrounding work.
(ii) *Proposition*: an independently useful result of smaller local weight.
(iii) *Lemma*: a subsidiary result introduced primarily to support a later proof.
(iv) *Corollary*: a direct consequence of an earlier proved result that requires little additional argument.

These names record local structural roles, not a universal ranking of importance.
Definitions stipulate objects or terminology, whereas conjectures record unproved claims.
Convention follows standard mathematical-writing guidance (MIT Department of Mathematics, 1998).

Process 8.1.2. *Theorem-Statement Construction and Audit*
Given a candidate conclusion and its supporting argument, produce a self-contained theorem-style statement that can be cited and proved without hidden context.

(i) *Purpose*: isolate the reusable conclusion and identify its role in the dependency chain.
(ii) *Objects*: bind every ambient set, structure, parameter, map, and index before its first use.
(iii) *Hypotheses*: retain only the argument's exact domain, regularity, finiteness, independence, and boundary conditions.
(iv) *Conclusion*: state existence, uniqueness, characterization, bound, convergence, or other claim explicitly.
(v) *Claim census*: itemize each conclusion whose omission would remove a distinct proof obligation.
(vi) *Role*: select theorem, proposition, lemma, or corollary by Convention 8.1.1.
(vii) *Locality*: move motivation, examples, and proof machinery outside the statement, and delete unused assumptions.
(viii) *Traceability*: attach a stable title and label, and cite the external owner of a nonoriginal result.

The audit passes only when the statement remains intelligible apart from the preceding paragraph and every symbol is defined at first use. These are central expository checks (Knuth et al., 1988).

Principle 8.1.3. *Claim–Authority Separation*
Match each formal block's environment and evidence to its epistemic status. A definition stipulates meaning; an assumption declares a condition; a conjecture remains unproved; a theorem-style result requires a proof; and an empirical statement requires data and a protocol. Only proved theorem-style results may serve as mathematical proof authority. A citation records prior ownership but does not discharge a proof obligation when the present work claims to establish the result.

Warning 8.1.4. *Statement–Proof Contamination*
If a theorem-style statement contains a proof plan, auxiliary construction, example, or claim of obviousness, its logical contract becomes obscure and later citation becomes unreliable. Move proof-specific objects into the proof, place explanatory material after the formal owner, and delete hypotheses unused by the proof.

8.2 Proof Construction and Revision

A proof is a checkable argument that derives the stated conclusion from declared hypotheses, definitions, and earlier proved results.

Proof Method 8.2.1. *Proof Construction and Audit*
Given an audited theorem-style statement, construct a proof with independently checkable dependencies and obligations.

(i) *Parse*: identify the objects, hypotheses, quantifier dependencies, and conclusion by Proof Method 2.4.1.
(ii) *Plan*: select a direct argument, cases, two inclusions, induction, contradiction, construction, or another fitting architecture.
(iii) *Dependencies*: record only earlier results whose hypotheses are verified at their citation points.
(iv) *Obligations*: assign each numbered claim, implication direction, existence witness, and uniqueness assertion to a proof unit.
(v) *Execution*: bind variables before use and justify every inference by a definition, calculation, hypothesis, or earlier result.
(vi) *Boundaries*: handle empty, zero, singular, endpoint, and other degenerate cases that affect an argument.
(vii) *Closure*: verify that the final sentence discharges the original conclusion rather than a nearby surrogate.
(viii) *Cold read*: test notation, citations, and transitions without relying on memory of the intended argument.

The audit passes only when every obligation is discharged and each dependency is valid at its point of use. When the route is not apparent, expose the plan while retaining every step needed for verification. Mathematical-writing guidance likewise emphasizes status, motivation, and conceptual structure (MIT Department of Mathematics, 1998; Knuth et al., 1988).

Process 8.2.2. *Proof Revision Passes*
Revise a completed proof in three ordered passes.

(i) *Validity*: check every dependency, implication, case split, witness, index, limit, and exceptional case.
(ii) *Architecture*: align proof units with statement claims and make long operative stages visible.
(iii) *Exposition*: remove repetition, bind notation before use, compress routine algebra, and expand decisive unfamiliar steps.

Review typography only after the mathematical and logical structure is stable.

Warning 8.2.3. *Evidence Is Not Proof*
Finite testing, a diagram, numerical agreement, or a successful program run supports only the cases and conditions examined. Such evidence cannot close an unrestricted quantified claim without an exhaustive argument and a proof that the tested cases cover the domain. Computation may discover a conjecture, locate a counterexample, or verify a formally specified finite case; otherwise provide a complete proof or narrow the claim to the verified scope.

Principle 8.2.4. *Proof Detail Proportionality*
Allocate detail where correctness or understanding depends on it. Expand novel constructions, fragile estimates, changes of quantifier, and boundary arguments; compress familiar algebra only while the remaining chain is checkable. The target is a reproducible mathematical argument, neither maximal formality nor persuasive rhetoric.

8.3 Methods, Algorithms, and Computational Artifacts

Procedural blocks separate conceptual guidance, algorithm specifications, implementations, and correctness or performance evidence.

Convention 8.3.1. *Procedural Environment Roles*
Use the following procedural roles throughout these notes.

(i) *Proof method*: a reusable proof route, reduction, recognition criterion, or obligation checklist.
(ii) *Computation method*: a reusable construction, calculation, approximation, diagnostic, or test.
(iii) *Method*: a short convention, classification workflow, or inseparable hybrid with no clearer subtype.
(iv) *Process*: an organizational workflow for research, writing, review, or documentation.
(v) *Algorithm*: executable pseudocode with declared inputs, outputs, control flow, and termination behavior when applicable.

Worked instances belong in examples, completed arguments in proofs, and implementation-specific behavior in code or artifact records.

Process 8.3.2. *Algorithm Specification and Audit*
Before publishing pseudocode, record its contract and associated verification obligations.

(i) *Problem*: state the task and the abstraction level at which the algorithm operates.
(ii) *Input*: specify types, dimensions, admissibility, representation, and malformed-input behavior.
(iii) *Output*: specify the returned object and a postcondition that characterizes success.
(iv) *State*: define initialization, mutable state, determinism or randomness, and any invariant used for correctness.
(v) *Steps*: write one unambiguous action per step with readable branches and loops.
(vi) *Termination*: state the stopping rule and prove termination when it is claimed.
(vii) *Correctness*: state the proof obligations and keep the proof of the postcondition separate from the pseudocode.
(viii) *Resources*: state complexity only with an input-size parameter, cost model, resource, and case type.
(ix) *Limits*: record numerical, memory, precision, platform, approximation, and applicability boundaries.

Labeled steps and explanatory invariants clarify the distinction between an algorithm and program text (Knuth et al., 1988).

Principle 8.3.3. *Formal and Informal Procedural Views*
Use complementary formal and informal views when a procedure is complex. The formal view fixes inputs, outputs, updates, and stopping conditions; the informal view explains the invariant, decomposition, or design choice. The views should not duplicate one another, and only a separate proof can establish mathematical correctness.

Process 8.3.4. *Computational Artifact Record*
For code supporting a reported result, preserve an auditable artifact record.

(i) *Identity*: archive the exact source revision and data identifiers.
(ii) *Environment*: record dependencies, runtime, platform, configuration, and random seeds.
(iii) *Access*: state licenses, unavailable components, and access restrictions.
(iv) *Execution*: record the commands and expected outputs.
(v) *Traceability*: map each reported figure, table, or metric to its generating command.

The 2026 NeurIPS checklist requests exact execution instructions and sufficient experimental detail for interpreting results (Neural Information Processing Systems Foundation, 2026).

Warning 8.3.5. *Algorithm Evidence Boundaries*
Passing tests does not establish correctness outside the tested cases, and an asymptotic bound does not predict wall-clock performance without a cost model and operating conditions. Keep specification, proof, implementation, benchmark, and interpretation distinct, and restrict each claim to the evidence that supports it.

8.4 Mathematical Research and Integrity

Mathematical research cycles through discovery, attempted refutation, proof, verification, and communication rather than proceeding linearly from question to theorem.

Process 8.4.1. *Mathematical Research Cycle*
For a proposed mathematical problem, maintain the following revisable cycle.

(i) *Question*: state the objects, scope, current assumptions, and a precise target that can be proved or refuted.
(ii) *Literature*: record definitions, earlier results, counterexamples, nearby formulations, and current open status.
(iii) *Exploration*: compute examples, boundary cases, and nonexamples while recording the tested domain.
(iv) *Conjecture*: state the strongest precise claim supported by exploration and include every suspected hypothesis.
(v) *Attack*: search independently for a proof and a counterexample; record failed routes and new obstructions.
(vi) *Revision*: weaken, strengthen, split, or reframe the conjecture when an example or proof obligation demands it.
(vii) *Verification*: check every proof dependency, compare with prior results, and seek independent criticism.
(viii) *Communication*: state novelty accurately, write a complete proof, cite provenance, and record limitations.

MIT's project laboratory similarly presents mathematical discovery through data generation, conjecture, experiment, proof or counterexample, and writing (MIT OpenCourseWare, 2013).

Process 8.4.2. *Research Claim Ledger*
Maintain one auditable row per substantive claim.

(i) *Claim*: record the exact wording and scope.
(ii) *Status*: record every applicable status: proved, conjectured, computed, observed, reproduced, replicated, or interpreted.
(iii) *Support*: record its evidence, dependencies, source locator, and unresolved checks.
(iv) *Revision*: update the row whenever a hypothesis, dataset, code revision, or conclusion changes.

The ledger prevents an exploratory observation from silently becoming a theorem or empirical conclusion.

Convention 8.4.3. *Reproducibility, Replicability, and Generalizability*
Adopt the National Academies' 2019 terminology (National Academies of Sciences, Engineering, and Medicine, 2019).

(i) *Reproducibility*: consistent computational results from the same data, code, methods, and analysis conditions.
(ii) *Replicability*: consistent results from a new study addressing the same question with newly obtained data.
(iii) *Generalizability*: applicability of a result to contexts or populations beyond those directly studied.

These are distinct evidential targets: a reproducible computation need not replicate on new data or generalize to a new setting.

Principle 8.4.4. *Research Integrity and Stewardship*
Maintain research records that permit scrutiny, correction, and appropriate reuse. Report methods and findings honestly; preserve accurate records; credit contributions and prior work; disclose conflicts; respect confidentiality, consent, licenses, and governing rules; and correct the record when an error is found. These duties reflect the honesty, accountability, fairness, and stewardship principles of the Singapore Statement (Second World Conference on Research Integrity, 2010).

Warning 8.4.5. *Novelty and Negative Results*
Novelty does not imply truth, and publication does not certify correctness. Search for prior and concurrent work before claiming priority, and report a failed proof, null result, or nonreplication with its actual protocol and uncertainty. A negative outcome may diagnose a false conjecture, an inadequate method, or a boundary of generality; the evidence must distinguish these possibilities.

8.5 Machine Learning and AI Research

Machine-learning and AI conclusions draw on statistical, computational, system, and social evidence; the study design must identify the authority supporting each conclusion.

Process 8.5.1. *Machine-Learning and AI Study Protocol*
Specify an empirical study before interpreting its scores.

(i) *Question*: state the population, task, context of use, intervention or comparison, and claimed outcome.
(ii) *Claim type*: classify each target as theoretical, empirical, causal, computational, system, or normative.
(iii) *Data*: record provenance, version, license, collection, consent or access status, labels, exclusions, and preprocessing.
(iv) *Evaluation boundary*: protect test data from training, model selection, prompt design, preprocessing, and human iteration.
(v) *Comparators*: choose relevant baselines, ablations, and resource-matched alternatives before inspecting final results.
(vi) *Protocol*: fix splits, metrics, hyperparameter selection, seeds, stopping rules, hardware, software, and compute budget.
(vii) *Uncertainty*: repeat stochastic runs and report how intervals, variation, or significance measures are calculated.
(viii) *Stress tests*: examine subgroups, shift, robustness, failure modes, sensitivity, and plausible misuse within scope.
(ix) *Artifacts*: preserve code, data handles, checkpoints, configurations, logs, and command-to-result mappings.
(x) *Report*: separate observed results from interpretation and state limitations, costs, risks, and unsupported generalizations.

The 2026 NeurIPS checklist and its reproducibility program emphasize assumptions, complete proofs, experimental detail, uncertainty, artifacts, ethics, and limitations (Neural Information Processing Systems Foundation, 2026; Pineau et al., 2021).

Process 8.5.2. *Dataset and Model Documentation*
Publish documentation alongside every durable ML artifact.

(i) *Dataset*: record motivation, version, provenance, composition, collection, preprocessing, labeling, license, intended uses, restrictions, and maintenance.
(ii) *Model*: record identity, version, intended and excluded uses, training data, evaluation protocol, subgroup results, and limitations.
(iii) *System*: record interfaces, human roles, deployment assumptions, monitoring, update policy, safeguards, and rollback conditions.

Dataset datasheets and model cards provide complementary documentation patterns for provenance, intended use, evaluation, and limitations (Gebru et al., 2021; Mitchell et al., 2019).

Principle 8.5.3. *Contextual AI Risk Assessment*
Accuracy alone is not a complete system claim. For intended context, select and justify tests for the following applicable characteristics.

(i) validity and reliability;
(ii) safety;
(iii) security and resilience;
(iv) accountability and transparency;
(v) explainability and interpretability;
(vi) privacy;
(vii) harmful-bias management.

These characteristics can conflict and require documented tradeoffs.
his versioned vocabulary follows the voluntary NIST AI RMF 1.0 (Tabassi, 2023).

Warning 8.5.4. *Adaptive Evaluation and Leakage*
Any choice influenced by protected evaluation data makes the final score adaptive. Repeated leaderboard queries, test-set debugging, manual example selection, and unreported prompt tuning can leak evaluation information.
Reserve a final assessment boundary; otherwise disclose the adaptation and narrow the claim.

Warning 8.5.5. *Benchmark Overclaiming*
A higher score supports a claim only under the reported dataset, metric, resources, selection procedure, and uncertainty.
It does not establish practical usefulness, causal benefit, robustness, fairness, safety, or superiority in another operating environment without matching evidence.

8.6 Reading and Synthesizing Research Papers

Choose reading depth according to the intended task: triage, comprehension, verification, reproduction, or synthesis.

Process 8.6.1. *Three-Pass Paper Reading*
Adapt the three-pass method of Keshav (2007) to the paper's evidence type and record the output of each completed pass.

(i) *Triage*: inspect the title, abstract, introduction, headings, conclusion, figures, and references. Record the paper's category, context, central claimed contribution, apparent assumptions, and relevance to the intended task.
(ii) *Comprehension*: read the body carefully, map each central claim to its stated assumptions and evidence, inspect figures and tables, and identify unfamiliar prerequisites or missing details.
(iii) *Reconstruction*: independently rebuild each decisive proof, derivation, algorithm, or experiment far enough to identify hidden assumptions, fragile steps, and reproducibility barriers.

Stop after the earliest pass whose recorded output resolves the intended task.
If a missing prerequisite prevents the next pass, identify and acquire it before resuming.

Process 8.6.2. *Paper Claim–Evidence Ledger*
Create one entry for each central claim and record its locator, exact scope, assumptions, evidence type, dependencies, result, uncertainty, and limitations. For a mathematical paper, reconstruct the definitions, hypotheses, cited dependencies, and proof obligations. For an empirical paper, reconstruct the sample, protocol, comparator, metric, variability, and analysis. For a systems paper, reconstruct the boundary, workload, implementation, platform, and failure model. After comparing the claim with that record, classify it as supported, partially supported, unsupported, or not assessed. Reputation and venue are metadata rather than claim-level evidence.

Process 8.6.3. *Literature Search and Synthesis*
Build a claim-centered literature map rather than a sequence of paper summaries.

(i) *Seed*: start from a precise research question, corresponding keywords, a recent survey, and several directly relevant papers.
(ii) *Backward search*: follow repeated foundational citations to original results, data, or methods.
(iii) *Forward search*: locate corrections, replications, extensions, and critiques that cite each key work.

(iv) *Cluster*: group papers by question, assumptions, method, evidence, and operating context rather than by author.
(v) *Compare*: normalize terminology and record agreements, contradictions, gaps, and incomparable protocols claim by claim.
(vi) *Saturate*: record the search date, sources, and inclusion boundary. Stop when another complete search and citation-tracing pass changes neither the paper clusters nor the identified agreements, contradictions, and gaps.

Keshav's survey procedure likewise expands seed papers through repeated authors, shared references, recent proceedings (Keshav, 2007).

Warning 8.6.4. *Version and Source Status*
Before treating a paper as evidence, check its version, venue, peer-review status, corrections, retractions, supplements, artifact links, and conflicts of interest. Cite the version examined and, when available, the original source for a result or dataset. Preprint and peer-review status affect provenance and scrutiny, but neither status establishes or refutes the paper's claims.

Principle 8.6.5. *Read Against the Claim*
For each central claim, compare its scope in the title, abstract, or conclusion with the assumptions and evidence in the body. Record what was established, under which assumptions, by which evidence, with what uncertainty, and within which population, system, or operating boundary. State the synthesized conclusion no more broadly than this comparison supports.

Part II
Structural Linear Algebra

1 Vector Spaces and Linear Structure

1.1 Vector Spaces

Vector spaces encode addition and scalar multiplication. Subspaces, spans, bases, dimension, and coordinates describe how linear structure is generated and measured.

Definition 1.1.1. *Vector Space*
Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set $\mathcal{V}$ with the following structure.
The operations are $+\colon \mathcal{V}\times\mathcal{V}\to\mathcal{V}$ and scalar multiplication $\mathbb{F}\times\mathcal{V}\to\mathcal{V}$. The scalar action is written $(a,\mathbf{v})\mapsto a\mathbf{v}$, and the axioms are:

(i) $(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{V})\;\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$;
(ii) $(\forall\,\mathbf{u},\mathbf{v},\mathbf{w}\in\mathcal{V})\;(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$;
(iii) an element $\mathbf{0}\in\mathcal{V}$ satisfies $(\forall\,\mathbf{v}\in\mathcal{V})\;\mathbf{0}+\mathbf{v}=\mathbf{v}$;
(iv) $(\forall\,\mathbf{v}\in\mathcal{V})(\exists\,\mathbf{w}\in\mathcal{V})\;\mathbf{v}+\mathbf{w}=\mathbf{0}$;
(v) $(\forall\,\mathbf{v}\in\mathcal{V})\;1\mathbf{v}=\mathbf{v}$;
(vi) $(\forall\,a,b\in\mathbb{F})(\forall\,\mathbf{v}\in\mathcal{V})\;(ab)\mathbf{v}=a(b\mathbf{v})$;
(vii) $(\forall\,a\in\mathbb{F})(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{V})\;a(\mathbf{u}+\mathbf{v})=a\mathbf{u}+a\mathbf{v}$;
(viii) $(\forall\,a,b\in\mathbb{F})(\forall\,\mathbf{v}\in\mathcal{V})\;(a+b)\mathbf{v}=a\mathbf{v}+b\mathbf{v}$.

If $\mathcal{V}=\{\mathbf{0}\}$, then $\mathcal{V}$ is a *zero vector space*; otherwise $\mathcal{V}$ is a *nonzero vector space*.

Theorem 1.1.2. *Basic Consequences of Vector Space Axioms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then:

(i) the additive identity is unique;
(ii) each $\mathbf{v}\in\mathcal{V}$ has a unique additive inverse, denoted $-\mathbf{v}$;
(iii) $(\forall\,\mathbf{v}\in\mathcal{V})\;0\mathbf{v}=\mathbf{0}$;
(iv) $(\forall\,a\in\mathbb{F})\;a\mathbf{0}=\mathbf{0}$;
(v) $(\forall\,a\in\mathbb{F})(\forall\,\mathbf{v}\in\mathcal{V})\;[a\mathbf{v}=\mathbf{0}\iff(a=0\vee\mathbf{v}=\mathbf{0})]$.

Proof.

(i) If $\mathbf{0}$ and $\mathbf{0}'$ are additive identities, then $\mathbf{0}=\mathbf{0}+\mathbf{0}'=\mathbf{0}'$.
(ii) If $\mathbf{w},\mathbf{w}'$ are additive inverses of $\mathbf{v}$, then $\mathbf{w}=\mathbf{w}+\mathbf{0}=\mathbf{w}+(\mathbf{v}+\mathbf{w}')=(\mathbf{w}+\mathbf{v})+\mathbf{w}'=\mathbf{0}+\mathbf{w}'=\mathbf{w}'$.
(iii) $0\mathbf{v}=(0+0)\mathbf{v}=0\mathbf{v}+0\mathbf{v}$; after adding $-(0\mathbf{v})$ to both sides, $0\mathbf{v}=\mathbf{0}$.
(iv) $a\mathbf{0}=a(\mathbf{0}+\mathbf{0})=a\mathbf{0}+a\mathbf{0}$; after adding $-(a\mathbf{0})$ to both sides, $a\mathbf{0}=\mathbf{0}$.
(v) ($\Rightarrow$) Suppose $a\mathbf{v}=\mathbf{0}$. If $a=0$, then $a=0\vee\mathbf{v}=\mathbf{0}$. If $a\neq0$, then $\mathbf{v}=1\mathbf{v}=(a^{-1}a)\mathbf{v}=a^{-1}(a\mathbf{v})=a^{-1}\mathbf{0}=\mathbf{0}$ by (iv).
(v) ($\Leftarrow$) If $a=0$ or $\mathbf{v}=\mathbf{0}$, then the conclusion follows from (iii) or (iv).

Structural Commentary: The proof derives cancellation, zero actions, and inverse uniqueness from the axioms.
Key Insight: Zero and inverse behavior is forced by the axioms. □

Corollary 1.1.3. *Scalar Multiplication by -1 Is Negation*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then $(\forall\,\mathbf{v}\in\mathcal{V})\;(-1)\mathbf{v}=-\mathbf{v}$.

Proof. For $\mathbf{v}\in\mathcal{V}$, $\mathbf{v}+(-1)\mathbf{v}=1\mathbf{v}+(-1)\mathbf{v}=(1-1)\mathbf{v}=0\mathbf{v}$.
By Theorem 1.1.2, $\mathbf{v}+(-1)\mathbf{v}=\mathbf{0}$ and additive inverses are unique, so $(-1)\mathbf{v}=-\mathbf{v}$.
Structural Commentary: The proof identifies $(-1)\mathbf{v}$ as the unique additive inverse.
Key Insight: Negation is scalar action by -1. □

Example 1.1.4. *Function-Space Model*
Let $\mathbb{F}$ be a field and let X be a set. Set $\mathbb{F}^X:=\{f:X\to\mathbb{F}\}$. For $f,g\in\mathbb{F}^X$ and $a\in\mathbb{F}$, define $f+g$ and af pointwise.
Thus $(f+g)(x)=f(x)+g(x)$ and $(af)(x)=af(x)$ for $x\in X$. Then $\mathbb{F}^X$ is an $\mathbb{F}$-vector space.
Evaluating each axiom at $x\in X$ reduces it to the corresponding field identity, and $\mathbf{0}$ is the constant 0 function.
Key Insight: Pointwise operations inherit linear structure from the scalar field.

1.2 Subspaces and Spans

Definition 1.2.1. *Subspace*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$.
A *subspace* of $\mathcal{V}$ is a subset $\mathcal{U}\subseteq\mathcal{V}$ that is an $\mathbb{F}$-vector space under the inherited operations; write $\mathcal{U}\leq\mathcal{V}$.

Theorem 1.2.2. *Subspace Test*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $\emptyset\neq\mathcal{U}\subseteq\mathcal{V}$. Then $\mathcal{U}\leq\mathcal{V}\iff(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{U})(\forall\,c\in\mathbb{F})\;c\mathbf{u}+\mathbf{v}\in\mathcal{U}$.

Proof.

($\Rightarrow$) $(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{U})(\forall\,c\in\mathbb{F})\;c\mathbf{u}+\mathbf{v}\in\mathcal{U}$ by inherited scalar and additive closure.
($\Leftarrow$) Let C denote the stated closure condition, and choose $\mathbf{u}_0\in\mathcal{U}$. The identity $\mathbf{0}=(-1)\mathbf{u}_0+\mathbf{u}_0\in\mathcal{U}$ holds by C and Corollary 1.1.3.
$(\forall\,\mathbf{u}\in\mathcal{U})\;-\mathbf{u}=(-1)\mathbf{u}+\mathbf{0}\in\mathcal{U}$ by C and Corollary 1.1.3. $(\forall\,\mathbf{u}\in\mathcal{U})(\forall\,c\in\mathbb{F})\;c\mathbf{u}=c\mathbf{u}+\mathbf{0}\in\mathcal{U}$ by C.
$(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{U})\;\mathbf{u}+\mathbf{v}=1\mathbf{u}+\mathbf{v}\in\mathcal{U}$ by C. The remaining vector-space axioms are inherited from $\mathcal{V}$, so $\mathcal{U}\leq\mathcal{V}$.

Structural Commentary: Nonemptiness supplies a zero; the single closure condition supplies inverses, scalar multiples, and sums.
Key Insight: One affine-looking closure test captures all subspace closure. □

Proposition 1.2.3. *Intersections of Subspaces*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$, let $\emptyset\neq I$, and let $\{\mathcal{U}_i\}_{i\in I}$ be a family of subspaces of $\mathcal{V}$. Then $\bigcap_{i\in I}\mathcal{U}_i\leq\mathcal{V}$.

Proof. Set $M := \bigcap_{i\in I} \mathcal{U}_i$. Then $\mathbf{0} \in M$ by $(\forall\, i \in I)\, \mathbf{0} \in \mathcal{U}_i$, so $M \neq \emptyset$.
$(\forall\, \mathbf{u}, \mathbf{v} \in M)(\forall\, c \in \mathbb{F})(\forall\, i \in I)\, c\mathbf{u} + \mathbf{v} \in \mathcal{U}_i$ by the definition of M and $\mathcal{U}_i \leq \mathcal{V}$.
Hence $(\forall\, \mathbf{u}, \mathbf{v} \in M)(\forall\, c \in \mathbb{F})\, c\mathbf{u} + \mathbf{v} \in M$, and $M \leq \mathcal{V}$ by Theorem 1.2.2.
Structural Commentary: The proof verifies nonemptiness and closure pointwise across the indexed family.
Key Insight: Shared closure survives intersection. □

Definition 1.2.4. *Lists, Linear Combinations, and Span*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

(i) For $m \in \mathbb{N}_0$, a *list* in $\mathcal{V}$ is $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$; the case $m = 0$ is the empty list $()$.

(ii) For $m \in \mathbb{N}_0$, $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$, and $(a_1, \ldots, a_m) \in \mathbb{F}^m$, a *linear combination* of β is $\sum_{j=1}^{m} a_j \mathbf{v}_j$.
The case $m = 0$ is the empty linear combination, whose value is $\mathbf{0}$.

(iii) For $E \subseteq \mathcal{V}$, its *span* is the set of all finite linear combinations of entries from E.
Precisely, $\operatorname{span}(E) := \left\{ \sum_{j=1}^{m} a_j \mathbf{v}_j : m \in \mathbb{N}_0,\ (a_1, \ldots, a_m) \in \mathbb{F}^m,\ (\mathbf{v}_1, \ldots, \mathbf{v}_m) \in E^m \right\}$; hence $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$.

(iv) For $m \in \mathbb{N}_0$ and $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$, define $\operatorname{span}(\beta) := \operatorname{span}(\{\mathbf{v}_j : 1 \leq j \leq m\})$. For the empty list, $\operatorname{span}(()) = \{\mathbf{0}\}$.

(v) $E \subseteq \mathcal{V}$ *spans* $\mathcal{V}$ iff $\operatorname{span}(E) = \mathcal{V}$. A list β *spans* $\mathcal{V}$ iff $\operatorname{span}(\beta) = \mathcal{V}$.

Theorem 1.2.5. *Span is the Smallest Containing Subspace*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $E \subseteq \mathcal{V}$. Then:

(i) $\operatorname{span}(E) \leq \mathcal{V}$;
(ii) $E \subseteq \operatorname{span}(E)$;
(iii) $(\forall\, \mathcal{U} \leq \mathcal{V})\, [E \subseteq \mathcal{U} \Rightarrow \operatorname{span}(E) \subseteq \mathcal{U}]$.

Proof.

(i) The empty combination has value $\mathbf{0}$, so $\mathbf{0} \in \operatorname{span}(E)$. Let $\mathbf{x}, \mathbf{y} \in \operatorname{span}(E)$ and $c \in \mathbb{F}$.
Choose $m, n \in \mathbb{N}_0$ and finite linear-combination representations from E, $\mathbf{x} = \sum_{j=1}^{m} a_j \mathbf{e}_j$ and $\mathbf{y} = \sum_{k=1}^{n} b_k \mathbf{f}_k$.
Then $c\mathbf{x} + \mathbf{y} = \sum_{j=1}^{m} (ca_j)\mathbf{e}_j + \sum_{k=1}^{n} b_k \mathbf{f}_k \in \operatorname{span}(E)$ by concatenating the two lists. Thus $\operatorname{span}(E) \leq \mathcal{V}$ by Theorem 1.2.2.

(ii) $(\forall\, \mathbf{v} \in E)\, \mathbf{v} = 1\mathbf{v} \in \operatorname{span}(E)$, so $E \subseteq \operatorname{span}(E)$.

(iii) Let $\mathcal{U} \leq \mathcal{V}$ and $E \subseteq \mathcal{U}$. Set $P(m) := [(\forall\, (a_1, \ldots, a_m) \in \mathbb{F}^m)(\forall\, (\mathbf{e}_1, \ldots, \mathbf{e}_m) \in E^m)\ \sum_{j=1}^{m} a_j \mathbf{e}_j \in \mathcal{U}]$.
The base case $P(0)$ holds by $\mathbf{0} \in \mathcal{U}$. Assume $m \in \mathbb{N}_0$, $P(m)$, $(a_1, \ldots, a_{m+1}) \in \mathbb{F}^{m+1}$, and $(\mathbf{e}_1, \ldots, \mathbf{e}_{m+1}) \in E^{m+1}$.
Set $\mathbf{s} := \sum_{j=1}^{m} a_j \mathbf{e}_j$. Then $\mathbf{s} \in \mathcal{U}$ by $P(m)$. $a_{m+1}\mathbf{e}_{m+1} \in \mathcal{U}$ by $E \subseteq \mathcal{U}$ and scalar closure in $\mathcal{U}$.
Hence $\sum_{j=1}^{m+1} a_j \mathbf{e}_j = \mathbf{s} + a_{m+1}\mathbf{e}_{m+1} \in \mathcal{U}$ by additive closure, so $P(m+1)$.
$(\forall\, m \in \mathbb{N}_0)\, P(m)$ by induction. Hence $\operatorname{span}(E) \subseteq \mathcal{U}$ by the definition of span.

Structural Commentary: The proof verifies closure, generator inclusion, and minimality in statement order.
Key Insight: Finite linear combinations are exactly the closure forced by E. □

Proposition 1.2.6. *Monotonicity of Span*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$, and let $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$. Then $\mathcal{U} \subseteq \mathcal{W} \Rightarrow \operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.

Proof. Assume $\mathcal{U} \subseteq \mathcal{W}$. Then $\operatorname{span}(\mathcal{W}) \leq \mathcal{V}$ and $\mathcal{W} \subseteq \operatorname{span}(\mathcal{W})$ by Theorem 1.2.5(i)–(ii).
Thus $\mathcal{U} \subseteq \operatorname{span}(\mathcal{W})$, and $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$ by Theorem 1.2.5(iii).
Structural Commentary: The proof compares both generating sets inside the same containing subspace.
Key Insight: Minimality converts generator inclusion into span inclusion. □

Computation Method 1.2.7. *Subspace and Span Diagnostic*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $S \subseteq \mathcal{V}$. Return the output matching the task:

(i) *Subspace certificate*: establish $S \neq \emptyset$ and $(\forall\, \mathbf{u}, \mathbf{v} \in S)(\forall\, c \in \mathbb{F})\, c\mathbf{u} + \mathbf{v} \in S$. Then $S \leq \mathcal{V}$ by Theorem 1.2.2.

(ii) *Failure certificate*: if $S = \emptyset$, then $S \not\leq \mathcal{V}$. Otherwise exhibit $\mathbf{u}, \mathbf{v} \in S$ and $c \in \mathbb{F}$ with $c\mathbf{u} + \mathbf{v} \notin S$.
Then $S \not\leq \mathcal{V}$ by Theorem 1.2.2. The condition $\mathbf{0} \in S$ alone is insufficient.

(iii) *Generated subspace*: return $\operatorname{span}(S)$, the set of all finite linear combinations from S.
Precisely, $\operatorname{span}(S) = \left\{ \sum_{i=1}^{n} a_i \mathbf{s}_i : n \in \mathbb{N}_0,\ (a_1, \ldots, a_n) \in \mathbb{F}^n,\ (\mathbf{s}_1, \ldots, \mathbf{s}_n) \in S^n \right\}$, with the empty sum $\mathbf{0}$ at $n = 0$.

1.3 Linear Independence and Bases

Throughout this subsection, let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

Definition 1.3.1. *Linear Independence, Bases, and Finite-Dimensionality*

(i) A nonempty finite list $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$ is *linearly independent* iff $(\forall\, (a_1, \ldots, a_m) \in \mathbb{F}^m)\, [\sum_{j=1}^{m} a_j \mathbf{v}_j = \mathbf{0} \Rightarrow a_1 = \cdots = a_m = 0]$.
The empty list is linearly independent. A finite list that is not linearly independent is *linearly dependent*.

(ii) A list β is a *basis* of $\mathcal{V}$ iff β is linearly independent and $\operatorname{span}(\beta) = \mathcal{V}$.

(iii) $\mathcal{V}$ is *finite-dimensional* iff $(\exists\, m \in \mathbb{N}_0)(\exists\, \beta \in \mathcal{V}^m)\ \operatorname{span}(\beta) = \mathcal{V}$. Otherwise $\mathcal{V}$ is *infinite-dimensional*.

Proposition 1.3.2. *Unique Coefficients for Independent Lists*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, $\mathbf{a} := (a_1, \ldots, a_m) \in \mathbb{F}^m$, $\mathbf{b} := (b_1, \ldots, b_m) \in \mathbb{F}^m$; $\sum_{j=1}^{m} a_j \mathbf{v}_j = \sum_{j=1}^{m} b_j \mathbf{v}_j$ iff $\mathbf{a} = \mathbf{b}$.

Proof.

($\Rightarrow$) $\sum_{j=1}^{m}(a_j - b_j)\mathbf{v}_j = \mathbf{0}$ by subtracting the two sums. Hence $(\forall\, j \in \{1,\dots,m\})\, a_j - b_j = 0$ by linear independence of β, so $\mathbf{a} = \mathbf{b}$.

($\Leftarrow$) The two sums are equal by equality of the coefficient tuples.

Structural Commentary: Subtracting two representations converts coefficient comparison into the defining zero-relation test.
Key Insight: Independence detects equality coefficient by coefficient. □

Proposition 1.3.3. *Unique Coordinate Representation*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a finite list in $\mathcal{V}$. Then β is a basis of $\mathcal{V}$ iff $(\forall\, \mathbf{v} \in \mathcal{V})(\exists!\,(a_1,\dots,a_n) \in \mathbb{F}^n)\, \mathbf{v} = \sum_{j=1}^{n} a_j\mathbf{v}_j$.

Proof.

($\Rightarrow$) Every $\mathbf{v} \in \mathcal{V}$ has a coefficient tuple by $\operatorname{span}(\beta) = \mathcal{V}$, and that tuple is unique by Proposition 1.3.2.

($\Leftarrow$) $\operatorname{span}(\beta) = \mathcal{V}$ by the existence clause. If $\sum_{j=1}^{n} a_j\mathbf{v}_j = \mathbf{0}$, then $\sum_{j=1}^{n} a_j\mathbf{v}_j = \sum_{j=1}^{n} 0\mathbf{v}_j$, so $(\forall\, j \in \{1,\dots,n\})\, a_j = 0$ by the uniqueness clause. Thus β is linearly independent and hence a basis.

Structural Commentary: Spanning supplies coordinate tuples, while independence separates any two such tuples.
Key Insight: A basis is precisely a list with universal and unique coordinates. □

Proposition 1.3.4. *Appending to a Dependent List*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ be linearly dependent in $\mathcal{V}$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\,(\mathbf{v}_1,\dots,\mathbf{v}_m,\mathbf{v})$ is linearly dependent.

Proof. Choose $(a_1,\dots,a_m) \in \mathbb{F}^m$ such that $\sum_{j=1}^{m} a_j\mathbf{v}_j = \mathbf{0}$ and $(\exists\, j \in \{1,\dots,m\})\, a_j \neq 0$ by dependence of β.

For $\mathbf{v} \in \mathcal{V}$, $\sum_{j=1}^{m} a_j\mathbf{v}_j + 0\mathbf{v} = \mathbf{0}$ is a nontrivial relation on $(\beta, \mathbf{v})$.

Structural Commentary: The original relation is embedded into the longer coefficient tuple by assigning zero to the appended entry.
Key Insight: An existing dependence relation survives every append. □

Definition 1.3.5. *Omitting a Vector from a List*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ be a nonempty list in $\mathcal{V}$. For $j \in \{1,\dots,m\}$, write $\beta_{\widehat{j}} := (\mathbf{v}_1,\dots,\widehat{\mathbf{v}_j},\dots,\mathbf{v}_m)$; if $m = 1$, then $\beta_{\widehat{1}} = ()$.

Proposition 1.3.6. *Omitting from an Independent List*

If $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ is a nonempty linearly independent list, then $(\forall\, j \in \{1,\dots,m\})$ $\beta_{\widehat{j}}$ is linearly independent.

Proof. Fix $j \in \{1,\dots,m\}$, and let $(a_i)_{i\neq j}$ satisfy $\sum_{i\neq j} a_i\mathbf{v}_i = \mathbf{0}$. Set $a_j := 0$.

Then $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$, so $(\forall\, i \in \{1,\dots,m\})\, a_i = 0$ by independence of β. Hence every coefficient in the original relation is zero.

Structural Commentary: Padding a sublist relation with a zero coefficient turns it into a relation on the original list.
Key Insight: Deletion cannot introduce a new cancellation. □

Proposition 1.3.7. *Adjoining and Removing Vectors from Lists*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ be a list in $\mathcal{V}$, and write $(\beta, \mathbf{v}) := (\mathbf{v}_1,\dots,\mathbf{v}_m,\mathbf{v})$.

(i) If β is linearly independent and $\mathbf{v} \notin \operatorname{span}(\beta)$, then $(\beta, \mathbf{v})$ is linearly independent.

(ii) If $(a_1,\dots,a_m) \in \mathbb{F}^m$, $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$, and $j \in \{1,\dots,m\}$ satisfies $a_j \neq 0$, then $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$.
Hence $\operatorname{span}(\beta) = \mathcal{V}$ implies $\operatorname{span}(\beta_{\widehat{j}}) = \mathcal{V}$.

Proof.

(i) Let $\sum_{i=1}^{m} a_i\mathbf{v}_i + a\mathbf{v} = \mathbf{0}$. If $a \neq 0$, then $\mathbf{v} = -a^{-1}\sum_{i=1}^{m} a_i\mathbf{v}_i \in \operatorname{span}(\beta)$, contrary to hypothesis.
Hence $a = 0$, and then $(\forall\, i \in \{1,\dots,m\})\, a_i = 0$ by independence of β.

(ii) $\mathbf{v}_j = -a_j^{-1}\sum_{i\neq j} a_i\mathbf{v}_i \in \operatorname{span}(\beta_{\widehat{j}})$ by $a_j \neq 0$ and the stated relation.
Thus every vector of β lies in $\operatorname{span}(\beta_{\widehat{j}})$, so $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{\widehat{j}})$. The reverse inclusion is immediate because $\beta_{\widehat{j}}$ is a sublist of β.

Structural Commentary: The first argument isolates the appended coefficient; the second solves for a nonzero-coefficient entry.
Key Insight: Append outside the current span, or delete an entry already controlled by the others. □

Theorem 1.3.8. *Basis Characterizations*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a finite list in $\mathcal{V}$. Then the following are equivalent:

(i) β is a basis of $\mathcal{V}$;

(ii) $(\forall\, \mathbf{x} \in \mathcal{V})(\exists!\,(a_1,\dots,a_n) \in \mathbb{F}^n)\, \mathbf{x} = \sum_{j=1}^{n} a_j\mathbf{v}_j$;

(iii) $\operatorname{span}(\beta) = \mathcal{V}$, and no proper sublist of β spans $\mathcal{V}$;

(iv) β is linearly independent, and $(\forall\, \mathbf{x} \in \mathcal{V})$ $(\beta, \mathbf{x})$ is linearly dependent.

Proof.

(*i*) $\Rightarrow$ (*ii*) by Proposition 1.3.3.

(*ii*) $\Rightarrow$ (*iii*). The equality $\operatorname{span}(\beta) = \mathcal{V}$ holds by the existence clause.
Suppose a proper sublist γ of β spans $\mathcal{V}$, and choose an occurrence $\mathbf{v}_j$ omitted from γ.
A representation of $\mathbf{v}_j$ by γ extends to a β-coefficient tuple whose jth entry is 0, whereas the standard coefficient tuple for $\mathbf{v}_j$ has jth entry 1. This contradicts the uniqueness clause.

(*iii*) $\Rightarrow$ (*iv*). If β were linearly dependent, choose a nontrivial relation $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}$ and an index j with $a_j \neq 0$.
Then $\beta_{\widehat{j}}$ would span $\mathcal{V}$ by Proposition 1.3.7(ii), contrary to minimality. Hence β is linearly independent.
For $\mathbf{x} \in \mathcal{V}$, choose $\mathbf{x} = \sum_{j=1}^{n} a_j \mathbf{v}_j$ by $\operatorname{span}(\beta) = \mathcal{V}$. Then $(a_1, \dots, a_n, -1)$ is a nonzero relation tuple for $(\beta, \mathbf{x})$.

(*iv*) $\Rightarrow$ (*i*). If $\operatorname{span}(\beta) \neq \mathcal{V}$, choose $\mathbf{x} \in \mathcal{V} \setminus \operatorname{span}(\beta)$. Then $(\beta, \mathbf{x})$ is linearly independent by Proposition 1.3.7(i), contrary to (iv). Thus $\operatorname{span}(\beta) = \mathcal{V}$, and β is a basis by the independence clause in (iv).

Structural Commentary: The implication cycle passes through unique coordinates, minimal spanning, and maximal independence.
Key Insight: A basis is both an irredundant spanning list and an independence-maximal list. □

Lemma 1.3.9. *Replacement Lemma*
Let $\mathcal{V} \neq \{\mathbf{0}\}$, let $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_r)$ span $\mathcal{V}$, and let $\mathbf{0} \neq \mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i$. For $j \in \{1, \dots, r\}$, set $\beta_{j \leftarrow \mathbf{v}} := (\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{v}, \mathbf{u}_{j+1}, \dots, \mathbf{u}_r)$.

(i) $(\exists\, j \in \{1, \dots, r\})\, c_j \neq 0$.
(ii) If $c_j \neq 0$, then $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$.
(iii) If $c_j \neq 0$ and β is a basis, then $\beta_{j \leftarrow \mathbf{v}}$ is a basis.
(iv) If $r \geq 2$, β is a basis, $k \in \{1, \dots, r-1\}$, and $\mathbf{v} \notin \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$, then $(\exists\, j \in \{k+1, \dots, r\})$ $\beta_{j \leftarrow \mathbf{v}}$ is a basis.

Proof.

(i) If $(\forall\, j \in \{1, \dots, r\})\, c_j = 0$, then $\mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i = \mathbf{0}$, contrary to hypothesis.

(ii) Assume $c_j \neq 0$. Then $\mathbf{u}_j = c_j^{-1} \left(\mathbf{v} - \sum_{i \neq j} c_i \mathbf{u}_i \right) \in \operatorname{span}(\beta_{j \leftarrow \mathbf{v}})$, so $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{j \leftarrow \mathbf{v}})$.
Conversely, $\mathbf{v} \in \operatorname{span}(\beta)$, so $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) \subseteq \operatorname{span}(\beta)$. Hence $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$.

(iii) Assume $c_j \neq 0$ and β is a basis. Let $\sum_{i<j} b_i \mathbf{u}_i + a\mathbf{v} + \sum_{i>j} b_i \mathbf{u}_i = \mathbf{0}$. Then $\sum_{i \neq j} (b_i + a c_i) \mathbf{u}_i + a c_j \mathbf{u}_j = \mathbf{0}$ by substituting $\mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i$.
Thus $a c_j = 0$ and $(\forall\, i \neq j)\, b_i + a c_i = 0$ by independence of β. Hence $a = 0$ by $c_j \neq 0$, and then $(\forall\, i \neq j)\, b_i = 0$.
Therefore $\beta_{j \leftarrow \mathbf{v}}$ is linearly independent; it spans $\mathcal{V}$ by (ii), so it is a basis.

(iv) If $(\forall\, j \in \{k+1, \dots, r\})\, c_j = 0$, then $\mathbf{v} = \sum_{i=1}^{k} c_i \mathbf{u}_i \in \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$, contrary to hypothesis.

(v) Choose $j \in \{k+1, \dots, r\}$ with $c_j \neq 0$. Then $\beta_{j \leftarrow \mathbf{v}}$ is a basis by (iii).

Structural Commentary: Solving for a nonzero-coefficient entry preserves span; comparing coefficients preserves independence.
Key Insight: A vector may replace precisely an entry on which its expansion genuinely depends. □

Lemma 1.3.10. *Linear Dependence Lemma*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ be linearly dependent with $\mathbf{v}_1 \neq \mathbf{0}$. Then $(\exists\, j \in \{2, \dots, m\})\, [\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}) \wedge \operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)]$.

Proof.

(i) *Terminal coefficient*: Choose a nontrivial relation $\sum_{i=1}^{m} a_i \mathbf{v}_i = \mathbf{0}$, and set $j := \max\{i : a_i \neq 0\}$.
The equality $j = 1$ would imply $\mathbf{v}_1 = \mathbf{0}$ by $a_1 \mathbf{v}_1 = \mathbf{0}$ and $a_1 \neq 0$, contrary to hypothesis. Hence $j \in \{2, \dots, m\}$.

(ii) *Earlier-span representation*: $(\forall\, i > j)\, a_i = 0$ by the choice of j, and therefore $\mathbf{v}_j = -a_j^{-1} \sum_{i=1}^{j-1} a_i \mathbf{v}_i \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$.

(iii) *Span preservation*: This representation rewrites every linear combination of β without $\mathbf{v}_j$, so $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{\widehat{j}})$.
The reverse inclusion holds because $\beta_{\widehat{j}}$ is a sublist of β. Hence $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$.

Structural Commentary: The last nonzero relation coefficient identifies a later vector in the earlier span that can be removed.
Key Insight: Choose the final active coefficient to preserve both order and span. □

Theorem 1.3.11. *Independent Lists Are No Longer Than Spanning Lists*
If $\mathcal{V}$ is finite-dimensional, $\alpha = (\mathbf{u}_1, \dots, \mathbf{u}_m)$ is linearly independent, and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ spans $\mathcal{V}$, then $m \leq n$.

Proof.

(i) *Contradictory regime*: Suppose $m > n$. Then $n \geq 1$: if $n = 0$, then $\mathbf{u}_1 \in \mathcal{V} = \operatorname{span}(\gamma) = \{\mathbf{0}\}$, contrary to independence of α.

(ii) *Successive replacement*: Construct spanning lists γ_k of length n with designated occurrences of $\mathbf{u}_1, \dots, \mathbf{u}_k$ for $0 \leq k \leq n$.
Set $\gamma_0 := \gamma$. Suppose $k < n$ and γ_k has been constructed. Then $\mathbf{u}_{k+1} \notin \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ by independence of α.
Expand $\mathbf{u}_{k+1}$ in the spanning list γ_k. Some coefficient at a nondesignated position is nonzero; otherwise the expansion would place $\mathbf{u}_{k+1}$ in $\operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$. The vector $\mathbf{u}_{k+1}$ is nonzero by independence of α. Replace the identified position by $\mathbf{u}_{k+1}$.
The resulting list γ_{k+1} spans $\mathcal{V}$ by Lemma 1.3.9(ii), has length n, and retains every earlier designated occurrence.

(iii) *Terminal list*: The designated occurrences of $\mathbf{u}_1, \dots, \mathbf{u}_n$ occupy all n positions of γ_n. Hence $\operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_n) = \operatorname{span}(\gamma_n) = \mathcal{V}$.

(iv) *Contradiction*: Since $m > n$, $\mathbf{u}_{n+1} \in \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_n)$ by the terminal equality, contrary to independence of α.

Structural Commentary: Each replacement reserves one independent vector while keeping the spanning-list length and span fixed.
Key Insight: After all spanning positions are reserved, no further independent direction can remain. □

Theorem 1.3.12. *Reducing Spanning Lists*
If $\mathcal{V}$ is finite-dimensional, then every spanning list in $\mathcal{V}$ contains a basis sublist.

Proof. Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a spanning list of $\mathcal{V}$. If β is independent, it is a basis. Otherwise remove one vector without changing span: omit $\mathbf{v}_1$ if $\mathbf{v}_1 = \mathbf{0}$; if $\mathbf{v}_1 \neq \mathbf{0}$, remove a suitable $\mathbf{v}_j$ by Lemma 1.3.10. The resulting sublist still spans $\mathcal{V}$ and has length one less. Repeating at most n times reaches a spanning independent sublist, hence a basis.
Structural Commentary: Each deletion shortens the list without changing span, so finite descent reaches an independent sublist.
Key Insight: Repeatedly remove a certified redundancy until only a basis remains. □

Theorem 1.3.13. *Extending Independent Lists*
If $\mathcal{V}$ is finite-dimensional, then every linearly independent list in $\mathcal{V}$ extends to a basis of $\mathcal{V}$.

Proof.

(i) *Setup*: Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, and choose a finite spanning list $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ of $\mathcal{V}$.
(ii) *Construction*: Starting from $\delta_0 := \beta$, inspect $\mathbf{w}_1, \dots, \mathbf{w}_n$ in order
At stage j, append $\mathbf{w}_j$ exactly when $\mathbf{w}_j \notin \operatorname{span}(\delta_{j-1})$, and call the resulting list δ_j.
Every δ_j is linearly independent by induction using Proposition 1.3.7(i) at each append.
(iii) *Verification*: $(\forall\, j \in \{1, \dots, n\})\ \mathbf{w}_j \in \operatorname{span}(\delta_n)$ because each $\mathbf{w}_j$ was either already spanned when inspected or appended, and subsequent stages only append. Thus $\operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_n) \subseteq \operatorname{span}(\delta_n)$ by Theorem 1.2.5(iii).
Hence $\operatorname{span}(\delta_n) = \mathcal{V}$, and the independent list δ_n is a basis extending β.

Structural Commentary: The finite scan preserves independence while absorbing each chosen spanning generator.
Key Insight: Append exactly the generators not already controlled by the current span. □

Corollary 1.3.14. *Bases and Subspaces in Finite Dimension*
If $\mathcal{V}$ is finite-dimensional, then:

(i) $\mathcal{V}$ has a basis;
(ii) $(\forall \mathcal{U} \leq \mathcal{V})$ $\mathcal{U}$ is finite-dimensional.

Proof. Choose a finite spanning list $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ of $\mathcal{V}$.

(i) A basis of $\mathcal{V}$ is contained in γ by Theorem 1.3.12.
(ii) Fix $\mathcal{U} \leq \mathcal{V}$, and set $\beta_0 := ()$. Whenever $\operatorname{span}(\beta_k) \neq \mathcal{U}$, choose $\mathbf{u}_{k+1} \in \mathcal{U} \setminus \operatorname{span}(\beta_k)$ and set $\beta_{k+1} := (\beta_k, \mathbf{u}_{k+1})$.
Every constructed list is linearly independent by induction using Proposition 1.3.7(i).
The construction cannot reach β_{n+1} by Theorem 1.3.11, because γ is a spanning list of length n.
Hence it stops at some $k \leq n$ with $\operatorname{span}(\beta_k) = \mathcal{U}$. Thus β_k is a finite basis of $\mathcal{U}$, so $\mathcal{U}$ is finite-dimensional.

Structural Commentary: Basis reduction handles $\mathcal{V}$; the ambient length bound forces every subspace construction to terminate.
Key Insight: Ambient finite spanning length bounds the size of every independent subspace list. □

Definition 1.3.15. *Independent Sets and Hamel Bases*
A subset $I \subseteq \mathcal{V}$ is *linearly independent* iff every finite list of distinct vectors in I is linearly independent.
A subset $\mathcal{B} \subseteq \mathcal{V}$ is a *Hamel basis* of $\mathcal{V}$ iff $\mathcal{B}$ is linearly independent and $\operatorname{span}(\mathcal{B}) = \mathcal{V}$.
For $a : \mathcal{B} \to \mathbb{F}$, define $\operatorname{supp}_{\mathcal{B}}(a) := \{\mathbf{b} \in \mathcal{B} : a(\mathbf{b}) \neq 0\}$. A *finite $\mathcal{B}$-expansion* of $\mathbf{v} \in \mathcal{V}$ is a function $a : \mathcal{B} \to \mathbb{F}$ with finite support such that $\mathbf{v} = \sum_{\mathbf{b} \in \operatorname{supp}_{\mathcal{B}}(a)} a(\mathbf{b})\mathbf{b}$. For a finite Hamel basis, any ordering of $\mathcal{B}$ is an ordered basis.

Proposition 1.3.16. *Hamel Coordinate Representation*
If $\mathcal{B}$ is a Hamel basis of $\mathcal{V}$, then every $\mathbf{v} \in \mathcal{V}$ has a unique finite $\mathcal{B}$-expansion.

Proof. Fix $\mathbf{v} \in \mathcal{V}$. Choose $m \in \mathbb{N}_0$, $(c_1, \dots, c_m) \in \mathbb{F}^m$, and $(\mathbf{b}_1, \dots, \mathbf{b}_m) \in \mathcal{B}^m$ such that $\mathbf{v} = \sum_{j=1}^{m} c_j \mathbf{b}_j$ by $\operatorname{span}(\mathcal{B}) = \mathcal{V}$.
For $\mathbf{b} \in \mathcal{B}$, set $J_{\mathbf{b}} := \{j \in \{1, \dots, m\} : \mathbf{b}_j = \mathbf{b}\}$, and define $a : \mathcal{B} \to \mathbb{F}$ by $a(\mathbf{b}) := \sum_{j \in J_{\mathbf{b}}} c_j$.
Then $\operatorname{supp}_{\mathcal{B}}(a) \subseteq \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$, and $\mathbf{v} = \sum_{\mathbf{b} \in \operatorname{supp}_{\mathcal{B}}(a)} a(\mathbf{b})\mathbf{b}$ by regrouping equal entries. Hence a is a finite $\mathcal{B}$-expansion of $\mathbf{v}$.
For uniqueness, let $a, a' : \mathcal{B} \to \mathbb{F}$ be finite $\mathcal{B}$-expansions of $\mathbf{v}$, and set $F := \operatorname{supp}_{\mathcal{B}}(a) \cup \operatorname{supp}_{\mathcal{B}}(a')$.
Then F is finite and $\sum_{\mathbf{b} \in F} (a(\mathbf{b}) - a'(\mathbf{b}))\mathbf{b} = \mathbf{0}$. Thus $(\forall\, \mathbf{b} \in F)\, a(\mathbf{b}) = a'(\mathbf{b})$ by independence of $\mathcal{B}$. Both vanish on $\mathcal{B} \setminus F$, so $a = a'$.
Structural Commentary: Existence consolidates repeated generators; uniqueness compares their combined finite support.
Key Insight: Hamel coordinates are ordinary independent-list coefficients indexed by finite support. □

Theorem 1.3.17. *Hamel Basis Extension*
Assume Zorn's lemma. Then:

(i) if $I \subseteq S \subseteq \mathcal{V}$, I is linearly independent, and $\operatorname{span}(S) = \mathcal{V}$, then $(\exists\, \mathcal{B})\, [I \subseteq \mathcal{B} \subseteq S \wedge \mathcal{B}$ is a Hamel basis of $\mathcal{V}]$;
(ii) $\mathcal{V}$ has a Hamel basis;
(iii) every linearly independent subset of $\mathcal{V}$ extends to a Hamel basis of $\mathcal{V}$;
(iv) every spanning subset of $\mathcal{V}$ contains a Hamel basis of $\mathcal{V}$.

Proof.

(i) Let $\mathscr{P} := \{E : I \subseteq E \subseteq S,\ E \text{ is linearly independent}\}$, ordered by inclusion.
Then $I \in \mathscr{P}$. Let $\mathcal{C} \subseteq \mathscr{P}$ be a nonempty chain, and set $U := \bigcup_{E \in \mathcal{C}} E$. Then $I \subseteq U \subseteq S$.
Let $m \in \mathbb{N}$, and consider a relation $\sum_{k=1}^{m} a_k \mathbf{u}_k = \mathbf{0}$ among distinct vectors of U.
Choose $E_1, \dots, E_m \in \mathcal{C}$ such that $(\forall\, k \in \{1, \dots, m\})\, \mathbf{u}_k \in E_k$.
Some $E_* \in \mathcal{C}$ contains $E_1 \cup \cdots \cup E_m$ by finite induction on the chain order.
Thus $(\forall\, k \in \{1, \dots, m\})\, a_k = 0$ by independence of E_*, so $U \in \mathscr{P}$.
Hence every nonempty chain has an upper bound in $\mathscr{P}$; the empty chain is bounded by I.
Choose a maximal $\mathcal{B} \in \mathscr{P}$ by Zorn's lemma. Suppose $s \in S \setminus \operatorname{span}(\mathcal{B})$.
In a finite relation among distinct vectors of $\mathcal{B} \cup \{s\}$, a nonzero coefficient of s would imply $s \in \operatorname{span}(\mathcal{B})$, while a zero coefficient leaves a relation in $\mathcal{B}$. Hence $\mathcal{B} \cup \{s\}$ is linearly independent, contrary to maximality of $\mathcal{B}$.
Therefore $S \subseteq \operatorname{span}(\mathcal{B})$, so $\operatorname{span}(\mathcal{B}) = \mathcal{V}$ by $\operatorname{span}(S) = \mathcal{V}$. Thus $I \subseteq \mathcal{B} \subseteq S$, and $\mathcal{B}$ is a Hamel basis.
(ii) $\mathcal{V}$ has a Hamel basis by (i) with the linearly independent set $I = \emptyset$ and the spanning set $S = \mathcal{V}$.
(iii) Every prescribed linearly independent subset extends to a Hamel basis by (i) with $S = \mathcal{V}$.
(iv) Every prescribed spanning subset contains a Hamel basis by (i) with $I = \emptyset$.

Structural Commentary: Chain unions preserve finite independence, and maximality rules out every unspanned element of S.
Key Insight: Finitary independence is stable under chain unions, so Zorn maximality forces a Hamel basis. □

1.4 Dimension and Coordinates

Throughout this subsection, let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

Theorem 1.4.1. *Basis Length Is Well-Defined*
If $\mathcal{V}$ is finite-dimensional and β_1, β_2 are bases of $\mathcal{V}$ of lengths m, n, respectively, then $m = n$.

Proof. The inequality $m \leq n$ holds by Theorem 1.3.11, with β_1 independent and β_2 spanning.
After interchanging the bases, $n \leq m$ holds by the same theorem. Hence $m = n$.
Structural Commentary: Using both basis roles proves the opposite length inequalities.
Key Insight: Symmetric comparison makes basis length intrinsic. □

Definition 1.4.2. *Dimension of a Finite-Dimensional Space*
Assume $\mathcal{V}$ is finite-dimensional. Choose a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ by Corollary 1.3.14, and define $\dim(\mathcal{V}) := n$.
This value is independent of β by Theorem 1.4.1, and $\mathcal{V}$ is called *n-dimensional*. Since $()$ is a basis of $\{\mathbf{0}\}$, $\dim(\{\mathbf{0}\}) = 0$.

Proposition 1.4.3. *Subspace Dimension Criterion*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U} \leq \mathcal{V}$. Then:
(i) $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$;
(ii) $\dim(\mathcal{U}) = \dim(\mathcal{V})$ *iff* $\mathcal{U} = \mathcal{V}$.

Proof. Choose a basis $\delta = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of $\mathcal{V}$ by Corollary 1.3.14.
(*i*) The list δ is independent in $\mathcal{V}$, while β spans $\mathcal{V}$. Hence $m \leq n$ by Theorem 1.3.11, so $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$.
(*ii*) ($\Rightarrow$) Suppose $\dim(\mathcal{U}) = \dim(\mathcal{V})$. Then $m = n$. Extend δ to a basis of $\mathcal{V}$ by Theorem 1.3.13.
The extension adds no entry because its length is n by Theorem 1.4.1. Thus $\mathcal{U} = \operatorname{span}(\delta) = \mathcal{V}$.
(*ii*) ($\Leftarrow$) If $\mathcal{U} = \mathcal{V}$, then δ is a basis of $\mathcal{V}$, so $m = n$ by Theorem 1.4.1. Hence $\dim(\mathcal{U}) = \dim(\mathcal{V})$.
Structural Commentary: Length comparison proves the bound; basis extension tests equality against a new ambient direction.
Key Insight: A subspace of full ambient dimension cannot be proper. □

Proposition 1.4.4. *Full-Dimension Basis Criteria*
Let $\mathcal{V}$ be finite-dimensional and set $n := \dim(\mathcal{V})$. Then:
(i) if γ is a spanning list in $\mathcal{V}$ of length n, then γ is a basis;
(ii) if α is a linearly independent list in $\mathcal{V}$ of length n, then α is a basis.

Proof.
(*i*) A basis sublist β of γ exists by Theorem 1.3.12. Its length is n by Definition 1.4.2 and Theorem 1.4.1.
Since γ also has length n, the sublist β equals γ. Hence γ is a basis.
(*ii*) Extend α to a basis β of $\mathcal{V}$ by Theorem 1.3.13. The length of β is n by Definition 1.4.2.
Hence the extension adds no entry to the length-n list α. Hence $\alpha = \beta$ is a basis.
Structural Commentary: At full length, basis reduction cannot delete and basis extension cannot append.
Key Insight: At the ambient dimension, either half of the basis condition forces the other. □

Theorem 1.4.5. *Dimension as Basis Cardinality*
Assume the Choice Principle 6.1.2. If $\mathcal{B}$ and $\mathcal{C}$ are Hamel bases of $\mathcal{V}$, then $|\mathcal{B}| = |\mathcal{C}|$.

Proof.
(i) *Finite case*: Assume $|\mathcal{B}| = n \in \mathbb{N}_0$, and order $\mathcal{B}$ as a spanning list β. Then $\mathcal{V}$ is finite-dimensional.
The set $\mathcal{C}$ cannot contain $n+1$ distinct vectors by Theorem 1.3.11, because they would form an independent list longer than β.
Thus $\mathcal{C}$ is finite; otherwise finite recursion would select such vectors. If γ orders $\mathcal{C}$ and has length k, then $k = n$ by Theorem 1.4.1, and $|\mathcal{C}| = k = n$ by Proposition 6.3.15(i). Interchanging the bases handles finite $\mathcal{C}$.
(ii) *Support incidence*: Assume $\mathcal{B}$ and $\mathcal{C}$ are infinite. For $\mathbf{c} \in \mathcal{C}$, let $a_{\mathbf{c}} : \mathcal{B} \to \mathbb{F}$ be its unique finite $\mathcal{B}$-expansion by Proposition 1.3.16, and set $S_{\mathbf{c}} := \operatorname{supp}_{\mathcal{B}}(a_{\mathbf{c}})$. Fix $\mathbf{b} \in \mathcal{B}$ and write $\mathbf{b} = \sum_{j=1}^{m} d_j \mathbf{c}_j$ with $\mathbf{c}_j \in \mathcal{C}$.
If $\mathbf{b} \notin \bigcup_{\mathbf{c} \in \mathcal{C}} S_{\mathbf{c}}$, then substituting the $\mathcal{B}$-expansions makes the coefficient at $\mathbf{b}$ equal to $\sum_{j=1}^{m} d_j a_{\mathbf{c}_j}(\mathbf{b}) = 0$, whereas the standard expansion of $\mathbf{b}$ has coefficient 1. This discrepancy is impossible by uniqueness in Proposition 1.3.16.
Hence $(\forall\, \mathbf{b} \in \mathcal{B})(\exists\, \mathbf{c} \in \mathcal{C})\, \mathbf{b} \in S_{\mathbf{c}}$. Set $D := \{(\mathbf{c}, \mathbf{b}) \in \mathcal{C} \times \mathcal{B} : \mathbf{b} \in S_{\mathbf{c}}\}$.
(iii) *Cardinal comparison*: Choose $(\mathbf{c}_{\mathbf{b}})_{\mathbf{b} \in \mathcal{B}}$ such that $(\forall\, \mathbf{b} \in \mathcal{B})\, [\mathbf{c}_{\mathbf{b}} \in \mathcal{C} \wedge \mathbf{b} \in S_{\mathbf{c}_{\mathbf{b}}}]$ by Principle 6.1.2.
Then $\mathbf{b} \mapsto (\mathbf{c}_{\mathbf{b}}, \mathbf{b})$ injects $\mathcal{B}$ into D. Fix a well-order of $\mathcal{B}$ by Principle 6.3.2(iv), let $p_{\mathbf{c}}(\mathbf{b}) \in \mathbb{N}$ be position of $\mathbf{b}$ in the induced ordering of the finite set $S_{\mathbf{c}}$. The map $(\mathbf{c}, \mathbf{b}) \mapsto (\mathbf{c}, p_{\mathbf{c}}(\mathbf{b}))$ injects D into $\mathcal{C} \times \mathbb{N}$. Hence $|\mathcal{B}| \leq |D| \leq |\mathcal{C} \times \mathbb{N}|$ by Proposition 6.5.2(i).
By Proposition 6.5.2(iv) and Proposition 6.3.15(iii), $|\mathcal{C} \times \mathbb{N}| = |\mathcal{C}|\, \aleph_0$.
Since $\mathcal{C}$ is infinite, $\aleph_0 \leq |\mathcal{C}|$ by Proposition 6.3.15(iv), so $|\mathcal{C}|\, \aleph_0 = |\mathcal{C}|$ by Theorem 6.6.3(ii). Therefore $|\mathcal{B}| \leq |\mathcal{C}|$.
(iv) *Symmetry*: Interchanging $\mathcal{B}$ and $\mathcal{C}$ proves $|\mathcal{C}| \leq |\mathcal{B}|$. These inequalities correspond to injections in both directions by Proposition 6.5.2(i), so $\mathcal{B} \sim_{\#} \mathcal{C}$ by Theorem 6.2.3. Hence $|\mathcal{B}| = |\mathcal{C}|$ by Proposition 6.3.14(ii).
Structural Commentary: Finite supports form an incidence relation that choice and well-ordering encode by cardinal injections.
Key Insight: Record each support occurrence by one basis vector and one finite position. □

Definition 1.4.6. *Hamel Dimension*
Assume Zorn's lemma and the Choice Principle 6.1.2. Choose a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.3.17, and define $\dim(\mathcal{V}) := |\mathcal{B}|$.
The value is independent of $\mathcal{B}$ by Theorem 1.4.5. If $\mathcal{V}$ has a finite ordered basis of length n, its underlying set is a Hamel basis of cardinality n by Proposition 6.3.15(i); hence this definition extends Definition 1.4.2.

1.5 Linear Maps and Operators

Linear maps preserve vector-space operations. Pointwise operations and composition form their coordinate-free algebra, while basis values determine finite-dimensional maps.

Throughout this subsection, $\mathcal{U}, \mathcal{V}, \mathcal{W}$ are vector spaces over the same fixed field $\mathbb{F}$.

Definition 1.5.1. *Linear Map*
A function $T : \mathcal{V} \to \mathcal{W}$ is *linear* iff $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})(\forall\, a, b \in \mathbb{F})\, T(a\mathbf{u} + b\mathbf{v}) = aT\mathbf{u} + bT\mathbf{v}$.
The collection of linear maps from $\mathcal{V}$ to $\mathcal{W}$ is denoted $\mathcal{L}(\mathcal{V}, \mathcal{W})$. Write $\mathcal{L}(\mathcal{V}) := \mathcal{L}(\mathcal{V}, \mathcal{V})$; its elements are *operators* on $\mathcal{V}$.

Definition 1.5.2. *Zero and Identity Maps*
The *zero map* from $\mathcal{V}$ to $\mathcal{W}$ is $0_{\mathcal{V},\mathcal{W}} : \mathcal{V} \to \mathcal{W}$ satisfying $(\forall\, \mathbf{v} \in \mathcal{V})\, 0_{\mathcal{V},\mathcal{W}}\mathbf{v} = \mathbf{0}_{\mathcal{W}}$.
The *identity map* on $\mathcal{V}$ is $I_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}$ satisfying $(\forall\, \mathbf{v} \in \mathcal{V})\, I_{\mathcal{V}}\mathbf{v} = \mathbf{v}$.

Proposition 1.5.3. *Linear Maps Send Zero to Zero*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$.

Proof. By linearity, $T\mathbf{0}_{\mathcal{V}} = T(\mathbf{0}_{\mathcal{V}} + \mathbf{0}_{\mathcal{V}}) = T\mathbf{0}_{\mathcal{V}} + T\mathbf{0}_{\mathcal{V}}$. Hence $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$ by cancellation.
Structural Commentary: Additivity converts zero preservation into a cancellation identity.
Key Insight: The additive identity is fixed by every linear map. □

Definition 1.5.4. *Operations on Linear Maps*
For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, define pointwise $(S + T)\mathbf{v} := S\mathbf{v} + T\mathbf{v}$ and $(aT)\mathbf{v} := a(T\mathbf{v})$ for $\mathbf{v} \in \mathcal{V}$.

Proposition 1.5.5. *Zero, Identity, and the Linear-Map Space*
(i) $0_{\mathcal{V},\mathcal{W}} \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(ii) $I_{\mathcal{V}} \in \mathcal{L}(\mathcal{V})$.
(iii) The operations above make $\mathcal{L}(\mathcal{V}, \mathcal{W})$ a vector space over $\mathbb{F}$.

Proof.
(i) For $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $0_{\mathcal{V},\mathcal{W}}(a\mathbf{u} + b\mathbf{v}) = \mathbf{0}_{\mathcal{W}} = a(0_{\mathcal{V},\mathcal{W}}\mathbf{u}) + b(0_{\mathcal{V},\mathcal{W}}\mathbf{v})$ by Theorem 1.1.2(iv). Hence $0_{\mathcal{V},\mathcal{W}} \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(ii) For $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $I_{\mathcal{V}}(a\mathbf{u} + b\mathbf{v}) = a\mathbf{u} + b\mathbf{v} = aI_{\mathcal{V}}\mathbf{u} + bI_{\mathcal{V}}\mathbf{v}$, so $I_{\mathcal{V}} \in \mathcal{L}(\mathcal{V})$.
(iii) For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a, b \in \mathbb{F}$, set $A := aS + bT$. For $c, d \in \mathbb{F}$, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $A(c\mathbf{u} + d\mathbf{v}) = a(cS\mathbf{u} + dS\mathbf{v}) + b(cT\mathbf{u} + dT\mathbf{v}) = cA\mathbf{u} + dA\mathbf{v}$, so $A \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Hence $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is closed under addition and scalar multiplication.
For $\mathbf{v} \in \mathcal{V}$, $(T + 0_{\mathcal{V},\mathcal{W}})\mathbf{v} = T\mathbf{v} = (0_{\mathcal{V},\mathcal{W}} + T)\mathbf{v}$, and $(T + (-1)T)\mathbf{v} = \mathbf{0}_{\mathcal{W}}$ by Corollary 1.1.3.
Thus $T + 0_{\mathcal{V},\mathcal{W}} = T = 0_{\mathcal{V},\mathcal{W}} + T$ and $T + (-1)T = 0_{\mathcal{V},\mathcal{W}}$ by Assumption 3.1.2.
The remaining axioms hold pointwise in $\mathcal{W}$ and hence for the maps by Assumption 3.1.2.
Structural Commentary: Closure is algebraic; function extensionality lifts codomain axioms pointwise.
Key Insight: Function spaces inherit vector-space structure from their codomain. □

Definition 1.5.6. *Product of Linear Maps*
For $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, define their *product* as the function $ST : \mathcal{U} \to \mathcal{W}$ satisfying $(\forall\, \mathbf{u} \in \mathcal{U})\, (ST)\mathbf{u} = S(T\mathbf{u})$.

Proposition 1.5.7. *Algebra of Linear Map Products*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(i) $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$.
(ii) If $\mathcal{X}$ is a vector space over $\mathbb{F}$ and $R \in \mathcal{L}(\mathcal{W}, \mathcal{X})$, then $(RS)T = R(ST)$.
(iii) $TI_{\mathcal{U}} = T = I_{\mathcal{V}}T$.
(iv) If $S_1, S_2 \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $(S_1 + S_2)T = S_1T + S_2T$.
(v) If $T_1, T_2 \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, then $S(T_1 + T_2) = ST_1 + ST_2$.

Proof.
(i) For $\mathbf{u}, \mathbf{x} \in \mathcal{U}$ and $a, b \in \mathbb{F}$, $(ST)(a\mathbf{u} + b\mathbf{x}) = S(aT\mathbf{u} + bT\mathbf{x}) = aS(T\mathbf{u}) + bS(T\mathbf{x}) = a(ST)\mathbf{u} + b(ST)\mathbf{x}$, so $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$.
(ii) For $\mathbf{u} \in \mathcal{U}$, $((RS)T)\mathbf{u} = R(S(T\mathbf{u})) = (R(ST))\mathbf{u}$. Hence $(RS)T = R(ST)$ by Assumption 3.1.2.
(iii) For $\mathbf{u} \in \mathcal{U}$, $(TI_{\mathcal{U}})\mathbf{u} = T\mathbf{u} = (I_{\mathcal{V}}T)\mathbf{u}$. Hence $TI_{\mathcal{U}} = T = I_{\mathcal{V}}T$ by Assumption 3.1.2.
(iv) For $\mathbf{u} \in \mathcal{U}$, $((S_1 + S_2)T)\mathbf{u} = S_1(T\mathbf{u}) + S_2(T\mathbf{u}) = (S_1T + S_2T)\mathbf{u}$. Hence $(S_1 + S_2)T = S_1T + S_2T$ by Assumption 3.1.2.
(v) For $\mathbf{u} \in \mathcal{U}$, $(S(T_1 + T_2))\mathbf{u} = S(T_1\mathbf{u} + T_2\mathbf{u}) = S(T_1\mathbf{u}) + S(T_2\mathbf{u}) = (ST_1 + ST_2)\mathbf{u}$.
Hence $S(T_1 + T_2) = ST_1 + ST_2$ by Assumption 3.1.2.
Structural Commentary: Pointwise evaluation reduces composition identities to codomain equalities.
Key Insight: Composition and pointwise addition form the algebra of linear maps. □

Proposition 1.5.8. *Linear Maps Are Determined by a Finite Basis*
If $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ and $\mathbf{w}_1, \ldots, \mathbf{w}_n \in \mathcal{W}$, then $(\exists!\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}))(\forall\, j \in \{1, \ldots, n\})\, T(\mathbf{v}_j) = \mathbf{w}_j$.

Proof. By Proposition 1.3.3, each $\mathbf{v} \in \mathcal{V}$ has a unique expression $\mathbf{v} = \sum_{j=1}^{n} a_j\mathbf{v}_j$. Define $T(\mathbf{v}) := \sum_{j=1}^{n} a_j\mathbf{w}_j$.
The map T is well-defined by coordinate uniqueness, and $(\forall\, j \in \{1, \ldots, n\})\, T\mathbf{v}_j = \mathbf{w}_j$ by the standard coordinate tuples.
For $\mathbf{x} = \sum_{j=1}^{n} a_j\mathbf{v}_j$, $\mathbf{y} = \sum_{j=1}^{n} b_j\mathbf{v}_j$, and $c, d \in \mathbb{F}$, $c\mathbf{x} + d\mathbf{y} = \sum_{j=1}^{n} (ca_j + db_j)\mathbf{v}_j$, so $T(c\mathbf{x} + d\mathbf{y}) = cT\mathbf{x} + dT\mathbf{y}$. Thus $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
For $A \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ with the prescribed basis values, $A\mathbf{v} = \sum_{j=1}^{n} a_jA\mathbf{v}_j = \sum_{j=1}^{n} a_j\mathbf{w}_j = T\mathbf{v}$ for $\mathbf{v} \in \mathcal{V}$. Hence $A = T$ by Assumption 3.1.2.
Structural Commentary: Coordinate uniqueness supports construction; function extensionality completes uniqueness.
Key Insight: Arbitrary basis images extend to exactly one linear map. □

Principle 1.5.9. *Coordinate-Free Principle*
By Proposition 1.5.8, basis images determine a linear map before coordinates encode it.
This viewpoint organizes finite-dimensional representations but does not make the map depend on the chosen basis.

1.6 Kernels, Ranges, and Map Properties

Throughout this subsection, $\mathcal{U}, \mathcal{V}, \mathcal{W}$ are vector spaces over the same fixed field $\mathbb{F}$.

Definition 1.6.1. *Kernel and Range of a Linear Map*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, define its *kernel*, or *nullspace*, by $\ker(T) := \{\mathbf{v} \in \mathcal{V} : T\mathbf{v} = \mathbf{0}_{\mathcal{W}}\}$ and its range by $\text{range}(T) := \{T\mathbf{v} : \mathbf{v} \in \mathcal{V}\}$.

Proposition 1.6.2. *Injectivity and the Kernel*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is injective iff $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$.

Proof.
($\Rightarrow$) Suppose T is injective. For $\mathbf{v} \in \ker(T)$, $T\mathbf{v} = \mathbf{0}_{\mathcal{W}} = T\mathbf{0}_{\mathcal{V}}$ by Proposition 1.5.3, so $\mathbf{v} = \mathbf{0}_{\mathcal{V}}$ by injectivity.
Moreover, $\mathbf{0}_{\mathcal{V}} \in \ker(T)$ by the displayed equality; hence $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$.
($\Leftarrow$) Suppose $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$. For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ with $T\mathbf{v} = T\mathbf{w}$, $T(\mathbf{v} - \mathbf{w}) = \mathbf{0}_{\mathcal{W}}$ by linearity, so $\mathbf{v} - \mathbf{w} \in \ker(T)$ and $\mathbf{v} = \mathbf{w}$.
Structural Commentary: The forward direction compares kernel vectors with zero; the reverse moves equal images into the kernel.
Key Insight: Differences convert image collisions into kernel vectors. □

Proposition 1.6.3. *Kernel and Range Are Subspaces*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then:
(i) $\ker(T)$ is a subspace of $\mathcal{V}$.
(ii) $\text{range}(T)$ is a subspace of $\mathcal{W}$.

Proof.
(i) The kernel contains $\mathbf{0}_{\mathcal{V}}$ by Proposition 1.5.3. For $\mathbf{u}, \mathbf{v} \in \ker(T)$ and $a \in \mathbb{F}$, $T(a\mathbf{u} + \mathbf{v}) = aT\mathbf{u} + T\mathbf{v} = \mathbf{0}_{\mathcal{W}}$, so $a\mathbf{u} + \mathbf{v} \in \ker(T)$. Therefore $\ker(T) \leq \mathcal{V}$ by Theorem 1.2.2.
(ii) The range contains $\mathbf{0}_{\mathcal{W}} = T\mathbf{0}_{\mathcal{V}}$ by Proposition 1.5.3. For $T\mathbf{u}, T\mathbf{v} \in \text{range}(T)$ and $a \in \mathbb{F}$, $aT\mathbf{u} + T\mathbf{v} = T(a\mathbf{u} + \mathbf{v}) \in \text{range}(T)$. Therefore $\text{range}(T) \leq \mathcal{W}$ by Theorem 1.2.2.
Structural Commentary: Each proof verifies nonemptiness and linear-combination closure before invoking the subspace test.
Key Insight: Linearity stabilizes the zero fiber and the image. □

Definition 1.6.4. *Invertible Linear Map and Isomorphism*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is *invertible* iff $(\exists\, S \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\,[ST = I_{\mathcal{V}} \wedge TS = I_{\mathcal{W}}]$.
Such an S is an *inverse* of T. An invertible linear map is an *isomorphism*; if an isomorphism $\mathcal{V} \to \mathcal{W}$ exists, write $\mathcal{V} \cong \mathcal{W}$.

Proposition 1.6.5. *Inverse Is Unique*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is invertible, then $(\exists!\, S \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\,[ST = I_{\mathcal{V}} \wedge TS = I_{\mathcal{W}}]$. Denote this map by T^{-1}.

Proof. An inverse exists by invertibility of T. Suppose $S_1, S_2 \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ satisfy $S_1T = I_{\mathcal{V}} = S_2T$ and $TS_1 = I_{\mathcal{W}} = TS_2$.
Then $S_1 = S_1I_{\mathcal{W}} = S_1(TS_2) = (S_1T)S_2 = I_{\mathcal{V}}S_2 = S_2$ by Proposition 1.5.7(ii)–(iii) and the candidate inverse identities.
Structural Commentary: Existence is definitional; associativity compares any two inverse candidates.
Key Insight: A two-sided inverse has no freedom. □

Proposition 1.6.6. *Inverse of a Product Reverses Order*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ be invertible.
(i) ST is invertible.
(ii) $(ST)^{-1} = T^{-1}S^{-1}$.

Proof.
(i) Set $R := T^{-1}S^{-1}$. Then $R \in \mathcal{L}(\mathcal{W}, \mathcal{U})$ by Proposition 1.5.7(i).
The identity $R(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = I_{\mathcal{U}}$ holds by Proposition 1.5.7(ii)–(iii) and the inverse identities.
Similarly, $(ST)R = S(TT^{-1})S^{-1} = SS^{-1} = I_{\mathcal{W}}$ by Proposition 1.5.7(ii)–(iii) and the inverse identities. Hence ST is invertible.
(ii) Since R is a two-sided inverse of ST, $(ST)^{-1} = T^{-1}S^{-1}$ by Proposition 1.6.5.
Structural Commentary: The reversed product is checked against both inverse identities, then identified by uniqueness.
Key Insight: Composition order reverses under inversion. □

Proposition 1.6.7. *Invertibility and Bijectivity*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the map T is invertible iff it is both injective and surjective.

Proof.
($\Rightarrow$) Let $S = T^{-1}$. If $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $T\mathbf{v} = T\mathbf{w}$, then $\mathbf{v} = (ST)\mathbf{v} = S(T\mathbf{v}) = S(T\mathbf{w}) = (ST)\mathbf{w} = \mathbf{w}$, so T is injective.
If $\mathbf{y} \in \mathcal{W}$, then $\mathbf{y} = (TS)\mathbf{y} = T(S\mathbf{y})$, so T is surjective.
($\Leftarrow$) Suppose T is injective and surjective. For $\mathbf{y} \in \mathcal{W}$, a unique $\mathbf{v} \in \mathcal{V}$ satisfies $T\mathbf{v} = \mathbf{y}$ by surjectivity and injectivity; define $S\mathbf{y} := \mathbf{v}$.
For $a, b \in \mathbb{F}$ and $\mathbf{y}_1, \mathbf{y}_2 \in \mathcal{W}$, $S(a\mathbf{y}_1 + b\mathbf{y}_2) = aS\mathbf{y}_1 + bS\mathbf{y}_2$ by injectivity because both sides have T-image $a\mathbf{y}_1 + b\mathbf{y}_2$.
Hence $S \in \mathcal{L}(\mathcal{W}, \mathcal{V})$. By construction, $(\forall\, \mathbf{y} \in \mathcal{W})\,(TS)\mathbf{y} = \mathbf{y}$, so $TS = I_{\mathcal{W}}$ by Assumption 3.1.2.
For $\mathbf{v} \in \mathcal{V}$, $T((ST)\mathbf{v}) = T\mathbf{v}$, whence $(ST)\mathbf{v} = \mathbf{v}$ by injectivity. Therefore $ST = I_{\mathcal{V}}$ by Assumption 3.1.2, and T is invertible.
Structural Commentary: Surjectivity constructs the reverse map; injectivity makes it linear and enforces both inverse identities.
Key Insight: Unique preimages turn a bijection into a linear inverse. □

Definition 1.6.8. *Morphism Terminology for Linear Maps*
Within vector spaces, the following terminology applies to $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$:
(i) T is a *homomorphism*.
(ii) T is an *endomorphism* iff $\mathcal{V} = \mathcal{W}$.
(iii) T is a *monomorphism*, or *embedding*, iff T is injective; equivalently, $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$ by Proposition 1.6.2.
(iv) T is an *epimorphism* iff T is surjective, equivalently $\text{range}(T) = \mathcal{W}$.
(v) T is an *isomorphism* iff T is invertible; equivalently, T is injective and surjective by Proposition 1.6.7.
(vi) T is an *automorphism* iff $\mathcal{V} = \mathcal{W}$ and T is invertible.

1.7 Rank-Nullity and Dimension Criteria

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Theorem 1.7.1. *Range from Basis Images*
Let $\mathcal{V}$ be finite-dimensional with basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) $\operatorname{range}(T) = \operatorname{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$.
(ii) $\operatorname{range}(T)$ *is finite-dimensional.*
(iii) $\dim(\operatorname{range}(T)) \leq n$.

Proof.

(i) ($\Rightarrow$) For $\mathbf{y} \in \operatorname{range}(T)$, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{y} = T\mathbf{v}$ and write $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{v}_j$.

Then $\mathbf{y} = \sum_{j=1}^{n} c_j T\mathbf{v}_j$ by linearity, so $\operatorname{range}(T) \subseteq \operatorname{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$.

(i) ($\Leftarrow$) $\sum_{j=1}^{n} a_j T\mathbf{v}_j = T(\sum_{j=1}^{n} a_j \mathbf{v}_j) \in \operatorname{range}(T)$ for $a_1, \dots, a_n \in \mathbb{F}$, so the reverse inclusion holds.

(ii) The finite list in (i) spans $\operatorname{range}(T)$, so the range is finite-dimensional.

(i) A basis sublist of the list in (i) exists by Theorem 1.3.12. Its length is at most n, and therefore $\dim(\operatorname{range}(T)) \leq n$.

Structural Commentary: Two inclusions identify the range; basis reduction then converts its generating list into a dimension bound.
Key Insight: A linear map transports every basis expansion to its image. □

Definition 1.7.2. *Rank and Nullity*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Define $\operatorname{rank}(T) := \dim(\operatorname{range}(T))$ and $\operatorname{nullity}(T) := \dim(\ker(T))$.
The rank is finite by Theorem 1.7.1(ii); the nullity is finite by Proposition 1.6.3 and Corollary 1.3.14.

Theorem 1.7.3. *Rank-Nullity Theorem*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then:

(i) $\operatorname{range}(T)$ *is finite-dimensional.*
(ii) $\dim(\mathcal{V}) = \operatorname{nullity}(T) + \operatorname{rank}(T)$.

Proof. The kernel has a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ by Proposition 1.6.3 and Corollary 1.3.14.
Extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r)$ of $\mathcal{V}$ by Theorem 1.3.13.
For $\mathbf{v} = \sum_{i=1}^{m} a_i \mathbf{u}_i + \sum_{j=1}^{r} b_j \mathbf{w}_j$, $T\mathbf{v} = \sum_{j=1}^{r} b_j T\mathbf{w}_j$ by linearity, so $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ spans $\operatorname{range}(T)$.
If $\sum_{j=1}^{r} c_j T\mathbf{w}_j = \mathbf{0}_{\mathcal{W}}$, then $\mathbf{x} := \sum_{j=1}^{r} c_j \mathbf{w}_j \in \ker(T) = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_m)$.
Thus $\mathbf{x}$ also has an extended-basis representation with every $\mathbf{w}_j$-coefficient zero, so $c_1 = \cdots = c_r = 0$ by Proposition 1.3.3.
Hence $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ is a basis of $\operatorname{range}(T)$.

(i) The basis $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ is finite, so $\operatorname{range}(T)$ is finite-dimensional.

(ii) The three bases have lengths m, r, and $m + r$, respectively. Hence $\operatorname{nullity}(T) = m$, $\operatorname{rank}(T) = r$, and $\dim(\mathcal{V}) = m + r$.

Structural Commentary: Extending a kernel basis produces complementary vectors whose images form a basis of the range.
Key Insight: Every domain basis direction is either killed or detected. □

Corollary 1.7.4. *Maps to Smaller Spaces Are Not Injective*
If finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ satisfy $\dim(\mathcal{V}) > \dim(\mathcal{W})$, then no $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is injective.

Proof. Fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Proposition 1.6.3 and Proposition 1.4.3, $\operatorname{rank}(T) \leq \dim(\mathcal{W})$.
By Theorem 1.7.3, $\operatorname{nullity}(T) = \dim(\mathcal{V}) - \operatorname{rank}(T) \geq \dim(\mathcal{V}) - \dim(\mathcal{W}) > 0$.
Thus $\ker(T) \neq \{\mathbf{0}_{\mathcal{V}}\}$, so T is not injective by Proposition 1.6.2.
Structural Commentary: The range bound feeds rank-nullity, forcing a nontrivial kernel.
Key Insight: Excess domain dimension must collapse. □

Corollary 1.7.5. *Maps to Larger Spaces Are Not Surjective*
If finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ satisfy $\dim(\mathcal{V}) < \dim(\mathcal{W})$, then no $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is surjective.

Proof. Fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 1.7.3, $\operatorname{rank}(T) = \dim(\mathcal{V}) - \operatorname{nullity}(T) \leq \dim(\mathcal{V}) < \dim(\mathcal{W})$. Since $\operatorname{range}(T) \leq \mathcal{W}$ by Proposition 1.6.3, $\operatorname{range}(T) \neq \mathcal{W}$ by Proposition 1.4.3. Hence T is not surjective.
Structural Commentary: Rank-nullity bounds the range below the codomain dimension.
Key Insight: A smaller domain cannot cover a larger codomain. □

Corollary 1.7.6. *Equal-Dimension Kernel-Range Criterion*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) = \dim(\mathcal{W})$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$ *iff* $\operatorname{range}(T) = \mathcal{W}$.
(ii) T *is injective iff T is surjective.*

Proof.

(i) ($\Rightarrow$) If $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$, then $\operatorname{nullity}(T) = 0$ and $\operatorname{rank}(T) = \dim(\mathcal{V}) = \dim(\mathcal{W})$ by Theorem 1.7.3.
Since $\operatorname{range}(T) \leq \mathcal{W}$ by Proposition 1.6.3, $\operatorname{range}(T) = \mathcal{W}$ by Proposition 1.4.3.

(i) ($\Leftarrow$) If $\operatorname{range}(T) = \mathcal{W}$, then $\operatorname{rank}(T) = \dim(\mathcal{W}) = \dim(\mathcal{V})$ and $\operatorname{nullity}(T) = 0$ by Theorem 1.7.3.
The zero subspace and $\ker(T)$ both have dimension zero, so they are equal by Proposition 1.4.3.

(ii) ($\Rightarrow$) If T is injective, then $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$ by Proposition 1.6.2. By (i), $\operatorname{range}(T) = \mathcal{W}$, so T is surjective.

(ii) ($\Leftarrow$) If T is surjective, then $\operatorname{range}(T) = \mathcal{W}$. By (i), $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$, so T is injective by Proposition 1.6.2.

Structural Commentary: Rank-nullity exchanges kernel loss with range deficit; the kernel criterion translates this to maps.
Key Insight: Equal dimensions leave no room for one-sided failure. □

Theorem 1.7.7. *Finite-Dimensional Invertibility Criteria*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then the following are equivalent:

(i) T *is invertible;*
(ii) T *is injective;*
(iii) T *is surjective.*

Proof.

$(i) \Rightarrow (ii)$ The map T is injective by Proposition 1.6.7.
$(ii) \Rightarrow (iii)$ The map T is surjective by Corollary 1.7.6, applied with the same domain and codomain.
$(iii) \Rightarrow (i)$ The map T is injective by Corollary 1.7.6, hence bijective and therefore invertible by Proposition 1.6.7.

Structural Commentary: Bijectivity supplies the outer implications; equal dimension links injectivity and surjectivity.
Key Insight: A nonzero kernel is the sole finite-dimensional obstruction to operator invertibility. □

Proposition 1.7.8. *Isomorphisms Preserve Bases*
Let $T : \mathcal{V} \to \mathcal{W}$ be an isomorphism and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a finite list in $\mathcal{V}$. Set $T\beta = (T\mathbf{v}_1, \dots, T\mathbf{v}_n)$. The following are equivalent:

(i) β *is a basis of* $\mathcal{V}$.
(ii) $T\beta$ *is a basis of* $\mathcal{W}$.

Proof. The map T is injective and surjective by Proposition 1.6.7.

$(i) \Rightarrow (ii)$ For $\mathbf{w} \in \mathcal{W}$, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{w} = T\mathbf{v}$ and write $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$. Then $\mathbf{w} = \sum_{j=1}^{n} a_j T\mathbf{v}_j$ by linearity, so $T\beta$ spans $\mathcal{W}$.
If $\sum_{j=1}^{n} a_j T\mathbf{v}_j = \mathbf{0}_{\mathcal{W}}$, then $T(\sum_{j=1}^{n} a_j \mathbf{v}_j) = \sum_{j=1}^{n} a_j T\mathbf{v}_j = \mathbf{0}_{\mathcal{W}} = T\mathbf{0}_{\mathcal{V}}$ by linearity and Proposition 1.5.3.
Injectivity and independence of β imply $a_1 = \cdots = a_n = 0$. Hence $T\beta$ is a basis of $\mathcal{W}$.
$(ii) \Rightarrow (i)$ For $\mathbf{v} \in \mathcal{V}$, write $T\mathbf{v} = \sum_{j=1}^{n} a_j T\mathbf{v}_j = T(\sum_{j=1}^{n} a_j \mathbf{v}_j)$ by linearity. By injectivity, $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$, so β spans $\mathcal{V}$.
If $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}_{\mathcal{V}}$, then $\sum_{j=1}^{n} a_j T\mathbf{v}_j = \mathbf{0}_{\mathcal{W}}$ by linearity and Proposition 1.5.3.
Independence of $T\beta$ implies $a_1 = \cdots = a_n = 0$. Hence β is a basis of $\mathcal{V}$.

Structural Commentary: Surjectivity transports spanning representations; injectivity preserves and reflects linear relations.
Key Insight: Isomorphisms preserve both halves of the basis condition. □

Theorem 1.7.9. *Isomorphism and Dimension*
Finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ are isomorphic iff $\dim(\mathcal{V}) = \dim(\mathcal{W})$.

Proof.

$(\Rightarrow)$ Let $T : \mathcal{V} \to \mathcal{W}$ be an isomorphism. Choose a basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ of $\mathcal{V}$ by Corollary 1.3.14.
Then $T\beta$ is a basis of $\mathcal{W}$ by Proposition 1.7.8, so $\dim(\mathcal{V}) = n = \dim(\mathcal{W})$.
$(\Leftarrow)$ Suppose $\dim(\mathcal{V}) = \dim(\mathcal{W}) = n$, and choose bases $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ by Corollary 1.3.14.
Choose $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ satisfying $(\forall\, j \in \{1, \dots, n\})\, T\mathbf{v}_j = \mathbf{w}_j$ by Proposition 1.5.8. By Theorem 1.7.1(i), $\operatorname{range}(T) = \operatorname{span}(\gamma) = \mathcal{W}$.
or $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j \in \ker(T)$, $\mathbf{0}_{\mathcal{W}} = T\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{w}_j$, so $a_1 = \cdots = a_n = 0$ by independence of γ.
Thus $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$, and T is injective by Proposition 1.6.2.
It is surjective by its range identity, hence invertible by Proposition 1.6.7 and therefore an isomorphism.

Structural Commentary: The forward direction transports a basis; the reverse prescribes an isomorphism on matching bases.
Key Insight: Dimension counts the coordinates preserved by an isomorphism. □

2 Direct Sums, Quotients, and Decomposition

2.1 Sums and Direct Sums

Sums combine subspaces inside a common ambient space, while products and external sums assemble separate spaces. Directness records when these decompositions are unique.

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 2.1.1. *Sums of Subspaces*
Let $m \in \mathbb{N}$ and let $\mathcal{U}_1, \dots, \mathcal{U}_m \leq \mathcal{V}$.

(i) Their *sum* is $\mathcal{U}_1 + \cdots + \mathcal{U}_m := \left\{ \sum_{j=1}^{m} \mathbf{u}_j : (\forall\, j \in \{1, \dots, m\})\, \mathbf{u}_j \in \mathcal{U}_j \right\}$.
(ii) For $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$, $\mathcal{U} + \mathcal{W} = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in \mathcal{U},\ \mathbf{w} \in \mathcal{W}\}$.
(iii) The sum is *direct* iff $(\forall\, \mathbf{v} \in \mathcal{U}_1 + \cdots + \mathcal{U}_m)(\exists!\, (\mathbf{u}_1, \dots, \mathbf{u}_m) \in \mathcal{U}_1 \times \cdots \times \mathcal{U}_m)\, \mathbf{v} = \sum_{j=1}^{m} \mathbf{u}_j$. A direct sum is denoted $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Proposition 2.1.2. *Sum of Subspaces Is the Smallest Containing Subspace*
Let $m \in \mathbb{N}$, let $\mathcal{U}_1, \dots, \mathcal{U}_m \leq \mathcal{V}$, and set $S := \sum_{j=1}^{m} \mathcal{U}_j$.

(i) $S \leq \mathcal{V}$, $(\forall\, j \in \{1, \dots, m\})\, \mathcal{U}_j \subseteq S$, *and* $(\forall\, \mathcal{W} \leq \mathcal{V})[(\forall\, j \in \{1, \dots, m\})\, \mathcal{U}_j \subseteq \mathcal{W} \Rightarrow S \subseteq \mathcal{W}]$.
(ii) $S = \operatorname{span}(\bigcup_{j=1}^{m} \mathcal{U}_j)$.

Proof.

(i) $(\forall\, j \in \{1,\dots,m\})\, \mathbf{0}_{\mathcal{V}} \in \mathcal{U}_j$, so $\mathbf{0}_{\mathcal{V}} \in S$. Let $\mathbf{x}, \mathbf{y} \in S$ and $c \in \mathbb{F}$.

Choose $(\mathbf{u}_1,\dots,\mathbf{u}_m), (\mathbf{v}_1,\dots,\mathbf{v}_m) \in \mathcal{U}_1 \times \cdots \times \mathcal{U}_m$ with $\mathbf{x} = \sum_{j=1}^{m} \mathbf{u}_j$ and $\mathbf{y} = \sum_{j=1}^{m} \mathbf{v}_j$.

Then $c\mathbf{x} + \mathbf{y} = \sum_{j=1}^{m} (c\mathbf{u}_j + \mathbf{v}_j) \in S$ by $(\forall\, j \in \{1,\dots,m\})\, c\mathbf{u}_j + \mathbf{v}_j \in \mathcal{U}_j$. Hence $S \leq \mathcal{V}$ by Theorem 1.2.2.

For $j \in \{1,\dots,m\}$ and $\mathbf{u} \in \mathcal{U}_j$, the sum with jth term $\mathbf{u}$ and all other terms $\mathbf{0}_{\mathcal{V}}$ belongs to S, so $\mathcal{U}_j \subseteq S$.
If $\mathcal{W} \leq \mathcal{V}$ and $(\forall\, j \in \{1,\dots,m\})\, \mathcal{U}_j \subseteq \mathcal{W}$, then each defining finite sum for S belongs to $\mathcal{W}$ by additive closure; hence $S \subseteq \mathcal{W}$.

(ii) Set $E := \bigcup_{j=1}^{m} \mathcal{U}_j$. Then $\operatorname{span}(E) \subseteq S$ by Theorem 1.2.5(iii) and part (i).

The inclusions $E \subseteq \operatorname{span}(E) \leq \mathcal{V}$ hold by Theorem 1.2.5(i)–(ii), so $S \subseteq \operatorname{span}(E)$ by part (i). Thus $S = \operatorname{span}(E)$.

Structural Commentary: Closure and minimality establish the sum; span minimality identifies the same subspace.
Key Insight: Both constructions are forced by the same generators. □

Perspective 2.1.3. *Sum Is Not Union*
Proposition 2.1.2 identifies $\mathcal{U} + \mathcal{W}$ with the linear closure of $\mathcal{U} \cup \mathcal{W}$. The union itself contains only vectors already in one summand.
If neither summand contains the other, choose $\mathbf{u} \in \mathcal{U} \setminus \mathcal{W}$ and $\mathbf{w} \in \mathcal{W} \setminus \mathcal{U}$.
Then $\mathbf{u} + \mathbf{w} \notin \mathcal{U} \cup \mathcal{W}$, since membership in either summand would place the other chosen vector there.
Thus the sum supplies exactly the mixed vectors missing from the union.

Theorem 2.1.4. *Finite Unions of Proper Subspaces*

Let $n \in \mathbb{N}$ and let $\mathcal{V} \neq \{\mathbf{0}_{\mathcal{V}}\}$ be over an infinite field $\mathbb{F}$. If $\mathcal{U}_1,\dots,\mathcal{U}_n \subsetneq \mathcal{V}$ are subspaces, then $\mathcal{V} \neq \bigcup_{j=1}^{n} \mathcal{U}_j$.

Proof.

(i) *Minimal cover*: Assume $\mathcal{V} = \bigcup_{j=1}^{n} \mathcal{U}_j$. Choose a minimal subcover $\mathcal{W}_1,\dots,\mathcal{W}_k$ and set $K := \{1,\dots,k\}$.

Then $(\forall\, i \in K)\, \mathcal{W}_i \not\subseteq \bigcup_{j \in K\setminus\{i\}} \mathcal{W}_j$ by minimality. The case $k = 1$ contradicts $\mathcal{W}_1 \subsetneq \mathcal{V}$; hence $k \geq 2$.

(ii) *Affine line*: Choose $\mathbf{w} \in \mathcal{W}_1 \setminus \bigcup_{j=2}^{k} \mathcal{W}_j$ and $\mathbf{v} \in \mathcal{V} \setminus \mathcal{W}_1$. Since $k \geq 2$ and $\mathbf{0}_{\mathcal{V}} \in \mathcal{W}_2$, $\mathbf{w} \neq \mathbf{0}_{\mathcal{V}}$. Set $A := \{\mathbf{v} + r\mathbf{w} : r \in \mathbb{F}\}$.

The parametrization $r \mapsto \mathbf{v} + r\mathbf{w}$ is injective, so A is infinite by the infinitude of $\mathbb{F}$.

(iii) *Intersections*: If $\mathbf{v} + r\mathbf{w} \in \mathcal{W}_1$, then $\mathbf{v} = (\mathbf{v} + r\mathbf{w}) - r\mathbf{w} \in \mathcal{W}_1$, a contradiction; hence $A \cap \mathcal{W}_1 = \emptyset$.
For $j \in \{2,\dots,k\}$, if $r_1 \neq r_2$ and $\mathbf{v} + r_1\mathbf{w}, \mathbf{v} + r_2\mathbf{w} \in \mathcal{W}_j$, then $(r_1 - r_2)\mathbf{w} \in \mathcal{W}_j$, so $\mathbf{w} \in \mathcal{W}_j$, a contradiction. Thus $|A \cap \mathcal{W}_j| \leq 1$.

(iv) *Contradiction*: The set $A \cap \bigcup_{j=1}^{k} \mathcal{W}_j$ is finite by part (iii), whereas $A \subseteq \mathcal{V} = \bigcup_{j=1}^{k} \mathcal{W}_j$ would make it equal to the infinite set A.

Therefore the assumed cover does not exist.

Structural Commentary: A minimal cover selects a direction whose affine line meets every summand finitely.
Key Insight: An infinite affine line defeats a finite cover. □

Definition 2.1.5. *Subspace Poset, Meet, and Join*

(i) The *subspace poset* is $\operatorname{Sub}(\mathcal{V}) := \{\mathcal{U} \subseteq \mathcal{V} : \mathcal{U} \leq \mathcal{V}\}$, ordered by inclusion.
(ii) For $\{\mathcal{U}_i\}_{i\in I} \subseteq \operatorname{Sub}(\mathcal{V})$, its *meet* is $\bigwedge_{i\in I} \mathcal{U}_i := \bigcap_{i\in I} \mathcal{U}_i$.
(iii) For $\{\mathcal{U}_i\}_{i\in I} \subseteq \operatorname{Sub}(\mathcal{V})$, its *join* is $\bigvee_{i\in I} \mathcal{U}_i := \sum_{i\in I} \mathcal{U}_i := \operatorname{span}(\bigcup_{i\in I} \mathcal{U}_i)$.
(iv) The empty meet is $\mathcal{V}$, and the empty join is $\{\mathbf{0}_{\mathcal{V}}\}$.

Proposition 2.1.6. *Subspace Meet and Join*
Let $\{\mathcal{U}_i\}_{i\in I} \subseteq \operatorname{Sub}(\mathcal{V})$.

(i) $\bigwedge_{i\in I} \mathcal{U}_i = \inf\{\mathcal{U}_i : i \in I\}$.

(ii) $\bigvee_{i\in I} \mathcal{U}_i = \sup\{\mathcal{U}_i : i \in I\}$.

(iii) $\bigvee_{i\in I} \mathcal{U}_i = \left\{ \sum_{i\in F} \mathbf{u}_i : F \subseteq I,\ |F| < \infty,\ (\forall\, i \in F)\, \mathbf{u}_i \in \mathcal{U}_i \right\}$, *where the empty sum is* $\mathbf{0}_{\mathcal{V}}$.

Proof.

(i) If $I = \emptyset$, the empty meet is $\mathcal{V}$, the greatest lower bound of the empty family. Assume $I \neq \emptyset$ and set $M := \bigcap_{i\in I} \mathcal{U}_i$.

Then $M \leq \mathcal{V}$ by Proposition 1.2.3, $(\forall\, i \in I)\, M \subseteq \mathcal{U}_i$, and $(\forall \mathcal{W} \leq \mathcal{V})[(\forall\, i \in I)\, \mathcal{W} \subseteq \mathcal{U}_i \Rightarrow \mathcal{W} \subseteq M]$. Hence $M = \inf\{\mathcal{U}_i : i \in I\}$.

(ii) Set $J := \operatorname{span}(\bigcup_{i\in I} \mathcal{U}_i)$. Then $J \leq \mathcal{V}$ and $(\forall\, i \in I)\, \mathcal{U}_i \subseteq J$ by Theorem 1.2.5(i)–(ii).

If $\mathcal{W} \leq \mathcal{V}$ and $(\forall\, i \in I)\, \mathcal{U}_i \subseteq \mathcal{W}$, then $\bigcup_{i\in I} \mathcal{U}_i \subseteq \mathcal{W}$, so $J \subseteq \mathcal{W}$ by Theorem 1.2.5(iii).

Thus J is the least upper bound, including $J = \{\mathbf{0}_{\mathcal{V}}\}$ when $I = \emptyset$.

(iii) Every stated finite sum is a linear combination of vectors in $\bigcup_{i\in I} \mathcal{U}_i$, so it belongs to J.

Conversely, each $\mathbf{x} \in J$ is a finite linear combination of vectors from that union.
Group terms assigned to the same index; linear closure of $\mathcal{U}_i$ replaces each nonempty group by one vector in $\mathcal{U}_i$.
The empty combination represents $\mathbf{0}_{\mathcal{V}}$, so the displayed representation set equals J.

Structural Commentary: Order bounds identify meet and join; grouping span terms establishes finite representation.
Key Insight: Intersection preserves constraints; span combines directions. □

Theorem 2.1.7. *Subspace Lattice*
The poset $\mathrm{Sub}(\mathcal{V})$ *has the following properties.*

(i) It is a complete lattice. For $\{\mathcal{U}_i\}_{i\in I} \subseteq \mathrm{Sub}(\mathcal{V})$, $\bot = \{\mathbf{0}_\mathcal{V}\}$, $\top = \mathcal{V}$, $\bigwedge_{i\in I} \mathcal{U}_i = \bigcap_{i\in I} \mathcal{U}_i$, *and* $\bigvee_{i\in I} \mathcal{U}_i = \sum_{i\in I} \mathcal{U}_i = \mathrm{span}(\bigcup_{i\in I} \mathcal{U}_i)$.

(ii) It is modular: $(\forall \mathcal{U},\mathcal{S},\mathcal{T} \leq \mathcal{V})[\mathcal{U} \leq \mathcal{S} \Rightarrow \mathcal{S} \cap (\mathcal{T}+\mathcal{U}) = (\mathcal{S}\cap\mathcal{T})+\mathcal{U}]$.

Proof.

(i) The arbitrary meet and join formulas hold by Proposition 2.1.6(i)–(ii).
The empty formulas identify $\top = \mathcal{V}$ and $\bot = \{\mathbf{0}_\mathcal{V}\}$, so $\mathrm{Sub}(\mathcal{V})$ is complete.

(ii) Let $\mathcal{U} \leq \mathcal{S}$. If $\mathbf{x} \in (\mathcal{S}\cap\mathcal{T})+\mathcal{U}$, write $\mathbf{x} = \mathbf{y}+\mathbf{u}$ with $\mathbf{y} \in \mathcal{S}\cap\mathcal{T}$ and $\mathbf{u} \in \mathcal{U}$. Then $\mathbf{x} \in \mathcal{S}\cap(\mathcal{T}+\mathcal{U})$.
Conversely, if $\mathbf{x} \in \mathcal{S}\cap(\mathcal{T}+\mathcal{U})$, write $\mathbf{x} = \mathbf{t}+\mathbf{u}$ with $\mathbf{t} \in \mathcal{T}$ and $\mathbf{u} \in \mathcal{U}$. Then $\mathbf{t} = \mathbf{x}-\mathbf{u} \in \mathcal{S}\cap\mathcal{T}$ by $\mathbf{x},\mathbf{u} \in \mathcal{S}$, so $\mathbf{x} \in (\mathcal{S}\cap\mathcal{T})+\mathcal{U}$.

Structural Commentary: Completeness follows from the bound formulas; two inclusions establish modularity.
Key Insight: A contained summand survives intersection unchanged. □

Proposition 2.1.8. *Two-Subspace Directness Criterion*
Let $\mathcal{U}$ *and* $\mathcal{W}$ *be subspaces of* $\mathcal{V}$. *Then* $\mathcal{U}+\mathcal{W}$ *is direct iff* $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$.

Proof.

(⇒) Assume $\mathcal{U}+\mathcal{W}$ is direct. For $\mathbf{x} \in \mathcal{U}\cap\mathcal{W}$, directness of $\mathbf{x} = \mathbf{x}+\mathbf{0}_\mathcal{V} = \mathbf{0}_\mathcal{V}+\mathbf{x}$ forces $\mathbf{x} = \mathbf{0}_\mathcal{V}$. Hence $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$.

(⇐) Assume $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$ and let $\mathbf{u}_1+\mathbf{w}_1 = \mathbf{u}_2+\mathbf{w}_2$, where $\mathbf{u}_1,\mathbf{u}_2 \in \mathcal{U}$ and $\mathbf{w}_1,\mathbf{w}_2 \in \mathcal{W}$.
Then $\mathbf{u}_1-\mathbf{u}_2 = \mathbf{w}_2-\mathbf{w}_1 \in \mathcal{U}\cap\mathcal{W}$, so $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$.
Every vector in the sum has a decomposition by its definition; therefore the sum is direct.

Structural Commentary: The two directions translate decomposition equality into intersection and back.
Key Insight: Trivial intersection prevents cross-summand cancellation. □

Perspective 2.1.9. *Sums Always Exist; Directness Is Extra*
For $\mathcal{U},\mathcal{W} \leq \mathcal{V}$, the sum $\mathcal{U}+\mathcal{W}$ exists as the smallest subspace containing both by Proposition 2.1.2.
The notation $\mathcal{U}\oplus\mathcal{W}$ records the additional equivalent conditions in Proposition 2.1.8: $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$ and uniqueness of $\mathbf{u}+\mathbf{w}$.
Thus $+$ records generation, whereas $\oplus$ records generation without ambiguity; this comparison is not proof authority.

Proposition 2.1.10. *Equivalent Conditions for a Direct Sum*
For $m \in \mathbb{N}$ *and subspaces* $\mathcal{U}_1,\ldots,\mathcal{U}_m \leq \mathcal{V}$, *the following are equivalent:*

(i) $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$;

(ii) $(\forall\, \mathbf{v} \in \mathcal{V})(\exists!\, (\mathbf{u}_1,\ldots,\mathbf{u}_m) \in \mathcal{U}_1\times\cdots\times\mathcal{U}_m)\, \mathbf{v} = \sum_{j=1}^{m} \mathbf{u}_j$;

(iii) $\mathcal{V} = \sum_{j=1}^{m} \mathcal{U}_j$ *and* $(\forall\, (\mathbf{u}_1,\ldots,\mathbf{u}_m) \in \mathcal{U}_1\times\cdots\times\mathcal{U}_m)[\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}_\mathcal{V} \Rightarrow (\forall\, j \in \{1,\ldots,m\})\, \mathbf{u}_j = \mathbf{0}_\mathcal{V}]$.

Proof.

(i) ⇒ *(ii)* Condition (i) asserts that the sum equals $\mathcal{V}$ and that its decompositions are unique, which is condition (ii).

(ii) ⇒ *(iii)* Existence in (ii) implies $\mathcal{V} = \sum_{j=1}^{m} \mathcal{U}_j$.

If $\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}_\mathcal{V}$, uniqueness identifies this representation with $\sum_{j=1}^{m} \mathbf{0}_\mathcal{V}$, so $(\forall\, j \in \{1,\ldots,m\})\, \mathbf{u}_j = \mathbf{0}_\mathcal{V}$.

(iii) ⇒ *(i)* The equality in (iii) supplies a decomposition of every $\mathbf{v} \in \mathcal{V}$. If $\sum_{j=1}^{m} \mathbf{u}_j = \sum_{j=1}^{m} \mathbf{w}_j$, then $\sum_{j=1}^{m} (\mathbf{u}_j-\mathbf{w}_j) = \mathbf{0}_\mathcal{V}$.

Hence $(\forall\, j \in \{1,\ldots,m\})\, \mathbf{u}_j = \mathbf{w}_j$ by the zero-relation condition, so the sum is direct and equals $\mathcal{V}$.

Structural Commentary: The implication cycle separates existence of decompositions from the zero-relation obstruction.
Key Insight: Uniqueness is equivalent to trivial zero relations. □

Definition 2.1.11. *Product of Vector Spaces*
Let $m \in \mathbb{N}$ and let $\mathcal{V}_1,\ldots,\mathcal{V}_m$ be $\mathbb{F}$-vector spaces.

(i) Their *product* is $\mathcal{V}_1\times\cdots\times\mathcal{V}_m := \{(\mathbf{v}_1,\ldots,\mathbf{v}_m) : (\forall\, j \in \{1,\ldots,m\})\, \mathbf{v}_j \in \mathcal{V}_j\}$.
(ii) Addition is $(\mathbf{u}_1,\ldots,\mathbf{u}_m)+(\mathbf{v}_1,\ldots,\mathbf{v}_m) := (\mathbf{u}_1+\mathbf{v}_1,\ldots,\mathbf{u}_m+\mathbf{v}_m)$.
(iii) Scalar multiplication is $a(\mathbf{v}_1,\ldots,\mathbf{v}_m) := (a\mathbf{v}_1,\ldots,a\mathbf{v}_m)$ for $a \in \mathbb{F}$.

Proposition 2.1.12. *Products Are Vector Spaces*
If $m \in \mathbb{N}$ *and* $\mathcal{V}_1,\ldots,\mathcal{V}_m$ *are vector spaces over* $\mathbb{F}$, *then* $\mathcal{V}_1\times\cdots\times\mathcal{V}_m$ *is a vector space under componentwise operations.*

Proof. Set $P := \mathcal{V}_1\times\cdots\times\mathcal{V}_m$ and let $\mathbf{0}_j$ be the zero vector of $\mathcal{V}_j$. The operations are closed on P by componentwise closure.
The tuple $(\mathbf{0}_1,\ldots,\mathbf{0}_m)$ is an additive identity, and $(-\mathbf{v}_1,\ldots,-\mathbf{v}_m)$ is the additive inverse of $(\mathbf{v}_1,\ldots,\mathbf{v}_m)$.
For arbitrary tuples $\mathbf{u},\mathbf{v},\mathbf{w} \in P$, scalars $a,b \in \mathbb{F}$, and $j \in \{1,\ldots,m\}$, the jth coordinates on both sides of each commutativity, associativity, scalar-identity, scalar-associativity, and distributivity axiom are equal by the corresponding axiom in $\mathcal{V}_j$.
Coordinatewise equality proves each tuple identity, so P is an $\mathbb{F}$-vector space.
Structural Commentary: Closure, identities, and all algebraic laws are verified coordinatewise.
Key Insight: Tuple equality reduces every axiom to its components. □

Theorem 2.1.13. *Dimension of a Product*
Let $m \in \mathbb{N}$ *and let* $\mathcal{V}_1,\ldots,\mathcal{V}_m$ *be finite-dimensional* $\mathbb{F}$*-vector spaces.*
Then $P := \mathcal{V}_1\times\cdots\times\mathcal{V}_m$ *is finite-dimensional and* $\dim(P) = \sum_{j=1}^{m} \dim(\mathcal{V}_j)$.

Proof. For $j \in \{1,\dots,m\}$, let $(\mathbf{v}_{j,1},\dots,\mathbf{v}_{j,n_j})$ be a basis of $\mathcal{V}_j$.
For $j \in \{1,\dots,m\}$ and $k \in \{1,\dots,n_j\}$, let $\mathbf{e}_{j,k} \in P$ have jth component $\mathbf{v}_{j,k}$ and zero in every other component.
The list of all $\mathbf{e}_{j,k}$ spans P because each component has an expansion in its chosen basis.
If $\sum_{j=1}^{m}\sum_{k=1}^{n_j} a_{j,k}\mathbf{e}_{j,k} = \mathbf{0}$, then $(\forall\, j \in \{1,\dots,m\})\ \sum_{k=1}^{n_j} a_{j,k}\mathbf{v}_{j,k} = \mathbf{0}_{\mathcal{V}_j}$.
Hence $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,n_j\})\, a_{j,k} = 0$ by linear independence of each factor basis.
Thus the list is a basis, and its length $\sum_{j=1}^{m} n_j = \sum_{j=1}^{m} \dim(\mathcal{V}_j)$ proves the formula.
Structural Commentary: Embedded factor bases span componentwise and are independent componentwise.
Key Insight: Each factor contributes its own coordinate directions. □

Definition 2.1.14. *Graph of a Function*
For $T : \mathcal{V} \to \mathcal{W}$, its *graph* is $\operatorname{graph}(T) := \{(\mathbf{v}, T\mathbf{v}) : \mathbf{v} \in \mathcal{V}\} \subseteq \mathcal{V} \times \mathcal{W}$.

Proposition 2.1.15. *Linearity and Graphs*
A function $T : \mathcal{V} \to \mathcal{W}$ is linear iff $\operatorname{graph}(T) \leq \mathcal{V} \times \mathcal{W}$.

Proof.
(⇒) Assume T is linear. Then $(\mathbf{0}_{\mathcal{V}}, \mathbf{0}_{\mathcal{W}}) \in \operatorname{graph}(T)$ by Proposition 1.5.3.
For $(\mathbf{u}, T\mathbf{u}), (\mathbf{v}, T\mathbf{v}) \in \operatorname{graph}(T)$ and $a, b \in \mathbb{F}$, $a(\mathbf{u}, T\mathbf{u}) + b(\mathbf{v}, T\mathbf{v}) = (a\mathbf{u} + b\mathbf{v}, T(a\mathbf{u} + b\mathbf{v})) \in \operatorname{graph}(T)$ by linearity.
Hence $\operatorname{graph}(T) \leq \mathcal{V} \times \mathcal{W}$ by Theorem 1.2.2.
(⇐) Assume $\operatorname{graph}(T) \leq \mathcal{V} \times \mathcal{W}$. For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $(a\mathbf{u} + b\mathbf{v}, aT\mathbf{u} + bT\mathbf{v}) = a(\mathbf{u}, T\mathbf{u}) + b(\mathbf{v}, T\mathbf{v}) \in \operatorname{graph}(T)$.
Membership in the graph forces $T(a\mathbf{u} + b\mathbf{v}) = aT\mathbf{u} + bT\mathbf{v}$, so T is linear.
Structural Commentary: Each direction translates the linear-combination law into graph closure.
Key Insight: Graph closure is the linearity identity in product form. □

Definition 2.1.16. *External Product*
Let $\{\mathcal{V}_i : i \in I\}$ be $\mathbb{F}$-vector spaces.
(i) The *external direct product* is $\prod_{i\in I} \mathcal{V}_i := \{(\mathbf{v}_i)_{i\in I} : (\forall i \in I)\, \mathbf{v}_i \in \mathcal{V}_i\}$.
(ii) Addition is defined by $(\forall\, i \in I)\, (\mathbf{u} + \mathbf{v})_i := \mathbf{u}_i + \mathbf{v}_i$.
(iii) Scalar multiplication is defined by $(\forall\, i \in I)\, (a\mathbf{v})_i := a\mathbf{v}_i$ for $a \in \mathbb{F}$.

Definition 2.1.17. *Support and Finite Sums of a Family*
Let I be an index set, and let $\mathcal{X}_i$ be a vector space with zero vector $\mathbf{0}_i$ for $i \in I$.
(i) The *support* of $\mathbf{x} = (\mathbf{x}_i)_{i\in I} \in \prod_{i\in I} \mathcal{X}_i$ is $\operatorname{supp}(\mathbf{x}) := \{i \in I : \mathbf{x}_i \neq \mathbf{0}_i\}$.
(ii) The family $\mathbf{x}$ is *finitely supported* iff $|\operatorname{supp}(\mathbf{x})| < \infty$.
(iii) If $(\forall\, i \in I)\, \mathcal{X}_i \leq \mathcal{V}$ and $\mathbf{x}$ is finitely supported, its *finite sum* is $\sum_{i\in I} \mathbf{x}_i := \sum_{i\in\operatorname{supp}(\mathbf{x})} \mathbf{x}_i$; the empty sum is $\mathbf{0}_{\mathcal{V}}$.

Definition 2.1.18. *External Direct Sum*
Let $\{\mathcal{V}_i : i \in I\}$ be $\mathbb{F}$-vector spaces. The *external direct sum* is $\bigoplus_{i\in I}^{\text{ext}} \mathcal{V}_i := \{\mathbf{v} \in \prod_{i\in I} \mathcal{V}_i : |\operatorname{supp}(\mathbf{v})| < \infty\}$.

Proposition 2.1.19. *External Sums Are Finite-Support Products*
Let $\{\mathcal{V}_i\}_{i\in I}$ be $\mathbb{F}$-vector spaces and set $P := \prod_{i\in I} \mathcal{V}_i$ and $S := \bigoplus_{i\in I}^{\text{ext}} \mathcal{V}_i$.
(i) P is an $\mathbb{F}$-vector space under componentwise operations.
(ii) $S \leq P$.
(iii) If $|I| < \infty$, then $S = P$.

Proof.
(i) Componentwise closure keeps addition and scalar multiplication in P.
The family $(\mathbf{0}_i)_{i\in I}$ is its zero, and $(-\mathbf{v}_i)_{i\in I}$ is the inverse of $\mathbf{v}$. For arbitrary elements, scalars, and $i \in I$, the relevant identity at coordinate i holds by the corresponding identity in $\mathcal{V}_i$; coordinatewise equality proves the identity in P.
(ii) The zero family has empty support. For $\mathbf{u}, \mathbf{v} \in S$ and $a \in \mathbb{F}$, $\operatorname{supp}(a\mathbf{u} + \mathbf{v}) \subseteq \operatorname{supp}(\mathbf{u}) \cup \operatorname{supp}(\mathbf{v})$, a finite set.
Hence $a\mathbf{u} + \mathbf{v} \in S$, so $S \leq P$ by Theorem 1.2.2.
(iii) If $|I| < \infty$ and $\mathbf{v} \in P$, then $\operatorname{supp}(\mathbf{v}) \subseteq I$ is finite. Thus $P \subseteq S$, while $S \subseteq P$ by construction, so $S = P$.
Structural Commentary: Coordinatewise axioms build the product; support containment proves the remaining claims.
Key Insight: Finite linear combinations preserve finite support. □

Definition 2.1.20. *Finitely Nonzero Sequence Space*
The *finitely nonzero sequence space* is $\mathbb{F}^{(\mathbb{N})} := \bigoplus_{j\in\mathbb{N}}^{\text{ext}} \mathbb{F} = \{(x_j)_{j\in\mathbb{N}} \in \mathbb{F}^{\mathbb{N}} : |\{j \in \mathbb{N} : x_j \neq 0\}| < \infty\}$, with componentwise operations.

Proposition 2.1.21. *Finitely Nonzero Sequences Form a Vector Space*
The set $\mathbb{F}^{(\mathbb{N})}$ is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication.

Proof. The space $\mathbb{F}^{(\mathbb{N})}$ is the external direct sum of the constant family $\mathcal{V}_j = \mathbb{F}$ by its definition, so it is a subspace of the product $\mathbb{F}^{\mathbb{N}}$ by Proposition 2.1.19(ii). Hence it is an $\mathbb{F}$-vector space under the inherited componentwise operations.
Structural Commentary: The result specializes the external direct-sum subspace construction to a constant countable family.
Key Insight: Finitely supported sequences are a canonical external sum. □

Set $\operatorname{Seq}(\mathbb{F}) := \mathbb{F}^{\mathbb{N}}$. Then $\mathbb{F}^{(\mathbb{N})} \leq \operatorname{Seq}(\mathbb{F})$ by Proposition 2.1.21.

Definition 2.1.22. *Internal Direct Sum of a Family*

For $\{\mathcal{U}_i\}_{i\in I} \subseteq \mathrm{Sub}(\mathcal{V})$, the internal sum $S := \sum_{i\in I} \mathcal{U}_i$ is *direct* iff $(\forall\, \mathbf{v} \in S)(\exists!\, \mathbf{u} \in \bigoplus_{i\in I}^{\text{ext}} \mathcal{U}_i)\, \mathbf{v} = \sum_{i\in I} \mathbf{u}_i$.
A direct internal sum is denoted $\bigoplus_{i\in I} \mathcal{U}_i$; the equality $\mathcal{V} = \bigoplus_{i\in I} \mathcal{U}_i$ additionally asserts $\mathcal{V} = S$.

Proposition 2.1.23. *Family Direct-Sum Criterion*
For $\{\mathcal{U}_i\}_{i\in I} \subseteq \mathrm{Sub}(\mathcal{V})$, the following are equivalent:

(i) $\sum_{i\in I} \mathcal{U}_i$ *is direct;*
(ii) $(\forall\, F \subseteq I)[|F| < \infty \Rightarrow (\forall\, (\mathbf{u}_i)_{i\in F} \in \prod_{i\in F} \mathcal{U}_i)(\sum_{i\in F} \mathbf{u}_i = \mathbf{0}_{\mathcal{V}} \Rightarrow (\forall\, i \in F)\, \mathbf{u}_i = \mathbf{0}_{\mathcal{V}})]$;
(iii) $(\forall\, i \in I)\mathcal{U}_i \cap \sum_{j\in I\setminus\{i\}} \mathcal{U}_j = \{\mathbf{0}_{\mathcal{V}}\}$.

Proof.

(i) $\Rightarrow$ *(ii)* Let $F \subseteq I$ be finite and let $(\mathbf{u}_i)_{i\in F} \in \prod_{i\in F} \mathcal{U}_i$ satisfy $\sum_{i\in F} \mathbf{u}_i = \mathbf{0}_{\mathcal{V}}$. Extend this family by zero outside F.
Directness identifies the resulting decomposition with the all-zero decomposition, so $(\forall\, i \in F)\, \mathbf{u}_i = \mathbf{0}_{\mathcal{V}}$.
(ii) $\Rightarrow$ *(iii)* Fix $i \in I$ and let $\mathbf{x} \in \mathcal{U}_i \cap \sum_{j\in I\setminus\{i\}} \mathcal{U}_j$.
Then $\mathbf{x} = \sum_{j\in F} \mathbf{u}_j$ for a finite $F \subseteq I \setminus \{i\}$ with $(\forall\, j \in F)\, \mathbf{u}_j \in \mathcal{U}_j$ by Proposition 2.1.6(iii).
The relation on $F \cup \{i\}$ with ith term $\mathbf{x}$ and jth term $-\mathbf{u}_j$ is zero. It is trivial by (ii), so $\mathbf{x} = \mathbf{0}_{\mathcal{V}}$.
(iii) $\Rightarrow$ *(i)* Let $F \subseteq I$ be finite and suppose $\sum_{i\in F} \mathbf{u}_i = \mathbf{0}_{\mathcal{V}}$ with $(\forall\, i \in F)\, \mathbf{u}_i \in \mathcal{U}_i$.
For $k \in F$, $\mathbf{u}_k = -\sum_{i\in F\setminus\{k\}} \mathbf{u}_i \in \mathcal{U}_k \cap \sum_{j\in I\setminus\{k\}} \mathcal{U}_j = \{\mathbf{0}_{\mathcal{V}}\}$ by (iii), so every finite zero relation is trivial.
Each vector in the sum has a finite decomposition by Proposition 2.1.6(iii).
The difference of two decompositions is a finite zero relation, so they coincide and the sum is direct.

Structural Commentary: The cycle converts unique decompositions into zero relations and collective intersection tests.
Key Insight: Directness forbids cancellation across distinct summands. □

Proposition 2.1.24. *Direct Sum Map*

Let $m \in \mathbb{N}$ and let $\mathcal{U}_1, \ldots, \mathcal{U}_m \leq \mathcal{V}$. Define $\Gamma : \prod_{j=1}^{m} \mathcal{U}_j \to \sum_{j=1}^{m} \mathcal{U}_j$ by $\Gamma((\mathbf{u}_j)_{j=1}^{m}) := \sum_{j=1}^{m} \mathbf{u}_j$. The following hold:

(i) $\Gamma \in \mathcal{L}(\mathcal{U}_1 \times \cdots \times \mathcal{U}_m, \mathcal{U}_1 + \cdots + \mathcal{U}_m)$;
(ii) $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ *is direct iff* Γ *is injective.*

Proof.

(i) For $\mathbf{u}, \mathbf{v} \in \mathcal{U}_1 \times \cdots \times \mathcal{U}_m$ and $a, b \in \mathbb{F}$, $\Gamma(a\mathbf{u} + b\mathbf{v}) = \sum_{j=1}^{m} (a\mathbf{u}_j + b\mathbf{v}_j) = a\Gamma(\mathbf{u}) + b\Gamma(\mathbf{v})$. Hence Γ is linear.
(ii) $(\Rightarrow)$ Assume sum is direct. If $\Gamma(\mathbf{u}) = \Gamma(\mathbf{v})$, then $\mathbf{u}$ and $\mathbf{v}$ are decompositions of same vector, so $\mathbf{u} = \mathbf{v}$ by directness. QED.
(ii) $(\Leftarrow)$ Assume Γ is injective. If $\sum_{j=1}^{m} \mathbf{u}_j = \sum_{j=1}^{m} \mathbf{v}_j$, then $\Gamma(\mathbf{u}) = \Gamma(\mathbf{v})$, so $\mathbf{u} = \mathbf{v}$ by injectivity.
Thus every vector in the sum has a unique decomposition, and the sum is direct.

Structural Commentary: Linearity is coordinatewise; injectivity translates equality of sums into equality of decompositions.
Key Insight: Directness is injectivity of the addition map. □

2.2 Quotient Spaces

Quotients collapse a prescribed subspace while retaining the linear structure that is independent of the collapsed directions.
Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 2.2.1. *Affine Subset Parallel to a Subspace*
Let $\mathcal{U} \leq \mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, the subset $\mathbf{v} + \mathcal{U} := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in \mathcal{U}\}$ is the *affine subset* through $\mathbf{v}$ parallel to $\mathcal{U}$.

Proposition 2.2.2. *Parallel Affine Subsets Are Equal or Disjoint*
Let $\mathcal{U} \leq \mathcal{V}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The following conditions are equivalent:

(i) $\mathbf{v} - \mathbf{w} \in \mathcal{U}$;
(ii) $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$;
(iii) $(\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U}) \neq \emptyset$.

Proof.

(i) $\Rightarrow$ *(ii)* Suppose $\mathbf{v} - \mathbf{w} \in \mathcal{U}$. For $\mathbf{u} \in \mathcal{U}$, $\mathbf{v} + \mathbf{u} = \mathbf{w} + ((\mathbf{v} - \mathbf{w}) + \mathbf{u}) \in \mathbf{w} + \mathcal{U}$, so $\mathbf{v} + \mathcal{U} \subseteq \mathbf{w} + \mathcal{U}$.
Since $\mathbf{w} - \mathbf{v} = -(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$, exchanging $\mathbf{v}$ and $\mathbf{w}$ establishes the reverse inclusion.
(ii) $\Rightarrow$ *(iii)* If $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$, then $\mathbf{v} \in (\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U})$.
(iii) $\Rightarrow$ *(i)* Choose $\mathbf{x} \in (\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U})$ and $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$ with $\mathbf{x} = \mathbf{v} + \mathbf{u}_1 = \mathbf{w} + \mathbf{u}_2$. Then $\mathbf{v} - \mathbf{w} = \mathbf{u}_2 - \mathbf{u}_1 \in \mathcal{U}$.

Structural Commentary: A cyclic implication proof connects differences, equality, and intersection.
Key Insight: A common point exposes the difference of representatives. □

Definition 2.2.3. *Underlying Quotient Set*
For $\mathcal{U} \leq \mathcal{V}$, the *quotient set* is $\mathcal{V}/\mathcal{U} := \{\mathbf{v} + \mathcal{U} : \mathbf{v} \in \mathcal{V}\}$.

Proposition 2.2.4. *Quotient Relation Is a Linear Congruence*
Let $\mathcal{U} \leq \mathcal{V}$ and define a relation on $\mathcal{V}$ by $\mathbf{v} \equiv_{\mathcal{U}} \mathbf{w} \iff \mathbf{v} - \mathbf{w} \in \mathcal{U}$.

(i) $(\forall\, \mathbf{v} \in \mathcal{V})\, \mathbf{v} \equiv_{\mathcal{U}} \mathbf{v}$.
(ii) $(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})[\mathbf{v} \equiv_{\mathcal{U}} \mathbf{w} \Rightarrow \mathbf{w} \equiv_{\mathcal{U}} \mathbf{v}]$.

(iii) $(\forall\, \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V})[(\mathbf{u} \equiv_{\mathcal{U}} \mathbf{v} \wedge \mathbf{v} \equiv_{\mathcal{U}} \mathbf{w}) \Rightarrow \mathbf{u} \equiv_{\mathcal{U}} \mathbf{w}]$.
(iv) $(\forall\, a, b \in \mathbb{F})(\forall\, \mathbf{v}_1, \mathbf{v}_2, \mathbf{w}_1, \mathbf{w}_2 \in \mathcal{V})[(\mathbf{v}_1 \equiv_{\mathcal{U}} \mathbf{v}_2 \wedge \mathbf{w}_1 \equiv_{\mathcal{U}} \mathbf{w}_2) \Rightarrow a\mathbf{v}_1 + b\mathbf{w}_1 \equiv_{\mathcal{U}} a\mathbf{v}_2 + b\mathbf{w}_2]$.
Thus $\equiv_{\mathcal{U}}$ is an equivalence relation compatible with linear combinations.

Proof.
(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} - \mathbf{v} = \mathbf{0} \in \mathcal{U}$.
(ii) If $\mathbf{v} - \mathbf{w} \in \mathcal{U}$, then $\mathbf{w} - \mathbf{v} = -(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$.
(iii) If $\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w} \in \mathcal{U}$, then $\mathbf{u} - \mathbf{w} = (\mathbf{u} - \mathbf{v}) + (\mathbf{v} - \mathbf{w}) \in \mathcal{U}$.
(iv) If $\mathbf{v}_1 - \mathbf{v}_2, \mathbf{w}_1 - \mathbf{w}_2 \in \mathcal{U}$, then $(a\mathbf{v}_1 + b\mathbf{w}_1) - (a\mathbf{v}_2 + b\mathbf{w}_2) = a(\mathbf{v}_1 - \mathbf{v}_2) + b(\mathbf{w}_1 - \mathbf{w}_2) \in \mathcal{U}$ for $a, b \in \mathbb{F}$.
Structural Commentary: The four items verify the equivalence axioms and linear compatibility.
Key Insight: Linear combinations preserve differences lying in the subspace. □

Perspective 2.2.5. *Cosets as Equivalence Classes*
The equivalence class of $\mathbf{v}$ under $\equiv_{\mathcal{U}}$ is $\mathbf{v} + \mathcal{U}$ by Proposition 2.2.2. Hence the quotient set in Definition 2.2.3 is the abstract quotient set from Definition 5.2.5. It remembers a vector only up to addition by an element of $\mathcal{U}$.

Proposition 2.2.6. *Quotient-Space Structure*
Let $\mathcal{U} \leq \mathcal{V}$. For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a \in \mathbb{F}$, set $(\mathbf{v} + \mathcal{U}) + (\mathbf{w} + \mathcal{U}) := (\mathbf{v} + \mathbf{w}) + \mathcal{U}$ and $a(\mathbf{v} + \mathcal{U}) := a\mathbf{v} + \mathcal{U}$.
(i) These operations are independent of the chosen representatives.
(ii) With these operations, $\mathcal{V}/\mathcal{U}$ is a vector space, called the quotient space of $\mathcal{V}$ by $\mathcal{U}$.

Proof.
(i) Suppose $\mathbf{v} + \mathcal{U} = \mathbf{v}' + \mathcal{U}$ and $\mathbf{w} + \mathcal{U} = \mathbf{w}' + \mathcal{U}$. Then $\mathbf{v} - \mathbf{v}', \mathbf{w} - \mathbf{w}' \in \mathcal{U}$ by Proposition 2.2.2.
Hence $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}') \in \mathcal{U}$ by Proposition 2.2.4(iv), and $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in \mathcal{U}$.
The resulting sum and scalar-multiple cosets are equal by Proposition 2.2.2.
(ii) Write $[\mathbf{v}] := \mathbf{v} + \mathcal{U}$. For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $([\mathbf{u}] + [\mathbf{v}]) + [\mathbf{w}] = [\mathbf{u} + \mathbf{v} + \mathbf{w}] = [\mathbf{u}] + ([\mathbf{v}] + [\mathbf{w}])$, $[\mathbf{v}] + [\mathbf{w}] = [\mathbf{w}] + [\mathbf{v}]$, $[\mathbf{0}] = \mathcal{U}$ is an additive identity, and $[-\mathbf{v}]$ is the additive inverse of $[\mathbf{v}]$.
The scalar identities $1[\mathbf{v}] = [\mathbf{v}]$, $a(b[\mathbf{v}]) = (ab)[\mathbf{v}]$, $a([\mathbf{v}] + [\mathbf{w}]) = a[\mathbf{v}] + a[\mathbf{w}]$, and $(a + b)[\mathbf{v}] = a[\mathbf{v}] + b[\mathbf{v}]$ hold by the vector-space identities in $\mathcal{V}$. Thus all vector-space axioms hold.
Structural Commentary: Representative independence precedes direct verification of the vector-space axioms.
Key Insight: Coset arithmetic inherits every identity from representatives. □

Definition 2.2.7. *Quotient Map*
For $\mathcal{U} \leq \mathcal{V}$, the *quotient map* is $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ defined by $\pi(\mathbf{v}) := \mathbf{v} + \mathcal{U}$.

Proposition 2.2.8. *Properties of the Quotient Map*
Let $\mathcal{U} \leq \mathcal{V}$ and define $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ by $\pi(\mathbf{v}) = \mathbf{v} + \mathcal{U}$.
(i) π is linear.
(ii) π is surjective.
(iii) $\ker(\pi) = \mathcal{U}$.

Proof.
(i) For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $\pi(a\mathbf{v} + b\mathbf{w}) = (a\mathbf{v} + b\mathbf{w}) + \mathcal{U} = a\pi(\mathbf{v}) + b\pi(\mathbf{w})$.
(ii) If $\mathbf{x} \in \mathcal{V}/\mathcal{U}$, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{x} = \mathbf{v} + \mathcal{U} = \pi(\mathbf{v})$ by the definition of $\mathcal{V}/\mathcal{U}$.
(iii) For $\mathbf{v} \in \mathcal{V}$, $\pi(\mathbf{v}) = \mathcal{U} \iff \mathbf{v} \in \mathcal{U}$ by Proposition 2.2.2. Therefore $\ker(\pi) = \mathcal{U}$.
Structural Commentary: The proof verifies linearity, surjectivity, and the kernel in statement order.
Key Insight: The zero coset is exactly the collapsed subspace. □

Theorem 2.2.9. *Dimension of a Quotient*
Let $\mathcal{V}$ be finite-dimensional and $\mathcal{U} \leq \mathcal{V}$.
(i) $\mathcal{V}/\mathcal{U}$ is finite-dimensional.
(ii) $\dim(\mathcal{V}/\mathcal{U}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. Let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map. Its range is $\mathcal{V}/\mathcal{U}$ and its kernel is $\mathcal{U}$ by Proposition 2.2.8(ii)–(iii).
(i) The range $\mathcal{V}/\mathcal{U}$ is finite-dimensional by Theorem 1.7.3(i).
(ii) The formula in (ii) holds by Theorem 1.7.3(ii) with $\ker(\pi) = \mathcal{U}$ and $\operatorname{range}(\pi) = \mathcal{V}/\mathcal{U}$.
Structural Commentary: Quotient-map properties followed by rank-nullity handle finiteness and dimension.
Key Insight: Quotient dimension is the rank of the quotient map. □

Theorem 2.2.10. *Subspace Correspondence for Quotients*
Let $\mathcal{U} \leq \mathcal{V}$, let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map, and set $\mathfrak{A} := \{\mathcal{T} \leq \mathcal{V} : \mathcal{U} \leq \mathcal{T}\}$ and $\mathfrak{B} := \{\mathcal{Y} : \mathcal{Y} \leq \mathcal{V}/\mathcal{U}\}$. For $\mathcal{T} \in \mathfrak{A}$ and $\mathcal{Y} \in \mathfrak{B}$, set $\Phi(\mathcal{T}) := \mathcal{T}/\mathcal{U} = \pi(\mathcal{T})$ and $\Psi(\mathcal{Y}) := \pi^{-1}(\mathcal{Y})$.
(i) $\Phi : \mathfrak{A} \to \mathfrak{B}$ and $\Psi : \mathfrak{B} \to \mathfrak{A}$ are well-defined.
(ii) Φ and Ψ are inverse bijections: $\Psi\Phi = I_{\mathfrak{A}}$ and $\Phi\Psi = I_{\mathfrak{B}}$.
(iii) Both maps preserve inclusion.

Proof.
(i) Let $\mathcal{T} \in \mathfrak{A}$. The set $\mathcal{T}/\mathcal{U}$ contains $\mathcal{U} = \mathbf{0} + \mathcal{U}$, and $a(\mathbf{t}_1 + \mathcal{U}) + b(\mathbf{t}_2 + \mathcal{U}) = (a\mathbf{t}_1 + b\mathbf{t}_2) + \mathcal{U} \in \mathcal{T}/\mathcal{U}$ for $\mathbf{t}_1, \mathbf{t}_2 \in \mathcal{T}$ and $a, b \in \mathbb{F}$.
Hence $\Phi(\mathcal{T}) \leq \mathcal{V}/\mathcal{U}$ by Theorem 1.2.2. Let $\mathcal{Y} \in \mathfrak{B}$. The set $\pi^{-1}(\mathcal{Y})$ contains $\mathbf{0}_{\mathcal{V}}$, and $\pi(a\mathbf{v} + b\mathbf{w}) = a\pi(\mathbf{v}) + b\pi(\mathbf{w}) \in \mathcal{Y}$ for $\mathbf{v}, \mathbf{w} \in \pi^{-1}(\mathcal{Y})$ and $a, b \in \mathbb{F}$; hence $\pi^{-1}(\mathcal{Y}) \leq \mathcal{V}$ by Theorem 1.2.2. Moreover, $\mathcal{U} = \ker(\pi) \subseteq \pi^{-1}(\mathcal{Y})$ by Proposition 2.2.8(iii).
(ii) Let $\mathcal{T} \in \mathfrak{A}$. For $\mathbf{t} \in \mathcal{T}$, $\pi(\mathbf{t}) \in \pi(\mathcal{T})$, so $\mathcal{T} \subseteq \pi^{-1}(\pi(\mathcal{T}))$. If $\mathbf{v} \in \pi^{-1}(\pi(\mathcal{T}))$, choose $\mathbf{t} \in \mathcal{T}$ with $\mathbf{v} + \mathcal{U} = \mathbf{t} + \mathcal{U}$.
Then $\mathbf{v} - \mathbf{t} \in \mathcal{U} \subseteq \mathcal{T}$ by Proposition 2.2.2, so $\mathbf{v} \in \mathcal{T}$. Thus $\Psi(\Phi(\mathcal{T})) = \mathcal{T}$. Let $\mathcal{Y} \in \mathfrak{B}$.
If $\mathbf{v} \in \pi^{-1}(\mathcal{Y})$, then $\pi(\mathbf{v}) \in \mathcal{Y}$, so $\pi(\pi^{-1}(\mathcal{Y})) \subseteq \mathcal{Y}$. If $\mathbf{y} \in \mathcal{Y}$, choose $\mathbf{v} \in \mathcal{V}$ with $\pi(\mathbf{v}) = \mathbf{y}$ by Proposition 2.2.8(ii).
Then $\mathbf{v} \in \pi^{-1}(\mathcal{Y})$, so $\mathbf{y} \in \pi(\pi^{-1}(\mathcal{Y}))$. Therefore $\Phi(\Psi(\mathcal{Y})) = \mathcal{Y}$.

(iii) If $\mathcal{T}_1 \subseteq \mathcal{T}_2$ in $\mathfrak{A}$, then $\pi(\mathcal{T}_1) \subseteq \pi(\mathcal{T}_2)$. If $\mathcal{Y}_1 \subseteq \mathcal{Y}_2$ in $\mathfrak{B}$, then $\pi^{-1}(\mathcal{Y}_1) \subseteq \pi^{-1}(\mathcal{Y}_2)$.

Structural Commentary: Well-defined assignments, inverse identities, and monotonicity establish the correspondence.
Key Insight: Images and preimages undo the quotient construction exactly. □

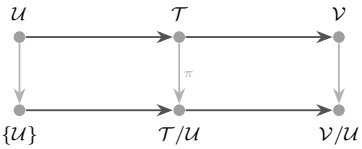

Figure 10: Intermediate subspaces above $\mathcal{U}$ correspond to subspaces of the quotient; horizontal arrows denote inclusions and vertical arrows denote quotient images.

Figure 10 illustrates Theorem 2.2.10: $\mathcal{U} \subseteq \mathcal{T} \subseteq \mathcal{V}$ descends to $\{\mathcal{U}\} \subseteq \mathcal{T}/\mathcal{U} \subseteq \mathcal{V}/\mathcal{U}$, while inverse images recover the original subspaces. The theorem, rather than the diagram, owns the correspondence.

Theorem 2.2.11. *Universal Property of the Quotient Map*
Let $\mathcal{U} \leq \mathcal{V}$, let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ satisfy $\mathcal{U} \leq \ker(T)$.

(i) $(\exists!\, \overline{T} \in \mathcal{L}(\mathcal{V}/\mathcal{U}, \mathcal{W}))[\overline{T}\pi = T \wedge (\forall\, \mathbf{v} \in \mathcal{V})\, \overline{T}(\mathbf{v}+\mathcal{U}) = T\mathbf{v}]$.
(ii) The map $\overline{T}$ from (i) satisfies $\ker(\overline{T}) = \ker(T)/\mathcal{U}$.
(iii) The map $\overline{T}$ from (i) satisfies $\operatorname{range}(\overline{T}) = \operatorname{range}(T)$.

Proof.

(i) Define $\overline{T}(\mathbf{v}+\mathcal{U}) := T\mathbf{v}$. If $\mathbf{v}+\mathcal{U} = \mathbf{w}+\mathcal{U}$, then $\mathbf{v}-\mathbf{w} \in \mathcal{U} \leq \ker(T)$ by Proposition 2.2.2, so $T\mathbf{v} - T\mathbf{w} = T(\mathbf{v}-\mathbf{w}) = \mathbf{0}$. Hence $\overline{T}$ is well-defined. For $a, b \in \mathbb{F}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\overline{T}(a(\mathbf{v}+\mathcal{U}) + b(\mathbf{w}+\mathcal{U})) = T(a\mathbf{v}+b\mathbf{w}) = aT\mathbf{v} + bT\mathbf{w}$, so $\overline{T}$ is linear. Also, $(\overline{T}\pi)(\mathbf{v}) = \overline{T}(\mathbf{v}+\mathcal{U}) = T\mathbf{v}$ for $\mathbf{v} \in \mathcal{V}$, so $\overline{T}\pi = T$. If $S \in \mathcal{L}(\mathcal{V}/\mathcal{U}, \mathcal{W})$ satisfies $S\pi = T$, then $(\forall\, \mathbf{v} \in \mathcal{V})\, S(\mathbf{v}+\mathcal{U}) = S\pi(\mathbf{v}) = T\mathbf{v} = \overline{T}(\mathbf{v}+\mathcal{U})$. Since $\mathcal{V}/\mathcal{U} = \{\mathbf{v}+\mathcal{U} : \mathbf{v} \in \mathcal{V}\}$, $S = \overline{T}$.
(ii) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v}+\mathcal{U} \in \ker(\overline{T}) \iff T\mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \ker(T)$. Hence $\ker(\overline{T}) = \{\mathbf{v}+\mathcal{U} : \mathbf{v} \in \ker(T)\} = \ker(T)/\mathcal{U}$.
(iii) The equality $\operatorname{range}(\overline{T}) = \{\overline{T}(\mathbf{v}+\mathcal{U}) : \mathbf{v} \in \mathcal{V}\} = \{T\mathbf{v} : \mathbf{v} \in \mathcal{V}\} = \operatorname{range}(T)$ is immediate from the defining formula.

Structural Commentary: Construction establishes factorisation; set calculations identify kernel and range.
Key Insight: Killing $\mathcal{U}$ is precisely the condition for descent. □

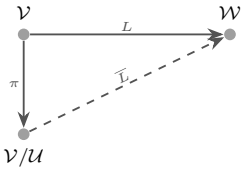

Figure 11: If $\mathcal{U} \subseteq \ker(L)$, the map L factors uniquely through the quotient map π; the dashed arrow is the induced map $\overline{L}$.

Figure 11 depicts Theorem 2.2.11, using L in place of T: collapsing $\mathcal{U}$ first does not change a map that already vanishes on $\mathcal{U}$. The theorem, rather than the diagram, owns existence and uniqueness.

Theorem 2.2.12. *Quotient by the Kernel*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and define $\widetilde{T}(\mathbf{v} + \ker(T)) := T\mathbf{v}$.

(i) The assignment defines a well-defined linear map $\widetilde{T} : \mathcal{V}/\ker(T) \to \mathcal{W}$.
(ii) $\widetilde{T}$ is injective.
(iii) $\operatorname{range}(\widetilde{T}) = \operatorname{range}(T)$.
(iv) The codomain restriction $\widehat{T} : \mathcal{V}/\ker(T) \to \operatorname{range}(T)$ is an isomorphism; hence $\mathcal{V}/\ker(T) \cong \operatorname{range}(T)$.

Proof.

(i) The claimed well-defined linear map exists by Theorem 2.2.11(i) with $\mathcal{U} = \ker(T)$.
(ii) The identity $\ker(\widetilde{T}) = \ker(T)/\ker(T) = \{\ker(T)\}$ holds by Theorem 2.2.11(ii). Thus $\widetilde{T}$ is injective by Proposition 1.6.2.
(iii) The range identity holds by Theorem 2.2.11(iii).
(iv) The codomain restriction $\widehat{T}$ is linear and injective by parts (i)–(ii), and it is surjective by part (iii). Therefore $\widehat{T}$ is invertible by Proposition 1.6.7, and $\mathcal{V}/\ker(T) \cong \operatorname{range}(T)$.

Structural Commentary: The universal property constructs the map; kernel and range identities produce the isomorphism.
Key Insight: Removing the kernel makes the induced map injective. □

Theorem 2.2.13. *Second Isomorphism Theorem*
If $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$, then $\mathcal{S}/(\mathcal{S} \cap \mathcal{T}) \cong (\mathcal{S}+\mathcal{T})/\mathcal{T}$.

Proof.

(i) *Construction*: The sum $\mathcal{S}+\mathcal{T}$ is a subspace containing $\mathcal{T}$ by Proposition 2.1.2(i). Define $L : \mathcal{S} \to (\mathcal{S}+\mathcal{T})/\mathcal{T}$ by $L\mathbf{s} := \mathbf{s}+\mathcal{T}$. For $\mathbf{s}, \mathbf{r} \in \mathcal{S}$ and $a, b \in \mathbb{F}$, $L(a\mathbf{s}+b\mathbf{r}) = (a\mathbf{s}+b\mathbf{r}) + \mathcal{T} = aL\mathbf{s} + bL\mathbf{r}$, so L is linear.
(ii) *Kernel*: For $\mathbf{s} \in \mathcal{S}$, $L\mathbf{s} = \mathcal{T} \iff \mathbf{s} \in \mathcal{T}$ by Proposition 2.2.2. Hence $\ker(L) = \mathcal{S} \cap \mathcal{T}$.
(iii) *Surjectivity*: Each coset equals $(\mathbf{s}+\mathbf{t}) + \mathcal{T} = \mathbf{s} + \mathcal{T} = L\mathbf{s}$ for suitable $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$; hence L is surjective.
(iv) *Conclusion*: The equality $\operatorname{range}(L) = (\mathcal{S}+\mathcal{T})/\mathcal{T}$ holds by part (iii). Therefore $\mathcal{S}/(\mathcal{S} \cap \mathcal{T}) \cong (\mathcal{S}+\mathcal{T})/\mathcal{T}$ by Theorem 2.2.12 with the linearity and kernel identity from parts (i)–(ii).

Structural Commentary: Construction, kernel, surjectivity, and the first isomorphism theorem form four operative stages.
Key Insight: The target forgets exactly $\mathcal{S} \cap \mathcal{T}$. □

Theorem 2.2.14. *Third Isomorphism Theorem*
If $\mathcal{U} \leq \mathcal{T} \leq \mathcal{V}$, then $(\mathcal{V}/\mathcal{U})/(\mathcal{T}/\mathcal{U}) \cong \mathcal{V}/\mathcal{T}$.

Proof.

(i) *Construction*: The inclusion $\mathcal{T}/\mathcal{U} \leq \mathcal{V}/\mathcal{U}$ holds by Theorem 2.2.10(i). Define $L : \mathcal{V}/\mathcal{U} \to \mathcal{V}/\mathcal{T}$ by $L(\mathbf{v}+\mathcal{U}) := \mathbf{v}+\mathcal{T}$.
(ii) *Well-Definedness and Linearity*: If $\mathbf{v}+\mathcal{U} = \mathbf{w}+\mathcal{U}$, then $\mathbf{v}-\mathbf{w} \in \mathcal{U} \subseteq \mathcal{T}$ by Proposition 2.2.2, so $\mathbf{v}+\mathcal{T} = \mathbf{w}+\mathcal{T}$ by the same proposition. For $a, b \in \mathbb{F}$, $L(a(\mathbf{v}+\mathcal{U}) + b(\mathbf{w}+\mathcal{U})) = (a\mathbf{v}+b\mathbf{w})+\mathcal{T} = aL(\mathbf{v}+\mathcal{U}) + bL(\mathbf{w}+\mathcal{U})$.
(iii) *Kernel and Range*: For $\mathbf{v} \in \mathcal{V}$, $L(\mathbf{v}+\mathcal{U}) = \mathcal{T} \iff \mathbf{v} \in \mathcal{T}$ by Proposition 2.2.2; hence $\ker(L) = \mathcal{T}/\mathcal{U}$.
Every $\mathbf{v}+\mathcal{T} \in \mathcal{V}/\mathcal{T}$ equals $L(\mathbf{v}+\mathcal{U})$, so L is surjective.
(iv) *Conclusion*: The equality $\operatorname{range}(L) = \mathcal{V}/\mathcal{T}$ holds by part (iii).
Therefore $(\mathcal{V}/\mathcal{U})/(\mathcal{T}/\mathcal{U}) \cong \mathcal{V}/\mathcal{T}$ by Theorem 2.2.12 with the linearity and kernel identity from parts (ii)–(iii).

Structural Commentary: Construction, kernel computation, and the first isomorphism theorem identify the iterated quotient.
Key Insight: The nested collapse has kernel $\mathcal{T}/\mathcal{U}$. □

Quotient spaces retain precisely the structure invariant under translation by a chosen subspace. Their correspondence, universal property, and isomorphism theorems formalise which data survive that collapse.

2.3 Decomposition and Complementation

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$. Dimension counts first detect directness; complements then realize quotients as concrete subspaces.

Theorem 2.3.1. *Dimension Formula for a Sum*
If $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ are finite-dimensional, then $\dim(\mathcal{U}+\mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} \cap \mathcal{W})$.

Proof. The subspace $\mathcal{U} \cap \mathcal{W}$ is finite-dimensional by Proposition 1.4.3. Choose basis $\epsilon = (\mathbf{e}_1, \ldots, \mathbf{e}_r)$ of intersection by Corollary 1.3.14.
Extend ϵ to bases $\beta_{\mathcal{U}} = (\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p)$ of $\mathcal{U}$ and $\beta_{\mathcal{W}} = (\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$ by Theorem 1.3.13.
Set $\gamma := (\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p, \mathbf{w}_1, \ldots, \mathbf{w}_q)$. The list γ spans $\mathcal{U}+\mathcal{W}$.
Suppose $\sum_{i=1}^{r} a_i\mathbf{e}_i + \sum_{j=1}^{p} b_j\mathbf{u}_j + \sum_{k=1}^{q} c_k\mathbf{w}_k = \mathbf{0}$. Then $\sum_{i=1}^{r} a_i\mathbf{e}_i + \sum_{j=1}^{p} b_j\mathbf{u}_j = -\sum_{k=1}^{q} c_k\mathbf{w}_k \in \mathcal{U} \cap \mathcal{W}$.
Its representation in $\beta_{\mathcal{U}}$ therefore has $b_1 = \cdots = b_p = 0$.
The remaining relation belongs to $\beta_{\mathcal{W}}$, so $a_1 = \cdots = a_r = c_1 = \cdots = c_q = 0$.
Thus γ is a basis of $\mathcal{U}+\mathcal{W}$, and $\dim(\mathcal{U}+\mathcal{W}) = r+p+q = (r+p)+(r+q)-r$.
Structural Commentary: An intersection basis is extended twice; concatenation then separates shared from new directions.
Key Insight: Subtract the dimension counted in both summands. □

Corollary 2.3.2. *Dimension of a Direct Sum*
If $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ are finite-dimensional and $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, then $\dim(\mathcal{U} \oplus \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W})$.

Proof. The sum is direct by Proposition 2.1.8. Since $\dim(\mathcal{U} \cap \mathcal{W}) = 0$, the asserted equality holds by Theorem 2.3.1.
Structural Commentary: The directness criterion removes the intersection term from the sum formula.
Key Insight: Trivial overlap makes dimensions additive. □

Corollary 2.3.3. *Large Subspaces Must Intersect*
Let $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$, where $\mathcal{V}$ is finite-dimensional, and let $k \in \mathbb{N}$.

(i) If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, then $\mathcal{U} \cap \mathcal{W}$ contains a nonzero vector.
(ii) If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, then $\mathcal{U} \cap \mathcal{W}$ contains k linearly independent vectors.

Proof. The inequality $\dim(\mathcal{U}+\mathcal{W}) \leq \dim(\mathcal{V})$ holds by Proposition 1.4.3.
Hence $\dim(\mathcal{U} \cap \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U}+\mathcal{W}) \geq \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{V})$ by Theorem 2.3.1.

(i) Under strict inequality, $\dim(\mathcal{U} \cap \mathcal{W}) \geq 1$. Choose basis of intersection by Corollary 1.3.14; any one of its vectors is nonzero.
(ii) Under second inequality, $\dim(\mathcal{U} \cap \mathcal{W}) \geq k$.
A basis of the intersection exists by Corollary 1.3.14, and any k of its vectors are linearly independent.

Structural Commentary: The sum formula converts excess ambient dimension into a lower bound on overlap.
Key Insight: Excess dimension must reappear in the intersection. □

Theorem 2.3.4. *Direct Sums and Dimension*
Let $m \in \mathbb{N}$, $\mathcal{V}$ be finite-dimensional, $\mathcal{U}_1, \ldots, \mathcal{U}_m \leq \mathcal{V}$. For $1 \leq j \leq m$, let β_j be an ordered basis of $\mathcal{U}_j$, $\beta = (\beta_1, \ldots, \beta_m)$ be their concatenation.

(i) The sum $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is direct iff $\dim(\mathcal{U}_1 + \cdots + \mathcal{U}_m) = \sum_{j=1}^{m} \dim(\mathcal{U}_j)$.
(ii) If the sum is direct, then β is an ordered basis of $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Proof. The list β spans $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ and has length $\sum_{j=1}^{m} \dim(\mathcal{U}_j)$.

(i) ($\Rightarrow$) Suppose the sum is direct. A linear relation among the vectors of β determines vectors $\mathbf{u}_j \in \mathcal{U}_j$ satisfying $\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}$.
Every $\mathbf{u}_j$ is zero by Proposition 2.1.23, and then every coefficient is zero by the independence of β_j.
Thus β is a basis, and its length proves the dimension equality.
(i) ($\Leftarrow$) Suppose dimension equality holds. The spanning list β has length $\dim(\mathcal{U}_1 + \cdots + \mathcal{U}_m)$, so it is a basis by Proposition 1.4.4.
If $\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$, expanding each $\mathbf{u}_j$ in β_j produces a relation in β.
Its coefficients, and hence every $\mathbf{u}_j$, are zero. The sum is therefore direct by Proposition 2.1.23.

(*ii*) Under directness, the linear-independence argument in the forward direction of part (i) applies to β.
Since β already spans the sum, it is an ordered basis of $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Structural Commentary: Concatenation converts directness into independence and dimension equality into full basis length.
Key Insight: Dimensions add exactly when concatenation has no redundancy. □

Definition 2.3.5. *Basis Adapted to a Direct Sum*
Let $m \in \mathbb{N}$, let $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$ be finite-dimensional, and let β be an ordered basis of $\mathcal{V}$.
The basis β is *adapted to the direct-sum decomposition* iff there are ordered bases β_j of $\mathcal{U}_j$ such that $\beta = (\beta_1, \ldots, \beta_m)$.

Proposition 2.3.6. *Complements Exist in Finite Dimensions*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \leq \mathcal{V}$, then $(\exists\, \mathcal{W} \leq \mathcal{V})\, \mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ by Corollary 1.3.14, and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$ by Theorem 1.3.13.
Set $\mathcal{W} := \operatorname{span}(\mathbf{w}_1, \ldots, \mathbf{w}_n)$. The extended basis spans $\mathcal{V}$, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$.
If $\mathbf{x} \in \mathcal{U} \cap \mathcal{W}$, subtracting its two representations produces a relation in the extended basis.
Linear independence forces every coefficient to vanish, so $\mathbf{x} = \mathbf{0}$. Therefore $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ by Proposition 2.1.8.
Structural Commentary: Basis extension isolates the old directions from the newly appended directions.
Key Insight: The appended basis vectors span a complement. □

Theorem 2.3.7. *Complements Exist*
Assume Zorn's lemma. If $\mathcal{U} \leq \mathcal{V}$, then $(\exists\, \mathcal{W} \leq \mathcal{V})\, \mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. Choose a Hamel basis $\mathcal{B}_\mathcal{U}$ of $\mathcal{U}$ and extend it to a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.3.17. Set $\mathcal{C} := \mathcal{B} \setminus \mathcal{B}_\mathcal{U}$ and $\mathcal{W} := \operatorname{span}(\mathcal{C})$.
Every finite $\mathcal{B}$-expansion separates into terms indexed by $\mathcal{B}_\mathcal{U}$ and terms indexed by $\mathcal{C}$, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$.
If $\mathbf{x} \in \mathcal{U} \cap \mathcal{W}$, then $\mathbf{x}$ has one $\mathcal{B}$-expansion supported in $\mathcal{B}_\mathcal{U}$ and another supported in the disjoint set $\mathcal{C}$.
These expansions agree by Proposition 1.3.16, so both have empty support and $\mathbf{x} = \mathbf{0}$. Therefore $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ by Proposition 2.1.8.
Structural Commentary: Hamel basis extension partitions all coordinates into subspace and complementary supports.
Key Insight: Disjoint coordinate supports produce a complement. □

Proposition 2.3.8. *Rank-Nullity as a Complement Isomorphism*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and suppose $\mathcal{V} = \ker(T) \oplus \mathcal{M}$.

(i) The restriction $T|_\mathcal{M} : \mathcal{M} \to \operatorname{range}(T)$ is an isomorphism.
(ii) If $\mathcal{V}$ is finite-dimensional, then $\dim(\mathcal{V}) = \dim(\ker(T)) + \dim(\operatorname{range}(T))$.

Proof.

(i) The restriction is linear and $\ker(T|_\mathcal{M}) = \mathcal{M} \cap \ker(T) = \{\mathbf{0}\}$, so it is injective by Proposition 1.6.2.
If $T\mathbf{v} \in \operatorname{range}(T)$, write $\mathbf{v} = \mathbf{k} + \mathbf{m}$ with $\mathbf{k} \in \ker(T)$ and $\mathbf{m} \in \mathcal{M}$. Then $T\mathbf{v} = T\mathbf{m}$, so the restriction is surjective onto $\operatorname{range}(T)$.
It is therefore an isomorphism by Proposition 1.6.7.

(ii) Suppose $\mathcal{V}$ is finite-dimensional. The subspaces $\ker(T)$ and $\mathcal{M}$ are finite-dimensional by Proposition 1.4.3, and $\operatorname{range}(T)$ is finite-dimensional by Theorem 1.7.1. Thus $\dim(\mathcal{V}) = \dim(\ker(T)) + \dim(\mathcal{M})$ by Corollary 2.3.2, while $\dim(\mathcal{M}) = \dim(\operatorname{range}(T))$ by part (i) and Theorem 1.7.9. Substitution proves the formula.

Structural Commentary: The restriction removes precisely the invisible kernel directions before dimensions are counted.
Key Insight: A kernel complement models the range exactly. □

Proposition 2.3.9. *One-Sided Inverses Split in Finite Dimensions*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) If T is injective, then $(\exists\, L \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\, LT = I_\mathcal{V}$.
(ii) If T is surjective, then $(\exists\, R \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\, TR = I_\mathcal{W}$.

Proof.

(i) Suppose T is injective. Choose $\mathcal{H} \leq \mathcal{W}$ with $\mathcal{W} = \operatorname{range}(T) \oplus \mathcal{H}$ by Proposition 2.3.6.
For the unique decomposition $\mathbf{w} = T\mathbf{v} + \mathbf{h}$, define $L\mathbf{w} := \mathbf{v}$.
This is well-defined: directness determines $T\mathbf{v}$ and $\mathbf{h}$, while injectivity determines $\mathbf{v}$.
If $\mathbf{w}_i = T\mathbf{v}_i + \mathbf{h}_i$ for $i = 1, 2$ and $a, b \in \mathbb{F}$, then $L(a\mathbf{w}_1 + b\mathbf{w}_2) = L(T(a\mathbf{v}_1 + b\mathbf{v}_2) + (a\mathbf{h}_1 + b\mathbf{h}_2)) = a\mathbf{v}_1 + b\mathbf{v}_2 = aL\mathbf{w}_1 + bL\mathbf{w}_2$.
Hence L is linear, and $LT\mathbf{v} = \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.

(ii) Suppose T is surjective. Choose $\mathcal{H} \leq \mathcal{V}$ with $\mathcal{V} = \ker(T) \oplus \mathcal{H}$ by Proposition 2.3.6.
Since $\operatorname{range}(T) = \mathcal{W}$, the restriction $T|_\mathcal{H} : \mathcal{H} \to \mathcal{W}$ is an isomorphism by Proposition 2.3.8(i).
Let $R := (T|_\mathcal{H})^{-1}$, regarded as a linear map from $\mathcal{W}$ to $\mathcal{V}$. Then $TR = I_\mathcal{W}$.

Structural Commentary: A range complement extends a left inverse; a kernel complement supports a right inverse.
Key Insight: Complements turn injectivity or surjectivity into a splitting. □

Proposition 2.3.10. *Quotient by a Complemented Subspace*
Let $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ with $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$. Then the map $J : \mathcal{W} \to \mathcal{V}/\mathcal{U}$ defined by $J(\mathbf{w}) := \mathbf{w} + \mathcal{U}$ is an isomorphism.
Consequently, every complement of $\mathcal{U}$ is isomorphic to $\mathcal{V}/\mathcal{U}$.

Proof.

(i) *Linearity*: For $a, b \in \mathbb{F}$, $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, equality $J(a\mathbf{w}_1 + b\mathbf{w}_2) = aJ(\mathbf{w}_1) + bJ(\mathbf{w}_2)$ holds by Proposition 2.2.6. Thus J is linear.

(ii) *Injectivity*: The equivalence $J(\mathbf{w}) = \mathcal{U} \Leftrightarrow \mathbf{w} \in \mathcal{U}$ holds by Proposition 2.2.2.
Hence $\ker(J) = \mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, and J is injective by Proposition 1.6.2.

(iii) *Surjectivity*: Let $\mathbf{v} + \mathcal{U} \in \mathcal{V}/\mathcal{U}$. Write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$.
Then $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U} = J(\mathbf{w})$ by Proposition 2.2.2, so J is surjective.

(iv) *Conclusion*: The map J is an isomorphism by Proposition 1.6.7.
Since $\mathcal{W}$ was an arbitrary complement of $\mathcal{U}$, this proves the consequent.

Structural Commentary: The proof checks that each coset has exactly one representative in the complement.
Key Insight: A complement is a concrete model of the quotient. □

Definition 2.3.11. *Codimension*
If $\mathcal{U} \leq \mathcal{V}$ and $\mathcal{V}/\mathcal{U}$ is finite-dimensional, then the *codimension of $\mathcal{U}$ in $\mathcal{V}$* is $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) := \dim(\mathcal{V}/\mathcal{U})$.

If $\mathcal{V}$ is finite-dimensional, then $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$ by Theorem 2.2.9. If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ and $\mathcal{V}/\mathcal{U}$ is finite-dimensional, then $\mathcal{W} \cong \mathcal{V}/\mathcal{U}$ by Proposition 2.3.10, so $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) = \dim(\mathcal{W})$ by Proposition 1.7.8.

Perspective 2.3.12. *Quotients Split over Fields*
In finite dimensions, a complement $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ exists by Proposition 2.3.6, and $\mathcal{V}/\mathcal{U} \cong \mathcal{W}$ by Proposition 2.3.10.
Thus the quotient forgets the $\mathcal{U}$-component and retains the complementary coordinate.

Proposition 2.3.13. *Quotient of a Direct Sum*
If $\mathcal{V} = \mathcal{V}_1 \oplus \mathcal{V}_2$ and $\mathcal{U}_j \leq \mathcal{V}_j$ for $j = 1, 2$, then $\mathcal{V}/(\mathcal{U}_1 \oplus \mathcal{U}_2) \cong (\mathcal{V}_1/\mathcal{U}_1) \times (\mathcal{V}_2/\mathcal{U}_2)$.

Proof.
(i) *Denominator*: Since $\mathcal{U}_1 \cap \mathcal{U}_2 \subseteq \mathcal{V}_1 \cap \mathcal{V}_2 = \{\mathbf{0}\}$, the sum $\mathcal{U}_1 + \mathcal{U}_2$ is direct by Proposition 2.1.8.
(ii) *Construction*: For the unique decomposition $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$, define $\Phi : \mathcal{V} \to (\mathcal{V}_1/\mathcal{U}_1) \times (\mathcal{V}_2/\mathcal{U}_2)$ by $\Phi(\mathbf{v}_1 + \mathbf{v}_2) := (\mathbf{v}_1 + \mathcal{U}_1, \mathbf{v}_2 + \mathcal{U}_2)$. Componentwise quotient operations show that Φ is linear by Proposition 2.2.6.
(iii) *Range and kernel*: Every pair $(\mathbf{x}_1 + \mathcal{U}_1, \mathbf{x}_2 + \mathcal{U}_2)$ equals $\Phi(\mathbf{x}_1 + \mathbf{x}_2)$, so Φ is surjective.
Moreover, $\Phi(\mathbf{v}_1 + \mathbf{v}_2) = (\mathcal{U}_1, \mathcal{U}_2)$ iff $\mathbf{v}_j \in \mathcal{U}_j$ for $j = 1, 2$ by Proposition 2.2.2. Hence $\ker(\Phi) = \mathcal{U}_1 \oplus \mathcal{U}_2$.
(iv) *Conclusion*: The claimed isomorphism holds by Theorem 2.2.12, because $\operatorname{range}(\Phi)$ is the whole product.
Structural Commentary: Direct-sum coordinates define the map; its componentwise kernel identifies the quotient.
Key Insight: Quotienting a direct sum acts coordinatewise. □

Perspective 2.3.14. *Quotients Depend on the Embedding*
A complement realizes a quotient, but the result depends on how the subspace is embedded.
Let $\mathcal{V}$ have countable basis $(\mathbf{e}_n)_{n\geq 1}$ and set $\mathcal{S} := \operatorname{span}(\mathbf{e}_n : n \geq 2)$.
The forward shift $F : \mathcal{V} \to \mathcal{S}$ defined by $F(\mathbf{e}_n) = \mathbf{e}_{n+1}$ is an isomorphism: unique finite basis expansions make it linear, and the backward shift $\mathbf{e}_{n+1} \mapsto \mathbf{e}_n$ is its inverse. Thus $\mathcal{S} \cong \mathcal{V}$.
Nevertheless, $\mathcal{V} = \mathcal{V} \oplus \{\mathbf{0}\}$ and $\mathcal{V} = \mathcal{S} \oplus \mathbb{F}\mathbf{e}_1$, so $\mathcal{V}/\mathcal{V} \cong \{\mathbf{0}\}$ while $\mathcal{V}/\mathcal{S} \cong \mathbb{F}\mathbf{e}_1$ by Proposition 2.3.10.
Hence isomorphic subspaces can have nonisomorphic quotients when embedded differently.

2.4 Algebraic Projections and Resolutions

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$. Projections encode direct-sum coordinates as operators, while resolutions package finite decompositions into mutually annihilating idempotents.

Definition 2.4.1. *Projection Along a Complement*
(i) An operator $P \in \mathcal{L}(\mathcal{V})$ is an *algebraic projection*, or *idempotent operator*, iff $P^2 = P$.
(ii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, the *projection onto $\mathcal{U}$ along $\mathcal{W}$* is the map $\rho_{\mathcal{U},\mathcal{W}} : \mathcal{V} \to \mathcal{V}$ defined by $(\forall\, (\mathbf{u}, \mathbf{w}) \in \mathcal{U} \times \mathcal{W})\, \rho_{\mathcal{U},\mathcal{W}}(\mathbf{u} + \mathbf{w}) := \mathbf{u}$.

Proposition 2.4.2. *Algebraic Projections and Direct-Sum Splittings*
The following hold.
(i) For $P \in \mathcal{L}(\mathcal{V})$, $P^2 = P$ iff $\mathcal{V} = \operatorname{range}(P) \oplus \ker(P)$ and $P = \rho_{\operatorname{range}(P),\ker(P)}$.
(ii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then $\rho_{\mathcal{U},\mathcal{W}} \in \mathcal{L}(\mathcal{V})$ and $\rho_{\mathcal{U},\mathcal{W}}^2 = \rho_{\mathcal{U},\mathcal{W}}$.
(iii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then $\operatorname{range}(\rho_{\mathcal{U},\mathcal{W}}) = \mathcal{U}$ and $\ker(\rho_{\mathcal{U},\mathcal{W}}) = \mathcal{W}$.

Proof.
(i) ($\Rightarrow$) Suppose $P^2 = P$. For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$, where $P\mathbf{v} \in \operatorname{range}(P)$ and $\mathbf{v} - P\mathbf{v} \in \ker(P)$.
If $\mathbf{x} \in \operatorname{range}(P) \cap \ker(P)$, choose $\mathbf{y} \in \mathcal{V}$ with $\mathbf{x} = P\mathbf{y}$. Then $\mathbf{x} = P^2\mathbf{y} = P\mathbf{x} = \mathbf{0}$, so the sum is direct by Proposition 2.1.8.
The displayed decomposition satisfies $(\forall\, \mathbf{v} \in \mathcal{V})\, \rho_{\operatorname{range}(P),\ker(P)}\mathbf{v} = P\mathbf{v}$; hence $P = \rho_{\operatorname{range}(P),\ker(P)}$.
(i) ($\Leftarrow$) Suppose $\mathcal{V} = \operatorname{range}(P) \oplus \ker(P)$ and $P = \rho_{\operatorname{range}(P),\ker(P)}$.
If $\mathbf{v} = \mathbf{x} + \mathbf{y}$ with $\mathbf{x} \in \operatorname{range}(P)$ and $\mathbf{y} \in \ker(P)$, then $P^2\mathbf{v} = P\mathbf{x} = \mathbf{x} = P\mathbf{v}$; hence $P^2 = P$.
(ii) Set $R := \rho_{\mathcal{U},\mathcal{W}}$ and write $\mathbf{v}_j = \mathbf{u}_j + \mathbf{w}_j$ with $(\mathbf{u}_j, \mathbf{w}_j) \in \mathcal{U} \times \mathcal{W}$ for $j = 1, 2$.
For $a, b \in \mathbb{F}$, $R(a\mathbf{v}_1 + b\mathbf{v}_2) = a\mathbf{u}_1 + b\mathbf{u}_2 = aR\mathbf{v}_1 + bR\mathbf{v}_2$ by uniqueness of direct-sum decompositions, so R is linear.
Also $R^2(\mathbf{u} + \mathbf{w}) = R\mathbf{u} = \mathbf{u} = R(\mathbf{u} + \mathbf{w})$, so R is idempotent.
(iii) The inclusion $\operatorname{range}(\rho_{\mathcal{U},\mathcal{W}}) \subseteq \mathcal{U}$ holds by the defining formula, while $\rho_{\mathcal{U},\mathcal{W}}\mathbf{u} = \mathbf{u}$ for $\mathbf{u} \in \mathcal{U}$; hence the range is $\mathcal{U}$.
For $\mathbf{v} = \mathbf{u} + \mathbf{w}$, $\rho_{\mathcal{U},\mathcal{W}}\mathbf{v} = \mathbf{0}$ iff $\mathbf{u} = \mathbf{0}$ iff $\mathbf{v} \in \mathcal{W}$, so the kernel is $\mathcal{W}$.
Structural Commentary: Decomposition handles characterization; coordinates verify linearity, idempotence, range, and kernel.
Key Insight: Every idempotent keeps its range and kills its kernel. □

Proposition 2.4.3. *Complementary Algebraic Projections*
If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then the following hold.
(i) $\rho_{\mathcal{U},\mathcal{W}} + \rho_{\mathcal{W},\mathcal{U}} = I_{\mathcal{V}}$.
(ii) $\rho_{\mathcal{U},\mathcal{W}}\rho_{\mathcal{W},\mathcal{U}} = \rho_{\mathcal{W},\mathcal{U}}\rho_{\mathcal{U},\mathcal{W}} = 0$.

Proof. Let $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $(\mathbf{u}, \mathbf{w}) \in \mathcal{U} \times \mathcal{W}$.
(i) $(\rho_{\mathcal{U},\mathcal{W}} + \rho_{\mathcal{W},\mathcal{U}})\mathbf{v} = \mathbf{u} + \mathbf{w} = \mathbf{v}$. Since this equality holds on $\mathcal{V}$, the sum is $I_{\mathcal{V}}$.
(ii) $\rho_{\mathcal{U},\mathcal{W}}\rho_{\mathcal{W},\mathcal{U}}\mathbf{v} = \rho_{\mathcal{U},\mathcal{W}}\mathbf{w} = \mathbf{0}$ and $\rho_{\mathcal{W},\mathcal{U}}\rho_{\mathcal{U},\mathcal{W}}\mathbf{v} = \rho_{\mathcal{W},\mathcal{U}}\mathbf{u} = \mathbf{0}$. Since both equalities hold on $\mathcal{V}$, both products are zero.
Structural Commentary: The two coordinate projections are evaluated on the same unique decomposition.
Key Insight: Complementary projections partition identity and annihilate each other. □

Definition 2.4.4. *Algebraic Orthogonality and Resolutions*
(i) Algebraic projections $\rho, \sigma \in \mathcal{L}(\mathcal{V})$ are *algebraically orthogonal*, written $\rho \perp \sigma$, iff $\rho\sigma = \sigma\rho = 0$.
(ii) A *resolution of identity* on $\mathcal{V}$ is a finite family of algebraic projections $(\rho_i)_{i=1}^k$ such that $k \in \mathbb{N}$,

$$(\forall\, i, j \in \{1, \ldots, k\})(i \neq j \Rightarrow \rho_i \perp \rho_j), \text{ and } \sum_{i=1}^{k} \rho_i = I_{\mathcal{V}}.$$

Theorem 2.4.5. *Resolution Ranges Form a Direct Sum*
Let $(\rho_i)_{i=1}^k$ be a resolution of identity on $\mathcal{V}$, and set $\mathcal{R}_i := \operatorname{range}(\rho_i)$ for $i \in \{1,\dots,k\}$. Then the following hold.

(i) $\mathcal{V} = \bigoplus_{i=1}^{k} \mathcal{R}_i$.

(ii) $(\forall\, i \in \{1,\dots,k\})\ \ker(\rho_i) = \bigoplus_{j\in\{1,\dots,k\}\setminus\{i\}} \mathcal{R}_j$.

Proof.

(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = \sum_{i=1}^{k} \rho_i \mathbf{v} \in \sum_{i=1}^{k} \mathcal{R}_i$. Suppose $\sum_{i=1}^{k} \mathbf{x}_i = \mathbf{0}$ with $\mathbf{x}_i \in \mathcal{R}_i$.
For $m \in \{1,\dots,k\}$, idempotence and algebraic orthogonality imply $\rho_m \mathbf{x}_m = \mathbf{x}_m$ and $(\forall\, i \in \{1,\dots,k\} \setminus \{m\})\ \rho_m \mathbf{x}_i = \mathbf{0}$.
Hence $\mathbf{x}_m = \rho_m \sum_{i=1}^{k} \mathbf{x}_i = \mathbf{0}$. The sum is direct by Proposition 2.1.23.

(ii) By part (i), write $\mathbf{v} = \sum_{j=1}^{k} \mathbf{x}_j$ uniquely with $\mathbf{x}_j \in \mathcal{R}_j$. The equality $\rho_i \mathbf{v} = \mathbf{x}_i$ holds by idempotence and algebraic orthogonality.
Hence $\mathbf{v} \in \ker(\rho_i)$ iff $\mathbf{x}_i = \mathbf{0}$. These are exactly the vectors in $\bigoplus_{j\in\{1,\dots,k\}\setminus\{i\}} \mathcal{R}_j$.

Structural Commentary: The identity establishes spanning; orthogonal idempotents isolate each component and identify every kernel.
Key Insight: Mutually annihilating projections recover unique direct-sum coordinates. □

Theorem 2.4.6. *Direct Sums Give Resolutions*
Let $k \in \mathbb{N}$, set $K := \{1,\dots,k\}$, and suppose $\mathcal{V} = \bigoplus_{i\in K} \mathcal{U}_i$. For $i \in K$, set $\rho_i := \rho_{\mathcal{U}_i, \sum_{j\in K\setminus\{i\}} \mathcal{U}_j}$. Then $(\rho_i)_{i\in K}$ is a resolution of identity.

Proof. For $i \in K$, set $\mathcal{W}_i := \sum_{j\in K\setminus\{i\}} \mathcal{U}_j$. Then $\mathcal{U}_i \cap \mathcal{W}_i = \{\mathbf{0}\}$ by Proposition 2.1.23(iii), while $\mathcal{V} = \mathcal{U}_i + \mathcal{W}_i$ by the assumed sum.
Hence $\mathcal{V} = \mathcal{U}_i \oplus \mathcal{W}_i$ by Proposition 2.1.8, and ρ_i is an algebraic projection by Proposition 2.4.2(ii).
For the unique decomposition $\mathbf{v} = \sum_{i\in K} \mathbf{u}_i$, $\rho_i \mathbf{v} = \mathbf{u}_i$; hence $(\sum_{i\in K} \rho_i)\mathbf{v} = \mathbf{v}$. This pointwise identity proves $\sum_{i\in K} \rho_i = I_{\mathcal{V}}$.
If $i \neq j$, then $\rho_i \rho_j \mathbf{v} = \rho_i \mathbf{u}_j = \mathbf{0}$ and $\rho_j \rho_i \mathbf{v} = \rho_j \mathbf{u}_i = \mathbf{0}$. Thus $\rho_i \perp \rho_j$, and the family is a resolution of identity.
Structural Commentary: Direct-sum uniqueness defines coordinate projections; pointwise identities verify the resolution conditions.
Key Insight: Coordinate projections isolate summands and reconstruct every vector. □

Theorem 2.4.7. *Algebra of Projections*
Let $\rho, \sigma \in \mathcal{L}(\mathcal{V})$ be algebraic projections, and assume $2 \neq 0$ in $\mathbb{F}$.

(i) The operator $\rho + \sigma$ is an algebraic projection iff $\rho \perp \sigma$.
Under these conditions, $\operatorname{range}(\rho+\sigma) = \operatorname{range}(\rho) \oplus \operatorname{range}(\sigma)$ *and* $\ker(\rho+\sigma) = \ker(\rho) \cap \ker(\sigma)$.

(ii) The operator $\rho - \sigma$ is an algebraic projection iff $\rho\sigma = \sigma\rho = \sigma$.
Under these conditions, $\operatorname{range}(\rho-\sigma) = \operatorname{range}(\rho) \cap \ker(\sigma)$ *and* $\ker(\rho-\sigma) = \ker(\rho) \oplus \operatorname{range}(\sigma)$.

(iii) If $\rho\sigma = \sigma\rho$, then $\rho\sigma$ is an algebraic projection, $\operatorname{range}(\rho\sigma) = \operatorname{range}(\rho) \cap \operatorname{range}(\sigma)$, *and* $\ker(\rho\sigma) = \ker(\rho) + \ker(\sigma)$.

Proof.

(i) ($\Rightarrow$) Set $A := \rho\sigma$, $B := \sigma\rho$, and $C := \rho\sigma\rho$. The identity $(\rho+\sigma)^2 = \rho+\sigma$ is equivalent to $A + B = 0$.
Then $A + C = \rho(A+B) = 0$ and $C + B = (A+B)\rho = 0$, so $A = B$ and $2A = 0$.
The scalar 2 is invertible by $2 \neq 0$ in the field $\mathbb{F}$; hence $A = B = 0$ and $\rho \perp \sigma$.

(i) ($\Leftarrow$) Suppose $\rho \perp \sigma$. Then $(\rho+\sigma)^2 = \rho+\sigma$, so the sum is an algebraic projection.
Its range is contained in $\operatorname{range}(\rho) + \operatorname{range}(\sigma)$, while $(\rho+\sigma)(\mathbf{x}+\mathbf{y}) = \mathbf{x}+\mathbf{y}$ for $(\mathbf{x},\mathbf{y}) \in \operatorname{range}(\rho) \times \operatorname{range}(\sigma)$.
If $\mathbf{z} \in \operatorname{range}(\rho) \cap \operatorname{range}(\sigma)$ and $\mathbf{z} = \sigma\mathbf{w}$, then $\mathbf{z} = \rho\mathbf{z} = \rho\sigma\mathbf{w} = \mathbf{0}$; hence the range sum is direct by Proposition 2.1.8.
$(\rho+\sigma)\mathbf{v} = \mathbf{0} \Rightarrow \rho\mathbf{v} = \rho(\rho+\sigma)\mathbf{v} = \mathbf{0}$ and $\sigma\mathbf{v} = \sigma(\rho+\sigma)\mathbf{v} = \mathbf{0}$. Reverse implication is immediate. Thus $\ker(\rho+\sigma) = \ker(\rho) \cap \ker(\sigma)$.

(ii) ($\Rightarrow$) Suppose $Q := \rho - \sigma$ is algebraic projection. Then $(I_{\mathcal{V}} - Q)^2 = I_{\mathcal{V}} - Q$ and $(I_{\mathcal{V}} - \rho)^2 = I_{\mathcal{V}} - \rho$ hold by $Q^2 = Q$ and $\rho^2 = \rho$.
Since $I_{\mathcal{V}} - Q = (I_{\mathcal{V}} - \rho) + \sigma$, $(I_{\mathcal{V}} - \rho) \perp \sigma$ by part (i). Therefore $(I_{\mathcal{V}} - \rho)\sigma = \sigma(I_{\mathcal{V}} - \rho) = 0$, which is equivalent to $\rho\sigma = \sigma\rho = \sigma$.

(ii) ($\Leftarrow$) Suppose $\rho\sigma = \sigma\rho = \sigma$ and set $Q := \rho - \sigma$. The operator $I_{\mathcal{V}} - \rho$ is an algebraic projection and is algebraically orthogonal to σ, so $I_{\mathcal{V}} - Q = (I_{\mathcal{V}} - \rho) + \sigma$ is an algebraic projection by part (i). Since $I_{\mathcal{V}} - Q$ is idempotent, $Q^2 = (I_{\mathcal{V}} - (I_{\mathcal{V}} - Q))^2 = I_{\mathcal{V}} - (I_{\mathcal{V}} - Q) = Q$.
Since $\rho Q = Q$ and $\sigma Q = 0$, $\operatorname{range}(Q) \subseteq \operatorname{range}(\rho) \cap \ker(\sigma)$; the reverse inclusion holds because $Q\mathbf{x} = \mathbf{x}$ when $\rho\mathbf{x} = \mathbf{x}$ and $\sigma\mathbf{x} = \mathbf{0}$.
The operator Q kills both $\ker(\rho)$ and $\operatorname{range}(\sigma)$. Their intersection is trivial: if $\mathbf{z} = \sigma\mathbf{w} \in \ker(\rho)$, then $\mathbf{z} = \rho\sigma\mathbf{w} = \rho\mathbf{z} = \mathbf{0}$.
Their sum is direct by Proposition 2.1.8, and $\ker(\rho) \oplus \operatorname{range}(\sigma) \subseteq \ker(Q)$ because Q kills both summands.
Conversely, if $\mathbf{v} \in \ker(Q)$, then $\rho\mathbf{v} = \sigma\mathbf{v}$ and $\mathbf{v} = (\mathbf{v} - \rho\mathbf{v}) + \sigma\mathbf{v} \in \ker(\rho) \oplus \operatorname{range}(\sigma)$. Hence $\ker(Q) = \ker(\rho) \oplus \operatorname{range}(\sigma)$.

(iii) Suppose $\rho\sigma = \sigma\rho$. Then $(\rho\sigma)^2 = \rho^2\sigma^2 = \rho\sigma$, so $\rho\sigma$ is an algebraic projection.
If $\mathbf{x} = \rho\sigma\mathbf{y}$, then $\rho\mathbf{x} = \mathbf{x} = \sigma\mathbf{x}$; conversely, $\rho\mathbf{x} = \mathbf{x} = \sigma\mathbf{x}$ implies $\mathbf{x} = \rho\sigma\mathbf{x}$. Thus $\operatorname{range}(\rho\sigma) = \operatorname{range}(\rho) \cap \operatorname{range}(\sigma)$.
If $\mathbf{x} \in \ker(\rho)$, then $\rho\sigma\mathbf{x} = \sigma\rho\mathbf{x} = \mathbf{0}$; if $\mathbf{x} \in \ker(\sigma)$, then $\rho\sigma\mathbf{x} = \mathbf{0}$. Hence $\ker(\rho) + \ker(\sigma) \subseteq \ker(\rho\sigma)$.
If $\rho\sigma\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = (\mathbf{v} - \sigma\mathbf{v}) + \sigma\mathbf{v}$ with $\mathbf{v} - \sigma\mathbf{v} \in \ker(\sigma)$ and $\sigma\mathbf{v} \in \ker(\rho)$; hence $\ker(\rho\sigma) = \ker(\rho) + \ker(\sigma)$.

Structural Commentary: Expanding idempotence isolates cross terms; fixed and killed subspaces determine each range and kernel.
Key Insight: Commuting idempotents encode intersections and sums. □

3 Matrix Representations, Change of Basis, Block Matrices, and Operations

3.1 Matrix Algebra and Linear Systems

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every size parameter is a positive integer.

Definition 3.1.1. *Matrices and Matrix Operations*

(i) *Matrix space and equality*: $\operatorname{Mat}(m,n,\mathbb{F}) := \{[A_{j,k}]_{1\le j\le m,\ 1\le k\le n} : A_{j,k} \in \mathbb{F}\}$.
For A, C in this space, $(A = C) \iff (\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,n\})\ A_{j,k} = C_{j,k}$.

(ii) *Entries, rows, and columns*: if $A = [A_{j,k}] \in \operatorname{Mat}(m,n,\mathbb{F})$, then $A_{j,k}$ is its (j,k)-entry, $A_{j,\cdot}$ is row j, and $A_{\cdot,k}$ is column k.

(iii) *Transpose*: for $A \in \operatorname{Mat}(m,n,\mathbb{F})$, the transpose $A^t \in \operatorname{Mat}(n,m,\mathbb{F})$ satisfies $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,n\})\ (A^t)_{k,j} := A_{j,k}$.

(iv) *Main diagonal and diagonal matrix*: if $A = [A_{j,k}] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $r = \min(m,n)$, its main diagonal is $(A_{1,1},\dots,A_{r,r})$. For $\lambda_1,\dots,\lambda_n \in \mathbb{F}$, $\mathrm{diag}(\lambda_1,\dots,\lambda_n) \in \mathrm{Mat}(n,n,\mathbb{F})$ has (j,j)-entry λ_j and all off-diagonal entries zero.
(v) *Symmetry*: a matrix $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is symmetric iff $A^t = A$, and skew-symmetric iff $A^t = -A$.
(vi) *Triangularity*: a matrix $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is upper triangular iff $(\forall\, j,k \in \{1,\dots,n\})(j > k \Rightarrow A_{j,k} = 0)$. Lower triangular iff $(\forall\, j,k \in \{1,\dots,n\})(j < k \Rightarrow A_{j,k} = 0)$. Diagonal iff $(\forall\, j,k \in \{1,\dots,n\})(j \neq k \Rightarrow A_{j,k} = 0)$.
(vii) *Linear operations*: for $A, C \in \mathrm{Mat}(m,n,\mathbb{F})$ and $a \in \mathbb{F}$, $(A+C)_{j,k} := A_{j,k} + C_{j,k}$ and $(aA)_{j,k} := aA_{j,k}$ for admissible j,k.
(viii) *Row-column pairing*: if $r_1,\dots,r_n,c_1,\dots,c_n \in \mathbb{F}$, $\mathbf{r} = (r_1,\dots,r_n)$, and $\mathbf{c} = (c_1,\dots,c_n)^t$, then $\mathbf{rc} := \sum_{s=1}^{n} r_s c_s$.
(ix) *Matrix product*: for $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(n,p,\mathbb{F})$, the product $AB \in \mathrm{Mat}(m,p,\mathbb{F})$ satisfies $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,p\})\,(AB)_{j,k} := \sum_{s=1}^{n} A_{j,s}B_{s,k}$.
(x) *Matrix-vector product*: identify $\mathbf{x} = (x_1,\dots,x_n)^t \in \mathbb{F}^n$ with its column matrix. For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, define $A\mathbf{x} \in \mathbb{F}^m$ by $(\forall\, j \in \{1,\dots,m\})\,(A\mathbf{x})_j := \sum_{k=1}^{n} A_{j,k}x_k$.
(xi) *Matrix units*: for $j \in \{1,\dots,m\}$ and $k \in \{1,\dots,n\}$, $E_{j,k} \in \mathrm{Mat}(m,n,\mathbb{F})$ has (j,k)-entry 1 and all other entries zero.
(xii) *Identity and inverse*: $I_n := \mathrm{diag}(1,\dots,1) = [\delta_{j,k}] \in \mathrm{Mat}(n,n,\mathbb{F})$, where $\delta_{j,k} = 1$ iff $j = k$. A matrix $B \in \mathrm{Mat}(n,n,\mathbb{F})$ is a two-sided inverse of $A \in \mathrm{Mat}(n,n,\mathbb{F})$ iff $AB = I_n = BA$; the matrix A is invertible iff it has a two-sided inverse.

Proposition 3.1.2. *Basic Matrix Product Laws*
Let $A, A' \in \mathrm{Mat}(m,n,\mathbb{F})$, $B, B' \in \mathrm{Mat}(n,p,\mathbb{F})$, $C \in \mathrm{Mat}(p,q,\mathbb{F})$, and $a \in \mathbb{F}$. Then:
(i) $(AB)C = A(BC)$;
(ii) $I_m A = A = AI_n$;
(iii) $(A+A')B = AB + A'B$ *and* $A(B+B') = AB + AB'$;
(iv) $(aA)B = a(AB) = A(aB)$.

Proof.
(i) For admissible j,k, $((AB)C)_{j,k} = \sum_{s=1}^{p}\sum_{r=1}^{n} A_{j,r}B_{r,s}C_{s,k} = \sum_{r=1}^{n}\sum_{s=1}^{p} A_{j,r}B_{r,s}C_{s,k} = (A(BC))_{j,k}$ by finite distributivity. Thus $(AB)C = A(BC)$ by entrywise equality.
(ii) For admissible j,k, $(I_m A)_{j,k} = \sum_{r=1}^{m} \delta_{j,r}A_{r,k} = A_{j,k} = \sum_{r=1}^{n} A_{j,r}\delta_{r,k} = (AI_n)_{j,k}$. Hence $I_m A = A = AI_n$.
(iii) For admissible j,k, finite distributivity implies $((A+A')B)_{j,k} = \sum_{r=1}^{n}(A_{j,r}+A'_{j,r})B_{r,k} = (AB+A'B)_{j,k}$. Also implies $(A(B+B'))_{j,k} = \sum_{r=1}^{n} A_{j,r}(B_{r,k}+B'_{r,k}) = (AB+AB')_{j,k}$. Entrywise equality proves both laws.
(iv) For admissible j,k, $((aA)B)_{j,k} = a\sum_{r=1}^{n} A_{j,r}B_{r,k} = (a(AB))_{j,k} = (A(aB))_{j,k}$; hence the matrices are equal.

Structural Commentary: Each law reduces to corresponding identities among finite scalar sums.
Key Insight: Entrywise equality transfers field arithmetic to compatible matrices. □

Lemma 3.1.3. *Uniqueness of Two-Sided Inverses*
Let $A, B, C \in \mathrm{Mat}(n,n,\mathbb{F})$. If $AB = I_n$ and $BC = I_n$, then $A = C$.

Proof. By Proposition 3.1.2 and the hypotheses, $A = AI_n = A(BC) = (AB)C = I_n C = C$.
Structural Commentary: A common middle inverse forces the left and right inverses to coincide.
Key Insight: Associativity turns $AB = I_n = BC$ into $A = C$. □

Definition 3.1.4. *Inverse Matrix Notation*
If $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, its unique two-sided inverse by Lemma 3.1.3 is denoted A^{-1}.
The set of invertible matrices in $\mathrm{Mat}(n,n,\mathbb{F})$ is denoted $\mathrm{Mat}(n,n,\mathbb{F})^{\times}$.

Proposition 3.1.5. *Matrix Spaces*
For $m,n \in \mathbb{Z}_{>0}$, the following hold.
(i) The entrywise operations make $\mathrm{Mat}(m,n,\mathbb{F})$ *a vector space over* $\mathbb{F}$.
(ii) The family $(E_{j,k})_{1\le j\le m,\ 1\le k\le n}$ *is a basis; hence* $\dim(\mathrm{Mat}(m,n,\mathbb{F})) = mn$.

Proof.
(i) The zero matrix and entrywise negatives supply the additive identity and inverses. Each remaining vector-space axiom holds in every entry by the corresponding field law, so it holds by matrix equality.
(ii) Every $A = [A_{j,k}]$ has the expansion $A = \sum_{j=1}^{m}\sum_{k=1}^{n} A_{j,k}E_{j,k}$. If $\sum_{j=1}^{m}\sum_{k=1}^{n} c_{j,k}E_{j,k} = 0$, then its (j,k)-entry is $c_{j,k} = 0$. Hence the family spans and is linearly independent. Its mn members form a basis, so the stated dimension is mn.

Structural Commentary: Entrywise field laws establish the space, and matrix units establish its basis.
Key Insight: Each matrix unit isolates one independent scalar coordinate. □

Proposition 3.1.6. *Matrix Product Entries Are Row Times Column*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(n,p,\mathbb{F})$, then $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,p\})\,(AB)_{j,k} = A_{j,\cdot}B_{\cdot,k}$.

Proof. Both sides equal $\sum_{r=1}^{n} A_{j,r}B_{r,k}$ by the definitions of matrix multiplication and row-column pairing.
Structural Commentary: Matrix multiplication is assembled entry-by-entry from row-column pairings.
Key Insight: Each entry of AB sees one row of A and one column of B. □

Definition 3.1.7. *Map Induced by a Matrix*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, the *map induced by A* is $T_A : \mathbb{F}^n \to \mathbb{F}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$.

Proposition 3.1.8. *Matrix-Induced Maps Are Linear*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$, then $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear.

Proof. For $\mathbf{x},\mathbf{y} \in \mathbb{F}^n$, $a,b \in \mathbb{F}$, and $j \in \{1,\dots,m\}$, $(T_A(a\mathbf{x}+b\mathbf{y}))_j = \sum_{k=1}^{n} A_{j,k}(ax_k + by_k) = a(T_A\mathbf{x})_j + b(T_A\mathbf{y})_j$.
Coordinatewise equality proves $T_A(a\mathbf{x}+b\mathbf{y}) = aT_A\mathbf{x} + bT_A\mathbf{y}$.
Structural Commentary: Matrix-vector multiplication preserves linear combinations.
Key Insight: Every matrix defines a canonical linear map between coordinate spaces. □

Proposition 3.1.9. *Matrix-Vector Products as Column Combinations*
If $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{x} = (x_1,\dots,x_n)^t \in \mathbb{F}^n$, then $A\mathbf{x} = \sum_{k=1}^{n} x_k\mathbf{a}_k$.

Proof. For $j \in \{1,\dots,m\}$, both vectors have jth entry $\sum_{k=1}^{n} A_{j,k}x_k$.
Structural Commentary: Multiplying by a column selects a linear combination of the columns of A.
Key Insight: The coordinates of $\mathbf{x}$ are precisely the column coefficients. □

Proposition 3.1.10. *Matrix Products by Columns*
If $A \in \mathrm{Mat}(m,r,\mathbb{F})$ and $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \in \mathrm{Mat}(r,n,\mathbb{F})$, then $AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_n]$.

Proof. For $j \in \{1,\dots,m\}$ and $k \in \{1,\dots,n\}$, the jth entries of the kth column of AB and of $A\mathbf{b}_k$ both equal $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k}$.
Structural Commentary: Matrix products apply A to columns of B.
Key Insight: The columns of AB are obtained by applying A to the columns of B. □

Proposition 3.1.11. *Matrix Products as Sums of Outer Products*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_r] \in \mathrm{Mat}(m,r,\mathbb{F})$ and $B \in \mathrm{Mat}(r,n,\mathbb{F})$. If the ℓth row of B is $\mathbf{b}_\ell^t$ with $\mathbf{b}_\ell \in \mathbb{F}^n$, then $AB = \sum_{\ell=1}^{r} \mathbf{a}_\ell\mathbf{b}_\ell^t$.

Proof. For admissible j,k, the (j,k)-entry of $\mathbf{a}_\ell\mathbf{b}_\ell^t$ is $A_{j,\ell}B_{\ell,k}$, so the corresponding entry of the stated sum is $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k} = (AB)_{j,k}$.
Structural Commentary: Column-row products decompose a matrix product into single column-row pieces.
Key Insight: Each shared index ℓ contributes one outer product. □

Definition 3.1.12. *Column Space, Matrix Rank, and Nullity*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$.
(i) $\mathrm{col}(A) := \mathrm{span}(\mathbf{a}_1,\dots,\mathbf{a}_n) \subseteq \mathbb{F}^m$, and $\mathrm{row}(A) := \mathrm{col}(A^t) \subseteq \mathbb{F}^n$.
(ii) $\ker(A) := \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{0}\}$ is the *nullspace*, and $\mathrm{nullity}(A) := \dim(\ker(A))$.
(iii) $\mathrm{rank}(A) := \dim(\mathrm{col}(A))$ is the column rank, while $\dim(\mathrm{row}(A))$ is the row rank.
(iv) The matrix A has *full rank* iff $\mathrm{rank}(A) = \min(m,n)$, *full column rank* iff $\mathrm{rank}(A) = n$, and *full row rank* iff $\mathrm{rank}(A) = m$.
$\mathrm{range}(T_A) = \mathrm{col}(A)$ and $\mathrm{range}(T_{A^t}) = \mathrm{row}(A)$ by Proposition 3.1.9, while $\ker(T_A) = \ker(A)$ by definition.
These three sets are subspaces by Proposition 3.1.8 and Proposition 1.6.3, so the stated dimensions are defined.

Proposition 3.1.13. *Solvability and Complete Solution Sets*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{b} \in \mathbb{F}^m$. Then:
(i) The system $A\mathbf{x} = \mathbf{b}$ with $\mathbf{x} \in \mathbb{F}^n$ is solvable iff $\mathbf{b} \in \mathrm{col}(A)$.
(ii) If $A\mathbf{x}_p = \mathbf{b}$, its solution set is $\mathbf{x}_p + \ker(A) = \{\mathbf{x}_p + \mathbf{x}_0 : \mathbf{x}_0 \in \ker(A)\}$.

Proof.

(i) ($\Rightarrow$) If $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x} = (x_1,\dots,x_n)^t$, then $\mathbf{b} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j \in \mathrm{col}(A)$ by Proposition 3.1.9.

(i) ($\Leftarrow$) If $\mathbf{b} \in \mathrm{col}(A)$, then $(\exists\, x_1,\dots,x_n \in \mathbb{F})\, \mathbf{b} = \sum_{j=1}^{n} x_j\mathbf{a}_j$. For $\mathbf{x} = (x_1,\dots,x_n)^t$, $A\mathbf{x} = \mathbf{b}$ by Proposition 3.1.9.

(ii) If $A\mathbf{x}_p = \mathbf{b}$, then $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = \mathbf{b} \iff A(\mathbf{x}-\mathbf{x}_p) = \mathbf{0} \iff \mathbf{x}-\mathbf{x}_p \in \ker(A) \iff \mathbf{x} \in \mathbf{x}_p + \ker(A)$ by Proposition 3.1.8.

Structural Commentary: Column expansion proves existence, while subtraction identifies the complete affine fiber.
Key Insight: Column space decides existence; kernel decides nonuniqueness. □

Proposition 3.1.14. *Elementary Transpose Rules*
Let $A, C \in \mathrm{Mat}(m,n,\mathbb{F})$ and $a \in \mathbb{F}$. Then:
(i) $(A^t)^t = A$;
(ii) $(A+C)^t = A^t + C^t$ and $(aA)^t = aA^t$;
(iii) $I_n^t = I_n$.

Proof.

(i) For admissible j,k, $((A^t)^t)_{j,k} = (A^t)_{k,j} = A_{j,k}$.
(ii) For admissible j,k, $((A+C)^t)_{j,k} = A_{k,j} + C_{k,j} = (A^t + C^t)_{j,k}$ and $((aA)^t)_{j,k} = aA_{k,j} = (aA^t)_{j,k}$.
(iii) For $j,k \in \{1,\dots,n\}$, $(I_n^t)_{j,k} = \delta_{k,j} = \delta_{j,k} = (I_n)_{j,k}$.

Structural Commentary: Each identity follows by swapping the two entry indices.
Key Insight: Transposition preserves entrywise operations and reverses index order. □

Example 3.1.15. *Symmetric and Skew-Symmetric Decomposition*
Assume $2 \neq 0$ in $\mathbb{F}$ and set $M := \mathrm{Mat}(n, n, \mathbb{F})$. Define $\mathrm{Sym}_n(\mathbb{F}) := \{A \in M : A^t = A\}$ and $\mathrm{Skew}_n(\mathbb{F}) := \{A \in M : A^t = -A\}$.
The map $\tau : M \to M$, $\tau(A) := A^t$, is linear by Proposition 3.1.14(ii). Thus $\tau - I_M, \tau + I_M \in \mathcal{L}(M)$ by Proposition 1.5.5.
Since $\mathrm{Sym}_n(\mathbb{F}) = \ker(\tau - I_M)$ and $\mathrm{Skew}_n(\mathbb{F}) = \ker(\tau + I_M)$, both are subspaces of M by Proposition 1.6.3(i).
For $A \in M$, set $S_A := \frac{1}{2}(A + A^t)$ and $K_A := \frac{1}{2}(A - A^t)$. Then $S_A^t = S_A$, $K_A^t = -K_A$, and $A = S_A + K_A$ by Proposition 3.1.14.
Hence $M = \mathrm{Sym}_n(\mathbb{F}) + \mathrm{Skew}_n(\mathbb{F})$. If $A \in \mathrm{Sym}_n(\mathbb{F}) \cap \mathrm{Skew}_n(\mathbb{F})$, then $A = A^t = -A$ and $2A = 0$, whence $A = 0$ because $2 \neq 0$ is invertible in $\mathbb{F}$. Therefore $M = \mathrm{Sym}_n(\mathbb{F}) \oplus \mathrm{Skew}_n(\mathbb{F})$ by Proposition 2.1.8.
Structural Commentary: Transpose splits a matrix into its even and odd parts.
Key Insight: The directness of the split uses the ability to divide by 2.

Proposition 3.1.16. *Transpose Reverses Products*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(n, p, \mathbb{F})$, then $(AB)^t = B^t A^t$.

Proof. For $j \in \{1, \ldots, p\}$ and $k \in \{1, \ldots, m\}$, $((AB)^t)_{j,k} = (AB)_{k,j} = \sum_{\ell=1}^{n} A_{k,\ell} B_{\ell,j}$.
The same sum equals $\sum_{\ell=1}^{n} (B^t)_{j,\ell} (A^t)_{\ell,k} = (B^t A^t)_{j,k}$, so the matrices are equal.
Structural Commentary: Transposition swaps the order of composition encoded by matrix multiplication.
Key Insight: Rows become columns, so the product order reverses. □

Corollary 3.1.17. *Matrix Inverses and Transposes Reverse Products*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be invertible. Then:
(i) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$;
(ii) A^t is invertible and $(A^t)^{-1} = (A^{-1})^t$.

Proof.
(i) For $D := B^{-1}A^{-1}$, $D(AB) = I_n = (AB)D$ by Proposition 3.1.2 and the inverse equations; hence $(AB)^{-1} = D$.
(ii) For $C := (A^{-1})^t$, $CA^t = I_n = A^t C$ by Proposition 3.1.16 and Proposition 3.1.14(iii); hence $(A^t)^{-1} = C$.
Structural Commentary: Each claim verifies the two inverse equations after reversing multiplication order.
Key Insight: Both inversion and transposition reverse products. □

3.2 Coordinates and Matrix Representations

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 3.2.1. *Coordinate Vector*
Let $n \geq 1$, let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of a vector space $\mathcal{V}$, and let $\mathbf{v} \in \mathcal{V}$. By Proposition 1.3.3, $(\exists!\, (c_1, \ldots, c_n) \in \mathbb{F}^n)\, \mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{v}_j$.
For this tuple, the *β-coordinate vector* of $\mathbf{v}$ is $[\mathbf{v}]_\beta := (c_1, \ldots, c_n)^t \in \mathbb{F}^n$.
The function $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ is the *β-coordinate map*. The order of the list β fixes the order of the coordinates.

Proposition 3.2.2. *Coordinate Maps Are Isomorphisms*
Let $n \geq 1$, and let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of a vector space $\mathcal{V}$. The coordinate map $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ has the following properties:
(i) It is linear.
(ii) It is injective.
(iii) It is surjective and hence an isomorphism.

Proof.
(i) Let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $c, d \in \mathbb{F}$, and write $[\mathbf{u}]_\beta = (a_1, \ldots, a_n)^t$ and $[\mathbf{v}]_\beta = (b_1, \ldots, b_n)^t$.
Then $[c\mathbf{u} + d\mathbf{v}]_\beta = (ca_1 + db_1, \ldots, ca_n + db_n)^t = c[\mathbf{u}]_\beta + d[\mathbf{v}]_\beta$, so $[\cdot]_\beta$ is linear.
(ii) If $[\mathbf{u}]_\beta = [\mathbf{v}]_\beta$, then $\mathbf{u}$ and $\mathbf{v}$ have the same coefficients in β, so $\mathbf{u} = \mathbf{v}$ by Proposition 1.3.3. Thus $[\cdot]_\beta$ is injective.
(iii) For $(c_1, \ldots, c_n)^t \in \mathbb{F}^n$, set $\mathbf{v} := \sum_{j=1}^{n} c_j \mathbf{v}_j$. Then $[\mathbf{v}]_\beta = (c_1, \ldots, c_n)^t$, so $[\cdot]_\beta$ is surjective.
It is therefore an isomorphism by Proposition 1.6.7.
Structural Commentary: The three checks compare vectors with their unique coefficient tuples.
Key Insight: An ordered basis identifies the vector space with its coordinate space without losing linear structure. □

Definition 3.2.3. *Matrix Representation*
Let $n, m \geq 1$, let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be a basis of $\mathcal{W}$.
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *matrix of T from β-coordinates to γ-coordinates* is $[T]_{\gamma \leftarrow \beta} := \big[[T(\mathbf{v}_1)]_\gamma \ \cdots \ [T(\mathbf{v}_n)]_\gamma\big] \in \mathrm{Mat}(m, n, \mathbb{F})$.
Thus its jth column is the uniquely determined coordinate vector $[T(\mathbf{v}_j)]_\gamma$.

Proposition 3.2.4. *Matrices Respect Linear Combinations of Maps*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be a basis of $\mathcal{W}$.
If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then $[aS + T]_{\gamma \leftarrow \beta} = a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.

Proof. By Proposition 1.5.5, $aS + T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. For $j \in \{1, \ldots, n\}$, $[(aS + T)\mathbf{v}_j]_\gamma = [aS\mathbf{v}_j + T\mathbf{v}_j]_\gamma = a[S\mathbf{v}_j]_\gamma + [T\mathbf{v}_j]_\gamma$ by Proposition 3.2.2(i). These are the respective jth columns of the two matrices, so the matrices are equal.
Structural Commentary: The proof compares the coordinate images of each domain basis vector.
Key Insight: Taking a matrix representation preserves the vector-space operations on linear maps. □

Proposition 3.2.5. *Matrix Representation Acts on Coordinates*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be a basis of $\mathcal{W}$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\, [T(\mathbf{v})]_\gamma = [T]_{\gamma \leftarrow \beta} [\mathbf{v}]_\beta$.

Proof. Let $\mathbf{v} \in \mathcal{V}$, and write $[\mathbf{v}]_\beta = (a_1, \ldots, a_n)^t$, so $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$. By linearity of T and Proposition 3.2.2(i), $[T(\mathbf{v})]_\gamma = \sum_{j=1}^{n} a_j [T(\mathbf{v}_j)]_\gamma$.
The summands are the columns of $[T]_{\gamma \leftarrow \beta}$, and this sum equals $[T]_{\gamma \leftarrow \beta} [\mathbf{v}]_\beta$ by Proposition 3.1.9.
Structural Commentary: The proof expands a vector in the domain basis and tracks that expansion through T.
Key Insight: Applying a linear map becomes matrix multiplication after both bases are fixed. □

3.3 Matrix-Induced Maps and Rank

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Corollary 3.3.1. *Matrix Rank-Nullity*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:
(i) $\dim(\ker(A)) + \dim(\mathrm{col}(A)) = n$; *equivalently,* $\mathrm{nullity}(A) + \mathrm{rank}(A) = n$.
(ii) If $m = n$, then $\ker(A) = \{\mathbf{0}\}$ iff $\mathrm{col}(A) = \mathbb{F}^n$.

Proof.
(i) Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. The map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear by Proposition 3.1.8.
For $\mathbf{x} = (x_1, \ldots, x_n)^t$, $T_A\mathbf{x} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j$ by Proposition 3.1.9.
Hence $\mathrm{range}(T_A) = \mathrm{col}(A)$, while $\ker(T_A) = \ker(A)$ holds by $T_A\mathbf{x} = A\mathbf{x}$; $\dim(\ker(A)) + \dim(\mathrm{col}(A)) = n$ by Theorem 1.7.3(ii).
The equivalent identity is the same equality under $\mathrm{nullity}(A) = \dim(\ker(A))$ and $\mathrm{rank}(A) = \dim(\mathrm{col}(A))$.
(ii) ($\Rightarrow$) Suppose $m = n$ and $\ker(A) = \{\mathbf{0}\}$. Then $\dim(\mathrm{col}(A)) = n = \dim(\mathbb{F}^n)$ by part (i).
Since $\mathrm{col}(A) \subseteq \mathbb{F}^n$, $\mathrm{col}(A) = \mathbb{F}^n$ by Proposition 1.4.3(ii).
(ii) ($\Leftarrow$) Suppose $m = n$ and $\mathrm{col}(A) = \mathbb{F}^n$. Then $\dim(\ker(A)) = 0$ by part (i).
The zero subspace and $\ker(A)$ have the same dimension, so $\ker(A) = \{\mathbf{0}\}$ by Proposition 1.4.3(ii).
Structural Commentary: Rank-nullity transfers through the induced map; the square case compares full-dimensional subspaces.
Key Insight: A matrix inherits rank-nullity from its coordinate-space map. □

Theorem 3.3.2. *Row Rank Equals Column Rank*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $\dim(\mathrm{row}(A)) = \dim(\mathrm{col}(A))$. Equivalently, $\mathrm{rank}(A^t) = \mathrm{rank}(A)$.

Proof. Let $C = [\mathbf{c}_1 \ \cdots \ \mathbf{c}_q] \in \mathrm{Mat}(p,q,\mathbb{F})$ be arbitrary, and set $s := \mathrm{rank}(C)$.
If $s = 0$, then $\dim(\mathrm{col}(C)) = 0 = \dim(\{\mathbf{0}\})$, so $\mathrm{col}(C) = \{\mathbf{0}\}$ by Proposition 1.4.3(ii). Hence $C = 0$ and $\mathrm{rank}(C^t) = 0$.
Suppose $s > 0$. A basis sublist $(\mathbf{c}_{j_1}, \ldots, \mathbf{c}_{j_s})$ of the columns exists by Theorem 1.3.12; set $B := [\mathbf{c}_{j_1} \ \cdots \ \mathbf{c}_{j_s}] \in \mathrm{Mat}(p,s,\mathbb{F})$.
The relation $(\forall\, k \in \{1,\ldots,q\})(\exists!\, \mathbf{x}_k \in \mathbb{F}^s)\, \mathbf{c}_k = B\mathbf{x}_k$ holds by Proposition 1.3.3.
Set $X := [\mathbf{x}_1 \ \cdots \ \mathbf{x}_q] \in \mathrm{Mat}(s,q,\mathbb{F})$. Then $C = BX$ by Proposition 3.1.10, and $C^t = X^tB^t$ by Proposition 3.1.16.
Each column of C^t lies in the span of the s columns of X^t by Proposition 3.1.10 and Proposition 3.1.9.
Hence $\mathrm{col}(C^t)$ has a basis of at most s members by Theorem 1.3.12, and $\mathrm{rank}(C^t) \le s = \mathrm{rank}(C)$.
Thus $\mathrm{rank}(C^t) \le \mathrm{rank}(C)$ for the arbitrary matrix C above. For $C = A$, $\mathrm{rank}(A^t) \le \mathrm{rank}(A)$.
For $C = A^t$, $\mathrm{rank}(A) \le \mathrm{rank}(A^t)$ because $(A^t)^t = A$ by Proposition 3.1.14(i).
Structural Commentary: A column-basis factorization proves one rank inequality; transposition supplies the reverse inequality.
Key Insight: Transposition preserves the number of independent matrix directions. □

Perspective 3.3.3. *Row and Column Spaces Have Different Hosts*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $\mathrm{row}(A) = \mathrm{col}(A^t) \subseteq \mathbb{F}^n$ and $\mathrm{col}(A) \subseteq \mathbb{F}^m$.
When $m \neq n$, Theorem 3.3.2 identifies their dimensions, not the subspaces in their different ambient spaces.

Theorem 3.3.4. *Four Fundamental Matrix Subspaces*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and let $r := \mathrm{rank}(A)$. Then:
(i) $\mathrm{row}(A), \ker(A) \subseteq \mathbb{F}^n$ *and* $\mathrm{col}(A), \ker(A^t) \subseteq \mathbb{F}^m$.
(ii) $\dim(\mathrm{col}(A)) = \dim(\mathrm{row}(A)) = r$, $\dim(\ker(A)) = n - r$, *and* $\dim(\ker(A^t)) = m - r$.

Proof.
(i) The row and column inclusions hold by their entry lengths, while the kernel inclusions hold by the domains of T_A and T_{A^t}.
(ii) The equality $\dim(\mathrm{col}(A)) = r$ holds by the definition of rank, and $\dim(\mathrm{row}(A)) = r = \mathrm{rank}(A^t)$ by Theorem 3.3.2.
Hence $\dim(\ker(A)) = n - r$ and $\dim(\ker(A^t)) = m - r$ by Corollary 3.3.1(i) for A and A^t, respectively.
Structural Commentary: Ambient sizes place the subspaces; row-column equality and rank-nullity determine their dimensions.
Key Insight: One rank parameter controls both image and kernel dimensions. □

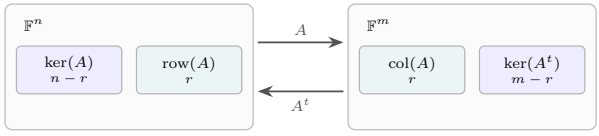

Figure 12: Hosts and dimensions of the four fundamental subspaces, with arrows for $A : \mathbb{F}^n \to \mathbb{F}^m$ and $A^t : \mathbb{F}^m \to \mathbb{F}^n$.

Figure 12 illustrates Theorem 3.3.4. Its boxes encode only ambient hosts and dimensions, not direct sums or orthogonality.

Corollary 3.3.5. *Homogeneous Systems with More Variables Than Equations*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $n > m$, then $(\exists\, \mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, A\mathbf{x} = \mathbf{0}$.

Proof. The map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear by Proposition 3.1.8 and is not injective by Corollary 1.7.4.
Thus $\ker(T_A) \neq \{\mathbf{0}\}$ by Proposition 1.6.2. Choose $\mathbf{x} \in \ker(T_A) \setminus \{\mathbf{0}\}$. Then $A\mathbf{x} = T_A\mathbf{x} = \mathbf{0}$.
Structural Commentary: The dimension obstruction transfers from the induced map to the homogeneous system.
Key Insight: More variables force a nontrivial nullspace. □

Corollary 3.3.6. *Inhomogeneous Systems with More Equations Than Variables*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $m > n$, then $(\exists\, \mathbf{b} \in \mathbb{F}^m)(\forall\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} \neq \mathbf{b}$.

Proof. The map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear by Proposition 3.1.8 and is not surjective by Corollary 1.7.5.
Choose $\mathbf{b} \in \mathbb{F}^m \setminus \mathrm{range}(T_A)$. Then $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = T_A\mathbf{x} \neq \mathbf{b}$.
Structural Commentary: The dimension obstruction leaves a right-hand side outside the induced map's range.
Key Insight: More equations force an unreachable right-hand side. □

3.4 Invertibility and the Matrix-Map Dictionary

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Theorem 3.4.1. *Columns of an Invertible Matrix Form a Basis*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$ *be invertible. Then* $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ *is a basis of* $\mathbb{F}^n$.

Proof. Fix $\mathbf{y} \in \mathbb{F}^n$ and set $\mathbf{x} = (x_1, \ldots, x_n)^t := A^{-1}\mathbf{y}$.
Then $\mathbf{y} = I_n\mathbf{y} = (AA^{-1})\mathbf{y} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j$ by Proposition 3.1.2, the inverse relation $AA^{-1} = I_n$, and Proposition 3.1.9.
If $\mathbf{c} = (c_1, \ldots, c_n)^t$ satisfies $\mathbf{y} = \sum_{j=1}^{n} c_j\mathbf{a}_j = A\mathbf{c}$, then $\mathbf{c} = I_n\mathbf{c} = (A^{-1}A)\mathbf{c} = A^{-1}\mathbf{y} = \mathbf{x}$ by Proposition 3.1.2 and $A^{-1}A = I_n$.
Hence $(\forall\, \mathbf{y} \in \mathbb{F}^n)(\exists!\, \mathbf{x} \in \mathbb{F}^n)\, \mathbf{y} = \sum_{j=1}^{n} x_j\mathbf{a}_j$, so $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis by Proposition 1.3.3.
Structural Commentary: The inverse constructs the unique coefficient column required by the basis criterion.
Key Insight: Invertibility turns matrix multiplication into unique coordinate representation. □

Theorem 3.4.2. *One-Sided Inverses of Square Matrices*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$. *If* $AB = I_n$, *then* $BA = I_n$.

Proof. The induced map $T_B : \mathbb{F}^n \to \mathbb{F}^n$ is linear by Proposition 3.1.8.
If $\mathbf{x} \in \ker(T_B)$, then $B\mathbf{x} = \mathbf{0}$ and $\mathbf{x} = I_n\mathbf{x} = (AB)\mathbf{x} = A(B\mathbf{x}) = \mathbf{0}$ by Proposition 3.1.2; hence T_B is injective by Proposition 1.6.2.
It is surjective by Theorem 1.7.7, $(ii) \Rightarrow (iii)$. Fix $\mathbf{y} \in \mathbb{F}^n$ and choose $\mathbf{x} \in \mathbb{F}^n$ satisfying $\mathbf{y} = B\mathbf{x}$.
Then $BA\mathbf{y} = BA(B\mathbf{x}) = B((AB)\mathbf{x}) = B\mathbf{x} = \mathbf{y}$ by Proposition 3.1.2.
Substituting the columns of I_n for $\mathbf{y}$, $BA = I_n$ by Proposition 3.1.10 and Proposition 3.1.2.
Structural Commentary: Kernel triviality and equal dimension convert a left inverse into the missing right inverse.
Key Insight: Surjectivity lets the known inverse relation reach every vector. □

Motivation 3.4.3. *Coordinate Encoding*
Fixing bases encodes a linear map by a matrix. The next results verify composition compatibility and exact matrix recovery.

Proposition 3.4.4. *Composition and Matrix Multiplication*
Let $\alpha = (\mathbf{u}_1, \ldots, \mathbf{u}_\ell)$, $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$, *and* $\delta = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ *be bases of* $\mathcal{U}$, $\mathcal{V}$, *and* $\mathcal{W}$, *respectively.*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ *and* $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, *then* $[ST]_{\delta\leftarrow\alpha} = [S]_{\delta\leftarrow\beta}[T]_{\beta\leftarrow\alpha}$.

Proof. $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$ by Proposition 1.5.7(i). Fix $j \in \{1, \ldots, \ell\}$.
Then $[(ST)\mathbf{u}_j]_\delta = [S(T\mathbf{u}_j)]_\delta = [S]_{\delta\leftarrow\beta}[T\mathbf{u}_j]_\beta$ by Proposition 3.2.5, with input $T\mathbf{u}_j$.
The first and last terms are jth columns of $[ST]_{\delta\leftarrow\alpha}$ and $[S]_{\delta\leftarrow\beta}[T]_{\beta\leftarrow\alpha}$, respectively, by Definition 3.2.3 and Proposition 3.1.10.
Since j was arbitrary, the matrices are equal by Definition 3.1.1(i).
Structural Commentary: Comparing basis-vector images identifies the corresponding columns of the composition and product matrices.
Key Insight: Composition order becomes matrix multiplication order. □

Theorem 3.4.5. *Linear Maps and Matrices Are the Same Data*
Let $\dim(\mathcal{V}) = n \geq 1$ *and* $\dim(\mathcal{W}) = m \geq 1$, *and fix bases* $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *and* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$.
Define $\Phi : \mathcal{L}(\mathcal{V}, \mathcal{W}) \to \mathrm{Mat}(m, n, \mathbb{F})$ *by* $\Phi(T) := [T]_{\gamma\leftarrow\beta}$. *Then:*

(i) Φ *is an isomorphism.*
(ii) $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = mn$.

Proof.

(i) Both spaces are vector spaces by Proposition 1.5.5 and Proposition 3.1.5(i), and Φ is linear by Proposition 3.2.4.
Suppose $\Phi(S) = \Phi(T)$. Then $(\forall\, j \in \{1, \ldots, n\})\, [S\mathbf{v}_j]_\gamma = [T\mathbf{v}_j]_\gamma$ by Definition 3.2.3, so $(\forall\, j \in \{1, \ldots, n\})\, S\mathbf{v}_j = T\mathbf{v}_j$ by Proposition 3.2.2(ii). Hence $S = T$ by Proposition 1.5.8, and Φ is injective.
Fix $A = [a_{i,j}] \in \mathrm{Mat}(m, n, \mathbb{F})$ and set $\mathbf{z}_j := \sum_{i=1}^{m} a_{i,j}\mathbf{w}_i$ $(j \in \{1, \ldots, n\})$.
The relation $(\exists!\, R \in \mathcal{L}(\mathcal{V}, \mathcal{W}))(\forall\, j \in \{1, \ldots, n\})\, R\mathbf{v}_j = \mathbf{z}_j$ holds by Proposition 1.5.8; fix this R.
Then $(\forall\, j \in \{1, \ldots, n\})\, [R\mathbf{v}_j]_\gamma = (a_{1,j}, \ldots, a_{m,j})^t$ by Definition 3.2.1, so $\Phi(R) = A$ by Definition 3.2.3.
Thus Φ is surjective and therefore an isomorphism by Proposition 1.6.7.

(ii) The matrix units form a basis of $\mathrm{Mat}(m, n, \mathbb{F})$ with mn members by Proposition 3.1.5(ii).
Their inverse images under Φ form a basis of $\mathcal{L}(\mathcal{V}, \mathcal{W})$ by part (i) and Proposition 1.7.8. Hence $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = mn$.

Structural Commentary: Basis images establish the isomorphism; transporting matrix units determines the dimension.
Key Insight: A matrix records exactly the images of a domain basis. □

Perspective 3.4.6. *Finite-Support Coordinate Model*
Proposition 3.2.2 and Theorem 3.4.5 identify finite-dimensional spaces and their maps with coordinate columns and matrices after bases are chosen. Definition 2.1.20 supplies the countable finite-support model $\mathbb{F}^{(\mathbb{N})}$; the analogy stops before infinite linear combinations.

3.5 Change of Basis and Coordinate Transformations

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 3.5.1. *Change-of-Basis Matrix*
Let $n \geq 1$, and let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be bases of the same vector space $\mathcal{V}$.
The *change-of-basis matrix from β-coordinates to γ-coordinates* is $C_{\gamma\leftarrow\beta} := [I_\mathcal{V}]_{\gamma\leftarrow\beta} \in \mathrm{Mat}(n, n, \mathbb{F})$.
The identity $(\forall\, \mathbf{v} \in \mathcal{V})\, [\mathbf{v}]_\gamma = C_{\gamma\leftarrow\beta}[\mathbf{v}]_\beta$ holds by Proposition 3.2.5, with $T = I_\mathcal{V}$.

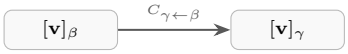

Figure 13: Direction of the coordinate rewrite implemented by $C_{\gamma\leftarrow\beta}$.

Figure 13 depicts $C_{\gamma\leftarrow\beta}$ carrying $[\mathbf{v}]_\beta$ to $[\mathbf{v}]_\gamma$; Definition 3.5.1 owns the exact identity.

Theorem 3.5.2. *Change-of-Basis Matrices*
Let $\mathcal{V}$ be n-dimensional, where $n \geq 1$, and let β be a basis of $\mathcal{V}$.

(i) If γ is a basis of $\mathcal{V}$, then $C_{\gamma\leftarrow\beta}$ is invertible with $(C_{\gamma\leftarrow\beta})^{-1} = C_{\beta\leftarrow\gamma}$.
(ii) If $S \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible, then $S = C_{\gamma\leftarrow\beta}$ for a basis γ of $\mathcal{V}$.

Proof.

(i) The identity $[I_\mathcal{V}]_{\beta\leftarrow\beta} = I_n = [I_\mathcal{V}]_{\gamma\leftarrow\gamma}$ holds by Definition 3.2.1 and Definition 3.2.3.
Hence $C_{\beta\leftarrow\gamma}C_{\gamma\leftarrow\beta} = [I_\mathcal{V} I_\mathcal{V}]_{\beta\leftarrow\beta} = I_n$ and $C_{\gamma\leftarrow\beta}C_{\beta\leftarrow\gamma} = [I_\mathcal{V} I_\mathcal{V}]_{\gamma\leftarrow\gamma} = I_n$ by Definition 3.5.1 and Proposition 3.4.4.
Thus $C_{\gamma\leftarrow\beta}$ is invertible with inverse $C_{\beta\leftarrow\gamma}$.
(ii) Write $S^{-1} = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_n]$. The relation $(\forall j \in \{1, \ldots, n\})(\exists!\, \mathbf{w}_j \in \mathcal{V})\, [\mathbf{w}_j]_\beta = \mathbf{s}_j$ holds by Proposition 3.2.2(ii),(iii).
Set $\gamma := (\mathbf{w}_1, \ldots, \mathbf{w}_n)$. The matrix S^{-1} is invertible with inverse S, so $(\mathbf{s}_1, \ldots, \mathbf{s}_n)$ is a basis of $\mathbb{F}^n$ by Theorem 3.4.1.
The map $[\cdot]_\beta^{-1} : \mathbb{F}^n \to \mathcal{V}$ is an isomorphism by Proposition 3.2.2(iii), so γ is a basis of $\mathcal{V}$ by Proposition 1.7.8.
The relation $(\forall j \in \{1, \ldots, n\})\, (C_{\beta\leftarrow\gamma})_{\cdot,j} = [\mathbf{w}_j]_\beta = \mathbf{s}_j$ holds by Definition 3.5.1 and Definition 3.2.3; hence $C_{\beta\leftarrow\gamma} = S^{-1}$ by Definition 3.1.1(i). Therefore $C_{\gamma\leftarrow\beta} = S$ by part (i) and Lemma 3.1.3.

Structural Commentary: Composition proves reciprocity; inverse-coordinate columns realize an arbitrary invertible matrix.
Key Insight: Coordinate isomorphisms transport bases in both directions. □

Theorem 3.5.3. *Change of Bases for a Linear Map*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\beta' = (\mathbf{v}'_1, \ldots, \mathbf{v}'_n)$ be bases of $\mathcal{V}$, and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ and $\gamma' = (\mathbf{w}'_1, \ldots, \mathbf{w}'_m)$ be bases of $\mathcal{W}$, where $n, m \geq 1$. Then $[T]_{\gamma'\leftarrow\beta'} = C_{\gamma'\leftarrow\gamma}[T]_{\gamma\leftarrow\beta}C_{\beta\leftarrow\beta'}$.

Proof. $T = I_\mathcal{W} T I_\mathcal{V}$ by Proposition 1.5.7(ii),(iii). Hence $[T]_{\gamma'\leftarrow\beta'} = [I_\mathcal{W} T I_\mathcal{V}]_{\gamma'\leftarrow\beta'} = [I_\mathcal{W}]_{\gamma'\leftarrow\gamma}[T]_{\gamma\leftarrow\beta}[I_\mathcal{V}]_{\beta\leftarrow\beta'}$ by Proposition 3.4.4 twice. The outer matrices are $C_{\gamma'\leftarrow\gamma}$ and $C_{\beta\leftarrow\beta'}$ by Definition 3.5.1.
Structural Commentary: Identity maps insert the codomain and domain coordinate transitions around the fixed representation.
Key Insight: Output coordinates transform on the left; input coordinates transform on the right. □

3.6 Rank and Invertibility Consequences

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Corollary 3.6.1. *Basis Columns Form an Invertible Matrix*
Let $n \geq 1$. If $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$, then $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible.

Proof. Let $\epsilon = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the ordered list of columns of I_n, and set $\alpha := (\mathbf{a}_1, \ldots, \mathbf{a}_n)$.
The matrix I_n is invertible with inverse I_n by Proposition 3.1.2(ii), so ϵ is a basis of $\mathbb{F}^n$ by Theorem 3.4.1.
The relation $(\forall j \in \{1, \ldots, n\})\, [\mathbf{a}_j]_\epsilon = \mathbf{a}_j$ holds by Definition 3.2.1, Proposition 3.1.9, and $I_n \mathbf{a}_j = \mathbf{a}_j$.
Thus $C_{\epsilon\leftarrow\alpha} = A$ by Definition 3.5.1 and Definition 3.2.3, and A is invertible by Theorem 3.5.2(i).
Structural Commentary: Standard coordinates identify the column matrix with a change-of-basis matrix.
Key Insight: A basis-to-basis coordinate rewrite is invertible. □

Corollary 3.6.2. *Invertibility and Full Rank*
Let $n \geq 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is invertible $\iff \mathrm{rank}(A) = n$.

Proof.

($\Rightarrow$) Suppose A is invertible. Its columns form basis of $\mathbb{F}^n$ by Theorem 3.4.1; hence $\mathrm{col}(A) = \mathbb{F}^n$, $\mathrm{rank}(A) = n$ by Definition 3.1.12.
($\Leftarrow$) Suppose $\mathrm{rank}(A) = n$. Then $\dim(\mathrm{col}(A)) = n = \dim(\mathbb{F}^n)$ by Definition 3.1.12, so $\mathrm{col}(A) = \mathbb{F}^n$ by Proposition 1.4.3(ii).
The n columns of A therefore form a basis of $\mathbb{F}^n$ by Proposition 1.4.4(i), and A is invertible by Corollary 3.6.1.

Structural Commentary: One direction reads rank from basis columns; the converse upgrades full dimension to a basis.
Key Insight: Square-matrix invertibility is equivalent to full column rank. □

Corollary 3.6.3. *Square Matrix Algebraic Invertibility Dictionary*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:

(i) A is invertible;
(ii) $\mathbf{a}_1, \ldots, \mathbf{a}_n$ form a basis of $\mathbb{F}^n$;
(iii) $\mathbf{a}_1, \ldots, \mathbf{a}_n$ are linearly independent;
(iv) $\mathbf{a}_1, \ldots, \mathbf{a}_n$ span $\mathbb{F}^n$;
(v) for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution;
(vi) the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the zero solution.

Proof.

(i) $\Rightarrow$ *(ii)* By Theorem 3.4.1, the columns of an invertible matrix form a basis.
(ii) $\Rightarrow$ *(iii)* A basis is linearly independent.
(iii) $\Rightarrow$ *(iv)* In $\mathbb{F}^n$, n linearly independent vectors form a basis, hence span.
(iv) $\Rightarrow$ *(v)* In $\mathbb{F}^n$, n spanning vectors form a basis, so every $\mathbf{b}$ has unique coordinates $\mathbf{x}$ in $A\mathbf{x} = \mathbf{b}$.
(v) $\Rightarrow$ *(vi)* Apply (v) to $\mathbf{b} = \mathbf{0}$; since $\mathbf{x} = \mathbf{0}$ is a solution, it is the only one.
(vi) $\Rightarrow$ *(i)* Condition (vi) says the columns are independent.
Hence they form a basis of $\mathbb{F}^n$, and Corollary 3.6.1 gives invertibility.

Structural Commentary: Square invertibility is column basishood.
Key Insight: Invertible columns give unique coordinates. □

Corollary 3.6.4. *Transpose Preserves Invertibility*
Let $n \geq 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is invertible $\iff$ A^t is invertible.

Proof.
($\Rightarrow$) Suppose A is invertible. Then $\mathrm{rank}(A) = n$ by Corollary 3.6.2, and $\mathrm{rank}(A^t) = \mathrm{rank}(A) = n$ by Theorem 3.3.2. Hence A^t is invertible by Corollary 3.6.2.
($\Leftarrow$) Suppose A^t is invertible. The matrix $(A^t)^t$ is invertible by the forward direction with A^t in place of A. Since $(A^t)^t = A$ by Proposition 3.1.14(i), A is invertible.

Structural Commentary: Rank invariance proves the forward direction; double transposition reverses it.
Key Insight: Transposition preserves full rank. □

Theorem 3.6.5. *Completing a Full-Column-Rank Matrix*
Let $1 \leq n < m$ and $X \in \mathrm{Mat}(m, n, \mathbb{F})$. If $\mathrm{rank}(X) = n$, then $(\exists\, X' \in \mathrm{Mat}(m, m-n, \mathbb{F}))$ $[\,X\ X'\,] \in \mathrm{Mat}(m, m, \mathbb{F})^{\times}$.

Proof. Write $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_n]$. Then $\dim(\mathrm{col}(X)) = n$ by Definition 3.1.12. The columns form a spanning list of length n in $\mathrm{col}(X)$. Hence they form a basis of $\mathrm{col}(X)$ by Proposition 1.4.4(i) and are linearly independent in $\mathbb{F}^m$.
Extend them to a basis $(\mathbf{x}_1, \ldots, \mathbf{x}_m)$ of $\mathbb{F}^m$ by Theorem 1.3.13, and set $X' := [\mathbf{x}_{n+1}\ \cdots\ \mathbf{x}_m] \in \mathrm{Mat}(m, m-n, \mathbb{F})$.
The matrix $[\,X\ X'\,]$ is invertible by Corollary 3.6.1.
Structural Commentary: Column rank identifies an independent list; basis extension supplies the missing columns.
Key Insight: Full-column-rank matrices are initial blocks of invertible matrices. □

Theorem 3.6.6. *Full-Rank Products Preserve Rank*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$. Then:
(i) If $X \in \mathrm{Mat}(p, m, \mathbb{F})$ and $Y \in \mathrm{Mat}(n, q, \mathbb{F})$ satisfy $\mathrm{rank}(X) = m$ and $\mathrm{rank}(Y) = n$, then $\mathrm{rank}(XAY) = \mathrm{rank}(A)$.
(ii) If $P \in \mathrm{Mat}(m, m, \mathbb{F})$ and $Q \in \mathrm{Mat}(n, n, \mathbb{F})$ are invertible, then $\mathrm{rank}(PAQ) = \mathrm{rank}(A)$.

Proof.
(i) The equality $\dim(\ker(X)) = 0$ holds by Corollary 3.3.1(i), so $\ker(X) = \{\mathbf{0}\}$ by Proposition 1.4.3(ii). Hence $\ker(XA) = \{\mathbf{z} \in \mathbb{F}^n : X(A\mathbf{z}) = \mathbf{0}\} = \{\mathbf{z} \in \mathbb{F}^n : A\mathbf{z} = \mathbf{0}\} = \ker(A)$, and $\mathrm{rank}(XA) = \mathrm{rank}(A)$ by Corollary 3.3.1(i). Since $\mathrm{rank}(Y^t) = \mathrm{rank}(Y) = n$ by Theorem 3.3.2, the equality $\mathrm{rank}(Y^t A^t) = \mathrm{rank}(A^t)$ holds by the preceding kernel argument with (X, A) replaced by (Y^t, A^t). The identity $(AY)^t = Y^t A^t$ holds by Proposition 3.1.16. Hence $\mathrm{rank}(AY) = \mathrm{rank}(Y^t A^t) = \mathrm{rank}(A^t) = \mathrm{rank}(A)$ by Theorem 3.3.2 and the preceding equality. Finally, $\mathrm{rank}(XAY) = \mathrm{rank}(AY) = \mathrm{rank}(A)$ by the preceding kernel argument with A replaced by AY.
(ii) Set $X = P$ and $Y = Q$. Then $\mathrm{rank}(P) = m$ and $\mathrm{rank}(Q) = n$ by Corollary 3.6.2, so $\mathrm{rank}(PAQ) = \mathrm{rank}(A)$ by part (i).

Structural Commentary: Kernel preservation handles left factors; transposition converts right factors to the same argument.
Key Insight: Full column and row rank prevent dimensional collapse. □

3.7 Block Matrix Algebra

Throughout this subsection, all vector spaces and matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Definition 3.7.1. *Block Partitions and Conformality*
Let $r, s \geq 1$, let $m_1, \ldots, m_r$ and $n_1, \ldots, n_s$ be positive integers, and let $A \in \mathrm{Mat}\left(\sum_{i=1}^{r} m_i, \sum_{j=1}^{s} n_j, \mathbb{F}\right)$.
An *$r \times s$ block partition* $A = [A_{ij}]$ groups consecutive rows according to $(m_1, \ldots, m_r)$ and consecutive columns according to $(n_1, \ldots, n_s)$, so $(\forall\, i \in \{1, \ldots, r\})(\forall\, j \in \{1, \ldots, s\})\ A_{ij} \in \mathrm{Mat}(m_i, n_j, \mathbb{F})$. The submatrices A_{ij} are the *blocks*.
(i) Two $r \times s$ block partitions $A = [A_{ij}]$ and $C = [C_{ij}]$ are *conformal for addition* if corresponding blocks have the same size.
(ii) An $r \times s$ block partition $A = [A_{ik}]$ and an $s \times t$ block partition $B = [B_{kj}]$, where $t \geq 1$, are *conformal for multiplication* if $\exists\, p_1, \ldots, p_t \in \mathbb{Z}_{>0}$ such that $A_{ik} \in \mathrm{Mat}(m_i, n_k, \mathbb{F})$ and $B_{kj} \in \mathrm{Mat}(n_k, p_j, \mathbb{F})$ $\forall i \in \{1, \ldots, r\}$, $k \in \{1, \ldots, s\}$, $j \in \{1, \ldots, t\}$.

Definition 3.7.2. *Block Matrix of a Linear Map*
Suppose $\mathcal{V} = \mathcal{U}_1 \oplus \mathcal{U}_2$ and $\mathcal{W} = \mathcal{Z}_1 \oplus \mathcal{Z}_2$, where $\dim(\mathcal{U}_j) = n_j \geq 1$ and $\dim(\mathcal{Z}_i) = m_i \geq 1$ for $i, j \in \{1, 2\}$. Choose ordered bases α_j of $\mathcal{U}_j$ and β_i of $\mathcal{Z}_i$. By Definition 2.3.5, $\alpha = (\alpha_1, \alpha_2)$ and $\beta = (\beta_1, \beta_2)$ are bases of $\mathcal{V}$ and $\mathcal{W}$ adapted to the decompositions.
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, its representation matrix has conformal partition $[T]_{\beta \leftarrow \alpha} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where $A_{ij} \in \mathrm{Mat}(m_i, n_j, \mathbb{F})$ for $i, j \in \{1, 2\}$.
The jth block column contains coordinate columns of images of basis α_j, while the ith block row retains their β_i-coordinate entries.

Proposition 3.7.3. *Block Matrix Arithmetic*
The following identities hold.
(i) If $A = [A_{ij}]$ and $C = [C_{ij}]$ are conformal $r \times s$ block partitions, then $A + C = [A_{ij} + C_{ij}]$.
In particular, conformal 2×2 partitions satisfy $A + C = \begin{bmatrix} A_{11}+C_{11} & A_{12}+C_{12} \\ A_{21}+C_{21} & A_{22}+C_{22} \end{bmatrix}$.
(ii) If $A = [A_{ik}]$ and $B = [B_{kj}]$ are conformal for multiplication, then $(AB)_{ij} = \sum_{k=1}^{s} A_{ik} B_{kj}$ for every admissible i, j.
In particular, conformal 2×2 partitions satisfy $AB = \begin{bmatrix} A_{11}B_{11}+A_{12}B_{21} & A_{11}B_{12}+A_{12}B_{22} \\ A_{21}B_{11}+A_{22}B_{21} & A_{21}B_{12}+A_{22}B_{22} \end{bmatrix}$.

Proof.
(i) By Definition 3.1.1, matrix addition is entrywise. Because conformality assigns the same entry positions to corresponding blocks, addition may be performed blockwise, so $A + C = [A_{ij} + C_{ij}]$. The case $r = s = 2$ is the stated specialization.
(ii) By Proposition 3.1.6, each entry of AB is the row-column sum over the shared index. Group that index into the s consecutive intervals determined by the common block sizes. The contribution from the kth interval is the corresponding entry of $A_{ik}B_{kj}$.
Therefore $(AB)_{ij} = \sum_{k=1}^{s} A_{ik} B_{kj}$. The case of two block rows and two block columns is the stated specialization.

Structural Commentary: Addition is entrywise; multiplication groups the shared index.
Key Insight: Block arithmetic is ordinary matrix arithmetic performed at a coarser scale. □

Proposition 3.7.4. *Stacks Intersect Kernels, Concatenations Sum Columns*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $B \in \mathrm{Mat}(p,n,\mathbb{F})$, $C \in \mathrm{Mat}(m,r,\mathbb{F})$, and $D \in \mathrm{Mat}(m,s,\mathbb{F})$. Then:

(i) $\ker\left(\left[\begin{smallmatrix} A \\ B \end{smallmatrix}\right]\right) = \ker(A) \cap \ker(B)$.
(ii) $\mathrm{col}([\,C\ D\,]) = \mathrm{col}(C) + \mathrm{col}(D)$.

Proof.

(i) By Proposition 3.7.3, $\left[\begin{smallmatrix} A \\ B \end{smallmatrix}\right]\mathbf{x} = \left[\begin{smallmatrix} A\mathbf{x} \\ B\mathbf{x} \end{smallmatrix}\right]$ for every $\mathbf{x} \in \mathbb{F}^n$.
By Definition 3.1.12, $\mathbf{x} \in \ker\left(\left[\begin{smallmatrix} A \\ B \end{smallmatrix}\right]\right)$ if and only if $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$, which is equivalent to $\mathbf{x} \in \ker(A) \cap \ker(B)$.
(ii) By Proposition 3.7.3, $[\,C\ D\,]\left[\begin{smallmatrix} \mathbf{u} \\ \mathbf{v} \end{smallmatrix}\right] = C\mathbf{u} + D\mathbf{v}$ for every $\mathbf{u} \in \mathbb{F}^r$ and $\mathbf{v} \in \mathbb{F}^s$.
By Definition 3.1.12, $\mathrm{col}([\,C\ D\,]) = \{C\mathbf{u} + D\mathbf{v} : \mathbf{u} \in \mathbb{F}^r,\ \mathbf{v} \in \mathbb{F}^s\} = \mathrm{col}(C) + \mathrm{col}(D)$.

Structural Commentary: The kernel and column-space definitions translate both block actions.
Key Insight: Vertical blocks intersect kernels; horizontal blocks add column spaces. □

Proposition 3.7.5. *Block Centrosymmetry and Commutation*
Let $n \geq 1$, let $A, B, C, D \in \mathrm{Mat}(n,n,\mathbb{F})$, and set $M = \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]$ and $L_{2n} = \left[\begin{smallmatrix} 0 & I_n \\ I_n & 0 \end{smallmatrix}\right]$ in $\mathrm{Mat}(2n,2n,\mathbb{F})$.
Then M has block centrosymmetric form, meaning $A = D$ and $B = C$, if and only if $L_{2n}M = ML_{2n}$.

Proof. By Proposition 3.7.3, $L_{2n}M = \left[\begin{smallmatrix} C & D \\ A & B \end{smallmatrix}\right]$ and $ML_{2n} = \left[\begin{smallmatrix} B & A \\ D & C \end{smallmatrix}\right]$.
(⇒) If M has block centrosymmetric form, then $A = D$ and $B = C$, so the two block products agree. Therefore $L_{2n}M = ML_{2n}$.
(⇐) If $L_{2n}M = ML_{2n}$, then corresponding blocks are equal by Definition 3.1.1.
Thus $C = B$ and $D = A$, so M has block centrosymmetric form.

Structural Commentary: Explicit block multiplication reduces commutation with L_{2n} to equality of opposite blocks.
Key Insight: The exchange matrix detects block centrosymmetry. □

Definition 3.7.6. *Direct Sum of Matrices*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(p,q,\mathbb{F})$, the *direct sum* of A and B is $A \oplus B := \left[\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right] \in \mathrm{Mat}(m+p, n+q, \mathbb{F})$, where the upper-right and lower-left zero blocks have sizes $m \times q$ and $p \times n$, respectively.

3.8 Matrix Operations and Identities

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Definition 3.8.1. *Trace and Determinant of a Matrix*
Let $n \geq 1$ and $A = [a_{jk}] \in \mathrm{Mat}(n,n,\mathbb{F})$.

(i) The *trace* of A is $\mathrm{tr}(A) := \sum_{j=1}^{n} a_{jj}$.

(ii) The *determinant* of A is $\det A := \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) \prod_{j=1}^{n} a_{\sigma(j),j}$.

Theorem 3.8.2. *Basic Determinant Laws*
Let $n \geq 1$ and $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$.

(i) $\det I_n = 1$.
(ii) For fixed $k \in \{1,\dots,n\}$ and $(A_{\cdot,j})_{j \neq k}$, the map $\mathbf{x} \mapsto \det(A_{\cdot,1},\dots,\mathbf{x},\dots,A_{\cdot,n})$ is linear.
(iii) Swapping two columns negates the determinant.
(iv) If two columns of A are equal, then $\det A = 0$.
(v) If A is triangular, then $\det A = \prod_{j=1}^{n} a_{jj}$.
(vi) $\det(AB) = \det(A)\det(B)$.
(vii) A is invertible if and only if $\det A \neq 0$.
(viii) If A is invertible, then $\det(A^{-1}) = (\det A)^{-1}$.
(ix) $\det(A^t) = \det(A)$.

Proof.

(i) In the permutation formula for I_n, only the identity permutation has a nonzero product, and its contribution is 1.
(ii) For $k \in \{1,\dots,n\}$, $\det(A_{\cdot,1},\dots,\mathbf{x},\dots,A_{\cdot,n}) = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) x_{\sigma(k)} \prod_{\ell \neq k} a_{\sigma(\ell),\ell}$, which is linear in $\mathbf{x}$.
(iii) Let τ transpose columns p and q, and let $B_{\cdot,j} = A_{\cdot,\tau(j)}$. Then $\det B = -\det A$ by reindexing the determinant sum with $\rho = \sigma \circ \tau$, because $\mathrm{sign}(\rho \circ \tau) = -\mathrm{sign}(\rho)$.
(iv) If columns p and q are equal, pair the term indexed by σ with the term indexed by $\sigma \circ (p\ q)$. Their products agree and their signs are opposite, so every pair sums to zero, including when $\mathrm{char}(\mathbb{F}) = 2$.
(v) Suppose A is upper triangular. A nonzero term satisfies $(\forall j \in \{1,\dots,n\})\, \sigma(j) \leq j$, hence $\sigma = \mathrm{id}$ and $\det A = \prod_{j=1}^{n} a_{jj}$. The lower-triangular case is analogous with $(\forall j \in \{1,\dots,n\})\, \sigma(j) \geq j$.
(vi) Let F be alternating and multilinear in its n column arguments. Standard-basis expansion produces terms indexed by $(i_1,\dots,i_n) \in \{1,\dots,n\}^n$; repeated indices vanish by alternation.
The surviving tuples are permutations, and $F(\mathbf{e}_{\sigma(1)},\dots,\mathbf{e}_{\sigma(n)}) = \mathrm{sign}(\sigma)F(I_n)$ by alternation. Hence $F(A_{\cdot,1},\dots,A_{\cdot,n}) = F(I_n)\det A$.
For fixed A, $H(B) := \det(AB)$ is alternating and multilinear in the columns of B, with $H(I_n) = \det A$. Thus $\det(AB) = H(B) = H(I_n)\det B = \det(A)\det(B)$.
(vii) (⇒) If A is invertible, then $\det A \neq 0$ because $1 = \det(AA^{-1}) = \det(A)\det(A^{-1})$ by parts (i) and (vi).
(vii) (⇐) If A were singular, one column would be a linear combination of the others, and $\det A = 0$ by parts (ii) and (iv), contrary to $\det A \neq 0$. Thus A is invertible.
(viii) If A is invertible, then $\det(A^{-1}) = (\det A)^{-1}$ by $1 = \det(AA^{-1}) = \det(A)\det(A^{-1})$ from parts (i), (vi), and (vii).

(ix) $\det(A^t) = \sum_{\sigma\in\mathfrak{S}_n} \operatorname{sign}(\sigma) \prod_{j=1}^{n} a_{j,\sigma(j)} = \sum_{\tau\in\mathfrak{S}_n} \operatorname{sign}(\tau) \prod_{j=1}^{n} a_{\tau(j),j} = \det A$ by reindexing with $\tau = \sigma^{-1}$.

Structural Commentary: The permutation formula establishes alternation and triangularity before multiplicativity and invertibility.
Key Insight: Alternating multilinearity reduces every column expansion to signed basis permutations. □

Lemma 3.8.3. *Identity Blocks Do Not Change Determinant*
Let $m \geq 1$, $n \geq 0$, and $A \in \operatorname{Mat}(m, m, \mathbb{F})$.

(i) $\det(I_n \oplus A) = \det A$;
(ii) $\det(A \oplus I_n) = \det A$.

Proof.

(i) If $n = 0$, the identity holds by the empty identity-block convention. If $n \geq 1$, every nonzero permutation term fixes $\{1, \ldots, n\}$ pointwise; relabeling the remaining indices produces exactly the determinant expansion of A.
(ii) If $n = 0$, the identity again holds by convention. If $n \geq 1$, every nonzero term fixes the last n indices pointwise, and the remaining sum is $\det A$.

Structural Commentary: Each block position reduces the permutation sum to the indices occupied by A.
Key Insight: An identity block forces its coordinates to remain fixed. □

Theorem 3.8.4. *Determinant of an Upper Block-Triangular Matrix*
Let $r, s \geq 1$, $A \in \operatorname{Mat}(r, r, \mathbb{F})$, $B \in \operatorname{Mat}(r, s, \mathbb{F})$, and $D \in \operatorname{Mat}(s, s, \mathbb{F})$.
Then $\det \left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right) = \det(A)\det(D)$.

Proof. $\left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right) = \left(\begin{smallmatrix} I_r & 0 \\ 0 & D \end{smallmatrix}\right) \left(\begin{smallmatrix} I_r & B \\ 0 & I_s \end{smallmatrix}\right) \left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$.
The outer determinants are $\det D$ and $\det A$ by Lemma 3.8.3; the middle determinant is 1 by Theorem 3.8.2(v).
The formula follows by Theorem 3.8.2(vi).
Structural Commentary: A three-factor decomposition isolates the two diagonal-block determinants.
Key Insight: The off-diagonal block lies in a unit upper-triangular factor. □

Definition 3.8.5. *Polynomial Evaluation on a Square Matrix*
Let $n \geq 1$ and $A \in \operatorname{Mat}(n, n, \mathbb{F})$. Define $A^0 := I_n$ and $(\forall\, j \in \mathbb{Z}_{\geq 0})\, A^{j+1} := A^j A$.
For $p(z) = \sum_{j=0}^{d} c_j z^j \in \mathbb{F}[z]$, where $d \geq 0$, the *evaluation of p at A* is $p(A) := \sum_{j=0}^{d} c_j A^j \in \operatorname{Mat}(n, n, \mathbb{F})$.

Proposition 3.8.6. *Algebra of Matrix Direct Sums*

(i) If $A, C \in \operatorname{Mat}(m, n, \mathbb{F})$, $B, D \in \operatorname{Mat}(r, s, \mathbb{F})$, and $a, b \in \mathbb{F}$, then $a(A \oplus B) + b(C \oplus D) = (aA + bC) \oplus (aB + bD)$.
(ii) If $A \in \operatorname{Mat}(m, n, \mathbb{F})$, $C \in \operatorname{Mat}(n, \ell, \mathbb{F})$, $B \in \operatorname{Mat}(r, s, \mathbb{F})$, and $D \in \operatorname{Mat}(s, t, \mathbb{F})$, then $(A \oplus B)(C \oplus D) = AC \oplus BD$.
(iii) If $A \in \operatorname{Mat}(m, m, \mathbb{F})$, $B \in \operatorname{Mat}(n, n, \mathbb{F})$, and $k \in \mathbb{Z}_{>0}$, then $(A \oplus B)^k = A^k \oplus B^k$.
(iv) If $A \in \operatorname{Mat}(m, m, \mathbb{F})$, $B \in \operatorname{Mat}(n, n, \mathbb{F})$, and $p \in \mathbb{F}[z]$, then $p(A \oplus B) = p(A) \oplus p(B)$.

Proof.

(i) By Definition 3.7.6, Definition 3.1.1, and Proposition 3.7.3, $a(A \oplus B) + b(C \oplus D) = \left[\begin{smallmatrix} aA+bC & 0 \\ 0 & aB+bD \end{smallmatrix}\right] = (aA + bC) \oplus (aB + bD)$.
(ii) By Definition 3.7.6 and Proposition 3.7.3, $(A \oplus B)(C \oplus D) = \left[\begin{smallmatrix} AC & 0 \\ 0 & BD \end{smallmatrix}\right] = AC \oplus BD$.
(iii) Set $P_k := ((A \oplus B)^k = A^k \oplus B^k)$. By Definition 3.8.5 and Proposition 3.1.2, P_1 holds.
If P_k holds, then $(A \oplus B)^{k+1} = (A \oplus B)^k (A \oplus B) = (A^k \oplus B^k)(A \oplus B) = A^{k+1} \oplus B^{k+1}$ by the recurrence defining powers, the induction hypothesis, and (ii). Therefore $(\forall\, k \in \mathbb{Z}_{>0})\, P_k$ by Theorem 2.6.2.
(iv) Write $p(z) = \sum_{j=0}^{d} c_j z^j$. The identity $(A \oplus B)^j = A^j \oplus B^j$ holds for $j = 0$ by Definition 3.8.5, Definition 3.7.6, for $j \geq 1$ by (iii).
By (i), scalar multiples, finite sums of these direct sums are componentwise. Hence $p(A \oplus B) = p(A) \oplus p(B)$ by Definition 3.8.5.

Structural Commentary: Block formulas extend by induction and linearity.
Key Insight: Direct-sum blocks evolve independently under matrix algebra. □

Proposition 3.8.7. *Diagonal Multiplication Scales Rows and Columns*
Let $A = [a_{ij}] \in \operatorname{Mat}(m, n, \mathbb{F})$, let $\lambda_1, \ldots, \lambda_n, \mu_1, \ldots, \mu_m \in \mathbb{F}$, and set $\Lambda := \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $M := \operatorname{diag}(\mu_1, \ldots, \mu_m)$. Then:

(i) $A\Lambda = [\lambda_j a_{ij}]$; right multiplication by Λ scales column j by λ_j.
(ii) $MA = [\mu_i a_{ij}]$; left multiplication by M scales row i by μ_i.

Proof.

(i) For admissible i, j, $(A\Lambda)_{ij} = \sum_{r=1}^{n} a_{ir}\Lambda_{rj} = a_{ij}\lambda_j = \lambda_j a_{ij}$ by Definition 3.1.1 and commutativity in $\mathbb{F}$.
(ii) For admissible i, j, $(MA)_{ij} = \sum_{r=1}^{m} M_{ir} a_{rj} = \mu_i a_{ij}$ by Definition 3.1.1.

Structural Commentary: Diagonal sparsity isolates one summand in each entry formula.
Key Insight: The multiplication side determines whether rows or columns are scaled. □

Proposition 3.8.8. *Inverse of an Upper Block-Triangular Matrix*
Let $Y \in \operatorname{Mat}(m, m, \mathbb{F})$ and $Z \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible, and let $X \in \operatorname{Mat}(m, n, \mathbb{F})$.
Then $\left[\begin{smallmatrix} Y & X \\ 0 & Z \end{smallmatrix}\right]^{-1} = \left[\begin{smallmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{smallmatrix}\right] \in \operatorname{Mat}(m+n, m+n, \mathbb{F})$, *where each lower-left zero block has size $n \times m$.*

Proof. Set $M := \left[\begin{smallmatrix} Y & X \\ 0 & Z \end{smallmatrix}\right]$ and $N := \left[\begin{smallmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{smallmatrix}\right]$.
By Proposition 3.7.3 and Proposition 3.1.2, $MN = \left[\begin{smallmatrix} I_m & -XZ^{-1}+XZ^{-1} \\ 0 & I_n \end{smallmatrix}\right] = I_{m+n}$ and $NM = \left[\begin{smallmatrix} I_m & Y^{-1}X-Y^{-1}X \\ 0 & I_n \end{smallmatrix}\right] = I_{m+n}$.
Thus $N = M^{-1}$ by Definition 3.1.1 and Lemma 3.1.3.
Structural Commentary: Two block products verify both inverse identities.
Key Insight: The off-diagonal correction cancels the coupling block. □

Theorem 3.8.9. *Inverse of an Upper Triangular Matrix*
Let $n \geq 1$ and let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ be upper triangular with $(\forall\, i \in \{1, \dots, n\})\, a_{ii} \neq 0$. Then:
(i) A is invertible.
(ii) A^{-1} is upper triangular.
(iii) The diagonal of A^{-1} is $(a_{11}^{-1}, \dots, a_{nn}^{-1})$.

Proof. Let P_n denote the conjunction of (i)–(iii) in size n under the stated hypotheses.
For $n = 1$, $A^{-1} = [a_{11}^{-1}]$ by $[a_{11}][a_{11}^{-1}] = I_1 = [a_{11}^{-1}][a_{11}]$, Definition 3.1.1, Lemma 3.1.3.
This inverse is upper triangular with diagonal (a_{11}^{-1}), so P_1 holds. Suppose $n > 1$ and P_{n-1} holds.
Write $A = \left[\begin{smallmatrix} B & \mathbf{x} \\ 0 & a_{nn} \end{smallmatrix}\right]$, where $B \in \mathrm{Mat}(n-1, n-1, \mathbb{F})$ is upper triangular with nonzero diagonal and $\mathbf{x} \in \mathbb{F}^{n-1}$.
(i) The blocks B, $[a_{nn}]$ invertible by induction hypothesis, Definition 3.1.1, respectively. Hence A invertible by Proposition 3.8.8.
(ii) The matrix B^{-1} is upper triangular by the induction hypothesis.
By Proposition 3.8.8, $A^{-1} = \left[\begin{smallmatrix} B^{-1} & -B^{-1}\mathbf{x}a_{nn}^{-1} \\ 0 & a_{nn}^{-1} \end{smallmatrix}\right]$, which is upper triangular by Definition 3.1.1.
(iii) By the induction hypothesis, the diagonal of B^{-1} is $(a_{11}^{-1}, \dots, a_{n-1,n-1}^{-1})$.
By the block formula in (ii), the diagonal of A^{-1} is $(a_{11}^{-1}, \dots, a_{n-1,n-1}^{-1}, a_{nn}^{-1})$.
Thus P_n holds, so $(\forall\, n \in \mathbb{Z}_{>0})\, P_n$ by Theorem 2.6.2.
Structural Commentary: Induction transfers all three claims through the block inverse formula.
Key Insight: Triangular inversion reduces dimension while preserving the zero pattern. □

4 Duality and Multilinear Structure

4.1 Dual Spaces and Dual Bases

Duality studies vectors through scalar-valued tests. This subsection develops algebraic duals, dual bases, coordinate functionals, separation by functionals, and their dimension consequences.

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.1.1. *Linear Functional and Dual Space*
(i) A *linear functional* on $\mathcal{V}$ is an element $\varphi \in \mathcal{L}(\mathcal{V}, \mathbb{F})$.
(ii) The *algebraic dual space* of $\mathcal{V}$ is $\mathcal{V}' := \mathcal{L}(\mathcal{V}, \mathbb{F})$.

Pointwise addition and scalar multiplication make $\mathcal{V}'$ a vector space over $\mathbb{F}$ by Proposition 1.5.5.

Example 4.1.2. *Evaluation and Integral Functionals*
(i) Define $\mathrm{ev}_0 : \mathbb{F}[x] \to \mathbb{F}$ by $\mathrm{ev}_0(p) := p(0)$.
The identity $(\forall\, \alpha, \beta \in \mathbb{F})(\forall\, p, q \in \mathbb{F}[x])\ \mathrm{ev}_0(\alpha p + \beta q) = \alpha\, \mathrm{ev}_0(p) + \beta\, \mathrm{ev}_0(q)$ proves $\mathrm{ev}_0 \in \mathcal{V}'$ for $\mathcal{V} = \mathbb{F}[x]$.
(ii) For $\mathbb{F} = \mathbb{R}$, let $C[a, b]$ be the real vector space of continuous functions on $[a, b]$ and define $I : C[a, b] \to \mathbb{R}$ by $I(f) := \int_a^b f(x)\, dx$.
By linearity of the integral, $(\forall\, \alpha, \beta \in \mathbb{R})(\forall\, f, g \in C[a, b])\, I(\alpha f + \beta g) = \alpha I(f) + \beta I(g)$, so $I \in C[a, b]'$.

Definition 4.1.3. *Dual Basis*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an ordered basis of $\mathcal{V}$, where $n \in \mathbb{N}_0$.
Its *dual basis* is the list $\beta' = (\varphi_1, \dots, \varphi_n)$ in $\mathcal{V}'$ characterized by $(\forall\, j, k \in \{1, \dots, n\})\, \varphi_j(\mathbf{v}_k) = \delta_{j,k}$.
If $n \geq 1$, each φ_j exists uniquely by Proposition 1.5.8; if $n = 0$, then $\beta' := ()$.

Theorem 4.1.4. *Dual Basis and Dimension of the Dual Space*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of the finite-dimensional space $\mathcal{V}$, and let $\beta' = (\varphi_1, \dots, \varphi_n)$ be its dual basis. Then:
(i) β' is a basis of $\mathcal{V}'$;
(ii) $\dim(\mathcal{V}') = \dim(\mathcal{V})$.

Proof.
(i) If $n = 0$, then $\mathcal{V} = \{\mathbf{0}\}$, and $\mathcal{V}' = \{0\}$ by Proposition 1.5.3; thus $()$ is a basis of $\mathcal{V}'$.
Suppose $n \geq 1$. If $\sum_{j=1}^{n} a_j \varphi_j = 0$, then $(\forall\, k \in \{1, \dots, n\})\, a_k = \sum_{j=1}^{n} a_j \varphi_j(\mathbf{v}_k) = 0$, so β' is linearly independent.
For $\varphi \in \mathcal{V}'$ and $\mathbf{v} = \sum_{k=1}^{n} c_k \mathbf{v}_k$, one has $\big(\sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j\big)(\mathbf{v}) = \sum_{j=1}^{n} \varphi(\mathbf{v}_j) c_j = \varphi(\mathbf{v})$ by Definition 4.1.3.
Hence $\varphi = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j$ by Assumption 3.1.2, and β' spans $\mathcal{V}'$.
(ii) Both bases have length n by (i), so $\dim(\mathcal{V}') = n = \dim(\mathcal{V})$ by Definition 1.4.2.
Structural Commentary: Evaluation on the original basis proves independence and reconstructs every functional.
Key Insight: Basis values are precisely the coordinates of a functional. □

Proposition 4.1.5. *Coordinate Functionals for a Hamel Basis*
Let $\mathcal{B} = \{\mathbf{b}_i\}_{i \in I}$ be Hamel basis of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, let $a_{\mathbf{v}} : I \to \mathbb{F}$ be unique finitely supported function satisfying $\mathbf{v} = \sum_{j \in \mathrm{supp}(a_{\mathbf{v}})} a_{\mathbf{v}}(j)\mathbf{b}_j$, and define $\varepsilon_i(\mathbf{v}) := a_{\mathbf{v}}(i)$ for $i \in I$. Then:
(i) $(\forall\, i \in I)\, \varepsilon_i \in \mathcal{V}'$;

(ii) $\{\varepsilon_i\}_{i\in I}$ *is linearly independent;*
(iii) $\operatorname{span}\{\varepsilon_i : i \in I\} = \mathcal{V}'$ *iff* $|I| < \infty$*;*
(iv) $|I| < \infty$ *iff* $\mathcal{V}$ *is finite-dimensional.*

Proof.

(i) For $\alpha, \beta \in \mathbb{F}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, the identity $a_{\alpha\mathbf{v}+\beta\mathbf{w}} = \alpha a_{\mathbf{v}} + \beta a_{\mathbf{w}}$ holds by uniqueness in Proposition 1.3.16.
Hence $(\forall\, i \in I)\, \varepsilon_i(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha\varepsilon_i(\mathbf{v}) + \beta\varepsilon_i(\mathbf{w})$, so $(\forall\, i \in I)\, \varepsilon_i \in \mathcal{V}'$ by Definition 4.1.1.

(ii) If $i_1, \dots, i_N \in I$ are distinct and $\sum_{r=1}^{N} c_r \varepsilon_{i_r} = 0$, then $(\forall\, s \in \{1, \dots, N\})\, c_s = \sum_{r=1}^{N} c_r \varepsilon_{i_r}(\mathbf{b}_{i_s}) = 0$.
Thus every finite relation is trivial, so $\{\varepsilon_i\}_{i\in I}$ is linearly independent.

(iii) $(\Rightarrow)$ Suppose $\operatorname{span}\{\varepsilon_i : i \in I\} = \mathcal{V}'$ and I is infinite. Define $\varphi(\mathbf{v}) := \sum_{j \in \operatorname{supp}(a_{\mathbf{v}})} a_{\mathbf{v}}(j)$.
By coordinate uniqueness, $(\forall\, \alpha, \beta \in \mathbb{F})(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\, \varphi(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha\varphi(\mathbf{v}) + \beta\varphi(\mathbf{w})$, so $\varphi \in \mathcal{V}'$
The equality $(\forall\, i \in I)\, \varphi(\mathbf{b}_i) = 1$ holds, whereas every element of $\operatorname{span}\{\varepsilon_i : i \in I\}$ vanishes on all but finitely many $\mathbf{b}_i$.
Thus $\varphi \notin \operatorname{span}\{\varepsilon_i : i \in I\}$, a contradiction, and $|I| < \infty$.

(iii) $(\Leftarrow)$ If I is finite, order $\mathcal{B}$. Then $\{\varepsilon_i\}_{i\in I}$ is its dual basis, so it spans $\mathcal{V}'$ by Theorem 4.1.4(i).

(iv) $(\Rightarrow)$ If I is finite, then any ordering of $\mathcal{B}$ is a finite basis, so $\mathcal{V}$ is finite-dimensional.

(iv) $(\Leftarrow)$ Suppose $\mathcal{V}$ has a spanning list of length n and I is infinite. Choose distinct $i_1, \dots, i_{n+1} \in I$.
The list $(\mathbf{b}_{i_1}, \dots, \mathbf{b}_{i_{n+1}})$ is linearly independent, so $n + 1 \le n$ by Theorem 1.3.11, a contradiction. Hence I is finite.

Structural Commentary: Coefficient uniqueness proves linearity and independence; finite support detects finite-dimensionality.
Key Insight: Finite coordinate support cannot represent a functional nonzero on every basis vector. □

Theorem 4.1.6. *Dimension of the Algebraic Dual*
Assume Zorn's lemma and the Choice Principle 6.1.2, and let $\mathcal{V}$ *be an* $\mathbb{F}$*-vector space. Then:*
(i) $\dim(\mathcal{V}) \le \dim(\mathcal{V}')$*;*
(ii) $\dim(\mathcal{V}) = \dim(\mathcal{V}')$ *iff* $\dim(\mathcal{V}) < \infty$*.*

Proof. Choose a Hamel basis $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ by Theorem 1.3.17(ii), set $\kappa := |I| = \dim(\mathcal{V})$ by Definition 1.4.6, and let $\{\varepsilon_i\}_{i\in I}$ denote the coordinate functionals associated with $\mathcal{B}$.

(i) The family $\{\varepsilon_i\}_{i\in I} \subseteq \mathcal{V}'$ is linearly independent by Proposition 4.1.5(ii).
Extend it to a Hamel basis $\mathcal{D}$ of $\mathcal{V}'$ by Theorem 1.3.17(iii).
The map $i \mapsto \varepsilon_i$ injects I into $\mathcal{D}$, so $\kappa = |I| \le |\mathcal{D}| = \dim(\mathcal{V}')$ by Proposition 6.5.2(i) and Definition 1.4.6.

(ii) $(\Rightarrow)$ Suppose $\dim(\mathcal{V}) = \dim(\mathcal{V}')$. Since $\kappa = \dim(\mathcal{V})$, assume for contradiction that κ is infinite.

(a) *Countable scalar core*: Set $D := \{(m, n) \in \mathbb{Z}^2 : n \cdot 1 \neq 0\}$ and $\mathbb{F}_0 := \{(m \cdot 1)(n \cdot 1)^{-1} : (m, n) \in D\} \subseteq \mathbb{F}$.
The elements $0, 1$ belong to $\mathbb{F}_0$. If $x = (m \cdot 1)(n \cdot 1)^{-1}$ and $y = (r \cdot 1)(s \cdot 1)^{-1}$, then $x - y = ((ms - rn) \cdot 1)((ns) \cdot 1)^{-1}$ and $xy = ((mr) \cdot 1)((ns) \cdot 1)^{-1}$; if $x \neq 0$, then $x^{-1} = (n \cdot 1)(m \cdot 1)^{-1}$.
Thus $\mathbb{F}_0$ is a subfield. The set D is at most countable by Proposition 6.2.10, Corollary 6.2.17(ii), and Proposition 6.2.12(i), so $\mathbb{F}_0$ is at most countable by Proposition 6.2.12(ii). The map $E \mapsto \mathbf{1}_E$ injects $\mathcal{P}(I)$ into $\mathbb{F}_0^I$.
The map $i \mapsto \{i\}$ injects I into $\mathcal{P}(I)$, while $I \not\sim_{\#} \mathcal{P}(I)$ by Theorem 6.4.1.
Hence $\kappa < |\mathcal{P}(I)| \le |\mathbb{F}_0^I|$ by Proposition 6.5.2(i),(ii).

(b) *Core-space dimension*: Suppose $\dim_{\mathbb{F}_0}(\mathbb{F}_0^I) \le \kappa$. Choose a Hamel basis $\mathcal{C}$ of $\mathbb{F}_0^I$ by Theorem 1.3.17(ii).
Since $|\mathcal{C}| = \dim_{\mathbb{F}_0}(\mathbb{F}_0^I) \le \kappa$, an injection $\iota : \mathcal{C} \to \kappa$ exists by Proposition 6.5.2(i).
By Proposition 1.3.16, the increasing list of ι-images of the support, paired with its nonzero coefficients, injectively encodes each $f \in \mathbb{F}_0^I$ in $\bigcup_{n<\omega} \big(\{n\} \times (\kappa \times \mathbb{F}_0)^n\big)$. Since $\mathbb{F}_0$ is at most countable and κ is infinite, each finite power injects into κ, and the tagged union injects into $\omega \times \kappa$, by Proposition 6.3.15(iv) and Theorem 6.6.3(ii).
Therefore $|\mathbb{F}_0^I| \le \kappa$ by Proposition 6.5.2(i), contrary to (a). Thus $\dim_{\mathbb{F}_0}(\mathbb{F}_0^I) > \kappa$.

(c) *Scalar transfer*: Choose an $\mathbb{F}_0$-basis $\mathcal{C}$ of $\mathbb{F}_0^I$ by Theorem 1.3.17(ii). For $\mathbf{v} \in \mathcal{V}$, a unique finitely supported $\mathcal{B}$-coefficient function $a_{\mathbf{v}} : I \to \mathbb{F}$ exists by Proposition 1.3.16. For $g \in \mathcal{C}$, define $\Phi_g : \mathcal{V} \to \mathbb{F}$ by $\Phi_g(\mathbf{v}) := \sum_{i \in \operatorname{supp}(a_{\mathbf{v}})} a_{\mathbf{v}}(i) g(i)$.
The identity $a_{\alpha\mathbf{v}+\beta\mathbf{w}} = \alpha a_{\mathbf{v}} + \beta a_{\mathbf{w}}$ holds by coordinate uniqueness.
Hence $(\forall\, \alpha, \beta \in \mathbb{F})(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\, \Phi_g(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha\Phi_g(\mathbf{v}) + \beta\Phi_g(\mathbf{w})$ and $\Phi_g \in \mathcal{V}'$.
Let $N \in \mathbb{N}$, let $g_1, \dots, g_N \in \mathcal{C}$ be distinct, and suppose $\sum_{r=1}^{N} c_r \Phi_{g_r} = 0$.
Equality $(\forall\, i \in I)\, \sum_{r=1}^{N} c_r g_r(i) = 0$ holds by evaluation at $\mathbf{b}_i$. Choose $\mathbb{F}_0$-basis $(e_1, \dots, e_s)$ of $\operatorname{span}_{\mathbb{F}_0}\{c_1, \dots, c_N\}$ by Corollary 1.3.14, and $d_{\ell r} \in \mathbb{F}_0$ satisfying $(\forall\, r \in \{1, \dots, N\})\, c_r = \sum_{\ell=1}^{s} d_{\ell r} e_\ell$. For $i \in I$, $0 = \sum_{r=1}^{N} c_r g_r(i) = \sum_{\ell=1}^{s} e_\ell \big(\sum_{r=1}^{N} d_{\ell r} g_r(i) \big)$.
Hence $(\forall\, \ell \in \{1, \dots, s\})(\forall\, i \in I)\, \sum_{r=1}^{N} d_{\ell r} g_r(i) = 0$ by independence of $(e_1, \dots, e_s)$.
$(\forall\, \ell \in \{1, \dots, s\})\, \sum_{r=1}^{N} d_{\ell r} g_r = 0$ by Assumption 3.1.2, $(\forall\, \ell \in \{1, \dots, s\})(\forall\, r \in \{1, \dots, N\})\, d_{\ell r} = 0$ by independence of $\mathcal{C}$.
Hence $(\forall\, r \in \{1, \dots, N\})\, c_r = 0$, so $g \mapsto \Phi_g$ is injective and $\{\Phi_g : g \in \mathcal{C}\}$ is $\mathbb{F}$-linearly independent.
Extend this set to a Hamel basis $\mathcal{E}$ of $\mathcal{V}'$ by Theorem 1.3.17(iii). By (b), $\dim_{\mathbb{F}}(\mathcal{V}') = |\mathcal{E}| \ge |\mathcal{C}| = \dim_{\mathbb{F}_0}(\mathbb{F}_0^I) > \kappa = \dim_{\mathbb{F}}(\mathcal{V})$ by Proposition 6.5.2(i) and Definition 1.4.6, contrary to the assumed equality.

(ii) $(\Leftarrow)$ If $\dim(\mathcal{V}) < \infty$, then $\dim(\mathcal{V}') = \dim(\mathcal{V})$ by Theorem 4.1.4(ii).

Structural Commentary: A countable scalar core transfers Cantor-sized independence from functions to functionals.
Key Insight: Scalar extension preserves independence of core-field coordinate functions. □

Proposition 4.1.7. *Functionals Separate Points*
Assume Zorn's lemma, and let $\mathcal{V}$ *be a vector space.*
(i) $(\forall\, \mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\})(\exists\, \varphi \in \mathcal{V}')\, \varphi(\mathbf{v}) \neq 0$.
(ii) $(\forall\, \mathbf{v} \in \mathcal{V})\, [\mathbf{v} = \mathbf{0} \iff (\forall\, \varphi \in \mathcal{V}')\, \varphi(\mathbf{v}) = 0]$.

(iii) *If* $\mathcal{U} \leq \mathcal{V}$ *and* $\mathbf{x} \in \mathcal{V} \setminus \mathcal{U}$, *then* $(\exists \varphi \in \mathcal{V}')\,[\varphi|_{\mathcal{U}} = 0 \wedge \varphi(\mathbf{x}) = 1]$.
(iv) *If* $\varphi \in \mathcal{V}'$ *and* $\mathbf{x} \in \mathcal{V}$ *satisfy* $\varphi(\mathbf{x}) \neq 0$, *then* $\mathcal{V} = \operatorname{span}(\mathbf{x}) \oplus \ker(\varphi)$.
(v) *If* $0 \neq \varphi, \psi \in \mathcal{V}'$ *and* $\ker(\varphi) = \ker(\psi)$, *then* $(\exists a \in \mathbb{F} \setminus \{0\})\,\varphi = a\psi$.

Proof.

(*i*) Fix $\mathbf{v} \neq \mathbf{0}$. The set $\{\mathbf{v}\}$ is linearly independent, so it extends to a Hamel basis $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ by Theorem 1.3.17(iii).
Let $\{\varepsilon_i\}_{i\in I}$ denote its coordinate functionals, choose $i_0 \in I$ with $\mathbf{b}_{i_0} = \mathbf{v}$. Then $\varepsilon_{i_0} \in \mathcal{V}'$ by Proposition 4.1.5(i), and $\varepsilon_{i_0}(\mathbf{v}) = 1$.
(*ii*) ($\Rightarrow$) If $\mathbf{v} = \mathbf{0}$, then $(\forall \varphi \in \mathcal{V}')\,\varphi(\mathbf{v}) = 0$ by Proposition 1.5.3.
(*ii*) ($\Leftarrow$) If $\mathbf{v} \neq \mathbf{0}$, then $(\exists \psi \in \mathcal{V}')\,\psi(\mathbf{v}) \neq 0$ by (i), contrary to the assumed vanishing. Hence $\mathbf{v} = \mathbf{0}$.
(*iii*) Choose a Hamel basis $\mathcal{B}_{\mathcal{U}}$ of $\mathcal{U}$ by Theorem 1.3.17(ii).
Since $\mathbf{x} \notin \operatorname{span}(\mathcal{B}_{\mathcal{U}})$, every finite relation on $\mathcal{B}_{\mathcal{U}} \cup \{\mathbf{x}\}$ has zero $\mathbf{x}$-coefficient and is then trivial by independence of $\mathcal{B}_{\mathcal{U}}$.
Thus $\mathcal{B}_{\mathcal{U}} \cup \{\mathbf{x}\}$ is linearly independent and extends to a Hamel basis $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ of $\mathcal{V}$ by Theorem 1.3.17(iii).
Let $\{\varepsilon_i\}_{i\in I}$ denote its coordinate functionals, choose $i_0 \in I$ with $\mathbf{b}_{i_0} = \mathbf{x}$, and set $\varphi := \varepsilon_{i_0}$.
Then $\varphi \in \mathcal{V}'$ by Proposition 4.1.5(i), $\varphi(\mathbf{x}) = 1$, and $\varphi|_{\mathcal{U}} = 0$ by $\mathcal{U} = \operatorname{span}(\mathcal{B}_{\mathcal{U}})$.
(*iv*) If $c\mathbf{x} \in \ker(\varphi)$, then $c\varphi(\mathbf{x}) = 0$, so $c = 0$; hence $\operatorname{span}(\mathbf{x}) \cap \ker(\varphi) = \{\mathbf{0}\}$. For $\mathbf{v} \in \mathcal{V}$, set $c_{\mathbf{v}} := \varphi(\mathbf{v})\varphi(\mathbf{x})^{-1}$.
Then $\mathbf{v} = c_{\mathbf{v}}\mathbf{x} + (\mathbf{v} - c_{\mathbf{v}}\mathbf{x})$, with $\mathbf{v} - c_{\mathbf{v}}\mathbf{x} \in \ker(\varphi)$. Thus $\mathcal{V} = \operatorname{span}(\mathbf{x}) \oplus \ker(\varphi)$ by Proposition 2.1.8.
(*v*) Set $\mathcal{K} := \ker(\varphi) = \ker(\psi)$. Since $\varphi \neq 0$, choose $\mathbf{x} \in \mathcal{V} \setminus \mathcal{K}$, set $a := \varphi(\mathbf{x})\psi(\mathbf{x})^{-1} \neq 0$. The identity $\mathcal{V} = \operatorname{span}(\mathbf{x}) \oplus \mathcal{K}$ holds by (iv).
Fix $\mathbf{v} = c\mathbf{x} + \mathbf{k}$ with $c \in \mathbb{F}$ and $\mathbf{k} \in \mathcal{K}$. Then $\varphi(\mathbf{v}) = c\varphi(\mathbf{x}) = ac\psi(\mathbf{x}) = a\psi(\mathbf{v})$. Hence $\varphi = a\psi$ by Assumption 3.1.2.

Structural Commentary: Basis extension constructs separating coordinates; kernel decomposition then compares nonzero functionals.
Key Insight: One normalized coordinate detects any direction outside a prescribed subspace. □

Corollary 4.1.8. *Kernels of Nonzero Functionals*
If $0 \neq \varphi \in \mathcal{V}'$, *then:*

(i) φ *is surjective;*
(ii) $\mathcal{V}/\ker(\varphi) \cong \mathbb{F}$;
(iii) $\operatorname{codim}_{\mathcal{V}}(\ker(\varphi)) = 1$.

Proof. Choose $\mathbf{x} \in \mathcal{V}$ with $\varphi(\mathbf{x}) \neq 0$.

(i) $(\forall a \in \mathbb{F})\,\varphi(a\varphi(\mathbf{x})^{-1}\mathbf{x}) = a$, so $\operatorname{range}(\varphi) = \mathbb{F}$.
(ii) The isomorphism $\mathcal{V}/\ker(\varphi) \cong \operatorname{range}(\varphi) = \mathbb{F}$ holds by Theorem 2.2.12 and (i).
(iii) Choose an isomorphism $T : \mathcal{V}/\ker(\varphi) \to \mathbb{F}$ by (ii). Since (1) is a basis of $\mathbb{F}$ over itself, $(T^{-1}(1))$ is a basis of $\mathcal{V}/\ker(\varphi)$ by Proposition 1.7.8. Hence $\operatorname{codim}_{\mathcal{V}}(\ker(\varphi)) = \dim(\mathcal{V}/\ker(\varphi)) = 1$ by Definition 2.3.11 and Definition 1.4.2.

Structural Commentary: Surjectivity identifies the quotient with the one-dimensional scalar field.
Key Insight: A nonzero functional removes exactly one scalar direction. □

4.2 Double Duals and Dual Maps

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.2.1. *Algebraic Double Dual and Natural Map*

(i) The *algebraic double dual* of $\mathcal{V}$ is $\mathcal{V}'' := (\mathcal{V}')'$.
(ii) The *natural evaluation map* $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ is defined by $(\forall \mathbf{v} \in \mathcal{V})(\forall \varphi \in \mathcal{V}')\,\Lambda_{\mathcal{V}}(\mathbf{v})(\varphi) := \varphi(\mathbf{v})$.
For $\mathbf{v} \in \mathcal{V}$, $(\forall a, b \in \mathbb{F})(\forall \varphi, \psi \in \mathcal{V}')\,\Lambda_{\mathcal{V}}(\mathbf{v})(a\varphi + b\psi) = a\varphi(\mathbf{v}) + b\psi(\mathbf{v})$ by pointwise operations in Proposition 1.5.5, so $\Lambda_{\mathcal{V}}(\mathbf{v}) \in \mathcal{V}''$.
(iii) The space $\mathcal{V}$ is *algebraically reflexive* iff $\Lambda_{\mathcal{V}}$ is surjective.

Theorem 4.2.2. *Algebraic Reflexivity*
Let $\mathcal{V}$ *be a vector space. Claims (iii)–(v) assume Zorn's lemma, and (v) additionally assumes the Choice Principle 6.1.2. Then:*

(i) $\Lambda_{\mathcal{V}} \in \mathcal{L}(\mathcal{V}, \mathcal{V}'')$;
(ii) *if* $\dim(\mathcal{V}) < \infty$, *then* $\Lambda_{\mathcal{V}}$ *is an isomorphism;*
(iii) $\Lambda_{\mathcal{V}}$ *is injective;*
(iv) $\Lambda_{\mathcal{V}}$ *is an isomorphism iff* $\mathcal{V}$ *is algebraically reflexive;*
(v) $\mathcal{V}$ *is algebraically reflexive iff* $\dim(\mathcal{V}) < \infty$.

Proof.

(*i*) For $a, b \in \mathbb{F}$, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, and $\varphi \in \mathcal{V}'$, one has $\Lambda_{\mathcal{V}}(a\mathbf{u} + b\mathbf{v})(\varphi) = \varphi(a\mathbf{u} + b\mathbf{v}) = \big(a\Lambda_{\mathcal{V}}(\mathbf{u}) + b\Lambda_{\mathcal{V}}(\mathbf{v})\big)(\varphi)$.
Hence $\Lambda_{\mathcal{V}}(a\mathbf{u} + b\mathbf{v}) = a\Lambda_{\mathcal{V}}(\mathbf{u}) + b\Lambda_{\mathcal{V}}(\mathbf{v})$ by Assumption 3.1.2, so $\Lambda_{\mathcal{V}} \in \mathcal{L}(\mathcal{V}, \mathcal{V}'')$.
(*ii*) Suppose $\dim(\mathcal{V}) < \infty$, let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis, and let $\beta' = (\varphi_1, \ldots, \varphi_n)$ be its dual basis.
If $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{v}_j \in \ker(\Lambda_{\mathcal{V}})$, then $(\forall k \in \{1, \ldots, n\})\,c_k = \Lambda_{\mathcal{V}}(\mathbf{v})(\varphi_k) = 0$; hence $\mathbf{v} = \mathbf{0}$, also for $n = 0$. Thus $\Lambda_{\mathcal{V}}$ is injective by Proposition 1.6.2. By Theorem 4.1.4(ii) for $\mathcal{V}$, $\dim(\mathcal{V}') = \dim(\mathcal{V}) < \infty$; by the same theorem for $\mathcal{V}'$, $\dim(\mathcal{V}'') = \dim(\mathcal{V}')$.
Therefore $\Lambda_{\mathcal{V}}$ is surjective by Corollary 1.7.6(ii) and is an isomorphism by Proposition 1.6.7.
(*iii*) Assume Zorn's lemma. By Proposition 1.6.2, it suffices to prove $\ker(\Lambda_{\mathcal{V}}) = \{\mathbf{0}\}$.
If $\mathbf{v} \in \ker(\Lambda_{\mathcal{V}})$, then $(\forall \varphi \in \mathcal{V}')\,\varphi(\mathbf{v}) = 0$, so $\mathbf{v} = \mathbf{0}$ by Proposition 4.1.7(ii).
(*iv*) ($\Rightarrow$) If $\Lambda_{\mathcal{V}}$ is an isomorphism, then it is surjective by Proposition 1.6.7; hence $\mathcal{V}$ is algebraically reflexive by Definition 4.2.1(iii).
(*iv*) ($\Leftarrow$) By Proposition 1.6.7, it suffices to prove that $\Lambda_{\mathcal{V}}$ is bijective. It is surjective by algebraic reflexivity and injective by (iii).
(*v*) ($\Rightarrow$) Suppose $\mathcal{V}$ is algebraically reflexive. Then $\Lambda_{\mathcal{V}}$ is an isomorphism by (iv).
If $\dim(\mathcal{V})$ were infinite, then $\dim(\mathcal{V}) < \dim(\mathcal{V}') < \dim(\mathcal{V}'')$ by Theorem 4.1.6(i),(ii), first for $\mathcal{V}$ and then for $\mathcal{V}'$.
Choose a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.3.17(ii). The set $\Lambda_{\mathcal{V}}(\mathcal{B})$ is a Hamel basis of $\mathcal{V}''$ by linearity, injectivity, and surjectivity of $\Lambda_{\mathcal{V}}$: linearity and injectivity preserve finite independence, while linearity and surjectivity preserve spanning.
Thus $\dim(\mathcal{V}) = \dim(\mathcal{V}'')$ by Definition 1.4.6, a contradiction. Hence $\dim(\mathcal{V}) < \infty$.
(*v*) ($\Leftarrow$) If $\dim(\mathcal{V}) < \infty$, then $\Lambda_{\mathcal{V}}$ is surjective by (ii) and Proposition 1.6.7; hence $\mathcal{V}$ is algebraically reflexive by Definition 4.2.1(iii).

Structural Commentary: Evaluation proves linearity, separation proves injectivity, and dual dimensions obstruct infinite reflexivity.
Key Insight: Canonical evaluation is surjective exactly before dual dimension grows. □

Perspective 4.2.3. *Why the Double Dual Is Natural*
An isomorphism $\mathcal{V} \cong \mathcal{V}'$ for finite-dimensional $\mathcal{V}$ generally depends on a chosen basis. In contrast, $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ is defined solely by evaluation and requires no basis.

Definition 4.2.4. *Dual Map*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, its *dual map* is $T' : \mathcal{W}' \to \mathcal{V}'$ defined by $(\forall\, \varphi \in \mathcal{W}')\, T'\varphi := \varphi T$.
The relation $(\forall\, \varphi \in \mathcal{W}')\, \varphi T \in \mathcal{L}(\mathcal{V}, \mathbb{F}) = \mathcal{V}'$ holds by Proposition 1.5.7(i). Equivalently, $(\forall\, \varphi \in \mathcal{W}')(\forall\, \mathbf{v} \in \mathcal{V})\, (T'\varphi)(\mathbf{v}) = \varphi(T\mathbf{v})$.

Proposition 4.2.5. *Dual Maps Are Linear*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$.

Proof. For $a, b \in \mathbb{F}$, $\varphi, \psi \in \mathcal{W}'$, and $\mathbf{v} \in \mathcal{V}$, one has $\big(T'(a\varphi + b\psi)\big)(\mathbf{v}) = (a\varphi + b\psi)(T\mathbf{v}) = \big(aT'\varphi + bT'\psi\big)(\mathbf{v})$.
Hence $T'(a\varphi + b\psi) = aT'\varphi + bT'\psi$ by Assumption 3.1.2, so $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$ by Definition 1.5.1.
Structural Commentary: Pointwise evaluation reduces linearity of the pullback to linearity of each functional.
Key Insight: Precomposition preserves linear combinations while reversing map direction. □

Theorem 4.2.6. *Algebra of Dual Maps*
(i) If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a, b \in \mathbb{F}$, then $(aS + bT)' = aS' + bT'$.
(ii) If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $(ST)' = T'S'$.
(iii) $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.

Proof.
(i) For $\varphi \in \mathcal{W}'$ and $\mathbf{v} \in \mathcal{V}$, $\big((aS + bT)'\varphi\big)(\mathbf{v}) = \varphi(aS\mathbf{v} + bT\mathbf{v}) = \big(aS'\varphi + bT'\varphi\big)(\mathbf{v})$.
Hence $(aS + bT)' = aS' + bT'$ by Assumption 3.1.2 on $\mathcal{V}$ and $\mathcal{W}'$.
(ii) For $\varphi \in \mathcal{W}'$ and $\mathbf{u} \in \mathcal{U}$, $\big((ST)'\varphi\big)(\mathbf{u}) = \varphi(S(T\mathbf{u})) = \big(T'(S'\varphi)\big)(\mathbf{u})$. Hence $(ST)' = T'S'$ by Assumption 3.1.2 on $\mathcal{U}$ and $\mathcal{W}'$.
(iii) For $\varphi \in \mathcal{V}'$ and $\mathbf{v} \in \mathcal{V}$, $\big((I_{\mathcal{V}})'\varphi\big)(\mathbf{v}) = \varphi(I_{\mathcal{V}}\mathbf{v}) = \varphi(\mathbf{v})$. Hence $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$ by Assumption 3.1.2 on $\mathcal{V}$ and $\mathcal{V}'$.
Structural Commentary: Function extensionality lifts pointwise linearity, composition, and identity calculations to dual maps.
Key Insight: Precomposition reverses products but preserves linear structure. □

Corollary 4.2.7. *Dual Map of an Inverse*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is an isomorphism, then:
(i) T' is an isomorphism;
(ii) $(T^{-1})' = (T')^{-1}$.

Proof. Set $S := (T^{-1})'$. Then $S \in \mathcal{L}(\mathcal{V}', \mathcal{W}')$ by Proposition 4.2.5.
(i) By Theorem 4.2.6(ii)–(iii), $T'S = (T^{-1}T)' = (I_{\mathcal{V}})' = I_{\mathcal{V}'}$ and $ST' = (TT^{-1})' = (I_{\mathcal{W}})' = I_{\mathcal{W}'}$.
Thus S is a two-sided inverse of T', and T' is an isomorphism by Definition 1.6.4.
(ii) The identity $S = (T')^{-1}$ holds by Proposition 1.6.5; hence $(T^{-1})' = (T')^{-1}$.
Structural Commentary: Reversed product identities construct the dual inverse; uniqueness identifies it.
Key Insight: Contravariance turns both inverse identities into dual identities. □

Corollary 4.2.8. *Double Dual Maps Identify with Original Maps*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and set $T'' := (T')'$. Then:
(i) $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}} T$;
(ii) if $\dim(\mathcal{V}), \dim(\mathcal{W}) < \infty$, then $T'' = \Lambda_{\mathcal{W}} T \Lambda_{\mathcal{V}}^{-1}$.

Proof. $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$ and $T'' \in \mathcal{L}(\mathcal{V}'', \mathcal{W}'')$ by Proposition 4.2.5, first for T and then for T'.
(i) For $\mathbf{v} \in \mathcal{V}$ and $\psi \in \mathcal{W}'$, $\big(T''\Lambda_{\mathcal{V}}(\mathbf{v})\big)(\psi) = \Lambda_{\mathcal{V}}(\mathbf{v})(T'\psi) = (T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = \Lambda_{\mathcal{W}}(T\mathbf{v})(\psi)$.
Hence $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}} T$ by Assumption 3.1.2 on $\mathcal{W}'$ and $\mathcal{V}$.
(ii) If $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional, then $\Lambda_{\mathcal{V}}$ and $\Lambda_{\mathcal{W}}$ are isomorphisms by Theorem 4.2.2(ii).
Thus $T'' = T''I_{\mathcal{V}''} = T''\Lambda_{\mathcal{V}}\Lambda_{\mathcal{V}}^{-1} = \Lambda_{\mathcal{W}} T \Lambda_{\mathcal{V}}^{-1}$ by (i) and Proposition 1.5.7(ii)–(iii).
Structural Commentary: Pointwise evaluation proves naturality; finite reflexivity transports T to its double dual.
Key Insight: Both routes evaluate each functional at the same image vector. □

4.3 Annihilators and Transposes

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.3.1. *Annihilator*
If $\mathcal{U} \subseteq \mathcal{V}$, the *annihilator of $\mathcal{U}$ in $\mathcal{V}'$* is $\mathcal{U}^0 := \{\varphi \in \mathcal{V}' : (\forall\, \mathbf{u} \in \mathcal{U})\, \varphi(\mathbf{u}) = 0\}$.
The ambient space $\mathcal{V}$ is understood from context; changing it changes the dual space containing $\mathcal{U}^0$.

Proposition 4.3.2. *Annihilators Are Subspaces*
If $\mathcal{U} \subseteq \mathcal{V}$, then $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.

Proof. The zero functional belongs to $\mathcal{U}^0$. If $\varphi, \psi \in \mathcal{U}^0$, $a \in \mathbb{F}$, and $\mathbf{u} \in \mathcal{U}$, then $(a\varphi + \psi)(\mathbf{u}) = a\varphi(\mathbf{u}) + \psi(\mathbf{u}) = 0$.
Hence $a\varphi + \psi \in \mathcal{U}^0$, and $\mathcal{U}^0 \leq \mathcal{V}'$ by Theorem 1.2.2.
Structural Commentary: The subspace test reduces closure to the preservation of pointwise vanishing.
Key Insight: Linear combinations preserve every annihilating constraint. □

Theorem 4.3.3. *Dimension of the Annihilator*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Then $\dim(\mathcal{U}) + \dim(\mathcal{U}^0) = \dim(\mathcal{V})$.

Proof. Choose basis $(\mathbf{u}_1,\dots,\mathbf{u}_m)$ of $\mathcal{U}$ by Corollary 1.3.14, extend to basis $\beta=(\mathbf{u}_1,\dots,\mathbf{u}_m,\mathbf{w}_1,\dots,\mathbf{w}_k)$ of $\mathcal{V}$ by Theorem 1.3.13.
Let $\beta'=(\varphi_1,\dots,\varphi_m,\psi_1,\dots,\psi_k)$ be the dual basis.
By Theorem 4.1.4(i) and Definition 4.1.3, every $\varphi\in\mathcal{V}'$ satisfies $\varphi=\sum_{j=1}^{m}\varphi(\mathbf{u}_j)\varphi_j+\sum_{\ell=1}^{k}\varphi(\mathbf{w}_\ell)\psi_\ell$. Thus $\varphi\in\mathcal{U}^0$ implies $\varphi=\sum_{\ell=1}^{k}\varphi(\mathbf{w}_\ell)\psi_\ell$.
Conversely, each ψ_ℓ vanishes on $\mathcal{U}=\operatorname{span}(\mathbf{u}_1,\dots,\mathbf{u}_m)$ by Definition 4.1.3.
Hence $(\psi_1,\dots,\psi_k)$ spans $\mathcal{U}^0$ and is linearly independent as a sublist of β', so it is a basis of $\mathcal{U}^0$.
The adapted and annihilator bases have lengths $m+k$ and k, respectively, so $\dim(\mathcal{U})+\dim(\mathcal{U}^0)=m+k=\dim(\mathcal{V})$.
Structural Commentary: An adapted basis identifies the annihilator with the dual coordinates outside $\mathcal{U}$.
Key Insight: Vanishing deletes precisely the dual coordinates indexed by $\mathcal{U}$. □

Theorem 4.3.4. *Annihilator Lattice Identities*
Let $\mathcal{V}$ be finite-dimensional, and let $\mathcal{S},\mathcal{T}$ be subspaces of $\mathcal{V}$.
(i) $\mathcal{S}\subseteq\mathcal{T}\Rightarrow\mathcal{T}^0\subseteq\mathcal{S}^0$.
(ii) $(\mathcal{S}+\mathcal{T})^0=\mathcal{S}^0\cap\mathcal{T}^0$.
(iii) $(\mathcal{S}\cap\mathcal{T})^0=\mathcal{S}^0+\mathcal{T}^0$.

Proof.
(i) If $\mathcal{S}\subseteq\mathcal{T}$ and $\varphi\in\mathcal{T}^0$, then $(\forall\,\mathbf{s}\in\mathcal{S})\,\varphi(\mathbf{s})=0$, so $\varphi\in\mathcal{S}^0$.
(ii) If $\varphi\in(\mathcal{S}+\mathcal{T})^0$, then $\varphi|_{\mathcal{S}}=0$ and $\varphi|_{\mathcal{T}}=0$, so $\varphi\in\mathcal{S}^0\cap\mathcal{T}^0$.
Conversely, if $\varphi\in\mathcal{S}^0\cap\mathcal{T}^0$, then $(\forall\,\mathbf{s}\in\mathcal{S})(\forall\,\mathbf{t}\in\mathcal{T})\,\varphi(\mathbf{s}+\mathbf{t})=\varphi(\mathbf{s})+\varphi(\mathbf{t})=0$, so $\varphi\in(\mathcal{S}+\mathcal{T})^0$.
(iii) By (i), $\mathcal{S}^0+\mathcal{T}^0\subseteq(\mathcal{S}\cap\mathcal{T})^0$. By (ii) and Theorem 2.3.1, $\dim(\mathcal{S}^0+\mathcal{T}^0)=\dim(\mathcal{S}^0)+\dim(\mathcal{T}^0)-\dim((\mathcal{S}+\mathcal{T})^0)$.
The identity $\dim(\mathcal{S}^0+\mathcal{T}^0)=\dim(\mathcal{V})-\dim(\mathcal{S}\cap\mathcal{T})=\dim((\mathcal{S}\cap\mathcal{T})^0)$ holds by Theorem 4.3.3 and Theorem 2.3.1.
Hence the inclusion is an equality by Proposition 1.4.3.
Structural Commentary: Two direct inclusions establish order reversal and sums; a dimension comparison recovers intersections.
Key Insight: Annihilation reverses the finite-dimensional subspace lattice. □

Theorem 4.3.5. *Double Annihilator*
Let $\dim(\mathcal{V})<\infty$ and $M\subseteq\mathcal{V}$, and set $M^{00}:=(M^0)^0\subseteq\mathcal{V}''$. Then:
(i) $M^{00}=\Lambda_{\mathcal{V}}(\operatorname{span}(M))$;
(ii) if $\mathcal{S}\leq\mathcal{V}$, then $\mathcal{S}^{00}=\Lambda_{\mathcal{V}}(\mathcal{S})$, and the corestriction $\Lambda_{\mathcal{V}}|_{\mathcal{S}}:\mathcal{S}\to\mathcal{S}^{00}$ is an isomorphism.

Proof.
(i) If $\mathbf{v}\in\operatorname{span}(M)$ and $\varphi\in M^0$, then $\Lambda_{\mathcal{V}}(\mathbf{v})(\varphi)=\varphi(\mathbf{v})=0$ by linearity of φ, so $\Lambda_{\mathcal{V}}(\operatorname{span}(M))\subseteq M^{00}$.
For the reverse inclusion, let $\eta\in M^{00}$. The relation $(\exists!\,\mathbf{v}\in\mathcal{V})\,\eta=\Lambda_{\mathcal{V}}(\mathbf{v})$ holds by Theorem 4.2.2(ii); fix this $\mathbf{v}$.
Suppose $\mathbf{v}\notin\operatorname{span}(M)$. Choose a basis $(\mathbf{u}_1,\dots,\mathbf{u}_m)$ of $\operatorname{span}(M)$ by Corollary 1.3.14.
The list $(\mathbf{u}_1,\dots,\mathbf{u}_m,\mathbf{v})$ is linearly independent by Proposition 1.3.7 and extends to a basis of $\mathcal{V}$ by Theorem 1.3.13.
Its dual-basis functional φ corresponding to $\mathbf{v}$ satisfies $\varphi|_{\operatorname{span}(M)}=0$ and $\varphi(\mathbf{v})=1$.
Thus $\varphi\in M^0$ but $\eta(\varphi)=1$, contrary to $\eta\in M^{00}$. Hence $M^{00}\subseteq\Lambda_{\mathcal{V}}(\operatorname{span}(M))$.
(ii) The identity $\mathcal{S}^{00}=\Lambda_{\mathcal{V}}(\mathcal{S})$ holds by (i) with $M=\mathcal{S}$. The corestriction is linear and injective by Theorem 4.2.2(i)–(ii), and it is surjective onto $\mathcal{S}^{00}$. Therefore it is an isomorphism by Proposition 1.6.7.
Structural Commentary: One inclusion extends vanishing; an adapted dual basis separates every vector outside the span.
Key Insight: Double annihilation remembers exactly the generated subspace. □

Theorem 4.3.6. *Direct Sums and Annihilators*
Suppose $\mathcal{V}=\mathcal{S}\oplus\mathcal{T}$, and define $E:\mathcal{T}'\to\mathcal{S}^0$ by $(\forall\,\varphi\in\mathcal{T}')(\forall\,\mathbf{s}\in\mathcal{S})(\forall\,\mathbf{t}\in\mathcal{T})\,(E\varphi)(\mathbf{s}+\mathbf{t}):=\varphi(\mathbf{t})$. Then:
(i) E is an isomorphism;
(ii) $\mathcal{V}'=\mathcal{S}^0\oplus\mathcal{T}^0$.

Proof.
(i) Each $\mathbf{v}\in\mathcal{V}$ has a unique representation $\mathbf{v}=\mathbf{s}+\mathbf{t}$ by Proposition 2.1.8, so E is well-defined.
If $\mathbf{v}_i=\mathbf{s}_i+\mathbf{t}_i$ for $i=1,2$ and $a,b\in\mathbb{F}$, then $(E\varphi)(a\mathbf{v}_1+b\mathbf{v}_2)=\varphi(a\mathbf{t}_1+b\mathbf{t}_2)=a(E\varphi)(\mathbf{v}_1)+b(E\varphi)(\mathbf{v}_2)$, so $E\varphi\in\mathcal{V}'$.
The relation $(\forall\,\mathbf{s}\in\mathcal{S})\,(E\varphi)(\mathbf{s})=0$ places $E\varphi$ in $\mathcal{S}^0$. For $a,b\in\mathbb{F}$ and $\varphi,\psi\in\mathcal{T}'$, the identity $E(a\varphi+b\psi)=aE\varphi+bE\psi$ holds pointwise and hence by Assumption 3.1.2, so $E\in\mathcal{L}(\mathcal{T}',\mathcal{S}^0)$. Define $R:\mathcal{S}^0\to\mathcal{T}'$ by $(\forall\,\eta\in\mathcal{S}^0)\,R\eta:=\eta|_{\mathcal{T}}$.
If $\eta\in\mathcal{S}^0$, $a,b\in\mathbb{F}$, and $\mathbf{t}_1,\mathbf{t}_2\in\mathcal{T}$, then $(R\eta)(a\mathbf{t}_1+b\mathbf{t}_2)=a(R\eta)(\mathbf{t}_1)+b(R\eta)(\mathbf{t}_2)$ by linearity of η, so $R\eta\in\mathcal{T}'$.
For $a,b\in\mathbb{F}$ and $\eta,\zeta\in\mathcal{S}^0$, the identity $R(a\eta+b\zeta)=aR\eta+bR\zeta$ holds pointwise, so $R\in\mathcal{L}(\mathcal{S}^0,\mathcal{T}')$ by Assumption 3.1.2.
The identities $(\forall\,\varphi\in\mathcal{T}')\,RE\varphi=\varphi$ and $(\forall\,\eta\in\mathcal{S}^0)(\forall\,\mathbf{s}\in\mathcal{S})(\forall\,\mathbf{t}\in\mathcal{T})\,(ER\eta)(\mathbf{s}+\mathbf{t})=\eta(\mathbf{t})=\eta(\mathbf{s}+\mathbf{t})$ hold by the definitions of E, R, and $\mathcal{S}^0$. Thus R is a two-sided inverse of E by Assumption 3.1.2, and E is an isomorphism.
(ii) For $\varphi\in\mathcal{V}'$, set $\varphi_{\mathcal{S}}:=\varphi\rho_{\mathcal{S},\mathcal{T}}$ and $\varphi_{\mathcal{T}}:=\varphi\rho_{\mathcal{T},\mathcal{S}}$. These functionals belong to $\mathcal{V}'$ by Proposition 2.4.2(ii) and Proposition 1.5.7(i).
Their respective membership in $\mathcal{T}^0$ and $\mathcal{S}^0$ holds by Proposition 2.4.2(iii).
The identity $\varphi=\varphi_{\mathcal{S}}+\varphi_{\mathcal{T}}$ holds by Proposition 2.4.3(i), so $\mathcal{V}'=\mathcal{S}^0+\mathcal{T}^0$.
If $\eta\in\mathcal{S}^0\cap\mathcal{T}^0$, then $(\forall\,\mathbf{s}\in\mathcal{S})(\forall\,\mathbf{t}\in\mathcal{T})\,\eta(\mathbf{s}+\mathbf{t})=0$, so $\eta=0_{\mathcal{V}'}$ by Assumption 3.1.2. Thus $\mathcal{V}'=\mathcal{S}^0\oplus\mathcal{T}^0$.
Structural Commentary: Extension and restriction identify one annihilator; component functionals split every element of the dual.
Key Insight: Each dual component ignores the opposite primal summand. □

Theorem 4.3.7. *Kernel and Range of the Dual Map*
Let $T\in\mathcal{L}(\mathcal{V},\mathcal{W})$. The following statements hold, with finite-dimensionality of $\mathcal{V}$ and $\mathcal{W}$ required only in (ii)–(iv):
(i) $\ker(T')=(\operatorname{range}(T))^0$;
(ii) $\dim(\ker(T'))=\dim(\ker(T))+\dim(\mathcal{W})-\dim(\mathcal{V})$;
(iii) $\operatorname{range}(T')=(\ker(T))^0$;
(iv) $\dim(\operatorname{range}(T'))=\dim(\operatorname{range}(T))$.

Proof. For (ii)–(iv), assume that $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional.
(i) For $\varphi\in\mathcal{W}'$, $T'\varphi=0$ iff $(\forall\,\mathbf{v}\in\mathcal{V})\,\varphi(T\mathbf{v})=0$ iff $\varphi\in(\operatorname{range}(T))^0$.
(ii) By (i), Theorem 4.3.3, and Theorem 1.7.3(ii), $\dim(\ker(T'))=\dim(\mathcal{W})-\dim(\operatorname{range}(T))=\dim(\ker(T))+\dim(\mathcal{W})-\dim(\mathcal{V})$.

(iii) If $\varphi = T'\psi$ and $\mathbf{v} \in \ker(T)$, then $\varphi(\mathbf{v}) = \psi(T\mathbf{v}) = 0$, so $\operatorname{range}(T') \subseteq (\ker(T))^0$.
By Theorem 1.7.3(ii), (ii), and Theorem 4.1.4(ii), $\dim(\operatorname{range}(T')) = \dim(\mathcal{W}') - \dim(\ker(T')) = \dim(\mathcal{V}) - \dim(\ker(T))$.
The latter dimension equals $\dim((\ker(T))^0)$ by Theorem 4.3.3, so the inclusion is an equality by Proposition 1.4.3.
(iv) By (iii), Theorem 4.3.3, and Theorem 1.7.3(ii), $\dim(\operatorname{range}(T')) = \dim(\mathcal{V}) - \dim(\ker(T)) = \dim(\operatorname{range}(T))$.

Structural Commentary: A direct kernel identity and finite-dimensional counts determine the dual range and both dimensions.
Key Insight: Pullback exchanges unreachable vectors with annihilating functionals. □

Corollary 4.3.8. *Surjectivity and Dual Injectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is surjective iff T' is injective.

Proof.
($\Rightarrow$) If T is surjective, then $\operatorname{range}(T) = \mathcal{W}$, so $\ker(T') = \mathcal{W}^0 = \{0_{\mathcal{W}'}\}$ by Theorem 4.3.7(i). Hence T' is injective by Proposition 1.6.2.
($\Leftarrow$) If T' is injective, then $(\operatorname{range}(T))^0 = \ker(T') = \{0_{\mathcal{W}'}\}$ by Proposition 1.6.2 and Theorem 4.3.7(i).
Thus $\dim(\operatorname{range}(T)) = \dim(\mathcal{W})$ by Theorem 4.3.3, so $\operatorname{range}(T) = \mathcal{W}$ by Proposition 1.4.3.

Structural Commentary: The dual-kernel identity translates surjectivity into a zero-kernel criterion.
Key Insight: A full range has no nonzero annihilator. □

Corollary 4.3.9. *Injectivity and Dual Surjectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is injective iff T' is surjective.

Proof.
($\Rightarrow$) If T is injective, then T' is surjective by Proposition 1.6.2, Theorem 4.3.7(iii), and Definition 4.3.1.
($\Leftarrow$) If T' is surjective, then $(\ker(T))^0 = \operatorname{range}(T') = \mathcal{V}'$ by Theorem 4.3.7(iii).
Thus $\dim(\ker(T)) = 0$ by Theorem 4.3.3, so $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$ by Proposition 1.4.3. Therefore T is injective by Proposition 1.6.2.

Structural Commentary: The dual-range identity converts injectivity into surjectivity of the pullback.
Key Insight: A trivial kernel leaves no functional-extension obstruction. □

Corollary 4.3.10. *Dual of a Quotient as an Annihilator*
Let $\mathcal{V}$ be finite-dimensional, let $\mathcal{U} \leq \mathcal{V}$, and let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map. Then:
(i) $\ker(\pi') = \{0_{(\mathcal{V}/\mathcal{U})'}\}$;
(ii) $\operatorname{range}(\pi') = \mathcal{U}^0$;
(iii) the corestriction $\pi' : (\mathcal{V}/\mathcal{U})' \to \mathcal{U}^0$ is an isomorphism, so $(\mathcal{V}/\mathcal{U})' \cong \mathcal{U}^0$.

Proof. The quotient is finite-dimensional by Theorem 2.2.9.
(i) The map π is surjective, so π' is injective by Corollary 4.3.8. Hence $\ker(\pi') = \{0_{(\mathcal{V}/\mathcal{U})'}\}$ by Proposition 1.6.2.
(ii) Since $\ker(\pi) = \mathcal{U}$, $\operatorname{range}(\pi') = (\ker(\pi))^0 = \mathcal{U}^0$ by Theorem 4.3.7(iii).
(iii) By (i)–(ii), the corestriction is a linear bijection and hence an isomorphism by Proposition 1.6.7.

Structural Commentary: Dual kernel and range identities make the quotient pullback a bijection onto the annihilator.
Key Insight: Quotient functionals are exactly functionals vanishing on the collapsed subspace. □

Theorem 4.3.11. *Range Membership Dual Alternative*
Let $\mathcal{V}, \mathcal{W}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and $\mathbf{w} \in \mathcal{W}$. Exactly one holds:
(i) $\mathbf{w} \in \operatorname{range}(T)$;
(ii) $(\exists\, \psi \in \mathcal{W}')[\psi T = 0 \wedge \psi(\mathbf{w}) \neq 0]$.

Proof.
(i) If (i) and (ii) hold, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{w} = T\mathbf{v}$ and $\psi \in \mathcal{W}'$ satisfying (ii).
Then $\psi(\mathbf{w}) = \psi(T\mathbf{v}) = 0$, contrary to $\psi(\mathbf{w}) \neq 0$. Hence (i) and (ii) cannot both hold.
(ii) Suppose (i) fails. The map $\Lambda_{\mathcal{W}}$ is injective by Theorem 4.2.2(ii), so $\Lambda_{\mathcal{W}}(\mathbf{w}) \notin \Lambda_{\mathcal{W}}(\operatorname{range}(T)) = (\operatorname{range}(T))^{00}$ by Theorem 4.3.5(i).
Hence $(\exists\, \psi \in (\operatorname{range}(T))^0)\, \Lambda_{\mathcal{W}}(\mathbf{w})(\psi) \neq 0$ by Definition 4.3.1. For such ψ, one has $\psi(\mathbf{w}) \neq 0$ and $(\forall\, \mathbf{v} \in \mathcal{V})\, (\psi T)(\mathbf{v}) = \psi(T\mathbf{v}) = 0$, so $\psi T = 0$ by Assumption 3.1.2. Thus (ii) holds whenever (i) fails.

Structural Commentary: Direct evaluation proves exclusion; double annihilation constructs every missing-range certificate.
Key Insight: A separating functional certifies every missing range vector. □

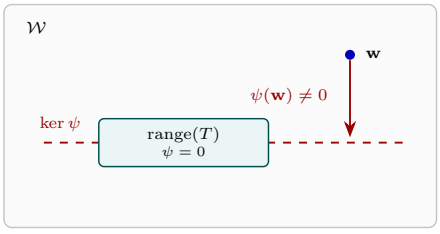

Figure 14: The dual certificate ψ vanishes on $\operatorname{range}(T)$ but detects $\mathbf{w}$ outside that range.

Figure 14 illustrates Theorem 4.3.11: $\operatorname{range}(T) \subseteq \ker(\psi)$ while $\mathbf{w} \notin \ker(\psi)$. It supplies no proof authority.

Proposition 4.3.12. *Standard Coordinate Dual Identification*
For $m \geq 1$, define $\Theta : \mathbb{F}^m \to (\mathbb{F}^m)'$ by $(\forall\, \mathbf{y}, \mathbf{z} \in \mathbb{F}^m)\, \Theta(\mathbf{y})(\mathbf{z}) := \mathbf{y}^t\mathbf{z}$. Then:
(i) Θ is an isomorphism;
(ii) $(\forall\, \psi \in (\mathbb{F}^m)')(\exists!\, \mathbf{y} \in \mathbb{F}^m)(\forall\, \mathbf{z} \in \mathbb{F}^m)\, \psi(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$.

Proof.

(i) For $a, b \in \mathbb{F}$ and $\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w} \in \mathbb{F}^m$, the identity $\Theta(\mathbf{y})(a\mathbf{z}+b\mathbf{w}) = a\Theta(\mathbf{y})(\mathbf{z})+b\Theta(\mathbf{y})(\mathbf{w})$ holds by Proposition 3.1.2, so $\Theta(\mathbf{y}) \in (\mathbb{F}^m)'$. The identity $\Theta(a\mathbf{x}+b\mathbf{y})(\mathbf{z}) = a\Theta(\mathbf{x})(\mathbf{z}) + b\Theta(\mathbf{y})(\mathbf{z})$ holds by Proposition 3.1.14 and Proposition 3.1.2; hence Θ is linear by Assumption 3.1.2. If $\Theta(\mathbf{y}) = 0$, then $(\forall\, j \in \{1, \dots, m\})\, 0 = \Theta(\mathbf{y})(\mathbf{e}_j) = y_j$, so $\mathbf{y} = \mathbf{0}$ and Θ is injective by Proposition 1.6.2. Since (1) is a basis of $\mathbb{F}$ over itself, $\dim(\mathbb{F}^m) = m = \dim((\mathbb{F}^m)')$ by Theorem 2.1.13 and Theorem 4.1.4(ii). Thus Θ is surjective by Corollary 1.7.6(ii) and is an isomorphism by Proposition 1.6.7.

(ii) Every $\psi \in (\mathbb{F}^m)'$ has a unique preimage $\mathbf{y} \in \mathbb{F}^m$ under Θ by (i), and $\psi(\mathbf{z}) = \Theta(\mathbf{y})(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$ by the defining formula.

Structural Commentary: Matrix identities prove linearity; coordinate evaluations and equal dimensions prove bijectivity.
Key Insight: Every coordinate functional is multiplication by one transposed column. □

Corollary 4.3.13. *Matrix Equation Alternative*
Let $m, n \geq 1$, $A \in \mathrm{Mat}(m, n, \mathbb{F})$, and $\mathbf{b} \in \mathbb{F}^m$. Exactly one holds:

(i) $(\exists\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = \mathbf{b}$;
(ii) $(\exists\, \mathbf{y} \in \mathbb{F}^m)[\mathbf{y}^t A = \mathbf{0}^t \wedge \mathbf{y}^t\mathbf{b} \neq 0]$.

Proof. Let $T_A : \mathbb{F}^n \to \mathbb{F}^m$ be the matrix-induced map.
Then $T_A \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ by Proposition 3.1.8, and both coordinate spaces are finite-dimensional by Theorem 2.1.13.

(i) The equivalence $\mathbf{b} \in \mathrm{range}(T_A) \iff (\exists\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = \mathbf{b}$ holds by $T_A\mathbf{x} = A\mathbf{x}$.

(ii) By Proposition 4.3.12(ii), each $\psi \in (\mathbb{F}^m)'$ has the unique form $\psi(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$. For this $\mathbf{y}$, $\psi T_A = 0$ iff $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, \mathbf{y}^t A\mathbf{x} = 0$ iff $\mathbf{y}^t A = \mathbf{0}^t$ by evaluation on the standard basis of $\mathbb{F}^n$; also, $\psi(\mathbf{b}) \neq 0$ iff $\mathbf{y}^t\mathbf{b} \neq 0$. Exactly one holds by Theorem 4.3.11.

Structural Commentary: Coordinate identification converts the abstract separating functional into a left-null vector.
Key Insight: Left-null vectors are coordinate dual certificates. □

Theorem 4.3.14. *Matrix of the Dual Map Is the Transpose*
Let $\dim(\mathcal{V}) = n \geq 1$ and $\dim(\mathcal{W}) = m \geq 1$, let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ be bases.
If β' and γ' are the corresponding dual bases, then $[T']_{\beta' \leftarrow \gamma'} = ([T]_{\gamma \leftarrow \beta})^t$.

Proof. Set $A := [T]_{\gamma \leftarrow \beta}$ and $C := [T']_{\beta' \leftarrow \gamma'}$. Write $\beta' = (\varphi_1, \dots, \varphi_n)$ and $\gamma' = (\psi_1, \dots, \psi_m)$. By Definition 3.2.3 and Definition 4.1.3, $(\forall\, k \in \{1, \dots, n\})(\forall\, j \in \{1, \dots, m\})\, C_{k,j} = (T'\psi_j)(\mathbf{v}_k) = \psi_j(T\mathbf{v}_k) = A_{j,k}$, where $T\mathbf{v}_k = \sum_{r=1}^{m} A_{r,k}\mathbf{w}_r$. Hence $C = A^t$.

Structural Commentary: Evaluating dual-basis expansions identifies each dual-matrix entry with the transposed primal entry.
Key Insight: Dual evaluation swaps the row and column indices. □

4.4 Multilinear Structure

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.4.1. *Bilinear Pairings, Matrices, and Induced Functions*

(i) A map $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ is *bilinear* if $(\forall\, a, b \in \mathbb{F})(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\, B(a\mathbf{u}+b\mathbf{v}, \mathbf{w}) = aB(\mathbf{u}, \mathbf{w}) + bB(\mathbf{v}, \mathbf{w})$, and $(\forall\, a, b \in \mathbb{F})(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{W})\, B(\mathbf{v}, a\mathbf{x}+b\mathbf{y}) = aB(\mathbf{v}, \mathbf{x}) + bB(\mathbf{v}, \mathbf{y})$.

(ii) Suppose $\dim(\mathcal{V}) = m \geq 1$ and $\dim(\mathcal{W}) = n \geq 1$, with ordered bases $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$. The *matrix of B in β, γ* is $[B]_{\beta,\gamma} := [B(\mathbf{v}_i, \mathbf{w}_j)]_{1 \leq i \leq m,\ 1 \leq j \leq n} \in \mathrm{Mat}(m, n, \mathbb{F})$.

(iii) For bilinear B, define Φ_B by $(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\, (\Phi_B\mathbf{v})(\mathbf{w}) := B(\mathbf{v}, \mathbf{w})$.
Define Ψ_B by $(\forall\, \mathbf{w} \in \mathcal{W})(\forall\, \mathbf{v} \in \mathcal{V})\, (\Psi_B\mathbf{w})(\mathbf{v}) := B(\mathbf{v}, \mathbf{w})$.

Proposition 4.4.2. *Coordinate Evaluation of a Bilinear Pairing*
Let $\dim(\mathcal{V}) = m \geq 1$ and $\dim(\mathcal{W}) = n \geq 1$, let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear.
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ be ordered bases of $\mathcal{V}$ and $\mathcal{W}$, respectively, and set $M := [B]_{\beta,\gamma}$.
For $\mathbf{v} \in \mathcal{V}$ and $\mathbf{w} \in \mathcal{W}$, set $\mathbf{x} := [\mathbf{v}]_\beta$ and $\mathbf{y} := [\mathbf{w}]_\gamma$. Then $B(\mathbf{v}, \mathbf{w}) = \sum_{i=1}^{m}\sum_{j=1}^{n} x_i B(\mathbf{v}_i, \mathbf{w}_j) y_j = \mathbf{x}^t M\mathbf{y}$.

Proof. The identities $\mathbf{v} = \sum_{i=1}^{m} x_i\mathbf{v}_i$ and $\mathbf{w} = \sum_{j=1}^{n} y_j\mathbf{w}_j$ hold by Definition 3.2.1.

Hence $B(\mathbf{v}, \mathbf{w}) = \sum_{i=1}^{m}\sum_{j=1}^{n} x_i y_j B(\mathbf{v}_i, \mathbf{w}_j)$ by Definition 4.4.1(i). The sum equals $\mathbf{x}^t M\mathbf{y}$ by Definition 4.4.1(ii) and Proposition 3.1.6.

Structural Commentary: Two slotwise expansions convert basis coefficients into a matrix product.
Key Insight: A pairing matrix records every basis-vector interaction. □

Proposition 4.4.3. *Bilinear Pairings Linearize through Duals*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear. Then:

(i) *$\Phi_B : \mathcal{V} \to \mathcal{W}'$ and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ are well-defined linear maps;*
(ii) *if $\dim(\mathcal{V}) = m \geq 1$ and $\dim(\mathcal{W}) = n \geq 1$, $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ are ordered bases of $\mathcal{V}$ and $\mathcal{W}$ with dual bases β', γ', and $M := [B]_{\beta,\gamma}$, then $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$ and $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.*

Proof.

(i) By Definition 4.4.1(i),(iii), $(\forall\, \mathbf{v} \in \mathcal{V})\, \Phi_B\mathbf{v} \in \mathcal{W}'$ and $(\forall\, \mathbf{w} \in \mathcal{W})\, \Psi_B\mathbf{w} \in \mathcal{V}'$.
For $a, b \in \mathbb{F}$, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, and $\mathbf{w} \in \mathcal{W}$, $(\Phi_B(a\mathbf{u}+b\mathbf{v}))(\mathbf{w}) = aB(\mathbf{u}, \mathbf{w}) + bB(\mathbf{v}, \mathbf{w}) = (a\Phi_B\mathbf{u} + b\Phi_B\mathbf{v})(\mathbf{w})$.
Hence $\Phi_B(a\mathbf{u}+b\mathbf{v}) = a\Phi_B\mathbf{u} + b\Phi_B\mathbf{v}$ by Assumption 3.1.2. For $\mathbf{x}, \mathbf{y} \in \mathcal{W}$, the identity $\Psi_B(a\mathbf{x}+b\mathbf{y}) = a\Psi_B\mathbf{x} + b\Psi_B\mathbf{y}$ holds by Definition 4.4.1(i) and Assumption 3.1.2. Thus both maps are linear.

(ii) By Definition 3.2.3, 4.1.3, and 4.4.1(ii), $(\forall\, i \in \{1, \dots, m\})(\forall\, j \in \{1, \dots, n\})\, ([\Phi_B]_{\gamma' \leftarrow \beta})_{j,i} = (\Phi_B\mathbf{v}_i)(\mathbf{w}_j) = B(\mathbf{v}_i, \mathbf{w}_j) = M_{i,j}$.
Hence $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$. Likewise, $(\forall\, i \in \{1, \dots, m\})(\forall\, j \in \{1, \dots, n\})\, ([\Psi_B]_{\beta' \leftarrow \gamma})_{i,j} = (\Psi_B\mathbf{w}_j)(\mathbf{v}_i) = M_{i,j}$ by Definition 3.2.3, 4.1.3, and 4.4.1(ii), so $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.

Structural Commentary: Frozen-slot linearity establishes the maps; dual-basis evaluation identifies their matrices.
Key Insight: Freezing one slot turns bilinearity into dual-valued linearity. □

Definition 4.4.4. *Radicals and Nonsingularity of a Bilinear Pairing*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear.

(i) The *left* and *right radicals* are $\text{rad}_L(B) := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{w} \in \mathcal{W})\, B(\mathbf{v}, \mathbf{w}) = 0\}$ and $\text{rad}_R(B) := \{\mathbf{w} \in \mathcal{W} : (\forall\, \mathbf{v} \in \mathcal{V})\, B(\mathbf{v}, \mathbf{w}) = 0\}$.
(ii) For $\mathcal{S} \subseteq \mathcal{V}$ and $\mathcal{T} \subseteq \mathcal{W}$, the *right* and *left orthogonals* are $\mathcal{S}^{\perp_R} := \{\mathbf{w} \in \mathcal{W} : (\forall\, \mathbf{s} \in \mathcal{S})\, B(\mathbf{s}, \mathbf{w}) = 0\}$ and ${}^{\perp_L}\mathcal{T} := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{t} \in \mathcal{T})\, B(\mathbf{v}, \mathbf{t}) = 0\}$.
(iii) B is *left nondegenerate* if $\text{rad}_L(B) = \{\mathbf{0}_\mathcal{V}\}$ and *right nondegenerate* if $\text{rad}_R(B) = \{\mathbf{0}_\mathcal{W}\}$.
(iv) B is *nonsingular* or *perfect* if $\Phi_B : \mathcal{V} \to \mathcal{W}'$ and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ are isomorphisms.

Proposition 4.4.5. *Radicals, Matrices, and Dual Maps*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear. Then:

(i) $\ker(\Phi_B) = \text{rad}_L(B)$ *and* $\ker(\Psi_B) = \text{rad}_R(B)$;
(ii) if $\dim(\mathcal{V}) = m \geq 1$ *and* $\dim(\mathcal{W}) = n \geq 1$, *with ordered bases* β *of* $\mathcal{V}$ *and* γ *of* $\mathcal{W}$ *and with* $M := [B]_{\beta,\gamma}$, *then* $\{[\mathbf{v}]_\beta : \mathbf{v} \in \text{rad}_L(B)\} = \ker(M^t)$ *and* $\{[\mathbf{w}]_\gamma : \mathbf{w} \in \text{rad}_R(B)\} = \ker(M)$;
(iii) under the hypotheses of (ii), $\dim(\text{rad}_L(B)) = m - \text{rank}(M)$ *and* $\dim(\text{rad}_R(B)) = n - \text{rank}(M)$.

Proof.

(i) By Definition 4.4.4(i), 4.4.1(iii), Assumption 3.1.2, $(\forall\, \mathbf{v} \in \mathcal{V})\, \mathbf{v} \in \ker(\Phi_B) \iff (\forall\, \mathbf{w} \in \mathcal{W})\, B(\mathbf{v}, \mathbf{w}) = 0 \iff \mathbf{v} \in \text{rad}_L(B)$.
By Definition 4.4.4(i), 4.4.1(iii), Assumption 3.1.2, $(\forall\, \mathbf{w} \in \mathcal{W})\, \mathbf{w} \in \ker(\Psi_B) \iff (\forall\, \mathbf{v} \in \mathcal{V})\, B(\mathbf{v}, \mathbf{w}) = 0 \iff \mathbf{w} \in \text{rad}_R(B)$
(ii) For $\mathbf{v} \in \mathcal{V}$ and $\mathbf{w} \in \mathcal{W}$, set $\mathbf{x} := [\mathbf{v}]_\beta$ and $\mathbf{y} := [\mathbf{w}]_\gamma$. By Proposition 3.2.5 and 4.4.3(ii), $[\Phi_B\mathbf{v}]_{\gamma'} = M^t\mathbf{x}$ and $[\Psi_B\mathbf{w}]_{\beta'} = M\mathbf{y}$.
Hence $\mathbf{v} \in \text{rad}_L(B) \iff \mathbf{x} \in \ker(M^t)$ and $\mathbf{w} \in \text{rad}_R(B) \iff \mathbf{y} \in \ker(M)$ by (i) and Proposition 3.2.2(ii).
These equivalences prove both coordinate-set identities.
(iii) The coordinate-map restrictions to the two radicals are linear and injective by Proposition 3.2.2(i)–(ii), and they are surjective onto $\ker(M^t)$ and $\ker(M)$ by (ii). Thus their corestrictions are isomorphisms by Proposition 1.6.7, so the radical dimensions equal the corresponding kernel dimensions.
By Corollary 3.3.1(i) and Theorem 3.3.2, $\dim(\ker(M^t)) = m - \text{rank}(M)$ and $\dim(\ker(M)) = n - \text{rank}(M)$.

Structural Commentary: Kernel identities pass through coordinate isomorphisms; rank-nullity then determines both dimensions.
Key Insight: Radicals are the two matrix nullspaces in intrinsic form. □

Proposition 4.4.6. *Square Pairing Nonsingularity Criteria*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear with $\dim(\mathcal{V}) = \dim(\mathcal{W}) = n \geq 1$, *let β and γ be ordered bases of $\mathcal{V}$ and $\mathcal{W}$, respectively.*
Set $M := [B]_{\beta,\gamma}$. The following are equivalent:

(i) B is left nondegenerate;
(ii) B is right nondegenerate;
(iii) M is invertible;
(iv) B is perfect.

Proof.

(i) $\Rightarrow$ *(ii)* If $\text{rad}_L(B) = \{\mathbf{0}_\mathcal{V}\}$, then $\dim(\text{rad}_R(B)) = n - \text{rank}(M) = \dim(\text{rad}_L(B)) = 0$ by Proposition 4.4.5(iii).
Hence $\text{rad}_R(B) = \{\mathbf{0}_\mathcal{W}\}$ by Proposition 1.4.3(ii).
(ii) $\Rightarrow$ *(iii)* If $\text{rad}_R(B) = \{\mathbf{0}_\mathcal{W}\}$, then $0 = \dim(\text{rad}_R(B)) = n - \text{rank}(M)$ by Proposition 4.4.5(iii); M invertible by Corollary 3.6.2.
(iii) $\Rightarrow$ *(iv)* If M is invertible, then $\text{rank}(M) = n$ by Corollary 3.6.2. Hence both radicals have dimension zero by Proposition 4.4.5(iii), so they are zero subspaces by Proposition 1.4.3(ii). Therefore $\ker(\Phi_B) = \{\mathbf{0}_\mathcal{V}\}$ and $\ker(\Psi_B) = \{\mathbf{0}_\mathcal{W}\}$ by Proposition 4.4.5(i). Their domains have dimension n by hypothesis, and their codomains have dimension n by Theorem 4.1.4(ii), so Φ_B and Ψ_B are isomorphisms by Corollary 1.7.6(i), Proposition 1.6.2, and Proposition 1.6.7. Thus B is perfect by Definition 4.4.4(iv).
(iv) $\Rightarrow$ *(i)* If B is perfect, then Φ_B is an isomorphism by Definition 4.4.4(iv) and hence injective by Proposition 1.6.7.
Thus $\ker(\Phi_B) = \{\mathbf{0}_\mathcal{V}\}$ by Proposition 1.6.2, and $\text{rad}_L(B) = \{\mathbf{0}_\mathcal{V}\}$ by Proposition 4.4.5(i).

Structural Commentary: Equal nullities link the radicals; full rank then identifies perfectness.
Key Insight: Square pairings have one shared rank obstruction. □

Corollary 4.4.7. *Perfect Pairings Represent Functionals*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be perfect. Then:

(i) $(\forall\, \psi \in \mathcal{W}')(\exists!\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\, \psi(\mathbf{w}) = B(\mathbf{v}, \mathbf{w})$;
(ii) $(\forall\, \varphi \in \mathcal{V}')(\exists!\, \mathbf{w} \in \mathcal{W})(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = B(\mathbf{v}, \mathbf{w})$.

Proof.

(i) The map $\Phi_B : \mathcal{V} \to \mathcal{W}'$ is an isomorphism by Definition 4.4.4(iv), and hence a bijection by Proposition 1.6.7.
Thus $(\forall\, \psi \in \mathcal{W}')(\exists!\, \mathbf{v} \in \mathcal{V})\, \psi = \Phi_B\mathbf{v}$. For this $\mathbf{v}$, $(\forall\, \mathbf{w} \in \mathcal{W})\, \psi(\mathbf{w}) = B(\mathbf{v}, \mathbf{w})$ by Definition 4.4.1(iii).
(ii) The map $\Psi_B : \mathcal{W} \to \mathcal{V}'$ is an isomorphism by Definition 4.4.4(iv), and hence a bijection by Proposition 1.6.7.
Thus $(\forall\, \varphi \in \mathcal{V}')(\exists!\, \mathbf{w} \in \mathcal{W})\, \varphi = \Psi_B\mathbf{w}$. For this $\mathbf{w}$, $(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = B(\mathbf{v}, \mathbf{w})$ by Definition 4.4.1(iii).

Structural Commentary: Bijectivity of the two induced maps supplies the two unique representatives.
Key Insight: Perfect pairings turn functionals into uniquely paired vectors. □

Definition 4.4.8. *Matrix Congruence*
For $n \geq 1$, matrices $M, N \in \text{Mat}(n, n, \mathbb{F})$ are *congruent* if $(\exists\, P \in \text{Mat}(n, n, \mathbb{F}))[P \text{ is invertible} \wedge N = P^t M P]$.

Proposition 4.4.9. *Two-Sided Basis Change for Pairings*
Let $\dim(\mathcal{U}) = p \geq 1$ *and* $\dim(\mathcal{W}) = q \geq 1$, *let $B : \mathcal{U} \times \mathcal{W} \to \mathbb{F}$ be bilinear, let $\alpha, \widetilde{\alpha}$ be ordered bases of $\mathcal{U}$, let $\beta, \widetilde{\beta}$ be ordered bases of $\mathcal{W}$.*
Set $M := [B]_{\alpha,\beta}$, $\widetilde{M} := [B]_{\widetilde{\alpha},\widetilde{\beta}}$, $P := C_{\alpha\leftarrow\widetilde{\alpha}}$, and $Q := C_{\beta\leftarrow\widetilde{\beta}}$. Then:

(i) $\widetilde{M} = P^t M Q$;
(ii) if $\mathcal{U} = \mathcal{W}$, $\alpha = \beta$, and $\widetilde{\alpha} = \widetilde{\beta}$, then $\widetilde{M}$ is congruent to M.

Proof.

(i) For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, set $\mathbf{x} := [\mathbf{u}]_{\widetilde{\alpha}}$ and $\mathbf{y} := [\mathbf{w}]_{\widetilde{\beta}}$. Then $[\mathbf{u}]_\alpha = P\mathbf{x}$ and $[\mathbf{w}]_\beta = Q\mathbf{y}$ by Definition 3.5.1.
By Proposition 4.4.2 and Proposition 3.1.16, $\mathbf{x}^t \widetilde{M} \mathbf{y} = B(\mathbf{u}, \mathbf{w}) = (P\mathbf{x})^t M (Q\mathbf{y}) = \mathbf{x}^t P^t M Q \mathbf{y}$.
The coordinate maps are surjective by Proposition 3.2.2(iii), so $(\forall\, \mathbf{x} \in \mathbb{F}^p)(\forall\, \mathbf{y} \in \mathbb{F}^q)\, \mathbf{x}^t \widetilde{M} \mathbf{y} = \mathbf{x}^t P^t M Q \mathbf{y}$.
The relation $(\forall\, i \in \{1, \ldots, p\})(\forall\, j \in \{1, \ldots, q\})\, \widetilde{M}_{i,j} = (P^t M Q)_{i,j}$ holds by substituting the standard coordinate columns.
Hence $\widetilde{M} = P^t M Q$ by Definition 3.1.1(i).

(ii) Under the stated equalities of spaces and bases, $P = Q$ and P is invertible by Theorem 3.5.2(i).
Thus $\widetilde{M} = P^t M P$ by (i), so $\widetilde{M}$ is congruent to M by Definition 4.4.8.

Structural Commentary: Coordinate evaluation transforms each slot; entry comparison proves the two-sided formula.
Key Insight: Each input slot contributes its own basis-change factor. □

Proposition 4.4.10. *Basis Change for a Square Bilinear Form*
Let $\dim(\mathcal{V}) = n \geq 1$, *let* $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ *be bilinear, and let* β, γ *be ordered bases of* $\mathcal{V}$.
Set $M_\beta := [B]_{\beta,\beta}$, $M_\gamma := [B]_{\gamma,\gamma}$, *and* $C := C_{\beta \leftarrow \gamma}$. *Then:*

(i) $M_\gamma = C^t M_\beta C$;
(ii) M_β *and* M_γ *are congruent.*

Proof.

(i) With $\mathcal{U} = \mathcal{W} = \mathcal{V}$, old basis β in both slots, and new basis γ in both slots, $M_\gamma = C^t M_\beta C$ by Proposition 4.4.9(i).

(ii) The matrix C is invertible by Theorem 3.5.2(i), and $M_\gamma = C^t M_\beta C$ by (i). Hence M_β and M_γ are congruent by Definition 4.4.8.

Structural Commentary: The square case specializes both slot changes to one invertible matrix.
Key Insight: Reusing one basis change produces congruence. □

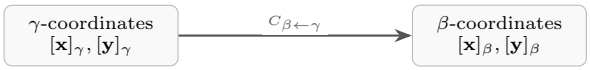

Figure 15: The same coordinate transition acts in both slots, producing $M_\gamma = C_{\beta \leftarrow \gamma}^t M_\beta C_{\beta \leftarrow \gamma}$.

Figure 15 depicts $C_{\beta \leftarrow \gamma}$ acting in both input slots; the formula belongs to Proposition 4.4.10, not to the illustration. The induced maps Φ_B and Ψ_B encode bilinear pairings in dual spaces. Part 4 develops real bilinear metric forms and complex sesquilinear inner products, while Part 8 constructs tensor products; neither later theory is used here.

Part III
Linear Maps and Matrix Representations

1 Linear Maps and Operations

1.1 Further Linear Map Operations and Examples

Operator polynomials encode powers and polynomial equations satisfied by a linear operator.

Definition 1.1.1. *Operator Powers*
Let $T \in \mathcal{L}(\mathcal{V})$. The *powers* of T are $T^m := \underbrace{T \cdots T}_{m \text{ times}}$ for $m \in \mathbb{N}$, with $T^0 := I_{\mathcal{V}}$. If T is invertible, set $T^{-m} := (T^{-1})^m$.

Definition 1.1.2. *Polynomial Applied to an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$. For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$, the *polynomial p applied to T* is $p(T) := \sum_{j=0}^{m} a_j T^j$.

Definition 1.1.3. *Annihilating Polynomial*
Let $T \in \mathcal{L}(\mathcal{V})$. A polynomial $p \in \mathbb{F}[z]$ is an *annihilating polynomial* for T if $p(T) = 0_{\mathcal{L}(\mathcal{V})}$; equivalently, p *annihilates* T.

Definition 1.1.4. *Product of Polynomials*
If $p, q \in \mathbb{F}[z]$, their product $pq \in \mathbb{F}[z]$ is defined by $(pq)(z) = p(z)q(z)$ for $z \in \mathbb{F}$.

Definition 1.1.5. *Monic Polynomial*
A nonzero polynomial is *monic* if its highest-degree coefficient is 1.

Proposition 1.1.6. *Multiplicative Properties of Operator Polynomials*
Let $p, q \in \mathbb{F}[z]$ and $T \in \mathcal{L}(\mathcal{V})$. Then
(i) $(pq)(T) = p(T)q(T)$;
(ii) $p(T)q(T) = q(T)p(T)$.

Proof.
(i) Write $p(z) = \sum_{j=0}^{m} a_j z^j$ and $q(z) = \sum_{k=0}^{n} b_k z^k$. Then $(pq)(z) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k z^{j+k}$.
Substituting T gives $(pq)(T) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k T^{j+k} = \left(\sum_{j=0}^{m} a_j T^j \right) \left(\sum_{k=0}^{n} b_k T^k \right) = p(T)q(T)$.
(ii) Since polynomial multiplication is commutative, $pq = qp$. By part (i), $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$.
Structural Commentary: Polynomial evaluation preserves products.
Key Insight: Expanding a polynomial product works the same with T as with z. □

Proposition 1.1.7. *Existence of Annihilating Polynomials*
Let $\mathcal{V}$ be finite-dimensional with $n = \dim(\mathcal{V}) > 0$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then T has a nonconstant annihilating polynomial of degree at most n^2.

Proof. By Theorem 3.4.5, the vector space $\mathcal{L}(\mathcal{V})$ has dimension n^2, so the $n^2 + 1$ operators $I_{\mathcal{V}}, T, T^2, \ldots, T^{n^2}$ are linearly dependent.
Hence there are scalars $a_0, \ldots, a_{n^2}$, not all zero, such that $\sum_{j=0}^{n^2} a_j T^j = 0$.
If $a_1 = \cdots = a_{n^2} = 0$, then $a_0 I_{\mathcal{V}} = 0$, forcing $a_0 = 0$ because $\mathcal{V} \neq \{\mathbf{0}\}$, a contradiction.
Thus $p(z) = \sum_{j=0}^{n^2} a_j z^j$ is a nonconstant annihilating polynomial.
Structural Commentary: Finite-dimensionality makes the powers of one operator linearly dependent inside $\mathcal{L}(\mathcal{V})$.
Key Insight: A dependence among $I_{\mathcal{V}}, T, \ldots, T^{n^2}$ is exactly a polynomial equation satisfied by T. □

Proposition 1.1.8. *Polynomial Evaluation on Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, let $p \in \mathbb{F}[z]$, and suppose $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $p(T)\mathbf{v} = p(\lambda)\mathbf{v}$.

Proof. Induct on j to get $T^j\mathbf{v} = \lambda^j\mathbf{v}$ $\forall j \geq 0$. If $p(z) = \sum_{j=0}^{m} a_j z^j$, then $p(T)\mathbf{v} = \sum_{j=0}^{m} a_j T^j\mathbf{v} = \sum_{j=0}^{m} a_j \lambda^j \mathbf{v} = p(\lambda)\mathbf{v}$.
Structural Commentary: On a one-dimensional eigenspace, T acts like the scalar λ.
Key Insight: Polynomial expressions in T reduce to the same polynomial evaluated at λ. □

Theorem 1.1.9. *Annihilating Polynomials Control the Spectrum*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let p be a nonconstant annihilating polynomial for T.
(i) Every eigenvalue of T is a root of p.
(ii) If $\mathbb{F} = \mathbb{C}$, then at least one root of p is an eigenvalue of T.

Proof.
(i) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then Proposition 1.1.8 gives $0 = p(T)\mathbf{v} = p(\lambda)\mathbf{v}$. Since $\mathbf{v} \neq \mathbf{0}$, $p(\lambda) = 0$.
(ii) Factor $p(z) = c(z - \lambda_1) \cdots (z - \lambda_m)$ over $\mathbb{C}$, with $c \neq 0$. Then $0 = p(T) = c(T - \lambda_1 I_{\mathcal{V}}) \cdots (T - \lambda_m I_{\mathcal{V}})$. At least one factor $T - \lambda_j I_{\mathcal{V}}$ is not injective; otherwise product would be injective. Hence λ_j is an eigenvalue.
Structural Commentary: Annihilating polynomial restricts spectrum to its roots; over $\mathbb{C}$, one root must appear.
Key Insight: Eigenvectors turn $p(T)$ into $p(\lambda)$. □

2 Kernels, Images, Rank, and Nullity

2.1 Shift Operators and Kernel-Range Examples

Fix a field $\mathbb{F}$, let $\mathcal{E} := \mathbb{F}^{(\mathbb{N})}$ be its finitely supported sequence space, and set $\mathbf{e}_1 := (1, 0, 0, \ldots)$ and $\mathcal{E}_0 := \{\mathbf{x} \in \mathcal{E} : x_1 = 0\}$.

Proposition 2.1.1. *Shift Operators on Finitely Supported Sequences*
Define $R, S : \mathcal{E} \to \mathcal{E}$ by $R(x_1, x_2, \ldots) := (0, x_1, x_2, \ldots)$ and $S(x_1, x_2, \ldots) := (x_2, x_3, \ldots)$. Then:

(i) $R, S \in \mathcal{L}(\mathcal{E})$.
(ii) $\ker(R) = \{\mathbf{0}_\mathcal{E}\}$.
(iii) $\text{range}(R) = \mathcal{E}_0$.
(iv) $\ker(S) = \text{span}(\mathbf{e}_1)$.
(v) $\text{range}(S) = \mathcal{E}$.
(vi) *R is injective but not surjective, whereas S is surjective but not injective.*
(vii) *$\mathcal{E}$ is infinite-dimensional.*

Proof.

(i) For $\mathbf{x} \in \mathcal{E}$, $|\text{supp}(R\mathbf{x})| = |\text{supp}(\mathbf{x})| < \infty$ and $|\text{supp}(S\mathbf{x})| \leq |\text{supp}(\mathbf{x})|$, so R and S are well-defined.
Coordinatewise operations establish $(\forall T \in \{R, S\})(\forall a, b \in \mathbb{F})(\forall \mathbf{x}, \mathbf{y} \in \mathcal{E})\, T(a\mathbf{x} + b\mathbf{y}) = aT\mathbf{x} + bT\mathbf{y}$; hence $R, S \in \mathcal{L}(\mathcal{E})$.
(ii) For $\mathbf{x} = (x_1, x_2, \ldots) \in \mathcal{E}$, $R\mathbf{x} = \mathbf{0}_\mathcal{E} \iff (\forall j \in \mathbb{N})\, x_j = 0 \iff \mathbf{x} = \mathbf{0}_\mathcal{E}$. Thus $\ker(R) = \{\mathbf{0}_\mathcal{E}\}$.
(iii) By definition, $\text{range}(R) \subseteq \mathcal{E}_0$. Conversely, $\mathbf{y} = (0, y_2, y_3, \ldots) \in \mathcal{E}_0$ has $\mathbf{x} := (y_2, y_3, \ldots) \in \mathcal{E}$ with $R\mathbf{x} = \mathbf{y}$; hence $\text{range}(R) = \mathcal{E}_0$.
(iv) For $\mathbf{x} = (x_1, x_2, \ldots) \in \mathcal{E}$, $S\mathbf{x} = \mathbf{0}_\mathcal{E} \iff (\forall j \in \mathbb{N} \setminus \{1\})\, x_j = 0 \iff \mathbf{x} = x_1\mathbf{e}_1$. Thus $\ker(S) = \text{span}(\mathbf{e}_1)$.
(v) The identity $(\forall \mathbf{y} \in \mathcal{E})\, S(R\mathbf{y}) = \mathbf{y}$ holds by the definitions, with $R\mathbf{y} \in \mathcal{E}$ by (i); hence $\text{range}(S) = \mathcal{E}$.
(vi) The map R is injective by Proposition 1.6.2 with (ii), and R is not surjective by (iii), since $\mathbf{e}_1 \notin \mathcal{E}_0$.
The map S is surjective by (v) and not injective by Proposition 1.6.2 with (iv), since $\mathbf{e}_1 \neq \mathbf{0}_\mathcal{E}$.
(vii) The assumption $\dim(\mathcal{E}) < \infty$ would make R surjective by Theorem 1.7.7 with (i) and injectivity from (vi), contradicting (vi).

Structural Commentary: Matched claim items verify linearity, fibers, images, map properties, and the dimension obstruction.
Key Insight: The shifts exchange a missing coordinate with a nontrivial kernel. □

Example 2.1.2. *Shift Operators Show Noncommutativity*
For the shifts above, $SR, RS \in \mathcal{L}(\mathcal{E})$ by Proposition 1.5.7(i). For $\mathbf{x} = (x_1, x_2, \ldots) \in \mathcal{E}$, $SR\mathbf{x} = \mathbf{x}$ and $RS\mathbf{x} = (0, x_2, x_3, \ldots)$.
Hence $SR = I_\mathcal{E}$ by Assumption 3.1.2. Since $RS\mathbf{e}_1 = \mathbf{0}_\mathcal{E} \neq \mathbf{e}_1 = SR\mathbf{e}_1$, $RS \neq SR$ and $\mathcal{L}(\mathcal{E})$ is noncommutative.

2.2 Further Rank-Nullity and Invertibility Results

2.3 Further Quotients and Isomorphism Theorems

2.4 Further Finite-Dimensional Splitting and Isomorphism Tests

3 Matrix Representations

3.1 Further Matrix Algebra and Linear Systems

3.2 Further Coordinates and Matrix Representations

3.3 Further Matrix-Induced Maps and Rank

3.4 Further Invertibility and the Matrix-Map Dictionary

4 Row Reduction and Elementary Matrices

4.1 Row Reduction and Elementary Matrices

Definition 4.1.1. *Elementary Row Operations and Elementary Matrices*
Let $m \geq 1$. For $i, j \in \{1, \ldots, m\}$, the *elementary row operations* are:

(i) *Row scaling*: $R_i \leftarrow cR_i$, where $c \in \mathbb{F}^\times$;
(ii) *Row swap*: $R_i \leftrightarrow R_j$, where $i \neq j$;
(iii) *Row replacement*: $R_i \leftarrow R_i + cR_j$, where $i \neq j$ and $c \in \mathbb{F}$.

For an elementary row operation ϵ on m rows, its *elementary matrix* is $E_\epsilon := \epsilon(I_m)$.

Proposition 4.1.2. *Elementary Matrices Perform Row Operations*
Let $A \in \text{Mat}(m, n, \mathbb{F})$, let ϵ be an elementary operation on m rows, and set $E_\epsilon := \epsilon(I_m)$. Then:

(i) $E_\epsilon A = \epsilon(A)$.
(ii) $E_\epsilon^{-1} = E_{\epsilon^{-1}}$; *in particular, every elementary matrix has an elementary inverse.*
(iii) *If η is an elementary operation on n rows and $F_\eta := \eta(I_n)^t$, then $AF_\eta = (\eta(A^t))^t$.*
Thus right multiplication by F_η performs the corresponding column operation.

Proof.

(i) The rows of E_ϵ identify the rows of $E_\epsilon A$ by matrix multiplication:
  (a) For scaling, $(E_\epsilon)_{i,\cdot} = c\mathbf{e}_i^t$, while every other row is standard; hence $(E_\epsilon A)_{i,\cdot} = cA_{i,\cdot}$ and the other rows are unchanged.
  (b) For a swap, $(E_\epsilon)_{i,\cdot} = \mathbf{e}_j^t$ and $(E_\epsilon)_{j,\cdot} = \mathbf{e}_i^t$; hence $E_\epsilon A$ exchanges rows i and j, with every other row fixed.
  (c) For replacement, $(E_\epsilon)_{i,\cdot} = \mathbf{e}_i^t + c\mathbf{e}_j^t$, so $(E_\epsilon A)_{i,\cdot} = A_{i,\cdot} + cA_{j,\cdot}$; the other coordinate rows preserve their rows.
(ii) The inverse of a scaling by c is scaling by c^{-1}, a swap is self-inverse, and the inverse of replacement by c is replacement by $-c$.
Therefore $E_{\epsilon^{-1}} E_\epsilon = \epsilon^{-1}(\epsilon(I_m)) = I_m$ and $E_\epsilon E_{\epsilon^{-1}} = \epsilon(\epsilon^{-1}(I_m)) = I_m$ by part (i).
(iii) $(AF_\eta)^t = F_\eta^t A^t = \eta(I_n)A^t = \eta(A^t)$ by Proposition 3.1.16, Proposition 3.1.14(i), and part (i).
After transposition, $(AF_\eta)^{tt} = AF_\eta$, so $AF_\eta = (\eta(A^t))^t$.

Structural Commentary: The proof translates row operations, their inverses, and transposition into multiplication identities.
Key Insight: Elementary matrices store row operations as left factors. □

Definition 4.1.3. *Row Equivalence and Reduced Row Echelon Form*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, $k=0$ interpreted as I_m; $A \sim_{\mathrm{row}} B \iff B = E_k \cdots E_1 A$ for $k \in \mathbb{N}_0$, elementary $E_1, \dots, E_k \in \mathrm{Mat}(m,m,\mathbb{F})$.
A matrix $R = [r_{i,j}] \in \mathrm{Mat}(m,n,\mathbb{F})$ is in *reduced row echelon form* if $(\exists s \in \{0,\dots,m\})(\exists p_1,\dots,p_s \in \{1,\dots,n\})$ such that:
(i) $(\forall i \in \{1,\dots,s\})\, R_{i,\cdot} \neq \mathbf{0}^t$ and $(\forall i \in \{s+1,\dots,m\})\, R_{i,\cdot} = \mathbf{0}^t$;
(ii) $(\forall i \in \{1,\dots,s\})\, [r_{i,p_i} = 1 \wedge (\forall j \in \{1,\dots,p_i-1\})\, r_{i,j} = 0]$;
(iii) $p_1 < \cdots < p_s$;
(iv) $(\forall i \in \{1,\dots,s\})(\forall \ell \in \{1,\dots,m\} \setminus \{i\})\, r_{\ell,p_i} = 0$.
The indices $p_1,\dots,p_s$ are the *pivot columns* of R.

Theorem 4.1.4. *Reduced Row Echelon Form*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then $(\exists!\, R \in \mathrm{Mat}(m,n,\mathbb{F}))\, [R$ is in RREF $\wedge\, A \sim_{\mathrm{row}} R]$. Write $\mathrm{rref}(A) := R$.

Proof.
(i) *Construction*: Set $i=1$ and $p_0 = 0$. Suppose rows $1,\dots,i-1$ have pivot columns $p_1 < \cdots < p_{i-1}$, these pivot columns are coordinate columns, and the remaining rows vanish in columns $1,\dots,p_{i-1}$.
If the remaining rows are zero, stop. Otherwise let p_i be the least column containing a nonzero entry in rows $i,\dots,m$.
Swap such an entry into row i, scale it to 1, and clear column p_i by row replacement.
The new pivot row vanishes before p_i, so these operations preserve the preceding pivot columns and extend the stated invariant.
(ii) *Verification*: Each stage increases both the pivot-row index and the pivot-column index, so the construction stops after at most $\min(m,n)$ stages. At termination, the invariant is exactly the four RREF conditions.
Each step is elementary, and hence the resulting matrix R satisfies $R = E_k \cdots E_1 A$ by Proposition 4.1.2. Thus $A \sim_{\mathrm{row}} R$.
(iii) *Leading Indices*: Let R and S be RREF matrices row-equivalent to A. Every elementary row operation replaces rows by linear combinations of rows, and its elementary inverse reverses the inclusion by Proposition 4.1.2(i)–(ii).
Hence $\mathrm{row}(R) = \mathrm{row}(A) = \mathrm{row}(S) := \mathcal{W}$. If $\mathcal{W} = \{\mathbf{0}\}$, then $R = S = 0$; assume $\mathcal{W} \neq \{\mathbf{0}\}$.
For $\mathbf{v} \in \mathcal{W} \setminus \{\mathbf{0}\}$, set $\ell(\mathbf{v}) := \min\{j : v_j \neq 0\}$. Let $X \in \{R, S\}$ have nonzero rows $X_{1,\cdot}, \dots, X_{h,\cdot}$ and pivot columns $q_1 < \cdots < q_h$.
$\mathbf{v} = \sum_{i=1}^{h} c_i X_{i,\cdot}^t$, set $a := \min\{i : c_i \neq 0\}$. The RREF conditions imply $(\forall j \in \{1,\dots,q_a-1\})\, v_j = 0$ and $v_{q_a} = c_a \neq 0$, so $\ell(\mathbf{v}) = q_a$.
Conversely, $\ell(X_{i,\cdot}^t) = q_i$. Therefore $\{\ell(\mathbf{v}) : \mathbf{v} \in \mathcal{W} \setminus \{\mathbf{0}\}\}$ is the pivot set of both R and S.
(iv) *Uniqueness*: Let $p_1 < \cdots < p_s$ be the common pivot columns.
For $i \in \{1,\dots,s\}$, the vector $\mathbf{d}_i := R_{i,\cdot}^t - S_{i,\cdot}^t$ belongs to $\mathcal{W}$ and $(\forall j \in \{1,\dots,s\})\, (\mathbf{d}_i)_{p_j} = 0$.
If $\mathbf{d}_i \neq \mathbf{0}$, then its leading index is a pivot column by part (iii), contradicting these zero pivot coordinates.
Thus $R_{i,\cdot} = S_{i,\cdot}$ for $i \leq s$, while every row with index greater than s is zero. Hence $R = S$.

Structural Commentary: Elimination constructs a representative; leading-index invariance proves uniqueness in the common row space.
Key Insight: A reduced row space determines every pivot row. □

Lemma 4.1.5. *Invertible Matrices Row-Reduce to the Identity*
Let $P \in \mathrm{Mat}(m,m,\mathbb{F})$ be invertible. Then:
(i) $\mathrm{rref}(P) = I_m$.
(ii) $I_m = E_k \cdots E_1 P$ *for a* $k \in \mathbb{N}_0$ *and elementary* $E_1,\dots,E_k \in \mathrm{Mat}(m,m,\mathbb{F})$.

Proof.
(i) Set $R = [r_{i,j}] := \mathrm{rref}(P)$. Then $R = E_k \cdots E_1 P$ by Theorem 4.1.4 and the definition of row equivalence.
Each E_i is invertible by Proposition 4.1.2(ii), so R is invertible by repeated use of Corollary 3.1.17(i). Consequently, $\ker(R) = \{\mathbf{0}\}$.
Let $p_1 < \cdots < p_s$ be the pivot columns of R. If $s < m$, choose $j \in \{1,\dots,m\} \setminus \{p_1,\dots,p_s\}$ and define $\mathbf{x} \in \mathbb{F}^m$ by $x_j = 1$, $(\forall i \in \{1,\dots,s\})\, x_{p_i} = -r_{i,j}$, and $(\forall \ell \in \{1,\dots,m\} \setminus \{j, p_1,\dots,p_s\})\, x_\ell = 0$.
Then $(\forall i \in \{1,\dots,s\})\, (R\mathbf{x})_i = x_{p_i} + r_{i,j} x_j = 0$ and $(\forall i \in \{s+1,\dots,m\})\, (R\mathbf{x})_i = 0$. This contradicts $\ker(R) = \{\mathbf{0}\}$, so $s = m$.
The strictly increasing pivot indices $p_1,\dots,p_m \in \{1,\dots,m\}$ satisfy $p_i = i$.
Every pivot column is therefore the corresponding coordinate column, so $R = I_m$.
(ii) By part (i), $P \sim_{\mathrm{row}} \mathrm{rref}(P) = I_m$. Thus $I_m = E_k \cdots E_1 P$ by the definition of row equivalence.

Structural Commentary: Kernel triviality excludes free columns; the square pivot set then forces the identity.
Key Insight: An invertible reduced matrix has no free variable. □

Corollary 4.1.6. *Row Equivalence Characterization*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, the following are equivalent:
(i) $A \sim_{\mathrm{row}} B$.
(ii) $\mathrm{rref}(A) = \mathrm{rref}(B)$.
(iii) $(\exists\, P \in \mathrm{Mat}(m,m,\mathbb{F})^{\times})\, B = PA$.

Proof.
(i) $\Rightarrow$ *(ii)* Write $B = G_q \cdots G_1 A$ and set $G := G_q \cdots G_1$.
The matrix G is invertible, $G^{-1} = G_1^{-1} \cdots G_q^{-1}$, and each G_j^{-1} is elementary by Proposition 4.1.2(ii) and Corollary 3.1.17(i).
For $R := \mathrm{rref}(A)$, write $R = E_k \cdots E_1 A$. Then $R = E_k \cdots E_1 G^{-1} B$, so $B \sim_{\mathrm{row}} R$ and $R = \mathrm{rref}(B)$ by Theorem 4.1.4.
(ii) $\Rightarrow$ *(iii)* Let $R := \mathrm{rref}(A) = \mathrm{rref}(B)$ and write $R = E_k \cdots E_1 A = F_\ell \cdots F_1 B$. For $E := E_k \cdots E_1$ and $F := F_\ell \cdots F_1$, the matrix $P := F^{-1}E$ is invertible by Proposition 4.1.2(ii) and Corollary 3.1.17(i), and $B = PA$.
(iii) $\Rightarrow$ *(i)* Let $B = PA$ with P invertible. Choose elementary matrices satisfying $E_k \cdots E_1 P = I_m$ by Lemma 4.1.5(ii).
Then $P = E_1^{-1} \cdots E_k^{-1}$, and each E_i^{-1} is elementary by Proposition 4.1.2(ii). Hence $A \sim_{\mathrm{row}} B$.

Structural Commentary: The implication cycle passes among canonical representatives, elementary products, and invertible left factors.
Key Insight: Row equivalence is exactly invertible left multiplication. □

Computation Method 4.1.7. *Inverse Computation by Gauss–Jordan Reduction*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, this procedure returns A^{-1} or certifies that A is not invertible.

(i) Form $[\,A \mid I_n\,]$ and perform each Gauss–Jordan row operation on the entire augmented matrix until the left block is $\mathrm{rref}(A)$.
(ii) If the result is $[\,I_n \mid B\,]$, choose $P \in \mathrm{Mat}(n,n,\mathbb{F})^\times$ such that $[\,I_n \mid B\,] = P[\,A \mid I_n\,] = [\,PA \mid P\,]$ by Corollary 4.1.6. Then $P = B$, $BA = I_n$, and $AB = I_n$ by Theorem 3.4.2; return $B = A^{-1}$.
(iii) If $\mathrm{rref}(A) \neq I_n$, then A is not invertible by Lemma 4.1.5(i).

5 Change of Basis

5.1 Further Change of Basis and Coordinate Transformations

Theorem 5.1.1. *Change of Basis Formula*
Let $n \geq 1$, let $\dim(\mathcal{V}) = n$, let $T \in \mathcal{L}(\mathcal{V})$, and fix a basis β of $\mathcal{V}$. Set $A := [T]_{\beta \leftarrow \beta}$.

(i) If γ is a basis of $\mathcal{V}$ and $S = C_{\gamma \leftarrow \beta}$, then $[T]_{\gamma \leftarrow \gamma} = SAS^{-1}$.
(ii) If $S \in \mathrm{Mat}(n,n,\mathbb{F})^\times$, then some basis γ of $\mathcal{V}$ satisfies $[T]_{\gamma \leftarrow \gamma} = SAS^{-1}$.

Proof.

(i) Suppose γ is a basis and $S = C_{\gamma \leftarrow \beta}$. Then $S^{-1} = C_{\beta \leftarrow \gamma}$ by Theorem 3.5.2(i). Hence $[T]_{\gamma \leftarrow \gamma} = C_{\gamma \leftarrow \beta} A C_{\beta \leftarrow \gamma} = SAS^{-1}$ by Theorem 3.5.3, with β as both original bases and γ as both new bases.
(ii) Let $S \in \mathrm{Mat}(n,n,\mathbb{F})^\times$. Choose a basis γ satisfying $S = C_{\gamma \leftarrow \beta}$ by Theorem 3.5.2(ii). Then $[T]_{\gamma \leftarrow \gamma} = SAS^{-1}$ by part (i).

Structural Commentary: The proof specializes two-sided transport, then realizes an arbitrary invertible coordinate map.
Key Insight: Conjugation records one operator after a simultaneous coordinate change. □

6 Matrix Equivalence and Similarity

6.1 Matrix Equivalence and Rank Normal Forms

Throughout this subsection, $m, n \geq 1$, and all vector spaces and matrices are over the same fixed field $\mathbb{F}$.

Definition 6.1.1. *Matrix Equivalence*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, write $A \sim_{\mathrm{eq}} B$ iff $(\exists\, P \in \mathrm{Mat}(m,m,\mathbb{F})^\times)(\exists\, Q \in \mathrm{Mat}(n,n,\mathbb{F})^\times)\; B = PAQ$.

Proposition 6.1.2. *Matrix Equivalence and Elementary Operations*
Let $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:

(i) The relation $\sim_{\mathrm{eq}}$ is an equivalence relation on $\mathrm{Mat}(m,n,\mathbb{F})$.
(ii) $A \sim_{\mathrm{eq}} B$ iff B can be obtained from A by finitely many elementary row and column operations.

Proof.

(i) The relation is reflexive by Definition 6.1.1, with $A = I_m A I_n$.
If $B = PAQ$ with P, Q invertible, then $A = P^{-1}BQ^{-1}$, so symmetry holds by Definition 6.1.1.
If $B = P_1 A Q_1$ and $C = P_2 B Q_2$, then $C = (P_2 P_1) A (Q_1 Q_2)$, and both products are invertible by Corollary 3.1.17(i).
Hence transitivity holds by Definition 6.1.1.
(ii) ($\Rightarrow$) Suppose $A \sim_{\mathrm{eq}} B$. Choose $P \in \mathrm{Mat}(m,m,\mathbb{F})^\times$ and $Q \in \mathrm{Mat}(n,n,\mathbb{F})^\times$ satisfying $B = PAQ$ by Definition 6.1.1.
The matrix Q^t is invertible by Corollary 3.1.17(ii). The matrices P and Q^t are products of elementary matrices by Lemma 4.1.5(ii) and Proposition 4.1.2(ii). Hence left multiplication by the factors of P performs elementary row operations, while right multiplication by the transposes of the factors of Q^t performs elementary column operations by Proposition 4.1.2(i),(iii).
Thus these operations transform A into $PAQ = B$.
(ii) ($\Leftarrow$) Suppose B is obtained from A by finitely many elementary row and column operations.
Each elementary row operation is left multiplication by an invertible elementary matrix by Proposition 4.1.2(i),(ii).
Each elementary column operation is right multiplication by the transpose of an invertible elementary matrix by Proposition 4.1.2(ii),(iii) and Corollary 3.1.17(ii). Associativity collects the left factors into P and the right factors into Q, so the operation sequence has the form $B = PAQ$ with P, Q invertible by Corollary 3.1.17(i). Hence $A \sim_{\mathrm{eq}} B$ by Definition 6.1.1.

Structural Commentary: Part (i) uses invertible-factor algebra; part (ii) translates elementary operations into factors.
Key Insight: Two-sided factors are exactly row and column operations. □

Theorem 6.1.3. *Equivalent Matrices Represent the Same Linear Map*
Let $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$, and $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$.
Then $A \sim_{\mathrm{eq}} B$ iff some $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ and bases β, β' of $\mathcal{V}$ and γ, γ' of $\mathcal{W}$ satisfy $A = [T]_{\gamma \leftarrow \beta}$ and $B = [T]_{\gamma' \leftarrow \beta'}$.

Proof.

($\Rightarrow$) Suppose $A \sim_{\mathrm{eq}} B$. Choose $P \in \mathrm{Mat}(m,m,\mathbb{F})^\times$ and $Q \in \mathrm{Mat}(n,n,\mathbb{F})^\times$ satisfying $B = PAQ$ by Definition 6.1.1.
Choose bases β, γ by Corollary 1.3.14, and choose $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ satisfying $[T]_{\gamma \leftarrow \beta} = A$ by Theorem 3.4.5(i).
Choose bases γ' and β' satisfying $P = C_{\gamma' \leftarrow \gamma}$ and $Q^{-1} = C_{\beta' \leftarrow \beta}$ by Theorem 3.5.2(ii).
Then $Q = C_{\beta \leftarrow \beta'}$ by Theorem 3.5.2(i), and $[T]_{\gamma' \leftarrow \beta'} = PAQ = B$ by Theorem 3.5.3.
($\Leftarrow$) Choose $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ and bases β, β' of $\mathcal{V}$ and γ, γ' of $\mathcal{W}$ satisfying $A = [T]_{\gamma \leftarrow \beta}$ and $B = [T]_{\gamma' \leftarrow \beta'}$ by the hypothesis.
Then $B = C_{\gamma' \leftarrow \gamma} A C_{\beta \leftarrow \beta'}$ by Theorem 3.5.3, so $A \sim_{\mathrm{eq}} B$ by Definition 6.1.1.

Structural Commentary: One direction realizes invertible factors as basis changes; the converse reads coordinate transport.
Key Insight: Domain and codomain coordinates account for both factors. □

Definition 6.1.4. *Matrix Rank Normal Form*
For $0 \leq r \leq \min(m,n)$, define $J_r^{m,n} \in \mathrm{Mat}(m,n,\mathbb{F})$ as follows:

(i) $J_0^{m,n} := 0$.
(ii) If $r \geq 1$, then $J_r^{m,n} := \begin{pmatrix} I_r & 0_{r \times (n-r)} \\ 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{pmatrix}$.

Theorem 6.1.5. *Rank Classifies Matrix Equivalence*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$:
(i) $A \sim_{\mathrm{eq}} J^{m,n}_{\mathrm{rank}(A)}$.
(ii) $A \sim_{\mathrm{eq}} B$ *iff* $\mathrm{rank}(A) = \mathrm{rank}(B)$.

Proof.
(*i*) Set $r := \mathrm{rank}(A)$ and $T_A(\mathbf{x}) := A\mathbf{x}$ for $\mathbf{x} \in \mathbb{F}^n$. Then $T_A \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ by Proposition 3.1.8.
Choose $\mathcal{M}_A \leq \mathbb{F}^n$ satisfying $\mathbb{F}^n = \ker(T_A) \oplus \mathcal{M}_A$ by Proposition 2.3.6.
The restriction $T_A|_{\mathcal{M}_A} : \mathcal{M}_A \to \mathrm{range}(T_A)$ is an isomorphism by Proposition 2.3.8(i). The equality $\mathrm{range}(T_A) = \mathrm{col}(A)$ holds by Proposition 3.1.9 and Definition 3.1.12; hence $\dim(\mathcal{M}_A) = r$ by Theorem 1.7.9 and Definition 3.1.12.
Choose a basis $\mathbf{u} := (\mathbf{u}_1, \ldots, \mathbf{u}_r)$ of $\mathcal{M}_A$ and a basis κ of $\ker(T_A)$ by Corollary 1.3.14, where $\mathbf{u}$ is empty if $r = 0$.
The list $(T_A\mathbf{u}_1, \ldots, T_A\mathbf{u}_r)$ is a basis of $\mathrm{range}(T_A)$ by Proposition 1.7.8; extend it to a basis γ_A of $\mathbb{F}^m$ by Theorem 1.3.13.
Equality $\mathbb{F}^n = \mathcal{M}_A \oplus \ker(T_A)$ holds by Proposition 2.1.8. Hence concatenation $\beta_A := (\mathbf{u}, \kappa)$ is basis of $\mathbb{F}^n$ by Theorem 2.3.4(ii).
The first r basis vectors map to the first r vectors of γ_A, while the remaining basis vectors lie in $\ker(T_A)$.
Thus $[T_A]_{\gamma_A \leftarrow \beta_A} = J^{m,n}_r$ by Definition 3.2.3 and Definition 6.1.4.
The standard-basis matrix of T_A is A by Definition 3.1.7 and Definition 3.2.3. Therefore $A \sim_{\mathrm{eq}} J^{m,n}_r$ by Theorem 6.1.3.
(*ii*) ($\Rightarrow$) If $A \sim_{\mathrm{eq}} B$, then $B = PAQ$ for invertible P, Q by Definition 6.1.1, and $\mathrm{rank}(B) = \mathrm{rank}(A)$ by Theorem 3.6.6(ii).
(*ii*) ($\Leftarrow$) If $\mathrm{rank}(A) = \mathrm{rank}(B) = r$, then $A \sim_{\mathrm{eq}} J^{m,n}_r \sim_{\mathrm{eq}} B$ and $A \sim_{\mathrm{eq}} B$ by part (i) and Proposition 6.1.2(i).
Structural Commentary: Complement bases construct the normal form; rank preservation proves classification.
Key Insight: Kernel and range dimensions determine the two-sided orbit. □

Corollary 6.1.6. *Rank Normal Form for Linear Maps*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$, and $\mathrm{rank}(T) = r$. Then some bases β of $\mathcal{V}$ and γ of $\mathcal{W}$ satisfy $[T]_{\gamma \leftarrow \beta} = J^{m,n}_r$.

Proof. Choose initial bases β_0, γ_0 by Corollary 1.3.14, and set $A := [T]_{\gamma_0 \leftarrow \beta_0}$.
Then $\{[\mathbf{y}]_{\gamma_0} : \mathbf{y} \in \mathrm{range}(T)\} = \{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \mathrm{col}(A)$ by Proposition 3.2.5, 3.2.2(iii), 3.1.9, and Definition 3.1.12.
The coordinate map is linear and injective by Proposition 3.2.2(i),(ii), so its restriction is an isomorphism from $\mathrm{range}(T)$ onto $\mathrm{col}(A)$.
Hence $\mathrm{rank}(A) = \mathrm{rank}(T) = r$ by Theorem 1.7.9 and the definitions of matrix and operator rank.
Thus $A \sim_{\mathrm{eq}} J^{m,n}_r$ by Theorem 6.1.5(i); choose invertible P, Q satisfying $J^{m,n}_r = PAQ$ by Definition 6.1.1.
Choose bases γ, β satisfying $P = C_{\gamma \leftarrow \gamma_0}$ and $Q^{-1} = C_{\beta \leftarrow \beta_0}$ by Theorem 3.5.2(ii).
Then $Q = C_{\beta_0 \leftarrow \beta}$ by Theorem 3.5.2(i), and $[T]_{\gamma \leftarrow \beta} = J^{m,n}_r$ by Theorem 3.5.3.
Structural Commentary: Matrix normal form supplies factors that are realized as coordinate changes.
Key Insight: Rank fixes the canonical coordinate matrix. □

Corollary 6.1.7. *Linear Maps Classified by Rank*
Let $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$. For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $S \sim_{\mathrm{bc}} T$ iff some bases β_S, β_T of $\mathcal{V}$ and γ_S, γ_T of $\mathcal{W}$ satisfy $[S]_{\gamma_S \leftarrow \beta_S} = [T]_{\gamma_T \leftarrow \beta_T}$. Then $S \sim_{\mathrm{bc}} T$ iff $\mathrm{rank}(S) = \mathrm{rank}(T)$.

Proof.
($\Rightarrow$) Suppose $S \sim_{\mathrm{bc}} T$. Choose bases β_S, β_T of $\mathcal{V}$ and γ_S, γ_T of $\mathcal{W}$ satisfying $[S]_{\gamma_S \leftarrow \beta_S} = [T]_{\gamma_T \leftarrow \beta_T}$ by the definition of $\sim_{\mathrm{bc}}$, and set $A := [S]_{\gamma_S \leftarrow \beta_S}$. For $R \in \{S, T\}$, interpret β_R and γ_R accordingly.
Then $\{[\mathbf{y}]_{\gamma_R} : \mathbf{y} \in \mathrm{range}(R)\} = \{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \mathrm{col}(A)$ by Proposition 3.2.5, 3.2.2(iii), 3.1.9, and Definition 3.1.12.
The restricted coordinate map is an isomorphism by Proposition 3.2.2(i),(ii).
Thus $(\forall R \in \{S, T\})\ \mathrm{rank}(R) = \dim(\mathrm{col}(A))$ by Theorem 1.7.9 and Definition 1.7.2, so $\mathrm{rank}(S) = \mathrm{rank}(T)$.
($\Leftarrow$) If $\mathrm{rank}(S) = \mathrm{rank}(T) = r$, choose bases satisfying $[S]_{\gamma_S \leftarrow \beta_S} = J^{m,n}_r = [T]_{\gamma_T \leftarrow \beta_T}$ by Corollary 6.1.6.
Structural Commentary: Both directions compare operator ranks with the shared rank normal form.
Key Insight: Equal ranks select the same canonical matrix. □

6.2 Similarity and Operator Coordinates

Throughout this subsection, $m, n \geq 1$, and all vector spaces and matrices are over the same fixed field $\mathbb{F}$.

Definition 6.2.1. *Similarity*
Matrices $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ are similar iff $(\exists\, P \in \mathrm{Mat}(n,n,\mathbb{F})^{\times})\ B = PAP^{-1}$.

Proposition 6.2.2. *Similarity Is an Equivalence Relation*
Similarity is an equivalence relation on $\mathrm{Mat}(n,n,\mathbb{F})$.

Proof. Let $A, B, C \in \mathrm{Mat}(n,n,\mathbb{F})$. Similarity is reflexive by Definition 6.2.1, with $P = I_n$.
If $B = PAP^{-1}$ for invertible P, then $A = P^{-1}BP$, so symmetry holds by Definition 6.2.1. If $B = PAP^{-1}$ and $C = QBQ^{-1}$ for invertible P, Q, then $C = (QP)A(QP)^{-1}$ and QP is invertible by Corollary 3.1.17(i), so transitivity holds by Definition 6.2.1.
Structural Commentary: Identity, inverse, and product witnesses verify reflexivity, symmetry, and transitivity.
Key Insight: Conjugation is an orbit relation under invertible matrices. □

Remark 6.2.3. *Similarity Convention*
The forms PAP^{-1} and $M^{-1}AM$ are the same convention with $P = M^{-1}$. In Theorem 5.1.1, this means choosing opposite coordinate-change directions.

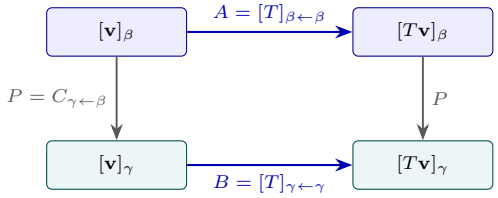

Figure 16: One operator in two bases: $PA = BP$, so $B = PAP^{-1}$.

In Figure 16, $A = [T]_{\beta\leftarrow\beta}$, $B = [T]_{\gamma\leftarrow\gamma}$, and $P = C_{\gamma\leftarrow\beta}$, so the coordinate square commutes as $PA = BP$. Theorem 5.1.1 owns the exact conjugation identity.

Corollary 6.2.4. *Similarity as One Operator in Two Bases*
For $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, the following are equivalent:

(i) A and B are similar.
(ii) Some n-dimensional vector space $\mathcal{V}$, bases β, γ of $\mathcal{V}$, and $T \in \mathcal{L}(\mathcal{V})$ satisfy $A = [T]_{\beta\leftarrow\beta}$ and $B = [T]_{\gamma\leftarrow\gamma}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Suppose A and B are similar. Choose $P \in \mathrm{Mat}(n, n, \mathbb{F})^\times$ satisfying $B = PAP^{-1}$ by Definition 6.2.1.
Let $\mathcal{V} := \mathbb{F}^n$, let β be its standard basis, and set $T_A(\mathbf{x}) := A\mathbf{x}$, which is linear by Proposition 3.1.8.
Then $[T_A]_{\beta\leftarrow\beta} = A$ by Definition 3.1.7 and Definition 3.2.3. Choose a basis γ satisfying $P = C_{\gamma\leftarrow\beta}$ by Theorem 3.5.2(ii).
Hence $[T_A]_{\gamma\leftarrow\gamma} = PAP^{-1} = B$ by Theorem 5.1.1(i).

(*ii*) $\Rightarrow$ (*i*) Suppose $A = [T]_{\beta\leftarrow\beta}$ and $B = [T]_{\gamma\leftarrow\gamma}$. Set $P := C_{\gamma\leftarrow\beta}$. Then P is invertible with $P^{-1} = C_{\beta\leftarrow\gamma}$ by Theorem 3.5.2(i), and $B = PAP^{-1}$ by Theorem 5.1.1(i). Thus A and B are similar by Definition 6.2.1.

Structural Commentary: One direction realizes conjugation as basis change; the converse reads the transport formula.
Key Insight: Similarity is one operator in two coordinate systems. □

Perspective 6.2.5. *Similarity as Same Operator, New Coordinates*
For matrices, similarity means that they represent the same operator in two bases, not merely that their entries are related.
This is the structural content behind change of basis.

Definition 6.2.6. *Similarity of Operators*
For $T \in \mathcal{L}(\mathcal{V})$ and $S \in \mathcal{L}(\mathcal{W})$, $T \sim S$ iff $(\exists\, \Phi : \mathcal{V} \xrightarrow{\sim} \mathcal{W})\, S = \Phi T \Phi^{-1}$.
When $\mathcal{V} = \mathcal{W}$, the witnesses lie in $\mathrm{Aut}(\mathcal{V})$, so similarity is conjugacy in $\mathcal{L}(\mathcal{V})$.

Proposition 6.2.7. *Operator Similarity and Common Matrix Representatives*
Let $\dim \mathcal{V} = \dim \mathcal{W} = n$, $T \in \mathcal{L}(\mathcal{V})$, and $S \in \mathcal{L}(\mathcal{W})$.
Then $T \sim S$ iff some bases β of $\mathcal{V}$ and γ of $\mathcal{W}$ satisfy $[T]_{\beta\leftarrow\beta} = [S]_{\gamma\leftarrow\gamma}$.

Proof.

($\Rightarrow$) Suppose $T \sim S$. Choose an isomorphism $\Phi : \mathcal{V} \to \mathcal{W}$ satisfying $S = \Phi T \Phi^{-1}$ by Definition 6.2.6.
Choose a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of $\mathcal{V}$ by Corollary 1.3.14. Set $\gamma := (\Phi\mathbf{v}_1, \ldots, \Phi\mathbf{v}_n)$, which is a basis of $\mathcal{W}$ by Proposition 1.7.8.
The identities $(\forall\, \mathbf{v} \in \mathcal{V})\, [\Phi\mathbf{v}]_\gamma = [\mathbf{v}]_\beta$ and $S\Phi = \Phi T$ hold by Definition 3.2.1 and Proposition 1.5.7(ii),(iii), respectively.
Consequently, each $j \in \{1, \ldots, n\}$ satisfies $[S(\Phi\mathbf{v}_j)]_\gamma = [T\mathbf{v}_j]_\beta$. These are the respective jth columns of $[S]_{\gamma\leftarrow\gamma}$ and $[T]_{\beta\leftarrow\beta}$ by Definition 3.2.3; hence the matrices are equal by Definition 3.1.1(i).

($\Leftarrow$) Suppose bases $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ satisfy $[T]_{\beta\leftarrow\beta} = [S]_{\gamma\leftarrow\gamma}$.
Choose $\Phi \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ satisfying $(\forall\, j \in \{1, \ldots, n\})\, \Phi\mathbf{v}_j = \mathbf{w}_j$ by Proposition 1.5.8. Its range is $\mathrm{span}(\gamma) = \mathcal{W}$ by Theorem 1.7.1(i), so Φ is surjective and hence injective by Corollary 1.7.6(ii). Therefore Φ is an isomorphism by Proposition 1.6.7.
The identity $(\forall\, \mathbf{v} \in \mathcal{V})\, [\Phi\mathbf{v}]_\gamma = [\mathbf{v}]_\beta$ holds by Definition 3.2.1.
Thus $(\forall\, j \in \{1, \ldots, n\})\, [\Phi T\mathbf{v}_j]_\gamma = [S\Phi\mathbf{v}_j]_\gamma$ by Definition 3.2.3, and $(\forall\, j \in \{1, \ldots, n\})\, \Phi T\mathbf{v}_j = S\Phi\mathbf{v}_j$ by Proposition 3.2.2(ii).
The compositions are linear by Proposition 1.5.7(i), so $\Phi T = S\Phi$ by Proposition 1.5.8.
Thus $S = \Phi T \Phi^{-1}$ by Proposition 1.5.7(ii),(iii), so $T \sim S$ by Definition 6.2.6.

Structural Commentary: A conjugating isomorphism transports a basis; a shared matrix reconstructs the isomorphism.
Key Insight: Matching basis actions force operator conjugacy. □

Theorem 6.2.8. *Similarity and Scalar Shifts*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) If A and B are similar, then $(\forall\, \lambda \in \mathbb{F})\, [A - \lambda I_n$ is similar to $B - \lambda I_n]$.
(ii) If $(\exists\, \lambda \in \mathbb{F})\, [A - \lambda I_n$ is similar to $B - \lambda I_n]$, then A and B are similar.

Proof.

(i) Choose $P \in \mathrm{Mat}(n, n, \mathbb{F})^\times$ satisfying $B = PAP^{-1}$ by the hypothesis and Definition 6.2.1.
For $\lambda \in \mathbb{F}$, $B - \lambda I_n = PAP^{-1} - \lambda PP^{-1} = P(A - \lambda I_n)P^{-1}$, so the shifted matrices are similar by Definition 6.2.1.

(ii) Choose $\lambda \in \mathbb{F}$ and $P \in \mathrm{Mat}(n, n, \mathbb{F})^\times$ satisfying $B - \lambda I_n = P(A - \lambda I_n)P^{-1}$ by the hypothesis and Definition 6.2.1.
After adding $\lambda I_n = \lambda PP^{-1}$, the equality becomes $B = PAP^{-1}$, so A and B are similar by Definition 6.2.1.

Structural Commentary: Both directions add or remove the central term λI_n inside a conjugation.
Key Insight: Scalar identity shifts commute with every basis change. □

Proposition 6.2.9. *Upper Block-Triangular Similarity Shift*
Let $B \in \mathrm{Mat}(m, m, \mathbb{F})$, $D \in \mathrm{Mat}(n, n, \mathbb{F})$, $C, X \in \mathrm{Mat}(m, n, \mathbb{F})$. Set $M := \left(\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right)$, $N := \left(\begin{smallmatrix} B & C+XD-BX \\ 0 & D \end{smallmatrix}\right)$. Then M and N are similar.

Proof. Set $U := \left(\begin{smallmatrix} I_m & X \\ 0 & I_n \end{smallmatrix}\right)$. By block multiplication, $U^{-1} = \left(\begin{smallmatrix} I_m & -X \\ 0 & I_n \end{smallmatrix}\right)$ and $UMU^{-1} = \left(\begin{smallmatrix} B & -BX+C+XD \\ 0 & D \end{smallmatrix}\right) = N$.
Thus M and N are similar by Definition 6.2.1.
Structural Commentary: Block multiplication verifies the inverse and computes the conjugated coupling block.
Key Insight: Unit upper conjugation preserves diagonal blocks and shears the coupling. □

Part IV
Geometric and Spectral Linear Algebra

1 Inner Product Geometry

1.1 Inner Products and Norms

Throughout, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. This subsection develops the norm and metric geometry determined by an inner product.

Definition 1.1.1. *Preliminary Terminology*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

(i) For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the *dot product* is $\mathbf{x} \cdot \mathbf{y} := x_1 y_1 + \cdots + x_n y_n$.
(ii) For $z = a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$, its *complex conjugate* and *modulus* are $\overline{z} := a - bi$ and $|z| := \sqrt{a^2 + b^2}$.
(iii) A map $f : \mathcal{V} \to \mathbb{F}$ is *conjugate linear* if $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})(\forall\, \lambda \in \mathbb{F})\ \big(f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v}) \wedge f(\lambda \mathbf{v}) = \overline{\lambda} f(\mathbf{v})\big)$.
(iv) A map $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is *sesquilinear* if it is linear in its first slot and conjugate linear in its second slot.

Motivation 1.1.2. *From Dot Products to Inner Products*
For $\mathbf{x} \in \mathbb{R}^n$, Euclidean squared length is $\mathbf{x} \cdot \mathbf{x} = \sum_{j=1}^{n} x_j^2$.

For $z \in \mathbb{C}$, the scalar z^2 need not be real, whereas $z\overline{z} = |z|^2 \geq 0$.
Conjugating one factor therefore preserves nonnegative squared length and motivates the following axioms.

Definition 1.1.3. *Inner Product*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. An *inner product* on $\mathcal{V}$ is a map $\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ satisfying, for $(\mathbf{u}, \mathbf{v}, \mathbf{w}, \lambda) \in \mathcal{V}^3 \times \mathbb{F}$:

(i) *positivity*: $\langle \mathbf{v}, \mathbf{v} \rangle \in [0, \infty)$;
(ii) *definiteness*: $\langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}$;
(iii) *additivity in the first slot*: $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$;
(iv) *homogeneity in the first slot*: $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle$;
(v) *conjugate symmetry*: $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$.

Definition 1.1.4. *Inner-Product Space*
An *inner-product space* is a vector space $\mathcal{V}$ together with an inner product on $\mathcal{V}$.

Henceforth, unless stated otherwise, $\mathcal{V}$ denotes an inner-product space over $\mathbb{F}$.

Definition 1.1.5. *Norm Induced by an Inner Product*
For $\mathbf{v} \in \mathcal{V}$, the *norm* of $\mathbf{v}$ is $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

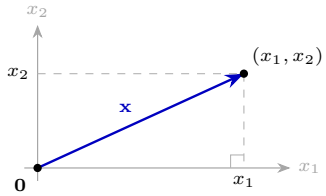

Figure 17: The norm is vector length, with $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$.

Figure 17 draws $\mathbf{x} \in \mathbb{R}^2$ with coordinate legs x_1, x_2, so its length realizes $\|\mathbf{x}\|^2 = x_1^2 + x_2^2$.

Example 1.1.6. *Standard Inner Product on $\mathbb{F}^n$*
For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, set $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{j=1}^{n} x_j \overline{y_j}$. Sesquilinearity and conjugate symmetry hold componentwise, while $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{j=1}^{n} |x_j|^2$ verifies positivity and definiteness. For $\mathbb{F} = \mathbb{R}$, this is the dot product.

Example 1.1.7. *Function and Matrix Inner Products*

(i) If $a < b$, then $\langle f, g \rangle := \int_a^b f(t)\overline{g(t)}\, dt$ is an inner product on $C_{\mathbb{F}}[a, b]$.
Positivity and definiteness follow from $\langle f, f \rangle = \int_a^b |f(t)|^2\, dt \geq 0$, with equality only for $f = 0$ by continuity.

(ii) On $\mathrm{Mat}(m, n, \mathbb{F})$, the *Frobenius inner product* is $\langle A, B \rangle_F := \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} \overline{b_{ij}}$.
Sesquilinearity holds entrywise, and $\langle A, A \rangle_F = \sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2$ is nonnegative and vanishes exactly when $A = 0$.

Example 1.1.8. *Square-Summable Sequences*
Set $\ell^2(\mathbb{F}) := \left\{ (x_j)_{j \geq 1} : \sum_{j=1}^{\infty} |x_j|^2 < \infty \right\}$ and $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{j=1}^{\infty} x_j \overline{y_j}$. The bound $|x_j \overline{y_j}| \leq \frac{1}{2}(|x_j|^2 + |y_j|^2)$ ensures absolute convergence.
Termwise algebra verifies sesquilinearity and conjugate symmetry, while $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{j=1}^{\infty} |x_j|^2$ verifies positivity and definiteness.
Thus $\ell^2(\mathbb{F})$ is an inner-product space containing $\mathbb{F}^{(\mathbb{N})}$.

Proposition 1.1.9. *Basic Inner-Product Identities*
For $(\mathbf{u}, \mathbf{v}, \mathbf{w}, \lambda) \in \mathcal{V}^3 \times \mathbb{F}$:

(i) for fixed $\mathbf{u}$, the map $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u} \rangle$ is linear;
(ii) $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ and $\langle \mathbf{u}, \mathbf{0} \rangle = 0$;

(iii) $\langle \mathbf{u}, \mathbf{v}+\mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$;
(iv) $\langle \mathbf{u}, \lambda\mathbf{v}\rangle = \overline{\lambda}\,\langle \mathbf{u}, \mathbf{v}\rangle$.

Proof.

(i) The map $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u}\rangle$ is linear by first-slot additivity and homogeneity.
(ii) $\langle \mathbf{0}, \mathbf{u}\rangle = 0$ by (i), and $\langle \mathbf{u}, \mathbf{0}\rangle = \overline{\langle \mathbf{0}, \mathbf{u}\rangle} = 0$ by conjugate symmetry.
(iii) $\langle \mathbf{u}, \mathbf{v}+\mathbf{w}\rangle = \overline{\langle \mathbf{v}+\mathbf{w}, \mathbf{u}\rangle} = \overline{\langle \mathbf{v}, \mathbf{u}\rangle} + \overline{\langle \mathbf{w}, \mathbf{u}\rangle} = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$ by conjugate symmetry and additivity.
(iv) $\langle \mathbf{u}, \lambda\mathbf{v}\rangle = \overline{\langle \lambda\mathbf{v}, \mathbf{u}\rangle} = \overline{\lambda}\,\overline{\langle \mathbf{v}, \mathbf{u}\rangle} = \overline{\lambda}\,\langle \mathbf{u}, \mathbf{v}\rangle$ by conjugate symmetry and homogeneity.

Structural Commentary: Conjugate symmetry transfers the first-slot axioms to the second slot.
Key Insight: One linear slot determines conjugate linearity in the other. □

Lemma 1.1.10. *Inner Products Separate Vectors*
For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$:
(i) $\big((\forall\, \mathbf{x} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{u}\rangle = \langle \mathbf{x}, \mathbf{v}\rangle\big) \Longrightarrow \mathbf{u} = \mathbf{v}$;
(ii) $\big((\forall\, \mathbf{x} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{w}\rangle = 0\big) \Longleftrightarrow \mathbf{w} = \mathbf{0}$.

Proof.

(i) With $\mathbf{x} = \mathbf{u} - \mathbf{v}$, $\|\mathbf{u}-\mathbf{v}\|^2 = \langle \mathbf{u}-\mathbf{v}, \mathbf{u}\rangle - \langle \mathbf{u}-\mathbf{v}, \mathbf{v}\rangle = 0$ by Proposition 1.1.9. The equality $\mathbf{u} = \mathbf{v}$ holds by definiteness.
(ii) $(\Rightarrow)$ $\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w}\rangle = 0$ by the hypothesis with $\mathbf{x} = \mathbf{w}$, and $\mathbf{w} = \mathbf{0}$ by definiteness.
(ii) $(\Leftarrow)$ If $\mathbf{w} = \mathbf{0}$, then $(\forall\, \mathbf{x} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{w}\rangle = 0$ by Proposition 1.1.9(ii).

Structural Commentary: Each claim tests a difference vector against a strategically chosen argument.
Key Insight: Testing a vector against itself detects whether it vanishes. □

Example 1.1.11. *Norms from Standard Inner Products*

(i) On $\mathbb{F}^n$, the standard inner product induces the *Euclidean norm* $\|(x_1, \ldots, x_n)\|_2 = \sqrt{\sum_{j=1}^{n} |x_j|^2}$.

(ii) On $C_{\mathbb{F}}[a,b]$ with $a < b$, the integral inner product induces the *L^2 norm* $\|f\|_2 = \sqrt{\int_a^b |f(t)|^2\, dt}$.

(iii) On $\mathrm{Mat}(m,n,\mathbb{F})$, the Frobenius inner product induces $\|A\|_F = \sqrt{\langle A, A\rangle_F} = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2}$ for $A = [a_{ij}]$.

This is the *Frobenius*, *Schur*, or *Hilbert–Schmidt norm*.

Proposition 1.1.12. *Basic Properties of the Derived Norm*
For $(\mathbf{v}, \lambda) \in \mathcal{V} \times \mathbb{F}$:
(i) $\|\mathbf{v}\|$ *is real and nonnegative;*
(ii) $\|\mathbf{v}\| = 0 \Longleftrightarrow \mathbf{v} = \mathbf{0}$;
(iii) $\|\lambda\mathbf{v}\| = |\lambda|\,\|\mathbf{v}\|$.

Proof.

(i) The scalar $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}$ is real and nonnegative by positivity.
(ii) $(\Rightarrow)$ If $\|\mathbf{v}\| = 0$, then $\langle \mathbf{v}, \mathbf{v}\rangle = 0$, and $\mathbf{v} = \mathbf{0}$ by definiteness.
(ii) $(\Leftarrow)$ If $\mathbf{v} = \mathbf{0}$, then $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ by definiteness and $\|\mathbf{v}\| = 0$ by the norm definition.
(iii) $\|\lambda\mathbf{v}\|^2 = \langle \lambda\mathbf{v}, \lambda\mathbf{v}\rangle = \lambda\overline{\lambda}\,\langle \mathbf{v}, \mathbf{v}\rangle = |\lambda|^2\,\|\mathbf{v}\|^2$ by sesquilinearity. The equality $\|\lambda\mathbf{v}\| = |\lambda|\,\|\mathbf{v}\|$ holds by nonnegative square roots.

Structural Commentary: Positivity controls the square root, while sesquilinearity controls scalar rescaling.
Key Insight: Squared length is easier to manipulate than length itself. □

Theorem 1.1.13. *Cauchy–Schwarz Inequality*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:
(i) $\big|\langle \mathbf{u}, \mathbf{v}\rangle\big| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$;
(ii) equality in (i) holds if and only if $\mathbf{u}, \mathbf{v}$ *are linearly dependent.*

Proof.

(i) If $\mathbf{v} = \mathbf{0}$, the inequality is an equality. Suppose $\mathbf{v} \neq \mathbf{0}$ and set $\lambda := \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{v}\|^2}$.

Then $0 \le \|\mathbf{u} - \lambda\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \frac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{\|\mathbf{v}\|^2}$ by positivity, sesquilinearity, and conjugate symmetry.

The inequality $\big|\langle \mathbf{u}, \mathbf{v}\rangle\big| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$ follows by multiplication by $\|\mathbf{v}\|^2$ and nonnegative square roots.
(ii) $(\Rightarrow)$ If equality holds and $\mathbf{v} = \mathbf{0}$, then $\mathbf{u}, \mathbf{v}$ are linearly dependent.
If $\mathbf{v} \neq \mathbf{0}$, then $\|\mathbf{u} - \lambda\mathbf{v}\| = 0$ by equality in (i) and the preceding identity, and $\mathbf{u} = \lambda\mathbf{v}$ by Proposition 1.1.12(ii).
(ii) $(\Leftarrow)$ If $\mathbf{v} = \mathbf{0}$, equality holds. Else linear dependence has form $\mathbf{u} = \lambda\mathbf{v}$; $\big|\langle \mathbf{u}, \mathbf{v}\rangle\big| = |\lambda|\,\|\mathbf{v}\|^2 = \|\mathbf{u}\|\,\|\mathbf{v}\|$ by Proposition 1.1.12(iii).

Structural Commentary: A least-residual scalar supplies the bound and characterizes equality.
Key Insight: Positivity controls the mixed term through optimal scalar comparison. □

Theorem 1.1.14. *Triangle Inequality*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:
(i) $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$;
(ii) equality in (i) holds if and only if one of $\mathbf{u}, \mathbf{v}$ *is a nonnegative real multiple of the other.*

Proof.

(i) $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\,\mathrm{Re}\,\langle \mathbf{u}, \mathbf{v}\rangle$ by sesquilinearity and conjugate symmetry.
By Theorem 1.1.13(i), this is at most $\big(\|\mathbf{u}\| + \|\mathbf{v}\|\big)^2$. The inequality $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ follows by nonnegative square roots.

(*ii*) (⇒) If either vector is $\mathbf{0}$, it is a nonnegative multiple of the other.
Otherwise equality requires $\operatorname{Re}\langle \mathbf{u}, \mathbf{v}\rangle = |\langle \mathbf{u}, \mathbf{v}\rangle|$ and equality in Cauchy–Schwarz.
Choose $a \in \mathbb{F}$ with $\mathbf{u} = a\mathbf{v}$ by Theorem 1.1.13(ii). Then $a \in [0, \infty)$ by $a\|\mathbf{v}\|^2 = \langle \mathbf{u}, \mathbf{v}\rangle \in [0, \infty)$.
(*ii*) (⇐) If $\mathbf{u} = a\mathbf{v}$ for $a \in [0, \infty)$, then $\|\mathbf{u} + \mathbf{v}\| = (a + 1)\|\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$. Interchanging $\mathbf{u}, \mathbf{v}$ in this calculation covers $\mathbf{v} = a\mathbf{u}$.

Structural Commentary: Cauchy–Schwarz bounds the cross term, and its equality case fixes the direction.
Key Insight: Equality requires aligned vectors with nonnegative relative scale. □

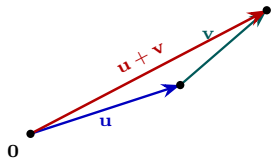

Figure 18: The direct displacement $\mathbf{u} + \mathbf{v}$ is no longer than the two-step path with legs $\mathbf{u}$ and $\mathbf{v}$.

Figure 18 identifies the three vectors compared by the triangle inequality; it does not supply the proof.

Proposition 1.1.15. *Reverse Triangle Inequality*
$(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ |\|\mathbf{u}\| - \|\mathbf{v}\|| \le \|\mathbf{u} - \mathbf{v}\|$.

Proof. $\|\mathbf{u}\| - \|\mathbf{v}\| \le \|\mathbf{u} - \mathbf{v}\|$ by Theorem 1.1.14(i) applied to $\mathbf{u} = (\mathbf{u} - \mathbf{v}) + \mathbf{v}$.
Likewise, $\|\mathbf{v}\| - \|\mathbf{u}\| \le \|\mathbf{u} - \mathbf{v}\|$ by the same result with $\mathbf{u}, \mathbf{v}$ interchanged.
The two bounds are equivalent to the asserted absolute-value inequality.
Structural Commentary: Two triangle-inequality bounds control both possible signs of the norm difference.
Key Insight: Distance bounds the change in length in either direction. □

Definition 1.1.16. *Norm and Normed Vector Space*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. A *norm* on $\mathcal{V}$ is a map $\|\cdot\| : \mathcal{V} \to [0, \infty)$ satisfying, for $(\mathbf{u}, \mathbf{v}, \lambda) \in \mathcal{V}^2 \times \mathbb{F}$:

(i) $\|\mathbf{u}\| \ge 0$;
(ii) $\|\mathbf{u}\| = 0 \iff \mathbf{u} = \mathbf{0}$;
(iii) $\|\lambda\mathbf{u}\| = |\lambda|\,\|\mathbf{u}\|$;
(iv) $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.

A *normed vector space* is a vector space equipped with a norm. Its *unit ball* is $\{\mathbf{v} \in \mathcal{V} : \|\mathbf{v}\| \le 1\}$.

Definition 1.1.17. *Distance Induced by an Inner Product*
The *distance induced by the inner product* is $d : \mathcal{V} \times \mathcal{V} \to [0, \infty)$ defined by $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$.

Proposition 1.1.18. *Inner-Product Distance Is a Metric*
For $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$ and $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$:

(i) $d(\mathbf{u}, \mathbf{v}) \ge 0$;
(ii) $d(\mathbf{u}, \mathbf{v}) = 0 \iff \mathbf{u} = \mathbf{v}$;
(iii) $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$;
(iv) $d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$.

Proof.

(*i*) $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| \ge 0$ by Proposition 1.1.12(i).
(*ii*) (⇒) If $d(\mathbf{u}, \mathbf{v}) = 0$, then $\mathbf{u} - \mathbf{v} = \mathbf{0}$ by Proposition 1.1.12(ii); equivalently, $\mathbf{u} = \mathbf{v}$.
(*ii*) (⇐) If $\mathbf{u} = \mathbf{v}$, then $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{0}\| = 0$ by Proposition 1.1.12(ii).
(*iii*) $d(\mathbf{u}, \mathbf{v}) = \|-(\mathbf{v} - \mathbf{u})\| = \|\mathbf{v} - \mathbf{u}\| = d(\mathbf{v}, \mathbf{u})$ by Proposition 1.1.12(iii) with $\lambda = -1$.
(*iv*) Since $\mathbf{u} - \mathbf{v} = (\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})$, $d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ by Theorem 1.1.14(i).

Structural Commentary: Each metric axiom reduces to the corresponding norm property on a difference vector.
Key Insight: Distances are norms of displacements. □

Perspective 1.1.19. *Metric Viewpoint*
Because $d(\mathbf{v}_n, \mathbf{v}) = \|\mathbf{v}_n - \mathbf{v}\|$, metric convergence is exactly norm convergence.
Closedness, continuity, and completeness require topological results developed later; no such conclusion is used as authority here.

Theorem 1.1.20. *Parallelogram Identity*
$(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ \|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\big(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\big)$.

Proof. For $\varepsilon \in \{1, -1\}$, $\|\mathbf{u} + \varepsilon\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\varepsilon \operatorname{Re}\langle \mathbf{u}, \mathbf{v}\rangle$ by sesquilinearity and conjugate symmetry.
The asserted identity is the sum of the cases $\varepsilon = 1, -1$.
Structural Commentary: Paired expansions cancel the cross terms while preserving both squared norms.
Key Insight: Opposite diagonals erase the mixed inner-product contribution. □

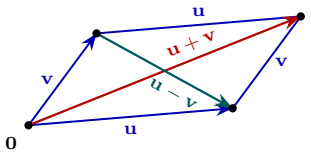

Figure 19: A parallelogram with side vectors $\mathbf{u}, \mathbf{v}$ and diagonal vectors $\mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v}$.

Figure 19 identifies the sides and diagonals whose squared lengths occur in the parallelogram identity.

Proposition 1.1.21. *Coordinate Norms on $\mathbb{F}^n$*
For $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}^n$, each of the following defines a norm on $\mathbb{F}^n$:

(i) $\|\mathbf{x}\|_1 := \sum_{j=1}^{n} |x_j|$;

(ii) $\|\mathbf{x}\|_2 := \left(\sum_{j=1}^{n} |x_j|^2 \right)^{1/2}$;

(iii) $\|\mathbf{x}\|_\infty := \max_{1 \le j \le n} |x_j|$.

Proof.

(i) Nonnegativity, definiteness, and absolute homogeneity of $\|\cdot\|_1$ hold by the corresponding scalar properties.
Its triangle inequality is $\|\mathbf{x}+\mathbf{y}\|_1 = \sum_{j=1}^{n} |x_j + y_j| \le \sum_{j=1}^{n} (|x_j| + |y_j|) = \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$ by the scalar triangle inequality.

(ii) Standard inner product induces $\|\cdot\|_2$, first 3 norm axioms hold (Proposition 1.1.12), triangle inequality hold (Theorem 1.1.14(i)).

(iii) Nonnegativity and absolute homogeneity of $\|\cdot\|_\infty$ hold by modulus, and $\|\mathbf{x}\|_\infty = 0 \iff (\forall\, j \in \{1, \dots, n\})\, x_j = 0 \iff \mathbf{x} = \mathbf{0}$.
Choose $k \in \{1, \dots, n\}$ with $\|\mathbf{x}+\mathbf{y}\|_\infty = |x_k + y_k|$.
Then $\|\mathbf{x}+\mathbf{y}\|_\infty \le |x_k| + |y_k| \le \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$ by the scalar triangle inequality and the definition of the maximum.

Structural Commentary: Scalar modulus handles the ℓ_1 and ℓ_∞ cases; inner-product geometry handles the ℓ_2 case.
Key Insight: Distinct coordinate aggregations satisfy the same norm axioms. □

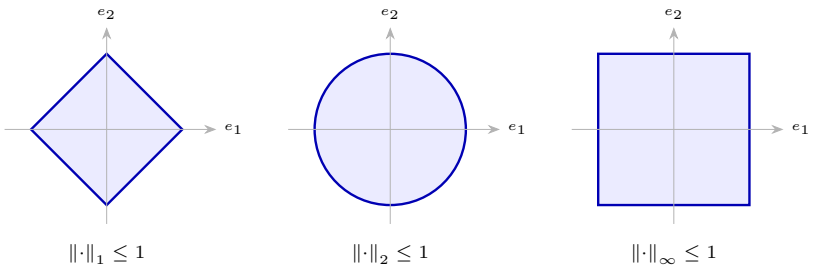

Figure 20: The ℓ_1, ℓ_2, and ℓ_∞ unit balls in $\mathbb{R}^2$ are respectively a diamond, a disk, and a square.

Figure 20 contrasts the three unit-ball shapes without serving as proof of the norm axioms.

Corollary 1.1.22. *ℓ_1 and ℓ_∞ Are Not Inner-Product Norms*
If $n \ge 2$, then the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $\mathbb{F}^n$ are not induced by any inner product.

Proof. For the first two coordinate vectors $\mathbf{e}_1, \mathbf{e}_2$, direct evaluation makes the two sides of the parallelogram identity 8 and 4 under $\|\cdot\|_1$, and 2 and 4 under $\|\cdot\|_\infty$. Neither norm is induced by an inner product by Theorem 1.1.20.
Structural Commentary: Two coordinate tests contradict the necessary parallelogram identity.
Key Insight: A failed parallelogram identity excludes every inducing inner product. □

Theorem 1.1.23. *Polarization Identities*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:

(i) If $\mathbb{F} = \mathbb{R}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 \right)$.

(ii) If $\mathbb{F} = \mathbb{C}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 + i\,\|\mathbf{u}+i\mathbf{v}\|^2 - i\,\|\mathbf{u}-i\mathbf{v}\|^2 \right)$.

Proof. Let $z := \langle \mathbf{u}, \mathbf{v} \rangle$.

(i) If $\mathbb{F} = \mathbb{R}$, then $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4 \operatorname{Re} z = 4z$ by sesquilinearity, which is equivalent to (i).

(ii) If $\mathbb{F} = \mathbb{C}$, then $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4 \operatorname{Re} z$ and $\|\mathbf{u}+i\mathbf{v}\|^2 - \|\mathbf{u}-i\mathbf{v}\|^2 = 4 \operatorname{Im} z$ by sesquilinearity.
Substitution into $z = \operatorname{Re} z + i \operatorname{Im} z$ is exactly (ii).

Structural Commentary: Norm differences recover the real part directly and the imaginary part after rotation.
Key Insight: The norm determines both components of every complex inner product. □

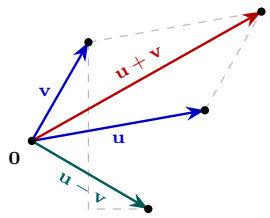

Figure 21: The sum and difference vectors isolate $4 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle$ through their squared-length difference.

Figure 21 illustrates the real-part identity; the complex identity also compares $\mathbf{u} \pm i\mathbf{v}$.

Proposition 1.1.24. *Operator Zero Tests from Inner Products*
Let $T \in \mathcal{L}(\mathcal{V})$.

(i) $\big((\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{w} \rangle = 0 \big) \Longrightarrow T = 0$.

(ii) *If* $\mathbb{F} = \mathbb{C}$, *then* $\big((\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle = 0\big) \Longrightarrow T = 0$.
(iii) *If* $\mathbb{F} = \mathbb{R}$, *the implication in (ii) can fail.*

Proof.

(i) $(\forall\, \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\|^2 = \langle T\mathbf{v}, T\mathbf{v}\rangle = 0$ by the hypothesis with $\mathbf{w} = T\mathbf{v}$; $T = 0$ by definiteness.
(ii) Set $Q(\mathbf{x}) := \langle T\mathbf{x}, \mathbf{x}\rangle$. For fixed $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, set $a := \langle T\mathbf{x}, \mathbf{y}\rangle$ and $b := \langle T\mathbf{y}, \mathbf{x}\rangle$.
For $r \in \mathbb{C}$, $0 = Q(r\mathbf{x} + \mathbf{y}) = ra + \overline{r}b$ by the hypothesis and sesquilinearity.
The choices $r = 1, i$ establish $a + b = a - b = 0$, and $a = b = 0$ by these equations.
The identity $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y}\rangle = 0$ holds by the arbitrary choice of $\mathbf{x}, \mathbf{y}$, and $T = 0$ by (i).
(iii) On standard $\mathbb{R}^2$, define $T(x_1, x_2) := (-x_2, x_1)$. Then $(\forall\, \mathbf{v} \in \mathbb{R}^2)\ \langle T\mathbf{v}, \mathbf{v}\rangle = -v_2v_1 + v_1v_2 = 0$, while $T(1, 0) = (0, 1) \neq \mathbf{0}$.

Structural Commentary: Self-testing handles (i), complex scalar probes handle (ii), and rotation refutes the real analogue.
Key Insight: Complex phases recover mixed terms that real diagonal tests miss. □

Principle 1.1.25. *Geometry from Algebra*
For inner-product norms, expanding $\|\mathbf{u} \pm \mathbf{v}\|^2$ exposes the cross terms that control angles, inequalities, and polarization.
This heuristic does not apply to arbitrary norms, as the ℓ_1 and ℓ_∞ countertests demonstrate.

1.2 Unit Vectors and Orthogonality

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Definition 1.2.1. *Unit Vector and Normalization*
A vector $\mathbf{u} \in \mathcal{V}$ is a *unit vector* if $\|\mathbf{u}\| = 1$. If $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is called the *normalization* of $\mathbf{v}$.

Proposition 1.2.2. *Properties of Normalization*
For $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$:
(i) $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ *is a unit vector;*
(ii) $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ *is a positive real scalar multiple of* $\mathbf{v}$.

Proof.

(i) Since $\mathbf{v} \neq \mathbf{0}$, $\|\mathbf{v}\| > 0$ by Proposition 1.1.12(i)–(ii). The identity $\left\|\frac{\mathbf{v}}{\|\mathbf{v}\|}\right\| = \frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1$ holds by Proposition 1.1.12(iii).
(ii) $\frac{\mathbf{v}}{\|\mathbf{v}\|} = \lambda\mathbf{v}$ for $\lambda := \frac{1}{\|\mathbf{v}\|} \in \mathbb{R}_{>0}$ by Proposition 1.1.12(i)–(ii).

Structural Commentary: Normalization changes scale but preserves the generated one-dimensional subspace.
Key Insight: Divide by the length to make the length equal to one. □

Definition 1.2.3. *Orthogonal Vectors and Finite Lists*
Vectors $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ are *orthogonal*, written $\mathbf{u} \perp \mathbf{v}$, if $\langle \mathbf{u}, \mathbf{v}\rangle = 0$.
A finite list $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$ is *mutually orthogonal* if $(\forall\, j, k \in \{1, \ldots, m\})[j \neq k \Longrightarrow \langle \mathbf{v}_j, \mathbf{v}_k\rangle = 0]$.

Proposition 1.2.4. *Orthogonality and the Zero Vector*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:
(i) $\mathbf{u} \perp \mathbf{v} \Longleftrightarrow \mathbf{v} \perp \mathbf{u}$.
(ii) $(\forall\, \mathbf{x} \in \mathcal{V})\,(\mathbf{0} \perp \mathbf{x} \wedge \mathbf{x} \perp \mathbf{0})$.
(iii) $\mathbf{v} \perp \mathbf{v} \Longrightarrow \mathbf{v} = \mathbf{0}$.

Proof.

(i) $(\Rightarrow)$ If $\mathbf{u} \perp \mathbf{v}$, then $\langle \mathbf{u}, \mathbf{v}\rangle = 0$ by Definition 1.2.3. Conjugate symmetry implies $\langle \mathbf{v}, \mathbf{u}\rangle = \overline{\langle \mathbf{u}, \mathbf{v}\rangle} = 0$, so $\mathbf{v} \perp \mathbf{u}$.
(i) $(\Leftarrow)$ If $\mathbf{v} \perp \mathbf{u}$, then $\langle \mathbf{v}, \mathbf{u}\rangle = 0$ by Definition 1.2.3. Conjugate symmetry implies $\langle \mathbf{u}, \mathbf{v}\rangle = \overline{\langle \mathbf{v}, \mathbf{u}\rangle} = 0$, so $\mathbf{u} \perp \mathbf{v}$.
(ii) $(\forall\, \mathbf{x} \in \mathcal{V})\ \big(\langle \mathbf{0}, \mathbf{x}\rangle = 0 \wedge \langle \mathbf{x}, \mathbf{0}\rangle = 0\big)$ by Proposition 1.1.9(ii).
(iii) If $\mathbf{v} \perp \mathbf{v}$, then $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ by Definition 1.2.3, and $\mathbf{v} = \mathbf{0}$ by definiteness.

Structural Commentary: Conjugate symmetry makes orthogonality symmetric, and definiteness isolates $\mathbf{0}$.
Key Insight: Conjugate symmetry preserves the equation "equals zero." □

Proposition 1.2.5. *Mutually Orthogonal Nonzero Vectors Are Independent*
Mutually orthogonal nonzero vectors $\mathbf{v}_1, \ldots, \mathbf{v}_m \in \mathcal{V}$ *are linearly independent.*

Proof. Suppose $\sum_{j=1}^{m} c_j\mathbf{v}_j = \mathbf{0}$ for $c_1, \ldots, c_m \in \mathbb{F}$, and fix $k \in \{1, \ldots, m\}$.

By Proposition 1.1.9(i)–(ii) and mutual orthogonality, $0 = \langle \mathbf{0}, \mathbf{v}_k\rangle = \left\langle \sum_{j=1}^{m} c_j\mathbf{v}_j, \mathbf{v}_k \right\rangle = c_k \langle \mathbf{v}_k, \mathbf{v}_k\rangle = c_k \|\mathbf{v}_k\|^2$.

Since $\mathbf{v}_k \neq \mathbf{0}$, $\|\mathbf{v}_k\| > 0$ by Proposition 1.1.12(i)–(ii), so $c_k = 0$. The index k was arbitrary, hence every coefficient vanishes.
Structural Commentary: Orthogonality isolates one coefficient at a time.
Key Insight: Pair a dependence relation with one vector from the list; every cross term disappears. □

1.3 Orthonormal Bases and Gram–Schmidt

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Definition 1.3.1. *Orthogonal and Orthonormal Families*
For an index set I, a family $(\mathbf{u}_i)_{i\in I}$ in $\mathcal{V}$ is *orthogonal* if $(\forall\, i,j\in I)[i\neq j \Longrightarrow \langle \mathbf{u}_i,\mathbf{u}_j\rangle = 0]$. It is *orthonormal* if $(\forall\, i,j\in I)\ \langle \mathbf{u}_i,\mathbf{u}_j\rangle = \delta_{ij}$.
Thus a finite list $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ is orthonormal exactly when $(\forall\, j,k\in\{1,\dots,m\})\ \langle \mathbf{e}_j,\mathbf{e}_k\rangle = \delta_{jk}$.
Equivalently, the vectors in the list are unit and mutually orthogonal.
If $(\mathbf{u}_i)_{i\in I}$ is orthogonal and $(\forall\, i\in I)\,\mathbf{u}_i\neq\mathbf{0}$, then $\left(\frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}\right)_{i\in I}$ is orthonormal.

Proposition 1.3.2. *Norm of an Orthonormal Linear Combination*
If $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ is orthonormal, then $(\forall\,(a_1,\dots,a_m)\in\mathbb{F}^m)\left[\left\|\sum_{j=1}^m a_j\mathbf{e}_j\right\|^2 = \sum_{j=1}^m |a_j|^2\right]$.

Proof. $\left\|\sum_{j=1}^m a_j\mathbf{e}_j\right\|^2 = \left\langle \sum_{j=1}^m a_j\mathbf{e}_j, \sum_{k=1}^m a_k\mathbf{e}_k\right\rangle = \sum_{j=1}^m\sum_{k=1}^m a_j\overline{a_k}\,\langle \mathbf{e}_j,\mathbf{e}_k\rangle = \sum_{j=1}^m |a_j|^2$ by sesquilinearity and orthonormality.
Structural Commentary: Orthonormality deletes every cross term in the squared-norm expansion.
Key Insight: In orthonormal coordinates, length is computed coefficient-by-coefficient. □

Corollary 1.3.3. *Orthonormal Lists Are Linearly Independent*
If $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ is orthonormal in $\mathcal{V}$, then it is linearly independent.

Proof. Suppose $\sum_{j=1}^m a_j\mathbf{e}_j = \mathbf{0}$ for $(a_1,\dots,a_m)\in\mathbb{F}^m$. Then $0 = \|\mathbf{0}\|^2 = \sum_{j=1}^m |a_j|^2$ by Proposition 1.3.2.
Each $a_j = 0$ by nonnegativity of the summands, so the list is linearly independent.
Structural Commentary: The norm identity converts a linear relation into a sum of nonnegative scalars.
Key Insight: A zero orthonormal linear combination can only have zero coefficients. □

Definition 1.3.4. *Orthonormal Basis*
An *orthonormal basis* of $\mathcal{V}$ is a basis of $\mathcal{V}$ that is an orthonormal list.

Proposition 1.3.5. *Right-Length Orthonormal Lists Are Bases*
If $\dim(\mathcal{V}) = n < \infty$ and $(\mathbf{e}_1,\dots,\mathbf{e}_n)$ is orthonormal, then it is an orthonormal basis of $\mathcal{V}$.

Proof. The list is linearly independent by Corollary 1.3.3. It is a basis by Proposition 1.4.4 because its length is $\dim(\mathcal{V})$.
Structural Commentary: Orthonormality supplies independence; the dimension criterion supplies spanning.
Key Insight: In finite dimensions, n independent vectors in an n-dimensional space already form a basis. □

Theorem 1.3.6. *Gram–Schmidt Process*
Let $m\geq 1$ and let $(\mathbf{v}_1,\dots,\mathbf{v}_m)$ be linearly independent in $\mathcal{V}$. Define $\mathbf{u}_1 := \mathbf{v}_1$ and $\mathbf{e}_1 := \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$.
For $2\leq j\leq m$, define $\mathbf{u}_j := \mathbf{v}_j - \sum_{k=1}^{j-1}\langle \mathbf{v}_j,\mathbf{e}_k\rangle\,\mathbf{e}_k$.
For $2\leq j\leq m$, set $\mathbf{e}_j := \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}$. Then:
(i) $(\forall\, j\in\{1,\dots,m\})[\operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j) = \operatorname{span}(\mathbf{e}_1,\dots,\mathbf{e}_j)]$;
(ii) $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ *is orthonormal.*

Proof.
(*i*) Put $V_j := \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j)$ and $E_j := \operatorname{span}(\mathbf{e}_1,\dots,\mathbf{e}_j)$. Induct on j.
Since $\mathbf{v}_1\neq\mathbf{0}$, $\mathbf{e}_1$ is a nonzero scalar multiple of $\mathbf{v}_1$, so $V_1 = E_1$. Suppose $V_{j-1} = E_{j-1}$.
The correction term in $\mathbf{u}_j$ lies in $E_{j-1} = V_{j-1}$, so $\mathbf{u}_j\in V_j$.
If $\mathbf{u}_j = \mathbf{0}$, then $\mathbf{v}_j\in V_{j-1}$, contrary to linear independence; hence $\mathbf{u}_j\neq\mathbf{0}$ and $\mathbf{e}_j\in V_j$, so $E_j\subseteq V_j$.
Conversely, $\mathbf{v}_j = \|\mathbf{u}_j\|\,\mathbf{e}_j + \sum_{k=1}^{j-1}\langle \mathbf{v}_j,\mathbf{e}_k\rangle\,\mathbf{e}_k\in E_j$, so $V_j\subseteq E_j$.
(*ii*) $(\forall\, j\in\{1,\dots,m\})\left[\|\mathbf{e}_j\| = \left\|\frac{1}{\|\mathbf{u}_j\|}\mathbf{u}_j\right\| = 1\right]$ by Proposition 1.1.12(iii) and nonvanishing established in (i). Induct on j to verify mutual orthogonality. If $(\mathbf{e}_1,\dots,\mathbf{e}_{j-1})$ is orthonormal and $k<j$, then $\langle \mathbf{u}_j,\mathbf{e}_k\rangle = \langle \mathbf{v}_j,\mathbf{e}_k\rangle - \sum_{i=1}^{j-1}\langle \mathbf{v}_j,\mathbf{e}_i\rangle\,\langle \mathbf{e}_i,\mathbf{e}_k\rangle = 0$.
Hence $\mathbf{e}_j\perp\mathbf{e}_k$ by first-slot homogeneity, and the reverse relation holds by Proposition 1.2.4(i). Thus the full list is orthonormal.
Structural Commentary: Prefix-span induction establishes well-definedness before orthogonality is verified.
Key Insight: Each remainder removes precisely the earlier orthonormal components. □

Theorem 1.3.7. *Gram–Schmidt for Arbitrary Finite Lists*
Let $m\geq 1$ and let $(\mathbf{v}_1,\dots,\mathbf{v}_m)$ be a finite list in $\mathcal{V}$. Set $\mathbf{u}_1 := \mathbf{v}_1$. For $2\leq k\leq m$, define $\mathcal{J}_{k-1} := \{j\in\{1,\dots,k-1\} : \mathbf{u}_j\neq\mathbf{0}\}$.
For $2\leq k\leq m$, set $\mathbf{u}_k := \mathbf{v}_k - \sum_{j\in\mathcal{J}_{k-1}}\frac{\langle \mathbf{v}_k,\mathbf{u}_j\rangle}{\langle \mathbf{u}_j,\mathbf{u}_j\rangle}\mathbf{u}_j$. Set $U_k := \operatorname{span}(\mathbf{u}_1,\dots,\mathbf{u}_k)$, $V_k := \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_k)$, with $U_0 = V_0 := \{\mathbf{0}\}$. Then:
(i) $(\mathbf{u}_1,\dots,\mathbf{u}_m)$ *is orthogonal;*
(ii) $(\forall\, k\in\{1,\dots,m\})[U_k = V_k]$.

Proof.
(*i*) Induct on k. A one-vector list is orthogonal. Suppose $(\mathbf{u}_1,\dots,\mathbf{u}_{k-1})$ is orthogonal and fix $i<k$.
If $\mathbf{u}_i = \mathbf{0}$, then $\mathbf{u}_k\perp\mathbf{u}_i$ by Proposition 1.2.4(ii). If $\mathbf{u}_i\neq\mathbf{0}$, then $i\in\mathcal{J}_{k-1}$ and $\langle \mathbf{u}_k,\mathbf{u}_i\rangle = \langle \mathbf{v}_k,\mathbf{u}_i\rangle - \sum_{j\in\mathcal{J}_{k-1}}\frac{\langle \mathbf{v}_k,\mathbf{u}_j\rangle}{\langle \mathbf{u}_j,\mathbf{u}_j\rangle}\langle \mathbf{u}_j,\mathbf{u}_i\rangle = 0$
by the induction hypothesis: all terms with $j\neq i$ vanish, and the $j=i$ term cancels the first term.
The reverse orthogonality relation holds by Proposition 1.2.4(i), so the full list is orthogonal.

(*ii*) Induct on k, starting from $U_0 = V_0$. Suppose $U_{k-1} = V_{k-1}$.
The correction term in $\mathbf{u}_k$ lies in $U_{k-1} = V_{k-1}$, so $\mathbf{u}_k \in V_k$ and $U_k \subseteq V_k$.
The recursion also expresses $\mathbf{v}_k$ as $\mathbf{u}_k$ plus a vector in U_{k-1}, so $\mathbf{v}_k \in U_k$ and $V_k \subseteq U_k$. Therefore $U_k = V_k$.

Structural Commentary: Orthogonality and prefix-span preservation follow by separate inductions.
Key Insight: Subtracting all nonzero earlier directions preserves the generated span. □

Corollary 1.3.8. *Zero Remainders Detect Dependence*
For the construction in Theorem 1.3.7, $(\forall\, k \in \{1, \dots, m\})[\mathbf{u}_k = \mathbf{0} \iff \mathbf{v}_k \in V_{k-1}]$.

Proof. Fix $k \in \{1, \dots, m\}$.
(⇒) Suppose $\mathbf{u}_k = \mathbf{0}$. If $k = 1$, then $\mathbf{v}_1 = \mathbf{0} \in V_0$. If $k \geq 2$, the recursion places $\mathbf{v}_k$ in $U_{k-1} = V_{k-1}$ by Theorem 1.3.7(ii).
(⇐) Suppose $\mathbf{v}_k \in V_{k-1}$. If $k = 1$, then $\mathbf{v}_1 = \mathbf{0} = \mathbf{u}_1$.
If $k \geq 2$, then $V_{k-1} = U_{k-1}$ by Theorem 1.3.7(ii), so write $\mathbf{v}_k = \sum_{j \in \mathcal{J}_{k-1}} a_j \mathbf{u}_j$ for a coefficient family $(a_j)_{j \in \mathcal{J}_{k-1}}$.
The nonzero vectors in this sum are orthogonal by Theorem 1.3.7(i), and $(\forall\, i \in \mathcal{J}_{k-1})\ \langle \mathbf{v}_k, \mathbf{u}_i \rangle = a_i \langle \mathbf{u}_i, \mathbf{u}_i \rangle$ by orthogonality.
Thus $(\forall\, i \in \mathcal{J}_{k-1})\, a_i = \frac{\langle \mathbf{v}_k, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle}$, so the correction term equals $\mathbf{v}_k$ and $\mathbf{u}_k = \mathbf{0}$ by the recursion.

Structural Commentary: Each direction compares the recursion with the preserved prefix span.
Key Insight: A zero remainder means the new vector adds no direction. □

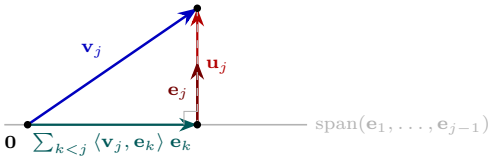

Figure 22: One Gram–Schmidt step: subtract the previous-span component, then normalize.

Figure 22 depicts the residual after removal of the previous-span component; normalization changes its length while preserving its generated one-dimensional subspace.

Example 1.3.9. *Legendre Polynomials from Gram–Schmidt*
Equip $\mathbb{R}[x]$ with $\langle p, q \rangle := \int_{-1}^{1} p(x) q(x)\, dx$ and perform Gram–Schmidt on $(1, x, x^2, x^3, \dots)$.
Parity annihilates mixed even–odd inner products. Since $\langle 1, 1 \rangle = 2$, $\langle x^2, 1 \rangle = \frac{2}{3}$, $\langle x, x \rangle = \frac{2}{3}$, and $\langle x^3, x \rangle = \frac{2}{5}$, the first remainders are $u_1(x) = 1$, $u_2(x) = x$, $u_3(x) = x^2 - \frac{1}{3}$, and $u_4(x) = x^3 - \frac{3}{5}x$.
These are nonzero scalar multiples of the first four Legendre polynomials.
Structural Commentary: Orthogonal polynomials are orthogonalized monomials.
Key Insight: Changing the inner product changes the polynomial family produced by Gram–Schmidt.

Lemma 1.3.10. *Gram–Schmidt on Orthogonal Lists*
Let $m \geq 1$ and let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a mutually orthogonal list of nonzero vectors. Then:
(i) $(\forall\, j \in \{1, \dots, m\})\, \mathbf{e}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$;
(ii) if the original list is orthonormal, then $(\forall\, j \in \{1, \dots, m\})\, \mathbf{e}_j = \mathbf{v}_j$.

Proof.
(*i*) The list is linearly independent by Proposition 1.2.5, so the construction is defined by Theorem 1.3.6. Induct on j.
The claim holds for $j = 1$. Suppose $(\forall\, i < j)\, \mathbf{e}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$. Then $(\forall\, i < j)\ \langle \mathbf{v}_j, \mathbf{e}_i \rangle = \frac{1}{\|\mathbf{v}_i\|} \langle \mathbf{v}_j, \mathbf{v}_i \rangle = 0$ by mutual orthogonality.
Thus $\mathbf{u}_j = \mathbf{v}_j$ in the recursion and $\mathbf{e}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$.
(*ii*) If the input is orthonormal, then $(\forall\, j \in \{1, \dots, m\})\ \|\mathbf{v}_j\| = 1$ by orthonormality, so $\mathbf{e}_j = \mathbf{v}_j$ by (i).

Structural Commentary: Induction verifies that every orthogonal correction term vanishes.
Key Insight: Orthogonal input already has no earlier-direction component. □

Corollary 1.3.11. *Existence of Orthonormal Bases*
If $\dim(\mathcal{V}) < \infty$, then $\mathcal{V}$ has an orthonormal basis.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, the empty list is an orthonormal basis. Suppose $\dim(\mathcal{V}) = n \geq 1$ and choose a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$.
By Theorem 1.3.6(i)–(ii), its Gram–Schmidt output is orthonormal and has span $\mathcal{V}$, so it is an orthonormal basis.
Structural Commentary: Gram–Schmidt orthonormalizes a basis without changing its span.
Key Insight: Finite-dimensional inner-product spaces always admit geometry-friendly coordinates. □

Corollary 1.3.12. *Extending an Orthonormal List*
If $\dim(\mathcal{V}) < \infty$ and $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ is orthonormal, then it extends to an orthonormal basis of $\mathcal{V}$.

Proof. If $m = 0$, an orthonormal basis exists by Corollary 1.3.11 and extends the empty list.
Suppose $m \geq 1$. The list is linearly independent by Corollary 1.3.3. $\exists\, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n$ such that $(\mathbf{e}_1, \dots, \mathbf{e}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ is a basis by Theorem 1.3.13. Let $(\mathbf{f}_1, \dots, \mathbf{f}_n)$ be its Gram–Schmidt output.
The first m steps depend only on the orthonormal prefix, so $(\forall\, j \in \{1, \dots, m\})\, \mathbf{f}_j = \mathbf{e}_j$ by Lemma 1.3.10(ii). The output is orthonormal by Theorem 1.3.6(ii), and it spans $\mathcal{V}$ by Theorem 1.3.6(i) with $j = n$. Thus $(\mathbf{f}_1, \dots, \mathbf{f}_n)$ is the required extension.
Structural Commentary: Extend linearly first, then orthonormalize while preserving the original prefix.
Key Insight: Gram–Schmidt leaves an orthonormal prefix fixed. □

1.4 Orthonormal Coordinates and Inner-Product Identities

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Theorem 1.4.1. *Coordinates in an Orthonormal Basis*
Let $\dim(\mathcal{V}) = n \geq 1$ *and let* $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *be an orthonormal basis. For* $\mathbf{v} \in \mathcal{V}$*:*

(i) $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \, \mathbf{e}_j$*;*

(ii) $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2$*;*

(iii) $[\mathbf{v}]_\beta = \big(\langle \mathbf{v}, \mathbf{e}_1 \rangle, \ldots, \langle \mathbf{v}, \mathbf{e}_n \rangle \big)^t$.

Proof.

(i) Write $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{e}_j$. For $k \in \{1, \ldots, n\}$, $\langle \mathbf{v}, \mathbf{e}_k \rangle = \sum_{j=1}^{n} a_j \langle \mathbf{e}_j, \mathbf{e}_k \rangle = a_k$ by orthonormality, which proves the expansion.

(ii) By (i) and Proposition 1.3.2, $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2$.

(iii) The coefficient computation in (i) means $[\mathbf{v}]_\beta = \big(\langle \mathbf{v}, \mathbf{e}_1 \rangle, \ldots, \langle \mathbf{v}, \mathbf{e}_n \rangle \big)^t$ by the coordinate-vector definition.

Structural Commentary: Pairing with each basis vector extracts the coefficients before the norm and coordinate formulas follow.
Key Insight: Orthonormality turns coefficient extraction into an inner-product calculation. □

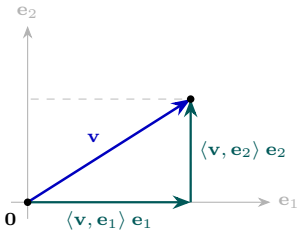

Figure 23: An orthonormal basis resolves a vector into perpendicular coordinate components.

Figure 23 illustrates Theorem 1.4.1(i) for $n = 2$: the two perpendicular coordinate components sum to $\mathbf{v}$.

Remark 1.4.2. *Cross Terms*
The mixed terms $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$ in $\|\mathbf{u} + \mathbf{v}\|^2$ are called *cross terms*; $\|\mathbf{u}\|^2$ and $\|\mathbf{v}\|^2$ are the pure squared-length terms.

Theorem 1.4.3. *Pythagorean Theorem*
$(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V}) \big[\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \big]$.

Proof. Fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ with $\mathbf{u} \perp \mathbf{v}$. Then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ by orthogonality and $\langle \mathbf{v}, \mathbf{u} \rangle = 0$ by Proposition 1.2.4(i).
Hence $\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
Structural Commentary: Orthogonality cancels the cross terms in the squared-norm expansion.
Key Insight: Perpendicular components add in squared length, not in length. □

Proposition 1.4.4. *Pythagorean Converse over* $\mathbb{R}$
If $\mathbb{F} = \mathbb{R}$*, then* $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V}) \big[\mathbf{u} \perp \mathbf{v} \iff \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \big]$.

Proof. Fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$. Conjugate symmetry over $\mathbb{R}$ implies $\langle \mathbf{v}, \mathbf{u} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$, so $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2 \langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$.
($\Rightarrow$) If $\mathbf{u} \perp \mathbf{v}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ by orthogonality, so the identity follows from the expansion.
($\Leftarrow$) If the identity holds, then $2 \langle \mathbf{u}, \mathbf{v} \rangle = 0$ by the expansion, so $\mathbf{u} \perp \mathbf{v}$ by orthogonality.
Structural Commentary: The two directions compare the norm identity with its single real cross term.
Key Insight: Over $\mathbb{R}$, vanishing length excess is equivalent to orthogonality. □

Theorem 1.4.5. *Bessel's Inequality*
Let $m \geq 1$ *and let* $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *be orthonormal in* $\mathcal{V}$*. Then* $(\forall\, \mathbf{v} \in \mathcal{V}) \sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2 \leq \|\mathbf{v}\|^2$.

Proof. Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{p} := \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \, \mathbf{e}_j$. For $k \in \{1, \ldots, m\}$, $\langle \mathbf{v} - \mathbf{p}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$ by orthonormality.
Hence $\langle \mathbf{v} - \mathbf{p}, \mathbf{p} \rangle = \sum_{j=1}^{m} \overline{\langle \mathbf{v}, \mathbf{e}_j \rangle} \langle \mathbf{v} - \mathbf{p}, \mathbf{e}_j \rangle = 0$, so $\mathbf{p} \perp (\mathbf{v} - \mathbf{p})$ by Proposition 1.2.4(i).
By Theorem 1.4.3 and Proposition 1.3.2, respectively, $\|\mathbf{v}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{v} - \mathbf{p}\|^2$ and $\|\mathbf{p}\|^2 = \sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2$. Thus $\sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2 \leq \|\mathbf{v}\|^2$.
Structural Commentary: The coordinate sum and its perpendicular residual split the squared norm by Pythagoras.
Key Insight: The residual norm is the nonnegative energy missing from selected coordinates. □

Corollary 1.4.6. *Coefficients Against an Orthonormal Sequence Vanish*
If $(\mathbf{e}_n)_{n \geq 1}$ *is orthonormal in* $\mathcal{V}$*, then* $(\forall\, \mathbf{v} \in \mathcal{V}) \lim_{n \to \infty} \langle \mathbf{v}, \mathbf{e}_n \rangle = 0$.

Proof. Fix $\mathbf{v} \in \mathcal{V}$ and $\epsilon > 0$, and set $S_\epsilon := \{j \in \mathbb{N}_{\geq 1} : |\langle \mathbf{v}, \mathbf{e}_j \rangle| \geq \epsilon\}$.
If S_ϵ were infinite, choose $N \in \mathbb{N}_{\geq 1}$ with $N\epsilon^2 > \|\mathbf{v}\|^2$ and distinct $j_1, \ldots, j_N \in S_\epsilon$.
For this orthonormal sublist, $N\epsilon^2 \leq \sum_{k=1}^{N} |\langle \mathbf{v}, \mathbf{e}_{j_k} \rangle|^2 \leq \|\mathbf{v}\|^2$ by Theorem 1.4.5, contradicting $N\epsilon^2 > \|\mathbf{v}\|^2$. Thus S_ϵ is finite.
Choose $N_0 \in \mathbb{N}_{\geq 1}$ greater than every member of S_ϵ, with $N_0 = 1$ when $S_\epsilon = \varnothing$. Then $(\forall\, n \geq N_0)\ |\langle \mathbf{v}, \mathbf{e}_n \rangle| < \epsilon$, so $\langle \mathbf{v}, \mathbf{e}_n \rangle \to 0$.
Structural Commentary: A contradiction argument applies Bessel to arbitrarily many allegedly large coefficients.
Key Insight: Finite total energy permits only finitely many coefficients above each threshold. □

Theorem 1.4.7. *Parseval's Identity*
Let $\dim(\mathcal{V}) = n \geq 1$ *and let* $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *be an orthonormal basis. For* $\mathbf{v}, \mathbf{w} \in \mathcal{V}$*:*

(i) $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle}$*;*

(ii) $\langle \mathbf{v}, \mathbf{w} \rangle = \big\langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \big\rangle_{\mathbb{F}^n}$*.*

Proof.

(i) Set $c_j := \langle \mathbf{v}, \mathbf{e}_j \rangle$ and $d_j := \langle \mathbf{w}, \mathbf{e}_j \rangle$ $(j \in \{1, \ldots, n\})$. Then $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{e}_j$ and $\mathbf{w} = \sum_{k=1}^{n} d_k \mathbf{e}_k$ by Theorem 1.4.1(i).
Hence $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \sum_{k=1}^{n} c_j \overline{d_k} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{n} c_j \overline{d_j}$ by sesquilinearity and orthonormality.

(ii) By Theorem 1.4.1(iii), $[\mathbf{v}]_\beta = (c_1, \ldots, c_n)^t$ and $[\mathbf{w}]_\beta = (d_1, \ldots, d_n)^t$. Therefore $\big\langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \big\rangle_{\mathbb{F}^n} = \sum_{j=1}^{n} c_j \overline{d_j} = \langle \mathbf{v}, \mathbf{w} \rangle$ by (i).

Structural Commentary: Basis expansions remove cross terms before the coordinate formula identifies the remaining sum.
Key Insight: Orthonormal coordinates preserve the inner product itself. □

Theorem 1.4.8. *Matrix Entries from an Orthonormal Codomain Basis*
Let $\dim(\mathcal{V}) = n \geq 1$ *and* $\dim(\mathcal{W}) = m \geq 1$*, with* $\mathcal{W}$ *an inner-product space.*
Fix a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *of* $\mathcal{V}$ *and an orthonormal basis* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ *of* $\mathcal{W}$*. Then:*

(i) $(\forall\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}),\ i \in \{1, \ldots, m\},\ j \in \{1, \ldots, n\})\ \big([T]_{\gamma \leftarrow \beta}\big)_{i,j} = \langle T\mathbf{v}_j, \mathbf{w}_i \rangle$*;*

(ii) $(\forall\, A \in \mathrm{Mat}(m, n, \mathbb{F}))(\exists!\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}))\ [T]_{\gamma \leftarrow \beta} = A$*.*

Proof.

(i) Fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $i \in \{1, \ldots, m\}$, and $j \in \{1, \ldots, n\}$. The jth column of $[T]_{\gamma \leftarrow \beta}$ is $[T\mathbf{v}_j]_\gamma = \big(\langle T\mathbf{v}_j, \mathbf{w}_1 \rangle, \ldots, \langle T\mathbf{v}_j, \mathbf{w}_m \rangle\big)^t$ by Theorem 1.4.1(iii). Therefore its ith entry is $\langle T\mathbf{v}_j, \mathbf{w}_i \rangle$.

(ii) The map $\Phi : \mathcal{L}(\mathcal{V}, \mathcal{W}) \to \mathrm{Mat}(m, n, \mathbb{F})$ defined by $\Phi(T) := [T]_{\gamma \leftarrow \beta}$ is an isomorphism by Theorem 3.4.5(i).
Hence every $A \in \mathrm{Mat}(m, n, \mathbb{F})$ has a unique preimage T under Φ.

Structural Commentary: Coordinate extraction identifies entries; the earlier encoding isomorphism supplies existence and uniqueness.
Key Insight: Entry (i, j) measures the jth basis image along output direction i. □

Figure 24 visualizes Theorem 1.4.8(i): applying T to $\mathbf{v}_j$ forms column j, and pairing with $\mathbf{w}_i$ reads entry (i, j).

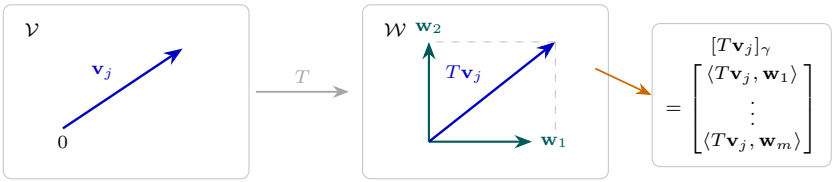

Figure 24: The jth column of $[T]_{\gamma \leftarrow \beta}$ is obtained by resolving $T\mathbf{v}_j$ against the orthonormal basis of $\mathcal{W}$.

Corollary 1.4.9. *Inner Products Through an Operator Matrix*
Let $\dim(\mathcal{V}) = n \geq 1$ *and let* $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *be an orthonormal basis.*
For $T \in \mathcal{L}(\mathcal{V})$*, set* $A := [T]_{\beta \leftarrow \beta}$*. Then* $(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{w} \rangle = \big\langle A[\mathbf{v}]_\beta, [\mathbf{w}]_\beta \big\rangle_{\mathbb{F}^n}$*.*

Proof. Fix $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The identity $[T\mathbf{v}]_\beta = A[\mathbf{v}]_\beta$ holds by Proposition 3.2.5. Hence $\langle T\mathbf{v}, \mathbf{w} \rangle = \big\langle A[\mathbf{v}]_\beta, [\mathbf{w}]_\beta \big\rangle_{\mathbb{F}^n}$ by Theorem 1.4.7(ii).
Structural Commentary: Matrix action computes the first coordinate vector before Parseval transfers the inner product.
Key Insight: Orthonormal coordinates convert operator geometry into matrix algebra. □

1.5 Gram Matrices

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Definition 1.5.1. *Gram Matrix*
For $n \geq 1$ and $\mathbf{u}_1, \ldots, \mathbf{u}_n \in \mathcal{V}$, the *Gram matrix* is $\mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := [\langle \mathbf{u}_j, \mathbf{u}_i \rangle]_{i,j=1}^{n} \in \mathrm{Mat}(n, n, \mathbb{F})$.

Definition 1.5.2. *Gram Determinant*
Let $n \in \mathbb{Z}_{>0}$, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $\mathbf{u}_1, \ldots, \mathbf{u}_n$ belong to an inner-product space $\mathcal{V}$ over $\mathbb{F}$.
Their *Gram determinant* is $\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \det \mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$.

Proposition 1.5.3. *Gram Matrix Quadratic Form*
Let $n \geq 1$*, let* $\mathbf{u}_1, \ldots, \mathbf{u}_n \in \mathcal{V}$*, and set* $G = [g_{ij}] := \mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$*. Then:*

(i) $(\forall\, i, j \in \{1, \ldots, n\})\ g_{ij} = \overline{g_{ji}}$*, so* G *is Hermitian;*

(ii) $(\forall\, \mathbf{c} = (c_1, \ldots, c_n)^t \in \mathbb{F}^n)\ \mathbf{c}^* G \mathbf{c} = \left\| \sum_{j=1}^{n} c_j \mathbf{u}_j \right\|^2 \geq 0$*;*

(iii) if $\mathbf{u}_1, \dots, \mathbf{u}_n$ are linearly independent, then $(\forall\, \mathbf{c} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, \mathbf{c}^* G \mathbf{c} > 0$.

Proof.

(i) For $i, j \in \{1, \dots, n\}$, $g_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle = \overline{\langle \mathbf{u}_i, \mathbf{u}_j \rangle} = \overline{g_{ji}}$ by conjugate symmetry.

(ii) Fix $\mathbf{c} = (c_1, \dots, c_n)^t \in \mathbb{F}^n$ and set $\mathbf{x} := \sum_{j=1}^{n} c_j \mathbf{u}_j$.

By sesquilinearity and definiteness, $\mathbf{c}^* G \mathbf{c} = \sum_{i=1}^{n} \sum_{j=1}^{n} \overline{c_i}\, \langle \mathbf{u}_j, \mathbf{u}_i \rangle\, c_j = \langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2 \geq 0$.

(iii) Suppose $\mathbf{u}_1, \dots, \mathbf{u}_n$ are linearly independent and fix $\mathbf{c} = (c_1, \dots, c_n)^t \neq \mathbf{0}$.

Then $\sum_{j=1}^{n} c_j \mathbf{u}_j \neq \mathbf{0}$ by linear independence, so $\mathbf{c}^* G \mathbf{c} > 0$ by (ii) and definiteness.

Structural Commentary: Entry symmetry and a norm-square expansion establish Hermitian positivity; independence makes it strict.
Key Insight: Gram quadratic forms measure norms of coefficient-weighted vector sums. □

1.6 Finite-Dimensional Riesz Representation

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Theorem 1.6.1. *Riesz Representation Theorem*
Suppose $\dim(\mathcal{V}) < \infty$ and let $\varphi \in \mathcal{V}'$. Then:
(i) $(\exists\, \mathbf{u} \in \mathcal{V})(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$;
(ii) $(\forall\, \mathbf{u}_1, \mathbf{u}_2 \in \mathcal{V})\big[(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle \implies \mathbf{u}_1 = \mathbf{u}_2\big]$.

Proof.

(i) If $\mathcal{V} = \{\mathbf{0}\}$, then $\varphi(\mathbf{0}) = 0$ by linearity, so $\mathbf{u} = \mathbf{0}$ represents φ.

Suppose $\dim(\mathcal{V}) = n \geq 1$ and choose an orthonormal basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ by Corollary 1.3.11. Set $\mathbf{u} := \sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)} \mathbf{e}_j$.

Fix $\mathbf{v} \in \mathcal{V}$. By Theorem 1.4.1(i), linearity of φ, and Proposition 1.1.9(iii)–(iv), $\varphi(\mathbf{v}) = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \varphi(\mathbf{e}_j) = \langle \mathbf{v}, \mathbf{u} \rangle$.

(ii) Suppose $(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle$. Then $\mathbf{u}_1 = \mathbf{u}_2$ by Lemma 1.1.10(i).

Structural Commentary: An orthonormal expansion constructs the representative; inner-product separation proves uniqueness.
Key Insight: Functional values become conjugated coordinates of one vector. □

Definition 1.6.2. *Riesz Vector*
If $\dim(\mathcal{V}) < \infty$ and $\varphi \in \mathcal{V}'$, the unique $\mathbf{u} \in \mathcal{V}$ satisfying $(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ is the *Riesz vector* representing φ.

Theorem 1.6.3. *Riesz Map*
Suppose $\dim(\mathcal{V}) < \infty$ and define $\tau : \mathcal{V} \to \mathcal{V}'$ by $\tau(\mathbf{u}) := \langle \cdot, \mathbf{u} \rangle$. Then:
(i) τ is conjugate linear: $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{u}, \mathbf{w} \in \mathcal{V})\, \tau(a\mathbf{u} + \mathbf{w}) = \overline{a}\, \tau(\mathbf{u}) + \tau(\mathbf{w})$;
(ii) τ is bijective;
(iii) $R := \tau^{-1}$ is conjugate linear: $(\forall\, a \in \mathbb{F})(\forall\, \varphi, \psi \in \mathcal{V}')\, R(a\varphi + \psi) = \overline{a}\, R(\varphi) + R(\psi)$;
(iv) if $\mathbb{F} = \mathbb{R}$, then τ and R are linear.

Proof.

(i) $(\forall\, \mathbf{u} \in \mathcal{V})\, \tau(\mathbf{u}) \in \mathcal{V}'$ by Proposition 1.1.9(i). Fix $a \in \mathbb{F}$, $\mathbf{u}, \mathbf{w}, \mathbf{v} \in \mathcal{V}$. Then $\tau(a\mathbf{u} + \mathbf{w})(\mathbf{v}) = \overline{a}\, \tau(\mathbf{u})(\mathbf{v}) + \tau(\mathbf{w})(\mathbf{v})$ by Proposition 1.1.9(iii)–(iv), so $\tau(a\mathbf{u} + \mathbf{w}) = \overline{a}\, \tau(\mathbf{u}) + \tau(\mathbf{w})$ by equality of functions.

(ii) If $\tau(\mathbf{u}) = \tau(\mathbf{w})$, then $\|\mathbf{u} - \mathbf{w}\|^2 = \tau(\mathbf{u} - \mathbf{w})(\mathbf{u} - \mathbf{w}) = 0$ by (i), so $\mathbf{u} = \mathbf{w}$ by definiteness.
Thus τ is injective. For $\varphi \in \mathcal{V}'$, $(\exists\, \mathbf{u} \in \mathcal{V})\, \tau(\mathbf{u}) = \varphi$ by Theorem 1.6.1(i), so τ is surjective.

(iii) Fix $a \in \mathbb{F}$ and $\varphi, \psi \in \mathcal{V}'$. By (i) and the definition of R, $\tau\big(\overline{a}\, R(\varphi) + R(\psi)\big) = a\varphi + \psi = \tau\big(R(a\varphi + \psi)\big)$.
Hence $R(a\varphi + \psi) = \overline{a}\, R(\varphi) + R(\psi)$ by injectivity from (ii).

(iv) If $\mathbb{F} = \mathbb{R}$, then $\overline{a} = a$ in (i) and (iii), so both maps are linear.

Structural Commentary: Conjugate-linearity, separation, and representation establish the bijection and its inverse law.
Key Insight: Conjugating the representing slot conjugates scalar multiplication. □

Figure 25 depicts Theorem 1.6.3: the vector $\mathbf{u}$ corresponds to the functional $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u} \rangle$.

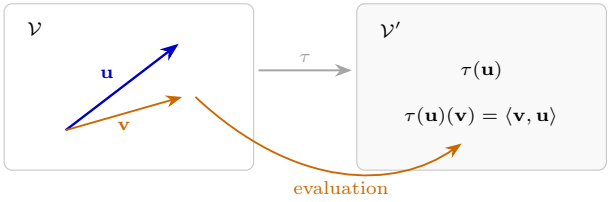

Figure 25: The map τ sends each vector to the functional it represents.

Perspective 1.6.4. *Riesz as Coordinate Duality*

For an orthonormal basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$, Theorem 1.6.1(i) constructs the Riesz vector of $\varphi \in \mathcal{V}'$ as $\sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)} \mathbf{e}_j$.

Although this formula uses the basis, uniqueness in part (ii) makes its value basis-independent.

2 Orthogonality, Projections, and Least Squares

2.1 Orthogonal Complements

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Orthogonal complements turn orthogonality into a subspace operation and organize canonical direct-sum decompositions.

Definition 2.1.1. *Orthogonality and Orthogonal Direct Sums*
Let $\mathcal{X}, \mathcal{Y} \subseteq \mathcal{V}$ and let $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$.

(i) $\mathcal{X}$ and $\mathcal{Y}$ are *orthogonal*, written $\mathcal{X} \perp \mathcal{Y}$, if $(\forall\, \mathbf{x} \in \mathcal{X})(\forall\, \mathbf{y} \in \mathcal{Y})\ \langle \mathbf{x}, \mathbf{y} \rangle = 0$.
(ii) $\mathcal{V}$ is the *orthogonal direct sum* of $\mathcal{S}$ and $\mathcal{T}$, written $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, if $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{S} \perp \mathcal{T}$.

Definition 2.1.2. *Orthogonal Complement*
For $\mathcal{U} \subseteq \mathcal{V}$, its *orthogonal complement* is $\mathcal{U}^{\perp} := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{u} \in \mathcal{U})\ \langle \mathbf{v}, \mathbf{u} \rangle = 0\}$.

Proposition 2.1.3. *Basic Properties of Orthogonal Complements*
For subsets $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$,

(i) $\mathcal{U}^{\perp}$ is a subspace of $\mathcal{V}$;
(ii) $\{\mathbf{0}\}^{\perp} = \mathcal{V}$;
(iii) $\mathcal{V}^{\perp} = \{\mathbf{0}\}$;
(iv) $\mathcal{U} \cap \mathcal{U}^{\perp} \subseteq \{\mathbf{0}\}$;
(v) if $\mathcal{U} \subseteq \mathcal{W}$, then $\mathcal{W}^{\perp} \subseteq \mathcal{U}^{\perp}$.

Proof.

(i) $\mathbf{0} \in \mathcal{U}^{\perp}$ by Proposition 1.1.9(ii). Fix $\mathbf{v}, \mathbf{w} \in \mathcal{U}^{\perp}$ and $a \in \mathbb{F}$.
Then $(\forall\, \mathbf{u} \in \mathcal{U})\ \langle a\mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = a\langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle = 0$ by linearity, so $a\mathbf{v} + \mathbf{w} \in \mathcal{U}^{\perp}$. Hence $\mathcal{U}^{\perp} \leq \mathcal{V}$ by Theorem 1.2.2.
(ii) $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{v}, \mathbf{0} \rangle = 0$ by Proposition 1.1.9(ii), so $\{\mathbf{0}\}^{\perp} = \mathcal{V}$.
(iii) $\mathbf{0} \in \mathcal{V}^{\perp}$ by (i). If $\mathbf{v} \in \mathcal{V}^{\perp}$, then $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$ by definiteness.
(iv) If $\mathbf{v} \in \mathcal{U} \cap \mathcal{U}^{\perp}$, then $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$ by definiteness.
(v) Suppose $\mathcal{U} \subseteq \mathcal{W}$ and fix $\mathbf{v} \in \mathcal{W}^{\perp}$. Then $(\forall\, \mathbf{u} \in \mathcal{U})\ \langle \mathbf{v}, \mathbf{u} \rangle = 0$, so $\mathbf{v} \in \mathcal{U}^{\perp}$.

Structural Commentary: Subspace closure, definiteness, and reversed constraint inclusion establish the five properties.
Key Insight: Larger tested sets impose stronger orthogonality constraints. □

Remark 2.1.4. *Definiteness Matters*
For a bilinear form $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ and $\mathcal{U} \subseteq \mathcal{V}$, set $\mathcal{U}^{\perp_B} := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{u} \in \mathcal{U})\, B(\mathbf{v}, \mathbf{u}) = 0\}$.
Bilinearity makes $\mathcal{U}^{\perp_B}$ a subspace, but without definiteness one may have $\mathcal{V}^{\perp_B} \neq \{\mathbf{0}\}$ or $\mathcal{U} \cap \mathcal{U}^{\perp_B} \neq \{\mathbf{0}\}$.
Thus $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp_B}$ is an inner-product conclusion, not a consequence of bilinearity alone.

Theorem 2.1.5. *Direct Sum with the Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$.

Proof. If $\mathcal{U} = \{\mathbf{0}\}$, then $\mathcal{U}^{\perp} = \mathcal{V}$ by Proposition 2.1.3(ii), and hence $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$.
Suppose $\dim(\mathcal{U}) = m \geq 1$ and choose an orthonormal basis $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ by Corollary 1.3.11.
Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{u} := \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \mathbf{e}_j \in \mathcal{U}$ and $\mathbf{w} := \mathbf{v} - \mathbf{u}$.
For $k \in \{1, \ldots, m\}$, $\langle \mathbf{w}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$ by orthonormality.
Fix $\mathbf{x} \in \mathcal{U}$ and write $\mathbf{x} = \sum_{k=1}^{m} a_k \mathbf{e}_k$. Then $\langle \mathbf{w}, \mathbf{x} \rangle = \sum_{k=1}^{m} \overline{a_k} \langle \mathbf{w}, \mathbf{e}_k \rangle = 0$ by Proposition 1.1.9(iii)–(iv), so $\mathbf{w} \in \mathcal{U}^{\perp}$ and $\mathbf{v} = \mathbf{u} + \mathbf{w}$.
Hence $\mathcal{V} = \mathcal{U} + \mathcal{U}^{\perp}$. The identity $\mathcal{U} \cap \mathcal{U}^{\perp} = \{\mathbf{0}\}$ holds by Proposition 2.1.3(i),(iv), so the sum is direct.
Structural Commentary: An orthonormal expansion constructs the component; definiteness makes the decomposition direct.
Key Insight: Every vector splits into a subspace component and orthogonal residual. □

Proposition 2.1.6. *Orthogonal Direct Sum Characterization*
For $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$, the following are equivalent:

(i) $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$.
(ii) $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$.

Proof.

(i) $\Rightarrow$ (ii) Suppose $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$. Then $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ by orthogonal-direct-sum definition, and $\mathcal{T} \subseteq \mathcal{S}^{\perp}$ by Proposition 1.2.4(i).
Fix $\mathbf{v} \in \mathcal{S}^{\perp}$ and write $\mathbf{v} = \mathbf{s} + \mathbf{t}$ with $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$.
Then $\mathbf{s} = \mathbf{v} - \mathbf{t} \in \mathcal{S}^{\perp}$, so $\mathbf{s} = \mathbf{0}$ by Proposition 2.1.3(iv). Hence $\mathbf{v} = \mathbf{t} \in \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$.
(ii) $\Rightarrow$ (i) Suppose $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$. Then $\mathcal{S} \perp \mathcal{T}$ by Proposition 1.2.4(i), so $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$ by orthogonal-direct-sum definition.

Structural Commentary: Direct-sum uniqueness forces an orthogonal partner to be maximal.
Key Insight: A full orthogonal summand is exactly the complement. □

Corollary 2.1.7. *Dimension of the Orthogonal Complement*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$, then $\dim(\mathcal{U}^{\perp}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. The subspace $\mathcal{U}$ is finite-dimensional by Corollary 1.3.14.
By Theorem 2.1.5, $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$, so $\dim(\mathcal{V}) = \dim(\mathcal{U}) + \dim(\mathcal{U}^{\perp})$ by Corollary 2.3.2. Rearranging proves the formula.
Structural Commentary: The direct-sum dimension formula converts decomposition into subtraction.
Key Insight: The complement contains precisely the unused dimensions. □

Corollary 2.1.8. *Double Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{U} = (\mathcal{U}^{\perp})^{\perp}$.

Proof. Fix $\mathbf{u} \in \mathcal{U}$. For $\mathbf{w} \in \mathcal{U}^\perp$, $\langle \mathbf{w}, \mathbf{u} \rangle = 0$ by membership, so $\langle \mathbf{u}, \mathbf{w} \rangle = 0$ by Proposition 1.2.4(i). Hence $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$.
Conversely, fix $\mathbf{v} \in (\mathcal{U}^\perp)^\perp$. By Theorem 2.1.5, write $\mathbf{v} = \mathbf{u}_0 + \mathbf{w}$ with $\mathbf{u}_0 \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^\perp$.
Since $(\mathcal{U}^\perp)^\perp$ is a subspace by Proposition 2.1.3(i), the established inclusion implies $\mathbf{w} = \mathbf{v} - \mathbf{u}_0 \in (\mathcal{U}^\perp)^\perp$.
Thus $\mathbf{w} \in \mathcal{U}^\perp \cap (\mathcal{U}^\perp)^\perp$, so $\mathbf{w} = \mathbf{0}$ by Proposition 2.1.3(iv), and $\mathbf{v} = \mathbf{u}_0 \in \mathcal{U}$.
Structural Commentary: Symmetry establishes one inclusion; the orthogonal decomposition eliminates every excess vector.
Key Insight: Double orthogonality adds no new finite-dimensional direction. □

Corollary 2.1.9. *Double Complement of a Finite Span*
If $\mathcal{X} \subseteq \mathcal{V}$ and $\operatorname{span}(\mathcal{X})$ is finite-dimensional, then $(\mathcal{X}^\perp)^\perp = \operatorname{span}(\mathcal{X})$.

Proof. Let $\mathcal{W} = \operatorname{span}(\mathcal{X})$. Since $\mathcal{X} \subseteq \mathcal{W}$, $\mathcal{W}^\perp \subseteq \mathcal{X}^\perp$ by Proposition 2.1.3(v).
Conversely, fix $\mathbf{v} \in \mathcal{X}^\perp$ and $\mathbf{w} \in \mathcal{W}$. Write $\mathbf{w} = \sum_{j=1}^{r} a_j \mathbf{x}_j$ with $\mathbf{x}_j \in \mathcal{X}$.
Then $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{r} \overline{a_j} \langle \mathbf{v}, \mathbf{x}_j \rangle = 0$ by Proposition 1.1.9(iii)–(iv), so $\mathbf{v} \in \mathcal{W}^\perp$.
Therefore $\mathcal{X}^\perp = \mathcal{W}^\perp$, and $(\mathcal{X}^\perp)^\perp = (\mathcal{W}^\perp)^\perp = \mathcal{W}$ by Corollary 2.1.8.
Structural Commentary: Span invariance reduces the double complement to the preceding corollary.
Key Insight: Orthogonality depends on the span of the tested set. □

Perspective 2.1.10. *Orthogonal Versus Complementary*
Let $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ with $\mathcal{U} \perp \mathcal{W}$. By Definitions 2.1.1 and 2.1.2, $\mathcal{W} \subseteq \mathcal{U}^\perp$ and $\mathcal{U} \subseteq \mathcal{W}^\perp$, not necessarily equality.
If $\mathcal{V}$ is finite-dimensional and $\dim(\mathcal{U}) + \dim(\mathcal{W}) = \dim(\mathcal{V})$, then $\dim(\mathcal{U}^\perp) = \dim(\mathcal{W})$ and $\dim(\mathcal{W}^\perp) = \dim(\mathcal{U})$ by Corollary 2.1.7.
The two inclusions are therefore equalities by Proposition 1.4.3(ii).

Figure 26 contrasts a proper inclusion $\mathcal{W} \subsetneq \mathcal{U}^\perp$ with the saturated decomposition $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$.

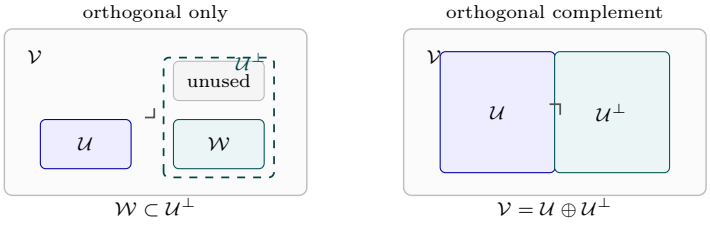

Figure 26: Orthogonal subspaces are complements only when their sum fills $\mathcal{V}$.

2.2 Orthogonal Projections

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Proposition 2.2.1. *Orthogonal Projection onto a Line*
Let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$, and set $c := \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$ and $\mathbf{w} := \mathbf{u} - c\mathbf{v}$. Then $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ and $\mathbf{w} \perp \mathbf{v}$.
The vector $\operatorname{proj}_{\mathbf{v}}(\mathbf{u}) := c\mathbf{v} = \left\langle \mathbf{u}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle \frac{\mathbf{v}}{\|\mathbf{v}\|}$ is the orthogonal projection of $\mathbf{u}$ onto $\operatorname{span}(\mathbf{v})$.

Proof. Since $\mathbf{v} \neq \mathbf{0}$, $\|\mathbf{v}\|^2 > 0$ by definiteness, so c is well-defined. The identity $\mathbf{w} = \mathbf{u} - c\mathbf{v}$ implies $\mathbf{u} = c\mathbf{v} + \mathbf{w}$.
By linearity, $\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - c \langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \|\mathbf{v}\|^2 = 0$, so $\mathbf{w} \perp \mathbf{v}$.
Since $\|\mathbf{v}\| \in \mathbb{R}_{>0}$, $\left\langle \mathbf{u}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} = c\mathbf{v}$ by Proposition 1.1.9(iv).
Structural Commentary: The chosen coefficient makes the residual orthogonal.
Key Insight: Projection isolates the parallel component and leaves an orthogonal residual. □

Figure 27 depicts the decomposition $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ with $c\mathbf{v} \in \operatorname{span}(\mathbf{v})$ and $\mathbf{w} \perp \mathbf{v}$.

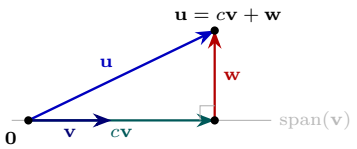

Figure 27: Writing $\mathbf{u}$ as a component parallel to $\mathbf{v}$ plus a component orthogonal to $\mathbf{v}$.

Definition 2.2.2. *Orthogonal Projection onto a Subspace*
Let $\mathcal{U}$ be finite-dimensional subspace of $\mathcal{V}$. By Theorem 2.1.5, every $\mathbf{v} \in \mathcal{V}$ has unique decomposition $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^\perp$.
The *orthogonal projection* of $\mathcal{V}$ onto $\mathcal{U}$ is the map $P_{\mathcal{U}} : \mathcal{V} \to \mathcal{V}$ defined by $P_{\mathcal{U}}\mathbf{v} := \mathbf{u}$.

Figure 28 illustrates the $\mathcal{U}$-component $P_{\mathcal{U}}\mathbf{v}$ and the residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^\perp$.

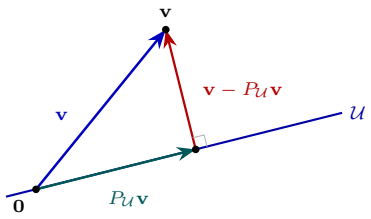

Figure 28: Orthogonal projection decomposes $\mathbf{v}$ into $P_{\mathcal{U}}\mathbf{v}$ plus an orthogonal residual.

Proposition 2.2.3. *Properties of Orthogonal Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. Then:

(i) $P_{\mathcal{U}} \in \mathcal{L}(\mathcal{V})$;
(ii) $(\forall\, \mathbf{u} \in \mathcal{U})\, P_{\mathcal{U}}\mathbf{u} = \mathbf{u}$;
(iii) $(\forall\, \mathbf{w} \in \mathcal{U}^{\perp})\, P_{\mathcal{U}}\mathbf{w} = \mathbf{0}$;
(iv) $\operatorname{range}(P_{\mathcal{U}}) = \mathcal{U}$;
(v) $(\forall\, \mathbf{v} \in \mathcal{V})\, \mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$;
(vi) $\ker(P_{\mathcal{U}}) = \mathcal{U}^{\perp}$;
(vii) $P_{\mathcal{U}}^2 = P_{\mathcal{U}}$;
(viii) $(\forall\, \mathbf{v} \in \mathcal{V})\, \|P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v}\|$;
(ix) if $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ is an orthonormal basis of $\mathcal{U}$, then $(\forall\, \mathbf{v} \in \mathcal{V})\, P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \mathbf{e}_j$.

Proof.

(i) Fix $a \in \mathbb{F}$ and $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$. By Theorem 2.1.5, write $\mathbf{v}_r = \mathbf{u}_r + \mathbf{w}_r$ with $\mathbf{u}_r \in \mathcal{U}$ and $\mathbf{w}_r \in \mathcal{U}^{\perp}$ for $r \in \{1, 2\}$.
Since $\mathcal{U}$ and $\mathcal{U}^{\perp}$ are subspaces, $a\mathbf{v}_1 + \mathbf{v}_2 = (a\mathbf{u}_1 + \mathbf{u}_2) + (a\mathbf{w}_1 + \mathbf{w}_2)$ is its orthogonal decomposition.
Uniqueness therefore implies $P_{\mathcal{U}}(a\mathbf{v}_1 + \mathbf{v}_2) = aP_{\mathcal{U}}\mathbf{v}_1 + P_{\mathcal{U}}\mathbf{v}_2$, so $P_{\mathcal{U}} \in \mathcal{L}(\mathcal{V})$.
(ii) Fix $\mathbf{u} \in \mathcal{U}$. The decomposition $\mathbf{u} = \mathbf{u} + \mathbf{0}$ has $\mathbf{0} \in \mathcal{U}^{\perp}$ by Proposition 2.1.3(i), so $P_{\mathcal{U}}\mathbf{u} = \mathbf{u}$.
(iii) Fix $\mathbf{w} \in \mathcal{U}^{\perp}$. The decomposition $\mathbf{w} = \mathbf{0} + \mathbf{w}$ has $\mathbf{0} \in \mathcal{U}$, so $P_{\mathcal{U}}\mathbf{w} = \mathbf{0}$.
(iv) The defining decomposition places every value of $P_{\mathcal{U}}$ in $\mathcal{U}$, so $\operatorname{range}(P_{\mathcal{U}}) \subseteq \mathcal{U}$. The reverse inclusion holds by (ii).
(v) Fix $\mathbf{v} \in \mathcal{V}$. The defining decomposition is $\mathbf{v} = P_{\mathcal{U}}\mathbf{v} + (\mathbf{v} - P_{\mathcal{U}}\mathbf{v})$ with $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$.
(vi) The inclusion $\mathcal{U}^{\perp} \subseteq \ker(P_{\mathcal{U}})$ holds by (iii). Conversely, if $P_{\mathcal{U}}\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = \mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$ by (v).
(vii) Fix $\mathbf{v} \in \mathcal{V}$. By (iv), $P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}$, so $P_{\mathcal{U}}(P_{\mathcal{U}}\mathbf{v}) = P_{\mathcal{U}}\mathbf{v}$ by (ii). Hence $P_{\mathcal{U}}^2 = P_{\mathcal{U}}$.
(viii) Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{p} := P_{\mathcal{U}}\mathbf{v}$ and $\mathbf{r} := \mathbf{v} - \mathbf{p}$. Then $\mathbf{p} \in \mathcal{U}$ by (iv) and $\mathbf{r} \in \mathcal{U}^{\perp}$ by (v).
Hence $\|\mathbf{v}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{r}\|^2 \ge \|\mathbf{p}\|^2$ by Theorem 1.4.3. Nonnegativity of norms implies $\|P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v}\|$.
(ix) Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{p} := \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \mathbf{e}_j \in \mathcal{U}$. For $k \in \{1, \dots, m\}$, $\langle \mathbf{v} - \mathbf{p}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$ by orthonormality.
If $\mathbf{x} \in \mathcal{U}$, write $\mathbf{x} = \sum_{k=1}^{m} a_k \mathbf{e}_k$; then $\langle \mathbf{v} - \mathbf{p}, \mathbf{x} \rangle = \sum_{k=1}^{m} \overline{a_k}\, \langle \mathbf{v} - \mathbf{p}, \mathbf{e}_k \rangle = 0$ by Proposition 1.1.9(iii)–(iv).
Thus $\mathbf{v} - \mathbf{p} \in \mathcal{U}^{\perp}$, and $P_{\mathcal{U}}\mathbf{v} = \mathbf{p}$ by uniqueness in Theorem 2.1.5.

Structural Commentary: The orthogonal decomposition organizes all nine arguments.
Key Insight: Projection preserves the $\mathcal{U}$-component and annihilates the $\mathcal{U}^{\perp}$-component. □

Definition 2.2.4. *Rank-One Projection Matrix*
Let $\mathbf{u} = (u_1, \dots, u_n)^t \in \mathbb{F}^n$ be a unit vector. The *rank-one projection matrix* onto $\operatorname{span}(\mathbf{u})$ is $P_{\mathbf{u}} := [u_i \overline{u_j}]_{i,j=1}^{n} \in \operatorname{Mat}(n, n, \mathbb{F})$.
It satisfies $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u}$.

Proposition 2.2.5. *Properties of Rank-One Projection Matrices*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. Then:

(i) $\operatorname{range}(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})$;
(ii) $\ker(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})^{\perp}$;
(iii) $(\forall, i, j \in \{1, \dots, n\})\, (P_{\mathbf{u}})_{ij} = \overline{(P_{\mathbf{u}})_{ji}}$;
(iv) $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.

Proof.

(i) For $\mathbf{x} \in \mathbb{F}^n$, $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u} \in \operatorname{span}(\mathbf{u})$, so $\operatorname{range}(P_{\mathbf{u}}) \subseteq \operatorname{span}(\mathbf{u})$. Since $P_{\mathbf{u}}\mathbf{u} = \langle \mathbf{u}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u} = \mathbf{u}$, the reverse inclusion also holds.
(ii) Fix $\mathbf{x} \in \mathbb{F}^n$. Since $\mathbf{u} \neq \mathbf{0}$, $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u} = \mathbf{0}$ if and only if $\langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} = 0$, which is equivalent to $\mathbf{x} \in \operatorname{span}(\mathbf{u})^{\perp}$.
(iii) For $i, j \in \{1, \dots, n\}$, $(P_{\mathbf{u}})_{ij} = u_i \overline{u_j} = \overline{u_j \overline{u_i}} = \overline{(P_{\mathbf{u}})_{ji}}$.
(iv) For $i, k \in \{1, \dots, n\}$, $(P_{\mathbf{u}}^2)_{ik} = \sum_{j=1}^{n} u_i \overline{u_j} u_j \overline{u_k} = u_i \overline{u_k} \sum_{j=1}^{n} |u_j|^2 = u_i \overline{u_k} = (P_{\mathbf{u}})_{ik}$ because $\mathbf{u}$ is a unit vector.

Structural Commentary: The projection formula proves all four matrix identities.
Key Insight: The matrix preserves the $\mathbf{u}$ direction and annihilates its orthogonal complement. □

Theorem 2.2.6. *Orthogonal Projection Minimizes Distance*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. If $\mathbf{v} \in \mathcal{V}$ and $\mathbf{u} \in \mathcal{U}$, then $\|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v} - \mathbf{u}\|$, with equality if and only if $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$.

Proof. Set $\mathbf{p} := P_{\mathcal{U}}\mathbf{v}$, $\mathbf{r} := \mathbf{v} - \mathbf{p}$, and $\mathbf{h} := \mathbf{p} - \mathbf{u}$. By Proposition 2.2.3(iv)–(v), $\mathbf{p}, \mathbf{h} \in \mathcal{U}$ and $\mathbf{r} \in \mathcal{U}^{\perp}$, so $\mathbf{r} \perp \mathbf{h}$.
Since $\mathbf{v} - \mathbf{u} = \mathbf{r} + \mathbf{h}$, $\|\mathbf{v} - \mathbf{u}\|^2 = \|\mathbf{r}\|^2 + \|\mathbf{h}\|^2 \ge \|\mathbf{r}\|^2 = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|^2$ by Theorem 1.4.3. Nonnegativity of norms proves the inequality.
($\Rightarrow$) If equality holds, then the preceding identity forces $\|\mathbf{h}\| = 0$, so $\mathbf{h} = \mathbf{0}$ by definiteness and $\mathbf{u} = \mathbf{p} = P_{\mathcal{U}}\mathbf{v}$.
($\Leftarrow$) If $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$, then $\mathbf{h} = \mathbf{0}$ and equality holds in the same identity.

Structural Commentary: Pythagoras isolates the nonnegative correction term.
Key Insight: The orthogonal projection is the unique nearest point in the subspace. □

Definition 2.2.7. *Distance to a Subspace*
Let $\mathcal{U}$ be a nonempty subset of $\mathcal{V}$. The *distance* from $\mathbf{v} \in \mathcal{V}$ to $\mathcal{U}$ is $d(\mathbf{v},\mathcal{U}) := \inf_{\mathbf{u}\in\mathcal{U}} \|\mathbf{v}-\mathbf{u}\|$.
If $\mathcal{U}$ is a finite-dimensional subspace, then $d(\mathbf{v},\mathcal{U}) = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|$ by Theorem 2.2.6.

Definition 2.2.8. *Fourier Coefficients and Expansion*
Let $\mathcal{O} = (\mathbf{e}_1,\dots,\mathbf{e}_m)$ be orthonormal list, $\mathbf{v} \in \mathcal{V}$. The *Fourier coefficients* of $\mathbf{v}$ relative to $\mathcal{O}$ are $c_j(\mathbf{v}) := \langle \mathbf{v}, \mathbf{e}_j\rangle$ for $j \in \{1,\dots,m\}$.
The *Fourier expansion* of $\mathbf{v}$ onto $\mathcal{O}$ is $\widehat{\mathbf{v}}_{\mathcal{O}} := \sum_{j=1}^{m} c_j(\mathbf{v})\mathbf{e}_j$.

Theorem 2.2.9. *Fourier Expansion onto a Finite Orthonormal Set*
Let $\mathcal{O} = (\mathbf{e}_1,\dots,\mathbf{e}_m)$ be orthonormal, let $\mathcal{S} = \operatorname{span}(\mathcal{O})$, and let $\mathbf{v} \in \mathcal{V}$. Then:
(i) $\widehat{\mathbf{v}}_{\mathcal{O}}$ is the unique $\mathbf{s} \in \mathcal{S}$ with $\mathbf{v} - \mathbf{s} \in \mathcal{S}^{\perp}$;
(ii) $(\forall\, \mathbf{s} \in \mathcal{S})\ \|\mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}\| \le \|\mathbf{v} - \mathbf{s}\|$, and $\widehat{\mathbf{v}}_{\mathcal{O}}$ is the only $\mathbf{s} \in \mathcal{S}$ for which equality holds;
(iii) $\|\widehat{\mathbf{v}}_{\mathcal{O}}\| \le \|\mathbf{v}\|$.

Proof.
(i) By construction, $\widehat{\mathbf{v}}_{\mathcal{O}} \in \mathcal{S}$. For $k \in \{1,\dots,m\}$, $\langle \mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}, \mathbf{e}_k\rangle = c_k(\mathbf{v}) - \sum_{j=1}^{m} c_j(\mathbf{v})\delta_{jk} = 0$.
If $\mathbf{x} \in \mathcal{S}$, write $\mathbf{x} = \sum_{k=1}^{m} a_k\mathbf{e}_k$; then $\langle \mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}, \mathbf{x}\rangle = \sum_{k=1}^{m} \overline{a_k}\langle \mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}, \mathbf{e}_k\rangle = 0$ by Proposition 1.1.9(iii)–(iv). Hence $\mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}} \in \mathcal{S}^{\perp}$.
If $\mathbf{s} \in \mathcal{S}$ also satisfies $\mathbf{v} - \mathbf{s} \in \mathcal{S}^{\perp}$, then $\mathbf{s} - \widehat{\mathbf{v}}_{\mathcal{O}} = (\mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}) - (\mathbf{v} - \mathbf{s}) \in \mathcal{S} \cap \mathcal{S}^{\perp}$, so $\mathbf{s} = \widehat{\mathbf{v}}_{\mathcal{O}}$ by Proposition 2.1.3(iv).
(ii) By (i), $\widehat{\mathbf{v}}_{\mathcal{O}} = P_{\mathcal{S}}\mathbf{v}$. The inequality and equality condition hold by Theorem 2.2.6.
(iii) If $m = 0$, then $\widehat{\mathbf{v}}_{\mathcal{O}} = \mathbf{0}$ and the claim is immediate. Suppose $m \ge 1$.
By Proposition 1.3.2 and Theorem 1.4.5, $\|\widehat{\mathbf{v}}_{\mathcal{O}}\|^2 = \sum_{j=1}^{m} |c_j(\mathbf{v})|^2 \le \|\mathbf{v}\|^2$, so $\|\widehat{\mathbf{v}}_{\mathcal{O}}\| \le \|\mathbf{v}\|$ by nonnegativity of norms.

Structural Commentary: Coordinate construction and nearest-point optimality prove all three claims.
Key Insight: Fourier expansion is orthogonal projection expressed in orthonormal coordinates. □

Theorem 2.2.10. *Characterizing Finite Orthonormal Bases*
Let $\mathcal{O} = (\mathbf{e}_1,\dots,\mathbf{e}_m)$ be orthonormal, $\mathcal{S} = \operatorname{span}(\mathcal{O})$. Write $c_j(\mathbf{v})$, $\widehat{\mathbf{v}} := \widehat{\mathbf{v}}_{\mathcal{O}}$ as in Definition 2.2.8. The following are equivalent:
(i) $\mathcal{S} = \mathcal{V}$.
(ii) $\mathcal{S}^{\perp} = \{\mathbf{0}\}$.
(iii) $(\forall\, \mathbf{v} \in \mathcal{V})\ \widehat{\mathbf{v}} = \mathbf{v}$.
(iv) $(\forall\, \mathbf{v},\mathbf{w} \in \mathcal{V})\ \langle \mathbf{v},\mathbf{w}\rangle = \sum_{j=1}^{m} c_j(\mathbf{v})\overline{c_j(\mathbf{w})}$.
(v) $(\forall\, \mathbf{v} \in \mathcal{V})\ \|\widehat{\mathbf{v}}\| = \|\mathbf{v}\|$.

Proof.
(i) ⇒ *(ii)* If $\mathcal{S} = \mathcal{V}$, then $\mathcal{S}^{\perp} = \mathcal{V}^{\perp} = \{\mathbf{0}\}$ by Proposition 2.1.3(iii).
(ii) ⇒ *(iii)* Fix $\mathbf{v} \in \mathcal{V}$. By Theorem 2.2.9(i), $\mathbf{v} - \widehat{\mathbf{v}} \in \mathcal{S}^{\perp} = \{\mathbf{0}\}$, so $\widehat{\mathbf{v}} = \mathbf{v}$.
(iii) ⇒ *(iv)* Fix $\mathbf{v},\mathbf{w} \in \mathcal{V}$. By (iii), $\mathbf{v} = \sum_{j=1}^{m} c_j(\mathbf{v})\mathbf{e}_j$ and $\mathbf{w} = \sum_{k=1}^{m} c_k(\mathbf{w})\mathbf{e}_k$.
Then $\langle \mathbf{v},\mathbf{w}\rangle = \sum_{j=1}^{m} c_j(\mathbf{v})\overline{c_j(\mathbf{w})}$ by Proposition 1.1.9(iii)–(iv) and orthonormality.
(iv) ⇒ *(v)* Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{w} = \mathbf{v}$ in (iv). Then $\|\mathbf{v}\|^2 = \sum_{j=1}^{m} |c_j(\mathbf{v})|^2 = \|\widehat{\mathbf{v}}\|^2$ by Proposition 1.3.2, so $\|\widehat{\mathbf{v}}\| = \|\mathbf{v}\|$.
(v) ⇒ *(i)* Fix $\mathbf{v} \in \mathcal{V}$. By Theorem 2.2.9(i), $\widehat{\mathbf{v}} \in \mathcal{S}$ and $\mathbf{v} - \widehat{\mathbf{v}} \in \mathcal{S}^{\perp}$. Hence $\|\mathbf{v}\|^2 = \|\widehat{\mathbf{v}}\|^2 + \|\mathbf{v} - \widehat{\mathbf{v}}\|^2$ by Theorem 1.4.3.
Condition (v) implies $\|\mathbf{v} - \widehat{\mathbf{v}}\| = 0$, so $\mathbf{v} = \widehat{\mathbf{v}} \in \mathcal{S}$ by definiteness. Thus $\mathcal{V} \subseteq \mathcal{S}$, while $\mathcal{S} \subseteq \mathcal{V}$ by construction, therefore $\mathcal{S} = \mathcal{V}$.

Structural Commentary: A five-step implication cycle proves equivalence.
Key Insight: An orthonormal list is a basis exactly when its projection loses no component. □

Perspective 2.2.11. *Why the Finite-Dimensional Hypothesis Matters*
In $\ell^2(\mathbb{F})$, let $\mathcal{S} = \mathbb{F}^{(\mathbb{N})} = \operatorname{span}\{\mathbf{e}_j : j \ge 1\}$. If $\mathbf{x} \in \mathcal{S}^{\perp}$, then $(\forall\, j \in \mathbb{N})\ x_j = \langle \mathbf{x}, \mathbf{e}_j\rangle = 0$, so $\mathcal{S}^{\perp} = \{\mathbf{0}\}$.
The vector $(2^{-j})_{j\ge1}$ belongs to $\ell^2(\mathbb{F})$ but not to $\mathcal{S}$, hence $\mathcal{S} \subsetneq \ell^2(\mathbb{F})$ and $\mathcal{S} \oplus \mathcal{S}^{\perp} \neq \ell^2(\mathbb{F})$.

2.3 Adjoints and Fundamental Subspaces

Adjoints express dual action in inner-product language. They become conjugate transposes in orthonormal coordinates and exchange kernels, ranges, and orthogonal complements.

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Definition 2.3.1. *Adjoint of a Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$.
A map $S : \mathcal{W} \to \mathcal{V}$ is an *adjoint* of T if $(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\ \langle T\mathbf{v},\mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, S\mathbf{w}\rangle_{\mathcal{V}}$.

Perspective 2.3.2. *Dual Map and Inner-Product Adjoint*
By Definition 4.2.4, the *dual map* of $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ is the pullback $T' : \mathcal{W}' \to \mathcal{V}'$, $T'\varphi = \varphi \circ T$.
This algebraic adjoint, sometimes written $T^{\times}$, is distinct from the inner-product adjoint T^*, whose construction requires inner products.

Lemma 2.3.3. *Uniqueness of Adjoints*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$. Then T has at most one adjoint.

Proof. Suppose $S_1, S_2 : \mathcal{W} \to \mathcal{V}$ are adjoints of T and fix $\mathbf{w} \in \mathcal{W}$.
Then $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{v}, S_1\mathbf{w}\rangle_{\mathcal{V}} = \langle \mathbf{v}, S_2\mathbf{w}\rangle_{\mathcal{V}}$ by the two adjoint identities.
Thus $S_1\mathbf{w} = S_2\mathbf{w}$ by Lemma 1.1.10(i), and $S_1 = S_2$ by pointwise equality.
Structural Commentary: Pointwise comparison reduces uniqueness to separation by inner products.
Key Insight: Distinct vectors cannot agree against every inner-product test. □

Theorem 2.3.4. *Existence of the Adjoint*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $\mathcal{W}$ be an inner-product space, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:
(i) T has an adjoint $T^* : \mathcal{W} \to \mathcal{V}$;
(ii) the adjoint T^* is unique;
(iii) $T^* \in \mathcal{L}(\mathcal{W}, \mathcal{V})$.

Proof.
(i) Fix $\mathbf{w} \in \mathcal{W}$. The map $\varphi_{\mathbf{w}} := \langle T(\cdot), \mathbf{w} \rangle_{\mathcal{W}}$ belongs to $\mathcal{V}'$ by linearity of T and the first inner-product slot.
By Theorem 1.6.1, $(\exists!\, \mathbf{u}_{\mathbf{w}} \in \mathcal{V})(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi_{\mathbf{w}}(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u}_{\mathbf{w}} \rangle_{\mathcal{V}}$.
Define $T^*\mathbf{w} := \mathbf{u}_{\mathbf{w}}$. Since $\mathbf{w}$ was arbitrary, $T^* : \mathcal{W} \to \mathcal{V}$ is an adjoint of T.
(ii) The adjoint T^* is unique by Lemma 2.3.3.
(iii) Fix $a \in \mathbb{F}$ and $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, and set $\mathbf{x} := aT^*\mathbf{w}_1 + T^*\mathbf{w}_2$.
For $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, T^*(a\mathbf{w}_1 + \mathbf{w}_2) - \mathbf{x} \rangle_{\mathcal{V}} = 0$ by the adjoint identity and conjugate linearity in the second slot.
Thus $T^*(a\mathbf{w}_1 + \mathbf{w}_2) = \mathbf{x}$ by Lemma 1.1.10(ii), so $T^* \in \mathcal{L}(\mathcal{W}, \mathcal{V})$.
Structural Commentary: Riesz representation constructs codomain-indexed vectors; separation verifies that their assignment is linear.
Key Insight: Riesz vectors assemble pointwise into the adjoint. □

Figure 29 records the inner-product identity that transfers T from the first slot to T^* in the second slot.

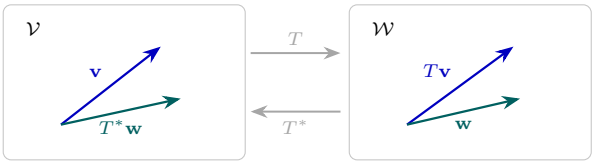

Figure 29: Adjoint relation: $\langle T\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w} \rangle_{\mathcal{V}}$.

Proposition 2.3.5. *Riesz Transport of the Dual Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces.
For $\mathcal{X} \in \{\mathcal{V}, \mathcal{W}\}$, let $R_{\mathcal{X}} : \mathcal{X}' \to \mathcal{X}$ be the Riesz map characterized by $(\forall\, \varphi \in \mathcal{X}')(\forall\, \mathbf{x} \in \mathcal{X})\, \varphi(\mathbf{x}) = \langle \mathbf{x}, R_{\mathcal{X}}\varphi \rangle_{\mathcal{X}}$.
Then $(\forall\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}))\, T' = (R_{\mathcal{V}})^{-1} \circ T^* \circ R_{\mathcal{W}}$, equivalently $R_{\mathcal{V}} T' = T^* R_{\mathcal{W}}$.

Proof. Fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $\varphi \in \mathcal{W}'$, and set $\mathbf{u} := T^* R_{\mathcal{W}} \varphi$ and $\psi := (R_{\mathcal{V}})^{-1}\mathbf{u}$.
For $\mathbf{v} \in \mathcal{V}$, $\psi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle_{\mathcal{V}} = \langle T\mathbf{v}, R_{\mathcal{W}}\varphi \rangle_{\mathcal{W}} = \varphi(T\mathbf{v}) = (T'\varphi)(\mathbf{v})$ by the Riesz-map, adjoint, and dual-map identities.
Thus $T' = (R_{\mathcal{V}})^{-1} T^* R_{\mathcal{W}}$ by pointwise equality, and composition with $R_{\mathcal{V}}$ proves the equivalent identity.
Structural Commentary: Pointwise evaluation identifies Riesz transport with the dual-map pullback.
Key Insight: Two conjugate-linear Riesz maps recover the linear dual map. □

Figure 30 displays the commuting square expressed by $R_{\mathcal{V}} T' = T^* R_{\mathcal{W}}$.

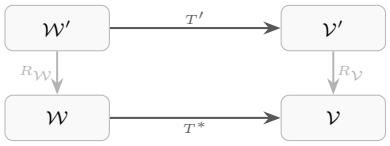

Figure 30: Riesz transports the dual map to the adjoint: $R_{\mathcal{V}} T' = T^* R_{\mathcal{W}}$.

Definition 2.3.6. *Conjugate Transpose*
For $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, the *conjugate transpose* of A is $A^* = [\overline{a_{ji}}] \in \mathrm{Mat}(n, m, \mathbb{F})$.
Over $\mathbb{R}$, the conjugate transpose is the ordinary transpose.

Theorem 2.3.7. *Matrix of the Adjoint*
Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ and $\gamma = (\mathbf{f}_1, \ldots, \mathbf{f}_m)$ be orthonormal bases of the finite-dimensional inner-product spaces $\mathcal{V}$ and $\mathcal{W}$, respectively.
Then $(\forall\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}))\, [T^*]_{\beta \leftarrow \gamma} = ([T]_{\gamma \leftarrow \beta})^*$.

Proof. Set $A := [T]_{\gamma \leftarrow \beta}$ and $B := [T^*]_{\beta \leftarrow \gamma}$.
If $n = 0$ or $m = 0$, then B and A^* are the unique $n \times m$ matrix, so $B = A^*$. Suppose $n, m \geq 1$.
For $i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, m\}$, $A_{j,i} = \langle T\mathbf{e}_i, \mathbf{f}_j \rangle_{\mathcal{W}}$ by Theorem 1.4.8(i).
By the same theorem for T^*, $B_{i,j} = \langle T^*\mathbf{f}_j, \mathbf{e}_i \rangle_{\mathcal{V}}$.
The identity $B_{i,j} = \overline{\langle \mathbf{e}_i, T^*\mathbf{f}_j \rangle_{\mathcal{V}}} = \overline{A_{j,i}}$ holds by conjugate symmetry and the adjoint identity. Since i, j were arbitrary, $B = A^*$.
Structural Commentary: Empty dimensions are immediate; orthonormal coordinates match each remaining entry after conjugation.
Key Insight: Moving an operator across an inner product becomes conjugate transposition in orthonormal bases. □

Proposition 2.3.8. *Basic Adjoint Identities*
Let $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$ be finite-dimensional inner-product spaces. Then:
(i) $(aS + T)^* = \overline{a}S^* + T^*$ for $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$;
(ii) $(SR)^* = R^*S^*$ for $R \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
(iii) $(T^*)^* = T$ for $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
(iv) $I_{\mathcal{V}}^* = I_{\mathcal{V}}$.

Proof. Each claim follows by Lemma 2.3.3 once the asserted adjoint verifies its defining identity.
(i) For $\mathbf{v} \in \mathcal{V}$ and $\mathbf{w} \in \mathcal{W}$, $\langle (aS + T)\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = \langle \mathbf{v}, (\overline{a}S^* + T^*)\mathbf{w} \rangle_{\mathcal{V}}$ by linearity and the adjoint identities.

(ii) For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, $\langle SR\mathbf{u}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{u}, R^*S^*\mathbf{w}\rangle_{\mathcal{U}}$ by the two adjoint identities.
(iii) The identity $(\forall\, \mathbf{w} \in \mathcal{W})(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T^*\mathbf{w}, \mathbf{v}\rangle_{\mathcal{V}} = \langle \mathbf{w}, T\mathbf{v}\rangle_{\mathcal{W}}$ holds by conjugate symmetry and the adjoint identity.
Thus T is the adjoint of T^*, so $(T^*)^* = T$.
(iv) For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle I_{\mathcal{V}}\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, I_{\mathcal{V}}\mathbf{w}\rangle$. Thus $I_{\mathcal{V}}^* = I_{\mathcal{V}}$.

Structural Commentary: Each algebraic operation is transported through the two inner-product slots and then settled by uniqueness.
Key Insight: Adjoint formation conjugates scalars, reverses products, and is involutive. □

Proposition 2.3.9. *Adjoints of Inverses and Polynomials*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space. For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$, set $p^{\#}(z) := \sum_{j=0}^{m} \overline{a_j} z^j$.

(i) If T is invertible, then $(T^*)^{-1} = (T^{-1})^*$.
(ii) $p(T)^* = p^{\#}(T^*)$.
(iii) $p \in \mathbb{R}[z] \Longrightarrow p(T)^* = p(T^*)$.

Proof.

(i) The identities $(T^{-1})^*T^* = (TT^{-1})^* = I_{\mathcal{V}}$ and $T^*(T^{-1})^* = (T^{-1}T)^* = I_{\mathcal{V}}$ hold by Proposition 2.3.8(ii),(iv).
Thus $(T^*)^{-1} = (T^{-1})^*$ by uniqueness of two-sided inverses.
(ii) The base identity $(T^0)^* = I_{\mathcal{V}} = (T^*)^0$ holds by Proposition 2.3.8(iv).
If $(T^j)^* = (T^*)^j$, then $(T^{j+1})^* = (T^jT)^* = T^*(T^j)^* = (T^*)^{j+1}$ by Proposition 2.3.8(ii).
Thus $(\forall\, j \in \{0, \dots, m\})\ (T^j)^* = (T^*)^j$ by induction.
The identity $p(T)^* = \sum_{j=0}^{m} \overline{a_j}(T^j)^* = \sum_{j=0}^{m} \overline{a_j}(T^*)^j = p^{\#}(T^*)$ holds by Proposition 2.3.8(i) and the power identity.
(iii) If $p \in \mathbb{R}[z]$, then $p^{\#} = p$, so $p(T)^* = p(T^*)$ by (ii).

Structural Commentary: The product law controls inverses and powers, while conjugate additivity controls polynomial coefficients.
Key Insight: Adjoint formation applies coefficientwise conjugation to polynomial expressions in an operator. □

Proposition 2.3.10. *Adjoint Kernel-Range Orthogonality*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) $\ker(T^*) = (\mathrm{range}(T))^{\perp}$;
(ii) $\ker(T) = (\mathrm{range}(T^*))^{\perp}$;
(iii) $\mathrm{range}(T^*) = (\ker(T))^{\perp}$;
(iv) $\mathrm{range}(T) = (\ker(T^*))^{\perp}$.

Equivalently, an adjoint annihilates exactly the vectors orthogonal to the range of the original map.

Proof.

(i) For $\mathbf{w} \in \mathcal{W}$, $\mathbf{w} \in \ker(T^*) \iff (\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}} = 0$ by Lemma 1.1.10(ii).
The latter condition is equivalent to $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{w}, T\mathbf{v}\rangle_{\mathcal{W}} = 0$ by the adjoint identity and conjugate symmetry.
Hence $\mathbf{w} \in \ker(T^*) \iff \mathbf{w} \in (\mathrm{range}(T))^{\perp}$ by the orthogonal-complement definition.
(ii) The identity $\ker(T) = \ker((T^*)^*) = (\mathrm{range}(T^*))^{\perp}$ holds by (i) for T^* and Proposition 2.3.8(iii).
(iii) Taking orthogonal complements in (ii), $(\ker(T))^{\perp} = ((\mathrm{range}(T^*))^{\perp})^{\perp} = \mathrm{range}(T^*)$ by Corollary 2.1.8.
(iv) Taking orthogonal complements in (i), $(\ker(T^*))^{\perp} = ((\mathrm{range}(T))^{\perp})^{\perp} = \mathrm{range}(T)$ by Corollary 2.1.8.

Structural Commentary: The kernel formula, involution, and double complementation establish all four identities.
Key Insight: Kernels and ranges are exchanged by taking an adjoint and an orthogonal complement. □

Proposition 2.3.11. *Adjoint Products and Algebraic Projections*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) $(\ker(T^*T), \ker(TT^*)) = (\ker(T), \ker(T^*))$.
(ii) $(\mathrm{range}(T^*T), \mathrm{range}(TT^*)) = (\mathrm{range}(T^*), \mathrm{range}(T))$.
(iii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$, then $\mathcal{V} = \mathcal{Z}^{\perp} \oplus \mathcal{U}^{\perp}$.
In this decomposition, $(\rho_{\mathcal{U},\mathcal{Z}})^* = \rho_{\mathcal{Z}^{\perp},\mathcal{U}^{\perp}}$.

Proof.

(i) For $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, T^*T\mathbf{v}\rangle_{\mathcal{V}} = \|T\mathbf{v}\|^2$ by the adjoint identity.
Thus $\mathbf{v} \in \ker(T^*T) \iff T\mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \ker(T)$ by this identity and definiteness.
The equality $\ker(TT^*) = \ker(T^*)$ follows by this result for T^* and Proposition 2.3.8(iii).
(ii) The identities $(T^*T)^* = T^*T$ and $(TT^*)^* = TT^*$ hold by Proposition 2.3.8(ii),(iii).
The identity $\mathrm{range}(T^*T) = (\ker(T^*T))^{\perp} = (\ker(T))^{\perp} = \mathrm{range}(T^*)$ holds by (i) and Proposition 2.3.10(iii).
The identity $\mathrm{range}(TT^*) = (\ker(TT^*))^{\perp} = (\ker(T^*))^{\perp} = \mathrm{range}(T)$ holds by (i) and Proposition 2.3.10(iv).
(iii) Suppose $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$ and set $P := \rho_{\mathcal{U},\mathcal{Z}}$.
Fix $\mathbf{c} \in \mathcal{Z}^{\perp} \cap \mathcal{U}^{\perp}$. For $\mathbf{x} \in \mathcal{V}$, write $\mathbf{x} = \mathbf{u} + \mathbf{z}$ with $(\mathbf{u}, \mathbf{z}) \in \mathcal{U} \times \mathcal{Z}$.
Then $\langle \mathbf{c}, \mathbf{x}\rangle = \langle \mathbf{c}, \mathbf{u}\rangle + \langle \mathbf{c}, \mathbf{z}\rangle = 0$, so $\mathbf{c} \in \mathcal{V}^{\perp} = \{\mathbf{0}\}$ by Proposition 2.1.3(iii).
Hence $\mathcal{Z}^{\perp} \cap \mathcal{U}^{\perp} = \{\mathbf{0}\}$.
By Corollaries 2.1.7 and 2.3.2, $\dim(\mathcal{Z}^{\perp} \oplus \mathcal{U}^{\perp}) = 2\dim(\mathcal{V}) - \dim(\mathcal{Z}) - \dim(\mathcal{U}) = \dim(\mathcal{V})$.
Thus $\mathcal{V} = \mathcal{Z}^{\perp} \oplus \mathcal{U}^{\perp}$ by Proposition 1.4.3(ii).
Set $Q := \rho_{\mathcal{Z}^{\perp},\mathcal{U}^{\perp}}$.
For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, write $(\mathbf{x}, \mathbf{y}) = (\mathbf{u} + \mathbf{z}, \mathbf{a} + \mathbf{b})$ with $(\mathbf{u}, \mathbf{z}, \mathbf{a}, \mathbf{b}) \in \mathcal{U} \times \mathcal{Z} \times \mathcal{Z}^{\perp} \times \mathcal{U}^{\perp}$.
Then $P\mathbf{x} = \mathbf{u}$, $Q\mathbf{y} = \mathbf{a}$, and $\langle \mathbf{u}, \mathbf{b}\rangle = \langle \mathbf{z}, \mathbf{a}\rangle = 0$ by the projection formulas and conjugate symmetry.
Consequently, $\langle P\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{u}, \mathbf{a}\rangle = \langle \mathbf{x}, Q\mathbf{y}\rangle$.
Thus $P^* = Q$ by Lemma 2.3.3.

Structural Commentary: Positivity fixes kernels, complementation fixes ranges, and decomposition fixes projection adjoints.
Key Insight: Adjoints preserve effective directions and orthogonally exchange projection complements. □

Corollary 2.3.12. *Matrix Fundamental Subspaces as Orthogonal Complements*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ act as $A : \mathbb{F}^n \to \mathbb{F}^m$ with the standard inner products. Then:

(i) $\ker(A) = \mathrm{col}(A^*)^\perp$;
(ii) $\mathrm{col}(A^*) = \ker(A)^\perp$;
(iii) $\ker(A^*) = \mathrm{col}(A)^\perp$;
(iv) $\mathrm{col}(A) = \ker(A^*)^\perp$.
(v) $\mathbb{F} = \mathbb{R} \Longrightarrow \mathbb{R}^n = \ker(A) \odot \mathrm{row}(A)$.
(vi) $\mathbb{F} = \mathbb{R} \Longrightarrow \mathbb{R}^m = \ker(A^t) \odot \mathrm{col}(A)$.

Proof. The induced map and its adjoint have ranges $\mathrm{col}(A)$ and $\mathrm{col}(A^*)$, respectively, by Theorem 2.3.7.

(i) The identity $\ker(A) = \mathrm{col}(A^*)^\perp$ holds by Proposition 2.3.10(ii).
(ii) The identity $\mathrm{col}(A^*) = \ker(A)^\perp$ holds by Proposition 2.3.10(iii).
(iii) The identity $\ker(A^*) = \mathrm{col}(A)^\perp$ holds by Proposition 2.3.10(i).
(iv) The identity $\mathrm{col}(A) = \ker(A^*)^\perp$ holds by Proposition 2.3.10(iv).
(v) If $\mathbb{F} = \mathbb{R}$, then $A^* = A^t$ and $\mathrm{col}(A^t) = \mathrm{row}(A)$.
Hence $\mathbb{R}^n = \ker(A) \odot \mathrm{row}(A)$ by (ii) and Theorem 2.1.5.
(vi) If $\mathbb{F} = \mathbb{R}$, then $A^* = A^t$.
Hence $\mathbb{R}^m = \ker(A^t) \odot \mathrm{col}(A)$ by (iv) and Theorem 2.1.5.

Structural Commentary: Standard coordinates convert operator formulas into matrix fundamental-subspace identities.
Key Insight: The row space complements $\ker(A)$, while the column space complements $\ker(A^*)$. □

2.4 Least Squares and Metric Optimization

Least squares is projection theory as optimization: normal equations, full-rank formulas, and minimum-norm solutions.

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Principle 2.4.1. *Least Squares as Orthogonal Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of an inner-product space $\mathcal{V}$, and let $\mathbf{v} \in \mathcal{V}$.
The identity $\arg\min_{\mathbf{u}\in\mathcal{U}} \|\mathbf{v}-\mathbf{u}\| = \{P_{\mathcal{U}}\mathbf{v}\}$ holds by Theorem 2.2.6. The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ belongs to $\mathcal{U}^\perp$ by Proposition 2.2.3(v).

Definition 2.4.2. *Matrix Least-Squares Solution*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$, define $\mathrm{LS}(A,\mathbf{y}) := \arg\min_{\mathbf{x}\in\mathbb{F}^n} \|A\mathbf{x}-\mathbf{y}\|_2$.

An element $\mathbf{x}_0 \in \mathrm{LS}(A,\mathbf{y})$ is a *least-squares solution* of $A\mathbf{x} = \mathbf{y}$. Its *fitted vector* is $A\mathbf{x}_0$, and its *residual* is $\mathbf{y} - A\mathbf{x}_0$.

Theorem 2.4.3. *Normal Equations for Projection*
Let $\mathcal{U} = \mathrm{span}(\mathbf{u}_1,\dots,\mathbf{u}_n)$ be a finite-dimensional subspace of an inner-product space $\mathcal{V}$, and fix $\mathbf{v} \in \mathcal{V}$.
For $\mathbf{c} = (c_1,\dots,c_n) \in \mathbb{F}^n$, set $\mathsf{N}(\mathbf{c}) := (\forall\, i \in \{1,\dots,n\}) \sum_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i\rangle = \langle \mathbf{v}, \mathbf{u}_i\rangle$. Then:

(i) $(\forall\, \mathbf{c} \in \mathbb{F}^n)\,[P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j\mathbf{u}_j \iff \mathsf{N}(\mathbf{c})]$;
(ii) $(\exists\, \mathbf{c} \in \mathbb{F}^n)\,\mathsf{N}(\mathbf{c})$;
(iii) if $(\mathbf{u}_1,\dots,\mathbf{u}_n)$ is linearly independent, then $(\exists!\, \mathbf{c} \in \mathbb{F}^n)\,\mathsf{N}(\mathbf{c})$.

Proof.

(i) ($\Rightarrow$) Fix $\mathbf{c} = (c_1,\dots,c_n) \in \mathbb{F}^n$ with $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j\mathbf{u}_j$. The residual belongs to $\mathcal{U}^\perp$ by Proposition 2.2.3(v).

Thus $(\forall\, i \in \{1,\dots,n\})\, 0 = \langle \mathbf{v} - P_{\mathcal{U}}\mathbf{v}, \mathbf{u}_i\rangle = \langle \mathbf{v}, \mathbf{u}_i\rangle - \sum_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i\rangle$, so $\mathsf{N}(\mathbf{c})$ holds.

(i) ($\Leftarrow$) Fix $\mathbf{c} = (c_1,\dots,c_n) \in \mathbb{F}^n$ with $\mathsf{N}(\mathbf{c})$, and set $\mathbf{q} := \sum_{j=1}^{n} c_j\mathbf{u}_j \in \mathcal{U}$.

For $\mathbf{u} = \sum_{i=1}^{n} a_i\mathbf{u}_i \in \mathcal{U}$, $\langle \mathbf{v}-\mathbf{q}, \mathbf{u}\rangle = \sum_{i=1}^{n} \overline{a_i}\,\langle \mathbf{v}-\mathbf{q}, \mathbf{u}_i\rangle = 0$ by $\mathsf{N}(\mathbf{c})$ and Proposition 1.1.9(iii)–(iv).
Hence $\mathbf{v}-\mathbf{q} \in \mathcal{U}^\perp$, and $\mathbf{q} = P_{\mathcal{U}}\mathbf{v}$ by uniqueness in Theorem 2.1.5.

(ii) Relation $P_{\mathcal{U}}\mathbf{v} \in \mathrm{span}(\mathbf{u}_1,\dots,\mathbf{u}_n)$ holds by Proposition 2.2.3(iv). Choose $\mathbf{c} \in \mathbb{F}^n$ with $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j\mathbf{u}_j$; then $\mathsf{N}(\mathbf{c})$ holds by (i).

(iii) Suppose $(\mathbf{u}_1,\dots,\mathbf{u}_n)$ is linearly independent. Existence holds by (ii).
If $\mathsf{N}(\mathbf{c})$ and $\mathsf{N}(\mathbf{d})$, then $\sum_{j=1}^{n} c_j\mathbf{u}_j = P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} d_j\mathbf{u}_j$ by (i), and hence $\mathbf{c} = \mathbf{d}$ by linear independence.

Structural Commentary: Projection coordinates construct solutions; residual orthogonality characterizes every coefficient vector.
Key Insight: Normal equations are the spanning-vector tests for an orthogonal residual. □

Figure 31 depicts the residual orthogonality tested against two spanning vectors.

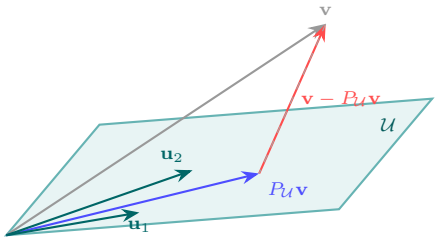

Figure 31: Normal equations impose $(\forall\, i \in \{1,\dots,n\})\ \langle \mathbf{v} - P_{\mathcal{U}}\mathbf{v}, \mathbf{u}_i\rangle = 0$.

Theorem 2.4.4. *Matrix Least Squares and Normal Equations*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ *and* $\mathbf{y} \in \mathbb{F}^m$. *Then:*

(i) $(\forall\, \mathbf{x}_0 \in \mathbb{F}^n)\,[\mathbf{x}_0 \in \mathrm{LS}(A,\mathbf{y}) \iff A^*A\mathbf{x}_0 = A^*\mathbf{y}]$;
(ii) $(\exists\, \mathbf{x}_0 \in \mathbb{F}^n)\, A^*A\mathbf{x}_0 = A^*\mathbf{y}$;
(iii) $[(\exists!\, \mathbf{x}_0 \in \mathbb{F}^n)\, A^*A\mathbf{x}_0 = A^*\mathbf{y}] \iff \mathrm{rank}(A) = n$.

Proof. Set $\mathcal{U} := \mathrm{col}(A)$.

(i) ($\Rightarrow$) Fix $\mathbf{x}_0 \in \mathrm{LS}(A,\mathbf{y})$. Then $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$ by Theorem 2.2.6.
The relation $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp = \ker(A^*)$ holds by Proposition 2.2.3(v) and Corollary 2.3.12(iii). Hence $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.
(i) ($\Leftarrow$) Suppose $A^*A\mathbf{x}_0 = A^*\mathbf{y}$. Then $\mathbf{y} - A\mathbf{x}_0 \in \ker(A^*) = \mathcal{U}^\perp$ by Corollary 2.3.12(iii).
Since $A\mathbf{x}_0 \in \mathcal{U}$, the identity $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$ holds by uniqueness in Theorem 2.1.5. Thus $\mathbf{x}_0 \in \mathrm{LS}(A,\mathbf{y})$ by Theorem 2.2.6.
(ii) The relation $P_{\mathcal{U}}\mathbf{y} \in \mathcal{U} = \mathrm{col}(A)$ holds by Proposition 2.2.3(iv). Choose $\mathbf{x}_0 \in \mathbb{F}^n$ with $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$; then $A^*A\mathbf{x}_0 = A^*\mathbf{y}$ by (i).
(iii) ($\Rightarrow$) Suppose the normal equations have the unique solution $\mathbf{x}_0$.
For $\mathbf{z} \in \ker(A)$, $A^*A(\mathbf{x}_0 + \mathbf{z}) = A^*A\mathbf{x}_0 = A^*\mathbf{y}$, so $\mathbf{z} = \mathbf{0}$ by uniqueness. Thus $\dim(\ker(A)) = 0$ and $\mathrm{rank}(A) = n$ by Corollary 3.3.1(i).
(iii) ($\Leftarrow$) Suppose $\mathrm{rank}(A) = n$. Existence holds by (ii), and $\ker(A) = \{\mathbf{0}\}$ by Corollary 3.3.1(i).
If $\mathbf{x}_0$ and $\mathbf{x}_1$ solve the normal equations, then $\mathbf{x}_0 - \mathbf{x}_1 \in \ker(A^*A) = \ker(A) = \{\mathbf{0}\}$ by Proposition 2.3.11(i).
Thus $\mathbf{x}_0 = \mathbf{x}_1$.

Structural Commentary: Projection converts optimality to residual orthogonality; kernel identities control existence and uniqueness.
Key Insight: The normal equations place the residual in $\ker(A^*) = \mathrm{col}(A)^\perp$. □

Figure 32 shows the fitted vector and its orthogonal residual in the codomain.

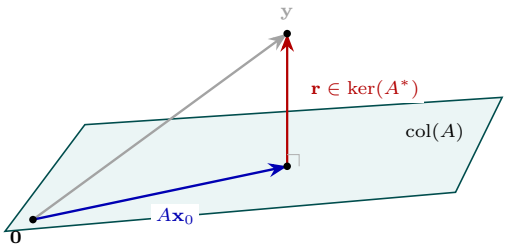

Figure 32: Fitted vector and residual: $A\mathbf{x}_0 = P_{\mathrm{col}(A)}\mathbf{y}$ and $\mathbf{y} - A\mathbf{x}_0 \in \mathrm{col}(A)^\perp = \ker(A^*)$.

Proposition 2.4.5. *Least-Squares Solution Set and Minimum Norm*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ *and* $\mathbf{y} \in \mathbb{F}^m$. *Fix* $\mathbf{x}_0 \in \mathrm{LS}(A,\mathbf{y})$ *and set* $\mathbf{s} := P_{(\ker(A))^\perp}\mathbf{x}_0$. *Then:*

(i) $\mathrm{LS}(A,\mathbf{y}) = \mathbf{x}_0 + \ker(A)$;
(ii) $\mathrm{LS}(A,\mathbf{y}) \cap (\ker(A))^\perp = \{\mathbf{s}\}$;
(iii) $\underset{\mathbf{x}\in\mathrm{LS}(A,\mathbf{y})}{\arg\min}\ \|\mathbf{x}\|_2 = \{\mathbf{s}\}$.

Proof.

(i) For $\mathbf{x} \in \mathbb{F}^n$, $\mathbf{x} \in \mathrm{LS}(A,\mathbf{y}) \iff A\mathbf{x} = P_{\mathrm{col}(A)}\mathbf{y}$ by Theorem 2.2.6.
The equality $A\mathbf{x}_0 = P_{\mathrm{col}(A)}\mathbf{y}$ holds by $\mathbf{x}_0 \in \mathrm{LS}(A,\mathbf{y})$ and Theorem 2.2.6.
Thus $\mathbf{x} \in \mathrm{LS}(A,\mathbf{y}) \iff A\mathbf{x} = A\mathbf{x}_0 \iff \mathbf{x} - \mathbf{x}_0 \in \ker(A)$, and hence $\mathrm{LS}(A,\mathbf{y}) = \mathbf{x}_0 + \ker(A)$.
(ii) The relation $\mathbf{s} \in (\ker(A))^\perp$ holds by Proposition 2.2.3(iv).
The identity $\mathbf{x}_0 - \mathbf{s} \in ((\ker(A))^\perp)^\perp = \ker(A)$ holds by Proposition 2.2.3(v) and Corollary 2.1.8. Hence $\mathbf{s} \in \mathrm{LS}(A,\mathbf{y})$ by (i).
If $\mathbf{u} \in \mathrm{LS}(A,\mathbf{y}) \cap (\ker(A))^\perp$, then $\mathbf{u} - \mathbf{s} \in \ker(A) \cap (\ker(A))^\perp = \{\mathbf{0}\}$ by (i) and Proposition 2.1.3(iv), so $\mathbf{u} = \mathbf{s}$.
(iii) Fix $\mathbf{x} \in \mathrm{LS}(A,\mathbf{y})$ and set $\mathbf{z} := \mathbf{x} - \mathbf{s} \in \ker(A)$ by (i)–(ii).
Then $\mathbf{s} \perp \mathbf{z}$, and $\|\mathbf{x}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{z}\|_2^2 \geq \|\mathbf{s}\|_2^2$ by Theorem 1.4.3, with equality iff $\mathbf{z} = \mathbf{0}$, equivalently $\mathbf{x} = \mathbf{s}$.

Structural Commentary: Kernel translation describes every minimizer; orthogonal decomposition isolates the shortest representative.
Key Insight: Norm excess is exactly the squared kernel-component norm. □

Corollary 2.4.6. *Projection Formula for Full Column Rank*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ *satisfy* $\mathrm{rank}(A) = n$. *Then:*

(i) A^*A *is invertible;*
(ii) $P_{\mathrm{col}(A)} = A(A^*A)^{-1}A^*$.

Proof.

(i) The identities $\ker(A^*A) = \ker(A) = \{\mathbf{0}\}$ hold by Proposition 2.3.11(i) and Corollary 3.3.1(i).
Hence A^*A is injective and therefore invertible by Theorem 1.7.7.
(ii) Fix $\mathbf{y} \in \mathbb{F}^m$ and set $\mathbf{x}_0 := (A^*A)^{-1}A^*\mathbf{y}$. Then $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.
The relation $\mathbf{x}_0 \in \mathrm{LS}(A,\mathbf{y})$ holds by Theorem 2.4.4(i). Thus $P_{\mathrm{col}(A)}\mathbf{y} = A\mathbf{x}_0 = A(A^*A)^{-1}A^*\mathbf{y}$ by Theorem 2.2.6.
The operator identity follows by arbitrariness of $\mathbf{y}$.

Structural Commentary: Kernel equality makes A^*A invertible; normal-equation coordinates identify the projection operator.
Key Insight: Full column rank turns the normal equations into an explicit projector. □

Definition 2.4.7. *Minimum-Norm Solution*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{y} \in \mathrm{col}(A)$. A vector $\mathbf{s} \in \mathbb{F}^n$ is a *minimum-norm solution* of $A\mathbf{x} = \mathbf{y}$ if:

(i) $A\mathbf{s} = \mathbf{y}$;
(ii) $(\forall\, \mathbf{x} \in \mathbb{F}^n)\,[A\mathbf{x} = \mathbf{y} \implies \|\mathbf{s}\|_2 \leq \|\mathbf{x}\|_2]$.

Theorem 2.4.8. *Minimum-Norm Solution of a Consistent Linear System*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, let $\mathbf{y} \in \mathbb{F}^m$, and suppose $A\mathbf{x} = \mathbf{y}$ is consistent. Then:

(i) $(\exists!\, \mathbf{s} \in \mathbb{F}^n)\,[A\mathbf{s} = \mathbf{y} \wedge (\forall\, \mathbf{x} \in \mathbb{F}^n)\,(A\mathbf{x} = \mathbf{y} \Longrightarrow \|\mathbf{s}\|_2 \leq \|\mathbf{x}\|_2)]$;
(ii) for the vector $\mathbf{s}$ *in (i),* $\{\mathbf{x} \in \mathrm{col}(A^*) : A\mathbf{x} = \mathbf{y}\} = \{\mathbf{s}\}$;
(iii) $(\exists\, \mathbf{u} \in \mathbb{F}^m)\, AA^*\mathbf{u} = \mathbf{y}$;
(iv) for the vector $\mathbf{s}$ *in (i),* $(\forall\, \mathbf{u}_0 \in \mathbb{F}^m)\,[AA^*\mathbf{u}_0 = \mathbf{y} \Longrightarrow \mathbf{s} = A^*\mathbf{u}_0]$.

Proof. The identity $(\mathrm{col}(A^*), \ker(A)) = ((\ker(A))^\perp, (\mathrm{col}(A^*))^\perp)$ holds by Corollary 2.3.12(ii),(i).
Hence $\mathbb{F}^n = \mathrm{col}(A^*) \odot \ker(A)$ by Theorem 2.1.5.

(i) Choose $\mathbf{x}_0 \in \mathbb{F}^n$ with $A\mathbf{x}_0 = \mathbf{y}$ by consistency.
Write $\mathbf{x}_0 = \mathbf{s} + \mathbf{v}$ with $\mathbf{s} \in \mathrm{col}(A^*)$ and $\mathbf{v} \in \ker(A)$ by the established orthogonal direct sum. Then $A\mathbf{s} = \mathbf{y}$.
For $\mathbf{x} \in \mathbb{F}^n$ with $A\mathbf{x} = \mathbf{y}$, set $\mathbf{w} := \mathbf{x} - \mathbf{s}$. Then $A\mathbf{w} = A\mathbf{x} - A\mathbf{s} = \mathbf{0}$, so $\mathbf{w} \in \ker(A)$.
Since $\mathbf{s} \perp \mathbf{w}$, $\|\mathbf{x}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{w}\|_2^2 \geq \|\mathbf{s}\|_2^2$ by Theorem 1.4.3.
Equality holds iff $\mathbf{w} = \mathbf{0}$, equivalently $\mathbf{x} = \mathbf{s}$; hence $\mathbf{s}$ is the unique minimum-norm solution.
(ii) The vector $\mathbf{s}$ from (i) is a solution in $\mathrm{col}(A^*)$.
For any solution $\mathbf{s}' \in \mathrm{col}(A^*)$, $\mathbf{s}' - \mathbf{s} \in \ker(A) \cap \mathrm{col}(A^*) = \{\mathbf{0}\}$ by the established direct sum, so $\mathbf{s}' = \mathbf{s}$.
(iii) Choose $\mathbf{u} \in \mathbb{F}^m$ with $\mathbf{s} = A^*\mathbf{u}$ by $\mathbf{s} \in \mathrm{col}(A^*)$. Then $AA^*\mathbf{u} = A\mathbf{s} = \mathbf{y}$.
(iv) If $AA^*\mathbf{u}_0 = \mathbf{y}$, then $A^*\mathbf{u}_0$ is a solution in $\mathrm{col}(A^*)$, so $A^*\mathbf{u}_0 = \mathbf{s}$ by (ii).

Structural Commentary: Orthogonal splitting removes the kernel component; the adjoint range supplies the AA^* representation.
Key Insight: $\mathrm{col}(A^*) = (\ker(A))^\perp$ contains the unique shortest feasible vector. □

Figure 33 depicts the orthogonal intersection of the solution set with $\mathrm{col}(A^*)$.

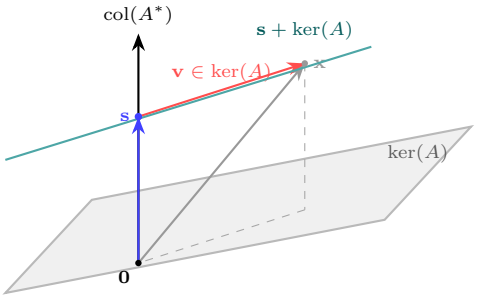

Figure 33: The solution set $\mathbf{s} + \ker(A)$ meets $\mathrm{col}(A^*) = (\ker(A))^\perp$ at its unique minimum-norm point $\mathbf{s}$.

Perspective 2.4.9. *Roadmap to the Pseudoinverse*
Theorem 2.4.4 identifies the fitted vector with the projection of $\mathbf{y}$ onto $\mathrm{col}(A)$.
Proposition 2.4.5 selects its minimum-norm preimage in $(\ker(A))^\perp = \mathrm{col}(A^*)$ by Theorem 2.4.8.
Later, Definition 5.2.4 and Theorem 5.2.9 package both choices as $A^\dagger\mathbf{y}$.

2.5 Self-Adjoint Operators and Orthogonal Projections

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Definition 2.5.1. *Self-Adjoint and Hermitian*

(i) Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F}$. An operator $T \in \mathcal{L}(\mathcal{V})$ is *self-adjoint* if $T^* = T$.
(ii) For $n \geq 1$, a matrix $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is *Hermitian* if $A^* = A$.

These conditions agree in orthonormal coordinates by Theorem 2.3.7.

Lemma 2.5.2. *Vanishing Complex Diagonal Values*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. If $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle = 0$, then $T = 0$.

Proof. Define $Q : \mathcal{V} \to \mathbb{C}$ by $Q(\mathbf{x}) := \langle T\mathbf{x}, \mathbf{x}\rangle$.
For $(\mathbf{u}, \mathbf{w}) \in \mathcal{V} \times \mathcal{V}$, $4\langle T\mathbf{u}, \mathbf{w}\rangle = Q(\mathbf{u}+\mathbf{w}) - Q(\mathbf{u}-\mathbf{w}) + iQ(\mathbf{u}+i\mathbf{w}) - iQ(\mathbf{u}-i\mathbf{w})$ by direct expansion.
Thus $(\forall (\mathbf{u}, \mathbf{w}) \in \mathcal{V} \times \mathcal{V})\ \langle T\mathbf{u}, \mathbf{w}\rangle = 0$ by the hypothesis. Taking $\mathbf{w} = T\mathbf{u}$, $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = 0$; hence $T = 0$ by positive definiteness.
Structural Commentary: Complex polarization recovers every mixed value from the vanishing diagonal map Q.
Key Insight: Diagonal vanishing forces the entire sesquilinear form to vanish. □

Proposition 2.5.3. *Self-Adjointness from Real Diagonal Values*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then $T = T^* \iff (\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$.

Proof.
($\Rightarrow$) Suppose $T = T^*$ and fix $\mathbf{v} \in \mathcal{V}$.
Then $\langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \overline{\langle T\mathbf{v}, \mathbf{v}\rangle}$ by the adjoint identity and conjugate symmetry; hence $\langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$.
($\Leftarrow$) Suppose $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$ and set $S := T - T^*$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle S\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle - \overline{\langle T\mathbf{v}, \mathbf{v}\rangle} = 0$.
Thus $S = 0$ by Lemma 2.5.2, and $T = T^*$ by the definition of S.
Structural Commentary: Adjoint symmetry proves reality; complex polarization annihilates the skew-adjoint part $T - T^*$.
Key Insight: Real diagonal values force the skew-adjoint part to vanish. □

Proposition 2.5.4. *Vanishing Diagonal Values for Self-Adjoint Operators*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$ satisfy $T^* = T$. If $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle = 0$, then $T = 0$.

Proof. If $\mathbb{F} = \mathbb{C}$, then $T = 0$ by Lemma 2.5.2. Suppose $\mathbb{F} = \mathbb{R}$.
For $(\mathbf{u}, \mathbf{w}) \in \mathcal{V} \times \mathcal{V}$, $\langle T\mathbf{w}, \mathbf{u}\rangle = \langle \mathbf{w}, T\mathbf{u}\rangle = \langle T\mathbf{u}, \mathbf{w}\rangle$ by self-adjointness and symmetry.
Hence $4\langle T\mathbf{u}, \mathbf{w}\rangle = \langle T(\mathbf{u}+\mathbf{w}), \mathbf{u}+\mathbf{w}\rangle - \langle T(\mathbf{u}-\mathbf{w}), \mathbf{u}-\mathbf{w}\rangle = 0$ by direct expansion and the hypothesis.
Taking $\mathbf{w} = T\mathbf{u}$, $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = 0$; hence $T = 0$ by positive definiteness.
Structural Commentary: Complex polarization handles $\mathbb{C}$; self-adjoint symmetry enables the real polarization identity.
Key Insight: Self-adjoint symmetry makes diagonal values determine every mixed value. □

Definition 2.5.5. *Orthogonal Projection Matrix*
For $n \geq 1$, a matrix $P \in \mathrm{Mat}(n, n, \mathbb{F})$ is an *orthogonal projection matrix* if $P^2 = P$ and $P^* = P$.

Theorem 2.5.6. *Orthogonal Projections as Self-Adjoint Idempotents*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $P \in \mathcal{L}(\mathcal{V})$. Then $P = P_{\mathrm{range}(P)} \iff [P^2 = P \wedge P^* = P]$.

Proof.
($\Rightarrow$) Set $\mathcal{U} := \mathrm{range}(P)$ and suppose $P = P_{\mathcal{U}}$. Then $P^2 = P$ by Proposition 2.2.3(vii).
The identity $P = \rho_{\mathcal{U},\mathcal{U}^\perp}$ holds by the definitions of orthogonal and algebraic projection.
The identity $P^* = \rho_{(\mathcal{U}^\perp)^\perp,\mathcal{U}^\perp}$ holds by Proposition 2.3.11(iii). Thus $P^* = \rho_{\mathcal{U},\mathcal{U}^\perp} = P$ by Corollary 2.1.8 and the preceding identity.
($\Leftarrow$) Set $\mathcal{U} := \mathrm{range}(P)$ and suppose $P^2 = P$ and $P^* = P$. Then $\mathcal{V} = \mathcal{U} \oplus \ker(P)$ and $P = \rho_{\mathcal{U},\ker(P)}$ by Proposition 2.4.2(i).
The identity $\ker(P) = \ker(P^*) = \mathcal{U}^\perp$ holds by $P^* = P$ and Proposition 2.3.10(i).
Therefore $P = \rho_{\mathcal{U},\mathcal{U}^\perp} = P_{\mathcal{U}}$ by the definitions of algebraic and orthogonal projection.

Structural Commentary: Idempotence supplies the range–kernel split; self-adjointness identifies $\ker(P)$ with $\mathcal{U}^\perp$.
Key Insight: For idempotent P, orthogonality is exactly $\ker(P) = (\mathrm{range}(P))^\perp$. □

2.6 Isometries and Unitary Matrices

Throughout this subsection, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 2.6.1. *Isometry and Isometric Isomorphism*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(i) T is an *isometry* if $(\forall\, \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\| = \|\mathbf{v}\|$.
(ii) T is an *isometric isomorphism* if it is a bijective isometry. Spaces $\mathcal{V}$ and $\mathcal{W}$ are *isometrically isomorphic* if such a map T exists.

Proposition 2.6.2. *Basic Properties of Isometries*
Let $\mathcal{V}$ be an inner-product space and let $S, T \in \mathcal{L}(\mathcal{V})$ be isometries.
(i) ST is an isometry;
(ii) T is injective;
(iii) if $\mathcal{V}$ is finite-dimensional, then T is invertible;
(iv) if $\mathcal{V}$ is finite-dimensional, then T^{-1} is an isometry.

Proof.
(i) $(\forall\, \mathbf{v} \in \mathcal{V})\ \|ST\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{v}\|$.
(ii) If $\mathbf{v} \in \ker(T)$, then $\|\mathbf{v}\| = \|T\mathbf{v}\| = 0$, so $\mathbf{v} = \mathbf{0}$ by positive definiteness. Hence $\ker(T) = \{\mathbf{0}\}$, and T is injective.
(iii) T is invertible by (ii) and Theorem 1.7.7.
(iv) Suppose $\dim(\mathcal{V}) < \infty$. Then T^{-1} exists by (iii). $(\forall\, \mathbf{u} \in \mathcal{V})\ \left\|T^{-1}\mathbf{u}\right\| = \left\|TT^{-1}\mathbf{u}\right\| = \|\mathbf{u}\|$.
Structural Commentary: Norm identities establish closure and injectivity before finite-dimensional invertibility controls the inverse.
Key Insight: Length preservation forces the kernel to vanish. □

Counterexample 2.6.3. *An Isometry Need Not Be Onto*
Define $T : \ell^2(\mathbb{F}) \to \ell^2(\mathbb{F})$ by $T(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$. The map T is linear by coordinatewise operations.
For $\mathbf{x} = (x_j)_{j\geq 1}$, $\|T\mathbf{x}\|_2^2 = \sum_{j=1}^{\infty} |x_j|^2 = \|\mathbf{x}\|_2^2$, so T is an isometry.
Every vector in $\mathrm{range}(T)$ has first coordinate 0, whereas $(1, 0, 0, \ldots) \notin \mathrm{range}(T)$.
Thus T is not surjective, so an infinite-dimensional isometry need not be an isomorphism.

Theorem 2.6.4. *Isometries Preserve Inner Products*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $(\forall\, \mathbf{x} \in \mathcal{V})\ \|T\mathbf{x}\| = \|\mathbf{x}\| \iff (\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{u}, T\mathbf{v}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}}$.

Proof.
($\Rightarrow$) Suppose T preserves norms and fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.
By linearity and norm preservation, the polarization formulas for $(T\mathbf{u}, T\mathbf{v})$ and $(\mathbf{u}, \mathbf{v})$ have equal norm terms.
Thus $\langle T\mathbf{u}, T\mathbf{v}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}}$ by Theorem 1.1.23.
($\Leftarrow$) If T preserves inner products, then $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{u}\rangle_{\mathcal{V}} = \|\mathbf{u}\|^2$.
Thus T preserves norms by nonnegativity.
Structural Commentary: Polarization transfers the norm identity to mixed inner products; diagonal evaluation reverses the transfer.
Key Insight: A norm determines its inducing inner product. □

Remark 2.6.5. *Isometry Conventions*
For linear maps, norm and inner-product preservation are equivalent by Theorem 2.6.4.
Thus either convention defines the same isometries.
For possibly degenerate bilinear forms, form preservation need not imply injectivity.
An isometry is therefore often required to be bijective and form-preserving.

Theorem 2.6.6. *Isometries and the Adjoint Product*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\| = \|\mathbf{v}\| \iff T^*T = I_{\mathcal{V}}$.

Proof.

($\Rightarrow$) Suppose T preserves norms and set $A := T^*T - I_{\mathcal{V}}$. Then $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{u}, A\mathbf{v}\rangle = 0$ by adjointness and Theorem 2.6.4.
Taking $\mathbf{u} = A\mathbf{v}$, $(\forall\, \mathbf{v} \in \mathcal{V})\ \|A\mathbf{v}\|^2 = 0$. Hence $A = 0$ by positive definiteness, so $T^*T = I_{\mathcal{V}}$.
($\Leftarrow$) If $T^*T = I_{\mathcal{V}}$, then $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle = \langle \mathbf{u}, T^*T\mathbf{u}\rangle = \|\mathbf{u}\|^2$ by the adjoint relation.
Thus T preserves norms by nonnegativity.

Structural Commentary: Adjointness converts preserved inner products into the vanishing of $T^*T - I_{\mathcal{V}}$.
Key Insight: The Gram operator of an isometry is the identity. □

Definition 2.6.7. *Unitary and Orthogonal*

(i) Let $\mathcal{V}$ be a finite-dimensional inner-product space. An operator $U \in \mathcal{L}(\mathcal{V})$ is *unitary* if $U^*U = UU^* = I_{\mathcal{V}}$.
(ii) Let $n \geq 1$. A matrix $U \in \mathrm{Mat}(n, n, \mathbb{F})$ is *unitary* if $U^*U = I_n$. If $\mathbb{F} = \mathbb{R}$, a unitary matrix is *orthogonal*.

Theorem 2.6.8. *Equivalent Conditions for a Unitary Matrix*
Let $n \geq 1$ and let $U = [\mathbf{u}_1\ \cdots\ \mathbf{u}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$. The following conditions are equivalent:

(i) $U^*U = I_n$;
(ii) *the columns* $\mathbf{u}_1, \ldots, \mathbf{u}_n$ *are orthonormal in* $\mathbb{F}^n$;
(iii) $UU^* = I_n$;
(iv) U^* *is unitary;*
(v) *the rows of* U *are orthonormal;*
(vi) U *is invertible and* $U^{-1} = U^*$;
(vii) $(\forall\, \mathbf{x} \in \mathbb{F}^n)\ \|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$;
(viii) $(\forall\, \mathbf{x}, \mathbf{y} \in \mathbb{F}^n)\ \langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} = \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n}$.

Proof.

(i) $\Rightarrow$ *(ii)* For $i, j \in J_n$, $(U^*U)_{ij} = \mathbf{u}_i^*\mathbf{u}_j = \langle \mathbf{u}_j, \mathbf{u}_i\rangle_{\mathbb{F}^n}$. Hence $U^*U = I_n$ if and only if the columns are orthonormal.
(ii) $\Rightarrow$ *(iii)* Column orthonormality implies $U^*U = I_n$, so $UU^* = I_n$ by Theorem 3.4.2.
(iii) $\Rightarrow$ *(iv)* The identity $(U^*)^*U^* = UU^* = I_n$ holds by conjugate-transpose involution, so U^* is unitary.
(iv) $\Rightarrow$ *(v)* Unitarity of U^* implies $UU^* = I_n$.
Let $\mathbf{r}_i$ denote row i of U. Then $(\forall\, i, j \in J_n)\ (UU^*)_{ij} = \langle \mathbf{r}_i, \mathbf{r}_j\rangle_{\mathbb{F}^n}$, so the rows are orthonormal.
(v) $\Rightarrow$ *(vi)* Row orthonormality implies $UU^* = I_n$. Thus U is invertible and $U^{-1} = U^*$ by Theorem 3.4.2.
(vi) $\Rightarrow$ *(vii)* The identity $U^*U = I_n$ holds by $U^{-1} = U^*$.
Hence $(\forall\, \mathbf{x} \in \mathbb{F}^n)\ \|U\mathbf{x}\|_2^2 = \langle \mathbf{x}, U^*U\mathbf{x}\rangle_{\mathbb{F}^n} = \|\mathbf{x}\|_2^2$ by the conjugate-transpose adjoint relation.
Thus U preserves norms by nonnegativity.
(vii) $\Rightarrow$ *(viii)* The map $\mathbf{x} \mapsto U\mathbf{x}$ preserves inner products by Theorem 2.6.4.
(viii) $\Rightarrow$ *(i)* Set $A := U^*U - I_n$. Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathbb{F}^n)\ \langle \mathbf{x}, A\mathbf{y}\rangle_{\mathbb{F}^n} = 0$ by the conjugate-transpose adjoint relation and (viii).
Taking $\mathbf{x} = A\mathbf{y}$, $(\forall\, \mathbf{y} \in \mathbb{F}^n)\ \|A\mathbf{y}\|_2^2 = 0$. Hence $A = 0$ by positive definiteness, so $U^*U = I_n$.

Structural Commentary: The implication cycle connects column and row Gram identities through inverse and invariance relations.
Key Insight: Both Gram products encode orthonormal coordinate frames. □

Figure 34 depicts the geometric invariance in Theorem 2.6.8.

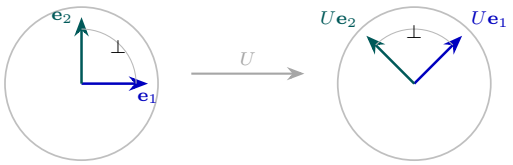

Figure 34: A unitary map preserves the lengths and right angle of an orthonormal pair.

Definition 2.6.9. *Unitary and Permutation Similarity*
Let $n \geq 1$ and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) The matrices A and B are *unitarily similar* if $A = UBU^*$ for a unitary $U \in \mathrm{Mat}(n, n, \mathbb{F})$.
(ii) They are *orthogonally similar* if $\mathbb{F} = \mathbb{R}$ and $A = UBU^*$ for an orthogonal $U \in \mathrm{Mat}(n, n, \mathbb{R})$.
(iii) A matrix $P = [p_{ij}] \in \mathrm{Mat}(n, n, \mathbb{R})$ is a *permutation matrix* if $(\forall\, i, j \in J_n)\ p_{ij} \in \{0, 1\}$ and $(\forall\, i \in J_n)\ \sum_{j=1}^{n} p_{ij} = \sum_{j=1}^{n} p_{ji} = 1$.
(iv) The matrices A and B are *permutation similar* if $A = PBP^*$ for a permutation matrix P.

Proposition 2.6.10. *Permutation Matrices Are Orthogonal*
Let $n \geq 1$, let $\sigma \in \mathfrak{S}_n$, and set $P := [\mathbf{e}_{\sigma(1)}\ \cdots\ \mathbf{e}_{\sigma(n)}] \in \mathrm{Mat}(n, n, \mathbb{R})$. Then $P^*P = PP^* = I_n$, so P is orthogonal.

Proof. The columns of P are a permutation of the standard orthonormal basis of $\mathbb{R}^n$.
Hence $P^*P = PP^* = I_n$ by Theorem 2.6.8(ii)–(iii). Therefore P is orthogonal by the real-field definition.
Structural Commentary: Standard-basis permutation converts column orthonormality directly into both Gram identities.
Key Insight: Reordering an orthonormal basis preserves its Gram matrix. □

Theorem 2.6.11. *Unitary Coordinate Form of an Orthogonal Projection*
Let $n \geq 1$, let $P \in \mathrm{Mat}(n, n, \mathbb{F})$ be an orthogonal projection matrix, and set $r := \mathrm{rank}(P)$.
Then $(\exists\, U \in \mathrm{Mat}(n, n, \mathbb{F}))\ [U^*U = I_n \wedge U^*PU = I_r \oplus 0_{n-r}]$. Equivalently, P is unitarily similar to $I_r \oplus 0_{n-r}$.

Proof.

(i) *Projection Operator*: Define $T_P : \mathbb{F}^n \to \mathbb{F}^n$ $T_P\mathbf{x} := P\mathbf{x}$.
Set $\mathcal{U} := \mathrm{range}(T_P) = \mathrm{col}(P)$. Then $T_P^2 = T_P = T_P^*$ by $P^2 = P = P^*$ and Theorem 2.3.7.
Hence $T_P = P_{\mathcal{U}}$ by Theorem 2.5.6. Therefore $(T_P|_{\mathcal{U}}, T_P|_{\mathcal{U}^\perp}) = (I_{\mathcal{U}}, 0)$ by Proposition 2.2.3(ii)–(iii).

(ii) *Adapted Basis*: Choose an orthonormal basis $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ of $\mathcal{U}$ by Corollary 1.3.11.
Extend it to an orthonormal basis $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ of $\mathbb{F}^n$ by Corollary 1.3.12.
Set $U := [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$. Then $U^*U = I_n$ by Theorem 2.6.8(i)–(ii).
(iii) *Block Form*: If $j \in J_r$, then $U^*PU\mathbf{e}_j = U^*P\mathbf{u}_j = U^*\mathbf{u}_j = \mathbf{e}_j$. If $j \in J_n \setminus J_r$, then $U^*PU\mathbf{e}_j = \mathbf{0}$.
Thus $U^*PU = I_r \oplus 0_{n-r}$ by equality on the standard basis.
(iv) *Similarity*: The identity $UU^* = I_n$ holds by Theorem 2.6.8(i),(iii).
Multiplication by U on the left and U^* on the right establishes $P = U(I_r \oplus 0_{n-r})U^*$.

Structural Commentary: The induced projection supplies an adapted orthonormal basis whose matrix action is block diagonal.
Key Insight: Projection is coordinate truncation in range–kernel coordinates. □

Figure 35 depicts the range–kernel coordinates in Theorem 2.6.11.

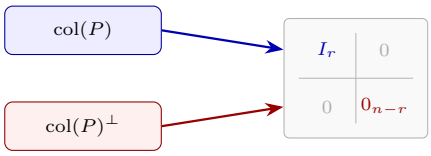

Figure 35: Unitary coordinates make an orthogonal projection the identity on $\mathrm{col}(P)$ and zero on $\mathrm{col}(P)^\perp$.

Proposition 2.6.12. *Algebra of Unitary Matrices*
Let $m, n \geq 1$, let $U, V \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $W \in \mathrm{Mat}(m, m, \mathbb{F})$. Let $V \oplus W$ denote the block diagonal matrix with blocks V and W.
(i) $[U^*U = I_n \wedge V^*V = I_n] \implies (UV)^*(UV) = I_n$;
(ii) $[(V \oplus W)^*(V \oplus W) = I_{n+m}] \iff [V^*V = I_n \wedge W^*W = I_m]$.

Proof.
(i) The identity $(UV)^*(UV) = V^*U^*UV = V^*V = I_n$ holds by the hypotheses.
(ii) ($\Rightarrow$) If $(V \oplus W)^*(V \oplus W) = I_{n+m}$, then $V^*V \oplus W^*W = I_n \oplus I_m$.
Hence $V^*V = I_n$ and $W^*W = I_m$ by equality of the diagonal blocks.
(ii) ($\Leftarrow$) If $V^*V = I_n$ and $W^*W = I_m$, then $(V \oplus W)^*(V \oplus W) = I_n \oplus I_m = I_{n+m}$.

Structural Commentary: Product reversal proves closure under multiplication; blockwise adjoints reduce direct sums componentwise.
Key Insight: Unitary identities multiply and split blockwise. □

Theorem 2.6.13. *Extending Isometric Columns to a Unitary Matrix*
Let $m, n \geq 1$ and let $X \in \mathrm{Mat}(m, n, \mathbb{F})$ satisfy $X^*X = I_n$. Then:
(i) $n \leq m$;
(ii) $(\exists\, X' \in \mathrm{Mat}(m, m-n, \mathbb{F}))\, [X \ X']^*[X \ X'] = I_m$.

Proof. Write $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$. For $i, j \in J_n$, $(X^*X)_{ij} = \langle \mathbf{x}_j, \mathbf{x}_i \rangle_{\mathbb{F}^m} = \delta_{ij}$ by hypothesis. Hence $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ is orthonormal in $\mathbb{F}^m$.
(i) The list is linearly independent by Corollary 1.3.3, so $n \leq m$ by Theorem 1.3.11.
(ii) Extend the list to an orthonormal basis $(\mathbf{x}_1, \dots, \mathbf{x}_m)$ of $\mathbb{F}^m$ by Corollary 1.3.12.
Set $X' := [\mathbf{x}_{n+1} \ \cdots \ \mathbf{x}_m]$. Then $[X \ X']^*[X \ X'] = I_m$ by Theorem 2.6.8(i)–(ii).

Structural Commentary: The Gram identity turns the columns into an orthonormal list, then basis extension completes it.
Key Insight: A partial orthonormal frame extends to a unitary frame. □

2.7 Householder Reflections

Throughout this subsection, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 2.7.1. *Householder Matrix*
Let $n \geq 1$ and $\mathbf{w} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$, and set $\mathbf{u} := \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$. The *Householder matrix* associated with $\mathbf{w}$ is $U_{\mathbf{w}} := I_n - 2P_{\mathbf{u}} = I_n - 2\frac{\mathbf{w}\mathbf{w}^*}{\|\mathbf{w}\|_2^2}$.
Equivalently, $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, U_{\mathbf{w}}\mathbf{x} = \mathbf{x} - 2\langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} \mathbf{u}$.

Theorem 2.7.2. *Householder Matrices Are Unitary Reflections*
Let $n \geq 1$ and let $\mathbf{w} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$. Then:
(i) $U_{\mathbf{w}}^* = U_{\mathbf{w}}$;
(ii) $U_{\mathbf{w}}^2 = I_n$;
(iii) $U_{\mathbf{w}}^* U_{\mathbf{w}} = I_n$, *so $U_{\mathbf{w}}$ is unitary;*
(iv) $(\forall\, \mathbf{x} \in \mathbf{w}^\perp)\, U_{\mathbf{w}}\mathbf{x} = \mathbf{x}$;
(v) $(\forall\, \mathbf{x} \in \mathrm{span}(\mathbf{w}))\, U_{\mathbf{w}}\mathbf{x} = -\mathbf{x}$;
(vi) $\mathbf{w} \in \mathbb{R}^n \implies U_{\mathbf{w}}$ *is symmetric and orthogonal over $\mathbb{R}$.*

Proof. Set $\mathbf{u} := \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$ and $P := P_{\mathbf{u}}$.
(i) The identity $P^* = P$ holds by Proposition 2.2.5(iii). Hence $U_{\mathbf{w}}^* = I_n - 2P = U_{\mathbf{w}}$.
(ii) The identity $P^2 = P$ holds by Proposition 2.2.5(iv). Hence $U_{\mathbf{w}}^2 = (I_n - 2P)^2 = I_n - 4P + 4P^2 = I_n$.
(iii) The identity $U_{\mathbf{w}}^* U_{\mathbf{w}} = U_{\mathbf{w}}^2 = I_n$ holds by (i)–(ii), so $U_{\mathbf{w}}$ is unitary.
(iv) Fix $\mathbf{x} \in \mathbf{w}^\perp = \mathrm{span}(\mathbf{u})^\perp$. Then $P\mathbf{x} = \mathbf{0}$ by Proposition 2.2.5(ii), so $U_{\mathbf{w}}\mathbf{x} = \mathbf{x}$.
(v) Fix $\mathbf{x} \in \mathrm{span}(\mathbf{w}) = \mathrm{span}(\mathbf{u})$, and choose $a \in \mathbb{F}$ with $\mathbf{x} = a\mathbf{u}$. Then $P\mathbf{x} = \langle a\mathbf{u}, \mathbf{u} \rangle\, \mathbf{u} = a\mathbf{u} = \mathbf{x}$ by $\|\mathbf{u}\|_2 = 1$, so $U_{\mathbf{w}}\mathbf{x} = -\mathbf{x}$.
(vi) Suppose $\mathbf{w} \in \mathbb{R}^n$. Then $\mathbf{u}$, P, and $U_{\mathbf{w}}$ have real entries.
The identities $U_{\mathbf{w}}^t = U_{\mathbf{w}}$ and $U_{\mathbf{w}}^t U_{\mathbf{w}} = I_n$ hold by (i) and (iii), respectively.

Structural Commentary: Self-adjoint idempotence proves unitarity; the range–kernel split identifies the two eigenspaces.
Key Insight: $I_n - 2P_{\mathbf{u}}$ fixes $\ker(P_{\mathbf{u}})$ and negates $\mathrm{range}(P_{\mathbf{u}})$. □

Figure 36 illustrates the two eigenspace actions in Theorem 2.7.2(iv)–(v).

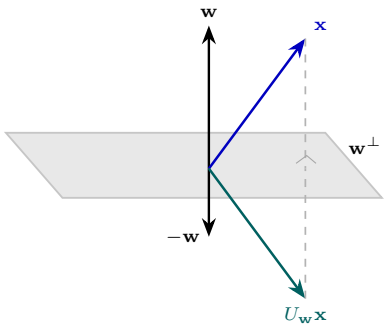

Figure 36: Householder reflection: $U_{\mathbf{w}}$ preserves the $\mathbf{w}^{\perp}$ component of $\mathbf{x}$ and negates its span($\mathbf{w}$) component.

Theorem 2.7.3. *Householder Map Between Equal-Norm Vectors*
Let $n \geq 1$ and let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ satisfy $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 \neq 0$. Set $z := \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n}$.
Choose $\sigma \in \mathbb{F}$ with $\sigma = 1$ for $z = 0$ and $\sigma = -\frac{\overline{z}}{|z|}$ for $z \neq 0$. Set $\mathbf{w} := \mathbf{y} - \sigma\mathbf{x}$. Then:

(i) $|\sigma| = 1$;
(ii) $\mathbf{w} \neq \mathbf{0}$;
(iii) $\sigma U_{\mathbf{w}} \mathbf{x} = \mathbf{y}$.

Proof. Set $r := \|\mathbf{x}\|_2 = \|\mathbf{y}\|_2$.
(i) The identity $|\sigma| = 1$ holds by the two defining cases.
(ii) The identity $\sigma z = -|z|$ holds by the two defining cases.
Thus $\|\mathbf{w}\|_2^2 = 2r^2 - 2\operatorname{Re}(\sigma z) = 2(r^2 + |z|) > 0$ by (i) and conjugate symmetry. Hence $\mathbf{w} \neq \mathbf{0}$ by positive definiteness.
(iii) By $\mathbf{y} = \mathbf{w} + \sigma\mathbf{x}$ and Definition 2.7.1, $\sigma U_{\mathbf{w}} \mathbf{x} = \mathbf{y} - \mathbf{w} - 2\sigma \frac{\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|_2^2} \mathbf{w}$.
The identity $2\sigma \langle \mathbf{x}, \mathbf{w} \rangle = 2\sigma z - 2r^2 = -\|\mathbf{w}\|_2^2$ holds by the definition of $\mathbf{w}$ and (i)–(ii). Therefore $\sigma U_{\mathbf{w}} \mathbf{x} = \mathbf{y}$.
Structural Commentary: The phase makes σz nonpositive real; the norm identity then cancels the reflected component.
Key Insight: $2\sigma \langle \mathbf{x}, \mathbf{w} \rangle = -\|\mathbf{w}\|_2^2$ annihilates the residual $\mathbf{w}$-coefficient. □

Corollary 2.7.4. *Householder Unitary Annihilator*
Let $n \geq 1$ and $\mathbf{x} = [x_1 \ \cdots \ x_n]^t \in \mathbb{F}^n \setminus \{\mathbf{0}\}$. Choose $\sigma \in \mathbb{F}$ with $\sigma = 1$ for $x_1 = 0$ and $\sigma = -\frac{\overline{x_1}}{|x_1|}$ for $x_1 \neq 0$.
Set $\mathbf{w} := \|\mathbf{x}\|_2 \mathbf{e}_1 - \sigma\mathbf{x}$. Then:

(i) $\mathbf{w} \neq \mathbf{0}$;
(ii) $\sigma U_{\mathbf{w}} \mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$;
(iii) $\|\mathbf{x}\|_2 = 1 \Longrightarrow \overline{\sigma} U_{\mathbf{w}}$ *is unitary with* $(\overline{\sigma} U_{\mathbf{w}}) \mathbf{e}_1 = \mathbf{x}$.

Proof. Set $\mathbf{y} := \|\mathbf{x}\|_2 \mathbf{e}_1$. Then $\|\mathbf{y}\|_2 = \|\mathbf{x}\|_2$ and $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 \|\mathbf{x}\|_2$.
Since $\|\mathbf{x}\|_2 > 0$, the defining cases for σ coincide with those determined by $\langle \mathbf{x}, \mathbf{y} \rangle$.
(i) The relation $\mathbf{w} \neq \mathbf{0}$ holds by Theorem 2.7.3(ii) with this $\mathbf{y}$.
(ii) The identity $\sigma U_{\mathbf{w}} \mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$ holds by Theorem 2.7.3(iii) with this $\mathbf{y}$.
(iii) Suppose $\|\mathbf{x}\|_2 = 1$ and set $A := \sigma U_{\mathbf{w}}$. The identity $A^* A = |\sigma|^2 U_{\mathbf{w}}^* U_{\mathbf{w}} = I_n$ holds by Theorem 2.7.3(i) and Theorem 2.7.2(iii).
Thus A^* is unitary by Theorem 2.6.8(i),(iv) with $U = A$. By (ii), $A\mathbf{x} = \mathbf{e}_1$, so $A^* \mathbf{e}_1 = A^* A \mathbf{x} = \mathbf{x}$.
The identity $A^* = \overline{\sigma} U_{\mathbf{w}}$ holds by Theorem 2.7.2(i).
Structural Commentary: Equal-norm reflection constructs $A\mathbf{x} = \mathbf{e}_1$; unitarity transfers this identity to the first column of A^*.
Key Insight: The adjoint reverses the annihilation map and installs $\mathbf{x}$ as a column. □

3 Bilinear and Quadratic Forms

3.1 Bilinear Form Types and Congruence Invariants

Metric vector spaces retain the algebraic structure of inner-product geometry without imposing positivity.
Unless stated otherwise, $\mathbb{F}$ is an arbitrary field and every vector space is finite-dimensional.
General bilinear pairings, their radicals, their matrices, and their basis-change law were established earlier.
Here the two input spaces coincide, and a fixed bilinear form B is written $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$.

Definition 3.1.1. *Symmetric, Skew-Symmetric, and Alternating Forms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ be bilinear.

(i) B is a *bilinear form* on $\mathcal{V}$; when fixed, write $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$.
(ii) $\operatorname{Sym}(B)$ means $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = B(\mathbf{y}, \mathbf{x})$; then B is *symmetric*.
(iii) $\operatorname{Skew}(B)$ means $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = -B(\mathbf{y}, \mathbf{x})$; then B is *skew-symmetric*.
(iv) $\operatorname{Alt}(B)$ means $(\forall\, \mathbf{x} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{x}) = 0$; then B is *alternating*.
(v) A *metric vector space* is a pair $(\mathcal{V}, B)$ such that $\operatorname{Sym}(B)$, $\operatorname{Skew}(B)$, or $\operatorname{Alt}(B)$ holds.
The symmetric and alternating cases define *orthogonal geometry* and *symplectic geometry*, respectively.

Proposition 3.1.2. *Characteristic Controls the Form Types*
Let $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$ be a bilinear form on $\mathcal{V}$.

(i) $\operatorname{Alt}(B) \Rightarrow \operatorname{Skew}(B)$.
(ii) If $\operatorname{char}(\mathbb{F}) \neq 2$, *then* $\operatorname{Sym}(B) \wedge \operatorname{Alt}(B) \Rightarrow B = 0$.

(iii) *If* $\operatorname{char}(\mathbb{F}) \neq 2$, *then* $\operatorname{Alt}(B) \iff \operatorname{Skew}(B)$.
(iv) *If* $\operatorname{Sym}(B)$, $\operatorname{Skew}(B)$, *or* $\operatorname{Alt}(B)$, *then* $\operatorname{rad}_L(B) = \operatorname{rad}_R(B)$.
(v) *If* $\operatorname{char}(\mathbb{F}) = 2$, *then* $\operatorname{Sym}(B) \iff \operatorname{Skew}(B)$.
(vi) *If* $\operatorname{char}(\mathbb{F}) = 2$, *then* $\operatorname{Alt}(B) \Rightarrow \operatorname{Sym}(B)$.

Proof.

(*i*) Under $\operatorname{Alt}(B)$, $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, 0 = B(\mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y}) = B(\mathbf{x}, \mathbf{y}) + B(\mathbf{y}, \mathbf{x})$ by bilinearity, so $\operatorname{Skew}(B)$.
(*ii*) Under $\operatorname{char}(\mathbb{F}) \neq 2$, $\operatorname{Sym}(B)$, and $\operatorname{Alt}(B)$, the relation $\operatorname{Skew}(B)$ holds by (i).
Thus $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, 2B(\mathbf{x}, \mathbf{y}) = 0$, and $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = 0$ by $2 \neq 0$; therefore $B = 0$.
(*iii*) ($\Rightarrow$) $\operatorname{Alt}(B) \Rightarrow \operatorname{Skew}(B)$ by (i).
(*iii*) ($\Leftarrow$) If $\operatorname{char}(\mathbb{F}) \neq 2$ and $\operatorname{Skew}(B)$, then $(\forall\, \mathbf{x} \in \mathcal{V})\, 2B(\mathbf{x}, \mathbf{x}) = 0$. Thus $(\forall\, \mathbf{x} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{x}) = 0$ by $2 \neq 0$, so $\operatorname{Alt}(B)$.
(*iv*) Under $\operatorname{Sym}(B) \vee \operatorname{Skew}(B)$, $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, [B(\mathbf{x}, \mathbf{y}) = 0 \iff B(\mathbf{y}, \mathbf{x}) = 0]$.
The same equivalence holds under $\operatorname{Alt}(B)$ by (i), so $\operatorname{rad}_L(B) = \operatorname{rad}_R(B)$ by Definition 4.4.4(i).
(*v*) ($\Rightarrow$) Under $\operatorname{Sym}(B)$, $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = B(\mathbf{y}, \mathbf{x}) = -B(\mathbf{y}, \mathbf{x})$, so $\operatorname{Skew}(B)$.
(*v*) ($\Leftarrow$) Under $\operatorname{Skew}(B)$, $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = -B(\mathbf{y}, \mathbf{x}) = B(\mathbf{y}, \mathbf{x})$, so $\operatorname{Sym}(B)$.
(*vi*) If $\operatorname{char}(\mathbb{F}) = 2$ and $\operatorname{Alt}(B)$, then $\operatorname{Skew}(B)$ by (i) and $\operatorname{Sym}(B)$ by (v).

Structural Commentary: Diagonal expansion controls form types; argument reversal preserves zeros and identifies radicals.
Key Insight: The scalar 2 is invertible off characteristic 2 and vanishes in characteristic 2. □

Henceforth, $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is symmetric, skew-symmetric, or alternating.
Its left and right radicals agree by Proposition 3.1.2(iv), so a single radical and a single orthogonality notation suffice.

Definition 3.1.3. *Matrix, Rank, and Discriminant of a Form*
Let $(\mathcal{V}, B)$ be a metric vector space with ordered basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$, where $n \geq 1$.

(i) The *matrix of the form* in β is $M_\beta := [\langle \mathbf{v}_i, \mathbf{v}_j \rangle]_{i,j}$.
(ii) The *rank* of the form is $\operatorname{rank}(M_\beta)$.
(iii) The *discriminant set* is $\Delta(\mathcal{V}) := \{r^2 \det(M_\beta) : r \in \mathbb{F}^\times\}$.
(iv) If $\det(M_\beta) \neq 0$, then $\Delta(\mathcal{V}) = \det(M_\beta)(\mathbb{F}^\times)^2$. This set represents the square class of $\det(M_\beta)$ in $\mathbb{F}^\times/(\mathbb{F}^\times)^2$.

Proposition 3.1.4. *Matrices and Invariants under Congruence*
Let $n = \dim(\mathcal{V}) \geq 1$ *and let* β, γ *be ordered bases of* $\mathcal{V}$. *Put* $C := C_{\beta \leftarrow \gamma}$.

(i) $\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\beta^t M_\beta [\mathbf{y}]_\beta$.
(ii) $M_\gamma = C^t M_\beta C$.
(iii) $\{M_\delta : \delta$ *is an ordered basis of* $\mathcal{V}\} = \{P^t M_\beta P : P \in \operatorname{Mat}(n, n, \mathbb{F})^\times\}$.
(iv) $\operatorname{rank}(M_\gamma) = \operatorname{rank}(M_\beta)$.
(v) $\Delta(\mathcal{V})$ *is independent of the ordered basis used in its definition.*

Proof.

(*i*) By Proposition 4.4.2, $\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\beta^t M_\beta [\mathbf{y}]_\beta$.
(*ii*) By Proposition 4.4.10(i), $M_\gamma = C^t M_\beta C$.
(*iii*) For $\subseteq$, let δ be an ordered basis of $\mathcal{V}$. Then $M_\delta = C_{\beta \leftarrow \delta}^t M_\beta C_{\beta \leftarrow \delta}$ by (ii), and $C_{\beta \leftarrow \delta} \in \operatorname{Mat}(n, n, \mathbb{F})^\times$ by Theorem 3.5.2(i).
For $\supseteq$, fix $P \in \operatorname{Mat}(n, n, \mathbb{F})^\times$. Choose an ordered basis δ satisfying $P^{-1} = C_{\delta \leftarrow \beta}$ by Theorem 3.5.2(ii).
Then $P = C_{\beta \leftarrow \delta}$ by Theorem 3.5.2(i), so $P^t M_\beta P = M_\delta$ by (ii).
(*iv*) The matrices C and C^t are invertible by Theorem 3.5.2(i) and Corollary 3.6.4, respectively.
Thus $\operatorname{rank}(M_\gamma) = \operatorname{rank}(M_\beta)$ by (ii) and Theorem 3.6.6(ii).
(*v*) Put $D_\eta := \{r^2 \det(M_\eta) : r \in \mathbb{F}^\times\}$ $(\eta \in \{\beta, \gamma\})$ and $c := \det(C)$. Then $c \in \mathbb{F}^\times$ by invertibility of C.
The identity $\det(M_\gamma) = \det(C^t) \det(M_\beta) \det(C) = c^2 \det(M_\beta)$ holds by (ii) and the determinant product and transpose laws.
Therefore $D_\gamma = D_\beta$ by the bijection $r \mapsto rc$ of $\mathbb{F}^\times$.

Structural Commentary: Congruence exhausts basis matrices; invertible factors preserve rank and square class.
Key Insight: Both arguments transform by the same basis-change matrix. □

3.2 Quadratic Forms and Polar Forms

Throughout this subsection, $\mathcal{V}$ is a vector space over an arbitrary field $\mathbb{F}$.

Definition 3.2.1. *Quadratic Forms and Polar Forms*
A *quadratic form* on $\mathcal{V}$ is a map $Q : \mathcal{V} \to \mathbb{F}$ satisfying:

(i) $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{v} \in \mathcal{V})\, Q(a\mathbf{v}) = a^2 Q(\mathbf{v})$;
(ii) the *polar form* $B_Q : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ defined by $B_Q(\mathbf{u}, \mathbf{v}) := Q(\mathbf{u}+\mathbf{v}) - Q(\mathbf{u}) - Q(\mathbf{v})$ is bilinear.

Proposition 3.2.2. *Quadratic Forms and Symmetric Bilinear Forms*

(i) *If* $Q : \mathcal{V} \to \mathbb{F}$ *is quadratic, then* $\operatorname{Sym}(B_Q)$.
(ii) *Let* $\operatorname{char}(\mathbb{F}) \neq 2$ *and let* $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ *be bilinear with* $\operatorname{Sym}(B)$. *Define* $Q_B(\mathbf{x}) := \frac{1}{2} B(\mathbf{x}, \mathbf{x})$; Q_B *is quadratic and* $B_{Q_B} = B$.
(iii) *Let* $\operatorname{char}(\mathbb{F}) \neq 2$ *and let* $Q : \mathcal{V} \to \mathbb{F}$ *be quadratic. Define* $Q_{B_Q}(\mathbf{x}) := \frac{1}{2} B_Q(\mathbf{x}, \mathbf{x})$. *Then* $Q_{B_Q} = Q$.

Proof.

(*i*) $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\, B_Q(\mathbf{u}, \mathbf{v}) = B_Q(\mathbf{v}, \mathbf{u})$ by commutativity of vector addition. $\operatorname{Sym}(B_Q)$ by this identity.
(*ii*) $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{x} \in \mathcal{V})\, Q_B(a\mathbf{x}) = a^2 Q_B(\mathbf{x})$ by bilinearity.
$(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\, B_{Q_B}(\mathbf{u}, \mathbf{v}) = \frac{1}{2}\big(B(\mathbf{u}, \mathbf{v}) + B(\mathbf{v}, \mathbf{u})\big) = B(\mathbf{u}, \mathbf{v})$ by bilinearity and symmetry.
$B_{Q_B} = B$ by this identity, and Q_B is quadratic by the homogeneity identity and bilinearity of B_{Q_B}.
(*iii*) $(\forall\, \mathbf{x} \in \mathcal{V})\, Q_{B_Q}(\mathbf{x}) = \frac{1}{2} B_Q(\mathbf{x}, \mathbf{x}) = Q(\mathbf{x})$ by quadratic homogeneity. $Q_{B_Q} = Q$ by this pointwise identity.

Structural Commentary: Symmetry places B_Q among symmetric forms; items (ii)–(iii) establish inverse reconstruction.
Key Insight: The identity $B_Q(\mathbf{x}, \mathbf{x}) = 2Q(\mathbf{x})$ makes diagonal evaluation invert polarization. □

Proposition 3.2.3. *Polar Forms in Characteristic Two*
Let $Q : \mathcal{V} \to \mathbb{F}$ be quadratic. If $\operatorname{char}(\mathbb{F}) = 2$, then $\operatorname{Alt}(B_Q)$.

Proof. $Q(\mathbf{0}) = 0$ by quadratic homogeneity at scalar 0.
$(\forall\, \mathbf{x} \in \mathcal{V})\, B_Q(\mathbf{x}, \mathbf{x}) = Q(2\mathbf{x}) - 2Q(\mathbf{x}) = Q(\mathbf{0}) = 0$ by $\operatorname{char}(\mathbb{F}) = 2$. $\operatorname{Alt}(B_Q)$ by the preceding identity.
Structural Commentary: Homogeneity supplies $Q(\mathbf{0}) = 0$; characteristic 2 then collapses the polar diagonal.
Key Insight: The identities $2\mathbf{x} = \mathbf{0}$ and $2Q(\mathbf{x}) = 0$ force diagonal vanishing. □

Proposition 3.2.4. *Square Perturbations in Characteristic Two*
Assume $\operatorname{char}(\mathbb{F}) = 2$.

(i) Let $Q : \mathcal{V} \to \mathbb{F}$ be quadratic and $L : \mathcal{V} \to \mathbb{F}$ linear, and define $L^2(\mathbf{v}) := L(\mathbf{v})^2$. Then $Q + L^2$ is quadratic and $B_{Q+L^2} = B_Q$.
(ii) On $\mathbb{F}^2$, define $Q_0(x, y) := xy$ and $Q_1(x, y) := xy + x^2$.
Then Q_0, Q_1 are quadratic, $Q_0 \neq Q_1$, and $B_{Q_0} = B_{Q_1} = B$ for $B((x, y), (u, v)) := xv + uy$.

Proof.

(i) $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{v} \in \mathcal{V})\, (Q+L^2)(a\mathbf{v}) = a^2(Q+L^2)(\mathbf{v})$ by homogeneity of Q and linearity of L. $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\, B_{Q+L^2}(\mathbf{u}, \mathbf{v}) - B_Q(\mathbf{u}, \mathbf{v}) = 2L(\mathbf{u})L(\mathbf{v}) = 0$ by linearity of L and characteristic 2.
$B_{Q+L^2} = B_Q$ by this identity, and $Q + L^2$ is quadratic by the homogeneity identity and bilinearity of B_Q.

(ii) Define $L : \mathbb{F}^2 \to \mathbb{F}$ by $L(x, y) := x$.
$(\forall\, a, x, y \in \mathbb{F})\, Q_0(ax, ay) = a^2 Q_0(x, y)$ and $B_{Q_0} = B$ by direct expansion.
Q_0 is quadratic by this homogeneity identity and bilinearity of B.
Q_1 is quadratic and $B_{Q_1} = B_{Q_0} = B$ by $Q_1 = Q_0 + L^2$ and (i). $Q_0 \neq Q_1$ by $Q_0(1, 0) = 0 \neq 1 = Q_1(1, 0)$.

Structural Commentary: Item (i) isolates the polarization kernel; item (ii) realizes it through the coordinate square x^2.
Key Insight: In characteristic 2, linear-functional squares are invisible to polarization. □

Example 3.2.5. *Alternating versus Nonalternating Symmetric Forms*
Assume $\operatorname{char}(\mathbb{F}) = 2$ and let $H = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$. The associated symmetric bilinear form on $\mathbb{F}^2$ is $B_H((x, y), (u, v)) = xv + yu$.
The form is alternating by $(\forall\, x, y \in \mathbb{F})\, B_H((x, y), (x, y)) = 2xy = 0$. The form is nonsingular by $H^2 = I_2$ and Proposition 4.4.6.
A diagonal matrix D congruent to H would represent B_H in an ordered basis by Proposition 3.1.4(iii).
Alternation would force $D = 0$. Congruence would instead force $\operatorname{rank}(D) = \operatorname{rank}(H) = 2$ by Proposition 3.1.4(iv).
H is not congruent to any diagonal matrix by this rank contradiction.
By contrast, [1] is symmetric and nonalternating because its basis vector has self-pairing 1.
Structural Commentary: The forms represented by H and [1] separate alternating from nonalternating symmetry.
Key Insight: Alternation forces a zero diagonal, but nonsingularity forbids the zero matrix.

3.3 Orthogonality, Radicals, and Nonsingular Metric Spaces

Throughout this subsection, metric vector spaces are finite-dimensional over an arbitrary field $\mathbb{F}$.

Definition 3.3.1. *Orthogonality, Isotropy, and Radical*
Let $\mathcal{V}$ be a metric vector space. Symmetric and skew-symmetric forms satisfy $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, [\langle \mathbf{x}, \mathbf{y} \rangle = 0 \iff \langle \mathbf{y}, \mathbf{x} \rangle = 0]$.
An alternating form is skew-symmetric by Proposition 3.1.2(i); thus left and right orthogonality agree in every case.

(i) $\mathbf{x} \perp \mathbf{y} \iff \langle \mathbf{x}, \mathbf{y} \rangle = 0$; then $\mathbf{x}, \mathbf{y}$ are *orthogonal*. For $S \subseteq \mathcal{V}$, set $S^\perp := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{s} \in S)\ \langle \mathbf{v}, \mathbf{s} \rangle = 0\}$.
(ii) A subspace $\mathcal{S} \leq \mathcal{V}$ is *totally isotropic* or *totally degenerate* if $\mathcal{S} \subseteq \mathcal{S}^\perp$. It is *nonsingular* if $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$.
(iii) A vector $\mathbf{x} \neq \mathbf{0}$ is *isotropic* or *null* if $\langle \mathbf{x}, \mathbf{x} \rangle = 0$, and *nonisotropic* otherwise.
The space $\mathcal{V}$ is *isotropic* if $(\exists\, \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\})\ \langle \mathbf{x}, \mathbf{x} \rangle = 0$. It is *anisotropic* if $(\forall\, \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\})\ \langle \mathbf{x}, \mathbf{x} \rangle \neq 0$.
(iv) A vector $\mathbf{x}$ is *degenerate* if $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{v} \rangle = 0$. The *radical* is $\operatorname{rad}(\mathcal{V}) := \mathcal{V}^\perp$.
(v) The space $\mathcal{V}$ is *nonsingular* or *nondegenerate* if $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, and *singular* otherwise.
(vi) The space $\mathcal{V}$ is *totally singular* or *totally degenerate* if $\operatorname{rad}(\mathcal{V}) = \mathcal{V}$.

Proposition 3.3.2. *Metric Radical and Perfection*
Let $\mathcal{V}$ be a metric vector space with form $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$.

(i) $\operatorname{rad}_L(B) = \operatorname{rad}_R(B) = \operatorname{rad}(\mathcal{V})$.
(ii) $\mathcal{V}$ is nonsingular $\iff$ B is perfect.

Proof.

(i) The identity $\operatorname{rad}(\mathcal{V}) = \operatorname{rad}_L(B)$ holds by the definitions of $\mathcal{V}^\perp$ and the left radical.
The equality $\operatorname{rad}_L(B) = \operatorname{rad}_R(B)$ holds by Proposition 3.1.2(iv).

(ii) ($\Rightarrow$) Suppose $\mathcal{V}$ is nonsingular. Both form-induced maps are linear by Proposition 4.4.3(i).
Their kernels equal $\{\mathbf{0}\}$ by (i) and Proposition 4.4.5(i). Thus both maps are injective by Proposition 1.6.2.
Their domains and codomains have equal dimension by Theorem 4.1.4(ii).
Hence both maps are surjective by Corollary 1.7.6(ii). They are isomorphisms by Proposition 1.6.7, so B is perfect.

(ii) ($\Leftarrow$) Suppose B is perfect. Then Φ_B is an isomorphism by the definition of perfection. It is injective by Proposition 1.6.7.
Hence $\ker(\Phi_B) = \{\mathbf{0}\}$ by Proposition 1.6.2. Thus $\operatorname{rad}(\mathcal{V}) = \operatorname{rad}_L(B) = \ker(\Phi_B) = \{\mathbf{0}\}$ by (i) and Proposition 4.4.5(i).

Structural Commentary: The common radical is the kernel obstruction; equal dimensions convert injectivity into perfection.
Key Insight: A metric form is perfect exactly when its radical vanishes. □

Proposition 3.3.3. *Coordinate Criteria for Nonsingularity*
Let $\mathcal{V}$ be a metric vector space with $n := \dim(\mathcal{V}) \geq 1$. Let β be an ordered basis, and set $M := M_\beta$.

(i) $\{[\mathbf{x}]_\beta : \mathbf{x} \in \operatorname{rad}(\mathcal{V})\} = \ker(M^t)$.
(ii) $\dim \operatorname{rad}(\mathcal{V}) = n - \operatorname{rank}(M)$.
(iii) $\mathcal{V}$ is nonsingular $\iff$ M is invertible.

Proof.

(*i*) For $\mathbf{x} \in \mathcal{V}$, set $X := [\mathbf{x}]_\beta$. The relation $(\forall\, Y \in \mathbb{F}^n)(\exists\, \mathbf{y} \in \mathcal{V})\, Y = [\mathbf{y}]_\beta$ holds by Proposition 3.2.2(i).
Thus $\mathbf{x} \in \operatorname{rad}(\mathcal{V}) \iff (\forall\, Y \in \mathbb{F}^n)\, X^t M Y = 0$ by Proposition 3.1.4(i).
The equivalence $(\forall\, Y \in \mathbb{F}^n)\, X^t M Y = 0 \iff M^t X = 0$ holds by evaluation at the standard coordinate columns. QED.

(*ii*) By (i) and Proposition 3.2.2(i), $\operatorname{rad}(\mathcal{V}) \cong \ker(M^t)$. Hence their dimensions agree by Theorem 1.7.9.
Thus $\dim \operatorname{rad}(\mathcal{V}) = n - \operatorname{rank}(M^t)$ by Corollary 3.3.1(i). Finally, $\operatorname{rank}(M^t) = \operatorname{rank}(M)$ by Theorem 3.3.2.

(*iii*) ($\Rightarrow$) Suppose $\mathcal{V}$ is nonsingular. Then $\operatorname{rank}(M) = n$ by (ii). Hence M is invertible by Corollary 3.6.2.

(*iii*) ($\Leftarrow$) Suppose M is invertible. Then $\operatorname{rank}(M) = n$ by Corollary 3.6.2.
Hence $\dim \operatorname{rad}(\mathcal{V}) = 0$ by (ii). Hence $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$ by Proposition 1.4.3(ii).

Structural Commentary: Coordinate evaluation identifies the radical kernel; rank-nullity and full rank test nonsingularity.
Key Insight: The form is nonsingular exactly when its matrix has full rank. □

Remark 3.3.4. *Subspace Radicals*
For $\mathcal{S} \leq \mathcal{V}$, the restricted form has radical $\operatorname{rad}(\mathcal{S}) = \mathcal{S} \cap \mathcal{S}^\perp$. In general, $\operatorname{rad}(\mathcal{S}) \neq \mathcal{S}^\perp$ because $\mathcal{S}^\perp$ is computed in $\mathcal{V}$.

Proposition 3.3.5. *Riesz Map Isomorphism Criterion*
Let $\mathcal{V}$ be a metric vector space and define $\tau : \mathcal{V} \to \mathcal{V}'$ by $\tau(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$.

(*i*) $\tau \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$.
(*ii*) $\ker(\tau) = \operatorname{rad}(\mathcal{V})$.
(*iii*) *τ is an isomorphism* $\iff \operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.

Proof. Set $B(\mathbf{x}, \mathbf{y}) := \langle \mathbf{x}, \mathbf{y} \rangle$.

(*i*) The identity $\tau = \Psi_B \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$ holds by Proposition 4.4.3(i).

(*ii*) The identity $\ker(\tau) = \operatorname{rad}_R(B)$ holds by Proposition 4.4.5(i). Hence $\ker(\tau) = \operatorname{rad}(\mathcal{V})$ by Proposition 3.3.2(i).

(*iii*) ($\Rightarrow$) If τ is an isomorphism, then it is injective by Proposition 1.6.7.
Thus $\ker(\tau) = \{\mathbf{0}\}$ by Proposition 1.6.2. Hence $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$ by (ii).

(*iii*) ($\Leftarrow$) If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, then $\ker(\tau) = \{\mathbf{0}\}$ by (ii). Thus τ is injective by Proposition 1.6.2.
The equality $\dim(\mathcal{V}) = \dim(\mathcal{V}')$ holds by Theorem 4.1.4(ii).
Thus τ is surjective by Corollary 1.7.6(ii) and is an isomorphism by Proposition 1.6.7.

Structural Commentary: The radical is the Riesz-map kernel; equal dimensions convert injectivity into invertibility.
Key Insight: The radical is precisely the kernel of the Riesz map. □

Theorem 3.3.6. *Riesz Map for Nonsingular Metric Spaces*
Let $\mathcal{V}$ be a nonsingular metric vector space. Then $(\forall\, f \in \mathcal{V}')(\exists!\, \mathbf{x} \in \mathcal{V})(\forall\, \mathbf{v} \in \mathcal{V})\, f(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$.

Proof. Define $\tau : \mathcal{V} \to \mathcal{V}'$ by $\tau(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$. The map τ is an isomorphism by Proposition 3.3.5(iii). Fix $f \in \mathcal{V}'$ and set $\mathbf{x} := \tau^{-1}(f)$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\, f(\mathbf{v}) = \tau(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$. If $\mathbf{y} \in \mathcal{V}$ satisfies $(\forall\, \mathbf{v} \in \mathcal{V})\, f(\mathbf{v}) = \langle \mathbf{v}, \mathbf{y} \rangle$, then $\tau(\mathbf{y}) = f = \tau(\mathbf{x})$; $\mathbf{y} = \mathbf{x}$ by injectivity of τ.
Structural Commentary: Riesz inversion constructs each representative; injectivity proves its uniqueness.
Key Insight: The radical is the sole obstruction to representing every functional. □

Definition 3.3.7. *Isometries and Classical Groups*
Let $\mathcal{V}, \mathcal{W}$ be metric vector spaces.

(i) An *isometry* $\sigma : \mathcal{V} \to \mathcal{W}$ is a bijective linear map satisfying $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle \sigma\mathbf{x}, \sigma\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$. Write $\mathcal{V} \cong \mathcal{W}$ if one exists.

(ii) The group of isometries $\mathcal{V} \to \mathcal{V}$ is $\operatorname{Iso}(\mathcal{V})$.
For nonsingular orthogonal $\mathcal{V}$, this is the *orthogonal group* $\mathcal{O}(\mathcal{V})$. For nonsingular symplectic $\mathcal{V}$, it is the *symplectic group* $\operatorname{Sp}(\mathcal{V})$.

Proposition 3.3.8. *Matrix Criterion for Isometries*
Let $\mathcal{V}$ be a metric vector space with $n := \dim(\mathcal{V}) \geq 1$, and let β be an ordered basis. Let $\sigma \in \mathcal{L}(\mathcal{V})$, and set $M := M_\beta$ and $A := [\sigma]_{\beta \leftarrow \beta}$.

(*i*) *$\sigma \in \operatorname{Iso}(\mathcal{V}) \iff \sigma$ is bijective and $A^t M A = M$.*
(*ii*) *If $\sigma \in \operatorname{Iso}(\mathcal{V})$ and M is invertible, then $\det A = \pm 1$.*

Proof.

(*i*) ($\Rightarrow$) Suppose $\sigma \in \operatorname{Iso}(\mathcal{V})$. Then σ is bijective by definition.
The identity $(\forall\, X, Y \in \mathbb{F}^n)\, X^t A^t M A Y = X^t M Y$ holds by metric preservation and Proposition 3.1.4(i). Hence $A^t M A = M$ by evaluation at pairs of standard coordinate columns.

(*i*) ($\Leftarrow$) Suppose σ is bijective and $A^t M A = M$.
Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle \sigma\mathbf{x}, \sigma\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ by Proposition 3.1.4(i).
Hence $\sigma \in \operatorname{Iso}(\mathcal{V})$ by definition.

(*ii*) Suppose $\sigma \in \operatorname{Iso}(\mathcal{V})$ and M is invertible. Then $A^t M A = M$ by (i).
Thus $\det(M) = \det(A^t M A) = (\det A)^2 \det(M)$ by determinant multiplicativity and transpose invariance.
Thus $(\det A)^2 = 1$ by cancellation of $\det(M) \neq 0$.
Hence $(\det A - 1)(\det A + 1) = 0$, so $\det A \in \{1, -1\}$ by the no-zero-divisor property of $\mathbb{F}$.

Structural Commentary: Coordinate evaluation identifies metric preservation with congruence; determinants constrain its stabilizer.
Key Insight: Isometries are exactly the invertible solutions of $A^t M A = M$. □

Definition 3.3.9. *Adjoint for a Nonsingular Bilinear Form*
Let $\mathcal{V}$ be nonsingular and $T \in \mathcal{L}(\mathcal{V})$. An operator $S \in \mathcal{L}(\mathcal{V})$ is a *metric adjoint* of $T \iff (\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, S\mathbf{y} \rangle$.
If this operator is unique, denote it by T^*.

Proposition 3.3.10. *Metric Adjoints and Isometries*
Let $\mathcal{V}$ be nonsingular with form $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$. Let $T \in \mathcal{L}(\mathcal{V})$, and set $n := \dim(\mathcal{V})$.

(*i*) *$(\exists!\, T^* \in \mathcal{L}(\mathcal{V}))(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, T^*\mathbf{y} \rangle$.*
(*ii*) *If $n \geq 1$ and β is an ordered basis, set $M := M_\beta$ and $A := [T]_\beta$; then $[T^*]_\beta = M^{-1} A^t M$.*
(*iii*) *T is an isometry $\iff T$ is bijective and $T^* T = I$.*
(*iv*) *If T is bijective, then $T^* T = I \iff T^* = T^{-1}$.*

Proof.

(*i*) Define $\tau : \mathcal{V} \to \mathcal{V}'$ by $\tau(\mathbf{y})(\mathbf{x}) = \langle \mathbf{x}, \mathbf{y} \rangle$. Define $F : \mathcal{V} \to \mathcal{V}'$ by $F(\mathbf{y})(\mathbf{x}) = \langle T\mathbf{x}, \mathbf{y} \rangle$.
Then τ is an isomorphism by Proposition 3.3.5(iii). The relation $F \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$ holds by bilinearity.
Set $T^* := \tau^{-1} \circ F \in \mathcal{L}(\mathcal{V})$. Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, T^*\mathbf{y} \rangle$ by the definitions of τ and F.
If $S \in \mathcal{L}(\mathcal{V})$ satisfies this identity, then $\tau \circ S = F = \tau \circ T^*$. Hence $S = T^*$ by injectivity of τ.

(*ii*) Assume $n \geq 1$. The matrix M is invertible by Proposition 3.3.3(iii).
Set $D := [T^*]_\beta$. Then $(\forall\, X, Y \in \mathbb{F}^n)\ X^t A^t M Y = X^t M D Y$ by the adjoint identity and Proposition 3.1.4(i).
Hence $A^t M = MD$ by evaluation at pairs of standard coordinate columns, and therefore $D = M^{-1} A^t M$.

(*iii*) ($\Rightarrow$) Suppose T is an isometry. Then T is bijective by definition.
For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, $\langle \mathbf{x}, (T^*T - I)\mathbf{y} \rangle = \langle T\mathbf{x}, T\mathbf{y} \rangle - \langle \mathbf{x}, \mathbf{y} \rangle$ by the adjoint identity.
This difference is 0 by metric preservation, so $(T^*T - I)\mathbf{y} \in \operatorname{rad}_R(B) = \{\mathbf{0}\}$ by Proposition 3.3.2(i); hence $T^*T = I$.

(*iii*) ($\Leftarrow$) Suppose T is bijective and $T^*T = I$.
Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, T\mathbf{y} \rangle = \langle \mathbf{x}, T^*T\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ by the adjoint identity. Hence T is an isometry by definition.

(*iv*) ($\Rightarrow$) If $T^*T = I$, then $T^* = T^{-1}$ by right-composition with T^{-1}.

(*iv*) ($\Leftarrow$) If $T^* = T^{-1}$, then $T^*T = T^{-1}T = I$.

Structural Commentary: Riesz inversion constructs the adjoint; the radical test converts metric preservation into $T^*T = I$.
Key Insight: Nonsingularity turns the adjoint relation into a unique operator identity. □

Proposition 3.3.11. *Isometry Is Matrix Congruence*
Let $(\mathcal{V}, B_\mathcal{V})$ and $(\mathcal{W}, B_\mathcal{W})$ be metric vector spaces with $\dim(\mathcal{V}) = \dim(\mathcal{W}) = n \geq 1$.
Let β, γ be ordered bases with respective form matrices M_β, N_γ. Then $\mathcal{V} \cong \mathcal{W} \iff M_\beta$ and N_γ are congruent.

Proof.

($\Rightarrow$) Let $\sigma : \mathcal{V} \to \mathcal{W}$ be an isometry and set $A := [\sigma]_{\gamma \leftarrow \beta}$. The map σ is invertible by Proposition 1.6.7.
By Proposition 3.4.4, $[\sigma^{-1}]_{\beta \leftarrow \gamma} A = I_n = A[\sigma^{-1}]_{\beta \leftarrow \gamma}$, so A is invertible.
The identity $(\forall\, X, Y \in \mathbb{F}^n)\ X^t M_\beta Y = X^t A^t N_\gamma A Y$ holds by metric preservation and Proposition 3.1.4(i).
Hence $M_\beta = A^t N_\gamma A$ by evaluation at pairs of standard coordinate columns, so the matrices are congruent.

($\Leftarrow$) Suppose M_β and N_γ are congruent. Fix an invertible $P \in \operatorname{Mat}(n, n, \mathbb{F})$ such that $N_\gamma = P^t M_\beta P$, and set $A := P^{-1}$.
Then $M_\beta = A^t N_\gamma A$. Define $\sigma : \mathcal{V} \to \mathcal{W}$ by $[\sigma\mathbf{x}]_\gamma = A[\mathbf{x}]_\beta$. The map σ is linear by Proposition 3.2.2(i) and Proposition 3.1.8.
The coordinate rule with A^{-1} defines its inverse, so σ is bijective.
For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$ with coordinate columns X, Y, $B_\mathcal{W}(\sigma\mathbf{x}, \sigma\mathbf{y}) = X^t A^t N_\gamma A Y$ by Proposition 3.1.4(i).
The identity $B_\mathcal{V}(\mathbf{x}, \mathbf{y}) = X^t M_\beta Y = X^t A^t N_\gamma A Y$ holds by Proposition 3.1.4(i) and the congruence identity; σ is an isometry.

Structural Commentary: Coordinate matrices translate metric preservation into congruence and reconstruct the isometry.
Key Insight: Isometry classes are exactly congruence classes of form matrices. □

Perspective 3.3.12. *Congruence versus Similarity*
Similarity $N = P^{-1}MP$ governs operator matrices. Congruence $N = P^tMP$ governs bilinear-form matrices because the same basis change acts in both slots. Over $\mathbb{R}$, transpose equals conjugate transpose, so real form classification uses congruence.

3.4 Radicals, Hyperbolic Spaces, and Witt Theory

Throughout this subsection, metric vector spaces are finite-dimensional over a fixed field $\mathbb{F}$.

Definition 3.4.1. *Orthogonal Direct Sum*
For $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$, write $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$ when $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{S} \perp \mathcal{T}$; this is the *orthogonal direct sum* of $\mathcal{S}$ and $\mathcal{T}$.

Theorem 3.4.2. *Dimensions of Orthogonal Complements*
Let $\mathcal{S} \leq \mathcal{V}$.

(i) If $\mathcal{V}$ or $\mathcal{S}$ is nonsingular, then $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = \dim(\mathcal{V})$.
(ii) If $\mathcal{S}$ is nonsingular, then $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^\perp$.
(iii) If $\mathcal{V}$ is nonsingular, then $\mathcal{S}^{\perp\perp} = \mathcal{S}$.
(iv) If $\mathcal{V}$ is nonsingular, then $\mathcal{S}$ is nonsingular $\iff \mathcal{S}^\perp$ is nonsingular.

Proof.

(*i*) Define $\rho : \mathcal{V} \to \mathcal{S}'$ by $\rho(\mathbf{x})(\mathbf{s}) = \langle \mathbf{s}, \mathbf{x} \rangle$. Then $\rho \in \mathcal{L}(\mathcal{V}, \mathcal{S}')$ and $\ker(\rho) = \mathcal{S}^\perp$ by bilinearity and the definition of $\mathcal{S}^\perp$.
Fix $f \in \mathcal{S}'$. If $\mathcal{V}$ is nonsingular, choose $\mathcal{V} = \mathcal{S} \oplus \mathcal{C}$ by Proposition 2.3.6 and define $\widetilde{f}(\mathbf{s} + \mathbf{c}) = f(\mathbf{s})$.
Then $\widetilde{f} \in \mathcal{V}'$ by the direct-sum decomposition. Fix $\mathbf{x} \in \mathcal{V}$ with $(\forall\, \mathbf{s} \in \mathcal{S})\ f(\mathbf{s}) = \widetilde{f}(\mathbf{s}) = \langle \mathbf{s}, \mathbf{x} \rangle$ by Theorem 3.3.6.
If $\mathcal{S}$ is nonsingular, fix $\mathbf{x} \in \mathcal{S}$ with $(\forall\, \mathbf{s} \in \mathcal{S})\ f(\mathbf{s}) = \langle \mathbf{s}, \mathbf{x} \rangle$ by Theorem 3.3.6 for $\mathcal{S}$. Thus ρ is surjective in either case.
The identity $\dim(\mathcal{V}) = \dim(\ker \rho) + \dim(\mathcal{S}') = \dim(\mathcal{S}^\perp) + \dim(\mathcal{S})$ holds by Theorem 1.7.3 and Theorem 4.1.4(ii).

(*ii*) If $\mathcal{S}$ is nonsingular, then $\mathcal{S} \cap \mathcal{S}^\perp = \operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$. Thus $\mathcal{V} = \mathcal{S} \oplus \mathcal{S}^\perp$ by (i), and $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^\perp$ by $\mathcal{S} \perp \mathcal{S}^\perp$.

(*iii*) Suppose that $\mathcal{V}$ is nonsingular. Then $\mathcal{S} \subseteq \mathcal{S}^{\perp\perp}$ by orthogonality.
The equality $\dim(\mathcal{S}^{\perp\perp}) = \dim(\mathcal{V}) - \dim(\mathcal{S}^\perp) = \dim(\mathcal{S})$ holds by (i). Hence $\mathcal{S}^{\perp\perp} = \mathcal{S}$ by Proposition 1.4.3(ii).

(*iv*) ($\Rightarrow$) If $\mathcal{S}$ is nonsingular, then $\operatorname{rad}(\mathcal{S}^\perp) = \mathcal{S}^\perp \cap \mathcal{S}^{\perp\perp} = \mathcal{S}^\perp \cap \mathcal{S} = \{\mathbf{0}\}$ by (iii).

(*iv*) ($\Leftarrow$) If $\mathcal{S}^\perp$ is nonsingular, then $\operatorname{rad}(\mathcal{S}) = \mathcal{S} \cap \mathcal{S}^\perp = \mathcal{S}^{\perp\perp} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$ by (iii).

Structural Commentary: The restriction map has kernel $\mathcal{S}^\perp$; rank–nullity and double complementation settle (i)–(iv).
Key Insight: Nonsingularity makes orthogonal complementation dimension-reversing and involutive. □

Theorem 3.4.3. *Splitting off the Radical*
Let $R = \operatorname{rad}(\mathcal{V})$. Then $\exists\, \mathcal{S} \leq \mathcal{V}$ such that $\mathcal{V} = R \odot \mathcal{S}$, $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, and R is totally singular.

Proof. Choose $\mathcal{S} \leq \mathcal{V}$ with $\mathcal{V} = R \oplus \mathcal{S}$ by Proposition 2.3.6. Since $R \perp \mathcal{V}$, $\mathcal{V} = R \odot \mathcal{S}$ and the form on R vanishes, so $\operatorname{rad}(R) = R$.
If $\mathbf{s} \in \operatorname{rad}(\mathcal{S})$, then $\mathbf{s} \perp \mathcal{S}$ and $\mathbf{s} \perp R$, whence $\mathbf{s} \in \operatorname{rad}(\mathcal{V}) \cap \mathcal{S} = R \cap \mathcal{S} = \{\mathbf{0}\}$. Thus $\mathcal{S}$ is nonsingular.
Structural Commentary: A linear complement is automatically orthogonal to the radical and has zero restricted radical.
Key Insight: All degeneracy lies in a zero-form orthogonal summand. □

Proposition 3.4.4. *Metric Quotients*
Let $\mathcal{U} \leq \mathcal{V}$ and consider $\langle \mathbf{x}+\mathcal{U}, \mathbf{y}+\mathcal{U}\rangle_{\mathcal{V}/\mathcal{U}} := \langle \mathbf{x}, \mathbf{y}\rangle$.

(i) The rule defines a metric form on $\mathcal{V}/\mathcal{U} \iff \mathcal{U} \leq \operatorname{rad}(\mathcal{V})$.
(ii) If $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$, then $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\mathcal{U}$.
(iii) The metric space $\mathcal{V}/\operatorname{rad}(\mathcal{V})$ is nonsingular.

Proof.

(i) ($\Rightarrow$) If the rule defines a metric form, then $(\forall\, \mathbf{u} \in \mathcal{U})(\forall\, \mathbf{y} \in \mathcal{V})\, 0 = \langle \mathbf{u}+\mathcal{U}, \mathbf{y}+\mathcal{U}\rangle_{\mathcal{V}/\mathcal{U}} = \langle \mathbf{u}, \mathbf{y}\rangle$, so $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$.
(i) ($\Leftarrow$) Suppose that $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$. For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, $\langle \mathbf{x}+\mathbf{u}, \mathbf{y}+\mathbf{v}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$ by Proposition 3.3.2(i) and bilinearity. Thus the rule is representative-independent; bilinearity and the form type descend from $\mathcal{V}$.
(ii) Suppose that $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$. Fix $\mathbf{x} \in \mathcal{V}$. $\mathbf{x}+\mathcal{U} \in \operatorname{rad}(\mathcal{V}/\mathcal{U}) \iff (\forall\, \mathbf{y} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{y}\rangle = 0 \iff \mathbf{x} \in \operatorname{rad}(\mathcal{V})$. Hence $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\mathcal{U}$.
(iii) With $\mathcal{U} = \operatorname{rad}(\mathcal{V})$, item (ii) becomes $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.

Structural Commentary: Item (i) tests representatives; item (ii) identifies the quotient radical; item (iii) removes it.
Key Insight: Precisely radical directions may be collapsed without changing pairings. □

Proposition 3.4.5. *Radicals and Orthogonal Direct Sums*
Suppose that $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, and put $R_{\mathcal{X}} := \operatorname{rad}(\mathcal{X})$ for $\mathcal{X} \in \{\mathcal{V}, \mathcal{S}, \mathcal{T}\}$.

(i) $R_{\mathcal{V}} = R_{\mathcal{S}} \odot R_{\mathcal{T}}$.
(ii) $\mathcal{V}/R_{\mathcal{V}} \cong (\mathcal{S}/R_{\mathcal{S}}) \odot (\mathcal{T}/R_{\mathcal{T}})$.
(iii) $\mathcal{V}$ is nonsingular $\iff$ $\mathcal{S}$ and $\mathcal{T}$ are nonsingular.

Proof.

(i) For $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$, $\mathbf{s}+\mathbf{t} \in R_{\mathcal{V}} \iff \mathbf{s} \in R_{\mathcal{S}} \wedge \mathbf{t} \in R_{\mathcal{T}}$ by $\mathcal{S} \perp \mathcal{T}$.
Uniqueness and orthogonality of the decomposition establish $R_{\mathcal{V}} = R_{\mathcal{S}} \odot R_{\mathcal{T}}$.
(ii) Define $\Phi((\mathbf{s}+\mathbf{t}) + R_{\mathcal{V}}) = (\mathbf{s}+R_{\mathcal{S}}, \mathbf{t}+R_{\mathcal{T}})$.
The map is well-defined by (i), with inverse $\Psi(\mathbf{s}+R_{\mathcal{S}}, \mathbf{t}+R_{\mathcal{T}}) = (\mathbf{s}+\mathbf{t}) + R_{\mathcal{V}}$, and hence is a linear bijection.
For $\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}$ and $\mathbf{t}_1, \mathbf{t}_2 \in \mathcal{T}$, $\langle \mathbf{s}_1+\mathbf{t}_1, \mathbf{s}_2+\mathbf{t}_2\rangle = \langle \mathbf{s}_1, \mathbf{s}_2\rangle + \langle \mathbf{t}_1, \mathbf{t}_2\rangle$ by orthogonality. Thus Φ is an isometry.
(iii) ($\Rightarrow$) If $\mathcal{V}$ is nonsingular, then $R_{\mathcal{S}} \odot R_{\mathcal{T}} = \{\mathbf{0}\}$ by (i), so $\mathcal{S}$ and $\mathcal{T}$ are nonsingular.
(iii) ($\Leftarrow$) If $\mathcal{S}$ and $\mathcal{T}$ are nonsingular, then $R_{\mathcal{V}} = R_{\mathcal{S}} \odot R_{\mathcal{T}} = \{\mathbf{0}\}$ by (i).

Structural Commentary: Item (i) splits the radical; item (ii) descends the splitting; item (iii) tests nonsingularity.
Key Insight: Orthogonality separates both pairings and radical directions. □

Corollary 3.4.6. *Recovering a Singular Space from Its Regular Quotient*
If $\dim(\mathcal{V}) = \dim(\mathcal{W})$ and $\mathcal{V}/\operatorname{rad}(\mathcal{V}) \cong \mathcal{W}/\operatorname{rad}(\mathcal{W})$, then $\mathcal{V} \cong \mathcal{W}$.

Proof.

(i) *Splitting*: Choose $\mathcal{V} = \operatorname{rad}(\mathcal{V}) \odot \mathcal{S}$ and $\mathcal{W} = \operatorname{rad}(\mathcal{W}) \odot \mathcal{T}$ with $\mathcal{S}, \mathcal{T}$ nonsingular by Theorem 3.4.3.
(ii) *Regular Parts*: The isometries $\mathcal{V}/\operatorname{rad}(\mathcal{V}) \cong \mathcal{S}$ and $\mathcal{W}/\operatorname{rad}(\mathcal{W}) \cong \mathcal{T}$ hold by Proposition 3.4.5(ii).
Hence $\mathcal{S} \cong \mathcal{T}$ by the assumed quotient isometry.
(iii) *Radicals*: The ambient and regular-part dimension equalities imply $\dim \operatorname{rad}(\mathcal{V}) = \dim \operatorname{rad}(\mathcal{W})$.
Fix a vector-space isomorphism between the radicals by Theorem 1.7.9; it is an isometry because both restricted forms vanish.
(iv) *Assembly*: The orthogonal direct sum of the radical isometry and an isometry $\mathcal{S} \to \mathcal{T}$ is an isometry $\mathcal{V} \to \mathcal{W}$.

Structural Commentary: Radical splittings reduce the classification to regular-part isometry and radical dimension.
Key Insight: A zero-form radical is classified solely by dimension. □

Definition 3.4.7. *Hyperbolic Pairs and Hyperbolic Spaces*

(i) A *hyperbolic pair* is $(\mathbf{u}, \mathbf{v})$ with $\langle \mathbf{u}, \mathbf{u}\rangle = \langle \mathbf{v}, \mathbf{v}\rangle = 0$ and $\langle \mathbf{u}, \mathbf{v}\rangle = 1$.
(ii) The plane $\operatorname{span}(\mathbf{u}, \mathbf{v})$ of a hyperbolic pair is a *hyperbolic plane*.
(iii) A *hyperbolic space* is an orthogonal direct sum of hyperbolic planes.
(iv) A basis $(\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_k, \mathbf{v}_k)$ is a *hyperbolic basis* if each $(\mathbf{u}_i, \mathbf{v}_i)$ is hyperbolic and the planes $\operatorname{span}(\mathbf{u}_i, \mathbf{v}_i)$ are mutually orthogonal. In the symplectic case it is a *symplectic basis*.

Theorem 3.4.8. *Capturing Isotropic Vectors*
Let $\mathcal{V}$ be nonsingular and either symplectic, or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
Then $(\forall\, \mathbf{u} \in \mathcal{V} \setminus \{\mathbf{0}\})(\langle \mathbf{u}, \mathbf{u}\rangle = 0 \Longrightarrow (\exists\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{u}, \mathbf{v}\rangle = 1 \wedge \langle \mathbf{v}, \mathbf{v}\rangle = 0)$.

Proof. Fix $\mathbf{u} \neq \mathbf{0}$ with $\langle \mathbf{u}, \mathbf{u}\rangle = 0$. Since $\mathbf{u} \notin \operatorname{rad}(\mathcal{V})$, fix $\mathbf{x} \in \mathcal{V}$ with $c := \langle \mathbf{u}, \mathbf{x}\rangle \neq 0$.
If $\mathcal{V}$ is symplectic, set $\mathbf{v} = c^{-1}\mathbf{x}$; then $\langle \mathbf{u}, \mathbf{v}\rangle = 1$ and $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ by alternation. If $\mathcal{V}$ is orthogonal, set $\mathbf{v} = c^{-1}\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{x}\rangle}{2c^2}\mathbf{u}$.
Then $\langle \mathbf{u}, \mathbf{v}\rangle = 1$ and $\langle \mathbf{v}, \mathbf{v}\rangle = c^{-2}\langle \mathbf{x}, \mathbf{x}\rangle - c^{-2}\langle \mathbf{x}, \mathbf{x}\rangle = 0$ by symmetry and $\langle \mathbf{u}, \mathbf{u}\rangle = 0$. Thus $(\mathbf{u}, \mathbf{v})$ is hyperbolic.
Structural Commentary: A nonorthogonal partner is rescaled, then corrected by $\mathbf{u}$ only in the orthogonal case.
Key Insight: The correction cancels self-pairing without changing the cross-pairing. □

Theorem 3.4.9. *Nonsingular Completions*
Let $\mathcal{U} \leq \mathcal{V}$, where $\mathcal{V}$ is nonsingular and either symplectic, or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
Write $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_r) \odot \mathcal{W}$, where $(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ is a basis of $\operatorname{rad}(\mathcal{U})$ and $\mathcal{W}$ is nonsingular.

(i) $(\exists\, \mathbf{v}_1, \ldots, \mathbf{v}_r \in \mathcal{V})$ such that $\mathcal{H}_r = \operatorname{span}(\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_r, \mathbf{v}_r)$ is hyperbolic and $\widehat{\mathcal{U}} = \mathcal{H}_r \odot \mathcal{W}$ contains $\mathcal{U}$ with $\dim(\widehat{\mathcal{U}}) = \dim(\mathcal{U}) + r$.
(ii) If $\mathcal{U} \leq \mathcal{X} \leq \mathcal{V}$ and $\mathcal{X}$ is nonsingular, then $\dim(\mathcal{X}) \geq \dim(\mathcal{U}) + r$; equality is attained, and all equality cases are isometric.
(iii) Let $\widetilde{\mathcal{V}}$ be nonsingular of same kind, $\mathcal{T} \leq \widetilde{\mathcal{V}}$, and $\sigma : \mathcal{U} \xrightarrow{\sim} \mathcal{T}$. Compatible equality-case extensions admit an isometry extending σ.

Proof.

(*i*) The decomposition $\mathcal{V} = \mathcal{W} \odot \mathcal{W}^\perp$ holds by Theorem 3.4.2(ii), and $(\forall\, i \in \{1,\dots,r\})\, \mathbf{u}_i \in \mathcal{W}^\perp$.
Set $\mathcal{H}_0 = \{\mathbf{0}\}$ and proceed by induction on $j \in \{1,\dots,r\}$. Given $\mathcal{H}_{j-1}$, set $\mathcal{S}_j = \operatorname{span}(\mathbf{u}_{j+1},\dots,\mathbf{u}_r) \odot \mathcal{W} \odot \mathcal{H}_{j-1}$.
In a relation $\mathbf{u}_j \in \mathcal{S}_j$, pairing with $\mathbf{u}_1,\dots,\mathbf{u}_{j-1}$ forces all $\mathbf{v}_i$-coefficients to vanish.
Independence of $\mathbf{u}_1,\dots,\mathbf{u}_r$ modulo $\mathcal{W}$ then contradicts the relation.
Hence $\mathbf{u}_j \notin \mathcal{S}_j$. The equality $\mathcal{S}_j^{\perp\perp} = \mathcal{S}_j$ holds by Theorem 3.4.2(iii). Since $\mathbf{u}_j \notin \mathcal{S}_j^{\perp\perp}$, fix $\mathbf{x} \in \mathcal{S}_j^\perp$ with $c := \langle \mathbf{u}_j, \mathbf{x}\rangle \neq 0$.
The inclusion $\mathbf{u}_j \in \mathcal{S}_j^\perp$ holds by the radical property and the preceding stages.
Set $\mathbf{v}_j = c^{-1}\mathbf{x}$ in the symplectic case and $\mathbf{v}_j = c^{-1}\mathbf{x} - \frac{\langle \mathbf{x},\mathbf{x}\rangle}{2c^2}\mathbf{u}_j$ in the orthogonal case.
Then $\mathbf{v}_j \in \mathcal{S}_j^\perp$, $\langle \mathbf{u}_j, \mathbf{v}_j\rangle = 1$, and $\langle \mathbf{v}_j, \mathbf{v}_j\rangle = 0$ by alternation or the correction calculation. Put $\mathcal{H}_j = \mathcal{H}_{j-1} \odot \operatorname{span}(\mathbf{u}_j, \mathbf{v}_j)$.
After r stages, $\mathcal{H}_r$ is hyperbolic, $\mathcal{U} \leq \widehat{\mathcal{U}} := \mathcal{H}_r \odot \mathcal{W}$, and $\dim(\widehat{\mathcal{U}}) = 2r + \dim(\mathcal{W}) = \dim(\mathcal{U}) + r$.

(*ii*) Choose $\mathcal{X} = \mathcal{U} \oplus \mathcal{Y}$ by Proposition 2.3.6.
The pairing $\operatorname{rad}(\mathcal{U}) \times \mathcal{Y} \to \mathbb{F}$ has zero left radical: any vector in it pairs trivially with $\mathcal{U} \oplus \mathcal{Y} = \mathcal{X}$ and therefore lies in $\operatorname{rad}(\mathcal{X}) = \{\mathbf{0}\}$.
Its matrix has rank r by Proposition 4.4.5(iii), so $\dim(\mathcal{Y}) \geq r$ and $\dim(\mathcal{X}) = \dim(\mathcal{U}) + \dim(\mathcal{Y}) \geq \dim(\mathcal{U}) + r$.
Equality is attained by (i). If equality holds, the construction in (i), performed inside $\mathcal{X}$, produces $\widehat{\mathcal{U}} \leq \mathcal{X}$ with $\dim(\widehat{\mathcal{U}}) = \dim(\mathcal{X})$.
Hence $\widehat{\mathcal{U}} = \mathcal{X}$ by Proposition 1.4.3(ii). Every equality-case extension is therefore isometric to $\mathcal{W} \odot \mathcal{H}_r$.

(*iii*) Choose source partners $\mathbf{v}_1,\dots,\mathbf{v}_r$ by (i). Put $\mathbf{u}_i' = \sigma\mathbf{u}_i$ and $\mathcal{W}' = \sigma\mathcal{W}$, so $\mathcal{T} = \operatorname{span}(\mathbf{u}_1',\dots,\mathbf{u}_r') \odot \mathcal{W}'$.
Choose target partners $\mathbf{v}_1',\dots,\mathbf{v}_r'$ by (i) in $\widetilde{\mathcal{V}}$, and extend σ by $(\forall\, i \in \{1,\dots,r\})\, \mathbf{v}_i \mapsto \mathbf{v}_i'$.
This basis-to-basis map preserves the form on the hyperbolic summands, on $\mathcal{W} \to \mathcal{W}'$, and across the orthogonal summands.
It is the required isometry between equality-case extensions.

Structural Commentary: Item (i) pairs radical directions inductively; item (ii) proves minimality; item (iii) transports partners.
Key Insight: Each radical basis vector requires exactly one hyperbolic partner. □

Definition 3.4.10. *Witt Extension and Cancellation Properties*
Let $\mathcal{V} \cong \mathcal{W}$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
Write $\tau : \mathcal{A} \xrightarrow{\sim} \mathcal{B}$ when τ is an isometry. The pair $(\mathcal{V}, \mathcal{W})$ has:

(i) the *Witt extension property* if $(\forall\, \mathcal{S} \leq \mathcal{V})(\forall\, \mathcal{T} \leq \mathcal{W})(\forall\, \tau : \mathcal{S} \xrightarrow{\sim} \mathcal{T})(\exists\, \overline{\tau} : \mathcal{V} \xrightarrow{\sim} \mathcal{W})\, \overline{\tau}|_{\mathcal{S}} = \tau$;
(ii) the *Witt cancellation property* if $(\forall\, \mathcal{S} \leq \mathcal{V})(\forall\, \mathcal{T} \leq \mathcal{W})\, [\operatorname{rad}(\mathcal{S}) = \operatorname{rad}(\mathcal{T}) = \{\mathbf{0}\} \wedge \mathcal{S} \cong \mathcal{T} \Longrightarrow \mathcal{S}^\perp \cong \mathcal{T}^\perp]$.

Proposition 3.4.11. *Witt Extension–Cancellation Bridge*
Let $\mathcal{V} \cong \mathcal{W}$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
Then $(\mathcal{V}, \mathcal{W})$ has the Witt extension property $\Longleftrightarrow$ it has the Witt cancellation property.

Proof.

($\Rightarrow$) Let $\mathcal{S} \leq \mathcal{V}$ and $\mathcal{T} \leq \mathcal{W}$ be nonsingular, fix $\sigma : \mathcal{S} \xrightarrow{\sim} \mathcal{T}$. Decompositions $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^\perp$ and $\mathcal{W} = \mathcal{T} \odot \mathcal{T}^\perp$ hold by Theorem 3.4.2(ii).
Fix an extension $\widehat{\sigma} : \mathcal{V} \xrightarrow{\sim} \mathcal{W}$ by Witt extension property. Then $\widehat{\sigma}(\mathcal{S}^\perp) = \mathcal{T}^\perp$ by form preservation and bijectivity, so $\mathcal{S}^\perp \cong \mathcal{T}^\perp$.

($\Leftarrow$) Fix $\sigma : \mathcal{S} \xrightarrow{\sim} \mathcal{T}$ for arbitrary $\mathcal{S} \leq \mathcal{V}$ and $\mathcal{T} \leq \mathcal{W}$.
Choose equality-case nonsingular extensions $\widehat{\mathcal{S}} \supseteq \mathcal{S}$ and $\widehat{\mathcal{T}} \supseteq \mathcal{T}$ and an extension $\widehat{\sigma} : \widehat{\mathcal{S}} \xrightarrow{\sim} \widehat{\mathcal{T}}$ by Theorem 3.4.9(iii).
The decompositions $\mathcal{V} = \widehat{\mathcal{S}} \odot \widehat{\mathcal{S}}^\perp$ and $\mathcal{W} = \widehat{\mathcal{T}} \odot \widehat{\mathcal{T}}^\perp$ hold by Theorem 3.4.2(ii).
Fix $\eta : \widehat{\mathcal{S}}^\perp \xrightarrow{\sim} \widehat{\mathcal{T}}^\perp$ by the Witt cancellation property. Then $\widehat{\sigma} \odot \eta : \mathcal{V} \xrightarrow{\sim} \mathcal{W}$ extends σ.

Structural Commentary: Extension identifies complements; cancellation pairs the complements of compatible completions.
Key Insight: Nonsingular completion converts arbitrary extension into orthogonal cancellation. □

Theorem 3.4.12. *Witt Extension Theorem*
Let $\mathcal{V} \cong \mathcal{W}$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
If $\mathcal{S} \leq \mathcal{V}$, $\mathcal{T} \leq \mathcal{W}$, and $\sigma : \mathcal{S} \xrightarrow{\sim} \mathcal{T}$, then $(\exists\, \widehat{\sigma} : \mathcal{V} \xrightarrow{\sim} \mathcal{W})\, \widehat{\sigma}|_{\mathcal{S}} = \sigma$.

Proof.

(i) *Symplectic Preparation*: Any two nonsingular symplectic spaces of equal dimension are isometric.
In dimension zero, the unique linear map is an isometry.
At positive dimension, a nonzero vector belongs to a hyperbolic plane by Theorem 3.4.8. The plane splits off by Theorem 3.4.2(ii).
Its orthogonal complement is nonsingular by Theorem 3.4.2(iv).
Induction constructs a hyperbolic basis, and matching hyperbolic bases defines the required isometry.

(ii) *Orthogonal Line Extension*: Let $\mathcal{V}, \mathcal{W}$ be orthogonal, fix $\theta : \mathcal{V} \xrightarrow{\sim} \mathcal{W}$, and suppose $\langle \mathbf{s}, \mathbf{s}\rangle = \langle \mathbf{t}, \mathbf{t}\rangle \neq 0$.
Put $\mathbf{x} = \theta\mathbf{s}$, $\mathbf{y} = \mathbf{t}$, and $q(\mathbf{z}) = \langle \mathbf{z}, \mathbf{z}\rangle$. For $q(\mathbf{a}) \neq 0$, define $r_{\mathbf{a}}(\mathbf{z}) = \mathbf{z} - \frac{2\langle \mathbf{z},\mathbf{a}\rangle}{q(\mathbf{a})}\mathbf{a}$.
Direct expansion establishes $\langle r_{\mathbf{a}}\mathbf{z}, r_{\mathbf{a}}\mathbf{w}\rangle = \langle \mathbf{z}, \mathbf{w}\rangle$, $r_{\mathbf{a}}^2 = I_{\mathcal{W}}$, and $r_{\mathbf{a}}\mathbf{a} = -\mathbf{a}$. Thus $r_{\mathbf{a}}$ is isometry. If $q(\mathbf{x}-\mathbf{y}) \neq 0$, then $r_{\mathbf{x}-\mathbf{y}}\mathbf{x} = \mathbf{y}$.
If $q(\mathbf{x}-\mathbf{y}) = 0$, then $q(\mathbf{x}+\mathbf{y}) = 4q(\mathbf{y}) \neq 0$ and $r_{\mathbf{y}} r_{\mathbf{x}+\mathbf{y}}\mathbf{x} = \mathbf{y}$. Thus $r_{\mathbf{x}-\mathbf{y}}\theta$ or $r_{\mathbf{y}} r_{\mathbf{x}+\mathbf{y}}\theta$ extends $\mathbf{s} \mapsto \mathbf{t}$.

(iii) *Nonsingular Subspaces*: Suppose that $\mathcal{S}, \mathcal{T}$ are nonsingular, and induct on $\dim(\mathcal{S})$.
The zero-dimensional case is the assumed isometry $\mathcal{V} \cong \mathcal{W}$.
At positive dimension, choose a hyperbolic plane $\mathcal{A} \leq \mathcal{S}$ in the symplectic case by Theorem 3.4.8.
In the orthogonal case, fix $\mathbf{s} \in \mathcal{S}$ with $\langle \mathbf{s}, \mathbf{s}\rangle \neq 0$ by Proposition 3.1.2(ii) and nonsingularity, and set $\mathcal{A} = \operatorname{span}(\mathbf{s})$.
In either case, put $\mathcal{B} = \sigma\mathcal{A}$. In the symplectic case, $\mathcal{A}^\perp$ and $\mathcal{B}^\perp$ are nonsingular and equal-dimensional by Theorem 3.4.2(i),(iv).
In that case, fix $\beta : \mathcal{A}^\perp \xrightarrow{\sim} \mathcal{B}^\perp$ by (i), and set $\alpha = (\sigma|_{\mathcal{A}}) \odot \beta$. In orthogonal case, fix ambient isometry α extending $\sigma|_{\mathcal{A}}$ by (ii).
Set $\mathcal{S}_0 = \mathcal{S} \cap \mathcal{A}^\perp$ and $\mathcal{T}_0 = \mathcal{T} \cap \mathcal{B}^\perp$; then $\sigma(\mathcal{S}_0) = \mathcal{T}_0$. Decompositions $\mathcal{S} = \mathcal{A} \odot \mathcal{S}_0$ and $\mathcal{T} = \mathcal{B} \odot \mathcal{T}_0$ hold by Theorem 3.4.2(ii).
Spaces $\mathcal{S}_0, \mathcal{T}_0$ are nonsingular by Theorem 3.4.2(iv). Decompositions $\mathcal{V} = \mathcal{A} \odot \mathcal{A}^\perp$ and $\mathcal{W} = \mathcal{B} \odot \mathcal{B}^\perp$ hold by Theorem 3.4.2(ii).
The restriction $\alpha|_{\mathcal{A}^\perp}$ identifies the ambient complements, and $\dim(\mathcal{S}_0) < \dim(\mathcal{S})$.
Fix $\eta : \mathcal{A}^\perp \xrightarrow{\sim} \mathcal{B}^\perp$ with $\eta|_{\mathcal{S}_0} = \sigma|_{\mathcal{S}_0}$ by induction. Then $(\sigma|_{\mathcal{A}}) \odot \eta : \mathcal{V} \xrightarrow{\sim} \mathcal{W}$ extends σ.

(iv) *General Subspaces*: Choose compatible nonsingular extensions $\widehat{\mathcal{S}}, \widehat{\mathcal{T}}$ and $\widehat{\sigma} : \widehat{\mathcal{S}} \xrightarrow{\sim} \widehat{\mathcal{T}}$ extending σ by Theorem 3.4.9(iii).
An ambient isometry extending $\widehat{\sigma}$ exists by (iii), and it also extends σ.

Structural Commentary: Reflections or hyperbolic planes drive induction on nonsingular completions of arbitrary subspaces.
Key Insight: Compatible completion reduces extension to nonsingular summand induction. □

Corollary 3.4.13. *Witt Cancellation*
Let $\mathcal{V} = \mathcal{S} \odot \mathcal{A}$, $\mathcal{W} = \mathcal{T} \odot \mathcal{B}$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
If $\mathcal{V} \cong \mathcal{W}$ and $\mathcal{S} \cong \mathcal{T}$, then $\mathcal{A} \cong \mathcal{B}$.

Proof. Fix $\sigma : \mathcal{S} \xrightarrow{\sim} \mathcal{T}$, and fix $\widehat{\sigma} : \mathcal{V} \xrightarrow{\sim} \mathcal{W}$ with $\widehat{\sigma}|_{\mathcal{S}} = \sigma$ by Theorem 3.4.12.
Identities $\mathcal{A} = \mathcal{S}^{\perp}$ and $\mathcal{B} = \mathcal{T}^{\perp}$ follow from orthogonal decompositions. Hence $\widehat{\sigma}(\mathcal{A}) = \widehat{\sigma}(\mathcal{S}^{\perp}) = (\widehat{\sigma}\mathcal{S})^{\perp} = \mathcal{T}^{\perp} = \mathcal{B}$, so $\widehat{\sigma}|_{\mathcal{A}} : \mathcal{A} \xrightarrow{\sim} \mathcal{B}$.
Structural Commentary: Witt extension sends the first summand to its mate and therefore identifies their complements.
Key Insight: An ambient isometry preserves orthogonal complements. □

Definition 3.4.14. *Witt Index*
Let $\mathcal{V}$ be nonsingular orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$. Its *Witt index* is $w(\mathcal{V}) := \max\{\dim(\mathcal{L}) : \mathcal{L} \leq \mathcal{V},\ \mathcal{L} \subseteq \mathcal{L}^{\perp}\}$.

Theorem 3.4.15. *Witt Index and Anisotropic Decomposition*
Let $\mathcal{V}$ be orthogonal over $\mathbb{F}$ with $\operatorname{char}(\mathbb{F}) \neq 2$.

(i) If $\mathcal{V}$ is nonsingular, every maximal totally isotropic subspace has dimension $w(\mathcal{V})$.
(ii) If $\mathcal{V}$ is nonsingular, every maximal hyperbolic subspace has dimension $2w(\mathcal{V})$.
(iii) Let $R = \operatorname{rad}(\mathcal{V})$ and $\mathcal{V} = R \oplus \mathcal{U}$ with $\mathcal{U}$ nonsingular.
Then $(\exists\, \mathcal{H}, \mathcal{S} \leq \mathcal{U})\ \mathcal{H}$ is hyperbolic, $\mathcal{S}$ is anisotropic, $\mathcal{V} = \mathcal{S} \odot \mathcal{H} \odot R$, and $\dim(\mathcal{H}) = 2w(\mathcal{U})$.

Proof.

(*i*) Assume that $\mathcal{V}$ is nonsingular, and let $\mathcal{L}$ be maximal totally isotropic.
Choose a totally isotropic $\mathcal{M}$ with $\dim(\mathcal{M}) = w(\mathcal{V})$ by the definition of $w(\mathcal{V})$.
As $\dim(\mathcal{L}) \leq \dim(\mathcal{M})$, choose $\mathcal{M}_0 \leq \mathcal{M}$ with $\dim(\mathcal{M}_0) = \dim(\mathcal{L})$. Fix vector-space isomorphism $\tau : \mathcal{L} \to \mathcal{M}_0$ by Theorem 1.7.9.
It is an isometry because both restricted forms vanish. Fix $\Phi \in \mathcal{O}(\mathcal{V})$ with $\Phi|_{\mathcal{L}} = \tau$ by Theorem 3.4.12.
Then $\mathcal{L} \subseteq \Phi^{-1}\mathcal{M}$, and $\Phi^{-1}\mathcal{M}$ is totally isotropic. Hence $\mathcal{L} = \Phi^{-1}\mathcal{M}$ by maximality, so $\dim(\mathcal{L}) = w(\mathcal{V})$.

(*ii*) Assume that $\mathcal{V}$ is nonsingular. If a hyperbolic subspace has k planes, the span of one first vector from each hyperbolic pair is a k-dimensional totally isotropic subspace; hence $k \leq w(\mathcal{V})$. Choose a totally isotropic $\mathcal{M}$ of dimension $w(\mathcal{V})$.
Since $\operatorname{rad}(\mathcal{M}) = \mathcal{M}$, a hyperbolic completion $\mathcal{H}_0 \supseteq \mathcal{M}$ with $\dim(\mathcal{H}_0) = 2w(\mathcal{V})$ exists by Theorem 3.4.9(i).
Suppose that a maximal hyperbolic $\mathcal{H}$ has dimension $2k < 2w(\mathcal{V})$, and choose the span $\mathcal{H}_1 \leq \mathcal{H}_0$ of k hyperbolic pairs.
Matching hyperbolic bases defines an isometry $\mathcal{H} \to \mathcal{H}_1$. Fix an extension $\Phi \in \mathcal{O}(\mathcal{V})$ by Theorem 3.4.12.
Then $\mathcal{H} \subsetneq \Phi^{-1}\mathcal{H}_0$, contrary to maximality. Therefore $\dim(\mathcal{H}) = 2w(\mathcal{V})$.

(*iii*) Since $R \perp \mathcal{V}$, the assumed complement satisfies $\mathcal{V} = R \odot \mathcal{U}$.
Choose a maximal hyperbolic $\mathcal{H} \leq \mathcal{U}$, and put $\mathcal{S} = \mathcal{H}^{\perp} \cap \mathcal{U}$. The hyperbolic basis makes $\mathcal{H}$ nonsingular.
The identities $\dim(\mathcal{H}) = 2w(\mathcal{U})$ and $\mathcal{U} = \mathcal{H} \odot \mathcal{S}$ hold by (ii) and Theorem 3.4.2(ii). Space $\mathcal{S}$ is nonsingular by Theorem 3.4.2(iv).
If $\mathcal{S}$ contained a nonzero isotropic vector, a hyperbolic plane $\mathcal{P} \leq \mathcal{S}$ would contain it by Theorem 3.4.8.
Then $\mathcal{H} \odot \mathcal{P}$ would contradict maximality of $\mathcal{H}$. Thus $\mathcal{S}$ is anisotropic and $\mathcal{V} = \mathcal{S} \odot \mathcal{H} \odot R$.

Structural Commentary: Witt extension fixes maximal isotropic and hyperbolic dimensions; splitting isolates the anisotropic residue.
Key Insight: Each hyperbolic plane contributes one maximal isotropic dimension. □

3.5 Symplectic Classification and Transvections

Throughout this subsection, symplectic geometries are finite-dimensional over a fixed arbitrary field $\mathbb{F}$.

Theorem 3.5.1. *Classification of Symplectic Geometries*
Let $\mathcal{V}$ be a symplectic geometry, $R = \operatorname{rad}(\mathcal{V})$, $n = \dim(\mathcal{V})$, and $r = \operatorname{rank}(\mathcal{V})$.

(i) If $R = \{\mathbf{0}\}$, then $\mathcal{V}$ is hyperbolic and n is even.
(ii) $\mathcal{V} = R \odot \mathcal{H}$ for a hyperbolic subspace $\mathcal{H} \leq \mathcal{V}$, and R is totally singular.
(iii) $r \in 2\mathbb{Z}_{\geq 0}$. If $r = 2k$, then $\mathcal{V}$ has a basis β. For these choices, $M_\beta = X_{2k,n-2k} := \operatorname{diag}(J_2^{\oplus k}, 0_{n-2k})$, where $J_2 := \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.
(iv) For symplectic geometries $\mathcal{V}, \mathcal{W}$, $\mathcal{V} \cong \mathcal{W} \iff \dim \mathcal{V} = \dim \mathcal{W}\ \wedge\ \operatorname{rank}(\mathcal{V}) = \operatorname{rank}(\mathcal{W})$.

Proof.

(*i*) The claim holds for $n = 0$. Suppose $n > 0$ and fix $\mathbf{0} \neq \mathbf{x} \in \mathcal{V}$.
The vector $\mathbf{x}$ is isotropic by alternation and lies in a hyperbolic plane $\mathcal{H}_1$ by Theorem 3.4.8.
By Theorem 3.4.2(ii),(iv), $\mathcal{V} = \mathcal{H}_1 \odot \mathcal{H}_1^{\perp}$ and $\mathcal{H}_1^{\perp}$ is nonsingular.
The complement $\mathcal{H}_1^{\perp}$ is hyperbolic by induction on n, so $\mathcal{V}$ is hyperbolic and n is even.

(*ii*) By Theorem 3.4.3, $\mathcal{V} = R \odot \mathcal{S}$ for a nonsingular $\mathcal{S}$, and R is totally singular.
The subspace $\mathcal{S}$ is hyperbolic by (i), so $\mathcal{H} := \mathcal{S}$ has the required properties.

(*iii*) By (ii), $\mathcal{V} = R \odot \mathcal{H}$ with $\mathcal{H}$ hyperbolic. Since $\dim R = n - r$, $\dim \mathcal{H} = r$.
Write $\mathcal{H}$ as an orthogonal sum of k hyperbolic planes; then $r = 2k$.
A hyperbolic basis of $\mathcal{H}$ followed by a basis of R is a basis β of $\mathcal{V}$ satisfying $M_\beta = X_{2k,n-2k}$.

(*iv*) ($\Rightarrow$) Let $\sigma : \mathcal{V} \to \mathcal{W}$ be an isometry. Then $\dim \mathcal{V} = \dim \mathcal{W}$ by bijectivity.
If β is a basis of $\mathcal{V}$, then $\sigma\beta$ is a basis of $\mathcal{W}$ and $M_{\sigma\beta} = M_\beta$ by form preservation; hence $\operatorname{rank}(\mathcal{V}) = \operatorname{rank}(\mathcal{W})$.

(*iv*) ($\Leftarrow$) Suppose $\dim \mathcal{V} = \dim \mathcal{W}$ and $\operatorname{rank}(\mathcal{V}) = \operatorname{rank}(\mathcal{W})$. By (iii), $\mathcal{V}$ and $\mathcal{W}$ have bases with the same matrix $X_{2k,n-2k}$.
The basis-to-basis isomorphism is an isometry by Proposition 3.1.4(i), so $\mathcal{V} \cong \mathcal{W}$.

Structural Commentary: Radical splitting transfers classification to hyperbolic planes, whose block matrix determines isometry.
Key Insight: Every nonsingular alternating summand is exhausted by hyperbolic planes. □

Definition 3.5.2. *Symplectic Transvection*
Let $\mathcal{V}$ be a nonsingular symplectic geometry, $\mathbf{0} \neq \mathbf{v} \in \mathcal{V}$, and $a \in \mathbb{F}$. Define $\tau_{\mathbf{v},a} : \mathcal{V} \to \mathcal{V}$ by $\tau_{\mathbf{v},a}(\mathbf{x}) = \mathbf{x} + a\,\langle \mathbf{x}, \mathbf{v} \rangle\,\mathbf{v}$.
This operator is the *symplectic transvection* with direction $\mathbf{v}$ and parameter a.

Proposition 3.5.3. *Basic Properties of Symplectic Transvections*
Let $\mathcal{V}$ be a nonsingular symplectic geometry and $\mathbf{0} \neq \mathbf{v} \in \mathcal{V}$. Fix $a, b \in \mathbb{F}$, $c \in \mathbb{F}^{\times}$, and $\sigma \in \operatorname{Sp}(\mathcal{V})$.

(i) $\tau_{\mathbf{v},a} \in \operatorname{Sp}(\mathcal{V})$;
(ii) $\tau_{\mathbf{v},a} = I_{\mathcal{V}} \iff a = 0$;
(iii) $\tau_{\mathbf{v},a}\tau_{\mathbf{v},b} = \tau_{\mathbf{v},a+b}$;

(iv) $\tau_{\mathbf{v},a}^{-1} = \tau_{\mathbf{v},-a}$;
(v) $\sigma\tau_{\mathbf{v},a}\sigma^{-1} = \tau_{\sigma\mathbf{v},a}$;
(vi) $\tau_{c\mathbf{v},a} = \tau_{\mathbf{v},ac^2}$.

Proof.
(i) Put $T := \tau_{\mathbf{v},a}$, $\lambda(\mathbf{x}) := \langle \mathbf{x}, \mathbf{v}\rangle$. Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, T\mathbf{y}\rangle - \langle \mathbf{x}, \mathbf{y}\rangle = a\lambda(\mathbf{y})\lambda(\mathbf{x}) - a\lambda(\mathbf{x})\lambda(\mathbf{y}) = 0$ by bilinearity and alternation.
$(\forall\, \mathbf{x} \in \mathcal{V})\ \tau_{\mathbf{v},-a}T(\mathbf{x}) = \mathbf{x} = T\tau_{\mathbf{v},-a}(\mathbf{x})$ by alternation. Thus T is an invertible form-preserving linear map, so $T \in \mathrm{Sp}(\mathcal{V})$.
(ii) ($\Rightarrow$) Suppose $\tau_{\mathbf{v},a} = I_{\mathcal{V}}$. If $a \neq 0$, fix $\mathbf{x}$ satisfying $\langle \mathbf{x}, \mathbf{v}\rangle \neq 0$ by nonsingularity.
Then $\tau_{\mathbf{v},a}\mathbf{x} - \mathbf{x} = a\langle \mathbf{x}, \mathbf{v}\rangle\, \mathbf{v} \neq \mathbf{0}$, a contradiction. Hence $a = 0$.
(ii) ($\Leftarrow$) If $a = 0$, then $\tau_{\mathbf{v},a} = I_{\mathcal{V}}$ by definition.
(iii) By alternation, $(\forall\, \mathbf{x} \in \mathcal{V})\ \tau_{\mathbf{v},a}\tau_{\mathbf{v},b}(\mathbf{x}) = \mathbf{x} + (a+b)\langle \mathbf{x}, \mathbf{v}\rangle\, \mathbf{v} = \tau_{\mathbf{v},a+b}(\mathbf{x})$. Hence $\tau_{\mathbf{v},a}\tau_{\mathbf{v},b} = \tau_{\mathbf{v},a+b}$.
(iv) $\tau_{\mathbf{v},a}^{-1} = \tau_{\mathbf{v},-a}$ by (ii),(iii) with $b = -a$.
(v) Fix $\mathbf{x} \in \mathcal{V}$. Then $\langle \sigma^{-1}\mathbf{x}, \mathbf{v}\rangle = \langle \mathbf{x}, \sigma\mathbf{v}\rangle$ by form preservation. Hence $\sigma\tau_{\mathbf{v},a}\sigma^{-1}(\mathbf{x}) = \mathbf{x} + a\langle \mathbf{x}, \sigma\mathbf{v}\rangle\, \sigma\mathbf{v} = \tau_{\sigma\mathbf{v},a}(\mathbf{x})$.
(vi) $(\forall\, \mathbf{x} \in \mathcal{V})\ \tau_{c\mathbf{v},a}(\mathbf{x}) = \mathbf{x} + ac^2\langle \mathbf{x}, \mathbf{v}\rangle\, \mathbf{v} = \tau_{\mathbf{v},ac^2}(\mathbf{x})$ by bilinearity.

Structural Commentary: Alternation controls preservation and composition; nonsingularity detects the zero parameter.
Key Insight: The rank-one square vanishes, and the two mixed form terms cancel. □

Theorem 3.5.4. *Symplectic Transvections Generate*
If $\mathcal{V}$ is nonsingular symplectic, then $\mathrm{Sp}(\mathcal{V}) = \langle \tau_{\mathbf{v},a} : \mathbf{0} \neq \mathbf{v} \in \mathcal{V},\ a \in \mathbb{F}\rangle$.

Proof.
(i) *Generated Subgroup*: Put $G := \langle \tau_{\mathbf{v},a} : \mathbf{0} \neq \mathbf{v} \in \mathcal{V},\ a \in \mathbb{F}\rangle$. Then $G \leq \mathrm{Sp}(\mathcal{V})$ by Proposition 3.5.3(i).
(ii) *Vector Transitivity*: Fix $\mathbf{0} \neq \mathbf{x}, \mathbf{y} \in \mathcal{V}$. If $\langle \mathbf{x}, \mathbf{y}\rangle \neq 0$, then $\tau_{\mathbf{y}-\mathbf{x},\langle \mathbf{x},\mathbf{y}\rangle^{-1}}(\mathbf{x}) = \mathbf{y}$.
Suppose $\langle \mathbf{x}, \mathbf{y}\rangle = 0$ and $\mathbf{y} = \lambda\mathbf{x}$ for $\lambda \in \mathbb{F}^\times$. Fix $\mathbf{z}$ satisfying $\langle \mathbf{x}, \mathbf{z}\rangle \neq 0$ by nonsingularity. Then $\langle \mathbf{z}, \mathbf{y}\rangle = -\lambda\langle \mathbf{x}, \mathbf{z}\rangle \neq 0$.
If $\mathbf{x}, \mathbf{y}$ are independent, choose $\mathcal{V} = \mathrm{span}(\mathbf{x}, \mathbf{y}) \oplus \mathcal{C}$ by Proposition 2.3.6. Define $f \in \mathcal{V}'$ by $f(\alpha\mathbf{x} + \beta\mathbf{y} + \mathbf{c}) = \alpha - \beta$ for $\mathbf{c} \in \mathcal{C}$.
Fix $\mathbf{z}$ satisfying $(\forall\, \mathbf{t} \in \mathcal{V})\ f(\mathbf{t}) = \langle \mathbf{t}, \mathbf{z}\rangle$ by Theorem 3.3.6. Then $\langle \mathbf{x}, \mathbf{z}\rangle = \langle \mathbf{z}, \mathbf{y}\rangle = 1$ by the definition of f and alternation.
In either case, put $c := \langle \mathbf{x}, \mathbf{z}\rangle$ and $d := \langle \mathbf{z}, \mathbf{y}\rangle$. Thus $c, d \neq 0$, $\tau_{\mathbf{z}-\mathbf{x},c^{-1}}(\mathbf{x}) = \mathbf{z}$, and $\tau_{\mathbf{y}-\mathbf{z},d^{-1}}(\mathbf{z}) = \mathbf{y}$; G is transitive on $\mathcal{V} \setminus \{\mathbf{0}\}$.
(iii) *Hyperbolic-Pair Transitivity*: Fix hyperbolic pairs $(\mathbf{u}, \mathbf{v})$ and $(\mathbf{u}', \mathbf{v}')$.
Choose $\gamma \in G$ satisfying $\gamma\mathbf{u} = \mathbf{u}'$ by (ii), and put $\mathbf{w} := \gamma\mathbf{v}$. Then $(\mathbf{u}', \mathbf{w})$ is hyperbolic. If $\mathbf{w} = \mathbf{v}'$, put $\delta := I_{\mathcal{V}}$.
If $p := \langle \mathbf{w}, \mathbf{v}'\rangle \neq 0$, put $\delta := \tau_{\mathbf{v}'-\mathbf{w},p^{-1}}$. Then $\delta\mathbf{u}' = \mathbf{u}'$ and $\delta\mathbf{w} = \mathbf{v}'$ by direct evaluation. Suppose $\mathbf{w} \neq \mathbf{v}'$ and $\langle \mathbf{w}, \mathbf{v}'\rangle = 0$.
If $a\mathbf{u}' + b\mathbf{w} + c\mathbf{v}' = \mathbf{0}$, then $b + c = 0$ and $a = 0$ by pairing with $\mathbf{u}'$ and $\mathbf{w}$.
Hence $b(\mathbf{w} - \mathbf{v}') = \mathbf{0}$, so $a = b = c = 0$ and $\mathbf{u}', \mathbf{w}, \mathbf{v}'$ are independent. Choose $\mathcal{V} = \mathrm{span}(\mathbf{u}', \mathbf{w}, \mathbf{v}') \oplus \mathcal{C}$ by Proposition 2.3.6.
Define $f \in \mathcal{V}'$ by $f(a\mathbf{u}' + b\mathbf{w} + c\mathbf{v}' + \mathbf{d}) = a + b - c$ for $\mathbf{d} \in \mathcal{C}$. Fix $\mathbf{z}$ satisfying $(\forall\, \mathbf{t} \in \mathcal{V})\ f(\mathbf{t}) = \langle \mathbf{t}, \mathbf{z}\rangle$ by Theorem 3.3.6.
Then $\langle \mathbf{u}', \mathbf{z}\rangle = \langle \mathbf{w}, \mathbf{z}\rangle = \langle \mathbf{z}, \mathbf{v}'\rangle = 1$ by the definition of f and alternation.
The transvections $\tau_{\mathbf{z}-\mathbf{w},1}$ and $\tau_{\mathbf{v}'-\mathbf{z},1}$ fix $\mathbf{u}'$. The identities $\tau_{\mathbf{z}-\mathbf{w},1}(\mathbf{w}) = \mathbf{z}$ and $\tau_{\mathbf{v}'-\mathbf{z},1}(\mathbf{z}) = \mathbf{v}'$ hold by direct evaluation.
Thus $\delta := \tau_{\mathbf{v}'-\mathbf{z},1}\tau_{\mathbf{z}-\mathbf{w},1}$ fixes $\mathbf{u}'$ and maps $\mathbf{w}$ to $\mathbf{v}'$. In every case, $\delta\gamma \in G$ maps $(\mathbf{u}, \mathbf{v})$ to $(\mathbf{u}', \mathbf{v}')$.
(iv) *Dimension Induction*: Fix $\sigma \in \mathrm{Sp}(\mathcal{V})$ and put $n := \dim \mathcal{V}$. We induct on n. If $n = 0$, then $\sigma = I_{\mathcal{V}} \in G$. Suppose $n > 0$.
Choose a hyperbolic plane $\mathcal{H} = \mathrm{span}(\mathbf{u}, \mathbf{v})$ by Theorem 3.5.1(i). By (iii), fix $\gamma \in G$ mapping $(\sigma\mathbf{u}, \sigma\mathbf{v})$ to $(\mathbf{u}, \mathbf{v})$; hence $\gamma\sigma|_{\mathcal{H}} = I_{\mathcal{H}}$.
By Theorem 3.4.2(ii),(iv), $\mathcal{V} = \mathcal{H} \odot \mathcal{H}^\perp$ and $\mathcal{H}^\perp$ is nonsingular.
The subspace $\mathcal{H}^\perp$ is $\gamma\sigma$-invariant by form preservation and pointwise fixation of $\mathcal{H}$.
Thus $(\gamma\sigma)|_{\mathcal{H}^\perp} \in \mathrm{Sp}(\mathcal{H}^\perp)$ and is a product of transvections by induction.
Each factor $\tau_{\mathbf{x},a}$ extends to the same transvection on $\mathcal{V}$. Since $\mathbf{x} \in \mathcal{H}^\perp$, it fixes $\mathcal{H}$ pointwise.
The ambient product agrees with $\gamma\sigma$ on both orthogonal summands. Therefore $\gamma\sigma \in G$ and $\sigma = \gamma^{-1}(\gamma\sigma) \in G$.

Structural Commentary: Vector transitivity upgrades to pair transitivity; hyperbolic-plane induction then factors every isometry.
Key Insight: Pair transitivity isolates a fixed hyperbolic plane for dimension induction. □

Proposition 3.5.5. *Symplectic Transformations Have Determinant One*
If $\mathcal{V}$ is nonsingular symplectic, then $(\forall\, \sigma \in \mathrm{Sp}(\mathcal{V}))\ \det(\sigma) = 1$.

Proof. Fix $\sigma \in \mathrm{Sp}(\mathcal{V})$. It is a product of transvections by Theorem 3.5.4.
By determinant multiplicativity, it suffices to prove $\det(\tau_{\mathbf{v},a}) = 1$ for $\mathbf{0} \neq \mathbf{v} \in \mathcal{V}$ and $a \in \mathbb{F}$.
The transvection $\tau_{\mathbf{v},a}$ fixes $\mathbb{F}\mathbf{v}$ by alternation and induces $I_{\mathcal{V}/\mathbb{F}\mathbf{v}}$ by its defining formula.
Thus its matrix in a basis beginning with $\mathbf{v}$ is upper triangular with unit diagonal, so $\det(\tau_{\mathbf{v},a}) = 1$.
Structural Commentary: Transvection generation reduces the determinant to one basis-adapted rank-one map.
Key Insight: Only the leading coordinate changes, so the matrix is unit upper triangular. □

3.6 Orthogonal Classification over Common Fields

Throughout this subsection, orthogonal geometries are finite-dimensional over a field $\mathbb{F}$; additional hypotheses are stated locally.

Theorem 3.6.1. *Orthogonal Bases and Diagonal Congruence*
Let $(\mathcal{V}, B)$ be an orthogonal geometry, $q(\mathbf{x}) := B(\mathbf{x}, \mathbf{x})$, $n = \dim \mathcal{V}$, and $r = \mathrm{rank}\, B$.
(i) An orthogonal basis β with $M_\beta = \mathrm{diag}(a_1, \ldots, a_r, 0_{n-r})$ and $a_1, \ldots, a_r \in \mathbb{F}^\times$ exists $\iff$ $\mathrm{char}(\mathbb{F}) \neq 2 \ \vee\ B = 0 \ \vee\ q \not\equiv 0$.
(ii) A symmetric $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is congruent to a diagonal matrix $\iff$ $\mathrm{char}(\mathbb{F}) \neq 2 \ \vee\ A = 0 \ \vee\ (\exists\, \mathbf{x} \in \mathbb{F}^n)\ \mathbf{x}^t A\mathbf{x} \neq 0$.

Proof.
(i) ($\Rightarrow$) Only $\mathrm{char}(\mathbb{F}) = 2$ and $q \equiv 0$ require proof. In this case every diagonal entry of M_β equals $q(\mathbf{v}) = 0$, so $M_\beta = 0$ and $B = 0$.
(i) ($\Leftarrow$) If $B = 0$, any basis works. Assume $B \neq 0$. Then $q \not\equiv 0$ by hypothesis if $\mathrm{char}(\mathbb{F}) = 2$, and by Proposition 3.1.2(ii) otherwise.
Splitting successive nonisotropic lines by Theorem 3.4.2(ii) produces $\mathcal{V} = \mathbb{F}\mathbf{u}_1 \odot \cdots \odot \mathbb{F}\mathbf{u}_s \odot \mathcal{W}$ with $q(\mathbf{u}_i) = a_i \in \mathbb{F}^\times$ and $q|_{\mathcal{W}} \equiv 0$.
If $\mathrm{char}(\mathbb{F}) \neq 2$, then $B|_{\mathcal{W}} = 0$ by Proposition 3.1.2(ii). Appending any basis of $\mathcal{W}$ produces a diagonal Gram matrix of rank r.
Assume $\mathrm{char}(\mathbb{F}) = 2$; then $B|_{\mathcal{W}}$ alternating. By Theorem 3.5.1(ii), $\mathcal{W} = R \odot \mathcal{H}_1 \odot \cdots \odot \mathcal{H}_t$, $B|_R = 0$, each $\mathcal{H}_j$ hyperbolic.
Fix a current $\mathbf{u}$ with $q(\mathbf{u}) = a \neq 0$. For a hyperbolic basis $(\mathbf{e}, \mathbf{f})$ of $\mathcal{H}_j$, rescale $\mathbf{f}$ so that $B(\mathbf{e}, \mathbf{f}) = a$.
The vectors $\mathbf{u} + \mathbf{e}$, $\mathbf{u} + \mathbf{f}$, and $\mathbf{u} + \mathbf{e} + \mathbf{f}$ are independent by coefficient comparison.
These vectors have norm a and mutual pairings $a + a = 0$, so they form an orthogonal basis of $\mathbb{F}\mathbf{u} \odot \mathcal{H}_j$.
Retain one new line for the next plane; after t replacements, append a basis of R.
The resulting diagonal matrix has exactly r nonzero entries by rank invariance.

(*ii*) ($\Rightarrow$) Let $D = P^tAP$ be diagonal with P invertible.
If $\mathrm{char}(\mathbb{F}) = 2$ and $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, \mathbf{x}^tA\mathbf{x} = 0$, then $(\forall\, j \in \{1, \dots, n\})\, D_{jj} = (P\mathbf{e}_j)^tA(P\mathbf{e}_j) = 0$. Thus $D = 0$ and $A = 0$.
(*ii*) ($\Leftarrow$) Define $B_A(\mathbf{x}, \mathbf{y}) := \mathbf{x}^tA\mathbf{y}$ on $\mathbb{F}^n$. Its norm map is $\mathbf{x} \mapsto \mathbf{x}^tA\mathbf{x}$, so the hypothesis is exactly that of (i).
Since $A = A^t$, the form B_A is symmetric, so choose an orthogonal basis by (i).
Its change-of-basis matrix diagonalizes A by Proposition 3.1.4(ii).

Structural Commentary: Split nonisotropic lines; in characteristic 2, absorb each residual hyperbolic plane into one retained line.
Key Insight: A nonzero alternating symmetric form in characteristic 2 is the sole obstruction. □

Proposition 3.6.2. *Recognizing Orthogonal Hyperbolic Planes*
Let $\mathcal{H}$ be a 2-dimensional nonsingular orthogonal geometry with $\mathrm{char}(\mathbb{F}) \neq 2$. *The following are equivalent:*

(i) $\mathcal{H}$ is a hyperbolic plane;
(ii) $(\exists\, \mathbf{u} \in \mathcal{H} \setminus \{\mathbf{0}\})\, \langle \mathbf{u}, \mathbf{u} \rangle = 0$;
(iii) $\Delta(\mathcal{H}) = -(\mathbb{F}^\times)^2$.

Proof.

(*i*) $\Rightarrow$ (*ii*) A hyperbolic basis contains a nonzero isotropic vector by the definition of a hyperbolic pair.
(*ii*) $\Rightarrow$ (*iii*) By Theorem 3.4.8, fix $\mathbf{v}$ such that $(\mathbf{u}, \mathbf{v})$ is hyperbolic.
Since $\dim \mathcal{H} = 2$, this pair is a basis. Its Gram matrix has determinant $\det \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) = -1$, so $\Delta(\mathcal{H}) = -(\mathbb{F}^\times)^2$.
(*iii*) $\Rightarrow$ (*i*) By Theorem 3.6.1(i), write $M = \mathrm{diag}(a, b)$ with $a, b \in \mathbb{F}^\times$. Choose $c \in \mathbb{F}^\times$ with $ab = -c^2$ by the discriminant hypothesis.
Then $a\left(\frac{c}{a}\right)^2 + b = 0$, so $\left(\frac{c}{a}, 1\right)$ is a nonzero isotropic coordinate vector.
It lies in a hyperbolic plane by Theorem 3.4.8, and that plane equals $\mathcal{H}$ by dimension.

Structural Commentary: Pass from a hyperbolic basis to determinant -1, then recover isotropy from the discriminant square class.
Key Insight: In a nonsingular orthogonal plane, representing zero is exactly hyperbolicity. □

Example 3.6.3. *Characteristic Two Orthogonal-Symplectic Plane*
Assume $\mathrm{char}(\mathbb{F}) = 2$ and let $(\mathcal{H}, B)$ have basis $(\mathbf{e}, \mathbf{f})$ with Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.
The matrix is symmetric and has determinant 1, while $(\forall\, a, b \in \mathbb{F})\, B(a\mathbf{e}+b\mathbf{f}, a\mathbf{e}+b\mathbf{f}) = 2ab = 0$; hence B is nonsingular and alternating.
For $\mathbf{0} \neq \mathbf{x} \in \mathcal{H}$, alternation implies $\mathbb{F}\mathbf{x} \leq (\mathbb{F}\mathbf{x})^\perp$, and both spaces have dimension 1 by Theorem 3.4.2(i).
Thus $(\mathbb{F}\mathbf{x})^\perp = \mathbb{F}\mathbf{x}$, so no second basis vector can be orthogonal to $\mathbf{x}$. This is precisely the obstruction in Theorem 3.6.1.

Computation Method 3.6.4. *Lagrange Orthogonalization*
Let $(\mathcal{V}, B)$ be an orthogonal geometry with $\mathrm{char}(\mathbb{F}) \neq 2$ and $q(\mathbf{x}) := B(\mathbf{x}, \mathbf{x})$.
Input: a finite spanning list L. Output: an orthogonal basis of $\mathcal{V}$.

(i) Initialize $\mathcal{W}_0 = \mathcal{V}$ and $L_0 = L$. At stage k, stop with $R = \mathcal{W}_k$ if $q|_{\mathcal{W}_k} \equiv 0$.
If $(\exists\, \mathbf{x} \in L_k)\, q(\mathbf{x}) \neq 0$, take such an $\mathbf{x}$ as $\mathbf{u}_{k+1}$. Otherwise write $L_k = (\mathbf{x}_1, \dots, \mathbf{x}_{\ell_k})$.
Then $(\exists\, i, j \in \{1, \dots, \ell_k\})\, B(\mathbf{x}_i, \mathbf{x}_j) \neq 0$, since the contrary would imply $B|_{\mathcal{W}_k} = 0$ by bilinearity and spanning.
Put $\mathbf{u}_{k+1} = \mathbf{x}_i + \mathbf{x}_j$; then $q(\mathbf{u}_{k+1}) = 2B(\mathbf{x}_i, \mathbf{x}_j) \neq 0$.
(ii) Put $\mathcal{W}_{k+1} = \mathcal{W}_k \cap (\mathbb{F}\mathbf{u}_{k+1})^\perp$. Let L_{k+1} be the nonzero vectors among $\left\{\mathbf{x} - \frac{B(\mathbf{x}, \mathbf{u}_{k+1})}{q(\mathbf{u}_{k+1})}\mathbf{u}_{k+1} : \mathbf{x} \in L_k\right\}$.
By Theorem 3.4.2(ii), $\mathcal{W}_k = \mathbb{F}\mathbf{u}_{k+1} \odot \mathcal{W}_{k+1}$. The displayed update is the projection onto $\mathcal{W}_{k+1}$, so its image L_{k+1} spans $\mathcal{W}_{k+1}$.
(iii) Since $\dim \mathcal{W}_{k+1} = \dim \mathcal{W}_k - 1$, the process stops. For any basis $(\mathbf{z}_1, \dots, \mathbf{z}_m)$ of R, output $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_s, \mathbf{z}_1, \dots, \mathbf{z}_m)$.
Selected lines are mutually orthogonal and orthogonal to R. Since $q|_R \equiv 0$, $B|_R = 0$ by Proposition 3.1.2(ii); hence β is orthogonal.

Definition 3.6.5. *Universal Orthogonal Geometry*
An orthogonal geometry $\mathcal{V}$ over $\mathbb{F}$ is *universal* if its norm map $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$ satisfies $q(\mathcal{V}) = \mathbb{F}$.

Proposition 3.6.6. *Finite-Field Square Classes and Universality*
Let $\mathbb{F}_q$ be a finite field and put $S = (\mathbb{F}_q^\times)^2$. For an orthogonal geometry $\mathcal{V}$, write $q_{\mathcal{V}}(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$.

(i) If $\mathrm{char}(\mathbb{F}_q) = 2$, then $x \mapsto x^2$ is a bijection of $\mathbb{F}_q$.
(ii) If $\mathrm{char}(\mathbb{F}_q) \neq 2$, then $|S| = \frac{q-1}{2}$.
(iii) If $\mathrm{char}(\mathbb{F}_q) \neq 2$, then $(\forall\, d \in \mathbb{F}_q^\times \setminus S)\, \mathbb{F}_q^\times \setminus S = dS$.
(iv) If $\mathcal{V}$ is orthogonal over odd $\mathbb{F}_q$ and contains a nonsingular subspace of dimension at least 2, then $q_{\mathcal{V}}(\mathcal{V}) = \mathbb{F}_q$.

Proof.

(i) If $x^2 = y^2$, then $(x - y)^2 = 0$ in characteristic 2, so $x = y$. Thus $x \mapsto x^2$ is injective and hence bijective on the finite set $\mathbb{F}_q$.
(ii) For $x, y \in \mathbb{F}_q^\times$, $x^2 = y^2 \iff (x - y)(x + y) = 0 \iff x = \pm y$.
Since $x \neq -x$, the $\frac{q-1}{2}$ pairs $\{x, -x\}$ are exactly the fibers of $x \mapsto x^2$, so $|S| = \frac{q-1}{2}$.
(iii) Fix $d \in \mathbb{F}_q^\times \setminus S$. If $dS \cap S \neq \varnothing$, choose $s_1, s_2 \in S$ with $ds_1 = s_2$. Then $d = s_2 s_1^{-1} \in S$, a contradiction.
Hence $dS \cap S = \varnothing$. Since $|dS| = |S| = \frac{q-1}{2}$ by (ii), $\mathbb{F}_q^\times \setminus S = dS$.
(iv) Put $Q = q_{\mathcal{V}}$. By Theorem 3.6.1(i), the nonsingular subspace contains orthogonal $\mathbf{u}, \mathbf{v}$. Put $a := Q(\mathbf{u}) \neq 0$ and $b := Q(\mathbf{v}) \neq 0$.
Let $T := \{\alpha^2 : \alpha \in \mathbb{F}_q\}$. By (ii), $|T| = \frac{q+1}{2}$. For fixed $c \in \mathbb{F}_q$, the sets aT and $c - bT$ both have $\frac{q+1}{2}$ elements, so they intersect.
Choose $\alpha, \beta \in \mathbb{F}_q$ with $a\alpha^2 = c - b\beta^2$. Orthogonality implies $Q(\alpha\mathbf{u} + \beta\mathbf{v}) = a\alpha^2 + b\beta^2 = c$, so $Q(\mathcal{V}) = \mathbb{F}_q$.

Structural Commentary: Count square-map fibers, identify the nonsquare coset, then intersect two affine square-value sets.
Key Insight: Two subsets of $\mathbb{F}_q$ of size $\frac{q+1}{2}$ must intersect. □

Theorem 3.6.7. *Canonical Orthogonal Forms over Common Fields*
Let $M = M^t \in \mathrm{Mat}(n, n, \mathbb{F})$, let $r = \mathrm{rank}\, M$, and write $M \sim N$ for congruence. If $\mathrm{char}(\mathbb{F}) = 2$, *assume M is not nonzero alternating.*

(i) If $\mathbb{F}$ is algebraically closed, then $M \sim I_r \oplus 0_{n-r}$, and rank is a complete congruence invariant.
(ii) If $\mathbb{F} = \mathbb{R}$, then $M \sim I_p \oplus (-I_m) \oplus 0_k$ for unique $p, m, k \in \mathbb{Z}_{\geq 0}$ satisfying $p + m + k = n$.
(iii) If $\mathbb{F} = \mathbb{F}_q$ has characteristic 2, then $M \sim I_r \oplus 0_{n-r}$, and rank is a complete congruence invariant among the admitted matrices.
(iv) If $\mathbb{F} = \mathbb{F}_q$ has odd characteristic and $d \in \mathbb{F}_q^\times$ is a fixed nonsquare, then $r = 0$ implies $M \sim 0_n$.
For $r > 0$, exactly one of $M \sim I_r \oplus 0_{n-r}$ and $M \sim I_{r-1} \oplus [d] \oplus 0_{n-r}$ holds.

Proof.

(i) By Theorem 3.6.1(ii), write $M \sim \operatorname{diag}(a_1, \dots, a_r, 0_{n-r})$.
By algebraic closure, choose $(s_1, \dots, s_r) \in (\mathbb{F}^\times)^r$ satisfying $(\forall\, i \in \{1, \dots, r\})\, a_i = s_i^2$.
Congruence by $\operatorname{diag}(s_1^{-1}, \dots, s_r^{-1}, I_{n-r})$ produces $I_r \oplus 0_{n-r}$.
Rank is congruence-invariant by Proposition 3.1.4(iv), so it is complete.

(ii) By Theorem 3.6.1(ii), diagonalization and rescaling produce $D = I_p \oplus (-I_m) \oplus 0_k$.
Let $E_{\le 0}$ be the coordinate subspace on which the form is nonpositive.
Every positive-definite subspace intersects $E_{\le 0}$ trivially, so its dimension is at most $\operatorname{codim} E_{\le 0} = p$.
The positive coordinate subspace has dimension p, so the maximum is p and is congruence-invariant.
The same argument for $-D$ determines m, and then $k = n - p - m$.

(iii) By Theorem 3.6.1(ii), write $M \sim \operatorname{diag}(a_1, \dots, a_r, 0_{n-r})$.
By Proposition 3.6.6(i), choose $(s_1, \dots, s_r) \in (\mathbb{F}_q^\times)^r$ satisfying $(\forall\, i \in \{1, \dots, r\})\, a_i = s_i^2$.
Congruence by $\operatorname{diag}(s_1^{-1}, \dots, s_r^{-1}, I_{n-r})$ produces $I_r \oplus 0_{n-r}$, and congruence invariance of rank proves completeness.

(iv) If $r = 0$, then $M = 0_n$. Assume $r > 0$ and put $S = (\mathbb{F}_q^\times)^2$.
By Theorem 3.6.1(ii), diagonalize M and let $\mathcal{U}$ be the nonsingular block, with norm map Q.
While $\dim \mathcal{U} \ge 2$, choose $\mathbf{u} \in \mathcal{U}$ with $Q(\mathbf{u}) = 1$ by Proposition 3.6.6(iv).
By Theorem 3.4.2(ii),(iv), split $\mathbb{F}\mathbf{u}$ with a nonsingular complement and continue there.
The nonsingular block becomes $I_{r-1} \oplus [a]$ with $a \in \mathbb{F}_q^\times$. By Proposition 3.6.6(ii),(iii), exactly one of $a \in S$ and $a \in dS$ holds.
Rescaling the final line produces the corresponding stated form.
Any congruence maps radicals onto radicals and induces an isometry of the nonsingular quotients.
The two quotients have discriminants S and dS, which are distinct because d is nonsquare.
Discriminant is isometry-invariant by Proposition 3.1.4(v), so the normal forms are not congruent.

Structural Commentary: Diagonalize, normalize coefficients by the field, then separate residual forms by inertia or discriminant.
Key Insight: Signs over $\mathbb{R}$ and square classes over finite fields survive rescaling. □

Definition 3.6.8. *Minkowski Space*
Minkowski space $\mathbb{R}^{3,1}$ is $\mathbb{R}^4$ with standard Gram matrix $\operatorname{diag}(1, 1, 1, -1)$.

Example 3.6.9. *Null Vectors in Minkowski Space*
In $\mathbb{R}^{3,1}$, $\mathcal{N} := \{\mathbf{x} \in \mathbb{R}^4 \setminus \{\mathbf{0}\} : x_1^2 + x_2^2 + x_3^2 = x_4^2\}$ is the null cone.
The vector $\mathbf{e}_1 + \mathbf{e}_4$ lies in $\mathcal{N}$. No isometry to Euclidean $\mathbb{R}^4$ exists: its matrix A would satisfy $A^t A = \operatorname{diag}(1, 1, 1, -1)$.
Taking determinants would force $(\det A)^2 = -1$, which is impossible over $\mathbb{R}$.

Example 3.6.10. *Two Rank-Two Orthogonal Planes over $\mathbb{F}_5$*
In $\mathbb{F}_5$, $(\mathbb{F}_5^\times)^2 = \{1, 4\}$ and the nonsquare class is $\{2, 3\}$. Let $q_1(x, y) := x^2 + y^2$ and $q_2(x, y) := x^2 + 2y^2$.

(i) The form q_1 has matrix I_2 and is hyperbolic by $q_1(1, 2) = 0$ and Proposition 3.6.2.

(ii) The form q_2 is anisotropic: if $q_2(x, y) = 0$ and $y = 0$, then $x = 0$; if $y \ne 0$, then $\left(\frac{x}{y}\right)^2 = 3$, contradicting that 3 is nonsquare.

Thus the two forms realize the nonsingular rank-2 classes in Theorem 3.6.7(iv).

Lemma 3.6.11. *Odd Finite-Field Isotropy Bound*
Let $\mathcal{V}$ be nonsingular orthogonal over odd $\mathbb{F}_q$ and put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$.
Write $\mathcal{H}^{\odot s}$ for an orthogonal sum of s hyperbolic planes.

(i) If $\dim \mathcal{V} \ge 3$, then $(\exists\, \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\})\, q(\mathbf{x}) = 0$.
(ii) $(\exists\, s \in \mathbb{Z}_{\ge 0})(\exists\, \mathcal{R} \le \mathcal{V})\, \mathcal{V} \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$, where $\mathcal{R}$ is nonsingular anisotropic and $\dim \mathcal{R} \le 2$.

Proof.

(i) By Theorem 3.6.1(i), choose orthogonal nonisotropic vectors $\mathbf{u}, \mathbf{v} \in \mathcal{V}$. Put $\mathcal{P} = \operatorname{span}(\mathbf{u}, \mathbf{v})$, which is nonsingular.
By Theorem 3.4.2(i), $\dim \mathcal{P}^\perp = \dim \mathcal{V} - 2 \ge 1$; choose $\mathbf{0} \ne \mathbf{z} \in \mathcal{P}^\perp$.
If $q(\mathbf{z}) = 0$, take $\mathbf{x} = \mathbf{z}$. Otherwise choose $\mathbf{p} \in \mathcal{P}$ with $q(\mathbf{p}) = -q(\mathbf{z})$ by Proposition 3.6.6(iv).
Since $\mathcal{P} \cap \mathcal{P}^\perp = \{\mathbf{0}\}$ and $\mathcal{P} \perp \mathcal{P}^\perp$, $\mathbf{p} + \mathbf{z} \ne \mathbf{0}$ and $q(\mathbf{p} + \mathbf{z}) = 0$.

(ii) Induct on $\dim \mathcal{V}$. If $\mathcal{V}$ is anisotropic, then $\dim \mathcal{V} \le 2$ by (i); take $s = 0$ and $\mathcal{R} = \mathcal{V}$.
Otherwise choose $\mathbf{0} \ne \mathbf{x} \in \mathcal{V}$ with $q(\mathbf{x}) = 0$. By Theorem 3.4.8, $\mathbf{x}$ lies in a hyperbolic plane $\mathcal{H}$.
By Theorem 3.4.2(ii),(iv), $\mathcal{V} = \mathcal{H} \odot \mathcal{H}^\perp$, where $\mathcal{H}^\perp$ is nonsingular and has dimension $\dim \mathcal{V} - 2$.
By the induction hypothesis, $\mathcal{H}^\perp \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$ with $\mathcal{R}$ nonsingular anisotropic and $\dim \mathcal{R} \le 2$. Hence $\mathcal{V} \cong \mathcal{H}^{\odot(s+1)} \odot \mathcal{R}$.

Structural Commentary: Cancel a represented value against an orthogonal line, then split the resulting hyperbolic plane inductively.
Key Insight: A nonsingular plane represents the negative norm needed to force isotropy. □

Theorem 3.6.12. *Odd Finite-Field Orthogonal Decomposition*
Let $\mathbb{F}_q$ have odd characteristic, fix a nonsquare $d \in \mathbb{F}_q^\times$, and let $\mathcal{V} \ne \{\mathbf{0}\}$ be nonsingular orthogonal.
Put $S = (\mathbb{F}_q^\times)^2$, write $\mathcal{H}^{\odot s}$ for s hyperbolic planes, and let $\langle a \rangle$ denote a line of value a. Set $\mathcal{A}_d := \langle 1 \rangle \odot \langle -d \rangle$.

(i) If $\dim \mathcal{V} = 2m + 1$, then $(\exists\, a \in \mathbb{F}_q^\times)\, \mathcal{V} \cong \mathcal{H}^{\odot m} \odot \langle a \rangle \ \wedge\ aS = (-1)^m \Delta(\mathcal{V})$.
(ii) If $\dim \mathcal{V} = 2m$, then $m \ge 1$ and exactly one of $\mathcal{V} \cong \mathcal{H}^{\odot m}$ and $\mathcal{V} \cong \mathcal{H}^{\odot(m-1)} \odot \mathcal{A}_d$ holds.
(iii) If $\dim \mathcal{V} = 2m$, then $\mathcal{V} \cong \mathcal{H}^{\odot m} \iff \Delta(\mathcal{V}) = (-1)^m S$.
(iv) If $\dim \mathcal{V} = 2m$, then $\mathcal{V} \cong \mathcal{H}^{\odot(m-1)} \odot \mathcal{A}_d \iff \Delta(\mathcal{V}) = (-1)^m dS$.

Proof.

(i) By Lemma 3.6.11(ii), write $\mathcal{V} \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$ with $\mathcal{R}$ anisotropic and $\dim \mathcal{R} \le 2$.
Since $2m + 1 = 2s + \dim \mathcal{R}$, parity implies $\dim \mathcal{R} = 1$ and $s = m$. Choose $a \in \mathbb{F}_q^\times$ with $\mathcal{R} \cong \langle a \rangle$.
Since $\Delta(\mathcal{H}) = -S$, $\Delta(\mathcal{V}) = (-1)^m aS$, and hence $aS = (-1)^m \Delta(\mathcal{V})$.

(ii) Since $\mathcal{V} \ne \{\mathbf{0}\}$, $m \ge 1$. By Lemma 3.6.11(ii), write $\mathcal{V} \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$ with $\dim \mathcal{R} \le 2$.
Since $2m = 2s + \dim \mathcal{R}$, $\dim \mathcal{R} \in \{0, 2\}$. If $\dim \mathcal{R} = 0$, then $s = m$ and $\mathcal{V} \cong \mathcal{H}^{\odot m}$.
Otherwise $s = m - 1$. Both $\mathcal{R}$ and $\mathcal{A}_d$ are anisotropic planes: for $\mathcal{A}_d$, $x^2 - dy^2 = 0$ forces $x = y = 0$ because d is nonsquare.
Both occupy the unique nonhyperbolic rank-2 class by Theorem 3.6.7(iv) and Proposition 3.6.2.
Hence $\mathcal{R} \cong \mathcal{A}_d$, so at least one alternative holds. Their discriminants are $(-1)^m S$ and $(-1)^m dS$.
Since d is nonsquare, these classes are distinct, so the alternatives are mutually exclusive.

(*iii*) ($\Rightarrow$) If $\mathcal{V} \cong \mathcal{H}^{\odot m}$, then $\Delta(\mathcal{V}) = (-1)^m S$ by $\Delta(\mathcal{H}) = -S$.
(*iii*) ($\Leftarrow$) By (ii), one alternative holds. Its discriminant is $(-1)^m S$ only in the first case, so $\mathcal{V} \cong \mathcal{H}^{\odot m}$.
(*iv*) ($\Rightarrow$) If $\mathcal{V} \cong \mathcal{H}^{\odot(m-1)} \odot \mathcal{A}_d$, then $\Delta(\mathcal{V}) = (-1)^m dS$ by $\Delta(\mathcal{A}_d) = -dS$.
(*iv*) ($\Leftarrow$) By (ii), one alternative holds. Its discriminant is $(-1)^m dS$ only in the second case, so $\mathcal{V} \cong \mathcal{H}^{\odot(m-1)} \odot \mathcal{A}_d$.

Structural Commentary: Split the anisotropic residue, determine its dimension by parity, then identify its class by discriminant.
Key Insight: Dimension parity fixes residue size; discriminant fixes its square class. □

3.7 Quadratic Refinements over $\mathbb{F}_2$

Throughout this subsection, vector spaces are finite-dimensional over $\mathbb{F}_2$.

Definition 3.7.1. *Quadratic Isometries and Arf Sums over $\mathbb{F}_2$*
Let $Q : \mathcal{V} \to \mathbb{F}_2$ be quadratic with nonsingular polar form B_Q. Let $\mathscr{B}_Q$ be the set of symplectic bases of B_Q.

(i) For quadratic $Q' : \mathcal{W} \to \mathbb{F}_2$, *quadratic isometry* $Q \to Q'$ is linear isomorphism $\sigma : \mathcal{V} \to \mathcal{W}$ satisfying $Q' \circ \sigma = Q$. Write $Q \cong Q'$.

(ii) For $\beta = (\mathbf{u}_i, \mathbf{v}_i)_{i=1}^m \in \mathscr{B}_Q$, its *Arf sum* is $\operatorname{Arf}_\beta(Q) := \sum_{i=1}^m Q(\mathbf{u}_i)Q(\mathbf{v}_i) \in \mathbb{F}_2$.

Example 3.7.2. *One Polar Form with Two Refinements*
On $\mathcal{V} = \mathbb{F}_2^2$, set $H(x, y) = xy$ and $A(x, y) = x^2 + xy + y^2$. Both have polar form $B((x, y), (u, v)) = xv + yu$.
The identities $B((x, y), (0, 1)) = x$ and $B((x, y), (1, 0)) = y$ imply $\operatorname{rad}(B) = \{\mathbf{0}\}$.
Moreover, $H^{-1}(0) = \{(0, 0), (1, 0), (0, 1)\}$, whereas $A^{-1}(0) = \{(0, 0)\}$. A quadratic isometry preserves zero sets, so $H \ncong A$.

Theorem 3.7.3. *Arf Classification over $\mathbb{F}_2$*
Let $Q : \mathcal{V} \to \mathbb{F}_2$ be quadratic with nonsingular polar form B_Q.
For $m \in \mathbb{Z}_{\geq 1}$ and $a \in \mathbb{F}_2$, define $Q_a^{(m)}(x_1, y_1, \ldots, x_m, y_m) := x_1y_1 + a(x_1^2 + y_1^2) + \sum_{i=2}^m x_iy_i$. Let $Q_0^{(0)}$ be the zero form on $\{\mathbf{0}\}$.

(i) $(\exists!\, m \in \mathbb{Z}_{\geq 0})\ \dim \mathcal{V} = 2m$.
(ii) For this m, $(\forall\, \beta \in \mathscr{B}_Q)\ Q \cong Q^{(m)}_{\operatorname{Arf}_\beta(Q)}$.
(iii) If $m \geq 1$, then $(\forall\, a \in \mathbb{F}_2)\ |(Q_a^{(m)})^{-1}(0)| = 2^{2m-1} + (-1)^a 2^{m-1}$.
(iv) If $m \geq 1$, then $Q_0^{(m)} \ncong Q_1^{(m)}$.
(v) $(\forall\, \beta, \gamma \in \mathscr{B}_Q)\ \operatorname{Arf}_\beta(Q) = \operatorname{Arf}_\gamma(Q)$*; denote this common value by* $\operatorname{Arf}(Q)$.
(vi) $Q \cong Q^{(m)}_{\operatorname{Arf}(Q)}$.
(vii) If $m \geq 1$, then $(\forall\, a \in \mathbb{F}_2)\ \operatorname{Arf}(Q_a^{(m)}) = a$.

Proof.

(i) The form B_Q is alternating by Proposition 3.2.3 and nonsingular by hypothesis.
By Theorem 3.5.1(i), $(\exists\, m \in \mathbb{Z}_{\geq 0})\ \dim \mathcal{V} = 2m$. Any such $m = \frac{1}{2}\dim \mathcal{V}$ is unique.

(ii) If $m = 0$, then $Q = Q_0^{(0)}$ by $Q(\mathbf{0}) = 0$. Assume $m \geq 1$ and fix $\beta = (\mathbf{u}_i, \mathbf{v}_i)_{i=1}^m \in \mathscr{B}_Q$. Put $r_i = Q(\mathbf{u}_i)$ and $s_i = Q(\mathbf{v}_i)$.
$(\forall\, x, y \in \mathbb{F}_2)\ Q(x\mathbf{u}_i + y\mathbf{v}_i) = r_ix^2 + xy + s_iy^2$ by the polar identity and symplectic pairings.
The form Q is the orthogonal sum of these plane restrictions by the vanishing cross-pairings.
If $(r_i, s_i) = (0, 0)$, the ith summand is $H(x, y) := xy$; if $(r_i, s_i) = (1, 1)$, it is $A(x, y) := x^2 + xy + y^2$.
The forms H and A are quadratic with the standard symplectic polar form by direct calculation.
For $(r_i, s_i) = (0, 1)$, the symplectic pair $(\mathbf{u}_i, \mathbf{u}_i + \mathbf{v}_i)$ has Q-values $(0, 0)$.
For $(r_i, s_i) = (1, 0)$, the symplectic pair $(\mathbf{u}_i + \mathbf{v}_i, \mathbf{v}_i)$ has Q-values $(0, 0)$.
Every basis plane is H or A after these changes. The number of A-planes modulo 2 is $\sum_{i=1}^m r_is_i = \operatorname{Arf}_\beta(Q)$.
Fix two A-planes with symplectic bases $(\mathbf{e}_1, \mathbf{f}_1)$ and $(\mathbf{e}_2, \mathbf{f}_2)$. Set $\mathbf{p}_1 = \mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{q}_1 = \mathbf{f}_1 + \mathbf{e}_2$.
Set $\mathbf{p}_2 = \mathbf{e}_1 + \mathbf{f}_1 + \mathbf{f}_2$ and $\mathbf{q}_2 = \mathbf{e}_1 + \mathbf{f}_1 + \mathbf{e}_2 + \mathbf{f}_2$.
The identity $(Q(\mathbf{p}_1), Q(\mathbf{q}_1), Q(\mathbf{p}_2), Q(\mathbf{q}_2)) = (0, 0, 0, 0)$ holds by direct calculation.
The equalities $B_Q(\mathbf{p}_1, \mathbf{q}_1) = B_Q(\mathbf{p}_2, \mathbf{q}_2) = 1$ hold by direct calculation.
Moreover, $B_Q(\mathbf{p}_1, \mathbf{p}_2) = B_Q(\mathbf{p}_1, \mathbf{q}_2) = B_Q(\mathbf{q}_1, \mathbf{p}_2) = B_Q(\mathbf{q}_1, \mathbf{q}_2) = 0$.
The tuple $(\mathbf{p}_1, \mathbf{q}_1, \mathbf{p}_2, \mathbf{q}_2)$ is a symplectic basis with zero Q-value on each basis vector by these identities.
Two A-planes are quadratically isometric to two H-planes by this basis change.
The isometry $Q \cong Q^{(m)}_{\operatorname{Arf}_\beta(Q)}$ holds by cancelling pairs of A-planes.

(iii) For $P : \mathcal{U} \to \mathbb{F}_2$, put $D(P) := \sum_{\mathbf{x} \in \mathcal{U}} (-1)^{P(\mathbf{x})}$.
If $P' : \mathcal{U}' \to \mathbb{F}_2$ and $R(\mathbf{x}, \mathbf{y}) = P(\mathbf{x}) + P'(\mathbf{y})$, then $D(R) = D(P)D(P')$ by factoring the two finite sums.
$D(H) = 1 + 1 + 1 - 1 = 2$ and $D(A) = 1 - 1 - 1 - 1 = -2$ by direct calculation.
The form $Q_0^{(m)}$ is the orthogonal sum of m copies of H by definition.
The form $Q_1^{(m)}$ is the orthogonal sum of one A and $m - 1$ copies of H by definition.
The equality $D(Q_a^{(m)}) = (-1)^a 2^m$ holds by multiplicativity, $D(H) = 2$, and $D(A) = -2$. Put $N_a = |(Q_a^{(m)})^{-1}(0)|$.
By $D(Q_a^{(m)}) = N_a - (4^m - N_a) = 2N_a - 4^m$, $N_a = 2^{2m-1} + (-1)^a 2^{m-1}$.

(iv) A quadratic isometry preserves zero-set cardinality by bijectivity.
The nonisometry $Q_0^{(m)} \ncong Q_1^{(m)}$ holds by (iii), since the two cardinalities differ by $2^m > 0$.

(v) Fix $\beta, \gamma \in \mathscr{B}_Q$. If $m = 0$, both Arf sums are empty and equal to zero.
For $m \geq 1$, put $a = \operatorname{Arf}_\beta(Q)$ and $b = \operatorname{Arf}_\gamma(Q)$. The isometry $Q_a^{(m)} \cong Q_b^{(m)}$ holds by (ii).
The equality $a = b$ holds by (iv), because $a \neq b$ would identify this isometry with $Q_0^{(m)} \cong Q_1^{(m)}$.
The invariant $\operatorname{Arf}(Q)$ is well-defined by these two cases.

(vi) The isometry $Q \cong Q^{(m)}_{\operatorname{Arf}(Q)}$ holds by (ii) and (v).

(vii) Fix $a \in \mathbb{F}_2$. The map $Q_a^{(m)}$ is quadratic by direct calculation from its defining formula.
Its polar form has standard symplectic basis $\beta_a = (\mathbf{e}_i, \mathbf{f}_i)_{i=1}^m$ by the same calculation.

By definition, $Q_a^{(m)}(\mathbf{e}_1) = Q_a^{(m)}(\mathbf{f}_1) = a$. $(\forall\, i \in \mathbb{Z},\ 2 \le i \le m)\, Q_a^{(m)}(\mathbf{e}_i) = Q_a^{(m)}(\mathbf{f}_i) = 0$ by definition.
The equality $\mathrm{Arf}_{\beta_a}(Q_a^{(m)}) = a^2 = a$ holds by these values. By (v), $\mathrm{Arf}(Q_a^{(m)}) = a$.

Structural Commentary: Plane normalization and A-pair cancellation reduce Q; exponential sums separate the two residues.
Key Insight: The parity of anisotropic planes is the basis-independent quadratic residue. □

3.8 Orthogonal Isometries and Symmetry Groups

Throughout, all spaces are finite-dimensional over a fixed field $\mathbb{F}$; characteristic hypotheses are stated locally.

Proposition 3.8.1. *Isometries from Quadratic Values*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, and put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$. For $T \in \mathrm{GL}(\mathcal{V})$, $T \in \mathcal{O}(\mathcal{V}) \iff (\forall\, \mathbf{x} \in \mathcal{V})\, q(T\mathbf{x}) = q(\mathbf{x})$.

Proof.
(⇒) If $T \in \mathcal{O}(\mathcal{V})$, then $(\forall\, \mathbf{x} \in \mathcal{V})\, q(T\mathbf{x}) = \langle T\mathbf{x}, T\mathbf{x} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle = q(\mathbf{x})$.
(⇐) Assume $q \circ T = q$. For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, $2\langle T\mathbf{x}, T\mathbf{y} \rangle = q(T(\mathbf{x}+\mathbf{y})) - q(T\mathbf{x}) - q(T\mathbf{y})$ by linearity and polarization.
Thus $2\langle T\mathbf{x}, T\mathbf{y} \rangle = q(\mathbf{x}+\mathbf{y}) - q(\mathbf{x}) - q(\mathbf{y}) = 2\langle \mathbf{x}, \mathbf{y} \rangle$ by $q \circ T = q$ and polarization.
Hence $\langle T\mathbf{x}, T\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ by $\mathrm{char}(\mathbb{F}) \neq 2$, and $T \in \mathcal{O}(\mathcal{V})$ by $T \in \mathrm{GL}(\mathcal{V})$.

Structural Commentary: Diagonal preservation and polarization transfer quadratic invariance to every cross-pairing.
Key Insight: Invertibility of 2 makes the diagonal determine the symmetric form. □

Definition 3.8.2. *Special Orthogonal Group and Symmetries*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, and put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$.

(i) The *special orthogonal group* is $\mathrm{SO}(\mathcal{V}) := \{\sigma \in \mathcal{O}(\mathcal{V}) : \det(\sigma) = 1\}$; its elements are *rotations*.
(ii) If $q(\mathbf{u}) \neq 0$, the *symmetry* with normal line $\mathbb{F}\mathbf{u}$ is the map $\sigma_{\mathbf{u}} : \mathcal{V} \to \mathcal{V}$. It is defined by $(\forall\, \mathbf{x} \in \mathcal{V})\, \sigma_{\mathbf{u}}(\mathbf{x}) := \mathbf{x} - 2\frac{\langle \mathbf{x}, \mathbf{u} \rangle}{q(\mathbf{u})}\mathbf{u}$.

Proposition 3.8.3. *Symmetries and Equal Quadratic Values*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, and put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$.

(i) If $q(\mathbf{w}) \neq 0$, then $\sigma_{\mathbf{w}}\mathbf{w} = -\mathbf{w}$ and $\sigma_{\mathbf{w}}|_{(\mathbb{F}\mathbf{w})^\perp} = I$.
(ii) If $q(\mathbf{w}) \neq 0$, then $\sigma_{\mathbf{w}} \in \mathcal{O}(\mathcal{V})$ and $\sigma_{\mathbf{w}}^2 = I$.
(iii) If $q(\mathbf{u}) = q(\mathbf{v}) \neq 0$, then $q(\mathbf{u}-\mathbf{v}) \neq 0$ or $q(\mathbf{u}+\mathbf{v}) \neq 0$. In the respective cases, $\sigma_{\mathbf{u}-\mathbf{v}}\mathbf{u} = \mathbf{v}$ and $\sigma_{\mathbf{u}+\mathbf{v}}\mathbf{u} = -\mathbf{v}$.

Proof.
(i) If $q(\mathbf{w}) \neq 0$, then $\sigma_{\mathbf{w}}\mathbf{w} = \mathbf{w} - 2\mathbf{w} = -\mathbf{w}$. For $\mathbf{z} \in (\mathbb{F}\mathbf{w})^\perp$, $\langle \mathbf{z}, \mathbf{w} \rangle = 0$, so $\sigma_{\mathbf{w}}\mathbf{z} = \mathbf{z}$.
(ii) The formula for $\sigma_{\mathbf{w}}$ is linear, and $\mathcal{V} = \mathbb{F}\mathbf{w} \odot (\mathbb{F}\mathbf{w})^\perp$ by Theorem 3.4.2(ii).
Thus $\sigma_{\mathbf{w}}(\alpha\mathbf{w} + \mathbf{z}) = -\alpha\mathbf{w} + \mathbf{z}$ by (i), so $\sigma_{\mathbf{w}}^2 = I$ and $q \circ \sigma_{\mathbf{w}} = q$. Hence $\sigma_{\mathbf{w}} \in \mathcal{O}(\mathcal{V})$ by Proposition 3.8.1.
(iii) Put $\mathbf{a} = \mathbf{u} - \mathbf{v}$ and $\mathbf{b} = \mathbf{u} + \mathbf{v}$. Then $\langle \mathbf{a}, \mathbf{b} \rangle = q(\mathbf{u}) - q(\mathbf{v}) = 0$.
The conjunction $q(\mathbf{a}) = q(\mathbf{b}) = 0$ would imply $0 = q(\mathbf{a}+\mathbf{b}) = q(2\mathbf{u}) = 4q(\mathbf{u})$, a contradiction.
Therefore $q(\mathbf{a}) \neq 0$ or $q(\mathbf{b}) \neq 0$. If $q(\mathbf{a}) \neq 0$, then $q(\mathbf{a}) = 2\langle \mathbf{u}, \mathbf{a} \rangle$ and $\sigma_{\mathbf{a}}\mathbf{u} = \mathbf{u} - \mathbf{a} = \mathbf{v}$.
If $q(\mathbf{b}) \neq 0$, then $q(\mathbf{b}) = 2\langle \mathbf{u}, \mathbf{b} \rangle$ and $\sigma_{\mathbf{b}}\mathbf{u} = \mathbf{u} - \mathbf{b} = -\mathbf{v}$.

Structural Commentary: The normal-line decomposition verifies a symmetry; the sum-difference pair then aligns equal quadratic values.
Key Insight: Orthogonal sum-difference directions cannot both be isotropic. □

Theorem 3.8.4. *Orthogonal Transformations Are Products of Symmetries*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, and put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$. Then $\mathcal{O}(\mathcal{V}) = \langle \sigma_{\mathbf{u}} : \mathbf{u} \in \mathcal{V},\ q(\mathbf{u}) \neq 0 \rangle$.

Proof.
(i) *Induction Basis*: Induct on $n = \dim \mathcal{V}$. If $n = 0$, then $\mathcal{O}(\mathcal{V}) = \{I\}$, generated by the empty product.
If $n = 1$, fix $\mathbf{0} \neq \mathbf{u} \in \mathcal{V}$. For $\tau \in \mathcal{O}(\mathcal{V})$, write $\tau\mathbf{u} = \lambda\mathbf{u}$.
Nonsingularity implies $q(\mathbf{u}) \neq 0$, and $\lambda^2 = 1$ by $q(\tau\mathbf{u}) = q(\mathbf{u})$. Thus $\tau = I$ or $\tau = \sigma_{\mathbf{u}}$ by Proposition 3.8.3(i).
(ii) *Nonisotropic Line*: Assume $n > 1$. If $(\forall\, \mathbf{x} \in \mathcal{V})\, q(\mathbf{x}) = 0$, then the form is alternating.
It is zero by Proposition 3.1.2(ii), contrary to nonsingularity. Thus $(\exists\, \mathbf{u} \in \mathcal{V})\, q(\mathbf{u}) \neq 0$; fix such a $\mathbf{u}$.
(iii) *Line Alignment*: Fix $\tau \in \mathcal{O}(\mathcal{V})$.
Since $q(\tau\mathbf{u}) = q(\mathbf{u})$, fix a symmetry σ and $\varepsilon \in \{1, -1\}$ satisfying $\sigma\tau\mathbf{u} = \varepsilon\mathbf{u}$ by Proposition 3.8.3(iii).
Put $\rho = \sigma$ if $\varepsilon = 1$, and $\rho = \sigma_{\mathbf{u}}\sigma$ if $\varepsilon = -1$. Then $\rho\tau\mathbf{u} = \mathbf{u}$ by Proposition 3.8.3(i).
(iv) *Complement Reduction*: Put $\mathcal{W} = (\mathbb{F}\mathbf{u})^\perp$. For $\mathbf{x} \in \mathcal{W}$, $\langle \rho\tau\mathbf{x}, \mathbf{u} \rangle = \langle \rho\tau\mathbf{x}, \rho\tau\mathbf{u} \rangle = \langle \mathbf{x}, \mathbf{u} \rangle = 0$.
Hence $\rho\tau\mathcal{W} \subseteq \mathcal{W}$, and equality holds by injectivity and finite dimensionality.
$\mathcal{V} = \mathbb{F}\mathbf{u} \odot \mathcal{W}$ and $\dim \mathcal{W} = n - 1$ by Theorem 3.4.2(i),(ii). The space $\mathcal{W}$ is nonsingular by Theorem 3.4.2(iv).
(v) *Factor Lifting*: By induction, $(\rho\tau)|_{\mathcal{W}}$ is a product of symmetries of $\mathcal{W}$.
Each factor with normal $\mathbf{w} \in \mathcal{W}$ extends to $\sigma_{\mathbf{w}} \in \mathcal{O}(\mathcal{V})$, which fixes $\mathbb{F}\mathbf{u}$ and has the required restriction.
The lifted product equals $\rho\tau$ on $\mathbb{F}\mathbf{u} \odot \mathcal{W}$, so $\rho\tau$ is a product of symmetries.
The map ρ^{-1} is also such a product by Proposition 3.8.3(ii). Therefore $\tau = \rho^{-1}(\rho\tau)$ is a product of symmetries.

Structural Commentary: Symmetries align one nonisotropic line; induction factors the induced complement isometry.
Key Insight: A fixed nonisotropic line exposes a lower-dimensional nonsingular complement. □

Proposition 3.8.5. *Fixed Spaces of Isometries*
Let $\mathcal{V}$ be nonsingular and let $\tau \in \mathrm{Iso}(\mathcal{V})$. Then $\ker(I - \tau) = \mathrm{im}(I - \tau)^\perp$.

Proof. If $\mathbf{x} \in \ker(I - \tau)$, then $(\forall\, \mathbf{y} \in \mathcal{V})\ \langle \mathbf{x}, (I - \tau)\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle - \langle \tau\mathbf{x}, \tau\mathbf{y} \rangle = 0$. Hence $\ker(I - \tau) \subseteq \mathrm{im}(I - \tau)^\perp$.
$\dim(\mathrm{im}(I-\tau)^\perp) = \dim \mathcal{V} - \dim(\mathrm{im}(I-\tau)) = \dim(\ker(I-\tau))$ by Theorem 3.4.2(i) and Theorem 1.7.3(ii). Thus $\ker(I-\tau) = \mathrm{im}(I-\tau)^\perp$.
Structural Commentary: Form preservation makes fixed vectors orthogonal to displacements; dimension equality closes the inclusion.
Key Insight: Fixed directions are exactly those orthogonal to every displacement. □

Proposition 3.8.6. *Symmetries and Fixed Hyperplanes*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$, and let $\tau \in \mathcal{O}(\mathcal{V})$ with $\tau \neq I$.

(i) If $q(\mathbf{u}) \neq 0$, then $\ker(I - \sigma_{\mathbf{u}}) = (\mathbb{F}\mathbf{u})^\perp$ is a nonsingular hyperplane.

(ii) The condition $(\exists\, \mathbf{u} \in \mathcal{V})\,(q(\mathbf{u}) \neq 0 \wedge \tau = \sigma_{\mathbf{u}})$ *holds if and only if* $\ker(I - \tau)$ *is a nonsingular hyperplane.*

Proof.

(i) The map $\sigma_{\mathbf{u}}$ fixes $(\mathbb{F}\mathbf{u})^{\perp}$ pointwise and sends $\mathbf{u}$ to $-\mathbf{u}$ by Proposition 3.8.3(i).
The decomposition $\mathcal{V} = \mathbb{F}\mathbf{u} \odot (\mathbb{F}\mathbf{u})^{\perp}$ holds by Theorem 3.4.2(ii). Thus $\ker(I - \sigma_{\mathbf{u}}) = (\mathbb{F}\mathbf{u})^{\perp}$ by $\operatorname{char}(\mathbb{F}) \neq 2$.
This complement is a nonsingular hyperplane by Theorem 3.4.2(i),(iv).
(ii) ($\Rightarrow$) If $\tau = \sigma_{\mathbf{u}}$ with $q(\mathbf{u}) \neq 0$, then $\ker(I - \tau)$ is a nonsingular hyperplane by (i).
(ii) ($\Leftarrow$) Suppose $\mathcal{H} := \ker(I - \tau)$ is a nonsingular hyperplane, and put $\mathcal{L} = \mathcal{H}^{\perp}$.
Then $\mathcal{V} = \mathcal{H} \odot \mathcal{L}$ and $\dim \mathcal{L} = 1$ by Theorem 3.4.2(i),(ii). The line $\mathcal{L}$ is nonsingular by Theorem 3.4.2(iv).
For $\mathbf{x} \in \mathcal{L}$ and $\mathbf{h} \in \mathcal{H}$, $\langle \tau\mathbf{x}, \mathbf{h}\rangle = \langle \tau\mathbf{x}, \tau\mathbf{h}\rangle = \langle \mathbf{x}, \mathbf{h}\rangle = 0$. Hence $\tau\mathcal{L} \subseteq \mathcal{L}$, and equality holds by injectivity.
Fix $\mathbf{0} \neq \mathbf{u} \in \mathcal{L}$ and write $\tau\mathbf{u} = \lambda\mathbf{u}$. Then $q(\mathbf{u}) \neq 0$ by nonsingularity of $\mathcal{L}$, while $q(\tau\mathbf{u}) = q(\mathbf{u})$ implies $\lambda^2 = 1$.
The alternative $\lambda = 1$ would imply $\tau = I$ on $\mathcal{H} \odot \mathcal{L}$, so $\lambda = -1$ by $\operatorname{char}(\mathbb{F}) \neq 2$. Finally, $\mathcal{H} = (\mathbb{F}\mathbf{u})^{\perp}$ by Theorem 3.4.2(iii).
The map $\sigma_{\mathbf{u}}$ acts as I on $\mathcal{H}$ and as $-I$ on $\mathcal{L}$ by Proposition 3.8.3(i), as does τ. Therefore $\tau = \sigma_{\mathbf{u}}$ by $\mathcal{V} = \mathcal{H} \odot \mathcal{L}$.

Structural Commentary: Item (i) reads the fixed space; item (ii) reconstructs the flipped orthogonal line.
Key Insight: A symmetry is determined by its fixed hyperplane and normal line. □

Proposition 3.8.7. *No Unipotent Isometries in Anisotropic Spaces*
Let $\mathcal{V}$ *be nonsingular anisotropic and let* $\tau \in \operatorname{Iso}(\mathcal{V})$. *If* $(\exists\, m \in \mathbb{Z}_{>0})\,(\tau - I)^m = 0$, *then* $\tau = I$.

Proof.

(i) *Top Displacement*: Put $N = \tau - I$ and suppose $N \neq 0$. By nilpotence, fix maximal $k \geq 1$ with $N^k \neq 0$.
Then $N^{k+1} = 0$; choose $\mathbf{x} \in \mathcal{V}$ with $\mathbf{u} := N^k\mathbf{x} \neq \mathbf{0}$.
(ii) *Adjoint Expansion*: $N^* = \tau^{-1} - I$ by the adjoint identity and Proposition 3.3.10(iv).
Since $\tau = I + N$ and $N^{k+1} = 0$, $N^* = -N + N^2 - \cdots + (-1)^k N^k$ by direct multiplication.
(iii) *Polynomial Annihilation*: Put $P(t) = -1 + t - \cdots + (-1)^k t^{k-1}$, so $N^* = NP(N)$.
Hence $(N^*)^k N^k = N^{2k} P(N)^k = 0$ by $2k \geq k + 1$ and $N^{k+1} = 0$.
(iv) *Contradiction*: By k iterations of the adjoint identity, $\langle \mathbf{u}, \mathbf{u}\rangle = \langle N^k\mathbf{x}, N^k\mathbf{x}\rangle = \langle \mathbf{x}, (N^*)^k N^k\mathbf{x}\rangle = 0$.
This contradicts anisotropy because $\mathbf{u} \neq \mathbf{0}$. Therefore $N = 0$ and $\tau = I$.

Structural Commentary: The adjoint polynomial annihilates the top displacement, forcing its self-pairing to vanish.
Key Insight: The terminal nilpotent image is forced to be isotropic. □

Ambient extension and orthogonal cancellation remain available by Theorem 3.4.12 and Corollary 3.4.13.

4 Affine, Projective, and Convex Geometry

4.1 Affine Flats, Hulls, and Incidence

Throughout, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$; affine geometry treats translated subspaces without choosing an origin.

Definition 4.1.1. *Flats*
Let $\mathcal{S} \leq \mathcal{V}$ and $\mathbf{x} \in \mathcal{V}$, and put $\mathbf{x} + \mathcal{S} := \{\mathbf{x} + \mathbf{s} : \mathbf{s} \in \mathcal{S}\}$.

(i) The affine subset $\mathbf{x} + \mathcal{S}$ from Definition 2.2.1 is a *flat* in $\mathcal{V}$; $\mathcal{S}$ is its *base*, and $\mathbf{x}$ is a *representative*.
(ii) The *affine geometry* on $\mathcal{V}$ is $\mathcal{A}(\mathcal{V}) := \{\mathbf{y} + \mathcal{T} : \mathbf{y} \in \mathcal{V},\ \mathcal{T} \leq \mathcal{V}\}$, with $\dim(\mathcal{A}(\mathcal{V})) := \dim(\mathcal{V})$.
(iii) The *dimension* of $\mathbf{x} + \mathcal{S}$ is $\dim(\mathcal{S})$.
A 0-flat is a *point*, a 1-flat is a *line*, a 2-flat is a *plane*, and a codimension-one flat is a *hyperplane*.

Proposition 4.1.2. *Flat Bases and Representatives*
Let $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$ *and* $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.

(i) $\mathbf{x} + \mathcal{S} = \mathbf{y} + \mathcal{T} \Rightarrow \mathcal{S} = \mathcal{T}$.
(ii) If $X = \mathbf{x} + \mathcal{S}$ *and* $\mathbf{z} \in X$, *then* $X = \mathbf{z} + \mathcal{S}$.

Proof.

(i) Suppose $\mathbf{x} + \mathcal{S} = \mathbf{y} + \mathcal{T}$. Then $\mathbf{x} - \mathbf{y} \in \mathcal{T}$, and $(\forall\, \mathbf{s} \in \mathcal{S})\,(\mathbf{x} - \mathbf{y}) + \mathbf{s} \in \mathcal{T}$, so $\mathcal{S} \subseteq \mathcal{T}$. $\mathcal{T} \subseteq \mathcal{S}$ by interchanging the representations.
(ii) Write $\mathbf{z} = \mathbf{x} + \mathbf{s}_0$ with $\mathbf{s}_0 \in \mathcal{S}$. Then $\mathbf{z} + \mathcal{S} = \mathbf{x} + (\mathbf{s}_0 + \mathcal{S}) = \mathbf{x} + \mathcal{S} = X$.

Structural Commentary: Base equality holds by two inclusions; representative invariance holds by subspace translation.
Key Insight: A flat has one base but many representatives. □

Definition 4.1.3. *Parallel Flats*
For flats $X = \mathbf{x} + \mathcal{S}$ and $Y = \mathbf{y} + \mathcal{T}$, define $X \parallel Y \iff (\mathcal{S} \subseteq \mathcal{T} \vee \mathcal{T} \subseteq \mathcal{S})$. The relation is representation-independent by Proposition 4.1.2(i).

Proposition 4.1.4. *Basic Incidence of Flats*
Let $X = \mathbf{x} + \mathcal{S}$ *and* $Y = \mathbf{y} + \mathcal{T}$ *be flats in* $\mathcal{V}$, *and write* $\mathbf{w} + X := \{\mathbf{w} + \mathbf{u} : \mathbf{u} \in X\}$.

(i) $(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w} + X \subseteq Y) \iff \mathcal{S} \subseteq \mathcal{T}$.
(ii) $(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w} + X = Y) \iff \mathcal{S} = \mathcal{T}$.
(iii) $X \cap Y \neq \emptyset \Rightarrow (X \subseteq Y \iff \mathcal{S} \subseteq \mathcal{T})$.
(iv) $X \cap Y \neq \emptyset \Rightarrow (X = Y \iff \mathcal{S} = \mathcal{T})$.
(v) $X \parallel Y \Rightarrow (X \subseteq Y \vee Y \subseteq X \vee X \cap Y = \emptyset)$.
(vi) $X \parallel Y \iff [(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w} + X \subseteq Y)] \vee [(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w} + Y \subseteq X)]$.

Proof.

(i) ($\Rightarrow$) Fix $\mathbf{w}$ with $\mathbf{w} + X \subseteq Y$ and choose $\mathbf{z} \in \mathbf{w} + X$. Then $\mathbf{w} + X = \mathbf{z} + \mathcal{S}$ and $Y = \mathbf{z} + \mathcal{T}$ by Proposition 4.1.2(ii).
Hence $(\forall\, \mathbf{s} \in \mathcal{S})\,\mathbf{z} + \mathbf{s} \in \mathbf{z} + \mathcal{T}$, so $\mathcal{S} \subseteq \mathcal{T}$.
(i) ($\Leftarrow$) If $\mathcal{S} \subseteq \mathcal{T}$, then $(\mathbf{y} - \mathbf{x}) + X = \mathbf{y} + \mathcal{S} \subseteq Y$.
(ii) ($\Rightarrow$) If $\mathbf{w} + X = Y$, then $\mathcal{S} = \mathcal{T}$ by Proposition 4.1.2(i).

(*ii*) ($\Leftarrow$) If $\mathcal{S} = \mathcal{T}$, then $(\mathbf{y} - \mathbf{x}) + X = Y$.
(*iii*) ($\Rightarrow$) Fix $\mathbf{z} \in X \cap Y$. Then $X = \mathbf{z} + \mathcal{S}$, $Y = \mathbf{z} + \mathcal{T}$ by Proposition 4.1.2(ii). Thus $X \subseteq Y \Rightarrow (\forall\, \mathbf{s} \in \mathcal{S})\, \mathbf{z} + \mathbf{s} \in \mathbf{z} + \mathcal{T}$, so $\mathcal{S} \subseteq \mathcal{T}$.
(*iii*) ($\Leftarrow$) Fix $\mathbf{z} \in X \cap Y$. By Proposition 4.1.2(ii), $X = \mathbf{z} + \mathcal{S} \subseteq \mathbf{z} + \mathcal{T} = Y$ whenever $\mathcal{S} \subseteq \mathcal{T}$.
(*iv*) ($\Rightarrow$) If $X = Y$, then $\mathcal{S} = \mathcal{T}$ by Proposition 4.1.2(i).
(*iv*) ($\Leftarrow$) Fix $\mathbf{z} \in X \cap Y$. If $\mathcal{S} = \mathcal{T}$, then $X = \mathbf{z} + \mathcal{S} = \mathbf{z} + \mathcal{T} = Y$ by Proposition 4.1.2(ii).
(*v*) By Definition 4.1.3, $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$. If $X \cap Y \neq \emptyset$, then $X \subseteq Y$ or $Y \subseteq X$ by (iii); otherwise $X \cap Y = \emptyset$.
(*vi*) ($\Rightarrow$) By Definition 4.1.3 and (i), a translate of X lies in Y or a translate of Y lies in X.
(*vi*) ($\Leftarrow$) By (i), either translate-containment condition implies $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$, so $X \parallel Y$ by Definition 4.1.3.

Structural Commentary: Claims (i)–(iv) reduce incidence by common representatives; (v)–(vi) then characterize parallelism.
Key Insight: A common representative converts affine incidence into subspace incidence. □

Definition 4.1.5. *Affine Combinations*

(i) Let $n \in \mathbb{Z}_{>0}$, $\mathbf{p} = (\mathbf{p}_1, \dots, \mathbf{p}_n) \in \mathcal{V}^n$, and $\mathbf{r} = (r_1, \dots, r_n) \in \mathbb{F}^n$. The vector $\sum_{j=1}^{n} r_j \mathbf{p}_j$ is an *affine combination* of $\mathbf{p}$ iff $\sum_{j=1}^{n} r_j = 1$.

(ii) A nonempty $E \subseteq \mathcal{V}$ is *affine closed* iff $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{p}, \mathbf{r}) \in E^n \times \mathbb{F}^n)\, [\sum_{j=1}^{n} r_j = 1 \Rightarrow \sum_{j=1}^{n} r_j \mathbf{p}_j \in E]$.

(iii) A nonempty $E \subseteq \mathcal{V}$ is *two-affine closed* iff $(\forall\, \mathbf{x}, \mathbf{y} \in E)(\forall\, r \in \mathbb{F})\, r\mathbf{x} + (1 - r)\mathbf{y} \in E$.

(iv) If $\mathbf{x} \neq \mathbf{y}$, their affine line is $\overline{\mathbf{xy}} := \{r\mathbf{x} + (1 - r)\mathbf{y} : r \in \mathbb{F}\} = \mathbf{y} + \operatorname{span}(\mathbf{x} - \mathbf{y})$.
Thus a nonempty $E \subseteq \mathcal{V}$ is two-affine closed iff $(\forall\, \mathbf{x}, \mathbf{y} \in E)\, [\mathbf{x} \neq \mathbf{y} \Rightarrow \overline{\mathbf{xy}} \subseteq E]$.

Theorem 4.1.6. *Two-Point Affine Closure*
Let $\emptyset \neq E \subseteq \mathcal{V}$ and assume $\operatorname{char}(\mathbb{F}) \neq 2$. Then E is affine closed iff E is two-affine closed.

Proof.

($\Rightarrow$) Two-affine closure is the $n = 2$ instance of affine closure.
($\Leftarrow$) Assume that E is two-affine closed, fix $\mathbf{x}_0 \in E$, and put $\mathcal{S} := E - \mathbf{x}_0$. Fix $\mathbf{s}, \mathbf{t} \in \mathcal{S}$ and $c \in \mathbb{F}$.
Put $\mathbf{a} := \mathbf{x}_0 + c\mathbf{s} = c(\mathbf{x}_0 + \mathbf{s}) + (1 - c)\mathbf{x}_0$; then $\mathbf{a} \in E$ by two-affine closure.
Put $\mathbf{b} := \frac{1}{2}\mathbf{a} + \frac{1}{2}(\mathbf{x}_0 + \mathbf{t}) = \mathbf{x}_0 + \frac{1}{2}(c\mathbf{s} + \mathbf{t})$; then $\mathbf{b} \in E$ by two-affine closure.
The identity $2\mathbf{b} - \mathbf{x}_0 = \mathbf{x}_0 + c\mathbf{s} + \mathbf{t} \in E$ holds by two-affine closure with coefficient 2.
Hence $(\forall\, \mathbf{s}, \mathbf{t} \in \mathcal{S})(\forall\, c \in \mathbb{F})\, c\mathbf{s} + \mathbf{t} \in \mathcal{S}$, so $\mathcal{S} \leq \mathcal{V}$ by Theorem 1.2.2.

Fix $n \in \mathbb{Z}_{>0}$, $\mathbf{p} = (\mathbf{p}_j)_{j=1}^n \in E^n$, and $\mathbf{r} = (r_j)_{j=1}^n \in \mathbb{F}^n$ with $\sum_{j=1}^{n} r_j = 1$. Choose $\mathbf{s} = (\mathbf{s}_j)_{j=1}^n \in \mathcal{S}^n$ with $\mathbf{p} = (\mathbf{x}_0 + \mathbf{s}_j)_{j=1}^n$.

Then $\sum_{j=1}^{n} r_j \mathbf{p}_j = \mathbf{x}_0 + \sum_{j=1}^{n} r_j \mathbf{s}_j \in \mathbf{x}_0 + \mathcal{S} = E$. Thus E is affine closed.

Structural Commentary: Affine closure follows by translation, subspace closure, and reconstruction of finite combinations.
Key Insight: Midpoints and dilation by 2 recover vector addition. □

Counterexample 4.1.7. *Failure in Characteristic Two*
The characteristic restriction in Theorem 4.1.6 cannot be omitted. In $\mathbb{F}_2^2$, put $E := \{(0,0), (1,0), (0,1)\}$.
Since $r \in \{0, 1\}$, every two-point affine combination equals one of its endpoints, so E is two-affine closed.
However, $(1,1) = (0,0) + (1,0) + (0,1) \notin E$ is an affine combination because $1 + 1 + 1 = 1$ in $\mathbb{F}_2$.

Theorem 4.1.8. *Flats Are the Affine-Closed Sets*
Let $\emptyset \neq E \subseteq \mathcal{V}$. Then $E \in \mathcal{A}(\mathcal{V})$ iff E is affine closed.

Proof.

($\Rightarrow$) Write $E = \mathbf{x} + \mathcal{S}$ with $\mathcal{S} \leq \mathcal{V}$. Fix $n \in \mathbb{Z}_{>0}$, $\mathbf{p} = (\mathbf{p}_j)_{j=1}^n \in E^n$, and $\mathbf{r} = (r_j)_{j=1}^n \in \mathbb{F}^n$ with $\sum_{j=1}^{n} r_j = 1$.

Choose $\mathbf{s} = (\mathbf{s}_j)_{j=1}^n \in \mathcal{S}^n$ with $\mathbf{p} = (\mathbf{x} + \mathbf{s}_j)_{j=1}^n$. Then $\sum_{j=1}^{n} r_j \mathbf{p}_j = \mathbf{x} + \sum_{j=1}^{n} r_j \mathbf{s}_j \in E$, so E is affine closed.

($\Leftarrow$) Assume that E is affine closed, fix $\mathbf{x} \in E$, and put $\mathcal{S} := E - \mathbf{x}$. Fix $\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}$ and $c \in \mathbb{F}$, and choose $\mathbf{x}_1, \mathbf{x}_2 \in E$ with $\mathbf{s}_j = \mathbf{x}_j - \mathbf{x}$.
Then $c\mathbf{s}_1 + \mathbf{s}_2 + \mathbf{x} = c\mathbf{x}_1 + \mathbf{x}_2 - c\mathbf{x} \in E$ by affine closure, so $c\mathbf{s}_1 + \mathbf{s}_2 \in \mathcal{S}$. Hence $\mathcal{S} \leq \mathcal{V}$ by Theorem 1.2.2, and $E = \mathbf{x} + \mathcal{S} \in \mathcal{A}(\mathcal{V})$.

Structural Commentary: Forward closure is inherited from the base; reverse closure becomes linear after translation.
Key Insight: Affine closure becomes linear closure after translation. □

Theorem 4.1.9. *Flats Are the Two-Affine-Closed Sets*
Let $\emptyset \neq E \subseteq \mathcal{V}$ and assume $\operatorname{char}(\mathbb{F}) \neq 2$. Then $E \in \mathcal{A}(\mathcal{V})$ iff E is two-affine closed.

Proof.

($\Rightarrow$) If $E \in \mathcal{A}(\mathcal{V})$, then E is affine closed by Theorem 4.1.8 and two-affine closed by Theorem 4.1.6.
($\Leftarrow$) If E is two-affine closed, then E is affine closed by Theorem 4.1.6 and $E \in \mathcal{A}(\mathcal{V})$ by Theorem 4.1.8.

Structural Commentary: Each direction follows by composing the preceding affine-closure characterizations.
Key Insight: Binary affine closure detects translated subspaces when 2 is invertible. □

Definition 4.1.10. *Affine Hull and Affine Span*
Let $\emptyset \neq E \subseteq \mathcal{V}$.

(i) The *affine hull* is $\operatorname{affhull}(E) := \bigcap \{X \in \mathcal{A}(\mathcal{V}) : E \subseteq X\}$; the defining family contains $\mathcal{V}$.

(ii) The *affine span* is $\operatorname{affspan}(E) := \{\sum_{j=1}^{n} r_j \mathbf{p}_j : n \in \mathbb{Z}_{>0},\ (\mathbf{p}, \mathbf{r}) \in E^n \times \mathbb{F}^n,\ \sum_{j=1}^{n} r_j = 1\}$.

Theorem 4.1.11. *Affine Hull Formula*
Let $\emptyset \neq E \subseteq \mathcal{V}$, fix $\mathbf{x} \in E$, and put $\mathcal{S}_{\mathbf{x}} := \operatorname{span}(E - \mathbf{x})$.

(i) $\operatorname{affhull}(E) = \operatorname{affspan}(E) = \mathbf{x} + \mathcal{S}_{\mathbf{x}}$.
(ii) $\dim(\operatorname{affhull}(E)) = \dim(\mathcal{S}_{\mathbf{x}})$.

Proof.

(i) Fix $\mathbf{z} \in \operatorname{affspan}(E)$ and choose an affine-combination representation $\mathbf{z} = \sum_{j=1}^{n} r_j \mathbf{e}_j$. Then $\mathbf{z} = \mathbf{x} + \sum_{j=1}^{n} r_j(\mathbf{e}_j - \mathbf{x}) \in \mathbf{x} + \mathcal{S}_{\mathbf{x}}$.

Conversely, write $\mathbf{z} \in \mathbf{x} + \mathcal{S}_{\mathbf{x}}$ as $\mathbf{z} = \mathbf{x} + \sum_{j=1}^{n} a_j(\mathbf{e}_j - \mathbf{x})$, where $\mathbf{e}_j \in E$.

Then $\mathbf{z} = (1 - \sum_{j=1}^{n} a_j)\mathbf{x} + \sum_{j=1}^{n} a_j \mathbf{e}_j \in \operatorname{affspan}(E)$. Hence $\operatorname{affspan}(E) = \mathbf{x} + \mathcal{S}_{\mathbf{x}}$, a flat containing E.

Every flat containing E contains $\operatorname{affspan}(E)$ by Theorem 4.1.8, so this flat equals $\operatorname{affhull}(E)$ by Definition 4.1.10(i).

(ii) By (i) and Definition 4.1.1(iii), $\dim(\operatorname{affhull}(E)) = \dim(\mathbf{x} + \mathcal{S}_{\mathbf{x}}) = \dim(\mathcal{S}_{\mathbf{x}})$.

Structural Commentary: The hull formula follows by two inclusions; its dimension is read from the common base.
Key Insight: Differences from one point encode every affine direction. □

Lemma 4.1.12. *Intersections of Flats*
Let $\emptyset \neq I$ and let $X_i = \mathbf{x}_i + \mathcal{S}_i$ be flats indexed by $i \in I$. If $\mathbf{z} \in \bigcap_{i\in I} X_i$, then $\bigcap_{i\in I} X_i = \mathbf{z} + \bigcap_{i\in I} \mathcal{S}_i$.

Proof. By Proposition 4.1.2(ii), $(\forall\, i \in I)\, X_i = \mathbf{z} + \mathcal{S}_i$. Hence $(\forall\, \mathbf{u} \in \mathcal{V})\,[\mathbf{u} \in \bigcap_{i\in I} X_i \iff \mathbf{u} - \mathbf{z} \in \bigcap_{i\in I} \mathcal{S}_i]$. Therefore $\bigcap_{i\in I} X_i = \mathbf{z} + \bigcap_{i\in I} \mathcal{S}_i$.

Structural Commentary: The intersection formula follows by synchronizing representatives and intersecting their bases.
Key Insight: Common incidence synchronizes all affine origins. □

Theorem 4.1.13. *Lattice of Flats*
Let $\mathcal{A}_0(\mathcal{V}) := \mathcal{A}(\mathcal{V}) \cup \{\emptyset\}$, ordered by inclusion.

(i) $(\forall\, \mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V}))\ \bigwedge \mathcal{C} = \bigcap_{X\in\mathcal{C}} X$.

(ii) If $\mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V})$ and $U := \bigcup_{X\in\mathcal{C}} X$, then $U \neq \emptyset \Rightarrow \bigvee \mathcal{C} = \operatorname{affhull}(U)$, while $U = \emptyset \Rightarrow \bigvee \mathcal{C} = \emptyset$.

(iii) $\mathcal{A}_0(\mathcal{V})$ is a complete lattice.

Proof.

(i) Fix $\mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V})$ and put $M := \bigcap_{X\in\mathcal{C}} X$. If $\mathcal{C} = \emptyset$, then $M = \mathcal{V}$; if $M = \emptyset$, then $M \in \mathcal{A}_0(\mathcal{V})$.

If $\mathcal{C} \neq \emptyset$ and $M \neq \emptyset$, choose $\mathbf{z} \in M$ and let $\mathcal{S}_X$ be the unique base of X. Then $M = \mathbf{z} + \bigcap_{X\in\mathcal{C}} \mathcal{S}_X$ by Lemma 4.1.12, so M is a flat.

Thus $M \in \mathcal{A}_0(\mathcal{V})$. It is a lower bound of $\mathcal{C}$, and any lower bound is contained in M, so $M = \bigwedge \mathcal{C}$.

(ii) Fix $\mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V})$ and put $U := \bigcup_{X\in\mathcal{C}} X$. If $U = \emptyset$, then $\emptyset$ is the least upper bound of $\mathcal{C}$.

Assume $U \neq \emptyset$. Then $\operatorname{affhull}(U)$ is a flat containing each $X \in \mathcal{C}$ by Theorem 4.1.11.
If $B \in \mathcal{A}_0(\mathcal{V})$ is an upper bound of $\mathcal{C}$, then $U \subseteq B$; since $U \neq \emptyset$, the set B is a flat.
Thus $\operatorname{affhull}(U) \subseteq B$ by Definition 4.1.10(i), so $\operatorname{affhull}(U) = \bigvee \mathcal{C}$.

(iii) $\mathcal{A}_0(\mathcal{V})$ is complete by (i)–(ii) and Definition 5.3.4(iv).

Structural Commentary: Completeness follows by arbitrary intersection meets and affine-hull joins.
Key Insight: Empty incidence must be adjoined to close arbitrary meets. □

Convention 4.1.14. *Empty Affine Hull Convention*
Within $\mathcal{A}_0(\mathcal{V})$, set $\operatorname{affhull}(\emptyset) := \emptyset$; then Theorem 4.1.13(ii) has the uniform form $\bigvee \mathcal{C} = \operatorname{affhull}(\bigcup_{X\in\mathcal{C}} X)$.

Corollary 4.1.15. *Join of Two Flats*
Let $X = \mathbf{x} + \mathcal{S}$ and $Y = \mathbf{y} + \mathcal{T}$ be flats.

(i) $X \vee Y = \mathbf{x} + \operatorname{span}(\{\mathbf{x} - \mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T})$.

(ii) $X \cap Y \neq \emptyset \Rightarrow X \vee Y = \mathbf{x} + (\mathcal{S} + \mathcal{T})$.

Proof.

(i) Put $\mathcal{W} := \operatorname{span}((X \cup Y) - \mathbf{x})$. Then $X \vee Y = \mathbf{x} + \mathcal{W}$ by Theorems 4.1.13(ii) and 4.1.11(i).
Since $(X \cup Y) - \mathbf{x} = \mathcal{S} \cup ((\mathbf{y} - \mathbf{x}) + \mathcal{T})$, the space $\mathcal{W}$ contains $\mathcal{S}$ and $\mathbf{x} - \mathbf{y}$ by $\mathbf{0} \in \mathcal{T}$ and scalar closure.
$(\forall\, \mathbf{t} \in \mathcal{T})\, \mathbf{t} = ((\mathbf{y} - \mathbf{x}) + \mathbf{t}) - (\mathbf{y} - \mathbf{x}) \in \mathcal{W}$, so $\mathcal{T} \subseteq \mathcal{W}$. Hence $\operatorname{span}(\{\mathbf{x} - \mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T}) \subseteq \mathcal{W}$ by Theorem 1.2.5(iii).
The reverse inclusion holds by $(X \cup Y) - \mathbf{x} \subseteq \operatorname{span}(\{\mathbf{x} - \mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T})$ and Theorem 1.2.5(iii).

(ii) Fix $\mathbf{z} = \mathbf{x} + \mathbf{s} = \mathbf{y} + \mathbf{t} \in X \cap Y$. Then $\mathbf{x} - \mathbf{y} = \mathbf{t} - \mathbf{s} \in \mathcal{S} + \mathcal{T}$, so the span in (i) equals $\mathcal{S} + \mathcal{T}$.

Structural Commentary: The join formula follows by a hull expansion; intersection then absorbs the displacement.
Key Insight: Intersection absorbs the displacement into the two bases. □

Corollary 4.1.16. *Dimension of the Join of Two Flats*
Let $X = \mathbf{x} + \mathcal{S}$ and $Y = \mathbf{y} + \mathcal{T}$ be flats.

(i) $X \cap Y \neq \emptyset \Rightarrow \dim(X \vee Y) = \dim(X) + \dim(Y) - \dim(X \cap Y)$.

(ii) $X \cap Y = \emptyset \Rightarrow \dim(X \vee Y) = \dim(\mathcal{S} + \mathcal{T}) + 1$.

Proof.

(i) Fix $\mathbf{z} \in X \cap Y$. By Corollary 4.1.15(ii) and Lemma 4.1.12, the bases of $X \vee Y$ and $X \cap Y$ are $\mathcal{S} + \mathcal{T}$ and $\mathcal{S} \cap \mathcal{T}$, respectively.
Hence $\dim(X \vee Y) = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T})$ by Theorem 2.3.1, which is the stated identity by Definition 4.1.1(iii).

(ii) Assume $X \cap Y = \emptyset$ and put $\mathcal{W} := \mathcal{S} + \mathcal{T}$. By Corollary 4.1.15(i), the base of $X \vee Y$ is $\mathcal{U} := \operatorname{span}(\{\mathbf{x} - \mathbf{y}\} \cup \mathcal{W})$.
If $\mathbf{x} - \mathbf{y} \in \mathcal{W}$, choose $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$ with $\mathbf{x} - \mathbf{y} = \mathbf{s} + \mathbf{t}$. Then $\mathbf{x} - \mathbf{s} = \mathbf{y} + \mathbf{t} \in X \cap Y$, a contradiction.
If $a(\mathbf{x} - \mathbf{y}) \in \mathcal{W}$ with $a \neq 0$, then $\mathbf{x} - \mathbf{y} \in \mathcal{W}$ by scalar closure. Thus $\operatorname{span}(\mathbf{x} - \mathbf{y}) \cap \mathcal{W} = \{\mathbf{0}\}$ and $\mathcal{U} = \mathcal{W} \oplus \operatorname{span}(\mathbf{x} - \mathbf{y})$.
Hence $\dim(X \vee Y) = \dim(\mathcal{W}) + 1 = \dim(\mathcal{S} + \mathcal{T}) + 1$ by Corollary 2.3.2.

Structural Commentary: The dimension formulas follow by the subspace sum formula or one independent displacement.
Key Insight: Disjointness makes the representative displacement linearly independent. □

4.2 Affine Independence and Barycentric Coordinates

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$.

Definition 4.2.1. *Affinely Independent Sets*
A set $E \subseteq \mathcal{V}$ is *affinely independent* iff $E \neq \emptyset$ and $(\forall\, \mathbf{x} \in E)\,(E - \mathbf{x}) \setminus \{\mathbf{0}\}$ is linearly independent.

Theorem 4.2.2. *Affine Independence Criteria*
Let $\emptyset \neq E \subseteq \mathcal{V}$ and, for $F \subseteq \mathcal{V}$ and $n \in \mathbb{Z}_{>0}$, put $F^n_{\neq} := \{\mathbf{x} \in F^n : \mathbf{x}_j \neq \mathbf{x}_k \ (1 \leq j < k \leq n)\}$.
The following conditions are equivalent.

(i) E *is affinely independent.*
(ii) $(\forall\, \mathbf{x} \in E)\, \mathbf{x} \notin \operatorname{affhull}(E \setminus \{\mathbf{x}\})$, *where* $\operatorname{affhull}(\emptyset) = \emptyset$.
(iii) $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{x}_1, \ldots, \mathbf{x}_n) \in E^n_{\neq})(\forall\, \mathbf{r} \in \mathbb{F}^n)\,[\sum_{j=1}^{n} r_j \mathbf{x}_j = \mathbf{0} \wedge \sum_{j=1}^{n} r_j = 0 \Rightarrow \mathbf{r} = \mathbf{0}]$.
(iv) $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{x}_1, \ldots, \mathbf{x}_n) \in E^n_{\neq})(\forall\, \mathbf{r}, \mathbf{s} \in \mathbb{F}^n)\,[\sum_{j=1}^{n} r_j \mathbf{x}_j = \sum_{j=1}^{n} s_j \mathbf{x}_j \wedge \sum_{j=1}^{n} r_j = \sum_{j=1}^{n} s_j = 1 \Rightarrow \mathbf{r} = \mathbf{s}]$.

Proof.

(i) $\Rightarrow$ *(ii)* Fix $\mathbf{x} \in E$. If $E = \{\mathbf{x}\}$, then $\mathbf{x} \notin \operatorname{affhull}(\emptyset)$ by the empty-hull convention. Suppose $E \neq \{\mathbf{x}\}$ and $\mathbf{x} \in \operatorname{affhull}(E \setminus \{\mathbf{x}\})$.
By Theorem 4.1.11(i), $\mathbf{x} \in \operatorname{affspan}(E \setminus \{\mathbf{x}\})$. Collect the repeated points in an affine representation of $\mathbf{x}$.
Fix $m \in \mathbb{Z}_{>0}$ and $(\mathbf{y}_1, \ldots, \mathbf{y}_m) \in (E \setminus \{\mathbf{x}\})^m_{\neq}$ as its distinct points.
The collected coefficients form $\mathbf{r} \in \mathbb{F}^m$ with $\mathbf{x} = \sum_{j=1}^{m} r_j \mathbf{y}_j$ and $\sum_{j=1}^{m} r_j = 1$.
The relation $\sum_{j=1}^{m} r_j (\mathbf{y}_j - \mathbf{x}) = \mathbf{0}$ with $\mathbf{r} \neq \mathbf{0}$ holds by these identities, contradicting (i).

(ii) $\Rightarrow$ *(iii)* Fix $n \in \mathbb{Z}_{>0}$ and $(\mathbf{x}_1, \ldots, \mathbf{x}_n) \in E^n_{\neq}$. Let $\mathbf{r} \in \mathbb{F}^n$ satisfy $\sum_{j=1}^{n} r_j \mathbf{x}_j = \mathbf{0}$ and $\sum_{j=1}^{n} r_j = 0$.
If $\mathbf{r} \neq \mathbf{0}$, choose k with $r_k \neq 0$. The inequality $n \geq 2$ holds by the contradiction $r_k = \sum_{j=1}^{n} r_j = 0$ when $n = 1$.
The identities $\mathbf{x}_k = \sum_{j \neq k} (-\frac{r_j}{r_k}) \mathbf{x}_j$ and $\sum_{j \neq k} -\frac{r_j}{r_k} = 1$ hold by rearrangement.
$\mathbf{x}_k \in \operatorname{affhull}(E \setminus \{\mathbf{x}_k\})$ by Theorem 4.1.11(i), contrary to (ii). $\mathbf{r} = \mathbf{0}$ by contradiction.

(iii) $\Rightarrow$ *(iv)* Subtraction produces $\sum_{j=1}^{n} (r_j - s_j) \mathbf{x}_j = \mathbf{0}$ and $\sum_{j=1}^{n} (r_j - s_j) = 0$. Thus $\mathbf{r} = \mathbf{s}$ by (iii).

(iv) $\Rightarrow$ *(i)* Fix $\mathbf{x} \in E$, $m \in \mathbb{Z}_{>0}$, and $(\mathbf{y}_1, \ldots, \mathbf{y}_m) \in (E \setminus \{\mathbf{x}\})^m_{\neq}$. Let $\mathbf{a} \in \mathbb{F}^m$ satisfy $\sum_{j=1}^{m} a_j (\mathbf{y}_j - \mathbf{x}) = \mathbf{0}$.
The tuple $(\mathbf{y}_1, \ldots, \mathbf{y}_m, \mathbf{x})$ lies in $E^{m+1}_{\neq}$. Put $\mathbf{c} := (a_1, \ldots, a_m, -\sum_{j=1}^{m} a_j)$ and $\mathbf{b} := (0, \ldots, 0, 1)$.
Both $\mathbf{b}$ and $\mathbf{b} + \mathbf{c}$ have coefficient sum 1 and represent $\mathbf{x}$, so $\mathbf{c} = \mathbf{0}$ by (iv).
$\mathbf{a} = \mathbf{0}$ by the definition of $\mathbf{c}$, and E is affinely independent by the arbitrariness of all choices.

Structural Commentary: Pass from excluded-point hulls to zero-sum relations, then identify them with coordinate differences.
Key Insight: Affine representations differ by a zero-sum linear relation. □

Corollary 4.2.3. *Dimension Criterion for Affine Independence*
Let $E \subseteq \mathcal{V}$ be finite and nonempty with $|E| = n$. Then E is affinely independent iff $\dim(\operatorname{affhull}(E)) = n - 1$.

Proof.

$(\Rightarrow)$ Fix $\mathbf{x} \in E$ and put $B_{\mathbf{x}} := (E - \mathbf{x}) \setminus \{\mathbf{0}\}$.
The set $B_{\mathbf{x}}$ has $n - 1$ elements, is linearly independent by the forward hypothesis, and spans $\operatorname{span}(E - \mathbf{x})$.
The identity $\dim(\operatorname{affhull}(E)) = \dim(\operatorname{span}(E - \mathbf{x})) = n - 1$ holds by Theorem 4.1.11(ii).
$(\Leftarrow)$ Fix $\mathbf{x} \in E$ and put $B_{\mathbf{x}} := (E - \mathbf{x}) \setminus \{\mathbf{0}\}$. By Theorem 4.1.11(ii), $\dim(\operatorname{span}(E - \mathbf{x})) = n - 1 = |B_{\mathbf{x}}|$.
Any ordering of $B_{\mathbf{x}}$ is a basis of $\operatorname{span}(E - \mathbf{x})$ by Proposition 1.4.4(i).
$B_{\mathbf{x}}$ is linearly independent by the basis property, and E is affinely independent by the arbitrariness of $\mathbf{x}$.

Structural Commentary: Compare each translated difference set's cardinality with the dimension of its span.
Key Insight: Translation removes one point but preserves all affine directions. □

Definition 4.2.4. *Affine Bases and Barycentric Coordinates*
Let $A = \mathbf{x} + \mathcal{S}$ be an n-flat.

(i) A finite subset $\mathcal{B} \subseteq A$ is an *affine basis* of A iff $\mathcal{B}$ is affinely independent and $\operatorname{affspan}(\mathcal{B}) = A$.
(ii) A tuple $(\mathbf{p}_1, \ldots, \mathbf{p}_m)$ is an *ordered affine basis* of A iff its entries are distinct and $\{\mathbf{p}_1, \ldots, \mathbf{p}_m\}$ is an affine basis of A.
(iii) Let $(\mathbf{p}_1, \ldots, \mathbf{p}_m)$ be an ordered affine basis of A and $\mathbf{v} \in A$.
A vector $\mathbf{r} = (r_1, \ldots, r_m) \in \mathbb{F}^m$ is a *barycentric coordinate vector* of $\mathbf{v}$ iff $\mathbf{v} = \sum_{j=1}^{m} r_j \mathbf{p}_j$ and $\sum_{j=1}^{m} r_j = 1$.

Theorem 4.2.5. *Affine Bases and Barycentric Coordinates*
Let $A = \mathbf{x} + \mathcal{S}$ be an n-flat.

(i) If $(\mathbf{s}_1, \ldots, \mathbf{s}_n)$ is an ordered basis of $\mathcal{S}$, then $(\mathbf{x} + \mathbf{s}_1, \ldots, \mathbf{x} + \mathbf{s}_n, \mathbf{x})$ is an ordered affine basis of A.
(ii) If $\mathcal{B}$ is an affine basis of A, then $|\mathcal{B}| = n + 1$.
(iii) If $(\mathbf{p}_1, \ldots, \mathbf{p}_{n+1})$ is an ordered affine basis of A, then $(\forall\, \mathbf{v} \in A)(\exists!\, \mathbf{r} \in \mathbb{F}^{n+1})\,[\mathbf{v} = \sum_{j=1}^{n+1} r_j \mathbf{p}_j \wedge \sum_{j=1}^{n+1} r_j = 1]$.

Proof.

(i) Put $\mathbf{p}_j := \mathbf{x} + \mathbf{s}_j$ for $1 \le j \le n$ and $\mathbf{p}_{n+1} := \mathbf{x}$. The entries of $(\mathbf{p}_1, \dots, \mathbf{p}_{n+1})$ are distinct by linear independence of $(\mathbf{s}_1, \dots, \mathbf{s}_n)$.
Every zero-sum relation on a subtuple extends by zero coefficients to the full tuple.
For $\mathbf{a} \in \mathbb{F}^{n+1}$ with $\sum_{j=1}^{n+1} a_j \mathbf{p}_j = \mathbf{0}$ and $\sum_{j=1}^{n+1} a_j = 0$, the relation $\sum_{j=1}^{n} a_j \mathbf{s}_j = \mathbf{0}$ holds by substitution.
The identities $a_1 = \cdots = a_n = 0$ and $a_{n+1} = 0$ hold by independence of the $\mathbf{s}_j$ and the coefficient sum, respectively.
The set $\{\mathbf{p}_1, \dots, \mathbf{p}_{n+1}\}$ is affinely independent by Theorem 4.2.2(iii).
By Theorem 4.1.11(i), $\operatorname{affspan}(\{\mathbf{p}_1, \dots, \mathbf{p}_{n+1}\}) = \mathbf{x} + \operatorname{span}(\mathbf{s}_1, \dots, \mathbf{s}_n) = A$.
The tuple $(\mathbf{p}_1, \dots, \mathbf{p}_{n+1})$ is an ordered affine basis of A by the preceding distinctness, independence, and spanning identities.

(ii) Put $m := |\mathcal{B}|$. By Corollary 4.2.3, $\dim(\operatorname{affhull}(\mathcal{B})) = m - 1$. The identity $\operatorname{affhull}(\mathcal{B}) = A$ holds by $\operatorname{affspan}(\mathcal{B}) = A$ and Theorem 4.1.11(i).
The equality $|\mathcal{B}| = m = n + 1$ holds by $n = \dim(A) = \dim(\operatorname{affhull}(\mathcal{B})) = m - 1$.

(iii) Fix $\mathbf{v} \in A$ and put $\mathcal{P} := \{\mathbf{p}_1, \dots, \mathbf{p}_{n+1}\}$. $\mathbf{v} \in \operatorname{affspan}(\mathcal{P})$ by $\operatorname{affspan}(\mathcal{P}) = A$.
Choose $\ell \in \mathbb{Z}_{>0}$, $\mathbf{q} \in \mathcal{P}^\ell$, and $\mathbf{c} \in \mathbb{F}^\ell$ from this affine representation.
The chosen data satisfy $\mathbf{v} = \sum_{k=1}^{\ell} c_k \mathbf{q}_k$ and $\sum_{k=1}^{\ell} c_k = 1$. For $1 \le j \le n+1$, put $I_j := \{k \in \{1, \dots, \ell\} : \mathbf{q}_k = \mathbf{p}_j\}$ and $r_j := \sum_{k \in I_j} c_k$.
The sets $I_1, \dots, I_{n+1}$ partition $\{1, \dots, \ell\}$. The identities $\mathbf{v} = \sum_{j=1}^{n+1} r_j \mathbf{p}_j$ and $\sum_{j=1}^{n+1} r_j = 1$ hold by this partition.
The vector $\mathbf{r} := (r_1, \dots, r_{n+1})$ is unique by Theorem 4.2.2(iv).

Structural Commentary: Adjoin an affine origin, compare dimensions, then group coefficients by basis point.
Key Insight: Barycentric uniqueness is the affine-independence uniqueness criterion. □

Example 4.2.6. *Barycentric Coordinates on a Plane*
Let $\mathbf{x}, \mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ with $(\mathbf{v}_1, \mathbf{v}_2)$ linearly independent, and put $A := \mathbf{x} + \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2)$. Put $\mathbf{p}_1 := \mathbf{x} + \mathbf{v}_1$, $\mathbf{p}_2 := \mathbf{x} + \mathbf{v}_2$, and $\mathbf{p}_3 := \mathbf{x}$.
The tuple $(\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3)$ is an ordered affine basis of A by Theorem 4.2.5(i). For $(t_1, t_2) \in \mathbb{R}^2$, put $\mathbf{a} := \mathbf{x} + t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2$.
Then $\mathbf{a} = t_1 \mathbf{p}_1 + t_2 \mathbf{p}_2 + (1 - t_1 - t_2)\mathbf{p}_3$ and $t_1 + t_2 + (1 - t_1 - t_2) = 1$.
Thus $(t_1, t_2, 1 - t_1 - t_2)$ is the barycentric coordinate vector of $\mathbf{a}$ by Theorem 4.2.5(iii).

4.3 Affine Maps and Affine Groups

Throughout, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$. For $n \in \mathbb{Z}_{>0}$, put $\mathcal{C}_n := \{\mathbf{r} = (r_1, \dots, r_n) \in \mathbb{F}^n : \sum_{j=1}^{n} r_j = 1\}$.

Definition 4.3.1. *Affine Transformation*
A map $f : \mathcal{V} \to \mathcal{V}$ is *affine* iff $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathcal{V}^n)(\forall\, \mathbf{r} \in \mathcal{C}_n)\, f(\sum_{j=1}^{n} r_j \mathbf{x}_j) = \sum_{j=1}^{n} r_j f(\mathbf{x}_j)$.

Theorem 4.3.2. *Two-Point Test for Affine Maps*
Assume $\operatorname{char}(\mathbb{F}) \neq 2$. *A map $f : \mathcal{V} \to \mathcal{V}$ is affine iff* $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})(\forall\, r \in \mathbb{F})\, f(r\mathbf{x} + (1 - r)\mathbf{y}) = rf(\mathbf{x}) + (1 - r)f(\mathbf{y})$.

Proof.

(⇒) The identity holds by Definition 4.3.1 with $n = 2$ and $(r, 1 - r) \in \mathcal{C}_2$.

(⇐) Put $\Gamma_f := \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in \mathcal{V}\} \subseteq \mathcal{V} \times \mathcal{V}$.
The relation $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})(\forall\, r \in \mathbb{F})\, r(\mathbf{x}, f(\mathbf{x})) + (1 - r)(\mathbf{y}, f(\mathbf{y})) \in \Gamma_f$ holds by the hypothesis, so Γ_f is two-affine closed.
The nonempty set Γ_f is affine closed by Theorem 4.1.6 in $\mathcal{V} \times \mathcal{V}$.
Hence $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathcal{V}^n)(\forall\, \mathbf{r} \in \mathcal{C}_n)\, (\sum_{j=1}^{n} r_j \mathbf{x}_j, \sum_{j=1}^{n} r_j f(\mathbf{x}_j)) \in \Gamma_f$.
By the definition of Γ_f, this membership is equivalent to the identity in Definition 4.3.1; hence f is affine.

Structural Commentary: Transfer binary preservation to the graph, then promote its two-affine closure to affine closure.
Key Insight: Binary preservation is two-affine closure of the graph. □

Definition 4.3.3. *Translations*
For $\mathbf{a} \in \mathcal{V}$, the *translation by* $\mathbf{a}$ is the map $T_{\mathbf{a}} : \mathcal{V} \to \mathcal{V}$ defined by $T_{\mathbf{a}}(\mathbf{x}) = \mathbf{x} + \mathbf{a}$.

Theorem 4.3.4. *Affine Maps Are Linear Maps Followed by Translations*
A map $f : \mathcal{V} \to \mathcal{V}$ is affine iff $(\exists!\, (\mathbf{a}, \tau) \in \mathcal{V} \times \mathcal{L}(\mathcal{V}))\, [\mathbf{a} = f(\mathbf{0}) \wedge \tau = T_{-\mathbf{a}} \circ f \wedge f = T_{\mathbf{a}} \circ \tau]$.

Proof.

(⇒) Put $\mathbf{a} := f(\mathbf{0})$ and $\tau := T_{-\mathbf{a}} \circ f$.
For $r, s \in \mathbb{F}$ and $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, $\tau(r\mathbf{x} + s\mathbf{y}) = r\tau(\mathbf{x}) + s\tau(\mathbf{y})$ by Definition 4.3.1 with $(\mathbf{x}, \mathbf{y}, \mathbf{0})$ and $(r, s, 1 - r - s) \in \mathcal{C}_3$.
Thus $\tau \in \mathcal{L}(\mathcal{V})$, $\mathbf{a} = f(\mathbf{0})$, $\tau = T_{-\mathbf{a}} \circ f$, and $f = T_{\mathbf{a}} \circ \tau$.
If $f = T_{\mathbf{b}} \circ \sigma$ with $(\mathbf{b}, \sigma) \in \mathcal{V} \times \mathcal{L}(\mathcal{V})$, then $\mathbf{b} = f(\mathbf{0}) = \mathbf{a}$ and $\sigma = T_{-\mathbf{b}} \circ f = \tau$, so the pair is unique.

(⇐) Suppose $(\mathbf{a}, \tau) \in \mathcal{V} \times \mathcal{L}(\mathcal{V})$ satisfies $f = T_{\mathbf{a}} \circ \tau$.
For $n \in \mathbb{Z}_{>0}$, $(\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathcal{V}^n$, and $\mathbf{r} \in \mathcal{C}_n$, $f(\sum_{j=1}^{n} r_j \mathbf{x}_j) = \mathbf{a} + \sum_{j=1}^{n} r_j \tau(\mathbf{x}_j) = \sum_{j=1}^{n} r_j f(\mathbf{x}_j)$. Therefore f affine by Definition 4.3.1.

Structural Commentary: Translate $f(\mathbf{0})$ to zero, verify linearity, and identify both factors by evaluation at zero.
Key Insight: The value $f(\mathbf{0})$ determines the translation factor. □

Definition 4.3.5. *Affine Group Data*

(i) The *general linear group* is $\operatorname{GL}(\mathcal{V}) := \{\tau \in \mathcal{L}(\mathcal{V}) : (\exists\, \sigma \in \mathcal{L}(\mathcal{V}))\, [\sigma\tau = I_{\mathcal{V}} = \tau\sigma]\}$.
(ii) The *affine group* is $\operatorname{Aff}(\mathcal{V}) := \{T_{\mathbf{a}} \circ \tau : \mathbf{a} \in \mathcal{V},\ \tau \in \operatorname{GL}(\mathcal{V})\}$.
(iii) The *translation group* is $\operatorname{Trans}(\mathcal{V}) := \{T_{\mathbf{a}} : \mathbf{a} \in \mathcal{V}\}$.
(iv) The *linear-part map* is $\Lambda : \operatorname{Aff}(\mathcal{V}) \to \operatorname{GL}(\mathcal{V})$, defined by $\Lambda(T_{\mathbf{a}} \circ \tau) = \tau$.

Well-definedness holds by Theorem 4.3.4.

Corollary 4.3.6. *Bijective Affine Maps*
Let $\mathbf{a} \in \mathcal{V}$ and $\tau \in \mathcal{L}(\mathcal{V})$. Then $T_{\mathbf{a}} \circ \tau$ is bijective iff $\tau \in \mathrm{GL}(\mathcal{V})$.

Proof.
($\Rightarrow$) The map $T_{-\mathbf{a}}$ is bijective with inverse $T_{\mathbf{a}}$. Thus $\tau = T_{-\mathbf{a}} \circ (T_{\mathbf{a}} \circ \tau)$ is bijective, so $\tau \in \mathrm{GL}(\mathcal{V})$ by Proposition 1.6.7.
($\Leftarrow$) If $\tau \in \mathrm{GL}(\mathcal{V})$, then $\tau^{-1} \circ T_{-\mathbf{a}}$ is a two-sided inverse of $T_{\mathbf{a}} \circ \tau$. Hence $T_{\mathbf{a}} \circ \tau$ is bijective by Proposition 1.6.7.

Structural Commentary: Cancel the translation forward and compose the linear and translation inverses backward.
Key Insight: Translation does not affect bijectivity. □

Theorem 4.3.7. *Affine Group and Translations*
(i) $(\forall \mathbf{a}, \mathbf{b} \in \mathcal{V})(\forall \tau, \sigma \in \mathcal{L}(\mathcal{V}))\ (T_{\mathbf{a}} \circ \tau) \circ (T_{\mathbf{b}} \circ \sigma) = T_{\mathbf{a}+\tau(\mathbf{b})} \circ (\tau\sigma)$.
(ii) $\mathrm{GL}(\mathcal{V})$ *is a group under composition.*
(iii) $\mathrm{Aff}(\mathcal{V})$ *is a group under composition, with identity* $T_{\mathbf{0}} \circ I_{\mathcal{V}}$ *and* $(T_{\mathbf{a}} \circ \tau)^{-1} = T_{-\tau^{-1}(\mathbf{a})} \circ \tau^{-1}$.
(iv) $\mathrm{Trans}(\mathcal{V}) \trianglelefteq \mathrm{Aff}(\mathcal{V})$.
(v) Λ *is a group homomorphism.*
(vi) Λ *is surjective.*
(vii) $\ker \Lambda = \mathrm{Trans}(\mathcal{V})$.
(viii) $\mathrm{Aff}(\mathcal{V})/\mathrm{Trans}(\mathcal{V}) \cong \mathrm{GL}(\mathcal{V})$.

Proof.
(i) For $\mathbf{x} \in \mathcal{V}$, $((T_{\mathbf{a}} \circ \tau) \circ (T_{\mathbf{b}} \circ \sigma))(\mathbf{x}) = \mathbf{a} + \tau(\mathbf{b}) + (\tau\sigma)(\mathbf{x})$. Hence $(T_{\mathbf{a}} \circ \tau) \circ (T_{\mathbf{b}} \circ \sigma) = T_{\mathbf{a}+\tau(\mathbf{b})} \circ (\tau\sigma)$ by Assumption 3.1.2.
(ii) The identity $I_{\mathcal{V}}$ belongs to $\mathrm{GL}(\mathcal{V})$, and composition is associative. Closure holds by Proposition 1.6.6(i).
If $\tau \in \mathrm{GL}(\mathcal{V})$, then $\tau^{-1} \in \mathrm{GL}(\mathcal{V})$ by Definition 1.6.4 and Proposition 1.6.5. Thus $\mathrm{GL}(\mathcal{V})$ is a group under composition.
(iii) Closure holds by (i)–(ii), associativity is inherited from composition, and the identity is $T_{\mathbf{0}} \circ I_{\mathcal{V}}$.
For $g = T_{\mathbf{a}} \circ \tau$, put $h := T_{-\tau^{-1}(\mathbf{a})} \circ \tau^{-1} \in \mathrm{Aff}(\mathcal{V})$. The identity $gh = hg = T_{\mathbf{0}} \circ I_{\mathcal{V}}$ holds by (i), so $h = g^{-1}$ and $\mathrm{Aff}(\mathcal{V})$ is a group.
(iv) The identities $T_{\mathbf{a}}T_{\mathbf{b}} = T_{\mathbf{a}+\mathbf{b}}$, $T_{\mathbf{0}} = I_{\mathcal{V}}$, and $T_{\mathbf{a}}^{-1} = T_{-\mathbf{a}}$ hold by (i), so $\mathrm{Trans}(\mathcal{V}) \leq \mathrm{Aff}(\mathcal{V})$.
If $g = T_{\mathbf{a}} \circ \tau$, then $(\forall\, \mathbf{b} \in \mathcal{V})\, gT_{\mathbf{b}}g^{-1} = T_{\tau(\mathbf{b})}$ by (i). The map τ is bijective by Proposition 1.6.7, so $g\,\mathrm{Trans}(\mathcal{V})g^{-1} = \mathrm{Trans}(\mathcal{V})$.
(v) For $g = T_{\mathbf{a}} \circ \tau$ and $h = T_{\mathbf{b}} \circ \sigma$, $\Lambda(gh) = \tau\sigma = \Lambda(g)\Lambda(h)$ by (i).
(vi) $(\forall\, \tau \in \mathrm{GL}(\mathcal{V}))\, \Lambda(T_{\mathbf{0}} \circ \tau) = \tau$, so Λ is surjective.
(vii) The identity $\ker \Lambda = \{T_{\mathbf{a}} \circ I_{\mathcal{V}} : \mathbf{a} \in \mathcal{V}\} = \mathrm{Trans}(\mathcal{V})$ holds by Definition 4.3.5.
(viii) Put $N := \mathrm{Trans}(\mathcal{V})$ and define $\overline{\Lambda}(gN) := \Lambda(g)$. If $gN = hN$, then $h^{-1}g \in N = \ker \Lambda$.
The identities $\Lambda(h^{-1}) = \Lambda(h)^{-1}$ and $\Lambda(h)^{-1}\Lambda(g) = I_{\mathcal{V}}$ hold by (v), so $\Lambda(g) = \Lambda(h)$ and $\overline{\Lambda}$ is well defined.
For $g, h \in \mathrm{Aff}(\mathcal{V})$, $\overline{\Lambda}((gN)(hN)) = \Lambda(gh) = \Lambda(g)\Lambda(h) = \overline{\Lambda}(gN)\overline{\Lambda}(hN)$ by (v), so $\overline{\Lambda}$ is a homomorphism.
It is surjective by (vi). If $\overline{\Lambda}(gN) = \overline{\Lambda}(hN)$, then $\Lambda(h^{-1}g) = I_{\mathcal{V}}$ by (v), so $h^{-1}g \in N$ by (vii) and $gN = hN$.
Hence $\overline{\Lambda}$ is a bijective homomorphism and $\mathrm{Aff}(\mathcal{V})/N \cong \mathrm{GL}(\mathcal{V})$.

Structural Commentary: Product and inverse laws establish normal translations before Λ descends injectively to their quotient.
Key Insight: The kernel of the linear-part map is exactly translation. □

4.4 Linear Projective Geometry and Affine Completion

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$.

Definition 4.4.1. *Projective Geometry of a Vector Space*
(i) The *projective geometry* of $\mathcal{V}$ is the inclusion poset $\mathrm{Sub}(\mathcal{V})$; its elements are *projective subspaces*.
Their projective dimension is $\mathrm{pdim}(\mathcal{U}) := \dim(\mathcal{U}) - 1$ for $\mathcal{U} \leq \mathcal{V}$.
(ii) A *projective point* is a 1-dimensional projective subspace.
A *projective line* is a 2-dimensional projective subspace.

Theorem 4.4.2. *Linear Span Map from Affine to Projective Geometry*
Let $H = \mathbf{x} + \mathcal{K}$ be an affine hyperplane with $\mathbf{0} \notin H$. Put $\mathcal{A}(H) := \{X \in \mathcal{A}(\mathcal{V}) : X \subseteq H\}$.
Define $P : \mathcal{A}(H) \to \mathrm{Sub}(\mathcal{V})$ by $P(X) := \mathrm{span}(X)$. Define $D : \mathcal{A}(H) \to \mathrm{Sub}(\mathcal{K})$ by $D(X) := P(X) \cap \mathcal{K}$.
(i) $(\forall\, X \in \mathcal{A}(H))\, P(X) \cap H = X$. *Consequently, P is injective and* $(\forall \mathcal{U} \in \mathrm{im}\, P)\, P^{-1}(\mathcal{U}) = \mathcal{U} \cap H$.
(ii) $\mathrm{im}\, P = \{\mathcal{U} \leq \mathcal{V} : \mathcal{U} \not\subseteq \mathcal{K}\}$.
(iii) $(\forall\, X, Y \in \mathcal{A}(H))\, [X \subseteq Y \iff P(X) \subseteq P(Y)]$.
(iv) $(\forall\, \mathcal{C} \subseteq \mathcal{A}(H))\, [\mathcal{C} \neq \emptyset \wedge \bigcap \mathcal{C} \neq \emptyset \Rightarrow P(\bigcap \mathcal{C}) = \bigcap_{X \in \mathcal{C}} P(X)]$.
(v) $(\forall\, \mathcal{C} \subseteq \mathcal{A}(H))\, [\mathcal{C} \neq \emptyset \Rightarrow P(\bigvee \mathcal{C}) = \bigvee_{X \in \mathcal{C}} P(X) = \sum_{X \in \mathcal{C}} P(X)]$. *Here $\bigvee \mathcal{C}$ is computed in $\mathcal{A}_0(\mathcal{V})$.*
(vi) $(\forall\, X \in \mathcal{A}(H))\ \mathrm{pdim}(P(X)) = \dim(X)$.
(vii) $(\forall\, X, Y \in \mathcal{A}(H))\, [X \parallel Y \iff (D(X) \subseteq D(Y) \vee D(Y) \subseteq D(X))]$.

Proof.
(i) Fix $X = \mathbf{u} + \mathcal{S} \in \mathcal{A}(H)$. Then $\mathcal{S} \leq \mathcal{K}$ by Proposition 4.1.4(iii). If $\mathbf{z} \in H \cap \mathcal{K}$, then $\mathbf{x} = \mathbf{z} - (\mathbf{z} - \mathbf{x}) \in \mathcal{K}$ and $\mathbf{0} \in H$, a contradiction.
Thus $H \cap \mathcal{K} = \emptyset$ and $\mathbf{u} \notin \mathcal{K}$. Since $\mathbf{u} \in X$ and $(\forall\, \mathbf{s} \in \mathcal{S})\, \mathbf{s} = (\mathbf{u} + \mathbf{s}) - \mathbf{u}$, $P(X) = \mathbb{F}\mathbf{u} \oplus \mathcal{S}$.
If $\mathbf{z} \in P(X) \cap H$, write $\mathbf{z} = a\mathbf{u} + \mathbf{s}$ with $(a, \mathbf{s}) \in \mathbb{F} \times \mathcal{S}$. Then $(a-1)\mathbf{u} = (\mathbf{z} - \mathbf{u}) - \mathbf{s} \in \mathcal{K}$, so $a = 1$ and $\mathbf{z} \in X$.
Thus $P(X) \cap H = X$; hence P is injective. For $\mathcal{U} \in \mathrm{im}\, P$, the same identity is $P^{-1}(\mathcal{U}) = \mathcal{U} \cap H$.
(ii) The relation $(\forall\, X \in \mathcal{A}(H))\, P(X) \not\subseteq \mathcal{K}$ holds by the decomposition in (i). Conversely, fix $\mathcal{U} \leq \mathcal{V}$ with $\mathcal{U} \not\subseteq \mathcal{K}$ and choose $\mathbf{u} \in \mathcal{U} \setminus \mathcal{K}$.
Let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{K}$ be the quotient map. The quotient has dimension 1 by Theorem 2.2.9(ii), and $\pi(\mathbf{x}), \pi(\mathbf{u}) \neq \mathbf{0}$.
Choose $r \in \mathbb{F}^{\times}$ with $\pi(\mathbf{x}) = r\pi(\mathbf{u})$, and put $\mathbf{h} := r\mathbf{u} \in \mathcal{U} \cap H$. For $\mathbf{z} \in \mathcal{V}$, $\mathbf{z} \in \mathcal{U} \cap H \iff \mathbf{z} - \mathbf{h} \in \mathcal{U} \cap \mathcal{K}$.
Hence $\mathcal{U} \cap H = \mathbf{h} + (\mathcal{U} \cap \mathcal{K}) \in \mathcal{A}(H)$.Since $\pi(\mathbf{h}) \neq \mathbf{0}$ spans $\mathcal{V}/\mathcal{K}$, $(\forall\, \mathbf{w} \in \mathcal{U})(\exists\, a \in \mathbb{F})\, \pi(\mathbf{w}) = a\pi(\mathbf{h})$.
Thus $\mathcal{U} = \mathbb{F}\mathbf{h} + (\mathcal{U} \cap \mathcal{K})$ by the preceding quotient representation. This sum is direct by $\mathbf{h} \notin \mathcal{K}$.
Therefore $P(\mathcal{U} \cap H) = \mathcal{U}$, which proves the image formula.
(iii) ($\Rightarrow$) For $X, Y \in \mathcal{A}(H)$, $X \subseteq Y \Rightarrow P(X) \subseteq P(Y)$ by monotonicity of linear span.
(iii) ($\Leftarrow$) For $X, Y \in \mathcal{A}(H)$, $P(X) \subseteq P(Y) \Rightarrow X = P(X) \cap H \subseteq P(Y) \cap H = Y$ by (i).

(iv) Fix $\emptyset \neq \mathcal{C} \subseteq \mathcal{A}(H)$ with $\bigcap \mathcal{C} \neq \emptyset$. Choose $\mathbf{z} \in \bigcap \mathcal{C}$. For $X \in \mathcal{C}$, let $\mathcal{S}_X$ be its base. Then $X = \mathbf{z} + \mathcal{S}_X$ by Proposition 4.1.2(ii).
The identity $\bigcap \mathcal{C} = \mathbf{z} + \bigcap_{X \in \mathcal{C}} \mathcal{S}_X$ holds by Lemma 4.1.12. The identity $(\forall\, X \in \mathcal{C})\, P(X) = \mathbb{F}\mathbf{z} \oplus \mathcal{S}_X$ holds by (i).
Fix $\mathbf{w} \in \bigcap_{X \in \mathcal{C}} P(X)$ and choose $X_0 \in \mathcal{C}$. Write $\mathbf{w} = a\mathbf{z} + \mathbf{s}_0$ with $\mathbf{s}_0 \in \mathcal{S}_{X_0}$. For $X \in \mathcal{C}$, write $\mathbf{w} = b_X \mathbf{z} + \mathbf{s}_X$ with $\mathbf{s}_X \in \mathcal{S}_X$.
Then $(a - b_X)\mathbf{z} = \mathbf{s}_X - \mathbf{s}_0 \in \mathcal{K}$, so $a = b_X$. Hence $\mathbf{w} - a\mathbf{z} \in \bigcap_{X \in \mathcal{C}} \mathcal{S}_X$.
Conversely, $(\forall\, X \in \mathcal{C})\, \mathbb{F}\mathbf{z} \oplus \bigcap_{Y \in \mathcal{C}} \mathcal{S}_Y \subseteq P(X)$. Therefore $\bigcap_{X \in \mathcal{C}} P(X) = \mathbb{F}\mathbf{z} \oplus \bigcap_{X \in \mathcal{C}} \mathcal{S}_X = P(\bigcap \mathcal{C})$.

(v) Fix $\emptyset \neq \mathcal{C} \subseteq \mathcal{A}(H)$ and put $E := \bigcup \mathcal{C}$. Since H is an upper bound, $\bigvee \mathcal{C} = \operatorname{affhull}(E) \subseteq H$ by Theorem 4.1.13(ii).
The inclusion $\operatorname{affhull}(E) = \operatorname{affspan}(E) \subseteq \operatorname{span}(E)$ holds by Theorem 4.1.11(i) and Definition 4.1.10(ii).
Since $E \subseteq \operatorname{affhull}(E)$, the reverse inclusion holds after taking linear spans.
Thus $P(\bigvee \mathcal{C}) = \operatorname{span}(E) = \sum_{X \in \mathcal{C}} P(X)$. This sum is $\bigvee_{X \in \mathcal{C}} P(X)$ by Theorem 2.1.7(i).

(vi) If $X = \mathbf{u} + \mathcal{S} \in \mathcal{A}(H)$, then $P(X) = \mathbb{F}\mathbf{u} \oplus \mathcal{S}$ by (i). Hence $\operatorname{pdim}(P(X)) = \dim(\mathbb{F}\mathbf{u} \oplus \mathcal{S}) - 1 = \dim \mathcal{S} = \dim X$.

(vii) ($\Rightarrow$) Fix $X = \mathbf{u} + \mathcal{S}, Y = \mathbf{v} + \mathcal{T} \in \mathcal{A}(H)$. The identities $D(X) = \mathcal{S}$ and $D(Y) = \mathcal{T}$ hold by (i).
Hence $X \parallel Y \Rightarrow D(X) \subseteq D(Y) \vee D(Y) \subseteq D(X)$ by Definition 4.1.3.

(vii) ($\Leftarrow$) Fix $X = \mathbf{u} + \mathcal{S}, Y = \mathbf{v} + \mathcal{T} \in \mathcal{A}(H)$. The identities $D(X) = \mathcal{S}$ and $D(Y) = \mathcal{T}$ hold by (i).
Thus $D(X) \subseteq D(Y) \vee D(Y) \subseteq D(X)$ implies $\mathcal{S} \subseteq \mathcal{T} \vee \mathcal{T} \subseteq \mathcal{S}$. Hence $X \parallel Y$ by Definition 4.1.3.

Structural Commentary: Trace–span inversion transports incidence and joins; intersection with $\mathcal{K}$ extracts each affine direction.
Key Insight: Codimension-one transversality makes the quotient coordinate unique. □

Perspective 4.4.3. *Projective Completion of an Affine Hyperplane*
Let $H = \mathbf{x} + \mathcal{K}$ be an affine hyperplane with $\mathbf{0} \notin H$. By Theorem 4.4.2, each affine flat $X \subseteq H$ corresponds to $P(X) = \operatorname{span}(X)$.
Its direction is $D(X) = P(X) \cap \mathcal{K}$. If $\dim H = 2$, distinct parallel affine lines X, Y satisfy $\dim D(X) = 1$ and $D(X) = D(Y)$.
This common projective point is their direction at infinity.
This completion is an incidence dictionary; the theorem, not the geometric picture, owns its formal claims.

Figure 37 illustrates the plane case.
Parallel affine lines are disjoint in H; their projective spans meet in the shared direction $D \leq \mathcal{K}$.

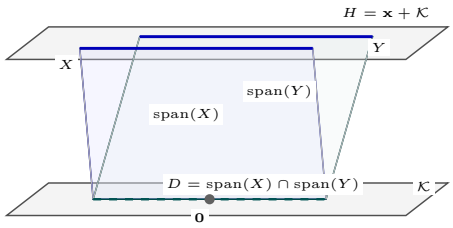

Figure 37: The affine plane $H = \mathbf{x} + \mathcal{K}$ contains the parallel lines X, Y. Their shaded spans meet along the dashed subspace $D = P(X) \cap \mathcal{K} = P(Y) \cap \mathcal{K}$. This projective point is their common direction at infinity.

4.5 Convex Combinations, Cones, and Hulls

Definition 4.5.1. *Convex Combinations, Cones, and Hulls*
Let $n, k \in \mathbb{Z}_{>0}$ and $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$. Put $\Delta_{k-1} := \{\mathbf{t} = (t_1, \dots, t_k) \in [0, \infty)^k : \sum_{i=1}^{k} t_i = 1\}$.

(i) The *convex combinations* of $\mathbf{x}_1, \dots, \mathbf{x}_k$ form $\{\sum_{i=1}^{k} t_i \mathbf{x}_i : \mathbf{t} \in \Delta_{k-1}\}$.
(ii) A set $C \subseteq \mathbb{R}^n$ is *convex* iff $(\forall\, \mathbf{x}, \mathbf{y} \in C)(\forall\, t \in [0,1])\, t\mathbf{x} + (1-t)\mathbf{y} \in C$.
(iii) A set $K \subseteq \mathbb{R}^n$ is a *cone* iff $(\forall\, \mathbf{x} \in K)(\forall\, a \in [0, \infty))\, a\mathbf{x} \in K$.
(iv) A *convex cone* is a subset of $\mathbb{R}^n$ that is both convex and a cone.
(v) For $S := \{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, let $\mathcal{F}_S$ be the family of convex subsets of $\mathbb{R}^n$ containing S.
The *convex hull* of S is $\operatorname{conv}(S) := \bigcap \mathcal{F}_S$; the family $\mathcal{F}_S$ contains $\mathbb{R}^n$.

Definition 4.5.2. *Euclidean Topological Scope*
Let $d \in \mathbb{Z}_{>0}$.

(i) For $\mathbf{a} \in \mathbb{R}^d$ and $r > 0$, put $B_r(\mathbf{a}) := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x} - \mathbf{a}\| < r\}$.
(ii) A set $G \subseteq \mathbb{R}^d$ is *open* iff $(\forall\, \mathbf{x} \in G)(\exists\, r > 0)\, B_r(\mathbf{x}) \subseteq G$, and *closed* iff $\mathbb{R}^d \setminus G$ is open.
(iii) A set $K \subseteq \mathbb{R}^d$ is *compact* iff every cover of K by open subsets of $\mathbb{R}^d$ has a finite subcover.

Proposition 4.5.3. *Finite Convex Hulls*
Let $n, k \in \mathbb{Z}_{>0}$, $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$, and $S := \{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

(i) $\operatorname{conv}(S) = \left\{ \sum_{i=1}^{k} t_i \mathbf{x}_i : \mathbf{t} \in \Delta_{k-1} \right\}$.
(ii) $\operatorname{conv}(S)$ *is compact.*

Proof. Define $T : \mathbb{R}^k \to \mathbb{R}^n$ by $T(\mathbf{t}) := \sum_{i=1}^{k} t_i \mathbf{x}_i$, and put $C := T(\Delta_{k-1})$.

(i) $(\forall\, i \in \{1, \dots, k\})\, [\mathbf{e}_i \in \Delta_{k-1} \wedge T(\mathbf{e}_i) = \mathbf{x}_i]$, so $S \subseteq C$. Let $K \subseteq \mathbb{R}^n$ be convex with $S \subseteq K$.
For $1 \leq q \leq k$, put $C_q := \{\sum_{i=1}^{q} t_i \mathbf{x}_i : \mathbf{t} \in \Delta_{q-1}\}$. We prove $C_q \subseteq K$ by induction on q. The base case is $C_1 = \{\mathbf{x}_1\} \subseteq K$.
Fix $2 \leq q \leq k$ and assume $C_{q-1} \subseteq K$. Fix $\mathbf{t} \in \Delta_{q-1}$ and put $\mathbf{z} := \sum_{i=1}^{q} t_i \mathbf{x}_i \in C_q$.

If $t_q = 1$, then $(\forall\, i < q)\, t_i = 0$ by nonnegativity, and $\mathbf{z} = \mathbf{x}_q \in K$. If $t_q < 1$, put $s := 1 - t_q > 0$ and $\mathbf{y} := \sum_{i=1}^{q-1} \frac{t_i}{s}\mathbf{x}_i$.

Then $(\frac{t_1}{s}, \dots, \frac{t_{q-1}}{s}) \in \Delta_{q-2}$ by $\mathbf{t} \in \Delta_{q-1}$ and $s = 1 - t_q$, so $\mathbf{y} \in C_{q-1}$.
Then $\mathbf{z} = s\mathbf{y} + t_q\mathbf{x}_q \in K$ by the induction hypothesis and convexity. Hence $C = C_k \subseteq K$. Thus $C \subseteq \operatorname{conv}(S)$ by Definition 4.5.1(v).
Conversely, $t\mathbf{r} + (1-t)\mathbf{s} \in \Delta_{k-1}$ for $\mathbf{r}, \mathbf{s} \in \Delta_{k-1}$ and $t \in [0,1]$ by nonnegativity and coefficient sums.
For $\mathbf{r}, \mathbf{s} \in \Delta_{k-1}$ and $t \in [0,1]$, $T(t\mathbf{r} + (1-t)\mathbf{s}) = tT(\mathbf{r}) + (1-t)T(\mathbf{s}) \in C$; hence C is convex.
Since $S \subseteq C$, $\operatorname{conv}(S) \subseteq C$ by Definition 4.5.1(v), and therefore $\operatorname{conv}(S) = C$.

(ii)

(a) *Coefficient compactness*: Let $\mathcal{U}$ be an open cover of Δ_{k-1} with no finite subcover, and put $Q_0 := [0,1]^k$.
Recursively, after Q_m is chosen, partition it into 2^k equal closed subcubes.
Choose Q_{m+1} among them so that $\Delta_{k-1} \cap Q_{m+1}$ has no finite $\mathcal{U}$-subcover.
At least one such subcube exists; otherwise finite subcovers of the 2^k intersections combine into a finite subcover of $\Delta_{k-1} \cap Q_m$.
Thus $(Q_m)_{m\ge 0}$ is nested, Q_m has side length 2^{-m}, and $\Delta_{k-1} \cap Q_m \neq \emptyset$.

Write $Q_m = \prod_{i=1}^{k} [a_{m,i}, b_{m,i}]$ and put $t_i := \sup\{a_{m,i} : m \ge 0\}$.

Nestedness makes $(a_{m,i})_{m\ge 0}$ nondecreasing, and $b_{m,i}$ is an upper bound of $\{a_{\ell,i} : \ell \ge 0\}$.
Hence $a_{m,i} \le t_i \le b_{m,i}$, so $\mathbf{t} := (t_1, \dots, t_k) \in \bigcap_{m\ge 0} Q_m$. For $m \ge 0$, choose $\mathbf{t}_m \in \Delta_{k-1} \cap Q_m$.

Then $\|\mathbf{t}_m - \mathbf{t}\| \le \sqrt{k}\, 2^{-m}$, so $\mathbf{t}_m \to \mathbf{t}$. Since $\mathbf{t} \in Q_0$, its coordinates are nonnegative.

The equality $\sum_{i=1}^{k} t_i = 1$ holds by $\left|\sum_{i=1}^{k} t_i - 1\right| \le k\, 2^{-m} \to 0$, so $\mathbf{t} \in \Delta_{k-1}$. Choose $U \in \mathcal{U}$ and $r > 0$ with $\mathbf{t} \in B_r(\mathbf{t}) \subseteq U$.

Choose m with $\sqrt{k}\, 2^{-m} < r$; then $Q_m \subseteq B_r(\mathbf{t}) \subseteq U$, contrary to the choice of Q_m. Therefore Δ_{k-1} is compact.

(b) *Open inverse images*: Put $L := 1 + \sum_{i=1}^{k} \|\mathbf{x}_i\| > 0$. For $\mathbf{r}, \mathbf{s} \in \mathbb{R}^k$, $\|T(\mathbf{r}) - T(\mathbf{s})\| \le \sum_{i=1}^{k} |r_i - s_i|\, \|\mathbf{x}_i\| \le L\, \|\mathbf{r} - \mathbf{s}\|$.

If $V \subseteq \mathbb{R}^n$ is open and $\mathbf{t} \in T^{-1}(V)$, choose $r > 0$ with $B_r(T(\mathbf{t})) \subseteq V$. Then $B_{\frac{r}{L}}(\mathbf{t}) \subseteq T^{-1}(V)$, so $T^{-1}(V)$ is open.

(c) *Finite image cover*: Let $\mathcal{V}$ be an open cover of C. By (b), $\{T^{-1}(V) : V \in \mathcal{V}\}$ consists of open subsets of $\mathbb{R}^k$.

The family covers Δ_{k-1} by $C = T(\Delta_{k-1})$. By (a), choose $m \in \mathbb{Z}_{>0}$ and $V_1, \dots, V_m \in \mathcal{V}$ with $\Delta_{k-1} \subseteq \bigcup_{j=1}^{m} T^{-1}(V_j)$.

Then $C \subseteq \bigcup_{j=1}^{m} V_j$, so $C = \operatorname{conv}(S)$ is compact by (i).

Structural Commentary: Induction identifies the hull; cube bisection and the coefficient map establish compactness.
Key Insight: A finite convex hull is the image of a compact coefficient simplex. □

Proposition 4.5.4. *Sums of Convex and Closed Sets*
Let $n \in \mathbb{Z}_{>0}$ and $X, Y \subseteq \mathbb{R}^n$. Put $X + Y := \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in X,\ \mathbf{y} \in Y\}$.

(i) If X and Y are convex, then $X + Y$ is convex.
(ii) If X is compact and Y is closed, then $X + Y$ is closed.

Proof.

(i) Assume X and Y are convex. Fix $\mathbf{z}_0, \mathbf{z}_1 \in X + Y$ and $t \in [0,1]$. For $r \in \{0,1\}$, choose $(\mathbf{x}_r, \mathbf{y}_r) \in X \times Y$ with $\mathbf{z}_r = \mathbf{x}_r + \mathbf{y}_r$.
By convexity, $t\mathbf{x}_0 + (1-t)\mathbf{x}_1 \in X$ and $t\mathbf{y}_0 + (1-t)\mathbf{y}_1 \in Y$. Hence $t\mathbf{z}_0 + (1-t)\mathbf{z}_1 = \big(t\mathbf{x}_0 + (1-t)\mathbf{x}_1\big) + \big(t\mathbf{y}_0 + (1-t)\mathbf{y}_1\big) \in X + Y$.

(ii) Assume X compact, Y closed. If $X = \emptyset$, then $X + Y = \emptyset$. Its complement $\mathbb{R}^n$ is open by $(\forall\, \mathbf{u} \in \mathbb{R}^n)\, B_1(\mathbf{u}) \subseteq \mathbb{R}^n$, so $\emptyset$ is closed.
Suppose $X \neq \emptyset$ and fix $\mathbf{z} \notin X + Y$. For $\mathbf{x} \in X$, choose $r_{\mathbf{x}} > 0$ with $B_{2r_{\mathbf{x}}}(\mathbf{z} - \mathbf{x}) \subseteq \mathbb{R}^n \setminus Y$ by openness of $\mathbb{R}^n \setminus Y$.
Fix $\mathbf{x} \in X$ and $\mathbf{u} \in B_{r_{\mathbf{x}}}(\mathbf{x})$, and put $\varepsilon := r_{\mathbf{x}} - \|\mathbf{u} - \mathbf{x}\| > 0$. Then $B_\varepsilon(\mathbf{u}) \subseteq B_{r_{\mathbf{x}}}(\mathbf{x})$ by Theorem 1.1.14(i).

Thus the balls $B_{r_{\mathbf{x}}}(\mathbf{x})$ form an open cover of X. Choose $m \in \mathbb{Z}_{>0}$ and $\mathbf{x}_1, \dots, \mathbf{x}_m \in X$ with $X \subseteq \bigcup_{j=1}^{m} B_{r_{\mathbf{x}_j}}(\mathbf{x}_j)$ by compactness.

Put $\delta := \min_{1\le j\le m} r_{\mathbf{x}_j} > 0$. Suppose $\mathbf{w} = \mathbf{x} + \mathbf{y} \in B_\delta(\mathbf{z}) \cap (X + Y)$ with $(\mathbf{x}, \mathbf{y}) \in X \times Y$. Choose j with $\mathbf{x} \in B_{r_{\mathbf{x}_j}}(\mathbf{x}_j)$.
By Theorem 1.1.14(i), $\|\mathbf{y} - (\mathbf{z} - \mathbf{x}_j)\| \le \|\mathbf{w} - \mathbf{z}\| + \|\mathbf{x} - \mathbf{x}_j\| < \delta + r_{\mathbf{x}_j} \le 2r_{\mathbf{x}_j}$. This contradicts $\mathbf{y} \in Y$, so $B_\delta(\mathbf{z}) \subseteq \mathbb{R}^n \setminus (X + Y)$.
Since $\mathbf{z} \notin X + Y$ was arbitrary, $\mathbb{R}^n \setminus (X + Y)$ is open and $X + Y$ is closed.

Structural Commentary: Part (i) separates summands; compactness makes the exclusion radii uniform in (ii).
Key Insight: A finite subcover converts local separation into one neighborhood radius. □

4.6 Hyperplanes, Nearest Points, and Separation

Definition 4.6.1. *Hyperplanes, Half-Spaces, and Separation*
Let $n \in \mathbb{Z}_{>0}$, $\mathbf{N} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, and $b \in \mathbb{R}$.

(i) The *affine hyperplane* with normal $\mathbf{N}$ and level b is $\mathcal{H}(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x}\rangle = b\}$.
Its affine decomposition is $\mathcal{H}(\mathbf{N}, b) = \frac{b}{\|\mathbf{N}\|^2}\mathbf{N} + \operatorname{span}(\mathbf{N})^\perp$.

(ii) The *linear hyperplane* with normal $\mathbf{N}$ is $\mathcal{H}(\mathbf{N}, 0) = \operatorname{span}(\mathbf{N})^\perp$.

(iii) The *closed positive half-space* is $\mathcal{H}_+(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x}\rangle \ge b\}$.
The *closed negative half-space* is $\mathcal{H}_-(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x}\rangle \le b\}$.
Replacing $\ge, \le$ by $>, <$ defines the corresponding open half-spaces.

(iv) For $X \subseteq \mathbb{R}^n$ and $\star \in \{<, \le, >, \ge\}$, the *set-inequality convention* is $\langle \mathbf{N}, X\rangle \star b \iff (\forall\, \mathbf{x} \in X)\ \langle \mathbf{N}, \mathbf{x}\rangle \star b$.
Symmetrically, $b \star \langle \mathbf{N}, X\rangle \iff (\forall\, \mathbf{x} \in X)\, b \star \langle \mathbf{N}, \mathbf{x}\rangle$.

(v) For $\sigma \in \{\pm 1\}$, $\sigma(\mathbf{N}, b) := (\sigma\mathbf{N}, \sigma b)$ is an *orientation* of $\mathcal{H}(\mathbf{N}, b)$.

(vi) Sets $X, Y \subseteq \mathbb{R}^n$ are *strictly separated* by $\mathcal{H}(\mathbf{N}, b)$ iff $(\exists\, \sigma \in \{\pm 1\})\ \langle \sigma\mathbf{N}, X\rangle < \sigma b < \langle \sigma\mathbf{N}, Y\rangle$.

(vii) Sets $X, Y \subseteq \mathbb{R}^n$ are *strongly separated* by $\mathcal{H}(\mathbf{N}, b)$ iff $(\exists\, \sigma \in \{\pm 1\})(\exists\, \varepsilon > 0)\ \langle \sigma\mathbf{N}, X\rangle < \sigma b - \varepsilon < \sigma b + \varepsilon < \langle \sigma\mathbf{N}, Y\rangle$.

Remark 4.6.2. *Basic Hyperplane Geometry*
The identity $\mathcal{H}_+(\mathbf{N}, b) \cap \mathcal{H}_-(\mathbf{N}, b) = \mathcal{H}(\mathbf{N}, b)$ holds.

$\mathbb{R}^n = \big(\mathcal{H}_-(\mathbf{N},b)\setminus\mathcal{H}(\mathbf{N},b)\big)\sqcup\mathcal{H}(\mathbf{N},b)\sqcup\big(\mathcal{H}_+(\mathbf{N},b)\setminus\mathcal{H}(\mathbf{N},b)\big)$.
The three terms are the negative open half-space, the hyperplane, and the positive open half-space.
Since $\mathbf{N}\in\mathcal{H}(\mathbf{N},\|\mathbf{N}\|^2)$, $(\forall\,\mathbf{x}\in\mathcal{H}(\mathbf{N},\|\mathbf{N}\|^2))\ \|\mathbf{N}\|^2=\langle\mathbf{N},\mathbf{x}\rangle\le\|\mathbf{N}\|\,\|\mathbf{x}\|$ by Theorem 1.1.13(i).
Hence $(\forall\,\mathbf{x}\in\mathcal{H}(\mathbf{N},\|\mathbf{N}\|^2))\ \|\mathbf{N}\|\le\|\mathbf{x}\|$, so $\mathbf{N}$ is a closest point of this hyperplane to $\mathbf{0}$.

Theorem 4.6.3. *Nearest Points and Separating a Point*
Let $C\subseteq\mathbb{R}^n$ be nonempty, closed, and convex.

(i) $(\exists!\,\mathbf{N}_C\in C)(\forall\,\mathbf{x}\in C)\ \|\mathbf{N}_C\|\le\|\mathbf{x}\|$.
(ii) If $\mathbf{b}\notin C$, put $D:=C-\mathbf{b}:=\{\mathbf{x}-\mathbf{b}:\mathbf{x}\in C\}$ and let $\mathbf{N}$ be its unique minimum-norm vector. Then $\mathbf{N}\neq\mathbf{0}$ and $\langle\mathbf{N},C\rangle\ge\|\mathbf{N}\|^2+\langle\mathbf{N},\mathbf{b}\rangle>\langle\mathbf{N},\mathbf{b}\rangle$.
(iii) Under the hypotheses and notation of (ii), set $\alpha:=\langle\mathbf{N},\mathbf{b}\rangle+\frac{1}{2}\|\mathbf{N}\|^2$. Then $\mathcal{H}(\mathbf{N},\alpha)$ strongly separates $\{\mathbf{b}\}$ and C.

Proof.

(i) Put $m:=\inf\{\|\mathbf{x}\|^2:\mathbf{x}\in C\}$. By the definition of infimum, choose $(\mathbf{x}_j)_{j\ge1}\subseteq C$ satisfying $m\le\|\mathbf{x}_j\|^2<m+\frac{1}{j}$.
For $j,\ell\ge1$, $\mathbf{w}_{j,\ell}:=\frac{1}{2}(\mathbf{x}_j+\mathbf{x}_\ell)\in C$ by convexity.
By Theorem 1.1.20, $\|\mathbf{x}_j-\mathbf{x}_\ell\|^2=2\|\mathbf{x}_j\|^2+2\|\mathbf{x}_\ell\|^2-4\|\mathbf{w}_{j,\ell}\|^2\le\frac{2}{j}+\frac{2}{\ell}$, so $(\mathbf{x}_j)_{j\ge1}$ is Cauchy.
Write $\mathbf{x}_j=(x_{j,1},\dots,x_{j,n})$. For $r\in\{1,\dots,n\}$ and $q\ge1$, put $a_{q,r}:=\inf\{x_{j,r}:j\ge q\}$ and $b_{q,r}:=\sup\{x_{j,r}:j\ge q\}$.
Since $(\forall\,j,\ell\ge1)\ |x_{j,r}-x_{\ell,r}|\le\|\mathbf{x}_j-\mathbf{x}_\ell\|$, each coordinate sequence is Cauchy.
Each coordinate sequence is bounded, so $a_{q,r},b_{q,r}\in\mathbb{R}$.
For $\eta>0$, choose $q_0\ge1$ with $(\forall\,q\ge q_0)\,0\le b_{q,r}-a_{q,r}\le\eta$ by the Cauchy property.
Thus $b_{q,r}-a_{q,r}\to0$, and the intervals $[a_{q,r},b_{q,r}]$ are nested. Put $N_r:=\sup\{a_{q,r}:q\ge1\}$.
Nestedness implies $(\forall\,q\ge1)\,[a_{q,r}\le N_r\le b_{q,r}\wedge(\forall\,j\ge q)\ |x_{j,r}-N_r|\le b_{q,r}-a_{q,r}]$, so $x_{j,r}\to N_r$.
For $\mathbf{N}_C:=(N_1,\dots,N_n)$, $\|\mathbf{x}_j-\mathbf{N}_C\|^2=\sum_{r=1}^{n}|x_{j,r}-N_r|^2\to0$.
If $\mathbf{N}_C\notin C$, choose $\rho>0$ with $B_\rho(\mathbf{N}_C)\subseteq\mathbb{R}^n\setminus C$ by openness of $\mathbb{R}^n\setminus C$.
$(\exists\,j\ge1)\,\mathbf{x}_j\in B_\rho(\mathbf{N}_C)\subseteq\mathbb{R}^n\setminus C$ by convergence. This contradicts $(\mathbf{x}_j)_{j\ge1}\subseteq C$, so $\mathbf{N}_C\in C$.
By Proposition 1.1.15, $\|\mathbf{x}_j\|\to\|\mathbf{N}_C\|$, and therefore $\|\mathbf{N}_C\|^2=m$.
If $\mathbf{u},\mathbf{v}\in C$ both have squared norm m, then $\mathbf{w}:=\frac{1}{2}(\mathbf{u}+\mathbf{v})\in C$ by convexity.
By Theorem 1.1.20, $m\le\|\mathbf{w}\|^2=m-\frac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2$, so $\mathbf{u}=\mathbf{v}$.

(ii) The singleton $\{-\mathbf{b}\}$ is convex by definition and compact by Proposition 4.5.3(i),(ii).
Since $C\neq\emptyset$, $D=\{-\mathbf{b}\}+C\neq\emptyset$, and D is closed and convex by Proposition 4.5.4(i),(ii).
Fix its unique minimum-norm vector $\mathbf{N}$ by (i). Since $\mathbf{b}\notin C\iff\mathbf{0}\notin D$, $\mathbf{N}\neq\mathbf{0}$.
Fix $\mathbf{z}\in D$ and put $\mathbf{h}:=\mathbf{z}-\mathbf{N}$. By convexity, $(\forall\,t\in(0,1])\,\mathbf{N}+t\mathbf{h}\in D$.
By minimality, $(\forall\,t\in(0,1])\,0\le\|\mathbf{N}+t\mathbf{h}\|^2-\|\mathbf{N}\|^2=2t\langle\mathbf{N},\mathbf{h}\rangle+t^2\|\mathbf{h}\|^2$.
If $a:=\langle\mathbf{N},\mathbf{h}\rangle<0$, then $\mathbf{h}\neq\mathbf{0}$ and the right-hand side is negative for $t\in\left(0,\min\left\{1,\frac{-2a}{\|\mathbf{h}\|^2}\right\}\right)$, a contradiction.
Hence $\langle\mathbf{N},\mathbf{z}\rangle\ge\|\mathbf{N}\|^2$. Substituting $\mathbf{z}=\mathbf{x}-\mathbf{b}$ proves $\langle\mathbf{N},C\rangle\ge\|\mathbf{N}\|^2+\langle\mathbf{N},\mathbf{b}\rangle>\langle\mathbf{N},\mathbf{b}\rangle$.

(iii) Put $\varepsilon:=\frac{1}{4}\|\mathbf{N}\|^2>0$. By (ii), $(\forall\,\mathbf{x}\in C)\ \langle\mathbf{N},\mathbf{b}\rangle<\alpha-\varepsilon<\alpha+\varepsilon<\langle\mathbf{N},\mathbf{x}\rangle$.
Hence $\mathcal{H}(\mathbf{N},\alpha)$ strongly separates $\{\mathbf{b}\}$ and C by Definition 4.6.1(vii) with $\sigma=1$.

Structural Commentary: Parallelogram control constructs the minimizer; translation and one-sided variations construct the separator.
Key Insight: Minimum norm forces every feasible displacement into one supporting half-space. □

Figure 38 illustrates Theorem 4.6.3(ii),(iii).
Here $\mathbf{p}=\mathbf{b}+\mathbf{N}\in C$ is nearest to $\mathbf{b}$. The vector $\mathbf{N}\in C-\mathbf{b}$ is both the displacement and the separating normal.

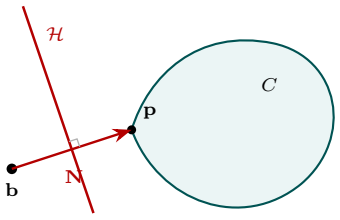

Figure 38: For $\mathbf{b}\notin C$, the labeled point $\mathbf{p}=\mathbf{b}+\mathbf{N}\in C$ is nearest to $\mathbf{b}$, the arrow $\mathbf{N}$ is the displacement from $\mathbf{b}$ to $\mathbf{p}$, and the labeled hyperplane lies strictly between $\mathbf{b}$ and the closed convex set C.

Theorem 4.6.4. *Separating a Compact Convex Set from a Subspace*
Let $C\subseteq\mathbb{R}^n$ be nonempty, compact, and convex, and let $S\le\mathbb{R}^n$ satisfy $C\cap S=\emptyset$. Then:

(i) $(\exists\,\mathbf{N}\in S^\perp\setminus\{\mathbf{0}\})\ \langle\mathbf{N},C\rangle\ge\|\mathbf{N}\|^2>0$.
(ii) For a vector $\mathbf{N}$ as in (i), $\mathcal{H}(\mathbf{N},\frac{1}{2}\|\mathbf{N}\|^2)$ strongly separates S and C.

Proof.

(i) The subspace S is convex. Put $Q:=P_{S^\perp}$. By Proposition 2.2.3(i),(vi) and Corollary 2.1.8, Q is linear and $\ker(Q)=S$.
By Proposition 2.2.3(viii), $(\forall\,\mathbf{v}\in\mathbb{R}^n)\ \|Q\mathbf{v}\|\le\|\mathbf{v}\|$. Fix $\mathbf{x}\notin S$ and put $\mathbf{q}:=Q\mathbf{x}\neq\mathbf{0}$.
If $\|\mathbf{y}-\mathbf{x}\|<\frac{1}{2}\|\mathbf{q}\|$, then $\|Q\mathbf{y}\|\ge\|\mathbf{q}\|-\|Q(\mathbf{y}-\mathbf{x})\|\ge\|\mathbf{q}\|-\|\mathbf{y}-\mathbf{x}\|>0$ by Proposition 1.1.15.
Thus $B_{\frac{1}{2}\|\mathbf{q}\|}(\mathbf{x})\subseteq\mathbb{R}^n\setminus S$, so S is closed. Put $D:=C+S$.
Since $\mathbf{0}\in S$, $C\subseteq D$ and $D\neq\emptyset$. The set D is closed and convex by Proposition 4.5.4(i),(ii).
If $\mathbf{0}\in D$, then $(\exists\,(\mathbf{c},\mathbf{s})\in C\times S)\,\mathbf{0}=\mathbf{c}+\mathbf{s}$, whence $\mathbf{c}=-\mathbf{s}\in C\cap S$, a contradiction.
Fix the unique minimum-norm vector $\mathbf{N}$ of D by Theorem 4.6.3(i).
By Theorem 4.6.3(ii) with D and $\mathbf{0}$, $\mathbf{N}\neq\mathbf{0}$ and $(\forall\,\mathbf{c}\in C)(\forall\,\mathbf{s}\in S)\ \langle\mathbf{N},\mathbf{c}+\mathbf{s}\rangle\ge\|\mathbf{N}\|^2$.
Fix $\mathbf{c}_0\in C$. For $\mathbf{s}\in S$, put $a:=\langle\mathbf{N},\mathbf{s}\rangle$. If $a\neq0$, set $\lambda:=\frac{\|\mathbf{N}\|^2-\langle\mathbf{N},\mathbf{c}_0\rangle-1}{a}$.

Then $\lambda\mathbf{s}+\mathbf{c}_0 \in D$ but $\langle \mathbf{N}, \lambda\mathbf{s}+\mathbf{c}_0\rangle = \|\mathbf{N}\|^2 - 1$, a contradiction.
Hence $(\forall\, \mathbf{s}\in S)\ \langle \mathbf{N},\mathbf{s}\rangle = 0$, so $\mathbf{N}\in S^\perp$. Taking $\mathbf{s}=\mathbf{0}$ in the preceding bound proves $\langle \mathbf{N}, C\rangle \geq \|\mathbf{N}\|^2 > 0$.

(ii) Put $\alpha := \frac{1}{2}\|\mathbf{N}\|^2$ and $\varepsilon := \frac{1}{4}\|\mathbf{N}\|^2$. By (i), $(\forall\, \mathbf{s}\in S)(\forall\, \mathbf{c}\in C)\ \langle \mathbf{N},\mathbf{s}\rangle = 0 < \alpha-\varepsilon < \alpha+\varepsilon < \|\mathbf{N}\|^2 \leq \langle \mathbf{N},\mathbf{c}\rangle$.
Hence $\mathcal{H}(\mathbf{N},\alpha)$ strongly separates S and C by Definition 4.6.1(vii) with $\sigma = 1$.

Structural Commentary: Minimum-norm separation of $C+S$ constructs the normal; unrestricted subspace scaling proves orthogonality.
Key Insight: Translation invariance along S forces the supporting normal into $S^\perp$. □

4.7 Positive Orthants, Alternatives, and Farkas's Lemma

This subsection uses real inner-product geometry to decide feasibility of positive linear systems.
A separating hyperplane converts infeasibility into a dual vector certificate.

Definition 4.7.1. *Positive Orthants*
Let $n\in\mathbb{Z}_{>0}$ and $\mathbf{x}=(x_1,\dots,x_n)\in\mathbb{R}^n$.

(i) $\mathbf{x}\geq\mathbf{0} \iff (\forall\, i\in\{1,\dots,n\})\, x_i\geq 0$ defines *coordinatewise nonnegativity*; $\mathbf{x}\leq\mathbf{0}\iff -\mathbf{x}\geq\mathbf{0}$.
(ii) $\mathbf{x}>\mathbf{0}\iff[\mathbf{x}\geq\mathbf{0}\wedge\mathbf{x}\neq\mathbf{0}]$ defines *nonzero nonnegativity*.
(iii) $\mathbf{x}\gg\mathbf{0}\iff(\forall\, i\in\{1,\dots,n\})\, x_i>0$ defines *coordinatewise positivity*.
(iv) $\mathbb{R}^n_{\geq 0} := \{\mathbf{y}\in\mathbb{R}^n : \mathbf{y}\geq\mathbf{0}\}$ is the *nonnegative orthant*.
(v) $\mathbb{R}^n_{+} := \{\mathbf{y}\in\mathbb{R}^n : \mathbf{y}>\mathbf{0}\} = \mathbb{R}^n_{\geq 0}\setminus\{\mathbf{0}\}$ is the *nonzero nonnegative orthant*.
(vi) $\mathbb{R}^n_{++} := \{\mathbf{y}\in\mathbb{R}^n : \mathbf{y}\gg\mathbf{0}\}$ is the *coordinatewise positive orthant*.

Motivation 4.7.2. *Positive Null Vectors as Probabilities*
Let $m,n\in\mathbb{Z}_{>0}$ and $A=[\mathbf{a}_1\ \cdots\ \mathbf{a}_n]\in\mathrm{Mat}(m,n,\mathbb{R})$. Suppose $\mathbf{x}=(x_1,\dots,x_n)>\mathbf{0}$ and $A\mathbf{x}=\mathbf{0}$.

(i) Set $s := x_1+\cdots+x_n>0$ and $\boldsymbol{\pi} := s^{-1}\mathbf{x}$.
(ii) $(\forall\, i\in\{1,\dots,n\})\,\pi_i\geq 0$ and $\sum_{i=1}^{n}\pi_i = 1$, so $\boldsymbol{\pi}$ is a probability vector.
(iii) The identity $A\boldsymbol{\pi} = s^{-1}A\mathbf{x} = \mathbf{0}$ is equivalent to $\sum_{i=1}^{n}\pi_i\mathbf{a}_i = \mathbf{0}$.
(iv) $\mathbf{x}\gg\mathbf{0}\Longrightarrow(\forall\, i\in\{1,\dots,n\})\,\pi_i>0$.

Thus a nonzero nonnegative null vector normalizes to a distribution whose probability-weighted column mean is $\mathbf{0}$.

Principle 4.7.3. *Positive Solutions Are Orthant Intersections*
Let $m,n\in\mathbb{Z}_{>0}$, $A\in\mathrm{Mat}(m,n,\mathbb{R})$, and $S := \mathrm{row}(A)$. Then $\ker(A) = S^\perp$ by Corollary 2.3.12(i).

(i) $\{\mathbf{x}\in\mathbb{R}^n : A\mathbf{x}=\mathbf{0}\wedge\mathbf{x}>\mathbf{0}\} = S^\perp\cap\mathbb{R}^n_{+}$.
(ii) $\{\mathbf{x}\in\mathbb{R}^n : A\mathbf{x}=\mathbf{0}\wedge\mathbf{x}\gg\mathbf{0}\} = S^\perp\cap\mathbb{R}^n_{++}$.

Thus positive null vectors are precisely the indicated orthant intersections with $\mathrm{row}(A)^\perp$.

Theorem 4.7.4. *Positive Orthant Alternatives for Subspaces*
Let $n\in\mathbb{Z}_{>0}$ and $S\leq\mathbb{R}^n$. Then:

(i) $S\cap\mathbb{R}^n_{+}=\emptyset\iff S^\perp\cap\mathbb{R}^n_{++}\neq\emptyset$.
(ii) $S\cap\mathbb{R}^n_{++}=\emptyset\iff S^\perp\cap\mathbb{R}^n_{+}\neq\emptyset$.

Proof.

(i) ($\Rightarrow$) Put $\Delta := \mathrm{conv}\{\mathbf{e}_1,\dots,\mathbf{e}_n\}$, which is convex by definition.
The identity $\Delta = \{(t_1,\dots,t_n)\in\mathbb{R}^n : (\forall\, i\in\{1,\dots,n\})\, t_i\geq 0,\ \sum_{i=1}^{n}t_i = 1\}$ holds by Proposition 4.5.3(i).
The set Δ is compact by Proposition 4.5.3(ii). Hence $\emptyset\neq\Delta\subseteq\mathbb{R}^n_{+}$ and $\Delta\cap S=\emptyset$.
Choose $\mathbf{N}\in S^\perp\setminus\{\mathbf{0}\}$ satisfying $(\forall\, i\in\{1,\dots,n\})\, N_i = \langle\mathbf{N},\mathbf{e}_i\rangle\geq\|\mathbf{N}\|^2>0$ by Theorem 4.6.4(i). Thus $\mathbf{N}\in S^\perp\cap\mathbb{R}^n_{++}$.

(i) ($\Leftarrow$) Suppose $\mathbf{N}\in S^\perp\cap\mathbb{R}^n_{++}$ and $\mathbf{x}\in S\cap\mathbb{R}^n_{+}$. Choose $j\in\{1,\dots,n\}$ with $x_j>0$ by $\mathbf{x}\geq\mathbf{0}$ and $\mathbf{x}\neq\mathbf{0}$.
Then $0=\langle\mathbf{N},\mathbf{x}\rangle = \sum_{i=1}^{n}N_ix_i\geq N_jx_j>0$, a contradiction. Hence $S\cap\mathbb{R}^n_{+}=\emptyset$.

(ii) ($\Rightarrow$) Suppose $S\cap\mathbb{R}^n_{++}=\emptyset$ and $S^\perp\cap\mathbb{R}^n_{+}=\emptyset$. Then $(S^\perp)^\perp\cap\mathbb{R}^n_{++}\neq\emptyset$ by (i) with $S^\perp$.
But $(S^\perp)^\perp = S$ by Corollary 2.1.8, contradicting $S\cap\mathbb{R}^n_{++}=\emptyset$.

(ii) ($\Leftarrow$) Suppose $\mathbf{N}\in S^\perp\cap\mathbb{R}^n_{+}$ and $\mathbf{x}\in S\cap\mathbb{R}^n_{++}$. Choose $j\in\{1,\dots,n\}$ with $N_j>0$ by $\mathbf{N}\geq\mathbf{0}$ and $\mathbf{N}\neq\mathbf{0}$.
Then $0=\langle\mathbf{N},\mathbf{x}\rangle = \sum_{i=1}^{n}N_ix_i\geq N_jx_j>0$, a contradiction. Hence $S\cap\mathbb{R}^n_{++}=\emptyset$.

Structural Commentary: Simplex separation proves (i); double orthogonal complementation converts (i) into (ii).
Key Insight: Orthant avoidance forces a strictly positive normal in the complementary subspace. □

Figure 39 illustrates Theorem 4.7.4(i) in $\mathbb{R}^2$.
The displayed $\mathbf{y}\in S^\perp\cap\mathbb{R}^2_{++}$ certifies $S\cap\mathbb{R}^2_{+}=\emptyset$; the theorem owns this implication.

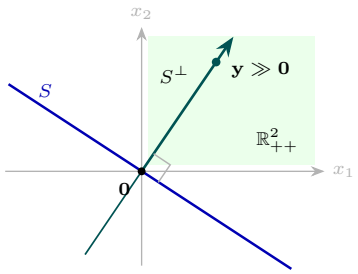

Figure 39: In $\mathbb{R}^2$, the negative-slope line S avoids $\mathbb{R}^2_+$, while $\mathbf{y} \in S^\perp \cap \mathbb{R}^2_{++}$.

Theorem 4.7.5. *Homogeneous Positive Alternatives*
Let $m, n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(m, n, \mathbb{R})$. Exactly one alternative holds in each pair:
(i) $\ker(A) \cap \mathbb{R}^n_{++} \neq \emptyset$ *or* $\mathrm{col}(A^t) \cap \mathbb{R}^n_+ \neq \emptyset$.
(ii) $\ker(A) \cap \mathbb{R}^n_+ \neq \emptyset$ *or* $\mathrm{col}(A^t) \cap \mathbb{R}^n_{++} \neq \emptyset$.

Proof. Put $S := \mathrm{row}(A) = \mathrm{col}(A^t)$. Then $\ker(A) = S^\perp$ by Corollary 2.3.12(i).
(i) The pair in (i) is equivalent to $S^\perp \cap \mathbb{R}^n_{++} \neq \emptyset$ or $S \cap \mathbb{R}^n_+ \neq \emptyset$.
Exactly one alternative holds by Theorem 4.7.4(i).
(ii) The pair in (ii) is equivalent to $S^\perp \cap \mathbb{R}^n_+ \neq \emptyset$ or $S \cap \mathbb{R}^n_{++} \neq \emptyset$.
Exactly one alternative holds by Theorem 4.7.4(ii).
Structural Commentary: The fundamental-subspace identity translates both subspace alternatives into matrix feasibility tests.
Key Insight: Row space and nullspace meet complementary positive orthants. □

Computation Method 4.7.6. *Testing Positive Homogeneous Systems*
Given $m, n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(m, n, \mathbb{R})$, set $S := \mathrm{row}(A) = \mathrm{col}(A^t)$. The following searches are exact by Theorem 4.7.5:
(i) Seek $\mathbf{v} \in \mathbb{R}^m$ with $A^t\mathbf{v} > \mathbf{0}$ to certify $\ker(A) \cap \mathbb{R}^n_{++} = \emptyset$.
(ii) Seek $\mathbf{v} \in \mathbb{R}^m$ with $A^t\mathbf{v} \gg \mathbf{0}$ to certify $\ker(A) \cap \mathbb{R}^n_+ = \emptyset$.
(iii) Seek $\mathbf{x} \in \ker(A) \cap \mathbb{R}^n_{++}$ to certify $S \cap \mathbb{R}^n_+ = \emptyset$.
(iv) Seek $\mathbf{x} \in \ker(A) \cap \mathbb{R}^n_+$ to certify $S \cap \mathbb{R}^n_{++} = \emptyset$.

Lemma 4.7.7. *Nonnegative Matrix Cone*
Let $m, n \in \mathbb{Z}_{>0}$ and $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{R})$. Define $C_A := A(\mathbb{R}^n_{\geq 0}) = \left\{ \sum_{i=1}^n t_i \mathbf{a}_i : (t_1, \ldots, t_n) \in \mathbb{R}^n_{\geq 0} \right\}$.
(i) C_A is a convex cone in $\mathbb{R}^m$.
(ii) C_A is closed in $\mathbb{R}^m$.

Proof.
(i) Let $\mathbf{x}_r = A\mathbf{y}_r \in C_A$ with $\mathbf{y}_r \geq \mathbf{0}$ for $r \in \{1, 2\}$. For $t \in [0, 1]$, $t\mathbf{x}_1 + (1-t)\mathbf{x}_2 = A(t\mathbf{y}_1 + (1-t)\mathbf{y}_2) \in C_A$.
For $a \in [0, \infty)$, $a\mathbf{x}_1 = A(a\mathbf{y}_1) \in C_A$. Thus C_A is convex and a cone.
(ii) It suffices to express C_A as a finite union of closed cones.
(a) *Independent supports*: Let $\mathcal{I}$ contain $I \subseteq \{1, \ldots, n\}$ exactly when $\{\mathbf{a}_i : i \in I\}$ is linearly independent; then $\emptyset \in \mathcal{I}$.
For $I \in \mathcal{I}$, set $K_I := \left\{ \sum_{i \in I} t_i \mathbf{a}_i : (\forall\, i \in I)\, t_i \geq 0 \right\}$.
Fix $\mathbf{x} \in C_A \setminus \{\mathbf{0}\}$ and choose $I \subseteq \{1, \ldots, n\}$ with $\mathbf{x} = \sum_{i \in I} t_i \mathbf{a}_i$, $(\forall\, i \in I)\, t_i > 0$, and minimal $|I|$.
If the indexed columns are dependent, choose $(d_i)_{i \in I} \in \mathbb{R}^I \setminus \{\mathbf{0}\}$ with $\sum_{i \in I} d_i \mathbf{a}_i = \mathbf{0}$.
After replacing $(d_i)_{i \in I}$ by its negative if necessary, $J := \{i \in I : d_i < 0\} \neq \emptyset$. Put $\lambda := \min_{i \in J} \frac{t_i}{-d_i} > 0$.
Then $(\forall\, i \in I)\, t_i + \lambda d_i \geq 0$, $(\exists\, j \in J)\, t_j + \lambda d_j = 0$, and $\mathbf{x} = \sum_{i \in I} (t_i + \lambda d_i)\mathbf{a}_i$.
Deleting the zero coefficients contradicts minimality of $|I|$, so $I \in \mathcal{I}$. Since $\mathbf{0} \in K_\emptyset$, $C_A = \bigcup_{I \in \mathcal{I}} K_I$.
(b) *Closed simplicial cones*: Fix $I \in \mathcal{I}$. If $I = \emptyset$, then $K_I = \{\mathbf{0}\}$. For $\mathbf{x} \neq \mathbf{0}$, $B_{\frac{1}{2}\|\mathbf{x}\|}(\mathbf{x}) \subseteq \mathbb{R}^m \setminus K_I$ by Proposition 1.1.15.
Hence K_I is closed when $I = \emptyset$. Suppose $I \neq \emptyset$ and put $U_I := \mathrm{span}\{\mathbf{a}_i : i \in I\}$.
Then $(\mathbf{a}_i)_{i \in I}$ is a basis of U_I. Its dual basis $(\varphi_i)_{i \in I}$ exists by Proposition 1.5.8.
Choose $(\mathbf{d}_i)_{i \in I} \subseteq U_I$ satisfying $(\forall\, \mathbf{u} \in U_I)(\forall\, i \in I)\, \varphi_i(\mathbf{u}) = \langle \mathbf{u}, \mathbf{d}_i \rangle$ by Theorem 1.6.1(i).
Hence $K_I = \{\mathbf{u} \in U_I : (\forall\, i \in I)\ \langle \mathbf{u}, \mathbf{d}_i \rangle \geq 0\}$. Identity $\ker(P_{U_I^\perp}) = U_I$ holds by Proposition 2.2.3(vi) and Corollary 2.1.8.
Fix $\mathbf{x} \notin K_I$. If $\mathbf{x} \notin U_I$, put $\mathbf{q} := P_{U_I^\perp}\mathbf{x} \neq \mathbf{0}$. For $\|\mathbf{y} - \mathbf{x}\| < \frac{1}{2}\|\mathbf{q}\|$, $\left\|P_{U_I^\perp}(\mathbf{y} - \mathbf{x})\right\| \leq \|\mathbf{y} - \mathbf{x}\|$ by Proposition 2.2.3(i),(viii).
Thus $\left\|P_{U_I^\perp}\mathbf{y}\right\| \geq \|\mathbf{q}\| - \left\|P_{U_I^\perp}(\mathbf{y} - \mathbf{x})\right\| > 0$ by Proposition 1.1.15. Hence $\mathbf{y} \notin U_I$.
If $\mathbf{x} \in U_I \setminus K_I$, choose $j \in I$ with $\langle \mathbf{x}, \mathbf{d}_j \rangle < 0$. Then $\mathbf{d}_j \neq \mathbf{0}$ because $\langle \mathbf{a}_j, \mathbf{d}_j \rangle = 1$. Put $r_j := -\frac{\langle \mathbf{x}, \mathbf{d}_j \rangle}{2\|\mathbf{d}_j\|} > 0$.
For $\|\mathbf{y} - \mathbf{x}\| < r_j$, $\left|\langle \mathbf{y} - \mathbf{x}, \mathbf{d}_j \rangle\right| < -\frac{1}{2}\langle \mathbf{x}, \mathbf{d}_j \rangle$ by Theorem 1.1.13(i).
Hence $\langle \mathbf{y}, \mathbf{d}_j \rangle \leq \langle \mathbf{x}, \mathbf{d}_j \rangle + \left|\langle \mathbf{y} - \mathbf{x}, \mathbf{d}_j \rangle\right| < \frac{1}{2}\langle \mathbf{x}, \mathbf{d}_j \rangle < 0$, so $\mathbf{y} \notin K_I$. Thus $\mathbb{R}^m \setminus K_I$ is open, and K_I is closed.
(c) *Finite union*: Fix $\mathbf{x} \notin C_A$. Choose $(r_I)_{I \in \mathcal{I}} \subseteq (0, \infty)$ satisfying $(\forall\, I \in \mathcal{I})\, B_{r_I}(\mathbf{x}) \subseteq \mathbb{R}^m \setminus K_I$ by (b).
Since $\mathcal{I}$ is finite and nonempty, put $r := \min_{I \in \mathcal{I}} r_I > 0$. Then $B_r(\mathbf{x}) \subseteq \mathbb{R}^m \setminus C_A$ by (a). Thus $\mathbb{R}^m \setminus C_A$ is open, so C_A is closed.
Structural Commentary: Minimal supports form a finite simplicial cover; dual coordinates prove each cone closed.
Key Insight: Independent generators convert cone membership into closed coordinate inequalities. □

Theorem 4.7.8. *Farkas's Lemma*
Let $m, n \in \mathbb{Z}_{>0}$ and $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{R})$. Fix $\mathbf{b} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$ and set $C_A := \{A\mathbf{y} : \mathbf{y} \in \mathbb{R}^n \wedge \mathbf{y} \geq \mathbf{0}\}$.
Exactly one of the following alternatives holds:

(i) $\mathbf{b} \in C_A$, *equivalently* $(\exists\, \mathbf{u} \in \mathbb{R}^n)\,[\mathbf{u} > \mathbf{0} \wedge A\mathbf{u} = \mathbf{b}]$.
(ii) $(\exists\, \mathbf{v} \in \mathbb{R}^m)\,[A^t\mathbf{v} \leq \mathbf{0} \wedge \langle \mathbf{v}, \mathbf{b}\rangle > 0]$.

Proof.

(i) The equivalence $\mathbf{b} \in C_A \iff (\exists\, \mathbf{u} \in \mathbb{R}^n)\,[\mathbf{u} \geq \mathbf{0} \wedge A\mathbf{u} = \mathbf{b}]$ holds by the definition of C_A.
Every such $\mathbf{u}$ is nonzero by $\mathbf{b} \neq \mathbf{0}$, proving the equivalent formulation in (i). If (i) and (ii) both hold, choose witnesses $\mathbf{u}$ and $\mathbf{v}$.
Then $0 < \langle \mathbf{v}, \mathbf{b}\rangle = \langle \mathbf{v}, A\mathbf{u}\rangle = \langle A^t\mathbf{v}, \mathbf{u}\rangle = \sum_{i=1}^{n} (A^t\mathbf{v})_i u_i \leq 0$ by the coordinate inequalities, a contradiction.

(ii) Suppose $\mathbf{b} \notin C_A$. Then $\mathbf{0} \in C_A$, and C_A is closed and convex by Lemma 4.7.7(i),(ii).
Choose $\mathbf{N} \neq \mathbf{0}$ satisfying $(\forall\, \mathbf{c} \in C_A)\ \langle \mathbf{N}, \mathbf{c}\rangle \geq \|\mathbf{N}\|^2 + \langle \mathbf{N}, \mathbf{b}\rangle$ by Theorem 4.6.3(ii).
Set $\mathbf{v} := -\mathbf{N}$. Then $(\forall\, \mathbf{c} \in C_A)\ \langle \mathbf{v}, \mathbf{c}\rangle \leq \langle \mathbf{v}, \mathbf{b}\rangle - \|\mathbf{v}\|^2$. The choice $\mathbf{c} = \mathbf{0}$ proves $\langle \mathbf{v}, \mathbf{b}\rangle \geq \|\mathbf{v}\|^2 > 0$. Fix $i \in \{1, \ldots, n\}$.
Then $(\forall\, \lambda \in [0, \infty))\ \lambda\mathbf{a}_i \in C_A$. Hence $(\forall\, \lambda \in [0, \infty))\ \lambda\langle \mathbf{v}, \mathbf{a}_i\rangle \leq \langle \mathbf{v}, \mathbf{b}\rangle - \|\mathbf{v}\|^2$. If $\gamma := \langle \mathbf{v}, \mathbf{a}_i\rangle > 0$, set $\lambda := \frac{\langle \mathbf{v}, \mathbf{b}\rangle - \|\mathbf{v}\|^2}{\gamma} + 1 \geq 1$.
Then $\lambda\gamma > \langle \mathbf{v}, \mathbf{b}\rangle - \|\mathbf{v}\|^2$, a contradiction. Thus $(\forall\, i \in \{1, \ldots, n\})\ \langle \mathbf{v}, \mathbf{a}_i\rangle \leq 0$.
Equivalently, $A^t\mathbf{v} \leq \mathbf{0}$. Therefore (ii) holds whenever (i) fails, so exactly one alternative holds.

Structural Commentary: Nearest-point separation constructs $\mathbf{v}$; ray scaling tests every coordinate of $A^t\mathbf{v}$.
Key Insight: A separating functional must be nonpositive on each cone generator. □

Counterexample 4.7.9. *A Certificate Need Not Be Strict*
Certificate in Theorem 4.7.8 need not satisfy $-A^t\mathbf{v} \gg \mathbf{0}$. Take $A = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{smallmatrix}\right)$, $\mathbf{b} = \left(\begin{smallmatrix} -1 \\ 0 \end{smallmatrix}\right)$. If $\mathbf{u} \geq \mathbf{0}$, then $(A\mathbf{u})_1 = u_1 \geq 0$, so $A\mathbf{u} \neq \mathbf{b}$.
The vector $\mathbf{v} = \left(\begin{smallmatrix} -1 \\ 0 \end{smallmatrix}\right)$ satisfies $A^t\mathbf{v} = (-1, 0, 0)^t \leq \mathbf{0}$ and $\langle \mathbf{v}, \mathbf{b}\rangle = 1 > 0$. For $\mathbf{w} = (r, s)^t \in \mathbb{R}^2$, $A^t\mathbf{w} = (r, s, -s)^t$.
Thus $-A^t\mathbf{w} \gg \mathbf{0}$ would require both $s < 0$ and $s > 0$, which is impossible. The weak coordinate inequality is therefore sharp.

5 Eigenstructure and Invariant Subspaces

5.1 Eigenvalues, Eigenspaces, and Spectral Criteria

Eigenstructure begins by identifying the scalars for which a shifted operator has a nontrivial kernel.
Throughout this subsection, $\mathcal{V}$ is a vector space over $\mathbb{F}$, and $n \in \mathbb{Z}_{>0}$ whenever an $n \times n$ matrix occurs.

Definition 5.1.1. *Eigenvalue and Eigenvector*
Let $T \in \mathcal{L}(\mathcal{V})$.

(i) A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of T iff $(\exists\, \mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\})\, T\mathbf{v} = \lambda\mathbf{v}$.
(ii) For $\lambda \in \mathbb{F}$, a vector $\mathbf{v} \in \mathcal{V}$ is an *eigenvector* of T corresponding to λ iff $\mathbf{v} \neq \mathbf{0} \wedge T\mathbf{v} = \lambda\mathbf{v}$.

Definition 5.1.2. *Eigenspace*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\lambda \in \mathbb{F}$. The *eigenspace* of T corresponding to λ is $E_\lambda(T) := \ker(T - \lambda I_\mathcal{V}) = \{\mathbf{v} \in \mathcal{V} : T\mathbf{v} = \lambda\mathbf{v}\}$.
Equivalently, λ is an eigenvalue of T iff $E_\lambda(T) \neq \{\mathbf{0}\}$.

Definition 5.1.3. *Spectrum and Geometric Multiplicity*
Let $T \in \mathcal{L}(\mathcal{V})$.

(i) The *spectrum* of T is $\mathrm{spec}(T) := \{\lambda \in \mathbb{F} : E_\lambda(T) \neq \{\mathbf{0}\}\}$.
(ii) If $\mathcal{V}$ is finite-dimensional and $\lambda \in \mathrm{spec}(T)$, then the *geometric multiplicity* of λ is $\dim E_\lambda(T)$.
(iii) For $A \in \mathrm{Mat}(n, n, \mathbb{F})$ and $\lambda \in \mathbb{F}$, set $E_\lambda(A) := \ker(A - \lambda I_n)$. Define $\mathrm{spec}(A) := \{\mu \in \mathbb{F} : E_\mu(A) \neq \{\mathbf{0}\}\}$.

Theorem 5.1.4. *Equivalent Conditions for an Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{F}$. The following are equivalent:

(i) λ *is an eigenvalue of* T;
(ii) $T - \lambda I_\mathcal{V}$ *is not injective;*
(iii) $T - \lambda I_\mathcal{V}$ *is not surjective;*
(iv) $T - \lambda I_\mathcal{V}$ *is not invertible.*

Proof. Set $S := T - \lambda I_\mathcal{V}$.

(i) $\Rightarrow$ *(ii)* Choose $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$ with $T\mathbf{v} = \lambda\mathbf{v}$ by (i).
The membership $\mathbf{v} \in \ker(S) \setminus \{\mathbf{0}\}$ holds by $S\mathbf{v} = T\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$, and S is not injective by Proposition 1.6.2.
(ii) $\Rightarrow$ *(iii)* The operator S is not surjective by Theorem 1.7.7(ii),(iii).
(iii) $\Rightarrow$ *(iv)* The operator S is not invertible by Theorem 1.7.7(iii),(i).
(iv) $\Rightarrow$ *(i)* The operator S is not injective by Theorem 1.7.7(i),(ii), and $\ker(S) \neq \{\mathbf{0}\}$ by Proposition 1.6.2.
Choose $\mathbf{v} \in \ker(S) \setminus \{\mathbf{0}\}$. The equality $T\mathbf{v} = \lambda\mathbf{v}$ holds by $S\mathbf{v} = \mathbf{0}$. The scalar λ is an eigenvalue of T by the choice of $\mathbf{v}$.

Structural Commentary: The kernel test links singular shifts to the finite-dimensional invertibility cycle.
Key Insight: A scalar is spectral exactly when its shifted operator has nonzero kernel. □

Theorem 5.1.5. *Eigenvectors for Distinct Eigenvalues Are Independent*
Let $m \in \mathbb{Z}_{>0}$ and $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are pairwise distinct eigenvalues of T with corresponding eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_m$. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. For $m = 1$, the list is independent by $\mathbf{v}_1 \neq \mathbf{0}$. Assume $m \geq 2$ and suppose $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is dependent.
Let k be least index for which $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ is dependent. Bounds $k \leq m$ and $k \geq 2$ hold by choice of k and $\mathbf{v}_1 \neq \mathbf{0}$, respectively.
The list $(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ is independent by minimality of k. Choose $c_1, \ldots, c_k \in \mathbb{F}$, not all zero, with $\sum_{i=1}^{k} c_i\mathbf{v}_i = \mathbf{0}$ by dependence.
The coefficient $c_k \neq 0$ by independence of the preceding prefix: $c_k = 0$ would force $c_1 = \cdots = c_{k-1} = 0$.
Set $a_i := -\frac{c_i}{c_k}$ $(1 \leq i < k)$. The representation $\mathbf{v}_k = \sum_{i=1}^{k-1} a_i\mathbf{v}_i$ holds by the chosen dependence relation.

The equality $\mathbf{0} = (T - \lambda_k I_{\mathcal{V}})\mathbf{v}_k = \sum_{i=1}^{k-1} a_i(\lambda_i - \lambda_k)\mathbf{v}_i$ holds by linearity and the eigenvector identities.
The relation $(\forall\, i \in \{1, \dots, k-1\})\, a_i(\lambda_i - \lambda_k) = 0$ holds by independence of the preceding prefix.
The relation $(\forall\, i \in \{1, \dots, k-1\})\, a_i = 0$ holds by pairwise distinctness of the eigenvalues.
The equality $\mathbf{v}_k = \mathbf{0}$ holds by its representation and the preceding coefficient identities, contrary to the eigenvector hypothesis.
The original list is linearly independent by contradiction.
Structural Commentary: Minimal dependence exposes the last coefficient; the spectral shift isolates the rest.
Key Insight: The shift by λ_k rescales every earlier eigenvector nontrivially. □

Corollary 5.1.6. *Number of Distinct Eigenvalues*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then $|\operatorname{spec}(T)| \leq \dim(\mathcal{V})$.

Proof. Set $d := \dim(\mathcal{V})$, and choose a basis β of length d by finite dimensionality and the definition of d.
Suppose $|\operatorname{spec}(T)| > d$. Choose distinct $\lambda_1, \dots, \lambda_{d+1} \in \operatorname{spec}(T)$.
Choose corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_{d+1}$ by the definition of the spectrum.
The list $(\mathbf{v}_1, \dots, \mathbf{v}_{d+1})$ is independent by Theorem 5.1.5, while β spans $\mathcal{V}$.
The inequality $d + 1 \leq d$ holds by Theorem 1.3.11, a contradiction. The claimed bound holds by contradiction.
Structural Commentary: Spectral representatives form an independent list bounded in length by a basis.
Key Insight: Each distinct eigenvalue requires a separate independent direction. □

Corollary 5.1.7. *Invertibility and the Zero Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is invertible iff $0 \notin \operatorname{spec}(T)$.

Proof.
(⇒) Suppose T is invertible. The scalar 0 is not an eigenvalue of T by Theorem 5.1.4(iv),(i).
The exclusion $0 \notin \operatorname{spec}(T)$ holds by the definition of $\operatorname{spec}(T)$.
(⇐) Suppose $0 \notin \operatorname{spec}(T)$. The scalar 0 is not an eigenvalue of T by the definition of $\operatorname{spec}(T)$.
The operator $T = T - 0I_{\mathcal{V}}$ is invertible by Theorem 5.1.4(i),(iv).
Structural Commentary: At zero, the singular-shift criterion equates invertibility with spectral exclusion.
Key Insight: The zero eigenspace is $\ker(T)$. □

Corollary 5.1.8. *Matrix Invertibility and the Spectrum*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$. Then A is invertible iff $0 \notin \operatorname{spec}(A)$.

Proof.
(⇒) Suppose A is invertible. The equality $\operatorname{rank}(A) = n$ holds by Corollary 3.6.2.
The equality $\dim(\ker(A)) = 0$ holds by Corollary 3.3.1(i).
The relations $\{\mathbf{0}\} \leq \ker(A)$ and $\dim(\{\mathbf{0}\}) = \dim(\ker(A)) = 0$ hold by $A\mathbf{0} = \mathbf{0}$ and the preceding equality.
The identity $\ker(A) = \{\mathbf{0}\}$ holds by Proposition 1.4.3(ii). The exclusion $0 \notin \operatorname{spec}(A)$ holds by the matrix-spectrum definition.
(⇐) Suppose $0 \notin \operatorname{spec}(A)$. The identity $\ker(A) = \{\mathbf{0}\}$ holds by the matrix-spectrum definition.
Equality $\dim(\ker(A)) = 0$ holds by $\ker(A) = \{\mathbf{0}\}$, $\operatorname{rank}(A) = n$ by Corollary 3.3.1(i). Matrix A is invertible by Corollary 3.6.2.
Structural Commentary: Rank–nullity carries full rank to the zero-kernel matrix-spectrum test.
Key Insight: Zero is spectral exactly when $\ker(A) \neq \{\mathbf{0}\}$. □

5.2 Invariant Subspaces, Reducing Pairs, and Quotient Operators

Throughout this subsection, $\mathcal{V}$ is a vector space over $\mathbb{F}$.
In statements involving inner products, adjoints, or unitary matrices, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 5.2.1. *Invariant Subspace*
Let $T \in \mathcal{L}(\mathcal{V})$. A subspace $\mathcal{U} \leq \mathcal{V}$ is *invariant under T* iff $T\mathcal{U} \subseteq \mathcal{U}$. The set of all T-invariant subspaces is $\operatorname{Inv}(T) := \{\mathcal{U} \leq \mathcal{V} : T\mathcal{U} \subseteq \mathcal{U}\}$.

Definition 5.2.2. *Restriction and Quotient Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathcal{U} \in \operatorname{Inv}(T)$.
(i) The *restriction operator* is $T|_{\mathcal{U}} : \mathcal{U} \to \mathcal{U}$, $T|_{\mathcal{U}}(\mathbf{u}) := T\mathbf{u}$.
(ii) The *quotient operator* is $T/\mathcal{U} : \mathcal{V}/\mathcal{U} \to \mathcal{V}/\mathcal{U}$, $(T/\mathcal{U})(\mathbf{v} + \mathcal{U}) := T\mathbf{v} + \mathcal{U}$.

Proposition 5.2.3. *Restriction and Quotient Operators Are Well-Defined*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathcal{U} \in \operatorname{Inv}(T)$.
(i) $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$.
(ii) $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$.

Proof.
(i) The inclusion $T\mathcal{U} \subseteq \mathcal{U}$ holds by Definition 5.2.1, so $T|_{\mathcal{U}}$ has codomain $\mathcal{U}$.
For $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, $T|_{\mathcal{U}}(a\mathbf{u} + b\mathbf{v}) = aT|_{\mathcal{U}}\mathbf{u} + bT|_{\mathcal{U}}\mathbf{v}$ by the linearity of T.
Therefore $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ by the codomain inclusion and linearity identity.
(ii) Suppose $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$. Then $\mathbf{v} - \mathbf{w} \in \mathcal{U}$ by Proposition 2.2.2.
The relation $T\mathbf{v} - T\mathbf{w} = T(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$ holds by linearity and invariance.
Consequently, $T\mathbf{v} + \mathcal{U} = T\mathbf{w} + \mathcal{U}$ by Proposition 2.2.2, so the quotient rule is independent of the representative.
Set $S := T/\mathcal{U}$ and $[\mathbf{x}] := \mathbf{x} + \mathcal{U}$. For $a, b \in \mathbb{F}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $S(a[\mathbf{v}] + b[\mathbf{w}]) = [T(a\mathbf{v} + b\mathbf{w})]$ by Proposition 2.2.6(ii).
The identity $[T(a\mathbf{v} + b\mathbf{w})] = [aT\mathbf{v} + bT\mathbf{w}] = aS[\mathbf{v}] + bS[\mathbf{w}]$ holds by linearity of T and the quotient rule.
Therefore $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$ by representative independence and the two linearity identities.
Structural Commentary: Invariance closes the restriction codomain and transports equality between quotient representatives.
Key Insight: Preserving $\mathcal{U}$ also preserves differences modulo $\mathcal{U}$. □

Proposition 5.2.4. *One-Dimensional Invariant Subspaces and Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$. Then $\operatorname{span}(\mathbf{v}) \in \operatorname{Inv}(T)$ iff $\mathbf{v}$ is an eigenvector of T.

Proof.

(⇒) Suppose $\operatorname{span}(\mathbf{v}) \in \operatorname{Inv}(T)$. The relation $(\exists\, \lambda \in \mathbb{F})\, T\mathbf{v} = \lambda\mathbf{v}$ holds by $T\mathbf{v} \in \operatorname{span}(\mathbf{v})$.
Therefore $\mathbf{v}$ is an eigenvector of T by $\mathbf{v} \neq \mathbf{0}$ and Definition 5.1.1.

(⇐) Suppose $\mathbf{v}$ is an eigenvector of T. Choose $\lambda \in \mathbb{F}$ satisfying $T\mathbf{v} = \lambda\mathbf{v}$ by Definition 5.1.1.
The inclusion $T(\operatorname{span}(\mathbf{v})) \subseteq \operatorname{span}(\mathbf{v})$ holds by $(\forall\, a \in \mathbb{F})\, T(a\mathbf{v}) = a\lambda\mathbf{v}$. Hence $\operatorname{span}(\mathbf{v}) \in \operatorname{Inv}(T)$ by Definition 5.2.1.

Structural Commentary: The two directions translate line invariance into the image relation for its generator.
Key Insight: A line is invariant exactly when its generator is rescaled. □

Figure 40 illustrates Proposition 5.2.4: T rescales $\mathbf{v}$ within $\operatorname{span}(\mathbf{v})$.
The proposition, rather than the diagram, owns the equivalence.

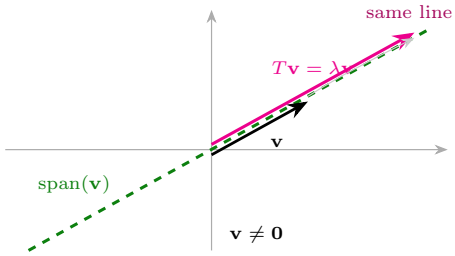

Figure 40: An eigenvector and its image lie on the same invariant line.

Proposition 5.2.5. *Invariant Subspaces and Block Form*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, suppose $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, and let β be a basis adapted to this decomposition.

(i) $\mathcal{U} \in \operatorname{Inv}(T) \Longrightarrow [T]_{\beta \leftarrow \beta}$ *is block upper triangular.*
(ii) $\mathcal{U}, \mathcal{W} \in \operatorname{Inv}(T) \Longrightarrow [T]_{\beta \leftarrow \beta}$ *is block diagonal.*

Proof. Write $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_p, \mathbf{w}_1, \ldots, \mathbf{w}_q)$, where $(\mathbf{u}_1, \ldots, \mathbf{u}_p)$ and $(\mathbf{w}_1, \ldots, \mathbf{w}_q)$ are bases of $\mathcal{U}$ and $\mathcal{W}$, respectively.

(i) Suppose $\mathcal{U} \in \operatorname{Inv}(T)$. The relation $(\forall\, j \in \{1, \ldots, p\})\, T\mathbf{u}_j \in \mathcal{U}$ holds by invariance.
Therefore the coordinates along $\mathbf{w}_1, \ldots, \mathbf{w}_q$ vanish, forming the zero lower-left block.

(ii) Suppose also that $\mathcal{W} \in \operatorname{Inv}(T)$. The relation $(\forall\, j \in \{1, \ldots, q\})\, T\mathbf{w}_j \in \mathcal{W}$ holds by invariance.
Therefore the coordinates along $\mathbf{u}_1, \ldots, \mathbf{u}_p$ vanish, forming the zero upper-right block.

Structural Commentary: Adapted-basis columns convert invariance of each summand into a vanishing off-diagonal block.
Key Insight: Each invariant summand eliminates the block pointing out of it. □

Definition 5.2.6. *Reducing Pair*
Let $T \in \mathcal{L}(\mathcal{V})$. A pair $(\mathcal{U}, \mathcal{W})$ of subspaces *reduces* T iff $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ and $\mathcal{U}, \mathcal{W} \in \operatorname{Inv}(T)$.
For such a pair, $T = T|_{\mathcal{U}} \oplus T|_{\mathcal{W}}$ means $(\forall\, (\mathbf{u}, \mathbf{w}) \in \mathcal{U} \times \mathcal{W})\, T(\mathbf{u} + \mathbf{w}) = T|_{\mathcal{U}}\mathbf{u} + T|_{\mathcal{W}}\mathbf{w}$.

Theorem 5.2.7. *Adjoints and Invariant Orthogonal Complements*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space and let $\mathcal{U} \leq \mathcal{V}$.

(i) $\mathcal{U} \in \operatorname{Inv}(T) \Longleftrightarrow \mathcal{U}^{\perp} \in \operatorname{Inv}(T^*)$.
(ii) $(\mathcal{U}, \mathcal{U}^{\perp})$ *reduces* $T \Longleftrightarrow \mathcal{U} \in \operatorname{Inv}(T) \cap \operatorname{Inv}(T^*)$.
(iii) If $(\mathcal{U}, \mathcal{U}^{\perp})$ *reduces* T, *then* $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$.

Proof.

(i) (⇒) Suppose $\mathcal{U} \in \operatorname{Inv}(T)$. For $\mathbf{z} \in \mathcal{U}^{\perp}$ and $\mathbf{u} \in \mathcal{U}$, $\langle T^*\mathbf{z}, \mathbf{u} \rangle = \langle \mathbf{z}, T\mathbf{u} \rangle = 0$ by conjugate symmetry, Definition 2.3.1, and $T\mathbf{u} \in \mathcal{U}$.
Thus $T^*\mathbf{z} \in \mathcal{U}^{\perp}$, and $\mathcal{U}^{\perp} \in \operatorname{Inv}(T^*)$ by Definition 5.2.1.

(i) (⇐) Suppose $\mathcal{U}^{\perp} \in \operatorname{Inv}(T^*)$. If $\mathbf{u} \in \mathcal{U}$ and $\mathbf{z} \in \mathcal{U}^{\perp}$, then $\langle T\mathbf{u}, \mathbf{z} \rangle = \langle \mathbf{u}, T^*\mathbf{z} \rangle = 0$ by Definition 2.3.1.
Therefore $T\mathbf{u} \in (\mathcal{U}^{\perp})^{\perp} = \mathcal{U}$ by Corollary 2.1.8, so $\mathcal{U} \in \operatorname{Inv}(T)$ by Definition 5.2.1.

(ii) (⇒) Suppose $(\mathcal{U}, \mathcal{U}^{\perp})$ reduces T. Then $\mathcal{U} \in \operatorname{Inv}(T)$ and $\mathcal{U}^{\perp} \in \operatorname{Inv}(T)$ by Definition 5.2.6.
Thus $\mathcal{U} \in \operatorname{Inv}(T^*)$ by (i) for T^* and Proposition 2.3.8(iii).

(ii) (⇐) Suppose $\mathcal{U} \in \operatorname{Inv}(T) \cap \operatorname{Inv}(T^*)$. Then $\mathcal{U}^{\perp} \in \operatorname{Inv}(T)$ by (i) for T^* and Proposition 2.3.8(iii).
The identity $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$ holds by Theorem 2.1.5. Hence $(\mathcal{U}, \mathcal{U}^{\perp})$ reduces T by Definition 5.2.6.

(iii) Suppose $(\mathcal{U}, \mathcal{U}^{\perp})$ reduces T. Then $\mathcal{U} \in \operatorname{Inv}(T) \cap \operatorname{Inv}(T^*)$ by (ii), and $T|_{\mathcal{U}}, (T^*)|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ by Proposition 5.2.3(i).
For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, $\langle T|_{\mathcal{U}}\mathbf{u}, \mathbf{v} \rangle_{\mathcal{U}} = \langle \mathbf{u}, (T^*)|_{\mathcal{U}}\mathbf{v} \rangle_{\mathcal{U}}$ by Definition 2.3.1. Identity $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$ follows by Definition 2.3.1, Lemma 2.3.3.

Structural Commentary: Adjoint duality transfers invariance to $\mathcal{U}^{\perp}$; orthogonal decomposition then characterizes reduction.
Key Insight: Reducing $\mathcal{U}$ is joint invariance under T and T^*. □

Proposition 5.2.8. *Reducing Pairs and Projections*
Let $T \in \mathcal{L}(\mathcal{V})$, let $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, and set $\rho := \rho_{\mathcal{U},\mathcal{W}}$. Then $(\mathcal{U}, \mathcal{W})$ reduces T iff $T\rho = \rho T$.

Proof.

(⇒) Suppose $(\mathcal{U}, \mathcal{W})$ reduces T. For $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $(\mathbf{u}, \mathbf{w}) \in \mathcal{U} \times \mathcal{W}$, $(T\rho)\mathbf{v} = T\mathbf{u} = (\rho T)\mathbf{v}$ by $T\mathbf{u} \in \mathcal{U}$ and $T\mathbf{w} \in \mathcal{W}$.
Therefore $T\rho = \rho T$ by pointwise equality.

(⇐) Suppose $T\rho = \rho T$. For $\mathbf{u} \in \mathcal{U}$, $\rho T\mathbf{u} = T\rho\mathbf{u} = T\mathbf{u}$, so $T\mathbf{u} \in \operatorname{range}(\rho) = \mathcal{U}$ by Proposition 2.4.2(iii).
For $\mathbf{w} \in \mathcal{W}$, $\rho T\mathbf{w} = T\rho\mathbf{w} = \mathbf{0}$, so $T\mathbf{w} \in \ker(\rho) = \mathcal{W}$ by Proposition 2.4.2(iii). Hence $(\mathcal{U}, \mathcal{W})$ reduces T by Definition 5.2.6.

Structural Commentary: Pointwise commutation preserves the range and kernel summands of the algebraic projection.
Key Insight: Commutation with ρ is simultaneous invariance of its range and kernel. □

Proposition 5.2.9. *Invariant Subspaces and Algebraic Projections*
Let $T \in \mathcal{L}(\mathcal{V})$, let $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, and set $\rho := \rho_{\mathcal{U},\mathcal{W}}$.

(i) $\mathcal{U} \in \mathrm{Inv}(T) \iff \rho T \rho = T\rho$.
(ii) If $\mathcal{U} \in \mathrm{Inv}(T)$, *then* $(\forall\, \mathcal{Z} \leq \mathcal{V})\,[\mathcal{V} = \mathcal{U} \oplus \mathcal{Z} \implies \rho_{\mathcal{U},\mathcal{Z}} T \rho_{\mathcal{U},\mathcal{Z}} = T\rho_{\mathcal{U},\mathcal{Z}}]$.

Proof.

(i) ($\Rightarrow$) Suppose $\mathcal{U} \in \mathrm{Inv}(T)$, and write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $(\mathbf{u}, \mathbf{w}) \in \mathcal{U} \times \mathcal{W}$.
The identity $(\rho T \rho)\mathbf{v} = \rho T \mathbf{u} = T\mathbf{u} = (T\rho)\mathbf{v}$ holds by $T\mathbf{u} \in \mathcal{U}$ and Definition 2.4.1. Therefore $\rho T \rho = T\rho$ by pointwise equality.
(i) ($\Leftarrow$) Suppose $\rho T \rho = T\rho$. If $\mathbf{u} \in \mathcal{U}$, then $\rho T \mathbf{u} = \rho T \rho \mathbf{u} = T\rho\mathbf{u} = T\mathbf{u}$.
Hence $T\mathbf{u} \in \mathrm{range}(\rho) = \mathcal{U}$ by Proposition 2.4.2(iii), so $\mathcal{U} \in \mathrm{Inv}(T)$ by Definition 5.2.1.
(ii) Suppose $\mathcal{U} \in \mathrm{Inv}(T)$ and fix $\mathcal{Z} \leq \mathcal{V}$ satisfying $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$. Set $\sigma := \rho_{\mathcal{U},\mathcal{Z}}$.
The identity $\sigma T \sigma = T\sigma$ holds by the forward implication in (i) for $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$. Claim (ii) follows by the arbitrary choice of $\mathcal{Z}$.

Structural Commentary: The compression identity tests range preservation and remains independent of the chosen complement.
Key Insight: $\rho T \rho = T\rho$ forbids T from leaving range(ρ). □

Theorem 5.2.10. *Invariant Subspaces, Blocks, and Projections*
Let $n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $\mathcal{U} \leq \mathbb{F}^n$, where $\dim(\mathcal{U}) = r$ and $1 \leq r \leq n-1$.
Let $Q = [Q_1\ Q_2]$ be unitary, where the columns of Q_1 and Q_2 are orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$, respectively.
Set $P := Q_1 Q_1^*$. The following are equivalent:

(i) $A\mathcal{U} \subseteq \mathcal{U}$.
(ii) The lower-left $(n-r) \times r$ block of $Q^* A Q$ is zero.
(iii) $PAP = AP$.

Proof. The identities $Q_1^* Q_1 = I_r$, $Q_2^* Q_1 = 0$, $P^* = P$, and $P^2 = Q_1 (Q_1^* Q_1) Q_1^* = P$ hold by unitarity of Q.
For $\mathbf{x} \in \mathbb{F}^n$ and $\mathbf{c} \in \mathbb{F}^r$, $P\mathbf{x} \in \mathrm{col}(Q_1)$ and $P(Q_1\mathbf{c}) = Q_1\mathbf{c}$, so $\mathrm{range}(P) = \mathrm{col}(Q_1) = \mathcal{U}$.
For $\mathbf{z} \in \mathcal{U}^\perp$, $Q_1^*\mathbf{z} = \mathbf{0}$ and $P\mathbf{z} = \mathbf{0}$. Therefore $P = \rho_{\mathcal{U},\mathcal{U}^\perp}$ by Theorem 2.1.5 and Definition 2.4.1.
The matrix P is an orthogonal projection matrix by Definition 2.5.5.
Set $M := Q^* A Q = \left[\begin{smallmatrix} B & X \\ Y & C \end{smallmatrix}\right]$, so $Y = Q_2^* A Q_1$, and set $E := Q^* P Q = I_r \oplus 0_{n-r}$.

(i) $\Rightarrow$ *(ii)* Suppose $A\mathcal{U} \subseteq \mathcal{U}$. Every column of AQ_1 belongs to $\mathcal{U} = (\mathcal{U}^\perp)^\perp$ by Corollary 2.1.8.
Since the columns of Q_2 span $\mathcal{U}^\perp$, $Y = Q_2^* A Q_1 = 0$.
(ii) $\Rightarrow$ *(iii)* Suppose $Y = 0$. Then $EME = \left[\begin{smallmatrix} B & 0 \\ 0 & 0 \end{smallmatrix}\right] = ME$, so $Q^*(PAP)Q = Q^*(AP)Q$.
Therefore $PAP = AP$ by multiplication by Q and Q^*.
(iii) $\Rightarrow$ *(i)* Suppose $PAP = AP$. Since P is the algebraic projection onto $\mathcal{U}$ along $\mathcal{U}^\perp$, $A\mathcal{U} \subseteq \mathcal{U}$ by Proposition 5.2.9(i).

Structural Commentary: Unitary coordinates identify the escape block; projection compression tests the same range preservation.
Key Insight: The lower-left block records precisely the component leaving $\mathcal{U}$. □

Corollary 5.2.11. *Reducing Subspaces and Commuting Projections*
Let $n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $\mathcal{U} \leq \mathbb{F}^n$, where $\dim(\mathcal{U}) = r$ and $1 \leq r \leq n-1$.
Let $Q = [Q_1\ Q_2]$ be unitary with $\mathrm{col}(Q_1) = \mathcal{U}$ and $\mathrm{col}(Q_2) = \mathcal{U}^\perp$. Set $P := Q_1 Q_1^*$. The following are equivalent:

(i) $A\mathcal{U} \subseteq \mathcal{U}$ *and* $A^*\mathcal{U} \subseteq \mathcal{U}$.
(ii) $Q^* A Q$ is block diagonal relative to $\mathbb{F}^r \oplus \mathbb{F}^{n-r}$.
(iii) $PA = AP$.

Proof. Set $M := Q^* A Q = \left[\begin{smallmatrix} B & X \\ Y & C \end{smallmatrix}\right]$ and $E := Q^* P Q = I_r \oplus 0_{n-r}$. Then $Q^* A^* Q = M^*$, $Q^*(PA)Q = EM$, and $Q^*(AP)Q = ME$.

(i) $\Rightarrow$ *(ii)* Suppose $A\mathcal{U} \subseteq \mathcal{U}$ and $A^*\mathcal{U} \subseteq \mathcal{U}$. Then $Y = 0$ by Theorem 5.2.10, (i)$\Rightarrow$(ii), for A.
The relation $X^* = 0$ holds by the same implication for A^* because X^* is the lower-left block of M^*.
Therefore $X = Y = 0$, so $M = B \oplus C$.
(ii) $\Rightarrow$ *(iii)* Suppose $M = B \oplus C$. Then $EM = ME$ by block multiplication, and $PA = AP$ by conjugation with Q.
(iii) $\Rightarrow$ *(i)* Suppose $PA = AP$. Then $EM = ME$. The identities $EM = \left[\begin{smallmatrix} B & X \\ 0 & 0 \end{smallmatrix}\right]$ and $ME = \left[\begin{smallmatrix} B & 0 \\ Y & 0 \end{smallmatrix}\right]$ imply $X = Y = 0$.
Therefore $A\mathcal{U} \subseteq \mathcal{U}$ by Theorem 5.2.10, (ii)$\Rightarrow$(i), for A.
Since $X^* = 0$ is the lower-left block of M^*, $A^*\mathcal{U} \subseteq \mathcal{U}$ by the same implication for A^*.

Structural Commentary: The invariant-block criterion for A and A^* eliminates opposite off-diagonal blocks.
Key Insight: Projection commutation is equivalent to simultaneous A- and A^*-invariance. □

Figure 41 summarizes the preceding results: invariance removes the lower-left block.
Projection commutation removes both off-diagonal blocks. Proposition 5.2.5, Theorem 5.2.10, and Corollary 5.2.11 own these claims.

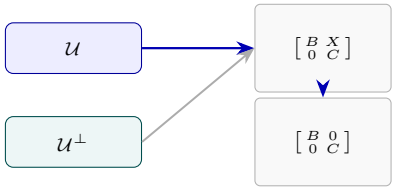

Figure 41: Adapted coordinates represent invariance by upper block form and reduction by block diagonal form.

Perspective 5.2.12. *Two Smaller Views of an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathcal{U} \in \mathrm{Inv}(T)$. Definition 5.2.2 separates the action inside $\mathcal{U}$ from the action modulo $\mathcal{U}$.
Proposition 5.2.3 validates both operators. In an adapted block matrix from Proposition 5.2.5(i), they determine the two diagonal blocks but not the upper-right block. Thus restriction and quotient simplify the study of T without necessarily determining T.

5.3 Triangularization over Complex and Real Fields

Definition 5.3.1. *Diagonal and Upper-Triangular Matrices*
Let $n \in \mathbb{Z}_{>0}$ and $A = [a_{ij}]_{i,j=1}^{n} \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) The *diagonal* of A is $(a_{11}, \dots, a_{nn})$.
(ii) The matrix A is *diagonal* iff $(\forall\, i, j \in \{1, \dots, n\})\, (i \neq j \Rightarrow a_{ij} = 0)$.
(iii) The matrix A is *upper triangular* iff $(\forall\, i, j \in \{1, \dots, n\})\, (i > j \Rightarrow a_{ij} = 0)$.

Theorem 5.3.2. *Complex Operators Have Eigenvalues*
Let $\mathcal{V}$ be a nonzero finite-dimensional complex vector space. Every $T \in \mathcal{L}(\mathcal{V})$ has an eigenvalue.

Proof. Fix $T \in \mathcal{L}(\mathcal{V})$. Choose a nonconstant annihilating polynomial for T by Proposition 1.1.7.
One of its complex roots is an eigenvalue of T by Theorem 1.1.9(ii).
Structural Commentary: An annihilator and complex factorization reduce existence to a singular linear factor.
Key Insight: A zero product of shifted operators contains a singular factor. □

Theorem 5.3.3. *Criteria for an Upper-Triangular Operator Matrix*
Let $n \in \mathbb{Z}_{>0}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$. The following are equivalent:

(i) $[T]_{\beta \leftarrow \beta}$ *is upper triangular;*
(ii) $(\forall\, j \in \{1, \dots, n\})\, T\mathbf{v}_j \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$;
(iii) $(\forall\, j \in \{1, \dots, n\})\ \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j) \in \mathrm{Inv}(T)$.

Proof.

(i) $\Rightarrow$ *(ii)* Fix $j \in \{1, \dots, n\}$. The jth column of $[T]_{\beta \leftarrow \beta}$ is $[T\mathbf{v}_j]_\beta$.
Its entries below row j vanish by (i), so $T\mathbf{v}_j \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$. Thus (ii) holds by the arbitrariness of j.

(ii) $\Rightarrow$ *(iii)* Fix $j \in \{1, \dots, n\}$ and $\mathbf{u} \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$.
Choose $a_1, \dots, a_j \in \mathbb{F}$ with $\mathbf{u} = \sum_{k=1}^{j} a_k \mathbf{v}_k$. Then $T\mathbf{u} = \sum_{k=1}^{j} a_k T\mathbf{v}_k \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ by (ii).
Hence the span is invariant under T by Definition 5.2.1, and (iii) follows.

(iii) $\Rightarrow$ *(i)* For $j \in \{1, \dots, n\}$, the vector $\mathbf{v}_j$ belongs to the T-invariant space $\mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ by (iii).
Thus $T\mathbf{v}_j \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ by Definition 5.2.1. Every matrix entry below the diagonal therefore vanishes, so (i) holds.

Structural Commentary: The implication cycle translates column support into preservation of every initial span.
Key Insight: Upper triangularity is the coordinate form of an invariant flag. □

Corollary 5.3.4. *Upper-Triangular Form in an Orthonormal Basis*
Let $\mathcal{V}$ be a nonzero finite-dimensional inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
If T has an upper-triangular matrix relative to a basis of $\mathcal{V}$, then it has one relative to an orthonormal basis.

Proof. Choose a basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ for which $[T]_{\beta \leftarrow \beta}$ is upper triangular, and set $\mathcal{U}_j := \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$.
Then $(\forall\, j \in \{1, \dots, n\})\, \mathcal{U}_j \in \mathrm{Inv}(T)$ by Theorem 5.3.3(i),(iii). Let $\gamma = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ be the Gram–Schmidt output for β.
List γ is orthonormal by Theorem 1.3.6(ii). Prefix identities $(\forall\, j \in \{1, \dots, n\})\ \mathrm{span}(\mathbf{e}_1, \dots, \mathbf{e}_j) = \mathcal{U}_j$ hold by Theorem 1.3.6(i).
Since $\mathrm{span}(\gamma) = \mathcal{U}_n = \mathcal{V}$, γ is an orthonormal basis. Its matrix for T is upper triangular by Theorem 5.3.3(iii),(i).
Structural Commentary: Gram–Schmidt transports the invariant flag to an orthonormal basis.
Key Insight: Triangularity depends on initial spans, not their chosen generators. □

Theorem 5.3.5. *Complex Upper Triangularization*
Let $\mathcal{V}$ be a nonzero finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$.
Then T has an upper-triangular matrix relative to a basis of $\mathcal{V}$.

Proof.

(i) *Induction*: Induct on $n := \dim_{\mathbb{C}}(\mathcal{V})$. For $n = 1$, every operator matrix is upper triangular. Assume $n > 1$ and that every operator on a nonzero complex vector space of dimension less than n has an upper-triangular matrix relative to a basis.
(ii) *Eigenline and Quotient*: Choose eigenvector $\mathbf{v}_1$ of T by Theorem 5.3.2, set $\mathcal{U} := \mathrm{span}(\mathbf{v}_1)$; $\mathcal{U} \in \mathrm{Inv}(T)$ by Proposition 5.2.4.
The relations $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$ and $\dim_{\mathbb{C}}(\mathcal{V}/\mathcal{U}) = n - 1 > 0$ hold by Proposition 5.2.3(ii) and Theorem 2.2.9(ii).
By induction, choose a quotient basis $(\mathbf{v}_2 + \mathcal{U}, \dots, \mathbf{v}_n + \mathcal{U})$ in which $T/\mathcal{U}$ is upper triangular.
(iii) *Lifted Basis*: Choose representatives $\mathbf{v}_2, \dots, \mathbf{v}_n \in \mathcal{V}$.
A zero relation among $\mathbf{v}_1, \dots, \mathbf{v}_n$ descends to the quotient basis, so the coefficients of $\mathbf{v}_2, \dots, \mathbf{v}_n$ vanish.
The remaining coefficient vanishes because $\mathbf{v}_1 \neq \mathbf{0}$. Thus $\beta := (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis by Proposition 1.4.4(ii).
(iv) *Triangularity*: The relation $T\mathbf{v}_1 \in \mathrm{span}(\mathbf{v}_1)$ holds by $\mathcal{U} \in \mathrm{Inv}(T)$. Fix $j \in \{2, \dots, n\}$.
Choose $c_{2,j}, \dots, c_{j,j} \in \mathbb{C}$ satisfying $(T/\mathcal{U})(\mathbf{v}_j + \mathcal{U}) = \sum_{k=2}^{j} c_{k,j}(\mathbf{v}_k + \mathcal{U})$ by Theorem 5.3.3(i),(ii).
The relation $T\mathbf{v}_j - \sum_{k=2}^{j} c_{k,j}\mathbf{v}_k \in \mathcal{U} = \mathrm{span}(\mathbf{v}_1)$ holds by Definition 5.2.2(ii).
Thus $(\forall\, j \in \{1, \dots, n\})\, T\mathbf{v}_j \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$, and $[T]_{\beta \leftarrow \beta}$ is upper triangular by Theorem 5.3.3(ii),(i).

Structural Commentary: Induction lifts a triangular quotient basis above a complex eigenline.
Key Insight: Each lifted quotient relation differs by a vector in the eigenline. □

Definition 5.3.6. *Complexification of a Real Vector Space*
Let $\mathcal{V}$ be a real vector space.

(i) The *complexification* of $\mathcal{V}$ is $\mathcal{V}_{\mathbb{C}} := \mathcal{V} \times \mathcal{V}$, with $(\mathbf{u}, \mathbf{v})$ denoted $\mathbf{u} + i\mathbf{v}$.
(ii) Addition is $(\mathbf{u}_1 + i\mathbf{v}_1) + (\mathbf{u}_2 + i\mathbf{v}_2) := (\mathbf{u}_1 + \mathbf{u}_2) + i(\mathbf{v}_1 + \mathbf{v}_2)$. For $a, b \in \mathbb{R}$, define $(a + bi)(\mathbf{u} + i\mathbf{v}) := (a\mathbf{u} - b\mathbf{v}) + i(b\mathbf{u} + a\mathbf{v})$.
(iii) The *complexification map* is $\mathrm{cpx} : \mathcal{V} \to \mathcal{V}_{\mathbb{C}}$, $\mathrm{cpx}(\mathbf{u}) := \mathbf{u} + i\mathbf{0}$. Its image is $\mathrm{im}(\mathrm{cpx}) = \{\mathbf{u} + i\mathbf{0} : \mathbf{u} \in \mathcal{V}\}$.
(iv) Define $\mathrm{Re}, \mathrm{Im} : \mathcal{V}_{\mathbb{C}} \to \mathcal{V}$ by $\mathrm{Re}(\mathbf{u} + i\mathbf{v}) := \mathbf{u}$ and $\mathrm{Im}(\mathbf{u} + i\mathbf{v}) := \mathbf{v}$. Then $\mathbf{z} = \mathbf{0}$ iff $\mathrm{Re}\,\mathbf{z} = \mathrm{Im}\,\mathbf{z} = \mathbf{0}$.

Proposition 5.3.7. *Complexification Preserves Bases*
Let $\mathcal{V}$ be a finite-dimensional real vector space and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a real basis of $\mathcal{V}$. Set $\beta_{\mathbb{C}} := \operatorname{cpx}(\beta)$. Then:

(i) $\mathcal{V}_{\mathbb{C}}$ is a complex vector space;
(ii) $\beta_{\mathbb{C}}$ is a complex basis of $\mathcal{V}_{\mathbb{C}}$;
(iii) $\dim_{\mathbb{C}}(\mathcal{V}_{\mathbb{C}}) = \dim_{\mathbb{R}}(\mathcal{V})$.

Proof.

(i) Addition makes $\mathcal{V}_{\mathbb{C}}$ an abelian group with zero $\mathbf{0} + i\mathbf{0}$ and inverse $-(\mathbf{u} + i\mathbf{v}) = (-\mathbf{u}) + i(-\mathbf{v})$ by the real vector-space axioms. Fix $\alpha = a + bi$, $\eta = c + di \in \mathbb{C}$ and $\mathbf{z} = \mathbf{u} + i\mathbf{v} \in \mathcal{V}_{\mathbb{C}}$. The real parts of $\alpha(\eta\mathbf{z})$ and $(\alpha\eta)\mathbf{z}$ both equal $(ac - bd)\mathbf{u} - (ad + bc)\mathbf{v}$. Imaginary parts both equal $(ad + bc)\mathbf{u} + (ac - bd)\mathbf{v}$, so scalar multiplication is associative. The identity $1\mathbf{z} = \mathbf{z}$ follows directly. The two distributive laws hold by Definition 5.3.6(ii) and real vector-space distributivity. Hence $\mathcal{V}_{\mathbb{C}}$ is a complex vector space.

(ii) For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, choose real coefficients satisfying $\mathbf{u} = \sum_{j=1}^{n} a_j \mathbf{v}_j$ and $\mathbf{v} = \sum_{j=1}^{n} b_j \mathbf{v}_j$.

Then $\mathbf{u} + i\mathbf{v} = \sum_{j=1}^{n} (a_j + ib_j) \operatorname{cpx}(\mathbf{v}_j)$, so $\beta_{\mathbb{C}}$ spans $\mathcal{V}_{\mathbb{C}}$. Suppose $\sum_{j=1}^{n} (c_j + id_j) \operatorname{cpx}(\mathbf{v}_j) = \mathbf{0}$.

The equality $\sum_{j=1}^{n} c_j \mathbf{v}_j = \sum_{j=1}^{n} d_j \mathbf{v}_j = \mathbf{0}$ holds by Definition 5.3.6(ii),(iv).

Thus $(\forall\, j \in \{1, \dots, n\})\, c_j = d_j = 0$ by independence of β, and $\beta_{\mathbb{C}}$ is a complex basis.

(iii) The basis $\beta_{\mathbb{C}}$ has $n = \dim_{\mathbb{R}}(\mathcal{V})$ members. Hence $\dim_{\mathbb{C}}(\mathcal{V}_{\mathbb{C}}) = \dim_{\mathbb{R}}(\mathcal{V})$.

Structural Commentary: Component identities establish scalar action; coefficient separation transports the basis and dimension.
Key Insight: Each complex coefficient is two independent real coefficients. □

Corollary 5.3.8. *Coordinates Survive Complexification*
Let $\mathcal{V}$ be a finite-dimensional real vector space and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$.
Set $\beta_{\mathbb{C}} := \operatorname{cpx}(\beta)$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\, [\operatorname{cpx}(\mathbf{v})]_{\beta_{\mathbb{C}}} = [\mathbf{v}]_{\beta} \in \mathbb{R}^n \subseteq \mathbb{C}^n$.

Proof. Fix $\mathbf{v} \in \mathcal{V}$ and write $[\mathbf{v}]_{\beta} = (a_1, \dots, a_n)^t \in \mathbb{R}^n$. Then $\operatorname{cpx}(\mathbf{v}) = \sum_{j=1}^{n} a_j \operatorname{cpx}(\mathbf{v}_j)$ by Definition 5.3.6(ii)–(iii).
Thus $[\operatorname{cpx}(\mathbf{v})]_{\beta_{\mathbb{C}}} = (a_1, \dots, a_n)^t = [\mathbf{v}]_{\beta}$ by Proposition 5.3.7(ii).
Structural Commentary: The embedding preserves one real expansion inside the complexified basis.
Key Insight: Embedded real vectors retain their real coordinate columns. □

Proposition 5.3.9. *Complexification as a Real Space*
Let $\mathcal{V}$ be finite-dimensional real vector space, $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be basis of $\mathcal{V}$. Regard $\mathcal{V}_{\mathbb{C}}$ as real vector space by restriction of scalars.
Set $\mathbf{e}_j := \operatorname{cpx}(\mathbf{v}_j)$ $(j \in \{1, \dots, n\})$. Define $\delta := (\mathbf{e}_1, \dots, \mathbf{e}_n, i\mathbf{e}_1, \dots, i\mathbf{e}_n)$. Then:

(i) δ is a real basis of $\mathcal{V}_{\mathbb{C}}$;
(ii) $\dim_{\mathbb{R}}(\mathcal{V}_{\mathbb{C}}) = 2 \dim_{\mathbb{R}}(\mathcal{V})$;
(iii) $\operatorname{cpx} : \mathcal{V} \to \mathcal{V}_{\mathbb{C}}$ is real-linear and injective.

Proof.

(i) For $\mathbf{u} + i\mathbf{w} \in \mathcal{V}_{\mathbb{C}}$, choose real coefficients with $\mathbf{u} = \sum_{j=1}^{n} a_j \mathbf{v}_j$ and $\mathbf{w} = \sum_{j=1}^{n} b_j \mathbf{v}_j$.

Then $\mathbf{u} + i\mathbf{w} = \sum_{j=1}^{n} a_j \mathbf{e}_j + \sum_{j=1}^{n} b_j (i\mathbf{e}_j)$, so δ spans over $\mathbb{R}$.

If this right side is zero, its real and imaginary parts vanish by Definition 5.3.6(iv).

Thus $(\forall\, j \in \{1, \dots, n\})\, a_j = b_j = 0$ by independence of β, and δ is a real basis.

(ii) The basis δ has $2n = 2 \dim_{\mathbb{R}}(\mathcal{V})$ members.

(iii) For $r, s \in \mathbb{R}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\operatorname{cpx}(r\mathbf{u} + s\mathbf{v}) = r \operatorname{cpx}(\mathbf{u}) + s \operatorname{cpx}(\mathbf{v})$ by Definition 5.3.6(ii)–(iii).
If $\operatorname{cpx}(\mathbf{u}) = \operatorname{cpx}(\mathbf{v})$, then $\operatorname{cpx}(\mathbf{u} - \mathbf{v}) = \mathbf{0}$ by linearity and $\mathbf{u} - \mathbf{v} = \mathbf{0}$ by Definition 5.3.6(iv). Hence cpx is real-linear and injective.

Structural Commentary: Real–imaginary separation builds the doubled basis and detects the embedding kernel.
Key Insight: Complexification is two independent real copies of $\mathcal{V}$. □

Definition 5.3.10. *Complexification of a Real Operator*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$. The *complexification* of T is $T_{\mathbb{C}} : \mathcal{V}_{\mathbb{C}} \to \mathcal{V}_{\mathbb{C}}$, defined by $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) := T\mathbf{u} + iT\mathbf{v}$.

Proposition 5.3.11. *Complexification Preserves Operator Data*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$.

(i) $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$.
(ii) $(\forall\, k \in \mathbb{Z}_{\geq 0})(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\, (T_{\mathbb{C}})^k(\mathbf{u} + i\mathbf{v}) = T^k\mathbf{u} + iT^k\mathbf{v}$;
(iii) For a real basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ of $\mathcal{V}$, set $\beta_{\mathbb{C}} := \operatorname{cpx}(\beta)$. Then $[T_{\mathbb{C}}]_{\beta_{\mathbb{C}} \leftarrow \beta_{\mathbb{C}}} = [T]_{\beta \leftarrow \beta}$ as complex matrices.

Proof.

(i) Let $\mathbf{z} = \mathbf{u} + i\mathbf{v}$, $\mathbf{w} = \mathbf{x} + i\mathbf{y}$, and $\alpha = a + bi \in \mathbb{C}$. Then $T_{\mathbb{C}}(\mathbf{z} + \mathbf{w}) = T_{\mathbb{C}}\mathbf{z} + T_{\mathbb{C}}\mathbf{w}$ by the additivity of T.
By the real linearity of T, $T_{\mathbb{C}}(\alpha\mathbf{z}) = T(a\mathbf{u} - b\mathbf{v}) + iT(b\mathbf{u} + a\mathbf{v}) = (aT\mathbf{u} - bT\mathbf{v}) + i(bT\mathbf{u} + aT\mathbf{v})$.
The last term equals $\alpha T_{\mathbb{C}}\mathbf{z}$ by Definition 5.3.6(ii). Hence $T_{\mathbb{C}}$ is complex-linear.

(ii) Fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$. The identity holds for $k = 0$. If it holds at k, then $(T_{\mathbb{C}})^{k+1}(\mathbf{u} + i\mathbf{v}) = T_{\mathbb{C}}(T^k\mathbf{u} + iT^k\mathbf{v}) = T^{k+1}\mathbf{u} + iT^{k+1}\mathbf{v}$.
Thus $(\forall\, k \in \mathbb{Z}_{\geq 0})\, (T_{\mathbb{C}})^k(\mathbf{u} + i\mathbf{v}) = T^k\mathbf{u} + iT^k\mathbf{v}$ by induction, and claim (ii) follows by the arbitrariness of $\mathbf{u}, \mathbf{v}$.

(iii) The list $\beta_{\mathbb{C}}$ is a complex basis by Proposition 5.3.7(ii). Fix $j \in \{1, \dots, n\}$ and write $T\mathbf{v}_j = \sum_{r=1}^{n} a_{r,j} \mathbf{v}_r$ with $a_{r,j} \in \mathbb{R}$.

Then $T_{\mathbb{C}} \operatorname{cpx}(\mathbf{v}_j) = \sum_{r=1}^{n} a_{r,j} \operatorname{cpx}(\mathbf{v}_r)$ by Definition 5.3.10. Hence these matrices have same columns by arbitrariness of j.

Structural Commentary: Scalar expansion proves linearity, induction controls powers, and basis images identify columns.
Key Insight: Complexification changes the scalar field without changing real coordinate columns. □

Figure 42 illustrates the componentwise action established in Proposition 5.3.11.
The proposition, rather than the diagram, owns complex linearity and coordinate compatibility.

Figure 42: Complexification packages two real vectors: $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = T\mathbf{u} + iT\mathbf{v}$.

Proposition 5.3.12. *Real Eigenvalues of a Complexification*
Let $\mathcal{V}$ be a finite-dimensional real vector space, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{R}$. Then λ is an eigenvalue of T iff it is an eigenvalue of $T_{\mathbb{C}}$.

Proof.

($\Rightarrow$) Suppose $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $\operatorname{cpx}(\mathbf{v}) \neq \mathbf{0}$ by Proposition 5.3.9(iii).
The identity $T_{\mathbb{C}} \operatorname{cpx}(\mathbf{v}) = \lambda \operatorname{cpx}(\mathbf{v})$ holds by Definition 5.3.10.
Hence λ is an eigenvalue of $T_{\mathbb{C}}$ by Definition 5.1.1.

($\Leftarrow$) Suppose $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ satisfy $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = \lambda(\mathbf{u} + i\mathbf{v})$ and $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$.
Then $T\mathbf{u} = \lambda\mathbf{u}$ and $T\mathbf{v} = \lambda\mathbf{v}$ by Definitions 5.3.6(ii),(iv) and 5.3.10.
The relation $\mathbf{u} \neq \mathbf{0} \vee \mathbf{v} \neq \mathbf{0}$ holds by Definition 5.3.6(iv).
Thus one of $\mathbf{u}, \mathbf{v}$ is an eigenvector of T for λ by Definition 5.1.1.

Structural Commentary: The embedding preserves a real eigenvector; component separation recovers one from its complexification.
Key Insight: A real eigenvalue acts identically on both real components. □

Computation Method 5.3.13. *Real Invariant Subspace from a Complex Eigenpair*
Given $n \in \mathbb{Z}_{>0}$ and $A \in \operatorname{Mat}(n, n, \mathbb{R})$, construct a nonzero subspace $\mathcal{U} \leq \mathbb{R}^n$ satisfying $A\mathcal{U} \subseteq \mathcal{U}$ and $\dim(\mathcal{U}) \in \{1, 2\}$:

(i) Let $T_A \in \mathcal{L}(\mathbb{C}^n)$ be defined by $T_A\mathbf{z} := A\mathbf{z}$.
(ii) Choose an eigenvalue $a + bi \in \mathbb{C}$ of T_A by Theorem 5.3.2.
Choose an associated eigenvector $\mathbf{0} \neq \mathbf{u} + i\mathbf{v} \in \mathbb{C}^n$ by Definition 5.1.1.
Then $A(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$ by the definition of T_A.
(iii) Separate components to obtain $A\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $A\mathbf{v} = b\mathbf{u} + a\mathbf{v}$.
(iv) Set $\mathcal{U} := \operatorname{span}(\mathbf{u}, \mathbf{v})$. Then $\mathcal{U} \neq \{\mathbf{0}\}$ by step (ii), and $A\mathcal{U} \subseteq \mathcal{U}$ by step (iii) and linearity.
A basis sublist of $(\mathbf{u}, \mathbf{v})$ exists by Theorem 1.3.12, so $\dim(\mathcal{U}) \in \{1, 2\}$.

Theorem 5.3.14. *Invariant Line or Plane over $\mathbb{R}$*
Let $\mathcal{V}$ be a nonzero finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
Then $(\exists \mathcal{U} \in \operatorname{Inv}(T))\ \dim(\mathcal{U}) \in \{1, 2\}$.

Proof.

(i) *Complex Eigenpair*: The space $\mathcal{V}_{\mathbb{C}}$ is nonzero and finite-dimensional over $\mathbb{C}$ by Proposition 5.3.7(i),(iii).
The complexified operator satisfies $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$ by Proposition 5.3.11(i). Choose an eigenvalue $\lambda := a + bi \in \mathbb{C}$ by Theorem 5.3.2.
Choose $\mathbf{z} := \mathbf{u} + i\mathbf{v} \in \mathcal{V}_{\mathbb{C}} \setminus \{\mathbf{0}\}$ satisfying $T_{\mathbb{C}}\mathbf{z} = \lambda\mathbf{z}$ by Definition 5.1.1.
(ii) *Invariant Real Span*: The component identities $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $T\mathbf{v} = b\mathbf{u} + a\mathbf{v}$ hold by Definitions 5.3.6(ii),(iv) and 5.3.10.
Set $\mathcal{U} := \operatorname{span}(\mathbf{u}, \mathbf{v})$. For $c, d \in \mathbb{R}$, $T(c\mathbf{u} + d\mathbf{v}) = (ca + db)\mathbf{u} + (-cb + da)\mathbf{v} \in \mathcal{U}$ by the component identities.
Thus $\mathcal{U} \in \operatorname{Inv}(T)$ by Definition 5.2.1.
(iii) *Dimension*: The relation $\mathcal{U} \neq \{\mathbf{0}\}$ holds by $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$ and Definition 5.3.6(iv).
A basis sublist of $(\mathbf{u}, \mathbf{v})$ exists by Theorem 1.3.12. Consequently, $\dim(\mathcal{U}) \in \{1, 2\}$.

Structural Commentary: A complex eigenpair separates into two generators whose span is closed under T.
Key Insight: The real and imaginary components form a nonzero invariant span. □

Figure 43 depicts the component separation in Theorem 5.3.14.
The theorem, rather than the diagram, owns the invariant-subspace conclusion.

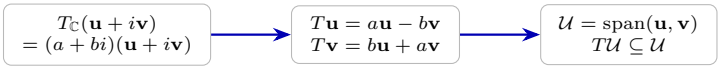

Figure 43: A complex eigenpair determines a nonzero invariant real line or plane.

Definition 5.3.15. *Almost Upper-Triangular Real Matrices*
Let $n \in \mathbb{Z}_{>0}$, $A \in \operatorname{Mat}(n, n, \mathbb{R})$. Matrix A is *almost upper triangular* iff it admits a block representation with following properties:

(i) The block count and sizes satisfy $m \in \mathbb{Z}_{>0}$ and $(d_1, \ldots, d_m) \in \{1, 2\}^m$.
(ii) One has $A = [A_{ij}]_{i,j=1}^{m}$, $A_{ij} \in \operatorname{Mat}(d_i, d_j, \mathbb{R})$, and $\sum_{j=1}^{m} d_j = n$.
(iii) $(\forall\, i, j \in \{1, \ldots, m\})\,(i > j \Rightarrow A_{ij} = 0)$.
(iv) Every diagonal block is either $A_{jj} = [\lambda_j]$ with $\lambda_j \in \mathbb{R}$, or $A_{jj} = \left[\begin{smallmatrix} a_j & -b_j \\ b_j & a_j \end{smallmatrix}\right]$ with $a_j, b_j \in \mathbb{R}$ and $b_j \neq 0$.

The matrix A is in *real Schur block form* iff it admits a block representation satisfying (i)–(iii), with no restriction beyond the prescribed sizes on its diagonal blocks.

Theorem 5.3.16. *Real Almost Upper Triangularization*
Let $\mathcal{V}$ be nonzero finite-dimensional real vector space, $T \in \mathcal{L}(\mathcal{V})$. Then T has almost-upper-triangular matrix relative to basis of $\mathcal{V}$.

Proof.

(i) *Induction*: Induct on $n := \dim_{\mathbb{R}}(\mathcal{V})$. If $n = 1$, every operator matrix is an admissible real 1×1 block. Assume $n \geq 2$ and that every operator on a nonzero real vector space of dimension less than n has an almost-upper-triangular matrix relative to a basis.

(ii) *Complex Eigenpair*: The space $\mathcal{V}_{\mathbb{C}}$ is a nonzero n-dimensional complex vector space by Proposition 5.3.7(i),(iii).
The complexified operator satisfies $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$ by Proposition 5.3.11(i). Choose an eigenvalue $\lambda := a + bi \in \mathbb{C}$ by Theorem 5.3.2.
Choose $\mathbf{z} := \mathbf{u} + i\mathbf{v} \in \mathcal{V}_{\mathbb{C}} \setminus \{\mathbf{0}\}$ satisfying $T_{\mathbb{C}}\mathbf{z} = \lambda\mathbf{z}$ by Definition 5.1.1.
Then $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $T\mathbf{v} = b\mathbf{u} + a\mathbf{v}$ by Definitions 5.3.6(ii),(iv) and 5.3.10.

(iii) *Initial Invariant Block*: Construct $\mathcal{U}$ and its basis α by cases.

(a) If $b = 0$, then $\mathbf{u} \neq \mathbf{0} \vee \mathbf{v} \neq \mathbf{0}$ by Definition 5.3.6(iv). Choose $\mathbf{x} \in \{\mathbf{u}, \mathbf{v}\} \setminus \{\mathbf{0}\}$. Set $\mathcal{U} := \operatorname{span}(\mathbf{x})$ and $\alpha := (\mathbf{x})$.
The component identities imply $T\mathbf{x} = a\mathbf{x}$, so $\mathcal{U} \in \operatorname{Inv}(T)$ and $[T|_{\mathcal{U}}]_{\alpha\leftarrow\alpha} = [a]$.

(b) Suppose $b \neq 0$. Implication $\mathbf{u} = \mathbf{0} \Rightarrow b\mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v} = \mathbf{0}$ contradicts chosen eigenvector by first component identity.
Hence $\mathbf{u} \neq \mathbf{0}$. If $\mathbf{v} = c\mathbf{u}$ for $c \in \mathbb{R}$, then $c(a - bc)\mathbf{u} = cT\mathbf{u} = T\mathbf{v} = (b + ac)\mathbf{u}$ by linearity and the component identities.
Thus $b(c^2 + 1) = 0$ by $\mathbf{u} \neq \mathbf{0}$, contrary to $b \neq 0$. Therefore $\mathbf{u}, \mathbf{v}$ are independent. Set $\mathcal{U} := \operatorname{span}(\mathbf{u}, \mathbf{v})$ and $\alpha := (\mathbf{u}, -\mathbf{v})$.
Then $\mathcal{U} \in \operatorname{Inv}(T)$ and $[T|_{\mathcal{U}}]_{\alpha\leftarrow\alpha} = \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$ by the component identities.

In both cases, α is a basis of $\mathcal{U}$, $\dim(\mathcal{U}) \in \{1, 2\}$, and $[T|_{\mathcal{U}}]_{\alpha\leftarrow\alpha}$ is an admissible first diagonal block.

(iv) *Quotient Induction*: If $\mathcal{U} = \mathcal{V}$, the basis α proves the claim. Suppose $\mathcal{U} \neq \mathcal{V}$. Then $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$ by Proposition 5.2.3(ii).
The inequality $0 < \dim_{\mathbb{R}}(\mathcal{V}/\mathcal{U}) = n - \dim_{\mathbb{R}}(\mathcal{U}) < n$ holds by Theorem 2.2.9(ii). Set $q := n - \dim_{\mathbb{R}}(\mathcal{U})$.
By induction, choose a quotient basis $\overline{\gamma} := (\mathbf{w}_1 + \mathcal{U}, \ldots, \mathbf{w}_q + \mathcal{U})$ in which $T/\mathcal{U}$ is almost upper triangular.

(v) *Lifted Block Form*: Set $\gamma := (\mathbf{w}_1, \ldots, \mathbf{w}_q)$ and $\beta := (\alpha, \gamma)$.
A zero relation in β descends to the quotient basis, so every coefficient of γ vanishes.
The coefficients of α then vanish because α is a basis of $\mathcal{U}$. Thus β is independent and has n members.
Hence β is a basis by Proposition 1.4.4(ii), and $\mathcal{V} = \mathcal{U} \oplus \operatorname{span}(\gamma)$.
The matrix of T in β is block upper triangular by Proposition 5.2.5(i).
Its first diagonal block is $[T|_{\mathcal{U}}]_{\alpha\leftarrow\alpha}$. Its lower-right block is $[T/\mathcal{U}]_{\overline{\gamma}\leftarrow\overline{\gamma}}$ by Definition 5.2.2(i)–(ii).
Therefore $[T]_{\beta\leftarrow\beta}$ is almost upper triangular by Definition 5.3.15.

Structural Commentary: Induction lifts an almost-triangular quotient above an invariant line or plane.
Key Insight: Complex eigenpairs supply the admissible real diagonal blocks. □

Corollary 5.3.17. *Orthonormal Real Schur Block Form*
Let $\mathcal{V}$ be nonzero finite-dimensional real inner-product space, $T \in \mathcal{L}(\mathcal{V})$.
Then T has real Schur block form relative to an orthonormal basis of $\mathcal{V}$.

Proof.

(i) *Invariant Block Flag*: Choose a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ in which T is almost upper triangular by Theorem 5.3.16.
Fix its block sizes $(d_1, \ldots, d_m) \in \{1, 2\}^m$. Set $s_k := \sum_{j=1}^{k} d_j$ and $\mathcal{U}_k := \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{s_k})$.
The zero blocks below the block diagonal imply $(\forall\, k \in \{1, \ldots, m\})\, \mathcal{U}_k \in \operatorname{Inv}(T)$ by Definitions 5.3.15(iii) and 5.2.1.

(ii) *Orthonormal Flag*: Let $\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the Gram–Schmidt output for β. The list γ is orthonormal by Theorem 1.3.6(ii).
The prefix identities $(\forall\, k \in \{1, \ldots, m\})\ \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{s_k}) = \mathcal{U}_k$ hold by Theorem 1.3.6(i).
Since $\operatorname{span}(\gamma) = \mathcal{U}_m = \mathcal{V}$, γ is an orthonormal basis.

(iii) *Block Form*: Relations $(\forall\, k \in \{1, \ldots, m\})\ \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{s_k}) \in \operatorname{Inv}(T)$ make $[T]_{\gamma\leftarrow\gamma}$ block upper triangular for sizes $(d_1, \ldots, d_m)$.
Hence it is in real Schur block form by Definition 5.3.15.

Structural Commentary: Gram–Schmidt preserves each subspace in the invariant block flag.
Key Insight: Orthonormalization changes block entries without changing block boundaries. □

Proposition 5.3.18. *Invertibility from an Upper-Triangular Matrix*
Let $\dim(\mathcal{V}) = n \in \mathbb{Z}_{>0}$ and $T \in \mathcal{L}(\mathcal{V})$. Suppose T has an upper-triangular matrix with diagonal $(\lambda_1, \ldots, \lambda_n)$ relative to a basis of $\mathcal{V}$.
Then T is invertible iff $(\forall\, j \in \{1, \ldots, n\})\ \lambda_j \neq 0$.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be such a basis.

($\Rightarrow$) Suppose T is invertible and $(\exists\, j \in \{1, \ldots, n\})\ \lambda_j = 0$; fix such a j. Set $\mathcal{U}_k := \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ and $\mathcal{U}_0 := \{\mathbf{0}\}$.
The first j matrix columns have zero entries in rows $j, \ldots, n$ by upper triangularity and $\lambda_j = 0$, so $T\mathcal{U}_j \subseteq \mathcal{U}_{j-1}$.
Map $S : \mathcal{U}_j \to \mathcal{U}_{j-1}$, $S\mathbf{u} := T\mathbf{u}$, is therefore linear. Since $\dim(\mathcal{U}_j) = j > j - 1 = \dim(\mathcal{U}_{j-1})$, S is not injective by Corollary 1.7.4.
Thus $(\exists\, \mathbf{u} \in \mathcal{U}_j \setminus \{\mathbf{0}\})\, T\mathbf{u} = \mathbf{0}$, contrary to the injectivity of T by Theorem 1.7.7(i),(ii). Hence $(\forall\, j \in \{1, \ldots, n\})\ \lambda_j \neq 0$.

($\Leftarrow$) Suppose $(\forall\, j \in \{1, \ldots, n\})\ \lambda_j \neq 0$, and set $M := [T]_{\beta\leftarrow\beta}$. Then M is invertible by Theorem 3.8.9(i).
Choose $S \in \mathcal{L}(\mathcal{V})$ satisfying $[S]_{\beta\leftarrow\beta} = M^{-1}$ by Theorem 3.4.5(i).
The identity $[ST]_{\beta\leftarrow\beta} = M^{-1}M = I_n = [I_{\mathcal{V}}]_{\beta\leftarrow\beta}$ holds by Proposition 3.4.4.
The identity $[TS]_{\beta\leftarrow\beta} = MM^{-1} = I_n = [I_{\mathcal{V}}]_{\beta\leftarrow\beta}$ holds by the same proposition.
Hence $ST = TS = I_{\mathcal{V}}$ by Theorem 3.4.5(i), so T is invertible by Definition 1.6.4.

Structural Commentary: A zero diagonal compresses an initial span; nonzero diagonals transport a matrix inverse.
Key Insight: Nonzero diagonal entries are exactly the invertible triangular layers. □

Proposition 5.3.19. *Eigenvalues from an Upper-Triangular Matrix*
Let $\dim(\mathcal{V}) = n \in \mathbb{Z}_{>0}$ and $T \in \mathcal{L}(\mathcal{V})$. Suppose T has an upper-triangular matrix with diagonal $(\lambda_1, \ldots, \lambda_n)$ relative to a basis of $\mathcal{V}$.
Then $\operatorname{spec}(T) = \{\lambda_1, \ldots, \lambda_n\}$.

Proof. Fix $\lambda \in \mathbb{F}$. The matrix of $T - \lambda I_{\mathcal{V}}$ in the same basis is upper triangular with diagonal $(\lambda_1 - \lambda, \ldots, \lambda_n - \lambda)$.
Thus $T - \lambda I_{\mathcal{V}}$ is not invertible iff $(\exists\, j \in \{1, \ldots, n\})\ \lambda = \lambda_j$ by Proposition 5.3.18.
The left condition is equivalent to $\lambda \in \operatorname{spec}(T)$ by Theorem 5.1.4(i),(iv) and Definition 5.1.3(i).
Hence $\operatorname{spec}(T) = \{\lambda_1, \ldots, \lambda_n\}$ by the arbitrariness of λ.
Structural Commentary: Scalar shifting reduces the spectrum calculation to triangular invertibility.
Key Insight: A shifted diagonal entry vanishes exactly at an eigenvalue. □

Perspective 5.3.20. *Why Triangular Form Matters*
Theorems 5.3.5 and 5.3.16 encode an operator by a basis-adapted flag.
Proposition 5.3.19 reads the spectrum from diagonal entries whenever triangular form exists.
Upper-block coupling records interactions that diagonalization would remove.
This viewpoint organizes spectral data without asserting diagonalizability.

5.4 Diagonalization, Commutants, and Simultaneous Forms

Definition 5.4.1. *Diagonalizable Operator*
Let $\mathcal{V}$ be a finite-dimensional vector space over $\mathbb{F}$. An operator $T \in \mathcal{L}(\mathcal{V})$ is *diagonalizable* iff $(\exists$ a basis β of $\mathcal{V})\,[T]_{\beta \leftarrow \beta}$ is diagonal.

Proposition 5.4.2. *Eigenspaces for Distinct Eigenvalues Form a Direct Sum*
Let $\mathcal{V}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_m$ be distinct eigenvalues of T, where $m \geq 1$. The following hold:

(i) The sum $E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)$ is direct.

(ii) $\sum_{j=1}^{m} \dim E_{\lambda_j}(T) \leq \dim(\mathcal{V})$.

Proof.

(i) Suppose $(\mathbf{u}_j)_{j=1}^m \in \prod_{j=1}^{m} E_{\lambda_j}(T)$ and $\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}$. Set $J := \{j \in \{1, \dots, m\} : \mathbf{u}_j \neq \mathbf{0}\}$.
If $J \neq \emptyset$, then $(\mathbf{u}_j)_{j \in J}$ is linearly independent by Theorem 5.1.5. This contradicts its zero relation with coefficients 1.
Thus $J = \emptyset$, so the sum is direct by Proposition 2.1.23(ii),(i).

(ii) $\dim(E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)) = \sum_{j=1}^{m} \dim E_{\lambda_j}(T)$ by Theorem 2.3.4(i). The left side is at most $\dim(\mathcal{V})$ by Proposition 1.4.3(i).

Structural Commentary: Removing zero components reduces every mixed relation to distinct-eigenvalue independence.
Key Insight: Distinct-eigenvalue independence forbids cross-eigenspace cancellation. □

Theorem 5.4.3. *Conditions Equivalent to Diagonalizability*
Let $\mathcal{V}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, and $n := \dim(\mathcal{V})$.
Let $(\lambda_1, \dots, \lambda_m)$ list the distinct elements of $\operatorname{spec}(T)$*, where $m \geq 0$. The following are equivalent:*

(i) T is diagonalizable;

(ii) $\mathcal{V}$ has a basis consisting of eigenvectors of T;

(iii) $(\exists\, (\mathcal{U}_j)_{j=1}^n \in \operatorname{Inv}(T)^n)\,[(\forall j \in \{1, \dots, n\})\ \dim(\mathcal{U}_j) = 1 \wedge \mathcal{V} = \bigoplus_{j=1}^{n} \mathcal{U}_j]$;

(iv) $\mathcal{V} = \bigoplus_{j=1}^{m} E_{\lambda_j}(T)$;

(v) $\dim(\mathcal{V}) = \sum_{j=1}^{m} \dim E_{\lambda_j}(T)$.

Proof.

(i) $\Rightarrow$ *(ii)* Choose a basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ such that $[T]_{\beta \leftarrow \beta}$ is diagonal.
Writing its jth column as $a_j \mathbf{e}_j$, $T\mathbf{v}_j = a_j \mathbf{v}_j$ by the matrix representation. Thus β is an eigenbasis.

(ii) $\Rightarrow$ *(iii)* Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an eigenbasis and set $\mathcal{U}_j := \operatorname{span}(\mathbf{v}_j)$.
Each $\mathcal{U}_j$ is one-dimensional and T-invariant by Proposition 5.2.4. The equality $\mathcal{V} = \sum_{j=1}^{n} \mathcal{U}_j$ holds by spanning.
Every zero relation across the summands is a basis relation, so the sum is direct by Proposition 2.1.23(ii),(i).

(iii) $\Rightarrow$ *(iv)* Choose $(\mathbf{v}_j)_{j=1}^n \in \prod_{j=1}^{n} (\mathcal{U}_j \setminus \{\mathbf{0}\})$. Each $\mathbf{v}_j$ is an eigenvector by Proposition 5.2.4.
Thus every $\mathcal{U}_j$ lies in an eigenspace of T, and the listed eigenspaces span $\mathcal{V}$.
If $m = 0$ and $n \geq 1$, then $\mathbf{v}_1$ contradicts $\operatorname{spec}(T) = \emptyset$; hence $m = 0 \Rightarrow n = 0$.
If $m = 0$, the asserted empty direct sum is $\mathcal{V} = \{\mathbf{0}\}$. If $m \geq 1$, the sum is direct by Proposition 5.4.2(i).

(iv) $\Rightarrow$ *(v)* If $m = 0$, condition (iv) states $\mathcal{V} = \{\mathbf{0}\}$, so both sides of (v) vanish. If $m \geq 1$, the dimensions add by Theorem 2.3.4(i).

(v) $\Rightarrow$ *(i)* If $m = 0$, then $n = 0$ by condition (v), and the empty basis diagonalizes T. Suppose $m \geq 1$.
Choose bases $(\beta_j)_{j=1}^m$ with β_j a basis of $E_{\lambda_j}(T)$, and let β be their concatenation.
The eigenspace sum is direct by Proposition 5.4.2(i). Hence β is a basis of that sum by Theorem 2.3.4(ii).
Its length is n by (v), so it is a basis of $\mathcal{V}$ by Proposition 1.4.4(ii).
Each member of β is an eigenvector; hence $[T]_{\beta \leftarrow \beta}$ is diagonal, and T is diagonalizable.

Structural Commentary: The implication cycle converts diagonal coordinates into invariant lines and exhaustive eigenspace dimension.
Key Insight: Diagonalizability is exactly exhaustion by distinct eigenspaces. □

Corollary 5.4.4. *Enough Eigenvalues Imply Diagonalizability*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. If $|\operatorname{spec}(T)| = \dim(\mathcal{V})$, then T is diagonalizable.

Proof. Set $n := \dim(\mathcal{V})$. If $n = 0$, the empty basis diagonalizes T. Suppose $n \geq 1$ and list $\operatorname{spec}(T)$ as $\lambda_1, \dots, \lambda_n$.
Choose $(\mathbf{v}_j)_{j=1}^n \in \prod_{j=1}^{n} (E_{\lambda_j}(T) \setminus \{\mathbf{0}\})$. The list $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is linearly independent by Theorem 5.1.5.
Its length is n, so it is a basis by Proposition 1.4.4(ii). Thus T is diagonalizable by Theorem 5.4.3(ii),(i).
Structural Commentary: Distinct eigenvalues supply an independent eigenvector list whose cardinality reaches the ambient dimension.
Key Insight: A full-length independent eigenvector list is already a diagonalizing basis. □

Perspective 5.4.5. *Diagonalization as Exhaustion by Eigenspaces*
Separation holds by Proposition 5.4.2(i). Spanning is the remaining condition by Theorem 5.4.3.
Diagonalization therefore records exhaustion of $\mathcal{V}$ by its distinct eigenspaces.

Theorem 5.4.6. *Algebraic Spectral Resolution*
Let $\mathcal{V}$ be nonzero and finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ be distinct, where $m \geq 1$.
Set $\Lambda := \{\lambda_1, \dots, \lambda_m\}$, and interpret an empty product below as $I_{\mathcal{V}}$. The following hold:

(i) T is diagonalizable with $\operatorname{spec}(T) = \Lambda$ *iff* $(\exists\, (\rho_i)_{i=1}^m \in (\mathcal{L}(\mathcal{V}) \setminus \{0\})^m)$ *such that* $(\forall i \in \{1, \dots, m\})\ \rho_i^2 = \rho_i$,
$(\forall i, j \in \{1, \dots, m\})\,[i \neq j \Rightarrow \rho_i \rho_j = 0]$, $\sum_{i=1}^{m} \rho_i = I_{\mathcal{V}}$, *and* $T = \sum_{i=1}^{m} \lambda_i \rho_i$.

(ii) *Every family* $(\rho_i)_{i=1}^m$ *satisfying the right side of part (i) obeys* $(\forall i \in \{1,\dots,m\})\ \operatorname{range}(\rho_i) = E_{\lambda_i}(T)$.
(iii) *Every such family obeys* $(\forall i \in \{1,\dots,m\})\ \ker(\rho_i) = \bigoplus_{j\in\{1,\dots,m\}\setminus\{i\}} E_{\lambda_j}(T)$.
(iv) *Every such family obeys* $(\forall i \in \{1,\dots,m\})\ \rho_i = \prod_{j\in\{1,\dots,m\}\setminus\{i\}} \frac{T-\lambda_j I_{\mathcal{V}}}{\lambda_i-\lambda_j}$.

Proof.

(*i*) ($\Rightarrow$) The decomposition $\mathcal{V} = \bigoplus_{i=1}^m E_{\lambda_i}(T)$ holds by Theorem 5.4.3(i),(iv).
For $i \in \{1,\dots,m\}$, set $\mathcal{W}_i := \bigoplus_{j\in\{1,\dots,m\}\setminus\{i\}} E_{\lambda_j}(T)$ and $\rho_i := \rho_{E_{\lambda_i}(T),\mathcal{W}_i}$.
Each ρ_i is idempotent by Proposition 2.4.2(ii), and $\rho_i \neq 0$ because $E_{\lambda_i}(T) \neq \{\mathbf{0}\}$.
If $\mathbf{x} = \sum_{k=1}^m \mathbf{x}_k$ with $\mathbf{x}_k \in E_{\lambda_k}(T)$, then $(\forall i \in \{1,\dots,m\})\ \rho_i\mathbf{x} = \mathbf{x}_i$ by the projection rule.
Thus $(\forall i,j \in \{1,\dots,m\})\ [i \neq j \Rightarrow \rho_i\rho_j = 0]$ and $\sum_{i=1}^m \rho_i = I_{\mathcal{V}}$.
The identity $(\forall \mathbf{x} \in \mathcal{V})\ T\mathbf{x} = \sum_{i=1}^m \lambda_i\mathbf{x}_i = \sum_{i=1}^m \lambda_i\rho_i\mathbf{x}$ holds by the eigenspace identities and projection rule, so $T = \sum_{i=1}^m \lambda_i\rho_i$.

(*i*) ($\Leftarrow$) Fix a family satisfying the right side, and set $\mathcal{V}_i := \operatorname{range}(\rho_i)$ for $i \in \{1,\dots,m\}$. The equality $\mathcal{V} = \sum_{i=1}^m \mathcal{V}_i$ holds by $\sum_{i=1}^m \rho_i = I_{\mathcal{V}}$.
Suppose $(\mathbf{x}_i)_{i=1}^m \in \prod_{i=1}^m \mathcal{V}_i$ and $\sum_{i=1}^m \mathbf{x}_i = \mathbf{0}$. Then $(\forall k \in \{1,\dots,m\})\ \mathbf{x}_k = \rho_k \sum_{i=1}^m \mathbf{x}_i = \mathbf{0}$ by idempotence and pairwise annihilation.
Hence $\mathcal{V} = \bigoplus_{i=1}^m \mathcal{V}_i$ by Proposition 2.1.23(ii),(i). For $i \in \{1,\dots,m\}$ and $\mathbf{x} \in \mathcal{V}_i$, $T\mathbf{x} = \lambda_i\mathbf{x}$ by idempotence and pairwise annihilation.
Thus $\mathcal{V}_i \subseteq E_{\lambda_i}(T)$. $\mathcal{V}_i \neq \{\mathbf{0}\}$ and $\lambda_i \in \operatorname{spec}(T)$ by $\rho_i \neq 0$ and $\mathcal{V}_i \subseteq E_{\lambda_i}(T)$.
Conversely, if $T\mathbf{x} = \mu\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, write $\mathbf{x} = \sum_{i=1}^m \mathbf{x}_i$ with $\mathbf{x}_i \in \mathcal{V}_i$.
Then $(\forall i \in \{1,\dots,m\})\ (\lambda_i - \mu)\mathbf{x}_i = \mathbf{0}$ by directness of the decomposition. Since some $\mathbf{x}_i \neq \mathbf{0}$, $\mu \in \Lambda$.
Thus $\operatorname{spec}(T) = \Lambda$. The eigenspaces span $\mathcal{V}$ by $\mathcal{V}_i \subseteq E_{\lambda_i}(T)$ and the direct-sum decomposition.
Hence T is diagonalizable by Theorem 5.4.3(iv),(i).

(*ii*) Fix a family satisfying the right side of part (i), set $\mathcal{V}_j := \operatorname{range}(\rho_j)$ for $j \in \{1,\dots,m\}$, and fix $i \in \{1,\dots,m\}$.
By the reverse direction of part (i), $\mathcal{V} = \bigoplus_{j=1}^m \mathcal{V}_j$ and $\mathcal{V}_i \subseteq E_{\lambda_i}(T)$. If $\mathbf{x} \in E_{\lambda_i}(T)$, write $\mathbf{x} = \sum_{j=1}^m \mathbf{x}_j$ with $\mathbf{x}_j \in \mathcal{V}_j$.
Then $(\forall j \in \{1,\dots,m\} \setminus \{i\})\ \mathbf{x}_j = \mathbf{0}$ by directness of $\sum_{j=1}^m (\lambda_j - \lambda_i)\mathbf{x}_j = \mathbf{0}$ and distinctness.
Hence $\mathbf{x} = \mathbf{x}_i \in \mathcal{V}_i$, and $\mathcal{V}_i = E_{\lambda_i}(T)$ by the two inclusions.

(*iii*) Fix a family satisfying the right side of part (i), set $\mathcal{V}_j := \operatorname{range}(\rho_j)$, and fix $i \in \{1,\dots,m\}$.
For $\mathbf{x} \in \mathcal{V}$, $\mathbf{x} = \sum_{j=1}^m \mathbf{x}_j$ with $\mathbf{x}_j \in \mathcal{V}_j$ by reverse direction of part (i). Then $\rho_i\mathbf{x} = \mathbf{x}_i$ by idempotence, pairwise annihilation.
Hence $\mathbf{x} \in \ker(\rho_i) \iff \mathbf{x}_i = \mathbf{0}$, so $\ker(\rho_i) = \bigoplus_{j\in\{1,\dots,m\}\setminus\{i\}} \mathcal{V}_j$. The stated identity holds by part (ii).

(*iv*) Fix a family satisfying the right side of part (i). For $i \in \{1,\dots,m\}$, set $\ell_i(z) := \prod_{j\in\{1,\dots,m\}\setminus\{i\}} \frac{z-\lambda_j}{\lambda_i-\lambda_j}$.
If $k \in \{1,\dots,m\}$ and $\mathbf{x} \in E_{\lambda_k}(T)$, then $\ell_i(T)\mathbf{x} = \ell_i(\lambda_k)\mathbf{x} = \delta_{ik}\mathbf{x} = \rho_i\mathbf{x}$ by part (ii), idempotence, and pairwise annihilation.
The eigenspaces span $\mathcal{V}$ by parts (i)–(ii), so $\ell_i(T) = \rho_i$ by equality on a spanning family.

Structural Commentary: Direct-sum coordinates construct the idempotents; Lagrange interpolation recovers them from T.
Key Insight: Lagrange polynomials isolate the individual eigenspaces. □

Lemma 5.4.7. *Diagonalizable Restrictions to Invariant Subspaces*
Let $\mathcal{V}$ *be a nonzero finite-dimensional vector space, let* $T \in \mathcal{L}(\mathcal{V})$ *be diagonalizable, and let* $\mathcal{U} \in \operatorname{Inv}(T)$.
For $\lambda \in \operatorname{spec}(T)$, *set* $\mathcal{U}_\lambda := \mathcal{U} \cap E_\lambda(T)$. *The following hold:*
(i) $\mathcal{U} = \bigoplus_{\lambda\in\operatorname{spec}(T)} \mathcal{U}_\lambda$.
(ii) *The restriction* $T|_{\mathcal{U}}$ *is diagonalizable.*

Proof. The family of polynomial spectral projections $(\rho_\lambda)_{\lambda\in\operatorname{spec}(T)}$ exists by Theorem 5.4.6(i),(iv); fix it.
The inclusion $T^0(\mathcal{U}) = \mathcal{U}$ is immediate. If $T^k(\mathcal{U}) \subseteq \mathcal{U}$, then $T^{k+1}(\mathcal{U}) \subseteq T(\mathcal{U}) \subseteq \mathcal{U}$ by T-invariance.
Thus $(\forall k \in \mathbb{Z}_{\geq 0})\ T^k(\mathcal{U}) \subseteq \mathcal{U}$ by induction, and $\rho_\lambda(\mathcal{U}) \subseteq \mathcal{U}$ by its polynomial formula.

(i) If $\mathbf{u} \in \mathcal{U}$, then $\mathbf{u} = \sum_{\lambda\in\operatorname{spec}(T)} \rho_\lambda\mathbf{u}$ by Theorem 5.4.6(i).
The membership $\rho_\lambda\mathbf{u} \in \mathcal{U}_\lambda$ holds by the preceding inclusion and Theorem 5.4.6(ii).
Hence $\mathcal{U} = \sum_{\lambda\in\operatorname{spec}(T)} \mathcal{U}_\lambda$. The sum is direct by Proposition 5.4.2(i).

(ii) If $\mathcal{U} = \{\mathbf{0}\}$, the empty basis diagonalizes $T|_{\mathcal{U}}$. Suppose $\mathcal{U} \neq \{\mathbf{0}\}$.
Concatenating bases of the nonzero summands forms a basis of $\mathcal{U}$ by part (i) and Theorem 2.3.4(ii).
Its members are eigenvectors of $T|_{\mathcal{U}}$, so the restriction is diagonalizable by Theorem 5.4.3(ii),(i).

Structural Commentary: Polynomial spectral projections preserve $\mathcal{U}$ and split it across the ambient eigenspaces.
Key Insight: T-invariance propagates to every polynomial in T. □

Theorem 5.4.8. *Simultaneous Diagonalization of Commuting Diagonalizable Operators*
Let $\mathcal{V}$ *be a nonzero finite-dimensional vector space, and let* $\emptyset \neq \mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ *be finite with each member diagonalizable.*
Then $(\forall S,T \in \mathcal{F})\ ST = TS$ *iff* $(\exists$ *a basis* β *of* $\mathcal{V})$ *such that* $(\forall T \in \mathcal{F})\ [T]_{\beta\leftarrow\beta}$ *is diagonal.*

Proof.

($\Rightarrow$) Assume $(\forall S,T\in\mathcal{F})\,ST=TS$, and proceed by strong induction on $n:=\dim(\mathcal{V})$.
If $n=1$, every member of $\mathcal{F}$ is scalar, so any basis is common. Suppose $n>1$ and result holds in every smaller positive dimension.
If all members of $\mathcal{F}$ are scalar, any basis is common. Otherwise choose a nonscalar $S\in\mathcal{F}$.
By Theorem 5.4.3(i),(iv), $\mathcal{V}=\bigoplus_{\lambda\in\operatorname{spec}(S)} E_\lambda(S)$.
If this decomposition had one summand, then $\mathcal{V}=E_\lambda(S)$ and $S=\lambda I_\mathcal{V}$, contrary to the choice of S.
Hence $(\forall\lambda\in\operatorname{spec}(S))\,1\le\dim E_\lambda(S)<n$. Fix $\lambda\in\operatorname{spec}(S)$.
If $T\in\mathcal{F}$ and $\mathbf{x}\in E_\lambda(S)$, then $S(T\mathbf{x})=T(S\mathbf{x})=\lambda T\mathbf{x}$ by commutation. Thus $E_\lambda(S)\in\operatorname{Inv}(T)$.
Set $\mathcal{F}_\lambda:=\{T|_{E_\lambda(S)}:T\in\mathcal{F}\}$. This family is finite, nonempty, and commuting.
Its members are diagonalizable by Lemma 5.4.7(ii). By induction, $E_\lambda(S)$ has a common eigenbasis for $\mathcal{F}_\lambda$.
Concatenating these bases over $\lambda\in\operatorname{spec}(S)$ forms a basis of $\mathcal{V}$ by Theorem 2.3.4(ii).
The concatenated basis diagonalizes every member of $\mathcal{F}$.

($\Leftarrow$) Choose a basis β satisfying the right side. For $S,T\in\mathcal{F}$, set $D_S:=[S]_{\beta\leftarrow\beta}$ and $D_T:=[T]_{\beta\leftarrow\beta}$.
As D_S and D_T are diagonal, $[ST]_{\beta\leftarrow\beta}=D_SD_T=D_TD_S=[TS]_{\beta\leftarrow\beta}$ by Proposition 3.4.4; $ST=TS$ by Theorem 3.4.5(i).

Structural Commentary: A nonscalar member supplies proper common invariant eigenspaces for dimensional induction.
Key Insight: Commutation preserves every eigenspace in the diagonalizable splitting. □

Definition 5.4.9. *Commutant*
Let $n\ge1$ and $\emptyset\ne\mathcal{F}\subseteq\operatorname{Mat}(n,n,\mathbb{F})$. The *commutant* of $\mathcal{F}$ is $\operatorname{Comm}(\mathcal{F}):=\{X\in\operatorname{Mat}(n,n,\mathbb{F}):(\forall A\in\mathcal{F})\,XA=AX\}$.
For a singleton family $\{A\}$, write $\operatorname{Comm}(A)$.

Lemma 5.4.10. *Commuting with a Block Scalar Diagonal Matrix*
Let $k,n_1,\dots,n_k\ge1$, $n:=\sum_{i=1}^k n_i$, and $\lambda_1,\dots,\lambda_k\in\mathbb{F}$. Set $\Lambda:=\lambda_1I_{n_1}\oplus\cdots\oplus\lambda_kI_{n_k}$.
Let $A=[A_{ij}]\in\operatorname{Mat}(n,n,\mathbb{F})$ be partitioned conformally, where $A_{ij}\in\operatorname{Mat}(n_i,n_j,\mathbb{F})$.

(i) If A is block diagonal, then $A\Lambda=\Lambda A$.
(ii) If $A\Lambda=\Lambda A$ and $\lambda_1,\dots,\lambda_k$ are distinct, then A is block diagonal.

Proof. By block multiplication, $(A\Lambda)_{ij}=\lambda_jA_{ij}$ and $(\Lambda A)_{ij}=\lambda_iA_{ij}$.

(i) If A is block diagonal, then $(\forall i,j\in\{1,\dots,k\})\,[i\ne j\Rightarrow A_{ij}=0]$. The two diagonal-block expressions coincide, so $A\Lambda=\Lambda A$.
(ii) If $A\Lambda=\Lambda A$, then $(\forall i,j\in\{1,\dots,k\})\,(\lambda_j-\lambda_i)A_{ij}=0$.
The relation $(\forall i,j\in\{1,\dots,k\})\,[i\ne j\Rightarrow A_{ij}=0]$ holds by distinctness, so A is block diagonal.

Structural Commentary: Block comparison reduces commutation to scalar off-diagonal equations.
Key Insight: Distinct block scalars force every off-diagonal block to vanish. □

Counterexample 5.4.11. *Repeated Block Scalars Do Not Force Block Diagonality*
Set $k=2$, $n_1=n_2=1$, $\lambda_1=\lambda_2=1$, and $A=\left[\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right]$. Then $\Lambda=I_2$, so $A\Lambda=\Lambda A$, but A is not block diagonal.
Thus the distinctness hypothesis in Lemma 5.4.10(ii) is necessary.

Theorem 5.4.12. *Commutant of a Diagonalizable Matrix*
Let $n\ge1$, $A\in\operatorname{Mat}(n,n,\mathbb{C})$ be diagonalizable. List its distinct eigenvalues as $\mu_1,\dots,\mu_d$, where $d\ge1$, set $n_i:=\dim_{\mathbb{C}}E_{\mu_i}(A)$.
Choose $S\in\operatorname{Mat}(n,n,\mathbb{C})^\times$ satisfying $A=S(\mu_1I_{n_1}\oplus\cdots\oplus\mu_dI_{n_d})S^{-1}$. Set $\mathcal{D}:=\{X_1\oplus\cdots\oplus X_d:(X_i)_{i=1}^d\in\prod_{i=1}^d\operatorname{Mat}(n_i,n_i,\mathbb{C})\}$. Then:

(i) $(\forall B\in\operatorname{Mat}(n,n,\mathbb{C}))\,[B\in\operatorname{Comm}(A)\iff S^{-1}BS\in\mathcal{D}]$.
(ii) $\dim_{\mathbb{C}}\operatorname{Comm}(A)=\sum_{i=1}^d n_i^2$.

Proof. Set $\Lambda:=\mu_1I_{n_1}\oplus\cdots\oplus\mu_dI_{n_d}$.

(*i*) ($\Rightarrow$) Suppose $B\in\operatorname{Comm}(A)$. Then $(S^{-1}BS)\Lambda=\Lambda(S^{-1}BS)$ by multiplying $BA=AB$ on the left by S^{-1} and on the right by S.
Thus $S^{-1}BS\in\mathcal{D}$ by Lemma 5.4.10(ii).
(*i*) ($\Leftarrow$) Suppose $S^{-1}BS\in\mathcal{D}$. Then $S^{-1}BS$ commutes with Λ by Lemma 5.4.10(i).
Thus $BA=AB$ by conjugating this equality by S, so $B\in\operatorname{Comm}(A)$.
(*ii*) Conjugation by S is a linear bijection from $\mathcal{D}$ onto $\operatorname{Comm}(A)$ by part (i), with inverse conjugation by S^{-1}.
Embedding the canonical matrix-unit bases of the d diagonal blocks forms a basis of $\mathcal{D}$.

Hence $\dim_{\mathbb{C}}\operatorname{Comm}(A)=\dim_{\mathbb{C}}\mathcal{D}=\sum_{i=1}^d n_i^2$ by Proposition 3.1.5(ii).

Structural Commentary: Conjugation identifies the commutant with freely chosen endomorphisms on the eigenspaces.
Key Insight: Commuting matrices preserve eigenspaces but act freely within each one. □

Theorem 5.4.13. *Common Eigenvector for a Commuting Family*
Let $\mathcal{V}$ be a nonzero finite-dimensional complex vector space, and let $\emptyset\ne\mathcal{F}\subseteq\mathcal{L}(\mathcal{V})$ satisfy $(\forall S,T\in\mathcal{F})\,ST=TS$.
Then $(\exists\,(\mathbf{v},(\lambda_T)_{T\in\mathcal{F}})\in(\mathcal{V}\setminus\{\mathbf{0}\})\times\mathbb{C}^{\mathcal{F}})\,(\forall T\in\mathcal{F})\,T\mathbf{v}=\lambda_T\mathbf{v}$.

Proof.

(i) *Finite family*: Suppose $\mathcal{F}=\{T_1,\dots,T_m\}$, and induct on $m\ge1$. The case $m=1$ holds by Theorem 5.3.2. Suppose $m>1$.
Choose $\lambda\in\operatorname{spec}(T_1)$ by Theorem 5.3.2, and set $\mathcal{U}:=E_\lambda(T_1)\ne\{\mathbf{0}\}$.
If $\mathbf{u}\in\mathcal{U}$ and $j\in\{2,\dots,m\}$, then $(T_1-\lambda I_\mathcal{V})T_j\mathbf{u}=T_j(T_1-\lambda I_\mathcal{V})\mathbf{u}=\mathbf{0}$ by commutation.
Thus $\mathcal{U}$ is invariant under $T_2,\dots,T_m$. The restrictions to $\mathcal{U}$ commute and form a nonempty family of cardinality at most $m-1$.
They have a common eigenvector $\mathbf{v}\in\mathcal{U}\setminus\{\mathbf{0}\}$ by induction, and $\mathbf{v}$ is an eigenvector of T_1 by $\mathbf{v}\in E_\lambda(T_1)$.
(ii) *Finite reduction*: Set $\mathcal{W}:=\operatorname{span}(\mathcal{F})$. This space is finite-dimensional by Theorem 3.4.5(ii).
If $\mathcal{W}=\{0\}$, then $(\forall T\in\mathcal{F})\,T=0$ by $\mathcal{F}\subseteq\mathcal{W}$. Thus any $\mathbf{v}\in\mathcal{V}\setminus\{\mathbf{0}\}$ is common with eigenvalues $\lambda_T=0$. Suppose $\mathcal{W}\ne\{0\}$.
Choose a basis $(R_1,\dots,R_q)$ of $\mathcal{W}$ by Corollary 1.3.14. Each R_j is a finite linear combination of members of $\mathcal{F}$.
The union of their finite supports is a finite nonempty set $\mathcal{F}_0\subseteq\mathcal{F}$ satisfying $\operatorname{span}(\mathcal{F}_0)=\mathcal{W}$.

(iii) *Spanning extension*: Suppose $\mathcal{W} \neq \{0\}$ and let $\mathcal{F}_0$ be as in part (ii).
By part (i), choose $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$ and $(\mu_T)_{T\in\mathcal{F}_0}$ satisfying $(\forall T \in \mathcal{F}_0)\, T\mathbf{v} = \mu_T \mathbf{v}$.
For $S \in \mathcal{F} \subseteq \mathcal{W}$, write $S = \sum_{T\in\mathcal{F}_0} a_T T$ and define $\lambda_S := \sum_{T\in\mathcal{F}_0} a_T \mu_T$.
Then $S\mathbf{v} = \lambda_S \mathbf{v}$, and λ_S is independent of the representation by $\mathbf{v} \neq \mathbf{0}$.

Structural Commentary: Finite induction finds a common eigenline; a finite spanning subfamily handles arbitrary $\mathcal{F}$.
Key Insight: Commutation preserves the first operator's chosen eigenspace. □

Theorem 5.4.14. *Simultaneous Schur Triangularization*
Let $n \geq 1$ and let $\emptyset \neq \mathcal{F} \subseteq \mathrm{Mat}(n,n,\mathbb{C})$ satisfy $(\forall A, B \in \mathcal{F})\, AB = BA$. The following hold:
(i) $(\exists$ *a unitary* $U \in \mathrm{Mat}(n,n,\mathbb{C}))$ *such that* $(\forall A \in \mathcal{F})\, U^*AU$ *is upper triangular.*
(ii) Suppose $\mathcal{F} \subseteq \mathrm{Mat}(n,n,\mathbb{R})$ *and* $(\forall A \in \mathcal{F})\ \mathrm{spec}(A) \subseteq \mathbb{R}$.
Then $(\exists$ *an orthogonal* $U \in \mathrm{Mat}(n,n,\mathbb{R}))$ *such that* $(\forall A \in \mathcal{F})\, U^tAU$ *is upper triangular.*

Proof. Proceed by strong induction on n. Both conclusions hold with $U = I_1$ when $n = 1$.
Suppose $n > 1$ and both conclusions hold in each smaller positive dimension.

(i) By Theorem 5.4.13, the operators on $\mathbb{C}^n$ have a common eigenvector.
Normalize it to $\mathbf{x}$ and extend it to a unitary matrix $V = [\mathbf{x}\ V_2]$ by Theorem 2.6.13(ii).
For $A \in \mathcal{F}$, write $M_A := V^*AV = \begin{bmatrix} \lambda_A & \mathbf{r}_A \\ 0 & \widehat{A} \end{bmatrix}$. The lower-left block vanishes by $A\mathbf{x} = \lambda_A \mathbf{x}$.
If $A, B \in \mathcal{F}$, then $M_AM_B = V^*ABV = V^*BAV = M_BM_A$, so $\widehat{A}\widehat{B} = \widehat{B}\widehat{A}$ by comparison of the lower-right blocks.
Thus $\widehat{\mathcal{F}} := \{\widehat{A} : A \in \mathcal{F}\}$ is nonempty and commuting.
By induction, a unitary W simultaneously upper triangularizes its members. Set $U := V(I_1 \oplus W)$.
The matrix U is unitary by Proposition 2.6.12(i)–(ii), and $(\forall A \in \mathcal{F})\, U^*AU$ is upper triangular.

(ii) Regard $\mathcal{F}$ as a commuting family on $\mathbb{C}^n$. A common eigenvector $\mathbf{z} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$ exists by Theorem 5.4.13; fix it.
Fix $(\lambda_A)_{A\in\mathcal{F}}$ satisfying $(\forall A \in \mathcal{F})\, A\mathbf{z} = \lambda_A \mathbf{z}$, and write $\mathbf{z} = \mathbf{u} + i\mathbf{v}$.
The relation $(\forall A \in \mathcal{F})\, \lambda_A \in \mathrm{spec}(A) \subseteq \mathbb{R}$ holds by the eigenvector identities and spectral hypothesis.
Because each A is real, $A\mathbf{u} = \lambda_A \mathbf{u}$ and $A\mathbf{v} = \lambda_A \mathbf{v}$ by comparison of real and imaginary parts.
At least one of $\mathbf{u}, \mathbf{v}$ is nonzero, so a common real unit eigenvector $\mathbf{x}$ exists by normalization.
Extend $\mathbf{x}$ to a real orthogonal matrix $V = [\mathbf{x}\ V_2]$ by Theorem 2.6.13(ii).
For $A \in \mathcal{F}$, write $M_A := V^tAV = \begin{bmatrix} \lambda_A & \mathbf{r}_A \\ 0 & \widehat{A} \end{bmatrix}$. The lower-right blocks are real and commute by the calculation in part (i).
Fix $A \in \mathcal{F}$ and $\mu \in \mathrm{spec}(\widehat{A})$, and choose $\mathbf{y} \in \mathbb{C}^{n-1} \setminus \{\mathbf{0}\}$ satisfying $\widehat{A}\mathbf{y} = \mu\mathbf{y}$.
If $\mu = \lambda_A$, then $\mu \in \mathbb{R}$. If $\mu \neq \lambda_A$, set $c := \frac{\mathbf{r}_A\mathbf{y}}{\mu - \lambda_A}$ and $\mathbf{w} := (c, \mathbf{y})^t$.
Then $\mathbf{w} \neq \mathbf{0}$ and $M_A\mathbf{w} = (\mu c, \mu\mathbf{y})^t = \mu\mathbf{w}$. Since $V\mathbf{w} \neq \mathbf{0}$ by orthogonality of V, it is an eigenvector of A for μ.
Hence $\mu \in \mathbb{R}$ by the spectral hypothesis. Thus $(\forall A \in \mathcal{F})\ \mathrm{spec}(\widehat{A}) \subseteq \mathbb{R}$.
By induction, a real orthogonal W simultaneously upper triangularizes $\widehat{\mathcal{F}}$.
Then $U := V(I_1 \oplus W)$ is real orthogonal by Proposition 2.6.12(i)–(ii), and $(\forall A \in \mathcal{F})\, U^tAU$ is upper triangular.

Structural Commentary: A common eigenline creates the first block, and commutation descends to the compression.
Key Insight: Real spectra turn a complex common eigenvector into a real one. □

Figure 44 depicts the complex induction in Theorem 5.4.14.
A common eigenline creates the first block, and pairwise commutation passes to the lower-right blocks.
The theorem, rather than the diagram, owns the simultaneous triangularization conclusion.

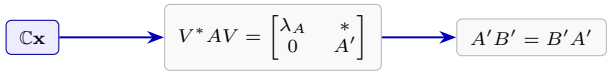

Figure 44: A shared eigenline determines a simultaneous first block; lower-right compressions remain commuting.

Corollary 5.4.15. *Eigenvalues of Sums and Products of Commuting Matrices*
Let $n \geq 1$ and let $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$ satisfy $AB = BA$. Then $(\exists\, ((\lambda_i)_{i=1}^n, (\mu_i)_{i=1}^n) \in (\mathbb{C}^n)^2)$ such that the following hold:
(i) $\mathrm{spec}(A) = \{\lambda_1, \ldots, \lambda_n\}$ *and* $\mathrm{spec}(B) = \{\mu_1, \ldots, \mu_n\}$;
(ii) $\mathrm{spec}(A + B) = \{\lambda_i + \mu_i : i \in \{1, \ldots, n\}\}$;
(iii) $\mathrm{spec}(AB) = \{\lambda_i\mu_i : i \in \{1, \ldots, n\}\}$.

Proof. Choose a unitary U by Theorem 5.4.14(i). Set $T := U^*AU$, $R := U^*BU$, $\lambda_i := t_{ii}$, $\mu_i := r_{ii}$. Fix $M \in \mathrm{Mat}(n,n,\mathbb{C})$ and $\theta \in \mathbb{C}$.
If $\mathbf{x} \neq \mathbf{0}$, then $(U^*MU)\mathbf{x} = \theta\mathbf{x} \iff M(U\mathbf{x}) = \theta(U\mathbf{x})$ and $U\mathbf{x} \neq \mathbf{0}$ by unitarity of U. Hence $\mathrm{spec}(U^*MU) = \mathrm{spec}(M)$.

(i) The matrices T and R are upper triangular with diagonals $(\lambda_i)_{i=1}^n$ and $(\mu_i)_{i=1}^n$.
Their spectra are underlying sets of these lists by Proposition 5.3.19. Identities in part (i) hold by preceding similarity calculation.
(ii) The matrix $T + R = U^*(A + B)U$ is upper triangular with diagonal $(\lambda_i + \mu_i)_{i=1}^n$.
Its spectrum is the asserted set by Proposition 5.3.19. The identity in part (ii) holds by the preceding similarity calculation.
(iii) The matrix $TR = U^*ABU$ is upper triangular with diagonal $(\lambda_i\mu_i)_{i=1}^n$.
Its spectrum is the asserted set by Proposition 5.3.19. The identity in part (iii) holds by the preceding similarity calculation.

Structural Commentary: Common triangular coordinates reduce both spectra to diagonal arithmetic.
Key Insight: A common diagonal pairing controls both resulting spectra. □

6 Spectral Theorems and Operator Functions

6.1 Normal, Self-Adjoint, and Spectral Theorems

Structured spectral theory studies operators whose geometry and spectrum align.
This subsection derives orthonormal eigenbases and spectral resolutions from normality and self-adjointness.
All inner-product spaces below are finite-dimensional over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ unless stated otherwise.

Definition 6.1.1. *Normal Operator*

(i) An operator $T \in \mathcal{L}(\mathcal{V})$ is *normal* iff $T^*T = TT^*$.
(ii) For $n \geq 1$, a matrix $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is *normal* iff $A^*A = AA^*$.

Theorem 6.1.2. *Normal Operators and Equal Adjoint Norms*
Let $T \in \mathcal{L}(\mathcal{V})$. Then T is normal iff $(\forall \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\| = \|T^*\mathbf{v}\|$.

Proof. Set $C := T^*T - TT^*$. Then $C^* = C$ by Proposition 2.3.8(i)–(iii).
The identity $(\forall \mathbf{v} \in \mathcal{V})\ \langle C\mathbf{v}, \mathbf{v}\rangle = \|T\mathbf{v}\|^2 - \|T^*\mathbf{v}\|^2$ holds by the adjoint identity and Proposition 2.3.8(iii).
($\Rightarrow$) If T is normal, then $C = 0$ by Definition 6.1.1(i), so the quadratic identity implies the norm identity.
($\Leftarrow$) The norm identity implies $(\forall \mathbf{v} \in \mathcal{V})\ \langle C\mathbf{v}, \mathbf{v}\rangle = 0$. Hence $C = 0$ by Proposition 2.5.4, and T is normal by Definition 6.1.1(i).
Structural Commentary: The adjoint defect becomes a quadratic form and is recovered from its diagonal values.
Key Insight: Equal norm functions detect $T^*T - TT^* = 0$. □

Theorem 6.1.3. *Normal Operators Share Eigenvectors with Their Adjoints*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal, let $\lambda \in \mathbb{F}$, and let $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$. If $T\mathbf{v} = \lambda\mathbf{v}$, then $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Proof. Set $S := T - \lambda I_{\mathcal{V}}$. Then $S^* = T^* - \overline{\lambda} I_{\mathcal{V}}$ by Proposition 2.3.8(i),(iv).
The identity $S^*S - SS^* = T^*T - TT^* = 0$ holds by direct expansion and normality of T, so S is normal.
Since $S\mathbf{v} = \mathbf{0}$, $\|S^*\mathbf{v}\| = \|S\mathbf{v}\| = 0$ by Theorem 6.1.2. Thus $S^*\mathbf{v} = \mathbf{0}$ by positive definiteness, equivalently $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.
Structural Commentary: Scalar shifting preserves normality, and the equal-norm criterion transfers the shifted kernel.
Key Insight: $\ker(T - \lambda I_{\mathcal{V}}) \subseteq \ker(T^* - \overline{\lambda} I_{\mathcal{V}})$. □

Figure 45 depicts the common eigenline in Theorem 6.1.3.

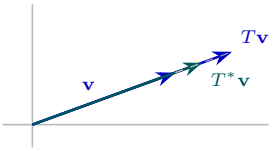

Figure 45: A normal operator and its adjoint preserve the same eigenline with conjugate eigenvalues.

Theorem 6.1.4. *Orthogonal Eigenvectors for Normal Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal and let $\alpha, \beta \in \mathrm{spec}(T)$ satisfy $\alpha \neq \beta$. Then $(\forall \mathbf{u} \in E_\alpha(T) \setminus \{\mathbf{0}\})(\forall \mathbf{v} \in E_\beta(T) \setminus \{\mathbf{0}\})\ \langle \mathbf{u}, \mathbf{v}\rangle = 0$.

Proof. Fix $\mathbf{u} \in E_\alpha(T) \setminus \{\mathbf{0}\}$ and $\mathbf{v} \in E_\beta(T) \setminus \{\mathbf{0}\}$. Then $T\mathbf{u} = \alpha\mathbf{u}$ by eigenspace membership and $T^*\mathbf{v} = \overline{\beta}\mathbf{v}$ by Theorem 6.1.3.
Hence $\alpha\langle \mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T^*\mathbf{v}\rangle = \beta\langle \mathbf{u}, \mathbf{v}\rangle$, so $\langle \mathbf{u}, \mathbf{v}\rangle = 0$ by $\alpha \neq \beta$.
Structural Commentary: The adjoint eigenvalue transports T across the inner product and isolates $\alpha - \beta$.
Key Insight: $(\alpha - \beta)\langle \mathbf{u}, \mathbf{v}\rangle = 0$ forces orthogonality. □

Theorem 6.1.5. *Eigenvalues of Self-Adjoint Operators Are Real*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint. Then $\mathrm{spec}(T) \subseteq \mathbb{R}$.

Proof. Fix $\lambda \in \mathrm{spec}(T)$ and $\mathbf{v} \in E_\lambda(T) \setminus \{\mathbf{0}\}$.
The identity $\lambda\|\mathbf{v}\|^2 = \langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \overline{\lambda}\|\mathbf{v}\|^2$ holds by the eigenvector relation and $T = T^*$. Thus $\lambda = \overline{\lambda}$ by $\|\mathbf{v}\|^2 > 0$, so $\lambda \in \mathbb{R}$.
Structural Commentary: Self-adjointness compares an eigenvalue with its conjugate on a nonzero eigenvector.
Key Insight: $\|\mathbf{v}\|^2 > 0$ forces $\lambda = \overline{\lambda}$. □

Proposition 6.1.6. *Invertible Quadratic Expressions*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint. If $b, c \in \mathbb{R}$ and $b^2 < 4c$, then $T^2 + bT + cI_{\mathcal{V}}$ is invertible.

Proof. Set $S := T^2 + bT + cI_{\mathcal{V}}$ and fix $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$. Set $x := \|T\mathbf{v}\|$, $y := \|\mathbf{v}\|$, and $a := \langle T\mathbf{v}, \mathbf{v}\rangle$.
The relations $a = \overline{a}$ and $\langle T^2\mathbf{v}, \mathbf{v}\rangle = x^2$ hold by self-adjointness, conjugate symmetry, and the adjoint identity, so $a \in \mathbb{R}$.
The bound $|a| \leq xy$ holds by Theorem 1.1.13(i), and $ba \geq -|b|\,|a| \geq -|b|\,xy$ by $a, b \in \mathbb{R}$.
Consequently, $\langle S\mathbf{v}, \mathbf{v}\rangle \geq x^2 - |b|\,xy + cy^2 = \left(x - \frac{|b|}{2}y\right)^2 + \left(c - \frac{b^2}{4}\right)y^2 > 0$ by $c - \frac{b^2}{4} > 0$ and $y > 0$.
Thus $S\mathbf{v} \neq \mathbf{0}$. Since $\mathbf{v}$ was arbitrary, S is injective and hence invertible by Theorem 1.7.7(ii),(i).
Structural Commentary: The quadratic form of S becomes a completed square in $\|T\mathbf{v}\|$ and $\|\mathbf{v}\|$.
Key Insight: The negative discriminant makes the completed square strictly positive. □

Lemma 6.1.7. *Self-Adjoint Operators Have Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real inner-product space. Every self-adjoint $T \in \mathcal{L}(\mathcal{V})$ has an eigenvalue.

Proof.

(i) *Hermitian Extension*: For $\mathbf{z} = \mathbf{u} + i\mathbf{y}$ and $\mathbf{w} = \mathbf{x} + i\mathbf{y}$ in $\mathcal{V}_{\mathbb{C}}$, define $B(\mathbf{z}, \mathbf{w}) := \langle \mathbf{u}, \mathbf{x}\rangle_{\mathcal{V}} + \langle \mathbf{v}, \mathbf{y}\rangle_{\mathcal{V}} + i\big(\langle \mathbf{v}, \mathbf{x}\rangle_{\mathcal{V}} - \langle \mathbf{u}, \mathbf{y}\rangle_{\mathcal{V}}\big)$.
Additivity and real homogeneity in the first slot hold componentwise, while $B(i\mathbf{z}, \mathbf{w}) = iB(\mathbf{z}, \mathbf{w})$ by $i\mathbf{z} = -\mathbf{v} + i\mathbf{u}$.
Thus the first slot is complex-linear, and $B(\mathbf{w}, \mathbf{z}) = \overline{B(\mathbf{z}, \mathbf{w})}$ by symmetry of the real inner product.
$B(\mathbf{z}, \mathbf{z}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \geq 0$ holds by direct substitution, with equality iff $\mathbf{z} = \mathbf{0}$. Hence B is complex inner product on $\mathcal{V}_{\mathbb{C}}$.
This space is finite-dimensional over $\mathbb{C}$ by Proposition 5.3.7(i)–(iii), and it is nonzero by the hypothesis and Proposition 5.3.9(iii).
(ii) *Adjoint Preservation*: The operator $T_{\mathbb{C}}$ is complex-linear by Proposition 5.3.11(i). Fix $\mathbf{z} = \mathbf{u} + i\mathbf{v}$ and $\mathbf{w} = \mathbf{x} + i\mathbf{y}$ in $\mathcal{V}_{\mathbb{C}}$.
Identity $B(T_{\mathbb{C}}\mathbf{z}, \mathbf{w}) = B(\mathbf{z}, T_{\mathbb{C}}\mathbf{w})$ holds by self-adjointness of T in each real component. Thus $T_{\mathbb{C}}$ self-adjoint by adjoint definition.
(iii) *Spectral Descent*: Fix $\lambda \in \mathrm{spec}(T_{\mathbb{C}})$ by Theorem 5.3.2. Then $\lambda \in \mathbb{R}$ by Theorem 6.1.5, and $\lambda \in \mathrm{spec}(T)$ by Proposition 5.3.12.

Structural Commentary: The Hermitian extension preserves the adjoint relation before complex eigenvalue existence descends.
Key Insight: Self-adjoint complexification forces a real eigenvalue shared by T. □

Proposition 6.1.8. *Self-Adjoint Operators and Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint, and let $\mathcal{U} \in \mathrm{Inv}(T)$. Then:
(i) $\mathcal{U}^\perp$ is invariant under T;
(ii) $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is self-adjoint;
(iii) $T|_{\mathcal{U}^\perp} \in \mathcal{L}(\mathcal{U}^\perp)$ is self-adjoint.

Proof.
(i) The subspace $\mathcal{U}^\perp$ is invariant under $T^* = T$ by the hypothesis and Theorem 5.2.7(i).
(ii) Pair $(\mathcal{U},\mathcal{U}^\perp)$ reduces T by (i), hypothesis, Theorem 2.1.5, Definition 5.2.6. Hence $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}} = T|_{\mathcal{U}}$ by Theorem 5.2.7(iii).
(iii) The restriction $T|_{\mathcal{U}^\perp}$ is self-adjoint by (ii) with the invariant subspace $\mathcal{U}^\perp$ from (i).
Structural Commentary: Adjoint-complement invariance reduces T and transfers self-adjointness to both restrictions.
Key Insight: $T = T^*$ preserves both summands of $\mathcal{U} \oplus \mathcal{U}^\perp$. □

Theorem 6.1.9. *Schur Triangularization*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
The space $\mathcal{V}$ has an orthonormal basis γ for which $[T]_{\gamma\leftarrow\gamma}$ is upper triangular.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, the empty orthonormal basis has the required property. Assume $n := \dim(\mathcal{V}) \geq 1$.
(i) *Coordinate Reduction*: Fix an orthonormal basis β by Corollary 1.3.11 and set $A := [T]_{\beta\leftarrow\beta}$.
(ii) *Unitary Triangularization*: Fix a unitary U with U^*AU upper triangular by Theorem 5.4.14(i) for $\{A\}$.
(iii) *Basis Transport*: Let γ be the ordered list whose β-coordinate columns are the columns of U.
These columns form an orthonormal basis of $\mathbb{C}^n$ by Theorems 2.6.8(i),(ii),(vi) and 3.4.1.
The coordinate map is an isomorphism by Proposition 3.2.2(iii), so γ is a basis by Proposition 1.7.8.
The basis γ is orthonormal by Theorem 1.4.7(ii).
(iv) *Matrix Identification*: The identity $C_{\beta\leftarrow\gamma} = U$ holds by Definition 3.5.1.
The identity $C_{\gamma\leftarrow\beta} = U^{-1} = U^*$ holds by Theorems 3.5.2(i) and 2.6.8(vi). Thus $[T]_{\gamma\leftarrow\gamma} = U^*AU$ by Theorem 3.5.3, so this matrix is upper triangular.
Structural Commentary: Unitary matrix triangularization is transported through coordinate columns to an orthonormal basis.
Key Insight: $C_{\beta\leftarrow\gamma} = U$ converts unitary similarity into basis change. □

Lemma 6.1.10. *Upper-Triangular Normal Matrices Are Diagonal*
Let $n \geq 1$. If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is upper triangular and normal, then A is diagonal.

Proof. Induct on n. If $n = 1$, then A is diagonal. Suppose $n \geq 2$ and the claim holds in dimension $n - 1$.
The $(1,1)$ entries of $A^*A = AA^*$ are $|a_{11}|^2$ and $|a_{11}|^2 + \sum_{j=2}^{n} |a_{1j}|^2$, respectively.
Hence $(\forall j \in \{2, \ldots, n\})\, a_{1j} = 0$, so $A = a_{11} \oplus B$ with B upper triangular.
The identity $B^*B = BB^*$ holds by comparison of the lower-right blocks, so B is normal.
Thus B is diagonal by the induction hypothesis, and A is diagonal.
Structural Commentary: The $(1,1)$ identity removes the first-row tail and exposes a smaller normal block.
Key Insight: Once $a_{12}, \ldots, a_{1n}$ vanish, block normality closes the induction. □

Theorem 6.1.11. *Complex Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:
(i) T is normal;
(ii) $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
(iii) T has a diagonal matrix in an orthonormal basis of $\mathcal{V}$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, the unique operator is normal and the empty basis is an orthonormal eigenbasis with an empty diagonal matrix.
Assume $n := \dim(\mathcal{V}) \geq 1$.
(*i*) $\Rightarrow$ (*ii*) Fix an orthonormal basis γ such that $A := [T]_{\gamma\leftarrow\gamma}$ is upper triangular by Theorem 6.1.9.
The identities $[T^*]_{\gamma\leftarrow\gamma} = A^*$, $[T^*T]_{\gamma\leftarrow\gamma} = A^*A$, and $[TT^*]_{\gamma\leftarrow\gamma} = AA^*$ hold by Theorem 2.3.7 and Proposition 3.4.4.
Thus $A^*A = AA^*$ by normality of T, so A is normal by Definition 6.1.1(ii).
The matrix A is diagonal by Lemma 6.1.10; hence γ consists of eigenvectors of T.
(*ii*) $\Rightarrow$ (*iii*) The matrix of T in an orthonormal eigenbasis is diagonal by Definition 3.2.3.
(*iii*) $\Rightarrow$ (*i*) Fix an orthonormal basis γ in which $A := [T]_{\gamma\leftarrow\gamma}$ is diagonal. Then $[T^*]_{\gamma\leftarrow\gamma} = A^*$ by Theorem 2.3.7.
Since $AA^* = A^*A$, $[TT^*]_{\gamma\leftarrow\gamma} = [T^*T]_{\gamma\leftarrow\gamma}$ by Proposition 3.4.4.
Therefore $TT^* = T^*T$ by Theorem 3.4.5(i), and T is normal by Definition 6.1.1(i).
Structural Commentary: Schur form reduces normality to the triangular lemma; diagonal coordinates verify the converse.
Key Insight: An upper-triangular normal matrix has no off-diagonal entries. □

Figure 46 depicts the forward mechanism in Theorem 6.1.11.

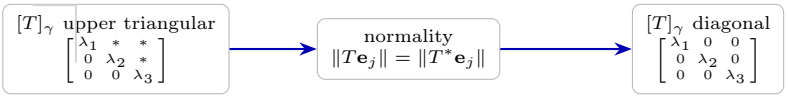

Figure 46: Normality turns Schur triangular form into diagonal form by forcing row tails to vanish.

Theorem 6.1.12. *Real Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:
(i) T is self-adjoint;
(ii) $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
(iii) T has a diagonal matrix in an orthonormal basis of $\mathcal{V}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Suppose $T = T^*$ and induct on $n := \dim(\mathcal{V})$. For $n = 0$, the empty basis suffices. For $n = 1$, fix $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$.
Then $\operatorname{span}(\mathbf{v}) = \mathcal{V} \in \operatorname{Inv}(T)$, so $\mathbf{v}$ is an eigenvector by Proposition 5.2.4. Hence $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is an orthonormal eigenbasis.
Suppose $n \geq 2$ and assume the implication in dimension $n-1$.
Fix $\lambda \in \operatorname{spec}(T)$ by Lemma 6.1.7, choose a unit $\mathbf{u} \in E_\lambda(T)$ by normalization, and set $\mathcal{U} := \operatorname{span}(\mathbf{u})$.
Then $\mathcal{U} \in \operatorname{Inv}(T)$ by Proposition 5.2.4, while $\mathcal{U}^\perp \in \operatorname{Inv}(T)$ and $T|_{\mathcal{U}^\perp}$ is self-adjoint by Proposition 6.1.8(i),(iii).
The identity $\dim(\mathcal{U}^\perp) = n - 1$ holds by Corollary 2.1.7, so $\mathcal{U}^\perp$ has an orthonormal eigenbasis by the induction hypothesis.
Adjoining $\mathbf{u}$ forms an orthonormal eigenbasis of $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$ by Theorem 2.1.5.

(*ii*) $\Rightarrow$ (*iii*) The matrix of T in an orthonormal eigenbasis is diagonal by Definition 3.2.3.

(*iii*) $\Rightarrow$ (*i*) Fix an orthonormal basis γ in which $A := [T]_{\gamma \leftarrow \gamma}$ is real diagonal. Then $[T^*]_{\gamma \leftarrow \gamma} = A^t = A$ by Theorem 2.3.7.
If $\mathcal{V} \neq \{\mathbf{0}\}$, then $T^* = T$ by Theorem 3.4.5(i).
If $\mathcal{V} = \{\mathbf{0}\}$, then T and T^* are both the unique operator on the zero space, so again $T^* = T$.

Structural Commentary: Real eigenvalue existence starts induction on the invariant self-adjoint orthogonal complement.
Key Insight: Removing one eigenline preserves self-adjointness on its complement. □

Theorem 6.1.13. *Spectral Resolution of a Normal Operator*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, $d \geq 1$, let $\mu_1, \dots, \mu_d \in \mathbb{C}$ be distinct. For $T \in \mathcal{L}(\mathcal{V})$, the following hold:

(i) T is normal with $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$ iff $(\exists (P_j)_{j=1}^d \in \mathcal{L}(\mathcal{V})^d)$ such that $(\forall j \in \{1, \dots, d\})\, P_j^* = P_j = P_j^2$,
$(\forall i, j \in \{1, \dots, d\})\, [i \neq j \Longrightarrow P_i P_j = 0]$, $\sum_{j=1}^d P_j = I_{\mathcal{V}}$, and $T = \sum_{j=1}^d \mu_j P_j$.

(ii) Every family $(P_1, \dots, P_d)$ satisfying the right side of (i) obeys $(\forall j \in \{1, \dots, d\})\, \operatorname{range}(P_j) = E_{\mu_j}(T)$.

Proof.

(*i*) ($\Rightarrow$) Fix an orthonormal eigenbasis of T by Theorem 6.1.11(i),(ii).
Decomposition $\mathcal{V} = \bigoplus_{j=1}^d E_{\mu_j}(T)$ holds by grouping basis by eigenvalue and assigning the zero subspace to absent eigenvalues.
Its summands are pairwise orthogonal by Theorem 6.1.4.
Set $P_j := P_{E_{\mu_j}(T)}$ by Definition 2.2.2. Then $(\forall j \in \{1, \dots, d\})\, \operatorname{range}(P_j) = E_{\mu_j}(T)$ by Proposition 2.2.3(iv).
Thus $(\forall j \in \{1, \dots, d\})\, P_j^* = P_j = P_j^2$ by Theorem 2.5.6.
Fix $\mathbf{x} \in \mathcal{V}$ and write $\mathbf{x} = \sum_{j=1}^d \mathbf{x}_j$ with $(\forall j \in \{1, \dots, d\})\, \mathbf{x}_j \in E_{\mu_j}(T)$. Then $(\forall j \in \{1, \dots, d\})\, P_j \mathbf{x} = \mathbf{x}_j$.
Consequently, $(\forall i, j \in \{1, \dots, d\})\, [i \neq j \Longrightarrow P_i P_j \mathbf{x} = \mathbf{0}]$ and $\sum_{j=1}^d P_j \mathbf{x} = \mathbf{x}$.
The identity $T\mathbf{x} = \sum_{j=1}^d \mu_j \mathbf{x}_j = \sum_{j=1}^d \mu_j P_j \mathbf{x}$ holds by $\mathbf{x}_j \in E_{\mu_j}(T)$. Since $\mathbf{x}$ was arbitrary, the required operator identities hold.

(*i*) ($\Leftarrow$) Fix a family satisfying the right side of (i).
The identities $T^* = \sum_{j=1}^d \overline{\mu_j} P_j$ and $T^* T = T T^* = \sum_{j=1}^d |\mu_j|^2 P_j$ hold by Proposition 2.3.8(i) and the product relations.
Thus T is normal by Definition 6.1.1(i). Fix $\lambda \in \operatorname{spec}(T)$ and $\mathbf{x} \in E_\lambda(T) \setminus \{\mathbf{0}\}$.
Then $\mathbf{0} = (T - \lambda I_{\mathcal{V}})\mathbf{x} = \sum_{j=1}^d (\mu_j - \lambda) P_j \mathbf{x}$. If $i \neq j$, then $\langle P_i \mathbf{x}, P_j \mathbf{x} \rangle = \langle \mathbf{x}, P_i P_j \mathbf{x} \rangle = 0$ by self-adjointness and pairwise annihilation.
Hence $0 = \sum_{j=1}^d |\mu_j - \lambda|^2 \|P_j \mathbf{x}\|^2$. Since $\sum_{j=1}^d P_j \mathbf{x} = \mathbf{x} \neq \mathbf{0}$, $(\exists k \in \{1, \dots, d\})\, P_k \mathbf{x} \neq \mathbf{0}$.
$\lambda = \mu_k$ by the nonnegative sum and $P_k \mathbf{x} \neq \mathbf{0}$, so $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$.

(*ii*) Fix a family satisfying the right side of (i), $j \in \{1, \dots, d\}$, and $\mathbf{x} \in \mathcal{V}$. Then $\mathbf{x} = \sum_{i=1}^d P_i \mathbf{x}$ and $(T - \mu_j I_{\mathcal{V}})\mathbf{x} = \sum_{i \neq j} (\mu_i - \mu_j) P_i \mathbf{x}$.
The vectors $P_i \mathbf{x}$ are pairwise orthogonal by self-adjointness and pairwise annihilation, so $\|(T - \mu_j I_{\mathcal{V}})\mathbf{x}\|^2 = \sum_{i \neq j} |\mu_i - \mu_j|^2 \|P_i \mathbf{x}\|^2$.
The equivalence $\mathbf{x} \in E_{\mu_j}(T) \Longleftrightarrow (\forall i \neq j)\, P_i \mathbf{x} = \mathbf{0} \Longleftrightarrow \mathbf{x} = P_j \mathbf{x}$ holds by distinctness and $\sum_{i=1}^d P_i = I_{\mathcal{V}}$.
The equivalence $\mathbf{x} = P_j \mathbf{x} \Longleftrightarrow \mathbf{x} \in \operatorname{range}(P_j)$ holds by $P_j^2 = P_j$.

Structural Commentary: Eigenspaces construct the projections; weighted norms recover normality, spectrum, and ranges.
Key Insight: The weighted norm identity separates every spectral coordinate. □

Figure 47 depicts the eigenspace resolution in Theorem 6.1.13.

Figure 47: Pairwise orthogonal spectral projections resolve $I_{\mathcal{V}}$ and reconstruct T by eigenvalue weights.

Perspective 6.1.14. *Spectral Resolutions Are Orthogonal Resolutions*
Part (ii) of Theorem 6.1.13 identifies $\operatorname{range}(P_j) = E_{\mu_j}(T)$.
Together with Theorem 2.5.6, this makes P_j the orthogonal projection onto $E_{\mu_j}(T)$.
Thus a spectral resolution records the orthogonal eigenspace decomposition without choosing bases in its summands. This viewpoint is confined here to finite-dimensional normal operators.

Corollary 6.1.15. *Functional Calculus from Spectral Resolution*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, let $d \geq 1$, and let $\mu_1, \dots, \mu_d \in \mathbb{C}$ be distinct.
Let $P_1, \dots, P_d \in \mathcal{L}(\mathcal{V})$ be orthogonal projections satisfying $(\forall i, j \in \{1, \dots, d\})\,[i \neq j \implies P_i P_j = 0]$, $\sum_{j=1}^{d} P_j = I_{\mathcal{V}}$; $T := \sum_{j=1}^{d} \mu_j P_j$. Then:

(i) $(\forall f \in \mathbb{C}[z])\, f(T) = \sum_{j=1}^{d} f(\mu_j) P_j$;
(ii) $(\forall j \in \{1, \dots, d\})(\exists p_j \in \mathbb{C}[z])\, p_j(T) = P_j$.

Proof. Each P_j is idempotent by Theorem 2.5.6.

(i) We prove $(\forall m \in \mathbb{Z}_{\geq 0})\, T^m = \sum_{j=1}^{d} \mu_j^m P_j$ by induction. The case $m = 0$ is $I_{\mathcal{V}} = \sum_{j=1}^{d} P_j$.

If the identity holds at m, then $T^{m+1} = \sum_{i,j=1}^{d} \mu_i^m \mu_j P_i P_j = \sum_{j=1}^{d} \mu_j^{m+1} P_j$ by the product relations.

Thus the power identity holds by induction. For $f(z) = \sum_{m=0}^{r} a_m z^m$, $f(T) = \sum_{m=0}^{r} a_m \sum_{j=1}^{d} \mu_j^m P_j = \sum_{j=1}^{d} f(\mu_j) P_j$.

(ii) Fix $j \in \{1, \dots, d\}$ and define $p_j(z) := \prod_{1 \leq i \leq d,\ i \neq j} \frac{z - \mu_i}{\mu_j - \mu_i}$, with the empty product interpreted as 1.

Then $p_j(\mu_i) = \delta_{ij}$ by distinctness of the μ_i. Hence $p_j(T) = \sum_{i=1}^{d} \delta_{ij} P_i = P_j$ by part (i).

Structural Commentary: Projector multiplication determines all powers; Lagrange interpolation then isolates each projection.
Key Insight: $p_j(\mu_i) = \delta_{ij}$ selects exactly P_j. □

6.2 Normal Operator Consequences

All inner-product spaces in this subsection are finite-dimensional over the stated field.

Perspective 6.2.1. *Normal Means Diagonal in Orthogonal Coordinates*
Theorem 6.1.11 identifies complex normality with orthonormal diagonal form.
Thus normality is the coordinate-free form of orthogonal diagonalizability.
Over $\mathbb{R}$, invariant planes replace the missing nonreal eigenvectors.

Proposition 6.2.2. *Basic Spectral Properties of Normal Operators*
Let $\mathcal{V}$ be a complex inner-product space, $T \in \mathcal{L}(\mathcal{V})$ be normal, and $p \in \mathbb{C}[z]$.

(i) $p(T)$ *is normal and commutes with* T.
(ii) T^* *is normal and* $(\forall \lambda \in \mathbb{C})\ E_\lambda(T) = E_{\overline{\lambda}}(T^*)$.
(iii) $\ker T = \ker T^*$ *and* $\operatorname{range}(T) = \operatorname{range}(T^*)$.

Proof. Set $n := \dim(\mathcal{V})$. By Theorem 6.1.11(i),(ii), fix an orthonormal basis $(\mathbf{e}_j)_{j=1}^n$ satisfying $(\forall j \in J_n)\, T\mathbf{e}_j = \mu_j \mathbf{e}_j$.

(i) $(\forall j \in J_n)\, p(T)\mathbf{e}_j = p(\mu_j)\mathbf{e}_j$ by polynomial evaluation. The operator $p(T)$ is normal by Theorem 6.1.11(iii),(i).
The identity $Tp(T) = p(T)T$ holds by Proposition 1.1.6.

(ii) $(\forall j \in J_n)\, T^*\mathbf{e}_j = \overline{\mu_j}\mathbf{e}_j$ by Theorem 6.1.3. Thus T^* is normal by Theorem 6.1.11(iii),(i).
For $\lambda \in \mathbb{C}$, $E_\lambda(T) \subseteq E_{\overline{\lambda}}(T^*)$ by Theorem 6.1.3.
The reverse inclusion holds by the same theorem for T^* and $(T^*)^* = T$ by Proposition 2.3.8(iii); hence equality.

(iii) $\ker T = E_0(T) = E_0(T^*) = \ker T^*$ by (ii).
$\operatorname{range}(T) = \operatorname{span}\{\mathbf{e}_j : \mu_j \neq 0\} = \operatorname{range}(T^*)$ by the common eigenbasis and $\mu_j \neq 0 \iff \overline{\mu_j} \neq 0$.

Structural Commentary: The spectral theorem transfers polynomial, adjoint, kernel, and range claims to one diagonal basis.
Key Insight: Adjunction conjugates eigenvalues while preserving the zero eigenspace. □

Lemma 6.2.3. *Adjoints of Normal Matrices as One Polynomial*
Let $k \in \mathbb{Z}_{\geq 1}$ and $(n_j)_{j=1}^k \in \mathbb{Z}_{\geq 1}^k$. Let $(A_j)_{j=1}^k$ satisfy $(\forall j \in J_k)\,[A_j \in \operatorname{Mat}(n_j, n_j, \mathbb{C}) \wedge A_j^* A_j = A_j A_j^*]$.
Then $(\exists p \in \mathbb{C}[z])(\forall j \in J_k)\, p(A_j) = A_j^*$.

Proof.

(i) *Spectral Set*: Set $\Omega := \bigcup_{j=1}^{k} \operatorname{spec}(A_j)$. By Theorem 5.3.2 and Corollary 5.1.6, $(\forall j \in J_k)\, 1 \leq \#\operatorname{spec}(A_j) \leq n_j$; hence $0 < \#\Omega < \infty$.

(ii) *Interpolation*: Define $p(z) := \sum_{\lambda \in \Omega} \overline{\lambda} \prod_{\nu \in \Omega \setminus \{\lambda\}} \frac{z - \nu}{\lambda - \nu}$. Then $(\forall \lambda \in \Omega)\, p(\lambda) = \overline{\lambda}$ by direct evaluation of the Lagrange factors.

(iii) *Spectral Evaluation*: Fix $j \in J_k$. The matrix A_j induces a normal operator on $\mathbb{C}^{n_j}$ by Theorem 2.3.7 and Proposition 3.4.4.
Fix $A_j = \sum_{\mu \in \operatorname{spec}(A_j)} \mu P_{j,\mu}$ by Theorem 6.1.13(i). By Corollary 6.1.15 and the interpolation identity, $p(A_j) = \sum_{\mu \in \operatorname{spec}(A_j)} \overline{\mu} P_{j,\mu}$.
By Proposition 2.3.8(i) and Theorem 2.5.6, $A_j^* = \sum_{\mu \in \operatorname{spec}(A_j)} \overline{\mu} P_{j,\mu}$. Thus $(\forall j \in J_k)\, p(A_j) = A_j^*$.

Structural Commentary: Spectral union and Lagrange interpolation precede evaluation on each normal matrix.
Key Insight: A finite spectral union admits one conjugating polynomial. □

Theorem 6.2.4. *Fuglede–Putnam Theorem*
Let $m, n \geq 1$, and let $A \in \operatorname{Mat}(m, m, \mathbb{C})$ and $B \in \operatorname{Mat}(n, n, \mathbb{C})$ be normal. Let $X \in \operatorname{Mat}(m, n, \mathbb{C})$. Then $AX = XB \iff A^*X = XB^*$.

Proof.

($\Rightarrow$) Assume $AX = XB$. Then $(\forall r \in \mathbb{Z}_{\geq 0})\, A^r X = XB^r$ by induction, so $(\forall p \in \mathbb{C}[z])\, p(A)X = Xp(B)$ by linearity.
Fix p with $p(A) = A^*$ and $p(B) = B^*$ by Lemma 6.2.3. Then $A^*X = p(A)X = Xp(B) = XB^*$.

($\Leftarrow$) Assume $A^*X = XB^*$. The matrices A^* and B^* are normal by $(A^*)^*A^* = AA^* = A^*A = A^*(A^*)^*$ and its analogue for B^*.
Thus $(\forall q \in \mathbb{C}[z])\, q(A^*)X = Xq(B^*)$ by induction on powers and linearity.
Fix q with $q(A^*) = A$ and $q(B^*) = B$ by Lemma 6.2.3 and Proposition 2.3.8(iii). Then $AX = q(A^*)X = Xq(B^*) = XB$.

Structural Commentary: Polynomial propagation and common interpolation form symmetric directional arguments.
Key Insight: Normal adjoints are polynomial functions of their matrices. □

Computation Method 6.2.5. *Finding a Spectral Decomposition of a Normal Matrix*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be normal. Its eigenspaces span $\mathbb{C}^n$ by Theorem 6.1.11(i),(ii).
Distinct eigenspaces are orthogonal by Theorem 6.1.4. The output is a unitary diagonalization $A = U\Lambda U^*$.

i. Find the distinct eigenvalues $\mu_1, \ldots, \mu_d$.
ii. Compute $(E_{\mu_j}(A))_{j=1}^d$ by solving $(A - \mu_j I_n)\mathbf{x} = 0$ for $j \in J_d$.
iii. Orthonormalize bases of $(E_{\mu_j}(A))_{j=1}^d$.
iv. Concatenate these bases as the columns of U, and set $\Lambda := U^*AU$.
v. Verify $U^*U = UU^* = I_n$ and $AU = U\Lambda$. Check that the diagonal of Λ repeats μ_j exactly $\dim E_{\mu_j}(A)$ times for $j \in J_d$.

Theorem 6.2.6. *Spectral Classification of Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$ be normal.

(i) T is unitary if and only if $(\forall \lambda \in \mathrm{spec}(T))\ |\lambda| = 1$.
(ii) T is self-adjoint if and only if $\mathrm{spec}(T) \subseteq \mathbb{R}$.
(iii) T is an orthogonal projection if and only if $\mathrm{spec}(T) \subseteq \{0, 1\}$.
(iv) $T^* = -T$ if and only if $\mathrm{spec}(T) \subseteq i\mathbb{R}$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, its unique operator satisfies all four operator conditions.
Its empty spectrum satisfies all four spectral conditions. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

List the distinct eigenvalues as $\mu_1, \ldots, \mu_d$, and fix $T = \sum_{j=1}^d \mu_j P_j$ by Theorem 6.1.13(i).

The identity $I_\mathcal{V} = \sum_{j=1}^d P_j$ holds by Theorem 6.1.13(i). The formula $T^* = \sum_{j=1}^d \overline{\mu_j} P_j$ holds by Proposition 2.3.8(i) and Theorem 2.5.6.

The formulas $T^*T = \sum_{j=1}^d |\mu_j|^2 P_j$ and $T^2 = \sum_{j=1}^d \mu_j^2 P_j$ hold by direct multiplication.

To verify linear independence, suppose $\sum_{j=1}^d \alpha_j P_j = 0$ and fix $k \in J_d$.

Fix $\mathbf{0} \neq \mathbf{x} \in E_{\mu_k}(T) = \mathrm{range}(P_k)$ by $\mu_k \in \mathrm{spec}(T)$ and Theorem 6.1.13(ii).

Then $0 = \sum_{j=1}^d \alpha_j P_j \mathbf{x} = \alpha_k \mathbf{x}$, so $\alpha_k = 0$. Thus $(P_j)_{j=1}^d$ is linearly independent.

(i) ($\Rightarrow$) If T is unitary, then $T^*T = I_\mathcal{V}$ by unitarity.
Hence $(\forall j \in J_d)\ |\mu_j|^2 = 1$ by linear independence of the P_j, and therefore $(\forall j \in J_d)\ |\mu_j| = 1$.

(i) ($\Leftarrow$) If $(\forall j \in J_d)\ |\mu_j| = 1$, then $T^*T = TT^* = \sum_{j=1}^d P_j = I_\mathcal{V}$. Thus T is unitary by the definition of unitarity.

(ii) ($\Rightarrow$) If $T = T^*$, then $\sum_{j=1}^d (\mu_j - \overline{\mu_j})P_j = 0$. Hence $(\forall j \in J_d)\, \mu_j = \overline{\mu_j}$ by linear independence of the P_j, so $\mathrm{spec}(T) \subseteq \mathbb{R}$.

(ii) ($\Leftarrow$) If $\mathrm{spec}(T) \subseteq \mathbb{R}$, then $T^* = \sum_{j=1}^d \overline{\mu_j} P_j = \sum_{j=1}^d \mu_j P_j = T$.

(iii) ($\Rightarrow$) If T is an orthogonal projection, then $T^* = T$ and $T^2 = T$ by Theorem 2.5.6.
Hence $(\forall j \in J_d)\, \mu_j = \overline{\mu_j}$ and $\mu_j^2 = \mu_j$ by linear independence of the P_j. Therefore $\mathrm{spec}(T) \subseteq \{0, 1\}$.

(iii) ($\Leftarrow$) If $\mathrm{spec}(T) \subseteq \{0, 1\}$, then $(\forall j \in J_d)\, \overline{\mu_j} = \mu_j^2 = \mu_j$. Hence $T^* = T^2 = T$ by the spectral formulas.
Thus T is an orthogonal projection by Theorem 2.5.6.

(iv) ($\Rightarrow$) If $T^* = -T$, then $\sum_{j=1}^d (\overline{\mu_j} + \mu_j)P_j = 0$. Hence $(\forall j \in J_d)\, \overline{\mu_j} = -\mu_j$ by linear independence of the P_j, so $\mathrm{spec}(T) \subseteq i\mathbb{R}$.

(iv) ($\Leftarrow$) If $\mathrm{spec}(T) \subseteq i\mathbb{R}$, then $(\forall j \in J_d)\, \overline{\mu_j} = -\mu_j$. Consequently, $T^* = \sum_{j=1}^d \overline{\mu_j} P_j = -\sum_{j=1}^d \mu_j P_j = -T$.

Structural Commentary: One spectral resolution converts four operator identities into coefficient comparisons.
Key Insight: Orthogonal spectral projections are linearly independent. □

Proposition 6.2.7. *Cartesian Decomposition and Normality*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.

(i) The operator T has a unique decomposition $T = H + iK$ with self-adjoint $H, K \in \mathcal{L}(\mathcal{V})$.
Its terms are $H = \frac{1}{2}(T + T^*)$ and $K = \frac{1}{2i}(T - T^*)$.
(ii) For these operators, T is normal if and only if $HK = KH$.

Proof.

(i) Set $H := \frac{1}{2}(T + T^*)$ and $K := \frac{1}{2i}(T - T^*)$. Then $H^* = H$, $K^* = K$, and $T = H + iK$ by Proposition 2.3.8(i),(iii).
If $T = X + iY$ for self-adjoint X, Y, then $T^* = X - iY$ by Proposition 2.3.8(i).
Consequently, $X = \frac{1}{2}(T + T^*) = H$ and $Y = \frac{1}{2i}(T - T^*) = K$.
(ii) ($\Rightarrow$) Suppose T is normal.
Then $HK = KH$ by the identities $T^*T = H^2 + K^2 + i(HK - KH)$, $TT^* = H^2 + K^2 - i(HK - KH)$, and $T^*T = TT^*$.
(ii) ($\Leftarrow$) If $HK = KH$, the same product identities satisfy $T^*T = TT^*$, so T is normal.

Structural Commentary: Adjoint symmetrization precedes a direct product comparison.
Key Insight: The normality defect is $2i(HK - KH)$. □

Figure 48 summarizes Proposition 6.2.7.

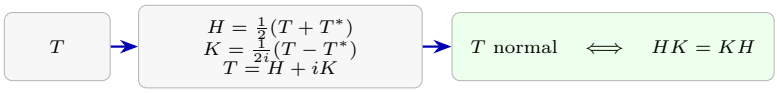

Figure 48: Cartesian decomposition: T is normal exactly when $HK = KH$.

Proposition 6.2.8. *Normal Block Upper-Triangular Matrices Collapse*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $r, s \geq 1$. Let $(B, C, X) \in \mathrm{Mat}(r, r, \mathbb{F}) \times \mathrm{Mat}(s, s, \mathbb{F}) \times \mathrm{Mat}(r, s, \mathbb{F})$.
Set $A := \left(\begin{smallmatrix} B & X \\ 0 & C \end{smallmatrix}\right) \in \mathrm{Mat}(r+s, r+s, \mathbb{F})$. Then A is normal if and only if $X = 0$ and both B and C are normal.

Proof.
($\Rightarrow$) Suppose A is normal; regard all matrices as complex when $\mathbb{F} = \mathbb{R}$.
By Theorem 6.1.9, fix unitary U_B, U_C such that $U_B^* B U_B$ and $U_C^* C U_C$ are upper triangular.
Set $Q := U_B \oplus U_C$, which is unitary by Proposition 2.6.12(ii). Then $Q^*Q = QQ^* = I_{r+s}$ by Theorem 2.6.8(i),(iii).
The matrix $Q^*AQ = \left(\begin{smallmatrix} U_B^* B U_B & U_B^* X U_C \\ 0 & U_C^* C U_C \end{smallmatrix}\right)$ is upper triangular.
The matrix Q^*AQ is normal by $(Q^*AQ)^*(Q^*AQ) = Q^*A^*AQ = Q^*AA^*Q = (Q^*AQ)(Q^*AQ)^*$.
Thus Q^*AQ is diagonal by Lemma 6.1.10. Hence $U_B^* X U_C = 0$ and $X = U_B(U_B^* X U_C)U_C^* = 0$.
Now $A = B \oplus C$, so $B^*B = BB^*$ and $C^*C = CC^*$ by blockwise normality.
($\Leftarrow$) If $X = 0$ and B, C are normal, then $A = B \oplus C$ and $A^*A = AA^*$ by blockwise multiplication.

Structural Commentary: Blockwise Schur reduction converts normality into the triangular-collapse lemma.
Key Insight: A normal triangular coupling cannot survive diagonalization. □

Figure 49 summarizes Proposition 6.2.8. Block-diagonal Schur coordinates reduce the claim to the earlier triangular normal-matrix lemma.

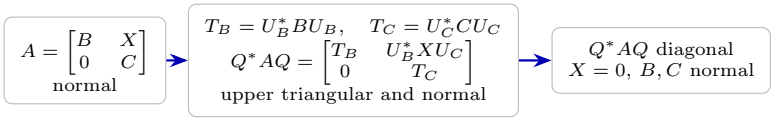

Figure 49: Normality removes the upper-right block in a block upper-triangular matrix.

Theorem 6.2.9. *Invariant Subspaces of Normal Matrices Reduce*
Let $n \geq 1$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be normal. Let $\mathcal{U} \leq \mathbb{F}^n$ be invariant under A, and let P be its orthogonal projection.
(i) The subspace $\mathcal{U}$ is invariant under A^*.
(ii) The identity $AP = PA$ holds.

Proof. If $\mathcal{U} = \{\mathbf{0}\}$, then $A^*\mathcal{U} \subseteq \mathcal{U}$ and $P = 0$, so both claims hold. If $\mathcal{U} = \mathbb{F}^n$, then $A^*\mathcal{U} \subseteq \mathcal{U}$ and $P = I_n$, so both claims hold.
Assume $r := \dim(\mathcal{U}) \in \{1, \ldots, n-1\}$. Choose orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$ by Corollary 1.3.11.
Let $Q = [Q_1 \; Q_2]$ be the unitary matrix obtained by concatenating these bases.
The identity $Q_2^* A Q_1 = 0$ holds by invariance of $\mathcal{U}$, so $Q^*AQ = \left(\begin{smallmatrix} B & X \\ 0 & C \end{smallmatrix}\right)$.
The matrix Q^*AQ is normal by $(Q^*AQ)^*(Q^*AQ) = Q^*A^*AQ$ and $(Q^*AQ)(Q^*AQ)^* = Q^*AA^*Q$. Hence $X = 0$ by Proposition 6.2.8.
(i) The inclusion $A^*\mathcal{U} \subseteq \mathcal{U}$ holds by $Q^*A^*Q = (Q^*AQ)^* = B^* \oplus C^*$.
(ii) The identity $Q^*PQ = I_r \oplus 0$ holds by Proposition 2.2.3(ii)–(iii).
The matrices Q^*AQ and Q^*PQ commute by blockwise multiplication; hence $AP = PA$ by conjugation with Q.

Structural Commentary: An adapted unitary basis reduces both conclusions to block multiplication.
Key Insight: Normality eliminates the off-diagonal coupling block. □

Corollary 6.2.10. *Invariant Subspaces of Normal Operators Reduce*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $\mathcal{V}$ be an inner-product space over $\mathbb{F}$, and let $T \in \mathcal{L}(\mathcal{V})$ be normal. Let $\mathcal{U} \leq \mathcal{V}$ be invariant under T.
(i) $\mathcal{U}^\perp$ is invariant under T;
(ii) $\mathcal{U}$ is invariant under T^*;
(iii) $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$;
(iv) $T|_{\mathcal{U}}$ and $T|_{\mathcal{U}^\perp}$ are normal.

Proof. If $\mathcal{U} = \{\mathbf{0}\}$ or $\mathcal{U} = \mathcal{V}$, all four claims hold by $T\mathbf{0} = T^*\mathbf{0} = \mathbf{0}$ and normality of the restrictions.
Assume $0 < r := \dim(\mathcal{U}) < n := \dim(\mathcal{V})$. Choose an orthonormal basis β adapted to $\mathcal{U} \oplus \mathcal{U}^\perp$ by Corollary 1.3.11 and Theorem 2.1.5.
Set $A := [T]_{\beta \leftarrow \beta}$ and $R := I_r \oplus 0_{n-r}$. The matrix A is normal by Theorem 2.3.7 and Proposition 3.4.4.
The coordinate subspace $\mathbb{F}^r \oplus \{\mathbf{0}\}$ is invariant under A, and R is its orthogonal projection.
Hence $AR = RA$ by Theorem 6.2.9(ii). The identity $A = B \oplus C$ holds by direct block multiplication with R.
(i) The inclusion $T(\mathcal{U}^\perp) \subseteq \mathcal{U}^\perp$ holds by $A = B \oplus C$.
(ii) The identity $[T^*]_{\beta \leftarrow \beta} = A^* = B^* \oplus C^*$ holds by Theorem 2.3.7. Hence $\mathcal{U}$ is invariant under T^*.
(iii) The subspaces $\mathcal{U}$ and $\mathcal{U}^\perp$ are invariant under T by the hypothesis and part (i).
The identity $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$ holds by Theorem 2.1.5. Thus $(\mathcal{U}, \mathcal{U}^\perp)$ reduces T, and $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$ by Theorem 5.2.7(iii).
(iv) The identity $A^*A = AA^*$ is equivalent blockwise to $B^*B = BB^*$ and $C^*C = CC^*$.
By Theorem 2.3.7 and Proposition 3.4.4, these are the adjoint-product identities for both restrictions.
Hence both restrictions are normal by Theorem 3.4.5(i).

Structural Commentary: Matrix reduction transfers four operator claims to two diagonal blocks.
Key Insight: Commutation with the projection is the reducing criterion. □

Theorem 6.2.11. *Simultaneous Unitary Diagonalization of Commuting Normal Operators*
Let $\mathcal{V}$ be a complex inner-product space, and let $\emptyset \neq \mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ consist of normal operators.
Then $\mathcal{F}$ is pairwise commuting if and only if $\mathcal{V}$ has an orthonormal basis of common eigenvectors for $\mathcal{F}$.

Proof.
($\Rightarrow$) Assume $\mathcal{F}$ is pairwise commuting and induct on $\dim(\mathcal{V})$.
If $(\forall A \in \mathcal{F})(\exists c_A \in \mathbb{C})\, A = c_A I_{\mathcal{V}}$, fix any orthonormal basis by Corollary 1.3.11.
It is common by the scalar condition, including the empty basis when $\mathcal{V} = \{\mathbf{0}\}$.
Otherwise, fix a nonscalar $A_0 \in \mathcal{F}$ and its spectral resolution $A_0 = \sum_{j=1}^{d} \mu_j P_j$ by Theorem 6.1.13(i).
The inequality $d \geq 2$ holds by nonscalarity. $(\forall j \in J_d)\, 0 < \dim(\text{range}(P_j)) < \dim(\mathcal{V})$ by Theorem 6.1.13(i)–(ii) and $d \geq 2$.
Each P_j is a polynomial in A_0 by Corollary 6.1.15(ii).
Fix $B \in \mathcal{F}$. Then $(\forall p \in \mathbb{C}[z])\, Bp(A_0) = p(A_0)B$ by $BA_0 = A_0B$, induction on powers, and linearity.
Hence $(\forall j \in J_d)\, BP_j = P_jB$. Fix $j \in J_d$ and $\mathbf{x} \in \text{range}(P_j)$. Then $P_jB\mathbf{x} = BP_j\mathbf{x} = B\mathbf{x}$, so $B\mathbf{x} \in \text{range}(P_j)$.
Thus $B|_{\text{range}(P_j)}$ is normal by Corollary 6.2.10(iv). The restricted family on each range is pairwise commuting and normal.
Fix a common orthonormal eigenbasis on each range by the induction hypothesis.
The identity $\mathcal{V} = \bigoplus_{j=1}^{d} \text{range}(P_j)$ holds by Theorem 6.1.13(i)–(ii).
Concatenating the restricted bases produces a common orthonormal eigenbasis of $\mathcal{V}$.
($\Leftarrow$) If $\mathcal{V} = \{\mathbf{0}\}$, its unique operator commutes with itself.
Otherwise, every $A \in \mathcal{F}$ has a diagonal matrix in the common orthonormal eigenbasis.
Hence the operators commute by Proposition 3.4.4 and Theorem 3.4.5(i).
Structural Commentary: Spectral reduction turns dimension induction into independent restricted families.
Key Insight: Commutants preserve all spectral subspaces. □

Figure 50 illustrates the spectral-subspace reduction in Theorem 6.2.11.

Figure 50: The spectral ranges of $A_0 = \sum_j \mu_j P_j$ reduce the commuting family for dimension induction.

Corollary 6.2.12. *Sums and Products of Commuting Normal Operators*
Let $\mathcal{V}$ be a complex inner-product space, and let $S, T \in \mathcal{L}(\mathcal{V})$ be commuting normal operators. Then $S+T$ and ST are normal.

Proof. Fix a common orthonormal eigenbasis for S and T by Theorem 6.2.11.
The operators $S+T$ and ST are diagonal in that basis and therefore normal by Theorem 6.1.11(iii),(i).
Structural Commentary: Simultaneous diagonalization reduces both constructions to scalar arithmetic.
Key Insight: Common diagonal coordinates are closed under sums and products. □

Perspective 6.2.13. *Real Normal Operators Use Planes*
Theorem 5.3.14 replaces complex eigenlines by invariant real lines and planes.
For normal operators, orthogonal reduction turns a nonreal conjugate pair into a rotation-dilation block.
This viewpoint is the real substitute for diagonalization; it does not make every invariant plane irreducible.

Proposition 6.2.14. *Two-Dimensional Real Normal Blocks*
Let $\mathcal{V}$ be a two-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
(i) T is normal and not self-adjoint if and only if every orthonormal-basis matrix has the form $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$. The corresponding parameters satisfy $a \in \mathbb{R}$ and $b \in \mathbb{R} \setminus \{0\}$.
(ii) If the equivalent conditions in (i) hold, then an orthonormal basis exists for which $b > 0$.

Proof.
(*i*) ($\Rightarrow$) Fix an arbitrary orthonormal basis and write $A = [T] = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right)$.
The identities $A^tA = AA^t$ and $A \neq A^t$ hold by Theorem 2.3.7 and Proposition 3.4.4.
Equality of the $(1,1)$ entries is $b^2 = c^2$. The relation $c \neq b$ holds by $A \neq A^t$, so $c = -b$ and $b \neq 0$.
Equality of the off-diagonal entries is $-ab + bd = ab - bd$, so $d = a$.
The stated form holds in every orthonormal basis by arbitrariness of the basis.
(*i*) ($\Leftarrow$) Fix an orthonormal basis with $A = [T] = \left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$ and $b \neq 0$.
The identities $A^tA = AA^t$ and $A \neq A^t$ hold by direct multiplication and $b \neq 0$.
Thus T is normal and not self-adjoint by Theorem 2.3.7 and Proposition 3.4.4.
(*ii*) Fix an orthonormal basis and its nonzero parameter b by part (i).
If $b < 0$, negate the second basis vector; the new parameter is $-b > 0$. If $b > 0$, retain the basis.
Structural Commentary: Entrywise normality determines the block, followed by an orientation choice.
Key Insight: Nonsymmetry selects $c = -b$ from $c^2 = b^2$. □

Theorem 6.2.15. *Real Normal Operator Block Form*
Let $\mathcal{V}$ be a real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. For $a \in \mathbb{R}$ and $b > 0$, set $N_{a,b} := \left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$.
Then T is normal if and only if some orthonormal basis makes $[T]$ a direct sum of scalar blocks and blocks $N_{a,b}$.

Proof.
($\Rightarrow$) Induct on $n := \dim(\mathcal{V})$. For $n = 0$, the empty orthonormal basis has the required form.
For $n = 1$, any unit vector produces a scalar block. Suppose $n \geq 2$. Fix a T-invariant $\mathcal{U}$ with $\dim(\mathcal{U}) \in \{1, 2\}$ by Theorem 5.3.14.
The subspace $\mathcal{U}^{\perp}$ is T-invariant, and both restrictions are normal, by Corollary 6.2.10(i),(iv).
If $\dim(\mathcal{U}) = 1$, a unit vector in $\mathcal{U}$ determines a scalar block.

If $\dim(\mathcal{U}) = 2$ and $T|_{\mathcal{U}}$ is self-adjoint, fix an orthonormal eigenbasis by Theorem 6.1.12(i),(ii). Its two blocks are scalar.
If $T|_{\mathcal{U}}$ is not self-adjoint, fix an orthonormal basis with matrix $N_{a,b}$ by Proposition 6.2.14(ii).
Fix the required basis on $\mathcal{U}^{\perp}$ by the induction hypothesis.
Their concatenation is an orthonormal basis of $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$ by Theorem 2.1.5.

($\Leftarrow$) Fix an orthonormal basis with the stated block form and set $A := [T]$.
Every allowed block D satisfies $D^*D = DD^*$ by direct multiplication, so $A^*A = AA^*$ blockwise.
The identity $T^*T = TT^*$ holds by Theorem 2.3.7, Proposition 3.4.4, and Theorem 3.4.5(i).
Thus T is normal by its defining identity.

Structural Commentary: Orthogonal reduction supports induction over invariant lines and planes.
Key Insight: Two-dimensional nonsymmetric blocks have rotation-dilation form. □

Figure 51 summarizes Theorem 6.2.15.

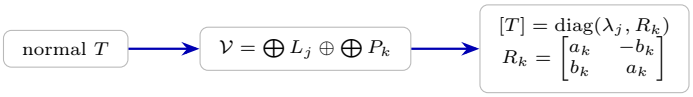

Figure 51: Orthogonal line/plane split into scalar and rotation blocks.

Theorem 6.2.16. *Description of Complex Isometries*
Let $\mathcal{V}$ be a complex inner-product space, $n := \dim(\mathcal{V})$, and $S \in \mathcal{L}(\mathcal{V})$.
Then S is an isometry if and only if $\mathcal{V}$ has an orthonormal eigenbasis $(\mathbf{e}_j)_{j=1}^n$ with eigenvalues $(\lambda_j)_{j=1}^n$ satisfying $(\forall j \in J_n)\ |\lambda_j| = 1$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, both conditions hold for the unique operator and empty orthonormal basis. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

($\Rightarrow$) Suppose S is an isometry. Then $S^*S = I_{\mathcal{V}}$ by Theorem 2.6.6.
The map S is injective by norm preservation and invertible by Theorem 1.7.7(ii),(i). Thus $S^* = S^{-1}$ and $SS^* = I_{\mathcal{V}}$, so S is normal.
Fix an orthonormal eigenbasis $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$ by Theorem 6.1.11(i),(ii).
Then $(\forall j \in J_n)\ |\lambda_j| = \|S\mathbf{e}_j\| = 1$ by norm preservation.

($\Leftarrow$) Suppose $(\mathbf{e}_j)_{j=1}^n$ is an orthonormal eigenbasis with eigenvalues $(\lambda_j)_{j=1}^n$ satisfying $(\forall j \in J_n)\ |\lambda_j| = 1$.
For $\mathbf{v} = \sum_{j=1}^n c_j \mathbf{e}_j$, $\|S\mathbf{v}\|^2 = \sum_{j=1}^n |\lambda_j c_j|^2 = \sum_{j=1}^n |c_j|^2 = \|\mathbf{v}\|^2$ by Theorem 1.4.7(i). Hence S is an isometry.

Structural Commentary: A forward normality reduction pairs with a reverse Parseval calculation.
Key Insight: Isometry becomes coefficientwise unit modulus. □

Example 6.2.17. *Plane Rotations*
For $\theta \in \mathbb{R}$, set $R_\theta := \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$. The identity $R_\theta^t R_\theta = I_2$ holds by direct multiplication.
Thus R_θ is an isometry of $\mathbb{R}^2$ by Theorem 2.3.7, Proposition 3.4.4, and Theorem 2.6.6.
If $\theta \notin \pi\mathbb{Z}$, its characteristic polynomial $x^2 - 2(\cos\theta)x + 1$ has discriminant $-4\sin^2\theta < 0$.
Hence R_θ has no real eigenvalue, which motivates two-dimensional blocks in the real isometry form.

Theorem 6.2.18. *Real Isometry Block Form*
Let $\mathcal{V}$ be a real inner-product space and let $S \in \mathcal{L}(\mathcal{V})$. For $\theta \in (0, \pi)$, set $R_\theta := \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$.
Then S is an isometry if and only if some orthonormal basis makes $[S]$ a direct sum of $[1]$, $[-1]$, and blocks R_θ.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, both conditions hold for the unique operator and empty orthonormal basis. Assume $\dim(\mathcal{V}) \geq 1$.

($\Rightarrow$) Suppose S is an isometry. Then $S^*S = I_{\mathcal{V}}$ by Theorem 2.6.6.
The map S is injective by norm preservation and invertible by Theorem 1.7.7(ii),(i).
Hence $S^* = S^{-1}$, so $SS^* = I_{\mathcal{V}}$ and S is normal. Fix a block basis by Theorem 6.2.15.
In this basis, $[S]^t[S] = I$ by Theorem 2.3.7 and Proposition 3.4.4. Every scalar block $[\lambda]$ satisfies $\lambda^2 = 1$, so $\lambda = \pm 1$.
Every two-dimensional block $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$ satisfies $a^2 + b^2 = 1$ with $b > 0$.
Hence a unique $\theta \in (0, \pi)$ satisfies $a = \cos\theta$ and $b = \sin\theta$, so the block equals R_θ.

($\Leftarrow$) Fix a basis with the stated block form. Every block D satisfies $D^tD = I$.
Thus $[S]^t[S] = I$, and $S^*S = I_{\mathcal{V}}$ by Theorem 2.3.7, Proposition 3.4.4, and Theorem 3.4.5(i).
Hence S is an isometry by Theorem 2.6.6.

Structural Commentary: The real normal form is filtered blockwise by the isometry identity.
Key Insight: Unit-length normal blocks are signs or rotations. □

Figure 52 illustrates the three block types in Theorem 6.2.18.

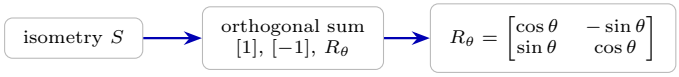

Figure 52: Normal form: fixed/flipped lines and rotation planes.

6.3 Operator Functions

This subsection develops finite spectral functions of normal operators and positive square roots.
It then characterizes commutativity through spectral projections and a single normal generator.

Definition 6.3.1. *Spectral Resolution Notation*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, and let $T \in \mathcal{L}(\mathcal{V})$ be normal. The associated spectral notation is:

(i) *Spectral Set*: $\Omega_T := \operatorname{spec}(T)$.
(ii) *Spectral Projections*: For $\lambda \in \Omega_T$, P_λ is the orthogonal projection onto $E_\lambda(T)$.

(iii) *Spectral Resolution*: $(P_\lambda)_{\lambda\in\Omega_T}$, with $T = \sum_{\lambda\in\Omega_T} \lambda P_\lambda$ and $I_\mathcal{V} = \sum_{\lambda\in\Omega_T} P_\lambda$.

For $\mathcal{V} = \{\mathbf{0}\}$, the spectral set and projection family are empty, and both sums are the unique zero operator.

Theorem 6.3.2. *Finite Spectral Functional Calculus for Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, and let $T \in \mathcal{L}(\mathcal{V})$ be normal.
Define $\Phi_T : \mathbb{C}^{\Omega_T} \to \mathcal{L}(\mathcal{V})$ by $\Phi_T(f) := f(T) := \sum_{\lambda\in\Omega_T} f(\lambda)P_\lambda$.

(i) $(\forall f \in \mathbb{C}^{\Omega_T})(\forall p \in \mathbb{C}[z])\,[p|_{\Omega_T} = f \implies p(T) = f(T)]$.
(ii) $(\forall f \in \mathbb{C}^{\Omega_T})(\exists p_f \in \mathbb{C}[z])\,[p_f|_{\Omega_T} = f \wedge p_f(T) = f(T)]$.

Proof. If $\Omega_T = \emptyset$, then $\mathcal{V} = \{\mathbf{0}\}$ by Theorem 6.1.11(i)–(ii).
The zero polynomial represents the unique $f \in \mathbb{C}^{\Omega_T}$, and all operator sums vanish; hence (i)–(ii).
Otherwise $\Omega_T \neq \emptyset$, and the following arguments apply.

(i) Fix $f \in \mathbb{C}^{\Omega_T}$ and $p \in \mathbb{C}[z]$ with $p|_{\Omega_T} = f$.
Then $p(T) = \sum_{\lambda\in\Omega_T} p(\lambda)P_\lambda = \sum_{\lambda\in\Omega_T} f(\lambda)P_\lambda = f(T)$ by Corollary 6.1.15(i) and the restriction identity.

(ii) Fix $f \in \mathbb{C}^{\Omega_T}$. For $\lambda \in \Omega_T$, set $\ell_\lambda(z) := \prod_{\mu\in\Omega_T\setminus\{\lambda\}} \frac{z-\mu}{\lambda-\mu}$.
Set $p_f(z) := \sum_{\lambda\in\Omega_T} f(\lambda)\ell_\lambda(z)$. For $\lambda_0 \in \Omega_T$, $(\forall \lambda \in \Omega_T)\,\ell_\lambda(\lambda_0) = \delta_{\lambda,\lambda_0}$ by distinctness.
Thus $p_f(\lambda_0) = f(\lambda_0)$, so $p_f|_{\Omega_T} = f$; moreover, $p_f(T) = f(T)$ by (i).

Structural Commentary: Polynomial spectral evaluation feeds an explicit Lagrange interpolation on Ω_T.
Key Insight: $\ell_\lambda(\mu) = \delta_{\lambda,\mu}$ encodes arbitrary finite spectral data. □

Theorem 6.3.3. *Rules for Normal Spectral Functions*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, and let $T \in \mathcal{L}(\mathcal{V})$ be normal.
For $f, g : \Omega_T \to \mathbb{C}$, the following identities and properties hold:

(i) $(f+g)(T) = f(T) + g(T)$;
(ii) $(fg)(T) = f(T)g(T)$;
(iii) $f(T)^* = \overline{f}(T)$, *where* $(\forall \lambda \in \Omega_T)\,\overline{f}(\lambda) := \overline{f(\lambda)}$;
(iv) $f(T)$ *is normal;*
(v) $\operatorname{spec}(f(T)) = f(\Omega_T)$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, all spectral sums and spectra are empty, so (i)–(v) hold. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(i) $(f+g)(T) = \sum_{\lambda\in\Omega_T} (f(\lambda)+g(\lambda))P_\lambda = f(T) + g(T)$ by the definition of the spectral functional calculus.

(ii) The relations $P_\lambda P_\mu = 0$ for $\lambda \neq \mu$ and $P_\lambda^2 = P_\lambda$ hold by Theorem 6.1.13(i)–(ii). Hence $f(T)g(T) = \sum_{\lambda\in\Omega_T} f(\lambda)g(\lambda)P_\lambda = (fg)(T)$.

(iii) $(\forall \lambda \in \Omega_T)\,P_\lambda^* = P_\lambda$ by Theorem 2.5.6. Thus $f(T)^* = \sum_{\lambda\in\Omega_T} \overline{f(\lambda)}P_\lambda = \overline{f}(T)$ by Proposition 2.3.8(i).

(iv) $f(T)^*f(T) = (\overline{f}f)(T) = (f\overline{f})(T) = f(T)f(T)^*$ by (ii)–(iii). Therefore $f(T)$ is normal by Definition 6.1.1(i).

(v) Set $n := \dim(\mathcal{V})$. Fix an orthonormal eigenbasis $(\mathbf{e}_j)_{j=1}^n$ for T by Theorem 6.1.11(i)–(ii), and write $T\mathbf{e}_j = \lambda_j\mathbf{e}_j$.
Then $f(T)\mathbf{e}_j = f(\lambda_j)\mathbf{e}_j$ by the spectral resolution. Thus $f(T)$ is diagonal, with diagonal-value set $f(\Omega_T)$, so $\operatorname{spec}(f(T)) = f(\Omega_T)$.

Structural Commentary: Projection arithmetic transfers pointwise algebra and conjugation before the eigenbasis reads the spectrum.
Key Insight: Every $E_\lambda(T)$ converts $f(T)$ into multiplication by $f(\lambda)$. □

Definition 6.3.4. *Positive Operator*
Let $\mathcal{V}$ be finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For $T \in \mathcal{L}(\mathcal{V})$, $T \geq 0$, T *positive* if $T = T^*$ and $(\forall \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle \geq 0$.

Definition 6.3.5. *Square Root of an Operator*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $T \in \mathcal{L}(\mathcal{V})$. An operator $R \in \mathcal{L}(\mathcal{V})$ is a *square root* of T if $R^2 = T$.

Theorem 6.3.6. *Characterization of Positive Operators*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

(i) T *is positive;*
(ii) $T = T^*$ *and* $\operatorname{spec}(T) \subseteq [0, \infty)$;
(iii) $(\exists S \in \mathcal{L}(\mathcal{V}))\,[S \geq 0 \wedge S^2 = T]$;
(iv) $(\exists R \in \mathcal{L}(\mathcal{V}))\,[R = R^* \wedge R^2 = T]$;
(v) $(\exists R \in \mathcal{L}(\mathcal{V}))\,T = R^*R$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, its unique operator satisfies all five conditions. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(*i*) $\Rightarrow$ (*ii*) The identity $T = T^*$ holds by Definition 6.3.4. Fix $\lambda \in \operatorname{spec}(T)$ and $\mathbf{v} \in E_\lambda(T) \setminus \{\mathbf{0}\}$.
Then $\lambda \in \mathbb{R}$ by Theorem 6.1.5, while $0 \leq \langle T\mathbf{v}, \mathbf{v}\rangle = \lambda\|\mathbf{v}\|^2$. Hence $\lambda \geq 0$.

(*ii*) $\Rightarrow$ (*iii*) Set $n := \dim(\mathcal{V})$. In the complex case, $T^*T = T^2 = TT^*$ by $T = T^*$, so T is normal.
Fix an orthonormal eigenbasis $(\mathbf{e}_j)_{j=1}^n$ by Theorem 6.1.12(i)–(ii) when $\mathbb{F} = \mathbb{R}$ and by Theorem 6.1.11(i)–(ii) when $\mathbb{F} = \mathbb{C}$.
Write $T\mathbf{e}_j = \lambda_j\mathbf{e}_j$, where $(\forall j \in \{1, \ldots, n\})\,\lambda_j \geq 0$. Fix $S \in \mathcal{L}(\mathcal{V})$ satisfying $(\forall j \in \{1, \ldots, n\})\,S\mathbf{e}_j = \sqrt{\lambda_j}\mathbf{e}_j$ by Proposition 1.5.8.
For $\mathbf{u} = \sum_{j=1}^n c_j\mathbf{e}_j$ and $\mathbf{v} = \sum_{j=1}^n d_j\mathbf{e}_j$, $\langle S\mathbf{u}, \mathbf{v}\rangle = \sum_{j=1}^n \sqrt{\lambda_j}c_j\overline{d_j} = \langle \mathbf{u}, S\mathbf{v}\rangle$.
$S = S^*$ by the preceding adjoint identity, and $\langle S\mathbf{v}, \mathbf{v}\rangle = \sum_{j=1}^n \sqrt{\lambda_j}\,|d_j|^2 \geq 0$. Hence $S \geq 0$ by Definition 6.3.4.
The identity $S^2 = T$ holds by Proposition 1.5.8 and the equality $(\forall j \in \{1, \ldots, n\})\,S^2\mathbf{e}_j = T\mathbf{e}_j$.

(*iii*) $\Rightarrow$ (*iv*) The relation $S = S^*$ holds by $S \geq 0$ and Definition 6.3.4, so S is a self-adjoint square root.

(*iv*) $\Rightarrow$ (*v*) The identity $T = R^*R$ holds by $T = R^2$ and $R = R^*$.

(v) $\Rightarrow$ (i) The identity $T^* = (R^*R)^* = R^*(R^*)^* = R^*R = T$ holds by Proposition 2.3.8(ii)–(iii).
The inequality $(\forall \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle = \|R\mathbf{v}\|^2 \geq 0$ holds by $T = R^*R$ and the defining adjoint identity.
Thus $T \geq 0$ by Definition 6.3.4 and $T^* = T$.

Structural Commentary: Spectral coordinates construct the positive root, and adjoint factorization closes the implication cycle.
Key Insight: $\lambda_j \geq 0$ permits the canonical multiplier $\sqrt{\lambda_j}$. □

Proposition 6.3.7. *Unique Positive Square Root*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For $T \in \mathcal{L}(\mathcal{V})$ with $T \geq 0$, $(\exists! R \in \mathcal{L}(\mathcal{V}))\,[R \geq 0 \wedge R^2 = T]$.

Proof.

(i) *Existence and Degenerate Case*: A positive square root exists by Theorem 6.3.6(i),(iii).
If $\mathcal{V} = \{\mathbf{0}\}$, its unique operator is its unique positive square root. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(ii) *Spectral Reduction*: Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T.
Then $T = T^*$ and $(\forall j \in \{1, \ldots, m\})\ \lambda_j \geq 0$ by Theorem 6.3.6(i),(ii). In the complex case, $T^*T = T^2 = TT^*$, so T is normal.
Hence $\mathcal{V} = E_{\lambda_1}(T) \odot \cdots \odot E_{\lambda_m}(T)$ by Theorem 6.1.12(i)–(ii) when $\mathbb{F} = \mathbb{R}$ and by Theorem 6.1.11(i)–(ii) when $\mathbb{F} = \mathbb{C}$.

(iii) *Restriction to One Eigenspace*: Let R be a positive square root of T.
The identity $RT = R^3 = TR$ holds by $R^2 = T$, so $(\forall j \in \{1, \ldots, m\})\ E_{\lambda_j}(T) \in \mathrm{Inv}(R)$.
For $j \in \{1, \ldots, m\}$, set $R_j := R|_{E_{\lambda_j}(T)}$. Then $R_j = R_j^*$ by Proposition 6.1.8(ii), and $(\forall \mathbf{x} \in E_{\lambda_j}(T))\ \langle R_j\mathbf{x}, \mathbf{x}\rangle \geq 0$ by $R \geq 0$.
Thus $R_j \geq 0$ by Definition 6.3.4, while $R_j^2 = \lambda_j I_{E_{\lambda_j}(T)}$ by $R^2 = T$. In the complex case, $R_j^*R_j = R_j^2 = R_jR_j^*$.
Fix an orthonormal eigenbasis of R_j by Theorem 6.1.12(i)–(ii) when $\mathbb{F} = \mathbb{R}$ and by Theorem 6.1.11(i)–(ii) when $\mathbb{F} = \mathbb{C}$.
Every corresponding eigenvalue ρ satisfies $\rho \geq 0$ by Theorem 6.3.6(i),(ii) and $\rho^2 = \lambda_j$ by the restriction identity.
Therefore $\rho = \sqrt{\lambda_j}$, so $R_j = \sqrt{\lambda_j} I_{E_{\lambda_j}(T)}$ by Proposition 1.5.8.

(iv) *Uniqueness*: Every positive square root R satisfies $(\forall j \in \{1, \ldots, m\})\ R|_{E_{\lambda_j}(T)} = \sqrt{\lambda_j} I_{E_{\lambda_j}(T)}$ by (iii).
Any two positive square roots agree on every summand by this identity and are therefore equal on $\mathcal{V}$ by the decomposition in (ii).

Structural Commentary: Spectral reduction localizes uniqueness to the restriction equation $R_j^2 = \lambda_j I$.
Key Insight: Positivity selects $\sqrt{\lambda_j}$ rather than $-\sqrt{\lambda_j}$. □

Definition 6.3.8. *Positive Square Root Notation*
Let $\mathcal{V}$ be finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For $T \in \mathcal{L}(\mathcal{V})$, $T \geq 0$, $\sqrt{T}$ denotes its unique positive square root.

Corollary 6.3.9. *Basic Functions from the Spectral Calculus*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, and let $T \in \mathcal{L}(\mathcal{V})$ be normal.

(i) $T^* = \sum_{\lambda \in \Omega_T} \overline{\lambda} P_\lambda$.

(ii) *If* $0 \notin \Omega_T$, *then* $T^{-1} = \sum_{\lambda \in \Omega_T} \lambda^{-1} P_\lambda$.

(iii) *If* $T \geq 0$, *then* $\sqrt{T} = \sum_{\lambda \in \Omega_T} \sqrt{\lambda} P_\lambda$.

Proof.

(i) $T^* = \sum_{\lambda \in \Omega_T} \overline{\lambda} P_\lambda$ by Theorem 6.3.3(iii) with the identity function on Ω_T.

(ii) Set $f(\lambda) := \lambda^{-1}$. Then $Tf(T) = f(T)T = \sum_{\lambda \in \Omega_T} P_\lambda = I_{\mathcal{V}}$ by the projection product relations and the resolution of $I_{\mathcal{V}}$.
Thus $T^{-1} = f(T) = \sum_{\lambda \in \Omega_T} \lambda^{-1} P_\lambda$ by uniqueness of the inverse.

(iii) The inclusion $\Omega_T \subseteq [0, \infty)$ holds by Theorem 6.3.6(i),(ii). Set $g(\lambda) := \sqrt{\lambda}$.
Then $g(T)^* = g(T)$, $g(T)^2 = T$, and $\mathrm{spec}(g(T)) = g(\Omega_T) \subseteq [0, \infty)$ by Theorem 6.3.3(ii)–(v).
Hence $g(T) \geq 0$ by Theorem 6.3.6(ii),(i), and $\sqrt{T} = g(T) = \sum_{\lambda \in \Omega_T} \sqrt{\lambda} P_\lambda$ by Proposition 6.3.7.

Structural Commentary: Conjugation, inversion, and positive square roots transfer pointwise through the spectral calculus.
Key Insight: Each scalar function acts independently on every spectral block. □

Theorem 6.3.10. *Commutativity via Spectral Projections*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, and let $T, S \in \mathcal{L}(\mathcal{V})$ be normal. Write $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$ and $S = \sum_{\mu \in \Omega_S} \mu Q_\mu$.

(i) *For* $R \in \mathcal{L}(\mathcal{V})$, $RT = TR \iff (\forall \lambda \in \Omega_T)\ RP_\lambda = P_\lambda R$.

(ii) $TS = ST \iff (\forall \lambda \in \Omega_T)(\forall \mu \in \Omega_S)\ P_\lambda Q_\mu = Q_\mu P_\lambda$.

(iii) *If* $f : \Omega_T \to \mathbb{C}$ *and* $g : \Omega_S \to \mathbb{C}$ *are injective, then* $f(T)g(S) = g(S)f(T) \iff TS = ST$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, both sides of every equivalence hold. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(*i*) ($\Rightarrow$) If $RT = TR$, then $(\forall p \in \mathbb{C}[z])\ Rp(T) = p(T)R$ by induction on powers and linearity.
Each P_λ is a polynomial in T by Corollary 6.1.15(ii), so $(\forall \lambda \in \Omega_T)\ RP_\lambda = P_\lambda R$.

(*i*) ($\Leftarrow$) If $(\forall \lambda \in \Omega_T)\ RP_\lambda = P_\lambda R$, then $RT = \sum_{\lambda \in \Omega_T} \lambda RP_\lambda = \sum_{\lambda \in \Omega_T} \lambda P_\lambda R = TR$.

(*ii*) ($\Rightarrow$) If $TS = ST$, then $(\forall \lambda \in \Omega_T)\ SP_\lambda = P_\lambda S$ by (i) with $R = S$. Hence $(\forall \lambda \in \Omega_T)(\forall \mu \in \Omega_S)\ P_\lambda Q_\mu = Q_\mu P_\lambda$ by (i) for S with $R = P_\lambda$.

(*ii*) ($\Leftarrow$) If all spectral projections commute pairwise, then $TS = \sum_{\lambda \in \Omega_T,\, \mu \in \Omega_S} \lambda\mu P_\lambda Q_\mu = \sum_{\lambda \in \Omega_T,\, \mu \in \Omega_S} \lambda\mu Q_\mu P_\lambda = ST$.

(*iii*) ($\Rightarrow$) Suppose $f(T)g(S) = g(S)f(T)$. Then $f(T)$ and $g(S)$ are normal by Theorem 6.3.3(iv).
Spectral coefficients are distinct by injectivity, so spectral projections are $(P_\lambda)_{\lambda \in \Omega_T}$ and $(Q_\mu)_{\mu \in \Omega_S}$ by Theorem 6.1.13(ii).
These projections commute by (ii) for $f(T)$ and $g(S)$; hence $TS = ST$ by (ii) for T and S.

(*iii*) ($\Leftarrow$) If $TS = ST$, then all P_λ commute with all Q_μ by (ii).
Thus $f(T)g(S) = \sum_{\lambda \in \Omega_T,\, \mu \in \Omega_S} f(\lambda)g(\mu) P_\lambda Q_\mu = \sum_{\lambda \in \Omega_T,\, \mu \in \Omega_S} f(\lambda)g(\mu) Q_\mu P_\lambda = g(S)f(T)$.

Structural Commentary: Polynomial recovery transfers operator commutation to pairwise commutation of spectral projections.
Key Insight: Commutation is equivalent to preserving every spectral block. □

Corollary 6.3.11. *A Commuting Normal Pair Comes from One Normal Operator*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, and let $T, S \in \mathcal{L}(\mathcal{V})$ be normal.
Let $\mathbb{C}[T,S]$ denote the finite sums of $c_{a,b}T^aS^b$ with $c_{a,b} \in \mathbb{C}$ and $a, b \in \mathbb{Z}_{\geq 0}$.
Then $TS = ST \iff (\exists H \in \mathbb{C}[T,S])(\exists p, q \in \mathbb{C}[z])\,[H^*H = HH^* \wedge T = p(H) \wedge S = q(H)]$.

Proof.
($\Rightarrow$) Suppose $TS = ST$. If $\mathcal{V} = \{\mathbf{0}\}$, the choices $H = T = S = 0$ and $p = q = 0$ satisfy the conclusion. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(a) *Joint Spectral Blocks*: Write $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$ and $S = \sum_{\mu \in \Omega_S} \mu Q_\mu$.
The projections commute by Theorem 6.3.10(ii). Set $\mathcal{I} := \{(\lambda, \mu) \in \Omega_T \times \Omega_S : P_\lambda Q_\mu \neq 0\}$.
For $(\lambda, \mu) \in \mathcal{I}$, $(P_\lambda Q_\mu)^2 = P_\lambda Q_\mu$ and $(P_\lambda Q_\mu)^* = P_\lambda Q_\mu$ by projection commutation and self-adjoint idempotence.
Thus $P_\lambda Q_\mu$ is an orthogonal projection by Theorem 2.5.6.
Distinct joint projections are orthogonal by pairwise annihilation and commutation.
After insertion of the zero products, $\sum_{(\lambda,\mu)\in\mathcal{I}} P_\lambda Q_\mu = \sum_{\lambda \in \Omega_T,\, \mu \in \Omega_S} P_\lambda Q_\mu$.
The latter sum equals $(\sum_{\lambda \in \Omega_T} P_\lambda)(\sum_{\mu \in \Omega_S} Q_\mu) = I_\mathcal{V}$ by the spectral resolutions.

(b) *Separating Operator*: Choose pairwise distinct $\alpha_{\lambda,\mu} \in \mathbb{C}$ for $(\lambda, \mu) \in \mathcal{I}$ and set $H := \sum_{(\lambda,\mu)\in\mathcal{I}} \alpha_{\lambda,\mu} P_\lambda Q_\mu$.
Then H is normal, with $\operatorname{spec}(H) = \{\alpha_{\lambda,\mu} : (\lambda, \mu) \in \mathcal{I}\}$, by Theorem 6.1.13(i)–(ii).

(c) *Generated Algebra*: Each P_λ is a polynomial in T and each Q_μ is a polynomial in S by Corollary 6.1.15(ii).
Therefore H is a finite sum of terms $c_{a,b}T^aS^b$, so $H \in \mathbb{C}[T,S]$.

(d) *Recovery of the Pair*: Define $f, g : \operatorname{spec}(H) \to \mathbb{C}$ by $f(\alpha_{\lambda,\mu}) := \lambda$ and $g(\alpha_{\lambda,\mu}) := \mu$.
Fix $p, q \in \mathbb{C}[z]$ satisfying $p|_{\operatorname{spec}(H)} = f$ and $q|_{\operatorname{spec}(H)} = g$ by Theorem 6.3.2(ii).
Then $p(H) = \sum_{(\lambda,\mu)\in\mathcal{I}} \lambda P_\lambda Q_\mu = T$ by the spectral resolution of T.
The identity $q(H) = \sum_{(\lambda,\mu)\in\mathcal{I}} \mu P_\lambda Q_\mu = S$ holds by the spectral resolution of S.

($\Leftarrow$) Suppose $T = p(H)$ and $S = q(H)$. Then $p(H)q(H) = q(H)p(H)$ by $H^aH^b = H^{a+b} = H^bH^a$ and distributivity, so $TS = ST$.

Structural Commentary: Joint spectral blocks build a separating normal generator before polynomial recovery of T and S.
Key Insight: Distinct labels encode both eigenvalue coordinates on each joint block. □

Part V
Polynomial Structure and Canonical Forms

1 Operator Polynomials

1.1 Polynomial Functional Calculus

Theorem 1.1.1. *Polynomial Functional Calculus for Diagonalizable Matrices*
Let $n \geq 1$, $\lambda_1, \dots, \lambda_n \in \mathbb{F}$, and set $\Lambda := \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ and $S \in \operatorname{Mat}(n, n, \mathbb{F})^\times$.
If $A = S\Lambda S^{-1}$, then $(\forall\, p \in \mathbb{F}[z])\, p(A) = S \operatorname{diag}(p(\lambda_1), \dots, p(\lambda_n))S^{-1}$.

Proof. Set $P_j := (A^j = S\Lambda^j S^{-1})$, $j \in \mathbb{Z}_{\geq 0}$. Base case P_0 holds by Definition 3.8.5 and inverse relations: $A^0 = I_n = SI_nS^{-1} = S\Lambda^0 S^{-1}$.
Fix $j \in \mathbb{Z}_{\geq 0}$ and suppose P_j. The identity $A^{j+1} = A^j A = S\Lambda^j S^{-1} S\Lambda S^{-1}$ holds by Definition 3.8.5 and P_j.
The identity $S\Lambda^j S^{-1} S\Lambda S^{-1} = S\Lambda^{j+1} S^{-1}$ holds by Proposition 3.1.2(i),(ii) and $S^{-1}S = I_n$.

Thus P_{j+1} holds, so $(\forall\, j \in \mathbb{Z}_{\geq 0})\, P_j$ by Theorem 2.6.2. Fix $p \in \mathbb{F}[z]$ and write $p(z) = \sum_{j=0}^{d} c_j z^j$ with $d \in \mathbb{Z}_{\geq 0}$ and $c_0, \dots, c_d \in \mathbb{F}$.

The identity $p(A) = \sum_{j=0}^{d} c_j S\Lambda^j S^{-1}$ holds by Definition 3.8.5 and the power identity.

The identity $\sum_{j=0}^{d} c_j S\Lambda^j S^{-1} = S\left(\sum_{j=0}^{d} c_j \Lambda^j\right) S^{-1}$ holds by Proposition 3.1.2(iii),(iv). Hence $p(A) = Sp(\Lambda)S^{-1}$ by Definition 3.8.5.
The identity $(\forall\, j \in \mathbb{Z}_{\geq 0})(\forall\, i, k \in J_n)\, (\Lambda^j)_{ik} = \delta_{ik}\lambda_i^j$ holds by induction on j and Definition 3.1.1(iv),(ix),(xii).
Hence $(\forall\, i, k \in J_n)\, p(\Lambda)_{ik} = \delta_{ik} p(\lambda_i)$ by Definition 3.8.5. Therefore $p(\Lambda) = \operatorname{diag}(p(\lambda_1), \dots, p(\lambda_n))$ by Definition 3.1.1(i),(iv).
The claimed identity holds by substituting this formula into $p(A) = Sp(\Lambda)S^{-1}$.
Structural Commentary: Power induction transfers conjugation to polynomial sums; diagonal powers reduce entrywise.
Key Insight: Conjugation cancels between powers, while diagonal entries evolve independently. □

2 Characteristic and Minimal Polynomials

2.1 Characteristic Polynomial Foundations

Motivation 2.1.1. *Two Spectral Polynomials*
Eigenvalues alone do not record algebraic repetition, and a matrix can satisfy many polynomial relations.
The characteristic polynomial records root multiplicities; the minimal polynomial selects the least-degree monic relation.

Definition 2.1.2. *Characteristic Polynomial*
Let $n \geq 1$ and $A \in \operatorname{Mat}(n, n, \mathbb{F})$. The *characteristic polynomial* of A is $p_A(z) := \det(zI_n - A) \in \mathbb{F}[z]$.

Theorem 2.1.3. *Basic Form of the Characteristic Polynomial*
Let $n \geq 1$ and $A = [a_{ij}] \in \operatorname{Mat}(n, n, \mathbb{F})$. The following hold:
(i) $p_A(z) = z^n - (\operatorname{tr} A)z^{n-1} + \cdots + (-1)^n \det A$*; in particular, p_A is monic of degree n.*
(ii) $\operatorname{spec}(A) = \{\lambda \in \mathbb{F} : p_A(\lambda) = 0\}$.

Proof.

(i) By Definitions 2.1.2 and 3.8.1, $p_A(z) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sign}(\sigma) \prod_{j=1}^{n} (z\delta_{\sigma(j),j} - a_{\sigma(j),j})$.
A nonidentity permutation moves at least two indices, so its summand has degree at most $n-2$.
The identity summand is $\prod_{j=1}^{n} (z - a_{jj}) = z^n - (\operatorname{tr} A)z^{n-1} + \cdots$ by Definition 3.8.1(i).
$p_A(0) = \det(-A) = (-1)^n \det A$ by Definition 2.1.2 and repeated column linearity from Theorem 3.8.2(ii). Thus (i) holds.

(ii) Fix $\lambda \in \mathbb{F}$. By Definition 2.1.2 and Theorem 3.8.2(vii), $p_A(\lambda) = 0 \iff \lambda I_n - A$ is singular.
By Corollary 5.1.8 and Definition 5.1.3(iii), this is equivalent to $\lambda \in \operatorname{spec}(A)$.

Structural Commentary: Permutation degrees determine the coefficients; shifted singularity identifies the root set.
Key Insight: The determinant vanishes exactly when the spectral shift has nonzero kernel. □

Definition 2.1.4. *Algebraic Multiplicity*
Let $n \geq 1$, $A \in \operatorname{Mat}(n, n, \mathbb{F})$, and $\lambda \in \mathbb{F}$. The *algebraic multiplicity* $m_A(\lambda)$ is the multiplicity of λ as a root of p_A.
If $\lambda \notin \operatorname{spec}(A)$, set $m_A(\lambda) := 0$. Unqualified multiplicity means algebraic multiplicity.

Figure 53 summarizes the passage from a matrix to its characteristic polynomial and roots.

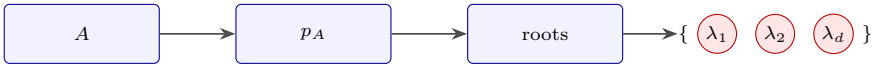

Figure 53: The chain $A \mapsto p_A \mapsto \{\lambda_1, \dots, \lambda_d\}$; m_A retains each root multiplicity.

Computation Method 2.1.5. *Characteristic Equation and Eigenspace Computation*
Given $n \geq 1$ and $A \in \operatorname{Mat}(n, n, \mathbb{F})$, compute its spectral data as follows:
(i) Form $p_A(z) = \det(zI_n - A)$ by Definition 2.1.2.
(ii) Compute $\{\lambda \in \mathbb{F} : p_A(\lambda) = 0\} = \operatorname{spec}(A)$ by Theorem 2.1.3(ii), recording each $m_A(\lambda)$.
(iii) $(\forall\, \lambda \in \operatorname{spec}(A))$, row-reduce $A - \lambda I_n$ and compute $E_\lambda(A) = \ker(A - \lambda I_n)$.
(iv) Choose a basis of each $E_\lambda(A)$ and return its dimension as the geometric multiplicity.
The output separates algebraic multiplicity from the nullspace dimension.

Proposition 2.1.6. *Upper-Triangular Eigenvalues*
Let $n \geq 1$, let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ be upper triangular, and set $J_n := \{1, \ldots, n\}$. Then:

(i) $p_A(z) = \prod_{j=1}^{n} (z - a_{jj})$;
(ii) $\mathrm{spec}(A) = \{a_{jj} : j \in J_n\}$;
(iii) $(\forall\, \lambda \in \mathbb{F})$, $m_A(\lambda) = \#\{j \in J_n : a_{jj} = \lambda\}$.

Proof.

(i) Matrix $zI_n - A$ is upper triangular with diagonal entries $z - a_{jj}$. By Theorem 3.8.2(v) and Definition 2.1.2, $p_A(z) = \prod_{j=1}^{n} (z - a_{jj})$.
(ii) By (i) and Theorem 2.1.3(ii), the roots of p_A and the elements of $\mathrm{spec}(A)$ are exactly the diagonal entries.
(iii) Fix $\lambda \in \mathbb{F}$. In the factorization from (i), the exponent of $z - \lambda$ is $\#\{j \in J_n : a_{jj} = \lambda\}$. This equals $m_A(\lambda)$ by Definition 2.1.4, including when $\lambda \notin \mathrm{spec}(A)$. Thus (iii) holds.

Structural Commentary: Triangularity reduces the determinant to diagonal factors; repeated factors encode algebraic multiplicity.
Key Insight: For an upper-triangular matrix, the entire characteristic root multiset is visible on the diagonal. □

Theorem 2.1.7. *Similarity Preserves Characteristic Data*
Let $n \geq 1$, let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $S \in \mathrm{Mat}(n, n, \mathbb{F})^{\times}$ satisfy $B = SAS^{-1}$. Then:

(i) $p_B = p_A$;
(ii) $(\forall\, \lambda \in \mathbb{F})$, $m_B(\lambda) = m_A(\lambda)$;
(iii) $(\forall\, \lambda \in \mathbb{F})$, *the restriction* $S|_{E_\lambda(A)} : E_\lambda(A) \to E_\lambda(B)$ *is an isomorphism, so* $\dim E_\lambda(B) = \dim E_\lambda(A)$.

Proof.

(i) By $zI_n - B = S(zI_n - A)S^{-1}$ and Theorem 3.8.2(vi),(viii), $p_B(z) = \det(S)\det(zI_n - A)\det(S^{-1}) = p_A(z)$.
(ii) Equal polynomials have the same roots with the same multiplicities. By (i) and Definition 2.1.4, $(\forall\, \lambda \in \mathbb{F})\, m_B(\lambda) = m_A(\lambda)$.
(iii) Fix $\lambda \in \mathbb{F}$. If $\mathbf{x} \in E_\lambda(A)$, then $(B - \lambda I_n)S\mathbf{x} = S(A - \lambda I_n)\mathbf{x} = \mathbf{0}$, so $S(E_\lambda(A)) \subseteq E_\lambda(B)$. The same calculation for $A = S^{-1}BS$ proves the reverse inclusion. Thus $S|_{E_\lambda(A)}$ is an isomorphism onto $E_\lambda(B)$, and the dimensions agree.

Structural Commentary: Conjugation preserves the shifted determinant and transports shifted kernels through one coordinate change.
Key Insight: Similarity preserves both algebraic multiplicity and geometric multiplicity. □

Proposition 2.1.8. *Block Upper-Triangular Characteristic Polynomial*
Let $r, s \geq 1$, let $B \in \mathrm{Mat}(r, r, \mathbb{F})$, $C \in \mathrm{Mat}(r, s, \mathbb{F})$, and $D \in \mathrm{Mat}(s, s, \mathbb{F})$, and set $A := \left(\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right)$. Then:

(i) $p_A = p_B p_D$;
(ii) $(\forall\, \lambda \in \mathbb{F})$, $m_A(\lambda) = m_B(\lambda) + m_D(\lambda)$.

Proof.

(i) The shifted matrix is $zI_{r+s} - A = \left(\begin{smallmatrix} zI_r - B & -C \\ 0 & zI_s - D \end{smallmatrix}\right)$. By Theorem 3.8.4 and Definition 2.1.2, $p_A = p_B p_D$.
(ii) Fix $\lambda \in \mathbb{F}$. By (i) and Definition 2.1.4, the multiplicity of λ in p_A is the sum of its multiplicities in p_B and p_D, including when λ is absent from one or both spectra.

Structural Commentary: The determinant splits across the diagonal blocks, so root orders add across the two characteristic factors.
Key Insight: The off-diagonal block affects eigenvectors but not the characteristic polynomial. □

Theorem 2.1.9. *Matrix Schur Triangularisation with Ordered Diagonal*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity and listed in a prescribed order, and let $A\mathbf{x} = \lambda_1 \mathbf{x}$ with $\|\mathbf{x}\|_2 = 1$. Then:

(i) $(\exists\, U_2 \in \mathrm{Mat}(n, n-1, \mathbb{C}))$ *such that* $[\mathbf{x}\ U_2]$ *is unitary and* $[\mathbf{x}\ U_2]^* A [\mathbf{x}\ U_2]$ *is upper triangular with diagonal* $\lambda_1, \ldots, \lambda_n$;
(ii) if $A \in \mathrm{Mat}(n, n, \mathbb{R})$, $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, *and* $\mathbf{x} \in \mathbb{R}^n$, *then the matrix in (i) may be chosen real orthogonal.*

Proof.

(i) We induct on n. The case $n = 1$ follows by taking $U = [\mathbf{x}]$.
Assume $n > 1$. Extend $\mathbf{x}$ to an orthonormal basis by Corollary 1.3.12, and write the resulting unitary matrix as $V = [\mathbf{x}\ V_2]$.
Since $A\mathbf{x} = \lambda_1 \mathbf{x}$, the first column of V^*AV is $\lambda_1 \mathbf{e}_1$, so write $V^*AV = \left(\begin{smallmatrix} \lambda_1 & \mathbf{b}^* \\ 0 & A' \end{smallmatrix}\right)$ with $\mathbf{b} \in \mathbb{C}^{n-1}$ and $A' \in \mathrm{Mat}(n-1, n-1, \mathbb{C})$.
By Theorem 2.1.7(i) and Proposition 2.1.8(i), $p_A(z) = (z - \lambda_1) p_{A'}(z)$. Thus $\lambda_2, \ldots, \lambda_n$ is the prescribed eigenvalue list of A'.
By the induction hypothesis, choose a unitary $W \in \mathrm{Mat}(n-1, n-1, \mathbb{C})$ such that $T' := W^* A' W$ is upper triangular with diagonal $\lambda_2, \ldots, \lambda_n$. Set $U := V(I_1 \oplus W)$. Then U is unitary, its first column is $\mathbf{x}$, and $U^*AU = \left(\begin{smallmatrix} \lambda_1 & \mathbf{b}^* W \\ 0 & T' \end{smallmatrix}\right)$ has the required form.

(ii) We repeat the induction within $\mathbb{R}^n$. The case $n = 1$ is immediate. For $n > 1$, extend the real unit vector $\mathbf{x}$ to a real orthonormal basis by Corollary 1.3.12. The resulting real orthogonal $V = [\mathbf{x}\ V_2]$ satisfies $V^{\mathsf{T}}AV = \left(\begin{smallmatrix} \lambda_1 & \mathbf{b}^{\mathsf{T}} \\ 0 & A' \end{smallmatrix}\right)$ with real blocks.
By the factorization in (i), the real matrix A' has the prescribed real eigenvalue list $\lambda_2, \ldots, \lambda_n$.
By the real induction hypothesis, choose a real orthogonal W with $W^{\mathsf{T}} A' W$ upper triangular in that order. Thus $U := V(I_1 \oplus W)$ is real orthogonal, begins with $\mathbf{x}$, and has the required triangularization.

Structural Commentary: Schur induction isolates the prescribed eigenline and passes the remaining ordered roots to the trailing block.
Key Insight: The first eigenvector fixes the first diagonal entry and leaves the remaining multiplicities for the next step. □

Figure 54 summarizes the induction by isolating the prescribed first eigenline.

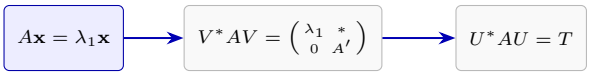

Figure 54: Schur induction: isolate an eigenline, then triangularise the compressed block.

Corollary 2.1.10. *Eigenvalues of the Inverse*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be invertible with eigenvalues $\lambda_1,\dots,\lambda_n$, counted with algebraic multiplicity. Then A^{-1} has eigenvalues $\lambda_1^{-1},\dots,\lambda_n^{-1}$, counted with algebraic multiplicity.

Proof. By Corollary 5.1.8, $(\forall\, j \in \{1,\dots,n\})\ \lambda_j \neq 0$. Choose $\mathbf{0} \neq \mathbf{y} \in E_{\lambda_1}(A)$ and set $\mathbf{x} := \frac{\mathbf{y}}{\|\mathbf{y}\|_2}$, so $\|\mathbf{x}\|_2 = 1$.
By Theorem 2.1.9(i), $A = UTU^*$ for a unitary U and an upper-triangular T with diagonal $\lambda_1,\dots,\lambda_n$.
Hence $A^{-1} = UT^{-1}U^*$, while T^{-1} is upper triangular with diagonal $\lambda_1^{-1},\dots,\lambda_n^{-1}$ by Theorem 3.8.9.
By Proposition 2.1.6(iii) and Theorem 2.1.7(ii), this reciprocal diagonal list is the eigenvalue multiset of A^{-1}.
Structural Commentary: Schur form turns inversion into inversion of a triangular matrix, whose diagonal entries invert individually.
Key Insight: Algebraic multiplicities are preserved when the nonzero spectral multiset is sent through $\lambda \mapsto \lambda^{-1}$. □

Theorem 2.1.11. *Geometric Multiplicity Is at Most Algebraic Multiplicity*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, and let $\lambda \in \mathrm{spec}(A)$. Then $\dim E_\lambda(A) \leq m_A(\lambda)$.

Proof. Set $k := \dim E_\lambda(A)$ and choose a basis $(\mathbf{x}_1,\dots,\mathbf{x}_k)$ of $E_\lambda(A)$.
If $k = n$, then $E_\lambda(A) = \mathbb{F}^n$, so $A = \lambda I_n$. By Proposition 2.1.6(i),(iii), $p_A(z) = (z-\lambda)^n$ and $m_A(\lambda) = n = k$. Suppose $k < n$.
Extend eigenspace basis to basis of $\mathbb{F}^n$ by Theorem 1.3.17, let S be invertible matrix whose columns are resulting basis vectors.
Since the first k columns are λ-eigenvectors, write $S^{-1}AS = \begin{pmatrix} \lambda I_k & C \\ 0 & D \end{pmatrix}$ with $C \in \mathrm{Mat}(k, n-k, \mathbb{F})$ and $D \in \mathrm{Mat}(n-k, n-k, \mathbb{F})$.
By Theorem 2.1.7(i) and Proposition 2.1.8(i), $p_A(z) = (z-\lambda)^k p_D(z)$. Hence $m_A(\lambda) \geq k = \dim E_\lambda(A)$ by Definition 2.1.4.
Structural Commentary: An eigenspace-first basis exposes $(z-\lambda)^k$, including the endpoint $k = n$.
Key Insight: Geometric multiplicity supplies a visible lower bound on the order of the corresponding characteristic root. □

Theorem 2.1.12. *Diagonalizability and Multiplicity*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then A is similar to a diagonal matrix if and only if $(\forall\, \lambda \in \mathrm{spec}(A))\ \dim E_\lambda(A) = m_A(\lambda)$.

Proof.
(⇒) Suppose A is similar to a diagonal matrix D. By Theorem 2.1.7(ii),(iii), $(\forall\, \lambda \in \mathbb{C})\ m_A(\lambda) = m_D(\lambda)$ and $\dim E_\lambda(A) = \dim E_\lambda(D)$. By Proposition 2.1.6(iii), $m_D(\lambda)$ counts the diagonal entries equal to λ, and the corresponding standard basis vectors form a basis of $E_\lambda(D)$. Thus the required equality holds.
(⇐) Assume the stated equality and let $T_A : \mathbb{C}^n \to \mathbb{C}^n$ be defined by $T_A(\mathbf{v}) = A\mathbf{v}$. By Theorem 5.3.5, T_A has an upper-triangular matrix B relative to a basis. By Corollary 6.2.4 and Theorem 2.1.7(ii), A and B have the same algebraic multiplicities.
List the distinct eigenvalues as $\mu_1,\dots,\mu_d$. By Proposition 2.1.6(ii),(iii), the n diagonal positions of B are partitioned by these eigenvalues and $\sum_{i=1}^{d} m_A(\mu_i) = n$. Thus the hypothesis implies $\sum_{i=1}^{d} \dim E_{\mu_i}(T_A) = \sum_{i=1}^{d} \dim E_{\mu_i}(A) = n$.
By Theorem 5.4.3(v),(i), T_A is diagonalizable. Its matrices in the standard basis and in a diagonalizing basis are therefore similar by Corollary 6.2.4, so A is similar to a diagonal matrix.
Structural Commentary: Triangularization accounts for the complete algebraic-multiplicity total, while the hypothesis converts it into a full eigenspace dimension count.
Key Insight: A complex matrix is diagonalizable exactly when no algebraic multiplicity is lost as an eigenspace dimension. □

Proposition 2.1.13. *Permutation Similarity Reorders Diagonal Entries*
Let $n \geq 1$, let $\lambda_1,\dots,\lambda_n \in \mathbb{C}$, and let $\sigma \in \mathfrak{S}_n$. Set $\Lambda := \mathrm{diag}(\lambda_1,\dots,\lambda_n)$ and $P := [\mathbf{e}_{\sigma(1)}\ \cdots\ \mathbf{e}_{\sigma(n)}]$. Then $P^*\Lambda P = \mathrm{diag}(\lambda_{\sigma(1)},\dots,\lambda_{\sigma(n)})$.

Proof. For $i,j \in \{1,\dots,n\}$, the (i,j) entry of $P^*\Lambda P$ is $\mathbf{e}_{\sigma(i)}^*\Lambda\mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)}\mathbf{e}_{\sigma(i)}^*\mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)}\delta_{ij}$.
Hence all off-diagonal entries vanish and the jth diagonal entry is $\lambda_{\sigma(j)}$, which proves the matrix identity.
Structural Commentary: The entry calculation tracks both the row and column reordering induced by the same permutation.
Key Insight: Permutation similarity reorders a diagonal list without changing its entries. □

Corollary 2.1.14. *Normal Matrices with the Same Spectrum Are Unitarily Similar*
Let $n \geq 1$. If $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$ are normal and have the same eigenvalue multiset, then A and B are unitarily similar.

Proof. By Theorem 6.1.11(i),(iii), write $A = U\Lambda U^*$ and $B = V\Gamma V^*$ with U, V unitary and Λ, Γ diagonal.
Since the eigenvalue multisets agree, choose $\sigma \in \mathfrak{S}_n$ that matches the diagonal of Λ to that of Γ, and set $P := [\mathbf{e}_{\sigma(1)}\ \cdots\ \mathbf{e}_{\sigma(n)}]$.
By Proposition 2.1.13, $\Gamma = P^*\Lambda P$. Set $Q := UPV^*$.
Then Q is unitary by Proposition 2.6.12, and $Q^*AQ = VP^*U^*(U\Lambda U^*)UPV^* = V\Gamma V^* = B$. Thus A and B are unitarily similar.
Structural Commentary: Spectral diagonalizations reduce the comparison to a unitary permutation of equal eigenvalue multisets.
Key Insight: For normal matrices, the eigenvalue multiset completely determines the unitary-similarity class. □

Definition 2.1.15. *Characteristic Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$, let $n := \dim(\mathcal{V})$, and let $T \in \mathcal{L}(\mathcal{V})$.
For a basis β of $\mathcal{V}$, set $q_{T,\beta}(z) := p_{[T]_{\beta\leftarrow\beta}}(z) = \det(zI_n - [T]_{\beta\leftarrow\beta})$.
This value is independent of β by Theorems 5.1.1 and 2.1.7(i). It is the *characteristic polynomial* of T and is denoted by q_T.

Proposition 2.1.16. *Degree and Roots of q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be the characteristic polynomial of T.
(i) $\deg q_T = \dim(\mathcal{V})$.
(ii) The roots of q_T are precisely the eigenvalues of T.

Proof. Choose a basis β of $\mathcal{V}$ and set $A = [T]_{\beta\leftarrow\beta}$.
(i) By Definition 2.1.15, $q_T = p_A$. Hence $\deg q_T = \dim(\mathcal{V})$ by Theorem 2.1.3(i).
(ii) Fix $\lambda \in \mathbb{F}$. By Proposition 3.2.5, $(\forall\, \mathbf{v} \in \mathcal{V})\ [T\mathbf{v}]_\beta = A[\mathbf{v}]_\beta$.
By Proposition 3.2.2(ii),(iii), $\lambda \in \mathrm{spec}(T) \iff \lambda \in \mathrm{spec}(A)$. By Theorem 2.1.3(ii), the latter set is the root set of $p_A = q_T$.
Structural Commentary: Coordinates transfer polynomial degree and the nonzero eigenvector equation between T and its matrix.
Key Insight: The operator characteristic polynomial has the expected dimension and detects exactly the operator eigenvalues. □

2.2 Minimal Polynomial Foundations

Theorem 2.2.1. *Existence of the Minimal Polynomial*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then:

(i) T has a monic annihilator of positive least degree;
(ii) the annihilator in (i) is unique.

Proof.

(i) By Proposition 1.1.7, choose a nonconstant annihilator h and replace it by its monic normalization.
Let $\mathcal{M}_T$ be the monic annihilators of T and set $\mathcal{D}_T := \{\deg p : p \in \mathcal{M}_T\}$; then $\varnothing \neq \mathcal{D}_T \subseteq \mathbb{N}$.
By well-ordering, set $d := \min \mathcal{D}_T$ and choose a monic annihilator p with $\deg p = d$.
A nonzero constant c cannot annihilate T because $c(T) = cI_{\mathcal{V}} \neq 0$ on $\mathcal{V} \neq \{\mathbf{0}\}$. Thus $d \geq 1$.

(ii) Let p and q be monic annihilators of degree d. If $p \neq q$, then $(p-q)(T) = p(T) - q(T) = 0$ by Definition 1.1.2.
Their leading terms cancel, so $\deg(p-q) < d$. Normalizing the nonzero polynomial $p - q$ therefore produces a monic annihilator of degree less than d, contrary to the definition of d. Hence $p = q$.

Structural Commentary: Well-ordering selects a least normalized annihilator; leading-term cancellation forces uniqueness.
Key Insight: Two minimal monic annihilators would differ by a smaller annihilator. □

Definition 2.2.2. *Minimal Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. The *minimal polynomial* μ_T is the unique monic polynomial of least degree satisfying $\mu_T(T) = 0$; it exists by Theorem 2.2.1.

Definition 2.2.3. *Minimal Polynomial of a Matrix*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $T_A \in \mathcal{L}(\mathbb{F}^n)$ be induced by A as in Definition 3.1.7. The *minimal polynomial* of A is $\mu_A := \mu_{T_A}$.
For $j \in \mathbb{Z}_{\geq 0}$, set $P_j := ((\forall\, \mathbf{v} \in \mathbb{F}^n)\, T_A^j \mathbf{v} = A^j \mathbf{v})$. Then P_0 holds by Definitions 1.1.1 and 3.1.7.
If P_j holds, then $(\forall\, \mathbf{v} \in \mathbb{F}^n)\, T_A^{j+1}\mathbf{v} = T_A(A^j\mathbf{v}) = A^{j+1}\mathbf{v}$ by Definition 3.1.7, so P_{j+1} holds. Hence $(\forall\, j \in \mathbb{Z}_{\geq 0})\, P_j$ by Theorem 2.6.2.
If $p(z) = \sum_{j=0}^{d} c_j z^j \in \mathbb{F}[z]$, then $(\forall\, \mathbf{v} \in \mathbb{F}^n)\, p(T_A)\mathbf{v} = \sum_{j=0}^{d} c_j A^j \mathbf{v} = p(A)\mathbf{v}$ by Definitions 1.1.2 and 3.8.5.
Thus μ_A is equivalently the unique monic polynomial of least positive degree satisfying $\mu_A(A) = 0$.

Theorem 2.2.4. *Annihilators Are Multiples of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $f \in \mathbb{F}[z]$ be nonzero. Then $f(T) = 0$ if and only if μ_T divides f.

Proof.

($\Rightarrow$) Suppose $f(T) = 0$ and set $d := \deg \mu_T$. Starting from $r_0 := f$ and $s_0 := 0$, suppose $\deg r_k \geq d$, set $e_k := \deg r_k - d$, and let $a_k \in \mathbb{F}^\times$ be the leading coefficient of r_k. Set $r_{k+1} := r_k - a_k z^{e_k}\mu_T$ and $s_{k+1} := s_k + a_k z^{e_k}$.
The identity $f = s_k\mu_T + r_k$ holds for $k = 0$ and is preserved by the recurrence.
Monicity of μ_T implies $r_{k+1} = 0$ or $\deg r_{k+1} < \deg r_k$. Degree descent terminates with $f = s\mu_T + r$, where $r = 0$ or $\deg r < d$.
By Definition 1.1.2 and Proposition 1.1.6(i), $r(T) = f(T) - s(T)\mu_T(T) = 0$. If $r \neq 0$, its monic normalization is an annihilator of degree less than d, which is impossible by Definition 2.2.2. Thus $r = 0$ and μ_T divides f.

($\Leftarrow$) If $f = \mu_T s$, then $f(T) = \mu_T(T)s(T) = 0$ by Proposition 1.1.6(i).

Structural Commentary: Leading-term cancellation constructs a remainder; minimality eliminates every nonzero remainder.
Key Insight: A nonzero remainder would normalize below $\deg \mu_T$. □

Theorem 2.2.5. *Similarity Preserves the Minimal Polynomial*
Let $n \geq 1$ and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be similar. Then:

(i) $p_A = p_B$;
(ii) $\mu_A = \mu_B$.

Proof. Choose $S \in \mathrm{Mat}(n, n, \mathbb{F})^\times$ satisfying $B = SAS^{-1}$ by Definition 6.2.1.

(i) $p_A = p_B$ by Theorem 2.1.7(i).

(ii) Set $P_j := (B^j = SA^jS^{-1})$ for $j \in \mathbb{Z}_{\geq 0}$. The identity P_0 holds because $B^0 = I_n = SI_nS^{-1} = SA^0S^{-1}$. If P_j holds, then $B^{j+1} = B^jB = SA^jS^{-1}SAS^{-1} = SA^{j+1}S^{-1}$, so P_{j+1} holds. Hence $(\forall\, j \in \mathbb{Z}_{\geq 0})\, P_j$ by Theorem 2.6.2.
By Definition 3.8.5, $(\forall\, f \in \mathbb{F}[z])\, f(B) = Sf(A)S^{-1}$. Therefore $f(B) = 0 \iff f(A) = 0$, so A and B have the same monic least-degree annihilator by Definition 2.2.3.

Structural Commentary: Power induction transports characteristic and minimal polynomial data across one conjugation.
Key Insight: Conjugation preserves every polynomial equation. □

3 Cayley-Hamilton and Annihilators

3.1 Cayley-Hamilton and Minimal Polynomials

Theorem 3.1.1. *Cayley–Hamilton Theorem*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have characteristic polynomial $p_A(z) = \det(zI_n - A)$. Then $p_A(A) = 0$.

Proof.

(i) *Triangular Reduction*: By Theorem 5.4.14(i) for $\{A\}$, choose a unitary U such that $T := U^*AU = [t_{ij}]$ is upper triangular.
Then $A = UTU^*$ by unitarity of U. Set $\lambda_j := t_{jj}$ for $j \in \{1, \ldots, n\}$.
Then $p_A = p_T = \prod_{j=1}^{n} (z - \lambda_j)$ by Theorem 2.1.7(i) and Proposition 2.1.6(i).

(ii) *Polynomial Transfer*: For $k \in \mathbb{Z}_{\geq 0}$, set $P_k := (A^k = UT^kU^*)$. The identity P_0 holds by unitarity of U.
If P_k holds, then $A^{k+1} = A^kA = UT^kU^*UTU^* = UT^{k+1}U^*$, so P_{k+1} holds. Hence $(\forall\, k \in \mathbb{Z}_{\geq 0})\, P_k$ by Theorem 2.6.2.
For $j \in \{1, \ldots, n\}$, set $Q_j := \prod_{i=1}^{j} (T - \lambda_i I_n)$. By Definition 3.8.5, part (i), and the identities P_k, $p_A(A) = Up_A(T)U^* = UQ_nU^*$.

(iii) *Column Induction*: For $j \in \{1, \dots, n\}$, set $R_j := ((\forall\, \ell \in \{1, \dots, j\})\, Q_j \mathbf{e}_\ell = \mathbf{0})$.
The predicate R_1 holds because the first column of $T - \lambda_1 I_n$ is zero. Suppose $j \in \{1, \dots, n-1\}$ and R_j holds.
By upper triangularity and the zero $(j+1, j+1)$ entry, $(\forall\, \ell \in \{1, \dots, j+1\})\, (T - \lambda_{j+1} I_n)\mathbf{e}_\ell \in \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$.
By R_j and linearity of Q_j, $(\forall\, \ell \in \{1, \dots, j+1\})\, Q_{j+1}\mathbf{e}_\ell = Q_j(T - \lambda_{j+1} I_n)\mathbf{e}_\ell = \mathbf{0}$, so R_{j+1} holds.
Hence $(\forall\, j \in \{1, \dots, n\})\, R_j$ by Theorem 2.6.2.
(iv) *Conclusion*: By R_n, every column of Q_n is zero, so $Q_n = 0$. Therefore $p_A(A) = U Q_n U^* = 0$ by part (ii).

Structural Commentary: Unitary triangularization transfers Cayley–Hamilton to a column-killing induction.
Key Insight: The jth factor extends the zero leading-column block from $j-1$ to j. □

Figure 55 depicts the invariant R_j: each new factor extends the zero leading-column block by one column.

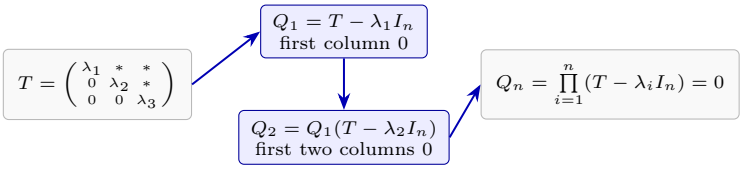

Figure 55: For upper-triangular T, the factors $T - \lambda_i I_n$ make the first j columns of Q_j vanish successively.

Theorem 3.1.2. *Cayley–Hamilton Theorem for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be the characteristic polynomial of T. Then $q_T(T) = 0$.

Proof. Choose a basis β of $\mathcal{V}$ and set $A = [T]_{\beta \leftarrow \beta}$. The identity $q_T = p_A$ holds by Definition 2.1.15.
The identity $[q_T(T)]_{\beta \leftarrow \beta} = q_T(A)$ holds by Propositions 3.2.4 and 3.4.4. Hence $[q_T(T)]_{\beta \leftarrow \beta} = p_A(A) = 0$ by Theorem 3.1.1.
Therefore $q_T(T) = 0$ by Theorem 3.4.5.
Structural Commentary: Matrix representation transfers operator polynomial evaluation to the matrix Cayley–Hamilton identity.
Key Insight: The coordinate matrix of $q_T(T)$ is exactly $q_T(A)$. □

Corollary 3.1.3. *Reduction Modulo the Characteristic Polynomial*
Let $n \geq 1$, let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and let $f, q, r \in \mathbb{C}[z]$. If $f = p_A q + r$ with $r = 0$ or $\deg r < n$, then $f(A) = r(A)$.

Proof. By the polynomial identity in the hypothesis, $f(A) = p_A(A)q(A) + r(A)$. Since $p_A(A) = 0$ by Theorem 3.1.1, $f(A) = r(A)$.
Structural Commentary: Evaluation transfers division by p_A to equality of the corresponding matrix polynomials.
Key Insight: The characteristic-polynomial term vanishes at A. □

Corollary 3.1.4. *Powers Span at Most n Dimensions*
Let $n \geq 1$, let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and set $\mathcal{W}_A := \operatorname{span}(I_n, A, \dots, A^{n-1})$. Then:
(i) $\operatorname{span}(I_n, A, A^2, \dots) = \mathcal{W}_A$;
(ii) $\dim(\mathcal{W}_A) \leq n$.

Proof.
(i) Write $p_A(z) = z^n + \sum_{j=0}^{n-1} c_j z^j$ by Theorem 2.1.3(i). Then $A^n = -\sum_{j=0}^{n-1} c_j A^j \in \mathcal{W}_A$ by Theorem 3.1.1.
If $0 \leq j < n-1$, then $AA^j = A^{j+1} \in \mathcal{W}_A$; the remaining generator satisfies $AA^{n-1} = A^n \in \mathcal{W}_A$.
Hence $A\mathcal{W}_A \subseteq \mathcal{W}_A$ by linearity. Set $P_k := (A^k \in \mathcal{W}_A)$ for $k \in \mathbb{Z}_{\geq 0}$.
The predicate P_0 holds by $A^0 = I_n \in \mathcal{W}_A$, and $P_k \Rightarrow P_{k+1}$ by $A\mathcal{W}_A \subseteq \mathcal{W}_A$. Thus $(\forall\, k \in \mathbb{Z}_{\geq 0})\, P_k$ by Theorem 2.6.2.
Therefore $\operatorname{span}(I_n, A, A^2, \dots) \subseteq \mathcal{W}_A$; the reverse inclusion holds by the definition of $\mathcal{W}_A$.
(ii) The list $(I_n, A, \dots, A^{n-1})$ spans $\mathcal{W}_A$ and has length n. It contains a basis of $\mathcal{W}_A$ by Theorem 1.3.12, so $\dim(\mathcal{W}_A) \leq n$.

Structural Commentary: The characteristic recurrence makes $\mathcal{W}_A$ invariant; a basis sublist supplies the dimension bound.
Key Insight: $A\mathcal{W}_A \subseteq \mathcal{W}_A$ propagates containment to every matrix power. □

Corollary 3.1.5. *Inverse from the Characteristic Polynomial*
Let $n \geq 1$, let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and write $p_A(z) = \sum_{j=0}^{n} c_j z^j$ with $c_n = 1$. If A is invertible, then $A^{-1} = -c_0^{-1} \sum_{j=1}^{n} c_j A^{j-1}$.

Proof. Since A is invertible, $0 \notin \operatorname{spec}(A)$ by Corollary 5.1.8. Hence $c_0 = p_A(0) \neq 0$ by Theorem 2.1.3(ii).
By Theorem 3.1.1, $c_0 I_n = -\sum_{j=1}^{n} c_j A^j$. By left multiplication by $c_0^{-1} A^{-1}$, $A^{-1} = -c_0^{-1} \sum_{j=1}^{n} c_j A^{j-1}$.
Structural Commentary: Cayley–Hamilton separates the constant term, while invertibility permits multiplication by $c_0^{-1} A^{-1}$.
Key Insight: The nonzero constant coefficient permits solving the characteristic identity for A^{-1}. □

Theorem 3.1.6. *Basic Properties of the Minimal Polynomial*
Let $n, d \geq 1$, let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and let $\lambda_1, \dots, \lambda_d$ be the distinct eigenvalues of A.
(i) $\mu_A \mid p_A$;
(ii) $\deg \mu_A \leq n$;
(iii) $(\exists\, q_1, \dots, q_d \in \mathbb{Z}_{>0})[(\forall\, i \in \{1, \dots, d\})\, q_i \leq m_A(\lambda_i) \ \wedge \ \mu_A(z) = \prod_{i=1}^{d} (z - \lambda_i)^{q_i}]$.

Proof.
(i) The polynomial p_A is nonzero by Theorem 2.1.3(i), and $p_A(A) = 0$ by Theorem 3.1.1.
Hence $\mu_A \mid p_A$ by Theorem 2.2.4 and Definition 2.2.3.
(ii) By (i) and $\deg p_A = n$ from Theorem 2.1.3(i), $\deg \mu_A \leq n$.

(iii) By Theorem 5.4.14(i) for $\{A\}$, choose a unitary U such that $R := U^*AU$ is upper triangular.

By Theorem 2.1.7(i),(ii) and Proposition 2.1.6(i),(iii), $p_A(z) = \prod_{i=1}^{d} (z-\lambda_i)^{m_A(\lambda_i)}$.

By factorization over $\mathbb{C}$ and monicity of μ_A, write μ_A as a product of monic linear factors.

By (i), each factor occurs in p_A with no greater exponent, so $\mu_A(z) = \prod_{i=1}^{d} (z-\lambda_i)^{q_i}$ for integers $0 \le q_i \le m_A(\lambda_i)$.

Fix $i \in \{1,\dots,d\}$ and choose $\mathbf{v} \neq \mathbf{0}$ satisfying $A\mathbf{v} = \lambda_i\mathbf{v}$ by the eigenvalue hypothesis.
Then $\mathbf{0} = \mu_A(A)\mathbf{v} = \mu_A(\lambda_i)\mathbf{v}$ by Proposition 1.1.8, so $\mu_A(\lambda_i) = 0$ and $q_i \ge 1$.
Since i was arbitrary, $(\forall\, i \in \{1,\dots,d\})\, q_i \ge 1$, which proves (iii).

Structural Commentary: Cayley–Hamilton supplies divisibility; Schur form and eigenvectors determine the exponents.
Key Insight: Every eigenvalue must occur in μ_A, but never above its multiplicity in p_A. □

Figure 56 depicts μ_A as a common divisor of p_A and every polynomial f satisfying $f(A) = 0$.

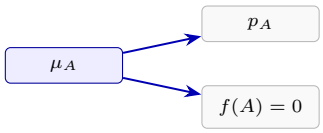

Figure 56: μ_A is the least monic annihilator; arrows indicate that μ_A divides p_A and every annihilator f.

3.2 Diagonalizability and Minimal Polynomial Tests

Theorem 3.2.1. *Diagonalizability and the Minimal Polynomial*
Let $n \ge 1$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have the distinct eigenvalues $\lambda_1,\dots,\lambda_d$, where $d \ge 1$. Interpret empty products below as 1 for scalars and as the identity operator for operators. Set $p(z) := \prod_{i=1}^{d}(z-\lambda_i)$. The following are equivalent:

(i) A is similar to a diagonal matrix;
(ii) $\mu_A = p$;
(iii) μ_A has no repeated root, or is square-free.

Proof.

(i) $\Rightarrow$ (ii) Suppose $A = S\Lambda S^{-1}$ for some $S \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ and diagonal $\Lambda = \mathrm{diag}(\nu_1,\dots,\nu_n)$.
For each k, the nonzero vector $S\mathbf{e}_k$ is an eigenvector of A with eigenvalue ν_k, so $p(\nu_k) = 0$.
Hence $p(A) = S\,\mathrm{diag}(p(\nu_1),\dots,p(\nu_n))S^{-1} = 0$ by Theorem 1.1.1. Divisibility $\mu_A \mid p$ holds by Theorem 2.2.4 and Definition 2.2.3.
By Theorem 3.1.6(iii), $\deg(\mu_A) \ge d$. Since $\mu_A \mid p$, $\deg(\mu_A) \le d$. Thus the monic polynomials μ_A and p are equal.

(ii) $\Rightarrow$ (iii) The roots $\lambda_1,\dots,\lambda_d$ are distinct, so $p = \prod_{i=1}^{d}(z-\lambda_i)$ has no repeated root.

(iii) $\Rightarrow$ (i) By Theorem 3.1.6(iii), there are integers $q_1,\dots,q_d \ge 1$ such that $\mu_A(z) = \prod_{i=1}^{d}(z-\lambda_i)^{q_i}$.
The absence of repeated roots forces $q_i = 1$ for every i, so $\mu_A = p$.
Let $T_A \in \mathcal{L}(\mathbb{C}^n)$ be the operator induced by A from Definition 3.1.7. Then $\mu_{T_A} = p$ and $p(T_A) = 0$ by Definition 2.2.3.
First, suppose $g \in \mathbb{C}[z]$ vanishes at every λ_i. Write $g(z) = \sum_{r=0}^{m} c_r z^r$.
If $g(\lambda) = 0$, then $g(z) = g(z) - g(\lambda)$; the identity $z^r - \lambda^r = (z-\lambda)\sum_{s=0}^{r-1} z^{r-1-s}\lambda^s$ therefore writes $g = (z-\lambda)h$ for some $h \in \mathbb{C}[z]$.
Write $g = (z-\lambda_1)g_1$. For $k \ge 2$, $0 = g(\lambda_k) = (\lambda_k - \lambda_1)g_1(\lambda_k)$ and $\lambda_k - \lambda_1 \neq 0$, so $g_1(\lambda_k) = 0$.
After d such factorizations, $g = ph$ for some $h \in \mathbb{C}[z]$, so $p \mid g$. For $i \in \{1,\dots,d\}$, set $\ell_i(z) := \prod_{j\neq i} \frac{z-\lambda_j}{\lambda_i-\lambda_j}$ and $\rho_i := \ell_i(T_A)$. Since $\ell_i(\lambda_k) = \delta_{ik}$, each polynomial $\ell_i\ell_j$ with $i \neq j$, $\sum_{i=1}^{d}\ell_i - 1$, and $z - \sum_{i=1}^{d}\lambda_i\ell_i$ vanishes at every λ_k and is therefore divisible by p.
Since $p(T_A) = 0$, the identities $\rho_i\rho_j = 0$ for $i \neq j$, $\sum_{i=1}^{d}\rho_i = I_{\mathbb{C}^n}$, and $T_A = \sum_{i=1}^{d}\lambda_i\rho_i$ hold by Definition 1.1.2 and Proposition 1.1.6.
Hence $\rho_i = \rho_i \sum_{j=1}^{d}\rho_j = \rho_i^2$ for every i. If $\rho_i = 0$, multiplication by the nonzero scalar $\prod_{j\neq i}(\lambda_i - \lambda_j)$ shows that the monic polynomial $\prod_{j\neq i}(z-\lambda_j)$ annihilates T_A, contradicting the degree-d minimality of $\mu_{T_A} = p$. Thus every ρ_i is nonzero.
By Theorem 5.4.6(i), T_A is diagonalizable. By Definition 5.4.1, choose a basis β of $\mathbb{C}^n$ for which $D := [T_A]_{\beta\leftarrow\beta}$ is diagonal.
Since A and D represent T_A in two bases, they are similar by Corollary 6.2.4.

Structural Commentary: Diagonal form plus degree bounds identify $\mu_A = p$; Lagrange projections transfer $\mu_A = p$ to diagonalization.
Key Insight: Multiplicity one in μ_A is exactly what makes the polynomial spectral projections complementary. □

Corollary 3.2.2. *Square-Free Annihilator Criterion*
Let $n \ge 1$ and $A \in \mathrm{Mat}(n,n,\mathbb{C})$. If $0 \neq f \in \mathbb{C}[z]$ is square-free and $f(A) = 0$, then A is similar to a diagonal matrix.

Proof. The divisibility $\mu_A \mid f$ holds by Theorem 2.2.4 and Definition 2.2.3. If μ_A had a repeated root λ, then $(z-\lambda)^2 \mid \mu_A$ and hence $(z-\lambda)^2 \mid f$, contrary to the hypothesis. Thus μ_A is square-free. By Theorem 3.2.1(iii),(i), A is similar to a diagonal matrix.
Structural Commentary: Divisibility transfers any repeated linear factor of the minimal polynomial to every annihilator.
Key Insight: A square-free annihilator excludes repeated roots from μ_A. □

Computation Method 3.2.3. *Computing μ_T from a Matrix*
Given an n-dimensional vector space $\mathcal{V}$ over $\mathbb{F}$, where $n \geq 1$, an operator $T \in \mathcal{L}(\mathcal{V})$, and a basis β, set $A := [T]_{\beta \leftarrow \beta} \in \mathrm{Mat}(n, n, \mathbb{F})$.
The output is μ_T. For every $p \in \mathbb{F}[z]$, the identity $[p(T)]_{\beta \leftarrow \beta} = p(A)$ holds by Proposition 3.2.4 and Proposition 3.4.4.
Hence $p(T) = 0$ if and only if $p(A) = 0$ by Theorem 3.4.5. Therefore $\mu_T = \mu_A$ by Definition 2.2.2 and Definition 2.2.3.

(i) For $1 \leq m \leq n^2$, set $p_m(z) := a_0 + a_1 z + \cdots + a_{m-1} z^{m-1} + z^m$, where $(a_0, \ldots, a_{m-1}) \in \mathbb{F}^m$ is unknown.
(ii) Compare matrix entries in $a_0 I_n + a_1 A + \cdots + a_{m-1} A^{m-1} = -A^m$ and solve the resulting linear system.
(iii) If the system is inconsistent, increase m. At the first solvable degree, choose a solution and return the corresponding p_m.

By Proposition 1.1.7 and Theorem 2.2.1, the search terminates by $m = n^2$. The returned polynomial is the least-degree monic annihilator of A and hence equals μ_T by the preceding equivalence and Definition 2.2.2.

Figure 57 summarizes the linear-system search for the least-degree monic annihilator.

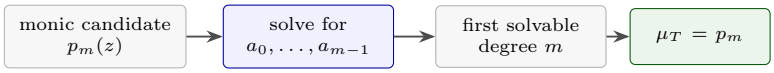

Figure 57: Solve $a_0 I_n + \cdots + a_{m-1} A^{m-1} = -A^m$; the first solvable monic candidate is μ_T.

Computation Method 3.2.4. *Testing a Minimal Polynomial from p_A*
Given $n, d \geq 1$, $A \in \mathrm{Mat}(n, n, \mathbb{C})$, known factorization $p_A(z) = \prod_{i=1}^{d} (z - \lambda_i)^{a_i}$, with distinct $\lambda_1, \ldots, \lambda_d$ and $a_1, \ldots, a_d \geq 1$, return μ_A.

i. Form the finite candidate family $m_{\mathbf{q}}(z) = \prod_{i=1}^{d} (z - \lambda_i)^{q_i}$, where $1 \leq q_i \leq a_i$.
ii. Test exponent vectors $\mathbf{q} = (q_i)_{i=1}^{d}$ in increasing total degree $\sum_{i=1}^{d} q_i$, using any order among vectors of equal total degree.
iii. For each $\mathbf{q}$, compute $m_{\mathbf{q}}(A) = \prod_{i=1}^{d} (A - \lambda_i I_n)^{q_i}$.
iv. Stop at the first candidate satisfying $m_{\mathbf{q}}(A) = 0$ and return it.

The search terminates because p_A is a candidate and $p_A(A) = 0$ by Theorem 3.1.1.
Since μ_A is a candidate by Theorem 3.1.6(iii), degree ordering and Definition 2.2.3 imply $\deg(m) = \deg(\mu_A)$ for returned candidate m.
The annihilator $m - \mu_A$ then has lower degree unless it is zero; hence $m = \mu_A$.

3.3 Annihilators and Spectral Control

Corollary 3.3.1. *q_T Is a Multiple of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then the characteristic polynomial q_T is a polynomial multiple of the minimal polynomial μ_T.

Proof. The polynomial q_T is nonzero because $\deg q_T = \dim(\mathcal{V}) \geq 1$ by Proposition 2.1.16(i).
The identity $q_T(T) = 0$ holds by Theorem 3.1.2. Hence $\mu_T \mid q_T$ by Theorem 2.2.4.
Structural Commentary: Cayley–Hamilton turns the characteristic polynomial into an annihilator.
Key Insight: The minimal polynomial divides every annihilator, hence divides q_T. □

Theorem 3.3.2. *Roots of μ_T Are Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $(\forall\, \lambda \in \mathbb{F})\, [\mu_T(\lambda) = 0 \iff \lambda \in \mathrm{spec}(T)]$.

Proof.

(⇒) Fix $\lambda \in \mathbb{F}$ with $\mu_T(\lambda) = 0$. Termwise, $\mu_T(z) - \mu_T(\lambda) \in (z - \lambda)\mathbb{F}[z]$ by $z^r - \lambda^r = (z - \lambda) \sum_{s=0}^{r-1} z^{r-1-s} \lambda^s$.
Thus $\mu_T(z) = (z - \lambda) g(z)$ for a monic g with $\deg g = \deg \mu_T - 1$.
Hence $g(T) \neq 0$ by Definition 2.2.2; choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{u} := g(T)\mathbf{v} \neq \mathbf{0}$.
Then $(T - \lambda I_{\mathcal{V}})\mathbf{u} = \mu_T(T)\mathbf{v} = \mathbf{0}$ by Proposition 1.1.6 and Definition 2.2.2. Thus $\lambda \in \mathrm{spec}(T)$ by Definition 5.1.3.
(⇐) Fix $\lambda \in \mathbb{F}$ and suppose $\lambda \in \mathrm{spec}(T)$. By Definition 5.1.3, choose $\mathbf{v} \neq \mathbf{0}$ satisfying $T\mathbf{v} = \lambda \mathbf{v}$.
Then $\mathbf{0} = \mu_T(T)\mathbf{v} = \mu_T(\lambda)\mathbf{v}$ by Definition 2.2.2 and Proposition 1.1.8. Since $\mathbf{v} \neq \mathbf{0}$, $\mu_T(\lambda) = 0$.

Structural Commentary: Root factorization supplies the forward eigenvector; evaluation proves the converse.
Key Insight: Minimality keeps the quotient operator nonzero. □

Theorem 3.3.3. *Minimal Polynomial of a Spectrally Disjoint Direct Sum*
Let $d, n_1, \ldots, n_d \geq 1$, let $A_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})$ for $i \in \{1, \ldots, d\}$, and set $A := A_1 \oplus \cdots \oplus A_d$.
If the spectra of $A_1, \ldots, A_d$ are pairwise disjoint, then $\mu_A = \mu_{A_1} \mu_{A_2} \cdots \mu_{A_d}$.

Proof. Set $f := \prod_{i=1}^{d} \mu_{A_i}$. The polynomial f is monic and hence nonzero.

(i) *Block Annihilation*: Since $\mu_{A_i} \mid f$, $(\forall\, i \in \{1, \ldots, d\})\, f(A_i) = 0$ by Theorem 2.2.4 and Definition 2.2.3.
Hence $f(A) = f(A_1) \oplus \cdots \oplus f(A_d) = 0$ by Proposition 3.8.6(iv), and $\mu_A \mid f$ by the same criterion.
(ii) *Block Restriction*: The identity $\mu_A(A) = \mu_A(A_1) \oplus \cdots \oplus \mu_A(A_d) = 0$ holds by Definition 2.2.3 and Proposition 3.8.6(iv).
Thus $(\forall\, i \in \{1, \ldots, d\})\, \mu_{A_i} \mid \mu_A$ by Theorem 2.2.4 and Definition 2.2.3.
(iii) *Coprime Factors*: The roots of μ_{A_i} are $\mathrm{spec}(A_i)$ by Theorem 3.3.2, so the polynomials $\mu_{A_1}, \ldots, \mu_{A_d}$ are pairwise coprime.
Hence $f \mid \mu_A$ by part (ii) and factorization over $\mathbb{C}$.
(iv) *Conclusion*: By parts (i) and (iii), $\mu_A \mid f$ and $f \mid \mu_A$. Monicity forces $\mu_A = f$.

Structural Commentary: Block evaluation establishes $\mu_A \mid f$; spectral disjointness establishes $f \mid \mu_A$.
Key Insight: Every spectral block contributes a distinct collection of factors to μ_A. □

Corollary 3.3.4. *Minimal and Characteristic Polynomials Have the Same Roots*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\{\lambda \in \mathbb{C} : \mu_T(\lambda) = 0\} = \{\lambda \in \mathbb{C} : q_T(\lambda) = 0\} = \mathrm{spec}(T)$.

Proof. By Theorem 3.3.2, $\{\lambda \in \mathbb{C} : \mu_T(\lambda) = 0\} = \operatorname{spec}(T)$. By Proposition 2.1.16, $\{\lambda \in \mathbb{C} : q_T(\lambda) = 0\} = \operatorname{spec}(T)$.
Structural Commentary: Different multiplicities, same spectral scalars.
Key Insight: Minimal polynomial roots and characteristic polynomial roots are both just eigenvalues. □

Perspective 3.3.5. *Characteristic vs. Minimal Polynomial*
By Corollary 3.3.4, q_T and μ_T have the same roots.
They encode multiplicity differently: q_T records algebraic multiplicity, whereas μ_T uses the least exponents that annihilate T.

3.4 Polynomial Spectral Mapping

Proposition 3.4.1. *Eigenvalues of a Polynomial in a Matrix*
Let $n \geq 1$, $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and $p \in \mathbb{C}[z]$.
If $(\lambda_i)_{i=1}^n$ lists the eigenvalues of A with algebraic multiplicity, then $(p(\lambda_i))_{i=1}^n$ lists the eigenvalues of $p(A)$ with algebraic multiplicity.

Proof.
(i) *Ordered Schur Form*: Choose a unit vector $\mathbf{x} \in E_{\lambda_1}(A)$.
By Theorem 2.1.9(i), $A = UTU^*$ for a unitary U and an upper-triangular T with diagonal $\lambda_1, \ldots, \lambda_n$.
(ii) *Polynomial Transfer*: By induction on k using $U^*U = UU^* = I_n$, $(\forall\, k \in \mathbb{Z}_{\geq 0})\, A^k = UT^kU^*$.
Hence $p(A) = Up(T)U^*$ by Definition 3.8.5.
(iii) *Triangular Evaluation*: By induction on k using Proposition 3.1.6, T^k is upper triangular with diagonal $\lambda_1^k, \ldots, \lambda_n^k$.
Hence $p(T)$ is upper triangular with diagonal $p(\lambda_1), \ldots, p(\lambda_n)$ by Definition 3.8.5.
(iv) *Spectral Multiplicities*: The diagonal of $p(T)$ is its eigenvalue list with algebraic multiplicity by Proposition 2.1.6(iii).
The same list applies to $p(A)$ by item (ii) and Theorem 2.1.7(ii).
Structural Commentary: Schur similarity transfers $p(A)$ to $p(T)$; triangularity determines its eigenvalue multiset.
Key Insight: Applying p to a triangular matrix applies p to every diagonal entry with its multiplicity. □

Corollary 3.4.2. *Polynomial Spectral Mapping*
Let $n \geq 1$, let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and let $p \in \mathbb{C}[z]$. Then $\operatorname{spec}(p(A)) = \{p(\lambda) : \lambda \in \operatorname{spec}(A)\}$.

Proof. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A, counted with algebraic multiplicity.
By Proposition 3.4.1, the eigenvalues of $p(A)$, counted with algebraic multiplicity, are $p(\lambda_1), \ldots, p(\lambda_n)$.
The claimed equality of the underlying spectral sets follows by Definition 5.1.3.
Structural Commentary: Polynomial evaluation pushes the spectrum forward through the scalar polynomial.
Key Insight: Once Schur form has exposed the diagonal, p acts entry by entry there. □

Corollary 3.4.3. *Polynomial Spectral Mapping for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $p \in \mathbb{C}[z]$. Then $\operatorname{spec}(p(T)) = \{p(\lambda) : \lambda \in \operatorname{spec}(T)\}$.

Proof. Choose a basis β and set $A = [T]_{\beta \leftarrow \beta}$.
By Proposition 3.2.4 and Proposition 3.4.4, $[p(T)]_{\beta \leftarrow \beta} = p(A)$.
For every $S \in \mathcal{L}(\mathcal{V})$, $\operatorname{spec}(S) = \operatorname{spec}([S]_{\beta \leftarrow \beta})$ by Definition 3.2.3, Proposition 3.2.2, and Definition 5.1.3.
Thus $\operatorname{spec}(T) = \operatorname{spec}(A)$ and $\operatorname{spec}(p(T)) = \operatorname{spec}(p(A))$; hence $\operatorname{spec}(p(T)) = \{p(\lambda) : \lambda \in \operatorname{spec}(T)\}$ by Corollary 3.4.2.
Structural Commentary: Coordinates reduce the operator statement to the matrix statement.
Key Insight: Polynomial spectral mapping is an operator fact seen through any convenient basis. □

Corollary 3.4.4. *Polynomial Invertibility Test*
Let $n \geq 1$, let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and let $p \in \mathbb{C}[z]$. If $p(\lambda) \neq 0$ for every $\lambda \in \operatorname{spec}(A)$, then $p(A)$ is invertible.

Proof. By Corollary 3.4.2, $0 \notin \operatorname{spec}(p(A))$. By Corollary 5.1.8, $p(A)$ is invertible.
Structural Commentary: Invertibility of $p(A)$ is checked on spectral values of A.
Key Insight: Avoiding zero on the spectrum prevents $p(A)$ from having a kernel. □

3.5 Polynomial Invariance and Block Separation

Proposition 3.5.1. *Kernel and Range of a Polynomial in an Operator Are Invariant*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $p \in \mathbb{F}[z]$. Then:
(i) $\ker(p(T))$ is invariant under T;
(ii) $\operatorname{range}(p(T))$ is invariant under T.

Proof. By Proposition 1.1.6(ii), with p and the polynomial z, the operators $p(T)$ and T commute.
(i) Let $\mathbf{v} \in \ker(p(T))$, so $p(T)\mathbf{v} = \mathbf{0}$. By commutativity, $p(T)(T\mathbf{v}) = T(p(T)\mathbf{v}) = \mathbf{0}$; $T\mathbf{v} \in \ker(p(T))$, so $\ker(p(T))$ invariant under T.
(ii) Let $\mathbf{v} \in \operatorname{range}(p(T))$. By this membership, choose $\mathbf{u} \in \mathcal{V}$ such that $\mathbf{v} = p(T)\mathbf{u}$.
By commutativity, $T\mathbf{v} = Tp(T)\mathbf{u} = p(T)(T\mathbf{u}) \in \operatorname{range}(p(T))$. Hence $\operatorname{range}(p(T))$ is invariant under T.
Structural Commentary: Commutativity transfers T-stability separately to the polynomial kernel and range.
Key Insight: The identity $Tp(T) = p(T)T$ preserves both subspaces. □

Theorem 3.5.2. *Sylvester Linear Matrix Equation*
Let $m, n \geq 1$, let $A \in \operatorname{Mat}(m, m, \mathbb{C})$, and let $B \in \operatorname{Mat}(n, n, \mathbb{C})$.
If $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$, then $(\forall C \in \operatorname{Mat}(m, n, \mathbb{C}))(\exists! X \in \operatorname{Mat}(m, n, \mathbb{C}))\, AX - XB = C$.

Proof.
(i) *Sylvester Map*: Set $\mathcal{M} := \operatorname{Mat}(m, n, \mathbb{C})$ and define $L : \mathcal{M} \to \mathcal{M}$ by $L(X) := AX - XB$.
The map L is linear by Proposition 3.1.2(iii)–(iv), and $\dim_{\mathbb{C}}(\mathcal{M}) = mn$ by Proposition 3.1.5.
It suffices to prove $\ker(L) = \{0\}$ by Proposition 1.6.2 and Corollary 1.7.6.
(ii) *Polynomial Intertwining*: Suppose $X \in \ker(L)$, so $AX = XB$. We prove $(\forall k \in \mathbb{Z}_{\geq 0})\, A^kX = XB^k$. For $k = 0$, this is $I_mX = XI_n$.
If $A^kX = XB^k$, then, by Proposition 3.1.2 and $AX = XB$, $A^{k+1}X = A(A^kX) = A(XB^k) = (AX)B^k = (XB)B^k = XB^{k+1}$.
The claim follows by mathematical induction. Therefore $(\forall p \in \mathbb{C}[z])\, p(A)X = Xp(B)$ by Definition 3.8.5.

(iii) *Kernel Vanishing*: By Theorem 3.1.1 and the polynomial-intertwining identity from (ii), $p_B(A)X = Xp_B(B) = 0$.
If $\lambda \in \operatorname{spec}(A)$, then $\lambda \notin \operatorname{spec}(B)$ by the disjointness hypothesis, so $p_B(\lambda) \neq 0$ by Theorem 2.1.3(ii).
Thus $p_B(A)$ is invertible by Corollary 3.4.4, and $X = p_B(A)^{-1}p_B(A)X = 0$. Hence $\ker(L) = \{0\}$.
(iv) *Existence and Uniqueness*: The map L is injective by (iii) and Proposition 1.6.2.
The map L is surjective by Corollary 1.7.6 and invertible by Proposition 1.6.7.
Consequently, $(\forall C \in \mathcal{M})(\exists! X \in \mathcal{M})\, L(X) = C$, equivalently, $(\forall C \in \mathcal{M})(\exists! X \in \mathcal{M})\, AX - XB = C$.

Structural Commentary: Polynomial intertwining combines Cayley–Hamilton for B with spectral invertibility for $p_B(A)$.
Key Insight: Disjoint spectra force every homogeneous Sylvester solution to vanish. □

Figure 58 summarizes the spectral-separation argument behind invertibility of the Sylvester map.

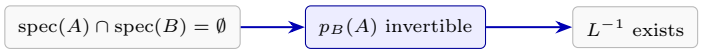

Figure 58: Disjoint spectra make $p_B(A)$ invertible, so the Sylvester map $L(X) = AX - XB$ is invertible.

Corollary 3.5.3. *Removing a Spectrally Separated Off-Diagonal Block*
Let $m, n \geq 1$, let $A \in \operatorname{Mat}(m, m, \mathbb{C})$, let $B \in \operatorname{Mat}(n, n, \mathbb{C})$, and let $C \in \operatorname{Mat}(m, n, \mathbb{C})$. Set $M := \left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right) \in \operatorname{Mat}(m+n, m+n, \mathbb{C})$.
If $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$, then M is similar to $A \oplus B$.

Proof. By Theorem 3.5.2, choose $X \in \operatorname{Mat}(m, n, \mathbb{C})$ such that $AX - XB = -C$. Set $P := \left(\begin{smallmatrix} I_m & -X \\ 0 & I_n \end{smallmatrix}\right)$ and $Q := \left(\begin{smallmatrix} I_m & X \\ 0 & I_n \end{smallmatrix}\right)$.
By Proposition 3.7.3(ii), $PQ = QP = I_{m+n}$, so P is invertible and $P^{-1} = Q$. By same proposition, $PMQ = \left(\begin{smallmatrix} A & AX+C-XB \\ 0 & B \end{smallmatrix}\right)$.
By $AX - XB = -C$, $PMQ = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right) = A \oplus B$. Thus $PMP^{-1} = A \oplus B$ by $P^{-1} = Q$, and M is similar to $A \oplus B$ by Definition 6.2.1.
Structural Commentary: A Sylvester solution determines the block shear that removes the coupling.
Key Insight: Spectral separation makes the upper-right block removable by similarity. □

Theorem 3.5.4. *Schur Form Grouped by Eigenvalues*
Let $n, d, n_1, \ldots, n_d \geq 1$, let $\lambda_1, \ldots, \lambda_d \in \mathbb{C}$ be distinct, and let $A \in \operatorname{Mat}(n, n, \mathbb{C})$.
Suppose $\sum_{i=1}^{d} n_i = n$, $\operatorname{spec}(A) = \{\lambda_1, \ldots, \lambda_d\}$, and $(\forall i \in \{1, \ldots, d\})\, m_A(\lambda_i) = n_i$.

(i) The matrix A is unitarily similar to a block upper-triangular matrix

$$T = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1d} \\ 0 & T_{22} & \cdots & T_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & T_{dd} \end{bmatrix}, \qquad T_{ii} \in \operatorname{Mat}(n_i, n_i, \mathbb{C}),$$

where each T_{ii} is upper triangular with every diagonal entry equal to λ_i.
(ii) The matrix A is similar to $T_{11} \oplus \cdots \oplus T_{dd}$.

Proof.
(i) Order the eigenvalue multiset of A by listing λ_i exactly n_i times, $i \in \{1, \ldots, d\}$. Choose a unit vector $\mathbf{x} \in E_{\lambda_1}(A)$.
By Theorem 2.1.9(i), $T := U^*AU$ is upper triangular with this ordered diagonal for some unitary U.
After grouping consecutive rows and columns according to $(n_1, \ldots, n_d)$, T has the stated block form and each T_{ii} has constant diagonal λ_i. Thus A is unitarily similar to T by Definition 2.6.9.
(ii) By part (i), Theorem 2.6.8(vi), and Definition 6.2.1, A is similar to T. If $d = 1$, then $T = T_{11}$.
Suppose $d \geq 2$. Set $R_0 := T$, and for $k \in \{1, \ldots, d-1\}$ let R_k be the trailing principal block with block indices $k+1, \ldots, d$.
Then $R_{k-1} = \left(\begin{smallmatrix} T_{kk} & C_k \\ 0 & R_k \end{smallmatrix}\right)$. By Proposition 2.1.6, $\operatorname{spec}(T_{kk}) = \{\lambda_k\}$ and $\operatorname{spec}(R_k) = \{\lambda_{k+1}, \ldots, \lambda_d\}$.
Hence R_{k-1} is similar to $T_{kk} \oplus R_k$ by Corollary 3.5.3.
Consequently, for every $k \in \{1, \ldots, d-1\}$, T is similar to $T_{11} \oplus \cdots \oplus T_{kk} \oplus R_k$ by mathematical induction, Proposition 3.8.6(ii), and transitivity in Proposition 6.2.2. Since $R_{d-1} = T_{dd}$, the conclusion follows by transitivity in Proposition 6.2.2.

Structural Commentary: Ordered Schur form groups eigenvalues; Sylvester shears then split successive spectral blocks.
Key Insight: Cross-group blocks are coordinate-dependent and removable by similarity. □

Corollary 3.5.5. *Commuting with Disjoint Blocks*
Let $d, n_1, \ldots, n_d \geq 1$ and let $A_{ii} \in \operatorname{Mat}(n_i, n_i, \mathbb{C})$, $i \in \{1, \ldots, d\}$, have pairwise disjoint spectra.
Set $n := \sum_{i=1}^{d} n_i$ and $A := A_{11} \oplus \cdots \oplus A_{dd} \in \operatorname{Mat}(n, n, \mathbb{C})$.
If a conformally partitioned $B = [B_{ij}] \in \operatorname{Mat}(n, n, \mathbb{C})$ satisfies $AB = BA$, then $B_{ij} = 0$ for all distinct $i, j \in \{1, \ldots, d\}$.

Proof. By $AB = BA$ and Proposition 3.7.3(ii), $(\forall i, j \in \{1, \ldots, d\})\, A_{ii}B_{ij} - B_{ij}A_{jj} = 0$.
Fix $i, j \in \{1, \ldots, d\}$ with $i \neq j$. The spectra of A_{ii} and A_{jj} are disjoint.
Hence $B_{ij} = 0$ by the uniqueness assertion in Theorem 3.5.2 with zero right-hand side.
Structural Commentary: Block commutation reduces every off-diagonal entry to a homogeneous Sylvester equation.
Key Insight: Homogeneous spectral separation forces each off-diagonal block to vanish. □

4 Generalized Eigenspaces

4.1 Generalized Eigenspaces and Nilpotent Structure

Throughout this subsection, $\mathcal{V}$ denotes a vector space over $\mathbb{F}$. Generalized eigenvectors enlarge ordinary eigendirections by allowing vectors annihilated by powers of shifted operators.

Definition 4.1.1. *Generalized Eigenvectors and Generalized Eigenspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\lambda \in \mathbb{F}$. Then:

(i) A nonzero $\mathbf{v} \in \mathcal{V}$ is a *generalized eigenvector* of T corresponding to λ iff $(\exists j \in \mathbb{Z}_{>0})\,(T - \lambda I_{\mathcal{V}})^j \mathbf{v} = \mathbf{0}$.
(ii) The *generalized eigenspace* corresponding to λ is $\mathcal{G}_\lambda(T) := \{\mathbf{v} \in \mathcal{V} : (\exists j \in \mathbb{Z}_{>0})\,(T - \lambda I_{\mathcal{V}})^j \mathbf{v} = \mathbf{0}\}$.

Every eigenvector is generalized by taking $j = 1$. For a nonzero generalized eigenvector, successive applications of $T - \lambda I_{\mathcal{V}}$ eventually reach the ordinary eigenspace.

Proposition 4.1.2. *Kernels of Powers Increase*
Let $T \in \mathcal{L}(\mathcal{V})$, with $T^0 := I_{\mathcal{V}}$. Then:

(i) $\ker(T^0) = \{\mathbf{0}\}$;
(ii) $(\forall k \in \mathbb{Z}_{\geq 0})\ \ker(T^k) \subseteq \ker(T^{k+1})$.

Proof.

(i) By $T^0 = I_{\mathcal{V}}$, $\ker(T^0) = \{\mathbf{0}\}$.
(ii) Fix $k \in \mathbb{Z}_{\geq 0}$ and $\mathbf{v} \in \ker(T^k)$. Then $T^k \mathbf{v} = \mathbf{0}$, so $T^{k+1}\mathbf{v} = T(T^k \mathbf{v}) = \mathbf{0}$. Hence $\mathbf{v} \in \ker(T^{k+1})$.

Structural Commentary: The identity power starts the filtration; one additional application preserves annihilation.
Key Insight: An operator cannot revive a vector already sent to zero. □

Proposition 4.1.3. *Equality Forces Stabilization*
Let $T \in \mathcal{L}(\mathcal{V})$ and $m \in \mathbb{Z}_{\geq 0}$. If $\ker(T^m) = \ker(T^{m+1})$, then $(\forall k \in \mathbb{Z}_{\geq 0})\ \ker(T^{m+k}) = \ker(T^m)$.

Proof. By Proposition 4.1.2(ii), it suffices to prove $(\forall k \in \mathbb{Z}_{\geq 0})\ \ker(T^{m+k+1}) \subseteq \ker(T^{m+k})$.
Fix $k \in \mathbb{Z}_{\geq 0}$ and $\mathbf{v} \in \ker(T^{m+k+1})$. Then $T^{m+1}(T^k \mathbf{v}) = T^{m+k+1}\mathbf{v} = \mathbf{0}$, so $T^k \mathbf{v} \in \ker(T^{m+1}) = \ker(T^m)$.
Thus $T^{m+k}\mathbf{v} = T^m(T^k \mathbf{v}) = \mathbf{0}$, so $\mathbf{v} \in \ker(T^{m+k})$. Hence every adjacent inclusion in the kernel tail is an equality.
Structural Commentary: The initial equality converts every later kernel inclusion into equality.
Key Insight: Shift a later kernel vector into the already-stable pair. □

Theorem 4.1.4. *Kernels Stop Growing by $\dim(\mathcal{V})$*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and set $n := \dim(\mathcal{V})$. Then $(\forall k \in \mathbb{Z}_{\geq 0})\ \ker(T^{n+k}) = \ker(T^n)$.

Proof. Suppose $\ker(T^n) \neq \ker(T^{n+1})$. By Proposition 4.1.3, every inclusion in $\{\mathbf{0}\} = \ker(T^0) \subsetneq \ker(T) \subsetneq \cdots \subsetneq \ker(T^{n+1})$ is strict.
Thus $n + 1 \leq \dim \ker(T^{n+1}) \leq n$ by Proposition 1.6.3(i) and Proposition 1.4.3(i)–(ii), a contradiction.
Hence $\ker(T^n) = \ker(T^{n+1})$, and the claim follows by Proposition 4.1.3.
Structural Commentary: A hypothetical late growth step forces too many strict subspace inclusions.
Key Insight: An n-dimensional space permits at most n strict kernel increases. □

Figure 59 records the dimension-limited growth and eventual stabilization of the kernel filtration.

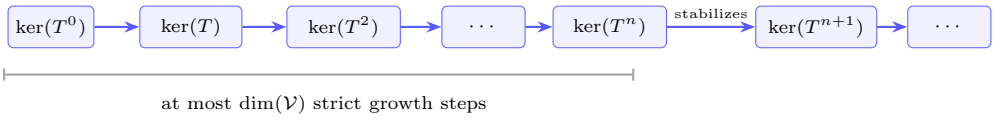

Figure 59: The kernel filtration grows until it stabilizes; after equality appears once, the tail is constant.

Theorem 4.1.5. *Stable Kernel-Range Decomposition*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and set $n := \dim(\mathcal{V})$. Then $\mathcal{V} = \ker(T^n) \oplus \operatorname{range}(T^n)$.

Proof. Let $\mathbf{v} \in \ker(T^n) \cap \operatorname{range}(T^n)$. Choose $\mathbf{u} \in \mathcal{V}$ with $\mathbf{v} = T^n \mathbf{u}$ by range membership.
Then $T^{2n}\mathbf{u} = T^n \mathbf{v} = \mathbf{0}$ by kernel membership. Thus $\mathbf{u} \in \ker(T^{2n}) = \ker(T^n)$ by Theorem 4.1.4, and $\mathbf{v} = T^n \mathbf{u} = \mathbf{0}$.
Hence $\ker(T^n) \cap \operatorname{range}(T^n) = \{\mathbf{0}\}$. By Theorem 1.7.3(ii), $\dim \ker(T^n) + \dim \operatorname{range}(T^n) = n$. The sum is direct by Proposition 2.1.8.
By Corollary 2.3.2, $\dim(\ker(T^n) \oplus \operatorname{range}(T^n)) = n$. Hence the direct sum equals $\mathcal{V}$ by Proposition 1.4.3(ii).
Structural Commentary: Kernel stabilization first kills the intersection; rank–nullity then fills the ambient dimension.
Key Insight: Stable image vectors cannot also lie in the stable kernel. □

Theorem 4.1.6. *Description of Generalized Eigenspaces*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, let $\lambda \in \mathbb{F}$, and set $n := \dim(\mathcal{V})$. Then:

(i) $\mathcal{G}_\lambda(T) = \ker((T - \lambda I_{\mathcal{V}})^n)$;
(ii) $\mathcal{G}_\lambda(T)$ *is a subspace of* $\mathcal{V}$.

Proof. Set $S := T - \lambda I_{\mathcal{V}}$.

(i) Let $\mathbf{v} \in \mathcal{G}_\lambda(T)$. By Definition 4.1.1(ii), choose $j \in \mathbb{Z}_{>0}$ such that $S^j \mathbf{v} = \mathbf{0}$.
If $j \leq n$, then $\mathbf{v} \in \ker(S^n)$ by Proposition 4.1.2(ii); if $j > n$, then $\mathbf{v} \in \ker(S^j) = \ker(S^n)$ by Theorem 4.1.4. Thus $\mathcal{G}_\lambda(T) \subseteq \ker(S^n)$.
Conversely, $S^{n+1}\mathbf{v} = \mathbf{0}$ for every $\mathbf{v} \in \ker(S^n)$, so $\ker(S^n) \subseteq \mathcal{G}_\lambda(T)$ by Definition 4.1.1(ii).
(ii) By (i) and Proposition 1.6.3(i), $\mathcal{G}_\lambda(T) = \ker(S^n)$ is a subspace of $\mathcal{V}$.

Structural Commentary: Kernel-chain stabilization replaces all admissible exponents by the single exponent n.
Key Insight: Every generalized eigenvector is detected by the stabilized shifted kernel. □

Proposition 4.1.7. *Generalized Eigenspaces Are Invariant*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then $(\forall \lambda \in \mathbb{F})\ T(\mathcal{G}_\lambda(T)) \subseteq \mathcal{G}_\lambda(T)$.

Proof. Fix $\lambda \in \mathbb{F}$, set $n := \dim(\mathcal{V})$, and let $p(z) := (z - \lambda)^n$. By Theorem 4.1.6(i) and Proposition 1.1.6(i), $\mathcal{G}_\lambda(T) = \ker(p(T))$.
Hence $\mathcal{G}_\lambda(T)$ is invariant under T by Proposition 3.5.1(i).
Structural Commentary: The stabilized shifted kernel is rewritten as a polynomial kernel in T.
Key Insight: Polynomial kernels remain invariant under their defining operator. □

Definition 4.1.8. *Nilpotent Operator*
An operator $N \in \mathcal{L}(\mathcal{V})$ is *nilpotent* iff $(\exists m \in \mathbb{Z}_{>0})\, N^m = 0$.

Corollary 4.1.9. *Nilpotent Power Bound*
Let $\mathcal{V}$ be finite-dimensional, let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent, and set $n := \dim(\mathcal{V})$. Then $N^n = 0$.

Proof. By nilpotence, choose $m \in \mathbb{Z}_{>0}$ such that $N^m = 0$, so $\ker(N^m) = \mathcal{V}$. If $m \leq n$, then $\ker(N^m) \subseteq \ker(N^n)$ by Proposition 4.1.2(ii).
If $m > n$, then $\ker(N^m) = \ker(N^n)$ by Theorem 4.1.4. Thus $\ker(N^n) = \mathcal{V}$ in both cases, and therefore $N^n = 0$.
Structural Commentary: Kernel growth or stabilization transfers a full nilpotent kernel to exponent n.
Key Insight: The dimension bound never requires a larger nilpotence exponent. □

Lemma 4.1.10. *Nonzero Scalar Plus Nilpotent Is Invertible*
Let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent and let $a \in \mathbb{F}$ with $a \neq 0$. Then $aI_{\mathcal{V}} + N$ is invertible.

Proof. Choose $m \in \mathbb{Z}_{>0}$ such that $N^m = 0$ by nilpotence, and set $X := -a^{-1}N$ and $B := a^{-1}\sum_{r=0}^{m-1} X^r$.
By Proposition 1.1.6(i)–(ii) and the finite geometric-series identity, $(aI_{\mathcal{V}}+N)B = B(aI_{\mathcal{V}}+N) = I_{\mathcal{V}} - X^m = I_{\mathcal{V}}$. Hence $B = (aI_{\mathcal{V}}+N)^{-1}$.
Structural Commentary: The finite geometric series verifies both inverse identities for the nilpotent perturbation.
Key Insight: Nilpotence makes the geometric-series remainder vanish exactly. □

Theorem 4.1.11. *Generalized Eigenvectors for Distinct Eigenvalues Are Independent*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $m \in \mathbb{Z}_{>0}$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding generalized eigenvectors.
Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof.
(i) *Candidate Relation*: Set $n := \dim(\mathcal{V})$ and suppose $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
(ii) *Polynomial Filter*: Fix $r \in \{1, \ldots, m\}$ and set $Q_r(T) := \prod_{i \in \{1,\ldots,m\}\setminus\{r\}} (T - \lambda_i I_{\mathcal{V}})^n$, with the empty product equal to $I_{\mathcal{V}}$.
(iii) *Non-Target Annihilation*: By Theorem 4.1.6(i) and commutativity in Proposition 1.1.6(ii), $Q_r(T)\mathbf{v}_i = \mathbf{0}$ for $i \neq r$.
Hence applying $Q_r(T)$ to the candidate relation leaves $a_r Q_r(T)\mathbf{v}_r = \mathbf{0}$.
(iv) *Target Invertibility*: Set $\mathcal{G}_r := \mathcal{G}_{\lambda_r}(T)$ and $N_r := (T - \lambda_r I_{\mathcal{V}})|_{\mathcal{G}_r}$.
By Proposition 4.1.7 and Theorem 4.1.6(i)–(ii), $N_r \in \mathcal{L}(\mathcal{G}_r)$ and $N_r^n = 0$.
For $i \neq r$, $(T - \lambda_i I_{\mathcal{V}})|_{\mathcal{G}_r} = N_r + (\lambda_r - \lambda_i) I_{\mathcal{G}_r}$ with $\lambda_r - \lambda_i \neq 0$, so this restriction is invertible by Lemma 4.1.10.
Thus $Q_r(T)|_{\mathcal{G}_r}$, a product of nth powers of these restrictions or the identity if $m = 1$, is invertible by Proposition 1.6.6(i).
(v) *Coefficient Vanishing*: Since $\mathbf{v}_r \in \mathcal{G}_r \setminus \{\mathbf{0}\}$, $Q_r(T)\mathbf{v}_r \neq \mathbf{0}$ by item (iv), and hence $a_r = 0$ by item (iii).
Since r was arbitrary, $a_1 = \cdots = a_m = 0$, so $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Structural Commentary: Polynomial filters annihilate non-target vectors while nilpotent-plus-scalar factors preserve the target.
Key Insight: Distinct spectral shifts are invertible on the remaining generalized eigenspace. □

Theorem 4.1.12. *Nilpotent Operators and Strictly Upper-Triangular Matrices*
Let $\mathcal{V}$ be nonzero and finite-dimensional, let $N \in \mathcal{L}(\mathcal{V})$, and set $n := \dim(\mathcal{V})$.
Then N is nilpotent iff some ordered basis β satisfies $[N]_\beta = [a_{ij}]$ and $(\forall i, j \in \{1, \ldots, n\})\, (i \geq j \implies a_{ij} = 0)$.

Proof.
(⇒) Suppose N is nilpotent. By Corollary 4.1.9 and Proposition 4.1.2(i)–(ii), $\{\mathbf{0}\} = \ker(N^0) \subseteq \ker(N) \subseteq \cdots \subseteq \ker(N^n) = \mathcal{V}$.
By Corollary 1.3.14(i)–(ii) and Theorem 1.3.13, choose an ordered basis β whose initial segments span the kernels in this chain.
If $\mathbf{v}$ first enters at level k, then $N\mathbf{v} \in \ker(N^{k-1})$, which is spanned by earlier vectors of β; hence $[N]_\beta$ is strictly upper triangular.
(⇐) Suppose $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $[N]_\beta$ is strictly upper triangular.
Set $F_0 := \{\mathbf{0}\}$ and $F_j := \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$ for $j \in \{1, \ldots, n\}$. Then $(\forall j \in \{1, \ldots, n\})\, N(F_j) \subseteq F_{j-1}$.
By iteration, $N^n(\mathcal{V}) = N^n(F_n) \subseteq F_0 = \{\mathbf{0}\}$, so $N^n = 0$ and N is nilpotent by Definition 4.1.8.

Structural Commentary: The forward direction adapts a basis to kernels; the reverse direction descends basis indices.
Key Insight: Strict index descent forces every basis vector to reach zero. □

Figure 60 illustrates how a basis adapted to the kernel filtration makes nilpotent action move toward earlier layers.

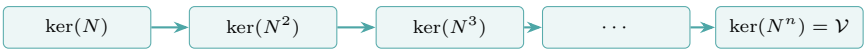

Figure 60: Extending bases through the kernel filtration makes N move vectors to earlier layers.

Perspective 4.1.13. *Eigenvalue Plus Nilpotent Part*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, let $\lambda \in \mathbb{F}$, and set $n := \dim(\mathcal{V})$. Define $\mathcal{G}_\lambda := \mathcal{G}_\lambda(T)$ and $N_\lambda := (T - \lambda I_{\mathcal{V}})|_{\mathcal{G}_\lambda}$.
By Proposition 4.1.7, $N_\lambda \in \mathcal{L}(\mathcal{G}_\lambda)$. By Theorem 4.1.6(i), $N_\lambda^n = 0$, and hence $N_\lambda^{n+1} = 0$. Thus N_λ is nilpotent and $T|_{\mathcal{G}_\lambda} = \lambda I_{\mathcal{G}_\lambda} + N_\lambda$.
Generalized eigenvectors therefore record the nilpotent component of T within the stabilized shifted kernel.

5 Primary Decomposition

5.1 Primary Decomposition and Spectral Splitting

Theorem 5.1.1. *Primary Decomposition over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, $T \in \mathcal{L}(\mathcal{V})$, $m \in \mathbb{Z}_{>0}$. Suppose $\lambda_1, \dots, \lambda_m$ are the distinct eigenvalues of T. Then:
(i) $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$;
(ii) $(\forall j \in \{1, \dots, m\})\, (T - \lambda_j I_{\mathcal{V}})|_{\mathcal{G}_{\lambda_j}(T)}$ *is nilpotent.*

Proof. Set $n := \dim(\mathcal{V})$. For $j \in \{1, \dots, m\}$, set $S_j := T - \lambda_j I_{\mathcal{V}}$ and $\mathcal{G}_j := \mathcal{G}_{\lambda_j}(T)$.

(*i*) We prove $\mathcal{V} = \mathcal{G}_1 \oplus \cdots \oplus \mathcal{G}_m$ by induction on n.
(a) *Base Case*: Suppose $n = 1$. Choose $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$ by $\mathcal{V} \neq \{\mathbf{0}\}$. Then $(\mathbf{v})$ is a basis by $n = 1$.
Choose $\lambda \in \mathbb{C}$ with $T\mathbf{v} = \lambda \mathbf{v}$ by spanning. Then $T = \lambda I_{\mathcal{V}}$ by linearity, so $m = 1$, $\lambda = \lambda_1$, and $\mathcal{V} = \mathcal{G}_1$.
Assume henceforth that $n > 1$ and that (i) holds in every smaller positive dimension.
(b) *Stable Split*: Set $\mathcal{U} := \operatorname{range}(S_1^n)$. By Theorem 4.1.5 and Theorem 4.1.6(i), $\mathcal{V} = \mathcal{G}_1 \oplus \mathcal{U}$.
The subspace $\mathcal{U}$ is T-invariant by Proposition 3.5.1(ii).
Since λ_1 is an eigenvalue, $\mathcal{G}_1 \neq \{\mathbf{0}\}$ by Definition 4.1.1(i); hence $\dim(\mathcal{U}) < n$ by Theorem 2.3.4(i).
(c) *Zero Range*: Suppose $\mathcal{U} = \{\mathbf{0}\}$. Then $\mathcal{V} = \mathcal{G}_1$ by item (b). Fix $\lambda \in \operatorname{spec}(T)$ and $\mathbf{v} \in E_\lambda(T) \setminus \{\mathbf{0}\}$.
Then $0 = S_1^n \mathbf{v} = (\lambda - \lambda_1)^n \mathbf{v}$, so $\lambda = \lambda_1$. Thus $\operatorname{spec}(T) = \{\lambda_1\}$, $m = 1$, and (i) holds.
(d) *Restricted Spectrum*: Suppose $\mathcal{U} \neq \{\mathbf{0}\}$. Then $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ by item (b), and $\operatorname{spec}(T|_{\mathcal{U}}) \neq \emptyset$ by Theorem 5.3.2.
$\operatorname{spec}(T|_{\mathcal{U}}) \subseteq \operatorname{spec}(T) \setminus \{\lambda_1\}$ by restriction and $\mathcal{U} \cap \ker(S_1) \subseteq \mathcal{U} \cap \mathcal{G}_1 = \{\mathbf{0}\}$. Hence $m \geq 2$.
Fix $j \in \{2, \dots, m\}$ and choose an eigenvector $\mathbf{v}_j$ for λ_j by $\lambda_j \in \operatorname{spec}(T)$.
Then $\mathbf{v}_j = (\lambda_j - \lambda_1)^{-n} S_1^n \mathbf{v}_j \in \mathcal{U}$, so $\operatorname{spec}(T|_{\mathcal{U}}) = \{\lambda_2, \dots, \lambda_m\}$.
(e) *Inductive Decomposition*: By items (b) and (d) and the induction hypothesis, $\mathcal{U} = \mathcal{G}_{\lambda_2}(T|_{\mathcal{U}}) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T|_{\mathcal{U}})$.
(f) *Eigenspace Identification*: Fix $j \in \{2, \dots, m\}$. By equality of restriction powers, $\mathcal{G}_{\lambda_j}(T|_{\mathcal{U}}) \subseteq \mathcal{G}_j$.
Conversely, let $\mathbf{v} \in \mathcal{G}_j$ and write $\mathbf{v} = \mathbf{g} + \mathbf{u}$ with $\mathbf{g} \in \mathcal{G}_1$ and $\mathbf{u} \in \mathcal{U}$ by item (b).
Choose $r \in \mathbb{Z}_{>0}$ with $S_j^r \mathbf{v} = \mathbf{0}$ by Definition 4.1.1(ii). Then $S_j^r \mathbf{g} \in \mathcal{G}_1$ and $S_j^r \mathbf{u} \in \mathcal{U}$ by Proposition 4.1.7 and item (b).
Both terms vanish by $S_j^r \mathbf{g} + S_j^r \mathbf{u} = \mathbf{0}$ and the directness in item (b). Thus $\mathbf{g} \in \mathcal{G}_1 \cap \mathcal{G}_j = \{\mathbf{0}\}$ by Theorem 4.1.11.
Hence $\mathbf{v} = \mathbf{u} \in \mathcal{G}_{\lambda_j}(T|_{\mathcal{U}})$. Therefore the generalized eigenspaces are equal, and items (b) and (e) prove (i).

(*ii*) Fix $j \in \{1, \dots, m\}$ and set $N_j := S_j|_{\mathcal{G}_j}$. By Theorem 4.1.6(ii) and Proposition 4.1.7, $N_j \in \mathcal{L}(\mathcal{G}_j)$.
By Theorem 4.1.6(i), $N_j^n = 0$, so N_j is nilpotent by Definition 4.1.8.

Structural Commentary: Stable kernel-range splitting isolates one primary summand; induction decomposes the invariant range.
Key Insight: Stable kernel-range splitting isolates one eigenvalue at a time. □

Figure 61 summarizes the primary summands and the nilpotent part of T on each one.

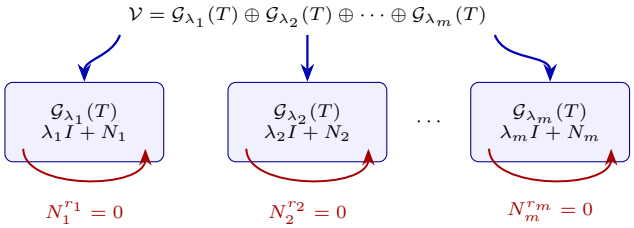

Figure 61: Primary decomposition: on $\mathcal{G}_{\lambda_j}(T)$, $T = \lambda_j I$ plus nilpotent.

Corollary 5.1.2. *Basis of Generalized Eigenvectors*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a generalized-eigenvector basis for T.

Proof. Enumerate the distinct eigenvalues as $\lambda_1, \dots, \lambda_m$ by Theorem 5.3.2 and Proposition 2.1.16(i)–(ii).
By Theorem 5.1.1(i), $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$.
For $j \in \{1, \dots, m\}$, the space $\mathcal{G}_{\lambda_j}(T)$ has an ordered basis β_j by Theorem 4.1.6(ii) and Corollary 1.3.14(i)–(ii).
The concatenation $\beta := (\beta_1, \dots, \beta_m)$ is an ordered basis of $\mathcal{V}$ by Theorem 2.3.4(ii).
Every vector in β_j is nonzero and belongs to $\mathcal{G}_{\lambda_j}(T)$, so it is a generalized eigenvector corresponding to λ_j by Definition 4.1.1(i).
Structural Commentary: Primary decomposition supplies finite-dimensional summands whose ordered bases concatenate by directness.
Key Insight: Every primary-summand basis vector is generalized for its eigenvalue. □

Theorem 5.1.3. *Block Diagonal Form with Upper-Triangular Primary Blocks*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Let $m \in \mathbb{Z}_{>0}$, let $(\lambda_j)_{j=1}^m$ list its distinct eigenvalues, and set $d_j := \dim \mathcal{G}_{\lambda_j}(T)$ for $j \in \{1, \dots, m\}$.
Then there are an ordered basis β of $\mathcal{V}$ and matrices $A_j \in \operatorname{Mat}(d_j, d_j, \mathbb{C})$, $j \in \{1, \dots, m\}$, such that $[T]_{\beta \leftarrow \beta} = A_1 \oplus \cdots \oplus A_m$.
Each A_j is upper triangular with diagonal $(\lambda_j, \dots, \lambda_j)$.

Proof.
(i) *Primary Summands*: Set $\mathcal{U}_j := \mathcal{G}_{\lambda_j}(T)$ and $N_j := (T - \lambda_j I_{\mathcal{V}})|_{\mathcal{U}_j}$ for $j \in \{1, \dots, m\}$.
By Theorem 5.1.1(i)–(ii), $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$ and each $N_j \in \mathcal{L}(\mathcal{U}_j)$ is nilpotent.
(ii) *Nilpotent Bases*: Fix $j \in \{1, \dots, m\}$. Choose ordered basis β_j; $[N_j]_{\beta_j}$ strictly upper triangular by item (i), Theorem 4.1.12.
Set $A_j := [T|_{\mathcal{U}_j}]_{\beta_j \leftarrow \beta_j}$. By Proposition 3.2.4, $A_j = \lambda_j I_{d_j} + [N_j]_{\beta_j}$, so A_j is upper triangular with diagonal $(\lambda_j, \dots, \lambda_j)$.
(iii) *Adapted Basis*: The concatenation $\beta := (\beta_1, \dots, \beta_m)$ is an ordered basis of $\mathcal{V}$ by Theorem 2.3.4(ii).
(iv) *Block Verification*: By items (i)–(iii) and Definition 3.2.3, $[T]_{\beta \leftarrow \beta} = A_1 \oplus \cdots \oplus A_m$.

Structural Commentary: Primary summands determine the blocks; nilpotent-adapted bases determine their triangular shape.
Key Insight: On each generalized eigenspace, T is eigenvalue plus nilpotent. □

Figure 62 illustrates the block form obtained from bases adapted to the primary summands.

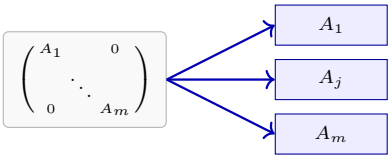

Figure 62: Primary blocks: $A_j = \lambda_j I + N_j$, with N_j strictly upper triangular.

Proposition 5.1.4. *Algebraic Multiplicity from the Generalized Eigenspace*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then $(\forall \lambda \in \operatorname{spec}(T))$, the multiplicity of λ as a root of q_T is $\dim \mathcal{G}_\lambda(T) = \dim \ker((T - \lambda I_\mathcal{V})^{\dim(\mathcal{V})})$.

Proof. Fix $\lambda \in \operatorname{spec}(T)$. By Theorem 5.1.3, $\exists$ ordered basis β, $A := [T]_{\beta \leftarrow \beta}$ upper triangular, diagonal has λ exactly $\dim \mathcal{G}_\lambda(T)$ times.
By Definition 2.1.15 and Proposition 2.1.6(iii), the multiplicity of λ as a root of q_T is $\dim \mathcal{G}_\lambda(T)$.
The identity $\mathcal{G}_\lambda(T) = \ker((T - \lambda I_\mathcal{V})^{\dim(\mathcal{V})})$ holds by Theorem 4.1.6(i).
Structural Commentary: Primary-block diagonal entries convert root multiplicity into generalized-eigenspace dimension.
Key Insight: Algebraic multiplicity is the dimension of the generalized eigenspace. □

Figure 63 records how the primary-summand dimensions determine the characteristic-polynomial exponents.

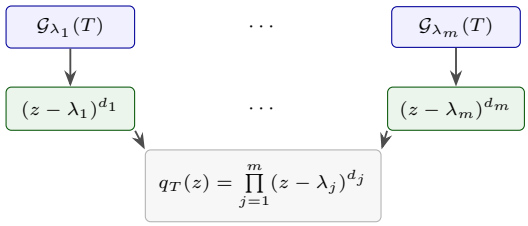

Figure 63: Each primary summand contributes $(z - \lambda_j)^{d_j}$, where $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.

Proposition 5.1.5. *Sum of Multiplicities*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$.
For $\lambda \in \operatorname{spec}(T)$, let $a(\lambda)$ be its multiplicity as a root of q_T. Then $\sum_{\lambda \in \operatorname{spec}(T)} a(\lambda) = \dim(\mathcal{V})$.

Proof. By Theorems 5.1.1(i) and 2.3.4(i), $\dim(\mathcal{V}) = \sum_{\lambda \in \operatorname{spec}(T)} \dim \mathcal{G}_\lambda(T)$.
By Proposition 5.1.4, $\dim \mathcal{G}_\lambda(T) = a(\lambda)$ for every $\lambda \in \operatorname{spec}(T)$, so the assertion follows.
Structural Commentary: Primary decomposition identifies algebraic multiplicities with summand dimensions.
Key Insight: Direct-sum dimensions add to $\dim(\mathcal{V})$. □

5.2 Real Spectral Data via Complexification

Lemma 5.2.1. *Conjugation Transfers Generalized Eigenspaces*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$. For $\mu \in \mathbb{C}$, set $S_\mu := T_\mathbb{C} - \mu I_{\mathcal{V}_\mathbb{C}}$.
Then $(\forall \lambda \in \mathbb{C})(\forall k \in \mathbb{Z}_{\geq 0})(\forall \mathbf{u}, \mathbf{v} \in \mathcal{V})\ \big(S_\lambda^k(\mathbf{u} + i\mathbf{v}) = \mathbf{0} \iff S_{\overline{\lambda}}^k(\mathbf{u} - i\mathbf{v}) = \mathbf{0}\big)$.

Proof. Define $\Gamma : \mathcal{V}_\mathbb{C} \to \mathcal{V}_\mathbb{C}$ by $\Gamma(\mathbf{a} + i\mathbf{b}) := \mathbf{a} - i\mathbf{b}$. By Definition 5.3.6, $(\forall \mathbf{x}, \mathbf{y} \in \mathcal{V}_\mathbb{C})\, \Gamma(\mathbf{x} + \mathbf{y}) = \Gamma\mathbf{x} + \Gamma\mathbf{y}$.
The same definition implies $(\forall \alpha \in \mathbb{C})(\forall \mathbf{x} \in \mathcal{V}_\mathbb{C})\, \Gamma(\alpha\mathbf{x}) = \overline{\alpha}\Gamma\mathbf{x}$ and $\Gamma^2 = I_{\mathcal{V}_\mathbb{C}}$, so Γ is a conjugate-linear bijection.
By Definition 5.3.10, $\Gamma T_\mathbb{C} = T_\mathbb{C}\Gamma$. Thus $(\forall \mu \in \mathbb{C})\, \Gamma S_\mu = S_{\overline{\mu}}\Gamma$. The corresponding power identity holds at exponent 0.
If it holds at $r \in \mathbb{Z}_{\geq 0}$, then it holds at $r + 1$ by composition with $\Gamma S_\lambda = S_{\overline{\lambda}}\Gamma$; $(\forall r \in \mathbb{Z}_{\geq 0})(\forall \mathbf{x} \in \mathcal{V}_\mathbb{C})\, \Gamma(S_\lambda^r \mathbf{x}) = S_{\overline{\lambda}}^r \Gamma\mathbf{x}$ by induction.

($\Rightarrow$) If $S_\lambda^k(\mathbf{u} + i\mathbf{v}) = \mathbf{0}$, then $S_{\overline{\lambda}}^k(\mathbf{u} - i\mathbf{v}) = \Gamma\mathbf{0} = \mathbf{0}$ by the power identity.
($\Leftarrow$) If $S_{\overline{\lambda}}^k(\mathbf{u} - i\mathbf{v}) = \mathbf{0}$, then $S_\lambda^k(\mathbf{u} + i\mathbf{v}) = \Gamma\mathbf{0} = \mathbf{0}$ by the power identity with $\overline{\lambda}$ in place of λ.
Structural Commentary: Conjugate-linearity intertwines every shifted power with its conjugate shift.
Key Insight: The involution Γ transports each shifted kernel to its conjugate kernel. □

Corollary 5.2.2. *Conjugate Eigenvalues Have Equal Multiplicity*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$. For $\lambda \in \operatorname{spec}(T_\mathbb{C})$:
(i) $\overline{\lambda} \in \operatorname{spec}(T_\mathbb{C})$;
(ii) $\dim_\mathbb{C} \mathcal{G}_\lambda(T_\mathbb{C}) = \dim_\mathbb{C} \mathcal{G}_{\overline{\lambda}}(T_\mathbb{C})$;
(iii) λ and $\overline{\lambda}$ have the same algebraic multiplicity.

Proof. For $\mu \in \mathbb{C}$, set $S_\mu := T_\mathbb{C} - \mu I_{\mathcal{V}_\mathbb{C}}$. Define $\Gamma : \mathcal{V}_\mathbb{C} \to \mathcal{V}_\mathbb{C}$ by $\Gamma(\mathbf{u} + i\mathbf{v}) := \mathbf{u} - i\mathbf{v}$. By Definition 5.3.6, $(\forall \mathbf{x}, \mathbf{y} \in \mathcal{V}_\mathbb{C})\, \Gamma(\mathbf{x} + \mathbf{y}) = \Gamma\mathbf{x} + \Gamma\mathbf{y}$.
The same definition implies $(\forall \alpha \in \mathbb{C})(\forall \mathbf{x} \in \mathcal{V}_\mathbb{C})\, \Gamma(\alpha\mathbf{x}) = \overline{\alpha}\Gamma\mathbf{x}$ and $\Gamma^2 = I_{\mathcal{V}_\mathbb{C}}$, so Γ is a conjugate-linear bijection.

(i) Choose a nonzero eigenvector $\mathbf{z} = \mathbf{u} + i\mathbf{v}$ for λ. By Lemma 5.2.1 with $k = 1$, $S_{\overline{\lambda}}\Gamma\mathbf{z} = \mathbf{0}$.
The vector $\Gamma\mathbf{z}$ is nonzero by $\Gamma^2 = I_{\mathcal{V}_\mathbb{C}}$, so $\overline{\lambda} \in \operatorname{spec}(T_\mathbb{C})$ by Definition 5.1.1.

(ii) Set $n := \dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}}$. By Theorem 4.1.6(i), the generalized eigenspaces are $\ker(S_\lambda^n)$ and $\ker(S_{\overline{\lambda}}^n)$.
By Lemma 5.2.1 with $k = n$ and $\Gamma^2 = I_{\mathcal{V}_{\mathbb{C}}}$, Γ restricts to a conjugate-linear bijection $\mathcal{G}_\lambda(T_{\mathbb{C}}) \to \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$.
The space $\mathcal{G}_\lambda(T_{\mathbb{C}})$ is finite-dimensional by Theorem 4.1.6(ii) and Corollary 1.3.14(ii).
Set $d := \dim_{\mathbb{C}} \mathcal{G}_\lambda(T_{\mathbb{C}})$ and choose a basis $(\mathbf{z}_1, \dots, \mathbf{z}_d)$ by Corollary 1.3.14(i).
If $\sum_{j=1}^{d} a_j \Gamma \mathbf{z}_j = \mathbf{0}$, then $\sum_{j=1}^{d} \overline{a_j} \mathbf{z}_j = \mathbf{0}$ by conjugate-linearity and $\Gamma^2 = I_{\mathcal{V}_{\mathbb{C}}}$. Hence $(\forall j \in \{1, \dots, d\})\, a_j = 0$ by independence.
Fix $\mathbf{y} \in \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$ and choose $b_1, \dots, b_d \in \mathbb{C}$ satisfying $\Gamma \mathbf{y} = \sum_{j=1}^{d} b_j \mathbf{z}_j$ by spanning.
Then $\mathbf{y} = \sum_{j=1}^{d} \overline{b_j} \Gamma \mathbf{z}_j$ by conjugate-linearity, $\Gamma^2 = I_{\mathcal{V}_{\mathbb{C}}}$. Thus $(\Gamma \mathbf{z}_1, \dots, \Gamma \mathbf{z}_d)$ is basis of $\mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$, its dimension is $d = \dim_{\mathbb{C}} \mathcal{G}_\lambda(T_{\mathbb{C}})$.

(iii) The algebraic multiplicities are the dimensions in (ii) by Proposition 5.1.4.

Structural Commentary: Kernel transport constructs an image basis; the multiplicity formula transfers its dimension equality.
Key Insight: Conjugation preserves generalized-eigenspace dimension. □

Corollary 5.2.3. *Odd-Dimensional Real Operators Have Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space with $\dim_{\mathbb{R}} \mathcal{V}$ odd. Then $(\forall T \in \mathcal{L}(\mathcal{V}))$ $\operatorname{spec}(T) \neq \varnothing$.

Proof. Fix $T \in \mathcal{L}(\mathcal{V})$ and set $n := \dim_{\mathbb{R}} \mathcal{V}$. Then $\dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}} = n$ by Proposition 5.3.7(iii).
The algebraic multiplicities of the eigenvalues of $T_{\mathbb{C}}$ sum to n by Proposition 5.1.5.
If $\operatorname{spec}(T_{\mathbb{C}}) \cap \mathbb{R} = \varnothing$, then the spectrum partitions into pairs $\{\lambda, \overline{\lambda}\}$.
Each pair has total equal to twice its common algebraic multiplicity by Corollary 5.2.2(i),(iii). Thus n would be even, a contradiction.
Thus $T_{\mathbb{C}}$ has a real eigenvalue, which is an eigenvalue of T by Proposition 5.3.12.
Structural Commentary: Parity compares the total complex multiplicity with the even totals of nonreal conjugate pairs.
Key Insight: An odd multiplicity sum cannot consist only of conjugate pairs. □

Theorem 5.2.4. *Characteristic Polynomial over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$. Let q_T be the characteristic polynomial of T.

(i) $q_T \in \mathbb{R}[z]$.
(ii) $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
(iii) The real zeros of q_T are precisely the eigenvalues of T.
(iv) $q_T(T) = 0$.

Proof.

(i) The relation $q_T \in \mathbb{R}[z]$ holds by Definition 2.1.15.
(ii) By Proposition 2.1.16(i), $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
(iii) By Proposition 2.1.16(ii), the zeros of q_T in $\mathbb{R}$ are precisely the eigenvalues of T.
(iv) Choose a basis β of $\mathcal{V}$ by Corollary 1.3.14(i), and set $\beta_{\mathbb{C}} := \operatorname{cpx}(\beta)$. The list $\beta_{\mathbb{C}}$ is a complex basis by Proposition 5.3.7(ii).
The identity $[T_{\mathbb{C}}]_{\beta_{\mathbb{C}} \leftarrow \beta_{\mathbb{C}}} = [T]_{\beta \leftarrow \beta}$ holds by Proposition 5.3.11(iii). Hence $q_{T_{\mathbb{C}}} = q_T$ by Definition 2.1.15.
By Theorem 3.1.2, $q_T(T_{\mathbb{C}}) = q_{T_{\mathbb{C}}}(T_{\mathbb{C}}) = 0$. Fix $\mathbf{u} \in \mathcal{V}$.
By (i), Proposition 5.3.11(ii), and linearity of polynomial evaluation, $\mathbf{0} = q_T(T_{\mathbb{C}})(\mathbf{u} + i\mathbf{0}) = q_T(T)\mathbf{u} + i\mathbf{0}$.
Thus $q_T(T)\mathbf{u} = \mathbf{0}$ by Definition 5.3.6(iv), so $q_T(T) = 0$ by the arbitrariness of $\mathbf{u}$.

Structural Commentary: Definition and root-degree properties settle (i)–(iii); unchanged coordinates transfer Cayley–Hamilton for (iv).
Key Insight: Complexification preserves the real coordinate matrix of T. □

The identity $q_{T_{\mathbb{C}}} = q_T$ holds by Proposition 5.3.11(iii) and Definition 2.1.15. If $\lambda \in \operatorname{spec}(T_{\mathbb{C}}) \setminus \mathbb{R}$ has algebraic multiplicity m, then $\overline{\lambda}$ has multiplicity m by Corollary 5.2.2(iii). Thus the paired factor is $(z - \lambda)^m (z - \overline{\lambda})^m = \left(z^2 - 2\operatorname{Re}(\lambda) z + |\lambda|^2\right)^m$.
Figure 64 records this real quadratic contribution.

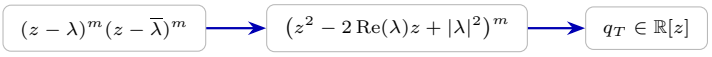

Figure 64: Conjugate factors pair into real quadratics.

Corollary 5.2.5. *q_T Is a Multiple of μ_T over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then:

(i) μ_T divides q_T in $\mathbb{R}[z]$;
(ii) $\deg \mu_T \leq \dim_{\mathbb{R}} \mathcal{V}$.

Proof. By Theorem 5.2.4(iv), $q_T(T) = 0$.

(i) The divisibility $\mu_T \mid q_T$ in $\mathbb{R}[z]$ holds by Theorem 2.2.4.
(ii) By (i) and Theorem 5.2.4(ii), $\deg \mu_T \leq \deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.

Structural Commentary: Real Cayley–Hamilton supplies an annihilator; minimal-polynomial divisibility then controls its degree.
Key Insight: The least monic annihilator divides q_T. □

Proposition 5.2.6. *Minimal Polynomial Survives Complexification*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$. Under $\mathbb{R}[z] \subseteq \mathbb{C}[z]$, $\mu_{T_{\mathbb{C}}} = \mu_T$.

Proof.

(i) *Extended Annihilation*: Fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.
Then $\mu_T(T_{\mathbb{C}})(\mathbf{u} + i\mathbf{v}) = \mathbf{0}$ by $\mu_T(T) = 0$, Proposition 5.3.11(ii), and linearity of polynomial evaluation.
Thus $\mu_T(T_{\mathbb{C}}) = 0$ by the arbitrariness of $\mathbf{u}, \mathbf{v}$, and $\mu_{T_{\mathbb{C}}}$ divides μ_T in $\mathbb{C}[z]$ by Theorem 2.2.4.

(ii) *Real-Part Descent*: Set $q := \mu_{T_{\mathbb{C}}} = r + is$, where $r, s \in \mathbb{R}[z]$. Since q is monic, r is monic and $\deg r = \deg q$.
Fix $\mathbf{u} \in \mathcal{V}$. By Definition 2.2.2, Proposition 5.3.11(ii), and linearity of polynomial evaluation, $\mathbf{0} = q(T_{\mathbb{C}})(\mathbf{u}+i\mathbf{0}) = r(T)\mathbf{u}+is(T)\mathbf{u}$.
Hence $r(T)\mathbf{u} = s(T)\mathbf{u} = \mathbf{0}$ by Definition 5.3.6(iv), so $r(T) = s(T) = 0$ by the arbitrariness of $\mathbf{u}$.
Therefore μ_T divides r in $\mathbb{R}[z]$ by Theorem 2.2.4.
(iii) *Degree Comparison*: By the first two stages and $\deg r = \deg q$, $\deg q \leq \deg \mu_T \leq \deg r = \deg q$; hence $\deg q = \deg \mu_T$.
(iv) *Monic Identification*: Choose $c \in \mathbb{C}^\times$ satisfying $\mu_T = cq$ by divisibility and degree equality.
Both polynomials are monic, so $c = 1$ and $\mu_{T_{\mathbb{C}}} = q = \mu_T$.

Structural Commentary: Extended annihilation and real-part descent force equal degrees before monicity identifies the divisors.
Key Insight: The real part of the complex minimal polynomial remains monic and annihilates T. □

5.3 Normal Primary Structure

Corollary 5.3.1. *Normal Operators Have No Nilpotent Primary Part*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$ be normal.
(i) $(\forall\, p \in \mathbb{C}[z])(\forall\, k \in \mathbb{Z}_{>0})\ \ker\big(p(T)^k\big) = \ker\big(p(T)\big)$.
(ii) $\mu_T(z) = \prod_{\lambda \in \operatorname{spec}(T)} (z - \lambda)$, *so μ_T is square-free.*

Proof. Set $n := \dim(\mathcal{V})$ and $J := \{1, \ldots, n\}$. By Theorem 6.1.11(i),(ii), choose an orthonormal eigenbasis $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$.
Fix $(\lambda_1, \ldots, \lambda_n) \in \mathbb{C}^n$ satisfying $(\forall\, j \in J)\, T\mathbf{e}_j = \lambda_j \mathbf{e}_j$.
(i) Fix $p \in \mathbb{C}[z]$ and $k \in \mathbb{Z}_{>0}$, and set $S := p(T)$. By Proposition 1.1.8, $(\forall\, j \in J)\, S\mathbf{e}_j = p(\lambda_j)\mathbf{e}_j$.
The identity $(\forall\, j \in J)\, S^k\mathbf{e}_j = p(\lambda_j)^k\mathbf{e}_j$ holds by induction on k.
Hence $\ker(S^k) = \operatorname{span}\{\mathbf{e}_j : j \in J,\ p(\lambda_j)^k = 0\} = \operatorname{span}\{\mathbf{e}_j : j \in J,\ p(\lambda_j) = 0\} = \ker(S)$ by uniqueness of β-coordinates and $k > 0$.
(ii) Set $A := [T]_{\beta \leftarrow \beta} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.
Its distinct eigenvalues are precisely the elements of $\operatorname{spec}(T)$ by Proposition 3.2.5 and Proposition 3.2.2(ii),(iii).
The identity $\mu_A(z) = \prod_{\lambda \in \operatorname{spec}(T)} (z - \lambda)$ holds by Theorem 3.2.1(i),(ii).
The identity $(\forall\, f \in \mathbb{C}[z])\, [f(T)]_{\beta \leftarrow \beta} = f(A)$ holds by Proposition 3.2.4 and Proposition 3.4.4.
Thus $(\forall\, f \in \mathbb{C}[z])\, [f(T) = 0 \iff f(A) = 0]$ by Theorem 3.4.5. Hence $\mu_T = \mu_A$ by Definitions 2.2.2 and 2.2.3.
Therefore $\mu_T(z) = \prod_{\lambda \in \operatorname{spec}(T)} (z - \lambda)$. Its linear factors are pairwise distinct, so μ_T is square-free.

Structural Commentary: The spectral basis controls polynomial kernels by scalar evaluation and identifies μ_T by diagonalizability.
Key Insight: Positive powers preserve the zero set of each scalar $p(\lambda)$. □

Proposition 5.3.2. *Coprime Normal Primary Pieces Are Orthogonal*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, and let $\mathcal{S}, \mathcal{R} \leq \mathcal{V}$ be T-invariant.
Let $(p, q, a, b) \in \mathbb{C}[z]^4$ satisfy $ap + bq = 1$, $p(T)|_{\mathcal{S}} = 0$, and $q(T)|_{\mathcal{R}} = 0$. Then $\mathcal{S} \perp \mathcal{R}$.

Proof. Fix $\mathbf{s} \in \mathcal{S}$ and $\mathbf{r} \in \mathcal{R}$.
(i) *Bézout Reduction*: By Proposition 1.1.6(i) and $ap + bq = 1$, $I_{\mathcal{V}} = a(T)p(T) + b(T)q(T)$. Hence $\mathbf{s} = b(T)q(T)\mathbf{s}$ by $p(T)|_{\mathcal{S}} = 0$.
(ii) *Adjoint Annihilation*: By Corollary 6.2.10(ii) and Proposition 2.3.9(ii), $b(T)^*\mathcal{R}, q(T)^*\mathcal{R} \subseteq \mathcal{R}$.
For $\mathbf{u}, \mathbf{v} \in \mathcal{R}$, $\langle q(T)^*\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, q(T)\mathbf{v}\rangle = 0$ by $q(T)|_{\mathcal{R}} = 0$. Hence $q(T)^*\mathcal{R} \subseteq \mathcal{R} \cap \mathcal{R}^\perp = \{\mathbf{0}\}$, so $q(T)^*|_{\mathcal{R}} = 0$.
(iii) *Orthogonality*: Since $b(T)^*\mathbf{r} \in \mathcal{R}$, $\langle \mathbf{s}, \mathbf{r}\rangle = \langle b(T)q(T)\mathbf{s}, \mathbf{r}\rangle = \langle \mathbf{s}, q(T)^*b(T)^*\mathbf{r}\rangle = 0$. Since $\mathbf{s}$ and $\mathbf{r}$ were arbitrary, $\mathcal{S} \perp \mathcal{R}$.

Structural Commentary: The Bézout identity transfers $\mathbf{s}$ to $q(T)$, while normality transfers $q(T)$-annihilation to $q(T)^*$.
Key Insight: Adjoint invariance and $q(T)|_{\mathcal{R}} = 0$ place $q(T)^*\mathcal{R}$ in $\mathcal{R} \cap \mathcal{R}^\perp$. □

6 Jordan and Canonical Forms

6.1 Jordan Chains and Nilpotent Blocks

Jordan chains convert nilpotent primary components into explicit blocks; each block records an eigenvalue and a chain length.

Definition 6.1.1. *Jordan Chains, Blocks, and Matrices*
(i) Let $n, k \in \mathbb{Z}_{>0}$, $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and $\lambda \in \mathbb{C}$. A *Jordan chain* for A at λ is $(\mathbf{x}_1, \ldots, \mathbf{x}_k) \in (\mathbb{C}^n)^k$ with $\mathbf{x}_1 \neq \mathbf{0}$.
For $\mathbf{x}_0 := \mathbf{0}$, its defining relations are $(\forall j \in \{1, \ldots, k\})\, (A - \lambda I_n)\mathbf{x}_j = \mathbf{x}_{j-1}$.
(ii) For $k \in \mathbb{Z}_{>0}$, the *nilpotent Jordan block* $J_k(0) \in \operatorname{Mat}(k, k, \mathbb{C})$ satisfies $(\forall i, j \in \{1, \ldots, k\})\, [J_k(0)]_{ij} = \delta_{j,i+1}$.
For $\lambda \in \mathbb{C}$, the *Jordan block* at λ is $J_k(\lambda) := \lambda I_k + J_k(0)$.
(iii) A *Jordan matrix* is a finite direct sum of Jordan blocks.
(iv) Let $\mathcal{V}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$.
An ordered basis β of $\mathcal{V}$ is a *Jordan basis* for T iff $[T]_{\beta \leftarrow \beta}$ is a Jordan matrix.

Proposition 6.1.2. *Jordan Chains as Matrix Columns*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. Then:
(i) Let $k \in \mathbb{Z}_{>0}$ and $\lambda \in \mathbb{C}$, and let $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$ be a Jordan chain for A at λ. Set $X := [\mathbf{x}_1\ \cdots\ \mathbf{x}_k]$; then $AX = XJ_k(\lambda)$.
(ii) Let $M \in \operatorname{Mat}(n, n, \mathbb{C})^\times$ be formed by concatenating Jordan chains for A, and let J be the matching ordered direct sum of blocks. Then $AM = MJ$ and $M^{-1}AM = J$.

Proof.
(i) Fix $j \in \{1, \ldots, k\}$. Then $(AX)_{\cdot,j} = \mathbf{x}_{j-1} + \lambda\mathbf{x}_j$ by the chain relation.
The identity $(XJ_k(\lambda))_{\cdot,j} = \mathbf{x}_{j-1} + \lambda\mathbf{x}_j$ holds by the entries of $J_k(\lambda)$. Hence $AX = XJ_k(\lambda)$ by equality of corresponding columns.
(ii) Write $M = [X_1\ \cdots\ X_s]$, where X_i is the column matrix of a chain of length k_i at λ_i.
The identities $(\forall i \in \{1, \ldots, s\})\, AX_i = X_iJ_{k_i}(\lambda_i)$ hold by part (i).
Thus $AM = MJ$ by block multiplication, and $M^{-1}AM = J$ by left multiplication with M^{-1}.

Structural Commentary: Column identities pass from individual chains to their ordered block concatenation.
Key Insight: Superdiagonal ones encode predecessor vectors in successive columns. □

Figure 65 depicts $N\mathbf{x}_1 = \mathbf{0}$ and $N\mathbf{x}_j = \mathbf{x}_{j-1}$ along a Jordan chain.

Figure 65: A Jordan chain for $N = A - \lambda I$; each arrow records the image under N.

Theorem 6.1.3. *Nilpotent Jordan Chain Basis*
Let $\mathcal{V}$ be a finite-dimensional vector space over $\mathbb{F}$, $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. For $\mathbf{v} \in \mathcal{V}$ and $m \in \mathbb{Z}_{\geq 0}$, set $\mathcal{C}_N(\mathbf{v}, m) := (N^{m-q}\mathbf{v})_{q=0}^{m}$.
For $r \in \mathbb{Z}_{\geq 0}$ and $(\mathbf{v}, \mathbf{m}) \in \mathcal{V}^r \times \mathbb{Z}_{\geq 0}^r$, let $\mathcal{B}_N(\mathbf{v}, \mathbf{m})$ denote the concatenation $(\mathcal{C}_N(\mathbf{v}_j, m_j))_{j=1}^{r}$.
Write $\mathsf{CB}_N(r, \mathbf{v}, \mathbf{m})$ when $\mathcal{B}_N(\mathbf{v}, \mathbf{m})$ is a basis of $\mathcal{V}$ and $(\forall j \in \{1, \dots, r\})\, N^{m_j+1}\mathbf{v}_j = \mathbf{0}$.
Then $(\exists r \in \mathbb{Z}_{\geq 0})(\exists \mathbf{v} \in \mathcal{V}^r)(\exists \mathbf{m} \in \mathbb{Z}_{\geq 0}^r)\, \mathsf{CB}_N(r, \mathbf{v}, \mathbf{m})$.

Proof. Set $n := \dim(\mathcal{V})$ and argue by strong induction on n, assuming the claim in every smaller dimension.

(i) *Zero Operator*: If $N = 0$, choose an ordered basis $(\mathbf{v}_1, \dots, \mathbf{v}_r)$ by Corollary 1.3.14(i).
Set $\mathbf{m} := (0, \dots, 0) \in \mathbb{Z}_{\geq 0}^r$; then $\mathcal{B}_N(\mathbf{v}, \mathbf{m}) = (\mathbf{v}_1, \dots, \mathbf{v}_r)$. The terminal relations $(\forall j \in \{1, \dots, r\})\, N\mathbf{v}_j = \mathbf{0}$ hold by $N = 0$.
Thus $\mathsf{CB}_N(r, \mathbf{v}, \mathbf{m})$ by its defining conditions; when $\mathcal{V} = \{\mathbf{0}\}$, $r = 0$ by linear independence of the chosen basis.

(ii) *Range Reduction*: Assume $N \neq 0$. Then $n > 0$ and $N^n = 0$ by Corollary 4.1.9.
If N were injective, then N^n would be injective by induction on n, contrary to $N^n = 0$ on the nonzero space $\mathcal{V}$.
Thus $\ker(N) \neq \{\mathbf{0}\}$ by noninjectivity, and $\dim \operatorname{range}(N) < n$ by Theorem 1.7.3(ii).
Set $\mathcal{W} := \operatorname{range}(N)$. The restriction $N|_{\mathcal{W}}$ is nilpotent by $N(\mathcal{W}) \subseteq \mathcal{W}$ and $(N|_{\mathcal{W}})^n = 0$.
Choose $t \in \mathbb{Z}_{\geq 0}$, $\mathbf{u} = (\mathbf{u}_j)_{j=1}^{t} \in \mathcal{W}^t$, and $\boldsymbol{\ell} = (\ell_j)_{j=1}^{t} \in \mathbb{Z}_{\geq 0}^t$ by the induction hypothesis for $N|_{\mathcal{W}}$.
Set $\mathcal{B}_{\mathcal{W}} := (\mathcal{C}_N(\mathbf{u}_j, \ell_j))_{j=1}^{t}$. This is a basis of $\mathcal{W}$ by equality of restriction powers and the induction hypothesis.
The identities $(\forall j \in \{1, \dots, t\})\, N^{\ell_j+1}\mathbf{u}_j = \mathbf{0}$ hold by the same facts.
For $j \in \{1, \dots, t\}$, choose $\mathbf{v}_j \in \mathcal{V}$ satisfying $N\mathbf{v}_j = \mathbf{u}_j$ by $\mathbf{u}_j \in \operatorname{range}(N)$. Set $\mathcal{L} := (\mathcal{C}_N(\mathbf{v}_j, \ell_j + 1))_{j=1}^{t}$.

(iii) *Lifted Independence*: Let $c_{j,q} \in \mathbb{F}$, where $1 \leq j \leq t$ and $0 \leq q \leq \ell_j + 1$.
Suppose $\sum_{j=1}^{t} \sum_{q=0}^{\ell_j+1} c_{j,q} N^q \mathbf{v}_j = \mathbf{0}$. The identities $(\forall j \in \{1, \dots, t\})\, N^{\ell_j+2}\mathbf{v}_j = N^{\ell_j+1}\mathbf{u}_j = \mathbf{0}$ hold by item (ii).
The image of the relation under N is $\sum_{j=1}^{t} \sum_{q=0}^{\ell_j} c_{j,q} N^q \mathbf{u}_j = \mathbf{0}$ by linearity and the terminal identities.
Thus $(\forall j \in \{1, \dots, t\})(\forall q \in \{0, \dots, \ell_j\})\, c_{j,q} = 0$ by linear independence of $\mathcal{B}_{\mathcal{W}}$.
The original relation reduces to $\sum_{j=1}^{t} c_{j,\ell_j+1} N^{\ell_j} \mathbf{u}_j = \mathbf{0}$ by these coefficient equalities.
Every remaining coefficient is zero by linear independence of $\mathcal{B}_{\mathcal{W}}$. Thus $\mathcal{L}$ is linearly independent by the vanishing of all coefficients.

(iv) *Kernel Completion*: The identities $(\forall j \in \{1, \dots, t\})(\forall q \in \{0, \dots, \ell_j\})\, N(N^q\mathbf{v}_j) = N^q\mathbf{u}_j$ hold by $N\mathbf{v}_j = \mathbf{u}_j$.
The inclusion $\mathcal{W} \subseteq N(\operatorname{span}(\mathcal{L}))$ holds by spanning of $\mathcal{B}_{\mathcal{W}}$. The reverse inclusion holds by $\mathcal{W} = \operatorname{range}(N)$, so $\mathcal{W} = N(\operatorname{span}(\mathcal{L}))$.
Choose $\mathbf{w}_1, \dots, \mathbf{w}_s$ such that $(\mathcal{L}, \mathbf{w}_1, \dots, \mathbf{w}_s)$ is a basis by Theorem 1.3.13.
Choose $(\mathbf{x}_i)_{i=1}^{s} \in \operatorname{span}(\mathcal{L})^s$ satisfying $(\forall i \in \{1, \dots, s\})\, N\mathbf{x}_i = N\mathbf{w}_i$ by $\mathcal{W} = N(\operatorname{span}(\mathcal{L}))$.
Set $\mathbf{z}_i := \mathbf{w}_i - \mathbf{x}_i$ for $i \in \{1, \dots, s\}$. Then $(\forall i \in \{1, \dots, s\})\, N\mathbf{z}_i = \mathbf{0}$ by construction.
The list $(\mathcal{L}, \mathbf{z}_1, \dots, \mathbf{z}_s)$ spans $\mathcal{V}$ by $\mathbf{w}_i = \mathbf{z}_i + \mathbf{x}_i$ and $\mathbf{x}_i \in \operatorname{span}(\mathcal{L})$. Suppose a linear combination of $(\mathcal{L}, \mathbf{z}_1, \dots, \mathbf{z}_s)$ is zero.
Substitute $\mathbf{z}_i = \mathbf{w}_i - \mathbf{x}_i$ in this relation. Every $\mathbf{w}_i$-coefficient is zero by linear independence of $(\mathcal{L}, \mathbf{w}_1, \dots, \mathbf{w}_s)$.
The remaining $\mathcal{L}$-coefficients are zero by linear independence of $\mathcal{L}$.
Thus $(\mathcal{L}, \mathbf{z}_1, \dots, \mathbf{z}_s)$ is linearly independent, hence a basis by the preceding spanning property.

(v) *Chain Verification*: Set $r := t + s$ and $\mathbf{v}' := (\mathbf{v}_1, \dots, \mathbf{v}_t, \mathbf{z}_1, \dots, \mathbf{z}_s)$. Set $\mathbf{m}' := (\ell_1 + 1, \dots, \ell_t + 1, \underbrace{0, \dots, 0}_{s \text{ entries}})$.
Then $\mathcal{B}_N(\mathbf{v}', \mathbf{m}') = (\mathcal{L}, \mathbf{z}_1, \dots, \mathbf{z}_s)$ is a basis by item (iv). The identities $(\forall j \in \{1, \dots, t\})\, N^{\ell_j+2}\mathbf{v}_j = \mathbf{0}$ hold by item (iii).
The identities $(\forall i \in \{1, \dots, s\})\, N\mathbf{z}_i = \mathbf{0}$ hold by item (iv). Thus $\mathsf{CB}_N(r, \mathbf{v}', \mathbf{m}')$ by its defining conditions.

Structural Commentary: Strong induction lifts a range-chain basis; kernel correction completes the lifted list.
Key Insight: The N-image of a lifted dependence lies in the known range basis. □

Figure 66 illustrates the nilpotent shift along one chain in the adapted basis order.

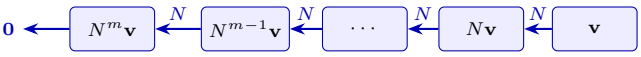

Figure 66: An ordered nilpotent chain shifts left under N and represents one Jordan block.

Computation Method 6.1.4. *Recursive Range-Lifting Jordan Construction*
Given $n \in \mathbb{Z}_{>0}$ and $A \in \operatorname{Mat}(n, n, \mathbb{C})$, construct basis matrix $M \in \operatorname{Mat}(n, n, \mathbb{C})^{\times}$. Construct Jordan matrix J with postcondition $AM = MJ$.
Fix $\lambda \in \operatorname{spec}(A)$ and set $\mathcal{G}_\lambda := \ker((A - \lambda I_n)^n)$. Set $N_\lambda := (A - \lambda I_n)|_{\mathcal{G}_\lambda}$.
The inclusion $(A - \lambda I_n)\mathcal{G}_\lambda \subseteq \mathcal{G}_\lambda$ holds by commutativity of powers of $A - \lambda I_n$. Thus $N_\lambda \in \mathcal{L}(\mathcal{G}_\lambda)$ and $N_\lambda^n = 0$ by the definition of $\mathcal{G}_\lambda$.

(i) If $N_\lambda = 0$, choose an ordered basis of $\mathcal{G}_\lambda$ and regard its vectors as singleton chains.
(ii) If $N_\lambda \neq 0$, recursively construct a chain basis for $N_\lambda|_{\operatorname{range}(N_\lambda)}$.
(iii) Given a returned chain $\mathcal{C}_{N_\lambda}(\mathbf{u}, \ell)$, choose $\mathbf{v} \in \mathcal{G}_\lambda$ with $N_\lambda\mathbf{v} = \mathbf{u}$. Replace the returned chain by $\mathcal{C}_{N_\lambda}(\mathbf{v}, \ell + 1)$.
(iv) Perform the kernel-completion correction from the proof of Theorem 6.1.3 on the lifted list.
Append the resulting kernel vectors as singleton chains.
(v) Repeat (i)–(iv) over $\lambda \in \operatorname{spec}(A)$, concatenate the resulting bases, and order the matching Jordan blocks identically.

If $N_\lambda \neq 0$, then N_λ is not injective because injectivity would make $N_\lambda^n = 0$ injective.
Thus its range dimension is strictly smaller by Theorem 1.7.3(ii), so the recursion terminates.
Steps (i)–(iv) produce a basis of $\mathcal{G}_\lambda$ by Theorem 6.1.3.
The concatenation in (v) is a basis by Theorem **??**(i).

Each returned $\mathcal{C}_{N_\lambda}(\mathbf{v}, m)$ satisfies $N_\lambda^{m+1}\mathbf{v} = \mathbf{0}$ by construction. It is a Jordan chain for A at λ by $N_\lambda = A - \lambda I_n$ on $\mathcal{G}_\lambda$. The postcondition $AM = MJ$ holds by Proposition 6.1.2(ii).

Proposition 6.1.5. *Powers of a Jordan Block*
Let $k \in \mathbb{Z}_{>0}$ and $\lambda \in \mathbb{C}$, using $0^0 := 1$ in (i). Then:

(i) $(\forall\, p \in \mathbb{Z}_{\geq 0})\; J_k(\lambda)^p = \sum_{r=0}^{\min(p,k-1)} \binom{p}{r} \lambda^{p-r} J_k(0)^r$.

(ii) $J_k(\lambda)^p \to 0$ *entrywise as* $p \to \infty$ *iff* $|\lambda| < 1$.

Proof.

(i) Fix $p \in \mathbb{Z}_{\geq 0}$, $R := J_k(0)$. For $0 \leq r < k$, matrix R^r has ones on rth superdiagonal, zeros elsewhere by induction on r.
The identity $(\forall r \in \mathbb{Z}_{\geq k})\; R^r = 0$ holds by the same induction.
By distributivity, each subset of the p factors in $(\lambda I_k + R)^p$ selects the factors contributing R.
For $r \in \{0, \ldots, p\}$, exactly $\binom{p}{r}$ subsets select r factors by Proposition 4.3.4.
The identity $(\lambda I_k + R)^p = \sum_{r=0}^{p} \binom{p}{r} \lambda^{p-r} R^r$ holds by commutativity and regrouping by subset size.
The identity in (i) holds by deleting the zero terms with $r \geq k$.

(ii) ($\Rightarrow$) The diagonal sequence satisfies $\lambda^p \to 0$ by entrywise convergence of $J_k(\lambda)^p$.
If $|\lambda| \geq 1$, then $(\forall p \in \mathbb{Z}_{\geq 0})\; |\lambda^p| \geq 1$, contrary to $\lambda^p \to 0$. Thus $|\lambda| < 1$ by contradiction.

(ii) ($\Leftarrow$) If $\lambda = 0$, then $(\forall p \in \mathbb{Z}_{\geq k})\; J_k(0)^p = 0$ by the power description in (i).
Assume $\rho := |\lambda| \in (0, 1)$ and fix $r \in \{0, \ldots, k-1\}$. Set $a_{p,r} := \binom{p}{r}\rho^{p-r}$ for $p \geq r$.
Set $\theta := \frac{1+\rho}{2} \in (\rho, 1)$. Choose $P \in \mathbb{Z}_{\geq r}$ satisfying $(\theta - \rho)(P+1) \geq \theta r$ by unboundedness of $\mathbb{Z}_{>0}$.
$(\forall p \in \mathbb{Z}_{\geq P})\; \frac{a_{p+1,r}}{a_{p,r}} = \frac{\rho(p+1)}{p+1-r} \leq \theta$ by factorial cancellation and the choice of P.
Hence $(\forall m \in \mathbb{Z}_{\geq 0})\; a_{P+m,r} \leq a_{P,r}\theta^m$ by induction on m. Fix $\varepsilon > 0$ and set $\delta := \theta^{-1} - 1 > 0$.
Choose $M \in \mathbb{Z}_{>0}$ satisfying $1 + M\delta > \frac{a_{P,r}}{\varepsilon}$ by unbounded $\mathbb{Z}_{>0}$. For $m \in \mathbb{Z}_{\geq M}$, $\theta^{-m} = (1+\delta)^m \geq 1 + m\delta$ by Proposition 4.3.7(i).
Thus $0 \leq a_{P+m,r} \leq \frac{a_{P,r}}{1+m\delta} < \varepsilon$, so $a_{p,r} \to 0$. Thus $(\forall r \in \{0, \ldots, k-1\})\; \binom{p}{r}\lambda^{p-r} \to 0$ as $p \to \infty$.
Each entry in the expansion from (i) is zero or $\binom{p}{r}\lambda^{p-r}$ for one $r \in \{0, \ldots, k-1\}$.
Its absolute value is therefore either 0 or $a_{p,r}$, and hence tends to 0. Thus $J_k(\lambda)^p \to 0$ by convergence of each entry.

Structural Commentary: Nilpotence truncates the expansion; diagonal entries and a geometric majorant resolve the limit criterion.
Key Insight: Each fixed superdiagonal has polynomial coefficients multiplied by geometric decay. □

6.2 Complex and Real Jordan Forms and Block Sizes

Jordan bases establish existence; shifted-power ranks recover block multiplicities and uniqueness.

Theorem 6.2.1. *Jordan Form over $\mathbb{C}$*
Let $\mathcal{V}$ be a complex vector space with $\dim_{\mathbb{C}} \mathcal{V} = n \in \mathbb{Z}_{>0}$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then $\exists$ an ordered basis $\mathcal{B}$ of $\mathcal{V}$ and a Jordan matrix J such that $[T]_{\mathcal{B}\leftarrow\mathcal{B}} = J$.

Proof.

(i) *Primary Splitting*: Let $\operatorname{spec}(T) = \{\lambda_1, \ldots, \lambda_m\}$ and set $G_j := \mathcal{G}_{\lambda_j}(T)$ for $j \in \{1, \ldots, m\}$. Set $N_j := (T - \lambda_j I_{\mathcal{V}})|_{G_j}$.
The identity $\mathcal{V} = \bigoplus_{j=1}^{m} G_j$ holds, and every N_j is nilpotent, by Theorem 5.1.1(i)–(ii).

(ii) *Chain Bases*: For $j \in \{1, \ldots, m\}$, choose an N_j-chain basis $\mathcal{B}_j$ of G_j by Theorem 6.1.3.

(iii) *Block Identification*: On a chain of length r in $\mathcal{B}_j$, the matrix of N_j is $J_r(0)$ by the chain recurrence.
The matrix of $T|_{G_j} = \lambda_j I_{G_j} + N_j$ on the same chain is $J_r(\lambda_j)$ by Definition 6.1.1(ii).

(iv) *Basis Assembly*: The concatenation $\mathcal{B} := (\mathcal{B}_1, \ldots, \mathcal{B}_m)$ is an ordered basis of $\mathcal{V}$ by the primary direct sum.
Matrix $[T]_{\mathcal{B}\leftarrow\mathcal{B}}$ is matching direct sum of Jordan blocks by item (iii). Hence it is Jordan matrix by Definition 6.1.1(iii).

Structural Commentary: Primary splitting isolates nilpotent shifts; chain bases concatenate into a Jordan basis.
Key Insight: Each primary restriction is $\lambda_j I_{G_j}$ plus its nilpotent shift. □

Figure 67 tracks primary summands through nilpotent chains to Jordan blocks.

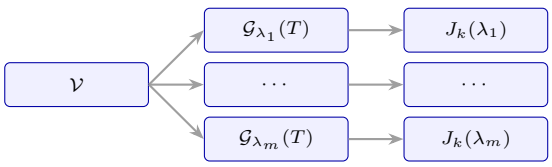

Figure 67: Primary pieces $\mathcal{G}_{\lambda_j}(T)$ decompose into nilpotent chains, which determine Jordan blocks.

Theorem 6.2.2. *Real Jordan Block Form*
Let $n \in \mathbb{Z}_{>0}$, $A \in \operatorname{Mat}(n, n, \mathbb{R})$. For $a \in \mathbb{R}$, $b > 0$, set $R(a,b) := \left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$. For $k \in \mathbb{Z}_{>0}$, define $\mathcal{J}_k(a,b) := [C_{rs}]_{r,s=1}^{k} \in \operatorname{Mat}(2k, 2k, \mathbb{R})$.
Its blocks satisfy $C_{rr} = R(a,b)$, $C_{r,r+1} = I_2$ for $r < k$, and $C_{rs} = 0_2$ otherwise. Then:

(i) *$\exists S \in \mathrm{GL}_n(\mathbb{R})$ such that $S^{-1}AS$ is a direct sum. Its scalar summands are $J_k(\lambda)$ with $(k, \lambda) \in \mathbb{Z}_{>0} \times \mathbb{R}$.*
Its remaining summands are $\mathcal{J}_k(a,b)$ with $(k, a, b) \in \mathbb{Z}_{>0} \times \mathbb{R} \times \mathbb{R}_{>0}$.

(ii) *For $k \in \mathbb{Z}_{>0}$, $a \in \mathbb{R}$, and $b > 0$, $\exists Q \in \mathrm{GL}_{2k}(\mathbb{C})$ such that $Q^{-1}\mathcal{J}_k(a,b)Q = J_k(a+ib) \oplus J_k(a-ib)$.*

Proof.

(*i*)

(a) *Primary Partition*: Regard A as an operator on $\mathbb{R}^n$ and let $A_{\mathbb{C}}$ be its complexification.
Set $\Lambda_{\mathbb{R}} := \operatorname{spec}(A_{\mathbb{C}}) \cap \mathbb{R}$, $\Lambda_+ := \{\lambda \in \operatorname{spec}(A_{\mathbb{C}}) : \operatorname{Im}(\lambda) > 0\}$. For $\mu \in \operatorname{spec}(A_{\mathbb{C}})$, set $G_\mu := \mathcal{G}_\mu(A_{\mathbb{C}})$. Set $\overline{\Lambda_+} := \{\overline{\lambda} : \lambda \in \Lambda_+\}$.
The spectrum is the disjoint union of $\Lambda_{\mathbb{R}}$, Λ_+, and $\overline{\Lambda_+}$ by Corollary 5.2.2(i).
Hence $\mathbb{C}^n = \bigoplus_{\lambda\in\Lambda_{\mathbb{R}}} G_\lambda \oplus \bigoplus_{\lambda\in\Lambda_+} (G_\lambda \oplus G_{\overline{\lambda}})$ by Theorem 5.1.1(i).

(b) *Real Primary Chains*: Fix $\lambda \in \Lambda_{\mathbb{R}}$ and set $G_\lambda^{\mathbb{R}} := \ker((A - \lambda I_n)^n)$.
The equality $(G_\lambda^{\mathbb{R}})_{\mathbb{C}} = \ker((A_{\mathbb{C}} - \lambda I_n)^n)$ holds by Proposition 5.3.11(ii) and equality of real and imaginary parts.
Thus $(G_\lambda^{\mathbb{R}})_{\mathbb{C}} = G_\lambda$ by Theorem 4.1.6(i). Subspace $G_\lambda^{\mathbb{R}}$ is invariant under $A - \lambda I_n$ by commutation with its nth power.
Set $N_\lambda := (A - \lambda I_n)|_{G_\lambda^{\mathbb{R}}}$; equality $N_\lambda^n = 0$ holds by definition of $G_\lambda^{\mathbb{R}}$. Choose an N_λ-chain basis $\mathcal{B}_\lambda$ by Theorem 6.1.3.
The corresponding matrix blocks of $A|_{G_\lambda^{\mathbb{R}}} = \lambda I_{G_\lambda^{\mathbb{R}}} + N_\lambda$ are $J_k(\lambda)$ by the chain recurrence and Definition 6.1.1.

(c) *Conjugate-Pair Chains*: Fix $\lambda = a + ib \in \Lambda_+$ and set $N_\lambda^{\mathbb{C}} := (A_{\mathbb{C}} - \lambda I_n)|_{G_\lambda}$.
This operator is nilpotent by Theorem 5.1.1(ii). Choose an $N_\lambda^{\mathbb{C}}$-chain basis by Theorem 6.1.3.
Each conjugate chain is a $\overline{\lambda}$-chain by conjugating its recurrence and using the real entries of A.
Conjugation is a conjugate-linear bijection $G_\lambda \to G_{\overline{\lambda}}$ by Lemma 5.2.1 and its involutivity.
Hence the conjugate chains form a basis of $G_{\overline{\lambda}}$ by coefficient conjugation.
Fix a chain $\mathcal{C} = (\mathbf{v}_1, \dots, \mathbf{v}_k)$, set $\mathbf{v}_0 := \mathbf{0}$, and write $\mathbf{v}_j = \mathbf{u}_j + i\mathbf{w}_j$ for $j \in \{0, \dots, k\}$.
Set $U_j := [\mathbf{u}_j\ \mathbf{w}_j]$ and $\mathcal{U}_{\mathcal{C}} := (\mathbf{u}_1, \mathbf{w}_1, \dots, \mathbf{u}_k, \mathbf{w}_k)$.
The lists $(\mathcal{C}, \overline{\mathcal{C}})$ and $\mathcal{U}_{\mathcal{C}}$ differ by an invertible change of coordinates by $\mathbf{u}_j = \frac{1}{2}(\mathbf{v}_j + \overline{\mathbf{v}}_j)$ and $\mathbf{w}_j = \frac{1}{2i}(\mathbf{v}_j - \overline{\mathbf{v}}_j)$.
Its inverse is $\mathbf{v}_j = \mathbf{u}_j + i\mathbf{w}_j$ and $\overline{\mathbf{v}}_j = \mathbf{u}_j - i\mathbf{w}_j$.
The identity $AU_j = U_j R(a, b) + U_{j-1}$ holds by separating the real and imaginary parts of the chain recurrence.
Thus the matrix of A on $\mathcal{U}_{\mathcal{C}}$ is $\mathcal{J}_k(a, b)$ by the block definition.

(d) *Real Basis Assembly*: Let $\mathcal{B}$ concatenate the bases $\mathcal{B}_\lambda$ for $\lambda \in \Lambda_{\mathbb{R}}$ and the lists $\mathcal{U}_{\mathcal{C}}$ over all chosen chains for $\lambda \in \Lambda_+$.
The complexified lists $\mathcal{B}_\lambda$ are bases of G_λ by (b) and Proposition 5.3.7(ii).
Hence $\mathcal{B}$ is a basis of $\mathbb{C}^n$ by the primary direct sum and the invertible replacements in (c).
The list $\mathcal{B}$ is a real basis of $\mathbb{R}^n$ by its complex linear independence and $\#\mathcal{B} = n$.
If S is its basis-column matrix, then $S \in \mathrm{GL}_n(\mathbb{R})$ and $S^{-1}AS$ has the asserted form by the block calculations in (b)–(c).

(*ii*) Fix $(k, a, b) \in \mathbb{Z}_{>0} \times \mathbb{R} \times \mathbb{R}_{>0}$ and let $(\mathbf{e}_j, \mathbf{f}_j)_{j=1}^k$ be the standard coordinate pairs of $\mathbb{C}^{2k}$, with $\mathbf{e}_0 = \mathbf{f}_0 := \mathbf{0}$.
Set $\lambda := a + ib$ and, for $j \in \{1, \dots, k\}$, set $\mathbf{v}_j := \mathbf{e}_j + i\mathbf{f}_j$.
The identity $(\forall j \in \{1, \dots, k\})\, (\mathcal{J}_k(a, b) - \lambda I_{2k})\mathbf{v}_j = \mathbf{v}_{j-1}$ holds by the block definition.
The conjugate identity $(\forall j \in \{1, \dots, k\})\, (\mathcal{J}_k(a, b) - \overline{\lambda} I_{2k})\overline{\mathbf{v}}_j = \overline{\mathbf{v}}_{j-1}$ holds by entrywise conjugation.
The list $(\mathbf{v}_1, \dots, \mathbf{v}_k, \overline{\mathbf{v}}_1, \dots, \overline{\mathbf{v}}_k)$ is a basis by the invertible pairwise relations $\mathbf{e}_j = \frac{1}{2}(\mathbf{v}_j + \overline{\mathbf{v}}_j)$ and $\mathbf{f}_j = \frac{1}{2i}(\mathbf{v}_j - \overline{\mathbf{v}}_j)$.
If Q is its basis-column matrix, then $Q^{-1}\mathcal{J}_k(a, b)Q = J_k(a + ib) \oplus J_k(a - ib)$ by the two chain recurrences.

Structural Commentary: Real and conjugate primary chains assemble the real blocks and their complex forms.
Key Insight: Pairing conjugate chains converts multiplication by $a + ib$ into $R(a, b)$. □

Corollary 6.2.3. *Diagonalizability in Jordan Form*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, and let $J = \bigoplus_{\alpha=1}^s J_{k_\alpha}(\mu_\alpha)$ be a Jordan form of A.

For $(\lambda, \alpha) \in \operatorname{spec}(A) \times \{1, \dots, s\}$, set $\delta_{\lambda\alpha} := \delta_{\lambda,\mu_\alpha}$, and set $b_J(\lambda) := \sum_{\alpha=1}^s \delta_{\lambda\alpha}$. Then:

(i) $(\forall \lambda \in \operatorname{spec}(A))\ \dim E_\lambda(A) = b_J(\lambda)$;
(ii) *A is diagonalizable if and only if* $(\forall \alpha \in \{1, \dots, s\})\, k_\alpha = 1$.

Proof. Choose $S \in \mathrm{GL}_n(\mathbb{C})$ such that $A = SJS^{-1}$. For $\lambda \in \operatorname{spec}(A)$, set $a_J(\lambda) := \sum_{\alpha=1}^s k_\alpha \delta_{\lambda\alpha}$.
The algebraic multiplicity of λ in A equals that in J by Theorem 2.1.7(ii). The latter multiplicity is $a_J(\lambda)$ by Proposition 2.1.6(iii).

(*i*) Fix $\lambda \in \operatorname{spec}(A)$. The map S restricts to an isomorphism $E_\lambda(J) \to E_\lambda(A)$ by $A = SJS^{-1}$.
If $\mu_\alpha = \lambda$, then $E_\lambda(J_{k_\alpha}(\mu_\alpha)) = \operatorname{span}(\mathbf{e}_1)$ by Definition 6.1.1(ii). If $\mu_\alpha \neq \lambda$, then $J_{k_\alpha}(\mu_\alpha) - \lambda I_{k_\alpha}$ is invertible by Lemma 4.1.10.
Thus $\dim E_\lambda(J_{k_\alpha}(\mu_\alpha)) = \delta_{\lambda\alpha}$ and $\dim E_\lambda(A) = \sum_{\alpha=1}^s \delta_{\lambda\alpha} = b_J(\lambda)$ by the direct-sum block structure.

(*ii*) ($\Rightarrow$) If A is diagonalizable, then $(\forall \lambda \in \operatorname{spec}(A))\, a_J(\lambda) = b_J(\lambda)$ by Theorem 2.1.12 and part (i).
Fix $\alpha \in \{1, \dots, s\}$ and set $\lambda := \mu_\alpha$. Then $k_\alpha = 1$ by $0 = a_J(\lambda) - b_J(\lambda) = \sum_{\beta=1}^s (k_\beta - 1)\delta_{\lambda\beta}$ and nonnegativity of its summands.

(*ii*) ($\Leftarrow$) Suppose $(\forall \alpha \in \{1, \dots, s\})\, k_\alpha = 1$. The matrix $A = SJS^{-1}$ is diagonalizable by diagonality of J.

Structural Commentary: Blockwise kernels count eigendirections; multiplicity equality then excludes every longer block.
Key Insight: A Jordan block is diagonal exactly when its length is one. □

Lemma 6.2.4. *Powers of a Nilpotent Jordan Block*
Let $k \in \mathbb{Z}_{>0}$, let $p \in \mathbb{Z}_{\geq 0}$, and set $N := J_k(0) \in \mathrm{Mat}(k, k, \mathbb{C})$. Then:

(i) $\operatorname{rank}(N^p) = \max(k - p, 0)$;
(ii) $\dim \ker(N^p) = \min(p, k)$.

Proof. Let $(\mathbf{e}_j)_{j=1}^k$ be the standard basis. Set $(q)_+ := \max(q, 0)$ and $\mathcal{E}_m := \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_m)$ for $m \in \{1, \dots, k\}$, with $\mathcal{E}_0 := \{\mathbf{0}\}$.
Set $\mathbf{e}_j := \mathbf{0}$ for $j \in \mathbb{Z}_{\leq 0}$. Then $(\forall j \in \{1, \dots, k\})\, N^p\mathbf{e}_j = \mathbf{e}_{j-p}$ by Definition 6.1.1(ii) and induction on p.

(i) If $p \geq k$, then $\operatorname{im}(N^p) = \mathcal{E}_0$ by the preceding power identity.
If $p < k$, then the nonzero images are exactly $\mathbf{e}_1, \dots, \mathbf{e}_{k-p}$, so $\operatorname{im}(N^p) = \mathcal{E}_{k-p}$.
Thus $\operatorname{im}(N^p) = \mathcal{E}_{(k-p)_+}$ and $\operatorname{rank}(N^p) = (k - p)_+ = \max(k - p, 0)$ in both cases.

(ii) The equality $\dim \ker(N^p) = k - (k - p)_+ = \min(p, k)$ holds by Corollary 3.3.1(i) and part (i).

Structural Commentary: The block shift identifies the surviving image basis; rank–nullity computes its complement.
Key Insight: The pth power deletes the first p chain positions. □

Definition 6.2.5. *Weyr Characteristic*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let $\lambda \in \mathrm{spec}(A)$.

(i) For $p \in \mathbb{Z}_{\geq 0}$, the *shifted-power rank* is $r_p(\lambda, A) := \mathrm{rank}((A-\lambda I_n)^p)$.
(ii) For $p \in \mathbb{Z}_{>0}$, the pth *Weyr number* is $w_p(\lambda, A) := r_{p-1}(\lambda, A) - r_p(\lambda, A)$.

Identity $(\forall p \in \mathbb{Z}_{>0})\, w_p(\lambda, A) = \dim\ker((A-\lambda I_n)^p) - \dim\ker((A-\lambda I_n)^{p-1}) \geq 0$ holds by Proposition 4.1.2(ii) and Corollary 3.3.1(i).
The inequality $w_1(\lambda, A) > 0$ holds by $\lambda \in \mathrm{spec}(A)$ and Corollary 3.3.1(i).
The equality $(\forall p \in \mathbb{Z}_{\geq n+1})\, w_p(\lambda, A) = 0$ holds by Theorem 4.1.4 and Corollary 3.3.1(i).
Thus $s_\lambda(A) := \max\{p \in \{1,\dots,n\} : w_p(\lambda, A) > 0\}$ is well-defined.
The *Weyr characteristic* of A at λ is $\big(w_p(\lambda, A)\big)_{p=1}^{s_\lambda(A)}$.

Theorem 6.2.6. *Jordan Block Sizes from Rank Drops*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $\lambda \in \mathrm{spec}(A)$, and let $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\mu_\alpha)$ be a Jordan form of A.
For $q \in \mathbb{Z}_{\geq 0}$, set $r_q := \mathrm{rank}\big((A-\lambda I_n)^q\big)$; for $q \in \mathbb{Z}_{>0}$, set $w_q := r_{q-1} - r_q$.
For $p \in \mathbb{Z}_{>0}$, set $I_p := \{\alpha \in \{1,\dots,s\} : \mu_\alpha = \lambda,\ k_\alpha \geq p\}$. Then:

(i) $w_p = \#I_p$;
(ii) $w_p - w_{p+1} = \#(I_p \setminus I_{p+1}) = \#\{\alpha : \mu_\alpha = \lambda,\ k_\alpha = p\}$.

Proof. Choose $S \in \mathrm{GL}_n(\mathbb{C})$ such that $A = SJS^{-1}$. The identity $(\forall q \in \mathbb{Z}_{\geq 0})\,(A-\lambda I_n)^q = S(J-\lambda I_n)^q S^{-1}$ holds by induction on q.
Hence $(\forall q \in \mathbb{Z}_{\geq 0})\, r_q = \mathrm{rank}((J-\lambda I_n)^q)$ by Theorem 3.6.6(ii). Set $B_\alpha := J_{k_\alpha}(\mu_\alpha)$ and $\rho_q(B) := \mathrm{rank}((B-\lambda I)^q)$.

Image of $(J-\lambda I_n)^q$ is direct sum of images of $(B_\alpha - \lambda I_{k_\alpha})^q$ by block-diagonal form. Hence $r_q = \sum_{\alpha=1}^{s} \rho_q(B_\alpha)$ by Theorem 2.3.4(i).

Set $\Delta_{p,\alpha} := \rho_{p-1}(B_\alpha) - \rho_p(B_\alpha)$. Then $w_p = \sum_{\alpha=1}^{s} \Delta_{p,\alpha}$ by subtracting the two block-rank sums.

(i) Fix $\alpha \in \{1,\dots,s\}$. If $\mu_\alpha \neq \lambda$, then $B_\alpha - \lambda I_{k_\alpha} = (\mu_\alpha - \lambda) I_{k_\alpha} + J_{k_\alpha}(0)$ is invertible by Lemma 4.1.10.
Thus $\Delta_{p,\alpha} = 0$ by full rank of every power. If $\mu_\alpha = \lambda$, then $\Delta_{p,\alpha} = 1$ exactly when $k_\alpha \geq p$ by Lemma 6.2.4(i).
Thus $\Delta_{p,\alpha} = \mathbf{1}_{I_p}(\alpha)$, where $\mathbf{1}_{I_p}$ is the indicator of I_p, and $w_p = \sum_{\alpha=1}^{s} \mathbf{1}_{I_p}(\alpha) = \#I_p$.
(ii) The equality $w_p - w_{p+1} = \#I_p - \#I_{p+1} = \#(I_p \setminus I_{p+1})$ holds by $I_{p+1} \subseteq I_p$ and part (i).
By the definitions of I_p and I_{p+1}, $I_p \setminus I_{p+1} = \{\alpha : \mu_\alpha = \lambda,\ k_\alpha = p\}$.

Structural Commentary: Similarity reduces shifted ranks to block sums; each block contributes a surviving-level indicator.
Key Insight: A λ-block contributes one rank drop through its final level. □

Figure 68 tracks shifted-rank drops to Jordan-block counts.

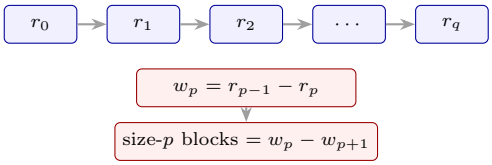

Figure 68: The pth shifted-rank drop counts chains of length at least p; consecutive differences count size-p blocks.

Corollary 6.2.7. *Uniqueness of Jordan Form*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let J, J' be Jordan matrices similar to A.
For $M \in \{J, J'\}$, let $\mathfrak{B}(M)$ be its multiset of Jordan blocks and set $m_M(\lambda, k) := \#\{B \in \mathfrak{B}(M) : B = J_k(\lambda)\}$. Then:

(i) $(\forall\, (\lambda, k) \in \mathrm{spec}(A) \times \mathbb{Z}_{>0})\, m_J(\lambda, k) = m_{J'}(\lambda, k)$;
(ii) $\exists$ *a permutation matrix P that permutes whole Jordan-block coordinate groups and satisfies $J' = P^{-1}JP$.*

Proof.

(i) Fix $(\lambda, k) \in \mathrm{spec}(A) \times \mathbb{Z}_{>0}$.
The equality $m_J(\lambda, k) = w_k(\lambda, A) - w_{k+1}(\lambda, A) = m_{J'}(\lambda, k)$ holds by Theorem 6.2.6(ii) with J and J' in turn.
(ii) The ordered block lists of J and J' admit a bijection matching equal blocks by part (i).
Let P be the coordinate permutation matrix that carries each block group of J to its matched group in J'.
Then $P^{-1}JP = J'$ by equality of the matched blocks.

Structural Commentary: Rank-drop multiplicities match equal blocks; a coordinate permutation aligns their ordered groups.
Key Insight: Every block multiplicity is intrinsic to the shifted-rank sequence. □

Computation Method 6.2.8. *Jordan Form from Shifted-Rank Drops*
Given $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n,n,\mathbb{C})$, the output is a Jordan matrix J similar to A. For $\lambda \in \mathrm{spec}(A)$, perform (i)–(iii).

(i) Compute $r_p(\lambda, A) := \mathrm{rank}((A-\lambda I_n)^p)$ through the first index $t_\lambda \in \mathbb{Z}_{>0}$ satisfying $r_{t_\lambda - 1}(\lambda, A) = r_{t_\lambda}(\lambda, A)$.
(ii) Set $w_p(\lambda, A) := r_{p-1}(\lambda, A) - r_p(\lambda, A)$ for $p \in \{1,\dots,t_\lambda\}$.
(iii) For $p \in \{1,\dots,t_\lambda - 1\}$, include $w_p(\lambda, A) - w_{p+1}(\lambda, A)$ copies of $J_p(\lambda)$.
(iv) Let J be the direct sum of the blocks obtained over $\lambda \in \mathrm{spec}(A)$.

The stopping index satisfies $t_\lambda \leq n+1$ by Theorem 4.1.4 and rank–nullity.
The stopping equality implies equality of adjacent shifted kernels by kernel inclusion and rank–nullity.
All later shifted-power ranks equal $r_{t_\lambda}(\lambda, A)$ by Proposition 4.1.3 and rank–nullity.
The matrix in (iv) is a Jordan form of A by the exact block counts in (iii) and Theorem 6.2.6(ii).

6.3 Products AB and BA and Jordan Structure

Proposition 6.3.1. *AB and BA Have the Same Characteristic Polynomial*
Let $\mathbb{F}$ be a field, let $n \in \mathbb{Z}_{>0}$, and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) $p_{AB} = p_{BA}$.
(ii) If A or B is invertible, then AB and BA are similar.
(iii) There exist singular $A_0, B_0 \in \mathrm{Mat}(2, 2, \mathbb{F})$ such that A_0B_0 and B_0A_0 are not similar.

Proof.

(i) Regard z as an indeterminate over $\mathbb{F}$ and set $M := \left(\begin{smallmatrix} I_n & -z^{-1}A \\ -B & I_n \end{smallmatrix}\right) \in \mathrm{Mat}(2n, 2n, \mathbb{F}(z))$.
By block multiplication, $\left(\begin{smallmatrix} I_n & 0 \\ B & I_n \end{smallmatrix}\right) M = \left(\begin{smallmatrix} I_n & -z^{-1}A \\ 0 & I_n - z^{-1}BA \end{smallmatrix}\right)$. Similarly, $\left(\begin{smallmatrix} I_n & z^{-1}A \\ 0 & I_n \end{smallmatrix}\right) M = \left(\begin{smallmatrix} I_n - z^{-1}AB & 0 \\ -B & I_n \end{smallmatrix}\right)$.
The two left factors have determinant 1 by Theorem 3.8.4 and Theorem 3.8.2(i),(ix).
The right sides have determinants $\det(I_n - z^{-1}BA)$ and $\det(I_n - z^{-1}AB)$, respectively, by Theorem 3.8.4 and Theorem 3.8.2(ix).
Thus $\det(I_n - z^{-1}AB) = \det M = \det(I_n - z^{-1}BA)$ by Theorem 3.8.2(vi).
For $T \in \{AB, BA\}$, $p_T(z) = \det(zI_n - T)$ by Definition 2.1.2.
The identity $\det(zI_n - T) = z^n \det(I_n - z^{-1}T)$ holds by repeated column linearity in Theorem 3.8.2(ii).
Hence $p_{AB} = p_{BA}$ by the preceding determinant equality.

(ii) If A is invertible, then $BA = A^{-1}(AB)A$. If B is invertible, then $AB = B^{-1}(BA)B$.
In either case, AB and BA are similar by Definition 6.2.1.

(iii) Set $A_0 := \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B_0 := \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Column spaces are one-dimensional, so both matrices have rank 1 by Definition 3.1.12.
Hence both are singular by Corollary 3.6.2. The identities $A_0B_0 = B_0$ and $B_0A_0 = 0$ hold by direct multiplication.
The products are not similar by their unequal ranks and Theorem 3.6.6(ii).

Structural Commentary: Elimination compares determinants; conjugation treats units; rank separates singular products.
Key Insight: The determinant identity survives singular factors, but conjugation need not. □

Theorem 6.3.2. *Nonzero Jordan Blocks of AB and BA*
Let $n \in \mathbb{Z}_{>0}$ and let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. For $\lambda \in \mathbb{C} \setminus \{0\}$, the Jordan blocks of AB and BA at λ coincide.

Proof. Fix $\lambda \in \mathbb{C} \setminus \{0\}$.

(i) *Auxiliary Similarity*: Set $X := \left(\begin{smallmatrix} AB & A \\ 0 & 0 \end{smallmatrix}\right)$ and $Y := \left(\begin{smallmatrix} 0 & A \\ 0 & BA \end{smallmatrix}\right)$. Set $P := \left(\begin{smallmatrix} I_n & 0 \\ B & I_n \end{smallmatrix}\right)$.
The identities $P^{-1} = \left(\begin{smallmatrix} I_n & 0 \\ -B & I_n \end{smallmatrix}\right)$ and $PXP^{-1} = Y$ hold by block multiplication. Hence X and Y are similar by Definition 6.2.1.

(ii) *Shifted Powers*: Set $C := AB - \lambda I_n$, $D := -\lambda I_n$, $G := BA - \lambda I_n$, and $E_1 = F_1 := A$.
For $p \in \mathbb{Z}_{>0}$, define $E_{p+1} := C^pA + E_pD$ and $F_{p+1} := D^pA + F_pG$.
For $p \in \mathbb{Z}_{>0}$, let $\mathcal{P}_p^X$ denote $(X - \lambda I_{2n})^p = \left(\begin{smallmatrix} C^p & E_p \\ 0 & D^p \end{smallmatrix}\right)$. Let $\mathcal{P}_p^Y$ denote $(Y - \lambda I_{2n})^p = \left(\begin{smallmatrix} D^p & F_p \\ 0 & G^p \end{smallmatrix}\right)$, and set $\mathcal{P}_p := (\mathcal{P}_p^X \wedge \mathcal{P}_p^Y)$.
The relations $\mathcal{P}_1$ and $(\forall\, p \in \mathbb{Z}_{>0})\,(\mathcal{P}_p \Longrightarrow \mathcal{P}_{p+1})$ hold by the definitions and block multiplication with the recurrences.
Hence $(\forall\, p \in \mathbb{Z}_{>0})\,\mathcal{P}_p$ by Theorem 2.6.2.

(iii) *Rank Splitting*: The matrix D is invertible by $\lambda \neq 0$. For $p \in \mathbb{Z}_{>0}$, set $U_p := \left(\begin{smallmatrix} I_n & -E_pD^{-p} \\ 0 & I_n \end{smallmatrix}\right)$. Set $V_p := \left(\begin{smallmatrix} I_n & -D^{-p}F_p \\ 0 & I_n \end{smallmatrix}\right)$.
Both matrices are invertible by Proposition 3.8.8.
Identity $U_p \left(\begin{smallmatrix} C^p & E_p \\ 0 & D^p \end{smallmatrix}\right) = C^p \oplus D^p$ holds by block multiplication. Identity $\left(\begin{smallmatrix} D^p & F_p \\ 0 & G^p \end{smallmatrix}\right) V_p = D^p \oplus G^p$ holds by block multiplication.
For $H, K \in \mathrm{Mat}(n, n, \mathbb{C})$, $\mathrm{im}(H \oplus K) = \mathrm{im}(H) \oplus \mathrm{im}(K)$ by the coordinate-block action.
The matrix D^p has inverse D^{-p} by induction on p, so $\mathrm{rank}(D^p) = n$ by Corollary 3.6.2.
The equality $\mathrm{rank}((X - \lambda I_{2n})^p) = \mathrm{rank}(C^p \oplus D^p)$ holds by item (ii), invertibility of U_p, and Theorem 3.6.6(ii).
Thus $\mathrm{rank}((X - \lambda I_{2n})^p) = n + \mathrm{rank}(C^p)$ by the image identity and Theorem 2.3.4(i).
The equality $\mathrm{rank}((Y - \lambda I_{2n})^p) = \mathrm{rank}(D^p \oplus G^p)$ holds by item (ii), invertibility of V_p, and Theorem 3.6.6(ii).
Thus $\mathrm{rank}((Y - \lambda I_{2n})^p) = n + \mathrm{rank}(G^p)$ by the image identity and Theorem 2.3.4(i).

(iv) *Jordan Recovery*: For $p \in \mathbb{Z}_{>0}$, $(Y - \lambda I_{2n})^p = P(X - \lambda I_{2n})^p P^{-1}$ by the identity in (i) and induction on p.
Thus $\mathrm{rank}(C^p) = \mathrm{rank}(G^p)$ by item (iii) and Theorem 3.6.6(ii). The equality also holds for $p = 0$ by $C^0 = G^0 = I_n$.
Polynomials p_{AB} and p_{BA} are equal by Proposition 6.3.1(i). Hence λ belongs to both spectra or to neither by Theorem 2.1.3(ii).
If λ is outside spectra, neither product has λ-block. If λ belongs to spectra, shifted-rank differences agree by rank equalities.
Hence each λ-block size has equal multiplicity by Theorem 6.2.6(ii).

Structural Commentary: Auxiliary similarity transfers rank data; Weyr differences recover each nonzero block.
Key Insight: The added diagonal block is invertible exactly when $\lambda \neq 0$. □

Figure 69 tracks the auxiliary similarity to the two shifted-rank sequences.

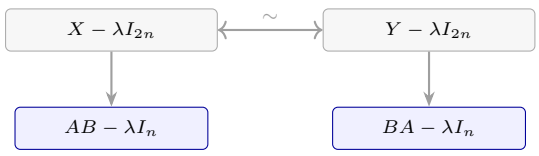

Figure 69: For $\lambda \neq 0$, the auxiliary similarity reduces the comparison to shifted ranks of AB and BA.

Corollary 6.3.3. *Similarity Criterion for AB and BA*
Let $n \in \mathbb{Z}_{>0}$ and let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. Then AB is similar to BA if and only if $(\forall\, p \in \{1, \ldots, n\})$ $\mathrm{rank}((AB)^p) = \mathrm{rank}((BA)^p)$.

Proof.

($\Rightarrow$) Suppose AB and BA are similar. Choose $S \in \mathrm{GL}_n(\mathbb{C})$ satisfying $BA = S(AB)S^{-1}$ by Definition 6.2.1.
For $p \in \mathbb{Z}_{>0}$, $(BA)^p = S(AB)^pS^{-1}$ by induction on p. Hence $\mathrm{rank}((AB)^p) = \mathrm{rank}((BA)^p)$ by Theorem 3.6.6(ii).

($\Leftarrow$) Assume the stated rank equalities. At exponent 0, both powers equal I_n and have rank n.
For $T \in \{AB, BA\}$, $(\operatorname{rank}(T^q))_{q=n}^{\infty}$ is constant by Theorem 4.1.4 and Corollary 3.3.1(i).
The two power-rank sequences agree by the assumed equalities, the exponent-zero identity, and stabilization.
The nonzero Jordan blocks coincide by Theorem 6.3.2. The characteristic polynomials agree by Proposition 6.3.1(i).
Hence $0 \in \operatorname{spec}(AB) \iff 0 \in \operatorname{spec}(BA)$ by Theorem 2.1.3(ii). Regard AB and BA as matrix-induced operators on $\mathbb{C}^n$. Choose Jordan matrices J_{AB} and J_{BA} similar to them by Theorem 6.2.1.
If 0 belongs to these spectra, the power-rank differences agree by the equal sequences.
In this case, the zero-block multiplicities agree by Theorem 6.2.6(ii) with J_{AB} and J_{BA} in turn.
If 0 lies outside spectra, neither product has zero-Jordan block. Block multisets agree by nonzero- and zero-block comparisons.
Choose a common block ordering and let J be its direct sum.
Coordinate permutation matrices $Q_{AB}, Q_{BA} \in \mathrm{GL}_n(\mathbb{C})$ exist by the matching block multisets.
They satisfy $Q_{AB}^{-1} J_{AB} Q_{AB} = J = Q_{BA}^{-1} J_{BA} Q_{BA}$ by block multiplication. Hence AB and BA are similar by Proposition 6.2.2.

Structural Commentary: Power-rank invariance handles one direction; zero-block recovery handles the converse.
Key Insight: Nonzero blocks already match, so ordinary power ranks detect the remainder. □

Computation Method 6.3.4. *Checking Whether AB and BA Are Similar*
Given $n \in \mathbb{Z}_{>0}$ and $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$, the output is a decision whether AB and BA are similar.

(i) Form $C := AB$ and $D := BA$.
(ii) Compute $\operatorname{rank}(C^p)$ and $\operatorname{rank}(D^p)$ for $p \in \{1, \ldots, n\}$.
(iii) Return "similar" if every computed rank pair agrees; otherwise return "not similar."
The first unequal pair certifies different zero-Jordan structures.

The nonzero Jordan blocks agree by Theorem 6.3.2. The output is correct by Corollary 6.3.3.

6.4 Nonderogatory Matrices and Commutants

Definition 6.4.1. *Nonderogatory Matrix*
Let $n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Matrix A *nonderogatory* if $(\forall \lambda \in \operatorname{spec}(A))$ its Jordan form contains exactly 1 block with eigenvalue λ. Equivalently, $(\forall \lambda \in \operatorname{spec}(A))\ \dim_{\mathbb{C}} E_\lambda(A) = 1$ by Corollary 6.2.3(i).

Lemma 6.4.2. *Commutant of One Jordan Block*
Let $m \in \mathbb{Z}_{>0}$, $\lambda \in \mathbb{C}$, $J := J_m(\lambda)$, and $X \in \mathrm{Mat}(m, m, \mathbb{C})$. If $XJ = JX$, then $(\exists a_0, \ldots, a_{m-1} \in \mathbb{C})\ X = \sum_{r=0}^{m-1} a_r J^r$.

Proof. Set $N := J_m(0)$; $J = \lambda I_m + N$, $XN = NX$ by $XJ = JX$. List $(\mathbf{e}_m, N\mathbf{e}_m, \ldots, N^{m-1}\mathbf{e}_m)$ is standard basis in reverse order.
Choose $b_0, \ldots, b_{m-1} \in \mathbb{C}$ satisfying $X\mathbf{e}_m = \sum_{r=0}^{m-1} b_r N^r \mathbf{e}_m$ by uniqueness of basis coordinates.
Set $p(z) := \sum_{r=0}^{m-1} b_r z^r$. For $s \in \{0, \ldots, m-1\}$, $p(N)N^s = N^s p(N)$ by Proposition 1.1.6(ii).
The equality $N^s X = X N^s$ holds by induction on s from $NX = XN$, and $(p(N) - X)N^s \mathbf{e}_m = 0$ by this equality and the choice of p.
Thus $p(N) = X$ by equality on a basis. Set $q(z) := p(z - \lambda)$.
By binomial expansion, $(\exists a_0, \ldots, a_{m-1} \in \mathbb{C})\ q(z) = \sum_{r=0}^{m-1} a_r z^r$. The equality $X = p(N) = q(J) = \sum_{r=0}^{m-1} a_r J^r$ holds by $J - \lambda I_m = N$.
Structural Commentary: Cyclic-basis equality propagates the image of $\mathbf{e}_m$ through every power of N.
Key Insight: Commutation determines the map from one cyclic vector. □

Theorem 6.4.3. *Nonderogatory Commutants Are Polynomials*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be nonderogatory. Then $\operatorname{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$.

Proof.

(i) *Polynomial Inclusion*: Fix $h \in \mathbb{C}[z]$ and regard A as an operator on $\mathbb{C}^n$.
The equality $h(A)A = Ah(A)$ holds by Proposition 1.1.6(ii) with the polynomials h and z. Hence $\{p(A) : p \in \mathbb{C}[z]\} \subseteq \operatorname{Comm}(A)$.

(ii) *Jordan Reduction*: By Theorem 6.2.1, choose $S \in \mathrm{GL}_n(\mathbb{C})$ and a Jordan matrix J such that $A = SJS^{-1}$.
By nonderogatoriness, choose $d, m_1, \ldots, m_d \in \mathbb{Z}_{>0}$ and pairwise distinct $\lambda_1, \ldots, \lambda_d \in \mathbb{C}$.
For $i \in \{1, \ldots, d\}$, set $J_i := J_{m_i}(\lambda_i)$. Then $J = \bigoplus_{i=1}^{d} J_i$ and $\sum_{i=1}^{d} m_i = n$.
Fix $B \in \operatorname{Comm}(A)$, set $X := S^{-1}BS$, and partition $X = [X_{ij}]$ conformally with J.
Then $XJ = JX$ by multiplying $BA = AB$ on the left by S^{-1} and on the right by S.
The equality $(\forall i \in \{1, \ldots, d\})\ \operatorname{spec}(J_i) = \{\lambda_i\}$ holds by Proposition 2.1.6(ii). Thus the spectra of $J_1, \ldots, J_d$ are pairwise disjoint.
Hence $(\forall i, j \in \{1, \ldots, d\})\ [i \neq j \Rightarrow X_{ij} = 0]$ by Corollary 3.5.5.
Thus $X = \bigoplus_{i=1}^{d} X_{ii}$. For $i \in \{1, \ldots, d\}$, choose $p_i \in \mathbb{C}[z]$ such that $X_{ii} = p_i(J_i)$ by Lemma 6.4.2.

(iii) *Congruence Projectors*: For $v, w, f \in \mathbb{C}[z]$ with $f \neq 0$, write $v \equiv w \pmod f$ if $f \mid (v - w)$.
For $i \in \{1, \ldots, d\}$, set $f_i(z) := (z - \lambda_i)^{m_i}$ and $F_i(z) := \prod_{j \neq i} f_j(z)$, where an empty product equals 1.
Set $c_i := F_i(\lambda_i) = \prod_{j \neq i} (\lambda_i - \lambda_j)^{m_j} \neq 0$ and $u_i(z) := c_i^{-1} F_i(z) - 1$.
The divisibility $z - \lambda_i \mid u_i(z)$ holds by $u_i(\lambda_i) = 0$ and expansion in powers of $z - \lambda_i$. Set $e_i(z) := c_i^{-1} F_i(z) \sum_{r=0}^{m_i - 1} (-u_i(z))^r$.
The identity $e_i = 1 - (-u_i)^{m_i}$ holds by $c_i^{-1} F_i = 1 + u_i$ and the finite geometric-sum identity. Hence $e_i \equiv 1 \pmod{f_i}$ by $z - \lambda_i \mid u_i$.
The relation $(\forall k \in \{1, \ldots, d\} \setminus \{i\})\ e_i \equiv 0 \pmod{f_k}$ holds by $f_k \mid F_i$.

(iv) *Block Interpolation*: Set $p := \sum_{i=1}^{d} p_i e_i$. Then $(\forall i \in \{1, \ldots, d\})\ p \equiv p_i \pmod{f_i}$ by item (iii).
For $i \in \{1, \ldots, d\}$, choose $q_i \in \mathbb{C}[z]$ satisfying $p - p_i = f_i q_i$. The identity $J_i - \lambda_i I_{m_i} = J_{m_i}(0)$ holds by Definition 6.1.1(ii).
Hence $f_i(J_i) = J_{m_i}(0)^{m_i} = 0$ by Lemma 6.2.4(i) and Definition 3.1.12.

The equality $(p-p_i)(J_i) = f_i(J_i)q_i(J_i) = 0$ holds by Proposition 1.1.6(i). Thus $p(J_i) = p_i(J_i) = X_{ii}$.

The identity $p(J) = \bigoplus_{i=1}^{d} p(J_i)$ holds by Proposition 3.8.6(iv) and induction on d. Hence $p(J) = \bigoplus_{i=1}^{d} X_{ii} = X$ by item (ii).

(v) *Conjugation Back*: The identity $(\forall r \in \mathbb{Z}_{\geq 0})\, A^r = SJ^rS^{-1}$ holds by induction on r from $A = SJS^{-1}$.
Thus $p(A) = Sp(J)S^{-1} = SXS^{-1} = B$ by polynomial evaluation.
The claimed equality follows by the resulting reverse inclusion and item (i).

Structural Commentary: Block separation isolates local polynomials; congruence projectors merge them before conjugation.
Key Insight: Truncated geometric sums realize every block congruence in one polynomial. □

Figure 70 tracks local commutant blocks into one polynomial satisfying every Jordan congruence.

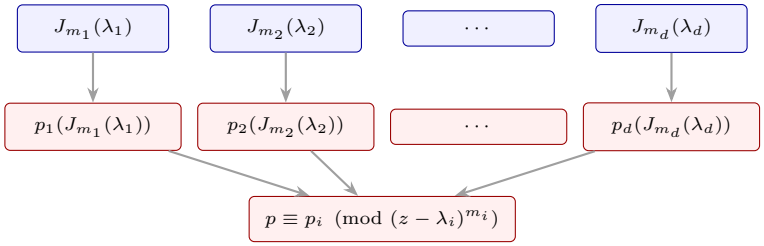

Figure 70: Blockwise polynomial data merge into one polynomial modulo the Jordan-block powers.

Corollary 6.4.4. *Simple Spectrum Commutants Are Polynomials*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have n distinct eigenvalues. Then:
(i) $\mathrm{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$;
(ii) $\dim_{\mathbb{C}} \mathrm{Comm}(A) = n$.

Proof. List the eigenvalues as $\lambda_1, \ldots, \lambda_n$ and choose corresponding eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$.
The list $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent by Theorem 5.1.5. It is a basis of $\mathbb{C}^n$ by Proposition 1.4.4(ii). Set $S := [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$.
Then $S \in \mathrm{Mat}(n,n,\mathbb{C})^{\times}$ by Corollary 3.6.1. The identity $A = S\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)S^{-1}$ holds by the eigenvector relations.
This Jordan form has one block per eigenvalue, so A is nonderogatory by Definition 6.4.1.
The equality $(\forall i \in \{1, \ldots, n\})\ \dim_{\mathbb{C}} E_{\lambda_i}(A) = 1$ holds by Corollary 6.2.3(i).

(i) The equality $\mathrm{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$ holds by Theorem 6.4.3.

(ii) The equality $\dim_{\mathbb{C}} \mathrm{Comm}(A) = \sum_{i=1}^{n} 1^2 = n$ holds by Theorem 5.4.12(ii).

Structural Commentary: A full eigenbasis supplies nonderogatoriness; its eigenline dimensions feed the commutant formula.
Key Insight: A simple spectrum contributes one commutant parameter per eigenline. □

6.5 Jordan Operator Functions and Square Roots

For $z \in \mathbb{C}$, set $e^z := \sum_{m=0}^{\infty} \frac{z^m}{m!}$.

Definition 6.5.1. *Matrix Exponential*
Let $n \in \mathbb{Z}_{>0}$ and $C \in \mathrm{Mat}(n,n,\mathbb{C})$. The *matrix exponential* of C is $e^C = \exp(C) := \sum_{m=0}^{\infty} \frac{C^m}{m!}$.

Proposition 6.5.2. *Entrywise Absolute Convergence of the Matrix Exponential*
Let $n \in \mathbb{Z}_{>0}$ and $C \in \mathrm{Mat}(n,n,\mathbb{C})$. Then $(\forall i,j \in \{1, \ldots, n\})\ \sum_{m=0}^{\infty} \frac{|(C^m)_{ij}|}{m!} < \infty$.

Proof. Set $M := \max_{1 \leq i,j \leq n} |C_{ij}|$. By induction, $(\forall m \in \mathbb{Z}_{>0})(\forall i,j \in \{1, \ldots, n\})\ |(C^m)_{ij}| \leq n^{m-1}M^m$.
For $(i,j) \in \{1, \ldots, n\}^2$, $|(C^0)_{ij}| \leq 1$ and $\sum_{m=0}^{\infty} \frac{|(C^m)_{ij}|}{m!} \leq 1 + \frac{e^{nM}-1}{n} < \infty$ by the scalar exponential series.
Structural Commentary: The entrywise power bound reduces matrix convergence to one scalar exponential majorant.
Key Insight: Each matrix multiplication contributes at most one factor of nM. □

The real scalar facts used below are $(\forall x \in \mathbb{R})\, e^x > 0$ and $(\forall x \in \mathbb{R}_{>0})(\exists \ell \in \mathbb{R})\, e^{\ell} = x$.
For complex scalars, $(\forall \lambda \in \mathbb{C} \setminus \{0\})(\exists \ell \in \mathbb{C})\, e^{\ell} = \lambda$ and $(\forall \lambda \in \mathbb{C})(\exists \alpha \in \mathbb{C})\, \alpha^2 = \lambda$.

Proposition 6.5.3. *Exponential of a Jordan Block*
Let $k \in \mathbb{Z}_{>0}$ and $\lambda, t \in \mathbb{C}$. Then $e^{tJ_k(\lambda)} = e^{\lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} J_k(0)^r$.

Proof. Set $N := J_k(0)$, so $J_k(\lambda) = \lambda I_k + N$ and $N^k = 0$. Adopt $\binom{m}{r} = 0$ for $r > m$.
For $m \in \mathbb{Z}_{\geq 0}$, $(tJ_k(\lambda))^m = \sum_{r=0}^{k-1} \binom{m}{r} (\lambda t)^{m-r} t^r N^r$ by induction and commutation of λI_k with N.
For $M \in \mathbb{Z}_{\geq 0}$ and $r \in \{0, \ldots, k-1\}$, the coefficient of N^r in the Mth exponential partial sum is $a_{M,r} := \sum_{m=r}^{M} \frac{1}{m!} \binom{m}{r} (\lambda t)^{m-r} t^r$.
The limit $\lim_{M \to \infty} a_{M,r} = \frac{t^r}{r!} \sum_{s=0}^{\infty} \frac{(\lambda t)^s}{s!} = \frac{t^r}{r!} e^{\lambda t}$ follows by setting $s = m - r$.
The asserted formula holds by taking the entrywise limit over the finite family $(N^r)_{r=0}^{k-1}$.
Structural Commentary: Jordan binomial expansion separates scalar powers from finitely many nilpotent powers before taking limits.
Key Insight: Nilpotence leaves finitely many scalar exponential coefficients. □

Proposition 6.5.4. *Commuting and Similarity Laws for the Exponential*
Let $n \in \mathbb{Z}_{>0}$, let $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$, and let $S \in \mathrm{Mat}(n,n,\mathbb{C})^\times$. Then:
(i) $AB = BA \implies e^{A+B} = e^A e^B$.
(ii) $e^{SAS^{-1}} = Se^A S^{-1}$.

Proof.
(i) Since A and B commute, $(A+B)^m = \sum_{r=0}^{m} \binom{m}{r} A^r B^{m-r}$ by induction on $m \in \mathbb{Z}_{\geq 0}$.
The Cauchy product is valid by entrywise absolute convergence. Hence $e^A e^B = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{r=0}^{m} \binom{m}{r} A^r B^{m-r} = e^{A+B}$ by the binomial identity.
(ii) Set $X := SAS^{-1}$. Then $(\forall m \in \mathbb{Z}_{\geq 0})\ X^m = SA^m S^{-1}$ by induction from $S^{-1}S = I_n$.
Thus $(\forall M \in \mathbb{Z}_{\geq 0})\ \sum_{m=0}^{M} \frac{X^m}{m!} = S\left(\sum_{m=0}^{M} \frac{A^m}{m!}\right) S^{-1}$. The equality $e^X = Se^A S^{-1}$ holds by entrywise limits.

Structural Commentary: Binomial powers control products; conjugated powers control similarity before entrywise limits.
Key Insight: Absolute convergence permits the required finite-degree regrouping. □

Theorem 6.5.5. *Analytic Power Series on Jordan Blocks*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let $f(z) = \sum_{m=0}^{\infty} c_m z^m$ have radius of convergence $R \in (0,\infty]$.
Set $\rho(A) := \max_{\lambda \in \mathrm{spec}(A)} |\lambda|$ and $D_R := \{z \in \mathbb{C} : |z| < R\}$, and suppose $\rho(A) < R$.
For $r \in \mathbb{Z}_{\geq 0}$ and $\lambda \in D_R$, set $f^{(r)}(\lambda) := \sum_{m=r}^{\infty} \frac{m!}{(m-r)!} c_m \lambda^{m-r}$. For $k \in \mathbb{Z}_{>0}$, set $N_k := J_k(0)$ and $F_k(\lambda) := \sum_{r=0}^{k-1} \frac{f^{(r)}(\lambda)}{r!} N_k^r$.
For $s, k_1, \ldots, k_s \in \mathbb{Z}_{>0}$ and $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$ with $\lambda_\alpha \in D_R$, set $f(J) := \bigoplus_{\alpha=1}^{s} F_{k_\alpha}(\lambda_\alpha)$.
Then:
(i) $(\forall (k,\lambda) \in \mathbb{Z}_{>0} \times D_R)\ \sum_{m=0}^{\infty} c_m J_k(\lambda)^m = F_k(\lambda)$.
(ii) The series $f(A) := \sum_{m=0}^{\infty} c_m A^m$ converges.
(iii) If $A = SJS^{-1}$ is a Jordan decomposition, then $f(A) = Sf(J)S^{-1}$.

Proof. For $M \in \mathbb{Z}_{\geq 0}$, set $P_M(X) := \sum_{m=0}^{M} c_m X^m$, and adopt $\binom{m}{r} = 0$ for $r > m$.
(i) Fix $(k,\lambda) \in \mathbb{Z}_{>0} \times D_R$ and set $N := N_k$.
Then $(\forall m \in \mathbb{Z}_{\geq 0})\ J_k(\lambda)^m = \sum_{r=0}^{k-1} \binom{m}{r} \lambda^{m-r} N^r$ by $N^k = 0$ and the binomial identity.
If $\lambda = 0$, only the term $m = r$ is nonzero in $f^{(r)}(0)$, and $f^{(r)}(0) = r! c_r$. Suppose $\lambda \neq 0$, and choose η with $|\lambda| < \eta < R$.
For fixed r, the sequence $m^r (|\lambda|/\eta)^m$ is bounded, and $\sum_{m=0}^{\infty} |c_m| \eta^m < \infty$.
For $m \geq r$, $\frac{m!}{(m-r)!} |c_m| |\lambda|^{m-r} \leq |\lambda|^{-r} m^r (|\lambda|/\eta)^m |c_m| \eta^m$.
Hence each differentiated series defining $F_k(\lambda)$ converges absolutely.
For $M \in \mathbb{Z}_{\geq 0}$, $P_M(J_k(\lambda)) = \sum_{r=0}^{k-1} \left(\sum_{m=r}^{M} c_m \binom{m}{r} \lambda^{m-r} \right) N^r$, with an empty inner sum interpreted as 0.
For $r \in \{0, \ldots, k-1\}$, the coefficient tends to $\frac{f^{(r)}(\lambda)}{r!}$. The outer sum is finite, so $P_M(J_k(\lambda)) \to F_k(\lambda)$.
(ii) By Theorem 6.2.1, choose $S \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ and $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$ such that $A = SJS^{-1}$.
Each $\lambda_\alpha \in D_R$ by $|\lambda_\alpha| \leq \rho(A) < R$. Thus $P_M(J) \to f(J)$ by part (i) and Proposition 3.8.6(iv).
By induction, $(\forall m \in \mathbb{Z}_{\geq 0})\ A^m = SJ^m S^{-1}$. Hence $P_M(A) = SP_M(J)S^{-1} \to Sf(J)S^{-1}$, so $f(A)$ converges.
(iii) By part (ii), $f(A) = \lim_{M\to\infty} P_M(A) = S\left(\lim_{M\to\infty} P_M(J)\right) S^{-1} = Sf(J)S^{-1}$.

Structural Commentary: Blockwise binomial limits establish convergence before similarity transfers the resulting function.
Key Insight: Derivatives are precisely the nilpotent-power coefficients. □

Proposition 6.5.6. *Closed Form for a 2×2 Exponential*
Let $a, b, c, d \in \mathbb{C}$ and set $A := \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Set $\delta := \frac{a+d}{2}$ and $B := A - \delta I_2$. Choose $\Delta \in \mathbb{C}$ with $\Delta^2 = \left(\frac{a-d}{2}\right)^2 + bc$.
Define $\cosh w := \sum_{m=0}^{\infty} \frac{w^{2m}}{(2m)!}$ and $\psi(w) := \sum_{m=0}^{\infty} \frac{w^{2m}}{(2m+1)!}$. Then $e^A = e^\delta \big(\cosh(\Delta) I_2 + \psi(\Delta) B\big)$.

Proof. The relations $\mathrm{tr}(B) = 0$ and $\det(B) = -\Delta^2$ hold by direct calculation. Hence $B^2 = \Delta^2 I_2$ by Theorem 3.1.1.
Thus $(\forall m \in \mathbb{Z}_{\geq 0})\ B^{2m} = \Delta^{2m} I_2$ and $B^{2m+1} = \Delta^{2m} B$.
Hence $e^B = \cosh(\Delta) I_2 + \psi(\Delta) B$ by separating the even and odd exponential terms.
The matrices δI_2 and B commute, so $e^A = e^{\delta I_2} e^B = e^\delta \big(\cosh(\Delta) I_2 + \psi(\Delta) B\big)$ by Proposition 6.5.4(i).
Structural Commentary: Cayley–Hamilton reduces all powers to even scalar terms and odd multiples of B.
Key Insight: The series ψ remains defined when $\Delta = 0$. □

Theorem 6.5.7. *Spectral Mapping and Jordan Structure of the Exponential*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let $A = SJS^{-1}$ be a Jordan decomposition with $S \in \mathrm{Mat}(n,n,\mathbb{C})^\times$.
Set $E := e^A$ and write $J = \bigoplus_{\alpha=1}^{s} B_\alpha$, where $s, k_1, \ldots, k_s \in \mathbb{Z}_{>0}$ and $B_\alpha := J_{k_\alpha}(\lambda_\alpha)$.

For $\rho \in \mathbb{C}$, set $I_\rho := \{\alpha \in \{1,\dots,s\} : e^{\lambda_\alpha} = \rho\}$ and $\Lambda_\rho(A) := \{\lambda \in \operatorname{spec}(A) : e^\lambda = \rho\}$.
Let $\mathcal{S}_\rho(M)$ denote the multiset of ρ-Jordan block sizes of M. Then:

(i) $\operatorname{spec}(E) = \{e^\lambda : \lambda \in \operatorname{spec}(A)\}$.
(ii) $(\forall\, \rho \in \operatorname{spec}(E))\ \mathcal{G}_\rho(E) = \bigoplus_{\lambda \in \Lambda_\rho(A)} \mathcal{G}_\lambda(A)$.
(iii) $(\forall\, \rho \in \operatorname{spec}(E))\ \mathcal{S}_\rho(E) = \{k_\alpha : \alpha \in I_\rho\}$ *as multisets.*
(iv) $\det(E) = e^{\operatorname{tr}(A)}$.

Proof. Set $\sigma(M) := \operatorname{spec}(M)$, $E_J := e^J$, and $E_\alpha := e^{B_\alpha}$. Then $E = SE_JS^{-1}$ by Proposition 6.5.4(ii).
The identity $E_J = \bigoplus_{\alpha=1}^{s} E_\alpha$ holds by Proposition 3.8.6(iv) for exponential partial sums and entrywise limits.
Let V_α be the coordinate space of B_α and set $K_\rho := \bigoplus_{\alpha \in I_\rho} V_\alpha$.
Write $B_\alpha = \lambda_\alpha I_{k_\alpha} + N_\alpha$, set $q_\alpha(z) := \sum_{r=0}^{k_\alpha - 1} \frac{z^r}{(r+1)!}$, and set $Q_\alpha := q_\alpha(N_\alpha)$. Then $E_\alpha - e^{\lambda_\alpha} I_{k_\alpha} = e^{\lambda_\alpha} N_\alpha Q_\alpha$ by Proposition 6.5.3.
Set $h_\alpha(z) := \sum_{r=1}^{k_\alpha - 1} \frac{z^{r-1}}{(r+1)!}$ and $M_\alpha := Q_\alpha - I_{k_\alpha} = N_\alpha h_\alpha(N_\alpha)$. Then $M_\alpha^{k_\alpha} = 0$.
Thus $Q_\alpha^{-1} = \sum_{j=0}^{k_\alpha - 1} (-M_\alpha)^j$ by direct multiplication. Therefore Q_α is invertible and commutes with N_α.
The identity $e^{\lambda_\alpha} e^{-\lambda_\alpha} = 1$ holds by Proposition 6.5.4(i) for 1×1 matrices, so $e^{\lambda_\alpha} \neq 0$.

(i) Each E_α is upper triangular with constant diagonal e^{λ_α} by Proposition 6.5.3.
Hence $\sigma(E_J) = \{e^{\lambda_\alpha} : \alpha \in \{1,\dots,s\}\}$ by Proposition 2.1.6(ii). The equality $\sigma(E) = \sigma(E_J)$ holds by Theorem 2.1.7(i).
Similarly, $\sigma(A) = \sigma(J) = \{\lambda_\alpha : \alpha \in \{1,\dots,s\}\}$ by Proposition 2.1.6(ii) and Theorem 2.1.7(i). Thus $\sigma(E) = \{e^\lambda : \lambda \in \sigma(A)\}$.

(ii) Fix $\rho \in \sigma(E)$. If $\alpha \in I_\rho$, then $(\forall r \in \mathbb{Z}_{\geq 0})\,(E_\alpha - \rho I_{k_\alpha})^r = \rho^r N_\alpha^r Q_\alpha^r$, so the k_αth shifted power is zero.
If $\alpha \notin I_\rho$, then $E_\alpha - \rho I_{k_\alpha}$ is upper triangular with nonzero diagonal and is invertible by Theorem 3.8.2(v),(vii).
Hence $\mathcal{G}_\rho(E_J) = K_\rho$. For $\lambda \in \mathbb{C}$ and $r \in \mathbb{Z}_{\geq 0}$, $(A - \lambda I_n)^r S = S(J - \lambda I_n)^r$ by induction from $A = SJS^{-1}$.
Similarly, $(\forall r \in \mathbb{Z}_{\geq 0})\,(E - \rho I_n)^r S = S(E_J - \rho I_n)^r$. Thus $S\mathcal{G}_\lambda(J) = \mathcal{G}_\lambda(A)$ and $S\mathcal{G}_\rho(E_J) = \mathcal{G}_\rho(E)$.
Consequently, $\mathcal{G}_\rho(E) = SK_\rho = \bigoplus_{\lambda \in \Lambda_\rho(A)} \mathcal{G}_\lambda(A)$.

(iii) Fix $\rho \in \sigma(E)$. For $\alpha \in I_\rho$ and $r \in \mathbb{Z}_{\geq 0}$, $\operatorname{rank}((E_\alpha - \rho I_{k_\alpha})^r) = \operatorname{rank}(N_\alpha^r)$ by Theorem 3.6.6(ii).
Thus the shifted rank drops of E_α equal those of $J_{k_\alpha}(\rho)$, and $\mathcal{S}_\rho(E_\alpha) = \{k_\alpha\}$ by Theorem 6.2.6.
If $\alpha \notin I_\rho$, then $\mathcal{S}_\rho(E_\alpha) = \varnothing$ by part (ii); $\mathcal{S}_\rho(E) = \{k_\alpha : \alpha \in I_\rho\}$ by direct-sum block decomposition, similarity of E_J and E.

(iv) For $X, Y \in \operatorname{Mat}(n,n,\mathbb{C})$, $\operatorname{tr}(XY) = \sum_{i=1}^{n}\sum_{j=1}^{n} X_{ij}Y_{ji} = \operatorname{tr}(YX)$ by Definition 3.8.1(i). Hence $\operatorname{tr}(A) = \operatorname{tr}(SJS^{-1}) = \operatorname{tr}(J) = \sum_{\alpha=1}^{s} k_\alpha \lambda_\alpha$.
The equality $\det(E) = \det(E_J) = \prod_{\alpha=1}^{s} e^{k_\alpha \lambda_\alpha}$ holds by Theorem 3.8.2(v),(vi),(viii), similarity, and triangularity of E_J.
This product equals $e^{\sum_{\alpha=1}^{s} k_\alpha \lambda_\alpha} = e^{\operatorname{tr}(A)}$ by repeated scalar instances of Proposition 6.5.4(i).

Structural Commentary: Similarity transfers block exponentials; invertible Q_α preserves their nilpotent rank profiles.
Key Insight: Exponentiation changes block labels but not block lengths. □

Definition 6.5.8. *Matrix Logarithm*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \operatorname{Mat}(n,n,\mathbb{C})$. A matrix $B \in \operatorname{Mat}(n,n,\mathbb{C})$ is a *matrix logarithm* of A if $e^B = A$.

Theorem 6.5.9. *Complex Matrix Logarithms*
Let $k, n \in \mathbb{Z}_{>0}$, let $\lambda \in \mathbb{C} \setminus \{0\}$, and choose $\ell \in \mathbb{C}$ satisfying $e^\ell = \lambda$.
Set $N := J_k(0)$, $X := \lambda^{-1}N$, and $L_k(\lambda,\ell) := \ell I_k + \sum_{r=1}^{k-1} \frac{(-1)^{r+1}}{r} X^r$, with an empty sum interpreted as 0. Let $A \in \operatorname{Mat}(n,n,\mathbb{C})$. Then:

(i) $e^{L_k(\lambda,\ell)} = J_k(\lambda)$.
(ii) $(\exists B \in \operatorname{Mat}(n,n,\mathbb{C}))\, e^B = A$ *if and only if* $A \in \operatorname{Mat}(n,n,\mathbb{C})^\times$.

Proof.

(i) Set $Y := \sum_{r=1}^{k-1} \frac{(-1)^{r+1}}{r} X^r$. If $k = 1$, then $X = Y = 0$ and $e^Y = I_k + X$. Suppose $k \geq 2$.
In $\mathbb{C}[[x]]$, set $y(x) := \sum_{r=1}^{k-1} \frac{(-1)^{r+1}}{r} x^r$ and write $e^{y(x)} = \sum_{q=0}^{\infty} b_q x^q$.
The coefficients through degree $k-2$ of $(1+x)(e^y)'$ and e^y agree by $(1+x)y'(x) = 1 + (-1)^k x^{k-1}$.
The relations $b_1 = 1$ and $(q+1)b_{q+1} = (1-q)b_q$ for $q \in \{1,\dots,k-2\}$ hold by $b_0 = 1$ and coefficient comparison.
Hence $b_q = 0$ for $q \in \{2,\dots,k-1\}$ by this recurrence. Set $h(z) := \sum_{r=1}^{k-1} \frac{(-1)^{r+1}}{r} z^{r-1}$. Then $Y = Xh(X)$, so $Y^k = 0$.
The equality $e^Y = \sum_{q=0}^{k-1} b_q X^q = I_k + X$ holds by regrouping its finite exponential by powers of X.
The summands ℓI_k and Y commute. Therefore $e^{L_k(\lambda,\ell)} = e^\ell(I_k + X) = \lambda I_k + N = J_k(\lambda)$ by Proposition 6.5.4(i).

(ii) ($\Rightarrow$) Suppose $A = e^B$. Then $Ae^{-B} = e^B e^{-B} = I_n = e^{-B}e^B = e^{-B}A$ by Proposition 6.5.4(i). Thus $A \in \operatorname{Mat}(n,n,\mathbb{C})^\times$.

(ii) ($\Leftarrow$) Suppose $A \in \operatorname{Mat}(n,n,\mathbb{C})^\times$. By Theorem 6.2.1, choose $S \in \operatorname{Mat}(n,n,\mathbb{C})^\times$ and $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$ such that $A = SJS^{-1}$.
For $\alpha \in \{1,\dots,s\}$, $\lambda_\alpha \neq 0$ by Corollary 5.1.8. Choose $\ell_\alpha \in \mathbb{C}$ satisfying $e^{\ell_\alpha} = \lambda_\alpha$, and set $L_\alpha := L_{k_\alpha}(\lambda_\alpha, \ell_\alpha)$.
Set $L_J := \bigoplus_{\alpha=1}^{s} L_\alpha$, $B := SL_JS^{-1}$. Then $e^{L_J} = \bigoplus_{\alpha=1}^{s} e^{L_\alpha} = J$ by part (i), Proposition 3.8.6(iv) for partial sums, entrywise limits.
Hence $e^B = Se^{L_J}S^{-1} = A$ by Proposition 6.5.4(ii).

Structural Commentary: A formal truncated logarithm handles each nilpotent block before direct-sum and similarity assembly.
Key Insight: Modulo X^k, the truncated logarithm exponentiates exactly to $I_k + X$. □

Proposition 6.5.10. *Real Matrix Logarithm Conditions*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n,n,\mathbb{R})^\times$. Then:

(i) If $(\exists B \in \mathrm{Mat}(n,n,\mathbb{R}))\, e^B = A$, then $\det(A) > 0$.
(ii) If $\mathrm{spec}(A) \cap \mathbb{R} \subset (0,\infty)$, then $(\exists B \in \mathrm{Mat}(n,n,\mathbb{R}))\, e^B = A$.
(iii) $(\exists C \in \mathrm{Mat}(n,n,\mathbb{R}))\, e^C = A^2$.

Proof. For $x, y \in \mathbb{R}$, set $R(x,y) := \left(\begin{smallmatrix} x & y \\ -y & x \end{smallmatrix}\right)$. For $m \in \mathbb{Z}_{>0}$ and $D = [d_{rs}] \in \mathrm{Mat}(m,m,\mathbb{C})$, set $\mathfrak{R}(D) := [R(\mathrm{Re}\, d_{rs}, \mathrm{Im}\, d_{rs})]_{r,s=1}^m$.
For $D, E \in \mathrm{Mat}(m,m,\mathbb{C})$, $\mathfrak{R}(D+E) = \mathfrak{R}(D) + \mathfrak{R}(E)$ and $\mathfrak{R}(DE) = \mathfrak{R}(D)\mathfrak{R}(E)$ by direct block multiplication.
Hence $e^{\mathfrak{R}(D)} = \mathfrak{R}(e^D)$ by exponential partial sums and entrywise limits.
For $\lambda = a + ib$ with $a \in \mathbb{R}$ and $b > 0$, $\mathfrak{R}(J_k(\lambda)) = \mathcal{J}_k(a,b)$ by the block definitions.

(i) If $A = e^B$ with $B \in \mathrm{Mat}(n,n,\mathbb{R})$, then $\det(A) = e^{\mathrm{tr}(B)} > 0$ by Theorem 6.5.7(iv) and positivity of the real scalar exponential.
(ii) By Theorem 6.2.2(i), choose $S \in \mathrm{GL}_n(\mathbb{R})$ such that $J_\mathbb{R} := S^{-1}AS$ is a direct sum of real Jordan blocks.
For a scalar summand $J_k(\lambda)$, $\lambda > 0$ by the hypothesis. Choose $\ell \in \mathbb{R}$ with $e^\ell = \lambda$.
Then $L_k(\lambda, \ell)$ is real, and $e^{L_k(\lambda,\ell)} = J_k(\lambda)$ by Theorem 6.5.9(i).
For a summand $\mathcal{J}_k(a,b)$, set $\lambda := a + ib$, choose $\ell \in \mathbb{C}$ with $e^\ell = \lambda$, and set $D := L_k(\lambda, \ell)$.
Then $e^{\mathfrak{R}(D)} = \mathfrak{R}(J_k(\lambda)) = \mathcal{J}_k(a,b)$ by the realification identities and Theorem 6.5.9(i).
Let $L_\mathbb{R}$ be the direct sum of these real block logarithms and set $B := SL_\mathbb{R}S^{-1}$.
Then $e^B = Se^{L_\mathbb{R}}S^{-1} = SJ_\mathbb{R}S^{-1} = A$ by Proposition 6.5.4(ii), exponential partial sums, and entrywise limits.
(iii) *Squared Block Sizes*: Fix $(k,\lambda) \in \mathbb{Z}_{>0} \times (\mathbb{C} \setminus \{0\})$, set $N := J_k(0)$, and set $X_\lambda := J_k(\lambda)^2$. Then $X_\lambda - \lambda^2 I_k = N(2\lambda I_k + N)$.
The matrix $2\lambda I_k + N$ is invertible by a finite geometric inverse and commutes with N.
Thus $(\forall r \in \mathbb{Z}_{\geq 0})$ $\mathrm{rank}((J_k(\lambda)^2 - \lambda^2 I_k)^r) = \mathrm{rank}(N^r)$ by Theorem 3.6.6(ii).
The matrix X_λ is upper triangular with constant diagonal λ^2, so its only eigenvalue is λ^2.
A Jordan form of X_λ exists by Theorem 6.2.1, and it consists of one block $J_k(\lambda^2)$ by Theorem 6.2.6 and the shifted-rank equality.
Real Assembly: Choose $S \in \mathrm{GL}_n(\mathbb{R})$ and a real Jordan block matrix $J_\mathbb{R} := S^{-1}AS$ by Theorem 6.2.2(i).
For scalar summand $J_k(\lambda)$, $\lambda \in \mathbb{R}\setminus\{0\}$. By squared-block step, Theorem 6.2.2(i), choose $Q \in \mathrm{GL}_k(\mathbb{R})$ s.t. $J_k(\lambda)^2 = QJ_k(\lambda^2)Q^{-1}$.
Choose $\ell \in \mathbb{R}$ with $e^\ell = \lambda^2$, $D := QL_k(\lambda^2, \ell)Q^{-1}$. Then D is real and $e^D = J_k(\lambda)^2$ by Theorem 6.5.9(i) and Proposition 6.5.4(ii).
For a summand $\mathcal{J}_k(a,b)$, set $\lambda := a + ib$. By the squared-block step, choose $Q_\lambda \in \mathrm{GL}_k(\mathbb{C})$ such that $X_\lambda = Q_\lambda J_k(\lambda^2)Q_\lambda^{-1}$.
Choose $\ell_\lambda \in \mathbb{C}$ with $e^{\ell_\lambda} = \lambda^2$, and set $D_\lambda := Q_\lambda L_k(\lambda^2, \ell_\lambda)Q_\lambda^{-1}$. Then $e^{D_\lambda} = X_\lambda$ by Theorem 6.5.9(i) and Proposition 6.5.4(ii).
Thus $e^{\mathfrak{R}(D_\lambda)} = \mathfrak{R}(X_\lambda) = \mathcal{J}_k(a,b)^2$ (realification identities). Let $C_\mathbb{R}$ be direct sum of resulting real log of all squared summands.
Set $C := SC_\mathbb{R}S^{-1}$. Then $e^C = Se^{C_\mathbb{R}}S^{-1} = SJ_\mathbb{R}^2S^{-1} = A^2$ by Proposition 6.5.4(ii), exponential partial sums, entrywise limits.

Structural Commentary: Realification handles nonreal blocks; shifted-rank profiles preserve block lengths after squaring.
Key Insight: The block realification map commutes with powers and exponentials. □

Proposition 6.5.11. *Jordan Resolvent and Primary Spectral Projectors*
Let $n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n,n,\mathbb{C})$; $\Sigma := \mathrm{spec}(A)$, $G_\nu := \mathcal{G}_\nu(A)$. For $m \in \mathbb{Z}_{>0}$, $M \in \mathrm{Mat}(m,m,\mathbb{C})$, $z \notin \mathrm{spec}(M)$, $R_M(z) := (zI_m - M)^{-1}$. The identity $\mathbb{C}^n = \bigoplus_{\nu\in\Sigma} G_\nu$ holds by Theorem 5.1.1. For $\mu \in \Sigma$, define $P_\mu \in \mathrm{Mat}(n,n,\mathbb{C})$ by $P_\mu|_{G_\nu} = \delta_{\mu\nu} I_{G_\nu}$.
When entries of matrix-valued function F admit Laurent expansions at μ, let $[F]_{\mu,-1}$ denote matrix of $(z-\mu)^{-1}$ coefficients. Then:

(i) For $k \in \mathbb{Z}_{>0}$, $\lambda \in \mathbb{C}$, $B := J_k(\lambda) = \lambda I_k + N$ with $N := J_k(0)$, and $z \neq \lambda$, $R_B(z) = \sum_{r=0}^{k-1} \frac{N^r}{(z-\lambda)^{r+1}}$.
(ii) $(\forall \mu \in \Sigma)\, [R_A]_{\mu,-1} = P_\mu$.
(iii) $(\forall \lambda, \mu \in \Sigma)\, P_\lambda P_\mu = \delta_{\lambda\mu} P_\mu$.
(iv) $(\forall \mu \in \Sigma)\, AP_\mu = P_\mu A$.
(v) $\sum_{\mu\in\Sigma} P_\mu = I_n$.

Proof.

(i) Set $a := (z-\lambda)^{-1}$. Since $N^k = 0$, $(I_k - aN)\sum_{r=0}^{k-1} a^r N^r = I_k - a^k N^k = I_k$ by direct multiplication.
Thus $(I_k - aN)^{-1} = \sum_{r=0}^{k-1} a^r N^r$. The equality $R_B(z) = a\sum_{r=0}^{k-1} a^r N^r = \sum_{r=0}^{k-1} \frac{N^r}{(z-\lambda)^{r+1}}$ holds by $zI_k - B = (z-\lambda)(I_k - aN)$.
(ii) By Theorem 6.2.1, choose $S \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ and $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$ such that $A = SJS^{-1}$.
The identity $R_A(z) = SR_J(z)S^{-1}$ holds by direct multiplication. Set $H_\nu := \mathcal{G}_\nu(J)$ and $C_\mu := [R_J]_{\mu,-1}$.
By part (i), C_μ is the identity on a block with eigenvalue μ and zero on the Laurent coefficient of every block regular at μ.
Hence $C_\mu|_{H_\nu} = \delta_{\mu\nu} I_{H_\nu}$. For $r \in \mathbb{Z}_{\geq 0}$, $(A - \nu I_n)^r S = S(J - \nu I_n)^r$ by induction, so $SH_\nu = G_\nu$. Thus $[R_A]_{\mu,-1} = SC_\mu S^{-1} = P_\mu$.
(iii) Let $\mathbf{v} = \sum_{\nu\in\Sigma} \mathbf{v}_\nu$ with $\mathbf{v}_\nu \in G_\nu$. Then $P_\lambda P_\mu \mathbf{v} = \delta_{\lambda\mu}\mathbf{v}_\mu = \delta_{\lambda\mu} P_\mu \mathbf{v}$.
(iv) Let $\mathbf{v}$ be as in part (iii). Since $AG_\nu \subseteq G_\nu$ by Proposition 4.1.7, $AP_\mu \mathbf{v} = A\mathbf{v}_\mu = P_\mu A\mathbf{v}$.
(v) For $\mathbf{v} = \sum_{\nu\in\Sigma} \mathbf{v}_\nu$ as in part (iii), $\left(\sum_{\mu\in\Sigma} P_\mu\right)\mathbf{v} = \sum_{\mu\in\Sigma} \mathbf{v}_\mu = \mathbf{v}$. Hence the operator sum equals I_n.

Structural Commentary: Jordan resolvents identify residues blockwise; primary decomposition then verifies projector algebra.
Key Insight: Only the matching Jordan blocks contribute a simple-pole coefficient. □

Definition 6.5.12. *Contracting, Neutral, and Expanding Primary Subspaces*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let P_λ be the primary projection onto $\mathcal{G}_\lambda(A)$ from Proposition 6.5.11.
Set $\Sigma_- := \{\lambda \in \mathrm{spec}(A) : |\lambda| < 1\}$, $\Sigma_0 := \{\lambda \in \mathrm{spec}(A) : |\lambda| = 1\}$, and $\Sigma_+ := \{\lambda \in \mathrm{spec}(A) : |\lambda| > 1\}$. Define:

(i) $\mathcal{G}^-(A) := \bigoplus_{\lambda\in\Sigma_-} \mathcal{G}_\lambda(A)$ and $P_A^- := \sum_{\lambda\in\Sigma_-} P_\lambda$, the *contracting* part.
(ii) $\mathcal{G}^0(A) := \bigoplus_{\lambda\in\Sigma_0} \mathcal{G}_\lambda(A)$ and $P_A^0 := \sum_{\lambda\in\Sigma_0} P_\lambda$, the *neutral* part.

(iii) $\mathcal{G}^+(A) := \bigoplus_{\lambda\in\Sigma_+} \mathcal{G}_\lambda(A)$ and $P_A^+ := \sum_{\lambda\in\Sigma_+} P_\lambda$, the *expanding* part.

Empty direct sum is understood as $\{\mathbf{0}\}$, empty projector sum as 0. Identity $\mathbb{C}^n = \mathcal{G}^-(A) \oplus \mathcal{G}^0(A) \oplus \mathcal{G}^+(A)$ holds by Theorem 5.1.1. Proposition 6.5.11(iii),(v) identifies P_A^-, P_A^0, and P_A^+ as the corresponding projections.

Lemma 6.5.13. *Identity Plus Nilpotent Has a Square Root*
Let $\mathcal{V}$ be a complex vector space and let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $\exists\, R \in \mathcal{L}(\mathcal{V})$ such that $R^2 = I_\mathcal{V} + N$.

Proof. Choose $m \in \mathbb{Z}_{>0}$ with $N^m = 0$ and set $a_0 := 1$. For $k \in \{1, \dots, m-1\}$, set $s_k := \sum_{r=1}^{k-1} a_r a_{k-r}$ and $a_k := \frac{1}{2}(\delta_{k1} - s_k)$.
Set $R := \sum_{k=0}^{m-1} a_k N^k$. For $k \in \{0, \dots, m-1\}$, the coefficient of N^k in the formal square is $c_k := \sum_{r=0}^{k} a_r a_{k-r}$.
The relations $c_0 = 1$ and $(\forall k \in \{1, \dots, m-1\})\, c_k = 2a_k + s_k = \delta_{k1}$ hold by the recursion.
Every term of degree at least m vanishes at N, so $R^2 = \sum_{k=0}^{m-1} c_k N^k = I_\mathcal{V} + N$.
Structural Commentary: Recursive coefficient matching constructs the square root before nilpotence truncates higher terms.
Key Insight: The recursion forces the formal square to equal $1 + x$. □

Figure 71: Recursive coefficients and nilpotence reduce the polynomial square to $R^2 = I + N$.

Figure 71 depicts the finite polynomial square-root construction.

Theorem 6.5.14. *Invertible Complex Operators Have Square Roots*
Let $\mathcal{V}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$ be invertible. Then $\exists\, R \in \mathcal{L}(\mathcal{V})$ such that $R^2 = T$.

Proof.
(i) *Primary Reduction*: If $\mathcal{V} = \{\mathbf{0}\}$, take $R = I_\mathcal{V}$. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.
List the distinct eigenvalues as $\lambda_1, \dots, \lambda_m$, where $m \in \mathbb{Z}_{>0}$, and set $\mathcal{U}_j := \mathcal{G}_{\lambda_j}(T)$.
Identity $\mathcal{V} = \bigoplus_{j=1}^{m} \mathcal{U}_j$ holds, $N_j := (T - \lambda_j I_\mathcal{V})|_{\mathcal{U}_j}$ is nilpotent, by Theorem 5.1.1. For $j \in \{1, \dots, m\}$, $\lambda_j \neq 0$ by Corollary 5.1.7.
(ii) *Local Roots*: Set $M_j := \lambda_j^{-1} N_j$, so $T|_{\mathcal{U}_j} = \lambda_j(I_{\mathcal{U}_j} + M_j)$. By Lemma 6.5.13, choose $B_j \in \mathcal{L}(\mathcal{U}_j)$ with $B_j^2 = I_{\mathcal{U}_j} + M_j$.
Choose $\alpha_j \in \mathbb{C}$ with $\alpha_j^2 = \lambda_j$ and set $R_j := \alpha_j B_j$. Then $R_j^2 = \lambda_j B_j^2 = T|_{\mathcal{U}_j}$.
(iii) *Assembly*: Define $R \in \mathcal{L}(\mathcal{V})$ by $R(\sum_{j=1}^{m} \mathbf{v}_j) := \sum_{j=1}^{m} R_j \mathbf{v}_j$ for $\mathbf{v}_j \in \mathcal{U}_j$. For $j \in \{1, \dots, m\}$, $(R^2)|_{\mathcal{U}_j} = R_j^2 = T|_{\mathcal{U}_j}$. Hence $R^2 = T$.

Structural Commentary: Primary decomposition reduces T to scalar multiples of identity-plus-nilpotent restrictions before assembly.
Key Insight: Each nonzero eigenvalue separates into scalar and nilpotent square roots. □

7 Transpose and Conjugate Similarity

7.1 Transpose Similarity and Conjugate Similarity

Theorem 7.1.1. *Conjugate Similarity Criterion*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Write $\sim$ for similarity on $\mathrm{Mat}(n, n, \mathbb{C})$.
For $(\lambda, k) \in \mathbb{C} \times \mathbb{Z}_{>0}$, let $m_A(\lambda, k)$ count the $J_k(\lambda)$ blocks in any Jordan form of A.
Then $A \sim \overline{A} \iff (\forall\, (\lambda, k) \in \mathbb{C} \times \mathbb{Z}_{>0})\, m_A(\lambda, k) = m_A(\overline{\lambda}, k)$.

Proof. The function m_A is well-defined by Corollary 6.2.7(i), with $m_A(\lambda, k) = 0$ for $\lambda \notin \mathrm{spec}(A)$.
By Theorem 6.2.1, choose $S \in \mathrm{Mat}(n, n, \mathbb{C})^\times$ and a Jordan matrix J satisfying $A = SJS^{-1}$.
Then $\overline{A} = \overline{S}\,\overline{J}\,\overline{S}^{-1}$, and the $J_k(\lambda)$-multiplicity in $\overline{J}$ is $m_A(\overline{\lambda}, k)$ by blockwise conjugation.
($\Rightarrow$) Assume $A \sim \overline{A}$. The chain $J \sim A \sim \overline{A} \sim \overline{J}$ holds by Proposition 6.2.2.
By Corollary 6.2.7(ii), choose a block-permutation matrix P satisfying $\overline{J} = P^{-1}JP$.
The equality $(\forall\, (\lambda, k) \in \mathbb{C} \times \mathbb{Z}_{>0})\, m_A(\lambda, k) = m_A(\overline{\lambda}, k)$ holds by $\overline{J} = P^{-1}JP$ and the conjugate-block identity.
($\Leftarrow$) Assume $(\forall\, (\lambda, k) \in \mathbb{C} \times \mathbb{Z}_{>0})\, m_A(\lambda, k) = m_A(\overline{\lambda}, k)$.
By the hypothesis and conjugate-block identity, choose a block-permutation matrix P satisfying $\overline{J} = P^{-1}JP$.
Set $Q := \overline{S}P^{-1}S^{-1} \in \mathrm{Mat}(n, n, \mathbb{C})^\times$. Then $\overline{A} = \overline{S}P^{-1}JP\overline{S}^{-1} = QAQ^{-1}$, so $A \sim \overline{A}$.
Structural Commentary: Conjugation relabels Jordan blocks; block-permutation uniqueness converts their counts into similarity.
Key Insight: $J_k(\lambda)$ becomes $J_k(\overline{\lambda})$ without changing k. □

Definition 7.1.2. *Reversal Matrix*
For $k \in \mathbb{Z}_{>0}$, the *reversal matrix* $R_k \in \mathrm{Mat}(k, k, \mathbb{C})$ is defined by $(\forall\, i, j \in \{1, \dots, k\})\, (R_k)_{ij} := \delta_{i+j,k+1}$.
The identities $R_k^t = R_k$ and $(R_k^2)_{ij} = \sum_{r=1}^{k} \delta_{i+r,k+1}\delta_{r+j,k+1} = \delta_{ij}$ hold by the entry rule. Hence $R_k^2 = I_k$.

Lemma 7.1.3. *Reversal Conjugates a Jordan Block to Its Transpose*
Let $k \in \mathbb{Z}_{>0}$ and $\lambda \in \mathbb{C}$. Then:
(i) $J_k(\lambda)^t = R_k J_k(\lambda) R_k$;
(ii) $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.

Proof. Set $N := J_k(0)$.
(i) For $i, j \in \{1, \dots, k\}$, $(R_k N R_k)_{ij} = N_{k+1-i,k+1-j} = \delta_{i,j+1} = (N^t)_{ij}$ by the entry rules for R_k and N.
Thus $R_k J_k(\lambda) R_k = \lambda R_k^2 + R_k N R_k = \lambda I_k + N^t = J_k(\lambda)^t$.

(ii) The identities $\big(R_k J_k(\lambda)\big)^t = J_k(\lambda)^t R_k = R_k J_k(\lambda)$ and $\big(J_k(\lambda) R_k\big)^t = R_k J_k(\lambda)^t = J_k(\lambda) R_k$ hold by part (i) and $R_k^2 = I_k$.

Structural Commentary: The reversal entry rule turns the superdiagonal shift into its transposed subdiagonal shift.
Key Insight: Reversal reads each Jordan chain in the opposite order. □

Theorem 7.1.4. *Transpose Similarity and Symmetric Factorisations*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then:
(i) $(\exists P \in \mathrm{Mat}(n,n,\mathbb{C})^\times)\ P^t = P \ \wedge\ PAP^{-1} = A^t$.
(ii) $(\exists B,C,D,E \in \mathrm{Mat}(n,n,\mathbb{C}))\ A = BC = DE$.
The witnesses satisfy $B, E \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ and $B^t = B$, $C^t = C$, $D^t = D$, and $E^t = E$.

Proof. By Theorem 6.2.1, choose $S \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ and a Jordan matrix J satisfying $A = SJS^{-1}$.
Let R be the direct sum of the reversal matrices matching the blocks of J.
The identities $J^t = RJR$, $R^t = R$, and $R^2 = I_n$ hold blockwise by Lemma 7.1.3(i), while RJ and JR are symmetric by Lemma 7.1.3(ii).
(i) Set $P := (S^{-1})^t R S^{-1}$. Then $P^t = P$ and $P^{-1} = SRS^t$ by $R^t = R = R^{-1}$.
The identity $PAP^{-1} = (S^{-1})^t RJRS^t = (S^{-1})^t J^t S^t = A^t$ holds by $A = SJS^{-1}$ and $J^t = RJR$.
(ii) Set $B := SRS^t$ and $C := (S^{-1})^t RJS^{-1}$. Set $D := SJRS^t$ and $E := (S^{-1})^t R S^{-1}$.
The matrices B, C, D, E are symmetric by congruence of R, RJ, JR, R, respectively, and B, E are invertible.
The identities $BC = SJS^{-1} = A$ and $DE = SJS^{-1} = A$ hold by $R^2 = I_n$.

Structural Commentary: Jordan reversal transfers through congruence to a symmetric similarity and two symmetric factor pairs.
Key Insight: The reversal involution both transposes blocks and cancels between factors. □

Figure 72 illustrates how reversing a Jordan chain converts the superdiagonal shift into its transpose.

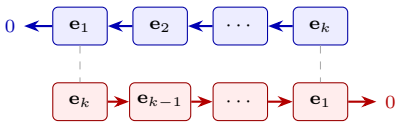

Figure 72: The identity $R_k J_k(0) R_k = J_k(0)^t$ records the same chain read backwards.

Corollary 7.1.5. *Hermitian Products and Conjugate Products*
Let $n \in \mathbb{Z}_{>0}$ and let $A, H, K \in \mathrm{Mat}(n,n,\mathbb{C})$, where H and K are Hermitian. Then:
(i) HK is similar to KH.
(ii) $A\overline{A}$ is similar to $\overline{A}A$.

Proof. For $M \in \mathrm{Mat}(n,n,\mathbb{C})$, $\ker(\overline{M}) = \{\overline{\mathbf{v}} : \mathbf{v} \in \ker(M)\}$ by $\overline{M}\,\overline{\mathbf{v}} = \overline{M\mathbf{v}}$ and involutivity of conjugation.
The identity $\mathrm{rank}(M^*) = \mathrm{rank}(\overline{M}^{\,t}) = \mathrm{rank}(\overline{M}) = \mathrm{rank}(M)$ holds by Theorem 3.3.2, Corollary 3.3.1(i), and the kernel identity.
(i) The equality $(\forall p \in \{1,\dots,n\})\ \big((HK)^p\big)^* = \big((HK)^*\big)^p = (KH)^p$ holds by $H^* = H$ and $K^* = K$.
Thus $(\forall p \in \{1,\dots,n\})\ \mathrm{rank}((HK)^p) = \mathrm{rank}((KH)^p)$, so HK is similar to KH by Corollary 6.3.3.
(ii) The equality $(\forall p \in \{1,\dots,n\})\ \overline{(A\overline{A})^p} = (\overline{A}A)^p$ holds by entrywise conjugation of products.
Thus $(\forall p \in \{1,\dots,n\})\ \mathrm{rank}((A\overline{A})^p) = \mathrm{rank}((\overline{A}A)^p)$, so $A\overline{A}$ is similar to $\overline{A}A$ by Corollary 6.3.3.

Structural Commentary: Adjoint and conjugation preserve every required power rank before the AB–BA criterion.
Key Insight: Equal power ranks convert reordered products into similar matrices. □

Theorem 7.1.6. *Conjugate Similarity and Hermitian Products*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. The following are equivalent:
(i) A is similar to $\overline{A}$.
(ii) A is similar to A^*.
(iii) $(\exists R \in \mathrm{Mat}(n,n,\mathbb{C})^\times)\ R^* = R \ \wedge\ A = RA^*R^{-1}$.
(iv) $(\exists H,K,L,M \in \mathrm{Mat}(n,n,\mathbb{C}))\ A = HK = LM$.
The witnesses satisfy $H, M \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ and $H^* = H$, $K^* = K$, $L^* = L$, and $M^* = M$.
(v) $(\exists H,K \in \mathrm{Mat}(n,n,\mathbb{C}))\ H^* = H \ \wedge\ K^* = K \ \wedge\ A = HK$.

Proof. Write $\sim$ for similarity on $\mathrm{Mat}(n,n,\mathbb{C})$, which is transitive by Proposition 6.2.2.
(i) $\Rightarrow$ *(ii)* $A \sim \overline{A} \sim \overline{A}^{\,t} = A^*$ by part (i) and Theorem 7.1.4(i).
(ii) $\Rightarrow$ *(iii)* Choose $S \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ satisfying $A = SA^*S^{-1}$, equivalently $AS = SA^*$, and set $Q := S^{-1}S^*$.
For $\theta \in \mathbb{R}$, set $S_\theta := e^{i\theta}S$ and $H_\theta := S_\theta + S_\theta^*$.
The identities $AH_\theta = H_\theta A^*$, $H_\theta^* = H_\theta$, and $S_\theta^{-1}H_\theta = I_n + e^{-2i\theta}Q$ hold by $AS = SA^*$, its adjoint, and the definitions.
For $\theta \in \mathbb{R}$, $H_\theta \notin \mathrm{Mat}(n,n,\mathbb{C})^\times \iff -e^{2i\theta} \in \mathrm{spec}(Q)$ by the last identity and invertibility of S_θ.
The existence $(\exists \theta \in \mathbb{R})\ -e^{2i\theta} \notin \mathrm{spec}(Q)$ holds by Corollary 5.1.6 and infinitude of $\{-e^{2i\phi} : \phi \in \mathbb{R}\}$.
For such θ, $H_\theta \in \mathrm{Mat}(n,n,\mathbb{C})^\times$, $H_\theta^* = H_\theta$, and $A = H_\theta A^* H_\theta^{-1}$ by the preceding identities.
(iii) $\Rightarrow$ *(iv)* Choose $R \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ satisfying $R^* = R$ and $A = RA^*R^{-1}$, and set $(H,K,L,M) := (R, A^*R^{-1}, RA^*, R^{-1})$.
$A = HK = LM$ and $H, M \in \mathrm{Mat}(n,n,\mathbb{C})^\times$ by the definitions and part (iii).
$(H^*, K^*, L^*, M^*) = (H,K,L,M)$ by $R^* = R$, $AR = RA^*$, and $R^{-1}A = A^*R^{-1}$.
(iv) $\Rightarrow$ *(v)* The witnesses (H,K) from part (iv) satisfy $H^* = H \ \wedge\ K^* = K \ \wedge\ A = HK$.
(v) $\Rightarrow$ *(i)* Choose $H, K \in \mathrm{Mat}(n,n,\mathbb{C})$ satisfying $H^* = H$, $K^* = K$, and $A = HK$ by part (v).
$A = HK \sim KH \sim (KH)^t = H^tK^t = \overline{H}\,\overline{K} = \overline{A}$ by Corollary 7.1.5(i), Theorem 7.1.4(i), and Hermitianity of H, K.

Structural Commentary: Transpose similarity starts the cycle; phase averaging Hermitianizes the adjoint-similarity witness.
Key Insight: A phase avoids the finitely many singular Hermitian averages. □

Figure 73 summarizes the implication cycle connecting conjugate similarity, adjoint similarity, and Hermitian factorization.

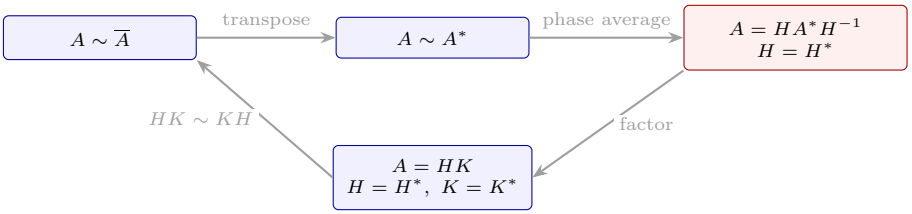

Figure 73: Conjugate similarity, adjoint similarity, and Hermitian product factorisations are equivalent.

Part VI
Matrix Analysis and Numerical Linear Algebra

Matrix classes package operator structure in coordinates. Trace, determinant, rank, positivity, idempotents, nilpotents, spectral radius, and structured products give computable invariants.

1 Trace, Determinant, Rank, and Matrix Invariants

1.1 Trace and Determinant

Proposition 1.1.1. *Trace Arithmetic*
Let $n \in \mathbb{Z}_{>0}$, $A = [a_{jk}], B = [b_{jk}] \in \mathrm{Mat}(n, n, \mathbb{F})$, and $r \in \mathbb{F}$. Then
(i) $\mathrm{tr}(rA) = r\,\mathrm{tr}(A)$;
(ii) $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$.

Proof.
(i) $\mathrm{tr}(rA) = \sum_{j=1}^{n} ra_{jj} = r \sum_{j=1}^{n} a_{jj} = r\,\mathrm{tr}(A)$ by Definition 3.8.1.
(ii) $\mathrm{tr}(A + B) = \sum_{j=1}^{n} (a_{jj} + b_{jj}) = \sum_{j=1}^{n} a_{jj} + \sum_{j=1}^{n} b_{jj} = \mathrm{tr}(A) + \mathrm{tr}(B)$ by Definition 3.8.1.
Structural Commentary: Each identity reduces trace to a finite diagonal sum.
Key Insight: Finite summation commutes with scalar multiplication and addition. □

Example 1.1.2. *Trace Functional and the Frobenius Inner Product*
Let $n \in \mathbb{Z}_{>0}$ and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and equip $\mathrm{Mat}(n, n, \mathbb{F})$ with the Frobenius inner product from Example 1.1.7.
Map $A \mapsto \mathrm{tr}(A)$ is a linear functional by Proposition 1.1.1. Riesz vector is I_n as $(\forall\, A = [a_{jk}] \in \mathrm{Mat}(n, n, \mathbb{F}))\ \langle A, I_n \rangle_F = \sum_{j=1}^{n} a_{jj} = \mathrm{tr}(A)$.

Proposition 1.1.3. *Trace Is Cyclic for Two Factors*
Let $m, n \in \mathbb{Z}_{>0}$, $A = [a_{jk}] \in \mathrm{Mat}(m, n, \mathbb{F})$, and $B = [b_{kj}] \in \mathrm{Mat}(n, m, \mathbb{F})$. Then $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.

Proof. $\mathrm{tr}(AB) = \sum_{j=1}^{m} \sum_{k=1}^{n} a_{jk} b_{kj} = \sum_{k=1}^{n} \sum_{j=1}^{m} b_{kj} a_{jk} = \mathrm{tr}(BA)$ by commutativity of $\mathbb{F}$ and interchange of finite sums.
Structural Commentary: Entry expansion identifies both traces after interchanging the finite indices.
Key Insight: Both traces sum the scalars $a_{jk} b_{kj}$. □

Proposition 1.1.4. *Cyclic Products*
Let $m \in \mathbb{Z}_{\geq 2}$ and $n_1, \ldots, n_m \in \mathbb{Z}_{>0}$, and set $n_{m+1} := n_1$. Suppose $(\forall\, j \in \{1, \ldots, m\})\ A_j \in \mathrm{Mat}(n_j, n_{j+1}, \mathbb{F})$. Then:
(i) $\mathrm{tr}(A_1 \cdots A_m) = \mathrm{tr}(A_m A_1 \cdots A_{m-1})$;
(ii) $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, A, B, C \in \mathrm{Mat}(n, n, \mathbb{F}))\ \mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$.

Proof.
(i) Set $X := A_1 \cdots A_{m-1} \in \mathrm{Mat}(n_1, n_m, \mathbb{F})$. Then $\mathrm{tr}(A_1 \cdots A_m) = \mathrm{tr}(X A_m) = \mathrm{tr}(A_m X)$ by Proposition 1.1.3.
(ii) Fix $n \in \mathbb{Z}_{>0}$ and $A, B, C \in \mathrm{Mat}(n, n, \mathbb{F})$. Then $\mathrm{tr}(ABC) = \mathrm{tr}(CAB) = \mathrm{tr}(BCA)$ by two applications of part (i).
Structural Commentary: Two-factor cyclicity proves one general rotation, whose repeated use proves the specialization.
Key Insight: Grouping $A_1 \cdots A_{m-1}$ reduces rotation to two factors. □

Example 1.1.5. *Cyclicity Is Not Arbitrary Commutativity*
Let $A = E_{1,2}$, $B = E_{2,1}$, and $C = E_{1,1}$ in $\mathrm{Mat}(2, 2, \mathbb{F})$. Then $\mathrm{tr}(ABC) = 1$, while $\mathrm{tr}(ACB) = 0$.
Thus trace permits cyclic rotations of factors, but not arbitrary rearrangements.

For real 2×2 matrices, Figure 74 illustrates the column geometry behind Theorem 3.8.2.

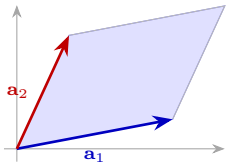

Figure 74: The columns $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{R}^2$ span a parallelogram whose signed area is their determinant.

Computation Method 1.1.6. *Determinant by Structure*
Given $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$, the output is $\det A$, computed by the first reduction that lowers the calculation:
(i) if A is triangular, multiply its diagonal entries;
(ii) if the columns are linearly dependent, including when two coincide, return 0;
(iii) if a column decomposition exposes triangular or dependent summands, expand that column by linearity;
(iv) if a column permutation exposes useful pivots, track the sign of the permutation;
(v) if $A = BC$ and the factors have simpler determinants, compute $\det B \det C$.
Each reduction and its output are valid by Theorem 3.8.2; verify that every column swap has been counted exactly once.

For real linear maps, Figure 75 summarizes the geometric scaling interpretation behind the determinant-by-structure method.

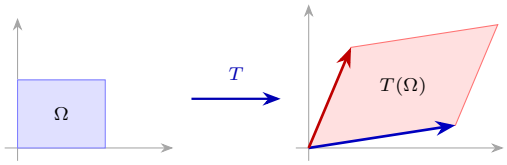

Figure 75: A real linear map scales oriented volume by its determinant; singular maps collapse volume to zero.

Definition 1.1.7. *Schur Complements*
Let $r, s \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $B \in \mathrm{Mat}(r, s, \mathbb{F})$, $C \in \mathrm{Mat}(s, r, \mathbb{F})$, and $D \in \mathrm{Mat}(s, s, \mathbb{F})$. Set $M := \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$.
(i) If A is invertible, the *Schur complement of A in M* is $M/A := D - CA^{-1}B$.
(ii) If D is invertible, the *Schur complement of D in M* is $M/D := A - BD^{-1}C$.

Proposition 1.1.8. *Schur Complement Determinant Formula*
Let $r, s \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $B \in \mathrm{Mat}(r, s, \mathbb{F})$, $C \in \mathrm{Mat}(s, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$, and $M := \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$.
If A is invertible, then $\det M = \det(A) \det(M/A)$.

Proof. Set $E := \left(\begin{smallmatrix} I_r & 0 \\ -CA^{-1} & I_s \end{smallmatrix}\right)$. Then $EM = \left(\begin{smallmatrix} A & B \\ 0 & M/A \end{smallmatrix}\right)$; $\det E = 1$ by Theorem 3.8.2(v).
Thus $\det M = \det(EM) = \det(A) \det(M/A)$ by Theorems 3.8.2(vi) and 3.8.4.
Structural Commentary: Unit block elimination replaces C by zero and exposes M/A.
Key Insight: The elimination factor has determinant one. □

Proposition 1.1.9. *Schur Complement of the Lower-Right Block*
Let $r, s \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $B \in \mathrm{Mat}(r, s, \mathbb{F})$, $C \in \mathrm{Mat}(s, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$, and $M := \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$.
If D is invertible, then $\det M = \det(D) \det(M/D)$.

Proof. Set $E := \left(\begin{smallmatrix} I_r & -BD^{-1} \\ 0 & I_s \end{smallmatrix}\right)$. Then $EM = \left(\begin{smallmatrix} M/D & 0 \\ C & D \end{smallmatrix}\right)$; $\det E = 1$ by Theorem 3.8.2(v). The transpose of EM is block upper triangular.
Thus $\det M = \det(EM) = \det(M/D) \det D$ by Theorems 3.8.2(vi),(ix) and 3.8.4.
Structural Commentary: Unit block elimination replaces B by zero before transposing the triangular form.
Key Insight: Transposition converts the lower block-triangular product to the proved upper case. □

Proposition 1.1.10. *Bordered Determinant Formulas*
Let $n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, and $c \in \mathbb{F}$.
(i) If $c \neq 0$, then $\det \left(\begin{smallmatrix} c & \mathbf{x}^t \\ \mathbf{y} & A \end{smallmatrix}\right) = c \det\left(A - c^{-1}\mathbf{y}\mathbf{x}^t\right)$.
(ii) If A is invertible, then $\det \left(\begin{smallmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{smallmatrix}\right) = \left(c - \mathbf{y}^t A^{-1}\mathbf{x}\right) \det A$.

Proof.
(i) Claim (i) holds by Proposition 1.1.8 with upper-left block $[c]$ and $M/[c] = A - c^{-1}\mathbf{y}\mathbf{x}^t$.
(ii) Claim (ii) holds by Proposition 1.1.8 with upper-left block A, $M/A = c - \mathbf{y}^t A^{-1}\mathbf{x}$, and commutativity of $\mathbb{F}$.
Structural Commentary: Each border orientation specializes the upper-left Schur determinant formula.
Key Insight: A scalar block turns its Schur complement into a rank-one correction. □

Theorem 1.1.11. *Sylvester Determinant Identity*
Let $m, n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n, m, \mathbb{F})$, and $B \in \mathrm{Mat}(m, n, \mathbb{F})$. Then $\det(I_n + AB) = \det(I_m + BA)$.

Proof. Set $M := \left(\begin{smallmatrix} I_n & -A \\ B & I_m \end{smallmatrix}\right)$. Then $\det M = \det(I_n + AB)$ by Proposition 1.1.9, while $\det M = \det(I_m + BA)$ by Proposition 1.1.8.
Structural Commentary: The two Schur determinant formulas evaluate one block matrix from opposite corners.
Key Insight: Both identity blocks have determinant one. □

Theorem 1.1.12. *Nonzero Eigenvalues of AB and BA*
Let $m, n \in \mathbb{Z}_{>0}$ with $n \geq m$, let $A \in \mathrm{Mat}(m, n, \mathbb{C})$, and let $B \in \mathrm{Mat}(n, m, \mathbb{C})$.
(i) $(\forall z \in \mathbb{C})\, p_{BA}(z) = z^{n-m} p_{AB}(z)$;
(ii) $(\forall \lambda \in \mathbb{C} \setminus \{0\})\, m_{BA}(\lambda) = m_{AB}(\lambda)$;
(iii) $m_{BA}(0) = m_{AB}(0) + n - m$.

Proof.
(i) For $z \neq 0$, $p_{BA}(z) = z^n \det(I_n - z^{-1}BA) = z^n \det(I_m - z^{-1}AB) = z^{n-m} p_{AB}(z)$ by Theorem 1.1.11 with $-z^{-1}B$ and A.
If $n = m$, then $p_{BA}(0) = (-1)^n \det(BA) = (-1)^n \det(AB) = p_{AB}(0)$ by Theorem 3.8.2(vi) and commutativity of $\mathbb{C}$.
If $n > m$, then $\mathrm{col}(BA) \subseteq \mathrm{col}(B)$ because $(\forall \mathbf{x} \in \mathbb{C}^n)\, BA\mathbf{x} = B(A\mathbf{x})$.
Thus $\mathrm{rank}(BA) \leq \mathrm{rank}(B) \leq m < n$ by Definition 3.1.12, Proposition 1.4.3(i), and Corollary 3.3.1(i).
BA singular by Corollary 3.6.2. Then $p_{BA}(0) = (-1)^n \det(BA) = 0 = 0^{n-m} p_{AB}(0)$ by Definition 2.1.2, Theorem 3.8.2(vii).
(ii) $m_{BA}(\lambda) = m_{AB}(\lambda)$ for $\lambda \neq 0$ by part (i), because z^{n-m} has multiplicity zero at λ.
(iii) $m_{BA}(0) = m_{AB}(0) + n - m$ by part (i), because z^{n-m} has multiplicity $n - m$ at 0.
Structural Commentary: Sylvester transfer proves the polynomial identity before root orders split into zero and nonzero cases.
Key Insight: The size difference appears only as the factor z^{n-m}. □

Theorem 1.1.13. *Trace and Determinant Are Similarity Invariants*
Let $n \in \mathbb{Z}_{>0}$ and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be similar.
(i) $\mathrm{tr}\, A = \mathrm{tr}\, B$;

(ii) $\det A = \det B$.

Proof. Choose an invertible P with $B = PAP^{-1}$ by Definition 6.2.1.

(i) $\operatorname{tr} B = \operatorname{tr}\big(P(AP^{-1})\big) = \operatorname{tr}\big((AP^{-1})P\big) = \operatorname{tr} A$ by Proposition 1.1.3.
(ii) $\det B = \det(P)\det(A)\det(P^{-1}) = \det A$ by Theorem 3.8.2(vi),(viii).

Structural Commentary: Cyclicity and multiplicativity remove the conjugating factors from the two invariants.
Key Insight: In both calculations, $P^{-1}P = I_n$. □

Proposition 1.1.14. *Trace and Determinant from Eigenvalues*
Let $n \in \mathbb{Z}_{>0}$, let $A \in \operatorname{Mat}(n,n,\mathbb{C})$, and let $\lambda_1, \dots, \lambda_n$ list its eigenvalues with algebraic multiplicity.

(i) $\operatorname{tr} A = \sum_{j=1}^{n} \lambda_j$;
(ii) $\det A = \prod_{j=1}^{n} \lambda_j$.

Proof. $p_A(z) = \prod_{j=1}^{n} (z - \lambda_j)$ by the algebraic-multiplicity list and monicity.
The identity $p_A(z) = z^n - (\operatorname{tr} A)z^{n-1} + \cdots + (-1)^n \det A$ holds by Theorem 2.1.3.

(i) $\operatorname{tr} A = \sum_{j=1}^{n} \lambda_j$ by comparison of the z^{n-1} coefficients.
(ii) $\det A = \prod_{j=1}^{n} \lambda_j$ by comparison of the constant coefficients.

Structural Commentary: Two coefficient comparisons transfer the root factorization to trace and determinant.
Key Insight: The next-to-leading and constant coefficients encode the root sum and product. □

Proposition 1.1.15. *Transpose, Conjugate, and Adjoint Spectra*
Let $n \in \mathbb{Z}_{>0}$, $A \in \operatorname{Mat}(n,n,\mathbb{C})$, and $\lambda \in \operatorname{spec}(A)$.

(i) $p_{A^t} = p_A$;
(ii) $(\forall z \in \mathbb{C})\, p_{\overline{A}}(z) = \overline{p_A(\overline{z})}$;
(iii) $(\forall z \in \mathbb{C})\, p_{A^*}(z) = \overline{p_A(\overline{z})}$;
(iv) $m_{A^t}(\lambda) = m_A(\lambda)$;
(v) $m_{\overline{A}}(\overline{\lambda}) = m_A(\lambda)$;
(vi) $m_{A^*}(\overline{\lambda}) = m_A(\lambda)$.

Proof.

(i) $(\forall z \in \mathbb{C})\, p_{A^t}(z) = \det((zI_n - A)^t) = p_A(z)$ by Theorem 3.8.2(ix).
(ii) $(\forall X \in \operatorname{Mat}(n,n,\mathbb{C}))\ \det(\overline{X}) = \overline{\det X}$ by conjugation of each term in the permutation formula.
Hence $(\forall z \in \mathbb{C})\, p_{\overline{A}}(z) = \det(\overline{z}I_n - \overline{A}) = \overline{p_A(\overline{z})}$.
(iii) $(\forall z \in \mathbb{C})\, p_{A^*}(z) = p_{\overline{A}}(z) = \overline{p_A(\overline{z})}$ by $A^* = \overline{A}^{\,t}$ and parts (i)–(ii).
(iv) $m_{A^t}(\lambda) = m_A(\lambda)$ by part (i).
(v) Set $k := m_A(\lambda)$. Choose $q \in \mathbb{C}[z]$ satisfying $p_A(z) = (z-\lambda)^k q(z)$ and $q(\lambda) \neq 0$ by Definition 2.1.4.
Identity $p_{\overline{A}}(z) = (z - \overline{\lambda})^k \overline{q(\overline{z})}$ holds by part (ii), while $\overline{q(\lambda)} \neq 0$ by choice of q. Thus $m_{\overline{A}}(\overline{\lambda}) = k = m_A(\lambda)$ by Definition 2.1.4.
(vi) $m_{A^*}(\overline{\lambda}) = m_A(\lambda)$ by parts (iii) and (v).

Structural Commentary: Determinant transposition preserves factors, while entrywise conjugation transports each root.
Key Insight: The adjoint adds no spectral change beyond complex conjugation. □

Perspective 1.1.16. *Similarity Invariants and Canonical Forms*
For square complex matrices, trace and determinant are similarity invariants by Theorem 1.1.13.
The characteristic and minimal polynomials are similarity invariants by Theorems 2.1.7 and 2.2.5.
Set $J := J_3(0) \oplus J_3(0)$ and $K := J_3(0) \oplus J_2(0) \oplus J_1(0)$.
Both have characteristic polynomial z^6 by Proposition 2.1.6(i).
Both are nilpotent of index 3 by Lemma 6.2.4 and Proposition 3.8.6(iii).
Hence $\mu_J = \mu_K = z^3$ by Theorem 2.2.4 and Definition 2.2.3.
They are not similar by Corollary 6.2.7(i), because their Jordan block multisets differ.
Thus Theorem 6.2.1 and that corollary make the Jordan block multiset a complete similarity invariant.

Theorem 1.1.17. *Trace and Determinant of an Operator Are Basis-Independent*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let β, γ be bases of $\mathcal{V}$.

(i) $\operatorname{tr}\big([T]_{\beta\leftarrow\beta}\big) = \operatorname{tr}\big([T]_{\gamma\leftarrow\gamma}\big)$;
(ii) $\det\big([T]_{\beta\leftarrow\beta}\big) = \det\big([T]_{\gamma\leftarrow\gamma}\big)$.

Proof. $[T]_{\gamma\leftarrow\gamma} = C_{\gamma\leftarrow\beta}[T]_{\beta\leftarrow\beta}C_{\beta\leftarrow\gamma}$ by Theorem 5.1.1(i). Thus the representing matrices are similar by Definition 6.2.1.

(i) The traces are equal by Theorem 1.1.13(i).
(ii) The determinants are equal by Theorem 1.1.13(ii).

Structural Commentary: Change of basis reduces both claims to the corresponding similarity invariants.
Key Insight: Every representing matrix lies in one conjugacy class. □

Figure 76 summarizes the conjugation relation that makes operator trace and determinant basis-independent.

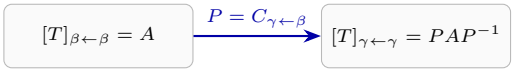

Figure 76: Basis changes conjugate matrices; trace and determinant are unchanged.

Definition 1.1.18. *Trace and Determinant of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and choose a basis β of $\mathcal{V}$.

(i) The *trace* of T is $\operatorname{tr} T := \operatorname{tr}\big([T]_{\beta\leftarrow\beta}\big)$.
(ii) The *determinant* of T is $\det T := \det\big([T]_{\beta\leftarrow\beta}\big)$.

Both quantities are independent of β by Theorem 1.1.17.

Proposition 1.1.19. *Unitary Matrices and Isometries Have Determinant of Modulus One*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

(i) If $n \in \mathbb{Z}_{>0}$ and $U \in \operatorname{Mat}(n, n, \mathbb{F})$ is unitary, then $|\det U| = 1$.
(ii) If $\mathcal{V} \neq \{\mathbf{0}\}$ is a finite-dimensional inner-product space over $\mathbb{F}$ and $S \in \mathcal{L}(\mathcal{V})$ is an isometry, then $|\det S| = 1$.

Proof.

(i) $\det(U^*) = \overline{\det U}$ by the permutation formula and Theorem 3.8.2(ix).
Taking determinants in $U^*U = I_n$, $1 = \overline{\det U}\det U = |\det U|^2$ by Theorem 3.8.2(i),(vi), so $|\det U| = 1$.
(ii) Set $n := \dim \mathcal{V}$. Choose an orthonormal basis β by Corollary 1.3.11 and set $U := [S]_{\beta\leftarrow\beta}$.
Then $U^*U = I_n$ by Theorems 2.6.6 and 2.3.7 and Proposition 3.4.4. Thus $|\det S| = |\det U| = 1$ by part (i) and Definition 1.1.18.

Structural Commentary: The matrix identity is proved first, then orthonormal coordinates transfer it to operators.
Key Insight: Determinants convert $U^*U = I_n$ into $|\det U|^2 = 1$. □

Proposition 1.1.20. *Trace and Determinant from q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $n := \dim \mathcal{V}$, let $T \in \mathcal{L}(\mathcal{V})$, and write $q_T(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$.

(i) $\operatorname{tr} T = -c_{n-1}$;
(ii) $\det T = (-1)^n c_0$;
(iii) if $\mathbb{F} = \mathbb{C}$ and $\lambda_1, \ldots, \lambda_n$ list the eigenvalues of T with algebraic multiplicity, then $\operatorname{tr} T = \sum_{j=1}^{n} \lambda_j$;
(iv) under the hypotheses of (iii), $\det T = \prod_{j=1}^{n} \lambda_j$.

Proof. Choose a basis β and set $A := [T]_{\beta\leftarrow\beta}$.
Then $q_T = p_A$ by Definition 2.1.15, while $\operatorname{tr} T = \operatorname{tr} A$ and $\det T = \det A$ by Definition 1.1.18.

(i) $\operatorname{tr} T = -c_{n-1}$ by Theorem 2.1.3.
(ii) $\det T = (-1)^n c_0$ by Theorem 2.1.3.
(iii) Under the hypotheses of (iii), $\lambda_1, \ldots, \lambda_n$ also list the eigenvalues of A with algebraic multiplicity by $q_T = p_A$.
Hence $\operatorname{tr} T = \sum_{j=1}^{n} \lambda_j$ by Proposition 1.1.14(i).
(iv) Under the same hypotheses, $\det T = \prod_{j=1}^{n} \lambda_j$ by Proposition 1.1.14(ii) with the list from part (iii).

Structural Commentary: A basis identifies q_T with p_A, after which matrix coefficient formulas transfer claimwise.
Key Insight: Operator trace and determinant are the corresponding matrix invariants in any basis. □

1.2 Minors, Cofactors, and Rank

Definition 1.2.1. *Minors, Cofactors, and Adjugate*
Let $n \geq 1$ and $A \in \operatorname{Mat}(n, n, \mathbb{F})$. Set $\det(\emptyset_{0\times 0}) := 1$.

(i) For $i, j \in \{1, \ldots, n\}$, the *minor matrix* $M_{ij}(A)$ is obtained by deleting row i and column j from A.
The corresponding *minor* is $\det(M_{ij}(A))$.
(ii) The (i,j)-*cofactor* is $C_{ij}(A) := (-1)^{i+j}\det(M_{ij}(A))$.
(iii) The *cofactor matrix* is $C(A) := [C_{ij}(A)]$, and the *adjugate* is $\operatorname{adj}(A) := C(A)^t$.

When A is fixed, write M_{ij} and C_{ij}.

Lemma 1.2.2. *Cofactor Expansion*
Let $n \geq 1$ and $A = [a_{ij}] \in \operatorname{Mat}(n, n, \mathbb{F})$.

(i) $(\forall i \in J_n)\ \det A = \sum_{k=1}^{n} a_{ik}C_{ik}$.
(ii) $(\forall j \in J_n)\ \det A = \sum_{k=1}^{n} a_{kj}C_{kj}$.

Proof.

(i) Fix $i \in J_n$. For $k \in J_n$, set $\mathfrak{S}_{i,k} := \{\sigma \in \mathfrak{S}_n : \sigma(i) = k\}$ and $T_{i,k} := \sum_{\sigma\in\mathfrak{S}_{i,k}} \operatorname{sign}(\sigma) \prod_{r=1}^{n} a_{r,\sigma(r)}$.

$(\forall k \in J_n)\, T_{i,k} = a_{ik}(-1)^{i+k}\det(M_{ik}) = a_{ik}C_{ik}$ by the order-preserving deletion bijection $\mathfrak{S}_{i,k} \xrightarrow{\sim} \mathfrak{S}_{n-1}$, $i + k - 2$ adjacent transpositions, the determinant and cofactor definitions, and Theorem 3.8.2(ix).

$\det A = \sum_{k\in J_n} T_{i,k} = \sum_{k=1}^{n} a_{ik}C_{ik}$ by $\mathfrak{S}_n = \bigsqcup_{k\in J_n} \mathfrak{S}_{i,k}$, Definition 3.8.1(ii), Theorem 3.8.2(ix), and the preceding identity.

(ii) Fix $j \in J_n$. $(\forall k \in J_n)\,[M_{jk}(A^t) = M_{kj}(A)^t \wedge C_{jk}(A^t) = C_{kj}(A)]$ by Definition 1.2.1 and Theorem 3.8.2(ix).
$\det A = \det(A^t) = \sum_{k=1}^{n} a_{kj} C_{jk}(A^t) = \sum_{k=1}^{n} a_{kj} C_{kj}(A)$ by Theorem 3.8.2(ix), part (i) for row j of A^t, and the preceding identities.

Structural Commentary: Permutation fibers prove the row formula; transposition transfers it to columns.
Key Insight: Deleting a fixed row and column leaves the corresponding minor sum. □

Theorem 1.2.3. *Adjugate Identity and Cramer's Rule*
Let $n \geq 1$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $\mathbf{b} \in \mathbb{F}^n$. For $j \in \{1, \ldots, n\}$, let $A_j(\mathbf{b})$ be obtained from A by replacing column j by $\mathbf{b}$.
(i) $A\,\mathrm{adj}(A) = \mathrm{adj}(A)A = (\det A) I_n$.
(ii) $(\forall j \in \{1, \ldots, n\}) \quad (\mathrm{adj}(A)\mathbf{b})_j = \det(A_j(\mathbf{b}))$.
(iii) If $\det A \neq 0$, then $A^{-1} = (\det A)^{-1}\,\mathrm{adj}(A)$.
(iv) If $\det A \neq 0$, then the unique solution of $A\mathbf{x} = \mathbf{b}$ satisfies $(\forall j \in \{1, \ldots, n\}) \quad x_j = \frac{\det(A_j(\mathbf{b}))}{\det A}$.

Proof.

(i) For $i, j \in J_n$, $(A\,\mathrm{adj}(A))_{ij} = \sum_{k=1}^{n} a_{ik} C_{jk}$ by Definition 1.2.1.
The identity $\sum_{k=1}^{n} a_{ik} C_{jk} = \delta_{ij} \det A$ holds by Lemma 1.2.2(i) and Theorem 3.8.2(iv),(ix) with $A_{j,\cdot} \leftarrow A_{i,\cdot}$ when $i \neq j$.
Likewise, $(\mathrm{adj}(A)A)_{ij} = \sum_{k=1}^{n} C_{ki} a_{kj}$ by Definition 1.2.1.
The identity $\sum_{k=1}^{n} C_{ki} a_{kj} = \delta_{ij} \det A$ holds by Lemma 1.2.2(ii) and Theorem 3.8.2(iv) with $A_{\cdot,i} \leftarrow A_{\cdot,j}$ when $i \neq j$.
Hence $A\,\mathrm{adj}(A) = \mathrm{adj}(A)A = (\det A) I_n$.

(ii) Fix $j \in J_n$; $(\forall i \in J_n)\, C_{ij}(A_j(\mathbf{b})) = C_{ij}(A)$ by Definition 1.2.1. Identity $(\mathrm{adj}(A)\mathbf{b})_j = \sum_{i=1}^{n} C_{ij}(A) b_i$ holds by Definition 1.2.1.
$\sum_{i=1}^{n} C_{ij}(A) b_i = \sum_{i=1}^{n} C_{ij}(A_j(\mathbf{b})) b_i = \det(A_j(\mathbf{b}))$ by the preceding cofactor identity and Lemma 1.2.2(ii).

(iii) If $\det A \neq 0$, then $A((\det A)^{-1}\,\mathrm{adj}(A)) = I_n = ((\det A)^{-1}\,\mathrm{adj}(A))A$ by part (i).
Hence $A^{-1} = (\det A)^{-1}\,\mathrm{adj}(A)$ by the definition of a matrix inverse.

(iv) If $\det A \neq 0$, then $\{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{b}\} = \{A^{-1}\mathbf{b}\}$ by part (iii).
For $\mathbf{x} = A^{-1}\mathbf{b}$, $(\forall j \in J_n)\, x_j = (\det A)^{-1}(\mathrm{adj}(A)\mathbf{b})_j = \frac{\det(A_j(\mathbf{b}))}{\det A}$ by parts (ii)–(iii).

Structural Commentary: Row and column replacements establish the adjugate identity before its inverse and solution consequences.
Key Insight: Off-diagonal cofactor sums create a repeated row or column. □

Proposition 1.2.4. *Adjugate Form of the Bordered Formula*
Let $n \geq 1$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, and $c \in \mathbb{F}$.
Then $\det \left[\begin{smallmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{smallmatrix}\right] = c \det(A) - \mathbf{y}^t\,\mathrm{adj}(A)\mathbf{x}$, *without any invertibility assumption on A.*

Proof. Set $D := \left[\begin{smallmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{smallmatrix}\right]$. $(\forall j \in J_n)$, $M_{n+1,j}(D) = [A_{\cdot,1} \;\cdots\; A_{\cdot,j-1}\; A_{\cdot,j+1} \;\cdots\; A_{\cdot,n}\; \mathbf{x}]$.
$(\forall j \in J_n)$, $\det(M_{n+1,j}(D)) = (-1)^{n-j} \det(A_j(\mathbf{x}))$ by $n - j$ column swaps and Theorem 3.8.2(iii).
$(\forall j \in J_n)$, $C_{n+1,j}(D) = (-1)^{n+1+j+n-j} \det(A_j(\mathbf{x})) = -\det(A_j(\mathbf{x}))$ by Definition 1.2.1.
$\det D = c \det A - \sum_{j=1}^{n} y_j \det(A_j(\mathbf{x}))$ by Lemma 1.2.2(i) for row $n+1$ and the preceding cofactor identity.
$\det D = c \det A - \mathbf{y}^t\,\mathrm{adj}(A)\mathbf{x}$ by Theorem 1.2.3(ii) and the preceding identity.
Structural Commentary: Last-row expansion converts each border entry into a one-column replacement determinant.
Key Insight: The cofactor and column-move signs combine to one minus sign. □

Theorem 1.2.5. *Cauchy–Binet Formula*
Let $1 \leq k \leq n$, $A \in \mathrm{Mat}(k, n, \mathbb{F})$, and $B \in \mathrm{Mat}(n, k, \mathbb{F})$. Let $\mathcal{J}_{k,n}$ be the set of k-element subsets of $\{1, \ldots, n\}$.
For $J = \{q_1 < \cdots < q_k\} \in \mathcal{J}_{k,n}$, let A^J consist of the columns of A indexed by J, and let B_J consist of the rows of B indexed by J.
Then $\det(AB) = \sum_{J \in \mathcal{J}_{k,n}} \det(A^J) \det(B_J)$.

Proof. For $\mathbf{j} = (j_1, \ldots, j_k) \in J_n^k$, set $B_{\mathbf{j}} := [b_{j_r,s}]_{r,s=1}^{k}$. Then $(\forall r \in J_k)\,(AB)_{r,\cdot} = \sum_{j \in J_n} a_{rj} B_{j,\cdot}$.
Hence $\det(AB) = \sum_{\mathbf{j} \in J_n^k} \left(\prod_{r=1}^{k} a_{rj_r} \right) \det(B_{\mathbf{j}})$ by Theorem 3.8.2(ii),(ix) and finite distributivity.
$[(\exists r, s \in J_k)\,(r \neq s \wedge j_r = j_s)] \Rightarrow \det(B_{\mathbf{j}}) = 0$ by Theorem 3.8.2(iv),(ix).
Fix $J = \{q_1 < \cdots < q_k\} \in \mathcal{J}_{k,n}$ and set $\mathbf{q}^\tau := (q_{\tau^{-1}(r)})_{r \in J_k}$ for $\tau \in \mathfrak{S}_k$.
Then $\{\mathbf{j} \in J_n^k : \{j_r : r \in J_k\} = J\} = \{\mathbf{q}^\tau : \tau \in \mathfrak{S}_k\}$, and $(\forall \tau \in \mathfrak{S}_k)\ \det(B_{\mathbf{q}^\tau}) = \mathrm{sign}(\tau) \det(B_J)$ by Theorem 3.8.2(iii),(ix).
$\sum_{\tau \in \mathfrak{S}_k} \mathrm{sign}(\tau) \left(\prod_{r=1}^{k} a_{\tau(r),q_r} \right) \det(B_J) = \det(A^J) \det(B_J)$ by Definition 3.8.1(ii).
Therefore $\det(AB) = \sum_{J \in \mathcal{J}_{k,n}} \det(A^J) \det(B_J)$ by the preceding expansion, vanishing implication, and identities.
Structural Commentary: Row multilinearity expands over J_n^k; alternation restricts the sum to $\mathcal{J}_{k,n}$.
Key Insight: Each surviving subset contributes one column minor times one row minor. □

Proposition 1.2.6. *Transpose Preserves Spectrum and Geometric Multiplicity*
Let $n \geq 1$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $\lambda \in \mathbb{F}$.
(i) $\dim(E_\lambda(A)) = \dim(E_\lambda(A^t))$.
(ii) $\lambda \in \mathrm{spec}(A)$ *iff* $\lambda \in \mathrm{spec}(A^t)$.

Proof.

(*i*) Since $(A-\lambda I_n)^t = A^t - \lambda I_n$, $\operatorname{rank}(A-\lambda I_n) = \operatorname{rank}(A^t-\lambda I_n)$ by Theorem 3.3.2.
Hence $\dim(E_\lambda(A)) = n - \operatorname{rank}(A-\lambda I_n) = n - \operatorname{rank}(A^t-\lambda I_n) = \dim(E_\lambda(A^t))$ by Corollary 3.3.1(i).
(*ii*) ($\Rightarrow$) If $\lambda \in \operatorname{spec}(A)$, then $\dim(E_\lambda(A)) > 0$ by the matrix-spectrum definition.
Hence $\dim(E_\lambda(A^t)) > 0$ by part (i), so $\lambda \in \operatorname{spec}(A^t)$ by the matrix-spectrum definition.
(*ii*) ($\Leftarrow$) If $\lambda \in \operatorname{spec}(A^t)$, then $\dim(E_\lambda(A^t)) > 0$ by the matrix-spectrum definition.
Hence $\dim(E_\lambda(A)) > 0$ by part (i), so $\lambda \in \operatorname{spec}(A)$ by the matrix-spectrum definition.

Structural Commentary: Transpose-invariant rank fixes shifted-kernel dimensions before the two spectral directions.
Key Insight: Spectrum and geometric multiplicity are nontriviality and dimension of shifted kernels. □

Corollary 1.2.7. *Nonsingularity by Rank, Determinant, and Spectrum*
Let $n \geq 1$ and $A \in \operatorname{Mat}(n,n,\mathbb{F})$. The following are equivalent:

(*i*) *A is invertible;*
(*ii*) $\operatorname{rank}(A) = n$;
(*iii*) $\det A \neq 0$;
(*iv*) $0 \notin \operatorname{spec}(A)$.

Proof.

(*i*) $\Rightarrow$ (*ii*) If A is invertible, then $\operatorname{rank}(A) = n$ by Corollary 3.6.2.
(*ii*) $\Rightarrow$ (*iii*) If $\operatorname{rank}(A) = n$, then A is invertible by Corollary 3.6.2. Hence $\det A \neq 0$ by Theorem 3.8.2(vii).
(*iii*) $\Rightarrow$ (*iv*) If $\det A \neq 0$, then A is invertible by Theorem 3.8.2(vii). Hence $0 \notin \operatorname{spec}(A)$ by Corollary 5.1.8.
(*iv*) $\Rightarrow$ (*i*) If $0 \notin \operatorname{spec}(A)$, then A is invertible by Corollary 5.1.8.

Structural Commentary: The cycle passes through full rank and determinant before returning through zero-spectrum exclusion.
Key Insight: Invertibility has equivalent range, determinant, and shifted-kernel tests. □

Theorem 1.2.8. *Rank Bounds for Products and Concatenations*
Let $A \in \operatorname{Mat}(m,k,\mathbb{F})$, $B \in \operatorname{Mat}(k,n,\mathbb{F})$, and $C \in \operatorname{Mat}(m,p,\mathbb{F})$. Set $D := [\,A\ C\,]$.

(*i*) $\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B))$.
(*ii*) $\operatorname{rank}(AB) = \operatorname{rank}(A) \iff \operatorname{col}(AB) = \operatorname{col}(A)$.
(*iii*) $\max(\operatorname{rank}(A), \operatorname{rank}(C)) \leq \operatorname{rank}(D)$.
(*iv*) $\operatorname{rank}(D) = \operatorname{rank}(A) \iff \operatorname{col}(C) \subseteq \operatorname{col}(A)$.

Proof. Column-space inclusion is rank-monotone, with equality iff the spaces coincide, by the rank definition and Proposition 1.4.3.
For matrices X, Y of compatible dimensions, $\operatorname{col}(XY) \subseteq \operatorname{col}(X)$ by Propositions 3.1.10 and 3.1.9.
The identities $(XY)^t = Y^tX^t$ and $\operatorname{rank}(X^t) = \operatorname{rank}(X)$ hold by Proposition 3.1.16 and Theorem 3.3.2, respectively.
The identity $\operatorname{col}(D) = \operatorname{col}(A) + \operatorname{col}(C)$ holds by the definition of D and the column-space definition.

(*i*) $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$ by the product-column inclusion and the rank criterion.
$\operatorname{rank}(AB) \leq \operatorname{rank}(B)$ by the transpose identities, the product-column inclusion for B^tA^t, and the rank criterion.
(*ii*) ($\Rightarrow$) $\operatorname{col}(AB) = \operatorname{col}(A)$ by $\operatorname{rank}(AB) = \operatorname{rank}(A)$, the product-column inclusion, and the rank criterion.
(*ii*) ($\Leftarrow$) $\operatorname{rank}(AB) = \operatorname{rank}(A)$ by $\operatorname{col}(AB) = \operatorname{col}(A)$ and the rank definition.
(*iii*) $\max(\operatorname{rank}(A), \operatorname{rank}(C)) \leq \operatorname{rank}(D)$ by $\operatorname{col}(A) \cup \operatorname{col}(C) \subseteq \operatorname{col}(D)$, the concatenation identity, and the rank criterion.
(*iv*) ($\Rightarrow$) $\operatorname{col}(C) \subseteq \operatorname{col}(D) = \operatorname{col}(A)$ by the concatenation identity, $\operatorname{rank}(D) = \operatorname{rank}(A)$, and the rank criterion.
(*iv*) ($\Leftarrow$) $\operatorname{col}(C) \subseteq \operatorname{col}(A) \Rightarrow \operatorname{col}(D) = \operatorname{col}(A)$ by the concatenation identity.
Hence $\operatorname{rank}(D) = \operatorname{rank}(A)$ by the rank definition.

Structural Commentary: Product and concatenation identities reduce every claim to nested column spaces.
Key Insight: Equal dimensions force equality after column-space inclusion is established. □

Theorem 1.2.9. *Sylvester Rank Inequality*
Let $A \in \operatorname{Mat}(m,k,\mathbb{F})$ and $B \in \operatorname{Mat}(k,n,\mathbb{F})$. Then $\operatorname{rank}(A) + \operatorname{rank}(B) - k \leq \operatorname{rank}(AB)$.

Proof. Set $R := \operatorname{col}(B)$ and define $S : R \to \mathbb{F}^m$ by $S(\mathbf{x}) := A\mathbf{x}$.
The identities $\operatorname{range}(S) = \operatorname{col}(AB)$ and $\ker(S) = R \cap \ker(A)$ hold by Proposition 3.1.10 and the definitions of R and S.
$\operatorname{rank}(AB) = \dim(R) - \dim(R \cap \ker(A)) \geq \operatorname{rank}(B) - \dim(\ker(A))$ by Theorem 1.7.3(ii), the rank definition, and Proposition 1.4.3(i).
$\operatorname{rank}(AB) \geq \operatorname{rank}(B) - (k - \operatorname{rank}(A)) = \operatorname{rank}(A) + \operatorname{rank}(B) - k$ by the preceding bound and Corollary 3.3.1(i).
Structural Commentary: Rank–nullity restricts A to $\operatorname{col}(B)$ and isolates the lost middle-space directions.
Key Insight: Only $\operatorname{col}(B) \cap \ker(A)$ can disappear under multiplication. □

Proposition 1.2.10. *Sylvester's Law of Nullity*
For a matrix X, write $\operatorname{nullity}(X) := \dim(\ker(X))$.

(*i*) *If $A \in \operatorname{Mat}(m,k,\mathbb{F})$ and $B \in \operatorname{Mat}(k,n,\mathbb{F})$, then $\operatorname{nullity}(AB) \leq \operatorname{nullity}(A) + \operatorname{nullity}(B)$.*
(*ii*) *If $A, B \in \operatorname{Mat}(n,n,\mathbb{F})$, then $\max(\operatorname{nullity}(A), \operatorname{nullity}(B)) \leq \operatorname{nullity}(AB)$.*

Proof.

(i) $\operatorname{nullity}(AB) \leq n - \operatorname{rank}(A) - \operatorname{rank}(B) + k = \operatorname{nullity}(A) + \operatorname{nullity}(B)$ by Theorem 1.2.9 and Corollary 3.3.1(i) for AB, A, and B.
(ii) $\operatorname{nullity}(AB) \geq \max(n - \operatorname{rank}(A), n - \operatorname{rank}(B)) = \max(\operatorname{nullity}(A), \operatorname{nullity}(B))$ by Theorem 1.2.8(i) and Corollary 3.3.1(i).

Structural Commentary: Rank–nullity reverses the lower and upper product-rank bounds into nullity bounds.
Key Insight: Subtracting rank from a fixed domain dimension reverses inequalities. □

Theorem 1.2.11. *Frobenius Rank Inequality*
Let $A \in \operatorname{Mat}(m,k,\mathbb{F})$, $B \in \operatorname{Mat}(k,p,\mathbb{F})$, and $C \in \operatorname{Mat}(p,n,\mathbb{F})$. Then $\operatorname{rank}(AB) + \operatorname{rank}(BC) \leq \operatorname{rank}(B) + \operatorname{rank}(ABC)$.

Proof. Set $R := \mathrm{col}(B)$ and $S := \mathrm{col}(BC)$, and define $T_A : \mathbb{F}^k \to \mathbb{F}^m$ by $T_A(\mathbf{x}) := A\mathbf{x}$.
The identities $S \subseteq R$, $T_A(R) = \mathrm{col}(AB)$, and $T_A(S) = \mathrm{col}(ABC)$ hold by Proposition 3.1.10.
$(\forall U \in \{R, S\})\ \dim(T_A(U)) = \dim(U) - \dim(U \cap \ker(A))$ by Theorem 1.7.3(ii) for $T_A|_U$ and the definition of restriction.
The Frobenius inequality holds by $S \cap \ker(A) \subseteq R \cap \ker(A)$, Proposition 1.4.3(i), the preceding identities, and the rank definition.
Structural Commentary: Rank–nullity compares A on the nested column spaces of BC and B.
Key Insight: The smaller space cannot meet $\ker(A)$ in more dimensions. □

Theorem 1.2.11 contains both two-factor bounds as degenerations: a zero outer factor recovers Theorem 1.2.8(i), while an identity middle factor recovers Theorem 1.2.9.

Theorem 1.2.12. *Rank Triangle Inequality*
Let $A, B \in \mathrm{Mat}(m, n, \mathbb{F})$.
(i) $\mathrm{rank}(A + B) \leq \mathrm{rank}(A) + \mathrm{rank}(B)$.
(ii) $|\mathrm{rank}(A) - \mathrm{rank}(B)| \leq \mathrm{rank}(A + B)$.

Proof.
(i) $\mathrm{col}(A + B) \subseteq \mathrm{col}(A) + \mathrm{col}(B)$ by adding corresponding columns.
$\mathrm{rank}(A + B) \leq \mathrm{rank}(A) + \mathrm{rank}(B)$ by Proposition 1.4.3(i), Theorem 2.3.1, and the rank definition.
(ii) $|\mathrm{rank}(A) - \mathrm{rank}(B)| \leq \mathrm{rank}(A + B)$ by part (i) with $A = (A + B) + (-B)$,
and $B = (A + B) + (-A)$, $(\forall X \in \{A, B\})\ \mathrm{col}(-X) = \mathrm{col}(X)$, and the rank and absolute-value definitions.
Structural Commentary: The column-space sum proves the upper bound; rearranging each summand proves both reverse bounds.
Key Insight: Rewriting either summand converts the upper bound into a lower bound. □

1.3 Commutators and Special Matrix Classes

Definition 1.3.1. *Commutator*
(i) For $S, T \in \mathcal{L}(\mathcal{V})$, their *commutator* is $[S, T] := ST - TS$.
(ii) For $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, their *matrix commutator* is $[A, B] := AB - BA$.

Proposition 1.3.2. *Trace Obstruction for Commutators*
Let $A, B, C \in \mathrm{Mat}(n, n, \mathbb{F})$.
(i) $\mathrm{tr}([A, B]) = 0$.
(ii) If C is similar to $[A, B]$, then $\mathrm{tr}(C) = 0$.

Proof.
(i) $\mathrm{tr}([A, B]) = \mathrm{tr}(AB) - \mathrm{tr}(BA) = 0$ by Proposition 1.1.1(i),(ii) and Proposition 1.1.3.
(ii) If C is similar to $[A, B]$, then $\mathrm{tr}(C) = \mathrm{tr}([A, B]) = 0$ by Theorem 1.1.13(i) and part (i).
Structural Commentary: Trace cyclicity annihilates a commutator; similarity invariance transfers the obstruction.
Key Insight: The two product traces cancel exactly. □

Corollary 1.3.3. *The Identity Is Not a Commutator*
Let $\mathcal{V}$ be finite-dimensional over $\mathbb{F}$ and $(\dim \mathcal{V}) \cdot 1_{\mathbb{F}} \neq 0$. Then $(\nexists S, T \in \mathcal{L}(\mathcal{V}))\ [S, T] = I_{\mathcal{V}}$.

Proof. Set $n := \dim \mathcal{V}$, and suppose $[S, T] = I_{\mathcal{V}}$. Choose a basis β of $\mathcal{V}$ and set $A := [S]_{\beta \leftarrow \beta}$ and $B := [T]_{\beta \leftarrow \beta}$.
Then $[A, B] = [ST - TS]_{\beta \leftarrow \beta} = I_n$ by Propositions 3.4.4 and 3.2.4.
By Proposition 1.3.2(i), $0 = \mathrm{tr}([A, B]) = \mathrm{tr}(I_n) = n \cdot 1_{\mathbb{F}}$, contrary to the hypothesis.
Structural Commentary: Coordinate matrices transfer the operator equation to the matrix trace obstruction.
Key Insight: The identity has trace $(\dim \mathcal{V}) \cdot 1_{\mathbb{F}}$. □

Lemma 1.3.4. *Scalar Matrices and One-Step Orbits*
Let $n \geq 2$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then $(\forall \mathbf{x} \in \mathbb{F}^n)$ $(\mathbf{x}, A\mathbf{x})$ is linearly dependent iff A is a scalar matrix.

Proof.
($\Rightarrow$) Suppose $(\forall \mathbf{x} \in \mathbb{F}^n)$ $(\mathbf{x}, A\mathbf{x})$ is linearly dependent. Since $\mathbf{e}_i \neq \mathbf{0}$, $(\exists\, a_i \in \mathbb{F})$ $A\mathbf{e}_i = a_i\mathbf{e}_i$ for $i \in \{1, \ldots, n\}$.
For $i \in \{2, \ldots, n\}$, the dependence of $\mathbf{e}_1 + \mathbf{e}_i$ and $A(\mathbf{e}_1 + \mathbf{e}_i) = a_1\mathbf{e}_1 + a_i\mathbf{e}_i$ implies $(\exists\, b_i \in \mathbb{F})$ $a_1\mathbf{e}_1 + a_i\mathbf{e}_i = b_i(\mathbf{e}_1 + \mathbf{e}_i)$.
Thus $a_1 = b_i = a_i$ by independence of $(\mathbf{e}_1, \mathbf{e}_i)$. Hence $(\forall\, i \in \{1, \ldots, n\})$ $A\mathbf{e}_i = a_1\mathbf{e}_i$, so $A = a_1 I_n$ by equality on standard basis.
($\Leftarrow$) If $A = cI_n$, then $(\forall \mathbf{x} \in \mathbb{F}^n)$ $A\mathbf{x} = c\mathbf{x}$, so each pair $(\mathbf{x}, A\mathbf{x})$ is linearly dependent.
Structural Commentary: Coordinate lines produce individual scalars; diagonal lines force those scalars to coincide.
Key Insight: Dependence along $\mathbf{e}_1 + \mathbf{e}_i$ equates the coordinate eigenvalues. □

Lemma 1.3.5. *Nonscalar Matrices Have a Zero Diagonal Entry After Similarity*
Let $n \geq 2$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. If A is not scalar, then A is similar over $\mathbb{F}$ to a matrix whose $(1, 1)$ entry is 0.

Proof. Choose $\mathbf{x} \in \mathbb{F}^n$ such that $(\mathbf{x}, A\mathbf{x})$ linearly independent by Lemma 1.3.4. Extend this pair to basis β of $\mathbb{F}^n$ by Theorem 1.3.13.
Let $T_A : \mathbb{F}^n \to \mathbb{F}^n$ be defined by $T_A(\mathbf{y}) := A\mathbf{y}$, and set $C := [T_A]_{\beta \leftarrow \beta}$. The first column of C is $[A\mathbf{x}]_\beta = (0, 1, 0, \ldots, 0)^t$.
Hence the $(1, 1)$ entry of C is zero, and C is similar to A by Theorem 5.1.1(i), with the standard basis as the original basis.
Structural Commentary: A noninvariant line begins a basis whose second vector is $A\mathbf{x}$.
Key Insight: The coordinate vector of $A\mathbf{x}$ begins with zero. □

Theorem 1.3.6. *Similarity to a Constant-Diagonal Matrix*
Let $n \geq 1$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $(\forall\, k \in \{1, \ldots, n\})$ $k \cdot 1_{\mathbb{F}} \neq 0$. Then A is similar to a matrix with constant diagonal $\frac{1}{n}\mathrm{tr}(A)$.

Proof.
(i) *Trace Shift*: Set $\tau := \frac{1}{n}\mathrm{tr}(A)$ and $B := A - \tau I_n$. Then $\mathrm{tr}(B) = \mathrm{tr}(A) - (n \cdot 1_{\mathbb{F}})\tau = 0$ by Proposition 1.1.1(i),(ii).
It suffices to show that every trace-zero matrix of size at most n is similar to a zero-diagonal matrix.
Indeed, adding τI_n preserves similarity by Theorem 6.2.8(i), with shift $-\tau$.

(ii) *Induction*: For $m \in \{1, \dots, n\}$, let $\mathcal{P}_m$ assert that every trace-zero matrix in $\mathrm{Mat}(m, m, \mathbb{F})$ is similar to a zero-diagonal matrix.
Property $\mathcal{P}_1$ holds because a trace-zero 1×1 matrix is $[0]$.
Fix $m \in \{2, \dots, n\}$, assume $\mathcal{P}_{m-1}$, and let $C \in \mathrm{Mat}(m, m, \mathbb{F})$ satisfy $\mathrm{tr}(C) = 0$.
(iii) *Scalar Case*: If $C = cI_m$, then $0 = \mathrm{tr}(C) = (m \cdot 1_{\mathbb{F}})c$. Thus $c = 0$ by $m \cdot 1_{\mathbb{F}} \neq 0$, so C already has zero diagonal.
(iv) *Nonscalar Case*: If C is not scalar, then C is similar to $D := \left[\begin{smallmatrix} 0 & \mathbf{u}^t \\ \mathbf{v} & C_1 \end{smallmatrix}\right]$ by Lemma 1.3.5.
The equality $\mathrm{tr}(C_1) = \mathrm{tr}(D) = \mathrm{tr}(C) = 0$ holds by Theorem 1.1.13(i) and the block diagonal entries.
Choose $S_1 \in \mathrm{Mat}(m-1, m-1, \mathbb{F})^{\times}$ such that $S_1^{-1} C_1 S_1$ has zero diagonal by $\mathcal{P}_{m-1}$. Set $R := I_1 \oplus S_1$.
Then $R^{-1} D R$ has zero diagonal because its lower-right block is $S_1^{-1} C_1 S_1$.
Thus C is similar to a zero-diagonal matrix by Proposition 6.2.2, so $\mathcal{P}_m$ holds.
Therefore $\mathcal{P}_n$ holds by induction, and the original assertion follows by the trace shift in part (i).

Structural Commentary: A trace shift reduces to zero diagonal; induction isolates one diagonal entry at a time.
Key Insight: The remaining block retains trace zero after the first entry vanishes. □

Theorem 1.3.7. *Shoda's Theorem in the Distinct-Diagonal Case*
Let $n \geq 1$, $(\forall\, k \in \{1, \dots, n\})\ k \cdot 1_{\mathbb{F}} \neq 0$, *and* $A \in \mathrm{Mat}(n, n, \mathbb{F})$.
(i) A *is a commutator in* $\mathrm{Mat}(n, n, \mathbb{F})$ *iff* $\mathrm{tr}(A) = 0$.
(ii) No nonzero scalar matrix in $\mathrm{Mat}(n, n, \mathbb{F})$ *is a commutator.*

Proof.
(i) ($\Rightarrow$) If $A = [U, V]$ for $U, V \in \mathrm{Mat}(n, n, \mathbb{F})$, then $\mathrm{tr}(A) = 0$ by Proposition 1.3.2(i).
(i) ($\Leftarrow$) Suppose $\mathrm{tr}(A) = 0$. Choose $S \in \mathrm{Mat}(n, n, \mathbb{F})^{\times}$ such that $B := S^{-1} A S = [b_{ij}]$ has zero diagonal by Theorem 1.3.6.
For $i \in \{1, \dots, n\}$, set $d_i := i \cdot 1_{\mathbb{F}}$, and let $X := \mathrm{diag}(d_1, \dots, d_n)$. Define $Y = [y_{ij}]$ by $y_{ii} := 0$ and $y_{ij} := (d_i - d_j)^{-1} b_{ij}$ for $i \neq j$.
For $i \neq j$, set $k := |i - j|$. Then $1 \leq k \leq n - 1$ and $d_i - d_j \in \{k \cdot 1_{\mathbb{F}}, -k \cdot 1_{\mathbb{F}}\} \subseteq \mathbb{F}^{\times}$ by the hypothesis.
Hence $[X, Y]_{ii} = 0 = b_{ii}$ and $[X, Y]_{ij} = (d_i - d_j) y_{ij} = b_{ij}$, so $B = [X, Y]$.
Therefore $A = [SXS^{-1}, SYS^{-1}]$ by direct multiplication, and A is a commutator.
(ii) If $c \in \mathbb{F} \setminus \{0\}$, then $\mathrm{tr}(cI_n) = (n \cdot 1_{\mathbb{F}})c \neq 0$. Thus cI_n is not a commutator by part (i).

Structural Commentary: Trace supplies necessity; constant-diagonal similarity and distinct scalars construct the converse.
Key Insight: Dividing each off-diagonal entry by $d_i - d_j$ realizes the commutator. □

Remark 1.3.8. *Field Hypothesis in the Diagonal Proof*
Theorem 1.3.7 uses the field hypothesis twice: the scalars $d_1, \dots, d_n$ must be distinct, and every block size in the induction must remain nonzero. In small characteristic this diagonal construction requires replacement.

Computation Method 1.3.9. *Trace–Determinant Invariant Computation*
Let $n \geq 1$, $A, C \in \mathrm{Mat}(n, n, \mathbb{F})$, and suppose $p_A(z) = \prod_{j=1}^{n} (z - \lambda_j)$ over a field containing $\mathbb{F}$. The following procedure extracts aggregate eigenvalue data and screens a commutator equation:
(i) If A represents an operator, choose a basis in which its representing matrix is easiest to compute.
(ii) Compute $\mathrm{tr}(A)$ from the diagonal entries and $\det(A)$ by a convenient determinant formula.
(iii) Record $\sum_{j=1}^{n} \lambda_j = \mathrm{tr}(A)$ and $\prod_{j=1}^{n} \lambda_j = \det(A)$ by Theorem 2.1.3(i).
(iv) For $[X, Y] = C$, first compute $\mathrm{tr}(C)$. If it is nonzero, no solution exists by Proposition 1.3.2(i).

Definition 1.3.10. *Idempotent and Nilpotent Matrices*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$.
(i) A is *idempotent* iff $A^2 = A$.
(ii) A is *nilpotent* iff $(\exists\, k \in \mathbb{Z}_{>0})\ A^k = 0$.

Proposition 1.3.11. *Basic Invariants of Idempotents*
Let $P \in \mathrm{Mat}(n, n, \mathbb{F})$ *satisfy* $P^2 = P$, *and set* $r := \mathrm{rank}(P)$.
(i) $\mathbb{F}^n = \mathrm{range}(P) \oplus \ker(P)$.
(ii) $(\forall\, \mathbf{y} \in \mathrm{range}(P))\ P\mathbf{y} = \mathbf{y}$ *and* $(\forall\, \mathbf{z} \in \ker(P))\ P\mathbf{z} = \mathbf{0}$.
(iii) P *is similar to* $I_r \oplus 0_{n-r}$.
(iv) P *is diagonalizable over* $\mathbb{F}$.
(v) $\mathrm{spec}(P) \subseteq \{0, 1\}$.
(vi) $\mathrm{tr}(P) = \mathrm{rank}(P)$.
(vii) If $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, *then* P *is the orthogonal projection onto* $\mathrm{range}(P)$ *iff* $P = P^*$.

Proof.
(i) For $\mathbf{x} \in \mathbb{F}^n$, write $\mathbf{x} = P\mathbf{x} + (\mathbf{x} - P\mathbf{x})$, where $P\mathbf{x} \in \mathrm{range}(P)$.
Since $P(\mathbf{x} - P\mathbf{x}) = P\mathbf{x} - P^2\mathbf{x} = \mathbf{0}$, the second term lies in $\ker(P)$.
If $\mathbf{x} \in \mathrm{range}(P) \cap \ker(P)$, write $\mathbf{x} = P\mathbf{u}$. Then $\mathbf{x} = P^2\mathbf{u} = P\mathbf{x} = \mathbf{0}$, so the sum is direct.
(ii) If $\mathbf{y} = P\mathbf{u} \in \mathrm{range}(P)$, then $P\mathbf{y} = P^2\mathbf{u} = P\mathbf{u} = \mathbf{y}$. The assertion on $\ker(P)$ follows from its definition.
(iii) Choose a basis $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ of $\mathrm{range}(P)$ and a basis $(\mathbf{v}_1, \dots, \mathbf{v}_{n-r})$ of $\ker(P)$ by Corollary 3.3.1(i).
Their concatenation is a basis by part (i).
By part (ii), the representing matrix in this basis is $I_r \oplus 0_{n-r}$. Thus P is similar to this matrix by Theorem 5.1.1(i).
(iv) The matrix $I_r \oplus 0_{n-r}$ is diagonal, and the similarity in part (iii) is a diagonalization of P.
(v) If $P\mathbf{x} = \lambda\mathbf{x}$ for $\mathbf{x} \neq \mathbf{0}$, then $\lambda^2\mathbf{x} = P^2\mathbf{x} = P\mathbf{x} = \lambda\mathbf{x}$. Hence $\lambda(\lambda - 1) = 0$, so $\lambda \in \{0, 1\}$.
(vi) By Theorem 1.1.13(i) and part (iii), $\mathrm{tr}(P) = \mathrm{tr}(I_r \oplus 0_{n-r}) = r = \mathrm{rank}(P)$.
(vii) ($\Rightarrow$) Regarding P as an operator on the standard inner-product space, $P = P^*$ by Theorem 2.5.6.
(vii) ($\Leftarrow$) Regarding P as an operator on the standard inner-product space, it is the orthogonal projection onto $\mathrm{range}(P)$ by the idempotence hypothesis, $P = P^*$, and Theorem 2.5.6.

Structural Commentary: Range–kernel coordinates determine invariants; self-adjointness characterizes orthogonal projection.
Key Insight: In an adapted basis, every idempotent is $I_r \oplus 0_{n-r}$. □

2 Positive Definite and Structured Matrices

2.1 Positive Semidefinite and Positive Definite Matrices

Definition 2.1.1. *Positive Semidefinite and Positive Definite Matrices*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) The matrix A is *positive semidefinite*, written $A \succeq 0$, iff $A = A^*$ and $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, \mathbf{x}^* A \mathbf{x} \geq 0$.
(ii) The matrix A is *positive definite*, written $A \succ 0$, iff $A = A^*$ and $(\forall\, \mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, \mathbf{x}^* A \mathbf{x} > 0$.
(iii) The matrix A is *negative semidefinite* iff $-A \succeq 0$.
(iv) The matrix A is *negative definite* iff $-A \succ 0$.

Proposition 2.1.2. *Rectangular Gram Products Are Positive Semidefinite*
Let $m, n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $R \in \mathrm{Mat}(m, n, \mathbb{F})$.

(i) $R^* R \succeq 0$.
(ii) $RR^* \succeq 0$.
(iii) $(\forall\, j \in \{1, \ldots, n\})\, (R^* R)_{j,j} = \|R_{\cdot,j}\|_2^2$.
(iv) $(\forall\, i \in \{1, \ldots, m\})\, (RR^*)_{i,i} = \|R_{i,\cdot}\|_2^2$.
(v) If $\mathrm{rank}(R) = n$, *then* $R^* R \succ 0$.
(vi) If $\mathrm{rank}(R) = m$, *then* $RR^* \succ 0$.

Proof. For $(A, d) \in \{(R, n), (R^*, m)\}$, $(A^* A)^* = A^* A$ by Proposition 2.3.8(ii),(iii).
For the same pairs, $(\forall\, \mathbf{z} \in \mathbb{F}^d)\, \mathbf{z}^* A^* A \mathbf{z} = \|A\mathbf{z}\|_2^2 \geq 0$.

(i) $R^* R \succeq 0$ by the preceding identities with $(A, d) = (R, n)$.
(ii) $RR^* \succeq 0$ by the preceding identities with $(A, d) = (R^*, m)$.
(iii) $(\forall\, j \in \{1, \ldots, n\})\, (R^* R)_{j,j} = \mathbf{e}_j^* R^* R \mathbf{e}_j = \|R\mathbf{e}_j\|_2^2 = \|R_{\cdot,j}\|_2^2$.
(iv) $(\forall\, i \in \{1, \ldots, m\})\, (RR^*)_{i,i} = \mathbf{e}_i^* R R^* \mathbf{e}_i = \|R^* \mathbf{e}_i\|_2^2 = \|R_{i,\cdot}\|_2^2$.
(v) If $\mathrm{rank}(R) = n$, then $\ker(R) = \{\mathbf{0}\}$ by Corollary 3.3.1(i). Thus $(\forall\, \mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, \mathbf{x}^* R^* R \mathbf{x} = \|R\mathbf{x}\|_2^2 > 0$, and $R^* R \succ 0$ by part (i).
(vi) If $\mathrm{rank}(R) = m$, then $\mathrm{col}(R) = \mathbb{F}^m$ by rank definition, Proposition 1.4.3(ii); $\ker(R^*) = \mathrm{col}(R)^\perp = \{\mathbf{0}\}$ by Proposition 2.3.10(i). Therefore $(\forall\, \mathbf{y} \in \mathbb{F}^m \setminus \{\mathbf{0}\})\, \mathbf{y}^* R R^* \mathbf{y} = \|R^* \mathbf{y}\|_2^2 > 0$, and $RR^* \succ 0$ by part (ii).

Structural Commentary: Adjoint identities establish self-adjointness; rank removes the kernels in the two squared-norm forms.
Key Insight: A Gram form is strictly positive exactly off its factor's kernel. □

Theorem 2.1.3. *Equivalent Forms of Positive Semidefiniteness*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:

(i) $A \succeq 0$.
(ii) $A = A^*$ *and* $\mathrm{spec}(A) \subseteq [0, \infty)$.
(iii) $(\exists\, C \in \mathrm{Mat}(n, n, \mathbb{F}))\, [C \succeq 0 \wedge C^2 = A]$.
(iv) $(\exists\, m \in \mathbb{Z}_{>0})(\exists\, B \in \mathrm{Mat}(m, n, \mathbb{F}))\, A = B^* B$.

Proof. Identify each square matrix with its induced operator on $\mathbb{F}^n$.

(i) $\Rightarrow$ *(ii)* $A = A^*$ and $\mathrm{spec}(A) \subseteq [0, \infty)$ by Theorem 6.3.6(i)$\Rightarrow$(ii).
(ii) $\Rightarrow$ *(iii)* Choose $C \succeq 0$ with $C^2 = A$ by Theorem 6.3.6(ii)$\Rightarrow$(iii).
(iii) $\Rightarrow$ *(iv)* If $C \succeq 0$ and $C^2 = A$, then $A = C^2 = C^* C$; take $(m, B) = (n, C)$.
(iv) $\Rightarrow$ *(i)* $A \succeq 0$ by Proposition 2.1.2(i) for $A = B^* B$.

Structural Commentary: Operator positivity supplies the spectral root; self-adjointness converts that root into a Gram factor.
Key Insight: For $C \succeq 0$, the identities $C^2 = C^* C$ coincide. □

Corollary 2.1.4. *Unique Positive Semidefinite Square Root*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A \succeq 0$.

(i) $(\exists!\, C \in \mathrm{Mat}(n, n, \mathbb{F}))\, [C \succeq 0 \wedge C^2 = A]$; *this matrix is denoted by* $A^{1/2}$.
(ii) If $A = U \,\mathrm{diag}(\lambda_1, \ldots, \lambda_n) U^*$, *where U is unitary and* $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j \geq 0$, *then* $A^{1/2} = U \,\mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}) U^*$.

Proof.

(i) Existence and uniqueness hold by Proposition 6.3.7 for the operator induced by A.
(ii) Set $C := U \,\mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}) U^*$. Then $C = C^*$, $\mathrm{spec}(C) \subseteq [0, \infty)$, and $C^2 = A$. Thus $C \succeq 0$ by Theorem 2.1.3(ii)$\Rightarrow$(i), and $C = A^{1/2}$ by part (i).

Structural Commentary: Spectral coordinates construct the root; operator uniqueness identifies every positive semidefinite root.
Key Insight: Each nonnegative eigenvalue has one nonnegative square root. □

Theorem 2.1.5. *Kernel, Rank, and Columns of $B^* B$*
Let $m, n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $B \in \mathrm{Mat}(m, n, \mathbb{F})$, and $A = B^* B$.

(i) $A \succeq 0$;
(ii) $\ker(A) = \ker(B)$;
(iii) $\mathrm{rank}(A) = \mathrm{rank}(B)$;
(iv) $\mathrm{col}(A) = \mathrm{col}(B^*)$;
(v) $A = 0$ *iff* $B = 0$.

Proof.

(i) $A \succeq 0$ by Proposition 2.1.2(i).
(ii) $\ker(B) \subseteq \ker(A)$ by $A = B^* B$. If $\mathbf{x} \in \ker(A)$, then $\|B\mathbf{x}\|_2^2 = \mathbf{x}^* A \mathbf{x} = 0$, so $\mathbf{x} \in \ker(B)$.
(iii) $\mathrm{rank}(A) = n - \dim \ker(A) = n - \dim \ker(B) = \mathrm{rank}(B)$ by part (ii) and Corollary 3.3.1(i).
(iv) $\mathrm{col}(A) = \ker(A)^\perp = \ker(B)^\perp = \mathrm{col}(B^*)$ by parts (i)–(ii) and Proposition 2.3.10(iii).
(v) ($\Rightarrow$) If $A = 0$, then $\mathrm{rank}(B) = \mathrm{rank}(A) = 0$ by part (iii), so $B = 0$.

(v) $(\Leftarrow)$ $B = 0 \Rightarrow A = B^*B = 0$.

Structural Commentary: Squared-norm identity fixes the kernel; rank–nullity and adjoint orthogonality determine rank and columns.
Key Insight: B^*B and B vanish on exactly the same vectors. □

Figure 77 summarizes the squared-norm argument that transfers kernel data from B^*B to B.

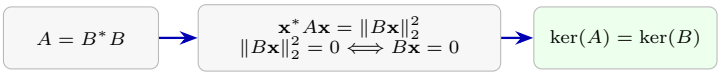

Figure 77: For $A = B^*B$, the squared-norm identity certifies $\ker(A) = \ker(B)$.

Corollary 2.1.6. *Zero Quadratic Form Detects the Kernel*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A \succeq 0$. Then $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, [\mathbf{x}^*A\mathbf{x} = 0 \iff A\mathbf{x} = \mathbf{0}]$.

Proof. Fix $\mathbf{x} \in \mathbb{F}^n$, and choose $m \in \mathbb{Z}_{>0}$ and $B \in \mathrm{Mat}(m, n, \mathbb{F})$ with $A = B^*B$ by Theorem 2.1.3(iv).
$(\Rightarrow)$ $\mathbf{x}^*A\mathbf{x} = 0 \Rightarrow \|B\mathbf{x}\|_2^2 = 0 \Rightarrow B\mathbf{x} = \mathbf{0} \Rightarrow A\mathbf{x} = \mathbf{0}$.
$(\Leftarrow)$ $A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x}^*A\mathbf{x} = 0$.

Structural Commentary: Gram factorization converts the quadratic value into a norm whose zero set is $\ker(A)$.
Key Insight: $\|B\mathbf{x}\|_2^2 = 0$ forces $B\mathbf{x} = \mathbf{0}$. □

Computation Method 2.1.7. *PSD Spectral Factorization*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A = A^*$.
The procedure decides whether $A \succeq 0$, constructs $A^{1/2}$ when it exists, and transfers kernel and rank questions to that factor.
(i) Compute a unitary diagonalization $A = U \operatorname{diag}(\lambda_1, \ldots, \lambda_n) U^*$ by the appropriate spectral theorem.
(ii) If $(\exists\, j \in \{1, \ldots, n\})\, \lambda_j < 0$, conclude that $A \not\succeq 0$. If $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j \geq 0$, conclude that $A \succeq 0$ by Theorem 2.1.3(ii).
(iii) In the positive semidefinite case, form $C = U \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}) U^*$. Then $C = A^{1/2}$ and $A = C^*C$ by Corollary 2.1.4(i),(ii).
(iv) Read the kernel, rank, column-space, and zero-quadratic-form data from C by Theorem 2.1.5 and Corollary 2.1.6.

Theorem 2.1.8. *Equivalent Forms of Positive Definiteness*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:
(i) $A \succ 0$.
(ii) $A = A^*$ *and* $\mathrm{spec}(A) \subseteq (0, \infty)$.
(iii) $A \succeq 0$ *and* A *is invertible.*
(iv) $(\exists\, B \in \mathrm{Mat}(n, n, \mathbb{F})^\times)\, A = B^*B$.
(v) A *is invertible and* $A^{-1} \succ 0$.

Proof.
(i) $\Rightarrow$ *(ii)* $A = A^*$ by positive definiteness. For $\lambda \in \mathrm{spec}(A)$ and $\mathbf{v} \in E_\lambda(A) \setminus \{\mathbf{0}\}$, $0 < \mathbf{v}^*A\mathbf{v} = \lambda \|\mathbf{v}\|_2^2$, so $\lambda > 0$.
(ii) $\Rightarrow$ *(iii)* $A \succeq 0$ by Theorem 2.1.3(ii)$\Rightarrow$(i), and A is invertible by Corollary 5.1.8.
(iii) $\Rightarrow$ *(iv)* Set $C := A^{1/2}$. Then $A = C^2 = C^*C$ by Corollary 2.1.4(i), while $\ker(C) \subseteq \ker(A) = \{\mathbf{0}\}$.
Thus C is invertible by Corollary 3.6.3(vi)$\Rightarrow$(i).
(iv) $\Rightarrow$ *(v)* $A = B^*B$ is invertible and $A^{-1} = B^{-1}(B^{-1})^*$ by Corollary 3.1.17(i) and Proposition 2.3.9(i).
The equality $\mathrm{rank}(B^{-1}) = n$ holds by Corollary 3.6.2, so $A^{-1} \succ 0$ by Proposition 2.1.2(vi) with $R = B^{-1}$.
(v) $\Rightarrow$ *(i)* $(A^{-1})^* = A^{-1}$ by positive definiteness, so $A^* = ((A^{-1})^*)^{-1} = A$ by Proposition 2.3.9(i).
For $\mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$, set $\mathbf{y} := A\mathbf{x} \neq \mathbf{0}$. Then $0 < \mathbf{y}^*A^{-1}\mathbf{y} = \mathbf{x}^*A\mathbf{x}$, so $A \succ 0$.

Structural Commentary: Spectral positivity removes zero, the square root supplies a Gram factor, and substitution reverses inversion.
Key Insight: $\mathbf{y} = A\mathbf{x}$ transfers strict positivity from A^{-1} to A. □

Proposition 2.1.9. *Trace and Determinant of Positive Semidefinite Matrices*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A \succeq 0$.
(i) $\mathrm{tr}(A) \geq 0$.
(ii) $\mathrm{tr}(A) = 0$ *iff* $A = 0$.
(iii) $\det(A) \geq 0$.
(iv) $\det(A) > 0$ *iff* $A \succ 0$.

Proof. Write $A = U \operatorname{diag}(\lambda_1, \ldots, \lambda_n) U^*$ by the appropriate one of Theorems 6.1.11 and 6.1.12.
The relation $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j \geq 0$ holds by Theorem 2.1.3(ii). Then $\mathrm{tr}(A) = \sum_{j=1}^{n} \lambda_j$ and $\det(A) = \prod_{j=1}^{n} \lambda_j$ by Theorem 1.1.13(i),(ii).

(i) $\mathrm{tr}(A) = \sum_{j=1}^{n} \lambda_j \geq 0$.
(ii) $(\Rightarrow)$ $\mathrm{tr}(A) = 0 \Rightarrow (\forall\, j \in \{1, \ldots, n\})\, \lambda_j = 0 \Rightarrow A = 0$.
(ii) $(\Leftarrow)$ $A = 0 \Rightarrow \mathrm{tr}(A) = 0$.
(iii) $\det(A) = \prod_{j=1}^{n} \lambda_j \geq 0$.
(iv) $(\Rightarrow)$ $\det(A) > 0 \Rightarrow (\forall\, j \in \{1, \ldots, n\})\, \lambda_j > 0$, so $A \succ 0$ by Theorem 2.1.8(ii).
(iv) $(\Leftarrow)$ $A \succ 0 \Rightarrow (\forall\, j \in \{1, \ldots, n\})\, \lambda_j > 0$ by Theorem 2.1.8(ii), so $\det(A) > 0$.

Structural Commentary: Unitary diagonalization transfers trace and determinant to a nonnegative eigenvalue sum and product.
Key Insight: The sum detects universal vanishing; the product detects any vanishing. □

Theorem 2.1.10. *Stability Rules for Positive Semidefinite Matrices*
Let $m, n, p \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $A, C \in \mathrm{Mat}(n, n, \mathbb{F})$, $B \in \mathrm{Mat}(m, m, \mathbb{F})$, $S \in \mathrm{Mat}(n, p, \mathbb{F})$. Fix $a, b \in \mathbb{R}_{\geq 0}$, suppose $A, C, B \succeq 0$.
(i) $aA + bC \succeq 0$.
(ii) $S^*AS \succeq 0$.

(iii) $A \oplus B \succeq 0$.
(iv) *If* $a, b > 0$ *and either* $A \succ 0$ *or* $C \succ 0$, *then* $aA + bC \succ 0$.
(v) $S^*AS \succ 0$ *iff* $(\forall\, \mathbf{x} \in \mathbb{F}^p \setminus \{\mathbf{0}\})\, S\mathbf{x} \notin \ker(A)$.
(vi) $A \oplus B \succ 0$ *iff* $A \succ 0$ *and* $B \succ 0$.

Proof. For a square matrix M and compatible vector $\mathbf{z}$, set $q_M(\mathbf{z}) := \mathbf{z}^*M\mathbf{z}$.
(i) $(aA + bC)^* = aA + bC$ and $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, q_{aA+bC}(\mathbf{x}) = aq_A(\mathbf{x}) + bq_C(\mathbf{x}) \geq 0$, so $aA + bC \succeq 0$.
(ii) $(S^*AS)^* = S^*AS$ by Proposition 2.3.8(ii),(iii). The identity $(\forall\, \mathbf{x} \in \mathbb{F}^p)\, q_{S^*AS}(\mathbf{x}) = q_A(S\mathbf{x}) \geq 0$ implies $S^*AS \succeq 0$.
(iii) $(A \oplus B)^* = A \oplus B$, and $(\forall\, (\mathbf{x}, \mathbf{y}) \in \mathbb{F}^n \times \mathbb{F}^m)\, q_{A\oplus B}(\mathbf{x} \oplus \mathbf{y}) = q_A(\mathbf{x}) + q_B(\mathbf{y})$. The right side is nonnegative, so $A \oplus B \succeq 0$.
(iv) If $a, b > 0$ and $A \succ 0$, then $(\forall\, \mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, aq_A(\mathbf{x}) > 0$ and $bq_C(\mathbf{x}) \geq 0$, so $aA + bC \succ 0$ by part (i).
If $a, b > 0$ and $C \succ 0$, then $(\forall\, \mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, aq_A(\mathbf{x}) \geq 0$ and $bq_C(\mathbf{x}) > 0$, so again $aA + bC \succ 0$.
(v) $(\Rightarrow)$ If $S^*AS \succ 0$ and $\mathbf{x} \in \mathbb{F}^p \setminus \{\mathbf{0}\}$, then $q_A(S\mathbf{x}) = q_{S^*AS}(\mathbf{x}) > 0$. Thus $S\mathbf{x} \notin \ker(A)$ by Corollary 2.1.6.
(v) $(\Leftarrow)$ If $\mathbf{x} \in \mathbb{F}^p \setminus \{\mathbf{0}\}$, then $S\mathbf{x} \notin \ker(A)$ by hypothesis.
Thus $q_{S^*AS}(\mathbf{x}) = q_A(S\mathbf{x}) > 0$ by $A \succeq 0$ and Corollary 2.1.6, and $S^*AS \succ 0$ by part (ii).
(vi) $(\Rightarrow)$ If $A \oplus B \succ 0$, then $(\forall\, \mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, q_A(\mathbf{x}) = q_{A\oplus B}(\mathbf{x} \oplus \mathbf{0}) > 0$.
Likewise, $(\forall\, \mathbf{y} \in \mathbb{F}^m \setminus \{\mathbf{0}\})\, q_B(\mathbf{y}) = q_{A\oplus B}(\mathbf{0} \oplus \mathbf{y}) > 0$, so $A \succ 0$ and $B \succ 0$.
(vi) $(\Leftarrow)$ If $A \succ 0$, $B \succ 0$, and $(\mathbf{x}, \mathbf{y}) \neq (\mathbf{0}, \mathbf{0})$, then $q_{A\oplus B}(\mathbf{x} \oplus \mathbf{y}) = q_A(\mathbf{x}) + q_B(\mathbf{y}) > 0$. Thus $A \oplus B \succ 0$ by part (iii).
Structural Commentary: The identities for q_M preserve nonnegativity and expose the zero sets of each construction.
Key Insight: Strict positivity is the absence of nonzero vectors in the transformed zero set. □

Proposition 2.1.11. *Explicit Square Root for a* 2×2 *Positive Semidefinite Matrix*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ *and* $A \in \mathrm{Mat}(2, 2, \mathbb{F})$ *satisfy* $A \succeq 0$ *and* $A \neq 0$. *Set* $\tau := \big(\mathrm{tr}(A) + 2\sqrt{\det(A)}\big)^{1/2}$.
(i) $\tau > 0$.
(ii) $A^{1/2} = \frac{1}{\tau}\big(A + \sqrt{\det(A)}\, I_2\big)$.

Proof.
(i) $\det(A) \geq 0$ and $\mathrm{tr}(A) > 0$ by Proposition 2.1.9(ii),(iii), so $\tau > 0$.
(ii) Set $C := \frac{1}{\tau}\big(A + \sqrt{\det(A)}\, I_2\big)$. Then $C \succeq 0$ by Theorem 2.1.10(i), since $I_2 \succeq 0$ and $\tau^{-1}, \tau^{-1}\sqrt{\det(A)} \geq 0$.
The identity $A^2 - \mathrm{tr}(A)A + \det(A)I_2 = 0$ holds by Theorems 2.1.3 and 3.1.1, regarding a real A as complex when necessary.
Thus $\tau^2 C^2 = A^2 + 2\sqrt{\det(A)}A + \det(A)I_2 = \tau^2 A$, so $C^2 = A$ and $C = A^{1/2}$ by Corollary 2.1.4(i).
Structural Commentary: Cayley–Hamilton reduces the squared linear candidate, and positive-root uniqueness identifies it.
Key Insight: $\tau^2 = \mathrm{tr}(A) + 2\sqrt{\det(A)}$ normalizes the square to A. □

Definition 2.1.12. *Directed Incidence Matrix*
Let $m, n \in \mathbb{Z}_{>0}$, and let $G = (V, E)$ be a directed graph with $V = \{1, \ldots, n\}$ and ordered edges $E = (e_k)_{k=1}^m$, where $e_k = u_k \to v_k$.
Its *directed incidence matrix* is the matrix $B \in \mathrm{Mat}(m, n, \mathbb{R})$ satisfying $(\forall\, k \in \{1, \ldots, m\})\, B_{k,\cdot} = -\mathbf{e}_{u_k}^t + \mathbf{e}_{v_k}^t$.

Figure 78 records the chosen edge orientations and their corresponding signed incidence rows.

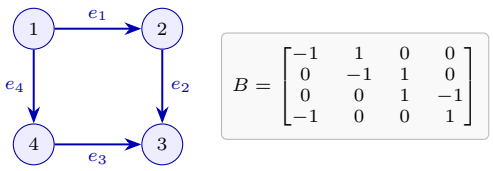

Figure 78: A directed incidence matrix records oriented edge differences from vertices to edges.

Example 2.1.13. *A Directed Incidence Matrix*
For Figure 78, the directed incidence matrix is

$$B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}.$$

For node potentials $\mathbf{x} = (x_1, x_2, x_3, x_4)^t$, the edge-difference vector is

$$B\mathbf{x} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_4 \\ x_4 - x_1 \end{bmatrix}.$$

The equivalence $B\mathbf{x} = \mathbf{0} \iff x_1 = x_2 = x_3 = x_4$ implies $\ker(B) = \mathrm{span}\{(1, 1, 1, 1)^t\}$.
Thus $\mathrm{rank}(B) = 3$ by Corollary 3.3.1(i). For $\mathbf{y} := (1, 1, -1, -1)^t$, $B^t\mathbf{y} = \mathbf{0}$ by direct multiplication.
The equality $\mathrm{rank}(B^t) = 3$ holds by Theorem 3.3.2, so $\ker(B^t) = \mathrm{span}\{\mathbf{y}\}$ by Corollary 3.3.1(i).
Thus $\mathbf{b} \in \mathrm{col}(B)$ iff $\mathbf{y}^t\mathbf{b} = b_1 + b_2 - b_3 - b_4 = 0$ by Proposition 2.3.10(iv). This is the Kirchhoff loop-compatibility condition.
Structural Commentary: The kernel records constant-potential freedom; the left kernel records edge-data compatibility.
Key Insight: The oriented rows encode both edge differences and the cycle relation that determines which edge data are realizable.

Perspective 2.1.14. *Incidence Data and Least Squares*
For an incidence matrix B as in Example 2.1.13, $B\mathbf{x} = \mathbf{b}$ seeks node potentials reproducing prescribed edge differences.
If loop compatibility fails, a least-squares solution projects $\mathbf{b}$ onto $\mathrm{col}(B)$ and satisfies $B^*(B\mathbf{x}_0 - \mathbf{b}) = \mathbf{0}$ by Theorem 2.4.4.
The constant-vector kernel makes the potentials unique only after fixing a reference node or selecting the minimum-norm solution.

Proposition 2.1.15. *Incidence Laplacians Are Positive Semidefinite*
Let $G = (V, E)$ be a directed graph, let $B \in \mathrm{Mat}(m, n, \mathbb{R})$ be its directed incidence matrix, and set $L := B^t B$. Let G_{u} be the underlying undirected graph obtained by forgetting edge orientations.

(i) $L \succeq 0$.
(ii) $(\forall\, \mathbf{x} \in \mathbb{R}^n)\, \mathbf{x}^t L \mathbf{x} = \sum_{e=(u\to v)\in E} (x_v - x_u)^2$.
(iii) $\ker(L)$ *is the subspace of vectors that are constant on each connected component of* G_{u}.

Proof.
(i) $L = B^* B \succeq 0$ by Proposition 2.1.2(i).
(ii) For $\mathbf{x} \in \mathbb{R}^n$, $(\forall\, e = (u \to v) \in E)\, (B\mathbf{x})_e = x_v - x_u$ by the incidence-row formula. Thus $\mathbf{x}^t L \mathbf{x} = \|B\mathbf{x}\|_2^2 = \sum_{e=(u\to v)\in E} (x_v - x_u)^2$.
(iii) $\ker(L) = \ker(B)$ by Theorem 2.1.5(ii). For $\mathbf{x} \in \mathbb{R}^n$, $B\mathbf{x} = \mathbf{0}$ iff $x_u = x_v$ on every edge.
These equalities propagate along paths and are equivalent to componentwise constancy on G_{u}.

Structural Commentary: The Gram identity converts the Laplacian form and kernel into edge-difference conditions.
Key Insight: Zero edge differences are exactly componentwise-constant vertex values. □

2.2 Principal Minors and Inertia

Definition 2.2.1. *Principal and Leading Principal Minors*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) If $\varnothing \neq I \subseteq \{1, \dots, n\}$, then $A[I] := [a_{ij}]_{i,j\in I}$ is the *principal submatrix indexed by* I.
Its determinant $\det A[I]$ is the corresponding *principal minor*.
(ii) If $k \in \{1, \dots, n\}$, then $\det A[\{1, \dots, k\}]$ is the kth *leading principal minor* of A.

Corollary 2.2.2. *Principal Blocks and Column Inclusion*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $k \in \{1, \dots, n\}$, and $A \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A \succeq 0$. Partition $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{22} \end{smallmatrix}\right)$ with $A_{11} \in \mathrm{Mat}(k, k, \mathbb{F})$.

(i) $A_{11} \succeq 0$.
(ii) $\mathrm{col}(A_{12}) \subseteq \mathrm{col}(A_{11})$.
(iii) $\mathrm{rank}(A_{12}) \leq \mathrm{rank}(A_{11})$.
(iv) If $A \succ 0$, then $A_{11} \succ 0$.

Proof. Choose $m \in \mathbb{Z}_{>0}$ and $B \in \mathrm{Mat}(m, n, \mathbb{F})$ with $A = B^* B$ by Theorem 2.1.3(iv).
Write $B = [B_1\ B_2]$ conformally; then $A_{11} = B_1^* B_1$ and $A_{12} = B_1^* B_2$.

(i) $A_{11} \succeq 0$ by Proposition 2.1.2(i).
(ii) $\mathrm{col}(A_{12}) \subseteq \mathrm{col}(B_1^*)$ by $A_{12} = B_1^* B_2$. The identity $\mathrm{col}(B_1^*) = \mathrm{col}(B_1^* B_1) = \mathrm{col}(A_{11})$ holds by Theorem 2.1.5(iv).
(iii) $\mathrm{rank}(A_{12}) = \dim \mathrm{col}(A_{12}) \leq \dim \mathrm{col}(A_{11}) = \mathrm{rank}(A_{11})$ by part (ii).
(iv) If $A \succ 0$ and $\mathbf{x} \in \mathbb{F}^k \setminus \{\mathbf{0}\}$, set $\mathbf{z} := \left(\begin{smallmatrix} \mathbf{x} \\ \mathbf{0} \end{smallmatrix}\right)$.
Then $\mathbf{z} \neq \mathbf{0}$ and $\mathbf{x}^* A_{11} \mathbf{x} = \mathbf{z}^* A \mathbf{z} > 0$, so $A_{11} \succ 0$ by part (i) and the definition of positive definiteness.

Structural Commentary: The Gram partition proves the block claims; zero padding transfers strict positivity.
Key Insight: $A_{11} = B_1^* B_1$ and $A_{12} = B_1^* B_2$ share the same left factor. □

Figure 79 highlights the principal and cross blocks involved in the positive-semidefinite column-inclusion result.

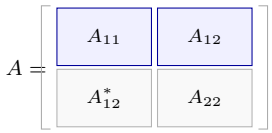

Figure 79: For $A \succeq 0$, factorisation forces $\mathrm{col}(A_{12}) \subseteq \mathrm{col}(A_{11})$.

Theorem 2.2.3. *Sylvester's Criterion*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A = A^*$. Then $A \succ 0$ iff $(\forall\, k \in \{1, \dots, n\})$ $\det A[\{1, \dots, k\}] > 0$.

Proof.
(⇒) The relation $A \succeq 0$ holds by positive definiteness. Fix $k \in \{1, \dots, n\}$ and set $A_k := A[\{1, \dots, k\}]$.
Then $A_k \succ 0$ by Corollary 2.2.2(iv), so $A_k \succeq 0$ by definition and $\det A_k > 0$ by Proposition 2.1.9(iv).
(⇐) Induct on n. If $n = 1$, then $A = [a]$ with $a > 0$, so $A \succ 0$ by definition. Assume $n \geq 2$ and the converse in dimension $n - 1$.
Write $A = \left(\begin{smallmatrix} B & \mathbf{y} \\ \mathbf{y}^* & a \end{smallmatrix}\right)$. Then $B = B^*$ and its leading principal minors are positive, so $B \succ 0$ by the induction hypothesis and B is invertible by Theorem 2.1.8(iii). Set $\delta := a - \mathbf{y}^* B^{-1} \mathbf{y}$. Then $\delta = \frac{\det A}{\det B} > 0$ by Proposition 1.1.8 and leading-minor hypothesis.
For $\mathbf{z} = \left(\begin{smallmatrix} \mathbf{u} \\ t \end{smallmatrix}\right)$, set $\mathbf{w} := \mathbf{u} + B^{-1} \mathbf{y} t$. Then $\mathbf{z}^* A \mathbf{z} = \mathbf{w}^* B \mathbf{w} + \delta\, |t|^2$ by direct expansion. Terms nonnegative by $B \succ 0$, $\delta > 0$.
For $\mathbf{z} \neq \mathbf{0}$, $t \neq 0 \Rightarrow \delta\, |t|^2 > 0$, while $t = 0 \Rightarrow \mathbf{w} = \mathbf{u} \neq \mathbf{0} \Rightarrow \mathbf{w}^* B \mathbf{w} > 0$. Thus $A \succ 0$ by definition.

Structural Commentary: Induction handles the leading block; the Schur complement supplies the final square coefficient.
Key Insight: Positive leading minors make every completed-square pivot positive. □

Computation Method 2.2.4. 2×2 *Quadratic Form Test*
Given a real symmetric matrix $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$, set $q(x, y) := ax^2 + 2bxy + cy^2$.
The output is the sign classification of q. Set $\Delta := ac - b^2$ and proceed as follows:

(i) If $a \neq 0$, complete the square as $q(x, y) = a\left(x + \frac{b}{a} y\right)^2 + \frac{\Delta}{a} y^2$.
(ii) If $a = 0$ and $c \neq 0$, interchange x and y and repeat part (i); if $a = c = 0$, inspect $q(x, y) = 2bxy$ directly.
(iii) Conclude that $A \succ 0$ iff $a > 0$ and $\Delta > 0$, while $-A \succ 0$ iff $a < 0$ and $\Delta > 0$.

(iv) Conclude that q takes both positive and negative values iff $\Delta < 0$.
(v) If $\Delta = 0$, conclude that $A \succeq 0$ iff $a, c \geq 0$, while $-A \succeq 0$ iff $a, c \leq 0$.

The strict classifications in part (iii) agree with Theorem 2.2.3.
The completed-square and zero-diagonal branches determine parts (iv) and (v).

Theorem 2.2.5. *Principal Axes for Real Quadratic Forms*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n,n,\mathbb{R})$ satisfy $A = A^t$. Set $q_A(\mathbf{x}) := \mathbf{x}^t A \mathbf{x}$.

(i) $(\exists\, Q \in \mathrm{Mat}(n,n,\mathbb{R}))(\exists\, \lambda_1, \ldots, \lambda_n \in \mathbb{R})[Q^t Q = I_n \wedge A = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n) Q^t]$. *Let $Q, \lambda_1, \ldots, \lambda_n$ be any choice satisfying these conditions.*
Then $(\forall\, \mathbf{x} \in \mathbb{R}^n)\, q_A(\mathbf{x}) = \sum_{j=1}^{n} \lambda_j (Q^t \mathbf{x})_j^2$. *Parts (ii)–(iv) are stated with respect to this choice.*
(ii) $A \succeq 0$ *iff* $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j \geq 0$.
(iii) $A \succ 0$ *iff* $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j > 0$.
(iv) q_A *takes both positive and negative values iff* $(\exists\, j, k \in \{1, \ldots, n\})[\lambda_j > 0 \wedge \lambda_k < 0]$.

Proof.
(i) Fix an orthogonal Q_0 and $\mu_1, \ldots, \mu_n \in \mathbb{R}$ with $A = Q_0\,\mathrm{diag}(\mu_1, \ldots, \mu_n) Q_0^t$ by Theorem 6.1.12(i)⇒(iii).
Then $(\forall\, \mathbf{x} \in \mathbb{R}^n)\, q_A(\mathbf{x}) = \sum_{j=1}^{n} \mu_j (Q_0^t \mathbf{x})_j^2$. For parts (ii)–(iv), fix any $Q, \lambda_1, \ldots, \lambda_n$ satisfying part (i).
(ii) (⇒) $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j = q_A(Q\mathbf{e}_j) \geq 0$ by part (i) and $A \succeq 0$.
(ii) (⇐) $(\forall\, \mathbf{x} \in \mathbb{R}^n)\, q_A(\mathbf{x}) \geq 0$ by part (i) and $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j \geq 0$. Thus $A \succeq 0$ by $A = A^t$ and definition.
(iii) (⇒) $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j = q_A(Q\mathbf{e}_j) > 0$ by part (i) and $A \succ 0$.
(iii) (⇐) If $\mathbf{x} \neq \mathbf{0}$, then $Q^t \mathbf{x} \neq \mathbf{0}$ and $q_A(\mathbf{x}) > 0$ by part (i) and $(\forall\, j \in \{1, \ldots, n\})\, \lambda_j > 0$. Thus $A \succ 0$ by $A = A^t$ and definition.
(iv) (⇒) If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ satisfy $q_A(\mathbf{u}) > 0 > q_A(\mathbf{v})$, then $(\exists\, j, k \in \{1, \ldots, n\})[\lambda_j > 0 \wedge \lambda_k < 0]$ by part (i).
(iv) (⇐) If $j, k \in \{1, \ldots, n\}$ satisfy $\lambda_j > 0 > \lambda_k$, then $q_A(Q\mathbf{e}_j) = \lambda_j > 0 > \lambda_k = q_A(Q\mathbf{e}_k)$.
Structural Commentary: Spectral coordinates diagonalise the form; coordinate vectors and the square sum classify its signs.
Key Insight: Each eigenvalue is the form value on its eigenvector. □

Computation Method 2.2.6. *Quadratic Form Principal-Axes Method*
Given a real quadratic form q in n variables, the output is an orthogonal coordinate representation of q and its sign classification.

(i) Write $q(\mathbf{x}) = \mathbf{x}^t A \mathbf{x}$ with $A = A^t$, assigning half of each cross-term coefficient to each of the symmetric entries a_{ij} and a_{ji}.
(ii) Compute an orthogonal diagonalisation $A = Q \Lambda Q^t$.
(iii) Set $\mathbf{y} = Q^t \mathbf{x}$ and rewrite $q(\mathbf{x}) = \sum_{j=1}^{n} \lambda_j y_j^2$.
(iv) Classify the form from the signs of the diagonal entries of Λ by Theorem 2.2.5(ii)–(iv).

Figure 80 illustrates the orthogonal change of coordinates underlying the principal-axes method.

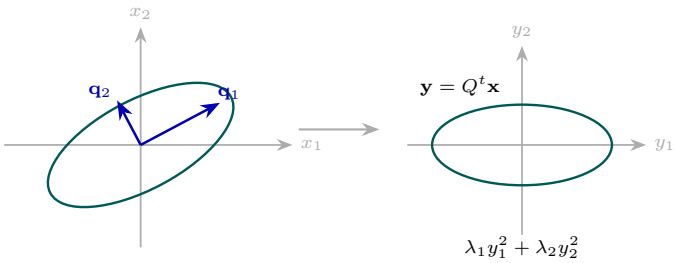

Figure 80: Orthogonal coordinates reveal the principal axes of a quadratic form.

Definition 2.2.7. *$*$-Congruence and Hermitian Inertia*
Let $n \in \mathbb{Z}_{>0}$ and $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$.

(i) The matrices A and B are *$*$-congruent* iff $(\exists\, S \in \mathrm{Mat}(n,n,\mathbb{C})^{\times})\, A = SBS^*$.
(ii) If $A = A^*$, its *inertia* is $\mathrm{in}(A) := (n_+(A), n_-(A), n_0(A))$.
The entries count the positive, negative, and zero eigenvalues, respectively, with algebraic multiplicity.

Lemma 2.2.8. *Positive Subspaces and Inertia*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ satisfy $A = A^*$.

(i) $\max\{\dim \mathcal{U} : \mathcal{U} \leq \mathbb{C}^n \wedge (\forall\, \mathbf{x} \in \mathcal{U} \setminus \{\mathbf{0}\})\, \mathbf{x}^* A \mathbf{x} > 0\} = n_+(A)$.
(ii) $\max\{\dim \mathcal{U} : \mathcal{U} \leq \mathbb{C}^n \wedge (\forall\, \mathbf{x} \in \mathcal{U} \setminus \{\mathbf{0}\})\, \mathbf{x}^* A \mathbf{x} < 0\} = n_-(A)$.

Proof.
(i) The matrix A is normal by $A = A^*$, and its eigenvalues are real by Theorem 6.1.5.
Let $\mathcal{E}_+$ and $\mathcal{E}_{\leq 0}$ be the orthogonal sums of its positive and nonpositive eigenspaces by Theorem 6.1.11(i)⇒(ii).
Then $\dim \mathcal{E}_+ = n_+(A)$ and $\dim \mathcal{E}_{\leq 0} = n - n_+(A)$.
The inequalities $(\forall\, \mathbf{x} \in \mathcal{E}_+ \setminus \{\mathbf{0}\})\, \mathbf{x}^* A \mathbf{x} > 0$ and $(\forall\, \mathbf{x} \in \mathcal{E}_{\leq 0})\, \mathbf{x}^* A \mathbf{x} \leq 0$ hold by the spectral decomposition.
If $\dim \mathcal{U} > n_+(A)$, then $\mathcal{U} \cap \mathcal{E}_{\leq 0} \neq \{\mathbf{0}\}$ by Corollary 2.3.3(i), so $\mathcal{U}$ is not positive. Thus $\mathcal{E}_+$ attains the maximal dimension $n_+(A)$.
(ii) A subspace is negative for A iff it is positive for $-A$.
Its maximal dimension is $n_+(-A) = n_-(A)$ by part (i) and $(-A)\mathbf{x} = \lambda \mathbf{x} \iff A\mathbf{x} = -\lambda \mathbf{x}$.
Structural Commentary: A spectral sign subspace attains the bound; its complementary sign range obstructs enlargement.
Key Insight: Excess dimension forces intersection with the wrong-sign spectral subspace. □

Theorem 2.2.9. *Sylvester's Inertia Theorem*
Let $n \in \mathbb{Z}_{>0}$ and let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian. Then A and B are $*$-congruent iff $n_+(A) = n_+(B)$ and $n_-(A) = n_-(B)$.

Proof.
($\Rightarrow$) Suppose $A = SBS^*$ with S invertible. The map S^* preserves subspace dimension and satisfies $(S^*\mathbf{x})^*B(S^*\mathbf{x}) = \mathbf{x}^*A\mathbf{x}$.
Thus $n_+(A) \le n_+(B)$ and $n_-(A) \le n_-(B)$ by Lemma 2.2.8.
The reverse inequalities hold by the same lemma and $B = S^{-1}A(S^{-1})^*$, so $n_+(A) = n_+(B)$ and $n_-(A) = n_-(B)$.
($\Leftarrow$) Set $p := n_+(A) = n_+(B)$, $q := n_-(A) = n_-(B)$, $r := n - p - q$, and $D := I_p \oplus (-I_q) \oplus 0_r$.
Fix $M \in \{A, B\}$. The matrix M is normal by $M = M^*$ and its eigenvalues are real by Theorem 6.1.5.
Order them as $\lambda_1^{(M)}, \dots, \lambda_n^{(M)}$ by positive, negative, and zero sign. Set $\Lambda_M := \mathrm{diag}(\lambda_1^{(M)}, \dots, \lambda_n^{(M)})$.
Fix a unitary U_M with $M = U_M \Lambda_M U_M^*$ by Theorem 6.1.11(i)$\Rightarrow$(iii). Set $d_j^{(M)} := \sqrt{\left|\lambda_j^{(M)}\right|}$ for $\lambda_j^{(M)} \neq 0$ and $d_j^{(M)} := 1$ otherwise.
Put $E_M := \mathrm{diag}(d_1^{(M)}, \dots, d_n^{(M)})$ and $R_M := U_M E_M$. Then R_M is invertible, $\Lambda_M = E_M D E_M$, and $M = R_M D R_M^*$.
The construction defines R_A, R_B. For $S := R_A R_B^{-1}$, $SBS^* = R_A D R_A^* = A$, so A and B are $*$-congruent.
Structural Commentary: Sign-subspace transport proves invariance; spectral rescaling constructs the shared signed diagonal form.
Key Insight: $I_p \oplus (-I_q) \oplus 0_r$ is the complete $*$-congruence invariant. □

Figure 81 summarizes the $*$-congruence reduction to positive, negative, and zero coordinate directions.

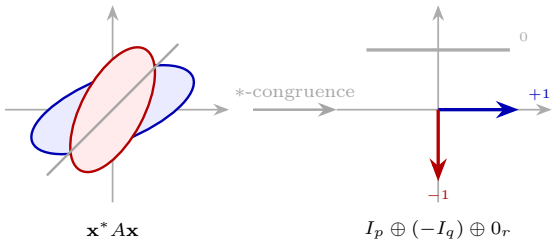

Figure 81: $*$-congruence reduces a Hermitian form to signed coordinate axes.

2.3 Hermitian and Normal Matrix Structure

Theorem 2.3.1. *Simultaneous Congruence with One Positive Definite Form*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A = A^*$, $B = B^*$, and $A \succ 0$.
Then $(\exists\, S \in \mathrm{Mat}(n, n, \mathbb{F})^\times)(\exists\, \lambda_1, \dots, \lambda_n \in \mathbb{R})$ such that, for $\Lambda := \mathrm{diag}(\lambda_1, \dots, \lambda_n)$:
(i) $A = SS^*$ and $B = S\Lambda S^*$.
(ii) $p_{A^{-1}B}(z) = \prod_{j=1}^{n} (z - \lambda_j)$.

Proof. Choose an invertible T with $A = TT^*$ by Theorem 2.1.8(iv), and set $C := T^{-1}B(T^*)^{-1}$.
The identity $C^* = C$ holds by Propositions 2.3.8(ii),(iii) and 2.3.9(i). Thus C is self-adjoint and normal.
If $\mathbb{F} = \mathbb{R}$, choose a unitary U and diagonal Λ with $C = U\Lambda U^*$ by Theorem 6.1.12(i)$\Rightarrow$(iii).
If $\mathbb{F} = \mathbb{C}$, make the same choice by Theorem 6.1.11(i)$\Rightarrow$(iii).
Its diagonal entries are real automatically over $\mathbb{R}$ and by Theorem 6.1.5 over $\mathbb{C}$.
Write $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, and set $S := TU$.
(i) The matrix S is invertible, and $SS^* = TUU^*T^* = A$ and $S\Lambda S^* = TCT^* = B$.
(ii) The identity $A^{-1}B = (S^*)^{-1}\Lambda S^*$ holds by part (i).
Hence $p_{A^{-1}B} = p_\Lambda$ by Theorem 2.1.7(i). The identity $p_\Lambda(z) = \prod_{j=1}^{n} (z - \lambda_j)$ holds by Proposition 2.1.6(i).
Structural Commentary: The Gram factor turns simultaneous congruence into one self-adjoint diagonalization.
Key Insight: Diagonalizing C once produces both S and the generalized eigenvalues. □

Corollary 2.3.2. *Products with One Positive Definite Factor*
Let $n \in \mathbb{Z}_{>0}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A \succ 0$.
(i) If $B = B^*$, then AB is diagonalizable over $\mathbb{F}$ and $\mathrm{spec}(AB) \subseteq \mathbb{R}$.
(ii) If $B \succeq 0$, then AB is diagonalizable over $\mathbb{F}$ and $\mathrm{spec}(AB) \subseteq [0, \infty)$.
(iii) If $B \succeq 0$ and $B \neq 0$, then $(\exists\, \lambda \in \mathrm{spec}(AB))\, \lambda > 0$.
(iv) If $B \succ 0$, then AB is diagonalizable over $\mathbb{F}$ and $\mathrm{spec}(AB) \subseteq (0, \infty)$.

Proof. The matrix $A^{-1} \succ 0$ by Theorem 2.1.8(i)$\Rightarrow$(v). If $B = B^*$, choose S and Λ by Theorem 2.3.1 for (A^{-1}, B).
Write $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$, where $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, S invertible. Then $A^{-1} = SS^*$ and $B = S\Lambda S^*$, so $AB = (S^*)^{-1}\Lambda S^*$.
(i) The identity $AB = (S^*)^{-1}\Lambda S^*$ diagonalizes AB over $\mathbb{F}$ with real eigenvalues $\lambda_1, \dots, \lambda_n$.
(ii) The identity $B = B^*$ holds by positive semidefiniteness, so part (i) applies.
Set $\mathbf{x}_j := (S^*)^{-1}\mathbf{e}_j$ for $j \in \{1, \dots, n\}$. Then $(\forall\, j \in \{1, \dots, n\})\, \lambda_j = \mathbf{x}_j^* B \mathbf{x}_j \ge 0$.
(iii) By parts (i) and (ii), $\mathrm{spec}(AB) = \{\lambda_1, \dots, \lambda_n\} \subseteq [0, \infty)$. Since $B = S\Lambda S^* \neq 0$, one has $\Lambda \neq 0$, so $(\exists\, \lambda \in \mathrm{spec}(AB))\, \lambda > 0$.
(iv) The identity $B = B^*$ holds by positive definiteness, so part (i) applies. Set $\mathbf{x}_j := (S^*)^{-1}\mathbf{e}_j$ for $j \in \{1, \dots, n\}$.
Since S is invertible, $(\forall\, j \in \{1, \dots, n\})[\mathbf{x}_j \neq \mathbf{0} \wedge \lambda_j = \mathbf{x}_j^* B \mathbf{x}_j > 0]$.
Structural Commentary: The simultaneous form of (A^{-1}, B) makes AB similar to the same real diagonal matrix.
Key Insight: Its eigenvalue signs are the quadratic-form signs of B on transformed coordinate vectors. □

Definition 2.3.3. *Phase Inertia Matrix*
Let $n \in \mathbb{Z}_{>0}$ and $D = [d_{jk}] \in \mathrm{Mat}(n, n, \mathbb{C})$. The matrix D is a *phase inertia matrix* iff D is diagonal and $(\forall\, j \in \{1, \dots, n\})\ |d_{jj}| \in \{0, 1\}$.

Proposition 2.3.4. *Phase Inertia Form for Normal Matrices*
Let $n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be normal. Let $\lambda_1,\dots,\lambda_n$ list its eigenvalues with algebraic multiplicity. For $j \in \{1,\dots,n\}$, set $d_j := 0$ if $\lambda_j = 0$ and $d_j := \frac{\lambda_j}{|\lambda_j|}$ otherwise. Put $D := \mathrm{diag}(d_1,\dots,d_n)$.

(i) D is a phase inertia matrix.
(ii) $(\exists\, S \in \mathrm{Mat}(n,n,\mathbb{C})^\times)\ A = SDS^*$.

Proof.
(i) By the definition of d_j, $(\forall\, j \in \{1,\dots,n\})\ |d_j| \in \{0,1\}$. Hence D is a phase inertia matrix.
(ii) Set $L := \mathrm{diag}(\lambda_1,\dots,\lambda_n)$. After permuting its eigenbasis, choose a unitary U with $A = ULU^*$ by Theorem 6.1.11(i)⇒(iii).
For $j \in \{1,\dots,n\}$, set $r_j := 1$ if $\lambda_j = 0$ and $r_j := |\lambda_j|$ otherwise. Then $(\forall\, j \in \{1,\dots,n\})[r_j > 0 \wedge r_j d_j = \lambda_j]$.
Set $S := U\,\mathrm{diag}(\sqrt{r_1},\dots,\sqrt{r_n})$. Then S is invertible and $SDS^* = U\,\mathrm{diag}(r_1 d_1,\dots,r_n d_n)U^* = A$.
Structural Commentary: Spectral diagonalization separates phase from positive radius, which the congruence factor absorbs.
Key Insight: Radial square roots provide the diagonal scaling in S. □

Perspective 2.3.5. *Hermitian Inertia as Real Phase Inertia*
Proposition 2.3.4 reduces a normal matrix to its eigenvalue phases under $*$-congruence.
Hermitian eigenvalues are real by Theorem 6.1.5, so their nonzero phases are ± 1;
Theorem 2.2.9 strengthens this reduction to a complete classification by sign multiplicities.

3 Matrix Products, Vectorization, and Spectral Radius

3.1 Kronecker Products and Vectorization

Throughout, $\mathbb{F}$ denotes a field.

Definition 3.1.1. *Kronecker Product*
Let $m,n,p,q \in \mathbb{Z}_{>0}$, $A = [a_{ij}] \in \mathrm{Mat}(m,n,\mathbb{F})$, and $B \in \mathrm{Mat}(p,q,\mathbb{F})$.
The *Kronecker product*, or *matrix tensor product*, is the block matrix $A \otimes B := [a_{ij}B]_{i\in J_m, j\in J_n} \in \mathrm{Mat}(mp,nq,\mathbb{F})$.

Proposition 3.1.2. *Basic Kronecker Identities*
Let $m,n,p,q,r,s \in \mathbb{Z}_{>0}$, $c \in \mathbb{F}$, and $A, A' \in \mathrm{Mat}(m,n,\mathbb{F})$.
Let $B, B' \in \mathrm{Mat}(p,q,\mathbb{F})$ and $C \in \mathrm{Mat}(r,s,\mathbb{F})$.

(i) $c(A\otimes B) = (cA)\otimes B = A\otimes(cB)$;
(ii) $(A+A')\otimes B = A\otimes B + A'\otimes B$ and $A\otimes(B+B') = A\otimes B + A\otimes B'$;
(iii) $(A\otimes B)\otimes C = A\otimes(B\otimes C)$;
(iv) $(A\otimes B)^t = A^t \otimes B^t$;
(v) $I_m \otimes I_p = I_{mp}$;
(vi) If $\mathbb{F} = \mathbb{C}$, then $\overline{A\otimes B} = \overline{A}\otimes\overline{B}$ and $(A\otimes B)^* = A^*\otimes B^*$.

Proof. Write $A = [a_{ij}]$ and $B = [b_{uv}]$.
(i) The corresponding blocks in the three matrices are $c(a_{ij}B) = (ca_{ij})B = a_{ij}(cB)$.
(ii) For $A' = [a'_{ij}]$, $(a_{ij}+a'_{ij})B = a_{ij}B + a'_{ij}B$ and $a_{ij}(B+B') = a_{ij}B + a_{ij}B'$.
(iii) Write $C = [c_{\alpha\beta}]$. Under lexicographic indexing, the $((i,u,\alpha),(j,v,\beta))$ entry of each side is $a_{ij}b_{uv}c_{\alpha\beta}$.
(iv) The corresponding (j,i) blocks are $(a_{ij}B)^t = a_{ij}B^t$.
(v) The diagonal blocks of $I_m \otimes I_p$ are I_p, and its other blocks are zero; hence $I_m\otimes I_p = I_{mp}$.
(vi) Suppose $\mathbb{F} = \mathbb{C}$. The first identity holds by blockwise conjugation. By part (iv), $(A\otimes B)^* = (\overline{A}\otimes\overline{B})^t = \overline{A}^{\,t}\otimes\overline{B}^{\,t} = A^*\otimes B^*$.
Structural Commentary: Entry factorization handles linearity, reindexing, and adjoints uniformly.
Key Insight: Each Kronecker entry is a product $a_{ij}b_{uv}$. □

Theorem 3.1.3. *Mixed Product Property*
Let $m,n,p,q,r,s \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Let $B \in \mathrm{Mat}(p,q,\mathbb{F})$ and $C \in \mathrm{Mat}(n,r,\mathbb{F})$.
Let $D \in \mathrm{Mat}(q,s,\mathbb{F})$. Then $(A\otimes B)(C\otimes D) = (AC)\otimes(BD)$.

Proof. Write $A = [a_{ik}]$ and $C = [c_{kj}]$, and fix $(i,j) \in J_m \times J_r$.
The (i,j) block of $(A\otimes B)(C\otimes D)$ is $\sum_{k=1}^{n}(a_{ik}B)(c_{kj}D) = \left(\sum_{k=1}^{n} a_{ik}c_{kj}\right)BD = (AC)_{ij}BD$.
This is the corresponding block of $(AC)\otimes(BD)$, so the matrices are equal.
Structural Commentary: Block multiplication separates the contracted index from the factor product BD.
Key Insight: The k-sum is the (i,j) entry of AC. □

Corollary 3.1.4. *Inverse of a Kronecker Product*
Let $m,n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(m,m,\mathbb{F})^\times$, and $B \in \mathrm{Mat}(n,n,\mathbb{F})^\times$. Then $A\otimes B \in \mathrm{Mat}(mn,mn,\mathbb{F})^\times$ and $(A\otimes B)^{-1} = A^{-1}\otimes B^{-1}$.

Proof. Set $K := A^{-1}\otimes B^{-1}$. The identity $(A\otimes B)K = (AA^{-1})\otimes(BB^{-1}) = I_{mn}$ holds by Theorem 3.1.3 and Proposition 3.1.2(v).
The identity $K(A\otimes B) = (A^{-1}A)\otimes(B^{-1}B) = I_{mn}$ holds by the same theorem and proposition.
Structural Commentary: The mixed-product law verifies both inverse equations factorwise.
Key Insight: The two-sided inverse is $A^{-1}\otimes B^{-1}$. □

Proposition 3.1.5. *Kronecker Product and Sum Spectra in the Diagonalizable Case*
Let $n,q \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n,n,\mathbb{F})$, and $B \in \mathrm{Mat}(q,q,\mathbb{F})$. Let $P \in \mathrm{Mat}(n,n,\mathbb{F})^\times$ and $Q \in \mathrm{Mat}(q,q,\mathbb{F})^\times$.
Let $\lambda_1,\dots,\lambda_n \in \mathbb{F}$ and $\mu_1,\dots,\mu_q \in \mathbb{F}$. Set $\Lambda := \mathrm{diag}(\lambda_1,\dots,\lambda_n)$ and $M := \mathrm{diag}(\mu_1,\dots,\mu_q)$. Suppose $A = P\Lambda P^{-1}$ and $B = QMQ^{-1}$.

(i) $A\otimes B \sim \Lambda\otimes M$.
(ii) $A\otimes I_q + I_n\otimes B \sim \Lambda\otimes I_q + I_n\otimes M$.
(iii) $\mathrm{spec}(A\otimes B) = \{\lambda_i\mu_j : (i,j) \in J_n\times J_q\}$.
(iv) $\mathrm{spec}(A\otimes I_q + I_n\otimes B) = \{\lambda_i+\mu_j : (i,j)\in J_n\times J_q\}$.

Proof. Set $R := P \otimes Q$. Then $R \in \mathrm{Mat}(nq, nq, \mathbb{F})^\times$ and $R^{-1} = P^{-1} \otimes Q^{-1}$ by Corollary 3.1.4.

(i) The identity $A \otimes B = R(\Lambda \otimes M)R^{-1}$ holds by Theorem 3.1.3. Hence $A \otimes B \sim \Lambda \otimes M$.
(ii) By distributivity and Theorem 3.1.3, $A \otimes I_q + I_n \otimes B = R(\Lambda \otimes I_q + I_n \otimes M)R^{-1}$. Hence $A \otimes I_q + I_n \otimes B \sim \Lambda \otimes I_q + I_n \otimes M$.
(iii) The matrix $\Lambda \otimes M$ is diagonal with entries $(\lambda_i \mu_j)_{(i,j)\in J_n \times J_q}$.
Thus $\mathrm{spec}(A \otimes B) = \{\lambda_i \mu_j : (i,j) \in J_n \times J_q\}$ by part (i) and Theorem 2.1.7(i).
(iv) The matrix $\Lambda \otimes I_q + I_n \otimes M$ is diagonal with entries $(\lambda_i + \mu_j)_{(i,j)\in J_n \times J_q}$.
Thus $\mathrm{spec}(A \otimes I_q + I_n \otimes B) = \{\lambda_i + \mu_j : (i,j) \in J_n \times J_q\}$ by part (ii) and Theorem 2.1.7(i).

Structural Commentary: The similarity $P \otimes Q$ transfers product and sum to diagonal coordinates.
Key Insight: Paired diagonal entries multiply for products and add for sums. □

For $r \in \mathbb{Z}_{>0}$, set $T_r := \mathrm{tridiag}(-1, 2, -1) \in \mathrm{Mat}(r, r, \mathbb{R})$.
For $(m, n) \in \mathbb{Z}^2_{>0}$, Figure 82 depicts the interior stencil of $T_m \otimes I_n + I_m \otimes T_n$.
The two summands act in the vertical and horizontal coordinate directions, respectively.

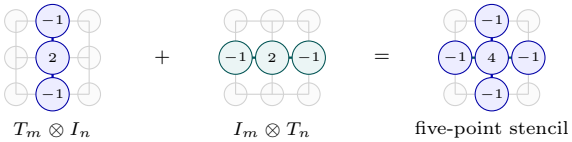

Figure 82: The two tridiagonal factors combine into center weight 4 and four neighboring weights −1.

Definition 3.1.6. *Vectorisation*
Let $m, n \in \mathbb{Z}_{>0}$ and $X = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_n] \in \mathrm{Mat}(m, n, \mathbb{F})$. The *vectorisation map* is $\mathrm{vec}\colon \mathrm{Mat}(m, n, \mathbb{F}) \to \mathbb{F}^{mn}$.
For X as above, $\mathrm{vec}(X) := (\mathbf{x}_1^t \ \cdots \ \mathbf{x}_n^t)^t$.

Theorem 3.1.7. *Vectorisation of a Triple Product*
Let $m, n, p, q \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(m, n, \mathbb{F})$. Let $X \in \mathrm{Mat}(n, p, \mathbb{F})$ and $B \in \mathrm{Mat}(p, q, \mathbb{F})$. Then $\mathrm{vec}(AXB) = (B^t \otimes A)\,\mathrm{vec}(X)$.

Proof. Write $B = [b_{ik}] = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_q]$ and $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_p]$.
For $k \in J_q$, the kth block on each side is $AX\mathbf{b}_k = \sum_{i=1}^{p} b_{ik} A\mathbf{x}_i$; hence the vectors are equal.

Structural Commentary: Column stacking matches the kth block with the kth column of AXB.
Key Insight: Both blocks equal $\sum_{i=1}^{p} b_{ik} A\mathbf{x}_i$. □

Perspective 3.1.8. *Vectorising a Minimal-Polynomial Test*
The candidate step in Method 3.2.3 admits a vectorised form. Fix $n \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $m \in \{1, \ldots, n^2\}$.
For $(a_0, \ldots, a_{m-1}) \in \mathbb{F}^m$, set $p_m(z) := a_0 + a_1 z + \cdots + a_{m-1} z^{m-1} + z^m$. Set $V_m(A) := [\mathrm{vec}(I_n) \ \mathrm{vec}(A) \ \cdots \ \mathrm{vec}(A^{m-1})] \in \mathrm{Mat}(n^2, m, \mathbb{F})$.
Set $\mathbf{a} := (a_0, \ldots, a_{m-1})^t$. Then $p_m(A) = 0 \iff V_m(A)\mathbf{a} = -\mathrm{vec}(A^m)$.
Structural Commentary: Vectorisation turns the candidate equation into an $n^2 \times m$ scalar system.
Key Insight: Its columns are vectorised matrix powers.

3.2 Hadamard Products

Throughout this subsection, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 3.2.1. *Hadamard Product*
Let $m, n \in \mathbb{Z}_{>0}$ and $A = [a_{ij}], B = [b_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$.
The *Hadamard product*, or *entrywise product*, of A and B is $A \circ B := [a_{ij} b_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$.

Proposition 3.2.2. *Basic Hadamard Identities*
Let $m, n \in \mathbb{Z}_{>0}$, $\gamma \in \mathbb{F}$, and $A = [a_{ij}], B = [b_{ij}], C = [c_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$. Set $E := [1]_{i\in J_m, j\in J_n} \in \mathrm{Mat}(m, n, \mathbb{F})$.

(i) $A \circ B = B \circ A$;
(ii) $A \circ (\gamma B + C) = \gamma(A \circ B) + A \circ C$;
(iii) $(A \circ B)^* = A^* \circ B^*$;
(iv) $A \circ E = A$;
(v) if $m = n$, $d_1, \ldots, d_n \in \mathbb{F}$, and $D = \mathrm{diag}(d_1, \ldots, d_n)$, then $A \circ D = \mathrm{diag}(a_{11}d_1, \ldots, a_{nn}d_n)$.

Proof. Fix $(i, j) \in J_m \times J_n$.

(i) $(A \circ B)_{ij} = a_{ij} b_{ij} = b_{ij} a_{ij} = (B \circ A)_{ij}$.
(ii) $\big(A \circ (\gamma B + C)\big)_{ij} = a_{ij}(\gamma b_{ij} + c_{ij}) = \gamma(A \circ B)_{ij} + (A \circ C)_{ij}$.
(iii) $\big((A \circ B)^*\big)_{ji} = \overline{a_{ij} b_{ij}} = \overline{a_{ij}}\,\overline{b_{ij}} = (A^* \circ B^*)_{ji}$.
(iv) $(A \circ E)_{ij} = a_{ij} = A_{ij}$.
(v) If $m = n$, then $(A \circ D)_{ij} = a_{ij} D_{ij}$ equals $a_{ii} d_i$ if $i = j$ and 0 otherwise.

Structural Commentary: A fixed entry reduces (i)–(iv) to scalar arithmetic; (v) holds by the support of D.
Key Insight: No summation couples distinct coordinate pairs. □

Theorem 3.2.3. *Schur Product Theorem*
Let $n \in \mathbb{Z}_{>0}$ and $A = [a_{ij}], B = [b_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A \succeq 0$ and $B \succeq 0$. Then $A \circ B \succeq 0$.

Proof. Set $R := A^{1/2}$ and $S := B^{1/2}$; then $A = R^2 = R^*R$ and $B = S^2 = S^*S$ by Corollary 2.1.4(i).
Write $R = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$, $S = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$, and set $(\mathbf{w}_j)_{j=1}^n := (\mathbf{u}_j \otimes \mathbf{v}_j)_{j=1}^n$. $(\forall\, i \in J_n)\ \mathbf{w}_i^* = \mathbf{u}_i^* \otimes \mathbf{v}_i^*$ by Proposition 3.1.2(iv),(vi).
$(\forall\, (i,j) \in J_n^2)\ \mathbf{w}_i^*\mathbf{w}_j = (\mathbf{u}_i^*\mathbf{u}_j)(\mathbf{v}_i^*\mathbf{v}_j) = a_{ij}b_{ij} = (A \circ B)_{ij}$ by Theorem 3.1.3.
$A \circ B = \operatorname{Gram}(\mathbf{w}_1, \dots, \mathbf{w}_n) \succeq 0$ by the preceding identity and Proposition 1.5.3(i),(ii).
Structural Commentary: Gram factorizations and the mixed-product law identify $A \circ B$ as a Gram matrix.
Key Insight: The jth tensor column multiplies the two ij-Gram entries. □

Figure 83 records the Gram-to-Gram transfer from paired tensor vectors to the Hadamard product.

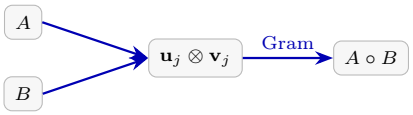

Figure 83: The paired vectors $\mathbf{u}_j \otimes \mathbf{v}_j$ form a Gram representation of $A \circ B$.

Corollary 3.2.4. *Positive Definite Hadamard Products*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \operatorname{Mat}(n,n,\mathbb{F})$, $B = [b_{ij}] \in \operatorname{Mat}(n,n,\mathbb{F})$ satisfy $A \succeq 0$ and $B \succeq 0$.
(i) If $A \succ 0$ and $(\forall\, j \in J_n)\ b_{jj} > 0$, then $A \circ B \succ 0$.
(ii) If $A \succ 0$ and $B \succ 0$, then $A \circ B \succ 0$.

Proof.
(i) By Theorem 2.1.8(ii), set $\lambda_{\min} := \min \operatorname{spec}(A) > 0$.
Then $A - \lambda_{\min} I_n \succeq 0$ by Theorem 2.1.3(ii) and $\operatorname{spec}(A - \lambda_{\min} I_n) = \{\lambda - \lambda_{\min} : \lambda \in \operatorname{spec}(A)\} \subseteq [0, \infty)$.
Set $C := (A - \lambda_{\min} I_n) \circ B$. Then $C \succeq 0$ by Theorem 3.2.3. Fix $\mathbf{x} = (x_1, \dots, x_n)^t \in \mathbb{F}^n \setminus \{\mathbf{0}\}$.
The identity $\mathbf{x}^*(A \circ B)\mathbf{x} = \mathbf{x}^* C \mathbf{x} + \lambda_{\min} \sum_{j=1}^n b_{jj} |x_j|^2$ holds by Proposition 3.2.2(i),(ii),(v).
Its right side is positive by $C \succeq 0$, $\lambda_{\min} > 0$, the diagonal hypothesis, and $\mathbf{x} \neq \mathbf{0}$.
$A \circ B \succ 0$ by Theorem 3.2.3, the preceding strict inequality, and arbitrariness of $\mathbf{x}$.
(ii) The hypothesis in (i), $(\forall\, j \in J_n)\ b_{jj} = \mathbf{e}_j^* B \mathbf{e}_j > 0$, holds by $B \succ 0$. $A \circ B \succ 0$ by (i).
Structural Commentary: A spectral-floor split separates a semidefinite Hadamard term from a positive diagonal term.
Key Insight: $I_n \circ B = \operatorname{diag}(b_{11}, \dots, b_{nn})$ transfers diagonal positivity. □

Proof Method 3.2.5. *Hadamard-Product Positivity Certificate*
Let $n \in \mathbb{Z}_{>0}$ and $X, Y \in \operatorname{Mat}(n,n,\mathbb{F})$ satisfy $X \succeq 0$ and $Y \succeq 0$. The following steps construct a Gram certificate for $X \circ Y \succeq 0$:
(i) set $R := X^{1/2}$ and $S := Y^{1/2}$; then $X = R^2 = R^*R$ and $Y = S^2 = S^*S$ by Corollary 2.1.4(i);
(ii) write $R = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$, $S = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$, and set $(\mathbf{w}_j)_{j=1}^n := (\mathbf{u}_j \otimes \mathbf{v}_j)_{j=1}^n$;
(iii) verify $(\forall\, (i,j) \in J_n^2)\ (X \circ Y)_{ij} = \mathbf{w}_i^*\mathbf{w}_j$ by Proposition 3.1.2(iv),(vi) and Theorem 3.1.3;
(iv) return $X \circ Y = \operatorname{Gram}(\mathbf{w}_1, \dots, \mathbf{w}_n) \succeq 0$ by Proposition 1.5.3(i),(ii).

3.3 Spectral Radius and Matrix Powers

Throughout this subsection, $\mathbb{F}$ is a field.

Theorem 3.3.1. *Ranks of Matrix Powers Stabilize*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \operatorname{Mat}(n,n,\mathbb{F})$. Then:
(i) $(\forall\, k \in \mathbb{Z}_{>0})\ \operatorname{rank}(A^k) \geq \operatorname{rank}(A^{k+1})$;
(ii) if $k \in \mathbb{Z}_{>0}$ and $\operatorname{rank}(A^k) = \operatorname{rank}(A^{k+1})$, then $(\forall\, p \in \mathbb{Z}_{>0})\ \big[\operatorname{col}(A^k) = \operatorname{col}(A^{k+p}) \ \wedge\ \operatorname{rank}(A^k) = \operatorname{rank}(A^{k+p})\big]$;
(iii) if A is not invertible, then $\min\{q \in \mathbb{Z}_{>0} : \operatorname{rank}(A^q) = \operatorname{rank}(A^{q+1})\} \in J_n$.

Proof.
(i) Fix $k \in \mathbb{Z}_{>0}$. Then $\operatorname{col}(A^{k+1}) \subseteq \operatorname{col}(A^k)$ by $A^{k+1} = A^k A$, and the rank inequality holds by taking dimensions.
(ii) Fix $k \in \mathbb{Z}_{>0}$ with $\operatorname{rank}(A^k) = \operatorname{rank}(A^{k+1})$, and set $U := \operatorname{col}(A^k)$.
The identities $U = \operatorname{col}(A^{k+1})$ and $A(U) = U$ hold by part (i), equality of dimensions, and $AA^k = A^{k+1}$.
$(\forall\, p \in \mathbb{Z}_{>0})\ \operatorname{col}(A^{k+p}) = U$ by induction; the rank identities hold by taking dimensions.
(iii) $\operatorname{rank}(A) \leq n - 1$ by Corollary 3.6.2 and noninvertibility.
If $(\forall\, q \in J_n)\ \operatorname{rank}(A^q) > \operatorname{rank}(A^{q+1})$, then $\operatorname{rank}(A^{n+1}) \leq \operatorname{rank}(A) - n \leq -1$, contradicting nonnegativity of rank.
The relation $(\exists\, q \in J_n)\ \operatorname{rank}(A^q) = \operatorname{rank}(A^{q+1})$ holds by this contradiction.
The minimum in (iii) belongs to J_n by this witness and well-ordering.
Structural Commentary: Nested column spaces and dimension equality create an invariant image; finite dimension forces stabilization.
Key Insight: Each strict rank drop consumes at least one dimension. □

Definition 3.3.2. *Index of a Matrix*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \operatorname{Mat}(n,n,\mathbb{F})$.
(i) If A is invertible, its *index* is 0.
(ii) If A is not invertible, its *index* is $\min\{q \in \mathbb{Z}_{>0} : \operatorname{rank}(A^q) = \operatorname{rank}(A^{q+1})\}$.

Definition 3.3.3. *Spectral Radius and Convergent Matrix*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \operatorname{Mat}(n,n,\mathbb{C})$.
(i) The *spectral radius* is $\rho(A) := \max_{\lambda \in \operatorname{spec}(A)} |\lambda|$.
(ii) The notation $A^p \to 0$ means $(\forall\, (i,j) \in J_n^2)\ \lim_{p \to \infty} (A^p)_{ij} = 0$.

Proposition 3.3.4. *Basic Invariants of Nilpotents*
Let $n \in \mathbb{Z}_{>0}$ and $N \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:

(i) *N is nilpotent;*
(ii) *the characteristic polynomial of N is z^n;*
(iii) *$N^n = 0$.*

Under these equivalent conditions, $\mathrm{tr}(N) = \det(N) = 0$ and, if $\mathbb{F} = \mathbb{C}$, $\rho(N) = 0$.

Proof.

(i) $\Rightarrow$ *(ii)* By nilpotency, choose $r \in \mathbb{Z}_{>0}$ satisfying $N^r = 0$.
Every eigenvalue λ over an algebraic closure equals 0 by $\lambda^r \mathbf{v} = N^r \mathbf{v} = \mathbf{0}$ for an associated $\mathbf{v} \neq \mathbf{0}$.
$p_N(z) = z^n$ by complete splitting and Theorem 2.1.3(i).
(ii) $\Rightarrow$ *(iii)* $p_N(N) = N^n = 0$ by Theorem 3.1.1.
(iii) $\Rightarrow$ *(i)* Matrix N is nilpotent by $N^n = 0$.

Assume the equivalent conditions. The identities $\mathrm{tr}(N) = \det(N) = 0$ hold by $p_N(z) = z^n$ and Theorem 2.1.3(i).
If $\mathbb{F} = \mathbb{C}$, then $\mathrm{spec}(N) = \{0\}$ by Theorem 2.1.3(ii), and $\rho(N) = 0$ by the spectral-radius formula.
Structural Commentary: Eigenvalue collapse identifies p_N; Cayley–Hamilton and its coefficients recover the remaining invariants.
Key Insight: Vanishing eigenvalues precede the uniform exponent $N^n = 0$. □

Theorem 3.3.5. *Matrix Powers Converge to Zero*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then $A^p \to 0$ if and only if $\rho(A) < 1$.

Proof. By Theorem 6.2.1, choose $S \in \mathrm{Mat}(n, n, \mathbb{C})^\times$ and a Jordan matrix J satisfying $A = SJS^{-1}$.
$A^p \to 0 \iff J^p \to 0$ by $A^p = SJ^pS^{-1}$, $J^p = S^{-1}A^pS$, and finite sums.
The convergence $J^p \to 0$ is equivalent to convergence of each Jordan block power by the block-diagonal structure of J.
$(\Rightarrow)$ $\rho(A) = \max_{\lambda \in \mathrm{spec}(A)} |\lambda| < 1$ by the block reduction and Proposition 6.1.5(ii).
$(\Leftarrow)$ Each Jordan block power tends to 0 by $\rho(A) < 1$ and Proposition 6.1.5(ii). $A^p \to 0$ by the two reductions above.
Structural Commentary: Similarity transfers entrywise convergence to block powers, which reduce it to eigenvalue moduli.
Key Insight: Geometric decay dominates each fixed polynomial superdiagonal factor. □

Figure 84 depicts the Jordan-block threshold at the unit circle.

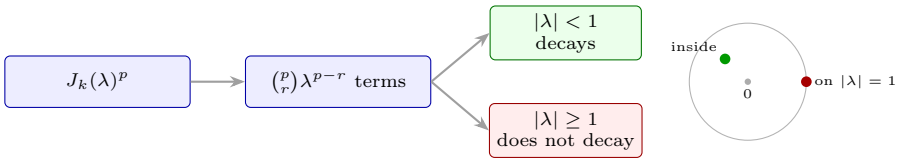

Figure 84: Jordan block powers decay for $|\lambda| < 1$, but not on or outside the gray unit circle.

Definition 3.3.6. *Power Bounded Matrix*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The matrix A is *power bounded* if $(\forall\, (i, j) \in J_n^2) \sup_{p \in \mathbb{Z}_{>0}} |(A^p)_{ij}| < \infty$.

Theorem 3.3.7. *Power Bounded Jordan Criterion*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) *A is power bounded.*
(ii) *$\rho(A) \leq 1$, and Jordan blocks associated with eigenvalues of modulus 1 have size 1.*
(iii) *$\rho(A) \leq 1$, and $(\forall\, \lambda \in \mathrm{spec}(A))\ \big(|\lambda| = 1 \Longrightarrow \dim E_\lambda(A) = m_A(\lambda)\big)$.*

Proof. By Theorem 6.2.1, choose $S \in \mathrm{Mat}(n, n, \mathbb{C})^\times$ and a Jordan matrix J satisfying $A = SJS^{-1}$.
Power boundedness of A and J is equivalent by $A^p = SJ^pS^{-1}$, $J^p = S^{-1}A^pS$, and entrywise fixed finite linear combinations.

(i) $\Rightarrow$ *(ii)* The matrix J is power bounded by (i) and the fixed-similarity equivalence.
A block with $|\lambda| > 1$ is excluded by its unbounded diagonal sequence $(\lambda^p)_{p \geq 1}$.
If $|\lambda| = 1$ and $k \geq 2$, then $J_k(\lambda)^p$ is unbounded by Proposition 6.1.5(i) and $\left|p\lambda^{p-1}\right| = p$. Condition (ii) holds by these exclusions.
(ii) $\Rightarrow$ *(iii)* Fix $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.
The dimension $\dim E_\lambda(A)$ is the number of Jordan blocks at λ by Corollary 6.2.3(i).
The multiplicity $m_A(\lambda)$ is the sum of their sizes by Theorem 2.1.7(ii) and Proposition 2.1.6(iii).
The desired equality holds by these two counting identities and the size-one condition in (ii).
(iii) $\Rightarrow$ *(i)* All unit-circle blocks have size 1 by (iii), the preceding counts, and positive integral block sizes.
Blocks with $|\lambda| < 1$ converge to 0 by Proposition 6.1.5(ii), while the size-one unit-circle blocks are bounded by $|\lambda^p| = 1$.
The matrix J is power bounded by $\rho(A) \leq 1$, these block bounds, and its block-diagonal structure.
The matrix A is power bounded by the fixed-similarity equivalence.

Structural Commentary: Jordan form separates exponential and boundary polynomial growth; block counts detect nilpotent parts.
Key Insight: A unit-circle block is bounded exactly when its size is one. □

4 Matrix Decompositions

Decompositions expose usable structure in a matrix. LU, QR, Cholesky, SVD, compact SVD, and polar decomposition separate elimination, geometry, positivity, rank, and stretching.

Perspective 4.0.1. *Matrix Factorisation Roadmap*
Factorisations expose structure by task: elimination gives LU and LDL^t (Theorem 4.1.4, Corollary 4.1.5); positivity gives Cholesky (Theorem 4.1.11); geometry gives QR (Theorem 4.1.7); singular directions give SVD, compact SVD, and the pseudoinverse (Theorems 4.2.9, 4.2.14, Definition 5.2.4); polar decomposition separates motion from stretch (Theorem 4.2.18). Spectral factorisations appeared earlier: diagonalisation, spectral theorems, Jordan form, and Schur form (Theorems 5.4.3, 6.1.11, 6.1.12, 6.2.1, 2.1.9).
Key Insight: Choose the factorisation that exposes the needed structure.

4.1 LU, QR, and Cholesky-Type Decompositions

Throughout this subsection, $\mathbb{F}$ is a field.

Definition 4.1.1. *Full-Rank Factorization*
Let $m, n, r \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(m, n, \mathbb{F})$.
A factorization $A = XY$, with $X \in \mathrm{Mat}(m, r, \mathbb{F})$ and $Y \in \mathrm{Mat}(r, n, \mathbb{F})$, is a *full-rank factorization* if $\mathrm{rank}(X) = \mathrm{rank}(Y) = r$.

Theorem 4.1.2. *Full-Rank Factorization*
Let $m, n \in \mathbb{Z}_{>0}$ and let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ have rank $r \geq 1$. Let the columns of $X \in \mathrm{Mat}(m, r, \mathbb{F})$ form a basis of $\mathrm{col}(A)$. Then:
(i) $(\exists!\, Y \in \mathrm{Mat}(r, n, \mathbb{F}))\ A = XY$;
(ii) for this matrix Y, $\mathrm{rank}(X) = \mathrm{rank}(Y) = r$.

Proof.
(i) Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. By the basis hypothesis, $(\forall\, j \in J_n)(\exists!\, \mathbf{y}_j \in \mathbb{F}^r)\, \mathbf{a}_j = X\mathbf{y}_j$.
For $Y := [\mathbf{y}_1 \ \cdots \ \mathbf{y}_n]$, one has $A = XY$. If $A = XZ$ with $Z = [\mathbf{z}_1 \ \cdots \ \mathbf{z}_n]$, then $Z = Y$ by coordinate uniqueness.
(ii) $\mathrm{rank}(X) = r$ by the basis hypothesis, and $r = \mathrm{rank}(XY) \leq \mathrm{rank}(Y) \leq r$ by Theorem 1.2.8(i) and the row count of Y.

Structural Commentary: Basis coordinates construct Y uniquely; the product-rank bound forces full row rank.
Key Insight: Column coordinates determine the right factor. □

Definition 4.1.3. *LU Factorisation*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$.
(i) An *LU factorisation* is a product $A = LU$ with L lower triangular and U upper triangular.
(ii) For $L = [l_{ij}]$, it is a *unit-lower-triangular LU factorisation* if $(\forall\, k \in J_n)\, l_{kk} = 1$.

Theorem 4.1.4. *LU Factorisation from Nonzero Leading Principal Minors*
Let $n \in \mathbb{Z}_{>0}$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Assume every leading principal minor of A is nonzero.
Then a unique pair $(L, U) \in \mathrm{Mat}(n, n, \mathbb{F})^2$ satisfies $A = LU$, with L unit lower triangular and U upper triangular.

Proof. Induct on n.
(i) *Base Case*: If $n = 1$, then $A = [a] = [1][a]$ uniquely, where $a \neq 0$ by hypothesis.
(ii) *Schur Reduction*: For $n \geq 2$, write $A = \left[\begin{smallmatrix} a & \mathbf{r} \\ \mathbf{c} & B \end{smallmatrix}\right]$ with $a \neq 0$, and set $\boldsymbol{\ell} := a^{-1}\mathbf{c}$ and $S := B - a^{-1}\mathbf{c}\mathbf{r}$.
For the leading blocks A_k and S_{k-1}, $\det(A_k) = a \det(S_{k-1})$ by Proposition 1.1.8.
Every leading principal minor of S is nonzero by this identity and the hypothesis. Hence $S = L_1U_1$ uniquely, with L_1 unit lower triangular and U_1 upper triangular, by the induction hypothesis.
(iii) *Construction*: Set $L := \left[\begin{smallmatrix} 1 & 0 \\ \boldsymbol{\ell} & L_1 \end{smallmatrix}\right]$ and $U := \left[\begin{smallmatrix} a & \mathbf{r} \\ 0 & U_1 \end{smallmatrix}\right]$. Then L and U have the required forms, and $LU = \left[\begin{smallmatrix} a & \mathbf{r} \\ \mathbf{c} & B \end{smallmatrix}\right] = A$.
(iv) *Uniqueness*: For another admissible pair, write $\widetilde{L} = \left[\begin{smallmatrix} 1 & 0 \\ \widetilde{\boldsymbol{\ell}} & \widetilde{L}_1 \end{smallmatrix}\right]$ and $\widetilde{U} = \left[\begin{smallmatrix} \widetilde{a} & \widetilde{\mathbf{r}} \\ 0 & \widetilde{U}_1 \end{smallmatrix}\right]$.
Then $(\widetilde{a}, \widetilde{\mathbf{r}}, \widetilde{\boldsymbol{\ell}}) = (a, \mathbf{r}, \boldsymbol{\ell})$ by the first row and column. Then $S = \widetilde{L}_1\widetilde{U}_1$, so $(\widetilde{L}_1, \widetilde{U}_1) = (L_1, U_1)$ by the induction hypothesis.

Structural Commentary: Schur reduction transfers the pivot hypothesis before block reconstruction and uniqueness.
Key Insight: The first pivot fixes the border blocks. □

Corollary 4.1.5. *Symmetric LU and an LDL^t Factorisation*
Let $n \in \mathbb{Z}_{>0}$ and let $A = A^t \in \mathrm{Mat}(n, n, \mathbb{R})$ have nonzero leading principal minors.
Then $(\exists\, L, D \in \mathrm{Mat}(n, n, \mathbb{R}))\ A = LDL^t$, where L is unit lower triangular and D is diagonal and invertible.

Proof. By Theorem 4.1.4, write $A = LU$ uniquely with L unit lower triangular and U upper triangular.
The relation $0 \neq \det(A) = \det(U) = \prod_{k=1}^{n} u_{kk}$ holds by Theorem 3.8.2(v)–(vi) and $\det(L) = 1$.
Thus $D := \mathrm{diag}(u_{11}, \ldots, u_{nn})$ is invertible. For $U_0 := D^{-1}U$, one has $A = LDU_0$, with U_0 unit upper triangular.
By symmetry, $A = U_0^t(DL^t)$ is another admissible LU factorisation; hence $U_0^t = L$ by Theorem 4.1.4. Thus $A = LDL^t$.
Structural Commentary: Pivot extraction normalises U; symmetry and LU uniqueness identify its unit factor.
Key Insight: Symmetry forces $U_0 = L^t$. □

Definition 4.1.6. *Wide and Narrow QR Factorisation*
Let $m, n \in \mathbb{Z}_{>0}$ satisfy $m \geq n$ and let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$.
(i) A *wide QR factorisation* is $A = V \left[\begin{smallmatrix} R \\ 0 \end{smallmatrix}\right]$, where $V \in \mathrm{Mat}(m, m, \mathbb{F})$ is unitary and $R \in \mathrm{Mat}(n, n, \mathbb{F})$ is upper triangular.
(ii) For $V = [Q\ Q']$ with $Q \in \mathrm{Mat}(m, n, \mathbb{F})$, the corresponding *narrow QR factorisation* is $A = QR$, where $Q^*Q = I_n$.

Theorem 4.1.7. *QR Factorisation*
Let $m, n \in \mathbb{Z}_{>0}$ satisfy $m \geq n$ and let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$.
(i) $(\exists\, V \in \mathrm{Mat}(m, m, \mathbb{F}))(\exists\, R \in \mathrm{Mat}(n, n, \mathbb{F}))\ A = V \left[\begin{smallmatrix} R \\ 0 \end{smallmatrix}\right]$. *Here V is unitary, $R = [r_{ij}]$ is upper triangular with $(\forall\, k \in J_n)\, r_{kk} \in \mathbb{R}_{\geq 0}$. For $V = [Q\ Q']$ with $Q \in \mathrm{Mat}(m, n, \mathbb{F})$, one has $A = QR$ and $Q^*Q = I_n$.*
(ii) If $\mathrm{rank}(A) = n$, then a unique pair (Q, R) satisfies $A = QR$ and $Q^*Q = I_n$.
Here $Q \in \mathrm{Mat}(m, n, \mathbb{F})$ and $R = [r_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ is upper triangular with $(\forall\, k \in J_n)\, r_{kk} > 0$.

Proof.

(i) Process $k \in J_n$ in increasing order, and let $\mathbf{x}_k \in \mathbb{F}^{m-k+1}$ be the active tail of column k.
For $\mathbf{x}_k = \mathbf{0}$, set $(H_k, r_{kk}) := (I_{m-k+1}, 0)$. Otherwise set $\mathbf{y}_k := \|\mathbf{x}_k\|_2 \mathbf{e}_1$ and choose $\sigma_k, \mathbf{w}_k$ by Theorem 2.7.3(i)–(iii).
Then $H_k := \sigma_k U_{\mathbf{w}_k}$ satisfies $H_k^* H_k = I_{m-k+1}$ by Theorem 2.7.3(i) and Theorem 2.7.2(iii).
It also satisfies $H_k \mathbf{x}_k = r_{kk}\mathbf{e}_1$ for $r_{kk} := \|\mathbf{x}_k\|_2$ by Theorem 2.7.3(iii).
In both cases, $\widehat{H}_k := I_{k-1} \oplus H_k$ preserves prior zeros by acting only on rows $k, \dots, m$.
Thus $WA = \left[\begin{smallmatrix} R \\ 0 \end{smallmatrix}\right]$ for $W := \widehat{H}_n \cdots \widehat{H}_1$, with R upper triangular and $(\forall\, k \in J_n)\, r_{kk} \geq 0$.
For $V := W^* = [Q\ Q']$, $A = V \left[\begin{smallmatrix} R \\ 0 \end{smallmatrix}\right] = QR$ and $Q^*Q = I_n$ by unitarity and block multiplication.

(ii) Assume $\operatorname{rank}(A) = n$. Then $\operatorname{rank}(R) = n$ by Theorem 3.6.6(ii), so R is invertible by Corollary 3.6.2.
Its nonnegative diagonal is positive by Theorem 3.8.2(v),(vii), proving existence.
If $A = Q_1R_1 = Q_2R_2$ are admissible, then each R_i is invertible by Theorem 3.8.2(v),(vii), and $R_1^*R_1 = A^*A = R_2^*R_2$.
For $D := R_1R_2^{-1}$, D is upper triangular by Theorem 3.8.9(ii) and the product-entry formula.
The identity $D^*D = (R_2^{-1})^*R_1^*R_1R_2^{-1} = I_n$ holds, and $DD^* = I_n$ by Theorem 2.6.8(i)⇒(iii).
Viewed over $\mathbb{C}$ when $\mathbb{F} = \mathbb{R}$, D is diagonal by Lemma 6.1.10.
Now $d_{kk} = (R_1)_{kk}(R_2)_{kk}^{-1} > 0$ and $|d_{kk}| = 1$, so $D = I_n$. Therefore $R_1 = R_2$ and $Q_1 = AR_1^{-1} = Q_2$.

Structural Commentary: Householder tail maps construct R; triangular normality reduces uniqueness to diagonal phases.
Key Insight: Positivity eliminates every unitary diagonal phase. □

Corollary 4.1.8. *Prefix Column Spaces in QR Factorisation*
Let $m, n \in \mathbb{Z}_{>0}$ satisfy $m \geq n$ and let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
Let $A = QR \in \operatorname{Mat}(m, n, \mathbb{F})$ satisfy $\operatorname{rank}(A) = n$ and $Q^*Q = I_n$, where R is upper triangular with positive diagonal.
For $k \in J_n$, let A_k and Q_k be the first k columns of A and Q. Then:

(i) $(\forall\, k \in J_n)\ \operatorname{col}(A_k) = \operatorname{col}(Q_k)$;
(ii) $(\forall\, k \in J_n)$, *the columns of Q_k form an orthonormal basis of* $\operatorname{col}(A_k)$.

Proof.

(i) Fix $k \in J_n$, and let R_k be the leading $k \times k$ block of R. Then R_k is invertible by Theorem 3.8.2(v),(vii).
Since $A_k = Q_kR_k$ and $Q_k = A_kR_k^{-1}$, one has $\operatorname{col}(A_k) = \operatorname{col}(Q_k)$.

(ii) Fix $k \in J_n$. The columns of Q_k are orthonormal by $Q_k^*Q_k = I_k$, and they span $\operatorname{col}(A_k)$ by part (i).

Structural Commentary: Invertible triangular prefixes transfer column spaces; $Q_k^*Q_k = I_k$ supplies orthonormality.
Key Insight: Each prefix uses only the first k orthonormal columns. □

Figure 85 depicts orthonormal column replacement and upper-triangular coordinates.

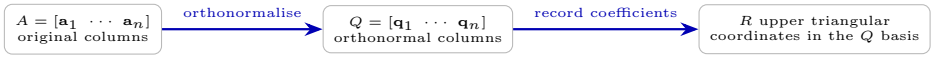

Figure 85: QR records the original columns in an orthonormal basis while preserving every prefix column space.

Perspective 4.1.9. *Gram–Schmidt and Householder QR*
For a full-column-rank matrix, Gram–Schmidt preserves prefix spans by Theorem 1.3.6(i)–(ii).
Householder transformations instead triangularise entire column tails in Theorem 4.1.7(i).
Both constructions reach the unique positive-diagonal factorisation in Theorem 4.1.7(ii).
Gram–Schmidt orthogonalises columns; Householder transformations act unitarily on rows.

Perspective 4.1.10. *Rank-Deficient QR from Gram–Schmidt*
Zero Gram–Schmidt remainders detect dependent prefix columns by Corollary 1.3.8.
The QR factorisation still exists by Theorem 4.1.7(i).
Rank defect blocks the positive-diagonal uniqueness hypothesis in Theorem 4.1.7(ii).

Theorem 4.1.11. *Cholesky Factorisation*
Let $n \in \mathbb{Z}_{>0}$, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A \in \operatorname{Mat}(n, n, \mathbb{F})$.

(i) If $A \succeq 0$, then $(\exists\, L \in \operatorname{Mat}(n, n, \mathbb{F}))\ A = LL^*$, where L is lower triangular with nonnegative diagonal entries.
(ii) If $A \succ 0$, then $\exists!\, L \in \operatorname{Mat}(n, n, \mathbb{F})$ satisfying $A = LL^*$, where L is lower triangular with positive diagonal entries.

Proof.

(i) Set $B := A^{1/2}$ by Corollary 2.1.4(i), so $B = B^*$ and $B^2 = A$.
Choose $B = QR$ with $Q^*Q = I_n$ and R upper triangular with nonnegative diagonal by Theorem 4.1.7(i).
Then $A = B^*B = R^*R$, so $L := R^*$ has the required form.

(ii) Assume $A \succ 0$. Then A is invertible by Theorem 2.1.8(i)⇒(iii).
For the factor L in (i), $n = \operatorname{rank}(A) \leq \operatorname{rank}(L) \leq n$ by Theorem 1.2.8(i).
Thus L is invertible by Corollary 3.6.2, so its nonnegative diagonal is positive by Theorem 3.8.2(v),(vii).
For uniqueness, set $B := A^{1/2}$ and suppose $A = L_1L_1^* = L_2L_2^*$ for admissible L_1, L_2.
Since $B^2 = A$ and A is invertible, $\ker(B) \subseteq \ker(A) = \{\mathbf{0}\}$; hence B is invertible by Corollary 3.6.3(vi)⇒(i).
For $R_i := L_i^*$, each R_i is upper triangular with positive diagonal and is invertible by Theorem 3.8.2(v),(vii). Set $Q_i := BR_i^{-1}$.
Then $B = Q_iR_i$ and $Q_i^*Q_i = (R_i^{-1})^*B^2R_i^{-1} = I_n$. Hence $R_1 = R_2$ by Theorem 4.1.7(ii), and therefore $L_1 = L_2$.

Structural Commentary: QR of $A^{1/2}$ constructs R^*; positive-diagonal QR uniqueness identifies competitors.
Key Insight: Cholesky is the adjoint triangular factor of $A^{1/2}$. □

Figure 86 depicts the Cholesky factor obtained from QR of the positive square root.

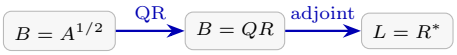

Figure 86: For $B = A^{1/2} = QR$, the choice $L = R^*$ satisfies $A = LL^*$; positive definiteness makes L unique.

4.2 Polar and Singular Value Decompositions

Throughout this subsection, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and all matrix dimensions are positive integers.
Every inner-product space is finite-dimensional over $\mathbb{F}$.

Theorem 4.2.1. *Polar Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$. Then $(\exists S \in \mathcal{L}(\mathcal{V}))\,[S$ is an isometry $\wedge\, T = S\sqrt{T^*T}]$.

Proof.
(i) *Positive Factor*: Set $P := \sqrt{T^*T}$ by Theorem 6.3.6(v)⇒(i) and Proposition 6.3.7.
$(\forall \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\|^2 = \langle P^2\mathbf{v}, \mathbf{v}\rangle = \|P\mathbf{v}\|^2$ by $P = P^*$. $\ker(P) = \ker(T)$ by this norm identity.
(ii) *Range Isometry*: The rule $S_1(P\mathbf{v}) := T\mathbf{v}$ defines a linear bijection $\operatorname{range}(P) \to \operatorname{range}(T)$ by (i).
S_1 is an isometry by $\|S_1(P\mathbf{v})\| = \|P\mathbf{v}\|$ from (i).
(iii) *Complement Extension*: $\dim \operatorname{range}(P)^\perp = \dim \operatorname{range}(T)^\perp$ by (i), Theorem 1.7.3, and Corollary 2.1.7.
Set $d := \dim \operatorname{range}(P)^\perp$. Choose orthonormal bases $(\mathbf{a}_j)_{j=1}^d$ and $(\mathbf{b}_j)_{j=1}^d$ of the two complements by Corollary 1.3.11.
Define $S_0\mathbf{a}_j := \mathbf{b}_j$. Then $(\forall i, j \in J_d)\ \langle S_0\mathbf{a}_i, S_0\mathbf{a}_j\rangle = \langle \mathbf{a}_i, \mathbf{a}_j\rangle = \delta_{ij}$. Thus S_0 is an isometry by sesquilinearity and Theorem 2.6.4.
(iv) *Global Isometry*: $\mathcal{V} = \operatorname{range}(P) \odot \operatorname{range}(P)^\perp$ by Theorem 2.1.5. Define $S(\mathbf{x} + \mathbf{y}) := S_1\mathbf{x} + S_0\mathbf{y}$.
S is an isometry by the orthogonal source and target decompositions. Finally, $(\forall \mathbf{v} \in \mathcal{V})\, SP\mathbf{v} = S_1(P\mathbf{v}) = T\mathbf{v}$.

Structural Commentary: The Gram norm identity builds a range isometry, extended across equidimensional complements.
Key Insight: $\ker\sqrt{T^*T} = \ker T$ synchronizes the two range dimensions. □

Corollary 4.2.2. *Uniqueness in Operator Polar Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$, and set $|T| := \sqrt{T^*T}$ and $|T^*| := \sqrt{TT^*}$.
(i) If $T = SP$, $S^*S = I_{\mathcal{V}}$, and $P \geq 0$, then $P = |T|$; hence the positive right polar factor is unique.
(ii) $(\exists R \in \mathcal{L}(\mathcal{V}))\,[R$ is an isometry $\wedge\, T = |T^*|\,R]$, and $|T^*|$ is the unique positive left polar factor.
(iii) If T is invertible, then both isometric factors are unique: $S = T\,|T|^{-1}$ and $R = |T^*|^{-1}\,T$.

Proof.
(i) If $T = SP$, then $T^*T = PS^*SP = P^2$, so $P = |T|$ by Proposition 6.3.7.
(ii) Fix $T^* = S_0\,|T^*|$ by Theorem 4.2.1.
Then $R := S_0^*$ is an isometry and $T = |T^*|\,R$ by Theorem 2.6.6 and Proposition 2.6.2(iii)–(iv).
If $T = Q\widetilde{R}$ with $Q \geq 0$ and $\widetilde{R}$ isometric, then $TT^* = Q^2$ by $\widetilde{R}\widetilde{R}^* = I_{\mathcal{V}}$; hence $Q = |T^*|$ by Proposition 6.3.7.
(iii) If T is invertible, then $\ker|T| = \ker|T^*| = \{\mathbf{0}\}$ by invertibility of their squares.
Thus both moduli are invertible by Theorem 1.7.7, and $S = T\,|T|^{-1}$, $R = |T^*|^{-1}\,T$.

Structural Commentary: Gram squaring fixes each modulus; invertibility then isolates its isometry.
Key Insight: Positivity selects one square root of each Gram operator. □

Figure 87 summarizes the range isometry used to separate the positive stretch from the isometric factor.

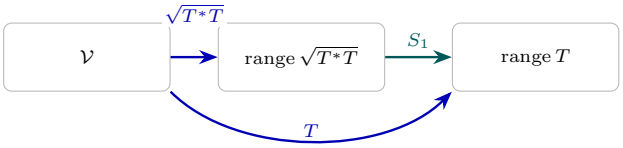

Figure 87: Polar decomposition builds S_1 using $\|T\mathbf{v}\| = \left\|\sqrt{T^*T}\mathbf{v}\right\|$.

Definition 4.2.3. *Singular Values*
For $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, the *singular values* of T are the eigenvalues of $\sqrt{T^*T} \in \mathcal{L}(\mathcal{U})$, listed with their eigenspace multiplicities.

Theorem 4.2.4. *Singular Value Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$, set $n := \dim \mathcal{V}$, and list its singular values as $s_1, \ldots, s_n$.
Then $\exists$ orthonormal bases $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ and $(\mathbf{f}_1, \ldots, \mathbf{f}_n)$ of $\mathcal{V}$ such that $(\forall \mathbf{v} \in \mathcal{V})\, T\mathbf{v} = \sum_{j=1}^{n} s_j \langle \mathbf{v}, \mathbf{e}_j\rangle \mathbf{f}_j$.

Proof. Set $|T| := \sqrt{T^*T}$. Fix an orthonormal eigenbasis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ of $|T|$ with $(\forall j \in J_n)\ |T|\,\mathbf{e}_j = s_j\mathbf{e}_j$ by Theorem 6.1.12 when $\mathbb{F} = \mathbb{R}$ and Theorem 6.1.11 when $\mathbb{F} = \mathbb{C}$. Fix $T = S\,|T|$ by Theorem 4.2.1, and set $(\forall j \in J_n)\, \mathbf{f}_j := S\mathbf{e}_j$.
The list $(\mathbf{f}_1, \ldots, \mathbf{f}_n)$ is an orthonormal basis by Theorem 2.6.4 and Proposition 2.6.2(iii). $(\forall \mathbf{v} \in \mathcal{V})\, T\mathbf{v} = \sum_{j=1}^{n} s_j \langle \mathbf{v}, \mathbf{e}_j\rangle \mathbf{f}_j$ by Theorem 1.4.1.
Structural Commentary: Spectral coordinates diagonalise $|T|$; the polar isometry transports its eigenbasis.
Key Insight: $T\mathbf{e}_j = s_j\mathbf{f}_j$ pairs stretch with direction. □

Figure 88 shows each right singular direction stretched and carried isometrically to its left singular direction.

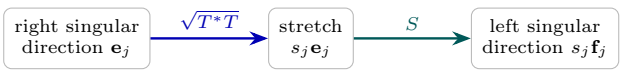

Figure 88: Since $T = S\sqrt{T^*T}$, each $\mathbf{e}_j$ maps to $s_j\mathbf{f}_j$.

Proposition 4.2.5. *Singular Values from T^*T*
The singular values of $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ are the nonnegative square roots of the eigenvalues of T^*T, with the same eigenspace multiplicities.

Proof. Set $n := \dim \mathcal{U}$ and $G := T^*T$. Then $G \geq 0$ by Theorem 6.3.6(v)⇒(i).
Fix an orthonormal eigenbasis $(\mathbf{e}_j)_{j=1}^n$ with $G\mathbf{e}_j = \lambda_j \mathbf{e}_j$ by Theorem 6.1.12 when $\mathbb{F} = \mathbb{R}$ and Theorem 6.1.11 when $\mathbb{F} = \mathbb{C}$.
$(\forall j \in J_n)\, \lambda_j \geq 0$ by Theorem 6.3.6(i)⇒(ii). Define $(\forall j \in J_n)\, R\mathbf{e}_j := \sqrt{\lambda_j}\mathbf{e}_j$.
Then $R \geq 0$ by Theorem 6.3.6(ii)⇒(i), and $R^2 = G$; hence $R = \sqrt{G}$ by Proposition 6.3.7.
The eigenspace multiplicities are preserved by $(\forall \lambda \geq 0)\, E_\lambda(G) = E_{\sqrt{\lambda}}(R)$.
Structural Commentary: One Gram eigenbasis constructs the root and identifies every eigenspace.
Key Insight: $\lambda \mapsto \sqrt{\lambda}$ is injective on the nonnegative spectrum. □

Theorem 4.2.6. *Singular Vectors for a Linear Map*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ have rank r. Set $n := \dim \mathcal{U}$, $m := \dim \mathcal{V}$, $\mathcal{U}_+ := \operatorname{range}(T^*)$, $\mathcal{U}_0 := \ker(T)$, $\mathcal{V}_+ := \operatorname{range}(T)$, and $\mathcal{V}_0 := \ker(T^*)$.
Then $\exists$ ordered orthonormal bases $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_n)$ and $\gamma = (\mathbf{v}_1, \dots, \mathbf{v}_m)$
with $\operatorname{span}\{\mathbf{u}_j : j \in J_r\} = \mathcal{U}_+$, $\operatorname{span}\{\mathbf{u}_j : j \in J_n \setminus J_r\} = \mathcal{U}_0$, $\operatorname{span}\{\mathbf{v}_j : j \in J_r\} = \mathcal{V}_+$, and $\operatorname{span}\{\mathbf{v}_j : j \in J_m \setminus J_r\} = \mathcal{V}_0$,
and positive scalars $(s_k)_{k \in J_r}$ such that $(\forall k \in J_r)\, T\mathbf{u}_k = s_k \mathbf{v}_k$ and $T^*\mathbf{v}_k = s_k \mathbf{u}_k$.
The s_k are the positive singular values; $\mathbf{u}_k$ and $\mathbf{v}_k$ are right and left singular vectors.

Proof.

(i) *Gram Reduction*: For $G := T^*T$, $G \geq 0$, $\ker(G) = \ker(T)$ by Theorem 6.3.6, Proposition 2.3.11; $\operatorname{rank}(G) = r$ by Theorem 1.7.3.
(ii) *Right Singular Basis*: Choose orthonormal eigenbasis $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ of G by Theorem 6.1.12 if $\mathbb{F} = \mathbb{R}$, Theorem 6.1.11 if $\mathbb{F} = \mathbb{C}$.
Order it so that $G\mathbf{u}_k = s_k^2 \mathbf{u}_k$ for $k \in J_r$ and $G\mathbf{u}_k = \mathbf{0}$ for $k \in J_n \setminus J_r$.
These groups span $\mathcal{U}_+$ and $\mathcal{U}_0$ by Proposition 2.3.10(iii), and the s_k are the positive singular values by Proposition 4.2.5.
(iii) *Left Singular Vectors*: Set $(\forall k \in J_r)\, \mathbf{v}_k := s_k^{-1} T\mathbf{u}_k$. Then $(\forall i, j \in J_r)\, \langle \mathbf{v}_i, \mathbf{v}_j \rangle = \frac{\langle \mathbf{u}_i, G\mathbf{u}_j \rangle}{s_i s_j} = \delta_{ij}$.
$(\forall k \in J_r)\, T\mathbf{u}_k = s_k \mathbf{v}_k$ and $T^*\mathbf{v}_k = s_k \mathbf{u}_k$ by the definitions of $\mathbf{v}_k$ and G.
(iv) *Codomain Completion*: The identity $\mathcal{V}_+ = T(\mathcal{U}_+) = \operatorname{span}\{\mathbf{v}_k : k \in J_r\}$ holds by $\mathcal{U} = \mathcal{U}_+ \odot \mathcal{U}_0$.
Complete $(\mathbf{v}_1, \dots, \mathbf{v}_r)$ orthonormally by Corollary 1.3.12. The remaining vectors lie in $\mathcal{V}_+^\perp = \mathcal{V}_0$ by Proposition 2.3.10(i).

Structural Commentary: The Gram spectrum splits the domain; normalized images build the active codomain basis.
Key Insight: $T^*T\mathbf{u}_k = s_k^2 \mathbf{u}_k$ forces both singular-vector relations. □

Figure 89 summarizes the paired active subspaces and kernels for T and T^*.

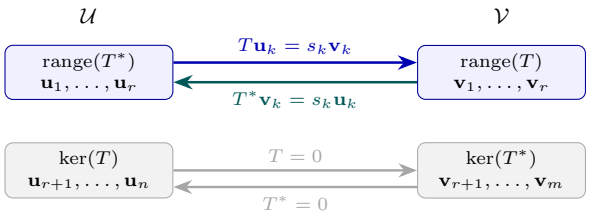

Figure 89: The singular-vector bases split the domain and codomain into active singular subspaces and kernels.

Definition 4.2.7. *Matrix Modulus*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$. The matrices A^*A and AA^* are positive semidefinite by Proposition 2.1.2(i)–(ii).
Their unique positive semidefinite square roots exist by Corollary 2.1.4(i).
The *moduli* of A and A^* are $|A| := (A^*A)^{1/2} \in \operatorname{Mat}(n, n, \mathbb{F})$ and $|A^*| := (AA^*)^{1/2} \in \operatorname{Mat}(m, m, \mathbb{F})$.

Definition 4.2.8. *Matrix Singular Values*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$, $q := \min(m, n)$, and $r := \operatorname{rank}(A)$.
The matrix $|A|$ has exactly r positive eigenvalues by Theorem 2.1.5(iii) and Corollary 2.1.4(ii).
These eigenvalues, counted with multiplicity, are the *positive singular values* of A.
For $r \geq 1$, list them as $\sigma_1 \geq \cdots \geq \sigma_r > 0$; for $r = 0$, this list is empty.
The full singular-value list has length q; if $r < q$, set $\sigma_{r+1} = \cdots = \sigma_q = 0$.

Theorem 4.2.9. *Matrix Singular Value Decomposition*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q := \min(m, n)$. Then $\exists$ unitary matrices $V \in \operatorname{Mat}(m, m, \mathbb{F})$ and $W \in \operatorname{Mat}(n, n, \mathbb{F})$ such that $A = V\Sigma W^*$, where the rectangular diagonal matrix $\Sigma \in \operatorname{Mat}(m, n, \mathbb{R})$ satisfies $(\forall j \in J_q)\, \Sigma_{jj} = \sigma_j$.

Proof.

(i) *Gram Spectrum*: Set $r := \operatorname{rank}(A)$. If $r = 0$, take $(V, W, \Sigma) = (I_m, I_n, 0)$; assume $r \geq 1$ below.
Choose a unitary eigenvector matrix $W = [\mathbf{w}_1 \ \cdots \ \mathbf{w}_n]$ for A^*A by Theorem 6.1.12 when $\mathbb{F} = \mathbb{R}$ and Theorem 6.1.11 when $\mathbb{F} = \mathbb{C}$.
Then $W^*A^*AW = \operatorname{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0)$ by Corollary 2.1.4(ii).
(ii) *Left Singular Vectors*: Set $(\forall j \in J_r)\, \mathbf{v}_j := \sigma_j^{-1} A\mathbf{w}_j$. Then $(\forall i, j \in J_r)\, \mathbf{v}_i^* \mathbf{v}_j = \frac{\mathbf{w}_i^* A^* A \mathbf{w}_j}{\sigma_i \sigma_j} = \delta_{ij}$.
(iii) *Unitary Completion*: Complete $(\mathbf{v}_1, \dots, \mathbf{v}_r)$ to an orthonormal basis by Corollary 1.3.12.
The matrix V with these columns is unitary by Theorem 2.6.8(ii)⇒(i).
(iv) *Factorisation*: For $j \in J_n \setminus J_r$, $\|A\mathbf{w}_j\|_2^2 = \mathbf{w}_j^* A^* A \mathbf{w}_j = 0$. $AW = V\Sigma$ by (ii) and the preceding identity, so $A = V\Sigma W^*$.

Structural Commentary: The Gram eigenbasis fixes W, Σ; normalized images construct V.
Key Insight: $\mathbf{v}_j = \sigma_j^{-1} A\mathbf{w}_j$ orthonormalizes the nonzero images. □

Figure 90 traces the unit ball through the three SVD factors.

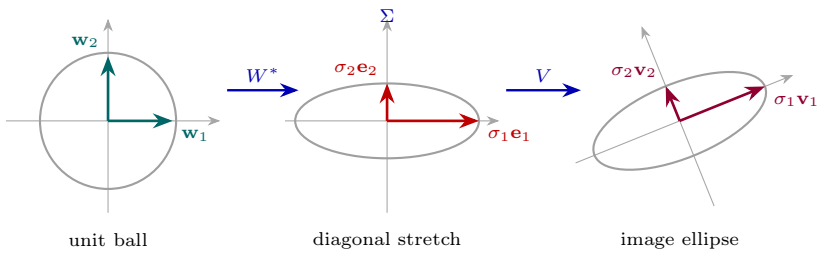

Figure 90: The factors W^*, Σ, and V rotate, stretch, and reorient the unit ball.

Perspective 4.2.10. *Operator-to-Matrix SVD Bridge*
In the bases from Theorem 4.2.6, the matrix of T is rectangular diagonal with positive entries $s_1, \dots, s_r$ and zeros elsewhere. Changing to standard orthonormal bases produces the matrix SVD in Theorem 4.2.9 without changing the singular data.

Proposition 4.2.11. *Basic Singular Value Identities*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q := \min(m,n)$ and $r := \mathrm{rank}(A)$. Let $c \in \mathbb{F}$.

(i) The positive eigenvalues of A^*A and AA^* are $\sigma_1^2, \dots, \sigma_r^2$.
(ii) $\sum_{j=1}^{q} \sigma_j^2 = \mathrm{tr}(A^*A) = \mathrm{tr}(AA^*) = \|A\|_F^2$, where $\|A\|_F^2 := \mathrm{tr}(A^*A)$.
(iii) A and A^* have the same singular values. Over $\mathbb{C}$, so do A^t and $\overline{A}$.
(iv) The singular values of cA are $|c|\,\sigma_1, \dots, |c|\,\sigma_q$.

Proof. Fix an SVD $A = V\Sigma W^*$ by Theorem 4.2.9. Then $A^*A = W(\Sigma^*\Sigma)W^*$ and $AA^* = V(\Sigma\Sigma^*)V^*$.

(i) The positive eigenvalues of both Gram matrices are $\sigma_1^2, \dots, \sigma_r^2$ by the two unitary similarities and Theorem 2.1.7(i)–(ii).

(ii) $\mathrm{tr}(A^*A) = \mathrm{tr}(AA^*) = \displaystyle\sum_{j=1}^{q} \sigma_j^2 = \|A\|_F^2$ by Theorem 1.1.13(i).

(iii) Claim for A^* holds by (i). Over $\mathbb{C}$, $(A^t)^*A^t = (AA^*)^t$, $(\overline{A})^*\overline{A} = (A^*A)^t$, with unchanged Gram spectra by Proposition 1.1.15.

(iv) The singular values of cA are $|c|\,\sigma_1, \dots, |c|\,\sigma_q$ by $(cA)^*(cA) = |c|^2\, A^*A$.

Structural Commentary: Unitary Gram reductions transfer spectra, traces, adjoints, and scalar scaling.
Key Insight: Both Gram spectra contain the squared entries of Σ. □

Corollary 4.2.12. *Square Invertibility and Singular Values*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, have singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. The following are equivalent:

(i) A is invertible;
(ii) $\mathrm{rank}(A) = n$;
(iii) $\sigma_n > 0$;
(iv) $A^*A \succ 0$.

Proof.

(i) $\Rightarrow$ (ii) $\mathrm{rank}(A) = n$ by Corollary 3.6.2.
(ii) $\Rightarrow$ (iii) $\sigma_n > 0$ by the definition of the full singular-value list.
(iii) $\Rightarrow$ (iv) Since all n singular values are positive, $\mathrm{rank}(A) = n$ and $A^*A \succ 0$ by Proposition 2.1.2(v).
(iv) $\Rightarrow$ (i) $\ker(A) = \{\mathbf{0}\}$ by $(\forall \mathbf{x} \in \mathbb{F}^n)\,[A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x}^*A^*A\mathbf{x} = 0 \Rightarrow \mathbf{x} = \mathbf{0}]$.
Thus $\mathrm{rank}(A) = n$ by Theorem 1.7.3, and A is invertible by Corollary 3.6.2.

Structural Commentary: Rank links invertibility and σ_n; the Gram form detects injectivity.
Key Insight: $\sigma_n > 0$ is exactly full rank for a square matrix. □

Definition 4.2.13. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$. Factorisation $A = X\Sigma_r Y^*$ is a *compact singular value decomposition* if $X \in \mathrm{Mat}(m,r,\mathbb{F})$, $Y \in \mathrm{Mat}(n,r,\mathbb{F})$, $X^*X = Y^*Y = I_r$, and $\Sigma_r = \mathrm{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

Theorem 4.2.14. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$.

(i) If $A = V\Sigma W^*$ is an SVD and X, Y are the first r columns of V, W, then $A = X\Sigma_r Y^*$.
(ii) Conversely, if $X \in \mathrm{Mat}(m,r,\mathbb{F})$ and $Y \in \mathrm{Mat}(n,r,\mathbb{F})$ have orthonormal columns and $\Sigma_r = \mathrm{diag}(\sigma_1, \dots, \sigma_r)$, with $\sigma_1 \geq \cdots \geq \sigma_r > 0$, then $X\Sigma_r Y^*$ has positive singular values $\sigma_1, \dots, \sigma_r$.

Proof.

(i) For $V = [X\ X_0]$ and $W = [Y\ Y_0]$, $A = V\Sigma W^* = X\Sigma_r Y^*$ by block multiplication.

(ii) Set $B := X\Sigma_r Y^*$. Then $B^*B = Y\Sigma_r^2 Y^*$ by $X^*X = I_r$. Complete Y to a unitary $\widetilde{Y} = [Y\ Y_0]$ by Theorem 2.6.13(ii). Then $\widetilde{Y}^*B^*B\widetilde{Y} = \mathrm{diag}(\Sigma_r^2, 0)$, so B has positive singular values $\sigma_1, \dots, \sigma_r$ by Corollary 2.1.4(ii).

Structural Commentary: Block truncation constructs the compact form; unitary completion recovers its Gram spectrum.
Key Insight: $B^*B = Y\Sigma_r^2 Y^*$ retains exactly the positive singular data. □

Theorem 4.2.15. *Column and Kernel Spaces from Compact SVD*
Let $A = X\Sigma_r Y^*$ be a compact SVD. Then:

(i) $\mathrm{col}(A) = \mathrm{col}(X) = (\ker A^*)^\perp$;
(ii) $\ker(A) = \mathrm{col}(Y)^\perp = (\mathrm{col}(A^*))^\perp$.

Proof.

(i) $\mathrm{col}(A) = \mathrm{col}(X)$ by $A = X\Sigma_r Y^*$ and $AY\Sigma_r^{-1} = X$, while $\mathrm{col}(A) = (\ker A^*)^\perp$ by Proposition 2.3.10(iv).

(ii) $(\forall \mathbf{z} \in \mathbb{F}^n)\, A\mathbf{z} = \mathbf{0} \iff Y^*\mathbf{z} = \mathbf{0} \iff \mathbf{z} \perp \mathrm{col}(Y)$ by $X^*X = I_r$ and invertibility of Σ_r. $\ker(A) = (\mathrm{col}(A^*))^\perp$ by Proposition 2.3.10(ii).

Structural Commentary: Invertible middle scaling transfers the image and isolates the right orthogonal kernel.
Key Insight: $X\Sigma_r$ is injective on the right singular coordinates. □

Figure 91 summarizes the active right singular subspace, its image, and the kernel annihilated by A.

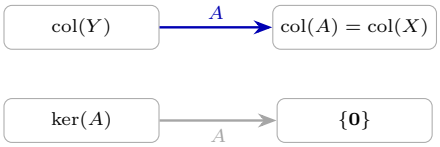

Figure 91: Compact SVD: A maps $\mathrm{col}(Y)$ onto $\mathrm{col}(X)$ and kills $\ker(A) = \mathrm{col}(Y)^\perp$.

Theorem 4.2.16. *Uniqueness of Compact SVD up to Singular Subspaces*
Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$ be compact SVDs with the same Σ_r.
Then $\exists$ unitary matrix $U \in \mathrm{Mat}(r, r, \mathbb{F})$, block diagonal on equal-value blocks of Σ_r, such that $X_1 = X_2U$, $Y_1 = Y_2U$, $U\Sigma_r = \Sigma_r U$.

Proof.
(i) *Coordinate Changes*: The equalities $\mathrm{col}(X_1) = \mathrm{col}(X_2)$ and $\mathrm{col}(Y_1) = \mathrm{col}(Y_2)$ hold by Theorem 4.2.15.
For $U := X_2^*X_1$ and $Z := Y_2^*Y_1$, $X_1 = X_2U$, $Y_1 = Y_2Z$, and U, Z are unitary by the common column spaces and orthonormality.
(ii) *Gram Coupling*: $\Sigma_r = U\Sigma_r Z^*$ by substitution, so $U\Sigma_r = \Sigma_r Z$. $U\Sigma_r^2U^* = \Sigma_r^2$ by this equality and its adjoint.
(iii) *Block Ambiguity*: The matrix U is block diagonal on the equal-value blocks of Σ_r by $U\Sigma_r^2 = \Sigma_r^2U$ and Lemma 5.4.10(ii).
$U\Sigma_r = \Sigma_r U$ by this block structure.
(iv) *Common Change*: $Z = U$ by $U\Sigma_r = \Sigma_r Z = \Sigma_r U$ and invertibility of Σ_r.

Structural Commentary: Common subspaces create two unitary changes; Gram coupling identifies them blockwise.
Key Insight: Commutation with Σ_r^2 confines motion to equal-value blocks. □

Proposition 4.2.17. *Modulus from SVD*
Let $A = V\Sigma W^*$ be an SVD. Then:
(i) $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$;
(ii) $|A^*| = V(\Sigma\Sigma^*)^{1/2}V^*$.

Proof. Set $D_R := (\Sigma^*\Sigma)^{1/2}$ and $D_L := (\Sigma\Sigma^*)^{1/2}$.
(i) The matrix WD_RW^* is positive semidefinite by Theorem 2.1.10(ii) and squares to A^*A. Hence $|A| = WD_RW^*$ by Corollary 2.1.4(i).
(ii) The matrix VD_LV^* is positive semidefinite by Theorem 2.1.10(ii) and squares to AA^*. Hence $|A^*| = VD_LV^*$ by Corollary 2.1.4(i).

Structural Commentary: Unitary conjugation transports each diagonal root to its Gram matrix.
Key Insight: The two moduli express Σ in opposite singular bases. □

Theorem 4.2.18. *Rectangular Polar Decomposition*
Let $A = V\Sigma W^* \in \mathrm{Mat}(m, n, \mathbb{F})$ be an SVD.
(i) If $m \geq n$, let V_n be the first n columns of V and set $U = V_nW^*$. Then $U^*U = I_n$ and $A = U|A|$.
(ii) If $m \leq n$, let W_m be the first m columns of W and set $U = VW_m^*$. Then $UU^* = I_m$ and $A = |A^*|U$.
(iii) If $m = n$, then $U = VW^*$ is unitary and $A = U|A| = |A^*|U$.

Proof.
(i) For $D_R := (\Sigma^*\Sigma)^{1/2}$, $V_nD_R = V\Sigma$; hence $U|A| = A$ by Proposition 4.2.17(i), while $U^*U = WV_n^*V_nW^* = I_n$.
(ii) By (i) with $A^* = W\Sigma^*V^*$, $A^* = U_0|A^*|$ for $U_0 := W_mV^*$ and $U_0^*U_0 = I_m$.
For $U = U_0^* = VW_m^*$, $A = |A^*|U$ and $UU^* = I_m$ by taking adjoints.
(iii) If $m = n$, then $V_n = V$, $W_m = W$, and $U = VW^*$ is unitary; both identities hold by (i)–(ii).

Structural Commentary: Each dimensional regime absorbs the compatible singular bases into a one-sided isometry.
Key Insight: VW^* preserves the diagonal stretch encoded by the modulus. □

Computation Method 4.2.19. *SVD and Polar Factor Computation*
Given $A \in \mathrm{Mat}(m, n, \mathbb{F})$, method constructs full SVD, its compact form when $A \neq 0$, and the dimension-compatible polar factorisation:
(i) Set $r := \mathrm{rank}(A)$ and diagonalise A^*A with a unitary eigenvector matrix $W = [\mathbf{w}_1 \ \cdots \ \mathbf{w}_n]$.
For $r \geq 1$, order its positive eigenvalues as $\sigma_1^2 \geq \cdots \geq \sigma_r^2 > 0$; for $r = 0$, the positive list is empty.
(ii) Set $(\forall j \in J_r)\, \mathbf{v}_j := \sigma_j^{-1}A\mathbf{w}_j$, and complete $(\mathbf{v}_1, \ldots, \mathbf{v}_r)$ to an orthonormal basis of $\mathbb{F}^m$. Let V contain this basis.
(iii) Form the rectangular diagonal Σ with diagonal $\sigma_1, \ldots, \sigma_r, 0, \ldots, 0$. Verify $AW = V\Sigma$, and hence $A = V\Sigma W^*$.
(iv) If $r \geq 1$, set $X := [\mathbf{v}_1 \ \cdots \ \mathbf{v}_r]$, $Y := [\mathbf{w}_1 \ \cdots \ \mathbf{w}_r]$, and $\Sigma_r := \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$. Then $A = X\Sigma_rY^*$ by Theorem 4.2.14(i).
(v) Compute the moduli by Proposition 4.2.17. For $m \geq n$, set $U := V_nW^*$ and obtain $A = U|A|$.
For $m \leq n$, set $U := VW_m^*$ and obtain $A = |A^*|U$ by Theorem 4.2.18.

Theorem 4.2.20. *Uniqueness of the Positive Polar Factor*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$.
(i) If $m \geq n$ and $A = UB$ with $U^*U = I_n$ and $B \succeq 0$, then $B = |A|$. If $\mathrm{rank}(A) = n$, then $U = A|A|^{-1}$ is unique.
(ii) If $m \leq n$ and $A = CU$ with $UU^* = I_m$ and $C \succeq 0$, then $C = |A^*|$. If $\mathrm{rank}(A) = m$, then $U = |A^*|^{-1}A$ is unique.

Proof.

(i) From $A^*A = BU^*UB = B^2$, $B = |A|$ by Corollary 2.1.4(i). If $\operatorname{rank}(A) = n$, then $A^*A \succ 0$ by Proposition 2.1.2(v). Thus $\ker |A| = \{\mathbf{0}\}$ and $U = A\,|A|^{-1}$ by Corollary 3.6.2.

(ii) From $AA^* = CUU^*C = C^2$, $C = |A^*|$ by Corollary 2.1.4(i). If $\operatorname{rank}(A) = m$, then $AA^* \succ 0$ by Proposition 2.1.2(vi). Thus $\ker |A^*| = \{\mathbf{0}\}$ and $U = |A^*|^{-1} A$ by Corollary 3.6.2.

Structural Commentary: Gram squaring fixes each positive factor; full rank makes it invertible.
Key Insight: An invertible modulus uniquely isolates the one-sided isometry. □

Theorem 4.2.21. *Normality and Polar Commutation*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ and suppose $A = U\,|A|$ with U unitary. Then A is normal if and only if $U\,|A| = |A|\,U$.

Proof.

($\Rightarrow$) $|A|^2 = A^*A = AA^* = U\,|A|^2\,U^*$ by normality. $U\,|A|\,U^*$ is positive semidefinite by Theorem 2.1.10(ii), and $(U\,|A|\,U^*)^2 = |A|^2$. Thus $U\,|A|\,U^* = |A|$ by Corollary 2.1.4(i), and $U\,|A| = |A|\,U$.

($\Leftarrow$) $AA^* = U\,|A|^2\,U^* = |A|^2 = A^*A$ by $U\,|A| = |A|\,U$.

Structural Commentary: Normality identifies positive roots; commutation reverses the Gram identity.
Key Insight: Root uniqueness converts Gram equality into polar commutation. □

Perspective 4.2.22. *Polar Decomposition Separates Stretch and Direction*
In square factorisation $A = U\,|A|$, positive factor records metric stretch, unitary factor records direction, paralleling scalar modulus and phase. For normal A, these factors commute by Theorem 4.2.21.
In rectangular case, Theorem 4.2.18 supplies only dimension-compatible one-sided isometry.

Proposition 4.2.23. *Absolute Determinant Comes from the Positive Polar Factor*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ and $T \in \mathcal{L}(\mathcal{V})$. Then $|\det T| = \det \sqrt{T^*T}$.

Proof. Set $P := \sqrt{T^*T}$ and fix an isometry S satisfying $T = SP$ by Theorem 4.2.1. Then $|\det S| = 1$ by Proposition 1.1.19(ii).
Fix an orthonormal eigenbasis of P by Theorem 6.1.12 when $\mathbb{F} = \mathbb{R}$ and Theorem 6.1.11 when $\mathbb{F} = \mathbb{C}$.
Its eigenvalues are nonnegative by Theorem 6.3.6(i)$\Rightarrow$(ii), so $\det P \geq 0$ by Theorem 3.8.2(v).
Thus $|\det T| = |\det S|\,|\det P| = \det P$ by Theorem 3.8.2(vi).
Structural Commentary: Polar multiplicativity separates unit-modulus motion from the nonnegative spectral product.
Key Insight: The positive factor alone carries the absolute determinant. □

5 Norms, Conditioning, Perturbation, and Stability

Matrix norms measure stretch, energy, approximation quality, and sensitivity. Singular values organize norms, low-rank approximation, pseudoinverses, conditioning, and error bounds.

Perturbation theory studies how matrix information changes under small errors. Bauer–Fike, Gersgorin, interlacing, Weyl inequalities, and pseudospectra measure robustness.

5.1 Matrix Norms

Throughout this subsection, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $m, n, k \in \mathbb{Z}_{>0}$ whenever they occur as dimensions.

Definition 5.1.1. *Spectral Norm and Extreme Singular Values*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ and $q := \min(m, n)$.

(i) The *spectral norm* of A is $\|A\|_2 := \sup_{\|\mathbf{x}\|_2 = 1} \|A\mathbf{x}\|_2$.

(ii) If $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$ are the singular values of A, set $(\forall j \in J_q)\,\sigma_j(A) := \sigma_j$ and $\sigma_{\max}(A) := \sigma_1$.

(iii) If $m = n$, set $\sigma_{\min}(A) := \sigma_n$.

Theorem 5.1.2. *Spectral Norm is the Largest Singular Value*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$.

(i) $\|A\|_2 = \sigma_{\max}(A)$.

(ii) If $A = V\Sigma W^*$ is an SVD and $\mathbf{w}_1$ is first column of W, then $\|A\mathbf{w}_1\|_2 = \sigma_1$. Hence supremum in Definition 5.1.1(i) is attained.

Proof. Fix an SVD $A = V\Sigma W^*$ by Theorem 4.2.9, and let $\mathbf{w}_1, \ldots, \mathbf{w}_n$ be the columns of W.

(i) For $\mathbf{x} = \sum_{j=1}^{n} c_j \mathbf{w}_j$ with $\|\mathbf{x}\|_2 = 1$, $\sum_{j=1}^{n} |c_j|^2 = 1$ by Theorem 2.6.8(vii). Thus $\|A\mathbf{x}\|_2^2 = \sum_{j=1}^{q} \sigma_j^2\,|c_j|^2 \leq \sigma_1^2$ by the SVD, so $\|A\|_2 \leq \sigma_1$.

(ii) The SVD has $A\mathbf{w}_1 = \sigma_1 \mathbf{v}_1$ with unit vectors $\mathbf{w}_1, \mathbf{v}_1$, so $\|A\mathbf{w}_1\|_2 = \sigma_1$. Hence $\|A\|_2 \geq \sigma_1$, attained at $\mathbf{w}_1$.

Structural Commentary: The SVD reduces unit-vector stretch to a weighted coefficient sum.
Key Insight: The coefficient bound is sharp at $\mathbf{w}_1$. □

Proposition 5.1.3. *Basic Properties of the Spectral Norm*
Let $A, B \in \operatorname{Mat}(m, n, \mathbb{F})$, $C \in \operatorname{Mat}(n, k, \mathbb{F})$, and $c \in \mathbb{F}$.

(i) $\|A\|_2 \geq 0$;
(ii) $\|A\|_2 = 0$ *if and only if* $A = 0$;
(iii) $\|cA\|_2 = |c|\,\|A\|_2$;
(iv) $\|A + B\|_2 \leq \|A\|_2 + \|B\|_2$;
(v) $\|AC\|_2 \leq \|A\|_2\,\|C\|_2$.

Proof. If $\mathbf{y} \neq \mathbf{0}$, then $\|A\mathbf{y}\|_2 \leq \|A\|_2\,\|\mathbf{y}\|_2$ by Definition 5.1.1(i), after normalization. For $\mathbf{y} = \mathbf{0}$, the same estimate is immediate.

(i) $\|A\|_2 \geq 0$ by Definition 5.1.1(i).
(ii) ($\Rightarrow$) If $\|A\|_2 = 0$, the estimate forces $(\forall \mathbf{y} \in \mathbb{F}^n)\,A\mathbf{y} = \mathbf{0}$, so $A = 0$.
(ii) ($\Leftarrow$) If $A = 0$, then $\|A\|_2 = 0$ by Definition 5.1.1(i).

(iii) $\|cA\|_2 = \sup_{\|\mathbf{x}\|_2=1} \|cA\mathbf{x}\|_2 = |c|\,\|A\|_2$ by Euclidean homogeneity.
(iv) For $\|\mathbf{x}\|_2 = 1$, $\|(A+B)\mathbf{x}\|_2 \le \|A\mathbf{x}\|_2 + \|B\mathbf{x}\|_2 \le \|A\|_2 + \|B\|_2$. Thus $\|A+B\|_2 \le \|A\|_2 + \|B\|_2$ by the unit-sphere supremum.
(v) For $\|\mathbf{x}\|_2 = 1$, $\|AC\mathbf{x}\|_2 \le \|A\|_2\,\|C\mathbf{x}\|_2 \le \|A\|_2\,\|C\|_2$. Thus $\|AC\|_2 \le \|A\|_2\,\|C\|_2$ by the unit-sphere supremum.

Structural Commentary: The scaled-vector estimate feeds definiteness, homogeneity, addition, and composition.
Key Insight: Euclidean control precedes the unit-sphere supremum. □

Definition 5.1.4. *Operator Norm and Induced Metric on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$*
For $A, B \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, define $\|A\|_{\mathrm{op}} := \sup_{\|\mathbf{h}\|_2 \le 1} \|A\mathbf{h}\|_2$ and $d(A,B) := \|A - B\|_{\mathrm{op}}$.
Under standard-basis identification, $\|A\|_{\mathrm{op}} = \|[A]\|_2$ by scaling; hence $(\forall \mathbf{h} \in \mathbb{R}^n)\ \|A\mathbf{h}\|_2 \le \|A\|_{\mathrm{op}}\,\|\mathbf{h}\|_2$.
The norm laws for sums and compatible compositions are Proposition 5.1.3.

Proposition 5.1.5. *Frobenius and Spectral Norm Comparison*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \ge \cdots \ge \sigma_q \ge 0$.

(i) $\|A\|_F^2 = \sum_{j=1}^{q} \sigma_j^2$.
(ii) $\|A\|_2 \le \|A\|_F$.
(iii) If $A \neq 0$, then $\|A\|_2 = \|A\|_F$ if and only if $\mathrm{rank}(A) = 1$.

Proof.
(i) The identity holds by Proposition 4.2.11(ii).
(ii) $\|A\|_2^2 = \sigma_1^2 \le \sum_{j=1}^{q} \sigma_j^2 = \|A\|_F^2$ by Theorem 5.1.2(i) and part (i).
(iii) ($\Rightarrow$) If $A \neq 0$ and $\|A\|_2 = \|A\|_F$, then $\sum_{j=2}^{q} \sigma_j^2 = \|A\|_F^2 - \|A\|_2^2 = 0$ by parts (i)–(ii). Thus $\mathrm{rank}(A) = 1$ by Definition 4.2.8.
(iii) ($\Leftarrow$) If $\mathrm{rank}(A) = 1$, then $\sigma_2 = \cdots = \sigma_q = 0$ by Definition 4.2.8; hence $\|A\|_2 = \|A\|_F$ by parts (i)–(ii).

Structural Commentary: Frobenius aggregates all squared singular values; spectral isolates their maximum.
Key Insight: Equality excludes every singular direction except one. □

Theorem 5.1.6. *Submultiplicativity of Frobenius and Spectral Norms*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(n,k,\mathbb{F})$.

(i) $\|AB\|_F \le \|A\|_2\,\|B\|_F \le \|A\|_F\,\|B\|_F$.
(ii) $\|AB\|_2 \le \|A\|_2\,\|B\|_2$.

Proof.
(i) Write $B = [\mathbf{b}_1\ \cdots\ \mathbf{b}_k]$. Then $(\forall j \in J_k)\ \|A\mathbf{b}_j\|_2 \le \|A\|_2\,\|\mathbf{b}_j\|_2$ by Definition 5.1.1(i) and scaling.
Hence $\|AB\|_F^2 = \sum_{j=1}^{k} \|A\mathbf{b}_j\|_2^2 \le \|A\|_2^2 \sum_{j=1}^{k} \|\mathbf{b}_j\|_2^2 = \|A\|_2^2\,\|B\|_F^2$; $\|AB\|_F \le \|A\|_2\,\|B\|_F \le \|A\|_F\,\|B\|_F$ by Proposition 5.1.5(ii).
(ii) The spectral-norm inequality holds by Proposition 5.1.3(v).

Structural Commentary: Columnwise spectral bounds sum to Frobenius control; norm comparison closes the chain.
Key Insight: Each column of AB inherits the same factor $\|A\|_2$. □

Proposition 5.1.7. *Unitary Invariance of Singular Values*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, and let $U \in \mathrm{Mat}(m,m,\mathbb{F})$ and $V \in \mathrm{Mat}(n,n,\mathbb{F})$ be unitary.

(i) A and UAV have the same singular values.
(ii) $\|UAV\|_2 = \|A\|_2$.
(iii) $\|UAV\|_F = \|A\|_F$.

Proof.
(i) Fix an SVD $A = X\Sigma Y^*$ by Theorem 4.2.9.
Since UX and V^*Y are unitary, $UAV = (UX)\Sigma(V^*Y)^*$ is an SVD with the same diagonal Σ.
(ii) $\|UAV\|_2 = \|A\|_2$ by part (i) and Theorem 5.1.2(i).
(iii) $\|UAV\|_F = \|A\|_F$ by part (i) and Proposition 4.2.11(ii).

Structural Commentary: Multiplying the two SVD bases absorbs the exterior unitary factors.
Key Insight: The diagonal singular factor Σ remains unchanged. □

Counterexample 5.1.8. *Similarity Need Not Preserve Singular Values*
The claim that arbitrary similarity preserves singular values fails in $\mathrm{Mat}(2,2,\mathbb{R})$. Set $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, and $D = \mathrm{diag}(2,1)$.
Then $B = DAD^{-1}$, so A and B are similar. Their Gram matrices are $A^*A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $B^*B = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 5 \end{smallmatrix}\right)$.
Both A and B are invertible, so all their Gram eigenvalues are positive. The Gram matrices have traces 3 and 6.
Hence their spectra, and therefore the singular values of A and B, differ by Proposition 4.2.11(i).
Thus Proposition 5.1.7 cannot be extended from unitary left-right equivalence to arbitrary similarity.

Theorem 5.1.9. *Schur's Inequality and Normality*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity.

(i) $\sum_{j=1}^{n} |\lambda_j|^2 \le \|A\|_F^2$.
(ii) Equality holds in (i) if and only if A is normal.

Proof. Fix a unitary U such that $T := U^*AU = [t_{ij}]$ is upper triangular by Theorem 5.4.14(i) with $\mathcal{F} = \{A\}$.
Relabel the eigenvalues so that $(\forall j \in J_n)\, t_{jj} = \lambda_j$ by Proposition 2.1.6(iii).

(*i*) $\|A\|_F^2 = \|T\|_F^2 = \sum_{i,j=1}^{n} |t_{ij}|^2$ by Propositions 5.1.7(iii) and 4.2.11(ii), with direct matrix multiplication.

Thus $\|A\|_F^2 = \sum_{j=1}^{n} |\lambda_j|^2 + \sum_{i<j} |t_{ij}|^2$ by triangularity, and (i) holds by nonnegativity of the final sum.

(*ii*) ($\Rightarrow$) Equality forces $\sum_{i<j} |t_{ij}|^2 = 0$ by part (i), so T is diagonal and $A = UTU^*$ is normal.

(*ii*) ($\Leftarrow$) If A is normal, then $T^*T = U^*A^*AU = U^*AA^*U = TT^*$, so T is diagonal by Lemma 6.1.10. Equality follows by part (i).

Structural Commentary: Unitary Schur form partitions Frobenius energy into diagonal spectral and strict-upper terms.
Key Insight: Equality is equivalent to vanishing strict-upper energy. □

Definition 5.1.10. *Defect from Normality*

Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$ with algebraic multiplicity. Its *defect from normality* is $\Delta(A) := \|A\|_F^2 - \sum_{j=1}^{n} |\lambda_j|^2$.

Then $\Delta(A) \geq 0$, and $\Delta(A) = 0$ if and only if A is normal, by Theorem 5.1.9(i)–(ii).

Theorem 5.1.11. *Normality Transfers from AB to BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. If A, B, and AB are normal, then BA is normal.

Proof. AB, BA have same eigenvalue multiset by Theorem 1.1.12(i) with equal sizes. Set $M := A^*A = AA^*$, $N := B^*B = BB^*$.
Then $\|BA\|_F^2 = \mathrm{tr}(A^*NA) = \mathrm{tr}(MN)$ and $\|AB\|_F^2 = \mathrm{tr}(B^*MB) = \mathrm{tr}(NM) = \mathrm{tr}(MN)$ by Propositions 4.2.11(ii) and 1.1.4(i).
Hence $\Delta(BA) = \Delta(AB)$ by Definition 5.1.10, $\Delta(AB) = 0$ by Theorem 5.1.9(ii); $\Delta(BA) = 0$, so BA is normal by Theorem 5.1.9(ii).
Structural Commentary: Common spectral data and a cyclic trace identity transfer the Schur defect from AB to BA.
Key Insight: Normal factors force $\|BA\|_F = \|AB\|_F$. □

Theorem 5.1.12. *Eigenvalues Versus Singular Values*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$ ordered so that $|\lambda_1| \geq \cdots \geq |\lambda_n|$.
Let $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$ be its singular values.

(*i*) $|\det A| = \prod_{j=1}^{n} |\lambda_j| = \prod_{j=1}^{n} \sigma_j$;
(*ii*) $\rho(A) \leq \|A\|_2$;
(*iii*) $\sigma_n \leq |\lambda_n|$;
(*iv*) *A is normal if and only if* $(\forall j \in J_n)\ |\lambda_j| = \sigma_j$;
(*v*) *A is positive semidefinite if and only if* $(\forall j \in J_n)\ \lambda_j = \sigma_j$.

Proof.

(*i*) $|\det A| = \prod_{j=1}^{n} |\lambda_j|$ by Proposition 1.1.14(ii).

For an SVD $A = V\Sigma W^*$, $|\det A| = |\det \Sigma| = \prod_{j=1}^{n} \sigma_j$ by Proposition 1.1.19(i) and Theorem 3.8.2(v)–(vi).

(*ii*) If $A\mathbf{x} = \lambda\mathbf{x}$ and $\|\mathbf{x}\|_2 = 1$, then $|\lambda| = \|A\mathbf{x}\|_2 \leq \|A\|_2$ by Definition 5.1.1(i); hence (ii) by maximizing over λ.

(*iii*) If A is singular, $|\lambda_n| = \sigma_n = 0$ by Corollaries 5.1.8 and 4.2.12.
If A is invertible, then $\rho(A^{-1}) = |\lambda_n|^{-1}$ by Corollary 2.1.10. For an SVD $A = V\Sigma W^*$, $A^{-1} = W\Sigma^{-1}V^*$, so $\left\|A^{-1}\right\|_2 = \sigma_n^{-1}$ by Theorem 5.1.2(i), after reversing the diagonal order. Thus $\sigma_n \leq |\lambda_n|$ by part (ii) for A^{-1}.

(*iv*) ($\Rightarrow$) After reordering, fix $A = U\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)U^*$ by Theorem 6.1.11(i)$\Rightarrow$(ii).
Then $(\forall j \in J_n)\, \sigma_j = |\lambda_j|$ by Proposition 5.1.7(i) and Definition 4.2.8.

(*iv*) ($\Leftarrow$) If $(\forall j \in J_n)\ |\lambda_j| = \sigma_j$, then $\sum_{j=1}^{n} |\lambda_j|^2 = \sum_{j=1}^{n} \sigma_j^2 = \|A\|_F^2$ by Proposition 4.2.11(ii). Thus A is normal by Theorem 5.1.9(ii).

(*v*) ($\Rightarrow$) If $A \succeq 0$, then A is Hermitian with nonnegative eigenvalues by Theorem 2.1.3(i)$\Rightarrow$(ii).
Hence A is normal and $(\forall j \in J_n)\, \lambda_j = \sigma_j$ by part (iv).

(*v*) ($\Leftarrow$) If $(\forall j \in J_n)\, \lambda_j = \sigma_j$, then A is normal by part (iv) and its eigenvalues are nonnegative.
By Theorem 6.1.11(i)$\Rightarrow$(ii), $A = U\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)U^*$ for a unitary U, so $A = A^*$ and $A \succeq 0$ by Theorem 2.1.3(ii)$\Rightarrow$(i).

Structural Commentary: Determinants, extremal norms, and Schur energy compare eigenvalues with singular values.
Key Insight: Equality of squared spectral masses detects normality. □

Theorem 5.1.13. *Trace Bound by Singular Values*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$, counted with algebraic multiplicity. Let $\sigma_1, \dots, \sigma_n$ be its singular values.

(*i*) $|\mathrm{tr}\, A| \leq \sum_{j=1}^{n} \sigma_j$;
(*ii*) $\sum_{j=1}^{n} |\lambda_j| \leq \sum_{j=1}^{n} \sigma_j$;
(*iii*) $\sum_{j=1}^{n} |a_{jj}| \leq \sum_{j=1}^{n} \sigma_j$.

Proof. Set $\omega : \mathbb{C} \to \mathbb{C}$ by $\omega(0) := 1$ and $\omega(z) := \overline{z}/\,|z|$ for $z \neq 0$; then $(\forall z \in \mathbb{C})\ |\omega(z)| = 1$ and $\omega(z)z = |z|$.

(i) Fix an SVD $A = V\Sigma W^*$ by Theorem 4.2.9, and write $\mathbf{v}_j, \mathbf{w}_j$ for the columns of V, W.
$\mathrm{tr}\, A = \sum_{j=1}^{n} \sigma_j \mathbf{w}_j^* \mathbf{v}_j$ by Proposition 1.1.3, while $(\forall j \in J_n)\ \left|\mathbf{w}_j^* \mathbf{v}_j\right| \leq 1$ by Theorem 1.1.13(i); $|\mathrm{tr}\, A| \leq \sum_{j=1}^{n} \sigma_j$ by triangle inequality.

(ii) Fix a Schur form $T := U^*AU = [t_{ij}]$ by Theorem 5.4.14(i) with $\mathcal{F} = \{A\}$.
Relabel so that $(\forall j \in J_n)\, t_{jj} = \lambda_j$ by Proposition 2.1.6(iii). For $D_\lambda := \mathrm{diag}(\omega(\lambda_1), \dots, \omega(\lambda_n))$, $\mathrm{tr}(D_\lambda T) = \sum_{j=1}^{n} |\lambda_j|$.
The matrices $D_\lambda T$ and A have the same singular values by Proposition 5.1.7(i), so (ii) holds by part (i) for $D_\lambda T$.

(iii) For $D_A := \operatorname{diag}(\omega(a_{11}), \dots, \omega(a_{nn}))$, $\operatorname{tr}(D_A A) = \sum_{j=1}^{n} |a_{jj}|$.

The matrices $D_A A$ and A have the same singular values by Proposition 5.1.7(i), so (iii) holds by part (i) for $D_A A$.

Structural Commentary: SVD controls trace; diagonal phase rotations transfer that bound to Schur and coordinate diagonals.
Key Insight: Choose phases so every selected diagonal contribution is nonnegative. □

5.2 Low-Rank Approximation and Pseudoinverses

Throughout this subsection, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$; every matrix dimension is positive.

Proposition 5.2.1. *Rank-One SVD Terms and Truncation Error*
Let $A = X\Sigma_r Y^* = \sum_{j=1}^{r} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$ be a compact SVD, with $\sigma_1 \geq \cdots \geq \sigma_r > 0$. For $k \in \{0, \dots, r\}$, set $A_k := \sum_{j=1}^{k} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$, where $A_0 := 0$.

(i) $(\forall i, j \in J_r)\ \left\langle \sigma_i \mathbf{x}_i \mathbf{y}_i^*, \sigma_j \mathbf{x}_j \mathbf{y}_j^* \right\rangle_F = \sigma_i^2 \delta_{ij}$.

(ii) $\|A\|_F^2 = \sum_{j=1}^{r} \sigma_j^2$ *and* $\|A - A_k\|_F^2 = \sum_{j=k+1}^{r} \sigma_j^2$.

(iii) $k < r \implies \|A - A_k\|_2 = \sigma_{k+1}$, *while* $k = r \implies \|A - A_k\|_2 = 0$.

Proof.

(i) $(\forall i, j \in J_r)\ \left\langle \sigma_i \mathbf{x}_i \mathbf{y}_i^*, \sigma_j \mathbf{x}_j \mathbf{y}_j^* \right\rangle_F = \sigma_i \sigma_j (\mathbf{y}_i^* \mathbf{y}_j)(\mathbf{x}_j^* \mathbf{x}_i) = \sigma_i^2 \delta_{ij}$ by orthonormality.

(ii) The first identity holds by Proposition 4.2.11(ii). For $k < r$, the residual is $A - A_k = \sum_{j=k+1}^{r} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$.

Its positive singular values are $\sigma_{k+1}, \dots, \sigma_r$ by Theorem 4.2.14(ii).
The second identity now holds by Proposition 4.2.11(ii); for $k = r$, it is $0 = 0$.

(iii) If $k < r$, then $\|A - A_k\|_2 = \sigma_{k+1}$ by (ii) and Theorem 5.1.2(i); if $k = r$, then $A - A_k = 0$.

Structural Commentary: Frobenius orthogonality separates rank-one terms; compact SVD identifies the residual spectrum.
Key Insight: Truncation leaves exactly the singular-value tail. □

Figure 92 separates the retained and discarded singular-value terms.

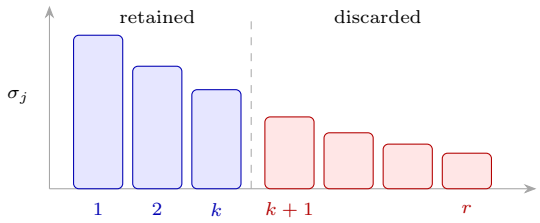

Figure 92: Bars left of the dashed divider form A_k; the right-hand bars satisfy $\|A - A_k\|_F^2 = \sum_{j=k+1}^{r} \sigma_j^2$.

Theorem 5.2.2. *Eckart–Young Low-Rank Approximation*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ have rank r, set $q := \min(m, n)$. Write $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$ for its singular values, fix a full SVD $A = V\Sigma W^*$. Write $\mathbf{v}_j, \mathbf{w}_j$ for the columns of V, W, and set $(\forall j \in J_n \setminus J_q)\ \sigma_j := 0$. For $0 \leq k < q$, set $A_k := \sum_{j=1}^{k} \sigma_j \mathbf{v}_j \mathbf{w}_j^*$.

If $B \in \operatorname{Mat}(m, n, \mathbb{F})$ and $\operatorname{rank}(B) \leq k$, then:

(i) $\|A - A_k\|_2 \leq \|A - B\|_2$;

(ii) $\|A - A_k\|_F \leq \|A - B\|_F$.

Proof. If $k \geq r$, then $A_k = A$ and $\|A - A_k\|_2 = \|A - A_k\|_F = 0$, so both claims hold.
For $k < r$, the first r columns of V, W form a compact SVD of A by Theorem 4.2.14(i). Its rank-k truncation is A_k.

(i) $\dim \ker(B) \geq n - k$ by Corollary 3.3.1(i). $\ker(B) \cap \operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_{k+1}) \neq \{\mathbf{0}\}$ by Corollary 2.3.3(i).

Fix a unit vector $\mathbf{x} = \sum_{i=1}^{k+1} c_i \mathbf{w}_i$ in this intersection. Then $\sum_{i=1}^{k+1} |c_i|^2 = 1$ and $\|A\mathbf{x}\|_2^2 = \sum_{i=1}^{k+1} \sigma_i^2 |c_i|^2 \geq \sigma_{k+1}^2$ by orthonormality.

$\|A - B\|_2 \geq \|(A - B)\mathbf{x}\|_2 = \|A\mathbf{x}\|_2 \geq \sigma_{k+1}$ by $B\mathbf{x} = \mathbf{0}$, spectral-norm definition. $\sigma_{k+1} = \|A - A_k\|_2$ by Proposition 5.2.1(iii).

(ii) $\dim \ker(B) \geq n - k$ by Corollary 3.3.1(i); choose $\mathcal{U} \subseteq \ker(B)$ with $\dim \mathcal{U} = n - k$.

An orthonormal basis $\mathbf{z}_1, \dots, \mathbf{z}_{n-k}$ of $\mathcal{U}$ exists by Corollary 1.3.11. Extend it to $\mathbf{z}_1, \dots, \mathbf{z}_n$ by Corollary 1.3.12.

For $Z := [\mathbf{z}_1 \ \cdots \ \mathbf{z}_n]$ and $P := P_{\mathcal{U}}$, Z is unitary by Theorem 2.6.8(i)–(ii). $\|A - B\|_F^2 = \|(A - B)Z\|_F^2$ by Proposition 5.1.7(iii).

$\|(A - B)Z\|_F^2 \geq \sum_{j=1}^{n-k} \|A\mathbf{z}_j\|_2^2$ by the column formula for $\|\cdot\|_F$ and $(\forall j \in J_{n-k})\ B\mathbf{z}_j = \mathbf{0}$.

Set $(\alpha_i)_{i \in J_n} := (\|P\mathbf{w}_i\|_2^2)_{i \in J_n}$. Then $\sum_{j=1}^{n-k} \|A\mathbf{z}_j\|_2^2 = \sum_{i=1}^{n} \sigma_i^2 \alpha_i$ by orthonormality and Proposition 2.2.3(ix).

$(\forall i \in J_n)\ 0 \leq \alpha_i \leq 1$ by Proposition 2.2.3(viii). $\sum_{i=1}^{n} \alpha_i = n - k$ by Theorem 1.4.7. Thus $\beta := \sum_{i=1}^{k} \alpha_i = \sum_{i=k+1}^{n} (1 - \alpha_i)$.

$\sum_{i=1}^{n} \sigma_i^2 \alpha_i - \sum_{i=k+1}^{n} \sigma_i^2 \geq \sigma_{k+1}^2 (\beta - \beta) = 0$ by singular-value ordering. $\|A - B\|_F^2 \geq \sum_{i=k+1}^{n} \sigma_i^2 = \|A - A_k\|_F^2$ by Proposition 5.2.1(ii).

Structural Commentary: Kernel intersection controls spectral loss; projection weights majorize the Frobenius tail.
Key Insight: Rank k preserves at most k singular directions. □

Computation Method 5.2.3. *Truncated SVD Approximation*
Given $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with compact SVD $A = X\Sigma_r Y^*$ and $k \in \{0,\dots,r-1\}$:

(i) set $\mathcal{R}_k := \{B \in \mathrm{Mat}(m,n,\mathbb{F}) : \mathrm{rank}(B) \le k\}$;

(ii) form $A_k := \sum_{j=1}^{k} \sigma_j \mathbf{x}_j \mathbf{y}_j^* \in \mathcal{R}_k$;

(iii) compute $\|A - A_k\|_F^2 = \sum_{j=k+1}^{r} \sigma_j^2$ and $\|A - A_k\|_2 = \sigma_{k+1}$;

(iv) certify $(\forall \nu \in \{2, F\})\ \|A - A_k\|_\nu = \min_{B \in \mathcal{R}_k} \|A - B\|_\nu$ by (ii) and Theorem 5.2.2.

The same compact SVD data defines a canonical inverse on the nonzero singular subspaces.

Definition 5.2.4. *Moore–Penrose Pseudoinverse*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, define $A^\dagger \in \mathrm{Mat}(n,m,\mathbb{F})$ by:

(i) $A = 0 \Longrightarrow A^\dagger := 0_{n,m}$;

(ii) $\mathrm{rank}(A) = r \ge 1 \Longrightarrow A^\dagger := Y\Sigma_r^{-1}X^*$ for a compact SVD $A = X\Sigma_r Y^*$.

The matrix $A^\dagger$ is the *pseudoinverse* or *Moore–Penrose pseudoinverse* of A.

Proposition 5.2.5. *Pseudoinverse is Well Defined*
The definition of $A^\dagger$ does not depend on the chosen compact SVD of A.

Proof. The assertion is immediate for $A = 0$. Suppose $\mathrm{rank}(A) = r \ge 1$. Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$ be compact SVDs.
Unitary U satisfies $X_1 = X_2U$, $Y_1 = Y_2U$, $U\Sigma_r = \Sigma_r U$ by Theorem 4.2.16. $U\Sigma_r^{-1}U^* = \Sigma_r^{-1}$; $Y_1\Sigma_r^{-1}X_1^* = Y_2U\Sigma_r^{-1}U^*X_2^* = Y_2\Sigma_r^{-1}X_2^*$.
Structural Commentary: Compact-SVD ambiguity commutes with reciprocal scaling and cancels between the singular bases.
Key Insight: Equal-value blocks also commute with Σ_r^{-1}. □

Corollary 5.2.6. *Pseudoinverse from a Full SVD*
Let $A = V\Sigma W^*$ be a full SVD of $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with rank r, and set $D_r := \{(j,j) : j \in J_r\}$.
Define $\Sigma^\dagger \in \mathrm{Mat}(n,m,\mathbb{F})$ by $(\forall j \in J_r)\,(\Sigma^\dagger)_{jj} := \sigma_j^{-1}$. $(\forall (i,j) \in (J_n \times J_m) \setminus D_r)\,(\Sigma^\dagger)_{ij} := 0$. Then $A^\dagger = W\Sigma^\dagger V^*$.

Proof. If $r = 0$, then $A = 0$, $\Sigma^\dagger = 0_{n,m}$, and $W\Sigma^\dagger V^* = A^\dagger = 0_{n,m}$. Suppose $r \ge 1$, and write $V = [X\ X_0]$ and $W = [Y\ Y_0]$.
Then $A = X\Sigma_r Y^*$ and $W\Sigma^\dagger V^* = Y\Sigma_r^{-1}X^*$ by block multiplication. Hence $A^\dagger = W\Sigma^\dagger V^*$ by the defining compact-SVD formula.
Structural Commentary: Full-SVD block multiplication isolates the compact factors and their reciprocal scaling.
Key Insight: Only positive diagonal entries are inverted. □

Theorem 5.2.7. *Basic Properties of the Pseudoinverse*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:
(i) $\mathrm{col}(A^\dagger) = \mathrm{col}(A^*)$ and $\ker(A^\dagger) = \ker(A^*)$;
(ii) $AA^\dagger$ is the orthogonal projection of $\mathbb{F}^m$ onto $\mathrm{col}(A)$;
(iii) $A^\dagger A$ is the orthogonal projection of $\mathbb{F}^n$ onto $\mathrm{col}(A^*)$;
(iv) $AA^\dagger A = A$ and $A^\dagger AA^\dagger = A^\dagger$;
(v) $(A^\dagger)^\dagger = A$;
(vi) if $m = n$ and A is invertible, then $A^\dagger = A^{-1}$.

Proof. If $A = 0$, then (i)–(v) hold directly and the hypothesis of (vi) fails.
Suppose $\mathrm{rank}(A) = r \ge 1$. Fix a compact SVD $A = X\Sigma_r Y^*$; then $A^\dagger = Y\Sigma_r^{-1}X^*$ by the defining formula.

(i) $\mathrm{col}(A^\dagger) = \mathrm{col}(Y)$ by $A^\dagger = Y\Sigma_r^{-1}X^*$ and $A^\dagger X\Sigma_r = Y$. $\ker(A^\dagger) = \ker(X^*)$ by $Y^*A^\dagger = \Sigma_r^{-1}X^*$ and invertibility of Σ_r.
$A^* = Y\Sigma_r X^*$ is a compact SVD. $\mathrm{col}(A^*) = \mathrm{col}(Y)$ and $\ker(A^*) = \ker(X^*)$ by Theorem 4.2.15(i)–(ii).

(ii) $AA^\dagger = XX^* = P_{\mathrm{col}(X)} = P_{\mathrm{col}(A)}$ by Proposition 2.2.3(ix) and Theorem 4.2.15(i).

(iii) $A^\dagger A = YY^* = P_{\mathrm{col}(Y)} = P_{\mathrm{col}(A^*)}$ by Proposition 2.2.3(ix) and Theorem 4.2.15(i) for $A^* = Y\Sigma_r X^*$.

(iv) $AA^\dagger A = XX^*X\Sigma_r Y^* = A$ and $A^\dagger AA^\dagger = YY^*Y\Sigma_r^{-1}X^* = A^\dagger$ by $X^*X = Y^*Y = I_r$.

(v) Let $R \in \mathrm{Mat}(r,r,\mathbb{F})$ be the reversal permutation matrix, so $(\forall j \in J_r)\,R\mathbf{e}_j = \mathbf{e}_{r+1-j}$.
Then $A^\dagger = (YR)(R^*\Sigma_r^{-1}R)(XR)^*$ is a compact SVD with decreasing positive diagonal.
Hence $(A^\dagger)^\dagger = (XR)(R^*\Sigma_r R)(YR)^* = X\Sigma_r Y^* = A$ by the defining compact-SVD formula.

(vi) If A is square and invertible, then A^* is invertible by Proposition 2.3.9(i).
$\mathrm{col}(A) = \mathrm{col}(A^*) = \mathbb{F}^n$ by Corollary 3.6.2. $AA^\dagger = A^\dagger A = I_n$ by (ii)–(iii), so $A^\dagger = A^{-1}$.

Structural Commentary: Compact-SVD multiplication identifies the two projectors; reversed reciprocal scaling recovers A.
Key Insight: Positive singular coordinates invert; null coordinates remain zero. □

Figure 93 identifies the active inverse pair and the annihilated kernels.

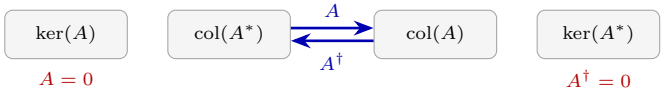

Figure 93: The arrows invert the active singular subspaces; the outer boxes mark $\ker(A)$ and $\ker(A^*)$.

Theorem 5.2.8. *Penrose Characterization of the Pseudoinverse*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $B \in \mathrm{Mat}(n,m,\mathbb{F})$. Set $\mathsf{P}_A(B) := [ABA = A \wedge BAB = B \wedge (AB)^* = AB \wedge (BA)^* = BA]$; $B = A^\dagger \iff \mathsf{P}_A(B)$.

Proof.
(⇒) Suppose $B = A^\dagger$. The first two conjuncts hold by Theorem 5.2.7(iv).
Matrices AB and BA are orthogonal projections by Theorem 5.2.7(ii)–(iii). $(AB)^* = AB$ and $(BA)^* = BA$ by Theorem 2.5.6.

($\Leftarrow$) Suppose $\mathsf{P}_A(B)$, and set $\mathcal{C} := \mathrm{col}(A)$ and $\mathcal{D} := \mathrm{col}(A^*)$. Then $(AB)^2 = ABAB = AB$ and $(AB)^* = AB$.
The relations $\mathrm{range}(AB) \subseteq \mathcal{C}$ and $(\forall \mathbf{x} \in \mathbb{F}^n)\, AB(A\mathbf{x}) = A\mathbf{x}$ establish $\mathrm{range}(AB) = \mathcal{C}$.
$AB = P_{\mathcal{C}}$ by Theorem 2.5.6. For BA, $(BA)^2 = BABA = BA$ and $(BA)^* = BA$.
$\ker(BA) \subseteq \ker(A)$ by $(\forall \mathbf{x} \in \mathbb{F}^n)\, [BA\mathbf{x} = \mathbf{0} \Rightarrow A\mathbf{x} = ABA\mathbf{x} = \mathbf{0}]$.
$\ker(A) \subseteq \ker(BA)$ by $(\forall \mathbf{x} \in \mathbb{F}^n)\, [A\mathbf{x} = \mathbf{0} \Rightarrow BA\mathbf{x} = \mathbf{0}]$; hence $\ker(BA) = \ker(A)$.
$\mathrm{range}(BA) = (\ker(BA))^\perp = (\ker A)^\perp = \mathcal{D}$ by Proposition 2.3.10(iii)–(iv). $BA = P_{\mathcal{D}}$ by Theorem 2.5.6.
The identities $AB = AA^\dagger$ and $BA = A^\dagger A$ hold by Theorem 5.2.7(ii)–(iii).
$B = BAB = B(AA^\dagger) = (BA)A^\dagger = (A^\dagger A)A^\dagger = A^\dagger$ by $\mathsf{P}_A(B)$ and Theorem 5.2.7(iv).

Structural Commentary: Self-adjoint idempotence identifies both range projections; their product identities collapse B.
Key Insight: $AB = AA^\dagger$ and $BA = A^\dagger A$ force $B = A^\dagger$. □

Theorem 5.2.9. *Pseudoinverse and Minimum-Norm Least Squares*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$, set $\mathbf{x}_0 := A^\dagger \mathbf{y}$.
(i) $\mathbf{x}_0 \in \mathrm{LS}(A, \mathbf{y})$ *and* $\operatorname{arg\,min}_{\mathbf{x} \in \mathrm{LS}(A,\mathbf{y})} \|\mathbf{x}\|_2 = \{\mathbf{x}_0\}$.
(ii) If $\mathbf{y} \in \mathrm{col}(A)$, *then* $A\mathbf{x}_0 = \mathbf{y}$ *and* $\operatorname{arg\,min}_{\mathbf{x} \in \mathbb{F}^n :\, A\mathbf{x} = \mathbf{y}} \|\mathbf{x}\|_2 = \{\mathbf{x}_0\}$.

Proof. Set $P := AA^\dagger = P_{\mathrm{col}(A)}$ by Theorem 5.2.7(ii); then $A\mathbf{x}_0 = P\mathbf{y}$.
$\mathbf{x}_0 \in \mathrm{col}(A^\dagger) = \mathrm{col}(A^*) = (\ker A)^\perp$ by Theorem 5.2.7(i) and Proposition 2.3.10(iii).
(i) $\mathbf{x}_0 \in \mathrm{LS}(A, \mathbf{y})$ by $A\mathbf{x}_0 = P\mathbf{y}$ and Theorem 2.2.6. $P_{(\ker A)^\perp}\mathbf{x}_0 = \mathbf{x}_0$ by Proposition 2.2.3(ii).
The minimum-norm identity holds by Proposition 2.4.5(iii).
(ii) If $\mathbf{y} \in \mathrm{col}(A)$, then $A\mathbf{x}_0 = P\mathbf{y} = \mathbf{y}$ by Proposition 2.2.3(ii).
Thus $\mathrm{LS}(A, \mathbf{y}) = \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{y}\}$ by the zero minimum residual, and the claim holds by (i).

Structural Commentary: $AA^\dagger$ fits the data; $\mathrm{col}(A^\dagger) = (\ker A)^\perp$ selects the shortest preimage.
Key Insight: The pseudoinverse removes every kernel component. □

Computation Method 5.2.10. *Minimum-Norm Least Squares by Pseudoinverse*
Given $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$, compute the minimum-norm least-squares solution as follows:
(i) if $A = 0$, set $A^\dagger = 0_{n,m}$;
(ii) if $\mathrm{rank}(A) = r \geq 1$, compute a compact SVD $A = X\Sigma_r Y^*$ and form $A^\dagger = Y\Sigma_r^{-1}X^*$;
(iii) set $\mathbf{x}_0 := A^\dagger \mathbf{y}$;
(iv) certify $\operatorname{arg\,min}_{\mathbf{x} \in \mathrm{LS}(A,\mathbf{y})} \|\mathbf{x}\|_2 = \{\mathbf{x}_0\}$ by Theorem 5.2.9(i);
(v) if $\mathbf{y} \in \mathrm{col}(A)$, report $A\mathbf{x}_0 = \mathbf{y}$ by Theorem 5.2.9(ii).

5.3 Conditioning

Throughout this subsection, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and all matrix dimensions are positive integers.

Definition 5.3.1. *Spectral Condition Number*
If $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible, its *spectral condition number* is $\kappa_2(A) := \|A\|_2 \left\|A^{-1}\right\|_2$. If A is singular, $\kappa_2(A)$ is undefined.

Proposition 5.3.2. *Properties of the Spectral Condition Number*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be invertible and let $c \in \mathbb{F} \setminus \{0\}$. Let $U, V \in \mathrm{Mat}(n, n, \mathbb{F})$ be unitary.
(i) $\kappa_2(AB) \leq \kappa_2(A)\kappa_2(B)$;
(ii) $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} \geq 1$, *with equality if and only if A is a nonzero scalar multiple of a unitary matrix;*
(iii) $\kappa_2(cA) = \kappa_2(A)$;
(iv) $\kappa_2(A) = \kappa_2(A^*) = \kappa_2(A^{-1})$; *over $\mathbb{C}$, these also equal $\kappa_2(\overline{A})$ and $\kappa_2(A^t)$;*
(v) $\kappa_2(A^*A) = \kappa_2(A)^2$;
(vi) $\kappa_2(UAV) = \kappa_2(A)$.

Proof.
(i) By Proposition 5.1.3(v), $\kappa_2(AB) = \|AB\|_2 \left\|B^{-1}A^{-1}\right\|_2 \leq \kappa_2(A)\kappa_2(B)$.
(ii) ($\Rightarrow$) Fix SVD $A = X\Sigma Y^*$ by Theorem 4.2.9. Since A invertible, $\sigma_n(A) > 0$ by Corollary 4.2.12(i)$\Rightarrow$(iii), $A^{-1} = Y\Sigma^{-1}X^*$.
Thus $\kappa_2(A) = \frac{\sigma_1(A)}{\sigma_n(A)} \geq 1$ by Theorem 5.1.2(i). For $s := \sigma_1(A) > 0$, equality forces $\Sigma = sI_n$, so $A = sXY^*$, a nonzero scalar multiple of a unitary matrix by Theorem 2.6.8(iv) and Proposition 2.6.12(i).
(ii) ($\Leftarrow$) If $A = dQ$ with $d \neq 0$ and Q unitary, then $Q^*Q = I_n$, so the singular values of Q are all 1.
Those of A are all $|d|$ by Proposition 4.2.11(iv); hence the quotient in part (ii) is 1.
(iii) Since $(cA)^{-1} = c^{-1}A^{-1}$, $\kappa_2(cA) = |c| \|A\|_2 |c|^{-1} \left\|A^{-1}\right\|_2 = \kappa_2(A)$ by Proposition 5.1.3(iii).
(iv) The singular values are unchanged under adjoint and, over $\mathbb{C}$, under transpose and conjugation by Proposition 4.2.11(iii).
Inversion reverses and reciprocates them by part (ii), so every stated condition number equals $\kappa_2(A)$ by part (ii).
(v) Matrix A^*A positive definite by Corollary 4.2.12(i)$\Leftrightarrow$(iv). Its eigenvalues are $\sigma_1(A)^2, \ldots, \sigma_n(A)^2$ by Proposition 4.2.11(i).
These are also its singular values by Theorems 6.1.12 and 6.1.11, as appropriate, and Proposition 5.1.7(i).
Hence $\kappa_2(A^*A) = \frac{\sigma_1(A)^2}{\sigma_n(A)^2} = \kappa_2(A)^2$ by part (ii).
(vi) The equality $\kappa_2(UAV) = \kappa_2(A)$ holds by Proposition 5.1.7(i) and part (ii).

Structural Commentary: Norm laws and singular-value transformations transfer directly to κ_2.
Key Insight: Only the ratio $\frac{\sigma_1(A)}{\sigma_n(A)}$ matters. □

Corollary 5.3.3. *Positive Definite Condition Number by Eigenvalues*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ satisfy $A \succ 0$ and have eigenvalues $0 < \lambda_1 \leq \cdots \leq \lambda_n$.
(i) $\|A\|_2 = \lambda_n$;
(ii) $\left\|A^{-1}\right\|_2 = \lambda_1^{-1}$;

(iii) $\kappa_2(A) = \frac{\lambda_n}{\lambda_1}$.

Proof.

(*i*) The matrix A is self-adjoint with positive eigenvalues by Theorem 2.1.8(i)⇒(ii).
Singular values are $\lambda_n, \dots, \lambda_1$ by Theorems 6.1.12, 6.1.11, as appropriate, Proposition 5.1.7(i); $\|A\|_2 = \lambda_n$ by Theorem 5.1.2(i).

(*ii*) The same diagonalization has diagonal entries $\lambda_1^{-1}, \dots, \lambda_n^{-1}$ for A^{-1}, so these are its singular values by Proposition 5.1.7(i).
Thus $\left\|A^{-1}\right\|_2 = \lambda_1^{-1}$ by Theorem 5.1.2(i).

(*iii*) The identity $\kappa_2(A) = \frac{\lambda_n}{\lambda_1}$ follows by parts (i)–(ii).

Structural Commentary: Diagonalization assigns the two operator norms to opposite endpoint eigenvalues.
Key Insight: Positive definiteness removes absolute values and makes λ_1 nonzero. □

Perspective 5.3.4. *Determinant is Not Conditioning*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ invertible, $c \in \mathbb{F} \setminus \{0\}$. Then $\kappa_2(cA) = \kappa_2(A)$ by Proposition 5.3.2(iii); $\det(cA) = c^n \det(A)$ by Theorem 3.8.2(ii). Thus determinant records volume scaling, not sensitivity under uniform rescaling.

Theorem 5.3.5. *Condition Number and Linear-System Error*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be invertible, $\mathbf{x}, \Delta\mathbf{x}, \Delta\mathbf{y} \in \mathbb{F}^n$, $\mathbf{y} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$. If $A\mathbf{x} = \mathbf{y}$ and $A\Delta\mathbf{x} = \Delta\mathbf{y}$, then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta\mathbf{y}\|_2}{\|\mathbf{y}\|_2}$.

Proof. Since A is invertible and $\mathbf{y} \neq \mathbf{0}$, $\mathbf{x} \neq \mathbf{0}$.
By defining norm bound, $\|\mathbf{y}\|_2 \le \|A\|_2 \|\mathbf{x}\|_2$, $\|\Delta\mathbf{x}\|_2 = \left\|A^{-1}\Delta\mathbf{y}\right\|_2 \le \left\|A^{-1}\right\|_2 \|\Delta\mathbf{y}\|_2$. Claim follows by division and multiplication.
Structural Commentary: The equations bound $\|\mathbf{x}\|_2$ through A and $\|\Delta\mathbf{x}\|_2$ through A^{-1}.
Key Insight: The opposing operator norms multiply to $\kappa_2(A)$. □

Theorem 5.3.6. *Condition Number and Matrix Perturbation*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be invertible, let $\Delta A \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $\mathbf{x}, \Delta\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$.
Suppose $A\mathbf{x} = \mathbf{y}$, $(A + \Delta A)(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{y}$, and $\mathbf{x} + \Delta\mathbf{x} \neq \mathbf{0}$. Then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}+\Delta\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2}$.

Proof. Set $\mathbf{z} := \mathbf{x} + \Delta\mathbf{x} \neq \mathbf{0}$. The system equations imply $\Delta\mathbf{x} = -A^{-1}\Delta A\mathbf{z}$.
Hence $\|\Delta\mathbf{x}\|_2 \le \left\|A^{-1}\right\|_2 \|\Delta A\|_2 \|\mathbf{z}\|_2$ by the defining norm bound and Proposition 5.1.3(v).
Since $\|A\|_2 > 0$ by Proposition 5.1.3(ii), the result follows by division by $\|\mathbf{z}\|_2$ and the defining formula for $\kappa_2(A)$.
Structural Commentary: Subtraction isolates $\Delta\mathbf{x}$ as $-A^{-1}\Delta A$ acting on the perturbed solution.
Key Insight: Matrix perturbation enters through the product $A^{-1}\Delta A$. □

Figure 94 depicts the ellipse determined by the extreme singular directions.

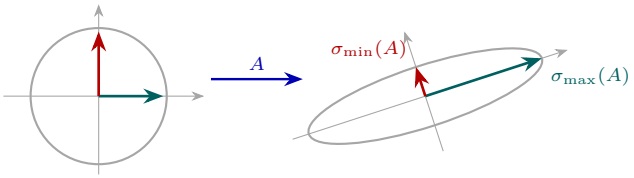

Figure 94: The unit circle maps to an ellipse with extreme-semiaxis ratio $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$.

5.4 Perturbation Bounds

Throughout this subsection, $n \in \mathbb{Z}_{>0}$.

Lemma 5.4.1. *Small Spectral-Norm Perturbations of the Identity*
If $E \in \mathrm{Mat}(n, n, \mathbb{C})$ and $\|E\|_2 < 1$, then $I_n + E$ is invertible.

Proof. $\rho(E) \le \|E\|_2 < 1$ by Theorem 5.1.12(ii), so $-1 \notin \mathrm{spec}(E)$.
For $p(z) = 1 + z$, $\mathrm{spec}(I_n + E) = \{1 + \lambda : \lambda \in \mathrm{spec}(E)\}$ by Corollary 3.4.2; $0 \notin \mathrm{spec}(I_n + E)$; $I_n + E$ invertible by Corollary 5.1.8.
Structural Commentary: The spectral-radius bound excludes -1, and spectral mapping transfers the exclusion to 0.
Key Insight: $\|E\|_2 < 1$ keeps -1 outside $\mathrm{spec}(E)$. □

Theorem 5.4.2. *Bauer–Fike Eigenvalue Perturbation Bound*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonalizable. Fix $S \in \mathrm{Mat}(n, n, \mathbb{C})^\times$ and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ satisfying $A = S\Lambda S^{-1}$.
If $\Delta A \in \mathrm{Mat}(n, n, \mathbb{C})$ and $\mu \in \mathrm{spec}(A + \Delta A)$, then $\min_{j \in J_n} |\mu - \lambda_j| \le \kappa_2(S) \|\Delta A\|_2$.

Proof.

(i) *Separation*: If $(\exists\, j \in J_n)\, \mu = \lambda_j$, bound is immediate. Otherwise set $\delta := \min_{j \in J_n} |\mu - \lambda_j| > 0$, so $\Lambda - \mu I_n$ is invertible.

(ii) *Shift Factorization*: Since $\mu \in \mathrm{spec}(A + \Delta A)$, the matrix $A + \Delta A - \mu I_n$ is singular. Set $F := (\Lambda - \mu I_n)^{-1} S^{-1} \Delta A S$.
The identity $S^{-1}(A + \Delta A - \mu I_n)S = (\Lambda - \mu I_n)(I_n + F)$ holds. Thus $I_n + F$ is singular because S and $\Lambda - \mu I_n$ are invertible.

(iii) *Norm Obstruction*: The inequality $\|F\|_2 \ge 1$ holds by the contrapositive of Lemma 5.4.1.

(iv) *Resolvent Estimate*: Diagonal resolvent has singular values $(|\lambda_j - \mu|^{-1})_{j \in J_n}$. Thus $\left\|(\Lambda - \mu I_n)^{-1}\right\|_2 = \delta^{-1}$ by Theorem 5.1.2(i).
By part (iii) and Proposition 5.1.3(v), $1 \le \|F\|_2 \le \delta^{-1} \left\|S^{-1}\right\|_2 \|\Delta A\|_2 \|S\|_2 = \delta^{-1}\kappa_2(S) \|\Delta A\|_2$. Multiply by δ for bound.

Structural Commentary: Shift factorization converts a perturbed eigenvalue into failure of the small-norm invertibility lemma.
Key Insight: Singularity forces $\left\|(\Lambda - \mu I_n)^{-1} S^{-1} \Delta A S\right\|_2 \ge 1$. □

Computation Method 5.4.3. *Bauer–Fike Perturbation Method*
For diagonalizable $A \in \mathrm{Mat}(n, n, \mathbb{C})$ and $\Delta A \in \mathrm{Mat}(n, n, \mathbb{C})$, construct a spectral enclosure:

(i) choose a diagonalization $A = S\Lambda S^{-1}$ with S invertible and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$;

(ii) compute $\|\Delta A\|_2$;
(iii) compute $\kappa_2(S)$ and set $r := \kappa_2(S)\,\|\Delta A\|_2$;
(iv) output $\operatorname{spec}(A+\Delta A) \subseteq \bigcup_{j\in J_n} \{z \in \mathbb{C} : |z-\lambda_j| \le r\}$ by Theorem 5.4.2.

Definition 5.4.4. *Gersgorin Disks*
Let $A = [a_{ij}] \in \operatorname{Mat}(n,n,\mathbb{C})$.
(i) The *deleted absolute row sums* are defined by $(\forall\, k \in J_n)\; R_k^{\text{off}}(A) := \sum_{j\in J_n\setminus\{k\}} |a_{kj}|$.
(ii) The *absolute row sums* are defined by $(\forall\, k \in J_n)\; R_k(A) := \sum_{j\in J_n} |a_{kj}|$.
(iii) The *maximum absolute row sum* is $R_{\max}(A) := \max_{k\in J_n} R_k(A)$.
(iv) The *Gersgorin disks* are $(\forall\, k \in J_n)\; \operatorname{Ger}_k(A) := \{z \in \mathbb{C} : |z-a_{kk}| \le R_k^{\text{off}}(A)\}$. Their union is $\operatorname{Ger}(A) := \bigcup_{k\in J_n} \operatorname{Ger}_k(A)$.

Theorem 5.4.5. *Gersgorin Disk Theorem*
Let $A = [a_{ij}] \in \operatorname{Mat}(n,n,\mathbb{C})$. *Then* $\operatorname{spec}(A) \subseteq \operatorname{Ger}(A) = \bigcup_{k\in J_n} \operatorname{Ger}_k(A)$.

Proof. Fix $\lambda \in \operatorname{spec}(A)$ and $\mathbf{0} \neq \mathbf{x} = [x_1 \ \cdots \ x_n]^t \in E_\lambda(A)$. Choose $k \in J_n$ with $|x_k| = \max_{j\in J_n} |x_j| > 0$.
The kth row of $A\mathbf{x} = \lambda\mathbf{x}$ is $(\lambda - a_{kk})x_k = \sum_{j\in J_n\setminus\{k\}} a_{kj}x_j$.
Therefore $|\lambda - a_{kk}| \le \sum_{j\in J_n\setminus\{k\}} |a_{kj}| \left|\frac{x_j}{x_k}\right| \le R_k^{\text{off}}(A)$ by the triangle inequality and maximality of $|x_k|$; $\lambda \in \operatorname{Ger}_k(A) \subseteq \operatorname{Ger}(A)$.
Structural Commentary: The maximal eigenvector component converts one row identity into the deleted-row-sum bound.
Key Insight: $(\forall\, j \in J_n)\; \left|\frac{x_j}{x_k}\right| \le 1$ closes the disk estimate. □

Corollary 5.4.6. *Row-Column Gersgorin Refinement*
Let $A = [a_{ij}] \in \operatorname{Mat}(n,n,\mathbb{C})$.
(i) $\operatorname{spec}(A) \subseteq \operatorname{Ger}(A) \cap \operatorname{Ger}(A^t)$.
(ii) $\operatorname{spec}(A) \subseteq \{z \in \mathbb{C} : |z| \le \min(R_{\max}(A), R_{\max}(A^t))\}$.

Proof.
(i) The inclusion $\operatorname{spec}(A) \subseteq \operatorname{Ger}(A)$ holds by Theorem 5.4.5.
Since $\operatorname{spec}(A) = \operatorname{spec}(A^t)$ by Proposition 1.2.6(ii), $\operatorname{spec}(A) \subseteq \operatorname{Ger}(A^t)$ by Theorem 5.4.5; intersection is (i).
(ii) For $B = [b_{ij}] \in \{A, A^t\}$, $k \in J_n$, and $z \in \operatorname{Ger}_k(B)$, $|z| \le |b_{kk}| + R_k^{\text{off}}(B) = R_k(B) \le R_{\max}(B)$ by the triangle inequality.
Thus $\operatorname{Ger}(B) \subseteq \{z \in \mathbb{C} : |z| \le R_{\max}(B)\}$. The enclosure in (ii) follows from the two radial bounds by part (i).
Structural Commentary: Gersgorin applied to A and A^t supplies intersecting row and column enclosures.
Key Insight: The triangle inequality places each disk inside its row-sum radius. □

Computation Method 5.4.7. *Gersgorin Localisation Method*
For $A = [a_{ij}] \in \operatorname{Mat}(n,n,\mathbb{C})$, construct a row-column spectral enclosure as follows:
(i) Compute $(\forall\, k \in J_n)\; R_k^{\text{off}}(A) = \sum_{j\in J_n\setminus\{k\}} |a_{kj}|$. Form $\operatorname{Ger}(A) = \bigcup_{k\in J_n} \operatorname{Ger}_k(A)$.
(ii) Record the certified row enclosure $\operatorname{spec}(A) \subseteq \operatorname{Ger}(A)$ by Theorem 5.4.5.
(iii) Compute the deleted row sums of A^t, which are the deleted column sums of A, and form $\operatorname{Ger}(A) \cap \operatorname{Ger}(A^t)$.
The inclusion $\operatorname{spec}(A) \subseteq \operatorname{Ger}(A) \cap \operatorname{Ger}(A^t)$ holds by Corollary 5.4.6(i).
(iv) Interpret the output as an intersection of two unions; do not replace it by $\bigcup_{k\in J_n} (\operatorname{Ger}_k(A) \cap \operatorname{Ger}_k(A^t))$.

Figure 95 compares the row and column disks whose unions form the enclosure in Corollary 5.4.6(i).

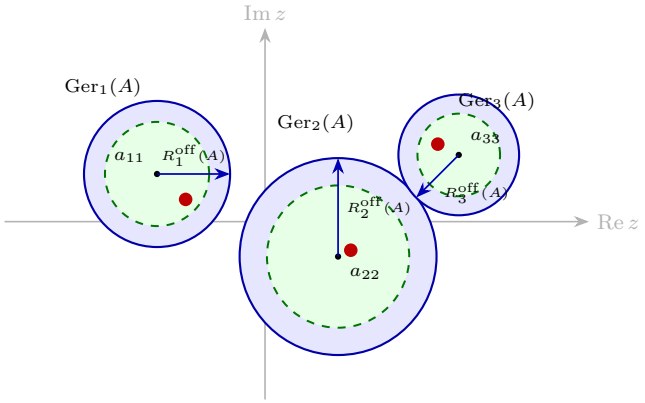

Figure 95: Solid blue row disks and dashed green column disks form $\operatorname{Ger}(A) \cap \operatorname{Ger}(A^t)$; red markers show candidate eigenvalues.

Definition 5.4.8. *Closed Spectral-Norm Pseudospectrum*
Let $\mathcal{M} := \operatorname{Mat}(n,n,\mathbb{C})$, $A \in \mathcal{M}$, and $\varepsilon \in \mathbb{R}_{\ge 0}$.
The *closed spectral-norm ε-pseudospectrum* of A is $\{z \in \mathbb{C} : (\exists\, \Delta A \in \mathcal{M})\; [\|\Delta A\|_2 \le \varepsilon \wedge z \in \operatorname{spec}(A+\Delta A)]\}$.

Let $A, \Delta A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $\varepsilon \in \mathbb{R}_{\geq 0}$, and fix a diagonalization $A = S\Lambda S^{-1}$.
If $\|\Delta A\|_2 \leq \varepsilon$, then the Bauer–Fike disk radius is at most $\kappa_2(S)\varepsilon$ by Theorem 5.4.2.
The inclusion $\mathrm{spec}(A+\Delta A) \subseteq \mathrm{Ger}(A+\Delta A)$ holds by Theorem 5.4.5.
These bounds distinguish spectral location from perturbation sensitivity.

5.5 Diagonal Dominance and Definiteness

Throughout this subsection, $n \in \mathbb{Z}_{>0}$.

Definition 5.5.1. *Diagonal Dominance*
Let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{C})$.

(i) The matrix A is *diagonally dominant* iff $(\forall\, k \in J_n)\ |a_{kk}| \geq R_k^{\mathrm{off}}(A)$.
(ii) The matrix A is *strictly diagonally dominant* iff $(\forall\, k \in J_n)\ |a_{kk}| > R_k^{\mathrm{off}}(A)$.

Corollary 5.5.2. *Strict Diagonal Dominance Implies Invertibility*
Let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{C})$.

(i) If A is strictly diagonally dominant, then A is invertible.
(ii) If $\mathrm{spec}(A) \subseteq \mathbb{R}$, $(\forall\, k \in J_n)\, a_{kk} \in \mathbb{R}_{\geq 0}$, and A is diagonally dominant, then $\mathrm{spec}(A) \subseteq [0,\infty)$.
(iii) If $\mathrm{spec}(A) \subseteq \mathbb{R}$, $(\forall\, k \in J_n)\, a_{kk} \in \mathbb{R}_{\geq 0}$, and A is strictly diagonally dominant, then $\mathrm{spec}(A) \subseteq (0,\infty)$.

Proof.

(i) $0 \notin \mathrm{Ger}(A)$ by Definition 5.4.4 and strict diagonal dominance.
The relation $0 \notin \mathrm{spec}(A)$ holds by Theorem 5.4.5, and A is invertible by Corollary 5.1.8.
(ii) $(\forall\, k \in J_n)(\forall\, z \in \mathrm{Ger}_k(A))\ \mathrm{Re}(z) \geq a_{kk} - |z - a_{kk}| \geq a_{kk} - R_k^{\mathrm{off}}(A) \geq 0$ by Definition 5.4.4 and diagonal dominance.
The inclusion $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathbb{R} \subseteq [0,\infty)$ holds by the preceding bound, Theorem 5.4.5, and the real-spectrum hypothesis.
(iii) $(\forall\, k \in J_n)(\forall\, z \in \mathrm{Ger}_k(A))\ \mathrm{Re}(z) \geq a_{kk} - R_k^{\mathrm{off}}(A) > 0$ by Definition 5.4.4 and strict diagonal dominance.
The inclusion $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathbb{R} \subseteq (0,\infty)$ holds by the preceding bound, Theorem 5.4.5, and the real-spectrum hypothesis.

Structural Commentary: Disk exclusion handles (i); closed and open half-plane bounds handle (ii)–(iii).
Key Insight: Each Gersgorin radius is controlled by its nonnegative real center. □

Theorem 5.5.3. *Weak Diagonal Dominance with One Strict Row*
Let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{C})$ be diagonally dominant and satisfy $(\forall\, i,j \in J_n)\,[i \neq j \Longrightarrow a_{ij} \neq 0]$.
If $(\exists\, k \in J_n)\ |a_{kk}| > R_k^{\mathrm{off}}(A)$, then A is invertible.

Proof. Suppose $\mathbf{0} \neq \mathbf{x} \in \ker(A)$, $M := \|\mathbf{x}\|_\infty$. Choose $p \in J_n$ satisfying $|x_p| = M > 0$ by Proposition 1.1.21(iii).
$|a_{pp}|\, M = \left| \sum_{j \in J_n \setminus \{p\}} a_{pj} x_j \right| \leq \sum_{j \in J_n \setminus \{p\}} |a_{pj}|\, |x_j| \leq R_p^{\mathrm{off}}(A) M$ by $A\mathbf{x} = \mathbf{0}$, the triangle inequality, and maximality of $|x_p|$.
$R_p^{\mathrm{off}}(A) M \leq |a_{pp}|\, M$ by diagonal dominance; all preceding inequalities are equalities by their common endpoints.
The relation $(\forall\, j \in J_n)\ |x_j| = M$ holds by equality in the weighted bound, $|x_p| = M$, and the nonzero off-diagonal hypothesis.
Fix $k \in J_n$ satisfying $|a_{kk}| > R_k^{\mathrm{off}}(A)$ by the strict-row hypothesis.
$|a_{kk}|\, M = \left| \sum_{j \in J_n \setminus \{k\}} a_{kj} x_j \right| \leq R_k^{\mathrm{off}}(A) M$ by $A\mathbf{x} = \mathbf{0}$ and the preceding modulus identity.
$R_k^{\mathrm{off}}(A) M < |a_{kk}|\, M$ by strict dominance and $M > 0$, a contradiction.
The equality $\ker(A) = \{\mathbf{0}\}$ holds by contradiction, and A is invertible by Corollary 3.6.3(vi)⇒(i).
Structural Commentary: A maximal kernel coordinate propagates equal moduli before the strict row closes the contradiction.
Key Insight: Positive off-diagonal weights propagate one maximal modulus to every coordinate. □

Corollary 5.5.4. *Diagonal Dominance and Positive Semidefiniteness*
Let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian.
Suppose $(\forall\, k \in J_n)\, a_{kk} \in \mathbb{R}_{\geq 0}$.

(i) If A is diagonally dominant, then A is positive semidefinite.
(ii) If A is diagonally dominant and invertible, then A is positive definite.
(iii) If A is strictly diagonally dominant, then A is positive definite.

Proof. The induced $T_A \in \mathcal{L}(\mathbb{C}^n)$ is self-adjoint by Proposition 3.1.8, Theorems 2.3.7 and 3.4.5(i), and $A = A^*$.
The relation $\mathrm{spec}(A) = \mathrm{spec}(T_A) \subseteq \mathbb{R}$ holds by Definitions 3.1.7, 5.1.2, and 5.1.3, and Theorem 6.1.5.

(i) $\mathrm{spec}(A) \subseteq [0,\infty)$ by Corollary 5.5.2(ii), so $A \succeq 0$ by Theorem 2.1.3(ii).
(ii) $A \succeq 0$ by part (i), so $A \succ 0$ by Theorem 2.1.8(iii) and the invertibility hypothesis.
(iii) $\mathrm{spec}(A) \subseteq (0,\infty)$ by Corollary 5.5.2(iii), so $A \succ 0$ by Theorem 2.1.8(ii).

Structural Commentary: Hermitian reality activates Gersgorin bounds; spectral criteria transfer eigenvalue signs to definiteness.
Key Insight: Invertibility removes zero from the nonnegative Hermitian spectrum. □

5.6 Rayleigh Quotients and Minimax

Throughout this subsection, $n \in \mathbb{Z}_{>0}$.

Definition 5.6.1. *Hermitian Eigenvalue Order and Rayleigh Quotient*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian. Matrix A has real eigenvalues and admits an orthonormal eigenbasis by Theorems 6.1.5 and 6.1.11.

(i) Write eigenvalues with algebraic multiplicity as $\lambda_1(A) \leq \lambda_2(A) \leq \cdots \leq \lambda_n(A)$. If A is fixed, abbreviate $\lambda_i(A)$ to λ_i.
(ii) For $\mathbf{x} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, the *Rayleigh quotient* is $\mathrm{Ray}_A(\mathbf{x}) := \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}}$. If $\|\mathbf{x}\|_2 = 1$, then $\mathrm{Ray}_A(\mathbf{x}) = \mathbf{x}^* A \mathbf{x}$.

Theorem 5.6.2. *Rayleigh Quotient on a Spectral Interval*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian with increasing eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$ and corresponding orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$.
If $p, q \in J_n$, $p \leq q$, and $\mathbf{x} \in \mathrm{span}(\mathbf{u}_p, \ldots, \mathbf{u}_q)$ satisfies $\|\mathbf{x}\|_2 = 1$, then $\lambda_p \leq \mathrm{Ray}_A(\mathbf{x}) \leq \lambda_q$.

Proof. Write $\mathbf{x} = \sum_{i=p}^{q} c_i \mathbf{u}_i$; then $(\forall\, i \in \{p, \dots, q\})\ |c_i|^2 \geq 0$ and $\sum_{i=p}^{q} |c_i|^2 = 1$ by orthonormality.

The identity $\mathrm{Ray}_A(\mathbf{x}) = \sum_{i=p}^{q} \lambda_i |c_i|^2$ holds by the eigenvector relations and Definition 5.6.1(ii).

The bound $\lambda_p \leq \mathrm{Ray}_A(\mathbf{x}) \leq \lambda_q$ holds by $\lambda_p \leq \lambda_i \leq \lambda_q$ and the preceding weight relations.

Structural Commentary: Eigenbasis expansion reduces the quotient to a weighted average over its spectral support.

Key Insight: The spectral weights $|c_i|^2$ are nonnegative and sum to one. □

Corollary 5.6.3. *Rayleigh Extrema and Diagonal Entry Bounds*

Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian with orthonormal eigenpairs $(\lambda_i, \mathbf{u}_i)_{i \in J_n}$, where $\lambda_1 \leq \cdots \leq \lambda_n$. For $p, q \in J_n$ with $p \leq q$, set $\mathcal{U}_{p,q} := \mathrm{span}(\mathbf{u}_p, \dots, \mathbf{u}_q)$.

(i) $\min_{\substack{\mathbf{x} \in \mathcal{U}_{p,q} \\ \|\mathbf{x}\|_2 = 1}} \mathrm{Ray}_A(\mathbf{x}) = \lambda_p$.

(ii) $\max_{\substack{\mathbf{x} \in \mathcal{U}_{p,q} \\ \|\mathbf{x}\|_2 = 1}} \mathrm{Ray}_A(\mathbf{x}) = \lambda_q$.

(iii) $(\forall\, \mathbf{x} \in \mathbb{C}^n)\ [\|\mathbf{x}\|_2 = 1 \implies \lambda_1 \leq \mathrm{Ray}_A(\mathbf{x}) \leq \lambda_n]$.

(iv) $(\forall\, i \in J_n)\ \lambda_1 \leq a_{ii} \leq \lambda_n$.

Proof.

(i) The lower bound and its attainment at $\mathbf{u}_p$ hold by Theorem 5.6.2 and $A\mathbf{u}_p = \lambda_p \mathbf{u}_p$, respectively.

(ii) The upper bound and its attainment at $\mathbf{u}_q$ hold by Theorem 5.6.2 and $A\mathbf{u}_q = \lambda_q \mathbf{u}_q$, respectively.

(iii) The asserted bound holds by (i)–(ii) with $(p, q) = (1, n)$ and $\mathcal{U}_{1,n} = \mathbb{C}^n$.

(iv) $(\forall\, i \in J_n)\ [\|\mathbf{e}_i\|_2 = 1 \wedge \mathrm{Ray}_A(\mathbf{e}_i) = a_{ii}]$ by Proposition 1.1.21(ii), Definition 5.6.1(ii), and coordinate multiplication. The relation $(\forall\, i \in J_n)\ \lambda_1 \leq a_{ii} \leq \lambda_n$ holds by the preceding identity and (iii).

Structural Commentary: Endpoint eigenvectors attain support bounds; coordinate vectors bound every diagonal entry.

Key Insight: Attainment turns spectral enclosures into exact extrema. □

Figure 96 places the quotient inside the eigenvalue interval determined by its spectral support, as asserted by Theorem 5.6.2.

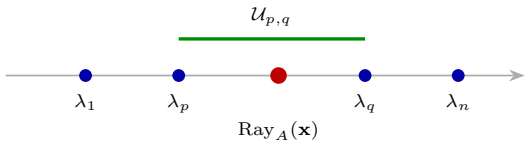

Figure 96: For ordered orthonormal eigenpairs of a Hermitian A, a unit vector supported on $\mathbf{u}_p, \dots, \mathbf{u}_q$ has $\mathrm{Ray}_A(\mathbf{x}) \in [\lambda_p, \lambda_q]$.

Computation Method 5.6.4. *Rayleigh Eigenbasis Method*

Given a Hermitian $A \in \mathrm{Mat}(n, n, \mathbb{C})$ and a unit vector $\mathbf{x} \in \mathbb{C}^n$, construct a certified spectral-support enclosure for $\mathrm{Ray}_A(\mathbf{x})$ as follows:

(i) compute an orthonormal eigenbasis $\mathbf{u}_1, \dots, \mathbf{u}_n$ in increasing eigenvalue order by Theorem 6.1.11 and Definition 5.6.1(i);

(ii) compute $c_i := \mathbf{u}_i^* \mathbf{x}$ for $i \in J_n$ and the nonempty support $I(\mathbf{x}) := \{i \in J_n : c_i \neq 0\}$;

(iii) evaluate $\mathrm{Ray}_A(\mathbf{x}) = \sum_{i \in J_n} \lambda_i |c_i|^2$;

(iv) set $p := \min I(\mathbf{x})$ and $q := \max I(\mathbf{x})$, and output the certified enclosure $\lambda_p \leq \mathrm{Ray}_A(\mathbf{x}) \leq \lambda_q$ by Theorem 5.6.2.

Theorem 5.6.5. *Rayleigh Minimax Principle*

Let $A = A^* \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$. For $k \in J_n$:

(i) $\lambda_k = \min_{\substack{\mathcal{S} \leq \mathbb{C}^n \\ \dim(\mathcal{S}) = k}} \max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{S}} \mathrm{Ray}_A(\mathbf{x})$.

(ii) $\lambda_k = \max_{\substack{\mathcal{T} \leq \mathbb{C}^n \\ \dim(\mathcal{T}) = n-k+1}} \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{T}} \mathrm{Ray}_A(\mathbf{x})$.

Proof. Fix an orthonormal eigenbasis $\mathbf{u}_1, \dots, \mathbf{u}_n$ in increasing eigenvalue order by Theorem 6.1.11 and Definition 5.6.1(i).
Set $E_k := \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ and $F_k := \mathrm{span}(\mathbf{u}_k, \dots, \mathbf{u}_n)$. $(\forall\, \mathbf{x} \in \mathbb{C}^n \setminus \{\mathbf{0}\})\ \mathrm{Ray}_A(\mathbf{x}) = \mathrm{Ray}_A(\mathbf{x}/\|\mathbf{x}\|_2)$ by Definition 5.6.1(ii).
Hence $\max_{\mathbf{0} \neq \mathbf{x} \in E_k} \mathrm{Ray}_A(\mathbf{x}) = \lambda_k$ and $\min_{\mathbf{0} \neq \mathbf{x} \in F_k} \mathrm{Ray}_A(\mathbf{x}) = \lambda_k$ by Corollary 5.6.3(ii),(i).

(i) If $\mathcal{S} \leq \mathbb{C}^n$ and $\dim(\mathcal{S}) = k$, then $\mathcal{S} \cap F_k \neq \{\mathbf{0}\}$ by Corollary 2.3.3(i) and $\dim(\mathcal{S}) + \dim(F_k) = n + 1$.
The bound $\max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{S}} \mathrm{Ray}_A(\mathbf{x}) \geq \lambda_k$ holds by the preceding intersection, Theorem 5.6.2, and scale invariance.
The outer minimum equals λ_k by this bound and $\mathcal{S} = E_k$ in the preceding extremum identity.

(ii) If $\mathcal{T} \leq \mathbb{C}^n$ and $\dim(\mathcal{T}) = n - k + 1$, then $\mathcal{T} \cap E_k \neq \{\mathbf{0}\}$ by Corollary 2.3.3(i) and $\dim(\mathcal{T}) + \dim(E_k) = n + 1$.
The bound $\min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{T}} \mathrm{Ray}_A(\mathbf{x}) \leq \lambda_k$ holds by the preceding intersection, Theorem 5.6.2, and scale invariance.
The outer maximum equals λ_k by this bound and $\mathcal{T} = F_k$ in the preceding extremum identity.

Structural Commentary: Spectral-flag intersections supply universal bounds; E_k and F_k attain them.

Key Insight: A dimension sum of $n + 1$ forces a shared comparison vector. □

5.7 Eigenvalue and Singular-Value Interlacing

Throughout this subsection, all matrix dimensions are positive integers.

Theorem 5.7.1. *Rank-One Positive Perturbation Interlacing*

Let $A, E \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian, with $E \succeq 0$ and $\mathrm{rank}(E) \leq 1$.

(i) $(\forall\, i \in J_n)\ \lambda_i(A) \leq \lambda_i(A + E)$.

(ii) $(\forall\, i \in J_{n-1})\ \lambda_i(A+E) \le \lambda_{i+1}(A)$.
(iii) $(\forall\, i \in J_n)\ \lambda_i(A-E) \le \lambda_i(A)$.
(iv) $(\forall\, i \in J_{n-1})\ \lambda_i(A) \le \lambda_{i+1}(A-E)$.

Proof. Fix ordered orthonormal eigenbases $(\mathbf{u}_j)_{j\in J_n}$ of A and $(\mathbf{v}_j)_{j\in J_n}$ of $A+E$ by Theorem 6.1.11 and Definition 5.6.1(i).

(i) Fix $i \in J_n$, and set $\mathcal{U} = \operatorname{span}(\mathbf{u}_i, \dots, \mathbf{u}_n)$ and $\mathcal{V} = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_i)$.
Fix a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V}$ by Corollary 2.3.3(i), normalization, and $\dim(\mathcal{U}) + \dim(\mathcal{V}) = n+1$.
The chain $\lambda_i(A) \le \mathbf{x}^*A\mathbf{x} \le \mathbf{x}^*(A+E)\mathbf{x} \le \lambda_i(A+E)$ holds by Theorem 5.6.2 and $E \succeq 0$.
(ii) Fix $i \in J_{n-1}$, and set $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$, $\mathcal{V} = \operatorname{span}(\mathbf{v}_i, \dots, \mathbf{v}_n)$, and $\mathcal{W} = \ker(E)$.
The bounds $\dim(\mathcal{U} \cap \mathcal{V}) \ge 2$ and $\dim(\mathcal{W}) \ge n-1$ hold by Corollaries 2.3.3(ii) and 3.3.1(i), respectively.
Fix a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$ by Corollary 2.3.3(i) and normalization.
The chain $\lambda_i(A+E) \le \mathbf{x}^*(A+E)\mathbf{x} = \mathbf{x}^*A\mathbf{x} \le \lambda_{i+1}(A)$ holds by Theorem 5.6.2.
(iii) $(\forall\, i \in J_n)\ \lambda_i(A-E) \le \lambda_i(A)$ by part (i) with $A-E$ in place of A.
(iv) $(\forall\, i \in J_{n-1})\ \lambda_i(A) \le \lambda_{i+1}(A-E)$ by part (ii) with $A-E$ in place of A.

Structural Commentary: Two flag intersections prove the $A+E$ bounds; substitution transfers them to $A-E$.
Key Insight: Membership in $\ker(E)$ removes the perturbation from the upper comparison. □

Corollary 5.7.2. *Signed Rank-Two Perturbation Interlacing*
Let $A, H \in \operatorname{Mat}(n,n,\mathbb{C})$ be Hermitian, with $n \ge 2$ and $\operatorname{rank}(H) \le 2$. Suppose that H has at most one positive and one negative eigenvalue.

(i) $(\forall\, i \in J_n \setminus \{1\})\ \lambda_{i-1}(A) \le \lambda_i(A+H)$.
(ii) $(\forall\, i \in J_{n-1})\ \lambda_i(A+H) \le \lambda_{i+1}(A)$.

Proof. Fix $H = Q\operatorname{diag}(\mu_1, \dots, \mu_n)Q^*$ with Q unitary and $\mu_1, \dots, \mu_n \in \mathbb{R}$ by Theorem 6.1.11.
Set $\mu_j^+ := \max\{\mu_j, 0\}$ and $\mu_j^- := \max\{-\mu_j, 0\}$ for $j \in J_n$, and set $E_\pm := Q\operatorname{diag}(\mu_1^\pm, \dots, \mu_n^\pm)Q^*$.
The relations $H = E_+ - E_-$, $E_\pm \succeq 0$, and $\operatorname{rank}(E_\pm) \le 1$ hold by construction and the sign hypothesis.
Set $B := A - E_-$, so $A = B + E_-$ and $A + H = B + E_+$.

(i) $(\forall\, i \in J_n \setminus \{1\})\ \lambda_{i-1}(A) \le \lambda_i(B) \le \lambda_i(A+H)$ by Theorem 5.7.1(ii),(i).
(ii) $(\forall\, i \in J_{n-1})\ \lambda_i(A+H) \le \lambda_{i+1}(B) \le \lambda_{i+1}(A)$ by Theorem 5.7.1(ii),(i).

Structural Commentary: The signed spectral split makes A and $A+H$ positive rank-one updates of a common base.
Key Insight: Comparison through B aligns the lower and upper index shifts. □

Theorem 5.7.3. *Cauchy Interlacing: One Principal Deletion*
Let $A \in \operatorname{Mat}(n+1, n+1, \mathbb{C})$ be Hermitian, and let $B \in \operatorname{Mat}(n,n,\mathbb{C})$ be its leading principal submatrix.

(i) $(\forall\, i \in J_n)\ \lambda_i(A) \le \lambda_i(B)$.
(ii) $(\forall\, i \in J_n)\ \lambda_i(B) \le \lambda_{i+1}(A)$.

Proof. Fix ordered orthonormal eigenbases $(\mathbf{u}_j)_{j\in J_{n+1}}$ of A and $(\mathbf{v}_j)_{j\in J_n}$ of B by Theorem 6.1.11 and Definition 5.6.1(i).
Define $\iota : \mathbb{C}^n \to \mathbb{C}^{n+1}$ by $\iota(\mathbf{z}) = (\mathbf{z}, 0)$. The identities $\|\iota(\mathbf{z})\|_2 = \|\mathbf{z}\|_2$ and $\iota(\mathbf{z})^*A\iota(\mathbf{z}) = \mathbf{z}^*B\mathbf{z}$ hold by coordinate multiplication.

(i) Fix $i \in J_n$, and set $\mathcal{S} = \operatorname{span}(\iota(\mathbf{v}_1), \dots, \iota(\mathbf{v}_i))$ and $\mathcal{T} = \operatorname{span}(\mathbf{u}_i, \dots, \mathbf{u}_{n+1})$.
Fix a unit $\mathbf{x} = \iota(\mathbf{z}) \in \mathcal{S} \cap \mathcal{T}$ by Corollary 2.3.3(i), normalization, and $\dim(\mathcal{S}) + \dim(\mathcal{T}) = n+2$.
The chain $\lambda_i(A) \le \mathbf{x}^*A\mathbf{x} = \mathbf{z}^*B\mathbf{z} \le \lambda_i(B)$ holds by Theorem 5.6.2.
(ii) Fix $i \in J_n$, and set $\mathcal{S} = \operatorname{span}(\iota(\mathbf{v}_i), \dots, \iota(\mathbf{v}_n))$ and $\mathcal{T} = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$.
Fix a unit $\mathbf{x} = \iota(\mathbf{z}) \in \mathcal{S} \cap \mathcal{T}$ by Corollary 2.3.3(i), normalization, and $\dim(\mathcal{S}) + \dim(\mathcal{T}) = n+2$.
The chain $\lambda_i(B) \le \mathbf{z}^*B\mathbf{z} = \mathbf{x}^*A\mathbf{x} \le \lambda_{i+1}(A)$ holds by Theorem 5.6.2.

Structural Commentary: The embedding preserves quadratic forms; complementary spectral flags establish both inequalities.
Key Insight: One coordinate deletion leaves the dimension sum $n+2$. □

Corollary 5.7.4. *Cauchy Interlacing for Principal Submatrices*
Let $m \in \mathbb{Z}_{>0}$, and let $B \in \operatorname{Mat}(n,n,\mathbb{C})$ be a principal submatrix of a Hermitian $A \in \operatorname{Mat}(n+m, n+m, \mathbb{C})$.

(i) $(\forall\, i \in J_n)\ \lambda_i(A) \le \lambda_i(B)$.
(ii) $(\forall\, i \in J_n)\ \lambda_i(B) \le \lambda_{i+m}(A)$.

Proof. Choose a permutation matrix P such that B is the leading principal submatrix of $\widetilde{A} := P^*AP$.
The matrices A and $\widetilde{A}$ have the same ordered eigenvalues by Theorem 2.1.7(i).
For $r \in \{0, \dots, m\}$, let $A^{(r)}$ be the leading principal submatrix of $\widetilde{A}$ of size $n+m-r$; thus $A^{(0)} = \widetilde{A}$ and $A^{(m)} = B$.

(i) $(\forall\, i \in J_n)\ \lambda_i(A^{(0)}) \le \lambda_i(A^{(1)}) \le \cdots \le \lambda_i(A^{(m)})$ by successive applications of Theorem 5.7.3(i).
(ii) $(\forall\, i \in J_n)\ \lambda_i(A^{(m)}) \le \lambda_{i+1}(A^{(m-1)}) \le \cdots \le \lambda_{i+m}(A^{(0)})$ by successive applications of Theorem 5.7.3(ii).

Structural Commentary: Permutation makes B leading; chained one-coordinate deletions accumulate the upper-index shift.
Key Insight: Each deletion raises the upper comparison index by one. □

Corollary 5.7.5. *Singular Values Interlace After Deleting a Row or Column*
Let $n \ge 2$, and let B be obtained from $A \in \operatorname{Mat}(n,n,\mathbb{C})$ by deleting one row or one column.
With singular values ordered decreasingly, $(\forall\, i \in J_{n-1})\ \sigma_i(A) \ge \sigma_i(B) \ge \sigma_{i+1}(A)$.

Proof.

(i) *Row Gram Reduction*: In the row case, choose a permutation matrix P with B as the first $n-1$ rows of PA. Set $G := (PA)(PA)^*$.
The matrix BB^* is its leading principal submatrix, and PA and A have the same singular values by Proposition 5.1.7(i).
(ii) *Index Identification*: $(\forall\, j \in J_n)\ \lambda_j(G) = \sigma_{n-j+1}(A)^2$ by Proposition 4.2.11(i) and Definition 4.2.8.
The identity $(\forall\, j \in J_{n-1})\ \lambda_j(BB^*) = \sigma_{n-j}(B)^2$ holds by Proposition 4.2.11(i) and Definition 4.2.8.
(iii) *Cauchy Transfer*: Fix $i \in J_{n-1}$ and set $j := n-i$. The chain $\sigma_{i+1}(A)^2 \le \sigma_i(B)^2 \le \sigma_i(A)^2$ holds by Theorem 5.7.3(i)–(ii).
The row-deletion claim follows by taking nonnegative square roots.
(iv) *Column Reduction*: If B is obtained by deleting a column, then B^* is obtained from A^* by deleting a row.
The stated inequalities hold by the row-deletion claim and Proposition 4.2.11(iii).

Structural Commentary: A principal Gram block proves row deletion; adjoints transfer the result to columns.
Key Insight: $j = n - i$ matches increasing Gram eigenvalues with decreasing singular values. □

Figure 97 compares the interlacing windows from Theorem 5.7.1 and Corollary 5.7.4.

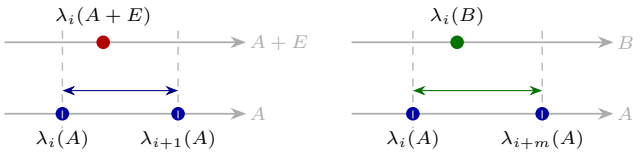

Figure 97: $\lambda_i(A+E)$ lies in $[\lambda_i(A), \lambda_{i+1}(A)]$, while $\lambda_i(B)$ lies in $[\lambda_i(A), \lambda_{i+m}(A)]$.

Theorem 5.7.6. *Schur Majorization of the Diagonal*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian, and let $d_1 \le \cdots \le d_n$ be its diagonal entries arranged increasingly.

(i) $(\forall\, k \in J_{n-1}) \sum_{i=1}^{k} \lambda_i(A) \le \sum_{i=1}^{k} d_i$.

(ii) $\sum_{i=1}^{n} \lambda_i(A) = \sum_{i=1}^{n} d_i$.

Proof. Choose a permutation matrix P such that the diagonal of $\widetilde{A} := P^* A P$ is $(d_1, \ldots, d_n)$.
The matrices A and $\widetilde{A}$ have the same ordered eigenvalues by Theorem 2.1.7(i).
Set A_k to be the leading $k \times k$ principal submatrix of $\widetilde{A}$ for $k \in J_n$.

(i) Fix $k \in J_{n-1}$. The inequalities $(\forall\, i \in J_k)\, \lambda_i(\widetilde{A}) \le \lambda_i(A_k)$ hold by Corollary 5.7.4(i).
The identity $\sum_{i=1}^{k} \lambda_i(A_k) = \mathrm{tr}(A_k) = \sum_{i=1}^{k} d_i$ holds by Proposition 1.1.14(i) and Definition 3.8.1.
Thus $\sum_{i=1}^{k} \lambda_i(A) \le \sum_{i=1}^{k} d_i$ by the preceding inequalities and the common ordered spectrum of A and $\widetilde{A}$.

(ii) $\sum_{i=1}^{n} \lambda_i(A) = \mathrm{tr}(A) = \sum_{i=1}^{n} d_i$ by Proposition 1.1.14(i) and Definition 3.8.1.

Structural Commentary: Cauchy bounds each leading-block eigenvalue before trace sums the inequalities.
Key Insight: Ordering the diagonal makes each partial sum a principal-block trace. □

Proof Method 5.7.7. *Eigenvalue Interlacing Proof Method*
Given a proposed Hermitian eigenvalue interlacing inequality, construct a proof route as follows:

(i) order every Hermitian spectrum increasingly and identify the requested lower and upper indices;
(ii) identify the lost or unaffected directions through perturbation rank, a kernel, or deleted coordinates;
(iii) choose lower and upper spectral flags whose dimensions force a nonzero intersection by Corollary 2.3.3;
(iv) normalize an intersection vector and certify the target eigenvalue chain by Theorem 5.6.2.

5.8 Weyl Eigenvalue Bounds

Throughout this subsection, $n \in \mathbb{Z}_{>0}$.

Theorem 5.8.1. *Weyl Eigenvalue Inequalities*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian.

(i) $(\forall\, i \in J_n)(\forall\, j \in J_{n-i} \cup \{0\})\, \lambda_i(A+B) \le \lambda_{i+j}(A) + \lambda_{n-j}(B)$.
(ii) $(\forall\, i \in J_n)(\forall\, j \in J_i)\, \lambda_{i-j+1}(A) + \lambda_j(B) \le \lambda_i(A+B)$.

Proof. Fix increasingly indexed orthonormal eigenbases $(\mathbf{u}_k)$, $(\mathbf{v}_k)$, and $(\mathbf{w}_k)$ of A, $A+B$, and B by Theorem 6.1.11, Definition 5.6.1(i).

(i) Fix $i \in J_n$ and $j \in J_{n-i} \cup \{0\}$. Set $\mathcal{U} = \mathrm{span}(\mathbf{u}_1, \ldots, \mathbf{u}_{i+j})$ and $\mathcal{V} = \mathrm{span}(\mathbf{v}_i, \ldots, \mathbf{v}_n)$.
Set $\mathcal{W} = \mathrm{span}(\mathbf{w}_1, \ldots, \mathbf{w}_{n-j})$. $\dim(\mathcal{U} \cap \mathcal{V}) \ge j + 1$ by Corollary 2.3.3(ii) with $\dim(\mathcal{U}) = i + j$ and $\dim(\mathcal{V}) = n - i + 1$.
Fix a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$ by Corollary 2.3.3(i), normalization, the preceding bound, and $\dim(\mathcal{W}) = n - j$.
The chain $\lambda_i(A+B) \le \mathbf{x}^*(A+B)\mathbf{x} = \mathbf{x}^* A \mathbf{x} + \mathbf{x}^* B \mathbf{x} \le \lambda_{i+j}(A) + \lambda_{n-j}(B)$ holds by Theorem 5.6.2.

(ii) Fix $i \in J_n$ and $j \in J_i$, and set $p := n - i + 1$ and $q := i - j$. $p \in J_n$ and $q \in J_{n-p} \cup \{0\}$ by index arithmetic.
$(\forall\, C \in \{A, B, A+B\})(\forall\, k \in J_n)\, \lambda_k(-C) = -\lambda_{n-k+1}(C)$ by Theorem 6.1.11 and Definition 5.6.1(i).
$-\lambda_i(A+B) \le -\lambda_j(B) - \lambda_{i-j+1}(A)$ by part (i) for $(-B, -A, p, q)$ and the reversal identity.
Claim (ii) holds by multiplying this inequality by -1.

Structural Commentary: Three-flag intersection establishes (i); spectral reversal and sign multiplication establish (ii).
Key Insight: $\dim(\mathcal{U}) + \dim(\mathcal{V}) + \dim(\mathcal{W}) = 2n + 1$ forces a common Rayleigh vector. □

Computation Method 5.8.2. *Weyl Eigenvalue-Bound Method*
Given Hermitian $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ and $i \in J_n$, compute a certified interval for $\lambda_i(A+B)$ as follows:

(i) compute the eigenvalues of A and B in increasing order;
(ii) set $U_i := \min_{j \in J_{n-i} \cup \{0\}} \{\lambda_{i+j}(A) + \lambda_{n-j}(B)\}$;
(iii) set $L_i := \max_{j \in J_i} \{\lambda_{i-j+1}(A) + \lambda_j(B)\}$;
(iv) output $[L_i, U_i]$, which satisfies $L_i \le \lambda_i(A+B) \le U_i$ by Theorem 5.8.1(i)–(ii);
(v) for structured B, simplify the candidates with any known sign or zero-eigenvalue data.
If $B \succeq 0$, then $\mathrm{spec}(B) \subseteq [0, \infty)$ by Theorem 2.1.3(ii).
If $\mathrm{rank}(B) = r$, then the algebraic multiplicity of 0 is $n - r$ by Corollary 3.3.1(i) and Theorem 6.1.11.

5.9 Brauer Rank-One Spectral Shifts

Throughout this subsection, $n \in \mathbb{Z}_{>0}$.

Theorem 5.9.1. *Brauer Rank-One Eigenvalue Shift*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, $\lambda \in \mathbb{C}$, and $\mathbf{x},\mathbf{y} \in \mathbb{C}^n$. Assume that $\mathbf{x} \neq \mathbf{0}$ and $A\mathbf{x} = \lambda\mathbf{x}$.
Let $\lambda_1,\dots,\lambda_n$ list the eigenvalues of A with algebraic multiplicity, where $\lambda_1 = \lambda$, and set $C := A + \mathbf{x}\mathbf{y}^*$.
(i) The eigenvalues of C, counted with algebraic multiplicity, are $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \dots, \lambda_n$.
(ii) $C\mathbf{x} = (\lambda + \mathbf{y}^*\mathbf{x})\mathbf{x}$.

Proof. Set $\mathbf{u} := \frac{\mathbf{x}}{\|\mathbf{x}\|_2}$. Then $\|\mathbf{u}\|_2 = 1$ and $A\mathbf{u} = \lambda\mathbf{u}$ by the hypotheses on $\mathbf{x}$.
Fix a unitary $U = [\mathbf{u}\ U_2]$ such that $T := U^*AU$ is upper triangular with diagonal $\lambda_1,\dots,\lambda_n$ by Theorem 2.1.9(i).
$U^*\mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$ by the definition of $\mathbf{u}$ and $U = [\mathbf{u}\ U_2]$.
For $R := U^*(\mathbf{x}\mathbf{y}^*)U$, only its first row may be nonzero and $R_{11} = \mathbf{y}^*\mathbf{x}$ by the preceding identity and direct multiplication.
$U^*CU = T + R$ is upper triangular with diagonal $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \dots, \lambda_n$ by the definitions of C, T, and R.
(i) The displayed diagonal is the eigenvalue list of C by Proposition 2.1.6(iii) and Theorem 2.1.7(ii).
(ii) $C\mathbf{x} = A\mathbf{x} + \mathbf{x}(\mathbf{y}^*\mathbf{x}) = (\lambda + \mathbf{y}^*\mathbf{x})\mathbf{x}$ by the definition of C and $A\mathbf{x} = \lambda\mathbf{x}$.
Structural Commentary: Ordered Schur form isolates the eigenline; triangular multiplicities transfer back after one diagonal shift.
Key Insight: $U^*\mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$ confines the update to the first Schur row. □

Figure 98 depicts the Schur-coordinate mechanism in Theorem 5.9.1.
The update replaces λ by $\lambda + \mathbf{y}^*\mathbf{x}$ and preserves the trailing Schur block T'.

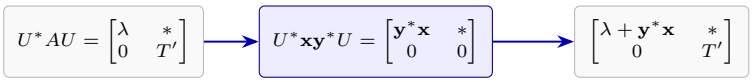

Figure 98: Brauer's aligned update replaces the leading Schur entry and preserves the trailing block T'.

Corollary 5.9.2. *Damped Rank-One Eigenvalue Reset*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, $\lambda,\tau \in \mathbb{C}$, and $\mathbf{x},\mathbf{y} \in \mathbb{C}^n$. Assume that $\mathbf{x} \neq \mathbf{0}$, $A\mathbf{x} = \lambda\mathbf{x}$, and $\mathbf{y}^*\mathbf{x} = 1$.
Let $\lambda_1,\dots,\lambda_n$ list the eigenvalues of A with algebraic multiplicity, where $\lambda_1 = \lambda$. Set $A_\tau := \tau A + (1-\tau)\lambda\mathbf{x}\mathbf{y}^*$.
Then the eigenvalues of A_τ are $\lambda, \tau\lambda_2, \dots, \tau\lambda_n$, counted with algebraic multiplicity.

Proof. The eigenvalues of τA are $\tau\lambda_1,\dots,\tau\lambda_n$ by Proposition 3.4.1 with $p(\zeta) = \tau\zeta$. Set $\mathbf{q} := (1-\overline{\tau})\overline{\lambda}\mathbf{y}$.
The identities $A_\tau = \tau A + \mathbf{x}\mathbf{q}^*$ and $(\tau A)\mathbf{x} = \tau\lambda\mathbf{x}$ hold by the definitions and $A\mathbf{x} = \lambda\mathbf{x}$. $\mathbf{q}^*\mathbf{x} = (1-\tau)\lambda\mathbf{y}^*\mathbf{x} = (1-\tau)\lambda$ by $\mathbf{y}^*\mathbf{x} = 1$.
The claimed eigenvalue list holds by Theorem 5.9.1(i) and $\tau\lambda + \mathbf{q}^*\mathbf{x} = \lambda$.
Structural Commentary: Scalar spectral mapping forms the damped spectrum; Brauer's aligned update restores its first entry.
Key Insight: $\mathbf{q}^*\mathbf{x} = (1-\tau)\lambda$ cancels the shift from λ to $\tau\lambda$. □

6 Numerical Linear Algebra

Numerical linear algebra turns matrix structure into algorithms for direct solves, least squares, eigenvalues, SVD, structured transforms, Krylov methods, iteration, and preconditioning.

6.1 Direct Linear-System Solves

Throughout this subsection, $n \in \mathbb{Z}_{>0}$ and $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$.

Perspective 6.1.1. *Solve with Factors Before Forming Inverses*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ be invertible and $\mathbf{b} \in \mathbb{F}^n$. LU and QR factorisations are defined in Definitions 4.1.3 and 4.1.6.
Cholesky factorisation is established in Theorem 4.1.11.
Solve $A\mathbf{x} = \mathbf{b}$ by triangular solves and, for QR, a unitary transformation; do not form A^{-1}.
Reuse the factors for additional right-hand sides; fill-in limits exploitable sparsity.

Computation Method 6.1.2. *Iterative Refinement by Residual Correction*
Let $k \in \mathbb{Z}_{\geq 0}$, let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ be invertible, and let $\mathbf{b} \in \mathbb{F}^n$. Given a current approximation $\mathbf{x}_k \in \mathbb{F}^n$, obtain $\mathbf{x}_{k+1}$ as follows:
(i) Set the residual $\mathbf{r}_k := \mathbf{b} - A\mathbf{x}_k$.
(ii) Compute a correction $\mathbf{d}_k$ for $A\mathbf{d}_k = \mathbf{r}_k$ with stored factors or a prescribed correction solver.
(iii) Set $\mathbf{x}_{k+1} := \mathbf{x}_k + \mathbf{d}_k$.

Proposition 6.1.3. *Exact Residual Correction*
Let $k \in \mathbb{Z}_{\geq 0}$, $A \in \mathrm{Mat}(n,n,\mathbb{F})$, and $\mathbf{b},\mathbf{x}_k,\mathbf{d}_k \in \mathbb{F}^n$. Set $\mathbf{r}_k := \mathbf{b} - A\mathbf{x}_k$ and $\mathbf{x}_{k+1} := \mathbf{x}_k + \mathbf{d}_k$. Then $A\mathbf{d}_k = \mathbf{r}_k \Rightarrow A\mathbf{x}_{k+1} = \mathbf{b}$.

Proof. If $A\mathbf{d}_k = \mathbf{r}_k$, then $A\mathbf{x}_{k+1} = A\mathbf{x}_k + A\mathbf{d}_k = A\mathbf{x}_k + \mathbf{r}_k = \mathbf{b}$ by the definitions of $\mathbf{x}_{k+1}$ and $\mathbf{r}_k$.
Structural Commentary: The update and correction substitutions reduce the claim to residual cancellation.
Key Insight: The correction exactly cancels the current residual. □

Perspective 6.1.4. *Finite-Precision Qualification*
Proposition 6.1.3 is the exact-correction case of Computation Method 6.1.2.
In finite precision, the computed residual and correction satisfy perturbed relations.
Improvement depends on residual evaluation, the correction solve, factor stability, and conditioning.
No quantitative convergence guarantee is asserted without a specified rounding-error model.

6.2 QR Least Squares

Throughout this subsection, $m, n \in \mathbb{Z}_{>0}$ satisfy $m \geq n$, and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Proposition 6.2.1. *QR Route for Least Squares*
Let $A = QR \in \mathrm{Mat}(m, n, \mathbb{F})$ be a narrow QR factorisation and let $\mathbf{y} \in \mathbb{F}^m$. Assume $\mathrm{rank}(A) = n$. Then:

(i) *R is invertible;*
(ii) *$(\forall\, \mathbf{x} \in \mathbb{F}^n)\,[\mathbf{x} \in \mathrm{LS}(A, \mathbf{y}) \iff R\mathbf{x} = Q^*\mathbf{y}]$;*
(iii) *$P_{\mathrm{col}(A)} = QQ^*$.*

Proof.

(i) Inclusion $\ker(R) \subseteq \ker(A) = \{\mathbf{0}\}$ holds by $A = QR$, rank hypothesis, Corollary 3.3.1(i); R invertible by Corollary 3.6.3(vi)⇒(i).
(ii) ($\Rightarrow$) For $\mathbf{x} \in \mathrm{LS}(A, \mathbf{y})$, $R^*(R\mathbf{x} - Q^*\mathbf{y}) = \mathbf{0}$ by $A = QR$, $Q^*Q = I_n$, Theorem 2.4.4(i); $R\mathbf{x} = Q^*\mathbf{y}$ by invertibile R^* from (i).
(ii) ($\Leftarrow$) For $\mathbf{x} \in \mathbb{F}^n$ with $R\mathbf{x} = Q^*\mathbf{y}$, $A^*A\mathbf{x} = R^*R\mathbf{x} = R^*Q^*\mathbf{y} = A^*\mathbf{y}$ by $A = QR$, $Q^*Q = I_n$; $\mathbf{x} \in \mathrm{LS}(A, \mathbf{y})$ by Theorem 2.4.4(i).
(iii) Set $\mathcal{U} := \mathrm{col}(A)$. The equality $\mathcal{U} = \mathrm{col}(Q)$ holds by $A = QR$, $Q = AR^{-1}$, and (i).
Write $Q = [\mathbf{q}_1, \ldots, \mathbf{q}_n]$; its columns form an orthonormal basis of $\mathcal{U}$ by Definition 4.1.6.
For $\mathbf{z} \in \mathbb{F}^m$, $P_{\mathcal{U}}\mathbf{z} = \sum_{j=1}^{n} \langle \mathbf{z}, \mathbf{q}_j \rangle_{\mathbb{F}^m}\, \mathbf{q}_j = QQ^*\mathbf{z}$ by Proposition 2.2.3(ix) and matrix multiplication.
Thus $P_{\mathrm{col}(A)} = QQ^*$ by the definition of $\mathcal{U}$ and arbitrariness of $\mathbf{z}$.

Structural Commentary: Kernel transfer proves (i); cancellation and the orthonormal-column formula prove (ii)–(iii).
Key Insight: Invertible R separates projection from triangular coordinate recovery. □

Perspective 6.2.2. *Why Normal Equations Can Be Numerically Fragile*
For $A = QR$ as in Proposition 6.2.1, $A^*A = R^*R$ and R is invertible.
The identity $\kappa_2(A^*A) = \kappa_2(R^*R) = \kappa_2(R)^2$ holds by the preceding equality and Proposition 5.3.2(v).
Forming the normal equations therefore squares the spectral condition number of R. A QR route omits A^*A and this squaring.
Its finite-precision stability still depends on the factorisation or orthogonalisation implementation and the rounding-error model.

Algorithm 1: Classical Gram–Schmidt Least Squares in Exact Arithmetic

Require: $N \in \mathbb{Z}_{>0}$ and $p \in \mathbb{Z}_{\geq 0}$
Require: $X = [\mathbf{x}_0, \ldots, \mathbf{x}_p] \in \mathrm{Mat}(N, p+1, \mathbb{R})$ with $\mathrm{rank}(X) = p + 1$
Require: $\mathbf{y} \in \mathbb{R}^N$
Ensure: $\mathrm{LS}(X, \mathbf{y}) = \{\widehat{\boldsymbol{\beta}}\}$
  for $j = 0, \ldots, p$ **do**
    $\mathbf{z}_j \leftarrow \mathbf{x}_j$
    for $\ell = 0, \ldots, j-1$ **do**
      $r_{\ell j} \leftarrow \mathbf{q}_\ell^t \mathbf{x}_j$
      $\mathbf{z}_j \leftarrow \mathbf{z}_j - r_{\ell j}\mathbf{q}_\ell$
    end for
    $r_{jj} \leftarrow \|\mathbf{z}_j\|_2$
    $\mathbf{q}_j \leftarrow \frac{\mathbf{z}_j}{r_{jj}}$
  end for
  Set $r_{\ell j} \leftarrow 0$ for $0 \leq j < \ell \leq p$
  Form $Q = [\mathbf{q}_0, \ldots, \mathbf{q}_p]$ and $R = [r_{\ell j}]_{\ell, j=0}^{p}$
  Solve $R\widehat{\boldsymbol{\beta}} = Q^t\mathbf{y}$ by back-substitution

The loop is well-defined, $Q^tQ = I_{p+1}$, and $(\forall\, j \in \{0, \ldots, p\})\, r_{jj} > 0$ by the rank contract and Theorem 1.3.6.
The relation $(\forall\, j \in \{0, \ldots, p\})\, \mathbf{x}_j = \sum_{\ell=0}^{j} r_{\ell j}\mathbf{q}_\ell$ holds by the update assignments.
The identity $X = QR$ with upper-triangular R holds by this relation and the zero lower entries.
The postcondition holds by Proposition 6.2.1(i)–(ii) and the final triangular solve.
Both loops terminate by their finite bounds; the inner loop is empty for $j = 0$. The algorithm asserts no finite-precision stability.

Perspective 6.2.3. *Solve Least Squares with Factors*
Let $N, p \in \mathbb{Z}_{>0}$, let $X \in \mathrm{Mat}(N, p, \mathbb{R})$ satisfy $\mathrm{rank}(X) = p$, and let $\mathbf{y} \in \mathbb{R}^N$. Fix a narrow QR factorisation $X = QR$.
Matrix R is invertible by Proposition 6.2.1(i). The identity $X^tX = R^tR$ holds by $X = QR$ and $Q^tQ = I_p$.
The equality $\kappa_2(X^tX) = \kappa_2(R^tR) = \kappa_2(R)^2$ holds by this identity and Proposition 5.3.2(v). Let $\widehat{\boldsymbol{\beta}}$ be the unique solution of $R\widehat{\boldsymbol{\beta}} = Q^t\mathbf{y}$.
Then $\mathrm{LS}(X, \mathbf{y}) = \{\widehat{\boldsymbol{\beta}}\}$ by Proposition 6.2.1(i)–(ii). Compute $\widehat{\boldsymbol{\beta}}$ by triangular substitution rather than forming X^tX.
Use Cholesky on X^tX only when its conditioning and the required accuracy justify that route.
The exact-arithmetic algorithm above alone supplies no finite-precision guarantee.

6.3 Banded and Sparse Direct Methods

Throughout this subsection, $\mathbb{F}$ is a field.

Definition 6.3.1. *Half-Bandwidth and Tridiagonal Matrices*
Let $m \in \mathbb{Z}_{>0}$, $A = [a_{ij}] \in \mathrm{Mat}(m, m, \mathbb{F})$, $w \in J_m$. Matrix A has *half-bandwidth at most w* if $(\forall\, i, j \in J_m)[|i - j| \geq w \implies a_{ij} = 0]$.
Its *half-bandwidth* is the least such w, which exists because $w = m$ is admissible.
Matrices of half-bandwidth at most 1 are precisely diagonal matrices. Matrices of half-bandwidth at most 2 are *tridiagonal*.

Computation Method 6.3.2. *Banded Elimination without Row Exchanges*
Let $n \in \mathbb{Z}_{>0}$, $w \in J_n$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$ have half-bandwidth at most w. Assume every leading principal minor of A is nonzero.
Compute the unique exact-arithmetic factors $A = LU$, with L unit lower triangular and U upper triangular, as follows:

(i) Initialize $L \leftarrow I_n$, $U \leftarrow 0$, and a current active array $B = [b_{ij}] \leftarrow A$.

(ii) For $k = 1, \dots, n$ in order, set $u_{kj} \leftarrow b_{kj}$ for $k \le j \le \min(n, k+w-1)$.
For $i = k+1, \dots, \min(n, k+w-1)$, set $l_{ik} \leftarrow \frac{b_{ik}}{u_{kk}}$ and $b_{ik} \leftarrow 0$.
Then set $b_{ij} \leftarrow b_{ij} - l_{ik}u_{kj}$ for $k < j \le \min(n, k+w-1)$.
(iii) Return (L, U).
Let A_k be the leading $k \times k$ block and set $\det(A_0) := 1$. Prior row additions preserve $\det(A_k)$ and triangularise its first $k-1$ columns.
Hence $\det(A_k) = \left(\prod_{s=1}^{k-1} u_{ss}\right) b_{kk}$. The minor hypothesis and induction from $k = 1$ imply $u_{kk} = b_{kk} \neq 0$, so every division is defined.
An update requires $0 < i-k < w$ and $0 < j-k < w$, so $|i-j| < w$.
Thus induction preserves the band, and L and U have half-bandwidth at most w.
Reversing the stored row operations establishes $A = LU$. The returned pair is the unique factorisation by Theorem 4.1.4.
At pivot k, set $r_k := \min(w-1, n-k)$; the work is r_k divisions and r_k^2 multiply-subtract updates.
Thus the worst-case scalar-field operation count is $O(nw^2)$ by $r_k \le w-1$.
If $w = 1$, the inner ranges are empty and $(L, U) = (I_n, A)$; if $w = 2$, the cost is $O(n)$.
The finite loops terminate, but without a rounding-error model the method makes no stability claim.
Outside the stated hypotheses, preserving the band does not justify suppressing required row exchanges.

For tridiagonal input, Figure 99 depicts the factors from Computation Method 6.3.2.
Here L and U are lower and upper bidiagonal, respectively, and L has unit diagonal.

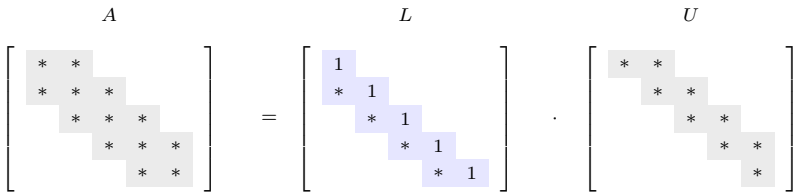

Figure 99: Possible nonzeros ($*$) in no-pivot tridiagonal elimination: L and U are bidiagonal.

Perspective 6.3.3. *Five-Point Stencils Are Sparse with Growing Bandwidth*
Let $N \in \mathbb{Z}_{\ge 2}$ and place the nodes of an $N \times N$ grid in row-major order.
Let $S_N \in \mathrm{Mat}(N^2, N^2, \mathbb{F})$ have nonzero entries precisely at self-couplings and grid-neighbour couplings.
Each row has at most five nonzeros, at offsets $0, \pm 1, \pm N$; boundary gaps remove horizontal wraparound.
Thus S_N has half-bandwidth $N+1$ by Definition 6.3.1. Its row sparsity is bounded independently of N.
An incomplete LU approximation may control fill by retaining a selected pattern or applying a drop rule.
Its quality depends on that rule and the ordering.

Figure 100 depicts the three offset classes.
Its ± 1 diagonals are schematic because row boundaries create periodic gaps that prevent wraparound couplings.

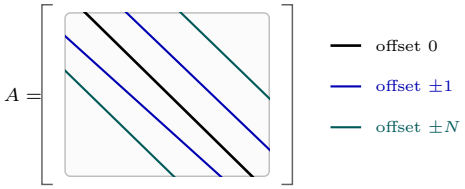

Figure 100: Row-major ordering places five-point couplings on the diagonals with offsets $0, \pm 1, \pm N$.

6.4 Krylov Subspace Methods

Throughout this subsection, $n \in \mathbb{Z}_{>0}$ and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 6.4.1. *Krylov Subspace*
Let $m \in \mathbb{Z}_{>0}$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $\mathbf{b} \in \mathbb{F}^n$. Set $\mathcal{K}_m(A, \mathbf{b}) := \mathrm{span}\{\mathbf{b}, A\mathbf{b}, \dots, A^{m-1}\mathbf{b}\}$.
This is the mth *Krylov subspace* generated by $(A, \mathbf{b})$.

Computation Method 6.4.2. *Arnoldi, Lanczos, CG, and GMRES Templates*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ and work in exact arithmetic.
(i) *Arnoldi*: Let $\mathbf{b} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$ and $m \in J_n$. Set $\mathbf{q}_1 := \mathbf{b}/\|\mathbf{b}\|_2$. For $j = 1, \dots, m$, set $\mathbf{v}_j := A\mathbf{q}_j$ and $h_{ij} := \mathbf{q}_i^* \mathbf{v}_j$ for $i \in J_j$.
Set $\mathbf{w}_j := \mathbf{v}_j - \sum_{i=1}^{j} h_{ij}\mathbf{q}_i$ and $\eta_j := \|\mathbf{w}_j\|_2$. If $\eta_j = 0$, set $s := j$ and stop; otherwise, set $h_{j+1,j} := \eta_j$ and $\mathbf{q}_{j+1} := \mathbf{w}_j/\eta_j$.
Write $Q_j := [\mathbf{q}_1 \ \cdots \ \mathbf{q}_j]$ and $H_j := [h_{i\ell}]_{i,\ell=1}^{j}$, where $h_{i\ell} := 0$ for $i > \ell + 1$.
The identities $\mathbf{q}_i^* \mathbf{w}_j = 0$ for $i \in J_j$ and $\mathbf{q}_{j+1} \in \mathcal{K}_{j+1}(A, \mathbf{b})$ imply inductively that $Q_j^* Q_j = I_j$ and $\mathrm{col}(Q_j) \subseteq \mathcal{K}_j(A, \mathbf{b})$.
Before breakdown, $\dim(\mathrm{col}(Q_j)) = j$ and $\dim(\mathcal{K}_j(A, \mathbf{b})) \le j$, so the inclusion is equality.
If no breakdown occurs through m, then $AQ_m = Q_m H_m + \eta_m \mathbf{q}_{m+1}\mathbf{e}_m^t$, and H_m is upper Hessenberg.
At breakdown s, the same relation becomes $AQ_s = Q_s H_s$, so $\mathcal{K}_s(A, \mathbf{b})$ is A-invariant and the construction terminates.
(ii) *Lanczos*: Suppose $A = A^*$ and Arnoldi does not break down through step m.
Then $H_m = Q_m^* A Q_m = H_m^*$, so its upper-Hessenberg structure is tridiagonal by Definition 6.3.1.
Set $\mathbf{q}_0 := \mathbf{0}$, $\beta_0 := 0$, $\alpha_j := h_{jj} \in \mathbb{R}$, and $\beta_j := \eta_j > 0$ for $j \in J_m$.
The Arnoldi relation becomes $(\forall\, j \in J_m)\ A\mathbf{q}_j = \beta_{j-1}\mathbf{q}_{j-1} + \alpha_j \mathbf{q}_j + \beta_j \mathbf{q}_{j+1}$.
At breakdown s, set $\beta_s := 0$; the recurrence through $j = s$ remains valid.

(iii) *CG, conjugate-direction form*: Suppose $A \succ 0$ and fix $\mathbf{b}, \mathbf{x}_0 \in \mathbb{F}^n$. Set $\mathbf{r}_0 := \mathbf{b} - A\mathbf{x}_0$. If $\mathbf{r}_0 = \mathbf{0}$, return $\mathbf{x}_0$.
Otherwise, let $k \in J_n$ satisfy $\dim(\mathcal{K}_k(A, \mathbf{r}_0)) = k$. Define $\langle \mathbf{u}, \mathbf{v} \rangle_A := \mathbf{v}^* A\mathbf{u}$, an inner product by $A \succ 0$.
Apply Gram–Schmidt in this inner product to $(\mathbf{r}_0, A\mathbf{r}_0, \dots, A^{k-1}\mathbf{r}_0)$, obtaining an A-orthogonal basis $\mathbf{p}_0, \dots, \mathbf{p}_{k-1}$ of $\mathcal{K}_k(A, \mathbf{r}_0)$.
For $i \in \{0, \dots, k-1\}$, set $c_i := \frac{\mathbf{p}_i^* \mathbf{r}_0}{\mathbf{p}_i^* A \mathbf{p}_i}$. The denominators are positive by $A \succ 0$. Set $\mathbf{x}_k := \mathbf{x}_0 + \sum_{i=0}^{k-1} c_i \mathbf{p}_i$ and $\mathbf{r}_k := \mathbf{b} - A\mathbf{x}_k$.
For $i \in \{0, \dots, k-1\}$, A-orthogonality reduces $\mathbf{p}_i^* \mathbf{r}_k$ to $\mathbf{p}_i^* \mathbf{r}_0 - c_i \mathbf{p}_i^* A \mathbf{p}_i = 0$. Hence $\mathbf{r}_k \perp \mathcal{K}_k(A, \mathbf{r}_0)$.
This correction is unique: if $\mathbf{z}, \mathbf{z}' \in \mathcal{K}_k(A, \mathbf{r}_0)$ both satisfy the residual condition, then $\mathbf{d} := \mathbf{z} - \mathbf{z}'$ lies in the same space.
Their residual difference is $-A\mathbf{d} \perp \mathcal{K}_k(A, \mathbf{r}_0)$, so $\mathbf{d}^* A \mathbf{d} = 0$ and $\mathbf{d} = \mathbf{0}$ by $A \succ 0$.

(iv) *GMRES*: Fix $\mathbf{b}, \mathbf{x}_0 \in \mathbb{F}^n$ and $k \in J_n$. Set $\mathbf{r}_0 := \mathbf{b} - A\mathbf{x}_0$. Set $\mathcal{A}_k := \mathbf{x}_0 + \mathcal{K}_k(A, \mathbf{r}_0)$.
If $\mathbf{r}_0 = \mathbf{0}$, return $\mathbf{x}_0$. Otherwise, set $s := \dim(\mathcal{K}_k(A, \mathbf{r}_0))$.
Choose an orthonormal basis matrix $Q_s \in \mathrm{Mat}(n, s, \mathbb{F})$ for this subspace and $\mathbf{y}_k \in \mathrm{LS}(AQ_s, \mathbf{r}_0)$ by Theorem 2.4.4(ii).
Set $\mathbf{x}_k := \mathbf{x}_0 + Q_s \mathbf{y}_k$. Since $\mathcal{A}_k = \{\mathbf{x}_0 + Q_s \mathbf{y} : \mathbf{y} \in \mathbb{F}^s\}$, $\mathbf{x}_k \in \operatorname{arg\,min}_{\mathbf{x} \in \mathcal{A}_k} \|\mathbf{b} - A\mathbf{x}\|_2$.

These templates access A through matrix-vector products. With stored products, a basis extension uses one new multiplication by A. The remaining work consists of inner products, vector updates, and reduced problems. The prescribed loops are finite because $m, k \le n$. In finite precision, orthogonality and convergence require a separate rounding-error analysis.

Perspective 6.4.3. *Projection Conditions across Krylov Methods*
Definition 6.4.1 supplies the common search spaces in Computation Method 6.4.2.
Arnoldi constructs orthonormal coordinates, and Lanczos reduces the recurrence when A is Hermitian.
CG uses A-orthogonal directions to impose residual orthogonality. GMRES instead minimizes the Euclidean residual norm.
These correspondences are exact-arithmetic statements; they do not supply finite-precision convergence guarantees.

6.5 Stationary Iterations and Splittings

Throughout this subsection, $n \in \mathbb{Z}_{>0}$.

Definition 6.5.1. *Matrix Splitting Iteration*
Let $A, S, T \in \mathrm{Mat}(n, n, \mathbb{C})$ satisfy $A = S - T$, with S invertible. Set $B := S^{-1}T$, and let $\mathbf{b}, \mathbf{x}_0 \in \mathbb{C}^n$.
The *splitting iteration* for $A\mathbf{x} = \mathbf{b}$ is $(\forall\, k \in \mathbb{Z}_{\ge 0})\, S\mathbf{x}_{k+1} = T\mathbf{x}_k + \mathbf{b}$. Equivalently, $(\forall\, k \in \mathbb{Z}_{\ge 0})\, \mathbf{x}_{k+1} = B\mathbf{x}_k + S^{-1}\mathbf{b}$.
If $A\mathbf{x}_* = \mathbf{b}$ and $\mathbf{e}_k := \mathbf{x}_k - \mathbf{x}_*$, then $(\forall\, k \in \mathbb{Z}_{\ge 0})\, \mathbf{e}_{k+1} = B\mathbf{e}_k$.

Theorem 6.5.2. *Splitting Convergence Criterion*
Let $A, S, T \in \mathrm{Mat}(n, n, \mathbb{C})$ satisfy $A = S - T$, with S invertible, and set $B := S^{-1}T$.
For $\mathbf{b}, \mathbf{x}_0 \in \mathbb{C}^n$, write $(\mathbf{x}_k(\mathbf{b}, \mathbf{x}_0))_{k \ge 0}$ for the splitting sequence in Definition 6.5.1.
Then $\rho(B) < 1$ if and only if A is invertible and $(\forall\, (\mathbf{b}, \mathbf{x}_0) \in (\mathbb{C}^n)^2)\, \mathbf{x}_k(\mathbf{b}, \mathbf{x}_0) \to A^{-1}\mathbf{b}$.

Proof.
($\Rightarrow$) By Theorem 3.3.5, $B^k \to 0$. If $(I_n - B)\mathbf{z} = \mathbf{0}$, then $B\mathbf{z} = \mathbf{z}$, so $\mathbf{z} = \mathbf{0}$ by $\rho(B) < 1$.
Thus $I_n - B$ is invertible by Corollary 3.6.3(vi)$\Rightarrow$(i), and $A = S(I_n - B)$ is invertible.
Fix $\mathbf{b}, \mathbf{x}_0 \in \mathbb{C}^n$, set $\mathbf{x}_* := A^{-1}\mathbf{b}$, $\mathbf{e}_k := \mathbf{x}_k(\mathbf{b}, \mathbf{x}_0) - \mathbf{x}_*$. Then $\mathbf{e}_k = B^k \mathbf{e}_0$ by Definition 6.5.1, induction, so $\mathbf{x}_k(\mathbf{b}, \mathbf{x}_0) \to A^{-1}\mathbf{b}$.
($\Leftarrow$) Fix $\mathbf{z} \in \mathbb{C}^n$, $(\mathbf{b}, \mathbf{x}_0) = (\mathbf{0}, \mathbf{z})$. Then $B^k \mathbf{z} = \mathbf{x}_k(\mathbf{0}, \mathbf{z}) \to \mathbf{0}$ by global-convergence hypothesis, Definition 6.5.1.
The convergence $B^k \to 0$ holds entrywise by specializing $\mathbf{z}$ to the standard basis vectors, and $\rho(B) < 1$ by Theorem 3.3.5.
Structural Commentary: The forward direction propagates errors by B^k; the reverse tests homogeneous iterations on a basis.
Key Insight: Arbitrary initial errors decay exactly when $B^k \to 0$. □

Corollary 6.5.3. *Singular Matrices Obstruct Splitting Convergence*
Let $A, S, T \in \mathrm{Mat}(n, n, \mathbb{C})$ satisfy $A = S - T$, with S invertible, and suppose A is singular. Then:
(i) $1 \in \mathrm{spec}(S^{-1}T)$;
(ii) $\rho(S^{-1}T) \ge 1$.

Proof.
(i) Choose $\mathbf{0} \ne \mathbf{x} \in \ker(A)$ by Corollary 3.6.3(i)$\Leftrightarrow$(vi). Then $(S - T)\mathbf{x} = \mathbf{0}$, so $S^{-1}T\mathbf{x} = \mathbf{x}$ and $1 \in \mathrm{spec}(S^{-1}T)$.
(ii) The inequality $\rho(S^{-1}T) \ge 1$ holds by (i) and Definition 3.3.3.
Structural Commentary: The identity $A\mathbf{x} = \mathbf{0}$ converts a null vector into a 1-eigenvector of $S^{-1}T$.
Key Insight: A fixed error mode prevents spectral-radius contraction. □

Computation Method 6.5.4. *Stationary Iterative Method*
Let $M \in \mathrm{Mat}(n, n, \mathbb{C})$ and $\mathbf{c}, \mathbf{x}_0 \in \mathbb{C}^n$.
(i) Generate $(\forall\, k \in \mathbb{Z}_{\ge 0})\, \mathbf{x}_{k+1} = M\mathbf{x}_k + \mathbf{c}$.
(ii) If $\rho(M) < 1$, then $I_n - M$ is invertible by Theorem 6.5.2, with $A = I_n - M$, $S = I_n$, and $T = M$.
The vector $\mathbf{x}_* := (I_n - M)^{-1}\mathbf{c}$ is the unique fixed point.
(iii) Under $\rho(M) < 1$, $\mathbf{x}_k - \mathbf{x}_* = M^k(\mathbf{x}_0 - \mathbf{x}_*) \to \mathbf{0}$ by Theorem 3.3.5.
This defines an infinite exact-arithmetic sequence. A finite implementation must additionally specify a stopping test and tolerance.

Computation Method 6.5.5. *Jacobi, Gauss–Seidel, and SOR Splittings*
Let $A = D + L + U \in \mathrm{Mat}(n, n, \mathbb{C})$, where D is diagonal and invertible, L is strictly lower triangular, and U is strictly upper triangular.
For a splitting $A = S - T$, use the iteration in Definition 6.5.1.
(i) *Jacobi*: $S = D$, $T = -(L + U)$, $B_J = -D^{-1}(L + U)$.
(ii) *Gauss–Seidel*: $S = D + L$, $T = -U$, $B_{\mathrm{GS}} = -(D + L)^{-1}U$.
(iii) *SOR*: For $\omega \in \mathbb{R} \setminus \{0\}$, $S_\omega = \frac{1}{\omega}D + L$, $T_\omega = \left(\frac{1}{\omega} - 1\right) D - U$, and $B_\omega = S_\omega^{-1} T_\omega$.
The transposes of $D + L$ and S_ω are upper triangular with nonzero diagonal entries.
The matrices $D + L$ and S_ω are invertible by Theorem 3.8.9(i) and Corollary 3.6.4.
Global convergence of listed scheme equivalent to $\rho(B) < 1$ for iteration matrix by Theorem 6.5.2. For $\omega = 1$, SOR is Gauss–Seidel.

Corollary 6.5.6. *Jacobi Convergence under Strict Diagonal Dominance*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ be strictly diagonally dominant, $\mathbf{b}, \mathbf{x}_0 \in \mathbb{C}^n$, $(\mathbf{x}_k)$ be Jacobi sequence. Then $\mathbf{x}_k \to A^{-1}\mathbf{b}$.

Proof. The inequalities $|a_{ii}| > R_i^{\text{off}}(A) \geq 0$ hold for $i \in J_n$ by Definition 5.5.1, so D is invertible by Theorem 3.8.9(i).
From $A = D + L + U$, $B_J = -D^{-1}(L+U) = I_n - D^{-1}A$ and $(\forall\, i \in J_n)\,[(B_J)_{ii} = 0, \sum_{j \in J_n \setminus \{i\}} |(B_J)_{ij}| = \frac{R_i^{\text{off}}(A)}{|a_{ii}|} < 1]$.
Hence $\operatorname{spec}(B_J) \subseteq \{z \in \mathbb{C} : |z| < 1\}$ by Theorem 5.4.5, so $\rho(B_J) < 1$. Convergence $\mathbf{x}_k \to A^{-1}\mathbf{b}$ holds by Theorem 6.5.2.
Structural Commentary: Strict dominance becomes Gersgorin disks for B_J centered at 0 with radii below 1.
Key Insight: Row dominance confines the Jacobi error spectrum to the unit disk. □

Example 6.5.7. *Two-by-Two Jacobi and Gauss–Seidel*
For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, write $A = D + L + U$.
The Jacobi matrix is $B_J = -D^{-1}(L+U) = \left(\begin{smallmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{smallmatrix}\right)$, with $\rho(B_J) = \frac{1}{2}$.
The Gauss–Seidel matrix is $B_{\text{GS}} = -(D+L)^{-1}U = \left(\begin{smallmatrix} 0 & -\frac{1}{2} \\ 0 & \frac{1}{4} \end{smallmatrix}\right)$, with $\rho(B_{\text{GS}}) = \frac{1}{4}$.
Both schemes converge globally by Theorem 6.5.2.
Thus $\rho(B_{\text{GS}}) < \rho(B_J) < 1$; this spectral-radius comparison does not assert uniformly smaller transient error.

Example 6.5.8. *Poisson Tridiagonal Stationary Rates*
Let $A_n = \operatorname{tridiag}(-1, 2, -1)$ and $\theta_k := \frac{k\pi}{n+1}$ for $k \in J_n$.
For $\mathbf{v}^{(k)} = [\sin(\theta_k)\ \cdots\ \sin(n\theta_k)]^t$, $A_n\mathbf{v}^{(k)} = (2 - 2\cos(\theta_k))\mathbf{v}^{(k)}$ by the sine addition formulas.
These n eigenvalues are distinct, so $B_J = I_n - \frac{1}{2}A_n$ has eigenvalues $\cos(\theta_k)$ and $\rho(B_J) = \cos\left(\frac{\pi}{n+1}\right)$.
For $m \in J_n$, write $A_m = D_m + L_m + U_m$, and let $B_{J,m}$ and $B_{\text{GS},m}$ be its iteration matrices.
Set $p_m(\mu) := \det(\mu I_m - B_{J,m})$ and $q_m(\lambda) := \det(\lambda I_m - B_{\text{GS},m})$, with $p_0 = q_0 := 1$.
Since $\det(D_m + L_m) = 2^m$ and $\lambda I_m - B_{\text{GS},m} = (D_m + L_m)^{-1}(\lambda(D_m + L_m) + U_m)$, $q_m(\lambda) = 2^{-m}\det(\lambda(D_m + L_m) + U_m)$.
By tridiagonal determinant expansion, $p_1(\mu) = \mu$, $q_1(\lambda) = \lambda$, and, for $2 \leq m \leq n$,
$p_m(\mu) = \mu p_{m-1}(\mu) - \frac{1}{4}p_{m-2}(\mu)$, and $q_m(\lambda) = \lambda q_{m-1}(\lambda) - \frac{\lambda}{4}q_{m-2}(\lambda)$. The identity $(\forall\, \mu \in \mathbb{C})\, q_m(\mu^2) = \mu^m p_m(\mu)$ holds for $m = 0, 1$.
If it holds at $m-1, m-2$, then $q_m(\mu^2) = \mu^m(\mu p_{m-1}(\mu) - \frac{1}{4}p_{m-2}(\mu)) = \mu^m p_m(\mu)$, so it holds for $0 \leq m \leq n$ by induction.
For $\mu \neq 0$, $q_n(\mu^2) = 0 \iff p_n(\mu) = 0$; hence $\operatorname{spec}(B_{\text{GS}}) \setminus \{0\} = \{\lambda^2 : \lambda \in \operatorname{spec}(B_J) \setminus \{0\}\}$. Thus, $\rho(B_{\text{GS}}) = \rho(B_J)^2 = \cos^2\left(\frac{\pi}{n+1}\right)$.
As $n \to \infty$, both radii tend to 1, so the iterations become asymptotically slow and motivate acceleration or preconditioning.

Perspective 6.5.9. *Preconditioning Preview*
Let $A, M \in \operatorname{Mat}(n, n, \mathbb{C})$, with M invertible, and let $\mathbf{b} \in \mathbb{C}^n$.
The left-preconditioned system $M^{-1}A\mathbf{x} = M^{-1}\mathbf{b}$ is equivalent to $A\mathbf{x} = \mathbf{b}$.
The right-preconditioned system $AM^{-1}\mathbf{y} = \mathbf{b}$ is equivalent under $\mathbf{x} = M^{-1}\mathbf{y}$.
Favorable conditioning or iteration spectra and inexpensive solves with M are design goals, not consequences of invertibility alone.

6.6 Hessenberg Reduction and QR Eigenvalue Methods

Throughout this subsection, $n \in \mathbb{Z}_{>0}$ and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 6.6.1. *Upper Hessenberg Matrix*
Matrix $A = [a_{ij}] \in \operatorname{Mat}(n, n, \mathbb{F})$ is *upper Hessenberg* iff $(\forall\, (i,j) \in J_n^2)\,[i > j+1 \implies a_{ij} = 0]$.

Numerical eigenvalue methods introduce structural zeros by unitary similarity.
Upper Hessenberg form is the standard preliminary reduction for QR iteration.
Figure 101 separates unrestricted entries from forced zeros below the first subdiagonal.

$$\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \end{bmatrix}$$

Figure 101: Upper Hessenberg pattern: asterisks are unrestricted and entries below the first subdiagonal are zero.

Theorem 6.6.2. *Unitary Upper Hessenberg Reduction*
Every $A \in \operatorname{Mat}(n, n, \mathbb{F})$ is unitarily similar to an upper Hessenberg matrix with real nonnegative subdiagonal entries.

Proof. For $n = 1$, take $U = I_1$. Assume $n \geq 2$ and set $A_0 := A$.
For $k \in J_{n-1}$ in increasing order, assume columns $1, \ldots, k-1$ have the required form.
This condition is vacuous for $k = 1$. Let $\mathbf{w}_k \in \mathbb{F}^{n-k}$ be the tail of column k of A_{k-1}.
If $\mathbf{w}_k = \mathbf{0}$, set $V_k := I_{n-k}$. Otherwise set $\mathbf{v}_k := \frac{\mathbf{w}_k}{\|\mathbf{w}_k\|_2}$; then $\|\mathbf{v}_k\|_2 = 1$.
Choose unitary V_k with $V_k\mathbf{e}_1 = \mathbf{v}_k$ by Corollary 2.7.4(iii) in the nonzero case.
In either case, $V_k^*\mathbf{w}_k = \|\mathbf{w}_k\|_2\,\mathbf{e}_1$ by the defining case and unitarity. Set $U_k := I_k \oplus V_k$ and $A_k := U_k^* A_{k-1} U_k$.
The identity $U_k^* U_k = I_n$ holds by block multiplication. The first $k-1$ columns are fixed by right multiplication with U_k.
They are fixed by left multiplication with U_k^* because only their zero tails are mixed.
Column k has the required zeros and subdiagonal entry $\|\mathbf{w}_k\|_2 \in \mathbb{R}_{\geq 0}$ by the tail identity.
For $U := U_1 \cdots U_{n-1}$, U is unitary by identities $U_k^* U_k = I_n$. Identity $A_{n-1} = U^* A U$ holds by iteration.
Matrix A_{n-1} has the required form by induction.
Structural Commentary: Trailing block similarities preserve completed columns while aligning the next column tail.
Key Insight: $V_k^*\mathbf{w}_k = \|\mathbf{w}_k\|_2\,\mathbf{e}_1$ installs both zeros and phase normalization. □

Perspective 6.6.3. *Why Hessenberg Form Matters*
Unitary similarity preserves p_A by Theorem 2.1.7(i), and hence the spectrum by Theorem 2.1.3(ii).
Upper Hessenberg form leaves only the first subdiagonal to eliminate in a QR factorisation.

Computation Method 6.6.4. *QR Eigenvalue Algorithm Preview*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. Choose unitary U such that $A_0 := U^*AU$ is upper Hessenberg by Theorem 6.6.2. For $k \in \mathbb{Z}_{\geq 0}$, perform:

(i) Choose $\mu_k \in \mathbb{F}$ and factor $A_k - \mu_k I_n = Q_k R_k$ by Theorem 4.1.7(i). Here Q_k is unitary and R_k is upper triangular.
(ii) Set $A_{k+1} = R_k Q_k + \mu_k I_n = Q_k^* A_k Q_k$.

$(\forall\, k \in \mathbb{Z}_{\geq 0})\ A_k \sim A$ by induction through the initialization and update similarities.
$(\forall\, k \in \mathbb{Z}_{\geq 0})\ \mathrm{spec}(A_k) = \mathrm{spec}(A)$ by the preceding relation, Theorem 2.1.7(i), and Theorem 2.1.3(ii).
No shift rule, deflation test, stopping criterion, or convergence claim is specified.

Computation Method 6.6.5. *Hessenberg QR Step by Plane Unitaries*
Assume $n \geq 2$. Let $H \in \mathrm{Mat}(n,n,\mathbb{C})$ be upper Hessenberg, and let $\mu \in \mathbb{C}$.

(i) Set $B_0 := H - \mu I_n$.
(ii) For $j \in J_{n-1}$, set $\mathbf{z}_j := \begin{pmatrix} (B_{j-1})_{jj} \\ (B_{j-1})_{j+1,j} \end{pmatrix}$. If $\mathbf{z}_j = \mathbf{0}$, set $\widehat{G}_j := I_2$. Otherwise set $\mathbf{u}_j := \frac{\mathbf{z}_j}{\|\mathbf{z}_j\|_2}$, so $\|\mathbf{u}_j\|_2 = 1$.
In the nonzero case, choose unitary $\widehat{G}_j$ with $\widehat{G}_j \mathbf{e}_1 = \mathbf{u}_j$ by Corollary 2.7.4(iii).
In either case, $\widehat{G}_j^* \mathbf{z}_j = \|\mathbf{z}_j\|_2 \mathbf{e}_1$ by the construction.
Let G_j act as $\widehat{G}_j$ on coordinates $j, j+1$ and as the identity elsewhere. Set $B_j := G_j^* B_{j-1}$.
(iii) Set $Q := G_1 \cdots G_{n-1}$ and $R := B_{n-1}$.
(iv) Define $H_+ := RQ + \mu I_n = Q^* H Q$.

Matrix B_0 is upper Hessenberg. At iteration j, the two mixed rows vanish in columns $1, \ldots, j-1$ by induction.
Their mixing preserves these zeros, and the tail identity annihilates entry $(j+1, j)$.
Thus B_j is upper Hessenberg with its first j columns upper triangular by induction.
Hence R is upper triangular and $H - \mu I_n = QR$ by $B_{n-1} = Q^*(H - \mu I_n)$.
Set $Q_0 := I_n$ and $Q_j := G_1 \cdots G_j$. Each Q_{j-1} acts only on the first j coordinates.
Thus $Q_{j-1}\mathbf{e}_{j+1} = \mathbf{e}_{j+1}$, and Q_j is upper Hessenberg by induction.
Matrix Q is unitary by the unitarity of each G_j. The identity $H_+ = Q^* H Q$ holds by the shifted factorisation.
For $i > j+1$, $(RQ)_{ij} = \sum_{\ell=1}^{n} r_{i\ell} q_{\ell j} = 0$ by $r_{i\ell} \neq 0 \Longrightarrow \ell \geq i$ and $q_{\ell j} \neq 0 \Longrightarrow \ell \leq j+1$. Hence $H_+ = RQ + \mu I_n$ is upper Hessenberg.
The identity $\mathrm{spec}(H_+) = \mathrm{spec}(H)$ holds by Theorem 2.1.7(i) and Theorem 2.1.3(ii).
The construction is exact-arithmetic; no floating-point stability or deflation claim is made.

6.7 Power and Shift-Invert Iterations

Throughout this subsection, $n \in \mathbb{Z}_{\geq 2}$.
For $\mathcal{E} \leq \mathbb{C}^n$, write $d(\mathbf{z}, \mathcal{E})$ for Euclidean distance, as in Definition 2.2.7.

Theorem 6.7.1. *Power Iteration toward a Dominant Eigenspace*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ bebasis of $\mathbb{C}^n$. Let $(\lambda_1, \ldots, \lambda_n) \in \mathbb{C}^n$ satisfy $(\forall\, j \in J_n)\ A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ and $|\lambda_1| > \max_{2 \leq j \leq n} |\lambda_j|$.
Let $(c_1, \ldots, c_n) \in \mathbb{C}^n$ satisfy $c_1 \neq 0$, and set $\mathbf{u}_0 := \sum_{j=1}^{n} c_j \mathbf{v}_j$ and $\mathcal{E}_1 := \mathrm{span}\{\mathbf{v}_1\}$. As $k \longrightarrow \infty$:

(i) $\lambda_1^{-k} A^k \mathbf{u}_0 \longrightarrow c_1 \mathbf{v}_1$;
(ii) $\frac{d(A^k \mathbf{u}_0, \mathcal{E}_1)}{\|A^k \mathbf{u}_0\|_2} \longrightarrow 0$.

Proof. The relation $\lambda_1 \neq 0$ holds by the modulus gap. Set $\rho := \max_{2 \leq j \leq n} \left|\frac{\lambda_j}{\lambda_1}\right| < 1$. For $k \in \mathbb{Z}_{\geq 0}$, set $\mathbf{r}_k := \sum_{j=2}^{n} c_j \left(\frac{\lambda_j}{\lambda_1}\right)^k \mathbf{v}_j$.
For $k \in \mathbb{Z}_{\geq 0}$, $\lambda_1^{-k} A^k \mathbf{u}_0 = c_1 \mathbf{v}_1 + \mathbf{r}_k$ by induction and linearity. Estimate $\|\mathbf{r}_k\|_2 \leq \rho^k \sum_{j=2}^{n} |c_j| \|\mathbf{v}_j\|_2$ holds by triangle inequality.
The convergence $\mathbf{r}_k \longrightarrow \mathbf{0}$ holds by $0 \leq \rho < 1$.

(i) Claim (i) holds by the scaled identity and $\mathbf{r}_k \longrightarrow \mathbf{0}$.
(ii) $(\forall\, k \in \mathbb{Z}_{\geq 0})\ A^k \mathbf{u}_0 \neq \mathbf{0}$ by the nonzero eigenbasis coordinate $c_1 \lambda_1^k$.
For $k \geq 0$, $\frac{d(A^k \mathbf{u}_0, \mathcal{E}_1)}{\|A^k \mathbf{u}_0\|_2} \leq \frac{\|\mathbf{r}_k\|_2}{\|c_1 \mathbf{v}_1 + \mathbf{r}_k\|_2}$ by distance homogeneity and $c_1 \mathbf{v}_1 \in \mathcal{E}_1$.
The denominator converges to $\|c_1 \mathbf{v}_1\|_2 > 0$ by the reverse triangle inequality.
Claim (ii) holds by the vanishing numerator and positive denominator limit.

Structural Commentary: Eigenbasis scaling isolates a geometric remainder; distance homogeneity transfers its decay to directions.
Key Insight: The spectral gap contracts every nondominant eigenvalue ratio. □

Computation Method 6.7.2. *Inverse, Shift-Invert, and Shifted Power Iteration*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis. Let $(\lambda_1, \ldots, \lambda_n) \in \mathbb{C}^n$ satisfy $(\forall\, j \in J_n)\ A\mathbf{v}_j = \lambda_j \mathbf{v}_j$.
Relabel target eigenpair as $(\lambda_1, \mathbf{v}_1)$. Let $(c_1, \ldots, c_n) \in \mathbb{C}^n$ satisfy $c_1 \neq 0$, set $\mathbf{x}_0 := \sum_{j=1}^{n} c_j \mathbf{v}_j$ and $\mathcal{E}_1 := \mathrm{span}\{\mathbf{v}_1\}$.
Set $\mathcal{N}(\mathbf{z}) := \frac{\mathbf{z}}{\|\mathbf{z}\|_2}$ for $\mathbf{z} \neq \mathbf{0}$. Define a transformed matrix B and eigenvalues $(\theta_j)_{j=1}^n$ by one branch:

(i) *Inverse*: Assume $0 \notin \mathrm{spec}(A)$ and $|\lambda_1| < \min_{2 \leq j \leq n} |\lambda_j|$. Matrix A is invertible by Corollary 5.1.8.
Set $B := A^{-1}$ and $(\theta_j)_{j=1}^n := (\lambda_j^{-1})_{j=1}^n$. Factor A once. For $k \in \mathbb{Z}_{\geq 0}$, solve $A\mathbf{y}_{k+1} = \mathbf{x}_k$ and set $\mathbf{x}_{k+1} := \mathcal{N}(\mathbf{y}_{k+1})$.
(ii) *Shift-Invert*: Choose $\sigma \in \mathbb{C} \setminus \mathrm{spec}(A)$ with $|\lambda_1 - \sigma| < \min_{2 \leq j \leq n} |\lambda_j - \sigma|$.
Matrix $A - \sigma I_n$ is invertible by Corollary 3.4.4 with $p(z) = z - \sigma$. Set $B := (A - \sigma I_n)^{-1}$ and $(\theta_j)_{j=1}^n := \big((\lambda_j - \sigma)^{-1}\big)_{j=1}^n$.
Factor $A - \sigma I_n$ once. For $k \in \mathbb{Z}_{\geq 0}$, solve $(A - \sigma I_n)\mathbf{y}_{k+1} = \mathbf{x}_k$ and set $\mathbf{x}_{k+1} := \mathcal{N}(\mathbf{y}_{k+1})$.
(iii) *Shifted Power*: Choose $\sigma \in \mathbb{C}$ with $|\lambda_1 - \sigma| > \max_{2 \leq j \leq n} |\lambda_j - \sigma|$. Set $B := A - \sigma I_n$ and $(\theta_j)_{j=1}^n := (\lambda_j - \sigma)_{j=1}^n$.
For $k \in \mathbb{Z}_{\geq 0}$, set $\mathbf{x}_{k+1} := \mathcal{N}\big((A - \sigma I_n)\mathbf{x}_k\big)$.

Each branch satisfies $(\forall\, j \in J_n)\, B\mathbf{v}_j = \theta_j \mathbf{v}_j$ by the original eigenpair identities.
It also satisfies $|\theta_1| > \max_{2 \le j \le n} |\theta_j|$ and $\theta_1 \neq 0$ by the branch hypothesis and the definition of θ_j.
$(\forall\, k \in \mathbb{Z}_{\ge 0})\, B^k \mathbf{x}_0 \neq \mathbf{0}$ by the nonzero eigenbasis coordinate $c_1 \theta_1^k$. $(\forall\, k \in \mathbb{Z}_{>0})(\exists\, \alpha_k \in \mathbb{C} \setminus \{0\})\, \mathbf{x}_k = \alpha_k B^k \mathbf{x}_0$ by induction.
The identity $(\forall\, k \in \mathbb{Z}_{>0})\ \|\mathbf{x}_k\|_2 = 1$ holds by the definition of $\mathcal{N}$.
The convergence $\frac{d(B^k \mathbf{x}_0, \mathcal{E}_1)}{\|B^k \mathbf{x}_0\|_2} \longrightarrow 0$ holds by Theorem 6.7.1(ii) with $A = B$ and $\lambda_j = \theta_j$.
Hence $d(\mathbf{x}_k, \mathcal{E}_1) \longrightarrow 0$ by homogeneity of distance and $\|\mathbf{x}_k\|_2 = 1$.
Branches (i) and (ii) reuse one factorization and never form the displayed inverse matrices.
These are exact-arithmetic iterations; no finite-precision stability, tolerance, or stopping rule is specified.

6.8 Numerical SVD Methods

Throughout this subsection, $m, n \in \mathbb{Z}_{>0}$ and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Computation Method 6.8.1. *Bidiagonal Reduction and SVD Reconstruction*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and set $q := \min\{m, n\}$.
A matrix $B = [b_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$ is *upper bidiagonal* iff $(\forall\, (i, j) \in J_m \times J_n)\, j \notin \{i, i+1\} \Longrightarrow b_{ij} = 0$.
The method constructs unitary $P \in \mathrm{Mat}(m, m, \mathbb{F})$ and $Q \in \mathrm{Mat}(n, n, \mathbb{F})$ such that $B = P^* A Q$ is upper bidiagonal.
If $m \ge n$ and $\mathrm{rank}(A) = n$, the positive eigenvalues of $A^* A$ are $\sigma_1(A)^2, \ldots, \sigma_n(A)^2$ by Proposition 4.2.11(i).
Thus forming $A^* A$ squares the extremal singular-value ratio; for square invertible A, this is $\kappa_2(A^* A) = \kappa_2(A)^2$ by Proposition 5.3.2(v).
The reduction instead acts directly on A:

(i) Set $B_0 := A$, $P_0 := I_m$, and $Q_0 := I_n$.
(ii) For $k \in J_q$, construct a left unitary L_k and a right unitary R_k as follows:
  (a) Let $\mathbf{x}_k \in \mathbb{F}^{m-k+1}$ be column k of B_{k-1} in rows $k, \ldots, m$.
  If $\mathbf{x}_k = \mathbf{0}$, set $\widehat{L}_k := I_{m-k+1}$. Otherwise, choose a unitary $\widehat{L}_k$ satisfying $\widehat{L}_k \mathbf{e}_1 = \frac{\mathbf{x}_k}{\|\mathbf{x}_k\|_2}$ by Corollary 2.7.4(iii).
  Then $\widehat{L}_k^* \mathbf{x}_k = \|\mathbf{x}_k\|_2 \mathbf{e}_1$. Let L_k act as $\widehat{L}_k$ on rows $k, \ldots, m$ and as the identity on earlier coordinates, and set $C_k := L_k^* B_{k-1}$.
  (b) If $k = n$, set $R_k := I_n$. Otherwise, let $\mathbf{w}_k^*$ be row k of C_k in columns $k+1, \ldots, n$.
  If $\mathbf{w}_k = \mathbf{0}$, set $\widehat{R}_k := I_{n-k}$. Otherwise, choose a unitary $\widehat{R}_k$ satisfying $\widehat{R}_k \mathbf{e}_1 = \frac{\mathbf{w}_k}{\|\mathbf{w}_k\|_2}$ by Corollary 2.7.4(iii).
  Then $\mathbf{w}_k^* \widehat{R}_k = \|\mathbf{w}_k\|_2 \mathbf{e}_1^*$. Let R_k act as $\widehat{R}_k$ on columns $k+1, \ldots, n$ and as the identity on earlier coordinates.
  Set $B_k := L_k^* B_{k-1} R_k$, $P_k := P_{k-1} L_k$, and $Q_k := Q_{k-1} R_k$.
  Writing $B_k = [b_{ij}^{(k)}]$, the column invariant is $(\forall\, j \in J_k)(\forall\, i \in J_m)\, i > j \Longrightarrow b_{ij}^{(k)} = 0$.
  The row invariant is $(\forall\, i \in J_k)(\forall\, j \in J_n)\, j > i+1 \Longrightarrow b_{ij}^{(k)} = 0$; trailing-coordinate actions preserve both.
(iii) Set $P := P_q$, $Q := Q_q$, and $B := B_q$. The matrices P, Q are unitary by Proposition 2.6.12(i), and $B = P^* A Q$ by induction.
The invariant makes B upper bidiagonal, and A, B have the same singular values by Proposition 5.1.7(i).
(iv) Choose a full SVD $B = \widetilde{U} \Sigma \widetilde{V}^*$ by Theorem 4.2.9.
Set $U := P\widetilde{U}$ and $V := Q\widetilde{V}$. Then U, V are unitary by Proposition 2.6.12(i), and $A = U \Sigma V^*$ follows from $B = P^* A Q$.
These singular triples give optimal low-rank approximations by Theorem 5.2.2, and $A^\dagger = V \Sigma^\dagger U^*$ by Corollary 5.2.6.
For $\mathbf{y} \in \mathbb{F}^m$, $A^\dagger \mathbf{y}$ is the unique minimum-norm least-squares solution by Theorem 5.2.9(i).

The bidiagonal SVD solver, stopping rule, and rounding-error model remain unspecified.
No finite-precision stability, accuracy, or convergence guarantee is asserted.

6.9 Discrete Fourier Transforms and Circulant Matrices

Throughout this subsection, $n \in \mathbb{Z}_{>0}$.

Definition 6.9.1. *Discrete Fourier Matrix*
Set $\omega_n := e^{2\pi i / n}$, $F_n := [\omega_n^{jk}]_{j,k=0}^{n-1} \in \mathrm{Mat}(n, n, \mathbb{C})$, and $Q_n := n^{-1/2} F_n$.
For $\mathbf{c} \in \mathbb{C}^n$, $F_n \mathbf{c}$ is its unnormalized DFT and $Q_n \mathbf{c}$ its normalized DFT.

Definition 6.9.2. *Circulant Matrix*
Let $(\mathbf{e}_0, \ldots, \mathbf{e}_{n-1})$ be the 0-indexed standard basis of $\mathbb{C}^n$.
The *cyclic shift* is the matrix $P_n \in \mathrm{Mat}(n, n, \mathbb{C})$ defined by $(\forall\, j \in \{0, \ldots, n-1\})\, P_n \mathbf{e}_j = \mathbf{e}_{(j-1) \bmod n}$.
A matrix $C \in \mathrm{Mat}(n, n, \mathbb{C})$ is *circulant* iff $(\exists\, (a_0, \ldots, a_{n-1}) \in \mathbb{C}^n)\, C = \sum_{r=0}^{n-1} a_r P_n^r$.
If $(a_0, \ldots, a_{n-1}) \in \mathbb{C}^n$ satisfies this identity, set $p(z) := \sum_{r=0}^{n-1} a_r z^r \in \mathbb{C}[z]$; then $C = p(P_n)$.
Here $p(P_n)$ denotes the polynomial evaluation from Definition 3.8.5.

Theorem 6.9.3. *Fourier Diagonalization of Cyclic Shifts and Circulants*
Let F_n, Q_n be as in Definition 6.9.1 and P_n as in Definition 6.9.2. Set $\Omega_n := \mathrm{diag}(1, \omega_n, \ldots, \omega_n^{n-1})$.
For $(a_0, \ldots, a_{n-1}) \in \mathbb{C}^n$, set $p(z) := \sum_{r=0}^{n-1} a_r z^r$ and $C := p(P_n)$. Then:

(i) $F_n^* F_n = n I_n$, $Q_n^* Q_n = I_n$, and Q_n is unitary;
(ii) $P_n F_n = F_n \Omega_n$ and $Q_n^* P_n Q_n = \Omega_n$;
(iii) $Q_n^* C Q_n = p(\Omega_n) = \mathrm{diag}\big(p(1), p(\omega_n), \ldots, p(\omega_n^{n-1})\big)$.

Proof. For $k \in \{0, \ldots, n-1\}$, set $\mathbf{f}_k := (\omega_n^{jk})_{j=0}^{n-1}$. Then $F_n = [\mathbf{f}_0\ \cdots\ \mathbf{f}_{n-1}]$ by Definition 6.9.1.

(i) For $k, \ell \in \{0, \ldots, n-1\}$, $(F_n^* F_n)_{k\ell} = \sum_{r=0}^{n-1} \omega_n^{r(\ell - k)}$. This sum is n if $k = \ell$.
If $k \neq \ell$, then $W := \omega_n^{\ell - k}$ satisfies $W \neq 1$ by $0 < |\ell - k| < n$ and $W^n = 1$, so $\sum_{r=0}^{n-1} W^r = \frac{1 - W^n}{1 - W} = 0$.
Thus $F_n^* F_n = n I_n$ and $Q_n^* Q_n = n^{-1} F_n^* F_n = I_n$. Hence Q_n is unitary by Theorem 2.6.8(i).

(ii) For $k, j \in \{0, \dots, n-1\}$, $(P_n \mathbf{f}_k)_j = (\mathbf{f}_k)_{(j+1) \bmod n} = \omega_n^k (\mathbf{f}_k)_j$ by $\omega_n^{nk} = 1$.
Hence $P_n F_n = F_n \Omega_n$ and $P_n Q_n = Q_n \Omega_n$ by column comparison. Therefore $Q_n^* P_n Q_n = \Omega_n$ by (i).
(iii) Set $D := \operatorname{diag}\big(p(1), p(\omega_n), \dots, p(\omega_n^{n-1})\big)$. By (i),(ii), $P_n = Q_n \Omega_n Q_n^*$.
The identity $C = Q_n D Q_n^*$ holds by Theorem 1.1.1 with $(A, S, \Lambda) = (P_n, Q_n, \Omega_n)$.
The identity $p(\Omega_n) = D$ holds by the same theorem with $(A, S, \Lambda) = (\Omega_n, I_n, \Omega_n)$. Hence $Q_n^* C Q_n = D = p(\Omega_n)$ by (i).

Structural Commentary: Root orthogonality proves unitarity; shift eigenvectors transfer polynomial evaluation to the diagonal.
Key Insight: The Fourier modes form an orthonormal eigenbasis of the cyclic shift. □

Figure 102 summarizes Theorem 6.9.3. Horizontal arrows conjugate by Q_n, and vertical arrows evaluate p.

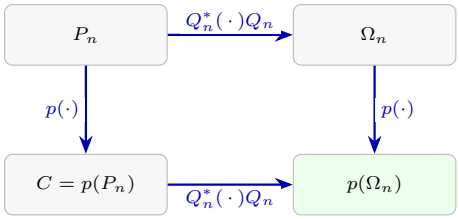

Figure 102: Fourier conjugation commutes with polynomial evaluation: P_n maps to Ω_n and $C = p(P_n)$ to $p(\Omega_n)$.

Proposition 6.9.4. *Inverse of the Fourier Matrix*
The Fourier matrix satisfies $F_n^{-1} = \frac{1}{n} F_n^*$.

Proof. The identity $\big(\frac{1}{n} F_n^*\big) F_n = I_n$ holds by Theorem 6.9.3(i). Hence $F_n^{-1} = \frac{1}{n} F_n^*$ by Theorem 3.4.2.
Structural Commentary: Fourier orthogonality supplies a left inverse; squareness identifies the two-sided inverse.
Key Insight: Normalization turns conjugate transpose into exact inversion. □

6.10 Fast Fourier Transform

Computation Method 6.10.1. *Radix-Two Fast Fourier Transform*
For $q \in \mathbb{Z}_{>0}$, let ω_q and F_q be as in Definition 6.9.1 with $n = q$. Let $\ell \in \mathbb{Z}_{\geq 0}$, set $n := 2^\ell$, and let $\mathbf{c} = (c_r)_{r=0}^{n-1} \in \mathbb{C}^n$.
Under exact complex arithmetic, the following recursion computes $\mathbf{y} = F_n \mathbf{c}$:

(i) If $\ell = 0$, set $\mathbf{y} := \mathbf{c}$ and stop.
(ii) If $\ell \geq 1$, set $m := \frac{n}{2} = 2^{\ell-1}$. Set $\mathbf{c}' := (c_{2r})_{r=0}^{m-1} \in \mathbb{C}^m$ and $\mathbf{c}'' := (c_{2r+1})_{r=0}^{m-1} \in \mathbb{C}^m$.
(iii) Recursively compute $\mathbf{y}' := F_m \mathbf{c}'$ and $\mathbf{y}'' := F_m \mathbf{c}''$.
(iv) Set $(\forall\, j \in \{0, \dots, m-1\})\, d_j := \omega_n^j$.
Set $(\forall\, j \in \{0, \dots, m-1\}) \big(y_j := y'_j + d_j y''_j,\ y_{j+m} := y'_j - d_j y''_j\big)$.

Termination: Each call decreases the exponent by 1, so the recursion reaches (i) after ℓ levels.
Correctness: If $\ell = 0$, then $F_1 = [1]$, and (i) returns $\mathbf{y} = F_1 \mathbf{c}$. Let $\ell \geq 1$, assume recursive calls return $\mathbf{y}' = F_m \mathbf{c}'$ and $\mathbf{y}'' = F_m \mathbf{c}''$.
Fix $j \in \{0, \dots, m-1\}$ and set $E_j := \sum_{r=0}^{m-1} \omega_m^{jr} c_{2r}$ and $O_j := \sum_{r=0}^{m-1} \omega_m^{jr} c_{2r+1}$. Then $E_j = y'_j$ and $O_j = y''_j$ by the recursive postconditions.
For $s \in \{0, 1\}$, the even-index contribution to $(F_n \mathbf{c})_{j+sm}$ is E_j by $\omega_n^2 = \omega_m$ and $\omega_n^{2m} = 1$.
The odd-index contribution is $(-1)^s d_j O_j$ by the same identities and $\omega_n^m = -1$. Hence $(F_n \mathbf{c})_{j+sm} = y'_j + (-1)^s d_j y''_j$.
Taking $s = 0, 1$ recovers (iv), so $\mathbf{y} = F_n \mathbf{c}$ by induction on ℓ.
Cost Model: Count unit-cost complex arithmetic with precomputed twiddle factors.
Exclude indexing, data movement, storage, and recursion overhead. Each butterfly reuses one multiplication in two additions.
Across each nonbase recursion depth, there are $\frac{n}{2}$ butterflies.
Thus the ℓ depths use $\frac{n\ell}{2}$ complex multiplications and $n\ell$ complex additions.
Dense multiplication by F_n uses n^2 complex multiplications and $n(n-1)$ complex additions.
Limitations: Only $n = 2^\ell$ is supported; no finite-precision accuracy or stability is asserted.

Let $\mathbf{c} = (c_0, c_1, c_2, c_3)^t \in \mathbb{C}^4$, and set $\mathbf{c}' := (c_0, c_2)^t$ and $\mathbf{c}'' := (c_1, c_3)^t$. Set $\mathbf{y}' := F_2 \mathbf{c}'$, $\mathbf{y}'' := F_2 \mathbf{c}''$, $\mathbf{y} := F_4 \mathbf{c}$, and $D := \operatorname{diag}(1, i)$.
Set $B_4(\mathbf{u}, \mathbf{v}) := \binom{\mathbf{u}+D\mathbf{v}}{\mathbf{u}-D\mathbf{v}}$ for $\mathbf{u}, \mathbf{v} \in \mathbb{C}^2$. Figure 103 instantiates the recurrence at $n = 4$.
It splits $\mathbf{c}$, applies two F_2 transforms, and recombines their outputs as $\mathbf{y} = B_4(\mathbf{y}', \mathbf{y}'')$.

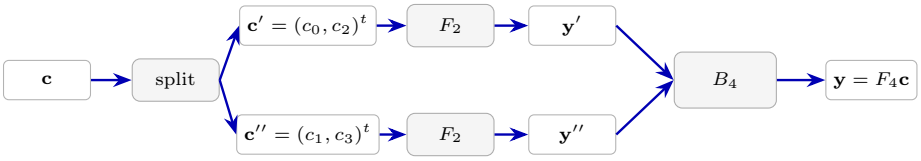

Figure 103: The $n = 4$ radix-two split: two F_2 transforms feed the butterfly B_4 and produce $F_4 \mathbf{c}$.

Part VII
Groups, Rings, and Fields

1 Algebraic Language and Categories

1.1 Universal Properties and Equivalence-Based Structure

1.2 Categories and Functors

1.3 Algebraic Structures as Objects

1.4 Exactness and Diagrams

2 Group Theory

2.1 Groups, Subgroups, and Cyclic Groups

Groups are the algebraic language of reversible operations. They appear here as the additive structure underlying rings and modules, and later as symmetry groups acting on mathematical objects.

Definition 2.1.1. *Groups*
A *group* is a nonempty set G with binary operation $(a,b)\mapsto ab$, identity element $e_G\in G$, and inversion map $g\mapsto g^{-1}$ such that:

(i) $(\forall\, a,b,c\in G)\,(ab)c=a(bc)$;
(ii) $(\forall\, g\in G)\,e_Gg=ge_G=g$;
(iii) $(\forall\, g\in G)\,gg^{-1}=g^{-1}g=e_G$.

The group is *abelian* if $(\forall\, a,b\in G)\,ab=ba$.
In an abelian group the operation is often written additively, with identity 0 and inverse $-a$.

Example 2.1.2. *Basic Groups*
Let S be a set, let $m,n\in\mathbb{N}$, and let $\mathbb{F}$ be a field.

(i) The bijections $S\to S$ form a group under composition; it is usually nonabelian.
(ii) $\mathrm{Mat}(m,n,\mathbb{F})$ is an abelian group under matrix addition.
(iii) $\mathrm{Mat}(n,n,\mathbb{F})^\times$, the invertible $n\times n$ matrices, is a group under multiplication.
(iv) $\mathbb{Z}/n\mathbb{Z}$ is a finite abelian group under addition modulo n.

Definition 2.1.3. *Subgroup*
Let G be a group. A subset $H\subseteq G$ is a *subgroup*, written $H\le G$, if $e_G\in H$, $(\forall\, a,b\in H)\,ab\in H$, and $(\forall\, a\in H)\,a^{-1}\in H$.

Proposition 2.1.4. *Subgroup Test*
Let G be a group and let $H\subseteq G$ be nonempty. Then $H\le G\iff(\forall\, a,b\in H)\,ab^{-1}\in H$.

Proof.
($\Rightarrow$) If $H\le G$ and $a,b\in H$, then $b^{-1}\in H$ and $ab^{-1}\in H$.
($\Leftarrow$) Assume $(\forall\, a,b\in H)\,ab^{-1}\in H$. Choose $h\in H$. Then $e_G=hh^{-1}\in H$. For $b\in H$, $b^{-1}=e_Gb^{-1}\in H$. For $a,c\in H$, since $c^{-1}\in H$, we have $ac=a(c^{-1})^{-1}\in H$. Hence $H\le G$.

Structural Commentary: Closure under one expression packages identity, inverses, and products.
Key Insight: In a group, subtraction-style closure is enough. □

Definition 2.1.5. *Generated and Cyclic Groups*
Let G be a group.

(i) For $a\in G$, define $a^0:=e_G$, $a^{n+1}:=a^na$ for $n\in\mathbb{N}_0$, and $a^{-n}:=(a^n)^{-1}$ for $n\in\mathbb{N}$.
(ii) For $a\in G$, the *subgroup generated by a* is $\langle a\rangle:=\{a^k:k\in\mathbb{Z}\}\le G$.
(iii) G is *cyclic* if $(\exists\, a\in G)\,G=\langle a\rangle$. If it exists, $\mathrm{ord}(a):=\min\{n\in\mathbb{N}:a^n=e_G\}$; then $\langle a\rangle=\{e_G,a,\dots,a^{\mathrm{ord}(a)-1}\}$.

2.2 Group Homomorphisms, Kernels, and Images

Definition 2.2.1. *Group Homomorphisms, Kernels, and Images*
Let G,H be groups.

(i) A *homomorphism* $\varphi:G\to H$ satisfies $(\forall\, a,b\in G)\,\varphi(ab)=\varphi(a)\varphi(b)$.
(ii) For a homomorphism $\varphi:G\to H$, $\ker\varphi:=\{g\in G:\varphi(g)=e_H\}$; and $\mathrm{im}\,\varphi:=\varphi(G)$. An *isomorphism* is a bijective homomorphism.

Proposition 2.2.2. *Basic Homomorphism Properties*
Let $\varphi:G\to H$ be a group homomorphism. Then:

(i) $\varphi(e_G)=e_H$;
(ii) $(\forall\, g\in G)\,\varphi(g^{-1})=\varphi(g)^{-1}$;
(iii) $\ker\varphi\le G$;
(iv) $\mathrm{im}\,\varphi\le H$.

Proof.
(i) Put $x:=\varphi(e_G)$. Then $x=x^2$, so $e_H=x^{-1}x=x^{-1}x^2=x$.
(ii) For $g\in G$, the identities $\varphi(g)\varphi(g^{-1})=e_H=\varphi(g^{-1})\varphi(g)$ hold by Definition 2.2.1. Thus $\varphi(g^{-1})=\varphi(g^{-1})\varphi(g)\varphi(g)^{-1}=\varphi(g)^{-1}$.
(iii) The identity $e_G\in\ker\varphi$ holds by (i). If $a,b\in\ker\varphi$, then $\varphi(ab^{-1})=\varphi(a)\varphi(b)^{-1}=e_H$ by Definition 2.2.1 and (ii). Hence $\ker\varphi\le G$ by Proposition 2.1.4.
(iv) The identity $e_H=\varphi(e_G)\in\mathrm{im}\,\varphi$ holds by (i). If $x=\varphi(a)$ and $y=\varphi(b)$ lie in $\mathrm{im}\,\varphi$, then $xy^{-1}=\varphi(ab^{-1})\in\mathrm{im}\,\varphi$ by Definition 2.2.1 and (ii). Hence $\mathrm{im}\,\varphi\le H$ by Proposition 2.1.4.

Structural Commentary: Homomorphisms preserve all group structure once they preserve multiplication.
Key Insight: $\varphi(ab) = \varphi(a)\varphi(b)$ preserves e, inverses, kernels, and images. □

Proposition 2.2.3. *Kernel Criterion for Injectivity*
Let $\varphi : G \to H$ be a group homomorphism. Then φ injective $\iff$ $\ker\varphi = \{e_G\}$.

Proof.
($\Rightarrow$) If φ injective, $g \in \ker\varphi$, then $\varphi(g) = e_H = \varphi(e_G)$ by Proposition 2.2.2, so $g = e_G$. Hence $\ker\varphi = \{e_G\}$.
($\Leftarrow$) If $\ker\varphi = \{e_G\}$ and $\varphi(a) = \varphi(b)$, then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H$ by Definition 2.2.1 and Proposition 2.2.2(ii). Thus $ab^{-1} = e_G$ and $a = (ab^{-1})b = b$.
Structural Commentary: Injectivity is detected by the identity fiber $\ker\varphi$.
Key Insight: The kernel measures exactly the failure of injectivity. □

2.3 Cosets, Normality, and Structure Theorems

2.4 Group Actions, Orbits, and Stabilizers

A group action packages the idea that elements of a group act as reversible motions of a set. The conjugation action is the finite-group ingredient used later in Wedderburn's theorem.

Definition 2.4.1. *Actions, Orbits, and Stabilizers*
Let G be a group and let $\mathcal{X}$ be a set.
(i) A *left action* of G on $\mathcal{X}$ is a map $G \times \mathcal{X} \to \mathcal{X}$, $(g, x) \mapsto gx$, satisfying $(\forall\, x \in \mathcal{X})\, e_G x = x$ and $(\forall\, g, h \in G)(\forall\, x \in \mathcal{X})\, (gh)x = g(hx)$.
(ii) The *orbit* of $x \in \mathcal{X}$ is $Gx := \{gx : g \in G\}$.
(iii) The *stabilizer* of $x \in \mathcal{X}$ is $G_x := \{g \in G : gx = x\}$.

Theorem 2.4.2. *Orbit-Stabilizer*
Let a finite group G act on $\mathcal{X}$. Then $(\forall\, x \in \mathcal{X})\, |Gx| = [G : G_x] = \frac{|G|}{|G_x|}$.

Proof. Define $\Phi : G/G_x \to Gx$, $\Phi(gG_x) = gx$. For $g, h \in G$, $gG_x = hG_x \iff h^{-1}g \in G_x \iff h^{-1}gx = x \iff gx = hx$. Thus Φ is well defined and injective. It is surjective by definition of Gx, hence bijective, so $|Gx| = |G/G_x| = [G : G_x]$.
Structural Commentary: Orbits are parametrized by cosets of the stabilizer.
Key Insight: Same orbit point $\iff$ same stabilizer coset. □

2.5 Conjugacy Classes and the Class Equation

Definition 2.5.1. *Conjugacy Classes and Centralizers*
Let G be a group.
(i) The *conjugation action* of G on itself is $g \cdot x := gxg^{-1}$.
(ii) The *conjugacy class* of $x \in G$ is $\{x\}^G := \{gxg^{-1} : g \in G\}$.
(iii) The *centralizer* of $x \in G$ is $C_G(x) := \{g \in G : gx = xg\}$.
(iv) The *center* of G is $Z(G) := \{g \in G : (\forall\, x \in G)\, gx = xg\}$.

Theorem 2.5.2. *Class Equation*
Let G be finite and let $S \subseteq G$ contain one representative of each noncentral conjugacy class. Then $|G| = |Z(G)| + \sum_{x \in S} [G : C_G(x)]$.

Proof. The conjugacy classes form a partition $G = Z(G) \sqcup \bigsqcup_{x \in S} \{x\}^G$. Indeed, $\left|\{x\}^G\right| = 1 \iff (\forall\, g \in G)\, gxg^{-1} = x \iff x \in Z(G)$.
For $x \in S$, $\left|\{x\}^G\right| = [G : C_G(x)]$ by Theorem 2.4.2 with conjugation action from Definition 2.5.1. Take cardinalities for formula.
Structural Commentary: G decomposes into central singleton classes and noncentral conjugacy classes.
Key Insight: Count conjugacy classes by orbit-stabilizer. □

2.6 Sylow Theory and Finite Groups

2.7 Representation-Theoretic Preview

3 Rings, Ideals, and Polynomial Rings

3.1 Rings, Identities, Fields, and Examples

Rings add multiplication to an abelian group.

Definition 3.1.1. *Rings, Identities, and Fields*
Let R be a set with binary operations $+$ and $\cdot$.
(i) R is a *ring* if $(R, +)$ is an abelian group, $(\forall\, a, b, c \in R)\, (ab)c = a(bc)$, and $(\forall\, a, b, c \in R)\, [a(b + c) = ab + ac \wedge (a + b)c = ac + bc]$. Denote the additive identity by 0_R and the additive inverse of a by $-a$.
(ii) A ring R is *commutative* if $(\forall\, a, b \in R)\, ab = ba$.
(iii) A ring R has *identity* if it is equipped with $1_R \in R$ satisfying $(\forall\, a \in R)\, 1_R a = a1_R = a$.
(iv) A *field* is a commutative ring with identity $1_R \neq 0_R$ equipped with an inversion map $R \setminus \{0_R\} \to R \setminus \{0_R\}$, $a \mapsto a^{-1}$, satisfying $(\forall\, a \in R \setminus \{0_R\})\, aa^{-1} = a^{-1}a = 1_R$.

Example 3.1.2. *Basic Rings*
Let $n \in \mathbb{N}$ and let $\mathbb{F}$ be a field.
(i) $(\mathbb{Z}, +, \cdot)$ and $(\mathbb{F}[x], +, \cdot)$ are commutative rings with identity 1.
(ii) $\mathrm{Mat}(n, n, \mathbb{F})$ is ing with identity under matrix addition and multiplication, with identity I_n; noncommutative for $n \geq 2$.
(iii) $2\mathbb{Z}$ is a commutative ring under inherited addition and multiplication, but has no identity.
(iv) $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring under addition and multiplication modulo n, with identity $1 + n\mathbb{Z}$.

3.2 Ring Homomorphisms and Subrings

Definition 3.2.1. *Ring Homomorphisms and Subrings*
Let R, S be rings with identity.

(i) A *ring homomorphism* $\varphi : R \to S$ is a map satisfying $(\forall\, a, b \in R)\,[\varphi(a+b) = \varphi(a) + \varphi(b) \wedge \varphi(ab) = \varphi(a)\varphi(b)]$ and $\varphi(1_R) = 1_S$.
(ii) A *subring* of R is a subset $T \subseteq R$, written $T \leq R$, that is a ring under the inherited operations and has identity 1_R.

Proposition 3.2.2. *Subring Test*
Let R be a ring with identity and let $S \subseteq R$. Then $S \leq R \iff [1_R \in S \wedge (\forall\, a, b \in S)\,(a - b \in S \wedge ab \in S)]$.

Proof.
($\Rightarrow$) If $S \leq R$, then $1_R \in S$ and S is closed under subtraction and multiplication.
($\Leftarrow$) Assume $1_R \in S$ and $(\forall\, a, b \in S)\,(a-b \in S \wedge ab \in S)$. Then S is nonempty, $(S, +) \leq (R, +)$ by Proposition 2.1.4, and multiplication restricts to S. All ring laws and the identity 1_R are inherited from R, so $S \leq R$.
Structural Commentary: A subring is a subset stable under the operations and containing the ambient unit.
Key Insight: Subtraction handles the additive group part in one stroke. □

3.3 Ideals and Generated Ideals

Ideals are subsets by which quotient rings can be formed, just as subspaces are the subsets by which quotient vector spaces can be formed.

Definition 3.3.1. *Ideals and Generated Ideals*
Let R be a ring.

(i) A nonempty subset $\mathcal{I} \subseteq R$ is a *two-sided ideal*, written $\mathcal{I} \trianglelefteq R$, if $(\forall\, a, b \in \mathcal{I})\, a - b \in \mathcal{I}$ and $(\forall\, a \in \mathcal{I})(\forall\, r \in R)\,[ra \in \mathcal{I} \wedge ar \in \mathcal{I}]$.
(ii) If R is commutative with identity and $S \subseteq R$, the *ideal generated by S* is $\langle S \rangle := \left\{ \sum_{j=1}^{n} r_j s_j : n \in \mathbb{N}_0,\ r_j \in R,\ s_j \in S \right\}$, where $n = 0$ denotes the empty sum 0_R.
(iii) *Finite generated ideal notation*: if $n \in \mathbb{N}$ and $S = \{s_1, \ldots, s_n\}$, write $\langle s_1, \ldots, s_n \rangle := \langle S \rangle$.

For example, if $\mathbb{F}$ is a field and $p(x) \in \mathbb{F}[x]$, then $\langle p(x) \rangle = \{q(x)p(x) : q(x) \in \mathbb{F}[x]\}$ is the *principal* ideal generated by $p(x)$.

Proposition 3.3.2. *Generated Ideals Are Smallest*
Let R be a commutative ring with identity, $S \subseteq R$. Then $\langle S \rangle$ is an ideal, $S \subseteq \langle S \rangle$, and $\forall$ ideal $\mathcal{J} \subseteq R$, $S \subseteq \mathcal{J} \implies \langle S \rangle \subseteq \mathcal{J}$.

Proof. Let $x = \sum_{j=1}^{m} r_j s_j$ and $y = \sum_{k=1}^{n} t_k u_k$ lie in $\langle S \rangle$. By empty-sum convention, $0_R \in \langle S \rangle$. Moreover, $x - y = \sum_{j=1}^{m} r_j s_j + \sum_{k=1}^{n} (-t_k) u_k \in \langle S \rangle$.
For every $a \in R$, $ax = \sum_{j=1}^{m} (ar_j) s_j \in \langle S \rangle$ and $xa = ax$ by commutativity, so $\langle S \rangle$ is an ideal.
For every $s \in S$, $s = 1_R s \in \langle S \rangle$, so $S \subseteq \langle S \rangle$. If $\mathcal{J}$ is an ideal with $S \subseteq \mathcal{J}$, then $r_j s_j \in \mathcal{J}$ for every summand, while $(\mathcal{J}, +) \leq (R, +)$ by Proposition 2.1.4. Thus each finite sum lies in $\mathcal{J}$, so $\langle S \rangle \subseteq \mathcal{J}$, QED.
Structural Commentary: Generated ideals close a set under exactly the operations ideals must allow.
Key Insight: To force S into an ideal, one must include all ring-linear combinations of elements of S. □

Proposition 3.3.3. *Intersections and Chain Unions of Ideals*
Let R be a ring.

(i) If $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ is a nonempty family of ideals of R, then $\bigcap_{\alpha \in A} \mathcal{I}_\alpha$ is an ideal of R.
(ii) If $A \neq \emptyset$ and $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ is a chain of ideals of R under inclusion, then $\bigcup_{\alpha \in A} \mathcal{I}_\alpha$ is an ideal of R.

Proof.
(i) Let $\mathcal{I} = \bigcap_{\alpha \in A} \mathcal{I}_\alpha$. For every $\alpha \in A$, $0_R \in \mathcal{I}_\alpha$ by Proposition 2.1.4, so $0_R \in \mathcal{I}$.
If $a, b \in \mathcal{I}$ and $r \in R$, then $a - b, ra, ar \in \mathcal{I}_\alpha$ for every $\alpha \in A$, hence $a - b, ra, ar \in \mathcal{I}$.
(ii) Let $\mathcal{U} = \bigcup_{\alpha \in A} \mathcal{I}_\alpha$. Since $A \neq \emptyset$ and 0_R belongs to every ideal by Proposition 2.1.4, $0_R \in \mathcal{U}$.
If $a, b \in \mathcal{U}$, then $a \in \mathcal{I}_\alpha$ and $b \in \mathcal{I}_\beta$ for some α, β. Since the family is a chain, one of $\mathcal{I}_\alpha, \mathcal{I}_\beta$ contains both a, b; hence $a - b \in \mathcal{U}$.
If $a \in \mathcal{U}$ and $r \in R$, then $a \in \mathcal{I}_\alpha$ for some α, so $ra, ar \in \mathcal{I}_\alpha \subseteq \mathcal{U}$.
Structural Commentary: Chains avoid the usual failure of unions to preserve closure.
Key Insight: Comparability places any finite test inside one ideal. □

3.4 Quotient Rings and Maximal Ideals

Definition 3.4.1. *Quotient Rings*
Let R be a ring and let $\mathcal{I} \trianglelefteq R$ be a two-sided ideal.

(i) The underlying set is the additive quotient group $R/\mathcal{I} = \{a + \mathcal{I} : a \in R\}$.
(ii) For $a, b \in R$, addition and multiplication are $(a + \mathcal{I}) + (b + \mathcal{I}) = (a + b) + \mathcal{I}$ and $(a + \mathcal{I})(b + \mathcal{I}) = ab + \mathcal{I}$.
(iii) With these operations, $R/\mathcal{I}$ is the *quotient ring* of R by $\mathcal{I}$.

Proposition 3.4.2. *Ideals Make Quotient Multiplication Well Defined*
Let R be a ring and let $\mathcal{I} \leq (R, +)$. The rule $(a + \mathcal{I})(b + \mathcal{I}) := ab + \mathcal{I}$ for $a, b \in R$ is well defined on $R/\mathcal{I}$ iff $\mathcal{I} \trianglelefteq R$.

Proof.
($\Rightarrow$) If the rule is well defined and $i \in \mathcal{I}$, then $i + \mathcal{I} = 0_R + \mathcal{I}$.
For $r \in R$, $ri + \mathcal{I} = r0_R + \mathcal{I}$, and $ir + \mathcal{I} = 0_R r + \mathcal{I}$, so $ri, ir \in \mathcal{I}$. Thus $\mathcal{I} \trianglelefteq R$.
($\Leftarrow$) If $\mathcal{I} \trianglelefteq R$ and $a', a, b', b \in R$ satisfy $a' = a + i$, $b' = b + j$ with $i, j \in \mathcal{I}$, then $a'b' - ab = aj + ib + ij \in \mathcal{I}$.
Hence $a'b' + \mathcal{I} = ab + \mathcal{I}$, so the product is independent of representatives.

Structural Commentary: Ideals are exactly the additive subgroups compatible with multiplication of cosets.
Key Insight: Coset multiplication is well defined exactly for ideals. □

Definition 3.4.3. *Maximal Ideals*
Let R be a ring and let $\mathcal{M} \trianglelefteq R$.

(i) $\mathcal{M}$ is *proper* if $\mathcal{M} \neq R$.
(ii) $\mathcal{M}$ is *maximal* if $\mathcal{M} \neq R$ and $(\forall\, \mathcal{J} \trianglelefteq R)\,[\mathcal{M} \subseteq \mathcal{J} \subseteq R \Rightarrow (\mathcal{J} = \mathcal{M} \vee \mathcal{J} = R)]$.

Theorem 3.4.4. *Maximal Ideals and Fields*
Let R be a commutative ring with identity and let $\mathcal{I} \trianglelefteq R$. Then $R/\mathcal{I}$ is a field $\iff$ $\mathcal{I}$ is maximal.

Proof.
($\Rightarrow$) Suppose $R/\mathcal{I}$ is a field. Then $1_R + \mathcal{I} \neq 0_R + \mathcal{I}$, so $\mathcal{I} \neq R$. Let $\mathcal{I} \subsetneq \mathcal{J} \trianglelefteq R$ and choose $a \in \mathcal{J} \setminus \mathcal{I}$.
The nonzero coset $a + \mathcal{I}$ has an inverse $r + \mathcal{I}$, so $ra + \mathcal{I} = 1_R + \mathcal{I}$. Thus $1_R - ra \in \mathcal{I} \subseteq \mathcal{J}$ and $ra \in \mathcal{J}$, whence $1_R \in \mathcal{J}$.
For every $x \in R$, $x = x1_R \in \mathcal{J}$, so $\mathcal{J} = R$ and $\mathcal{I}$ is maximal.
($\Leftarrow$) Suppose $\mathcal{I}$ is maximal and let $a + \mathcal{I} \in R/\mathcal{I}$ satisfy $a + \mathcal{I} \neq 0_R + \mathcal{I}$. Then $a \notin \mathcal{I}$.
Set $\mathcal{K} := \langle \mathcal{I} \cup \{a\} \rangle$. Then $\mathcal{K} \trianglelefteq R$ and $\mathcal{I} \cup \{a\} \subseteq \mathcal{K}$ by Proposition 3.3.2, so $\mathcal{I} \subsetneq \mathcal{K}$. Maximality forces $\mathcal{K} = R$.
By Definition 3.3.1, an expression for $1_R \in \mathcal{K}$ can be regrouped as $1_R = i + ra$ for some $i \in \mathcal{I}$ and $r \in R$.
Hence $(r + \mathcal{I})(a + \mathcal{I}) = 1_R + \mathcal{I}$. The quotient ring is commutative with identity, and $\mathcal{I} \neq R$ because it is maximal.
Thus its zero and identity cosets are distinct, while every nonzero coset is invertible; hence $R/\mathcal{I}$ is a field.

Structural Commentary: Maximal ideals are exactly the proper ideals whose quotient rings are fields.
Key Insight: Maximality makes every nonzero coset invertible. □

Theorem 3.4.5. *Existence of Maximal Ideals*
If R is a commutative ring with identity and $1_R \neq 0_R$, then R has a maximal ideal.

Proof. Let $\mathcal{P} := \{\mathcal{I} \trianglelefteq R : \mathcal{I} \neq R\}$, ordered by inclusion. Then $\mathcal{P}$ is a poset by Proposition 6.1.5.
Also, $\{0_R\} = \langle \emptyset \rangle \trianglelefteq R$ by Proposition 3.3.2; as $1_R \neq 0_R$, $\{0_R\} \in \mathcal{P}$. Let $\mathcal{C} \subseteq \mathcal{P}$ be a chain. If $\mathcal{C} = \emptyset$, then $\{0_R\}$ upper bound for $\mathcal{C}$ in $\mathcal{P}$.
Assume $\mathcal{C} \neq \emptyset$ and set $\mathcal{U} := \bigcup_{\mathcal{I} \in \mathcal{C}} \mathcal{I}$. Then $\mathcal{U} \trianglelefteq R$ by Proposition 3.3.3.
Moreover, $\mathcal{U} \neq R$: if $1_R \in \mathcal{U}$, then $1_R \in \mathcal{I}$ for some $\mathcal{I} \in \mathcal{C}$, and $x = x1_R \in \mathcal{I}$ for every $x \in R$, contradicting $\mathcal{I} \in \mathcal{P}$.
Thus $\mathcal{U} \in \mathcal{P}$ is upper bound for $\mathcal{C}$. Every chain in $\mathcal{P}$ hence has an upper bound in $\mathcal{P}$, so $\mathcal{P}$ has a maximal element by Principle 6.1.3.
By Definition 3.4.3, this element is a maximal ideal of R.
Structural Commentary: Maximal ideals are produced by maximality, not by an explicit formula in general.
Key Insight: Chain unions keep properness because containing 1 is a finite obstruction. □

3.5 Integral Domains and Fields of Quotients

Definition 3.5.1. *Zero Divisors and Integral Domains*
Let R be a ring.

(i) A *zero divisor* is an element $a \in R \setminus \{0_R\}$ such that $(\exists\, b \in R \setminus \{0_R\})\,(ab = 0_R \vee ba = 0_R)$.
(ii) An *integral domain* is a commutative ring with identity $1_R \neq 0_R$ and no zero divisors.

Proposition 3.5.2. *Cancellation in Integral Domains*
Let R be a commutative ring with identity $1_R \neq 0_R$.
Then R is an integral domain $\iff$ $(\forall\, r, x, y \in R)\,[(r \neq 0_R \wedge rx = ry) \Rightarrow x = y]$.

Proof.
($\Rightarrow$) If $rx = ry$ and $r \neq 0_R$, then $r(x - y) = 0_R$. Since R has no zero divisors, $x - y = 0_R$, so $x = y$.
($\Leftarrow$) If $ab = 0_R$ and $a \neq 0_R$, then $ab = a0_R$, so the stated cancellation condition forces $b = 0_R$.
Commutativity therefore excludes zero divisors.

Structural Commentary: Domains are exactly cancellative commutative rings.
Key Insight: Zero divisors obstruct cancellation. □

Definition 3.5.3. *Field of Quotients*
Let R be an integral domain. For $a, c \in R$ and $b, d \in R \setminus \{0_R\}$, set $(a, b) \sim (c, d) \iff ad = bc$.
The *field of quotients* of R is $(R \times (R \setminus \{0_R\}))/\sim$; write $[(a, b)]_\sim = a/b$. Its operations are $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ and $\frac{a}{b}\frac{c}{d} = \frac{ac}{bd}$.

Proposition 3.5.4. *Domains Embed in Fields*
Let R be an integral domain with field of quotients K.
Then the map $\iota : R \to K$, $\iota(a) = a/1_R$, is an injective ring homomorphism.

Proof. By Definition 3.5.3, for $a, b \in R$, $\iota(a + b) = \frac{a+b}{1_R} = \frac{a}{1_R} + \frac{b}{1_R}$, $\iota(ab) = \frac{ab}{1_R} = \frac{a}{1_R}\frac{b}{1_R}$, and $\iota(1_R) = \frac{1_R}{1_R}$.
Thus ι is a ring homomorphism. If $\iota(a) = \iota(b)$, then $a/1_R = b/1_R$, so $a1_R = b1_R$ by Definition 3.5.3; hence $a = b$, and ι is injective.
Structural Commentary: The field of quotients formally adds denominators to a domain.
Key Insight: Integral domains are exactly the setting where fraction arithmetic is consistent. □

3.6 Ring Characteristic

Definition 3.6.1. *Characteristic*
Let R be a ring with identity. For $n \in \mathbb{N}$, define $n1_R := \underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}$.
The *characteristic* of R is $\operatorname{char}(R) := 0$ if $(\forall\, n \in \mathbb{N})\, n1_R \neq 0_R$; otherwise, $\operatorname{char}(R) := \min\{n \in \mathbb{N} : n1_R = 0_R\}$.

Proposition 3.6.2. *Characteristic of Finite Rings and Domains*
Let R be a finite ring with identity.

(i) $\operatorname{char}(R) > 0$.
(ii) If R is an integral domain, then $\operatorname{char}(R)$ *is prime.*

Proof.

(i) Choose $m \in \mathbb{N}_0$ and a bijection $e : R \to J_m$ by Definition 6.2.4. Define $f : \mathbb{N} \to R$ by $f(n) := n1_R$.
The composite $e \circ f : \mathbb{N} \to J_m$ is not injective by Proposition 6.2.5(i), so f is not injective.
Choose $i, j \in \mathbb{N}$ with $i < j$ and $i1_R = j1_R$. Then $(j-i)1_R = 0_R$, and $\operatorname{char}(R) > 0$ by Definition 3.6.1.

(ii) Suppose R is an integral domain and set $c := \operatorname{char}(R)$. Then $c > 1$ by (i), Definition 3.6.1, and $1_R \neq 0_R$.
If c is composite, choose $a, b \in \mathbb{N}$ with $c = ab$ and $1 < a, b < c$.
Then $(a1_R)(b1_R) = ab1_R = c1_R = 0_R$, whereas $a1_R \neq 0_R$ and $b1_R \neq 0_R$ by minimality of c.
c is prime by this contradiction with the absence of zero divisors in R.

Structural Commentary: Finiteness creates an additive period; zero-divisor exclusion forces primality.
Key Insight: A composite characteristic factors zero through smaller nonzero multiples. □

3.7 Units, Divisibility, and Factorization Domains

Definition 3.7.1. *Units, Associates, Prime Elements, and Irreducible Elements*
Let R be an integral domain.

(i) *Units*: $R^\times := \{u \in R : (\exists v \in R)\, uv = 1_R\}$.
(ii) *Divisibility*: for $a, b \in R$, $a \mid b \iff (\exists c \in R)\, b = ac$.
(iii) *Associates*: for $a, b \in R \setminus \{0_R\}$, $a \sim b \iff (\exists u \in R^\times)\, a = ub$.
(iv) *Prime element*: $p \in R \setminus (R^\times \cup \{0_R\})$ is prime precisely when $(\forall a, b \in R)\,[p \mid ab \Rightarrow (p \mid a \vee p \mid b)]$.
(v) *Irreducible element*: $r \in R \setminus (R^\times \cup \{0_R\})$ is irreducible precisely when $(\forall a, b \in R)\,[r = ab \Rightarrow (a \in R^\times \vee b \in R^\times)]$.

Proposition 3.7.2. *Divisibility and Principal Ideals*
Let R be an integral domain and let $r, s \in R \setminus \{0_R\}$.

(i) $r \in R^\times \iff \langle r \rangle = R$.
(ii) $r \sim s \iff \langle r \rangle = \langle s \rangle$.
(iii) $r \mid s \iff \langle s \rangle \subseteq \langle r \rangle$.

Proof.

(i) ($\Rightarrow$) If $r \in R^\times$, choose $v \in R$ with $rv = 1_R$. Then $1_R \in \langle r \rangle$, so $\langle r \rangle = R$.
(i) ($\Leftarrow$) If $\langle r \rangle = R$, choose $a \in R$ with $1_R = ar = ra$. Then $r \in R^\times$ by Definition 3.7.1(i).
(ii) ($\Rightarrow$) If $r = us$ with $u \in R^\times$, choose $v \in R$ with $uv = 1_R$. Then $r = us \in \langle s \rangle$ and $s = (vu)s = v(us) = vr \in \langle r \rangle$, so $\langle r \rangle = \langle s \rangle$.
(ii) ($\Leftarrow$) If $\langle r \rangle = \langle s \rangle$, choose $a, b \in R$ with $r = as$ and $s = br$. The equality $r = (ab)r$ and $r \neq 0_R$ imply $ab = 1_R$ by Proposition 3.5.2.
Thus $a \in R^\times$ and $r \sim s$ by Definition 3.7.1(i), (iii).
(iii) ($\Rightarrow$) If $r \mid s$, choose $a \in R$ with $s = ar$ by Definition 3.7.1(ii). Then $\langle s \rangle = \{xs : x \in R\} = \{xar : x \in R\} \subseteq \langle r \rangle$.
(iii) ($\Leftarrow$) If $\langle s \rangle \subseteq \langle r \rangle$, then $s \in \langle r \rangle = \{ar : a \in R\}$, so $r \mid s$ by Definition 3.7.1(ii).

Structural Commentary: Generators prove containment; cancellation detects units and associates.
Key Insight: A principal ideal records its generator up to multiplication by a unit. □

3.8 Polynomial Rings, Division, and GCDs

Polynomial rings over fields support division with remainder, Euclidean gcds, and Bezout identities.

Definition 3.8.1. *Polynomials over a Field*
Let $\mathbb{F}$ be a field.

(i) $\mathbb{F}[x]$ denotes the ring of polynomials in one variable over $\mathbb{F}$.

(ii) If $n \in \mathbb{N}_0$, $a_0, \ldots, a_n \in \mathbb{F}$, $a_n \neq 0$, and $p(x) = \sum_{j=0}^{n} a_j x^j$, set $\operatorname{lc}(p) := a_n$ and $\deg p := n$.
The polynomial p is *monic* iff $\operatorname{lc}(p) = 1$; set $\deg 0 = -\infty$.

(iii) A polynomial $p \in \mathbb{F}[x] \setminus \{0\}$ *splits over* $\mathbb{F}$ iff $(\exists m \in \mathbb{N}_0)(\exists \alpha \in \mathbb{F}^\times)(\exists \lambda_1, \ldots, \lambda_m \in \mathbb{F})\, p(x) = \alpha \prod_{i=1}^{m} (x - \lambda_i)$.

Theorem 3.8.2. *Polynomial Division Algorithm*
Let $\mathbb{F}$ be a field and let $f, g \in \mathbb{F}[x]$ with $g \neq 0$.

(i) $(\exists q, r \in \mathbb{F}[x])\,[f = qg + r \wedge (r = 0 \vee \deg r < \deg g)]$.
(ii) Suppose $f = qg + r = q'g + r'$ for $(q, r), (q', r') \in \mathbb{F}[x]^2$. If $(r = 0 \vee \deg r < \deg g) \wedge (r' = 0 \vee \deg r' < \deg g)$, then $(q, r) = (q', r')$.

Proof. Set $d := \deg g$ and $\ell := \operatorname{lc}(g)$. Call (q, r) admissible for $h \in \mathbb{F}[x]$ if $h = qg + r$ and $(r = 0 \vee \deg r < d)$.

(i) The pair $(0, 0)$ is admissible for 0. Suppose some polynomial has no admissible pair.
The shift $k \mapsto k + 1$ sends the nonzero counterexample degrees into $\mathbb{N}$.
Choose a counterexample f of minimal degree by Assumption 2.6.1; set $n := \deg f$.
If $n < d$, then $(0, f)$ is admissible for f, so $n \geq d$. Set $a := \operatorname{lc}(f)$ and $f_1 := f - a\ell^{-1}x^{n-d}g$.
Coefficient of x^n in f_1 is $a - a\ell^{-1}\ell = 0$, so $f_1 = 0$ or $\deg f_1 < n$. Admissible pair (q_1, r) for f_1 exists by zero case or minimality.
Set $q := q_1 + a\ell^{-1}x^{n-d}$. Then $f = qg + r$, so (q, r) is admissible for f, a contradiction.

(ii) Suppose (q, r) and (q', r') are admissible for f. Then $(q - q')g = r' - r$.
If $q \neq q'$, then $\operatorname{lc}(q - q')\ell \neq 0$, so $\deg((q - q')g) = \deg(q - q') + d \geq d$.
Since both remainders are zero or have degree below d, $\deg(r' - r) < d$, a contradiction.
Therefore $q = q'$, and $(q - q')g = r' - r$ implies $r = r'$.

Structural Commentary: Minimal-degree descent proves existence; degree comparison proves uniqueness.
Key Insight: The inverse leading coefficient cancels exactly one highest-degree term. □

Definition 3.8.3. *Divisibility, GCDs, and Coprimality*
Let $\mathbb{F}$ be a field.

(i) *Divisibility*: for $p, q \in \mathbb{F}[x]$, $p \mid q \iff (\exists h \in \mathbb{F}[x])\, q = hp$.

(ii) *Greatest common divisor*: for $(f,g) \in \mathbb{F}[x]^2 \setminus \{(0,0)\}$, a monic $d \in \mathbb{F}[x]$ is a gcd iff
  (a) $d \mid f$ and $d \mid g$;
  (b) $(\forall h \in \mathbb{F}[x])\,[(h \mid f \wedge h \mid g) \Rightarrow h \mid d]$.

(iii) *Coprimality*: $f, g \in \mathbb{F}[x]$ are relatively prime iff 1 is a greatest common divisor of f and g.

Theorem 3.8.4. *Polynomial GCD and Bezout Identity*
Let $\mathbb{F}$ be a field and let $f, g \in \mathbb{F}[x]$ be not both zero.

(i) A monic greatest common divisor of f and g exists.
(ii) Any two monic greatest common divisors of f and g are equal.
(iii) Writing $d := \gcd(f,g)$ for this unique polynomial, $(\exists a, b \in \mathbb{F}[x])\, d = af + bg$.
(iv) For $d := \gcd(f,g)$, $d = 1 \iff (\exists a, b \in \mathbb{F}[x])\, af + bg = 1$.

Proof.

(i) If exactly one of f, g is zero, let h be the nonzero polynomial and set $d := \mathrm{lc}(h)^{-1}h$.
Then d is monic, $h = \mathrm{lc}(h)d$, $d \mid 0$, and every common divisor of f, g divides d. Now assume $f, g \neq 0$ and set $(r_0, r_1) := (f, g)$.
For $r_i \neq 0$, choose (q_i, r_{i+1}) by Theorem 3.8.2(i) with inputs (r_{i-1}, r_i).
Then $r_{i-1} = q_i r_i + r_{i+1}$ and $r_{i+1} = 0$ or $\deg r_{i+1} < \deg r_i$. The nonzero remainder degrees strictly decrease in $\mathbb{N}_0$.
Hence the process terminates with $r_N \neq 0$ and $r_{N+1} = 0$ by Assumption 2.6.1.
Since $r_{N-1} = q_N r_N$, $r_N \mid r_{N-1}$. Backward induction in recurrence proves $r_N \mid f$ and $r_N \mid g$. Let c be a common divisor of f, g.
If $c \mid r_{i-1}$ and $c \mid r_i$, then $c \mid r_{i+1} = r_{i-1} - q_i r_i$. Thus $c \mid r_N$ by induction on i from $c \mid r_0$ and $c \mid r_1$.
Thus $d := \mathrm{lc}(r_N)^{-1} r_N$ is a monic greatest common divisor.

(ii) Let d, e be monic greatest common divisors. Then $d \mid e$ and $e \mid d$ by Definition 3.8.3(ii).
Choose $h, k \in \mathbb{F}[x]$ with $e = hd$ and $d = ke$. Since $d, e \neq 0$, $\deg e = \deg h + \deg d$ and $\deg d = \deg k + \deg e$.
Thus $\deg h = \deg k = 0$ and $\deg d = \deg e$. Since d, e are monic, $\mathrm{lc}(h) = 1$, so $h = 1$ and $e = d$.

(iii) If one input is zero, the construction in (i) is an $\mathbb{F}[x]$-linear combination of f and g.
For $f, g \neq 0$, each r_i is an $\mathbb{F}[x]$-linear combination of f, g by induction on i.
The induction step uses $r_{i+1} = r_{i-1} - q_i r_i$. Therefore $(\exists a, b \in \mathbb{F}[x])\, d = \mathrm{lc}(r_N)^{-1} r_N = af + bg$.

(iv) ($\Rightarrow$) If $d = 1$, then $(\exists a, b \in \mathbb{F}[x])\, af + bg = 1$ by (iii).

(iv) ($\Leftarrow$) Suppose $a, b \in \mathbb{F}[x]$ satisfy $af + bg = 1$. Then $d \mid 1$ because $d \mid f$ and $d \mid g$; write $1 = hd$.
Then $\deg h + \deg d = 0$, so $\deg d = 0$; monicity forces $d = 1$.

Structural Commentary: Descent constructs the gcd; back-substitution constructs its Bezout form.
Key Insight: Euclidean remainders preserve common divisors until the last nonzero term. □

3.9 Irreducible Polynomials and Unique Factorization

Definition 3.9.1. *Irreducible Polynomials*
Let $\mathbb{F}$ be a field and let $p \in \mathbb{F}[x]$ be nonconstant.
The polynomial p is *irreducible over $\mathbb{F}$* iff $(\forall a, b \in \mathbb{F}[x])\,[p = ab \Rightarrow (a \in \mathbb{F}[x]^\times \vee b \in \mathbb{F}[x]^\times)]$.

Proposition 3.9.2. *Irreducibility and the Prime Property in $\mathbb{F}[x]$*
Let $\mathbb{F}$ be a field and let $p \in \mathbb{F}[x]$ be nonconstant. Then p is irreducible over $\mathbb{F}$ iff $(\forall f, g \in \mathbb{F}[x])\,[p \mid fg \Rightarrow (p \mid f \vee p \mid g)]$.

Proof.

($\Rightarrow$) Assume p is irreducible and $p \mid fg$. If $p \mid f$, the conclusion holds.
Otherwise set $d := \gcd(p, f)$ and choose $A, B \in \mathbb{F}[x]$ with $d = Ap + Bf$ by Theorem 3.8.4(iii).
By Definition 3.8.3(ii), $d \mid p$ and $d \mid f$; choose $e \in \mathbb{F}[x]$ with $p = de$. Either d or e is a unit by Definition 3.9.1.
If e is a unit, choose $v \in \mathbb{F}[x]$ with $ev = 1$. Then $d = vp$, so $d \mid f$ implies $p \mid f$, a contradiction. Thus d is a unit.
Choose $u \in \mathbb{F}[x]$ with $ud = 1$. Now $1 = (uA)p + (uB)f$, so $g = (uA)pg + (uB)fg$. Since $p \mid (uA)pg$ and $p \mid (uB)fg$, $p \mid g$.

($\Leftarrow$) Assume the divisibility property and write $p = ab$. Then $p \mid a$ or $p \mid b$.
If $p \mid a$, choose $h \in \mathbb{F}[x]$ with $a = ph$. Then $(1 - hb)p = 0$. The nonzero-product property and $p \neq 0$ imply $hb = 1$, so $b \in \mathbb{F}[x]^\times$.
If $p \mid b$, choose $k \in \mathbb{F}[x]$ with $b = pk$. Then $(1 - ak)p = 0$. The nonzero-product property and $p \neq 0$ imply $ak = 1$, so $a \in \mathbb{F}[x]^\times$.
Thus p is irreducible by Definition 3.9.1.

Structural Commentary: GCD reduction proves primality; polynomial cancellation proves the converse.
Key Insight: Bezout converts coprimality with f into divisibility of g. □

Theorem 3.9.3. *Unique Factorization in $\mathbb{F}[x]$*
Let $\mathbb{F}$ be a field and let $f \in \mathbb{F}[x]$ be nonconstant.

(i) f admits a factorization $f = c\prod_{i=1}^{m} p_i$ with $m \in \mathbb{N}$, $c \in \mathbb{F}^\times$, and monic irreducibles $p_1, \ldots, p_m \in \mathbb{F}[x]$.

(ii) Suppose $f = c\prod_{i=1}^{m} p_i = d\prod_{j=1}^{n} q_j$, where both factorizations have the form in (i).
Then $m = n$, $c = d$, and $(\exists \sigma \in \mathfrak{S}_m)(\forall i \in \{1, \ldots, m\})\, p_i = q_{\sigma(i)}$.

Proof.

(i) Suppose a nonconstant polynomial without a factorization in (i) exists. Its degree lies in $\mathbb{N}$.
Choose a counterexample w of minimal degree by Assumption 2.6.1. If w is irreducible, set $c := \mathrm{lc}(w)$ and $p_1 := c^{-1}w$.
Scalar c is a unit, so p_1 is monic, irreducible; hence $w = cp_1$, a contradiction. Thus $w = uv$ with $u, v \notin \mathbb{F}[x]^\times$ by Definition 3.9.1.
Since nonzero constants are units, $\deg u, \deg v > 0$. Since $w = uv \neq 0$, $\deg w = \deg u + \deg v$, so $\deg u, \deg v < \deg w$.
By minimality, u and v have factorizations of the form in (i). Their product is such a factorization of w, a contradiction.

(ii) Suppose $f = c\prod_{i=1}^{m} p_i = d\prod_{j=1}^{n} q_j$ has two factorizations of the form in (i). Since $c, d \in \mathbb{F}^\times$, $p_1 \mid \prod_{j=1}^{n} q_j$.
Then $(\exists j \in \{1, \ldots, n\})\, p_1 \mid q_j$ by induction on n with Proposition 3.9.2.
Write $q_j = p_1 h$. Since q_j is irreducible and p_1 is not a unit, $h \in \mathbb{F}[x]^\times$. Choose $k \in \mathbb{F}[x]$ with $hk = 1$.
Since $hk = 1$, $\deg h = 0$ by degree additivity; monicity of p_1, q_j then forces $h = 1$, so $q_j = p_1$.
Reindex the q-factors to place q_j first; the nonzero-product property permits cancellation of p_1.

Repeating this step removes one factor from each side.
A nonempty product of irreducibles has positive degree, so neither side retains factors after the other is exhausted.
Thus $m = n$; the matched indices define $\sigma \in \mathfrak{S}_m$, and the remaining scalar equality is $c = d$.

Structural Commentary: Degree descent proves existence; prime-factor cancellation proves uniqueness.
Key Insight: Primality matches each irreducible factor with one factor on the other side. □

3.10 Principal Ideal Domains and Unique Factorization

Definition 3.10.1. *Principal Ideal Domains and Unique Factorization Domains*
Let R be an integral domain.

(i) The ring R is a *principal ideal domain (PID)* iff $(\forall\, \mathcal{I} \trianglelefteq R)(\exists\, a \in R)\, \mathcal{I} = \langle a \rangle$.
(ii) The ring R is a *unique factorization domain (UFD)* iff both conditions hold:
  (a) Every $a \in R \setminus (R^\times \cup \{0_R\})$ admits a factorization $a = \prod_{i=1}^{m} p_i$ with $m \in \mathbb{N}$ and irreducible $p_1, \dots, p_m \in R$.
  (b) If $a = \prod_{i=1}^{m} p_i = \prod_{j=1}^{n} q_j$ are two such factorizations, then $m = n$ and $(\exists\, \sigma \in \mathfrak{S}_m)(\forall\, i \in \{1, \dots, m\})\, p_i \sim q_{\sigma(i)}$.

Theorem 3.10.2. *$\mathbb{Z}$ and $\mathbb{F}[x]$ Are Principal Ideal Domains*
Let $\mathbb{F}$ be a field.

(i) Every ideal of $\mathbb{Z}$ is principal.
(ii) If $\{0\} \neq \mathcal{I} \trianglelefteq \mathbb{Z}$, then $\mathcal{I}$ has the unique positive generator $d = \min(\mathcal{I} \cap \mathbb{N})$.
(iii) Every ideal of $\mathbb{F}[x]$ is principal.
(iv) If $\{0\} \neq \mathcal{I} \trianglelefteq \mathbb{F}[x]$, then its unique monic generator is the unique monic element of $\mathcal{I}$ having minimal degree.

Proof.

(i) The zero ideal is $\langle 0 \rangle$. Let $\{0\} \neq \mathcal{I} \trianglelefteq \mathbb{Z}$. Then $\mathcal{I} \cap \mathbb{N} \neq \emptyset$, because $\mathcal{I}$ contains the positive one of $a, -a$ whenever $0 \neq a \in \mathcal{I}$.
Choose $d = \min(\mathcal{I} \cap \mathbb{N})$ by Assumption 2.6.1. For $a \in \mathcal{I}$, choose $q, r \in \mathbb{Z}$ with $a = qd + r$ and $0 \leq r < d$.
Since $r = a - qd \in \mathcal{I}$, $r = 0$ by minimality of d. Thus $\mathcal{I} \subseteq \langle d \rangle$. The reverse inclusion holds by $d \in \mathcal{I}$.
(ii) Let e be a positive generator of $\mathcal{I}$. Then $d \leq e$ because $e \in \mathcal{I} \cap \mathbb{N}$.
Since $d \in \langle e \rangle$, choose $k \in \mathbb{Z}$ with $d = ke$. Then $k \in \mathbb{N}$ by $d, e > 0$, so $e \leq d$. Hence $e = d$.
(iii) The zero ideal is $\langle 0 \rangle$. Let $\{0\} \neq \mathcal{I} \trianglelefteq \mathbb{F}[x]$.
A monic element of $\mathcal{I}$ exists by scaling a nonzero element by the inverse of its leading coefficient.
Choose a monic $m \in \mathcal{I}$ of minimal degree by Assumption 2.6.1 for the shifted degrees.
For $p \in \mathcal{I}$, write $p = qm + r$, where $r = 0$ or $\deg r < \deg m$, by Theorem 3.8.2(i).
Since $r = p - qm \in \mathcal{I}$, a nonzero r could be scaled to a monic element of smaller degree.
Hence $r = 0$ by minimality of m, so $\mathcal{I} \subseteq \langle m \rangle$; the reverse inclusion holds by $m \in \mathcal{I}$.
(iv) Let m' be another monic element of $\mathcal{I}$ with $\deg m' = \deg m$.
Then $m - m' \in \mathcal{I}$ is zero or has degree less than $\deg m$, so $m = m'$ by minimality of m.
If g is a monic generator of $\mathcal{I}$, then $\deg m \leq \deg g$ by minimality of m.
Choose $h \in \mathbb{F}[x]$ with $m = gh$. Then $\deg m = \deg g + \deg h$, so $\deg g \leq \deg m$. Thus $g = m$ by the preceding uniqueness.

Structural Commentary: Minimal representatives and division remainders identify ideal generators.
Key Insight: A nonzero smaller remainder contradicts the selected representative. □

Corollary 3.10.3. *GCDs Generate Polynomial Ideals*
Let $\mathbb{F}$ be a field, let $n \in \mathbb{N}$, and let $0 \neq p_1, \dots, p_n \in \mathbb{F}[x]$. Set $\mathcal{I} := \langle p_1, \dots, p_n \rangle$.

(i) A monic $m \in \mathbb{F}[x]$ with $\mathcal{I} = \langle m \rangle$ exists.
(ii) The monic generator m in (i) is unique.
(iii) $(\forall\, i \in \{1, \dots, n\})\, m \mid p_i$.
(iv) If $q \in \mathbb{F}[x]$ satisfies $(\forall\, i \in \{1, \dots, n\})\, q \mid p_i$, then $q \mid m$. Consequently, $m = \gcd(p_1, \dots, p_n)$.

Proof.

(i) The ideal $\mathcal{I}$ is nonzero because $p_1 \in \mathcal{I}$. A monic m with $\mathcal{I} = \langle m \rangle$ exists by Theorem 3.10.2(iv).
(ii) The monic generator is unique by Theorem 3.10.2(iv).
(iii) Let $i \in \{1, \dots, n\}$. Since $p_i \in \mathcal{I} = \langle m \rangle$, $m \mid p_i$ by Definition 3.7.1(ii).
(iv) Suppose $(\forall\, i \in \{1, \dots, n\})\, q \mid p_i$. Then $q \neq 0$ by $p_1 \neq 0$ and $q \mid p_1$.
The inclusions $(\forall\, i \in \{1, \dots, n\})\, \langle p_i \rangle \subseteq \langle q \rangle$ hold by Proposition 3.7.2(iii), so $\mathcal{I} \subseteq \langle q \rangle$.
Thus $m \in \langle q \rangle$, and $q \mid m$ by Definition 3.7.1(ii). Therefore m has the asserted gcd property.

Structural Commentary: Principal generation constructs the gcd; containment verifies universality.
Key Insight: Divisibility reverses inclusion of the corresponding principal ideals. □

Counterexample 3.10.4. *A Polynomial Ring That Is Not a PID*
The PID conclusion for $\mathbb{F}[x]$ in Theorem 3.10.2(iii) does not extend to $\mathbb{F}[x, y]$. Let $\mathbb{F}$ be a field and set $\mathcal{I} := \langle x, y \rangle \trianglelefteq \mathbb{F}[x, y]$.
Suppose $\mathcal{I} = \langle p \rangle$. Since $x \in \mathcal{I}$ and $x \neq 0$, $p \neq 0$. Choose $q, r \in \mathbb{F}[x, y]$ with $x = pq$ and $y = pr$.
As both products nonzero, $0 = \deg_y p + \deg_y q$, $0 = \deg_x p + \deg_x r$. Thus $p \in \mathbb{F}[x] \cap \mathbb{F}[y] = \mathbb{F}$. Since $p \neq 0$, it is a unit, so $\mathcal{I} = \mathbb{F}[x, y]$.
This contradicts $1 \notin \mathcal{I}$, because every element of $\mathcal{I}$ has zero constant term. Hence $\mathbb{F}[x, y]$ is not a PID.

Proposition 3.10.5. *PIDs Satisfy the Ascending Chain Condition*
If R is a PID and $(\mathcal{I}_j)_{j \in \mathbb{N}}$ is an ascending chain of ideals, then $(\exists\, N \in \mathbb{N})(\forall\, j \in \mathbb{N})\, [j \geq N \Rightarrow \mathcal{I}_j = \mathcal{I}_N]$.

Proof. Set $\mathcal{I} := \bigcup_{j \in \mathbb{N}} \mathcal{I}_j$. The set $\mathcal{I}$ is an ideal by Proposition 3.3.3. Choose $a \in R$ with $\mathcal{I} = \langle a \rangle$ by Definition 3.10.1(i).
Choose $N \in \mathbb{N}$ with $a \in \mathcal{I}_N$. Then $\mathcal{I} = \langle a \rangle \subseteq \mathcal{I}_N \subseteq \mathcal{I}$. If $j \geq N$, then $\mathcal{I}_N \subseteq \mathcal{I}_j \subseteq \mathcal{I} = \mathcal{I}_N$, so $\mathcal{I}_j = \mathcal{I}_N$.
Structural Commentary: The union is principal, and its generator enters one finite stage.
Key Insight: That stage contains the entire union and stabilizes the chain. □

Theorem 3.10.6. *Prime, Irreducible, and Bezout Properties in a PID*
Let R be a principal ideal domain.

(i) For $r \in R \setminus (R^\times \cup \{0_R\})$, r is irreducible iff $\langle r \rangle$ is a maximal ideal of R.
(ii) Every irreducible element of R is prime.
(iii) For $a, b \in R$, $(\forall\, h \in R)\,[(h \mid a \wedge h \mid b) \Rightarrow h \in R^\times]$ iff $(\exists\, x, y \in R)\, xa + yb = 1_R$.

Proof.

(i) ($\Rightarrow$) Suppose r is irreducible. The ideal $\langle r \rangle$ is proper by Proposition 3.7.2(i).
Let $\langle r \rangle \subseteq \mathcal{I} \trianglelefteq R$, and choose $a \in R$ with $\mathcal{I} = \langle a \rangle$ by Definition 3.10.1(i).
Choose $x \in R$ with $r = xa$. Then x or a is a unit by Definition 3.7.1(v).
If x is a unit, then $\langle r \rangle = \langle a \rangle = \mathcal{I}$ by Proposition 3.7.2(ii). If a is a unit, then $\mathcal{I} = R$ by Proposition 3.7.2(i). Thus $\langle r \rangle$ is maximal.
(i) ($\Leftarrow$) Suppose $\langle r \rangle$ is maximal and write $r = ab$. Then $\langle r \rangle \subseteq \langle a \rangle$, so $\langle a \rangle = \langle r \rangle$ or $\langle a \rangle = R$.
In the second case, a is a unit by Proposition 3.7.2(i). In the first case, $a \sim r$ by Proposition 3.7.2(ii).
Choose $u \in R^\times$ with $a = ur$ by Definition 3.7.1(iii); $r = urb$, $ub = 1_R$ by Proposition 3.5.2. Hence b is a unit, so r is irreducible.
(ii) Let r be irreducible and suppose $r \mid ab$. Set $\mathcal{K} := \langle r, a \rangle$. The ideal $\langle r \rangle$ is maximal by (i), so $\mathcal{K} = \langle r \rangle$ or $\mathcal{K} = R$.
In the first case, $a \in \langle r \rangle$, so $r \mid a$ by Definition 3.7.1(ii). In the second case, choose $x, y \in R$ with $1_R = xr + ya$.
Then $b = xrb + yab$. Since $r \mid xrb$ and $r \mid yab$, $r \mid b$. Hence r is prime by Definition 3.7.1(iv).
(iii) ($\Rightarrow$) Suppose $(\forall\, h \in R)\,[(h \mid a \wedge h \mid b) \Rightarrow h \in R^\times]$. Choose $d \in R$ with $\langle a, b \rangle = \langle d \rangle$ by Definition 3.10.1(i).
Since $a, b \in \langle d \rangle$, $d \mid a$, $d \mid b$ by Definition 3.7.1(ii). Thus d is a unit, so $\langle a, b \rangle = R$ by Proposition 3.7.2(i); $(\exists\, x, y \in R)\, xa + yb = 1_R$.
(iii) ($\Leftarrow$) Choose $x, y \in R$ with $xa + yb = 1_R$, and let $h \in R$ satisfy $h \mid a$ and $h \mid b$. Then $h \mid 1_R$, so h is a unit by Definition 3.7.1(i).

Structural Commentary: Maximality-to-primality meets Bezout identities from two-generator ideals.
Key Insight: Principal ideal containment converts factors and common divisors into units. □

Theorem 3.10.7. *Every PID Is a UFD*
Let R be a principal ideal domain.

(i) Every $a \in R \setminus (R^\times \cup \{0_R\})$ admits a factorization $a = \prod_{i=1}^{m} p_i$ with $m \in \mathbb{N}$ and irreducible $p_1, \ldots, p_m \in R$.

(ii) If $a = \prod_{i=1}^{m} p_i = \prod_{j=1}^{n} q_j$ are two such factorizations, then $m = n$ and $(\exists\, \sigma \in \mathfrak{S}_m)(\forall\, i \in \{1, \ldots, m\})\, p_i \sim q_{\sigma(i)}$.

Proof.

(i) Suppose a has no irreducible factorization and set $s_1 := a$.
Recursively suppose s_j is a nonzero nonunit without an irreducible factorization.
Since an irreducible element is itself a factorization, s_j is not irreducible.
Choose nonunits $u_j, v_j \in R$ with $s_j = u_j v_j$ by Definition 3.7.1(v).
At least one of u_j, v_j is nonfactorable; otherwise their factorizations concatenate to one of s_j.
Let s_{j+1} be a nonfactorable factor and let t_j be the other factor, so $s_j = s_{j+1} t_j$. Both factors are nonzero because $s_j \neq 0_R$.
Then $\langle s_j \rangle \subseteq \langle s_{j+1} \rangle$ by Proposition 3.7.2(iii). If equality held, then $s_{j+1} \sim s_j$ by Proposition 3.7.2(ii).
Choose $w \in R^\times$ with $s_{j+1} = w s_j$ by Definition 3.7.1(iii). Now $s_{j+1} = w s_{j+1} t_j$, so $1_R = w t_j$ by Proposition 3.5.2.
This contradicts that t_j is a nonunit. Hence $\langle s_1 \rangle \subsetneq \langle s_2 \rangle \subsetneq \cdots$, which is impossible by Proposition 3.10.5.
Therefore every nonzero nonunit has an irreducible factorization.

(ii) The proof is by induction on m. The element p_1 is prime by Theorem 3.10.6(ii).
Then $(\exists\, j \in \{1, \ldots, n\})\, p_1 \mid q_j$ by induction on n with the prime property. Reindex q-factors, choose $u \in R^\times$ with $q_1 = u p_1$.
Choose $v \in R$ with $uv = 1_R$ by Definition 3.7.1(i). Suppose first that $m = 1$. Equality $1_R = u \prod_{j=2}^{n} q_j$ holds by Proposition 3.5.2.
If $n > 1$, then $1_R = q_2 \big(u \prod_{j=3}^{n} q_j\big)$, so $q_2 \in R^\times$ by Definition 3.7.1(i), a contradiction. Thus $n = 1$ and $p_1 \sim q_1$.
Now let $m > 1$ and suppose the conclusion holds for factorizations with $m - 1$ factors on the left.
If $n = 1$, then $\prod_{i=2}^{m} p_i = u$ by Proposition 3.5.2. Then $1_R = p_2 \big(v \prod_{i=3}^{m} p_i\big)$, so p_2 is a unit by Definition 3.7.1(i), a contradiction.
Hence $n > 1$. Then $\prod_{i=2}^{m} p_i = (uq_2) \prod_{j=3}^{n} q_j$ by Proposition 3.5.2.
The element uq_2 is a nonzero nonunit, because otherwise $q_2 = v(uq_2)$ would be zero or a unit.
If $uq_2 = bc$, then $q_2 = (vb)c$. Thus vb or c is a unit by Definition 3.7.1(v).
If vb is a unit, then $b = u(vb)$ is a unit. Hence uq_2 is irreducible by Definition 3.7.1(v).
By the induction hypothesis, $m - 1 = n - 1$, and the tail factors match up to permutation and associates.
Therefore $m = n$, and the first match combined with the tail permutation defines $\sigma \in \mathfrak{S}_m$ with $(\forall\, i \in \{1, \ldots, m\})\, p_i \sim q_{\sigma(i)}$.

Structural Commentary: ACC terminates factor descent; primality matches irreducible factors.
Key Insight: Strict principal-ideal growth obstructs endless factor refinement. □

3.11 Roots of Unity and Cyclotomic Polynomials

Definition 3.11.1. *Roots of Unity, Primitive Roots, and Euler's Totient*
Let $n \in \mathbb{N}$.

(i) The complex *nth roots of unity* are $U_n := \{z \in \mathbb{C}^\times : z^n = 1\}$.
(ii) A root $\omega \in U_n$ is *primitive* iff $\operatorname{ord}(\omega) = n$ in $\mathbb{C}^\times$; set $\Omega_n := \{\omega \in U_n : \operatorname{ord}(\omega) = n\}$.
(iii) *Euler's totient* is $\varphi(n) := |\{k \in \{1, \ldots, n\} : \gcd(k, n) = 1\}|$.

Proposition 3.11.2. *Cyclic Structure of Roots of Unity*
Let $n \in \mathbb{N}$ and set $\zeta_n := \exp\left(\frac{2\pi i}{n}\right)$.

(i) $U_n = \langle \zeta_n \rangle$.
(ii) $|U_n| = n$.
(iii) $\Omega_n = \{\zeta_n^k : k \in \{1, \ldots, n\},\ \gcd(k, n) = 1\}$.
(iv) $|\Omega_n| = \varphi(n)$.

(v) $U_n = \coprod_{d|n} \Omega_d$.

Proof.

(i) The roots of $x^n - 1$ in $\mathbb{C}$ are precisely ζ_n^k for $k \in \{1, \dots, n\}$. Hence $U_n = \{\zeta_n^k : k \in \{1, \dots, n\}\} = \langle \zeta_n \rangle$.

(ii) Suppose $1 \leq j < k \leq n$ and $\zeta_n^j = \zeta_n^k$. Then $\exp\left(\frac{2\pi i(k-j)}{n}\right) = 1$, so $\frac{k-j}{n} \in \mathbb{Z}$, contrary to $0 < k - j < n$.
Thus the n representatives in (i) are distinct, so $|U_n| = n$.

(iii) Fix $k \in \{1, \dots, n\}$, set $g := \gcd(k, n)$, and write $k = gk_0$ and $n = gn_0$.
Then $(\zeta_n^k)^{n_0} = 1$. If $(\zeta_n^k)^m = 1$ for $m \in \mathbb{N}$, then $n \mid km$, so $n_0 \mid k_0 m$.
The ring $\mathbb{Z}$ is a PID by Theorem 3.10.2(i) and Definition 3.10.1(i).
Since $\gcd(k_0, n_0) = 1$, choose $a, b \in \mathbb{Z}$ with $ak_0 + bn_0 = 1$ by Theorem 3.10.6(iii).
Then $n_0 \mid m$ by $m = ak_0 m + bn_0 m$. Therefore $\operatorname{ord}(\zeta_n^k) = n_0 = \frac{n}{\gcd(k,n)}$.
Hence $\zeta_n^k \in \Omega_n \iff \gcd(k, n) = 1$, which is the asserted description of Ω_n.

(iv) $|\Omega_n| = \varphi(n)$ by (iii) and Definition 3.11.1(iii).

(v) Let $z \in U_n$ and set $d := \operatorname{ord}(z)$. Choose $q, r \in \mathbb{Z}$ with $n = qd + r$ and $0 \leq r < d$.
Then $1 = z^n = (z^d)^q z^r = z^r$, so $r = 0$ by minimality of d. Thus $d \mid n$ and $z \in \Omega_d$.
Conversely, if $d \mid n$ and $\omega \in \Omega_d$, then $\omega^n = 1$, so $\omega \in U_n$. The union is disjoint by uniqueness of element order.

Structural Commentary: Root parametrization proves cyclicity; orders prove counting and partition.
Key Insight: $\gcd(k, n)$ determines the order of ζ_n^k. □

Definition 3.11.3. *Cyclotomic Polynomials*
Let $n \in \mathbb{N}$. The set Ω_n is finite by Proposition 3.11.2(iii). The *nth cyclotomic polynomial* is $\Phi_n(x) := \prod_{\omega \in \Omega_n} (x - \omega) \in \mathbb{C}[x]$.

Proposition 3.11.4. *Cyclotomic Factorization of $x^n - 1$*
For $n \in \mathbb{N}$, $x^n - 1 = \prod_{d|n} \Phi_d(x)$ in $\mathbb{C}[x]$.

Proof. The roots of $x^n - 1$ are U_n by Definition 3.11.1(i). Their number is n by Proposition 3.11.2(ii).
Since $x^n - 1$ is monic of degree n, $x^n - 1 = \prod_{z \in U_n} (x - z)$. The equality $\prod_{z \in U_n} (x - z) = \prod_{d|n} \prod_{\omega \in \Omega_d} (x - \omega)$ holds by Proposition 3.11.2(v).
Each inner product equals $\Phi_d(x)$ by Definition 3.11.3, so $x^n - 1 = \prod_{d|n} \Phi_d(x)$.

Structural Commentary: The order partition regroups linear factors into cyclotomic factors.
Key Insight: Each root belongs to exactly one primitive-order class. □

Proposition 3.11.5. *Cyclotomic Polynomials Have Integer Coefficients*
For $n \in \mathbb{N}$:

(i) Φ_n is monic.
(ii) $\Phi_n \in \mathbb{Z}[x]$.

Proof.

(i) The polynomial Φ_n is monic by Definition 3.11.3, because every factor $x - \omega$ is monic.

(ii) Suppose some Φ_n does not lie in $\mathbb{Z}[x]$, and choose the least such n by Assumption 2.6.1.
The identity $\Phi_1(x) = x - 1$ holds by Proposition 3.11.2(iii) and Definition 3.11.3, so $n > 1$. Set $D_n := \{d \in \mathbb{N} : d \mid n,\ d < n\}$.
If $d \in D_n$, then $\Phi_d \in \mathbb{Z}[x]$ by minimality and Φ_d is monic by (i). Hence $P_n(x) := \prod_{d \in D_n} \Phi_d(x)$ is a monic polynomial in $\mathbb{Z}[x]$.
The identity $x^n - 1 = \Phi_n(x) P_n(x)$ holds by Proposition 3.11.4.
Since P_n is monic, leading-term cancellation stays in $\mathbb{Z}[x]$, strictly lowers degree. Process terminates by Assumption 2.6.1.
Thus $h, r \in \mathbb{Z}[x]$ satisfy $x^n - 1 = hP_n + r$ with $r = 0$ or $\deg r < \deg P_n$.
Over $\mathbb{C}[x]$, $(\Phi_n, 0)$ is another quotient-remainder pair by the factorization identity.
Therefore $(h, r) = (\Phi_n, 0)$ by Theorem 3.8.2(ii). Thus $\Phi_n = h \in \mathbb{Z}[x]$, contrary to the choice of n.

Structural Commentary: Minimal divisors construct an integral quotient; uniqueness identifies Φ_n.
Key Insight: A monic divisor requires no coefficient division outside $\mathbb{Z}$. □

Lemma 3.11.6. *A Cyclotomic Size Estimate*
Let $n \in \mathbb{N}$ with $n > 1$ and let $z \in \mathbb{Z}$ with $z \geq 2$. Then $|\Phi_n(z)| > z - 1$.

Proof. Let $\omega \in \Omega_n$. Then $\omega \neq 1$ by $\operatorname{ord}(\omega) = n > 1$, $|\omega| = 1$ by $\omega^n = 1$. Since a point on unit circle has real part 1 only at 1, $\operatorname{Re}(\omega) < 1$.
Therefore $|z - \omega|^2 = z^2 - 2z\operatorname{Re}(\omega) + 1 > (z - 1)^2$, so $|z - \omega| > z - 1$. Relation $|\Omega_n| = \varphi(n) \geq 1$ holds by Proposition 3.11.2(iv), Definition 3.11.1(iii). By Definition 3.11.3, $|\Phi_n(z)| = \prod_{\omega \in \Omega_n} |z - \omega| > (z - 1)^{\varphi(n)} \geq z - 1$.

Structural Commentary: Unit-circle geometry bounds each factor; root counting bounds their product.
Key Insight: A primitive nth root with $n > 1$ cannot equal 1. □

4 Fields and Galois Theory

4.1 Field Extensions

4.2 Splitting Fields and Normality

4.3 Galois Groups

4.4 Finite Fields and Characteristic

Perspective 4.4.1. *Finite Fields and Characteristic*
Let F be a finite field. Then $\operatorname{char}(F)$ is prime by Proposition 3.6.2(ii).
If $\operatorname{char}(F) = 2$, then $(\forall\, a \in F)\, a + a = 0_F$, equivalently $(\forall\, a \in F)\, a = -a$.
Thus characteristic two identifies addition and subtraction in F-linear calculations.

Definition 4.4.2. *Linear Code and Hamming Weight*
Let $n, q \in \mathbb{N}$ and let $\mathbb{F}_q$ be a field with $|\mathbb{F}_q| = q$.

(i) Set $V(n,q) := \mathbb{F}_q^n$, the *ambient code space*.
(ii) A *linear code* of length n over $\mathbb{F}_q$ is a subspace $C \leq V(n,q)$.
(iii) The *Hamming weight* of $\mathbf{x} \in V(n,q)$ is $\mathrm{wt}(\mathbf{x}) := |\{j \in \{1,\dots,n\} : x_j \neq 0\}|$.

Example 4.4.3. *Even-Weight Code*
Let $n \in \mathbb{N}$ and define $\pi_n : V(n,2) \to \mathbb{F}_2$ by $\pi_n(\mathbf{x}) := \sum_{j=1}^{n} x_j$.

Then $(\forall\, \mathbf{x}, \mathbf{y} \in V(n,2))(\forall\, c \in \mathbb{F}_2)\, \pi_n(c\mathbf{x} + \mathbf{y}) = c\pi_n(\mathbf{x}) + \pi_n(\mathbf{y})$, so π_n is linear. Set $E_n := \ker(\pi_n) = \{\mathbf{x} \in V(n,2) : \sum_{j=1}^{n} x_j = 0\}$.

Then $E_n \leq V(n,2)$ by Proposition 1.6.3(i), so E_n is a linear code by Definition 4.4.2(ii).
Since $\mathbb{F}_2 = \{0,1\}$ and $1+1=0$, $(\forall\, \mathbf{x} \in V(n,2))\, [\mathbf{x} \in E_n \iff 2 \mid \mathrm{wt}(\mathbf{x})]$.
Structural Commentary: $E_n = \ker(\pi_n)$ supplies closure; binary addition supplies parity.
Key Insight: Over $\mathbb{F}_2$, coordinate summation records Hamming-weight parity.

4.5 Classical Applications

Part VIII
Modules, Multilinear Algebra, and Homological Methods

1 Module Theory

1.1 Modules and Submodules

Throughout this subsection, R is a nonzero commutative ring with identity 1_R.
Modules replace field scalars by ring scalars.
Nonzero scalars may therefore be noninvertible or annihilate nonzero vectors.

Definition 1.1.1. *R-Module*
An *R-module* is an abelian group $(M,+)$ equipped with an action $R\times M\to M$, $(r,u)\mapsto ru$.
For $(r,s,u,v)\in R^2\times M^2$, the following identities hold:

(i) $r(u+v)=ru+rv$.
(ii) $(r+s)u=ru+su$.
(iii) $(rs)u=r(su)$.
(iv) $1_Ru=u$.

The ring R is the *base ring* of M; denote the additive identity of M by 0_M.

Example 1.1.2. *Basic Module Models*
Fix $m,n\in\mathbb{N}$.

(i) R^n is an R-module under componentwise addition and scalar multiplication.
(ii) $\mathrm{Mat}(m,n,R)$ is an R-module under entrywise operations.
(iii) R is an R-module over itself by ring multiplication.
(iv) Every vector space over a field $\mathbb{F}$ is an $\mathbb{F}$-module.

The module axioms hold componentwise in (i) and entrywise in (ii). They are ring laws in (iii) and vector-space axioms in (iv).

Perspective 1.1.3. *The Base Ring Controls the Geometry*
The module axioms separate the additive group from the scalar ring.
Over a field, scalar division supports the usual theory of bases, dimension, and complements.
Over a general ring, nonunits and zero divisors can obstruct division and cancellation.
Consequently, bases, dimension, and complements need not retain their vector-space behavior.

Definition 1.1.4. *Submodule*
Let M be an R-module. A *submodule* $N\le M$ is a subset $N\subseteq M$ that is an R-module under the restricted operations.

Theorem 1.1.5. *Submodule Test*
Let M be an R-module and $\emptyset\neq N\subseteq M$. Then $N\le M\iff(\forall\,(r,s,u,v)\in R^2\times N^2)\,ru+sv\in N$.

Proof.
($\Rightarrow$) Assume $N\le M$. Fix $(r,s,u,v)\in R^2\times N^2$. Then $ru,sv,ru+sv\in N$ by scalar and additive closure.
($\Leftarrow$) Assume $(\forall\,(r,s,u,v)\in R^2\times N^2)\,ru+sv\in N$. $(\forall\,x\in M)\,0_Rx=0_M$ by $0_Rx=(0_R+0_R)x=0_Rx+0_Rx$, cancellation in $(M,+)$.
$(\forall\,x\in M)\,(-1_R)x=-x$ by $(-1_R)x+x=0_Rx=0_M$. Fix $n\in N$. Then $0_M=0_Rn+0_Rn\in N$ by the assumed closure.
$(\forall\,(x,y)\in N^2)\,x+y=1_Rx+1_Ry\in N$ by the assumed closure. $(\forall\,x\in N)\,-x=(-1_R)x+0_Rx\in N$ by the assumed closure.
$(\forall\,(r,x)\in R\times N)\,rx=rx+0_Rx\in N$ by the assumed closure. $N\le(M,+)$ by the preceding zero, addition, and inverse relations.
The set N is closed under R-scalar multiplication by the final relation.
$N\le M$ by Definition 1.1.4, since the remaining module axioms are inherited from M.

Structural Commentary: The reverse direction extracts every submodule operation from two-term closure.
Key Insight: Two-term R-linear combinations encode all submodule closure laws. □

Proposition 1.1.6. *Intersections and Sums of Submodules*
Let M be an R-module and $A,B\le M$. Define $\mathcal{S}_{A,B}:=\{N\le M:A\cup B\subseteq N\}$.

(i) $A\cap B\le M$.
(ii) $A+B:=\{a+b:(a,b)\in A\times B\}\le M$.
(iii) $A+B=\bigcap_{N\in\mathcal{S}_{A,B}}N$.

Proof.
(i) $0_M\in A\cap B$ by $A,B\le M$. For $(r,s,x,y)\in R^2\times(A\cap B)^2$, $rx+sy\in A\cap B$ by closure of A, B; $A\cap B\le M$ by Theorem 1.1.5.
(ii) $0_M=0_M+0_M\in A+B$. Fix $x=a+b$ and $y=a'+b'$ with $(a,a',b,b')\in A^2\times B^2$.
For $(r,s)\in R^2$, $rx+sy=(ra+sa')+(rb+sb')\in A+B$ by closure of A and B. Hence $A+B\le M$ by Theorem 1.1.5.
(iii) $(\forall\,a\in A)\,a=a+0_M\in A+B$ and $(\forall\,b\in B)\,b=0_M+b\in A+B$. $A\cup B\subseteq A+B$ by these inclusions, so $A+B\in\mathcal{S}_{A,B}$ by (ii).
$(\forall\,N\in\mathcal{S}_{A,B})\,A+B\subseteq N$ by additive closure in N. Hence $A+B\subseteq\bigcap_{N\in\mathcal{S}_{A,B}}N$ by the definition of intersection.
Conversely, $\bigcap_{N\in\mathcal{S}_{A,B}}N\subseteq A+B$ by $A+B\in\mathcal{S}_{A,B}$.

Structural Commentary: The closure test constructs both submodules; universal containment identifies their join.
Key Insight: $A+B$ contains exactly the additions forced by A and B. □

Proposition 1.1.7. *Ideals Are Submodules of the Regular Module*
Equip $(R,+)$ with the action $R\times R\to R$, $(r,x)\mapsto rx$. Write ${}_RR$ for this regular module. Then $(\forall\,N\subseteq R)\,[N\le{}_RR\iff N\trianglelefteq R]$.

Proof. The group $(R,+)$ is abelian by Definition 3.1.1. The ring laws make ${}_RR$ an R-module by Definition 1.1.1.
($\Rightarrow$) Assume $N\le{}_RR$. Then $N\neq\emptyset$ and $(\forall\,a,b\in N)\,a-b\in N$ by Definition 1.1.4.
$(\forall\,(r,n)\in R\times N)\,rn,nr\in N$ by scalar closure and commutativity of R. Hence $N\trianglelefteq R$ by Definition 3.3.1.

($\Leftarrow$) Assume $N \trianglelefteq R$. Then $N \leq (R,+)$ by Proposition 2.1.4 and Definition 3.3.1.
For $(r,s,u,v) \in R^2 \times N^2$, $ru, sv \in N$ by ideal absorption and $ru+sv \in N$ by additive closure. Hence $N \leq {}_RR$ by Theorem 1.1.5.

Structural Commentary: The regular action identifies scalar closure with ideal absorption; commutativity adds right absorption.
Key Insight: Submodules of ${}_RR$ and ideals share the same closure laws. □

1.2 Generated, Cyclic, and Finitely Generated Modules

Throughout this subsection, R is a nonzero commutative ring with identity 1_R. Fix an R-module M.

Definition 1.2.1. *Generated Submodule*

(i) For $S \subseteq M$, set $\langle\langle S \rangle\rangle_R := \left\{ \sum_{j=1}^{n} r_j s_j : n \in \mathbb{N}_0,\ (r_1,\dots,r_n) \in R^n,\ (s_1,\dots,s_n) \in S^n \right\}$.
Call this the *submodule generated by S*; the term $n=0$ denotes the empty sum 0_M. Hence $\langle\langle \emptyset \rangle\rangle_R = \{0_M\}$.
(ii) A subset $S \subseteq M$ *spans* or *generates* M if $\langle\langle S \rangle\rangle_R = M$.
(iii) For $n \in \mathbb{N}$ and $(s_1,\dots,s_n) \in M^n$, write $\langle\langle s_1,\dots,s_n \rangle\rangle_R := \langle\langle \{s_1,\dots,s_n\} \rangle\rangle_R$.

Proposition 1.2.2. *Generated Submodule Is Smallest*
Let $S \subseteq M$, $G := \langle\langle S \rangle\rangle_R$, and $\mathcal{S}_S := \{N \leq M : S \subseteq N\}$.
(i) $G \leq M$.
(ii) $S \subseteq G$.
(iii) $G = \bigcap_{N \in \mathcal{S}_S} N$.

Proof.
(i) $0_M \in G$ by the empty-sum convention.
Fix $x, y \in G$ and $a, b \in R$. Choose representations $x = \sum_{j=1}^{m} r_j s_j$ and $y = \sum_{k=1}^{n} t_k u_k$ by Definition 1.2.1(i).
Then $ax + by = \sum_{j=1}^{m} (ar_j)s_j + \sum_{k=1}^{n} (bt_k)u_k \in G$ by distributivity and Definition 1.2.1(i). Hence $G \leq M$ by Theorem 1.1.5.
(ii) $S \subseteq G$ by $(\forall\, s \in S)\, s = 1_R s \in G$.
(iii) $G \in \mathcal{S}_S$ by (i) and (ii). Fix $N \in \mathcal{S}_S$. Then $G \subseteq N$ by $S \subseteq N$ and scalar and additive closure in N.
Hence $G \subseteq \bigcap_{N \in \mathcal{S}_S} N$ by the arbitrariness of N. Conversely, $\bigcap_{N \in \mathcal{S}_S} N \subseteq G$ by $G \in \mathcal{S}_S$.

Structural Commentary: Closure is by the submodule test; minimality is by unit coefficients and containment.
Key Insight: Every containing submodule absorbs all finite R-linear combinations. □

Definition 1.2.3. *Cyclic and Finitely Generated Modules*

(i) For $v \in M$, the *cyclic submodule generated by v* is $\langle\langle v \rangle\rangle_R = Rv = \{rv : r \in R\}$.
(ii) M is *finitely generated* if $(\exists\, n \in \mathbb{N})(\exists\, (v_1,\dots,v_n) \in M^n)\, M = \langle\langle v_1,\dots,v_n \rangle\rangle_R$.
(iii) M is *cyclic* if $(\exists\, v \in M)\, M = Rv$.

Counterexample 1.2.4. *A Cyclic Module with a Non-Finitely Generated Submodule*
A submodule of a finitely generated module need not be finitely generated. Let $\mathbb{F}$ be a field and $R := \mathbb{F}[x_1, x_2, \dots]$.
This is the polynomial ring in countably many commuting indeterminates.
Every $f \in R$ involves only finitely many indeterminates, and ${}_RR = \langle\langle 1_R \rangle\rangle_R$ is cyclic.
Set $I := \langle \{x_j : j \in \mathbb{N}\} \rangle$ and $\mathcal{Z} := \{f \in R : f(0,0,\dots) = 0\}$. $I \trianglelefteq R$ by Proposition 3.3.2.
Hence $I \leq {}_RR$ by Proposition 1.1.7. $I \subseteq \mathcal{Z}$ by the zero constant terms of the generators.
Every $f \in \mathcal{Z}$ is a finite sum of nonconstant monomials.
Each such monomial lies in I because it is divisible by some x_j; hence $\mathcal{Z} \subseteq I$ and $I = \mathcal{Z}$. Assume I is finitely generated.
Choose $n \in \mathbb{N}$ and $(p_1,\dots,p_n) \in I^n$ satisfying $I = \langle\langle p_1,\dots,p_n \rangle\rangle_R$ by Definition 1.2.3(ii).
Only finitely many indeterminates occur among $p_1,\dots,p_n$. Choose $k \in \mathbb{N}$ such that x_k occurs in none of them.
Define $\varphi_k : R \to \mathbb{F}[x_k]$ by $\varphi_k|_{\mathbb{F}} = \mathrm{id}_{\mathbb{F}}$, $\varphi_k(x_k) = x_k$, and $(\forall\, j \in \mathbb{N} \setminus \{k\})\, \varphi_k(x_j) = 0$.
The map φ_k preserves sums, products, and identity by substitution. Hence φ_k is a ring homomorphism by Definition 3.2.1(i).
Each p_i has zero constant term by $p_i \in I = \mathcal{Z}$, and no p_i contains x_k. Hence $(\forall\, i \in \{1,\dots,n\})\, \varphi_k(p_i) = 0$.
$(\forall\, (r_1,\dots,r_n) \in R^n)\, \varphi_k\left(\sum_{i=1}^{n} r_i p_i\right) = \sum_{i=1}^{n} \varphi_k(r_i)\varphi_k(p_i) = 0$. $\varphi_k(I) = \{0\}$ by $I = \langle\langle p_1,\dots,p_n \rangle\rangle_R$ and the preceding identity.
However, $x_k \in I$ and $\varphi_k(x_k) = x_k \neq 0$, a contradiction. Therefore I is a non-finitely generated submodule of the cyclic module ${}_RR$.

1.3 Linear Independence in Modules

Throughout this subsection, R is a nonzero commutative ring with identity 1_R.
Fix an R-module M with additive identity 0_M.

Definition 1.3.1. *Linear Independence in a Module*

(i) $S \subseteq M$ is *R-linearly independent* if $(\forall\, F \subseteq S,\ |F| < \infty)(\forall\, (r_v)_{v \in F} \in R^F) \left[\sum_{v \in F} r_v v = 0_M \Rightarrow (\forall\, v \in F)\, r_v = 0_R \right]$.
(ii) $S \subseteq M$ is *R-linearly dependent* if it is not R-linearly independent.

Counterexample 1.3.2. *Dependence Without Solving for One Vector*
For distinct $u, v \in M$, dependence of $\{u, v\}$ need not imply $u \in Rv \vee v \in Ru$.
In the $\mathbb{Z}$-module $\mathbb{Z}$, the distinct elements $2, 3$ satisfy $3 \cdot 2 + (-2) \cdot 3 = 0$.
The set $\{2, 3\}$ is dependent by Definition 1.3.1(i)–(ii) with the nonzero coefficients $3, -2$.
Yet $2 \notin 3\mathbb{Z}$ and $3 \notin 2\mathbb{Z}$, so neither element is an integer multiple of the other.

Proposition 1.3.3. *Scaling Independent Sets over a Domain*
Assume R is integral domain. Let $S \subseteq M$ be finite, R-linearly independent, $r \in R \setminus \{0_R\}$. Then $rS := \{rv : v \in S\}$ is R-linearly independent.

Proof.

(i) *Reduction*: If $S = \emptyset$, then $rS = \emptyset$ is R-linearly independent by Definition 1.3.1(i).
Assume $S \neq \emptyset$ and write $S = \{v_1, \dots, v_n\}$ with $n \in \mathbb{N}$ and pairwise distinct $v_1, \dots, v_n$.

(ii) *Injective scaling*: If $rv_i = rv_j$ for $i \neq j$, then $rv_i + (-r)v_j = 0_M$.
Then $r = 0_R$ by Definition 1.3.1(i), contradicting $r \neq 0_R$; hence $rv_1, \dots, rv_n$ are distinct.

(iii) *Relation transfer*: A relation on $T \subseteq rS$ extends to rS by zero coefficients. Fix $(a_1, \dots, a_n) \in R^n$ satisfying $\sum_{i=1}^{n} a_i(rv_i) = 0_M$.
$0_M = \sum_{i=1}^{n} a_i(rv_i) = \sum_{i=1}^{n} (ra_i)v_i$ by Definition 1.1.1(iii), commutativity of R. $(\forall\, i \in \{1, \dots, n\})\, ra_i = 0_R$ by Definition 1.3.1(i).

(iv) *Coefficient cancellation*: $(\forall\, i \in \{1, \dots, n\})\, a_i = 0_R$ by Proposition 3.5.2 with $ra_i = 0_R = r0_R$ and $r \neq 0_R$.
Thus rS is R-linearly independent by Definition 1.3.1(i).

Structural Commentary: Relations on rS reduce to relations on S, followed by coefficient cancellation in R.
Key Insight: Each transferred coefficient has the cancellable factor $r \neq 0_R$. □

1.4 Module Homomorphisms, Kernels, and Images

Throughout this subsection, R is a nonzero commutative ring with identity 1_R.

Definition 1.4.1. *Module Homomorphism*
Let M, N be R-modules and let $\tau : M \to N$.

(i) The map τ is *R-linear* if $(\forall\, (r, s, u, v) \in R^2 \times M^2)\, \tau(ru + sv) = r\tau(u) + s\tau(v)$. An R-linear map is an *R-homomorphism*.
(ii) The set of all R-homomorphisms $M \to N$ is $\mathrm{Hom}_R(M, N)$.
(iii) An R-homomorphism $M \to M$ is an *endomorphism*.
(iv) If τ is an R-homomorphism, then:
  (a) τ is a *monomorphism* $\iff$ τ is injective;
  (b) τ is an *epimorphism* $\iff$ τ is surjective;
  (c) τ is an *isomorphism* $\iff$ τ is bijective.

Proposition 1.4.2. *Hom, Kernel, and Image*
Let M, N be R-modules with additive identities $0_M, 0_N$, and let $\tau \in \mathrm{Hom}_R(M, N)$.

(i) For $(\sigma, \rho, m) \in \mathrm{Hom}_R(M, N)^2 \times M$, set $(\sigma + \rho)(m) := \sigma(m) + \rho(m)$.
For $(a, \sigma, m) \in R \times \mathrm{Hom}_R(M, N) \times M$, set $(a\sigma)(m) := a\sigma(m)$. These pointwise operations make $\mathrm{Hom}_R(M, N)$ an R-module.
(ii) $\ker \tau := \tau^{-1}(\{0_N\}) \leq M$.
(iii) $\mathrm{im}\, \tau := \tau(M) \leq N$.
(iv) τ is a monomorphism $\iff \ker \tau = \{0_M\}$.

Proof. $(\forall\, a \in R)\, a0_N = 0_N$ by additive cancellation in $a0_N = a(0_N + 0_N) = a0_N + a0_N$.
$\tau(0_M) = 0_N$ by Definition 1.4.1(i) with $(r, s, u, v) = (1_R, 1_R, 0_M, 0_M)$ and additive cancellation in N.

(i) Set $\mathcal{F}(M, N) := \{f : f : M \to N\}$ and equip it with the pointwise operations in the statement.
Zero map $0_{\mathcal{F}} : M \to N$, $m \mapsto 0_N$, is additive identity. Module axioms hold pointwise in N and in $\mathcal{F}(M, N)$ by Assumption 3.1.2.
Thus $\mathcal{F}(M, N)$ is an R-module by Definition 1.1.1. The zero map is R-linear by $(\forall\, r, s \in R)\, r0_N + s0_N = 0_N$.
Fix $\sigma, \rho \in \mathrm{Hom}_R(M, N)$, $a, b, r, s \in R$, and $u, v \in M$.
$(a\sigma + b\rho)(ru + sv) = r(a\sigma + b\rho)(u) + s(a\sigma + b\rho)(v)$ by R-linearity of σ, ρ and ring commutativity.
Hence $a\sigma + b\rho \in \mathrm{Hom}_R(M, N)$ by Definition 1.4.1(i). Therefore $\mathrm{Hom}_R(M, N) \leq \mathcal{F}(M, N)$ by Theorem 1.1.5.

(ii) $0_M \in \ker \tau$ by $\tau(0_M) = 0_N$. For $(a, b, x, y) \in R^2 \times (\ker \tau)^2$, $\tau(ax + by) = a\tau(x) + b\tau(y) = a0_N + b0_N = 0_N$.
Thus $\ker \tau \leq M$ by Theorem 1.1.5.

(iii) $0_N \in \mathrm{im}\, \tau$ by $0_N = \tau(0_M)$. Fix $(a, b, y_1, y_2) \in R^2 \times (\mathrm{im}\, \tau)^2$.
Choose $x_1, x_2 \in M$ with $(y_1, y_2) = (\tau(x_1), \tau(x_2))$ by the definition of the image.
Then $ay_1 + by_2 = \tau(ax_1 + bx_2) \in \mathrm{im}\, \tau$ by Definition 1.4.1(i). Thus $\mathrm{im}\, \tau \leq N$ by Theorem 1.1.5.

(iv) ($\Rightarrow$) Assume τ is a monomorphism, hence injective by Definition 1.4.1(iv).
$(\forall\, m \in \ker \tau)\, m = 0_M$ by injectivity and $\tau(m) = 0_N = \tau(0_M)$. Therefore $\ker \tau = \{0_M\}$ by $\ker \tau \subseteq \{0_M\}$ and (ii).

(iv) ($\Leftarrow$) Assume $\ker \tau = \{0_M\}$ and fix $m, m' \in M$ with $\tau(m) = \tau(m')$. Then $\tau(m - m') = 0_N$ by Definition 1.4.1(i).
Thus $m - m' \in \ker \tau = \{0_M\}$ and $m = m'$. Thus τ is injective and hence a monomorphism by Definition 1.4.1(iv).

Structural Commentary: Hom closes pointwise; kernels use differences; images use lifted representatives.
Key Insight: R-linearity transports combinations and detects equality through kernel differences. □

1.5 Quotient Modules, Correspondence, and Isomorphism

Throughout this subsection, R is a nonzero commutative ring with identity 1_R.
Fix an R-module M with additive identity 0_M.

Definition 1.5.1. *Quotient Module*
Let $S \leq M$. For $(u, v) \in M^2$, define $u \equiv v \pmod{S} \iff u - v \in S$.
For $u \in M$, set $u + S := \{u + s : s \in S\}$, and set $M/S := \{u + S : u \in M\}$.
For $(u, v) \in M^2$, set $(u + S) + (v + S) := (u + v) + S$. For $(r, u) \in R \times M$, set $r(u + S) := ru + S$.

Proposition 1.5.2. *Quotient Operations Are Well-Defined*
Let $S \leq M$.

(i) For $(u, v) \in M^2$, $u + S = v + S \iff u - v \in S$. Consequently, congruence modulo S is an equivalence relation.
(ii) The operations of Definition 1.5.1 are well-defined and make M/S an R-module.
(iii) The additive identity is $0_{M/S} = S$. Quotient map $\pi : M \twoheadrightarrow M/S$, $\pi(u) := u + S$, is epimorphism with $\ker \pi = S$.

Proof.

(i) ($\Rightarrow$) Assume $u + S = v + S$. Then $u \in u + S = v + S$ by $0_M \in S$. Thus $(\exists\, s \in S)\, u = v + s$ by Definition 1.5.1, hence $u - v \in S$.

(i) ($\Leftarrow$) Assume $u - v \in S$ and fix $x \in u + S$. Choose $s \in S$ with $x = u + s$ by Definition 1.5.1.
$x = v + \big((u - v) + s\big) \in v + S$ by closure of S, so $u + S \subseteq v + S$. Fix $y \in v + S$, choose $t \in S$ with $y = v + t$ by Definition 1.5.1.
Then $y = u + \big((v - u) + t\big) \in u + S$ by $v - u = -(u - v) \in S$ and closure of S. Thus $v + S \subseteq u + S$ and $u + S = v + S$.
Congruence modulo S is an equivalence relation by this characterization and equality of cosets.
(ii) Assume $u + S = u' + S$ and $v + S = v' + S$. Then $u - u', v - v' \in S$ by (i). $(u + v) - (u' + v') = (u - u') + (v - v') \in S$.
For $r \in R$, $ru - ru' = r(u - u') \in S$. Hence $(u + v) + S = (u' + v') + S$ and $ru + S = ru' + S$ by (i).
Thus both operations are representative-independent.
Every module axiom holds on representatives by the corresponding axiom in M. Therefore M/S is an R-module.
(iii) The additive identity is $0_{M/S} = S$ by $0_M + S = S$.
For $(r, s, u, v) \in R^2 \times M^2$, $\pi(ru + sv) = (ru + sv) + S = r(u + S) + s(v + S) = r\pi(u) + s\pi(v)$.
Thus $\pi \in \mathrm{Hom}_R(M, M/S)$ by Definition 1.4.1(i). The map π is surjective by $M/S = \{u + S : u \in M\}$.
Hence π is an epimorphism by Definition 1.4.1(iv). Finally, $\ker \pi = \{u \in M : u + S = S\} = \{u \in M : u \in S\} = S$ by (i).

Structural Commentary: Coset equality controls representatives; module axioms descend from M.
Key Insight: Representatives agree exactly when their difference lies in S. □

Theorem 1.5.3. *Correspondence Theorem for Modules*
Let $S \leq M$ and let $\pi : M \twoheadrightarrow M/S$ be the quotient map. Set $\mathcal{A} := \{T \leq M : S \leq T\}$ and $\mathcal{B} := \{U : U \leq M/S\}$.
For $T \in \mathcal{A}$, define $\Phi(T) := T/S = \pi(T)$. For $U \in \mathcal{B}$, define $\Psi(U) := \pi^{-1}(U)$.

(i) $(\forall\, T \in \mathcal{A})\, \Phi(T) \in \mathcal{B}$ *and* $(\forall\, U \in \mathcal{B})\, \Psi(U) \in \mathcal{A}$.
(ii) $\Psi\Phi = \mathrm{id}_{\mathcal{A}}$ *and* $\Phi\Psi = \mathrm{id}_{\mathcal{B}}$.
(iii) Both Φ and Ψ preserve inclusion.

Consequently, $\Phi : \mathcal{A} \to \mathcal{B}$ is an order isomorphism with inverse Ψ.

Proof.

(i) Fix $T \in \mathcal{A}$; $0_{M/S} = S \in T/S$ by $S \leq T$. For $(r, s, t_1, t_2) \in R^2 \times T^2$, $r(t_1 + S) + s(t_2 + S) = (rt_1 + st_2) + S$ by Definition 1.5.1.
The coset $(rt_1 + st_2) + S$ lies in T/S by closure of T. Thus $\Phi(T) = T/S \leq M/S$ by Theorem 1.1.5. Fix $U \in \mathcal{B}$.
Then $0_M \in \pi^{-1}(U)$ by $\pi(0_M) = 0_{M/S} \in U$. Fix $(r, s, u, v) \in R^2 \times \pi^{-1}(U)^2$.
Then $\pi(ru + sv) = r\pi(u) + s\pi(v) \in U$ by Proposition 1.5.2(iii) and closure of U. Hence $\pi^{-1}(U) \leq M$ by Theorem 1.1.5.
$S = \ker \pi \subseteq \pi^{-1}(U)$ by Proposition 1.5.2(iii) and $0_{M/S} \in U$. Hence $\Psi(U) = \pi^{-1}(U) \in \mathcal{A}$ by the definition of $\mathcal{A}$.
(ii) Fix $T \in \mathcal{A}$. Then $(\forall\, t \in T)\, \pi(t) = t + S$ by Proposition 1.5.2(iii). Hence $T \subseteq \pi^{-1}(T/S)$ by the definition of T/S.
If $m \in \pi^{-1}(T/S)$, choose $t \in T$ with $m + S = t + S$ by the definition of T/S.
Then $m - t \in S \subseteq T$ by Proposition 1.5.2(i), so $m \in T$. Therefore $\Psi\Phi(T) = \pi^{-1}(T/S) = T$. Fix $U \in \mathcal{B}$.
Inclusion $\pi(\pi^{-1}(U)) \subseteq U$ holds by definition of preimage. If $x \in U$, choose $m \in M$ with $x = \pi(m)$ by Proposition 1.5.2(iii).
Then $m \in \pi^{-1}(U)$ and $x \in \pi(\pi^{-1}(U))$. Therefore $\Phi\Psi(U) = \pi(\pi^{-1}(U)) = U$.
(iii) If $T_1 \subseteq T_2$ in $\mathcal{A}$, then $\Phi(T_1) = \pi(T_1) \subseteq \pi(T_2) = \Phi(T_2)$. If $U_1 \subseteq U_2$ in $\mathcal{B}$, then $\Psi(U_1) = \pi^{-1}(U_1) \subseteq \pi^{-1}(U_2) = \Psi(U_2)$.
Thus both assignments preserve inclusion.

Thus $\Phi : \mathcal{A} \to \mathcal{B}$ is an order isomorphism with inverse Ψ by (ii)–(iii).
Structural Commentary: Closure constructs Φ, Ψ; lifting identifies inverses; set maps preserve order.
Key Insight: Preimages recover quotient images by surjectivity and kernel containment. □

Theorem 1.5.4. *Module Isomorphism Theorems*
The following canonical maps are R-module isomorphisms.

(i) Let N be R-module with additive identity 0_N, $\tau \in \mathrm{Hom}_R(M, N)$. For $m \in M$, $\overline{\tau}(m + \ker \tau) := \tau(m)$. Then $\overline{\tau} : M/\ker \tau \xrightarrow{\sim} \mathrm{im}\, \tau$.
(ii) For $S, T \leq M$ and $s \in S$, define $\theta\big(s + (S \cap T)\big) := s + T$. Then $\theta : S/(S \cap T) \xrightarrow{\sim} (S + T)/T$.
(iii) For $S \leq T \leq M$ and $m \in M$, define $\eta\big((m + S) + T/S\big) := m + T$. Then $\eta : (M/S)/(T/S) \xrightarrow{\sim} M/T$.

Proof.

(i) Assume $m + \ker \tau = m' + \ker \tau$. Then $m - m' \in \ker \tau$ by Proposition 1.5.2(i).
$\tau(m) - \tau(m') = \tau(m - m') = 0_N$ by Definition 1.4.1(i). Thus $\tau(m) = \tau(m')$ and $\overline{\tau}$ is well-defined.
Fix $(r, s, m, m') \in R^2 \times M^2$. Then $\overline{\tau}\big(r(m + \ker \tau) + s(m' + \ker \tau)\big) = \tau(rm + sm') = r\tau(m) + s\tau(m')$.
Therefore $\overline{\tau} \in \mathrm{Hom}_R(M/\ker \tau, \mathrm{im}\, \tau)$ by Definition 1.4.1(i). $\mathrm{im}\, \overline{\tau} = \mathrm{im}\, \tau$ by the definition of $\overline{\tau}$.
$\ker \overline{\tau} = \{m + \ker \tau : m \in \ker \tau\} = \{\ker \tau\} = \{0_{M/\ker \tau}\}$. Thus $\overline{\tau}$ is a monomorphism by Proposition 1.4.2(iv).
It is an epimorphism by $\mathrm{im}\, \overline{\tau} = \mathrm{im}\, \tau$ and Definition 1.4.1(iv). Hence $\overline{\tau}$ is an isomorphism by Definition 1.4.1(iv).
(ii) Assume $s + (S \cap T) = s' + (S \cap T)$. Then $s - s' \in S \cap T \subseteq T$ by Proposition 1.5.2(i).
Thus $s + T = s' + T$ by Proposition 1.5.2(i), so θ is well-defined. Fix $(r, q, s_1, s_2) \in R^2 \times S^2$.
Then $\theta\big(r(s_1 + (S \cap T)) + q(s_2 + (S \cap T))\big) = r(s_1 + T) + q(s_2 + T)$. Hence $\theta \in \mathrm{Hom}_R(S/(S \cap T), (S + T)/T)$ by Definition 1.4.1(i).
Fix $x \in (S + T)/T$, $y \in S + T$ with $x = y + T$ by Definition 1.5.1. Choose $(s, t) \in S \times T$ with $y = s + t$ by Proposition 1.1.6(ii).
Then $x = s + T = \theta\big(s + (S \cap T)\big)$, so θ is surjective. $\ker \theta = \{s + (S \cap T) : s \in S \cap T\} = \{S \cap T\}$ by Proposition 1.5.2(i).
Hence $\ker \theta = \{0_{S/(S \cap T)}\}$ by $0_{S/(S \cap T)} = S \cap T$. Thus θ is an isomorphism by Proposition 1.4.2(iv) and Definition 1.4.1(iv).
(iii) Assume $(m + S) + T/S = (m' + S) + T/S$. Then $(m - m') + S \in T/S$ by Proposition 1.5.2(i).
Choose $t \in T$ with $(m - m') + S = t + S$ by the definition of T/S. $m - m' - t \in S \subseteq T$ by Proposition 1.5.2(i), so $m - m' \in T$.
Thus $m + T = m' + T$ by Proposition 1.5.2(i), and η is well-defined. Fix $(r, s, m, m') \in R^2 \times M^2$.
Then $\eta\big(r((m + S) + T/S) + s((m' + S) + T/S)\big) = r(m + T) + s(m' + T)$; $\eta \in \mathrm{Hom}_R((M/S)/(T/S), M/T)$ by Definition 1.4.1(i).
η is surjective by $(\forall\, m \in M)\, \eta\big((m + S) + T/S\big) = m + T$. $\ker \eta = \{(m + S) + T/S : m \in T\} = \{T/S\}$ by Proposition 1.5.2(i).
Hence $\ker \eta = \{0_{(M/S)/(T/S)}\}$ by $0_{(M/S)/(T/S)} = T/S$. Thus η is isomorphism by Proposition 1.4.2(iv), Definition 1.4.1(iv).

Structural Commentary: Coset checks define maps; kernels and representatives certify bijectivity.
Key Insight: Each denominator is exactly the ambiguity annihilated by its canonical map. □

1.6 Torsion and Annihilators

Throughout this subsection, R is a nonzero commutative ring with identity 1_R and additive identity 0_R.
Fix an R-module M with additive identity 0_M.

Definition 1.6.1. *Torsion*

(i) An element $v \in M \setminus \{0_M\}$ is a *torsion element* if $(\exists\, r \in R \setminus \{0_R\})\, rv = 0_M$.
(ii) The module M is *torsion-free* if $(\forall\, (r, v) \in (R \setminus \{0_R\}) \times M)\, [rv = 0_M \Rightarrow v = 0_M]$.
(iii) The module M is a *torsion module* if $(\forall\, v \in M \setminus \{0_M\})(\exists\, r \in R \setminus \{0_R\})\, rv = 0_M$.
(iv) The *torsion set* of M is $M_{\text{tor}} := \{v \in M : (\exists\, r \in R \setminus \{0_R\})\, rv = 0_M\}$.

Example 1.6.2. *$\mathbb{Z}/n\mathbb{Z}$ as a Torsion Module*
Let $n \in \mathbb{N}$ with $n > 1$. Quotient $\mathbb{Z}/n\mathbb{Z}$ is $\mathbb{Z}$-module with $k\overline{a} := \overline{ka}$. $(\forall\, \overline{a} \in \mathbb{Z}/n\mathbb{Z})\, n\overline{a} = \overline{0}$; $\mathbb{Z}/n\mathbb{Z}$ is torsion module by Definition 1.6.1(iii). If $\overline{a} \neq \overline{0}$, then $n\overline{a} = \overline{0}$ with $n \neq 0_{\mathbb{Z}}$. Thus $\{\overline{a}\}$ is linearly dependent by Definition 1.3.1(i)–(ii).

Proposition 1.6.3. *Torsion Submodule over an Integral Domain*
Assume R is an integral domain.

(i) $M_{\text{tor}} \leq M$.
(ii) $(M/M_{\text{tor}})_{\text{tor}} = \{0_{M/M_{\text{tor}}}\} = \{M_{\text{tor}}\}$; *equivalently, M/M_{tor} is torsion-free.*

Proof.

(i) $0_M \in M_{\text{tor}}$ by $1_R 0_M = 0_M$ and Definition 1.6.1(iv). Fix $(r, s, x, y) \in R^2 \times M_{\text{tor}}^2$.
Choose $(a, b) \in (R \setminus \{0_R\})^2$ with $ax = by = 0_M$ by Definition 1.6.1(iv).
By Definition 3.5.1(ii), $ab \neq 0_R$. $ab(rx + sy) = rb(ax) + sa(by) = 0_M$.
Thus $rx + sy \in M_{\text{tor}}$ by Definition 1.6.1(iv). Hence $M_{\text{tor}} \leq M$ by Theorem 1.1.5.
(ii) Set $T := M_{\text{tor}}$ and fix $x \in (M/T)_{\text{tor}}$. Choose $m \in M$ with $x = m + T$ by Definition 1.5.1.
Choose $r \in R \setminus \{0_R\}$ with $rx = T$ by Definition 1.6.1(iv). Then $rm \in T$ by Proposition 1.5.2(i).
Choose $s \in R \setminus \{0_R\}$ with $srm = 0_M$ by Definition 1.6.1(iv). By Definition 3.5.1(ii), $sr \neq 0_R$.
Hence $m \in T$ by Definition 1.6.1(iv), so $x = T$ by Proposition 1.5.2(i).
Therefore $(M/T)_{\text{tor}} \subseteq \{0_{M/T}\}$. The reverse inclusion holds by Definition 1.6.1(iv).
Thus $(M/T)_{\text{tor}} = \{0_{M/T}\} = \{T\}$. Equivalently, M/T is torsion-free by Definition 1.6.1(ii).

Structural Commentary: Products of annihilators prove closure and eliminate quotient torsion.
Key Insight: Domain multiplication preserves nonzero annihilators. □

Definition 1.6.4. *Annihilators*

(i) For $v \in M$, the *annihilator* of v is $\text{Ann}_R(v) := \{r \in R : rv = 0_M\}$.
(ii) For $N \leq M$, the *annihilator* of N is $\text{Ann}_R(N) := \{r \in R : (\forall\, n \in N)\, rn = 0_M\}$.

Proposition 1.6.5. *Annihilator Properties*
Let $v \in M$ and $N \leq M$.

(i) $\text{Ann}_R(v) \trianglelefteq R$ *and* $\text{Ann}_R(N) \trianglelefteq R$.
(ii) v *is torsion* $\iff v \neq 0_M$ *and* $\text{Ann}_R(v) \neq \{0_R\}$.
(iii) If $A \leq B \leq M$, *then* $\text{Ann}_R(B) \subseteq \text{Ann}_R(A)$.

Proof.

(i) $0_R \in \text{Ann}_R(v)$ by $0_R v = 0_M$ and Definition 1.6.4(i). Fix $(r, s, a, b) \in R^2 \times \text{Ann}_R(v)^2$.
Then $(ra + sb)v = r(av) + s(bv) = 0_M$ by Definition 1.6.4(i). Thus $\text{Ann}_R(v) \leq {}_R R$ by Theorem 1.1.5.
Hence $\text{Ann}_R(v) \trianglelefteq R$ by Proposition 1.1.7. $(\forall\, n \in N)\, 0_R n = 0_M$, so $0_R \in \text{Ann}_R(N)$ by Definition 1.6.4(ii).
Fix $(r, s, a, b) \in R^2 \times \text{Ann}_R(N)^2$. $(\forall\, n \in N)\, [(ra + sb)n = r(an) + s(bn) = 0_M]$ by Definition 1.6.4(ii).
Thus $\text{Ann}_R(N) \leq {}_R R$ by Theorem 1.1.5. Hence $\text{Ann}_R(N) \trianglelefteq R$ by Proposition 1.1.7.
(ii) ($\Rightarrow$) Suppose v is torsion. Then $v \neq 0_M$, $(\exists\, r \in R \setminus \{0_R\})\, rv = 0_M$ by Definition 1.6.1(i); $\text{Ann}_R(v) \neq \{0_R\}$ by Definition 1.6.4(i).
(ii) ($\Leftarrow$) Suppose $v \neq 0_M$ and $\text{Ann}_R(v) \neq \{0_R\}$. Choose $r \in \text{Ann}_R(v) \setminus \{0_R\}$.
Then $rv = 0_M$ by Definition 1.6.4(i). Hence v is torsion by Definition 1.6.1(i).
(iii) Fix $A, B \leq M$ with $A \leq B$, and let $r \in \text{Ann}_R(B)$. Then $(\forall\, a \in A)\, ra = 0_M$ by $A \subseteq B$ and Definition 1.6.4(ii).
Hence $r \in \text{Ann}_R(A)$ by Definition 1.6.4(ii).

Structural Commentary: Closure establishes ideals; annihilators detect torsion and reverse containment.
Key Insight: Annihilation conditions strengthen as the tested subset grows. □

Proposition 1.6.6. *A Nonzero Annihilator from Torsion Generators over a Domain*
Assume R is an integral domain. Let $n \in \mathbb{N}$ and $M = \langle\!\langle u_1, \ldots, u_n \rangle\!\rangle_R$. If $(u_1, \ldots, u_n) \in M_{\text{tor}}^n$, then $\text{Ann}_R(M) \neq \{0_R\}$.

Proof. Choose $(a_1, \ldots, a_n) \in (R \setminus \{0_R\})^n$ with $(\forall\, i \in \{1, \ldots, n\})\, a_i u_i = 0_M$ by Definition 1.6.1(iv).
Set $a := \prod_{i=1}^{n} a_i$. By repeated application of Definition 3.5.1(ii), $a \neq 0_R$.
Fix $i \in \{1, \ldots, n\}$ and choose $b_i \in R$ with $a = b_i a_i$ by commutativity of R. Then $au_i = b_i(a_i u_i) = 0_M$.
Thus $(\forall\, i \in \{1, \ldots, n\})\, au_i = 0_M$. Fix $m \in M$. Choose $(r_1, \ldots, r_n) \in R^n$ with $m = \sum_{i=1}^{n} r_i u_i$ by Definition 1.2.1(i),(iii).
By commutativity of R, $am = \sum_{i=1}^{n} r_i(au_i) = 0_M$. Thus $a \in \text{Ann}_R(M) \setminus \{0_R\}$ by Definition 1.6.4(ii).

Structural Commentary: Generator annihilators multiply to a scalar killing all finite combinations.
Key Insight: Finite products stay nonzero because R is a domain. □

1.7 Direct Sums and Complements

Throughout this subsection, R denotes a nonzero commutative ring with identity 1_R.
The symbol M denotes an R-module with additive identity 0_M.

Definition 1.7.1. *Direct Sums and Direct Summands*

(i) For $m = (m_i)_{i \in I} \in \prod_{i \in I} M_i$, set $\text{supp}(m) := \{i \in I : m_i \neq 0_{M_i}\}$.
The *external direct sum* is $\bigoplus_{i \in I} M_i := \{m \in \prod_{i \in I} M_i : |\text{supp}(m)| < \infty\}$, with componentwise operations.

(ii) For a family $(S_i)_{i\in I}$ of submodules of M, define $\sum_{i\in I} S_i := \left\langle\left\langle \bigcup_{i\in I} S_i \right\rangle\right\rangle_R$.
Equivalently, $\sum_{i\in I} S_i = \{\sum_{i\in F} s_i : F \subseteq I,\ |F|<\infty,\ (s_i)_{i\in F} \in \prod_{i\in F} S_i\}$.
Here the empty sum is 0_M. The equivalence follows by absorbing coefficients into each submodule and collecting like indices.
Sum satisfies $\sum_{i\in I} S_i \le M$ by Proposition 1.2.2(i). It is *direct* if $(\forall\, k\in I)\, S_k \cap \sum_{i\in I\setminus\{k\}} S_i = \{0_M\}$.
If it is direct and equals M, write $M = \bigoplus_{i\in I} S_i$.

(iii) A submodule $S \le M$ is a *direct summand* if $(\exists\, T \le M)\, M = S \oplus T$. Such a T is a *complement* of S in M.

Proposition 1.7.2. *Direct Sum Characterizations*
For a family $(S_i)_{i\in I}$ of submodules of M, the following conditions are equivalent:

(i) $\sum_{i\in I} S_i$ *is direct.*

(ii) $(\forall\, F\subseteq I)\left[|F|<\infty \Rightarrow (\forall\,(x_i)_{i\in F}\in \prod_{i\in F} S_i)\left[\sum_{i\in F} x_i = 0_M \Rightarrow (\forall\, i\in F)\, x_i = 0_M\right]\right]$.

(iii) $(\forall\, x\in \sum_{i\in I} S_i)(\exists!\,(x_i)_{i\in I}\in \bigoplus_{i\in I} S_i)\, x = \sum_{i\in I} x_i$.

Proof.

(i) $\Rightarrow$ *(ii)* Assume (i). Fix $F\subseteq I$ with $|F|<\infty$ and $(x_i)_{i\in F}\in \prod_{i\in F} S_i$ with $\sum_{i\in F} x_i = 0_M$.
For $k\in F$, $x_k = -\sum_{i\in F\setminus\{k\}} x_i \in S_k \cap \sum_{i\in I\setminus\{k\}} S_i$. This intersection is $\{0_M\}$ by Definition 1.7.1(ii); hence $(\forall\, k\in F)\, x_k = 0_M$.

(ii) $\Rightarrow$ *(iii)* Fix $x\in \sum_{i\in I} S_i$. Choose $F\subseteq I$ with $|F|<\infty$ and $(x_i)_{i\in F}\in \prod_{i\in F} S_i$ with $x = \sum_{i\in F} x_i$ by Definition 1.7.1(ii).
Extend this family by 0_M. The resulting $(x_i)_{i\in I}$ lies in $\bigoplus_{i\in I} S_i$ by Definition 1.7.1(i), so a decomposition exists.
Suppose $(x_i)_{i\in I}, (y_i)_{i\in I}\in \bigoplus_{i\in I} S_i$ both sum to x, and set $H := \{i\in I : x_i\neq 0_M\}\cup\{i\in I : y_i\neq 0_M\}$.
Then $|H|<\infty$ by Definition 1.7.1(i), and $\sum_{i\in H}(x_i - y_i) = 0_M$.
Thus $(\forall\, i\in H)\, x_i = y_i$ by (ii), while $(\forall\, i\in I\setminus H)\, x_i = y_i = 0_M$ by the definition of H. Hence $(x_i)_{i\in I} = (y_i)_{i\in I}$.

(iii) $\Rightarrow$ *(i)* Fix $k\in I$ and $x\in S_k \cap \sum_{i\in I\setminus\{k\}} S_i$. Suppose $x\neq 0_M$.
Choose $F\subseteq I\setminus\{k\}$ with $|F|<\infty$ and $(y_i)_{i\in F}\in \prod_{i\in F} S_i$ with $x = \sum_{i\in F} y_i$ by Definition 1.7.1(ii).
Extend $(y_i)_{i\in F}$ by 0_M, and define $(z_i)_{i\in I}$ by $z_k = x$ and $(\forall\, i\in I\setminus\{k\})\, z_i = 0_M$.
These are distinct elements of $\bigoplus_{i\in I} S_i$ whose coordinates both sum to x, contrary to (iii).
Hence $x = 0_M$, so $(\forall\, k\in I)\, S_k \cap \sum_{i\in I\setminus\{k\}} S_i = \{0_M\}$.

Structural Commentary: Zero-sum rigidity turns the intersection criterion into coordinate uniqueness.
Key Insight: Decompositions are unique exactly when cross-summand cancellation is impossible. □

Counterexample 1.7.3. *Not Every Submodule Is Complemented*
The $\mathbb{Z}$-module $\mathbb{Z}$ refutes the assertion that every submodule of a module is complemented.
Each $S \le {}_{\mathbb{Z}}\mathbb{Z}$ is an ideal by Proposition 1.1.7. The zero submodule is $0\mathbb{Z}$.
Each nonzero S has a unique positive generator by Theorem 3.10.2(i),(ii). Hence $(\forall\, S\le {}_{\mathbb{Z}}\mathbb{Z})(\exists!\, n\in\mathbb{N}_0)\, S = n\mathbb{Z}$.
For $(m,n)\in\mathbb{N}^2$, $0\neq mn\in m\mathbb{Z}\cap n\mathbb{Z}$, so no two nonzero submodules form a direct sum.
Thus, for $k\in\mathbb{N}_0$, $k\mathbb{Z}$ is complemented only if $k\in\{0,1\}$; conversely, $\mathbb{Z} = \{0\}\oplus\mathbb{Z} = \mathbb{Z}\oplus\{0\}$.
Hence the complemented submodules of $\mathbb{Z}$ are exactly $\{0\}$ and $\mathbb{Z}$.

Theorem 1.7.4. *Complements and Quotients*
Let $S\le M$ and let $\pi : M \twoheadrightarrow M/S$ be the quotient map.

(i) If $T\le M$ and $M = S\oplus T$, then $\pi|_T : T \xrightarrow{\sim} M/S$, $t\mapsto t+S$.

(ii) If $T_1, T_2 \le M$ satisfy $M = S\oplus T_1 = S\oplus T_2$, then $T_1\cong T_2$.

Proof.

(i) Fix $T\le M$ and assume $M = S\oplus T$. Then $\pi|_T\in \mathrm{Hom}_R(T, M/S)$ by Proposition 1.5.2(iii).
The identity $\ker(\pi|_T) = T\cap S = \{0_M\}$ holds by Proposition 1.5.2(i) and Definition 1.7.1(ii).
Hence $\pi|_T$ is a monomorphism by Proposition 1.4.2(iv). Fix $m+S\in M/S$.
Choose $(s,t)\in S\times T$ with $m = s+t$ by $M = S+T$. Then $m+S = t+S = \pi|_T(t)$, so $\pi|_T$ is epimorphism by Definition 1.4.1(iv).
Therefore $\pi|_T$ is an isomorphism by Definition 1.4.1(iv).

(ii) Assume $M = S\oplus T_1 = S\oplus T_2$. Then $T_1\cong M/S\cong T_2$ by (i), and hence $T_1\cong T_2$.

Structural Commentary: Kernel and image tests transfer each complement to the common quotient model.
Key Insight: Its kernel on a complement is precisely the intersection with S. □

1.8 Retracts, One-Sided Inverses, and Split Maps

Throughout this subsection, R denotes a nonzero commutative ring with identity 1_R.
All modules are R-modules.

Theorem 1.8.1. *Direct Summands Extend Epimorphisms*
Let M, M_1 be R-modules, let $N\le M$, and let $\sigma\in\mathrm{Hom}_R(N, M_1)$ be an epimorphism.
Set $\mathcal{C}_N(M) := \{H\le M : M = N\oplus H\}$. Set $\mathcal{E}_\sigma := \{\tau\in\mathrm{Hom}_R(M,M_1) : \tau|_N = \sigma,\ \mathrm{im}\,\tau = M_1\}$.

(i) $(\forall\, H\in\mathcal{C}_N(M))(\exists!\,\overline{\sigma}_H\in\mathrm{Hom}_R(M,M_1))(\forall\,(n,h)\in N\times H)\,\overline{\sigma}_H(n+h) = \sigma(n)$.

(ii) $(\forall\, H\in\mathcal{C}_N(M))\,\overline{\sigma}_H\in\mathcal{E}_\sigma$.

(iii) $(\forall\, H \in \mathcal{C}_N(M))\ \ker \overline{\sigma}_H = \ker \sigma \oplus H$.
(iv) *If σ is an isomorphism, then $H \mapsto \overline{\sigma}_H$ is a bijection $\mathcal{C}_N(M) \to \mathcal{E}_\sigma$ with inverse $\tau \mapsto \ker \tau$.*

Proof.

(i) Fix $H \in \mathcal{C}_N(M)$. If $n + h = n' + h'$ for $(n, n', h, h') \in N^2 \times H^2$, then $(n - n') + (h - h') = 0_M$.
Hence $n = n'$ and $h = h'$ by Proposition 1.7.2(ii), so $\overline{\sigma}_H(n + h) := \sigma(n)$ is well-defined.
Fix $(r, s, n, n', h, h') \in R^2 \times N^2 \times H^2$. Then $\overline{\sigma}_H(r(n + h) + s(n' + h')) = \sigma(rn + sn') = r\sigma(n) + s\sigma(n')$.
Thus $\overline{\sigma}_H \in \mathrm{Hom}_R(M, M_1)$ by Definition 1.4.1(i). The extension is unique by the defining formula and Proposition 1.7.2(iii).
(ii) Fix $H \in \mathcal{C}_N(M)$. Then $\overline{\sigma}_H|_N = \sigma$ by the defining formula with $h = 0_M$.
The identity $\operatorname{im} \overline{\sigma}_H = \operatorname{im} \sigma = M_1$ holds by the same formula, so $\overline{\sigma}_H \in \mathcal{E}_\sigma$.
(iii) Fix $H \in \mathcal{C}_N(M)$. Then $\ker \overline{\sigma}_H = \{n + h : (n, h) \in N \times H,\ \sigma(n) = 0_{M_1}\} = \ker \sigma + H$.
Since $\ker \sigma \le N$ and $N \cap H = \{0_M\}$, $(\ker \sigma) \cap H = \{0_M\}$. Therefore $\ker \overline{\sigma}_H = \ker \sigma \oplus H$ by Definition 1.7.1(ii).
(iv) Suppose σ is an isomorphism. For $H \in \mathcal{C}_N(M)$, $\ker \overline{\sigma}_H = H$ by (iii) and $\ker \sigma = \{0_M\}$.
Conversely, fix $\tau \in \mathcal{E}_\sigma$. If $x \in N \cap \ker \tau$, then $0_{M_1} = \tau(x) = \sigma(x)$ and $x = 0_M$ by injectivity of σ.
For $m \in M$, set $n := \sigma^{-1}(\tau(m)) \in N$. Then $\tau(m - n) = 0_{M_1}$, so $m = n + (m - n) \in N + \ker \tau$.
Thus $\ker \tau \in \mathcal{C}_N(M)$ by Definition 1.7.1(ii). Starting with H recovers H by the first sentence.
Starting with τ, write $m = n + k$ with $(n, k) \in N \times \ker \tau$. Then $\tau(m) = \sigma(n) = \overline{\sigma}_{\ker \tau}(m)$, so $\overline{\sigma}_{\ker \tau} = \tau$.

Structural Commentary: Coordinates extend maps; kernels classify extensions when σ is invertible.
Key Insight: An isomorphic restriction makes the extension kernel recover its complement. □

Definition 1.8.2. *Retract*
Let $N \le M$.
(i) A *retraction* onto N is a map $\rho \in \mathrm{Hom}_R(M, N)$ such that $\rho|_N = \mathrm{id}_N$.
(ii) The submodule N is a *retract* of M if $(\exists\, \rho \in \mathrm{Hom}_R(M, N))\ \rho|_N = \mathrm{id}_N$.

Corollary 1.8.3. *Retracts Are Direct Summands*
For $N \le M$, N is a retract of $M \iff (\exists\, H \le M)\ M = N \oplus H$.

Proof.

(⇒) Choose $\rho \in \mathrm{Hom}_R(M, N)$ with $\rho|_N = \mathrm{id}_N$ by Definition 1.8.2(ii).
For $m \in M$, $\rho(m - \rho(m)) = \rho(m) - \rho(\rho(m)) = 0_N$ by $\rho(m) \in N$, $\rho|_N = \mathrm{id}_N$; $m = \rho(m) + (m - \rho(m)) \in N + \ker \rho$, so $M = N + \ker \rho$.
If $x \in N \cap \ker \rho$, then $x = \rho(x) = 0_N$; hence $N \cap \ker \rho = \{0_M\}$. Therefore $M = N \oplus \ker \rho$ by Definition 1.7.1(ii).
(⇐) Fix $H \le M$ with $M = N \oplus H$. Define $\rho : M \to N$ by $\rho(n + h) := n$. The map is well-defined by Proposition 1.7.2(ii).
For $(r, s, n, n', h, h') \in R^2 \times N^2 \times H^2$, $\rho(r(n + h) + s(n' + h')) = rn + sn' = r\rho(n + h) + s\rho(n' + h')$.
Thus $\rho \in \mathrm{Hom}_R(M, N)$ by Definition 1.4.1(i), and $\rho|_N = \mathrm{id}_N$. Hence N is a retract of M by Definition 1.8.2(ii).

Structural Commentary: Retraction kernels are complements; projections reverse the construction.
Key Insight: A retraction separates every element into image and kernel components. □

Definition 1.8.4. *One-Sided Inverses*
Let $\tau \in \mathrm{Hom}_R(A, B)$.
(i) A *left inverse* of τ is $\lambda \in \mathrm{Hom}_R(B, A)$ such that $\lambda\tau = \mathrm{id}_A$.
(ii) A *right inverse* of τ is $\rho \in \mathrm{Hom}_R(B, A)$ such that $\tau\rho = \mathrm{id}_B$.
(iii) A *two-sided inverse* of τ is $\upsilon \in \mathrm{Hom}_R(B, A)$ such that $\upsilon\tau = \mathrm{id}_A$ and $\tau\upsilon = \mathrm{id}_B$.

Theorem 1.8.5. *One-Sided Inverses and Complements*
Let $\sigma \in \mathrm{Hom}_R(M, M_1)$. For an R-module P and $S \le P$, set $\mathcal{C}_S(P) := \{H \le P : P = S \oplus H\}$.
Set $\mathcal{L}_\sigma := \{\lambda \in \mathrm{Hom}_R(M_1, M) : \lambda\sigma = \mathrm{id}_M\}$. Set $\mathcal{R}_\sigma := \{\rho \in \mathrm{Hom}_R(M_1, M) : \sigma\rho = \mathrm{id}_{M_1}\}$.
(i) If σ is injective, then $\lambda \mapsto \ker \lambda$ is a bijection $\mathcal{L}_\sigma \to \mathcal{C}_{\operatorname{im}\sigma}(M_1)$.
(ii) If σ is surjective, then $\rho \mapsto \operatorname{im} \rho$ is a bijection $\mathcal{R}_\sigma \to \mathcal{C}_{\ker\sigma}(M)$.

Proof.

(i) Assume σ is injective. Fix $H \in \mathcal{C}_{\operatorname{im}\sigma}(M_1)$. For $y \in M_1$, write $y = u + h$ with $(u, h) \in \operatorname{im}\sigma \times H$.
This pair is unique by Proposition 1.7.2(iii). Choose unique $m \in M$ with $u = \sigma(m)$ by $u \in \operatorname{im}\sigma$, injectivity of σ.
Define $\lambda_H(y) := m$. For $y = \sigma(m) + h$, $z = \sigma(m') + h'$, and $(r, s) \in R^2$, $\lambda_H(ry + sz) = rm + sm' = r\lambda_H(y) + s\lambda_H(z)$.
Thus $\lambda_H \in \mathrm{Hom}_R(M_1, M)$ by Definition 1.4.1(i). Identities $\lambda_H\sigma = \mathrm{id}_M$ and $\ker \lambda_H = H$ hold by defining formula, so $\lambda_H \in \mathcal{L}_\sigma$.
Conversely, fix $\lambda \in \mathcal{L}_\sigma$. If $y \in \operatorname{im}\sigma \cap \ker \lambda$, write $y = \sigma(m)$. Then $m = \lambda\sigma(m) = 0_M$, so $y = 0_{M_1}$.
For $y \in M_1$, $y = \sigma\lambda(y) + (y - \sigma\lambda(y))$ and $y - \sigma\lambda(y) \in \ker \lambda$. Hence $\ker \lambda \in \mathcal{C}_{\operatorname{im}\sigma}(M_1)$ by Definition 1.7.1(ii).
Start with H, recover H by $\ker \lambda_H = H$. Start with λ, recover λ by decomposition $y = \sigma\lambda(y) + (y - \sigma\lambda(y))$, injectivity of σ.
(ii) Assume σ is surjective. Fix $H \in \mathcal{C}_{\ker\sigma}(M)$ and set $\sigma_H := \sigma|_H : H \to M_1$.
The map σ_H is an R-homomorphism with $\ker \sigma_H = H \cap \ker \sigma = \{0_M\}$.
Fix $y \in M_1$. Choose $m \in M$ with $\sigma(m) = y$, and write $m = k + h$ with $(k, h) \in \ker\sigma \times H$.
Then $y = \sigma_H(h)$, so σ_H is an isomorphism by Proposition 1.4.2(iv) and Definition 1.4.1(iv).
For $y \in M_1$, define $\rho_H(y) := \sigma_H^{-1}(y) \in H \subseteq M$. For $(r, s, y, z) \in R^2 \times M_1^2$, $\sigma_H(r\rho_H(y) + s\rho_H(z)) = ry + sz = \sigma_H(\rho_H(ry + sz))$.
Hence $\rho_H(ry + sz) = r\rho_H(y) + s\rho_H(z)$ by injectivity of σ_H. Thus $\rho_H \in \mathrm{Hom}_R(M_1, M)$ by Definition 1.4.1(i).
The identities $\sigma\rho_H = \mathrm{id}_{M_1}$ and $\operatorname{im}\rho_H = H$ hold by construction, so $\rho_H \in \mathcal{R}_\sigma$.
Conversely, fix $\rho \in \mathcal{R}_\sigma$. The map ρ is injective by $\rho(y) = 0_M \Rightarrow y = \sigma\rho(y) = 0_{M_1}$.
If $x \in \operatorname{im}\rho \cap \ker\sigma$, write $x = \rho(y)$. Then $y = \sigma\rho(y) = \sigma(x) = 0_{M_1}$, so $x = 0_M$.
For $m \in M$, $m = (m - \rho\sigma(m)) + \rho\sigma(m)$ with $m - \rho\sigma(m) \in \ker\sigma$ and $\rho\sigma(m) \in \operatorname{im}\rho$. Thus $\operatorname{im}\rho \in \mathcal{C}_{\ker\sigma}(M)$ by Definition 1.7.1(ii).
Starting with H recovers H by $\operatorname{im}\rho_H = H$. Starting with ρ recovers ρ by $(\sigma|_{\operatorname{im}\rho})^{-1}(y) = \rho(y)$.

Structural Commentary: Coordinates construct inverses; decompositions recover their complements.
Key Insight: Left inverses split images; right inverses split kernels. □

2 Noetherian Modules and PID Structure Theorems

This section begins with the structure facts that make free modules and finite generation usable: rank, splitting maps onto free modules, Noetherian modules. It then develops the classification of finitely generated modules over a principal ideal domain. The guiding model is the abelian-group theorem over $\mathbb{Z}$ and, later, the polynomial-module classification behind rational canonical form.

2.1 Free Modules, Bases, and Universal Property

Throughout this subsection, R denotes a nonzero commutative ring with identity 1_R.
The symbol M denotes an R-module with additive identity 0_M.

Definition 2.1.1. *Free Module and Basis*

(i) A subset $\mathcal{B} \subseteq M$ is a *basis* of M if $\mathcal{B}$ is R-linearly independent and $\langle\langle \mathcal{B} \rangle\rangle_R = M$.
(ii) An R-module M is *free* if M has a basis.
(iii) If $\mathcal{B}$ is a basis of M, then M is *free on* $\mathcal{B}$.

The empty set is independent and generates $\{0_M\}$, so the zero module is free on $\emptyset$.

Theorem 2.1.2. *Basis Means Unique Finite Coordinates*
For $\mathcal{B} \subseteq M$, define $\Sigma_{\mathcal{B}} : \bigoplus_{b\in\mathcal{B}} R \to M$ by $\Sigma_{\mathcal{B}}((r_b)_{b\in\mathcal{B}}) := \sum_{b\in\mathcal{B}} r_b b$. Then $\mathcal{B}$ is a basis of M if and only if $\Sigma_{\mathcal{B}}$ is bijective.

Proof. The sum defining $\Sigma_{\mathcal{B}}$ is finite by membership in $\bigoplus_{b\in\mathcal{B}} R$, so $\Sigma_{\mathcal{B}}$ is well-defined.

($\Rightarrow$) Assume $\mathcal{B}$ is a basis. Then $\Sigma_{\mathcal{B}}$ is surjective by $\langle\langle \mathcal{B} \rangle\rangle_R = M$.
Fix $r = (r_b)_{b\in\mathcal{B}}$, $s = (s_b)_{b\in\mathcal{B}}$ with $\Sigma_{\mathcal{B}}(r) = \Sigma_{\mathcal{B}}(s)$, set $H := \operatorname{supp}(r) \cup \operatorname{supp}(s)$. Then $|H| < \infty$ and $\sum_{b\in H} (r_b - s_b)b = 0_M$.
Thus $(\forall\, b \in H)\, r_b = s_b$ by the R-linear independence of $\mathcal{B}$. Both families vanish on $\mathcal{B} \setminus H$, so $r = s$ and $\Sigma_{\mathcal{B}}$ is injective.
($\Leftarrow$) Assume $\Sigma_{\mathcal{B}}$ is bijective. Then $M = \operatorname{im} \Sigma_{\mathcal{B}} = \langle\langle \mathcal{B} \rangle\rangle_R$ by surjectivity and Definition 1.2.1(i), so $\mathcal{B}$ spans M.
Fix finite $F \subseteq \mathcal{B}$ and $(r_b)_{b\in F} \in R^F$ with $\sum_{b\in F} r_b b = 0_M$, and extend this family by 0_R to $\bar{r} \in \bigoplus_{b\in\mathcal{B}} R$.
Then $\Sigma_{\mathcal{B}}(\bar{r}) = 0_M = \Sigma_{\mathcal{B}}(0)$, so $\bar{r} = 0$ by injectivity. Thus $(\forall\, b \in F)\, r_b = 0_R$, so $\mathcal{B}$ is R-linearly independent by Definition 1.3.1(i).
Therefore $\mathcal{B}$ is a basis by Definition 2.1.1(i).

Structural Commentary: Surjectivity encodes spanning, while injectivity encodes rigidity of finite relations.
Key Insight: Coordinate families are the fibers of the summation map. □

Proposition 2.1.3. *Bases Are Minimal Spanning and Maximal Independent*
If $\mathcal{B}$ is a basis of M, then:
(i) $(\forall b \in \mathcal{B})\ \langle\langle \mathcal{B} \setminus \{b\} \rangle\rangle_R \neq M$;
(ii) $(\forall v \in M \setminus \mathcal{B})\ \mathcal{B} \cup \{v\}$ *is R-linearly dependent.*

Proof.

(i) Fix $b \in \mathcal{B}$ and suppose $b \in \langle\langle \mathcal{B} \setminus \{b\} \rangle\rangle_R$.
Then b has a coordinate family vanishing at b, whereas the singleton family with coefficient 1_R at b represents b.
These distinct families both map to b under $\Sigma_{\mathcal{B}}$, contrary to injectivity by Theorem 2.1.2. Hence $b \notin \langle\langle \mathcal{B} \setminus \{b\} \rangle\rangle_R$.
(ii) Fix $v \in M \setminus \mathcal{B}$. There is $r = (r_b)_{b\in\mathcal{B}} \in \bigoplus_{b\in\mathcal{B}} R$ with $\Sigma_{\mathcal{B}}(r) = v$ by Theorem 2.1.2.
Thus $1_R v - \sum_{b\in\mathcal{B}} r_b b = 0_M$ is a finite relation in $\mathcal{B} \cup \{v\}$ with nonzero coefficient 1_R at v.
Hence $\mathcal{B} \cup \{v\}$ is R-linearly dependent by Definition 1.3.1(ii).

Structural Commentary: Coordinate uniqueness obstructs deletion; coordinate existence creates dependence after enlargement.
Key Insight: Removing or adjoining one element destroys one basis condition. □

Theorem 2.1.4. *Maps Out of a Free Module*
Let M be free on $\mathcal{B}$, let N be an R-module, and let $f : \mathcal{B} \to N$. Then:
(i) $(\exists\, \widetilde{f} \in \operatorname{Hom}_R(M, N))\ \widetilde{f}|_{\mathcal{B}} = f$.
(ii) If $\phi, \psi \in \operatorname{Hom}_R(M, N)$ satisfy $\phi|_{\mathcal{B}} = \psi|_{\mathcal{B}} = f$, then $\phi = \psi$.

Proof.

(i) For $r = (r_b)_{b\in\mathcal{B}} \in \bigoplus_{b\in\mathcal{B}} R$, define $\widetilde{f}(\Sigma_{\mathcal{B}}(r)) := \sum_{b\in\mathcal{B}} r_b f(b)$.
This defines $\widetilde{f} : M \to N$ by the bijectivity in Theorem 2.1.2 and the finite support of r.
For $r, s \in \bigoplus_{b\in\mathcal{B}} R$ and $a, c \in R$, $\Sigma_{\mathcal{B}}(ar + cs) = a\Sigma_{\mathcal{B}}(r) + c\Sigma_{\mathcal{B}}(s)$ by the module laws.
Thus $\widetilde{f}(a\Sigma_{\mathcal{B}}(r) + c\Sigma_{\mathcal{B}}(s)) = a\widetilde{f}(\Sigma_{\mathcal{B}}(r)) + c\widetilde{f}(\Sigma_{\mathcal{B}}(s))$, so $\widetilde{f} \in \operatorname{Hom}_R(M, N)$.
The singleton coordinate families satisfy $(\forall\, b \in \mathcal{B})\ \widetilde{f}(b) = f(b)$, so $\widetilde{f}|_{\mathcal{B}} = f$.
(ii) Let $\phi, \psi \in \operatorname{Hom}_R(M, N)$ satisfy $\phi|_{\mathcal{B}} = \psi|_{\mathcal{B}} = f$. For $r = (r_b)_{b\in\mathcal{B}}$, $\phi(\Sigma_{\mathcal{B}}(r)) = \sum_{b\in\mathcal{B}} r_b f(b) = \psi(\Sigma_{\mathcal{B}}(r))$ by R-linearity.
Thus $\phi = \psi$ by the surjectivity of $\Sigma_{\mathcal{B}}$ and Assumption 3.1.2.

Structural Commentary: Coordinate evaluation constructs the extension; summation-map surjectivity globalizes basis agreement.
Key Insight: Agreement on a basis propagates through finite coordinate sums. □

Counterexample 2.1.5. *A Free Module with a Nonfree Submodule*
A submodule of a free module need not be free. Let $R = \mathbb{Z} \times \mathbb{Z}$ with componentwise operations, so $1_R = (1, 1)$ and $0_R = (0, 0)$.
Regular module R is free on $\{(1, 1)\}$ as $(\forall\, x \in R)(\exists!\, r \in R)\, x = r(1, 1)$. Set $N := \mathbb{Z} \times \{0\}$.
Then $N \trianglelefteq R$, so $N \leq {}_R R$ by Proposition 1.1.7. The nonzero scalar $(0, 1)$ satisfies $(\forall\, v \in N)\, (0, 1)v = 0_R$.
For $\emptyset \neq S \subseteq N$, fix $v \in S$; then $(0, 1)v = 0_R$ is a relation with nonzero coefficient.
Thus S is dependent by Definition 1.3.1(ii). The set $\emptyset$ does not span N because $(1, 0) \in N \setminus \{0_R\}$.
Hence N has no basis and is not free. Therefore $\emptyset$ is maximal independent but not spanning.

Counterexample 2.1.6. *A Quotient of a Free Module Need Not Be Free*
A quotient of a free module by a free submodule need not be free. Fix $n \in \mathbb{N}$ with $n > 1$.
As a $\mathbb{Z}$-module, $\mathbb{Z}$ is free on $\{1\}$, and $n\mathbb{Z}$ is free on $\{n\}$.
Set $Q := \mathbb{Z}/n\mathbb{Z}$ and $\overline{1} := 1 + n\mathbb{Z}$. Then $Q = \langle\!\langle \overline{1} \rangle\!\rangle_{\mathbb{Z}} \neq \{0_Q\}$, so $\{\overline{1}\}$ minimally spans Q.
However, $(\forall\, q \in Q)\, nq = 0_Q$ with $n \neq 0_{\mathbb{Z}}$. Thus each $\emptyset \neq S \subseteq Q$ is linearly dependent by Definition 1.3.1(ii).
Hence Q has no basis and is not free; the minimally spanning set $\{\overline{1}\}$ is therefore not a basis.

Principle 2.1.7. *Modules Are Not Just Vector Spaces with New Letters*
For a general ring R, the following vector-space implications fail:

(i) $N \leq M \nRightarrow (\exists\, H \leq M)\ M = N \oplus H$.
(ii) M finitely generated and $N \leq M \nRightarrow N$ finitely generated.
(iii) $M \neq \{0_M\} \nRightarrow (\exists\, \emptyset \neq S \subseteq M)\ S$ is linearly independent.
(iv) S minimally spans $M \nRightarrow S$ is a basis.
(v) S is maximal linearly independent in $M \nRightarrow S$ is a basis.
(vi) M free and $N \leq M \nRightarrow N$ free.
(vii) F free and $S \leq F \nRightarrow F/S$ free.

Counterexamples 1.7.3 and 1.2.4 witness (i) and (ii). Counterexample 2.1.5 witnesses (iii), (v), and (vi).
Counterexample 2.1.6 witnesses (iv) and (vii). Over a field, these obstructions vanish; algebraically, $R \setminus \{0_R\} \subseteq R^{\times}$.

2.2 Rank and Coordinate Models of Free Modules

Throughout this subsection, assume Principles 6.1.3 and 6.1.2.

Perspective 2.2.1. *Dimension as the Field Case of Rank*
An $\mathbb{F}$-vector space is a free $\mathbb{F}$-module by Theorem 1.3.17(ii) and Definition 2.1.1.
Finite basis cardinality is invariant by Theorem 1.4.1. Arbitrary Hamel-basis cardinality is invariant by Theorem 1.4.5.
Over general scalar rings, rank invariance requires additional hypotheses.

Theorem 2.2.2. *Rank Is Well Defined for Free Modules*
Let A be a nonzero commutative ring with identity 1_A, and let M be a free A-module.

(i) If B, B' are A-bases of M, then $|B| = |B'|$.
(ii) If B is an A-basis of M and $S \subseteq M$ satisfies $\langle\!\langle S \rangle\!\rangle_A = M$, then $|B| \leq |S|$.

Proof. Choose a maximal ideal $\mathfrak{m} \trianglelefteq A$ by Theorem 3.4.5. Set $\mathfrak{m}M := \langle\!\langle \{ax : (a, x) \in \mathfrak{m} \times M\} \rangle\!\rangle_A$, $K := A/\mathfrak{m}$, and $V := M/\mathfrak{m}M$.
The ring K is a field by Theorem 3.4.4. If $a - a' \in \mathfrak{m}$ and $x - x' \in \mathfrak{m}M$, then $ax - a'x' = (a - a')x + a'(x - x') \in \mathfrak{m}M$.
Hence $(a + \mathfrak{m})(x + \mathfrak{m}M) := ax + \mathfrak{m}M$ is well-defined, and the inherited module identities make V a K-vector space.
Fix an A-basis B of M, set $X_B := \{ \sum_{b \in B} i_b b : (i_b)_{b \in B} \in \bigoplus_{b \in B} \mathfrak{m} \}$. Inclusion $X_B \leq_A M$ holds by coefficientwise ideal closure.
If $i \in \mathfrak{m}$ and $x = \sum_{b \in B} r_b b$, then $ix = \sum_{b \in B} (ir_b)b \in X_B$. Thus $\mathfrak{m}M \subseteq X_B$ by Proposition 1.2.2(iii).
Conversely, $X_B \subseteq \mathfrak{m}M$ by Definition 1.2.1(i), so $\mathfrak{m}M = X_B$. Write $\overline{x} := x + \mathfrak{m}M$ for $x \in M$ and set $\overline{B} := \{\overline{b} : b \in B\}$.
The set $\overline{B}$ spans V by spanning of B. Let $F \subseteq B$ be finite and $(a_b)_{b \in F} \in A^F$ satisfy $\sum_{b \in F} (a_b + \mathfrak{m})\overline{b} = 0_V$.
Then $\sum_{b \in F} a_b b \in \mathfrak{m}M = X_B$. Consequently, $(\forall b \in F)\ a_b \in \mathfrak{m}$ by coordinate uniqueness in Theorem 2.1.2.
Thus $\overline{B}$ is K-linearly independent by Definition 1.3.1. If $\overline{b} = \overline{b'}$ for distinct $b, b' \in B$, then $b - b' \in X_B$.
The same coordinate uniqueness forces $1_A \in \mathfrak{m}$, contrary to $\mathfrak{m} \neq A$.
The map $b \mapsto \overline{b}$ is a bijection $B \to \overline{B}$ by the preceding injectivity and the definition of $\overline{B}$.
Hence $\overline{B}$ is a K-basis of V by spanning and independence.
Because B was arbitrary, the construction applies to every A-basis of M.

(i) If B, B' are A-bases of M, then $\overline{B}, \overline{B'}$ are K-bases of V. Thus $\left|\overline{B}\right| = \left|\overline{B'}\right|$ by Theorem 1.4.5.
$|B| = \left|\overline{B}\right| = \left|\overline{B'}\right| = |B'|$ by bijections above.

(ii) Fix an A-basis B of M and $S \subseteq M$ with $\langle\!\langle S \rangle\!\rangle_A = M$. Then $\overline{S} := \{\overline{s} : s \in S\}$ spans V by spanning of S.
Choose a K-basis $C \subseteq \overline{S}$ by Theorem 1.3.17(iv). Since $\overline{B}, C$ are K-bases, $\left|\overline{B}\right| = |C|$ by Theorem 1.4.5; $|B| = \left|\overline{B}\right| = |C| \leq \left|\overline{S}\right| \leq |S|$.

Structural Commentary: Reduction modulo $\mathfrak{m}$ transfers each A-basis to an equipotent K-basis for vector-space comparison.
Key Insight: Basis coordinates detect exactly which cosets vanish modulo $\mathfrak{m}$. □

Definition 2.2.3. *Rank of a Free Module*
Let A be a nonzero commutative ring with identity, let M be a free A-module, and fix an A-basis B of M.
Define the *rank* of M by $\mathrm{rk}_A(M) := |B|$. This value is independent of B by Theorem 2.2.2(i).
The empty basis of the zero module satisfies $\mathrm{rk}_A(\{0_M\}) = 0$.

Counterexample 2.2.4. *Why Commutativity Matters for Rank*
Commutativity cannot be omitted from Theorem 2.2.2. Here *basis* and *free* mean the left-module analogues of Definition 2.1.1.
Let $\mathbb{F}$ be a field, let $V = \bigoplus_{k \in \mathbb{N}_0} \mathbb{F}e_k$, and set $E := \mathcal{L}(V)$ with multiplication given by composition.
Define $p, u \in E$ by $p(e_0) = e_0$, $(\forall k \in \mathbb{N})\ p(e_k) = 0_V$, and $(\forall k \in \mathbb{N}_0)\ u(e_k) = e_{k+1}$. These operators exist uniquely by Theorem 2.1.4.
Ring E has identity id_V, $\mathrm{id}_V(e_0) = e_0 \neq 0_V$, so $\mathrm{id}_V \neq 0_E$. Since $(pu)(e_0) = 0_V \neq e_1 = (up)(e_0)$, ring E is noncommutative.
The left regular E-module E is free on $\{\mathrm{id}_V\}$ because $(\forall h \in E)(\exists! \alpha \in E)\ h = \alpha\, \mathrm{id}_V$.
Fix $n \in \mathbb{N}$ and define $\beta_0, \ldots, \beta_{n-1} \in E$ on the basis by $(\forall (k, s) \in \mathbb{N}_0 \times \{0, \ldots, n-1\})\ \beta_s(e_{kn+s}) = e_k$.
On the complementary residue classes, $(\forall (k, s, t) \in \mathbb{N}_0 \times \{0, \ldots, n-1\}^2)\, [s \neq t \Rightarrow \beta_s(e_{kn+t}) = 0_V]$.
These operators exist uniquely by Theorem 2.1.4, since Euclidean division indexes each basis vector exactly once.
If $(\alpha_s)_{s=0}^{n-1} \in E^n$ and $\sum_{s=0}^{n-1} \alpha_s \beta_s = 0_E$, then $(\forall (k, t) \in \mathbb{N}_0 \times \{0, \ldots, n-1\})\ \alpha_t(e_k) = 0_V$.
Hence $(\forall t \in \{0, \ldots, n-1\})\ \alpha_t = 0_E$ by uniqueness in Theorem 2.1.4. Thus $\{\beta_0, \ldots, \beta_{n-1}\}$ is E-linearly independent.
Fix $h \in E$ and define $\alpha_0, \ldots, \alpha_{n-1} \in E$ by $(\forall (k, s) \in \mathbb{N}_0 \times \{0, \ldots, n-1\})\ \alpha_s(e_k) := h(e_{kn+s})$.

These operators exist uniquely by Theorem 2.1.4. Then $(\forall (k,t) \in \mathbb{N}_0 \times \{0,\dots,n-1\})\ (\sum_{s=0}^{n-1} \alpha_s\beta_s)(e_{kn+t}) = h(e_{kn+t})$.
Hence $h = \sum_{s=0}^{n-1} \alpha_s\beta_s$ by the same uniqueness. Thus $\{\beta_0,\dots,\beta_{n-1}\}$ is an E-basis of E.
If $s,t \in \{0,\dots,n-1\}$ and $s \neq t$, then $\beta_s(e_s) = e_0 \neq 0_V = \beta_t(e_s)$. Therefore $|\{\beta_0,\dots,\beta_{n-1}\}| = n$.
Since $n \in \mathbb{N}$ was arbitrary, ${}_EE$ has bases of every positive finite cardinality. Thus rank invariance fails without commutativity.

Theorem 2.2.5. *Finite-Support Coordinate Modules*
Let A be a nonzero commutative ring with identity, let B be a set, and equip A^B with pointwise operations. Using Definition 3.1.4, set $A^{(B)} := \{f \in A^B : |\mathrm{supp}(f)| < \infty\}$. For $b \in B$, define $\delta_b \in A^{(B)}$ by $\delta_b(b) = 1_A$, $(\forall x \in B \setminus \{b\})\ \delta_b(x) = 0_A$.

(i) $A^{(B)} \leq_A A^B$.
(ii) The set $\{\delta_b : b \in B\}$ is an A-basis of $A^{(B)}$; equivalently, $A^{(B)} = \bigoplus_{b\in B} A\delta_b$.
(iii) $\mathrm{rk}_A(A^{(B)}) = |B|$.

Proof.

(i) The module identities in A^B hold pointwise, and equality is by Assumption 3.1.2. The zero function has empty support. $(\forall (a,c,f,g) \in A^2 \times (A^{(B)})^2)\ \mathrm{supp}(af+cg) \subseteq \mathrm{supp}(f) \cup \mathrm{supp}(g)$. Thus $A^{(B)} \leq_A A^B$ by Theorem 1.1.5.

(ii) $(\forall b \in B)\ \mathrm{supp}(\delta_b) = \{b\}$, so $\{\delta_b : b \in B\} \subseteq A^{(B)}$. The identity $f = \sum_{b\in\mathrm{supp}(f)} f(b)\delta_b$ holds for $f \in A^{(B)}$ by Assumption 3.1.2.
Hence the δ_b span $A^{(B)}$. Let $F \subseteq B$ be finite and $(a_b)_{b\in F} \in A^F$ satisfy $\sum_{b\in F} a_b\delta_b = 0_{A^{(B)}}$.
Then $(\forall x \in F)\ a_x = 0_A$ by evaluation at x. Hence $\{\delta_b : b \in B\}$ is independent by Definition 1.3.1.
If $x_b \in A\delta_b$ for $b \in F$ and $\sum_{b\in F} x_b = 0_{A^{(B)}}$, write $x_b = a_b\delta_b$. Then $(\forall b \in F)\ x_b = 0_{A^{(B)}}$ by independence of $\{\delta_b : b \in B\}$.
Therefore $\sum_{b\in B} A\delta_b$ is direct by Proposition 1.7.2(ii). $A^{(B)} = \bigoplus_{b\in B} A\delta_b$ by the spanning established above.

(iii) The map $b \mapsto \delta_b$ is surjective onto $\{\delta_b : b \in B\}$ by definition. If $b \neq b'$, then $\delta_b(b) = 1_A \neq 0_A = \delta_{b'}(b)$, so the map is injective.
Thus $\mathrm{rk}_A(A^{(B)}) = |B|$ by Definition 2.2.3.

Structural Commentary: Support closure defines the submodule; delta evaluation proves independence, directness, and cardinality.
Key Insight: A delta coefficient equals the function value at its index. □

Proposition 2.2.6. *Free Modules Are Coordinate Modules*
Let A be a nonzero commutative ring with identity.

(i) If M is a free A-module with A-basis B, then $b \mapsto \delta_b$ extends to an isomorphism $\Phi_B : M \xrightarrow{\sim} A^{(B)}$.
(ii) If M, N are free A-modules, then $M \cong N \iff \mathrm{rk}_A(M) = \mathrm{rk}_A(N)$.

Proof.

(i) Assignment $b \mapsto \delta_b$ extends uniquely to $\Phi_B \in \mathrm{Hom}_A(M, A^{(B)})$ by Theorem 2.1.4. Set $\{\delta_b : b \in B\}$ is basis by Theorem 2.2.5(ii).
Thus $\delta_b \mapsto b$ extends uniquely to $\Psi_B \in \mathrm{Hom}_A(A^{(B)}, M)$ by Theorem 2.1.4.
The composites $\Psi_B\Phi_B$ and $\Phi_B\Psi_B$ restrict to the identity on their respective bases.
Therefore they equal identity maps by uniqueness in Theorem 2.1.4. Hence Φ_B is isomorphism with inverse Ψ_B.

(ii) $(\Rightarrow)$ Let $\psi : M \xrightarrow{\sim} N$ and fix an A-basis C of M. The set $\psi(C)$ spans N by surjectivity.
If $F \subseteq C$ is finite and $(a_c)_{c\in F} \in A^F$ satisfies $\sum_{c\in F} a_c\psi(c) = 0_N$, then $\sum_{c\in F} a_c c = 0_M$ by injectivity.
Thus $(\forall c \in F)\ a_c = 0_A$ by independence of C. Set $\psi(C)$ is A-basis of N by preceding spanning, independence.
Since $\psi|_C : C \to \psi(C)$ is a bijection, $\mathrm{rk}_A(M) = |C| = |\psi(C)| = \mathrm{rk}_A(N)$.

(ii) $(\Leftarrow)$ Assume $\mathrm{rk}_A(M) = \mathrm{rk}_A(N)$ and fix A-bases C, D. Then $|C| = |D|$ by Definition 2.2.3; choose bijection $u : C \to D$.
The maps u, u^{-1} extend to homomorphisms $\Phi : M \to N$ and $\Psi : N \to M$ by Theorem 2.1.4.
Their composites restrict to the identity on C, D, respectively.
Hence $\Psi\Phi = \mathrm{id}_M$ and $\Phi\Psi = \mathrm{id}_N$ by uniqueness in Theorem 2.1.4, so Φ is an isomorphism.

Structural Commentary: Basis-map extension constructs coordinate isomorphisms, while basis transport proves rank invariance.
Key Insight: A basis bijection extends to mutually inverse module maps. □

Theorem 2.2.7. *Independent Sets Are Bounded by Rank over a Domain*
Let A be integral domain with field of quotients Q, M be free A-module. If $S \subseteq M$ is A-linearly independent, then $|S| \leq \mathrm{rk}_A(M)$.

Proof. Fix an A-basis B of M and identify $M \cong A^{(B)}$ by Proposition 2.2.6(i).
The coordinate isomorphism preserves finite relations, so S remains A-linearly independent.
Identify A with its image in Q by Proposition 3.5.4; then $A^{(B)} \subseteq Q^{(B)}$. Fix $n \in \mathbb{N}$ and distinct $s_1,\dots,s_n \in S$.
Let $(q_1,\dots,q_n) \in Q^n$ satisfy $\sum_{i=1}^{n} q_i s_i = 0_{Q^{(B)}}$. For $i \in \{1,\dots,n\}$, choose $(c_i,e_i) \in A \times (A \setminus \{0_A\})$ with $q_i = \frac{c_i}{e_i}$ by Definition 3.5.3.
Set $d := \prod_{i=1}^{n} e_i$. Then $d \neq 0_A$ by integral-domain hypothesis, $(\forall i \in \{1,\dots,n\})\ dq_i \in A$.
Relation $\sum_{i=1}^{n} (dq_i)s_i = 0_{Q^{(B)}}$ holds by multiplication by d. Both sides lie in $A^{(B)}$, so the same equality holds in $A^{(B)}$.
Thus $(\forall i \in \{1,\dots,n\})\ dq_i = 0_A$ by A-linear independence. Since $d \neq 0_Q$, $(\forall i \in \{1,\dots,n\})\ q_i = 0_Q$.
Hence S is Q-linearly independent in $Q^{(B)}$ by Definition 1.3.1. Choose Q-basis C of $Q^{(B)}$ with $S \subseteq C$ by Theorem 1.3.17(iii).
Set $\{\delta_b : b \in B\}$ is Q-basis by Theorem 2.2.5(ii); $|C| = |B|$ by Theorem 1.4.5; $|S| \leq |C| = |B| = \mathrm{rk}_A(M)$ by $S \subseteq C$, Definition 2.2.3.
Structural Commentary: Coordinate embedding and denominator clearing preserve independence before vector-space basis comparison.
Key Insight: One common denominator converts each finite Q-relation into an A-relation. □

2.3 Free Quotients and Splittings

Theorem 2.3.1. *Free Quotients Split*
Assume the Choice Principle 6.1.2. Let A be a nonzero commutative ring with identity, and M, F be A-modules.

(i) If $\sigma : M \twoheadrightarrow F$ and F is free, then $(\exists\, \tau \in \mathrm{Hom}_A(F, M))[\sigma\tau = \mathrm{id}_F \wedge M = \ker\sigma \oplus \mathrm{im}\,\tau \wedge \mathrm{im}\,\tau \cong F]$.
(ii) If $S \leq M$ and M/S is free, then $(\exists\, T \leq M)[M = S \oplus T \wedge T \cong M/S]$ and $M \cong S \oplus M/S$.
(iii) If $S \leq M$ and $M, S, M/S$ are free, then $\mathrm{rk}_A(M) = \mathrm{rk}_A(S) + \mathrm{rk}_A(M/S)$, where $+$ denotes cardinal addition. If $\mathrm{rk}_A(M) \in \mathbb{N}_0$, then $\mathrm{rk}_A(M/S) = \mathrm{rk}_A(M) - \mathrm{rk}_A(S)$.

Proof.
(i) Fix A-basis B of F. Relation $(\forall b \in B)\, \sigma^{-1}(\{b\}) \neq \emptyset$ holds by surjectivity. Fix $(m_b)_{b\in B} \in \prod_{b\in B} \sigma^{-1}(\{b\})$ by Choice Principle.
The assignment $b \mapsto m_b$ extends uniquely to $\tau \in \mathrm{Hom}_A(F, M)$ by Theorem 2.1.4.
The equality $\sigma\tau = \mathrm{id}_F$ holds by $(\forall b \in B)\, \sigma\tau(b) = \sigma(m_b) = b$ and uniqueness in Theorem 2.1.4.
Decomposition $M = \ker\sigma \oplus \mathrm{im}\,\tau$ holds by Theorem 1.8.5(ii). Map τ is injective by $(\forall x \in F)[\tau(x) = 0_M \Rightarrow x = \sigma\tau(x) = 0_F]$.
Its corestriction $F \to \mathrm{im}\,\tau$ is an isomorphism by linearity and bijectivity.
(ii) Let $\pi : M \twoheadrightarrow M/S$ be the quotient map; $\ker\pi = S$ by Proposition 1.5.2(iii).
Fix a section $\tau \in \mathrm{Hom}_A(M/S, M)$ of π by (i), and set $T := \mathrm{im}\,\tau$. Relations $M = S \oplus T$, $T \cong M/S$ hold by (i) and $\ker\pi = S$.
Define $\Phi : S \oplus M/S \to M$ by $\Phi(s, q) := s + \tau(q)$. Map Φ is A-linear by linearity of τ, surjective by $M = S \oplus T$.
Identity $\ker\Phi = \{(0_M, 0_{M/S})\}$ holds by $S \cap T = \{0_M\}$, injectivity of τ. Map Φ injective by Proposition 1.4.2(iv), kernel identity.
It is an isomorphism by injectivity, surjectivity, and Definition 1.4.1(iv)(c). Relation $M \cong S \oplus M/S$ holds by $\Phi^{-1} : M \xrightarrow{\sim} S \oplus M/S$.
(iii) Fix $T \leq M$ with $M = S \oplus T$ and $\theta : T \xrightarrow{\sim} M/S$ by (ii). Fix A-bases B_S of S and B_Q of M/S, and set $B_T := \theta^{-1}(B_Q)$.
The set B_T is an A-basis of T, and $\theta|_{B_T} : B_T \to B_Q$ is a bijection, by bijectivity and linearity of θ.
Relation $B_S \cap B_T = \emptyset$ holds by $S \cap T = \{0_M\}$, $0_M \notin B_S \cup B_T$; $B_S \cup B_T$ spans M by $M = S \oplus T$, spanning properties of B_S, B_T.
It is A-linearly independent by Proposition 1.7.2(ii) and the independence of B_S, B_T.
The union $B_S \cup B_T$ is an A-basis of M by its spanning and independence.
The identities $|B_S \cup B_T| = |B_S| + |B_T|$ and $|B_T| = |B_Q|$ hold by disjointness and the bijection $\theta|_{B_T}$.
Rank equality in (iii) holds by Definition 2.2.3, identities. Under $\mathrm{rk}_A(M) \in \mathbb{N}_0$, both summand ranks lie in $\mathbb{N}_0$ by rank equality.
The formula $\mathrm{rk}_A(M/S) = \mathrm{rk}_A(M) - \mathrm{rk}_A(S)$ holds by subtraction in $\mathbb{N}_0$.

Structural Commentary: Choice lifts a free target basis to a section; direct-sum bases then add ranks.
Key Insight: The section image is simultaneously a kernel complement and quotient model. □

2.4 Noetherian Modules

Throughout this subsection, A denotes a nonzero commutative ring with identity 1_A.

Definition 2.4.1. *Noetherian Modules and Rings*
(i) An *ascending chain* in an A-module M is a sequence $(M_n)_{n\in\mathbb{N}}$ of submodules satisfying $(\forall\, n \in \mathbb{N})\, M_n \leq M_{n+1}$.
It *stabilizes* if $(\exists\, n_0 \in \mathbb{N})(\forall\, n \in \mathbb{N})\, [n \geq n_0 \Rightarrow M_n = M_{n_0}]$.
The module M is *Noetherian*, or satisfies the *ascending chain condition (ACC)*, if all its ascending chains stabilize.
(ii) The ring A is *Noetherian* if the regular A-module ${}_AA$ is Noetherian.
By Proposition 1.1.7, this means that all ascending chains of ideals of A stabilize.

Theorem 2.4.2. *Noetherian Means Finitely Generated Submodules*
Let M be an A-module.

(i) M is Noetherian $\iff (\forall\, N \leq M)\, N$ is finitely generated.
(ii) A is Noetherian $\iff (\forall\, \mathcal{I} \trianglelefteq A)\, \mathcal{I}$ is finitely generated as an A-module.

Proof.
(i) ($\Rightarrow$) Assume M is Noetherian and fix a non-finitely generated $N \leq M$. Set $\mathcal{C} := \{X \subseteq N : X \neq \emptyset\}$.
Fix $c : \mathcal{C} \to N$ satisfying $(\forall\, X \in \mathcal{C})\, c(X) \in X$ by Principle 6.1.2. Set $u_1 := c(N)$ and $N_1 := \langle\!\langle u_1 \rangle\!\rangle_A$.
Given $u_1, \ldots, u_n$, set $N_n := \langle\!\langle u_1, \ldots, u_n \rangle\!\rangle_A$. $N \setminus N_n \neq \emptyset$ by non-finite generation of N and finite generation of N_n.
Set $u_{n+1} := c(N \setminus N_n)$. These recursive clauses define $(u_n)_{n\in\mathbb{N}}$ by Principle 6.3.5(i).
$(\forall\, n \in \mathbb{N})\, N_n \leq N$ by Proposition 1.2.2(i). $(\forall\, n \in \mathbb{N})\, u_{n+1} \in N_{n+1} \setminus N_n$ by Definition 1.2.1(i),(iii).
$(\forall\, n \in \mathbb{N})\, N_n < N_{n+1} \leq M$ by the preceding containment and nonmembership. This contradicts ACC.
(i) ($\Leftarrow$) Suppose $(\forall\, N \leq M)\, N$ is finitely generated. Let $(M_n)_{n\in\mathbb{N}}$ be an ascending chain in M, and set $U := \bigcup_{n\in\mathbb{N}} M_n$.
$0_M \in U$ by $0_M \in M_1$. Fix $(a, b, x, y) \in A^2 \times U^2$. Choose $i, j \in \mathbb{N}$ with $x \in M_i$ and $y \in M_j$ by the definition of U.
For $k := \max\{i, j\}$, $ax + by \in M_k \subseteq U$ by $M_i, M_j \subseteq M_k$ and submodule closure.
Hence $U \leq M$ by Theorem 1.1.5. Fix $r \in \mathbb{N}$ and $(u_1, \ldots, u_r) \in U^r$ satisfying $U = \langle\!\langle u_1, \ldots, u_r \rangle\!\rangle_A$ by finite generation of U.
Fix $(k_1, \ldots, k_r) \in \mathbb{N}^r$ satisfying $(\forall\, j \in \{1, \ldots, r\})\, u_j \in M_{k_j}$. Set $k := \max\{k_1, \ldots, k_r\}$. Then $U \subseteq M_k$ by Proposition 1.2.2(iii).
$M_k \subseteq U$ by construction. $(\forall\, n \in \mathbb{N})\, [n \geq k \Rightarrow M_n = M_k]$ by $M_k \subseteq M_n \subseteq U \subseteq M_k$. Hence M is Noetherian by Definition 2.4.1(i).
(ii) ($\Rightarrow$) If A is Noetherian, then ${}_AA$ is Noetherian by Definition 2.4.1(ii). All ideals of A finitely generated by (i), Proposition 1.1.7.
(ii) ($\Leftarrow$) Suppose $(\forall\, \mathcal{I} \trianglelefteq A)\, \mathcal{I}$ is finitely generated as an A-module.
All submodules of ${}_AA$ are finitely generated by the supposition and Proposition 1.1.7.
Thus ${}_AA$ is Noetherian by (i), and A is Noetherian by Definition 2.4.1(ii).

Structural Commentary: Generator adjunction contradicts ACC; finite generation of the chain union forces stabilization.
Key Insight: Non-finite generation supplies a generator outside every current stage. □

2.5 Orders and Cyclic Modules over a PID

Convention 2.5.1. *Standing PID Notation*
In this and the subsequent PID-structure subsections, R denotes a principal ideal domain unless otherwise stated.
For $r \in R$, write $\langle r \rangle = Rr$; for an R-module M and $v \in M$, write $\langle\!\langle v \rangle\!\rangle_R = Rv$.

Motivation 2.5.2. *Scalar Arithmetic over a PID*
For a field F, $F \setminus \{0_F\} = F^\times$, so every nonzero scalar is invertible.
A PID need not permit universal inversion; divisibility, gcds, and lcms replace it. Module order records this scalar arithmetic.

Definition 2.5.3. *Order of a Submodule or Element*
Let M be R-module, $N \leq M$. An *order* of N is any $\alpha \in R$ satisfying $\operatorname{Ann}_R(N) = \{r \in R : (\forall\, n \in N)\, rn = 0_M\} = \langle \alpha \rangle$.
Write $o(N) = \alpha$; nonzero choices are understood up to associates. For $v \in M$, define $o(v) := o(\langle\!\langle v \rangle\!\rangle_R)$.

Proposition 2.5.4. *Orders and Direct Sums*
Let $n \in \mathbb{N}$, and let $M_1, \ldots, M_n$ be R-modules with $o(M_i) = \alpha_i$. Then $o(\bigoplus_{i=1}^{n} M_i) = \operatorname{lcm}(\alpha_1, \ldots, \alpha_n)$.

Proof. Set $I := \{1, \ldots, n\}$, $\lambda := \operatorname{lcm}(\alpha_1, \ldots, \alpha_n)$. For $r \in R$, $r \in \operatorname{Ann}_R(\bigoplus_{i \in I} M_i) \iff (\forall\, i \in I)\, \alpha_i \mid r \iff \lambda \mid r$.

Hence $\operatorname{Ann}_R(\bigoplus_{i=1}^{n} M_i) = \langle \lambda \rangle$. Thus $o(\bigoplus_{i=1}^{n} M_i) = \lambda$.
Structural Commentary: The direct-sum annihilator is the intersection of the summand annihilators.
Key Insight: Common annihilators are exactly multiples of the least common multiple. □

Theorem 2.5.5. *Cyclic Modules over a PID*
Let M be an R-module and let $v \in M$.

(i) If $\operatorname{Ann}_R(v) = \langle \alpha \rangle$, define $\phi_v : R/\langle \alpha \rangle \to \langle\!\langle v \rangle\!\rangle_R$ by $\phi_v(r + \langle \alpha \rangle) = rv$. Then ϕ_v is an R-module isomorphism.
(ii) If $C = \langle\!\langle v \rangle\!\rangle_R$ and $S \leq C$, then $(\exists\, s \in R)\, S = \langle\!\langle sv \rangle\!\rangle_R$.
(iii) Let $o(v) = \alpha$ and $\beta \in R$. If $(\alpha, \beta) \neq (0_R, 0_R)$, then $o(\beta v) = \frac{\alpha}{\gcd(\alpha, \beta)}$. If $\alpha = \beta = 0_R$, then $o(\beta v) = 1_R$.
(iv) Let $o(v) = \alpha$ and $\beta \in R$ satisfy $(\alpha, \beta) \neq (0_R, 0_R)$. Then $\langle\!\langle \beta v \rangle\!\rangle_R = \langle\!\langle v \rangle\!\rangle_R \iff \gcd(\alpha, \beta) \sim 1_R$.

Proof.

(i) Set $\psi : R \to \langle\!\langle v \rangle\!\rangle_R$, $\psi(r) = rv$. 1qThen $\operatorname{im} \psi = \langle\!\langle v \rangle\!\rangle_R$ and $\ker \psi = \operatorname{Ann}_R(v) = \langle \alpha \rangle$.
The induced map ϕ_v is an R-module isomorphism by Theorem 1.5.4(i).
(ii) Set $\psi : R \to C$, $\psi(r) = rv$, $I := \psi^{-1}(S) = \{r \in R : rv \in S\}$; $0_R \in I$ and $(\forall\, (a, b, x, y) \in R^2 \times I^2)\, \psi(ax + by) = a\psi(x) + b\psi(y) \in S$.
Thus $I \leq {}_R R$ by Theorem 1.1.5. Choose $s \in R$ with $I = \langle s \rangle$ by Definition 3.10.1(i).
The equality $S = \psi(I)$ holds by $S \leq \operatorname{im} \psi$, so $S = \psi(Rs) = R(sv) = \langle\!\langle sv \rangle\!\rangle_R$.
(iii) If $\alpha = \beta = 0_R$, then $\beta v = 0_M$ and $\operatorname{Ann}_R(\beta v) = R = \langle 1_R \rangle$, so $o(\beta v) = 1_R$. Suppose $(\alpha, \beta) \neq (0_R, 0_R)$, $d := \gcd(\alpha, \beta) \neq 0_R$.
Choose $\alpha', \beta' \in R$ with $\alpha = d\alpha'$, $\beta = d\beta'$, and $\gcd(\alpha', \beta') \sim 1_R$. Choose $x, y \in R$ with $x\alpha' + y\beta' = 1_R$ by Theorem 3.10.6(iii).
Fix $r \in R$. If $\alpha' \mid r\beta'$, then $\alpha' \mid (xr\alpha' + yr\beta') = r$; the reverse divisibility holds by multiplication by β'.
Thus $r\beta v = 0_M \iff \alpha \mid r\beta \iff \alpha' \mid r$ by Proposition 3.5.2 with $d \neq 0_R$. Hence $\operatorname{Ann}_R(\beta v) = \langle \alpha' \rangle$ and $o(\beta v) = \alpha' = \frac{\alpha}{d}$.
(iv) ($\Rightarrow$) Suppose $\langle\!\langle \beta v \rangle\!\rangle_R = \langle\!\langle v \rangle\!\rangle_R$. Choose $c \in R$ with $v = c\beta v$. Then $\alpha \mid (1_R - c\beta)$, so $(\exists\, q \in R)\, q\alpha + c\beta = 1_R$.
Hence $\gcd(\alpha, \beta) \sim 1_R$ by Theorem 3.10.6(iii).
(iv) ($\Leftarrow$) Suppose $\gcd(\alpha, \beta) \sim 1_R$. Choose $q, c \in R$ with $q\alpha + c\beta = 1_R$ by Theorem 3.10.6(iii).
Then $v = (q\alpha + c\beta)v = c\beta v$, so $\langle\!\langle v \rangle\!\rangle_R \leq \langle\!\langle \beta v \rangle\!\rangle_R$. plaThe reverse inclusion holds by $\beta v \in \langle\!\langle v \rangle\!\rangle_R$.

Structural Commentary: Kernel quotients and preimage ideals transfer cyclic questions to ideal arithmetic.
Key Insight: Annihilators convert cyclic-module questions into divisibility in R. □

Theorem 2.5.6. *Composing and Decomposing Cyclic Modules*
Let M be an R-module, let $n \in \mathbb{N}$, and set $I := \{1, \ldots, n\}$.
Let $(\alpha_1, \ldots, \alpha_n) \in (R \setminus \{0_R\})^n$ satisfy $(\forall\, i, j \in I)\, [i \neq j \Rightarrow \gcd(\alpha_i, \alpha_j) \sim 1_R]$.

(i) If $(u_1, \ldots, u_n) \in M^n$ and $(\forall\, i \in I)\, o(u_i) = \alpha_i$, then $o(\sum_{i=1}^{n} u_i) \sim \prod_{i=1}^{n} \alpha_i$.

(ii) Let $(u_1, \ldots, u_n) \in M^n$ satisfy $(\forall\, i \in I)\, o(u_i) = \alpha_i$. Then $\bigoplus_{i=1}^{n} \langle\!\langle u_i \rangle\!\rangle_R = \langle\!\langle \sum_{i=1}^{n} u_i \rangle\!\rangle_R$.

(iii) If $v \in M$ and $o(v) = \alpha = \prod_{i=1}^{n} \alpha_i$, then there is a tuple $(v_1, \ldots, v_n) \in M^n$ satisfying

(a) $v = \sum_{i=1}^{n} v_i$;
(b) $(\forall\, i \in I)\, o(v_i) \sim \alpha_i$;
(c) $\langle\!\langle v \rangle\!\rangle_R = \bigoplus_{i=1}^{n} \langle\!\langle v_i \rangle\!\rangle_R$.

Proof. Set $\alpha := \prod_{i=1}^{n} \alpha_i$ and $\beta_i := \frac{\alpha}{\alpha_i} = \prod_{j \in I \setminus \{i\}} \alpha_j$. Then $(\forall\, i \in I)\, \gcd(\alpha_i, \beta_i) \sim 1_R$ by Theorem 3.10.7(ii).

(i) Set $u := \sum_{i=1}^{n} u_i$. Then $\alpha u = 0_M$, so $\alpha \in \operatorname{Ann}_R(u)$. Suppose $ru = 0_M$, $i \in I$. Then $r\beta_i u_i = 0_M$ by $(\forall\, j \in I \setminus \{i\})\, \alpha_j \mid \beta_i$.
Thus $\alpha_i \mid r\beta_i$. Choose $a_i, b_i \in R$ with $a_i\alpha_i + b_i\beta_i = 1_R$ by Theorem 3.10.6(iii). Then $\alpha_i \mid r(a_i\alpha_i + b_i\beta_i) = r$.
Hence $(\forall\, i \in I)\, \alpha_i \mid r$. Thus $\alpha \mid r$ by Theorem 3.10.7(ii). Therefore $\operatorname{Ann}_R(u) = \langle \alpha \rangle$ and $o(u) \sim \alpha$.
(ii) Choose $((a_i, b_i))_{i=1}^{n} \in (R^2)^n$ with $(\forall\, i \in I)\, a_i\alpha_i + b_i\beta_i = 1_R$ by Theorem 3.10.6(iii).
Suppose $\sum_{i=1}^{n} x_i = 0_M$ with $(\forall\, i \in I)\, x_i \in \langle\!\langle u_i \rangle\!\rangle_R$, and fix $i \in I$.
Then $\alpha_i x_i = 0_M$ and $\beta_i x_i = 0_M$, the latter by multiplying $\sum_{j=1}^{n} x_j = 0_M$ by β_i.
Hence $x_i = (a_i\alpha_i + b_i\beta_i)x_i = 0_M$, and the sum is internal.
Set $u := \sum_{i=1}^{n} u_i$. The inclusion $\langle\!\langle u \rangle\!\rangle_R \leq \sum_{i=1}^{n} \langle\!\langle u_i \rangle\!\rangle_R$ holds by u belonging to the sum.
$(\forall\, i \in I)\, u_i = b_i\beta_i u$ by $\alpha_i u_i = 0_M$ and $(\forall\, j \in I \setminus \{i\})\, \beta_i u_j = 0_M$. Thus $\sum_{i=1}^{n} \langle\!\langle u_i \rangle\!\rangle_R \leq \langle\!\langle u \rangle\!\rangle_R$.

(iii) If $n = 1$, then $(\forall\, h \in R)\,[h \mid \beta_1 \Rightarrow h \in R^\times]$ by $\beta_1 = 1_R$. Suppose $n \geq 2$ and fix $h \in R \setminus R^\times$ satisfying $(\forall\, i \in I)\, h \mid \beta_i$.
Then $h \neq 0_R$ by $h \mid \beta_1 \neq 0_R$. Choose an irreducible $p \mid h$ by Theorem 3.10.7(i). Then p is prime by Theorem 3.10.6(ii).
By the prime property and $p \mid \beta_1$, choose $j \neq 1$ with $p \mid \alpha_j$. By the prime property and $p \mid \beta_j$, choose $k \neq j$ with $p \mid \alpha_k$.
This contradicts $\gcd(\alpha_j, \alpha_k) \sim 1_R$. Thus $(\forall\, h \in R)\,[(\forall\, i \in I)\, h \mid \beta_i \Rightarrow h \in R^\times]$.
Choose $d \in R$ with $\langle \beta_1, \ldots, \beta_n \rangle = \langle d \rangle$ by Definition 3.10.1(i).
The element d is a common divisor of the β_i, so $d \in R^\times$ and $\langle \beta_1, \ldots, \beta_n \rangle = R$.
Choose $c_1, \ldots, c_n \in R$ with $\sum_{i=1}^{n} c_i \beta_i = 1_R$, and set $v_i := c_i \beta_i v$.

(a) The identity $v = \sum_{i=1}^{n} v_i$ holds by $\sum_{i=1}^{n} c_i \beta_i = 1_R$.

(b) Fix $i \in I$. If $h \mid c_i$ and $h \mid \alpha_i$, then $(\forall\, j \in I)\, h \mid c_j \beta_j$, because $\alpha_i \mid \beta_j$ when $j \neq i$. Hence $h \mid 1_R$ and $\gcd(c_i, \alpha_i) \sim 1_R$.
Therefore $\gcd(\alpha, c_i \beta_i) \sim \beta_i$ by Theorem 3.10.7(ii). Thus $o(v_i) \sim \alpha_i$ by Theorem 2.5.5(iii).

(c) The equality $\langle\!\langle v \rangle\!\rangle_R = \bigoplus_{i=1}^{n} \langle\!\langle v_i \rangle\!\rangle_R$ holds by (ii), $v = \sum_{i=1}^{n} v_i$, and $(\forall\, i \in I)\, o(v_i) \sim \alpha_i$.

Structural Commentary: Bézout projections compose the summands; complementary products construct the reverse decomposition.
Key Insight: Coprime orders isolate each cyclic component from the total sum. □

Proposition 2.5.7. *Prime-Order Cyclic Submodules from Prime Divisors*
Let M be a finitely generated torsion R-module with $o(M) = \mu$. If $p \mid \mu$ is prime, then $(\exists\, w \in M)\,[o(w) \sim p \wedge Rw \cong R/\langle p \rangle]$.

Proof. $\mathrm{Ann}_R(M) \neq \{0_R\}$ by Proposition 1.6.6, so $\mu \neq 0_R$. Choose $q \in R$ with $\mu = pq$. If $q \in \langle \mu \rangle$, choose $a \in R$ with $q = \mu a = pqa$.
Then $1_R = pa$ by Proposition 3.5.2 with $q \neq 0_R$, contrary to $p \notin R^\times$. Thus $q \notin \mathrm{Ann}_R(M)$.
Choose $v \in M$ with $qv \neq 0_M$ by the definition of $\mathrm{Ann}_R(M)$, and set $w := qv$. Then $pw = \mu v = 0_M$, so $o(w) \mid p$.
The element $o(w)$ is a nonunit by $w \neq 0_M$. Write $p = o(w)c$. Then $p \mid o(w)$ or $p \mid c$ by primality of p.
If $p \mid c$, write $c = pe$; then $1_R = o(w)e$ by Proposition 3.5.2 in $p = o(w)pe$, contrary to $o(w) \notin R^\times$.
Thus $p \mid o(w)$, and $o(w) \sim p$ by Proposition 3.7.2(ii)–(iii). Finally, $Rw \cong R/\langle p \rangle$ by Theorem 2.5.5(i).
Structural Commentary: A complementary factor constructs a nonzero element annihilated by the selected prime.
Key Insight: The complementary factor is not a global annihilator. □

2.6 Submodules of Finite Free PID Modules

Definition 2.6.1. *Minimum Number of Module Generators*
For an R-module M, define $\mu_R(M) := \min\{|X| : X \subseteq M \wedge \langle\!\langle X \rangle\!\rangle_R = M\}$.

Theorem 2.6.2. *Submodules of Finite Free PID Modules*
Let $n \in \mathbb{N}_0$, let F be a free R-module with $\mathrm{rk}_R(F) = n$, and let $N \leq F$. Then $(\exists\, m \in \mathbb{N}_0)\,[m \leq n \wedge N \cong R^m]$.

Proof. Identify F with R^n by Proposition 2.2.6(ii).

(i) *Coordinate Ideals*: Set $J := \{1, \ldots, n\}$. For $k \in J$, set $E_k := R^k \times \{0_R\}^{n-k}$ and $N_k := N \cap E_k$.
The relation $E_k \leq R^n$ holds by coordinatewise closure, and $N_k \leq R^n$ by Proposition 1.1.6(i).
Let $\pi_k \in \mathrm{Hom}_R(N_k, R)$ be kth-coordinate map, set $I_k := \mathrm{im}(\pi_k)$. Then $I_k \trianglelefteq R$ by Propositions 1.4.2(iii) and 1.1.7.
Choose $(r_k)_{k \in J} \in R^J$ satisfying $(\forall k \in J)\, I_k = \langle r_k \rangle$ by Definition 3.10.1(i).

(ii) *Triangular Independence*: Set $K := \{k \in J : r_k \neq 0_R\}$.
For $k \in K$, choose $u_k \in N_k$ with $(u_k)_k = r_k$ by $r_k \in I_k = \mathrm{im}(\pi_k)$, and set $U := \{u_k : k \in K\}$.
Let $(c_k)_{k \in K} \in R^K$ satisfy $\sum_{k \in K} c_k u_k = 0_{R^n}$, and set $C := \{k \in K : c_k \neq 0_R\}$. If $C \neq \emptyset$, set $j := \max C$.
The equality $c_j r_j = 0_R$ holds in coordinate j by $c_k = 0_R$ for $k > j$ and $(u_k)_j = 0_R$ for $k < j$.
Then $c_j = 0_R$ by Proposition 3.5.2, $r_j \neq 0_R$, contrary to $j \in C$; $C = \emptyset$, so U is R-linearly independent by Definition 1.3.1.

(iii) *Height Descent*: For $x \in N$, set $\ell(x) := \max(\mathrm{supp}(x) \cup \{0\}) \in \mathbb{N}_0$.
The implication $\ell(x) = 0 \Rightarrow x = 0_{R^n} \in \langle\!\langle U \rangle\!\rangle_R$ holds by Definition 3.1.4.
For the inductive step, fix $x \in N$ with $\ell(x) = j > 0$ and assume $(\forall y \in N)\,[\ell(y) < j \Rightarrow y \in \langle\!\langle U \rangle\!\rangle_R]$.
Then $x \in N_j$ and $x_j \neq 0_R$ by the definition of ℓ. The membership $j \in K$ holds by $x_j \in I_j = \langle r_j \rangle$ and $x_j \neq 0_R$.
Choose $c \in R$ with $x_j = cr_j$ by $x_j \in I_j$. Then $x - cu_j \in N$ and $\ell(x - cu_j) < j$ by $u_j \in N_j$ and $(u_j)_j = r_j$.
The membership $x - cu_j \in \langle\!\langle U \rangle\!\rangle_R$ holds by the induction hypothesis. Then $x = (x - cu_j) + cu_j \in \langle\!\langle U \rangle\!\rangle_R$.
Therefore $N = \langle\!\langle U \rangle\!\rangle_R$ by strong induction and $U \subseteq N$.

(iv) *Rank Conclusion*: Set $m := |K| \in \mathbb{N}_0$ and enumerate $K = \{k_1 < \cdots < k_m\}$. Define $\Phi : R^m \to N$ by $\Phi((a_i)_{i=1}^m) := \sum_{i=1}^{m} a_i u_{k_i}$.
The map Φ is an isomorphism by R-linearity, independence, and spanning of U.
The inequalities $m \leq |J| = n$ hold by $K \subseteq J$, so $m \leq n$ and $N \cong R^m$.

Structural Commentary: Coordinate ideals construct triangular pivots; height descent proves they span N.
Key Insight: The largest nonzero coordinate isolates each pivot coefficient. □

Corollary 2.6.3. *Submodules Need No More Generators over a PID*
Let $n \in \mathbb{N}_0$, let M be an R-module, and let $S \leq M$. If $\mu_R(M) \leq n$, then $\mu_R(S) \leq n$.

Proof.

(i) *Free Cover*: Set $d := \mu_R(M) \in \mathbb{N}_0$; the inequality $d \leq n$ holds by hypothesis.
Choose $(x_i)_{i=1}^d \in M^d$ with $M = \langle\!\langle x_1, \ldots, x_d \rangle\!\rangle_R$ by Definition 2.6.1.
Extend this tuple to $(x_i)_{i=1}^n$ by $(\forall i \in \{1, \ldots, n\})\,[i > d \Rightarrow x_i = 0_M]$.
Then $M = \langle\!\langle x_1, \ldots, x_n \rangle\!\rangle_R$ by the preceding spanning relation. Define $\pi : R^n \to M$ by $\pi((a_i)_{i=1}^n) := \sum_{i=1}^{n} a_i x_i$.
The map π is an epimorphism by the module laws and the spanning relation.

(ii) *Pullback*: Set $P := \pi^{-1}(S)$. Then $0_{R^n} \in P$ and $(\forall (a, b, u, v) \in R^2 \times P^2)\, \pi(au + bv) = a\pi(u) + b\pi(v) \in S$.
Thus $P \leq R^n$ by Theorem 1.1.5. Choose $m \in \mathbb{N}_0$ and $\theta : R^m \xrightarrow{\sim} P$ with $m \leq n$ by Theorem 2.6.2.

(iii) *Generator Transfer*: Let $(e_i)_{i=1}^m$ be the standard basis of R^m and set $v_i := \theta(e_i)$.
The equality $P = \langle\!\langle v_1, \dots, v_m \rangle\!\rangle_R$ holds by surjectivity of θ. The equality $\pi(P) = S$ holds by surjectivity of π and $P = \pi^{-1}(S)$.
Thus $S = \langle\!\langle \pi(v_1), \dots, \pi(v_m) \rangle\!\rangle_R$ by the preceding equalities. The bound $\mu_R(S) \le m \le n$ holds by Definition 2.6.1.

Structural Commentary: Pullback to R^n transfers finite-free rank control through a surjection.
Key Insight: Images of a pullback basis generate S. □

Corollary 2.6.4. *Finitely Generated PID Modules Are Noetherian*
Let M be an R-module. If $\mu_R(M) \in \mathbb{N}_0$, then M is Noetherian.

Proof. Set $n := \mu_R(M)$. Then $(\forall\, N \le M)\, \mu_R(N) \le n$ by Corollary 2.6.3.
Every submodule of M is finitely generated by Definition 2.6.1 and the bound. Thus M is Noetherian by Theorem 2.4.2(i).
Structural Commentary: The generator bound verifies that every submodule is finitely generated.
Key Insight: One finite bound controls every submodule of M. □

Perspective 2.6.5. *Noetherian Modules over a PID*
Theorem 2.6.2 restores finite-rank control for submodules of finite free modules over a PID.
Corollary 2.6.3 transfers that control through free covers.
Corollary 2.6.4 deduces Noetherianity; neither result asserts a submodule complement.

Proposition 2.6.6. *A Full-Size Spanning Set Is a Basis*
Let $n \in \mathbb{N}_0$ and let F be a free R-module with $\mathrm{rk}_R(F) = n$. If $F = \langle\!\langle s_1, \dots, s_n \rangle\!\rangle_R$, then $(s_1, \dots, s_n)$ is an R-basis of F.

Proof.

(i) *Surjective Endomorphism*: Choose an R-basis $(b_i)_{i=1}^n$ of F.
The assignment $b_i \mapsto s_i$ extends to $\tau \in \mathrm{End}_R(F)$ by Theorem 2.1.4. The map τ is surjective by $F = \langle\!\langle s_1, \dots, s_n \rangle\!\rangle_R$.

(ii) *Splitting*: Choose $(a_i)_{i=1}^n \in F^n$ satisfying $(\forall i \in \{1, \dots, n\})\, \tau(a_i) = b_i$ by surjectivity of τ.
The assignment $b_i \mapsto a_i$ extends to $\rho \in \mathrm{End}_R(F)$ by Theorem 2.1.4.
The equality $\tau\rho = \mathrm{id}_F$ holds by basis agreement and uniqueness in Theorem 2.1.4.
Then $F = \ker\tau \oplus \mathrm{im}\,\rho$ by Theorem 1.8.5(ii). The map ρ is injective by $\tau\rho = \mathrm{id}_F$.
Its corestriction $F \to \mathrm{im}\,\rho$ is an isomorphism by R-linearity, injectivity, and the definition of image.

(iii) *Rank Cancellation*: Choose $r \in \mathbb{N}_0$ with $r \le n$ and $\ker\tau \cong R^r$ by Theorem 2.6.2.
The chain $R^n \cong F \cong \ker\tau \oplus \mathrm{im}\,\rho \cong R^r \oplus R^n$ holds by the preceding isomorphisms.
Thus $R^n \cong R^{r+n}$ by coordinate concatenation. Thus $n = r + n$ by Proposition 2.2.6(ii), so $r = 0$.

(iv) *Basis Transport*: The equality $\ker\tau = \{0_F\}$ holds by $r = 0$ and $\ker\tau \cong R^r$.
The map τ is injective by Proposition 1.4.2(iv). It is an isomorphism by Definition 1.4.1(iv) and surjectivity.
Therefore $(s_i)_{i=1}^n = (\tau(b_i))_{i=1}^n$ is an R-basis by bijectivity and linearity of τ.

Structural Commentary: A splitting isolates the kernel; finite-free rank comparison forces it to vanish.
Key Insight: The equality $n = r + n$ leaves only $r = 0$. □

2.7 Smith Normal Form and Module Presentations

Definition 2.7.1. *Finite Free Presentations and Cokernels*
Let $m, n \in \mathbb{N}_0$ and let $A \in \mathrm{Mat}(m, n, R)$ act by $A : R^n \to R^m$. The *cokernel* of A is $\mathrm{coker}(A) := R^m / \mathrm{im}(A)$.
The matrix A is *finite presentation matrix* for R-module M if $M \cong \mathrm{coker}(A)$. Notation $R^n \xrightarrow{A} R^m \to M \to 0$ records this presentation.

Proposition 2.7.2. *Finitely Generated PID Modules Are Finitely Presented*
Let M be a f.g. R-module. Then there are $m, n \in \mathbb{N}_0$ and $A \in \mathrm{Mat}(m, n, R)$ such that $M \cong \mathrm{coker}(A)$.

Proof.

(i) *Free Cover*: Choose generators $u_1, \dots, u_m$ of M and define $\varepsilon : R^m \to M$ by $\varepsilon((a_i)_{i=1}^m) = \sum_{i=1}^m a_i u_i$.
Its image contains every u_i, so it equals M by Proposition 1.2.2(iii). Thus ε is an epimorphism by Definition 1.4.1(iv).

(ii) *Relation Module*: Set $K := \ker\varepsilon \le R^m$ by Proposition 1.4.2(ii).
By Theorem 2.6.2, choose $n \in \mathbb{N}_0$ with $n \le m$ and an isomorphism $\theta : R^n \xrightarrow{\sim} K$.

(iii) *Presentation Matrix*: Let $A \in \mathrm{Mat}(m, n, R)$ be the matrix of the composite $R^n \xrightarrow{\theta} K \hookrightarrow R^m$ in the standard bases.
Then $\mathrm{im}(A) = K$, and $M \cong R^m / K = \mathrm{coker}(A)$ by Theorem 1.5.4(i).

Structural Commentary: A finite generating map has a finite free kernel over a PID.
Key Insight: The inclusion of the relation module is a finite presentation matrix. □

Definition 2.7.3. *Matrix Equivalence and Determinantal Ideals over a PID*
Let $m, n \in \mathbb{N}_0$, set $q := \min(m, n)$, and let $A, B \in \mathrm{Mat}(m, n, R)$.

(i) The matrices A and B are *equivalent*, written $A \sim_{\mathrm{eq}} B$.
This relation is defined by $(\exists P \in \mathrm{Mat}(m, m, R)^\times)(\exists Q \in \mathrm{Mat}(n, n, R)^\times)\, [B = PAQ]$.

(ii) Set $\Delta_0(A) := R$. For $1 \le k \le q$, the *kth determinantal ideal* $\Delta_k(A) \trianglelefteq R$ is generated by all $k \times k$ minors of A.

Theorem 2.7.4. *Smith Normal Form over a PID*
Let $m, n \in \mathbb{N}_0$, $A \in \mathrm{Mat}(m, n, R)$, $q := \min(m, n)$. For $0 \le r \le q$ and $\mathbf{d} = (d_i)_{i=1}^r$, set $D_{\mathbf{d}} := \mathrm{diag}(d_1, \dots, d_r, 0, \dots, 0) \in \mathrm{Mat}(m, n, R)$.

(i) $\exists\ 0 \le r \le q$ and $d_1, \dots, d_r \in R \setminus \{0_R\}$ such that $d_1 \mid \cdots \mid d_r$ and $A \sim_{\mathrm{eq}} D_{\mathbf{d}}$. For these data, $r = \mathrm{rk}_R(\mathrm{im}(A))$.

(ii) Suppose $A \sim_{\mathrm{eq}} D_{\mathbf{d}}$ and $A \sim_{\mathrm{eq}} D_{\mathbf{e}}$ for nonzero divisibility chains $\mathbf{d} = (d_i)_{i=1}^r$ and $\mathbf{e} = (e_i)_{i=1}^s$.
Then $r = s$ and $(\forall i \in \{1, \dots, r\})\, [\langle d_i \rangle = \langle e_i \rangle]$.

Proof.

(i)

(a) *Bézout Pivot*: The existence assertion is proved by induction on q.
If $q = 0$ or $A = 0$, take $r = 0$ and $D_{\mathbf{d}} = 0$. Then $A \sim_{\mathrm{eq}} D_{\mathbf{d}}$ and $\mathrm{rk}_R(\mathrm{im}(A)) = 0$.
Otherwise, invertible permutation matrices move a nonzero entry to position $(1, 1)$; denote the resulting pivot by p.
Suppose an entry a in the first column is not divisible by p.

Choose $g \in R$ with $\langle p, a\rangle = \langle g\rangle$ by Definition 3.10.1(i), and write $p = gu$ and $a = gv$.
Since $g \in \langle p, a\rangle$, choose $x, y \in R$ with $g = xp + ya = g(xu + yv)$. Then $xu + yv = 1_R$ by $g \neq 0_R$ and Proposition 3.5.2.
Set $U := \left(\begin{smallmatrix} x & y \\ -v & u \end{smallmatrix}\right)$. The matrix $U^{-1} = \left(\begin{smallmatrix} u & -y \\ v & x \end{smallmatrix}\right)$ is its inverse, and $U\left(\begin{smallmatrix} p \\ a \end{smallmatrix}\right) = \left(\begin{smallmatrix} g \\ 0_R \end{smallmatrix}\right)$.
Embedding U into the corresponding two rows replaces the pivot by g.
Right multiplication by the embedded transpose U^{T} is the analogous first-row reduction. Finally, $\langle p\rangle \subsetneq \langle p, a\rangle = \langle g\rangle$ as $p \nmid a$.

(b) *Termination*: For a fixed pivot, entries divisible by it are cleared from its row and column by invertible row and column additions. If an interior entry is not divisible by the pivot after this clearing, adding its row to the first row exposes it to the Bézout pivot operation. Every such pivot improvement strictly enlarges the principal pivot ideal.
Only finitely many improvements occur by Proposition 3.10.5. At termination, the pivot p divides every entry, so the first row and column can be cleared to obtain a matrix equivalent to $\operatorname{diag}(p, B)$. Every entry of B remains divisible by p.

(c) *Block Induction*: The complementary matrix B has $\min(m-1, n-1) = q - 1$. By the induction hypothesis, invertible operations internal to this block put it in Smith form with nonzero diagonal entries $d_2 \mid \cdots \mid d_r$.
Those operations take R-linear combinations of entries of B, so $(\forall i \in \{2, \dots, r\})\,[p \mid d_i]$.
Setting $d_1 := p$ establishes $d_1 \mid \cdots \mid d_r$ and $A \sim_{\mathrm{eq}} D_{\mathbf{d}}$.

(d) *Rank*: If $D_{\mathbf{d}} = PAQ$, then $\operatorname{im}(D_{\mathbf{d}}) = P(\operatorname{im}(A))$ because Q is surjective. The vectors $d_1 e_1, \dots, d_r e_r$ span $\operatorname{im}(D_{\mathbf{d}})$.
They are linearly independent because R is an integral domain and every d_i is nonzero.
Hence their inverse images under P form a basis of $\operatorname{im}(A)$, so $\operatorname{rk}_R(\operatorname{im}(A)) = r$.

(ii) Let $B = PAQ$ with P and Q invertible. Two applications of the Cauchy–Binet expansion express each $k \times k$ minor of B as an R-linear combination of the $k \times k$ minors of A. Thus $\Delta_k(B) \subseteq \Delta_k(A)$.
Same argument applied to $A = P^{-1}BQ^{-1}$ proves reverse inclusion, so matrix equivalence preserves every determinantal ideal.
For $1 \leq k \leq r$, each nonzero $k \times k$ minor of $D_{\mathbf{d}}$ is a product $d_{i_1} \cdots d_{i_k}$ with $i_j \geq j$.
It is therefore divisible by $d_1 \cdots d_k$, while the leading $k \times k$ minor equals this latter product.
Consequently, $\Delta_k(D_{\mathbf{d}}) = \langle d_1 \cdots d_k\rangle$ for $k \leq r$, while $\Delta_k(D_{\mathbf{d}}) = \{0_R\}$ for $r < k \leq q$. The same formulas hold for $D_{\mathbf{e}}$.
Since R is an integral domain, the nonzero determinantal ideals occur precisely for $k \leq r$ and for $k \leq s$, respectively; hence $r = s$.
If $r = 0$, the uniqueness assertion follows. Assume $r \geq 1$. Thus $\langle d_1\rangle = \langle e_1\rangle$ by equality of the first determinantal ideals.
Assume inductively that these principal ideals agree through index $k - 1$.
By Proposition 3.7.2(ii), the corresponding generators are associates.
After absorbing their unit factors, equality of the kth determinantal ideals becomes $\langle cd_k\rangle = \langle ce_k\rangle$, where $c = d_1 \cdots d_{k-1} \neq 0_R$.
Thus $\langle d_k\rangle = \langle e_k\rangle$ by mutual divisibility and Proposition 3.5.2, completing the induction.

Structural Commentary: Bézout pivots establish existence; invertible row and column operations preserve determinantal ideals.
Key Insight: The ideals $\Delta_k(A)$ recover the successive invariant factors up to associates. □

Theorem 2.7.5. *Adapted Bases for Submodules of Free Modules*
Let $n \in \mathbb{N}_0$, F be free R-module with $\operatorname{rk}_R(F) = n$, $N \leq F$. There are $k \in \mathbb{N}_0$ with $k \leq n$ and an R-basis $(v_i)_{i=1}^n$ of F. For these choices, there are $r_1, \dots, r_k \in R \setminus \{0_R\}$ such that $r_1 \mid \cdots \mid r_k$ and $(r_i v_i)_{i=1}^k$ is an R-basis of N.

Proof.

(i) *Free Setup*: By Theorem 2.6.2, choose $k \in \mathbb{N}_0$ with $k \leq n$ and $N \cong R^k$.
Choose an R-basis $(w_j)_{j=1}^k$ of N and an R-basis $(f_i)_{i=1}^n$ of F.
Let $A \in \operatorname{Mat}(n, k, R)$ be the matrix of the inclusion $\iota : N \hookrightarrow F$ in these bases.

(ii) *Zero Case*: If $k = 0$, then $N = \{0_F\}$, chosen basis of F together with empty basis of N satisfies conclusion. Assume $k \geq 1$.

(iii) *Smith Reduction*: The Smith diagonal of A has k nonzero entries by Theorem 2.7.4(i), since $\operatorname{im}(\iota) = N$ has rank k.
Denote these entries by $r_1 \mid \cdots \mid r_k$ and set $D := \operatorname{diag}(r_1, \dots, r_k, 0, \dots, 0)$.
Choose $P \in \operatorname{Mat}(n, n, R)^\times$ and $Q \in \operatorname{Mat}(k, k, R)^\times$ with $PAQ = D$ by Definition 2.7.3(i).

(iv) *Basis Transport*: Define $w'_j := \sum_{\ell=1}^{k} Q_{\ell j} w_\ell$ for $1 \leq j \leq k$. Define $v_i := \sum_{\ell=1}^{n} (P^{-1})_{\ell i} f_\ell$ for $1 \leq i \leq n$.
Since Q and P^{-1} are invertible, $(w'_j)_{j=1}^k$ and $(v_i)_{i=1}^n$ are bases of N and F, respectively.
Identity $AQ = P^{-1}D$ implies $(\forall j \in \{1, \dots, k\})\,[\iota(w'_j) = r_j v_j]$. Because ι is inclusion, $(r_j v_j)_{j=1}^k = (w'_j)_{j=1}^k$ is R-basis of N.

Structural Commentary: Smith reduction changes the domain and codomain bases of the inclusion.
Key Insight: In the transported bases, the inclusion sends the jth basis vector to $r_j v_j$. □

Corollary 2.7.6. *Smith Form Presents Finitely Generated PID Modules*
Let $A \in \operatorname{Mat}(m, n, R)$ be a finite presentation matrix for M. Suppose $A \sim_{\mathrm{eq}} D$, where $D := \operatorname{diag}(d_1, \dots, d_r, 0, \dots, 0)$. Assume $0 \leq r \leq \min(m, n)$, $d_1 \mid \cdots \mid d_r$ are nonzero. Set $I := \{i \in \{1, \dots, r\} : d_i \notin R^\times\}$. Then $M \cong R^{m-r} \oplus \bigoplus_{i \in I} R/\langle d_i\rangle$.

Proof.

(i) *Equivalent Cokernels*: Choose invertible P, Q with $D = PAQ$ by Definition 2.7.3(i). As $Q(R^n) = R^n$, thus $\operatorname{im}(D) = P(\operatorname{im}(A))$.
Consequently, $x + \operatorname{im}(A) \mapsto Px + \operatorname{im}(D)$ is an isomorphism $\operatorname{coker}(A) \xrightarrow{\sim} \operatorname{coker}(D)$; its inverse is induced by P^{-1}.

(ii) *Diagonal Cokernel*: With $(e_i)_{i=1}^m$ the standard basis of R^m, $\operatorname{im}(D) = \bigoplus_{i=1}^r \langle d_i\rangle e_i$.
Thus, coordinatewise quotienting produces $\operatorname{coker}(D) \cong \left(\bigoplus_{i=1}^r R/\langle d_i\rangle\right) \oplus R^{m-r}$.

(iii) *Unit Factors*: If $d_i \in R^\times$, then $R/\langle d_i\rangle = \{0_R\}$.
Removing these zero summands and using $M \cong \operatorname{coker}(A)$ proves the asserted decomposition.

Structural Commentary: Smith form turns presentations into direct sums.
Key Insight: Diagonal relations create cyclic torsion summands, and zero rows create free summands. □

Perspective 2.7.7. *Smith Form as Diagonalized Relations*
Proposition 2.7.2 encodes generators and relations as a finite matrix cokernel.
Theorem 2.7.4 changes bases to isolate those relations.
Corollary 2.7.6 reads off the resulting cyclic and free summands.
This mechanism concerns finite presentations; it does not classify arbitrary non-f.g. modules.

Example 2.7.8. *Smith Form for a Finite Abelian Group*
Let $M = \mathbb{Z}^2/\langle\!\langle (6, 0), (0, 10), (2, 4)\rangle\!\rangle_{\mathbb{Z}}$. Its three displayed relations are the columns of the finite presentation matrix $A = \left(\begin{smallmatrix} 6 & 0 & 2 \\ 0 & 10 & 4 \end{smallmatrix}\right)$.
The entries generate $\Delta_1(A) = \langle 6, 10, 2, 4\rangle = \langle 2\rangle$.
The 2×2 minors from the three column pairs are 60, 24, and -20, so $\Delta_2(A) = \langle 60, 24, -20\rangle = \langle 4\rangle$.

The Smith form has two nonzero invariant factors by the determinantal-ideal formulas in the proof of Theorem 2.7.4.
Those factors satisfy $d_1 \sim 2$ and $d_1 d_2 \sim 4$ by same formulas. Hence $d_2 \sim 2$ by cancellation; $M \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ by Corollary 2.7.6.
Key Insight: The first two determinantal ideals recover the two cyclic summands.

2.8 Finitely Generated PID Modules

Theorem 2.8.1. *Finitely Generated Torsion-Free Modules Are Free*
Let M be a f.g. R-module. Then M is free $\iff$ M is torsion-free.

Proof.
($\Rightarrow$) Let $\mathcal{B}$ be an R-basis of M, $a \in R \setminus \{0_R\}$, $am = 0_M$. Choose finite $H \subseteq \mathcal{B}$, $(r_b)_{b\in H} \in R^H$ such that $m = \sum_{b\in H} r_b b$.
Then $\sum_{b\in H} (ar_b)b = 0_M$, so $(\forall b \in H)\,[ar_b = 0_R]$ by the independence of $\mathcal{B}$.
Since R is an integral domain and $a \neq 0_R$, $(\forall b \in H)\,[r_b = 0_R]$. Thus $m = 0_M$, and M is torsion-free by Definition 1.6.1(ii).
($\Leftarrow$) Assume M is torsion-free.
(a) *Independent Core*: Choose a finite generating set $G = \{u_1, \dots, u_n\}$ of M.
Choose an inclusion-maximal R-linearly independent subset $S = \{v_1, \dots, v_k\} \subseteq G$.
Set $L := \langle\!\langle S \rangle\!\rangle_R$. Then S is an R-basis of L, so L is finite free of rank k.
(b) *Common Scalar*: Set $W := G \setminus S$. Maximality makes $S \cup \{w\}$ dependent for $w \in W$.
The independence of S forces the coefficient of w in each nontrivial relation to be nonzero.
Choose coefficients $a_w \in R \setminus \{0_R\}$ and $b_{w,i} \in R$, indexed by $w \in W$, such that $(\forall w \in W)\,[a_w w = \sum_{i=1}^{k} b_{w,i} v_i]$.
Set $a := \prod_{w\in W} a_w$, with the empty product equal to 1_R. Then $a \neq 0_R$ because R is an integral domain.
Set $aM := \{am : m \in M\}$. For $w \in W$, write $a = c_w a_w$; then $aw = \sum_{i=1}^{k} c_w b_{w,i} v_i \in L$.
Together with $av_i \in L$ for $1 \le i \le k$, these relations imply $(\forall u \in G)\,[au \in L]$. Since G generates M, one has $aM \subseteq L$.
(c) *Free Image*: Define $\mu_a : M \to M$ by $\mu_a(m) := am$. Then $\mu_a \in \mathrm{Hom}_R(M, M)$ and $aM = \mathrm{im}(\mu_a)$.
Hence $aM \le L$ by Proposition 1.4.2(iii) and the containment $aM \subseteq L$.
The module aM is free by Theorem 2.6.2, since L is finite free.
(d) *Transport*: The corestriction $\mu_a : M \to aM$ is surjective by the definition of aM.
It is injective because $a \neq 0_R$ and M is torsion-free. Thus $M \cong aM$, so M is free.

Structural Commentary: A maximal independent core produces a nonzero scalar sending M into a finite free span.
Key Insight: Multiplication by that nonzero product identifies M with the free submodule aM. □

Theorem 2.8.2. *Free Plus Torsion Decomposition*
Let M be a f.g. R-module and set $T := M_{\mathrm{tor}}$.
(i) There is a f.g. free submodule $F \le M$ such that $M = T \oplus F$ and $F \cong M/T$.
(ii) If $M = T' \oplus G$, where $T' \le M$ is torsion and $G \le M$ is free, then $T' = T$ and $G \cong M/T$.

Proof.
(i) Set $Q := M/T$. Then $T \le M$ and Q is torsion-free by Proposition 1.6.3(i)–(ii). Choose generators $u_1, \dots, u_n$ of M.
Then $Q = \langle\!\langle u_1 + T, \dots, u_n + T \rangle\!\rangle_R$, so Q is f.g. Hence Q is free by Theorem 2.8.1. Let $\pi : M \twoheadrightarrow Q$ be the quotient map.
Choose a section $\sigma : Q \to M$ of π by Theorem 2.3.1(i). Set $F := \mathrm{im}(\sigma)$. The identity $\ker \pi = T$ holds by Proposition 1.5.2(iii).
Hence $M = T \oplus F$ and $F \cong Q = M/T$ by Theorem 2.3.1(i). Since Q is free and $F \cong Q$, the module F is free.
The images under σ of the generators $u_i + T$ generate F, so F is f.g.
(ii) Assume $M = T' \oplus G$, where T' is torsion, G is free. Then $T' \subseteq T$ by Definition 1.6.1(iv). Fix $x \in T$, $x = t' + g$ with $(t', g) \in T' \times G$.
Since $T \le M$ and $x, t' \in T$, one has $g = x - t' \in T \cap G$. Let $\rho : M \twoheadrightarrow G$, $t' + g \mapsto g$, be the coordinate projection.
The image under ρ of a finite generating set of M generates G, so G is f.g. Hence G is torsion-free by Theorem 2.8.1.
Since $g \in T \cap G$, one has $g = 0_M$. Thus $x = t' \in T'$, and hence $T = T'$. Finally, $G \cong M/T$ by Theorem 1.7.4(i).

Structural Commentary: The torsion-free quotient is free and splits; a free summand contains no nonzero torsion.
Key Insight: Every free complement is isomorphic to the common quotient model M/T. □

2.9 Primary Decomposition and Cyclic Primary Modules

Definition 2.9.1. *Primary Module*
Let $p \in R$ be prime. A torsion R-module M is *p-primary* if $(\exists e \in \mathbb{N})\,[\mathrm{Ann}_R(M) = \langle p^e \rangle]$.
Equivalently, $(\exists e \in \mathbb{N})\,[o(M) \sim p^e]$ by Definition 2.5.3.

Theorem 2.9.2. *Primary Decomposition*
Let $n \in \mathbb{N}$, $M \neq \{0_M\}$ be f.g. torsion, $p_1, \dots, p_n \in R$ be distinct nonassociate primes. Suppose $e_1, \dots, e_n \in \mathbb{N}$ and $o(M) \sim \mu := \prod_{i=1}^{n} p_i^{e_i}$.
Set $I := \{1, \dots, n\}$. For $i \in I$, set $q_i := p_i^{e_i}$, $\mu_i := \prod_{j\in I\setminus\{i\}} q_j$, and $M_{p_i} := \mu_i M$.
(i) $M = \bigoplus_{i=1}^{n} M_{p_i}$ and $(\forall i \in I)\,[M_{p_i} = \{v \in M : q_i v = 0_M\}]$.
(ii) Each M_{p_i} is f.g. and p_i-primary, with $o(M_{p_i}) \sim q_i$.
(iii) The unordered family $\{(p_i, M_{p_i}) : i \in I\}$, with primes taken up to associates, is determined by M.

Proof.
(i)
(a) *Spanning*: The elements $\mu_1, \dots, \mu_n$ have no common nonunit divisor by the distinct-prime factorization of μ.
Choose $a_1, \dots, a_n \in R$ with $\sum_{i=1}^{n} a_i \mu_i = 1_R$ by induction on n from Theorem 3.10.6(iii).

Thus $(\forall x \in M)\,[x = \sum_{i=1}^{n} a_i\mu_i x]$, so $M = \sum_{i=1}^{n} M_{p_i}$.

(b) *Kernel Description*: Since $q_i\mu_i = \mu$ and $\mu M = \{0_M\}$, $(\forall i \in I)\,[q_i M_{p_i} = \{0_M\}]$.
Fix $i \in I$ and $v \in M$ with $q_i v = 0_M$. The elements μ_i and q_i have no common nonunit divisor.
Choose $a, b \in R$ with $a\mu_i + bq_i = 1_R$ by Theorem 3.10.6(iii).
Then $v = a\mu_i v + bq_i v = a\mu_i v \in M_{p_i}$. Hence $M_{p_i} = \{v \in M : q_i v = 0_M\}$.

(c) *Directness*: Let $(x_i)_{i=1}^n \in \prod_{i=1}^{n} M_{p_i}$ satisfy $\sum_{i=1}^{n} x_i = 0_M$, and fix $i \in I$.
If $j \neq i$, then $q_j \mid \mu_i$ and $q_j x_j = 0_M$, so $\mu_i x_j = 0_M$. By multiplying the displayed sum by μ_i, $\mu_i x_i = 0_M$.
Together with $q_i x_i = 0_M$ and the Bézout identity for μ_i, q_i, this implies $x_i = 0_M$. Thus the sum is direct.

(ii) The image $M_{p_i} = \mu_i M$ is f.g. because the μ_i-multiples of a finite generating set of M generate it. For $i \in I$, let $\alpha_i := o(M_{p_i})$.
Since $q_i M_{p_i} = \{0_M\}$, $(\forall i \in I)\,[\alpha_i \mid q_i]$. By Theorem 3.10.7(i)–(ii), choose $f_i \in \{0, \dots, e_i\}$ with $\alpha_i \sim p_i^{f_i}$ $(i \in I)$.
By part (i), Proposition 2.5.4, $o(M) \sim \operatorname{lcm}(\alpha_1, \dots, \alpha_n)$. Hence $\operatorname{lcm}(\alpha_1, \dots, \alpha_n) \sim \prod_{i=1}^{n} q_i$.
Unique factorization forces $(\forall i \in I)\,[f_i = e_i]$. Since $q_i \neq 0_R$ and $q_i M_{p_i} = \{0_M\}$, the module M_{p_i} is torsion.
Thus $o(M_{p_i}) \sim q_i$, so M_{p_i} is p_i-primary by Definition 2.9.1.

(iii) The prime-power factors q_i of $o(M)$ are determined up to associates and permutation by Theorem 3.10.7(ii).
If $q_i' \sim q_i$, then $\{v \in M : q_i' v = 0_M\} = \{v \in M : q_i v = 0_M\}$ because the associate factor is a unit.
Part (i) therefore makes each primary component intrinsic.

Structural Commentary: Bézout isolates prime-power kernels; their order lcm recovers every primary exponent.
Key Insight: Each intrinsic component is exactly $\ker(q_i : M \to M)$. □

Definition 2.9.3. *p-Power and p-Torsion Submodules*
Let $p \in R$ be prime and let M be an R-module.

(i) For $j \in \mathbb{N}_0$, the *jth p-power submodule* is $p^j M := \{p^j v : v \in M\} = \operatorname{im}(p^j \operatorname{id}_M) \leq M$.
(ii) The *first p-torsion layer* is $M[p] := \ker(p \operatorname{id}_M) = \{v \in M : pv = 0_M\} \leq M$.

Lemma 2.9.4. *The First p-Torsion Layer*
Let $p \in R$ be prime, let $\kappa := R/\langle p\rangle$, and let M be an R-module.

(i) If N is an R-module and $pN = \{0_N\}$, then $(r + \langle p\rangle)n := rn$ $(r \in R, n \in N)$ defines a κ-vector-space structure on N. In particular, $M[p]$ is a κ-vector space.
(ii) If $M = S \oplus T$, then $M[p] = S[p] \oplus T[p]$.

Proof.

(i) The prime element p is irreducible, so $\langle p\rangle$ is maximal by Theorem 3.10.6(i). Hence κ is a field by Theorem 3.4.4.
Suppose $r + \langle p\rangle = r' + \langle p\rangle$ and fix $n \in N$. Choose $c \in R$ satisfying $r - r' = cp$. Then $(r - r')n = c(pn) = 0_N$.
Thus the scalar action is well-defined. The vector-space axioms follow from the R-module axioms and the quotient-ring operations.
The submodule $M[p]$ satisfies $p(M[p]) = \{0_M\}$ by Definition 2.9.3(ii), so the particular case follows.

(ii) Let $M = S \oplus T$ and fix $v \in M[p]$. Write $v = s + t$ with $(s, t) \in S \times T$. Then $ps + pt = 0_M$.
Since $ps \in S$, $pt \in T$, and $S \cap T = \{0_M\}$, one has $ps = pt = 0_M$. Hence $v \in S[p] + T[p]$.
Conversely, $s \in S[p]$ and $t \in T[p]$ imply $p(s + t) = 0_M$. Finally, $S[p] \cap T[p] \subseteq S \cap T = \{0_M\}$, so the sum is direct.

Structural Commentary: Annihilation by p permits residue-class scalars; direct-sum intersections separate the p-killed coordinates.
Key Insight: The functor $M \mapsto M[p]$ preserves finite internal direct sums. □

Definition 2.9.5. *Primary Layer Dimensions*
Let $p \in R$ be prime, let $\kappa := R/\langle p\rangle$, and let M be an R-module. For $k \in \mathbb{N}$, one has $p^k M \leq p^{k-1} M$ because $(\forall v \in M)\,[p^k v = p^{k-1}(pv)]$.
Every class $p^{k-1}v + p^k M$ is annihilated by p. Thus $p^{k-1}M/p^k M$ is a κ-vector space by Lemma 2.9.4(i).
Define the *kth primary layer dimension* by $\lambda_k(M) := \dim_\kappa(p^{k-1}M/p^k M)$.

Proposition 2.9.6. *Layer Counts Recover Primary Exponents*
Let $p \in R$ be prime, $\kappa := R/\langle p\rangle$, $m \in \mathbb{N}$, $e_1 \geq \cdots \geq e_m \geq 1$. Suppose $M \cong \bigoplus_{i=1}^{m} R/\langle p^{e_i}\rangle$.

(i) $(\forall k \in \mathbb{N})\,[\lambda_k(M) = \#\{i \in \{1, \dots, m\} : e_i \geq k\}]$.
(ii) $(\forall k \in \mathbb{N})\,[\#\{i \in \{1, \dots, m\} : e_i = k\} = \lambda_k(M) - \lambda_{k+1}(M)]$.

Proof.

(i) Fix $(e, k) \in \mathbb{N}^2$ and set $C_e := R/\langle p^e\rangle$ and $L_{e,k} := p^{k-1}C_e/p^k C_e$. If $k > e$, then $L_{e,k} = \{0_{L_{e,k}}\}$.
Assume $k \leq e$ and define $\phi_{e,k} : \kappa \to L_{e,k}$ by $\phi_{e,k}(r + \langle p\rangle) := p^{k-1}(r + \langle p^e\rangle) + p^k C_e$.
If $r - r' = ps$, then the two images differ by $p^k(s + \langle p^e\rangle) \in p^k C_e$, so $\phi_{e,k}$ is well-defined.
The map $\phi_{e,k}$ is κ-linear and surjective by its formula. If $\phi_{e,k}(r + \langle p\rangle) = 0$, choose $s, t \in R$ satisfying $p^{k-1}(r - ps) = p^e t$.
By cancellation in the domain, $r = ps + p^{e-k+1}t \in \langle p\rangle$. Thus $\phi_{e,k}$ is injective, and consequently $L_{e,k} \cong \kappa$ for $k \leq e$.
An R-module isomorphism maps p^j-power submodules isomorphically, induces isomorphisms on corresponding primary layers.
Thus λ_k is invariant under R-module isomorphism.
For the displayed cyclic direct sum, p^j-power submodules and their quotients are coordinatewise.
Set $I_k := \{i \in \{1, \dots, m\} : e_i \geq k\}$. Hence its kth primary layer is $\bigoplus_{i \in I_k} \kappa$, so $\lambda_k(M) = \#I_k$.

(ii) Fix $k \in \mathbb{N}$ and set $I_k := \{i \in \{1, \dots, m\} : e_i \geq k\}$. Then $I_{k+1} \subseteq I_k$ and $I_k \setminus I_{k+1} = \{i \in \{1, \dots, m\} : e_i = k\}$.
By part (i), $\#(I_k \setminus I_{k+1}) = \#I_k - \#I_{k+1} = \lambda_k(M) - \lambda_{k+1}(M)$.

Structural Commentary: A cyclic block contributes one residue-field dimension until its exponent is exhausted.
Key Insight: Consecutive layer-dimension differences count blocks ending at each exponent. □

Theorem 2.9.7. *Cyclic Decomposition of a Primary Module*
Let $p \in R$ be prime, let $e \in \mathbb{N}$, and let M be a f.g. p-primary module with $o(M) \sim p^e$.

(i) There are $m \in \mathbb{N}$, $v_1, \dots, v_m \in M$, and $e = e_1 \geq e_2 \geq \cdots \geq e_m \geq 1$. The vectors satisfy $M = \bigoplus_{i=1}^{m} \langle\langle v_i \rangle\rangle_R$.
Their annihilators satisfy $(\forall i \in \{1, \dots, m\})\,[\mathrm{Ann}_R(v_i) = \langle p^{e_i} \rangle]$.
(ii) The exponent multiset $\{e_1, \dots, e_m\}$ is determined by M. Consequently, so is the ordered chain $\langle p^{e_1} \rangle \subseteq \cdots \subseteq \langle p^{e_m} \rangle$.

Proof.
(i)

(a) *Diagonal Presentation*: Choose $R^a \xrightarrow{A} R^b \to M \to 0$ by Proposition 2.7.2. Choose Smith form D of A by Theorem 2.7.4.
Choose $s, t \in \mathbb{N}_0$ and nonunits $d_1, \dots, d_t \in R$ with $M \cong R^s \oplus \bigoplus_{\ell=1}^{t} R/\langle d_\ell \rangle$ by Corollary 2.7.6.
If $s > 0$, the free summand R^s contains a nonzero torsion-free element, contradicting that M is torsion.
Hence $s = 0$. Since $M \neq \{0_M\}$, one has $t \geq 1$.
(b) *Prime Powers*: The order identity $\mathrm{lcm}(d_1, \dots, d_t) \sim o(M) \sim p^e$ holds by Proposition 2.5.4.
Every prime factor of d_ℓ is associate to p. By Theorem 3.10.7(i)–(ii), choose $e_\ell \in \mathbb{N}$ with $d_\ell \sim p^{e_\ell}$ $(1 \leq \ell \leq t)$.
Set $m := t$ and reorder so $e_1 \geq \cdots \geq e_m$. The least-common-multiple identity implies $e_1 = e$.
(c) *Internal Summands*: Set $C := \bigoplus_{i=1}^{m} R/\langle p^{e_i} \rangle$. Since $\langle d_i \rangle = \langle p^{e_i} \rangle$, choose an isomorphism $\Psi : C \to M$.
Let $\mathbf{u}_i$ be the standard generator of the ith coordinate summand and set $v_i := \Psi(\mathbf{u}_i)$.
Since Ψ is an isomorphism, $M = \bigoplus_{i=1}^{m} \langle\langle v_i \rangle\rangle_R$. The standard generator of $R/\langle p^{e_i} \rangle$ has annihilator $\langle p^{e_i} \rangle$.
An isomorphism preserves annihilators, so $\mathrm{Ann}_R(v_i) = \langle p^{e_i} \rangle$.

(ii) The dimensions $\lambda_k(M)$ are intrinsic because module isomorphisms induce isomorphisms on the layer quotients.
By Proposition 2.9.6(ii), the multiplicity of the exponent k is $\lambda_k(M) - \lambda_{k+1}(M)$.
Thus M determines every exponent multiplicity and hence the exponent multiset.
Its decreasing order determines the annihilator chain.

Structural Commentary: Smith form supplies prime-power blocks; primary-layer differences recover their multiplicities.
Key Insight: The differences $\lambda_k(M) - \lambda_{k+1}(M)$ recover each exponent multiplicity. □

2.10 Elementary Divisors and Invariant Factors

Definition 2.10.1. *Elementary Divisors*
Set $\mathrm{ElemDiv}(0) := \emptyset$. Let $0 \neq M$ be f.g. torsion. Choose its intrinsic primary components by Theorem 2.9.2.
Choose their intrinsic exponent chains by Theorem 2.9.7. Suppose the resulting cyclic summands have annihilators $\langle p_i^{e_{i,j}} \rangle$.
Define the *elementary-divisor multiset* by $\mathrm{ElemDiv}(M) := \{p_i^{e_{i,j}}\}_{i,j}$, with entries taken up to associates.

Theorem 2.10.2. *Primary Cyclic Decomposition*
Let M be a f.g. torsion R-module.

(i) There is a finite set I, together with distinct nonassociate primes $(p_i)_{i \in I}$ and $(k_i)_{i \in I} \in \mathbb{N}^I$.
There are exponents $e_{i,1} \geq \cdots \geq e_{i,k_i} \geq 1$ and vectors $v_{i,j} \in M$ $(i \in I,\ 1 \leq j \leq k_i)$. These data satisfy $M = \bigoplus_{i \in I} \bigoplus_{j=1}^{k_i} \langle\langle v_{i,j} \rangle\rangle_R$.
Their annihilators satisfy $\mathrm{Ann}_R(v_{i,j}) = \langle p_i^{e_{i,j}} \rangle$ $(i \in I,\ 1 \leq j \leq k_i)$. For $M = \{0_M\}$, take $I = \emptyset$.
(ii) If N is f.g. torsion, then $M \cong N \iff \mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$.

Proof.
(i) If $M = \{0_M\}$, the empty direct sum has the asserted form. Suppose $M \neq \{0_M\}$.
By Theorem 2.9.2, there is a finite family of f.g. p_i-primary components. It satisfies $M = \bigoplus_{i \in I} M_{p_i}$.
By Theorem 2.9.7(i), each M_{p_i} has an internal cyclic decomposition. Its exponent chain and annihilators have the asserted form.
Substitution into the primary decomposition proves the displayed internal direct sum.
(ii) $(\Rightarrow)$ Suppose $\Phi : M \xrightarrow{\sim} N$. The map Φ identifies the intrinsic primary components by Theorem 2.9.2(iii).
Their exponent multisets agree by Theorem 2.9.7(ii), so $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$.
(ii) $(\Leftarrow)$ Suppose $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$. Set $C := \bigoplus_{q \in \mathrm{ElemDiv}(M)} R/\langle q \rangle$. Then $M \cong C \cong N$ by part (i) and Theorem 2.5.5(i).

Structural Commentary: Intrinsic primary kernels and exponent layers make the prime-power blocks independent of choices.
Key Insight: The elementary-divisor multiset determines the common quotient-sum model. □

Proposition 2.10.3. *Elementary Divisors of a Direct Sum*
Let A and B be f.g. torsion, and let $\uplus$ denote multiset union. Then $\mathrm{ElemDiv}(A \oplus B) = \mathrm{ElemDiv}(A) \uplus \mathrm{ElemDiv}(B)$.

Proof. Choose primary cyclic decompositions of A and B by Theorem 2.10.2(i). Write $A = \bigoplus_{i \in I} A_i$ and $B = \bigoplus_{j \in J} B_j$.
Then $A \oplus B = (\bigoplus_{i \in I} A_i) \oplus (\bigoplus_{j \in J} B_j)$. This is a primary cyclic decomposition, including when I or J is empty.
By Definition 2.10.1, its prime-power annihilators form the required multiset. That multiset is $\mathrm{ElemDiv}(A) \uplus \mathrm{ElemDiv}(B)$.
Structural Commentary: Concatenating primary cyclic decompositions concatenates their annihilator multisets.
Key Insight: Multiset union retains repeated prime-power blocks. □

Definition 2.10.4. *Invariant Factors*
Let M be f.g. torsion. If $\mathrm{ElemDiv}(M) = \emptyset$, define $\mathrm{Inv}(M) := ()$. Suppose $\mathrm{ElemDiv}(M) \neq \emptyset$.
Group it as $\{p_i^{e_{i,j}} : i \in I,\ 1 \leq j \leq k_i\}$; p_i are distinct nonassociate primes. Order each exponent row by $e_{i,1} \geq \cdots \geq e_{i,k_i} \geq 1$.
Set $m := \max_{i \in I} k_i$ and extend every row by $e_{i,j} := 0$ for $k_i < j \leq m$. Set $d_j := \prod_{i \in I} p_i^{e_{i,j}}$ $(1 \leq j \leq m)$.
Invariant-factor tuple is $\mathrm{Inv}(M) := (d_1, \dots, d_m)$, with entries taken up to associates. Entries are nonunits and satisfy $d_m \mid \cdots \mid d_1$.

Method 2.10.5. *From Elementary Divisors to Invariant Factors*

(i) If $\mathrm{ElemDiv}(M) = \emptyset$, return $\mathrm{Inv}(M) = ()$. Otherwise, group the input multiset by associate prime.
(ii) Write the ith group as $p_i^{e_{i,1}}, \dots, p_i^{e_{i,k_i}}$ with $e_{i,1} \geq \cdots \geq e_{i,k_i}$. Pad every exponent row by zeros to length $m := \max_i k_i$.
(iii) Form $d_j := \prod_i p_i^{e_{i,j}}$ for $1 \leq j \leq m$.
(iv) Return $\mathrm{Inv}(M) = (d_1, \dots, d_m)$ and verify $d_{j+1} \mid d_j$ from $e_{i,j+1} \leq e_{i,j}$.
Recover the input by extracting its maximal prime-power factors from each d_j.

Theorem 2.10.6. *Invariant Factor Decomposition*
Let M be f.g. torsion and write $\mathrm{Inv}(M) = (d_1, \dots, d_m)$, where $m \in \mathbb{N}_0$.

(i) $M \cong \bigoplus_{j=1}^{m} R/\langle d_j \rangle$. If $m \geq 1$, then $d_j \notin R^\times$ $(1 \leq j \leq m)$ and $d_{j+1} \mid d_j$ $(1 \leq j < m)$.
(ii) If N is f.g. torsion, then $M \cong N \iff \mathrm{Inv}(M) = \mathrm{Inv}(N)$.

Proof.

(*i*) If $m = 0$, then $\mathrm{ElemDiv}(M) = \emptyset$ by Definition 2.10.4.
Hence $M = \{0_M\}$ by Theorem 2.10.2(i), so the empty direct sum proves the claim. Suppose $m \geq 1$.
By Definition 2.10.4, $\mathrm{ElemDiv}(M) = \{p_i^{e_{i,j}} : i \in I,\ 1 \leq j \leq k_i\}$. Pad exponent rows by zeros, set $d_j = \prod_{i \in I} p_i^{e_{i,j}}$.
By Theorem 2.10.2(i), M is a direct sum of the associated primary cyclic blocks.
Each block is isomorphic to its quotient model by Theorem 2.5.5(i).
Hence $M \cong \bigoplus_{i \in I} \bigoplus_{j=1}^{k_i} R/\langle p_i^{e_{i,j}} \rangle$. For $1 \leq j \leq m$, set $C_j := \bigoplus_{\substack{i \in I \\ e_{i,j} > 0}} R/\langle p_i^{e_{i,j}} \rangle$. By finite reindexing, $M \cong \bigoplus_{j=1}^{m} C_j$.
The nonunit orders in C_j are pairwise coprime. By Theorem 2.5.6(i)–(ii), C_j is cyclic and has order associated to d_j.
Hence $C_j \cong R/\langle d_j \rangle$ by Theorem 2.5.5(i). Substitution proves the asserted decomposition.
The nonunit and divisibility properties hold by Definition 2.10.4.
(*ii*) ($\Rightarrow$) Suppose $M \cong N$. Then $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$ by Theorem 2.10.2(ii); $\mathrm{Inv}(M) = \mathrm{Inv}(N)$ by Definition 2.10.4.
(*ii*) ($\Leftarrow$) Suppose $\mathrm{Inv}(M) = \mathrm{Inv}(N)$. Factoring invariant factors recovers equal prime-power multisets by Theorem 3.10.7(ii).
Thus $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$. Hence $M \cong N$ by Theorem 2.10.2(ii).

Structural Commentary: Coprime cyclic composition packs a single block from each prime into a divisibility column.
Key Insight: Factoring the invariant factors reverses the column packing. □

Example 2.10.7. *Packing Elementary Divisors into Invariant Factors*
Let $R = \mathbb{R}[z]$ and $\mathrm{ElemDiv}(M) = \{(z-1)^3, z-1, (z^2+1)^2, z^2+1\}$. The polynomials $z-1$ and z^2+1 are nonassociate primes in R.
By Method 2.10.5, $d_1 = (z-1)^3(z^2+1)^2$ and $d_2 = (z-1)(z^2+1)$. Thus $\mathrm{Inv}(M) = (d_1, d_2)$ with $d_2 \mid d_1$.
By Theorem 2.10.6(i), $M \cong R/\langle d_1 \rangle \oplus R/\langle d_2 \rangle$.
Structural Commentary: Separate prime rows become the two invariant-factor columns.
Key Insight: Align exponents by height before multiplying across primes.

Example 2.10.8. *Recombining Elementary Divisors*
As a $\mathbb{Z}$-module, let $G := \mathbb{Z}/12\mathbb{Z} \oplus \mathbb{Z}/18\mathbb{Z}$. By the primary decompositions of its cyclic summands, $\mathrm{ElemDiv}(G) = \{2^2, 2, 3^2, 3\}$.
The exponent rows are $(2,1)$ for both primes, so $d_1 = 2^2 \cdot 3^2 = 36$ and $d_2 = 2 \cdot 3 = 6$.
Thus $\mathrm{Inv}(G) = (36, 6)$ and $G \cong \mathbb{Z}/36\mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z}$ by Theorem 2.10.6(i), with $6 \mid 36$.
Structural Commentary: The 2- and 3-primary rows recombine into the divisibility chain $6 \mid 36$.
Key Insight: Matching exponent ranks determines which prime powers multiply together.

Corollary 2.10.9. *Classification of Finitely Generated PID Modules*
Let M and N be f.g. R-modules. Set $T_X := X_{\mathrm{tor}}$ and $\rho_X := \mathrm{rk}_R(X/T_X)$ $(X \in \{M, N\})$.

(i) There are $r, m \in \mathbb{N}_0$ and nonunits $d_1, \dots, d_m \in R$ with $d_{j+1} \mid d_j$ $(1 \leq j < m)$. For these data, $M \cong R^r \oplus \bigoplus_{j=1}^{m} R/\langle d_j \rangle$.
(ii) $M \cong N \iff [\rho_M = \rho_N \wedge \mathrm{Inv}(T_M) = \mathrm{Inv}(T_N)]$.
(iii) $M \cong N \iff [\rho_M = \rho_N \wedge \mathrm{ElemDiv}(T_M) = \mathrm{ElemDiv}(T_N)]$.

Proof.

(*i*) By Theorem 2.8.2(i), choose f.g. free $F \leq M$ with $M = T_M \oplus F$, $F \cong M/T_M$, $r := \mathrm{rk}_R(F) \in \mathbb{N}_0$; $F \cong R^r$ by Proposition 2.2.6(i).
Coordinate projection $M \twoheadrightarrow T_M$ maps finite generating set of M onto generating set of T_M, so T_M is f.g.
Write $\mathrm{Inv}(T_M) = (d_1, \dots, d_m)$ with $m \in \mathbb{N}_0$. By Theorem 2.10.6(i), these data have the asserted properties for T_M.
Substitution proves the claim.
(*ii*) ($\Rightarrow$) Suppose $\Phi : M \xrightarrow{\sim} N$ and fix $x \in T_M$. Choose $a \in R \setminus \{0_R\}$ with $ax = 0_M$. Then $a\Phi(x) = 0_N$, so $\Phi(T_M) \subseteq T_N$.
The same argument for Φ^{-1} proves $\Phi(T_M) = T_N$. Thus Φ restricts to $T_M \xrightarrow{\sim} T_N$ and induces $M/T_M \xrightarrow{\sim} N/T_N$.
Hence $\rho_M = \rho_N$ by Proposition 2.2.6(ii). The invariant-factor tuples agree by Theorem 2.10.6(ii).
(*ii*) ($\Leftarrow$) Suppose $\rho_M = \rho_N$ and $\mathrm{Inv}(T_M) = \mathrm{Inv}(T_N)$. Then $M/T_M \cong N/T_N$ by Proposition 2.2.6(ii).
Torsion modules satisfy $T_M \cong T_N$ by Theorem 2.10.6(ii). As $M \cong T_M \oplus M/T_M$, $N \cong T_N \oplus N/T_N$ by Theorem 2.8.2(i), $M \cong N$.
(*iii*) ($\Rightarrow$) Suppose $M \cong N$. Then $\rho_M = \rho_N$ by part (ii). The given module isomorphism induces $T_M \cong T_N$.
Hence $\mathrm{ElemDiv}(T_M) = \mathrm{ElemDiv}(T_N)$ by Theorem 2.10.2(ii).
(*iii*) ($\Leftarrow$) Suppose $\rho_M = \rho_N$ and $\mathrm{ElemDiv}(T_M) = \mathrm{ElemDiv}(T_N)$. Then $M/T_M \cong N/T_N$ by Proposition 2.2.6(ii).
The torsion modules satisfy $T_M \cong T_N$ by Theorem 2.10.2(ii).
The free-plus-torsion decompositions therefore imply $M \cong N$ by Theorem 2.8.2(i).

Structural Commentary: The intrinsic torsion summand carries invariant factors; the free quotient carries the rank.
Key Insight: Isomorphism requires equality of both independent data sets. □

Method 2.10.10. *Reading a Module from Smith Form*

Let $R^n \xrightarrow{A} R^m \to M \to 0$ be a finite presentation. Set $D := \operatorname{diag}(d_1, \dots, d_r, 0, \dots, 0)$.
Suppose $A \sim_{\text{eq}} D$ and $0 \le r \le \min\{m, n\}$. If $r \ge 1$, suppose $(d_1, \dots, d_r) \in (R \setminus \{0_R\})^r$ and $d_1 \mid \cdots \mid d_r$.

(i) Record the torsion-coordinate quotients $(R/\langle d_i \rangle)_{i=1}^r$ and the free-coordinate module R^{m-r}.
(ii) Delete the zero quotients indexed by $d_i \in R^\times$. Remaining list, together with R^{m-r}, is decomposition from Corollary 2.7.6.
(iii) If every d_i is a unit, return $\operatorname{Inv}(M_{\text{tor}}) = ()$. Otherwise, if nonunit entries are $d_s, \dots, d_r$, return $\operatorname{Inv}(M_{\text{tor}}) = (d_r, \dots, d_s)$.
(iv) In nonempty case, factor each d_i into pairwise nonassociate maximal prime powers. Retain multiplicities for $\operatorname{ElemDiv}(M_{\text{tor}})$.

Key Insight: Reverse the nonunit Smith entries for the repository's largest-first invariant-factor tuple.

Perspective 2.10.11. *Largest Invariant Factor and Product*

Let $0 \ne M$ be f.g. torsion with $\operatorname{Inv}(M) = (d_1, \dots, d_m)$. Decomposition $M \cong \bigoplus_{j=1}^m R/\langle d_j \rangle$ holds by Theorem 2.10.6(i).

Thus $\operatorname{Ann}_R(M) = \langle \operatorname{lcm}(d_1, \dots, d_m) \rangle$ by Proposition 2.5.4. Since $d_m \mid \cdots \mid d_1$, this ideal is $\langle d_1 \rangle$.

The column construction in Definition 2.10.4 uses each elementary divisor once. Hence $\prod_{j=1}^m d_j \sim \prod_{q \in \operatorname{ElemDiv}(M)} q$.

Thus the largest invariant factor controls the annihilator, whereas the total product records every elementary-divisor block.

Theorem 2.10.12. *Characterizing Cyclic Torsion Modules*
Let $n \in \mathbb{N}$, let $0 \ne M$ be f.g. torsion, and let $p_1, \dots, p_n$ be distinct nonassociate primes.
Suppose $e_1, \dots, e_n \in \mathbb{N}$ and $o(M) \sim \mu := \prod_{i=1}^n p_i^{e_i}$. Set $I := \{1, \dots, n\}$. The following are equivalent:

(i) M is cyclic.
(ii) The primary components M_{p_i} are cyclic ($i \in I$).
(iii) $\operatorname{ElemDiv}(M) = \{p_i^{e_i} : i \in I\}$.
(iv) $M \cong R/\langle \mu \rangle$.

Proof.

(i) $\Rightarrow$ (ii) Suppose $M = \langle\!\langle x \rangle\!\rangle_R$ and choose the order representative $o(x) = \mu$. By Theorem 2.5.6(iii), choose $x_1, \dots, x_n \in M$.
These vectors satisfy $M = \bigoplus_{i=1}^n \langle\!\langle x_i \rangle\!\rangle_R$. These vectors satisfy $o(x_i) \sim p_i^{e_i}$ ($i \in I$).
Hence $M_{p_i} = \langle\!\langle x_i \rangle\!\rangle_R$ by Theorem 2.9.2(iii), so every primary component is cyclic.

(ii) $\Rightarrow$ (iii) By Theorem 2.9.2(i)–(ii), the primary decomposition is indexed by I. It satisfies $M = \bigoplus_{i=1}^n M_{p_i}$ and $o(M_{p_i}) \sim p_i^{e_i}$ ($i \in I$).
Choose generators $(y_i)_{i \in I}$ with $M_{p_i} = \langle\!\langle y_i \rangle\!\rangle_R$. Then $\operatorname{Ann}_R(y_i) = \operatorname{Ann}_R(M_{p_i}) = \langle p_i^{e_i} \rangle$ ($i \in I$).
Hence $M_{p_i} \cong R/\langle p_i^{e_i} \rangle$ by Theorem 2.5.5(i). Thus $\operatorname{ElemDiv}(M_{p_i}) = \{p_i^{e_i}\}$ by Definition 2.10.1.
By Proposition 2.10.3, iterated over the primary components, $\operatorname{ElemDiv}(M) = \{p_i^{e_i} : i \in I\}$.

(iii) $\Rightarrow$ (iv) By Definition 2.10.4, $\operatorname{Inv}(M)$ has unique entry d_1. It satisfies $d_1 \sim \prod_{i=1}^n p_i^{e_i} \sim \mu$; $M \cong R/\langle d_1 \rangle = R/\langle \mu \rangle$ by Theorem 2.10.6(i).

(iv) $\Rightarrow$ (i) The module $R/\langle \mu \rangle$ is generated by $1 + \langle \mu \rangle$. Hence its image under any isomorphism to M generates M.

Structural Commentary: Coprime primary generators combine into single generator exactly when each prime contributes a single block.
Key Insight: Repeated same-prime blocks obstruct cyclicity. □

Definition 2.10.13. *Indecomposable Module*
A nonzero R-module M is *indecomposable* if $(\forall\, A, B \le M)\,[M = A \oplus B \Rightarrow A = \{0_M\} \vee B = \{0_M\}]$.

Theorem 2.10.14. *Indecomposable Finitely Generated Torsion Modules*
Let $0 \ne M$ be f.g. torsion. The following are equivalent:

(i) M is indecomposable.
(ii) M is primary cyclic.
(iii) $(\exists\, p \in R)(\exists\, e \in \mathbb{N})$ such that p is prime and $\operatorname{ElemDiv}(M) = \{p^e\}$.

Proof.

(i) $\Rightarrow$ (ii) By Theorem 2.10.2(i), M has a finite nonempty primary cyclic decomposition. Its summands are nonzero.
After finite reindexing, write $M \cong \bigoplus_{\ell=1}^m C_\ell$ with $m \in \mathbb{N}$. Transport the coordinate summands into M.
If $m \ge 2$, a coordinate summand and the sum of the others form a nontrivial decomposition, contrary to indecomposability.
Hence $m = 1$, so M is primary cyclic.
(ii) $\Rightarrow$ (iii) Choose a prime $p \in R$ and $e \in \mathbb{N}$ such that M is cyclic, p-primary, and $o(M) \sim p^e$.
Then $M \cong R/\langle p^e \rangle$ by Theorem 2.5.5(i). Thus $\operatorname{ElemDiv}(M) = \{p^e\}$ by Definition 2.10.1.
(iii) $\Rightarrow$ (i) Suppose $M = A \oplus B$. Coordinate projections map a finite generating set of M onto generating sets of A and B.
Hence both summands are f.g.; they are torsion as submodules of M.
By Proposition 2.10.3, $\{p^e\} = \operatorname{ElemDiv}(A) \uplus \operatorname{ElemDiv}(B)$. Thus $\operatorname{ElemDiv}(A) = \emptyset$ or $\operatorname{ElemDiv}(B) = \emptyset$.
The corresponding summand is zero by Theorem 2.10.2(i), so M is indecomposable.

Structural Commentary: Transported primary cyclic summands split M unless the elementary-divisor multiset is a singleton.
Key Insight: A singleton elementary divisor is exactly an indecomposable block. □

3 Linear Operators as Modules

3.1 Polynomial Modules from Operators

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over $\mathbb{F}$ unless explicitly stated otherwise.

Definition 3.1.1. *The Module Attached to an Operator*
For $T \in \mathcal{L}(\mathcal{V})$, the *operator module* attached to T is the $\mathbb{F}[z]$-module $\mathcal{V}_T$ whose underlying $\mathbb{F}$-vector space is $\mathcal{V}$. Its scalar action is $p(z) \cdot \mathbf{v} := p(T)\mathbf{v}$. For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, define $(\mathbb{F}^n)_A$ by $p(z) \cdot \mathbf{x} := p(A)\mathbf{x}$.

Proposition 3.1.2. *Operator Modules Are Finitely Generated Torsion Modules*
Let $0 < \dim \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$.
(i) $\mathcal{V}_T$ is finitely generated over $\mathbb{F}[z]$.
(ii) $\mathcal{V}_T$ is a torsion $\mathbb{F}[z]$-module.

Proof.
(i) Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an $\mathbb{F}$-basis of $\mathcal{V}$. Then $\mathcal{V}_T = \langle\!\langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle\!\rangle_{\mathbb{F}[z]}$ by $\mathbb{F} \subseteq \mathbb{F}[z]$.
(ii) Choose $0 \neq p \in \mathbb{F}[z]$ with $p(T) = 0$ by Proposition 1.1.7; $(\forall\, \mathbf{v} \in \mathcal{V})\, p \cdot \mathbf{v} = p(T)\mathbf{v} = \mathbf{0}$, so $\mathcal{V}_T$ is torsion by Definition 1.6.1(iii).
Structural Commentary: An $\mathbb{F}$-basis handles generation; one operator annihilator handles torsion.
Key Insight: One nonzero polynomial annihilates every vector. □

Theorem 3.1.3. *Submodules Are Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathcal{S} \subseteq \mathcal{V}$. Then $\mathcal{S} \leq_{\mathbb{F}[z]} \mathcal{V}_T \iff (\mathcal{S} \leq_{\mathbb{F}} \mathcal{V} \wedge T\mathcal{S} \subseteq \mathcal{S})$.

Proof.
($\Rightarrow$) Suppose $\mathcal{S} \leq_{\mathbb{F}[z]} \mathcal{V}_T$. Then $\mathcal{S} \leq_{\mathbb{F}} \mathcal{V}$ by $\mathbb{F} \subseteq \mathbb{F}[z]$. Hence $T\mathcal{S} \subseteq \mathcal{S}$ by $(\forall\, \mathbf{s} \in \mathcal{S})\, T\mathbf{s} = z \cdot \mathbf{s} \in \mathcal{S}$.
($\Leftarrow$) Suppose $\mathcal{S} \leq_{\mathbb{F}} \mathcal{V}$ and $T\mathcal{S} \subseteq \mathcal{S}$. Then $(\forall\, j \in \mathbb{Z}_{\geq 0})\, T^j\mathcal{S} \subseteq \mathcal{S}$ by induction on j.
For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$ and $\mathbf{s} \in \mathcal{S}$, $p \cdot \mathbf{s} = \sum_{j=0}^{m} a_j T^j \mathbf{s} \in \mathcal{S}$. Thus $\mathcal{S} \leq_{\mathbb{F}[z]} \mathcal{V}_T$.
Structural Commentary: Restriction to constants recovers linearity; closure under z extends to every polynomial.
Key Insight: The scalar z acts exactly as T. □

Theorem 3.1.4. *Similarity Is Module Isomorphism*
Let $T, S \in \mathcal{L}(\mathcal{V})$.
(i) For $\Phi \in \mathrm{GL}(\mathcal{V})$, $\Phi \in \mathrm{Hom}_{\mathbb{F}[z]}(\mathcal{V}_T, \mathcal{V}_S) \iff \Phi T = S\Phi$.
(ii) $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S \iff T \sim S$.

Proof.
(i) ($\Rightarrow$) If $\Phi : \mathcal{V}_T \to \mathcal{V}_S$ is $\mathbb{F}[z]$-linear, then $(\forall\, \mathbf{v} \in \mathcal{V})\, \Phi T\mathbf{v} = \Phi(z \cdot \mathbf{v}) = z \cdot \Phi\mathbf{v} = S\Phi\mathbf{v}$. Hence $\Phi T = S\Phi$.
(i) ($\Leftarrow$) Suppose $\Phi T = S\Phi$. Then $(\forall\, j \in \mathbb{Z}_{\geq 0})\, \Phi T^j = S^j\Phi$ by induction on j.
Thus $(\forall\, (p, \mathbf{v}) \in \mathbb{F}[z] \times \mathcal{V})\, \Phi(p \cdot \mathbf{v}) = \Phi(p(T)\mathbf{v}) = p(S)\Phi\mathbf{v} = p \cdot \Phi\mathbf{v}$, so Φ is $\mathbb{F}[z]$-linear.
(ii) ($\Rightarrow$) Let $\Phi : \mathcal{V}_T \xrightarrow{\sim} \mathcal{V}_S$ be an $\mathbb{F}[z]$-module isomorphism. Then $\Phi \in \mathrm{GL}(\mathcal{V})$ by $\mathbb{F} \subseteq \mathbb{F}[z]$, and $\Phi T = S\Phi$ by part (i).
Hence $S = \Phi T \Phi^{-1}$, so $T \sim S$ by Definition 6.2.6.
(ii) ($\Leftarrow$) Choose $\Phi \in \mathrm{GL}(\mathcal{V})$ with $S = \Phi T \Phi^{-1}$ by Definition 6.2.6.
Then $\Phi T = S\Phi$, so Φ is $\mathbb{F}[z]$-linear by part (i). Since Φ is bijective, $\Phi : \mathcal{V}_T \xrightarrow{\sim} \mathcal{V}_S$.
Structural Commentary: Intertwining the action of z extends to every polynomial and characterizes conjugacy.
Key Insight: Module linearity is exactly the equation $\Phi T = S\Phi$. □

Corollary 3.1.5. *Matrix Similarity Is Matrix-Module Isomorphism*
Let $n \geq 1$ and $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.
(i) For $X \in \mathrm{GL}_n(\mathbb{F})$, $T_X \in \mathrm{Hom}_{\mathbb{F}[z]}((\mathbb{F}^n)_A, (\mathbb{F}^n)_B) \iff XA = BX$.
(ii) $(\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B \iff A \sim B$.

Proof.
(i) ($\Rightarrow$) If T_X is $\mathbb{F}[z]$-linear, then $T_X T_A = T_B T_X$ by Theorem 3.1.4(i). Hence $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, XA\mathbf{x} = BX\mathbf{x}$, so $XA = BX$.
(i) ($\Leftarrow$) If $XA = BX$, then $T_X T_A = T_B T_X$ by Definition 3.1.7. Thus T_X is $\mathbb{F}[z]$-linear by Theorem 3.1.4(i).
(ii) ($\Rightarrow$) Let $\Psi : (\mathbb{F}^n)_A \xrightarrow{\sim} (\mathbb{F}^n)_B$. Choose $X, Y \in \mathrm{Mat}(n, n, \mathbb{F})$ with $\Psi = T_X$ and $\Psi^{-1} = T_Y$ by Theorem 3.4.5(i).
For $\mathbf{x} \in \mathbb{F}^n$, $XY\mathbf{x} = YX\mathbf{x} = \mathbf{x}$ by the inverse identities and Definition 3.1.7. Hence $XY = YX = I_n$ and $X \in \mathrm{GL}_n(\mathbb{F})$.
Then $XA = BX$ by part (i), and $A \sim B$ by $B = XAX^{-1}$ and Definition 6.2.1.
(ii) ($\Leftarrow$) Choose $X \in \mathrm{GL}_n(\mathbb{F})$ with $B = XAX^{-1}$ by Definition 6.2.1.
Then $XA = BX$, so the bijection T_X is $\mathbb{F}[z]$-linear by part (i) and is an $\mathbb{F}[z]$-module isomorphism.
Structural Commentary: Coordinate intertwiners translate module isomorphisms into the conjugation equation $XA = BX$.
Key Insight: Matrix similarity is polynomial-module isomorphism in coordinates. □

Definition 3.1.6. *Operator-Module Annihilator*
Let $0 < \dim \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$. The *annihilator* of $\mathcal{V}_T$ is $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) := \{p \in \mathbb{F}[z] : p(T) = 0_{\mathcal{L}(\mathcal{V})}\}$.

Proposition 3.1.7. *Minimal Polynomial as Module Order*
Let $0 < \dim \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$.
(i) $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$. Equivalently, μ_T is unique monic polynomial of minimal degree in $\{p \in \mathbb{F}[z] : p(T) = 0\}$.
(ii) If $S \in \mathcal{L}(\mathcal{V})$ and $S \sim T$, then $\mu_S = \mu_T$.
(iii) If $n \geq 1$ and $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ are similar, then $\mu_A = \mu_B$.

Proof.
(i) $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \{p \in \mathbb{F}[z] : p(T) = 0\}$ by Definition 3.1.6.
For $p \neq 0$, $p(T) = 0 \iff \mu_T \mid p$ by Theorem 2.2.4; the zero polynomial belongs to both ideals.
Hence $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$, and the characterization of μ_T follows by Definition 2.2.2.
(ii) Choose $\Phi \in \mathrm{GL}(\mathcal{V})$ with $S = \Phi T \Phi^{-1}$ by Definition 6.2.6. Then $(\forall\, p \in \mathbb{F}[z])\, p(S) = \Phi p(T) \Phi^{-1}$, so $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_S) = \mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T)$.
Hence $\mu_S = \mu_T$ by part (i).
(iii) $\mu_A = \mu_B$ by Theorem 2.2.5(ii).

Structural Commentary: Annihilator divisibility identifies μ_T; conjugation transports the resulting principal ideal.
Key Insight: The module order is invariant under intertwiners. □

Definition 3.1.8. *Cyclic Subspace and Vector Annihilator*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V}$.

(i) The *T-cyclic subspace generated by $\mathbf{v}$* is $\mathcal{C}_T(\mathbf{v}) := \operatorname{span}\{T^j\mathbf{v} : j \in \mathbb{Z}_{\geq 0}\}$.
(ii) The vector $\mathbf{v}$ is a *cyclic vector* for T if $\mathcal{C}_T(\mathbf{v}) = \mathcal{V}$.
(iii) The *annihilator of $\mathbf{v}$ for T* is $\operatorname{ann}_T(\mathbf{v}) := \{p \in \mathbb{F}[z] : p(T)\mathbf{v} = \mathbf{0}\}$.

Proposition 3.1.9. *Cyclic Subspaces Are Smallest Invariant Hosts*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V}$.

(i) $\mathcal{C}_T(\mathbf{v}) = \{p(T)\mathbf{v} : p \in \mathbb{F}[z]\}$.
(ii) $\mathcal{C}_T(\mathbf{v})$ *is the smallest T-invariant subspace containing* $\mathbf{v}$.
(iii) $\operatorname{ann}_T(\mathbf{v}) \trianglelefteq \mathbb{F}[z]$.

Proof.

(i) For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$, $p(T)\mathbf{v} = \sum_{j=0}^{m} a_j T^j\mathbf{v}$. Both inclusions hold by this identity.
(ii) $T(p(T)\mathbf{v}) = (zp)(T)\mathbf{v} \in \mathcal{C}_T(\mathbf{v})$ by part (i), so $\mathcal{C}_T(\mathbf{v})$ is T-invariant.
Let $\mathcal{U} \leq_{\mathbb{F}} \mathcal{V}$ contain $\mathbf{v}$ and satisfy $T\mathcal{U} \subseteq \mathcal{U}$. By induction, $(\forall\, j \in \mathbb{Z}_{\geq 0})\, T^j\mathbf{v} \in \mathcal{U}$, so $\mathcal{C}_T(\mathbf{v}) \subseteq \mathcal{U}$.
(iii) The zero polynomial belongs to $\operatorname{ann}_T(\mathbf{v})$.
For $a, b \in \mathbb{F}[z]$ and $p, q \in \operatorname{ann}_T(\mathbf{v})$, $(ap + bq)(T)\mathbf{v} = a(T)p(T)\mathbf{v} + b(T)q(T)\mathbf{v} = \mathbf{0}$ by Proposition 1.1.6(i).
Thus $\operatorname{ann}_T(\mathbf{v}) \leq_{\mathbb{F}[z]} \mathbb{F}[z]$ by Theorem 1.1.5, and $\operatorname{ann}_T(\mathbf{v}) \trianglelefteq \mathbb{F}[z]$ by Proposition 1.1.7.

Structural Commentary: Orbit-polynomial equality controls minimal invariance; evaluation closure controls the ideal property.
Key Insight: Polynomial evaluation packages every finite orbit combination. □

Perspective 3.1.10. *An Operator as a Polynomial Action*
Definition 3.1.1 interprets z as T. Proposition 3.1.9(i) identifies the cyclic module generated by $\mathbf{v}$ with $\mathcal{C}_T(\mathbf{v})$.
Figure 104 illustrates how iteration under T forms this cyclic span; the cited formal owners establish the identities.

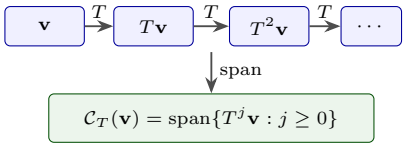

Figure 104: Iteration under T forms the orbit whose finite linear span is $\mathcal{C}_T(\mathbf{v})$.

Definition 3.1.11. *Order of a Vector under an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$. Then $\{0\} \neq \operatorname{ann}_T(\mathbf{v}) \trianglelefteq \mathbb{F}[z]$ by Proposition 3.1.2(ii), Definition 1.6.1(iii), Proposition 3.1.9(iii).
This ideal has a unique monic generator by Theorem 3.10.2(iii)–(iv).
Specializing Definition 2.5.3, the *order of $\mathbf{v}$ under T* is this generator $m_{\mathbf{v}}$; equivalently, $\operatorname{ann}_T(\mathbf{v}) = \langle m_{\mathbf{v}} \rangle$.

Definition 3.1.12. *Companion Matrix*
Let $n \geq 1$ and let $f(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$ be monic. For $n \geq 2$, the *companion matrix* of f is

$$C_f := \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}.$$

For $n = 1$, set $C_f = [-c_0]$.

Theorem 3.1.13. *Companion Matrix Realises a Polynomial*
Let $n \geq 1$, let $f(z) = z^n + \sum_{i=0}^{n-1} c_i z^i$ be monic, and let C_f be its companion matrix.

(i) $\mu_{C_f} = f$.
(ii) $p_{C_f} = f$.

Proof.

(i) Let $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the standard basis. Then $(\forall\, j \in \{0, \ldots, n-1\})\, C_f^j\mathbf{e}_1 = \mathbf{e}_{j+1}$.
The last column satisfies $C_f^n\mathbf{e}_1 = -\sum_{i=0}^{n-1} c_i C_f^i\mathbf{e}_1$, so $f(C_f)\mathbf{e}_1 = \mathbf{0}$.
For $j \in \{1, \ldots, n\}$, $f(C_f)\mathbf{e}_j = C_f^{j-1}f(C_f)\mathbf{e}_1 = \mathbf{0}$ by $\mathbf{e}_j = C_f^{j-1}\mathbf{e}_1$ and commutation with C_f. Thus $f(C_f) = 0$.
If $0 \neq g(z) = \sum_{i=0}^{m} b_i z^i$ with $m < n$, then $g(C_f)\mathbf{e}_1 = \sum_{i=0}^{m} b_i\mathbf{e}_{i+1} \neq \mathbf{0}$ by linear independence of the standard basis.
Therefore no nonzero polynomial of degree less than n annihilates C_f.
Since f is a monic degree-n annihilator, $\mu_{C_f} = f$ by Definition 2.2.3.
(ii) Work over the rational-function field $\mathbb{F}(z)$ and set $M(z) := zI_n - C_f$.
Row operation $R_1 \leftarrow R_1 + \sum_{k=2}^{n} z^{k-1}R_k$ preserves $\det M(z)$ by Theorem 3.8.2(ii)–(iv),(ix). New first row is $(0, \ldots, 0, f(z))$.
Moving it below the other rows uses $n-1$ swaps and produces an upper-triangular matrix with diagonal $(-1, \ldots, -1, f(z))$.
Hence $(-1)^{n-1}\det M(z) = (-1)^{n-1}f(z)$ by Theorem 3.8.2(iii),(v), so $p_{C_f}(z) = \det M(z) = f(z)$ by Definition 2.1.2.

Structural Commentary: The orbit basis fixes the least annihilator; determinant-preserving row reduction computes p_{C_f}.
Key Insight: One last-column relation controls both polynomial invariants. □

Theorem 3.1.14. *Cyclic Subspaces as Polynomial Quotients*

Let $\dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, *and* $0 \neq \mathbf{v} \in \mathcal{V}$. *Let* $m_{\mathbf{v}}(z) = z^d + \sum_{i=0}^{d-1} c_i z^i$ *be the order of* $\mathbf{v}$, *where* $d \geq 1$.

Define $\Phi_{\mathbf{v}} : \mathbb{F}[z] \to \mathcal{C}_T(\mathbf{v})$ *by* $\Phi_{\mathbf{v}}(p) = p(T)\mathbf{v}$. *Set* $\mathcal{B}_{\mathbf{v}} := (\mathbf{v}, T\mathbf{v}, \dots, T^{d-1}\mathbf{v})$. *Then*

(i) $\Phi_{\mathbf{v}}$ *is a surjective* $\mathbb{F}[z]$*-module homomorphism with* $\ker \Phi_{\mathbf{v}} = \langle m_{\mathbf{v}} \rangle$, *and* $\mathbb{F}[z]/\langle m_{\mathbf{v}} \rangle \cong \mathcal{C}_T(\mathbf{v})$;
(ii) $\mathcal{B}_{\mathbf{v}}$ *is a basis of* $\mathcal{C}_T(\mathbf{v})$;
(iii) $[T|_{\mathcal{C}_T(\mathbf{v})}]_{\mathcal{B}_{\mathbf{v}} \leftarrow \mathcal{B}_{\mathbf{v}}} = C_{m_{\mathbf{v}}}$.

Proof.

(i) For $a, b, p, q \in \mathbb{F}[z]$, $\Phi_{\mathbf{v}}(ap + bq) = a \cdot \Phi_{\mathbf{v}}(p) + b \cdot \Phi_{\mathbf{v}}(q)$ by Proposition 1.1.6(i).
Thus $\Phi_{\mathbf{v}}$ is $\mathbb{F}[z]$-linear by Definition 1.4.1(i). Its image is $\mathcal{C}_T(\mathbf{v})$ by Proposition 3.1.9(i).
Its kernel is $\operatorname{ann}_T(\mathbf{v}) = \langle m_{\mathbf{v}} \rangle$ by Definition 3.1.11. The quotient isomorphism follows by Theorem 1.5.4(i).

(ii) If $\sum_{j=0}^{d-1} a_j T^j \mathbf{v} = \mathbf{0}$, set $q(z) := \sum_{j=0}^{d-1} a_j z^j \in \langle m_{\mathbf{v}} \rangle$. If $q \neq 0$, then $m_{\mathbf{v}} \mid q$ and $d \leq \deg q < d$, a contradiction.
Thus $q = 0$, so $\mathcal{B}_{\mathbf{v}}$ is independent. For $p \in \mathbb{F}[z]$, choose $h, r \in \mathbb{F}[z]$ with $p = hm_{\mathbf{v}} + r$ and $\deg r < d$ or $r = 0$ by Theorem 3.8.2(i).
Then $p(T)\mathbf{v} = r(T)\mathbf{v} \in \operatorname{span} \mathcal{B}_{\mathbf{v}}$, so $\mathcal{B}_{\mathbf{v}}$ spans.

(iii) For $0 \leq j < d-1$, $T(T^j\mathbf{v}) = T^{j+1}\mathbf{v}$. The relation $T^d\mathbf{v} = -\sum_{i=0}^{d-1} c_i T^i \mathbf{v}$ holds by $m_{\mathbf{v}}(T)\mathbf{v} = \mathbf{0}$.
These are precisely the columns of $C_{m_{\mathbf{v}}}$ by Definitions 3.2.3 and 3.1.12.

Structural Commentary: The evaluation kernel forms the quotient; division selects the basis; the order relation fixes the matrix.
Key Insight: The first orbit relation becomes the companion column. □

Perspective 3.1.15. *The Translation Dictionary*
Definitions 3.1.1, 3.1.8, and 3.1.11 identify the module objects.
Proposition 3.1.7 and Theorems 3.1.3 and 3.1.14 establish the corresponding operator structures:

$\mathbb{F}[z]$-module $\mathcal{V}_T$	$\mathbb{F}$-vector space $\mathcal{V}$
$p(z) \cdot \mathbf{v}$	$p(T)\mathbf{v}$
submodule	T-invariant subspace
$\operatorname{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$	minimal polynomial of T
$\langle\langle \mathbf{v} \rangle\rangle_{\mathbb{F}[z]}$	$\mathcal{C}_T(\mathbf{v})$
$\mathbb{F}[z]/\langle m_{\mathbf{v}} \rangle$	cyclic block with companion matrix $C_{m_{\mathbf{v}}}$

This dictionary is local to $\mathcal{V}_T$; classification of arbitrary operator modules requires the subsequent PID decomposition results.

3.2 Invariant Factors

Definition 3.2.1. *Invariant Factors and Elementary Divisors*
Let $0 < \dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}_T$ is a finitely generated torsion $\mathbb{F}[z]$-module by Proposition 3.1.2(i)–(ii).

(i) For $r \in \mathbb{N}$, a *cyclic decomposition* of T is $\mathcal{V} = \bigoplus_{j=1}^{r} \mathcal{C}_T(\mathbf{v}_j)$ with $(\mathbf{v}_1, \dots, \mathbf{v}_r) \in (\mathcal{V} \setminus \{\mathbf{0}\})^r$.
(ii) The *invariant factors* of T are the unique monic representatives of $\operatorname{Inv}(\mathcal{V}_T)$ from Definition 2.10.4.
Write $\operatorname{InvFact}(T) = (f_1, \dots, f_r)$. Then each f_j is monic and nonconstant, and $f_r \mid \cdots \mid f_1$.
(iii) The *elementary divisors* of T are the unique monic representatives of $\operatorname{ElemDiv}(\mathcal{V}_T)$ from Definition 2.10.1.
Denote the resulting multiset by $\operatorname{ElemDiv}(T)$.
(iv) Let $\mathcal{I}$ be finite, $(\mathbf{v}_\alpha)_{\alpha \in \mathcal{I}} \in (\mathcal{V} \setminus \{\mathbf{0}\})^{\mathcal{I}}$, $\mathcal{V} = \bigoplus_{\alpha \in \mathcal{I}} \mathcal{C}_T(\mathbf{v}_\alpha)$. For $\alpha \in \mathcal{I}$, set $d_\alpha := \deg m_{\mathbf{v}_\alpha}$ and $\mathcal{B}_\alpha := (\mathbf{v}_\alpha, T\mathbf{v}_\alpha, \dots, T^{d_\alpha - 1}\mathbf{v}_\alpha)$.
After choosing an ordering of $\mathcal{I}$, the corresponding concatenation of the $\mathcal{B}_\alpha$ is an *elementary divisor basis* if $\{m_{\mathbf{v}_\alpha} : \alpha \in \mathcal{I}\} = \operatorname{ElemDiv}(T)$ as multisets.
(v) For $n \geq 1$ and $A \in \operatorname{Mat}(n, n, \mathbb{F})$, set $\operatorname{InvFact}(A) := \operatorname{InvFact}(T_A)$. Set $\operatorname{ElemDiv}(A) := \operatorname{ElemDiv}(T_A)$.

Theorem 3.2.2. *Invariant-Factor Skeleton for Operators*
Let $0 < \dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, $r \in \mathbb{N}$, *and* $\operatorname{InvFact}(T) = (f_1, \dots, f_r)$.

(i) $(\exists\, \mathbf{v}_1, \dots, \mathbf{v}_r \in \mathcal{V} \setminus \{\mathbf{0}\})$ *such that, for* $\mathcal{C}_j := \mathcal{C}_T(\mathbf{v}_j)$, $\mathcal{V} = \bigoplus_{j=1}^{r} \mathcal{C}_j$. *Then* $m_{\mathbf{v}_j} = f_j$, $\mathcal{C}_j \cong \mathbb{F}[z]/\langle f_j \rangle$ *for* $j \in \{1, \dots, r\}$.

(ii) Set $\mathcal{B}_j := (\mathbf{v}_j, T\mathbf{v}_j, \dots, T^{\deg f_j - 1}\mathbf{v}_j)$ *and* $\mathcal{B} := (\mathcal{B}_1, \dots, \mathcal{B}_r)$. *Then* $\mathcal{B}$ *is a basis of* $\mathcal{V}$ *and* $[T]_{\mathcal{B} \leftarrow \mathcal{B}} = \bigoplus_{j=1}^{r} C_{f_j}$.

(iii) $\mu_T = f_1$.

(iv) $q_T = \prod_{j=1}^{r} f_j$.

Proof.

(i) The module $\mathcal{V}_T$ is finitely generated and torsion by Proposition 3.1.2(i)–(ii).
Choose an $\mathbb{F}[z]$-module isomorphism $\Psi : \bigoplus_{j=1}^{r} \mathbb{F}[z]/\langle f_j \rangle \to \mathcal{V}_T$ by Theorem 2.10.6(i).
Let $\mathbf{e}_j$ have jth coordinate $1 + \langle f_j \rangle$ and all other coordinates zero. Set $\mathbf{v}_j := \Psi(\mathbf{e}_j)$ and $\mathcal{C}_j := \Psi(\mathbb{F}[z]\mathbf{e}_j)$.
Since f_j is nonconstant and Ψ is injective, $\mathbf{e}_j \neq \mathbf{0}$ and $\mathbf{v}_j \neq \mathbf{0}$. For $p \in \mathbb{F}[z]$, $\Psi(p\mathbf{e}_j) = p(T)\mathbf{v}_j$ by $\mathbb{F}[z]$-linearity.
$\mathcal{C}_j = \mathcal{C}_T(\mathbf{v}_j)$ by the definitions of $\mathcal{C}_j$ and $\mathcal{C}_T(\mathbf{v}_j)$. Since Ψ is bijective, $\mathcal{V} = \bigoplus_{j=1}^{r} \mathcal{C}_j$ and $\mathcal{C}_j \cong \mathbb{F}[z]/\langle f_j \rangle$.
The annihilator of $\mathbf{e}_j$ is $\langle f_j \rangle$, so $\operatorname{ann}_T(\mathbf{v}_j) = \langle f_j \rangle$ by Ψ. Hence $m_{\mathbf{v}_j} = f_j$ by Definition 3.1.11.

(ii) For $j \in \{1,\dots,r\}$, $\mathcal{B}_j$ is a basis of $\mathcal{C}_j$ by Theorem 3.1.14(ii). $[T|_{\mathcal{C}_j}]_{\mathcal{B}_j \leftarrow \mathcal{B}_j} = C_{f_j}$ by Theorem 3.1.14(iii).
The concatenation $\mathcal{B}$ is a basis of $\mathcal{V}$ by part (i) and Definition 2.3.5.

Since $(\forall\, j \in \{1,\dots,r\})\, T\mathcal{C}_j \subseteq \mathcal{C}_j$, all off-diagonal blocks vanish. Thus $[T]_{\mathcal{B}\leftarrow\mathcal{B}} = \bigoplus_{j=1}^{r} C_{f_j}$.

(iii) For $j \in \{1,\dots,r\}$, $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{C}_j) = \langle f_j \rangle$ because $\mathcal{C}_j = \mathbb{F}[z]\mathbf{v}_j$ and $m_{\mathbf{v}_j} = f_j$ by part (i).
$\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mathrm{lcm}(f_1,\dots,f_r) \rangle$ by part (i), Proposition 2.5.4. Since $f_r \mid \cdots \mid f_1$, $\mathrm{lcm}(f_1,\dots,f_r) = f_1$ and $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle f_1 \rangle$.
$\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$ by Proposition 3.1.7(i). Since f_1 and μ_T are monic, $f_1 = \mu_T$.

(iv) Set $A := [T]_{\mathcal{B}\leftarrow\mathcal{B}}$. Then $q_T = p_A$ by Definition 2.1.15.

$A = \bigoplus_{j=1}^{r} C_{f_j}$ by part (ii), so $p_A = \prod_{j=1}^{r} p_{C_{f_j}}$ by repeated application of Proposition 2.1.8(i).

For $j \in \{1,\dots,r\}$, $p_{C_{f_j}} = f_j$ by Theorem 3.1.13(ii). Hence $q_T = \prod_{j=1}^{r} f_j$.

Structural Commentary: PID quotient generators produce cyclic bases; annihilators and determinants recover μ_T and q_T.
Key Insight: Cyclic quotient generators turn invariant factors into companion blocks. □

3.3 Elementary Divisors

Theorem 3.3.1. *Primary Cyclic Decomposition for an Operator*
Let $0 < \dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, $s \in \mathbb{N}$. Set $I := \{1,\dots,s\}$. Let $(p_i)_{i\in I} \in \mathbb{F}[z]^I$ be distinct monic irreducibles and $(e_i)_{i\in I} \in \mathbb{N}^I$. Suppose $\mu_T = \prod_{i\in I} p_i^{e_i}$ and set $\mathcal{V}_{p_i} := \ker p_i(T)^{e_i}$ for $i \in I$.

(i) $\mathcal{V}_T = \bigoplus_{i\in I} \mathcal{V}_{p_i}$.

(ii) For $i \in I$, there are $k_i \in \mathbb{N}$ and exponents $e_i = e_{i,1} \geq \cdots \geq e_{i,k_i} \geq 1$. There are vectors $(\mathbf{v}_{i,j})_{j=1}^{k_i} \in (\mathcal{V}_{p_i} \setminus \{\mathbf{0}\})^{k_i}$.

These data satisfy $\mathcal{V}_{p_i} = \bigoplus_{j=1}^{k_i} \mathcal{C}_T(\mathbf{v}_{i,j})$. They satisfy $m_{\mathbf{v}_{i,j}} = p_i^{e_{i,j}}$ for $j \in \{1,\dots,k_i\}$.

(iii) For $i \in I$ and $j \in \{1,\dots,k_i\}$, set $d_{i,j} := \deg(p_i^{e_{i,j}})$. Set $\mathcal{B}_{i,j} := (\mathbf{v}_{i,j}, T\mathbf{v}_{i,j}, \dots, T^{d_{i,j}-1}\mathbf{v}_{i,j})$.
Then $\mathrm{ElemDiv}(T) = \{p_i^{e_{i,j}} : i \in I,\ 1 \leq j \leq k_i\}$ as multisets.
The lexicographically ordered concatenation of the $\mathcal{B}_{i,j}$ is an elementary divisor basis.

Proof.

(i) Since $\dim \mathcal{V} > 0$, $\mathcal{V}_T \neq \{\mathbf{0}\}$. It is finitely generated and torsion by Proposition 3.1.2(i)–(ii).
Its order satisfies $o(\mathcal{V}_T) \sim \mu_T$ by Proposition 3.1.7(i). For $i \in I$, set $M_i := \{\mathbf{v} \in \mathcal{V} : p_i^{e_i} \cdot \mathbf{v} = \mathbf{0}\}$.
Then M_i is the p_i-primary component and $\mathcal{V}_T = \bigoplus_{i\in I} M_i$ by Theorem 2.9.2(i).

The identity $p_i^{e_i} \cdot \mathbf{v} = p_i(T)^{e_i}\mathbf{v}$ holds by Definition 3.1.1. Hence $M_i = \ker p_i(T)^{e_i} = \mathcal{V}_{p_i}$.

(ii) For $i \in I$, $\mathcal{V}_{p_i}$ is finitely generated and p_i-primary by Theorem 2.9.2(ii). Its order satisfies $o(\mathcal{V}_{p_i}) \sim p_i^{e_i}$ by the same theorem.
Choose k_i, $(\mathbf{v}_{i,j})_{j=1}^{k_i}$, and $(e_{i,j})_{j=1}^{k_i}$ by Theorem 2.9.7(i).

They satisfy the exponent chain and $\mathcal{V}_{p_i} = \bigoplus_{j=1}^{k_i} \mathbb{F}[z]\mathbf{v}_{i,j}$ by the same theorem.

Their annihilators are $\langle p_i^{e_{i,j}} \rangle$ by the same theorem. Each $\mathbf{v}_{i,j}$ is nonzero because $\langle p_i^{e_{i,j}} \rangle \neq \mathbb{F}[z]$.
$\mathbb{F}[z]\mathbf{v}_{i,j} = \mathcal{C}_T(\mathbf{v}_{i,j})$ by Proposition 3.1.9(i). Thus $m_{\mathbf{v}_{i,j}} = p_i^{e_{i,j}}$ by Definition 3.1.11 and monicity.

(iii) $\mathrm{ElemDiv}(\mathcal{V}_T) = \{p_i^{e_{i,j}} : i \in I,\ 1 \leq j \leq k_i\}$ by part (ii) and Definition 2.10.1.
The multiset identity in (iii) follows by Definition 3.2.1(iii) and monicity. Each $\mathcal{B}_{i,j}$ is a basis of $\mathcal{C}_T(\mathbf{v}_{i,j})$ by Theorem 3.1.14(ii).
Their lexicographically ordered concatenation is basis of $\mathcal{V}$ by parts (i)–(ii). It is elementary divisor basis by Definition 3.2.1(iv).

Structural Commentary: Primary decomposition identifies kernels; cyclic decomposition refines each into annihilator blocks.
Key Insight: Each elementary divisor is one cyclic summand's monic annihilator. □

Theorem 3.3.2. *Reducing Decompositions Combine the Data*
Let $0 < \dim \mathcal{V} < \infty$, let $T \in \mathcal{L}(\mathcal{V})$, and suppose $\mathcal{V} = \mathcal{A} \oplus \mathcal{B}$.
Assume $\mathcal{A}$ and $\mathcal{B}$ are nonzero T-invariant subspaces. Set $T_{\mathcal{A}} := T|_{\mathcal{A}}$ and $T_{\mathcal{B}} := T|_{\mathcal{B}}$.

(i) $\mu_T = \mathrm{lcm}(\mu_{T_{\mathcal{A}}}, \mu_{T_{\mathcal{B}}})$.
(ii) $\mathrm{ElemDiv}(T) = \mathrm{ElemDiv}(T_{\mathcal{A}}) \uplus \mathrm{ElemDiv}(T_{\mathcal{B}})$. *Here $\uplus$ denotes multiset union.*
(iii) The multiset in (ii) is grouped and zero-padded as in Definition 2.10.4. Its column products form $\mathrm{InvFact}(T)$.

Proof.

(i) The subspaces $\mathcal{A}$ and $\mathcal{B}$ are $\mathbb{F}[z]$-submodules by Theorem 3.1.3. Hence $\mathcal{V}_T = \mathcal{A}_{T_{\mathcal{A}}} \oplus \mathcal{B}_{T_{\mathcal{B}}}$ by the assumed direct sum.
The order of this module sum is the least common multiple of the summand orders by Proposition 2.5.4.
They are μ_T, $\mu_{T_{\mathcal{A}}}$, and $\mu_{T_{\mathcal{B}}}$ by Proposition 3.1.7(i), proving (i).

(ii) Set $M_{\mathcal{A}} := \mathcal{A}_{T_{\mathcal{A}}}$ and $M_{\mathcal{B}} := \mathcal{B}_{T_{\mathcal{B}}}$. $\mathcal{V}_T = M_{\mathcal{A}} \oplus M_{\mathcal{B}}$ by the assumed direct sum and Theorem 3.1.3.
Hence $\mathrm{ElemDiv}(\mathcal{V}_T) = \mathrm{ElemDiv}(M_{\mathcal{A}} \oplus M_{\mathcal{B}})$. $\mathrm{ElemDiv}(M_{\mathcal{A}} \oplus M_{\mathcal{B}}) = \mathrm{ElemDiv}(M_{\mathcal{A}}) \uplus \mathrm{ElemDiv}(M_{\mathcal{B}})$ by Proposition 2.10.3.
The identity in (ii) follows by Definition 3.2.1(iii) after taking monic representatives.

(iii) Grouping, padding, and column multiplication produce $\mathrm{Inv}(\mathcal{V}_T)$ from (ii) by Definition 2.10.4.
Taking its unique monic representatives produces $\mathrm{InvFact}(T)$ by Definition 3.2.1(ii).

Structural Commentary: The reducing sum becomes a module sum; order and elementary-divisor laws separate its data.
Key Insight: Direct sums take least common multiples and multiset unions. □

Definition 3.3.3. *Characteristic Polynomial from Elementary Divisors*
Let $0 < \dim \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$. Define $\chi_T(z) := \prod_{q \in \mathrm{ElemDiv}(T)} q(z)$, retaining multiset multiplicities.
For $n \geq 1$ and $A \in \mathrm{Mat}(n,n,\mathbb{F})$, set $\chi_A := \chi_{T_A}$.

Theorem 3.3.4. *Minimal and Characteristic Polynomials from Elementary Divisors*
Let $0 < \dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, $s \in \mathbb{N}$. $I := \{1, \dots, s\}$. *Let* $(k_i)_{i\in I} \in \mathbb{N}^I$, $(p_i)_{i\in I} \in \mathbb{F}[z]^I$ *be distinct monic irreducibles.*
Let $e_{i,j} \in \mathbb{N}$ *for* $i \in I$, $j \in \{1, \dots, k_i\}$. *Let* $\mathrm{ElemDiv}(T) = \{p_i^{e_{i,j}} : i \in I,\ 1 \le j \le k_i\}$ *be multisets. For* $i \in I$, *set* $m_i := \max_{1\le j\le k_i} e_{i,j}$.

(i) $\mu_T = \prod_{i\in I} p_i^{m_i}$.

(ii) $q_T = \chi_T = \prod_{i\in I} \prod_{j=1}^{k_i} p_i^{e_{i,j}}$.

(iii) $\deg \chi_T = \deg q_T = \dim \mathcal{V}$.

(iv) $\mu_T \mid q_T$, *and* μ_T, q_T, *and* χ_T *have the same monic irreducible factors.*

Proof.

(i) The module $\mathcal{V}_T$ is finitely generated and torsion by Proposition 3.1.2(i)–(ii).
Choose a primary cyclic decomposition of $\mathcal{V}_T$ by Theorem 2.10.2(i).
Its cyclic orders form $\mathrm{ElemDiv}(\mathcal{V}_T)$ by Definition 2.10.1. Their monic representatives form $\mathrm{ElemDiv}(T)$ by Definition 3.2.1(iii).
Set $\lambda := \mathrm{lcm}\{p_i^{e_{i,j}} : i \in I,\ 1 \le j \le k_i\}$. Then $o(\mathcal{V}_T) \sim \lambda$ by repeated application of Proposition 2.5.4.
The monic representative of this order is μ_T by Proposition 3.1.7(i).
Since the p_i are distinct and monic, this least common multiple is $\prod_{i\in I} p_i^{m_i}$.

(ii) Write $\mathrm{InvFact}(T) = (f_1, \dots, f_r)$. Then $q_T = \prod_{\ell=1}^{r} f_\ell$ by Theorem 3.2.2(iv).
The column construction uses every elementary divisor once by Definition 2.10.4.
Its monic column products are the f_ℓ by Definition 3.2.1(ii)–(iii).
Hence $\prod_{\ell=1}^{r} f_\ell = \prod_{i\in I} \prod_{j=1}^{k_i} p_i^{e_{i,j}}$. The latter product is χ_T by Definition 3.3.3.

(iii) $\deg q_T = \dim \mathcal{V}$ by Proposition 2.1.16(i). The equality $\deg \chi_T = \deg q_T$ holds by part (ii).

(iv) For $i \in I$, set $b_i := \sum_{j=1}^{k_i} e_{i,j}$. Then m_i is exponent of p_i in μ_T by part (i), b_i is its exponent in $q_T = \chi_T$ by part (ii).
Since $e_{i,j} \ge 1$, $1 \le m_i \le b_i$. Thus $\mu_T \mid q_T$, and all three polynomials have the monic irreducible factors $(p_i)_{i\in I}$.

Structural Commentary: Module orders take primewise maxima; invariant-factor products retain every primary exponent.
Key Insight: Minimal exponents take maxima; characteristic exponents take sums. □

Definition 3.3.5. *Nonderogatory Operators*
Let $0 < \dim \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$. The operator T is *nonderogatory* if $\mu_T = q_T$.
The equivalence $\mu_T = q_T \iff \mu_T = \chi_T$ holds by Theorem 3.3.4(ii).
The equivalence $\mu_T = q_T \iff \deg \mu_T = \dim \mathcal{V}$ holds by Theorem 3.3.4(iii)–(iv).
For $n \ge 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$, A is nonderogatory if T_A is.

Theorem 3.3.6. *Cyclic and Nonderogatory Are the Same*
Let $0 < \dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, $s \in \mathbb{N}$, $I := \{1, \dots, s\}$. *Let* $(p_i)_{i\in I} \in \mathbb{F}[z]^I$ *be distinct monic irreducibles and* $(e_i)_{i\in I} \in \mathbb{N}^I$.
Suppose $\mu_T = \prod_{i\in I} p_i^{e_i}$. *The following are equivalent:*

(i) $\mathcal{V}_T$ *is cyclic.*

(ii) T *has a cyclic vector.*

(iii) $\mathrm{ElemDiv}(T) = \{p_1^{e_1}, \dots, p_s^{e_s}\}$ *as multisets.*

(iv) T *is nonderogatory.*

Proof.

(i) $\Rightarrow$ *(ii)* Choose $\mathbf{v} \in \mathcal{V}$ with $\mathcal{V}_T = \mathbb{F}[z]\mathbf{v}$ by Definition 1.2.3(iii). Then $\mathcal{V} = \mathcal{C}_T(\mathbf{v})$ by Proposition 3.1.9(i).
Thus $\mathbf{v}$ is cyclic for T by Definition 3.1.8(ii).

(ii) $\Rightarrow$ *(iii)* Let $\mathbf{v}$ be cyclic for T. Then $\mathcal{V} = \mathcal{C}_T(\mathbf{v})$ by Definition 3.1.8(ii). Thus $\mathcal{V}_T = \mathbb{F}[z]\mathbf{v}$ by Proposition 3.1.9(i).
Since $\dim \mathcal{V} > 0$, $\mathcal{V}_T \ne \{\mathbf{0}\}$. It is finitely generated and torsion by Proposition 3.1.2(i)–(ii).
Its order satisfies $o(\mathcal{V}_T) \sim \mu_T$ by Proposition 3.1.7(i). Hence $\mathrm{ElemDiv}(\mathcal{V}_T) = \{p_i^{e_i} : i \in I\}$ by Theorem 2.10.12(i),(iii).
$\mathrm{ElemDiv}(T) = \{p_i^{e_i} : i \in I\}$ by Definition 3.2.1(iii) and monicity.

(iii) $\Rightarrow$ *(iv)* The identity $\mu_T = q_T = \prod_{i\in I} p_i^{e_i}$ holds by Theorem 3.3.4(i)–(ii). Thus T is nonderogatory by Definition 3.3.5.

(iv) $\Rightarrow$ *(i)* Choose $k_i \in \mathbb{N}$ and $e_i = e_{i,1} \ge \cdots \ge e_{i,k_i} \ge 1$ by Theorem 3.3.1(ii).
Then $\mathrm{ElemDiv}(T) = \{p_i^{e_{i,j}} : i \in I,\ 1 \le j \le k_i\}$ by Theorem 3.3.1(iii). Identity $\mu_T = q_T$ holds by condition (iv), Definition 3.3.5.
For $i \in I$, $e_i = \sum_{j=1}^{k_i} e_{i,j}$ by Theorem 3.3.4(i)–(ii). The inequalities $e_{i,j} \ge 1$ and $e_i = e_{i,1}$ force $k_i = 1$.
Hence $\mathrm{ElemDiv}(T) = \{p_i^{e_i} : i \in I\}$. $\mathrm{ElemDiv}(\mathcal{V}_T) = \{p_i^{e_i} : i \in I\}$ by Definition 3.2.1(iii); $\mathcal{V}_T$ is cyclic by Theorem 2.10.12(i),(iii).

Structural Commentary: Module generators become cyclic vectors; exponent sums detect multiple primary blocks.
Key Insight: Cyclicity fails exactly when one prime contributes multiple blocks. □

Perspective 3.3.7. *Elementary Divisors and Jordan Blocks*
Let $0 < \dim_{\mathbb{C}} \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$. Choose a Jordan form of T by Theorem 6.2.1.
Multiplication by z on $\mathbb{C}[z]/\langle (z-\lambda)^k \rangle$ has matrix $J_k(\lambda)$ in the reversed shifted-power basis.
Hence each Jordan block is cyclic with order $(z-\lambda)^k$. The Jordan direct sum is a primary cyclic decomposition.
Its elementary-divisor multiset is multiset of $(z-\lambda)^k$ by Definition 2.10.1; $\mathrm{ElemDiv}(T)$ records pairs (λ, k) by Definition 3.2.1(iii).
The prime-power polynomials and Jordan blocks encode corresponding data; they are not the same objects.

3.4 Operator Classification

Perspective 3.4.1. *Operator Modules and Similarity*
Let $0 < \dim \mathcal{V} < \infty$ and $S, T \in \mathcal{L}(\mathcal{V})$. Then $S \sim T \iff \mathcal{V}_S \cong_{\mathbb{F}[z]} \mathcal{V}_T$ by Theorem 3.1.4.
A cyclic summand of order f is isomorphic to $\mathbb{F}[z]/\langle f \rangle$, and its operator restriction has matrix C_f in a cyclic basis, by Theorem 3.1.14.

Theorem 3.4.2. *Complete Similarity Invariants*
Let $0 < \dim \mathcal{V} < \infty$, let $S, T \in \mathcal{L}(\mathcal{V})$, let $n \geq 1$, and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) $T \sim S \iff \mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S \iff \mathrm{ElemDiv}(T) = \mathrm{ElemDiv}(S) \iff \mathrm{InvFact}(T) = \mathrm{InvFact}(S)$.
(ii) $A \sim_{\mathbb{F}} B \iff (\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B \iff \mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B) \iff \mathrm{InvFact}(A) = \mathrm{InvFact}(B)$.
(iii) If $\mathcal{B}$ is a basis of $\mathcal{V}$ and $A = [T]_{\mathcal{B} \leftarrow \mathcal{B}}$, then $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(T)$ *and* $\mathrm{InvFact}(A) = \mathrm{InvFact}(T)$.

Proof.

(i) ($\Rightarrow$) Suppose $T \sim S$. Then $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S$ by Theorem 3.1.4.
Both divisor data sets agree by Theorems 2.10.2(ii) and 2.10.6(ii) with Definition 3.2.1(ii)–(iii).
(i) ($\Leftarrow$) Suppose either $\mathrm{ElemDiv}(T) = \mathrm{ElemDiv}(S)$ or $\mathrm{InvFact}(T) = \mathrm{InvFact}(S)$. Then $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S$ by Theorem 2.10.2(ii) or Theorem 2.10.6(ii), respectively, with Definition 3.2.1(ii)–(iii).
Hence $T \sim S$ and $\mathrm{ElemDiv}(T) = \mathrm{ElemDiv}(S)$ by Theorems 3.1.4 and 2.10.2(ii) with Definition 3.2.1(iii).
(ii) ($\Rightarrow$) Suppose $A \sim_{\mathbb{F}} B$. Then $(\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$ by Corollary 3.1.5.
Both divisor data sets agree by Theorems 2.10.2(ii) and 2.10.6(ii) with Definition 3.2.1(ii)–(iii).
(ii) ($\Leftarrow$) Suppose either $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$ or $\mathrm{InvFact}(A) = \mathrm{InvFact}(B)$.
Then $(\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$ by Theorem 2.10.2(ii) or Theorem 2.10.6(ii), respectively, with Definition 3.2.1(ii)–(iii).
Hence $A \sim_{\mathbb{F}} B$ and $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$ by Corollary 3.1.5, Theorem 2.10.2(ii), and Definition 3.2.1(iii).
(iii) Set $\Phi(\mathbf{v}) := [\mathbf{v}]_{\mathcal{B}}$. The map Φ is bijective, and $(\forall\, (p, \mathbf{v}) \in \mathbb{F}[z] \times \mathcal{V})\, \Phi(p(T)\mathbf{v}) = p(A)\Phi(\mathbf{v})$ by $A = [T]_{\mathcal{B} \leftarrow \mathcal{B}}$.
Thus $\Phi : \mathcal{V}_T \to (\mathbb{F}^n)_A$ is an $\mathbb{F}[z]$-isomorphism by Definitions 3.1.1 and 1.4.1.
Both divisor data sets agree by Theorems 2.10.2(ii) and 2.10.6(ii) with Definition 3.2.1(ii)–(iii).

Structural Commentary: Similarity identifies the operator module; PID uniqueness fixes both divisor data sets.
Key Insight: Either divisor datum determines the module and its similarity class. □

Definition 3.4.3. *Rational Canonical Block Forms*
Let $\mathcal{P}$ be the set of monic nonconstant polynomials in $\mathbb{F}[z]$.
Let $\mathcal{Q}$ be the set of positive powers of monic irreducible polynomials in $\mathbb{F}[z]$. Let $n \geq 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) Matrix A is in *invariant-factor rational canonical form* if $(\exists\, r \in \mathbb{N})(\exists\, (s_1, \ldots, s_r) \in \mathcal{P}^r)\, [A = \mathrm{diag}(C_{s_1}, \ldots, C_{s_r}) \wedge s_r \mid \cdots \mid s_1]$.
(ii) Matrix A is in *elementary-divisor rational canonical form* if $(\exists\, m \in \mathbb{N})(\exists\, (q_1, \ldots, q_m) \in \mathcal{Q}^m)\, A = \mathrm{diag}(C_{q_1}, \ldots, C_{q_m})$.

Theorem 3.4.4. *Rational Canonical Form Skeleton*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and write $\mathrm{InvFact}(A) = (s_1, \ldots, s_r)$. *Set $R_A := \mathrm{diag}(C_{s_1}, \ldots, C_{s_r})$. Then:*

(i) $r \in \mathbb{N}$, the s_j are monic and nonconstant, and $s_r \mid \cdots \mid s_1$.
(ii) $A \sim_{\mathbb{F}} R_A$.
(iii) $\mu_A = s_1$ and $p_A = \prod_{j=1}^{r} s_j$.

Proof.

(i) The tuple properties hold by Definition 3.2.1(ii).
(ii) Let $T_A : \mathbb{F}^n \to \mathbb{F}^n$, $\mathbf{x} \mapsto A\mathbf{x}$. There is a basis $\mathcal{B}$ with $[T_A]_{\mathcal{B} \leftarrow \mathcal{B}} = R_A$ by Theorem 3.2.2(ii). Hence $A \sim_{\mathbb{F}} R_A$ by Corollary 6.2.4.
(iii) Identities $\mu_{T_A} = s_1$ and $q_{T_A} = \prod_{j=1}^{r} s_j$ hold by Theorem 3.2.2(iii)–(iv). Thus $\mu_A = s_1$, $p_A = \prod_{j=1}^{r} s_j$ by Definitions 2.2.3, 2.1.15.

Structural Commentary: The invariant-factor decomposition supplies a cyclic basis whose coordinate blocks represent A.
Key Insight: The divisibility-ordered tuple fixes the companion-block representative. □

Theorem 3.4.5. *Rational Canonical Form: Two Versions*
Let $0 < \dim \mathcal{V} < \infty$, let $T \in \mathcal{L}(\mathcal{V})$, let $n \geq 1$, and let $A \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) If $\mathrm{ElemDiv}(T) = \{q_1, \ldots, q_m\}$, *then $(\exists\, \mathcal{R})\, [T]_{\mathcal{R} \leftarrow \mathcal{R}} = \mathrm{diag}(C_{q_1}, \ldots, C_{q_m})$ for a basis $\mathcal{R}$ of $\mathcal{V}$.*
(ii) The elementary-divisor form in (i) is unique up to block permutation.
(iii) If $\mathrm{InvFact}(T) = (s_1, \ldots, s_r)$, *then $(\exists\, \mathcal{B})\, [T]_{\mathcal{B} \leftarrow \mathcal{B}} = \mathrm{diag}(C_{s_1}, \ldots, C_{s_r})$ for a basis $\mathcal{B}$ of $\mathcal{V}$.*
(iv) The invariant-factor form in (iii) is unique under the order $s_r \mid \cdots \mid s_1$.
(v) The class $[A]_{\sim_{\mathbb{F}}}$ contains both forms, with the uniqueness in (ii) and (iv).

Proof.

(i) Basis $\mathcal{R}$ with displayed block matrix is obtained by concatenating cyclic bases in Theorem 3.3.1(ii)–(iii) and Theorem 3.1.14(iii).
(ii) Each block C_q is cyclic with order q by Definition 3.1.12, Theorem 3.1.13(i), and Proposition 3.1.7(i).
Hence an elementary-divisor block matrix has the multiset of its block orders by Proposition 2.10.3 and Definition 3.2.1(iii).
Every coordinate matrix of T has divisor data equal to those of T by Theorem 3.4.2(iii), so only the block order can vary.
(iii) A basis $\mathcal{B}$ with the displayed block matrix exists by Theorem 3.2.2(i)–(ii).
(iv) The companion blocks in an invariant-factor form are cyclic with their defining orders by Definition 3.1.12 and Theorem 3.1.13(i).
Their divisibility-ordered list is the invariant-factor tuple of the coordinate module by Theorem 2.10.6(i)–(ii).
It equals $\mathrm{InvFact}(T)$ by Theorem 3.4.2(iii), so the list is unique.
(v) Both forms for $T_A : \mathbb{F}^n \to \mathbb{F}^n$, $\mathbf{x} \mapsto A\mathbf{x}$, exist with the stated uniqueness by parts (i)–(iv).
Each resulting coordinate matrix is similar to A by Corollary 6.2.4.

Structural Commentary: Primary and invariant-factor decompositions concatenate cyclic bases in two intrinsic block orders.
Key Insight: The two forms preserve one module while exposing different divisor groupings. □

Computation Method 3.4.6. *Constructing an Operator from Elementary Divisors*

Given $m \in \mathbb{N}$ and a multiset $Q = \{q_1, \dots, q_m\}$ of monic nonconstant prime powers in $\mathbb{F}[z]$, set $n := \sum_{j=1}^{m} \deg q_j$.

The output is a matrix in $\mathrm{Mat}(n, n, \mathbb{F})$ with elementary-divisor multiset Q and its invariant-factor form.

(i) Set $R_Q := \mathrm{diag}(C_{q_1}, \dots, C_{q_m}) \in \mathrm{Mat}(n, n, \mathbb{F})$.

(ii) $(\forall\, j \in \{1, \dots, m\})\, \mathcal{C}_{T_{C_{q_j}}}(\mathbf{e}_1) = \mathbb{F}^{\deg q_j}$ by Definition 3.1.12. Identities $(\forall\, j \in \{1, \dots, m\})\, \mu_{C_{q_j}} = q_j$ hold by Theorem 3.1.13(i).
These polynomials represent the corresponding module orders by Proposition 3.1.7(i).
Hence $(\forall\, j \in \{1, \dots, m\})$ $\mathrm{ElemDiv}(T_{C_{q_j}}) = \{q_j\}$ by Theorem 2.10.12(i),(iii) and Definition 3.2.1(iii).
Therefore $\mathrm{ElemDiv}(T_{R_Q}) = Q$ by repeated application of Proposition 2.10.3.

(iii) Recombine Q into $\mathrm{InvFact}(T_{R_Q}) = (s_1, \dots, s_r)$ by Method 2.10.5. Then $\mathrm{diag}(C_{s_1}, \dots, C_{s_r}) \sim_{\mathbb{F}} R_Q$ by Theorem 3.4.4(ii).

Key Insight: Companion blocks realise cyclic summands; block sums realise direct sums.

Example 3.4.7. *Rational Canonical Possibilities from Degree Counts*

Let $T \in \mathcal{L}(\mathbb{R}^7)$ satisfy $\mu_T = (z-1)(z^2+1)^2$, and set $Q := \mathrm{ElemDiv}(T)$.
Then $\mathrm{lcm}\, Q = \mu_T$ and $\sum_{q \in Q} \deg q = 7$ by Theorem 3.3.4(i),(iii) and Definition 3.3.3.
Let a, b, c be the multiplicities in Q of $z-1$, z^2+1, and $(z^2+1)^2$, respectively.
The constraints are $a, c \geq 1$, $b \geq 0$, and $a + 2b + 4c = 7$, so $c = 1$ and $(a, b) \in \{(1,1), (3,0)\}$.
Thus $Q_1 = \{z-1, z^2+1, (z^2+1)^2\}$ or $Q_2 = \{z-1, z-1, z-1, (z^2+1)^2\}$.
The two possible elementary-divisor forms are $R_i = \bigoplus_{q \in Q_i} C_q$ $(i \in \{1, 2\})$, up to block order.

Theorem 3.4.8. *Determinant Form of the Characteristic Polynomial*

Let $n \geq 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) $\chi_A = p_A$, so $\chi_A(z) = \det(zI_n - A)$.

(ii) If $0 < \dim \mathcal{V} = n$, $T \in \mathcal{L}(\mathcal{V})$, and $A = [T]_{\mathcal{B} \leftarrow \mathcal{B}}$ for a basis $\mathcal{B}$, then $\chi_T = q_T = \chi_A$.

Proof.

(i) $\chi_A = \chi_{T_A}$ by Definition 3.3.3. $\chi_{T_A} = q_{T_A} = p_A$ and $p_A(z) = \det(zI_n - A)$ hold by Theorem 3.3.4(ii), Definitions 2.1.15, 2.1.2.

(ii) The identity $\chi_T = q_T = p_A$ holds by Theorem 3.3.4(ii) and Definition 2.1.15. Hence $p_A = \chi_A$ by part (i).

Structural Commentary: Elementary-divisor products equal operator polynomial, which coordinates identify with shifted determinant.
Key Insight: The abstract block product is the concrete characteristic determinant. □

Theorem 3.4.9. *Changing the Base Field*

Let $\mathbb{F} \subseteq \mathbb{K}$ be fields, $n \geq 1$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Subscripts on ElemDiv *and* InvFact *indicate the coefficient field.*
Let $s \in \mathbb{N}$, $(k_i)_{i=1}^{s} \in \mathbb{N}^s$, $p_i \in \mathbb{F}[z]$ be distinct monic irreducibles; $\mathrm{ElemDiv}_{\mathbb{F}}(A) = \{p_i^{e_{i,j}} : 1 \leq i \leq s,\ 1 \leq j \leq k_i\}$, *where $e_{i,j} \in \mathbb{N}$.*
For $i \in \{1, \dots, s\}$, let $m_i \in \mathbb{N}$, $(d_{i,\ell})_{\ell=1}^{m_i} \in \mathbb{N}^{m_i}$, $a_{i,\ell} \in \mathbb{K}[z]$ be pairwise distinct monic irreducibles. Suppose $p_i = \prod_{\ell=1}^{m_i} a_{i,\ell}^{d_{i,\ell}}$. Then:

(i) $\mathrm{ElemDiv}_{\mathbb{K}}(A) = \{a_{i,\ell}^{d_{i,\ell} e_{i,j}} : 1 \leq i \leq s,\ 1 \leq j \leq k_i,\ 1 \leq \ell \leq m_i\}$ as multisets;

(ii) $\mathrm{InvFact}_{\mathbb{K}}(A) = \mathrm{InvFact}_{\mathbb{F}}(A)$ under the coefficient inclusion $\mathbb{F}[z] \subseteq \mathbb{K}[z]$.

Proof.

(i) Set $R_{\mathbb{F}} := \bigoplus_{i=1}^{s} \bigoplus_{j=1}^{k_i} C_{p_i^{e_{i,j}}}$. The similarities $A \sim_{\mathbb{F}} R_{\mathbb{F}}$ and $A \sim_{\mathbb{K}} R_{\mathbb{F}}$ hold by Theorem 3.4.5(v) and $\mathrm{GL}_n(\mathbb{F}) \subseteq \mathrm{GL}_n(\mathbb{K})$.
Fix (i, j). The first standard vector is cyclic for $C_{p_i^{e_{i,j}}}$ by Definition 3.1.12.
Its order over $\mathbb{K}[z]$ is $p_i^{e_{i,j}}$ by Theorem 3.1.13(i) and Proposition 3.1.7(i).
The factors $a_{i,\ell}^{d_{i,\ell} e_{i,j}}$ are pairwise coprime, so $C_{p_i^{e_{i,j}}} \sim_{\mathbb{K}} \bigoplus_{\ell=1}^{m_i} C_{a_{i,\ell}^{d_{i,\ell} e_{i,j}}}$ by Theorems 2.5.6(iii) and 3.1.14(iii).
Replacing every block defines an elementary-divisor form $R_{\mathbb{K}}$ with $A \sim_{\mathbb{K}} R_{\mathbb{K}}$ by the direct sum of the blockwise similarity matrices.
Each block of $R_{\mathbb{K}}$ is cyclic with its displayed order by Definition 3.1.12, Theorem 3.1.13(i), and Proposition 3.1.7(i).
Thus the elementary divisors of $R_{\mathbb{K}}$ are its block orders by Proposition 2.10.3 and Definition 3.2.1(iii).
The asserted multiset is therefore $\mathrm{ElemDiv}_{\mathbb{K}}(A)$ by Theorem 3.4.2(ii).

(ii) Write $\mathrm{InvFact}_{\mathbb{F}}(A) = (s_1, \dots, s_r)$. Then $A \sim_{\mathbb{F}} \mathrm{diag}(C_{s_1}, \dots, C_{s_r})$ by Theorem 3.4.4(ii).
The same similarity holds over $\mathbb{K}$, and the s_j remain monic and nonconstant with $s_r \mid \cdots \mid s_1$, by coefficient inclusion.
This block matrix is invariant-factor form over $\mathbb{K}$ by Definition 3.4.3(i); claimed tuple equality holds by Theorem 3.4.5(iv)–(v).

Structural Commentary: Coprime factorization refines each elementary block; the original divisibility-ordered form remains valid.
Key Insight: Scalar extension splits prime powers without changing invariant-factor blocks. □

Corollary 3.4.10. *Nonderogatory Means One Companion Block*

Let $n \geq 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:

(i) A is nonderogatory;

(ii) $(\mathbb{F}^n)_A$ is cyclic;

(iii) $A \sim_{\mathbb{F}} C_{\chi_A}$.

Proof.

(i) ⇒ *(ii)* The operator T_A is nonderogatory by condition (i) and Definition 3.3.5; $(\mathbb{F}^n)_A$ is cyclic by Theorem 3.3.6(iv),(i).

(ii) ⇒ *(iii)* $\exists\, s \in \mathbb{F}[z]$ with $\mathrm{InvFact}(A) = (s)$ by condition (ii), Theorem 2.10.12(i),(iv), Theorem 2.10.6(i)–(ii), and Definition 3.2.1(ii).
The identities $A \sim_{\mathbb{F}} C_s$ and $s = q_{T_A} = p_A = \chi_A$ hold by Theorem 3.2.2(ii)–(iv), Theorem 3.4.8(i), and Definition 2.1.15.

(iii) ⇒ *(i)* The identity $\mu_A = \mu_{C_{\chi_A}} = \chi_A$ holds by condition (iii), Theorem 3.1.13(i), and Proposition 3.1.7(iii).
Thus A is nonderogatory by Theorem 3.4.8(i) and Definitions 2.1.15 and 3.3.5.

Structural Commentary: Cyclic-module characterization reduces nonderogatoriness to one invariant-factor companion block.
Key Insight: A cyclic generator produces a single companion block. □

Corollary 3.4.11. *When the Elementary-Divisor Form Is Diagonal*
Let $0 < \dim \mathcal{V} < \infty$, *let* $T \in \mathcal{L}(\mathcal{V})$, *set* $Q := \operatorname{ElemDiv}(T)$, *and set* $R_{\mathrm{ed}} := \bigoplus_{q \in Q} C_q$. *The following are equivalent:*

(i) R_{ed} *is diagonal;*
(ii) $(\forall\, q \in Q)\ \deg q = 1$*;*
(iii) $(\exists\, \Lambda \subseteq \mathbb{F})\,[0 < |\Lambda| < \infty \wedge \mu_T = \prod_{\lambda \in \Lambda} (z - \lambda)]$.

Proof.

(i) $\Rightarrow$ *(ii)* Every companion block C_q is diagonal by condition (i).
Hence $(\forall\, q \in Q)\ \deg q = 1$ by the nonzero subdiagonal in Definition 3.1.12 whenever $\deg q \geq 2$.
(ii) $\Rightarrow$ *(iii)* Polynomial μ_T is least common multiple of monic linear polynomials in Q by condition (ii) and Theorem 3.3.4(i).
Hence condition (iii) holds for the finite nonempty set of their roots.
(iii) $\Rightarrow$ *(i)* The divisibility $(\forall\, q \in Q)\, q \mid \mu_T$ holds by condition (iii) and Theorem 3.3.4(i).
Since the elements of Q are prime powers and μ_T is square-free, $(\forall\, q \in Q)(\exists\, \lambda \in \Lambda)\, q = z - \lambda$.
Thus every C_q is a 1×1 diagonal matrix by Definition 3.1.12, so R_{ed} is diagonal.

Structural Commentary: Companion matrices of degree at least two have a nonzero subdiagonal; μ_T detects the remainder.
Key Insight: Diagonal companion blocks are exactly the linear elementary divisors. □

Example 3.4.12. *The Same Polynomial over Different Fields*
Set $\mu := (z^2 + 1)(z^2 - 2)$, $A := C_\mu \in \operatorname{Mat}(4, 4, \mathbb{Q})$. Then $\mu_A = p_A = \mu$ by Theorem 3.1.13(i)–(ii).
Hence A is nonderogatory by Definitions 2.1.15 and 3.3.5.
The matrix $A = C_\mu$ is an invariant-factor form by Definition 3.4.3(i), so $\operatorname{InvFact}_{\mathbb{Q}}(A) = (\mu)$ by Theorem 3.4.5(iv)–(v).
Thus the invariant-factor form over K is C_μ for $K \in \{\mathbb{Q}, \mathbb{R}, \mathbb{C}\}$ by Theorem 3.4.9(ii). The elementary divisors are:

K	$\operatorname{ElemDiv}_K(A)$
$\mathbb{Q}$	$\{z^2 + 1,\ z^2 - 2\}$
$\mathbb{R}$	$\{z^2 + 1,\ z - \sqrt{2},\ z + \sqrt{2}\}$
$\mathbb{C}$	$\{z - i,\ z + i,\ z - \sqrt{2},\ z + \sqrt{2}\}$

The elementary-divisor form over K is $\bigoplus_{q \in \operatorname{ElemDiv}_K(A)} C_q$.
Scalar extension refines these blocks while preserving C_μ; over $\mathbb{C}$, the elementary-divisor form is diagonal.

Perspective 3.4.13. *Crosswalk with the Spectral Presentation*
Let $0 < \dim_{\mathbb{C}} \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$. The module and spectral presentations encode the same data at different splitting levels:
(i) T is diagonalizable exactly when $(\forall\, q \in \operatorname{ElemDiv}(T))\ \deg q = 1$ by Theorem 3.2.1 and Corollary 3.4.11;
(ii) the elementary divisor $(z - \lambda)^k$ corresponds to the Jordan block $J_k(\lambda)$ by Theorem 6.2.1 and Perspective 3.3.7;
(iii) the rank drops of the $(z - \lambda)$-primary component recover the block exponents k by Proposition 2.9.6.
Key Insight: Rational form is Jordan form before irreducibles split.

Computation Method 3.4.14. *Moving Between Module and Spectral Data*
Let $\mathbb{F} \subseteq \mathbb{C}$, $0 < \dim_{\mathbb{F}} \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$. Choose basis $\mathcal{B}$, set $A := [T]_{\mathcal{B} \leftarrow \mathcal{B}}$, and let $\mathbb{K} \subseteq \mathbb{C}$ be the splitting field of μ_T over $\mathbb{F}$.
The outputs are a rational canonical form over $\mathbb{F}$ and the Jordan-block data of A over $\mathbb{K}$.

(i) Factor $\mu_T = \prod_{i=1}^{s} p_i^{e_i}$ into distinct monic irreducibles over $\mathbb{F}$, and set $\mathcal{V}_{p_i} := \ker p_i(T)^{e_i}$.
(ii) Decompose each $\mathcal{V}_{p_i}$ into cyclic summands of orders $p_i^{e_{i,j}}$ by Theorem 3.3.1(ii).
These orders form $\operatorname{ElemDiv}(T)$ by part (iii) of that theorem.
(iii) Assemble the blocks $C_{p_i^{e_{i,j}}}$ into an elementary-divisor rational canonical form over $\mathbb{F}$ by Theorem 3.4.5(i).
(iv) Factor the p_i over $\mathbb{K}$ and refine the elementary divisors into powers $(z - \lambda)^k$ by Theorem 3.4.9(i) for A.
In the reversed shifted-power basis of $\mathbb{K}[z]/\langle (z - \lambda)^k \rangle$, multiplication by z has matrix $J_k(\lambda)$.
Hence each refined cyclic block is similar over $\mathbb{K}$ to $J_k(\lambda)$ by Theorem 3.1.14(i),(iii).
(v) If T is diagonalizable over $\mathbb{F}$, a diagonal coordinate matrix is a direct sum of blocks $C_{z-\lambda}$.
Elementary divisors linear by Proposition 2.10.3, Definition 3.2.1(iii); $(\forall\, q \in \operatorname{ElemDiv}(T))\ \deg q = 1$ by Theorem 3.4.2(iii).
Conversely, if every elementary divisor has degree one, form in (iii) is diagonal by Corollary 3.4.11; T is diagonalizable over $\mathbb{F}$.

Key Insight: Rational and Jordan forms encode the same elementary-divisor data over different fields.

Figure 105 tracks a similarity class through its operator module, divisor data, and companion-block representative.

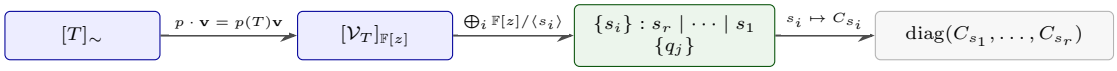

Figure 105: Similarity classes pass through operator modules and divisor data to companion-block representatives.

4 Associative Algebras

4.1 Associative Algebra Structures

The module viewpoint in Definition 3.1.1 encodes one operator at a time.
An algebra records a vector space of compatible products or operators.
The guiding example is $\mathcal{L}(\mathcal{V})$, whose vector-space operations are pointwise and whose multiplication is composition.

Definition 4.1.1. *Algebra over a Field*
Let $\mathbb{F}$ be a field. An *$\mathbb{F}$-algebra* is an $\mathbb{F}$-vector space $\mathcal{A}$ with multiplication $\mathcal{A} \times \mathcal{A} \to \mathcal{A}$, $(a, b) \mapsto ab$.
The structure $(\mathcal{A}, +, \cdot)$ is a ring. Scalar compatibility is $(\forall\, \lambda \in \mathbb{F})(\forall\, a, b \in \mathcal{A})\, \lambda(ab) = (\lambda a)b = a(\lambda b)$.

Definition 4.1.2. *Unital Algebras and Basic Properties*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra.

(i) $\mathcal{A}$ is a *unital associative $\mathbb{F}$-algebra* if its underlying ring is equipped with an identity $1_{\mathcal{A}}$.
(ii) $\mathcal{A}$ is *finite-dimensional* iff $\dim_{\mathbb{F}} \mathcal{A} < \infty$.
(iii) $\mathcal{A}$ is *commutative* iff $(\forall\, a, b \in \mathcal{A})\, ab = ba$.
(iv) If $\mathcal{A}$ is unital, then $a \in \mathcal{A}$ is *invertible* iff $(\exists\, b \in \mathcal{A})\, ab = ba = 1_{\mathcal{A}}$.

Convention 4.1.3. *Algebra Convention*
For a fixed field $\mathbb{F}$, *algebra* means unital associative $\mathbb{F}$-algebra unless stated otherwise.

Example 4.1.4. *Basic Algebras*

(i) If $\mathbb{F} \subseteq \mathbb{E}$ are fields, then $\mathbb{E}$ is an $\mathbb{F}$-algebra by $r \cdot x = rx$ and multiplication in $\mathbb{E}$.
(ii) $\mathbb{F}[x]$ is a commutative $\mathbb{F}$-algebra with unit 1 and scalar embedding $r \mapsto r$.
(iii) If $n \geq 1$, then $\mathrm{Mat}(n, n, \mathbb{F})$ is an $\mathbb{F}$-algebra with unit I_n and product AB.
(iv) If $\mathcal{V}$ is an $\mathbb{F}$-vector space, then $\mathcal{L}(\mathcal{V}) = \mathrm{End}_{\mathbb{F}}(\mathcal{V})$ is an $\mathbb{F}$-algebra. Its unit is $I_{\mathcal{V}}$, and its product is composition $S \circ T$.

Definition 4.1.5. *Subalgebras and Generated Subalgebras*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra.

(i) A *subalgebra*, written $\mathcal{B} \leq_{\mathbb{F}\text{-alg}} \mathcal{A}$, is an $\mathbb{F}$-linear subspace $\mathcal{B} \leq_{\mathbb{F}} \mathcal{A}$ containing $1_{\mathcal{A}}$. It satisfies $(\forall\, b, b' \in \mathcal{B})\, bb' \in \mathcal{B}$.
(ii) For $X \subseteq \mathcal{A}$, the *subalgebra generated by X* is $\langle X \rangle_{\mathrm{subalg}} := \bigcap \{\mathcal{B} \leq_{\mathbb{F}\text{-alg}} \mathcal{A} : X \subseteq \mathcal{B}\}$.

Definition 4.1.6. *Center and Central Algebras*

(i) The *center* of an $\mathbb{F}$-algebra $\mathcal{A}$ is $Z(\mathcal{A}) := \{a \in \mathcal{A} : (\forall\, x \in \mathcal{A})\, ax = xa\}$.
(ii) $\mathcal{A}$ is *central over $\mathbb{F}$* iff $Z(\mathcal{A}) = \mathbb{F}1_{\mathcal{A}} := \{r1_{\mathcal{A}} : r \in \mathbb{F}\}$.

Proposition 4.1.7. *Centers and Central Scalar Embeddings*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra.

(i) $Z(\mathcal{A}) \leq_{\mathbb{F}\text{-alg}} \mathcal{A}$.
(ii) If $1_{\mathcal{A}} \neq 0$, then $\iota_{\mathcal{A}} : \mathbb{F} \hookrightarrow Z(\mathcal{A})$, $r \mapsto r1_{\mathcal{A}}$, is an injective unital ring homomorphism.
(iii) If R is a unital ring and $\mathbb{F} \leq Z(R)$ is a subfield, then R is an $\mathbb{F}$-algebra via $r \cdot x := rx$.

Proof.

(i) Identity belongs to $Z(\mathcal{A})$. If $(r, s, a, b, x) \in \mathbb{F}^2 \times Z(\mathcal{A})^2 \times \mathcal{A}$, then $(ra + sb)x = r(ax) + s(bx) = r(xa) + s(xb) = x(ra + sb)$.
Thus $Z(\mathcal{A}) \leq_{\mathbb{F}} \mathcal{A}$. For $a, b \in Z(\mathcal{A})$ and $x \in \mathcal{A}$, $(ab)x = a(bx) = a(xb) = (ax)b = (xa)b = x(ab)$. Thus $Z(\mathcal{A}) \leq_{\mathbb{F}\text{-alg}} \mathcal{A}$.
(ii) Set $\iota := \iota_{\mathcal{A}}$. For $r, s \in \mathbb{F}$, $\iota(r + s) = \iota(r) + \iota(s)$, $\iota(rs) = \iota(r)\iota(s)$, and $\iota(1) = 1_{\mathcal{A}}$ by the algebra axioms.
For $(r, a) \in \mathbb{F} \times \mathcal{A}$, $\iota(r)a = (r1_{\mathcal{A}})a = ra = a(r1_{\mathcal{A}}) = a\iota(r)$, so $\operatorname{im} \iota \subseteq Z(\mathcal{A})$.
If $\iota(r) = 0$ and $r \neq 0$, then $1_{\mathcal{A}} = \iota(r^{-1})\iota(r) = 0$, contradiction; hence $\ker \iota = \{0\}$.
(iii) Set $r \cdot x := rx$. Then R is an $\mathbb{F}$-vector space by the field laws in $\mathbb{F} \subseteq R$.
Scalar compatibility $(\forall\, r \in \mathbb{F})(\forall\, x, y \in R)\, r \cdot (xy) = (r \cdot x)y = x(r \cdot y)$ holds by $\mathbb{F} \subseteq Z(R)$. Thus R is an $\mathbb{F}$-algebra.

Structural Commentary: Center closure supports the scalar embedding, while a central subfield reconstructs the scalar action.
Key Insight: Scalar multiplication is multiplication by the corresponding central element. □

Definition 4.1.8. *Ideals, Simple Algebras, and Quotients*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra.

(i) A *two-sided ideal*, written $\mathcal{I} \trianglelefteq \mathcal{A}$, is an $\mathbb{F}$-linear subspace $\mathcal{I} \leq_{\mathbb{F}} \mathcal{A}$. It satisfies $(\forall\, a \in \mathcal{A})(\forall\, u \in \mathcal{I})\, au, ua \in \mathcal{I}$.
(ii) For $X \subseteq \mathcal{A}$, the *ideal generated by X* is $\langle X \rangle_{\mathrm{ideal}} := \bigcap \{\mathcal{I} \trianglelefteq \mathcal{A} : X \subseteq \mathcal{I}\}$.
(iii) $\mathcal{A}$ is *simple* iff $\mathcal{A} \neq \{0_{\mathcal{A}}\}$ and its only two-sided ideals are $\{0_{\mathcal{A}}\}$ and $\mathcal{A}$.
(iv) If $\mathcal{I} \trianglelefteq \mathcal{A}$, *quotient algebra* $\mathcal{A}/\mathcal{I}$ is quotient vector space. Product and identity are $(a + \mathcal{I})(b + \mathcal{I}) := ab + \mathcal{I}$ and $1_{\mathcal{A}/\mathcal{I}} := 1_{\mathcal{A}} + \mathcal{I}$.

Proposition 4.1.9. *Concrete Generated Substructures*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and let $X \subseteq \mathcal{A}$. Set $\mathcal{W}_X := \{1_{\mathcal{A}}\} \cup \{x_1 \cdots x_m : m \in \mathbb{N},\ (x_1, \ldots, x_m) \in X^m\}$.
Set $\mathcal{I}_X := \{\sum_{j=1}^{n} a_j x_j b_j : n \in \mathbb{N}_0,\ (a_j, b_j, x_j)_{j=1}^{n} \in (\mathcal{A} \times \mathcal{A} \times X)^n\}$, with the empty sum at $n = 0$.

(i) $\langle X \rangle_{\mathrm{subalg}} = \mathrm{span}_{\mathbb{F}}(\mathcal{W}_X)$.
(ii) $\langle X \rangle_{\mathrm{ideal}} = \mathcal{I}_X$.

Proof.

(i) Products of members of $\mathcal{W}_X$ belong to $\mathcal{W}_X$, so $\mathrm{span}_{\mathbb{F}}(\mathcal{W}_X)$ is a subalgebra containing X by bilinearity.
If $\mathcal{B} \leq_{\mathbb{F}\text{-alg}} \mathcal{A}$ and $X \subseteq \mathcal{B}$, then $\mathcal{W}_X \subseteq \mathcal{B}$ and $\mathrm{span}_{\mathbb{F}}(\mathcal{W}_X) \subseteq \mathcal{B}$. The equality follows by Definition 4.1.5(ii).
(ii) $\mathcal{I}_X \leq_{\mathbb{F}} \mathcal{A}$ by closure of finite sums under $\mathbb{F}$-linear combinations, and $X \subseteq \mathcal{I}_X$ by $x = 1_{\mathcal{A}} x 1_{\mathcal{A}}$.
For $(a, b) \in \mathcal{A}^2$ and $u = \sum_{j=1}^{n} a_j x_j b_j \in \mathcal{I}_X$, both $au = \sum_{j=1}^{n} (aa_j) x_j b_j$ and $ub = \sum_{j=1}^{n} a_j x_j (b_j b)$ belong to $\mathcal{I}_X$.
Thus $\mathcal{I}_X \trianglelefteq \mathcal{A}$. If $\mathcal{J} \trianglelefteq \mathcal{A}$ and $X \subseteq \mathcal{J}$, then $(\forall\, a, b \in \mathcal{A})(\forall\, x \in X)\, axb \in \mathcal{J}$, so $\mathcal{I}_X \subseteq \mathcal{J}$. The equality follows by Definition 4.1.8(ii).

Structural Commentary: Closure of words constructs the generated subalgebra; two-sided products construct the generated ideal.
Key Insight: Intersections agree with the explicit closures forced by their defining operations. □

Definition 4.1.10. *Algebra Homomorphisms*
Let $\mathcal{A}, \mathcal{B}$ be $\mathbb{F}$-algebras.

(i) An *algebra homomorphism* is an $\mathbb{F}$-linear map $\sigma \in \mathcal{L}(\mathcal{A}, \mathcal{B})$. It satisfies $(\forall\, a, b \in \mathcal{A})\, \sigma(ab) = \sigma(a)\sigma(b)$ and $\sigma(1_{\mathcal{A}}) = 1_{\mathcal{B}}$.
(ii) An algebra homomorphism is a *monomorphism* iff it is injective.
(iii) An algebra homomorphism is an *isomorphism* iff it is bijective.

Proposition 4.1.11. *Kernel, Image, and First Isomorphism Theorem*
Let $\sigma : \mathcal{A} \to \mathcal{B}$ be an $\mathbb{F}$-algebra homomorphism and set $K := \ker \sigma$.

(i) $K \trianglelefteq \mathcal{A}$.

(ii) $\operatorname{im}\sigma \leq_{\mathbb{F}\text{-}alg} \mathcal{B}$.
(iii) *The map* $\overline{\sigma} : \mathcal{A}/K \to \operatorname{im}\sigma$, $a + K \mapsto \sigma(a)$, *is an* $\mathbb{F}$*-algebra isomorphism.*

Proof.
(i) $K \leq_{\mathbb{F}} \mathcal{A}$ by linearity. If $(a, u) \in \mathcal{A} \times K$, then $\sigma(au) = \sigma(a)\sigma(u) = 0 = \sigma(u)\sigma(a) = \sigma(ua)$, so $au, ua \in K$ and $K \triangleleft \mathcal{A}$.
(ii) The image is an $\mathbb{F}$-linear subspace by linearity and contains $1_{\mathcal{B}} = \sigma(1_{\mathcal{A}})$.
It is closed under multiplication by $(\forall\, a, b \in \mathcal{A})\, \sigma(a)\sigma(b) = \sigma(ab)$. Thus $\operatorname{im}\sigma \leq_{\mathbb{F}\text{-alg}} \mathcal{B}$.
(iii) Define $\overline{\sigma}(a + K) := \sigma(a)$. The equivalence $a + K = a' + K \iff a - a' \in K$ holds by Proposition 1.5.2(i).
The equivalence $a - a' \in K \iff \sigma(a) = \sigma(a')$ holds by $K = \ker\sigma$.
Thus $\overline{\sigma}$ is well defined and injective; it is surjective by its codomain.
For $(a, a', r) \in \mathcal{A}^2 \times \mathbb{F}$, $\overline{\sigma}((a + K) + (a' + K)) = \sigma(a) + \sigma(a')$ and $\overline{\sigma}(r(a + K)) = r\sigma(a)$.
Multiplicativity is $\overline{\sigma}((a + K)(a' + K)) = \sigma(a)\sigma(a')$.
The identity $\overline{\sigma}(1_{\mathcal{A}} + K) = 1_{\mathcal{B}}$ completes the verification that $\overline{\sigma}$ is an $\mathbb{F}$-algebra isomorphism.

Structural Commentary: Kernel absorption forms the quotient, while image closure supplies the induced map's algebra codomain.
Key Insight: Cosets agree exactly when their representatives have the same image. □

Definition 4.1.12. *Left Regular Representation*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra. For $a \in \mathcal{A}$, the *left multiplication map* is $L_a : \mathcal{A} \to \mathcal{A}$, $x \mapsto ax$.
The map $L_{\mathcal{A}} : \mathcal{A} \to \mathcal{L}(\mathcal{A})$, $a \mapsto L_a$, is the *left regular representation* of $\mathcal{A}$.

Theorem 4.1.13. *Left Regular Representation*
If $\mathcal{A}$ *is an* $\mathbb{F}$*-algebra, then* $L_{\mathcal{A}} : \mathcal{A} \hookrightarrow \mathcal{L}(\mathcal{A})$, $a \mapsto L_a$, *is an* $\mathbb{F}$*-algebra monomorphism.*

Proof. For $(a, x, y, r) \in \mathcal{A}^3 \times \mathbb{F}$, $L_a(x + y) = L_a(x) + L_a(y)$ and $L_a(rx) = rL_a(x)$, so $L_a \in \mathcal{L}(\mathcal{A})$.
For $(a, b, r, x) \in \mathcal{A}^3 \times \mathbb{F}$, $L_{a+b}(x) = L_a(x) + L_b(x)$ and $L_{ra}(x) = rL_a(x)$. Hence $L_{\mathcal{A}}$ is $\mathbb{F}$-linear by Assumption 3.1.2.
For $a, b, x \in \mathcal{A}$, $L_{ab}(x) = a(bx) = (L_aL_b)(x)$ and $L_{1_{\mathcal{A}}}(x) = x$. Hence $L_{ab} = L_aL_b$ and $L_{1_{\mathcal{A}}} = I_{\mathcal{A}}$ by Assumption 3.1.2.
If $L_a = 0$, then $a = L_a(1_{\mathcal{A}}) = 0$, so $L_{\mathcal{A}}$ is injective.
Structural Commentary: Bilinearity defines the algebra map, while unit evaluation proves injectivity.
Key Insight: The identity recovers each element by $L_a(1_{\mathcal{A}}) = a$. □

Figure 106 records the multiplicativity and unit-evaluation identities used in the proof.

Figure 106: The left regular representation sends $a \in \mathcal{A}$ to $L_a : x \mapsto ax$; the identities $L_{ab} = L_aL_b$ and $L_a(1_{\mathcal{A}}) = a$ encode multiplicativity and injectivity.

Computation Method 4.1.14. *Building Multiplication from Structure Constants*
Input: A vector space $V = \bigoplus_{i \in I} \mathbb{F}b_i$ and coefficients $c_{ij}^k \in \mathbb{F}$. The support condition is $(\forall\, i, j \in I)\ \left|\{k \in I : c_{ij}^k \neq 0\}\right| < \infty$.
Output: A bilinear multiplication on V, together with coordinate tests for associativity, commutativity, and a basis identity.
(i) Define $b_ib_j := \sum_{k \in I} c_{ij}^k b_k$. Extend bilinearly by $(\sum_i r_ib_i)(\sum_j s_jb_j) := \sum_{i,j,k} r_is_jc_{ij}^k b_k$.
(ii) The product is associative iff $(\forall\, i, j, k, m \in I)\ \sum_{\ell \in I} c_{ij}^\ell c_{\ell k}^m = \sum_{\ell \in I} c_{jk}^\ell c_{i\ell}^m$.
(iii) The product is commutative iff $(\forall\, i, j, k \in I)\, c_{ij}^k = c_{ji}^k$.
(iv) A basis vector b_e is an identity iff $(\forall\, j, m \in I)\, c_{ej}^m = \delta_j^m = c_{je}^m$.

Definition 4.1.15. *Group Algebra*
For a group G and field $\mathbb{F}$, the *group algebra* is $\mathbb{F}[G] := \{\sum_{g \in G} r_g g : r_g \in \mathbb{F},\ |\{g \in G : r_g \neq 0\}| < \infty\}$.
Its vector-space operations are coefficientwise, and $(\sum_g r_g g)(\sum_h s_h h) := \sum_{g,h} r_gs_h(gh)$.

Proposition 4.1.16. *Structure of a Group Algebra*
Let G *be a group and let* $\mathbb{F}$ *be a field.*
(i) G *is an* $\mathbb{F}$*-basis of* $\mathbb{F}[G]$.
(ii) With the multiplication in Definition 4.1.15, $\mathbb{F}[G]$ *is an* $\mathbb{F}$*-algebra with identity* e_G.
(iii) $\mathbb{F}[G]$ *is commutative iff* G *is abelian.*
(iv) If G *is finite, then* $\dim_{\mathbb{F}} \mathbb{F}[G] = |G|$.

Proof.
(i) G is an $\mathbb{F}$-basis by the unique finite-support coefficient expansions in Definition 4.1.15.
(ii) The product is bilinear by construction. The two associative products are trilinear and agree on basis triples by $(gh)k = g(hk)$.
The identity relation follows by bilinearity and $(\forall\, g \in G)\, e_Gg = g = ge_G$.
(iii) $(\Rightarrow)$ If $\mathbb{F}[G]$ is commutative, then $(\forall\, g, h \in G)\, gh = hg$, so G is abelian.
(iii) $(\Leftarrow)$ If G is abelian, then $\mathbb{F}[G]$ is commutative by bilinearity and $(\forall\, g, h \in G)\, gh = hg$.
(iv) The dimension equals the cardinality of the basis G by part (i).

Structural Commentary: Basis multiplication transfers the group law to the algebra, while basis equality detects commutativity.
Key Insight: The group elements remain linearly independent inside $\mathbb{F}[G]$. □

Example 4.1.17. *Cyclic Group Algebras and Circulant Shifts*
Let $G = \langle\alpha\rangle$ have order n, and let $\mathcal{B} = (1, \alpha, \ldots, \alpha^{n-1})$ be the ordered $\mathbb{F}$-basis of $\mathbb{F}[G]$.

With exponents read modulo n, $(\forall\, j,k \in \{0,\dots,n-1\})\, L_{\alpha^k}(\alpha^j) = \alpha^{j+k}$, so L_{α^k} is the k-fold cyclic shift of $\mathcal{B}$.
Set $P := [L_\alpha]_\mathcal{B}$. Then $(\forall\, k \in \{0,\dots,n-1\})\, [L_{\alpha^k}]_\mathcal{B} = P^k$. For $n \geq 5$, the following coordinate form displays the cyclic shift.

$$P = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \\ & & \ddots & & \\ 0 & 1 & & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$$

Thus $[L_{\sum_{k=0}^{n-1} c_k \alpha^k}]_\mathcal{B} = \sum_{k=0}^{n-1} c_k P^k$, and these are precisely the circulant matrices.

Proposition 4.1.18. *Endomorphisms and Matrices Are the Same Algebra after Choosing a Basis*
Let $\dim \mathcal{V} = n \geq 1$ and fix an ordered basis $\mathcal{B}$ of $\mathcal{V}$. Then $\mu_\mathcal{B} : \mathcal{L}(\mathcal{V}) \to \mathrm{Mat}(n,n,\mathbb{F})$, $T \mapsto [T]_\mathcal{B}$, is an $\mathbb{F}$-algebra isomorphism.

Proof. By Theorem 3.4.5, $\mu_\mathcal{B}$ is an $\mathbb{F}$-linear bijection. For $S,T \in \mathcal{L}(\mathcal{V})$, $\mu_\mathcal{B}(ST) = \mu_\mathcal{B}(S)\mu_\mathcal{B}(T)$ by Proposition 3.4.4.
Since $\mu_\mathcal{B}(I_\mathcal{V}) = I_n$, $\mu_\mathcal{B}$ is an $\mathbb{F}$-algebra isomorphism.
Structural Commentary: Linear coordinate encoding becomes algebra isomorphism by composition compatibility, identity preservation.
Key Insight: A fixed basis converts composition exactly into matrix multiplication. □

Theorem 4.1.19. *Center and Ideals of an Endomorphism Algebra*
Assume Zorn's lemma, and let $0 \neq \mathcal{V}$ be an $\mathbb{F}$-vector space. Set $\mathcal{F}(\mathcal{V}) := \{T \in \mathcal{L}(\mathcal{V}) : \dim \operatorname{im} T < \infty\}$.

(i) $Z(\mathcal{L}(\mathcal{V})) = \mathbb{F}I_\mathcal{V}$, *so* $\mathcal{L}(\mathcal{V})$ *is central.*
(ii) $\mathcal{F}(\mathcal{V}) \lhd \mathcal{L}(\mathcal{V})$.
(iii) $(\forall\, \mathcal{J} \lhd \mathcal{L}(\mathcal{V}))\, [\mathcal{J} \neq \{0\} \Rightarrow \mathcal{F}(\mathcal{V}) \subseteq \mathcal{J}]$.
(iv) $\mathcal{L}(\mathcal{V})$ *is simple iff* $\dim \mathcal{V} < \infty$.

Proof.

(*i*) For $(v,\varphi) \in \mathcal{V} \times \mathcal{V}'$, write $v \otimes \varphi \in \mathcal{L}(\mathcal{V})$ for $x \mapsto \varphi(x)v$. Fix $T \in Z(\mathcal{L}(\mathcal{V}))$.
For $0 \neq v \in \mathcal{V}$, choose $\varphi \in \mathcal{V}'$ with $\varphi(v) = 1$ by Proposition 4.1.7(iii), with $\{0\} \leq \mathcal{V}$.
Set $\lambda_v := \varphi(Tv)$. Then $Tv = T(v \otimes \varphi)(v) = (v \otimes \varphi)T(v) = \lambda_v v$. Fix $0 \neq v_0 \in \mathcal{V}$. If $v = cv_0 \neq 0$, then $\lambda_v = \lambda_{v_0}$ by linearity.
If v, v_0 are independent, then $\lambda_{v+v_0}(v+v_0) = \lambda_v v + \lambda_{v_0} v_0$, so $\lambda_v = \lambda_{v_0}$ by independence. Hence $T = \lambda_{v_0} I_\mathcal{V}$ by Assumption 3.1.2.
Since scalar operators commute with every endomorphism, $Z(\mathcal{L}(\mathcal{V})) = \mathbb{F}I_\mathcal{V}$.

(*ii*) The zero map belongs to $\mathcal{F}(\mathcal{V})$. For $(r,s,T,U) \in \mathbb{F}^2 \times \mathcal{F}(\mathcal{V})^2$, $\operatorname{im}(rT + sU) \subseteq \operatorname{im} T + \operatorname{im} U$.
Thus $\mathcal{F}(\mathcal{V}) \leq_\mathbb{F} \mathcal{L}(\mathcal{V})$. For $R,S \in \mathcal{L}(\mathcal{V})$ and $T \in \mathcal{F}(\mathcal{V})$, $\operatorname{im}(RTS) \subseteq R(\operatorname{im} T)$; $\mathcal{F}(\mathcal{V}) \lhd \mathcal{L}(\mathcal{V})$.

(*iii*) Let $\{0\} \neq \mathcal{J} \lhd \mathcal{L}(\mathcal{V})$. Choose $0 \neq S \in \mathcal{J}$, $e \in \mathcal{V}$ with $Se \neq 0$. Choose $\eta \in \mathcal{V}'$ with $\eta(Se) = 1$ by Proposition 4.1.7(iii), $\{0\} \leq \mathcal{V}$.
For $(v,\varphi) \in \mathcal{V} \times \mathcal{V}'$, $v \otimes \varphi = (v \otimes \eta)S(e \otimes \varphi) \in \mathcal{J}$, so $\mathcal{J}$ contains every rank-one map.
If $T \in \mathcal{F}(\mathcal{V})$ and $(w_1,\dots,w_m)$ is a basis of $\operatorname{im} T$, let ψ_j be its coordinate functionals.
For $x \in \mathcal{V}$, $T(x) = \sum_{j=1}^m \psi_j(Tx) w_j$. The identity $T = \sum_{j=1}^m w_j \otimes (\psi_j \circ T)$ holds by Assumption 3.1.2.
Each summand belongs to $\mathcal{J}$, so $T \in \mathcal{J}$ and $\mathcal{F}(\mathcal{V}) \subseteq \mathcal{J}$.

(*iv*) ($\Rightarrow$) Suppose $\mathcal{L}(\mathcal{V})$ is simple, $\dim \mathcal{V} = \infty$. Choose $0 \neq v \in \mathcal{V}$, $\varphi \in \mathcal{V}'$ with $\varphi(v) = 1$ by Proposition 4.1.7(iii), $\{0\} \leq \mathcal{V}$.
Then $0 \neq v \otimes \varphi \in \mathcal{F}(\mathcal{V})$, whereas $I_\mathcal{V} \notin \mathcal{F}(\mathcal{V})$; $\mathcal{F}(\mathcal{V})$ is nonzero proper ideal by part (ii), contradiction. Hence $\dim \mathcal{V} < \infty$.

(*iv*) ($\Leftarrow$) If $\dim \mathcal{V} < \infty$, then $\mathcal{F}(\mathcal{V}) = \mathcal{L}(\mathcal{V})$, so every nonzero two-sided ideal equals $\mathcal{L}(\mathcal{V})$ by part (iii).
Since $I_\mathcal{V} \neq 0$, the algebra $\mathcal{L}(\mathcal{V})$ is nonzero and therefore simple.

Structural Commentary: Rank-one probes force scalar central maps, while rank-one factorizations control every nonzero ideal.
Key Insight: Sandwiching one nonzero operator produces every rank-one operator. □

Perspective 4.1.20. *Operator Connection*
$T \in \mathcal{L}(\mathcal{V})$ determines an $\mathbb{F}[z]$-module by $z \cdot v = T(v)$.
The left regular map $L_\mathcal{A} : \mathcal{A} \hookrightarrow \mathcal{L}(\mathcal{A})$ identifies $\mathcal{A}$ with a subalgebra of its endomorphism algebra and supplies a faithful representation.

4.2 Algebraic Elements and Minimal Polynomials

Definition 4.2.1. *Evaluation and Algebraic Elements*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and let $a \in \mathcal{A}$.

(i) For $p(x) = \sum_{j=0}^n c_j x^j \in \mathbb{F}[x]$, set $p(a) := \sum_{j=0}^n c_j a^j$, where $a^0 := 1_\mathcal{A}$. The *evaluation map at a* is $\mathrm{ev}_a : \mathbb{F}[x] \to \mathcal{A}$, $p \mapsto p(a)$.

(ii) The element a is *algebraic over $\mathbb{F}$* iff $(\exists\, p \in \mathbb{F}[x] \setminus \{0\})\, p(a) = 0$. Equivalently, a is algebraic over $\mathbb{F}$ iff $\ker \mathrm{ev}_a \neq \{0\}$.

Proposition 4.2.2. *Evaluation Is an Algebra Homomorphism*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and let $a \in \mathcal{A}$.

(i) The map $\mathrm{ev}_a : \mathbb{F}[x] \to \mathcal{A}$ is an $\mathbb{F}$-algebra homomorphism.
(ii) If a is algebraic over $\mathbb{F}$, then $\ker \mathrm{ev}_a \lhd \mathbb{F}[x]$ is nonzero.

Proof.

(*i*) Let $p(x) = \sum_{i=0}^m c_i x^i$ and $q(x) = \sum_{j=0}^n d_j x^j$. Bilinearity and scalar centrality imply $(pq)(a) = \sum_{i=0}^m \sum_{j=0}^n c_i d_j a^{i+j} = p(a)q(a)$.
For $r \in \mathbb{F}$, the evaluation definition implies $(p+q)(a) = p(a) + q(a)$ and $(rp)(a) = rp(a)$.
Since $\mathrm{ev}_a(1) = 1_\mathcal{A}$, ev_a is an $\mathbb{F}$-algebra homomorphism by Definition 4.1.10(i).

(*ii*) If a is algebraic, then $\ker \mathrm{ev}_a \neq \{0\}$ by Definition 4.2.1(ii). $\ker \mathrm{ev}_a \lhd \mathbb{F}[x]$ by Proposition 4.1.11(i).

Structural Commentary: Multiplication verifies evaluation; its kernel is an ideal by kernel absorption.
Key Insight: Powers of one element commute, so evaluation preserves polynomial products in any algebra. □

Definition 4.2.3. *Minimal Polynomial of an Algebraic Element*
Let $\mathcal{A}$ be $\mathbb{F}$-algebra, $a \in \mathcal{A}$ algebraic. Ideal $\ker \mathrm{ev}_a$ nonzero by Proposition 4.2.2(ii). Has unique monic generator by Theorem 3.10.2(iv).

(i) The *minimal polynomial of a* is this generator $m_a \in \mathbb{F}[x]$.
Equivalently, m_a is the unique monic polynomial of least degree satisfying $m_a(a) = 0$.
(ii) The *polynomial annihilator of a* is $\mathrm{ann}_{\mathbb{F}[x]}(a) := \{f \in \mathbb{F}[x] : f(a) = 0\} = \ker \mathrm{ev}_a = \langle m_a \rangle$.

Theorem 4.2.4. *Finite-Dimensional Algebras Are Algebraic*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra with $\dim_{\mathbb{F}} \mathcal{A} < \infty$. Then $(\forall\, a \in \mathcal{A})\, a$ is algebraic over $\mathbb{F}$.

Proof. Let $n := \dim_{\mathbb{F}} \mathcal{A}$ and let $a \in \mathcal{A}$. The $n+1$ vectors $1_{\mathcal{A}}, a, \ldots, a^n$ are linearly dependent.
Thus $(\exists\, (c_0, \ldots, c_n) \in \mathbb{F}^{n+1} \setminus \{0\})\ \sum_{j=0}^{n} c_j a^j = 0$; $p(x) := \sum_{j=0}^{n} c_j x^j$ is nonzero, satisfies $p(a) = 0$; a algebraic by Definition 4.2.1(ii).
Structural Commentary: Linear dependence among initial powers constructs an annihilating polynomial for arbitrary a.
Key Insight: Any list of more than $\dim_{\mathbb{F}} \mathcal{A}$ powers must contain a nontrivial linear relation. □

Example 4.2.5. *An Algebraic Infinite-Dimensional Algebra*
Let $\overline{\mathbb{Q}}_{\mathbb{C}} := \{z \in \mathbb{C} : (\exists\, p \in \mathbb{Q}[x] \setminus \{0\})\, p(z) = 0\}$. For $\alpha, \beta \in \overline{\mathbb{Q}}_{\mathbb{C}}$, choose nonzero $f, g \in \mathbb{Q}[x]$ with $f(\alpha) = g(\beta) = 0$.
Set $m := \deg f$ and $n := \deg g$. Reduction by $f(\alpha) = g(\beta) = 0$ proves $\mathbb{Q}[\alpha, \beta] = \mathrm{span}_{\mathbb{Q}}\{\alpha^i \beta^j : 0 \le i < m,\ 0 \le j < n\}$.
For $r \in \mathbb{Q}$, $\alpha + \beta, \alpha\beta, r\alpha \in \overline{\mathbb{Q}}_{\mathbb{C}}$ by Theorem 4.2.4. Since $1 \in \overline{\mathbb{Q}}_{\mathbb{C}}$, this set is a $\mathbb{Q}$-algebra.
For $n \in \mathbb{N}$, the polynomial $x^n - 2$ is Eisenstein at 2 and hence irreducible over $\mathbb{Q}$.
Any root $\alpha_n \in \mathbb{C}$ has $[\mathbb{Q}(\alpha_n) : \mathbb{Q}] = n$ and belongs to $\overline{\mathbb{Q}}_{\mathbb{C}}$. Thus $(\forall\, n \in \mathbb{N})\ \dim_{\mathbb{Q}} \overline{\mathbb{Q}}_{\mathbb{C}} \ge n$.
Hence this algebra is infinite-dimensional although each of its elements is algebraic.

Proposition 4.2.6. *The Algebra Generated by One Algebraic Element*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra, let $a \in \mathcal{A}$ be algebraic, and set $d := \deg m_a$ and $\mathbb{F}[a] := \langle \{a\} \rangle_{\mathrm{subalg}}$.

(i) $\mathbb{F}[a] = \mathrm{im}\, \mathrm{ev}_a = \{p(a) : p \in \mathbb{F}[x]\}$.
(ii) $(\forall\, b \in \mathbb{F}[a])(\exists!\, r \in \mathbb{F}[x])\, [b = r(a) \wedge (r = 0 \vee \deg r < d)]$.
(iii) Set $\overline{\mathrm{ev}}_a : \mathbb{F}[x]/\langle m_a \rangle \to \mathbb{F}[a]$, $p + \langle m_a \rangle \mapsto p(a)$. Then $\overline{\mathrm{ev}}_a$ is an $\mathbb{F}$-algebra isomorphism.

Proof.

(*i*) The identity $\mathbb{F}[a] = \mathrm{span}_{\mathbb{F}}\{1_{\mathcal{A}}, a, a^2, \ldots\}$ holds by Proposition 4.1.9(i) with $X = \{a\}$.
This span equals $\{p(a) : p \in \mathbb{F}[x]\} = \mathrm{im}\, \mathrm{ev}_a$ by Definition 4.2.1(i).
(*ii*) For $p \in \mathbb{F}[x]$, choose $q, r \in \mathbb{F}[x]$ by Theorem 3.8.2(i) with $p = qm_a + r$ and $r = 0$ or $\deg r < d$.
Since $m_a(a) = 0$, $p(a) = r(a)$, so the reduced representation exists.
Suppose r, s satisfy the degree condition and $r(a) = s(a)$. Then $r - s \in \ker \mathrm{ev}_a = \langle m_a \rangle$.
If $r \ne s$, then $\deg(r - s) < d$, contradicting $\deg(hm_a) \ge d$ for $0 \ne h \in \mathbb{F}[x]$. Thus $r = s$.
(*iii*) $\ker \mathrm{ev}_a = \langle m_a \rangle$ by Definition 4.2.3(ii), and $\mathrm{im}\, \mathrm{ev}_a = \mathbb{F}[a]$ by (i). Asserted induced isomorphism holds by Proposition 4.1.11(iii).

Structural Commentary: Generated powers identify the image; division selects representatives; the kernel excludes duplicates.
Key Insight: Reduction modulo m_a converts polynomial division into unique normal forms for elements of $\mathbb{F}[a]$. □

Theorem 4.2.7. *Invertibility from the Constant Term*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra, $a \in \mathcal{A}$ be algebraic, m_a be its minimal polynomial. Then $a \in \mathcal{A}^\times \iff m_a(0) \ne 0$.

Proof.

(⇒) Suppose $a \in \mathcal{A}^\times$ and $m_a(0) = 0$. Then $m_a = xq$ for a nonzero $q \in \mathbb{F}[x]$ with $\deg q < \deg m_a$.
Equality $0 = m_a(a) = aq(a)$ implies $q(a) = 0$ by multiplication by a^{-1}, contradicting minimality of $\deg m_a$, so $m_a(0) \ne 0$.
(⇐) Suppose $m_a(0) \ne 0$, $m_a(x) = xq(x) + m_a(0)$. Then $0 = m_a(a) = aq(a) + m_a(0)1_{\mathcal{A}}$, so $b := -m_a(0)^{-1} q(a)$ satisfies $ab = 1_{\mathcal{A}}$.
Since $q(a) \in \mathbb{F}[a]$, $aq(a) = q(a)a$; hence $ba = ab = 1_{\mathcal{A}}$ and $a \in \mathcal{A}^\times$.

Structural Commentary: Factoring handles a zero constant term; separating a nonzero constant constructs the inverse.
Key Insight: A nonzero constant term constructs a^{-1}; a zero term lowers the annihilator degree. □

4.3 Zero Divisors and Spectra of Algebraic Elements

Definition 4.3.1. *Zero Divisors*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and let $0 \ne a \in \mathcal{A}$.

(i) a is a *left zero divisor* iff $(\exists\, b \in \mathcal{A} \setminus \{0\})\, ab = 0$.
(ii) a is a *right zero divisor* iff $(\exists\, c \in \mathcal{A} \setminus \{0\})\, ca = 0$.

Theorem 4.3.2. *Invertibility and Zero Divisors*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and let $0 \ne a \in \mathcal{A}$ be algebraic. The following conditions are equivalent.

(i) $a \in \mathcal{A}^\times$.
(ii) a is not a left zero divisor.
(iii) a is not a right zero divisor.

Proof.

(*i*) ⇒ (*ii*) Suppose $a \in \mathcal{A}^\times$. If $ab = 0$, then $b = a^{-1}ab = 0$, so a is not a left zero divisor.
(*ii*) ⇒ (*iii*) Suppose a is not a left zero divisor. If $a \notin \mathcal{A}^\times$, then $m_a(0) = 0$ by Theorem 4.2.7.
Write $m_a = xp$ with $0 \ne p \in \mathbb{F}[x]$, $\deg p < \deg m_a$. Then $p(a) \ne 0$ by minimality, while $0 = m_a(a) = ap(a) = p(a)a$, contradiction.
Hence $a \in \mathcal{A}^\times$. If $ca = 0$, then $c = caa^{-1} = 0$, so a is not a right zero divisor.
(*iii*) ⇒ (*i*) Suppose a is not a right zero divisor. If $a \notin \mathcal{A}^\times$, then $m_a(0) = 0$ by Theorem 4.2.7.
Write $m_a = xp$ with $0 \ne p \in \mathbb{F}[x]$ and $\deg p < \deg m_a$.
Then $p(a) \ne 0$ by minimality, while $0 = m_a(a) = ap(a) = p(a)a$, a contradiction. Hence $a \in \mathcal{A}^\times$.

Structural Commentary: The implication cycle combines unit cancellation with minimal-polynomial factorization.
Key Insight: A zero constant term creates one nonzero polynomial witness annihilated by a on both sides. □

Corollary 4.3.3. *One-Sided Inverses Become Two-Sided*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and let $a \in \mathcal{A}$ be algebraic. If $(\exists\, b \in \mathcal{A})\,[ba = 1_{\mathcal{A}} \vee ab = 1_{\mathcal{A}}]$, then $a \in \mathcal{A}^{\times}$.

Proof. If $1_{\mathcal{A}} = 0$, then $\mathcal{A} = \{0\}$ and $a = 1_{\mathcal{A}} \in \mathcal{A}^{\times}$. Assume $1_{\mathcal{A}} \neq 0$ and choose $b \in \mathcal{A}$ with $ba = 1_{\mathcal{A}}$ or $ab = 1_{\mathcal{A}}$.
If $ba = 1_{\mathcal{A}}$, then $a \neq 0$, and $ac = 0$ implies $c = bac = 0$. Thus a is not a left zero divisor, so $a \in \mathcal{A}^{\times}$ by Theorem 4.3.2.
If $ab = 1_{\mathcal{A}}$, then $a \neq 0$, and $ca = 0$ implies $c = cab = 0$. Thus a is not a right zero divisor, so $a \in \mathcal{A}^{\times}$ by Theorem 4.3.2.
Structural Commentary: The two cases convert a left or right inverse into the matching cancellation law.
Key Insight: Algebraicity upgrades either one-sided cancellation law to two-sided invertibility. □

Definition 4.3.4. *Spectrum*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and let $a \in \mathcal{A}$ be algebraic. The *spectrum of a* is $\operatorname{spec}_{\mathcal{A}}(a) := \{r \in \mathbb{F} : m_a(r) = 0\}$.
An element of $\operatorname{spec}_{\mathcal{A}}(a)$ is an *algebraic eigenvalue of a*.

Theorem 4.3.5. *Spectrum Detected by Invertibility*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra, let $a \in \mathcal{A}$ be algebraic, and let $r \in \mathbb{F}$.
(i) $r \in \operatorname{spec}_{\mathcal{A}}(a) \iff a - r1_{\mathcal{A}} \notin \mathcal{A}^{\times}$.
(ii) $a \in \mathcal{A}^{\times} \iff 0 \notin \operatorname{spec}_{\mathcal{A}}(a)$.

Proof. Choose $q \in \mathbb{F}[x]$ and $c \in \mathbb{F}$ by Theorem 3.8.2(i) with $m_a = (x - r)q + c$. Then $c = m_a(r)$ by evaluation at r.
(i) ($\Rightarrow$) Suppose $r \in \operatorname{spec}_{\mathcal{A}}(a)$. Then $c = m_a(r) = 0$ by Definition 4.3.4; $m_a = (x - r)q$, where $q \neq 0$ and $\deg q < \deg m_a$.
The relation $0 = m_a(a) = (a - r1_{\mathcal{A}})q(a) = q(a)(a - r1_{\mathcal{A}})$ holds, and $q(a) \neq 0$ by minimality. Hence $a - r1_{\mathcal{A}} \notin \mathcal{A}^{\times}$.
(i) ($\Leftarrow$) Suppose $a - r1_{\mathcal{A}} \notin \mathcal{A}^{\times}$ and $r \notin \operatorname{spec}_{\mathcal{A}}(a)$. Then $c = m_a(r) \neq 0$ by Definition 4.3.4. Set $v := -c^{-1}q(a)$.
Since $m_a(a) = 0$ and $q(a)$ commutes with $a - r1_{\mathcal{A}}$, $v(a - r1_{\mathcal{A}}) = 1_{\mathcal{A}} = (a - r1_{\mathcal{A}})v$. This contradicts $a - r1_{\mathcal{A}} \notin \mathcal{A}^{\times}$, so $r \in \operatorname{spec}_{\mathcal{A}}(a)$.
(ii) ($\Rightarrow$) Suppose $a \in \mathcal{A}^{\times}$. Then $0 \notin \operatorname{spec}_{\mathcal{A}}(a)$ by (i) with $r = 0$.
(ii) ($\Leftarrow$) Suppose $0 \notin \operatorname{spec}_{\mathcal{A}}(a)$. Then $a \in \mathcal{A}^{\times}$ by (i) with $r = 0$.
Structural Commentary: Division isolates $m_a(r)$; zero supplies an annihilator, while nonzero constructs an inverse.
Key Insight: The scalar remainder is exactly the obstruction to invertibility of $a - r1_{\mathcal{A}}$. □

Theorem 4.3.6. *Polynomial Spectral Mapping for Algebraic Elements*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and suppose $(\forall\, f \in \mathbb{F}[x])\,[\deg f \geq 1 \Rightarrow f$ splits over $\mathbb{F}]$. Let $a \in \mathcal{A}$ be algebraic and let $p \in \mathbb{F}[x]$.
(i) $p(a)$ *is algebraic over* $\mathbb{F}$.
(ii) $\operatorname{spec}_{\mathcal{A}}(p(a)) = p(\operatorname{spec}_{\mathcal{A}}(a)) := \{p(r) : r \in \operatorname{spec}_{\mathcal{A}}(a)\}$.

Proof.
(i) Set $d := \deg m_a$. Then $p(a) \in \mathbb{F}[a]$ and $\dim_{\mathbb{F}} \mathbb{F}[a] \leq d$ by Proposition 4.2.6(i)–(ii).
Hence $p(a)$ is algebraic by Theorem 4.2.4 for the $\mathbb{F}$-algebra $\mathbb{F}[a]$.
(ii)
(a) *Degenerate Cases*: If $1_{\mathcal{A}} = 0$, then $\mathcal{A} = \{0\}$ and $\ker \operatorname{ev}_a = \mathbb{F}[x] = \ker \operatorname{ev}_{p(a)}$. Thus $m_a = m_{p(a)} = 1$ by Definition 4.2.3(i).
Thus both spectra are empty, and the equality holds. Assume $1_{\mathcal{A}} \neq 0$.
If $p = c \in \mathbb{F}$, then m_a is nonconstant as $m_a(a) = 0$, $1_{\mathcal{A}} \neq 0$. It has a root in $\mathbb{F}$ by splitting hypothesis, so $\operatorname{spec}_{\mathcal{A}}(a) \neq \emptyset$.
If $\lambda \neq c$, then $(c - \lambda)^{-1}1_{\mathcal{A}}$ is the inverse of $(c - \lambda)1_{\mathcal{A}}$; if $\lambda = c$, then 0 is noninvertible because $1_{\mathcal{A}} \neq 0$.
Thus $\operatorname{spec}_{\mathcal{A}}(p(a)) = \{c\} = p(\operatorname{spec}_{\mathcal{A}}(a))$ by Theorem 4.3.5(i).
(b) *Forward Inclusion*: Suppose p is nonconstant, $\lambda \in \operatorname{spec}_{\mathcal{A}}(p(a))$. Then $p(a) - \lambda 1_{\mathcal{A}}$ noninvertible by Theorem 4.3.5(i).
Choose $N \in \mathbb{N}$, $c \in \mathbb{F}^{\times}$, and $r_1, \ldots, r_N \in \mathbb{F}$ by the splitting hypothesis with $p(x) - \lambda = c \prod_{j=1}^{N} (x - r_j)$.
If $(\forall\, j \in \{1, \ldots, N\})\, a - r_j 1_{\mathcal{A}} \in \mathcal{A}^{\times}$, then $p(a) - \lambda 1_{\mathcal{A}}$ is invertible. Hence $(\exists\, j \in \{1, \ldots, N\})\, a - r_j 1_{\mathcal{A}} \notin \mathcal{A}^{\times}$.
For such j, $r_j \in \operatorname{spec}_{\mathcal{A}}(a)$ by Theorem 4.3.5(i), and $p(r_j) = \lambda$. Therefore $\operatorname{spec}_{\mathcal{A}}(p(a)) \subseteq p(\operatorname{spec}_{\mathcal{A}}(a))$.
(c) *Reverse Inclusion*: Suppose p is nonconstant and let $r \in \operatorname{spec}_{\mathcal{A}}(a)$.
Choose $q \in \mathbb{F}[x]$ and $s \in \mathbb{F}$ by Theorem 3.8.2(i) with $p(x) - p(r) = (x - r)q(x) + s$.
$s = 0$ by evaluation at r, so $p(x) - p(r) = (x - r)q(x)$. The element $a - r1_{\mathcal{A}}$ is algebraic by (i) with the polynomial $x - r$.
If $p(a) - p(r)1_{\mathcal{A}} = (a - r1_{\mathcal{A}})q(a)$ were invertible with inverse w, then $q(a)w$ would be a right inverse of $a - r1_{\mathcal{A}}$.
Then $a - r1_{\mathcal{A}}$ would be invertible by Corollary 4.3.3. Yet $a - r1_{\mathcal{A}} \notin \mathcal{A}^{\times}$ by Theorem 4.3.5(i), a contradiction.
Thus $p(a) - p(r)1_{\mathcal{A}}$ is noninvertible; $p(r) \in \operatorname{spec}_{\mathcal{A}}(p(a))$ by Theorem 4.3.5(i). Hence $p(\operatorname{spec}_{\mathcal{A}}(a)) \subseteq \operatorname{spec}_{\mathcal{A}}(p(a))$.
Structural Commentary: Finite-dimensional reduction handles algebraicity; splitting and one-sided rigidity handle opposite inclusions.
Key Insight: Linear factors recover spectral preimages, while $x - r$ carries spectral membership forward. □

Proposition 4.3.7. *Spectra of ab and ba*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra with $\dim_{\mathbb{F}} \mathcal{A} < \infty$, $a, b \in \mathcal{A}$. Then $\operatorname{spec}_{\mathcal{A}}(ab) = \operatorname{spec}_{\mathcal{A}}(ba)$.

Proof. Every element of $\mathcal{A}$ is algebraic by Theorem 4.2.4. Let $r \in \mathbb{F}^{\times}$, $ba - r1_{\mathcal{A}} \in \mathcal{A}^{\times}$. Set $u := a(ba - r1_{\mathcal{A}})^{-1}b - 1_{\mathcal{A}}$.
Then $(ab - r1_{\mathcal{A}})u = a(ba - r1_{\mathcal{A}})(ba - r1_{\mathcal{A}})^{-1}b - ab + r1_{\mathcal{A}} = r1_{\mathcal{A}}$. Thus $r^{-1}u$ is a right inverse of $ab - r1_{\mathcal{A}}$.
The element $ab - r1_{\mathcal{A}}$ is invertible by Corollary 4.3.3. Hence $r \notin \operatorname{spec}_{\mathcal{A}}(ba) \Rightarrow r \notin \operatorname{spec}_{\mathcal{A}}(ab)$ by Theorem 4.3.5(i).
$\operatorname{spec}_{\mathcal{A}}(ab) \setminus \{0\} = \operatorname{spec}_{\mathcal{A}}(ba) \setminus \{0\}$ by symmetry. Suppose $ba \in \mathcal{A}^{\times}$ with inverse y. Then $b(ay) = 1_{\mathcal{A}}$ and $(yb)a = 1_{\mathcal{A}}$.
Thus $a, b \in \mathcal{A}^{\times}$ by Corollary 4.3.3, so $ab \in \mathcal{A}^{\times}$. The converse holds by symmetry, so $ab \in \mathcal{A}^{\times} \iff ba \in \mathcal{A}^{\times}$.
Thus $0 \in \operatorname{spec}_{\mathcal{A}}(ab) \iff 0 \in \operatorname{spec}_{\mathcal{A}}(ba)$ by Theorem 4.3.5(ii). Therefore $\operatorname{spec}_{\mathcal{A}}(ab) = \operatorname{spec}_{\mathcal{A}}(ba)$.
Structural Commentary: The resolvent calculation handles nonzero scalars; one-sided inverse rigidity handles the zero shift.
Key Insight: The resolvent correction $a(ba - r1_{\mathcal{A}})^{-1}b - 1_{\mathcal{A}}$ transfers invertibility across reversed products. □

4.4 Division Algebras

Definition 4.4.1. *Division Algebras*
An $\mathbb{F}$-algebra $\mathcal{D}$ is a *division algebra* iff $1_{\mathcal{D}} \neq 0$ and $\mathcal{D}^{\times} = \mathcal{D} \setminus \{0\}$.

Example 4.4.2. *Hamilton's Quaternions*
Hamilton's quaternion algebra is associative real algebra $\mathbb{H} := \mathbb{R}1 \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$. Multiplication is $\mathbb{R}$-bilinear, and 1 is the identity.
The basis-unit products satisfy $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, and $ki = j = -ik$.
The algebra is noncommutative by $ij = k = -ji$ and $k \neq 0$. For $q = r_0 + r_1 i + r_2 j + r_3 k$, set $\overline{q} := r_0 - r_1 i - r_2 j - r_3 k$.
Then $q\overline{q} = \overline{q}q = r_0^2 + r_1^2 + r_2^2 + r_3^2 := N(q)$. If $q \neq 0$, then $N(q) > 0$ and $q^{-1} = N(q)^{-1}\overline{q}$; hence $\mathbb{H}$ is a real division algebra.

Figure 107 records the oriented multiplication rules for the imaginary basis units.

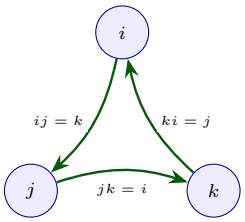

Figure 107: Quaternion basis products. The cycle $i \to j \to k \to i$ encodes $ij = k$, $jk = i$, and $ki = j$; reversed order negates each product. Each imaginary basis unit squares to -1, and 1 is the identity.

Theorem 4.4.3. *Division Algebras over Algebraically Closed Fields*
Let $\mathcal{D}$ be a division $\mathbb{F}$-algebra with $\dim_{\mathbb{F}} \mathcal{D} < \infty$. Suppose $(\forall\, f \in \mathbb{F}[x])\,[\deg f \geq 1 \Rightarrow f$ splits over $\mathbb{F}]$. Then $\mathcal{D} = \mathbb{F}1_{\mathcal{D}}$.

Proof. Fix $a \in \mathcal{D}$. The element a is algebraic by Theorem 4.2.4; let m_a be its minimal polynomial.
The polynomial m_a is nonconstant because $1_{\mathcal{D}} \neq 0$. Suppose $m_a = pq$ with $0 < \deg p, \deg q < \deg m_a$.
Then $p(a), q(a) \neq 0$ by minimality, whereas $q(a) = p(a)^{-1} m_a(a) = 0$, a contradiction.
Thus m_a is irreducible, and the splitting hypothesis forces $\deg m_a = 1$. Monicity fixes $m_a = x - r$ with $r \in \mathbb{F}$; hence $a = r1_{\mathcal{D}}$.
Thus $\mathcal{D} \subseteq \mathbb{F}1_{\mathcal{D}}$, while the reverse inclusion holds by the scalar action.
Structural Commentary: Algebraicity supplies a minimal polynomial; division makes it irreducible; splitting makes it linear.
Key Insight: A nontrivial factorization would create nonzero factors with zero product. □

Theorem 4.4.4. *Wedderburn's Theorem*
Every division algebra with finite underlying set is a field.

Proof.

(i) *Centralizer Data*: Suppose $\mathcal{D}$ is noncommutative and set $F := Z(\mathcal{D})$.
The set F is a finite field: every nonzero central element has a central inverse. Algebra $\mathcal{D}$ is F-algebra by Proposition 4.1.7(iii).
Set $z := |F| \geq 2$, $n := \dim_F \mathcal{D} > 1$, and $G := \mathcal{D}^{\times}$. Let B represent the noncentral conjugacy classes of G.
For $\beta \in B$, set $C_\beta := \{d \in \mathcal{D} : d\beta = \beta d\}$ and $k_\beta := \dim_F C_\beta$. Noncentrality implies $F \subsetneq C_\beta \subsetneq \mathcal{D}$.
The set C_β is a division subalgebra. Choose a right C_β-basis of $\mathcal{D}$ with r_β members.
Then $n = r_\beta k_\beta$ by expansion against F-basis of C_β. Thus $1 < k_\beta < n$, $k_\beta \mid n$. Identities $Z(G) = F^{\times}$, $C_G(\beta) = C_\beta^{\times}$ hold directly.
Finite vector-space counting therefore implies $|G| = z^n - 1$, $|Z(G)| = z - 1$, and $|C_G(\beta)| = z^{k_\beta} - 1$.

(ii) *Class Equation*: By Theorem 2.5.2, $z^n - 1 = z - 1 + \sum_{\beta \in B} \frac{z^n - 1}{z^{k_\beta} - 1}$.

(iii) *Cyclotomic Divisor*: Fix $\beta \in B$. Since $k_\beta \mid n$ and $k_\beta < n$, each divisor of k_β divides n but $n \nmid k_\beta$.
Thus $\Phi_n(z) \mid \frac{z^n - 1}{z^{k_\beta} - 1}$ by Propositions 3.11.4 and 3.11.5. The same propositions imply $\Phi_n(z) \mid z^n - 1$.

(iv) *Contradiction*: The class equation and (iii) imply $\Phi_n(z) \mid z - 1$. Yet $|\Phi_n(z)| > z - 1$ by Lemma 3.11.6, because $n > 1$ and $z \geq 2$.
This contradicts divisibility in $\mathbb{Z}$. Hence $\mathcal{D}$ is commutative and therefore is a finite field.

Structural Commentary: Centralizer dimensions convert the class equation into divisibility; the cyclotomic estimate excludes it.
Key Insight: The common cyclotomic divisor cannot divide the smaller integer $z - 1$. □

Theorem 4.4.5. *Frobenius Classification over $\mathbb{R}$*
If $\mathcal{D}$ is a finite-dimensional real division algebra, then $\mathcal{D} \cong \mathbb{R}$, $\mathcal{D} \cong \mathbb{C}$, or $\mathcal{D} \cong \mathbb{H}$.

Proof.

(i) *A Complex Unit*: The relation $\mathbb{R}1_{\mathcal{D}} \cong \mathbb{R}$ holds by Proposition 4.1.7(ii). If $\mathcal{D} = \mathbb{R}1_{\mathcal{D}}$, then $\mathcal{D} \cong \mathbb{R}$.
Otherwise choose $a \in \mathcal{D} \setminus \mathbb{R}1_{\mathcal{D}}$. The element a is algebraic by Theorem 4.2.4; let m_a be its minimal polynomial.
Suppose $m_a = pq$, $0 < \deg p, \deg q < \deg m_a$. Then $p(a), q(a) \neq 0$ by minimality, whereas $q(a) = p(a)^{-1} m_a(a) = 0$, contradiction.
Thus m_a is irreducible. Set $d := \deg m_a$, regard C_{m_a} as operator on $\mathbb{C}^d$. Characteristic polynomial is m_a by Theorem 3.1.13(ii).
Choose eigenvalue $\lambda \in \mathbb{C}$ by Theorem 5.3.2; $m_a(\lambda) = 0$ by Proposition 2.1.16(ii). If $\lambda \in \mathbb{R}$, then $x - \lambda \mid m_a$ by polynomial division.
Irreducibility and monicity would make $m_a = x - \lambda$ and $a = \lambda 1_{\mathcal{D}}$, contradiction. Thus $\lambda \notin \mathbb{R}$.
Set $h(x) := (x - \lambda)(x - \overline{\lambda}) \in \mathbb{R}[x]$. Then $h \mid m_a$ by polynomial division at the roots $\lambda, \overline{\lambda}$.
Hence $m_a = h$ by irreducibility and monicity. Set $r := \operatorname{Re}\lambda$ and $s := \operatorname{Im}\lambda$.
Then $s \neq 0$ and $m_a(x) = (x - r)^2 + s^2$. Set $i := s^{-1}(a - r1_{\mathcal{D}})$; then $i^2 = -1_{\mathcal{D}}$.

(ii) *Commuting and Anticommuting Parts*: Set $K := \mathbb{R}1_{\mathcal{D}} \oplus \mathbb{R}i$; then $K \cong \mathbb{C}$. Define the $\mathbb{R}$-linear map $\tau : \mathcal{D} \to \mathcal{D}$ by $\tau(x) := -ixi$.
The identity $\tau^2 = I_{\mathcal{D}}$ holds by $i^2 = -1_{\mathcal{D}}$. Set $\mathcal{D}_+ := \ker(\tau - I_{\mathcal{D}})$ and $\mathcal{D}_- := \ker(\tau + I_{\mathcal{D}})$. For $x \in \mathcal{D}$, $x = \frac{1}{2}(x + \tau(x)) + \frac{1}{2}(x - \tau(x))$.
The two summands lie in $\mathcal{D}_+$ and $\mathcal{D}_-$, respectively. Since $\mathcal{D}_+ \cap \mathcal{D}_- = \{0\}$, $\mathcal{D} = \mathcal{D}_+ \oplus \mathcal{D}_-$.
Equivalently, $\mathcal{D}_+ = \{x \in \mathcal{D} : xi = ix\}$ and $\mathcal{D}_- = \{x \in \mathcal{D} : xi = -ix\}$. The set $\mathcal{D}_+$ is a division subalgebra, and $K \subseteq Z(\mathcal{D}_+)$.
For $y \in \mathcal{D}_+$, the map L_y is complex-linear on the finite-dimensional complex space $\mathcal{D}_+$.
Choose $\mu \in K$ and $0 \neq x \in \mathcal{D}_+$ with $yx = \mu x$ by Theorem 5.3.2. Since $x^{-1} \in \mathcal{D}_+$, $y = \mu \in K$. Thus $\mathcal{D}_+ = K$.

(iii) *The Anticommuting Line*: If $\mathcal{D}_- = \{0\}$, then $\mathcal{D} = K \cong \mathbb{C}$. Otherwise choose $0 \neq j_0 \in \mathcal{D}_-$.
For $x \in \mathcal{D}_-$, $(xj_0^{-1})i = x(-ij_0^{-1}) = ixj_0^{-1}$; hence $xj_0^{-1} \in \mathcal{D}_+ = K$.
Conversely, $Kj_0 \subseteq \mathcal{D}_-$ by $j_0 i = -ij_0$, so $\mathcal{D}_- = Kj_0$. The relation $j_0^2 i = ij_0^2$ places j_0^2 in $\mathcal{D}_+ = K$.
Write $j_0^2 = u1_{\mathcal{D}} + vi$, $u, v \in \mathbb{R}$. Relations $j_0 i = -ij_0$ and $j_0 j_0^2 = j_0^2 j_0$ imply $0 = -2vij_0$, so $v = 0$; $j_0^2 = u1_{\mathcal{D}}$ with $u \neq 0$.
Suppose $u > 0$. Then $(j_0 - \sqrt{u}1_{\mathcal{D}})(j_0 + \sqrt{u}1_{\mathcal{D}}) = 0$. One factor vanishes by the absence of zero divisors, so j_0 is scalar.
This contradicts $\mathcal{D}_+ \cap \mathcal{D}_- = \{0\}$. Hence $u < 0$. Set $j := (-u)^{-1/2} j_0$; then $j^2 = -1_{\mathcal{D}}$ and $\mathcal{D}_- = Kj$.

(iv) *Quaternion Identification*: Now $\mathcal{D} = K \oplus Kj$ has real basis $(1_{\mathcal{D}}, i, j, ij)$. Set $k := ij$.
The relations $i^2 = j^2 = -1_{\mathcal{D}}$ and $ji = -ij$ imply $k^2 = -1_{\mathcal{D}}$.
They also imply $jk = i$ and $ki = j$; the reversed products are negated. Let $\varphi : \mathbb{H} \to \mathcal{D}$ send $(1, i, j, k)$ to $(1_{\mathcal{D}}, i, j, k)$ $\mathbb{R}$-linearly.
The map φ is bijective by the basis and multiplicative by the relations; hence $\mathcal{D} \cong \mathbb{H}$.

Structural Commentary: Quadratic minimal polynomial constructs K; conjugation splits $\mathcal{D}$ into commuting, anticommuting parts.
Key Insight: The anticommuting eigenspace is at most one-dimensional over K. □

5 Tensor Products

5.1 Tensor Products and Universal Properties

Tensor products linearize bilinear data.
Their construction combines free extension from Theorem 2.1.4 with quotient descent from Theorem 2.2.11.

Definition 5.1.1. *Bilinear Maps*
Let $\mathcal{U}, \mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces and $f : \mathcal{U} \times \mathcal{V} \to \mathcal{W}$.

(i) The map f is *bilinear* if $(\forall\, \mathbf{v} \in \mathcal{V})\, f(\,\cdot\,, \mathbf{v}) \in \mathcal{L}(\mathcal{U}, \mathcal{W})$ and $(\forall\, \mathbf{u} \in \mathcal{U})\, f(\mathbf{u}, \,\cdot\,) \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(ii) $\mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W}) := \{f : \mathcal{U} \times \mathcal{V} \to \mathcal{W} : f \text{ is bilinear}\}$.
(iii) If $\mathcal{W} = \mathbb{F}$, then $\mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathbb{F})$ is the set of *bilinear forms* from Definition 4.4.1.

Example 5.1.2. *Basic Bilinear Maps*

(i) For an $\mathbb{F}$-algebra A, define $\mu : A \times A \to A$ by $\mu(a, b) := ab$.
The map μ is bilinear by distributivity and compatibility of scalar and algebra multiplication.
(ii) Define $\mathrm{ev} : \mathcal{L}(\mathcal{V}, \mathcal{W}) \times \mathcal{V} \to \mathcal{W}$ by $\mathrm{ev}(T, \mathbf{v}) := T\mathbf{v}$. The map ev is bilinear by pointwise linearity in T and the linearity of T in $\mathbf{v}$.
(iii) For $\varphi \in \mathcal{V}'$, $\psi \in \mathcal{W}'$, define $B_{\varphi,\psi} : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ by $B_{\varphi,\psi}(\mathbf{v}, \mathbf{w}) := \varphi(\mathbf{v})\psi(\mathbf{w})$. Map $B_{\varphi,\psi}$ is bilinear by linearity of φ, ψ.

Definition 5.1.3. *Tensor Product by Universal Property*
Let $\mathcal{U}, \mathcal{V}, \mathcal{T}$ be $\mathbb{F}$-vector spaces and $t \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{T})$.
Given any $\mathbb{F}$-vector space $\mathcal{W}$, define $t^*_{\mathcal{W}} : \mathcal{L}(\mathcal{T}, \mathcal{W}) \to \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ by $t^*_{\mathcal{W}}(L) := L \circ t$.

(i) The pair $(\mathcal{T}, t)$ is a *tensor product* of $\mathcal{U}$ and $\mathcal{V}$ if every map $t^*_{\mathcal{W}}$ above is bijective.
(ii) For $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$, the map $\widetilde{f} := (t^*_{\mathcal{W}})^{-1}(f)$ is the *mediating map* for f; equivalently, $\widetilde{f} \circ t = f$.
(iii) For a chosen tensor product, write $\mathcal{T} = \mathcal{U} \otimes \mathcal{V}$ and $\mathbf{u} \otimes \mathbf{v} := t(\mathbf{u}, \mathbf{v})$. A tensor of the form $\mathbf{u} \otimes \mathbf{v}$ is *decomposable*.

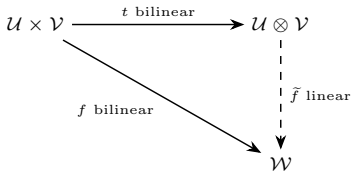

Figure 108: A tensor product is universal for bilinear maps out of $\mathcal{U} \times \mathcal{V}$.

Figure 108 depicts $f = \widetilde{f} \circ t$ from Definition 5.1.3. The definition owns existence and uniqueness of $\widetilde{f}$.

Theorem 5.1.4. *Quotient Construction of the Tensor Product*
Let $\mathcal{U}, \mathcal{V}$ be $\mathbb{F}$-vector spaces, $\mathcal{F} := \bigoplus^{\mathrm{ext}}_{(\mathbf{u},\mathbf{v}) \in \mathcal{U} \times \mathcal{V}} \mathbb{F}$. For $(\mathbf{u}, \mathbf{v}) \in \mathcal{U} \times \mathcal{V}$, let $[\mathbf{u}, \mathbf{v}] \in \mathcal{F}$ be coordinate family supported at $(\mathbf{u}, \mathbf{v})$ with value 1.
In the following relation families, $r, s \in \mathbb{F}$, $\mathbf{u}, \mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$, and $\mathbf{v}, \mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$.
Let $\mathcal{R}_{\mathcal{U}} \leq \mathcal{F}$ be spanned by all $r[\mathbf{u}_1, \mathbf{v}] + s[\mathbf{u}_2, \mathbf{v}] - [r\mathbf{u}_1 + s\mathbf{u}_2, \mathbf{v}]$. Let $\mathcal{R}_{\mathcal{V}} \leq \mathcal{F}$ be spanned by all $r[\mathbf{u}, \mathbf{v}_1] + s[\mathbf{u}, \mathbf{v}_2] - [\mathbf{u}, r\mathbf{v}_1 + s\mathbf{v}_2]$.
Set $\mathcal{R} := \mathcal{R}_{\mathcal{U}} + \mathcal{R}_{\mathcal{V}}$, $\mathcal{U} \otimes \mathcal{V} := \mathcal{F}/\mathcal{R}$, and $\mathbf{u} \otimes \mathbf{v} := [\mathbf{u}, \mathbf{v}] + \mathcal{R}$. Define $t : \mathcal{U} \times \mathcal{V} \to \mathcal{U} \otimes \mathcal{V}$ by $t(\mathbf{u}, \mathbf{v}) := \mathbf{u} \otimes \mathbf{v}$. Then:
(i) $\mathcal{U} \otimes \mathcal{V} = \mathrm{span}\{\mathbf{u} \otimes \mathbf{v} : \mathbf{u} \in \mathcal{U},\ \mathbf{v} \in \mathcal{V}\}$.
(ii) $(\mathcal{U} \otimes \mathcal{V}, t)$ is a tensor product of $\mathcal{U}$ and $\mathcal{V}$.

Proof. $\mathcal{F}$ is an $\mathbb{F}$-vector space by Proposition 2.1.19(ii). Let $\mathcal{B} := \{[\mathbf{u}, \mathbf{v}] : (\mathbf{u}, \mathbf{v}) \in \mathcal{U} \times \mathcal{V}\}$.
Every $\mathbf{x} = (x_{\mathbf{u},\mathbf{v}}) \in \mathcal{F}$ has a unique finite expansion $\mathbf{x} = \sum_{(\mathbf{u},\mathbf{v}) \in \mathrm{supp}(\mathbf{x})} x_{\mathbf{u},\mathbf{v}}[\mathbf{u}, \mathbf{v}]$ by Definition 2.1.18.
The set $\mathcal{B}$ is a basis by this expansion, and $\mathcal{F}$ is free on $\mathcal{B}$ by Definition 2.1.1.
Let $j : \mathcal{U} \times \mathcal{V} \to \mathcal{F}$ satisfy $j(\mathbf{u}, \mathbf{v}) := [\mathbf{u}, \mathbf{v}]$. Let $\pi : \mathcal{F} \to \mathcal{U} \otimes \mathcal{V}$ be the quotient map.

(i) Set $\mathcal{D} := \{\mathbf{u} \otimes \mathbf{v} : \mathbf{u} \in \mathcal{U},\ \mathbf{v} \in \mathcal{V}\}$. Then $\pi(\mathcal{B}) = \mathcal{D}$ by the definitions of π and decomposable tensors.
The equality $\mathcal{U} \otimes \mathcal{V} = \pi(\mathcal{F}) = \mathrm{span}(\pi(\mathcal{B})) = \mathrm{span}(\mathcal{D})$ holds by the linearity and surjectivity of π.
(ii) Fix $r, s \in \mathbb{F}$, $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$, and $\mathbf{v} \in \mathcal{V}$. The identity $t(r\mathbf{u}_1 + s\mathbf{u}_2, \mathbf{v}) = rt(\mathbf{u}_1, \mathbf{v}) + st(\mathbf{u}_2, \mathbf{v})$ holds by the definition of $\mathcal{R}_{\mathcal{U}}$.
Fix $\mathbf{u} \in \mathcal{U}$ and $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$. The identity $t(\mathbf{u}, r\mathbf{v}_1 + s\mathbf{v}_2) = rt(\mathbf{u}, \mathbf{v}_1) + st(\mathbf{u}, \mathbf{v}_2)$ holds by the definition of $\mathcal{R}_{\mathcal{V}}$.
The map $t = \pi \circ j$ is bilinear by these identities. Fix an $\mathbb{F}$-vector space $\mathcal{W}$ and $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$.
The assignment $[\mathbf{u}, \mathbf{v}] \mapsto f(\mathbf{u}, \mathbf{v})$ extends uniquely to $\alpha \in \mathcal{L}(\mathcal{F}, \mathcal{W})$ by Theorem 2.1.4.
Fix a displayed generator $x_{\mathcal{U}} := r[\mathbf{u}_1, \mathbf{v}] + s[\mathbf{u}_2, \mathbf{v}] - [r\mathbf{u}_1 + s\mathbf{u}_2, \mathbf{v}]$ of $\mathcal{R}_{\mathcal{U}}$.
Then $\alpha(x_{\mathcal{U}}) = rf(\mathbf{u}_1, \mathbf{v}) + sf(\mathbf{u}_2, \mathbf{v}) - f(r\mathbf{u}_1 + s\mathbf{u}_2, \mathbf{v}) = 0$ by bilinearity of f.
Fix a displayed generator $x_{\mathcal{V}} := r[\mathbf{u}, \mathbf{v}_1] + s[\mathbf{u}, \mathbf{v}_2] - [\mathbf{u}, r\mathbf{v}_1 + s\mathbf{v}_2]$ of $\mathcal{R}_{\mathcal{V}}$.

Then $\alpha(x_{\mathcal{V}}) = rf(\mathbf{u},\mathbf{v}_1) + sf(\mathbf{u},\mathbf{v}_2) - f(\mathbf{u}, r\mathbf{v}_1 + s\mathbf{v}_2) = 0$ by bilinearity of f.
$\mathcal{R} \leq \ker \alpha$ by the linearity of α and the two generator calculations.
Unique $\widetilde{f} \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$ satisfies $\widetilde{f} \circ \pi = \alpha$ by Theorem 2.2.11(i). Equality $\widetilde{f} \circ t = f$ holds by $(\widetilde{f} \circ t)(\mathbf{u},\mathbf{v}) = \alpha[\mathbf{u},\mathbf{v}] = f(\mathbf{u},\mathbf{v})$.
If $L \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$ satisfies $L \circ t = f$, then $L = \widetilde{f}$ by agreement on the spanning decomposable tensors from (i).

Structural Commentary: Free extension kills both relation subspaces and descends through their quotient.
Key Insight: The two relation subspaces encode linearity in the tensor factors. □

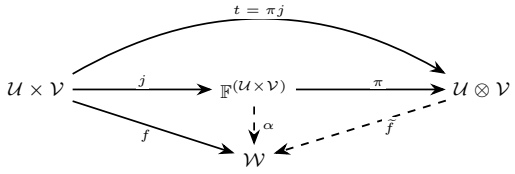

Figure 109: The free linear map α kills $\mathcal{R}$, so it descends uniquely to $\widetilde{f} : \mathcal{U} \otimes \mathcal{V} \to \mathcal{W}$.

Figure 109 records $j(\mathbf{u},\mathbf{v}) = [\mathbf{u},\mathbf{v}]$, $t = \pi \circ j$, and $\widetilde{f} \circ \pi = \alpha$ from Theorem 5.1.4.
The theorem owns the spanning and factorisation claims.

Corollary 5.1.5. *Tensor Products Are Unique up to Isomorphism*
If $(\mathcal{T}, t)$ and $(\mathcal{S}, s)$ are tensor products of $\mathcal{U}$ and $\mathcal{V}$, then a unique isomorphism $\Phi : \mathcal{T} \to \mathcal{S}$ satisfies $\Phi \circ t = s$.

Proof. Unique mediating maps $\Phi : \mathcal{T} \to \mathcal{S}$ and $\Psi : \mathcal{S} \to \mathcal{T}$ exist by Definition 5.1.3(i). They satisfy $\Phi \circ t = s$ and $\Psi \circ s = t$.
The identity $\Psi\Phi = I_{\mathcal{T}}$ holds by mediator uniqueness because $(\Psi\Phi) \circ t = \Psi \circ s = t = I_{\mathcal{T}} \circ t$.
Likewise, $\Phi\Psi = I_{\mathcal{S}}$ by mediator uniqueness because $(\Phi\Psi) \circ s = \Phi \circ t = s = I_{\mathcal{S}} \circ s$.
The map Φ is an isomorphism with inverse Ψ by these two identities.
If $\Theta : \mathcal{T} \to \mathcal{S}$ is linear and $\Theta \circ t = s$, then $\Theta = \Phi$ by uniqueness of the mediating map for s.
Structural Commentary: The two universal factorisations compose to the unique mediating identity maps.
Key Insight: Mediator uniqueness forces the comparison maps to be mutual inverses. □

Theorem 5.1.6. *Bilinearity Equals Linearity on a Tensor Product*
Let $\mathcal{U}, \mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces. Fix a tensor product $(\mathcal{U} \otimes \mathcal{V}, t)$ of $\mathcal{U}$ and $\mathcal{V}$.

(i) The set $\mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ is an $\mathbb{F}$-vector space under pointwise addition and scalar multiplication.
(ii) The assignment $f \mapsto \widetilde{f}$ defines an isomorphism $\Phi : \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W}) \to \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$.
For $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$, the map $\Phi(f) = \widetilde{f}$ is characterized by $\widetilde{f} \circ t = f$.

Proof.

(i) The zero map is bilinear by Definition 5.1.1(i). Fix $a, b \in \mathbb{F}$ and $f, g \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$.
For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{v} \in \mathcal{V}$, $(af + bg)(\,\cdot\,, \mathbf{v}) \in \mathcal{L}(\mathcal{U}, \mathcal{W})$ and $(af + bg)(\mathbf{u}, \,\cdot\,) \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ by linearity of the corresponding partial maps.
$af + bg \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ by Definition 5.1.1(i). The vector-space axioms hold by their pointwise counterparts in $\mathcal{W}$.

(ii) The assignment $\Phi(f) = \widetilde{f}$ is well-defined by Definition 5.1.3(i). Define $\Psi : \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W}) \to \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ by $\Psi(L) := L \circ t$.
The map Ψ is well-defined by linearity of L and bilinearity of t.
The identity $\Psi(\Phi(f)) = f$ holds by $\widetilde{f} \circ t = f$, and $\Phi(\Psi(L)) = L$ by mediator uniqueness for $L \circ t$.
Hence $\Psi = \Phi^{-1}$ by these identities. Fix $a, b \in \mathbb{F}$ and bilinear f, g.
The equality $(a\Phi(f) + b\Phi(g)) \circ t = af + bg = \Phi(af + bg) \circ t$ holds by the definition of Φ.
Thus $\Phi(af + bg) = a\Phi(f) + b\Phi(g)$ by mediator uniqueness. Map Φ is linear isomorphism by this linearity and inverse above.

Structural Commentary: Mediator uniqueness constructs the inverse and transfers pointwise linear combinations through t.
Key Insight: Precomposition with t inverts bilinear linearization. □

Computation Method 5.1.7. *Constructing Maps from Tensor Products*
Fix $\mathbb{F}$-vector spaces $\mathcal{U}, \mathcal{V}, \mathcal{W}$ and a tensor product $(\mathcal{U} \otimes \mathcal{V}, t)$. To construct $L \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$, proceed as follows:

(i) specify a rule $f : \mathcal{U} \times \mathcal{V} \to \mathcal{W}$;
(ii) verify $(\forall\, \mathbf{v} \in \mathcal{V})\, f(\,\cdot\,, \mathbf{v}) \in \mathcal{L}(\mathcal{U}, \mathcal{W})$ and $(\forall\, \mathbf{u} \in \mathcal{U})\, f(\mathbf{u}, \,\cdot\,) \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
(iii) set $L := \Phi(f)$; then $L \circ t = f$ by Theorem 5.1.6(ii).

The decomposable tensors span $\mathcal{U} \otimes \mathcal{V}$ by Theorem 5.1.4(i), but are generally linearly dependent.
Assignments on decomposable tensors must therefore respect their linear relations.

5.2 Tensor Product Bases and Maps

Definition 5.2.1. *Finite-Coordinate Tensor Product Model*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be ordered bases of $\mathcal{V}$ and $\mathcal{W}$.
The *coordinate tensor model* is $\mathcal{V} \otimes_{\beta,\gamma} \mathcal{W} := \mathrm{Mat}(m, n, \mathbb{F})$. For $(\mathbf{v},\mathbf{w}) \in \mathcal{V} \times \mathcal{W}$, define $\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w} := [\mathbf{v}]_\beta [\mathbf{w}]_\gamma^{\mathsf{T}}$.
The identity $\mathbf{v}_i \otimes_{\beta,\gamma} \mathbf{w}_j = E_{i,j}$ holds by the coordinate columns of the basis vectors.

Proposition 5.2.2. *Tensor Model Basis and Bilinear Evaluation*
In the coordinate tensor model associated to bases $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$:

(i) The matrices $E_{i,j} = \mathbf{v}_i \otimes_{\beta,\gamma} \mathbf{w}_j$ form a basis.
(ii) The map $(\mathbf{v},\mathbf{w}) \mapsto \mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}$ is bilinear.
(iii) Let $\mathcal{X}$ be $\mathbb{F}$-vector space and let $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{V}, \mathcal{W}; \mathcal{X})$.
A unique $\Theta_f \in \mathcal{L}(\mathrm{Mat}(m, n, \mathbb{F}), \mathcal{X})$ satisfies $(\forall\, (\mathbf{v},\mathbf{w}) \in \mathcal{V} \times \mathcal{W})\, \Theta_f(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) = f(\mathbf{v},\mathbf{w})$.

Proof.

(i) The matrices $E_{i,j}$ form the entry-matrix basis of $\mathrm{Mat}(m, n, \mathbb{F})$ by Proposition 3.1.5(ii).

(ii) Fix $a, b \in \mathbb{F}$, $\mathbf{v}, \mathbf{v}' \in \mathcal{V}$, and $\mathbf{w}, \mathbf{w}' \in \mathcal{W}$. The identity $(a\mathbf{v} + b\mathbf{v}') \otimes_{\beta,\gamma} \mathbf{w} = a(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) + b(\mathbf{v}' \otimes_{\beta,\gamma} \mathbf{w})$ holds by $\mathbb{F}$-linearity of $[\cdot]_\beta$. Likewise, $\mathbf{v} \otimes_{\beta,\gamma} (a\mathbf{w} + b\mathbf{w}') = a(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) + b(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}')$ by $\mathbb{F}$-linearity of $[\cdot]_\gamma$.
(iii) The basis-value assignment $E_{i,j} \mapsto f(\mathbf{v}_i, \mathbf{w}_j)$ extends uniquely to $\Theta_f \in \mathcal{L}(\mathrm{Mat}(m, n, \mathbb{F}), \mathcal{X})$ by Proposition 1.5.8.
Write $\mathbf{v} = \sum_{i=1}^{m} x_i \mathbf{v}_i$ and $\mathbf{w} = \sum_{j=1}^{n} y_j \mathbf{w}_j$. The identity $\Theta_f(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j f(\mathbf{v}_i, \mathbf{w}_j)$ holds by linearity of Θ_f.
Last sum equals $f(\mathbf{v}, \mathbf{w})$ by bilinearity of f. Uniqueness holds by agreement on the basis matrices $E_{i,j}$ and Proposition 1.5.8.

Structural Commentary: Entry matrices form the basis; coordinate expansion proves factorisation.
Key Insight: Each basis pair occupies one independent matrix coordinate. □

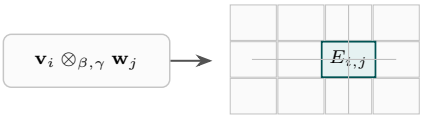

Figure 110: A basis pair occupies the (i, j) matrix slot.

Figure 110 illustrates the basis correspondence in Proposition 5.2.2(i). The proposition owns the basis and bilinearity claims.

Theorem 5.2.3. *Tensor Bases, Dimension, and Coordinate Rank*
Let $\mathcal{U}$ and $\mathcal{V}$ be finite-dimensional with bases $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ and $\gamma = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$.
(i) The tensors $\mathbf{u}_i \otimes \mathbf{v}_j$ form a basis of $\mathcal{U} \otimes \mathcal{V}$.
(ii) $\dim(\mathcal{U} \otimes \mathcal{V}) = \dim(\mathcal{U}) \dim(\mathcal{V})$.
(iii) Every $\mathbf{z} \in \mathcal{U} \otimes \mathcal{V}$ admits unique coefficients $r_{ij} \in \mathbb{F}$. They satisfy $\mathbf{z} = \sum_{i=1}^{m} \sum_{j=1}^{n} r_{ij}(\mathbf{u}_i \otimes \mathbf{v}_j)$.
(iv) Define $R_{\beta,\gamma}(\mathbf{z}) := [r_{ij}]$ from (iii). The value $\operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$ is independent of the bases β, γ.

Proof.
(i) The coordinate tensor model is a tensor product by Proposition 5.2.2(i)–(iii) and Definition 5.1.3(i).
A unique comparison isomorphism $\Xi : \mathcal{U} \otimes \mathcal{V} \to \mathrm{Mat}(m, n, \mathbb{F})$ exists by Corollary 5.1.5.
It satisfies $\Xi(\mathbf{u}_i \otimes \mathbf{v}_j) = E_{i,j}$ by the same corollary. The tensors $\mathbf{u}_i \otimes \mathbf{v}_j$ form a basis by bijectivity of Ξ and Proposition 5.2.2(i).
(ii) $\dim(\mathcal{U} \otimes \mathcal{V}) = mn = \dim(\mathcal{U}) \dim(\mathcal{V})$ by the mn-element basis in (i).
(iii) The coefficients r_{ij} exist uniquely by the basis in (i).
(iv) Let β', γ' be other ordered bases and set $P := [I_{\mathcal{U}}]_{\beta' \leftarrow \beta}$ and $Q := [I_{\mathcal{V}}]_{\gamma' \leftarrow \gamma}$.
The identity $R_{\beta',\gamma'}(\mathbf{z}) = P R_{\beta,\gamma}(\mathbf{z}) Q^{\mathsf{T}}$ holds by factorwise coordinate substitution.
Matrices P, Q^{T} invertible by Theorem 3.5.2(i), Corollary 3.6.4. Equality $\operatorname{rank} R_{\beta',\gamma'}(\mathbf{z}) = \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$ holds by Theorem 3.6.6(ii).

Structural Commentary: Coordinates transfer the basis; invertible basis changes preserve rank.
Key Insight: Basis changes multiply coordinate matrices by invertible factors. □

Definition 5.2.4. *Tensor Coordinate Matrix and Rank*
Let $\mathcal{U}, \mathcal{V}$ be finite-dimensional, with bases $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ and $\gamma = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$.
(i) For $\mathbf{z} \in \mathcal{U} \otimes \mathcal{V}$, let r_{ij} be the unique coefficients from Theorem 5.2.3(iii). The *coordinate matrix* is $R_{\beta,\gamma}(\mathbf{z}) := [r_{ij}]$.
(ii) The *rank of the tensor* is $\operatorname{rank}(\mathbf{z}) := \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$. It is independent of β, γ by Theorem 5.2.3(iv).

Theorem 5.2.5. *Zero Tests and Decomposable Rank*
Let $\mathcal{U}, \mathcal{V}$ be $\mathbb{F}$-vector spaces.
(i) $(\forall\, \mathbf{u} \in \mathcal{U})(\forall\, \mathbf{v} \in \mathcal{V})\ \mathbf{0} \otimes \mathbf{v} = \mathbf{u} \otimes \mathbf{0} = \mathbf{0}$.
(ii) Assume Zorn's lemma, let $r \in \mathbb{Z}_{>0}$, and let $(\mathbf{u}_i)_{i=1}^{r} \in \mathcal{U}^r$ be linearly independent.
Let $(\mathbf{v}_i)_{i=1}^{r} \in \mathcal{V}^r$. If $\sum_{i=1}^{r} \mathbf{u}_i \otimes \mathbf{v}_i = \mathbf{0}$, then $\mathbf{v}_1 = \cdots = \mathbf{v}_r = \mathbf{0}$.
(iii) If $\dim \mathcal{U}, \dim \mathcal{V} < \infty$ and $\mathbf{z} \in (\mathcal{U} \otimes \mathcal{V}) \setminus \{\mathbf{0}\}$, then $\mathbf{z}$ is decomposable $\iff \operatorname{rank}(\mathbf{z}) = 1$.
(iv) Assume $\dim \mathcal{U}, \dim \mathcal{V} < \infty$ and let $\mathbf{z} \in \mathcal{U} \otimes \mathcal{V}$. For $r \in \mathbb{Z}_{\geq 0}$, define $\mathcal{D}_r := \{\sum_{i=1}^{r} \mathbf{x}_i \otimes \mathbf{y}_i : (\mathbf{x}_i, \mathbf{y}_i)_{i=1}^{r} \in (\mathcal{U} \times \mathcal{V})^r\}$.
Then $\operatorname{rank}(\mathbf{z}) = \min\{r \in \mathbb{Z}_{\geq 0} : \mathbf{z} \in \mathcal{D}_r\}$.

Proof.
(i) The tensor map is linear in each slot by Definition 5.1.3(i). The equality $\mathbf{0} \otimes \mathbf{v} = \mathbf{u} \otimes \mathbf{0} = \mathbf{0}$ holds by Proposition 1.5.3.
(ii) Fix $k \in \{1, \ldots, r\}$ and set $\mathcal{S}_k := \operatorname{span}\{\mathbf{u}_i : i \neq k\}$. The inclusion $\mathbf{u}_k \notin \mathcal{S}_k$ holds by linear independence.
A functional $\varphi_k \in \mathcal{U}'$ satisfies $\varphi_k|_{\mathcal{S}_k} = 0$ and $\varphi_k(\mathbf{u}_k) = 1$ by Proposition 4.1.7(iii).
The bilinear map $(\mathbf{u}, \mathbf{v}) \mapsto \varphi_k(\mathbf{u})\mathbf{v}$ induces $T_k \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{V})$ by Theorem 5.1.6(ii).
Then $\mathbf{0} = T_k(\sum_{i=1}^{r} \mathbf{u}_i \otimes \mathbf{v}_i) = \sum_{i=1}^{r} \delta_{ki} \mathbf{v}_i = \mathbf{v}_k$. The equality $\mathbf{v}_1 = \cdots = \mathbf{v}_r = \mathbf{0}$ holds by the arbitrary choice of k.
(iii) ($\Rightarrow$) Fix ordered bases β, γ and write $\mathbf{z} = \mathbf{x} \otimes \mathbf{y} \neq \mathbf{0}$. The vectors $\mathbf{x}, \mathbf{y}$ are nonzero by (i).
The identity $R_{\beta,\gamma}(\mathbf{z}) = [\mathbf{x}]_\beta [\mathbf{y}]_\gamma^{\mathsf{T}}$ holds by Definition 5.2.4(i) and bilinearity.
The equality $\operatorname{rank}(\mathbf{z}) = 1$ holds by Definition 5.2.4(ii) and $\operatorname{col}(R_{\beta,\gamma}(\mathbf{z})) = \operatorname{span}([\mathbf{x}]_\beta) \neq \{\mathbf{0}\}$.
(iii) ($\Leftarrow$) Fix ordered bases β, γ and assume $\operatorname{rank} R_{\beta,\gamma}(\mathbf{z}) = 1$. Choose a nonzero column $\mathbf{a}$ spanning $\operatorname{col}(R_{\beta,\gamma}(\mathbf{z}))$.
A row $\mathbf{b}^{\mathsf{T}}$ satisfies $R_{\beta,\gamma}(\mathbf{z}) = \mathbf{a}\mathbf{b}^{\mathsf{T}}$ by Theorem 4.1.2(i). Let $[\mathbf{x}]_\beta = \mathbf{a}$ and $[\mathbf{y}]_\gamma = \mathbf{b}$. Then $\mathbf{z} = \mathbf{x} \otimes \mathbf{y}$ by Theorem 5.2.3(iii).
(iv) Fix ordered bases β, γ, set $R := R_{\beta,\gamma}(\mathbf{z})$, $r := \operatorname{rank}(R)$. If $r = 0$, then $\operatorname{col}(R) = \{\mathbf{0}\}$ by Definition 3.1.12(iii), Proposition 1.4.3(ii).
Then $R = 0$ by $\operatorname{col}(R) = \{\mathbf{0}\}$. Equalities $\mathbf{z} = \mathbf{0}$, $\mathbf{z} \in \mathcal{D}_0$ hold by Theorem 5.2.3(iii), empty-sum convention respectively.
Suppose $r \geq 1$, and let the columns of $X \in \mathrm{Mat}(\dim \mathcal{U}, r, \mathbb{F})$ form a basis of $\operatorname{col}(R)$. A matrix Y satisfies $R = XY$ by Theorem 4.1.2(i).
If $\mathbf{a}_i$ is column i of X and $\mathbf{b}_i^{\mathsf{T}}$ is row i of Y, then $R = \sum_{i=1}^{r} \mathbf{a}_i \mathbf{b}_i^{\mathsf{T}}$ by matrix multiplication.

Choose $[\mathbf{x}_i]_\beta = \mathbf{a}_i$ and $[\mathbf{y}_i]_\gamma = \mathbf{b}_i$. Then $\mathbf{z} = \sum_{i=1}^{r} \mathbf{x}_i \otimes \mathbf{y}_i$ by Theorem 5.2.3(iii).

Membership $\mathbf{z} \in \mathcal{D}_r$ holds by definition. If $\mathbf{z} \in \mathcal{D}_s$, choose $\mathbf{z} = \sum_{i=1}^{s} \mathbf{x}_i \otimes \mathbf{y}_i$. Every column of R lies in $\operatorname{span}\{[\mathbf{x}_1]_\beta, \dots, [\mathbf{x}_s]_\beta\}$.

The inequality $r \le s$ holds by Definition 3.1.12(iii), Proposition 1.4.3(i), and Theorem 1.3.11.
The minimum formula holds by $\mathbf{z} \in \mathcal{D}_r$ and $(\forall\, s \in \mathbb{Z}_{\ge 0})\,[\mathbf{z} \in \mathcal{D}_s \Rightarrow r \le s]$.

Structural Commentary: Linearity proves zeros; functionals isolate terms; matrix factors fix rank.
Key Insight: Tensor rank is the minimum number of decomposable summands. □

Theorem 5.2.6. *Tensor Product of Linear Maps*
Let $A \in \mathcal{L}(\mathcal{U},\mathcal{U}')$ and $B \in \mathcal{L}(\mathcal{V},\mathcal{V}')$.

(i) A unique $A \otimes B \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{U}' \otimes \mathcal{V}')$ exists. It satisfies $(\forall\, \mathbf{u} \in \mathcal{U})(\forall\, \mathbf{v} \in \mathcal{V})\,(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$.
(ii) Define $\tau : \mathcal{L}(\mathcal{U},\mathcal{U}') \times \mathcal{L}(\mathcal{V},\mathcal{V}') \to \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{U}' \otimes \mathcal{V}')$ by $\tau(A,B) := A \otimes B$. The map τ is bilinear.
(iii) Let $A_i \in \mathcal{L}(\mathcal{U}_{i-1},\mathcal{U}_i)$ and $B_i \in \mathcal{L}(\mathcal{V}_{i-1},\mathcal{V}_i)$ for $i \in \{1,2\}$. Then $(A_2A_1) \otimes (B_2B_1) = (A_2 \otimes B_2)(A_1 \otimes B_1)$.
(iv) $I_\mathcal{U} \otimes I_\mathcal{V} = I_{\mathcal{U}\otimes\mathcal{V}}$.
(v) Suppose $\mathcal{U},\mathcal{V},\mathcal{U}',\mathcal{V}'$ are finite-dimensional with ordered bases $\beta,\gamma,\beta',\gamma'$, respectively.
For the lexicographically ordered tensor-product bases, $[A \otimes B]_{\beta'\otimes\gamma' \leftarrow \beta\otimes\gamma} = [A]_{\beta' \leftarrow \beta} \otimes [B]_{\gamma' \leftarrow \gamma}$.
The right-hand $\otimes$ denotes the matrix Kronecker product.

Proof.

(i) The map $(\mathbf{u},\mathbf{v}) \mapsto A\mathbf{u} \otimes B\mathbf{v}$ is bilinear by linearity of A, B and bilinearity of the tensor map.
The unique linear map $A \otimes B$ exists by Theorem 5.1.6(ii).

(ii) Fix $a,b \in \mathbb{F}$, $A, C \in \mathcal{L}(\mathcal{U},\mathcal{U}')$, and $B, D \in \mathcal{L}(\mathcal{V},\mathcal{V}')$.
On decomposable tensors, $\tau(aA + bC, B) = a\tau(A,B) + b\tau(C,B)$ by the linearity of A, C and the tensor map.
Likewise, $\tau(A, aB + bD) = a\tau(A,B) + b\tau(A,D)$ on decomposable tensors.
The identities extend to $\mathcal{U} \otimes \mathcal{V}$ by Theorem 5.1.4(i), and τ is bilinear by these identities.

(iii) For $(\mathbf{u},\mathbf{v}) \in \mathcal{U}_0 \times \mathcal{V}_0$, both sides send $\mathbf{u} \otimes \mathbf{v}$ to $A_2A_1\mathbf{u} \otimes B_2B_1\mathbf{v}$.
The composition identity holds by decomposable spanning in Theorem 5.1.4(i).

(iv) Both $I_\mathcal{U} \otimes I_\mathcal{V}$ and $I_{\mathcal{U}\otimes\mathcal{V}}$ fix every decomposable tensor. Their equality holds by decomposable spanning in Theorem 5.1.4(i).

(v) Fix expansions $A\mathbf{u}_j = \sum_i a_{ij}\mathbf{u}'_i$ and $B\mathbf{v}_\ell = \sum_k b_{k\ell}\mathbf{v}'_k$. Then $(A \otimes B)(\mathbf{u}_j \otimes \mathbf{v}_\ell) = \sum_{i,k} a_{ij}b_{k\ell}(\mathbf{u}'_i \otimes \mathbf{v}'_k)$ by bilinearity.
The claimed matrix identity holds by this Kronecker block-entry rule and Definition 3.2.3.

Structural Commentary: Linearization builds $A \otimes B$; spanning proves composition; bases fix the matrix.
Key Insight: Factorwise action becomes Kronecker multiplication in coordinates. □

Theorem 5.2.7. *Tensor Product of Spaces of Linear Maps*
Assume Zorn's lemma, and let $\mathcal{U},\mathcal{V},\mathcal{X},\mathcal{Y}$ be $\mathbb{F}$-vector spaces.

(i) A unique linear map $\Theta : \mathcal{L}(\mathcal{U},\mathcal{X}) \otimes \mathcal{L}(\mathcal{V},\mathcal{Y}) \to \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{X} \otimes \mathcal{Y})$ exists. It is characterized by $\Theta(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$.
(ii) The map Θ is injective.
(iii) If $\mathcal{U},\mathcal{V},\mathcal{X},\mathcal{Y}$ are finite-dimensional, then Θ is an isomorphism.
(iv) Define $\Delta(\varphi \otimes \psi)(\mathbf{u} \otimes \mathbf{v}) := \varphi(\mathbf{u})\psi(\mathbf{v})$. This rule determines an injective map $\Delta : \mathcal{U}' \otimes \mathcal{V}' \to (\mathcal{U} \otimes \mathcal{V})'$.
(v) If $\mathcal{U},\mathcal{V}$ are finite-dimensional, then Δ is an isomorphism.

Proof.

(i) The assignment $(A,B) \mapsto A \otimes B$ is bilinear by Theorem 5.2.6(ii).
The unique linear map Θ exists by Theorem 5.1.6(ii). Its characterization holds by Theorem 5.2.6(i).

(ii) Fix $0 \ne Z = \sum_{i=1}^{r} A_i \otimes B_i$ with r minimal. The relations $(\forall\, i \in \{1,\dots,r\})\,[A_i \ne 0 \wedge B_i \ne 0]$ hold by minimality.

If $(A_i)_{i=1}^r$ is dependent, reindex with $A_r = \sum_{i=1}^{r-1} c_iA_i$. Then $Z = \sum_{i=1}^{r-1} A_i \otimes (B_i + c_iB_r)$, contrary to minimality.

The family $(A_i)_{i=1}^r$ is independent by contradiction. If $(B_i)_{i=1}^r$ is dependent, reindex with $B_r = \sum_{i=1}^{r-1} d_iB_i$.

Then $Z = \sum_{i=1}^{r-1} (A_i + d_iA_r) \otimes B_i$, contrary to minimality. The family $(B_i)_{i=1}^r$ is independent by contradiction.

Choose $\mathbf{u} \in \mathcal{U}$ with $A_1\mathbf{u} \ne 0$. After reindexing, let $A_1\mathbf{u}, \dots, A_m\mathbf{u}$ be a basis of $\operatorname{span}\{A_i\mathbf{u} : 1 \le i \le r\}$.

For $m < j \le r$, write $A_j\mathbf{u} = \sum_{i=1}^{m} c_{ij}A_i\mathbf{u}$. Set $C_i := B_i + \sum_{j=m+1}^{r} c_{ij}B_j$ for $1 \le i \le m$. Then $C_i \ne 0$ by independence of $(B_i)_{i=1}^r$.

For $\mathbf{v} \in \mathcal{V}$, $\Theta(Z)(\mathbf{u} \otimes \mathbf{v}) = \sum_{i=1}^{m} (A_i\mathbf{u}) \otimes C_i\mathbf{v}$. If $\Theta(Z) = 0$, then $(\forall\, i \in \{1,\dots,m\})(\forall\, \mathbf{v} \in \mathcal{V})\, C_i\mathbf{v} = \mathbf{0}$ by Theorem 5.2.5(ii).

The conclusion $(\forall\, i \in \{1,\dots,m\})\, C_i = 0$ holds by Assumption 3.1.2, contrary to the preceding nonvanishing.
The inequality $\Theta(Z) \ne 0$ holds by contradiction, and Θ is injective.

(iii) Both spaces have dimension $\dim\mathcal{U}\dim\mathcal{V}\dim\mathcal{X}\dim\mathcal{Y}$ by Theorem 5.2.3(ii) and Theorem 3.4.5(ii).
The map Θ is an isomorphism by (ii) and Corollary 1.7.6(ii).

(iv) Set $\mathcal{X} = \mathcal{Y} = \mathbb{F}$. The multiplication map induces $\lambda : \mathbb{F} \otimes \mathbb{F} \to \mathbb{F}$ with $\lambda(a \otimes b) = ab$ by Theorem 5.1.6(ii).
The map λ is an isomorphism by the basis correspondence $\{1 \otimes 1\} \mapsto \{1\}$ and Theorem 5.2.3(i).
Postcomposition with λ is an isomorphism $\lambda_* : \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathbb{F} \otimes \mathbb{F}) \to (\mathcal{U} \otimes \mathcal{V})'$. Its inverse is postcomposition with λ^{-1}.
The identity $\Delta = \lambda_* \circ \Theta$ holds on decomposable tensors by the defining formulas. The map Δ is injective by (ii).

(v) If $\mathcal{U},\mathcal{V}$ are finite-dimensional, then Θ is an isomorphism by (iii).
The map Δ is an isomorphism by $\Delta = \lambda_* \circ \Theta$ and the isomorphisms λ_*, Θ.

Structural Commentary: Minimality proves injection; dimensions and multiplication prove isomorphism.
Key Insight: Independent operator factors are detected on one decomposable tensor. □

Proposition 5.2.8. *Scalar Extension by Tensor Product*
Let $K/\mathbb{F}$ be a field extension, regard K as an $\mathbb{F}$-vector space, and let $\mathcal{V}_\mathbb{F}$ be an $\mathbb{F}$-vector space.

(i) The $\mathbb{F}$-vector space $K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F}$ is a K-vector space via $\alpha \cdot (\beta \otimes \mathbf{v}) := (\alpha\beta) \otimes \mathbf{v}$.
(ii) Define $\mu : \mathcal{V}_\mathbb{F} \to K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F}$ by $\mu(\mathbf{v}) := 1 \otimes \mathbf{v}$. Let $\mathcal{Y}$ be a K-vector space and let $f : \mathcal{V}_\mathbb{F} \to \mathcal{Y}$ be $\mathbb{F}$-linear. A unique K-linear $\widetilde{f} : K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F} \to \mathcal{Y}$ satisfies $\widetilde{f} \circ \mu = f$.

Proof.

(i) For $\alpha \in K$, set $m_\alpha(\beta, \mathbf{v}) := (\alpha\beta) \otimes \mathbf{v}$. The map m_α is $\mathbb{F}$-bilinear. It induces $M_\alpha \in \mathcal{L}_\mathbb{F}(K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F})$ by Theorem 5.1.6(ii).
For $\alpha, \lambda \in K$, the identities $M_0 = 0$ and $M_{\alpha+\lambda} = M_\alpha + M_\lambda$ hold on decomposable tensors.
Identities $M_{\alpha\lambda} = M_\alpha M_\lambda$ and $M_1 = I$ likewise hold there. They extend to tensor product by Theorem 5.1.4(i).
The map $\alpha \mapsto M_\alpha$ is a unital ring homomorphism $K \to \mathrm{End}_\mathbb{F}(K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F})$ by these identities.
The action $\alpha \cdot \mathbf{z} := M_\alpha \mathbf{z}$ defines the claimed K-vector space by this unital ring homomorphism.
(ii) The map μ is $\mathbb{F}$-linear by bilinearity of the tensor map. Set $g(\beta, \mathbf{v}) := \beta f(\mathbf{v})$.
The map g is $\mathbb{F}$-bilinear by the $\mathbb{F}$-linearity of f and scalar multiplication on $\mathcal{Y}$.
A unique $\mathbb{F}$-linear map $\widetilde{f}$ satisfies $\widetilde{f}(\beta \otimes \mathbf{v}) = \beta f(\mathbf{v})$ by Theorem 5.1.6(ii).
For $\alpha, \beta \in K$ and $\mathbf{v} \in \mathcal{V}_\mathbb{F}$, $\widetilde{f}(\alpha \cdot (\beta \otimes \mathbf{v})) = \alpha\beta f(\mathbf{v}) = \alpha \widetilde{f}(\beta \otimes \mathbf{v})$.
The map $\widetilde{f}$ is K-linear by Theorem 5.1.4(i). The identity $\widetilde{f} \circ \mu = f$ holds by $\widetilde{f}(1 \otimes \mathbf{v}) = f(\mathbf{v})$.
If h is K-linear, $h \circ \mu = f$, then $h(\beta \otimes \mathbf{v}) = \beta h(1 \otimes \mathbf{v}) = \beta f(\mathbf{v}) = \widetilde{f}(\beta \otimes \mathbf{v})$. Equality $h = \widetilde{f}$ holds by Theorem 5.1.4(i).

Structural Commentary: Field-action maps define K-scalars; bilinear extension proves universality.
Key Insight: The factor K freely supplies the missing scalars. □

5.3 Iterated Tensor Products and Tensor Spaces

The tensor-product universal property extends from bilinear maps to multilinear maps.
Its iterated form leads naturally to tensor spaces and contractions.

Definition 5.3.1. *Multilinear Maps*
Let $n \in \mathbb{Z}_{>0}$ and let $\mathcal{V}_1, \dots, \mathcal{V}_n, \mathcal{W}$ be $\mathbb{F}$-vector spaces.

(i) Fix $k \in \{1, \dots, n\}$ and all entries except the kth. For $f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to \mathcal{W}$, set $f_k(\mathbf{v}) := f(\mathbf{v}_1, \dots, \mathbf{v}, \dots, \mathbf{v}_n)$.
The map f is *multilinear*, or *n-linear*, if $f_k \in \mathcal{L}(\mathcal{V}_k, \mathcal{W})$ in this setup.
(ii) The set of such maps is $\mathrm{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{W})$.
(iii) If $\mathcal{W} = \mathbb{F}$, these maps are *multilinear forms*.

Example 5.3.2. *Basic Multilinear Maps*

(i) Let A be an associative $\mathbb{F}$-algebra and let $n \in \mathbb{Z}_{>0}$. Set $m_n(a_1, \dots, a_n) := a_1 \cdots a_n$.
Distributivity and homogeneity in A prove that $m_n : A^n \to A$ is n-linear.
(ii) Suppose $\dim \mathcal{V} = n \geq 1$ and β is an ordered basis.
The determinant of the β-coordinate columns is n-linear by Theorem 3.8.2(ii). It is alternating by parts (iii)–(iv).

Definition 5.3.3. *Iterated Tensor Product by Universal Property*
Let $n \in \mathbb{Z}_{>0}$ and let $\mathcal{V}_1, \dots, \mathcal{V}_n$ be $\mathbb{F}$-vector spaces.

(i) A pair $(\mathcal{T}, t)$ is an *iterated tensor product* of $\mathcal{V}_1, \dots, \mathcal{V}_n$ if both conditions hold.
  (a) $t \in \mathrm{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{T})$.
  (b) Let $\mathcal{Y}$ be an $\mathbb{F}$-vector space. Set $\mathcal{M}_\mathcal{Y} := \mathrm{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{Y})$. $(\forall\, f \in \mathcal{M}_\mathcal{Y})(\exists!\, \widetilde{f} \in \mathcal{L}(\mathcal{T}, \mathcal{Y}))\ f = \widetilde{f} \circ t$.
(ii) Write $\mathcal{T} = \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$. The notation $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n$ abbreviates $t(\mathbf{v}_1, \dots, \mathbf{v}_n)$.

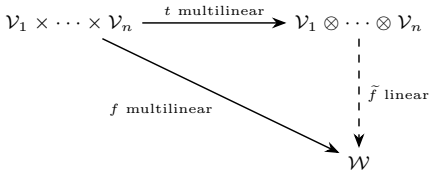

Figure 111: The iterated tensor product is universal for multilinear maps.

Figure 111 depicts the factorization in Definition 5.3.3.
The definition, rather than the diagram, owns the universal property.

Theorem 5.3.4. *Construction and Basic Laws for Iterated Tensor Products*
Let $n \in \mathbb{Z}_{>0}$ and let $\mathcal{V}_1, \dots, \mathcal{V}_n, \mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces. For $1 \leq a \leq b \leq n$, set $\mathcal{V}_{a:b}^{\otimes} := \mathcal{V}_a \otimes \cdots \otimes \mathcal{V}_b$.
Write $\mathbf{v}_{a:b}^{\otimes} := \mathbf{v}_a \otimes \cdots \otimes \mathbf{v}_b$. Set $\mathcal{M}_\mathcal{W} := \mathrm{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{W})$.

(i) The following construction and uniqueness statements hold.
  (a) Every binary parenthesization of $\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$ is an iterated tensor product.
  (b) Every iterated tensor product is spanned by its decomposable tensors.
  (c) Any two iterated tensor products are uniquely isomorphic by an isomorphism preserving decomposable tensors.
(ii) Pointwise operations make $\mathcal{M}_\mathcal{W}$ an $\mathbb{F}$-vector space. Map $\Phi : \mathcal{M}_\mathcal{W} \to \mathcal{L}(\mathcal{V}_{1:n}^{\otimes}, \mathcal{W})$ defined by $\Phi(f) := \widetilde{f}$ and is an isomorphism.
(iii) If $1 \leq r < n$, set $\mathcal{A}_r := \mathcal{V}_{1:r}^{\otimes} \otimes \mathcal{V}_{r+1:n}^{\otimes}$. The canonical isomorphism $\alpha_r : \mathcal{A}_r \xrightarrow{\sim} \mathcal{V}_{1:n}^{\otimes}$ satisfies $\alpha_r(\mathbf{v}_{1:r}^{\otimes} \otimes \mathbf{v}_{r+1:n}^{\otimes}) = \mathbf{v}_{1:n}^{\otimes}$.
(iv) If $\pi \in S_n$, set $\mathcal{C}_\pi := \mathcal{V}_{\pi(1)} \otimes \cdots \otimes \mathcal{V}_{\pi(n)}$. The canonical isomorphism $\tau_\pi : \mathcal{V}_{1:n}^{\otimes} \xrightarrow{\sim} \mathcal{C}_\pi$ satisfies $\tau_\pi(\mathbf{v}_{1:n}^{\otimes}) = \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}$.
(v) The isomorphism $\lambda : \mathbb{F} \otimes \mathcal{V} \xrightarrow{\sim} \mathcal{V}$ is determined by $\lambda(r \otimes \mathbf{v}) = r\mathbf{v}$. The isomorphism $\rho : \mathcal{V} \otimes \mathbb{F} \xrightarrow{\sim} \mathcal{V}$ is determined by $\rho(\mathbf{v} \otimes r) = r\mathbf{v}$.
(vi) If $(\forall\, i \in \{1, \dots, n\})\ \dim \mathcal{V}_i < \infty$, then $\dim(\mathcal{V}_{1:n}^{\otimes}) = \prod_{i=1}^{n} \dim(\mathcal{V}_i)$.

Proof.

(i)

(a) For $n = 1$, take $\mathcal{T} = \mathcal{V}_1$ and $t = I_{\mathcal{V}_1}$. The unique mediator of a linear map $f : \mathcal{V}_1 \to \mathcal{Y}$ is f itself.
Assume the assertion for fewer than n factors and split a binary parenthesization as $\mathcal{A} \otimes \mathcal{B}$.
By induction, $\mathcal{A}$ and $\mathcal{B}$ satisfy the appropriate iterated universal properties.
Let f be a multilinear map from the original factors to $\mathcal{Y}$.
Fixing the factors represented by $\mathcal{B}$ and linearizing the remaining variables produces an element of $\mathcal{L}(\mathcal{A}, \mathcal{Y})$.
This assignment is multilinear in the factors represented by $\mathcal{B}$ because the factorizations through $\mathcal{A}$ are unique.
Linearizing again produces $L : \mathcal{B} \to \mathcal{L}(\mathcal{A}, \mathcal{Y})$. Set $b_f(\mathbf{a}, \mathbf{b}) := L(\mathbf{b})(\mathbf{a})$.
Then map b_f is bilinear, and induces unique map $\mathcal{A} \otimes \mathcal{B} \to \mathcal{Y}$ by Theorem 5.1.6(ii).
Any mediator determines the same two intermediate factorizations and hence the same induced map.
Thus the chosen parenthesization is an iterated tensor product.

(b) Let $(\mathcal{T}, t)$ be an iterated tensor product. Set $\mathcal{S} := \operatorname{span}(t(\mathcal{V}_1 \times \cdots \times \mathcal{V}_n))$.
The quotient map $\pi : \mathcal{T} \to \mathcal{T}/\mathcal{S}$ is linear by Proposition 2.2.8(i).
Both π and the zero map mediate the zero multilinear map because $\pi \circ t = 0$.
Thus $\pi = 0$ by Definition 5.3.3(i). Since π is surjective by Proposition 2.2.8(ii), $\mathcal{T}/\mathcal{S} = 0$ and $\mathcal{S} = \mathcal{T}$.

(c) Let $(\mathcal{T}, t)$ and $(\mathcal{U}, u)$ be iterated tensor products. The universal properties produce a unique $F : \mathcal{T} \to \mathcal{U}$ with $F \circ t = u$.
They produce a unique $G : \mathcal{U} \to \mathcal{T}$ with $G \circ u = t$. Both GF and $I_{\mathcal{T}}$ mediate t, while both FG and $I_{\mathcal{U}}$ mediate u.
Uniqueness implies $GF = I_{\mathcal{T}}$, $FG = I_{\mathcal{U}}$. Any isomorphism preserving decomposable tensors also mediates u, so it equals F.

(ii) The zero map is multilinear, and $af + g$ is multilinear when $a \in \mathbb{F}$ and f, g are multilinear.
The remaining vector-space axioms hold pointwise because they hold in $\mathcal{W}$. Let t be the canonical multilinear map.
Set $\Phi(f) := \widetilde{f}$ by Definition 5.3.3(i). Set $\Psi(T) := T \circ t$. Defining factorisation implies $\Psi\Phi = I$, while uniqueness implies $\Phi\Psi = I$.
The maps $\Phi(af + g)$ and $a\Phi(f) + \Phi(g)$ both mediate $af + g$, so uniqueness proves that Φ is linear. Hence Φ is an isomorphism.

(iii) Set $\mathcal{B} := \mathcal{V}_{1:n}^{\otimes}$. Both $\mathcal{A}_r$ and $\mathcal{B}$ are iterated tensor products by (i)(a).
Their universal properties produce comparison maps $\alpha_r : \mathcal{A}_r \to \mathcal{B}$ and $\beta_r : \mathcal{B} \to \mathcal{A}_r$.
These maps have the displayed actions on decomposable tensors. These maps are inverse isomorphisms by (i)(c).

(iv) The map $p_\pi : \prod_{i=1}^{n} \mathcal{V}_i \to \mathcal{C}_\pi$ satisfies $p_\pi(\mathbf{v}_1, \ldots, \mathbf{v}_n) := \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}$ and is multilinear.
Its unique mediating map is τ_π by Definition 5.3.3(i). The analogous map for the inverse reordering is inverse to τ_π by (i)(c).

(v) The map $m_L : \mathbb{F} \times \mathcal{V} \to \mathcal{V}$ defined by $m_L(r, \mathbf{v}) := r\mathbf{v}$ is bilinear. The map $m_R : \mathcal{V} \times \mathbb{F} \to \mathcal{V}$ defined by $m_R(\mathbf{v}, r) := r\mathbf{v}$ is bilinear.
Hence λ and ρ exist by Theorem 5.1.6(ii). Define $u_L : \mathcal{V} \to \mathbb{F} \otimes \mathcal{V}$ by $u_L(\mathbf{v}) := 1 \otimes \mathbf{v}$. Define $u_R : \mathcal{V} \to \mathcal{V} \otimes \mathbb{F}$ by $u_R(\mathbf{v}) := \mathbf{v} \otimes 1$.
Both maps are linear by bilinearity of the canonical tensor maps. Direct computation proves $\lambda u_L = I_{\mathcal{V}}$ and $\rho u_R = I_{\mathcal{V}}$.
Maps $u_L\lambda$ and $u_R\rho$ fix decomposable tensors by bilinearity, hence are identity maps by (i)(b). Hence λ and ρ are isomorphisms.

(vi) Choose finite bases B_i of $\mathcal{V}_i$. Set $\mathcal{D} := \{\mathbf{b}_1 \otimes \cdots \otimes \mathbf{b}_n : \mathbf{b}_i \in B_i\}$.

The set $\mathcal{D}$ is a basis by repeated application of Theorem 5.2.3(i). Consequently, $\dim(\mathcal{V}_{1:n}^{\otimes}) = \#\mathcal{D} = \prod_{i=1}^{n} \#B_i = \prod_{i=1}^{n} \dim \mathcal{V}_i$.

Structural Commentary: Induction builds products; quotients prove spanning; uniqueness fixes maps.
Key Insight: Every parenthesization realizes the same multilinear universal property. □

Definition 5.3.5. *Tensor Spaces of Type (p, q)*
Let $\dim \mathcal{V} < \infty$.

(i) Set $\mathcal{V}^{\otimes 0} := \mathbb{F}$. If $k \in \mathbb{Z}_{>0}$, let $\mathcal{V}^{\otimes k}$ denote the iterated tensor product of k copies of $\mathcal{V}$.

(ii) Let $(p, q) \in \mathbb{Z}_{\geq 0}^2 \setminus \{(0, 0)\}$. Set $\mathcal{T}_q^p(\mathcal{V}) := \mathcal{V}^{\otimes p} \otimes (\mathcal{V}')^{\otimes q}$.
Parentheses are suppressed by Theorem 5.3.4(iii). Unit factors are identified by part (v).

(iii) Space $\mathcal{T}_q^p(\mathcal{V})$ is *tensor space of type (p, q)*. Its p factors from $\mathcal{V}$ are contravariant, q factors from $\mathcal{V}'$ are covariant. $\mathcal{T}_0^0(\mathcal{V}) := \mathbb{F}$.

(iv) If $p + q \geq 1$, set $\mathcal{M}_{p,q}(\mathcal{V}) := \operatorname{Mult}_{\mathbb{F}}((\mathcal{V}')^{\times p} \times \mathcal{V}^{\times q}; \mathbb{F})$.
Its elements are multilinear forms on p arguments from $\mathcal{V}'$ followed by q arguments from $\mathcal{V}$.

Proposition 5.3.6. *Tensor Spaces as Multilinear Functionals*
Let $\dim \mathcal{V} = n < \infty$ and $p, q \in \mathbb{Z}_{\geq 0}$. Write $\mathbf{v}^{\otimes} := \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p$. Write $\boldsymbol{\varphi}^{\otimes} := \varphi_1 \otimes \cdots \otimes \varphi_q$.
Set $\boldsymbol{\eta} := (\eta_1, \ldots, \eta_p)$ and $\mathbf{x} := (\mathbf{x}_1, \ldots, \mathbf{x}_q)$. In an argument list, $(\boldsymbol{\eta}, \mathbf{x})$ denotes the concatenation of these tuples.
Set $P_{\mathbf{v}}(\boldsymbol{\eta}) := \prod_{a=1}^{p} \eta_a(\mathbf{v}_a)$ and $Q_{\boldsymbol{\varphi}}(\mathbf{x}) := \prod_{b=1}^{q} \varphi_b(\mathbf{x}_b)$. An empty tensor or product below equals 1.

(i) $\dim \mathcal{T}_q^p(\mathcal{V}) = n^{p+q}$.

(ii) If $p + q \geq 1$, a canonical isomorphism $\Theta_{p,q} : \mathcal{T}_q^p(\mathcal{V}) \xrightarrow{\sim} \mathcal{M}_{p,q}(\mathcal{V})$ is determined by $\Theta_{p,q}(\mathbf{v}^{\otimes} \otimes \boldsymbol{\varphi}^{\otimes})(\boldsymbol{\eta}, \mathbf{x}) = P_{\mathbf{v}}(\boldsymbol{\eta}) Q_{\boldsymbol{\varphi}}(\mathbf{x})$.

(iii) Suppose $p, q \geq 1$ and $(i, j) \in \{1, \ldots, p\} \times \{1, \ldots, q\}$. A subscript $\widehat{\imath}$ or $\widehat{\jmath}$ denotes omission from the corresponding tensor.
Set $\mathbf{z}_{ij} := \mathbf{v}_{\widehat{\imath}}^{\otimes} \otimes \boldsymbol{\varphi}_{\widehat{\jmath}}^{\otimes}$. A unique linear map $C_{ij} : \mathcal{T}_q^p(\mathcal{V}) \to \mathcal{T}_{q-1}^{p-1}(\mathcal{V})$ satisfies $C_{ij}(\mathbf{v}^{\otimes} \otimes \boldsymbol{\varphi}^{\otimes}) = \varphi_j(\mathbf{v}_i)\mathbf{z}_{ij}$.

(iv) If $n \geq 1$, an isomorphism $R : \mathcal{V} \otimes \mathcal{V}' \xrightarrow{\sim} \mathcal{L}(\mathcal{V})$ satisfies $R(\mathbf{v} \otimes \varphi)(\mathbf{x}) := \varphi(\mathbf{x})\mathbf{v}$. On $\mathcal{T}_1^1(\mathcal{V})$, $C_{11} = \operatorname{tr} \circ R$.

Proof.

(i) The equality $\dim \mathcal{V}' = n$ holds by Theorem 4.1.4(ii). If $p = q = 0$, then $\mathcal{T}_0^0(\mathcal{V}) = \mathbb{F}$ by Definition 5.3.5(iv).
Hence $\dim \mathcal{T}_0^0(\mathcal{V}) = 1 = n^0$, where $n^0 := 1$. If $p + q \geq 1$, then $\mathcal{T}_q^p(\mathcal{V})$ has p factors of dimension n and q factors of dimension n.
Its dimension is n^{p+q} by Theorem 5.3.4(vi) and Definition 5.3.5(ii).

(ii) Target of $\Theta_{p,q}$ is vector space by Theorem 5.3.4(ii). Displayed formula defines multilinear form in $\eta_1, \ldots, \eta_p, \mathbf{x}_1, \ldots, \mathbf{x}_q$.
Varying $\mathbf{v}_1, \ldots, \mathbf{v}_p, \varphi_1, \ldots, \varphi_q$ defines a multilinear assignment into that target.
Unique linear map $\Theta_{p,q}$ therefore exists by Theorem 5.3.4(ii). Set $\mathcal{X} := (\mathcal{V}')^{\otimes p} \otimes \mathcal{V}^{\otimes q}$.
The codomain is canonically isomorphic to $\mathcal{X}'$ by Theorem 5.3.4(ii). The domain has dimension n^{p+q} by part (i).
The equality $\dim \mathcal{X} = n^{p+q}$ holds by Theorem 5.3.4(vi). The codomain has dimension $\dim \mathcal{X}' = \dim \mathcal{X}$ by Theorem 4.1.4(ii).
If $n = 0$, this dimension is zero because $p + q \geq 1$, so $\Theta_{p,q}$ is an isomorphism.
Suppose $n \geq 1$. Choose a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ and its dual basis $(\varepsilon_1, \ldots, \varepsilon_n)$.
For $I = (i_1, \ldots, i_p)$, set $\mathbf{e}_I^{\otimes} := \mathbf{e}_{i_1} \otimes \cdots \otimes \mathbf{e}_{i_p}$. For $J = (j_1, \ldots, j_q)$, set $\varepsilon_J^{\otimes} := \varepsilon_{j_1} \otimes \cdots \otimes \varepsilon_{j_q}$. Set $\mathbf{b}_{I,J} := \mathbf{e}_I^{\otimes} \otimes \varepsilon_J^{\otimes}$.
The elements $\mathbf{b}_{I,J}$ form a basis by repeated application of Theorem 5.2.3(i).
For $K = (k_1, \ldots, k_p)$, set $\delta_{K,I} := \prod_{a=1}^{p} \delta_{k_a, i_a}$. For $L = (\ell_1, \ldots, \ell_q)$, set $\delta_{J,L} := \prod_{b=1}^{q} \delta_{j_b, \ell_b}$.

Set $\boldsymbol{\varepsilon}_K := (\varepsilon_{k_1}, \dots, \varepsilon_{k_p})$. Set $\mathbf{e}_L := (\mathbf{e}_{\ell_1}, \dots, \mathbf{e}_{\ell_q})$. The defining formula implies $\Theta_{p,q}(\mathbf{b}_{I,J})(\boldsymbol{\varepsilon}_K, \mathbf{e}_L) = \delta_{K,I}\delta_{J,L}$.
If $z \in \ker \Theta_{p,q}$, evaluation at $(\boldsymbol{\varepsilon}_I, \mathbf{e}_J)$ equals both zero and the coefficient of $\mathbf{b}_{I,J}$ in z.
Thus every coefficient is zero, so $\Theta_{p,q}$ is injective. It is an isomorphism by Corollary 1.7.6(ii).

(iii) Let Γ_{ij} be the rule in part (iii). In the selected variables $\mathbf{v}_i$ and φ_j, linearity follows from bilinearity of evaluation.
In every remaining variable, linearity holds by Theorem 5.3.4(i)(a).
Thus Γ_{ij} is multilinear, and a unique linear map C_{ij} with the required action exists by part (ii) of that theorem.

(iv) The rule $(\mathbf{v}, \varphi) \mapsto (\mathbf{x} \mapsto \varphi(\mathbf{x})\mathbf{v})$ is bilinear. Thus R exists by Theorem 5.1.6(ii).
Choose a basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ and its dual basis $(\varepsilon_1, \dots, \varepsilon_n)$. The map R sends $\mathbf{e}_a \otimes \varepsilon_b$ to the operator whose matrix is $E_{a,b}$.
The source elements form a basis by Theorem 5.2.3(i). The matrices $E_{a,b}$ form a basis by Proposition 3.1.5(ii).
Matrix representation identifies this basis with an operator basis by Theorem 3.4.5(i).

Hence R is an isomorphism. Write $\mathbf{v} = \sum_{a=1}^{n} x_a \mathbf{e}_a$. The diagonal entries of $[R(\mathbf{v} \otimes \varphi)]$ are $x_a\varphi(\mathbf{e}_a)$.

The identity $\operatorname{tr}(R(\mathbf{v} \otimes \varphi)) = \sum_{a=1}^{n} x_a\varphi(\mathbf{e}_a)$ holds by Definition 1.1.18(i).

Sum equals $\varphi(\mathbf{v}) = C_{11}(\mathbf{v} \otimes \varphi)$ by linearity and part (iii). Maps $\operatorname{tr} \circ R$ and C_{11} are linear by Proposition 1.1.1 and part (iii).
They agree on the spanning decomposable tensors by Theorem 5.3.4(i)(b), so they are equal.

Structural Commentary: Bases establish dimension and duality; evaluation defines contractions.
Key Insight: Contraction evaluates one vector–covector pair and preserves the remaining slots. □

Definition 5.3.7. *Contraction*
Let $\dim \mathcal{V} < \infty$ and let $p, q \in \mathbb{Z}_{>0}$. The maps C_{ij} in Proposition 5.3.6(iii) are the *contractions* of $\mathcal{T}^p_q(\mathcal{V})$.
The *standard contraction* is $C := C_{11}$.

Example 5.3.8. *Rank-One Operators and Trace*
Let $n := \dim \mathcal{V} \in \mathbb{Z}_{>0}$ and let $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ be a basis. Let $(\varepsilon_1, \dots, \varepsilon_n)$ be the dual basis.
By Proposition 5.3.6(iv), $R(\mathbf{e}_a \otimes \varepsilon_b)$ has matrix $E_{a,b}$ for $a, b \in \{1, \dots, n\}$. Its contraction records the trace: $C(\mathbf{e}_a \otimes \varepsilon_b) = \delta_{a,b} = \operatorname{tr}(E_{a,b})$.

6 Symmetric and Exterior Algebras

6.1 Symmetric and Exterior Powers

Definition 6.1.1. *Symmetric, Skew-Symmetric, and Alternating Maps*
Let $n \in \mathbb{Z}_{>0}$, let $\mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces, and set $P_n := \{(i,j) \in \{1, \dots, n\}^2 : i < j\}$. Let $f \in \operatorname{Mult}_{\mathbb{F}}(\mathcal{V}, \dots, \mathcal{V}; \mathcal{W})$, with n copies of $\mathcal{V}$.
For $\sigma \in \mathfrak{S}_n$, set $f^\sigma(\mathbf{v}_1, \dots, \mathbf{v}_n) := f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(n)})$. For $(i,j) \in P_n$, let s_{ij} transpose i and j.

(i) The map f is *symmetric* if $(\forall\,(i,j) \in P_n)\, f^{s_{ij}} = f$.
(ii) The map f is *skew-symmetric* if $(\forall\,(i,j) \in P_n)\, f^{s_{ij}} = -f$.
(iii) The map f is *alternating* if $(\forall\, \mathbf{v} \in \mathcal{V}^{\times n})[(\exists\,(i,j) \in P_n)\, \mathbf{v}_i = \mathbf{v}_j \Rightarrow f(\mathbf{v}) = 0]$.

Proposition 6.1.2. *Alternation, Skew-Symmetry, and Characteristic*
Let $n \in \mathbb{Z}_{>0}$, let $\mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces, and set $P_n := \{(i,j) \in \{1, \dots, n\}^2 : i < j\}$. Let $f \in \operatorname{Mult}_{\mathbb{F}}(\mathcal{V}, \dots, \mathcal{V}; \mathcal{W})$, with n copies of $\mathcal{V}$.

(i) If f is alternating, then f is skew-symmetric.
(ii) If $\operatorname{char}(\mathbb{F}) \neq 2$ and f is skew-symmetric, then f is alternating.
(iii) If $\operatorname{char}(\mathbb{F}) = 2$, then the symmetric and skew-symmetric conditions coincide.

Proof.

(i) Fix $(i,j) \in P_n$ and the entries outside slots i, j, and let $g : \mathcal{V}^2 \to \mathcal{W}$ be the resulting bilinear map.
Then $(\forall\, \mathbf{u} \in \mathcal{V})\, g(\mathbf{u}, \mathbf{u}) = 0$ by alternation. For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $0 = g(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v}) = g(\mathbf{u}, \mathbf{v}) + g(\mathbf{v}, \mathbf{u})$.
Hence $g(\mathbf{v}, \mathbf{u}) = -g(\mathbf{u}, \mathbf{v})$. The fixed data were arbitrary, so $(\forall\,(i,j) \in P_n)\, f^{s_{ij}} = -f$.

(ii) Fix $\mathbf{v} \in \mathcal{V}^{\times n}$ and $(i,j) \in P_n$ with $\mathbf{v}_i = \mathbf{v}_j$. Then $f(\mathbf{v}) = f^{s_{ij}}(\mathbf{v}) = -f(\mathbf{v})$ by skew-symmetry, whence $2f(\mathbf{v}) = 0$.
The scalar 2 is invertible in $\mathbb{F}$ by $\operatorname{char}(\mathbb{F}) \neq 2$, so $f(\mathbf{v}) = 0$.

(iii) The equality $-1 = 1$ holds in $\mathbb{F}$ by $\operatorname{char}(\mathbb{F}) = 2$, so the two conditions coincide.

Structural Commentary: Diagonal polarization establishes skew-symmetry; invertibility of 2 recovers alternation.
Key Insight: Characteristic 2 removes the sign distinction but not diagonal vanishing. □

Definition 6.1.3. *Permutation Action on Tensor Powers*
Let $\mathcal{V}$ be an $\mathbb{F}$-vector space and $p \in \mathbb{Z}_{\geq 0}$. Set $T_0 := \mathbb{F}$, $T_p := \mathcal{V}^{\otimes p}$ for $p > 0$, and $G_p := \mathfrak{S}_p$.
For $p = 0$, let $t_0 : \{()\} \to T_0$ satisfy $t_0(()) = 1$ and set $\lambda_{\mathrm{id}} := I_{T_0}$.
For $p > 0$, let $t_p : \mathcal{V}^{\times p} \to T_p$ be canonical and set $\lambda_\sigma := \tau_{\sigma^{-1}} \in \mathcal{L}(T_p)$ for $\sigma \in G_p$ by Theorem 5.3.4(iv).
Thus $\lambda_\sigma t_p(\mathbf{v}_1, \dots, \mathbf{v}_p) = t_p(\mathbf{v}_{\sigma^{-1}(1)}, \dots, \mathbf{v}_{\sigma^{-1}(p)})$.

Proposition 6.1.4. *Permutation Representation on Tensor Powers*
Let $\mathcal{V}$ be an $\mathbb{F}$-vector space and $p \in \mathbb{Z}_{\geq 0}$. Let T_p, G_p, and $(\lambda_\sigma)_{\sigma \in G_p}$ be as in Definition 6.1.3.

(i) $\lambda_{\mathrm{id}} = I_{T_p}$.
(ii) $(\forall\, \sigma, \tau \in G_p)\, \lambda_\sigma\lambda_\tau = \lambda_{\sigma\tau}$.

Proof. Both claims hold by Definition 6.1.3 when $p = 0$. Assume $p > 0$.

(i) $(\forall\, \mathbf{v} \in \mathcal{V}^{\times p})\, \lambda_{\mathrm{id}} t_p(\mathbf{v}) = t_p(\mathbf{v})$. Hence $\lambda_{\mathrm{id}} = I_{T_p}$ by Theorem 5.3.4(i)(b).

(ii) For $(\sigma, \tau, \mathbf{v}) \in G_p^2 \times \mathcal{V}^{\times p}$, $\lambda_\sigma\lambda_\tau t_p(\mathbf{v}) = t_p(\mathbf{v}_{\tau^{-1}\sigma^{-1}(1)}, \dots, \mathbf{v}_{\tau^{-1}\sigma^{-1}(p)}) = \lambda_{\sigma\tau} t_p(\mathbf{v})$.
Thus $(\forall\, \sigma, \tau \in G_p)\, \lambda_\sigma\lambda_\tau = \lambda_{\sigma\tau}$ by Theorem 5.3.4(i)(b).

Structural Commentary: Decomposable evaluation verifies each law; spanning extends it to T_p.
Key Insight: Inverse indexing converts factor permutations into a left action. □

Definition 6.1.5. *Symmetric and Exterior Powers*
Let $\mathcal{V}$ be an $\mathbb{F}$-vector space, let $p \in \mathbb{Z}_{\geq 0}$, and use the notation of Definition 6.1.3.
Set $P_p := \{(i,j) \in \{1, \dots, p\}^2 : i < j\}$ and $\Delta_p(\mathcal{V}) := \{\mathbf{v} \in \mathcal{V}^{\times p} : (\exists\,(i,j) \in P_p)\, \mathbf{v}_i = \mathbf{v}_j\}$.

(i) The *symmetric tensors* are $\mathrm{ST}^p(\mathcal{V}) := \{\mathbf{z} \in T_p : (\forall\, \sigma \in G_p)\, \lambda_\sigma \mathbf{z} = \mathbf{z}\}$.

(ii) The *antisymmetric tensors* are $\mathrm{AT}^p(\mathcal{V}) := \{\mathbf{z} \in T_p : (\forall\, \sigma \in G_p)\, \lambda_\sigma \mathbf{z} = \mathrm{sign}(\sigma)\mathbf{z}\}$.
(iii) Set $R_{\vee,p} := \mathrm{span}\{\lambda_\sigma \mathbf{z} - \mathbf{z} : \sigma \in G_p,\ \mathbf{z} \in T_p\}$.
(iv) Set $R_{\wedge,p} := \mathrm{span}\{t_p(\mathbf{v}) : \mathbf{v} \in \Delta_p(\mathcal{V})\}$.
(v) The *symmetric power* is $\mathrm{Sym}^p(\mathcal{V}) := T_p/R_{\vee,p}$. For the quotient map $q_{\vee,p}$, set $q_{\vee,p}t_p(\mathbf{v}) := \mathbf{v}_1 \vee \cdots \vee \mathbf{v}_p$.
(vi) The *exterior power* is $\bigwedge^p \mathcal{V} := T_p/R_{\wedge,p}$. For the quotient map $q_{\wedge,p}$, set $q_{\wedge,p}t_p(\mathbf{v}) := \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_p$.

For $p = 0$, each empty product denotes $1 \in \mathbb{F}$.

Theorem 6.1.6. *Universal Properties of Symmetric and Exterior Powers*
Let $\mathcal{V},\mathcal{U}$ be $\mathbb{F}$-vector spaces and $p \in \mathbb{Z}_{>0}$. Set $M := \mathrm{Mult}_{\mathbb{F}}(\mathcal{V},\dots,\mathcal{V};\mathcal{U})$, with p copies of $\mathcal{V}$.
Let $t_p : \mathcal{V}^{\times p} \to T_p$ be canonical. For $f \in M$ and $\sigma \in G_p$, set $f^\sigma(\mathbf{v}_1,\dots,\mathbf{v}_p) := f(\mathbf{v}_{\sigma(1)},\dots,\mathbf{v}_{\sigma(p)})$.
(i) $M_\vee := \{f \in M : (\forall\, \sigma \in G_p)\, f^\sigma = f\}$. Map $\Phi_\vee : \mathcal{L}(\mathrm{Sym}^p(\mathcal{V}),\mathcal{U}) \to M_\vee$ defined by $\Phi_\vee(T) := T \circ q_{\vee,p} \circ t_p$ is linear isomorphism.
(ii) $M_\wedge := \{f \in M : (\forall\, \mathbf{v} \in \Delta_p(\mathcal{V}))\, f(\mathbf{v}) = 0\}$. Map $\Phi_\wedge : \mathcal{L}(\bigwedge^p \mathcal{V},\mathcal{U}) \to M_\wedge$ defined by $\Phi_\wedge(T) := T \circ q_{\wedge,p} \circ t_p$ is linear isomorphism.

Proof.
(i) For $T \in \mathcal{L}(\mathrm{Sym}^p(\mathcal{V}),\mathcal{U})$, $(\forall\, \sigma \in G_p)\, q_{\vee,p}\lambda_\sigma = q_{\vee,p}$ by the definition of $R_{\vee,p}$. Hence $\Phi_\vee(T) \in M_\vee$.
Map $\Phi_\vee$ is linear by pointwise operations, linear composition. For $f \in M_\vee$, $(\exists!\, \widetilde{f} \in \mathcal{L}(T_p,\mathcal{U}))\, f = \widetilde{f} \circ t_p$ by Theorem 5.3.4(ii).
The identity $(\forall\, \sigma \in G_p)\, \widetilde{f}\lambda_\sigma = \widetilde{f}$ holds on decomposable tensors by symmetry and on T_p by Theorem 5.3.4(i)(b).
Hence $R_{\vee,p} \le \ker(\widetilde{f})$. Thus $(\exists!\, \overline{f} \in \mathcal{L}(\mathrm{Sym}^p(\mathcal{V}),\mathcal{U}))\, \widetilde{f} = \overline{f} \circ q_{\vee,p}$ by Theorem 2.2.11(i).
For $T \in \mathcal{L}(\mathrm{Sym}^p(\mathcal{V}),\mathcal{U})$, the construction applied to $\Phi_\vee(T)$ has $\widetilde{\Phi_\vee(T)} = T \circ q_{\vee,p}$. Thus $\overline{\Phi_\vee(T)} = T$ by uniqueness.
For $f \in M_\vee$, $\Phi_\vee(\overline{f}) = \overline{f} \circ q_{\vee,p} \circ t_p = \widetilde{f} \circ t_p = f$. Thus $\Phi_\vee$ is an isomorphism.
(ii) For $T \in \mathcal{L}(\bigwedge^p \mathcal{V},\mathcal{U})$, $\Phi_\wedge(T) \in M_\wedge$ by the definition of $R_{\wedge,p}$. The map $\Phi_\wedge$ is linear by pointwise operations and linear composition.
For $f \in M_\wedge$, $(\exists!\, \widetilde{f} \in \mathcal{L}(T_p,\mathcal{U}))\, f = \widetilde{f} \circ t_p$ by Theorem 5.3.4(ii).
The inclusion $R_{\wedge,p} \le \ker(\widetilde{f})$ holds by alternation. Thus $(\exists!\, \overline{f} \in \mathcal{L}(\bigwedge^p \mathcal{V},\mathcal{U}))\, \widetilde{f} = \overline{f} \circ q_{\wedge,p}$ by Theorem 2.2.11(i).
For $T \in \mathcal{L}(\bigwedge^p \mathcal{V},\mathcal{U})$, the construction applied to $\Phi_\wedge(T)$ has $\widetilde{\Phi_\wedge(T)} = T \circ q_{\wedge,p}$. Thus $\overline{\Phi_\wedge(T)} = T$ by uniqueness.
For $f \in M_\wedge$, $\Phi_\wedge(\overline{f}) = \overline{f} \circ q_{\wedge,p} \circ t_p = \widetilde{f} \circ t_p = f$. Thus $\Phi_\wedge$ is an isomorphism.

Structural Commentary: Tensor universality linearizes each map; quotient descent removes its relation subspace.
Key Insight: Kernel containment is the exact descent criterion for both tensor quotients. □

Theorem 6.1.7. *Dimensions of Symmetric and Exterior Powers*
Let $\dim \mathcal{V} = n \in \mathbb{Z}_{>0}$, let $p \in \mathbb{Z}_{\ge 0}$, and let $E = (\mathbf{e}_1,\dots,\mathbf{e}_n)$ be an ordered basis. If $p = 0$, set $I_\vee = I_\wedge := \{()\}$.
If $p > 0$, set $I_\vee := \{(i_1,\dots,i_p) \in \{1,\dots,n\}^p : i_1 \le \cdots \le i_p\}$ and $I_\wedge := \{(i_1,\dots,i_p) \in \{1,\dots,n\}^p : i_1 < \cdots < i_p\}$.
For $I = (i_1,\dots,i_p)$, set $\mathbf{e}_I^\vee := \mathbf{e}_{i_1} \vee \cdots \vee \mathbf{e}_{i_p}$ and $\mathbf{e}_I^\wedge := \mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_p}$, with empty products equal to $1 \in \mathbb{F}$.
Set $B_\vee := \{\mathbf{e}_I^\vee : I \in I_\vee\}$ and $B_\wedge := \{\mathbf{e}_I^\wedge : I \in I_\wedge\}$.
(i) The following symmetric-power conclusions hold.
(a) The set $B_\vee$ is a basis of $\mathrm{Sym}^p(\mathcal{V})$.
(b) $\dim \mathrm{Sym}^p(\mathcal{V}) = \#I_\vee = \binom{n+p-1}{p}$.
(ii) The following exterior-power conclusions hold, with $\binom{n}{p} := 0$ for $p > n$.
(a) The set $B_\wedge$ is a basis of $\bigwedge^p \mathcal{V}$.
(b) $\dim \bigwedge^p \mathcal{V} = \#I_\wedge = \binom{n}{p}$.

Proof. For $p > 0$, let $t_p : \mathcal{V}^{\times p} \to T_p$ be canonical and, for finite S, set $\mathcal{P}_p(S) := \{A \subseteq S : \#A = p\}$.
For $p > 0$, the tensor words $\mathbf{e}_{j_1} \otimes \cdots \otimes \mathbf{e}_{j_p}$ form a basis of T_p by repeated applications of Theorem 5.2.3(i).
(i)
(a) If $p = 0$, then $R_{\vee,0} = \{0\}$ and $B_\vee = \{1\}$, so $B_\vee$ is a basis.
Assume $p > 0$. Each tensor word is congruent modulo $R_{\vee,p}$ to its nondecreasing rearrangement; hence $B_\vee$ spans.
For $I \in I_\vee$, define $\chi_I \in T_p'$ on the tensor-word basis by $\chi_I(\mathbf{e}_{j_1} \otimes \cdots \otimes \mathbf{e}_{j_p}) = 1$ when $(j_1,\dots,j_p)$ is a rearrangement of I, and 0 otherwise. The inclusion $R_{\vee,p} \le \ker(\chi_I)$ holds by $(\forall\, \sigma \in G_p)\, \chi_I\lambda_\sigma = \chi_I$.
Thus $(\exists!\, \overline{\chi}_I \in (\mathrm{Sym}^p(\mathcal{V}))')\, \chi_I = \overline{\chi}_I \circ q_{\vee,p}$ by Theorem 2.2.11(i).
For $I, J \in I_\vee$, $\overline{\chi}_I(\mathbf{e}_J^\vee) = \delta_{I,J}$; hence $B_\vee$ is linearly independent and therefore a basis.
(b) If $p = 0$, then $\dim \mathrm{Sym}^0(\mathcal{V}) = \#I_\vee = 1 = \binom{n-1}{0}$ by Proposition 4.3.6(i).
For $p > 0$, the map $(i_1,\dots,i_p) \mapsto \{i_1, i_2+1, \dots, i_p+p-1\}$ is a bijection from $I_\vee$ to $\mathcal{P}_p(\{1,\dots,n+p-1\})$.
Inverse sends $\{a_1 < \cdots < a_p\}$ to $(a_1, a_2-1, \dots, a_p-p+1)$. Thus $\dim \mathrm{Sym}^p(\mathcal{V}) = \#I_\vee = \binom{n+p-1}{p}$ by (a), Proposition 4.3.4.

(ii)
(a) If $p = 0$, then $R_{\wedge,0} = \{0\}$ and $B_\wedge = \{1\}$, so $B_\wedge$ is a basis.
Assume $p > 0$. If $p \ge 2$, fix two slots and all other factors, and write $w(\mathbf{u},\mathbf{v})$ for the resulting tensor word.
Then $w(\mathbf{u},\mathbf{v}) + w(\mathbf{v},\mathbf{u}) = w(\mathbf{u}+\mathbf{v},\mathbf{u}+\mathbf{v}) - w(\mathbf{u},\mathbf{u}) - w(\mathbf{v},\mathbf{v}) \in R_{\wedge,p}$. Thus a transposition changes sign modulo $R_{\wedge,p}$.
Repeated basis words lie in $R_{\wedge,p}$ by definition, and every other basis word is congruent up to sign to an increasing word; hence $B_\wedge$ spans. Let $E' = (\varepsilon_1,\dots,\varepsilon_n)$ be the dual basis. For $I = (i_1,\dots,i_p) \in I_\wedge$, set $a_I(\mathbf{v}_1,\dots,\mathbf{v}_p) := \det[\varepsilon_{i_r}(\mathbf{v}_s)]_{r,s=1}^p$.
The form a_I is alternating by Theorem 3.8.2(ii)–(iv). Thus $(\exists!\, \ell_I \in (\bigwedge^p \mathcal{V})')\, a_I = \ell_I \circ q_{\wedge,p} \circ t_p$ by Theorem 6.1.6(ii).
For $I, J \in I_\wedge$, $\ell_I(\mathbf{e}_J^\wedge) = \delta_{I,J}$; hence $B_\wedge$ is linearly independent and therefore a basis.
(b) If $p = 0$, then $\dim \bigwedge^0 \mathcal{V} = \#I_\wedge = 1 = \binom{n}{0}$ by Proposition 4.3.6(i).
For $0 < p \le n$, the map $(i_1,\dots,i_p) \mapsto \{i_1,\dots,i_p\}$ is a bijection from $I_\wedge$ to $\mathcal{P}_p(\{1,\dots,n\})$.
Thus $\dim \bigwedge^p \mathcal{V} = \#I_\wedge = \binom{n}{p}$ by (a) and Proposition 4.3.4.
If $p > n$, then $I_\wedge = \emptyset$ and $\dim \bigwedge^p \mathcal{V} = 0 = \binom{n}{p}$ by (a) and the stated convention.

Structural Commentary: Orbit-constant functionals separate symmetric words; determinants separate increasing exterior words.
Key Insight: Dual separators turn spanning normal forms into bases. □

Proposition 6.1.8. *Symmetrization and Antisymmetrization in Characteristic Zero*
Let $\mathcal{V}$ be $\mathbb{F}$-vector space, $p \in \mathbb{Z}_{\geq 0}$, $\operatorname{char}(\mathbb{F}) = 0$. The scalar $p!$ is nonzero in $\mathbb{F}$.
Define $S := \frac{1}{p!} \sum_{\sigma \in G_p} \lambda_\sigma \in \mathcal{L}(T_p)$; $A := \frac{1}{p!} \sum_{\sigma \in G_p} \operatorname{sign}(\sigma) \lambda_\sigma \in \mathcal{L}(T_p)$.

(i) $\operatorname{im} S = \mathrm{ST}^p(\mathcal{V})$.
(ii) $S^2 = S$.
(iii) $\operatorname{im} A = \mathrm{AT}^p(\mathcal{V})$.
(iv) $A^2 = A$.

Proof.

(i) For $\tau \in G_p$, $\lambda_\tau S = \frac{1}{p!} \sum_{\sigma \in G_p} \lambda_{\tau\sigma} = \frac{1}{p!} \sum_{\rho \in G_p} \lambda_\rho = S$ by the reindexing $\rho = \tau\sigma$. Hence $\operatorname{im} S \subseteq \mathrm{ST}^p(\mathcal{V})$.
If $\mathbf{z} \in \mathrm{ST}^p(\mathcal{V})$, then $S\mathbf{z} = \frac{1}{p!} \sum_{\sigma \in G_p} \mathbf{z} = \mathbf{z}$ by Proposition 4.3.3(iii). Thus $\mathrm{ST}^p(\mathcal{V}) \subseteq \operatorname{im} S$; $\operatorname{im} S = \mathrm{ST}^p(\mathcal{V})$.

(ii) For $\mathbf{x} \in T_p$, $S\mathbf{x} \in \mathrm{ST}^p(\mathcal{V})$ by (i), so $S^2\mathbf{x} = S\mathbf{x}$ by the defining average and Proposition 4.3.3(iii). Hence $S^2 = S$.

(iii) For $p > 0$, let $(\mathbf{e}_1, \ldots, \mathbf{e}_p)$ be the standard basis of $\mathbb{F}^p$, and let P_σ have columns $\mathbf{e}_{\sigma(1)}, \ldots, \mathbf{e}_{\sigma(p)}$ for $\sigma \in G_p$.
Then $\det P_\sigma = \operatorname{sign}(\sigma)$ by Definition 3.8.1(ii), and $P_\sigma P_\tau = P_{\sigma\tau}$.
Hence $\operatorname{sign}(\sigma\tau) = \operatorname{sign}(\sigma)\operatorname{sign}(\tau)$ by Theorem 3.8.2(vi). The identity also holds for $p = 0$.
For $\tau \in G_p$, $\lambda_\tau A = \frac{1}{p!} \sum_{\sigma \in G_p} \operatorname{sign}(\sigma) \lambda_{\tau\sigma} = \operatorname{sign}(\tau) A$ by the reindexing $\rho = \tau\sigma$. Hence $\operatorname{im} A \subseteq \mathrm{AT}^p(\mathcal{V})$.
If $\mathbf{z} \in \mathrm{AT}^p(\mathcal{V})$, then $A\mathbf{z} = \frac{1}{p!} \sum_{\sigma \in G_p} \operatorname{sign}(\sigma)^2 \mathbf{z} = \mathbf{z}$ by Proposition 4.3.3(iii). Thus $\mathrm{AT}^p(\mathcal{V}) \subseteq \operatorname{im} A$; $\operatorname{im} A = \mathrm{AT}^p(\mathcal{V})$.

(iv) For $\mathbf{x} \in T_p$, $A\mathbf{x} \in \mathrm{AT}^p(\mathcal{V})$ by (iii), so $A^2\mathbf{x} = A\mathbf{x}$ by the defining average and Proposition 4.3.3(iii). Hence $A^2 = A$.

Structural Commentary: Reindexed character averages identify the images; fixed-point identities establish idempotence.
Key Insight: Normalisation by $p!$ turns each character average into a projection. □

Definition 6.1.9. *Tensor, Symmetric, and Exterior Algebras*
Let $\mathcal{V}$ be an $\mathbb{F}$-vector space.

(i) The *tensor algebra* is $\mathcal{T}(\mathcal{V}) := \bigoplus_{p \geq 0} T_p$. Unit is $1 \in T_0 = \mathbb{F}$, homogeneous multiplication is tensor concatenation.

(ii) The *symmetric algebra* is $\mathrm{Sym}(\mathcal{V}) := \bigoplus_{p \geq 0} \mathrm{Sym}^p(\mathcal{V})$. For $\mathbf{x} \in T_p$ and $\mathbf{y} \in T_q$, set $q_{\vee,p}(\mathbf{x}) \vee q_{\vee,q}(\mathbf{y}) := q_{\vee,p+q}(\mathbf{x} \otimes \mathbf{y})$.

(iii) The *exterior algebra* is $\bigwedge \mathcal{V} := \bigoplus_{p \geq 0} \bigwedge^p \mathcal{V}$. For $\mathbf{x} \in T_p$ and $\mathbf{y} \in T_q$, set $q_{\wedge,p}(\mathbf{x}) \wedge q_{\wedge,q}(\mathbf{y}) := q_{\wedge,p+q}(\mathbf{x} \otimes \mathbf{y})$.

Proposition 6.1.10. *Graded Algebra Laws*
Let $\mathcal{V}$ be an $\mathbb{F}$-vector space, with the three graded constructions and products from Definition 6.1.9.

(i) Tensor concatenation makes $\mathcal{T}(\mathcal{V})$ an associative unital graded $\mathbb{F}$-algebra.
(ii) The symmetric product is well-defined and bilinear.
(iii) The symmetric product makes $\mathrm{Sym}(\mathcal{V})$ an associative unital graded $\mathbb{F}$-algebra.
(iv) The exterior product is well-defined and bilinear.
(v) The exterior product makes $\bigwedge \mathcal{V}$ an associative unital graded $\mathbb{F}$-algebra.

Proof.

(i) Tensor concatenation is bilinear and associative under the canonical identifications of iterated tensor products.
Its unit is $1 \in T_0$, and $(\forall\, p, q \in \mathbb{Z}_{\geq 0})\, T_p T_q \subseteq T_{p+q}$ by construction.

(ii) Let $p, q \in \mathbb{Z}_{\geq 0}$. Extend $\sigma \in G_p$ to $\widehat{\sigma} \in G_{p+q}$ by fixing the last q indices.
Extend $\tau \in G_q$ to $\widehat{\tau} \in G_{p+q}$ by fixing the first p indices.
For $\mathbf{x} \in T_p$ and $\mathbf{y} \in T_q$, $(\lambda_\sigma \mathbf{x} - \mathbf{x}) \otimes \mathbf{y} = \lambda_{\widehat{\sigma}}(\mathbf{x} \otimes \mathbf{y}) - \mathbf{x} \otimes \mathbf{y} \in R_{\vee,p+q}$.
Similarly, $\mathbf{x} \otimes (\lambda_\tau \mathbf{y} - \mathbf{y}) = \lambda_{\widehat{\tau}}(\mathbf{x} \otimes \mathbf{y}) - \mathbf{x} \otimes \mathbf{y} \in R_{\vee,p+q}$.
Hence $R_{\vee,p} \otimes T_q + T_p \otimes R_{\vee,q} \subseteq R_{\vee,p+q}$.
The quotient formula is therefore independent of representatives, and bilinearity descends from concatenation.

(iii) Let $p, q, r \in \mathbb{Z}_{\geq 0}$ and $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in T_p \times T_q \times T_r$. Set $\mathbf{a} := q_{\vee,p}(\mathbf{x})$, $\mathbf{b} := q_{\vee,q}(\mathbf{y})$, and $\mathbf{c} := q_{\vee,r}(\mathbf{z})$.
Then $(\mathbf{a} \vee \mathbf{b}) \vee \mathbf{c} = q_{\vee,p+q+r}((\mathbf{x} \otimes \mathbf{y}) \otimes \mathbf{z}) = q_{\vee,p+q+r}(\mathbf{x} \otimes (\mathbf{y} \otimes \mathbf{z})) = \mathbf{a} \vee (\mathbf{b} \vee \mathbf{c})$.
The identities $1 \vee \mathbf{a} = \mathbf{a} = \mathbf{a} \vee 1$ hold by the tensor unit identifications.
For $\mathbf{a} \in \mathrm{Sym}^p(\mathcal{V})$ and $\mathbf{b} \in \mathrm{Sym}^q(\mathcal{V})$, $\mathbf{a} \vee \mathbf{b} \in \mathrm{Sym}^{p+q}(\mathcal{V})$ by definition.

(iv) Let $p, q \in \mathbb{Z}_{\geq 0}$. For $(\mathbf{v}, \mathbf{w}) \in \mathcal{V}^{\times p} \times \mathcal{V}^{\times q}$, $t_p(\mathbf{v}) \otimes t_q(\mathbf{w}) \in R_{\wedge,p+q}$ if $\mathbf{v} \in \Delta_p(\mathcal{V})$ or $\mathbf{w} \in \Delta_q(\mathcal{V})$. For $k = 0$, T_k is spanned by $t_k(())$.
For $k > 0$, decomposable tensors span T_k by Theorem 5.3.4(i)(b). Hence $R_{\wedge,p} \otimes T_q + T_p \otimes R_{\wedge,q} \subseteq R_{\wedge,p+q}$.
The quotient formula is therefore independent of representatives, and bilinearity descends from concatenation.

(v) Let $p, q, r \in \mathbb{Z}_{\geq 0}$ and $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in T_p \times T_q \times T_r$ Set $\mathbf{a} := q_{\wedge,p}(\mathbf{x})$, $\mathbf{b} := q_{\wedge,q}(\mathbf{y})$, and $\mathbf{c} := q_{\wedge,r}(\mathbf{z})$.
Then $(\mathbf{a} \wedge \mathbf{b}) \wedge \mathbf{c} = q_{\wedge,p+q+r}((\mathbf{x} \otimes \mathbf{y}) \otimes \mathbf{z}) = q_{\wedge,p+q+r}(\mathbf{x} \otimes (\mathbf{y} \otimes \mathbf{z})) = \mathbf{a} \wedge (\mathbf{b} \wedge \mathbf{c})$.
The identities $1 \wedge \mathbf{a} = \mathbf{a} = \mathbf{a} \wedge 1$ hold by the tensor unit identifications. For $\mathbf{a} \in \bigwedge^p \mathcal{V}$ and $\mathbf{b} \in \bigwedge^q \mathcal{V}$, $\mathbf{a} \wedge \mathbf{b} \in \bigwedge^{p+q} \mathcal{V}$ by definition.

Structural Commentary: Block-relation stability descends concatenation; tensor identities transfer the algebra laws.
Key Insight: Adjoining tensor factors preserves both relation subspaces. □

6.2 Determinants from Exterior Powers

Corollary 6.2.1. *Determinant from the Exterior Universal Property*
Let $\dim \mathcal{V} = n \in \mathbb{Z}_{>0}$, $E = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be ordered basis. Set $\Omega_E := \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n$.
Define $A_E : \mathcal{V}^{\times n} \to \mathrm{Mat}(n, n, \mathbb{F})$ by $A_E(\mathbf{v}) := ([\mathbf{v}_j]_i)_{i,j=1}^n$ for $\mathbf{v} = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

(i) $(\exists!\, \ell_E \in (\bigwedge^n \mathcal{V})')\, \ell_E(\Omega_E) = 1$.
(ii) Define $d_E : \mathcal{V}^{\times n} \to \mathbb{F}$ by $d_E(\mathbf{v}) := \ell_E(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n)$. The map d_E is the unique alternating form satisfying $d_E(E) = 1$.
(iii) $(\forall\, \mathbf{v} \in \mathcal{V}^{\times n})\, d_E(\mathbf{v}) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sign}(\sigma) \prod_{j=1}^n [\mathbf{v}_j]_{\sigma(j)} = \det A_E(\mathbf{v})$.

Proof.
(i) Singleton $\{\Omega_E\}$ is basis of $\bigwedge^n \mathcal{V}$ by Theorem 6.1.7(ii)(a). A unique $\ell_E \in (\bigwedge^n \mathcal{V})'$ satisfies $\ell_E(\Omega_E) = 1$ by Proposition 1.5.8.
(ii) The form d_E is alternating by Theorem 6.1.6(ii), $d_E(E) = \ell_E(\Omega_E) = 1$. Let $g : \mathcal{V}^{\times n} \to \mathbb{F}$ be alternating with $g(E) = 1$.
A unique $\ell \in (\bigwedge^n \mathcal{V})'$ satisfies $g(\mathbf{v}) = \ell(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n)$ by Theorem 6.1.6(ii).
$\ell(\Omega_E) = g(E) = 1$, so $\ell = \ell_E$ by (i) and $g = d_E$ by the two factorizations.
(iii) Define $g : \mathcal{V}^{\times n} \to \mathbb{F}$ by $g(\mathbf{v}) := \det A_E(\mathbf{v})$. Map g is multilinear and alternating by coordinate linearity and Theorem 3.8.2(ii),(iv).
The identity $g(E) = \det I_n = 1$ holds by Theorem 3.8.2(i). Hence $g = d_E$ by (ii).
The permutation identity in (iii) holds by Definition 3.8.1(ii).

Structural Commentary: Top-degree normalization fixes ℓ_E; exterior universality transfers uniqueness.
Key Insight: One basis value determines a normalized top-degree alternating form. □

Corollary 6.2.2. *Classical Laws for the Exterior Determinant*
Let $n \in \mathbb{Z}_{>0}$ and $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$. For the standard basis E of $\mathbb{F}^n$, define $d : \mathrm{Mat}(n, n, \mathbb{F}) \to \mathbb{F}$ by $d(C) := d_E(C_{\cdot,1}, \ldots, C_{\cdot,n})$.
(i) $d(A^t) = d(A)$.
(ii) $d(AB) = d(A)d(B)$.
(iii) $A \in \mathrm{GL}(n, \mathbb{F}) \iff d(A) \neq 0$.

Proof. The identity $(\forall\, C \in \mathrm{Mat}(n, n, \mathbb{F}))\, d(C) = \det C$ holds by Corollary 6.2.1(iii).
(i) $d(A^t) = \det(A^t) = \det A = d(A)$ by Theorem 3.8.2(ix).
(ii) $d(AB) = \det(AB) = \det(A)\det(B) = d(A)d(B)$ by Theorem 3.8.2(vi).
(iii) $(\Rightarrow)$ $A \in \mathrm{GL}(n, \mathbb{F}) \Rightarrow \det A \neq 0 \Rightarrow d(A) \neq 0$ by Theorem 3.8.2(vii) and $d(A) = \det A$.
(iii) $(\Leftarrow)$ $d(A) \neq 0 \Rightarrow \det A \neq 0 \Rightarrow A \in \mathrm{GL}(n, \mathbb{F})$ by $d(A) = \det A$ and Theorem 3.8.2(vii).

Structural Commentary: The identity $d = \det$ transfers transpose, product, and invertibility laws.
Key Insight: Exterior normalization and the permutation determinant define the same scalar. □

6.3 Multilinear Algebra Revisited

7 Chain Complexes and Homology

7.1 Chain Complexes

Throughout this subsection, R denotes a nonzero commutative ring with identity 1_R.
All modules are R-modules.

Definition 7.1.1. *Chain Complex*
A *chain complex* $C_\bullet = (C_n, d_n^C)_{n\in\mathbb{Z}}$ consists of R-modules C_n. Its *differentials* are homomorphisms $d_n^C : C_n \to C_{n-1}$.
They satisfy $(\forall\, n \in \mathbb{Z})\, d_{n-1}^C \circ d_n^C = 0$.

Proposition 7.1.2. *Successive Differentials*
Let $C_\bullet$ be a chain complex. Then $(\forall\, n \in \mathbb{Z})$ $\operatorname{im} d_{n+1}^C \leq \ker d_n^C$.

Proof. Fix $n \in \mathbb{Z}$ and $y \in \operatorname{im} d_{n+1}^C$. Choose $x \in C_{n+1}$ with $y = d_{n+1}^C(x)$ by definition of the image.
Then $d_n^C(y) = d_n^C d_{n+1}^C(x) = 0$ by Definition 7.1.1, so $y \in \ker d_n^C$.
Structural Commentary: Differential composition sends image representatives into the next kernel.
Key Insight: Consecutive differentials compose to zero. □

Definition 7.1.3. *Chain Maps and Chain Isomorphisms*
Let $C_\bullet, D_\bullet$ be chain complexes.
(i) A degreewise homomorphism $f_\bullet : C_\bullet \to D_\bullet$ is a family $(f_n)_{n\in\mathbb{Z}}$ with $f_n \in \mathrm{Hom}_R(C_n, D_n)$.
It is a *chain map* if $(\forall\, n \in \mathbb{Z})\, d_n^D f_n = f_{n-1} d_n^C$.
(ii) A chain map $g_\bullet : D_\bullet \to C_\bullet$ is an *inverse* of $f_\bullet$ if $g_\bullet f_\bullet = \mathrm{id}_{C_\bullet}$ and $f_\bullet g_\bullet = \mathrm{id}_{D_\bullet}$.
The map $f_\bullet$ is a *chain isomorphism* if it has an inverse.

Proposition 7.1.4. *Composition and Degreewise Inversion*
Let $C_\bullet, D_\bullet, E_\bullet$ be chain complexes.
(i) $(\mathrm{id}_{C_n})_{n\in\mathbb{Z}}$ *is a chain map* $C_\bullet \to C_\bullet$.
(ii) Let $f_\bullet : C_\bullet \to D_\bullet$ and $g_\bullet : D_\bullet \to E_\bullet$ be chain maps. Then $(g_n f_n)_{n\in\mathbb{Z}}$ is a chain map $C_\bullet \to E_\bullet$.
(iii) Let $f_\bullet : C_\bullet \to D_\bullet$ be a chain map.
Then $f_\bullet$ is a chain isomorphism if and only if $(\forall\, n \in \mathbb{Z})$ f_n is an R-module isomorphism.

Proof.
(i) $(\forall\, n \in \mathbb{Z})\, d_n^C \mathrm{id}_{C_n} = \mathrm{id}_{C_{n-1}} d_n^C = d_n^C$. Identity family is chain map by Definition 7.1.3(i).
(ii) For $n \in \mathbb{Z}$, $d_n^E g_n f_n = g_{n-1} d_n^D f_n = g_{n-1} f_{n-1} d_n^C$ by the chain-map identities for $g_\bullet, f_\bullet$; $(g_n f_n)_{n\in\mathbb{Z}}$ is a chain map.
(iii) $(\Rightarrow)$ Let $g_\bullet$ be an inverse of $f_\bullet$. Then $(\forall\, n \in \mathbb{Z})\, g_n f_n = \mathrm{id}_{C_n}$ and $f_n g_n = \mathrm{id}_{D_n}$ by Definition 7.1.3(ii).
Thus each f_n is an R-module isomorphism.
(iii) $(\Leftarrow)$ Assume each f_n is an R-module isomorphism and set $g_n := f_n^{-1}$.
Fix $n \in \mathbb{Z}$, $(a, b) \in R^2$, and $(y, z) \in D_n^2$. For $x = g_n(y)$ and $w = g_n(z)$, $f_n(ax + bw) = ay + bz$.
Hence $g_n(ay + bz) = ax + bw = a g_n(y) + b g_n(z)$ by bijectivity of f_n, so $g_n \in \mathrm{Hom}_R(D_n, C_n)$.
$d_n^C g_n = g_{n-1} f_{n-1} d_n^C g_n = g_{n-1} d_n^D f_n g_n = g_{n-1} d_n^D$ by the chain-map identity for $f_\bullet$.
Thus $g_\bullet$ is a chain map, and $g_\bullet f_\bullet = \mathrm{id}_{C_\bullet}$ and $f_\bullet g_\bullet = \mathrm{id}_{D_\bullet}$ degreewise.

Structural Commentary: Commuting squares persist under composition and componentwise inversion.
Key Insight: Inverting every component reverses the chain-map square. □

Definition 7.1.5. *Subcomplex*
Let $C_\bullet$ be a chain complex. A *subcomplex* $S_\bullet \leq C_\bullet$ consists of submodules $S_n \leq C_n$ indexed by $n \in \mathbb{Z}$.
They satisfy $(\forall\, n \in \mathbb{Z})\, d_n^C(S_n) \subseteq S_{n-1}$, and their differentials are $d_n^S := d_n^C|_{S_n}$.

Proposition 7.1.6. *Kernel and Image Subcomplexes*
Let $f_\bullet : C_\bullet \to D_\bullet$ be a chain map.

(i) $(\ker f_n)_{n\in\mathbb{Z}}$ *is a subcomplex of* $C_\bullet$.
(ii) $(\operatorname{im} f_n)_{n\in\mathbb{Z}}$ *is a subcomplex of* $D_\bullet$.

Proof.

(i) Each $\ker f_n \le C_n$ by Proposition 1.4.2(ii). Fix $x \in \ker f_n$; $f_{n-1}(d_n^C x) = d_n^D f_n(x) = 0$ by Definition 7.1.3(i), so $d_n^C x \in \ker f_{n-1}$.
(ii) Each $\operatorname{im} f_n \le D_n$ by Proposition 1.4.2(iii). Fix $y \in \operatorname{im} f_n$. Choose $x \in C_n$ with $y = f_n(x)$ by the definition of the image. Then $d_n^D(y) = f_{n-1}(d_n^C x) \in \operatorname{im} f_{n-1}$ by Definition 7.1.3(i).

Structural Commentary: Commuting squares restrict to kernels and transport images.
Key Insight: Commuting with differentials preserves both vanishing and representability. □

Proposition 7.1.7. *Quotient Complex*
Let $C_\bullet$ be a chain complex and let $S_\bullet \le C_\bullet$ be a subcomplex. Set $(C/S)_n := C_n/S_n$.
For $n \in \mathbb{Z}$ and $c \in C_n$, set $\overline{d}_n(c + S_n) := d_n^C(c) + S_{n-1}$.

(i) Each $\overline{d}_n : (C/S)_n \to (C/S)_{n-1}$ *is a well-defined R-homomorphism.*
(ii) $C_\bullet/S_\bullet := ((C/S)_n, \overline{d}_n)_{n\in\mathbb{Z}}$ *is a chain complex.*
(iii) The quotient maps $\pi_n : C_n \twoheadrightarrow C_n/S_n$ form a chain map $\pi_\bullet : C_\bullet \to C_\bullet/S_\bullet$.
(iv) $(\forall\, n \in \mathbb{Z})\ \ker \pi_n = S_n$.

Proof.

(i) If $c + S_n = c' + S_n$, then $c - c' \in S_n$ by Proposition 1.5.2(i). Thus $d_n^C(c - c') \in S_{n-1}$ by Definition 7.1.5. Hence $d_n^C(c) + S_{n-1} = d_n^C(c') + S_{n-1}$ by Proposition 1.5.2(i). Hence $\overline{d}_n$ is well-defined. Fix $(a, b, c, c') \in R^2 \times C_n^2$. $\overline{d}_n\big(a(c + S_n) + b(c' + S_n)\big) = a\overline{d}_n(c + S_n) + b\overline{d}_n(c' + S_n)$ by R-linearity of d_n^C. Thus $\overline{d}_n$ is R-linear.
(ii) For $n \in \mathbb{Z}$ and $c \in C_n$, $\overline{d}_{n-1}\overline{d}_n(c + S_n) = d_{n-1}^C d_n^C(c) + S_{n-2} = S_{n-2}$ by Definition 7.1.1. Hence $\overline{d}_{n-1}\overline{d}_n = 0$.
(iii) For $n \in \mathbb{Z}$ and $c \in C_n$, $\overline{d}_n \pi_n(c) = d_n^C(c) + S_{n-1} = \pi_{n-1} d_n^C(c)$. Thus $\pi_\bullet$ is a chain map by Definition 7.1.3(i).
(iv) $(\forall\, n \in \mathbb{Z})\ \ker \pi_n = S_n$ by Proposition 1.5.2(iii).

Structural Commentary: Subcomplex stability descends differentials and preserves projection kernels.
Key Insight: Differential images of equivalent representatives remain equivalent. □

Definition 7.1.8. *Exact Sequences and Exact Complexes*

(i) Homomorphisms $M' \xrightarrow{u} M \xrightarrow{v} M''$ are *exact at M* if $\operatorname{im} u = \ker v$.
(ii) A sequence of modules and homomorphisms is *exact* if it is exact at each module. An exact sequence $0 \to A \xrightarrow{u} B \xrightarrow{v} C \to 0$ is *short exact*.
(iii) A chain complex $C_\bullet$ is *exact* if $(\forall\, n \in \mathbb{Z})\ \operatorname{im} d_{n+1}^C = \ker d_n^C$.
(iv) Let $u_\bullet : A_\bullet \to B_\bullet$ and $v_\bullet : B_\bullet \to C_\bullet$ be chain maps.
The sequence $0 \to A_\bullet \xrightarrow{u_\bullet} B_\bullet \xrightarrow{v_\bullet} C_\bullet \to 0$ is *short exact* if $(\forall\, n \in \mathbb{Z})$, $0 \to A_n \xrightarrow{u_n} B_n \xrightarrow{v_n} C_n \to 0$ is exact.

Proposition 7.1.9. *Short Exact Sequence Criterion*
Let $u \in \operatorname{Hom}_R(A, B)$ and $v \in \operatorname{Hom}_R(B, C)$. The following conditions are equivalent:

(i) The sequence $0 \to A \xrightarrow{u} B \xrightarrow{v} C \to 0$ is exact.
(ii) The map u is a monomorphism, $\operatorname{im} u = \ker v$, and v is an epimorphism.

Proof.

(i) $\Rightarrow$ *(ii)* Assume (i). Exactness at A implies $\ker u = \operatorname{im}(0 \to A) = \{0_A\}$. Thus u is a monomorphism by Proposition 1.4.2(iv). Exactness at B is $\operatorname{im} u = \ker v$. Exactness at C implies $\operatorname{im} v = \ker(C \to 0) = C$, so v is an epimorphism by Definition 1.4.1(iv).
(ii) $\Rightarrow$ *(i)* Assume (ii). Then $\operatorname{im}(0 \to A) = \{0_A\} = \ker u$ by Proposition 1.4.2(iv). Middle equality is an assumption, and $\operatorname{im} v = C = \ker(C \to 0)$ by surjectivity of v; sequence is exact by Definition 7.1.8(i)–(ii).

Structural Commentary: Endpoints test injectivity and surjectivity; the middle compares image and kernel.
Key Insight: Zero endpoints convert exactness into kernel and image tests. □

Corollary 7.1.10. *Degreewise Criterion for Short Exact Complexes*
Let $u_\bullet : A_\bullet \to B_\bullet$ and $v_\bullet : B_\bullet \to C_\bullet$ be chain maps. The following conditions are equivalent:

(i) The sequence $0 \to A_\bullet \xrightarrow{u_\bullet} B_\bullet \xrightarrow{v_\bullet} C_\bullet \to 0$ is short exact.
(ii) $(\forall\, n \in \mathbb{Z})$, u_n is a monomorphism, $\operatorname{im} u_n = \ker v_n$, and v_n is an epimorphism.

Proof.

(i) $\Rightarrow$ *(ii)* Assume (i) and fix $n \in \mathbb{Z}$. The component sequence is exact by Definition 7.1.8(iv). Thus u_n is a monomorphism, $\operatorname{im} u_n = \ker v_n$, and v_n is an epimorphism by Proposition 7.1.9.
(ii) $\Rightarrow$ *(i)* Assume (ii) and fix $n \in \mathbb{Z}$. Then $0 \to A_n \xrightarrow{u_n} B_n \xrightarrow{v_n} C_n \to 0$ is exact by Proposition 7.1.9. Hence the sequence of chain maps is short exact by Definition 7.1.8(iv).

Structural Commentary: The module criterion transfers short exactness degree by degree.
Key Insight: Short exactness of complexes is componentwise. □

Example 7.1.11. *An Exact Integer Complex*
Take $R = \mathbb{Z}$. Let $u : \mathbb{Z} \to \mathbb{Z}$, $u(n) := 2n$, $\pi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ be quotient map. Then u is injective, $\operatorname{im} u = 2\mathbb{Z} = \ker \pi$, and π is surjective. Hence $0 \to \mathbb{Z} \xrightarrow{u} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \to 0$ is short exact by Proposition 7.1.9. Set $C_2 = C_1 = \mathbb{Z}$, $C_0 = \mathbb{Z}/2\mathbb{Z}$, and $(\forall\, n \in \mathbb{Z} \setminus \{0, 1, 2\})\, C_n = 0$. Set $d_2^C = u$, $d_1^C = \pi$, and all other differentials equal to zero.
Three checks above and the zero terms imply $(\forall\, n \in \mathbb{Z})\ \operatorname{im} d_{n+1}^C = \ker d_n^C$. Hence $C_\bullet$ is an exact chain complex by Definition 7.1.8(iii).

7.2 Homology

Throughout this subsection, R denotes a nonzero commutative ring with identity 1_R. All modules are R-modules.

Definition 7.2.1. *Cycles, Boundaries, and Homology*
Let $C_\bullet$ be a chain complex and let $n \in \mathbb{Z}$.

(i) The *n-cycle module* is $Z_n(C) := \ker d_n^C$.
(ii) The *n-boundary module* is $B_n(C) := \operatorname{im} d_{n+1}^C$.
(iii) $B_n(C) \leq Z_n(C)$ by Proposition 7.1.2. The *nth homology module* is $H_n(C) := Z_n(C)/B_n(C)$. For $z \in Z_n(C)$, $[z]_C := z + B_n(C)$.

Example 7.2.2. *Homology of a Two-Term Complex*
Take $R = \mathbb{Z}$ and $m \in \mathbb{N}$. Set $C_1 = C_0 = \mathbb{Z}$ and $(\forall\, n \in \mathbb{Z} \setminus \{0,1\})\, C_n = 0$. Set $d_1^C(k) := mk$ and all other differentials equal to zero.
Every consecutive composite is zero, so $C_\bullet$ is a chain complex by Definition 7.1.1.
Then $Z_1(C) = B_1(C) = \{0\}$, $Z_0(C) = \mathbb{Z}$, and $B_0(C) = m\mathbb{Z}$. Hence $H_1(C) = \{0\}$, $H_0(C) = \mathbb{Z}/m\mathbb{Z}$, and $(\forall\, n \in \mathbb{Z} \setminus \{0,1\})\, H_n(C) = \{0\}$.

Proposition 7.2.3. *Exactness Is Vanishing Homology*
Let $C_\bullet$ be a chain complex. The following conditions are equivalent:

(i) The complex $C_\bullet$ is exact.
(ii) $(\forall\, n \in \mathbb{Z})\, H_n(C) = \{0\}$.

Proof.

(i) $\Rightarrow$ (ii) Assume (i), $n \in \mathbb{Z}$. Then $B_n(C) = Z_n(C)$ by Definition 7.1.8(iii); $H_n(C) = Z_n(C)/B_n(C) = \{0\}$ by Definition 7.2.1(iii).
(ii) $\Rightarrow$ (i) Assume (ii), $n \in \mathbb{Z}$. For $z \in Z_n(C)$, $[z]_C = 0_{H_n(C)} = B_n(C)$. Thus $z \in B_n(C)$ by Proposition 1.5.2(i), so $Z_n(C) \subseteq B_n(C)$. The reverse inclusion holds by Proposition 7.1.2. Therefore $Z_n(C) = B_n(C)$, and $C_\bullet$ is exact by Definition 7.1.8(iii).

Structural Commentary: Quotient zero turns boundary containment into degreewise equality.
Key Insight: Zero homology is degreewise exactness. □

Proposition 7.2.4. *Functoriality of Homology*
Let $f_\bullet : C_\bullet \to D_\bullet$ be a chain map and let $n \in \mathbb{Z}$.

(i) $f_n(Z_n(C)) \subseteq Z_n(D)$.
(ii) $f_n(B_n(C)) \subseteq B_n(D)$.
(iii) The rule $H_n(f)([z]_C) := [f_n(z)]_D$ defines $H_n(f) \in \operatorname{Hom}_R(H_n(C), H_n(D))$.
(iv) $H_n(\mathrm{id}_{C_\bullet}) = \mathrm{id}_{H_n(C)}$.
(v) For a chain map $g_\bullet : D_\bullet \to E_\bullet$, $H_n(g_\bullet f_\bullet) = H_n(g)H_n(f)$.
(vi) If $f_\bullet$ is a chain isomorphism, then $H_n(f)$ is an R-module isomorphism.

Proof.

(i) Fix $z \in Z_n(C)$, so $d_n^C(z) = 0$. Then $d_n^D f_n(z) = f_{n-1} d_n^C(z) = 0$ by Definition 7.1.3(i), so $f_n(z) \in Z_n(D)$.
(ii) Fix $b \in B_n(C)$, choose $x \in C_{n+1}$ with $b = d_{n+1}^C(x)$ by Definition 7.2.1(ii); $f_n(b) = d_{n+1}^D f_{n+1}(x) \in B_n(D)$ by Definition 7.1.3(i).
(iii) If $[z]_C = [z']_C$, then $z - z' \in B_n(C)$ by Proposition 1.5.2(i). Thus $f_n(z) - f_n(z') \in B_n(D)$ by (ii), so $[f_n(z)]_D = [f_n(z')]_D$. Hence $H_n(f)$ is well-defined. Fix $(a,b) \in R^2$ and $([z]_C, [z']_C) \in H_n(C)^2$.
$H_n(f)(a[z]_C + b[z']_C) = aH_n(f)([z]_C) + bH_n(f)([z']_C)$ by R-linearity of f_n.
(iv) $(\forall\, z \in Z_n(C))\, H_n(\mathrm{id}_{C_\bullet})([z]_C) = [z]_C$ by the defining rule in (iii).
(v) For $z \in Z_n(C)$, $H_n(g_\bullet f_\bullet)([z]_C) = [g_n f_n(z)]_E = H_n(g)H_n(f)([z]_C)$ by (iii).
(vi) Let $g_\bullet : D_\bullet \to C_\bullet$ be an inverse of $f_\bullet$. Then $H_n(g)H_n(f) = \mathrm{id}_{H_n(C)}$ and $H_n(f)H_n(g) = \mathrm{id}_{H_n(D)}$ by (iv)–(v). The map $H_n(f)$ is bijective by these identities. Hence it is an R-module isomorphism by Definition 1.4.1(iv).

Structural Commentary: Differential commutation descends maps and preserves composition on homology.
Key Insight: Cycle and boundary preservation makes homology functorial. □

Theorem 7.2.5. *Long Exact Sequence in Homology*
Let $0 \to A_\bullet \xrightarrow{u_\bullet} E_\bullet \xrightarrow{v_\bullet} C_\bullet \to 0$ be a short exact sequence of chain complexes.
For $n \in \mathbb{Z}$ and $c \in Z_n(C)$, choose $e \in E_n$ with $v_n(e) = c$. Choose $a \in A_{n-1}$ with $u_{n-1}(a) = d_n^E(e)$, and set $\partial_n([c]_C) := [a]_A$.

(i) The required e, a exist, and $a \in Z_{n-1}(A)$.
(ii) The class $\partial_n([c]_C)$ is independent of the representative c and the choices e, a.
(iii) $\partial_n \in \operatorname{Hom}_R(H_n(C), H_{n-1}(A))$.
(iv) $(\forall\, n \in \mathbb{Z})\ \operatorname{im} \partial_{n+1} = \ker H_n(u)$.
(v) $(\forall\, n \in \mathbb{Z})\ \operatorname{im} H_n(u) = \ker H_n(v)$.
(vi) $(\forall\, n \in \mathbb{Z})\ \operatorname{im} H_n(v) = \ker \partial_n$.

Proof.

(i) Fix $n \in \mathbb{Z}$ and $c \in Z_n(C)$, so $d_n^C(c) = 0$. The map v_n is surjective by Corollary 7.1.10 and Definition 1.4.1(iv). Choose $e \in E_n$ with $v_n(e) = c$. Then $v_{n-1} d_n^E(e) = d_n^C v_n(e) = d_n^C(c) = 0$ by the chain-map identity for $v_\bullet$. Thus $d_n^E(e) \in \ker v_{n-1} = \operatorname{im} u_{n-1}$ by Corollary 7.1.10. Choose $a \in A_{n-1}$ with $u_{n-1}(a) = d_n^E(e)$. $u_{n-2} d_{n-1}^A(a) = d_{n-1}^E d_n^E(e) = 0$ by the chain-map and chain-complex identities. The map u_{n-2} is injective by Corollary 7.1.10 and Definition 1.4.1(iv). Hence $d_{n-1}^A(a) = 0$ and $a \in Z_{n-1}(A)$.
(ii) Fix $n \in \mathbb{Z}$ and let $c, c' \in Z_n(C)$ with $[c]_C = [c']_C$. Let (e, a) and (e'', a'') be construction data for c, c', respectively. Then $c' - c \in B_n(C)$ by Proposition 1.5.2(i). Choose $t \in C_{n+1}$ with $c' - c = d_{n+1}^C(t)$ by Definition 7.2.1(ii). The map v_{n+1} is surjective by Corollary 7.1.10 and Definition 1.4.1(iv). Choose $s \in E_{n+1}$ with $v_{n+1}(s) = t$. Set $e' := e + d_{n+1}^E(s)$. Then $v_n(e') = c + d_{n+1}^C(t) = c'$ by the chain-map identity for $v_\bullet$. $d_n^E(e') = d_n^E(e)$ by Definition 7.1.1, so (e', a) is construction data for c'. Now $e'' - e' \in \ker v_n = \operatorname{im} u_n$ by Corollary 7.1.10. Choose $x \in A_n$ with $u_n(x) = e'' - e'$. $u_{n-1}(a'' - a - d_n^A x) = 0$ by construction data, differential commutation, $u_n(x) = e'' - e'$. The map u_{n-1} is injective by Corollary 7.1.10 and Definition 1.4.1(iv). Thus $a'' - a = d_n^A x \in B_{n-1}(A)$ and $[a'']_A = [a]_A$.
(iii) Fix $n \in \mathbb{Z}$, $(r,s) \in R^2$, and $([c]_C, [c']_C) \in H_n(C)^2$. Let (e, a) and (e', a') be construction data for c, c'. By R-linearity, $(re + se', ra + sa')$ is construction data for $rc + sc'$. Hence $\partial_n(r[c]_C + s[c']_C) = r\partial_n([c]_C) + s\partial_n([c']_C)$ by (ii).

(iv) Fix $n \in \mathbb{Z}$ and $[c]_C \in H_{n+1}(C)$ with construction data (e, a). $H_n(u)\partial_{n+1}([c]_C) = [u_n(a)]_E = [d^E_{n+1}(e)]_E = 0_{H_n(E)}$.
Thus $\operatorname{im}\partial_{n+1} \subseteq \ker H_n(u)$. Let $[a]_A \in \ker H_n(u)$ with $a \in Z_n(A)$. Choose $e \in E_{n+1}$ with $u_n(a) = d^E_{n+1}(e)$ by $[u_n(a)]_E = 0_{H_n(E)}$.
Set $c := v_{n+1}(e)$. Then $d^C_{n+1}(c) = v_n d^E_{n+1}(e) = v_n u_n(a) = 0$, so $c \in Z_{n+1}(C)$.
The pair (e, a) is construction data for c, hence $\partial_{n+1}([c]_C) = [a]_A$.

(v) Fix $n \in \mathbb{Z}$ and $[a]_A \in H_n(A)$. $H_n(v)H_n(u)([a]_A) = [v_n u_n(a)]_C = 0_{H_n(C)}$ by exactness at E_n.
Thus $\operatorname{im} H_n(u) \subseteq \ker H_n(v)$. Let $[e]_E \in \ker H_n(v)$ with $e \in Z_n(E)$. Choose $c \in C_{n+1}$ with $d^C_{n+1}(c) = v_n(e)$ by $[v_n(e)]_C = 0_{H_n(C)}$.
The map v_{n+1} is surjective by Corollary 7.1.10 and Definition 1.4.1(iv). Choose $e' \in E_{n+1}$ with $v_{n+1}(e') = c$.
$v_n(e - d^E_{n+1}(e')) = v_n(e) - d^C_{n+1}(c) = 0$ by chain-map identity for $v_\bullet$. Thus $e - d^E_{n+1}(e') \in \ker v_n = \operatorname{im} u_n$ by Corollary 7.1.10.
Choose $a \in A_n$ with $u_n(a) = e - d^E_{n+1}(e')$. Then $u_{n-1}d^A_n(a) = d^E_n u_n(a) = 0$ by $e \in Z_n(E)$, Definition 7.1.1.
Map u_{n-1} is monomorphism by Corollary 7.1.10; injective by Definition 1.4.1(iv), so $a \in Z_n(A)$; $[e]_E = [u_n(a)]_E = H_n(u)([a]_A)$.

(vi) Fix $n \in \mathbb{Z}$ and $[e]_E \in H_n(E)$. The construction of $\partial_n(H_n(v)([e]_E))$ may use $(e, 0_{A_{n-1}})$, so $\partial_n H_n(v)([e]_E) = 0_{H_{n-1}(A)}$.
Thus $\operatorname{im} H_n(v) \subseteq \ker \partial_n$. Let $[c]_C \in \ker \partial_n$ with construction data (e, a).
Choose $x \in A_n$ with $a = d^A_n(x)$ by $[a]_A = 0_{H_{n-1}(A)}$. Set $e_0 := e - u_n(x)$.
Then $d^E_n(e_0) = d^E_n(e) - u_{n-1}d^A_n(x) = 0$, and $v_n(e_0) = v_n(e) = c$ by exactness at E_n. Hence $e_0 \in Z_n(E)$ and $[c]_C = H_n(v)([e_0]_E)$.

Structural Commentary: Lifts define connecting classes; exactness matches three image–kernel pairs.
Key Insight: Failed cycle lifts become boundaries one degree lower. □

7.3 Derived Functors Preview

Throughout this subsection, R denotes a nonzero commutative ring with identity 1_R. All modules are R-modules.

Definition 7.3.1. *Projective and Injective Modules*
Let P and I be R-modules.

(i) The module P is *projective* if each epimorphism $q : E \to N$ and homomorphism $f : P \to N$ admit $\widetilde{f} \in \operatorname{Hom}_R(P, E)$ with $q\widetilde{f} = f$.
The map $\widetilde{f}$ is a *lift* of f through q.

(ii) The module I is *injective* if each monomorphism $j : A \to B$ and homomorphism $f : A \to I$ admit $\widetilde{f} \in \operatorname{Hom}_R(B, I)$ with $\widetilde{f}j = f$.
The map $\widetilde{f}$ is an *extension* of f along j.

Theorem 7.3.2. *Basic Projective Constructions*
Assume the Choice Principle 6.1.2.

(i) Every free R-module is projective.
(ii) Every retract of a projective R-module is projective.
(iii) Every direct summand of a free R-module is projective.

Proof.

(i) Let F be free with basis $\mathcal{B}$, let $q : E \to N$ be an epimorphism, and let $f : F \to N$ be a homomorphism.
Then $(\forall\, b \in \mathcal{B})\, q^{-1}(\{f(b)\}) \neq \emptyset$ by surjectivity of q. Choose $(e_b)_{b\in\mathcal{B}} \in \prod_{b\in\mathcal{B}} q^{-1}(\{f(b)\})$ by the Choice Principle.
The assignment $b \mapsto e_b$ extends uniquely to $\widetilde{f} \in \operatorname{Hom}_R(F, E)$ by Theorem 2.1.4.
Maps $q\widetilde{f}$ and f agree on $\mathcal{B}$, so $q\widetilde{f} = f$ by the uniqueness clause of that theorem; F is projective by Definition 7.3.1(i).

(ii) Let Q be a retract of a projective module P, with $i : Q \to P$, $r : P \to Q$, and $ri = \mathrm{id}_Q$.
Let $q : E \to N$ be an epimorphism and let $f : Q \to N$ be a homomorphism.
Choose $h : P \to E$ with $qh = fr$ by projectivity of P; $q(hi) = f(ri) = f$, so hi lifts f; Q is projective by Definition 7.3.1(i).

(iii) Let Q be a direct summand of a free module F. Then Q is a retract of F by Corollary 1.8.3.
The module F is projective by (i), so Q is projective by (ii).

Structural Commentary: Basis lifts establish freeness, and retractions transfer lifting problems to a projective ambient module.
Key Insight: Projectivity is stable under retracts. □

Proposition 7.3.3. *Injective Submodules Split*
Let I be injective and let $j : I \to E$ be a monomorphism.

(i) The map j has a left inverse $r : E \to I$.
(ii) The submodule $\operatorname{im} j$ is a direct summand of E.

Proof.

(i) Choose $r : E \to I$ with $rj = \mathrm{id}_I$ by injectivity of I, applied to $\mathrm{id}_I : I \to I$.

(ii) We prove $E = \operatorname{im} j \oplus \ker r$. For $e \in E$, $e = j(r(e)) + (e - j(r(e)))$ and $r(e - j(r(e))) = r(e) - (rj)(r(e)) = 0$.
Hence $E = \operatorname{im} j + \ker r$. If $x \in \operatorname{im} j \cap \ker r$, choose $a \in I$ with $x = j(a)$.
Then $a = (rj)(a) = r(x) = 0$, so $x = 0$. Thus the sum is direct.

Structural Commentary: Extending the identity of I produces a retraction, whose kernel is a complementary summand.
Key Insight: Injective modules split every monomorphism out of them. □

Definition 7.3.4. *Cochain Complexes and Cohomology*

(i) A *cochain complex* $K^\bullet = (K^n, \delta^n_K)_{n\in\mathbb{Z}}$ consists of homomorphisms $\delta^n_K : K^n \to K^{n+1}$ satisfying $(\forall\, n \in \mathbb{Z})\, \delta^{n+1}_K \delta^n_K = 0$.

(ii) The associated chain complex $K_\sharp$ is defined by $(K_\sharp)_m := K^{-m}$ and $d^{K_\sharp}_m := \delta^{-m}_K$.

(iii) Set $Z^n(K) := \ker \delta^n_K$ and $B^n(K) := \operatorname{im} \delta^{n-1}_K$. Then $B^n(K) \leq Z^n(K)$ by Proposition 7.1.2 applied to $K_\sharp$.
The *nth cohomology module* is $H^n(K) := Z^n(K)/B^n(K) = H_{-n}(K_\sharp)$.

Definition 7.3.5. *Projective and Injective Resolutions*

(i) A *projective resolution* of M consists of a chain complex $P_\bullet = (P_n, d_n)_{n\in\mathbb{Z}}$ and an epimorphism $\varepsilon : P_0 \to M$.
It satisfies $(\forall\, n < 0)\, P_n = 0$, $\ker\varepsilon = \operatorname{im} d_1$, $(\forall\, n \in \mathbb{N})\ \operatorname{im} d_{n+1} = \ker d_n$, and each P_n for $n \in \mathbb{N}_0$ is projective.

(ii) An *injective resolution* of N consists of a cochain complex $I^\bullet = (I^n, \delta^n_I)_{n\in\mathbb{Z}}$ and a monomorphism $\eta : N \to I^0$.
It satisfies $(\forall\, n < 0)\, I^n = 0$, $\operatorname{im}\eta = \ker\delta^0_I$, $(\forall\, n \in \mathbb{N})\ \operatorname{im}\delta^{n-1}_I = \ker\delta^n_I$, and each I^n for $n \in \mathbb{N}_0$ is injective.

Proposition 7.3.6. *The Hom Cochain Complex*
Let M be an R-module and let $I^\bullet = (I^n, \delta_I^n)_{n\in\mathbb{Z}}$ be a cochain complex. For $n \in \mathbb{Z}$, set $C^n := \mathrm{Hom}_R(M, I^n)$ and $\Delta^n(\varphi) := \delta_I^n \varphi$.

(i) Each $\Delta^n : C^n \to C^{n+1}$ is an R-module homomorphism.
(ii) The family $C^\bullet = (C^n, \Delta^n)_{n\in\mathbb{Z}}$ is a cochain complex.

Proof.

(i) Fix $(a, b) \in R^2$ and $(\varphi, \psi) \in (C^n)^2$. Then $\Delta^n(a\varphi + b\psi) = a\Delta^n(\varphi) + b\Delta^n(\psi)$ by R-linearity of δ_I^n.

(ii) For $\varphi \in C^n$, $\Delta^{n+1}\Delta^n(\varphi) = \delta_I^{n+1}\delta_I^n\varphi = 0$ by Definition 7.3.4(i). Hence $C^\bullet$ is a cochain complex by that definition.

Structural Commentary: The covariant variable of $\mathrm{Hom}_R(M, -)$ preserves the direction of the coboundary maps.
Key Insight: Applying $\mathrm{Hom}_R(M, -)$ to an injective resolution produces a cochain complex. □

Definition 7.3.7. *Chosen-Resolution Model for Ext*
Let M, N be R-modules, and let $(I^\bullet, \eta)$ be an injective resolution of N. For $n \in \mathbb{N}_0$, set $\mathrm{Ext}_R^n(M, N; I^\bullet) := H^n\big(\mathrm{Hom}_R(M, I^\bullet)\big)$.
The resolution remains in the notation because resolution independence is not part of this definition.

Perspective 7.3.8. *Resolution Models for Tor and Ext*
After tensor products of arbitrary R-modules and their induced maps are developed, a projective resolution $P_\bullet \to M$ produces a tensor complex with terms $P_n \otimes_R N$. Its homology is the projective-resolution model for $\mathrm{Tor}_n^R(M, N)$, parallel to Definition 7.3.7.
A comparison theorem is still required to identify the models from different resolutions by canonical isomorphisms.
Until that theorem and the module-level tensor construction are available, the abbreviations $\mathrm{Tor}_n^R(M, N)$ and $\mathrm{Ext}_R^n(M, N)$ are forward notation rather than proof authority.

Part IX
Commutative Algebra

1 Prime Ideals and Spectra

1.1 Prime Ideals

Definition 1.1.1. *Prime Ideals*
Let R be a commutative ring with identity. An ideal $\mathfrak{p} \trianglelefteq R$ is *prime* if $\mathfrak{p} \neq R$ and $(\forall\, a, b \in R)\,[ab \in \mathfrak{p} \Rightarrow (a \in \mathfrak{p} \vee b \in \mathfrak{p})]$.

Theorem 1.1.2. *Prime Ideals and Quotient Domains*
Let R be a commutative ring with identity and $\mathfrak{p} \trianglelefteq R$. Then $\mathfrak{p}$ is prime $\iff$ $R/\mathfrak{p}$ is an integral domain.

Proof.
($\Rightarrow$) Assume $\mathfrak{p}$ is prime. The ring $R/\mathfrak{p}$ is commutative with identity $1_R + \mathfrak{p}$ by Definition 3.4.1.
The relation $1_R \notin \mathfrak{p}$ holds by $\mathfrak{p} \neq R$ and ideal absorption. The relation $1_R + \mathfrak{p} \neq \mathfrak{p}$ holds by additive-coset equality.
Fix $a, b \in R$ and assume $(a+\mathfrak{p})(b+\mathfrak{p}) = \mathfrak{p}$. The relation $ab \in \mathfrak{p}$ holds by quotient multiplication and additive-coset equality.
The relation $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$ holds by Definition 1.1.1. The relation $a+\mathfrak{p} = \mathfrak{p}$ or $b+\mathfrak{p} = \mathfrak{p}$ holds by additive-coset equality.
The ring $R/\mathfrak{p}$ has no zero divisors by this zero-product test. It is an integral domain by Definition 3.5.1.
($\Leftarrow$) Assume $R/\mathfrak{p}$ is an integral domain. The relation $\mathfrak{p} \neq R$ holds by its distinct zero and identity cosets.
Fix $a, b \in R$ and assume $ab \in \mathfrak{p}$. The equality $(a+\mathfrak{p})(b+\mathfrak{p}) = \mathfrak{p}$ holds by quotient multiplication and additive-coset equality.
The relation $a+\mathfrak{p} = \mathfrak{p}$ or $b+\mathfrak{p} = \mathfrak{p}$ holds by absence of zero divisors. Relation $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$ holds by additive-coset equality.
The ideal $\mathfrak{p}$ is prime by Definition 1.1.1.
Structural Commentary: Quotient multiplication transfers the zero-divisor test to product membership in $\mathfrak{p}$.
Key Insight: A zero product of cosets is precisely a product in $\mathfrak{p}$. □

Corollary 1.1.3. *Maximal Ideals Are Prime*
Let R be a commutative ring with identity. Every maximal ideal of R is prime.

Proof. Let $\mathfrak{m} \trianglelefteq R$ be maximal. The quotient $R/\mathfrak{m}$ is a field by Theorem 3.4.4.
Fix $x, y \in R/\mathfrak{m}$ with $xy = 0$ and $x \neq 0$. The equality $y = x^{-1}(xy) = 0$ holds by field inversion.
The quotient $R/\mathfrak{m}$ is an integral domain by Definition 3.5.1. The ideal $\mathfrak{m}$ is prime by Theorem 1.1.2.
Structural Commentary: The quotient-field theorem feeds the quotient-domain criterion.
Key Insight: Invertibility excludes zero divisors in every field. □

Definition 1.1.4. *Radicals and Radical Ideals*
Let R be a commutative ring with identity and $\mathcal{I} \trianglelefteq R$.
The *radical* of $\mathcal{I}$ is $\operatorname{rad}(\mathcal{I}) := \sqrt{\mathcal{I}} := \{r \in R : (\exists\, n \in \mathbb{N})\, r^n \in \mathcal{I}\}$. The ideal $\mathcal{I}$ is *radical* if $\sqrt{\mathcal{I}} = \mathcal{I}$.

Proposition 1.1.5. *Radical Closure*
Let R be a commutative ring with identity and $\mathcal{I} \trianglelefteq R$.
(i) $\sqrt{\mathcal{I}} \trianglelefteq R$.
(ii) $\mathcal{I} \subseteq \sqrt{\mathcal{I}}$.
(iii) The ideal $\sqrt{\mathcal{I}}$ is radical: $\operatorname{rad}(\operatorname{rad}(\mathcal{I})) = \operatorname{rad}(\mathcal{I})$.
(iv) If $\mathcal{J} \trianglelefteq R$ is radical and $\mathcal{I} \subseteq \mathcal{J}$, then $\sqrt{\mathcal{I}} \subseteq \mathcal{J}$.

Proof.
(i) The relation $0_R \in \sqrt{\mathcal{I}}$ holds by $0_R^1 \in \mathcal{I}$. Fix $x, y \in \sqrt{\mathcal{I}}$ and choose $m, n \in \mathbb{N}$ with $x^m, y^n \in \mathcal{I}$.
For $0 \leq k \leq m+n$, either $k \geq m$ or $m+n-k \geq n$; the relation $x^k(-y)^{m+n-k} \in \mathcal{I}$ holds by ideal absorption.
The relation $(x-y)^{m+n} \in \mathcal{I}$ holds by distributivity, ideal absorption, and additive closure; $x - y \in \sqrt{\mathcal{I}}$ by Definition 1.1.4.
For $r \in R$, $(rx)^m = r^m x^m \in \mathcal{I}$ by commutativity and ideal absorption. The relation $rx = xr \in \sqrt{\mathcal{I}}$ holds by Definition 1.1.4.
The conclusion $\sqrt{\mathcal{I}} \trianglelefteq R$ holds by Definition 3.3.1.
(ii) Fix $x \in \mathcal{I}$; $x^1 = x \in \mathcal{I}$. The relation $x \in \sqrt{\mathcal{I}}$ holds by Definition 1.1.4. The inclusion $\mathcal{I} \subseteq \sqrt{\mathcal{I}}$ holds by this element test.
(iii) Fix $x \in \operatorname{rad}(\mathcal{I})$; $x^1 \in \operatorname{rad}(\mathcal{I})$. The relation $x \in \operatorname{rad}(\operatorname{rad}(\mathcal{I}))$ holds by Definition 1.1.4.
Inclusion $\operatorname{rad}(\mathcal{I}) \subseteq \operatorname{rad}(\operatorname{rad}(\mathcal{I}))$ holds by this element test. If $x \in \operatorname{rad}(\operatorname{rad}(\mathcal{I}))$, choose $m \in \mathbb{N}$ with $x^m \in \operatorname{rad}(\mathcal{I})$.
Relation $(\exists\, n \in \mathbb{N})\, x^{mn} = (x^m)^n \in \mathcal{I}$ holds by Definition 1.1.4, exponent laws. Relation $x \in \operatorname{rad}(\mathcal{I})$ holds by Definition 1.1.4.
Equality $\operatorname{rad}(\operatorname{rad}(\mathcal{I})) = \operatorname{rad}(\mathcal{I})$ holds by mutual inclusion. Ideal $\sqrt{\mathcal{I}}$ is radical by Definition 1.1.4.
(iv) Let $\mathcal{J} \trianglelefteq R$ be radical with $\mathcal{I} \subseteq \mathcal{J}$. If $x \in \sqrt{\mathcal{I}}$, choose $n \in \mathbb{N}$ with $x^n \in \mathcal{I} \subseteq \mathcal{J}$.
The relation $x \in \sqrt{\mathcal{J}} = \mathcal{J}$ holds by Definition 1.1.4. The inclusion $\sqrt{\mathcal{I}} \subseteq \mathcal{J}$ holds by this element test.
Structural Commentary: Power witnesses establish ideal closure, idempotence, and minimality under radical containment.
Key Insight: A sufficiently large power absorbs every term in a binomial expansion. □

1.2 Spectra

Definition 1.2.1. *Spectrum, Vanishing Sets, and Principal Complements*
Let R be a commutative ring with identity.
(i) The *prime spectrum* of R is $\operatorname{Spec} R := \{\mathfrak{p} \trianglelefteq R : \mathfrak{p} \text{ is prime}\}$.
(ii) For $\mathcal{I} \trianglelefteq R$, set $V(\mathcal{I}) := \{\mathfrak{p} \in \operatorname{Spec} R : \mathcal{I} \subseteq \mathfrak{p}\}$.
(iii) For $f \in R$, set $D(f) := \operatorname{Spec} R \setminus V(\langle f \rangle)$.

Proposition 1.2.2. *Vanishing-Set Calculus*
Let R be a commutative ring with identity, $\mathcal{I}, \mathcal{J} \trianglelefteq R$. Let $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ be a nonempty family of ideals of R.
(i) $V(\langle 0_R \rangle) = \operatorname{Spec} R$.
(ii) $V(R) = \emptyset$.

(iii) $\bigcap_{\alpha\in A} V(\mathcal{I}_\alpha) = V\left(\left\langle \bigcup_{\alpha\in A} \mathcal{I}_\alpha \right\rangle\right)$.
(iv) $V(\mathcal{I}) \cup V(\mathcal{J}) = V(\mathcal{I}\cap\mathcal{J})$.

Proof.

(i) Fix $\mathfrak{p} \in \operatorname{Spec} R$. The relation $0_R \in \mathfrak{p}$ holds by the ideal property. The inclusion $\langle 0_R\rangle \subseteq \mathfrak{p}$ holds by Proposition 3.3.2.
The equality $V(\langle 0_R\rangle) = \operatorname{Spec} R$ holds by Definition 1.2.1(ii) and set extensionality.
(ii) Fix $\mathfrak{p} \in \operatorname{Spec} R$. The relation $R \nsubseteq \mathfrak{p}$ holds by Definition 1.1.1. The equality $V(R) = \emptyset$ holds by Definition 1.2.1(ii).
(iii) Set $\mathcal{K} := \left\langle \bigcup_{\alpha\in A} \mathcal{I}_\alpha \right\rangle$. Fix $\mathfrak{p} \in \operatorname{Spec} R$. Membership in $\bigcap_{\alpha\in A} V(\mathcal{I}_\alpha)$ is equivalent to $\bigcup_{\alpha\in A} \mathcal{I}_\alpha \subseteq \mathfrak{p}$ by Definition 1.2.1(ii).
The latter inclusion is equivalent to $\mathcal{K} \subseteq \mathfrak{p}$ by Proposition 3.3.2.
The asserted equality holds by Definition 1.2.1(ii) and set extensionality.
(iv) The inclusions $\mathcal{I}\cap\mathcal{J} \subseteq \mathcal{I}$ and $\mathcal{I}\cap\mathcal{J} \subseteq \mathcal{J}$ hold by definition of intersection.
The inclusion $V(\mathcal{I}) \cup V(\mathcal{J}) \subseteq V(\mathcal{I}\cap\mathcal{J})$ holds by Definition 1.2.1(ii). Fix $\mathfrak{p} \in V(\mathcal{I}\cap\mathcal{J})$ and assume $\mathfrak{p} \notin V(\mathcal{I}) \cup V(\mathcal{J})$.
Choose $a \in \mathcal{I} \setminus \mathfrak{p}$ and $b \in \mathcal{J} \setminus \mathfrak{p}$ by the failed containments. The relation $ab \in \mathcal{I}\cap\mathcal{J} \subseteq \mathfrak{p}$ holds by ideal absorption.
This contradicts primeness by Definition 1.1.1. The reverse inclusion holds by contradiction.

Structural Commentary: Generated unions control intersections; primality converts ideal intersections into finite unions.
Key Insight: A product in $\mathcal{I}\cap\mathcal{J}$ detects which ideal a prime contains. □

Proposition 1.2.3. *Principal-Complement Calculus*
Let R be a commutative ring with identity and let $f, g \in R$.

(i) $D(0_R) = \emptyset$.
(ii) $D(1_R) = \operatorname{Spec} R$.
(iii) $D(f) \cap D(g) = D(fg)$.

Proof.

(i) The equality $D(0_R) = \operatorname{Spec} R \setminus V(\langle 0_R\rangle) = \emptyset$ holds by Proposition 1.2.2(i).
(ii) The equality $\langle 1_R\rangle = R$ holds by Definition 3.3.1. The equality $D(1_R) = \operatorname{Spec} R \setminus V(R) = \operatorname{Spec} R$ holds by Proposition 1.2.2(ii).
(iii) The relation $(\forall\,(\mathfrak{p}, h) \in \operatorname{Spec} R \times R)\,[\mathfrak{p} \in D(h) \iff h \notin \mathfrak{p}]$ holds by Definition 1.2.1(iii) and Proposition 3.3.2.
Fix $\mathfrak{p} \in \operatorname{Spec} R$. The equivalence $\mathfrak{p} \in D(f)\cap D(g) \iff (f \notin \mathfrak{p} \wedge g \notin \mathfrak{p})$ holds by the preceding test.
The equivalence $(f \notin \mathfrak{p} \wedge g \notin \mathfrak{p}) \iff fg \notin \mathfrak{p}$ holds by ideal absorption and Definition 1.1.1.
The equivalence $fg \notin \mathfrak{p} \iff \mathfrak{p} \in D(fg)$ holds by the preceding test. The equality $D(f)\cap D(g) = D(fg)$ holds by set extensionality.

Structural Commentary: Complementation transfers boundary values and primality to principal-complement intersections.
Key Insight: The product fg avoids a prime exactly when both factors do. □

2 Localization and Local Rings

2.1 Localization

Definition 2.1.1. *Multiplicative Sets and Ring Localization*
Let R be a commutative ring with identity.

(i) A subset $S \subseteq R$ is *multiplicative* if $1_R \in S$ and $(\forall\, s, t \in S)\, st \in S$.
(ii) On $R \times S$, define $(r, s) \sim (r', s') \iff (\exists\, u \in S)\, u(s'r - sr') = 0_R$.
(iii) The *localization* of R at S is $S^{-1}R := (R \times S)/\!\sim$; write $[(r, s)]_\sim = r/s$.
(iv) For $(r, s), (r', s') \in R \times S$, define $\frac{r}{s} + \frac{r'}{s'} := \frac{s'r + sr'}{ss'}$ and $\frac{r}{s}\frac{r'}{s'} := \frac{rr'}{ss'}$. Set $0_{S^{-1}R} := 0_R/1_R$ and $1_{S^{-1}R} := 1_R/1_R$.

Proposition 2.1.2. *Ring Localization Construction*
Let R be a commutative ring with identity and let $S \subseteq R$ be multiplicative.

(i) The relation $\sim$ on $R \times S$ is an equivalence relation.
(ii) The fraction operations are well defined and make $S^{-1}R$ a commutative ring with identity.
(iii) The map $\lambda_S : R \to S^{-1}R$, $\lambda_S(r) := r/1_R$, is a ring homomorphism.
(iv) $\lambda_S(S) \subseteq (S^{-1}R)^\times$.
(v) $\ker(\lambda_S) = \{r \in R : (\exists\, s \in S)\, sr = 0_R\}$.

Proof.

(i) Fix $(r, s), (r', s') \in R \times S$. Then $(r, s) \sim (r, s)$ with witness $1_R \in S$.
If $(r, s) \sim (r', s')$ with witness $u \in S$, then $u(sr' - s'r) = -u(s'r - sr') = 0_R$, so $(r', s') \sim (r, s)$.
For transitivity, assume $(a, s) \sim (b, t)$ via $u \in S$ and $(b, t) \sim (c, q)$ via $v \in S$.
Then $uvt(qa - sc) = uvq(ta - sb) + uvs(qb - tc) = 0_R$ by the two witness identities. Thus $(a, s) \sim (c, q)$ with witness $uvt \in S$.
(ii) Assume $a/s = a'/s'$ with witness $u \in S$ and $b/t = b'/t'$ with witness $v \in S$.
The cross-multiplied sum difference is $tt'(s'a - sa') + ss'(t'b - tb')$ and vanishes after multiplication by uv.
The cross-multiplied product difference is $t'b(s'a - sa') + sa'(t'b - tb')$ and likewise vanishes after multiplication by uv.
Since $uv \in S$, both operations are well defined by Definition 2.1.1(ii).
Fix $(a, s), (b, t), (c, q) \in R \times S$. Then $(a/s + b/t) + c/q = (tqa + sqb + stc)/(stq) = a/s + (b/t + c/q)$.
$((a/s)(b/t))(c/q) = abc/(stq) = (a/s)((b/t)(c/q))$. Fraction operations are commutative by commutativity of R.
Distributivity holds by $(a/s)(b/t + c/q) = (aqb + atc)/(stq) = ab/(st) + ac/(sq)$.
The classes $0_R/1_R$ and $1_R/1_R$ are the identities, while $-(a/s) = (-a)/s$; hence $S^{-1}R$ is a commutative ring with identity.
(iii) Fix $a, b \in R$. The identities $\lambda_S(a + b) = \lambda_S(a) + \lambda_S(b)$, $\lambda_S(ab) = \lambda_S(a)\lambda_S(b)$, and $\lambda_S(1_R) = 1$ hold by Definition 2.1.1(iv).
(iv) Fix $s \in S$. Then $\lambda_S(s) \in (S^{-1}R)^\times$ by $\lambda_S(s)(1_R/s) = 1$.
(v) For $r \in R$, $\lambda_S(r) = 0 \iff (\exists\, u \in S)\, ur = 0_R$ by Definition 2.1.1(ii).

Structural Commentary: Witness products secure equivalence and operations; representatives verify ring laws.
Key Insight: One denominator witness certifies every equality of localized fractions. □

Theorem 2.1.3. *Universal Property of Ring Localization*
Let R, A be commutative rings with identity, $S \subseteq R$ be multiplicative, $\varphi : R \to A$ ring homomorphism satisfying $\varphi(S) \subseteq A^\times$.

(i) The rule $\widetilde{\varphi} : S^{-1}R \to A$, $r/s \mapsto \varphi(r)\varphi(s)^{-1}$, is a ring homomorphism. It satisfies $\widetilde{\varphi} \circ \lambda_S = \varphi$.
(ii) The homomorphism $\widetilde{\varphi}$ is unique subject to $\widetilde{\varphi} \circ \lambda_S = \varphi$.

Proof.

(i) If $r/s = r'/s'$, choose $u \in S$ with $u(s'r - sr') = 0_R$ by Definition 2.1.1(ii).
Then $\varphi(s')\varphi(r) = \varphi(s)\varphi(r')$ by the homomorphism property and cancellation of the unit $\varphi(u)$.
The equality $\widetilde{\varphi}(r/s) = \widetilde{\varphi}(r'/s')$ holds by multiplication by $\varphi(s)^{-1}\varphi(s')^{-1}$. Thus the rule is well defined. Fix $(r,s),(r',s') \in R \times S$.
$\widetilde{\varphi}(r/s + r'/s') = \varphi(s'r + sr')\varphi(ss')^{-1} = \widetilde{\varphi}(r/s) + \widetilde{\varphi}(r'/s')$; $\widetilde{\varphi}((r/s)(r'/s')) = \varphi(rr')\varphi(ss')^{-1} = \widetilde{\varphi}(r/s)\widetilde{\varphi}(r'/s')$.
The map $\widetilde{\varphi}$ is a ring homomorphism by these identities and $\widetilde{\varphi}(1_R/1_R) = 1_A$. Equality $\widetilde{\varphi} \circ \lambda_S = \varphi$ holds by $\widetilde{\varphi}(r/1_R) = \varphi(r)$.
(ii) Let $\psi : S^{-1}R \to A$ be a ring homomorphism with $\psi \circ \lambda_S = \varphi$. Fix $(r,s) \in R \times S$.
$\psi(r/s) = \psi(\lambda_S(r))\psi(\lambda_S(s))^{-1} = \varphi(r)\varphi(s)^{-1} = \widetilde{\varphi}(r/s)$. Thus $\psi = \widetilde{\varphi}$ by function extensionality.

Structural Commentary: Unit cancellation descends fractions; denominator factorization forces uniqueness.
Key Insight: Images of numerators and inverted denominators determine every localized homomorphism. □

Definition 2.1.4. *Module Localization*
Let R be a commutative ring with identity, let $S \subseteq R$ be multiplicative, and let M be an R-module.

(i) On $M \times S$, define $(m,s) \sim (n,t) \iff (\exists\, u \in S)\, u(tm - sn) = 0_M$.
(ii) Set $S^{-1}M := (M \times S)/\sim$ and write $[(m,s)]_\sim = m/s$.
(iii) For $(m,s),(n,t) \in M \times S$, define $\frac{m}{s} + \frac{n}{t} := \frac{tm+sn}{st}$. For $(r,q) \in R \times S$ and $(m,s) \in M \times S$, define $\frac{r}{q}\frac{m}{s} := \frac{rm}{qs}$.

Proposition 2.1.5. *Module Localization Construction*
Let R be a commutative ring with identity, let $S \subseteq R$ be multiplicative, and let M be an R-module.

(i) The relation $\sim$ on $M \times S$ is an equivalence relation.
(ii) The fraction operations are well defined and make $S^{-1}M$ an $S^{-1}R$-module.
(iii) The map $\eta_S : M \to S^{-1}M$, $m \mapsto m/1_R$, is R-linear via $r \cdot x := \lambda_S(r)x$.
(iv) $\ker(\eta_S) = \{m \in M : (\exists\, s \in S)\, sm = 0_M\}$.

Proof.

(i) Fix $(m,s),(n,t) \in M \times S$. Then $(m,s) \sim (m,s)$ with witness $1_R \in S$.
If $(m,s) \sim (n,t)$ with witness $u \in S$, then $u(sn - tm) = -u(tm - sn) = 0_M$, so $(n,t) \sim (m,s)$.
Assume $(m,s) \sim (n,t)$ via $u \in S$ and $(n,t) \sim (p,q)$ via $v \in S$.
Then $uvt(qm - sp) = uvq(tm - sn) + uvs(qn - tp) = 0_M$ by the two witness identities. Thus $(m,s) \sim (p,q)$ with witness $uvt \in S$.
(ii) Assume $m/s = m'/s'$ with witness $u \in S$ and $n/t = n'/t'$ with witness $v \in S$.
The cross-multiplied sum difference is $tt'(s'm - sm') + ss'(t'n - tn')$ and vanishes after multiplication by uv.
Thus addition is well defined by Definition 2.1.4(i). Assume $a/q = a'/q'$ in $S^{-1}R$ with witness $w \in S$.
The scalar-product difference is $s'm(q'a - qa') + qa'(s'm - sm')$ and vanishes after multiplication by uw.
Thus scalar multiplication is well defined by Definitions 2.1.1(ii) and 2.1.4(i).
For $(m,s),(n,t),(p,u) \in M \times S$, $(m/s + n/t) + p/u = (tum + sun + stp)/(stu) = m/s + (n/t + p/u)$.
Localized addition is commutative by commutativity of M; $0_M/1_R$ and $(-m)/s$ are its identity and inverses.
Fix $a/q, b/t \in S^{-1}R$ and $m/s, n/u \in S^{-1}M$. Then $((a/q)(b/t))(m/s) = abm/(qts) = (a/q)((b/t)(m/s))$,
$((a/q) + (b/t))(m/s) = (tam + qbm)/(qts) = (a/q)(m/s) + (b/t)(m/s)$,
$(a/q)(m/s + n/u) = (aum + asn)/(qsu) = (a/q)(m/s) + (a/q)(n/u)$, and $(1_R/1_R)(m/s) = m/s$; $S^{-1}M$ is an $S^{-1}R$-module.
(iii) Fix $r \in R$ and $m,n \in M$. The identities $\eta_S(m+n) = \eta_S(m) + \eta_S(n)$ and $\eta_S(rm) = \lambda_S(r)\eta_S(m)$ hold by Definition 2.1.4(iii).
(iv) For $m \in M$, $\eta_S(m) = 0 \iff (\exists\, s \in S)\, sm = 0_M$ by Definition 2.1.4(i).

Structural Commentary: Witness products descend addition and scalars; representatives verify module laws.
Key Insight: Local vanishing is equivalent to annihilation by one denominator. □

Theorem 2.1.6. *Universal Property of Module Localization*
Let R be a commutative ring with identity, $S \subseteq R$ be multiplicative, M be an R-module.
Let N be an $S^{-1}R$-module, $\psi : M \to N$ be R-linear for action $r \cdot x := \lambda_S(r)x$.

(i) The rule $\widetilde{\psi} : S^{-1}M \to N$, $m/s \mapsto (1_R/s)\psi(m)$, is $S^{-1}R$-linear. It satisfies $\widetilde{\psi} \circ \eta_S = \psi$.
(ii) The map $\widetilde{\psi}$ is unique subject to $\widetilde{\psi} \circ \eta_S = \psi$.

Proof.

(i) If $m/s = n/t$, choose $u \in S$, $u(tm - sn) = 0_M$ by Definition 2.1.4(i); $\lambda_S(u)[\lambda_S(t)\psi(m) - \lambda_S(s)\psi(n)] = 0_N$ by R-linearity of ψ.
The equality $(1_R/s)\psi(m) = (1_R/t)\psi(n)$ holds by cancelling the unit $\lambda_S(u)$ and scaling by $\lambda_S(s)^{-1}\lambda_S(t)^{-1}$.
Thus the rule is well defined. Fix $(m,s),(n,t) \in M \times S$ and $r/q \in S^{-1}R$.
$\widetilde{\psi}(m/s + n/t) = \frac{1_R}{st}\psi(tm + sn) = \frac{1_R}{s}\psi(m) + \frac{1_R}{t}\psi(n)$; $\widetilde{\psi}((r/q)(m/s)) = \frac{1_R}{qs}\psi(rm) = \frac{r}{q}\frac{1_R}{s}\psi(m) = \frac{r}{q}\widetilde{\psi}(m/s)$.
The map $\widetilde{\psi}$ is $S^{-1}R$-linear by the preceding identities. The equality $\widetilde{\psi} \circ \eta_S = \psi$ holds by $\widetilde{\psi}(m/1_R) = \psi(m)$.
(ii) Let $\theta : S^{-1}M \to N$ be $S^{-1}R$-linear with $\theta \circ \eta_S = \psi$. Fix $(m,s) \in M \times S$.
$\theta(m/s) = (1_R/s)\theta(m/1_R) = (1_R/s)\psi(m) = \widetilde{\psi}(m/s)$. Thus $\theta = \widetilde{\psi}$ by function extensionality.

Structural Commentary: Unit cancellation descends module fractions; factorization forces uniqueness.
Key Insight: Denominator-one values determine every localized module map. □

Definition 2.1.7. *Extension and Contraction under Localization*
Let R be a commutative ring with identity and let $S \subseteq R$ be multiplicative.

(i) The *extension* of $\mathcal{I} \trianglelefteq R$ is $\mathcal{I}^{\mathrm{e}} := S^{-1}\mathcal{I} = \{a/s : a \in \mathcal{I},\, s \in S\}$.
(ii) The *contraction* of $\mathcal{J} \trianglelefteq S^{-1}R$ is $\mathcal{J}^{\mathrm{c}} := \lambda_S^{-1}(\mathcal{J})$.

Proposition 2.1.8. *Localization Extension–Contraction Calculus*
Let R be a commutative ring with identity, $S \subseteq R$ be multiplicative. Let $\mathcal{I} \trianglelefteq R$ and $\mathcal{J} \trianglelefteq S^{-1}R$.

(i) $\mathcal{I}^{\mathrm{e}} \trianglelefteq S^{-1}R$.
(ii) $(\forall\,(r,s) \in R \times S)\,[r/s \in \mathcal{I}^{\mathrm{e}} \iff (\exists\,u \in S)\,ur \in \mathcal{I}]$.
(iii) $\mathcal{J}^{\mathrm{c}} \trianglelefteq R$.
(iv) $\mathcal{I}^{\mathrm{e}} = S^{-1}R \iff \mathcal{I} \cap S \neq \emptyset$.

Proof.

(i) $0_R/1_R \in \mathcal{I}^{\mathrm{e}}$ holds by $0_R \in \mathcal{I}$. Fix $(a,s),(b,t) \in \mathcal{I} \times S$ and $r/q \in S^{-1}R$; then $(a/s) - (b/t) = (ta - sb)/(st) \in \mathcal{I}^{\mathrm{e}}$ by $ta - sb \in \mathcal{I}$. The membership $(r/q)(a/s) = ra/(qs) \in \mathcal{I}^{\mathrm{e}}$ holds by ideal absorption. Thus $\mathcal{I}^{\mathrm{e}} \trianglelefteq S^{-1}R$ by the ideal axioms.
(ii) ($\Rightarrow$) Assume $r/s \in \mathcal{I}^{\mathrm{e}}$, $(a,t) \in \mathcal{I} \times S$ with $r/s = a/t$. Choose $u \in S$ satisfying $u(tr - sa) = 0_R$ by Definition 2.1.1(ii). Then $(ut)r = u(sa) \in \mathcal{I}$ by ideal absorption, and $ut \in S$ by multiplicativity.
(ii) ($\Leftarrow$) Choose $u \in S$ satisfying $ur \in \mathcal{I}$. Then $r/s = (ur)/(us) \in \mathcal{I}^{\mathrm{e}}$ by Definitions 2.1.1(ii) and 2.1.7(i).
(iii) The relation $0_R \in \mathcal{J}^{\mathrm{c}}$ holds by $\lambda_S(0_R) = 0 \in \mathcal{J}$. Fix $a,b \in \mathcal{J}^{\mathrm{c}}$ and $r \in R$. $\lambda_S(a-b) = \lambda_S(a) - \lambda_S(b) \in \mathcal{J}$ and $\lambda_S(ra) = \lambda_S(r)\lambda_S(a) \in \mathcal{J}$ by ideal closure; $\mathcal{J}^{\mathrm{c}} \trianglelefteq R$ by Definition 2.1.7(ii) and ideal axioms.
(iv) ($\Rightarrow$) Assume $\mathcal{I}^{\mathrm{e}} = S^{-1}R$. Choose $u \in S \cap \mathcal{I}$ by $1_R/1_R \in \mathcal{I}^{\mathrm{e}}$ and (ii).
(iv) ($\Leftarrow$) Choose $u \in \mathcal{I} \cap S$. Then $1_R/1_R \in \mathcal{I}^{\mathrm{e}}$ by (ii), so $\mathcal{I}^{\mathrm{e}} = S^{-1}R$ by ideal absorption.

Structural Commentary: Ideal closure handles both transfers; numerator membership detects the unit ideal.
Key Insight: Extension membership depends only on a denominator multiple of the numerator. □

Theorem 2.1.9. *Prime Ideals under Localization*
Let R be a commutative ring with identity and let $S \subseteq R$ be multiplicative. Set $\mathcal{X}_S := \{\mathfrak{p} \in \operatorname{Spec} R : \mathfrak{p} \cap S = \emptyset\}$. The assignment $\mathfrak{q} \mapsto \mathfrak{q}^{\mathrm{c}}$ is a bijection $\operatorname{Spec}(S^{-1}R) \to \mathcal{X}_S$; its inverse is $\mathfrak{p} \mapsto \mathfrak{p}^{\mathrm{e}}$.

Proof.

(i) *Contraction*: Fix $\mathfrak{q} \in \operatorname{Spec}(S^{-1}R)$. Then $\mathfrak{q}^{\mathrm{c}} \trianglelefteq R$ by Proposition 2.1.8(iii). The ideal $\mathfrak{q}^{\mathrm{c}}$ is proper by $\lambda_S(1_R) = 1 \notin \mathfrak{q}$. If $ab \in \mathfrak{q}^{\mathrm{c}}$, then $\lambda_S(a)\lambda_S(b) = \lambda_S(ab) \in \mathfrak{q}$. Hence $a \in \mathfrak{q}^{\mathrm{c}}$ or $b \in \mathfrak{q}^{\mathrm{c}}$ by primeness of $\mathfrak{q}$; $\mathfrak{q}^{\mathrm{c}} \in \operatorname{Spec} R$ by Definition 1.1.1. Fix $s \in S$. The element $\lambda_S(s)$ is a unit by Proposition 2.1.2(iv). The relation $\lambda_S(s) \notin \mathfrak{q}$ holds by properness and ideal absorption. Therefore $\mathfrak{q}^{\mathrm{c}} \cap S = \emptyset$ by Definition 2.1.7(ii).
(ii) *Extension*: Fix $\mathfrak{p} \in \mathcal{X}_S$. Then $\mathfrak{p}^{\mathrm{e}} \trianglelefteq S^{-1}R$ by Proposition 2.1.8(i). The ideal $\mathfrak{p}^{\mathrm{e}}$ is proper by Proposition 2.1.8(iv). Fix $(a,s),(b,t) \in R \times S$ and suppose $(a/s)(b/t) \in \mathfrak{p}^{\mathrm{e}}$. Choose $u \in S$ with $uab \in \mathfrak{p}$ by Proposition 2.1.8(ii). The relation $ab \in \mathfrak{p}$ holds by $u \notin \mathfrak{p}$ and primeness; hence $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$ by primeness. Thus $a/s \in \mathfrak{p}^{\mathrm{e}}$ or $b/t \in \mathfrak{p}^{\mathrm{e}}$ by Definition 2.1.7(i). Therefore $\mathfrak{p}^{\mathrm{e}} \in \operatorname{Spec}(S^{-1}R)$ by Definition 1.1.1.
(iii) *Contracting an Extension*: Fix $\mathfrak{p} \in \mathcal{X}_S$ and $r \in R$. The equivalence $r \in (\mathfrak{p}^{\mathrm{e}})^{\mathrm{c}} \iff r/1_R \in \mathfrak{p}^{\mathrm{e}}$ holds by Definition 2.1.7(ii). The latter condition is equivalent to $(\exists\,u \in S)\,ur \in \mathfrak{p}$ by Proposition 2.1.8(ii). If $ur \in \mathfrak{p}$ for $u \in S$, then $u \notin \mathfrak{p}$ and $r \in \mathfrak{p}$ by primeness. Converse holds with $u = 1_R$. Hence $(\mathfrak{p}^{\mathrm{e}})^{\mathrm{c}} = \mathfrak{p}$ by set extensionality.
(iv) *Extending a Contraction*: Fix $\mathfrak{q} \in \operatorname{Spec}(S^{-1}R)$. If $r \in \mathfrak{q}^{\mathrm{c}}$ and $s \in S$, then $r/s = (r/1_R)(1_R/s) \in \mathfrak{q}$; $(\mathfrak{q}^{\mathrm{c}})^{\mathrm{e}} \subseteq \mathfrak{q}$. Conversely, fix $(r,s) \in R \times S$ with $r/s \in \mathfrak{q}$. Then $r/1_R = (s/1_R)(r/s) \in \mathfrak{q}$, so $r \in \mathfrak{q}^{\mathrm{c}}$ by Definition 2.1.7(ii). Thus $r/s \in (\mathfrak{q}^{\mathrm{c}})^{\mathrm{e}}$ by Definition 2.1.7(i); $(\mathfrak{q}^{\mathrm{c}})^{\mathrm{e}} = \mathfrak{q}$ by mutual inclusion. Asserted bijection follows by (i)–(iv).

Structural Commentary: Primality survives both transfers; numerator tests prove mutual inversion.
Key Insight: A localized prime records exactly which original elements avoid all denominators. □

Definition 2.1.10. *Principal Denominator Set*
Let R be a commutative ring with identity and let $f \in R$. Define $S_f := \{f^n : n \in \mathbb{N}_0\}$.

Proposition 2.1.11. *Canonical Denominator Sets*
Let R be a commutative ring with identity, let $f \in R$, and let $\mathfrak{p} \in \operatorname{Spec} R$.

(i) The set S_f is multiplicative.
(ii) The set $R \setminus \mathfrak{p}$ is multiplicative.

Proof.

(i) Multiplicativity holds by $1_R = f^0 \in S_f$ and $(\forall\,m,n \in \mathbb{N}_0)\,f^m f^n = f^{m+n} \in S_f$.
(ii) The relation $1_R \notin \mathfrak{p}$ holds by properness of $\mathfrak{p}$. If $a,b \notin \mathfrak{p}$, then $ab \notin \mathfrak{p}$ by the contrapositive of primeness.

Structural Commentary: Powers and prime complements form the two denominator systems.
Key Insight: Primeness is precisely what makes a complement closed under multiplication. □

Definition 2.1.12. *Principal and Prime Localizations*
Let R be a commutative ring with identity and let M be an R-module. Fix $f \in R$ and $\mathfrak{p} \in \operatorname{Spec} R$.

(i) The *principal localizations* are $R_f := S_f^{-1}R$ and $M_f := S_f^{-1}M$.
(ii) The *localizations at* $\mathfrak{p}$ are $R_{\mathfrak{p}} := (R \setminus \mathfrak{p})^{-1}R$ and $M_{\mathfrak{p}} := (R \setminus \mathfrak{p})^{-1}M$.

Proposition 2.1.13. *Canonical Localization Consequences*
Let R be a commutative ring with identity and let M be an R-module.

(i) $(\forall\,f \in R)$*, contraction induces a bijection* $\operatorname{Spec}(R_f) \to D(f)$.
(ii) $(\forall\,(\mathfrak{p},m) \in \operatorname{Spec} R \times M)\,[m/1_R = 0_{M_{\mathfrak{p}}} \iff (\exists\,s \in R \setminus \mathfrak{p})\,sm = 0_M]$.
(iii) $(\forall\,m \in M)\,[m = 0_M \iff (\forall\,\mathfrak{p} \in \operatorname{Spec} R)\,m/1_R = 0_{M_{\mathfrak{p}}}]$.
(iv) If R is an integral domain with field of quotients K, then $(R \setminus \{0_R\})^{-1}R \cong K$ canonically.

Proof.

(i) Fix $f \in R$ and $\mathfrak{p} \in \operatorname{Spec} R$. The equivalence $\mathfrak{p} \cap S_f = \emptyset \iff f \notin \mathfrak{p}$ holds by primeness and induction. Equivalence $f \notin \mathfrak{p} \iff \mathfrak{p} \in D(f)$ holds by Proposition 1.2.3(iii). Asserted bijection follows by Theorem 2.1.9.
(ii) ($\Rightarrow$) Fix $(\mathfrak{p},m) \in \operatorname{Spec} R \times M$ and assume $m/1_R = 0_{M_{\mathfrak{p}}}$. The set $R \setminus \mathfrak{p}$ is multiplicative by Proposition 2.1.11(ii). Choose $s \in R \setminus \mathfrak{p}$ satisfying $sm = 0_M$ by Proposition 2.1.5(iv).
(ii) ($\Leftarrow$) Fix $(\mathfrak{p},m) \in \operatorname{Spec} R \times M$, $s \in R \setminus \mathfrak{p}$ satisfying $sm = 0_M$. Equality $m/1_R = 0_{M_{\mathfrak{p}}}$ holds by Proposition 2.1.5(iv).

(*iii*) ($\Rightarrow$) Fix $m \in M$, $m = 0_M$. Relation $(\forall \mathfrak{p} \in \operatorname{Spec} R)\, m/1_R = 0_{M_\mathfrak{p}}$ holds by (ii) with $s = 1_R$.
(*iii*) ($\Leftarrow$) Fix $m \in M$. For contrapositive, suppose $m \neq 0_M$, $\mathcal{A} := \operatorname{Ann}_R(m)$. The set $\mathcal{A}$ is an ideal by Proposition 1.6.5(i).
It is proper by $m \neq 0_M$, $1_R m = m$, and Definition 1.6.4(i). Set $\mathcal{P} := \{\mathcal{I} \trianglelefteq R : \mathcal{A} \subseteq \mathcal{I},\ \mathcal{I} \neq R\}$, ordered by inclusion.
Then $\mathcal{A} \in \mathcal{P}$, and $\mathcal{P}$ is a nonempty poset by Proposition 6.1.5. The ideal $\mathcal{A}$ is an upper bound for the empty chain.
Fix a nonempty chain $\mathcal{C} \subseteq \mathcal{P}$, $\mathcal{U} := \bigcup_{\mathcal{I} \in \mathcal{C}} \mathcal{I}$; $\mathcal{U} \trianglelefteq R$ by Proposition 3.3.3. Inclusion $\mathcal{A} \subseteq \mathcal{U}$ holds by $\mathcal{C} \neq \emptyset$, $(\forall \mathcal{I} \in \mathcal{C})\, \mathcal{A} \subseteq \mathcal{I}$.
Relation $1_R \notin \mathcal{U}$ holds by $(\forall \mathcal{I} \in \mathcal{C})\, 1_R \notin \mathcal{I}$; $\mathcal{U} \in \mathcal{P}$ is upper bound for $\mathcal{C}$; chain-upper-bound hypothesis of Principle 6.1.3 holds.
Choose a maximal $\mathfrak{m} \in \mathcal{P}$ by Principle 6.1.3. If $\mathfrak{m} \subseteq \mathcal{J} \trianglelefteq R$ and $\mathcal{J} \neq R$, then $\mathcal{J} \in \mathcal{P}$; $\mathfrak{m}$ is maximal in R by Definition 3.4.3.
Ideal $\mathfrak{m}$ prime by Corollary 1.1.3. If $m/1_R = 0_{M_\mathfrak{m}}$, choose $s \in R \setminus \mathfrak{m}$ with $sm = 0_M$ by (ii); contradicts $s \in \operatorname{Ann}_R(m) = \mathcal{A} \subseteq \mathfrak{m}$.
The inequality $m/1_R \neq 0_{M_\mathfrak{m}}$ holds by contradiction; the contrapositive follows by this prime $\mathfrak{m}$.
(*iv*) The set $R \setminus \{0_R\}$ is multiplicative by the integral-domain axioms Fix $r, r' \in R$ and $s, s' \in R \setminus \{0_R\}$.
The relation $(r, s) \sim (r', s')$ is equivalent to $s'r = sr'$ by cancellation.
Thus the localization and field-of-quotients equivalence relations and fraction operations agree by Definition 3.5.3.
The identity on $R \times (R \setminus \{0_R\})$ therefore descends to the canonical ring isomorphism.

Structural Commentary: Prime correspondence handles denominators; maximal annihilators detect nonvanishing.
Key Insight: Vanishing in all prime localizations detects the zero element. □

2.2 Local Rings

Definition 2.2.1. *Local Rings*
A commutative ring R with identity is *local* if it has a unique maximal ideal $\mathfrak{m}$. This unique ideal is the *maximal ideal* of R.

Definition 2.2.2. *Localized Prime Ideal*
Let R be a commutative ring with identity and let $\mathfrak{p} \in \operatorname{Spec} R$. The *localized prime ideal* is $\mathfrak{p}R_\mathfrak{p} := (R \setminus \mathfrak{p})^{-1}\mathfrak{p}$.

Proposition 2.2.3. *Elements of a Prime Localization*
Let R be a commutative ring with identity, $\mathfrak{p} \in \operatorname{Spec} R$. Let $(a, s) \in R \times (R \setminus \mathfrak{p})$.

(i) $a/s \in \mathfrak{p}R_\mathfrak{p} \iff a \in \mathfrak{p}$.
(ii) $a/s \notin \mathfrak{p}R_\mathfrak{p} \implies a \in R \setminus \mathfrak{p}$.
(iii) If $a \in R \setminus \mathfrak{p}$, then s/a is a multiplicative inverse of a/s in $R_\mathfrak{p}$.

Proof.

(*i*) ($\Rightarrow$) Assume $a/s \in \mathfrak{p}R_\mathfrak{p}$. Choose $u \in R \setminus \mathfrak{p}$ with $ua \in \mathfrak{p}$ by Proposition 2.1.8(ii); $a \in \mathfrak{p}$ by $u \notin \mathfrak{p}$ and primeness.
(*i*) ($\Leftarrow$) Assume $a \in \mathfrak{p}$. Then $a/s \in \mathfrak{p}R_\mathfrak{p}$ by Definition 2.2.2.
(*ii*) The implication holds by the contrapositive of (i).
(*iii*) The membership $s/a \in R_\mathfrak{p}$ holds by $a \in R \setminus \mathfrak{p}$. The identity $(a/s)(s/a) = 1_{R_\mathfrak{p}}$ holds by Definition 2.1.1(ii),(iv).

Structural Commentary: Extension tests numerators; swapping fractions constructs inverses.
Key Insight: Numerators outside $\mathfrak{p}$ become denominators for explicit inverses. □

Theorem 2.2.4. *Prime Localizations Are Local*
Let R be a commutative ring with identity and let $\mathfrak{p} \in \operatorname{Spec} R$.

(i) The ideal $\mathfrak{p}R_\mathfrak{p}$ is the unique maximal ideal of $R_\mathfrak{p}$.
(ii) The ring $R_\mathfrak{p}$ is local.
(iii) The quotient ring $R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}$ is a field.

Proof.

(i) The relation $\mathfrak{p}R_\mathfrak{p} \trianglelefteq R_\mathfrak{p}$ holds by Proposition 2.1.8(i). The ideal is proper by $\mathfrak{p} \cap (R \setminus \mathfrak{p}) = \emptyset$ and Proposition 2.1.8(iv).
Fix a proper ideal $\mathcal{J} \trianglelefteq R_\mathfrak{p}$ and assume $\mathcal{J} \not\subseteq \mathfrak{p}R_\mathfrak{p}$. Choose $x \in \mathcal{J} \setminus \mathfrak{p}R_\mathfrak{p}$ by the failed inclusion.
Choose $(a, s) \in R \times (R \setminus \mathfrak{p})$ with $x = a/s$ by Definition 2.1.1(iii); $a \in R \setminus \mathfrak{p}$ and $(a/s)(s/a) = 1_{R_\mathfrak{p}}$ by Proposition 2.2.3(ii),(iii).
The membership $1_{R_\mathfrak{p}} = x(s/a) \in \mathcal{J}$ contradicts properness by ideal absorption. Thus $\mathcal{J} \subseteq \mathfrak{p}R_\mathfrak{p}$.
Fix $\mathcal{K} \trianglelefteq R_\mathfrak{p}$ with $\mathfrak{p}R_\mathfrak{p} \subseteq \mathcal{K}$ and $\mathcal{K} \neq R_\mathfrak{p}$. Then $\mathcal{K} \subseteq \mathfrak{p}R_\mathfrak{p}$ by the proper-ideal containment.
Hence $\mathcal{K} = \mathfrak{p}R_\mathfrak{p}$, so $\mathfrak{p}R_\mathfrak{p}$ is maximal by Definition 3.4.3. If $\mathfrak{m} \trianglelefteq R_\mathfrak{p}$ is maximal, then $\mathfrak{m} \subseteq \mathfrak{p}R_\mathfrak{p}$ by the proper-ideal containment.
Therefore $\mathfrak{m} = \mathfrak{p}R_\mathfrak{p}$ by maximality of $\mathfrak{m}$ and properness of $\mathfrak{p}R_\mathfrak{p}$.
(ii) The ring $R_\mathfrak{p}$ is local by (i) and Definition 2.2.1.
(iii) The quotient is a field by (i) and Theorem 3.4.4.

Structural Commentary: Explicit inverses force proper ideals into $\mathfrak{p}R_\mathfrak{p}$; quotient theory finishes.
Key Insight: Fractions outside $\mathfrak{p}R_\mathfrak{p}$ invert by swapping numerator and denominator. □

Definition 2.2.5. *Residue Fields*
Let R be a commutative ring with identity.

(i) If R is local with maximal ideal $\mathfrak{m}$, its *residue field* is $R/\mathfrak{m}$.
(ii) For $\mathfrak{p} \in \operatorname{Spec} R$, the *residue field at $\mathfrak{p}$* is $k(\mathfrak{p}) := R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}$.

3 Noetherian Rings and Modules

3.1 Noetherian Rings

Theorem 3.1.1. *Hilbert Basis Theorem*
Let A be a nonzero commutative ring with identity. If A is Noetherian, then $A[x]$ is Noetherian.

Proof.

(i) *Reduction*: By Theorem 2.4.2(ii), to prove $(\forall \mathcal{J} \trianglelefteq A[x])\, \mathcal{J}$ is f.g. Fix $\mathcal{J} \trianglelefteq A[x]$. If $\mathcal{J} = 0$, then $\mathcal{J} = \langle 0 \rangle$; assume $\mathcal{J} \neq 0$.
For $g = \sum_{j=0}^{n} b_j x^j \in A[x]$ with $n \in \mathbb{N}_0$ and $b_n \neq 0$, set $\deg g := n$ and $\operatorname{lc}(g) := b_n$.

(ii) *Leading-Coefficient Generators*: Set $L := \langle \mathrm{lc}(g) : g \in \mathcal{J} \setminus \{0\} \rangle \trianglelefteq A$.
The ideal L is f.g. by Noetherianity of A and Theorem 2.4.2(ii). Choose generators $\ell_1, \dots, \ell_s$ of L by finite generation of L.
Each ℓ_j is a finite A-linear combination of displayed leading coefficients by Definition 3.3.1(ii).
Write the collected coefficients as $\mathrm{lc}(f_1), \dots, \mathrm{lc}(f_m)$ with $(f_1, \dots, f_m) \in (\mathcal{J} \setminus \{0\})^m$.
$L = \langle \mathrm{lc}(f_i) : 1 \le i \le m \rangle$ by the generator expansions and the definition of L.
Set $d := \max\limits_{1 \le i \le m} \deg f_i$ and replace each f_i with $x^{d - \deg f_i} f_i$.
$(\forall\, i \in \{1, \dots, m\})\, [f_i \in \mathcal{J} \wedge \deg f_i = d]$ by ideal absorption and the definition of d.
$L = \langle \mathrm{lc}(f_i) : 1 \le i \le m \rangle$ by invariance of leading coefficients under these monomial shifts.

(iii) *Low-Degree Generators*: For $0 \le k < d$, set $L_k := \langle \mathrm{lc}(g) : g \in \mathcal{J},\ \deg g = k \rangle$. Each L_k is f.g. by Theorem 2.4.2(ii).
Choose finite generators of L_k by its finite generation.
Each generator is a finite A-linear combination of those coefficients by Definition 3.3.1(ii).
Let $P_k \subseteq \{p \in \mathcal{J} : \deg p = k\}$ be the corresponding finite polynomial set.
Then $L_k = \langle \mathrm{lc}(p) : p \in P_k \rangle$ by the generator expansions and the definition of L_k.
Set $G := \{f_1, \dots, f_m\} \cup \bigcup\limits_{0 \le k < d} P_k$ and $\mathcal{I} := \langle G \rangle \trianglelefteq A[x]$.
The set G is finite by finiteness of the index range and each constituent. $\mathcal{I} \subseteq \mathcal{J}$ by Proposition 3.3.2.

(iv) *Degree Descent*: Suppose $\mathcal{I} \neq \mathcal{J}$. Set $D := \{\deg u + 1 : u \in \mathcal{J} \setminus \mathcal{I}\} \subseteq \mathbb{N}$. $D \neq \emptyset$ by $\mathcal{I} \neq \mathcal{J}$.
Choose $g \in \mathcal{J} \setminus \mathcal{I}$ with $\deg g + 1 = \min D$ by Assumption 2.6.1. Set $r := \deg g$. If $r < d$, then $\mathrm{lc}(g) \in L_r$ by the definition of L_r.
Choose $(c_p)_{p \in P_r} \in A^{P_r}$ by generation of L_r by the leading coefficients indexed by P_r.
$\mathrm{lc}(g) = \sum\limits_{p \in P_r} c_p \,\mathrm{lc}(p)$ by the choice of $(c_p)_{p \in P_r}$. In this case, set $h := \sum\limits_{p \in P_r} c_p p$. If $r \ge d$, then $\mathrm{lc}(g) \in L$ by the definition of L.
Choose $c_1, \dots, c_m \in A$ by $L = \langle \mathrm{lc}(f_i) : 1 \le i \le m \rangle$. $\mathrm{lc}(g) = \sum\limits_{i=1}^{m} c_i \,\mathrm{lc}(f_i)$ by the choice of $c_1, \dots, c_m$.
In this case, set $h := \sum\limits_{i=1}^{m} c_i x^{r-d} f_i$. In both cases, $h \in \mathcal{I}$, $\deg h = r$, and $\mathrm{lc}(h) = \mathrm{lc}(g)$ by construction.
If $g - h \in \mathcal{I}$, then $g = (g - h) + h \in \mathcal{I}$, a contradiction. Hence $g - h \in \mathcal{J} \setminus \mathcal{I}$ by $g, h \in \mathcal{J}$ and the contradiction above.
$\deg(g - h) < r$ by cancellation of the degree-r coefficient.
$\deg(g - h) + 1 \in D$ by $g - h \in \mathcal{J} \setminus \mathcal{I}$, while $\deg(g - h) + 1 < \min D$ by the preceding inequality, a contradiction.
$\mathcal{J} = \mathcal{I} = \langle G \rangle$ by contradiction; hence $\mathcal{J}$ is f.g. $A[x]$ is Noetherian by arbitrariness of $\mathcal{J}$ and Theorem 2.4.2(ii).

Structural Commentary: Finite leading-coefficient data span $\mathcal{J}$ by degree descent.
Key Insight: Matching the leading coefficient cancels the highest-degree term. □

3.2 Noetherian Modules

Theorem 3.2.1. *Finitely Generated Modules over Noetherian Rings*
Let A be a nonzero commutative ring with identity. Then A is Noetherian $\iff$ $(\forall\, M)\, [M$ is an f.g. A-module $\Rightarrow M$ is Noetherian$]$.

Proof.

($\Rightarrow$) Assume A is Noetherian.

(a) *Finite Free Modules*: For $n \in \mathbb{N}$, set $Q(n) := (\forall\, S \le A^n)\, S$ is f.g.; $Q(1)$ holds by Definition 2.4.1(ii), Theorem 2.4.2(i).
Fix $n \in \mathbb{N}$, $Q(n)$, $S \le A^{n+1}$. Let $\pi \in \mathrm{Hom}_A(S, A)$ be last-coordinate projection. Set $K := \{v \in A^n : (v, 0_A) \in S\} \subseteq A^n$.
$0_{A^n} \in K$ by $(0_{A^n}, 0_A) = 0_{A^{n+1}} \in S$. $(\forall\, (a, b, v, w) \in A^2 \times K^2)\, av + bw \in K$ by $(av + bw, 0_A) = a(v, 0_A) + b(w, 0_A) \in S$.
$K \le A^n$ by Theorem 1.1.5. $\ker \pi \le S$ by Proposition 1.4.2(ii). Map $v \mapsto (v, 0_A)$ identifies K with $\ker \pi$ by definition of π.
Thus K is f.g. by $Q(n)$. The image $\pi(S) \le {}_A A$ by Proposition 1.4.2(iii). The module $\pi(S)$ is f.g. by $Q(1)$.
Choose $p \in \mathbb{N}$ and $(k_1, \dots, k_p) \in K^p$ by finite generation of K. $K = \langle\!\langle k_1, \dots, k_p \rangle\!\rangle_A$ by the choice of $(k_1, \dots, k_p)$.
Choose $q \in \mathbb{N}$ and $(t_1, \dots, t_q) \in \pi(S)^q$ by finite generation of $\pi(S)$. $\pi(S) = \langle\!\langle t_1, \dots, t_q \rangle\!\rangle_A$ by the choice of $(t_1, \dots, t_q)$.
Choose $(s_1, \dots, s_q) \in S^q$ satisfying $(\forall\, j \in \{1, \dots, q\})\, \pi(s_j) = t_j$ by the definition of $\pi(S)$.
Fix $s \in S$. Choose $(a_1, \dots, a_q) \in A^q$ satisfying $\pi(s) = \sum\limits_{j=1}^{q} a_j t_j$ by generation of $\pi(S)$.
Set $u := s - \sum\limits_{j=1}^{q} a_j s_j \in \ker \pi$ by A-linearity of π and the preceding equality. Write $u = (k, 0_A)$ with $k \in K$.
Choose $(b_1, \dots, b_p) \in A^p$ satisfying $k = \sum\limits_{i=1}^{p} b_i k_i$ by generation of K.
$s = \sum\limits_{j=1}^{q} a_j s_j + \sum\limits_{i=1}^{p} b_i (k_i, 0_A)$ by the defining equations for u, k, and $(b_i)_{i=1}^{p}$.
Thus S is f.g. by arbitrariness of s, closure of S, and Definition 1.2.3(ii).
$Q(n+1)$ holds by arbitrariness of S; $(\forall\, n \in \mathbb{N})\, Q(n)$ by Theorem 2.6.2; $(\forall\, n \in \mathbb{N})\, [A^n$ is Noetherian$]$ by Theorem 2.4.2(i).

(b) *Finite Free Cover*: Fix an f.g. A-module M. Choose $n \in \mathbb{N}$ and $(v_1, \dots, v_n) \in M^n$ by finite generation of M.
$M = \langle\!\langle v_1, \dots, v_n \rangle\!\rangle_A$ by choice of $(v_1, \dots, v_n)$. Define $\varphi : A^n \to M$ by $\varphi(r_1, \dots, r_n) := \sum\limits_{i=1}^{n} r_i v_i$.
The map φ is A-linear by the module laws and surjective by generation of M.

(c) *Submodule Preimages*: Fix $N \le M$ and set $P := \varphi^{-1}(N) \subseteq A^n$. $0_{A^n} \in P$ by $\varphi(0_{A^n}) = 0_M \in N$.
$(\forall\, (a, b, x, y) \in A^2 \times P^2)\, ax + by \in P$ by A-linearity of φ and closure of N. $P \le A^n$ by Theorem 1.1.5.
Module P is f.g. by Noetherianity of A^n, Theorem 2.4.2(i). Choose $m \in \mathbb{N}$ and $(p_1, \dots, p_m) \in P^m$ by finite generation of P.
$P = \langle\!\langle p_1, \dots, p_m \rangle\!\rangle_A$ by the choice of $(p_1, \dots, p_m)$. $N = \varphi(P)$ by surjectivity of φ and the definition of P.
$N = \langle\!\langle \varphi(p_1), \dots, \varphi(p_m) \rangle\!\rangle_A$ by A-linearity of φ and generation of P. Hence M is Noetherian by Theorem 2.4.2(i).
The forward implication holds by arbitrariness of M.

($\Leftarrow$) Assume $(\forall\, M)\, [M$ is an f.g. A-module $\Rightarrow M$ is Noetherian$]$. Regular module ${}_A A = \langle\!\langle 1_A \rangle\!\rangle_A$ is f.g. by Definition 1.2.3(ii).
It is Noetherian by the hypothesis. Hence A is Noetherian by Definition 2.4.1(ii).

Structural Commentary: Induction handles A^n; free covers transfer Noetherianity to quotients.
Key Insight: Generators of $\varphi^{-1}(N)$ map to generators of N. □

4 Integral Dependence and Dimension

4.1 Integral Dependence

4.2 Dimension Theory

5 Regularity and Singularities

5.1 Regularity

Definition 5.1.1. *Ideal Multiples and the Local Cotangent Module*
Let R be a commutative ring with identity, let $\mathcal{I} \trianglelefteq R$, and let M be an R-module.

(i) The *ideal multiple* of M is $\mathcal{I}M := \langle\!\langle ax : (a,x) \in \mathcal{I} \times M \rangle\!\rangle_R$.
(ii) If R is local with maximal ideal $\mathfrak{m}$, set $\mathfrak{m}^2 := \mathfrak{m}\mathfrak{m}$. The quotient R-module $\mathfrak{m}/\mathfrak{m}^2$ is the *local cotangent module* of R.

Proposition 5.1.2. *Units in a Local Ring*
Let R be nonzero commutative local ring with identity and maximal ideal $\mathfrak{m}$. Then $(\forall\, a \in R)\,[a \notin \mathfrak{m} \iff (\exists\, b \in R)\, ab = 1_R]$.

Proof.
($\Rightarrow$) Fix $a \notin \mathfrak{m}$ and suppose $(\forall\, b \in R)\, ab \neq 1_R$. Then $\mathcal{I} := \langle a \rangle$ is proper. The quotient $S := R/\mathcal{I}$ is nonzero.
Choose a maximal ideal $\mathfrak{n} \trianglelefteq S$ by Theorem 3.4.5. Let $q : R \twoheadrightarrow S$ be the quotient map and set $\mathfrak{p} := q^{-1}(\mathfrak{n})$.
The set $\mathfrak{p}$ is a proper ideal by closure of $\mathfrak{n}$ and $1_S \notin \mathfrak{n}$. Let $\mathfrak{p} \subseteq \mathcal{J} \trianglelefteq R$. The image $q(\mathcal{J}) \trianglelefteq S$ by surjectivity of q and ideal closure.
Equality $q(\mathfrak{p}) = \mathfrak{n}$ holds by surjectivity of q, so $\mathfrak{n} \subseteq q(\mathcal{J})$. Thus $q(\mathcal{J}) = \mathfrak{n}$ or $q(\mathcal{J}) = S$ by maximality.
In first case, $\mathcal{J} = \mathfrak{p}$ by $\ker q = \mathcal{I} \subseteq \mathfrak{p}$. In second case, choose $j \in \mathcal{J}$ with $q(j) = 1_S$; then $1_R - j \in \mathcal{I} \subseteq \mathcal{J}$, so $\mathcal{J} = R$.
Hence $\mathfrak{p}$ is maximal by Definition 3.4.3, and $\mathfrak{p} = \mathfrak{m}$ by locality. But $a \in \mathcal{I} \subseteq \mathfrak{p} = \mathfrak{m}$, a contradiction.
($\Leftarrow$) If $ab = 1_R$ and $a \in \mathfrak{m}$, then $1_R = ab \in \mathfrak{m}$ by ideal absorption, contradicting properness of $\mathfrak{m}$.
Structural Commentary: Maximal ideals above a nonunit principal ideal reduce invertibility to membership in $\mathfrak{m}$.
Key Insight: Every nonunit lies in the unique maximal ideal. □

Theorem 5.1.3. *Nakayama's Lemma*
Let R be a nonzero commutative local ring with identity and maximal ideal $\mathfrak{m}$.
If M is a finitely generated R-module and $\mathfrak{m}M = M$, then $M = \{0_M\}$.

Proof.
(i) *Minimal Generators*: Suppose $M \neq \{0_M\}$.
Choose a generating tuple $(x_i)_{i=1}^n$ of minimal positive length by finite generation and Assumption 2.6.1.
(ii) *Coefficient Relation*: The tuple $(x_i)_{i=1}^n$ generates M.
Thus $(\forall\, y \in \mathfrak{m}M)(\exists\, (a_i)_{i=1}^n \in \mathfrak{m}^n)\, y = \sum_{i=1}^n a_i x_i$ by distributivity and ideal absorption.
Choose such $a_1, \ldots, a_n$ for $y = x_n$ by $x_n \in M = \mathfrak{m}M$. Then $(1_R - a_n)x_n = \sum_{i=1}^{n-1} a_i x_i$.
(iii) *Unit Reduction*: The relation $1_R - a_n \notin \mathfrak{m}$ holds because $\mathfrak{m}$ is proper and $a_n \in \mathfrak{m}$.
Choose $b \in R$ with $b(1_R - a_n) = 1_R$ by Proposition 5.1.2. Multiplication by b places x_n in $\langle\!\langle x_1, \ldots, x_{n-1} \rangle\!\rangle_R$.
(iv) *Contradiction*: Thus $M = \langle\!\langle x_1, \ldots, x_{n-1} \rangle\!\rangle_R$, contrary to minimality of n. Hence $M = \{0_M\}$.
Structural Commentary: Minimal generators convert $\mathfrak{m}M = M$ into a removable last generator.
Key Insight: A coefficient congruent to 1_R modulo $\mathfrak{m}$ is invertible. □

Proposition 5.1.4. *Residue-Field Structure on the Local Cotangent Module*
Let R be a nonzero commutative local ring with identity, maximal ideal $\mathfrak{m}$, and residue field $k := R/\mathfrak{m}$.
The formula $(a + \mathfrak{m})(x + \mathfrak{m}^2) := ax + \mathfrak{m}^2$ makes $\mathfrak{m}/\mathfrak{m}^2$ a k-vector space.

Proof. The inclusion $\mathfrak{m}^2 \subseteq \mathfrak{m}$ holds by ideal absorption and Definition 5.1.1. The ring k is a field by Theorem 3.4.4.
If $a' - a \in \mathfrak{m}$ and $x' - x \in \mathfrak{m}^2$, then $a'x' - ax = a'(x' - x) + (a' - a)x \in \mathfrak{m}^2$.
Thus the scalar action is well defined. The vector-space axioms hold by the R-module axioms and quotient operations.
Structural Commentary: Ideal absorption makes scalar representatives modulo $\mathfrak{m}$ act consistently modulo $\mathfrak{m}^2$.
Key Insight: Products of two maximal-ideal elements vanish in the cotangent quotient. □

Definition 5.1.5. *Embedding Dimension*
Let R be a nonzero commutative local ring with maximal ideal $\mathfrak{m}$ and residue field $k = R/\mathfrak{m}$.
The *embedding dimension* of R is $\operatorname{edim}(R) := \dim_k(\mathfrak{m}/\mathfrak{m}^2)$.

Proposition 5.1.6. *Embedding Dimension Counts Minimal Generators*
Let R be a nonzero Noetherian commutative local ring with identity and maximal ideal $\mathfrak{m}$.

(i) $\operatorname{edim}(R) \in \mathbb{N}_0$.
(ii) $\operatorname{edim}(R) = \min\{n \in \mathbb{N}_0 : (\exists\, (x_i)_{i=1}^n \in \mathfrak{m}^n)\, \mathfrak{m} = \langle x_1, \ldots, x_n \rangle\}$.

Proof. The ideal $\mathfrak{m}$ is finitely generated by Theorem 2.4.2(ii). Choose $r \in \mathbb{N}_0$ and $x_1, \ldots, x_r \in \mathfrak{m}$ with $\mathfrak{m} = \langle x_1, \ldots, x_r \rangle$.
(i) Cotangent space is finite-dimensional by Definition 1.3.1(iii) with spanning list $(x_i + \mathfrak{m}^2)_{i=1}^r$; $\operatorname{edim}(R) \in \mathbb{N}_0$ by Definition 1.4.2.
(ii) Set $d := \operatorname{edim}(R)$. If $\mathfrak{m} = \langle z_1, \ldots, z_s \rangle$, then $d \leq s$ by Theorem 1.3.11 with spanning list $(z_i + \mathfrak{m}^2)_{i=1}^s$.
Choose a k-basis $(y_i + \mathfrak{m}^2)_{i=1}^d$ with $y_i \in \mathfrak{m}$, $N := \langle y_1, \ldots, y_d \rangle$. Then $\mathfrak{m} = N + \mathfrak{m}^2$ by spanning property. Set $Q := \mathfrak{m}/N$.
The module Q is finitely generated by the images of $x_1, \ldots, x_r$. The equality $\mathfrak{m}Q = (\mathfrak{m}^2 + N)/N = Q$ holds by $\mathfrak{m} = N + \mathfrak{m}^2$.
Thus $Q = \{0_Q\}$ by Theorem 5.1.3. Hence $\mathfrak{m} = N$, so d generators attain the displayed minimum.
Structural Commentary: Cotangent spanning yields a quotient vanishing by Nakayama; arbitrary generators descend to spanning lists.
Key Insight: Modulo $\mathfrak{m}^2$, precisely the irredundant generator directions remain. □

Definition 5.1.7. *Krull Dimension and Regular Local Rings*
Let R be a nonzero commutative local ring with identity.

(i) Let $\mathscr{C}(R)$ be the set of finite strictly increasing chains in $\operatorname{Spec} R$. For $\mathcal{P} = (\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n) \in \mathscr{C}(R)$, set $\ell(\mathcal{P}) := n$. The *Krull dimension* is $\dim R := \sup_{\mathcal{P} \in \mathscr{C}(R)} \ell(\mathcal{P})$. Its value is ∞ if these lengths are unbounded.

(ii) The ring R is *regular local* if it is Noetherian and $\dim R = \operatorname{edim}(R)$.

5.2 Singularities Preview

Definition 5.2.1. *Singular Local Rings and Embedding-Dimension Excess*
Let R be a nonzero Noetherian commutative local ring with identity.

(i) The ring R is *singular* if it is not regular. Equivalently, $\dim R \neq \operatorname{edim}(R)$ by Definition 5.1.7(ii).

(ii) Assume $\dim R \in \mathbb{N}_0$ and $\dim R \leq \operatorname{edim}(R)$. The *embedding-dimension excess* is $\operatorname{exc}(R) := \operatorname{edim}(R) - \dim R$.

Proposition 5.2.2. *Positive Embedding-Dimension Excess Detects Singularity*
Let R be a nonzero Noetherian commutative local ring with identity.
If $\dim R \in \mathbb{N}_0$ and $\dim R \leq \operatorname{edim}(R)$, then R is singular $\iff \operatorname{exc}(R) > 0$.

Proof.

($\Rightarrow$) If R is singular, then $\dim R \neq \operatorname{edim}(R)$ by Definition 5.2.1(i). Thus $\dim R < \operatorname{edim}(R)$ by the dimension-order hypothesis. The inequality $\operatorname{exc}(R) > 0$ holds by Definition 5.2.1(ii).

($\Leftarrow$) If $\operatorname{exc}(R) > 0$, then $\dim R < \operatorname{edim}(R)$ by Definition 5.2.1(ii). Hence R is singular by Definition 5.2.1(i).

Structural Commentary: The dimension order converts failure of regularity equality into positivity of the excess.
Key Insight: The excess vanishes exactly when Krull and embedding dimensions coincide. □

Example 5.2.3. *A Square-Zero Singular Local Ring*
Let k be a field and set $R := k[x,y]/\langle x^2, xy, y^2\rangle$. Write $\bar{x}, \bar{y} \in R$ for the residue classes of x, y, and set $\mathfrak{m} := \langle \bar{x}, \bar{y}\rangle$.
Relation $(\forall\, r \in R)(\exists!\, (a,b,c) \in k^3)\, r = a + b\bar{x} + c\bar{y}$ holds by polynomial reduction. Equality $\mathfrak{m}^2 = \{0_R\}$ holds by quotient relations.
For $(a,u) \in (k \setminus \{0_k\}) \times \mathfrak{m}$, $(a+u)(a^{-1} - a^{-2}u) = 1_R$. Thus $(\forall\, r \in R \setminus \mathfrak{m})(\exists\, s \in R)\, rs = 1_R$.
The quotient $R/\mathfrak{m} \cong k$ is a field, so $\mathfrak{m}$ is maximal by Theorem 3.4.4.
The implication $(\forall\, \mathcal{J} \trianglelefteq R)\, [\mathcal{J} \neq R \Rightarrow \mathcal{J} \subseteq \mathfrak{m}]$ holds because a proper ideal contains no invertible element.
Hence R is local with maximal ideal $\mathfrak{m}$. The equality $\dim_k R = 3$ holds by the unique normal form. Fix $\mathcal{J} \trianglelefteq R$.
Since $\mathcal{J}$ is a k-subspace of R, choose a finite k-basis by Corollary 1.3.14(i)–(ii). This basis generates $\mathcal{J}$ as an R-module.
Hence R is Noetherian by Theorem 2.4.2(ii), because $\mathcal{J}$ was arbitrary.
The inclusion $(\forall\, \mathfrak{p} \in \operatorname{Spec} R)\, \{\bar{x}, \bar{y}\} \subseteq \mathfrak{p}$ holds by $\bar{x}^2 = \bar{y}^2 = 0_R$ and primeness. Thus $\operatorname{Spec} R = \{\mathfrak{m}\}$.
The equality $\dim R = 0$ holds by Definition 5.1.7(i). The tuple $(\bar{x} + \mathfrak{m}^2, \bar{y} + \mathfrak{m}^2)$ is a k-basis of $\mathfrak{m}/\mathfrak{m}^2$.
Thus $\operatorname{edim}(R) = 2$ by Definition 5.1.5. Therefore $\operatorname{exc}(R) = 2 > 0$, and R is singular by Proposition 5.2.2.

Part X
Algebraic Geometry

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1.1 Affine Algebraic Sets

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7 Divisors, Line Bundles, and Curves

7.1 Divisors

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Part XI
Real and Topological Analysis

1 Number Systems and Completeness

1.1 Ordered Fields, Suprema, and Real Structure

$\mathbb{Q}$ has gaps that completeness repairs in $\mathbb{R}$, while $\mathbb{C}$ retains field algebra without a compatible order.

Definition 1.1.1. *Ordered Fields*
Let F be a field equipped with a total order $<$. Then F is an *ordered field* iff

(i) $(\forall x, y, z \in F)\,[y < z \Rightarrow x + y < x + z]$;
(ii) $(\forall x, y \in F)\,[(0 < x \wedge 0 < y) \Rightarrow 0 < xy]$.

Proposition 1.1.2. *Basic Consequences of the Field Axioms*
Let F be a field. Then $(\forall a, b, c, x, y \in F)$:

(i) $a + c = b + c \Rightarrow a = b$, *and* $c \neq 0 \Rightarrow (ac = bc \Rightarrow a = b)$.
(ii) *The identities* $0, 1$ *and the inverses* $-a$, a^{-1} *for* $a \neq 0$ *are unique.*
(iii) $0x = 0$.
(iv) $xy = 0 \Rightarrow (x = 0 \vee y = 0)$.
(v) $(-x)y = -(xy) = x(-y)$ *and* $(-x)(-y) = xy$.

Proof.

(i) $a = b$ by adding $-c$ to $a + c = b + c$. If $c \neq 0$, then $a = b$ by multiplying $ac = bc$ by c^{-1}.
(ii) In a group $(G, *, e)$, any identity e' satisfies $e = e * e' = e'$.
If $u * a = e = a * v$, then $u = u * e = u * (a * v) = (u * a) * v = v$. These calculations apply to $(F, +, 0)$ and $(F \setminus \{0\}, \cdot, 1)$.
(iii) $0x = 0$ by cancellation from $0x + 0x = (0 + 0)x = 0x$.
(iv) If $x = 0$, then $x = 0 \vee y = 0$. If $xy = 0$ and $x \neq 0$, then $y = 0$ by multiplication by x^{-1}.
(v) $(-x)y = -(xy)$ by $xy + (-x)y = (x + (-x))y = 0$, and $x(-y) = -(xy)$ by the same calculation.
The identity $(-x)(-y) = -(x(-y)) = xy$ follows by the first equality.

Structural Commentary: Group cancellation fixes identities and inverses; distributivity then controls zero and signs.
Key Insight: Uniqueness turns zero-sum identities into sign laws. □

Proposition 1.1.3. *Basic Inequality Rules in Ordered Fields*
Let F be an ordered field. Then $(\forall c, x, y \in F)$:

(i) $x > 0 \iff -x < 0$.
(ii) $(c > 0 \wedge x < y) \Rightarrow cx < cy$.
(iii) $(c < 0 \wedge x < y) \Rightarrow cy < cx$.
(iv) $x \neq 0 \Rightarrow x^2 > 0$; *in particular,* $1 > 0$.
(v) $0 < x < y \Rightarrow 0 < y^{-1} < x^{-1}$.

Proof.

(i) ($\Rightarrow$) $-x < 0$ by adding $-x$ to $0 < x$.
(i) ($\Leftarrow$) $0 < x$ by adding x to $-x < 0$.
(ii) The relations $0 < y - x$ and $0 < c(y - x) = cy - cx$ hold by additive order invariance and Definition 1.1.1(ii), respectively.
The inequality $cx < cy$ follows by adding cx to $0 < cy - cx$.
(iii) $-c > 0$ by (i), and $(-c)x < (-c)y$ by (ii). The inequality $cy < cx$ follows by adding $cx + cy$ to $-cx < -cy$.
(iv) If $x > 0$, then $x^2 > 0$ by Definition 1.1.1(ii). If $x < 0$, then $-x > 0$ by (i), so $x^2 = (-x)^2 > 0$.
One case holds by $x \neq 0$ and totality. For $x = 1$, $1 = 1^2 > 0$.
(v) $x^{-1} = x(x^{-1})^2 > 0$ and $y^{-1} = y(y^{-1})^2 > 0$ by (iv) and Definition 1.1.1(ii).
$x^{-1}y^{-1} > 0$ by Definition 1.1.1(ii), and $y^{-1} < x^{-1}$ by (ii) with multiplier $x^{-1}y^{-1}$.

Structural Commentary: Additive order invariance reduces every inequality rule to multiplication by a positive element.
Key Insight: A negative multiplier is the negative of a positive one. □

Motivation 1.1.4. *Completeness Becomes a Theorem*
Dedekind cuts encode a real number by its rational lower side.
The construction turns rational arithmetic into cut arithmetic and bounded suprema into unions.

Definition 1.1.5. *Dedekind Cuts and Their Order*
A *cut* is a set $\alpha \subseteq \mathbb{Q}$ such that

(i) $\emptyset \neq \alpha \neq \mathbb{Q}$;
(ii) $(\forall p \in \alpha)(\forall q \in \mathbb{Q})\,[q < p \Rightarrow q \in \alpha]$;
(iii) $(\forall p \in \alpha)(\exists r \in \alpha)\, p < r$.

Let $\mathscr{D}$ be the set of cuts. For $\alpha, \beta \in \mathscr{D}$, define $\alpha \leq \beta \iff \alpha \subseteq \beta$ and $\alpha < \beta \iff \alpha \subsetneq \beta$.
Within the construction, $p, q, r, \ldots \in \mathbb{Q}$ and $\alpha, \beta, \gamma, \ldots \in \mathscr{D}$. A cut is a proper rational initial segment without a largest element.

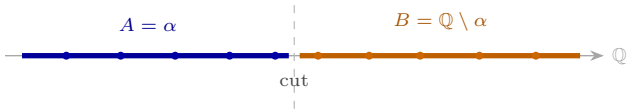

Figure 112: A Dedekind cut splits $\mathbb{Q}$ into α and $\mathbb{Q} \setminus \alpha$, with no last rational in α.

Figure 112 depicts the two sides determined by Definition 1.1.5.

Remark 1.1.6. *Left Cuts Versus Two-Sided Cuts*
Two-sided notation writes $A \mid B$ with $A = \alpha$ and $B = \mathbb{Q} \setminus \alpha$.
Definition 1.1.5 retains A; its lack of a largest element prevents duplicate boundary representations.

Definition 1.1.7. *Arithmetic of Cuts*
For $\alpha, \beta \in \mathscr{D}$, define $\alpha + \beta := \{r + s : r \in \alpha,\ s \in \beta\}$ and $0^* := \{q \in \mathbb{Q} : q < 0\}$.
Define $-\alpha := \{p \in \mathbb{Q} : (\exists r \in \mathbb{Q})\,[r > 0 \wedge -p - r \notin \alpha]\}$ and $\mathscr{D}^+ := \{\alpha \in \mathscr{D} : 0^* < \alpha\}$.
For $\alpha, \beta \in \mathscr{D}^+$, define $\alpha\beta := \{p \in \mathbb{Q} : (\exists r \in \alpha)(\exists s \in \beta)\,[r > 0 \wedge s > 0 \wedge p \le rs]\}$. Set $1^* := \{q \in \mathbb{Q} : q < 1\}$.
For $\alpha \in \mathscr{D}^+$, define $\alpha^{-1} := \{p \in \mathbb{Q} : p \le 0 \vee [p > 0 \wedge (\exists q \in \mathbb{Q} \setminus \alpha)\,(q > 0 \wedge p < q^{-1})]\}$.
Extend multiplication to zero factors by $\alpha 0^* = 0^*\alpha = 0^*$.

$$\alpha\beta := \begin{cases} (-\alpha)(-\beta), & \alpha < 0^*,\ \beta < 0^*, \\ -\big((-\alpha)\beta\big), & \alpha < 0^*,\ \beta > 0^*, \\ -\big(\alpha(-\beta)\big), & \alpha > 0^*,\ \beta < 0^*. \end{cases}$$

Positive rational representatives multiply without reversing order.
The sign rules reduce every remaining product to the positive case.

Definition 1.1.8. *Rational Cuts*
For $r \in \mathbb{Q}$, define the *rational cut* $r^* := \{p \in \mathbb{Q} : p < r\}$. Define $\iota : \mathbb{Q} \to \mathscr{D}$ by $\iota(r) := r^*$.

Theorem 1.1.9. *Existence of the Real Field*
The Dedekind-cut construction satisfies the following claims.

(i) $(\mathscr{D}, +, \cdot, <)$ is an ordered field with the least-upper-bound property.
(ii) The map $\iota : \mathbb{Q} \to \mathscr{D}$, $r \mapsto r^*$, is an order-preserving field embedding.

Proof.

(i) *Field and completeness*:

(a) *Order and suprema*: Inclusion is reflexive, antisymmetric, and transitive. If $\alpha \not\subseteq \beta$, choose $p \in \alpha \setminus \beta$.
The relation $(\forall q \in \beta)\, q < p$ holds by downward closure of β, and $\beta \subseteq \alpha$ by downward closure of α. Hence inclusion is total.
Let $\emptyset \neq A \subseteq \mathscr{D}$ have upper bound β, and set $\gamma := \bigcup_{\alpha \in A} \alpha$. The set γ is nonempty, and $\gamma \subseteq \beta \neq \mathbb{Q}$.
Downward closure and the absence of a largest element pass from the member of A containing a chosen $p \in \gamma$; hence $\gamma \in \mathscr{D}$.
Every $\alpha \in A$ satisfies $\alpha \subseteq \gamma$, while any upper bound δ of A contains every $\alpha \in A$ and hence contains γ.
These bound relations prove $\gamma = \sup A$.

(b) *Addition*: Set $\alpha + \beta$ is nonempty. Choose $r' \notin \alpha$, $s' \notin \beta$. Relation $r + s < r' + s'$ holds for $r \in \alpha$, $s \in \beta$, so $r' + s' \notin \alpha + \beta$.
Downward closure follows from $q < r + s \Rightarrow q = r + (s + q - r - s)$ because the final summand lies in β.
If $p = r + s$, choose $r_1 \in \alpha$ with $r < r_1$; then $p < r_1 + s \in \alpha + \beta$. These properties prove $\alpha + \beta \in \mathscr{D}$.
Commutativity and associativity follow from the rational identities. Inclusion $\alpha + 0^* \subseteq \alpha$ holds by downward closure.
The reverse inclusion follows from $p = r + (p - r)$ with $p < r \in \alpha$, so $\alpha + 0^* = \alpha$.
The set $-\alpha$ is nonempty by any $q \notin \alpha$ and the witness $p = -q - 1$. It is proper because $-a \notin -\alpha$ for $a \in \alpha$.
Downward closure follows from upward closure of $\mathbb{Q} \setminus \alpha$, and $p \in -\alpha$ with witness $r > 0$ satisfies $p < p + \frac{r}{2} \in -\alpha$.
Thus $-\alpha$ is a cut. For $s \in -\alpha$, relation $-s \notin \alpha$ follows by its witness, so $(\forall r \in \alpha)\, r + s < 0$ and $\alpha + (-\alpha) \subseteq 0^*$.
For the reverse inclusion, let $v < 0$ and set $w := -\frac{v}{2} > 0$. Rational multiples of w are unbounded in both directions.
Indeed, if $w = \frac{A}{B}$ and $q = \frac{C}{D}$ with $A, B, D \in \mathbb{N}$, then $n = B(|C| + 1)$ satisfies $nw > q$.
The same calculation for $-q$ supplies a negative multiple below q.
Choose $a \in \alpha$ and $b \notin \alpha$. Select $j < k$ in $\mathbb{Z}$ with $jw < a$ and $kw > b$ by this unboundedness.
The finite set $\{\ell \in \mathbb{Z} : j \le \ell \le k,\ \ell w \in \alpha\}$ has a largest element n, with $(n + 1)w \notin \alpha$.
The relations $nw \in \alpha$, $-(n + 2)w \in -\alpha$ with witness w, and $v = nw - (n + 2)w$ prove $v \in \alpha + (-\alpha)$. Hence $\alpha + (-\alpha) = 0^*$.

(c) *Positive multiplication*: For $\alpha, \beta > 0^*$, the product $\alpha\beta$ is nonempty because both cuts contain positive rationals.
Positive $r' \notin \alpha$ and $s' \notin \beta$ satisfy $\alpha\beta \subseteq \{p : p < r's'\}$, so the product is proper.
Downward closure follows from the defining inequality. If $p \le rs$ is a defining witness, choose $p < q < rs$ when $p < rs$.
If $p = rs$, choose $r < r_1 \in \alpha$ and then $p < q < r_1 s$. Hence $\alpha\beta$ has no largest element.
Commutativity follows from $rs = sr$. If $p \in (\alpha\beta)\gamma$, then $p \le ut \le rst$ for positive $u \in \alpha\beta$, $r \in \alpha$, $s \in \beta$, and $t \in \gamma$.
Since $st \in \beta\gamma$, $p \in \alpha(\beta\gamma)$. The reverse inclusion is symmetric, so multiplication is associative.
If $p \in \alpha(\beta + \gamma)$, choose positive $r \in \alpha$, $u = s_0 + t_0 \in \beta + \gamma$ with $p \le ru$, and then positive $s \in \beta$, $t \in \gamma$ with $s > s_0$, $t > t_0$.
The membership $p \in \alpha\beta + \alpha\gamma$ follows from $p < r(s + t) = rs + rt$.
Conversely, if $p = u + v$ with $u \in \alpha\beta$ and $v \in \alpha\gamma$, choose positive witnesses $u \le r_1 s$, $v \le r_2 t$, and $r \in \alpha$ with $r > \max\{r_1, r_2\}$.
Then $p < r(s + t)$ with $s + t \in \beta + \gamma$, so $p \in \alpha(\beta + \gamma)$. Thus positive multiplication distributes over addition.
If $p \le rs$ with $r \in \alpha$, $0 < s < 1$, then $p < r$ and $p \in \alpha$. Conversely, for $p \in \alpha$, choose $p < r \in \alpha$ and $0 < s < 1$ with $p \le rs$.
Such an s exists when $p > 0$, while any $s > 0$ works when $p \le 0$. Hence $\alpha 1^* = \alpha$, and $\alpha, \beta > 0^* \Rightarrow \alpha\beta > 0^*$.

(d) *Positive reciprocals*: For $\alpha > 0^*$, the set α^{-1} contains every nonpositive rational and is nonempty.
It is proper because $r^{-1} \notin \alpha^{-1}$ for any positive $r \in \alpha$.
Downward closure is immediate. A larger member follows from a positive defining witness by taking a midpoint.
A nonpositive member is smaller than $\frac{1}{2q} \in \alpha^{-1}$ for any positive $q \notin \alpha$. These properties prove $\alpha^{-1} \in \mathscr{D}^+$.
If $p \in \alpha\alpha^{-1}$, then $p \le rs$ for positive $r \in \alpha$ and $s < q^{-1}$ with positive $q \notin \alpha$. Since $r < q$, $p < \frac{r}{q} < 1$, so $\alpha\alpha^{-1} \subseteq 1^*$.
For the reverse inclusion, fix $0 < c < 1$ and positive $r_0 \in \alpha$, $q_0 \notin \alpha$.
Repeated bisection produces $r_k \in \alpha$, $q_k \notin \alpha$ with $0 < r_0 \le r_k < q_k$ and $q_k - r_k = 2^{-k}(q_0 - r_0)$.
Choose k with $q_k - r_k < \frac{(1-c)r_0}{c}$; such a k exists because $2^k \ge k + 1$ and positive rational multiples are unbounded.
Then $cq_k < r_k$. Choose $s \in \mathbb{Q}$ with $\frac{c}{r_k} < s < \frac{1}{q_k}$. The relations $s \in \alpha^{-1}$ and $c < r_k s$ prove $c \in \alpha\alpha^{-1}$.
Every $c \le 0$ belongs to the product by any positive pair of witnesses. Therefore $\alpha\alpha^{-1} = 1^*$.

(e) *Signs and order compatibility*: Addition by a fixed cut preserves inclusion.
If $\beta < \gamma$, then $\alpha + \beta \subseteq \alpha + \gamma$; equality would imply $\beta = \gamma$ by adding $-\alpha$. Hence $\alpha + \beta < \alpha + \gamma$ and $\alpha > 0^* \iff -\alpha < 0^*$.
Every nonzero cut has the form εa with $a > 0^*$ and $\varepsilon \in \{1, -1\}$.
The signed definition makes products commutative and associative and sets $(-a)^{-1} = -a^{-1}$.
For distributivity, $a(b - c) = ab - ac$ when $a, b, c > 0^*$.
If $b > c$, then $a(b - c) + ac = ab$ by positive distributivity and additive cancellation.

The cases $b = c$ and $b < c$ follow by zero and sign reversal, so this identity covers all sign patterns.
Since $0^* \neq 1^*$, all field axioms hold. Strict addition compatibility and $\alpha, \beta > 0^* \Rightarrow \alpha\beta > 0^*$ make the field ordered.

(ii) *Rational embedding*: Each r^* is a cut. For $r, s \in \mathbb{Q}$, one inclusion in $r^* + s^* = (r+s)^*$ follows from $u < r$ and $v < s$.
For $p < r + s$, set $t := \frac{r+s-p}{2} > 0$. Then $r - t \in r^*$, $s - t \in s^*$, and $p = (r - t) + (s - t)$, so $p \in r^* + s^*$.
For $r, s > 0$, the forward inclusion in $r^*s^* = (rs)^*$ follows from $0 < u < r$, $0 < v < s \Rightarrow uv < rs$.
Conversely, if $0 < p < rs$, choose $0 < \delta < \min\{r, s, \frac{rs-p}{r+s}\}$. Then $p < (r - \delta)(s - \delta)$, so $p \in r^*s^*$.
Nonpositive p belongs by any positive witnesses, and the signed cases follow from the definitions.
Finally, $r < s \iff r^* < s^*$. The forward implication follows from $r \in s^* \setminus r^*$.
The reverse implication follows because $r^* < s^*$ supplies $p \in s^* \setminus r^*$ with $r \leq p < s$.
Hence ι is injective and preserves $0, 1, +, \cdot$, and order.

Structural Commentary: Cut inclusion supplies suprema; rational witnesses transfer field operations and reciprocals.
Key Insight: Bounded unions and boundary approximation create completeness and inverses. □

Write $\mathbb{R} := \mathscr{D}$ and identify $\mathbb{Q}$, together with its subsets $\mathbb{Z}$ and $\mathbb{N}$, with its image under ι.

Definition 1.1.10. *Irrational and Dense Subsets*
A real number x is *irrational* iff $x \in \mathbb{R} \setminus \mathbb{Q}$. A set $D \subseteq \mathbb{R}$ is *dense in* $\mathbb{R}$ iff $(\forall a, b \in \mathbb{R})\,[a < b \Rightarrow (\exists d \in D)\, a < d < b]$.

Theorem 1.1.11. *Existence and Uniqueness of Positive nth Roots*
Let $x \in \mathbb{R}$ with $x > 0$, and let $n \in \mathbb{N}$.
(i) $(\exists y \in \mathbb{R})\,[y > 0 \wedge y^n = x]$.
(ii) $(\forall y, z \in \mathbb{R})\,[(y > 0 \wedge z > 0 \wedge y^n = x \wedge z^n = x) \Rightarrow y = z]$.
Denote this unique root by $x^{1/n}$.

Proof.
(i) *Existence*:
(a) *Supremum candidate*: Set $E := \{t \in \mathbb{R} : t > 0,\ t^n < x\}$. For $t_0 := \frac{x}{1+x}$, $0 < t_0 < 1$ and $t_0^n \leq t_0 < x$, so $E \neq \emptyset$.
Every $t \in E$ satisfies $t \leq 1 + x$ because $t > 1 + x$ would imply $t^n \geq t > x$.
Hence $y := \sup E$ exists by Theorem 1.1.9, and $y \geq t_0 > 0$.
(b) *Lower perturbation*: For $0 < a < b$, the factorization satisfies $0 < b^n - a^n \leq n(b - a)b^{n-1}$.
Suppose $y^n < x$ and choose $0 < h < \min\{1, \frac{x-y^n}{n(y+1)^{n-1}}\}$.
The bound $(y + h)^n - y^n \leq hn(y + h)^{n-1} < x - y^n$ places $y + h$ in E, contradicting that y is an upper bound.
(c) *Upper perturbation*: Suppose $y^n > x$ and set $k := \frac{y^n - x}{ny^{n-1}}$. Then $0 < k < \frac{y}{n} \leq y$, and $y^n - (y - k)^n \leq kny^{n-1} = y^n - x$.
The inequality $(y - k)^n \geq x$ makes $y - k$ an upper bound of E by monotonicity of positive integer powers.
This contradicts $y = \sup E$. Hence $y^n = x$ by trichotomy.
(ii) *Uniqueness*: If $0 < y < z$, then $z^n - y^n = (z - y)(z^{n-1} + z^{n-2}y + \cdots + y^{n-1}) > 0$.
Equal positive nth powers therefore have equal bases by trichotomy.

Structural Commentary: Completeness selects a supremum; two perturbations exclude powers below and above x.
Key Insight: The supremum cannot survive either strict power inequality. □

Corollary 1.1.12. *Product Rule for Positive Roots*
For $a, b \in \mathbb{R}$ with $a, b > 0$ and $n \in \mathbb{N}$, $(ab)^{1/n} = a^{1/n}b^{1/n}$.

Proof. Set $\alpha := a^{1/n}$ and $\beta := b^{1/n}$. Then $(\alpha\beta)^n = \alpha^n\beta^n = ab$ by commutativity; $(ab)^{1/n} = \alpha\beta$ holds by Theorem 1.1.11(ii).
Structural Commentary: Multiplicativity of nth powers transfers the product to the uniqueness claim.
Key Insight: Equal positive nth powers have equal bases. □

Proposition 1.1.13. *The Rational Gap at $\sqrt{2}$*
Let $A := \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}$ and $B := \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}$.
(i) $(\nexists p \in \mathbb{Q})\, p^2 = 2$.
(ii) The set A has no largest element, and B has no smallest element.
(iii) The nonempty set A is bounded above in $\mathbb{Q}$ but has no supremum in $\mathbb{Q}$.

Proof.
(i) Suppose $p = \frac{m}{n} \in \mathbb{Q}$ satisfies $p^2 = 2$, where $m, n \in \mathbb{Z}$ and $n \neq 0$. Then $m^2 = 2n^2$, with $m, n \neq 0$.
The conflicting parity of the exponent of 2 contradicts unique factorization in $\mathbb{Z}$ by Theorems 3.10.2 and 3.10.7.
(ii) For $p > 0$, set $q := p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2}$. Then $q - p = -\frac{p^2-2}{p+2}$ and $q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2}$.
Thus $p \in A \Rightarrow q \in A$ with $q > p$, while $p \in B \Rightarrow q \in B$ with $0 < q < p$.
(iii) Relations $1 \in A$ and $(\forall p \in A)\, p < 2$ make A nonempty, bounded above. Suppose $s = \sup_{\mathbb{Q}} A$. Then $s \geq 1$, and $s^2 \neq 2$ by (i).
If $s^2 < 2$, the element constructed in (ii) belongs to A and exceeds s.
If $s^2 > 2$, the element q from (ii) satisfies $0 < q < s$ and is an upper bound of A.
Any $a \in A$ with $a \geq q$ would satisfy $a^2 \geq q^2 > 2$. Both cases contradict $s = \sup_{\mathbb{Q}} A$.

Structural Commentary: Unique factorization excludes the boundary; a rational correction moves toward it from either side.
Key Insight: Every rational candidate can be improved without crossing the gap. □

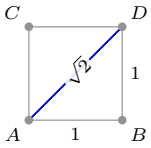

Figure 113: The diagonal of a unit square has length $\sqrt{2}$, so the rational gap is geometric as well as algebraic.

Figure 113 illustrates Proposition 1.1.13 geometrically.

Remark 1.1.14. *Density Does Not Mean Completeness*
Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. For $r, s \in \mathbb{Q}$ with $r < s$, the midpoint $\frac{r+s}{2}$ proves that $\mathbb{Q}$ is order-dense in itself.
$\mathbb{Q}$ nevertheless lacks the least-upper-bound property by Proposition 1.1.13(iii). The field $\mathbb{R}$ has this property by Theorem 1.1.9(i).

Corollary 1.1.15. *Infimum by Reflection*
Let $\emptyset \neq B \subseteq \mathbb{R}$ be bounded below and set $-B := \{-b : b \in B\}$. Then $-B$ is bounded above and $\inf B = -\sup(-B)$.

Proof. If m is a lower bound of B, then $-m$ is an upper bound of $-B$. Hence $s := \sup(-B)$ exists by Theorem 1.1.9.
The element $-s$ is a lower bound of B by $(\forall b \in B)\,(-b \leq s \Rightarrow -s \leq b)$.
If t is a lower bound of B, then $-t$ is an upper bound of $-B$, so $s \leq -t$ and $t \leq -s$. Thus $-s = \inf B$ by the definition of an infimum.
Structural Commentary: Order reversal converts lower bounds of B into upper bounds of $-B$.
Key Insight: Negating the least upper bound makes the greatest lower bound. □

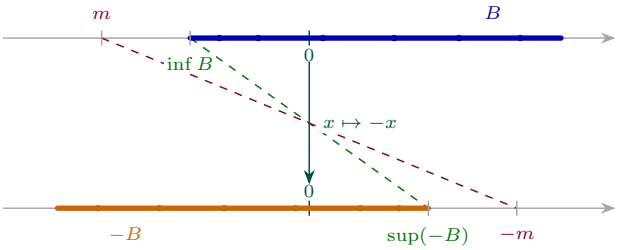

Figure 114: Reflecting B through 0 turns lower bounds of B into upper bounds of $-B$, so $\inf B = -\sup(-B)$.

Figure 114 illustrates the order reversal used in the proof.

Definition 1.1.16. *Real Absolute Value and Distance*
For $x \in \mathbb{R}$, define $|x| := \max\{x, -x\}$. For $a, b \in \mathbb{R}$, define their real-line distance by $d(a, b) := |a - b|$.

Proposition 1.1.17. *Real Absolute-Value Laws*
For $x, y, a, b, c \in \mathbb{R}$:

(i) $|x| \geq 0$, $|x| = 0 \iff x = 0$, *and* $-|x| \leq x \leq |x|$;
(ii) $|xy| = |x|\,|y|$;
(iii) $|x + y| \leq |x| + |y|$;
(iv) $|a - b| \leq |a - c| + |c - b|$.

Proof.

(*i*) ($\Rightarrow$) At least one of x and $-x$ is nonnegative by totality, so $|x| = \max\{x, -x\} \geq 0$.
Both $x \leq |x|$ and $-x \leq |x|$ hold by the definition of a maximum, and the second inequality is equivalent to $-|x| \leq x$.
If $|x| = 0$, then $x = 0$ in each of the cases $x \geq 0$ and $x < 0$ by Definition 1.1.16.
(*i*) ($\Leftarrow$) $|0| = 0$ by Definition 1.1.16.
(*ii*) If $xy > 0$, then $|xy| = xy = |x|\,|y|$ when $x, y > 0$ or $x, y < 0$. If $xy < 0$, then $|xy| = -xy = |x|\,|y|$; the case $xy = 0$ follows directly.
(*iii*) The inequalities $x + y \leq |x| + |y|$ and $-(x + y) \leq |x| + |y|$ follow by (i). Their maximum is $|x + y|$, and therefore $|x + y| \leq |x| + |y|$.
(*iv*) $|a - b| = |(a - c) + (c - b)| \leq |a - c| + |c - b|$ by (iii).
Structural Commentary: The maximum representation supplies two-sided bounds; addition then transfers them to sums and distances.
Key Insight: Bounding a number and its negative bounds its absolute value. □

Perspective 1.1.18. *Number Systems from Representatives*
The constructions of $\mathbb{Z}$ and $\mathbb{Q}$ by equivalence classes require operations and order to descend from representatives.
Thus $x \sim x'$ and $y \sim y'$ must imply $x \star y \sim x' \star y'$, with analogous compatibility for order.
Dedekind cuts use subsets rather than equivalence classes, so closure of cut operations replaces representative independence.
Key Insight: Each construction must transfer algebra from its underlying data.

Theorem 1.1.19. *Archimedean Property and Density of $\mathbb{Q}$*

(i) $(\forall x, y \in \mathbb{R})\,[x > 0 \Rightarrow (\exists n \in \mathbb{N})\, nx > y]$.
(ii) $(\forall x, y \in \mathbb{R})\,[x < y \Rightarrow (\exists p \in \mathbb{Q})\, x < p < y]$.

Proof.

(i) Fix $x > 0$ and set $A := \{nx : n \in \mathbb{N}\}$. If A were bounded above, then $\alpha := \sup A$ would exist by Theorem 1.1.9.
Since $\alpha - x < \alpha$, element $\alpha - x$ is not upper bound of A. Choose $m \in \mathbb{N}$ with $\alpha - x < mx$; $\alpha < (m + 1)x \in A$, contradiction.
(ii) Fix $x < y$. Choose $n \in \mathbb{N}$ with $n(y - x) > 1$ by (i), and choose $N \in \mathbb{N}$ with $N > |nx| + 1$ by (i) with unit step.
The finite set $S := \{k \in \mathbb{Z} : -N \leq k \leq N,\ k \leq nx\}$ is nonempty; let $m := \max S$.
Since $nx < N$, inequalities $m \leq nx < m + 1$ hold by maximality; $nx < m + 1 \leq nx + 1 < ny$, so $x < \frac{m+1}{n} < y$ with $\frac{m+1}{n} \in \mathbb{Q}$.

Structural Commentary: Completeness forbids finite supremum for integer multiples; finite integer grid then locates rational point.
Key Insight: A mesh smaller than $y - x$ must cross the interval. □

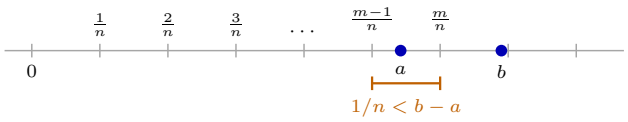

Figure 115: Choose n so the spacing $\frac{1}{n}$ is smaller than $b - a$; then one grid point $\frac{m}{n}$ falls inside (a, b).

Figure 115 illustrates the grid step used in Theorem 1.1.19(ii).

Corollary 1.1.20. *Density of Rational and Irrational Points*
For $x, y \in \mathbb{R}$ with $x < y$, $(\exists r \in \mathbb{Q})(\exists s \in \mathbb{R} \setminus \mathbb{Q})\,[x < r < y \wedge x < s < y]$.

Proof. Choose $r \in \mathbb{Q} \cap (x, y)$ by Theorem 1.1.19(ii). The number $\eta := 2^{1/2} > 0$ exists by Theorem 1.1.11(i) with $x = n = 2$.
It satisfies $\eta \notin \mathbb{Q}$ by Proposition 1.1.13(i). Choose $n \in \mathbb{N}$ with $n(y - x) > \eta$ by Theorem 1.1.19(i), $q \in \mathbb{Q} \cap (x - \frac{\eta}{n}, y - \frac{\eta}{n})$ by part (ii).
Then $s := q + \frac{\eta}{n}$ lies in (x, y), and $s \notin \mathbb{Q}$ because $s \in \mathbb{Q}$ would imply $\eta = n(s - q) \in \mathbb{Q}$.
Structural Commentary: Rational density is applied before and after translating the interval by a small irrational.
Key Insight: A rational translate of a nonzero irrational remains irrational. □

Definition 1.1.21. *Extended Real Number System*
The *extended real line* is $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$. It retains the usual order on $\mathbb{R}$ and satisfies $(\forall x \in \mathbb{R})\ -\infty < x < +\infty$.
For $E \subseteq \mathbb{R}$, extend the extrema as follows.

(i) $E = \emptyset \Rightarrow \sup E = -\infty$; if E is unbounded above, $\sup E := +\infty$; otherwise retain the real supremum.
(ii) $E = \emptyset \Rightarrow \inf E = +\infty$; if E is unbounded below, $\inf E := -\infty$; otherwise retain the real infimum.

Definition 1.1.22. *Extended Arithmetic Conventions*
For $x \in \mathbb{R}$, define the following non-indeterminate operations:

(i) $(+\infty) + (+\infty) = +\infty$ and $(-\infty) + (-\infty) = -\infty$.
(ii) $x + (+\infty) = (+\infty) + x = +\infty$ and $x + (-\infty) = (-\infty) + x = -\infty$.
(iii) $x > 0 \Rightarrow [x(+\infty) = (+\infty)x = +\infty \wedge x(-\infty) = (-\infty)x = -\infty]$.
(iv) $x < 0 \Rightarrow [x(+\infty) = (+\infty)x = -\infty \wedge x(-\infty) = (-\infty)x = +\infty]$.

The forms $(+\infty) + (-\infty)$ and $0(\pm\infty)$ are undefined, so $\overline{\mathbb{R}}$ is not a field.

Definition 1.1.23. *Unbounded Real Intervals*
For $a, b \in \mathbb{R}$:

(i) $(a, +\infty) := \{x \in \mathbb{R} : a < x\}$ and $[a, +\infty) := \{x \in \mathbb{R} : a \leq x\}$.
(ii) $(-\infty, b) := \{x \in \mathbb{R} : x < b\}$ and $(-\infty, b] := \{x \in \mathbb{R} : x \leq b\}$.

Define $(-\infty, +\infty) := \mathbb{R}$.

1.2 Complex Numbers and Absolute Value

Definition 1.2.1. *Complex Numbers*
Let $\mathbb{C} := \mathbb{R}^2$. For $x = (a, b)$ and $y = (c, d)$ in $\mathbb{C}$:

(i) $x = y \iff (a = c \wedge b = d)$.
(ii) $x + y := (a + c, b + d)$.
(iii) $xy := (ac - bd, ad + bc)$.

Elements of $\mathbb{C}$ are *complex numbers*, and $i := (0, 1) \in \mathbb{C}$ is the *complex unit*.

Theorem 1.2.2. *The Complex Numbers Form a Field*
Let $\mathbb{C} = \mathbb{R}^2$ with $(a, b) + (c, d) = (a + c, b + d)$ and $(a, b)(c, d) = (ac - bd, ad + bc)$.

(i) The additive and multiplicative identities are $(0, 0)$ and $(1, 0)$.
(ii) Every $x = (a, b) \neq (0, 0)$ has inverse $x^{-1} = \left(\frac{a}{a^2+b^2}, -\frac{b}{a^2+b^2}\right)$.
(iii) With these operations, $\mathbb{C}$ is a field.

Proof.

(i) For $x = (a, b)$, $x + (0, 0) = (0, 0) + x = x$ and $x(1, 0) = (1, 0)x = x$ by direct calculation.
(ii) At least one of a, b is nonzero, so $a^2 + b^2 > 0$ by Proposition 1.1.3(iv).
Set $q := \left(\frac{a}{a^2+b^2}, -\frac{b}{a^2+b^2}\right)$. Direct multiplication verifies $xq = qx = \left(\frac{a^2+b^2}{a^2+b^2}, 0\right) = (1, 0)$.
(iii) Write $x = (a, b)$, $y = (c, d)$, and $u = (e, f)$. The coordinate formulas lie in $\mathbb{R}^2$, so both operations are closed.
Additive associativity is $(x + y) + u = (a + c + e, b + d + f) = x + (y + u)$.
Additive commutativity is $x + y = (c + a, d + b) = y + x$.
Together with (i) and $-(a, b) = (-a, -b)$, these identities make $(\mathbb{C}, +)$ an abelian group.
Multiplicative commutativity is $xy = (ac - bd, ad + bc) = (ca - db, cb + da) = yx$.
Multiplicative associativity is $(xy)u = (ace - adf - bcf - bde, acf + ade + bce - bdf) = x(yu)$.
The identity $x(y + u) = (a(c + e) - b(d + f), a(d + f) + b(c + e)) = xy + xu$ proves left distributivity.
Right distributivity holds by commutativity. Parts (i)–(ii) and $(0, 0) \neq (1, 0)$ complete the field axioms.

Structural Commentary: Coordinate identities verify the ring laws; $a^2 + b^2 > 0$ constructs each inverse.
Key Insight: The relation $a^2 + b^2 > 0$ makes conjugate-over-norm division possible. □

Proposition 1.2.3. *Real Numbers Inside Complex Numbers*
Let $\iota : \mathbb{R} \to \mathbb{C}$ *be* $\iota(a) := (a, 0)$.
(i) $(\forall a, b \in \mathbb{R})\,[\iota(a+b) = \iota(a) + \iota(b) \wedge \iota(ab) = \iota(a)\iota(b)]$.
(ii) ι *is injective,* $\iota(0) = (0,0)$, *and* $\iota(1) = (1,0)$.

Proof.
(i) $\iota(a) + \iota(b) = (a,0) + (b,0) = (a+b,0) = \iota(a+b)$, and $\iota(a)\iota(b) = (a,0)(b,0) = (ab,0) = \iota(ab)$.
(ii) If $\iota(a) = \iota(b)$, then $(a,0) = (b,0)$, so $a = b$. The identities $\iota(0) = (0,0)$ and $\iota(1) = (1,0)$ hold directly.
Structural Commentary: Zero second coordinates are stable under both operations; pair equality proves injectivity.
Key Insight: The map $a \mapsto (a,0)$ realizes real arithmetic on the first coordinate. □

Convention 1.2.4. *Real Scalars Inside* $\mathbb{C}$
Henceforth identify $a \in \mathbb{R}$ with its image $\iota(a) = (a,0) \in \mathbb{C}$.

Proposition 1.2.5. *Coordinate Form and* i^2
For $i = (0,1) \in \mathbb{C}$*:*
(i) $i^2 = -1$.
(ii) $(\forall a, b \in \mathbb{R})\,[(a,b) = a + bi]$.

Proof.
(i) $i^2 = (0,1)(0,1) = (-1,0) = -1$.
(ii) $a + bi = (a,0) + (b,0)(0,1) = (a,0) + (0,b) = (a,b)$.
Structural Commentary: The products $(0,1)^2 = (-1,0)$ and $(b,0)(0,1) = (0,b)$ convert pair coordinates into $a + bi$.
Key Insight: Multiplication by $i = (0,1)$ transfers a real scalar to the second coordinate. □

Definition 1.2.6. *Conjugate, Real Part, and Imaginary Part*
Let $z = a + bi$ with $a, b \in \mathbb{R}$.
(i) The *complex conjugate* of z is $\overline{z} := a - bi$.
(ii) The *real part* and *imaginary part* of z are $\mathrm{Re}(z) := a$ and $\mathrm{Im}(z) := b$.

Proposition 1.2.7. *Basic Conjugation Identities*
If $z, w \in \mathbb{C}$, *then*
(i) $\overline{z+w} = \overline{z} + \overline{w}$;
(ii) $\overline{zw} = \overline{z}\,\overline{w}$;
(iii) $z + \overline{z} = 2\,\mathrm{Re}(z)$ *and* $z - \overline{z} = 2i\,\mathrm{Im}(z)$;
(iv) $z\overline{z} \in [0,\infty)$; $z \neq 0 \Rightarrow z\overline{z} > 0$, *and* $z = 0 \Rightarrow z\overline{z} = 0$.

Proof. Write $z = a + bi$ and $w = c + di$.
(i) $\overline{z+w} = (a+c) - (b+d)i = \overline{z} + \overline{w}$.
(ii) $\overline{zw} = (ac - bd) - (ad + bc)i = (a - bi)(c - di) = \overline{z}\,\overline{w}$.
(iii) $z + \overline{z} = 2a = 2\,\mathrm{Re}(z)$ and $z - \overline{z} = 2bi = 2i\,\mathrm{Im}(z)$.
(iv) $z\overline{z} = a^2 + b^2 \in [0,\infty)$. For $z \neq 0$, $a^2 + b^2 > 0$ by Proposition 1.1.3(iv), as at least one coordinate nonzero. For $z = 0$, $z\overline{z} = 0\overline{0} = 0$.
Structural Commentary: Coordinate sign reversal preserves both operations and turns $z\overline{z}$ into a sum of squares.
Key Insight: $z\overline{z} = a^2 + b^2 \geq 0$. □

Definition 1.2.8. *Absolute Value of a Complex Number*
For $z = a + bi \in \mathbb{C}$, define $|z| := \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}$ as the unique $r \in [0,\infty)$ with $r^2 = a^2 + b^2$.
For $z \neq 0$, existence and uniqueness hold by Proposition 1.2.7(iv) and Theorem 1.1.11 with $n = 2$.
For $z = 0$, the unique nonnegative solution is $r = 0$ by Proposition 1.1.3(iv).

Theorem 1.2.9. *Absolute Value Identities*
For $z, w \in \mathbb{C}$,
(i) $|z| \geq 0$; $z = 0 \Rightarrow |z| = 0$, *and* $z \neq 0 \Rightarrow |z| > 0$;
(ii) $|\overline{z}| = |z|$;
(iii) $|zw| = |z|\,|w|$;
(iv) $|\mathrm{Re}(z)| \leq |z|$;
(v) $|z + w| \leq |z| + |w|$.

Proof. For $r, s \in [0,\infty)$, $r^2 \leq s^2 \Rightarrow r \leq s$, since $s < r$ would imply $0 < (r-s)(r+s) = r^2 - s^2$.
For $t \in \mathbb{R}$ with $t \geq 0$, $t = |t|$ by equality of their nonnegative squares. For $t < 0$, $t < 0 \leq |t|$; thus $(\forall t \in \mathbb{R})\, t \leq |t|$.
(i) The relations $|z| \geq 0$ and $|0| = 0$ hold by Definition 1.2.8. For $z \neq 0$, $|z|^2 = z\overline{z} > 0$ by Proposition 1.2.7(iv).
The inequality $|z| > 0$ holds by $|z|^2 > 0$ and nonnegativity.
(ii) Since $\overline{\overline{z}} = z$ by definition, $|\overline{z}|^2 = \overline{z}z = z\overline{z} = |z|^2$. Then $|\overline{z}| = |z|$ holds by preliminary square comparison in both directions.
(iii) $|zw|^2 = (zw)\overline{zw} = (z\overline{z})(w\overline{w})$ by Proposition 1.2.7(ii) and commutativity. $|zw|^2 = |z|^2\,|w|^2 = (|z|\,|w|)^2$.
Then $|zw| = |z|\,|w|$ holds by the preliminary square comparison and nonnegativity.
(iv) For $z = a + bi$, $|\mathrm{Re}(z)|^2 = a^2 \leq a^2 + b^2 = |z|^2$. Then $|\mathrm{Re}(z)| \leq |z|$ holds by preliminary square comparison and nonnegativity.
(v) Set $q := z\overline{w}$. The identity $\overline{q} = \overline{z}w$ holds by Proposition 1.2.7(ii) and the definition of conjugation.
The bound $\mathrm{Re}(q) \leq |\mathrm{Re}(q)| \leq |q|$ holds by the real estimate and (iv). The identity $|q| = |z|\,|w|$ holds by (iii).
By Proposition 1.2.7(i), $|z+w|^2 = (z+w)(\overline{z} + \overline{w})$. By distributivity, $(z+w)(\overline{z} + \overline{w}) = |z|^2 + q + \overline{q} + |w|^2$.
Identity $q + \overline{q} = 2\,\mathrm{Re}(q)$ holds by Proposition 1.2.7(iii). $|z+w|^2 \leq |z|^2 + 2\,|z|\,|w| + |w|^2 = (|z| + |w|)^2$ by preceding bounds.
$|z + w| \leq |z| + |w|$ by the preliminary square comparison and nonnegativity.
Structural Commentary: Squared moduli prove (ii)–(iv); the expansion in (v) reduces its cross term to (iii)–(iv).
Key Insight: The identity $z\overline{z} = a^2 + b^2$ converts complex algebra into ordered real estimates. □

2 Real Sequences and Completeness Methods

2.1 Sequences, Cauchy Criteria, and Completeness

Sequences test how limits interact with algebra, order, and completeness. This subsection develops limit algebra and subsequences. It then derives monotone convergence, Bolzano–Weierstrass, and the Cauchy criterion.
Divergence and upper and lower limits appear in the next subsection.

Definition 2.1.1. *Convergent Real and Complex Sequences*
Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, let (s_n) be a sequence in $\mathbb{K}$, and let $s \in \mathbb{K}$.

(i) (s_n) *converges* to s iff $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\ |s_n - s| < \varepsilon$.
(ii) Write $s_n \to s$ or $\lim_{n\to\infty} s_n = s$. The sequence *diverges* iff no $s \in \mathbb{K}$ satisfies (i).

For $\mathbb{K} = \mathbb{R}$, $|\cdot|$ is from Definition 1.1.16. For $\mathbb{K} = \mathbb{C}$, it is from Definition 1.2.8.

Figure 116: For a real sequence, the tail $\{s_n : n \geq N\}$ stays inside $(s - \varepsilon, s + \varepsilon)$.

For real sequences, Figure 116 illustrates the tail condition in the definition.

Definition 2.1.2. *Range and Boundedness of a Real Sequence*
Let (s_n) be a real sequence.

(i) The *range* of (s_n) is $\text{range}(s_n) := \{s_n : n \in \mathbb{N}\}$.
(ii) (s_n) is *bounded* iff $(\exists m, M \in \mathbb{R})(\forall n \in \mathbb{N})\ m \leq s_n \leq M$.

Convention 2.1.3. *Eventual Properties*
For statements $P(n)$ indexed by $n \in \mathbb{N}$, $P(n)$ holds *eventually* iff $(\exists N \in \mathbb{N})(\forall n \geq N)\ P(n)$.

Proposition 2.1.4. *Basic Consequences of Sequence Convergence*
Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and let (s_n) and (t_n) be sequences in $\mathbb{K}$.

(i) If $s_n \to s$ and $s_n \to t$, then $s = t$.
(ii) If $s_n \to s$, then $\{|s_n| : n \in \mathbb{N}\}$ is bounded.
(iii) If $\mathbb{K} = \mathbb{R}$, $s_n \leq t_n$ eventually, $s_n \to s$, and $t_n \to t$, then $s \leq t$.
If $\mathbb{K} = \mathbb{R}$ and $c \in \mathbb{R}$, then $s_n \leq c$ eventually and $s_n \to s$ imply $s \leq c$.

Proof. For $\mathbb{K} = \mathbb{R}$, positivity, triangle inequality hold by Proposition 1.1.17(i),(iii). For $\mathbb{K} = \mathbb{C}$, these hold by Theorem 1.2.9(i),(v).

(i) Suppose $s \neq t$ and set $d := |s - t| > 0$. For $\varepsilon := \frac{d}{3}$, choose N such that $|s_n - s| < \varepsilon$ and $|s_n - t| < \varepsilon$ for $n \geq N$.
Then $d \leq |s - s_n| + |s_n - t| < 2\varepsilon = \frac{2d}{3}$, a contradiction.
(ii) Choose $N \geq 2$ such that $|s_n - s| < 1$ for $n \geq N$, and set $M := \max\{|s_1|, \ldots, |s_{N-1}|, |s| + 1\}$.
For $n \geq N$, $|s_n| \leq |s_n - s| + |s| < |s| + 1 \leq M$. The set $\{|s_n| : n \in \mathbb{N}\}$ is bounded by $(\forall n \in \mathbb{N})\ 0 \leq |s_n| \leq M$.
(iii) Suppose $s > t$ and set $\varepsilon := \frac{s-t}{3}$. Choose N such that $n \geq N$ implies $s_n \leq t_n$, $|s_n - s| < \varepsilon$, and $|t_n - t| < \varepsilon$.
Then $s_n > s - \varepsilon > t + \varepsilon > t_n$, contradicting $s_n \leq t_n$. Final claim is preceding result with constant sequence $t_n = t = c$.

Structural Commentary: The triangle inequality separates limits, bounds tails, and transfers eventual order.
Key Insight: A common late index turns simultaneous estimates into one contradiction or bound. □

Theorem 2.1.5. *Limit Algebra for Real and Complex Sequences*
Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and let (s_n) and (t_n) be sequences in $\mathbb{K}$. Assume $s_n \to s$ and $t_n \to t$, and let $c \in \mathbb{K}$.

(i) $s_n + t_n \to s + t$;
(ii) $cs_n \to cs$ and $c + s_n \to c + s$;
(iii) $s_n t_n \to st$;
(iv) if $s \neq 0$ and $(\forall n \in \mathbb{N})\ s_n \neq 0$, then $\frac{1}{s_n} \to \frac{1}{s}$.

Proof. For $\mathbb{K} = \mathbb{R}$, required modulus laws hold by Proposition 1.1.17(i)–(iii). For $\mathbb{K} = \mathbb{C}$, these hold by Theorem 1.2.9(i),(iii),(v).

(i) Given $\varepsilon > 0$, choose N with $(\forall n \geq N)\ [|s_n - s| < \frac{\varepsilon}{2} \wedge |t_n - t| < \frac{\varepsilon}{2}]$.
Then $|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \varepsilon$ by triangle inequality.
(ii) If $c = 0$, then $(\forall n \in \mathbb{N})\ cs_n = 0 = cs$. If $c \neq 0$, then for $\varepsilon > 0$ choose N such that $|s_n - s| < \frac{\varepsilon}{|c|}$ for $n \geq N$.
The inequality $|cs_n - cs| = |c|\,|s_n - s| < \varepsilon$ holds by multiplicativity.
$c + s_n \to c + s$ by (i) and convergence of the constant sequence with value c.
(iii) Choose $M > 0$ with $(\forall n \in \mathbb{N})\ |s_n| \leq M$ by Proposition 2.1.4(ii).
Given $\varepsilon > 0$, choose N with $(\forall n \geq N)\ [|t_n - t| < \frac{\varepsilon}{2M} \wedge |s_n - s| < \frac{\varepsilon}{2(|t|+1)}]$. The identity $s_n t_n - st = s_n(t_n - t) + t(s_n - s)$ holds.
The bound $|s_n t_n - st| \leq M\,|t_n - t| + |t|\,|s_n - s| < \varepsilon$ holds by multiplicativity and the triangle inequality.
(iv) The inequality $|s| > 0$ holds by the relevant positivity law. Choose N_0 such that $|s_n - s| < \frac{|s|}{2}$ for $n \geq N_0$.
The inequality $|s| \leq |s - s_n| + |s_n|$ holds by triangle inequality. For $n \geq N_0$, $|s_n| > \frac{|s|}{2}$ by this inequality, choice of N_0.
Given $\varepsilon > 0$, choose N_1 such that $|s_n - s| < \frac{\varepsilon |s|^2}{2}$ for $n \geq N_1$.
For $u \in \mathbb{K} \setminus \{0\}$, $\left|u^{-1}\right| = \frac{1}{|u|}$ because $1 = |1| = |u|\left|u^{-1}\right|$ by multiplicativity. Hence $\left|\frac{1}{s_n} - \frac{1}{s}\right| = \frac{|s_n - s|}{|s_n||s|}$.
For $n \geq \max\{N_0, N_1\}$, this quotient is less than $\frac{2}{|s|^2}\,|s_n - s| < \varepsilon$.

Structural Commentary: Uniform bounds control products; reverse triangle bounds control reciprocals.
Key Insight: Every algebraic limit law reduces to an error identity and a late uniform bound. □

Definition 2.1.6. *Subsequences and Subsequential Limits*
Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and let $p : \mathbb{N} \to \mathbb{K}$, $n \mapsto p_n$, be a sequence.

(i) If $\sigma : \mathbb{N} \to \mathbb{N}$ is strictly increasing, then $p \circ \sigma : k \mapsto p_{\sigma(k)}$ is a *subsequence* of (p_n).

(ii) $q \in \mathbb{K}$ is a *subsequential limit* iff $(\exists\, \sigma : \mathbb{N} \to \mathbb{N})\,[(\forall k \in \mathbb{N})\, \sigma(k+1) > \sigma(k) \wedge p_{\sigma(k)} \to q]$.

Proposition 2.1.7. *Subsequences of Convergent Sequences*
Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, (p_n) be a sequence in $\mathbb{K}$, $p \in \mathbb{K}$. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be strictly increasing.

(i) $p_n \to p \Rightarrow p_{\sigma(k)} \to p$.
(ii) If each subsequence of (p_n) converges to p, then $p_n \to p$.

Proof.

(i) Strict increase implies $\sigma(1) \geq 1$ and $\sigma(k+1) \geq \sigma(k) + 1$. The bound $(\forall k \in \mathbb{N})\, \sigma(k) \geq k$ holds by Theorem 2.6.2.
Given $\varepsilon > 0$, choose N such that $|p_n - p| < \varepsilon$ for $n \geq N$. If $k \geq N$, then $\sigma(k) \geq k \geq N$, so $|p_{\sigma(k)} - p| < \varepsilon$.
(ii) Apply the hypothesis to the identity selection map $\sigma(k) = k$.

Structural Commentary: Increasing selection maps place late subsequence indices in the original tail.
Key Insight: The bound $\sigma(k) \geq k$ transfers every convergence threshold. □

Definition 2.1.8. *Monotone Real Sequence*
Let (s_n) be a real sequence.

(i) (s_n) is *monotonically increasing* iff $(\forall n \in \mathbb{N})\, s_n \leq s_{n+1}$.
(ii) (s_n) is *monotonically decreasing* iff $(\forall n \in \mathbb{N})\, s_n \geq s_{n+1}$.

The sequence is *monotonic* iff it is increasing or decreasing.

Definition 2.1.9. *Dominant Index*
For a real sequence (s_n), an index $n \in \mathbb{N}$ is *dominant* iff $(\forall m > n)\, s_n \geq s_m$.

Theorem 2.1.10. *Monotone Subsequence Theorem*
Every real sequence has a monotone subsequence.

Proof.

(i) *Dominant set*: Let $D := \{n \in \mathbb{N} : (\forall m > n)\, s_n \geq s_m\}$.
(ii) *Infinite case*: If D is infinite, set $n_1 := \min D$ by Assumption 2.6.1.
The set $\{n \in D : n > n_k\}$ is nonempty at each stage because D is infinite.
Recursively set $n_{k+1} := \min\{n \in D : n > n_k\}$ by the same assumption.
The subsequence (s_{n_k}) is decreasing by $n_1 < n_2 < \cdots$ and $s_{n_1} \geq s_{n_2} \geq \cdots$.
(iii) *Finite case*: If D is finite, choose N such that no $n \geq N$ belongs to D, and set $n_1 := N$.
Given $n_k \geq N$, choose $n_{k+1} > n_k$ with $s_{n_{k+1}} > s_{n_k}$ by the nondominance of n_k.
The subsequence (s_{n_k}) is increasing by $n_1 < n_2 < \cdots$ and $s_{n_1} < s_{n_2} < \cdots$.

Structural Commentary: Dominant indices decrease; recursive nondominance beyond them increases.
Key Insight: Dominance splits the index set into the two constructions needed for monotonicity. □

Theorem 2.1.11. *Monotone Convergence Theorem for Sequences*
A monotonic real sequence converges iff it is bounded.

Proof.

(⇒) The sequence is bounded by Proposition 2.1.4(ii).
(⇐) Let $E := \{s_n : n \in \mathbb{N}\}$. If (s_n) is increasing, then E is bounded above and $s := \sup E$ exists by Theorem 1.1.9.
Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound of E, so choose N with $s - \varepsilon < s_N \leq s$.
For $n \geq N$, $0 \leq s - s_n \leq s - s_N < \varepsilon$, so $|s_n - s| < \varepsilon$ by Definition 1.1.16.
If (s_n) is decreasing, then E is bounded below and $s := \inf E$ exists by Corollary 1.1.15.
Given $\varepsilon > 0$, $s + \varepsilon$ is not a lower bound of E, so choose N with $s \leq s_N < s + \varepsilon$.
For $n \geq N$, $0 \leq s_n - s \leq s_N - s < \varepsilon$, so $|s_n - s| < \varepsilon$ by Definition 1.1.16.

Structural Commentary: Boundedness supplies extrema; monotonicity traps each tail against one term.
Key Insight: An extremum becomes the limit because no ε-shift remains a bound. □

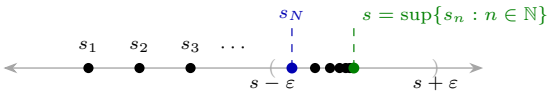

Figure 117: For an increasing bounded sequence, $s = \sup_{n \in \mathbb{N}} s_n$ is the limit.

Figure 117 illustrates the supremum mechanism in the increasing case.

Theorem 2.1.12. *Bolzano–Weierstrass for Real Sequences*
Every bounded real sequence has a convergent subsequence.

Proof. Choose a monotone subsequence (s_{n_k}) by Theorem 2.1.10.
Subsequence (s_{n_k}) is bounded because its range is contained in the bounded range of (s_n). Subsequence converges by Theorem 2.1.11.
Structural Commentary: Monotone subsequences inherit boundedness and therefore converge.
Key Insight: Boundedness survives passage to the monotone subsequence. □

Definition 2.1.13. *Cauchy Real Sequence*
A real sequence (s_n) is *Cauchy* iff $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall m, n \geq N)\ |s_n - s_m| < \varepsilon$.

Lemma 2.1.14. *Cauchy Plus One Convergent Subsequence*
If a real Cauchy sequence has a subsequence converging to $s \in \mathbb{R}$, then the whole sequence converges to s.

Proof.

(i) *Index growth*: The inequalities $n_1 \geq 1$ and $n_{j+1} \geq n_j + 1$ hold. The bound $(\forall j \in \mathbb{N})\, n_j \geq j$ holds by Theorem 2.6.2.
(ii) *Tail thresholds*: Given $\varepsilon > 0$, choose N_0 such that $m, n \geq N_0 \Rightarrow |s_m - s_n| < \frac{\varepsilon}{2}$. Choose J such that $j \geq J \Rightarrow \left|s_{n_j} - s\right| < \frac{\varepsilon}{2}$.
(iii) *Common anchor*: Fix $j \geq \max\{J, N_0\}$. Then $n_j \geq j \geq N_0$.
(iv) *Final estimate*: The required triangle inequality holds by Proposition 1.1.17(iv). For $n \geq N_0$, $|s_n - s| \leq \left|s_n - s_{n_j}\right| + \left|s_{n_j} - s\right| < \varepsilon$.

Structural Commentary: A term common to both tails anchors every later original term.
Key Insight: A single shared late index joins the two $\frac{\varepsilon}{2}$ estimates. □

Theorem 2.1.15. *Real Cauchy Criterion*
A real sequence converges in $\mathbb{R}$ iff it is Cauchy.

Proof.

(⇒) Given $\varepsilon > 0$, choose N such that $|s_n - s| < \frac{\varepsilon}{2}$ for $n \geq N$.
For $m, n \geq N$, $|s_n - s_m| \leq |s_n - s| + |s - s_m| < \varepsilon$ by Proposition 1.1.17(iv).
(⇐) Choose $N \geq 2$ such that $m, n \geq N \Rightarrow |s_m - s_n| < 1$.
The sequence is bounded because its tail lies in $(s_N - 1, s_N + 1)$ and its finite initial set is bounded.
Choose a subsequence converging to some $s \in \mathbb{R}$ by Theorem 2.1.12. The whole sequence converges by Lemma 2.1.14.

Structural Commentary: Convergence controls late pairs; Cauchy boundedness supplies an anchoring limit.
Key Insight: Bolzano–Weierstrass locates a candidate limit, and the Cauchy lemma propagates it. □

2.2 Divergence, Upper and Lower Limits, and Sequence Estimates

Definition 2.2.1. *Divergence to $\pm\infty$*
Let (s_n) be a real sequence.

(i) $s_n \to +\infty \iff (\forall M \in \mathbb{R})(\exists N \in \mathbb{N})(\forall n \geq N)\, s_n \geq M$.
(ii) $s_n \to -\infty \iff (\forall M \in \mathbb{R})(\exists N \in \mathbb{N})(\forall n \geq N)\, s_n \leq M$.

Proposition 2.2.2. *Monotone Sequences Have Extended Limits*
Let (s_n) be a real sequence.

(i) If (s_n) is increasing, then $s_n \to \sup\{s_n : n \in \mathbb{N}\}$ in $\overline{\mathbb{R}}$.
(ii) If (s_n) is decreasing, then $s_n \to \inf\{s_n : n \in \mathbb{N}\}$ in $\overline{\mathbb{R}}$.
(iii) In either case, the extended limit lies in $\mathbb{R}$ iff (s_n) is bounded.

Proof.

(*i*) Set $a := \sup\{s_n : n \in \mathbb{N}\} \in \overline{\mathbb{R}}$. Since the range is nonempty, $a \neq -\infty$.
If $a \in \mathbb{R}$, fix $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $a - \varepsilon < s_N \leq a$ by the defining property of a.
Then $(\forall n \geq N)\, a - \varepsilon < s_n \leq a$ by monotonicity, so $s_n \to a$.
If $a = +\infty$, fix $M \in \mathbb{R}$ and choose $N \in \mathbb{N}$ with $s_N \geq M$ by the defining property of a.
Then $(\forall n \geq N)\, s_n \geq M$ by monotonicity, so $s_n \to +\infty$.
(*ii*) Set $b := \inf\{s_n : n \in \mathbb{N}\} \in \overline{\mathbb{R}}$. Since the range is nonempty, $b \neq +\infty$.
If $b \in \mathbb{R}$, fix $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $b \leq s_N < b + \varepsilon$ by the defining property of b.
Then $(\forall n \geq N)\, b \leq s_n < b + \varepsilon$ by monotonicity, so $s_n \to b$.
If $b = -\infty$, fix $M \in \mathbb{R}$ and choose $N \in \mathbb{N}$ with $s_N \leq M$ by the defining property of b.
Then $(\forall n \geq N)\, s_n \leq M$ by monotonicity, so $s_n \to -\infty$.
(*iii*) (⇒) In the increasing case, a finite limit a bounds the range above, while s_1 bounds it below.
In the decreasing case, a finite limit b bounds the range below, while s_1 bounds it above. Thus the range is bounded.
(*iii*) (⇐) If the range is bounded, its supremum in (i) or infimum in (ii) is real by the extended-extremum definitions.

Structural Commentary: Extended extrema split the finite ε-trap and infinite threshold cases.
Key Insight: Monotonicity propagates one extremal witness through the entire tail. □

Proposition 2.2.3. *Unbounded Real Sequences Have Infinite Subsequences*
A real sequence is unbounded iff some subsequence tends to $+\infty$ or $-\infty$.

Proof.

(⇒) An unbounded real sequence is unbounded above or below. If it is unbounded above, every tail is unbounded above.
Set $n_0 := 0$ and recursively choose $n_k > n_{k-1}$ with $s_{n_k} \geq k$. For $M \in \mathbb{R}$, choose $K \in \mathbb{N}$ with $K \geq M$ by Theorem 1.1.19(i).
Then $(\forall k \geq K)\, s_{n_k} \geq k \geq M$, so $s_{n_k} \to +\infty$. If (s_n) is unbounded below, same construction with $s_{n_k} \leq -k$ satisfies $s_{n_k} \to -\infty$.
(⇐) If $s_{n_k} \to +\infty$, then $(\forall M \in \mathbb{R})(\exists k \in \mathbb{N})\, s_{n_k} \geq M$, so (s_n) is unbounded above.
If $s_{n_k} \to -\infty$, then $(\forall M \in \mathbb{R})(\exists k \in \mathbb{N})\, s_{n_k} \leq M$, so (s_n) is unbounded below.

Structural Commentary: Tail unboundedness supports a recursive threshold subsequence with levels $\pm k$.
Key Insight: Increasing indices convert unbounded tails into extended divergence. □

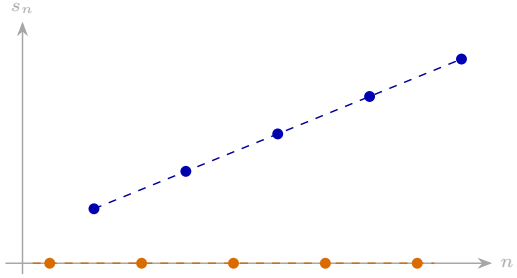

Figure 118: For $s_n = [(-1)^n + 1]n$, odd terms (orange) equal 0, while even terms (blue) equal $2n$ and tend to $+\infty$.

Figure 118 exhibits finite and infinite subsequential limits in one sequence.

Definition 2.2.4. *Upper and Lower Limits*
Let (s_n) be a real sequence and set $E := E(s) := \{x \in \overline{\mathbb{R}} : (\exists\, n_1 < n_2 < \cdots)\ s_{n_k} \to x\}$.

(i) The *upper limit* is $\limsup_{n\to\infty} s_n := s^* := \sup_{\overline{\mathbb{R}}} E$.

(ii) The *lower limit* is $\liminf_{n\to\infty} s_n := s_* := \inf_{\overline{\mathbb{R}}} E$.

Theorem 2.2.5. *Characterization of Upper and Lower Limits*
Let (s_n) be a real sequence and set $E := \{x \in \overline{\mathbb{R}} : (\exists\, n_1 < n_2 < \cdots)\ s_{n_k} \to x\}$. Set $s^* := \sup_{\overline{\mathbb{R}}} E$ and $s_* := \inf_{\overline{\mathbb{R}}} E$.

For $y \in \overline{\mathbb{R}}$, set $U(y) := (\forall x \in \mathbb{R})[x > y \Rightarrow (\exists N \in \mathbb{N})(\forall n \geq N)\, s_n < x]$. Set $L(y) := (\forall x \in \mathbb{R})[x < y \Rightarrow (\exists N \in \mathbb{N})(\forall n \geq N)\, s_n > x]$.

(i) $s^* \in E$.
(ii) $U(s^*)$.
(iii) $\{y \in E : U(y)\} = \{s^*\}$.
(iv) $s_* \in E$.
(v) $L(s_*)$.
(vi) $\{y \in E : L(y)\} = \{s_*\}$.

Proof.

(i) Set $n_0 := 0$ and distinguish the value of s^*.
  (a) If $s^* = +\infty$, choose $y_k \in E$ with $y_k > k$. Choose $n_k > n_{k-1}$ with $s_{n_k} > k$; this is possible by $y_k \in E$ and $y_k > k$.
  For $M \in \mathbb{R}$, choose $K \in \mathbb{N}$ with $K > M$ by Theorem 1.1.19(i). Then $(\forall k \geq K)\, s_{n_k} > k > M$, so $s_{n_k} \to +\infty$ and $+\infty \in E$.
  (b) If $s^* \in \mathbb{R}$, choose $y_k \in E \cap (s^* - k^{-1}, s^*]$ and then $n_k > n_{k-1}$ with $|s_{n_k} - y_k| < k^{-1}$.
  The bound $|s_{n_k} - s^*| < 2k^{-1}$ holds by Proposition 1.1.17(iv). For $\varepsilon > 0$, choose $K > 2\varepsilon^{-1}$ by Theorem 1.1.19(i).
  Then $(\forall k \geq K)\ |s_{n_k} - s^*| < \varepsilon$, so $s_{n_k} \to s^*$ and $s^* \in E$.
  (c) Suppose $s^* = -\infty$ and $-\infty \notin E$. Some tail is bounded below.
  Otherwise the recursive construction with $s_{n_k} \leq -k$ would place $-\infty$ in E.
  If this tail is bounded above, it has a finite subsequential limit by Theorem 2.1.12.
  If it is unbounded above, it has an infinite subsequential limit by Proposition 2.2.3.
  The infinite limit is $+\infty$ because the tail is bounded below.
  Either case places an element greater than $-\infty$ in E, contradicting $s^* = \sup_{\overline{\mathbb{R}}} E = -\infty$.

(ii) Fix $x \in \mathbb{R}$ with $x > s^*$. If $s_n \geq x$ infinitely often, choose a subsequence with $(\forall k \in \mathbb{N})\, s_{n_k} \geq x$.
If subsequence bounded, a further subsequence converges by Theorem 2.1.12. Its limit y satisfies $y \geq x$ by Proposition 2.1.4(iii).
If the subsequence is unbounded, it is bounded below by x. A further subsequence tends to $+\infty$ by Proposition 2.2.3.
Thus $E \cap [x, +\infty] \neq \emptyset$, contradicting $x > s^* = \sup_{\overline{\mathbb{R}}} E$. Therefore $s_n < x$ eventually.

(iii) Let $A := \{y \in E : U(y)\}$. Then $s^* \in A$ by (i)–(ii). If $p, q \in A$ with $p < q$, choose $x \in \mathbb{R}$ with $p < x < q$.
Then $s_n < x$ eventually by $U(p)$. Since $q \in E$, $q > x$, a subsequence is eventually greater than x, contradiction. Thus $A = \{s^*\}$.

(iv) For $r_n := -s_n$, its subsequential-limit set is $-E$, upper limit is $-s_*$.
Relation $-s_* \in -E$ holds by (i) for (r_n); equivalently, $s_* \in E$.

(v) Fix $x \in \mathbb{R}$ with $x < s_*$. Then $-x > -s_*$, so $-s_n < -x$ eventually by (ii) for (r_n). Equivalently, $s_n > x$ eventually.

(vi) The set $\{y \in E : L(y)\}$ is the negative of the upper-characterizing set for (r_n), so it equals $\{s_*\}$ by (iii).

Structural Commentary: Diagonal subsequences attain each extremum; sign reversal transfers the upper characterization downward.
Key Insight: Failure of an eventual bound creates a competing subsequential limit. □

Proposition 2.2.6. *Tail-Envelope Formulae*
For a real sequence (s_n) and $N \in \mathbb{N}$, set $T_N := \{s_n : n \geq N\}$. Set $u_N := \sup_{\overline{\mathbb{R}}} T_N$ and $\ell_N := \inf_{\overline{\mathbb{R}}} T_N$.

(i) $(\forall N \in \mathbb{N})\, [u_{N+1} \leq u_N \wedge \ell_{N+1} \geq \ell_N]$.
(ii) $\limsup_{n\to\infty} s_n = \inf_{N\in\mathbb{N}} u_N$ *and* $\liminf_{n\to\infty} s_n = \sup_{N\in\mathbb{N}} \ell_N$.

Proof.

(i) The inclusion $T_{N+1} \subseteq T_N$ implies $u_{N+1} \leq u_N$ and $\ell_{N+1} \geq \ell_N$.

(ii) Set $s^* := \limsup s_n$ and $u := \inf_{N\in\mathbb{N}} u_N$. If $x \in \mathbb{R}$ and $x > s^*$, then $u \leq x$ by Theorem 2.2.5(ii) and the definition of u.
If $y \in \mathbb{R}$ and $y < s^*$, each tail contains a term greater than y by Theorem 2.2.5(i). Thus $(\forall N \in \mathbb{N})\, u_N \geq y$ and $u \geq y$.
If $s^* = -\infty$, then $(\forall x \in \mathbb{R})\, u \leq x$, so $u = -\infty$. If $s^* = +\infty$, then $(\forall y \in \mathbb{R})\, u \geq y$, so $u = +\infty$.
If $s^* \in \mathbb{R}$, then $(\forall \varepsilon > 0)\, s^* - \varepsilon \leq u \leq s^* + \varepsilon$, so $u = s^*$. For $r_n := -s_n$, its upper tail extrema are $-\ell_N$ and $\limsup r_n = -s_*$.
Thus $-s_* = \inf_{N\in\mathbb{N}} (-\ell_N) = -\sup_{N\in\mathbb{N}} \ell_N$, so $\sup_{N\in\mathbb{N}} \ell_N = s_*$.

Structural Commentary: Nested tails make extrema monotone; subsequential thresholds identify their terminal bounds.
Key Insight: Every extremal subsequence returns arbitrarily far into each tail. □

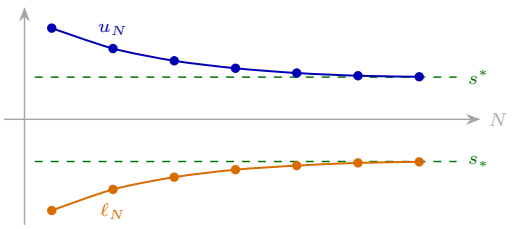

Figure 119: The upper tail extrema u_N decrease to $s^* = \limsup s_n$, while the lower extrema ℓ_N increase to $s_* = \liminf s_n$.

Figure 119 visualizes the two monotone envelopes in Proposition 2.2.6.

Theorem 2.2.7. *Convergence and Equal Upper/Lower Limits*
For a real sequence (s_n) and $s \in \mathbb{R}$, $s_n \to s \iff \limsup_{n\to\infty} s_n = \liminf_{n\to\infty} s_n = s$.

Proof.
($\Rightarrow$) Suppose $s_n \to s$ and set $E := E(s)$. Every subsequence converges to s by Proposition 2.1.7(i).
Choose $M > 0$, $(\forall n \in \mathbb{N})\ |s_n| \le M$ by Proposition 2.1.4(ii). Then $(\forall n \in \mathbb{N})\ -M \le s_n \le M$, no subsequence tends to $\pm\infty$.
Thus $E = \{s\}$ and $\limsup s_n = \liminf s_n = s$.
($\Leftarrow$) Suppose $\limsup s_n = \liminf s_n = s$, $\varepsilon > 0$. Inequalities $s_n < s + \varepsilon$ and $s_n > s - \varepsilon$ hold eventually by Theorem 2.2.5(ii),(v).
Therefore $|s_n - s| < \varepsilon$ eventually, so $s_n \to s$.
Structural Commentary: Subsequence inheritance proves necessity; the two eventual extremal bounds prove sufficiency.
Key Insight: Limsup supplies an eventual ceiling; liminf supplies an eventual floor. □

Theorem 2.2.8. *Order Comparison for Upper and Lower Limits*
Let (s_n) and (t_n) be real sequences with $s_n \le t_n$ eventually.
(i) $\limsup_{n\to\infty} s_n \le \limsup_{n\to\infty} t_n$.
(ii) $\liminf_{n\to\infty} s_n \le \liminf_{n\to\infty} t_n$.

Proof. Choose $N \in \mathbb{N}$ such that $(\forall n \ge N)\ s_n \le t_n$ by the definition of eventual order.
(i) $(\forall k \ge N)\ \sup_{n\ge k} s_n \le \sup_{n\ge k} t_n$. Then $\limsup s_n = \inf_{k\ge N} \sup_{n\ge k} s_n \le \inf_{k\ge N} \sup_{n\ge k} t_n = \limsup t_n$ by Proposition 2.2.6(i)–(ii).
(ii) $(\forall k \ge N)\ \inf_{n\ge k} s_n \le \inf_{n\ge k} t_n$. Then $\liminf s_n = \sup_{k\ge N} \inf_{n\ge k} s_n \le \sup_{k\ge N} \inf_{n\ge k} t_n = \liminf t_n$ by Proposition 2.2.6(i)–(ii).
Structural Commentary: Tailwise extrema transfer eventual pointwise order to both envelope limits.
Key Insight: Deleting finitely many tail extrema preserves their terminal infimum and supremum. □

Theorem 2.2.9. *Pinching Theorem for Real Sequences*
Let (a_n), (s_n), and (b_n) be real sequences, and let $s \in \mathbb{R}$. If $a_n \le s_n \le b_n$ eventually and $a_n, b_n \to s$, then $s_n \to s$.

Proof. Fix $\varepsilon > 0$, N_0 such that $(\forall n \ge N_0)\ a_n \le s_n \le b_n$. Choose N_1, N_2 such that $(\forall n \ge N_1)\ s - \varepsilon < a_n$, $(\forall n \ge N_2)\ b_n < s + \varepsilon$.
Set $N := \max\{N_0, N_1, N_2\}$. Then $(\forall n \ge N)\ s - \varepsilon < s_n < s + \varepsilon$, so $s_n \to s$.
Structural Commentary: Three eventual thresholds combine at their maximum to trap the middle sequence.
Key Insight: $N = \max\{N_0, N_1, N_2\}$ synchronizes both convergent bounds. □

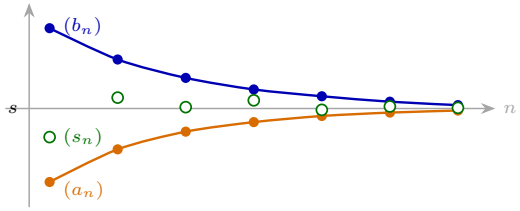

Figure 120: With $a_n \le s_n \le b_n$ and $a_n, b_n \to s$, the middle sequence also tends to s.

Figure 120 shows the middle sequence trapped between the two convergent bounds.

Theorem 2.2.10. *Special Sequence Limits*
(i) If $p > 0$, then $\lim_{n\to\infty} n^{-p} = 0$.
(ii) If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.
(iii) $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
(iv) If $p > 0$ and $\alpha \in \mathbb{R}$, then $\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$.
(v) If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.

Proof. All positive roots below exist uniquely by Theorem 1.1.11.
(i) Fix $\varepsilon > 0$ and choose $N > \varepsilon^{-p^{-1}}$ by Theorem 1.1.19(i). Then $(\forall n \ge N)\ 0 < n^{-p} < \varepsilon$.
(ii) Suppose $p > 1$ and set $x_n := \sqrt[n]{p} - 1$. Then $x_n > 0$ and $1 + nx_n \le (1 + x_n)^n = p$ by Proposition 4.3.7(i), so $0 < x_n \le (p-1)n^{-1}$.
Upper bound tends to 0 by (i) and Theorem 2.1.5(ii). Thus $x_n \to 0$ by Theorem 2.2.9; $\sqrt[n]{p} = 1 + x_n \to 1$ by Theorem 2.1.5(ii).
If $p = 1$, then $(\forall n \in \mathbb{N})\ \sqrt[n]{p} = 1$. If $0 < p < 1$, set $q := p^{-1} > 1$. Then $\sqrt[n]{q} \to 1$ by the preceding case.
The identity $\sqrt[n]{p} = (\sqrt[n]{q})^{-1}$ holds by Corollary 1.1.12. Therefore $\sqrt[n]{p} \to 1$ by Theorem 2.1.5(iv).
(iii) Set $x_n := \sqrt[n]{n} - 1 \ge 0$. For $n \ge 2$, $n = (1 + x_n)^n \ge \binom{n}{2} x_n^2 = \frac{n(n-1)}{2} x_n^2$ by Proposition 4.3.7(i). Thus $0 \le x_n^2 \le \frac{2}{n-1}$.
Fix $\varepsilon > 0$. Choose $N > 1 + 2\varepsilon^{-2}$ by Theorem 1.1.19(i). For $n \ge \max\{2, N\}$, $0 \le x_n < \varepsilon$. Hence $x_n \to 0$ and $\sqrt[n]{n} = 1 + x_n \to 1$.
(iv) Choose $k \in \mathbb{N}$ with $k > \alpha$ by Theorem 1.1.19(i). If $n > 2k$, then $(\forall j \in \{0, \dots, k-1\})\ n - j > \frac{n}{2}$.
Thus $(1 + p)^n \ge \binom{n}{k} p^k > \frac{n^k p^k}{2^k k!}$ by Proposition 4.3.7(i); $0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{-(k-\alpha)}$.
Upper bound tends to 0 by (i), Theorem 2.1.5(ii). Quotient tends to 0 by Theorem 2.2.9.
(v) If $x \ne 0$, set $p := |x|^{-1} - 1 > 0$. Then $|x| = (1 + p)^{-1}$ and $|x|^n \to 0$ by (iv) with $\alpha = 0$.
The bounds $-|x|^n \le x^n \le |x|^n$ imply $x^n \to 0$ by Theorem 2.2.9. The case $x = 0$ is immediate.
Structural Commentary: Reciprocal powers and binomial lower bounds reduce roots and exponential quotients to pinching.
Key Insight: A fixed binomial term dominates every lower polynomial power. □

Proposition 2.2.11. *Decimal Truncations Squeeze a Real*
Let $x \geq 0$ and set $S_n := \{k \in \mathbb{Z}_{\geq 0} : k10^{-n} \leq x\}$ for $n \in \mathbb{N}_0$.

(i) The maximum $s_n := \max S_n$ exists and satisfies $s_n 10^{-n} \leq x < (s_n+1)10^{-n}$ for $n \in \mathbb{N}_0$.
(ii) $(\forall n \in \mathbb{N}_0)\, 0 \leq x - s_n 10^{-n} < 10^{-n}$, and $s_n 10^{-n} \to x$.
(iii) $(\forall n \in \mathbb{N}_0)(\exists! a_{n+1} \in \{0,\dots,9\})\, s_{n+1} = 10s_n + a_{n+1}$.

Proof.

(i) Fix $n \in \mathbb{N}_0$ and set $A_n := \{j \in \mathbb{N} : j10^{-n} > x\}$. The set A_n is nonempty by Theorem 1.1.19(i).
Let m_n be its least element by the standing well-ordering assumption.
Since $x \geq 0$, $m_n \geq 1$, and minimality implies $m_n - 1 \in S_n$. Every $k \in S_n$ satisfies $k < m_n$, so $s_n = m_n - 1 = \max S_n$.
The inequalities $s_n 10^{-n} \leq x < (s_n+1)10^{-n}$ follow from $s_n \in S_n$ and $s_n + 1 = m_n \in A_n$.
(ii) The inequalities in (i) imply $(\forall n \in \mathbb{N}_0)\, 0 \leq x - s_n 10^{-n} < 10^{-n}$. The limit $10^{-n} \to 0$ holds by Theorem 2.2.10(v) with $x = 10^{-1}$.
Thus $x - 10^{-n} \to x$ by Theorem 2.1.5(ii). The bounds $x - 10^{-n} \leq s_n 10^{-n} \leq x$ imply $s_n 10^{-n} \to x$ by Theorem 2.2.9.
(iii) The relations $10s_n \in S_{n+1}$ and $10s_n + 10 \in A_{n+1}$ imply $10s_n \leq s_{n+1} < 10s_n + 10$.
Thus $a_{n+1} := s_{n+1} - 10s_n \in \{0,\dots,9\}$, and this formula makes a_{n+1} unique.

Structural Commentary: Archimedean grids define each truncation; nested mesh bounds then support the squeeze.
Key Insight: Refinement by a factor of 10 leaves one digit as the integer remainder. □

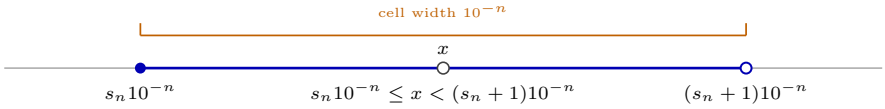

Figure 121: At precision 10^{-n}, x lies in the half-open cell $[s_n 10^{-n}, (s_n+1)10^{-n})$ of width 10^{-n}.

Figure 121 illustrates the shrinking interval in Proposition 2.2.11(i).

Proof Method 2.2.12. *Real-Sequence Proof Patterns*

(i) *Convergence*: Fix $\varepsilon > 0$ and construct N such that $(\forall n \geq N)\ |s_n - s| < \varepsilon$.
(ii) *Uniqueness*: Synchronize two $\frac{\varepsilon}{2}$ tail bounds and compare the proposed limits by the triangle inequality.
(iii) *Subsequences*: Recursively choose $n_k > n_{k-1}$ so the kth term meets a threshold or an error bound of k^{-1}.
(iv) *Cauchy transfer*: Anchor an arbitrary late term to a sufficiently late convergent subsequence term.
(v) *Monotone convergence*: Choose the relevant supremum or infimum and propagate one extremal witness through its tail.

Key Insight: Synchronized tail indices reduce global limit claims to one controlling estimate.

3 Metric Spaces and Euclidean Topology

3.1 Metrics, Open Sets, and Euclidean Topology

Metric topology studies sets through balls: openness, closure, and limit points come from distance.
Euclidean estimates rely on norms, Cauchy–Schwarz, and the triangle inequality.

Definition 3.1.1. *Metric Space*
A *metric space* is a pair (X, d), where X is a set and $d : X \times X \to [0, \infty)$. For all $p, q, r \in X$, the following conditions hold:

i. $d(p,q) = 0 \iff p = q$;
ii. $d(p,q) = d(q,p)$;
iii. $d(p,q) \leq d(p,r) + d(r,q)$.

The function d is called a *distance function* or *metric*.

Definition 3.1.2. *Metric Topology Terminology*
Let (X, d) be a metric space. All points and sets below are understood to be elements and subsets of X, respectively.

(i) For $p \in X$ and $r > 0$, the *open ball* centered at p with radius r is $B_r(p) := \{q \in X : d(p,q) < r\}$.
The corresponding *closed ball* is $\overline{B}_r(p) := \{q \in X : d(p,q) \leq r\}$. The *sphere* of radius r about p is $\{q \in X : d(p,q) = r\}$.
(ii) A point p is an *interior point* of E iff $(\exists\, r > 0)\, B_r(p) \subseteq E$.
(iii) A set E is *open* if every point of E is an interior point of E.
(iv) A *neighborhood* of p is an open set $U \subseteq X$ with $p \in U$.
(v) A point p is a *limit point* of E iff $(\forall\, r > 0)\, B_r(p) \cap (E \setminus \{p\}) \neq \emptyset$.
(vi) If $p \in E$ and p is not a limit point of E, then p is an *isolated point* of E.
(vii) A set E is *closed* iff every limit point of E belongs to E.
(viii) The *complement* of E is $E^c := X \setminus E$.
(ix) A set E is *dense* in X iff every point of X belongs to E, is a limit point of E, or both.

The *metric topology* induced by d is the collection τ_d of all open subsets of X.
For $X = \mathbb{R}$, $B_\varepsilon(a) = (a - \varepsilon, a + \varepsilon)$. Open balls supply the local tests for the metric topology.
Neighborhoods are the open sets containing the point.

Theorem 3.1.3. *Every Open Ball Is Open*
Every open ball in a metric space is an open set.

Proof. Let $E = B_r(p)$, and choose $q \in E$. Put $h := r - d(p,q) > 0$. If $s \in B_h(q)$, then $d(p,s) \leq d(p,q) + d(q,s) < d(p,q) + h = r$.
Thus $B_h(q) \subseteq E$, so q is an interior point of E. Since q was arbitrary, E is open.
Structural Commentary: For $q \in B_r(p)$, positive slack $h = r - d(p,q)$ defines smaller ball. Triangle inequality keeps $B_h(q)$ inside $B_r(p)$.
Key Insight: The triangle inequality turns radius slack into openness. □

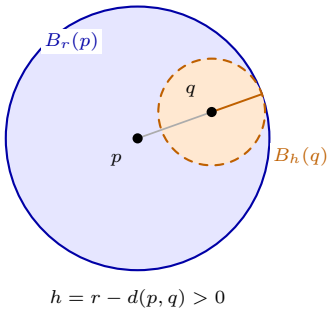

Figure 122: Every point inside a metric ball has a smaller ball still contained in the original ball.

Figure 122 illustrates the smaller-ball construction in the proof.

Perspective 3.1.4. *ε-Closeness Becomes Metric Language*
For the usual metric $d(x,y) := |x-y|$ on $\mathbb{R}$, $|x-y| < \varepsilon \iff y \in B_\varepsilon(x)$.
Metric balls extend this notion of ε-closeness to arbitrary metric spaces.
Key Insight: Absolute value is distance on $\mathbb{R}$.

Example 3.1.5. *Euclidean Metric and Subspaces*
For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, define $d(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_2$. Nonnegativity and definiteness hold by Proposition 1.1.12(i)–(ii).
Symmetry holds by Proposition 1.1.12(iii) with scalar -1. Write $\mathbf{x}-\mathbf{z} = (\mathbf{x}-\mathbf{y}) + (\mathbf{y}-\mathbf{z})$.
The metric triangle inequality holds by Theorem 1.1.14(i). Thus the Euclidean space $\mathbb{R}^k$ is a metric space.
If $Y \subseteq X$ for a metric space (X,d), then $d|_{Y\times Y}$ is a metric on Y.

Example 3.1.6. *Discrete Metric*
On any set X, define $d(x,y) := 0$ when $x = y$ and $d(x,y) := 1$ when $x \neq y$. Definiteness and symmetry are immediate.
If $x \neq z$, then at least one of $x \neq y$ or $y \neq z$ holds, so $d(x,z) = 1 \leq d(x,y) + d(y,z)$.
Thus d is the *discrete metric*: distinct points are all distance 1 apart.

Example 3.1.7. *Hamming Distance*
For $x, y \in \{0,1\}^n$, define $h(x,y) := |\{i \in \{1,\dots,n\} : x_i \neq y_i\}|$. Then h is a metric.

Proof. Value $h(x,y)$ nonnegative, satisfies $h(x,y) = h(y,x)$. Moreover, $h(x,y) = 0$ iff $(\forall\, i \in \{1,\dots,n\})\, x_i = y_i$, equivalently $x = y$.
Put $D(u,v) := \{i \in \{1,\dots,n\} : u_i \neq v_i\}$. Then $D(x,y) \subseteq D(x,z) \cup D(z,y)$; $h(x,y) \leq h(x,z) + h(z,y)$ by cardinality of finite unions.
Structural Commentary: Definiteness uses $D(x,y) = \emptyset$; inclusion proves the triangle inequality.
Key Insight: Each disagreement between x and y must be paid for through z in at least one leg. □

Definition 3.1.8. *Euclidean Intervals, Cells, Balls, and Convexity*
(i) In $\mathbb{R}$, $(a,b) := \{x : a < x < b\}$, $[a,b] := \{x : a \leq x \leq b\}$, with $[a,b)$ and $(a,b]$ defined by mixed inequalities.
(ii) A set $J \subseteq \mathbb{R}$ is an *interval* iff $(\forall\, x, y \in J)\, [x < y \Rightarrow [x,y] \subseteq J]$.
(iii) For $a_i < b_i$ with $1 \leq i \leq k$, a *k-cell* is $\prod_{i=1}^{k} [a_i, b_i]$.
(iv) For $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, $B_r(\mathbf{x}) := \{\mathbf{y} : \|\mathbf{y}-\mathbf{x}\|_2 < r\}$. The corresponding closed ball is $\overline{B}_r(\mathbf{x}) := \{\mathbf{y} : \|\mathbf{y}-\mathbf{x}\|_2 \leq r\}$.
(v) For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, the points $\lambda\mathbf{x} + (1-\lambda)\mathbf{y}$ with $\lambda \in [0,1]$ are the two-point convex combinations.
They form the *line segment* $[\mathbf{x},\mathbf{y}] := \{\lambda\mathbf{x} + (1-\lambda)\mathbf{y} : \lambda \in [0,1]\}$.
(vi) A set $C \subseteq \mathbb{R}^k$ is *convex* iff $(\forall\, \mathbf{x}, \mathbf{y} \in C)\, [\mathbf{x},\mathbf{y}] \subseteq C$.
(vii) The *integer lattice* is $\mathbb{Z}^k$, and the *first orthant* is $\{\mathbf{x} \in \mathbb{R}^k : (\forall\, i \in \{1,\dots,k\})\, x_i \geq 0\}$.
(viii) The *unit ball* is $B^k := \{\mathbf{x} \in \mathbb{R}^k : \|\mathbf{x}\|_2 \leq 1\}$, and the *unit sphere* is $S^{k-1} := \{\mathbf{x} \in \mathbb{R}^k : \|\mathbf{x}\|_2 = 1\}$.

Proposition 3.1.9. *Balls and Cells Are Convex*
(i) Every open or closed ball in $\mathbb{R}^k$ is convex.
(ii) Every k-cell in $\mathbb{R}^k$ is convex.

Proof.
(i) Let $\mathbf{y}, \mathbf{z} \in B_r(\mathbf{x})$, let $\lambda \in [0,1]$, and put $\mathbf{w} := \lambda\mathbf{y} + (1-\lambda)\mathbf{z}$.
Then $\|\mathbf{w}-\mathbf{x}\|_2 \leq \lambda\|\mathbf{y}-\mathbf{x}\|_2 + (1-\lambda)\|\mathbf{z}-\mathbf{x}\|_2 < r$ by norm homogeneity and the triangle inequality.
Hence $B_r(\mathbf{x})$ is convex. For $\mathbf{y}, \mathbf{z} \in \overline{B}_r(\mathbf{x})$ and $\lambda \in [0,1]$, define $\mathbf{w}$ as above. Same estimate ends with $\leq r$, so $\overline{B}_r(\mathbf{x})$ is convex.
(ii) Let $\mathbf{y}, \mathbf{z} \in \prod_{i=1}^{k} [a_i, b_i]$ and $\lambda \in [0,1]$. For every $1 \leq i \leq k$, $a_i \leq \lambda y_i + (1-\lambda) z_i \leq b_i$ by $a_i \leq y_i, z_i \leq b_i$.
Thus $\lambda\mathbf{y} + (1-\lambda)\mathbf{z}$ belongs to the cell.
Structural Commentary: Norm bounds preserve balls; coordinate bounds preserve cells.
Key Insight: Norm subadditivity preserves balls, while order intervals preserve each coordinate. □

Proposition 3.1.10. *Bounded Compression of a Metric*
Let (X,d) be a metric space and set $\rho(x,y) := \frac{d(x,y)}{1+d(x,y)}$.
(i) The function ρ is a bounded metric on X.
(ii) The induced topologies satisfy $\tau_\rho = \tau_d$.

Proof.

(i) Put $f(t) := \frac{t}{1+t}$ for $t \geq 0$. Then $0 \leq f(t) < 1$, so ρ is bounded; definiteness and symmetry follow from d.
For $a, b \geq 0$, $\frac{a}{1+a} + \frac{b}{1+b} - \frac{a+b}{1+a+b} = \frac{ab(a+b+2)}{(1+a)(1+b)(1+a+b)} \geq 0$.
For $0 \leq u \leq v$, $\frac{v}{1+v} - \frac{u}{1+u} = \frac{v-u}{(1+u)(1+v)} \geq 0$. Hence f is nondecreasing and subadditive.
Consequently, $\rho(x,z) = f(d(x,z)) \leq f(d(x,y) + d(y,z)) \leq \rho(x,y) + \rho(y,z)$ by the triangle inequality for d.

(ii) Write $B_s^d(x)$ and $B_s^\rho(x)$ for the balls associated with the indicated metrics.
Since f is strictly increasing, $B_\eta^\rho(x) = B^d_{\frac{\eta}{1-\eta}}(x)$ for $0 < \eta < 1$. Similarly, $B^\rho_{\frac{r}{1+r}}(x) = B_r^d(x)$ for $r > 0$.
A local radius may be reduced below 1, so the two ball identities imply $\tau_\rho = \tau_d$.

Structural Commentary: Scalar monotonicity, subadditivity prove metric claim. Inverse radius formulas prove topology claim.
Key Insight: Compression changes distances but exactly reparametrizes small balls. □

Definition 3.1.11. *Isometries and Pseudometrics*
Let (X, ρ) and (Y, σ) be metric spaces, and let $F : X \to Y$.

(i) The map F is an *isometry* iff $(\forall\, x, y \in X)\, \sigma(Fx, Fy) = \rho(x, y)$.
(ii) The spaces (X, ρ) and (Y, σ) are *isometric*, written $(X, \rho) \cong (Y, \sigma)$, iff a bijective isometry $F : X \to Y$ exists.

For a set Z, a *pseudometric* on Z is a map $d : Z \times Z \to [0, \infty)$.
For all $x, y, z \in Z$, it satisfies $d(x, x) = 0$, $d(x, y) = d(y, x)$, and $d(x, z) \leq d(x, y) + d(y, z)$.
Unlike a metric, a pseudometric may satisfy $d(x, y) = 0$ for distinct points $x, y \in Z$.
Isometries preserve distance; pseudometrics permit zero-distance identification.

Proposition 3.1.12. *Metric Quotient of a Pseudometric*
Let X be set, ρ be pseudometric on X. Define $x \sim_\rho y$ iff $\rho(x, y) = 0$, and $\widetilde{\rho} : X/\!\sim_\rho \times X/\!\sim_\rho \to [0, \infty)$ by $\widetilde{\rho}([x], [y]) := \rho(x, y)$.

(i) $\sim_\rho$ is an equivalence relation on X.
(ii) $\widetilde{\rho}$ is a well-defined metric on $X/\!\sim_\rho$.

Proof.

(i) Reflexivity and symmetry of $\sim_\rho$ follow from $\rho(x, x) = 0$ and symmetry of ρ.
If $x \sim_\rho y$ and $y \sim_\rho z$, then $0 \leq \rho(x, z) \leq \rho(x, y) + \rho(y, z) = 0$ by the pseudometric triangle inequality.
Hence $x \sim_\rho z$, so $\sim_\rho$ is an equivalence relation.

(ii) Suppose $x \sim_\rho x'$ and $y \sim_\rho y'$. Then $\rho(x, y) \leq \rho(x', y')$ by two triangle inequalities and the zero-distance hypotheses.
Interchanging the primed and unprimed representatives proves the reverse inequality.
Thus $\widetilde{\rho}$ is well-defined. For $[x], [y], [z] \in X/\!\sim_\rho$, nonnegativity and symmetry follow from ρ.
Moreover, $\widetilde{\rho}([x], [z]) \leq \widetilde{\rho}([x], [y]) + \widetilde{\rho}([y], [z])$ by the pseudometric triangle inequality.
Finally, $\widetilde{\rho}([x], [y]) = 0$ iff $\rho(x, y) = 0$ iff $[x] = [y]$, so $\widetilde{\rho}$ is definite and therefore a metric.

Structural Commentary: The triangle inequality makes zero distance transitive.
It also makes quotient distance independent of representatives.
Key Insight: Collapsing zero-distance classes restores definiteness and preserves the other axioms. □

Example 3.1.13. *Open, Closed, Both, or Neither*
In $\mathbb{R}$, both $\emptyset$ and $\mathbb{R}$ are open and closed. If $a < b$, then $[a, b)$ is not open because a is not an interior point.
It is not closed because b is a limit point not in the set. If d is discrete on X, then $B_{\frac{1}{2}}(x) = \{x\}$ for every $x \in X$.
Thus every $E \subseteq X$ and its complement are open, so every subset of X is also closed.

Theorem 3.1.14. *Limit Points Force Infinitely Many Nearby Points*
Let (X, d) be metric space, $E \subseteq X$, $p \in X$. If p is limit point of E, then $N \cap E$ is infinite for every neighborhood N of p.

Proof. Let N be a neighborhood of p and suppose $N \cap E$ is finite. Since N is open and $p \in N$, choose $\eta > 0$ with $B_\eta(p) \subseteq N$.
Put $F := (N \cap E) \setminus \{p\}$. If $F = \emptyset$, then $B_\eta(p) \cap (E \setminus \{p\}) = \emptyset$, contrary to the limit-point hypothesis.
Suppose $F \neq \emptyset$, and set $\delta := \min(\{\eta\} \cup \{d(p, q) : q \in F\}) > 0$. If $q \in B_\delta(p) \cap (E \setminus \{p\})$, then $q \in F$ and $d(p, q) < \delta \leq d(p, q)$, impossible.
Thus $B_\delta(p) \cap (E \setminus \{p\}) = \emptyset$, again contrary to the limit-point hypothesis.
Structural Commentary: A finite list of nearby exceptions has a smallest positive distance from p.
Key Insight: Limit points cannot be witnessed by only finitely many neighboring points. □

Corollary 3.1.15. *Finite Point Sets Have No Limit Points*
A finite point set in a metric space has no limit points.

Proof. Let $E \subseteq X$ be finite, and suppose p is a limit point of E.
Every neighborhood of p then contains infinitely many distinct points of E by Theorem 3.1.14.
This contradicts finiteness of E, so E has no limit points.
Structural Commentary: One limit point would force infinitely many members of finite E.
Key Insight: A limit point requires infinitely many nearby witnesses. □

Theorem 3.1.16. *Open Sets and Closed Complements*
Let (X, d) be a metric space and $E \subseteq X$. Then E is open if and only if its complement E^c is closed.

Proof.

(⇒) Suppose E is open. If $x \in E$, then $B_r(x) \subseteq E$ for some $r > 0$, so x is not a limit point of E^c.
Every limit point of E^c therefore belongs to E^c, and E^c is closed.

(⇐) Suppose E^c is closed, and fix $x \in E$. Then $x \notin E^c$ and x is not a limit point of E^c.
Some $r > 0$ therefore satisfies $B_r(x) \cap (E^c \setminus \{x\}) = \emptyset$.
Since $x \notin E^c$, this reduces to $B_r(x) \cap E^c = \emptyset$, so $B_r(x) \subseteq E$. Thus E is open.

Structural Commentary: Ball in E excludes accumulation from E^c. Failure of accumulation supplies the reverse ball.
Key Insight: The same local separation test expresses openness of E and closedness of E^c. □

Corollary 3.1.17. *Closed Sets and Open Complements*
Let (X, d) be a metric space and $F \subseteq X$. Then F is closed if and only if F^c is open.

Proof.
($\Rightarrow$) If F is closed, then $(F^c)^c = F$ is closed; by Theorem 3.1.16 with $E = F^c$, F^c is open.
($\Leftarrow$) If F^c is open, then $(F^c)^c$ is closed by Theorem 3.1.16; since $(F^c)^c = F$, F is closed.
Structural Commentary: Applying the open-complement theorem to F^c exchanges set and complement.
Key Insight: Complement involution turns the open-set criterion into the closed-set criterion. □

Corollary 3.1.18. *Closed Balls Are Closed*
Let (X, d) be a metric space. For every $p \in X$ and $r > 0$, the closed ball $\overline{B}_r(p)$ is closed.

Proof. Let $q \notin \overline{B}_r(p)$ and put $h := d(p, q) - r > 0$. If $s \in B_h(q)$, then $d(p, s) \geq d(p, q) - d(q, s) > r$ by the triangle inequality. Thus $B_h(q) \subseteq (\overline{B}_r(p))^c$, so the complement of $\overline{B}_r(p)$ is open. The closed ball $\overline{B}_r(p)$ is therefore closed by Corollary 3.1.17.
Structural Commentary: At an exterior point, the excess $d(p, q) - r$ defines a ball outside $\overline{B}_r(p)$.
Key Insight: The triangle inequality converts positive boundary separation into an open complement. □

Theorem 3.1.19. *Basic Topological Calculus*
In a metric space:
(i) any union of open sets is open;
(ii) any intersection of closed sets is closed;
(iii) any finite intersection of open sets is open;
(iv) any finite union of closed sets is closed.

Proof.
(i) Let $G := \bigcup_{\alpha \in A} G_\alpha$ for open sets G_α. If $x \in G$, then $x \in G_\alpha$ for some $\alpha \in A$.
Since G_α is open, $B_r(x) \subseteq G_\alpha \subseteq G$ for some $r > 0$. Hence G is open.
(ii) Let $F := \bigcap_{\alpha \in A} F_\alpha$. By Theorem 1.5.2, $F^c = \bigcup_{\alpha \in A} F_\alpha^c$.
Each F_α^c is open by Corollary 3.1.17, so F^c is open by (i). Hence F is closed by Corollary 3.1.17.
(iii) The empty intersection is X, which is open. For $n \geq 1$, let $H := \bigcap_{i=1}^{n} G_i$.
If $x \in H$, choose $r_i > 0$ with $B_{r_i}(x) \subseteq G_i$ for $1 \leq i \leq n$, and set $r := \min_{1 \leq i \leq n} r_i$. Then $B_r(x) \subseteq H$, so H is open.
(iv) The empty union is $\emptyset$, which is closed. For $n \geq 1$, let $F := \bigcup_{i=1}^{n} F_i$. By Theorem 1.5.2, $F^c = \bigcap_{i=1}^{n} F_i^c$.
Each F_i^c is open by Corollary 3.1.17, so F^c is open by (iii). Hence F is closed by Corollary 3.1.17.
Structural Commentary: Open unions retain one ball; finite intersections use a minimum radius.
Complement duality transfers both arguments to closed sets.
Key Insight: Radius minima explain why only finite intersections preserve openness. □

Theorem 3.1.20. *Open Sets on the Real Line Split into Intervals*
Let $\emptyset \neq G \subseteq \mathbb{R}$ be open.
(i) $G = \bigsqcup_{\alpha \in A} I_\alpha$ for a family of pairwise disjoint open intervals $I_\alpha \subseteq \mathbb{R}$, possibly unbounded.
(ii) The index set A is at most countable.

Proof.
(i) For $a, b \in \mathbb{R}$, set $I(a, b) := [\min\{a, b\}, \max\{a, b\}]$.
For $x, y \in G$, set $x \sim y$ iff $I(x, y) \subseteq G$, and let $C_x := [x]_\sim$. Reflexivity and symmetry are immediate.
If $x \sim y$ and $y \sim z$, then $I(x, z) \subseteq I(x, y) \cup I(y, z) \subseteq G$, so $x \sim z$. Thus $\sim$ is an equivalence relation.
If $y \in C_x$ and $z \in I(x, y)$, then $I(x, z) \subseteq I(x, y) \subseteq G$, so $z \in C_x$; hence C_x is an interval.
If $y \in C_x$, choose $r > 0$ with $(y - r, y + r) \subseteq G$. Every $z \in (y - r, y + r)$ satisfies $I(x, z) \subseteq I(x, y) \cup (y - r, y + r) \subseteq G$, so $z \in C_x$.
Thus each C_x is open, and the equivalence classes form a disjoint union of G.
(ii) Let $\mathcal{C} := G/\sim$. Choose a bijection $j \mapsto q_j$ from $\mathbb{N}$ onto $\mathbb{Q}$ by Corollary 6.2.19.
Each $C \in \mathcal{C}$ contains a bounded nonempty open interval, so $C \cap \mathbb{Q} \neq \emptyset$ by Theorem 1.1.19(ii).
By well-ordering, the nonempty set $\{j \in \mathbb{N} : q_j \in C\}$ has a least element; define $\phi(C) := q_{\min\{j : q_j \in C\}}$.
The map $\phi : \mathcal{C} \to \mathbb{Q}$ is injective by disjointness, so $\mathcal{C}$ is at most countable.
Structural Commentary: Segment containment partitions G into open intervals; their least-indexed rationals embed partition into $\mathbb{Q}$.
Key Insight: One rational witness per disjoint component turns interval geometry into countability. □

Definition 3.1.21. *Closure*
Let (X, d) be a metric space and $E \subseteq X$, and E^{acc} denote set of all limit points of E in X. The *closure* of E is $\overline{E} := E \cup E^{\text{acc}}$.

Theorem 3.1.22. *Closure Is the Smallest Closed Superset*
If X is a metric space and $E \subseteq X$, then:
(i) $\overline{E}$ is closed;
(ii) $\overline{E} = E$ if and only if E is closed;
(iii) $\overline{E} \subseteq F$ for every closed set $F \subseteq X$ such that $E \subseteq F$. Consequently, $\overline{E}$ is the smallest closed subset of X containing E.

Proof.
(i) Let $p \notin \overline{E}$. Then $p \notin E$ and p is not a limit point of E, so $B_r(p) \cap E = \emptyset$ for some $r > 0$.
If $q \in B_r(p)$, then $q \notin E$ and $B_{r - d(p,q)}(q) \subseteq B_r(p)$, so $q \notin E^{\text{acc}}$. Hence $B_r(p) \subseteq (\overline{E})^c$, and $\overline{E}$ is closed by Corollary 3.1.17.
(ii) ($\Rightarrow$) If $\overline{E} = E$, then E is closed by (i).
(ii) ($\Leftarrow$) If E is closed, then $E^{\text{acc}} \subseteq E$, so $\overline{E} = E \cup E^{\text{acc}} = E$.

(*iii*) Let F be closed with $E \subseteq F$. Then $E^{\mathrm{acc}} \subseteq F^{\mathrm{acc}} \subseteq F$ by $E \subseteq F$ and closedness of F. Thus $\overline{E} = E \cup E^{\mathrm{acc}} \subseteq F$.
Together with $E \subseteq \overline{E}$ and (i), this proves the asserted minimality.

Structural Commentary: Exterior balls avoid E and its limit points; limit-point monotonicity puts $\overline{E}$ in every closed superset.
Key Insight: The closure is closed, and every closed superset must contain its added limit points. □

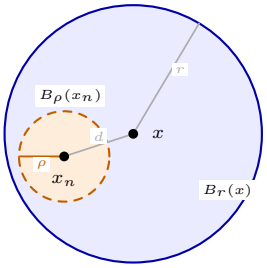

Figure 123: If $d = d(x, x_n) < r$ and $0 < \rho < r - d$, then $B_\rho(x_n) \subseteq B_r(x)$.

Figure 123 depicts the inner-ball containment used in part (i), with $x = p$ and $x_n = q$.

Definition 3.1.23. *Adherent Points*
Let (X, d) be a metric space and $E \subseteq X$. A point $p \in X$ is *adherent* to E iff $(\forall\, r > 0)\, B_r(p) \cap E \neq \emptyset$.
Equivalently, p is adherent to E iff $p \in \overline{E}$. Ball condition is automatic for $p \in E$ and is limit-point condition for $p \notin E$.

Figure 124: Adherent endpoints need not lie in E; isolated points are adherent but not limit points.

Figure 124 distinguishes adherent endpoints and isolated points from limit points.

Definition 3.1.24. *Interior, Exterior, and Boundary*
Let (X, d) be a metric space and $E \subseteq X$.

(i) $E^\circ := \{p \in E : (\exists\, r > 0)\, B_r(p) \subseteq E\}$ is the *interior*.
(ii) $\mathrm{ext}(E) := (E^c)^\circ = \{p \in X : (\exists\, r > 0)\, B_r(p) \cap E = \emptyset\}$ is the *exterior*.
(iii) $\mathrm{bd}(E) := \overline{E} \setminus E^\circ = \{p \in X : (\forall\, r > 0)\, [B_r(p) \cap E \neq \emptyset \wedge B_r(p) \cap E^c \neq \emptyset]\}$ is the *boundary*.

The boundary ball test follows by Definition 3.1.23 applied to E and E^c.

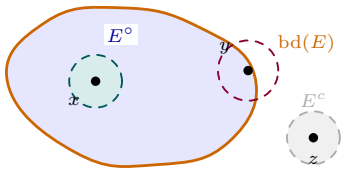

Figure 125: Interior, exterior, and boundary points are distinguished by how small balls meet E and E^c.

Figure 125 compares the ball tests for interior, exterior, and boundary points.

Definition 3.1.25. *Point-Set and Set-Set Distance*
Let (X, d) be a metric space and let $E, F \subseteq X$ be nonempty.

(i) For $p \in X$, the *point-set distance* is $\mathrm{dist}(p, E) := \inf_{q \in E} d(p, q)$.
(ii) The *set-set distance* is $\mathrm{dist}(E, F) := \inf_{(x,y) \in E \times F} d(x, y)$.

Both infima exist in $\mathbb{R}$ by Corollary 1.1.15. The corresponding families of distances are nonempty and bounded below by 0.

Proposition 3.1.26. *Distance-to-Set Closure Criterion*
Let (X, d) be a metric space and $\emptyset \neq E \subseteq X$. For $p \in X$, $p \in \overline{E} \iff \mathrm{dist}(p, E) = 0$.

Proof.

($\Rightarrow$) Suppose $p \in \overline{E}$. If $p \in E$, then $0 \leq \mathrm{dist}(p, E) \leq d(p, p) = 0$.
If $p \notin E$, then p is a limit point of E, so for every $\varepsilon > 0$ there exists $q_\varepsilon \in E$ with $d(p, q_\varepsilon) < \varepsilon$.
Thus $0 \leq \mathrm{dist}(p, E) \leq \varepsilon$ for every $\varepsilon > 0$, and $\mathrm{dist}(p, E) = 0$.
($\Leftarrow$) Suppose $\mathrm{dist}(p, E) = 0$. If $p \notin \overline{E}$, then some $r > 0$ satisfies $B_r(p) \cap E = \emptyset$.
Then $d(p, q) \geq r$ for every $q \in E$, so r is a positive lower bound for $\{d(p, q) : q \in E\}$. Contradicts infimum 0, so $p \in \overline{E}$.

Structural Commentary: Adherence supplies points at every positive scale. Failed adherence supplies a positive lower bound.
Key Insight: Zero distance and intersection with every centered ball both characterize closure. □

Remark 3.1.27. *Set Distance Is Not a Metric on Subsets*
The function $\mathrm{dist}(E, F)$ is useful, but it is not a metric on nonempty subsets.
In $\mathbb{R}$, for example, $\mathrm{dist}((0, 1), (1, 2)) = 0$ although the sets are distinct and disjoint.

3.2 Relative Topology

Definition 3.2.1. *Relative Openness*
Let (X,d) be a metric space and $E \subseteq Y \subseteq X$. The set E is *open relative to Y* iff $(\forall\, p \in E)(\exists\, r>0)\, B_r(p) \cap Y \subseteq E$.

Definition 3.2.2. *Relative Closedness*
Let (X,d) be a metric space and $F \subseteq Y \subseteq X$. The set F is *closed relative to Y* iff $Y \setminus F$ is open relative to Y.

Theorem 3.2.3. *Relative Opens Are Intersections*
Let (X,d) be a metric space, $Y \subseteq X$, $E \subseteq Y$. Then E is open relative to Y iff $(\exists\, G \in \tau_d)\, E = Y \cap G$.

Proof.
($\Rightarrow$) Suppose E is open relative to Y. Let $\mathcal{B} := \{B_r(p) : p \in E,\ r>0,\ B_r(p) \cap Y \subseteq E\}$.
Set $G := \bigcup_{B \in \mathcal{B}} B$. Every $B \in \mathcal{B}$ is open by Theorem 3.1.3, so $G \in \tau_d$ by Theorem 3.1.19(i).
For every $p \in E$, relative openness supplies $B_r(p) \in \mathcal{B}$ containing p, so $E \subseteq G \cap Y$.
If $q \in G \cap Y$, then $q \in B_r(p) \cap Y \subseteq E$ for some $B_r(p) \in \mathcal{B}$. Thus $E = G \cap Y$.
($\Leftarrow$) Suppose $E = G \cap Y$ for some $G \in \tau_d$. For $p \in E$, choose $r>0$ with $B_r(p) \subseteq G$.
Then $B_r(p) \cap Y \subseteq G \cap Y = E$, so E is open relative to Y.
Structural Commentary: Witnessing ambient balls form G; intersecting G with Y recovers E.
Key Insight: Relative openness is exactly ambient openness restricted to Y. □

Corollary 3.2.4. *Relative Closed Sets Are Intersections*
Let (X,d) be a metric space, $Y \subseteq X$, $F \subseteq Y$. Then F is closed relative to Y iff $F = Y \cap C$ for some closed set $C \subseteq X$.

Proof.
($\Rightarrow$) Suppose F is closed relative to Y. Then $Y \setminus F$ is open relative to Y. By Theorem 3.2.3, $Y \setminus F = Y \cap G$ for some $G \in \tau_d$.
Put $C := X \setminus G$. The set C is closed in X by Corollary 3.1.17, and $F = Y \setminus (Y \cap G) = Y \cap C$.
($\Leftarrow$) Suppose $F = Y \cap C$ for a closed set $C \subseteq X$. Then $X \setminus C$ is open by Corollary 3.1.17.
The set $Y \setminus F = Y \cap (X \setminus C)$ is open relative to Y by Theorem 3.2.3. Thus F is closed relative to Y.
Structural Commentary: Complementing $Y \cap G$ in Y replaces open G by closed $X \setminus G$.
Key Insight: Relative closedness is the complement-dual of relative openness. □

Example 3.2.5. *Relative Topology Depends on the Ambient Space*
Let $X := \{(x,0) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the inherited Euclidean metric, and set $E := \{(x,0) : -1 < x < 1\}$.
No ball in $\mathbb{R}^2$ centered at a point of E lies in the x-axis, so E is not open in $\mathbb{R}^2$.
However, $E = X \cap \big((-1,1) \times (-1,1)\big)$, so E is open relative to X. Set $[0,1)$ is not closed in $\mathbb{R}$ because 1 is a limit point not in the set.
Yet $[0,1) = (-1,1) \cap [0,\infty)$, and $[0,\infty)$ is closed in $\mathbb{R}$. Hence $[0,1)$ is closed relative to $(-1,1)$ by Corollary 3.2.4.

3.3 Density, Separability, and Closure Tests

Proposition 3.3.1. *Density and Closure*
Let (X,d) be a metric space and $E \subseteq X$. Then E is dense in X iff $\overline{E} = X$.

Proof.
($\Rightarrow$) Suppose E dense in X. Every $p \in X$ belongs to E or is limit point of E, so $p \in E \cup E^{\text{acc}} = \overline{E}$. Thus $X \subseteq \overline{E} \subseteq X$, $\overline{E} = X$.
($\Leftarrow$) Suppose $\overline{E} = X$. Then every $p \in X$ belongs to $E \cup E^{\text{acc}}$. Hence p belongs to E or is a limit point of E, so E is dense in X.
Structural Commentary: The density alternative is membership in $E \cup E^{\text{acc}}$.
Key Insight: Closure packages the points of E together with all its limit points. □

Proposition 3.3.2. *Real-Line Density Interval Test*
Let $E \subseteq \mathbb{R}$. Then E is dense in $\mathbb{R}$ iff every nonempty open interval (a,b) satisfies $(a,b) \cap E \neq \emptyset$.

Proof.
($\Rightarrow$) Suppose E is dense in $\mathbb{R}$. Let $(a,b) \neq \emptyset$, choose $p \in (a,b)$, and set $r := \min\{p-a, b-p\} > 0$.
If $p \in E$, then $p \in (a,b) \cap E$. If $p \notin E$, then p is a limit point of E, so choose $q \in B_r(p) \cap (E \setminus \{p\})$.
Since $B_r(p) \subseteq (a,b)$, the point q belongs to $(a,b) \cap E$.
($\Leftarrow$) Suppose every nonempty open interval meets E, and fix $p \in \mathbb{R} \setminus E$.
For every $\varepsilon > 0$, choose $q \in (p-\varepsilon, p+\varepsilon) \cap E$. Then $q \neq p$ and $q \in B_\varepsilon(p) \cap (E \setminus \{p\})$.
Thus every $p \in \mathbb{R} \setminus E$ is a limit point of E, so E is dense in $\mathbb{R}$.
Structural Commentary: On $\mathbb{R}$, balls and open intervals test the same local condition.
Key Insight: Density means no interval can avoid the set. □

Definition 3.3.3. *Separable Metric Space*
A metric space (X,d) is *separable* iff it contains an at most countable dense subset.

Example 3.3.4. *Separable and Nonseparable Metric Spaces*
(i) Let $k \in \mathbb{Z}_{>0}$. The set $\mathbb{Q}^k$ is countable by Theorem 6.2.18 and Corollary 6.2.19.
For $\mathbf{x} \in \mathbb{R}^k$ and $\varepsilon > 0$, choose $\mathbf{q} \in \mathbb{Q}^k$ satisfying $(\forall\, j \in \{1,\dots,k\})\ |x_j - q_j| < \frac{\varepsilon}{\sqrt{k}}$ by Theorem 1.1.19(ii).
Then $\|\mathbf{x}-\mathbf{q}\|_2 = \sqrt{\sum_{j=1}^{k} |x_j - q_j|^2} < \varepsilon$, so $\mathbb{Q}^k$ is dense in $\mathbb{R}^k$. Hence $\mathbb{R}^k$ is separable; in particular, $\mathbb{R}$ is separable.
The standard isometry $\mathbb{C}^k \cong \mathbb{R}^{2k}$ identifies $(\mathbb{Q}+i\mathbb{Q})^k$ with $\mathbb{Q}^{2k}$, so $\mathbb{C}^k$ is separable.
(ii) If discrete X is at most countable, then X itself is a dense witness to separability.
Conversely, let $D \subseteq X$ be an at most countable dense subset. For every $x \in X$, the ball $B_{\frac{1}{2}}(x) = \{x\}$ meets D, so $x \in D$.
Thus $D = X$, and X is at most countable. Therefore X is separable iff X is at most countable.

Structural Commentary: Rational coordinate grids are dense in Euclidean spaces. Singleton balls force $D = X$ in discrete spaces.
Key Insight: Separability depends on whether countable data can meet every open ball.

Theorem 3.3.5. *Suprema Belong to Closures*
Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above and let $y := \sup E$.

(i) $y \in \overline{E}$.
(ii) If E is closed, then $y \in E$.

Proof.

(i) If $y \in E$, then $y \in \overline{E}$. Suppose instead that $y \notin E$.
Fix $h > 0$. If $y - h$ were an upper bound for E, then minimality of $y = \sup E$ would imply $y \leq y - h$.
This contradicts $h > 0$, so $y - h$ is not an upper bound for E.
Choose $x_h \in E$ with $y - h < x_h \leq y$. Then $x_h \neq y$, $|x_h - y| < h$, and $B_h(y) \cap (E \setminus \{y\}) \neq \emptyset$. Hence $y \in E^{\text{acc}} \subseteq \overline{E}$.
(ii) If E is closed, then $y \in \overline{E}$ by (i) and $\overline{E} = E$ by Theorem 3.1.22(ii). Hence $y \in E$.

Structural Commentary: Minimality of $y = \sup E$ produces points of E arbitrarily close below y.
Key Insight: A supremum is either attained or approached. □

3.4 Bounded Sets and Metric Sequences

Definition 3.4.1. *Bounded and Perfect Metric Sets*
Let (X, d) be a metric space and let $E \subseteq X$.

(i) The set E is *bounded* iff $(\exists q \in X)(\exists M > 0)\, E \subseteq B_M(q)$.
(ii) The set E is *perfect* iff E is closed and $(\forall p \in E)\, p \in E^{\text{acc}}$.

Definition 3.4.2. *Metric Convergence*
Let (p_n) be a sequence in a metric space (X, d). It *converges* to $p \in X$ iff $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\, d(p_n, p) < \varepsilon$.
Write $p_n \to p$ or $\lim_{n\to\infty} p_n = p$. The sequence *diverges* iff $(\nexists p \in X)\, p_n \to p$.

Definition 3.4.3. *Metric Sequence Terminology*
Let $p : \mathbb{N} \to X$, $n \mapsto p_n$, be a sequence in a metric space (X, d).

(i) The *range* of (p_n) is $\text{ran}(p_n) := \{p_n : n \in \mathbb{N}\}$.
(ii) The sequence (p_n) is *bounded* iff $\text{ran}(p_n)$ is bounded.
(iii) If $\sigma : \mathbb{N} \to \mathbb{N}$ satisfies $(\forall k \in \mathbb{N})\, \sigma(k+1) > \sigma(k)$, then $(p_{\sigma(k)})_{k\in\mathbb{N}}$ is a *subsequence* of (p_n).
(iv) A point $q \in X$ is a *subsequential limit* of (p_n) iff $(\exists\, \sigma : \mathbb{N} \to \mathbb{N})\, [(\forall k \in \mathbb{N})\, \sigma(k+1) > \sigma(k) \wedge p_{\sigma(k)} \to q]$.

Definition 3.4.4. *Eventual and Frequent Membership*
Let X be a set, let (p_n) be a sequence in X, and let $E \subseteq X$.

(i) The sequence (p_n) is *eventually in E* iff $(\exists N \in \mathbb{N})(\forall n \geq N)\, p_n \in E$.
(ii) The sequence (p_n) is *frequently in E* iff $(\forall N \in \mathbb{N})(\exists n \geq N)\, p_n \in E$.

Perspective 3.4.5. *Sequences as Metric Tests*
Definitions 3.4.2 and 3.4.4 separate tail control from arbitrarily late recurrence.
In metric spaces, sequences convert local ball conditions into countable tests; this viewpoint does not replace the ball definitions.

Theorem 3.4.6. *Basic Properties of Metric Convergence*
Let (p_n) be a sequence in a metric space X, let $E \subseteq X$, and let $p, p' \in X$.

(i) $p_n \to p \iff (\forall r > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\, p_n \in B_r(p)$.
(ii) $p_n \to p \wedge p_n \to p' \implies p = p'$.
(iii) $p \in E^{\text{acc}} \implies (\exists q : \mathbb{N} \to E \setminus \{p\})\, q_n \to p$.

Proof.

(i) ($\Rightarrow$) Let $r > 0$. Some $N \in \mathbb{N}$ satisfies $(\forall n \geq N)\, p_n \in B_r(p)$ by Definition 3.4.2 with $\varepsilon = r$.
(i) ($\Leftarrow$) Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $(\forall n \geq N)\, d(p_n, p) < \varepsilon$ by the ball condition at $r = \varepsilon$. Then $p_n \to p$ by Definition 3.4.2.
(ii) Suppose $p_n \to p$, $p_n \to p'$, and $p \neq p'$. Set $\delta := d(p, p') > 0$.
Choose $N, N' \in \mathbb{N}$ with $(\forall n \geq N)\, d(p_n, p) < \frac{\delta}{2}$ and $(\forall n \geq N')\, d(p_n, p') < \frac{\delta}{2}$.
For $n \geq \max\{N, N'\}$, $\delta \leq d(p, p_n) + d(p_n, p') < \delta$ by the triangle inequality, a contradiction.
(iii) Suppose $p \in E^{\text{acc}}$. Choose (q_n) with $(\forall n \in \mathbb{N})\, q_n \in (E \setminus \{p\}) \cap B_{\frac{1}{n}}(p)$.
Given $\varepsilon > 0$, choose $N \in \mathbb{N}$, $N\varepsilon > 1$ by Theorem 1.1.19(i). Then $(\forall n \geq N)\, d(q_n, p) < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$, so $q_n \to p$ by Definition 3.4.2.

Structural Commentary: Ball membership restates the metric bound; shrinking punctured balls construct the approximating sequence.
Key Insight: The triangle inequality forbids two distinct limits. □

Corollary 3.4.7. *Subsequences Preserve Metric Limits*
Let (p_{n_k}) be a subsequence of (p_n) in a metric space. If $p_n \to p$, then $p_{n_k} \to p$.

Proof. The bound $(\forall k \in \mathbb{N})\, n_k \geq k$ holds by induction from strict increase. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $(\forall n \geq N)\, d(p_n, p) < \varepsilon$.
Then $(\forall k \geq N)\, n_k \geq k \geq N$, so $d(p_{n_k}, p) < \varepsilon$ and $p_{n_k} \to p$ by Definition 3.4.2.
Structural Commentary: Strictly increasing indices place every subsequence tail inside the corresponding original tail.
Key Insight: Deleting terms preserves every eventual metric bound. □

Proposition 3.4.8. *Subsequential Limit Criterion*
Let (p_n) be a sequence in a metric space X and let $p \in X$. The following are equivalent:

(i) $(\forall r > 0)(\forall N \in \mathbb{N})(\exists n \geq N)\, p_n \in B_r(p)$.
(ii) $(\exists\, \sigma : \mathbb{N} \to \mathbb{N})\, [(\forall k \in \mathbb{N})\, \sigma(k+1) > \sigma(k) \wedge p_{\sigma(k)} \to p]$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Set $n_0 := 0$. Given n_{k-1}, let $I_k := \{n \geq n_{k-1}+1 : p_n \in B_{\frac{1}{k}}(p)\}$.
The set I_k is nonempty by (i); define $n_k := \min I_k$ by well-ordering. Thus $n_k > n_{k-1}$ and $d(p_{n_k}, p) < \frac{1}{k}$.
Given $\varepsilon > 0$, choose $K \in \mathbb{N}$, $K\varepsilon > 1$ by Theorem 1.1.19(i). Then $(\forall k \geq K)\, d(p_{n_k}, p) < \frac{1}{k} \leq \frac{1}{K} < \varepsilon$, so $p_{n_k} \to p$ by Definition 3.4.2.

(*ii*) $\Rightarrow$ (*i*) Write $n_k := \sigma(k)$. The bound $(\forall k \in \mathbb{N})\, n_k \geq k$ holds by induction from strict increase.
Given $r > 0$ and $N \in \mathbb{N}$, choose $K \in \mathbb{N}$ with $(\forall k \geq K)\, d(p_{n_k}, p) < r$. For $k \geq \max\{K, N\}$, $n_k \geq k \geq N$ and $p_{n_k} \in B_r(p)$.

Structural Commentary: Repeated late ball visits support increasing indices with errors bounded by k^{-1}.
Key Insight: Increasing indices turn recurrence at every scale into subsequential convergence. □

Proposition 3.4.9. *Discrete-Metric Convergence*
Let X carry the discrete metric, let (p_n) be a sequence in X, and let $p \in X$. Then $p_n \to p \iff (\exists N \in \mathbb{N})(\forall n \geq N)\, p_n = p$.

Proof.

($\Rightarrow$) Suppose $p_n \to p$. With $\varepsilon = \frac{1}{2}$, choose $N \in \mathbb{N}$ such that $(\forall n \geq N)\, d(p_n, p) < \frac{1}{2}$.
Since $d(X \times X) \subseteq \{0, 1\}$, $(\forall n \geq N)\, d(p_n, p) = 0$, so $(\forall n \geq N)\, p_n = p$.

($\Leftarrow$) Choose $N \in \mathbb{N}$ with $(\forall n \geq N)\, p_n = p$. Then $(\forall \varepsilon > 0)(\forall n \geq N)\, d(p_n, p) = 0 < \varepsilon$, so $p_n \to p$ by Definition 3.4.2.

Structural Commentary: The threshold $\frac{1}{2}$ converts the convergence tail into a zero-distance tail.
Key Insight: In the discrete metric, $d(x, y) < 1 \iff x = y$. □

Definition 3.4.10. *Equivalent Metrics*
Let X be a set and let $\rho, \sigma : X \times X \to [0, \infty)$ be metrics.
They are *equivalent*, written $\rho \sim \sigma$, iff $(\exists c_1, c_2 > 0)(\forall x, y \in X)\, c_1\sigma(x, y) \leq \rho(x, y) \leq c_2\sigma(x, y)$.

Proposition 3.4.11. *Equivalent Metrics Preserve Topological Tests*
Let X be a set, $\rho \sim \sigma$ be metrics on X, τ_ρ, τ_σ be their induced topologies. Let $E \subseteq X$, and (x_n) be a sequence in X with $x \in X$.

(*i*) $\tau_\rho = \tau_\sigma$.
(*ii*) *E is ρ-closed iff E is σ-closed.*
(*iii*) $\rho(x_n, x) \to 0$ *iff* $\sigma(x_n, x) \to 0$.

Proof. Choose $c_1, c_2 > 0$ with $(\forall u, v \in X)\, c_1\sigma(u, v) \leq \rho(u, v) \leq c_2\sigma(u, v)$. Then $(\forall z \in X)(\forall r > 0)\, B^\sigma_{\frac{r}{c_2}}(z) \subseteq B^\rho_r(z) \subseteq B^\sigma_{\frac{r}{c_1}}(z)$.

(*i*) If $U \in \tau_\rho$ and $z \in U$, choose $r > 0$ with $B^\rho_r(z) \subseteq U$. Then $B^\sigma_{\frac{r}{c_2}}(z) \subseteq U$, so $U \in \tau_\sigma$.
If $U \in \tau_\sigma$ and $z \in U$, choose $s > 0$ with $B^\sigma_s(z) \subseteq U$. Since $B^\rho_{c_1 s}(z) \subseteq B^\sigma_s(z)$, $U \in \tau_\rho$.

(*ii*) ($\Rightarrow$) If E is ρ-closed, then $E^c \in \tau_\rho = \tau_\sigma$ by (i). Thus E is σ-closed by Corollary 3.1.17.
(*ii*) ($\Leftarrow$) If E is σ-closed, then $E^c \in \tau_\sigma = \tau_\rho$ by (i). Thus E is ρ-closed by Corollary 3.1.17.

(*iii*) ($\Rightarrow$) Suppose $\rho(x_n, x) \to 0$. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $(\forall n \geq N)\, \rho(x_n, x) < c_1\varepsilon$.
Then $(\forall n \geq N)\, \sigma(x_n, x) \leq \frac{\rho(x_n, x)}{c_1} < \varepsilon$, so $\sigma(x_n, x) \to 0$.

(*iii*) ($\Leftarrow$) Suppose $\sigma(x_n, x) \to 0$. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $(\forall n \geq N)\, \sigma(x_n, x) < \frac{\varepsilon}{c_2}$.
Then $(\forall n \geq N)\, \rho(x_n, x) \leq c_2\sigma(x_n, x) < \varepsilon$, so $\rho(x_n, x) \to 0$.

Structural Commentary: Two-sided comparison nests metric balls, then transfers closedness, convergence through same constants.
Key Insight: Each metric ball contains a scaled ball for the other metric. □

Example 3.4.12. *Coordinate Norm Metrics on $\mathbb{R}^k$*
Let $k \in \mathbb{N}$ and set $d_j(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_j$ for $j \in \{1, 2, \infty\}$. Each $\|\cdot\|_j$ is a norm by Proposition 1.1.21.
Its norm axioms imply nonnegativity, definiteness, symmetry, and the triangle inequality for d_j; hence d_1, d_2, d_∞ are metrics on $\mathbb{R}^k$.
For $\mathbf{v} \in \mathbb{R}^k$, $\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1$. The bound $\|\mathbf{v}\|_1 \leq \sqrt{k}\,\|\mathbf{v}\|_2$ follows by Theorem 1.1.13(i), applied to $(|v_i|)_{i=1}^k$ and $(1)_{i=1}^k$.
Since $\|\mathbf{v}\|_2^2 \leq k\,\|\mathbf{v}\|_\infty^2$, $\|\mathbf{v}\|_2 \leq \sqrt{k}\,\|\mathbf{v}\|_\infty$. Thus d_1, d_2, d_∞ are pairwise equivalent by Definition 3.4.10.
Their convergent sequences coincide by Proposition 3.4.11(iii).

Theorem 3.4.13. *Coordinatewise Convergence in $\mathbb{R}^k$*
Let $k \in \mathbb{N}$, and let $\mathbf{x}_n = (\alpha_{j,n})_{j=1}^k$ and $\mathbf{y}_n = (\gamma_{j,n})_{j=1}^k$ be sequences in $\mathbb{R}^k$.
Let $\mathbf{x} = (\alpha_j)_{j=1}^k$ and $\mathbf{y} = (\gamma_j)_{j=1}^k$ lie in $\mathbb{R}^k$. Let (β_n) be a sequence in $\mathbb{R}$ and let $\beta \in \mathbb{R}$.

(*i*) $\mathbf{x}_n \to \mathbf{x} \iff (\forall j \in \{1, \ldots, k\})\, \alpha_{j,n} \to \alpha_j$.
(*ii*) $\mathbf{x}_n \to \mathbf{x} \wedge \mathbf{y}_n \to \mathbf{y} \Longrightarrow \mathbf{x}_n + \mathbf{y}_n \to \mathbf{x} + \mathbf{y}$.
(*iii*) $\mathbf{x}_n \to \mathbf{x} \wedge \mathbf{y}_n \to \mathbf{y} \Longrightarrow \mathbf{x}_n \cdot \mathbf{y}_n \to \mathbf{x} \cdot \mathbf{y}$.
(*iv*) $\beta_n \to \beta \wedge \mathbf{x}_n \to \mathbf{x} \Longrightarrow \beta_n\mathbf{x}_n \to \beta\mathbf{x}$.

Proof.

(*i*) ($\Rightarrow$) Fix $j \in \{1, \ldots, k\}$ and let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $(\forall n \geq N)\, \|\mathbf{x}_n - \mathbf{x}\|_2 < \varepsilon$.
Then $(\forall n \geq N)\, |\alpha_{j,n} - \alpha_j| \leq \|\mathbf{x}_n - \mathbf{x}\|_2 < \varepsilon$, so $\alpha_{j,n} \to \alpha_j$ by Definition 3.4.2.

(*i*) ($\Leftarrow$) Given $\varepsilon > 0$, choose $N_j \in \mathbb{N}$ with $(\forall n \geq N_j)\, |\alpha_{j,n} - \alpha_j| < \frac{\varepsilon}{\sqrt{k}}$ for $j \in \{1, \ldots, k\}$.
For $N := \max_{1 \leq j \leq k} N_j$ and $n \geq N$, $\|\mathbf{x}_n - \mathbf{x}\|_2^2 < \sum_{j=1}^k \frac{\varepsilon^2}{k} = \varepsilon^2$, so $\mathbf{x}_n \to \mathbf{x}$.

(*ii*) The coordinate limits $(\forall j \in \{1, \ldots, k\})\, \alpha_{j,n} + \gamma_{j,n} \to \alpha_j + \gamma_j$ hold by Theorem 2.1.5(i). The vector convergence follows by (i).

(*iii*) The product limits $(\forall j \in \{1, \ldots, k\})\, \alpha_{j,n}\gamma_{j,n} \to \alpha_j\gamma_j$ hold by Theorem 2.1.5(iii).
The convergence $\sum_{j=1}^k \alpha_{j,n}\gamma_{j,n} \to \sum_{j=1}^k \alpha_j\gamma_j$ follows by repeated application of Theorem 2.1.5(i).

(*iv*) The coordinate limits $(\forall j \in \{1, \ldots, k\})\, \beta_n\alpha_{j,n} \to \beta\alpha_j$ hold by Theorem 2.1.5(iii). The vector convergence follows by (i).

Structural Commentary: Coordinate bounds prove (i); scalar limit algebra transfers sums, products, and scalar multiples.
Key Insight: A finite maximum synchronizes all coordinate tails. □

Definition 3.4.14. *Product Distance Functions*
Let (X, d_X) and (Y, d_Y) be metric spaces. For $p = (x, y)$ and $p' = (x', y')$ in $X \times Y$, define:

(i) $d_E(p, p') := \sqrt{d_X(x, x')^2 + d_Y(y, y')^2}$;
(ii) $d_{\max}(p, p') := \max\{d_X(x, x'), d_Y(y, y')\}$;
(iii) $d_{\text{sum}}(p, p') := d_X(x, x') + d_Y(y, y')$.

These are the Euclidean, maximum, and sum product distance functions.

Figure 126 contrasts the maximum and sum product distance functions on $\mathbb{R}^2$.

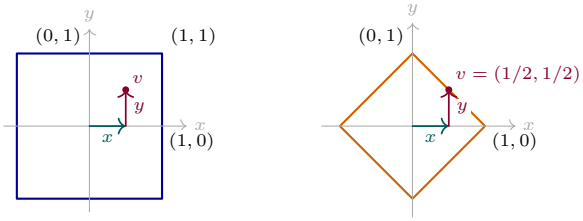

Figure 126: The maximum-metric unit ball is square; the sum-metric unit ball is diamond-shaped.

Proposition 3.4.15. *Product Metrics and Comparisons*
Let (X, d_X) and (Y, d_Y) be metric spaces. The product distance functions $d_E, d_{\max}, d_{\text{sum}}$ satisfy:

(i) $d_E, d_{\max}, d_{\text{sum}}$ are metrics on $X \times Y$;
(ii) $d_{\max} \le d_E \le d_{\text{sum}} \le 2d_{\max}$;
(iii) they are pairwise equivalent and have the same convergent sequences.

Proof.

(i) Let $p = (x, y)$, $p' = (x', y')$, and $p'' = (x'', y'')$, and set $(a, b, c, e) := (d_X(x, x'), d_Y(y, y'), d_X(x', x''), d_Y(y', y''))$.
All three functions are nonnegative and symmetric, and each vanishes iff $a = b = 0$, equivalently $p = p'$.
By the component triangle inequalities, $d_{\text{sum}}(p, p'') \le a + b + c + e = d_{\text{sum}}(p, p') + d_{\text{sum}}(p', p'')$.
Likewise, $d_{\max}(p, p'') \le \max\{a + c, b + e\} \le \max\{a, b\} + \max\{c, e\}$ by the component triangle inequalities.
By Theorem 1.1.14(i), $d_E(p, p'') \le \sqrt{(a + c)^2 + (b + e)^2} \le \sqrt{a^2 + b^2} + \sqrt{c^2 + e^2}$.
(ii) For $p = (x, y)$ and $p' = (x', y')$, set $a := d_X(x, x')$ and $b := d_Y(y, y')$. Then $\max\{a, b\} \le \sqrt{a^2 + b^2} \le a + b \le 2\max\{a, b\}$.
(iii) The metrics are pairwise equivalent by (ii) and Definition 3.4.10. Their convergent sequences coincide by Proposition 3.4.11(iii).

Structural Commentary: Component triangle inequalities prove the metric axioms; uniform comparison transfers convergence.
Key Insight: Maximum and sum bounds trap the Euclidean product distance. □

Proposition 3.4.16. *Sequential Closure Criterion*
Let X be a metric space, let $E \subseteq X$, and let $p \in X$. Then $p \in \overline{E} \iff (\exists u : \mathbb{N} \to E)\, u_n \to p$.

Proof.

($\Rightarrow$) Suppose $p \in \overline{E}$. By Definition 3.1.23, $(\forall n \in \mathbb{N})\, E \cap B_{\frac{1}{n}}(p) \neq \emptyset$. Choose (p_n) with $(\forall n \in \mathbb{N})\, p_n \in E \cap B_{\frac{1}{n}}(p)$.
Given $\varepsilon > 0$, choose $N \in \mathbb{N}$, $N\varepsilon > 1$ by Theorem 1.1.19(i). Then $(\forall n \ge N)\, d(p_n, p) < \frac{1}{n} \le \frac{1}{N} < \varepsilon$, so $p_n \to p$ by Definition 3.4.2.
($\Leftarrow$) Suppose $(\forall n \in \mathbb{N})\, p_n \in E$ and $p_n \to p$. Given $r > 0$, choose $N \in \mathbb{N}$ with $d(p_N, p) < r$.
Then $p_N \in E \cap B_r(p)$, so $p \in \overline{E}$ by Definition 3.1.23.

Structural Commentary: Adherence selects points from shrinking balls; convergence returns a witness to each ball.
Key Insight: The radii n^{-1} encode all positive metric scales. □

Theorem 3.4.17. *Closed Sets and Limit Points via Sequences*
Let X be a metric space and $E \subseteq X$.

(i) $(\forall p \in X)\,[p \in E^{\text{acc}} \iff (\exists u : \mathbb{N} \to E \setminus \{p\})\, u_n \to p]$.
(ii) E is closed iff $(\forall p \in X)(\forall u : \mathbb{N} \to E)\,[u_n \to p \implies p \in E]$.

Proof.

(i) ($\Rightarrow$) Fix $p \in E^{\text{acc}}$. A sequence (p_n) in $E \setminus \{p\}$ with $p_n \to p$ exists by Theorem 3.4.6(iii).
(i) ($\Leftarrow$) Fix $p \in X$ and suppose $(\forall n \in \mathbb{N})\, p_n \in E \setminus \{p\}$ with $p_n \to p$.
Given $r > 0$, choose $N \in \mathbb{N}$ with $p_N \in B_r(p)$. Then $p_N \in B_r(p) \cap (E \setminus \{p\})$, so $p \in E^{\text{acc}}$.
(ii) ($\Rightarrow$) Suppose E closed, $(\forall n \in \mathbb{N})\, p_n \in E$, $p_n \to p \in X$; $p \in \overline{E}$ by Proposition 3.4.16. Since $\overline{E} = E$ by Theorem 3.1.22(ii), $p \in E$.
(ii) ($\Leftarrow$) Assume the sequential limit condition, and fix $p \in \overline{E}$.
By Proposition 3.4.16, some sequence (p_n) in E converges to p, so $p \in E$ by the assumed condition.
Thus $\overline{E} \subseteq E$, while $E \subseteq \overline{E}$ by Definition 3.1.21. Therefore $E = \overline{E}$, and E is closed by Theorem 3.1.22(ii).

Structural Commentary: Punctured-ball sequences characterize accumulation; ordinary ball sequences characterize closure.
Key Insight: Closedness is exactly retention of limits of internal sequences. □

Theorem 3.4.18. *Subsequential Limits Form a Closed Set*
For a sequence (p_n) in a metric space X, the set of all subsequential limits is closed.

Proof.

(i) *Approximation*: Let E^* be set of subsequential limits, $q \in (E^*)^{\text{acc}}$. Choose (x_i) with $(\forall i \in \mathbb{N})\, x_i \in (E^* \setminus \{q\}) \cap B_{\frac{1}{2i}}(q)$.
(ii) *Diagonal selection*: Each x_i satisfies condition (i) by Proposition 3.4.8, direction $(ii) \Rightarrow (i)$.
Set $n_0 := 0$ and recursively choose $n_i > n_{i-1}$ with $d(p_{n_i}, x_i) < \frac{1}{2i}$ by condition (i).

(iii) *Closure*: By the triangle inequality, $(\forall i \in \mathbb{N})\, d(p_{n_i}, q) < \frac{1}{i}$. Given $\varepsilon > 0$, choose $I \in \mathbb{N}$ with $I\varepsilon > 1$ by Theorem 1.1.19(i).
Then $(\forall i \geq I)\, d(p_{n_i}, q) < \frac{1}{i} \leq \frac{1}{I} < \varepsilon$, so $p_{n_i} \to q$ by Definition 3.4.2. Thus $q \in E^*$; $(E^*)^{\mathrm{acc}} \subseteq E^*$, E^* closed by Definition 3.1.2.

Structural Commentary: Shrinking approximants and increasing original indices combine into one diagonal subsequence.
Key Insight: Each diagonal term is simultaneously later and closer to q. □

Proof Method 3.4.19. *Metric Proof Patterns*

(i) To prove openness of G, fix $x \in G$ and construct $r > 0$ with $B_r(x) \subseteq G$; then G is open by Definition 3.1.2.
(ii) To prove closedness of F, establish $F^{\mathrm{acc}} \subseteq F$ and conclude by Definition 3.1.2.
Alternatively, prove F^c open and conclude by Corollary 3.1.17.
(iii) To identify $C = \overline{E}$, verify $E \subseteq C$, $C^{\mathrm{acc}} \subseteq C$, and $(\forall F \subseteq X)\,[(E \subseteq F \wedge F^{\mathrm{acc}} \subseteq F) \Longrightarrow C \subseteq F]$; $C = \overline{E}$ by Theorem 3.1.22.
(iv) For $\emptyset \neq E \subseteq X$, prove $\operatorname{dist}(p, E) = 0$ and conclude $p \in \overline{E}$ by Proposition 3.1.26.
Alternatively, construct (p_n) in E with $p_n \to p$ and conclude by Proposition 3.4.16.

4 Compactness and Completeness

4.1 Compactness, Completeness, and Total Boundedness

Compactness extracts finite or convergent structure from infinite behavior;
completeness asks whether Cauchy approximations converge internally.

Definition 4.1.1. *Open Covers and Compact Sets*
Let (X, d) be a metric space.

(i) A family $(G_\alpha)_{\alpha \in A}$ of open subsets of X is an *open cover* of $E \subseteq X$ iff $E \subseteq \bigcup_{\alpha \in A} G_\alpha$.

(ii) A set $K \subseteq X$ is *compact* iff every open cover of K contains a finite subcover.

Definition 4.1.2. *Finite Intersection Property*
Let X be set, $\mathcal{F} \subseteq \mathcal{P}(X)$. Family $\mathcal{F}$ has *finite intersection property (FIP)* iff $(\forall\, \mathcal{F}_0 \subseteq \mathcal{F})\,[0 < |\mathcal{F}_0| < \infty \Rightarrow \bigcap_{F \in \mathcal{F}_0} F \neq \emptyset]$.

Proposition 4.1.3. *Closed-Set Compactness Test*
A metric space (X, d) is compact iff every nonempty family $\mathcal{F}$ of closed subsets of X with FIP satisfies $\bigcap_{F \in \mathcal{F}} F \neq \emptyset$.

Proof.

($\Rightarrow$) Assume X is compact, and let $\emptyset \neq \mathcal{F} \subseteq \mathcal{P}(X)$ be a family of closed subsets with the finite intersection property.
If $X = \emptyset$, then $\mathcal{F} = \{\emptyset\}$, contrary to its finite intersection property. Suppose $X \neq \emptyset$ and $\bigcap_{F \in \mathcal{F}} F = \emptyset$.
The family $\{X \setminus F : F \in \mathcal{F}\}$ is an open cover of X.
By compactness, choose $n \in \mathbb{N}$ and $F_1, \ldots, F_n \in \mathcal{F}$ such that $X = \bigcup_{j=1}^{n} (X \setminus F_j)$.
Since $X \neq \emptyset$, $n \geq 1$, whereas $\bigcap_{j=1}^{n} F_j = \emptyset$. This contradicts the finite intersection property.

($\Leftarrow$) Assume the closed-family condition, and let $(G_\alpha)_{\alpha \in A}$ be an open cover of X.
If $X = \emptyset$, the empty subfamily covers X. Suppose $X \neq \emptyset$ and no finite subcover exists.
Then $A \neq \emptyset$, and $\mathcal{F} := \{X \setminus G_\alpha : \alpha \in A\}$ is a nonempty family of closed subsets. Let $\emptyset \neq \mathcal{F}_0 \subseteq \mathcal{F}$ be finite.
For $C \in \mathcal{F}_0$, choose $\alpha_C \in A$ with $C = X \setminus G_{\alpha_C}$. Set $A_0 := \{\alpha_C : C \in \mathcal{F}_0\}$.
The finite set A_0 does not index a cover of X, so $\bigcap_{C \in \mathcal{F}_0} C \neq \emptyset$. Thus $\mathcal{F}$ has the finite intersection property.
By the closed-family condition, $\bigcap_{\alpha \in A} (X \setminus G_\alpha) \neq \emptyset$, contrary to $X \subseteq \bigcup_{\alpha \in A} G_\alpha$.

Structural Commentary: Taking complements exchanges finite subcovers with finite intersections.
Key Insight: An empty total intersection is already witnessed finitely. □

Figure 127 depicts the finite extraction required by compactness.

Figure 127: An arbitrary dashed cover of K (left) is reduced to a finite solid subcover (right).

Counterexample 4.1.4. *An Open Cover Without a Finite Subcover*
For $x \in (0, 1)$, set $O_x := (\frac{x}{2}, 1)$. The family $(O_x)_{x \in (0,1)}$ covers $(0, 1)$ because $(\forall\, y \in (0, 1))\, y \in O_y$.
The empty subfamily does not cover $(0, 1)$. Given $x_1, \ldots, x_n \in (0, 1)$ with $n \geq 1$, put $x' := \min_{1 \leq i \leq n} x_i$.
Then $(\forall\, i \in \{1, \ldots, n\})\, O_{x_i} \subseteq O_{x'}$, while $(0, \frac{x'}{2}] \cap O_{x'} = \emptyset$. Hence no finite subfamily covers $(0, 1)$.

Figure 128 shows why the preceding cover of $(0, 1)$ has no finite subcover.

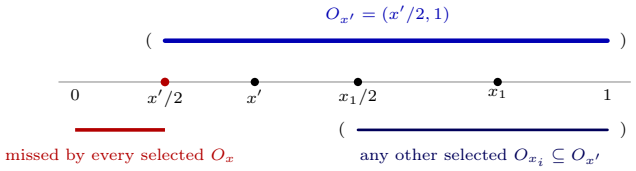

Figure 128: For $x' = \min_i x_i$, $(\forall\, i \in \{1, \dots, n\})\, O_{x_i} \subseteq O_{x'}$, while $(0, \frac{x'}{2}]$ remains uncovered.

Theorem 4.1.5. *Compactness Is Independent of the Ambient Space*
Let (X, d) be a metric space, and suppose $K \subseteq Y \subseteq X$. Then K is compact in X iff it is compact in the subspace $(Y, d|_{Y\times Y})$.

Proof.
($\Rightarrow$) Let $(V_\alpha)_{\alpha\in A}$ be a subspace-open cover of K in Y. Set $\mathcal{G} := \{G \in \tau_d : (\exists\, \alpha \in A)\, Y \cap G = V_\alpha\}$.
The family $\mathcal{G}$ covers K by Theorem 3.2.3. By compactness in X, choose a finite $\mathcal{G}_0 \subseteq \mathcal{G}$ that covers K.
For $G \in \mathcal{G}_0$, choose $\alpha(G) \in A$ with $Y \cap G = V_{\alpha(G)}$. Since $K \subseteq Y$, the finite family $(V_{\alpha(G)})_{G\in\mathcal{G}_0}$ covers K.
($\Leftarrow$) Let $(G_\alpha)_{\alpha\in A}$ be an open cover of K in X. The family $(Y \cap G_\alpha)_{\alpha\in A}$ is a subspace-open cover of K by Theorem 3.2.3.
By compactness in Y, some finite $F \subseteq A$ satisfies $K \subseteq \bigcup_{\alpha\in F} (Y \cap G_\alpha) \subseteq \bigcup_{\alpha\in F} G_\alpha$.

Structural Commentary: Intersecting ambient opens with Y transfers covers in both directions.
Key Insight: Finite subcovers survive restriction to the subspace. □

Theorem 4.1.6. *Compact Sets Are Closed*
Let (X, d) be a metric space. Every compact set $K \subseteq X$ is closed in X.

Proof.
(i) *Pointwise separation*: The claim holds for $K = \emptyset$ by Theorem 3.1.19(iv). Suppose $K \neq \emptyset$, and fix $p \in X \setminus K$.
For $q \in K$, set $r_q := \frac{1}{2}d(p, q) > 0$, $V_q := B_{r_q}(p)$, and $W_q := B_{r_q}(q)$.
If $x \in V_q \cap W_q$, then $d(p, q) \leq d(p, x) + d(x, q) < 2r_q = d(p, q)$, a contradiction. Thus $V_q \cap W_q = \emptyset$.
(ii) *Finite consolidation*: The family $(W_q)_{q\in K}$ covers K.
By compactness, choose $n \in \mathbb{N}$ and $q_1, \dots, q_n \in K$ such that $K \subseteq \bigcup_{i=1}^{n} W_{q_i}$.
The set $V := \bigcap_{i=1}^{n} V_{q_i}$ is open by Theorem 3.1.19(iii), contains p, and is disjoint from K.
(iii) *Complement*: Thus $(\forall\, p \in X \setminus K)(\exists\, V \in \tau_d)\, p \in V \subseteq X \setminus K$. Hence $X \setminus K$ is open, so K is closed.
Structural Commentary: A finite cover converts pointwise disjoint balls into one complement neighborhood.
Key Insight: Half-distance balls around p and q cannot intersect. □

Theorem 4.1.7. *Closed Subsets of Compact Sets Are Compact*
Let (X, d) be a metric space, and suppose $F \subseteq K \subseteq X$. If F is closed in X and K is compact, then F is compact.

Proof. Let $(V_\alpha)_{\alpha\in A}$ be an open cover of F. Since $X \setminus F$ is open, $(V_\alpha)_{\alpha\in A} \cup \{X \setminus F\}$ covers K.
By compactness of K, some finite $F_0 \subseteq A$ satisfies $K \subseteq (X \setminus F) \cup \bigcup_{\alpha\in F_0} V_\alpha$. Therefore $F \subseteq \bigcup_{\alpha\in F_0} V_\alpha$.
Structural Commentary: The open complement extends a cover of F to one of K.
Key Insight: Discarding the complement leaves a finite cover of F. □

Corollary 4.1.8. *Closed Intersections with Compact Sets*
Let (X, d) be a metric space. If $F \subseteq X$ is closed and $K \subseteq X$ is compact, then $F \cap K$ is compact.

Proof. The set K is closed by Theorem 4.1.6. Thus $F \cap K$ is closed by Theorem 3.1.19(ii).
The set $F \cap K$ is compact by Theorem 4.1.7 with $F \cap K \subseteq K$.
Structural Commentary: Closed-set calculus makes $F \cap K$ closed; the closed-subset theorem transfers compactness.
Key Insight: The intersection remains a closed subset of compact K. □

Theorem 4.1.9. *Finite Intersection Property for Compact Sets*
Let (X, d) be a metric space, $(K_\alpha)_{\alpha\in A}$ be compact subsets of X, $A \neq \emptyset$. If $(\forall\, F \subseteq A)\, [0 < |F| < \infty \Rightarrow \bigcap_{\alpha\in F} K_\alpha \neq \emptyset]$, then $\bigcap_{\alpha\in A} K_\alpha \neq \emptyset$.

Proof. Choose $\alpha_0 \in A$ and suppose $\bigcap_{\alpha\in A} K_\alpha = \emptyset$. Then $K_{\alpha_0} \subseteq \bigcup_{\alpha\in A} K_\alpha^c$. Each K_α^c is open by Theorem 4.1.6.
By compactness of K_{α_0}, some finite $F \subseteq A$ satisfies $K_{\alpha_0} \subseteq \bigcup_{\alpha\in F} K_\alpha^c$. Hence $\bigcap_{\alpha\in F\cup\{\alpha_0\}} K_\alpha = \emptyset$, a contradiction.
Structural Commentary: Complements turn failure into a finite cover.
Key Insight: For compact sets, failure of intersection is already witnessed finitely. □

Corollary 4.1.10. *Nested Nonempty Compact Sets Intersect*
Let (X, d) be a metric space. If $K_1 \supseteq K_2 \supseteq \cdots$ are nonempty compact subsets of X, then $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

Proof. Let $\emptyset \neq F \subseteq \mathbb{N}$ be finite, and put $m := \max_{n\in F} n$. Then $\bigcap_{n\in F} K_n = K_m \neq \emptyset$ by nestedness.
Thus $(K_n)_{n\in\mathbb{N}}$ has the finite intersection property. Hence $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$ by Theorem 4.1.9.
Structural Commentary: Nestedness reduces finite intersections to one term.
Key Insight: A nested compact sequence has the finite intersection property. □

Theorem 4.1.11. *Infinite Subsets of Compact Sets Have Limit Points*
Let (X,d) be a metric space, let $K \subseteq X$ be compact, and let $E \subseteq K$ be infinite. Then $E^{\mathrm{acc}} \cap K \neq \emptyset$.

Proof. Suppose $E^{\mathrm{acc}} \cap K = \emptyset$, and set $\mathcal{U} := \{B_r(q) : q \in K,\ r > 0,\ B_r(q) \cap (E \setminus \{q\}) = \emptyset\}$.
The family $\mathcal{U}$ covers K because no $q \in K$ is an accumulation point of E.
By compactness, choose $n \in \mathbb{N}$ and $B_{r_i}(q_i) \in \mathcal{U}$, $1 \le i \le n$, such that $K \subseteq \bigcup_{i=1}^{n} B_{r_i}(q_i)$.
Each selected ball meets E in at most one point, so $|E| \le n$, contrary to the infinitude of E.
Structural Commentary: Compactness reduces locally isolated intersections with E to finitely many balls.
Key Insight: Finitely many one-point intersections cannot cover infinite E. □

Definition 4.1.12. *Sequential Compactness*
A metric space (X,d) is *sequentially compact* iff every sequence in X has a subsequence converging in X.

Theorem 4.1.13. *Compactness Produces Convergent Subsequences*
If (p_n) is a sequence in a compact metric space (X,d), then some subsequence (p_{n_i}) converges to a point $p \in X$.

Proof. Let $E := \{p_n : n \in \mathbb{N}\}$. If E is finite and $(\forall\, q \in E)\,|\{n \in \mathbb{N} : p_n = q\}| < \infty$, then $\mathbb{N}$ is finite, a contradiction.
Thus some $p \in E$ occurs infinitely often when E is finite, and $(\forall\, r > 0)(\forall\, N \in \mathbb{N})(\exists\, n \ge N)\, p_n \in B_r(p)$.
Suppose E is infinite. Some $p \in E^{\mathrm{acc}} \cap X$ exists by Theorem 4.1.11 with $K = X$.
Fix $r > 0$ and $N \in \mathbb{N}$. The set $B_r(p) \cap E$ is infinite by Theorem 3.1.14, whereas $\{p_1, \dots, p_{N-1}\}$ is finite.
Therefore $(\exists\, n \ge N)\, p_n \in B_r(p)$. In either case, a subsequence converges to p by Proposition 3.4.8, $(i) \Rightarrow (ii)$.
Structural Commentary: Finite ranges repeat; infinite compact ranges contain an accumulation point.
Key Insight: Repetition and accumulation both create a convergent subsequence. □

Definition 4.1.14. *Cauchy Sequence*
A sequence (p_n) in a metric space (X,d) is *Cauchy* if $(\forall\, \varepsilon > 0)(\exists\, N \in \mathbb{N})(\forall\, m,n \ge N)\, d(p_m,p_n) < \varepsilon$.

Definition 4.1.15. *Diameter*
Let (X,d) be a metric space and $\emptyset \neq E \subseteq X$. The *diameter* of E is $\operatorname{diam} E := \sup_{\overline{\mathbb{R}}}\{d(p,q) : p,q \in E\} \in [0,+\infty]$.

Proposition 4.1.16. *Basic Diameter Properties*
Let (X,d) be a metric space and let $\emptyset \neq E, F \subseteq X$.
(i) E is bounded iff $\operatorname{diam} E < \infty$.
(ii) $\operatorname{diam} E = 0 \iff |E| = 1$.
(iii) $E \subseteq F \implies \operatorname{diam} E \le \operatorname{diam} F$.
(iv) E, F bounded $\implies E \cup F$ bounded.

Proof.
(i) ($\Rightarrow$) If E is bounded, choose $q \in X$ and $M > 0$ such that $E \subseteq B_M(q)$.
Then $(\forall\, x,y \in E)\, d(x,y) < 2M$ by the triangle inequality, so $\operatorname{diam} E \le 2M < \infty$.

(i) ($\Leftarrow$) If $\operatorname{diam} E < \infty$ and $q \in E$, then $E \subseteq B_{\operatorname{diam} E+1}(q)$, so E is bounded.
(ii) ($\Rightarrow$) If $\operatorname{diam} E = 0$, then $(\forall\, x,y \in E)\, d(x,y) = 0$. Thus all points of the nonempty set E are equal, and $|E| = 1$.

(ii) ($\Leftarrow$) If $|E| = 1$, then $(\forall\, x,y \in E)\, d(x,y) = 0$, so $\operatorname{diam} E = 0$.
(iii) Since $\{d(x,y) : x,y \in E\} \subseteq \{d(x,y) : x,y \in F\}$, $\operatorname{diam} E \le \operatorname{diam} F$ by monotonicity of the supremum.
(iv) Choose $e \in E$, $f \in F$, and set $D := \operatorname{diam} E + d(e,f) + \operatorname{diam} F < \infty$.
If x,y lie in the same one of E, F, then $d(x,y) \le D$. Otherwise, after swapping x,y, assume $x \in E$ and $y \in F$.
Then $d(x,y) \le d(x,e) + d(e,f) + d(f,y) \le D$. Thus $\operatorname{diam}(E \cup F) \le D < \infty$, so $E \cup F$ is bounded by (i).
Structural Commentary: Ball bounds control diameters; bridge points control cross-distances in unions.
Key Insight: One bridge distance joins two bounded sets. □

Proposition 4.1.17. *Cauchy Sequences and Tail Diameters*
For (p_n) in a metric space X, set $E_N := \{p_j : j \ge N\}$. Then (p_n) is Cauchy if and only if $\operatorname{diam} E_N \to 0$.

Proof.
($\Rightarrow$) Given $\varepsilon > 0$, choose N_0 so that $m,n \ge N_0$ implies $d(p_m,p_n) < \frac{\varepsilon}{2}$.
If $N \ge N_0$, then $(\forall\, m,n \ge N)\, d(p_m,p_n) < \frac{\varepsilon}{2}$, so $\operatorname{diam} E_N \le \frac{\varepsilon}{2} < \varepsilon$.
($\Leftarrow$) Given $\varepsilon > 0$, choose N_0 with $\operatorname{diam} E_{N_0} < \varepsilon$. If $m,n \ge N_0$, then $p_m, p_n \in E_{N_0}$, hence $d(p_m,p_n) \le \operatorname{diam} E_{N_0} < \varepsilon$.
Structural Commentary: The Cauchy quantifiers match convergence of the decreasing tail diameters.
Key Insight: Tail diameter measures mutual closeness without naming a limit. □

Definition 4.1.18. *Contracting Sequence of Sets*
Let (X,d) be a metric space and let $(F_n)_{n\ge1}$ satisfy $\emptyset \neq F_n \subseteq X$.
(i) The sequence (F_n) is *descending* iff $(\forall\, n \ge 1)\, F_{n+1} \subseteq F_n$.
(ii) The sequence (F_n) is *contracting* iff it is descending and $\operatorname{diam} F_n \to 0$.

Figure 129 visualizes the tail-diameter criterion for Cauchy sequences.

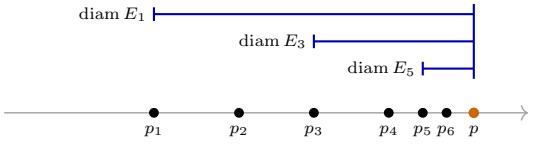

Figure 129: Tail diameters $\operatorname{diam} E_N$ shrink toward 0.

Theorem 4.1.19. *Nested Compact Sets with Shrinking Diameter*
Let (X,d) be a metric space.

(i) If $\emptyset \neq E \subseteq X$, then $\operatorname{diam}\overline{E} = \operatorname{diam} E$.

(ii) If $K_1 \supseteq K_2 \supseteq \cdots$ are nonempty compact subsets of X and $\operatorname{diam} K_n \to 0$, then $\left|\bigcap_{n=1}^{\infty} K_n\right| = 1$.

Proof.

(i) Since $E \subseteq \overline{E}$, $\operatorname{diam} E \leq \operatorname{diam}\overline{E}$ by Proposition 4.1.16(iii). If $\operatorname{diam} E = +\infty$, equality follows.
Suppose $\operatorname{diam} E < \infty$. For $p,q \in \overline{E}$ and $\varepsilon > 0$, choose $p',q' \in E$ with $d(p,p'), d(q,q') < \frac{\varepsilon}{2}$.
Then $d(p,q) < \operatorname{diam} E + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $d(p,q) \leq \operatorname{diam} E$; hence $\operatorname{diam}\overline{E} \leq \operatorname{diam} E$.

(ii) Set $I := \bigcap_{n=1}^{\infty} K_n$. Then $I \neq \emptyset$ by Corollary 4.1.10.
For $x,y \in I$, $(\forall\, n \in \mathbb{N})\, d(x,y) \leq \operatorname{diam} K_n$, so $d(x,y) = 0$ by $\operatorname{diam} K_n \to 0$. Thus $x = y$, and $|I| = 1$.

Structural Commentary: Closure preserves diameter; nested compactness and shrinking diameters split existence from uniqueness.
Key Insight: Every two intersection points have distance zero. □

Theorem 4.1.20. *Cauchy Criterion and Completeness Results*

(i) Every convergent sequence in a metric space is Cauchy.
(ii) Every Cauchy sequence in a compact metric space X converges in X.

Proof.

(i) If $p_n \to p$, then for $\varepsilon > 0$ choose N with $d(p_n,p) < \frac{\varepsilon}{2}$ for $n \geq N$. For $m,n \geq N$, $d(p_m,p_n) \leq d(p_m,p) + d(p,p_n) < \varepsilon$.

(ii) Let (p_n) be Cauchy in compact X, and set $E_N := \{p_j : j \geq N\}$ and $F_N := \overline{E_N}$. Each F_N is nonempty.
Since $E_{N+1} \subseteq E_N \subseteq F_N$, one has $F_{N+1} \subseteq F_N$ by Theorem 3.1.22(iii).
Each F_N is closed by Theorem 3.1.22(i), hence compact by Theorem 4.1.7.
The convergence $\operatorname{diam} F_N = \operatorname{diam} E_N \to 0$ holds by Proposition 4.1.17 and Theorem 4.1.19(i).
The relation $(\exists\, p \in X) \bigcap_{N=1}^{\infty} F_N = \{p\}$ holds by Theorem 4.1.19(ii).
Given $\varepsilon > 0$, choose N with $\operatorname{diam} F_N < \varepsilon$. For $n \geq N$, both $p_n, p \in F_N$, so $d(p_n,p) < \varepsilon$.

Structural Commentary: Cauchy tails create a contracting nest whose compact intersection supplies the limit.
Key Insight: Shrinking closed tails trap every late term near one point. □

Lemma 4.1.21. *A Cauchy Sequence with a Convergent Subsequence Converges*
Let (p_n) be a Cauchy sequence in a metric space X. If some subsequence (p_{n_j}) converges to $p \in X$, then $p_n \to p$.

Proof. Given $\varepsilon > 0$, choose N_0 so that $m,n \geq N_0$ implies $d(p_m,p_n) < \frac{\varepsilon}{2}$.
Choose $j \geq N_0$ with $d(p_{n_j},p) < \frac{\varepsilon}{2}$; then $n_j \geq j \geq N_0$. For $n \geq N_0$, $d(p_n,p) \leq d(p_n,p_{n_j}) + d(p_{n_j},p) < \varepsilon$.
Structural Commentary: A late subsequence term links the Cauchy tail to the subsequential limit.
Key Insight: One convergent thread fixes the destination of the entire tail. □

Proposition 4.1.22. *Boundedness and Distances of Cauchy Sequences*
Let (X,d) be a metric space.

(i) If (p_n) is Cauchy, then $\operatorname{ran}(p_n)$ is bounded.
(ii) If (p_n) and (q_n) are Cauchy, then $(d(p_n,q_n))$ converges in $\mathbb{R}$.
(iii) A bounded real sequence need not be Cauchy.

Proof.

(i) Choose $N \in \mathbb{N}$ with $m,n \geq N \Longrightarrow d(p_m,p_n) < 1$, and set $R := 1 + \max_{1 \leq j \leq N} d(p_j,p_N)$.
If $n \leq N$, then $d(p_n,p_N) < R$; if $n \geq N$, then $d(p_n,p_N) < 1 \leq R$. Hence $\operatorname{ran}(p_n) \subseteq B_R(p_N)$.

(ii) Let (p_n), (q_n) be Cauchy. For $m,n \in \mathbb{N}$, by triangle inequality, $|d(p_n,q_n) - d(p_m,q_m)| \leq d(p_n,p_m) + d(q_n,q_m)$.
Given $\varepsilon > 0$, both terms on the right are less than $\frac{\varepsilon}{2}$ for sufficiently large m,n.
Hence $(d(p_n,q_n))$ is Cauchy in $\mathbb{R}$ and converges by Theorem 2.1.15.

(iii) Let $a_n := (-1)^n$. Then $(\forall\, n \in \mathbb{N})\, |a_n| \leq 1$, so (a_n) is bounded.
For $N \in \mathbb{N}$, choose even $m \geq N$ and odd $n \geq N$; then $|a_m - a_n| = 2$. Thus (a_n) is not Cauchy.

Structural Commentary: A bounded tail controls the range; the four-point estimate controls pairwise distances.
Key Insight: Cauchy control persists under the metric map. □

Definition 4.1.23. *Complete Metric Space*
A metric space (X,d) is *complete* iff every Cauchy sequence in X converges in X.

Corollary 4.1.24. *Compact Metric Spaces Are Complete*
Every compact metric space is complete.

Proof. Every Cauchy sequence converges to a point of the space by Theorem 4.1.20(ii); hence the space is complete.
Structural Commentary: The compact Cauchy criterion discharges the defining convergence requirement.
Key Insight: Compactness keeps every Cauchy limit inside the space. □

Remark 4.1.25. *Intrinsic and Ambient Metric Properties*
For $E \subseteq X$ in a metric space (X,d), boundedness, compactness, and completeness are intrinsic to $d|_{E\times E}$.
In contrast, openness, closedness, and perfection depend on the ambient metric space.

Proposition 4.1.26. *Completeness and Closed Subspaces*
Let (X,d) be a metric space, let $E \subseteq X$, and set $d_E := d|_{E\times E}$.

(i) If (E,d_E) is complete, then E is closed in X, equivalently $E = \overline{E}^{\,X}$.

(ii) If (X,d) is complete, then (E,d_E) is complete if and only if E is closed in X, equivalently $E=\overline{E}^{\,X}$.

Proof.

(i) Assume (E,d_E) is complete, and let (p_n) be a sequence in E with $p_n\to p\in X$.
The sequence is Cauchy in X by Theorem 4.1.20(i), hence it is Cauchy in (E,d_E).
Choose $q\in E$ with $p_n\to q$ in (E,d_E) by completeness. The same convergence holds in X because $d_E=d|_{E\times E}$.
Equality $p=q$ holds by Theorem 3.4.6(ii); E is closed by Theorem 3.4.17(ii). Equality $E=\overline{E}^{\,X}$ holds by Theorem 3.1.22(ii).

(ii) ($\Rightarrow$) Assume (X,d) and (E,d_E) are complete. Then $E=\overline{E}^{\,X}$ by (i).

(ii) ($\Leftarrow$) Assume (X,d) is complete and $E=\overline{E}^{\,X}$. Then E is closed by Theorem 3.1.22(ii).
If (p_n) is Cauchy in (E,d_E), then it is Cauchy in X, and $p_n\to p\in X$ by completeness.
The membership $p\in E$ holds by Theorem 3.4.17(ii). Thus every Cauchy sequence in (E,d_E) converges in (E,d_E).

Structural Commentary: Subspace completeness forces closure; ambient completeness plus closure returns subspace limits.
Key Insight: Closedness is exactly the limit-retention condition. □

Theorem 4.1.27. *Cantor Intersection Theorem*
Let (X,d) be a metric space. Then X is complete iff every contracting sequence $(F_n)_{n\geq 1}$ of closed subsets of X satisfies $\left|\bigcap_{n=1}^{\infty}F_n\right|=1$.

Proof.

($\Rightarrow$) Assume X is complete, and let (F_n) satisfy the stated hypotheses. Choose (x_n) with $(\forall\, n\in\mathbb{N})\,x_n\in F_n$.
Given $\varepsilon>0$, choose N with $\operatorname{diam}F_N<\varepsilon$. For $m,n\geq N$, descendingness places x_m,x_n in F_N, so $d(x_m,x_n)<\varepsilon$.
The sequence (x_n) is Cauchy, and $x_n\to x\in X$ by completeness.
For $N\in\mathbb{N}$, the tail $(x_n)_{n\geq N}$ lies in closed F_N, so $x\in F_N$ by Theorem 3.4.17(ii).
Hence $x\in\bigcap_{N=1}^{\infty}F_N$. If $y\in\bigcap_{N=1}^{\infty}F_N$, then $(\forall\, N\in\mathbb{N})\,d(x,y)\leq\operatorname{diam}F_N$, so $x=y$ by $\operatorname{diam}F_N\to 0$.

($\Leftarrow$) Assume the intersection property, and let (x_n) be Cauchy. Set $E_N:=\{x_n : n\geq N\}$ and $F_N:=\overline{E_N}$ for $N\in\mathbb{N}$.
Each F_N is nonempty and closed by Theorem 3.1.22(i).
The inclusions $E_{N+1}\subseteq E_N\subseteq F_N$ imply $F_{N+1}\subseteq F_N$ by Theorem 3.1.22(iii).
The convergence $\operatorname{diam}F_N=\operatorname{diam}E_N\to 0$ holds by Proposition 4.1.17 and Theorem 4.1.19(i).
Choose $x\in X$ with $\bigcap_{N=1}^{\infty}F_N=\{x\}$ by the intersection property.
Given $\varepsilon>0$, choose N with $\operatorname{diam}F_N<\varepsilon$. If $n\geq N$, then $x_n,x\in F_N$, so $d(x_n,x)<\varepsilon$. Thus $x_n\to x$, and X is complete.

Structural Commentary: Complete limits intersect closed nests; closed Cauchy tails reconstruct limits.
Key Insight: Contracting tails encode both Cauchy control and a unique destination. □

We now turn from contracting closed families to the finite-resolution side of compact metric spaces.

Definition 4.1.28. *Total Boundedness*
Let (X,d) be a metric space and let $E\subseteq X$.

(i) For $\varepsilon>0$, a finite set $F\subseteq X$ is an *ε-net* for E iff $E\subseteq\bigcup_{x\in F}B_\varepsilon(x)$.

(ii) The set E is *totally bounded* iff $(\forall\,\varepsilon>0)(\exists\, F\subseteq X)\,[|F|<\infty\wedge E\subseteq\bigcup_{x\in F}B_\varepsilon(x)]$.

(iii) The metric space (X,d) is *totally bounded* iff condition (ii) holds with $E=X$.

Proposition 4.1.29. *Centers of Finite Nets*
Let (X,d) be a metric space and let $\emptyset\neq E\subseteq X$.
Then E is totally bounded iff every $\varepsilon>0$ admits a finite ε-net $F\subseteq E$.

Proof.

($\Rightarrow$) Suppose E is totally bounded, $\varepsilon>0$. Choose finite $\frac{\varepsilon}{2}$-net $A\subseteq X$ for E, $A':=\{a\in A : E\cap B_{\frac{\varepsilon}{2}}(a)\neq\emptyset\}$.
For $a\in A'$, choose $x_a\in E\cap B_{\frac{\varepsilon}{2}}(a)$, and set $F:=\{x_a : a\in A'\}\subseteq E$. The set F is finite.
Given $y\in E$, some $a\in A'$ satisfies $y\in B_{\frac{\varepsilon}{2}}(a)$. Then $d(y,x_a)<\varepsilon$ by the triangle inequality, so F is an ε-net for E.

($\Leftarrow$) Fix $\varepsilon>0$, and choose a finite ε-net $F\subseteq E$ by stated condition. Since $F\subseteq X$, the defining condition for total boundedness holds.

Structural Commentary: Half-radius nets move ambient centers into E without losing coverage.
Key Insight: Two points in one half-radius ball are ε-close. □

Figure 130 compares finite nets at two resolutions.

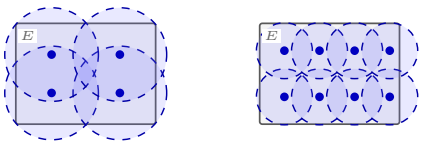

Figure 130: Dots mark a finite ε-net (left) and finer δ-net (right); their balls cover E, with $0<\delta<\varepsilon$.

Theorem 4.1.30. *Total Boundedness Implies Boundedness*
Let (X,d) be a metric space. If X is totally bounded, then X is bounded.

Proof. If $X=\emptyset$, it is bounded. Suppose $X\neq\emptyset$, choose finite 1-net $F\subseteq X$. Then $F\neq\emptyset$; fix $x_0\in F$ and put $R:=1+\max_{a\in F}d(a,x_0)$.
For $x\in X$, some $a\in F$ satisfies $d(x,a)<1$, so $d(x,x_0)\leq d(x,a)+d(a,x_0)<R$. Hence $X\subseteq B_R(x_0)$.
Structural Commentary: A finite unit-ball cover converts finitely many center distances into one radius.
Key Insight: One fixed scale already supplies a global bound. □

Counterexample 4.1.31. *Bounded Need Not Mean Totally Bounded*
In $\ell^2(\mathbb{F})$, let $B := \{\mathbf{x} \in \ell^2(\mathbb{F}) : \|\mathbf{x}\|_2 \leq 1\}$. Then $B \subseteq B_2(0)$, so B is bounded.
For the standard unit vectors (e_n), $(\forall\, n \in \mathbb{N})\, e_n \in B$ and $(\forall\, m \neq n)\ \|e_n - e_m\|_2 = \sqrt{2}$.
Any $\frac{1}{2}$-ball contains at most one e_n, because two points in such a ball have distance less than 1.
Thus B has no finite $\frac{1}{2}$-net, so B is not totally bounded.

Definition 4.1.32. *Lebesgue Number*
Let (X, d) be a metric space, $\mathcal{U}$ be an open cover of X. A number $\delta > 0$ is a *Lebesgue number* for $\mathcal{U}$ iff $(\forall\, x \in X)(\exists\, U \in \mathcal{U})\, B_\delta(x) \subseteq U$.

Lemma 4.1.33. *Lebesgue Covering Lemma*
Let (X, d) be a metric space.

(i) If X is sequentially compact, then every open cover of X has a Lebesgue number.
(ii) If X is compact, then every open cover of X has a Lebesgue number.

Proof.

(i) Let $\mathcal{U} = (U_\alpha)_{\alpha \in A}$ be an open cover of sequentially compact X.
If $\mathcal{U}$ has no Lebesgue number, choose (x_n) in X satisfying $(\forall\, n \in \mathbb{N})(\forall\, \alpha \in A)\, B_{\frac{1}{n}}(x_n) \not\subseteq U_\alpha$.
Choose a subsequence $x_{n_k} \to x \in X$. Since $\mathcal{U}$ covers X, choose $\alpha_0 \in A$ and $r > 0$ such that $x \in U_{\alpha_0}$ and $B_r(x) \subseteq U_{\alpha_0}$.
For sufficiently large k, both $d(x_{n_k}, x) < \frac{r}{2}$ and $\frac{1}{n_k} < \frac{r}{2}$. If $y \in B_{\frac{1}{n_k}}(x_{n_k})$, then $d(y, x) < r$, so $B_{\frac{1}{n_k}}(x_{n_k}) \subseteq U_{\alpha_0}$, a contradiction.

(ii) If X is compact, then it is sequentially compact by Theorem 4.1.13. Hence every open cover of X has a Lebesgue number by (i).

Structural Commentary: Bad-radius sequences contradict subsequential convergence; compactness supplies sequential compactness.
Key Insight: Near the limit, one cover set absorbs every small ball. □

Theorem 4.1.34. *Compactness as Complete Finite Resolution*
Let (X, d) be a metric space. The following are equivalent:

(i) X is compact;
(ii) X is complete and totally bounded;
(iii) X is sequentially compact.

Proof.

(i) $\Rightarrow$ *(ii)* Every Cauchy sequence in X converges by Theorem 4.1.20(ii), so X is complete. Fix $\varepsilon > 0$.
Family $\{B_\varepsilon(x) : x \in X\}$ is open cover of X. By compactness, it has finite subcover, whose centers form a finite ε-net. QED.

(ii) $\Rightarrow$ *(iii)* Let (x_n) be a sequence in X, and put $I_0 := \mathbb{N}$ and $n_0 := 0$.
Recursively, suppose I_{j-1} is infinite and choose a finite $\frac{1}{j}$-net $A_j \subseteq X$.
If each ball centered in A_j contained x_n for only finitely many $n \in I_{j-1}$, then I_{j-1} would be finite.
Hence some $B_{\frac{1}{j}}(a_j)$ contains x_n for infinitely many $n \in I_{j-1}$. Set $I_j := \{n \in I_{j-1} : x_n \in B_{\frac{1}{j}}(a_j)\}$, which is infinite.
Choose $n_j \in I_j$ with $n_j > n_{j-1}$. If $k, \ell \geq j$, then $x_{n_k}, x_{n_\ell} \in B_{\frac{1}{j}}(a_j)$, so $d(x_{n_k}, x_{n_\ell}) < \frac{2}{j}$.
Given $\varepsilon > 0$, choose J with $\frac{2}{J} < \varepsilon$. If $k, \ell \geq J$, then $d(x_{n_k}, x_{n_\ell}) < \varepsilon$ by the preceding bound.
Thus (x_{n_j}) is Cauchy and converges in X by completeness.

(iii) $\Rightarrow$ *(i)* Suppose X is not totally bounded, so some $\varepsilon > 0$ admits no finite ε-net.
Then $X \neq \emptyset$ as $\emptyset$ is an ε-net for itself. Choose $x_1 \in X$, and recursively choose $x_{n+1} \notin \bigcup_{j=1}^{n} B_\varepsilon(x_j)$.
Then $(\forall\, j < n)\, d(x_n, x_j) \geq \varepsilon$, so no subsequence is Cauchy.
No subsequence converges by Theorem 4.1.20(i), contrary to sequential compactness. Thus X is totally bounded.
Let $(U_\alpha)_{\alpha \in A}$ be an open cover of X. Choose a Lebesgue number $\delta > 0$ by Lemma 4.1.33(i).
Choose a finite $\frac{\delta}{2}$-net $F \subseteq X$. For $z \in F$, choose $\alpha(z) \in A$ with $B_\delta(z) \subseteq U_{\alpha(z)}$.
If $x \in X$, then $(\exists\, z \in F)\, x \in B_{\frac{\delta}{2}}(z) \subseteq U_{\alpha(z)}$. Hence the finite family $(U_{\alpha(z)})_{z \in F}$ covers X, so X is compact.

Structural Commentary: Finite nets create Cauchy subsequences; Lebesgue numbers convert nets into finite subcovers.
Key Insight: Completeness turns finite-resolution control into actual convergence. □

Figure 131 summarizes the three equivalent compactness tests.

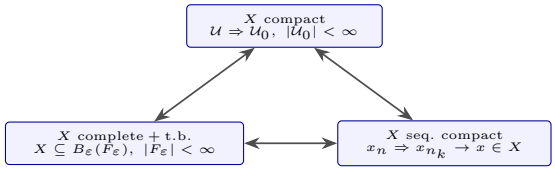

Figure 131: The proof cycle connects finite subcovers, completeness with finite ε-nets, and convergent subsequences.

Proposition 4.1.35. *Compact Metric Spaces Are Separable*
Let (X, d) be a compact metric space. Then X contains an at most countable set D such that $\overline{D}^X = X$.

Proof. The space X is totally bounded by Theorem 4.1.34, direction $(i) \Rightarrow (ii)$.
Choose finite $\frac{1}{n}$-nets $F_n \subseteq X$ for $n \in \mathbb{N}$. Set $D := \bigcup_{n=1}^{\infty} F_n$. The set D is at most countable by Corollary 6.2.16.
Let $x \in X$ and $r > 0$, and choose $n \in \mathbb{N}$ with $\frac{1}{n} < r$. Some $a \in F_n$ satisfies $d(x, a) < \frac{1}{n} < r$, so $a \in B_r(x) \cap D$.
Thus every open ball centered at x meets D, so $x \in \overline{D}^X$. Therefore $\overline{D}^X = X$, and X is separable.
Structural Commentary: Finite nets at reciprocal scales form an at most countable dense union.
Key Insight: One sufficiently fine net meets every prescribed ball. □

4.2 Heine–Borel and Euclidean Compactness

Theorem 4.2.1. *Nested Closed Intervals Intersect*

Let $(I_n)_{n\in\mathbb{N}}$ be nonempty closed intervals in $\mathbb{R}$, where $I_n = [a_n, b_n]$. If $(\forall n \in \mathbb{N})\, I_{n+1} \subseteq I_n$, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. Set $A := \{a_n : n \in \mathbb{N}\}$, and fix $m \in \mathbb{N}$. If $n \geq m$, then $a_n \in I_n \subseteq I_m$, so $a_n \leq b_m$.
If $n < m$, then $b_m \in I_m \subseteq I_n$, so $a_n \leq b_m$. Thus b_m is an upper bound for A.
Since $A \neq \emptyset$ and A is bounded above, $x := \sup A$ exists by Theorem 1.1.9(i).

Inequalities $a_m \leq x \leq b_m$ follow from $a_m \in A$ and the upper-bound property of b_m. Therefore $(\forall m \in \mathbb{N})\, x \in I_m$, so $x \in \bigcap_{n=1}^{\infty} I_n$.

Structural Commentary: Nestedness makes each right endpoint bound all left endpoints; real completeness supplies their supremum.
Key Insight: Every interval contains $\sup\{a_n : n \in \mathbb{N}\}$. □

Figure 132 depicts the common point selected by the supremum construction.

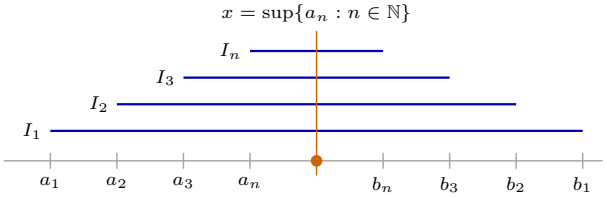

Figure 132: The nested segments represent $I_n = [a_n, b_n]$; the marked point $x = \sup\{a_n : n \in \mathbb{N}\}$ lies between every endpoint pair.

Theorem 4.2.2. *Nested k-Cells Intersect*

Let $k \in \mathbb{N}$, $(I_n)_{n\in\mathbb{N}}$ be sequence of nonempty k-cells, where $I_n = \prod_{j=1}^{k} [a_{n,j}, b_{n,j}] \subseteq \mathbb{R}^k$. If $(\forall n \in \mathbb{N})\, I_{n+1} \subseteq I_n$, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. For $n \in \mathbb{N}$ and $1 \leq j \leq k$, coordinate projection of $I_{n+1} \subseteq I_n$ establishes $[a_{n+1,j}, b_{n+1,j}] \subseteq [a_{n,j}, b_{n,j}]$.
Choose $x_j^* \in \bigcap_{n=1}^{\infty} [a_{n,j}, b_{n,j}]$ for $1 \leq j \leq k$ by Theorem 4.2.1. Then $\mathbf{x}^* := (x_1^*, \ldots, x_k^*)$ satisfies $(\forall n \in \mathbb{N})\, \mathbf{x}^* \in I_n$, so $\mathbf{x}^* \in \bigcap_{n=1}^{\infty} I_n$.
Structural Commentary: Coordinate projection reduces a nested cell sequence to k nested interval sequences.
Key Insight: One persistent point in each coordinate defines a persistent vector. □

Figure 133 illustrates the product structure of the first three cell dimensions.

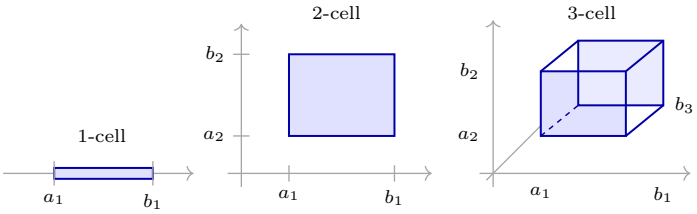

Figure 133: One-, two-, and three-factor cells are an interval, rectangle, and box; dashes mark a hidden edge.

Theorem 4.2.3. *k-Cells Are Compact*

Let $k \in \mathbb{N}$, and let $I = \prod_{j=1}^{k} [a_j, b_j] \subseteq \mathbb{R}^k$ be a nonempty k-cell. Then I is compact.

Proof.

(i) *Contradictory cover*: Put $\delta := \|(b_1 - a_1, \ldots, b_k - a_k)\|_2$. Suppose an open cover $\mathcal{G} = (G_\alpha)_{\alpha\in A}$ of I has no finite subcover.

(ii) *Nested construction*: Set $I_0 := I$. Bisecting every coordinate interval covers I_n by 2^k nonempty subcells.
If each subcell had a finite $\mathcal{G}$-subcover, their finite union would cover I_n. Thus some subcell has no finite subcover.
Choose I_{n+1} as the lexicographically first such subcell. This recursion defines $I_0 \supseteq I_1 \supseteq \cdots$.

(iii) *Diameter control*: The jth side of I_n has length $2^{-n}(b_j - a_j)$. Hence $(\forall\, \mathbf{p}, \mathbf{q} \in I_n)\ \|\mathbf{p} - \mathbf{q}\|_2 \leq 2^{-n}\delta$, so $\operatorname{diam} I_n \leq 2^{-n}\delta$.

(iv) *Contradiction*: Choose $\mathbf{x}^* \in \bigcap_{n=1}^{\infty} I_n$ by Theorem 4.2.2. Choose $\alpha \in A$ with $\mathbf{x}^* \in G_\alpha$, and $r > 0$ with $B_r(\mathbf{x}^*) \subseteq G_\alpha$.
The bound $(\forall n \in \mathbb{N})\, 2^n \geq n + 1$ holds by induction: $2^1 = 2$, and $2^n \geq n + 1$ implies $2^{n+1} \geq 2(n + 1) \geq n + 2$.
Choose $n \in \mathbb{N}$ with $n + 1 > \frac{\delta}{r}$ by Theorem 1.1.19(i). Then $2^{-n}\delta < r$.
Because $\mathbf{x}^* \in I_n$, the diameter bound implies $I_n \subseteq B_r(\mathbf{x}^*) \subseteq G_\alpha$. This one-member subcover contradicts the construction of I_n.

Structural Commentary: Recursive bisection combines nested-cell intersection with geometric diameter decay.
Key Insight: An open neighborhood of the trapped point eventually contains an entire bad cell. □

Figure 134 depicts the nested subcells and the terminal neighborhood contradiction.

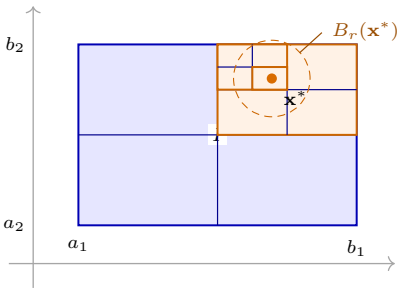

Figure 134: Successive outlined subcells form a nested chain; the dashed ball about $\mathbf{x}^*$ eventually contains a selected subcell.

Lemma 4.2.4. *Bounded Euclidean Sets Fit in k-Cells*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$ be bounded. Then E is contained in a k-cell.

Proof. If $E = \emptyset$, then $E \subseteq \prod_{j=1}^{k} [-1, 1]$. Suppose $E \neq \emptyset$, and choose $\mathbf{q} = (q_1, \ldots, q_k) \in \mathbb{R}^k$ and $M > 0$ with $E \subseteq B_M(\mathbf{q})$.

Set $I := \prod_{j=1}^{k} [q_j - M, q_j + M]$. If $\mathbf{p} = (p_1, \ldots, p_k) \in E$, then $(\forall j \in \{1, \ldots, k\})\ |p_j - q_j| \leq \|\mathbf{p} - \mathbf{q}\|_2 < M$. Hence $\mathbf{p} \in I$, $E \subseteq I$.
Structural Commentary: A containing Euclidean ball transfers its radius bound to every coordinate interval.
Key Insight: Euclidean distance dominates each coordinate difference. □

Theorem 4.2.5. *Heine–Borel in $\mathbb{R}^k$*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$. The following are equivalent:

(i) E is closed and bounded;
(ii) E is compact;
(iii) $(\forall\, S \subseteq E)\,[S$ infinite $\implies S^{\mathrm{acc}} \cap E \neq \emptyset]$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Choose a k-cell I with $E \subseteq I$ by Lemma 4.2.4. The set I is compact by Theorem 4.2.3.
Since E is closed in $\mathbb{R}^k$, it is compact by Theorem 4.1.7 with $E \subseteq I$.

(*ii*) $\Rightarrow$ (*iii*) If $S \subseteq E$ is infinite, then $S^{\mathrm{acc}} \cap E \neq \emptyset$ by Theorem 4.1.11.

(*iii*) $\Rightarrow$ (*i*)

(a) *Boundedness*: Suppose E is unbounded. Choose $(\mathbf{x}_n) \subseteq E$ with $(\forall n \in \mathbb{N})\ \|\mathbf{x}_n\|_2 > n$, $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$.
The set S is infinite because a finite subset of $\mathbb{R}^k$ has bounded norms. If $\mathbf{x}_n \in B_1(\mathbf{y})$, then $n < \|\mathbf{x}_n\|_2 < \|\mathbf{y}\|_2 + 1$.
Thus $(\forall\, \mathbf{y} \in \mathbb{R}^k)\,[S \cap B_1(\mathbf{y})$ is finite$]$. Hence $S^{\mathrm{acc}} = \emptyset$ by contrapositive of Theorem 3.1.14.
This contradicts (iii), and hence E is bounded.

(b) *Closedness*: Suppose E is not closed, and choose $\mathbf{x}_0 \in \overline{E} \setminus E$. Then $\mathbf{x}_0 \in E^{\mathrm{acc}}$ by Definition 3.1.21.
Choose $(\mathbf{x}_n) \subseteq E \setminus \{\mathbf{x}_0\}$ with $\mathbf{x}_n \to \mathbf{x}_0$ by Theorem 3.4.17(i).
Set $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$. If S were finite, then $\rho := \min_{\mathbf{x} \in S} \|\mathbf{x} - \mathbf{x}_0\|_2 > 0$.
Since $(\forall n \in \mathbb{N})\ \mathbf{x}_n \in S$, this contradicts $\mathbf{x}_n \to \mathbf{x}_0$; hence S is infinite. Fix $\mathbf{y} \neq \mathbf{x}_0$, and put $r := \frac{1}{3}\|\mathbf{y} - \mathbf{x}_0\|_2$.
The balls $B_r(\mathbf{y})$ and $B_r(\mathbf{x}_0)$ are disjoint, while all but finitely many $\mathbf{x}_n$ lie in $B_r(\mathbf{x}_0)$.
Hence $B_r(\mathbf{y}) \cap S$ is finite, so $\mathbf{y} \notin S^{\mathrm{acc}}$ by the contrapositive of Theorem 3.1.14.
Therefore $S^{\mathrm{acc}} \subseteq \{\mathbf{x}_0\}$ and $S^{\mathrm{acc}} \cap E = \emptyset$, contradicting (iii); hence E is closed.

Structural Commentary: Compact cells prove sufficiency; escaping and boundary-convergent sequences prove necessity.
Key Insight: Failure of either bound or closure creates an infinite set without internal accumulation. □

Theorem 4.2.6. *Weierstrass Limit-Point Theorem*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$ be bounded and infinite. Then $E^{\mathrm{acc}} \neq \emptyset$.

Proof. Choose a k-cell I with $E \subseteq I$ by Lemma 4.2.4. Set I is compact by Theorem 4.2.3, so $E^{\mathrm{acc}} \cap I \neq \emptyset$ by Theorem 4.1.11.
Structural Commentary: Cell containment transfers the abstract compact accumulation theorem to bounded Euclidean sets.
Key Insight: A bounded infinite set is an infinite subset of one compact cell. □

Corollary 4.2.7. *Bounded Euclidean Sequences Have Convergent Subsequences*
Let $k \in \mathbb{N}$ and let $(\mathbf{x}_n)$ be a bounded sequence in $\mathbb{R}^k$. Then $(\mathbf{x}_n)$ has a convergent subsequence.

Proof. Set $R := \{\mathbf{x}_n : n \in \mathbb{N}\}$. Choose a k-cell I with $R \subseteq I$ by Lemma 4.2.4.
The set I is compact by Theorem 4.2.3; hence $(\mathbf{x}_n)$ has a convergent subsequence by Theorem 4.1.13.
Structural Commentary: Range containment moves the sequence into a compact cell before subsequence extraction.
Key Insight: One compact cell contains every term. □

Corollary 4.2.8. *Euclidean Cauchy Convergence*
Let $k \in \mathbb{N}$ and let $(\mathbf{x}_n)$ be a Cauchy sequence in $\mathbb{R}^k$. Then $(\mathbf{x}_n)$ converges.

Proof. The range of $(\mathbf{x}_n)$ is bounded by Proposition 4.1.22(i). Thus $(\mathbf{x}_n)$ has a convergent subsequence by Corollary 4.2.7.
Therefore $(\mathbf{x}_n)$ converges by Lemma 4.1.21.
Structural Commentary: Cauchy boundedness supplies a convergent subsequence, whose limit controls the full sequence.
Key Insight: A Cauchy sequence cannot diverge from a convergent subsequence. □

Corollary 4.2.9. *Euclidean Compactness Tests*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$, with the subspace metric. The following are equivalent:

(i) *E is closed and bounded;*
(ii) *E is compact;*
(iii) $(\forall\, (\mathbf{x}_n) \subseteq E)(\exists\, n_1 < n_2 < \cdots)(\exists\, \mathbf{x} \in E)\, \mathbf{x}_{n_j} \to \mathbf{x}$.

Proof.

(i) $\Rightarrow$ *(ii)* The set E is compact by Theorem 4.2.5, direction $(i) \Rightarrow (ii)$.
(ii) $\Rightarrow$ *(iii)* Condition (iii) holds by Theorem 4.1.13.
(iii) $\Rightarrow$ *(i)* The metric space E is sequentially compact by (iii). Hence E is compact by Theorem 4.1.34, direction $(iii) \Rightarrow (i)$. Therefore E is closed and bounded by Theorem 4.2.5, direction $(ii) \Rightarrow (i)$.

Structural Commentary: Heine–Borel and metric sequential compactness connect geometric, cover, and subsequence criteria.
Key Insight: In Euclidean space, closed bounded geometry is exactly sequential compactness. □

Corollary 4.2.10. *Heine–Borel on the Real Line*
Let $E \subseteq \mathbb{R}$, with the usual metric. The following are equivalent:

(i) *E is closed and bounded;*
(ii) *E is compact;*
(iii) $(\forall\, (x_n) \subseteq E)(\exists\, n_1 < n_2 < \cdots)(\exists\, x \in E)\, x_{n_j} \to x$.

Proof.

(i) $\Rightarrow$ *(ii)* The set E is compact by Corollary 4.2.9, direction $(i) \Rightarrow (ii)$, with $k = 1$.
(ii) $\Rightarrow$ *(iii)* Condition (iii) holds by Corollary 4.2.9, direction $(ii) \Rightarrow (iii)$, with $k = 1$.
(iii) $\Rightarrow$ *(i)* The set E is closed and bounded by Corollary 4.2.9, direction $(iii) \Rightarrow (i)$, with $k = 1$.

Structural Commentary: Specializing the Euclidean compactness cycle to $k = 1$ transfers all three implications.
Key Insight: Real-line compactness is the one-dimensional Euclidean case. □

Corollary 4.2.11. *Nested Closed Bounded Real Sets Intersect*
Let $(F_n)_{n\in\mathbb{N}}$ be nonempty closed subsets of $\mathbb{R}$ with $(\forall n \in \mathbb{N})\, F_{n+1} \subseteq F_n$. If F_1 is bounded, then $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$.

Proof. The set F_1 is compact by Corollary 4.2.10, direction $(i) \Rightarrow (ii)$.

For $n \in \mathbb{N}$, $F_n \subseteq F_1$ by nestedness, so F_n is compact by Theorem 4.1.7. Hence $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$ by Corollary 4.1.10.

Structural Commentary: Heine–Borel makes F_1 compact; closed-subset compactness transfers this to the nested tail.
Key Insight: Nested nonempty compact sets have the finite intersection property. □

4.3 Baire Category

Definition 4.3.1. *F_σ, G_δ, and Nowhere-Dense Sets*
Let (X, d) be a metric space.

(i) A set $A \subseteq X$ is an *F_σ set* iff $A = \bigcup_{n=1}^{\infty} F_n$ for a sequence (F_n) of closed subsets of X.
(ii) A set $B \subseteq X$ is a *G_δ set* iff $B = \bigcap_{n=1}^{\infty} G_n$ for a sequence (G_n) of open subsets of X.
(iii) A set $E \subseteq X$ is *nowhere dense* iff $(\overline{E})^\circ = \emptyset$; equivalently, no nonempty open set is contained in $\overline{E}$.

For $X = \mathbb{R}$, this says exactly that $\overline{E}$ contains no nonempty open interval.

Proposition 4.3.2. *Open Real Sets Are F_σ*
Let $G, A \subseteq \mathbb{R}$.

(i) *If G is open, then G is F_σ.*
(ii) *If A is closed, then A is G_δ.*

Proof.

(i)

(a) *Whole-line case*: If $G = \mathbb{R}$, then $G = \bigcup_{m=1}^{\infty} [-m, m]$ by Theorem 1.1.19(i). Each interval closed by Corollary 3.1.18.
(b) *Distance control*: Suppose $G \neq \mathbb{R}$, and set $C := \mathbb{R} \setminus G \neq \emptyset$.
For $m \in \mathbb{N}$, define $D_m := \{x \in \mathbb{R} : \operatorname{dist}(x, C) \geq \frac{1}{m}\}$ and $F_m := [-m, m] \cap D_m$.
For $x, y \in \mathbb{R}$ and $c \in C$, $|x - c| \leq |x - y| + |y - c|$ by the triangle inequality.
Taking the infimum over $c \in C$, $\operatorname{dist}(x, C) \leq |x - y| + \operatorname{dist}(y, C)$ by Definition 3.1.25(i).
After interchanging x and y, $|\operatorname{dist}(x, C) - \operatorname{dist}(y, C)| \leq |x - y|$.
(c) *Closed approximants*: If $x \notin D_m$, put $\eta := \frac{1}{m} - \operatorname{dist}(x, C) > 0$. Then $y \in B_\eta(x)$ implies $\operatorname{dist}(y, C) < \frac{1}{m}$, so $B_\eta(x) \subseteq D_m^c$.
Thus D_m is closed by Corollary 3.1.17. The interval $[-m, m]$ is closed, so F_m is closed by Theorem 3.1.19(ii).
(d) *Exhaustion*: If $x \in C$, then $\operatorname{dist}(x, C) = 0$; hence $F_m \cap C = \emptyset$ and $F_m \subseteq G$. Fix $x \in G$, $r > 0$ with $(x - r, x + r) \subseteq G$.
Then $(\forall c \in C)\ |x - c| \geq r$, so $\operatorname{dist}(x, C) \geq r$. Choose $m \in \mathbb{N}$, $m \geq |x|$, $\frac{1}{m} \leq r$ by Theorem 1.1.19(i); $x \in F_m$, so $G = \bigcup_{m=1}^{\infty} F_m$.

(ii) If A is closed, then $A^c = \bigcup_{m=1}^{\infty} F_m$ for a sequence (F_m) of closed sets by (i). Therefore $A = \bigcap_{m=1}^{\infty} F_m^c$ is G_δ.

Structural Commentary: Distance from the complement constructs a closed exhaustion; complementation transfers it to closed sets.
Key Insight: Positive distance converts local openness into closed approximants. □

Definition 4.3.3. *First Category*

Let (X,d) be metric space, $E \subseteq X$. Set E is *meager* in X iff $E \subseteq \bigcup_{n=1}^{\infty} N_n$ for sequence (N_n) of nowhere-dense subsets of X.

The terms *of first category* and *thin* are synonymous. The ambient space X is part of the property.

Theorem 4.3.4. *Dense Open Intersections in $\mathbb{R}$*

Let $(G_n)_{n\in\mathbb{N}}$ be dense open subsets of $\mathbb{R}$. Then $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$.

Proof.

(i) *Initialization*: Choose $x_1 \in G_1 \cap (0,1)$ by Proposition 3.3.2. By openness, choose $s_1 > 0$ with $B_{s_1}(x_1) \subseteq G_1 \cap (0,1)$.
Set $\rho_1 := \min\{\frac{s_1}{2}, \frac{1}{4}\}$ and $I_1 := [x_1 - \rho_1, x_1 + \rho_1]$. Then $I_1 \subseteq G_1 \cap (0,1)$ and $I_1^\circ \neq \emptyset$.

(ii) *Recursion*: Suppose $I_n = [a_n, b_n] \subseteq G_n$ and $a_n < b_n$. Choose $x_{n+1} \in G_{n+1} \cap (a_n, b_n)$ by Proposition 3.3.2.
By openness, choose $s_{n+1} > 0$ with $B_{s_{n+1}}(x_{n+1}) \subseteq G_{n+1} \cap (a_n, b_n)$.
Put $\rho_{n+1} := \frac{s_{n+1}}{2}$ and $I_{n+1} := [x_{n+1} - \rho_{n+1}, x_{n+1} + \rho_{n+1}]$. Then $I_{n+1} \subseteq I_n \cap G_{n+1}$ and $I_{n+1}^\circ \neq \emptyset$.

(iii) *Intersection*: Choose $x \in \bigcap_{n=1}^{\infty} I_n$ by Theorem 4.2.1. Since $(\forall n \in \mathbb{N})\, I_n \subseteq G_n$, one has $x \in \bigcap_{n=1}^{\infty} G_n$.

Structural Commentary: Dense openness constructs nested intervals; real completeness supplies their common point.
Key Insight: Each interval can be trapped inside the next dense open set. □

Theorem 4.3.5. *Dense Open Intersections in Complete Metric Spaces*
Let (X,d) be complete, and let $(O_n)_{n\in\mathbb{N}}$ be dense open subsets of X.

(i) If $\emptyset \neq U \subseteq X$ is open, then $U \cap \bigcap_{n=1}^{\infty} O_n \neq \emptyset$.

(ii) The set $\bigcap_{n=1}^{\infty} O_n$ is dense in X.

Proof.

(i)

(a) *Initialization*: Fix a nonempty open set $U \subseteq X$. Choose $x_1 \in U \cap O_1$ by density of O_1.
By openness, choose $s_1 > 0$, $B_{s_1}(x_1) \subseteq U \cap O_1$. Set $r_1 := \min\{\frac{s_1}{2}, \frac{1}{4}\}$, $C_1 := \overline{B}_{r_1}(x_1)$; $C_1 \subseteq U \cap O_1$ and $0 < r_1 \leq \frac{1}{2}$.

(b) *Recursion*: Suppose $C_n = \overline{B}_{r_n}(x_n) \subseteq U \cap \bigcap_{j=1}^{n} O_j$, $0 < r_n \leq \frac{1}{2n}$. Since O_{n+1} is dense, choose $x_{n+1} \in B_{r_n}(x_n) \cap O_{n+1}$.
By openness, choose $s_{n+1} > 0$ with $B_{s_{n+1}}(x_{n+1}) \subseteq B_{r_n}(x_n) \cap O_{n+1}$.
Set $r_{n+1} := \min\{\frac{s_{n+1}}{2}, \frac{1}{2(n+1)}\}$ and $C_{n+1} := \overline{B}_{r_{n+1}}(x_{n+1})$. Then $C_{n+1} \subseteq C_n \cap O_{n+1}$ and $0 < r_{n+1} \leq \frac{1}{2(n+1)}$.

(c) *Diameter decay*: Each C_n is nonempty and closed by Corollary 3.1.18. For $p, q \in C_n$, $d(p,q) \leq 2r_n \leq \frac{1}{n}$, so $\operatorname{diam} C_n \leq \frac{1}{n}$.
Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$ by Theorem 1.1.19(i). Then $(\forall n \geq N)\, \operatorname{diam} C_n \leq \frac{1}{n} < \varepsilon$, so $\operatorname{diam} C_n \to 0$.

(d) *Intersection*: Choose $x \in X$ with $\bigcap_{n=1}^{\infty} C_n = \{x\}$ by Theorem 4.1.27, direction ($\Rightarrow$). As $C_n \subseteq U \cap O_n$, $x \in U \cap \bigcap_{n=1}^{\infty} O_n$.

(ii) Set $D := \bigcap_{n=1}^{\infty} O_n$. If $p \in X \setminus D$ and $r > 0$, then the nonempty open ball $B_r(p)$ meets D by (i).
Hence $B_r(p) \cap (D \setminus \{p\}) \neq \emptyset$, so $p \in D^{\mathrm{acc}}$. Thus $X = D \cup D^{\mathrm{acc}} = \overline{D}$, and D is dense by Proposition 3.3.1.

Structural Commentary: Dense openness builds contracting closed balls; Cantor intersection supplies the common point.
Key Insight: Completeness prevents a nested sequence of shrinking admissible balls from vanishing. □

Figure 135 illustrates the nested-ball construction in the complete-metric proof.

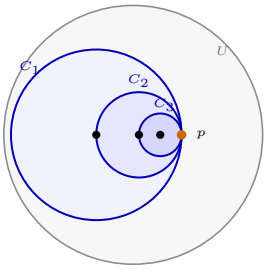

Figure 135: Nested closed balls $C_1 \supseteq C_2 \supseteq C_3$ lie in U; filled dots are their centers, and p is their common point.

Definition 4.3.6. *Residual, Thick, and Generic Sets*
Let (X,d) be a metric space.

(i) A set $G \subseteq X$ is *residual* in X iff it contains a countable intersection of dense open subsets of X. Term *thick* is synonymous.
(ii) A property P is *generic* in X iff $\{x \in X : P(x)\}$ is residual in X.

Thus meager sets are topologically small, whereas residual sets contain a countable dense-open core.

Corollary 4.3.7. *Residual Subsets of Complete Metric Spaces Are Dense*
Let (X,d) be complete, and let $G \subseteq X$ be residual. Then G is dense in X.

Proof. Choose dense open sets (O_n) in X with $\bigcap_{n=1}^{\infty} O_n \subseteq G$ by Definition 4.3.6(i).

Set $D := \bigcap_{n=1}^{\infty} O_n$. The set D is dense by Theorem 4.3.5(ii), so $\overline{D} = X$ by Proposition 3.3.1.

Since $D \subseteq G$, one has $X = \overline{D} \subseteq \overline{G} \subseteq X$; hence G is dense by Proposition 3.3.1.
Structural Commentary: Baire makes the dense-open core dense; closure monotonicity transfers density to its residual superset.
Key Insight: Every superset of a dense set is dense. □

Theorem 4.3.8. *Baire Category Theorem for Complete Metric Spaces*

Let (X, d) be a nonempty complete metric space, and let $(E_n)_{n\in\mathbb{N}}$ be nowhere-dense subsets of X. Then $X \neq \bigcup_{n=1}^{\infty} E_n$.

Proof. Suppose $X = \bigcup_{n=1}^{\infty} E_n$, $O_n := X \setminus \overline{E_n}$. Then (O_n) is sequence of open subsets of X by Theorem 3.1.22(i), Corollary 3.1.17.
Fix $n \in \mathbb{N}$, $p \in X \setminus O_n$, $r > 0$. Since $(\overline{E_n})^\circ = \emptyset$, ball $B_r(p)$ is not contained in $\overline{E_n}$. Hence $B_r(p) \cap O_n \neq \emptyset$, so $p \in O_n^{\mathrm{acc}}$.
Thus (O_n) is a sequence of dense open subsets of X by Definition 3.1.2.
Choose $x \in \bigcap_{n=1}^{\infty} O_n$ by Theorem 4.3.5(i) with $U = X$. Then $(\forall n \in \mathbb{N})\, x \notin E_n$, contradicting $X = \bigcup_{n=1}^{\infty} E_n$.
Structural Commentary: Nowhere-dense closures have dense open complements; their Baire intersection avoids the proposed cover.
Key Insight: One point can simultaneously escape every nowhere-dense layer. □

Theorem 4.3.9. *Baire's Theorem for $\mathbb{R}$*

If $(E_n)_{n\in\mathbb{N}}$ are nowhere-dense subsets of $\mathbb{R}$, then $\mathbb{R} \neq \bigcup_{n=1}^{\infty} E_n$.

Proof. Suppose $\mathbb{R} = \bigcup_{n=1}^{\infty} E_n$, $U_n := \mathbb{R} \setminus \overline{E_n}$. Then (U_n) is a sequence of open subsets of $\mathbb{R}$ by Theorem 3.1.22(i), Corollary 3.1.17.
Fix $n \in \mathbb{N}$ and a nonempty open interval I. The set I meets U_n because $I \not\subseteq \overline{E_n}$.
Hence (U_n) is a sequence of dense open subsets of $\mathbb{R}$ by Proposition 3.3.2.
Choose $x \in \bigcap_{n=1}^{\infty} U_n$ by Theorem 4.3.4. Then $(\forall n \in \mathbb{N})\, x \notin E_n$, contradicting $\mathbb{R} = \bigcup_{n=1}^{\infty} E_n$.
Structural Commentary: Real-line interval tests turn nowhere-dense complements into the dense-open intersection theorem.
Key Insight: Countably many thin sets leave a common point uncovered. □

Corollary 4.3.10. *Rationals and Irrationals by Category*

(i) The set $\mathbb{Q}$ is F_σ.
(ii) The set $\mathbb{R} \setminus \mathbb{Q}$ is G_δ.
(iii) The set $\mathbb{Q}$ is not G_δ.
(iv) The set $\mathbb{R} \setminus \mathbb{Q}$ is not F_σ.

Proof.

(i) Choose an enumeration $\mathbb{Q} = \{q_1, q_2, \ldots\}$ by Corollary 6.2.19.
For $n \in \mathbb{N}$, the singleton $\{q_n\}$ is closed by Corollary 3.1.15 and Definition 3.1.2. Hence $\mathbb{Q} = \bigcup_{n=1}^{\infty} \{q_n\}$ is F_σ.

(ii) Taking complements in (i), $\mathbb{R} \setminus \mathbb{Q} = \bigcap_{n=1}^{\infty} (\mathbb{R} \setminus \{q_n\})$ is G_δ.

(iii)

(a) *Dense layers*: Suppose $\mathbb{Q} = \bigcap_{n=1}^{\infty} G_n$, where (G_n) is a sequence of open subsets of $\mathbb{R}$. Fix $n \in \mathbb{N}$.
The set G_n is dense by $\mathbb{Q} \subseteq G_n$, Theorem 1.1.19(ii), and Proposition 3.3.2.

(b) *Punctured layers*: For $n \in \mathbb{N}$, set $H_n := G_n \setminus \{q_n\}$.
Each H_n is open by Corollary 3.1.17 and Theorem 3.1.19(iii). Fix $n \in \mathbb{N}$ and a nonempty open interval (a, b).
Choose $y \in (a, b) \cap G_n$ by density of G_n. By openness, choose $\varepsilon > 0$ with $(y - \varepsilon, y + \varepsilon) \subseteq (a, b) \cap G_n$.
At least one of $y - \frac{\varepsilon}{2}$ and $y + \frac{\varepsilon}{2}$ differs from q_n, so $(a, b) \cap H_n \neq \emptyset$. Thus H_n is dense by Proposition 3.3.2.

(c) *Baire contradiction*: Therefore (H_n) is a sequence of dense open subsets of $\mathbb{R}$.
Choose $x \in \bigcap_{n=1}^{\infty} H_n$ by Theorem 4.3.4. Since $(\forall n \in \mathbb{N})\, H_n \subseteq G_n$, one has $x \in \bigcap_{n=1}^{\infty} G_n = \mathbb{Q}$.
Choose $k \in \mathbb{N}$, $x = q_k$ by $\mathbb{Q} = \{q_1, q_2, \ldots\}$. Then $x \notin H_k$, whereas $x \in H_k$ by choice of x, contradiction.

(iv) If $\mathbb{R} \setminus \mathbb{Q}$ were F_σ, then its complement $\mathbb{Q}$ would be G_δ, contrary to (iii).

Structural Commentary: Singleton decompositions establish the positive classes; Baire excludes the reversed classifications.
Key Insight: Removing the enumerated rational from each dense open layer forces an impossible intersection. □

Corollary 4.3.11. *Closed Cover Interior Test*

Let X be a complete metric space and let $(F_n)_{n\in\mathbb{N}}$ be closed subsets of X.

(i) If $\emptyset \neq U \subseteq X$ is open and $U \subseteq \bigcup_{n=1}^{\infty} F_n$, then $(\exists N \in \mathbb{N})\, F_N^\circ \neq \emptyset$.

(ii) If $X \neq \emptyset$ and $(\forall n \in \mathbb{N})\, F_n^\circ = \emptyset$, then $X \neq \bigcup_{n=1}^{\infty} F_n$.

Proof.

(i) Suppose conclusion fails, and $O_n := X \setminus F_n$. Then (O_n) is sequence of open subsets of X by Corollary 3.1.17.
Fix $n \in \mathbb{N}$, $p \in F_n$, and $r > 0$. Then $B_r(p) \not\subseteq F_n$ because $F_n^\circ = \emptyset$. Thus $B_r(p) \cap O_n \neq \emptyset$, so $p \in O_n^{\mathrm{acc}}$.
Hence (O_n) is a sequence of dense open subsets of X by Definition 3.1.2.
Choose $x \in U \cap \bigcap_{n=1}^{\infty} O_n$ by Theorem 4.3.5(i). Then $x \in U \subseteq \bigcup_{n=1}^{\infty} F_n$ while $(\forall n \in \mathbb{N})\, x \notin F_n$, a contradiction.

(ii) If $X = \bigcup_{n=1}^{\infty} F_n$, then (i) with $U = X$ implies $(\exists N \in \mathbb{N})\, F_N^\circ \neq \emptyset$, contrary to the hypothesis.

Structural Commentary: Empty interiors create dense open complements; their Baire intersection contradicts the closed cover.
Key Insight: A complete space cannot be covered locally by closed sets with empty interior. □

Perspective 4.3.12. *Category in Function Spaces*
Theorems 4.3.5 and 4.3.8 depend on metric completeness rather than the order structure of $\mathbb{R}$.
They therefore transfer to later function spaces after completeness of the chosen function-space metric has been established.

Proof Method 4.3.13. *Baire-Category Proof Patterns*

(i) *Dense-open target*: to prove $\bigcap_{n=1}^{\infty} O_n$ dense, fix a nonempty open set U. Construct a witness in $U \cap \bigcap_{n=1}^{\infty} O_n$.
(ii) *Complete-metric construction*: choose nonempty closed balls $C_{n+1} \subseteq C_n \cap O_{n+1}$ with $\operatorname{diam} C_n \to 0$.
The common point exists by Theorem 4.1.27.
(iii) *Nowhere-dense input*: replace exceptional sets E_n by the dense open complements $X \setminus \overline{E_n}$.
(iv) *Meager exceptional set*: find nowhere-dense (E_n) with $E \subseteq \bigcup_{n=1}^{\infty} E_n$.
Then choose a point in the complementary dense-open intersection.
(v) *Residual property*: exhibit dense open sets (O_n) with $\bigcap_{n=1}^{\infty} O_n$ contained in the truth set.
(vi) *F_σ/G_δ duality*: complement a countable union of closed sets by De Morgan's laws.

4.4 Perfect Sets and the Cantor Set

Theorem 4.4.1. *Euclidean Perfect Sets Are Uncountable*
Let $k \in \mathbb{N}$, and let $\emptyset \neq P \subseteq \mathbb{R}^k$ be perfect. Then P is uncountable.

Proof.

(i) *Enumeration*: Suppose P is at most countable.
By Proposition 6.2.13, choose $(\mathbf{x}_n)$ satisfying $P = \{\mathbf{x}_n : n \in \mathbb{N}\}$. Set $\mathbf{c}_1 := \mathbf{x}_1$ and $r_1 := 1$.
(ii) *Recursive construction*: Suppose $\mathbf{c}_n \in P$ and $r_n > 0$ have been defined.
The point $\mathbf{c}_n$ belongs to P^{acc} by Definition 3.4.1(ii). The ball $B_{r_n}(\mathbf{c}_n)$ is a neighborhood of $\mathbf{c}_n$ by Theorem 3.1.3.
Hence $P \cap B_{r_n}(\mathbf{c}_n)$ is infinite by Theorem 3.1.14. Set $A_n := \{j \in \mathbb{N} : \mathbf{x}_j \in B_{r_n}(\mathbf{c}_n),\ \mathbf{x}_j \neq \mathbf{x}_n\} \neq \emptyset$.
The minimum $m_n := \min A_n$ exists by Assumption 2.6.1. Put $\mathbf{y}_n := \mathbf{x}_{m_n}$ and $\mathbf{c}_{n+1} := \mathbf{y}_n$.
Set $\delta_n := \min\{r_n - \|\mathbf{y}_n - \mathbf{c}_n\|_2, \|\mathbf{y}_n - \mathbf{x}_n\|_2\}$ and $r_{n+1} := \frac{\delta_n}{2}$.
The relations $\|\mathbf{y}_n - \mathbf{c}_n\|_2 < r_n$ and $\|\mathbf{y}_n - \mathbf{x}_n\|_2 > 0$ imply $\delta_n > 0$ and $r_{n+1} > 0$.
(iii) *Containment and avoidance*: Fix $\mathbf{u} \in \overline{B}_{r_{n+1}}(\mathbf{c}_{n+1})$.
Then $\|\mathbf{u} - \mathbf{c}_n\|_2 \leq r_{n+1} + \|\mathbf{y}_n - \mathbf{c}_n\|_2 \leq \frac{r_n + \|\mathbf{y}_n - \mathbf{c}_n\|_2}{2} < r_n$ by the triangle inequality.
Thus $\overline{B}_{r_{n+1}}(\mathbf{c}_{n+1}) \subseteq B_{r_n}(\mathbf{c}_n)$. Hence $\|\mathbf{x}_n - \mathbf{c}_{n+1}\|_2 = \|\mathbf{x}_n - \mathbf{y}_n\|_2 \geq \delta_n = 2r_{n+1} > r_{n+1}$, so $\mathbf{x}_n \notin \overline{B}_{r_{n+1}}(\mathbf{c}_{n+1})$.
(iv) *Compact intersection*: Put $K_n := \overline{B}_{r_n}(\mathbf{c}_n) \cap P$; then $\mathbf{c}_n \in K_n$.
The set P is closed by Definition 3.4.1(ii). The ball $\overline{B}_{r_n}(\mathbf{c}_n)$ is closed by Corollary 3.1.18.
Set K_n is closed by Theorem 3.1.19(ii) and bounded. Hence K_n is compact by Theorem 4.2.5, direction $(i) \Rightarrow (ii)$.
The inclusion $K_{n+1} \subseteq K_n$ holds by stage (iii). Choose $\mathbf{z} \in \bigcap_{n=1}^{\infty} K_n$ by Corollary 4.1.10.
Then $\mathbf{z} \in K_1 \subseteq P$ and $(\forall n \in \mathbb{N})\,[\mathbf{z} \in K_{n+1} \wedge \mathbf{x}_n \notin K_{n+1}]$, so $(\forall n \in \mathbb{N})\,\mathbf{z} \neq \mathbf{x}_n$.
This contradicts $P = \{\mathbf{x}_n : n \in \mathbb{N}\}$. Hence P is uncountable by Definition 6.2.4(iv)–(v).

Structural Commentary: The enumeration determines nested compact balls whose nth refinement excludes $\mathbf{x}_n$.
Key Insight: Perfectness selects centers; compactness preserves one point through every exclusion. □

Figure 136 depicts stage (iii) on the real line.

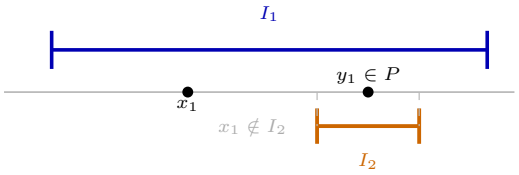

Figure 136: A closed interval $I_2 \subseteq I_1$ centered at $y_1 \in P$ avoids x_1.

Corollary 4.4.2. *Nondegenerate Intervals and the Real Line Are Uncountable*
Let $J \subseteq \mathbb{R}$ be an interval.

(i) If J contains two distinct points, then J is uncountable.
(ii) $\mathbb{R}$ is uncountable.

Proof.

(i) Choose $u < v$ in J, $I := [u, v] \subseteq J$ by Definition 3.1.8(ii). Set I is nonempty 1-cell, hence compact by Theorem 4.2.3.
It is closed by Theorem 4.1.6. Fix $x \in I$ and $\varepsilon > 0$. If $x < v$, put $y := x + \frac{1}{2}\min\{\varepsilon, v - x\}$.
If $x = v$, put $y := v - \frac{1}{2}\min\{\varepsilon, v - u\}$. In either case, $y \in I \cap (B_\varepsilon(x) \setminus \{x\})$. Thus I is perfect by Definition 3.4.1(ii).
The set I is uncountable by Theorem 4.4.1 with $k = 1$.
At-most-countability of J imply that of $I \subseteq J$ by Proposition 6.2.12(i), contradiction; J uncountable by Definition 6.2.4(iv)–(v).
(ii) Since $\mathbb{R}$ is an interval and contains the distinct points 0 and 1, it is uncountable by (i).

Structural Commentary: A closed subinterval transfers Euclidean perfect-set uncountability to its ambient interval.
Key Insight: Each endpoint is approached from within the closed interval. □

Definition 4.4.3. *Cantor Set*
Set $\mathcal{I}_0 := \{[0,1]\}$. For $n \in \mathbb{N}_0$, define $\mathcal{I}_{n+1} := \bigcup_{[a,b]\in\mathcal{I}_n} \left\{\left[a, \frac{2a+b}{3}\right], \left[\frac{a+2b}{3}, b\right]\right\}$. Put $E_n := \bigcup_{I\in\mathcal{I}_n} I$. The *Cantor set* is $P := \bigcap_{n=0}^{\infty} E_n$.

Figure 137 displays the first four stages in Definition 4.4.3.

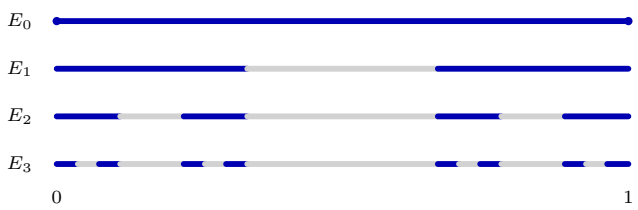

Figure 137: The retained intervals shrink from E_0 through E_3, while the middle-third gaps persist.

Proposition 4.4.4. *Basic Topological Properties of the Cantor Set*
Let $(\mathcal{I}_n)_{n\in\mathbb{N}_0}$, $(E_n)_{n\in\mathbb{N}_0}$, and P be as in Definition 4.4.3.
(i) $(\forall n \in \mathbb{N}_0)$, *the family $\mathcal{I}_n$ consists of 2^n pairwise disjoint compact intervals of length 3^{-n}.*
(ii) $(\forall n \in \mathbb{N}_0)\, E_{n+1} \subseteq E_n$.
(iii) The set P is nonempty and compact.
(iv) The set P contains no interval with two distinct points.
(v) The set P is perfect.

Proof.
(i) For $m \in \mathbb{N}$, let Q_m denote claim (i) at $n = m-1$. The statement Q_1 holds by $\mathcal{I}_0 = \{[0,1]\}$.
Under Q_m, each $[a,b] \in \mathcal{I}_{m-1}$ has two disjoint compact children in $\mathcal{I}_m$, each of length $\frac{b-a}{3} = 3^{-m}$.
Children of distinct parents are disjoint, so $\#\mathcal{I}_m = 2 \cdot 2^{m-1} = 2^m$ and $Q_m \Rightarrow Q_{m+1}$.
Therefore $(\forall m \in \mathbb{N})\, Q_m$ by Theorem 2.6.2, which establishes (i).
(ii) $(\forall n \in \mathbb{N}_0)\, E_{n+1} \subseteq E_n$ by Definition 4.4.3.
(iii) Fix $n \in \mathbb{N}_0$. The set E_n is nonempty by (i) and Definition 4.4.3. Each $I \in \mathcal{I}_n$ is closed by (i) and Theorem 4.1.6.
Hence E_n is closed by Theorem 3.1.19(iv). The inclusion $E_n \subseteq E_0 = [0,1]$ holds by (ii), so E_n is bounded.
Hence E_n is compact by Theorem 4.2.5, direction $(i) \Rightarrow (ii)$. Since $E_1 \subseteq E_0$, $P = \bigcap_{n=1}^{\infty} E_n \neq \emptyset$ by Corollary 4.1.10.
The set P is closed by Theorem 3.1.19(ii). Since $P \subseteq E_0$ and E_0 is compact, P is compact by Theorem 4.1.7.
(iv) Suppose $J \subseteq P$ is an interval containing $u < v$. Choose $n \in \mathbb{N}$ with $3^{-n} < v - u$ by Theorem 2.2.10(v) with $x = \frac{1}{3}$.
Choose $I_u, I_v \in \mathcal{I}_n$ with $u \in I_u$ and $v \in I_v$ by Definition 4.4.3. Intervals I_u and I_v are distinct by (i), $3^{-n} < v - u$.
Order $\mathcal{I}_n$ from left to right; then I_u precedes I_v by $u < v$. (i). Choose adjacent $[a,b], [c,d]$ in block from I_u to I_v with $b < c$.
Then $w := \frac{b+c}{2} \in (u,v) \setminus E_n$. But $w \in J \subseteq P \subseteq E_n$ by Definition 3.1.8(ii), a contradiction.
(v) Fix $x \in P$ and $\varepsilon > 0$. Choose $n \in \mathbb{N}$ with $3^{-n} < \varepsilon$ by Theorem 2.2.10(v) with $x = \frac{1}{3}$.
The point x lies in a unique $I_n(x) \in \mathcal{I}_n$ by (i) and Definition 4.4.3. Choose an endpoint $y \neq x$ of $I_n(x)$.
One has $(\forall m \in \mathbb{N}_0)\, [m \leq n \Rightarrow y \in E_m]$ by (ii). Each parent endpoint remains in one child by Definition 4.4.3.
Hence $(\forall m \in \mathbb{N}_0)\, [m \geq n \Rightarrow y \in E_m]$ by Theorem 2.6.2 with index shift $j = m - n + 1$.
Thus $y \in P$ and $0 < |x - y| \leq 3^{-n} < \varepsilon$ by (i). Hence $x \in P^{\mathrm{acc}}$ by Definition 3.1.2(v).
The set P is closed by (iii) and Theorem 4.1.6. Hence P is perfect by Definition 3.4.1(ii).
Structural Commentary: Induction establishes stage geometry; nested compactness and endpoint persistence control the limit.
Key Insight: Each stage shrinks components while preserving endpoints and opening gaps. □

4.5 Complete Metric Spaces and Completions

Completeness asks whether Cauchy approximations converge inside the ambient space.
This subsection records standard complete spaces, the contraction principle, and completion by Cauchy sequences.

Example 4.5.1. *Finite-Dimensional Complete Metric Spaces*
Let $k \in \mathbb{N}$. For $\mathbf{z}, \mathbf{w} \in \mathbb{C}^k$, set $d_2(\mathbf{z}, \mathbf{w}) := \left(\sum_{j=1}^{k} |z_j - w_j|^2\right)^{1/2}$. Then $(\mathbb{R}^k, d_2)$, $(\mathbb{C}^k, d_2)$ are complete; case $k = 1$ includes $\mathbb{R}$ and $\mathbb{C}$.

Proof. Space $\mathbb{R}^k$ is complete by Corollary 4.2.8. Write $z_j = x_j + iy_j$, $w_j = u_j + iv_j$, define $F : \mathbb{C}^k \to \mathbb{R}^{2k}$ by $F\mathbf{z} := (x_1, y_1, \ldots, x_k, y_k)$.
$d_2(F\mathbf{z}, F\mathbf{w})^2 = \sum_{j=1}^{k} [(x_j - u_j)^2 + (y_j - v_j)^2] = d_2(\mathbf{z}, \mathbf{w})^2$ by Definition 1.2.8.
The coordinate map F is bijective, and the distance identity makes it an isometry by Definition 3.1.11(i).
If $(\mathbf{z}_n)$ is Cauchy in $\mathbb{C}^k$, then $(F\mathbf{z}_n)$ is Cauchy in $\mathbb{R}^{2k}$ and converges to some $\mathbf{u} \in \mathbb{R}^{2k}$.
The identity $d_2(\mathbf{z}_n, F^{-1}\mathbf{u}) = d_2(F\mathbf{z}_n, \mathbf{u}) \to 0$ implies $\mathbf{z}_n \to F^{-1}\mathbf{u}$. Hence $\mathbb{C}^k$ is complete by Definition 4.1.23.
Structural Commentary: The coordinate isometry transfers real finite-dimensional completeness to complex coordinates.
Key Insight: Separating real and imaginary parts preserves the Euclidean distance. □

Definition 4.5.2. *Contraction*
Let (X, d) be a metric space. A map $\varphi : X \to X$ is a *contraction* iff $(\exists c \in [0,1))(\forall x, y \in X)\, d(\varphi(x), \varphi(y)) \leq c\, d(x, y)$.

Theorem 4.5.3. *Contraction Principle*
If X is a nonempty complete metric space and $\varphi : X \to X$ is a contraction, then $(\exists!\, x \in X)\, \varphi(x) = x$.

Proof.

(i) *Iteration estimate*: Choose $x_0 \in X$ and a contraction constant $c \in [0,1)$ by Definition 4.5.2.
Set $x_{n+1} := \varphi(x_n)$. The identity $d(x_1,x_0) = c^0 d(x_1,x_0)$ is the induction basis.
If $d(x_{n+1},x_n) \le c^n d(x_1,x_0)$, then $d(x_{n+2},x_{n+1}) \le c\, d(x_{n+1},x_n) \le c^{n+1} d(x_1,x_0)$ by the contraction inequality.
Hence $(\forall n \in \mathbb{N}_0)\, d(x_{n+1},x_n) \le c^n d(x_1,x_0)$ by Theorem 2.6.2 with index $n+1$.

(ii) *Cauchy control*: If $0 \le n < m$, then $d(x_n,x_m) \le \sum_{j=n}^{m-1} d(x_j,x_{j+1})$ by the triangle inequality.
By (i), $\sum_{j=n}^{m-1} d(x_j,x_{j+1}) \le d(x_1,x_0) \sum_{j=n}^{m-1} c^j$. The bound $d(x_n,x_m) \le \frac{c^n}{1-c} d(x_1,x_0)$ follows by $(1-c)\sum_{j=n}^{m-1} c^j = c^n - c^m \le c^n$.
The right side tends to 0 by Theorem 2.2.10(v) and Theorem 2.1.5(ii). Hence (x_n) is Cauchy by Definition 4.1.14.

(iii) *Fixed-point existence*: By Definition 4.1.23, choose $x \in X$ with $x_n \to x$. Bound $d(\varphi(x_n),\varphi(x)) \le c\, d(x_n,x)$ implies $\varphi(x_n) \to \varphi(x)$.
The shifted sequence satisfies $x_{n+1} \to x$ by Corollary 3.4.7. Since $\varphi(x_n) = x_{n+1}$, $\varphi(x) = x$ by Theorem 3.4.6(ii).

(iv) *Uniqueness*: If $\varphi(x) = x$ and $\varphi(y) = y$, then $d(x,y) \le c\, d(x,y)$, hence $d(x,y) = 0$ and $x = y$.

Structural Commentary: Iteration turns the contraction estimate into a geometric Cauchy estimate.
Key Insight: Completeness supplies the limiting point; the strict constant $c < 1$ forces uniqueness. □

Figure 138 combines contraction of distances with convergence of the orbit.

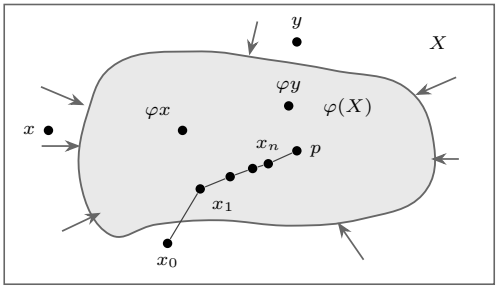

Figure 138: The iterates approach the fixed point p while φ contracts pairwise distances.

Perspective 4.5.4. *Fixed Points as Existence Proofs*
Theorem 4.5.3 solves $x = \varphi(x)$ by convergence of iterates in a complete space.
This viewpoint later motivates local inverse-function arguments but does not replace their hypotheses or proof.

Example 4.5.5. *A Noncomplete Metric Space*
The metric space $(\mathbb{Q},d)$ with $d(x,y) := |x-y|$ is not complete.

Proof. Let (r_n) be decimal truncations $1.4, 1.41, 1.414, 1.4142, \dots$ of $\sqrt{2}$. Then $(\forall n \in \mathbb{N})\, r_n \in \mathbb{Q}$, $r_n \to \sqrt{2}$ by Proposition 2.2.11(ii).
The sequence (r_n) is Cauchy in $\mathbb{R}$ by Theorem 4.1.20(i), hence also in $\mathbb{Q}$ with the inherited metric.
If $(\mathbb{Q},d)$ were complete, choose $r \in \mathbb{Q}$ with $r_n \to r$ by Definition 4.1.23. The same convergence holds in $\mathbb{R}$.
Thus $r = \sqrt{2}$ by Proposition 2.1.4(i), contrary to Proposition 1.1.13(i).
Structural Commentary: The decimal approximants are Cauchy before their missing limit is available inside $\mathbb{Q}$.
Key Insight: The inherited metric preserves Cauchy control but not the irrational limit. □

Perspective 4.5.6. *Cauchy Construction as a Parallel Model*
Example 4.5.5 exhibits a rational Cauchy sequence with a missing endpoint.
For the set $\mathcal{C}_{\mathbb{Q}}$ of rational Cauchy sequences, define $(a_n) \sim (b_n) \iff |a_n - b_n| \to 0$.
The quotient $\mathcal{C}_{\mathbb{Q}}/\!\sim$ motivates the construction below; only the proved completion theorem supplies later authority.

Figure 139 depicts the quotient relation and the embedded constant sequences.

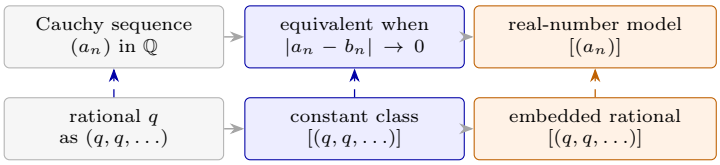

Figure 139: Equivalent rational Cauchy sequences determine one formal limit, while constant sequences embed $\mathbb{Q}$.

Definition 4.5.7. *Metric Completion*
A *completion* of a metric space (X,d) is a triple (Y,d_Y,τ) satisfying:

(i) (Y,d_Y) is complete;
(ii) $\tau : X \to Y$ is an isometry;
(iii) $\overline{\tau(X)}^{\,Y} = Y$.

Theorem 4.5.8. *Existence and Uniqueness of Metric Completions*
Let (X,d) be a metric space.

(i) The space (X,d) has a completion.
(ii) Let (Y,d_Y,τ) and (Z,d_Z,σ) be completions of (X,d). Then $(\exists!\, \Phi : Y \to Z)$ such that Φ is a bijective isometry and $\Phi \circ \tau = \sigma$.

Proof.

(i) *Existence*:

(a) *Pseudometric*: Let $\mathrm{CS}(X)$ be the set of Cauchy sequences in X.
For $a=(a_n), b=(b_n)\in \mathrm{CS}(X)$, set $D(a,b):=\lim_{n\to\infty} d(a_n,b_n)$. This limit exists by Proposition 4.1.22(ii).
The map D has zero diagonal, is nonnegative, and is symmetric by the corresponding properties of d.
Fix $a,b,c\in\mathrm{CS}(X)$. Then $(\forall n\in\mathbb{N})\, d(a_n,c_n)\le d(a_n,b_n)+d(b_n,c_n)$ by triangle inequality.
Hence $D(a,c)\le D(a,b)+D(b,c)$ by Proposition 2.1.4(iii) and Theorem 2.1.5(i). Thus D is a pseudometric.

(b) *Quotient and embedding*: Define $a\sim b \iff D(a,b)=0$ and set $\widehat{X}:=\mathrm{CS}(X)/\!\sim$. Put $\widehat{d}([a],[b]):=D(a,b)$.
Then $\sim$ is an equivalence relation and $\widehat{d}$ is a well-defined metric by Proposition 3.1.12.
For $x\in X$, let $\iota(x)$ be the class of the constant sequence with value x.
The identity $\widehat{d}(\iota(x),\iota(y))=d(x,y)$ makes $\iota: X\to\widehat{X}$ an isometry by Definition 3.1.11(i).

(c) *Density*: Fix $[a]\in\widehat{X}$ and $\varepsilon>0$. Choose $N_0\in\mathbb{N}$ with $(\forall m,n\ge N_0)\, d(a_m,a_n)<\frac{\varepsilon}{2}$.
For $N\ge N_0$, $\widehat{d}(\iota(a_N),[a])\le\frac{\varepsilon}{2}<\varepsilon$ by Proposition 2.1.4(iii). Hence $\iota(a_N)\to[a]$, and $\overline{\iota(X)}^{\widehat{X}}=\widehat{X}$ by Proposition 3.4.16.

(d) *Completeness*: Let (ξ_k) be Cauchy in $\widehat{X}$. By (c), choose $x_k\in X$ with $\widehat{d}(\iota(x_k),\xi_k)<\frac{1}{k}$.
The bound $d(x_k,x_l)\le\frac{1}{k}+\widehat{d}(\xi_k,\xi_l)+\frac{1}{l}$ follows by the triangle inequality.
Hence (x_k) is Cauchy by Definition 4.1.14 and Theorem 2.2.10(i).
Set $a:=(x_k)\in\mathrm{CS}(X)$. Then $\widehat{d}(\xi_k,[a])<\frac{1}{k}+\lim_{j\to\infty} d(x_k,x_j)$ by the triangle inequality.
Given $\varepsilon>0$, choose $K\in\mathbb{N}$ with $\frac{1}{K}<\frac{\varepsilon}{2}$ and $(\forall k,j\ge K)\, d(x_k,x_j)<\frac{\varepsilon}{2}$.
For $k\ge K$, the final limit is at most $\frac{\varepsilon}{2}$ by Proposition 2.1.4(iii). Hence $\widehat{d}(\xi_k,[a])<\varepsilon$.
Thus $\xi_k\to[a]$, and $(\widehat{X},\widehat{d},\iota)$ is a completion of (X,d) by Definitions 4.1.23 and 4.5.7.

(ii) *Uniqueness*:

(a) *Dense-copy isometry*: Let (Y,d_Y,τ) and (Z,d_Z,σ) be completions of (X,d).
If $\tau x=\tau x'$, then $d(x,x')=d_Y(\tau x,\tau x')=0$, so $x=x'$; similarly, σ is injective.
Define $\rho_0:\tau(X)\to\sigma(X)$ by $\rho_0(\tau x):=\sigma x$. Iinjectivity of τ makes ρ_0 well defined, injectivity of σ makes ρ_0 injective.
The identity $\rho_0(\tau x)=\sigma x$ implies $\rho_0(\tau(X))=\sigma(X)$, so ρ_0 is surjective.
For $x,x'\in X$, $d_Z(\rho_0(\tau x),\rho_0(\tau x'))=d(x,x')=d_Y(\tau x,\tau x')$. Hence ρ_0 is a bijective isometry by Definition 3.1.11(i).

(b) *Extension*: Fix $y\in Y$. Choose $(y_n)\subseteq\tau(X)$ with $y_n\to y$ by Proposition 3.4.16.
The sequence $(\rho_0(y_n))$ is Cauchy in Z by the isometry property, so it converges by completeness of Z.
Define $\Phi(y):=\lim_{n\to\infty}\rho_0(y_n)$. Suppose $(z_n)\subseteq\tau(X)$ also satisfies $z_n\to y$.
The sequence $(\rho_0(z_n))$ is Cauchy by the isometry property and converges by completeness of Z.
Set $p:=\lim_{n\to\infty}\rho_0(y_n)$ and $q:=\lim_{n\to\infty}\rho_0(z_n)$. Then $d_Z(\rho_0(y_n),\rho_0(z_n))=d_Y(y_n,z_n)\le d_Y(y_n,y)+d_Y(y,z_n)\to 0$.
By triangle inequality, $d_Z(p,q)\le d_Z(p,\rho_0(y_n))+d_Z(\rho_0(y_n),\rho_0(z_n))+d_Z(\rho_0(z_n),q)\to 0$. Thus $p=q$, so Φ is well defined.

(c) *Distance preservation*: For $y,y'\in Y$, choose $(y_n),(y_n')\subseteq\tau(X)$ converging to y,y', respectively.
Set $u_n:=\rho_0(y_n)$, $v_n:=\rho_0(y_n')$. Inequality $|d_Y(y_n,y_n')-d_Y(y,y')|\le d_Y(y_n,y)+d_Y(y_n',y')$ implies $d_Y(y_n,y_n')\to d_Y(y,y')$.
$|d_Z(u_n,v_n)-d_Z(\Phi(y),\Phi(y'))|\le d_Z(u_n,\Phi(y))+d_Z(v_n,\Phi(y'))$ by triangle inequality. Hence $d_Z(u_n,v_n)\to d_Z(\Phi(y),\Phi(y'))$.
$(\forall n\in\mathbb{N})\, d_Z(u_n,v_n)=d_Y(y_n,y_n')$. Hence $d_Z(\Phi(y),\Phi(y'))=d_Y(y,y')$ by Proposition 2.1.4(i).
Thus Φ is an isometry by Definition 3.1.11(i).
For $x\in X$, the constant sequence $y_n=\tau x$ in the extension satisfies $\Phi(\tau x)=\sigma x$. Hence $\Phi\circ\tau=\sigma$.

(d) *Surjectivity*: If $(\Phi(y_n))$ is Cauchy in $\Phi(Y)$, then (y_n) is Cauchy in Y and converges to some $y\in Y$.
The identity $d_Z(\Phi(y_n),\Phi(y))=d_Y(y_n,y)\to 0$ implies $\Phi(y_n)\to\Phi(y)$, so $\Phi(Y)$ is complete.
Hence $\Phi(Y)$ is closed in Z by Proposition 4.1.26(i).
The inclusions $\sigma(X)\subseteq\Phi(Y)$ and $Z=\overline{\sigma(X)}^{Z}$ imply $Z\subseteq\Phi(Y)$ by Theorem 3.1.22(iii).
Thus Φ is surjective; since every isometry is injective, it is bijective.

(e) *Uniqueness*: Let $\Psi: Y\to Z$ be another isometry satisfying $\Psi\circ\tau=\sigma$, and fix $y\in Y$.
Choose $(y_n)\subseteq\tau(X)$ with $y_n\to y$ by Proposition 3.4.16.
Then $\Phi(y_n)=\Psi(y_n)$ and $d_Z(\Phi(y),\Psi(y))\le d_Y(y,y_n)+d_Y(y_n,y)\to 0$, so $\Phi(y)=\Psi(y)$. Hence $\Phi=\Psi$.

Structural Commentary: The quotient installs formal limits; density extends the copy of X uniquely between completions.
Key Insight: Cauchy classes provide existence; dense agreement forces uniqueness. □

Figure 140 summarizes the unique extension between two completions.

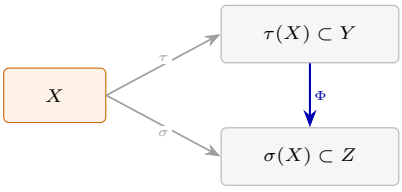

Figure 140: The dense-copy isometry $\sigma\tau^{-1}$ extends uniquely to $\Phi: Y\cong Z$; adapted from Roman (2008, Ch. 12, Fig. 12.4).

5 Connectedness and Continuity

5.1 Connectedness and Continuity

Connectedness forbids separated decompositions; continuity transfers local and global topological structure.
This subsection develops metric limits, compact-image results, the intermediate value theorem, and paths.

Definition 5.1.1. *Separated and Connected Sets*
Let X be a metric space.

(i) Subsets $A,B\subseteq X$ are *separated* iff $A\cap\overline{B}=\overline{A}\cap B=\emptyset$.

(ii) A subset $E \subseteq X$ is *connected* iff no nonempty separated $A, B \subseteq E$ satisfy $E = A \cup B$.
(iii) A subset $U \subseteq X$ is *clopen* iff it is both open and closed.

Theorem 5.1.2. *Connected Subsets of the Real Line*
A set $E \subseteq \mathbb{R}$ is connected iff $(\forall x, y \in E)\,[x < y \Longrightarrow [x, y] \subseteq E]$.

Proof.
($\Rightarrow$) If the interval property fails, choose $x, y \in E$ and $x < z < y$ with $z \notin E$. Set $A := E \cap (-\infty, z)$ and $B := E \cap (z, \infty)$.
Then $A, B \neq \emptyset$ and $E = A \cup B$. The inclusions $\overline{A} \subseteq (-\infty, z]$ and $\overline{B} \subseteq [z, \infty)$ imply $A \cap \overline{B} = \overline{A} \cap B = \emptyset$.
Thus A, B separate E, contrary to connectedness.
($\Leftarrow$) Assume the interval property and suppose $E = A \cup B$ for nonempty separated A, B.
Choose $x \in A$ and $y \in B$; interchange A, B if necessary so that $x < y$. Set $S := A \cap [x, y]$ and $z := \sup S$.
Then $z \in \overline{A}$ by Theorem 3.3.5(i). Hence $z \notin B$ by separatedness, and $z < y$ because $y \in B$.
If $z \notin A$, then $z \in [x, y] \setminus E$, contrary to the interval property. If $z \in A$, then $z \notin \overline{B}$; choose $r > 0$ with $B_r(z) \cap B = \emptyset$.
Set $z_1 := z + \frac{1}{2}\min\{r, y - z\}$. Then $z < z_1 < y$, $z_1 \notin B$, and $z_1 \notin A$ by $z = \sup S$.
Thus $z_1 \in [x, y] \setminus E$, again contrary to the interval property.

Structural Commentary: A gap induces a separation; a separation produces a gap at the supremum of one side.
Key Insight: The supremum locates the missing boundary point between separated pieces. □

Method 5.1.3. *Compactness and Connectedness Proof Patterns*
(i) *Open-cover compactness*: to prove K compact, start with open cover $\{G_\alpha\}_{\alpha \in A}$. Find finite $F \subseteq A$ with $K \subseteq \bigcup_{\alpha \in F} G_\alpha$.
(ii) *Euclidean closed-bounded compactness*: to prove a closed bounded set $E \subseteq \mathbb{R}^k$ compact, choose a compact k-cell $I \supseteq E$. Then E is compact by Theorem 4.1.7.
(iii) *Connectedness on $\mathbb{R}$*: replace connectedness of $E \subseteq \mathbb{R}$ by $(\forall x, y \in E)\,[x < y \Longrightarrow [x, y] \subseteq E]$.

Definition 5.1.4. *Limit of a Function Between Metric Spaces*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, let $E \subseteq X$, let $f : E \to Y$, and let p be a limit point of E.
For $q \in Y$, $\lim_{x \to p} f(x) = q$ iff $(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in E)\,[0 < d_X(x, p) < \delta \Longrightarrow d_Y(f(x), q) < \varepsilon]$.
The point p need not belong to E, and $f(p)$ is irrelevant when it does.
Key Insight: A punctured interval determines whether approach is two-sided, right-sided, or left-sided.

Figure 141 illustrates the punctured-neighborhood test in the definition.

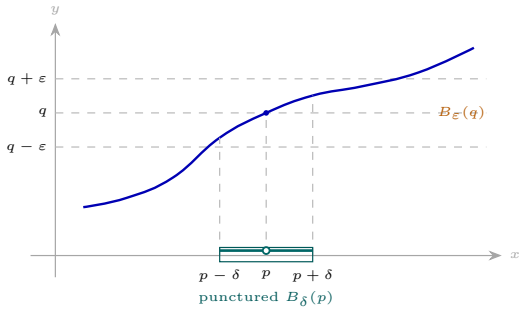

Figure 141: The epsilon–delta geometry for $\lim_{x \to p} f(x) = q$.

Theorem 5.1.5. *Sequential Characterization of Function Limits*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, let $E \subseteq X$, let $f : E \to Y$, and let p be a limit point of E.
For $q \in Y$, $\lim_{x \to p} f(x) = q$ iff $f(p_n) \to q$ whenever $(p_n) \subseteq E \setminus \{p\}$ and $p_n \to p$.

Proof.
($\Rightarrow$) Let $(p_n) \subseteq E \setminus \{p\}$ satisfy $p_n \to p$, and fix $\varepsilon > 0$.
Choose $\delta > 0$ with $(\forall x \in E)\,[0 < d_X(x, p) < \delta \Longrightarrow d_Y(f(x), q) < \varepsilon]$ by Definition 5.1.4.
Eventually $0 < d_X(p_n, p) < \delta$, hence eventually $d_Y(f(p_n), q) < \varepsilon$. Thus $f(p_n) \to q$.
($\Leftarrow$) If the limit fails, choose $\varepsilon > 0$ with $(\forall \delta > 0)(\exists x \in E)\,[0 < d_X(x, p) < \delta \wedge d_Y(f(x), q) \geq \varepsilon]$.
Then p is a limit point of $A_\varepsilon := \{x \in E \setminus \{p\} : d_Y(f(x), q) \geq \varepsilon\}$.
Choose $(x_n) \subseteq A_\varepsilon$ with $x_n \to p$ by Theorem 3.4.6(iii). Since $(\forall n \in \mathbb{N})\, d_Y(f(x_n), q) \geq \varepsilon$, $f(x_n) \not\to q$.

Structural Commentary: The forward direction transfers a δ-bound; the reverse direction selects from the bad set.
Key Insight: Failure of one ε creates a sequence that defeats convergence. □

Corollary 5.1.6. *Uniqueness of Function Limits*
Let X, Y be metric spaces, $E \subseteq X$, $f : E \to Y$, p be a limit point of E. If f has a limit at p, then that limit is unique.

Proof. Let q and r be limits of f at p. Choose $(p_n) \subseteq E \setminus \{p\}$ with $p_n \to p$ by Theorem 3.4.6(iii).
Then $f(p_n) \to q$ and $f(p_n) \to r$ by Theorem 5.1.5. Hence $q = r$ by Theorem 3.4.6(ii).
Structural Commentary: Limit uniqueness is inherited from metric sequence uniqueness.
Key Insight: One approaching sequence cannot have two different limits. □

Theorem 5.1.7. *Algebra of Function Limits*
Let X be a metric space, $E \subseteq X$, p be a limit point of E. For $f, g : E \to \mathbb{C}$, assume $\lim_{x \to p} f(x) = A$ and $\lim_{x \to p} g(x) = B$. Then:

(i) $\lim_{x \to p} (f + g)(x) = A + B$;

(ii) $\lim_{x\to p}(fg)(x) = AB$;

(iii) *if* $B \neq 0$, *then* $\lim_{x\to p;\, g(x)\neq 0} \frac{f(x)}{g(x)} = \frac{A}{B}$;

(iv) *Let* $u, v : E \to \mathbb{R}^k$, $\lambda \in \mathbb{R}$. *Assume* $\lim_{x\to p} u(x) = a$, $\lim_{x\to p} v(x) = b$. *Then* $u + v \to a + b$, $\lambda u \to \lambda a$, *and* $u \cdot v \to a \cdot b$ *as* $x \to p$.

Proof. Let $(p_n) \subseteq E \setminus \{p\}$ satisfy $p_n \to p$. Then $f(p_n) \to A$ and $g(p_n) \to B$ by Theorem 5.1.5.

(i) $(f+g)(p_n) \to A + B$ by Theorem 2.1.5(i). Hence (i) by Theorem 5.1.5.

(ii) $(fg)(p_n) \to AB$ by Theorem 2.1.5(iii). Hence (ii) by Theorem 5.1.5.

(iii) Set $U := \{x \in E : g(x) \neq 0\}$. Choose $r > 0$ with $(\forall x \in E)\,[0 < d_X(x,p) < r \Longrightarrow |g(x) - B| < \frac{|B|}{2}]$ by $\lim_{x\to p} g(x) = B$.
Then $E \cap B_r(p) \setminus \{p\} \subseteq U$ by the reverse triangle inequality, so p is a limit point of U.
If $(q_n) \subseteq U \setminus \{p\}$ and $q_n \to p$, then $f(q_n) \to A$ and $g(q_n) \to B$ by Theorem 5.1.5.
Thus $\frac{f(q_n)}{g(q_n)} \to \frac{A}{B}$ by Theorem 2.1.5(iii),(iv). Hence (iii) by Theorem 5.1.5.

(iv) Under the hypotheses of (iv), $u(p_n) \to a$ and $v(p_n) \to b$ by Theorem 5.1.5.
Three vector limits hold by Theorem 3.4.13(ii)–(iv), with constant scalar sequence (λ). Hence (iv) by Theorem 5.1.5.

Structural Commentary: Sequential characterization transfers scalar and vector sequence algebra to function limits.
Key Insight: Every approaching sequence inherits the corresponding algebraic limit. □

Proposition 5.1.8. *Function Limits Are Local*
Let X, Y *be metric spaces,* $E \subseteq X$, $f : E \to Y$, p *be limit point of* E, $q \in Y$. $(\forall r > 0)\,[\lim_{x\to p;\, x\in E} f(x) = q \Longleftrightarrow \lim_{x\to p;\, x\in E\cap B_r(p)} f(x) = q]$.

Proof.

(⇒) Any δ witnessing the limit on E witnesses the limit on $E \cap B_r(p)$ by $E \cap B_r(p) \subseteq E$.

(⇐) Assume the restricted limit and fix $\varepsilon > 0$. Choose $\delta_0 > 0$ by the restricted limit.
Then $(\forall x \in E \cap B_r(p))\,[0 < d_X(x,p) < \delta_0 \Longrightarrow d_Y(f(x), q) < \varepsilon]$. Let $\delta := \min\{\delta_0, r\}$.
If $x \in E$ and $0 < d_X(x,p) < \delta$, then $x \in E \cap B_r(p)$, so $d_Y(f(x), q) < \varepsilon$.

Structural Commentary: Values away from p cannot affect the limit at p.
Key Insight: Shrink δ until every tested point lies in the chosen local neighborhood. □

Definition 5.1.9. *Continuity at a Point and on a Set*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, $E \subseteq X$, and $f : E \to Y$.

(i) For $p \in E$, f is *continuous at* p iff $(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in E)\,[d_X(x,p) < \delta \Longrightarrow d_Y(f(x), f(p)) < \varepsilon]$.

(ii) The map f is *continuous on* E iff $(\forall p \in E)$, f is continuous at p.

Corollary 5.1.10. *Every Map Is Continuous at an Isolated Point*
Let $(X, d_X), (Y, d_Y)$ *be metric spaces,* $E \subseteq X$, $f : E \to Y$, p *be isolated in* E. *Then* f *is continuous at* p.

Proof. Given $\varepsilon > 0$, choose $\delta > 0$ with $E \cap B_\delta(p) = \{p\}$. If $x \in E$ and $d_X(x,p) < \delta$, then $x = p$, so $d_Y(f(x), f(p)) = 0 < \varepsilon$.
Structural Commentary: Isolation reduces every sufficiently small domain neighborhood to the singleton $\{p\}$.
Key Insight: No distinct domain point can approach an isolated point. □

Corollary 5.1.11. *Discrete Domains Make Every Map Continuous*
Let X, Y *be metric spaces, and assume that* X *carries the discrete metric. Then every map* $f : X \to Y$ *is continuous.*

Proof. Fix $p \in X$. Then $B_{1/2}(p) = \{p\}$, p isolated; f continuous at p by Corollary 5.1.10; f continuous on X by Definition 5.1.9(ii).
Structural Commentary: In the discrete metric, no distinct point can approach p.
Key Insight: Pointwise continuity is automatic when small enough balls are singletons. □

For the case $E = X$, Figure 142 expresses continuity through nested input and output balls.

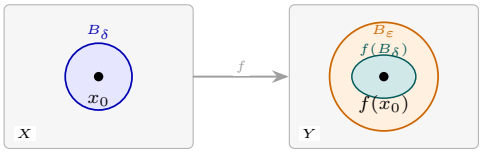

Figure 142: Nested balls encode continuity at x_0 for $E = X$; adapted from Roman (2008, Ch. 12, Fig. 12.2).

Theorem 5.1.12. *Continuity as Equality with the Function Limit*
Let X, Y *be metric spaces,* $E \subseteq X$, $p \in E$, *and* $f : E \to Y$. *If* p *is a limit point of* E, *then* f *is continuous at* p *iff* $\lim_{x\to p} f(x) = f(p)$.

Proof.

(⇒) Fix $\varepsilon > 0$, $\delta > 0$ by continuity at p. If $x \in E$, $0 < d_X(x,p) < \delta$, then $d_Y(f(x), f(p)) < \varepsilon$. Thus $\lim_{x\to p} f(x) = f(p)$.

(⇐) Fix $\varepsilon > 0$, $\delta > 0$ by $\lim_{x\to p} f(x) = f(p)$. If $x \in E \setminus \{p\}$, $d_X(x,p) < \delta$, then $d_Y(f(x), f(p)) < \varepsilon$; if $x = p$, distance is 0.

Structural Commentary: Continuity is the compatibility of the value at p with nearby values.
Key Insight: Limits ignore the point; continuity ties the point back in. □

Remark 5.1.13. *Adherent-Point Limit Convention*
The adherent-point convention takes $p \in \overline{E}$ and allows $x = p$ in the δ-test.
Definition 5.1.4 instead assumes p is a limit point and requires $0 < d_X(x,p)$.
If $p \in E$, the unpunctured convention tests $f(p)$ and forces $q = f(p)$.
For punctured limits, this condition is continuity at p by Theorem 5.1.12.
Key Insight: Punctured limits study nearby values; continuity adds compatibility with the value at the point.

Remark 5.1.14. *The Sequence-First View*
At limit points of a real domain, continuity is equivalent to $x_n \to p \Longrightarrow f(x_n) \to f(p)$.
This equivalence holds by Theorems 5.1.5 and 5.1.12. Continuity at isolated points holds by Corollary 5.1.10.

Theorem 5.1.15. *Sequential Characterization of Continuity*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, $f : X \to Y$, and $p \in X$. Then f is continuous at p iff $(x_n) \subseteq X$ and $x_n \to p$ imply $f(x_n) \to f(p)$.

Proof.
($\Rightarrow$) Assume f is continuous at p and let $x_n \to p$.
Given $\varepsilon > 0$, choose $\delta > 0$ such that $(\forall\, x \in X)\,[d_X(x,p) < \delta \Longrightarrow d_Y(f(x), f(p)) < \varepsilon]$.
Since $x_n \to p$, eventually $d_X(x_n, p) < \delta$, hence eventually $d_Y(f(x_n), f(p)) < \varepsilon$. Thus $f(x_n) \to f(p)$.
($\Leftarrow$) Suppose the sequential condition holds and f is not continuous at p.
Choose $\varepsilon_0 > 0$ with $(\forall \delta > 0)(\exists x \in X)\,[d_X(x,p) < \delta \wedge d_Y(f(x), f(p)) \geq \varepsilon_0]$.
Then p is a limit point of $A := \{x \in X : d_Y(f(x), f(p)) \geq \varepsilon_0\}$.
Choose $(x_n) \subseteq A$ with $x_n \to p$ by Theorem 3.4.6(iii). Since $(\forall n \in \mathbb{N})\, d_Y(f(x_n), f(p)) \geq \varepsilon_0$, $f(x_n) \nrightarrow f(p)$, a contradiction.
Structural Commentary: Failure of epsilon–delta continuity manufactures a bad sequence.
Key Insight: In metric spaces, sequences detect continuity at points. □

Theorem 5.1.16. *Composition of Continuous Functions*
Let $E \subseteq X$ and let $f : E \to Y$, $g : f(E) \to Z$ be maps between metric spaces.
If f is continuous at $p \in E$ and g is continuous at $f(p)$, then $g \circ f$ is continuous at p.

Proof. Given $\varepsilon > 0$, choose $\eta > 0$ by continuity of g at $f(p)$. Choose $\delta > 0$ by continuity of f at p with tolerance η.
If $x \in E$ and $d_X(x,p) < \delta$, then $d_Y(f(x), f(p)) < \eta$, hence $d_Z(g(f(x)), g(f(p))) < \varepsilon$.
Structural Commentary: The output tolerance for g becomes the input tolerance for f.
Key Insight: Continuity composes by passing epsilon through an intermediate eta. □

Theorem 5.1.17. *The Distance Function Is Continuous*
Let (X, d) be a metric space and $D : X \times X \to \mathbb{R}$, $D(x,y) := d(x,y)$.
For $d_\times \in \{d_{\text{sum}}, d_E, d_{\max}\}$ on $X \times X$, $D : (X \times X, d_\times) \to \mathbb{R}$ is continuous.
Equivalently, $x_n \to x$ and $y_n \to y$ in X imply $d(x_n, y_n) \to d(x,y)$.

Proof. For $x, y, u, v \in X$, $|D(x,y) - D(u,v)| \leq d(x,u) + d(y,v)$ by triangle inequality. Thus $x_n \to x$, $y_n \to y$ imply $D(x_n, y_n) \to D(x,y)$.
Convergence under d_{sum} is equivalent to coordinate convergence because both summands are nonnegative.
Continuity for d_{sum} follows by Theorem 5.1.15. Other product metrics have same convergent sequences by Proposition 3.4.15(iii).
Structural Commentary: Distance changes continuously when both endpoints move.
Key Insight: The triangle inequality controls distance error by endpoint errors. □

Theorem 5.1.18. *Continuity via Open Preimages*
For metric spaces X, Y, a map $f : X \to Y$ is continuous iff $f^{-1}(V)$ is open in X whenever $V \subseteq Y$ is open.

Proof.
($\Rightarrow$) If $p \in f^{-1}(V)$, V is open, choose $\varepsilon > 0$ with $B_\varepsilon(f(p)) \subseteq V$. By continuity, $\exists\ \delta > 0$ with $B_\delta(p) \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open.
($\Leftarrow$) For $p \in X$ and $\varepsilon > 0$, the ball $V = B_\varepsilon(f(p))$ is open by Theorem 3.1.3.
Since $p \in f^{-1}(V)$ and $f^{-1}(V)$ is open, some $B_\delta(p) \subseteq f^{-1}(V)$; this is continuity at p.
Structural Commentary: Continuity is exactly preservation of openness under inverse image.
Key Insight: Images are not Boolean-exact, but preimages are; continuity belongs to preimages. □

Proposition 5.1.19. *Continuity on a Relative Domain*
Let (X, d_X) and (Y, d_Y) be metric spaces, let $E \subseteq X$, and let $f : E \to Y$. The following are equivalent:
(i) $f : (E, d_X|_{E\times E}) \to (Y, d_Y)$ is continuous.
(ii) The set $f^{-1}(V)$ is open relative to E whenever $V \subseteq Y$ is open.
(iii) If $V \subseteq Y$ is open, then $f^{-1}(V) = E \cap G$ for an open set $G \subseteq X$.

Proof. Every constant map $h : X \to \mathbb{R}$ is continuous by Definition 5.1.9, since $|h(x) - h(p)| = 0$.
(i) $\Rightarrow$ *(ii)* For open $V \subseteq Y$, the set $f^{-1}(V)$ is open in the metric space E by Theorem 5.1.18.
(ii) $\Rightarrow$ *(iii)* For open $V \subseteq Y$, choose open $G \subseteq X$ with $f^{-1}(V) = E \cap G$ by Theorem 3.2.3.
(iii) $\Rightarrow$ *(i)* For open $V \subseteq Y$, $f^{-1}(V) = E \cap G$ with G open in X, hence $f^{-1}(V)$ is open in E.
Therefore f is continuous by Theorem 5.1.18 on E.
Structural Commentary: Open preimages are relative opens in E.
Key Insight: Relative continuity says ordinary open sets in X are seen through intersection with E. □

Figure 143 visualizes the inverse-image characterization of continuity.

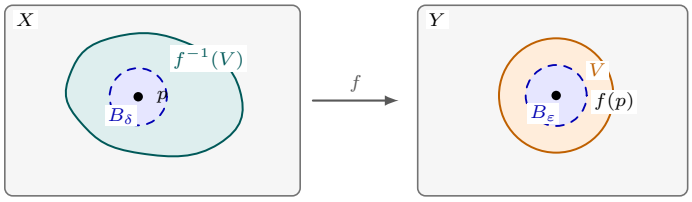

Figure 143: Continuity pulls open sets back to open sets.

Corollary 5.1.20. *Continuity via Closed Preimages*
Let X, Y be metric spaces, $f : X \to Y$. Then f is continuous iff $f^{-1}(C)$ is closed in X whenever $C \subseteq Y$ is closed.

Proof.
($\Rightarrow$) For closed $C \subseteq Y$, $X \setminus f^{-1}(C) = f^{-1}(Y \setminus C)$ is open by Theorem 5.1.18. Hence $f^{-1}(C)$ is closed.
($\Leftarrow$) For open $V \subseteq Y$, $X \setminus f^{-1}(V) = f^{-1}(Y \setminus V)$ is closed by hypothesis; hence $f^{-1}(V)$ is open.
Structural Commentary: Closed-set continuity is the complement form of open-set continuity.
Key Insight: $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ transfers open and closed formulations. □

Perspective 5.1.21. *Three Views of Continuity*
For metric-space maps, continuity has three equivalent forms: ε–δ control, sequence preservation, and open inverse images.
The owners are Definition 5.1.9 and Theorems 5.1.15, 5.1.18.
Key Insight: Continuity sends near points forward and pulls open sets back.

Proposition 5.1.22. *Distance-to-Set Is Continuous*
Let (X, d) be a metric space, let $\emptyset \neq E \subseteq X$, and define $f : X \to \mathbb{R}$ by $f(p) := \operatorname{dist}(p, E)$.
(i) f is 1-Lipschitz: $(\forall\, p, q \in X)\ |f(p) - f(q)| \leq d(p, q)$.
(ii) The map f is continuous.
(iii) $f^{-1}(\{0\}) = \overline{E}$.

Proof.
(i) For $p, q \in X$ and $e \in E$, $d(p, e) \leq d(p, q) + d(q, e)$. Then $f(p) - f(q) \leq d(p, q)$ by taking $\inf_{e \in E}$.
Interchanging p, q, $f(q) - f(p) \leq d(p, q)$. Thus $|f(p) - f(q)| \leq d(p, q)$.
(ii) Fix $p \in X$ and $\varepsilon > 0$, and set $\delta := \varepsilon$. If $d(p, q) < \delta$, then $|f(p) - f(q)| < \varepsilon$ by (i). Hence f is continuous on X.
(iii) For $p \in X$, $p \in f^{-1}(\{0\})$ iff $\operatorname{dist}(p, E) = 0$ iff $p \in \overline{E}$ by Proposition 3.1.26.
Structural Commentary: Distance to a fixed set changes no faster than the point itself moves.
Key Insight: The triangle inequality turns nearest-set distance into a continuous detector of closure. □

Corollary 5.1.23. *Open and Closed Sets by Distance Functions*
Let X be a metric space.
(i) A set $O \subseteq X$ is open iff $O = \{x \in X : f(x) > 0\}$ for a continuous map $f : X \to \mathbb{R}$.
(ii) A set $F \subseteq X$ is closed iff $F = g^{-1}(\{0\})$ for a continuous map $g : X \to \mathbb{R}$.

Proof.
(i) ($\Rightarrow$) If $O = X$, take $f \equiv 1$; this map is continuous and $O = \{x \in X : f(x) > 0\}$.
If $O \neq X$, set $f(x) := \operatorname{dist}(x, O^c)$. Then f is continuous by Proposition 5.1.22(ii).
Since O^c is closed, $f^{-1}(\{0\}) = O^c$ by Proposition 5.1.22(iii). Hence $O = \{x \in X : f(x) > 0\}$.
(i) ($\Leftarrow$) If $O = \{x \in X : f(x) > 0\}$ for continuous $f : X \to \mathbb{R}$, then $O = f^{-1}((0, \infty))$ is open by Theorem 5.1.18.
(ii) ($\Rightarrow$) If $F = \emptyset$, take $g \equiv 1$; this map is continuous and $F = g^{-1}(\{0\})$. If $F \neq \emptyset$, set $g(x) := \operatorname{dist}(x, F)$.
Then g is continuous by Proposition 5.1.22(ii). Since F is closed, $g^{-1}(\{0\}) = \overline{F} = F$ by Proposition 5.1.22(iii).
(ii) ($\Leftarrow$) If $F = g^{-1}(\{0\})$ for a continuous $g : X \to \mathbb{R}$, then F is closed by Corollary 5.1.20.
Structural Commentary: Metric spaces encode open and closed sets by continuous real-valued distance functions.
Key Insight: A set is closed exactly when its distance-zero locus has no extra points. □

Definition 5.1.24. *Homeomorphism*
Let X, Y be metric spaces. A *homeomorphism* from X to Y is a bijection $f : X \to Y$ such that both f and f^{-1} are continuous.
If such a map exists, X and Y are *homeomorphic*, written $X \simeq Y$.

Proposition 5.1.25. *Homeomorphisms Preserve Topology*
If $f : X \to Y$ is a homeomorphism, then $U \mapsto f(U)$ bijects open subsets of X with open subsets of Y.

Proof. If $U \subseteq X$ is open, then $f(U) = (f^{-1})^{-1}(U)$ is open in Y by continuity of f^{-1}.
If $V \subseteq Y$ is open, then $f^{-1}(V)$ is open in X by continuity of f. These assignments are inverse because f is bijective.
Structural Commentary: A homeomorphism preserves open sets in both directions.
Key Insight: Topological sameness is continuity plus continuous reversibility. □

Theorem 5.1.26. *Basic Properties of Metric Isometries*
Let $f : (X, d_X) \to (Y, d_Y)$ be an isometry.
(i) f is injective.
(ii) f is continuous.
(iii) $f^{-1} : (f(X), d_Y|_{f(X) \times f(X)}) \to (X, d_X)$ is an isometry.

Proof.
(i) If $f(x) = f(y)$, then $d_X(x, y) = d_Y(f(x), f(y)) = 0$, hence $x = y$.
(ii) If $x_n \to x$ in X, then $d_Y(f(x_n), f(x)) = d_X(x_n, x) \to 0$. Hence f is continuous by Theorem 5.1.15.
(iii) For $u = f(x), v = f(y) \in f(X)$, $d_X(f^{-1}(u), f^{-1}(v)) = d_X(x, y) = d_Y(f(x), f(y)) = d_Y(u, v)$.
Structural Commentary: An isometry preserves all distances, so injectivity and continuity are automatic.
Key Insight: Distance preservation is stronger than topological equivalence. □

Remark 5.1.27. *Metric Versus Linear Isometries*
Metric isometry is purely distance-theoretic. If $T : V \to W$ is linear between inner-product spaces, then
$(\forall v \in V)\ \|Tv\| = \|v\| \iff (\forall u, v \in V)\, d_W(Tu, Tv) = d_V(u, v)$.

Counterexample 5.1.28. *Continuous Bijection Need Not Be a Homeomorphism*
Let $S^1 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ and define $f : [0, 2\pi) \to S^1$ by $f(t) := (\cos t, \sin t)$.
This standard parametrization is a continuous bijection. Set $t_n := 2\pi - \frac{1}{n}$ and $z_n := f(t_n)$.
Then $z_n \to (1, 0) = f(0)$, while $f^{-1}(z_n) = t_n \to 2\pi \neq 0 = f^{-1}(1, 0)$.
Thus f^{-1} is not continuous by Theorem 5.1.15, so f is not a homeomorphism.

Figure 144 shows the discontinuity created by cutting the circle at one point.

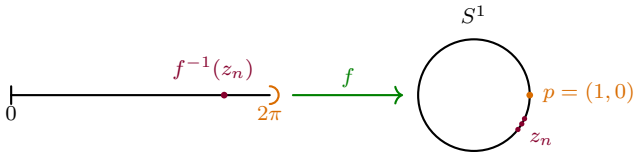

Figure 144: Parametrization $[0, 2\pi) \to S^1$ wraps continuously and bijectively, but inverse jumps at $(1, 0)$.

Remark 5.1.29. *Continuity Only Uses the Domain Metric*
For $f : E \to Y$ with $E \subseteq X$, continuity on E depends only on the restricted metric on E, not on $X \setminus E$.

Definition 5.1.30. *Product Map*
If $f : X \to Y$ and $g : X \to Z$ are maps between metric spaces, their *product map*, or direct sum, is $f \oplus g : X \to Y \times Z$.
It is defined by $(f \oplus g)(x) := (f(x), g(x))$. The product $Y \times Z$ carries any product metric from Definition 3.4.14.

Proposition 5.1.31. *Continuity of Product Maps*
Let X, Y, Z be metric spaces, let $f : X \to Y$, let $g : X \to Z$, and let $x_0 \in X$.

(i) The product map $f \oplus g : X \to Y \times Z$ is continuous at x_0 iff f and g are continuous at x_0.
(ii) The map $f \oplus g$ is continuous on X iff both f and g are continuous on X.

Proof. By Proposition 3.4.15(iii), it suffices to use d_{sum}.

(*i*) ($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ from continuity of $f \oplus g$ at x_0.
If $d_X(x, x_0) < \delta$, then $d_Y(f(x), f(x_0)), d_Z(g(x), g(x_0)) \le d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) < \varepsilon$.
(*i*) ($\Leftarrow$) Given $\varepsilon > 0$, choose δ_f, δ_g for f, g at tolerance $\frac{\varepsilon}{2}$. Set $\delta := \min\{\delta_f, \delta_g\}$.
If $d_X(x, x_0) < \delta$, then $d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) = d_Y(f(x), f(x_0)) + d_Z(g(x), g(x_0)) < \varepsilon$.
(*ii*) ($\Rightarrow$) If $f \oplus g$ is continuous on X, then $(\forall x_0 \in X)$, both f and g are continuous at x_0 by (i).
Hence both are continuous on X by Definition 5.1.9(ii).
(*ii*) ($\Leftarrow$) If f and g are continuous on X, then $(\forall x_0 \in X)$, $f \oplus g$ is continuous at x_0 by (i).
Hence $f \oplus g$ is continuous on X by Definition 5.1.9(ii).

Structural Commentary: Product continuity is exactly componentwise continuity.
Key Insight: A product metric packages two output estimates into one. □

Theorem 5.1.32. *Continuous Algebra and Components*
Let X be a metric space.

(i) If $f, g : X \to \mathbb{C}$ are continuous, then $f + g$ and fg are continuous on X, while f/g is continuous on $U := \{x \in X : g(x) \neq 0\}$.
(ii) For $f = (f_1, \dots, f_k) : X \to \mathbb{R}^k$, f is continuous iff $(\forall j \in \{1, \dots, k\})$ f_j is continuous.
(iii) If $f, g : X \to \mathbb{R}^k$ are continuous, so are $f + g$ and $f \cdot g$.

Proof.

(*i*) If $x_n \to x$ in X, then $f(x_n) \to f(x)$ and $g(x_n) \to g(x)$ by Theorem 5.1.15.
Thus $(f + g)(x_n) \to (f + g)(x)$ and $(fg)(x_n) \to (fg)(x)$ by Theorem 2.1.5(i),(iii).
If $x \in U$ and $(x_n) \subseteq U$, then $(f/g)(x_n) \to (f/g)(x)$ by Theorem 2.1.5(iii),(iv).
The three continuity conclusions follow by Theorem 5.1.15.
(*ii*) ($\Rightarrow$) If f is continuous and $x_n \to x$, then $f(x_n) \to f(x)$ by Theorem 5.1.15.
Hence $(\forall j \in \{1, \dots, k\})$ $f_j(x_n) \to f_j(x)$ by Theorem 3.4.13(i), so each f_j is continuous.
(*ii*) ($\Leftarrow$) If each f_j is continuous and $x_n \to x$, then $(\forall j \in \{1, \dots, k\})$ $f_j(x_n) \to f_j(x)$.
Hence $f(x_n) \to f(x)$ by Theorem 3.4.13(i), so f is continuous.
(*iii*) Each coordinate of $f + g$ is continuous by (i), so $f + g$ is continuous by (ii).
The map $f \cdot g = \sum_{j=1}^{k} f_j g_j$ is continuous by repeated scalar sums and products in (i).

Structural Commentary: Continuity is stable under the algebraic operations already stable for limits.
Key Insight: Componentwise control is equivalent to Euclidean control in finite dimensions. □

Theorem 5.1.33. *Pointwise Bounded Families Are Locally Uniformly Bounded*
Let (X, d) be a nonempty complete metric space, $\mathcal{F}$ be a family of continuous maps $f : X \to \mathbb{R}$.
Assume $(\forall x \in X)(\exists M_x > 0)(\forall f \in \mathcal{F})$ $|f(x)| \le M_x$.
Then there are a nonempty open set $O \subseteq X$ and $M > 0$ such that $(\forall x \in O)(\forall f \in \mathcal{F})$ $|f(x)| \le M$.

Proof. For $n \in \mathbb{N}$, set $E_n := \{x \in X : (\forall f \in \mathcal{F})\ |f(x)| \le n\}$. The interval $[-n, n] = \overline{B}_n(0)$ is closed by Corollary 3.1.18.
Hence $E_n = \bigcap_{f \in \mathcal{F}} f^{-1}([-n, n])$ closed by Corollary 5.1.20, Theorem 3.1.19(ii). $X = \bigcup_{n=1}^{\infty} E_n$ by pointwise bound, Theorem 1.1.19(i).
Choose $N \in \mathbb{N}$ with $E_N^\circ \neq \emptyset$ by Corollary 4.3.11(ii). Set $O := E_N^\circ$ and $M := N$. Then $(\forall x \in O)(\forall f \in \mathcal{F})$ $|f(x)| \le M$.
Structural Commentary: Closed inverse images form a countable cover, and Baire supplies one member with interior.
Key Insight: One interior level set bounds the entire family locally. □

Theorem 5.1.34. *Dense Continuity Set for Pointwise Limits*
Let (X, d) be a complete metric space, let $f : X \to \mathbb{R}$, and let $(f_n)_{n \in \mathbb{N}}$ be continuous maps $X \to \mathbb{R}$.
Assume $(\forall x \in X)$ $f_n(x) \to f(x)$. Then $(\exists D \subseteq X)$ satisfying:

(i) D is dense in X;
(ii) $(\forall x_0 \in D)(\forall \varepsilon > 0)(\exists \delta > 0)(\forall n \in \mathbb{N})(\forall x \in X)\,[d(x, x_0) < \delta \Longrightarrow |f_n(x) - f_n(x_0)| < \varepsilon]$;
(iii) $(\forall x_0 \in D)$, f is continuous at x_0.

Proof. For $m,n\in\mathbb{N}$, set $I_m:=[-m^{-1},m^{-1}]$ and $E_{m,n}:=\{x\in X:(\forall j,k\geq n)\,f_j(x)-f_k(x)\in I_m\}$.
For $j,k\geq n$, the map f_j-f_k is continuous by Theorem 5.1.32(i). The interval I_m is closed by Corollary 3.1.18.
Thus $E_{m,n}=\bigcap_{j,k\geq n}(f_j-f_k)^{-1}(I_m)$ is closed by Corollary 5.1.20 and Theorem 3.1.19(ii). Fix $m\in\mathbb{N}$ and $x\in X$.

Then $(f_n(x))$ is Cauchy by Theorem 4.1.20(i). Choose $N\in\mathbb{N}$ with $x\in E_{m,N}$ by Definition 4.1.14. Hence $X=\bigcup_{n=1}^{\infty}E_{m,n}$.

For $E\subseteq X$, $\operatorname{bd}(E)=\overline{E}\cap\overline{E^c}$ by Definitions 3.1.23 and 3.1.24(iii). Thus $\operatorname{bd}(E)$ is closed by Theorem 3.1.22(i) and Theorem 3.1.19(ii).
If E closed, O open with $O\subseteq\operatorname{bd}(E)=E\setminus E^\circ$, then $O\subseteq E$. Hence $O\subseteq E^\circ$ by Definition 3.1.24(i), assumed inclusion forces $O=\emptyset$.
Hence boundary of closed set is nowhere dense by Definition 4.3.1(iii). Hence $X\setminus\operatorname{bd}(E_{m,n})$ is dense and open.
Set $D:=X\setminus\bigcup_{m,n\in\mathbb{N}}\operatorname{bd}(E_{m,n})$. Then $D=\bigcap_{m,n\in\mathbb{N}}\big(X\setminus\operatorname{bd}(E_{m,n})\big)$ by Theorem 1.5.2.

(*i*) The set $\mathbb{N}^2$ is countable by Corollary 6.2.17(ii). Hence D is dense in X by Theorem 4.3.5.

(*ii*) Fix $x_0\in D$, $\varepsilon>0$. Choose $m\in\mathbb{N}$ with $\frac{3}{m}<\varepsilon$ by Theorem 1.1.19(i). Choose $N\in\mathbb{N}$ with $x_0\in E_{m,N}$ by $X=\bigcup_{n=1}^{\infty}E_{m,n}$.
Since $x_0\notin\operatorname{bd}(E_{m,N})$, $x_0\in E_{m,N}^\circ$; choose $r>0$ with $B_r(x_0)\subseteq E_{m,N}$.
For $1\leq i\leq N$, choose $\delta_i>0$ by continuity of f_i at x_0 with tolerance $\frac{1}{m}$.
Set $\delta:=\min\{r,\delta_1,\dots,\delta_N\}>0$. If $d(x,x_0)<\delta$ and $j>N$, then $x,x_0\in E_{m,N}$.
By the triangle inequality, $|f_j(x)-f_j(x_0)|\leq|f_j(x)-f_N(x)|+|f_N(x)-f_N(x_0)|+|f_N(x_0)-f_j(x_0)|$.
The three terms are bounded by m^{-1},m^{-1},m^{-1}, with the middle bound strict; hence $|f_j(x)-f_j(x_0)|<\frac{3}{m}<\varepsilon$.
The same bound holds for $1\leq j\leq N$ by the choice of δ. Hence (ii).

(*iii*) Fix $x_0\in D$, $\varepsilon>0$, $\eta:=\frac{\varepsilon}{2}$. By (ii), choose $\delta>0$ with $(\forall n\in\mathbb{N})(\forall x\in X)\,[d(x,x_0)<\delta\Longrightarrow|f_n(x)-f_n(x_0)|<\eta]$.
If $d(x,x_0)<\delta$, then $f_n(x)-f_n(x_0)\to f(x)-f(x_0)$ by Theorem 2.1.5(i),(ii).
Hence $|f_n(x)-f_n(x_0)|\to|f(x)-f(x_0)|$ by reverse triangle inequality; $|f(x)-f(x_0)|\leq\eta<\varepsilon$ by Proposition 2.1.4(iii).

Structural Commentary: Baire avoids every tail-Cauchy boundary; one interior tail set controls all indices.
Key Insight: Tail Cauchy bounds plus finitely many continuous terms create equicontinuity. □

Example 5.1.35. *Coordinate, Polynomial, Rational, and Norm Maps*
Coordinate maps on $\mathbb{R}^k$ are continuous; hence polynomials are continuous, and rational functions are continuous where defined.
The reverse triangle inequality is $\big|\|\mathbf{x}\|_2-\|\mathbf{y}\|_2\big|\leq\|\mathbf{x}-\mathbf{y}\|_2$. Hence $\mathbf{x}\mapsto\|\mathbf{x}\|_2$ is continuous by Definition 5.1.9.
Thus $p\mapsto\|f(p)\|_2$ is continuous for continuous $f:X\to\mathbb{R}^k$ by Theorem 5.1.16.

Definition 5.1.36. *Bounded Mapping*
A mapping $f:E\to\mathbb{R}^k$ is *bounded* iff $(\exists M\in\mathbb{R})(\forall x\in E)\ \|f(x)\|_2\leq M$.

Theorem 5.1.37. *Continuous Images of Compact Sets Are Compact*
Let X and Y be metric spaces. If $f:X\to Y$ is continuous and X is compact, then $f(X)$ is compact.

Proof. Let $\{V_\alpha\}$ be an open cover of $f(X)$. Then $\{f^{-1}(V_\alpha)\}$ is an open cover of X by Theorem 5.1.18.
Choose $\alpha_1,\dots,\alpha_n$ with $X\subseteq\bigcup_{i=1}^{n}f^{-1}(V_{\alpha_i})$; then $f(X)\subseteq\bigcup_{i=1}^{n}V_{\alpha_i}$.
Structural Commentary: Pull back, take a finite subcover, push forward.
Key Insight: Compactness is preserved by continuous images because inverse images preserve openness. □

Corollary 5.1.38. *Continuous Images into $\mathbb{R}^k$ Are Closed and Bounded*
Let X be a metric space. If $f:X\to\mathbb{R}^k$ is continuous, X is compact, then $f(X)$ is closed and bounded; consequently, f is bounded.

Proof. By Theorem 5.1.37, $f(X)$ is compact. By Theorem 4.2.5, compact subsets of $\mathbb{R}^k$ are closed and bounded; hence f is bounded.
Structural Commentary: Compactness in the domain becomes closed-bounded geometry in Euclidean range.
Key Insight: Continuous functions on compact spaces cannot escape to infinity. □

Theorem 5.1.39. *Extreme Value Theorem*
Let X be a nonempty compact metric space and let $f:X\to\mathbb{R}$ be continuous.
If $M:=\sup_{p\in X}f(p)$ and $m:=\inf_{p\in X}f(p)$, then $(\exists p,q\in X)\,[f(p)=M\wedge f(q)=m]$.

Proof. By Corollary 5.1.38, $f(X)$ is nonempty, closed, and bounded in $\mathbb{R}$. By Theorem 3.3.5(ii), $M=\sup f(X)$ belongs to $f(X)$.
Fix $\varepsilon>0$. Since $m+\varepsilon$ is not a lower bound for $f(X)$, choose $y_\varepsilon\in f(X)$ with $m\leq y_\varepsilon<m+\varepsilon$.
Hence $m\in\overline{f(X)}=f(X)$. Thus $(\exists p,q\in X)\,[M=f(p)\wedge m=f(q)]$.
Structural Commentary: Compactness turns suprema and infima into attained values.
Key Insight: Closed bounded images contain their endpoints. □

Theorem 5.1.40. *Compact Bijective Maps Have Continuous Inverses*
Let X and Y be metric spaces. If $f:X\to Y$ is continuous bijection, X is compact, then $f^{-1}:Y\to X$ is continuous.

Proof. By Theorem 5.1.18, it suffices that $f(V)$ be open in Y whenever $V\subseteq X$ is open.
For open $V\subseteq X$, V^c is compact by Theorem 4.1.7.
Then $f(V^c)$ is compact by Theorem 5.1.37 and closed by Theorem 4.1.6. Since f is bijective, $f(V)=Y\setminus f(V^c)$ is open.
Structural Commentary: Compactness turns closed complements into closed images.
Key Insight: A continuous bijection from compact domain has no room for inverse discontinuity. □

Remark 5.1.41. *Preimages, Not Forward Images*
Theorem 5.1.18 characterizes continuity through inverse images.
A continuous map need not preserve open or closed forward images; compactness recovers closedness through Theorem 5.1.37.

Theorem 5.1.42. *Continuous Images of Connected Sets Are Connected*
Let X and Y be metric spaces. If $f:X\to Y$ is continuous and $E\subseteq X$ is connected, then $f(E)$ is connected.

Proof. Suppose $f(E) = A \cup B$ for nonempty separated A, B, and set $G := E \cap f^{-1}(A)$ and $H := E \cap f^{-1}(B)$.
Then $G, H \neq \emptyset$ and $E = G \cup H$. The set $f^{-1}(\overline{A})$ is closed by Corollary 5.1.20. Since $G \subseteq f^{-1}(\overline{A})$, $\overline{G} \subseteq f^{-1}(\overline{A})$, so $\overline{G} \cap H = \emptyset$.
Likewise $G \cap \overline{H} = \emptyset$. Thus G, H separate E, contrary to connectedness.
Structural Commentary: A separation of the image pulls back to a separation of the domain.
Key Insight: Continuous maps cannot create a gap in the image of a connected set. □

Theorem 5.1.43. *Intermediate Value Theorem*
If $f : [a, b] \to \mathbb{R}$ is continuous and c lies strictly between $f(a)$ and $f(b)$, then $(\exists\, x \in (a, b))\ f(x) = c$.

Proof. The interval $[a, b]$ is connected by Theorem 5.1.2. Hence $f([a, b])$ is connected by Theorem 5.1.42.
The membership $c \in f([a, b])$ holds by Theorem 5.1.2. Choose $x \in [a, b]$ with $f(x) = c$. Since $c \notin \{f(a), f(b)\}$, $x \in (a, b)$.
Structural Commentary: The intermediate value theorem is connectedness of intervals transported by continuity.
Key Insight: Continuous real functions on intervals cannot jump over a value. □

Method 5.1.44. *Bisection Proof of the Intermediate Value Theorem*
Let $f : [a, b] \to \mathbb{R}$ be continuous, c lie between $f(a)$ and $f(b)$. Replacing (f, c) by $(-f, -c)$ if needed, assume $f(a) \leq c \leq f(b)$.
If $c = f(a)$ or $c = f(b)$, take the corresponding endpoint; otherwise $f(a) < c < f(b)$.

(i) Set $[a_1, b_1] := [a, b]$.
(ii) Given $[a_n, b_n]$ with $f(a_n) \leq c \leq f(b_n)$, set $m_n := \frac{a_n + b_n}{2}$.
If $f(m_n) \geq c$, set $[a_{n+1}, b_{n+1}] := [a_n, m_n]$; otherwise set $[a_{n+1}, b_{n+1}] := [m_n, b_n]$.

Then $(\forall n \in \mathbb{N})\, [a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \wedge f(a_n) \leq c \leq f(b_n)$. The length identity is $(\forall n \in \mathbb{N})\, b_n - a_n = \frac{b-a}{2^{n-1}}$.
Choose $x_0 \in \bigcap_{n=1}^{\infty} [a_n, b_n]$ by Theorem 4.2.1. Interval lengths tend to 0 by Theorem 2.2.10(v), Theorem 2.1.5(ii).
Since $0 \leq x_0 - a_n, b_n - x_0 \leq b_n - a_n$, $a_n, b_n \to x_0$ by Theorem 2.2.9. If $y \in \bigcap_{n=1}^{\infty} [a_n, b_n]$, then $(\forall n \in \mathbb{N})\ |y - x_0| \leq b_n - a_n$, so $y = x_0$.
By Theorem 5.1.15, $f(a_n), f(b_n) \to f(x_0)$, so $f(x_0) \leq c \leq f(x_0)$; hence $f(x_0) = c$. If $f(a) < c < f(b)$, then $x_0 \in (a, b)$.
Key Insight: Bisection keeps the target value bracketed while nested intervals collapse to one point.

Definition 5.1.45. *Path-Connected Set*
A *path* in E from p to q is a continuous map $\gamma : [0, 1] \to E$ satisfying $\gamma(0) = p$ and $\gamma(1) = q$.
The set E is *path-connected* iff any $p, q \in E$ are joined by a path in E.

Theorem 5.1.46. *Path-Connected Sets Are Connected*
Let E be a path-connected subset of a metric space. Then E is connected.

Proof. Suppose $E = A \cup B$ with A, B nonempty separated. Choose $p \in A$, $q \in B$, and a path $\gamma : [0, 1] \to E$ from p to q.
The interval $[0, 1]$ is connected by Theorem 5.1.2. Hence $\gamma([0, 1])$ is connected by Theorem 5.1.42.
Set $G := \gamma([0, 1]) \cap A$ and $H := \gamma([0, 1]) \cap B$. Then $p \in G$, $q \in H$, and $\gamma([0, 1]) = G \cup H$.
Since $G \subseteq A$ and $H \subseteq B$, $\overline{G} \cap H = G \cap \overline{H} = \emptyset$. Thus G, H separate $\gamma([0, 1])$, a contradiction.
Structural Commentary: A path transports connectedness from an interval to the set.
Key Insight: To connect two points by a path, the path image cannot be split into two separated pieces. □

Corollary 5.1.47. *Convex Sets Are Path-Connected*
Let $E \subseteq \mathbb{R}^k$ be convex. Then E is path-connected.

Proof. For $\mathbf{x}, \mathbf{y} \in E$, set $\gamma(t) := (1 - t)\mathbf{x} + t\mathbf{y}$ on $[0, 1]$. By convexity, $\gamma([0, 1]) \subseteq E$, with $\gamma(0) = \mathbf{x}$ and $\gamma(1) = \mathbf{y}$.
For $s, t \in [0, 1]$, $\|\gamma(t) - \gamma(s)\|_2 = |t - s|\, \|\mathbf{y} - \mathbf{x}\|_2$. Hence γ is continuous by Definition 5.1.9, so E is path-connected.
Structural Commentary: Convexity supplies the whole straight path between any two points.
Key Insight: In Euclidean space, line segments are continuous paths. □

5.2 Uniform Continuity and Compactness

Definition 5.2.1. *Uniform Continuity*
Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$.
The map f is *uniformly continuous* iff $(\forall\, \varepsilon > 0)(\exists\, \delta > 0)(\forall\, p, q \in X)\, [d_X(p, q) < \delta \implies d_Y(f(p), f(q)) < \varepsilon]$.
Key Insight: The choice of δ may depend on ε, but not on either point in the pair.

Principle 5.2.2. *Challenge–Response Order*
Limits and continuity depend on quantifier order. Definition 5.1.4 answers a range challenge ε by a domain radius δ at the base point. Definition 5.2.1 strengthens this pattern by requiring δ to be independent of the domain point.

Proposition 5.2.3. *Sequential Test for Uniform Continuity*
Let X, Y be metric spaces, $f : X \to Y$. Then f uniformly continuous iff $d_Y(f(p_n), f(q_n)) \to 0$ when $(p_n), (q_n) \subseteq X$, $d_X(p_n, q_n) \to 0$.

Proof.
($\Rightarrow$) Let $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ such that $x, y \in X$ and $d_X(x, y) < \delta$ imply $d_Y(f(x), f(y)) < \varepsilon$.
Since $d_X(p_n, q_n) \to 0$, eventually $d_X(p_n, q_n) < \delta$, and hence eventually $d_Y(f(p_n), f(q_n)) < \varepsilon$.
($\Leftarrow$) Prove contrapositive. If f not uniformly continuous, $(\exists \varepsilon_0 > 0)(\forall n \geq 1)(\exists p_n, q_n \in X)\, [d_X(p_n, q_n) < \frac{1}{n} \wedge d_Y(f(p_n), f(q_n)) \geq \varepsilon_0]$.
$\frac{1}{n} \to 0$ by Theorem 2.2.10(i), $d_X(p_n, q_n) \to 0$ by Theorem 2.2.9; $d_Y(f(p_n), f(q_n)) \not\to 0$, contradicting sequential condition.
Structural Commentary: The forward estimate transfers to pairs; the negation selects a separated pair at each scale.
Key Insight: Failed uniformity produces close inputs whose outputs stay apart. □

Definition 5.2.4. *Lipschitz Map*
Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$, $L \geq 0$. The map f is *L-Lipschitz* if $(\forall\, p, q \in X)\ d_Y(f(p), f(q)) \leq L d_X(p, q)$.
It is *Lipschitz* if $(\exists\, L \geq 0)$ such that it is L-Lipschitz.
Key Insight: Lipschitz control is one global linear modulus of continuity.

Proposition 5.2.5. *Lipschitz Implies Uniform Continuity*
Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$. If f is Lipschitz, then f is uniformly continuous and continuous.

Proof. Choose $L \geq 0$ such that f is L-Lipschitz by Definition 5.2.4, and let $\varepsilon > 0$.
If $L = 0$, set $\delta := 1$; then $(\forall\, p, q \in X)\ d_Y(f(p), f(q)) = 0 < \varepsilon$.
If $L > 0$, take $\delta := \frac{\varepsilon}{L}$; if $p, q \in X$ and $d_X(p, q) < \delta$, then $d_Y(f(p), f(q)) \leq L d_X(p, q) < \varepsilon$. Thus f is uniformly continuous.
Fix $a \in X$. If $p \in X$ and $d_X(p, a) < \delta$, the uniform estimate with $q = a$ verifies continuity at a.
Structural Commentary: The cases $L = 0$ and $L > 0$ provide a point-independent δ; specialization proves continuity.
Key Insight: The linear bound supplies a global modulus. □

Theorem 5.2.6. *Continuity on Compact Sets Is Uniform*
Let (X, d_X), (Y, d_Y) be metric spaces. If $f : X \to Y$ is continuous and X is compact, then f is uniformly continuous.

Proof. If $X = \emptyset$, the conclusion is vacuous. Assume $X \neq \emptyset$, and fix $\varepsilon > 0$. For $a \in X$, set $U_a := f^{-1}(B_{\frac{\varepsilon}{2}}(f(a)))$.
Each U_a is open by Theorems 3.1.3 and 5.1.18, while $a \in U_a$. Hence $\mathcal{U} := \{U_a : a \in X\}$ is an open cover of X.
Choose Lebesgue number $\delta > 0$ for $\mathcal{U}$ by Lemma 4.1.33(ii). If $p, q \in X$, $d_X(p, q) < \delta$, choose $a \in X$, $B_\delta(p) \subseteq U_a$ by Definition 4.1.32.
Then $p, q \in U_a$, so $d_Y(f(p), f(a)) < \frac{\varepsilon}{2}$ and $d_Y(f(q), f(a)) < \frac{\varepsilon}{2}$. Thus $d_Y(f(p), f(q)) < \varepsilon$ by the triangle inequality.
Structural Commentary: Output balls induce an inverse-image cover; its Lebesgue number controls every input pair.
Key Insight: One cover-scale replaces all point-dependent continuity radii. □

Counterexample 5.2.7. *The Reverse Implications Fail*

(i) The identity on $\mathbb{R}$ is an isometry, so it is continuous by Theorem 5.1.26(ii).
Thus $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^2$, is continuous by Theorem 5.1.32(i).
For $p_n := n$ and $q_n := n + \frac{1}{n}$, $|p_n - q_n| = \frac{1}{n} \to 0$, whereas $|f(q_n) - f(p_n)| = 2 + \frac{1}{n^2} \not\to 0$.
Thus f is not uniformly continuous by Proposition 5.2.3.

(ii) Define $g : [0, 1] \to \mathbb{R}$ by $g(x) := \sqrt{x}$. For $\varepsilon > 0$, the inequality $\left|\sqrt{x} - \sqrt{y}\right| \leq \sqrt{|x - y|}$ shows that $\delta := \varepsilon^2$ verifies continuity.
The set $[0, 1]$ is compact by Theorem 4.2.5. Hence g is uniformly continuous by Theorem 5.2.6.
If g were L-Lipschitz, then $(\forall n \in \mathbb{N})\ \frac{1}{n} = \left|g(\frac{1}{n^2}) - g(0)\right| \leq \frac{L}{n^2}$, so $n \leq L$.
Choose $n \in \mathbb{N}$ with $n > L$ by Theorem 1.1.19(i), a contradiction. Hence g is not Lipschitz.

Structural Commentary: The quadratic breaks uniformity; the square root has no linear modulus.
Key Insight: Compactness supplies a modulus, but not necessarily a linear one.

Theorem 5.2.8. *Compactness Hypotheses Are Essential*
Let $E \subseteq \mathbb{R}$ be noncompact in the subspace metric. Then:

(i) $\exists f : E \to \mathbb{R}$ continuous and unbounded;
(ii) $\exists g : E \to \mathbb{R}$ continuous and bounded with no maximum;
(iii) if E is bounded, then $\exists h : E \to \mathbb{R}$ continuous and not uniformly continuous.

Proof. The inclusion $\iota : E \to \mathbb{R}$ is an isometry, so it is continuous by Theorem 5.1.26(ii).
For $c \in \mathbb{R}$, define $\kappa_c : E \to \mathbb{R}$ by $\kappa_c(x) := c$. Since $(\forall p, x \in E)\ |\kappa_c(x) - \kappa_c(p)| = 0$, κ_c is continuous by Definition 5.1.9.
Their denominators are nonzero on E, so the rational expressions below are continuous by Theorem 5.1.32(i).

(i) If E is unbounded, $f := \iota$ is continuous and unbounded. Suppose E is bounded.
Then E not closed by Theorem 4.2.5; choose $x_0 \in \overline{E} \setminus E$. Point x_0 is limit point of E by Definition 3.1.21.
Define $f(x) := \frac{1}{x - x_0}$. For $M > 0$, choose $x \in E \cap B_{\frac{1}{M}}(x_0)$ by limit-point property of x_0. Then $|f(x)| > M$, so f is unbounded.

(ii) Suppose first that E is unbounded, and define $g(x) := \frac{x^2}{1 + x^2}$. Then $(\forall x \in E)\ 0 \leq g(x) < 1$.
Given $\varepsilon > 0$, choose $n \in \mathbb{N}$ with $n > \varepsilon^{-\frac{1}{2}}$ by Theorem 1.1.19(i). Choose $x \in E$ with $|x| > n$ by unboundedness.
Then $1 - g(x) = \frac{1}{1 + x^2} < \varepsilon$. Since $g < 1$ on E, $\sup_E g = 1$, which is not attained. Suppose next that E is bounded.
Choose $x_0 \in \overline{E} \setminus E$ as in (i), define $g(x) := \frac{1}{1 + (x - x_0)^2}$. Then $(\forall x \in E)\ 0 < g(x) < 1$.
Given $\varepsilon > 0$, $x \in E \cap B_{\sqrt{\varepsilon}}(x_0)$ by limit-point property of x_0; $1 - g(x) = \frac{(x - x_0)^2}{1 + (x - x_0)^2} < \varepsilon$. Hence $\sup_E g = 1$, which is not attained.

(iii) Suppose E is bounded. Choose $x_0 \in \overline{E} \setminus E$ as in (i), and define $h(x) := \frac{1}{x - x_0}$.
For $\delta > 0$, $x \in E \cap B_{\frac{\delta}{2}}(x_0)$ by limit-point property of x_0. Choose $y \in E$, $|y - x_0| < \min\{\frac{\delta}{2}, \frac{1}{|h(x)| + 1}\}$ by same property.
Then $|x - y| < \delta$ and $|h(y)| > |h(x)| + 1$, so $|h(y) - h(x)| > 1$ by the reverse triangle inequality.
Thus h is not uniformly continuous by Definition 5.2.1.

Structural Commentary: Unbounded sets support escape functions; bounded nonclosed sets support singular boundary functions.
Key Insight: Every noncompact subset of $\mathbb{R}$ has an escape or boundary obstruction. □

Figure 145 illustrates the boundary obstruction.
Near 0, a fixed output tolerance for $x \mapsto \frac{1}{x^2}$ requires progressively smaller input radii.

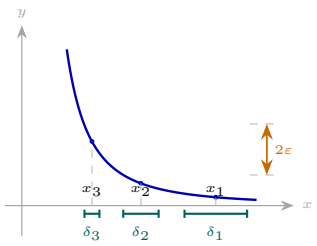

Figure 145: Dimensionless $y = \frac{1}{x^2}$ graph for $x > 0$: input radii δ_i shrink as $x_i \to 0$; orange marks the fixed 2ε output tolerance.

Proof Method 5.2.9. *Continuity Proof Patterns*
Let (X, d_X) and (Y, d_Y) be metric spaces, $E \subseteq X$, $f : E \to Y$, $p \in X$ and $q \in Y$.

(i) *ε–δ at p*: for $p \in E$, prove $(\forall\, \varepsilon > 0)(\exists\, \delta > 0)(\forall\, x \in E)\,[d_X(x,p) < \delta \Longrightarrow d_Y(f(x), f(p)) < \varepsilon]$.
(ii) *Limit at p*: for $p \in E^{\mathrm{acc}}$, replace antecedent by $0 < d_X(x,p) < \delta$ and target by q.
Equivalently, test sequences with $x_n \to p$ and $x_n \neq p$.
(iii) *Topological continuity*: prove that open subsets of Y have relatively open preimages in E.
Equivalently, prove that closed subsets of Y have relatively closed preimages.
(iv) *Compact-image argument*: for compact E, pull an open cover $\{V_\alpha\}$ of $f(E)$ back to $\{f^{-1}(V_\alpha)\}$ of E.
Extract a finite subcover of E and push it forward to $f(E)$.
(v) *Uniform continuity*: for compact E and $\varepsilon > 0$, form $U_a := f^{-1}(B_{\frac{\varepsilon}{2}}(f(a)))$ for $a \in E$.
Choose a Lebesgue number $\delta > 0$ by Lemma 4.1.33(ii).

Key Insight: Match the target property to pointwise, sequential, inverse-image, or compact-cover control.

5.3 One-Sided Limits, Monotone Functions, and Extended Limits

Definition 5.3.1. *One-Sided Limits and Types of Discontinuity*
Let $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$, $x_0 \in \mathbb{R}$, $q \in \mathbb{R}$, $X_+ := X \cap (x_0, \infty)$, and $X_- := X \cap (-\infty, x_0)$.

(i) If x_0 is a limit point of X_+, then $f(x_0+) = q$ means $\lim\limits_{x \to x_0} (f|_{X_+})(x) = q$.
(ii) If x_0 is a limit point of X_-, then $f(x_0-) = q$ means $\lim\limits_{x \to x_0} (f|_{X_-})(x) = q$.

For an interval I, $f : I \to \mathbb{R}$, and an interior discontinuity $x_0 \in I^\circ$:

(i) The discontinuity is of the *first kind*, or *simple*, iff both $f(x_0+)$ and $f(x_0-)$ exist.
(ii) A first-kind discontinuity is a *jump* iff $f(x_0+) \neq f(x_0-)$.
(iii) A first-kind discontinuity is *removable* iff $f(x_0+) = f(x_0-) \neq f(x_0)$.
(iv) The discontinuity is of the *second kind* iff at least one of $f(x_0+)$ and $f(x_0-)$ fails to exist.

Proposition 5.3.2. *Two-Sided Limits from One-Sided Limits*
Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $x_0 \in \mathbb{R}$, and set $X_+ := X \cap (x_0, \infty)$ and $X_- := X \cap (-\infty, x_0)$.
If x_0 is a limit point of both X_+ and X_-, then, for $q \in \mathbb{R}$, $\lim\limits_{x \to x_0} f(x) = q$ iff $f(x_0-) = f(x_0+) = q$.

Proof.
(⇒) Assume $\lim\limits_{x \to x_0} f(x) = q$, $\varepsilon > 0$. Choose $\delta > 0$ such that $t \in X$ and $0 < |t - x_0| < \delta$ imply $|f(t) - q| < \varepsilon$.
Restricting this estimate to X_+ and X_- proves $f(x_0+) = f(x_0-) = q$ by Definitions 5.1.4 and 5.3.1.
(⇐) Assume $f(x_0+) = f(x_0-) = q$, and fix $\varepsilon > 0$. Choose $\delta_+, \delta_- > 0$ by the right and left limits, and set $\delta := \min\{\delta_+, \delta_-\}$.
If $t \in X$ and $0 < |t - x_0| < \delta$, then $t \in X_+ \cup X_-$, so $|f(t) - q| < \varepsilon$. Hence $\lim\limits_{x \to x_0} f(x) = q$ by Definition 5.1.4.

Structural Commentary: The punctured domain splits into its right and left approach sets.
Key Insight: A common one-sided value controls every punctured approach. □

Example 5.3.3. *Basic Discontinuity Types*
Let $D := \mathbf{1}_{\mathbb{Q}}$. Rational and irrational approaches have values 1 and 0, respectively.
Thus D has a second-kind discontinuity throughout $\mathbb{R}$ by Corollary 1.1.20.
Let $h(x) := x$ for $x \in \mathbb{Q}$ and $h(x) := 0$ otherwise. Since $|h(x)| \leq |x|$, the function h is continuous at 0.
At $c \neq 0$, rational and irrational approaches tend to c and 0, so both one-sided limits fail.
A piecewise linear function with unequal finite one-sided values has a jump and hence a first-kind discontinuity.

Figure 146 depicts both values arbitrarily near c, precluding either one-sided limit of $\mathbf{1}_{\mathbb{Q}}$.

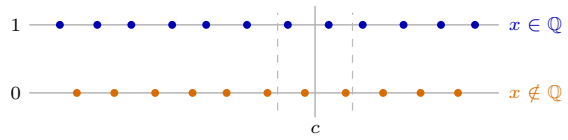

Figure 146: For $g = \mathbf{1}_{\mathbb{Q}}$, rational inputs map to 1 and irrational inputs to 0 in every interval about c.

Definition 5.3.4. *Monotone Functions*
Let $f : (a, b) \to \mathbb{R}$.

(i) f is *monotonically increasing* iff $(\forall x, y \in (a,b))\,[x < y \Rightarrow f(x) \leq f(y)]$.
(ii) f is *monotonically decreasing* iff $(\forall x, y \in (a,b))\,[x < y \Rightarrow f(x) \geq f(y)]$.
(iii) f is *monotone* iff f is monotonically increasing or monotonically decreasing.

Theorem 5.3.5. *One-Sided Limits of Monotone Functions*
Let $f : (a, b) \to \mathbb{R}$.

(i) If f is increasing, then $(\forall x \in (a,b))\, f(x-) = \sup\limits_{a<t<x} f(t) \leq f(x) \leq \inf\limits_{x<t<b} f(t) = f(x+)$.
(ii) If f is increasing, then $(\forall x, y \in (a,b))\,[x < y \Rightarrow f(x+) \leq f(y-)]$.
(iii) If f is decreasing, then $(\forall x \in (a,b))\, f(x-) = \inf\limits_{a<t<x} f(t) \geq f(x) \geq \sup\limits_{x<t<b} f(t) = f(x+)$.
(iv) If f is decreasing, then $(\forall x, y \in (a,b))\,[x < y \Rightarrow f(x+) \geq f(y-)]$.

Proof.
(i) Fix $x \in (a,b)$, $L_x := \{f(t) : a < t < x\}$, $R_x := \{f(t) : x < t < b\}$. Sets are nonempty as $f(\frac{a+x}{2}) \in L_x$ and $f(\frac{x+b}{2}) \in R_x$.
Monotonicity makes $f(x)$ an upper bound of L_x and a lower bound of R_x.

Thus $A := \sup L_x \in \mathbb{R}$ by Theorem 1.1.9, and $B := \inf R_x \in \mathbb{R}$ by Corollary 1.1.15.
For $\varepsilon > 0$, choose $s \in (a, x)$ with $A - \varepsilon < f(s) \le A$. If $s < t < x$, then $A - \varepsilon < f(t) \le A$, so $f(x-) = A$.
Likewise, choose $u \in (x, b)$ with $B \le f(u) < B + \varepsilon$. If $x < t < u$, then $B \le f(t) < B + \varepsilon$, so $f(x+) = B$.
Finally, $A \le f(x) \le B$ by monotonicity. Since x was arbitrary, (i) follows.

(ii) If $a < x < y < b$ and $t := \frac{x+y}{2}$, then $f(x+) \le f(t) \le f(y-)$ by (i).

(iii) Set $h := -f$. Then h is increasing, so its one-sided limits exist by (i).
Since $f = -h$, one has $f(x\pm) = -h(x\pm)$ by Theorem 5.1.7. Identities in (iii) follow by (i) applied to h, Corollary 1.1.15.

(iv) By (ii), $h(x+) \le h(y-)$ whenever $x < y$. By (iii), multiplication by -1 establishes $f(x+) \ge f(y-)$.

Structural Commentary: Suprema and infima identify the increasing limits; reflection transfers both conclusions to decreasing maps.
Key Insight: Order traps each one-sided tail at its extremal value. □

Corollary 5.3.6. *Monotone Functions Have No Second-Kind Discontinuities*
Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Every discontinuity at $x \in I^\circ$ is a jump, and hence is not of the second kind.

Proof. Fix discontinuity $x \in I^\circ$, choose $\rho > 0$ with $(x - \rho, x + \rho) \subseteq I$. Both one-sided limits of $f|_{(x-\rho,x+\rho)}$ exist by Theorem 5.3.5.
They equal one-sided limits of f at x by Proposition 5.1.8, common value would equal $f(x)$ by Theorem 5.3.5(i),(iii).
Then $\lim_{t\to x} f(t) = f(x)$ by Proposition 5.3.2. The function f would be continuous at x by Theorem 5.1.12, a contradiction.
Thus the one-sided limits are unequal, and x is a jump by Definition 5.3.1(ii).
Structural Commentary: Local one-sided extrema combine into continuity unless their finite values form a jump.
Key Insight: Monotonicity leaves unequal side limits as the only discontinuity mechanism. □

Theorem 5.3.7. *Monotone Continuity and Interval Images*
Let $f : (a, b) \to \mathbb{R}$ be monotone. Then f is continuous iff $f((a, b))$ is an interval.

Proof. Put $I = (a, b)$.

(⇒) The image $f(I)$ is connected by Theorem 5.1.42. It is therefore an interval by Theorem 5.1.2.

(⇐) Prove contrapositive. Suppose f is discontinuous at $x_0 \in I$. After replacing f by $-f$ if necessary, assume f is increasing.
Reflection preserves discontinuity, interval property of image. Then $f(x_0-) < f(x_0+)$ by Corollary 5.3.6, Theorem 5.3.5(i).
Choose $y \in (f(x_0-), f(x_0+)) \setminus \{f(x_0)\}$. If $x < x_0$, then $f(x) \le f(x_0-) < y$; if $x > x_0$, then $f(x) \ge f(x_0+) > y$; and $f(x_0) \ne y$.
So $y \notin f(I)$ while $f(I)$ has values below and above y. Hence $f(I)$ is not an interval.

Structural Commentary: Monotone discontinuities create image gaps.
Key Insight: Interval images exclude jumps. □

Figure 147 depicts the range gap created by a monotone jump.

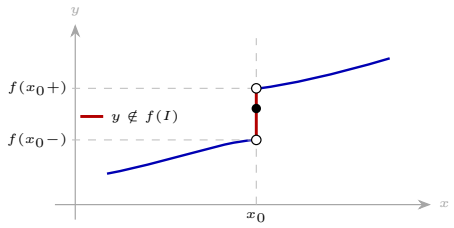

Figure 147: A jump of an increasing function omits $y \in (f(x_0-), f(x_0+))$ from $f(I)$, so the image is not an interval.

Theorem 5.3.8. *Monotone Discontinuities Are Countable*
If f is monotone on (a, b), then the set of points of (a, b) at which f is discontinuous is at most countable.

Proof.

(i) *Jump gaps*: Assume f is increasing, E be its discontinuity set. For $x \in E$, side limits unequal by Corollary 5.3.6.
By Theorem 5.3.5(i), $f(x-) \le f(x+)$, so $f(x-) < f(x+)$.

(ii) *Canonical markers*: Fix a bijection $n \mapsto q_n$ from $\mathbb{N}$ onto $\mathbb{Q}$ by Corollary 6.2.19.
For $x \in E$, the set $J_x := \{n \in \mathbb{N} : f(x-) < q_n < f(x+)\}$ is nonempty by Corollary 1.1.20.
Set $m(x) := \min J_x$ by Assumption 2.6.1, and put $r(x) := q_{m(x)}$.

(iii) *Injection*: If $x_1, x_2 \in E$ and $x_1 < x_2$, then $r(x_1) < f(x_1+) \le f(x_2-) < r(x_2)$ by Theorem 5.3.5(ii).
Thus $r : E \to \mathbb{Q}$ is injective. The set $r(E) \subseteq \mathbb{Q}$ is at most countable by Proposition 6.2.12(i) and Corollary 6.2.19.
The corestriction $r : E \to r(E)$ is bijective, so E is at most countable by Definitions 6.2.1 and 6.2.4.

(iv) *Decreasing case*: If f is decreasing, then $-f$ is increasing, has same discontinuity set. Result follows by increasing case.

Structural Commentary: Each jump contains a rational marker, and monotonicity keeps the markers distinct.
Key Insight: Countability enters by injecting discontinuities into $\mathbb{Q}$. □

Counterexample 5.3.9. *Dense Monotone Discontinuities*
Let $a < b$. The set $\mathbb{Q} \cap (a, b)$ is at most countable by Proposition 6.2.12(i) and Corollary 6.2.19.
It is infinite by Corollary 1.1.20, and hence countably infinite by Definition 6.2.4. Fix a bijection $n \mapsto q_n$ from $\mathbb{N}$ onto $\mathbb{Q} \cap (a, b)$.

For $N \in \mathbb{N}$, define $F_N : (a, b) \to \mathbb{R}$ by $F_N(x) := \sum_{n=1}^{N} 2^{-n} \mathbf{1}_{(-\infty,x)}(q_n)$. Define $F : (a, b) \to \mathbb{R}$ by $F(x) := \sup_{N\in\mathbb{N}} F_N(x)$.

The identity $\sum_{n=1}^{N} 2^{-n} = 1 - 2^{-N}$ makes $\{F_N(x) : N \in \mathbb{N}\}$ nonempty and bounded above, so $F(x) \in \mathbb{R}$ by Theorem 1.1.9.

If $x < y$, then $(\forall N \in \mathbb{N})\, F_N(x) \le F_N(y)$, so $F(x) \le F(y)$. Thus F is increasing. For $M > N$, $0 \le F_M(x) - F_N(x) \le \sum_{n=N+1}^{M} 2^{-n} < 2^{-N}$.

Taking the supremum over M proves $0 \le F(x) - F_N(x) \le 2^{-N}$. Fix $x \notin \mathbb{Q}$ and $\varepsilon > 0$. Choose N with $2^{1-N} < \varepsilon$ by Theorem 2.2.10(v).
Choose $\delta > 0$ such that $(x - \delta, x + \delta) \cap \{q_1, \ldots, q_N\} = \emptyset$. If $|y - x| < \delta$, then $F_N(y) = F_N(x)$, so $|F(y) - F(x)| \le 2^{1-N} < \varepsilon$.

Hence F continuous at x. Fix $k \in \mathbb{N}$, $\varepsilon > 0$. Choose $N \geq k$ with $2^{1-N} < \varepsilon$. Choose $\delta > 0$ such that $(q_k - \delta, q_k + \delta) \cap \{q_1, \ldots, q_N\} = \{q_k\}$.
If $q_k - \delta < y < q_k$, then $F_N(y) = F_N(q_k)$; if $q_k < y < q_k + \delta$, then $F_N(y) = F_N(q_k) + 2^{-k}$.
The tail bound therefore proves $F(q_k-) = F(q_k)$ and $F(q_k+) = F(q_k) + 2^{-k}$.
Thus the discontinuity set of F is $\mathbb{Q} \cap (a, b)$, which is dense by Corollary 1.1.20. This countable set has no isolated points.

Definition 5.3.10. *Extended Limits and Limits at Infinity*
On $\overline{\mathbb{R}}$ from Definition 1.1.21, retain the usual neighborhoods at finite points.
For $c \in \mathbb{R}$, take $(c, +\infty]$ and $[-\infty, c)$ as basic neighborhoods of $+\infty$ and $-\infty$, respectively.
Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, let $x \in \overline{\mathbb{R}}$ be a limit point of E, and let $A \in \overline{\mathbb{R}}$.
The notation $f(t) \to A$ as $t \to x$ means $(\forall U \ni A)(\exists V \ni x)(\forall t \in (V \cap E) \setminus \{x\})\, f(t) \in U$.
Here U and V range over neighborhoods in $\overline{\mathbb{R}}$. If $A, x \in \mathbb{R}$, this agrees with Definition 5.1.4.
Key Insight: The same neighborhood test permits $\pm\infty$ as input or output limits.

Figure 148 depicts the opposite extended side limits of x^{-3} at 0.

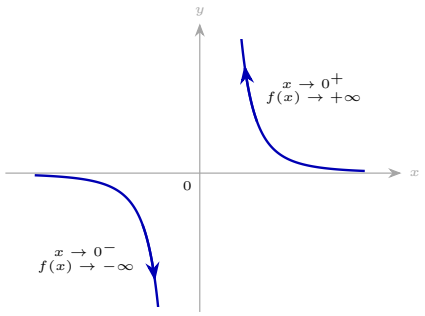

Figure 148: Qualitative dimensionless plot: a positive vertical rescaling of $y = x^{-3}$ tends to $-\infty$ as $x \to 0^-$ and to $+\infty$ as $x \to 0^+$.

Theorem 5.3.11. *Algebra of Extended Limits*
Let $E \subseteq \mathbb{R}$, let $x \in \overline{\mathbb{R}}$ be a limit point of E, and let $f, g : E \to \mathbb{R}$ satisfy $f(t) \to A$ and $g(t) \to B$ as $t \to x$, where $A, B \in \overline{\mathbb{R}}$.

(i) if $A' \in \overline{\mathbb{R}}$ and $f(t) \to A'$ as $t \to x$, then $A' = A$;
(ii) if $A + B$ is defined by ordinary real arithmetic or Definition 1.1.22, then $(f + g)(t) \to A + B$;
(iii) if AB is defined by ordinary real arithmetic or Definition 1.1.22, then $(fg)(t) \to AB$;
(iv) if $A, B \in \mathbb{R}$ and $B \neq 0$, then $\frac{f(t)}{g(t)} \to \frac{A}{B}$ on the domain where $g(t) \neq 0$.

Theorem makes no claim for $(+\infty) + (-\infty)$, $0(\pm\infty)$, product of two infinite factors, zero denominator, or quotient involving $\pm\infty$.

Proof. For $C := A \star B$, define $\mathcal{P}_\star(A, B, C) := (\forall W \ni C)(\exists U \ni A)(\exists V \ni B)\, (U \cap \mathbb{R}) \star (V \cap \mathbb{R}) \subseteq W$.
If $A, B \in \mathbb{R}$, the addition property follows from $|(u + v) - (A + B)| \leq |u - A| + |v - B|$.
The product property follows from $|uv - AB| \leq |u|\,|v - B| + |B|\,|u - A|$ by taking u near A and v near B.
For $B \neq 0$, restrict to $|v| \geq \frac{|B|}{2}$ and use $\left|\frac{u}{v} - \frac{A}{B}\right| \leq \frac{|u-A|}{|v|} + \frac{|A||v-B|}{|v||B|}$. Thus the quotient property holds with $0 \notin V$.
If $A + B = +\infty$ and $W \ni +\infty$, choose $c \in \mathbb{R}$ with $(c, +\infty] \subseteq W$. Choose $\alpha < A$ and $\beta < B$ in $\mathbb{R}$ with $\alpha + \beta > c$.
For $U = (\alpha, +\infty]$ and $V = (\beta, +\infty]$, one has $(U \cap \mathbb{R}) + (V \cap \mathbb{R}) \subseteq (c, +\infty] \subseteq W$. The case $A + B = -\infty$ is symmetric.
If $AB = \pm\infty$, exactly one factor is infinite and the other is finite and nonzero by Definition 1.1.22.
Let d be the finite factor, and restrict its neighboring values v by $vd > 0$ and $|v| \geq \frac{|d|}{2}$.
For a target tail with threshold c, restrict the infinite factor to the required sign and $|u| > \frac{2(|c|+1)}{|d|}$; then uv lies in that tail.
Thus $\mathcal{P}_\star(A, B, C)$ holds in (ii)–(iv).

(i) If $A' \neq A$, choose disjoint neighborhoods U of A and U' of A'.
By $f(t) \to A$, choose a neighborhood P of x with $f((P \cap E) \setminus \{x\}) \subseteq U$.
By $f(t) \to A'$, choose a neighborhood P' of x with $f((P' \cap E) \setminus \{x\}) \subseteq U'$.
Since x is a limit point of E, $(P \cap P' \cap E) \setminus \{x\} \neq \emptyset$, contradicting $U \cap U' = \emptyset$.
(ii) Let W be a neighborhood of $A + B$, and choose U, V by $\mathcal{P}_+(A, B, A + B)$.
By $f(t) \to A$, choose a neighborhood P of x with $f((P \cap E) \setminus \{x\}) \subseteq U$.
By $g(t) \to B$, choose a neighborhood Q of x with $g((Q \cap E) \setminus \{x\}) \subseteq V$. Then $(f + g)(t) \in W$ on $(P \cap Q \cap E) \setminus \{x\}$.
(iii) Let W be a neighborhood of AB, and choose U, V by $\mathcal{P}.(A, B, AB)$.
Intersecting the corresponding approach neighborhoods produces a punctured neighborhood on which $(fg)(t) \in W$.
(iv) Let W be a neighborhood of $\frac{A}{B}$, and choose U, V by $\mathcal{P}_{\text{div}}(A, B, \frac{A}{B})$, with $0 \notin V$.
Intersecting the corresponding approach neighborhoods produces a punctured neighborhood on which $g(t) \neq 0$ and $\frac{f(t)}{g(t)} \in W$.

Structural Commentary: Operation-compatible neighborhoods transfer two input tails, while disjoint neighborhoods force uniqueness.
Key Insight: Finite sign separation controls every defined infinite product and excludes the indeterminate cases. □

6 General Topology for Analysis

6.1 Topological Spaces, Bases, and Separation

General topology retains the open-set structure underlying metric limits and continuity.
The basic objects here are topological spaces, bases, subspaces, closure operations, and separation axioms.

Definition 6.1.1. *Topological Space*
A *topology* on a set X is a collection $\mathcal{T} \subseteq 2^X$ such that:

(i) $\emptyset, X \in \mathcal{T}$;
(ii) if $\{\mathcal{O}_\alpha\}_{\alpha \in A} \subseteq \mathcal{T}$, then $\bigcup_{\alpha \in A} \mathcal{O}_\alpha \in \mathcal{T}$;

(iii) if $\mathcal{O}_1, \dots, \mathcal{O}_n \in \mathcal{T}$ with $n \geq 1$, then $\bigcap_{j=1}^{n} \mathcal{O}_j \in \mathcal{T}$.

Then $(X, \mathcal{T})$ is a *topological space*.

Definition 6.1.2. *Open, Closed, and Neighborhoods*
Let $(X, \mathcal{T})$ be a topological space.

(i) A set $U \subseteq X$ is *open* iff $U \in \mathcal{T}$.
(ii) A set $F \subseteq X$ is *closed* iff $X \setminus F \in \mathcal{T}$.
(iii) For $x \in X$, a *neighborhood* of x is an open set $U \subseteq X$ with $x \in U$.
(iv) For $A \subseteq X$, a *neighborhood* of A is an open set $U \subseteq X$ with $A \subseteq U$.

Proposition 6.1.3. *Open Sets by Neighborhoods*
Let $(X, \mathcal{T})$ be a topological space and let $E \subseteq X$. Then $E \in \mathcal{T}$ iff $(\forall x \in E)(\exists U \in \mathcal{T})\,[x \in U \subseteq E]$.

Proof.
($\Rightarrow$) If $E \in \mathcal{T}$, take $U = E$ for $x \in E$.
($\Leftarrow$) Set $\mathcal{U}_E := \{U \in \mathcal{T} : U \subseteq E\}$. The hypothesis implies $E = \bigcup \mathcal{U}_E$, so $E \in \mathcal{T}$ by Definition 6.1.1(ii).
Structural Commentary: The union axiom assembles pointwise open witnesses into the entire set.
Key Insight: Local open containment is equivalent to global openness. □

Proposition 6.1.4. *Metric Open Sets Form a Topology*
Let (X, d) be a metric space, τ_d be collection of metric-open sets from Definition 3.1.2. Then τ_d is a topology on X.

Proof. $\emptyset, X \in \tau_d$ by Definition 3.1.2. Collection τ_d is closed under arbitrary unions and finite intersections by Theorem 3.1.19(i),(iii). Hence τ_d is a topology by Definition 6.1.1.
Structural Commentary: Metric union and finite-intersection calculus verifies the three topology axioms.
Key Insight: General topology keeps the open-set calculus and forgets the distance function. □

Example 6.1.5. *Discrete and Trivial Topologies*
On a nonempty set X, the *discrete topology* is 2^X, so every subset is open and closed.
It is induced by the discrete metric. The *trivial topology* is $\{\emptyset, X\}$, so the only neighborhood of any point is X.

Definition 6.1.6. *Closure, Interior, and Boundary*
Let $(X, \mathcal{T})$ be a topological space and let $E \subseteq X$.

(i) The *closure* is $\overline{E} := \{x \in X : (\forall U \in \mathcal{T})\,[x \in U \Rightarrow U \cap E \neq \emptyset]\}$.
(ii) The *interior* is $\operatorname{int}(E) := \bigcup \{U \in \mathcal{T} : U \subseteq E\}$.
(iii) The *exterior* is $\operatorname{ext}(E) := \operatorname{int}(X \setminus E)$.
(iv) The *boundary* of E is $\operatorname{bd}(E) := \overline{E} \setminus \operatorname{int}(E)$.

Proposition 6.1.7. *Closure, Interior, and Boundary Tests*
Let $(X, \mathcal{T})$ be a topological space and let $E \subseteq X$.

(i) $\operatorname{int}(E)$ *is the largest open subset of* E.
(ii) E *is open iff* $E = \operatorname{int}(E)$.
(iii) $\overline{E}$ *is the smallest closed subset of* X *containing* E.
(iv) E *is closed iff* $E = \overline{E}$.
(v) For $x \in X$, $x \in \operatorname{bd}(E)$ *iff* $(\forall U \in \mathcal{T})\,[x \in U \Rightarrow U \cap E \neq \emptyset \wedge U \cap (X \setminus E) \neq \emptyset]$.
(vi) $\operatorname{bd}(E) = \overline{E} \cap \overline{X \setminus E}$.
(vii) $\operatorname{bd}(E)$ *is closed.*

Proof.
(i) The set $\operatorname{int}(E) = \bigcup \{U \in \mathcal{T} : U \subseteq E\}$ is open by Definition 6.1.1(ii).
It contains every open subset of E and is contained in E, so it is the largest open subset of E.
(ii) ($\Rightarrow$) If E is open, then $E \subseteq \operatorname{int}(E) \subseteq E$, so $E = \operatorname{int}(E)$.
(ii) ($\Leftarrow$) If $E = \operatorname{int}(E)$, then E is open by (i).
(iii) Every $x \in E$ belongs to $\overline{E}$ by Definition 6.1.6(i). If $x \notin \overline{E}$, choose neighborhood U of x with $U \cap E = \emptyset$ by same definition.
Then $U \subseteq X \setminus \overline{E}$. Thus $X \setminus \overline{E}$ open by Proposition 6.1.3, so $\overline{E}$ closed.
If F is closed and $E \subseteq F$, then $X \setminus F$ is an open neighborhood of each $x \notin F$ and is disjoint from E.
Hence $\overline{E} \subseteq F$, proving the smallest-closed-superset claim.
(iv) ($\Rightarrow$) If E is closed, then $\overline{E} \subseteq E$ by (iii), while $E \subseteq \overline{E}$. Hence $E = \overline{E}$.
(iv) ($\Leftarrow$) If $E = \overline{E}$, then E is closed by (iii).
(v) ($\Rightarrow$) If $x \in \operatorname{bd}(E)$, then $x \in \overline{E} \setminus \operatorname{int}(E)$ by Definition 6.1.6(iv). Thus every neighborhood of x meets E by Definition 6.1.6(i).
No neighborhood of x lies in E, so each neighborhood also meets $X \setminus E$.
(v) ($\Leftarrow$) If every neighborhood of x meets both E and $X \setminus E$, then $x \in \overline{E}$ and $x \notin \operatorname{int}(E)$. Hence $x \in \operatorname{bd}(E)$ by Definition 6.1.6(iv).
(vi) By (v), $x \in \operatorname{bd}(E)$ iff $x \in \overline{E} \cap \overline{X \setminus E}$. Thus $\operatorname{bd}(E) = \overline{E} \cap \overline{X \setminus E}$.
(vii) Both factors in (vi) are closed by (iii), intersection is closed because its complement is the union of their open complements.
Structural Commentary: Open complements make closure minimal; two-sided neighborhood contact identifies the boundary.
Key Insight: Boundary points persist under approaches from both E and its complement. □

Definition 6.1.8. *Bases and Subbases*
Let X be a set.

(i) Let $\mathcal{T}$ be a topology on X, let $x \in X$, and let $\mathcal{B}_x$ be a collection of neighborhoods of x.
Then $\mathcal{B}_x$ is a *neighborhood basis at* x iff $(\forall U \in \mathcal{T})\,[x \in U \Rightarrow (\exists B \in \mathcal{B}_x)\, B \subseteq U]$.
(ii) A collection $\mathcal{B} \subseteq 2^X$ is a *basis* for a topology on X iff:
  (a) $X = \bigcup_{B \in \mathcal{B}} B$;
  (b) $(\forall B_1, B_2 \in \mathcal{B})(\forall x \in B_1 \cap B_2)(\exists B_3 \in \mathcal{B})\, x \in B_3 \subseteq B_1 \cap B_2$.

(iii) If $\mathcal{B}$ is a basis on X, the collection generated by $\mathcal{B}$ is $\mathcal{T}_{\mathcal{B}} := \{\bigcup_{B\in\mathcal{U}} B : \mathcal{U} \subseteq \mathcal{B}\}$.

(iv) For $\mathcal{S} \subseteq 2^X$, set $\mathcal{B}_{\mathcal{S}} := \{\bigcap_{j=1}^{n} S_j : n \in \mathbb{N},\ S_1, \dots, S_n \in \mathcal{S}\}$. The collection $\mathcal{S}$ is a *subbasis* on X iff $\bigcup \mathcal{S} = X$ and $\mathcal{B}_{\mathcal{S}}$ is a basis on X.

Key Insight: Bases generate open sets by unions; subbases generate bases by finite intersections.

Proposition 6.1.9. *Basis Characterization*
Let X be a set and let $\mathcal{B} \subseteq 2^X$.
(i) If $\mathcal{B}$ is a basis on X, then $\mathcal{T}_{\mathcal{B}} := \{\bigcup \mathcal{U} : \mathcal{U} \subseteq \mathcal{B}\}$ is a topology on X.
(ii) Let $\mathcal{T}$ be a topology on X with $\mathcal{B} \subseteq \mathcal{T}$ and $(\forall U \in \mathcal{T} \setminus \{\emptyset\})(\exists \mathcal{U} \subseteq \mathcal{B})\, U = \bigcup \mathcal{U}$. Then $\mathcal{B}$ is a basis on X.
(iii) Under the hypotheses of (ii), $\mathcal{T} = \mathcal{T}_{\mathcal{B}}$.

Proof.
(i) Set $O_{\mathcal{U}} := \bigcup \mathcal{U}$ for $\mathcal{U} \subseteq \mathcal{B}$. Then $X = O_{\mathcal{B}}$, $\emptyset = O_{\emptyset}$, and $\bigcup_{\alpha} O_{\mathcal{U}_\alpha} = O_{\bigcup_\alpha \mathcal{U}_\alpha}$.
If $x \in O_{\mathcal{U}} \cap O_{\mathcal{V}}$, choose $B_1 \in \mathcal{U}$ and $B_2 \in \mathcal{V}$ with $x \in B_1 \cap B_2$.
Choose $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$ by the basis refinement property. Hence $O_{\mathcal{U}} \cap O_{\mathcal{V}} = \bigcup\{B \in \mathcal{B} : B \subseteq O_{\mathcal{U}} \cap O_{\mathcal{V}}\}$.
Closure under finite intersections follows from this binary identity by Theorem 2.6.2. Thus $\mathcal{T}_{\mathcal{B}}$ is a topology.
(ii) If $X \neq \emptyset$, the hypothesis for $U = X$ supplies $\mathcal{U} \subseteq \mathcal{B}$ with $X = \bigcup \mathcal{U}$. If $X = \emptyset$, then $\bigcup \mathcal{B} = X$.
If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, then $B_1 \cap B_2 \in \mathcal{T} \setminus \{\emptyset\}$. Choose $\mathcal{V} \subseteq \mathcal{B}$ with $B_1 \cap B_2 = \bigcup \mathcal{V}$ by the hypothesis.
Then choose $B_3 \in \mathcal{V}$ with $x \in B_3 \subseteq B_1 \cap B_2$. Hence $\mathcal{B}$ is a basis on X by Definition 6.1.8(ii).
(iii) The inclusion $\mathcal{T}_{\mathcal{B}} \subseteq \mathcal{T}$ follows by $\mathcal{B} \subseteq \mathcal{T}$ and closure under arbitrary unions.
Every nonempty $U \in \mathcal{T}$ belongs to $\mathcal{T}_{\mathcal{B}}$ by the hypothesis. Since $\emptyset = \bigcup \emptyset \in \mathcal{T}_{\mathcal{B}}$, one has $\mathcal{T} \subseteq \mathcal{T}_{\mathcal{B}}$ and therefore $\mathcal{T} = \mathcal{T}_{\mathcal{B}}$.

Structural Commentary: Basis refinements rewrite binary intersections as basis unions; topology axioms complete the generation.
Key Insight: Refinement is precisely the intersection mechanism missing from arbitrary unions. □

Example 6.1.10. *Metric Balls as a Basis*
In metric space (X, d), every point x lies in $B_1(x)$. If $z \in B_r(x) \cap B_s(y)$, $B_\varepsilon(z) \subseteq B_r(x) \cap B_s(y)$ for $\varepsilon := \min\{r - d(x,z), s - d(y,z)\} > 0$. Hence $\{B_r(x) : x \in X,\ r > 0\}$ is a basis. The family $\{B_r(q) : q \in \mathbb{Q}^k,\ r \in \mathbb{Q}_{>0}\}$ is countable by Theorem 6.2.18 and Corollary 6.2.19. Coordinatewise rational density and the triangle inequality make this family a basis on $\mathbb{R}^k$ by Theorem 1.1.19(ii).
If $a < b$, then $\{[a, c) : a < c < b\} \cup \{(c, b] : a < c < b\}$ is a subbasis for the usual topology on $[a, b]$.

Definition 6.1.11. *Subspace Topology*
Let $(X, \mathcal{T})$ be topological space, $E \subseteq X$. The *subspace topology* on E is $\mathcal{T}_E := \{E \cap U : U \in \mathcal{T}\}$, and $(E, \mathcal{T}_E)$ is a *subspace* of X.

Proposition 6.1.12. *The Subspace Collection Is a Topology*
Let $(X, \mathcal{T})$ be a topological space and let $E \subseteq X$. Then $\mathcal{T}_E$ from Definition 6.1.11 is a topology on E.

Proof. The identities $\emptyset = E \cap \emptyset$ and $E = E \cap X$ put $\emptyset, E$ in $\mathcal{T}_E$. For $(U_\alpha)_{\alpha\in A} \subseteq \mathcal{T}$, $\bigcup_{\alpha\in A} (E \cap U_\alpha) = E \cap \bigcup_{\alpha\in A} U_\alpha \in \mathcal{T}_E$.
For $U_1, \dots, U_n \in \mathcal{T}$ with $n \geq 1$, $\bigcap_{j=1}^{n} (E \cap U_j) = E \cap \bigcap_{j=1}^{n} U_j \in \mathcal{T}_E$. Thus $\mathcal{T}_E$ is a topology on E by Definition 6.1.1.

Structural Commentary: Intersecting ambient opens with E preserves the topology axioms.
Key Insight: Unions and finite intersections commute with restriction to E. □

Definition 6.1.13. *Separation Properties*
Let $(X, \mathcal{T})$ be a topological space.
Disjoint sets $A, B \subseteq X$ are *separated by disjoint neighborhoods* iff $(\exists U, V \in \mathcal{T})\,[A \subseteq U \wedge B \subseteq V \wedge U \cap V = \emptyset]$. Space X is:
(i) T_1 iff $(\forall x \in X)\,\{x\}$ is closed;
(ii) *Hausdorff* iff $\{x\}$ and $\{y\}$ are separated by disjoint neighborhoods whenever $x, y \in X$ and $x \neq y$;
(iii) *regular* iff X is T_1 and $\{x\}, F$ can be separated by disjoint neighborhoods when $F \subseteq X$ is closed and $x \in X \setminus F$;
(iv) *normal* iff X is T_1 and any disjoint closed $F, G \subseteq X$ can be separated by disjoint neighborhoods.

Key Insight: Successive axioms separate distinct points, point–closed-set pairs, and closed-set pairs.

Proposition 6.1.14. *T_1 Spaces and Closed Points*
Let $(X, \mathcal{T})$ be a topological space. Then X is T_1 iff $(\forall x, y \in X)\,[x \neq y \Rightarrow (\exists U \in \mathcal{T})\,[x \in U \wedge y \notin U]]$.

Proof.
($\Rightarrow$) Assume X is T_1, and fix $x, y \in X$ with $x \neq y$. Then $\{y\}$ is closed, so $U = X \setminus \{y\} \in \mathcal{T}$ satisfies $x \in U$ and $y \notin U$.
($\Leftarrow$) Assume the stated condition. Fix $y \in X$ and set $\mathcal{U}_y := \{U \in \mathcal{T} : y \notin U\}$.
Every $x \in X \setminus \{y\}$ belongs to a member of $\mathcal{U}_y$ by the condition, while no member contains y.
Hence $X \setminus \{y\} = \bigcup \mathcal{U}_y \in \mathcal{T}$ by Definition 6.1.1(ii), so $\{y\}$ is closed. Thus X is T_1 by Definition 6.1.13(i).

Structural Commentary: Point exclusion makes singleton complements open.
Key Insight: T_1 means each point can be omitted near every other point. □

Proposition 6.1.15. *Metric Spaces Are Normal*
Let (X, d) be a metric space with metric topology τ_d.
(i) $(\forall x \in X)\,\{x\}$ is closed.
(ii) If $F_1, F_2 \subseteq X$ are closed and $F_1 \cap F_2 = \emptyset$, then $(\exists O_1, O_2 \in \tau_d)\,[F_1 \subseteq O_1 \wedge F_2 \subseteq O_2 \wedge O_1 \cap O_2 = \emptyset]$.
(iii) X is normal.

Proof.
(i) Fix $x \in X$. For $y \in X \setminus \{x\}$, $B_{\frac{d(x,y)}{2}}(y) \subseteq X \setminus \{x\}$. Thus $X \setminus \{x\}$ is open and $\{x\}$ is closed.
(ii)
(a) *Empty cases*: If $F_1 = \emptyset$, take $(O_1, O_2) = (\emptyset, X)$. If $F_1 \neq \emptyset$, $F_2 = \emptyset$, take $(O_1, O_2) = (X, \emptyset)$. Hence assume $F_1, F_2 \neq \emptyset$.

(b) *Distance construction*: For $i = 1, 2$, set $d_i(x) := \operatorname{dist}(x, F_i)$ and $O_i := \{x \in X : d_i(x) < d_{3-i}(x)\}$.
The complement of F_i lies in τ_d by Definition 6.1.2(ii). It is therefore metric-open by Definition 3.1.2(iii).
Thus F_i is metric-closed by Corollary 3.1.17. The map d_i is continuous by Proposition 5.1.22(ii).
Its zero set satisfies $d_i^{-1}(\{0\}) = \overline{F_i} = F_i$ by Proposition 5.1.22(iii) and Theorem 3.1.22(ii).
(c) *Open containment*: The map $d_i - d_{3-i}$ is continuous by Theorem 5.1.32(i).
Hence O_i is open by Theorem 5.1.18. If $x \in F_i$, then $d_i(x) = 0 < d_{3-i}(x)$ because $x \notin F_{3-i} = d_{3-i}^{-1}(\{0\})$. Thus $F_i \subseteq O_i$.
(d) *Disjointness*: The intersection $O_1 \cap O_2$ is empty because $d_1(x) < d_2(x)$ and $d_2(x) < d_1(x)$ cannot both hold.

(iii) The space X is normal by (i), (ii), and Definition 6.1.13(iv).

Structural Commentary: Distance zero sets and strict distance comparisons construct the required disjoint open neighborhoods.
Key Insight: No point can be closer to each closed set than to the other. □

Proposition 6.1.16. *Normality by Nested Neighborhoods*
Let X be a T_1 topological space. Then X is normal iff each closed–open inclusion $F \subseteq U$ admits an open O with $F \subseteq O \subseteq \overline{O} \subseteq U$.

Proof.

(⇒) Assume X is normal. Let $F \subseteq X$ be closed and let $U \subseteq X$ be open with $F \subseteq U$.
Choose disjoint open sets O, V with $F \subseteq O$ and $X \setminus U \subseteq V$ by Definition 6.1.13(iv).
Then $O \subseteq X \setminus V \subseteq U$. Since $X \setminus V$ is closed, $\overline{O} \subseteq X \setminus V \subseteq U$ by Proposition 6.1.7(iii).
(⇐) Assume the nested-neighborhood property. Let $A, B \subseteq X$ be closed with $A \cap B = \emptyset$.
By the property with $F = A$ and $U = X \setminus B$, choose open O with $A \subseteq O \subseteq \overline{O} \subseteq X \setminus B$.
The set $\overline{O}$ is closed by Proposition 6.1.7(iii). Hence O and $X \setminus \overline{O}$ are disjoint open neighborhoods of A and B, respectively.
Since X is T_1, X is normal by Definition 6.1.13(iv).

Structural Commentary: Disjoint neighborhoods place $\overline{O}$ inside U; taking its complement reverses the construction.
Key Insight: Separating F from $X \setminus U$ is equivalent to closing O within U. □

Figure 149 depicts the nested-neighborhood criterion inside X.

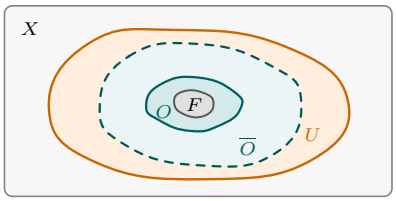

Figure 149: Normality thickens closed $F \subseteq U$ to an open O with $F \subseteq O \subseteq \overline{O} \subseteq U$.

6.2 Continuity and Topological Constructions

Inverse images organize topological constructions; homeomorphisms preserve open-set structure.
Products, quotients, and weak topologies are characterized by the maps they make continuous.

Definition 6.2.1. *Continuity Between Topological Spaces*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and let $f : X \to Y$.
For $x \in X$, set $\mathcal{N}_X(x) := \{U \in \mathcal{T}_X : x \in U\}$. For $y \in Y$, set $\mathcal{N}_Y(y) := \{O \in \mathcal{T}_Y : y \in O\}$.

(i) For $x_0 \in X$, f is *continuous at x_0* iff $(\forall O \in \mathcal{N}_Y(f(x_0)))(\exists U \in \mathcal{N}_X(x_0))\, f(U) \subseteq O$.
(ii) f is *continuous* iff $(\forall x \in X)$, f is continuous at x.

Proposition 6.2.2. *Continuity by Open Preimages*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and let $f : X \to Y$.

(i) f is continuous iff $(\forall O \in \mathcal{T}_Y)\, f^{-1}(O) \in \mathcal{T}_X$.
(ii) Let $A \subseteq X$ carry the subspace topology $\mathcal{T}_A$. If f is continuous, then $f|_A : (A, \mathcal{T}_A) \to (Y, \mathcal{T}_Y)$ is continuous.
(iii) Let $(Z, \mathcal{T}_Z)$ be topological space, $g : (Y, \mathcal{T}_Y) \to (Z, \mathcal{T}_Z)$ continuous. If f continuous, then $g \circ f : (X, \mathcal{T}_X) \to (Z, \mathcal{T}_Z)$ continuous.

Proof.

(i) (⇒) For $O \in \mathcal{T}_Y$ and $x \in f^{-1}(O)$, choose $U \in \mathcal{N}_X(x)$ with $f(U) \subseteq O$ by continuity at x.
The relations $x \in U \subseteq f^{-1}(O)$ imply $f^{-1}(O) \in \mathcal{T}_X$ by Proposition 6.1.3.
(i) (⇐) Assume $(\forall O \in \mathcal{T}_Y)\, f^{-1}(O) \in \mathcal{T}_X$. For $x_0 \in X$ and $O \in \mathcal{N}_Y(f(x_0))$, set $U := f^{-1}(O)$.
Then $U \in \mathcal{N}_X(x_0)$ and $f(U) \subseteq O$, so f is continuous at x_0 by Definition 6.2.1(i). Thus f is continuous by Definition 6.2.1(ii).
(ii) For $O \in \mathcal{T}_Y$, $f^{-1}(O) \in \mathcal{T}_X$ by (i), $(f|_A)^{-1}(O) = A \cap f^{-1}(O) \in \mathcal{T}_A$ by definition of $\mathcal{T}_A$; $f|_A$ is continuous by (i).
(iii) For $W \in \mathcal{T}_Z$, $g^{-1}(W) \in \mathcal{T}_Y$ by (i), $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)) \in \mathcal{T}_X$ by (i) for f; $g \circ f$ continuous by (i).

Structural Commentary: Neighborhood witnesses build opens; intersection and nesting preserve them.
Key Insight: Continuity is exactly preservation of openness under inverse images. □

Corollary 6.2.3. *Metric and Topological Continuity Agree*
Let $(X, d_X), (Y, d_Y)$ be metric spaces with metric topologies τ_X, τ_Y, and let $f : X \to Y$.
Then f is metrically continuous iff $f : (X, \tau_X) \to (Y, \tau_Y)$ is topologically continuous.

Proof.

(⇒) If f is metrically continuous, then $(\forall O \in \tau_Y)\, f^{-1}(O) \in \tau_X$ by Theorem 5.1.18.
Hence $f : (X, \tau_X) \to (Y, \tau_Y)$ is continuous by Proposition 6.2.2(i).
(⇐) Assume $f : (X, \tau_X) \to (Y, \tau_Y)$ is continuous. Then $(\forall O \in \tau_Y)\, f^{-1}(O) \in \tau_X$ by Proposition 6.2.2(i).
Hence f is metrically continuous by Theorem 5.1.18.

Structural Commentary: The common open-preimage condition identifies the two notions.
Key Insight: Their open-preimage conditions coincide. □

Proposition 6.2.4. *Continuity Tested on a Subbasis*
Let $(X, \mathcal{T}_X)$ be topological space, $\mathcal{S}$ be a subbasis generating $\mathcal{T}_Y$ on Y. For $f : X \to Y$, f is continuous iff $(\forall S \in \mathcal{S})\, f^{-1}(S) \in \mathcal{T}_X$.

Proof.
(⇒) For $S \in \mathcal{S} \subseteq \mathcal{T}_Y$, $f^{-1}(S) \in \mathcal{T}_X$ by Proposition 6.2.2(i).
(⇐) Assume $(\forall S \in \mathcal{S})\, f^{-1}(S) \in \mathcal{T}_X$, and let $\mathcal{B}_{\mathcal{S}}$ be the basis of finite intersections from $\mathcal{S}$.
For $B \in \mathcal{B}_{\mathcal{S}}$, choose $n \in \mathbb{N}$ and $S_1, \ldots, S_n \in \mathcal{S}$ with $B = \bigcap_{j=1}^{n} S_j$ by Definition 6.1.8(iv).
Then $f^{-1}(B) = \bigcap_{j=1}^{n} f^{-1}(S_j) \in \mathcal{T}_X$ by the hypothesis and Definition 6.1.1(iii).
For $O \in \mathcal{T}_Y$, choose $\mathcal{U} \subseteq \mathcal{B}_{\mathcal{S}}$ with $O = \bigcup \mathcal{U}$ by Definition 6.1.8(iii),(iv).
Then $f^{-1}(O) = \bigcup_{B \in \mathcal{U}} f^{-1}(B) \in \mathcal{T}_X$ by Definition 6.1.1(ii). Hence f is continuous by Proposition 6.2.2(i).
Structural Commentary: Finite intersections reach the basis; unions reach the topology.
Key Insight: Inverse images preserve both topology-generating operations. □

Definition 6.2.5. *Homeomorphism*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces.
(i) A bijection $f : X \to Y$ is a *homeomorphism* iff f and f^{-1} are continuous.
(ii) The spaces X and Y are *homeomorphic*, written $X \simeq Y$, iff a homeomorphism $f : X \to Y$ exists.

Proposition 6.2.6. *Homeomorphic Spaces Have the Same Open-Set Structure*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, let $f : X \to Y$ be a homeomorphism, and let $E \subseteq X$.
(i) $E \in \mathcal{T}_X$ iff $f(E) \in \mathcal{T}_Y$.
(ii) E is closed in X iff $f(E)$ is closed in Y.

Proof. The map f is bijective, and f, f^{-1} are continuous by Definition 6.2.5(i).
(i) (⇒) If $E \in \mathcal{T}_X$, then $f(E) = (f^{-1})^{-1}(E) \in \mathcal{T}_Y$ by Proposition 6.2.2(i) for f^{-1}.
(i) (⇐) If $f(E) \in \mathcal{T}_Y$, then $E = f^{-1}(f(E))$ by bijectivity. Hence $E \in \mathcal{T}_X$ by Proposition 6.2.2(i) for f.
(ii) (⇒) If E is closed in X, then $f(X \setminus E) \in \mathcal{T}_Y$ by (i).
The identity $f(X \setminus E) = Y \setminus f(E)$ holds by bijectivity, so $f(E)$ is closed in Y by Definition 6.1.2(ii).
(ii) (⇐) If $f(E)$ is closed in Y, then $Y \setminus f(E) \in \mathcal{T}_Y$ by Definition 6.1.2(ii). Thus $X \setminus E = f^{-1}(Y \setminus f(E)) \in \mathcal{T}_X$ by Proposition 6.2.2(i).
Hence E is closed in X by Definition 6.1.2(ii).
Structural Commentary: Continuity of f, f^{-1} transfers opens; bijectivity transfers complements.
Key Insight: A homeomorphism preserves and reflects the topology. □

Definition 6.2.7. *Product Topology*
Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ and $(X_i, \mathcal{T}_i)$, $i \in I$, be topological spaces.
(i) Set $\mathcal{B}_{X\times Y} := \{U \times V : U \in \mathcal{T}_X,\ V \in \mathcal{T}_Y\}$. The *product topology* on $X \times Y$ is generated by $\mathcal{B}_{X\times Y}$.
(ii) Set $\mathcal{B}_{\Pi} := \{\prod_{i\in I} U_i : (\forall i \in I)\, U_i \in \mathcal{T}_i,\ |\{i \in I : U_i \neq X_i\}| < \infty\}$. The *product topology* on $\prod_{i\in I} X_i$ is generated by $\mathcal{B}_{\Pi}$.

Proposition 6.2.8. *Product Collections Are Bases*
Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ and $(X_i, \mathcal{T}_i)$, $i \in I$, be topological spaces.
(i) $\mathcal{B}_{X\times Y}$ is a basis on $X \times Y$.
(ii) $\mathcal{B}_{\Pi}$ is a basis on $\prod_{i\in I} X_i$.

Proof.
(i) The rectangle $X \times Y$ belongs to $\mathcal{B}_{X\times Y}$. For $(x, y) \in (U_1 \times V_1) \cap (U_2 \times V_2)$, set $W := (U_1 \cap U_2) \times (V_1 \cap V_2)$.
Then $W \in \mathcal{B}_{X\times Y}$ by Definitions 6.1.1(iii) and 6.2.7(i).
Refinement relation $(x, y) \in W \subseteq (U_1 \times V_1) \cap (U_2 \times V_2)$ holds by definition of W. Hence $\mathcal{B}_{X\times Y}$ is a basis by Definition 6.1.8(ii).
(ii) The full product $\prod_{i\in I} X_i$ belongs to $\mathcal{B}_{\Pi}$. For boxes $U = \prod_{i\in I} U_i$ and $V = \prod_{i\in I} V_i$ in $\mathcal{B}_{\Pi}$ and $x \in U \cap V$, set $W_i := U_i \cap V_i$.
Then $(\forall i \in I)\, W_i \in \mathcal{T}_i$ by Definition 6.1.1(iii). The index sets satisfy $\{i \in I : W_i \neq X_i\} \subseteq \{i \in I : U_i \neq X_i\} \cup \{i \in I : V_i \neq X_i\}$.
Right-hand side finite by $U, V \in \mathcal{B}_{\Pi}$; $\prod_{i\in I} W_i \in \mathcal{B}_{\Pi}$ by Definition 6.2.7(ii), $x \in \prod_{i\in I} W_i \subseteq U \cap V$; $\mathcal{B}_{\Pi}$ is basis by Definition 6.1.8(ii).
Structural Commentary: Box intersections preserve finite support.
Key Insight: Intersecting product boxes introduces only finitely many restrictions. □

Proposition 6.2.9. *Product Topology and Coordinate Maps*
Let $(Z, \mathcal{T}_Z), (X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ be topological spaces, and equip $X \times Y$ with product topology $\mathcal{T}_{X\times Y}$.
Define $\pi_X : X \times Y \to X$ by $\pi_X(x, y) := x$. Define $\pi_Y : X \times Y \to Y$ by $\pi_Y(x, y) := y$.
For $f : Z \to X \times Y$, f is continuous iff $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous.

Proof.
(⇒) For $U \in \mathcal{T}_X$, $U \times Y \in \mathcal{T}_{X\times Y}$ by Definition 6.2.7(i). Thus $(\pi_X \circ f)^{-1}(U) = f^{-1}(U \times Y) \in \mathcal{T}_Z$ by Proposition 6.2.2(i) for f.
For $V \in \mathcal{T}_Y$, $X \times V \in \mathcal{T}_{X\times Y}$ by Definition 6.2.7(i). Thus $(\pi_Y \circ f)^{-1}(V) = f^{-1}(X \times V) \in \mathcal{T}_Z$ by Proposition 6.2.2(i) for f.
Hence both coordinate maps are continuous by Proposition 6.2.2(i).
(⇐) Let $U \times V \in \mathcal{B}_{X\times Y}$. Both coordinate preimages $(\pi_X \circ f)^{-1}(U)$ and $(\pi_Y \circ f)^{-1}(V)$ are open by Proposition 6.2.2(i).
Intersection $= f^{-1}(U \times V)$, lies in $\mathcal{T}_Z$ by Definition 6.1.1(iii). For $O \in \mathcal{T}_{X\times Y}$, $\mathcal{U} \subseteq \mathcal{B}_{X\times Y}$ with $O = \bigcup \mathcal{U}$ by Definition 6.2.7(i).
Then $f^{-1}(O) = \bigcup_{B \in \mathcal{U}} f^{-1}(B) \in \mathcal{T}_Z$ by Definition 6.1.1(ii). Hence f is continuous by Proposition 6.2.2(i).

Structural Commentary: Coordinate preimages handle rectangles; unions handle all product opens.
Key Insight: A product map is continuous exactly coordinatewise. □

Definition 6.2.10. *Quotient Topology*
Let $q : (X, \tau_X) \twoheadrightarrow Y$. The *quotient topology* on Y induced by q is $\tau_q := \{V \subseteq Y : q^{-1}(V) \in \tau_X\}$.

Proposition 6.2.11. *The Quotient Collection Is a Topology*
Let $q : (X, \tau_X) \twoheadrightarrow Y$, and let τ_q be the quotient topology induced by q.

(i) τ_q is a topology on Y.
(ii) $q : (X, \tau_X) \to (Y, \tau_q)$ is continuous.
(iii) If σ is a topology on Y and $q : (X, \tau_X) \to (Y, \sigma)$ is continuous, then $\sigma \subseteq \tau_q$.

Proof.

(i) The identities $q^{-1}(\emptyset) = \emptyset$ and $q^{-1}(Y) = X$ imply $\emptyset, Y \in \tau_q$ by Definitions 6.1.1(i) and 6.2.10.
For $(V_\alpha)_{\alpha\in A} \subseteq \tau_q$, $q^{-1}(\bigcup_{\alpha\in A} V_\alpha) = \bigcup_{\alpha\in A} q^{-1}(V_\alpha) \in \tau_X$ by Definition 6.1.1(ii). Hence $\bigcup_{\alpha\in A} V_\alpha \in \tau_q$ by Definition 6.2.10.
If $n \geq 1$ and $(V_j)_{j=1}^n \subseteq \tau_q$, then $q^{-1}(\bigcap_{j=1}^n V_j) = \bigcap_{j=1}^n q^{-1}(V_j) \in \tau_X$ by Definition 6.1.1(iii).
Hence $\bigcap_{j=1}^n V_j \in \tau_q$ by Definition 6.2.10. Hence τ_q is a topology by Definition 6.1.1.
(ii) For $V \in \tau_q$, $q^{-1}(V) \in \tau_X$ by Definition 6.2.10. Hence q is continuous by Proposition 6.2.2(i).
(iii) For $V \in \sigma$, $q^{-1}(V) \in \tau_X$ by Proposition 6.2.2(i), so $V \in \tau_q$ by Definition 6.2.10. Thus $\sigma \subseteq \tau_q$.

Structural Commentary: Inverse-image identities prove the axioms and compare admissible topologies.
Key Insight: Quotient-open sets are exactly those with open pullbacks. □

Proposition 6.2.12. *Quotient Universal Property*
Let $q : (X, \tau_X) \twoheadrightarrow Y$, and equip Y with the quotient topology τ_Y induced by q.
For $g : Y \to (Z, \tau_Z)$, $g : (Y, \tau_Y) \to (Z, \tau_Z)$ is continuous iff $g \circ q : (X, \tau_X) \to (Z, \tau_Z)$ is continuous.

Proof.

($\Rightarrow$) If g is continuous, then q is continuous by Proposition 6.2.11(ii). Hence $g \circ q$ is continuous by Proposition 6.2.2(iii).
($\Leftarrow$) Assume $g \circ q$ is continuous. For $W \in \tau_Z$, $q^{-1}(g^{-1}(W)) = (g \circ q)^{-1}(W) \in \tau_X$ by Proposition 6.2.2(i).
Thus $g^{-1}(W) \in \tau_Y$ by Definition 6.2.10, so g is continuous by Proposition 6.2.2(i).

Structural Commentary: Precomposition with q converts continuity into the quotient test.
Key Insight: Continuity descends precisely through quotient-open pullbacks. □

Definition 6.2.13. *Weak Topology Induced by Maps*
Let X be a set.

(i) For topologies $\mathcal{T}_1, \mathcal{T}_2$ on X, $\mathcal{T}_2$ is *weaker* than $\mathcal{T}_1$ iff $\mathcal{T}_2 \subseteq \mathcal{T}_1$. Equivalently, $\mathcal{T}_1$ is *stronger* than $\mathcal{T}_2$.
(ii) Let $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha\in A}$ be topological spaces, and let $(f_\alpha : X \to X_\alpha)_{\alpha\in A}$ be maps.
Set $\mathcal{S} := \{X\} \cup \{f_\alpha^{-1}(O) : \alpha \in A,\ O \in \mathcal{T}_\alpha\}$. The *weak topology induced by* $\{f_\alpha\}_{\alpha\in A}$ is the topology generated by $\mathcal{S}$.

Proposition 6.2.14. *Weak Topology Universal Property*
Let X be a set, let $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha\in A}$ be topological spaces, and let $(f_\alpha : X \to X_\alpha)_{\alpha\in A}$ be maps.
Let $\mathcal{T}_w$ be generated by $\mathcal{S} := \{X\} \cup \{f_\alpha^{-1}(O) : \alpha \in A,\ O \in \mathcal{T}_\alpha\}$.

(i) $\mathcal{T}_w$ is a topology on X.
(ii) $(\forall \alpha \in A)(\forall O \in \mathcal{T}_\alpha)\ f_\alpha^{-1}(O) \in \mathcal{T}_w$.
(iii) Let $\mathcal{T}$ be a topology on X satisfying $(\forall \alpha \in A)(\forall O \in \mathcal{T}_\alpha)\ f_\alpha^{-1}(O) \in \mathcal{T}$. Then the weak topology satisfies $\mathcal{T}_w \subseteq \mathcal{T}$.

Proof.

(i) Let $\mathcal{B}_w$ be finite intersections of members of $\mathcal{S}$. Then $X \in \mathcal{B}_w$ and $(\forall B_1, B_2 \in \mathcal{B}_w)\, B_1 \cap B_2 \in \mathcal{B}_w$ by definition of $\mathcal{B}_w$.
$\mathcal{B}_w$ is basis by Definition 6.1.8(ii); $\mathcal{T}_w = \{\bigcup \mathcal{U} : \mathcal{U} \subseteq \mathcal{B}_w\}$ by Definition 6.1.8(iii),(iv); $\mathcal{T}_w$ is topology by Proposition 6.1.9(i).
(ii) For $\alpha \in A$ and $O \in \mathcal{T}_\alpha$, $f_\alpha^{-1}(O) \in \mathcal{S} \subseteq \mathcal{T}_w$ by the definitions of $\mathcal{S}$ and $\mathcal{T}_w$.
(iii) If $\mathcal{T}$ satisfies the stated predicate, then $\mathcal{S} \subseteq \mathcal{T}$.
Every member of $\mathcal{B}_w$, every union of members lies in $\mathcal{T}$ by Definition 6.1.1(ii),(iii). Hence $\mathcal{T}_w \subseteq \mathcal{T}$ by union representation in (i).

Structural Commentary: Finite intersections and unions generate the least admissible topology.
Key Insight: Only subbasic preimages are forced to be open. □

6.3 Topological Compactness and Normality Tools

Compactness retains the open-cover principle without requiring a metric.
Together with Hausdorff and normal separation, it also supports strong closed-set and continuous-function tools.

Definition 6.3.1. *Compactness in Topological Spaces*
Let $(X, \mathcal{T})$ be a topological space.

(i) Let $E \subseteq X$, $(O_\alpha)_{\alpha\in A}$ be family of subsets of X. It is an *open cover* of E iff $(\forall \alpha \in A)\, O_\alpha \in \mathcal{T}$ and $E \subseteq \bigcup_{\alpha\in A} O_\alpha$.
(ii) The space X is *compact* iff $(\forall A)(\forall\, (O_\alpha)_{\alpha\in A})\, [((\forall \alpha \in A)\, O_\alpha \in \mathcal{T} \wedge X \subseteq \bigcup_{\alpha\in A} O_\alpha) \Longrightarrow (\exists\, J \subseteq A)\, (|J| < \infty \wedge X \subseteq \bigcup_{\alpha\in J} O_\alpha)]$.
(iii) For $K \subseteq X$, let $\mathcal{T}_K$ be subspace topology from Definition 6.1.11. The set K is *compact* iff $(K, \mathcal{T}_K)$ is compact.

Proposition 6.3.2. *Ambient Covers Test Subspace Compactness*
Let $(X, \mathcal{T})$ be topological space, $K \subseteq X$. Then K is compact iff every cover of K by members of $\mathcal{T}$ has a finite subcover.

Proof.

(⇒) Let $(O_\alpha)_{\alpha\in A} \subseteq \mathcal{T}$ cover K. By Definition 6.1.11, $(K \cap O_\alpha)_{\alpha\in A}$ is an open cover of $(K, \mathcal{T}_K)$.
By compactness of K, choose a finite $J \subseteq A$ such that $K = \bigcup_{\alpha\in J} (K \cap O_\alpha) \subseteq \bigcup_{\alpha\in J} O_\alpha$.

(⇐) Assume every cover of K by members of $\mathcal{T}$ has a finite subcover. Let $(V_\alpha)_{\alpha\in A} \subseteq \mathcal{T}_K$ cover K.
For $\alpha \in A$, set $O_\alpha := \bigcup\{O \in \mathcal{T} : K \cap O \subseteq V_\alpha\}$. By Definitions 6.1.1(ii) and 6.1.11, $O_\alpha \in \mathcal{T}$ and $K \cap O_\alpha = V_\alpha$.
Thus $(O_\alpha)_{\alpha\in A}$ covers K. By the hypothesis, choose a finite $J \subseteq A$ for which $(O_\alpha)_{\alpha\in J}$ covers K.
Hence $(V_\alpha)_{\alpha\in J}$ covers K, and K is compact by Definition 6.3.1(ii),(iii).

Structural Commentary: Intersecting ambient opens with K preserves exactly the cover seen by the subspace.
Key Insight: Subspace compactness can be tested without changing the ambient open sets. □

Proposition 6.3.3. *Finite-Intersection Compactness Test*
Let $(X, \mathcal{T})$ be topological space, $\mathfrak{C}_X$ be collection of nonempty families of closed subsets of X with finite intersection property. The finite intersection property is as in Definition 4.1.2. Then X is compact iff $(\forall \mathcal{F} \in \mathfrak{C}_X) \bigcap_{F\in\mathcal{F}} F \neq \emptyset$.

Proof.

(⇒) Assume X is compact. Let $\emptyset \neq \mathcal{F} \subseteq \mathcal{P}(X)$ be a family of closed subsets with finite intersection property.
Suppose $\bigcap_{F\in\mathcal{F}} F = \emptyset$. By Definition 6.1.2(ii) and De Morgan's law, $(X \setminus F)_{F\in\mathcal{F}}$ is an open cover of X.
By compactness, choose $n \in \mathbb{N}$ and $F_1, \ldots, F_n \in \mathcal{F}$ such that $X = \bigcup_{j=1}^{n} (X \setminus F_j)$.
By De Morgan's law, $\bigcap_{j=1}^{n} F_j = \emptyset$, contradicting the finite intersection property.

(⇐) Assume the closed-family condition, and let $(O_\alpha)_{\alpha\in A}$ be an open cover of X.
If $X = \emptyset$, then the empty subfamily is a finite subcover. Suppose $X \neq \emptyset$ and no finite subfamily covers X.
Then $A \neq \emptyset$, and $\mathcal{F} := \{X \setminus O_\alpha : \alpha \in A\}$ is a nonempty family of closed subsets.
By the absence of a finite subcover and De Morgan's law, $\mathcal{F}$ has the finite intersection property.
By hypothesis, $\emptyset \neq \bigcap_{\alpha\in A} (X \setminus O_\alpha) = X \setminus \bigcup_{\alpha\in A} O_\alpha$. This contradicts that $(O_\alpha)_{\alpha\in A}$ covers X.

Structural Commentary: Complements exchange covers and intersections.
Key Insight: Compactness forbids closed FIP families from missing every point. □

Proposition 6.3.4. *Closed Subsets of Compact Spaces Are Compact*
Let $(X, \mathcal{T})$ be compact and let $K \subseteq X$ be closed. Then K is compact in the subspace topology.

Proof. Let $(O_\alpha)_{\alpha\in A} \subseteq \mathcal{T}$ cover K. By closedness of K, $X \setminus K \in \mathcal{T}$, so $(O_\alpha)_{\alpha\in A} \cup \{X \setminus K\}$ is an open cover of X.
By compactness of X, this cover has a finite subcover. After discarding $X \setminus K$, the remaining members still cover K.
Hence K is compact by Proposition 6.3.2.
Structural Commentary: Add the open complement to cover X.
Key Insight: Compactness passes to closed subspaces. □

Proposition 6.3.5. *Compact Sets Separate from Points and Are Closed*
Let $(X, \mathcal{T})$ be Hausdorff and let $K \subseteq X$ be compact in the subspace topology.

(i) If $y \in X \setminus K$, then $(\exists U, V \in \mathcal{T})\,[y \in U \wedge K \subseteq V \wedge U \cap V = \emptyset]$.
(ii) K is closed in X.

Proof.

(i) If $K = \emptyset$, then $U = X$, $V = \emptyset$. Suppose $K \neq \emptyset$.
For $x \in K$, choose disjoint $V_x, U_x \in \mathcal{T}$ such that $x \in V_x$ and $y \in U_x$ by the Hausdorff property.
By Proposition 6.3.2, choose $x_1, \ldots, x_n \in K$ such that $K \subseteq \bigcup_{j=1}^{n} V_{x_j}$. Set $V := \bigcup_{j=1}^{n} V_{x_j}$ and $U := \bigcap_{j=1}^{n} U_{x_j}$.
By Definitions 6.1.1(ii),(iii), $U, V \in \mathcal{T}$. By construction, $y \in U$, $K \subseteq V$, and $U \cap V = \emptyset$.

(ii) For $y \in X \setminus K$, choose $U_y \in \mathcal{T}$ as in (i); $X \setminus K = \bigcup_{y\in X\setminus K} U_y \in \mathcal{T}$ by Definition 6.1.1(ii); K closed by Definition 6.1.2(ii).

Structural Commentary: Compactness turns pointwise Hausdorff separations into a finite one.
Key Insight: Every exterior point has an open neighborhood avoiding K. □

Theorem 6.3.6. *Compact Hausdorff Spaces Are Normal*
Let $(X, \mathcal{T})$ be compact and Hausdorff. Then X is normal.

Proof. For distinct $x, y \in X$, choose open neighborhood of x that omits y by Hausdorff property; X is T_1 by Proposition 6.1.14.
Let $A, B \subseteq X$ be disjoint closed sets. By Proposition 6.3.4, A and B are compact. If $A = \emptyset$ or $B = \emptyset$, then $\emptyset$, X separate them.
Suppose $A, B \neq \emptyset$. For $b \in B$, choose disjoint $U_b, V_b \in \mathcal{T}$ such that $b \in U_b$ and $A \subseteq V_b$ by Proposition 6.3.5(i).
By Proposition 6.3.2, choose $b_1, \ldots, b_n \in B$ such that $B \subseteq \bigcup_{j=1}^{n} U_{b_j}$. Set $U := \bigcup_{j=1}^{n} U_{b_j}$ and $V := \bigcap_{j=1}^{n} V_{b_j}$.
Then $U, V \in \mathcal{T}$, $B \subseteq U$, $A \subseteq V$, and $U \cap V = \emptyset$. Thus X is normal by Definition 6.1.13.
Structural Commentary: Compactness makes finitely many Hausdorff separations suffice.
Key Insight: Disjoint closed compact sets have disjoint neighborhoods. □

Theorem 6.3.7. *Continuous Images of Compact Spaces Are Compact*
Let $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ be continuous. If X is compact, then $f(X)$ is compact in the subspace topology.

Proof. Let $(V_\alpha)_{\alpha\in A} \subseteq \mathcal{T}_Y$ cover $f(X)$. By Proposition 6.2.2(i), $(f^{-1}(V_\alpha))_{\alpha\in A} \subseteq \mathcal{T}_X$, and this family covers X.
By compactness, choose finite $J \subseteq A$ such that $X = \bigcup_{\alpha\in J} f^{-1}(V_\alpha)$. Then $f(X) \subseteq \bigcup_{\alpha\in J} V_\alpha$, so $f(X)$ is compact by Proposition 6.3.2.
Structural Commentary: Pull the cover back to the compact domain.
Key Insight: Finite subcovers descend through the image. □

Corollary 6.3.8. *Extreme Values on Compact Spaces*
Let X be nonempty compact topological space, $f : X \to \mathbb{R}$ be continuous.
Then $(\exists x_{\min}, x_{\max} \in X)\,[f(x_{\min}) = \inf f(X) \wedge f(x_{\max}) = \sup f(X)]$.

Proof. By Theorem 6.3.7, $f(X)$ is nonempty and compact in its subspace topology.
By Proposition 6.3.2 and Definition 4.1.1, $f(X)$ is compact as a metric subspace of $\mathbb{R}$.
Let $\iota : f(X) \to \mathbb{R}$ be the inclusion map. Since ι preserves the induced metric, it is continuous by Theorem 5.1.26(ii).
By Theorem 5.1.39, choose $m, M \in f(X)$ such that $m = \min f(X)$ and $M = \max f(X)$.
Choose $x_{\min}, x_{\max} \in X$ such that $f(x_{\min}) = m$ and $f(x_{\max}) = M$.
Structural Commentary: The image converts topological compactness into the metric hypothesis of the earlier extreme-value theorem.
Key Insight: Applying the metric result to the inclusion $f(X) \hookrightarrow \mathbb{R}$ locates both endpoint values in the image. □

Theorem 6.3.9. *Compact-to-Hausdorff Bijection Criterion*
Let $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ be a continuous bijection. If X is compact and Y is Hausdorff, then f is a homeomorphism.

Proof. Let $O \in \mathcal{T}_X$, and set $F := X \setminus O$. By Proposition 6.3.4, F is compact. By Theorem 6.3.7, $f(F)$ is compact.
By Proposition 6.3.5(ii), $f(F)$ is closed in Y. By bijectivity of f, $(f^{-1})^{-1}(O) = f(O) = Y \setminus f(F) \in \mathcal{T}_Y$.
Thus f^{-1} is continuous by Proposition 6.2.2(i). Hence f is a homeomorphism by Definition 6.2.5(i).
Structural Commentary: Closed sets map compact, then closed.
Key Insight: Compact-to-Hausdorff bijections have continuous inverses. □

Proposition 6.3.10. *Compact Hausdorff Topologies Are Rigid*
Let X be a set and let $\mathcal{T}, \mathcal{T}_1, \mathcal{T}_2$ be topologies on X.

(i) If $(X, \mathcal{T})$ is compact and $\mathcal{T}_1 \subseteq \mathcal{T}$, then $(X, \mathcal{T}_1)$ is compact.
(ii) If $(X, \mathcal{T})$ is Hausdorff and $\mathcal{T} \subseteq \mathcal{T}_2$, then $(X, \mathcal{T}_2)$ is Hausdorff.
(iii) If $(X, \mathcal{T})$ is compact Hausdorff, $\mathcal{T}_1 \subseteq \mathcal{T}$, and $(X, \mathcal{T}_1)$ is Hausdorff, then $\mathcal{T}_1 = \mathcal{T}$.
(iv) If $(X, \mathcal{T})$ is compact Hausdorff, $\mathcal{T} \subseteq \mathcal{T}_2$, and $(X, \mathcal{T}_2)$ is compact, then $\mathcal{T}_2 = \mathcal{T}$.

Proof.

(i) Every $\mathcal{T}_1$-open cover is a $\mathcal{T}$-open cover. Hence it has a finite subcover by compactness of $(X, \mathcal{T})$.
(ii) For distinct $x, y \in X$, choose disjoint $\mathcal{T}$-open neighborhoods by Hausdorff property. They are $\mathcal{T}_2$-open because $\mathcal{T} \subseteq \mathcal{T}_2$.
(iii) The map $\mathrm{id} : (X, \mathcal{T}) \to (X, \mathcal{T}_1)$ is bijective. It is continuous by $\mathcal{T}_1 \subseteq \mathcal{T}$ and Proposition 6.2.2(i).
By Theorem 6.3.9, it is a homeomorphism. Hence $\mathcal{T}_1 = \mathcal{T}$ by Proposition 6.2.6(i).
(iv) The map $\mathrm{id} : (X, \mathcal{T}_2) \to (X, \mathcal{T})$ is bijective. It is continuous by $\mathcal{T} \subseteq \mathcal{T}_2$ and Proposition 6.2.2(i).
By Theorem 6.3.9, it is a homeomorphism. Hence $\mathcal{T}_2 = \mathcal{T}$ by Proposition 6.2.6(i).

Structural Commentary: Weakening preserves covers; strengthening preserves separations.
Key Insight: Compact Hausdorff topologies cannot be enlarged or shrunk compatibly. □

Lemma 6.3.11. *Maximal Finite-Intersection Families*
Assume Zorn's lemma. Let X be a set, and let $\mathcal{A} \subseteq \mathcal{P}(X)$ have the finite intersection property.
Then $\mathcal{A}$ is contained in a maximal family $\mathcal{M} \subseteq \mathcal{P}(X)$ with the finite intersection property.

Proof.

(i) *Extension poset*: Write $\mathrm{FIP}(\mathcal{B})$ when $\mathcal{B}$ has the finite intersection property.
Set $\mathfrak{P} := \{\mathcal{B} \subseteq \mathcal{P}(X) : \mathcal{A} \subseteq \mathcal{B} \wedge \mathrm{FIP}(\mathcal{B})\}$. Then $\mathcal{A} \in \mathfrak{P}$, and inclusion partially orders $\mathfrak{P}$ by Proposition 6.1.5.
(ii) *Chain upper bounds*: The empty chain has upper bound $\mathcal{A}$. Let $\emptyset \neq \mathfrak{C} \subseteq \mathfrak{P}$ be a chain, and set $\mathcal{U} := \bigcup \mathfrak{C}$.
For a finite nonempty family $\{E_1, \dots, E_n\} \subseteq \mathcal{U}$, choose $\mathcal{B}_j \in \mathfrak{C}$ such that $E_j \in \mathcal{B}_j$.
Because $\mathfrak{C}$ is a chain, one of $\mathcal{B}_1, \dots, \mathcal{B}_n$ contains all the others.
Thus $\bigcap_{j=1}^n E_j \neq \emptyset$ by the finite intersection property of that member. Hence $\mathcal{U} \in \mathfrak{P}$ and $\mathcal{U}$ is an upper bound of $\mathfrak{C}$.
(iii) *Maximal extension*: By the Zorn hypothesis, $\mathfrak{P}$ has a maximal member $\mathcal{M}$.
By the definition of $\mathfrak{P}$, $\mathcal{A} \subseteq \mathcal{M}$ and $\mathcal{M}$ has the finite intersection property.
Maximality in $\mathfrak{P}$ is exactly maximality among families in $\mathcal{P}(X)$ with that property.

Structural Commentary: Chain unions preserve FIP because tests are finite.
Key Insight: Zorn extends any FIP family to a maximal one. □

Lemma 6.3.12. *Closure Properties of Maximal FIP Families*
Let X be a nonempty set, and let $\mathcal{M} \subseteq \mathcal{P}(X)$ be maximal with respect to the finite intersection property.

(i) $X \in \mathcal{M}$.
(ii) If $n \in \mathbb{N}$ and $M_1, \dots, M_n \in \mathcal{M}$, then $\bigcap_{j=1}^{n} M_j \in \mathcal{M}$.
(iii) If $C \subseteq X$ and $(\forall n \in \mathbb{N})(\forall M_1, \dots, M_n \in \mathcal{M})\, C \cap \bigcap_{j=1}^{n} M_j \neq \emptyset$, then $C \in \mathcal{M}$.

Proof.

(i) Since $X \neq \emptyset$, adjoining X to $\mathcal{M}$ preserves every nonempty finite intersection.
Thus $\mathcal{M} \cup \{X\}$ has the finite intersection property, so $X \in \mathcal{M}$ by maximality.
(ii) Set $N := \bigcap_{j=1}^{n} M_j$. Every nonempty finite intersection from $\mathcal{M} \cup \{N\}$ is a finite intersection of members of $\mathcal{M}$.
Hence $\mathcal{M} \cup \{N\}$ has the finite intersection property, and $N \in \mathcal{M}$ by maximality.
(iii) By (i) and the hypothesis with $n = 1$ and $M_1 = X$, $C \neq \emptyset$.
Every nonempty finite intersection from $\mathcal{M} \cup \{C\}$ is nonempty by the hypothesis and the finite intersection property of $\mathcal{M}$.
Hence $\mathcal{M} \cup \{C\}$ has the finite intersection property, and $C \in \mathcal{M}$ by maximality.

Structural Commentary: Maximality absorbs every FIP-compatible set.
Key Insight: Finite intersections are already forced inside $\mathcal{M}$. □

Theorem 6.3.13. *Tychonoff Product Theorem*
Assume the axiom of choice and Zorn's lemma. Let $(X_\lambda)_{\lambda\in\Lambda}$ be a family of compact topological spaces. Then $X := \prod_{\lambda\in\Lambda} X_\lambda$ is compact in the product topology.

Proof.

(i) *Maximal family*: If $X = \emptyset$, then its empty subfamily is a finite subcover of any open cover.
Suppose $X \neq \emptyset$, and let $\mathcal{F}$ be a nonempty family of closed subsets of X with the finite intersection property.
By Lemma 6.3.11, choose a maximal family $\mathcal{M} \supseteq \mathcal{F}$ with the finite intersection property.

(ii) *Coordinate choice*: For $\lambda \in \Lambda$, define $\pi_\lambda : X \to X_\lambda$ by $\pi_\lambda(x) := x_\lambda$. For $n \in \mathbb{N}$ and $M_1, \dots, M_n \in \mathcal{M}$, set $P := \bigcap_{j=1}^{n} M_j$.
By the finite intersection property, $P \neq \emptyset$ and $\pi_\lambda(P) \subseteq \bigcap_{j=1}^{n} \overline{\pi_\lambda(M_j)}$.
Thus $(\overline{\pi_\lambda(M)})_{M\in\mathcal{M}}$ is a nonempty family of closed subsets of X_λ with the finite intersection property.
Set $I_\lambda := \bigcap_{M\in\mathcal{M}} \overline{\pi_\lambda(M)}$. Then $I_\lambda \neq \emptyset$ by compactness of X_λ and Proposition 6.3.3.
By the choice hypothesis, select $(x_\lambda)_{\lambda\in\Lambda} \in \prod_{\lambda\in\Lambda} I_\lambda$. Set $x := (x_\lambda)_{\lambda\in\Lambda} \in X$.

(iii) *Coordinate cylinders*: Let $\lambda \in \Lambda$, and let $O \subseteq X_\lambda$ be an open neighborhood of x_λ. Set $N := \pi_\lambda^{-1}(O)$.
For $n \in \mathbb{N}$ and $M_1, \dots, M_n \in \mathcal{M}$, set $P := \bigcap_{j=1}^{n} M_j$. By Lemma 6.3.12(ii), $P \in \mathcal{M}$.
Since $x_\lambda \in \overline{\pi_\lambda(P)}$, one has $O \cap \pi_\lambda(P) \neq \emptyset$, and hence $N \cap P \neq \emptyset$. Therefore $N \in \mathcal{M}$ by Lemma 6.3.12(iii).

(iv) *Basic neighborhoods*: Let $B = \prod_{\lambda\in\Lambda} U_\lambda$ be a basic neighborhood of x. Set $S := \{\lambda \in \Lambda : U_\lambda \neq X_\lambda\}$.
If $S = \emptyset$, then $B = X \in \mathcal{M}$ by Lemma 6.3.12(i). If $S = \{\lambda_1, \dots, \lambda_n\}$, then $B = \bigcap_{j=1}^{n} \pi_{\lambda_j}^{-1}(U_{\lambda_j})$.
By (iii), $(\forall j \in \{1, \dots, n\})\, \pi_{\lambda_j}^{-1}(U_{\lambda_j}) \in \mathcal{M}$. Hence $B \in \mathcal{M}$ by Lemma 6.3.12(ii).

(v) *Common point*: Fix $F \in \mathcal{F}$, and let U be a neighborhood of x.
By Definition 6.2.7(ii) and Proposition 6.2.8(ii), U contains a basic neighborhood B of x.
By (iv), $B \in \mathcal{M}$, while $F \in \mathcal{M}$ by $\mathcal{F} \subseteq \mathcal{M}$. By the finite intersection property, $U \cap F \supseteq B \cap F \neq \emptyset$.
Thus $x \in \overline{F} = F$ by Proposition 6.1.7(iv). Therefore $x \in \bigcap_{F\in\mathcal{F}} F$, and X is compact by Proposition 6.3.3.

Structural Commentary: Coordinate compactness selects a candidate point.
Key Insight: Maximal FIP forces all basic neighborhoods of that point to remain compatible. □

Definition 6.3.14. *Compact-Open Topology*
Let X, Y be topological spaces, and set $Y^X := \{f : X \to Y\}$. For compact $K \subseteq X$ and open $O \subseteq Y$, set $\mathcal{U}_{K,O} := \{f \in Y^X : f(K) \subseteq O\}$.
Let $\mathcal{S}_{\text{co}}$ be the collection of all such sets $\mathcal{U}_{K,O}$. The *compact-open topology* on Y^X is the topology generated by the subbasis $\mathcal{S}_{\text{co}}$.

Proposition 6.3.15. *The Compact-Open Collection Is a Subbasis*
Let X, Y be topological spaces, $Y^X := \{f : X \to Y\}$. For compact $K \subseteq X$ and open $O \subseteq Y$, set $\mathcal{U}_{K,O} := \{f \in Y^X : f(K) \subseteq O\}$. Then the collection $\mathcal{S}_{\text{co}}$ of all such sets $\mathcal{U}_{K,O}$ is a subbasis on Y^X.

Proof. The set $\emptyset \subseteq X$ is compact, $\emptyset \subseteq Y$ is open, and $\mathcal{U}_{\emptyset,\emptyset} = Y^X$. Thus $Y^X \in \mathcal{S}_{\text{co}}$.
Let $\mathcal{B}_{\text{co}}$ be collection of all nonempty finite intersections of members of $\mathcal{S}_{\text{co}}$. Then $Y^X \in \mathcal{B}_{\text{co}}$, $\mathcal{B}_{\text{co}}$ is closed under finite intersections.
Hence $\mathcal{B}_{\text{co}}$ is a basis by Definition 6.1.8(ii). Thus $\mathcal{S}_{\text{co}}$ is a subbasis by Definition 6.1.8(iv).
Structural Commentary: Finite compact-open constraints remain closed under finite intersection.
Key Insight: The empty-domain constraint supplies the full function space. □

Remark 6.3.16. *Compact-Open Versus Product Topology*
On $Y^X = \prod_{x\in X} Y$, the product topology tests finitely many coordinates $f(x)$.
The compact-open topology tests compact sets at once through $\mathcal{U}_{K,O} = \{f \in Y^X : f(K) \subseteq O\}$.
When K is infinite, one subbasic constraint can restrict infinitely many coordinates.

Definition 6.3.17. *Normally Ascending Families*
Let X be a topological space, $\Lambda \subseteq \mathbb{R}$, $(O_\lambda)_{\lambda\in\Lambda}$ be family of open subsets of X.
It is *normally ascending* iff $(\forall \lambda_1, \lambda_2 \in \Lambda)\, [\lambda_1 < \lambda_2 \Rightarrow \overline{O_{\lambda_1}} \subseteq O_{\lambda_2}]$.
Key Insight: Later open sets contain the closures of earlier ones.

Lemma 6.3.18. *Normally Ascending Families Define Continuous Functions*
Let X be a topological space, let $a < b$, let $\Lambda \subseteq (a, b)$ be dense, and let $(O_\lambda)_{\lambda\in\Lambda}$ be normally ascending. For $x \in X$, set $\Lambda_x := \{\lambda \in \Lambda : x \in O_\lambda\}$ and $g(x) := \inf \Lambda_x$, with $\inf \emptyset := b$. Then $g : X \to [a, b]$ is continuous.

Proof. Fix $x \in X$. If $\Lambda_x \neq \emptyset$, then Λ_x is bounded below by a. By Corollary 1.1.15, $\inf \Lambda_x$ exists, and $a \leq \inf \Lambda_x < b$.
Together with the empty-set convention, this proves that $g : X \to [a, b]$ is well-defined. Fix $c \in (a, b)$.
By the defining property of the infimum, $\{x \in X : g(x) < c\} = \bigcup_{\substack{\lambda\in\Lambda \\ \lambda<c}} O_\lambda$.
If $\lambda \in \Lambda$, $\lambda > c$, and $x \notin \overline{O_\lambda}$, then $(\forall \mu \in \Lambda)\, [\mu < \lambda \Rightarrow x \notin O_\mu]$ by normal ascent. Hence $g(x) \geq \lambda > c$.
Conversely, if $g(x) > c$, choose $\mu, \lambda \in \Lambda$ by density such that $c < \mu < \lambda < g(x)$.
Then $x \notin O_\lambda$, while $\overline{O_\mu} \subseteq O_\lambda$ by normal ascent. Thus $x \notin \overline{O_\mu}$, and therefore $\{x \in X : g(x) > c\} = \bigcup_{\substack{\lambda\in\Lambda \\ \lambda>c}} (X \setminus \overline{O_\lambda})$.
Both threshold sets are open. Set $\mathcal{R} := \{[a, c) : c \in (a, b)\} \cup \{(c, b] : c \in (a, b)\}$.
Its finite intersections form the relative-interval basis, so $\mathcal{R}$ is a subbasis for the metric topology on $[a, b]$.
For $c \in (a, b)$, $g^{-1}([a, c)) = \{g < c\}$ and $g^{-1}((c, b]) = \{g > c\}$ are open. Hence g is continuous by Proposition 6.2.4.
Structural Commentary: Threshold preimages reduce to unions of nested open or closed-complement sets.
Key Insight: Normal ascent makes upper thresholds open. □

Lemma 6.3.19. *Dyadic Neighborhood Ladders in Normal Spaces*
Let X be normal, let $F \subseteq X$ be closed, let U be an open neighborhood of F, and let $a < b$.
Then dense set $\Lambda \subseteq (a,b)$ and normally ascending family $(O_\lambda)_{\lambda\in\Lambda}$ of open sets exist, chosen so that $(\forall \lambda \in \Lambda)\, F \subseteq O_\lambda \subseteq \overline{O_\lambda} \subseteq U$.

Proof. The space X is T_1 by Definition 6.1.13.

(i) *Index set*: First construct the family on $(0,1)$. For $n \in \mathbb{N}$, set $D_n := \{m2^{-n} : 0 < m < 2^n\}$, and set $D := \bigcup_{n\in\mathbb{N}} D_n$.

(ii) *Base*: By Proposition 6.1.16, choose an open set $O_{1/2}$. The choice satisfies $F \subseteq O_{1/2} \subseteq \overline{O_{1/2}} \subseteq U$.

(iii) *Dyadic insertion*: Suppose the sets indexed by D_n have been chosen with the required containments between adjacent indices.
Let $r = m2^{-(n+1)}$ be a new index, where m is odd. Set $p := (m-1)2^{-(n+1)}$ and $q := (m+1)2^{-(n+1)}$.
Set $L_p := F$ when $p = 0$ and $L_p := \overline{O_p}$ otherwise. Set $W_q := U$ when $q = 1$ and $W_q := O_q$ otherwise.
By the induction hypothesis, L_p is closed, W_q is open, and $L_p \subseteq W_q$.
By Proposition 6.1.16, choose an open set O_r such that $L_p \subseteq O_r \subseteq \overline{O_r} \subseteq W_q$.
Perform the finitely many insertions at level $n+1$; all adjacent containments remain valid.

(iv) *Conclusion*: Fix $0 < s < t < 1$. By Theorem 2.2.10(v), choose $n \in \mathbb{N}$ such that $2^{-n} < t - s$.
The finite set $S := \{k \in \mathbb{Z} : 0 \le k \le 2^n,\ k \le 2^n s\}$ is nonempty; set $m := \max S$.
Then $s < (m+1)2^{-n} < s + 2^{-n} < t$, so $(m+1)2^{-n} \in D \cap (s,t)$. Thus D is dense in $(0,1)$.
If $\lambda_1, \lambda_2 \in D$ and $\lambda_1 < \lambda_2$, choose $n \in \mathbb{N}$ such that $\lambda_1, \lambda_2 \in D_n$. Then $\overline{O_{\lambda_1}} \subseteq O_{\lambda_2}$ by the adjacent containment chain.
By the endpoint containments, $(\forall \lambda \in D)\, F \subseteq O_\lambda \subseteq \overline{O_\lambda} \subseteq U$. Set $\Lambda := \{a + (b-a)r : r \in D\}$ and assign $O_{a+(b-a)r} := O_r$.
Since $r \mapsto a + (b-a)r$ is an order-preserving bijection from $(0,1)$ to (a,b), the transported family has all the required properties.

Structural Commentary: Normality inserts closed-neighborhood buffers.
Key Insight: Dyadic induction builds the whole ascending ladder. □

Theorem 6.3.20. *Urysohn's Lemma*
Let X be normal, $A, B \subseteq X$ be disjoint closed sets, $a < b$ in $\mathbb{R}$.
Then continuous map $f : X \to [a,b]$ exists such that $f|_A = a$ and $f|_B = b$.

Proof.

(i) *Open ladder*: The set $X \setminus B$ is an open neighborhood of A.
By Lemma 6.3.19, choose dense $\Lambda \subseteq (a,b)$, normally ascending family $(O_\lambda)_{\lambda\in\Lambda}$. Choice satisfies $(\forall \lambda \in \Lambda)\, A \subseteq O_\lambda \subseteq \overline{O_\lambda} \subseteq X \setminus B$.

(ii) *Separator*: Define $f(x) := \inf\{\lambda \in \Lambda : x \in O_\lambda\}$, with $\inf \emptyset := b$. By Lemma 6.3.18, $f : X \to [a,b]$ is continuous.

(iii) *Boundary values*: If $x \in A$, then $(\forall \lambda \in \Lambda)\, x \in O_\lambda$. Since Λ is dense in (a,b), $\inf \Lambda = a$, and hence $f(x) = a$.
If $x \in B$, then $(\forall \lambda \in \Lambda)\, x \notin O_\lambda$, so $f(x) = b$ by the empty-set convention. Thus $f|_A = a$ and $f|_B = b$.

Structural Commentary: A nested open ladder becomes a continuous separator.
Key Insight: Normality separates closed sets by a real-valued function. □

Proposition 6.3.21. *Real-Valued Continuity under Sums and Uniform Error Bounds*
Let X be a topological space.

(i) If $u, v : X \to \mathbb{R}$ are continuous and $c, d \in \mathbb{R}$, then $cu + dv$ is continuous.

(ii) Let (s_n) be a sequence of continuous maps from X to $\mathbb{R}$, and let $g : X \to \mathbb{R}$.
If $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall n \ge N)(\forall x \in X)\ |s_n(x) - g(x)| < \varepsilon$, then g is continuous.

Proof.

(i) Define $A : \mathbb{R}^2 \to \mathbb{R}$ and $M_c : \mathbb{R} \to \mathbb{R}$ by $A(r,s) := r + s$ and $M_c(r) := cr$. Fix $(r,s) \in \mathbb{R}^2$ and an open neighborhood W of $r + s$.
Choose $\varepsilon > 0$ such that $B_\varepsilon(r+s) \subseteq W$. Then $A\big((r - \frac{\varepsilon}{2}, r + \frac{\varepsilon}{2}) \times (s - \frac{\varepsilon}{2}, s + \frac{\varepsilon}{2})\big) \subseteq W$.
Thus A is continuous in the product topology by Definition 6.2.1. The map M_0 is continuous because it is constant.
Suppose $c \neq 0$, fix $r \in \mathbb{R}$, and let W_c be an open neighborhood of cr. Choose $\eta > 0$ such that $B_\eta(cr) \subseteq W_c$.
Then $M_c(B_{\frac{\eta}{|c|}}(r)) \subseteq B_\eta(cr) \subseteq W_c$, so M_c is continuous at r by Definition 6.2.1(i). Thus M_c is continuous by Definition 6.2.1(ii).
Hence $M_c \circ u$ and $M_d \circ v$ are continuous by Proposition 6.2.2(iii). Their pair is continuous by Proposition 6.2.9.
Therefore $cu + dv = A \circ (M_c \circ u, M_d \circ v)$ is continuous by Proposition 6.2.2(iii).

(ii) Fix $x_0 \in X$ and an open neighborhood O of $g(x_0)$. Choose $\varepsilon > 0$ such that $B_\varepsilon(g(x_0)) \subseteq O$.
By the uniform-convergence hypothesis, choose $n \in \mathbb{N}$ such that $(\forall x \in X)\ |s_n(x) - g(x)| < \frac{\varepsilon}{3}$. Set $U := s_n^{-1}(B_{\frac{\varepsilon}{3}}(s_n(x_0)))$.
Then U is an open neighborhood of x_0 by continuity of s_n and Proposition 6.2.2(i).
For $x \in U$, $|g(x) - g(x_0)| \le |g(x) - s_n(x)| + |s_n(x) - s_n(x_0)| + |s_n(x_0) - g(x_0)| < \varepsilon$.
Thus $g(U) \subseteq O$, so g is continuous by Definition 6.2.1.

Structural Commentary: Product continuity controls finite sums; uniform error transfers neighborhoods through approximants.
Key Insight: A domain-wide error bound lets one continuous approximant control the limit throughout X. □

Theorem 6.3.22. *Tietze Extension Theorem, Bounded Form*
Let X be normal, let $F \subseteq X$ be closed, let $a < b$ in $\mathbb{R}$, and let $f : F \to [a,b]$ be continuous.
Then a continuous map $g : X \to [a,b]$ exists with $g|_F = f$.

Proof.

(i) *Correction step*: Constant maps on F are continuous because their open preimages belong to $\{\emptyset, F\}$.
Hence $\widetilde{f} := \frac{2f - a - b}{b - a} : F \to [-1,1]$ is continuous by Proposition 6.3.21(i). Let $A > 0$, and let $h : F \to [-A, A]$ be continuous.
Set $C_- := \{x \in F : h(x) \le -\frac{A}{3}\}$ and $C_+ := \{x \in F : h(x) \ge \frac{A}{3}\}$.
The sets are disjoint, and $F \setminus C_- = h^{-1}((-\frac{A}{3}, A])$ and $F \setminus C_+ = h^{-1}([-A, \frac{A}{3}))$.
Both complements are open in F by Proposition 6.2.2(i), so C_- and C_+ are closed in F.
If $C \subseteq F$ is closed in F, choose an open $O \subseteq X$ such that $F \setminus C = F \cap O$. Then $X \setminus C = (X \setminus F) \cup O$ is open, so C is closed in X.
Thus C_- and C_+ are closed in X. By Theorem 6.3.20, choose a continuous $u : X \to [-\frac{A}{3}, \frac{A}{3}]$.
Choice satisfies $u|_{C_-} = -\frac{A}{3}$ and $u|_{C_+} = \frac{A}{3}$. By cases $h(x) \le -\frac{A}{3}$, $|h(x)| < \frac{A}{3}$, and $h(x) \ge \frac{A}{3}$, $(\forall x \in F)\ |h(x) - u(x)| \le \frac{2A}{3}$.

(ii) *Recursive approximation*: Set $s_0 := 0$, which is continuous.
For $n \in \mathbb{N}$, suppose $s_{n-1} : X \to \mathbb{R}$ is continuous and $(\forall x \in F)\ \left|\widetilde{f}(x) - s_{n-1}(x)\right| \le (\frac{2}{3})^{n-1}$.
By Proposition 6.2.2(ii), $s_{n-1}|_F$ is continuous. Set $A_n := (\frac{2}{3})^{n-1}$ and $h := \widetilde{f} - s_{n-1}|_F$.
Then $h : F \to [-A_n, A_n]$ is continuous by the induction bound and Proposition 6.3.21(i).
By the correction step with $A = A_n$, choose a continuous $u_n : X \to \mathbb{R}$.
The domain-wide bound $(\forall x \in X)\ |u_n(x)| \le \frac{1}{3}(\frac{2}{3})^{n-1}$ holds.
The residual bound $(\forall x \in F)\ \left|\widetilde{f}(x) - s_{n-1}(x) - u_n(x)\right| \le (\frac{2}{3})^n$ also holds.
Set $s_n := s_{n-1} + u_n$. By Proposition 6.3.21(i), s_n is continuous. Sequence (s_n) follows by induction.

(iii) *Limit and continuity*: Fix $x \in X$. If $m < n$, then $|s_n(x) - s_m(x)| \le \sum_{k=m+1}^{n} \frac{1}{3}(\frac{2}{3})^{k-1} \le (\frac{2}{3})^m$.
By Theorem 2.2.10(v), $(\frac{2}{3})^m \to 0$. Thus $(s_n(x))$ is Cauchy by Definition 2.1.13.
By Theorem 2.1.15, set $\widetilde{g}(x) := \lim_{n\to\infty} s_n(x) \in \mathbb{R}$. Fix $m \in \mathbb{N}$, $x \in X$, and $\delta > 0$.
Choose $n > m$ such that $|\widetilde{g}(x) - s_n(x)| < \delta$. Then $|\widetilde{g}(x) - s_m(x)| < \delta + (\frac{2}{3})^m$.
Since $\delta > 0$ is arbitrary, $(\forall m \in \mathbb{N})(\forall x \in X)\ |\widetilde{g}(x) - s_m(x)| \le (\frac{2}{3})^m$.
Hence $s_m \to \widetilde{g}$ uniformly, and $\widetilde{g}$ is continuous by Proposition 6.3.21(ii).

(iv) *Extension and range*: For $x \in F$, $\left|\widetilde{f}(x) - s_n(x)\right| \le (\frac{2}{3})^n \to 0$, so $\widetilde{g}|_F = \widetilde{f}$. For $x \in X$, $|s_n(x)| \le \sum_{k=1}^{n} \frac{1}{3}(\frac{2}{3})^{k-1} = 1 - (\frac{2}{3})^n$.
By the uniform error bound, $|\widetilde{g}(x)| \le |\widetilde{g}(x) - s_n(x)| + |s_n(x)| \le 1$. Hence $\widetilde{g}(X) \subseteq [-1, 1]$.
Set $g := \frac{(b-a)\widetilde{g}+a+b}{2} : X \to [a, b]$. By Proposition 6.3.21(i), g is continuous. By the definition of $\widetilde{f}$, $g|_F = f$.

Structural Commentary: Urysohn corrections shrink the residual geometrically.
Key Insight: Geometric domain-wide error decay turns approximate extensions into an exact one. □

6.4 Sequences, Countability, and Metrization

Countability axioms control when topological questions reduce to sequences and countable tests.
They also separate metric phenomena from genuinely non-metric topologies.

Definition 6.4.1. *Sequence Convergence*
Let X be a topological space, and let $\mathcal{N}_X(x)$ denote the neighborhoods of $x \in X$.
For $(x_n) \subseteq X$ and $x \in X$, the sequence (x_n) *converges* to x, written $x_n \to x$, iff $(\forall U \in \mathcal{N}_X(x))(\exists N \in \mathbb{N})(\forall n \ge N)\, x_n \in U$.

Definition 6.4.2. *Sequential Compactness*
A topological space X is *sequentially compact* iff $(\forall (x_n) \subseteq X)(\exists (n_k) \subseteq \mathbb{N})(\exists x \in X)\, [n_1 < n_2 < \cdots \wedge x_{n_k} \to x]$.
Key Insight: Sequential compactness is compactness tested by subsequences.

Proposition 6.4.3. *Product Convergence Is Pointwise*
Let $X^\Lambda = \prod_{\lambda\in\Lambda} X_\lambda$ carry the product topology from $\{(X_\lambda, \mathcal{T}_\lambda)\}_{\lambda\in\Lambda}$. For $\lambda \in \Lambda$, define $\pi_\lambda : X^\Lambda \to X_\lambda$ by $\pi_\lambda(z) := z(\lambda)$.
For $(x_n) \subseteq X^\Lambda$ and $x \in X^\Lambda$, $x_n \to x$ iff $(\forall \lambda \in \Lambda)\, x_n(\lambda) \to x(\lambda)$ in X_λ.

Proof.
(⇒) Fix $\lambda \in \Lambda$ and let $U \in \mathcal{T}_\lambda$ contain $x(\lambda)$. Cylinder $\pi_\lambda^{-1}(U)$ is basic neighborhood of x by Proposition 6.2.8(ii).
If $x_n \to x$, then $x_n \in \pi_\lambda^{-1}(U)$ eventually, equivalently $x_n(\lambda) \in U$ eventually. Hence $x_n(\lambda) \to x(\lambda)$.
(⇐) Assume $(\forall \lambda \in \Lambda)\, x_n(\lambda) \to x(\lambda)$, and let $G \in \mathcal{N}_{X^\Lambda}(x)$.
Choose a basic neighborhood $W = \prod_{\lambda\in\Lambda} U_\lambda$ with $x \in W \subseteq G$ by Proposition 6.2.8(ii).
Set $F := \{\lambda \in \Lambda : U_\lambda \ne X_\lambda\}$; then F is finite. For $\lambda \in F$, choose $N_\lambda \in \mathbb{N}$ such that $(\forall n \ge N_\lambda)\, x_n(\lambda) \in U_\lambda$.
If $F = \emptyset$, set $N := 1$; otherwise set $N := \max_{\lambda\in F} N_\lambda$. Then $(\forall n \ge N)\, x_n \in W \subseteq G$, so $x_n \to x$.

Structural Commentary: Product neighborhoods impose finitely many coordinate constraints.
Key Insight: Coordinate convergence satisfies every finite constraint. □

Proposition 6.4.4. *Hausdorff Sequence Limits Are Unique*
Let X be a topological space and let $(x_n) \subseteq X$.
(i) If X is Hausdorff, then (x_n) has at most one limit.
(ii) If X has the trivial topology $\{\emptyset, X\}$, then $(\forall x \in X)\, x_n \to x$.

Proof.
(*i*) If $x_n \to x$ and $x_n \to y$ with $x \ne y$, choose disjoint neighborhoods $U \in \mathcal{N}_X(x)$ and $V \in \mathcal{N}_X(y)$.
Eventually $x_n \in U$ and $x_n \in V$, contradicting $U \cap V = \emptyset$.
(*ii*) In the trivial topology, $(\forall x \in X)\, \mathcal{N}_X(x) = \{X\}$, and every sequence is eventually in X.

Structural Commentary: Hausdorff neighborhoods separate competing limits.
Key Insight: A sequence tail cannot lie in two disjoint neighborhoods. □

Definition 6.4.5. *Countability Axioms*
Let (X, τ) be a topological space.
(i) X is *first countable* iff each $x \in X$ has an at most countable neighborhood basis.
(ii) X is *second countable* iff τ has an at most countable basis.
(iii) X is *separable* iff $(\exists D \subseteq X)[D$ is at most countable $\wedge\ \overline{D} = X]$. Equivalently, such a D satisfies $(\forall U \in \tau \setminus \{\emptyset\})\, U \cap D \ne \emptyset$.
Key Insight: Countability axioms reduce topology to at most countably many local, global, or dense tests.

Proposition 6.4.6. *Metric Spaces Are First Countable*
Let (X, d) be a metric space. Then X is first countable.

Proof. Let $x \in X$. The collection $\mathcal{B}_x := \{B_{1/n}(x) : n \in \mathbb{N}\}$ is at most countable.
If U is a neighborhood of x, choose $r > 0$ with $B_r(x) \subseteq U$ and $n \in \mathbb{N}$ with $n > r^{-1}$ by Theorem 1.1.19(i).
Then $B_{1/n}(x) \subseteq B_r(x) \subseteq U$, so $\mathcal{B}_x$ is a neighborhood basis at x.
Structural Commentary: Archimedean reciprocal radii refine every metric neighborhood.
Key Insight: Reciprocal-radius balls form one local test sequence. □

Proposition 6.4.7. *Second Countable Implies First Countable*
Let (X, τ) be second countable. Then X is first countable.

Proof. Let $\mathcal{B}$ be an at most countable basis for τ. For $x \in X$, set $\mathcal{B}_x := \{B \in \mathcal{B} : x \in B\}$.
Then $\mathcal{B}_x$ is at most countable, and $(\forall U \in \mathcal{N}_X(x))(\exists B \in \mathcal{B}_x)\, x \in B \subseteq U$.
Structural Commentary: Restrict the global basis to sets containing x.
Key Insight: At most countably many global tests become at most countably many local tests. □

Proposition 6.4.8. *First Countable Sequential Closure*
Let X be first countable, let $E \subseteq X$, and let $x \in X$.

(i) $x \in \overline{E}$ iff $\exists (x_n) \subseteq E$ with $x_n \to x$.
(ii) E is closed iff $(\forall (x_n) \subseteq E)(\forall y \in X)\, [x_n \to y \Rightarrow y \in E]$.

Proof.

(*i*) ($\Rightarrow$) The neighborhood basis at x is nonempty, so write it as $\{B_n : n \in \mathbb{N}\}$ by Proposition 6.2.13.
Since $x \in \overline{E}$ and $\bigcap_{j=1}^{n} B_j \in \mathcal{N}_X(x)$, choose $x_n \in E \cap \bigcap_{j=1}^{n} B_j$.
If $U \in \mathcal{N}_X(x)$, choose $m \in \mathbb{N}$ with $B_m \subseteq U$. Then $(\forall n \geq m)\, x_n \in B_m \subseteq U$, so $x_n \to x$.
(*i*) ($\Leftarrow$) If $(x_n) \subseteq E$ and $x_n \to x$, then every neighborhood of x contains some $x_n \in E$, so $x \in \overline{E}$.
(*ii*) ($\Rightarrow$) If E is closed and $(x_n) \subseteq E$ with $x_n \to x$, then $x \in \overline{E} = E$ by (*i*).
(*ii*) ($\Leftarrow$) Assume $(\forall (x_n) \subseteq E)(\forall y \in X)\, [x_n \to y \Rightarrow y \in E]$. Then $\overline{E} \subseteq E$ by (*i*). Hence $E = \overline{E}$.

Structural Commentary: Countable local bases produce one testing sequence.
Key Insight: In first countable spaces, closure is sequential closure. □

Proposition 6.4.9. *Second Countable Consequences*
Let (X, τ) be second countable.

(i) An at most countable $D \subseteq X$ exists with $\overline{D} = X$.
(ii) $(\forall Y \subseteq X)$, the subspace Y is second countable.

Proof.

(i) Let $\mathcal{B} = \{B_n : n \in I\}$ be an at most countable basis, where $I \subseteq \mathbb{N}$.
For $n \in I$ with $B_n \neq \emptyset$, choose $x_n \in B_n$ and set $D := \{x_n : n \in I,\ B_n \neq \emptyset\}$.
Then D is at most countable by Proposition 6.2.12(i),(ii).
If $U \in \tau \setminus \{\emptyset\}$, choose $x \in U$ and $B_n \in \mathcal{B}$ with $x \in B_n \subseteq U$. Then $x_n \in D \cap U$, so $\overline{D} = X$.
(ii) If $Y \subseteq X$, then $\{B_n \cap Y : n \in I\}$ is an at most countable basis for the subspace topology on Y.

Structural Commentary: Basis representatives establish density; basis restrictions preserve second countability.
Key Insight: One basis indexed within $\mathbb{N}$ controls both dense points and subspaces. □

Proposition 6.4.10. *Separable Metric Spaces and Countable Bases*
Let (X, d) be a metric space, and let τ_d be its metric topology. The following are equivalent:

(i) An at most countable $D \subseteq X$ exists with $\overline{D}^{\,X} = X$.
(ii) An at most countable basis $\mathcal{B} \subseteq \tau_d$ exists for τ_d.

Proof.

(*i*) $\Rightarrow$ (*ii*) Let $D \subseteq X$ be at most countable with $\overline{D}^{\,X} = X$, and set $\mathcal{B} := \{B_q(x) : x \in D,\ q \in \mathbb{Q},\ q > 0\}$.
The parameter set $D \times \mathbb{Q}_{>0}$ is at most countable by Proposition 6.2.12(i) and Corollary 6.2.17(i).
Hence $\mathcal{B}$ is at most countable by Proposition 6.2.12(ii).
For $U \in \tau_d$ and $y \in U$, choose $r > 0$ with $B_r(y) \subseteq U$ and $q \in \mathbb{Q}$ with $0 < q < r/2$ by Theorem 1.1.19(ii).
Then choose $x \in D \cap B_q(y)$ by $\overline{D}^{\,X} = X$. If $z \in B_q(x)$, then $d(z, y) < 2q < r$, so $y \in B_q(x) \subseteq B_r(y) \subseteq U$.
(*ii*) $\Rightarrow$ (*i*) Let $\mathcal{B} = \{U_n : n \in I\}$ be an at most countable basis for τ_d, where $I \subseteq \mathbb{N}$.
For $n \in I$ with $U_n \neq \emptyset$, choose $x_n \in U_n$ and set $D := \{x_n : n \in I,\ U_n \neq \emptyset\}$.
Then D is at most countable by Proposition 6.2.12(i),(ii).
If $\emptyset \neq U \in \tau_d$, choose $y \in U$ and $n \in I$ with $y \in U_n \subseteq U$; then $x_n \in D \cap U$.
Hence every nonempty open set meets D, so $\overline{D}^{\,X} = X$.

Structural Commentary: In metric spaces, a dense set indexed within $\mathbb{N}$ produces all rational-radius local tests.
Key Insight: Separability and second countability coincide for metric spaces. □

Proposition 6.4.11. *Subspaces of Separable Metric Spaces*
Let (X, d) be a separable metric space and let $E \subseteq X$ with the subspace metric $d|_{E \times E}$. Then E is separable.

Proof. The metric topology τ_d has an at most countable basis $\mathcal{B}$ by Proposition 6.4.10.
Set $\mathcal{B}_E := \{B \cap E : B \in \mathcal{B}\}$. Then $\mathcal{B}_E$ is at most countable by Proposition 6.2.12(ii).
If U is open in E and $x \in U$, choose $G \in \tau_d$ with $U = E \cap G$ and $B \in \mathcal{B}$ with $x \in B \subseteq G$.
Hence $x \in B \cap E \subseteq U$, so $\mathcal{B}_E$ is an at most countable basis for E. Therefore E is separable by Proposition 6.4.10.
Structural Commentary: An ambient basis indexed within $\mathbb{N}$ restricts to a subspace basis indexed within $\mathbb{N}$.
Key Insight: Metric separability passes to all subspaces through countable bases. □

Corollary 6.4.12. *Euclidean Subsets Are Separable*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$ have the subspace Euclidean metric. Then E is separable.

Proof. The set $\mathbb{Q}^k$ is countable by Theorem 6.2.18 and Corollary 6.2.19.
For $x \in \mathbb{R}^k$ and $r > 0$, choose $q = (q_1, \dots, q_k) \in \mathbb{Q}^k$ with $(\forall i \in \{1, \dots, k\})\ |x_i - q_i| < r/\sqrt{k}$, so $\|x - q\|_2 < r$.
Hence $\mathbb{Q}^k$ is dense in $\mathbb{R}^k$. Thus $\mathbb{R}^k$ is separable, and E is separable by Proposition 6.4.11.
Structural Commentary: Rational grids are countable dense nets.
Key Insight: Euclidean separability passes to subspaces. □

Counterexample 6.4.13. *T_1 Spaces Need Not Be First Countable*
Let X be uncountable and let $\tau := \{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$.

(i) τ is a topology on X.
(ii) (X, τ) is T_1.
(iii) (X, τ) is not first countable.

Proof.

(*i*) One has $\emptyset, X \in \tau$, and a finite intersection of cofinite sets is cofinite.
If $\{U_\alpha\}_{\alpha \in A} \subseteq \tau$ and some U_{α_0} is nonempty, then $X \setminus \bigcup_{\alpha \in A} U_\alpha \subseteq X \setminus U_{\alpha_0}$ is finite.
If every U_α is empty, then the union is empty. Hence τ is a topology.
(*ii*) For $x \in X$, $X \setminus \{x\} \in \tau$, so $\{x\}$ is closed. Hence (X, τ) is T_1.
(*iii*) Fix $x \in X$. If an at most countable neighborhood basis existed at x, write it as $\{B_n : n \in \mathbb{N}\}$ by Proposition 6.2.13.
For $n \in \mathbb{N}$, choose $O_n \in \tau$ with $x \in O_n \subseteq B_n$; then $X \setminus B_n \subseteq X \setminus O_n$ is finite.
Thus $C := \{x\} \cup \bigcup_{n \in \mathbb{N}} (X \setminus B_n)$ is at most countable by Corollary 6.2.16.
Choose $y \in X \setminus C$. Then $U := X \setminus \{y\} \in \mathcal{N}_X(x)$, but $(\forall n \in \mathbb{N})\, y \in B_n$, so no B_n is contained in U.

Key Insight: Countably many finite complements miss a point needed to refine one cofinite neighborhood. □

Definition 6.4.14. *Lindelöf Spaces*
Let (X, τ) be a topological space. The space X is *Lindelöf* iff every open cover has an at most countable subcover.
Key Insight: Lindelöfness reduces arbitrary covers to at most countable subcovers.

Definition 6.4.15. *Countably Compact Spaces*
Let (X, τ) be a topological space. The space X is *countably compact* iff every at most countable open cover has a finite subcover.
Key Insight: Countable compactness reduces at most countable covers to finite subcovers.

Proposition 6.4.16. *Fréchet Intersection Test for Countable Compactness*
Let X be a topological space.
Then X is countably compact iff every descending sequence of nonempty closed subsets of X has nonempty intersection.

Proof.

(⇒) Assume X is countably compact. If $\bigcap_{n \geq 1} F_n = \emptyset$, then $\{X \setminus F_n : n \geq 1\}$ is an at most countable open cover.
A finite subcover exists; since F_n descends, some $X \setminus F_N$ covers X, contradicting $F_N \neq \emptyset$.
(⇐) Suppose an at most countable open cover has no finite subcover. It has the form $\{O_n : n \in \mathbb{N}\}$ by Proposition 6.2.13.
Then $F_n := X \setminus \bigcup_{j=1}^{n} O_j$ is nonempty, closed, and descending, while $\bigcap_{n \geq 1} F_n = \emptyset$, a contradiction.

Structural Commentary: Complements turn at most countable covers into descending closed sequences.
Key Insight: Countable compactness is nested closed intersection. □

Theorem 6.4.17. *Dini's Theorem on Countably Compact Spaces*
Let X be countably compact, and let $f_n : X \to \mathbb{R}$ be continuous with $(\forall x \in X)\, f_n(x) \downarrow 0$.
Then $f_n \to 0$ uniformly: $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall x \in X)(\forall n \geq N)\, 0 \leq f_n(x) < \varepsilon$.

Proof. Fix $\varepsilon > 0$ and set $O_n := \{x \in X : f_n(x) < \varepsilon\}$. The set O_n is open by Proposition 6.2.2(i), while $O_n \subseteq O_{n+1}$ and $\bigcup_{n \geq 1} O_n = X$.
Choose $m, n_1, \dots, n_m \in \mathbb{N}$ with $X = \bigcup_{j=1}^{m} O_{n_j}$ by countable compactness. Set $N := \max_{1 \leq j \leq m} n_j$; $O_N = X$ by increasingness of (O_n).
Hence $(\forall x \in X)(\forall n \geq N)\, 0 \leq f_n(x) \leq f_N(x) < \varepsilon$.
Structural Commentary: Increasing sublevel covers reduce finite subcovers to one index.
Key Insight: Monotone pointwise convergence becomes uniform convergence. □

Proposition 6.4.18. *Second Countable Implies Lindelöf*
Let (X, τ) be second countable. Then X is Lindelöf.

Proof. Let $\mathcal{B} = \{B_n : n \in I\}$ be an at most countable basis with $I \subseteq \mathbb{N}$, and let $\mathcal{U} \subseteq \tau$ cover X.
Set $I_0 := \{n \in I : (\exists U \in \mathcal{U})\, B_n \subseteq U\}$, and choose $U_n \in \mathcal{U}$ with $B_n \subseteq U_n$ for $n \in I_0$.
The family $\{U_n : n \in I_0\}$ is at most countable by Proposition 6.2.12(i),(ii).
If $x \in X$, choose $U \in \mathcal{U}$ with $x \in U$ and $B_n \in \mathcal{B}$ with $x \in B_n \subseteq U$; then $n \in I_0$ and $x \in U_n$.
Structural Commentary: Basis elements refine arbitrary open covers.
Key Insight: An at most countable basis produces an at most countable subcover. □

Proposition 6.4.19. *Second Countable Countable Compactness Is Compactness*
Let (X, τ) be second countable. Then X is compact iff X is countably compact.

Proof.

(⇒) If an at most countable $\mathcal{U} \subseteq \tau$ covers X, then a finite $\mathcal{V} \subseteq \mathcal{U}$ with $X = \bigcup \mathcal{V}$ exists by compactness.

(⇐) Let $\mathcal{U} \subseteq \tau$ cover X. An at most countable $\mathcal{V} \subseteq \mathcal{U}$ with $X = \bigcup \mathcal{V}$ exists by Proposition 6.4.18.
A finite $\mathcal{W} \subseteq \mathcal{V}$ with $X = \bigcup \mathcal{W}$ exists by countable compactness.

Structural Commentary: Second countability reduces covers to at most countable subcovers.
Key Insight: Countable compactness then makes the subcover finite. □

Theorem 6.4.20. *Second Countable Compactness and Sequential Compactness*
Let X be a second countable topological space. Then X is compact iff X is sequentially compact.

Proof.
(⇒) Assume X is compact. Let $(x_n) \subseteq X$ and set $F_n := \overline{\{x_k : k \geq n\}}$.
The sets $F_1 \supseteq F_2 \supseteq \cdots$ are nonempty, closed, and have the finite intersection property.
Hence $\exists x \in \bigcap_{n \geq 1} F_n$ by Proposition 6.3.3. The space is first countable by Proposition 6.4.7.

Write a neighborhood basis at x as $\{C_j : j \in \mathbb{N}\}$ by Proposition 6.2.13, and set $B_j := \bigcap_{i=1}^{j} C_i$.

With $n_0 := 0$, recursively choose $n_j > n_{j-1}$ with $x_{n_j} \in B_j$ by $x \in F_{n_{j-1}+1}$ and $B_j \cap \{x_k : k \geq n_{j-1} + 1\} \neq \emptyset$.
If $U \in \mathcal{N}_X(x)$, choose $i \in \mathbb{N}$ with $C_i \subseteq U$. Then $(\forall j \geq i)\, x_{n_j} \in B_j \subseteq C_i \subseteq U$, so $x_{n_j} \to x$.
(⇐) Assume X is sequentially compact. Countable compactness suffices by Proposition 6.4.19.
Suppose an at most countable open cover has no finite subcover.

It has the form $\{O_n : n \in \mathbb{N}\}$ by Proposition 6.2.13; choose $x_n \in X \setminus \bigcup_{j=1}^{n} O_j$.

There are a subsequence (x_{n_k}) and $x \in X$ with $x_{n_k} \to x$ by sequential compactness. Choose $N \in \mathbb{N}$ with $x \in O_N$.
Eventually $n_k \geq N$ and $x_{n_k} \in O_N$, contradicting $x_{n_k} \notin O_N$.

Structural Commentary: Compact closed tails locate a cluster point; an enumerated local basis extracts its subsequence.
Key Insight: At most countable covers and bases make compactness sequential. □

Definition 6.4.21. *Metrizable Space*
A topological space is *metrizable* if its topology is induced by some metric.

Definition 6.4.22. *Sorgenfrey Line*
Let $\mathcal{B}_S := \{[a, b) : a < b\}$ on $\mathbb{R}$. The *Sorgenfrey line* is $\mathbb{R}_S := (\mathbb{R}, \tau_S)$, where τ_S is the topology generated by $\mathcal{B}_S$.

Counterexample 6.4.23. *First Countable Separable Spaces Need Not Be Second Countable*
The Sorgenfrey line $\mathbb{R}_S$ satisfies:

(i) $\mathcal{B}_S$ is a basis for τ_S;
(ii) $\mathbb{R}_S$ is first countable;
(iii) $\mathbb{R}_S$ is separable;
(iv) $\mathbb{R}_S$ is not second countable;
(v) $\mathbb{R}_S$ is not metrizable.

Proof.
(i) Every $x \in \mathbb{R}$ belongs to $[x, x + 1)$. If $x \in [a, b) \cap [c, d)$, choose e with $x < e \leq \min\{b, d\}$.
Then $x \in [x, e) \subseteq [a, b) \cap [c, d)$, so $\mathcal{B}_S$ is a basis by Definition 6.1.8(ii).
(ii) Fix $x \in \mathbb{R}$. If $x \in [a, b)$, choose $n \in \mathbb{N}$ with $n > (b - x)^{-1}$ by Theorem 1.1.19(i).
Then $[x, x + 1/n) \subseteq [a, b)$, so $\{[x, x + 1/n) : n \in \mathbb{N}\}$ is a neighborhood basis at x by (i).
(iii) Every $[a, b) \in \mathcal{B}_S$ meets $\mathbb{Q}$ by Theorem 1.1.19(ii). Hence $\overline{\mathbb{Q}}^{\mathbb{R}_S} = \mathbb{R}_S$ by (i), and $\mathbb{R}_S$ is separable by Corollary 6.2.19.
(iv) Let $\mathcal{C}$ be a basis for τ_S. For $x \in \mathbb{R}$, choose $C_x \in \mathcal{C}$ with $x \in C_x \subseteq [x, x + 1)$.
If $x < y$, then $x \in C_x \setminus C_y$, so $C_x \neq C_y$. Thus $x \mapsto C_x$ is injective. The space $\mathbb{R}$ is uncountable by Corollary 4.4.2.
Hence $\mathcal{C}$ is not at most countable by Proposition 6.2.12(i), so $\mathbb{R}_S$ is not second countable.
(v) If $\mathbb{R}_S$ were metrizable, then it would be second countable by (iii) and Proposition 6.4.10, contradicting (iv).

Structural Commentary: Half-open neighborhoods assign distinct left-anchored refinements to every real point.
Key Insight: Local countability coexists with an uncountable global basis obstruction. □

Definition 6.4.24. *Moore Plane*
Let $\mathbb{M} := \{(x, y) \in \mathbb{R}^2 : y \geq 0\}$ and $L := \mathbb{R} \times \{0\}$. Let $\mathcal{B}_{\mathbb{M}}$ consist of:

(i) $B_{\text{euc}}(p, r) \subseteq \mathbb{M}$ for $p \in \mathbb{M} \setminus L$ and $r > 0$;
(ii) $N(x, r) := \{(x, 0)\} \cup B_{\text{euc}}((x, r), r)$ for $(x, 0) \in L$ and $r > 0$.

The *Moore plane* is $(\mathbb{M}, \tau_{\mathbb{M}})$, where $\tau_{\mathbb{M}}$ is generated by $\mathcal{B}_{\mathbb{M}}$.

Figure 150 depicts the two generators from Definition 6.4.24: interior balls and boundary points augmented by tangent disks.

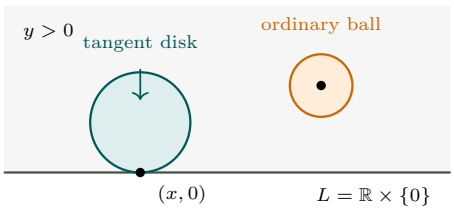

Figure 150: The Moore-plane generators are ordinary interior balls and tangent disks augmented by their boundary points.

Counterexample 6.4.25. *Separable Hausdorff Spaces Need Not Be Metrizable*
The Moore plane $(\mathbb{M}, \tau_{\mathbb{M}})$ satisfies:

(i) $\mathcal{B}_{\mathbb{M}}$ is a basis for $\tau_{\mathbb{M}}$;

(ii) $(\mathbb{M}, \tau_{\mathbb{M}})$ is Hausdorff;
(iii) $(\mathbb{M}, \tau_{\mathbb{M}})$ is separable;
(iv) $(\mathbb{M}, \tau_{\mathbb{M}})$ is not second countable;
(v) $(\mathbb{M}, \tau_{\mathbb{M}})$ is not metrizable.

Proof.

(i) Every interior point belongs to a sufficiently small Euclidean ball contained in $\mathbb{M}$, and $(x,0) \in N(x,1)$ for $(x,0) \in L$.
If an interior point lies in two members of $\mathcal{B}_{\mathbb{M}}$, their intersection contains a sufficiently small Euclidean ball around it.
Every member of $\mathcal{B}_{\mathbb{M}}$ containing $(x,0) \in L$ has the form $N(x,r)$, and $N(x, \min\{r,s\}) \subseteq N(x,r) \cap N(x,s)$.
Hence $\mathcal{B}_{\mathbb{M}}$ is a basis by Definition 6.1.8(ii).

(ii) Distinct interior points have disjoint Euclidean balls contained in $\mathbb{M}$. If $(x,0),(y,0) \in L$ with $x \neq y$, then $N(x,r) \cap N(y,r) = \emptyset$ when $0 < 2r < |x-y|$. Let $p = (x,0) \in L$, $q = (u,v) \in \mathbb{M} \setminus L$, $\delta := \|p-q\|_2$. Choose $0 < r < \delta/4$, $0 < s < \min\{v, \delta/2\}$.
Then $N(x,r) \subseteq B^{\mathrm{euc}}_{\delta/2}(p)$ and $B_{\mathrm{euc}}(q,s) \in \mathcal{B}_{\mathbb{M}}$ are disjoint neighborhoods of p and q. Hence $(\mathbb{M}, \tau_{\mathbb{M}})$ is Hausdorff by (i).

(iii) Set $D := \mathbb{Q} \times (\mathbb{Q} \cap (0,\infty))$. Then D is at most countable by Proposition 6.2.12(i) and Corollary 6.2.17(i).
Every member of $\mathcal{B}_{\mathbb{M}}$ contains some $B_{\mathrm{euc}}((a,b),s)$ with $b > s > 0$.
Choose $q_1, q_2 \in \mathbb{Q}$ with $|a - q_1| < s/\sqrt{2}$ and $|b - q_2| < s/\sqrt{2}$ by Theorem 1.1.19(ii).
Then $(q_1,q_2) \in D \cap B_{\mathrm{euc}}((a,b),s)$. Hence every basis set meets D, so $\overline{D}^{\mathbb{M}} = \mathbb{M}$ by (i), and $(\mathbb{M}, \tau_{\mathbb{M}})$ is separable.

(iv) For $(x,0) \in L$, $N(x,r) \cap L = \{(x,0)\}$, so L is discrete. Map $x \mapsto (x,0)$ is a bijection $\mathbb{R} \to L$, so L is uncountable by Corollary 4.4.2.
A dense subset of the discrete space L equals L, so L is not separable.
If $\mathbb{M}$ were second countable, then L would be second countable by Proposition 6.4.9(ii).
Then L would be separable by Proposition 6.4.9(i), a contradiction.

(v) If $\mathbb{M}$ were metrizable, then it would be second countable by (iii) and Proposition 6.4.10, contradicting (iv).

Structural Commentary: Rational interior points establish density, while singleton boundary traces force a nonseparable subspace.
Key Insight: The uncountable discrete boundary obstructs every at most countable basis. □

Theorem 6.4.26. *Urysohn Metrization Theorem*
Let (X,τ) be second countable. Then τ is induced by a metric iff (X,τ) is normal.

Proof.

($\Rightarrow$) Let d induce τ. Then (X,τ) is normal by Proposition 6.1.15.

($\Leftarrow$) Assume (X,τ) is normal.

(a) *Separating family*: Let $\{U_n : n \in I\}$ be an at most countable basis with $I \subseteq \mathbb{N}$, and set $A := \{(n,m) \in I^2 : \overline{U_n} \subseteq U_m\}$.
For $(n,m) \in A$, the closed sets $\overline{U_n}$ and $X \setminus U_m$ are disjoint.
Choose a continuous $f_{n,m} : X \to [0,1]$ such that $f_{n,m}|_{\overline{U_n}} = 0$ and $f_{n,m}|_{X \setminus U_m} = 1$ by Theorem 6.3.20.

(b) *Weighted limit*: For $k \in \mathbb{N}$, set $A_k := A \cap \{1,\dots,k\}^2$ and $\rho_k(x,y) := \sum_{(n,m) \in A_k} 2^{-(n+m)} |f_{n,m}(x) - f_{n,m}(y)|$.
For $x,y \in X$ and $(n,m) \in A$, one has $0 \le |f_{n,m}(x) - f_{n,m}(y)| \le 1$.
Thus $(\rho_k(x,y))$ is increasing and $0 \le \rho_k(x,y) \le (\sum_{j=1}^{k} 2^{-j})^2 = (1 - 2^{-k})^2 < 1$.
Hence $\rho(x,y) := \lim_{k \to \infty} \rho_k(x,y)$ exists by Theorem 2.1.11.
Let $k, \ell \in \mathbb{N}$ with $\ell > k$, and let $x,y \in X$. Every pair in $A_\ell \setminus A_k$ has at least one coordinate exceeding k.
Thus $0 \le \rho_\ell(x,y) - \rho_k(x,y) \le 2 \sum_{j=k+1}^{\ell} 2^{-j} = 2^{1-k}(1 - 2^{k-\ell}) \le 2^{1-k}$.
For fixed k,x,y, one has $\rho_\ell(x,y) - \rho_k(x,y) \to \rho(x,y) - \rho_k(x,y)$ by Theorem 2.1.5(ii).
Hence $(\forall k \in \mathbb{N})(\forall x,y \in X)\, 0 \le \rho(x,y) - \rho_k(x,y) \le 2^{1-k}$ by Proposition 2.1.4(iii).

(c) *Pseudometric and continuity*: Each summand defining ρ_k is nonnegative, vanishes on diagonal by Proposition 1.1.17(i).
It is symmetric by Proposition 1.1.17(ii) with $|-1| = 1$.
It satisfies the triangle inequality by Proposition 1.1.17(iv). Hence ρ_k is a pseudometric.
The diagonal and symmetry identities pass directly to ρ. For $x,y,z \in X$, $\rho_k(x,z) \le \rho_k(x,y) + \rho_k(y,z)$.
The right side converges to $\rho(x,y) + \rho(y,z)$ by Theorem 2.1.5(ii).
Thus $\rho(x,z) \le \rho(x,y) + \rho(y,z)$ by Proposition 2.1.4(iii), so ρ is a pseudometric.
Fix $x \in X$. For $(n,m) \in A$, set $h_{n,m,x}(y) := |f_{n,m}(x) - f_{n,m}(y)|$.
The inequality $|h_{n,m,x}(y) - h_{n,m,x}(z)| \le |f_{n,m}(y) - f_{n,m}(z)|$ holds by two instances of Proposition 1.1.17(iv).
For $y \in X$ and $\varepsilon > 0$, choose a neighborhood W of y such that $(\forall z \in W)\ |f_{n,m}(y) - f_{n,m}(z)| < \varepsilon$ by continuity of $f_{n,m}$.
Hence $h_{n,m,x}$ is continuous. Therefore $\rho_k(x,\cdot)$ is continuous by Proposition 6.3.21(i).
The map $\rho(x,\cdot)$ is continuous by Proposition 6.3.21(ii) with the estimate in (b) and $2^{1-k} \to 0$ by Theorem 2.2.10(v).

(d) *Local metric balls*: Let $G \in \tau$ and $x \in G$. Since normality includes T_1, the set $\{x\}$ is closed.
Choose open O with $x \in O \subseteq \overline{O} \subseteq G$ by Proposition 6.1.16, and choose $m \in I$ with $x \in U_m \subseteq O$.
Choose open V with $x \in V \subseteq \overline{V} \subseteq U_m$ by same proposition, $n \in I$ with $x \in U_n \subseteq V$. Then $\overline{U_n} \subseteq \overline{V} \subseteq U_m$, so $(n,m) \in A$.
If $y \notin U_m$, then $f_{n,m}(x) = 0$ and $f_{n,m}(y) = 1$, so $\rho(x,y) \ge 2^{-(n+m)}$. Hence $B_\rho(x, 2^{-(n+m+1)}) \subseteq U_m \subseteq G$.

(e) *Definiteness and topology*: If $x \neq y$, take $G = X \setminus \{y\}$ in (d); then $\rho(x,y) > 0$. Thus ρ is a metric.
Let τ_ρ be its metric topology. Then $\tau \subseteq \tau_\rho$ by (d).
Conversely, $B_\rho(x,r) = \rho(x,\cdot)^{-1}((-\infty,r)) \in \tau$ by (c) and Proposition 6.2.2(i), so $\tau_\rho \subseteq \tau$. Hence $\tau_\rho = \tau$.

Structural Commentary: Urysohn separators form a uniformly convergent pseudometric whose local balls recover the basis topology.
Key Insight: Nested basis pairs both distinguish points and refine every open neighborhood. □

6.5 Topological Connectedness and Paths

Connectedness is an open-set obstruction to splitting a space into two visible pieces. In general topology it is most naturally phrased by separations and then transported through continuous maps.

Definition 6.5.1. *Connected Topological Spaces*
Let (X, τ) be a topological space, and let $E \subseteq X$.

(i) A set $A \subseteq X$ is *clopen* iff $A \in \tau$ and $X \setminus A \in \tau$.
(ii) A *separation* of X is a pair (U, V) with $U, V \in \tau$, $U, V \neq \emptyset$, $U \cap V = \emptyset$, and $X = U \cup V$.
(iii) X is *connected* iff no separation of X exists.
(iv) E is *connected* iff (E, τ_E) is connected, where $\tau_E := \{E \cap U : U \in \tau\}$.
Equivalently, E is disconnected iff $\exists U, V \in \tau$ with $U \cap E \neq \emptyset$, $V \cap E \neq \emptyset$, $E \subseteq U \cup V$, and $E \cap U \cap V = \emptyset$.

Key Insight: Connectedness forbids a nonempty disjoint open split.

Proposition 6.5.2. *Metric and Topological Connectedness Agree*
Let (X, d) be a metric space with its metric topology, and let $E \subseteq X$.
Then E is connected by Definition 5.1.1 iff it is connected as a topological subspace by Definition 6.5.1.

Proof.
($\Rightarrow$) Assume E is connected in the metric sense. If (U, V) separates the topological subspace E, then $U, V \neq \emptyset$ and $E = U \cup V$.
For $u \in U$, choose an open $O_u \subseteq X$ with $u \in O_u$ and $E \cap O_u \subseteq U$. Then $O_u \cap V = \emptyset$, so $u \notin \overline{V}$.
Hence $U \cap \overline{V} = \emptyset$, and similarly $\overline{U} \cap V = \emptyset$. Thus U, V are separated in the metric sense, a contradiction.
($\Leftarrow$) Assume E is connected as a topological subspace. If nonempty $A, B \subseteq E$ are separated in the metric sense and $E = A \cup B$, then $A = E \cap (X \setminus \overline{B})$ and $B = E \cap (X \setminus \overline{A})$. Thus A, B are open in E, so (A, B) separates the topological subspace E, a contradiction.

Structural Commentary: Ambient closures and relative open complements translate through subspace neighborhoods.
Key Insight: Each side avoids the opposite ambient closure exactly when it is relatively open. □

Proposition 6.5.3. *Connectedness and Clopen Sets*
Let (X, τ) be a topological space. Then X is connected iff $(\forall A \subseteq X)\,[A$ clopen $\Rightarrow (A = \emptyset \vee A = X)]$.

Proof.
($\Rightarrow$) If $A \subseteq X$ is clopen, nonempty, and proper, then $(A, X \setminus A)$ is a separation of X, impossible.
($\Leftarrow$) If (U, V) separates X, then $U = X \setminus V$ is clopen, nonempty, and proper, impossible.

Structural Commentary: Complements convert separations into clopen sets.
Key Insight: Connected means no nontrivial clopen subset. □

Proposition 6.5.4. *Continuous Images of Connected Spaces Are Connected*
Let X, Y be topological spaces and let $f : X \to Y$ be continuous. If X is connected, then $f(X)$ is connected in the subspace topology.

Proof. Let $g : X \to f(X)$ be f with codomain restricted to $f(X)$. If W is open in $f(X)$, choose O open in Y with $W = f(X) \cap O$. Then $g^{-1}(W) = f^{-1}(O)$ is open in X by Proposition 6.2.2(i), so g is continuous.
If (U, V) separates $f(X)$, then $g^{-1}(U)$, $g^{-1}(V)$ nonempty disjoint open subsets of X with union X, contrary to connectedness of X.
Structural Commentary: Restrict the codomain to make the image map surjective.
Key Insight: Separations pull back by continuity. □

Definition 6.5.5. *Intermediate Value Property*
A topological space X has the *intermediate value property* iff $(\forall f : X \to \mathbb{R})\,[f$ continuous $\Rightarrow f(X)$ is an interval$]$.

Theorem 6.5.6. *Connectedness and the Intermediate Value Property*
Let X be a topological space. Then X is connected iff X has the intermediate value property.

Proof.
($\Rightarrow$) If $f : X \to \mathbb{R}$ is continuous, then $f(X)$ is topologically connected by Proposition 6.5.4 and connected in the metric sense by Proposition 6.5.2. Hence $f(X)$ is an interval by Theorem 5.1.2.
($\Leftarrow$) We prove the contrapositive. If (U, V) separates X, define $f : X \to \mathbb{R}$ by $f|_U = 0$ and $f|_V = 1$.
For an open $O \subseteq \mathbb{R}$, $f^{-1}(O) \in \{\emptyset, U, V, X\}$, so f is continuous by Proposition 6.2.2(i). However, $f(X) = \{0, 1\}$ is not an interval.

Structural Commentary: Separations produce continuous two-valued tests.
Key Insight: Intermediate values are connectedness seen by $\mathbb{R}$. □

Proposition 6.5.7. *Unions Through a Common Point*
Let $\{C_\alpha\}_{\alpha \in A}$ be connected subsets of a topological space X with $\bigcap_{\alpha \in A} C_\alpha \neq \emptyset$. Then $C := \bigcup_{\alpha \in A} C_\alpha$ is connected.

Proof. If $A = \emptyset$, then $C = \emptyset$ is connected. Assume $A \neq \emptyset$ and choose $p \in \bigcap_{\alpha \in A} C_\alpha$.
Suppose (U, V) separates C and, after interchanging the sides if necessary, $p \in U$.
For $\alpha \in A$, the pair $(U \cap C_\alpha, V \cap C_\alpha)$ would separate C_α if $C_\alpha \cap V \neq \emptyset$.
Hence $(\forall \alpha \in A)\, C_\alpha \cap V = \emptyset$, so $V \cap C = \emptyset$, contrary to the nonemptiness of V.
Structural Commentary: Each connected piece must stay on the side containing the shared point.
Key Insight: A common point blocks separation. □

Proposition 6.5.8. *Connectedness Between a Set and Its Closure*
Let (X, τ) be a topological space and let $A \subseteq B \subseteq \overline{A}$. If A is connected, then:

(i) B is connected;
(ii) $\overline{A}$ is connected.

Proof.
(i) Suppose (P, Q) separates B. Since A is connected, $A \subseteq P$ or $A \subseteq Q$; assume $A \subseteq P$.
If $q \in Q$, choose $O \in \tau$ with $q \in O$ and $O \cap B \subseteq Q$.
Since $q \in B \subseteq \overline{A}$, $O \cap A \neq \emptyset$, contradicting $A \subseteq P$ and $P \cap Q = \emptyset$. Thus $Q = \emptyset$, contrary to the definition of a separation.
(ii) The set $\overline{A}$ is connected by (i) with $B = \overline{A}$.

Structural Commentary: Relative open separation near $\overline{A}$ must meet the connected core A.
Key Insight: Adding only closure points cannot create a new component. □

Definition 6.5.9. *Path-Connected Spaces*
A topological space X is *path-connected* iff $(\forall u, v \in X)(\exists \gamma : [0,1] \to X)\,[\gamma \text{ continuous} \wedge \gamma(0) = u \wedge \gamma(1) = v]$.

Theorem 6.5.10. *Path-Connected Spaces Are Connected*
Let X be path-connected. Then X is connected.

Proof. Suppose (U, V) separates X. Choose $u \in U$, $v \in V$, and a path $\gamma : [0,1] \to X$ from u to v.
The interval $[0,1]$ is connected in the metric sense by Theorem 5.1.2 and topologically connected by Proposition 6.5.2.
Hence $\gamma([0,1])$ is connected by Proposition 6.5.4. However, $(U \cap \gamma([0,1]), V \cap \gamma([0,1]))$ separates it, a contradiction.
Structural Commentary: A path imports interval connectedness into X.
Key Insight: Paths cannot cross a separation. □

Proposition 6.5.11. *Connected Open Sets in Euclidean Space Are Path-Connected*
Let $n \in \mathbb{N}$, and let $O \subseteq \mathbb{R}^n$ be open and connected. Then any $x, y \in O$ can be joined in O by a finite polygonal path.

Proof. Fix $x_0 \in O$. Write $a \sim b$ if $a, b \in O$ are joined in O by a finite polygonal path.
Constant paths, path reversal, and concatenation make $\sim$ an equivalence relation. Set $C := \{x \in O : x_0 \sim x\}$.
Then $x_0 \in C$. Let $x \in O$ and choose $r > 0$ with $B(x, r) \subseteq O$. Any two points of $B(x, r)$ are joined by a segment.
Thus $x \in C$ implies $B(x, r) \subseteq C$. If $x \notin C$ and $B(x, r) \cap C \neq \emptyset$, a segment from a point of C to x would imply $x \in C$.
Hence $B(x, r) \cap C = \emptyset$. Therefore C and $O \setminus C$ are open in O. Since $C \neq \emptyset$, one has $C = O$ by Proposition 6.5.3.
Thus all points of O lie in one $\sim$-equivalence class, so any two are joined by a finite polygonal path.
Structural Commentary: Polygonal reachability is both open and closed in an open set.
Key Insight: Local line segments make one component all of O. □

Counterexample 6.5.12. *Connected Need Not Mean Path-Connected*
The converse of Theorem 6.5.10 fails. Let $G := \{(x, \sin(1/x)) : 0 < x \leq 1\}$, $L := \{0\} \times [-1, 1]$, and $T := G \cup L \subseteq \mathbb{R}^2$.

(i) T is connected.
(ii) T is not path-connected.

Proof.

(i) The interval $(0, 1]$ is connected in the metric sense by Theorem 5.1.2 and topologically connected by Proposition 6.5.2.
Map $h : (0, 1] \to \mathbb{R}^2$, $h(x) := (x, \sin(1/x))$, is continuous in metric sense, hence in topological sense by Corollary 6.2.3.
Thus $G = h((0, 1])$ is connected by Proposition 6.5.4. For $y \in [-1, 1]$, choose $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ with $\sin\theta = y$ and set $x_n := \frac{1}{2\pi n + \theta}$.
Then $(x_n, y) = h(x_n) \to (0, y)$, so $L \subseteq \overline{G}$ by Proposition 3.4.16. Hence $T \subseteq \overline{G}$.
Conversely, fix $p = (x, y) \in \overline{G}$ and choose $(x_n, y_n) \subseteq G$ with $(x_n, y_n) \to p$ by Proposition 3.4.16.
Then $x_n \to x$, $y_n \to y$ by Theorem 3.4.13(i). Since $(x_n) \subseteq (0, 1]$, $(y_n) \subseteq [-1, 1]$, then $x \in [0, 1]$, $y \in [-1, 1]$ by Theorem 3.4.17(ii).
If $x = 0$, then $p \in L$. If $x > 0$, then $h(x_n) \to h(x)$ by Theorem 5.1.15; hence $p = h(x) \in G$ by Theorem 3.4.6(ii).
Thus $\overline{G} \subseteq T$, so $T = \overline{G}$ and T is connected by Proposition 6.5.8(ii).

(ii) Suppose T is path-connected. Choose a path $\gamma : [0, 1] \to T$ from $(0, 0)$ to $(1, \sin 1)$ and write $\gamma(t) = (a(t), b(t))$.
The coordinate maps a, b are topologically continuous by Proposition 6.2.9 and metrically continuous by Corollary 6.2.3.
Set $Z := \{t \in [0, 1] : a(t) = 0\}$. Since $[0, 1] \setminus Z = a^{-1}(\mathbb{R} \setminus \{0\})$ is open by Proposition 6.2.2(i), Z is nonempty and closed in $[0, 1]$.
Interval $[0, 1]$ is compact in its usual topology by Theorem 4.2.5, Proposition 6.3.2; hence Z is compact by Proposition 6.3.4.
The identity on Z attains its maximum t_0 by Corollary 6.3.8. Since $a(1) = 1$, one has $t_0 < 1$.
One has $(\forall t \in (t_0, 1])\, a(t) > 0$ by $a([0, 1]) \subseteq [0, 1]$ and maximality of t_0. Set $t_j := t_0 + \frac{1 - t_0}{j+1}$.
For $j \in \mathbb{N}$, choose $N_j \in \mathbb{N}$ so large that $s_j^+ := \frac{1}{\pi/2 + 2\pi N_j} < a(t_j)$ and $s_j^- := \frac{1}{3\pi/2 + 2\pi N_j} < a(t_j)$ by Theorem 1.1.19(i).
By Theorem 5.1.43, there are $u_j^+, u_j^- \in (t_0, t_j)$ with $a(u_j^\pm) = s_j^\pm$. Since $u_j^\pm > t_0$, $\gamma(u_j^\pm) \in G$, so $b(u_j^+) = 1$ and $b(u_j^-) = -1$.
Convergence $\frac{1 - t_0}{j+1} \to 0$ follows by Theorem 2.2.10(i), Proposition 2.1.7(i), Theorem 2.1.5(ii), so $t_j \to t_0$ by Theorem 2.1.5(ii).
Hence $u_j^\pm \to t_0$ by Theorem 2.2.9 with $t_0 < u_j^\pm < t_j$. Thus $b(u_j^\pm) \to b(t_0)$ by Theorem 5.1.15.
Since $b(u_j^+) = 1$, $b(u_j^-) = -1$, uniqueness of real sequence limits by Proposition 2.1.4(i) forces $b(t_0) = 1$, $b(t_0) = -1$, contradiction.

Structural Commentary: Closure transfers connectedness from G, while the last zero forces incompatible vertical limits.
Key Insight: Infinite oscillation near $x = 0$ prevents a path from leaving L. □

Figure 151 plots the oscillatory graph G approaching the vertical segment L, the accumulation geometry behind the distinction between connectedness and path-connectedness.

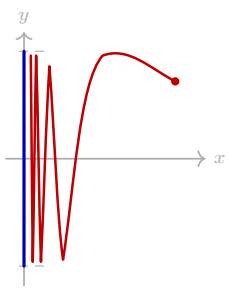

Figure 151: Topologist's sine curve: $y = \sin(1/x)$ accumulates on $\{0\} \times [-1, 1]$; connected, not path-connected.

7 Finite-Dimensional Topological Vector Spaces

7.1 Natural Topology and Continuity

Nonzero finite-dimensional real vector spaces carry a canonical topology: choose coordinates, use the standard Euclidean topology on $\mathbb{R}^n$, and then forget that coordinates were chosen. All coordinate changes are homeomorphisms. Throughout this subsection, $\mathbb{R}$ carries its usual topology and $\mathbb{R}^n$ carries its Euclidean topology.

Definition 7.1.1. *Topological Vector Space*
Let $\mathcal{V}$ be a real vector space and let $\mathcal{T}$ be a topology on its underlying set.
Define $A : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ by $A(\mathbf{u}, \mathbf{v}) := \mathbf{u} + \mathbf{v}$ and $M : \mathbb{R} \times \mathcal{V} \to \mathcal{V}$ by $M(a, \mathbf{v}) := a\mathbf{v}$.
The pair $(\mathcal{V}, \mathcal{T})$ is a *topological vector space* iff A and M are continuous, where both domains carry the product topology.

Proposition 7.1.2. *The Standard Topology on $\mathbb{R}^n$*
Let $n \in \mathbb{N}$. With the Euclidean topology, $\mathbb{R}^n$ is a topological vector space.

Proof. Fix $(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^n \times \mathbb{R}^n$ and an open neighborhood O of $\mathbf{x} + \mathbf{y}$.
Choose $\varepsilon > 0$ with $B_\varepsilon(\mathbf{x} + \mathbf{y}) \subseteq O$. If $\mathbf{u} \in B_{\varepsilon/2}(\mathbf{x})$ and $\mathbf{v} \in B_{\varepsilon/2}(\mathbf{y})$, then $\|(\mathbf{u} + \mathbf{v}) - (\mathbf{x} + \mathbf{y})\|_2 \le \|\mathbf{u} - \mathbf{x}\|_2 + \|\mathbf{v} - \mathbf{y}\|_2 < \varepsilon$.
Hence $A(B_{\varepsilon/2}(\mathbf{x}) \times B_{\varepsilon/2}(\mathbf{y})) \subseteq O$, so A continuous at $(\mathbf{x}, \mathbf{y})$ by Definition 6.2.1. Fix $(a, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^n$, open neighborhood W of $a\mathbf{x}$.
Choose $\varepsilon > 0$ with $B_\varepsilon(a\mathbf{x}) \subseteq W$, and choose $\delta \in (0, 1)$ with $\delta(\|\mathbf{x}\|_2 + |a| + 1) < \varepsilon$.
If $b \in (a - \delta, a + \delta)$ and $\mathbf{y} \in B_\delta(\mathbf{x})$, then $|b| < |a| + 1$ and $\|b\mathbf{y} - a\mathbf{x}\|_2 \le |b| \, \|\mathbf{y} - \mathbf{x}\|_2 + |b - a| \, \|\mathbf{x}\|_2 < \delta(|a| + 1 + \|\mathbf{x}\|_2) < \varepsilon$.
Hence $M((a - \delta, a + \delta) \times B_\delta(\mathbf{x})) \subseteq W$, M continuous at $(a, \mathbf{x})$ by Definition 6.2.1; $\mathbb{R}^n$ is topological vector space by Definition 7.1.1.
Structural Commentary: Product neighborhoods split the output error between the input coordinates.
Key Insight: The Euclidean norm estimates both vector operations locally. □

Proposition 7.1.3. *Real Matrix Maps Are Continuous*
Let $m, n \in \mathbb{N}$, let $A = [a_{j,k}] \in \mathrm{Mat}(m, n, \mathbb{R})$, and define $T_A : \mathbb{R}^n \to \mathbb{R}^m$ by $T_A(\mathbf{x}) := A\mathbf{x}$.

(i) The map T_A is continuous for the Euclidean topologies.
(ii) If $m = n$ and A is invertible, then T_A is a homeomorphism.
(iii) $(\mathbb{R}^n)' \subseteq C(\mathbb{R}^n, \mathbb{R})$.

Proof.

(i) Fix $k \in \{1, \dots, n\}$. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $|\pi_k(\mathbf{x}) - \pi_k(\mathbf{y})| \le \|\mathbf{x} - \mathbf{y}\|_2$, so π_k is 1-Lipschitz by Definition 5.2.4.
Hence π_k is continuous by Proposition 5.2.5. Fix $j \in \{1, \dots, m\}$.
The jth coordinate $\sum_{k=1}^{n} a_{j,k}\pi_k$ of T_A is continuous by Theorem 5.1.32(i), since constant maps are continuous.
Therefore T_A is continuous in metric sense by Theorem 5.1.32(ii). It is continuous for Euclidean topologies by Corollary 6.2.3.
(ii) If A is invertible, then $(T_A)^{-1} = T_{A^{-1}}$, which is continuous by (i). Thus T_A is a homeomorphism by Definition 6.2.5.
(iii) Fix $f \in (\mathbb{R}^n)'$ and let $B \in \mathrm{Mat}(1, n, \mathbb{R})$ be its matrix. Then $f = T_B$ by Proposition 3.2.5, so f is continuous by (i).

Structural Commentary: Coordinate continuity handles matrix maps; inverse matrices, row representations handle remaining claims.
Key Insight: Matrix coordinates transfer continuity to inverses and linear functionals. □

Theorem 7.1.4. *Finite-Dimensional Linear Maps Are Lipschitz*
Let $m, n \in \mathbb{N}$ and let $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$.

(i) $(\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n) \ \|A\mathbf{x} - A\mathbf{y}\|_2 \le \|A\|_{op} \|\mathbf{x} - \mathbf{y}\|_2$; hence A is $\|A\|_{op}$-Lipschitz.
(ii) If $m = n$ and A is invertible, then $(\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n) \ \frac{1}{\|A^{-1}\|_{op}} \|\mathbf{x} - \mathbf{y}\|_2 \le \|A\mathbf{x} - A\mathbf{y}\|_2 \le \|A\|_{op} \|\mathbf{x} - \mathbf{y}\|_2$.
Hence $A : \mathbb{R}^n \to \mathbb{R}^n$ is a homeomorphism.

Proof.

(i) The bound $(\forall \mathbf{z} \in \mathbb{R}^n) \ \|A\mathbf{z}\|_2 \le \|A\|_{\mathrm{op}} \|\mathbf{z}\|_2$ holds by Definition 5.1.4. With $\mathbf{z} = \mathbf{x} - \mathbf{y}$, linearity of A proves the claim.
(ii) Since $\|A^{-1}\|_{\mathrm{op}} > 0$, $\|\mathbf{x} - \mathbf{y}\|_2 = \|A^{-1}(A\mathbf{x} - A\mathbf{y})\|_2 \le \|A^{-1}\|_{\mathrm{op}} \|A\mathbf{x} - A\mathbf{y}\|_2$ by part (i) for A^{-1}.
This inequality and part (i) are the lower and upper bounds.
The maps A and A^{-1} are continuous by part (i) and Proposition 5.2.5, so A is a homeomorphism by Definition 6.2.5.

Structural Commentary: A^{-1} converts upper control into lower control.
Key Insight: Invertible linear maps are bi-Lipschitz. □

Theorem 7.1.5. *Openness of Invertibility and Continuity of Inversion*
Let $n \in \mathbb{N}$, equip $\mathcal{L}(\mathbb{R}^n)$ with the operator-norm metric, and let Ω be its set of invertible operators.

(i) If $A \in \Omega$, $B \in \mathcal{L}(\mathbb{R}^n)$, and $\|B - A\|_{op} \|A^{-1}\|_{op} < 1$, then $B \in \Omega$.
(ii) Ω is open in $\mathcal{L}(\mathbb{R}^n)$.
(iii) The inversion map $\iota : \Omega \to \Omega$, $\iota(A) := A^{-1}$, is continuous.

Proof.

(i) Set $\alpha := \frac{1}{\|A^{-1}\|_{\mathrm{op}}}$ and $\beta := \|B - A\|_{\mathrm{op}}$; then $0 \le \beta < \alpha$.
For $\mathbf{x} \in \mathbb{R}^n$, $\alpha \|\mathbf{x}\|_2 \le \|A\mathbf{x}\|_2$ by Theorem 7.1.4(ii), while $\|A\mathbf{x}\|_2 \le \|(A - B)\mathbf{x}\|_2 + \|B\mathbf{x}\|_2 \le \beta \|\mathbf{x}\|_2 + \|B\mathbf{x}\|_2$ by Definition 5.1.4.
Hence $(\alpha - \beta) \|\mathbf{x}\|_2 \le \|B\mathbf{x}\|_2$, so $\ker B = \{\mathbf{0}\}$ and $B \in \Omega$ by Theorem 1.7.7.
For $\mathbf{y} \in \mathbb{R}^n$, set $\mathbf{x} := B^{-1}\mathbf{y}$; then $\|B^{-1}\mathbf{y}\|_2 \le \frac{1}{\alpha - \beta} \|\mathbf{y}\|_2$, so $\|B^{-1}\|_{\mathrm{op}} \le \frac{1}{\alpha - \beta}$ by Definition 5.1.4.
(ii) For $A \in \Omega$, the ball $\{B \in \mathcal{L}(\mathbb{R}^n) : \|B - A\|_{\mathrm{op}} < \frac{1}{\|A^{-1}\|_{\mathrm{op}}}\}$ lies in Ω by (i); hence Ω is open.
(iii) Fix $A \in \Omega$, let $(B_k) \subseteq \Omega$ satisfy $B_k \to A$. Set $\alpha := \frac{1}{\|A^{-1}\|_{\mathrm{op}}}$, $\beta_k := \|B_k - A\|_{\mathrm{op}}$. Eventually $\beta_k < \frac{\alpha}{2}$, so $\|B_k^{-1}\|_{\mathrm{op}} \le \frac{2}{\alpha}$ by (i).
The bound $\|B_k^{-1} - A^{-1}\|_{\mathrm{op}} \le \frac{2}{\alpha} \beta_k \|A^{-1}\|_{\mathrm{op}} \to 0$ holds by $B_k^{-1} - A^{-1} = B_k^{-1}(A - B_k)A^{-1}$ and Proposition 5.1.3(v).
Thus inversion is continuous at A by Theorem 5.1.15. Since $A \in \Omega$ was arbitrary, ι is continuous.

Structural Commentary: The lower stretch controls inverse norms; the resolvent identity transfers perturbations to inverses.
Key Insight: Uniform inverse bounds make $B^{-1}(A-B)A^{-1}$ vanish with $B-A$. □

Definition 7.1.6. *Natural Topology from a Basis*
Let $n \in \mathbb{N}$, let $\mathcal{V}$ be an n-dimensional real vector space, let β be a basis, and let $[\cdot]_\beta : \mathcal{V} \to \mathbb{R}^n$ be the coordinate isomorphism.
The *natural topology induced by* β is $\mathcal{T}_\beta := \{[\cdot]_\beta^{-1}(O) : O \subseteq \mathbb{R}^n \text{ is open}\}$. Equivalently, $(\forall U \subseteq \mathcal{V})\,[U \in \mathcal{T}_\beta \iff [\cdot]_\beta(U) \text{ is open in } \mathbb{R}^n]$.

Proposition 7.1.7. *Natural Topology Is Basis-Independent*
Let $n \in \mathbb{N}$, let $\mathcal{V}$ be an n-dimensional real vector space, and let β and γ be bases of $\mathcal{V}$. Then $\mathcal{T}_\beta = \mathcal{T}_\gamma$.

Proof. Set $H_{\gamma\leftarrow\beta} := [\cdot]_\gamma \circ [\cdot]_\beta^{-1} : \mathbb{R}^n \to \mathbb{R}^n$. The identity $H_{\gamma\leftarrow\beta}\mathbf{x} = C_{\gamma\leftarrow\beta}\mathbf{x}$ holds by Definition 3.5.1.
Matrix $C_{\gamma\leftarrow\beta}$ invertible with inverse $C_{\beta\leftarrow\gamma}$ by Theorem 3.5.2(i); $H_{\gamma\leftarrow\beta}$ is homeomorphism with inverse $H_{\beta\leftarrow\gamma}$ by Proposition 7.1.3(ii).
For $U \subseteq \mathcal{V}$, $U \in \mathcal{T}_\beta \iff [\cdot]_\beta(U)$ open $\iff H_{\gamma\leftarrow\beta}([\cdot]_\beta(U)) = [\cdot]_\gamma(U)$ open $\iff U \in \mathcal{T}_\gamma$ by Definition 7.1.6, homeomorphism property.
Structural Commentary: Basis changes are homeomorphisms.
Key Insight: The natural topology is carried by the vector-space structure, not by a preferred basis. □

Theorem 7.1.8. *Characterization of the Natural Topology*
Let $\mathcal{V}$ be a real vector space, set $n := \dim_{\mathbb{R}} \mathcal{V}$, assume $1 \le n < \infty$, let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis, and let $\mathcal{T}$ be a topology on $\mathcal{V}$.

(i) $(\mathcal{V}, \mathcal{T}_\beta)$ is a topological vector space.
(ii) $\mathcal{V}' \subseteq C((\mathcal{V}, \mathcal{T}_\beta), \mathbb{R})$.
(iii) If $(\mathcal{V}, \mathcal{T})$ is a topological vector space and $\mathcal{V}' \subseteq C((\mathcal{V}, \mathcal{T}), \mathbb{R})$, then $\mathcal{T} = \mathcal{T}_\beta$.

Proof. Put $\Phi_\beta := [\cdot]_\beta$.

(i) The map $\Phi_\beta : (\mathcal{V}, \mathcal{T}_\beta) \to \mathbb{R}^n$ is a homeomorphism by Definition 7.1.6.
The product maps $\Phi_\beta \times \Phi_\beta$ and $\mathrm{id}_{\mathbb{R}} \times \Phi_\beta$, together with their product inverses, are continuous by Propositions 6.2.9 and 6.2.2(iii).
Hence both product maps are homeomorphisms by Definition 6.2.5.
Let $A_{\mathcal{V}}, M_{\mathcal{V}}$ and $A_{\mathbb{R}^n}, M_{\mathbb{R}^n}$ denote the vector operations on $\mathcal{V}$ and $\mathbb{R}^n$, respectively.
Identity $A_{\mathcal{V}} = \Phi_\beta^{-1} \circ A_{\mathbb{R}^n} \circ (\Phi_\beta \times \Phi_\beta)$ holds by Proposition 3.2.2(i). Also, $M_{\mathcal{V}} = \Phi_\beta^{-1} \circ M_{\mathbb{R}^n} \circ (\mathrm{id}_{\mathbb{R}} \times \Phi_\beta)$ by Proposition 3.2.2(i).
The maps $A_{\mathcal{V}}$ and $M_{\mathcal{V}}$ are continuous by Propositions 7.1.2 and 6.2.2(iii).

(ii) If $\varphi \in \mathcal{V}'$, then $\varphi \circ \Phi_\beta^{-1} \in (\mathbb{R}^n)'$, continuous by Proposition 7.1.3(iii); $\varphi = (\varphi \circ \Phi_\beta^{-1}) \circ \Phi_\beta$ is continuous by Proposition 6.2.2(iii).

(iii) Suppose $(\mathcal{V}, \mathcal{T})$ is a topological vector space and $\mathcal{V}' \subseteq C((\mathcal{V}, \mathcal{T}), \mathbb{R})$.

(a) *Coordinate Functionals*: For $i \in \{1, \dots, n\}$, set $\varphi_i := \pi_i \circ \Phi_\beta \in \mathcal{V}'$. Each φ_i is continuous by the dual-continuity hypothesis.

(b) *Forward Coordinate Map*: Fix $\mathbf{v} \in \mathcal{V}$ and an open neighborhood O of $\Phi_\beta(\mathbf{v})$.
Choose $\varepsilon > 0$ with $B_\varepsilon(\Phi_\beta(\mathbf{v})) \subseteq O$, and set $I_i := \left(\varphi_i(\mathbf{v}) - \frac{\varepsilon}{\sqrt{n}}, \varphi_i(\mathbf{v}) + \frac{\varepsilon}{\sqrt{n}}\right)$ for $i \in \{1, \dots, n\}$.
The set $U := \bigcap_{i=1}^{n} \varphi_i^{-1}(I_i)$ is an open neighborhood of $\mathbf{v}$ by Definition 6.1.1 and Proposition 6.2.2(i).
For $\mathbf{u} \in U$, $\left\|\Phi_\beta(\mathbf{u}) - \Phi_\beta(\mathbf{v})\right\|_2 < \varepsilon$, so $\Phi_\beta(U) \subseteq O$. Hence $\Phi_\beta : (\mathcal{V}, \mathcal{T}) \to \mathbb{R}^n$ is continuous by Definition 6.2.1.

(c) *Coordinate Synthesis*: The inverse of Φ_β is $\Psi : \mathbb{R}^n \to \mathcal{V}$, $\Psi(\mathbf{a}) := \sum_{i=1}^{n} \pi_i(\mathbf{a})\mathbf{v}_i$, by Definition 3.2.1.
Each π_i is continuous by Proposition 7.1.3(iii). Each constant map $\mathbf{a} \mapsto \mathbf{v}_i$ is continuous by Proposition 6.2.2(i), because every open-set preimage is either $\emptyset$ or $\mathbb{R}^n$. Hence $\mathbf{a} \mapsto (\pi_i(\mathbf{a}), \mathbf{v}_i)$ is continuous by Proposition 6.2.9, and $\mathbf{a} \mapsto \pi_i(\mathbf{a})\mathbf{v}_i$ is continuous by the topological-vector-space hypothesis and Proposition 6.2.2(iii).
Finite sums of these maps continuous by same hypothesis, Proposition 6.2.9, Proposition 6.2.2(iii). Thus Ψ is continuous.

(d) *Conclusion*: The map Φ_β is a homeomorphism by (b)–(c) and Definition 6.2.5. Hence $\mathcal{T} = \mathcal{T}_\beta$ by Definition 7.1.6.

Structural Commentary: Coordinate functionals recover the forward map; vector operations recover its inverse.
Key Insight: Functionals and vector operations determine finite-dimensional coordinates topologically. □

Corollary 7.1.9. *Finite-Dimensional Linear Maps Are Continuous*
Let $\mathcal{V}$ and $\mathcal{W}$ be nonzero finite-dimensional real vector spaces with their natural topologies.

(i) $\mathcal{L}(\mathcal{V}, \mathcal{W}) \subseteq C(\mathcal{V}, \mathcal{W})$.
(ii) If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is an isomorphism, then T is a homeomorphism.

Proof.

(i) Choose bases β of $\mathcal{V}$ and γ of $\mathcal{W}$, and fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Coordinate map $[\cdot]_\gamma \circ T \circ [\cdot]_\beta^{-1}$ equals $\mathbf{x} \mapsto [T]_{\gamma\leftarrow\beta}\mathbf{x}$ by Proposition 3.2.5, and it is continuous by Proposition 7.1.3(i).
Coordinate maps $[\cdot]_\beta$, $[\cdot]_\gamma$ are homeomorphisms by Definition 7.1.6, Proposition 7.1.7; T is continuous by Proposition 6.2.2(iii).

(ii) If T is an isomorphism, then $T^{-1} \in \mathcal{L}(\mathcal{W}, \mathcal{V})$, so T^{-1} is continuous by (i); T is a homeomorphism by Definition 6.2.5.

Structural Commentary: Coordinate homeomorphisms conjugate linear maps to continuous real matrix maps.
Key Insight: Natural coordinates transfer matrix continuity back to the vector spaces. □

Perspective 7.1.10. *Finite-Dimensional Topology Is Rigid*
Natural topology is independent of coordinates by Proposition 7.1.7, recoverable from vector operations and continuous algebraic dual by Theorem 7.1.8. Every linear map is continuous and every linear isomorphism is a homeomorphism by Corollary 7.1.9.
Infinite-dimensional topological vector spaces need not share this rigidity.

Part XII
Calculus, Series, and Approximation

1 One-Variable Differentiation and Riemann Integration

1.1 Differentiation and Basic Theorems

This subsection develops the derivative as a relative-domain limit and first-order approximation.
It then derives algebraic, mean-value, inverse, higher-order, and vector-valued consequences.

Definition 1.1.1. *Derivative of a Real Function*
Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $x_0 \in X^{\mathrm{acc}}$. Define $\phi_{x_0} : X \setminus \{x_0\} \to \mathbb{R}$ by $\phi_{x_0}(x) := \frac{f(x)-f(x_0)}{x-x_0}$.
If limit exists, define $f'(x_0) := \lim_{x \to x_0} \phi_{x_0}(x)$ as *derivative at x_0 relative to X*. Map f is then *differentiable at x_0 relative to X*.
For $E \subseteq X^{\mathrm{acc}}$, f is *differentiable on E relative to X* $\iff (\forall x \in E)\, f'(x)$ exists relative to X.

Perspective 1.1.2. *Derivative Depends on the Domain*
The punctured approach set $X \setminus \{x_0\}$ determines the derivative in Definition 1.1.1.
Interior interval points produce two-sided derivatives, whereas interval endpoints produce one-sided derivatives.
The limit-point hypothesis excludes isolated points.

Figure 152 contrasts an isolated point with a half-line endpoint.
Only the endpoint admits a punctured approach within its displayed domain.

Figure 152: Left: x_0 is isolated. Right: $X = [x_0, \infty)$ permits right-sided approach to x_0.

Figure 153 represents a difference quotient by a secant slope. Its limiting value is the tangent slope $f'(x)$.

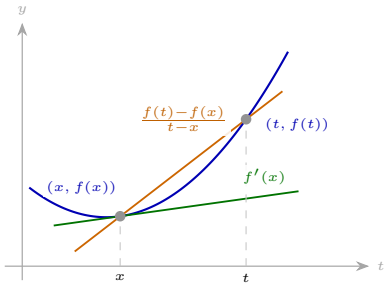

Figure 153: The secant through $(x, f(x))$ and $(t, f(t))$ has slope $\frac{f(t)-f(x)}{t-x}$; its limit is $f'(x)$.

Proposition 1.1.3. *First-Order Approximation Form*
Let $X \subseteq \mathbb{R}$, let $x_0 \in X^{\mathrm{acc}}$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. Set $R_L(x) := f(x) - f(x_0) - L(x - x_0)$.
Then $f'(x_0) = L \iff (\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in X \setminus \{x_0\})\, [|x - x_0| < \delta \implies |R_L(x)| < \varepsilon\, |x - x_0|]$.

Proof. For $x \in X \setminus \{x_0\}$, $\frac{|R_L(x)|}{|x-x_0|} = \left| \frac{f(x)-f(x_0)}{x-x_0} - L \right|$.
($\Rightarrow$) Assume $f'(x_0) = L$, $\varepsilon > 0$. On some punctured δ-neighborhood, the right-hand side is less than ε by Definition 1.1.1.
Multiplication by $|x - x_0|$ proves the estimate.
($\Leftarrow$) Assume the estimate, and $\varepsilon > 0$. Choose $\delta > 0$ satisfying the estimate by the hypothesis.
Division by $|x - x_0|$ makes right-hand side less than ε on same punctured neighborhood. Hence $f'(x_0) = L$ by Definition 1.1.1.
Structural Commentary: The two directions rescale the same remainder by $|x - x_0|$.
Key Insight: The normalized first-order error is the derivative-limit error. □

Theorem 1.1.4. *Differentiability Implies Continuity*
Let $X \subseteq \mathbb{R}$, let $x_0 \in X^{\mathrm{acc}}$, and let $f : X \to \mathbb{R}$. If f is differentiable at x_0 relative to X, then f is continuous at x_0.

Proof. For $t \in X \setminus \{x_0\}$, $f(t) - f(x_0) = \frac{f(t)-f(x_0)}{t-x_0}(t - x_0)$. Factors tend to $f'(x_0)$ and 0, so $f(t) - f(x_0) \to 0$ by Theorem 5.1.7(ii).
Thus $f(t) \to f(x_0)$ by Theorem 5.1.7(i), and continuity holds by Theorem 5.1.12.
Structural Commentary: Limit algebra transfers difference-quotient convergence to the function increment.
Key Insight: A finite quotient multiplied by a vanishing increment tends to zero. □

Theorem 1.1.5. *Algebra Rules for Derivatives*
Let $X \subseteq \mathbb{R}$, let $x_0 \in X^{\mathrm{acc}}$, and let $f, g : X \to \mathbb{R}$ be differentiable at x_0 relative to X.
(i) $(f + g)'(x_0) = f'(x_0) + g'(x_0)$.
(ii) $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$.
(iii) If $g(x_0) \neq 0$, then $\left(\frac{f}{g}\right)'(x_0) = \frac{g(x_0)f'(x_0)-f(x_0)g'(x_0)}{g(x_0)^2}$.

Proof. For $h \in \{f, g\}$ and $t \in X \setminus \{x_0\}$, set $Q_h(t) := \frac{h(t)-h(x_0)}{t-x_0}$; then $Q_h(t) \to h'(x_0)$ as $t \to x_0$.

(i) Since $Q_{f+g} = Q_f + Q_g$, the sum rule holds by Theorem 5.1.7(i).
(ii) The identity $Q_{fg}(t) = f(t)Q_g(t) + g(x_0)Q_f(t)$ holds.
The limit $f(t) \to f(x_0)$ holds by Theorem 1.1.4, so the product rule follows by Theorem 5.1.7(i),(ii).
(iii) Assume $g(x_0) \neq 0$. Then $g(t) \neq 0$ near x_0 by Theorem 1.1.4 and the assumption.
On this punctured neighborhood, $Q_{f/g}(t) = \frac{g(x_0)Q_f(t)-f(x_0)Q_g(t)}{g(t)g(x_0)}$. The quotient rule follows by Theorem 5.1.7(i)–(iii).

Structural Commentary: Difference-quotient identities reduce all three claims to limit algebra and continuity.
Key Insight: Expand increments before passing to their limits. □

Theorem 1.1.6. *Chain Rule*
Let $X, Y \subseteq \mathbb{R}$, $f : X \to Y$, $g : Y \to \mathbb{R}$, $x_0 \in X^{\mathrm{acc}}$. Set $y_0 := f(x_0)$ and assume $y_0 \in Y^{\mathrm{acc}}$.
If $f'(x_0)$ and $g'(y_0)$ exist relative to X and Y, then $(g \circ f)'(x_0) = g'(y_0)f'(x_0)$ relative to X.

Proof. Define $u : X \to \mathbb{R}$ by $u(x_0) := 0$ and $u(t) := \frac{f(t)-f(x_0)}{t-x_0} - f'(x_0)$ for $t \neq x_0$.
Define $v : Y \to \mathbb{R}$ by $v(y_0) := 0$ and $v(s) := \frac{g(s)-g(y_0)}{s-y_0} - g'(y_0)$ for $s \neq y_0$. Then $u(t) \to 0$ and $v(s) \to 0$ by Definition 1.1.1.
The limit $f(t) \to y_0$ holds by Theorem 1.1.4. The definition $v(y_0) = 0$ and punctured limit $v(s) \to 0$ therefore imply $v(f(t)) \to 0$.
For $t \in X \setminus \{x_0\}$, $\frac{g(f(t))-g(f(x_0))}{t-x_0} = \big(g'(y_0) + v(f(t))\big)\big(f'(x_0) + u(t)\big)$. Right-hand side tends to $g'(y_0)f'(x_0)$ by Theorem 5.1.7(i),(ii).
Structural Commentary: Composed first-order expansions reduce the derivative to a product of two limit factors.
Key Insight: The linear terms multiply while both normalized remainders vanish. □

Example 1.1.7. *Oscillation at a Point*
Assume the usual derivative of sin, and define $f_1, f_2 : \mathbb{R} \to \mathbb{R}$ by

$$f_1(x) = \begin{cases} x \sin(\frac{1}{x}), & x \neq 0, \\ 0, & x = 0, \end{cases}$$

$$f_2(x) = \begin{cases} x^2 \sin(\frac{1}{x}), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Quotient $\frac{f_1(t)-f_1(0)}{t} = \sin(\frac{1}{t})$ has no limit, so $f_1'(0)$ does not exist. Since $\left|\frac{f_2(t)-f_2(0)}{t}\right| \leq |t|$, $f_2'(0) = 0$.
For $x \neq 0$, $f_2'(x) = 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})$. Along $x_n = (2\pi n)^{-1}$ and $y_n = ((2n+1)\pi)^{-1}$, $f_2'(x_n) = -1$ and $f_2'(y_n) = 1$.
Thus f_2 is differentiable at 0, but f_2' is not continuous there.

Definition 1.1.8. *Local Extrema*
Let (X, d) be a metric space, let $f : X \to \mathbb{R}$, and let $p \in X$.

(i) The map f has a *local maximum at p* $\iff (\exists \delta > 0)(\forall q \in X)\,[d(p, q) < \delta \implies f(q) \leq f(p)]$.
(ii) The map f has a *local minimum at p* $\iff (\exists \delta > 0)(\forall q \in X)\,[d(p, q) < \delta \implies f(q) \geq f(p)]$.

Remark 1.1.9. *Local and Global Extrema*
Every global extremum is local, whereas every isolated domain point is both a local maximum and minimum.
Derivative tests therefore require an interior interval point.
Continuous real maps on nonempty compact spaces attain global extrema by Theorem 5.1.39.

Theorem 1.1.10. *Interior Extremum Has Zero Derivative*
Let $a < b$, $f : (a, b) \to \mathbb{R}$, $x \in (a, b)$. If f has a local extremum at x and is differentiable there, then $f'(x) = 0$.

Proof. At a local maximum, $\frac{f(t)-f(x)}{t-x} \geq 0$ for $t < x$ near x and $\frac{f(t)-f(x)}{t-x} \leq 0$ for $t > x$ near x.
The derivative limit therefore satisfies $f'(x) \geq 0$ and $f'(x) \leq 0$, so $f'(x) = 0$. At a local minimum, $-f$ has a local maximum.
The identity $(-f)'(x) = -f'(x)$ holds by Definition 1.1.1, so $f'(x) = 0$ by the maximum case.
Structural Commentary: The two approach sides impose opposite sign bounds on the same derivative limit.
Key Insight: A two-sided limit compatible with both quotient signs must be zero. □

Theorem 1.1.11. *Rolle's Theorem*
Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, then $(\exists x \in (a, b))\, f'(x) = 0$.

Proof. If f is constant, $f'(\frac{a+b}{2}) = 0$ by Definition 1.1.1. Otherwise, $[a, b]$ is compact by Theorem 4.2.5.
The map f attains a maximum and a minimum by Theorem 5.1.39.
At least one extremal value differs from $f(a) = f(b)$, so its attaining point x lies in (a, b). Then $f'(x) = 0$ by Theorem 1.1.10.
Structural Commentary: Compactness supplies extrema, and equal endpoint values force a nonconstant extremum into the interior.
Key Insight: An interior attained extremum is stationary. □

Theorem 1.1.12. *Cauchy Mean Value Theorem*
Let $a < b$, $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$, differentiable on (a, b). Then $(\exists x \in (a, b))\,\big(f(b) - f(a)\big)g'(x) = \big(g(b) - g(a)\big)f'(x)$.

Proof. Set $h(t) := \big(f(b) - f(a)\big)g(t) - \big(g(b) - g(a)\big)f(t)$; then $h(b) - h(a) = 0$. The map h is continuous on $[a, b]$ by Theorem 5.1.32(i).
It is differentiable on (a, b) by Theorem 1.1.5(i),(ii). Choose $x \in (a, b)$ with $h'(x) = 0$ by Theorem 1.1.11.
Since $h'(x) = \big(f(b) - f(a)\big)g'(x) - \big(g(b) - g(a)\big)f'(x)$, the claimed identity follows.
Structural Commentary: A determinant-like linear combination converts the desired identity into Rolle's endpoint condition.
Key Insight: Cancel the endpoint increments before differentiating. □

Corollary 1.1.13. *Mean Value Theorem*
Let $a < b$, $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then $(\exists x \in (a, b))\, f(b) - f(a) = (b - a)f'(x)$.

Proof. Set $g(t) := t$. Then $g'(x) = 1$ and $g(b) - g(a) = b - a$. Choose $x \in (a, b)$ with $f(b) - f(a) = (b - a)f'(x)$ by Theorem 1.1.12.
Structural Commentary: Specializing Cauchy's auxiliary comparison to the identity map isolates the secant increment.
Key Insight: The identity map converts Cauchy's relation into slope equality. □

Figure 154 depicts Corollary 1.1.13. The tangent at an interior point c is parallel to the endpoint secant.

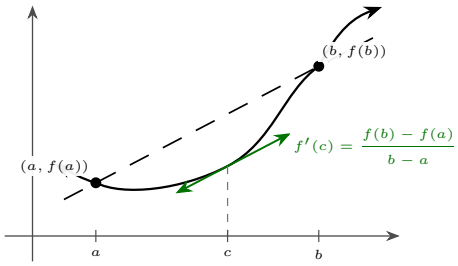

Figure 154: The Mean Value Theorem finds an interior tangent slope equal to the endpoint secant slope.

Corollary 1.1.14. *Lipschitz Control from a Bounded Derivative*
Let $a < b$, let $M \geq 0$, and let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). If $(\forall t \in (a,b))\ |f'(t)| \leq M$, then:
(i) $(\forall x, y \in [a,b])\ |f(y) - f(x)| \leq M\,|y - x|$;
(ii) f is uniformly continuous on $[a,b]$.

Proof.
(i) If $x < y$, choose $c \in (x,y)$ with $f(y) - f(x) = (y-x)f'(c)$ by Corollary 1.1.13.
Hence $|f(y) - f(x)| \leq M\,|y - x|$. Interchanging x, y covers $x > y$, and $x = y$ is immediate.
(ii) Given $\varepsilon > 0$, set $\delta := \frac{\varepsilon}{M+1}$. For $x, y \in [a,b]$, $|x - y| < \delta$ implies $|f(x) - f(y)| \leq M\,|x - y| < \varepsilon$ by (i).
Structural Commentary: The mean-value identity transfers the derivative bound to secants, which supplies a uniform modulus.
Key Insight: One global slope bound controls every pair of points. □

Theorem 1.1.15. *Derivative Sign and Monotonicity*
Let $a < b$ and let $f : (a,b) \to \mathbb{R}$ be differentiable.
(i) If $(\forall x \in (a,b))\ f'(x) \geq 0$, then f is monotonically increasing.
(ii) If $(\forall x \in (a,b))\ f'(x) = 0$, then f is constant.
(iii) If $(\forall x \in (a,b))\ f'(x) \leq 0$, then f is monotonically decreasing.

Proof. Fix $x_1, x_2 \in (a,b)$ with $x_1 < x_2$. The map f is continuous on $[x_1, x_2]$ by Theorem 1.1.4.
Choose $c \in (x_1, x_2)$ with $f(x_2) - f(x_1) = (x_2 - x_1)f'(c)$ by Corollary 1.1.13.
(i) If $f' \geq 0$, then $f(x_2) - f(x_1) \geq 0$.
(ii) If $f' = 0$, then $f(x_2) - f(x_1) = 0$.
(iii) If $f' \leq 0$, then $f(x_2) - f(x_1) \leq 0$.
Structural Commentary: The mean-value identity transfers each derivative-sign hypothesis to every ordered pair.
Key Insight: Tangent-sign control becomes secant-sign control. □

Theorem 1.1.16. *Derivatives Have the Intermediate Value Property*
Let $a < b$, $f : [a,b] \to \mathbb{R}$ be differentiable relative to $[a,b]$. If $\lambda \in \mathbb{R}$ and $\big(f'(a) - \lambda\big)\big(f'(b) - \lambda\big) < 0$, then $(\exists x \in (a,b))\ f'(x) = \lambda$.

Proof. The transformation $(f, \lambda) \mapsto (-f, -\lambda)$ interchanges the endpoint orders and preserves the conclusion.
It therefore suffices to assume $f'(a) < \lambda < f'(b)$. Set $g(t) := f(t) - \lambda t$. Then $g'(a) < 0 < g'(b)$ by Definition 1.1.1.
Choose $t_1, t_2 \in (a,b)$ with $g(t_1) < g(a)$, $g(t_2) < g(b)$ by two endpoint derivative limits. Map g is continuous on $[a,b]$ by Theorem 1.1.4.
The interval $[a,b]$ is compact by Theorem 4.2.5, so g attains a minimum by Theorem 5.1.39.
Neither endpoint is a minimizer, so a minimizer x lies in (a,b). Then $0 = g'(x) = f'(x) - \lambda$ by Theorem 1.1.10.
Structural Commentary: Linear tilting turns the target derivative value into an interior-minimum condition.
Key Insight: Opposite endpoint slopes force the tilted minimum away from both endpoints. □

Perspective 1.1.17. *Sliding Secants*
Let f, λ be as in Theorem 1.1.16, and fix $0 < h < b - a$. Define $\sigma_h : [a, b-h] \to \mathbb{R}$ by $\sigma_h(x) := \frac{f(x+h) - f(x)}{h}$.
Map f is continuous by Theorem 1.1.4. Thus σ_h is continuous by Theorems 5.1.16 and 5.1.32(i).
As $h \to 0^+$, $\sigma_h(a) \to f'(a)$ and $\sigma_h(b-h) \to f'(b)$. For sufficiently small h, λ lies strictly between these endpoint values.
Choose $x \in (a, b-h)$ with $\lambda = \sigma_h(x)$ by Theorem 5.1.43. Choose $c \in (x, x+h)$ with $\lambda = f'(c)$ by Corollary 1.1.13.
Key Insight: Slide a fixed-width secant, then convert its slope to a tangent slope.

Figure 155 illustrates the fixed-width secants in the preceding perspective.
Their horizontal separation remains h while the base point moves.

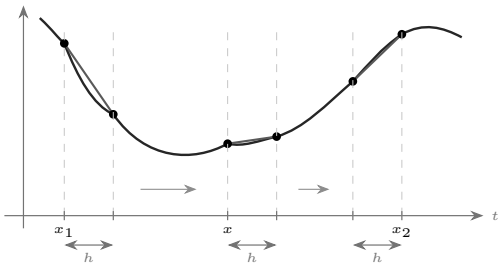

Figure 155: Fixed-width secants slide across the graph while retaining horizontal separation h.

Corollary 1.1.18. *Derivatives Have No Simple Discontinuities*
Let $a < b$, $f : [a, b] \to \mathbb{R}$ be differentiable relative to $[a, b]$. Then f' has no simple discontinuity in (a, b).

Proof. Fix $c \in (a, b)$, let L be either existing one-sided limit $f'(c-)$ or $f'(c+)$. If $L \neq f'(c)$, choose λ strictly between them.
The function f' remains strictly on the L-side of λ in some one-sided neighborhood of c by Definition 5.3.1.
Fix x in that neighborhood. The endpoint derivatives of restriction to the segment with endpoints x, c are $f'(x)$ and $f'(c)$.
Choose z strictly between x and c with $f'(z) = \lambda$ by Theorem 1.1.16, a contradiction.
Thus every existing one-sided limit of f' at c equals $f'(c)$. If c a simple discontinuity, both one-sided limits exist by Definition 5.3.1.
Then $f'(t) \to f'(c)$ by Proposition 5.3.2. Thus f' would be continuous at c by Theorem 5.1.12.
Structural Commentary: Darboux interpolation forces each existing one-sided derivative limit to equal the central derivative.
Key Insight: Any gap from $f'(c)$ would force the forbidden value arbitrarily nearby. □

Theorem 1.1.19. *Derivative of an Inverse Function*
Let $I \subseteq \mathbb{R}$ be an open interval, $f : I \to \mathbb{R}$ be injective, $x_0 \in I$. Set $y_0 := f(x_0)$ and $g := f^{-1} : f(I) \to I$.
If $f'(x_0)$ exists and is nonzero and g is continuous at y_0, then $g'(y_0) = \frac{1}{f'(x_0)}$ relative to $f(I)$.

Proof. Choose $(x_n) \subseteq I \setminus \{x_0\}$ with $x_n \to x_0$. Then $f(x_n) \to y_0$ by Theorem 1.1.4, and $f(x_n) \neq y_0$ by injectivity.
Thus $y_0 \in f(I)^{\mathrm{acc}}$. For $y \in f(I) \setminus \{y_0\}$, set $x := g(y)$. Then $\frac{g(y)-g(y_0)}{y-y_0} = \frac{x-x_0}{f(x)-f(x_0)} = \left(\frac{f(x)-f(x_0)}{x-x_0}\right)^{-1}$.
As $y \to y_0$, continuity of g implies $x = g(y) \to x_0$. The reciprocal therefore tends to $\frac{1}{f'(x_0)}$ by Theorem 5.1.7(iii).
Structural Commentary: Inverse continuity transports the input limit, while injectivity makes the matched quotients reciprocal.
Key Insight: Exchange input and output increments before taking the limit. □

Corollary 1.1.20. *One-Dimensional Inverse Function Theorem*
Let $a < b$, $c < d$, $f : (a, b) \to (c, d)$ be a differentiable bijection. Assume $(\forall x \in (a, b))\, f'(x) \neq 0$.

(i) The map f is a homeomorphism.
(ii) The inverse $g := f^{-1}$ is differentiable and satisfies $(\forall y \in (c, d))\, g'(y) = \frac{1}{f'(g(y))}$.

Proof. The derivative f' has constant sign by Theorem 1.1.16 and the nonvanishing hypothesis. Fix $x_1, x_2 \in (a, b)$ with $x_1 < x_2$.
Choose $\xi \in (x_1, x_2)$ with $f(x_2) - f(x_1) = (x_2 - x_1)f'(\xi)$ by Corollary 1.1.13. Hence f is strictly monotone.

(i) The inverse $g : (c, d) \to (a, b)$ is monotone, and its image is the interval (a, b).
Thus g is continuous by Theorem 5.3.7. The map f is continuous by Theorem 1.1.4, so f is a homeomorphism.
(ii) Fix $y \in (c, d)$ and set $x := g(y)$. Then $g'(y) = \frac{1}{f'(x)} = \frac{1}{f'(g(y))}$ by Theorem 1.1.19.

Structural Commentary: Darboux fixes the derivative sign, monotonicity controls the inverse, and the local quotient formula finishes.
Key Insight: A nonvanishing derivative prevents both turning and reciprocal-quotient failure. □

Theorem 1.1.21. *L'Hospital's Rule*
Let $-\infty \leq a < b \leq +\infty$, $A \in \overline{\mathbb{R}}$, $f, g : (a, b) \to \mathbb{R}$ be differentiable. Assume $(\forall x \in (a, b))\, g'(x) \neq 0$.

(i) If $\frac{f'(x)}{g'(x)} \to A$ and $f(x), g(x) \to 0$ as $x \to a^+$, then $\frac{f(x)}{g(x)} \to A$.
(ii) If $\frac{f'(x)}{g'(x)} \to A$ and $|g(x)| \to +\infty$ as $x \to a^+$, then $\frac{f(x)}{g(x)} \to A$.
(iii) If $\frac{f'(x)}{g'(x)} \to A$ as $x \to b^-$, the conclusions in (i)–(ii) hold with b^- in place of a^+.

Proof. The derivative g' has constant sign by Theorem 1.1.16 and the nonvanishing hypothesis.
For $r < s$, choose $t \in (r, s)$ with $g(s) - g(r) = (s - r)g'(t)$ by Corollary 1.1.13. Hence g is strictly monotone.
For $a < r < s < b$, set $Q(r, s) := \frac{f(s)-f(r)}{g(s)-g(r)}$. Choose $t \in (r, s)$ with $Q(r, s) = \frac{f'(t)}{g'(t)}$ by Theorem 1.1.12.
For (i)–(ii), every eventual order bound on $\frac{f'}{g'}$ near a also holds for $Q(r, s)$ when r, s are sufficiently near a.

(i) After restricting to a smaller right-neighborhood of a, $g \neq 0$ by strict monotonicity and $g(x) \to 0$.
Fix s there and let $r \to a^+$. Then $Q(r, s) \to \frac{f(s)}{g(s)}$.
Each tightened eventual bound for $\frac{f'}{g'}$ therefore transfers to $\frac{f(s)}{g(s)}$. Thus $\frac{f(x)}{g(x)} \to A$ by Definition 5.3.10.
(ii) The limit $|g(r)| \to +\infty$ makes $g(r) \neq 0$ near a. Fix s sufficient near a.
For $r \in (a, s)$, $\frac{f(r)}{g(r)} = Q(r, s)\frac{g(r)-g(s)}{g(r)} + \frac{f(s)}{g(r)}$.
As $r \to a^+$, the second factor tends to 1 and the last term tends to 0.
For finite A, use a half-width interval; for $A = \pm\infty$, use a farther ray threshold.
Each tightened bound remains inside prescribed neighborhood after these perturbations. Thus $\frac{f(r)}{g(r)} \to A$ by Definition 5.3.10.
(iii) Define $F(u) := f(-u)$ and $G(u) := g(-u)$ on $(-b, -a)$. Then $\frac{F'(u)}{G'(u)} = \frac{f'(-u)}{g'(-u)}$.
The left-endpoint conclusions for F, G hold by (i)–(ii) and translate to the stated limits as $x \to b^-$.

Structural Commentary: Cauchy's quotient identity transfers derivative bounds; endpoint limits remove or dominate the base terms.
Key Insight: Control every two-point quotient before sending one endpoint to the boundary. □

Definition 1.1.22. *Higher Derivatives*
Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. Set $f^{(0)} := f$.
For $n \in \mathbb{N}$, define $f^{(n)} := \left(f^{(n-1)}\right)'$ relative to I whenever this derivative exists on I.
The map f is *n times differentiable* on $I \iff (\forall j \in \{1, \ldots, n\})\, f^{(j)}$ exists on I.

Definition 1.1.23. *Smoothness Classes*
Let $I \subseteq \mathbb{R}$ be an interval.

(i) $C^0(I)$ is the class of continuous maps $f : I \to \mathbb{R}$.
(ii) For $r \geq 1$, $C^r(I)$ is the class of r times differentiable maps $f : I \to \mathbb{R}$ with $f^{(r)}$ continuous.
(iii) $C^\infty(I) := \bigcap_{r=0}^{\infty} C^r(I)$ is the class of *smooth* maps on I.

Theorem 1.1.24. *Higher-Order Leibniz Product Rule*
Let $a < b$, $r \in \mathbb{N}$, $f, g : (a,b) \to \mathbb{R}$ be r times differentiable. Then $(\forall x \in (a,b))\,(fg)^{(r)}(x) = \sum_{k=0}^{r} \binom{r}{k} f^{(k)}(x) g^{(r-k)}(x)$.

Proof. For $r = 1$, formula holds by Theorem 1.1.5(ii). Assume it holds at order r, f, g be $(r+1)$ times differentiable.
The differentiated identity is $(fg)^{(r+1)} = \sum_{k=0}^{r} \binom{r}{k} f^{(k+1)} g^{(r-k)} + \sum_{k=0}^{r} \binom{r}{k} f^{(k)} g^{(r-k+1)}$ by Theorem 1.1.5(ii).
After reindexing the first sum by $j = k+1$, the coefficient of $f^{(j)} g^{(r+1-j)}$ is $\binom{r}{j-1} + \binom{r}{j} = \binom{r+1}{j}$.
The endpoint coefficients at $j = 0, r+1$ are 1, so the order-$(r+1)$ formula follows.
Structural Commentary: Differentiation creates two shifted sums, and reindexing aligns their common derivative orders.
Key Insight: Pascal's identity merges the two coefficient contributions. □

Theorem 1.1.25. *Taylor's Theorem with Lagrange Remainder*
Let $a < b$, $n \in \mathbb{N}$, $f : [a,b] \to \mathbb{R}$, $f^{(n-1)}$ continuous on $[a,b]$, $\exists\, f^{(n)}$ on (a,b). For distinct $\alpha, \beta \in [a,b]$, $J_{\alpha,\beta} := (\min\{\alpha,\beta\}, \max\{\alpha,\beta\})$. Define $P(t) := \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t-\alpha)^k$. Then $(\exists x \in J_{\alpha,\beta})\, f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta-\alpha)^n$.

Proof.
(i) *Normalization*: Set $M := \frac{f(\beta)-P(\beta)}{(\beta-\alpha)^n}$ and $G(t) := f(t) - P(t) - M(t-\alpha)^n$.
(ii) *Endpoint cancellation*: For $0 \le j \le n-1$, $P^{(j)}(\alpha) = f^{(j)}(\alpha)$, so $G^{(j)}(\alpha) = 0$. Identity $G(\beta) = 0$ holds by definition of M.
(iii) *Iterated Rolle step*: Set $c_0 := \beta$ and $J(u,v) := (\min\{u,v\}, \max\{u,v\})$.
For $0 \le j < n$, $G^{(j)}$ is continuous on $[a,b]$ by the regularity hypotheses and Theorem 1.1.4.
It is differentiable on (a,b) by the regularity hypotheses. Fix $0 \le j < n$ with $G^{(j)}(c_j) = G^{(j)}(\alpha) = 0$.
Choose $c_{j+1} \in J(\alpha, c_j)$ with $G^{(j+1)}(c_{j+1}) = 0$ by Theorem 1.1.11. This recursion reaches $c_n \in J_{\alpha,\beta}$.
(iv) *Remainder identification*: For $x := c_n$, $0 = G^{(n)}(x) = f^{(n)}(x) - n!M$. Thus $M = \frac{f^{(n)}(x)}{n!}$, which is the claimed formula.
Structural Commentary: Endpoint cancellation feeds an n-step Rolle chain whose last zero identifies the normalized remainder.
Key Insight: Force the error to vanish once at β and n times at α. □

Definition 1.1.26. *Vector-Valued Derivative*
Let $k \in \mathbb{N}$, $X \subseteq \mathbb{R}$, $x_0 \in X^{\mathrm{acc}}$, $\mathbf{f} : X \to \mathbb{R}^k$. For $\mathbf{L} \in \mathbb{R}^k$ and $t \in X \setminus \{x_0\}$, set $\mathbf{q}_{\mathbf{L}}(t) := \frac{\mathbf{f}(t)-\mathbf{f}(x_0)}{t-x_0} - \mathbf{L}$.
Then $\mathbf{f}'(x_0) = \mathbf{L}$ relative to X means $\lim_{t \to x_0} \|\mathbf{q}_{\mathbf{L}}(t)\|_2 = 0$.

Proposition 1.1.27. *Componentwise Characterization of the Vector Derivative*
Let $k \in \mathbb{N}$, $X \subseteq \mathbb{R}$, $\mathbf{f} = (f_1, \dots, f_k) : X \to \mathbb{R}^k$, $x_0 \in X^{\mathrm{acc}}$. Let $\mathbf{L} = (L_1, \dots, L_k) \in \mathbb{R}^k$; all derivatives are relative to X. Then $\mathbf{f}'(x_0) = \mathbf{L} \iff (\forall i \in \{1, \dots, k\})\, f_i'(x_0) = L_i$.

Proof. For $t \in X \setminus \{x_0\}$, set $\mathbf{q}(t) := \frac{\mathbf{f}(t)-\mathbf{f}(x_0)}{t-x_0} - \mathbf{L}$.
($\Rightarrow$) If $\|\mathbf{q}(t)\|_2 \to 0$, then $(\forall i \in \{1, \dots, k\})\ |q_i(t)| \le \|\mathbf{q}(t)\|_2 \to 0$. Hence $(\forall i \in \{1, \dots, k\})\, f_i'(x_0) = L_i$ by Definition 1.1.1.
($\Leftarrow$) If $(\forall i \in \{1, \dots, k\})\, q_i(t) \to 0$, then $\|\mathbf{q}(t)\|_2 \le \sqrt{k} \max_{1 \le i \le k} |q_i(t)| \to 0$. Hence $\mathbf{f}'(x_0) = \mathbf{L}$ by Definition 1.1.26.
Structural Commentary: Coordinate bounds compare the vector quotient error with its finitely many scalar errors.
Key Insight: Finite-dimensional norm convergence is exactly coordinatewise convergence. □

Theorem 1.1.28. *Vector-Valued Mean Value Inequality*
Let $k \in \mathbb{N}$, $a < b$, $\mathbf{f} : [a,b] \to \mathbb{R}^k$ be continuous on $[a,b]$, differentiable on (a,b). Then $(\exists x \in (a,b))\ \|\mathbf{f}(b) - \mathbf{f}(a)\|_2 \le (b-a)\,\|\mathbf{f}'(x)\|_2$.

Proof. Set $\mathbf{z} := \mathbf{f}(b) - \mathbf{f}(a)$ and $\varphi(t) := \mathbf{z} \cdot \mathbf{f}(t)$. Map φ is continuous on $[a,b]$ by Theorem 5.1.32(ii),(iii).
For $t \in (a,b)$ and $s \in [a,b] \setminus \{t\}$, $\frac{\varphi(s)-\varphi(t)}{s-t} = \mathbf{z} \cdot \frac{\mathbf{f}(s)-\mathbf{f}(t)}{s-t}$. Thus $\varphi'(t) = \mathbf{z} \cdot \mathbf{f}'(t)$ by Theorem 5.1.7(iv).
Choose $x \in (a,b)$ with $\varphi(b) - \varphi(a) = (b-a)\mathbf{z} \cdot \mathbf{f}'(x)$ by Corollary 1.1.13. The identity $\varphi(b) - \varphi(a) = \|\mathbf{z}\|_2^2$ holds.
Hence $\|\mathbf{z}\|_2^2 \le (b-a)\,\|\mathbf{z}\|_2\,\|\mathbf{f}'(x)\|_2$ by Theorem 1.1.13(i). If $\mathbf{z} \neq \mathbf{0}$, cancellation proves the claim; if $\mathbf{z} = \mathbf{0}$, every $x \in (a,b)$ satisfies it.
Structural Commentary: Scalar projection creates a mean-value identity, and Cauchy–Schwarz converts it back to norms.
Key Insight: Project along the endpoint displacement itself. □

Proof Method 1.1.29. *Differentiation Proof Patterns*
(i) *Domain*: Bind $f : X \to \mathbb{R}$ and $x_0 \in X^{\mathrm{acc}}$; isolated points admit no derivative by Definition 1.1.1.
(ii) *Definition*: Factor $f(t) - f(x) = (t-x)Q_x(t)$ with $Q_x(t) \to f'(x)$.
(iii) *Approximation*: Replace the quotient limit by a first-order error bound by Proposition 1.1.3.
(iv) *Algebra*: Establish exact difference-quotient identities before passing to limits through continuity.
(v) *Mean values*: Construct h with $h(a) = h(b)$; an interior stationary point then exists by Theorem 1.1.11.
(vi) *Taylor*: Set $G = f - P - M(t-\alpha)^n$ and iterate Rolle's theorem through successive derivatives.
Key Insight: Normalize local errors or engineer endpoint cancellation before invoking earlier results.

1.2 Riemann Integration

Ordinary Riemann integration compares Darboux envelopes and approximates their common value by tagged sums. Riemann–Stieltjes integration and the fundamental theorems follow in the next section.

Definition 1.2.1. *Partitions, Darboux Sums, and the Riemann Integral*
(i) Fix $a, b \in \mathbb{R}$ with $a < b$ and $n \ge 1$. A *partition* of $[a,b]$ is $\Pi = \{x_0, \dots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$.
Set $\Delta x_i := x_i - x_{i-1}$ for $1 \le i \le n$.
(ii) For bounded $f : [a,b] \to \mathbb{R}$, set $M_i := \sup_{x \in [x_{i-1}, x_i]} f(x)$. Set $m_i := \inf_{x \in [x_{i-1}, x_i]} f(x)$.
(iii) The *upper Darboux sum* is $U(\Pi, f) := \sum_{i=1}^{n} M_i \Delta x_i$. The *lower Darboux sum* is $L(\Pi, f) := \sum_{i=1}^{n} m_i \Delta x_i$.

(iv) The *upper Darboux integral* is $\overline{\int_a^b f\,dx} := \inf_\Pi U(\Pi, f)$. The *lower Darboux integral* is $\underline{\int_a^b f\,dx} := \sup_\Pi L(\Pi, f)$.
Both extrema range over all partitions of $[a, b]$.

(v) The function f is *Riemann integrable* on $[a, b]$ iff $\overline{\int_a^b f\,dx} = \underline{\int_a^b f\,dx}$.
In this case, write $f \in \mathcal{R}[a, b]$ and let $\int_a^b f\,dx$ denote their common value.

Key Insight: Integrability means that the least upper envelope area equals the greatest lower envelope area.

Figure 156 depicts the lower and upper step envelopes from one partition.

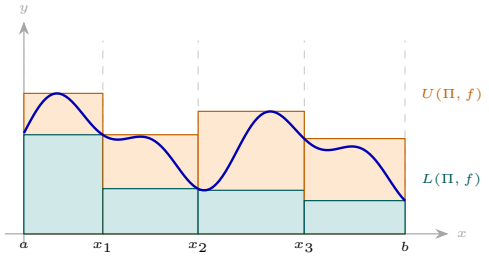

Figure 156: On each partition interval, the infimum and supremum of f determine the lower and upper rectangle heights.

Definition 1.2.2. *Tagged Partitions and Riemann Sums*
Let $\Pi = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. Set $\operatorname{mesh}(\Pi) := \max_{1 \le i \le n} \Delta x_i$.

(i) A *tagging* of Π is a tuple $c = (c_i)_{i=1}^n$ satisfying $c_i \in [x_{i-1}, x_i]$ for $1 \le i \le n$. The pair (Π, c) is a *tagged partition*.

(ii) For $f : [a, b] \to \mathbb{R}$, the associated *Riemann sum* is $R(f, \Pi, c) := \sum_{i=1}^n f(c_i)\Delta x_i$.

Figure 157 identifies the tags that determine the rectangle heights.

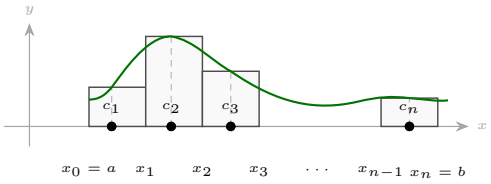

Figure 157: A tagged Riemann sum uses the sampled height $f(c_i)$ on each subinterval $[x_{i-1}, x_i]$.

Proposition 1.2.3. *Darboux and Tagged-Sum Bounds*
Let $f : [a, b] \to \mathbb{R}$ be bounded, $m, M \in \mathbb{R}$ satisfy $(\forall x \in [a, b])\, m \le f(x) \le M$. For a tagged partition (Π, c), the following hold:

(i) $m(b - a) \le L(\Pi, f) \le U(\Pi, f) \le M(b - a)$.

(ii) $L(\Pi, f) \le R(f, \Pi, c) \le U(\Pi, f)$.

(iii) $m(b - a) \le \underline{\int_a^b f\,dx} \le \int_a^b f\,dx \le M(b - a)$.

Proof.

(i) For $1 \le i \le n$, $m \le m_i \le M_i \le M$ by the definitions of m_i and M_i.
Multiplication by $\Delta x_i > 0$ preserves these inequalities. Their sum is Part (i) by $\sum_{i=1}^n \Delta x_i = b - a$.

(ii) For $1 \le i \le n$, $m_i \le f(c_i) \le M_i$ by $c_i \in [x_{i-1}, x_i]$.
Multiplication by Δx_i and summation establish $L(\Pi, f) \le R(f, \Pi, c) \le U(\Pi, f)$.

(iii) The outer bounds follow by Part (i). For the middle inequality, let Π' insert $y \in (x_{j-1}, x_j)$ into a partition Π.
Let m_j', m_j'' and M_j', M_j'' be the infima and suprema of f on the two resulting intervals.
Then $m_j', m_j'' \ge m_j$, so $m_j'(y - x_{j-1}) + m_j''(x_j - y) \ge m_j \Delta x_j$.
Also $M_j', M_j'' \le M_j$, hence $M_j'(y - x_{j-1}) + M_j''(x_j - y) \le M_j \Delta x_j$.
All other summands are unchanged; hence $L(\Pi, f) \le L(\Pi', f)$ and $U(\Pi', f) \le U(\Pi, f)$.
For arbitrary partitions Π_1, Π_2, set $\Pi^* := \Pi_1 \cup \Pi_2$. Then $L(\Pi_1, f) \le L(\Pi^*, f)$ and $U(\Pi^*, f) \le U(\Pi_2, f)$ by iteration.
Since $L(\Pi^*, f) \le U(\Pi^*, f)$, $L(\Pi_1, f) \le U(\Pi_2, f)$.
Taking the supremum over Π_1 and then the infimum over Π_2 proves the middle inequality.

Structural Commentary: Refinement compares distinct Darboux sums; subinterval extrema control tagged sums.
Key Insight: Subinterval infima and suprema trap every admissible rectangle height. □

Perspective 1.2.4. *Tagged Sums and Gauges*
Relative to Definition 1.2.2, gauges replace one uniform mesh tolerance with tag-dependent tolerances.
This comparison motivates stronger integration frameworks, but no gauge-integral result is used here.

2 Riemann-Stieltjes and Improper Integration

2.1 Riemann–Stieltjes Integrability

Definition 2.1.1. *Riemann–Stieltjes Integral*
Let $\alpha : [a,b] \to \mathbb{R}$ be increasing, and let $\Pi = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$.
Let $f : [a,b] \to \mathbb{R}$ be bounded. Set $M_i := \sup_{x \in [x_{i-1}, x_i]} f(x)$ and $m_i := \inf_{x \in [x_{i-1}, x_i]} f(x)$.

(i) The *Stieltjes increment* on $[x_{i-1}, x_i]$ is $\Delta\alpha_i := \alpha(x_i) - \alpha(x_{i-1}) \geq 0$ for $1 \leq i \leq n$.

(ii) The *upper Riemann–Stieltjes sum* is $U(\Pi, f, \alpha) := \sum_{i=1}^{n} M_i \Delta\alpha_i$.
The *lower Riemann–Stieltjes sum* is $L(\Pi, f, \alpha) := \sum_{i=1}^{n} m_i \Delta\alpha_i$.

(iii) The *upper Riemann–Stieltjes integral* is $\overline{\int_a^b f \, d\alpha} := \inf_\Pi U(\Pi, f, \alpha)$.
The *lower Riemann–Stieltjes integral* is $\underline{\int_a^b f \, d\alpha} := \sup_\Pi L(\Pi, f, \alpha)$.
Both extrema range over all partitions of $[a,b]$.

(iv) For a tagging $t = (t_i)_{i=1}^n$ of Π, the *tagged Riemann–Stieltjes sum* is $S_\alpha(f; \Pi, t) := \sum_{i=1}^{n} f(t_i) \Delta\alpha_i$.

(v) The function f is *Riemann integrable with respect to* α iff $\overline{\int_a^b f \, d\alpha} = \underline{\int_a^b f \, d\alpha}$.
In this case, write $f \in \mathcal{R}(\alpha)$ and let $\int_a^b f \, d\alpha$ denote their common value.

For $\alpha(x) = x$, these objects reduce to the ordinary Riemann objects in Definitions 1.2.1 and 1.2.2.

Definition 2.1.2. *Refinements*
Let Π, Π^*, Π_1, Π_2 be partitions of $[a,b]$.

(i) Π^* is a *refinement* of Π iff $\Pi \subseteq \Pi^*$.

(ii) The *common refinement* of Π_1 and Π_2 is $\Pi_1 \cup \Pi_2$.

Theorem 2.1.3. *Refinement Monotonicity*
Let $\alpha : [a,b] \to \mathbb{R}$ be increasing, let $f : [a,b] \to \mathbb{R}$ be bounded, and let Π^* refine Π. Then:

(i) $L(\Pi, f, \alpha) \leq L(\Pi^*, f, \alpha)$.

(ii) $U(\Pi^*, f, \alpha) \leq U(\Pi, f, \alpha)$.

Proof. Prove both claims when Π^* inserts one point $y \in (x_{j-1}, x_j)$ into Π; every refinement obtained by finitely many such insertions.
Let m_j', m_j'' and M_j', M_j'' be the infima and suprema of f on $[x_{j-1}, y]$ and $[y, x_j]$, respectively.

(i) As $m_j', m_j'' \geq m_j$, α increasing, $L(\Pi^*, f, \alpha) - L(\Pi, f, \alpha) = (m_j' - m_j)\big[\alpha(y) - \alpha(x_{j-1})\big] + (m_j'' - m_j)\big[\alpha(x_j) - \alpha(y)\big] \geq 0$

(ii) As $M_j', M_j'' \leq M_j$, α increasing, $U(\Pi, f, \alpha) - U(\Pi^*, f, \alpha) = (M_j - M_j')\big[\alpha(y) - \alpha(x_{j-1})\big] + (M_j - M_j'')\big[\alpha(x_j) - \alpha(y)\big] \geq 0$.

Structural Commentary: One-point insertion compares local weighted extrema; finite iteration handles every refinement.
Key Insight: Subdivision raises infima and lowers suprema while preserving total Stieltjes weight. □

Perspective 2.1.4. *Piecewise-Constant Envelopes*
Definition 2.1.1 retains the ordinary lower and upper step heights but replaces interval widths by Stieltjes increments.
Comparability of these weighted envelopes under subdivision holds by Theorem 2.1.3.

Theorem 2.1.5. *Lower Integral Below Upper Integral*
If $\alpha : [a,b] \to \mathbb{R}$ is increasing and $f : [a,b] \to \mathbb{R}$ is bounded, then $\underline{\int_a^b f \, d\alpha} \leq \overline{\int_a^b f \, d\alpha}$.

Proof. Let Π_1, Π_2 be partitions of $[a,b]$, and set $\Pi^* := \Pi_1 \cup \Pi_2$.
The inequalities $L(\Pi_1, f, \alpha) \leq L(\Pi^*, f, \alpha) \leq U(\Pi^*, f, \alpha) \leq U(\Pi_2, f, \alpha)$ hold by Theorem 2.1.3.
Taking the supremum over Π_1 and then the infimum over Π_2 proves the claim.
Structural Commentary: A common refinement places arbitrary lower and upper sums in one comparison chain.
Key Insight: Every lower sum lies below every upper sum. □

Theorem 2.1.6. *Upper–Lower Criterion*
Let $\alpha : [a,b] \to \mathbb{R}$ be increasing, $f : [a,b] \to \mathbb{R}$ be bounded. Then $f \in \mathcal{R}(\alpha)$ iff $(\forall \varepsilon > 0)(\exists \Pi)\ U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.
Here Π ranges over partitions of $[a,b]$.

Proof.

($\Rightarrow$) Set $I := \int_a^b f \, d\alpha$, and fix $\varepsilon > 0$. Choose Π_L with $I - \frac{\varepsilon}{2} < L(\Pi_L, f, \alpha)$ by the definition of supremum.
Choose Π_U with $U(\Pi_U, f, \alpha) < I + \frac{\varepsilon}{2}$ by the definition of infimum.
For $\Pi := \Pi_L \cup \Pi_U$, $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) \leq U(\Pi_U, f, \alpha) - L(\Pi_L, f, \alpha) < \varepsilon$ by Theorem 2.1.3.

($\Leftarrow$) Assume right-hand condition, and set $D := \overline{\int_a^b f \, d\alpha} - \underline{\int_a^b f \, d\alpha} \geq 0$ by Theorem 2.1.5. For $\varepsilon > 0$, choose corresponding partition Π.
Then $D \leq U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$ by the definitions of the Darboux integrals. Hence $D = 0$.

Structural Commentary: Near-extremal sums merge through refinement; the converse bounds the integral gap by one partition gap.
Key Insight: The Darboux gap is the exact quantitative defect of integrability. □

Theorem 2.1.7. *Tagged Sums Controlled by the Darboux Gap*
Let $\alpha : [a,b] \to \mathbb{R}$ be increasing, let $f : [a,b] \to \mathbb{R}$ be bounded, and let $\varepsilon > 0$.
Suppose that a partition $\Pi = \{x_0, \dots, x_n\}$ satisfies $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$, and let s, t be taggings of Π. Then:

(i) $\sum_{i=1}^{n} |f(s_i) - f(t_i)| \, \Delta\alpha_i < \varepsilon$.

(ii) If $f \in \mathcal{R}(\alpha)$, then $\left| S_\alpha(f; \Pi, t) - \int_a^b f \, d\alpha \right| < \varepsilon$.

Proof.

(i) As $|f(s_i) - f(t_i)| \le M_i - m_i$, $\Delta\alpha_i \ge 0$ for $1 \le i \le n$, $\sum_{i=1}^{n} |f(s_i) - f(t_i)|\,\Delta\alpha_i \le \sum_{i=1}^{n} (M_i - m_i)\Delta\alpha_i = U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$

(ii) The inequalities $L(\Pi, f, \alpha) \le S_\alpha(f; \Pi, t) \le U(\Pi, f, \alpha)$ hold by the tag conditions.
The integral lies in the same interval by the definitions of the lower and upper integrals.
Its length is less than ε, so the asserted distance is less than ε.

Structural Commentary: Intervalwise oscillation bounds control tag changes and locate the integral inside the same Darboux bracket.
Key Insight: A small Darboux gap makes every compatible tagged sum close. □

Perspective 2.1.8. *Darboux Gaps and Tagged Sums*
For $\alpha(x) = x$, Theorem 2.1.7 traps every compatible tagged Riemann sum near the integral.
For general α, the same mechanism measures oscillation with the weights $\Delta\alpha_i$.

Theorem 2.1.9. *Continuous Functions Are Riemann–Stieltjes Integrable*
If $f : [a, b] \to \mathbb{R}$ is continuous and $\alpha : [a, b] \to \mathbb{R}$ is increasing, then $f \in \mathcal{R}(\alpha)$.

Proof. The interval $[a, b]$ is compact by Theorem 4.2.5, so f is bounded by Theorem 5.1.39. Set $A := \alpha(b) - \alpha(a)$, and fix $\varepsilon > 0$.
If $A = 0$, every Darboux gap is zero, so $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.6. Assume $A > 0$ and set $\eta := \varepsilon/(2A)$.
The function f is uniformly continuous by Theorem 5.2.6. Choose $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \eta$.
Choose partition Π, $\operatorname{mesh}(\Pi) < \delta$. Extrema on every partition interval attained by Theorem 5.1.39, $M_i - m_i < \eta$ for $1 \le i \le n$.
Hence $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \sum_{i=1}^{n} (M_i - m_i)\Delta\alpha_i < \eta A < \varepsilon$. Therefore $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.

Structural Commentary: Uniform continuity bounds local oscillations; total Stieltjes weight controls the Darboux gap.
Key Insight: Uniformly small oscillations remain small after finite Stieltjes weighting. □

Theorem 2.1.10. *Monotone Functions and Continuous Integrators*
If $f : [a, b] \to \mathbb{R}$ is monotone and $\alpha : [a, b] \to \mathbb{R}$ is continuous and increasing, then $f \in \mathcal{R}(\alpha)$.

Proof. The function f is bounded. Set $V := |f(b) - f(a)|$, and fix $\varepsilon > 0$.
If $V = 0$, monotonicity makes f constant, so $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.
Assume $V > 0$. The function α is uniformly continuous by Theorem 5.2.6.
Choose $\delta > 0$ such that $|x - y| < \delta$ implies $|\alpha(x) - \alpha(y)| < \varepsilon/V$. Choose a partition Π with $\operatorname{mesh}(\Pi) < \delta$.
If f is increasing, then $M_i - m_i = f(x_i) - f(x_{i-1})$; if f is decreasing, then $M_i - m_i = f(x_{i-1}) - f(x_i)$.
In either case, $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \sum_{i=1}^{n} (M_i - m_i)\Delta\alpha_i < \frac{\varepsilon}{V} \sum_{i=1}^{n} (M_i - m_i) = \varepsilon$. Hence $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.

Structural Commentary: Uniform continuity of α limits every weight; monotonicity telescopes the total value oscillation.
Key Insight: Small Stieltjes weights distribute finite monotone variation below any gap threshold. □

Theorem 2.1.11. *Finitely Many Discontinuities*
Let $\alpha : [a, b] \to \mathbb{R}$ be increasing and let $f : [a, b] \to \mathbb{R}$ be bounded.
If $E := \{x \in [a, b] : f \text{ is discontinuous at } x\}$ is finite and α is continuous at all points of E, then $f \in \mathcal{R}(\alpha)$.

Proof. Set $A := \alpha(b) - \alpha(a)$ and $B := \sup_{x \in [a,b]} |f(x)|$. Fix $\varepsilon > 0$ and set $\eta := \varepsilon/(A + 2B + 1)$.
If $E = \emptyset$, the claim follows by Theorem 2.1.9. Assume $E \neq \emptyset$ and write $E = \{c_1, \dots, c_N\}$.

(i) *Isolation*: Choose closed intervals $I_j = [u_j, v_j] \subseteq [a, b]$ with pairwise disjoint relative interiors.
Require $c_j \in \operatorname{int}_{[a,b]} I_j$, $\alpha(v_j) - \alpha(u_j) < \frac{\eta}{N}$ for $1 \le j \le N$. These choices exist by finiteness of E, continuity of α at each c_j.

(ii) *Compact Complement*: Set $K := [a, b] \setminus \bigcup_{j=1}^{N} \operatorname{int}_{[a,b]} I_j$.
Then K is compact and contains no point of E, so $f|_K$ is uniformly continuous by Theorem 5.2.6.
Choose $\delta > 0$ such that $x, y \in K$ and $|x - y| < \delta$ imply $|f(x) - f(y)| < \eta$.

(iii) *Adapted Partition*: Choose a partition Π containing every u_j, v_j, and leave each I_j unsubdivided.
Subdivide each component of K into intervals of length less than δ.
Every partition interval is then either some I_j or is contained in K.

(iv) *Gap Estimate*: On intervals contained in K, $M_i - m_i \le \eta$; on each I_j, $M_i - m_i \le 2B$.
Hence $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) \le \eta A + 2B \sum_{j=1}^{N} \big[\alpha(v_j) - \alpha(u_j)\big] < \eta(A + 2B) < \varepsilon$. Therefore $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.

Structural Commentary: Small Stieltjes neighborhoods isolate discontinuities; uniform continuity controls the compact complement.
Key Insight: Discontinuities carry arbitrarily small weight when α is continuous there. □

Figure 158 depicts the isolation and compact-complement stages of Theorem 2.1.11.

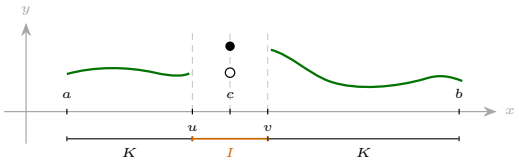

Figure 158: A short interval $I = [u, v]$ isolates a discontinuity c, while the compact complement K supports uniform control.

Example 2.1.12. *Dirichlet's Function Is Not Riemann Integrable*
Let $g = \mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$. Every nondegenerate subinterval contains rational and irrational points by Corollary 1.1.20.
Thus $M_i = 1$ and $m_i = 0$ on every partition interval. Therefore $(\forall \Pi)\ U(\Pi, g) = 1$ and $L(\Pi, g) = 0$, so $g \notin \mathcal{R}[0, 1]$.

Theorem 2.1.13. *Continuous Composition Preserves Integrability*
Let $\alpha : [a, b] \to \mathbb{R}$ be increasing, let $f \in \mathcal{R}(\alpha)$, and suppose $(\forall x \in [a, b])\ m \le f(x) \le M$.
If $\phi : [m, M] \to \mathbb{R}$ is continuous and $h := \phi \circ f$, then $h \in \mathcal{R}(\alpha)$.

Proof. The interval $[m, M]$ is compact by Theorem 4.2.5, so $C := \sup_{y\in[m,M]} |\phi(y)|$ is finite by Theorem 5.1.39.
Set $A := \alpha(b) - \alpha(a)$, fix $\varepsilon > 0$, and set $\eta := \varepsilon/(A + 2C + 1)$.

(i) *Uniform Value Control*: The function ϕ is uniformly continuous by Theorem 5.2.6.
Choose $\gamma > 0$ such that $|s - t| < \gamma$ implies $|\phi(s) - \phi(t)| < \eta$ for $s, t \in [m, M]$.

(ii) *Oscillation Split*: Choose a partition Π satisfying $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \gamma\eta$ by Theorem 2.1.6.
Let M_i^h, m_i^h be the suprema and infima of h on $[x_{i-1}, x_i]$, and set $I := \{i : M_i - m_i < \gamma\}$ and $J := \{i : M_i - m_i \geq \gamma\}$.

(iii) *Weight Control*: For $i \in I$, $M_i^h - m_i^h \leq \eta$ by uniform continuity; for $i \in J$, $M_i^h - m_i^h \leq 2C$.
The large-oscillation indices satisfy $\gamma \sum_{i\in J} \Delta\alpha_i \leq \sum_{i\in J} (M_i - m_i)\Delta\alpha_i < \gamma\eta$, so $\sum_{i\in J} \Delta\alpha_i < \eta$.

(iv) *Gap Estimate*: Darboux gap $U(\Pi, h, \alpha) - L(\Pi, h, \alpha) \leq \eta \sum_{i\in I} \Delta\alpha_i + 2C \sum_{i\in J} \Delta\alpha_i < \eta(A + 2C) < \varepsilon$; $h \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.

Structural Commentary: An oscillation split converts uniform continuity of ϕ into a weighted Darboux-gap estimate.
Key Insight: Intervals with large f-oscillation have small total Stieltjes weight. □

Theorem 2.1.14. *Basic Properties of the Integral*
Let $\alpha, \alpha_1, \alpha_2 : [a, b] \to \mathbb{R}$ be increasing, and let $f, g : [a, b] \to \mathbb{R}$ be bounded.

(i) Linearity in the integrand: If $f, g \in \mathcal{R}(\alpha)$, $c \in \mathbb{R}$, then $f + g, cf \in \mathcal{R}(\alpha)$; $\int_a^b (f + g)\, d\alpha = \int_a^b f\, d\alpha + \int_a^b g\, d\alpha$; $\int_a^b cf\, d\alpha = c\int_a^b f\, d\alpha$.

(ii) Order: Assume $f, g \in \mathcal{R}(\alpha)$ and $(\forall x \in [a, b])\, f(x) \leq g(x)$. Then $\int_a^b f\, d\alpha \leq \int_a^b g\, d\alpha$.

(iii) Interval additivity: Assume $f \in \mathcal{R}(\alpha)$, $d \in (a, b)$. Then $f|_{[a,d]} \in \mathcal{R}(\alpha|_{[a,d]})$ and $f|_{[d,b]} \in \mathcal{R}(\alpha|_{[d,b]})$; $\int_a^d f\, d\alpha + \int_d^b f\, d\alpha = \int_a^b f\, d\alpha$.

(iv) Uniform bound: Assume $f \in \mathcal{R}(\alpha)$ and $M \in [0, \infty)$. If $(\forall x \in [a, b])\, |f(x)| \leq M$, then $\left|\int_a^b f\, d\alpha\right| \leq M[\alpha(b) - \alpha(a)]$.

(v) Linearity in the integrator: If $f \in \mathcal{R}(\alpha_1) \cap \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$. In this case, $\int_a^b f\, d(\alpha_1 + \alpha_2) = \int_a^b f\, d\alpha_1 + \int_a^b f\, d\alpha_2$.
If $\lambda > 0$ and $f \in \mathcal{R}(\alpha)$, then $f \in \mathcal{R}(\lambda\alpha)$ and $\int_a^b f\, d(\lambda\alpha) = \lambda \int_a^b f\, d\alpha$.

Proof. For a partition Π, set $G_\beta^\Pi(h) := U(\Pi, h, \beta) - L(\Pi, h, \beta)$.
Finitely many integrable pairs admit common partitions with arbitrarily small gaps by Theorems 2.1.6 and 2.1.3.
An exact identity among their tagged sums passes to their integrals by Theorem 2.1.7(ii) and the triangle inequality.

(i) *Linearity in the integrand*: $(\forall\Pi)\, G_\alpha^\Pi(f + g) \leq G_\alpha^\Pi(f) + G_\alpha^\Pi(g)$. $(\forall\Pi)\, G_\alpha^\Pi(cf) = |c|\, G_\alpha^\Pi(f)$; $f + g, cf \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.
The identities $S_\alpha(f + g; \Pi, t) = S_\alpha(f; \Pi, t) + S_\alpha(g; \Pi, t)$ and $S_\alpha(cf; \Pi, t) = cS_\alpha(f; \Pi, t)$ pass to the asserted integral identities.

(ii) *Order*: Pointwise order implies $(\forall\Pi)\, L(\Pi, f, \alpha) \leq L(\Pi, g, \alpha)$. Taking suprema proves $\int_a^b f\, d\alpha \leq \int_a^b g\, d\alpha$ by Definition 2.1.1.

(iii) *Interval additivity*: For $\varepsilon > 0$, choose $\Pi \ni d$ with $G_\alpha^\Pi(f) < \varepsilon$ by the common-refinement observation.
Set $\Pi^- := \Pi \cap [a, d]$, $\Pi^+ := \Pi \cap [d, b]$. Then $G_\alpha^\Pi(f) = G_\alpha^{\Pi^-}(f) + G_\alpha^{\Pi^+}(f)$, both restrictions integrable by Theorem 2.1.6.
The tagged sum over Π is the sum of its restrictions to Π^- and Π^+; the integral identity follows by the preceding observation.

(iv) *Uniform bound*: The constant function q has integral $q[\alpha(b) - \alpha(a)]$ by Definition 2.1.1.
The asserted bound follows by part (ii) with $-M \leq f \leq M$.

(v) *Linearity in the integrator*: $(\forall\Pi)\, G_{\alpha_1+\alpha_2}^\Pi(f) = G_{\alpha_1}^\Pi(f) + G_{\alpha_2}^\Pi(f)$. $(\forall\Pi)\, G_{\lambda\alpha}^\Pi(f) = \lambda G_\alpha^\Pi(f)$.
The claimed integrability follows by common refinement and Theorem 2.1.6.
The corresponding tagged-sum identities pass to the asserted integral identities by the preceding observation.

Structural Commentary: Gap identities establish integrability; tagged-sum identities then transfer to integral values.
Key Insight: Exact finite-sum algebra survives arbitrarily small Darboux gaps. □

Theorem 2.1.15. *Products and Absolute Values*
Let $\alpha : [a, b] \to \mathbb{R}$ be increasing and let $f, g \in \mathcal{R}(\alpha)$. Then:

(i) $fg \in \mathcal{R}(\alpha)$.
(ii) $|f| \in \mathcal{R}(\alpha)$.
(iii) $\left|\int_a^b f\, d\alpha\right| \leq \int_a^b |f|\, d\alpha$.

Proof. The functions f, g are bounded by Definition 2.1.1; hence $f \pm g$ are bounded.

(i) The functions $f \pm g$ belong to $\mathcal{R}(\alpha)$ by Theorem 2.1.14(i).
Choose a compact interval containing both ranges. Then $(f \pm g)^2 \in \mathcal{R}(\alpha)$ by Theorem 2.1.13 with $\phi(t) = t^2$.
Hence $fg \in \mathcal{R}(\alpha)$ by $4fg = (f + g)^2 - (f - g)^2$ and Theorem 2.1.14(i).

(ii) Choose a compact interval containing $f([a, b])$. Then $|f| \in \mathcal{R}(\alpha)$ by Theorem 2.1.13 with $\phi(t) = |t|$.

(iii) Bounds $-\int_a^b |f|\, d\alpha \leq \int_a^b f\, d\alpha \leq \int_a^b |f|\, d\alpha$ hold by Theorem 2.1.14(i)–(ii), $-|f| \leq f \leq |f|$. Equivalent to claim.

Structural Commentary: Continuous composition handles squares and absolute values; linearity and order complete the reductions.
Key Insight: Products reduce to squares, while $-|f| \leq f \leq |f|$ controls the integral. □

2.2 Fundamental Theorems of Calculus

Theorem 2.2.1. *Integral Functions*
Let $f \in \mathcal{R}[a, b]$, and define $F : [a, b] \to \mathbb{R}$ by $F(a) := 0$ and $F(x) := \int_a^x f(t)\, dt$ for $a < x \leq b$. Then:

(i) F is continuous on $[a, b]$.
(ii) If f is continuous at $x_0 \in (a, b)$, then $F'(x_0) = f(x_0)$.
At the endpoints, continuity of f implies $F'_+(a) = f(a)$ and $F'_-(b) = f(b)$.

Proof. All integral identities and bounds below hold by Theorem 2.1.14(i),(iii)–(iv) with $\alpha(t) = t$.

(i) Choose $M \geq 0$ such that $(\forall t \in [a, b])\, |f(t)| \leq M$. For $a \leq x < y \leq b$, $|F(y) - F(x)| = \left|\int_x^y f(t)\, dt\right| \leq M(y - x)$.
Thus F is Lipschitz and therefore continuous on $[a, b]$.

(ii) Fix a continuity point $x_0 \in [a, b]$.
For $x \in [a, b] \setminus \{x_0\}$, set $J_x := [\min\{x, x_0\}, \max\{x, x_0\}]$.
Then $\left|\frac{F(x)-F(x_0)}{x-x_0} - f(x_0)\right| \leq \sup_{u\in J_x} |f(u) - f(x_0)|$.
For $\varepsilon > 0$, choose $\delta > 0$ such that $(\forall u \in [a, b])\, |u - x_0| < \delta \implies |f(u) - f(x_0)| < \varepsilon$.

If $0 < |x - x_0| < \delta$, then $J_x \subseteq (x_0 - \delta, x_0 + \delta)$, so the right-hand side is at most ε.
Thus the difference quotient tends to $f(x_0)$ two-sidedly in (a, b), right-sidedly at a, and left-sidedly at b.

Structural Commentary: A global Lipschitz bound proves continuity; local average bounds identify the derivative.
Key Insight: Difference quotients of F are averages over shrinking intervals. □

Theorem 2.2.2. *Fundamental Theorem of Calculus*
Let $f \in \mathcal{R}[a, b]$, and let $F : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b).
If $(\forall x \in (a, b))\, F'(x) = f(x)$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

Proof. Fix $\varepsilon > 0$. Choose a partition $\Pi = \{x_0, \ldots, x_n\}$ with $U(\Pi, f) - L(\Pi, f) < \varepsilon$ by Theorem 2.1.6 with $\alpha(x) = x$.
For $1 \le i \le n$, choose $t_i \in (x_{i-1}, x_i)$ satisfying $F(x_i) - F(x_{i-1}) = f(t_i)\Delta x_i$ by Corollary 1.1.13.
Then $F(b) - F(a) = \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})] = \sum_{i=1}^{n} f(t_i)\Delta x_i$. Thus $\left|F(b) - F(a) - \int_a^b f(x)\,dx\right| < \varepsilon$ by Theorem 2.1.7(ii) with $\alpha(x) = x$.
Since $\varepsilon > 0$ is arbitrary, the claimed identity holds.
Structural Commentary: Mean-value increments form a tagged sum; the Darboux gap identifies its integral limit.
Key Insight: Antiderivative increments are exact tagged rectangles. □

Theorem 2.2.3. *Integration by Parts*
Let $F, G : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Suppose $f, g \in \mathcal{R}[a, b]$ satisfy $F' = f$ and $G' = g$ on (a, b).
Then $\int_a^b F(x)g(x)\,dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x)\,dx$.

Proof. The functions F, G belong to $\mathcal{R}[a, b]$ by Theorem 2.1.9 with $\alpha(x) = x$. Hence $Fg, fG \in \mathcal{R}[a, b]$ by Theorem 2.1.15(i).
Set $H := FG$ and $h := fG + Fg$. The function H is continuous on $[a, b]$ by Theorem 5.1.32(i).
$H' = h$ on (a, b) by Theorem 1.1.5(ii), $h \in \mathcal{R}[a, b]$ by Theorem 2.1.14(i). Identity $H(b) - H(a) = \int_a^b h(x)\,dx$ holds by Theorem 2.2.2.
The claim follows by $H = FG$, $h = fG + Fg$, Theorem 2.1.14(i), and rearrangement.
Structural Commentary: The product rule creates an integrable derivative; the fundamental theorem supplies its boundary increment.
Key Insight: $H' = fG + Fg$ packages both integration-by-parts terms. □

2.3 Step Integrators and Change of Variable

Definition 2.3.1. *Unit Step Function*
The *unit step function* is $I : \mathbb{R} \to \mathbb{R}$ with $I(x) := 0$ for $x \le 0$ and $I(x) := 1$ for $x > 0$.

Theorem 2.3.2. *Single Jump Integrator*
Let $a < s < b$, let $f : [a, b] \to \mathbb{R}$ be bounded and continuous at s, and set $\alpha(x) := I(x - s)$. Then:
(i) $f \in \mathcal{R}(\alpha)$.
(ii) $\int_a^b f\,d\alpha = f(s)$.

Proof. Fix $\varepsilon > 0$. Choose $\delta > 0$, $(\forall x \in [a, b])\ |x - s| < \delta \implies |f(x) - f(s)| < \frac{\varepsilon}{2}$. Choose $x_2 \in (s, \min\{b, s + \delta\})$, $\Pi := \{a, s, x_2, b\}$.
The sole nonzero Stieltjes increment is the unit increment on $[s, x_2]$. Hence $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \sup_{[s,x_2]} f - \inf_{[s,x_2]} f < \varepsilon$.

(i) The function f belongs to $\mathcal{R}(\alpha)$ by Theorem 2.1.6.
(ii) Choose tagging t whose tag on $[s, x_2]$ is s. Then $S_\alpha(f; \Pi, t) = f(s)$; $\left|\int_a^b f\,d\alpha - f(s)\right| < \varepsilon$ by Theorem 2.1.7(ii), $\varepsilon > 0$. QED.

Structural Commentary: The unit jump reduces Darboux and tagged sums to the interval containing s.
Key Insight: Only the jump interval carries Stieltjes mass. □

Remark 2.3.3. *Stieltjes Increments as α-Lengths*
Relative to Definition 2.1.1, the increments $\Delta\alpha_i$ are interval lengths measured by α.
For $\alpha(x) = x$, they are ordinary lengths; at a jump, they concentrate weight at one point as in Theorem 2.3.2.

Theorem 2.3.4. *Series from Pure Step Integrators*
Let $(c_n)_{n\ge1} \subseteq [0, \infty)$ satisfy $\sum_{n=1}^{\infty} c_n < \infty$, $(s_n)_{n\ge1}$ be distinct points of (a, b), $\alpha(x) := \sum_{n=1}^{\infty} c_n I(x - s_n)$, $f : [a, b] \to \mathbb{R}$ continuous.
(i) The series defining α converges on $[a, b]$, and α is increasing.
(ii) $f \in \mathcal{R}(\alpha)$.
(iii) $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$.

Proof.

(i) If $p > q$, $(\forall x \in [a, b])\ 0 \le \sum_{n=q+1}^{p} c_n I(x - s_n) \le \sum_{n=q+1}^{p} c_n$. Scalar tail tends to zero, so defining series converges uniformly on $[a, b]$.
If $x \le y$, then $(\forall n \ge 1)\ I(x - s_n) \le I(y - s_n)$; hence $\alpha(x) \le \alpha(y)$.
(ii) The claim follows by Theorem 2.1.9.
(iii) For $N \ge 1$, set $\alpha_N(x) := \sum_{n=1}^{N} c_n I(x - s_n)$ and $\beta_N := \alpha - \alpha_N$.
Both are increasing by $c_n \ge 0$ and monotonicity of I, and $f \in \mathcal{R}(\beta_N)$ by Theorem 2.1.9.
The finite identity $\int_a^b f\,d\alpha_N = \sum_{n=1}^{N} c_n f(s_n)$ holds by Theorems 2.3.2 and 2.1.14(v).
Terms with $c_n = 0$ vanish by the Darboux sums. Set $M := \sup_{x\in[a,b]} |f(x)|$.
The identity $\left|\int_a^b f\,d\alpha - \sum_{n=1}^{N} c_n f(s_n)\right| = \left|\int_a^b f\,d\beta_N\right|$ holds by Theorem 2.1.14(v). Since $a < s_n < b$, $\beta_N(b) - \beta_N(a) = \sum_{n>N} c_n$.
Hence $\left|\int_a^b f\,d\beta_N\right| \le M \sum_{n>N} c_n \to 0$ by Theorem 2.1.14(iv); $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$.

Structural Commentary: Finite jump identities determine partial sums; the uniform bound controls the summable tail.
Key Insight: Each jump contributes $c_n f(s_n)$. □

Figure 159 depicts the first three jumps of the pure-step integrator.
The jump c_i at s_i corresponds to the summand $c_i f(s_i)$ in Theorem 2.3.4(iii).

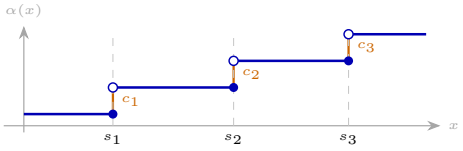

Figure 159: A pure-step integrator has jump c_i at s_i; closed lower and open upper markers record $I(0) = 0$.

Theorem 2.3.5. *Differentiable Integrators Reduce to Riemann Integrals*
Let $\alpha : [a,b] \to \mathbb{R}$ be continuous and increasing on $[a,b]$ and differentiable on (a,b).
Let $q \in \mathcal{R}[a,b]$ satisfy $q|_{(a,b)} = \alpha'$, and let $f : [a,b] \to \mathbb{R}$ be bounded. Then:

(i) $f \in \mathcal{R}(\alpha) \iff fq \in \mathcal{R}[a,b]$.
(ii) Under these conditions, $\int_a^b f\,d\alpha = \int_a^b f(x)q(x)\,dx$.

Proof. Set $M := \sup_{x\in[a,b]} |f(x)|$. For partition Π, bounded h, set $D_\alpha(\Pi,h) := U(\Pi,h,\alpha) - L(\Pi,h,\alpha)$, $D(\Pi,h) := U(\Pi,h) - L(\Pi,h)$.
Set $E(\Pi) := MD(\Pi,q)$. For tagging t, choose $u_i \in (x_{i-1},x_i)$ with $\Delta\alpha_i = q(u_i)\Delta x_i$ for $1 \le i \le n$ by Corollary 1.1.13.
Then $|S_\alpha(f;\Pi,t) - R(fq,\Pi,t)| \le E(\Pi)$ by the intervalwise oscillation of q.
Hence $|D_\alpha(\Pi,f) - D(\Pi,fq)| \le 2E(\Pi)$ by taking suprema over pairs of taggings.
The gaps $D_\alpha(\Pi,f)$ and $D(\Pi,fq)$ decrease under refinement by Theorem 2.1.3.
The quantity $E(\Pi) = MD(\Pi,q)$ decreases by the same theorem with integrator x.
The relation $(\forall\delta > 0)(\exists\Pi)\, E(\Pi) < \delta$ holds by Theorem 2.1.6 with integrator x.

(i) ($\Rightarrow$) Fix $\varepsilon > 0$. Choose Π with $\max\{2D_\alpha(\Pi,f), 4E(\Pi)\} < \varepsilon$ by the integrability criterion and refinement.
Then $D(\Pi,fq) \le D_\alpha(\Pi,f) + 2E(\Pi) < \varepsilon$, so $fq \in \mathcal{R}[a,b]$ by Theorem 2.1.6 with integrator x.
(i) ($\Leftarrow$) Fix $\varepsilon > 0$. Choose Π with $\max\{2D(\Pi,fq), 4E(\Pi)\} < \varepsilon$ by the integrability criterion and refinement.
Then $D_\alpha(\Pi,f) \le D(\Pi,fq) + 2E(\Pi) < \varepsilon$, so $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.
(ii) Assume the equivalent conditions, fix $\varepsilon > 0$, and set $I_\alpha := \int_a^b f\,d\alpha$ and $I_q := \int_a^b fq\,dx$.
Choose Π with $\max\{3D_\alpha(\Pi,f), 3D(\Pi,fq), 3E(\Pi)\} < \varepsilon$ by the integrability criterion and refinement.
For any tagging t, each tagged sum is within $\frac{\varepsilon}{3}$ of its integral by Theorem 2.1.7(ii).
Hence $|I_\alpha - I_q| < \frac{2\varepsilon}{3} + E(\Pi) < \varepsilon$ by the tagged-sum comparison. Thus $I_\alpha = I_q$.

Structural Commentary: Mean-value points compare tagged sums; derivative gaps transfer integrability and identify limits.
Key Insight: The sole comparison error is the Darboux gap of q. □

Theorem 2.3.6. *Change of Variable*
Let $\varphi : [A,B] \to [a,b]$ be strictly increasing, continuous, and onto. Let $\alpha : [a,b] \to \mathbb{R}$ be increasing and $f \in \mathcal{R}(\alpha)$.
Set $\beta := \alpha \circ \varphi$ and $g := f \circ \varphi$. Then:

(i) β *is increasing on* $[A,B]$.
(ii) $g \in \mathcal{R}(\beta)$.
(iii) $\int_A^B g\,d\beta = \int_a^b f\,d\alpha$.
(iv) Suppose $\alpha = \mathrm{id}_{[a,b]}$ *and* φ *is differentiable on* (A,B). *If* $q \in \mathcal{R}[A,B]$ *satisfies* $q|_{(A,B)} = \varphi'$, *then* $gq \in \mathcal{R}[A,B]$.
(v) Under the hypotheses of Part (iv), $\int_a^b f(x)\,dx = \int_A^B f(\varphi(y))q(y)\,dy$.

Proof.

(i) The composition $\beta = \alpha \circ \varphi$ is increasing by the monotonicity of α and φ.
(ii) Fix $\varepsilon > 0$. Choose partition $\Pi = \{x_0,\dots,x_n\}$ with $U(\Pi,f,\alpha) - L(\Pi,f,\alpha) < \varepsilon$ by Theorem 2.1.6. Set $Q := \{\varphi^{-1}(x_0),\dots,\varphi^{-1}(x_n)\}$.
The map φ sends each interval of Q onto the corresponding interval of Π. Hence g and f have the same range there and $\Delta\beta_i = \Delta\alpha_i$.
Hence $U(Q,g,\beta) = U(\Pi,f,\alpha)$ and $L(Q,g,\beta) = L(\Pi,f,\alpha)$. Thus $g \in \mathcal{R}(\beta)$ by Theorem 2.1.6.
(iii) Upper-sum, lower-sum families coincide by bijection $\Pi \mapsto \varphi^{-1}(\Pi)$ from Part (ii). Hence $\int_A^B g\,d\beta = \int_a^b f\,d\alpha$ by Part (ii), $f \in \mathcal{R}(\alpha)$.
(iv) Here $\beta = \varphi$ and $g \in \mathcal{R}(\varphi)$ by Part (ii). Hence $gq \in \mathcal{R}[A,B]$ by Theorem 2.3.5(i).
(v) The equality $\int_a^b f(x)\,dx = \int_A^B g\,d\varphi = \int_A^B f(\varphi(y))q(y)\,dy$ holds by Part (iii) and Theorem 2.3.5(ii).

Structural Commentary: Pullback identifies Darboux sums; derivative reduction replaces $d\varphi$ by $q\,dy$.
Key Insight: Corresponding intervals have identical function ranges and integrator increments. □

2.4 Improper Integrals

Definition 2.4.1. *Improper Riemann Integrals*
Let $a,b \in \overline{\mathbb{R}}$ satisfy $a < b$, and let $f : (a,b) \to \mathbb{R}$. Suppose $(\forall c,d \in \mathbb{R})\,[a < c < d < b \Rightarrow f|_{[c,d]} \in \mathcal{R}[c,d]]$.
Fix $\xi \in (a,b)$. A one-sided integral below is defined whenever its limit exists in $\overline{\mathbb{R}}$.

(i) $\int_\xi^b f(x)\,dx := \lim_{d\uparrow b} \int_\xi^d f(x)\,dx$, and $\int_a^\xi f(x)\,dx := \lim_{c\downarrow a} \int_c^\xi f(x)\,dx$.
(ii) If their sum is defined in $\overline{\mathbb{R}}$, set $\int_a^b f(x)\,dx := \int_a^\xi f(x)\,dx + \int_\xi^b f(x)\,dx$.
This value is independent of ξ by Theorem 2.1.14(iii) with the identity integrator.
(iii) The improper integral *converges* iff both one-sided integrals lie in $\mathbb{R}$; it *diverges* otherwise.

Figure 160 depicts the two right-endpoint truncations in Definition 2.4.1.
The left panel has $b \in \mathbb{R}$, while the right panel has $b = +\infty$.

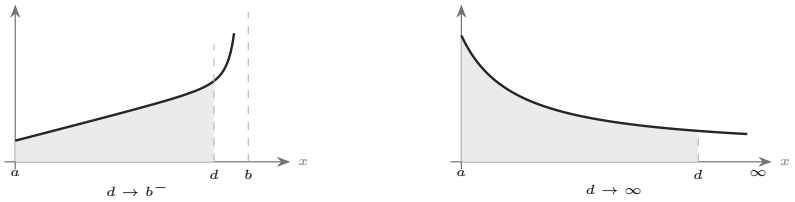

Figure 160: Compact truncations approach a finite singular endpoint on the left and $+\infty$ on the right.

Theorem 2.4.2. *Nonnegative Improper Integrals*
Let $a \in \mathbb{R}$, $b \in \overline{\mathbb{R}}$, $a < b$, and let $f : [a, b) \to [0, +\infty)$ satisfy $(\forall d \in (a, b))\, f|_{[a,d]} \in \mathcal{R}[a, d]$.
Set $H(d) := \int_a^d f(x)\, dx$ for $d \in (a, b)$ and $S := \sup_{\overline{\mathbb{R}}} H((a, b))$. Then:

(i) $\int_a^b f(x)\, dx = \lim_{d \uparrow b} H(d) = S \in [0, +\infty]$.

(ii) The integral $\int_a^b f(x)\, dx$ converges iff $S \in \mathbb{R}$.

Proof.

(*i*) Inequalities $0 \le H(d_1) \le H(d_2)$ hold for $a < d_1 < d_2 < b$ by Theorem 2.1.14(ii)–(iii) with $\alpha(t) = t$; H increasing, $S \in [0, +\infty]$.
If $S \in \mathbb{R}$, then $(\forall \varepsilon > 0)(\exists d_0 \in (a, b))(\forall d \in [d_0, b))\, S - \varepsilon < H(d) \le S$ by the definition of S and monotonicity; hence $H(d) \to S$.
If $S = +\infty$, then $(\forall K \in \mathbb{R})(\exists d_0 \in (a, b))(\forall d \in [d_0, b))\, H(d) > K$ by the same properties; hence $H(d) \to +\infty$.
Fix $\xi \in (a, b)$. The limit $\lim_{c \downarrow a} \int_c^\xi f(x)\, dx = H(\xi)$ holds by Theorems 2.2.1(i) and 2.1.14(iii).
For $\xi < d < b$, $\int_\xi^d f(x)\, dx = H(d) - H(\xi)$ by Theorem 2.1.14(iii) with $\alpha(t) = t$.
Thus $\int_\xi^b f(x)\, dx = S - H(\xi)$ by Definition 2.4.1(i) and Theorem 5.3.11(ii); $\int_a^b f(x)\, dx = H(\xi) + [S - H(\xi)] = S$ by Definition 2.4.1(ii).
(*ii*) ($\Rightarrow$) If the integral converges, then $S \in \mathbb{R}$ by Part (i) and Definition 2.4.1(iii).
(*ii*) ($\Leftarrow$) If $S \in \mathbb{R}$, then the integral converges by Part (i) and Definition 2.4.1(iii).

Structural Commentary: Monotonicity fixes the extended limit; additivity transfers it to the improper integral.
Key Insight: Bounded truncation areas are exactly finite improper mass. □

Perspective 2.4.3. *Principal Value Is Different*
Definition 2.4.1 requires separate one-sided limits on $(-\infty, +\infty)$. For $A > 0$, $\int_{-A}^A \sin x\, dx = 0$, so the Cauchy principal value is 0.
However, $\int_0^d \sin x\, dx = 1 - \cos d$ has no limit as $d \to +\infty$, so the improper integral diverges.

Definition 2.4.4. *Improper Riemann–Stieltjes Integrals on $\mathbb{R}$*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing. Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $(\forall a, b \in \mathbb{R})\, [a < b \Rightarrow f|_{[a,b]} \in \mathcal{R}(\alpha|_{[a,b]})]$.

(i) Set $\alpha(-\infty) := \inf_{\overline{\mathbb{R}}} \alpha(\mathbb{R})$ and $\alpha(+\infty) := \sup_{\overline{\mathbb{R}}} \alpha(\mathbb{R})$.

(ii) When the limits exist in $\overline{\mathbb{R}}$, set $I_-(f, \alpha) := \lim_{a \to -\infty} \int_a^0 f\, d\alpha$. Under the same condition, set $I_+(f, \alpha) := \lim_{b \to +\infty} \int_0^b f\, d\alpha$.

(iii) If $I_-(f, \alpha) + I_+(f, \alpha)$ is defined in $\overline{\mathbb{R}}$, set $\int_{-\infty}^{+\infty} f\, d\alpha := I_-(f, \alpha) + I_+(f, \alpha)$.

(iv) The function f is *α-integrable on $\mathbb{R}$* iff $I_-(f, \alpha), I_+(f, \alpha) \in \mathbb{R}$.

Theorem 2.4.5. *Nonnegative and Bounded-Mass Stieltjes Integrals*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing. Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $(\forall a, b \in \mathbb{R})\, [a < b \Rightarrow f|_{[a,b]} \in \mathcal{R}(\alpha|_{[a,b]})]$.
Set $S_+ := \sup_{\overline{\mathbb{R}}} \{\int_0^b f\, d\alpha : b > 0\}$ and $S_- := \sup_{\overline{\mathbb{R}}} \{\int_a^0 f\, d\alpha : a < 0\}$.
For Parts (i)–(iii), assume $(\forall x \in \mathbb{R})\, f(x) \ge 0$; Part (iv) has no sign hypothesis. Then:

(i) $I_-(f, \alpha) = S_-$ and $I_+(f, \alpha) = S_+$ belong to $[0, +\infty]$.
(ii) $\int_{-\infty}^{+\infty} f\, d\alpha = S_- + S_+ \in [0, +\infty]$.
(iii) The function f is α-integrable on $\mathbb{R}$ iff $S_-, S_+ \in \mathbb{R}$.
(iv) If $\alpha(-\infty), \alpha(+\infty) \in \mathbb{R}$ and f is bounded, then f is α-integrable on $\mathbb{R}$.

Proof.

(*i*) Set $H_+(t) := \int_0^t f\, d\alpha$ and $H_-(t) := \int_{-t}^0 f\, d\alpha$ for $t > 0$. For $0 < s < t$, $0 \le H_\pm(s) \le H_\pm(t)$ by Theorem 2.1.14(ii)–(iii).
Fix $\sigma \in \{-, +\}$. If $S_\sigma \in \mathbb{R}$, then $(\forall \varepsilon > 0)(\exists t_0 > 0)(\forall t \ge t_0)\, S_\sigma - \varepsilon < H_\sigma(t) \le S_\sigma$ by the supremum property and monotonicity.
If $S_\sigma = +\infty$, then $(\forall K \in \mathbb{R})(\exists t_0 > 0)(\forall t \ge t_0)\, H_\sigma(t) > K$ by the same properties.
Thus $H_\sigma(t) \to S_\sigma$, and $I_\pm(f, \alpha) = S_\pm$ by Definition 2.4.4(ii), with $a = -t$ for $\sigma = -$.
(*ii*) The identity $\int_{-\infty}^{+\infty} f\, d\alpha = S_- + S_+$ holds by Part (i) and Definition 2.4.4(iii).
(*iii*) ($\Rightarrow$) If f is α-integrable, then $S_-, S_+ \in \mathbb{R}$ by Part (i) and Definition 2.4.4(iv).
(*iii*) ($\Leftarrow$) If $S_-, S_+ \in \mathbb{R}$, then f is α-integrable by Part (i) and Definition 2.4.4(iv).
(*iv*) Choose $M > 0$ with $(\forall x \in \mathbb{R})\, |f(x)| \le M$, and set $g_\pm := M \pm f$.
Each $g_\pm$ is locally Riemann–Stieltjes integrable by Definition 2.1.1 and Theorem 2.1.14(i).
The bounds $(\forall x \in \mathbb{R})\, 0 \le g_\pm(x) \le 2M$ hold by the choice of M.
For $g \in \{g_-, g_+\}$, $0 \le I_+(g, \alpha) \le 2M[\alpha(+\infty) - \alpha(0)] \in \mathbb{R}$ by Part (i) and Theorem 2.1.14(iv).
For $g \in \{g_-, g_+\}$, $0 \le I_-(g, \alpha) \le 2M[\alpha(0) - \alpha(-\infty)] \in \mathbb{R}$ by the same part and theorem.
Thus $I_\pm(f, \alpha) = \frac{1}{2}[I_\pm(g_+, \alpha) - I_\pm(g_-, \alpha)] \in \mathbb{R}$ by Theorems 2.1.14(i), 5.3.11(ii)–(iii); f α-integrable by Definition 2.4.4(iv).

Structural Commentary: Suprema identify nonnegative limits; the shifts $M \pm f$ reduce bounded integrands to them.
Key Insight: Finite integrator mass bounds both positive shifts uniformly. □

2.5 Vector-Valued Integrals and Curves

Definition 2.5.1. *Vector-Valued Riemann–Stieltjes Integral*
Let $a, b \in \mathbb{R}$ satisfy $a < b$, $k \in \mathbb{N}$, $\alpha : [a,b] \to \mathbb{R}$ increasing. For $\mathbf{f} = (f_1, \dots, f_k) : [a,b] \to \mathbb{R}^k$, $\mathbf{f} \in \mathcal{R}(\alpha)$ iff $(\forall j \in \{1, \dots, k\})\, f_j \in \mathcal{R}(\alpha)$.
In that case, set $\int_a^b \mathbf{f}\, d\alpha := \left(\int_a^b f_1\, d\alpha, \dots, \int_a^b f_k\, d\alpha\right)$. For the identity integrator $t \mapsto t$, write $\mathbf{f} \in \mathcal{R}[a,b]$ and $\int_a^b \mathbf{f}(t)\, dt$.

Theorem 2.5.2. *Vector-Valued Integral Results*
Let $a, b \in \mathbb{R}$ satisfy $a < b$, $k \in \mathbb{N}$. For Parts (ii)–(iii), let $\alpha : [a,b] \to \mathbb{R}$ be increasing and let $\mathbf{f} \in \mathcal{R}(\alpha)$. Then:

(i) Let $\mathbf{g}, \mathbf{F} : [a,b] \to \mathbb{R}^k$, where $\mathbf{g} \in \mathcal{R}[a,b]$ and $\mathbf{F}$ is continuous on $[a,b]$ and differentiable on (a,b).
If $(\forall t \in (a,b))\, \mathbf{F}'(t) = \mathbf{g}(t)$, then $\int_a^b \mathbf{g}(t)\, dt = \mathbf{F}(b) - \mathbf{F}(a)$.
(ii) The map $t \mapsto \|\mathbf{f}(t)\|$ belongs to $\mathcal{R}(\alpha)$.
(iii) $\left\|\int_a^b \mathbf{f}\, d\alpha\right\| \le \int_a^b \|\mathbf{f}(t)\|\, d\alpha$.

Proof.

(i) Write $\mathbf{g} = (g_1, \dots, g_k)$ and $\mathbf{F} = (F_1, \dots, F_k)$. By Definition 2.5.1, $(\forall j \in \{1, \dots, k\})\, g_j \in \mathcal{R}[a,b]$.
Maps $F_1, \dots, F_k$ continuous on $[a,b]$ by Theorem 5.1.32(ii). $(\forall j \in \{1, \dots, k\})(\forall t \in (a,b))\, F_j'(t) = g_j(t)$ by Proposition 1.1.27.
Thus $(\forall j \in \{1, \dots, k\})\, \int_a^b g_j(t)\, dt = F_j(b) - F_j(a)$ by Theorem 2.2.2. The vector identity follows by Definition 2.5.1.

(ii) Write $\mathbf{f} = (f_1, \dots, f_k)$, $q := \sum_{j=1}^{k} f_j^2$. Then $q \in \mathcal{R}(\alpha)$ by Theorem 2.1.15(i) and Theorem 2.1.14(i).
Choose $M \ge 0$ with $q([a,b]) \subseteq [0, M]$ by Definition 2.1.1. Since $(\forall t \in [a,b])\, \|\mathbf{f}(t)\| = \sqrt{q(t)}$, claim follows by Theorem 2.1.13.

(iii) Set $\mathbf{y} := \int_a^b \mathbf{f}\, d\alpha$. If $\mathbf{y} = \mathbf{0}$, claim holds by Part (ii) and Theorem 2.1.14(ii) with zero function.
Suppose $\mathbf{y} \neq \mathbf{0}$, $\mathbf{u} := \mathbf{y}/\|\mathbf{y}\|$, $h(t) := \mathbf{u} \cdot \mathbf{f}(t)$. Then $h \in \mathcal{R}(\alpha)$ by Definition 2.5.1 and Theorem 2.1.14(i).
Since $\|\mathbf{u}\| = 1$, $(\forall t \in [a,b])\, h(t) \le |h(t)| \le \|\mathbf{f}(t)\|$ by Theorem 1.1.13(i).
Moreover, $\int_a^b h\, d\alpha = \mathbf{u} \cdot \mathbf{y} = \|\mathbf{y}\|$ by Definition 2.5.1, Theorem 2.1.14(i). Claim follows by Theorem 2.1.14(ii).

Structural Commentary: Coordinates prove the antiderivative formula; norm composition and unit projection prove the norm claims.
Key Insight: Normalize the integral vector before applying scalar order. □

Definition 2.5.3. *Curves and Rectifiability*
Let $a, b \in \mathbb{R}$ satisfy $a < b$, and let $k \in \mathbb{N}$.

(i) A *curve* in $\mathbb{R}^k$ is a continuous map $\gamma : [a,b] \to \mathbb{R}^k$.
The curve γ is an *arc* iff γ is injective; it is a *closed curve* iff $\gamma(a) = \gamma(b)$.

(ii) For a partition $\Pi = \{a = x_0 < \cdots < x_n = b\}$, set $\ell(\Pi, \gamma) := \sum_{i=1}^{n} \|\gamma(x_i) - \gamma(x_{i-1})\|$.

(iii) The *length* of γ is $\ell(\gamma) := \sup_{\Pi} \ell(\Pi, \gamma) \in [0, +\infty]$, where Π ranges over all partitions of $[a,b]$. The curve γ is *rectifiable* iff $\ell(\gamma) < +\infty$.

Perspective 2.5.4. *Length Is Not Just Shape*
For $N \in \mathbb{N} \setminus \{0\}$, set $v_{2i} := (\frac{i}{N}, \frac{i}{N})$ for $0 \le i \le N$ and $v_{2i+1} := (\frac{i+1}{N}, \frac{i}{N})$ for $0 \le i < N$.
Let $\gamma_N : [0,1] \to \mathbb{R}^2$ satisfy $\gamma_N(\frac{m}{2N}) = v_m$ for $0 \le m \le 2N$ and be affine on each intervening subinterval.
Its $2N$ segments have length $\frac{1}{N}$, so $\ell(\gamma_N) = 2$, while its range lies within distance $\frac{1}{N}$ of the diagonal chord.
The chord has length $\sqrt{2}$; closeness of curve ranges therefore does not control length.

Figure 161 depicts the case $N = 6$ from Perspective 2.5.4.

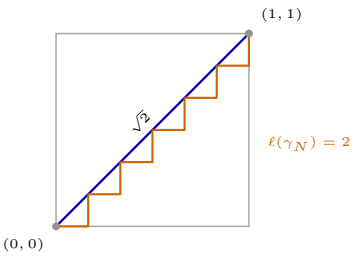

Figure 161: The six-step staircase has length 2, while the diagonal chord has length $\sqrt{2}$.

Theorem 2.5.5. *Length of a Continuously Differentiable Curve*
Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $k \in \mathbb{N}$, and let $\gamma : [a,b] \to \mathbb{R}^k$ be a curve differentiable relative to $[a,b]$.
If $\gamma' : [a,b] \to \mathbb{R}^k$ is continuous, then γ is rectifiable, with $\ell(\gamma) = \int_a^b \|\gamma'(t)\|\, dt$.

Proof.

(i) *Integrability*: Write $\gamma' = (g_1, \dots, g_k)$. Then $(\forall j \in \{1, \dots, k\})\, g_j$ is continuous by Theorem 5.1.32(ii).
Thus $(\forall j \in \{1, \dots, k\})\, g_j \in \mathcal{R}[a,b]$ by Theorem 2.1.9 with the identity integrator, so $\gamma' \in \mathcal{R}[a,b]$ by Definition 2.5.1.
Set $\rho := (t \mapsto \|\gamma'(t)\|)$. Then $\rho \in \mathcal{R}[a,b]$ by Theorem 2.5.2(ii); set $L := \int_a^b \rho(t)\, dt$.

(ii) *Upper Bound*: Fix a partition $\Pi = \{a = x_0 < \cdots < x_n = b\}$ and set $\Delta\gamma_i := \gamma(x_i) - \gamma(x_{i-1})$.
Then $(\forall i \in \{1, \dots, n\})\, \|\Delta\gamma_i\| \le \int_{x_{i-1}}^{x_i} \rho(t)\, dt$ by Theorem 2.5.2(i),(iii).
Thus $\ell(\Pi, \gamma) \le L$ by Theorem 2.1.14(iii) with the identity integrator. Hence $\ell(\gamma) \le L < +\infty$ by Definition 2.5.3, so γ is rectifiable.

(iii) *Local Reverse Bound*: Fix $\varepsilon > 0$. Choose $\delta > 0$ such that $(\forall s, t \in [a,b])\, [|s - t| < \delta \Rightarrow \|\gamma'(s) - \gamma'(t)\| < \varepsilon]$ by Theorem 5.2.6.
Choose a partition $\Pi = \{a = x_0 < \cdots < x_n = b\}$ with $\operatorname{mesh}(\Pi) < \delta$. Set $\mathbf{v}_i := \gamma'(x_{i-1})$ and $\Delta\gamma_i := \gamma(x_i) - \gamma(x_{i-1})$ for $1 \le i \le n$.
Then $(\forall i \in \{1, \dots, n\})(\forall t \in [x_{i-1}, x_i])\, \|\gamma'(t) - \mathbf{v}_i\| < \varepsilon$ by the choice of δ.
The identities $(\forall i \in \{1, \dots, n\})\, \mathbf{v}_i \Delta x_i - \Delta\gamma_i = \int_{x_{i-1}}^{x_i} [\mathbf{v}_i - \gamma'(t)]\, dt$ hold by Theorem 2.5.2(i).
Hence $(\forall i \in \{1, \dots, n\})\, \|\mathbf{v}_i \Delta x_i - \Delta\gamma_i\| \le \varepsilon \Delta x_i$ by Theorem 2.5.2(iii), Theorem 2.1.14(ii), and the pointwise bound.

The bounds $(\forall i \in \{1,\dots,n\})(\forall t \in [x_{i-1},x_i])\,\rho(t) \le \|\mathbf{v}_i\| + \varepsilon$ hold by Theorem 1.1.14(i).
Thus $(\forall i \in \{1,\dots,n\})\ \int_{x_{i-1}}^{x_i} \rho(t)\,dt \le \|\Delta\gamma_i\| + 2\varepsilon\Delta x_i$ by Theorem 2.1.14(ii), Theorem 1.1.14(i), and the preceding bound.

(iv) *Conclusion*: $L \le \ell(\Pi,\gamma) + 2\varepsilon(b-a) \le \ell(\gamma) + 2\varepsilon(b-a)$ by Theorem 2.1.14(iii) with the identity integrator and Definition 2.5.3.
Since $\varepsilon > 0$ is arbitrary, $L \le \ell(\gamma) \le L$, so $\ell(\gamma) = L$.

Structural Commentary: Vector integration bounds polygonal sums; uniform derivative control reverses the estimate on fine partitions.
Key Insight: Short chords approximate their constant-velocity integrals. □

2.6 Further Topics in Integration

Proof Method 2.6.1. *Riemann–Stieltjes Proof Patterns*
Use this method for Riemann–Stieltjes integrability, estimates, calculus identities, and improper convergence.
Record the interval, integrator, integrand, and target; for Darboux arguments, use (i) and, if needed, (ii) before (iii)–(vii).

(i) *Darboux Setup*: Let $a,b \in \mathbb{R}$ satisfy $a < b$, let $\alpha : [a,b] \to \mathbb{R}$ be increasing, and let $f : [a,b] \to \mathbb{R}$ be bounded.
Let $\mathfrak{P}[a,b]$ denote the partitions of $[a,b]$. For $\Pi \in \mathfrak{P}[a,b]$, set $G_\alpha(\Pi; f) := U(\Pi, f, \alpha) - L(\Pi, f, \alpha)$.

(ii) *Common Refinement*: Let $r \in \mathbb{N}$ and $\Pi_1,\dots,\Pi_r \in \mathfrak{P}[a,b]$. Set $\Pi := \bigcup_{j=1}^{r} \Pi_j$.
Then $(\forall j \in \{1,\dots,r\})\,G_\alpha(\Pi; f) \le G_\alpha(\Pi_j; f)$ by Theorem 2.1.3.

(iii) *Integrability Criterion*: $f \in \mathcal{R}(\alpha) \iff (\forall \varepsilon > 0)(\exists \Pi \in \mathfrak{P}[a,b])\,G_\alpha(\Pi; f) < \varepsilon$ by Theorem 2.1.6.
To prove membership, fix $\varepsilon > 0$ and construct such a partition Π.

(iv) *Finite Discontinuities*: Let $E := \{x \in [a,b] : f \text{ discontinuous at } x\}$ be finite, α continuous on E; $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.11.
For a direct Darboux proof with $E = \{c_1,\dots,c_N\} \neq \emptyset$, fix $\eta > 0$ and choose $J_j = [u_j, v_j] \subseteq [a,b]$.
Require pairwise disjoint relative interiors, $(\forall j \in \{1,\dots,N\})\,c_j \in \operatorname{int}_{[a,b]} J_j$, and $\sum_{j=1}^{N} [\alpha(v_j) - \alpha(u_j)] < \eta$.
On $K := [a,b] \setminus \bigcup_{j=1}^{N} \operatorname{int}_{[a,b]} J_j$, control the remaining gap by uniform continuity of $f|_K$.

(v) *Integral Algebra*: Prove linear-combination and order bounds by Theorem 2.1.14(i)–(ii).
Prove interval decompositions and uniform estimates by Parts (iii)–(iv) of the same theorem.

(vi) *Calculus Formula*: Let $f \in \mathcal{R}[a,b]$, and let F be continuous on $[a,b]$, differentiable on (a,b), with $(\forall t \in (a,b))\,F'(t) = f(t)$.
Fix $\varepsilon > 0$. Choose $\Pi = \{a = x_0 < \cdots < x_n = b\} \in \mathfrak{P}[a,b]$ with $U(\Pi, f) - L(\Pi, f) < \varepsilon$.
Choose tags $(t_i)_{i=1}^n$ satisfying $(\forall i \in \{1,\dots,n\})\,F(x_i) - F(x_{i-1}) = f(t_i)\Delta x_i$ by Corollary 1.1.13.
After telescoping, $\left|F(b) - F(a) - \int_a^b f(t)\,dt\right| < \varepsilon$ by Theorem 2.1.7(ii) with the identity integrator.
Hence $F(b) - F(a) = \int_a^b f(t)\,dt$ because $\varepsilon > 0$ is arbitrary.

(vii) *Improper Limits*: Verify local integrability and form the separate one-sided limits required by Definitions 2.4.1 and 2.4.4.
Let $a \in \mathbb{R}$ and $b \in \overline{\mathbb{R}}$ satisfy $a < b$, $f : [a,b) \to [0,+\infty)$. Assume $(\forall d \in (a,b))\,f|_{[a,d]} \in \mathcal{R}[a,d]$, and set $H(d) := \int_a^d f(x)\,dx$.
Then $\int_a^b f(x)\,dx$ converges iff $\sup_{\overline{\mathbb{R}}} H((a,b)) \in \mathbb{R}$ by Theorem 2.4.2.
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing with $\alpha(-\infty), \alpha(+\infty) \in \mathbb{R}$, and let $f : \mathbb{R} \to \mathbb{R}$ be bounded.
If $(\forall c,d \in \mathbb{R})\,[c < d \Rightarrow f|_{[c,d]} \in \mathcal{R}(\alpha|_{[c,d]})]$, then f is α-integrable on $\mathbb{R}$ by Theorem 2.4.5(iv).

Verification: Confirm that the selected route proves the target and that every cited hypothesis holds.
Check that all restrictions, local integrals, endpoint limits, and extended-real sums are defined.
Key Insight: Darboux gaps control finite intervals; monotone truncations or finite integrator mass control tails.

3 Several-Variable Differentiation

3.1 Partial and Total Derivatives

Several-variable differentiation treats derivatives as best linear approximations, with operator norms controlling the remainder.
Determinants and invertibility encode the resulting local geometry.

Definition 3.1.1. *Total Derivative*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} : E \to \mathbb{R}^m$, $\mathbf{x} \in E$. Write $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ for space of linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ (Definition 1.5.1).
The map $\mathbf{f}$ is *differentiable at* $\mathbf{x}$ iff $(\exists\, A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))\ \lim_{\mathbf{h}\to\mathbf{0}} \frac{\|\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - A\mathbf{h}\|_2}{\|\mathbf{h}\|_2} = 0$.
Any such A is a *total derivative* of $\mathbf{f}$ at $\mathbf{x}$; when it exists, it is denoted $D\mathbf{f}(\mathbf{x})$ or $\mathbf{f}'(\mathbf{x})$.
Equivalently, with $\mathbf{r}(\mathbf{h}) := \mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}$, one has $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
The map $\mathbf{f}$ is *differentiable on* E iff $(\forall\, \mathbf{x} \in E)\,D\mathbf{f}(\mathbf{x})$ exists.

Perspective 3.1.2. *Derivative First, Matrix Second*
The total derivative is a linear map by Definition 3.1.1; its coordinate matrix is secondary.
The intrinsic control is $(\forall\, A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))(\forall\, \mathbf{h} \in \mathbb{R}^n)\ \|A\mathbf{h}\|_2 \le \|A\|_{\mathrm{op}}\,\|\mathbf{h}\|_2$ by Definition 5.1.4.

Figure 162 separates the input increment $\mathbf{h}$ from its image $D\mathbf{f}(\mathbf{x})\mathbf{h}$. The dashed vector is the smaller remainder $\mathbf{r}(\mathbf{h})$.

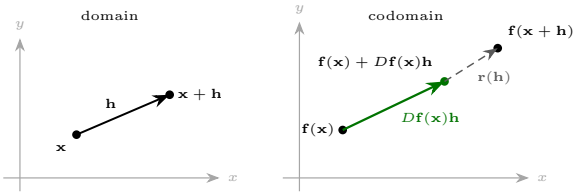

Figure 162: The solid vector $D\mathbf{f}(\mathbf{x})\mathbf{h}$ reaches the linear approximation; the dashed $\mathbf{r}(\mathbf{h}) = o(\|\mathbf{h}\|_2)$ completes the actual increment.

Theorem 3.1.3. *Basic Consequences of Total Differentiability*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} : E \to \mathbb{R}^m$, and let $\mathbf{x} \in E$. If $\mathbf{f}$ is differentiable at $\mathbf{x}$, then:
(i) the total derivative $D\mathbf{f}(\mathbf{x})$ is unique;
(ii) $\mathbf{f}$ is continuous at $\mathbf{x}$.

Proof.
(i) Let A_1, A_2 be total derivatives at $\mathbf{x}$, $B := A_1 - A_2$. If $\mathbf{r}_j(\mathbf{h}) := \mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - A_j\mathbf{h}$, then $B\mathbf{h} = \mathbf{r}_2(\mathbf{h}) - \mathbf{r}_1(\mathbf{h})$.
$\|B\mathbf{h}\|_2 = o(\|\mathbf{h}\|_2)$ by Definition 3.1.1, triangle inequality. For $\mathbf{u} \neq \mathbf{0}$ and $t \neq 0$, $\frac{\|B(t\mathbf{u})\|_2}{\|t\mathbf{u}\|_2} = \frac{\|B\mathbf{u}\|_2}{\|\mathbf{u}\|_2}$ by linearity, homogeneity.
For $\mathbf{u} \neq \mathbf{0}$, $\|B\mathbf{u}\|_2 = 0$ by this equality and its limit as $t \to 0$; $B\mathbf{0} = \mathbf{0}$ by linearity. Hence $B = 0$.
(ii) Set $A := D\mathbf{f}(\mathbf{x})$ an $\mathbf{r}(\mathbf{h}) := \mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - A\mathbf{h}$.
Then $\|\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x})\|_2 \leq \|A\|_{\text{op}} \|\mathbf{h}\|_2 + \|\mathbf{r}(\mathbf{h})\|_2 \to 0$ by Definitions 5.1.4 and 3.1.1.
Thus $\mathbf{f}(\mathbf{x}+\mathbf{h}) \to \mathbf{f}(\mathbf{x})$ as $\mathbf{h} \to \mathbf{0}$, so $\mathbf{f}$ is continuous at $\mathbf{x}$.

Structural Commentary: Scaling eliminates a second linear part; the operator bound transfers the remainder estimate to continuity.
Key Insight: A nonzero linear error cannot be $o(\|\mathbf{h}\|_2)$. □

Example 3.1.4. *Derivative of a Linear Map*
Let $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ and set $\mathbf{f}(\mathbf{x}) := A\mathbf{x}$.
Then $(\forall \mathbf{x}, \mathbf{h} \in \mathbb{R}^n)\, \mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - A\mathbf{h} = \mathbf{0}$, so $(\forall \mathbf{x} \in \mathbb{R}^n)\, D\mathbf{f}(\mathbf{x}) = A$ by Definition 3.1.1.

Theorem 3.1.5. *Chain Rule*
Let $E \subseteq \mathbb{R}^n$ and $G \subseteq \mathbb{R}^m$ be open, and let $\mathbf{x}_0 \in E$.
Let $\mathbf{f} : E \to \mathbb{R}^m$ and $\mathbf{g} : G \to \mathbb{R}^k$, with $\mathbf{f}(E) \subseteq G$.
If $\mathbf{f}$ is differentiable at $\mathbf{x}_0$ and $\mathbf{g}$ is differentiable at $\mathbf{f}(\mathbf{x}_0)$, then:
(i) $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}_0$;
(ii) $D(\mathbf{g} \circ \mathbf{f})(\mathbf{x}_0) = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))D\mathbf{f}(\mathbf{x}_0)$.

Proof. Set $A := D\mathbf{f}(\mathbf{x}_0)$ and $B := D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))$.
Then $BA \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^k)$ by Proposition 1.5.7(i).
Set $\mathbf{u}(\mathbf{h}) := \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0) - A\mathbf{h}$ and $\mathbf{q}(\mathbf{h}) := \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0)$.
Finally, set $\mathbf{v}(\mathbf{q}) := \mathbf{g}(\mathbf{f}(\mathbf{x}_0)+\mathbf{q}) - \mathbf{g}(\mathbf{f}(\mathbf{x}_0)) - B\mathbf{q}$.
(i) The relation $\|\mathbf{u}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$ holds by Definition 3.1.1.
The relation $\|\mathbf{v}(\mathbf{q})\|_2 = o(\|\mathbf{q}\|_2)$ as $\mathbf{q} \to \mathbf{0}$ holds by the same definition.
Eventually $\|\mathbf{u}(\mathbf{h})\|_2 \leq \|\mathbf{h}\|_2$ as $\mathbf{h} \to \mathbf{0}$ by the first estimate.
Thus $\|\mathbf{q}(\mathbf{h})\|_2 \leq (\|A\|_{\text{op}} + 1)\|\mathbf{h}\|_2$ eventually by $\mathbf{q}(\mathbf{h}) = A\mathbf{h} + \mathbf{u}(\mathbf{h})$ and Definition 5.1.4.
Hence $\|\mathbf{v}(\mathbf{q}(\mathbf{h}))\|_2 = o(\|\mathbf{h}\|_2)$.
The identity $(\mathbf{g} \circ \mathbf{f})(\mathbf{x}_0+\mathbf{h}) - (\mathbf{g} \circ \mathbf{f})(\mathbf{x}_0) - BA\mathbf{h} = B\mathbf{u}(\mathbf{h}) + \mathbf{v}(\mathbf{q}(\mathbf{h}))$ holds by the definitions of $\mathbf{u}$, $\mathbf{q}$, and $\mathbf{v}$.
The right side is $o(\|\mathbf{h}\|_2)$ by the preceding estimates and the operator-norm bound, so $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}_0$.
(ii) The derivative candidate in (i) is BA.
Thus $D(\mathbf{g} \circ \mathbf{f})(\mathbf{x}_0) = BA = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))D\mathbf{f}(\mathbf{x}_0)$ by Theorem 3.1.3(i).

Structural Commentary: The inner increment is $O(\|\mathbf{h}\|_2)$, allowing the outer remainder to compose with it.
Key Insight: Both composed error terms remain $o(\|\mathbf{h}\|_2)$. □

Theorem 3.1.6. *Componentwise Differentiability*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} = (f_1, \ldots, f_m) : E \to \mathbb{R}^m$, and let $\mathbf{x} \in E$.
For $i \in J_m$, let $\pi_i \in \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ be the ith coordinate projection.
(i) $\mathbf{f}$ is differentiable at $\mathbf{x}$ iff $(\forall\, i \in J_m)\, f_i$ is differentiable at $\mathbf{x}$.
(ii) If $\mathbf{f}$ is differentiable at $\mathbf{x}$, then $(\forall\, i \in J_m)\, Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$.

Proof.
(i) ($\Rightarrow$) Let $\mathbf{r}(\mathbf{h}) := \mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}$.
For $i \in J_m$, set $r_i(\mathbf{h}) := \pi_i(\mathbf{r}(\mathbf{h}))$.
The map $\pi_i \circ D\mathbf{f}(\mathbf{x})$ is linear by Proposition 1.5.7(i).
Then $f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) - (\pi_i \circ D\mathbf{f}(\mathbf{x}))\mathbf{h} = r_i(\mathbf{h})$ and $|r_i(\mathbf{h})| \leq \|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Thus all component functions are differentiable at $\mathbf{x}$ by Definition 3.1.1.
(*i*) ($\Leftarrow$) Suppose $(\forall\, i \in J_m)\, f_i$ is differentiable at $\mathbf{x}$, and set $A_i := Df_i(\mathbf{x})$.
Define $A : \mathbb{R}^n \to \mathbb{R}^m$ by $A\mathbf{h} := (A_1\mathbf{h}, \ldots, A_m\mathbf{h})$.
Then $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ by coordinatewise linearity.
For $i \in J_m$, set $r_i(\mathbf{h}) := f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) - A_i\mathbf{h}$.
Then $(\forall\, i \in J_m)\ |r_i(\mathbf{h})| = o(\|\mathbf{h}\|_2)$.
For $\mathbf{r}(\mathbf{h}) := (r_1(\mathbf{h}), \ldots, r_m(\mathbf{h}))$, $\|\mathbf{r}(\mathbf{h})\|_2 \leq \sum_{i=1}^{m} |r_i(\mathbf{h})| = o(\|\mathbf{h}\|_2)$ by finiteness of J_m.
Hence $\mathbf{f}$ is differentiable at $\mathbf{x}$ with derivative A by Definition 3.1.1.
(*ii*) For $i \in J_m$, the derivative candidate in (*i*), ($\Rightarrow$), is $\pi_i \circ D\mathbf{f}(\mathbf{x})$.
Hence $(\forall\, i \in J_m)\, Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$ by Theorem 3.1.3(i).

Structural Commentary: Coordinate projections prove necessity; finite assembly of scalar remainders proves sufficiency.
Key Insight: Finitely many scalar $o(\|\mathbf{h}\|_2)$ terms form one vector remainder. □

Definition 3.1.7. *Partial, Gradient, and Directional Derivatives*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} = (f_1, \ldots, f_m) : E \to \mathbb{R}^m$, and let $\mathbf{x} \in E$.
Let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis of $\mathbb{R}^n$.
(i) For $(i, j) \in J_m \times J_n$, set $\phi_{ij}(t) := f_i(\mathbf{x} + t\mathbf{e}_j)$ whenever $\mathbf{x} + t\mathbf{e}_j \in E$.
When the limit exists, the *partial derivative* is $(D_j f_i)(\mathbf{x}) := \lim_{t \to 0} \frac{\phi_{ij}(t) - \phi_{ij}(0)}{t}$.

(ii) For $\mathbf{u} \in \mathbb{R}^n$, set $\gamma_{\mathbf{u}}(t) := \mathbf{f}(\mathbf{x} + t\mathbf{u})$ whenever $\mathbf{x} + t\mathbf{u} \in E$.
When the limit exists, the *directional derivative* is $(D_{\mathbf{u}}\mathbf{f})(\mathbf{x}) := \lim_{t\to 0} \frac{\gamma_{\mathbf{u}}(t)-\gamma_{\mathbf{u}}(0)}{t}$.
When only direction matters, take $\|\mathbf{u}\|_2 = 1$.

(iii) If $m = 1$, write $f = f_1$; if all $(D_j f)(\mathbf{x})$ exist, the *gradient* of f at $\mathbf{x}$ is $(\nabla f)(\mathbf{x}) := \sum_{j=1}^{n} (D_j f)(\mathbf{x})\mathbf{e}_j$.

Perspective 3.1.8. *Partials Are Coordinate Tests*
Partial and directional derivatives in Definition 3.1.7 test selected lines.
The total derivative in Definition 3.1.1 controls every small increment by one linear map.
The matrix theorem below extracts the line tests; the next subsection proves a continuity-based converse.

Theorem 3.1.9. *Matrix of the Total Derivative*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} = (f_1, \dots, f_m) : E \to \mathbb{R}^m$, and let $\mathbf{x} \in E$.
If $\mathbf{f}$ is differentiable at $\mathbf{x}$, then:

(i) all $(D_j f_i)(\mathbf{x})$ exist, and $(\forall\, (i,j) \in J_m \times J_n)\, [D\mathbf{f}(\mathbf{x})]_{ij} = (D_j f_i)(\mathbf{x})$;
(ii) if $m = 1$ and $f = f_1$, then $(\forall\, \mathbf{u} \in \mathbb{R}^n)\, (D_{\mathbf{u}} f)(\mathbf{x}) = (\nabla f)(\mathbf{x}) \cdot \mathbf{u}$.

Proof. Set $\mathbf{r}(\mathbf{h}) := \mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}$; then $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$ by Definition 3.1.1.

(i) For $j \in J_n$, set $\mathbf{h} = t\mathbf{e}_j$.
Then $\frac{\mathbf{f}(\mathbf{x}+t\mathbf{e}_j)-\mathbf{f}(\mathbf{x})}{t} = D\mathbf{f}(\mathbf{x})\mathbf{e}_j + \frac{\mathbf{r}(t\mathbf{e}_j)}{t} \to D\mathbf{f}(\mathbf{x})\mathbf{e}_j$ as $t \to 0$ by Definition 3.1.1.
Thus all component limits exist, and their values form column j of $[D\mathbf{f}(\mathbf{x})]$ by Proposition 3.2.5.

(ii) Suppose $m = 1$, and fix $\mathbf{u} = (u_1, \dots, u_n) \in \mathbb{R}^n$.
The identity $(D_{\mathbf{u}} f)(\mathbf{x}) = Df(\mathbf{x})\mathbf{u}$ holds by Definition 3.1.1 with $\mathbf{h} = t\mathbf{u}$.
By part (i) and linearity, $Df(\mathbf{x})\mathbf{u} = \sum_{j=1}^{n} u_j Df(\mathbf{x})\mathbf{e}_j = \sum_{j=1}^{n} u_j (D_j f)(\mathbf{x}) = (\nabla f)(\mathbf{x}) \cdot \mathbf{u}$.

Structural Commentary: Line restrictions identify $D\mathbf{f}(\mathbf{x})\mathbf{e}_j$; matrix coordinates and linearity recover both claims.
Key Insight: The standard basis converts derivative action into partial-derivative columns. □

Definition 3.1.10. *Jacobian*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} = (y_1, \dots, y_n) : E \to \mathbb{R}^n$, and suppose $\mathbf{f}$ is differentiable at $\mathbf{x} \in E$.
The *Jacobian determinant* of $\mathbf{f}$ at $\mathbf{x}$ is $J_{\mathbf{f}}(\mathbf{x}) := \det D\mathbf{f}(\mathbf{x}) = \frac{\partial(y_1,\dots,y_n)}{\partial(x_1,\dots,x_n)}(\mathbf{x})$.

Corollary 3.1.11. *Nonvanishing Jacobian*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} : E \to \mathbb{R}^n$ be differentiable at $\mathbf{x} \in E$, and let $J_{\mathbf{f}}(\mathbf{x})$ be its Jacobian determinant.
Then $D\mathbf{f}(\mathbf{x})$ is invertible iff $J_{\mathbf{f}}(\mathbf{x}) \neq 0$.

Proof.
(⇒) If $D\mathbf{f}(\mathbf{x})$ is invertible, then $\det D\mathbf{f}(\mathbf{x}) \neq 0$ by Theorem 3.8.2(vii).
Hence $J_{\mathbf{f}}(\mathbf{x}) \neq 0$ by Definition 3.1.10.
(⇐) If $J_{\mathbf{f}}(\mathbf{x}) \neq 0$, then $\det D\mathbf{f}(\mathbf{x}) \neq 0$ by Definition 3.1.10.
Hence $D\mathbf{f}(\mathbf{x})$ is invertible by Theorem 3.8.2(vii).

Structural Commentary: The Jacobian definition transfers the determinant criterion to the derivative matrix.
Key Insight: Nonzero determinant is exactly linear invertibility. □

For the map in Definition 3.1.10, $|J_{\mathbf{f}}(\mathbf{x})|$ is the volume scale of the linear approximation $D\mathbf{f}(\mathbf{x})$.

3.2 Differentiability and Linear Approximation

Theorem 3.2.1. *Mean-Value Inequality on Convex Sets*
Let $E \subseteq \mathbb{R}^n$ be open and convex, and let $\mathbf{f} : E \to \mathbb{R}^m$ be differentiable.

(i) If $M \geq 0$ and $(\forall\, \mathbf{x} \in E)\ \|D\mathbf{f}(\mathbf{x})\|_{op} \leq M$, then $(\forall\, \mathbf{a}, \mathbf{b} \in E)\ \|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 \leq M \|\mathbf{b} - \mathbf{a}\|_2$.
(ii) If $(\forall\, \mathbf{x} \in E)\, D\mathbf{f}(\mathbf{x}) = 0$, then $\mathbf{f}$ is constant on E.

Proof.

(i) Fix $\mathbf{a}, \mathbf{b} \in E$, set $\gamma(t) := (1-t)\mathbf{a} + t\mathbf{b}$, and put $\mathbf{g} := \mathbf{f} \circ \gamma$.
Then $\gamma([0,1]) \subseteq E$ by convexity, and $\mathbf{g}'(t) = D\mathbf{f}(\gamma(t))(\mathbf{b} - \mathbf{a})$ by Theorem 3.1.5.
Hence $\|\mathbf{g}'(t)\|_2 \leq M \|\mathbf{b} - \mathbf{a}\|_2$ by the operator-norm hypothesis.
Therefore $\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 \leq M \|\mathbf{b} - \mathbf{a}\|_2$ by Theorem 1.1.28.

(ii) If $(\forall\, \mathbf{x} \in E)\, D\mathbf{f}(\mathbf{x}) = 0$, then $(\forall\, \mathbf{a}, \mathbf{b} \in E)\ \|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 = 0$ by part (i) with $M = 0$.
Thus $(\forall\, \mathbf{a}, \mathbf{b} \in E)\, \mathbf{f}(\mathbf{a}) = \mathbf{f}(\mathbf{b})$.

Structural Commentary: Convexity retains the segment; the chain rule transfers the bound.
Key Insight: A vector mean-value bound converts derivative control into secant control. □

Definition 3.2.2. *C^1 Mapping*
Let $E \subseteq \mathbb{R}^n$ be open, and let $\mathbf{f} : E \to \mathbb{R}^m$ be differentiable.
The map $\mathbf{f}$ is *continuously differentiable*, or C^1, if $D\mathbf{f} : E \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is continuous in the operator-norm metric.

Perspective 3.2.3. *C^1 Uses the Operator-Norm Metric*
Definition 3.2.2 makes $D\mathbf{f}$ itself the object whose continuity is measured.
For matrices $A, B \in \mathrm{Mat}(m, n, \mathbb{R})$, $|A_{ij} - B_{ij}| \leq \|A - B\|_2 \leq \|A - B\|_F$.
Moreover, $\|A - B\|_F \leq \sqrt{mn} \max_{i,j} |A_{ij} - B_{ij}|$.
Thus operator-norm continuity and entrywise continuity are equivalent in these finite dimensions.

Figure 163 decomposes $\mathbf{h}$ into coordinate steps.
This is the path used below to telescope from $\mathbf{x}$ to $\mathbf{x}+\mathbf{h}$.

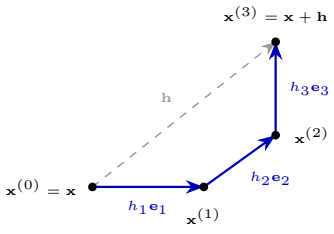

Figure 163: The solid coordinate steps sum to the dashed direct increment $\mathbf{h}$ from $\mathbf{x}$ to $\mathbf{x}+\mathbf{h}$.

Theorem 3.2.4. *C^1 Means Continuous Partials*
Let $E \subseteq \mathbb{R}^n$ be open, and let $\mathbf{f} = (f_1, \dots, f_m) : E \to \mathbb{R}^m$.
Then $\mathbf{f}$ is C^1 on E iff $(\forall\,(i,j) \in J_m \times J_n)\, D_j f_i$ exists and is continuous on E.

Proof.
($\Rightarrow$) Suppose $\mathbf{f}$ is C^1 on E.
For $(\mathbf{x}, i, j) \in E \times J_m \times J_n$, $(D_j f_i)(\mathbf{x}) = [D\mathbf{f}(\mathbf{x})]_{ij}$ by Theorem 3.1.9.
Since $|A_{ij} - B_{ij}| \le \|A - B\|_2$, each $D_j f_i$ is continuous on E by continuity of $D\mathbf{f}$.
($\Leftarrow$) Fix $\mathbf{x} \in E$ and $i \in J_m$, and choose $\rho > 0$ with $B_\rho(\mathbf{x}) \subseteq E$ by openness.

For $\|\mathbf{h}\|_2 < \rho$, set $\mathbf{x}^{(0)} := \mathbf{x}$ and $\mathbf{x}^{(j)} := \mathbf{x} + \sum_{k=1}^{j} h_k \mathbf{e}_k$ for $j \in J_n$.

Each segment $[\mathbf{x}^{(j-1)}, \mathbf{x}^{(j)}]$ lies in $B_\rho(\mathbf{x})$ because every point on it is at distance at most $\|\mathbf{h}\|_2$ from $\mathbf{x}$.
For $j \in J_n$ with $h_j \neq 0$, let ξ_j be a mean-value point on $[\mathbf{x}^{(j-1)}, \mathbf{x}^{(j)}]$ by Corollary 1.1.13.
Then $f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) = h_j (D_j f_i)(\xi_j)$.
For $j \in J_n$ with $h_j = 0$, set $\xi_j := \mathbf{x}^{(j)}$; the identity holds trivially.

Define $A_{i,\mathbf{x}}\mathbf{h} := \sum_{j=1}^{n} h_j (D_j f_i)(\mathbf{x})$; then $A_{i,\mathbf{x}} \in \mathcal{L}(\mathbb{R}^n, \mathbb{R})$ by coordinatewise linearity.

The bound $|f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) - A_{i,\mathbf{x}}\mathbf{h}| \le \sum_{j=1}^{n} |h_j|\,|(D_j f_i)(\xi_j) - (D_j f_i)(\mathbf{x})|$ holds by telescoping.

The right-hand side is at most $\sqrt{n}\,\|\mathbf{h}\|_2 \max_{j \in J_n} |(D_j f_i)(\xi_j) - (D_j f_i)(\mathbf{x})|$ by Cauchy–Schwarz.

Here $\xi_j \to \mathbf{x}$ as $\mathbf{h} \to \mathbf{0}$, and the maximum tends to 0 by continuity and finiteness of J_n.
Hence the displayed bound is $o(\|\mathbf{h}\|_2)$.
Thus every f_i is differentiable at every $\mathbf{x}$, so $\mathbf{f}$ is differentiable on E by Theorem 3.1.6(i).
Its derivative entries are $D_j f_i$ by Theorem 3.1.9.
Their continuity and $\|A - B\|_2 \le \|A - B\|_F \le \sqrt{mn} \max_{i,j} |A_{ij} - B_{ij}|$ make $D\mathbf{f}$ continuous.

Hence $\mathbf{f}$ is C^1 on E by Definition 3.2.2.

Structural Commentary: Telescoping builds the components; norm comparison assembles them.
Key Insight: Continuous partials control the finitely many coordinate remainders. □

Higher differentiation views $D\mathbf{f}$ as a map into the finite-dimensional space $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$.
The second derivative is therefore a bilinear map, not merely an array of second partials.

Definition 3.2.5. *Higher Derivatives and C^r*
Let $E \subseteq \mathbb{R}^n$ be open, and let $\mathbf{f} : E \to \mathbb{R}^m$.
For $q \in \mathbb{Z}_{\ge 1}$, let $\mathcal{L}^q(\mathbb{R}^n, \mathbb{R}^m)$ denote the space of q-linear maps $(\mathbb{R}^n)^q \to \mathbb{R}^m$.
For $T \in \mathcal{L}^q(\mathbb{R}^n, \mathbb{R}^m)$, set $\|T\|_{\mathrm{op}} := \sup_{\|\mathbf{v}_s\|_2 \le 1,\ s \in J_q} \|T(\mathbf{v}_1, \dots, \mathbf{v}_q)\|_2$.

(i) Set $D^1\mathbf{f} := D\mathbf{f}$.
For $q \ge 2$, suppose $D^{q-1}\mathbf{f}$ is differentiable at $\mathbf{x}$.
The *qth derivative* is $D^q\mathbf{f}(\mathbf{x})(\mathbf{v}_1, \dots, \mathbf{v}_q) := \big(D(D^{q-1}\mathbf{f})(\mathbf{x})\mathbf{v}_1\big)(\mathbf{v}_2, \dots, \mathbf{v}_q)$.
(ii) For real-valued $f : E \to \mathbb{R}$, define the mixed second partial $D_{ij} f := D_i(D_j f)$ wherever the right-hand side exists.
(iii) For $r \in \mathbb{Z}_{\ge 1}$, the map $\mathbf{f}$ is *C^r* on E iff $(\forall\, q \in J_r)\, D^q\mathbf{f} : E \to \mathcal{L}^q(\mathbb{R}^n, \mathbb{R}^m)$ exists and is continuous.
The map $\mathbf{f}$ is *C^∞* on E iff $(\forall\, r \in \mathbb{Z}_{\ge 1})\, \mathbf{f}$ is C^r on E.

Theorem 3.2.6. *Second Derivatives and Second Partials*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} = (f_1, \dots, f_m) : E \to \mathbb{R}^m$, and let $p \in E$.
Set $\mathcal{I} := J_n \times J_n \times J_m$.

(i) If $D^2\mathbf{f}(p)$ exists, then $(\forall\,(i,j,\ell) \in \mathcal{I})\, D_{ij} f_\ell(p)$ exists.
For these indices, $D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j) = D_{ij} f_\ell(p)$.
(ii) Suppose $\mathbf{f}$ is differentiable on a neighborhood $U \subseteq E$ of p.
If $(\forall\,(i,j,\ell) \in \mathcal{I})\, D_{ij} f_\ell$ exists on U and is continuous at p, then $D^2\mathbf{f}(p)$ exists.

Proof.
(i) Identify $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ with $\mathbb{R}^{mn}$ by standard bases.
The Frobenius and operator norms are equivalent because $\|M\|_2 \le \|M\|_F \le \sqrt{mn}\,\|M\|_2$.
The entries of $D\mathbf{f}$ are $D_j f_\ell$ by Theorem 3.1.9.
Each entry is differentiable at p by Theorem 3.1.6(i).

Its ith partial exists by Theorem 3.1.9(i).
Consequently Df_ℓ is differentiable at p by Theorem 3.1.6(i).
The identity $D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j) = D_i(D_j f_\ell)(p) = D_{ij} f_\ell(p)$ holds by Definition 3.2.5(i).

(ii) Set $F := D\mathbf{f} : U \to \mathbb{R}^{mn}$ under the same identification, so $F_{\ell j} = D_j f_\ell$ by Theorem 3.1.9.

Define $A : \mathbb{R}^n \to \mathbb{R}^{mn}$ by $[A\mathbf{h}]_{\ell j} := \sum_{i=1}^{n} h_i D_{ij} f_\ell(p)$; then A is linear.

Choose $\rho > 0$ with $B_\rho(p) \subseteq U$ by openness, and define the coordinate polygon $p^{(i)} := p + \sum_{q=1}^{i} h_q \mathbf{e}_q$.

Set $S_i := [p^{(i-1)}, p^{(i)}]$; if $\|\mathbf{h}\|_2 < \rho$, then $S_i \subseteq B_\rho(p)$.
For $(i, j, \ell) \in \mathcal{I}$ with $h_i \neq 0$, let $\eta_{i\ell j}$ be a mean-value point for $F_{\ell j}$ on S_i by Corollary 1.1.13.
Then $F_{\ell j}(p^{(i)}) - F_{\ell j}(p^{(i-1)}) = h_i D_{ij} f_\ell(\eta_{i\ell j})$.
For $(i, j, \ell) \in \mathcal{I}$ with $h_i = 0$, set $\eta_{i\ell j} := p^{(i)}$; the identity holds trivially.

For each (ℓ, j), $\left|[F(p+\mathbf{h}) - F(p) - A\mathbf{h}]_{\ell j}\right| \leq \sum_{i=1}^{n} |h_i| \left|D_{ij} f_\ell(\eta_{i\ell j}) - D_{ij} f_\ell(p)\right|$ by telescoping.

This is at most $\sqrt{n}\,\|\mathbf{h}\|_2 \max_{i \in J_n} \left|D_{ij} f_\ell(\eta_{i\ell j}) - D_{ij} f_\ell(p)\right|$ by Cauchy–Schwarz.

Since $\eta_{i\ell j} \to p$ as $\mathbf{h} \to \mathbf{0}$, the right-hand side is $o(\|\mathbf{h}\|_2)$ by continuity at p.
Therefore $\|F(p+\mathbf{h}) - F(p) - A\mathbf{h}\|_2 = o(\|\mathbf{h}\|_2)$ by finiteness of $J_n \times J_m$.
This is also an operator-norm remainder by the preceding norm equivalence.
Thus $F = D\mathbf{f}$ is differentiable at p, and $D^2\mathbf{f}(p)$ exists by Definition 3.2.5(i).

Structural Commentary: Entrywise estimates combine into one derivative-matrix remainder.
Key Insight: Pointwise continuity suffices because only finitely many coordinates occur. □

Figure 164 records the signed four-vertex difference used below.
Exchanging $\mathbf{v}$ and $\mathbf{w}$ preserves both the vertices and their signs.

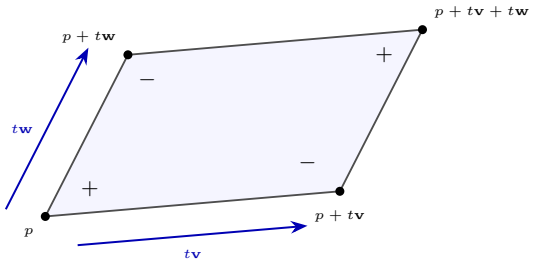

Figure 164: The $+, -, -, +$ vertex pattern is invariant under exchanging $\mathbf{v}$ and $\mathbf{w}$.

Theorem 3.2.7. *Symmetry of the Second Derivative*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} : E \to \mathbb{R}^m$, and let $p \in E$.
If $D^2\mathbf{f}(p)$ exists, then $(\forall\, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n)\, D^2\mathbf{f}(p)(\mathbf{v}, \mathbf{w}) = D^2\mathbf{f}(p)(\mathbf{w}, \mathbf{v})$.

Proof.

(i) *Scalar Reduction*: Component projections commute with both derivatives by Theorem 3.1.6.
It therefore suffices to prove symmetry for a scalar-valued f.
The map Df is defined on a neighborhood of p by Definition 3.2.5(i).

(ii) *Symmetric Increment*: Fix $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$.
For t sufficiently close to 0, the four arguments below lie in E by openness.
Set $\Delta(t, \mathbf{v}, \mathbf{w}) := f(p + t\mathbf{v} + t\mathbf{w}) - f(p + t\mathbf{v}) - f(p + t\mathbf{w}) + f(p)$.
This expression is unchanged when $\mathbf{v}$ and $\mathbf{w}$ are exchanged.

(iii) *Limit Formula*: Set $R(\mathbf{z}) := Df(p + \mathbf{z}) - Df(p) - D^2 f(p)(\mathbf{z}, \cdot)$.
Then $\|R(\mathbf{z})\|_{\mathrm{op}} = o(\|\mathbf{z}\|_2)$ by Definition 3.2.5(i).
For $t \neq 0$, define $\phi_t(s) := f(p + t\mathbf{v} + s\mathbf{w}) - f(p + s\mathbf{w})$.
Let ξ_t be a mean-value point for ϕ_t between 0 and t by Corollary 1.1.13.
Then $\Delta(t, \mathbf{v}, \mathbf{w}) = t[Df(p + t\mathbf{v} + \xi_t\mathbf{w}) - Df(p + \xi_t\mathbf{w})]\mathbf{w}$.
Set $\mathbf{z}_0 := \xi_t\mathbf{w}$ and $\mathbf{z}_1 := t\mathbf{v} + \xi_t\mathbf{w}$.
Then $[Df(p + \mathbf{z}_1) - Df(p + \mathbf{z}_0)]\mathbf{w} = tD^2 f(p)(\mathbf{v}, \mathbf{w}) + [R(\mathbf{z}_1) - R(\mathbf{z}_0)]\mathbf{w}$ by the definition of R.
Both remainder terms are $o(|t|)$ because $\|\mathbf{z}_0\|_2, \|\mathbf{z}_1\|_2 = O(|t|)$.
Therefore $D^2 f(p)(\mathbf{v}, \mathbf{w}) = \lim_{t \to 0} \frac{\Delta(t, \mathbf{v}, \mathbf{w})}{t^2}$.

(iv) *Symmetry*: The same formula with $\mathbf{v}$ and $\mathbf{w}$ exchanged has the same numerator by part (ii).
Hence $D^2 f(p)(\mathbf{v}, \mathbf{w}) = D^2 f(p)(\mathbf{w}, \mathbf{v})$.

Structural Commentary: A symmetric four-point increment encodes either bilinear order.
Key Insight: One geometric numerator computes both orders of directional differentiation. □

Corollary 3.2.8. *Mixed Partials from D^2*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} = (f_1, \ldots, f_m) : E \to \mathbb{R}^m$, and let $p \in E$.
If $D^2\mathbf{f}(p)$ exists, then $(\forall\, (i, j, \ell) \in J_n \times J_n \times J_m)\, D_{ij} f_\ell(p) = D_{ji} f_\ell(p)$.

Proof. For $(i, j, \ell) \in J_n \times J_n \times J_m$, $D_{ij} f_\ell(p) = D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j)$ by Theorem 3.2.6(i).
Likewise, $D_{ji} f_\ell(p) = D^2 f_\ell(p)(\mathbf{e}_j, \mathbf{e}_i)$ by the same part.
The two right-hand sides are equal by Theorem 3.2.7.
Structural Commentary: Second partials are standard-basis values of one bilinear derivative.
Key Insight: Symmetry of $D^2 f_\ell(p)$ identifies the two mixed entries. □

Corollary 3.2.9. *Symmetry of Higher Derivatives*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} : E \to \mathbb{R}^m$, let $p \in E$, and let $r \in \mathbb{Z}_{\geq 2}$.
Let $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_r) \in (\mathbb{R}^n)^r$.
For $\sigma \in \mathfrak{S}_r$, set $\mathbf{v}^\sigma := (\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(r)})$.
If $D^r\mathbf{f}(p)$ exists, then $(\forall\, \sigma \in \mathfrak{S}_r)\, D^r\mathbf{f}(p)(\mathbf{v}^\sigma) = D^r\mathbf{f}(p)(\mathbf{v})$.

Proof.

(i) *Base Case*: The case $r = 2$ holds by Theorem 3.2.7.
(ii) *Trailing Slots*: Let $r > 2$ and assume the claim for order $r - 1$.
Since $D^r\mathbf{f}(p)$ exists, the map $D^{r-1}\mathbf{f}$ is defined on a neighborhood of p.
Near p, each value of $D^{r-1}\mathbf{f}$ is symmetric by the induction hypothesis.
The symmetric $(r-1)$-linear maps form a closed finite-dimensional subspace.
Thus $D(D^{r-1}\mathbf{f})(p)\mathbf{v}_1$ is symmetric in slots $2, \dots, r$ by closure under limits.
(iii) *First Two Slots*: Identify the finite-dimensional codomain of $D^{r-2}\mathbf{f}$ with a Euclidean space.
Its second derivative at p exists by the existence of $D^r\mathbf{f}(p)$ and is symmetric by Theorem 3.2.7.
After evaluation at $\mathbf{v}_3, \dots, \mathbf{v}_r$, $D^r\mathbf{f}(p)$ is symmetric in slots 1 and 2 by the preceding identity.
(iv) *Permutation Step*: Invariance under every adjacent transposition holds by parts (ii)–(iii).
Since adjacent transpositions generate $\mathfrak{S}_r$, every permutation fixes $D^r\mathbf{f}(p)$.

Structural Commentary: Induction controls the tail; D^2 symmetry swaps the first pair.
Key Insight: Adjacent transpositions generate every permutation of the derivative slots. □

The next argument uses coordinate rectangles to compare the two iterated partial derivatives. Its local continuity hypothesis is weaker than a global C^2 assumption.

Theorem 3.2.10. *Rectangle Mean Value Formula*
Let $E \subseteq \mathbb{R}^2$ be open, let $f : E \to \mathbb{R}$, and suppose $D_1 f$ and $D_{21} f = D_2(D_1 f)$ exist on E.
Let $h, k \in \mathbb{R} \setminus \{0\}$.
Let $Q \subseteq E$ be the closed coordinate rectangle with opposite vertices (a, b) and $(a + h, b + k)$.
Set $\Delta(f, Q) := f(a + h, b + k) - f(a + h, b) - f(a, b + k) + f(a, b)$.
Then $(\exists\, (x, y) \in \operatorname{int} Q)\, \Delta(f, Q) = hk(D_{21} f)(x, y)$.

Proof. Let I and J be the open intervals with endpoints $a, a + h$ and $b, b + k$, respectively.
Set $u(t) := f(t, b + k) - f(t, b)$.
The map u is differentiable on I and continuous on its closure by existence of $D_1 f$.
Let $x \in I$ be a mean-value point for u by Corollary 1.1.13.
Then $\Delta(f, Q) = h[(D_1 f)(x, b + k) - (D_1 f)(x, b)]$.
The map $s \mapsto (D_1 f)(x, s)$ is differentiable on J and continuous on its closure by existence of $D_{21} f$.
Let $y \in J$ be a mean-value point for $s \mapsto (D_1 f)(x, s)$ by the same corollary.
Then $(D_1 f)(x, b + k) - (D_1 f)(x, b) = k(D_{21} f)(x, y)$.
Then $(x, y) \in I \times J = \operatorname{int} Q$ and $\Delta(f, Q) = hk(D_{21} f)(x, y)$.
Structural Commentary: Two mean-value steps factor the rectangle increment.
Key Insight: The four-corner difference samples the vertical change of $D_1 f$. □

Theorem 3.2.11. *Equality of Mixed Partials*
Let $E \subseteq \mathbb{R}^2$ be open, let $f : E \to \mathbb{R}$, and let $(a, b) \in E$.
Suppose $D_1 f$, $D_2 f$, and $D_{21} f$ exist on E, and $D_{21} f$ is continuous at (a, b). Then:
(i) The mixed partial $(D_{12} f)(a, b)$ exists.
(ii) $(D_{12} f)(a, b) = (D_{21} f)(a, b)$.

Proof. Set $A := (D_{21} f)(a, b)$.

(i) Fix $\varepsilon > 0$, and choose $\delta > 0$ with $B_\delta(a, b) \subseteq E$ by openness.
After shrinking δ, $(\forall\, \mathbf{z} \in B_\delta(a, b))\ |(D_{21} f)(\mathbf{z}) - A| < \frac{\varepsilon}{2}$ by continuity at (a, b).
For $h, k \neq 0$ with $\sqrt{h^2 + k^2} < \delta$, let $Q_{h,k}$ be the coordinate rectangle.
Its opposite vertices are (a, b) and $(a + h, b + k)$.
Then $Q_{h,k} \subseteq B_\delta(a, b) \subseteq E$.
Let $\mathbf{z}_{h,k} \in \operatorname{int} Q_{h,k}$ be a rectangle mean-value point by Theorem 3.2.10.
Then $\frac{\Delta(f, Q_{h,k})}{hk} = (D_{21} f)(\mathbf{z}_{h,k})$.
Hence $\left| \frac{\Delta(f, Q_{h,k})}{hk} - A \right| < \frac{\varepsilon}{2}$.
Fix h with $0 < |h| < \delta$ and let $k \to 0$.
Since $D_2 f$ exists at $(a + h, b)$ and (a, b), $\frac{\Delta(f, Q_{h,k})}{hk} \to \frac{(D_2 f)(a+h,b) - (D_2 f)(a,b)}{h}$.
Therefore the last quotient differs from A by at most $\frac{\varepsilon}{2} < \varepsilon$.
This holds for every h with $0 < |h| < \delta$, so the quotient tends to A as $h \to 0$.
Thus $(D_{12} f)(a, b)$ exists.
(ii) The limit in part (i) equals $A = (D_{21} f)(a, b)$ by the definition of A.

Structural Commentary: Continuity traps each shrinking rectangle quotient near one value.
Key Insight: The vertical limit recovers $D_2 f$; the horizontal limit produces $D_{12} f$. □

Corollary 3.2.12. *C^2 Mixed Partials Commute*
Let $E \subseteq \mathbb{R}^2$ be open, and let $f : E \to \mathbb{R}$ be C^2.
Then $(\forall\, (i, j) \in J_2 \times J_2)\, D_{ij} f = D_{ji} f$ on E.

Proof. Fix $p \in E$ and $(i, j) \in J_2 \times J_2$.
The derivative $D^2 f(p)$ exists by Definition 3.2.5(iii).
Hence $D_{ij} f(p) = D_{ji} f(p)$ by Corollary 3.2.8.
Structural Commentary: The C^2 hypothesis ensures second differentiability at every point.
Key Insight: Pointwise symmetry of $D^2 f$ makes the coordinate partials commute globally. □

Perspective 3.2.13. *Mixed Partials Need Hypotheses*
The quantities $D_{12}f$ and $D_{21}f$ arise from different iterated limits.
Existence of $D^2 f$ guarantees equality by Corollary 3.2.8.
Continuity of one mixed partial guarantees it by Theorem 3.2.11.

4 Inverse, Implicit, and Rank Theorems

4.1 Inverse Function Theorem

Perspective 4.1.1. *Linear Inverse Estimate*
Let $A \in \mathcal{L}(\mathbb{R}^n)$ be invertible, and set $c := \frac{1}{\|A^{-1}\|_{\mathrm{op}}} > 0$.
Then $(\forall\, \mathbf{h} \in \mathbb{R}^n)\, c\,\|\mathbf{h}\|_2 \le \|A\mathbf{h}\|_2 \le \|A\|_{\mathrm{op}}\,\|\mathbf{h}\|_2$ by Theorem 7.1.4(ii).
The inverse function theorem retains this lower stretch when the derivative varies little.

Figure 165 depicts the bound: $A(S^{n-1}) \cap B_c(\mathbf{0}) = \varnothing$.

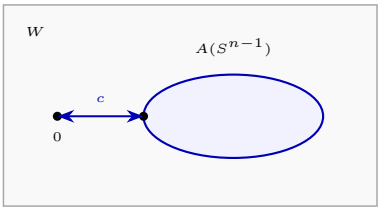

Figure 165: The image $A(S^{n-1})$ stays outside $B_c(\mathbf{0})$, where $c = \frac{1}{\|A^{-1}\|_{\mathrm{op}}}$.

Theorem 4.1.2. *Inverse Function Theorem*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} : E \to \mathbb{R}^n$ be C^1, let $\mathbf{a} \in E$, and set $\mathbf{b} := \mathbf{f}(\mathbf{a})$.
Assume $A := D\mathbf{f}(\mathbf{a})$ is invertible.
Then open neighborhoods $U \ni \mathbf{a}$ and $V \ni \mathbf{b}$ can be chosen so that:

(i) The restriction $\mathbf{f}|_U : U \to V$ is bijective; write $\mathbf{g} := (\mathbf{f}|_U)^{-1} : V \to U$.
(ii) The map $\mathbf{g}$ is differentiable on V, and $(\forall\, \mathbf{y} \in V)\, D\mathbf{g}(\mathbf{y}) = \big(D\mathbf{f}(\mathbf{g}(\mathbf{y}))\big)^{-1}$.
(iii) The map $\mathbf{g}$ is C^1 on V; equivalently, $\mathbf{f}|_U$ is a C^1 diffeomorphism.

Proof. Fix $\theta \in (0,1)$, and choose $r > 0$ with $\overline{B}_{2r}(\mathbf{a}) \subseteq E$ by openness.
After shrinking r, $(\forall\, \mathbf{x} \in \overline{B}_{2r}(\mathbf{a}))\ \big\|I - A^{-1}D\mathbf{f}(\mathbf{x})\big\|_{\mathrm{op}} \le \theta$ by continuity of $D\mathbf{f}$ at $\mathbf{a}$.
Set $V := \{\mathbf{y} \in \mathbb{R}^n : \big\|A^{-1}(\mathbf{y} - \mathbf{b})\big\|_2 < (1-\theta)r\}$.
The set V is an open neighborhood of $\mathbf{b}$ by continuity of the Euclidean norm and Theorem 7.1.4(i).
For $\mathbf{x} \in E$, set $\mathbf{R}_{\mathbf{x}}(\mathbf{h}) := \mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}$.
Then $\|\mathbf{R}_{\mathbf{x}}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$ by Definition 3.1.1.

(i) Fix $\mathbf{y} \in V$, set $q(\mathbf{x}) := \mathbf{x} - A^{-1}\mathbf{f}(\mathbf{x})$, and define $\varphi_{\mathbf{y}}(\mathbf{x}) := q(\mathbf{x}) + A^{-1}\mathbf{y}$.
The identity $q(\mathbf{x}+\mathbf{h}) - q(\mathbf{x}) - [I - A^{-1}D\mathbf{f}(\mathbf{x})]\mathbf{h} = -A^{-1}\mathbf{R}_{\mathbf{x}}(\mathbf{h})$ holds by the definition of q.
Moreover, $\big\|A^{-1}\mathbf{R}_{\mathbf{x}}(\mathbf{h})\big\|_2 \le \big\|A^{-1}\big\|_{\mathrm{op}}\,\|\mathbf{R}_{\mathbf{x}}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Thus $Dq(\mathbf{x}) = I - A^{-1}D\mathbf{f}(\mathbf{x})$ by Definition 3.1.1.
For $\mathbf{x}, \mathbf{z} \in \overline{B}_r(\mathbf{a})$, $\|\varphi_{\mathbf{y}}(\mathbf{x}) - \varphi_{\mathbf{y}}(\mathbf{z})\|_2 \le \theta\,\|\mathbf{x} - \mathbf{z}\|_2$ by Theorem 3.2.1 on $B_{2r}(\mathbf{a})$.
Since $\varphi_{\mathbf{y}}(\mathbf{a}) = \mathbf{a} + A^{-1}(\mathbf{y} - \mathbf{b})$, $(\forall\, \mathbf{x} \in \overline{B}_r(\mathbf{a}))\ \|\varphi_{\mathbf{y}}(\mathbf{x}) - \mathbf{a}\|_2 < r$ by the preceding bound and the definition of V.
Thus $\varphi_{\mathbf{y}} : \overline{B}_r(\mathbf{a}) \to \overline{B}_r(\mathbf{a})$ is a contraction.
Every Cauchy sequence in the closed ball converges in $\mathbb{R}^n$ by Corollary 4.2.8.
Its limit remains in the ball by continuity of the Euclidean norm, so $\overline{B}_r(\mathbf{a})$ is complete.
Fix the unique point $\mathbf{x}_{\mathbf{y}} \in \overline{B}_r(\mathbf{a})$ satisfying $\varphi_{\mathbf{y}}(\mathbf{x}_{\mathbf{y}}) = \mathbf{x}_{\mathbf{y}}$ by Theorem 4.5.3.
Then $\mathbf{x}_{\mathbf{y}} \in B_r(\mathbf{a})$ and $\mathbf{f}(\mathbf{x}_{\mathbf{y}}) = \mathbf{y}$ by the self-map estimate and the fixed-point identity.
The map $\mathbf{f}$ is continuous on E by Theorem 3.1.3(ii).
Set $U := B_r(\mathbf{a}) \cap \mathbf{f}^{-1}(V)$; it is open, and $\mathbf{a} \in U$ because $\mathbf{b} \in V$.
Every $\mathbf{y} \in V$ has the preimage $\mathbf{x}_{\mathbf{y}} \in U$, and any two preimages in U are fixed points of $\varphi_{\mathbf{y}}$.
Hence $\mathbf{f}|_U : U \to V$ is bijective by fixed-point uniqueness.

(ii) Fix $\mathbf{y} \in V$ and $\mathbf{k} \in \mathbb{R}^n$ with $\mathbf{y} + \mathbf{k} \in V$.
Set $\mathbf{x} := \mathbf{g}(\mathbf{y})$ and $\mathbf{h} := \mathbf{g}(\mathbf{y}+\mathbf{k}) - \mathbf{x}$; then $\mathbf{x}, \mathbf{x}+\mathbf{h} \in U \subseteq B_r(\mathbf{a})$.
The bound $\big\|\mathbf{h} - A^{-1}\mathbf{k}\big\|_2 \le \theta\,\|\mathbf{h}\|_2$ holds by Theorem 3.2.1 for q on $B_{2r}(\mathbf{a})$.
Therefore $(1-\theta)\,\|\mathbf{h}\|_2 \le \big\|A^{-1}\big\|_{\mathrm{op}}\,\|\mathbf{k}\|_2$, so $\mathbf{h} \to \mathbf{0}$ as $\mathbf{k} \to \mathbf{0}$.
The operator $A^{-1}D\mathbf{f}(\mathbf{x})$ is invertible by Theorem 7.1.5(i), since $\big\|A^{-1}D\mathbf{f}(\mathbf{x}) - I\big\|_{\mathrm{op}} \le \theta < 1$.
Hence $D\mathbf{f}(\mathbf{x})$ is invertible.
Then $\mathbf{g}(\mathbf{y}+\mathbf{k}) - \mathbf{g}(\mathbf{y}) - \big(D\mathbf{f}(\mathbf{x})\big)^{-1}\mathbf{k} = -\big(D\mathbf{f}(\mathbf{x})\big)^{-1}\mathbf{R}_{\mathbf{x}}(\mathbf{h})$ by $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = \mathbf{k}$.
The right-hand side is $o(\|\mathbf{k}\|_2)$ by Definition 3.1.1 and the preceding bound on $\mathbf{h}$.
Thus $D\mathbf{g}(\mathbf{y}) = \big(D\mathbf{f}(\mathbf{x})\big)^{-1} = \big(D\mathbf{f}(\mathbf{g}(\mathbf{y}))\big)^{-1}$ by Definition 3.1.1.

(iii) The map $\mathbf{g}$ is $\frac{\|A^{-1}\|_{\mathrm{op}}}{1-\theta}$-Lipschitz by the displacement bound in part (ii).
Hence $\mathbf{g}$ is continuous by Proposition 5.2.5.
Let $\iota(B) := B^{-1}$ on the open set of invertible operators.
The identity $D\mathbf{g} = \iota \circ D\mathbf{f} \circ \mathbf{g}$ holds by part (ii).
This composition is continuous by continuity of $D\mathbf{f}$ and $\mathbf{g}$ and Theorem 7.1.5(iii).
Therefore $\mathbf{g}$ is C^1 by Definition 3.2.2.
The restriction $\mathbf{f}|_U$ and its inverse $\mathbf{g}$ are C^1, so $\mathbf{f}|_U$ is a C^1 diffeomorphism by definition.

Structural Commentary: Contraction constructs inverse values; the same bound controls inverse remainders.
Key Insight: Closeness of $A^{-1}D\mathbf{f}$ to I controls local injectivity and $D\mathbf{g}$. □

Corollary 4.1.3. *Local Diffeomorphisms Are Open*
Let $E \subseteq \mathbb{R}^n$ be open, and let $\mathbf{f} : E \to \mathbb{R}^n$ be C^1.
Suppose $(\forall\, \mathbf{x} \in E)\, D\mathbf{f}(\mathbf{x})$ is invertible.

(i) The map $\mathbf{f}$ is open.
(ii) The map $\mathbf{f}$ is locally one-to-one.

Proof.

(i) Let $W \subseteq E$ be open, fix $\mathbf{y} \in \mathbf{f}(W)$, and choose $\mathbf{x} \in W$ with $\mathbf{f}(\mathbf{x}) = \mathbf{y}$.
Open neighborhoods $U \subseteq W$ of $\mathbf{x}$ and V of $\mathbf{y}$ with $\mathbf{f}(U) = V$ exist by Theorem 4.1.2(i) for $\mathbf{f}|_W$.
Thus $V \subseteq \mathbf{f}(W)$, so $\mathbf{y}$ is an interior point of $\mathbf{f}(W)$.
Hence $\mathbf{f}(W)$ is open because $\mathbf{y} \in \mathbf{f}(W)$ was arbitrary.
(ii) Fix $\mathbf{x} \in E$; an open neighborhood $U \ni \mathbf{x}$ on which $\mathbf{f}|_U$ is one-to-one exists by Theorem 4.1.2(i).

Structural Commentary: Image neighborhoods prove openness; inverse charts prove local injectivity.
Key Insight: An invertible derivative creates an open, one-to-one chart around every point. □

4.2 Implicit Function Theorem

Definition 4.2.1. *Block Splitting of a Linear Map*
For $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$ and $(\mathbf{h}, \mathbf{k}) \in \mathbb{R}^n \times \mathbb{R}^m$, define $A_x\mathbf{h} := A(\mathbf{h}, \mathbf{0})$ and $A_y\mathbf{k} := A(\mathbf{0}, \mathbf{k})$.
Then $A_x \in \mathcal{L}(\mathbb{R}^n)$, $A_y \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$, and $(\forall\, (\mathbf{h}, \mathbf{k}) \in \mathbb{R}^n \times \mathbb{R}^m)\, A(\mathbf{h}, \mathbf{k}) = A_x\mathbf{h} + A_y\mathbf{k}$.
If $E \subseteq \mathbb{R}^{n+m}$ is open and $\mathbf{f} : E \to \mathbb{R}^n$ is differentiable at $(\mathbf{x}, \mathbf{y}) \in E$, write $D_1\mathbf{f}(\mathbf{x}, \mathbf{y}) := (D\mathbf{f}(\mathbf{x}, \mathbf{y}))_x$ and $D_2\mathbf{f}(\mathbf{x}, \mathbf{y}) := (D\mathbf{f}(\mathbf{x}, \mathbf{y}))_y$.

Theorem 4.2.2. *Linear Implicit Function Theorem*
Let $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$ and suppose A_x is invertible. Then $(\forall\, \mathbf{k} \in \mathbb{R}^m)(\exists!\, \mathbf{h} \in \mathbb{R}^n)\, A(\mathbf{h}, \mathbf{k}) = \mathbf{0}$, and this solution is $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k}$.

Proof. Fix $\mathbf{k} \in \mathbb{R}^m$ and set $\mathbf{h}_0 := -(A_x)^{-1}A_y\mathbf{k}$.
By Definition 4.2.1, $A(\mathbf{h}_0, \mathbf{k}) = A_x\mathbf{h}_0 + A_y\mathbf{k} = -A_y\mathbf{k} + A_y\mathbf{k} = \mathbf{0}$, so a solution exists.
If $\mathbf{h} \in \mathbb{R}^n$ also satisfies $A(\mathbf{h}, \mathbf{k}) = \mathbf{0}$, then $A_x\mathbf{h} + A_y\mathbf{k} = \mathbf{0}$, and therefore $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k} = \mathbf{h}_0$.
Thus the solution exists and is unique.
Structural Commentary: Splitting A separates the unknown variable from the parameter.
Key Insight: Invertibility of A_x turns the constraint $A(\mathbf{h}, \mathbf{k}) = \mathbf{0}$ into a uniquely solvable linear equation for $\mathbf{h}$. □

The implicit theorem below is the nonlinear counterpart of the block equation theorem. The invertible block is the set of variables one can solve for, just as in Theorem 4.2.2.
Figure 166 depicts the local graph form: writing the solved variable as $\mathbf{x} = \mathbf{g}(\mathbf{y})$ and marking the parameter value $\mathbf{b}$, the zero locus is parameterized by $\mathbf{y} \mapsto (\mathbf{g}(\mathbf{y}), \mathbf{y})$ near $(\mathbf{g}(\mathbf{b}), \mathbf{b})$.

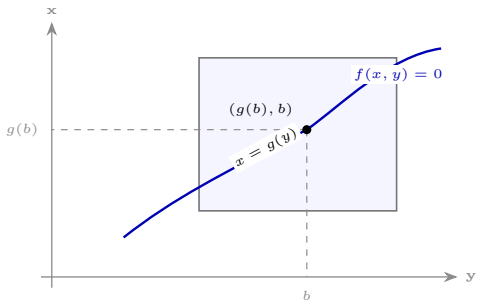

Figure 166: Inside the shaded product neighborhood, the blue zero locus is the graph $\mathbf{x} = \mathbf{g}(\mathbf{y})$ over the parameter neighborhood.

Theorem 4.2.3. *Implicit Function Theorem*
Let $E \subseteq \mathbb{R}^{n+m}$ be open, $\mathbf{f} \in C^1(E, \mathbb{R}^n)$, $(\mathbf{a}, \mathbf{b}) \in E$, and $\mathbf{f}(\mathbf{a}, \mathbf{b}) = \mathbf{0}$.
Put $A := D\mathbf{f}(\mathbf{a}, \mathbf{b})$, with blocks $A_x = D_1\mathbf{f}(\mathbf{a}, \mathbf{b})$ and $A_y = D_2\mathbf{f}(\mathbf{a}, \mathbf{b})$; assume A_x is invertible.
Then there exist open neighborhoods U of $\mathbf{a}$ and V of $\mathbf{b}$, and a unique map $\mathbf{g} : V \to U$, such that:

(i) $\{(\mathbf{x}, \mathbf{y}) \in U \times V : \mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{0}\} = \{(\mathbf{g}(\mathbf{y}), \mathbf{y}) : \mathbf{y} \in V\}$ and $\mathbf{g}(\mathbf{b}) = \mathbf{a}$.
(ii) $\mathbf{g} \in C^1(V, \mathbb{R}^n)$ and $D\mathbf{g}(\mathbf{y}) = -\big(D_1\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})\big)^{-1} D_2\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})$ for $\mathbf{y} \in V$; in particular, $D\mathbf{g}(\mathbf{b}) = -(A_x)^{-1}A_y$.

Proof. Let $\Phi(\mathbf{x}, \mathbf{y}) := (\mathbf{f}(\mathbf{x}, \mathbf{y}), \mathbf{y})$. Then Φ is C^1 and $D\Phi(\mathbf{a}, \mathbf{b})(\mathbf{h}, \mathbf{k}) = (A_x\mathbf{h} + A_y\mathbf{k}, \mathbf{k})$ by Definition 4.2.1. If this derivative sends $(\mathbf{h}, \mathbf{k})$ to $\mathbf{0}$, then $\mathbf{k} = \mathbf{0}$ and $A_x\mathbf{h} = \mathbf{0}$, so $\mathbf{h} = \mathbf{0}$. Thus $D\Phi(\mathbf{a}, \mathbf{b})$ is invertible by Theorem 1.7.7.
Open neighborhoods W of $(\mathbf{a}, \mathbf{b})$ and Z of $(\mathbf{0}, \mathbf{b})$ for which $\Phi : W \to Z$ is a C^1 diffeomorphism exist by Theorem 4.1.2.
The map $D_1\mathbf{f}$ is continuous, and the invertible operators form an open set by Theorem 7.1.5(ii). Hence open neighborhoods U of $\mathbf{a}$ and V_0 of $\mathbf{b}$ can be chosen so that $U \times V_0 \subseteq W$ and $D_1\mathbf{f}(\mathbf{x}, \mathbf{y})$ is invertible for every $(\mathbf{x}, \mathbf{y}) \in U \times V_0$.
By openness of Z, choose an open neighborhood V_1 of $\mathbf{b}$ such that $\{\mathbf{0}\} \times V_1 \subseteq Z$. The map $\mathbf{q} : V_1 \to W$ defined by $\mathbf{q}(\mathbf{y}) := \Phi^{-1}(\mathbf{0}, \mathbf{y})$ is C^1 and satisfies $\mathbf{q}(\mathbf{b}) = (\mathbf{a}, \mathbf{b})$. After replacing V_1 by a smaller neighborhood contained in V_0, one has $\mathbf{q}(V_1) \subseteq U \times V_0$. Set $V := V_1$; since the second component of $\mathbf{q}(\mathbf{y})$ is $\mathbf{y}$, one has $\mathbf{q}(V) \subseteq U \times V$.

(i) Define $\mathbf{g} : V \to U$ by $\mathbf{q}(\mathbf{y}) = (\mathbf{g}(\mathbf{y}), \mathbf{y})$. For $(\mathbf{x}, \mathbf{y}) \in U \times V$, one has $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ if and only if $\Phi(\mathbf{x}, \mathbf{y}) = (\mathbf{0}, \mathbf{y})$, which holds if and only if $(\mathbf{x}, \mathbf{y}) = \mathbf{q}(\mathbf{y}) = (\mathbf{g}(\mathbf{y}), \mathbf{y})$. Thus the zero set in $U \times V$ is the graph of $\mathbf{g}$. Moreover, $\mathbf{g}(\mathbf{b}) = \mathbf{a}$ because $\mathbf{q}(\mathbf{b}) = (\mathbf{a}, \mathbf{b})$, and the graph identity determines $\mathbf{g}$ uniquely.

(ii) The map $\mathbf{g}$ is C^1 as the first component of $\mathbf{q}$. Fix $\mathbf{y} \in V$ and $\mathbf{k} \in \mathbb{R}^m$. The derivative at $\mathbf{y}$ of the map $\mathbf{z} \mapsto (\mathbf{g}(\mathbf{z}), \mathbf{z})$ sends $\mathbf{k}$ to $(D\mathbf{g}(\mathbf{y})\mathbf{k}, \mathbf{k})$. Since $\mathbf{f}(\mathbf{g}(\mathbf{z}), \mathbf{z}) = \mathbf{0}$ for every $\mathbf{z} \in V$, the identity $D_1\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})D\mathbf{g}(\mathbf{y})\mathbf{k} + D_2\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})\mathbf{k} = \mathbf{0}$ holds by Theorem 3.1.5 and Definition 4.2.1.
The block $D_1\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})$ is invertible because $(\mathbf{g}(\mathbf{y}), \mathbf{y}) \in U \times V$. Therefore $D\mathbf{g}(\mathbf{y})\mathbf{k} = -\big(D_1\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})\big)^{-1} D_2\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})\mathbf{k}$ by Theorem 4.2.2, applied to $D\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})$.
Since $\mathbf{k}$ was arbitrary, this is the claimed operator identity. At $\mathbf{y} = \mathbf{b}$, it becomes $D\mathbf{g}(\mathbf{b}) = -(A_x)^{-1}A_y$ by $\mathbf{g}(\mathbf{b}) = \mathbf{a}$ and the definitions of A_x and A_y.

Structural Commentary: Adjoining the parameter as an output coordinate converts a level-set problem into a local inversion problem.
Key Insight: Invertibility of the solved-variable block both isolates the local graph and determines its derivative. □

4.3 Rank Theorem

Perspective 4.3.1. *Rank Theorem Linear Model*
Theorem 2.2.12 identifies the quotient by the kernel with the range, the linear kernel–range model.
The constant-rank theorem is the local C^1 analogue after coordinate changes.

Theorem 4.3.2. *Rank Theorem*
Let $E \subseteq \mathbb{R}^n$ be open, let $r \in \mathbb{N}_0$, let $\mathbf{F} \in C^1(E, \mathbb{R}^m)$ satisfy $(\forall\, \mathbf{x} \in E)\ \operatorname{rank} D\mathbf{F}(\mathbf{x}) = r$, and fix $\mathbf{a} \in E$.
Set $A := D\mathbf{F}(\mathbf{a})$ and $Y := \operatorname{range} A$, and let $P : \mathbb{R}^m \to \mathbb{R}^m$ be the orthogonal projection onto Y of Definition 2.2.2.
Then one can choose $\mathbf{x}_0 \in \mathbb{R}^n$, open neighborhoods $V_0 \ni \mathbf{x}_0$ and $U \ni \mathbf{a}$, a C^1 diffeomorphism $H : V_0 \to U$, and $\varphi \in C^1(A(V_0), \ker P)$ such that $\mathbf{F} \circ H = A + \varphi \circ A$ on V_0.

Proof.

(i) *Zero-Rank Case*: If $r = 0$, choose an open ball B about $\mathbf{a}$ contained in E. Since $D\mathbf{F} = 0$ on B, the map $\mathbf{F}$ is constant on B by Theorem 3.2.1(ii).
Take $V_0 = U = B$, $\mathbf{x}_0 = \mathbf{a}$, and $H = I_B$. Here $A = 0$, $Y = \{\mathbf{0}\}$, and $P = 0$; define $\varphi(\mathbf{0}) := \mathbf{F}(\mathbf{a})$. Then $\mathbf{F} \circ H = A + \varphi \circ A$ on V_0.

(ii) *Coordinate Construction*: Assume $r > 0$. Choose a basis $(\mathbf{y}_1, \dots, \mathbf{y}_r)$ of Y and vectors $\mathbf{s}_j \in \mathbb{R}^n$ with $A\mathbf{s}_j = \mathbf{y}_j$. By Proposition 1.5.8, the assignments $S\mathbf{y}_j = \mathbf{s}_j$ determine a linear map $S : Y \to \mathbb{R}^n$, and $AS = I_Y$. Define $G(\mathbf{x}) := \mathbf{x} + SP\big(\mathbf{F}(\mathbf{x}) - A\mathbf{x}\big)$.
Then $DG(\mathbf{a}) = I + SP\big(D\mathbf{F}(\mathbf{a}) - A\big) = I$. Open neighborhoods $U \ni \mathbf{a}$ and $V_0 \ni \mathbf{x}_0 := G(\mathbf{a})$ for which $G : U \to V_0$ is a C^1 diffeomorphism exist by Theorem 4.1.2(i),(iii); write $H := G^{-1}$.

(iii) *Rank Control*: Let $\Psi := \mathbf{F} \circ H$. Since $AS = I_Y$ and P fixes Y by Proposition 2.2.3(ii), $(\forall\, \mathbf{x} \in U)\ AG(\mathbf{x}) = A\mathbf{x} + P\big(\mathbf{F}(\mathbf{x}) - A\mathbf{x}\big) = P\mathbf{F}(\mathbf{x})$. Hence $P\Psi = A$ on V_0, and $(\forall\, \mathbf{x} \in V_0)\ PD\Psi(\mathbf{x}) = A$ by differentiating this identity.
Fix $\mathbf{x} \in V_0$. The operator $DH(\mathbf{x})$ is invertible by Theorem 4.1.2(ii), and $D\Psi(\mathbf{x}) = D\mathbf{F}(H(\mathbf{x}))DH(\mathbf{x})$ by Theorem 3.1.5. Thus $\operatorname{range} D\Psi(\mathbf{x}) = \operatorname{range} D\mathbf{F}(H(\mathbf{x}))$ and $\operatorname{rank} D\Psi(\mathbf{x}) = r$.
The restriction $P|_{\operatorname{range} D\Psi(\mathbf{x})} : \operatorname{range} D\Psi(\mathbf{x}) \to Y$ is surjective because $\operatorname{range}(PD\Psi(\mathbf{x})) = \operatorname{range} A = Y$. Its domain and codomain both have dimension r, so it is injective by Corollary 1.7.6(ii). Consequently, $(\forall\, \mathbf{v} \in \ker A)\ D\Psi(\mathbf{x})\mathbf{v} = \mathbf{0}$.

(iv) *Fiber Constancy*: For $\mathbf{v} \in \mathbb{R}^n$, one has $\mathbf{v} = SA\mathbf{v} + (\mathbf{v} - SA\mathbf{v})$, where $SA\mathbf{v} \in \operatorname{range} S$ and $\mathbf{v} - SA\mathbf{v} \in \ker A$. If $S\mathbf{y} \in \ker A$, then $\mathbf{y} = AS\mathbf{y} = \mathbf{0}$; hence $\mathbb{R}^n = \operatorname{range} S \oplus \ker A$.
The addition map $\operatorname{range} S \times \ker A \to \mathbb{R}^n$ is a linear isomorphism, hence a homeomorphism by Corollary 7.1.9(ii). Choose open convex neighborhoods C of $\mathbf{0}$ in $\operatorname{range} S$ and K of $\mathbf{0}$ in $\ker A$ such that $\mathbf{x}_0 + C + K \subseteq V_0$. Replace V_0 by the open product neighborhood $\mathbf{x}_0 + C + K$ and U by $H(V_0)$.
If $\ker A \neq \{\mathbf{0}\}$, each fiber $V_0 \cap (\mathbf{x} + \ker A)$ is relatively open and convex in the affine space $\mathbf{x} + \ker A$. The derivative of Ψ along the fiber is zero by part (iii), so the restriction of Ψ to the fiber is constant by Theorem 3.2.1(ii), after an affine identification of $\ker A$ with a Euclidean space. If $\ker A = \{\mathbf{0}\}$, every fiber is a singleton and the same conclusion holds.

(v) *Graph Representation*: If $A\mathbf{x}_1 = A\mathbf{x}_2$ for $\mathbf{x}_1, \mathbf{x}_2 \in V_0$, then $\mathbf{x}_1 - \mathbf{x}_2 \in \ker A$, so $\Psi(\mathbf{x}_1) = \Psi(\mathbf{x}_2)$ by part (iv). Thus $\varphi(A\mathbf{x}) := (I - P)\Psi(\mathbf{x})$ defines a map $\varphi : A(V_0) \to \ker P$ by Proposition 2.2.3(vii).
Since $P\Psi = A$, one has $\Psi = A + \varphi \circ A$ on V_0. Moreover, $A(V_0) = A\mathbf{x}_0 + A(C)$ is open in Y: the isomorphism $A|_{\operatorname{range} S} : \operatorname{range} S \to Y$ is a homeomorphism by Corollary 7.1.9(ii), and its inverse is S.
For $\mathbf{u} \in A(V_0)$, set $\sigma(\mathbf{u}) := \mathbf{x}_0 + S(\mathbf{u} - A\mathbf{x}_0)$. Then $\sigma(\mathbf{u}) \in \mathbf{x}_0 + C \subseteq V_0$ and $A\sigma(\mathbf{u}) = \mathbf{u}$, so $\varphi = (I - P)\Psi \circ \sigma$. Hence $\varphi \in C^1(A(V_0), \ker P)$.

Structural Commentary: Local inversion fixes the projected component; rank equality then forces constancy along the complementary fibers.
Key Insight: Constant rank converts the nonlinear map into a graph over the range of its derivative at the base point. □

Rank-nullity (Theorem 1.7.3) counts rank and kernel directions by $n = \operatorname{rank} A + \dim(\ker A)$. The rank theorem upgrades this dimension count: after local C^1 coordinate changes, a constant-rank map separates those directions.
Figure 167 contrasts the original curved fibers and image with the straight $\ker A$-fibers and projection image in local normal-form coordinates.

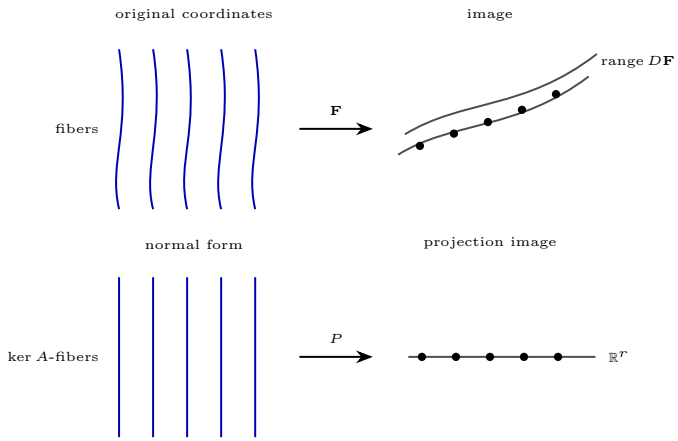

Figure 167: The upper panels depict curved fibers and image before normalization; the lower panels depict straight $\ker A$-fibers and projection onto the rank directions.

Theorem 4.3.3. *One-Constraint Lagrange Multiplier*
Let $E \subseteq \mathbb{R}^n$ be open, let $f, g : E \to \mathbb{R}$ be C^1, let $c \in \mathbb{R}$, and set $S = \{\mathbf{x} \in E : g(\mathbf{x}) = c\}$.
If $p \in S$, $\nabla g(p) \neq \mathbf{0}$, and $f|_S$ has a local extremum at p, then $(\exists\, \lambda \in \mathbb{R})\, \nabla f(p) = \lambda \nabla g(p)$.

Proof.

(i) *Reduction*: The standard-basis row of $Dg(p)$ is $\nabla g(p)^t$ by Theorem 3.1.9, so $Dg(p) \neq 0$. If $Df(p) \in \operatorname{span}\{Dg(p)\}$, choose $\lambda \in \mathbb{R}$ with $Df(p) = \lambda Dg(p)$; then $\nabla f(p) = \lambda \nabla g(p)$ by the same theorem, which proves the claim in this case. Otherwise $Df(p) \notin \operatorname{span}\{Dg(p)\}$, so $Dg(p), Df(p)$ are linearly independent.

(ii) *Coordinate Construction*: The dual space $(\mathbb{R}^n)'$ has dimension n by Theorem 4.1.4(ii). A basis $\big(Dg(p), Df(p), \ell_3, \dots, \ell_n\big)$ of $(\mathbb{R}^n)'$ extending $Dg(p), Df(p)$ exists by Theorem 1.3.13. Define $\Phi : E \to \mathbb{R}^n$ by $\Phi(\mathbf{x}) := \big(g(\mathbf{x}) - c,\ f(\mathbf{x}) - f(p),\ \ell_3(\mathbf{x} - p), \dots, \ell_n(\mathbf{x} - p)\big)$.

(iii) *Invertibility*: For $\mathbf{v} \in \mathbb{R}^n$, one has $D\Phi(p)\mathbf{v} = \big(Dg(p)\mathbf{v}, Df(p)\mathbf{v}, \ell_3(\mathbf{v}), \dots, \ell_n(\mathbf{v})\big)$. If this vector is zero, every member of the displayed dual-space basis vanishes at $\mathbf{v}$, and therefore every member of $(\mathbb{R}^n)'$ vanishes at $\mathbf{v}$. In particular, $0 = \mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|_2^2$. Thus $D\Phi(p)$ is injective and therefore invertible by Theorem 1.7.7.

(iv) *Contradiction and Conclusion*: By Theorem 4.1.2(i),(iii), Φ is a C^1 diffeomorphism from a neighborhood U of p onto an open neighborhood V of $\mathbf{0} = \Phi(p)$. Shrink U inside a neighborhood witnessing the local extremum and replace V by $\Phi(U)$. Choose $\varepsilon > 0$ so small that $(0, \pm\varepsilon, 0, \dots, 0) \in V$. Their inverse images lie in $S \cap U$ and satisfy $f - f(p) = \pm\varepsilon$, contradicting the local extremum.
The contradiction proves $Df(p) \in \operatorname{span}\{Dg(p)\}$. Choose $\lambda \in \mathbb{R}$ with $Df(p) = \lambda Dg(p)$; then $\nabla f(p) = \lambda \nabla g(p)$ by Theorem 3.1.9.

Structural Commentary: Independent differentials would make the constraint and objective separate local coordinates.
Key Insight: Extremality excludes objective variation transverse to the constraint differential, forcing $Df(p) \in \operatorname{span}\{Dg(p)\}$. □

4.4 Further Topics in Local Differential Theory

Proof Method 4.4.1. *Several-Variable Differentiation Proof Patterns*
Let $m, n \in \mathbb{N}$, let $E \subseteq \mathbb{R}^n$ be open, and let $\mathbf{f} : E \to \mathbb{R}^m$.
Select the pattern matching the target and verify its hypotheses.

(i) *Linear approximation*: At $\mathbf{x} \in E$ where $\mathbf{f}$ is differentiable, set $\mathbf{r}_{\mathbf{x}}(\mathbf{h}) := \mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}$.
The relation $\|\mathbf{r}_{\mathbf{x}}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$ holds by Definition 3.1.1.

(ii) *Norm control*: For $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, $(\forall\, \mathbf{h} \in \mathbb{R}^n)\ \|A\mathbf{h}\|_2 \le \|A\|_{\mathrm{op}} \|\mathbf{h}\|_2$ by Definition 5.1.4.
For $k \in \mathbb{N}$ and $C \in \mathcal{L}(\mathbb{R}^k, \mathbb{R}^n)$, $\|AC\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}} \|C\|_{\mathrm{op}}$ by Proposition 5.1.3(v) under standard-basis identification.

(iii) *Coordinate extraction*: At $\mathbf{x} \in E$ where $\mathbf{f} = (f_1, \dots, f_m)$ is differentiable, use the standard bases.
Then $(\forall\, (i, j) \in J_m \times J_n)\, [D\mathbf{f}(\mathbf{x})]_{ij} = (D_j f_i)(\mathbf{x})$ by Theorem 3.1.9(i).

(iv) *Line-segment reduction*: If E is convex and $\mathbf{f}$ is differentiable, fix $\mathbf{a}, \mathbf{b} \in E$.
Set $\gamma(t) := (1 - t)\mathbf{a} + t\mathbf{b}$; then $\gamma([0, 1]) \subseteq E$ by convexity.
For $M \ge 0$, assume $(\forall\, \mathbf{x} \in E)\ \|D\mathbf{f}(\mathbf{x})\|_{\mathrm{op}} \le M$.
Then $\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 \le M \|\mathbf{b} - \mathbf{a}\|_2$ by Theorem 3.2.1(i).
For $t \in [0, 1]$, $(\mathbf{f} \circ \gamma)'(t) = D\mathbf{f}(\gamma(t))(\mathbf{b} - \mathbf{a})$ by Theorem 3.1.5(ii).

(v) *Higher derivatives*: Fix $p \in E$ and $q \in \mathbb{Z}_{\ge 1}$.
If $D^q\mathbf{f}(p)$ exists, then $D^q\mathbf{f}(p) \in \mathcal{L}^q(\mathbb{R}^n, \mathbb{R}^m)$ by Definition 3.2.5(i).
If $D^2\mathbf{f}(p)$ exists, use the standard bases.
Then $(\forall\, (i, j, \ell) \in J_n \times J_n \times J_m)\, D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j) = D_{ij} f_\ell(p)$ by Theorem 3.2.6(i).

(vi) *Local inversion*: If $m = n$ and $\mathbf{f} \in C^1(E, \mathbb{R}^n)$, fix $\mathbf{a} \in E$.
Set $A := D\mathbf{f}(\mathbf{a})$ and suppose A is invertible. Choose a closed ball $B \subseteq E$ about $\mathbf{a}$.
It is complete by Theorem 4.2.5 and Corollary 4.1.24.
Choose $\theta \in (0, 1)$ so that $(\forall\, \mathbf{x} \in B)\ \big\|I - A^{-1} D\mathbf{f}(\mathbf{x})\big\|_{\mathrm{op}} \le \theta$.
For $\mathbf{y}$ near $\mathbf{f}(\mathbf{a})$, verify that $\varphi_{\mathbf{y}}(\mathbf{x}) := \mathbf{x} + A^{-1}(\mathbf{y} - \mathbf{f}(\mathbf{x}))$ is a self-map and a θ-contraction.
The unique fixed point $\mathbf{x}_{\mathbf{y}}$ exists by Theorem 4.5.3.
The identity $\mathbf{f}(\mathbf{x}_{\mathbf{y}}) = \mathbf{y}$ holds by the fixed-point equation.
Local bijectivity follows by Theorem 4.1.2(i).

(vii) *Implicit linearization*: For $L \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$, form L_x and L_y by Definition 4.2.1.
If L_x is invertible, fix $\mathbf{k} \in \mathbb{R}^m$.
The unique solution of $L(\mathbf{h}, \mathbf{k}) = \mathbf{0}$ is $\mathbf{h} = -(L_x)^{-1} L_y \mathbf{k}$ by Theorem 4.2.2.

(viii) *Rank normal form*: If $\mathbf{f} \in C^1(E, \mathbb{R}^m)$ has constant rank, fix $\mathbf{a} \in E$.
Set $A := D\mathbf{f}(\mathbf{a})$ and $Y := \operatorname{range} A$.
The isomorphism $\mathbb{R}^n / \ker A \cong Y$ holds by Theorem 2.2.12(iv).
Let P be the orthogonal projection onto Y of Definition 2.2.2.
A point $\mathbf{x}_0 \in \mathbb{R}^n$ and open neighborhoods $V_0 \ni \mathbf{x}_0$ and $U \ni \mathbf{a}$ exist by Theorem 4.3.2.
A C^1 diffeomorphism $H : V_0 \to U$ and $\varphi \in C^1(A(V_0), \ker P)$ exist by Theorem 4.3.2.
It satisfies $\mathbf{f} \circ H = A + \varphi \circ A$ on V_0 by Theorem 4.3.2.

Key Insight: Remainder bounds and derivative geometry control local nonlinear structure.

5 Differential Forms and Stokes Theory

5.1 Multiple Integration and Change of Variables

This subsection develops integration on cells, compactly supported integrals, and the determinant factor in change of variables. These constructions support the differential-form and manifold integrals that follow.

Definition 5.1.1. *k-Cells and Multiple Integrals*
Fix $k \in \mathbb{Z}_{\geq 1}$ and real endpoints satisfying $(\forall\, i \in J_k)\, a_i \leq b_i$. Use the convention $\int_a^a h = 0$.

(i) Set $I^j := \prod_{i=1}^{j} [a_i, b_i] \subseteq \mathbb{R}^j$ for $0 \leq j \leq k$, with I^0 a point. The *k-cell* is I^k.

(ii) For $f \in C(I^k)$, set $f_k := f$ and $f_{j-1}(x_1, \dots, x_{j-1}) := \int_{a_j}^{b_j} f_j(x_1, \dots, x_j)\, dx_j$, $j = k, \dots, 1$.
The *multiple integral* is $\int_{I^k} f(\mathbf{x})\, d\mathbf{x} := f_0$, also written $\int_{I^k} f$.

For bounded functions on rectangles, the Darboux envelopes from Definition 1.2.1 generalize to grid rectangles.

Definition 5.1.2. *Rectangular Grids and Riemann Sums*
Fix $m, n \in \mathbb{N}$. Let $R = [a, b] \times [c, d] \subseteq \mathbb{R}^2$ and let $f : R \to \mathbb{R}$ be bounded.
Let $P : a = x_0 < \cdots < x_m = b$, let $Q : c = y_0 < \cdots < y_n = d$, and set $G := P \times Q$.

(i) The *grid rectangles* are $R_{ij} := [x_{i-1}, x_i] \times [y_{j-1}, y_j]$, with $|R_{ij}| := (x_i - x_{i-1})(y_j - y_{j-1})$.

(ii) A *tagging* is $S = \big((s_{ij}, t_{ij})\big)_{(i,j) \in J_m \times J_n}$ with $(\forall\, (i, j) \in J_m \times J_n)\, (s_{ij}, t_{ij}) \in R_{ij}$.
The *Riemann sum* is $R(f, G, S) := \sum_{i=1}^{m} \sum_{j=1}^{n} f(s_{ij}, t_{ij})\, |R_{ij}|$.

(iii) Set $(m_{ij}, M_{ij}) := (\inf_{R_{ij}} f, \sup_{R_{ij}} f)$.
Set $L(f, G) := \sum_{i=1}^{m} \sum_{j=1}^{n} m_{ij}\, |R_{ij}|$ and $U(f, G) := \sum_{i=1}^{m} \sum_{j=1}^{n} M_{ij}\, |R_{ij}|$.

(iv) The *lower* and *upper integrals* are $\underline{\int_R} f := \sup_G L(f, G)$ and $\overline{\int_R} f := \inf_G U(f, G)$.
The extrema range over rectangular grids. If they agree, f is Riemann integrable on R, and $\int_R f$ denotes the common value.

Figure 168 marks a grid rectangle and its tag, the local Riemann-sum data.

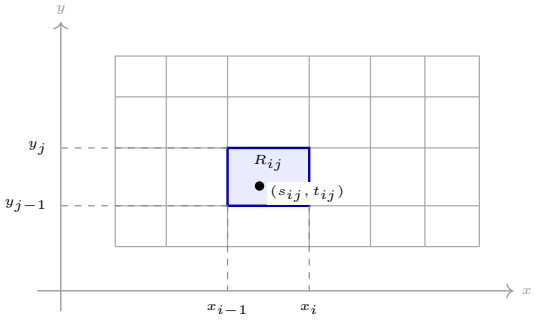

Figure 168: A grid rectangle R_{ij} and sample point (s_{ij}, t_{ij}).

Theorem 5.1.3. *Fubini on a Rectangle, Darboux Form*
Let $a < b$, $c < d$, $R = [a, b] \times [c, d]$, and let $f : R \to \mathbb{R}$ be Riemann integrable. Set $f_y(x) := f(x, y)$.
For $y \in [c, d]$, define $\underline{F}(y) := \sup_P L(P, f_y)$ and $\overline{F}(y) := \inf_P U(P, f_y)$.
Here P ranges over partitions of $[a, b]$.

(i) $\underline{F}, \overline{F}$ are Riemann integrable on $[c, d]$.

(ii) $\int_R f = \int_c^d \underline{F}(y)\, dy = \int_c^d \overline{F}(y)\, dy$.

(iii) If $(\forall\, y \in [c, d])\, f_y \in \mathcal{R}[a, b]$, then $y \mapsto \int_a^b f(x, y)\, dx$ is Riemann integrable.
In that case, $\int_R f = \int_c^d \int_a^b f(x, y)\, dx\, dy$.

Proof.

(i) Fix $\varepsilon > 0$ and write $I := \int_R f$.
Choose grids G_L, G_U with $I - L(f, G_L) < \frac{\varepsilon}{2}$ and $U(f, G_U) - I < \frac{\varepsilon}{2}$ by the Darboux definitions.
The common-refinement grid $G = P \times Q$ satisfies $U(f, G) - L(f, G) < \varepsilon$ because subdivision raises lower sums and lowers upper sums.
Both $\underline{F}$ and $\overline{F}$ are bounded between $\inf_R f$ and $\sup_R f$ by their definitions.

The slice definitions imply $L(f,G) \le L(Q,\underline{F}) \le U(Q,\overline{F}) \le U(f,G)$.
Since $\underline{F} \le \overline{F}$, both Darboux gaps are at most $U(Q,\overline{F}) - L(Q,\underline{F}) < \varepsilon$.
Thus $\underline{F}, \overline{F} \in \mathcal{R}[c,d]$ by Theorem 2.1.6 with $\alpha(t) = t$.
The inequality $0 \le \int_c^d \overline{F} - \int_c^d \underline{F} \le U(Q,\overline{F}) - L(Q,\underline{F}) < \varepsilon$ holds.
Thus their integrals are equal because $\varepsilon > 0$ was arbitrary.

(ii) Write I for the common integral in (i), and fix a product grid G.
Then $L(f,G) \le I \le U(f,G)$ by the slice inequalities and the definitions of the Darboux integrals.
Hence $I = \int_R f$ by the equality of the lower and upper rectangular integrals.

(iii) Under the slice hypothesis, $\underline{F} = \overline{F} = \left(y \mapsto \int_a^b f(x,y)\,dx\right)$ on $[c,d]$ by the one-variable Darboux definitions.
The claimed integrability and iterated-integral identity follow by (i)–(ii).

Structural Commentary: One product-grid gap controls both slice gaps and their common integral.
Key Insight: The cross-gap dominates both one-variable integrability defects. □

Theorem 5.1.4. *Order Independence on Cells*
Fix $k \in \mathbb{Z}_{\ge 1}$ and a k-cell I^k as in Definition 5.1.1.
For $f \in C(I^k)$, all iterated orders exist and equal $\int_{I^k} f$.

Proof. If $(\exists\, i \in J_k)\, a_i = b_i$, every iterated integral is zero; assume $(\forall\, i \in J_k)\, a_i < b_i$.
The function f is uniformly continuous by Theorem 5.2.6.
The defining estimate preserves continuity under one-coordinate integration. Thus every iterated order exists.
Fix an adjacent pair in one order and hold the other variables fixed.
Its two-variable slice is continuous, so the pair can be interchanged by Theorem 5.1.3(iii).
Every permutation reaches the defining order by finitely many adjacent swaps.
Move its first variable into position, then repeat on the remaining positions. Hence every iterated order equals $\int_{I^k} f$.
Structural Commentary: Continuity preserves intermediate integrals; adjacent swaps transfer two-variable Fubini to every order.
Key Insight: Adjacent transpositions reduce all orders to rectangular Fubini. □

Perspective 5.1.5. *Order Interchange Requires Hypotheses*
Finite sums are order-free; limiting processes are not.
Theorems 5.1.3 and 5.1.4 state hypotheses for interchanging integral order.

Definition 5.1.6. *Support and Simplex Integrals*
Fix $k \in \mathbb{Z}_{\ge 1}$ and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Integrals of complex-valued functions are taken componentwise.
For $f : \mathbb{R}^k \to \mathbb{F}$, define $\operatorname{supp} f := \overline{\{\mathbf{x} \in \mathbb{R}^k : f(\mathbf{x}) \ne 0\}}$.
Set $C_c(\mathbb{R}^k, \mathbb{F}) := \{f \in C(\mathbb{R}^k, \mathbb{F}) : \operatorname{supp} f \text{ is compact}\}$.

(i) For $f \in C_c(\mathbb{R}^k, \mathbb{F})$ and a cell $I^k \supseteq \operatorname{supp} f$, define $\int_{\mathbb{R}^k} f := \int_{I^k} f$.
This value is independent of I^k by repeated one-variable additivity.

(ii) Let Q^0 be a point. For $j \in J_k$, set $Q^j := \{\mathbf{x} \in \mathbb{R}^j : (\forall\, i \in J_j)\, x_i \ge 0,\ \sum_{i=1}^{j} x_i \le 1\}$.

The *standard k-simplex* is Q^k.
For $f \in C(Q^k, \mathbb{F})$, set $f_k := f$.

For $j = k, \dots, 1$, set $f_{j-1}(x_1, \dots, x_{j-1}) := \int_0^{1 - \sum_{i=1}^{j-1} x_i} f_j(x_1, \dots, x_j)\,dx_j$.
Define $\int_{Q^k} f := f_0$.

Theorem 5.1.7. *Linear Determinant Change of Variables*
Fix $n \in \mathbb{N}$. Let $T \in \mathcal{L}(\mathbb{R}^n)$ be invertible and let $f \in C_c(\mathbb{R}^n, \mathbb{R})$.
Then $\int_{\mathbb{R}^n} f(\mathbf{y})\,d\mathbf{y} = |\det T| \int_{\mathbb{R}^n} f(T\mathbf{x})\,d\mathbf{x}$.

Proof. Set $A := [T]$. Elementary matrices $E_1, \dots, E_q$ with $I = E_q \cdots E_1 A$ exist by Lemma 4.1.5(ii).
Hence $A = E_1^{-1} \cdots E_q^{-1}$, and every E_j^{-1} is elementary by Proposition 4.1.2(ii).
Let S be the linear map of one elementary matrix. Its integral identity follows by cases:

(a) Suppose S scales coordinate i by $c > 0$, and integrate in x_i first.
Then $\int_{\mathbb{R}^n} f(S\mathbf{x})\,d\mathbf{x} = \frac{1}{c} \int_{\mathbb{R}^n} f(\mathbf{y})\,d\mathbf{y}$ by Theorem 2.3.6(v).
If $c < 0$, reflected partitions reduce the active coordinate to scaling by $|c|$.
In both cases, $|\det S| = |c|$ by Theorem 3.8.2(v).

(b) If S swaps two coordinates, its integral is unchanged by Theorem 5.1.4.
The identity $|\det S| = 1$ holds by Theorem 3.8.2(i),(iii),(ix).

(c) Suppose $S\mathbf{x} = \mathbf{x} + cx_j\mathbf{e}_i$ with $i \ne j$, and integrate in x_i first.
For fixed remaining coordinates, $y_i = x_i + cx_j$ is a translation, so the integral is unchanged by Theorem 2.3.6(v).
The identity $\det S = 1$ holds by Theorem 3.8.2(v).

Thus the formula holds for the elementary factors.
The determinant identity $\det(PQ) = \det(P)\det(Q)$ holds by Theorem 3.8.2(vi).
Repeated substitution through $A = E_1^{-1} \cdots E_q^{-1}$ therefore establishes the stated identity.
Structural Commentary: Elementary factorization reduces the integral identity to scaling, swapping, and shear substitutions.
Key Insight: Elementary Jacobian factors multiply to $|\det T|$. □

Figure 169 depicts the linear density factor in Theorem 5.1.7.
In the illustrated two-dimensional case, T carries a cell to a parallelogram.

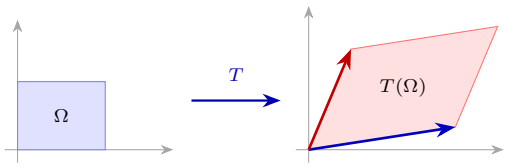

Figure 169: The image parallelogram has the determinant density factor $|\det T|$.

Definition 5.1.8. *Primitive Mappings and Flips*
Fix $n \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^n$ be open.

(i) For $m \in J_n$, a map $G : E \to \mathbb{R}^n$ is *primitive in coordinate m* if $(\exists\, g : E \to \mathbb{R})(\forall\, \mathbf{x} \in E)\, G(\mathbf{x}) = \mathbf{x} + (g(\mathbf{x}) - x_m)\mathbf{e}_m$.
(ii) For distinct $p, q \in J_n$, let $\tau = (p\ q) \in \mathfrak{S}_n$. A *flip* is the unique $B \in \mathcal{L}(\mathbb{R}^n)$ satisfying $(\forall\, i \in J_n)\, B\mathbf{e}_i = \mathbf{e}_{\tau(i)}$.
Existence and uniqueness hold by Proposition 1.5.8.

Proposition 5.1.9. *Derivative and Jacobian of a Primitive Map*
Fix $n \in \mathbb{N}$. Let $E \subseteq \mathbb{R}^n$ be open and let $m \in J_n$.
Let $G : E \to \mathbb{R}^n$ be primitive in coordinate m, with component function $g : E \to \mathbb{R}$.
Suppose g is differentiable at $\mathbf{a} \in E$.

(i) $DG(\mathbf{a})$ is I_n with row m replaced by $\nabla g(\mathbf{a})^t$.
(ii) $J_G(\mathbf{a}) = (D_m g)(\mathbf{a})$.

Proof.

(i) The components of G are coordinate maps except for g.
Thus G is differentiable at $\mathbf{a}$ by Theorem 3.1.6(i).
The stated matrix holds by Theorem 3.1.9(i).
(ii) Only the identity permutation contributes to the determinant of the matrix in (i) by Definition 3.8.1.
Its contribution is $(D_m g)(\mathbf{a})$.

Structural Commentary: Componentwise differentiation replaces one identity row; the permutation formula isolates its diagonal entry.
Key Insight: Every off-diagonal entry conflicts with an unchanged coordinate row. □

Theorem 5.1.10. *Local Decomposition into Primitive Mappings*
Fix $n \in \mathbb{N}$. Let $E \subseteq \mathbb{R}^n$ be open and let $\mathbf{F} \in C^1(E, \mathbb{R}^n)$.
Assume $\mathbf{0} \in E$, $\mathbf{F}(\mathbf{0}) = \mathbf{0}$, and $D\mathbf{F}(\mathbf{0})$ is invertible.
Then U, $(G_i)_{i \in J_n}$, and $(B_i)_{i \in J_{n-1}}$ exist such that:

(i) For $i \in J_n$, G_i is primitive and C^1 near $\mathbf{0}$, with $G_i(\mathbf{0}) = \mathbf{0}$ and $\det DG_i(\mathbf{0}) \neq 0$.
For $i \in J_{n-1}$, $B_i \in \mathcal{L}(\mathbb{R}^n)$ is a flip or I.
(ii) The set $U \subseteq E$ is open and $\mathbf{0} \in U$.
The identity $\mathbf{F} = B_1 \circ \cdots \circ B_{n-1} \circ G_n \circ \cdots \circ G_1$ holds on U.
The B-prefix is omitted when $n = 1$.

Proof. For $0 \leq m \leq n$, set $P_m\mathbf{x} := \sum_{i=1}^{m} x_i \mathbf{e}_i$, and set $\mathbf{F}_1 := \mathbf{F}$.

(i) Suppose $1 \leq m < n$, $P_{m-1} \circ \mathbf{F}_m = P_{m-1}$, $\mathbf{F}_m(\mathbf{0}) = \mathbf{0}$, and $D\mathbf{F}_m(\mathbf{0})$ is invertible.
Write $\mathbf{F}_m = P_{m-1} + \sum_{i=m}^{n} \alpha_i \mathbf{e}_i$.
If $(\forall\, k \in \{m, \ldots, n\})\, (D_m \alpha_k)(\mathbf{0}) = 0$, then $D\mathbf{F}_m(\mathbf{0})\mathbf{e}_m = \mathbf{0}$, contrary to invertibility.
Choose $k \in \{m, \ldots, n\}$ with $(D_m \alpha_k)(\mathbf{0}) \neq 0$.
Let $B_m = I$ when $k = m$ and otherwise let B_m be the flip of coordinates m, k.
Set $G_m(\mathbf{x}) := \mathbf{x} + (\alpha_k(\mathbf{x}) - x_m)\mathbf{e}_m$.
Then G_m is primitive and C^1, and $G_m(\mathbf{0}) = \mathbf{0}$.
Its Jacobian satisfies $J_{G_m}(\mathbf{0}) = (D_m \alpha_k)(\mathbf{0}) \neq 0$ by Proposition 5.1.9(ii).
Thus $DG_m(\mathbf{0})$ is invertible by Theorem 3.8.2(vii).
The map G_m is a C^1 diffeomorphism between neighborhoods of $\mathbf{0}$ by Theorem 4.1.2(i),(iii).
On these neighborhoods, set $\mathbf{F}_{m+1} := B_m \circ \mathbf{F}_m \circ G_m^{-1}$.
The first m components of $\mathbf{F}_{m+1}$ are the coordinate maps by the definitions of B_m and G_m.
The recurrence is C^1 and satisfies $\mathbf{F}_{m+1}(\mathbf{0}) = \mathbf{0}$.
Its derivative is $D\mathbf{F}_{m+1}(\mathbf{0}) = B_m D\mathbf{F}_m(\mathbf{0}) D(G_m^{-1})(\mathbf{0})$ by Theorem 3.1.5(ii).
Each factor is invertible by the definition of a flip, the induction hypothesis, and Theorem 4.1.2(ii).
Thus $D\mathbf{F}_{m+1}(\mathbf{0})$ is invertible.
Iteration for $m = 1, \ldots, n-1$ is valid after shrinking to common neighborhoods.
The final map $G_n := \mathbf{F}_n$ is primitive in coordinate n, fixes $\mathbf{0}$, and has invertible derivative.
Its determinant is nonzero by Theorem 3.8.2(vii), so Claim (i) follows by this induction.
(ii) Since $B_m^2 = I$ and $\mathbf{F}_{m+1} = B_m \circ \mathbf{F}_m \circ G_m^{-1}$, one has $\mathbf{F}_m = B_m \circ \mathbf{F}_{m+1} \circ G_m$.
Restrict the maps to a common open neighborhood $U \subseteq E$ of $\mathbf{0}$.
Successive substitution with $G_n = \mathbf{F}_n$ establishes $\mathbf{F} = B_1 \circ \cdots \circ B_{n-1} \circ G_n \circ \cdots \circ G_1$ on U.

Structural Commentary: Each primitive inverse fixes one more coordinate; the flip moves a nonzero pivot into place.
Key Insight: Invertibility prevents the next derivative column from vanishing. □

Theorem 5.1.11. *Partition of Unity on a Compact Set*
Fix $n \in \mathbb{N}$. Let $K \subseteq \mathbb{R}^n$ be nonempty and compact, and let $(V_\alpha)_{\alpha \in \mathcal{A}}$ be an open cover of K.
Then there exist $s \in \mathbb{N}$ and $\psi_1, \ldots, \psi_s \in C^1(\mathbb{R}^n)$ such that

(i) $(\forall\, i \in J_s)\, 0 \leq \psi_i \leq 1$;
(ii) $(\forall\, i \in J_s)(\exists\, \alpha \in \mathcal{A})\ \operatorname{supp}(\psi_i) \subseteq V_\alpha$;

(iii) $\sum_{i=1}^{s} \psi_i = 1$ *on* K.

(iv) If $f \in C(\mathbb{R}^n)$ *and* $\operatorname{supp}(f) \subseteq K$, *then* $f = \sum_{i=1}^{s} \psi_i f$.
The summands satisfy $(\forall\, i \in J_s)(\exists\, \alpha \in \mathcal{A})\ \operatorname{supp}(\psi_i f) \subseteq V_\alpha$.

Proof. Choose $\alpha(\mathbf{x}) \in \mathcal{A}$ and $0 < r_{\mathbf{x}} < R_{\mathbf{x}}$ with $\overline{B}_{R_{\mathbf{x}}}(\mathbf{x}) \subseteq V_{\alpha(\mathbf{x})}$ for $\mathbf{x} \in K$ by openness.
Points $\mathbf{x}_1, \dots, \mathbf{x}_s \in K$ with $K \subseteq \bigcup_{i=1}^{s} B_{r_i}(\mathbf{x}_i)$ exist by Definition 4.1.1.
Set $r_i := r_{\mathbf{x}_i}$ and $R_i := R_{\mathbf{x}_i}$ for $i \in J_s$.
Let $\eta(t)$ equal 1 for $t \le 0$, $1 - 3t^2 + 2t^3$ for $0 \le t \le 1$, and 0 for $t \ge 1$.
Then $\eta \in C^1(\mathbb{R})$, $0 \le \eta \le 1$, and $\eta'(0) = \eta'(1) = 0$.
Set $\varphi_i(\mathbf{x}) := \eta\left(\frac{\|\mathbf{x}-\mathbf{x}_i\|_2^2 - r_i^2}{R_i^2 - r_i^2}\right)$.
Set $\psi_i := \prod_{j=1}^{i-1}(1-\varphi_j)\varphi_i$.

(i) If $i \in J_s$, then $\varphi_i, \psi_i \in C^1(\mathbb{R}^n)$ and $0 \le \psi_i \le 1$ by the construction and $0 \le \eta \le 1$.
(ii) If $i \in J_s$, then $\operatorname{supp}(\psi_i) \subseteq \operatorname{supp}(\varphi_i) \subseteq \overline{B}_{R_i}(\mathbf{x}_i) \subseteq V_{\alpha(\mathbf{x}_i)}$ by the definition of η.
(iii) The telescoping identity $\sum_{i=1}^{s} \psi_i = 1 - \prod_{i=1}^{s}(1-\varphi_i)$ holds.
If $\mathbf{x} \in K$, then $(\exists\, i \in J_s)\, \varphi_i(\mathbf{x}) = 1$ by the finite inner-ball cover, so the product vanishes.
(iv) Multiplication of (iii) by f establishes $f = \sum_{i=1}^{s} \psi_i f$ on K.
Both sides vanish outside K.
If $i \in J_s$, then $\operatorname{supp}(\psi_i f) \subseteq V_{\alpha(\mathbf{x}_i)}$ by (ii).
Structural Commentary: Finite inner balls localize cubic cutoffs; telescoping products assemble the exact sum.
Key Insight: Coverage makes one cutoff equal 1, annihilating the residual product. □

Theorem 5.1.7 identifies the determinant factor for linear maps.
Local primitive decompositions recover this factor, and compact support globalizes it.
Figure 170 depicts the corresponding local grid deformation.

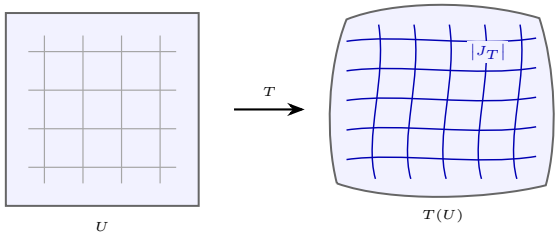

Figure 170: A C^1 change of variables warps a grid; $|J_T|$ records local volume scaling.

Theorem 5.1.12. *Change of Variables in* $\mathbb{R}^k$
Fix $k \in \mathbb{N}$. *Let* $E \subseteq \mathbb{R}^k$ *be open and let* $T : E \to \mathbb{R}^k$ *be one-to-one and* C^1.
Assume $(\forall\, \mathbf{x} \in E)\, J_T(\mathbf{x}) \neq 0$.
Let $f \in C(\mathbb{R}^k)$ *have compact support contained in* $T(E)$.
Then $\int_{\mathbb{R}^k} f(\mathbf{y})\, d\mathbf{y} = \int_{\mathbb{R}^k} f(T(\mathbf{x}))\, |J_T(\mathbf{x})|\, d\mathbf{x}$.
The right-hand integrand on E *is extended by* 0 *to* $\mathbb{R}^k$.

Proof. Set $K := \operatorname{supp}(f)$.
If $f = 0$, the identity is immediate; assume $f \neq 0$, so K is nonempty.
For $\mathbf{x} \in E$, $DT(\mathbf{x})$ is invertible by Theorem 3.8.2(vii).
Therefore T is open by Corollary 4.1.3(i), so injectivity makes $T^{-1} : T(E) \to E$ continuous.
Hence $T^{-1}(K)$ is compact by Theorem 5.1.37.
Set $\rho(\mathbf{x}) := f(T(\mathbf{x}))\, |J_T(\mathbf{x})|$ on E. Then ρ is continuous and vanishes on $E \setminus T^{-1}(K)$.
Since $T^{-1}(K)$ is a compact subset of open E, the zero extension of ρ is continuous with compact support.

(i) *Model Maps*: Let $\mathcal{C}$ be the class of local C^1 diffeomorphisms satisfying the formula for continuous test functions compactly supported in their images.
All integrands use zero extensions. For a primitive coordinate-m map G, restrict to an open box U where $(D_m g)$ has constant sign.
Given a test function h, the set $G^{-1}(\operatorname{supp} h)$ is compact by Theorem 5.1.37 and lies inside a closed cell in U.
The active-coordinate map is strictly monotone by Corollary 1.1.13.
Integrate in x_m first; compact support permits zero extension to containing closed cells.
The formula follows by Theorem 2.3.6(v) and Theorem 5.1.4.
For negative sign, reflected partitions reduce to the increasing case.
Flips belong to $\mathcal{C}$ by Theorem 5.1.4.
Coordinate translations are primitive maps with Jacobian 1.
(ii) *Composition*: Let $Q : U \to V$ and $P : V \to W$ belong to $\mathcal{C}$, and let h be continuous with compact support in W.
The function $h(P(\cdot))\, |J_P|$ is continuous and compactly supported in V because P^{-1} is continuous.
Apply the identity for P to h and that for Q to $h(P(\cdot))\, |J_P|$.
For the zero extensions, $\int_{\mathbb{R}^k} h(\mathbf{y})\, d\mathbf{y} = \int_{\mathbb{R}^k} h(P(Q(\mathbf{x})))\, |J_P(Q(\mathbf{x}))|\, \big|J_Q(\mathbf{x})\big|\, d\mathbf{x}$.
Thus $P \circ Q \in \mathcal{C}$ because $J_{P\circ Q}(\mathbf{x}) = J_P(Q(\mathbf{x}))J_Q(\mathbf{x})$ by Theorem 3.1.5(ii) and Theorem 3.8.2(vi).

(iii) *Localization*: Fix $\mathbf{a} \in E$ and set $E_{\mathbf{a}} := \{\mathbf{h} \in \mathbb{R}^k : \mathbf{a} + \mathbf{h} \in E\}$.
Define $\mathbf{F}_{\mathbf{a}} : E_{\mathbf{a}} \to \mathbb{R}^k$ by $\mathbf{F}_{\mathbf{a}}(\mathbf{h}) := T(\mathbf{a} + \mathbf{h}) - T(\mathbf{a})$.
Then $\mathbf{F}_{\mathbf{a}}(\mathbf{0}) = \mathbf{0}$ and $D\mathbf{F}_{\mathbf{a}}(\mathbf{0}) = DT(\mathbf{a})$ is invertible by Theorem 3.8.2(vii).
A factorization of $\mathbf{F}_{\mathbf{a}}$ into primitive maps and flips exists near $\mathbf{0}$ by Theorem 5.1.10.
Choose an open neighborhood $U_{\mathbf{a}} \ni \mathbf{a}$ small enough that Parts (i)–(ii) and the translations place $T|_{U_{\mathbf{a}}}$ in $\mathcal{C}$.
Since T is open, $(T(U_{\mathbf{a}}))_{\mathbf{a} \in E}$ is an open cover of K.

(iv) *Globalization*: Functions $\psi_1, \dots, \psi_s$ with $f = \sum_{i=1}^{s} \psi_i f$ exist by Theorem 5.1.11(iv).
Choose associated $\mathbf{a}_i$ so that $\operatorname{supp}(\psi_i f) \subseteq T(U_{\mathbf{a}_i})$.
Apply the local formula to the summands $\psi_i f$ and add.
Because the right-hand integrand vanishes outside $T^{-1}(K)$, the sum is the stated zero-extended integral over $\mathbb{R}^k$.

Structural Commentary: Primitive substitutions compose; a finite partition carries them across compact support.
Key Insight: Compact support converts compatible local Jacobian identities into one global integral. □

Computation Method 5.1.13. *Change-of-Variables Jacobian Method*
Fix $n \in \mathbb{N}$. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $\sigma : \Omega \to \mathbb{R}^n$ be one-to-one and C^1.
Assume $(\forall\, \mathbf{x} \in \Omega)\ \det D\sigma(\mathbf{x}) \neq 0$.
For continuous f with compact support in $\sigma(\Omega)$, the procedure rewrites the $\mathbf{y}$-integral as an integral over $\mathbf{x} \in \Omega$.

(i) Compute $D\sigma(\mathbf{x})$ and $J_\sigma(\mathbf{x}) = \det D\sigma(\mathbf{x})$.
(ii) Substitute $\mathbf{y} = \sigma(\mathbf{x})$ in the integrand, producing $f(\sigma(\mathbf{x}))$.
(iii) Multiply by the local volume factor $|J_\sigma(\mathbf{x})|$.
(iv) Express $\sigma^{-1}(\operatorname{supp} f)$ and the integration bounds in the $\mathbf{x}$-coordinates.
(v) Verify that the transformed integrand has compact support in Ω.
Then $\int_{\mathbb{R}^n} f(\mathbf{y})\, d\mathbf{y} = \int_{\mathbb{R}^n} f(\sigma(\mathbf{x}))\, |J_\sigma(\mathbf{x})|\, d\mathbf{x}$ by Theorem 5.1.12.

Example 5.1.14. *Polar and Spherical Jacobian Factors*
On $(0,\infty) \times (0, 2\pi)$, set $\sigma_{\mathrm{p}}(r,\theta) := (r\cos\theta, r\sin\theta)$.
Then $D\sigma_{\mathrm{p}}(r,\theta) = \left(\begin{smallmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{smallmatrix}\right)$ and $|\det D\sigma_{\mathrm{p}}(r,\theta)| = r$.
Thus $d\mathbf{x} = r\, dr\, d\theta$ on this coordinate patch by Theorem 5.1.12.
Use the spherical patch $(0,\infty) \times (0,\pi) \times (0, 2\pi)$.
Set $\sigma_{\mathrm{s}}(\rho,\varphi,\theta) := (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi)$.
The identity $\det D\sigma_{\mathrm{s}}(\rho,\varphi,\theta) = \rho^2 \sin\varphi > 0$ holds by direct differentiation.
Hence $d\mathbf{x} = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta$ by Theorem 5.1.12.

Proof Method 5.1.15. *Multiple-Integration Proof Patterns*
For a multiple-integration identity, select the reduction matching the obstruction and close the resulting equality.

(i) *Order swaps*: Reduce an integration-order permutation to adjacent swaps.
Equality follows by Theorem 5.1.4.
(ii) *Grid squeeze*: Compare the slice sums with one product-grid lower and upper sum.
Slice integrability and equality then follow by Theorem 5.1.3(i)–(ii).
(iii) *Compact support*: Enclose the support in a cell.
Remove the cell choice by Definition 5.1.6(i).
(iv) *Local form*: A C^1 coordinate change factors into primitive maps and flips by Theorem 5.1.10.
(v) *Jacobian product*: Combine local factors through $J_{P\circ Q} = (J_P \circ Q) J_Q$.
This identity holds by Theorem 3.1.5(ii) and Theorem 3.8.2(vi).
(vi) *Globalization*: Choose a finite partition subordinate to the local images by Theorem 5.1.11(iv).
Apply the local identity to the summands and add.

Key Insight: Adjacent swaps control order; primitive factors and compact support control coordinates.

5.2 Differential Forms and Manifolds

Differential forms package oriented integration on parametrized cells.
Definitions 6.1.5 and 6.1.9 construct their exterior powers.
Proposition 6.1.10 supplies the algebraic laws. Definition 3.8.1 fixes determinant signs.
Theorems 3.8.2, 1.2.5 and 5.1.12 supply the integration laws.

Perspective 5.2.1. *Forms as Functionals*
The notation $\int f\, dx$ treats dx as directed coordinate variation.
A 1-form such as $f\, dx + g\, dy$ assigns a scalar to a tangent direction;
its integral assigns a number to an oriented path.
The factor dx records signed x-variation, and $f\, dx$ weights it by f.

Figure 171 contrasts oriented paths C, C' from p to q.
The sign of y changes the contribution of $y\, dx$.

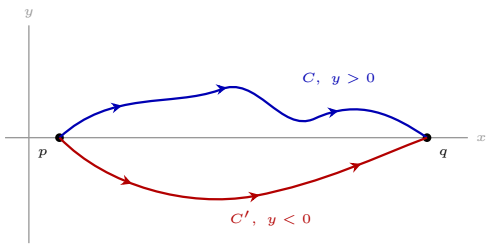

Figure 171: $y\, dx$ records signed, y-weighted x-variation.

Definition 5.2.2. *k-Surfaces and Differential Forms*
Fix $n \in \mathbb{N}$ and $k \in J_n$, and let $E \subseteq \mathbb{R}^n$ be open.
Let $(dx_1, \dots, dx_n)$ be the dual basis to the standard basis of $\mathbb{R}^n$ from Definition 4.1.3.
Regard these covectors as degree-one elements of the exterior algebra in Definition 6.1.9.

(i) A compact *parameter domain* $D \subseteq \mathbb{R}^k$ is either a nondegenerate k-cell I^k or Q^k.
These domains are defined in Definitions 5.1.1 and 5.1.6(ii), respectively.
For $r \in \mathbb{Z}_{\geq 0}$, $C^r(D)$ consists of maps extendable as C^r to an open neighborhood of D.
(ii) A *k-surface* in E is a C^1 map $\Phi : D \to E$ from a compact parameter domain $D \subseteq \mathbb{R}^k$.
(iii) For an increasing k-index $I = (i_1 < \cdots < i_k)$, set $dx_I := dx_{i_1} \wedge \cdots \wedge dx_{i_k}$.
A *k-form* on E is an expression $\omega = \sum_I b_I(\mathbf{x})\, dx_I$ over increasing k-indices, where $b_I \in C(E, \mathbb{R})$.
This expression is its *standard presentation*.
(iv) A 0-form is a continuous function on E.
A form is C^r iff every coefficient function in its standard presentation is C^r.

Remark 5.2.3. *Parametrized Cells, Not Just Images*
Parametrizations may fail to be injective or regular.
Forms therefore act on parametrized cells, not just images;
this distinction retains orientation, multiplicity, and cancellation.

Proposition 5.2.4. *Basic Forms and Standard Presentation*
Fix $n \in \mathbb{N}$ and $k \in J_n$, and let $E \subseteq \mathbb{R}^n$ be open.

(i) Swapping neighboring factors in $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ changes sign.
Any repeated index makes the word 0.
(ii) Let $\mathcal{A} \subseteq J_n^k$ be finite and $(a_A)_{A \in \mathcal{A}} \subseteq C(E, \mathbb{R})$.
Then $\sum_{A=(a_1,\dots,a_k)\in\mathcal{A}} a_A(\mathbf{x})\, dx_{a_1} \wedge \cdots \wedge dx_{a_k}$ reduces to a unique standard presentation.

Proof.

(i) A word with a repeated factor is 0 by Definition 6.1.5(iv),(vi).
For $p \neq q$, bilinearity implies $0 = (dx_p + dx_q) \wedge (dx_p + dx_q) = dx_p \wedge dx_q + dx_q \wedge dx_p$.
Exterior multiplication is associative by Proposition 6.1.10(iv),(v).
Thus the two-factor identity proves the neighboring-swap rule.
(ii) By (i), every word either vanishes or sorts to an increasing word with the sign of its sorting permutation.
The increasing words form a basis of $\bigwedge^k((\mathbb{R}^n)')$ by Theorem 6.1.7(ii)(a).
Hence the resulting coefficients are unique.

Structural Commentary: Exterior-algebra relations reduce arbitrary coordinate words to ordered basis words.
Key Insight: Antisymmetry converts every nonzero coordinate word to one increasing basis word. □

Definition 5.2.5. *Integration of a Differential Form*
Fix $n \in \mathbb{N}$ and $k \in J_n$, and let $E \subseteq \mathbb{R}^n$ be open.
Let $\Phi = (\Phi_1, \dots, \Phi_n) : D \to E$ be a C^1 k-surface.
Let $\omega = \sum_I b_I(\mathbf{x})\, dx_I$ be a k-form in standard presentation.

For an increasing k-index $I = (i_1, \dots, i_k)$, set $J_I\Phi(\mathbf{u}) := \det[(D_j\Phi_{i_r})(\mathbf{u})]_{r,j=1}^k$.
Define $\int_\Phi \omega := \int_D \sum_I b_I(\Phi(\mathbf{u}))J_I\Phi(\mathbf{u})\, d\mathbf{u}$.

Theorem 5.2.6. *Naturality Under Reparametrization*
Fix $n \in \mathbb{N}$ and $k \in J_n$, and let $E \subseteq \mathbb{R}^n$ be open.
Let $\Phi : D \to E$ be a C^1 k-surface, and let $D' \subseteq \mathbb{R}^k$ be a parameter domain.
Let $U', U \subseteq \mathbb{R}^k$ be open neighborhoods of D', D, respectively.
Let $\widetilde{R} : U' \to U$ be a C^1 diffeomorphism such that $\widetilde{R}(D') = D$.
Set $R := \widetilde{R}|_{D'}$ and suppose $\epsilon := \operatorname{sgn}\det D\widetilde{R}$ is constant on D'.
If ω is a k-form on E, then $\int_{\Phi\circ R}\omega = \epsilon\int_\Phi \omega$.

Proof.

(i) *Reduction*: By linearity, take $\omega = f\, dx_I$.
Set $h(\mathbf{v}) := f(\Phi(\mathbf{v}))J_I\Phi(\mathbf{v})$.
By Definition 5.2.2, Φ has a C^1 extension $\Phi_N : N \to E$ on an open set $D \Subset N \Subset U$.
The formula $h_N := (f \circ \Phi_N)J_I\Phi_N$ defines a continuous extension of h to N.
Compactness supplies $\rho \in C_c(\mathbb{R}^k, [0,1])$ such that $\rho|_D = 1$.
Choose $\operatorname{supp}(\rho) \subseteq N$.
Extending ρh_N by 0 produces $\widetilde{h} \in C_c(\mathbb{R}^k, \mathbb{R})$.
It satisfies $\widetilde{h}|_D = h$ and $\operatorname{supp}(\widetilde{h}) \subseteq U$.
(ii) *Jacobian and Cutoffs*: The composition-minor identity holds by Theorem 1.2.5.
Explicitly, $J_I(\Phi \circ R)(\mathbf{u}) = J_I\Phi(R(\mathbf{u}))\det D\widetilde{R}(\mathbf{u})$.
For $\delta > 0$, set $\chi_\delta(\mathbf{v}) := \min\{1, \operatorname{dist}(\mathbf{v}, \partial D)/\delta\}$ on D.
Extend χ_δ by 0 on $\mathbb{R}^k \setminus D$.
Set $\beta_\delta(\mathbf{v}) := \max\{0, 1 - \operatorname{dist}(\mathbf{v}, \partial D)/\delta\}$.
Then χ_δ is continuous and $\operatorname{supp}(\chi_\delta) = D \Subset U$.
The cutoff inequality is $(\forall\, \mathbf{v} \in D)\, 0 \leq 1 - \chi_\delta(\mathbf{v}) \leq \beta_\delta(\mathbf{v})$.
(iii) *Compactly Supported Substitution*: Set $F_\delta := \widetilde{h}\chi_\delta$.
The change-of-variables hypotheses hold because $\operatorname{supp}(F_\delta) \subseteq D \Subset U$ and $\widetilde{R}^{-1}(D) = D'$.
Change of variables for F_δ and $\widetilde{R}$ is valid by Theorem 5.1.12.
Consequently, $\int_D F_\delta = \int_{D'}(F_\delta \circ R)\left|\det D\widetilde{R}\right|$.

On D', $\det D\widetilde{R} = \epsilon \left|\det D\widetilde{R}\right|$.
Thus the preceding Cauchy–Binet identity contains the signed Jacobian.

(iv) *Boundary Limit*: The boundary of a cell or simplex is a finite union of flat faces.
Thus the δ-neighborhood of ∂D has volume $O(\delta)$.
Since $\widetilde{h}$ is bounded and $0 \le \beta_\delta \le 1$, $\int_{\mathbb{R}^k} \left|\widetilde{h}\right| \beta_\delta \to 0$ as $\delta \downarrow 0$.
This integral bounds the left-hand cutoff error.
The same bound controls the right-hand cutoff error by Theorem 5.1.12 for $\left|\widetilde{h}\right| \beta_\delta$.
Indeed, restrict its transformed integral to D'.
Consequently, $\int_{\Phi\circ R} \omega = \epsilon \int_\Phi \omega$ by Definition 5.2.5 and the cutoff limit.

Structural Commentary: Cauchy–Binet isolates the parameter determinant; cutoffs justify compact-support substitution.
Key Insight: The absolute Jacobian changes density, while its constant sign records orientation. □

Definition 5.2.7. *Product and Exterior Derivative*
Fix $n \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^n$ be open.
The *wedge product* is the pointwise exterior-algebra product.
It is bilinear over coefficient functions.
In coordinates, $dx_I \wedge dx_J$ is obtained by concatenating I, J and reducing by Proposition 5.2.4.

(i) For $f \in C^1(E)$, define $df := \sum_{i=1}^{n} (D_i f)(\mathbf{x})\, dx_i$.

(ii) Fix $k \in J_n$ and a C^1 k-form $\omega = \sum_I b_I(\mathbf{x})\, dx_I$.
Set $d\omega := \sum_I (db_I) \wedge dx_I$.

Theorem 5.2.8. *Exterior Derivative Laws*
Fix $n \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^n$ be open.

(i) Let $k, \ell \in \{0\} \cup J_n$, let ω be a C^1 k-form, and let λ be a C^1 ℓ-form. Then $d(\omega \wedge \lambda) = (d\omega) \wedge \lambda + (-1)^k \omega \wedge d\lambda$.

(ii) If ω is a C^2 form, then $d^2\omega = 0$.

Proof.

(i) By bilinearity, take $\omega = f\alpha$ and $\lambda = g\beta$.
Each of α, β is 1 in degree 0 or an increasing coordinate word in positive degree.
Since $d\alpha = d\beta = 0$, $d(\omega \wedge \lambda) = d(fg) \wedge \alpha \wedge \beta$.
The identity $d(fg) = f\, dg + g\, df$ holds by the ordinary product rule.
The $g\, df$ term equals $(d\omega) \wedge \lambda$.
The $f\, dg$ term equals $(-1)^k \omega \wedge d\lambda$ by Proposition 5.2.4(i), after moving dg past the k factors in α.

(ii) For a C^2 0-form f, $d^2 f = \sum_{i,j} (D_{ij} f)(\mathbf{x})\, dx_i \wedge dx_j$.
This sum is 0 by Theorem 3.2.11 and Proposition 5.2.4(i).
For a positive-degree form $\omega = \sum_I b_I\, dx_I$, part (i) and $d(dx_I) = 0$ imply $d^2\omega = \sum_I (d^2 b_I) \wedge dx_I = 0$.

Structural Commentary: The exterior derivative is built to make mixed partial cancellation visible.
Key Insight: Symmetric second derivatives meet antisymmetric wedge products, so $d^2 = 0$. □

Definition 5.2.9. *Change of Variables for Forms*
Fix $n, m \in \mathbb{N}$.
Let $E \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open, and let $T = (t_1, \dots, t_m) : E \to V$ be C^1.

(i) Set $dt_i := d(t_i) = \sum_{j=1}^{n} (D_j t_i)(\mathbf{x})\, dx_j$ for $i \in J_m$.

(ii) For a 0-form $f \in C(V, \mathbb{R})$, define its *pullback* by $T^* f := f_T := f \circ T$.

(iii) Fix $k \in J_m$ and a k-form $\omega = \sum_I b_I(\mathbf{y})\, dy_I$ on V.
Define $T^*\omega := \omega_T := \sum_I b_I(T(\mathbf{x}))\, dt_{i_1} \wedge \cdots \wedge dt_{i_k}$.
If $k \in J_n$, reduce this expression to standard presentation by Proposition 5.2.4.
If $k > n$, every coordinate word vanishes.

Theorem 5.2.10. *Pullback Commutes with the Basic Operations*
Fix $n, m, p \in \mathbb{N}$.
Let $E \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$, and $W \subseteq \mathbb{R}^p$ be open, and let $T : E \to V$ be C^1.

(i) If ω, λ are forms of the same degree on V, then $(\omega + \lambda)_T = \omega_T + \lambda_T$.

(ii) If ω, λ are forms on V, then $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$.

(iii) If ω is a C^1 form on V and T is C^2, then $d(\omega_T) = (d\omega)_T$.

(iv) If $S : V \to W$ is C^1 and η is a form on W, then $(\eta_S)_T = \eta_{S\circ T}$.

(v) Let $k \in J_{\min\{m,n\}}$, let Φ be a k-surface in E, and let ω be a k-form on V. Then $\int_{T\circ\Phi} \omega = \int_\Phi \omega_T$.

Proof.

(i) Addition follows termwise from $b_I \mapsto b_I \circ T$ and $dy_i \mapsto dt_i$.

(ii) Pullback replaces coefficients by their compositions with T and each dy_i by dt_i.
Thus $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$ by bilinearity.

(iii) If ω is a 0-form, the identity holds by Theorem 3.1.5.
For positive degree, write $\omega = \sum_I b_I\, dy_I$.
The identity $d(b_I \circ T) = (db_I)_T$ holds by Theorem 3.1.5.
The identity $d(dt_i) = d^2 t_i = 0$ holds by Theorem 5.2.8(ii).
Hence $d(\omega_T) = (d\omega)_T$ by repeated use of Theorem 5.2.8(i).
(iv) If η is a 0-form, the identity is associativity of composition.
For positive degree, let $S = (s_1, \dots, s_p)$. The coefficient functions agree by associativity of composition.
Each dz_q has the same replacement in $(\eta_S)_T$ and $\eta_{S\circ T}$ by Theorem 3.1.5.
(v) Write $\omega = \sum_I b_I\, dy_I$, where I ranges over increasing k-indices in J_m.
Let J range over increasing k-indices in J_n.
Set $M_{I,J}(\mathbf{x}) := \det[(D_{j_s} t_{i_r})(\mathbf{x})]_{r,s=1}^{k}$.
The pullback expansion is valid by multilinearity and Proposition 5.2.4.
Thus $\omega_T = \sum_{I,J} b_I(T(\mathbf{x}))M_{I,J}(\mathbf{x})\, dx_J$.
The composition-minor identity holds by Theorem 1.2.5.
Thus $J_I(T \circ \Phi)(\mathbf{u}) = \sum_J M_{I,J}(\Phi(\mathbf{u}))J_J\Phi(\mathbf{u})$.
Therefore, $\int_\Phi \omega_T = \int_{T\circ\Phi} \omega$ by Definition 5.2.5 and the preceding identities.

Structural Commentary: Pullback composes coefficients before Cauchy–Binet recombines coordinate minors.
Key Insight: The chain rule makes algebraic pullback exactly dual to composition of surfaces. □

Pullback is the differential-form analogue of the dual map in Definition 4.2.4.
Cells move forward by composition with T, while forms move backward.
The two motions have the same integral by Theorem 5.2.10(v).
Figure 172 records this duality with $T_*\varphi = T \circ \varphi$.

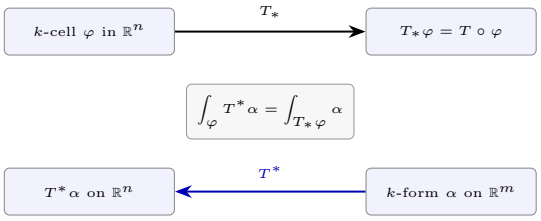

Figure 172: Cells push forward with T, forms pull back with T^*, and their integrals agree by Theorem 5.2.10(v).

5.3 Stokes Theory

Stokes theory relates exterior differentiation to oriented boundary.
Green's theorem, classical Stokes, and the divergence theorem are its coordinate forms.

Definition 5.3.1. *Chains and Boundaries*
Fix $n \in \mathbb{N}$ and $k \in \mathbb{Z}_{\geq 0}$, set $Q^0 := \{0\}$, and let $V \subseteq \mathbb{R}^n$ be open.
(i) Fix $\mathbf{p}_0, \dots, \mathbf{p}_k \in \mathbb{R}^n$.
Set $\sigma(\mathbf{u}) := \mathbf{p}_0 + \sum_{i=1}^{k} u_i(\mathbf{p}_i - \mathbf{p}_0)$.
This map represents the oriented affine *k-simplex* $[\mathbf{p}_0, \dots, \mathbf{p}_k]$.
Work in the real vector space generated by these symbols.
For a permutation π of $\{0, \dots, k\}$, impose $[\mathbf{p}_{\pi(0)}, \dots, \mathbf{p}_{\pi(k)}] = \operatorname{sgn}(\pi)[\mathbf{p}_0, \dots, \mathbf{p}_k]$.
(ii) A *C^2 oriented k-simplex* in V is a map $\Phi \in C^2(Q^k, V)$.
A *C^2 k-chain* is a finite sum $\Psi = \sum_\ell c_\ell \Phi_\ell$ with $c_\ell \in \mathbb{R}$.
(iii) If $k \geq 1$ and ω is a k-form on V, set $\int_\Psi \omega := \sum_\ell c_\ell \int_{\Phi_\ell} \omega$.
If $k = 0$ and f is a 0-form, set $\int_\Phi f := f(\Phi(0))$ and extend linearly.
(iv) Let $\mathbf{q}_0 = \mathbf{0}$ and $\mathbf{q}_i = \mathbf{e}_i$ for $1 \leq i \leq k$.
For $k \geq 1$ and $j \in \{0, \dots, k\}$, let $\iota_j : Q^{k-1} \to Q^k$ be the affine map underlying $[\mathbf{q}_0, \dots, \widehat{\mathbf{q}_j}, \dots, \mathbf{q}_k]$.
Set $\partial_{\mathrm{ch}}\Phi := \sum_{j=0}^{k} (-1)^j (\Phi \circ \iota_j)$ for $k \geq 1$ and extend linearly.
Set $\partial_{\mathrm{ch}}\Psi := 0$ on 0-chains.

Theorem 5.3.2. *Stokes' Theorem*
Fix $m \in \mathbb{N}$ and $k \in J_m$.
Let $V \subseteq \mathbb{R}^m$ be open, let Ψ be a C^2 k-chain in V, and let ω be a C^1 $(k-1)$-form on V.
Then $\int_\Psi d\omega = \int_{\partial_{\mathrm{ch}}\Psi} \omega$.

Proof.
(i) *Linearity Reduction*: By linearity, take $\Psi = \Phi$.
Choose a C^2 extension $T : U \to V$ on an open neighborhood U of Q^k.
Let $\sigma : Q^k \to U$ be the standard inclusion; then $\Phi = T \circ \sigma$.
(ii) *Pullback*: Set $\lambda := T^*\omega$.
The identity $\int_\Phi d\omega = \int_\sigma d\lambda$ holds by Theorem 5.2.10(iii),(v).
For $k \geq 2$, pullback naturality holds facewise by Theorem 5.2.10(v).

Thus $\int_{\partial_{\mathrm{ch}}\Phi}\omega=\int_{\partial_{\mathrm{ch}}\sigma}\lambda$.
For $k=1$, the same boundary identity holds directly by the definitions of pullback and integration of 0-forms.

(iii) *Coordinate Form*: By standard presentation and linearity, take one basic summand.
It has the form $\lambda=f\,dx_1\wedge\cdots\wedge\widehat{dx_r}\wedge\cdots\wedge dx_k$ for $r\in J_k$.
For $k=1$, interpret the empty wedge as 1.
Coefficientwise expansion is valid by Definition 5.2.7.
Reordering contributes $(-1)^{r-1}$ by Proposition 5.2.4(i).
Therefore, $d\lambda=(-1)^{r-1}(D_rf)\,dx_1\wedge\cdots\wedge dx_k$.

(iv) *Boundary Faces*: Write $\partial_{\mathrm{ch}}\sigma=\sum_{j=0}^{k}(-1)^j\iota_j$.
The pullback of λ vanishes when $j\notin\{0,r\}$.
Indeed, $x_j\circ\iota_j=0$ while dx_j occurs in λ.
The $j=0$ pullback has sign $(-1)^{r-1}$ and restricts to $x_r=1-\sum_{i\neq r}x_i$.
The $j=r$ term has boundary sign $(-1)^r$ and restricts to $x_r=0$.

(v) *Conclusion*: If $k=1$, both sides equal $f(1)-f(0)$ by Theorem 2.2.2.
Suppose $k\geq 2$, and write $\mathbf{x}_{\widehat{r}}$ for the coordinates other than x_r.
Set $s_r(\mathbf{x}_{\widehat{r}}):=1-\sum_{i\neq r}x_i$.
Set $g_r(\mathbf{x}_{\widehat{r}}):=f(\mathbf{x}_{\widehat{r}},s_r(\mathbf{x}_{\widehat{r}}))-f(\mathbf{x}_{\widehat{r}},0)$.
The boundary-face integral equals $(-1)^{r-1}\int_{Q^{k-1}}g_r$.
The interior integral has the same value by Theorem 2.2.2 on the x_r-slices.

Structural Commentary: Pullback moves the chain to Q^k; face restrictions isolate the missing coordinate's endpoints.
Key Insight: The wedge sign $(-1)^{r-1}$ matches the orientation difference between the slanted face and $x_r=0$. □

Definition 5.3.3. *Closed and Exact Forms*
Fix $n\in\mathbb{N}$ and $k\in J_n$.
Let $E\subseteq\mathbb{R}^n$ be open, and let ω be a C^1 k-form on E.
(i) ω is *exact* iff $(\exists\,\lambda\in C^1(E,\bigwedge^{k-1}((\mathbb{R}^n)')))\ \omega=d\lambda$.
(ii) ω is *closed* iff $d\omega=0$.

Perspective 5.3.4. *Closed Is Local, Exact Is Global*
If $\omega=d\lambda$ with $\lambda\in C^2$, then ω is closed by Theorem 5.2.8(ii).
The converse depends on the domain.
On $\mathbb{R}^2\setminus\{\mathbf{0}\}$, write $(x,y):=(x_1,x_2)$ and $(dx,dy):=(dx_1,dx_2)$.
The form $\omega=\frac{-y}{x^2+y^2}\,dx+\frac{x}{x^2+y^2}\,dy$ satisfies $d\omega=0$.
Set $\gamma(u):=(\cos(2\pi u),\sin(2\pi u))$ for $u\in Q^1=[0,1]$.
The calculation $\int_\gamma\omega=\int_0^1 2\pi\,du=2\pi$ holds by Definition 5.2.5.
Since $\partial_{\mathrm{ch}}\gamma=0$, exactness would imply $\int_\gamma\omega=0$ by Theorem 5.3.2.
Holes can obstruct global antiderivatives even when local derivative tests vanish.

Theorem 5.3.5. *Poincaré Lemma on Convex Sets*
Fix $n\in\mathbb{N}$ and $k\in J_n$.
(i) Let $E\subseteq\mathbb{R}^n$ be open and convex, and let ω be a closed C^1 k-form on E. Then ω is exact.
(ii) Let $C,E\subseteq\mathbb{R}^n$ be open, let C be convex, and let $\Phi:C\to E$ be a C^2 diffeomorphism. If ω is a closed C^1 k-form on E, then ω is exact.

Proof.
(i) Fix $\mathbf{a}\in E$ and set $H_t(\mathbf{x}):=\mathbf{a}+t(\mathbf{x}-\mathbf{a})$ for $t\in[0,1]$.
Write $\omega=\sum_I b_I\,dx_I$.
For $I=(i_1<\cdots<i_k)$, let $I\setminus i_r$ denote the increasing index obtained by omitting i_r.
For $k=1$, set $dx_{I\setminus i_1}:=1$.
Set $A_I(\mathbf{x}):=\int_0^1 t^{k-1}b_I(H_t(\mathbf{x}))\,dt$.
Define $K\omega:=\sum_I\sum_{r=1}^{k}(-1)^{r-1}(x_{i_r}-a_{i_r})A_I(\mathbf{x})\,dx_{I\setminus i_r}$, and use the same formula in each positive degree.
The coordinate difference-quotient bound holds by Corollary 1.1.13.
The derivatives are uniformly continuous on a compact tube around the segment from $\mathbf{a}$ to $\mathbf{x}$ by Theorem 5.2.6.
Hence the coordinate difference quotients of $b_I(H_t(\mathbf{x}))$ converge uniformly in t.
Consequently, $D_jA_I(\mathbf{x})=\int_0^1 t^k(D_jb_I)(H_t(\mathbf{x}))\,dt$ by Theorem 2.1.14(i),(iv).
By linearity, consider one basic summand $b_I\,dx_I$.
The undifferentiated factors in $dK(b_I\,dx_I)$ contribute kA_I to the dx_I coefficient.
The terms differentiated in the i_r direction contribute $\sum_{r=1}^{k}(x_{i_r}-a_{i_r})D_{i_r}A_I$.
Their sum is $\int_0^1\frac{d}{dt}[t^kb_I(H_t(\mathbf{x}))]\,dt=b_I(\mathbf{x})$ by Theorem 2.2.2.
The coefficients containing dx_j with $j\notin I$ cancel against those of $K(d(b_I\,dx_I))$.
Their signs are opposite by Proposition 5.2.4(i).
Hence $dK\omega+Kd\omega=\omega$.
If $d\omega=0$, then $K\omega\in C^1$ and $\omega=d(K\omega)$, so ω is exact.
(ii) Let $\Phi:C\to E$ be as stated, and let ω be a closed C^1 k-form on E.
The identity $d(\Phi^*\omega)=\Phi^*(d\omega)=0$ holds by Theorem 5.2.10(iii).
Choose a C^1 $(k-1)$-form η on C such that $\Phi^*\omega=d\eta$ by (i).
Then $\omega=(\Phi^{-1})^*(\Phi^*\omega)=d((\Phi^{-1})^*\eta)$ by Theorem 5.2.10(iii),(iv).

Structural Commentary: The contraction satisfies $dK + Kd = \mathrm{id}$; pullback transports it through a diffeomorphism.
Key Insight: If $d\omega = 0$, the homotopy identity reduces to $\omega = d(K\omega)$. □

Definition 5.3.6. *Vector Fields as Differential Forms*
Let $E \subseteq \mathbb{R}^3$ be open, and write $(x, y, z) := (x_1, x_2, x_3)$ and $(dx, dy, dz) := (dx_1, dx_2, dx_3)$.
Fix $\mathbf{H} = (H_1, H_2, H_3) \in C^1(E, \mathbb{R}^3)$.
Set $\nabla \cdot \mathbf{H} := D_1H_1 + D_2H_2 + D_3H_3$.
Set $\nabla \times \mathbf{H} := (D_2H_3 - D_3H_2, D_3H_1 - D_1H_3, D_1H_2 - D_2H_1)$.
Set $\lambda_{\mathbf{H}} := H_1\,dx + H_2\,dy + H_3\,dz$.
Set $\omega_{\mathbf{H}} := H_1\,dy \wedge dz + H_2\,dz \wedge dx + H_3\,dx \wedge dy$.

Theorem 5.3.7. *Vector Analysis Dictionary in $\mathbb{R}^3$*
Let $E \subseteq \mathbb{R}^3$ be open, and use the notation of Definition 5.3.6.
Fix $\mathbf{F} \in C^1(E, \mathbb{R}^3)$.

(i) *For $u \in C^1(E)$, $\mathbf{F} = \nabla u$ iff $\lambda_{\mathbf{F}} = du$.*
(ii) $d\lambda_{\mathbf{F}} = \omega_{\nabla\times\mathbf{F}}$.
(iii) $\nabla \times \mathbf{F} = 0$ *iff* $d\lambda_{\mathbf{F}} = 0$.
(iv) *For $\mathbf{G} \in C^1(E, \mathbb{R}^3)$, $\mathbf{F} = \nabla \times \mathbf{G}$ iff $\omega_{\mathbf{F}} = d\lambda_{\mathbf{G}}$.*
(v) $d\omega_{\mathbf{F}} = (\nabla \cdot \mathbf{F})\,dx \wedge dy \wedge dz$.
(vi) $\nabla \cdot \mathbf{F} = 0$ *iff* $d\omega_{\mathbf{F}} = 0$.
(vii) *Assume E is C^2-diffeomorphic to a convex open set and $\nabla \times \mathbf{F} = 0$.*
Then $\exists u \in C^1(E) : \mathbf{F} = \nabla u$.
(viii) *Assume E is C^2-diffeomorphic to a convex open set and $\nabla \cdot \mathbf{F} = 0$.*
Then $\exists \mathbf{G} \in C^1(E, \mathbb{R}^3) : \mathbf{F} = \nabla \times \mathbf{G}$.

Proof.

(i) ($\Rightarrow$) If $\mathbf{F} = \nabla u$, then $\lambda_{\mathbf{F}} = (D_1u)\,dx + (D_2u)\,dy + (D_3u)\,dz = du$.
(i) ($\Leftarrow$) If $\lambda_{\mathbf{F}} = du$, then $\mathbf{F} = \nabla u$ by equality of the dx, dy, dz coefficients.
(ii) The $dy \wedge dz$ coefficient of $d\lambda_{\mathbf{F}}$ is $D_2F_3 - D_3F_2$.
Cyclic permutation produces the other two curl coefficients, so $d\lambda_{\mathbf{F}} = \omega_{\nabla\times\mathbf{F}}$.
(iii) ($\Rightarrow$) If $\nabla \times \mathbf{F} = 0$, then $d\lambda_{\mathbf{F}} = \omega_{\nabla\times\mathbf{F}} = 0$ by (ii).
(iii) ($\Leftarrow$) If $d\lambda_{\mathbf{F}} = 0$, then $\omega_{\nabla\times\mathbf{F}} = 0$ by (ii).
Hence $\nabla \times \mathbf{F} = 0$ by Proposition 5.2.4(ii).
(iv) ($\Rightarrow$) If $\mathbf{F} = \nabla \times \mathbf{G}$, then $\omega_{\mathbf{F}} = \omega_{\nabla\times\mathbf{G}}$.
This form equals $d\lambda_{\mathbf{G}}$ by (ii).
(iv) ($\Leftarrow$) If $\omega_{\mathbf{F}} = d\lambda_{\mathbf{G}}$, then $\omega_{\mathbf{F}} = \omega_{\nabla\times\mathbf{G}}$ by (ii).
Hence $\mathbf{F} = \nabla \times \mathbf{G}$ by Proposition 5.2.4(ii).
(v) Only derivatives in the missing coordinate survive in $d\omega_{\mathbf{F}}$.
Thus $d\omega_{\mathbf{F}} = (D_1F_1 + D_2F_2 + D_3F_3)\,dx \wedge dy \wedge dz = (\nabla \cdot \mathbf{F})\,dx \wedge dy \wedge dz$.
(vi) ($\Rightarrow$) If $\nabla \cdot \mathbf{F} = 0$, then $d\omega_{\mathbf{F}} = 0$ by (v).
(vi) ($\Leftarrow$) If $d\omega_{\mathbf{F}} = 0$, then its $dx \wedge dy \wedge dz$ coefficient is 0.
Hence $\nabla \cdot \mathbf{F} = 0$ by (v).
(vii) If $\nabla \times \mathbf{F} = 0$, then $d\lambda_{\mathbf{F}} = 0$ by (iii).
Choose $u \in C^1(E)$ satisfying $\lambda_{\mathbf{F}} = du$ by Theorem 5.3.5(ii).
Then $\mathbf{F} = \nabla u$ by (i).
(viii) If $\nabla \cdot \mathbf{F} = 0$, then $d\omega_{\mathbf{F}} = 0$ by (vi).
Choose a C^1 1-form $\eta = g_1\,dx + g_2\,dy + g_3\,dz$ satisfying $\omega_{\mathbf{F}} = d\eta$ by Theorem 5.3.5(ii).
Set $\mathbf{G} := g_1\mathbf{e}_1 + g_2\mathbf{e}_2 + g_3\mathbf{e}_3$.
Then $\eta = \lambda_{\mathbf{G}}$ and $\mathbf{F} = \nabla \times \mathbf{G}$ by (iv).

Structural Commentary: Coefficient expansion realizes d as curl and divergence; the Poincaré lemma supplies potentials.
Key Insight: Closed 1- and 2-forms correspond respectively to curl-free and divergence-free fields. □

Definition 5.3.8. *Vector-Calculus Integral Notation*
In $\mathbb{R}^2$, write $(x, y) := (x_1, x_2)$ and $(dx, dy) := (dx_1, dx_2)$.
In $\mathbb{R}^3$, use $\lambda_{\mathbf{F}}$ and $\omega_{\mathbf{F}}$ from Definition 5.3.6.

(i) Let $V \subseteq \mathbb{R}^3$ be open, let Γ be a C^2 1-chain in V, and let $\mathbf{F} \in C^1(V, \mathbb{R}^3)$.
Set $\int_\Gamma \mathbf{F} \cdot \mathbf{t}\,ds := \int_\Gamma \lambda_{\mathbf{F}}$.
(ii) Let $V \subseteq \mathbb{R}^3$ be open, let Σ be a C^2 2-chain in V, and let $\mathbf{F} \in C^1(V, \mathbb{R}^3)$.
Set $\int_\Sigma \mathbf{F} \cdot \mathbf{n}\,dA := \int_\Sigma \omega_{\mathbf{F}}$.
(iii) Let $V \subseteq \mathbb{R}^2$ be open, let Σ be a C^2 2-chain in V, and let $g \in C(V)$.
Set $\int_\Sigma g\,dA := \int_\Sigma g\,dx \wedge dy$.
(iv) Let $V \subseteq \mathbb{R}^3$ be open, let Ω be a C^2 3-chain in V, and let $g \in C(V)$.
Set $\int_\Omega g\,dV := \int_\Omega g\,dx \wedge dy \wedge dz$.

Theorem 5.3.9. *Green, Classical Stokes, and Divergence*

(i) *Let $V \subseteq \mathbb{R}^2$ be open, let Σ be a C^2 2-chain in V, and let $\alpha, \beta \in C^1(V)$.*
Then $\int_{\partial_{\mathrm{ch}}\Sigma}(\alpha\,dx + \beta\,dy) = \int_\Sigma (D_1\beta - D_2\alpha)\,dA$.
(ii) *Let $V \subseteq \mathbb{R}^3$ be open, let Σ be a C^2 2-chain in V, and let $\mathbf{F} \in C^1(V, \mathbb{R}^3)$.*
Then $\int_{\partial_{\mathrm{ch}}\Sigma} \mathbf{F} \cdot \mathbf{t}\,ds = \int_\Sigma (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dA$.
(iii) *Let $V \subseteq \mathbb{R}^3$ be open, let Ω be a C^2 3-chain in V, and let $\mathbf{F} \in C^1(V, \mathbb{R}^3)$.*
Then $\int_{\partial_{\mathrm{ch}}\Omega} \mathbf{F} \cdot \mathbf{n}\,dA = \int_\Omega (\nabla \cdot \mathbf{F})\,dV$.

Proof.

(i) Set $\lambda := \alpha\,dx + \beta\,dy$.
Then $d\lambda = (D_1\beta - D_2\alpha)\,dx \wedge dy$.
The stated identity holds by Theorem 5.3.2 and Definition 5.3.8(iii).

(ii) Set $\lambda := \lambda_{\mathbf{F}}$.
The identity $d\lambda = \omega_{\nabla\times\mathbf{F}}$ holds by Theorem 5.3.7(ii).
The stated identity holds by Theorem 5.3.2 and Definition 5.3.8(i),(ii).
(iii) Set $\omega := \omega_{\mathbf{F}}$.
The identity $d\omega = (\nabla\cdot\mathbf{F})\,dx\wedge dy\wedge dz$ holds by Theorem 5.3.7(v).
The stated identity holds by Theorem 5.3.2 and Definition 5.3.8(ii),(iv).

Structural Commentary: Stokes transfers the three exterior-derivative identities to chain integrals.
Key Insight: The same Stokes identity becomes circulation, flux, or volume balance in degrees $1, 2, 3$. □

5.4 Further Topics in Differential Forms and Stokes Theory

Proof Method 5.4.1. *Differential-Forms Proof Patterns*
Use When: A proof mixes form algebra with parametrization or boundary integration.
Select the branch matching the obstruction and verify its hypotheses.

(i) *Signs*: Reduce wedge permutations by Proposition 5.2.4(i).
Track determinant swaps by Theorem 3.8.2(iii).
(ii) *Standard presentation*: Delete repeated-index terms and sort the remaining factors.
Existence and uniqueness of the ordered result hold by Proposition 5.2.4.
(iii) *Pullback*: Replace dy_i by dt_i by Definition 5.2.9.
Then commute pullback with the required operation by Theorem 5.2.10.
(iv) *Stokes*: Identify λ with $d\lambda$ equal to the target interior form, and verify the boundary orientation.
Then $\int_\Psi d\lambda = \int_{\partial_{\mathrm{ch}}\Psi}\lambda$ by Theorem 5.3.2.
For a coordinate proof, reduce by linearity to one chain and one basic summand.
Pull back to Q^k and retain only the two faces associated with the missing differential.
Close each slice in that coordinate by Theorem 2.2.2.
(v) *Closed versus exact*: If $\omega = d\lambda$ with $\lambda\in C^2$, then $d\omega = 0$ by Theorem 5.2.8(ii).
On a convex open set, a closed C^1 positive-degree ω is exact by Theorem 5.3.5(i).
On its C^2-diffeomorphic image, the same conclusion holds by Theorem 5.3.5(ii).
(vi) *Vector analysis*: Encode vector fields as forms by Definition 5.3.6.
Identify exterior derivatives with curl or divergence by Theorem 5.3.7.
Convert them to boundary identities by Theorem 5.3.2.
Translate the integrals by Definition 5.3.8.

Key Insight: Normal form fixes signs; pullback changes variables; Stokes moves d to the boundary.

6 Infinite Series and Summability

6.1 Series Tests

Series are sequences of partial sums. Comparison, condensation, root and ratio tests, rearrangements, products, and double sums all refine that viewpoint.

Definition 6.1.1. *Series and Partial Sums*
Let $(a_n)_{n\ge1}\subseteq\mathbb{C}$. Summation indices are bound variables and may be renamed within unchanged scope.

(i) For integers $1\le p\le q$, $\sum_{n=p}^{q} a_n := a_p + a_{p+1} + \cdots + a_q$.

(ii) The *series* $\sum_{n=1}^{\infty} a_n$ is the *partial-sum sequence* (s_N), where $s_N := \sum_{n=1}^{N} a_n$.

(iii) The series *converges to* s, written $\sum_{n=1}^{\infty} a_n = s$, iff $s_N\to s$; otherwise it *diverges*. The same definitions apply with initial index 0.

Key Insight: A series is a sequence first; its sum is a limit when that limit exists.

Definition 6.1.2. *Absolute and Nonabsolute Convergence*
The series $\sum a_n$ *converges absolutely* if $\sum|a_n|$ converges.
If $\sum a_n$ converges but $\sum|a_n|$ diverges, then $\sum a_n$ converges *nonabsolutely*.

Theorem 6.1.3. *Cauchy Criterion for Series*
The series $\sum a_n$ *converges iff* $(\forall\varepsilon>0)(\exists N\in\mathbb{N})(\forall m,n\in\mathbb{N})\,[m\ge n\ge N\Rightarrow\left|\sum_{k=n}^{m} a_k\right|\le\varepsilon]$.

Proof.
($\Rightarrow$) If $s_n := \sum_{k=1}^{n} a_k$ converges, then (s_n) is Cauchy. Thus, for $m\ge n$ large enough, $\left|\sum_{k=n}^{m} a_k\right| = |s_m - s_{n-1}|\le\varepsilon$.
($\Leftarrow$) For $m>n$ sufficiently large, the tail condition is $|s_m - s_n| = \left|\sum_{k=n+1}^{m} a_k\right|\le\varepsilon$.
Hence (s_n) is Cauchy in $\mathbb{C}\cong\mathbb{R}^2$ and converges by Corollary 4.2.8.
Structural Commentary: A series converges exactly when all sufficiently late finite tails are small.
Key Insight: The Cauchy test measures tails, not the eventual value of the sum. □

Corollary 6.1.4. *Terms of a Convergent Series Tend to Zero*
If $\sum a_n$ *converges, then* $a_n\to0$.

Proof. Let $\varepsilon>0$. By Theorem 6.1.3, there is N such that $m\ge n\ge N$ implies $\left|\sum_{k=n}^{m} a_k\right|\le\varepsilon$.
Taking $m=n$ shows $(\forall n\ge N)\ |a_n|\le\varepsilon$, hence $a_n\to0$.
Structural Commentary: A single term is the shortest possible tail.
Key Insight: The condition $a_n\to0$ is necessary, but not sufficient for convergence. □

Theorem 6.1.5. *Nonnegative Series and Bounded Partial Sums*
A series of nonnegative real terms converges if and only if its partial sums form a bounded sequence.

Proof.
($\Rightarrow$) Every convergent sequence is bounded, so the partial sums are bounded.
($\Leftarrow$) Since $a_n \geq 0$, $s_n = \sum_{k=1}^{n} a_k$ increases; if bounded, it converges by Theorem 2.1.11.
Structural Commentary: Nonnegative series replace cancellation by monotonicity.
Key Insight: For nonnegative terms, convergence is exactly bounded accumulation. □

Theorem 6.1.6. *Comparison Test*
(i) If $|a_n| \leq c_n$ for $n \geq N_0$, and if $\sum c_n$ converges, then $\sum a_n$ converges.
(ii) If $a_n \geq d_n \geq 0$ for $n \geq N_0$, and if $\sum d_n$ diverges, then $\sum a_n$ diverges.

Proof.
(i) Given $\varepsilon > 0$, choose $N \geq N_0$ so that $m \geq n \geq N$ implies $\sum_{k=n}^{m} c_k \leq \varepsilon$.
Then $\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k| \leq \sum_{k=n}^{m} c_k \leq \varepsilon$, so $\sum a_n$ converges by Theorem 6.1.3.
(ii) If $\sum a_n$ converged, then (i) applied to $d_n \leq a_n$ would force $\sum d_n$ to converge, a contradiction.
Structural Commentary: Absolute tail domination transfers Cauchy convergence.
Key Insight: Comparison is a tail test; finitely many early terms do not matter. □

Theorem 6.1.7. *Geometric Series*
(i) If $0 \leq x < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$.
(ii) If $x \geq 1$, the series diverges.

Proof.
(i) For $0 \leq x < 1$, $\sum_{k=0}^{n} x^k = \frac{1-x^{n+1}}{1-x}$ and $x^{n+1} \to 0$ by Theorem 2.2.10.
(ii) If $x = 1$, the partial sums are $n+1$, so they diverge. If $x > 1$, then $x^n \not\to 0$, so the series diverges by Corollary 6.1.4.
Structural Commentary: The finite geometric identity converts convergence into the limit of x^{n+1}.
Key Insight: The geometric series is the model comparison series for ratio-like decay. □

Figure 173 represents the case $x = \frac{2}{3}$ by adjacent strip widths whose total length is 3.

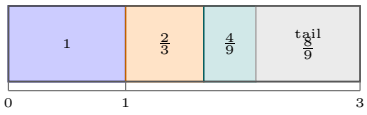

Figure 173: At $x = 2/3$, the strip widths sum to $1 + \frac{2}{3} + \frac{4}{9} + \cdots = 3$.

Definition 6.1.8. *Real Decimals as Series*
For $N \in \mathbb{Z}_{\geq 0}$ and digits $b_N, \ldots, b_0, a_1, a_2, \ldots \in \{0, \ldots, 9\}$, with $b_N \neq 0$ unless the integer part is 0,
$b_N \cdots b_0.a_1a_2\cdots := \sum_{i=0}^{N} b_i 10^i + \sum_{j=1}^{\infty} a_j 10^{-j}$. The fractional series converges absolutely by Theorems 6.1.6(i) and 6.1.7(i), since $0 \leq a_j 10^{-j} \leq 10^{1-j}$. A minus sign denotes the negative of this value.

Proposition 6.1.9. *Terminal Nines Are Nonunique*
The same real number can have two decimal representations: $1.000\ldots = 0.999\ldots$.

Proof. $1.000\ldots = 1 + \sum_{j=1}^{\infty} 0 \cdot 10^{-j} = 1$, while $0.999\ldots = \sum_{j=1}^{\infty} 9 \cdot 10^{-j} = 9 \cdot \frac{10^{-1}}{1-10^{-1}} = 1$ by Theorem 6.1.7(i).
Structural Commentary: The two digit strings have the same partial-sum limit.
Key Insight: Uniqueness requires excluding terminal 9 tails. □

Perspective 6.1.10. *Decimal Expansions Are Series*
Proposition 2.2.11 constructs decimal truncations converging to a real number, while Definition 6.1.8 interprets the infinite digit string as a series. Proposition 6.1.9 identifies the standard terminal-9 ambiguity.

Theorem 6.1.11. *Cauchy Condensation Test*
Let $(a_n)_{n\geq 1}$ satisfy $a_1 \geq a_2 \geq \cdots \geq 0$. Then $\sum_{n=1}^{\infty} a_n$ converges iff $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Proof. Let $s_n := \sum_{j=1}^{n} a_j$ and $t_k := \sum_{j=0}^{k} 2^j a_{2^j}$. Both partial-sum sequences are nondecreasing.

($\Rightarrow$) If (s_n) is bounded, then $t_m \leq 2 \sum_{j=1}^{2^m} a_j = 2s_{2^m}$, so (t_m) is bounded and the condensed series converges by Theorem 6.1.5.

($\Leftarrow$) If (t_k) is bounded, then for $2^m \leq n < 2^{m+1}$, $s_n \leq \sum_{j=0}^{m} 2^j a_{2^j} = t_m$; hence (s_n) is bounded and the original series converges by Theorem 6.1.5.

Structural Commentary: Dyadic blocks compare to endpoint terms.
Key Insight: Monotone tails are detected on dyadic scales. □

Corollary 6.1.12. *p-Series Test*
For $p \in \mathbb{R}$:

(i) If $p > 1$, then $\sum_{n=1}^{\infty} n^{-p}$ converges.

(ii) If $p \le 1$, then $\sum_{n=1}^{\infty} n^{-p}$ diverges.

Proof.

(i) If $p > 1$, the condensed series is $\sum_{k=0}^{\infty} 2^k (2^k)^{-p} = \sum_{k=0}^{\infty} 2^{(1-p)k}$ by Theorem 6.1.11; it converges by Theorem 6.1.7(i) because $2^{1-p} < 1$.

(ii) If $p \le 0$, then $n^{-p} \not\to 0$, so the series diverges by Corollary 6.1.4. If $0 < p \le 1$, the condensed geometric series has ratio $2^{1-p} \ge 1$, so divergence follows by Theorems 6.1.11 and 6.1.7(ii).

Structural Commentary: Condensation turns n-powers into a geometric series.
Key Insight: The threshold is the ratio $2^{1-p} = 1$. □

Theorem 6.1.13. *Integral Test*
Let $N \in \mathbb{N}$ and let $f : [N, \infty) \to [0, \infty)$ be decreasing and Riemann integrable on compact subintervals.

(i) For $M \ge N$, $\int_N^{M+1} f(x)\,dx \le \sum_{k=N}^{M} f(k) \le f(N) + \int_N^M f(x)\,dx$.

(ii) $\sum_{k=N}^{\infty} f(k)$ converges iff $\int_N^{\infty} f(x)\,dx$ converges.

Proof.

(i) For $x \in [k, k+1]$, $f(x) \le f(k)$, so $\int_N^{M+1} f \le \sum_{k=N}^{M} f(k)$.

For $k \ge N+1$ and $x \in [k-1, k]$, $f(k) \le f(x)$, so $\sum_{k=N}^{M} f(k) \le f(N) + \int_N^M f$.

(ii) ($\Rightarrow$) If $\sum_{k=N}^{\infty} f(k)$ converges, its partial sums are bounded. The nondecreasing truncations $\int_N^{M+1} f$ are bounded by (i), so their finite limit exists by Theorem 2.1.11; hence $\int_N^{\infty} f$ converges.

(ii) ($\Leftarrow$) If $\int_N^{\infty} f$ converges, its truncations are bounded. The partial sums $\sum_{k=N}^{M} f(k)$ are bounded by (i), so the series converges by Theorem 6.1.5.

Structural Commentary: Unit intervals compare samples to area.
Key Insight: Same monotone tail, same boundedness. □

Figure 174 compares the graph of decreasing f with left- and right-endpoint rectangles that produce the bounds in Theorem 6.1.13(i).

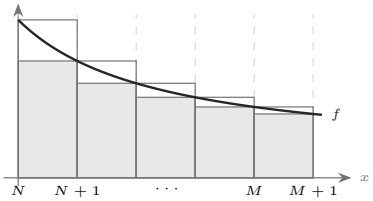

Figure 174: Left and right endpoint estimates for decreasing $f \ge 0$.

Definition 6.1.14. *The Number e*
Define $e := \sum_{n=0}^{\infty} \frac{1}{n!}$, where $0! := 1$. The defining series converges by Theorems 6.1.6(i) and 6.1.7(i), since $1/n! \le 2^{1-n}$ for $n \ge 1$.

Theorem 6.1.15. *A Limit Formula for e*
The limit formula is $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$.

Proof. Let $s_m := \sum_{k=0}^{m} \frac{1}{k!}$ and $t_n := (1 + \frac{1}{n})^n$. By Proposition 4.3.7(i), $t_n \le s_n$, hence $\limsup t_n \le e$ by Theorems 2.2.8(i) and 2.2.7.
For fixed $m \le n$, the first m terms in the same binomial expansion satisfy $t_n \ge 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{m!}\left(1 - \frac{1}{n}\right)\cdots\left(1 - \frac{m-1}{n}\right)$.
With m fixed, $n \to \infty$ implies $\liminf t_n \ge s_m$ by Theorems 2.2.8(ii) and 2.2.7; then $m \to \infty$ implies $e \le \liminf t_n$.
Structural Commentary: Bound above by the factorial series; bound below by fixed binomial terms.
Key Insight: Limsup and liminf squeeze the binomial limit to the series definition. □

Proposition 6.1.16. *Remainder Bound for the Series of e*
If $n \ge 1$ and $s_n = \sum_{k=0}^{n} 1/k!$, then $0 < e - s_n < \frac{1}{n!\,n}$.

Proof. The tail satisfies $e - s_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \cdots < \frac{1}{(n+1)!}\left(1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots\right) = \frac{1}{n!\,n}$ by Theorem 6.1.7(i).
Structural Commentary: The factorial tail is bounded by a geometric tail.
Key Insight: Rapid convergence of $\sum \frac{1}{n!}$ permits arithmetic control of e. □

Theorem 6.1.17. *e Is Irrational*
The number e is irrational.

Proof. Suppose $e = p/q$ with p, q positive integers. By Proposition 6.1.16, $0 < q!(e - s_q) < \frac{1}{q}$.
But $q!e$ and $q!s_q$ are integers, so $q!(e - s_q)$ is an integer strictly between 0 and 1, impossible.
Structural Commentary: The sharp tail bound turns rationality into an impossible integer.
Key Insight: A positive integer cannot be smaller than 1. □

Theorem 6.1.18. *Root Test*
For a series $\sum a_n$, put $\alpha := \limsup_{n\to\infty} \sqrt[n]{|a_n|}$. Then:
(i) if $\alpha < 1$, then $\sum a_n$ converges absolutely;
(ii) if $\alpha > 1$, then $\sum a_n$ diverges;
(iii) if $\alpha = 1$, the test is inconclusive.

Proof.
(i) Choose $\alpha < \beta < 1$. Eventually $\sqrt[n]{|a_n|} < \beta$ by Theorem 2.2.5(ii), hence eventually $|a_n| < \beta^n$. Thus $\sum |a_n|$ converges by Theorems 6.1.6(i) and 6.1.7(i).
(ii) Since $\alpha > 1$, $|a_n| > 1$ for infinitely many n by Theorem 2.2.5(i), so $a_n \not\to 0$ and the series diverges by Corollary 6.1.4.
(iii) $\sum 1/n$ and $\sum 1/n^2$ both have $\alpha = 1$, but behave differently by Corollary 6.1.12.
Structural Commentary: The root test compares term size directly to geometric decay.
Key Insight: Values below 1 force absolute convergence; values above 1 prevent terms from tending to 0. □

Theorem 6.1.19. *Ratio Test*
Assume $(\exists N_0 \in \mathbb{N})(\forall n \geq N_0)\, a_n \neq 0$.
(i) If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum a_n$ converges absolutely.
(ii) If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$, then $\sum a_n$ diverges.
(iii) In the remaining cases, the test is inconclusive.

Proof.
(i) Choose $\beta < 1$ with $|a_{n+1}/a_n| < \beta$ eventually by Theorem 2.2.5(ii).
Choose N such that $(\forall n \geq N)\ |a_{n+1}| \leq \beta\,|a_n|$. Then $(\forall n \geq N)\ |a_n| \leq |a_N|\,\beta^{n-N}$, so $\sum |a_n|$ converges by Theorems 6.1.6(i), 6.1.7(i), and 2.1.5.
(ii) Choose $\beta > 1$ with $|a_{n+1}/a_n| > \beta$ eventually by Theorem 2.2.5(v).
Then the tail $|a_n|$ is eventually increasing and bounded away from 0, so the series diverges by Corollary 6.1.4.
(iii) The series $\sum 1/n$ and $\sum 1/n^2$ both have ratio limit 1, but behave differently by Corollary 6.1.12.
Structural Commentary: Ratios turn successive terms into a geometric majorant.
Key Insight: Ratios below 1 force geometric decay; ratios above 1 prevent terms from tending to zero. □

Theorem 6.1.20. *Root and Ratio Test Comparison*
For any sequence (c_n) of positive numbers:
(i) $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} \sqrt[n]{c_n}$;
(ii) $\liminf_{n\to\infty} \sqrt[n]{c_n} \leq \limsup_{n\to\infty} \sqrt[n]{c_n}$;
(iii) $\limsup_{n\to\infty} \sqrt[n]{c_n} \leq \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$.

Proof.
(i) Let $\ell = \liminf c_{n+1}/c_n$. If $\ell = 0$, the inequality is automatic. If $\gamma < \ell$ with $\gamma > 0$, choose N such that $(\forall n \geq N)\, c_{n+1} \geq \gamma c_n$. Then $(\forall n \geq N)\, c_n \geq c_N \gamma^{n-N}$. After taking nth roots, $\liminf \sqrt[n]{c_n} \geq \gamma$; let $\gamma \uparrow \ell$.
(ii) This is the general fact that $\liminf a_n \leq \limsup a_n$.
(iii) Let $\alpha = \limsup c_{n+1}/c_n$. If $\alpha < +\infty$ and $\beta > \alpha$, choose N such that $(\forall n \geq N)\, c_{n+1} \leq \beta c_n$. Then $(\forall n \geq N)\, c_n \leq c_N \beta^{n-N}$. After taking nth roots, $\limsup \sqrt[n]{c_n} \leq \beta$; let $\beta \downarrow \alpha$. If $\alpha = +\infty$, the inequality is automatic.
Structural Commentary: A tail ratio bound multiplies into a root bound.
Key Insight: Whenever the ratio test proves convergence, the root test does too. □

Theorem 6.1.21. *Summation by Parts*
Given two sequences (a_n) and (b_n), put $A_n := \sum_{k=0}^{n} a_k$ and $A_{-1} := 0$.
If $0 \leq p \leq q$, then $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$.

Proof. Since $a_n = A_n - A_{n-1}$, $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1} = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$.
Structural Commentary: This is the discrete analogue of integration by parts.
Key Insight: Move the difference from a_n onto b_n by rewriting $a_n = A_n - A_{n-1}$. □

Theorem 6.1.22. *Dirichlet Test*
If the partial sums A_n of $\sum a_n$ are bounded, $b_n \downarrow 0$, and $b_n \geq 0$, then $\sum a_n b_n$ converges.

Proof. Fix $\varepsilon > 0$ and choose $M \geq 0$ with $(\forall n \in \mathbb{N})\ |A_n| \leq M$. If $M = 0$, the result is immediate. Otherwise choose N with $b_N \leq \varepsilon/(2M)$.
For $N \leq p \leq q$, $\left|\sum_{n=p}^{q} a_n b_n\right| \leq M\left(\sum_{n=p}^{q-1}(b_n - b_{n+1}) + b_q + b_p\right) = 2Mb_p \leq 2Mb_N \leq \varepsilon$ by Theorem 6.1.21. Thus convergence follows by Theorem 6.1.3.
Structural Commentary: Bounded oscillation in A_n is damped by a monotone scalar tending to 0.
Key Insight: Summation by parts converts cancellation into a total variation estimate for b_n. □

Corollary 6.1.23. *Alternating Series Test*
Suppose $|c_1| \geq |c_2| \geq |c_3| \geq \cdots$, the signs alternate, and $c_n \to 0$. Then $\sum c_n$ converges.

Proof. After replacing (c_n) by $(-c_n)$ if necessary, $c_n = (-1)^{n+1}|c_n|$. The sequence $|c_n| \downarrow 0$, while the partial sums of $((-1)^{n+1})$ are bounded, so convergence follows by Theorem 6.1.22.
Structural Commentary: Alternation supplies bounded sums; decreasing magnitudes damp them.
Key Insight: Alternating convergence is Dirichlet convergence in its simplest form. □

Perspective 6.1.24. *Alternating Partial Sums as Nested Control*
Set $a_n := |c_n|$. Figure 175 depicts the ordered intervals $I_j = [a_{2j}, a_{2j-1}]$, whose interiors are pairwise disjoint and whose lengths form the paired terms in the even partial sums. The interval picture explains the boundedness in Corollary 6.1.23: the even partial sums collect these lengths inside $[0, a_1]$, while the odd partial sums differ from the even ones by the next term, which tends to 0.

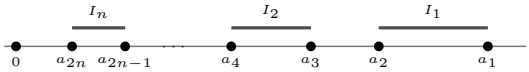

Figure 175: For $a_n := |c_n|$, the ordered intervals $I_j = [a_{2j}, a_{2j-1}]$ have pairwise disjoint interiors in $[0, a_1]$.

Set $s := \sum_{k=1}^{\infty}(-1)^{k+1}/k$, whose existence follows by Corollary 6.1.23. Figure 176 plots the dimensionless partial sums $s_n = \sum_{k=1}^{n}(-1)^{k+1}/k$ vertically against the integer index n horizontally for $1 \leq n \leq 10$, with dashed reference level s.

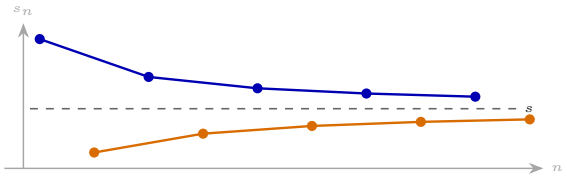

Figure 176: Odd partial sums (blue) decrease and even partial sums (orange) increase toward the series sum s.

Theorem 6.1.25. *Absolute Convergence Implies Convergence*
If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

Proof. For $m \geq n$, $\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k|$. The tails on the right are arbitrarily small by Theorem 6.1.3, so $\sum a_n$ converges by the same theorem.
Structural Commentary: Absolute convergence removes cancellation from the convergence question.
Key Insight: Small absolute tails force small signed or complex tails. □

Theorem 6.1.26. *Addition and Scalar Multiplication of Series*
(i) If $\sum a_n = A$ and $\sum b_n = B$, then $\sum(a_n + b_n) = A + B$.
(ii) If $\sum a_n = A$ and $c \in \mathbb{C}$, then $\sum ca_n = cA$.

Proof.
(i) The sums of the corresponding partial sums converge to $A + B$ by Theorem 2.1.5.
(ii) The scalar multiples of the corresponding partial sums converge to cA by Theorem 2.1.5.
Structural Commentary: Linearity of convergent series is just limit algebra for partial sums.
Key Insight: Add series by adding their partial-sum sequences. □

Definition 6.1.27. *Cauchy Product*
For series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, set $c_n := \sum_{k=0}^{n} a_k b_{n-k}$ for $n \in \mathbb{Z}_{\geq 0}$. The series $\sum_{n=0}^{\infty} c_n$ is their *Cauchy product*.

Figure 177 groups the array entries $a_i b_j$ by diagonals $i + j = n$, matching the definition of each coefficient c_n.

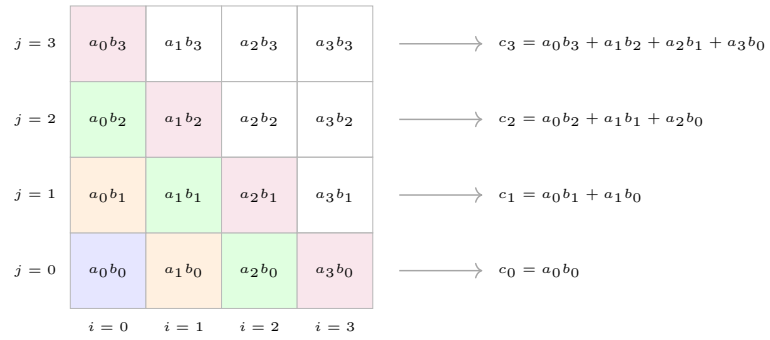

Figure 177: Diagonal sums of (a_ib_j) form the Cauchy product; finite rectangular sums are order-free.

Theorem 6.1.28. *Mertens' Theorem for Cauchy Products*
Suppose $\sum a_n$ *converges absolutely,* $\sum a_n = A$, $\sum b_n = B$, *and* $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. *Then* $\sum_{n=0}^{\infty} c_n = AB$.

Proof.
(i) *Setup*: Let A_n, B_n, C_n be the partial sums and set $\beta_n := B_n - B$.
(ii) *Error Decomposition*: Then $C_n = A_nB + \gamma_n$, and $\gamma_n := a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$. Since $A_nB \to AB$, it suffices to show $\gamma_n \to 0$.
(iii) *Tail Estimate*: Put $\alpha := \sum |a_n|$. If $|\beta_j| \le \varepsilon$ for $j \ge N$, then $|\gamma_n| \le |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon\alpha$.
(iv) *Conclusion*: Let $n \to \infty$ with N fixed. The first term tends to 0 by Corollary 6.1.4, so $\limsup |\gamma_n| \le \varepsilon\alpha$; let $\varepsilon \downarrow 0$.
Structural Commentary: Absolute convergence of one factor controls the convolution error.
Key Insight: One absolute factor makes the Cauchy product converge to AB. □

Perspective 6.1.29. *Diagonal Ordering in a Product*
The coefficient c_n is the diagonal sum of the products a_ib_j with $i + j = n$. Theorem 6.1.28 controls this diagonal ordering when one factor converges absolutely; it does not assert invariance under arbitrary reorderings of the double array.

Definition 6.1.30. *Rearrangement of a Series*
If (k_n) lists each positive integer exactly once and $a'_n := a_{k_n}$, then $\sum a'_n$ is a *rearrangement* of $\sum a_n$.

Theorem 6.1.31. *Riemann Rearrangement Theorem*
Let $\sum a_n$ *be a real series converging but not absolutely, and let* $\alpha, \beta \in [-\infty, \infty]$ *with* $\alpha \le \beta$.
A bijection $\pi : \mathbb{N} \to \mathbb{N}$ *can be chosen so that, for* $s'_N := \sum_{n=1}^{N} a_{\pi(n)}$:

(i) π *rearranges every term of* $\sum a_n$ *exactly once.*
(ii) $\liminf_{N\to\infty} s'_N = \alpha$.
(iii) $\limsup_{N\to\infty} s'_N = \beta$.

Proof. Write $a_n = p_n - q_n$, where $p_n := \max(a_n, 0)$ and $q_n := \max(-a_n, 0)$. Set $u_N := \sum_{n=1}^{N} p_n$ and $v_N := \sum_{n=1}^{N} q_n$. The sequence $(u_N - v_N)$ converges, while $(u_N + v_N)$ is unbounded because $\sum |a_n|$ diverges. Hence neither nondecreasing sequence (u_N) nor (v_N) is bounded, so $\sum p_n = \sum q_n = +\infty$. The limits $p_n, q_n \to 0$ hold by Corollary 6.1.4. List the nonzero p_n as $P_1, P_2, \ldots$ and the nonzero q_n as $Q_1, Q_2, \ldots$, preserving order.
Choose finite targets $L_N < U_N$ with $L_N \to \alpha$, $U_N \to \beta$, and $L_N < U_{N+1}$, taking (L_N) nondecreasing when $\alpha > -\infty$ and (U_N) nonincreasing when $\beta < +\infty$.
(i) Alternately add unused P terms until $s' > U_N$, then unused $-Q$ terms until $s' < L_N$.
Since $\sum P_j = \sum Q_j = +\infty$, all stops are finite and every nonzero term is used; insert zeros anywhere.
(ii) At the Nth down-crossing, $L_N - Q_j \le s'_k < L_N$ with $Q_j \to 0$. Thus lower crossing subsequences tend to α, and the crossing bounds exclude a lower subsequential limit below α; hence $\liminf s'_N = \alpha$.
(iii) At the Nth up-crossing, $U_N < s'_k \le U_N + P_j$ with $P_j \to 0$. Thus upper crossing subsequences tend to β, and the crossing bounds exclude an upper subsequential limit above β; hence $\limsup s'_N = \beta$.
Structural Commentary: Divergent one-sided masses steer the partial sums.
Key Insight: Vanishing terms make the overshoot vanish. □

Theorem 6.1.32. *Absolutely Convergent Series Are Rearrangement-Invariant*
Every rearrangement of an absolutely convergent complex series converges to the same sum.

Proof. Let s_n, s'_n be the original and rearranged partial sums.
Given $\varepsilon > 0$, choose N so that $\sum_{i=N}^{m} |a_i| \le \varepsilon$ for $m \ge N$, then choose p so $a_1, \ldots, a_N$ occur among $a'_1, \ldots, a'_p$.
For $n > p$, $s_n - s'_n$ uses only tail terms, hence $|s_n - s'_n| \le \varepsilon$. Thus s'_n has the same limit as s_n.
Structural Commentary: Absolute tails control any finite reordering of the remaining terms.
Key Insight: Absolute convergence is exactly the protection needed against rearrangement. □

Motivation 6.1.33. *Partial Sums Remember Order*
An infinite sum is a limit of partial sums. Alternating partial sums converge under the hypotheses of Corollary 6.1.23. Nonabsolute convergence remains sensitive to ordering by Theorem 6.1.31, whereas absolute convergence is protected by Theorem 6.1.32.

Perspective 6.1.34. *Formal Algebra Needs Convergence*
Finite sums are algebraic; infinite sums are partial-sum limits. Algebraic manipulations are justified only by convergence control, as in Theorem 6.1.3, Corollary 6.1.4, and Theorems 6.1.28, 6.1.32.
Key Insight: The symbol $\sum_{n=1}^{\infty}$ is a limit claim before it is an algebraic object.

Definition 6.1.35. *Absolute Summation over Countable Sets*
Let X be countable, $f : X \to \mathbb{C}$, and $g : \mathbb{N} \to X$ be a bijection, $g_n := g(n)$.
If $\sum_{n=1}^{\infty} |f(g_n)| < \infty$, then f is *absolutely summable*, and $\sum_{x\in X} f(x) := \sum_{n=1}^{\infty} f(g_n)$.

Theorem 6.1.36. *Countable Sums Are Order-Free Under Absolute Convergence*
Let X, Y be countable, let $f : X \to \mathbb{C}$ be absolutely summable, and let $h : Y \to X$ be a bijection.
(i) $f \circ h : Y \to \mathbb{C}$ *is absolutely summable.*
(ii) $\sum_{y\in Y} f(h(y)) = \sum_{x\in X} f(x)$.
(iii) Countable absolutely convergent sums are independent of enumeration.

Proof. Let $g : \mathbb{N} \to X$ and $k : \mathbb{N} \to Y$ be enumerations, and set $\sigma := g^{-1} \circ h \circ k : \mathbb{N} \to \mathbb{N}$.

(i) As σ is bijective, $h(k_n) = g_{\sigma(n)}$; $\sum_{n=1}^{\infty} |f(h(k_n))| = \sum_{n=1}^{\infty} \left|f(g_{\sigma(n)})\right| < \infty$ by Theorem 6.1.32.

(ii) By Theorem 6.1.32, $\sum_{y\in Y} f(h(y)) = \sum_{n=1}^{\infty} f(g_{\sigma(n)}) = \sum_{n=1}^{\infty} f(g_n) = \sum_{x\in X} f(x)$.

(iii) Take $Y = \mathbb{N}$ and let h be any enumeration of X.

Structural Commentary: Bijections only reorder an absolutely summable list.
Key Insight: Absolute tails make enumeration irrelevant. □

Theorem 6.1.37. *Fubini for Absolutely Summable Arrays*
Let $(a_{mn})_{m,n\geq 1} \subseteq \mathbb{C}$ and $L := \sum_{(m,n)\in\mathbb{N}^2} |a_{mn}| < \infty$. Define $S := \sum_{(m,n)\in\mathbb{N}^2} a_{mn}$ when summed absolutely over $\mathbb{N}^2$.
(i) Each row $\sum_{n=1}^{\infty} a_{mn}$ *and column* $\sum_{m=1}^{\infty} a_{mn}$ *converges absolutely.*
(ii) With $r_m := \sum_{n=1}^{\infty} a_{mn}$ *and* $c_n := \sum_{m=1}^{\infty} a_{mn}$, $\sum_m |r_m| \leq L$ *and* $\sum_n |c_n| \leq L$.
(iii) $\sum_{m=1}^{\infty} r_m = S = \sum_{n=1}^{\infty} c_n$.

Proof.

(i) For fixed m, N, $\sum_{n=1}^{N} |a_{mn}| \leq L$; similarly $\sum_{m=1}^{M} |a_{mn}| \leq L$. Hence every row and column is absolutely summable by Theorem 6.1.5.

(ii) For $M \geq 1$, $\sum_{m=1}^{M} |r_m| \leq \sum_{m=1}^{M} \sum_{n=1}^{\infty} |a_{mn}| \leq L$. Hence $\sum_m |r_m|$ converges with sum at most L by Theorem 6.1.5. The column estimate is identical.

(iii) Given $\varepsilon > 0$, choose finite $F \subseteq \mathbb{N}^2$ with $\sum_{\mathbb{N}^2\setminus F} |a_{mn}| < \varepsilon$, then $R = \{1, \ldots, M\} \times \{1, \ldots, N\} \supseteq F$.
Each of S, $\sum_m r_m$, and $\sum_n c_n$ differs from $S_R := \sum_{(m,n)\in R} a_{mn}$ by at most $\sum_{\mathbb{N}^2\setminus R} |a_{mn}| \leq \varepsilon$. Let $\varepsilon \downarrow 0$.

Structural Commentary: Rectangles plus absolute tails justify Fubini.
Key Insight: Absolute summability makes order irrelevant. □

Method 6.1.38. *Series Proof Patterns*
(i) *Partial sums*: translate assertions about $\sum a_n$ into assertions about $s_n = \sum_{k=1}^{n} a_k$ by Definition 6.1.1.
(ii) *Cauchy tails*: prove convergence from an estimate of $\left|\sum_{k=n}^{m} a_k\right|$ for $m \geq n$ by Theorem 6.1.3.
(iii) *Nonnegative terms*: replace convergence by boundedness of partial sums by Theorem 6.1.5.
(iv) *Comparison*: dominate only the tail by Theorem 6.1.6; finitely many early terms are irrelevant.
(v) *Absolute convergence*: pass to $\sum |a_n|$ by Theorem 6.1.25 when cancellation should not matter.
(vi) *Countable index sets*: choose a bijection $g : \mathbb{N} \to X$; the value of $\sum_{x\in X} a_x$ is independent of g by Theorem 6.1.36.
(vii) *Double sums*: prove identities on $R_{M,N} := \{1, \ldots, M\} \times \{1, \ldots, N\}$, then pass to infinite arrays by Theorem 6.1.37.
(viii) *Conditional convergence*: control the original order by Theorem 6.1.21 and possible rearrangements by Theorem 6.1.31.
Key Insight: Series arguments are sequence arguments with special notation for tails.

7 Uniform Convergence

7.1 Uniform Limits

Uniform convergence asks when limits of functions preserve structure. The key distinction is whether N may depend on x, which controls continuity, integration, differentiation, and the supremum metric.

Perspective 7.1.1. *Function Limits Before Function-Sequence Limits*
$\lim_{t\to x} f(t)$ denotes the metric-space function limit of Definition 5.1.4.
Uniform convergence is the hypothesis that lets the limits in n and t commute.

Definition 7.1.2. *Pointwise Limits of Functions*
Let (Y, d_Y) be a metric space and let $f_n, f : E \to Y$.
(i) $f_n \to f$ *pointwise* on E if $(\forall x \in E)\ d_Y(f_n(x), f(x)) \to 0$.
(ii) If $Y \in \{\mathbb{R}, \mathbb{C}\}$, set $s_N(x) := \sum_{n=1}^{N} f_n(x)$. Then $\sum f_n \to f$ *pointwise* on E if $(\forall x \in E)\ s_N(x) \to f(x)$.

No metric on E is required; topology matters only for preserving continuity, limits, or integrals.

Example 7.1.3. *Pointwise Limits Can Lose Structure*

(i) For $m, n \in \mathbb{N}$, $s_{m,n} := m/(m+n)$ satisfies $\lim_n \lim_m s_{m,n} = 1$ but $\lim_m \lim_n s_{m,n} = 0$.
(ii) On $\mathbb{R}$, set $f_n(x) = x^2/(1+x^2)^n$. Then $\sum_{n=0}^{\infty} f_n$ equals 0 at 0 and $1 + x^2$ otherwise, by Theorem 6.1.7.
(iii) On $\mathbb{R}$, $f_n(x) = \sin(nx)/\sqrt{n}$ satisfies $f_n \to 0$ pointwise but $f_n'(0) = \sqrt{n} \to +\infty$.
(iv) $f_n(x) = n^2 x(1-x^2)^n$ on $[0,1]$ satisfies $f_n \to 0$ pointwise, while $\int_0^1 f_n\,dx = n^2/(2n+2) \to +\infty$.

Key Insight: Pointwise control can miss global spikes and oscillations.

Definition 7.1.4. *Uniform Convergence*
Let (Y, d_Y) be a metric space and let $f_n, f : E \to Y$. Then $f_n \rightrightarrows f$ on E if $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)(\forall x \in E)\ d_Y(f_n(x), f(x)) \leq \varepsilon$.
In the scalar-valued case, take $d_Y(u,v) = |u - v|$; $\sum f_n$ converges uniformly on E if its partial sums do.

Perspective 7.1.5. *One Tail, All Points*
Pointwise convergence lets N depend on x; uniform convergence makes N independent of x.
Equivalently, late graphs lie in one ε-strip around f, as illustrated in Figure 178.

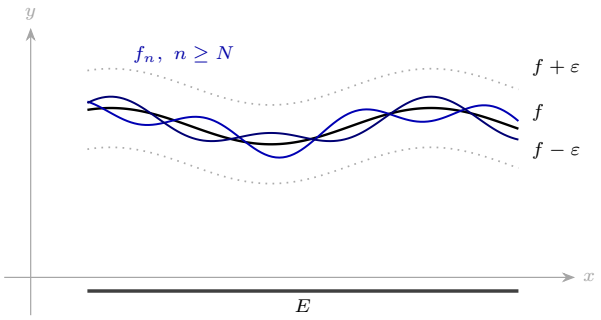

Figure 178: One index N puts all later graphs inside the same ε-strip around f on E.

Example 7.1.6. *x^n Converges Pointwise But Not Uniformly*
On $[0,1]$, let $g_n(x) = x^n$. Then $g_n \to g$ pointwise, where $g(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$
The convergence is not uniform: $(\forall n \geq 1)(\exists x \in [0,1))\ |x^n - g(x)| = x^n > 1/2$. Figure 179 plots the dimensionless functions x, x^2, x^4, x^8 vertically against $x \in [0,1]$ horizontally, showing the persistent endpoint transition.

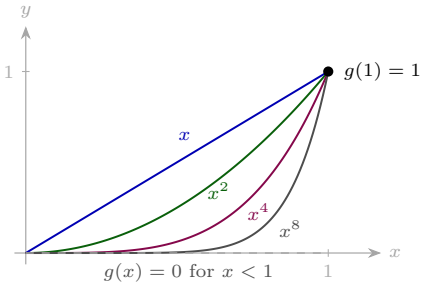

Figure 179: $x^n \to 0$ pointwise on $[0,1)$, but the endpoint prevents uniform convergence on $[0,1]$.

Theorem 7.1.7. *Cauchy Criterion for Uniform Convergence*
For $f_n : E \to \mathbb{C}$, f_n converges uniformly on E if and only if

$$(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall m, n \geq N)(\forall x \in E)\ |f_n(x) - f_m(x)| \leq \varepsilon.$$

Proof.
($\Rightarrow$) Fix $\varepsilon > 0$. If $f_n \rightrightarrows f$, choose N so that $(k \geq N,\ x \in E) \Rightarrow |f_k(x) - f(x)| \leq \varepsilon/2$. Then $(m, n \geq N,\ x \in E) \Rightarrow |f_n(x) - f_m(x)| \leq \varepsilon$.
($\Leftarrow$) For $x \in E$, the criterion makes $(f_n(x))$ Cauchy in $\mathbb{C} \cong \mathbb{R}^2$, so it converges by Corollary 4.2.8; set $f(x) := \lim_n f_n(x)$.
Fix $\varepsilon > 0$ and choose N from the criterion. If $n \geq N$ and $x \in E$, then letting $m \to \infty$ in $|f_n(x) - f_m(x)| \leq \varepsilon$ produces $|f_n(x) - f(x)| \leq \varepsilon$. Hence $f_n \rightrightarrows f$.

Structural Commentary: The same N controls the pointwise Cauchy tails simultaneously across E.
Key Insight: Uniform Cauchy tails are exactly uniform convergence. □

Theorem 7.1.8. *Supremum Test for Uniform Convergence*
Let $\emptyset \neq E$ and $f_n, f : E \to \mathbb{C}$ satisfy $(\forall x \in E)\ f_n(x) \to f(x)$, and set $M_n := \sup_{\overline{\mathbb{R}}}\{|f_n(x) - f(x)| : x \in E\} \in [0, +\infty]$. Then $f_n \rightrightarrows f$ on E if and only if $M_n \to 0$.

Proof.
($\Rightarrow$) Suppose $f_n \rightrightarrows f$ and fix $\varepsilon > 0$. Choose N with $(\forall n \geq N)(\forall x \in E)\ |f_n(x) - f(x)| \leq \varepsilon$. Then $(\forall n \geq N)\ M_n \leq \varepsilon$.
($\Leftarrow$) Suppose $M_n \to 0$ and fix $\varepsilon > 0$. Choose N with $(\forall n \geq N)\ M_n \leq \varepsilon$. Then $(\forall n \geq N)(\forall x \in E)\ |f_n(x) - f(x)| \leq \varepsilon$.

Structural Commentary: Uniform convergence is convergence in the supremum error.
Key Insight: The pointwise errors must vanish as one global quantity. □

Theorem 7.1.9. *Weierstrass M-Test*
Let $f_n : E \to \mathbb{C}$ and $(M_n) \subseteq [0, +\infty)$ satisfy $(\forall n \in \mathbb{N})(\forall x \in E)\ |f_n(x)| \le M_n$.
If $\sum M_n$ converges, then $\sum f_n$ converges uniformly on E.

Proof. Set $s_N := \sum_{k=1}^{N} f_k$. If $m > n$ and $x \in E$, then $|s_m(x) - s_n(x)| \le \sum_{k=n+1}^{m} M_k$.
By Theorem 6.1.3, $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall m > n \ge N)\ \sum_{k=n+1}^{m} M_k \le \varepsilon$.
Thus (s_N) is uniformly Cauchy and converges uniformly by Theorem 7.1.7.
Structural Commentary: Absolute numerical tails dominate functional tails.
Key Insight: A summable uniform majorant ensures uniform convergence of a function series. □

Theorem 7.1.10. *Uniform Convergence Interchanges Limits*
Let (X, d) be a metric space, let $E \subseteq X$, let p be a limit point of E, and let $f_n, f : E \to \mathbb{C}$ satisfy $f_n \rightrightarrows f$ on E.
Suppose $A_n := \lim_{t \to p} f_n(t) \in \mathbb{C}$ for $n \in \mathbb{N}$. Then:

(i) (A_n) converges;
(ii) $\lim_{t \to p} f(t) = \lim_{n \to \infty} A_n$.

Proof.

(i) Fix $\varepsilon > 0$ and choose N with $(\forall m, n \ge N)(\forall t \in E)\ |f_n(t) - f_m(t)| \le \varepsilon$ by Theorem 7.1.7. Letting $t \to p$ produces $(\forall m, n \ge N)\ |A_n - A_m| \le \varepsilon$. Thus (A_n) is Cauchy in $\mathbb{C} \cong \mathbb{R}^2$ and converges by Corollary 4.2.8; set $A := \lim_n A_n$.
(ii) For $n \in \mathbb{N}$, $|f(t) - A| \le |f(t) - f_n(t)| + |f_n(t) - A_n| + |A_n - A|$. Given $\varepsilon > 0$, choose n so that $(\forall t \in E)\ |f(t) - f_n(t)| \le \varepsilon/4$ and $|A_n - A| < \varepsilon/4$. Since $A_n = \lim_{t \to p} f_n(t)$, choose $\delta > 0$ so that $(t \in E,\ 0 < d(t, p) < \delta) \Rightarrow |f_n(t) - A_n| < \varepsilon/4$. The three-term estimate is then $< 3\varepsilon/4$, so $\lim_{t \to p} f(t) = A$.

Structural Commentary: Uniform convergence controls the outer two terms in the three-term estimate.
Key Insight: Uniformity is precisely what allows the $n \to \infty$ limit to move past the $t \to p$ limit. □

Theorem 7.1.11. *Uniform Limits Preserve Continuity*
Let E be a subspace of a metric space and let $f_n, f : E \to \mathbb{C}$. If (f_n) consists of continuous functions and $f_n \rightrightarrows f$ on E, then f is continuous on E.

Proof. Let $x \in E$. If x is isolated, f is continuous at x.
If x is a limit point, then $(\forall n \in \mathbb{N})\ \lim_{t \to x} f_n(t) = f_n(x)$; by Theorem 7.1.10, $\lim_{t \to x} f(t) = \lim_{n \to \infty} f_n(x) = f(x)$.
Structural Commentary: Uniform convergence passes continuity to the limit.
Key Insight: Uniform convergence protects continuity from the failure modes in Example 7.1.3. □

Theorem 7.1.12. *Dini's Monotone Uniform Convergence Theorem*
Let K be a compact metric space and let $f_n, f : K \to \mathbb{R}$ be continuous. If $(\forall x \in K)\ f_n(x) \downarrow f(x)$, then $f_n \rightrightarrows f$ on K.

Proof. Set $g_n := f_n - f$. Each g_n is continuous by Theorem 5.1.32(i), and $(\forall x \in K)\ g_n(x) \downarrow 0$. Given $\varepsilon > 0$, let $K_n := g_n^{-1}([\varepsilon, +\infty))$.
By Corollary 5.1.20, K_n is closed in K, hence compact by Theorem 4.1.7. The sequence (K_n) is nested and $\bigcap_n K_n = \emptyset$. If $(\forall n \in \mathbb{N})\ K_n \neq \emptyset$, then $\bigcap_n K_n \neq \emptyset$ by Corollary 4.1.10, a contradiction. Thus $(\exists N \in \mathbb{N})\ K_N = \emptyset$.
Nestedness now implies $(\forall n \ge N)(\forall x \in K)\ 0 \le f_n(x) - f(x) < \varepsilon$.
Structural Commentary: Monotone convergence turns the bad sets into a nested compact sequence.
Key Insight: Compactness turns pointwise eventual escape from the nested bad sets into a uniform escape. □

Definition 7.1.13. *The Supremum Metric on Bounded Continuous Functions*
Let X be a nonempty metric space.

(i) $C_b(X)$ is the set of bounded continuous maps $X \to \mathbb{C}$.
(ii) For $f \in C_b(X)$, $\|f\|_\infty := \sup_{x \in X} |f(x)|$.
(iii) The *supremum metric* on $C_b(X)$ is $d_\infty(f, g) := \|f - g\|_\infty$.
(iv) For $(f_n) \subseteq C_b(X)$ and $f \in C_b(X)$, $f_n \to f$ in $d_\infty \iff \|f_n - f\|_\infty \to 0 \iff f_n \rightrightarrows f$ on X.

Key Insight: Uniform convergence is metric convergence in $C_b(X)$.

Perspective 7.1.14. *A Metric Space Whose Points Are Functions*
Uniform convergence is sequence convergence in the metric space $C_b(X)$.

Perspective 7.1.15. *Four Limit Notions*
The four limit notions are sequence limits, function limits, pointwise function limits, and uniform function limits. By Theorem 7.1.8, the last is ordinary sequence convergence in the supremum metric.

7.2 Interchange of Limits

Theorem 7.2.1. *Uniform Convergence and Riemann–Stieltjes Integration*
Let $a < b$, let $\alpha : [a, b] \to \mathbb{R}$ be increasing, and let $f_n, f : [a, b] \to \mathbb{R}$. Assume $(\forall n \in \mathbb{N})\ f_n \in \mathcal{R}(\alpha)$ and $f_n \rightrightarrows f$ on $[a, b]$. Then:

(i) $f \in \mathcal{R}(\alpha)$;
(ii) $\int_a^b f\, d\alpha = \lim_{n \to \infty} \int_a^b f_n\, d\alpha$.

Proof.

(i) Choose n_0 with $(\forall x \in [a, b])\ |f(x) - f_{n_0}(x)| \le 1$. Since f_{n_0} is bounded by Definition 2.1.1, so is f. Set $A := \alpha(b) - \alpha(a)$ and $\varepsilon_n := \sup_{x \in [a,b]} |f_n(x) - f(x)|$. Then $\varepsilon_n \to 0$ by Theorem 7.1.8.
Fix $\delta > 0$ and choose n with $2A\varepsilon_n < \delta/2$. Choose a partition Π with $U(\Pi, f_n, \alpha) - L(\Pi, f_n, \alpha) < \delta/2$ by Theorem 2.1.6. The pointwise bounds $f_n - \varepsilon_n \le f \le f_n + \varepsilon_n$ imply $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) \le U(\Pi, f_n, \alpha) - L(\Pi, f_n, \alpha) + 2A\varepsilon_n < \delta$. Thus $f \in \mathcal{R}(\alpha)$ by Theorem 2.1.6.

(ii) The function $f - f_n$ is integrable by Theorem 2.1.14(i), and $(\forall x \in [a,b])\ |f(x) - f_n(x)| \le \varepsilon_n$. Hence $\left|\int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha\right| \le A\varepsilon_n \to 0$ by Theorem 2.1.14(i),(iv).

Structural Commentary: Uniform error becomes an integral error multiplied by the total α-variation.
Key Insight: Uniform convergence is strong enough to pass limits through the Riemann–Stieltjes integral. □

Corollary 7.2.2. *Termwise Integration of Uniformly Convergent Series*
Let $a < b$, let $\alpha : [a,b] \to \mathbb{R}$ be increasing, and let $f_n : [a,b] \to \mathbb{R}$ satisfy $(\forall n \in \mathbb{N})\ f_n \in \mathcal{R}(\alpha)$. If $\sum_{n=1}^{\infty} f_n$ converges uniformly to f on $[a,b]$, then $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$.

Proof. Let $s_N := \sum_{n=1}^{N} f_n$. Then $s_N \in \mathcal{R}(\alpha)$ and $\int_a^b s_N\,d\alpha = \sum_{n=1}^{N} \int_a^b f_n\,d\alpha$ by Theorem 2.1.14(i).
Since $s_N \rightrightarrows f$, $\int_a^b f\,d\alpha = \lim_{N\to\infty} \int_a^b s_N\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$ by Theorem 7.2.1(ii).

Structural Commentary: Reduce the series to its uniformly convergent partial sums.
Key Insight: Uniform convergence lets integration be performed term by term. □

Theorem 7.2.3. *Differentiation Under a Riemann–Stieltjes Integral*
Let $a < b$, $c < d$, let $\alpha : [a,b] \to \mathbb{R}$ be increasing, let $\varphi : [a,b] \times [c,d] \to \mathbb{R}$, and write $\varphi^t(x) := \varphi(x,t)$ and $(D_2\varphi)^t(x) := D_2\varphi(x,t)$ where defined. Fix $s \in (c,d)$ and let $\eta > 0$ satisfy $(s-\eta, s+\eta) \subseteq (c,d)$.
Assume $(\forall t \in [c,d])\ \varphi^t \in \mathcal{R}(\alpha)$, $D_2\varphi$ exists on $[a,b] \times (s-\eta, s+\eta)$, and $\sup_{x\in[a,b]} |D_2\varphi(x,t) - D_2\varphi(x,s)| \to 0$ as $t \to s$. Set $F(t) := \int_a^b \varphi(x,t)\,d\alpha(x)$. Then:
(i) $(D_2\varphi)^s \in \mathcal{R}(\alpha)$;
(ii) $F'(s) = \int_a^b (D_2\varphi)(x,s)\,d\alpha(x)$.

Proof. For $0 < |t-s| < \eta$, set $\psi_t := (\varphi^t - \varphi^s)/(t-s) \in \mathcal{R}(\alpha)$ by Theorem 2.1.14(i). For fixed $x \in [a,b]$, the map $u \mapsto \varphi(x,u)$ is continuous on the interval with endpoints s,t by Theorem 1.1.4. Choose $\tau_x(t)$ between s and t such that $\psi_t(x) = D_2\varphi(x,\tau_x(t))$ by Corollary 1.1.13.
Since $|\tau_x(t) - s| \le |t-s|$, the uniform slice hypothesis implies $\sup_{x\in[a,b]} |\psi_t(x) - D_2\varphi(x,s)| \to 0$ as $t \to s$.

(i) Set $t_j := s + \eta/(j+2)$. Then $\psi_{t_j} \rightrightarrows (D_2\varphi)^s$ on $[a,b]$, so $(D_2\varphi)^s \in \mathcal{R}(\alpha)$ by Theorem 7.2.1(i).
(ii) Set $A := \alpha(b) - \alpha(a)$. By Theorem 2.1.14(i),(iv),

$$\left|\frac{F(t)-F(s)}{t-s} - \int_a^b (D_2\varphi)^s\,d\alpha\right| = \left|\int_a^b \left[\psi_t - (D_2\varphi)^s\right] d\alpha\right| \le A \sup_{x\in[a,b]} |\psi_t(x) - D_2\varphi(x,s)| \to 0.$$

Thus $F'(s) = \int_a^b (D_2\varphi)(x,s)\,d\alpha(x)$.

Structural Commentary: The mean-value theorem transfers uniform derivative-slice control to integrable difference quotients.
Key Insight: Uniform convergence permits $D_t \int = \int D_t$. □

Theorem 7.2.4. *Uniform Convergence and Differentiation*
Let $a < b$, let $f_n : [a,b] \to \mathbb{R}$ be differentiable relative to $[a,b]$, let $g : [a,b] \to \mathbb{R}$, and fix $x_0 \in [a,b]$. Assume $f_n(x_0)$ converges and $f_n' \rightrightarrows g$. Then $(\exists f : [a,b] \to \mathbb{R})$ such that:
(i) $f_n \rightrightarrows f$ on $[a,b]$;
(ii) f is differentiable relative to $[a,b]$ and $f' = g$.

Proof. For $m,n \in \mathbb{N}$, the function $f_n - f_m$ is differentiable relative to $[a,b]$ by Theorem 1.1.5(i), hence continuous by Theorem 1.1.4.

(i) For $m,n \in \mathbb{N}$ and $x \in [a,b]$, $|f_n(x) - f_m(x)| \le |f_n(x_0) - f_m(x_0)| + (b-a) \sup_{t\in[a,b]} |f_n'(t) - f_m'(t)|$ by Corollary 1.1.13. The anchor sequence is Cauchy by Theorem 2.1.15, and (f_n') is uniformly Cauchy by Theorem 7.1.7. Thus (f_n) is uniformly Cauchy. For $x \in [a,b]$, set $f(x) := \lim_n f_n(x) \in \mathbb{R}$ by Theorem 2.1.15. Then $f_n \rightrightarrows f$ by Theorem 7.1.7.
(ii) Fix $x \in [a,b]$, set $E := [a,b] \setminus \{x\}$, and define $\phi_n, \phi : E \to \mathbb{R}$ by $\phi_n(t) := \frac{f_n(t)-f_n(x)}{t-x}$ and $\phi(t) := \frac{f(t)-f(x)}{t-x}$. For $t \in E$, $|\phi_n(t) - \phi_m(t)| \le \sup_{u\in[a,b]} |f_n'(u) - f_m'(u)|$ by Corollary 1.1.13. Hence (ϕ_n) is uniformly Cauchy. Its pointwise limit is ϕ by (i), so $\phi_n \rightrightarrows \phi$ by Theorem 7.1.7.
Since $a < b$, x is a limit point of E. Since $\lim_{t\to x} \phi_n(t) = f_n'(x)$, $\lim_{t\to x} \phi(t) = \lim_{n\to\infty} f_n'(x) = g(x)$ by Theorem 7.1.10. Thus $f'(x) = g(x)$ by Definition 1.1.1.

Structural Commentary: Derivative control anchors functions, then quotients.
Key Insight: Uniform derivative convergence plus one anchor passes derivatives to the limit. □

7.3 Further Topics in Uniform Convergence

Method 7.3.1. *Uniform Convergence Proof Patterns*
Let $\emptyset \ne E$ and let $f_n, f : E \to \mathbb{C}$.

(i) *Pointwise*: fix $(x,\varepsilon) \in E \times (0,+\infty)$, then choose $N = N(x,\varepsilon)$ with $(n \ge N) \Rightarrow |f_n(x) - f(x)| \le \varepsilon$ by Definition 7.1.2.
(ii) *Uniform*: fix $\varepsilon > 0$, then choose $N = N(\varepsilon)$ with $(n \ge N,\ x \in E) \Rightarrow |f_n(x) - f(x)| \le \varepsilon$ by Definition 7.1.4. After pointwise convergence is known, equivalently set $M_n := \sup_{\mathbb{R}}\{|f_n(x) - f(x)| : x \in E\}$ and prove $M_n \to 0$ by Theorem 7.1.8.
(iii) *Uniform Cauchy*: verify $(\forall \varepsilon > 0)(\exists N \in \mathbb{N})(\forall m,n \ge N)(\forall x \in E)\ |f_n(x) - f_m(x)| \le \varepsilon$; uniform convergence follows by Theorem 7.1.7.
(iv) *Preservation*: when E is a metric subspace, split the target error into uniform-approximation, fixed-index local, and limiting-value errors by Theorem 7.1.10; continuity follows by Theorem 7.1.11.
(v) *Series of functions*: find $(M_n) \subseteq [0,+\infty)$ with $(\forall n \in \mathbb{N})(\forall x \in E)\ |f_n(x)| \le M_n$ and $\sum M_n$ convergent; uniform convergence follows by Theorem 7.1.9.

(vi) *Termwise operations*: for $a < b$, increasing $\alpha : [a,b] \to \mathbb{R}$, and real-valued f_n, pass uniformly convergent series through the Riemann–Stieltjes integral by Corollary 7.2.2. Pass derivatives to the limit from one convergent anchor by Theorem 7.2.4, or differentiate under an integral from uniform derivative-slice control by Theorem 7.2.3.

Key Insight: Uniformity is the quantifier order $N = N(\varepsilon)$ before $x \in E$.

8 Function-Space Compactness

8.1 Equicontinuity

Function-space compactness treats functions as points. Supremum metrics preview Banach spaces, Baire category describes generic behavior, and equicontinuity leads to Arzela–Ascoli.

Theorem 8.1.1. *Completeness of the Supremum Metric*
Let X be a nonempty metric space. Then $C_b(X)$ with $d_\infty(f,g) = \|f-g\|_\infty$ is complete.

Proof. Let (f_n) be d_∞-Cauchy in $C_b(X)$. By Definition 7.1.13, (f_n) satisfies the uniform Cauchy criterion, so $f_n \rightrightarrows f$ by Theorem 7.1.7, where $f : X \to \mathbb{C}$.
The function f is continuous by Theorem 7.1.11. Choose N with $(\forall x \in X)\ |f(x) - f_N(x)| \le 1$. Since f_N is bounded, so is f, hence $f \in C_b(X)$. Finally, $d_\infty(f_n, f) = \|f_n - f\|_\infty \to 0$ by Definition 7.1.13.
Structural Commentary: Uniform Cauchy convergence builds the limit inside the same function space.
Key Insight: Completeness of $\mathbb{C}$ plus uniform convergence makes $C_b(X)$ complete. □

Definition 8.1.2. *Bounded Target Metric for All Functions*
Let (X, d_X) and (Y, d_Y) be nonempty metric spaces, let $\mathcal{F} := Y^X$, and set $\rho(y,z) := \frac{d_Y(y,z)}{1+d_Y(y,z)}$ and $d_\rho(f,g) := \sup_{x\in X} \rho(f(x), g(x))$. The function ρ is a bounded metric on Y by Proposition 3.1.10. Taking the supremum of its pointwise triangle inequality makes d_ρ a finite metric on $\mathcal{F}$. The radius identity $d_Y(y,z) < r \iff \rho(y,z) < r/(1+r)$ for $r > 0$ makes d_ρ-convergence equivalent to uniform convergence measured by d_Y.

Theorem 8.1.3. *Completeness and Closed Subspaces for d_ρ*
Let (X, d_X) and (Y, d_Y) be nonempty metric spaces, and let $\mathcal{F}$, ρ, and d_ρ be as in Definition 8.1.2. If (Y, d_Y) is complete, then:
(i) $(\mathcal{F}, d_\rho)$ is complete;
(ii) $\mathcal{F}_b := \{f \in \mathcal{F} : f(X)$ is d_Y-bounded$\}$ is closed in $\mathcal{F}$;
(iii) $C(X,Y)$ is closed in $\mathcal{F}$.

Proof.
(i) Let (f_n) be d_ρ-Cauchy. Fix $(x,\varepsilon) \in X \times (0,+\infty)$ and set $\eta := \varepsilon/(1+\varepsilon)$. Eventually $d_\rho(f_n, f_m) < \eta$, hence $d_Y(f_n(x), f_m(x)) < \varepsilon$ by Definition 8.1.2. Thus $(f_n(x))$ is Cauchy and converges in Y by completeness; set $f(x) := \lim_n f_n(x)$.
Fix $0 < \varepsilon < 1$ and choose N with $(m, n \ge N) \Rightarrow d_\rho(f_n, f_m) < \varepsilon/2$. For $n \ge N$, let $m \to \infty$ pointwise and use continuity of ρ to obtain $(\forall x \in X)\ \rho(f_n(x), f(x)) \le \varepsilon/2$. Hence $d_\rho(f_n, f) \le \varepsilon/2 < \varepsilon$, so $(\mathcal{F}, d_\rho)$ is complete.
(ii) Let $(f_n) \subseteq \mathcal{F}_b$ and $d_\rho(f_n, f) \to 0$. Choose N with $d_\rho(f_N, f) < 1/2$. Then $(\forall x \in X)\ d_Y(f_N(x), f(x)) < 1$ by Definition 8.1.2. Since $f_N(X)$ is bounded, so is $f(X)$. Therefore $\mathcal{F}_b$ is closed by Theorem 3.4.17(ii).
(iii) Let $(f_n) \subseteq C(X,Y)$ and $d_\rho(f_n, f) \to 0$. Then $f_n \to f$ uniformly in d_Y by Definition 8.1.2. Fix $x_0 \in X$ and $\varepsilon > 0$. Choose N with $(\forall x \in X)\ d_Y(f_N(x), f(x)) < \varepsilon/3$, and then choose $\delta > 0$ with $d_X(x, x_0) < \delta \Rightarrow d_Y(f_N(x), f_N(x_0)) < \varepsilon/3$. By the triangle inequality, $d_Y(f(x), f(x_0)) < \varepsilon$, so f is continuous. Therefore $C(X,Y)$ is closed by Theorem 3.4.17(ii).

Structural Commentary: Radius inversion recovers target Cauchy control; uniform limits retain boundedness and continuity.
Key Insight: Completeness and uniform control make boundedness and continuity closed. □

Definition 8.1.4. *Boundedness and Equicontinuity of Function Families*
Let (X, d_X) and (Y, d_Y) be metric spaces, let $E \subseteq X$, let $\mathcal{F} \subseteq Y^E$, and let $(f_n) \subseteq \mathcal{F}$.
(i) (f_n) is *pointwise bounded* on E if $(\forall x \in E)\ \{f_n(x) : n \in \mathbb{N}\}$ is bounded in Y.
(ii) If $Y \in \{\mathbb{R}, \mathbb{C}\}$, then (f_n) is *uniformly bounded* on E if $(\exists M > 0)(\forall n \in \mathbb{N})(\forall x \in E)\ |f_n(x)| \le M$.
(iii) The family $\mathcal{F}$ is *equicontinuous* on E if $(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x, y \in E)(\forall f \in \mathcal{F})\ [d_X(x,y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon]$.
(iv) The family $\mathcal{F}$ is *pointwise equicontinuous* on E if $(\forall p \in E)(\forall \varepsilon > 0)(\exists \delta > 0)(\forall y \in E)(\forall f \in \mathcal{F})\ [d_X(p,y) < \delta \Rightarrow d_Y(f(p), f(y)) < \varepsilon]$.

Figure 180 depicts the common scale: one δ about p controls $(\forall f \in \mathcal{F})\ d_Y(f(p), f(y)) < \varepsilon$ when $d_X(p,y) < \delta$.

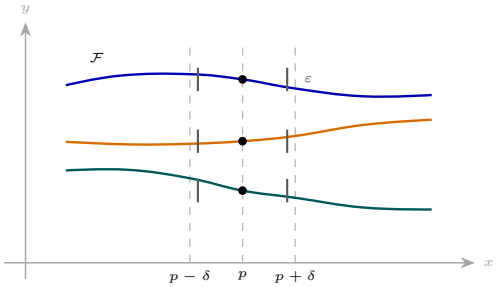

Figure 180: One input scale δ controls every curve in $\mathcal{F}$ within tolerance ε.

Theorem 8.1.5. *Pointwise Equicontinuity Becomes Uniform on Compact Sets*
Let K be a compact metric space and let $\mathcal{F} \subseteq C(K, \mathbb{C})$ be pointwise equicontinuous. Then $\mathcal{F}$ is equicontinuous.

Proof. If $\mathcal{F}$ is not equicontinuous, choose $\varepsilon > 0$, $f_n \in \mathcal{F}$, and $x_n, t_n \in K$ with $d(x_n, t_n) < 1/n$ and $|f_n(x_n) - f_n(t_n)| \geq \varepsilon$. Pass to a subsequence with $x_n \to x_0$ by Theorem 4.1.13; then $t_n \to x_0$ by the triangle inequality.
Choose $\delta > 0$ with $(y \in K,\ d(x_0, y) < \delta,\ h \in \mathcal{F}) \Rightarrow |h(x_0) - h(y)| < \varepsilon/2$ by Definition 8.1.4(iv). Eventually $x_n, t_n \in B_\delta(x_0)$, so $|f_n(x_n) - f_n(t_n)| < \varepsilon$ by the triangle inequality, a contradiction.
Structural Commentary: Compactness turns moving counterexamples into one base point.
Key Insight: The local $\delta(x, \varepsilon)$ becomes one uniform δ only after compactness prevents escape. □

Theorem 8.1.6. *Uniform Convergence Implies Equicontinuity on Compact Sets*
Let K be a nonempty compact metric space and let $(f_n) \subseteq C_b(K)$. If $f_n \rightrightarrows f$ on K, then $\{f_n : n \in \mathbb{N}\}$ is equicontinuous.

Proof. Fix $\varepsilon > 0$.

(i) *Tail Anchor*: Choose N with $(\forall n > N)\ \|f_n - f_N\|_\infty < \varepsilon/3$ by Theorem 7.1.7.
(ii) *Finite Initial Family*: By Theorem 5.2.6 and finiteness, choose $\delta > 0$ with $(\forall j \in \{1, \dots, N\})(\forall x, y \in K)\ [d(x, y) < \delta \Rightarrow |f_j(x) - f_j(y)| < \varepsilon/3]$.
(iii) *Tail Estimate*: If $n > N$ and $d(x, y) < \delta$, then $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \varepsilon$.
(iv) *Conclusion*: The preceding bounds establish equicontinuity for $\{f_n : n \in \mathbb{N}\}$.

Structural Commentary: A uniformly convergent tail stays close to one uniformly continuous function.
Key Insight: Compactness handles finitely many functions; uniform convergence handles the tail. □

Example 8.1.7. *Uniform Boundedness Alone Is Not Enough*
On $[0, 1]$, let $f_n(x) := \frac{x^2}{x^2 + (1 - nx)^2}$. Then $0 \leq f_n \leq 1$, $f_n \to 0$ pointwise, and $f_n(1/n) = 1$, so $\|f_n\|_\infty = 1$. Every subsequence has pointwise limit 0 and therefore fails the supremum test for uniform convergence in Theorem 7.1.8. Moreover, $|f_n(1/n) - f_n(0)| = 1$ while $1/n \to 0$, so $\{f_n : n \in \mathbb{N}\}$ is not equicontinuous.

Theorem 8.1.8. *Diagonal Selection on a Countable Set*
Let E be countable and let $f_n : E \to \mathbb{C}$ satisfy $(\forall x \in E)\ \{f_n(x) : n \in \mathbb{N}\}$ bounded. Then (f_n) has a subsequence converging pointwise on E.

Proof. If $E = \emptyset$, the claim is immediate. Choose a sequence (x_j) with range E, repeating elements when E is finite. Set $S_0 := (f_n)$. Inductively choose a subsequence S_j of S_{j-1} whose values at x_j converge by Corollary 4.2.7, with $\mathbb{C} \cong \mathbb{R}^2$.
Write $S_j = (f_k^{(j)})_{k \in \mathbb{N}}$ and set $g_j := f_j^{(j)}$. For fixed k, $(g_j)_{j \geq k}$ is a subsequence of S_k, so $g_j(x_k)$ converges. Since $\{x_j : j \in \mathbb{N}\} = E$, (g_j) converges pointwise on E.
Structural Commentary: The diagonal keeps the ith convergence property after the ith step.
Key Insight: Countability permits diagonal convergence. □

8.2 Arzela-Ascoli-Type Compactness

Theorem 8.2.1. *Arzela–Ascoli Selection Theorem*
Let K be a nonempty compact metric space and let $(f_n) \subseteq C_b(K)$. If $(\forall x \in K)\ \sup_{n \in \mathbb{N}} |f_n(x)| < +\infty$ and $\{f_n : n \in \mathbb{N}\}$ is equicontinuous on K, then:

(i) $(\exists M > 0)(\forall n \in \mathbb{N})(\forall x \in K)\ |f_n(x)| \leq M$;
(ii) *(f_n) contains a uniformly convergent subsequence.*

Proof.

(i) Choose $\delta > 0$ so that $(n \in \mathbb{N},\ x, y \in K,\ d(x, y) < \delta) \Rightarrow |f_n(x) - f_n(y)| < 1$ by Definition 8.1.4(iii). By compactness, choose $p_1, \dots, p_r \in K$ with $K \subseteq \bigcup_{j=1}^r B_\delta(p_j)$, and set $M_j := \sup_{n \in \mathbb{N}} |f_n(p_j)| < +\infty$. If $x \in B_\delta(p_j)$, then $|f_n(x)| < 1 + M_j$. Thus (i) holds with $M := 1 + \max_{1 \leq j \leq r} M_j$.
(ii) Choose an at most countable dense set $E \subseteq K$ by Proposition 4.1.35. A subsequence (g_i) converging pointwise on E exists by Theorem 8.1.8 and pointwise boundedness. Fix $\varepsilon > 0$, and choose $\delta > 0$ from equicontinuity with tolerance $\varepsilon/4$. The cover $\{B_\delta(a) : a \in E\}$ of K has a finite subcover with centers $x_1, \dots, x_r \in E$. Choose $I \in \mathbb{N}$ such that $(i, j \geq I,\ 1 \leq s \leq r) \Rightarrow |g_i(x_s) - g_j(x_s)| < \varepsilon/4$ by pointwise convergence at these centers.
For $x \in K$, choose s with $d(x, x_s) < \delta$. If $i, j \geq I$, then $|g_i(x) - g_j(x)| < 3\varepsilon/4$. Hence $d_\infty(g_i, g_j) \leq 3\varepsilon/4 < \varepsilon$, and (g_i) converges uniformly by Theorem 8.1.1.

Structural Commentary: Finite domain nets globalize pointwise bounds; dense convergence and a second finite net produce a uniform Cauchy subsequence.
Key Insight: Equicontinuity turns finitely many sampled bounds and limits into uniform control. □

Figure 181 illustrates conclusion (ii) of Theorem 8.2.1: the selected functions f_{n_k} converge uniformly to f.

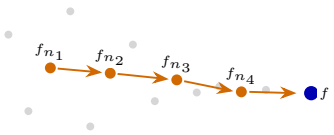

Figure 181: Arzela–Ascoli selects a uniformly convergent subsequence.

Theorem 8.2.2. *Compact Subsets of $C(K)$*
Let K be a nonempty compact metric space and let $C(K, \mathbb{R})$ carry the supremum metric.
For $\mathcal{F} \subseteq C(K, \mathbb{R})$, the following are equivalent:

(i) *$\mathcal{F}$ is compact.*
(ii) *The family $\mathcal{F}$ is closed and equicontinuous, and $(\exists M > 0)(\forall f \in \mathcal{F})(\forall x \in K)\ |f(x)| \leq M$.*

Proof.

(*i*) $\Rightarrow$ (*ii*) The set $\mathcal{F}$ is closed by Theorem 4.1.6, and it is totally bounded by Theorem 4.1.34, direction (*i*) $\Rightarrow$ (*ii*). If $\mathcal{F} = \emptyset$, the remaining conditions are vacuous; assume $\mathcal{F} \neq \emptyset$. The family is bounded by Theorem 4.1.30; choose $g \in \mathcal{F}$ and $R > 0$ with $(\forall f \in \mathcal{F})\ \|f - g\|_\infty < R$. Then $(\forall f \in \mathcal{F})(\forall x \in K)\ |f(x)| < R + \|g\|_\infty$.
Fix $\varepsilon > 0$. Choose a finite $\varepsilon/3$-net $g_1, \dots, g_r \in \mathcal{F}$ by Proposition 4.1.29. Choose $\delta > 0$ such that $(1 \le i \le r,\ x, y \in K,\ d(x,y) < \delta) \Rightarrow |g_i(x) - g_i(y)| < \varepsilon/3$ by Theorem 5.2.6 and finiteness. For $f \in \mathcal{F}$, choose i with $\|f - g_i\|_\infty < \varepsilon/3$. Then $(x, y \in K,\ d(x,y) < \delta) \Rightarrow |f(x) - f(y)| < \varepsilon$, so $\mathcal{F}$ is equicontinuous.

(*ii*) $\Rightarrow$ (*i*) Each function in $C(K, \mathbb{R})$ is bounded by Corollary 5.1.38, so $C(K, \mathbb{R}) \subseteq C_b(K)$. A supremum-Cauchy sequence in $C(K, \mathbb{R})$ converges in $C_b(K)$ by Theorem 8.1.1, and its pointwise limits remain real-valued. Thus $C(K, \mathbb{R})$ is complete, and the closed subspace $\mathcal{F}$ is complete by Proposition 4.1.26(ii).
Fix $\varepsilon > 0$, choose $\delta > 0$ from equicontinuity with tolerance $\varepsilon/4$, and choose a finite δ-net $x_1, \dots, x_r \in K$ by Theorem 4.1.34, direction (*i*) $\Rightarrow$ (*ii*). Choose $M > 0$ from (ii), and partition $[-M, M]$ into finitely many sets of diameter less than $\varepsilon/4$. Classify $f \in \mathcal{F}$ by the cells containing $(f(x_1), \dots, f(x_r))$, and choose one representative from each realized class. If f and g lie in the same class and $d(x, x_i) < \delta$, then $|f(x) - g(x)| < 3\varepsilon/4$. Thus the finite representative set is an ε-net for $\mathcal{F}$, so $\mathcal{F}$ is totally bounded. Therefore $\mathcal{F}$ is compact by Theorem 4.1.34, direction (*ii*) $\Rightarrow$ (*i*).

Structural Commentary: Compactness in $C(K)$ is finite-resolution control plus closedness.
Key Insight: Equicontinuity lets finitely many sample values control whole functions. □

Example 8.2.3. *Hölder Balls Have Compact Closure*
Fix $0 < \alpha \le 1$. For $f \in C([0,1], \mathbb{R})$, set $[f]_\alpha := \sup_{0 \le x < y \le 1} \frac{|f(x) - f(y)|}{|x - y|^\alpha} \in [0, +\infty]$ and $\mathcal{H}_\alpha := \{f \in C([0,1], \mathbb{R}) : \|f\|_\infty + [f]_\alpha \le 1\}$.

(i) $(\forall f \in \mathcal{H}_\alpha)\ \|f\|_\infty \le 1$.
(ii) $(\forall f \in \mathcal{H}_\alpha)(\forall x, y \in [0,1])\ |f(x) - f(y)| \le |x - y|^\alpha$, so $\mathcal{H}_\alpha$ is equicontinuous.
(iii) If $(f_m) \subseteq \mathcal{H}_\alpha$ and $f_m \rightrightarrows f$, then $\|f\|_\infty \le 1$ and $(\forall x, y \in [0,1])\ |f(x) - f(y)| \le |x - y|^\alpha$.

Thus $\overline{\mathcal{H}_\alpha}$ is closed and equicontinuous and satisfies $(\forall f \in \overline{\mathcal{H}_\alpha})(\forall x \in [0,1])\ |f(x)| \le 1$. It is compact by Theorem 8.2.2, direction (*ii*) $\Rightarrow$ (*i*).

Definition 8.2.4. *σ-Compact Metric Space*
A metric space X is *σ-compact* if $(\exists (K_j)_{j \in \mathbb{N}})\ [X = \bigcup_{j=1}^{\infty} K_j \wedge (\forall j \in \mathbb{N})\ K_j \subseteq X \text{ is compact}]$.

Theorem 8.2.5. *Arzela–Ascoli on σ-Compact Spaces*
Let (X, d_X) be a σ-compact metric space, let (Y, d_Y) be a metric space, and let $(f_n) \subseteq Y^X$. Suppose $\{f_n : n \in \mathbb{N}\}$ is pointwise equicontinuous and $(\forall x \in X)\ \overline{\{f_n(x) : n \in \mathbb{N}\}}^Y$ is compact. Then (f_n) has a subsequence (g_m) converging uniformly on every compact $K \subseteq X$.

Proof. If $X = \emptyset$, the conclusion is vacuous. Assume $X \neq \emptyset$.

(i) *Compact-Set Selection*: Let $\emptyset \neq L \subseteq X$ be compact and let (u_n) be a subsequence of (f_n). If (u_n) is not equicontinuous on L, choose $\varepsilon > 0$, $h_i \in \{u_n : n \in \mathbb{N}\}$, and $x_i, t_i \in L$ with $d_X(x_i, t_i) < 1/i$ and $d_Y(h_i(x_i), h_i(t_i)) \ge \varepsilon$. Pass to a subsequence with $x_i \to p \in L$ by Theorem 4.1.13; then $t_i \to p$ by the triangle inequality. By pointwise equicontinuity at p with tolerance $\varepsilon/2$, $d_Y(h_i(x_i), h_i(t_i)) < \varepsilon$ eventually, a contradiction.
Choose an at most countable dense set $E \subseteq L$ by Proposition 4.1.35, and enumerate it with repetitions when finite. A diagonal subsequence (v_i) converging pointwise on E exists by successive applications of Theorem 4.1.13 to the compact orbit closures at the enumerated points. Fix $\varepsilon > 0$, choose $\delta > 0$ from equicontinuity with tolerance $\varepsilon/4$, and extract a finite subcover $L \subseteq \bigcup_{s=1}^{r} B_\delta(a_s)$ with $a_s \in E$. Choose $I \in \mathbb{N}$ with $(i, j \ge I,\ 1 \le s \le r) \Rightarrow d_Y(v_i(a_s), v_j(a_s)) < \varepsilon/4$ by pointwise convergence at the centers. The three-term estimate implies $(i, j \ge I) \Rightarrow \sup_{x \in L} d_Y(v_i(x), v_j(x)) \le 3\varepsilon/4$.
For $x \in L$, $(v_i(x))$ is Cauchy and has a convergent subsequence in $\overline{\{f_n(x) : n \in \mathbb{N}\}}^Y$ by Theorem 4.1.13. Hence $(v_i(x))$ converges by Lemma 4.1.21; write its limit as $v(x)$. Letting $j \to \infty$ pointwise in the uniform Cauchy estimate shows $v_i \rightrightarrows v$ on L. For $L = \emptyset$, uniform convergence is vacuous.

(ii) *Exhaustion and Diagonalization*: Choose compact sets (L_j) with $X = \bigcup_{j=1}^{\infty} L_j$ by Definition 8.2.4, and set $K_j := \bigcup_{q=1}^{j} L_q$. Each K_j is compact by finite-subcover consolidation across the finite union, so (K_j) is an increasing compact cover of X. By (i), recursively choose nested subsequences $S_j = (u_k^{(j)})_{k \in \mathbb{N}}$ such that S_j converges uniformly on K_j. Set $g_m := u_m^{(m)}$. For fixed j, $(g_m)_{m \ge j}$ is a subsequence of S_j, so it converges uniformly on K_j. Since $X = \bigcup_j K_j$, define $g(x) := \lim_m g_m(x)$ on X.

(iii) *Arbitrary Compact Set*: Let $\emptyset \neq K \subseteq X$ be compact. The family $(g_m|_K)$ is equicontinuous by the first paragraph of (i). Fix $\varepsilon > 0$, choose $\delta > 0$ with tolerance $\varepsilon/4$, and choose a finite δ-net $x_1, \dots, x_r \in K$ by compactness. Choose $M \in \mathbb{N}$ such that $(m, n \ge M,\ 1 \le s \le r) \Rightarrow d_Y(g_m(x_s), g_n(x_s)) < \varepsilon/4$ by pointwise convergence at the centers. By the three-term estimate, $(m, n \ge M) \Rightarrow \sup_{x \in K} d_Y(g_m(x), g_n(x)) \le 3\varepsilon/4$. Letting $n \to \infty$ pointwise results in $\sup_{x \in K} d_Y(g_m(x), g(x)) \le 3\varepsilon/4 < \varepsilon$, so $g_m \rightrightarrows g$ on K. Uniform convergence on $K = \emptyset$ is vacuous.

Structural Commentary: Compact orbit closures supply pointwise subsequences; equicontinuity upgrades dense convergence before exhaustion diagonalization.
Key Insight: A common diagonal handles the exhaustion, and a final finite-net estimate reaches compact sets not contained in one exhaustion stage. □

Corollary 8.2.6. *Real-Line Local Uniform Subsequence*
Let $(f_n) \subseteq \mathbb{R}^{\mathbb{R}}$. Suppose $\{f_n : n \in \mathbb{N}\}$ is pointwise equicontinuous and $(\exists x_0 \in \mathbb{R})\ \sup_{n \in \mathbb{N}} |f_n(x_0)| < +\infty$. Then (f_n) has a subsequence converging uniformly on every compact $K \subseteq \mathbb{R}$.

Proof. Fix $x \in \mathbb{R}$ and set $I := [\min\{x_0, x\}, \max\{x_0, x\}]$. Each $f_n|_I$ is continuous by Definition 8.1.4(iv), and $\{f_n|_I : n \in \mathbb{N}\}$ is equicontinuous by Theorem 8.1.5. Choose $\delta > 0$ with step-size tolerance 1, choose $m \in \mathbb{N}$ with $|x - x_0|/m < \delta$, and set $t_j := x_0 + j(x - x_0)/m$ for $0 \le j \le m$. Since $|t_j - t_{j-1}| < \delta$ for $1 \le j \le m$, $(\forall n \in \mathbb{N})\ |f_n(x)| \le |f_n(x_0)| + m$, so $\{f_n(x) : n \in \mathbb{N}\}$ is bounded. Its closure is closed and bounded, hence compact by Theorem 4.2.5, direction (*i*) $\Rightarrow$ (*ii*). The identity $\mathbb{R} = \bigcup_{j=1}^{\infty} [-j, j]$ makes $\mathbb{R}$ σ-compact by Definition 8.2.4, so the desired subsequence exists by Theorem 8.2.5.
Structural Commentary: One bounded orbit propagates by finite chains.
Key Insight: Equicontinuity turns one anchor into compact-local control. □

8.3 Further Topics in Function-Space Compactness

Proof Method 8.3.1. *Function-Space Compactness Proof Patterns*

(i) *Completeness*: convert metric-Cauchy control into a uniform Cauchy estimate, construct the pointwise limit in the complete target, and use uniform convergence to retain boundedness or continuity; compare Theorems 8.1.1 and 8.1.3.
(ii) *Baire category*: first verify completeness of the chosen function-space metric, then construct dense open sets whose countable intersection lies in the desired property; compare Proof Method 4.3.13.
(iii) *Compactness/equicontinuity*: use equicontinuity and finite domain nets to obtain uniform bounds or finite-resolution control, then add closedness to retain uniform limits; compare Theorems 8.2.1 and 8.2.2.
(iv) *Diagonal selection*: select pointwise convergence on a countable dense set, then use a three-term equicontinuity estimate to make the subsequence uniformly Cauchy; compare Theorems 8.1.8 and 8.2.1.
(v) *Locally uniform*: combine compact point-orbit closures with equicontinuity on an increasing compact exhaustion, diagonalize the nested subsequences, and use a final finite-net estimate on an arbitrary compact set; compare Theorem 8.2.5.

Key Insight: Finite-net control and equicontinuity produce total boundedness; completeness and closedness convert that finite-resolution control into compactness.

9 Power Series and Approximation

9.1 Power Series

Power series are controlled infinite polynomials, and approximation is uniform closure. This section covers radii, Taylor structure, Abel behavior, elementary analytic functions, the gamma function, Weierstrass approximation, and Stone–Weierstrass.

Definition 9.1.1. *Core Objects: Power Series, Approximation, and Gamma*

(i) A real *power series about a* is $\sum_{n=0}^{\infty} c_n(x-a)^n$; a local representation of f by one makes f *analytic* at a.
(ii) A *polynomial approximant* to f on a compact set K is a polynomial P chosen so that $\|f-P\|_{\infty,K}$ is small.
(iii) For $x>0$, the *gamma function* is $\Gamma(x) := \int_0^\infty t^{x-1}e^{-t}\,dt$ when the improper integral converges.

Key Insight: Power series are local analytic objects; approximation theorems turn algebraic families into dense function families; Γ extends factorials by integration.

Definition 9.1.2. *Power Series and Radius of Convergence*
Let $(c_n)_{n\ge0} \subseteq \mathbb{C}$ and $\alpha := \limsup_{n\to\infty} \sqrt[n]{|c_n|} \in [0,+\infty]$.

(i) The series $\sum_{n=0}^{\infty} c_n z^n$ is a *power series*.
(ii) The *radius of convergence* is $R := \alpha^{-1} \in [0,+\infty]$, with $0^{-1}=+\infty$ and $(+\infty)^{-1}=0$.

Theorem 9.1.3. *Convergence Inside the Circle*
Let $(c_n)_{n\ge0} \subseteq \mathbb{C}$, and let R be the radius of $\sum_{n=0}^{\infty} c_n z^n$ from Definition 9.1.2. Then:
(i) the series converges absolutely if $|z|<R$;
(ii) the series diverges if $|z|>R$.

Proof. Set $\alpha := \limsup_{n\to\infty} \sqrt[n]{|c_n|}$. For $a_n := c_n z^n$, $\limsup_{n\to\infty} \sqrt[n]{|a_n|} = |z|\,\alpha$.
(i) If $|z|<R$, then $|z|\,\alpha<1$, including $\alpha=0$ when $R=+\infty$. Thus the series converges absolutely by Theorem 6.1.18(i).
(ii) If $|z|>R$, then $|z|\,\alpha>1$, including $\alpha=+\infty$ when $R=0$. Thus the series diverges by Theorem 6.1.18(ii).
Structural Commentary: Radius controls the interior and exterior, not the boundary.
Key Insight: The root test is the radius-of-convergence theorem in disguise. □

Theorem 9.1.4. *A Boundary Test for Power Series*
Let $(c_n)_{n\ge0} \subseteq [0,+\infty)$ satisfy $c_n \downarrow 0$, and suppose $\sum_{n=0}^{\infty} c_n z^n$ has radius 1. Then $(\forall z\in\mathbb{C})\ [(|z|=1 \wedge z\neq 1) \Rightarrow \sum_{n=0}^{\infty} c_n z^n$ converges$]$.

Proof. For $|z|=1$ and $z\neq 1$, put $a_n = z^n$ and $b_n = c_n$.
The bound $\left|\sum_{m=0}^{n} z^m\right| = \left|\frac{1-z^{n+1}}{1-z}\right| \le \frac{2}{|1-z|}$ holds for $n\ge0$. Hence $\sum_{n=0}^{\infty} z^n c_n$ converges by Theorem 6.1.22.
Structural Commentary: Boundary convergence away from $z=1$ comes from bounded geometric partial sums.
Key Insight: Monotone decay permits conditional boundary convergence. □

Theorem 9.1.5. *Abel's Endpoint Theorem*
Let $(c_n)_{n\ge0} \subseteq \mathbb{C}$ and $s_n := \sum_{k=0}^{n} c_k \to s \in \mathbb{C}$. For $x\in(-1,1)$, set $f(x) := \sum_{n=0}^{\infty} c_n x^n$. Then $\lim_{x\to1^-} f(x) = s$.

Proof.
(i) *Abel Identity*: Set $s_{-1} := 0$. The sequence (s_n) is bounded, so $\sum_{n=0}^{\infty} s_n x^n$ converges absolutely for $|x|<1$. Since $c_n = s_n - s_{n-1}$, $f(x) = \sum_{n=0}^{\infty}(s_n - s_{n-1})x^n = (1-x)\sum_{n=0}^{\infty} s_n x^n$.
(ii) *Error Split*: Thus $f(x) - s = (1-x)\sum_{n\ge0}(s_n - s)x^n$. Given $\varepsilon>0$, choose N with $|s_n - s| < \varepsilon/2$ for $n>N$.
(iii) *Estimate*: For $0<x<1$, $|f(x)-s| \le (1-x)\sum_{n=0}^{N} |s_n - s|\,x^n + \frac{\varepsilon}{2}(1-x)\sum_{n>N} x^n$.
(iv) *Limit*: The first term tends to 0 as $x\to1^-$, and the second is $\le \varepsilon/2$; hence $f(x)\to s$.
Structural Commentary: Endpoint convergence is one-sided Abel continuity.
Key Insight: $(1-x)\sum s_n x^n$ is a geometric average of partial sums. □

Theorem 9.1.6. *Abel Product Theorem for Cauchy Products*
Let $(a_n),(b_n) \subseteq \mathbb{C}$ satisfy $\sum_{n=0}^{\infty} a_n = A$ and $\sum_{n=0}^{\infty} b_n = B$. Define $c_n := \sum_{k=0}^{n} a_k b_{n-k}$, and suppose $\sum_{n=0}^{\infty} c_n = C$. Then $C = AB$.

Proof. For $x \in (-1,1)$, set $f(x) := \sum_{n=0}^{\infty} a_n x^n$, $g(x) := \sum_{n=0}^{\infty} b_n x^n$, and $h(x) := \sum_{n=0}^{\infty} c_n x^n$. Convergence of the three unweighted series makes (a_n), (b_n), and (c_n) bounded, so the weighted series converge absolutely. Their Cauchy-product coefficients are $c_n x^n$, and $f(x)g(x) = h(x)$ by Theorem 6.1.28.
By Theorem 9.1.5, $f(x) \to A$, $g(x) \to B$, and $h(x) \to C$ as $x \to 1^-$. Therefore $C = \lim_{x\to 1^-} h(x) = \lim_{x\to 1^-} f(x)g(x) = AB$.
Structural Commentary: Regularize by x, multiply inside, pass to $x = 1$.
Key Insight: Abel moves the safe product identity to the endpoint. □

Perspective 9.1.7. *Centered Series Are Translated Series*
$\sum c_n(x-a)^n$ is $\sum c_n y^n$ with $y = x - a$ and radius R: absolute convergence for $|x-a| < R$, divergence for $|x-a| > R$, and separate endpoint tests for $|x-a| = R$.

Figure 182 marks the convergence interval centered at a and, for $0 < \rho < R$, the compact subinterval $[a-\rho, a+\rho]$ used for locally uniform estimates.

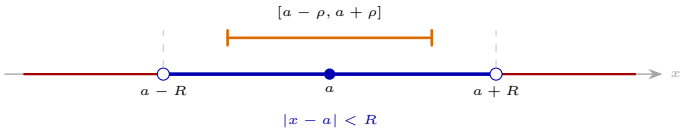

Figure 182: Inside $|x-a| < R$: absolute and locally uniform; outside divergence; endpoints undecided.

Definition 9.1.8. *Real Analytic at a Point*
Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $a \in \operatorname{int} E$. The function f is *real analytic at a* if $(\exists r > 0)(\exists (c_n)_{n\geq 0} \subseteq \mathbb{R})$ such that:
(i) $(a-r, a+r) \subseteq E$.
(ii) The power series $\sum c_n(x-a)^n$ has radius of convergence $\geq r$.
(iii) $(\forall x \in (a-r, a+r))\ f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$.
If E is open and $(\forall a \in E)$ f is real analytic at a, then f is *real analytic on E*. Analytic functions form the class C^ω.

Theorem 9.1.9. *Power Series May Be Differentiated Term by Term*
Let $(c_n)_{n\geq 0} \subseteq \mathbb{C}$, let $f(x) := \sum_{n=0}^{\infty} c_n x^n$ have radius $R > 0$, and set $I_R := (-R, R)$ and $K_\rho := [-\rho, \rho]$ for $0 < \rho < R$.
(i) $(\forall \rho \in (0,R))$ $\sum c_n x^n$ *converges uniformly on* K_ρ.
(ii) The function f lies in $C^\infty(I_R)$, and $(\forall k \in \mathbb{N})(\forall x \in I_R)$ $f^{(k)}(x) = \sum_{n\geq k} \frac{n!}{(n-k)!} c_n x^{n-k}$; *in particular,* $(\forall k \in \mathbb{N})$ $f^{(k)}(0) = k! c_k$.

Proof.
(i) If $0 < \rho < R$, then $\sum_{n=0}^{\infty} |c_n| \rho^n < \infty$ by Theorem 9.1.3(i). Since $(x \in K_\rho) \Rightarrow |c_n x^n| \leq |c_n| \rho^n$, uniform convergence follows by Theorem 7.1.9.
(ii) For fixed $k \geq 1$, $\frac{n!}{(n-k)!} \sim n^k$, so $\limsup_{n\to\infty} \sqrt[n]{\left|\frac{n!}{(n-k)!} c_n\right|} = \limsup_{n\to\infty} \sqrt[n]{|c_n|}$. Hence every formal derivative series has radius R and converges uniformly on K_ρ by (i).
Let $S_N(x) := \sum_{n=0}^{N} c_n x^n$. On K_ρ, $(S_N(0))$ converges and (S_N') converges uniformly, so $S_N \rightrightarrows f$ and $f' = \sum_{n=1}^{\infty} n c_n x^{n-1}$ by Theorem 7.2.4, applied to real and imaginary parts. Iteration proves the formula for $k \in \mathbb{N}$, and $I_R = \bigcup_{0<\rho<R} K_\rho$ implies $f \in C^\infty(I_R)$. At $x = 0$, the formula becomes $f^{(k)}(0) = k! c_k$.
Structural Commentary: Work on $K_\rho \Subset I_R$ to justify termwise operations.
Key Insight: Polynomial coefficient growth does not change the radius. □

Corollary 9.1.10. *Termwise Calculus for Centered Power Series*
Let $(c_n)_{n\geq 0} \subseteq \mathbb{C}$ and let $f(x) := \sum_{n=0}^{\infty} c_n(x-a)^n$ have radius $R > 0$.
(i) The series converges locally uniformly on $(a-R, a+R)$ and defines a continuous function.
(ii) $(\forall x \in (a-R, a+R))$ $f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$, *locally uniformly.*
(iii) If $[y,z] \subset (a-R, a+R)$, then $\int_y^z f(x)\,dx = \sum_{n=0}^{\infty} c_n \frac{(z-a)^{n+1} - (y-a)^{n+1}}{n+1}$.

Proof. Set $g(u) := \sum c_n u^n$ and $T(x) := x - a$, so $f = g \circ T$ on $(a-R, a+R)$.
(i) Local uniform convergence of $\sum c_n u^n$ on $(-R, R)$ implies local uniform convergence of $\sum c_n (Tx)^n$ on $(a-R, a+R)$. The function f is continuous by Theorem 7.1.11, applied on each compact subinterval.
(ii) Since $T'(x) = 1$, $f'(x) = g'(Tx) = \sum_{n=1}^{\infty} n c_n (Tx)^{n-1}$ by Theorem 9.1.9(ii), with local uniform convergence by the same theorem.
(iii) If $[y,z] \subset (a-R, a+R)$, choose $\rho < R$ with $[y,z] \subset [a-\rho, a+\rho]$.
By Corollary 7.2.2, applied componentwise with integrator $\alpha(x) = x$, $\int_y^z f(x)\,dx = \sum_{n=0}^{\infty} c_n \int_y^z (Tx)^n\,dx$. Evaluating the polynomial integrals proves the stated formula.
Structural Commentary: Work on compact intervals strictly inside the radius.
Key Insight: Translation reduces centered series to ordinary power series. □

Corollary 9.1.11. *Analytic Functions Are Their Taylor Series*
Let $(c_n)_{n\geq 0} \subseteq \mathbb{C}$, let $a \in \mathbb{R}$ and $r > 0$, and suppose $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ when $|x-a| < r$.
(i) $f \in C^\infty(a-r, a+r)$.
(ii) $(\forall k \in \mathbb{N}_0)$ $c_k = f^{(k)}(a)/k!$.
(iii) The power-series expansion of f about a is unique.
(iv) $(\forall x \in (a-r, a+r))$ $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$.

Proof.

(i) The smoothness follows by Corollary 9.1.10 and Theorem 9.1.9(ii).
(ii) If $k \in \mathbb{N}_0$, then $f^{(k)}(x) = \sum_{n \geq k} \frac{n!}{(n-k)!} c_n (x-a)^{n-k}$; at $x = a$, this identity becomes $f^{(k)}(a) = k! c_k$.
(iii) If $f = \sum b_n (x-a)^n$ also, then $(\forall k \in \mathbb{N}_0)\ b_k = f^{(k)}(a)/k! = c_k$ by (ii).
(iv) Replace c_n by $f^{(n)}(a)/n!$ in $f(x) = \sum c_n (x-a)^n$.

Structural Commentary: Derivatives determine every coefficient.
Key Insight: Analytic representation forces its own Taylor series. □

Theorem 9.1.12. *Re-Expansion and Identity for Power Series*
Let $(c_n)_{n\geq 0} \subseteq \mathbb{C}$, let $f(x) = \sum_{n\geq 0} c_n x^n$ have radius $R > 0$, and set $S = (-R, R)$.

(i) If $a \in S$, then $(\forall x \in (a - (R - |a|), a + (R - |a|)))\ f(x) = \sum_{m \geq 0} \frac{f^{(m)}(a)}{m!} (x-a)^m$.

(ii) Let $(a_n), (b_n) \subseteq \mathbb{C}$, and suppose $F(x) = \sum_{n\geq 0} a_n x^n$ and $G(x) = \sum_{n\geq 0} b_n x^n$ converge on S.
If $\mathcal{Z}_{F,G} := \{x \in S : F(x) = G(x)\}$ has a limit point in S, then $(\forall n \in \mathbb{N}_0)\ a_n = b_n$, and hence $F = G$ on S.

Proof.

(i) Put $\rho := |x-a| + |a| < R$. The binomial double sum is absolutely summable because $\sum_{n\geq 0} |c_n| \sum_{m=0}^{n} \binom{n}{m} |a|^{n-m} |x-a|^m = \sum_{n\geq 0} |c_n| \rho^n < \infty$.
Rearrangement is valid by Theorem 6.1.37; hence $f(x) = \sum_{m\geq 0} A_m (x-a)^m$, where $A_m := \sum_{n \geq m} \binom{n}{m} c_n a^{n-m}$. By Theorem 9.1.9(ii), termwise differentiation shows $f^{(m)}(a) = \sum_{n \geq m} \frac{n!}{(n-m)!} c_n a^{n-m} = m! A_m$.

(ii) Set $H := F - G$ and $Z := \{x \in S : H(x) = 0\}$. For any $x_0 \in S$, part (i) expands H as $H(x) = \sum_{n\geq 0} e_n (x - x_0)^n$ near x_0. If zeros of H accumulate at x_0 and $k = \min\{n : e_n \neq 0\}$ existed, then $H(x) = (x-x_0)^k g(x)$ with g continuous and $g(x_0) = e_k \neq 0$; this would isolate x_0 in Z, a contradiction. Hence all e_n vanish and $H = 0$ on a neighborhood of x_0.
Let $A := \{x_0 \in S : H = 0 \text{ on a neighborhood of } x_0\}$. The assumed limit point of Z and the preceding argument show that $A \neq \emptyset$, and the definition makes A open in S. If $x_0 \in \overline{A}^{\,S}$, then either $x_0 \in A$ or zeros of H accumulate at x_0; in either case the same local argument puts x_0 in A. Thus A is also closed in the connected interval S, so $A = S$ and $H = 0$ on S. Finally, $a_n - b_n = H^{(n)}(0)/n! = 0$ by Corollary 9.1.11(ii).

Structural Commentary: Re-expand locally; connectedness propagates zeros.
Key Insight: Accumulating zeros force every local coefficient to vanish. □

Perspective 9.1.13. *Regularity Hierarchy*
The inclusions are $C^\omega \subseteq C^\infty \subseteq \cdots \subseteq C^2 \subseteq C^1 \subseteq C^0$.
The inclusions follow from Theorem 9.1.9 and Definition 1.1.23; they are generally proper.
Key Insight: Higher regularity is extra structure, not just a different notation for continuity.

Example 9.1.14. *Smooth but Not Analytic*
Define $\phi(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$. Induction proves $(\forall r \in \mathbb{N}_0)(\exists p_r \in \mathbb{R}[u])(\forall x > 0)\ \phi^{(r)}(x) = p_r(1/x) e^{-1/x}$. After the substitution $u = 1/x$, $(\forall m \in \mathbb{N}_0)\ \lim_{x \to 0^+} x^{-m} e^{-1/x} = \lim_{u \to +\infty} u^m e^{-u} = 0$ by repeated applications of Theorem 1.1.21. Induction across $x = 0$ therefore proves $\phi \in C^\infty$ and $(\forall r \in \mathbb{N}_0)\ \phi^{(r)}(0) = 0$.
Its Taylor series at 0 is zero, but $\phi(x) > 0$ for $x > 0$, so ϕ is not analytic at 0 by Definition 9.1.8.
Key Insight: A function can be infinitely tangent to the x-axis without being locally equal to its Taylor series.

Perspective 9.1.15. *Taylor Coefficients Are Only the First Half*
Power-series coefficients are forced by derivatives (Corollary 9.1.11); for a function not already known to be analytic, proving the Taylor series represents the function needs a remainder estimate (Theorem 1.1.25).

Theorem 9.1.16. *Exponential and Logarithm from the Power Series*
Define $\mathsf{E}(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}$ for $z \in \mathbb{C}$.

(i) $(\forall z, w \in \mathbb{C})\ \mathsf{E}(z+w) = \mathsf{E}(z)\mathsf{E}(w)$.
(ii) The restriction $\mathsf{E}|_{\mathbb{R}}$ is positive, satisfies $\mathsf{E}' = \mathsf{E}$, and is a strictly increasing bijection from $\mathbb{R}$ onto $(0, \infty)$. Write $e^x := \mathsf{E}(x)$ for $x \in \mathbb{R}$ and $e := \mathsf{E}(1)$.
(iii) The inverse $\mathsf{L} := (\mathsf{E}|_{\mathbb{R}})^{-1} : (0, \infty) \to \mathbb{R}$ satisfies $(\forall y > 0)\ \mathsf{L}'(y) = 1/y$ and $(\forall u, v > 0)\ \mathsf{L}(uv) = \mathsf{L}(u) + \mathsf{L}(v)$. Write $\log y := \mathsf{L}(y)$.
(iv) For $\alpha \in \mathbb{R}$ and $x > 0$, set $x^\alpha := \mathsf{E}(\alpha \mathsf{L}(x))$; then the function $x \mapsto x^\alpha$ has derivative $\alpha x^{\alpha - 1}$.

Proof. For $z \neq 0$, the ratio of successive absolute terms is $|z|/(n+1) \to 0$, so the defining series converges absolutely by Theorem 6.1.19(i); convergence at $z = 0$ is immediate.

(i) By Theorem 6.1.28 and the binomial theorem, absolute convergence permits $\mathsf{E}(z)\mathsf{E}(w) = \sum_{n\geq 0} \sum_{j=0}^{n} \frac{z^j w^{n-j}}{j!(n-j)!} = \sum_{n\geq 0} \frac{(z+w)^n}{n!} = \mathsf{E}(z+w)$.
(ii) The identity $\mathsf{E}' = \mathsf{E}$ on $\mathbb{R}$ holds by Theorem 9.1.9(ii). Since $\mathsf{E}(x)\mathsf{E}(-x) = 1$ by (i), E has no real zero; continuity and $\mathsf{E}(0) = 1$ imply $\mathsf{E} > 0$. Consequently $\mathsf{E}' > 0$, and E is strictly increasing. For $x \geq 0$, $\mathsf{E}(x) \geq 1 + x$, so $\mathsf{E}(x) \to +\infty$ as $x \to +\infty$; then $\mathsf{E}(x) = 1/\mathsf{E}(-x) \to 0$ as $x \to -\infty$ by (i). Therefore the range is $(0, \infty)$ by Theorem 5.1.43.
(iii) By Corollary 1.1.20, applied to the bijection in (ii), $\mathsf{L}'(y) = 1/\mathsf{E}'(\mathsf{L}(y)) = 1/y$.
If $u = \mathsf{E}(s)$ and $v = \mathsf{E}(t)$, then $\mathsf{L}(uv) = \mathsf{L}(\mathsf{E}(s+t)) = s + t = \mathsf{L}(u) + \mathsf{L}(v)$.
(iv) By Theorem 1.1.6, $\frac{d}{dx} \mathsf{E}(\alpha \mathsf{L}(x)) = \mathsf{E}(\alpha \mathsf{L}(x)) \alpha \mathsf{L}'(x) = x^\alpha \alpha / x = \alpha x^{\alpha - 1}$.

Structural Commentary: Addition, inversion, powers.
Key Insight: $\mathsf{E}(z+w) = \mathsf{E}(z)\mathsf{E}(w)$ creates the logarithm law. □

Perspective 9.1.17. *Real Powers by Rational Approximation*
One route defines x^α for $x > 0$ by choosing $q_n \in \mathbb{Q}$ with $q_n \to \alpha$ and setting $x^\alpha = \lim_{n\to\infty} x^{q_n}$ after proving existence and independence of (q_n). The power-series construction sets $x^\alpha = \mathsf{E}(\alpha \mathsf{L}(x))$; both routes must agree with rational powers and be choice-independent.

Perspective 9.1.18. *Logarithmic Integral Route*
With the Riemann integral, set $L(y) := \int_1^y \frac{1}{t}\,dt$ for $y > 0$. The identity $L'(y) = 1/y$ follows from Theorem 2.2.1; endpoint limits make L invertible, and Theorem 1.1.6 then recovers $\mathsf{E}' = \mathsf{E}$ and the product laws.

Theorem 9.1.19. *Trigonometric Functions from the Exponential*
For $x \in \mathbb{R}$, set $\mathsf{C}(x) := \frac{\mathsf{E}(ix)+\mathsf{E}(-ix)}{2}$, $\mathsf{S}(x) := \frac{\mathsf{E}(ix)-\mathsf{E}(-ix)}{2i}$, and $\gamma(t) := \mathsf{E}(it)$.

(i) $\mathsf{E}(ix) = \mathsf{C}(x) + i\mathsf{S}(x)$, $\mathsf{C}^2 + \mathsf{S}^2 = 1$, $\mathsf{C}' = -\mathsf{S}$, *and* $\mathsf{S}' = \mathsf{C}$.
(ii) $x_0 := \min\{x > 0 : \mathsf{C}(x) = 0\}$ *exists; with* $\pi := 2x_0$, $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, *and* $(\forall z \in \mathbb{C})\ \mathsf{E}(z + 2\pi i) = \mathsf{E}(z)$.
(iii) C, S *are 2π-periodic,* $\gamma([0, 2\pi]) = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$, *and γ is injective on $[0, 2\pi)$.*

Proof.

(i) Since the power-series coefficients are real, $\overline{\mathsf{E}(ix)} = \mathsf{E}(-ix)$. Hence $\mathsf{E}(ix) = \mathsf{C}(x) + i\mathsf{S}(x)$ and $|\mathsf{E}(ix)|^2 = \mathsf{E}(ix)\mathsf{E}(-ix) = 1$ by Theorem 9.1.16(i).
Therefore $\mathsf{C}^2 + \mathsf{S}^2 = 1$; by Theorem 9.1.9(ii), differentiation of the defining power series establishes $\mathsf{C}' = -\mathsf{S}$ and $\mathsf{S}' = \mathsf{C}$.

(ii) If C has no positive zero, then $\mathsf{C} > 0$ on $(0, \infty)$, hence $\mathsf{S}' > 0$ and $\mathsf{S}(x) > 0$ for $x > 0$.
For $0 < x < y$, $\mathsf{S}(x)(y - x) < \int_x^y \mathsf{S}(t)\,dt = \mathsf{C}(x) - \mathsf{C}(y) \le 2$, which is impossible as $y \to \infty$. Thus C has a positive zero. Choose one such zero z_1. The zero set of C in $[0, z_1]$ is nonempty and compact, so it has a minimum $x_0 > 0$. On $(0, x_0)$, $\mathsf{C} > 0$, and hence S increases from 0 to $\mathsf{S}(x_0) = 1$.
The definition $\pi = 2x_0$ now implies $\mathsf{E}(\pi i/2) = i$. By repeated applications of Theorem 9.1.16(i), $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, and $\mathsf{E}(z + 2\pi i) = \mathsf{E}(z)$.

(iii) Periodicity of C, S follows from (ii). On $[0, \pi/2]$, C decreases from 1 to 0 and S increases from 0 to 1; by Theorem 5.1.43 and $\mathsf{C}^2 + \mathsf{S}^2 = 1$, γ traverses the first-quadrant arc exactly once. Since $\gamma(t + m\pi/2) = \gamma(t)i^m$ for $m \in \{0, 1, 2, 3\}$, the four rotated arcs cover the unit circle and meet only at their endpoints. Hence $\gamma([0, 2\pi]) = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$ and γ is injective on $[0, 2\pi)$.

Structural Commentary: $\mathsf{E}(it)$ supplies trigonometry.
Key Insight: $\mathsf{E}(it)\mathsf{E}(-it) = 1$ puts the path on the unit circle. □

Figure 183 depicts the path $\gamma(t) = \mathsf{E}(it)$ and its coordinate projections $\mathsf{C}(t)$ and $\mathsf{S}(t)$ from Theorem 9.1.19.

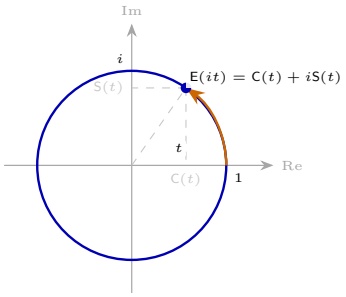

Figure 183: $\mathsf{E}(it) = \mathsf{C}(t) + i\mathsf{S}(t)$ traces the unit circle, defining sine and cosine analytically.

Theorem 9.1.20. *Algebraic Completeness of $\mathbb{C}$*
If $P \in \mathbb{C}[z]$ and $\deg P \ge 1$, then $(\exists z_0 \in \mathbb{C})\ P(z_0) = 0$.

Proof.

(i) *Minimum Setup*: Dividing by the leading coefficient does not change the roots, so write $P(z) = \sum_{j=0}^n a_j z^j$ with $n \ge 1$ and $a_n = 1$. Since $|P(z)| \to \infty$ as $|z| \to \infty$, choose $R > 0$ so that $|P(z)| > |P(0)|$ when $|z| \ge R$. The closed disk $\overline{D}(0, R) \subseteq \mathbb{C} \cong \mathbb{R}^2$ is compact by Theorem 4.2.5, and the continuous function $|P|$ attains its minimum there by Theorem 5.1.39. Since this minimum is at most $|P(0)|$, it is also the global minimum. Choose z_0 with $\mu := |P(z_0)| = \min_{\mathbb{C}} |P|$. If $\mu = 0$, the conclusion holds.

(ii) *Normalized Expansion*: Assume $\mu > 0$ and set $Q(w) := P(z_0 + w)/P(z_0)$. Then $Q(0) = 1$ and $(\forall w \in \mathbb{C})\ |Q(w)| \ge 1$. Write $Q(w) = 1 + b_k w^k + w^{k+1} H(w)$ with $k \ge 1$, $b_k \ne 0$, and H polynomial.

(iii) *Decreasing Direction*: Choose $\theta \in \mathbb{R}$ with $b_k e^{ik\theta} = -|b_k|$. For $r > 0$, $Q(re^{i\theta}) = 1 - r^k A(r)$, where $A(r) := |b_k| - re^{i(k+1)\theta} H(re^{i\theta})$. For small r, $\Re A(r) > |b_k|/2$, hence $\left|1 - r^k A(r)\right|^2 = 1 - 2r^k \Re A(r) + r^{2k} |A(r)|^2 < 1$, contradicting $|Q| \ge 1$.

Structural Commentary: Minimize $|P|$, translate, perturb.
Key Insight: A phase choice makes the first nonzero term decrease $|Q|$. □

Perspective 9.1.21. *From Exponentials to Factorials*
Theorems 9.1.16, 2.2.3 supply real-power laws and integration-by-parts pattern used for Γ in Theorem 9.1.22.

Theorem 9.1.22. *Gamma Function, Beta Function, and Stirling's Formula*
For $x, y > 0$, the following improper integrals converge and are positive: $\Gamma(x) := \int_0^\infty t^{x-1} e^{-t}\,dt$ *and* $\mathrm{B}(x, y) := \int_0^1 t^{x-1}(1 - t)^{y-1}\,dt$.

(i) $(\forall x > 0)\ \Gamma(x + 1) = x\Gamma(x)$, $(\forall n \in \mathbb{N})\ \Gamma(n + 1) = n!$, *and* $\log \Gamma$ *is convex on* $(0, \infty)$.
(ii) The function Γ is the unique $f : (0, \infty) \to (0, \infty)$ such that $f(1) = 1$, $(\forall x > 0)\ f(x + 1) = xf(x)$, and $\log f$ is convex.
(iii) $(\forall x, y > 0)\ \mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.
(iv) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, $\int_{-\infty}^\infty e^{-s^2}\,ds = \sqrt{\pi}$, $(\forall x > 0)\ \Gamma(x) = \frac{2^{x-1}}{\sqrt{\pi}}\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{2}\right)$, *and* $\lim_{x\to\infty} \frac{\Gamma(x+1)}{(x/e)^x\sqrt{2\pi x}} = 1$.

Proof. On $(0,1]$, $0 < t^{x-1}e^{-t} \le t^{x-1}$, whose integral is $1/x$. On $[1,\infty)$, choose $m \in \mathbb{N}$ with $m > x$. Since $e^t = \sum_{n\ge 0} t^n/n! \ge t^m/m!$, one has $t^{x-1}e^{-t} \le m! t^{x-1-m}$, which is integrable at infinity. Thus $\Gamma(x)$ converges. For $\mathrm{B}(x,y)$, split at $1/2$: the factor $(1-t)^{y-1}$ is bounded on $(0,1/2]$, and after the substitution $s = 1-t$ the other half has the same form with x, y interchanged. Positivity is immediate.

(i) By integration by parts on $[\varepsilon, R]$ and the endpoint bounds above, $\Gamma(x+1) = \int_0^\infty t^x e^{-t}\,dt = x\Gamma(x)$. The identity $\Gamma(1) = \lim_{R\to\infty}\int_0^R e^{-t}\,dt = \lim_{R\to\infty}(1-e^{-R}) = 1$ then proves the factorial formula by induction.
Fix $x, y > 0$ and $0 < \lambda < 1$, and set $p(t) := t^{x-1}e^{-t}/\Gamma(x)$ and $q(t) := t^{y-1}e^{-t}/\Gamma(y)$. The weighted arithmetic–geometric mean inequality implies $p(t)^\lambda q(t)^{1-\lambda} \le \lambda p(t) + (1-\lambda)q(t)$. After integration, $\Gamma(\lambda x + (1-\lambda)y) \le \Gamma(x)^\lambda \Gamma(y)^{1-\lambda}$, which is the convexity inequality for $\log\Gamma$.

(ii) Let $\varphi := \log f$. For $0 < x \le 1$ and $n \in \mathbb{N}$, the secant-slope inequalities for the convex function φ, together with $\varphi(t+1) = \varphi(t) + \log t$, imply

$$0 \le \varphi(x) - \log\frac{n!\,n^x}{x(x+1)\cdots(x+n)} \le x\log\left(1+\frac{1}{n}\right).$$

The width of this interval tends to zero, so the recurrence and convexity determine f on $(0,1]$ and then on $(0,\infty)$. The function Γ has the required properties by (i); hence it is the unique such function.

(iii) Fix $y > 0$ and set $F_y(x) := \frac{\Gamma(x+y)}{\Gamma(y)}\mathrm{B}(x,y)$. Direct integration establishes $\mathrm{B}(1,y) = 1/y$, while integration by parts together with $\mathrm{B}(x,y) = \mathrm{B}(x+1,y) + \mathrm{B}(x,y+1)$ establishes $\mathrm{B}(x+1,y) = \frac{x}{x+y}\mathrm{B}(x,y)$. Therefore $F_y(1) = 1$ and $F_y(x+1) = xF_y(x)$.
The function $x \mapsto \log \mathrm{B}(x,y)$ is convex by the normalized weighted arithmetic–geometric mean argument from (i), applied to the beta integrands. The function $x \mapsto \log\Gamma(x+y)$ is convex by (i), so $\log F_y$ is convex. Hence $F_y = \Gamma$ by (ii), which is the beta–gamma identity.

(iv) By the substitution $t = \mathsf{S}(\theta)^2$ on $[0,\pi/2]$ and Theorem 9.1.19(i), $\mathrm{B}(\frac12,\frac12) = 2\int_0^{\pi/2} d\theta = \pi$. By (iii) and $\Gamma(1) = 1$, $\Gamma(\frac12)^2 = \pi$; positivity therefore implies $\Gamma(\frac12) = \sqrt{\pi}$. The substitution $t = s^2$ then shows $\Gamma(\frac12) = 2\int_0^\infty e^{-s^2}\,ds = \int_{-\infty}^\infty e^{-s^2}\,ds$.
For $u > 0$, the substitutions $t = (1+s)/2$ and then $r = s^2$ show $\mathrm{B}(u,u) = 2^{1-2u}\mathrm{B}(u,\frac12)$. By (iii) and cancellation of $\Gamma(u) > 0$, $\Gamma(2u) = \frac{2^{2u-1}}{\sqrt{\pi}}\Gamma(u)\Gamma(u+\frac12)$, which is the stated duplication formula after setting $x = 2u$.
Finally, in $\Gamma(x+1) = \int_0^\infty t^x e^{-t}\,dt$, put $t = x + u\sqrt{x}$. Then

$$\Gamma(x+1) = \left(\frac{x}{e}\right)^x \sqrt{x}\,I_x, \qquad I_x := \int_{-\sqrt{x}}^\infty e^{\,x[\log(1+u/\sqrt{x}) - u/\sqrt{x}]}\,du.$$

Fix $A > 0$. By Theorem 1.1.25, applied to $\log(1+s)$ near $s = 0$, the exponent converges uniformly to $-u^2/2$ on $[-A, A]$. The inequalities $\log(1-v)+v \le -v^2/2$ for $0 \le v < 1$ and $\log(1+v)-v \le -v^2/(2(1+v))$ for $v \ge 0$ bound the left tail by $\int_A^\infty e^{-u^2/2}\,du$ and, when $x \ge A^2$, the right tail by $\int_A^\infty e^{-u^2/4}\,du + (4/\sqrt{x})e^{-x/4}$. Hence $I_x \to \int_{-\infty}^\infty e^{-u^2/2}\,du = \sqrt{2\pi}$, where the last equality follows from the already proved Gaussian integral by $u = \sqrt{2}\,s$. Substitution in the displayed formula proves Stirling's limit.

Structural Commentary: Recurrence plus log-convexity determines the function uniquely.
Key Insight: Bohr–Mollerup turns beta identities into gamma identities. □

Proof Method 9.1.23. *Gamma Function Proof Pattern*

(i) *Convergence and recurrence*: control the endpoints before integrating by parts.
(ii) *Uniqueness*: combine $f(x+1) = xf(x)$ with convex secant-slope bounds.
(iii) *Identity transport*: make a beta–gamma expression satisfy the same recurrence and log-convexity.
(iv) *Special values and asymptotics*: use substitutions for $\Gamma(\frac12)$ and duplication, then center the Gamma integral at its dominant point for Stirling.

9.2 Approximation Methods

Figure 184 motivates uniform approximation: the polygonal interpolants P_2 and P_8 approach $f(x) = \sqrt{1-x}$ on $[0,1]$, while Theorem 9.2.1 replaces polygonal interpolants by polynomials.

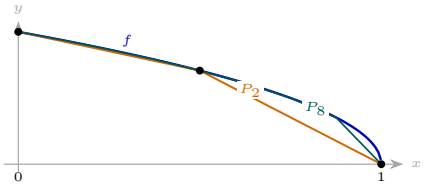

Figure 184: The polygonal interpolants P_2 and P_8 approach $f(x) = \sqrt{1-x}$ uniformly on $[0,1]$.

Theorem 9.2.1. *Weierstrass Polynomial Approximation*
Let $a < b$ and $f \in C([a,b],\mathbb{C})$. Then:
(i) $(\exists (P_n) \subseteq \mathbb{C}[x])$ $P_n \rightrightarrows f$ *on* $[a,b]$*;*
(ii) if $f([a,b]) \subset \mathbb{R}$*, then* (P_n) *may be chosen in* $\mathbb{R}[x]$*.*

Proof.

(i) By an affine change of variable and subtraction of the endpoint interpolant, it suffices to treat $[0,1]$ with $f(0) = f(1) = 0$.
Extend f by zero to $[-1,2]\setminus[0,1]$; the extension is continuous and therefore uniformly continuous by Theorem 5.2.6. For $n \in \mathbb{N}$, choose $c_n > 0$ so that $Q_n(t) := c_n(1-t^2)^n$ satisfies $\int_{-1}^1 Q_n(t)\,dt = 1$. If $|t| \le 1/(n+1)$, an induction proves $(1-t^2)^n \ge 1 - nt^2 > 1/2$; hence $\int_{-1}^1 (1-t^2)^n\,dt > 1/(n+1)$ and $c_n < n+1$. Thus $(\delta \le |t| \le 1) \Rightarrow Q_n(t) \le (n+1)(1-\delta^2)^n$.
Set $P_n(x) := \int_{-1}^1 f(x+t)Q_n(t)\,dt$.
Since $P_n(x) = \int_0^1 f(u)Q_n(u-x)\,du$, $P_n \in \mathbb{C}[x]$.

Given $\varepsilon > 0$, choose $0 < \delta \le 1$ such that $(u, v \in [-1,2] \wedge |u - v| < \delta) \Rightarrow |f(u) - f(v)| < \varepsilon/2$, and set $M := \sup_{[-1,2]} |f|$.
For $x \in [0,1]$, positivity and normalization of Q_n imply $|P_n(x) - f(x)| \le \int_{-1}^{1} |f(x+t) - f(x)|\, Q_n(t)\, dt \le \frac{\varepsilon}{2} + 4M(n+1)(1-\delta^2)^n$.
The final term tends to zero, so $\|P_n - f\|_\infty \to 0$ on $[0,1]$; reversing the affine change and restoring the endpoint interpolant proves (i).

(ii) If f is real-valued, then the same construction has $P_n \in \mathbb{R}[x]$.

Structural Commentary: Polynomial kernels approximate the identity.
Key Insight: Uniform continuity handles near mass; kernel concentration removes far mass. □

Figure 185 shows the proof's concentration mechanism: for $n > m$, the solid Q_n is narrower than the dashed Q_m, while the shaded interval has width 2δ and each kernel retains integral 1.

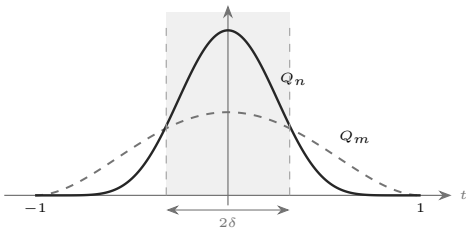

Figure 185: The normalized kernels $Q_k(t) = c_k(1-t^2)^k$ concentrate near 0: for $n > m$, the solid Q_n is narrower than the dashed Q_m, and the central window is $[-\delta, \delta]$.

Corollary 9.2.2. *Polynomial Approximation of $|x|$*
Let $a > 0$. Then $(\exists (P_n) \subseteq \mathbb{R}[x])\ [(\forall n \in \mathbb{N})\ P_n(0) = 0 \wedge P_n \rightrightarrows |x| \text{ on } [-a,a]]$.

Proof. By Theorem 9.2.1, choose $Q_n \in \mathbb{R}[x]$ with $Q_n \rightrightarrows |x|$ on $[-a,a]$. Set $P_n(x) := Q_n(x) - Q_n(0)$.
Then $P_n \in \mathbb{R}[x]$, $P_n(0) = 0$, and $Q_n(0) \to 0$, so $\|P_n - |x|\|_\infty \le \|Q_n - |x|\|_\infty + |Q_n(0)| \to 0$.
Structural Commentary: Normalize at 0.
Key Insight: The constant correction vanishes uniformly. □

Definition 9.2.3. *Algebras and Uniform Closure*
Let E be a set and let $\mathcal{A}$ be a family of functions $E \to \mathbb{C}$.

(i) The family $\mathcal{A}$ is an *algebra* if $(\forall f, g \in \mathcal{A})(\forall \lambda \in \mathbb{C})\ f + g, fg, \lambda f \in \mathcal{A}$; replace $\mathbb{C}$ by $\mathbb{R}$ in the real case.
(ii) The family $\mathcal{A}$ is *uniformly closed* if $[(f_n) \subseteq \mathcal{A} \wedge f : E \to \mathbb{C} \wedge f_n \rightrightarrows f \text{ on } E] \Rightarrow f \in \mathcal{A}$.
(iii) The *uniform closure* of $\mathcal{A}$ is $\overline{\mathcal{A}}^{\,u} := \{f : E \to \mathbb{C} : (\exists (f_n) \subseteq \mathcal{A})\ f_n \rightrightarrows f \text{ on } E\}$.

Theorem 9.2.4. *Uniform Closures Are Closed Algebras*
Let $\mathcal{A}$ be an algebra of bounded functions and let $\mathcal{B}$ be its uniform closure. Then:

(i) $\mathcal{B}$ is an algebra;
(ii) $\mathcal{B}$ is uniformly closed.

Proof.

(i) If $f, g \in \mathcal{B}$ and $c \in \mathbb{C}$, choose $(f_n), (g_n) \subseteq \mathcal{A}$ with $f_n \rightrightarrows f$ and $g_n \rightrightarrows g$. Then $f_n + g_n \rightrightarrows f + g$ and $cf_n \rightrightarrows cf$. The sequence (f_n) is uniformly bounded eventually, and $\|f_n g_n - fg\|_\infty \le \|f_n\|_\infty \|g_n - g\|_\infty + \|g\|_\infty \|f_n - f\|_\infty \to 0$. Thus $\mathcal{B}$ is an algebra.
(ii) If $(h_n) \subseteq \mathcal{B}$ and $h_n \rightrightarrows h$, choose $a_n \in \mathcal{A}$ with $\|a_n - h_n\|_\infty < 1/n$. Then $\|a_n - h\|_\infty \le 1/n + \|h_n - h\|_\infty \to 0$, so $h \in \mathcal{B}$.

Structural Commentary: Uniform limits preserve algebraic operations for bounded functions.
Key Insight: The uniform closure of an algebra has already absorbed all further uniform limits. □

Definition 9.2.5. *Separation, Nonvanishing, and Self-Adjointness*
Let E be a set and let $\mathcal{A}$ be a family of scalar-valued functions on E.

(i) The family $\mathcal{A}$ *separates points* on E if $(\forall x_1, x_2 \in E)\ x_1 \neq x_2 \Rightarrow (\exists f \in \mathcal{A})\ f(x_1) \neq f(x_2)$.
(ii) The family $\mathcal{A}$ *vanishes at no point* of E if $(\forall x \in E)(\exists g \in \mathcal{A})\ g(x) \neq 0$.
(iii) A complex algebra $\mathcal{A}$ is *self-adjoint* if $(\forall f \in \mathcal{A})\ \overline{f} \in \mathcal{A}$, where $\overline{f}(x) := \overline{f(x)}$.

Theorem 9.2.6. *Two-Point Interpolation in a Separating Algebra*
Let $\mathbf{F} \in \{\mathbb{R}, \mathbb{C}\}$, let E be a set, and let $\mathcal{A} \subseteq \mathbf{F}^E$ be an algebra that separates points and vanishes nowhere. If $x_1, x_2 \in E$, $x_1 \neq x_2$, and $c_1, c_2 \in \mathbf{F}$, then $(\exists f \in \mathcal{A})(\forall i \in \{1,2\})\ f(x_i) = c_i$.

Proof. Choose $g, h, k \in \mathcal{A}$ with $g(x_1) \neq g(x_2)$, $h(x_1) \neq 0$, $k(x_2) \neq 0$. Set $u := gk - g(x_1)k$ and $v := gh - g(x_2)h$.
Then $u(x_1) = v(x_2) = 0$, while $u(x_2) = (g(x_2) - g(x_1))k(x_2) \neq 0$ and $v(x_1) = (g(x_1) - g(x_2))h(x_1) \neq 0$.
Hence $f := \frac{c_1 v}{v(x_1)} + \frac{c_2 u}{u(x_2)} \in \mathcal{A}$ and $(\forall i \in \{1,2\})\ f(x_i) = c_i$.
Structural Commentary: Build point-isolating functions.
Key Insight: Isolation reduces interpolation to linear combination. □

Perspective 9.2.7. *Interpolation Is Linear Algebra*
The two-point interpolation step can also be written as a 2×2 linear system. Its invertibility is exactly the finite-dimensional criterion from Theorem 1.7.7, with the determinant test supplied by Theorem 3.8.2. The proof above chooses explicit point-isolating functions instead of reopening the calculation.

Theorem 9.2.8. *Stone–Weierstrass Theorem, Real Form*
Let K be a compact Hausdorff space, and let $\mathcal{A} \subseteq C(K, \mathbb{R})$ be a point-separating, nowhere-vanishing real algebra. Then $\overline{\mathcal{A}}^{\|\cdot\|_\infty} = C(K, \mathbb{R})$.

Proof.

(i) *Closure and Lattice Operations*: Put $\mathcal{B} := \overline{\mathcal{A}}^{\|\cdot\|_\infty}$. The set $\mathcal{B}$ is a uniformly closed algebra by Theorem 9.2.4, and it inherits separation and nonvanishing because $\mathcal{A} \subseteq \mathcal{B}$. If $r \in \mathcal{B}$ and $a := \|r\|_\infty > 0$, choose $(p_n) \subseteq \mathbb{R}[x]$ with $p_n(0) = 0$ and $p_n \rightrightarrows |t|$ on $[-a, a]$ by Corollary 9.2.2. Since p_n has zero constant term, $p_n(r) \in \mathcal{B}$, and $p_n(r) \rightrightarrows |r|$ on K. The case $a = 0$ is immediate. Thus $|r| \in \mathcal{B}$, and $(\forall r, s \in \mathcal{B})\ r \vee s = \frac{r+s+|r-s|}{2} \in \mathcal{B}$ and $r \wedge s = \frac{r+s-|r-s|}{2} \in \mathcal{B}$.

(ii) *Pointwise Interpolation*: Fix $f \in C(K, \mathbb{R})$ and $\varepsilon > 0$. If $x, y \in K$ and $x \neq y$, choose $h_{x,y} \in \mathcal{B}$ with $h_{x,y}(x) = f(x)$ and $h_{x,y}(y) = f(y)$ by Theorem 9.2.6. If $x = y$, choose $q \in \mathcal{B}$ with $q(x) \neq 0$ and set $h_{x,x} := f(x)q/q(x)$. For fixed x, define $V_y := \{z \in K : h_{x,y}(z) > f(z) - \varepsilon\}$.

(iii) *Lower Approximation*: Each V_y is open and contains y. By compactness, choose $y_1, \dots, y_m \in K$ such that $K = \bigcup_{i=1}^m V_{y_i}$, and put $g_x := \bigvee_{i=1}^m h_{x,y_i} \in \mathcal{B}$. Then $f - \varepsilon < g_x$ on K and $g_x(x) = f(x)$.

(iv) *Uniform Approximation*: Define $U_x := \{z \in K : g_x(z) < f(z) + \varepsilon\}$. Each U_x is open and contains x. By compactness, choose $x_1, \dots, x_\ell \in K$ such that $K = \bigcup_{j=1}^\ell U_{x_j}$, and put $h := \bigwedge_{j=1}^\ell g_{x_j} \in \mathcal{B}$. Then $f - \varepsilon < h < f + \varepsilon$ on K, so $\|h - f\|_\infty < \varepsilon$. Since $\varepsilon > 0$ was arbitrary and $\mathcal{B}$ is uniformly closed, $f \in \mathcal{B}$.

Structural Commentary: Finite joins/meets globalize interpolation.
Key Insight: Compactness upgrades pointwise to uniform. □

Theorem 9.2.9. *Stone–Weierstrass Theorem, Complex Form*
Let K be a compact Hausdorff space and let $\mathcal{A} \subseteq C(K, \mathbb{C})$ be a self-adjoint, point-separating, nowhere-vanishing complex algebra. Then $\overline{\mathcal{A}}^{\|\cdot\|_\infty} = C(K, \mathbb{C})$.

Proof. Set $\mathcal{A}_\mathbb{R} := \mathcal{A} \cap C(K, \mathbb{R})$. If $f \in \mathcal{A}$, then $\operatorname{Re} f = (f + \overline{f})/2 \in \mathcal{A}_\mathbb{R}$ and $\operatorname{Im} f = (f - \overline{f})/(2i) \in \mathcal{A}_\mathbb{R}$ by self-adjointness. If $f(x) \neq f(y)$ or $f(x) \neq 0$, then one of $\operatorname{Re} f, \operatorname{Im} f$ witnesses the same condition. Thus $\mathcal{A}_\mathbb{R}$ is point-separating and nowhere-vanishing.
By Theorem 9.2.8, $\overline{\mathcal{A}_\mathbb{R}}^{\|\cdot\|_\infty} = C(K, \mathbb{R})$. For $F = u + iv \in C(K, \mathbb{C})$, take $(u_n), (v_n) \subseteq \mathcal{A}_\mathbb{R}$ with $u_n \rightrightarrows u$ and $v_n \rightrightarrows v$.
Then $u_n + iv_n \in \mathcal{A}$ and $u_n + iv_n \rightrightarrows F$.
Structural Commentary: Conjugation closure contains the real and imaginary parts.
Key Insight: Approximate real and imaginary parts. □

Theorem 9.2.10. *Riesz Separability Criterion for $C(K)$*
Let K be a compact Hausdorff space. Then $(C(K, \mathbb{R}), \|\cdot\|_\infty)$ is separable iff K is metrizable.

Proof.

(⇒) Let (g_n) be dense in $C(K, \mathbb{R})$ and define $O_n := \{x \in K : g_n(x) > 1/2\}$. Fix an open set $O \subseteq K$ and $x \in O$. The space K is normal by Theorem 6.3.6; choose $g \in C(K, \mathbb{R})$ with $0 \leq g \leq 1$, $g(x) = 1$, and $g|_{K \setminus O} = 0$ by Theorem 6.3.20. Choose n with $\|g - g_n\|_\infty < 1/3$. Then $x \in O_n \subseteq O$, so (O_n) is a countable basis. Therefore K is metrizable by Theorem 6.4.26.

(⇐) Let d metrize K. Choose a dense sequence $(x_n) \subseteq K$ by Proposition 4.1.35, and put $f_0 := 1$ and $f_n(x) := d(x, x_n)$.
If $x, y \in K$ and $x \neq y$, choose n with $d(x, x_n) < d(x, y)/3$. Then $f_n(x) < d(x, y)/3 < 2d(x, y)/3 < f_n(y)$, so $\{f_n : n \in \mathbb{N}_0\}$ separates points and vanishes nowhere. Hence $\operatorname{alg}(f_n : n \in \mathbb{N}_0)$ is dense in $C(K, \mathbb{R})$ by Theorem 9.2.8. The polynomials in finitely many f_n with rational coefficients form a countable dense subalgebra, so $C(K, \mathbb{R})$ is separable.

Structural Commentary: Dense functions determine a basis; dense points determine algebraic generators.
Key Insight: Countability moves between K and $C(K)$. □

Theorem 9.2.11. *A Continuous Nowhere Differentiable Function Exists*
$(\exists f \in C(\mathbb{R}, \mathbb{R}))(\forall x \in \mathbb{R})$ *$f'(x)$ does not exist.*

Proof.

(i) *Construction and Continuity*: Let φ be the 2-periodic extension of $|x|$ on $[-1, 1]$. Then $0 \leq \varphi \leq 1$ and $|\varphi(s) - \varphi(t)| \leq |s - t|$. Define $f(x) := \sum_{n=0}^{\infty} (\frac{3}{4})^n \varphi(4^n x)$. Since $\left|(\frac{3}{4})^n \varphi(4^n x)\right| \leq (\frac{3}{4})^n$, the series converges uniformly by Theorem 7.1.9, and f is continuous by Theorem 7.1.11.

(ii) *Scale Choice*: Fix x and $m \in \mathbb{N}$. Choose $\delta_m = \pm\frac{1}{2}4^{-m}$ so that no integer lies between $4^m x$ and $4^m(x + \delta_m)$. Set $\gamma_{n,m} := \frac{\varphi(4^n(x+\delta_m)) - \varphi(4^n x)}{\delta_m}$.

(iii) *Difference Quotient Bound*: If $n > m$, then $4^n \delta_m$ is an even integer, so $\gamma_{n,m} = 0$. If $0 \leq n \leq m$, then $|\gamma_{n,m}| \leq 4^n$. The segment between $4^m x$ and $4^m(x + \delta_m)$ lies in one linearity interval of φ, so $|\gamma_{m,m}| = 4^m$. Consequently $\left|\frac{f(x+\delta_m) - f(x)}{\delta_m}\right| = \left|\sum_{n=0}^m (\frac{3}{4})^n \gamma_{n,m}\right| \geq 3^m - \sum_{n=0}^{m-1} 3^n = \frac{1}{2}(3^m + 1)$.

(iv) *Conclusion*: Since $\delta_m \to 0$, the difference quotients are unbounded near x; since x was arbitrary, f is nowhere differentiable.

Structural Commentary: Uniform summability preserves continuity; scaled cusps force unbounded quotients.
Key Insight: Uniform convergence preserves continuity, not differentiability. □

Figure 186 plots the first three summands $\sigma_k(x) := (3/4)^k \varphi(4^k x)$: their amplitudes decrease while their oscillation frequencies and slopes increase.

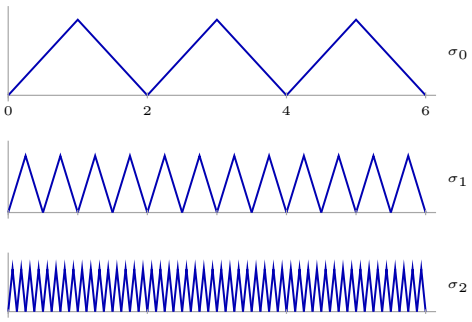

Figure 186: The first compressed sawtooth terms $\sigma_k(x) = (3/4)^k \varphi(4^k x)$: heights shrink, slopes grow.

Perspective 9.2.12. *Explicit Versus Generic Pathology*
The sawtooth series proves existence; Baire category shows nowhere differentiability is residual in $C([a,b],\mathbb{R})$.

Theorem 9.2.13. *Generic Nowhere Differentiability on Compact Intervals*
Let $a < b$, and let $C = C([a,b],\mathbb{R})$ with the supremum metric. A residual set $\mathcal{G} \subseteq C$ can be chosen such that:

(i) $(\forall f \in \mathcal{G})(\forall x \in [a,b])$ *f has no available one-sided derivative at x;*
(ii) $(\forall f \in \mathcal{G})$ *f is not monotone on any nondegenerate subinterval of $[a,b]$;*
(iii) the set $\{f \in C : f$ is differentiable at some point of $[a,b]\}$ is first category.

Proof. For $n, k \in \mathbb{N}$ with $1/k < b - a$, define

$$A^{+}_{n,k} := \left\{ f \in C : (\exists x \in [a, b-1/k])(\forall h \in (0,1/k]) \ \left| \frac{f(x+h)-f(x)}{h} \right| \le n \right\},$$
$$A^{-}_{n,k} := \left\{ f \in C : (\exists x \in [a+1/k, b])(\forall h \in [-1/k,0)) \ \left| \frac{f(x+h)-f(x)}{h} \right| \le n \right\}.$$

These sets are closed. Indeed, if $f_j \rightrightarrows f$ with $f_j \in A^{+}_{n,k}$ and witnesses $x_j \in [a, b-1/k]$, compactness provides a subsequence with $x_j \to x$. For fixed $h \in (0,1/k]$, uniform convergence and continuity imply $|f(x+h)-f(x)| \le nh$, so $f \in A^{+}_{n,k}$; the negative-side argument is identical.
They have empty interior. Fix $f \in C$ and $\varepsilon > 0$, and choose $p \in \mathbb{R}[x]$ with $\|p - f\|_\infty < \varepsilon/2$ by Theorem 9.2.1. Choose a piecewise-linear sawtooth w with $\|w\|_\infty < \varepsilon/2$ whose segment slopes alternate between $\pm S$, where $S > n + \|p'\|_\infty$. On a sufficiently short segment to either side of any available point, the corresponding difference quotient of $p + w$ has absolute value greater than n by Corollary 1.1.13. Thus $p + w \notin A^{+}_{n,k} \cup A^{-}_{n,k}$ and $\|p + w - f\|_\infty < \varepsilon$. Hence each $A^{\pm}_{n,k}$ is nowhere dense.
Set $D := \mathbb{Q} \cap [a,b]$. For $r, s \in D$ with $r < s$, let $M^{\uparrow}_{r,s}$ and $M^{\downarrow}_{r,s}$ be the sets of functions in C that are respectively nondecreasing and nonincreasing on $[r,s]$. Both sets are closed under uniform convergence. Given $f \in C$ and $\varepsilon > 0$, choose p as above and a sawtooth w with $\|w\|_\infty < \varepsilon/2$, two teeth contained in (r,s), and segment slopes of absolute value greater than $\|p'\|_\infty$. Then $p + w$ has both an increase and a decrease and lies within ε of f, so $M^{\uparrow}_{r,s}$ and $M^{\downarrow}_{r,s}$ have empty interior and are nowhere dense.
Let $\mathcal{N}$ be the union of all $A^{\pm}_{n,k}$ and all $M^{\uparrow}_{r,s}, M^{\downarrow}_{r,s}$ over the indicated countable index sets, and set $\mathcal{G} := C \setminus \mathcal{N}$. The set $\mathcal{N}$ is first category, while $\mathcal{G}$ is a countable intersection of dense open sets and is therefore residual by Definition 4.3.6. The space C is a closed subspace of $C_b([a,b])$ and is complete by Theorem 8.1.1 and Proposition 4.1.26(ii); hence $\mathcal{G}$ is dense by Corollary 4.3.7.

(i) If $f \in \mathcal{G}$ had a finite right derivative at some $x < b$, then its right difference quotients would be bounded by some n on $(0,1/k]$ with $x \le b - 1/k$, placing f in $A^{+}_{n,k}$. A finite left derivative similarly places f in some $A^{-}_{n,k}$. Both contradict $f \notin \mathcal{N}$.
(ii) If $f \in \mathcal{G}$ were monotone on a nondegenerate interval $J \subseteq [a,b]$, density of D would provide $r, s \in D$ with $r < s$ and $[r,s] \subseteq J$. Then $f \in M^{\uparrow}_{r,s} \cup M^{\downarrow}_{r,s} \subseteq \mathcal{N}$, a contradiction.
(iii) Differentiability at a point entails an available finite one-sided derivative. Thus the differentiable-somewhere set is contained in $\mathcal{N}$ and is first category.

Structural Commentary: Baire category turns dense sawtooth perturbability into a generic regularity failure.
Key Insight: The explicit construction shows existence; the category proof shows typicality. □

Proof Method 9.2.14. *Power-Series and Approximation Proof Patterns*

(i) *Power series*: work on $K_\rho \Subset I_R$; use uniform convergence before termwise limits or derivatives.
(ii) *Endpoint theorems*: prove identities inside the radius and pass to endpoints with Abel-type control.
(iii) *Approximate identities*: use kernels $Q_n \ge 0$, $\int Q_n = 1$, with tails $\to 0$ off 0.
(iv) *Uniform closure*: show $a_n \in \mathcal{A}$ and $a_n \rightrightarrows f$.
(v) *Lattice closure*: derive $|f|$, then $\max, \min$ from algebraic formulas.
(vi) *Stone–Weierstrass*: interpolate locally; use compactness and finite max/min to globalize.

Key Insight: Uniform convergence and compactness are what make infinite algebra behave like finite algebra.

9.3 The Umbral Calculus

The umbral calculus is the formal linear algebra of polynomial sequences. It identifies a formal power series with three related objects: an algebra element, a linear functional on $\mathbb{C}[x]$, and an operator on $\mathbb{C}[x]$. This refines the duality of Definition 4.2.4 and the polynomial operator calculus of Definition 1.1.2, without invoking convergence.

Definition 9.3.1. *Formal Power Series and Order*
Let $\mathscr{F} := \mathbb{C}[[t]]$.

(i) Each $f \in \mathscr{F}$ has a unique expression $f(t) = \sum_{k \ge 0} a_k t^k$, where $(a_k)_{k \ge 0} \subseteq \mathbb{C}$; addition is coefficientwise and multiplication is the Cauchy product.
(ii) The *order* of f is $o(f) := \min\{k : a_k \neq 0\}$, where $o(0) := +\infty$.

(iii) A series is *invertible* if it has a multiplicative inverse in $\mathscr{F}$, and is a *delta series* if $o(f) = 1$.
(iv) A *compositional inverse* of f is a series h satisfying $f \circ h = h \circ f = t$; when unique, it is denoted $\overline{f}$.
(v) The *lower factorial polynomials* are $(x)_0 := 1$ and $(x)_n := x(x-1)\cdots(x-n+1)$ $(n \in \mathbb{N},\ n \geq 1)$.

Proposition 9.3.2. *Formal Order Calculus*
For $f, g \in \mathscr{F}$:
(i) $f \in \mathscr{F}^\times \iff o(f) = 0$;
(ii) $o(fg) = o(f) + o(g)$ *and* $o(f+g) \geq \min\{o(f), o(g)\}$;
(iii) if $g(t) = \sum_{k\geq 0} b_k t^k$ *and* $o(f) \geq 1$, *then* $g \circ f := g(f(t)) := \sum_{k\geq 0} b_k f(t)^k \in \mathscr{F}$;
(iv) if $1 \leq o(f) < \infty$ *and* $g \neq 0$, *then* $o(g \circ f) = o(g)o(f)$;
(v) if $o(f) = 1$, *then* $(\forall g \in \mathscr{F})(\exists!\,(c_k)_{k\geq 0} \subseteq \mathbb{C})\ g(t) = \sum_{k\geq 0} c_k f(t)^k$;
(vi) if $o(f) = 1$, *then* $(\exists!\,\overline{f} \in \mathscr{F} : o(\overline{f}) = 1)\ f \circ \overline{f} = \overline{f} \circ f = t$.

Proof.
(i) ($\Rightarrow$) If $f \in \mathscr{F}^\times$, then $[t^0]f \in \mathbb{C}^\times$, hence $o(f) = 0$.
(i) ($\Leftarrow$) If $o(f) = 0$, then $[t^0]f \in \mathbb{C}^\times$, hence $f \in \mathscr{F}^\times$.
(ii) If $f = 0$ or $g = 0$, the product identity follows from the convention $r + \infty = \infty + r = \infty$; otherwise, let $r = o(f)$ and $s = o(g)$. Then $[t^{r+s}]fg = [t^r]f\,[t^s]g \neq 0$, and all lower coefficients vanish, so $o(fg) = r+s$. In all cases, $[t^m](f+g) = 0$ when $m < \min\{o(f), o(g)\}$, giving the stated lower bound.
(iii) Since $o(f) \geq 1$, $o(f^k) \to \infty$; hence, for a fixed $n \in \mathbb{N}_0$, the coefficient $[t^n]\sum_{k\geq 0} b_k f^k$ has only finitely many nonzero contributions.
(iv) Let $s = o(g)$. Then $g \circ f = f^s \left(b_s + \sum_{k>s} b_k f^{k-s} \right)$, where the second factor has order 0; hence $o(g \circ f) = s\,o(f)$.
(v) If $o(f) = 1$ and $f(t) = a_1 t + \cdots$ with $a_1 \neq 0$, then $f^k = a_1^k t^k +$ higher terms. Thus $\{t^k\}_{k\geq 0} \mapsto \{f^k\}_{k\geq 0}$ is triangular with nonzero diagonal, so existence and uniqueness follow by recursive coefficient matching.
(vi) By (v) with $g = t$, fix the unique h satisfying $h \circ f = t$. By (iv), $o(h) = 1$. By associativity, $(f \circ h) \circ f = f \circ (h \circ f) = f = t \circ f$; uniqueness in (v) therefore implies $f \circ h = t$. If q is another compositional inverse, then $q = q \circ (f \circ h) = (q \circ f) \circ h = h$.

Structural Commentary: Delta series behave like a formal change of coordinate at $t = 0$.
Key Insight: Order replaces convergence as the bookkeeping device. □

Definition 9.3.3. *Derivation*
Let $\mathcal{A}$ be an algebra over a field $\mathbb{K}$. A $\mathbb{K}$-linear operator $\partial : \mathcal{A} \to \mathcal{A}$ is a *derivation* if $(\forall a, b \in \mathcal{A})\ \partial(ab) = (\partial a)b + a(\partial b)$.

Proposition 9.3.4. *Formal Differentiation*
Define $\partial_t : \mathscr{F} \to \mathscr{F}$ *by* $\partial_t \left(\sum_{k\geq 0} a_k t^k \right) := \sum_{k\geq 1} k a_k t^{k-1}$. *Then:*
(i) ∂_t *is* $\mathbb{C}$*-linear.*
(ii) $(\forall f, g \in \mathscr{F})\ \partial_t(fg) = (\partial_t f)g + f(\partial_t g)$; *hence* $\partial_t \in \mathrm{Der}_{\mathbb{C}}(\mathscr{F})$.

Proof.
(i) Let $f = \sum a_k t^k$, $g = \sum b_k t^k$, and $\alpha, \beta \in \mathbb{C}$. Then $\partial_t(\alpha f + \beta g) = \sum_{k\geq 1} k(\alpha a_k + \beta b_k)t^{k-1} = \alpha \partial_t f + \beta \partial_t g$.
(ii) Let $f(t) = \sum_{k\geq 0} a_k t^k$ and $g(t) = \sum_{k\geq 0} b_k t^k$. For $n \geq 0$, $[t^n]\partial_t(fg) = (n+1)\sum_{j=0}^{n+1} a_j b_{n+1-j}$.
The coefficient $[t^n]\big((\partial_t f)g + f(\partial_t g)\big) = \sum_{j=1}^{n+1} j a_j b_{n+1-j} + \sum_{j=0}^{n} (n+1-j) a_j b_{n+1-j}$ equals the preceding coefficient.

Structural Commentary: Formal differentiation is the canonical derivation of $\mathbb{C}[[t]]$.
Key Insight: Leibniz is a Cauchy-product coefficient identity. □

Definition 9.3.5. *The Umbral Algebra*
(i) *Polynomial algebra*: set $\mathscr{P} := \mathbb{C}[x]$ and let $\mathscr{P}' := \mathrm{Hom}_{\mathbb{C}}(\mathscr{P}, \mathbb{C})$.
(ii) *Umbral pairing*: for $L \in \mathscr{P}'$ and $p(x) \in \mathscr{P}$, write $\langle L \mid p(x) \rangle := L(p(x))$.
(iii) *Exponential normalization*: write each $f \in \mathscr{F}$ uniquely as $f(t) = \sum_{k\geq 0} \frac{a_k}{k!} t^k$, where $(a_k)_{k\geq 0} \subseteq \mathbb{C}$.
(iv) *Associated functional*: set $\langle f(t) \mid x^n \rangle := a_n$ $(n \in \mathbb{N}_0)$, extending $\mathbb{C}$-linearly.

Theorem 9.3.6. *Formal Series Represent All Algebraic Functionals*
Define $\Phi : \mathscr{P}' \to \mathscr{F}$ *by* $\Phi(L) := \sum_{k\geq 0} \frac{\langle L \mid x^k \rangle}{k!} t^k$. *Then:*
(i) Φ *is a* $\mathbb{C}$*-linear isomorphism with inverse* $\Phi^{-1}\left(\sum_{k\geq 0} \frac{a_k}{k!} t^k \right)\left(\sum_{k=0}^{d} c_k x^k \right) := \sum_{k=0}^{d} c_k a_k$.
(ii) $\langle t^k \mid x^n \rangle = n! \delta_{n,k}$ *and* $\langle t^k \mid p \rangle = p^{(k)}(0)$ *for* $p \in \mathscr{P}$; *hence* $t^0 = \mathrm{ev}_0$ *and* $t^k = D^k|_0$.

Proof.
(i) Let $T : \mathscr{P}' \to \mathbb{C}^{\mathbb{N}}$ be $T(L) = (a_k)_{k\geq 0}$, $a_k = \langle L \mid x^k \rangle$.
Since $\{x^k\}_{k\geq 0}$ is a Hamel basis, T is a $\mathbb{C}$-linear bijection with $T^{-1}(a)\left(\sum_{k=0}^{d} c_k x^k \right) = \sum_{k=0}^{d} c_k a_k$.
The map $E : \mathbb{C}^{\mathbb{N}} \to \mathscr{F}$, $E(a) = \sum_{k\geq 0} \frac{a_k}{k!} t^k$, is a $\mathbb{C}$-linear bijection, hence $\Phi = E \circ T$ and $\Phi^{-1} = T^{-1} \circ E^{-1}$.
(ii) As $t^k = k!\,\frac{t^k}{k!}$, coefficient sequence is $(n!\delta_{n,k})_{n\geq 0}$. For $p = \sum_{n=0}^{d} c_n x^n$, by linearity, $\langle t^k \mid p \rangle = k! c_k = p^{(k)}(0)$.

Structural Commentary: Functionals are monomial-coordinate sequences.
Key Insight: Exponential normalization makes t^k equal $D^k|_0$. □

Definition 9.3.7. *Delta and Invertible Functionals*
Under the identification $\mathscr{F} \simeq \mathscr{P}'$:

(i) *Delta functional*: a functional represented by $f \in \mathscr{F}$ with $o(f) = 1$.
(ii) *Invertible functional*: a functional represented by $f \in \mathscr{F}^\times$, equivalently $o(f) = 0$.

Proposition 9.3.8. *Basic Umbral Identities*
For $f, g \in \mathscr{F}$, $p \in \mathscr{P}$, and $n \geq 0$:

(i) $f(t) = \sum_{k\geq 0} \frac{\langle f(t) \mid x^k\rangle}{k!} t^k$;

(ii) $p(x) = \sum_{k=0}^{\deg p} \frac{\langle t^k \mid p\rangle}{k!} x^k$;

(iii) $\langle fg \mid x^n\rangle = \sum_{k=0}^{n} \binom{n}{k} \langle f \mid x^k\rangle \langle g \mid x^{n-k}\rangle$;

(iv) if $o(f) > \deg p$, then $\langle f \mid p\rangle = 0$;

(v) if $(f_k)_{k\geq 0} \subseteq \mathscr{F}$ satisfies $(\forall k \in \mathbb{N}_0)\ o(f_k) = k$ and $(a_k)_{k\geq 0} \subseteq \mathbb{C}$, then $\left\langle \sum_{k\geq 0} a_k f_k \middle| p \right\rangle = \sum_{k=0}^{\deg p} a_k \langle f_k \mid p\rangle$, with $\sum_{k\geq 0} a_k f_k$ well-defined in $\mathscr{F}$.

Proof.
(i) If $f(t) = \sum_{k\geq 0} a_k t^k/k!$, then $a_k = \langle f \mid x^k\rangle$ by definition.
(ii) Since $\langle t^k \mid p\rangle = p^{(k)}(0)$, this is Taylor's formula for the polynomial p.
(iii) Write $f = \sum a_k t^k/k!$ and $g = \sum b_k t^k/k!$.
The Cauchy product is $fg = \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \right) \frac{t^n}{n!}$, and $a_k = \langle f \mid x^k\rangle$, $b_k = \langle g \mid x^k\rangle$.
(iv) If $p = \sum_{j=0}^{d} c_j x^j$ and $o(f) > d$, then $\langle f \mid x^j\rangle = 0$ for $0 \leq j \leq d$.
(v) Since $o(f_k) = k$, each coefficient of $\sum_{k\geq 0} a_k f_k$ has only finitely many contributions.
With $d = \deg p$, $\langle f_k \mid p\rangle = 0$ for $k > d$ by (iv), proving the finite formula.
Structural Commentary: The umbral pairing dualizes coefficient calculus.
Key Insight: Order makes infinite formal sums act finitely on each polynomial. □

Theorem 9.3.9. *Multiplication by x Moves to Formal Differentiation*
Let $M_x : \mathscr{P} \to \mathscr{P}$ be $M_x p = xp$, and let $\partial_t : \mathscr{F} \to \mathscr{F}$ be formal differentiation. Then:

(i) $(\forall f \in \mathscr{F})(\forall p \in \mathscr{P})\ \langle f \mid M_x p\rangle = \langle \partial_t f \mid p\rangle$.
(ii) The algebraic transpose $M_x^\times$, defined by $\langle M_x^\times f \mid p\rangle = \langle f \mid M_x p\rangle$, satisfies $M_x^\times = \partial_t$.

Proof.
(i) By linearity in p, take $p = x^n$.
For $f = \sum_{k\geq 0} \frac{a_k}{k!} t^k$, $\partial_t f = \sum_{k\geq 0} \frac{a_{k+1}}{k!} t^k$, hence $\langle \partial_t f \mid x^n\rangle = a_{n+1} = \langle f \mid x^{n+1}\rangle = \langle f \mid M_x x^n\rangle$.
(ii) By (i), $\langle M_x^\times f \mid p\rangle = \langle \partial_t f \mid p\rangle$ for $f \in \mathscr{F}$ and $p \in \mathscr{P}$. By Theorem 9.3.6, the pairing is nondegenerate in its first argument, so $M_x^\times f = \partial_t f$ and $M_x^\times = \partial_t$.
Structural Commentary: M_x and ∂_t are adjoint under the umbral pairing.
Key Insight: Exponential normalization makes multiplication by x a coefficient shift. □

Definition 9.3.10. *Basic Umbral Functionals*
For $a, b \in \mathbb{C}$, $p \in \mathscr{P}$, and $n \in \mathbb{N}_0$:

(i) *Evaluation*: $\epsilon_a = e^{at}$ and $\langle e^{at} \mid p\rangle = p(a)$; hence $\epsilon_a \epsilon_b = \epsilon_{a+b}$.
(ii) *Forward difference*: $\Delta_a = e^{at} - 1$ and $\langle \Delta_a \mid p\rangle = p(a) - p(0)$.
(iii) *Abel functional*: te^{at} and $\langle te^{at} \mid p\rangle = p'(a)$.
(iv) *Gamma functional*: $(1-t)^{-1}$ and $\left\langle (1-t)^{-1} \middle| p \right\rangle = \int_0^\infty p(u) e^{-u}\, du$.
(v) *Integral functional*: $\left\langle \frac{e^{at}-1}{t} \middle| p \right\rangle = \int_0^a p(u)\, du$.
(vi) *Bernoulli numbers*: for $a = 1$, $B_n = \left\langle \frac{t}{e^t - 1} \middle| x^n \right\rangle$.

Definition 9.3.11. *Formal Series as Polynomial Operators*
For $f(t) = \sum_{k\geq 0} a_k t^k/k! \in \mathscr{F}$:

(i) *Operator action*: define $f(t) \in \mathrm{End}_{\mathbb{C}}(\mathscr{P})$ by $f(t)p(x) := \sum_{k\geq 0} \frac{a_k}{k!} p^{(k)}(x)$, a finite sum for $p \in \mathscr{P}$.
(ii) *Monomial action*: $f(t)x^n = \sum_{k=0}^{n} \binom{n}{k} a_k x^{n-k}$ $(n \in \mathbb{N}_0)$.

Theorem 9.3.12. *Functional-Operator Compatibility*
For $f, g \in \mathscr{F}$ and $p \in \mathscr{P}$:

(i) The representation $\rho : \mathscr{F} \to \mathrm{End}_{\mathbb{C}}(\mathscr{P})$, $f \mapsto f(t)$, is a unital $\mathbb{C}$-algebra homomorphism satisfying $\rho(t) = D$.
(ii) $\langle fg \mid p\rangle = \langle f \mid g(t)p\rangle$.

(iii) $\langle f \mid p\rangle = \langle t^0 \mid f(t)p\rangle$.

Proof.

(i) Write $f = \sum_{i\ge 0} a_i t^i/i!$ and $g = \sum_{j\ge 0} b_j t^j/j!$. On a fixed $p \in \mathscr{P}$ all derivative sums are finite, and after regrouping by total degree, $f(t)g(t)p = \sum_{i,j\ge 0} a_i b_j D^{i+j}p/(i!j!) = (fg)(t)p$. The constant series acts as I, and the series t acts as D.

(ii) By linearity in p, take $p = x^n$. Write $f = \sum_i \frac{a_i}{i!}t^i$ and $g = \sum_j \frac{b_j}{j!}t^j$.

Then $\langle fg \mid x^n\rangle = \sum_{j=0}^{n} \binom{n}{j} a_{n-j}b_j = \left\langle f \middle| \sum_{j=0}^{n}\binom{n}{j} b_j x^{n-j}\right\rangle = \langle f \mid g(t)x^n\rangle$.

(iii) In (ii), set $(f,g) = (t^0, F)$ with $F \in \mathscr{F}$. Since $t^0F = F$ in $\mathscr{F}$, $\langle F \mid p\rangle = \langle t^0F \mid p\rangle = \langle t^0 \mid F(t)p\rangle$.

Structural Commentary: Multiplication is adjoint to action.
Key Insight: One finite coefficient sum appears on both sides. □

Definition 9.3.13. *Basic Umbral Operators*
For $a \in \mathbb{C}$ and $p \in \mathscr{P}$:

(i) *Translation*: $e^{at}p(x) = p(x+a)$.
(ii) *Forward difference*: $(e^{at}-1)p(x) = p(x+a) - p(x)$.
(iii) *Abel operator*: $te^{at}p(x) = p'(x+a)$.
(iv) *Gamma operator*: $(1-t)^{-1}p(x) = \int_0^\infty p(x+u)e^{-u}\,du$.
(v) *Integral operator*: $\frac{e^{at}-1}{t}p(x) = \int_x^{x+a} p(u)\,du$.

Theorem 9.3.14. *Shift-Invariant Operators*
For $T \in \mathrm{End}_{\mathbb{C}}(\mathscr{P})$, the following are equivalent:

(i) $(\exists f \in \mathscr{F})\ T = f(t)$;
(ii) $[T,t] = 0$;
(iii) $(\forall h \in \mathscr{F} : o(h) = 1)\ [T, h(t)] = 0$;
(iv) $(\forall a \in \mathbb{C})\ [T, e^{at}] = 0$.

Proof.

(i) $\Rightarrow$ *(ii)* If $T = f(t)$, then $[T,t] = [f(t),t] = 0$ by Theorem 9.3.12(i) and the commutativity of $\mathscr{F}$.

(ii) $\Rightarrow$ *(iii)* Set $g(t) := \sum_{k\ge 0} \frac{\langle t^0 \mid Tx^k\rangle}{k!}t^k$ and $a_k := \langle t^0 \mid Tx^k\rangle$.

Since $[T,t] = 0$, for $m,k \ge 0$, $\langle t^m \mid Tx^k\rangle = \langle t^0 \mid t^mTx^k\rangle = \langle t^0 \mid Tt^mx^k\rangle = (k)_m a_{k-m}$, with $a_{k-m} = 0$ if $m > k$. Here $g(t) = \sum_{r\ge 0}\frac{a_r}{r!}t^r$ and $t^rx^k = (k)_r x^{k-r}$.

Then $\langle t^m \mid g(t)x^k\rangle = \sum_{r=0}^{k}\binom{k}{r}a_r\langle t^m \mid x^{k-r}\rangle = (k)_m a_{k-m}$.

Thus $(\forall m,k \in \mathbb{N}_0)\ \langle t^m \mid Tx^k - g(t)x^k\rangle = 0$. Since $\{t^m : m \ge 0\}$ separates $\mathscr{P}$, $T = g(t)$; if $o(h) = 1$, then $[T,h(t)] = 0$ by Theorem 9.3.12(i).

(iii) $\Rightarrow$ *(iv)* For $a \ne 0$, $e^{at}-1$ is a delta series, so $[T,e^{at}] = [T, e^{at}-1] = 0$; for $a = 0$, the commutator vanishes trivially.

(iv) $\Rightarrow$ *(i)* For fixed $p \in \mathscr{P}$, the expression $[T,e^{at}]p$ is a polynomial in a. Its vanishing on $\mathbb{C}$ implies $0 = [a^1]\,[T,e^{at}]p = [T,t]p$, so $[T,t] = 0$. The construction in *(ii)* $\Rightarrow$ *(iii)* then identifies $T = g(t)$ with $g \in \mathscr{F}$.

Structural Commentary: The commutant of t on $\mathscr{P}$ is $\mathscr{F}$.
Key Insight: The derivative is enough to recover every coefficient of the operator. □

Definition 9.3.15. *Sheffer, Associated, and Appell Sequences*
Let f be a delta series, let g be an invertible series, and let $(s_n)_{n\ge 0} \subseteq \mathscr{P}$ satisfy $(\forall n \in \mathbb{N}_0)\ \deg s_n = n$.

(i) *Sheffer sequence*: $(s_n)_{n\ge 0}$ is Sheffer for (g,f) if $(\forall n,k \in \mathbb{N}_0)\ \langle g(t)f(t)^k \mid s_n(x)\rangle = n!\delta_{n,k}$.
(ii) *Associated sequence*: a Sheffer sequence for $(1,f)$.
(iii) *Appell sequence*: a Sheffer sequence for (g,t).

Theorem 9.3.16. *Existence and Expansion Theorems*
Let f be a delta series, g an invertible series, and $q_k := g(t)f(t)^k$.

(i) $(\exists!\,(s_n)_{n\ge 0} \subseteq \mathscr{P})(\forall n,k \in \mathbb{N}_0)\ \deg s_n = n$ *and* $\langle q_k \mid s_n\rangle = n!\delta_{n,k}$.
(ii) If $p_n := g(t)s_n$, then (s_n) is Sheffer for (g,f) iff (p_n) is associated with f.
(iii) $(\forall h \in \mathscr{F})\ h = \sum_{k\ge 0}\frac{\langle h \mid s_k\rangle}{k!}q_k$.
(iv) $(\forall p \in \mathscr{P})\ p = \sum_{k=0}^{\deg p}\frac{\langle q_k \mid p\rangle}{k!}s_k$.

Proof.

(i) Since $o(q_k) = k$, the matrix $A_n = (\langle q_i \mid x^j\rangle)_{0\le i,j\le n}$ is triangular with nonzero diagonal.
Hence the equations $\langle q_k \mid s_n\rangle = n!\delta_{n,k}$ have a unique solution $s_n \in \mathscr{P}_{\le n}$. The equation at $k = n$ has nonzero right-hand side, so $\deg s_n = n$; the equations with $k > n$ hold because $o(q_k) > n$.

(ii) $(\Rightarrow)$ If (s_n) is Sheffer for (g,f), then $\langle f(t)^k \mid p_n\rangle = \langle f(t)^k \mid g(t)s_n\rangle = \langle q_k \mid s_n\rangle = n!\delta_{n,k}$ by Theorem 9.3.12, so (p_n) is associated with f.

(ii) $(\Leftarrow)$ If (p_n) is associated with f, then $\langle q_k \mid s_n\rangle = \langle f(t)^k \mid g(t)s_n\rangle = \langle f(t)^k \mid p_n\rangle = n!\delta_{n,k}$ by Theorem 9.3.12, so (s_n) is Sheffer for (g,f).

(iii) Since $o(q_k) = k$, every $h \in \mathscr{F}$ has a unique expansion $h = \sum c_k q_k$.
Pairing with s_n, $\langle h \mid s_n\rangle = c_n n!$, so $c_n = \langle h \mid s_n\rangle/n!$.

(*iv*) Let $d = \deg p$ and $r := p - \sum_{k=0}^{d} \langle q_k \mid p\rangle s_k/k!$.

Then $\langle q_j \mid r\rangle = 0$ when $j \le d$, and the same equality holds when $j > d$ because $o(q_j) > d$. The triangular coordinate argument from (iii) shows that the q_j separate $\mathscr{P}$, so $r = 0$.

Structural Commentary: (s_n) is dual to $q_k = gf^k$.
Key Insight: The order filtration makes the orthogonality equations triangular. □

Theorem 9.3.17. *Generating and Operator Characterizations*
Let f be a delta series with compositional inverse $\overline{f}$, let $g \in \mathscr{F}^\times$, and let $(p_n), (s_n)$ be polynomial sequences with $(\forall n \in \mathbb{N}_0)$ $\deg p_n = \deg s_n = n$. Set $P(y,u) := \sum_{n\ge0} \frac{p_n(y)}{n!}u^n$ and $S(y,u) := \sum_{n\ge0} \frac{s_n(y)}{n!}u^n$ in $\mathbb{C}[y][[u]]$.

(i) (p_n) is associated with $f \iff P(y,u) = e^{y\overline{f}(u)}$.
(ii) (s_n) is Sheffer for $(g,f) \iff S(y,u) = g(\overline{f}(u))^{-1}e^{y\overline{f}(u)}$.
(iii) (p_n) is associated with $f \iff (\forall n \in \mathbb{N}_0)$ $p_n(0) = \delta_{n,0}$ and $f(t)p_n = np_{n-1}$, where $p_m := 0$ $(m < 0)$.
(iv) $[(\exists \gamma \in \mathscr{F}^\times)$ (s_n) is Sheffer for $(\gamma, f)] \iff (\forall n \in \mathbb{N}_0)$ $f(t)s_n = ns_{n-1}$, where $s_m := 0$ $(m < 0)$.
(v) If (p_n) is associated with f, then $(\forall n \in \mathbb{N}_0)$ $p_n(x) = \sum_{k=0}^{n} \frac{\langle \overline{f}(t)^k \mid x^n\rangle}{k!}x^k$.
(vi) If (s_n) is Sheffer for (g,f), then $(\forall n \in \mathbb{N}_0)$ $s_n(x) = \sum_{k=0}^{n} \frac{1}{k!}\left\langle g(\overline{f}(t))^{-1}\overline{f}(t)^k \middle| x^n \right\rangle x^k$.

Proof.

(*i*) ($\Rightarrow$) By Theorem 9.3.16, $e^{yt} = \sum_{n\ge0} \frac{p_n(y)}{n!}f(t)^n$. Substitute $t = \overline{f}(u)$ to obtain $P(y,u) = e^{y\overline{f}(u)}$.

(*i*) ($\Leftarrow$) If $P(x,u) = e^{x\overline{f}(u)}$, then $(\forall k \in \mathbb{N}_0)$ $\sum_{n\ge0} \frac{\langle f(t)^k \mid p_n\rangle}{n!}u^n = \langle f(t)^k \mid P(x,u)\rangle = f(\overline{f}(u))^k = u^k$.

Thus $\langle f(t)^k \mid p_n\rangle = n!\delta_{n,k}$.

(*ii*) ($\Rightarrow$) By Theorem 9.3.16, $e^{yt} = \sum_{n\ge0} \frac{s_n(y)}{n!}g(t)f(t)^n$. Substitute $t = \overline{f}(u)$ and divide by $g(\overline{f}(u))$.

(*ii*) ($\Leftarrow$) If $S(x,u) = g(\overline{f}(u))^{-1}e^{x\overline{f}(u)}$, then $\sum_{n\ge0} \frac{\langle g(t)f(t)^k \mid s_n\rangle}{n!}u^n = \langle g(t)f(t)^k \mid S(x,u)\rangle = u^k$,

so $\langle g(t)f(t)^k \mid s_n\rangle = n!\delta_{n,k}$.

(*iii*) ($\Rightarrow$) If (p_n) is associated, then $p_n(0) = \langle 1 \mid p_n\rangle = \delta_{n,0}$ and $(\forall k \in \mathbb{N}_0)$ $\langle f(t)^k \mid f(t)p_n - np_{n-1}\rangle = 0$.
Since $\{f(t)^k : k \ge 0\}$ separates $\mathscr{P}$, $f(t)p_n = np_{n-1}$.

(*iii*) ($\Leftarrow$) By the lowering relation, $f(t)^k p_n = (n)_k p_{n-k}$.
Hence $\langle f(t)^k \mid p_n\rangle = \langle 1 \mid f(t)^k p_n\rangle = (n)_k p_{n-k}(0) = n!\delta_{n,k}$.

(*iv*) ($\Rightarrow$) If (s_n) is Sheffer for (γ, f), then $(\forall k \in \mathbb{N}_0)$ $\langle \gamma(t)f(t)^k \mid f(t)s_n - ns_{n-1}\rangle = 0$.
Since $\{\gamma(t)f(t)^k : k \ge 0\}$ separates $\mathscr{P}$, $f(t)s_n = ns_{n-1}$.

(*iv*) ($\Leftarrow$) Since (s_n) is a degree-graded basis of $\mathscr{P}$, define $L \in \mathscr{P}'$ by $L(s_0) = 1$ and $L(s_n) = 0$ for $n > 0$, and put $\gamma := \Phi(L)$. Write $s_0 = c \in \mathbb{C}^\times$; then $\langle \gamma \mid 1\rangle = L(1) = c^{-1} \ne 0$, so $\gamma \in \mathscr{F}^\times$. By the lowering relation, $f(t)^k s_n = (n)_k s_{n-k}$ when $k \le n$ and 0 when $k > n$, so $\langle \gamma(t)f(t)^k \mid s_n\rangle = n!\delta_{n,k}$.

(*v*) From (i), $e^{x\overline{f}(u)} = \sum_{k\ge0} \frac{\overline{f}(u)^k}{k!}x^k$. The stated formula for p_n follows after extracting $n![u^n]$.

(*vi*) From (ii), $g(\overline{f}(u))^{-1}e^{x\overline{f}(u)} = \sum_{k\ge0} \frac{g(\overline{f}(u))^{-1}\overline{f}(u)^k}{k!}x^k$. The stated formula for s_n follows after extracting $n![u^n]$.

Structural Commentary: $\overline{f}$ converts f-lowering to exponential generating functions.
Key Insight: Generating functions package the whole triangular duality into one formal identity. □

Definition 9.3.18. *Binomial Type and Sheffer Identity*
Let (p_n) and (s_n) be polynomial sequences.

(i) *Binomial type*: (p_n) is of binomial type if $(\forall n \in \mathbb{N}_0)$ $p_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} p_k(y)p_{n-k}(x)$.

(ii) *Sheffer identity*: (s_n) satisfies the Sheffer identity relative to (p_n) if $(\forall n \in \mathbb{N}_0)$ $s_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} p_k(y)s_{n-k}(x)$.

Theorem 9.3.19. *Action and Binomial Identities*
Let (s_n) be Sheffer for (g,f), let (p_n) be associated with f, let (r_n) satisfy $(\forall n \in \mathbb{N}_0)$ $\deg r_n = n$, and write $E_y := e^{yt}$. Then:

(i) $(\forall h \in \mathscr{F})(\forall n \in \mathbb{N}_0)$ $h(t)s_n = \sum_{k=0}^{n} \binom{n}{k}\langle h \mid s_k\rangle p_{n-k}$;
(ii) $[(\exists u \in \mathscr{F} : o(u) = 1)$ (r_n) is associated with $u] \iff (r_n)$ is of binomial type; the delta series u is unique;
(iii) $[(\exists \gamma \in \mathscr{F}^\times)$ (r_n) is Sheffer for $(\gamma, f)] \iff (r_n)$ satisfies the Sheffer identity relative to (p_n).

Proof.

(*i*) By Theorem 9.3.16, $h = \sum_{k\ge0} \frac{\langle h \mid s_k\rangle}{k!}g(t)f(t)^k$.

Since $g(t)s_n = p_n$ and $f(t)^k p_n = (n)_k p_{n-k}$, $h(t)s_n = \sum_{k=0}^{n} \frac{\langle h \mid s_k\rangle}{k!}(n)_k p_{n-k}$.

(*ii*) ($\Rightarrow$) Suppose (r_n) is associated with a delta series u. The formula in (i) holds with $(s_n, p_n, g, f) = (r_n, r_n, 1, u)$ and $h = E_y$. Since $\langle E_y \mid r_k\rangle = r_k(y)$ and $E_y r_n(x) = r_n(x+y)$, (r_n) is of binomial type.

(*ii*) ($\Leftarrow$) Suppose (r_n) is of binomial type. Its degree assumptions and the $n = 0$ identity imply $r_0 = 1$; induction after setting $y = 0$ establishes $r_n(0) = \delta_{n,0}$. Define $Qr_n := nr_{n-1}$. By the binomial identity, $(\forall n \in \mathbb{N}_0)(\forall y \in \mathbb{C})$ $QE_y r_n = E_y Q r_n$, so $[Q, E_y] = 0$. By Theorem 9.3.14, $Q = u(t)$ with $u \in \mathscr{F}$.
Now $Q1 = 0$ implies $[t^0]u = 0$, while $Qr_1 = 1$ excludes $o(u) \ge 2$; hence $o(u) = 1$. By Theorem 9.3.17(iii), (r_n) is associated with u. By the reconstruction in Theorem 9.3.14, u is uniquely determined by Q.

(*iii*) ($\Rightarrow$) If (r_n) is Sheffer for (γ, f), the formula in (i) holds with $(s_n, g) = (r_n, \gamma)$ and $h = E_y$.
Since $\langle E_y \mid r_k\rangle = r_k(y)$ and $E_y r_n(x) = r_n(x+y)$, $r_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} r_k(y) p_{n-k}(x)$.
Relabel and swap x, y to get the Sheffer identity relative to (p_n).
(*iii*) ($\Leftarrow$) If the Sheffer identity holds, act by $f(t)$ in the y-variable and set $y = 0$.
Since $f(t)p_k(y)|_{y=0} = kp_{k-1}(0) = k\delta_{k,1}$, $f(t)r_n = nr_{n-1}$.
By Theorem 9.3.17(iv), (r_n) is Sheffer for (γ, f) with $\gamma \in \mathscr{F}^\times$.

Structural Commentary: Translation turns the operator formula into addition laws for polynomial sequences.
Key Insight: Binomial identities are the visible polynomial shadow of a delta operator. □

Example 9.3.20. *Standard Sheffer Examples*

(i) *Monomials*: $p_n(x) = x^n$ is associated with $f(t) = t$; the binomial identity is the usual binomial theorem.
(ii) *Lower factorials*: $(x)_n$ is associated with $f(t) = e^t - 1$, with $(1+t)^y = e^{y\log(1+t)} = \sum_{k\geq 0} \frac{(y)_k}{k!} t^k$,
and Vandermonde identity $(x+y)_n = \sum_{k=0}^{n} \binom{n}{k} (x)_k (y)_{n-k}$.
(iii) *Abel polynomials*: for $a \in \mathbb{C}$, the sequence $A_0(x;a) = 1$ and $A_n(x;a) = x(x-an)^{n-1}$ $(n \geq 1)$ is associated with $f(t) = te^{at}$.
(iv) *Hermite polynomials*: for $g(t) = e^{t^2/2}$, the polynomial is $H_n(x) = e^{-t^2/2} x^n$, where $e^{yt - t^2/2} = \sum_{k\geq 0} \frac{H_k(y)}{k!} t^k$.
(v) *Laguerre polynomials*: for $\alpha \in \mathbb{C}$, the order-α sequence is Sheffer for $g(t) = (1-t)^{-\alpha-1}$ and $f(t) = \frac{t}{t-1}$, with $\frac{1}{(1-t)^{\alpha+1}} e^{yt/(t-1)} = \sum_{k\geq 0} \frac{L_k^{(\alpha)}(y)}{k!} t^k$.

Definition 9.3.21. *Umbral Operators and Shifts*
Let (s_n) be Sheffer for (g, f), and let (p_n) be associated with f.
(i) *Sheffer operator*: $\lambda_{g,f} \in \mathcal{L}(\mathscr{P})$ is defined by $\lambda_{g,f}(x^n) = s_n(x)$.
(ii) *Umbral operator*: $\lambda_f \in \mathcal{L}(\mathscr{P})$ is defined by $\lambda_f(x^n) = p_n(x)$.
(iii) *Sheffer/umbral shift*: $\theta_{g,f} s_n = s_{n+1}$; in the associated case, $\theta_f p_n = p_{n+1}$.

Definition 9.3.22. *Continuous Operators on the Umbral Algebra*
An operator $T \in \mathcal{L}(\mathscr{F})$ is *continuous* iff $o(f_k) \to \infty$ implies $o(Tf_k) \to \infty$. Equivalently, if $o(f_k) \to \infty$ and $(a_k)_{k\geq 0} \subseteq \mathbb{C}$, then $T\left(\sum_{k\geq 0} a_k f_k\right) = \sum_{k\geq 0} a_k T(f_k)$.

Definition 9.3.23. *Algebraic Adjoints on the Umbral Algebra*
For $\tau \in \mathcal{L}(\mathscr{P})$, its *algebraic adjoint* $\tau^\times \in \mathcal{L}(\mathscr{F})$ is defined by $(\forall h \in \mathscr{F})(\forall p \in \mathscr{P})\ \langle \tau^\times h \mid p\rangle = \langle h \mid \tau p\rangle$. Under the isomorphism of Theorem 9.3.6, the dual map from Definition 4.2.4 determines this unique operator; it is not an inner-product adjoint.

Theorem 9.3.24. *Continuous Operators Are Exactly Operator Adjoints*
Let $\mathcal{L}_{\mathrm{cont}}(\mathscr{F})$ denote the continuous linear operators on $\mathscr{F}$. Then $(-)^\times : \mathcal{L}(\mathscr{P}) \xrightarrow{\sim} \mathcal{L}_{\mathrm{cont}}(\mathscr{F})$, $\tau \longmapsto \tau^\times$, is a vector-space isomorphism.

Proof.
(i) *Well-Definedness and Injectivity*: Let $\tau \in \mathcal{L}(\mathscr{P})$ and $o(h_j) \to \infty$. With $n \in \mathbb{N}_0$ fixed, $\tau x^n \in \mathscr{P}$, so $\langle \tau^\times h_j \mid x^n\rangle = \langle h_j \mid \tau x^n\rangle = 0$ once j is sufficiently large. Thus $o(\tau^\times h_j) \to \infty$, so $\tau^\times \in \mathcal{L}_{\mathrm{cont}}(\mathscr{F})$.
If $\tau^\times = 0$, then $(\forall h \in \mathscr{F})(\forall p \in \mathscr{P})\ \langle h \mid \tau p\rangle = 0$, hence $\tau = 0$; the map is injective.
(ii) *Surjectivity*: Let $T \in \mathcal{L}_{\mathrm{cont}}(\mathscr{F})$. Define $\tau_T x^n := \sum_{k\geq 0} \frac{\langle Tt^k \mid x^n\rangle}{k!} x^k$. Since $o(Tt^k) \to \infty$, the sum is finite when $n \in \mathbb{N}_0$ is fixed.
For $m, n \in \mathbb{N}_0$, $\langle \tau_T^\times t^m \mid x^n\rangle = \langle t^m \mid \tau_T x^n\rangle = \langle Tt^m \mid x^n\rangle$. Thus $(\forall m \in \mathbb{N}_0)\ \tau_T^\times t^m = Tt^m$. If $h = \sum_{m\geq 0} a_m t^m/m!$, then $\tau_T^\times h = Th$ by continuity of both operators, so the map is surjective.

Structural Commentary: Continuity is the finiteness condition for dualising polynomial operators.
Key Insight: Finiteness on each polynomial degree is the hidden topology. □

Theorem 9.3.25. *Umbral Operators and Adjoint Automorphisms*
Let f be a delta series, let $\overline{f}$ be its compositional inverse, let (p_n) be associated with f, and set $\lambda_f(x^n) = p_n(x)$.
For $C_{\overline{f}}(h) := h(\overline{f})$ and $\lambda \in \mathcal{L}(\mathscr{P})$:

(i) $\lambda = \lambda_f$ iff $\lambda^\times = C_{\overline{f}}$; in this case $\lambda_f^\times f = t$.
(ii) λ is umbral iff $\lambda^\times \in \mathrm{Aut}_{\mathbb{C}\text{-}alg}(\mathscr{F})$.

Proof.
(*i*) ($\Rightarrow$) If $\lambda = \lambda_f$, then $\langle \lambda^\times f^k \mid x^n\rangle = \langle f^k \mid p_n\rangle = n!\delta_{n,k} = \langle t^k \mid x^n\rangle$, so $\lambda^\times f^k = t^k$. For $h \in \mathscr{F}$, put $q := h(\overline{f})$, so $h = q(f)$. Then $\lambda^\times h = q(t) = h(\overline{f}) = C_{\overline{f}} h$ by continuity. Hence $\lambda^\times = C_{\overline{f}}$ and $\lambda_f^\times f = t$.
(*i*) ($\Leftarrow$) If $\lambda^\times = C_{\overline{f}}$, set $r_n := \lambda x^n$. Then $\langle f^k \mid r_n\rangle = \langle \lambda^\times f^k \mid x^n\rangle = \langle t^k \mid x^n\rangle = n!\delta_{n,k} = \langle f^k \mid p_n\rangle$.
Since $\{f^k : k \geq 0\}$ separates $\mathscr{P}$, $r_n = p_n$ and $\lambda = \lambda_f$.
(*ii*) ($\Rightarrow$) If λ is umbral, then $\lambda = \lambda_u$ with u a delta series. By (i), $\lambda^\times = C_{\overline{u}} \in \mathrm{Aut}_{\mathbb{C}\text{-alg}}(\mathscr{F})$.
(*ii*) ($\Leftarrow$) Let $A := \lambda^\times \in \mathrm{Aut}_{\mathbb{C}\text{-alg}}(\mathscr{F})$ and set $u := A^{-1}t$. The ideal (t) is the unique maximal ideal of $\mathscr{F}$, so $A^{-1}((t)) = (t)$ and $A^{-1}((t)^2) = (t)^2$. Hence $u \in (t) \setminus (t)^2$ and $o(u) = 1$. Let (q_n) be associated with u and set $r_n := \lambda x^n$. Then $\langle u^k \mid r_n\rangle = \langle Au^k \mid x^n\rangle = n!\delta_{n,k} = \langle u^k \mid q_n\rangle$.
Since $\{u^k : k \geq 0\}$ separates $\mathscr{P}$, $r_n = q_n$ and $\lambda = \lambda_u$ is umbral.

Structural Commentary: Umbral adjoints are substitutions.
Key Insight: The delta series is $A^{-1}t$. □

Definition 9.3.26. *Umbral Composition*

For polynomial sequences (p_n) and (q_n) with $p_n(x) = \sum_{k=0}^{n} p_{n,k}x^k$, define $(p \circ_U q)_n(x) := \sum_{k=0}^{n} p_{n,k}q_k(x)$.

Theorem 9.3.27. *Group Law for Umbral Operators*

Let f, g be delta series, and let (p_n) and (q_n) be associated with f and g. Then:

(i) $\lambda_g\lambda_f = \lambda_{f\circ g}$.
(ii) $\lambda_f^{-1} = \lambda_{\overline{f}}$.
(iii) $(p \circ_U q)_n = (\lambda_g\lambda_f)x^n$; *hence associated sequences form a group under* $\circ_U$.

Proof.

(i) By $(AB)^\times = B^\times A^\times$ and Theorem 9.3.25, $(\lambda_g\lambda_f)^\times h = \lambda_f^\times\lambda_g^\times h = h(\overline{g}(\overline{f})) = h(\overline{f\circ g})$ for $h \in \mathscr{F}$; hence $\lambda_g\lambda_f = \lambda_{f\circ g}$.

(ii) Put $g = \overline{f}$ in (i). Then $\lambda_{\overline{f}}\lambda_f = \lambda_t = I$ and $\lambda_f\lambda_{\overline{f}} = \lambda_t = I$, so $\lambda_f^{-1} = \lambda_{\overline{f}}$.

(iii) If $p_n = \sum_{k=0}^{n} p_{n,k}x^k$, then $(\lambda_g\lambda_f)x^n = \lambda_g p_n = \sum_{k=0}^{n} p_{n,k}q_k = (p \circ_U q)_n$.
By (i) and (ii), the operation inherits associativity, identity, and inverses from operator composition.

Structural Commentary: Umbral composition is substitution of delta series.
Key Insight: Adjoints reverse the operator composition. □

Proposition 9.3.28. *Sheffer Operators*

Let f, k be delta series, $g, h \in \mathscr{F}^\times$, (s_n) Sheffer for (g, f), and (p_n) associated with f. Then:

(i) $s_n = g(t)^{-1}p_n$ *and* $\lambda_{g,f} = g(t)^{-1}\lambda_f$.
(ii) $\lambda_{g,f}\lambda_{h,k} = \lambda_{g(h\circ f),\,k\circ f}$.
(iii) $\lambda_{g,f}^{-1} = \lambda_{g^{-1}(\overline{f}),\,\overline{f}}$.

Proof.

(i) By Theorem 9.3.16(ii), $(g(t)s_n)$ is associated with f; uniqueness in part (i) therefore implies $g(t)s_n = p_n$. Hence $s_n = g(t)^{-1}p_n$ and $\lambda_{g,f} = g(t)^{-1}\lambda_f$.

(ii) For $a \in \mathscr{F}$, $\lambda_f a(t) = a(f(t))\lambda_f$ by Theorem 9.3.25.
Thus $\lambda_{g,f}\lambda_{h,k} = g(t)^{-1}\lambda_f h(t)^{-1}\lambda_k = g(t)^{-1}h(f(t))^{-1}\lambda_{k\circ f} = \lambda_{g(h\circ f),\,k\circ f}$.

(iii) In (ii), take $(h, k) = (g^{-1}(\overline{f}), \overline{f})$ to get $\lambda_{g,f}\lambda_{g^{-1}(\overline{f}),\,\overline{f}} = \lambda_{1,t} = I$.
Reversing the factors, $\lambda_{g^{-1}(\overline{f}),\,\overline{f}}\lambda_{g,f} = \lambda_{g^{-1}(\overline{f})(g\circ\overline{f}),\,f\circ\overline{f}} = \lambda_{1,t} = I$.

Structural Commentary: Weight then umbral substitution.
Key Insight: $(g, f)(h, k) = (g(h \circ f), k \circ f)$. □

Theorem 9.3.29. *Umbral Shift Derivations*

Let f be a delta series and let θ_f be its umbral shift.
For $h \in \mathscr{F}$, let $(c_k)_{k\geq 0}$ be the unique coordinate sequence from Proposition 9.3.2(v), so $h = \sum_{k\geq 0} c_k f^k$, and define $\partial_f h := \sum_{k\geq 1} kc_k f^{k-1}$. Then:

(i) $\partial_f \in \mathrm{Der}_{\mathbb{C}}(\mathscr{F})$ *and* ∂_f *is continuous.*
(ii) $\theta_f^\times = \partial_f$.
(iii) $\partial_f f = 1$.
(iv) If g is a delta series, then $\partial_g = (\partial_g f)\partial_f$.

Proof.

(i) The coordinate formula is linear and satisfies $o(\partial_f h) \geq o(h) - 1$, so it defines a continuous linear operator. On coordinate monomials, $\partial_f(f^j f^k) = (j + k)f^{j+k-1} = (\partial_f f^j)f^k + f^j(\partial_f f^k)$; bilinearity and continuity extend the Leibniz identity to $\mathscr{F}$.

(ii) Let (p_n) be associated with f. For $n \in \mathbb{N}_0$, both $\langle\theta_f^\times 1 \mid p_n\rangle$ and $\langle 0 \mid p_n\rangle$ vanish. For $k \geq 1$ and $n \geq 0$, $\langle\theta_f^\times f^k \mid p_n\rangle = \langle f^k \mid p_{n+1}\rangle = (n + 1)!\delta_{n+1,k} = \langle kf^{k-1} \mid p_n\rangle$.
Hence $\theta_f^\times f^k = \partial_f(f^k)$ $(k \in \mathbb{N}_0)$, so $\theta_f^\times = \partial_f$ by continuity.

(iii) By definition of ∂_f, $\partial_f f = \partial_f(f^1) = 1 \cdot f^0 = 1$.

(iv) Both sides vanish on $f^0 = 1$. For $k \geq 1$, continuity reduces the identity to $(\partial_g f)\partial_f(f^k) = k(\partial_g f)f^{k-1} = \partial_g(f^k)$.

Structural Commentary: Umbral shift is delta-coordinate differentiation.
Key Insight: The associated basis turns shifts into derivatives. □

Theorem 9.3.30. *Umbral Shifts and Surjective Derivations*

For $\theta \in \mathcal{L}(\mathscr{P})$, θ is an umbral shift $\iff$ $\theta^\times$ is a surjective derivation of $\mathscr{F}$.

Proof.

($\Rightarrow$) If $\theta = \theta_f$, then $\theta^\times = \partial_f$ by Theorem 9.3.29. Since $\partial_f\left(\frac{f^{k+1}}{k+1}\right) = f^k$ $(k \in \mathbb{N}_0)$, ∂_f is a surjective derivation.

($\Leftarrow$) Let $D := \theta^\times$ be a surjective derivation. By Theorem 9.3.24, D is continuous. Put $a := Dt$. The identity $D(t^n) = nt^{n-1}a$ follows inductively from the derivation law, and $Dh = a\partial_t h$ for $h \in \mathscr{F}$ follows by continuity. Choose $u \in \mathscr{F}$ with $Du = 1$ and set $f := u - [t^0]u$.
Then $Df = 1$, so f is nonconstant. Put $r := o(f)$; then $1 \leq r < \infty$ and $0 = o(1) = o(a) + r - 1$. Hence $o(a) = 0$ and $r = 1$, so f is a delta series. Let (p_n) be associated with f.
For $n \in \mathbb{N}_0$, the equality $\langle 1 \mid \theta p_n\rangle = \langle 1 \mid p_{n+1}\rangle = 0$ follows from $D1 = 0$. For $k \geq 1$ and $n \geq 0$, $\langle f^k \mid \theta p_n\rangle = \langle Df^k \mid p_n\rangle = \langle kf^{k-1} \mid p_n\rangle = \langle f^k \mid p_{n+1}\rangle$.
The functionals f^k separate $\mathscr{P}$, so $\theta p_n = p_{n+1}$ and $\theta = \theta_f$.

Structural Commentary: Surjective derivations are derivatives in delta coordinates.
Key Insight: Surjectivity chooses the coordinate whose derivative is 1. □

Figure 187 summarizes the adjoint correspondences established by Theorems 9.3.25 and 9.3.30: umbral operators correspond to algebra automorphisms, while umbral shifts correspond to surjective derivations.

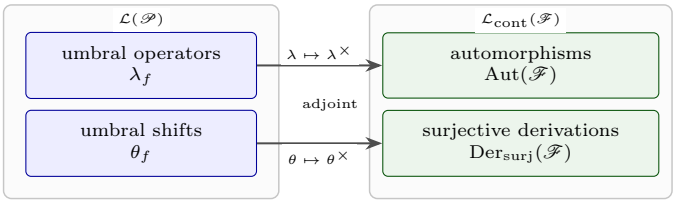

Figure 187: Adjoints identify $\mathcal{L}(\mathscr{P})$ with $\mathcal{L}_{\mathrm{cont}}(\mathscr{F})$ and map the umbral classes shown.

9.4 Umbral Recurrences and Variants

Lemma 9.4.1. *Pincherle Commutator*
For $a(t) \in \mathscr{F}$, $xa(t) = a(t)x - a'(t)$ as operators on $\mathscr{P}$.

Proof. Since t acts as differentiation and x as multiplication, $tx = xt + 1$. The case $k = 0$ is immediate, and induction establishes $xt^k = t^k x - kt^{k-1}$ $(k \geq 1)$. For $a(t) = \sum_{k\geq 0} a_k t^k/k!$, the operator sum is finite on each polynomial, so $xa(t) = a(t)x - a'(t)$.
Structural Commentary: The basic commutator extends coefficientwise from monomials to formal series.
Key Insight: The relation $tx - xt = 1$ differentiates operator series. □

Theorem 9.4.2. *Recurrence and Transfer Formulas*
Let (p_n) be associated with the delta series f, set $u(t) := f(t)/t \in \mathscr{F}^{\times}$, and write $f' := \partial_t f \in \mathscr{F}^{\times}$. Then:

(i) $p_{n+1} = x(f')^{-1} p_n$ $(n \in \mathbb{N}_0)$.
(ii) $p_n = f' u^{-n-1} x^n = x u^{-n} x^{n-1}$ $(n \in \mathbb{N},\ n \geq 1)$.

Proof.

(i) By Theorem 9.3.29, $\partial_t = f'\partial_f$. By Theorem 9.3.12(ii), $(f'(t))^{\times} h = f'h$, so $(\theta_f f'(t))^{\times} = f'\partial_f = \partial_t = \theta_t^{\times}$. Injectivity of the adjoint correspondence therefore implies $\theta_t = \theta_f f'(t)$. Since $\theta_t = x$, $\theta_f = x(f')^{-1}$, and $p_{n+1} = x(f')^{-1} p_n$.

(ii) Put $q_n := f' u^{-n-1} x^n$ $(n \in \mathbb{N}_0)$. Since $[t^0]f' = [t^0]u$, $q_0 = f'u^{-1}1 = 1 = p_0$.
By $f' = u + tu'$ and Lemma 9.4.1, $q_{n+1} = u^{-n-1}x^{n+1} + (n+1)u^{-n-2}u'x^n = xu^{-n-1}x^n = x(f')^{-1}q_n$. Thus $q_n = p_n$ by (i).
By $f' = u + tu'$ and Lemma 9.4.1 with $a = u^{-n}$, $xu^{-n}x^{n-1} = u^{-n}x^n + nu^{-n-1}u'x^{n-1} = f'u^{-n-1}x^n$.

Structural Commentary: The recurrence determines (p_n); transfer solves it using $f = tu$.
Key Insight: The Pincherle commutator converts the shift recurrence into closed form. □

Corollary 9.4.3. *Sheffer Shift Formula*
Let (s_n) be Sheffer for (g, f), where f is delta and $g \in \mathscr{F}^{\times}$. Write $f' := \partial_t f$ and $g' := \partial_t g$. Then:

(i) $\theta_{g,f} = g^{-1}\theta_f g = \left(x - \frac{g'}{g}\right)(f')^{-1}$.
(ii) $s_{n+1} = \left(x - \frac{g'}{g}\right)(f')^{-1} s_n$ $(n \in \mathbb{N}_0)$.

Proof.

(i) Put $p_n := g(t)s_n$, which is associated with f by Theorem 9.3.16(ii). Then $g^{-1}\theta_f g\, s_n = g^{-1}\theta_f p_n = g^{-1}p_{n+1} = s_{n+1} = \theta_{g,f} s_n$. Since (s_n) is a basis, $\theta_{g,f} = g^{-1}\theta_f g$. By Theorem 9.4.2, $\theta_f = x(f')^{-1}$.
By Lemma 9.4.1, $g' = gx - xg$. Since f' and g commute, $g^{-1}\theta_f g = g^{-1}x(f')^{-1}g = g^{-1}xg(f')^{-1} = \left(x - \frac{g'}{g}\right)(f')^{-1}$.

(ii) By definition of the Sheffer shift, $\theta_{g,f} s_n = s_{n+1}$ $(n \in \mathbb{N}_0)$. By (i), $s_{n+1} = \theta_{g,f} s_n = \left(x - \frac{g'}{g}\right)(f')^{-1} s_n$.

Structural Commentary: Sheffer shifts are conjugated associated shifts.
Key Insight: Conjugation by g contributes the correction g'/g. □

Method 9.4.4. *Building Sheffer Sequences*
To construct the Sheffer sequence for (g, f):

(i) Verify $o(f) = 1$ and $o(g) = 0$.
(ii) If $\overline{f}$ is explicit, compute $S(y,t) := g(\overline{f}(t))^{-1} e^{y\overline{f}(t)}$, where $s_n(y) = n![t^n]S(y,t)$.
(iii) If $\overline{f}$ is unavailable, set $s_0 = g(0)^{-1}$ and use $s_{n+1} = (x - g'/g)(f')^{-1} s_n$ from Corollary 9.4.3.
(iv) For associated sequences, set $g = 1$ and compute p_n by Theorem 9.4.2.

Definition 9.4.5. *Stirling and Exponential Polynomials*

(i) *Stirling numbers of the second kind*: define $S(n,k)$ uniquely by $x^n = \sum_{k=0}^{n} S(n,k)(x)_k$ $(n \in \mathbb{N}_0)$.

(ii) *Exponential polynomials*: define $\phi_n(x) := \sum_{k=0}^{n} S(n,k)x^k$ $(n \in \mathbb{N}_0)$.

(iii) *Boundary convention*: $S(n,k) = 0$ if $n \in \mathbb{N}_0$, $k \in \mathbb{Z}$, and either $k < 0$ or $k > n$.

Example 9.4.6. *Lower Factorials and Stirling Numbers*
For $f(t) = e^t - 1$, $f'(t) = e^t$, hence $p_{n+1}(x) = xe^{-t}p_n(x) = xp_n(x-1)$. With $p_0 = 1$, therefore $p_n = (x)_n$ $(n \in \mathbb{N}_0)$.
By Theorem 9.3.16(iv), $S(n,k) = \langle (e^t-1)^k \mid x^n \rangle / k!$. Hence $e^{x(e^t-1)} = \sum_{n\geq 0} \phi_n(x) t^n/n!$. Since $e^t - 1$ is the compositional inverse of $\log(1+t)$, (ϕ_n) is associated with $f(t) = \log(1+t)$ by Theorem 9.3.17(i).
Thus $x^n = \sum_{k=0}^{n} S(n,k)(x)_k$ and $\phi_n(x) = \sum_{k=0}^{n} S(n,k)x^k$ are inverse umbral relations.

By the recurrence formula, $\phi_{n+1}(x) = x(1+t)\phi_n(x) = x\big(\phi_n(x) + \phi_n'(x)\big)$.
Thus $S(n+1,k) = S(n,k-1) + kS(n,k)$ $(n,k \in \mathbb{N}_0)$ by coefficient comparison.

Definition 9.4.7. *Gaussian Coefficients*

For an indeterminate q and integers $0 \le k \le n$, define $\binom{n}{k}_q := \prod_{j=1}^{n}(1-q^j) \Big/ \left(\prod_{j=1}^{k}(1-q^j)\prod_{j=1}^{n-k}(1-q^j)\right) \in \mathbb{Q}(q)$, with an empty product equal to 1.

Remark 9.4.8. *Further Directions*
Classical umbral methods extend to many polynomial families and operations:

(i) *Families*: lower factorial, Stirling, exponential, Gould, central factorial, Abel, Mittag–Leffler, Bessel, Bell, Hermite, Bernoulli, Euler, Laguerre, Poisson–Charlier, actuarial, Meixner, Pidduck, Narumi, Boole, Peters, Mahler, and Mott polynomials.
(ii) *Operations*: connection constants, duplication formulas, Lagrange inversion, cross sequences, Steffensen sequences, operational formulas, inverse relations, recurrence solving, and binomial convolution.
(iii) *q-umbral calculus*: the deformation $\binom{n}{k} \rightsquigarrow \binom{n}{k}_q$ through the Gaussian coefficients of Definition 9.4.7.
(iv) *Logarithmic umbral calculus*: the logarithmic-type variant based on harmonic logarithms.

10 Fourier Structure

10.1 Fourier Orthogonality

Fourier structure studies trigonometric series through orthogonality, projection, kernels, and convergence, with inner-product geometry reappearing throughout.

Definition 10.1.1. *Core Fourier Objects*

(i) A *complex trigonometric polynomial* is $a_0 + \sum_{n=1}^{N}(a_n\cos nx + b_n \sin nx)$, equivalently $\sum_{n=-N}^{N}\alpha_n e^{inx}$, where $N \in \mathbb{N}_0$ and $a_n, b_n, \alpha_n \in \mathbb{C}$.
(ii) If $f : [-\pi,\pi] \to \mathbb{C}$ is Riemann integrable, its *Fourier coefficients* are $\widehat{f}(m) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-imx}\,dx$ $(m \in \mathbb{Z})$, and $f(x) \sim_{\text{coef}} \sum_{m\in\mathbb{Z}}\widehat{f}(m)e^{imx}$ records coefficients, not convergence.
(iii) Let $a,b \in \mathbb{R}$ with $a<b$, and let $(\phi_n)_{n\ge 0} \subseteq C([a,b];\mathbb{C})$. The sequence is *orthonormal* if $(\forall m,n \in \mathbb{N}_0)$ $\int_a^b \phi_n(x)\overline{\phi_m(x)}\,dx = \delta_{n,m}$. If $f : [a,b] \to \mathbb{C}$ is Riemann integrable, its coefficients relative to (ϕ_n) are $c_n := \int_a^b f(t)\overline{\phi_n(t)}\,dt$ $(n \in \mathbb{N}_0)$.

Key Insight: Fourier coefficients are orthonormal coordinates; convergence is a separate analytic question.

Definition 10.1.2. *Period-One Fourier Normalization*
Let $C(\mathbb{R}/\mathbb{Z};\mathbb{C})$ denote the continuous functions $f : \mathbb{R} \to \mathbb{C}$ satisfying $(\forall x \in \mathbb{R})$ $f(x+1) = f(x)$.
For $f,g \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, set $\langle f,g\rangle_{\mathbb{T}} := \int_0^1 f(x)\overline{g(x)}\,dx$; $\|f\|_{L^2(\mathbb{T})} := \sqrt{\langle f,f\rangle_{\mathbb{T}}}$; and $e_n(x) := e^{2\pi inx}$ for $n \in \mathbb{Z}$.
The *period-one Fourier coefficient* is $\widehat{f}_{\mathbb{T}}(n) := \langle f, e_n\rangle_{\mathbb{T}} = \int_0^1 f(x)e^{-2\pi inx}\,dx$ $(n \in \mathbb{Z})$.
A period-one trigonometric polynomial is $\sum_{n=-N}^{N} c_n e_n$, where $N \in \mathbb{N}_0$ and $c_{-N},\dots,c_N \in \mathbb{C}$.

Perspective 10.1.3. *Two Equivalent Fourier Clocks*
The period-one and 2π-periodic conventions differ by $t = 2\pi x$; only normalizing constants change.
Key Insight: Changing Fourier clocks changes constants, not the projection geometry.

Proposition 10.1.4. *Characters Are an Orthonormal System*
For $m,n \in \mathbb{Z}$ and $N \in \mathbb{N}_0$:

(i) $\langle e_n, e_m\rangle_{\mathbb{T}} = \delta_{n,m}$.
(ii) If $P = \sum_{k=-N}^{N} c_k e_k$ *with* $(c_k)_{k=-N}^{N} \subseteq \mathbb{C}$, *then* $\langle P, e_n\rangle_{\mathbb{T}} = c_n$ *when* $-N \le n \le N$, *and* $\|P\|_{L^2(\mathbb{T})}^2 = \sum_{k=-N}^{N}|c_k|^2$.

Proof.

(i) $\langle e_n,e_m\rangle_{\mathbb{T}} = \int_0^1 e^{2\pi i(n-m)x}\,dx$. This equals 1 if $n=m$ and $\frac{e^{2\pi i(n-m)}-1}{2\pi i(n-m)} = 0$ if $n \ne m$.
(ii) By (i), $\langle P, e_n\rangle_{\mathbb{T}} = \sum_{k=-N}^{N} c_k\delta_{k,n} = c_n$, and $\|P\|_{L^2(\mathbb{T})}^2 = \langle P,P\rangle_{\mathbb{T}} = \sum_{j,k=-N}^{N} c_j\overline{c_k}\delta_{j,k} = \sum_{k=-N}^{N}|c_k|^2$.

Structural Commentary: The Gram matrix of $\{e_{-N},\dots,e_N\}$ is the identity.
Key Insight: Character orthogonality kills all cross terms. □

10.2 Fourier Approximation

Perspective 10.2.1. *Fourier and Plancherel Roadmap*
Path: orthonormal characters, Fejer density, best L^2 projection, then Parseval/Plancherel.
Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

Definition 10.2.2. *Periodic Convolution and Fejer Kernels*
For $f,g \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, define their *periodic convolution* as $(f*g)(x) := \int_0^1 f(y)g(x-y)\,dy$.
For $N \in \mathbb{N}$ with $N \ge 1$, the *Fejer kernel* is $F_N(x) := \sum_{n=-N}^{N}\left(1-\frac{|n|}{N}\right)e_n(x) = \frac{1}{N}\left|\sum_{n=0}^{N-1} e_n(x)\right|^2$.

Figure 188 schematically compares two concentration profiles with $N > M$: the larger-index kernel is narrower around 0, and the shaded window marks a central interval of width 2δ. The curves are qualitative rather than exact plots of F_M and F_N.

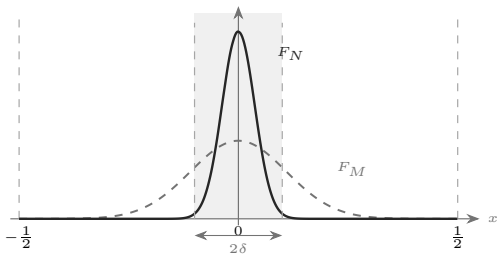

Figure 188: Qualitative concentration of period-one Fejer kernels near 0: for $N > M$, the F_N profile is narrower than F_M, and the central window has width 2δ.

Theorem 10.2.3. *Fejer Approximation on $\mathbb{R}/\mathbb{Z}$*
Let $(F_N)_{N\geq 1}$ be the period-one Fejer kernels. Then:

(i) $(\forall N \in \mathbb{N})$ *F_N is a nonnegative trigonometric polynomial and* $\int_0^1 F_N(x)\,dx = 1$.
(ii) If $0 < \delta < 1/2$, *then* $\sup_{\delta\leq x\leq 1-\delta} F_N(x) \to 0$ *as* $N \to \infty$.
(iii) If $f \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, *then* $(\forall N \in \mathbb{N})$ *$f * F_N$ is a trigonometric polynomial.*
(iv) If $f \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, *then* $f * F_N \rightrightarrows f$ *on* $\mathbb{R}/\mathbb{Z}$ *as* $N \to \infty$.

Proof.

(i) By definition, $F_N = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e_n$ is trigonometric, and $F_N = \frac{1}{N}\left|\sum_{n=0}^{N-1} e_n\right|^2 \geq 0$.
Since $\int_0^1 e_n(x)\,dx = 0$ for $n \neq 0$, $\int_0^1 F_N(x)\,dx = 1$ by the e_0 coefficient.

(ii) For $x \notin \mathbb{Z}$, $F_N(x) = \frac{\sin^2(\pi N x)}{N\sin^2(\pi x)}$. If $m_\delta := \min_{\delta\leq x\leq 1-\delta} \sin^2(\pi x) > 0$, then $0 \leq F_N(x) \leq \frac{1}{N m_\delta}$ on $[\delta, 1-\delta]$.

(iii) Since $f * e_n = \widehat{f_{\mathbb{T}}}(n) e_n$ and F_N is a finite linear combination of the e_n, $f * F_N$ is a trigonometric polynomial.

(iv) Write $(f * F_N)(x) = \int_0^1 f(x-t)F_N(t)\,dt$. Let $M = \|f\|_\infty$ and $\varepsilon > 0$.
By uniform continuity, choose $0 < \delta < 1/2$ such that $|f(x-t) - f(x)| < \varepsilon$ for $x \in \mathbb{R}/\mathbb{Z}$ and $t \in [0,\delta] \cup [1-\delta, 1]$.
By (i), $F_N \geq 0$ and $\int_0^1 F_N(t)\,dt = 1$; hence $|(f * F_N)(x) - f(x)| \leq \varepsilon + 2M \sup_{\delta\leq t\leq 1-\delta} F_N(t)$.
By (ii), $\limsup_{N\to\infty} \|f * F_N - f\|_\infty \leq \varepsilon$. Since ε is arbitrary, $f * F_N \rightrightarrows f$.

Structural Commentary: Fejer kernels are positive approximate identities.
Key Insight: Normalized mass concentrating at 0 forces uniform convergence. □

Corollary 10.2.4. *Uniform Approximation by Trigonometric Polynomials*
Let $f \in C(\mathbb{R};\mathbb{C})$ be 2π-periodic. Then $(\forall \varepsilon > 0)(\exists N \in \mathbb{N}_0)(\exists (c_n)_{n=-N}^{N} \subseteq \mathbb{C})$ $\|f - P\|_\infty < \varepsilon$, *where* $P(x) := \sum_{n=-N}^{N} c_n e^{inx}$.

Proof. Let $\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}$ and define $F : \mathbb{T} \to \mathbb{C}$ by $F(e^{ix}) := f(x)$.
Periodicity makes F well-defined and continuous.
The algebra $\mathcal{T} := \operatorname{span}\{z^n : n \in \mathbb{Z}\} \subset C(\mathbb{T};\mathbb{C})$ is self-adjoint, separates points, and vanishes nowhere.
By Theorem 9.2.9, choose $Q(z) = \sum_{n=-N}^{N} c_n z^n \in \mathcal{T}$ with $\|F - Q\|_{C(\mathbb{T})} < \varepsilon$.
Set $P(x) := Q(e^{ix}) = \sum_{n=-N}^{N} c_n e^{inx}$; then $\|f - P\|_\infty = \|F - Q\|_{C(\mathbb{T})} < \varepsilon$.
Structural Commentary: Trigonometric polynomials are Laurent polynomials pulled back from $\mathbb{T}$.
Key Insight: Uniform periodic approximation follows from Stone–Weierstrass on $\mathbb{T}$. □

Perspective 10.2.5. *Fourier Coefficients as Projections*
Fourier coefficients are projections onto orthonormal directions, modeled finitely by Theorems 2.1.5, 1.4.1, and 1.4.5; infinite directions make convergence separate.

Theorem 10.2.6. *Best Mean-Square Approximation by Fourier Coefficients*
Let $a, b \in \mathbb{R}$ with $a < b$, let $f : [a,b] \to \mathbb{C}$ and $\phi_m : [a,b] \to \mathbb{C}$ $(m \in \mathbb{N})$ be Riemann integrable, and set $\langle u, v\rangle := \int_a^b u(x)\overline{v(x)}\,dx$ and $\|u\|_2^2 := \langle u, u\rangle$. Assume $(\forall j,k \in \mathbb{N})$ $\langle \phi_j, \phi_k\rangle = \delta_{j,k}$.
For $n \in \mathbb{N}$, set $c_m := \langle f, \phi_m\rangle$, $s_n := \sum_{m=1}^{n} c_m \phi_m$, and $t_n := \sum_{m=1}^{n} \gamma_m \phi_m$, where $\gamma_1, \ldots, \gamma_n \in \mathbb{C}$. Then:

(i) $\|f - s_n\|_2 \leq \|f - t_n\|_2$, *with equality* $\iff (\forall m \in \{1,\ldots,n\})$ $\gamma_m = c_m$.
(ii) $\sum_{m=1}^{\infty} |c_m|^2 \leq \|f\|_2^2$, *hence* $c_m \to 0$.

Proof.

(i) $(\Rightarrow)$ By orthonormality, $\|f - t_n\|_2^2 = \|f\|_2^2 - 2\operatorname{Re}\sum_{m=1}^{n} c_m\overline{\gamma_m} + \sum_{m=1}^{n} |\gamma_m|^2 = \|f\|_2^2 - \sum_{m=1}^{n} |c_m|^2 + \sum_{m=1}^{n} |\gamma_m - c_m|^2$. Thus $\|f - t_n\|_2^2 \geq \|f - s_n\|_2^2$; equality implies $\gamma_m = c_m$ for $1 \leq m \leq n$.
(i) $(\Leftarrow)$ If $\gamma_m = c_m$ for $1 \leq m \leq n$, then $t_n = s_n$, so equality holds.
(ii) By the identity in (i) with $\gamma_m = c_m$, $\|f - s_n\|_2^2 = \|f\|_2^2 - \sum_{m=1}^{n} |c_m|^2 \geq 0$.
Hence $\sum_{m=1}^{n} |c_m|^2 = \|s_n\|_2^2 \leq \|f\|_2^2$ $(n \in \mathbb{N})$. Letting $n \to \infty$, $\sum_{m=1}^{\infty} |c_m|^2 \leq \|f\|_2^2$.
Since $(|c_m|^2)$ is summable, $|c_m|^2 \to 0$, hence $c_m \to 0$.

Structural Commentary: Orthogonality diagonalizes the finite error.
Key Insight: Fourier coefficients are projection coordinates. □

Theorem 10.2.7. *Dirichlet Kernel and Local Fourier Convergence*
Let $f : \mathbb{R} \to \mathbb{C}$ be Riemann integrable on compact intervals and 2π-periodic, with $\widehat{f}(m) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-imt}\,dt$ $(m \in \mathbb{Z})$. For $N \in \mathbb{N}_0$, set $s_N(f;x) := \sum_{m=-N}^{N} \widehat{f}(m)e^{imx}$ and $D_N(t) := \sum_{m=-N}^{N} e^{imt}$. Then:

(i) $D_N(t) = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}$ for $t \notin 2\pi\mathbb{Z}$, with $D_N(0) = 2N+1$, and $s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)\,dt$.
(ii) If $x \in \mathbb{R}$ and $(\exists M, \delta > 0)(\forall t \in \mathbb{R})\ |t| < \delta \Rightarrow |f(x+t) - f(x)| \le M|t|$, then $s_N(f;x) \to f(x)$ as $N \to \infty$.
(iii) If $g : \mathbb{R} \to \mathbb{C}$ is Riemann integrable on compact intervals and 2π-periodic, and $(\exists \eta > 0)(\forall t \in \mathbb{R})\ |t - x| < \eta \Rightarrow f(t) = g(t)$, then $s_N(f;x) - s_N(g;x) \to 0$ as $N \to \infty$.

Proof.

(i) For $z = e^{it} \neq 1$, $D_N(t) = z^{-N}\frac{1-z^{2N+1}}{1-z} = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}$, and $D_N(0) = \sum_{m=-N}^{N} 1 = 2N+1$.
After substituting $\widehat{f}(m)$ and interchanging the finite sum with the integral, $s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(u)\sum_{m=-N}^{N} e^{im(x-u)}\,du = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)\,dt$, where periodicity permits shifting the transformed interval back to $[-\pi,\pi]$.
(ii) By (i) and $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N(t)\,dt = 1$, $s_N(f;x) - f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi}(f(x-t) - f(x))D_N(t)\,dt$.
Set $q_x(t) := \frac{f(x-t)-f(x)}{\sin(t/2)}$ for $0 < |t| \le \pi$ and $q_x(0) := 0$. The hypothesis makes q_x bounded near 0, while q_x is Riemann integrable on each closed subinterval not containing 0. A partition with an arbitrarily short central interval verifies the upper–lower criterion for both components, so $q_x \in \mathcal{R}[-\pi,\pi]$. Hence $s_N(f;x) - f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} q_x(t)\sin((N+\frac{1}{2})t)\,dt$.
Since $\sin((N+\frac{1}{2})t) = \sin Nt\cos(t/2) + \cos Nt\sin(t/2)$, by Theorem 10.2.6(ii), applied to $q_x(t)\cos(t/2)$ against $(\pi^{-1/2}\sin Nt)_{N\ge1}$ and to $q_x(t)\sin(t/2)$ against $(\pi^{-1/2}\cos Nt)_{N\ge1}$, both integrals tend to 0.
(iii) For $h := f - g$, $h = 0$ near x, so $s_N(h;x) \to h(x) = 0$ by (ii).
By linearity, $s_N(h;x) = s_N(f;x) - s_N(g;x)$.

Structural Commentary: Dirichlet kernels express partial sums as oscillatory local averages.
Key Insight: Local regularity makes the oscillatory error vanish. □

Theorem 10.2.8. *Parseval's Theorem for Trigonometric Fourier Series*
Let $f, g : \mathbb{R} \to \mathbb{C}$ be Riemann integrable on compact intervals and 2π-periodic.
Set $\langle u, v\rangle_2 := \frac{1}{2\pi}\int_{-\pi}^{\pi} u(x)\overline{v(x)}\,dx$ and $\|u\|_2^2 := \langle u, u\rangle_2$. If $u : \mathbb{R} \to \mathbb{C}$ is Riemann integrable on compact intervals and 2π-periodic, set $\widehat{u}(n) := \frac{1}{2\pi}\int_{-\pi}^{\pi} u(x)e^{-inx}\,dx$ $(n \in \mathbb{Z})$ and $s_N(u) := \sum_{|n|\le N} \widehat{u}(n)e^{inx}$ $(N \in \mathbb{N}_0)$. Then:

(i) $\|f - s_N(f)\|_2 \to 0$ as $N \to \infty$.
(ii) $\langle f, g\rangle_2 = \sum_{n\in\mathbb{Z}} \widehat{f}(n)\overline{\widehat{g}(n)}$, and the series converges absolutely.
(iii) $\|f\|_2^2 = \sum_{n\in\mathbb{Z}} \left|\widehat{f}(n)\right|^2$.

Proof.

(i) Put $d_N(u) := \|u - s_N(u)\|_2$. If u is continuous and $\varepsilon > 0$, then by Corollary 10.2.4, choose $P(x) = \sum_{|n|\le K} a_n e^{inx}$ with $\|u - P\|_2 \le \|u - P\|_\infty < \varepsilon$.
For $N \ge K$, by Theorem 10.2.6(i), applied to the normalized exponentials $\{(2\pi)^{-1/2}e^{inx} : |n| \le N\}$, $d_N(u) \le \|u - P\|_2 < \varepsilon$; hence $d_N(u) \to 0$.
For Riemann-integrable f and $\varepsilon > 0$, by Theorem 2.1.6 with $\alpha(x) = x$, choose partitions for the real and imaginary parts whose oscillation gaps are sufficiently small. Selecting one value on each interval defines a periodic step function v with $\|f - v\|_2 < \varepsilon/2$. Choose transition intervals around the finitely many jumps of v, including the endpoint seam, with sufficiently small total length. Linear interpolation across them constructs a continuous 2π-periodic function h with $\|v - h\|_2 < \varepsilon/2$, so $\|f - h\|_2 < \varepsilon$.
By Bessel's inequality, $\|s_N(h - f)\|_2 \le \|h - f\|_2$, so $d_N(f) \le 2\|f - h\|_2 + d_N(h)$. Hence $\limsup_{N\to\infty} d_N(f) \le 2\varepsilon$, and letting $\varepsilon \downarrow 0$ proves the claim.
(ii) By orthogonality, $\langle e^{inx}, g\rangle_2 = \overline{\widehat{g}(n)}$. Hence $\langle s_N(f), g\rangle_2 = \sum_{|n|\le N} \widehat{f}(n)\langle e^{inx}, g\rangle_2 = \sum_{|n|\le N} \widehat{f}(n)\overline{\widehat{g}(n)}$.
By (i) and Cauchy–Schwarz, $\langle s_N(f), g\rangle_2 \to \langle f, g\rangle_2$. By Theorem 10.2.6(ii), applied after enumerating $((2\pi)^{-1/2}e^{inx})_{n\in\mathbb{Z}}$, $(\widehat{f}(n))_{n\in\mathbb{Z}}$ and $(\widehat{g}(n))_{n\in\mathbb{Z}}$ are square summable.
By Cauchy–Schwarz for series, $\sum_{n\in\mathbb{Z}} \left|\widehat{f}(n)\overline{\widehat{g}(n)}\right| < \infty$, so the symmetric partial sums converge to the asserted absolutely convergent series.
(iii) Set $g = f$ in (ii). Then $\widehat{g}(n) = \widehat{f}(n)$ and $\langle f, f\rangle_2 = \|f\|_2^2$, so $\|f\|_2^2 = \sum_{n\in\mathbb{Z}} \left|\widehat{f}(n)\right|^2$.

Structural Commentary: Density turns finite projection identities into L^2 identities.
Key Insight: Partial Fourier projections converge in L^2. □

Method 10.2.9. *Fourier Proof Patterns*
(i) *Characters*: use orthogonality of e_n to compute coefficients and finite-dimensional projections.
(ii) *Kernels*: rewrite partial sums or averages as convolutions with Dirichlet or Fejer kernels.
(iii) *Approximation*: use Fejer positivity or Stone–Weierstrass to approximate continuous periodic functions by trigonometric polynomials.
(iv) *L^2 control*: project onto finite orthonormal spans, then pass to limits by density.
(v) *Localization*: express the error as an oscillatory integral and use local regularity near the point.

Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

10.3 Further Topics in Fourier Analysis

Example 10.3.1. *A Gaussian Fourier Integral*

For $t \in \mathbb{R}$, set $f(t) := \int_{-\infty}^{\infty} e^{-x^2}\cos(xt)\,dx$ and $g(t) := -\int_{-\infty}^{\infty} xe^{-x^2}\sin(xt)\,dx$. Both integrals converge absolutely by Theorem 9.1.22(iv) and $2\int_0^{\infty} xe^{-x^2}\,dx = 1$.

For $R > 0$, set $f_R, g_R : \mathbb{R} \to \mathbb{R}$ by $f_R(t) := \int_{-R}^{R} e^{-x^2}\cos(xt)\,dx$ and $g_R(t) := -\int_{-R}^{R} xe^{-x^2}\sin(xt)\,dx$. By Theorem 7.2.3 with $\alpha(x) = x$, $f_R' = g_R$.

The tail estimates $\sup_{t\in\mathbb{R}} |f_R(t) - f(t)| \le 2\int_R^{\infty} e^{-x^2}\,dx \to 0$ and $\sup_{t\in\mathbb{R}} |g_R(t) - g(t)| \le 2\int_R^{\infty} xe^{-x^2}\,dx = e^{-R^2} \to 0$ hold as $R \to \infty$.

Fix $T > 0$ and take $R = N \in \mathbb{N}$. By Theorem 7.2.4 on $[-T, T]$, $f' = g$ there. Since T is arbitrary, $f' = g$ on $\mathbb{R}$.

For $R > 0$ and $t \in \mathbb{R}$, by Theorem 2.2.3 on $[-R, R]$, $2g_R(t) + tf_R(t) = 2e^{-R^2}\sin(Rt)$. Taking $R \to \infty$, $2g(t) + tf(t) = 0$, hence $2f'(t) + tf(t) = 0$.

By Theorem 9.1.22(iv), $f(0) = \sqrt{\pi}$. The differential equation implies $\frac{d}{dt}\left(e^{t^2/4}f(t)\right) = 0$, so $f(t) = \sqrt{\pi}\exp\left(-\frac{t^2}{4}\right)$ $(t \in \mathbb{R})$.

Part XIII
Measure Foundations and Extension

1 Sigma-Algebras and Measurable Spaces

1.1 Sigma-Algebras and Measurable Structures

Measure theory first builds measurable sets, then measurable functions and integrals. Outer measure covers from outside; measurability splits test sets without losing measure.

Perspective 1.1.1. *Why Abstract Measure Comes First*
Lebesgue integration is built so that, under hypotheses such as domination, $f_n \to f$ implies $\int f_n\,d\mu \to \int f\,d\mu$. The abstract data separate into a measurable space $(X, \mathcal{M})$, a countably additive measure $\mu : \mathcal{M} \to [0, +\infty]$, and an integral defined by approximation from simple functions.
Key Insight: Countable additivity is the set-level structure behind the convergence theorems.

Definition 1.1.2. *σ-Algebras and Measurable Spaces*
Let X be a set. A family $\mathcal{M} \subseteq \mathcal{P}(X)$ is a *σ-algebra* on X if:

(i) $\emptyset, X \in \mathcal{M}$;
(ii) $E \in \mathcal{M}$ implies $X \setminus E \in \mathcal{M}$;
(iii) $E_n \in \mathcal{M}$ for $n \in \mathbb{N}$ implies $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is a *measurable space*; the members of $\mathcal{M}$ are *measurable sets*.

Proposition 1.1.3. *Closure and Generation for σ-Algebras*
Let $\mathcal{M} \subseteq \mathcal{P}(X)$ be a σ-algebra.

(i) $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M} \Rightarrow \bigcap_{n=1}^{\infty} E_n \in \mathcal{M}$.
(ii) If $(\mathcal{M}_\alpha)_{\alpha\in A}$ is a nonempty family of σ-algebras on X, then $\bigcap_{\alpha\in A} \mathcal{M}_\alpha$ is a σ-algebra on X.
(iii) For $\mathcal{A} \subseteq \mathcal{P}(X)$, let $\mathscr{S}_{\mathcal{A}}$ be the family of σ-algebras $\mathcal{N}$ on X with $\mathcal{A} \subseteq \mathcal{N}$. Then $\sigma(\mathcal{A}) := \bigcap_{\mathcal{N}\in\mathscr{S}_{\mathcal{A}}} \mathcal{N}$ is the smallest σ-algebra containing $\mathcal{A}$.

Proof.

(i) By Theorem 1.5.2, $\bigcap_n E_n = X \setminus \bigcup_n (X \setminus E_n) \in \mathcal{M}$ since $(X \setminus E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M}$.
(ii) Set $\mathcal{N} := \bigcap_{\alpha\in A} \mathcal{M}_\alpha$. Then $\emptyset, X \in \mathcal{N}$, and $E \in \mathcal{N}$ implies $X \setminus E \in \mathcal{N}$. Likewise $(E_n) \subseteq \mathcal{N}$ implies $\bigcup_n E_n \in \mathcal{N}$.
(iii) The family $\mathscr{S}_{\mathcal{A}}$ is nonempty because $\mathcal{P}(X) \in \mathscr{S}_{\mathcal{A}}$. By (ii), its intersection is a σ-algebra containing $\mathcal{A}$, and $(\forall \mathcal{N} \in \mathscr{S}_{\mathcal{A}})\ \sigma(\mathcal{A}) \subseteq \mathcal{N}$.

Structural Commentary: Generation is intersection over all admissible σ-algebras.
Key Insight: Intersections preserve universal closure properties. □

Example 1.1.4. *Basic σ-Algebras*
Standard models:

(i) Extremes: If $\mathcal{M}$ is a σ-algebra on X, then $\{\emptyset, X\} \subseteq \mathcal{M} \subseteq \mathcal{P}(X)$.
(ii) For $E \subseteq X$, $\sigma(\{E\}) = \{\emptyset, E, X \setminus E, X\}$, with repetitions omitted.
(iii) On $\mathbb{N}$, if $O := \{2n-1 : n \in \mathbb{N}\}$, then $\sigma(\{O\}) = \{\emptyset, O, \mathbb{N} \setminus O, \mathbb{N}\}$.
(iv) If X is uncountable, $\mathcal{C}_X := \{E \subseteq X : E \text{ is countable or } X \setminus E \text{ is countable}\}$ is the countable–cocountable σ-algebra.

Key Insight: A σ-algebra records exactly which distinctions the theory is allowed to observe.

Definition 1.1.5. *Borel and Extended Borel σ-Algebras*

(i) For $n \in \mathbb{N}$, the *Borel σ-algebra* on $\mathbb{R}^n$ is $\mathcal{B}(\mathbb{R}^n) := \sigma(\{U \subseteq \mathbb{R}^n : U \text{ is open}\})$.
(ii) The *extended Borel σ-algebra* on $\overline{\mathbb{R}}$ is $\overline{\mathcal{B}} := \{E \cup F : E \in \mathcal{B}(\mathbb{R}),\ F \subseteq \{-\infty, +\infty\}\}$.

Proposition 1.1.6. *Generators of the Borel σ-Algebra on $\mathbb{R}$*
Set $\mathcal{I}_{\mathrm{o}} := \{(a,b) : a,b \in \mathbb{R},\ a < b\}$, $\mathcal{I}_{\mathrm{c}} := \{[a,b] : a,b \in \mathbb{R},\ a < b\}$, $\mathcal{I}_{\mathrm{h}} := \{(a,b] : a,b \in \mathbb{R},\ a < b\}$, and $\mathcal{H} := \{(a,+\infty) : a \in \mathbb{R}\}$. Then:

(i) $\sigma(\mathcal{I}_{\mathrm{o}}) = \mathcal{B}(\mathbb{R})$;
(ii) $\sigma(\mathcal{I}_{\mathrm{c}}) = \mathcal{B}(\mathbb{R})$;
(iii) $\sigma(\mathcal{I}_{\mathrm{h}}) = \mathcal{B}(\mathbb{R})$;
(iv) $\sigma(\mathcal{H}) = \mathcal{B}(\mathbb{R})$.

Proof.

(i) Since $\mathcal{I}_{\mathrm{o}} \subseteq \mathcal{B}(\mathbb{R})$, $\sigma(\mathcal{I}_{\mathrm{o}}) \subseteq \mathcal{B}(\mathbb{R})$. If $U \subseteq \mathbb{R}$ is open, then U is the union of the intervals $(p,q) \subseteq U$ with $p, q \in \mathbb{Q}$. This family is countable by Corollaries 6.2.19 and 6.2.17, and it covers U by Theorem 1.1.19(ii). Hence $U \in \sigma(\mathcal{I}_{\mathrm{o}})$.
(ii) Closed intervals are Borel. Given $a < b$, choose $N \in \mathbb{N}$ with $2/N < b - a$; then $(a,b) = \bigcup_{n=N}^{\infty} [a + 1/n, b - 1/n]$. Thus $\mathcal{I}_{\mathrm{o}} \subseteq \sigma(\mathcal{I}_{\mathrm{c}}) \subseteq \mathcal{B}(\mathbb{R})$, and (i) proves the claim.
(iii) Half-open intervals are Borel. Given $a < b$, choose $N \in \mathbb{N}$ with $1/N < b-a$; then $(a,b) = \bigcup_{n=N}^{\infty} (a, b-1/n]$. Thus $\mathcal{I}_{\mathrm{o}} \subseteq \sigma(\mathcal{I}_{\mathrm{h}}) \subseteq \mathcal{B}(\mathbb{R})$, and (i) proves the claim.
(iv) Open half-rays are Borel, and $[b, +\infty) = \bigcap_{n=1}^{\infty} (b - 1/n, +\infty)$. Hence $(a,b) = (a,+\infty) \setminus [b,+\infty) \in \sigma(\mathcal{H})$, so $\mathcal{I}_{\mathrm{o}} \subseteq \sigma(\mathcal{H}) \subseteq \mathcal{B}(\mathbb{R})$. Together with (i), these inclusions prove the claim.

Structural Commentary: Countable rational intervals generate open sets; elementary limits transfer generation to the other interval families.
Key Insight: To compare generators, express one family through countable set operations on another. □

Definition 1.1.7. *Monotone Classes*
Let X be a set. A nonempty $\mathcal{C} \subseteq \mathcal{P}(X)$ is a *monotone class* iff $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{C},\ E_n \uparrow E \Rightarrow E \in \mathcal{C}$ and $(F_n)_{n\in\mathbb{N}} \subseteq \mathcal{C},\ F_n \downarrow F \Rightarrow F \in \mathcal{C}$.
For $\mathcal{A} \subseteq \mathcal{P}(X)$, set $\mathfrak{M}_{\mathcal{A}} := \{\mathcal{C} \subseteq \mathcal{P}(X) : \mathcal{C} \text{ is a monotone class, } \mathcal{A} \subseteq \mathcal{C}\}$ and $\operatorname{mon}(\mathcal{A}) := \bigcap \mathfrak{M}_{\mathcal{A}}$.

Proposition 1.1.8. *Minimality of Monotone Closure*
Let X be a set and $\mathcal{A} \subseteq \mathcal{P}(X)$. Then:

(i) $\operatorname{mon}(\mathcal{A})$ *is the least monotone class containing* $\mathcal{A}$*;*
(ii) $\operatorname{mon}(\mathcal{A}) \subseteq \sigma(\mathcal{A})$.

Proof.

(i) Since $\mathcal{P}(X) \in \mathfrak{M}_{\mathcal{A}}$, this indexing family is nonempty, and its intersection contains $\mathcal{A}$. If $(E_n)_{n\in\mathbb{N}} \subseteq \bigcap \mathfrak{M}_{\mathcal{A}}$ and $E_n \uparrow E$, then $(\forall \mathcal{C} \in \mathfrak{M}_{\mathcal{A}})\ E \in \mathcal{C}$, so $E \in \bigcap \mathfrak{M}_{\mathcal{A}}$; the decreasing case is identical. Thus the intersection is monotone. If $\mathcal{C}$ is monotone and $\mathcal{A} \subseteq \mathcal{C}$, then $\mathcal{C} \in \mathfrak{M}_{\mathcal{A}}$, so $\operatorname{mon}(\mathcal{A}) \subseteq \mathcal{C}$.
(ii) The σ-algebra $\sigma(\mathcal{A})$ is a monotone class containing $\mathcal{A}$, so (i) implies $\operatorname{mon}(\mathcal{A}) \subseteq \sigma(\mathcal{A})$.

Structural Commentary: Intersection preserves monotone limits and lies inside every admissible class.
Key Insight: Monotone closure captures limits, not finite unions. □

Definition 1.1.9. *Rings and σ-Rings of Sets*
Let X be a set and $\mathcal{A} \subseteq \mathcal{P}(X)$.

(i) $\mathcal{A}$ is a *ring of sets* iff $\emptyset \in \mathcal{A}$ and $A, B \in \mathcal{A} \Rightarrow A \cup B,\ A \setminus B \in \mathcal{A}$.
(ii) $\mathcal{A}$ is a *σ-ring* iff it is a ring and $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

Key Insight: Rings support finite operations; σ-rings support countable unions.

Definition 1.1.10. *Algebras of Sets and Premeasures*
Let X be a set.

(i) An *algebra of sets* (or *field of sets*) on X is $\mathcal{A} \subseteq \mathcal{P}(X)$ such that $\emptyset, X \in \mathcal{A}$; $E \in \mathcal{A} \Rightarrow X \setminus E \in \mathcal{A}$; and $(\forall n \in \mathbb{N})\ E_1, \ldots, E_n \in \mathcal{A} \Rightarrow \bigcup_{j=1}^{n} E_j \in \mathcal{A}$.
Equivalently, $\mathcal{A}$ is a ring of sets containing X.
(ii) A *premeasure* (or *measure on an algebra*) on $\mathcal{A}$ is $\mu : \mathcal{A} \to [0, +\infty]$ such that $\mu(\emptyset) = 0$ and, whenever $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ is pairwise disjoint with $\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$, then $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n)$.

Key Insight: Premeasures are countably additive where the algebra permits unions.

Theorem 1.1.11. *Monotone Class Theorem for Algebras*
Let X be a set, let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra, and set $\mathcal{C} := \operatorname{mon}(\mathcal{A})$. Then $\mathcal{C} = \sigma(\mathcal{A})$.

Proof.

(i) *Setup*: By Proposition 1.1.8, $\mathcal{A} \subseteq \mathcal{C} \subseteq \sigma(\mathcal{A})$. It remains to prove that $\mathcal{C}$ is a σ-algebra.
(ii) *Complements*: Set $\mathcal{D} := \{E \in \mathcal{C} : E^c \in \mathcal{C}\}$. If $E_n \uparrow E$ in $\mathcal{D}$, then $E_n^c \downarrow E^c$; if $E_n \downarrow E$, then $E_n^c \uparrow E^c$. Thus $\mathcal{D}$ is monotone. By algebra complement closure, $\mathcal{A} \subseteq \mathcal{D}$; hence $\mathcal{C} \subseteq \mathcal{D}$ by Proposition 1.1.8(i), so $\mathcal{C}$ is closed under complements.
(iii) *Finite Intersections*: For $B \in \mathcal{C}$, set $\mathcal{D}_B := \{E \in \mathcal{C} : E \cap B \in \mathcal{C}\}$. The identities $E_n \cap B \uparrow E \cap B$ when $E_n \uparrow E$ and $E_n \cap B \downarrow E \cap B$ when $E_n \downarrow E$ show that $\mathcal{D}_B$ is monotone. If $B \in \mathcal{A}$, then $\mathcal{A} \subseteq \mathcal{D}_B$ by algebra closure, hence $\mathcal{C} \subseteq \mathcal{D}_B$ by Proposition 1.1.8(i). Therefore $E \cap B \in \mathcal{C}$ for $E \in \mathcal{C}$ and $B \in \mathcal{A}$. Fixing $E \in \mathcal{C}$, the preceding conclusion implies $\mathcal{A} \subseteq \mathcal{D}_E$, hence $\mathcal{C} \subseteq \mathcal{D}_E$ by Proposition 1.1.8(i). Thus $\mathcal{C}$ is closed under finite intersections and under finite unions by complement closure.
(iv) *Countable Unions*: If $(E_n) \subseteq \mathcal{C}$, then $F_N := \bigcup_{n=1}^{N} E_n \in \mathcal{C}$ and $F_N \uparrow \bigcup_{n=1}^{\infty} E_n$, so $\bigcup_{n=1}^{\infty} E_n \in \mathcal{C}$.
(v) *Conclusion*: Hence $\mathcal{C}$ is a σ-algebra containing $\mathcal{A}$, so $\sigma(\mathcal{A}) \subseteq \mathcal{C}$ by Proposition 1.1.3(iii).

Structural Commentary: Monotone lifting establishes Boolean closure, then countable unions.
Key Insight: Show each closure property holds on a monotone class containing $\mathcal{A}$. □

Corollary 1.1.12. *Monotone-Class Test*
Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra and let $\mathcal{C} \subseteq \mathcal{P}(X)$ be a monotone class. If $\mathcal{A} \subseteq \mathcal{C}$, then $\sigma(\mathcal{A}) \subseteq \mathcal{C}$.

Proof. By Proposition 1.1.8(i), $\operatorname{mon}(\mathcal{A}) \subseteq \mathcal{C}$. By Theorem 1.1.11, $\operatorname{mon}(\mathcal{A}) = \sigma(\mathcal{A})$.
Structural Commentary: Algebra checks plus monotone stability control $\sigma(\mathcal{A})$.
Key Insight: Prove on the algebra; extend by monotone closure. □

2 Measures, Charges, and Basic Examples

2.1 Measures, Charges, and Basic Constructions

Definition 2.1.1. *Set Functions and Additivity*
Let X be a set and $\mathcal{A} \subseteq \mathcal{P}(X)$ with $\emptyset \in \mathcal{A}$.

(i) A *set function* on $\mathcal{A}$ is $\phi : \mathcal{A} \to [-\infty, +\infty]$ with $\{+\infty, -\infty\} \nsubseteq \phi(\mathcal{A})$.
(ii) ϕ is *additive* iff $\phi(\emptyset) = 0$ and, whenever $A, B \in \mathcal{A}$ are disjoint with $A \cup B \in \mathcal{A}$, $\phi(A \cup B) = \phi(A) + \phi(B)$.
(iii) ϕ is *countably additive* iff $\phi(\emptyset) = 0$ and, whenever $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ is pairwise disjoint with $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$, the series $\sum_{n=1}^{\infty} \phi(A_n)$ converges in $[-\infty, +\infty]$ and equals $\phi\left(\bigcup_{n=1}^{\infty} A_n\right)$.

Definition 2.1.2. *Measures on Measurable Spaces*
If $(a_n)_{n\in\mathbb{N}} \subseteq [0, +\infty]$, set $\sum_{n=1}^{\infty} a_n := \lim_{N\to\infty} \sum_{n=1}^{N} a_n \in [0, +\infty]$, and use $0 \cdot (+\infty) := 0$.
Let $(X, \mathcal{M})$ be a measurable space. A *measure* on $(X, \mathcal{M})$ is a map $\mu : \mathcal{M} \to [0, +\infty]$ such that:

(i) $\mu(\emptyset) = 0$;
(ii) if $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M}$ is pairwise disjoint, then $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n)$.

The triple $(X, \mathcal{M}, \mu)$ is a *measure space*.
Key Insight: Measure is countably additive size on the chosen observable sets.

Definition 2.1.3. *Finite and σ-Finite Measures*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) μ is *finite* iff $\mu(X) < +\infty$.
(ii) μ is *σ-finite* iff $(\exists (E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M})\ X = \bigcup_{n=1}^{\infty} E_n$ and $(\forall n \in \mathbb{N})\ \mu(E_n) < +\infty$.
(iii) A measurable set $E \in \mathcal{M}$ has *finite measure* iff $\mu(E) < +\infty$.
(iv) A measurable set $E \in \mathcal{M}$ is *σ-finite* iff $(\exists (E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M})\ E = \bigcup_{n=1}^{\infty} E_n$ and $(\forall n \in \mathbb{N})\ \mu(E_n) < +\infty$.

Key Insight: σ-finiteness replaces one finite global bound by a countable finite-measure exhaustion.

Example 2.1.4. *First Abstract Measures*
Let $(X, \mathcal{M})$ be a measurable space.

(i) The zero set function $\mu(E) = 0$ on $\mathcal{M}$ is a finite measure.
(ii) If $X \neq \emptyset$ and $\mathcal{M} = 2^X$, the assignments $\mu(\emptyset) = 0$ and $\mu(E) = +\infty$ for $E \neq \emptyset$ define a measure, neither finite nor σ-finite.
(iii) For $p \in X$, the *unit mass* $\delta_p(E) := \mathbf{1}_E(p)$ on 2^X is a finite measure.
(iv) On $2^{\mathbb{N}}$, *counting measure* is $\#(E)$ for finite E and $+\infty$ otherwise; it is σ-finite.
More generally, $(a_n)_{n\in\mathbb{N}} \subseteq [0, +\infty]$ defines a measure by $\mu(E) := \sum_{n=1}^{\infty} a_n \mathbf{1}_E(n)$. Conversely, if ρ is a measure on $2^{\mathbb{N}}$ and $a_n := \rho(\{n\})$, then $\rho(E) = \sum_{n=1}^{\infty} a_n \mathbf{1}_E(n)$, so the representation is unique.
(v) If X is uncountable and $\mathcal{M} = 2^X$, $\mu(E) = 0$ for countable E and $\mu(E) = +\infty$ otherwise defines a measure.
(vi) If X is uncountable and $\mathcal{C}_X = \{E \subseteq X : E \text{ is countable or } X \setminus E \text{ is countable}\}$, then $\mu(E) = 0$ on countable sets and $\mu(E) = 1$ on cocountable sets defines a measure on $\mathcal{C}_X$.
(vii) On $2^{\mathbb{N}}$, $\mu(E) = 0$ for finite E and $\mu(E) = +\infty$ otherwise is not a measure, since $\mathbb{N} = \bigcup_{n=1}^{\infty} \{n\}$ has left side $+\infty$ and right side 0 in the countable-additivity identity; the constant $+\infty$ rule fails at $\emptyset$.

Structural Commentary: Measures may be zero, atomic, counting-type, or cardinality-based.
Key Insight: Countable additivity turns disjoint decompositions into nonnegative series.

Proposition 2.1.5. *Elementary Measure Constructions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) If μ is a measure on $\mathcal{M}$ and $A \in \mathcal{M}$, then $\mu_A(E) := \mu(A \cap E)$ defines a measure μ_A on $\mathcal{M}$.
(ii) If $m \in \mathbb{N}$, $\mu_1, \dots, \mu_m$ are measures on $\mathcal{M}$, and $a_1, \dots, a_m \in [0, +\infty)$, then $\lambda := \sum_{j=1}^{m} a_j \mu_j$ is a measure on $\mathcal{M}$.
(iii) If $(\mu_n)_{n\in\mathbb{N}}$ are measures on $\mathcal{M}$ with $(\forall n \in \mathbb{N})\ \mu_n(X) = 1$, then $\lambda := \sum_{n=1}^{\infty} 2^{-n} \mu_n$ is a measure on $\mathcal{M}$ and $\lambda(X) = 1$.

Proof.

(i) $\mu_A(\emptyset) = 0$. If $(E_k)_{k\in\mathbb{N}} \subseteq \mathcal{M}$ is pairwise disjoint, then $A \cap \bigsqcup_k E_k = \bigsqcup_k (A \cap E_k)$, whence $\mu_A(\bigsqcup_k E_k) = \sum_k \mu_A(E_k)$.
(ii) $\lambda(\emptyset) = 0$. If $(E_k)_{k\in\mathbb{N}} \subseteq \mathcal{M}$ is pairwise disjoint, then $\lambda(\bigsqcup_k E_k) = \sum_{j=1}^{m} a_j \sum_k \mu_j(E_k) = \sum_k \lambda(E_k)$ by finite summation.
(iii) $\lambda(\emptyset) = 0$. If $(E_k)_{k\in\mathbb{N}} \subseteq \mathcal{M}$ is pairwise disjoint, then $\lambda(\bigsqcup_k E_k) = \sum_n 2^{-n} \sum_k \mu_n(E_k) = \sum_k \lambda(E_k)$ because both iterated sums are suprema of the same finite rectangular subsums.
The normalization is $\lambda(X) = \sum_n 2^{-n} \mu_n(X) = \sum_n 2^{-n} = 1$.

Structural Commentary: New measures come from restriction, superposition, and countable convex mixtures.
Key Insight: Nonnegative series make the bookkeeping order-independent. □

Proposition 2.1.6. *Basic Laws for Measures*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) If $n \in \mathbb{N}$ and $E_1, \dots, E_n \in \mathcal{M}$ are pairwise disjoint, then $\mu\left(\bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} \mu(E_k)$.
(ii) If $A, B \in \mathcal{M}$ and $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
(iii) If $A, B \in \mathcal{M}$, $A \subseteq B$, and $\mu(A) < +\infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.
If $\mu(A) = 0$, then $\mu(B \setminus A) = \mu(B)$.
(iv) If $E \in \mathcal{M}$ and $(E_k)_{k\in\mathbb{N}} \subseteq \mathcal{M}$ satisfy $E \subseteq \bigcup_{k=1}^{\infty} E_k$, then $\mu(E) \leq \sum_{k=1}^{\infty} \mu(E_k)$.

Proof.

(i) Define $G_k = E_k$ for $1 \leq k \leq n$ and $G_k = \emptyset$ for $k > n$. Then (G_k) is pairwise disjoint and $\mu\left(\bigcup_{k=1}^{n} E_k\right) = \mu\left(\bigcup_{k=1}^{\infty} G_k\right) = \sum_{k=1}^{\infty} \mu(G_k) = \sum_{k=1}^{n} \mu(E_k)$ by countable additivity.
(ii) Since $B = A \sqcup (B \setminus A)$, $\mu(B) = \mu(A) + \mu(B \setminus A)$ by (i). Hence $\mu(A) \leq \mu(B)$ since $\mu(B \setminus A) \geq 0$.
(iii) Since $B = A \sqcup (B \setminus A)$, $\mu(B) = \mu(A) + \mu(B \setminus A)$ by (i). Because $\mu(A) < +\infty$, $\mu(B \setminus A) = \mu(B) - \mu(A)$ by subtraction. If $\mu(A) = 0$, this becomes $\mu(B \setminus A) = \mu(B)$.
(iv) Set $F_1 = E \cap E_1$ and $F_k = E \cap \left(E_k \setminus \bigcup_{j=1}^{k-1} E_j\right)$ for $k \geq 2$. Then $(F_k)_{k\in\mathbb{N}} \subseteq \mathcal{M}$ is pairwise disjoint, $\bigcup_{k=1}^{\infty} F_k = E$, and $(\forall k \in \mathbb{N})\ F_k \subseteq E_k$. Hence $\mu(E) = \sum_{k=1}^{\infty} \mu(F_k) \leq \sum_{k=1}^{\infty} \mu(E_k)$ by countable additivity and (ii).

Structural Commentary: Finite additivity establishes order and excision; disjointization turns covers into sums.
Key Insight: Measure disjoint pieces, then compare. □

Theorem 2.1.7. *Continuity from Below*
Let X be a set, let $\mathcal{A}$ be a ring of subsets of X, and let ϕ be a countably additive set function on $\mathcal{A}$.
If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ and $A \in \mathcal{A}$ satisfy $(\forall n \in \mathbb{N})\ A_n \subseteq A_{n+1}$ and $A = \bigcup_{n=1}^{\infty} A_n$, then $\lim_{N\to\infty} \phi(A_N) = \phi(A)$.

Proof. Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Since $\mathcal{A}$ is a ring, $B_n \in \mathcal{A}$ $(n \in \mathbb{N})$.
The family $(B_n)_{n\in\mathbb{N}}$ is pairwise disjoint, $A_N = \bigsqcup_{n=1}^{N} B_n$, and $A = \bigsqcup_{n=1}^{\infty} B_n$.
By countable additivity, applied to the finite union by appending empty sets, $\phi(A_N) = \sum_{n=1}^{N} \phi(B_n)$, while $\phi(A) = \sum_{n=1}^{\infty} \phi(B_n)$ by countable additivity. Thus $\lim_{N\to\infty} \phi(A_N) = \phi(A)$.
Structural Commentary: Increasing chains split into disjoint increments.
Key Insight: Continuity is convergence of partial sums. □

Definition 2.1.8. *Charges*
Let $(X, \mathcal{M})$ be a measurable space. A *charge* is a map $\lambda : \mathcal{M} \to \mathbb{R}$ such that:

(i) $\lambda(\emptyset) = 0$.

(ii) If $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M}$ is pairwise disjoint, then $\sum_{n=1}^{\infty} \lambda(E_n)$ converges in $\mathbb{R}$ and $\lambda\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \lambda(E_n)$.

Key Insight: A charge is a real-valued countably additive set function.

Definition 2.1.9. *Variation Candidates of a Charge*
Let $(X, \mathcal{M})$ be a measurable space and let λ be a charge on $\mathcal{M}$. For $E \in \mathcal{M}$, define

(i) $\mathfrak{P}_{\mathcal{M}}(E) := \{(A_j)_{j=1}^n : n \in \mathbb{N}_0,\ E = \bigsqcup_{j=1}^n A_j,\ A_j \in \mathcal{M}\}$, the finite measurable partitions of E;
(ii) $\pi_\lambda(E) := \sup\{\lambda(A) : A \subseteq E,\ A \in \mathcal{M}\}$, the *positive variation* candidate;
(iii) $\nu_\lambda(E) := \sup\{\sum_{j=1}^n |\lambda(A_j)| : (A_j)_{j=1}^n \in \mathfrak{P}_{\mathcal{M}}(E)\}$, the *total variation* candidate.

For $E = \emptyset$, the empty partition is allowed and contributes 0.

Proposition 2.1.10. *Charge Operations and Variation Measures*
Let $(X, \mathcal{M})$ be a measurable space, let $m \in \mathbb{N}$, and let $\lambda, \lambda_1, \dots, \lambda_m$ be charges on $\mathcal{M}$.

(i) If $a_1, \dots, a_m \in \mathbb{R}$ and $\Lambda : \mathcal{M} \to \mathbb{R}$ is defined by $\Lambda(E) := \sum_{j=1}^{m} a_j \lambda_j(E)$, then Λ is a charge.
(ii) $\pi_\lambda : \mathcal{M} \to [0, +\infty]$ is a measure.
(iii) $\nu_\lambda : \mathcal{M} \to [0, +\infty]$ is a measure.

Proof.
(i) $\Lambda(\emptyset) = 0$. If $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M}$ is pairwise disjoint and $E = \bigsqcup_n E_n$, then $\sum_n \Lambda(E_n)$ converges as the finite linear combination $\sum_{j=1}^m a_j \sum_n \lambda_j(E_n)$, and $\Lambda(E) = \sum_{j=1}^m a_j \sum_n \lambda_j(E_n) = \sum_n \Lambda(E_n)$.
(ii) $\pi_\lambda(\emptyset) = 0$ and $\pi_\lambda \geq 0$. Let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M}$ be pairwise disjoint and $E = \bigsqcup_n E_n$. If $A \in \mathcal{M}$ and $A \subseteq E$, then $\lambda(A) = \sum_n \lambda(A \cap E_n) \leq \sum_n \pi_\lambda(E_n)$, so $\pi_\lambda(E) \leq \sum_n \pi_\lambda(E_n)$.
For $N \in \mathbb{N}$, set $S_N := \sum_{n=1}^N \pi_\lambda(E_n)$. If $S_N < +\infty$ and $\varepsilon > 0$, choose $A_n \in \mathcal{M}$ with $A_n \subseteq E_n$ and $\lambda(A_n) > \pi_\lambda(E_n) - \varepsilon/N$ $(1 \leq n \leq N)$. For $A^{(N)} := \bigsqcup_{n=1}^N A_n$, finite additivity implies $\pi_\lambda(E) \geq \lambda(A^{(N)}) > S_N - \varepsilon$, hence $\pi_\lambda(E) \geq S_N$. If $S_N = +\infty$, some $\pi_\lambda(E_\ell) = +\infty$, so $\pi_\lambda(E) = +\infty$. Therefore $(\forall N \in \mathbb{N})\ \pi_\lambda(E) \geq S_N$, and letting $N \to \infty$ proves countable additivity.
(iii) $\nu_\lambda(\emptyset) = 0$ and $\nu_\lambda \geq 0$. Let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M}$ be pairwise disjoint, set $E := \bigsqcup_n E_n$, and put $v((A_j)_{j=1}^r) := \sum_{j=1}^r |\lambda(A_j)|$. If $P = (A_j)_{j=1}^r \in \mathfrak{P}_{\mathcal{M}}(E)$, then $v(P) \leq \sum_{j=1}^r \sum_n |\lambda(A_j \cap E_n)| \leq \sum_n \nu_\lambda(E_n)$, so $\nu_\lambda(E) \leq \sum_n \nu_\lambda(E_n)$.
For $N \in \mathbb{N}$, set $T_N := \sum_{n=1}^N \nu_\lambda(E_n)$. If $T_N < +\infty$ and $\varepsilon > 0$, choose $P_n \in \mathfrak{P}_{\mathcal{M}}(E_n)$ with $v(P_n) > \nu_\lambda(E_n) - \varepsilon/N$ $(1 \leq n \leq N)$. Concatenating $P_1, \dots, P_N$ and adjoining the tail $\bigcup_{n>N} E_n$ produces a finite partition of E, so $\nu_\lambda(E) > T_N - \varepsilon$ and hence $\nu_\lambda(E) \geq T_N$.
If $T_N = +\infty$, some $\nu_\lambda(E_\ell) = +\infty$, and adjoining $E \setminus E_\ell$ extends partitions of E_ℓ to E, so $\nu_\lambda(E) = +\infty$. Therefore $(\forall N \in \mathbb{N})\ \nu_\lambda(E) \geq T_N$, and letting $N \to \infty$ proves countable additivity.

Structural Commentary: Suprema preserve additivity after refinement.
Key Insight: Refine first, then take suprema. □

3 Outer Measures and Caratheodory Extension

3.1 Outer Measures and Extension Theorems

Definition 3.1.1. *Abstract Outer Measure and Clean Division*
Let M be a set.

(i) An *abstract outer measure* on M is a map $\omega : \mathcal{P}(M) \to [0, +\infty]$ such that $\omega(\emptyset) = 0$, $(\forall A, B \subseteq M)\ A \subseteq B \Rightarrow \omega(A) \leq \omega(B)$, and $(\forall (A_k)_{k\in\mathbb{N}} \subseteq \mathcal{P}(M))\ \omega(\bigcup_{k=1}^\infty A_k) \leq \sum_{k=1}^\infty \omega(A_k)$.
(ii) For an outer measure ω, $E \subseteq M$ is *ω-measurable* iff $(\forall X \subseteq M)\ \omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$.

Definition 3.1.2. *Outer Measure Induced by a Covering Family*
Let $\mathcal{S} \subseteq \mathcal{P}(X)$ and let $\rho : \mathcal{S} \to [0, +\infty]$.

(i) For $E \subseteq X$, $\mathcal{C}_{\mathcal{S}}(E) := \{(S_k)_{k\geq 1} \in \mathcal{S}^{\mathbb{N}} : E \subseteq \bigcup_{k=1}^{\infty} S_k\}$.

(ii) The induced map $\rho^* : \mathcal{P}(X) \to [0, +\infty]$ is defined by $\rho^*(\emptyset) = 0$ and $\rho^*(E) := \inf_{(S_k)\in\mathcal{C}_{\mathcal{S}}(E)} \sum_{k=1}^{\infty} \rho(S_k)$ $(\emptyset \neq E \subseteq X)$, with $\inf \emptyset = +\infty$.

The map ρ^* is the *outer measure induced by ρ and $\mathcal{S}$*.
Key Insight: Minimize total ρ-cost over countable $\mathcal{S}$-covers.

Proposition 3.1.3. *Cover-Induced Outer Measure*
For $\mathcal{S} \subseteq \mathcal{P}(X)$ and $\rho : \mathcal{S} \to [0, +\infty]$, the cover-induced map $\rho^* : \mathcal{P}(X) \to [0, +\infty]$ is an outer measure on X.

Proof. By construction, $\rho^*(\emptyset) = 0$ and $\rho^* \geq 0$. If $A \subseteq B$, then $\mathcal{C}_{\mathcal{S}}(B) \subseteq \mathcal{C}_{\mathcal{S}}(A)$, hence $\rho^*(A) \leq \rho^*(B)$.
Let $E = \bigcup_k E_k$ and $I = \{k : E_k \neq \emptyset\}$. If $I = \emptyset$, then $E = \emptyset$; if $\sum_k \rho^*(E_k) = +\infty$, the subadditivity inequality is automatic. Otherwise, fix $\varepsilon > 0$ and choose $(S_{k,j})_j \in \mathcal{C}_{\mathcal{S}}(E_k)$ for $k \in I$ with $\sum_j \rho(S_{k,j}) < \rho^*(E_k) + \varepsilon 2^{-k}$.
After enumerating the double family, $(S_{k,j})_{k\in I, j\geq 1}$ is a countable cover of E, so $\rho^*(E) \leq \sum_{k\in I} \sum_j \rho(S_{k,j}) \leq \sum_k \rho^*(E_k) + \varepsilon$; let $\varepsilon \downarrow 0$.

Structural Commentary: Inclusion orders covers; regrouping covers sums.
Key Insight: Covers generate subadditivity. □

Lemma 3.1.4. *Finite-Measure Test for Caratheodory Measurability*
Let $\omega : \mathcal{P}(M) \to [0, +\infty]$ be an outer measure. Then $E \subseteq M$ is ω-measurable iff $(\forall X \subseteq M)\ \omega(X) < +\infty \Longrightarrow \omega(X) \geq \omega(X \cap E) + \omega(X \cap E^c)$.

Proof.

($\Rightarrow$) If E is ω-measurable, then $(\forall X \subseteq M)\ \omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$, hence the finite-test inequality holds.

($\Leftarrow$) Fix $X \subseteq M$. Then $\omega(X) \leq \omega(X \cap E) + \omega(X \cap E^c)$ by $X = (X \cap E) \sqcup (X \cap E^c)$ and subadditivity. If $\omega(X) < +\infty$, the reverse inequality holds by hypothesis; if $\omega(X) = +\infty$, it is automatic in $[0, +\infty]$. Thus E is ω-measurable.

Structural Commentary: To verify a clean split, only finite outer-measure test sets need real work.
Key Insight: Subadditivity supplies one inequality for free. □

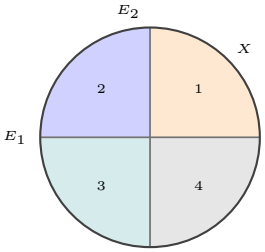

Figure 189: Two cuts split X into four regions; $E_1 \setminus E_2$ is region 2.

Figure 189 records the four intersections used when two measurable cuts are combined.

Definition 3.1.5. *Zero Sets*
For outer measure $\omega : \mathcal{P}(M) \to [0, +\infty]$, *$\omega$-zero set* is $Z \subseteq M$ with $\omega(Z) = 0$; $\mathcal{Z}_\omega := \{Z \subseteq M : \omega(Z) = 0\}$.

Theorem 3.1.6. *Caratheodory Measurable Sets*
Let $\omega : \mathcal{P}(M) \to [0, +\infty]$ be an outer measure.
Let $\mathcal{M}_\omega := \{E \subseteq M : (\forall X \subseteq M)\ \omega(X) = \omega(X \cap E) + \omega(X \cap E^c)\}$ and $\mathcal{Z}_\omega := \{Z \subseteq M : \omega(Z) = 0\}$. Then:

(i) $\mathcal{Z}_\omega \subseteq \mathcal{M}_\omega$;
(ii) $\mathcal{M}_\omega$ is a σ-algebra;
(iii) $\omega|_{\mathcal{M}_\omega}$ is countably additive;
(iv) if $E \in \mathcal{M}_\omega$ and $Z \in \mathcal{Z}_\omega$, then $E \cup Z, E \setminus Z, E \triangle Z \in \mathcal{M}_\omega$ and $\omega(E \cup Z) = \omega(E \setminus Z) = \omega(E \triangle Z) = \omega(E)$.

Proof.

(i) If $Z \in \mathcal{Z}_\omega$ and $X \subseteq M$, then $\omega(X) \leq \omega(X \cap Z) + \omega(X \cap Z^c) \leq 0 + \omega(X)$, so equality holds and $Z \in \mathcal{M}_\omega$.

(ii) Complements are immediate. If $E, F \in \mathcal{M}_\omega$, with $E^1 = E$, $E^0 = E^c$, $F^1 = F$, and $F^0 = F^c$, then $\omega(X) = \sum_{\epsilon,\eta \in \{0,1\}} \omega(X \cap E^\epsilon \cap F^\eta)$; hence finite Boolean combinations are measurable.
For pairwise disjoint $E_i \in \mathcal{M}_\omega$, set $E = \bigsqcup_i E_i$ and $F_N = (\bigcup_{i=1}^N E_i)^c$.
By finite splitting and $E^c \subseteq F_N$, $\omega(X) \geq \sum_{i=1}^N \omega(X \cap E_i) + \omega(X \cap E^c)$ $(N \in \mathbb{N})$. Hence $\omega(X) \geq \sum_{i=1}^\infty \omega(X \cap E_i) + \omega(X \cap E^c)$ by taking the supremum over N.
By subadditivity, $\omega(X) \leq \omega(X \cap E) + \omega(X \cap E^c) \leq \sum_{i=1}^\infty \omega(X \cap E_i) + \omega(X \cap E^c)$. Together with the preceding reverse inequality, all three quantities are equal. Hence $\omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$ and $E \in \mathcal{M}_\omega$.
General countable unions follow from $H_1 = E_1$ and $H_n = E_n \setminus \bigcup_{i<n} E_i$.

(iii) If $E = \bigsqcup_{i=1}^\infty E_i$ with $E_i \in \mathcal{M}_\omega$, then $\omega(E) = \sum_{i=1}^\infty \omega(E_i)$ by the equality in (ii) with $X = E$.

(iv) By (i)–(ii), $E \cup Z, E \setminus Z, E \triangle Z \in \mathcal{M}_\omega$.
Since $E \cap Z$ and $Z \setminus E$ are null, $\omega(E \setminus Z) = \omega(E \cup Z) = \omega(E \triangle Z) = \omega(E)$ by additivity.

Structural Commentary: Measurable sets preserve countable additivity.
Key Insight: Measurable cuts turn subadditivity into equality. □

Corollary 3.1.7. *Disjointization of Countable Measurable Unions*
Let $\omega : \mathcal{P}(M) \to [0, +\infty]$ be an outer measure and let $(E_k)_{k \geq 1} \subseteq \mathcal{M}_\omega$.
Set $F_1 := E_1$ and $F_k := E_k \setminus \bigcup_{i=1}^{k-1} E_i$ for $k \geq 2$.
Then (F_k) is pairwise disjoint in $\mathcal{M}_\omega$, $\bigcup_{k \geq 1} F_k = \bigcup_{k \geq 1} E_k$, and $\omega(\bigcup_{k \geq 1} E_k) = \sum_{k \geq 1} \omega(F_k)$ in $[0, +\infty]$.

Proof. Since $\mathcal{M}_\omega$ is a σ-algebra, $F_k = E_k \cap (\bigcup_{i<k} E_i)^c \in \mathcal{M}_\omega$. If $i < k$, then $F_i \cap F_k \subseteq E_i \cap (\bigcup_{j<k} E_j)^c = \emptyset$.
For $x \in \bigcup_{k \geq 1} E_k$, $x \in F_{\min\{j : x \in E_j\}}$; hence $\bigcup_{k \geq 1} F_k = \bigcup_{k \geq 1} E_k$, and $\omega(\bigcup_{k \geq 1} E_k) = \sum_{k \geq 1} \omega(F_k)$ by countable additivity.
Structural Commentary: Disjointization turns unions into sums.
Key Insight: Peel off earlier pieces so a countable union becomes a disjoint sum. □

Definition 3.1.8. *Countable Cover Classes*
Let $\mathcal{S} \subseteq \mathcal{P}(X)$.

(i) *Countable-union class*: $\mathcal{S}_\sigma := \left\{ \bigcup_{n=1}^\infty S_n : (S_n)_{n \geq 1} \in \mathcal{S}^{\mathbb{N}} \right\}$.

(ii) *Countable-intersection class*: $\mathcal{S}_{\sigma\delta} := \left\{ \bigcap_{n=1}^\infty A_n : (A_n)_{n \geq 1} \in (\mathcal{S}_\sigma)^{\mathbb{N}} \right\}$.

Key Insight: $\mathcal{S}_\sigma$ closes under countable unions; $\mathcal{S}_{\sigma\delta}$ then closes once under countable intersections.

Proposition 3.1.9. *Finite Outer-Measure Envelope*
Let $\mathcal{S} \subseteq \mathcal{P}(X)$, let $\rho : \mathcal{S} \to [0, +\infty]$, and let ρ^* be the cover-induced map.
If $\emptyset \neq E \subseteq X$ and $\rho^*(E) < +\infty$, then $(\exists A \in \mathcal{S}_{\sigma\delta})$ such that:

(i) $E \subseteq A$ and $\rho^*(A) = \rho^*(E)$.

(ii) If E is ρ^*-measurable and $(\forall S \in \mathcal{S})$ S is ρ^*-measurable, then A is ρ^*-measurable and $\rho^*(A \setminus E) = 0$.

Proof. $\forall n \geq 1$, choose $(S_{n,k})_{k\geq 1} \in \mathcal{C}_{\mathcal{S}}(E)$ with $\sum_{k=1}^{\infty} \rho(S_{n,k}) < \rho^*(E) + \frac{1}{n}$. Set $A_n := \bigcup_{k=1}^{\infty} S_{n,k}$, $A := \bigcap_{n=1}^{\infty} A_n$.
Then $A_n \in \mathcal{S}_\sigma$, $A \in \mathcal{S}_{\sigma\delta}$, $E \subseteq A \subseteq A_n$, and $\rho^*(A_n) < \rho^*(E) + \frac{1}{n}$.

(i) By monotonicity, $\rho^*(E) \leq \rho^*(A) \leq \rho^*(A_n) < \rho^*(E) + \frac{1}{n}$ $(n \in \mathbb{N})$, hence $\rho^*(A) = \rho^*(E)$.
(ii) If E is ρ^*-measurable and $(\forall S \in \mathcal{S})$ S is ρ^*-measurable, then the ρ^*-measurable sets form a σ-algebra, so A_n and A are ρ^*-measurable. Since $E \subseteq A$ and $\rho^*(E) < +\infty$, $\rho^*(A \setminus E) = \rho^*(A) - \rho^*(E) = 0$ by Proposition 2.1.6(iii).

Structural Commentary: Finite cost produces a countable-intersection envelope.
Key Insight: Cover within $\frac{1}{n}$, then intersect. □

Definition 3.1.10. *Outer Measure Generated by a Premeasure*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$.

(i) *Algebra covers*: For $B \subseteq X$, set $\mathcal{C}_{\mathcal{A}}(B) := \{(E_j)_{j\geq 1} \in \mathcal{A}^{\mathbb{N}} : B \subseteq \bigcup_{j=1}^{\infty} E_j\}$.

(ii) *Generated outer measure*: Define $\mu^* : \mathcal{P}(X) \to [0, +\infty]$ by $\mu^*(\emptyset) = 0$ and, for $B \neq \emptyset$, $\mu^*(B) := \inf_{(E_j)\in\mathcal{C}_{\mathcal{A}}(B)} \sum_{j=1}^{\infty} \mu(E_j)$, with $\inf \emptyset = +\infty$.

Key Insight: Cover arbitrary sets by algebra sets and minimize total premeasure.

Lemma 3.1.11. *Generated Outer-Measure Properties*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, and let μ^* be the generated map from Definition 3.1.10. Then:

(i) $\mu^*(\emptyset) = 0$.
(ii) If $C \subseteq D \subseteq X$, then $\mu^*(C) \leq \mu^*(D)$.
(iii) If $(B_n)_{n\geq 1} \subseteq \mathcal{P}(X)$, then $\mu^*(\bigcup_{n=1}^{\infty} B_n) \leq \sum_{n=1}^{\infty} \mu^*(B_n)$.
(iv) If $E \in \mathcal{A}$, then $\mu^*(E) = \mu(E)$.

Proof.

(i) This is the assigned value of $\mu^*(\emptyset)$.
(ii) If $C \subseteq D$, then $\mathcal{C}_{\mathcal{A}}(D) \subseteq \mathcal{C}_{\mathcal{A}}(C)$; hence $\mu^*(C) \leq \mu^*(D)$ by taking infima.
(iii) Let $B = \bigcup_n B_n$ and $I = \{n : B_n \neq \emptyset\}$. If $I = \emptyset$ or $\sum_n \mu^*(B_n) = +\infty$, the claim is immediate.
Otherwise, for $\varepsilon > 0$ choose $(E_{n,k})_{k\geq 1} \in \mathcal{C}_{\mathcal{A}}(B_n)$ for $n \in I$ with $\sum_k \mu(E_{n,k}) < \mu^*(B_n) + \varepsilon 2^{-n}$.
Then $(E_{n,k})_{n\in I, k\geq 1}$ covers B, so $\mu^*(B) \leq \sum_{n\in I}\sum_k \mu(E_{n,k}) \leq \sum_n \mu^*(B_n) + \varepsilon$; let $\varepsilon \downarrow 0$.
(iv) By the cover $E, \emptyset, \emptyset, \ldots$, $\mu^*(E) \leq \mu(E)$. Conversely, fix $(F_n) \in \mathcal{C}_{\mathcal{A}}(E)$ and set $G_n := E \cap \left(F_n \setminus \bigcup_{k<n} F_k\right)$.
Then (G_n) is disjoint in $\mathcal{A}$, $\bigcup_n G_n = E$, and $G_n \subseteq F_n$. The inequality $\mu(G_n) \leq \mu(F_n)$ holds by premeasure additivity on the finite decomposition $F_n = G_n \sqcup (F_n \setminus G_n)$, so $\mu(E) = \sum_n \mu(G_n) \leq \sum_n \mu(F_n)$.
Hence $\mu(E) \leq \mu^*(E)$ by taking infima.

Structural Commentary: Cover containment and regrouping establish the outer-measure axioms.
Key Insight: Disjointize algebra covers to recover μ. □

Proposition 3.1.12. *Single-Set Envelope Comparison*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, let μ^* be the generated outer measure, and define $\mu^\sharp : \mathcal{P}(X) \to [0, +\infty]$ by $\mu^\sharp(B) := \inf\{\mu(C) : C \in \mathcal{A},\ B \subseteq C\}$ with $\inf \emptyset = +\infty$.

(i) $(\forall E \in \mathcal{A})$ $\mu^\sharp(E) = \mu(E)$.
(ii) $(\forall B \subseteq X)$ $\mu^*(B) \leq \mu^\sharp(B)$.

Proof.

(i) If $E \in \mathcal{A}$, then E is admissible, so $\mu^\sharp(E) \leq \mu(E)$.
Conversely, if $E \subseteq C \in \mathcal{A}$, then $C = E \sqcup (C \setminus E)$ and $\mu(E) \leq \mu(C)$ by premeasure additivity on this finite decomposition; hence $\mu(E) \leq \mu^\sharp(E)$.
(ii) If no $C \in \mathcal{A}$ contains B, then $\mu^\sharp(B) = +\infty$. If $C \in \mathcal{A}$ and $B \subseteq C$, then $\mu^*(B) \leq \mu(C)$ by the algebra cover $C, \emptyset, \emptyset, \ldots$. Hence $\mu^*(B) \leq \mu^\sharp(B)$ by taking infima over such C.

Structural Commentary: One algebra superset is a special countable cover.
Key Insight: The single-set envelope is larger. □

Counterexample 3.1.13. *Single-Set Envelopes Need Not Be Outer Measures*
Let X be uncountable, let $\mathcal{A}$ be the finite–cofinite algebra on X, and set $\mu(E) := 0$ for finite E, $\mu(E) := +\infty$ for cofinite E. If a disjoint algebra family contains a cofinite member, its sum and the value of its union are $+\infty$; otherwise, a countable union of its finite members cannot be cofinite. Hence μ is a premeasure. A set has a zero-cost countable $\mathcal{A}$-cover iff it is countable, whereas it has a zero-cost single $\mathcal{A}$-superset iff it is finite. Thus $\mu^*(B) = 0$ iff B is countable, while $\mu^\sharp(B) = 0$ iff B is finite.
If $S \subseteq X$ is countably infinite, then $\mu^\sharp(S) = +\infty$ while $S = \bigcup_{x\in S}\{x\}$ and $\sum_{x\in S} \mu^\sharp(\{x\}) = 0$, so $\mu^\sharp$ is not countably subadditive.
Structural Commentary: Countable covers can be strictly smaller than one-set covers.
Key Insight: Outer measure is built from countable covers precisely to recover countable subadditivity.

Theorem 3.1.14. *Generated Caratheodory Extension*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, let μ^* be the generated map from Definition 3.1.10, and set $\mathcal{C}_\mu := \{E \subseteq X : (\forall B \subseteq X)\ \mu^*(B) = \mu^*(B \cap E) + \mu^*(B \setminus E)\}$. Then:

(i) $\mathcal{A} \subseteq \mathcal{C}_\mu$ and $\mu^*|_{\mathcal{A}} = \mu$.

(ii) $\mathcal{C}_\mu$ *is a* σ*-algebra on* X*.*
(iii) $\mu^*|_{\mathcal{C}_\mu}$ *is a complete measure extending* μ*.*

Proof. By Lemma 3.1.11, μ^* is an outer measure and $\mu^*|_{\mathcal{A}} = \mu$.

(i) Let $E \in \mathcal{A}$ and $B \subseteq X$. By subadditivity, $\mu^*(B) \le \mu^*(B \cap E) + \mu^*(B \setminus E)$. If $(F_n) \in \mathcal{C}_{\mathcal{A}}(B)$, then $F_n \cap E, F_n \setminus E \in \mathcal{A}$ and $\mu(F_n) = \mu(F_n \cap E) + \mu(F_n \setminus E)$, so $\mu^*(B \cap E) + \mu^*(B \setminus E) \le \sum_n \mu(F_n)$.
Infimizing over (F_n) establishes equality; hence $E \in \mathcal{C}_\mu$, while $\mu^*|_{\mathcal{A}} = \mu$ by Lemma 3.1.11.
(ii) The class $\mathcal{C}_\mu$ is exactly the μ^*-measurable class $\mathcal{M}_{\mu^*}$.
By Theorem 3.1.6(ii), $\mathcal{C}_\mu$ is a σ-algebra on X.
(iii) By Theorem 3.1.6(iii), $\mu^*|_{\mathcal{C}_\mu}$ is countably additive. If $A \subseteq Z \in \mathcal{C}_\mu$ and $\mu^*(Z) = 0$, then $\mu^*(A) = 0$ by monotonicity and $A \in \mathcal{C}_\mu$ by Theorem 3.1.6(i). Thus $\mu^*|_{\mathcal{C}_\mu}$ is complete, and it extends μ by (i).

Structural Commentary: Caratheodory equality is the additivity domain.
Key Insight: Algebra sets split every algebra cover. □

Proposition 3.1.15. *Caratheodory Splitters for Finitely Subadditive Size*
Let $\alpha : \mathcal{P}(X) \to [0, +\infty)$ *satisfy* $(\forall E, F \subseteq X)\ 0 \le \alpha(E) \le \alpha(E \cup F) \le \alpha(E) + \alpha(F)$.
Set $\mathcal{S} := \{E \subseteq X : (\forall A \subseteq X)\ \alpha(A) = \alpha(A \cap E) + \alpha(A \setminus E)\}$.

(i) If $\mathcal{S} \neq \emptyset$*, then* $\mathcal{S}$ *is an algebra on* X*.*
(ii) If $E, F \in \mathcal{S}$ *and* $E \cap F = \emptyset$*, then* $\alpha(E \cup F) = \alpha(E) + \alpha(F)$*.*
(iii) The hypothesis $\mathcal{S} \neq \emptyset$ *in (i) is necessary: if* $\alpha \equiv 1$*, then* $\mathcal{S} = \emptyset$*.*

Proof.

(i) Choose $E_0 \in \mathcal{S}$. By splitting $\emptyset$, $\alpha(\emptyset) = 0$, hence $\emptyset, X \in \mathcal{S}$.
By complement symmetry, $E \in \mathcal{S} \Rightarrow X \setminus E \in \mathcal{S}$. For $E, F \in \mathcal{S}$ and $A \subseteq X$, put $A_{11} = A \cap E \cap F$, $A_{10} = A \cap E \setminus F$, $A_{01} = A \cap F \setminus E$, and $A_{00} = A \setminus (E \cup F)$. By successive splitting,
$\alpha(A) = \alpha(A_{11}) + \alpha(A_{10}) + \alpha(A_{01}) + \alpha(A_{00})$; $\alpha(A \setminus (E \cap F)) = \alpha(A_{10}) + \alpha(A_{01}) + \alpha(A_{00})$.
Thus $\alpha(A) = \alpha(A \cap E \cap F) + \alpha(A \setminus (E \cap F))$, so $E \cap F \in \mathcal{S}$. Finite unions follow by complements.
(ii) Since $F \subseteq X \setminus E$, $\alpha(E \cup F) = \alpha(E) + \alpha(F)$ by the splitting identity for E with $A = E \cup F$.
(iii) If $\alpha \equiv 1$ and $E \in \mathcal{S}$, then $1 = 1 + 1$ by the splitting identity with $A = \emptyset$, which is impossible.

Structural Commentary: Splitters are Boolean once one splitter exists.
Key Insight: Empty-set splitting forces $\alpha(\emptyset) = 0$. □

Theorem 3.1.16. *σ-Finite Generated Extensions Are Unique*
Let $\mu : \mathcal{A} \to [0, +\infty]$ *be a premeasure on an algebra* $\mathcal{A} \subseteq \mathcal{P}(X)$*, let* μ^* *be generated by* μ *as in Definition 3.1.10, and put* $\mathcal{C}_\mu := \{E \subseteq X : (\forall B \subseteq X)\ \mu^*(B) = \mu^*(B \cap E) + \mu^*(B \setminus E)\}$.
Assume $(F_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$*,* $X = \bigcup_n F_n$*, and* $(\forall n \in \mathbb{N})\ \mu(F_n) < +\infty$.

(i) $\mu^*|_{\mathcal{C}_\mu}$ *is a measure on* $\mathcal{C}_\mu$ *and extends* μ*.*
(ii) If ν *is a measure on* $\mathcal{C}_\mu$ *with* $\nu|_{\mathcal{A}} = \mu$*, then* $\nu = \mu^*|_{\mathcal{C}_\mu}$*.*

Proof.

(i) By Theorem 3.1.14, $\mathcal{C}_\mu$ is a σ-algebra, $\mathcal{A} \subseteq \mathcal{C}_\mu$, and $\mu^*|_{\mathcal{C}_\mu}$ is a complete measure with $(\mu^*|_{\mathcal{C}_\mu})|_{\mathcal{A}} = \mu$.

(ii) Let ν be a measure on $\mathcal{C}_\mu$ with $\nu|_{\mathcal{A}} = \mu$, fix $E \in \mathcal{C}_\mu$, and put $H_N := \bigcup_{j=1}^{N} F_j$. Then $H_N \in \mathcal{A}$, $H_N \uparrow X$, and $\mu(H_N) < +\infty$.
If $G \in \mathcal{C}_\mu$ and $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$ satisfies $G \subseteq \bigcup_n A_n$, then $\nu(G) \le \sum_n \mu(A_n)$; hence $\nu(G) \le \mu^*(G)$ by taking infima.
The sets $E \cap H_N$ and $H_N \setminus E$ lie in $\mathcal{C}_\mu$ and partition H_N, so $\nu(E \cap H_N) + \nu(H_N \setminus E) = \mu(H_N) = \mu^*(E \cap H_N) + \mu^*(H_N \setminus E)$.
Because the two ν-terms are bounded above by the corresponding μ^*-terms and the total is finite, $\nu(E \cap H_N) = \mu^*(E \cap H_N)$.
Since $E \cap H_N \uparrow E$, $\nu(E) = \lim_{N \to \infty} \nu(E \cap H_N) = \lim_{N \to \infty} \mu^*(E \cap H_N) = \mu^*(E)$ by Theorem 2.1.7 for the two measures.

Structural Commentary: Finite uniqueness localizes along a finite-mass exhaustion.
Key Insight: Compare on each finite piece, then pass upward. □

Corollary 3.1.17. *Uniqueness on the Generated σ-Algebra*
Let $\mu : \mathcal{A} \to [0, +\infty]$ *be a premeasure on an algebra* $\mathcal{A} \subseteq \mathcal{P}(X)$*, and assume* $(F_n)_{n \in \mathbb{N}} \subseteq \mathcal{A}$*,* $X = \bigcup_n F_n$*, and* $(\forall n \in \mathbb{N})\ \mu(F_n) < +\infty$*. If measures* ν_1, ν_2 *on* $\sigma(\mathcal{A})$ *extend* μ*, then* $\nu_1 = \nu_2$*.*

Proof. Put $H_n := \bigcup_{j=1}^{n} F_j$. Then $H_n \in \mathcal{A}$, $H_n \uparrow X$, and $\nu_i(H_n) = \mu(H_n) < +\infty$.
Fix $n \in \mathbb{N}$ and set $\mathcal{D}_n := \{E \in \sigma(\mathcal{A}) : \nu_1(E \cap H_n) = \nu_2(E \cap H_n)\}$. Equality passes through increasing unions by Theorem 2.1.7. If $E_k \downarrow E$ in $\mathcal{D}_n$, then $(E_1 \setminus E_k) \cap H_n \uparrow (E_1 \setminus E) \cap H_n$; equality passes to the limit by finite subtraction on $E_1 \cap H_n$ and the same theorem. Thus $\mathcal{D}_n$ is a monotone class.
If $A \in \mathcal{A}$, then $A \cap H_n \in \mathcal{A}$ and $\nu_1(A \cap H_n) = \mu(A \cap H_n) = \nu_2(A \cap H_n)$, so $\mathcal{A} \subseteq \mathcal{D}_n$. Hence $\mathcal{D}_n = \sigma(\mathcal{A})$ by Corollary 1.1.12. Fixing $E \in \sigma(\mathcal{A})$, one has $(\forall n \in \mathbb{N})\ \nu_1(E \cap H_n) = \nu_2(E \cap H_n)$ and therefore $\nu_1(E) = \nu_2(E)$ by Theorem 2.1.7.
Structural Commentary: Finite exhaustions reduce uniqueness to monotone closure.
Key Insight: Compare on H_n, then pass upward. □

Example 3.1.18. *Why σ-Finiteness Matters*
Let $X = \mathbb{Q} \cap (0, 1]$, and let $\mathcal{A}$ be the algebra of finite unions of sets $(a, b] \cap X$, where $0 \le a < b \le 1$ and $a, b \in \mathbb{Q}$.

(i) *Infinite algebra pieces*: $(\forall A \in \mathcal{A} \setminus \{\emptyset\})$ A is infinite.
(ii) *Generated σ-algebra*: Fix $r \in X$ and choose $a_n \in \mathbb{Q} \cap [0, r)$ with $a_n \uparrow r$.
Then $\{r\} = \bigcap_{n=1}^{\infty} ((a_n, r] \cap X) \in \sigma(\mathcal{A})$, hence $\sigma(\mathcal{A}) = \mathcal{P}(X)$.
(iii) *Nonunique extensions*: Let μ_1 be counting measure and $\mu_2 = 2\mu_1$ on $\mathcal{P}(X)$.
By (i), $\mu_1|_{\mathcal{A}} = \mu_2|_{\mathcal{A}}$, but $\mu_1(\{r\}) = 1 \neq 2 = \mu_2(\{r\})$ $(r \in X)$.

Thus uniqueness may fail without σ-finiteness.
Key Insight: Infinite algebra pieces cannot determine singleton scale.

3.2 Euclidean Cells and Cell Outer Measure

Definition 3.2.1. *Cells, Intervals, and Volume in $\mathbb{R}^p$*
Fix $p \geq 1$.

(i) *Cells in $\mathbb{R}$.* For $a \leq b$, a cell with endpoints a, b is $I \in \{(a,b), [a,b], (a,b], [a,b)\}$.
If $a = b$, only $[a,a]$ is nonempty.
(ii) *Length.* If I is a cell with endpoints $a \leq b$, then $\ell(I) := b - a$; unbounded intervals $(-\infty, a)$, $(-\infty, a]$, $[a, +\infty)$, $(a, +\infty)$, and $\mathbb{R}$ are not cells and have length $+\infty$.
(iii) *Cells in $\mathbb{R}^p$.* A cell is $I = \prod_{i=1}^p I_i$, each I_i a cell in $\mathbb{R}$; $\prod_i (a_i, b_i)$ is open, $\prod_i [a_i, b_i]$ is closed, $\prod_i [a_i, b_i)$ is *southwest*, $\prod_i (a_i, b_i]$ is *northeast*, and I is a *cube* iff $\ell(I_1) = \cdots = \ell(I_p)$.
(iv) *Volume.* For a cell $I = \prod_{i=1}^p I_i$ with endpoints $a_i \leq b_i$, $\mathrm{vol}_p(I) := \prod_{i=1}^p \ell(I_i) = \prod_{i=1}^p (b_i - a_i)$.
Also $\mathrm{vol}_p(\emptyset) := 0$ and $\mathrm{vol}_p(I^\circ) = \mathrm{vol}_p(\overline{I}) = \mathrm{vol}_p(I)$.
(v) *Unbounded products.* For $I = \prod_{i=1}^p I_i$ with possibly unbounded intervals, $\mathrm{vol}_p(I) = 0$ if $\exists j : \ell(I_j) = 0$, $\mathrm{vol}_p(I) = +\infty$ if $\ell(I_i) > 0$ for all i and $\exists j : \ell(I_j) = +\infty$, and $\mathrm{vol}_p(I) = \prod_{i=1}^p \ell(I_i)$ otherwise.

Remark 3.2.2. *Cells versus Intervals*
Cells are precisely bounded intervals with finite endpoints, so every cell is an interval but not conversely.
Open cells are open in $\mathbb{R}^p$, closed cells are closed in $\mathbb{R}^p$, and nonempty half-open cells are neither.
In $\mathbb{R}$, $(-\infty, a)$ and $(b, +\infty)$ are open, $(-\infty, a]$ and $[b, +\infty)$ are closed, and $\mathbb{R} = (-\infty, +\infty)$ is clopen.
Key Insight: Topology likes open and closed endpoints; measure bookkeeping often likes half-open endpoints.

Proposition 3.2.3. *Half-Open Cell Calculus*
Fix $\sigma \in \{$southwest, northeast$\}$, and let $\mathcal{H}_\sigma$ be the half-open cells in $\mathbb{R}^p$ with orientation σ.

(i) Any finite Boolean combination of $H_1, \ldots, H_n \in \mathcal{H}_\sigma$ is $\bigsqcup_{r=1}^N K_r$ with $K_r \in \mathcal{H}_\sigma$.
(ii) If $H \in \mathcal{H}_\sigma$, then $\mathbb{R}^p \setminus H = \bigsqcup_{r=1}^\infty K_r$ for some $K_r \in \mathcal{H}_\sigma$.
(iii) For $H \in \mathcal{H}_\sigma$ and $\delta > 0$, $H = \bigsqcup_{r=1}^N K_r$ for some $K_r \in \mathcal{H}_\sigma$ with $\operatorname{diam} K_r < \delta$.

Proof.

(i) For coordinate endpoints $E_j = \{c_{j,0} < \cdots < c_{j,m_j}\}$ of $H_1, \ldots, H_n$, set $[a,b)_\sigma = [a,b)$ if $\sigma =$ *southwest* and $[a,b)_\sigma = (a,b]$ if $\sigma =$ *northeast*, $\mathcal{P}_j = \{[c_{j,q-1}, c_{j,q})_\sigma : 1 \leq q \leq m_j\}$, and $\mathcal{G} = \{\prod_{j=1}^p P_j : P_j \in \mathcal{P}_j\}$.
Every finite Boolean combination is $\bigsqcup_{G \in \mathcal{G}'} G$, $\mathcal{G}' \subseteq \mathcal{G}$.
(ii) Write $H = \prod_{j=1}^p H_j$ and $D_s = (\prod_{j<s} H_j) \times (\mathbb{R} \setminus H_s) \times \mathbb{R}^{p-s}$; then $\mathbb{R}^p \setminus H = \bigsqcup_{s=1}^p D_s$. Decomposing each one-dimensional ray and $\mathbb{R}$ into countably many bounded σ-half-open intervals and taking products produces the required union. By Corollary 6.2.17, the product family is countable.
(iii) Choose m with $2^{-m}(\sum_{j=1}^p \ell(H_j)^2)^{1/2} < \delta$. After m bisections per coordinate, each subcell has diameter $< \delta$; disjointness and volume preservation follow coordinatewise.

Structural Commentary: Half-open cells behave like grid atoms: finite set operations become finite grid selection.
Key Insight: Half-open endpoints prevent overlap while still partitioning boxes cleanly. □

Definition 3.2.4. *Translation of a Set*
If $A \subseteq \mathbb{R}^p$ and $t \in \mathbb{R}^p$, the translate of A by t is $t + A := \{t + a : a \in A\}$.

Proposition 3.2.5. *Cell Volume Is Translation Invariant*
Let $I = \prod_{i=1}^p I_i$ be a cell in $\mathbb{R}^p$ and $t = (t_i)_{i=1}^p \in \mathbb{R}^p$.
Then $t + I = \prod_{i=1}^p (t_i + I_i)$ is a cell with the same endpoint pattern as I, and $\mathrm{vol}_p(t + I) = \mathrm{vol}_p(I)$.

Proof. Let $a_i \leq b_i$ be the endpoints of I_i. Then $t_i + I_i$ has endpoints $t_i + a_i \leq t_i + b_i$, unchanged endpoint inclusion, and $\ell(t_i + I_i) = (t_i + b_i) - (t_i + a_i) = \ell(I_i)$. Hence $\mathrm{vol}_p(t + I) = \prod_{i=1}^p \ell(t_i + I_i) = \prod_{i=1}^p \ell(I_i) = \mathrm{vol}_p(I)$.
Structural Commentary: Translation moves location but not side lengths.
Key Insight: Coordinate volume depends only on side lengths. □

Definition 3.2.6. *Elementary Sets and Regularity in $\mathbb{R}^p$*

(i) *Elementary sets.* $\mathcal{E} := \{\bigcup_{r=1}^n I_r : n \in \mathbb{N}\}$, where each $I_r \subseteq \mathbb{R}^p$ is a bounded cell.
(ii) *Elementary volume.* If $A = \bigsqcup_{r=1}^n I_r$ with cells I_r, set $m(A) := \sum_{r=1}^n \mathrm{vol}_p(I_r)$.
(iii) *Regularity.* A nonnegative additive $\mu : \mathcal{E} \to [0, +\infty]$ is regular iff $\forall A \in \mathcal{E}$, $\forall \varepsilon > 0$, $\exists F, G \in \mathcal{E}$ with $F \subseteq A \subseteq G$, F closed, G open, and $\mu(G) - \varepsilon \leq \mu(A) \leq \mu(F) + \varepsilon$.

$\mathcal{E}$ is a ring; every $A \in \mathcal{E}$ admits a finite disjoint cell decomposition; m is well defined, additive, and regular.

Perspective 3.2.7. *Box Volume and Linear Algebra*
For a box $B = \prod_i (a_i, b_i)$ or $\prod_i [a_i, b_i]$, $m(B) = \prod_i (b_i - a_i)$, the diagonal case of determinant volume scaling.
Measure extends this product rule beyond boxes.
Key Insight: Lebesgue measure starts from determinant-style coordinate volume.

Proposition 3.2.8. *Lattice Counts Recover Cell Volume*
Fix $p \geq 1$. For $q \geq 1$ and $A \subseteq \mathbb{R}^q$, write $N_\mathbb{Z}(A) := \#(A \cap \mathbb{Z}^q)$, and for $\lambda > 0$ set $D_\lambda A := \{\lambda x : x \in A\}$.
If $I = \prod_{j=1}^p I_j$ is a bounded cell in $\mathbb{R}^p$, then $\lim_{\lambda \to +\infty} \lambda^{-p} N_\mathbb{Z}(D_\lambda I) = \mathrm{vol}_p(I)$.

Proof. For a bounded interval $J \subseteq \mathbb{R}$ with length L, $\max\{0, \lambda L - 1\} \leq N_\mathbb{Z}(D_\lambda J) \leq \lambda L + 1$.
Hence $\lambda^{-1} N_\mathbb{Z}(D_\lambda J) \to L$ if $L > 0$, and $N_\mathbb{Z}(D_\lambda J) \leq 1$ if $L = 0$.
Let $L_j := \ell(I_j)$. Since $D_\lambda I = \prod_j D_\lambda I_j$, $N_\mathbb{Z}(D_\lambda I) = \prod_j N_\mathbb{Z}(D_\lambda I_j)$.

If every $L_j > 0$, then $\lambda^{-p} N_\mathbb{Z}(D_\lambda I) = \prod_j \lambda^{-1} N_\mathbb{Z}(D_\lambda I_j) \to \prod_j L_j = \mathrm{vol}_p(I)$.

If some $L_j = 0$, then $N_\mathbb{Z}(D_\lambda I) = O(\lambda^{p-1})$, so $\lambda^{-p} N_\mathbb{Z}(D_\lambda I) \to 0 = \mathrm{vol}_p(I)$.

Structural Commentary: Coordinate lattice density multiplies.
Key Insight: Endpoint effects are lower order. □

Perspective 3.2.9. *Outer Measure as Overestimation*
Elementary $A = \bigsqcup_{r=1}^{n} I_r$ have $m(A) = \sum_{r=1}^{n} \operatorname{vol}_p(I_r)$.
General $E \subseteq \mathbb{R}^p$, e.g. $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$, need not be a finite disjoint cell union.
Replace decompositions by countable cell covers $E \subseteq \bigcup_{k=1}^{\infty} I_k$ and minimize $\sum_{k=1}^{\infty} \operatorname{vol}_p(I_k)$.
Key Insight: Outer covers size every $E \subseteq \mathbb{R}^p$ before measurability.

Definition 3.2.10. *Cell-Cover Outer Measure*
For $E \subseteq \mathbb{R}^p$, let $\mathcal{C}(E)$ be the countable cell covers $(I_k)_{k\geq 1}$ with $E \subseteq \bigcup_{k=1}^{\infty} I_k$. Define $m_p^*(E) := \inf_{(I_k)\in\mathcal{C}(E)} \sum_{k=1}^{\infty} \operatorname{vol}_p(I_k)$.

Here $\mathcal{C}(E) \neq \emptyset$ since $E \subseteq \bigcup_{j=1}^{\infty} [-j, j]^p$, and the sum is a nonnegative extended-real series, hence order-independent.

Proposition 3.2.11. *Equivalent Covering Families*
Let $E \subseteq \mathbb{R}^p$, $\delta > 0$, and fix a half-open orientation σ. Let $\mathcal{C}_{\mathsf{cl}}(E), \mathcal{C}_{\mathsf{op}}(E), \mathcal{C}_{\sigma}(E), \mathcal{C}_{\delta}(E) \subseteq \mathcal{C}(E)$ denote covers by closed cells, open cells, σ-half-open cells, and cells with diameter $< \delta$, respectively.
For $\mathcal{F} \subseteq \mathcal{C}(E)$, put $\alpha(\mathcal{F}) := \inf_{(I_k)\in\mathcal{F}} \sum_k \operatorname{vol}_p(I_k)$. Then $m_p^*(E) = \alpha(\mathcal{C}_{\mathsf{cl}}(E)) = \alpha(\mathcal{C}_{\mathsf{op}}(E)) = \alpha(\mathcal{C}_{\sigma}(E)) = \alpha(\mathcal{C}_{\delta}(E))$.

Proof. Let $\mathfrak{C} := \{\mathcal{C}_{\mathsf{cl}}(E), \mathcal{C}_{\mathsf{op}}(E), \mathcal{C}_{\sigma}(E), \mathcal{C}_{\delta}(E)\}$. Since each $\mathcal{F} \in \mathfrak{C}$ satisfies $\mathcal{F} \subseteq \mathcal{C}(E)$, $\alpha(\mathcal{F}) \geq m_p^*(E)$.
Conversely, fix $(I_k) \in \mathcal{C}(E)$ and $\varepsilon > 0$. The chain $I_k \subseteq \overline{I_k} \subseteq J_k$, with J_k open and $\sum_k \operatorname{vol}_p(J_k) \leq \sum_k \operatorname{vol}_p(I_k) + \varepsilon$, followed by disjoint σ-half-open decomposition and finite δ-subdivision, produces covers in every $\mathcal{F} \in \mathfrak{C}$ of total volume $\leq \sum_k \operatorname{vol}_p(I_k) + \varepsilon$. Infimize over (I_k) and let $\varepsilon \downarrow 0$.
Structural Commentary: Cover normalizations preserve the infimum.
Key Insight: Change endpoints, then refine. □

Proposition 3.2.12. *Basic Properties of Cell-Cover Outer Measure*
For $E, F, E_n \subseteq \mathbb{R}^p$,

(i) Null empty set. $0 \leq m_p^*(E) \leq +\infty$ and $m_p^*(\emptyset) = 0$.
(ii) Monotonicity. $E \subseteq F \Longrightarrow m_p^*(E) \leq m_p^*(F)$.
(iii) Subadditivity. $m_p^*(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} m_p^*(E_n)$.

Proof.
(i) Since all cover sums lie in $[0, +\infty]$, $0 \leq m_p^*(E) \leq +\infty$; moreover, $m_p^*(\emptyset) = 0$ by the cover $(\emptyset, \emptyset, \ldots)$.
(ii) If $E \subseteq F$, then $\mathcal{C}(F) \subseteq \mathcal{C}(E)$, hence $\inf_{\mathcal{C}(E)} \leq \inf_{\mathcal{C}(F)}$.
(iii) If some $m_p^*(E_n) = +\infty$, the claim is trivial.
Otherwise, for $\varepsilon > 0$, choose $(I_k^n)_{k\geq 1} \in \mathcal{C}(E_n)$ with $\sum_k \operatorname{vol}_p(I_k^n) \leq m_p^*(E_n) + \varepsilon 2^{-n}$.
Then $(I_k^n)_{(n,k)\in\mathbb{N}^2}$ countable, covers $\bigcup_n E_n$, and $m_p^*(\bigcup_n E_n) \leq \sum_{n,k} \operatorname{vol}_p(I_k^n) \leq \sum_n m_p^*(E_n) + \varepsilon$.
Structural Commentary: Countable subadditivity is proved by merging near-optimal covers.
Key Insight: Nonnegative double sums may be scanned in one sequence. □

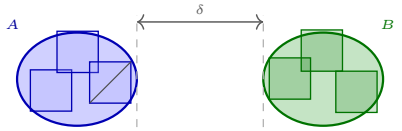

Figure 190: Positive distance separates cells with $\operatorname{diam} I_k < \delta$.

Figure 190 shows why a cell of diameter below the gap δ cannot meet both sets.

Theorem 3.2.13. *Additivity at Positive Distance*
Let $A, B \subseteq \mathbb{R}^p$ be nonempty and disjoint, and suppose $\operatorname{dist}(A, B) := \inf\{\|a - b\|_2 : a \in A,\ b \in B\} > 0$.
Then $m_p^*(A \cup B) = m_p^*(A) + m_p^*(B)$.

Proof. By Proposition 3.2.12, $m_p^*(A \cup B) \leq m_p^*(A) + m_p^*(B)$. For the reverse inequality, set $\delta := \operatorname{dist}(A, B)$. If $m_p^*(A \cup B) = +\infty$, done.
Otherwise, given $\varepsilon > 0$, choose $(I_n) \in \mathcal{C}(A \cup B)$ with $\operatorname{diam} I_n < \delta$ and $\sum_n \operatorname{vol}_p(I_n) \leq m_p^*(A \cup B) + \varepsilon$.
Put $N_A := \{n : I_n \cap A \neq \emptyset\}$ and $N_B := \{n : I_n \cap B \neq \emptyset\}$. Since $N_A \cap N_B = \emptyset$, $(I_n)_{n\in N_A}$ covers A, $(I_n)_{n\in N_B}$ covers B, and $m_p^*(A) + m_p^*(B) \leq \sum_{n\in N_A} \operatorname{vol}_p(I_n) + \sum_{n\in N_B} \operatorname{vol}_p(I_n) \leq \sum_n \operatorname{vol}_p(I_n) \leq m_p^*(A \cup B) + \varepsilon$. Let $\varepsilon \downarrow 0$.
Structural Commentary: Positive separation lets a sufficiently fine cover be sorted unambiguously.
Key Insight: Countable subadditivity becomes equality when no small cell can touch both pieces. □

Theorem 3.2.14. *Outer Measure of a Cell*
If I is a cell in $\mathbb{R}^p$, then $m_p^*(I) = \operatorname{vol}_p(I)$.

Proof. By the cover $(I, \emptyset, \ldots)$, $m_p^*(I) \leq \operatorname{vol}_p(I)$. For the reverse inequality, the case $\operatorname{vol}_p(I) = 0$ follows by nonnegativity.
Assume $\operatorname{vol}_p(I) > 0$ and fix $\varepsilon > 0$. Choose a closed cell $J \subseteq I$ with $\operatorname{vol}_p(I) - \varepsilon < \operatorname{vol}_p(J)$, taking $J = I$ when I is closed. For any open-cell cover (G_k) of I, $J \subseteq \bigcup_{k=1}^{m} G_k$ for some m by compactness.
Let $\mathcal{Q}$ be the finite rectangular grid determined by the faces of $J, \overline{G_1}, \ldots, \overline{G_m}$. Up to zero-volume boundary atoms, each $Q \in \mathcal{Q}$ with $Q \subseteq J$ lies in some G_k, hence $\operatorname{vol}_p(J) \leq \sum_{k=1}^{m} \operatorname{vol}_p(G_k) \leq \sum_{k=1}^{\infty} \operatorname{vol}_p(G_k)$.
By Proposition 3.2.11, the infimum of the open-cell cover sums is $m_p^*(I)$, so $\operatorname{vol}_p(I) - \varepsilon < \operatorname{vol}_p(J) \leq m_p^*(I)$. Let $\varepsilon \downarrow 0$.
Structural Commentary: Compact inner cells reduce covers to finite grids.
Key Insight: Outer measure is calibrated so that cells keep their original volume. □

Corollary 3.2.15. *Outer Measure of Real Intervals*
Let $J \subseteq \mathbb{R}$ be a nonempty interval, and set $\ell(J) = +\infty$ when J is unbounded. Then $m_1^*(J) = \ell(J)$.

Proof. If J is bounded, then J is a one-dimensional cell, so $m_1^*(J) = \mathrm{vol}_1(J) = \ell(J)$ by Theorem 3.2.14.
If J is unbounded, then for every $N \in \mathbb{N}$ there exists a bounded interval $I_N \subseteq J$ with $\ell(I_N) = N$.
Hence $m_1^*(J) \geq m_1^*(I_N) = N$ by monotonicity and the bounded case. Thus $m_1^*(J) = +\infty = \ell(J)$.
Structural Commentary: Cells calibrate bounded intervals; monotonicity handles unbounded ones.
Key Insight: Infinite length is forced by arbitrarily long bounded subintervals. □

Theorem 3.2.16. *Translation Invariance of Outer Measure*
For $E \subseteq \mathbb{R}^p$ and $t \in \mathbb{R}^p$, $m_p^*(t + E) = m_p^*(E)$.

Proof. The map $\mathcal{C}(E) \to \mathcal{C}(t + E)$, $(I_k) \mapsto (t + I_k)$, is bijective with inverse $(J_k) \mapsto (-t + J_k)$.
By Proposition 3.2.5, $\sum_k \mathrm{vol}_p(t + I_k) = \sum_k \mathrm{vol}_p(I_k)$, so the two infima are equal.
Structural Commentary: Translate covers forward and backward.
Key Insight: Translation invariance of cell volume lifts directly to translation invariance of outer measure. □

3.3 Caratheodory Measurability and Lebesgue Measure

Definition 3.3.1. *Outer Measure and μ-Measurable Sets*
Let $\mathcal{E}$ and $\mathcal{E}_{\mathrm{op}}$ denote the elementary and open elementary subsets of $\mathbb{R}^p$, let $\mu : \mathcal{E} \to [0, \infty)$ be additive and regular, and set $A \triangle B := (A \setminus B) \cup (B \setminus A)$.

(i) For $E \subseteq \mathbb{R}^p$, $\mu^*(E) := \inf\{\sum_{n=1}^{\infty} \mu(A_n) : E \subseteq \bigcup_{n=1}^{\infty} A_n,\ A_n \in \mathcal{E}_{\mathrm{op}}\}$.
(ii) $d_\mu(A, B) := \mu^*(A \triangle B)$.
(iii) $\mathcal{M}_F(\mu)$ is the class of *finitely μ-measurable* sets: $E \in \mathcal{M}_F(\mu)$ iff $\exists (E_k)_{k \geq 1} \subseteq \mathcal{E}$ with $d_\mu(E_k, E) \to 0$.
(iv) $\mathcal{M}(\mu)$ is the class of *μ-measurable* sets: $E \in \mathcal{M}(\mu)$ iff $E = \bigcup_{k=1}^{\infty} E_k$ for some $E_k \in \mathcal{M}_F(\mu)$.

Perspective 3.3.2. *Outer Measure as Outside Approximation*
Outer measure covers A from outside; $d_\mu(A, B) = \mu^*(A \triangle B)$ measures closeness.
Key Insight: Countable subadditivity collects near-optimal covers.

Theorem 3.3.3. *Elementary Approximation Extension*
Let $\mathcal{E}$, $\mathcal{E}_{\mathrm{op}}$, μ^*, d_μ, $\mathcal{M}_F(\mu)$, and $\mathcal{M}(\mu)$ be as in Definition 3.3.1. Then:

(i) $\mu^*(A) = \mu(A)$ for every $A \in \mathcal{E}$;
(ii) μ^* is countably subadditive on $2^{\mathbb{R}^p}$;
(iii) $\mathcal{M}(\mu)$ is a σ-ring;
(iv) $\mu^*|_{\mathcal{M}(\mu)}$ is a measure.

Proof.

(i) Let $A \in \mathcal{E}$. For $\varepsilon > 0$, choose $G \in \mathcal{E}_{\mathrm{op}}$ with $A \subseteq G$ and $\mu(G) \leq \mu(A) + \varepsilon$ by regularity; then $\mu^*(A) \leq \mu(A)$.
Conversely, choose closed elementary $F \subseteq A$ with $\mu(F) \geq \mu(A) - \varepsilon$ by regularity. Every open elementary cover of A covers F; some finite subfamily covers F by compactness, and its total μ-mass is at least $\mu(F)$ by additivity.
After taking infima and letting $\varepsilon \downarrow 0$, $\mu(A) \leq \mu^*(A)$.
(ii) Let $E_j \subseteq \mathbb{R}^p$ for $j \geq 1$, and let $\varepsilon > 0$. For each j, choose an open elementary cover $(A_{j,n})_{n \geq 1}$ of E_j with $\sum_{n=1}^{\infty} \mu(A_{j,n}) \leq \mu^*(E_j) + \varepsilon 2^{-j}$. Then $(A_{j,n})_{j,n}$ covers $\bigcup_{j=1}^{\infty} E_j$, so $\mu^*(\bigcup_j E_j) \leq \sum_j \mu^*(E_j) + \varepsilon$; let $\varepsilon \downarrow 0$.
(iii) If $A_k \to A$ and $B_k \to B$ in d_μ, with $A_k, B_k \in \mathcal{E}$, then $d_\mu(A_k \diamond B_k, A \diamond B) \leq d_\mu(A_k, A) + d_\mu(B_k, B)$ for $\diamond \in \{\cup, \cap, \setminus\}$. Hence $\mathcal{M}_F(\mu)$ is a ring. Since difference distributes over countable unions, $\mathcal{M}(\mu)$ is a σ-ring.
(iv) If $A = \bigsqcup_{j=1}^{\infty} A_j$ with $A_j \in \mathcal{M}_F(\mu)$, then $\mu^*(A) \leq \sum_j \mu^*(A_j)$ by subadditivity.
By finite additivity on $\mathcal{M}_F(\mu)$, $\mu^*(A) \geq \sum_{j=1}^{N} \mu^*(A_j)$ for every N, hence $\mu^*(A) \geq \sum_j \mu^*(A_j)$.
Disjointize countable $\mathcal{M}_F(\mu)$-decompositions.

Structural Commentary: Elementary sets are completed under the distance measured by symmetric difference.
Key Insight: Approximate by elementary sets, prove identities there, then pass through $d_\mu(A, B)$. □

Definition 3.3.4. *Extended and Lebesgue Measure*
Let μ^* be the outer measure induced by μ and let $\mathcal{M}(\mu)$ be its measurable class.

(i) The *extended measure* of μ is $\mu_{\mathrm{ext}} := \mu^*|_{\mathcal{M}(\mu)}$, usually denoted again by μ.
(ii) If $\mu = m$ is elementary volume on $\mathbb{R}^p$, then $m^*|_{\mathcal{M}(m)}$ is *Lebesgue measure* on $\mathbb{R}^p$.

By Proposition 3.2.11, the outer measure m_n^* from Definition 3.2.10 equals the infimum of total box volume over countable open-box covers.

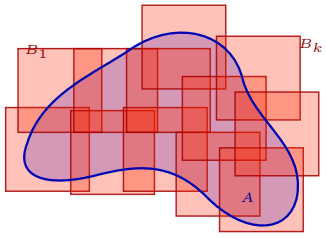

Figure 191: Cover $A \subseteq \bigcup_k B_k$ by squares; minimize $\sum_k |B_k|$.

Figure 191 depicts the open-box minimization used in the first construction step.

Method 3.3.5. *Lebesgue Measure Construction*

(i) *Cover from outside*
For $A \subseteq \mathbb{R}^n$, define $m_n^*(A) := \inf\left\{\sum_{j\in J} \mathrm{vol}_n(B_j) : (B_j)_{j\in J} \text{ covers } A,\ J \text{ at most countable}\right\}$.
This is defined for every set, but it is only countably subadditive in general.

(ii) *Keep the clean splitters*
Call A measurable when every test set $X \subseteq \mathbb{R}^n$ splits additively: $m_n^*(X) = m_n^*(X \cap A) + m_n^*(X \cap A^c)$.
These measurable sets form a σ-algebra.

(iii) *Restrict the outer measure*
Define Lebesgue measure on measurable sets by $m(A) := m_n^*(A)$.
On this restricted class, m is countably additive.

Key Insight: Outer measure sizes all sets; clean splitting restores additivity.

Definition 3.3.6. *Lebesgue σ-Algebra via Caratheodory*

(i) $\mathcal{L}^n := \{E \subseteq \mathbb{R}^n : E \text{ is } m_n^*\text{-measurable}\}$ is the *Lebesgue σ-algebra*.
(ii) $m := m_n^*|_{\mathcal{L}^n}$ is *Lebesgue measure*; for $E \in \mathcal{L}^n$, write $m(E) = m_n^*(E)$.

Proposition 3.3.7. *Outer-Null Sets Are Lebesgue Measurable*
If $Z \subseteq \mathbb{R}^n$ and $m_n^*(Z) = 0$, then:

(i) $Z \in \mathcal{L}^n$;
(ii) $m(Z) = 0$;
(iii) $\mathcal{P}(Z) \subseteq \mathcal{L}^n$ *and* $m(W) = 0$ *for every* $W \subseteq Z$.

Proof.

(i) For $A \subseteq \mathbb{R}^n$, $m_n^*(A \cap Z) = 0$ by monotonicity, so $m_n^*(A) \le m_n^*(A \cap Z) + m_n^*(A \cap Z^c) = m_n^*(A \cap Z^c) \le m_n^*(A)$; hence Z satisfies Caratheodory's criterion.
(ii) Since $Z \in \mathcal{L}^n$, $m(Z) = m_n^*(Z) = 0$.
(iii) If $W \subseteq Z$, then $m_n^*(W) \le m_n^*(Z) = 0$ by monotonicity; by (i)–(ii) applied to W, $W \in \mathcal{L}^n$ and $m(W) = 0$.

Structural Commentary: Lebesgue measure is complete because outer-null subsets split every test set trivially.
Key Insight: A zero piece cannot disturb additivity. □

Example 3.3.8. *Countable Real Sets Are Outer-Null*
Let $C \subseteq \mathbb{R}$ be finite or countable, and choose $(c_k)_{k\ge1}$ with $C \subseteq \{c_k : k \ge 1\}$. For $\varepsilon > 0$, choose open intervals $I_k \ni c_k$ with $\ell(I_k) < \varepsilon 2^{-k}$. Then $C \subseteq \bigcup_{k\ge1} I_k$, so $0 \le m_1^*(C) \le \sum_{k=1}^{\infty} \ell(I_k) < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $m_1^*(C) = 0$. By Proposition 3.3.7, $C \in \mathcal{L}^1$ and $m(C) = 0$.
Key Insight: A countable list can spend a rapidly decreasing measure budget around its points.

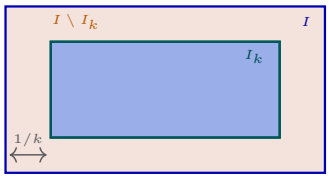

Figure 192: Inner cells I_k exhaust I; the collar $I \setminus I_k$ has vanishing volume.

Figure 192 depicts the shrinking inner cells and vanishing collar used in the positive-volume case.

Proposition 3.3.9. *Cells Are Lebesgue Measurable*
Let $I \subseteq \mathbb{R}^n$ be a coordinate cell. Then $I \in \mathcal{L}^n$ and $m(I) = \mathrm{vol}_n(I)$.

Proof. If $\mathrm{vol}_n(I) = 0$, then $m_n^*(I) = 0$ by Theorem 3.2.14, and $I \in \mathcal{L}^n$ with $m(I) = 0$ by Proposition 3.3.7.
Assume $\mathrm{vol}_n(I) > 0$. For small $r > 0$, let I_r be the inward r-shrink of I.
Then $\mathrm{dist}(I_r, \mathbb{R}^n \setminus I) > 0$ and $\eta_r := m_n^*(I \setminus I_r) \to 0$ by finite slab covers.
Fix $A \subseteq \mathbb{R}^n$ with $m_n^*(A) < +\infty$. The sets $A \cap I_r$ and $A \setminus I$ have positive distance whenever both are nonempty, so $m_n^*(A) \ge m_n^*(A \cap I_r) + m_n^*(A \setminus I)$ by Theorem 3.2.13; the empty cases are immediate.
Since $A \cap I \subseteq (A \cap I_r) \cup (I \setminus I_r)$, $m_n^*(A \cap I) \le m_n^*(A \cap I_r) + \eta_r$; hence $m_n^*(A) \ge m_n^*(A \cap I) + m_n^*(A \setminus I) - \eta_r$.
After letting $r \downarrow 0$, the finite-measure test inequality holds, so $I \in \mathcal{L}^n$ by Lemma 3.1.4.
By Theorem 3.2.14, $m(I) = m_n^*(I) = \mathrm{vol}_n(I)$.
Structural Commentary: Cells are the bridge between coordinate volume and Lebesgue measure.
Key Insight: Shrink, split, let the collar vanish. □

3.4 Caratheodory-Hahn Extension

Definition 3.4.1. *Premeasures on Set Families*
Let $\emptyset \ne \mathcal{S} \subseteq \mathcal{P}(X)$ and $\mu : \mathcal{S} \to [0, +\infty]$.

(i) μ is *finitely additive on $\mathcal{S}$* if $\mu(\bigcup_{k=1}^{n} E_k) = \sum_{k=1}^{n} \mu(E_k)$ $\forall$ finite pairwise disjoint family $E_k \in \mathcal{S}$ with $\bigcup_{k=1}^{n} E_k \in \mathcal{S}$.

(ii) μ is *countably monotone on $\mathcal{S}$* if, for all $E, E_1, E_2, \ldots \in \mathcal{S}$ with $E \subseteq \bigcup_{k=1}^{\infty} E_k$, $\mu(E) \le \sum_{k=1}^{\infty} \mu(E_k)$.

(iii) μ is a *premeasure on $\mathcal{S}$* if μ is finitely additive on $\mathcal{S}$, countably monotone on $\mathcal{S}$, and $\emptyset \in \mathcal{S} \Rightarrow \mu(\emptyset) = 0$.

Structural Commentary: No closure property of $\mathcal{S}$ is assumed; additivity is imposed only for finite disjoint unions which remain in $\mathcal{S}$, and countable monotonicity only for countable $\mathcal{S}$-covers of members of $\mathcal{S}$.

Proposition 3.4.2. *Necessary Conditions for Extension*
Let $\emptyset \neq \mathcal{S} \subseteq \mathcal{P}(X)$, let $\mu : \mathcal{S} \to [0, +\infty]$, let μ^* be the outer measure induced by μ, and put $\overline{\mu} := \mu^*|_{\mathcal{M}_{\mu^*}}$.
If $\mathcal{S} \subseteq \mathcal{M}_{\mu^*}$ and $\overline{\mu}(E) = \mu(E)$ for all $E \in \mathcal{S}$, then the following hold:

(i) $\emptyset \in \mathcal{S} \Rightarrow \mu(\emptyset) = 0$.
(ii) μ *is finitely additive on* $\mathcal{S}$.
(iii) μ *is countably monotone on* $\mathcal{S}$.

Hence μ is a premeasure on $\mathcal{S}$.

Proof. By the extension hypotheses, $\mu(F) = \overline{\mu}(F) = \mu^*(F)$ for every $F \in \mathcal{S}$.

(i) If $\emptyset \in \mathcal{S}$, then $\mu(\emptyset) = \overline{\mu}(\emptyset) = 0$.

(ii) Let $n \geq 1$, let $E_1, \ldots, E_n \in \mathcal{S}$ be pairwise disjoint, and assume $F := \bigcup_{k=1}^{n} E_k \in \mathcal{S}$.

Since $\overline{\mu}$ is a measure, $\mu(F) = \overline{\mu}(F) = \sum_{k=1}^{n} \overline{\mu}(E_k) = \sum_{k=1}^{n} \mu(E_k)$.

(iii) Let $E, E_1, E_2, \ldots \in \mathcal{S}$ and $E \subseteq \bigcup_{k=1}^{\infty} E_k$. By monotonicity and countable subadditivity of μ^*,

$\mu(E) = \mu^*(E) \leq \mu^*(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} \mu^*(E_k) = \sum_{k=1}^{\infty} \mu(E_k)$.

Structural Commentary: The measure extension handles finite disjoint unions; μ^* handles countable covers.
Key Insight: Outer subadditivity supplies the inequality even when $\bigcup_k E_k \notin \mathcal{S}$. □

Definition 3.4.3. *Relative Complements and Semirings*
Let $\emptyset \neq \mathcal{S} \subseteq \mathcal{P}(X)$.

(i) $\mathcal{S}$ is *closed under relative complements* if $A, B \in \mathcal{S} \Rightarrow A \setminus B \in \mathcal{S}$.

(ii) $\mathcal{S}$ is a *semiring of sets* if, for all $A, B \in \mathcal{S}$, $A \cap B \in \mathcal{S}$ and $A \setminus B = \bigsqcup_{j=1}^{n} C_j$ for some $n \geq 0$ and $C_j \in \mathcal{S}$.

Structural Commentary: Relative-complement closure gives $\emptyset = A \setminus A \in \mathcal{S}$ and $A \cap B = A \setminus (A \setminus B) \in \mathcal{S}$.

Theorem 3.4.4. *Extension from Relative-Complement Families*
Let $\emptyset \neq \mathcal{S} \subseteq \mathcal{P}(X)$ be closed under relative complements, let $\mu : \mathcal{S} \to [0, +\infty]$ be a premeasure, and let μ^* be the outer measure induced by μ. Then:

(i) $\mu^*(E) = \mu(E)$ *for every* $E \in \mathcal{S}$.
(ii) $\mathcal{S} \subseteq \mathcal{M}_{\mu^*}$.

Proof.

(i) Fix $E \in \mathcal{S}$. The one-set cover gives $\mu^*(E) \leq \mu(E)$. If $(E_k)_{k \geq 1} \subseteq \mathcal{S}$ covers E, then countable monotonicity gives $\mu(E) \leq \sum_{k=1}^{\infty} \mu(E_k)$; taking the infimum over all such covers gives $\mu(E) \leq \mu^*(E)$.

(ii) Fix $A \in \mathcal{S}$. By Lemma 3.1.4, it suffices to prove $\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \mu^*(B)$ for every $B \subseteq X$ with $\mu^*(B) < +\infty$. Let $\varepsilon > 0$ and choose an $\mathcal{S}$-cover $(E_k)_{k \geq 1}$ of B with $\sum_{k=1}^{\infty} \mu(E_k) < \mu^*(B) + \varepsilon$.

For each k, relative-complement closure gives $E_k \setminus A \in \mathcal{S}$ and $E_k \cap A = E_k \setminus (E_k \setminus A) \in \mathcal{S}$. By finite additivity, $\mu(E_k) = \mu(E_k \cap A) + \mu(E_k \setminus A)$. Since $(E_k \cap A)_{k \geq 1}$ covers $B \cap A$ and $(E_k \setminus A)_{k \geq 1}$ covers $B \setminus A$, $\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \sum_{k=1}^{\infty} \mu(E_k \cap A) + \sum_{k=1}^{\infty} \mu(E_k \setminus A) = \sum_{k=1}^{\infty} \mu(E_k) < \mu^*(B) + \varepsilon$. Let $\varepsilon \downarrow 0$.

Structural Commentary: Countable monotonicity identifies μ with μ^*; finite additivity splits covers.
Key Insight: Relative complements turn one cover into inside/outside covers. □

Remark 3.4.5. *Two Roles of a Premeasure*
In Theorem 3.4.4, finite additivity is used to split each covering set across A and A^c, while countable monotonicity is used to force $\mu^*(E) = \mu(E)$ on the original family.
Key Insight: Measurability comes from finite splitting; extension equality comes from countable cover control.

Proposition 3.4.6. *Finite-Union Extension from a Semiring*
Let $\mathcal{S}$ be a semiring of subsets of X, and set $\mathcal{S}' := \{\bigsqcup_{k=1}^{n} A_k : n \geq 0,\ A_1, \ldots, A_n \in \mathcal{S}\}$. If $\mu : \mathcal{S} \to [0, +\infty]$ is a premeasure, define $\mu' : \mathcal{S}' \to [0, +\infty]$ by $\mu'(\bigsqcup_{k=1}^{n} A_k) := \sum_{k=1}^{n} \mu(A_k)$ for any disjoint representation. Then:

(i) $\mathcal{S}'$ *is closed under relative complements.*
(ii) μ' *is well-defined and extends* μ.
(iii) μ' *is a premeasure on* $\mathcal{S}'$.
(iv) μ' *is the unique premeasure on* $\mathcal{S}'$ *extending* μ.

Proof.

(i) Let $E = \bigsqcup_{i=1}^{r} A_i$ and $F = \bigsqcup_{j=1}^{s} B_j$ in $\mathcal{S}'$.

For each i, repeated semiring differences decompose $A_i \setminus F$ into finitely many disjoint $\mathcal{S}$-sets.
These decompositions, over $i = 1, \ldots, r$, form a finite disjoint $\mathcal{S}$-decomposition of $E \setminus F$, so $E \setminus F \in \mathcal{S}'$.

(ii) If $E = \bigsqcup_{i=1}^{r} A_i = \bigsqcup_{j=1}^{s} B_j$, then $A_i = \bigsqcup_{j=1}^{s} (A_i \cap B_j)$ and $B_j = \bigsqcup_{i=1}^{r} (A_i \cap B_j)$ with all intersections in $\mathcal{S}$.
Finite additivity of μ gives $\sum_{i=1}^{r} \mu(A_i) = \sum_{i=1}^{r} \sum_{j=1}^{s} \mu(A_i \cap B_j) = \sum_{j=1}^{s} \mu(B_j)$, so μ' is well-defined.
If $A \in \mathcal{S}$, the one-piece representation gives $\mu'(A) = \mu(A)$.

(iii) Since $\emptyset$ is the empty union, $\mu'(\emptyset) = 0$.
Finite additivity of μ' follows by refining finitely many disjoint $\mathcal{S}'$-sets to their disjoint $\mathcal{S}$-pieces.
For countable monotonicity, let $E = \bigsqcup_{i=1}^{r} A_i \in \mathcal{S}'$, let $F_m = \bigsqcup_{j=1}^{N_m} B_{mj} \in \mathcal{S}'$, assume $E \subseteq \bigcup_{m=1}^{\infty} F_m$.
For each i, $A_i \subseteq \bigcup_{m,j} (A_i \cap B_{mj})$ with $A_i \cap B_{mj} \in \mathcal{S}$, so $\mu(A_i) \leq \sum_{m,j} \mu(A_i \cap B_{mj})$.
Summing over i gives $\mu'(E) \leq \sum_m \sum_j \sum_i \mu(A_i \cap B_{mj})$.
For fixed m, j, repeated semiring differences decompose $B_{mj} \setminus \bigcup_i (A_i \cap B_{mj})$ into finitely many disjoint $\mathcal{S}$-sets, so finite additivity gives $\sum_i \mu(A_i \cap B_{mj}) \leq \mu(B_{mj})$. Thus $\mu'(E) \leq \sum_m \sum_j \mu(B_{mj}) = \sum_m \mu'(F_m)$.

(iv) If ν is a premeasure on $\mathcal{S}'$ extending μ, then finite additivity gives
$\nu(\bigsqcup_{k=1}^{n} A_k) = \sum_{k=1}^{n} \nu(A_k) = \sum_{k=1}^{n} \mu(A_k) = \mu'(\bigsqcup_{k=1}^{n} A_k)$ for every disjoint representation. Hence $\nu = \mu'$.

Structural Commentary: Finite disjoint unions close a semiring under relative complements.
Key Insight: All estimates reduce to intersections of semiring pieces. □

Theorem 3.4.7. *Caratheodory-Hahn Extension from a Semiring*
Let $\mathcal{S}$ be a semiring of subsets of X, let $\mu : \mathcal{S} \to [0, +\infty]$ be a premeasure, let μ^* be the outer measure induced by μ, and set $\overline{\mu} := \mu^*|_{\mathcal{M}_{\mu^*}}$. Then:

(i) $\mathcal{S} \subseteq \mathcal{M}_{\mu^*}$ and $\overline{\mu}(A) = \mu(A)$ for every $A \in \mathcal{S}$.
(ii) If $X = \bigcup_{k \geq 1} S_k$ with $S_k \in \mathcal{S}$ and $\mu(S_k) < +\infty$, then $\overline{\mu}$ is σ-finite.
(iii) Under the hypothesis in (ii), if ν is a measure on $\mathcal{M}_{\mu^*}$ with $\nu(A) = \mu(A)$ for every $A \in \mathcal{S}$, then $\nu = \overline{\mu}$.

Proof.

(i) Let $(\mathcal{S}', \mu')$ be the finite-union extension from Proposition 3.4.6. The $\mathcal{S}$- and $\mathcal{S}'$-cover infima coincide, so μ^* is also induced by μ'. Theorem 3.4.4 gives $\mathcal{S}' \subseteq \mathcal{M}_{\mu^*}$ and $\mu^*|_{\mathcal{S}'} = \mu'$. Since $\mu'|_{\mathcal{S}} = \mu$, the claim follows.

(ii) Assume $X = \bigcup_{k \geq 1} S_k$, $S_k \in \mathcal{S}$, $\mu(S_k) < +\infty$. By (i), each $S_k \in \mathcal{M}_{\mu^*}$, $\overline{\mu}(S_k) = \mu(S_k) < +\infty$, so $\overline{\mu}$ is σ-finite.

(iii) Assume (ii), and let ν be a measure on $\mathcal{M}_{\mu^*}$ extending μ. By finite additivity, $\nu = \overline{\mu} = \mu'$ on $\mathcal{S}'$.
Hence every $\mathcal{S}'$-cover of $E \in \mathcal{M}_{\mu^*}$ gives $\nu(E) \leq$ its μ'-mass, so $\nu(E) \leq \overline{\mu}(E)$. Put $H_n := \bigcup_{k=1}^{n} S_k \in \mathcal{S}'$.
Then $H_n \uparrow X$ and $\nu(H_n) = \overline{\mu}(H_n) = \mu'(H_n) < +\infty$.
Applying $\nu \leq \overline{\mu}$ to $E \cap H_n$ and $H_n \setminus E$ forces $\nu(E \cap H_n) = \overline{\mu}(E \cap H_n)$, because both measures split the same finite mass H_n.
Since $E \cap H_n \uparrow E$, continuity from below gives $\nu(E) = \overline{\mu}(E)$.

Structural Commentary: Pass to finite disjoint unions, then exhaust by finite-mass pieces.
Key Insight: $\mathcal{S}'$ supplies relative complements and the cover infimum for uniqueness. □

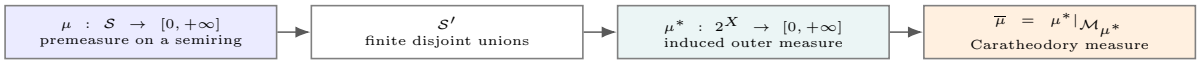

Figure 193: Caratheodory-Hahn: pass to $\mathcal{S}'$, form μ^* by cover infima, then $\overline{\mu} := \mu^*|_{\mathcal{M}_{\mu^*}}$ with $\overline{\mu}|_{\mathcal{S}} = \mu$.

Corollary 3.4.8. *Agreement on a σ-Finite Semiring*
Let $\mathcal{S}$ be a semiring of subsets of X, let $\mathcal{B} := \sigma(\mathcal{S})$, and let ν_1, ν_2 be measures on $\mathcal{B}$ with $\nu_1|_{\mathcal{S}} = \nu_2|_{\mathcal{S}}$.
If $X = \bigcup_{k \geq 1} S_k$ for some $S_k \in \mathcal{S}$ with $\nu_1(S_k) = \nu_2(S_k) < +\infty$ for all k, then $\nu_1 = \nu_2$.

Proof. Let $\mathcal{S}'$ be the finite-disjoint-union closure of $\mathcal{S}$. Finite additivity gives $\nu_1 = \nu_2$ on $\mathcal{S}'$.
Put $H_n := \bigcup_{k=1}^{n} S_k \in \mathcal{S}'$; then $H_n \uparrow X$ and $\nu_i(H_n) < +\infty$. For fixed n, $\mathcal{D}_n := \{E \in \mathcal{B} : \nu_1(E \cap H_n) = \nu_2(E \cap H_n)\}$ is a Dynkin class containing the π-system $\mathcal{S}$, since $A \cap H_n \in \mathcal{S}'$ for $A \in \mathcal{S}$. Hence $\mathcal{D}_n = \mathcal{B}$.
Thus $\nu_1(E \cap H_n) = \nu_2(E \cap H_n)$ $\forall E \in \mathcal{B}$ and $n \geq 1$; since $E \cap H_n \uparrow E$, continuity from below gives $\nu_1(E) = \nu_2(E)$.
Structural Commentary: Localize to finite-mass H_n, then apply π-λ.
Key Insight: Agreement on $\mathcal{S}$ propagates through $\mathcal{S}'$ and the exhaustion. □

Example 3.4.9. *Bounded Intervals and Rectangles Are Semirings*
Let $\mathcal{I}$ be the bounded intervals of $\mathbb{R}$, and let $\mathcal{R}_n := \{I_1 \times \cdots \times I_n : I_j \in \mathcal{I}\}$.

(i) $\mathcal{I}$ is a semiring: if $I, J \in \mathcal{I}$, then $I \cap J \in \mathcal{I}$ and $I \setminus J$ is a disjoint union of at most two members of $\mathcal{I}$.
(ii) $\mathcal{R}_n$ is a semiring in $\mathbb{R}^n$: intersections are coordinatewise products, and $R \setminus Q$ decomposes by coordinate cuts into finitely many disjoint rectangles in $\mathcal{R}_n$.

Thus length on $\mathcal{I}$ and rectangular volume on $\mathcal{R}_n$ fit the Caratheodory-Hahn extension pattern.

3.5 Metric Outer Measures and Hausdorff Measure

Metric outer measures are outer measures whose additivity recognizes positive separation. They abstract the positive-distance additivity of Euclidean cell-cover outer measure from Theorem 3.2.13.

Definition 3.5.1. *Metric/Caratheodory Outer Measure*
Let (X,d) be a metric space and let $\omega : 2^X \to [0,+\infty]$ be an outer measure.

(i) For nonempty $A, B \subseteq X$, $\operatorname{dist}(A,B) := \inf\{d(a,b) : a \in A,\ b \in B\}$.
(ii) ω is *metric outer measure* if $\operatorname{dist}(A,B) > 0 \Rightarrow \omega(A \cup B) = \omega(A) + \omega(B)$ $\forall$ nonempty $A, B \subseteq X$.

Key Insight: Positive metric separation forces outer-measure additivity.

Definition 3.5.2. *Separation by a Real-Valued Function*
Let X be a set and let $\varphi : X \to \mathbb{R}$. Sets $A, B \subseteq X$ are *φ-separated* if, for some $r < s$, one of A, B is contained in $\{\varphi \le r\}$ and the other is contained in $\{\varphi \ge s\}$.

Proposition 3.5.3. *Separation by One Real-Valued Function*
Let X be a set, let $\omega : 2^X \to [0,+\infty]$ be an outer measure, and let $\varphi : X \to \mathbb{R}$. If $\omega(A \cup B) = \omega(A) + \omega(B)$ for every φ-separated pair $A, B \subseteq X$, then $\varphi : (X, \mathcal{M}_\omega) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable.

Proof. Fix $t \in \mathbb{R}$, put $E_t := \{\varphi > t\}$, and fix $A \subseteq X$ with $\omega(A) < +\infty$. Set $B := A \cap E_t$ and $C := A \setminus E_t$.
For $n \ge 1$, let $B_n := A \cap \{\varphi > t + \frac{1}{n}\}$, $B_0 := \emptyset$, and $R_n := B_n \setminus B_{n-1}$.
Same-parity strips are pairwise φ-separated, so $\sum_{n \ge 1} \omega(R_n) < +\infty$. Given $\varepsilon > 0$, choose n with $\sum_{k>n} \omega(R_k) < \varepsilon$.
Since $B \subseteq B_n \cup \bigcup_{k>n} R_k$ and B_n, C are φ-separated, $\omega(B) + \omega(C) \le \omega(B_n \cup C) + \varepsilon \le \omega(A) + \varepsilon$.
Let $\varepsilon \downarrow 0$, then $\omega(A) \ge \omega(A \cap E_t) + \omega(A \setminus E_t)$; Lemma 3.1.4 gives $E_t \in \mathcal{M}_\omega$ for every t, hence φ is measurable.
Structural Commentary: Measurability reduces to a finite-mass split of each level set.
Key Insight: Summable level strips leave an arbitrarily small boundary layer. □

Theorem 3.5.4. *Metric Outer Measures Measure Borel Sets*
Let (X,d) be a metric space and let $\omega : 2^X \to [0,+\infty]$ be a metric outer measure. Then:

(i) Every closed $F \subseteq X$ belongs to $\mathcal{M}_\omega$.
(ii) $\mathcal{B}(X) \subseteq \mathcal{M}_\omega$.

Proof.

(i) The case $F = \emptyset$ is immediate. For nonempty closed F, set $\varphi(x) := \operatorname{dist}(\{x\}, F)$. Proposition 5.1.22 gives φ 1-Lipschitz and $F = \varphi^{-1}(\{0\})$, so φ-separated pairs have positive distance. Metric additivity verifies Proposition 3.5.3; hence φ is $\mathcal{M}_\omega$-measurable and $F \in \mathcal{M}_\omega$.
(ii) By (i), all closed sets lie in the σ-algebra $\mathcal{M}_\omega$; since they generate $\mathcal{B}(X)$, $\mathcal{B}(X) \subseteq \mathcal{M}_\omega$.

Structural Commentary: Distance functions turn closed sets into real-valued separation tests.
Key Insight: Closed measurability plus generation gives Borel measurability. □

Definition 3.5.5. *Hausdorff Outer Measure*
Let (X,d) be a metric space, $\alpha > 0$, $\varepsilon > 0$.

(i) $\mathcal{H}^*_{\alpha,\varepsilon}(\emptyset) := 0$; $\mathcal{H}^*_{\alpha,\varepsilon}(E) := \inf\{\sum_{k=1}^{\infty} (\operatorname{diam} A_k)^\alpha : E \subseteq \bigcup_{k=1}^{\infty} A_k,\ \emptyset \ne A_k \subseteq X,\ \operatorname{diam} A_k < \varepsilon\}$, nonempty $E \subseteq X$
(ii) If $0 < \varepsilon_1 < \varepsilon_2$, then $\mathcal{H}^*_{\alpha,\varepsilon_1}(E) \ge \mathcal{H}^*_{\alpha,\varepsilon_2}(E)$.
(iii) $\mathcal{H}^*_\alpha(E) := \sup_{\varepsilon>0} \mathcal{H}^*_{\alpha,\varepsilon}(E) = \lim_{\varepsilon \downarrow 0} \mathcal{H}^*_{\alpha,\varepsilon}(E)$.

Key Insight: $\mathcal{H}^*_\alpha$ is the limiting cost of countable small-diameter covers.

Figure 194: $\operatorname{dist}(E,F) = \delta$ and $\operatorname{diam} A_k < \delta$: each A_k meets at most one of E, F.

Theorem 3.5.6. *Hausdorff Measure on Borel Sets*
Let (X,d) be a metric space and let $\alpha > 0$. Then:

(i) $\mathcal{H}^*_\alpha : 2^X \to [0,+\infty]$ is a metric outer measure.
(ii) $\mathcal{H}_\alpha := \mathcal{H}^*_\alpha|_{\mathcal{B}(X)}$ is a measure on $\mathcal{B}(X)$, called the α-dimensional Hausdorff measure on X.

Proof.

(i) For each $\varepsilon > 0$, $\mathcal{H}^*_{\alpha,\varepsilon}(\emptyset) = 0$, monotonicity is immediate from admissible covers, and countable subadditivity follows by joining admissible ε-covers; hence $\mathcal{H}^*_\alpha(\bigcup_i E_i) = \sup_{\varepsilon>0} \mathcal{H}^*_{\alpha,\varepsilon}(\bigcup_i E_i) \le \sum_i \mathcal{H}^*_\alpha(E_i)$.
Thus $\mathcal{H}^*_\alpha$ is an outer measure. If $\operatorname{dist}(E,F) = \delta > 0$ and $0 < \varepsilon < \delta$, every admissible ε-cover of $E \cup F$ splits into disjoint subcovers of E and F, so $\mathcal{H}^*_{\alpha,\varepsilon}(E \cup F) \ge \mathcal{H}^*_{\alpha,\varepsilon}(E) + \mathcal{H}^*_{\alpha,\varepsilon}(F)$.
Letting $\varepsilon \downarrow 0$ and using subadditivity gives metric additivity.
(ii) By (i), $\mathcal{H}^*_\alpha$ is a metric outer measure; Theorem 3.5.4 gives $\mathcal{B}(X) \subseteq \mathcal{M}_{\mathcal{H}^*_\alpha}$.
Since $\mathcal{H}^*_\alpha|_{\mathcal{M}_{\mathcal{H}^*_\alpha}}$ is a measure, its restriction $\mathcal{H}^*_\alpha|_{\mathcal{B}(X)} = \mathcal{H}_\alpha$ is a measure on $\mathcal{B}(X)$.

Structural Commentary: Cover infima give the outer measure; separated cover splitting gives metric additivity.
Key Insight: Small-diameter covers cannot straddle positive gaps. □

Remark 3.5.7. *One-Dimensional Calibration*
On $\mathbb{R}$, interval covers give $\mathcal{H}_1^*((a,b]) \le b-a$, and every diameter cover of $(a,b]$ enlarges to interval covers with arbitrarily small extra total length, giving $\mathcal{H}_1^*((a,b]) \ge b-a$. Hence $\mathcal{H}_1^*((a,b]) = b-a$, and uniqueness of the Borel extension from half-open intervals gives $\mathcal{H}_1 = m$ on $\mathcal{B}(\mathbb{R})$.

Example 3.5.8. *Finite Sets Have Zero Positive-Dimensional Hausdorff Measure*
If $E = \{x_1, \dots, x_N\} \subseteq X$ and $\alpha > 0$, then $\mathcal{H}_\alpha^*(E) = 0$: for every $\varepsilon > 0$, the singleton cover $E \subseteq \bigcup_{j=1}^{N} \{x_j\}$ is admissible and has total α-content $\sum_{j=1}^{N} (\operatorname{diam}\{x_j\})^\alpha = 0$.
Key Insight: Finite sets are covered by zero-diameter singletons.

Proposition 3.5.9. *Hausdorff Measures Decrease Through Dimension*
Let (X,d) be a metric space, let $E \subseteq X$, and let $0 < \alpha < \beta$. If $\mathcal{H}_\alpha^*(E) < +\infty$, then $\mathcal{H}_\beta^*(E) = 0$.

Proof. Fix $\varepsilon > 0$. Choose admissible ε-cover (A_k) of E with $\sum_k (\operatorname{diam} A_k)^\alpha \le \mathcal{H}_{\alpha,\varepsilon}^*(E) + 1 \le \mathcal{H}_\alpha^*(E) + 1$.
Since $\operatorname{diam} A_k < \varepsilon$, $\sum_k (\operatorname{diam} A_k)^\beta = \sum_k (\operatorname{diam} A_k)^{\beta-\alpha} (\operatorname{diam} A_k)^\alpha \le \varepsilon^{\beta-\alpha}(\mathcal{H}_\alpha^*(E) + 1)$.
Hence $\mathcal{H}_{\beta,\varepsilon}^*(E) \le \varepsilon^{\beta-\alpha}(\mathcal{H}_\alpha^*(E) + 1)$. Letting $\varepsilon \downarrow 0$ gives $\mathcal{H}_\beta^*(E) = 0$.
Structural Commentary: Compare one small cover at dimensions α and β.
Key Insight: The factor $\varepsilon^{\beta-\alpha}$ kills finite α-content. □

Definition 3.5.10. *Hausdorff Dimension*
For $E \subseteq X$, the *Hausdorff dimension* is $\dim_H(E) := \inf\{\beta > 0 : \mathcal{H}_\beta^*(E) = 0\}$, with $\inf \emptyset := +\infty$.

Remark 3.5.11. *Dimension as a Critical Exponent*
For $E \subseteq X$, set $Z_E := \{\beta > 0 : \mathcal{H}_\beta^*(E) = 0\}$. Proposition 3.5.9 gives $Z_E = (\dim_H(E), +\infty)$ or $[\dim_H(E), +\infty)$ inside $(0, +\infty)$, so $\dim_H(E)$ is the zero-measure threshold for the Hausdorff scale.

4 Borel, Lebesgue, and Stieltjes Measures

4.1 Borel, Lebesgue, and Stieltjes Measure Constructions

Proposition 4.1.1. *Half-Open Interval Algebra on $\mathbb{R}$*
Let $\mathcal{F}$ be the algebra of finite disjoint unions of $(a,b]$, $(-\infty,b]$, $(a,+\infty)$ $(a,b \in \mathbb{R})$, and $\mathbb{R}$. Set $\ell((a,b]) = b-a$ and $\ell((-\infty,b]) = \ell((a,+\infty)) = \ell(\mathbb{R}) = +\infty$, extending additively over disjoint unions. Then
(i) $\mathcal{F}$ is an algebra on $\mathbb{R}$ and $\sigma(\mathcal{F}) = \mathcal{B}(\mathbb{R})$.
(ii) ℓ is a premeasure on $\mathcal{F}$.

Proof.
(i) Generator complements lie in $\mathcal{F}$, hence $\mathcal{F}$ is an algebra.
Since $\mathcal{F} \subseteq \mathcal{B}(\mathbb{R})$ and $(a,b) = \bigcup_{n > 1/(b-a)} (a, b-1/n]$, $\sigma(\mathcal{F}) = \mathcal{B}(\mathbb{R})$.
(ii) Normal forms make ℓ well defined and finitely additive.
For countable additivity, reduce to $(a,b] = \bigsqcup_{j\ge1} (a_j, b_j]$: finite subunions give $\sum_{j=1}^{N} (b_j - a_j) \le b-a$.
Conversely, choose $\varepsilon_j > 0$ with $\sum_{j\ge1} \varepsilon_j < \varepsilon$; by compactness, $\exists$ finite $K \subseteq \mathbb{N}$ s.t. $[a+\delta, b] \subseteq \bigcup_{j\in K} (a_j, b_j + \varepsilon_j)$, hence $b - a - \delta \le \sum_{j\in K} (b_j - a_j + \varepsilon_j) \le \sum_{j\ge1} (b_j - a_j) + \varepsilon$.
Let $\delta, \varepsilon \downarrow 0$. Truncation handles half-lines; finite disjoint unions follow componentwise.
Structural Commentary: Lebesgue measure starts as length on a finite interval algebra, not on all subsets.
Key Insight: Compactness supplies the reverse inequality in countable additivity of interval length. □

Remark 4.1.2. *Why the Half-Open Convention Matters*
Finite unions generated by (a,b), $(-\infty,b)$, $(a,+\infty)$, and $\mathbb{R}$ still generate $\mathcal{B}(\mathbb{R})$, but are not an algebra, where $(0,1)^c = (-\infty,0] \cup [1,+\infty)$. Half-open endpoints make complements remain in the starting family.

Definition 4.1.3. *Borel–Stieltjes and Lebesgue–Stieltjes Measures*
Let $g : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, and set $g(\pm\infty) := \lim_{x\to\pm\infty} g(x) \in \overline{\mathbb{R}}$.
Let $\mathcal{I} := \{(a,b] : a < b\} \cup \{(-\infty,b] : b \in \mathbb{R}\} \cup \{(a,+\infty) : a \in \mathbb{R}\} \cup \{\mathbb{R}\}$ and $\mathcal{F} := \operatorname{alg}(\mathcal{I})$.
(i) Define G_g on $\mathcal{I}$ by endpoint increments and set $\mu_g(\bigsqcup_{j=1}^m I_j) := \sum_{j=1}^m G_g(I_j)$ for disjoint $I_j \in \mathcal{I}$.
(ii) The unique extension of μ_g to $\mathcal{B}(\mathbb{R})$ is the *Borel–Stieltjes measure generated by g*.
(iii) The completion of this Borel measure is the *Lebesgue–Stieltjes measure generated by g*.

Theorem 4.1.4. *Borel–Stieltjes Measure Existence*
Let $g : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, and put $g(\pm\infty) := \lim_{x\to\pm\infty} g(x) \in \overline{\mathbb{R}}$.
Set $\mathcal{I} := \{(a,b] : a < b\} \cup \{(-\infty,b] : b \in \mathbb{R}\} \cup \{(a,+\infty) : a \in \mathbb{R}\} \cup \{\mathbb{R}\}$ and $\mathcal{F} := \operatorname{alg}(\mathcal{I})$.
Define G_g on $\mathcal{I}$ by endpoint increments and set $\mu_g(\bigsqcup_{j=1}^m I_j) := \sum_{j=1}^m G_g(I_j)$ for disjoint $I_j \in \mathcal{I}$. Then:
(i) μ_g is a σ-finite premeasure on $\mathcal{F}$ and has a unique extension to $\mathcal{B}(\mathbb{R})$, again denoted μ_g.
(ii) If $g = \mathrm{id}_{\mathbb{R}}$, then μ_g is Lebesgue measure.
(iii) If g is continuous, then $\mu_g(J) = g(b) - g(a)$ for $J \in \{(a,b),[a,b),(a,b],[a,b]\}$, $a < b$.

Proof.

(i) Normal forms give well-defined finite additivity: G_g-increments telescope and $G_g \geq 0$.
If $(a,b] = \bigsqcup_{j\geq 1}(a_j,b_j]$, finite subunions give $\sum_{j=1}^{N} G_g((a_j,b_j]) \leq G_g((a,b])$.
Conversely, compactness gives finite K with $[a+\varepsilon,b] \subseteq \bigcup_{j\in K}(a_j,b_j+\varepsilon_j)$. Hence
$g(b)-g(a+\varepsilon) \leq \sum_{j\in K}(g(b_j+\varepsilon_j)-g(a_j))$. Let $\varepsilon,\varepsilon_j \downarrow 0$, hence countable additivity on bounded generators.
Half-lines follow by truncation, finite unions componentwise, so μ_g is a premeasure. Since $\mathbb{R} = \bigcup_{n\geq 1}(-n,n]$ and $\mu_g((-n,n]) = g(n)-g(-n) < +\infty$, μ_g is σ-finite; Theorem 3.1.16 gives uniqueness.
(ii) If $g = \mathrm{id}_{\mathbb{R}}$, then $G_g((a,b]) = b-a$, so μ_g agrees with half-open length on $\mathcal{F}$.
Uniqueness identifies its Borel extension with Lebesgue measure.
(iii) If g is continuous, then $(x-1/n,x] \downarrow \{x\}$ gives $\mu_g(\{x\}) = \lim_n (g(x)-g(x-1/n)) = 0$.

Structural Commentary: Stieltjes measure replaces ordinary length $b-a$ by the increment $g(b)-g(a)$.
Key Insight: Right-continuity aligns the half-open interval convention with countable covers. □

Proposition 4.1.5. *Finite Borel Measures Give Stieltjes Generators*
Let ν be finite on $\mathcal{B}(\mathbb{R})$ and set $g(x) := \nu((-\infty,x])$. Then:

(i) g is increasing and right-continuous;
(ii) $g(-\infty) = 0$ and $g(+\infty) = \nu(\mathbb{R})$;
(iii) $\nu((a,b]) = g(b)-g(a)$ for $a \leq b$;
(iv) $\nu = \mu_g$ on $\mathcal{B}(\mathbb{R})$.

Proof.

(i) If $a \leq b$, then $(-\infty,a] \subseteq (-\infty,b]$, so $g(a) \leq g(b)$. If $x_n \downarrow x$, then $(-\infty,x_n] \downarrow (-\infty,x]$.
Finite continuity from above gives $g(x_n) \downarrow g(x)$.
(ii) Since $(-\infty,-n] \downarrow \emptyset$ and $(-\infty,n] \uparrow \mathbb{R}$, continuity from above/below gives $g(-\infty) = 0$ and $g(+\infty) = \nu(\mathbb{R})$.
(iii) For $a \leq b$, $(-\infty,b] = (-\infty,a] \sqcup (a,b]$, hence $\nu((a,b]) = g(b)-g(a)$.
(iv) By (ii)–(iii), ν and μ_g agree on $\mathcal{I}$, hence on $\mathcal{F}$. Theorem 4.1.4 gives $\nu = \mu_g$ on $\mathcal{B}(\mathbb{R})$.

Structural Commentary: A finite real-line measure is encoded by its cumulative mass function.
Key Insight: Right-continuity of distribution functions is continuity from above for measures. □

Example 4.1.6. *Basic Borel–Stieltjes Generators*
The following generators illustrate the interval rule in Theorem 4.1.4.

(i) If $g(x) = 2x$, then $\mu_g = 2m$ on $\mathcal{B}(\mathbb{R})$.
(ii) If $g(x) = \arctan x$, then $\mu_g((a,b]) = \arctan b - \arctan a = \int_a^b (1+x^2)^{-1}\,dx$ and $\mu_g(\mathbb{R}) = \pi$.
(iii) If $g(x) = 0$ for $x < 0$ and $g(x) = 1$ for $x \geq 0$, then $\mu_g = \delta_0$.
(iv) If $g(x) = 0$ for $x < 0$ and $g(x) = x$ for $x \geq 0$, then $\mu_g(E) = m(E \cap [0,+\infty))$ for Borel E.

Lemma 4.1.7. *Open Sets Are Countable Unions of Rational Boxes*
For every open $G \subseteq \mathbb{R}^n$, there is a countable $\Sigma \subseteq \{\prod_{i=1}^n (a_i,b_i) : a_i,b_i \in \mathbb{Q},\ a_i < b_i\}$ such that $G = \bigcup_{B\in\Sigma} B$.

Proof. Let $\mathcal{B}_{\mathbb{Q}} := \{\prod_{i=1}^n (a_i,b_i) : a_i,b_i \in \mathbb{Q},\ a_i < b_i\}$; then $\mathcal{B}_{\mathbb{Q}}$ is countable since $\mathbb{Q}^{2n}$ is countable.
Set $\Sigma := \{B \in \mathcal{B}_{\mathbb{Q}} : B \subseteq G\}$. The inclusion $\bigcup_{B\in\Sigma} B \subseteq G$ is immediate.
If $x \in G$, choose $r > 0$ with $B_r(x) \subseteq G$, then choose $a_i,b_i \in \mathbb{Q}$ with $x_i - r/n < a_i < x_i < b_i < x_i + r/n$.
Thus $x \in B := \prod_i (a_i,b_i) \subseteq B_r(x) \subseteq G$, so $x \in \bigcup_{B\in\Sigma} B$.
Structural Commentary: Rational boxes give a countable Euclidean base.
Key Insight: Openness supplies a ball; density of $\mathbb{Q}$ supplies rational box endpoints inside it. □

Theorem 4.1.8. *Lebesgue Measure Basic Properties*
Let $m = m_n^*|_{\mathcal{L}^n}$.

(i) $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{L}^n$, and $\mathcal{L}^n$ is closed under $E \mapsto E^c$, countable $\cup$, and countable $\cap$.
(ii) For $E,F,E_k \in \mathcal{L}^n$, $0 \leq m(E)$, $E \subseteq F \Rightarrow m(E) \leq m(F)$, and $m(\bigcup_k E_k) \leq \sum_k m(E_k)$.
(iii) If $(E_k) \subseteq \mathcal{L}^n$ are pairwise disjoint, then $m(\bigsqcup_k E_k) = \sum_k m(E_k)$.
(iv) $m([0,1]^n) = 1$, and $E \in \mathcal{L}^n$, $t \in \mathbb{R}^n$ imply $E+t \in \mathcal{L}^n$ and $m(E+t) = m(E)$.

Proof.

(i) By Theorem 3.1.6, $\mathcal{L}^n$ is a σ-algebra, hence closed under complements and countable unions/intersections. Open boxes are cells, hence in $\mathcal{L}^n$ by Proposition 3.3.9; by Lemma 4.1.7, each open $G = \bigcup_j B_j$ with rational open boxes B_j, so $G \in \mathcal{L}^n$. Thus $\mathcal{B}(\mathbb{R}^n) = \sigma(\mathcal{O}_{\mathbb{R}^n}) \subseteq \mathcal{L}^n$.
(ii) Since $m = m_n^*|_{\mathcal{L}^n}$, nonnegativity follows from $m_n^* \geq 0$. If $E \subseteq F$, then $m(E) \leq m(F)$ by outer-measure monotonicity. Also $m(\bigcup_k E_k) = m_n^*(\bigcup_k E_k) \leq \sum_k m_n^*(E_k) = \sum_k m(E_k)$ by outer-measure subadditivity.
(iii) If $(E_k) \subseteq \mathcal{L}^n$ pairwise disjoint, by Theorem 3.1.6, m countably additive on $\mathcal{L}^n$, so $m(\bigsqcup_k E_k) = \sum_k m(E_k)$.
(iv) By Theorem 3.2.14, since $[0,1]^n$ is a cell, $m([0,1]^n) = \mathrm{vol}_n([0,1]^n) = 1$.
For translation, let $A \subseteq \mathbb{R}^n$. By Theorem 3.2.16 and
$E \in \mathcal{L}^n$, $m_n^*(A) = m_n^*(A-t) = m_n^*((A-t)\cap E) + m_n^*((A-t)\cap E^c) = m_n^*(A\cap(E+t)) + m_n^*(A\cap(E+t)^c)$; hence $E+t \in \mathcal{L}^n$, and $m(E+t) = m_n^*(E+t) = m_n^*(E) = m(E)$.

Structural Commentary: The wish list is recovered only after restricting outer measure to measurable sets.
Key Insight: The box-cover construction gives size; the measurable-set condition restores additivity. □

Theorem 4.1.9. *Uniqueness from Open Cells*
Let ν be a measure on $\mathcal{L}^n$ with $\nu(I) = \mathrm{vol}_n(I)$ for every open cell $I \subseteq \mathbb{R}^n$. Then $\nu = m$ on $\mathcal{L}^n$.

Proof. Fix $N \in \mathbb{N}$, $Q_N = (-N, N)^n$. If $E \in \mathcal{L}^n$ and $E \subseteq Q_N$, then any open-cell cover $E \subseteq \bigcup_j J_j$ gives $\nu(E) \leq \sum_j \nu(J_j) = \sum_j \mathrm{vol}_n(J_j)$, hence $\nu(E) \leq m(E)$ by infimum.
Applying this to E and $Q_N \setminus E$, and using $\nu(Q_N) = m(Q_N) < +\infty$, yields $\nu(E) + \nu(Q_N \setminus E) = m(E) + m(Q_N \setminus E)$ with each left summand $\leq$ its right counterpart; hence $\nu(E) = m(E)$.
For arbitrary $E \in \mathcal{L}^n$, set $A_1 = E \cap Q_1$, $A_N = E \cap (Q_N \setminus Q_{N-1})$ for $N > 1$.
Then $E = \bigsqcup_N A_N$, each $A_N \subseteq Q_N$, and countable additivity gives $\nu(E) = \sum_N \nu(A_N) = \sum_N m(A_N) = m(E)$.
Structural Commentary: Agreement on coordinate cells determines the whole Lebesgue measure.
Key Insight: Covers give $\nu \leq m$ locally; finite complements force equality. □

Method 4.1.10. *Translation Invariance by Moving the Test Set*

(i) For $t \in \mathbb{R}^n$ and $A \subseteq \mathbb{R}^n$, translated box covers give $m_n^*(A + t) = m_n^*(A)$.
(ii) If $E \in \mathcal{L}^n$, then for all $A \subseteq \mathbb{R}^n$, $(A \cap (E + t)) - t = (A - t) \cap E$ and $(A \cap (E + t)^c) - t = (A - t) \cap E^c$.
(iii) Apply the Caratheodory split for E to $A - t$ to get $E + t \in \mathcal{L}^n$ and $m(E + t) = m(E)$.

Key Insight: Translation invariance of outer measure transports the clean-splitting identity.

Proposition 4.1.11. *First Borel Hierarchy Examples*
In $\mathbb{R}^p$:

(i) If $G = \bigcap_{n \geq 1} U_n$ with U_n open or $F = \bigcup_{n \geq 1} C_n$ with C_n closed, then $G, F \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$.
(ii) Any set obtained from open or closed sets by countably many $\cup, \cap, {}^c$ operations is Borel.
(iii) In $\mathbb{R}$, $\{0\} = \bigcap_{n \geq 1} (-\frac{1}{n}, \frac{1}{n})$ is G_δ but not open.
(iv) In $\mathbb{R}$, $(0, 1) = \bigcup_{n \geq 2} [\frac{1}{n}, 1 - \frac{1}{n}]$ is F_σ but not closed.

Proof.

(i) Since $\mathcal{B}(\mathbb{R}^p) = \sigma(\mathcal{O}_{\mathbb{R}^p})$, it contains open sets, closed sets by complements, and countable $\cap, \cup$ by σ-algebra closure; Theorem 4.1.8 gives $\mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$.
(ii) This is Proposition 1.1.3 applied to $\mathcal{B}(\mathbb{R}^p)$.
(iii) For $x \in \mathbb{R}$, $x = 0 \iff |x| < \frac{1}{n}$ for all $n \geq 1$; the singleton $\{0\}$ is not open.
(iv) For $x \in \mathbb{R}$, $0 < x < 1 \iff \exists n \geq 2 : \frac{1}{n} \leq x \leq 1 - \frac{1}{n}$; the interval $(0, 1)$ is not closed.

Structural Commentary: Borel sets are generated by countable topological operations.
Key Insight: Countable intersections and unions quickly leave the class of open and closed sets. □

Perspective 4.1.12. *Why Not Measure Every Subset?*
Vitali-type sets show m_1^* is not additive on $2^{\mathbb{R}}$: disjoint unions and refinements can break additivity.
Thus measure is m^* on clean splitters E: $m^*(X) = m^*(X \cap E) + m^*(X \cap E^c)$ for all X.
Key Insight: Outer measure is universal; measure lives on clean splitters.

Definition 4.1.13. *Null Sets and Real-Line Measure Zero*
Let $(X, \mathcal{A}, \mu)$ be a measure space and let $A \subseteq \mathbb{R}$.

(i) $N \subseteq X$ is *null* iff $N \in \mathcal{A}$ and $\mu(N) = 0$.
(ii) A has *measure zero* iff $\forall \varepsilon > 0$, $\exists$ open intervals (O_n) with $A \subseteq \bigcup_{n=1}^\infty O_n$ and $\sum_{n=1}^\infty |O_n| \leq \varepsilon$.
(iii) If $A \subseteq [a, b]$, then A has *content zero* iff the condition in (ii) holds with finitely many intervals.

For Lebesgue measure on $\mathbb{R}$, null sets are also *zero sets*.

Proposition 4.1.14. *Basic Measure-Zero Examples on $\mathbb{R}$*

(i) Every finite $A \subseteq \mathbb{R}$ has content 0.
(ii) Every countable $A \subseteq \mathbb{R}$ has measure 0.
(iii) The Cantor set P has content 0.

Proof.

(i) For $A = \{a_1, \ldots, a_N\}$ and $\varepsilon > 0$, choose open $I_j \ni a_j$ with $|I_j| < \frac{\varepsilon}{N}$; then $A \subseteq \bigcup_{j=1}^N I_j$ and $\sum_{j=1}^N |I_j| < \varepsilon$. The case $A = \emptyset$ is trivial.
(ii) For $A = \{a_j : j \in \mathbb{N}\}$ and $\varepsilon > 0$, choose open $I_j \ni a_j$ with $|I_j| < \frac{\varepsilon}{2^j}$; then $A \subseteq \bigcup_{j \geq 1} I_j$ and $\sum_{j \geq 1} |I_j| < \varepsilon$.
(iii) Given $\varepsilon > 0$, choose N with $\left(\frac{2}{3}\right)^N < \frac{\varepsilon}{2}$. If $E_N = \bigcup_{k=1}^{2^N} J_k$, $|J_k| = 3^{-N}$, choose open $I_k \supseteq J_k$ with $|I_k| < 3^{-N} + \frac{\varepsilon}{2^{N+1}}$; then $P \subseteq \bigcup_{k=1}^{2^N} I_k$ and $\sum_{k=1}^{2^N} |I_k| < \left(\frac{2}{3}\right)^N + \frac{\varepsilon}{2} < \varepsilon$.

Structural Commentary: Zero size means arbitrarily small interval covers.
Key Insight: Lists split ε by terms; Cantor stages have length $(\frac{2}{3})^N \to 0$. □

Corollary 4.1.15. *Uncountable Null Sets Exist*
There exists $P \subseteq \mathbb{R}$ such that:

(i) $|P| = 2^{\aleph_0}$;
(ii) $P \in \mathcal{L}^1$ and $m(P) = 0$.

Proof. Let $P := \{\sum_{k=1}^\infty \varepsilon_k 3^{-k} : \varepsilon_k \in \{0, 2\}\}$.

(i) The map $(\varepsilon_k) \mapsto \sum_k \varepsilon_k 3^{-k}$ identifies P with $\{0, 2\}^{\mathbb{N}}$, so $|P| = 2^{\aleph_0}$.
(ii) At stage N, P is covered by 2^N closed intervals of length 3^{-N}, hence $m_1^*(P) \leq (2/3)^N$ for all N.
Letting $N \to \infty$ gives $m_1^*(P) = 0$, so outer-null completeness gives $P \in \mathcal{L}^1$ and $m(P) = 0$.

Structural Commentary: Cantor coding separates cardinality from length.
Key Insight: Null need not mean countable. □

Corollary 4.1.16. *Real-Line Lebesgue Measure Checks*
Let m be Lebesgue measure on $\mathbb{R}$.

(i) $a < b \Rightarrow m((a,b)) = m((a,b]) = m([a,b)) = m([a,b]) = b - a$.
(ii) $A \subseteq \mathbb{R}$ *countable* $\Rightarrow A \in \mathcal{B}(\mathbb{R})$ *and* $m(A) = 0$.
(iii) $\emptyset \neq G \subseteq \mathbb{R}$ *open* $\Rightarrow m(G) > 0$.
(iv) $K \subseteq \mathbb{R}$ *compact* $\Rightarrow m(K) < +\infty$.

Proof.

(i) By cell volume, $m([a,b]) = b - a$; endpoint sets finite null; add or delete $\{a\}, \{b\}$ does not change measure.
(ii) If $A = \{a_k : k \in J\}$ with J finite or countable, then $A = \bigcup_{k\in J}\{a_k\} \in \mathcal{B}(\mathbb{R})$; $m(A) \leq \sum_{k\in J} m(\{a_k\}) = 0$.
(iii) Choose $(a,b) \subseteq G$ with $a < b$; by monotonicity and (i), $m(G) \geq m((a,b)) = b - a > 0$.
(iv) Compact K is bounded, so $K \subseteq [-R,R]$ for some $R > 0$; hence $m(K) \leq m([-R,R]) = 2R < +\infty$.

Structural Commentary: On $\mathbb{R}$, length, null endpoints, and compact boundedness give quick measure estimates.
Key Insight: Nonempty open sets contain intervals; compact sets fit inside one. □

Example 4.1.17. *A Positive-Measure Cantor-Type Set with No Interval*
Let $S_0 = [0,1]$ and, for $n \geq 1$, let R_n be the union of the middle open intervals of length 2^{-2n} removed from the 2^{n-1} components of S_{n-1}; set $S_n := S_{n-1} \setminus R_n$ and $S := \bigcap_{n=0}^{\infty} S_n$.
Then $m(R_n) = 2^{-n-1}$, $\sum_{n\geq 1} m(R_n) = 1/2$, so $S = [0,1] \setminus \bigcup_n R_n$ is compact with $m(S) = 1/2$.
Since every component of S_n has length $\leq 2^{-n}$, no interval (a,b) can lie in S; hence $\operatorname{int} S = \emptyset$.
Structural Commentary: Summable removals leave measure; repeated splitting kills interior.
Key Insight: Total removed length controls measure; repeated splitting controls interior.

5 Continuity and Regularity of Measures

5.1 Measure Continuity and Regularity

Corollary 5.1.1. *Measure Continuity*
Let μ be a measure, and let (E_k) and (F_k) be measurable sequences.

(i) If $E_k \uparrow E$, then $\mu(E_k) \uparrow \mu(E)$.
(ii) If $F_k \downarrow F$ and $\mu(F_1) < +\infty$, then $\mu(F_k) \downarrow \mu(F)$.

Proof.

(i) Set $G_1 = E_1$ and $G_k = E_k \setminus E_{k-1}$ for $k \geq 2$.
Then $E_k = \bigsqcup_{j=1}^{k} G_j$ and $E = \bigsqcup_{j\geq 1} G_j$, so $\mu(E_k) = \sum_{j=1}^{k} \mu(G_j) \uparrow \sum_{j\geq 1} \mu(G_j) = \mu(E)$.
(ii) Apply (i) to $H_k = F_1 \setminus F_k \uparrow F_1 \setminus F$. Since $\mu(F_1) < +\infty$, $\mu(F_k) = \mu(F_1) - \mu(H_k) \downarrow \mu(F_1) - \mu(F_1 \setminus F) = \mu(F)$.

Structural Commentary: Monotone set limits are disjoint series or complements of such series.
Key Insight: Downward continuity needs one finite anchor so subtraction is legal. □

Definition 5.1.2. *Limit Inferior and Superior of Sets*
For $(E_n) \subseteq \mathcal{P}(X)$, define $\limsup_n E_n := \bigcap_{m=1}^{\infty} \bigcup_{n\geq m} E_n$ and $\liminf_n E_n := \bigcup_{m=1}^{\infty} \bigcap_{n\geq m} E_n$.

(i) *Infinitely often*: $x \in \limsup_n E_n$ iff $\{n : x \in E_n\}$ is infinite;
(ii) *Eventually always*: $x \in \liminf_n E_n$ iff $\{n : x \notin E_n\}$ is finite;
(iii) *Set limit*: if $\limsup_n E_n = \liminf_n E_n$, write $\lim_n E_n$ for the common set.

Proposition 5.1.3. *Basic Laws for Set Limits*
Let $(E_n) \subseteq \mathcal{P}(X)$.

(i) $\emptyset \subseteq \liminf_n E_n \subseteq \limsup_n E_n \subseteq X$.
(ii) $E_n \uparrow E \Longrightarrow \liminf_n E_n = \limsup_n E_n = E = \bigcup_n E_n$.
(iii) $E_n \downarrow E \Longrightarrow \liminf_n E_n = \limsup_n E_n = E = \bigcap_n E_n$.
(iv) $(E_n) \subseteq \mathcal{M} \Longrightarrow \liminf_n E_n, \limsup_n E_n \in \mathcal{M}$.

Proof.

(i) Since $\bigcap_m \bigcup_{n\geq m} E_n \subseteq \bigcup_{n\geq 1} E_n \subseteq X$ and $\bigcap_{n\geq m} E_n \subseteq \bigcup_{n\geq m} E_n$ for all m, the inclusions follow.
(ii) If $E_n \uparrow E$, then $\{n : x \in E_n\}$ is nonempty iff it is cofinite, hence $\liminf_n E_n = \limsup_n E_n = \bigcup_n E_n = E$.
(iii) If $E_n \downarrow E$, then $\{n : x \in E_n\}$ is infinite iff it is all of $\mathbb{N}$, hence $\liminf_n E_n = \limsup_n E_n = \bigcap_n E_n = E$.
(iv) If $(E_n) \subseteq \mathcal{M}$, then $\bigcup_{n\geq m} E_n$, $\bigcap_{n\geq m} E_n \in \mathcal{M}$ $\forall m$; countable closure gives $\limsup_n E_n, \liminf_n E_n \in \mathcal{M}$.

Structural Commentary: Set limsup means infinitely often; set liminf means eventually always.
Key Insight: Translate the definitions into pointwise membership statements. □

Proposition 5.1.4. *Measure Bounds for Set Limits*
Let (E_n) be measurable in a measure space $(X, \mathcal{M}, \mu)$.

(i) $\mu(\liminf_n E_n) \leq \liminf_n \mu(E_n)$.
(ii) $\mu(\bigcup_{n=1}^{\infty} E_n) < +\infty \Longrightarrow \limsup_n \mu(E_n) \leq \mu(\limsup_n E_n)$.

Proof.

(i) Set $A_m := \bigcap_{n\geq m} E_n$. Then $A_m \uparrow \liminf_n E_n$ and $A_m \subseteq E_n$ for all $n \geq m$, so $\mu(A_m) \leq \inf_{n\geq m} \mu(E_n)$.
By continuity from below, $\mu(\liminf_n E_n) = \lim_m \mu(A_m) \leq \sup_m \inf_{n\geq m} \mu(E_n) = \liminf_n \mu(E_n)$.

(ii) Set $B_m := \bigcup_{n\geq m} E_n$. $B_m \downarrow \limsup_n E_n$, $\mu(B_1) < +\infty$, $E_n \subseteq B_m$ $\forall n \geq m$, so $\sup_{n\geq m} \mu(E_n) \leq \mu(B_m)$.
By continuity from above, $\limsup_n \mu(E_n) = \inf_m \sup_{n\geq m} \mu(E_n) \leq \lim_m \mu(B_m) = \mu(\limsup_n E_n)$.

Structural Commentary: Tail intersections increase to liminf; tail unions decrease to limsup.
Key Insight: Compare each tail envelope to the corresponding tail bounds of the measures. □

Definition 5.1.5. *Almost Everywhere*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, and let $P(x)$ be a property on E.

(i) P holds *μ-almost everywhere on E* iff $\exists N \in \mathcal{M}$ with $N \subseteq E$, $\mu(N) = 0$, and P holds on $E \setminus N$.
(ii) For Lebesgue measure, write *almost everywhere*.

Theorem 5.1.6. *First Borel–Cantelli Lemma*
Let $(X, \mathcal{M}, \mu)$ be measure space, $(E_n) \subseteq \mathcal{M}$, set $L := \limsup_n E_n = \bigcap_{m\geq 1} \bigcup_{n\geq m} E_n$. Assume $\sum_{n=1}^{\infty} \mu(E_n) < +\infty$.

(i) $L \in \mathcal{M}$ and $\mu(L) = 0$.
(ii) μ-almost every $x \in X$ belongs to at most finitely many E_n.

Proof.

(i) Set $B_m := \bigcup_{n\geq m} E_n$. Then $B_m \downarrow L$ and $\mu(B_m) \leq \sum_{n\geq m} \mu(E_n)$.
Since $\mu(B_1) \leq \sum_{n\geq 1} \mu(E_n) < +\infty$, continuity from above gives $\mu(L) = \lim_{m\to\infty} \mu(B_m) \leq \lim_{m\to\infty} \sum_{n\geq m} \mu(E_n) = 0$.

(ii) Since $x \in L \iff \forall m\, \exists n \geq m$ with $x \in E_n$, L is the set of points lying in infinitely many E_n.
By (i), outside the null set L, membership occurs only finitely often.

Structural Commentary: Summable tails force the limsup set to have zero measure.
Key Insight: Points appearing infinitely often must lie in every tail union. □

Remark 5.1.7. *Borel Sets, Null Sets, and Regular Approximation*
Open, closed, and Borel subsets of $\mathbb{R}^p$ are measurable. Each $A \in \mathcal{M}(\mu)$ admits closed–open approximations $F \subseteq A \subseteq G$ with $\mu(G \setminus A)$ and $\mu(A \setminus F)$ arbitrarily small. In Lebesgue measure, one may take $F_\sigma \subseteq A \subseteq G_\delta$ with $\mu(G_\delta \setminus F_\sigma) = 0$. Null sets form a σ-ring, and countable subsets of $\mathbb{R}^p$ are null.
Structural Commentary: Measurable sets are Borel sets modified by null sets.
Key Insight: Regularity explains why open and closed approximations are enough for integration.

Proposition 5.1.8. *Local Measurability on Bounded Cubes*
For arbitrary $E \subseteq \mathbb{R}^p$, $E \in \mathcal{L}^p \iff E \cap [-n, n]^p \in \mathcal{L}^p$ for all $n \in \mathbb{N}$.

Proof.

($\Rightarrow$) If $E \in \mathcal{L}^p$, then $E \cap [-n, n]^p \in \mathcal{L}^p$ for all n, since $[-n, n]^p$ is Borel.
($\Leftarrow$) If $E \cap [-n, n]^p \in \mathcal{L}^p$ for all n, then $E = \bigcup_{n=1}^{\infty} (E \cap [-n, n]^p) \in \mathcal{L}^p$.

Structural Commentary: Nested Borel cubes exhaust $\mathbb{R}^p$.
Key Insight: Measurability is local along a countable exhaustion. □

Proposition 5.1.9. *Local Measurability on Unit Intervals*
Let $I_n := (n, n+1]$, $n \in \mathbb{Z}$. For $E \subseteq \mathbb{R}$, $E \in \mathcal{L}^1$ iff $E \cap I_n \in \mathcal{L}^1$ for all $n \in \mathbb{Z}$.

Proof.

($\Rightarrow$) Since $I_n \in \mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}^1$, $E \in \mathcal{L}^1$ implies $E \cap I_n \in \mathcal{L}^1$ for all n.
($\Leftarrow$) If $E \cap I_n \in \mathcal{L}^1$ for all n, then $E = \bigcup_{n\in\mathbb{Z}} (E \cap I_n) \in \mathcal{L}^1$.

Structural Commentary: A countable Borel exhaustion localizes measurability.
Key Insight: Local measurable pieces patch by countable unions. □

Proposition 5.1.10. *Finite Measure Meeting Every Unit Interval*
With $I_n := (n, n+1]$, there exists $E \in \mathcal{B}(\mathbb{R})$ such that $0 < m(E \cap I_n)$ for all $n \in \mathbb{Z}$ and $m(E) < +\infty$.

Proof. Set $\varepsilon_n := 2^{-|n|-3}$ and $E := \bigcup_{n\in\mathbb{Z}} (n, n + \varepsilon_n]$. Then $(n, n + \varepsilon_n] \subseteq I_n$ are pairwise disjoint Borel intervals, so $E \in \mathcal{B}(\mathbb{R})$ and $m(E \cap I_n) = \varepsilon_n > 0$. Countable additivity gives $m(E) = \sum_{n\in\mathbb{Z}} \varepsilon_n = \frac{3}{8} < +\infty$.
Structural Commentary: Finite measure can be spread over countably many cells.
Key Insight: Positive local masses can be summable globally. □

Proposition 5.1.11. *Null Sets Form a Sigma-Ideal*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$.
Then $\mathcal{Z}$ is a σ-ring, and $N \in \mathcal{Z}$, $E \in \mathcal{M}$ imply $N \cap E \in \mathcal{Z}$.

Proof. If $(N_k) \subseteq \mathcal{Z}$, then $\mu(\bigcup_k N_k) \leq \sum_k \mu(N_k) = 0$, so $\bigcup_k N_k \in \mathcal{Z}$.
If $A, B \in \mathcal{Z}$, then $A \setminus B \subseteq A$, hence $A \setminus B \in \mathcal{Z}$. Finally, $N \cap E \subseteq N$ gives $N \cap E \in \mathcal{Z}$.
Structural Commentary: Null sets are stable under countable unions and measurable cuts.
Key Insight: Monotonicity makes every measurable piece of a null set null. □

Proposition 5.1.12. *Completed Sigma-Algebra*
Let $(X, \mathcal{M}, \mu)$ be measure space, $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$; $\overline{\mathcal{M}}_\mu := \{A \subseteq X : \exists E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ A \triangle E \subseteq N\}$.

(i) $\mathcal{M} \subseteq \overline{\mathcal{M}}_\mu$, and $\overline{\mathcal{M}}_\mu$ is a σ-algebra;

(ii) $A \subseteq N \in \mathcal{Z} \Longrightarrow A \in \overline{\mathcal{M}}_\mu$;
(iii) $\overline{\mathcal{M}}_\mu = \{E \cup Z : E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ Z \subseteq N\}$.

Proof.

(i) For $E \in \mathcal{M}$, $E \triangle E = \emptyset$, so $\mathcal{M} \subseteq \overline{\mathcal{M}}_\mu$. If $A \triangle E \subseteq N$, then $A^c \triangle E^c = A \triangle E \subseteq N$.
If $A_k \triangle E_k \subseteq N_k$, then $(\bigcup_k A_k) \triangle (\bigcup_k E_k) \subseteq \bigcup_k N_k \in \mathcal{Z}$. Hence $\overline{\mathcal{M}}_\mu$ is a σ-algebra.
(ii) If $A \subseteq N \in \mathcal{Z}$, then $A \triangle \emptyset \subseteq N$, so $A \in \overline{\mathcal{M}}_\mu$.
(iii) If $A \triangle E \subseteq N$, then $A = (E \setminus N) \cup (A \cap N)$, with $E \setminus N \in \mathcal{M}$ and $A \cap N \subseteq N$.
Conversely, if $A = E \cup Z$ and $Z \subseteq N \in \mathcal{Z}$, then $A \triangle E \subseteq Z \subseteq N$.

Structural Commentary: Completion adds all null modifications of measurable sets.
Key Insight: Equality modulo null sets is the membership criterion. □

Proposition 5.1.13. *Completed Measure*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$, and $\overline{\mathcal{M}}_\mu := \{A \subseteq X : \exists E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ A \triangle E \subseteq N\}$. For $A \in \overline{\mathcal{M}}_\mu$, define $\overline{\mu}(A) := \mu(E)$ whenever $E \in \mathcal{M}$ and $A \triangle E \subseteq N \in \mathcal{Z}$. Then $\overline{\mu}$ is a well-defined complete measure on $\overline{\mathcal{M}}_\mu$, and $\overline{\mu}|_{\mathcal{M}} = \mu$.

Proof. If $A \triangle E \subseteq N_1$ and $A \triangle F \subseteq N_2$, with $E, F \in \mathcal{M}$ and $N_1, N_2 \in \mathcal{Z}$, then $E \triangle F \subseteq N_1 \cup N_2$, so $\mu(E) = \mu(F)$. Hence $\overline{\mu}$ is well defined; since $E \triangle E = \emptyset$, $\overline{\mu}|_{\mathcal{M}} = \mu$.
For disjoint $(A_k) \subseteq \overline{\mathcal{M}}_\mu$, choose $E_k \in \mathcal{M}$ and $N_k \in \mathcal{Z}$ with $A_k \triangle E_k \subseteq N_k$, and set $N := \bigcup_k N_k$.
Then $E_i \cap E_j \subseteq N$ for $i \neq j$. Put $F_1 := E_1$ and $F_k := E_k \setminus \bigcup_{j<k} E_j$.
Then (F_k) disjoint, $A_k \triangle F_k \subseteq N$, $(\bigcup_k A_k) \triangle (\bigcup_k F_k) \subseteq N$, so $\overline{\mu}\left(\bigcup_k A_k\right) = \mu\left(\bigcup_k F_k\right) = \sum_k \mu(F_k) = \sum_k \overline{\mu}(A_k)$.
If $A \subseteq B \in \overline{\mathcal{M}}_\mu$ and $\overline{\mu}(B) = 0$, choose $E \in \mathcal{M}$ and $N \in \mathcal{Z}$ with $B \triangle E \subseteq N$.
Then $\mu(E) = 0$, so $A \subseteq B \subseteq E \cup N \in \mathcal{Z}$; hence $A \in \overline{\mathcal{M}}_\mu$ and $\overline{\mu}(A) = 0$.
Structural Commentary: The completed measure ignores null errors.
Key Insight: Null perturbations do not change countable additivity. □

5.2 Radon Measures on Locally Compact Spaces

Radon measure theory is where topology and measure start determining one another. The topological input is a locally compact Hausdorff space; the measure-theoretic output is a Borel measure that can be recovered from open neighborhoods, compact subsets, and compactly supported continuous test functions.

Definition 5.2.1. *Locally Compact Borel Setup*
Let $(X, \mathcal{T})$ be a topological space.

(i) X is *locally compact* if every $x \in X$ admits $U \in \mathcal{T}$ with $x \in U$ and $\overline{U}$ compact.
(ii) The *Borel σ-algebra* is $\mathcal{B}(X) := \sigma(\mathcal{T})$.
(iii) For $f : X \to \mathbb{R}$, define $\operatorname{supp} f := \overline{\{x \in X : f(x) \neq 0\}}$.
(iv) $C_c(X)$ is the subspace of $C(X, \mathbb{R})$ consisting of functions with compact support.

Proposition 5.2.2. *Compact Neighborhood Shrinking*
Let X be locally compact and Hausdorff, $K \subseteq X$ compact, and $O \subseteq X$ open with $K \subseteq O$. Then there is an open $V \subseteq X$ such that $K \subseteq V$, $\overline{V} \subseteq O$, and $\overline{V}$ is compact.

Proof. For each $x \in K$, choose open $U_x \ni x$ with $\overline{U_x}$ compact.
Normality of the compact Hausdorff space $\overline{U_x}$ gives a relatively open $W_x \ni x$ with $\overline{W_x}^{\overline{U_x}} \subseteq O \cap U_x$.
Choose open $A_x \subseteq X$ with $W_x = A_x \cap \overline{U_x}$ and set $V_x := A_x \cap U_x$. Then $x \in V_x$ and $\overline{V_x} \subseteq \overline{W_x}^{\overline{U_x}} \subseteq O \cap U_x$.
The family $(V_x)_{x \in K}$ covers K, so $K \subseteq \bigcup_{i=1}^n V_{x_i}$ for some $x_1, \ldots, x_n \in K$.
Set $V := \bigcup_{i=1}^n V_{x_i}$. Then $\overline{V} = \bigcup_{i=1}^n \overline{V_{x_i}} \subseteq O$, and $\overline{V}$ is compact as a finite union of compact sets.
Structural Commentary: Shrink each point neighborhood inside O, then take a finite union.
Key Insight: Compactness converts local shrinkings into one global compact neighborhood. □

Proposition 5.2.3. *Compactly Supported Cutoff*
Let X be locally compact and Hausdorff, $K \subseteq X$ compact, and $O \subseteq X$ open with $K \subseteq O$. Then there is $f \in C_c(X)$ such that $0 \leq f \leq 1$ on X, $f|_K = 1$, and $\operatorname{supp} f \subseteq O$.

Proof. Choose open $V \subseteq X$ as in Proposition 5.2.2, so $K \subseteq V$, $\overline{V}$ is compact, and $\overline{V} \subseteq O$.
The disjoint sets K and $\overline{V} \setminus V$ are closed in the compact Hausdorff space $\overline{V}$.
By Urysohn's lemma, Theorem 6.3.20, there is $f_0 : \overline{V} \to [0,1]$ with $f_0|_K = 1$ and $f_0|_{\overline{V} \setminus V} = 0$.
Define $f : X \to [0,1]$ by $f|_{\overline{V}} = f_0$ and $f|_{X \setminus V} = 0$.
Closed-set pasting gives $f \in C_c(X)$ with $f|_K = 1$ and $\operatorname{supp} f \subseteq \overline{V} \subseteq O$.
Structural Commentary: Apply Urysohn separation on $\overline{V}$ and extend by zero.
Key Insight: The compact neighborhood forces the extension to have compact support. □

Proposition 5.2.4. *Finite Partition of Unity on a Compact Set*
Let X be locally compact and Hausdorff, $K \subseteq X$ compact, and $O_1, \ldots, O_n \subseteq X$ open with $K \subseteq \bigcup_{i=1}^n O_i$. Then there are $\varphi_1, \ldots, \varphi_n \in C_c(X)$ such that $0 \leq \varphi_i \leq 1$, $\operatorname{supp} \varphi_i \subseteq O_i$ for every i, and $(\sum_{i=1}^n \varphi_i)|_K = 1$.

Proof. For each $x \in K$, choose $i(x)$ with $x \in O_{i(x)}$ and, by Proposition 5.2.2, an open $V_x \ni x$ with $\overline{V_x}$ compact and $\overline{V_x} \subseteq O_{i(x)}$. Choose $x_1, \ldots, x_m \in K$ with $K \subseteq \bigcup_{j=1}^m V_{x_j}$, write $i_j := i(x_j)$, and set $U_i := \bigcup_{\{j : i_j = i\}} V_{x_j}$.
Then $K \subseteq \bigcup_i U_i$, each U_i is open, and $\overline{U_i}$ is compact with $\overline{U_i} \subseteq O_i$.
For $U_i \neq \emptyset$, Proposition 5.2.3 gives $f_i \in C_c(X)$ with $0 \leq f_i \leq 1$, $f_i|_{\overline{U_i}} = 1$, $\operatorname{supp} f_i \subseteq O_i$; set $f_i = 0$ otherwise. Applying the same proposition to $K \subseteq \bigcup_i U_i$ gives $h \in C_c(X)$ with $0 \leq h \leq 1$, $h|_K = 1$, and $\operatorname{supp} h \subseteq \bigcup_i U_i$.
Set $g := \sum_{i=1}^n f_i$ and $d := g + 1 - h$. Since $g \geq 1$ on $\operatorname{supp} h$ and $h = 0$ off $\operatorname{supp} h$, $d \geq 1$. Hence $\varphi_i := f_i / d$ belongs to $C_c(X)$, satisfies $0 \leq \varphi_i \leq 1$, and has $\operatorname{supp} \varphi_i \subseteq O_i$. On K, $h = 1$ and $g \geq 1$, so $d = g$ and $(\sum_{i=1}^n \varphi_i)|_K = 1$.

Structural Commentary: Build finitely many local cutoffs, then normalize their sum.
Key Insight: The auxiliary cutoff keeps the denominator positive where local cutoffs vanish. □

Theorem 5.2.5. *Locally Compact Separation and Cutoff Package*
Let X be Hausdorff. The following are equivalent.

(i) X is locally compact.
(ii) For every compact $K \subseteq O \in \mathcal{T}$, there is open $V \supseteq K$ with $\overline{V}$ compact and $\overline{V} \subseteq O$.
(iii) For every compact $K \subseteq O \in \mathcal{T}$, there is $f \in C_c(X)$ with $0 \le f \le 1$, $f|_K = 1$, and $f|_{X \setminus O} = 0$.

Proof.

$(i) \Rightarrow (ii)$ Proposition 5.2.2.
$(i) \Rightarrow (iii)$ Proposition 5.2.3.
$(ii) \Rightarrow (i)$ For $x \in X$, apply (ii) to $K = \{x\}$ and $O = X$; the resulting open $V \ni x$ has compact closure.
$(iii) \Rightarrow (i)$ For $x \in X$, apply (iii) to $K = \{x\}$ and $O = X$, and set $U := f^{-1}((1/2, 3/2))$.
Then $x \in U$ and $\overline{U} \subseteq f^{-1}([1/2, 3/2]) \subseteq \operatorname{supp} f$, so $\overline{U}$ is compact.

Structural Commentary: Compact cutoffs recover compact neighborhoods through level sets.
Key Insight: Compact-neighborhood shrinking and compactly supported cutoffs are equivalent local forms of local compactness. □

Method 5.2.6. *Metric Cutoff Formula*
Let X be locally compact and Hausdorff, assume that ρ induces its topology.
Let $K \subseteq O \subseteq X$ with K compact and O open, and choose open $U \subseteq X$ with $K \subseteq U$, $\overline{U}$ compact, and $\overline{U} \subseteq O$.
For every nonempty $A \subseteq X$, set $\operatorname{dist}_A(x) := \inf_{a \in A} \rho(x, a)$.

(i) *Empty Set*: If $K = \emptyset$, set $f := 0$.
(ii) *Trivial Complement*: If $K \neq \emptyset$ and $X \setminus U = \emptyset$, set $f := 1$.
(iii) *Distance Ratio*: Otherwise, set $f(x) := \frac{\operatorname{dist}_{X \setminus U}(x)}{\operatorname{dist}_{X \setminus U}(x) + \operatorname{dist}_K(x)}$.

In case (iii), K and $X \setminus U$ are disjoint closed sets, so the denominator is positive; continuity of distance functions gives $f \in C(X, [0, 1])$.
In all cases, $f \in C_c(X)$, $f|_K = 1$, $f|_{X \setminus U} = 0$, and $\operatorname{supp} f \subseteq \overline{U} \subseteq O$.
Key Insight: Metric distances give an explicit compactly supported cutoff.

Proposition 5.2.7. *Alexandroff One-Point Compactification*
Let $(X, \mathcal{T})$ be locally compact and Hausdorff, let $\omega \notin X$, and write $\mathcal{K}(X)$ for the compact subsets of X.
Set $X^* := X \cup \{\omega\}$ and $\mathcal{T}^* := \mathcal{T} \cup \{X^* \setminus K : K \in \mathcal{K}(X)\}$. Let $\iota : X \hookrightarrow X^*$ be the inclusion.

(i) $\mathcal{T}^*$ is a topology on X^*, $(X^*, \mathcal{T}^*)$ is compact Hausdorff, and $\iota : X \to X^* \setminus \{\omega\}$ is a homeomorphism.
(ii) Every $h \in C(K, \mathbb{R})$ with $K \subseteq X$ compact has an extension $\widetilde{h} \in C(X, \mathbb{R})$ satisfying $\widetilde{h}|_K = h$.

Proof.

(i) Compact subsets of the Hausdorff space X are closed.
Thus $U \setminus C \in \mathcal{T}$ and $(X^* \setminus C) \cap (X^* \setminus D) = X^* \setminus (C \cup D)$ for $U \in \mathcal{T}$ and $C, D \in \mathcal{K}(X)$.
If $\omega \notin \bigcup_\alpha U_\alpha$, then $\bigcup_\alpha U_\alpha \in \mathcal{T}$; otherwise choose $U_{\alpha_0} = X^* \setminus C$, so $X^* \setminus \bigcup_\alpha U_\alpha$ is closed in C and hence compact. Thus $\mathcal{T}^*$ is a topology.
Any open cover contains $X^* \setminus C$ for some $C \in \mathcal{K}(X)$, and finitely many remaining members cover C; hence X^* is compact. Points of X are already separated.
For $x \in X$, choose open $V \ni x$ with $\overline{V}$ compact; then V and $X^* \setminus \overline{V}$ separate x and ω.
Finally, $\mathcal{T}^*|_X = \mathcal{T}$ since $X \setminus C \in \mathcal{T}$ for $C \in \mathcal{K}(X)$, so ι is a homeomorphism onto $X^* \setminus \{\omega\}$.
(ii) By (i), X^* is compact Hausdorff and hence normal. The compact set K is closed in X^*, and h is bounded.
Tietze extension, Theorem 6.3.22, gives $H \in C(X^*, \mathbb{R})$ with $H|_K = h$. Set $\widetilde{h} := H|_X$.

Structural Commentary: Neighborhoods of ω are complements of compact subsets of X.
Key Insight: Compactification transfers extension problems to a compact Hausdorff space. □

Definition 5.2.8. *Radon Measures and Radon Premeasures*
Let $(X, \mathcal{T})$ be a topological space, and write $\mathcal{K}(X)$ for its compact subsets.

(i) $\nu : \mathcal{B}(X) \to [0, +\infty]$ is a *Radon measure* if $\nu(K) < +\infty$ $\forall K \in \mathcal{K}(X)$, $\nu(E) = \inf\{\nu(O) : E \subseteq O,\ O \in \mathcal{T}\}$ $\forall E \in \mathcal{B}(X)$, and $\nu(O) = \sup\{\nu(K) : K \subseteq O,\ K \in \mathcal{K}(X)\}$ $\forall O \in \mathcal{T}$.
(ii) $\mu : \mathcal{T} \to [0, +\infty]$ is a *Radon premeasure* if $\mu(U) < +\infty$ $\forall U \in \mathcal{T}$ with $\overline{U} \in \mathcal{K}(X)$ and $\mu(O) = \sup\{\mu(U) : U \in \mathcal{T},\ \overline{U} \in \mathcal{K}(X),\ \overline{U} \subseteq O\}$ $\forall O \in \mathcal{T}$.

Lemma 5.2.9. *Open Envelope for a Topological Premeasure*
Let $\mu : \mathcal{T} \to [0, +\infty]$ be a premeasure. For $E \subseteq X$, define $\mu^*(E) := \inf\{\sum_{j=1}^\infty \mu(U_j) : E \subseteq \bigcup_{j=1}^\infty U_j,\ U_j \in \mathcal{T}\}$.
Then $\mu^*(E) = \inf\{\mu(U) : E \subseteq U,\ U \in \mathcal{T}\}$.

Proof.

$(\Rightarrow)$ Set $I(E) := \inf\{\mu(U) : E \subseteq U,\ U \in \mathcal{T}\}$. Every $U \in \mathcal{T}$ with $E \subseteq U$ is a one-set cover, so $\mu^*(E) \le I(E)$.
$(\Leftarrow)$ If $(U_j)_{j \ge 1} \subseteq \mathcal{T}$ covers E, then $U := \bigcup_{j=1}^\infty U_j$ belongs to $\mathcal{T}$ and satisfies $\mu(U) \le \sum_{j=1}^\infty \mu(U_j)$.
Hence $I(E) \le \sum_{j=1}^\infty \mu(U_j)$; taking the infimum over covers gives $I(E) \le \mu^*(E)$.

Structural Commentary: Replace each countable cover by its union.
Key Insight: Subadditivity makes the union no more expensive than the cover. □

Lemma 5.2.10. *Open-Set Test for Caratheodory Measurability*
Let $\mu : \mathcal{T} \to [0, +\infty]$ be a premeasure, let μ^* be the outer measure induced by open covers, and let $E \subseteq X$.
Assume $\mu(O) \ge \mu^*(O \cap E) + \mu^*(O \setminus E)$ for every $O \in \mathcal{T}$ with $\mu(O) < +\infty$. Then E is μ^*-measurable.

Proof. Fix $A \subseteq X$. Outer subadditivity gives $\mu^*(A) \le \mu^*(A \cap E) + \mu^*(A \setminus E)$.
If $\mu^*(A) = +\infty$, the reverse inequality is automatic. Otherwise, for $\varepsilon > 0$, Lemma 5.2.9 gives $O \in \mathcal{T}$ with $A \subseteq O$ and $\mu(O) < \mu^*(A) + \varepsilon$, so $\mu^*(A \cap E) + \mu^*(A \setminus E) \le \mu^*(O \cap E) + \mu^*(O \setminus E) \le \mu(O) < \mu^*(A) + \varepsilon$.
Letting $\varepsilon \downarrow 0$ gives the reverse inequality, hence the Caratheodory criterion.
Structural Commentary: Approximate A by an open envelope of nearly equal outer mass.
Key Insight: Open-set splitting propagates to all sets through outer regularity. □

Proposition 5.2.11. *Open Sets from a Radon Premeasure Are Measurable*
Let X be locally compact and Hausdorff, $\mu : \mathcal{T} \to [0, +\infty]$ Radon premeasure, μ^* open-cover outer measure.

(i) $\mu^*|_{\mathcal{T}} = \mu$.
(ii) Every $V \in \mathcal{T}$ is μ^*-measurable.

Proof.

(i) The one-set cover gives $\mu^*(V) \le \mu(V)$ for $V \in \mathcal{T}$.
Every $(U_j)_{j \ge 1} \subseteq \mathcal{T}$ covering V satisfies $\mu(V) \le \mu(\bigcup_j U_j) \le \sum_j \mu(U_j)$, so $\mu(V) \le \mu^*(V)$.

(ii) Let $V, O \in \mathcal{T}$ with $\mu(O) < +\infty$.
For $\varepsilon > 0$, choose $U \in \mathcal{T}$ with $\overline{U}$ compact, $\overline{U} \subseteq O \cap V$, and $\mu(U) > \underline{\mu}(O \cap V) - \varepsilon$.
Since U and $O \setminus \overline{U}$ are disjoint open subsets of O and $O \setminus V \subseteq O \setminus \overline{U}$,
$\mu(O) \ge \mu(U) + \mu(O \setminus \overline{U}) \ge \mu^*(O \cap V) - \varepsilon + \mu^*(O \setminus V)$ by (i). Let $\varepsilon \downarrow 0$ and apply Lemma 5.2.10.

Structural Commentary: Inner approximation leaves an open remainder containing $O \setminus V$.
Key Insight: Open inner regularity yields Caratheodory splitting. □

Theorem 5.2.12. *Radon Premeasure Extension*
Let X be locally compact and Hausdorff, let $\mu : \mathcal{T} \to [0, +\infty]$ be a Radon premeasure, let μ^* be its open-cover outer measure, write $\mathcal{K}(X)$ for the compact subsets of X, and set $\widehat{\mu} := \mu^*|_{\mathcal{B}(X)}$.

(i) $\widehat{\mu}|_{\mathcal{T}} = \mu$.
(ii) $\widehat{\mu}$ is a Radon measure on $\mathcal{B}(X)$.

Proof.

(i) For $O \in \mathcal{T} \subseteq \mathcal{B}(X)$, $\widehat{\mu}(O) = \mu^*(O) = \mu(O)$ by Proposition 5.2.11; hence $\widehat{\mu}|_{\mathcal{T}} = \mu$.

(ii) Let $\mathcal{C}_{\mu^*}$ be the Caratheodory σ-algebra of μ^*. Proposition 5.2.11 gives $\mathcal{T} \subseteq \mathcal{C}_{\mu^*}$, hence $\mathcal{B}(X) = \sigma(\mathcal{T}) \subseteq \mathcal{C}_{\mu^*}$.
As $\mu^*|_{\mathcal{C}_{\mu^*}}$ is measure, so is $\widehat{\mu}$. Lemma 5.2.9, (i) give $\widehat{\mu}(E) = \inf\{\widehat{\mu}(O) : E \subseteq O,\ O \in \mathcal{T}\}$ for $E \in \mathcal{B}(X)$.
For $K \in \mathcal{K}(X)$, Proposition 5.2.2 gives $K \subseteq U \in \mathcal{T}$ with $\overline{U} \in \mathcal{K}(X)$, so $\widehat{\mu}(K) \le \widehat{\mu}(U) = \mu(U) < +\infty$.
For $O \in \mathcal{T}$, inner regularity and (i) give
$\widehat{\mu}(O) = \sup\{\mu(U) : U \in \mathcal{T},\ \overline{U} \in \mathcal{K}(X),\ \overline{U} \subseteq O\} \le \sup\{\widehat{\mu}(K) : K \in \mathcal{K}(X),\ K \subseteq O\} \le \widehat{\mu}(O)$.

Structural Commentary: $\mathcal{T} \subseteq \mathcal{C}_{\mu^*}$ yields $\widehat{\mu}$ on $\mathcal{B}(X)$; $K \in \mathcal{K}(X) \Rightarrow \widehat{\mu}(K) < +\infty$.
Key Insight: Open inner regularity becomes compact inner regularity after extension. □

Theorem 5.2.13. *Radon Measures Determined by $C_c(X)$*
Let X be locally compact and Hausdorff, and let μ_1, μ_2 be Radon measures on $\mathcal{B}(X)$.
If $\int_X f\, d\mu_1 = \int_X f\, d\mu_2$ for every $f \in C_c(X)$, then $\mu_1 = \mu_2$.

Proof. Write $\mathcal{K}(X)$ for the compact subsets of X. Let $K \in \mathcal{K}(X)$ and $\varepsilon > 0$.
By outer regularity, choose $O \in \mathcal{T}$ with $K \subseteq O$ and $\mu_i(O \setminus K) < \varepsilon$ for $i = 1, 2$.
Choose $f \in C_c(X)$ with $0 \le f \le 1$, $f|_K = 1$, and $\operatorname{supp} f \subseteq O$, and set $r_i := \int_{O \setminus K} f\, d\mu_i$.
Then $0 \le r_i < \varepsilon$ and $\int_X f\, d\mu_i = \mu_i(K) + r_i$, so $|\mu_1(K) - \mu_2(K)| = |r_1 - r_2| < \varepsilon$.
Hence $\mu_1 = \mu_2$ on $\mathcal{K}(X)$, and inner then outer regularity give $\mu_1|_{\mathcal{T}} = \mu_2|_{\mathcal{T}}$ and $\mu_1 = \mu_2$ on $\mathcal{B}(X)$.
Structural Commentary: $C_c(X)$ identifies μ_1 and μ_2 on $\mathcal{K}(X)$.
Key Insight: $\mathcal{K}(X) \to \mathcal{T} \to \mathcal{B}(X)$ by Radon regularity. □

Figure 195: Riesz–Markov: local compactness supplies cutoffs; orange denotes Caratheodory extension.

Theorem 5.2.14. *Riesz–Markov Theorem on $C_c(X)$*
Let X be locally compact and Hausdorff, $I : C_c(X) \to \mathbb{R}$ linear with $I(f) \ge 0$ $\forall f \in C_c(X)$ satisfying $f \ge 0$.
Then $\exists$ a unique Radon measure $\widehat{\mu}$ on $\mathcal{B}(X)$ such that $I(f) = \int_X f\, d\widehat{\mu}$ $\forall f \in C_c(X)$.

Proof. For $O \in \mathcal{T}$, define $\mu(O) := \sup\{I(f) : f \in C_c(X),\ 0 \le f \le 1,\ \operatorname{supp} f \subseteq O\}$ and $\mu(\emptyset) := 0$.

(i) *Premeasure*: If $O \subseteq \bigcup_{j=1}^{\infty} O_j$ and f is admissible for $\mu(O)$, a finite partition $(\varphi_\ell)_{\ell=1}^m$ subordinate to a finite subcover of $\operatorname{supp} f$ gives $f = \sum_{\ell=1}^m f\varphi_\ell$ and $I(f) \le \sum_{\ell=1}^m \mu(O_{j_\ell}) \le \sum_{j=1}^{\infty} \mu(O_j)$, so μ is countably subadditive. For disjoint $O_1, O_2 \in \mathcal{T}$, countable subadditivity gives one inequality and admissible f_1, f_2 give
$\mu(O_1) + \mu(O_2) \le \mu(O_1 \cup O_2)$, hence finite additivity.

(ii) *Radon Regularity*: For $a < \mu(O)$, choose admissible f with $I(f) > a$ and $U \in \mathcal{T}$ with $\operatorname{supp} f \subseteq U \subseteq \overline{U} \subseteq O$ and $\overline{U}$ compact. Then $\mu(U) > a$, proving inner regularity. If $\overline{U}$ is compact, a cutoff $g \in C_c(X)$ with $g|_{\overline{U}} = 1$ dominates every admissible f for $\mu(U)$, so $\mu(U) \le I(g) < +\infty$. Thus μ is a Radon premeasure.

(iii) *Extension*: Stages (i)–(ii) make μ a Radon premeasure on $\mathcal{T}$. Theorem 5.2.12 applies to its open-cover outer measure and yields the Radon measure $\widehat{\mu} := \mu^*|_{\mathcal{B}(X)}$ with $\widehat{\mu}|_{\mathcal{T}} = \mu$.

(iv) *Identification*: Let $0 \le f \le 1$ belong to $C_c(X)$, choose $O \in \mathcal{T}$ with $\operatorname{supp} f \subseteq O$ and $\overline{O}$ compact, and set $\psi_k := \min\{1, \max\{0, nf - (k-1)\}\}$, $O_0 := O$, $O_{n+1} := \emptyset$, $O_k := \{x \in O : f(x) > \frac{k-1}{n}\}$ for $1 \le k \le n$.

Then $f = \frac{1}{n}\sum_{k=1}^{n} \psi_k$, $\operatorname{supp}\psi_k \subseteq O_{k-1}$, and $\psi_k|_{O_{k+1}} = 1$, so $\mu(O_{k+1}) \le I(\psi_k) \le \mu(O_{k-1})$

and $\mu(O_{k+1}) \le \int_X \psi_k \, d\widehat{\mu} \le \mu(O_{k-1})$. Therefore $|I(f) - \int_X f \, d\widehat{\mu}| \le \frac{1}{n}\sum_{k=1}^{n} (\mu(O_{k-1}) - \mu(O_{k+1})) \le \frac{2\mu(O)}{n}$.

Letting $n \to \infty$, scaling, and $f = f^+ - f^-$ give $I(f) = \int_X f \, d\widehat{\mu}$ for every $f \in C_c(X)$.
Uniqueness follows from Theorem 5.2.13.

Structural Commentary: Open-set contents and compactly supported cutoffs build the Radon measure before identifying its integral.
Key Insight: Layer functions recover I from $\widehat{\mu}$. □

Theorem 5.2.15. *Lusin Approximation for Radon Measures*
Let X be locally compact Hausdorff, let μ be Radon on $\mathcal{B}(X)$, let $f : X \to \mathbb{R}$ be Borel, and set $A := \{f \ne 0\}$ with $\mu(A) < +\infty$. For every $\varepsilon > 0$, there exist $X_0 \in \mathcal{B}(X)$ and $g \in C_c(X)$ such that:

(i) $g = f$ on X_0.
(ii) $\mu(X \setminus X_0) < \varepsilon$.

Proof. Set $\delta := \varepsilon/4$. Choose $M > 0$ with $\mu(A_M) < \delta$, where $A_M := A \cap \{|f| > M\}$, and put $B := A \setminus A_M$.
For $n \ge 1$, choose finite-valued Borel $s_n : B \to [-M, M]$ with $\|s_n - f\|_{\infty,B} \le \frac{1}{n}$.
Let $s_n(B) = \{a_{n,1}, \dots, a_{n,m_n}\}$ and $E_{n,j} := s_n^{-1}(\{a_{n,j}\})$. Inner regularity gives compact $K_{n,j} \subseteq E_{n,j}$, $\mu(E_{n,j} \setminus K_{n,j}) < \frac{\delta}{2^n m_n}$. Set $F_n := \bigcup_{j=1}^{m_n} K_{n,j}$ and $F := \bigcap_{n=1}^{\infty} F_n$.
Then F is compact, $B \setminus F \subseteq \bigcup_{n\ge1}\bigcup_{j=1}^{m_n} (E_{n,j} \setminus K_{n,j})$, and $\mu(B \setminus F) \le \delta$.
For each n, $F \cap E_{n,j} = F \cap K_{n,j}$ is closed in F, so $s_n|_F \in C(F)$; since $s_n|_F \to f|_F$ uniformly, $f|_F \in C(F)$.
Choose $W \in \mathcal{T}$ with $F \subseteq W$ and $\mu(W \setminus F) < \delta$, then choose $O \in \mathcal{T}$ by Proposition 5.2.2 with $F \subseteq O \subseteq \overline{O} \subseteq W$ and $\overline{O}$ compact.
Extend $f|_F$ to $H \in C(X)$ by Proposition 5.2.7, choose $q \in C_c(X)$ with $q|_F = 1$ and $\operatorname{supp} q \subseteq O$ by Proposition 5.2.3, and set $g := qH$, $X_0 := F \cup (X \setminus (A \cup O))$. Then $g \in C_c(X)$ and $X_0 \in \mathcal{B}(X)$.

(i) On F, $g = qH = f$; on $X \setminus (A \cup O)$, $f = q = 0$.
(ii) $X \setminus X_0 \subseteq (A \setminus F) \cup (O \setminus F)$, $\mu(A \setminus F) \le \mu(A_M) + \mu(B \setminus F) < 2\delta$, $\mu(O \setminus F) \le \mu(W \setminus F) < \delta$; $\mu(X \setminus X_0) < 3\delta < \varepsilon$.

Structural Commentary: Compact core plus cutoff extension.
Key Insight: Discard a small set to make Borel data continuous. □

5.3 Invariant Measures on Compact Groups

Compact groups are the meeting point of topological symmetry and Radon measure theory. The compactness supplies fixed points in dual spaces, and Riesz–Markov converts those fixed functionals into invariant Borel measures.

Definition 5.3.1. *Compact Groups and Haar Measures*
For a group G, $A \subseteq G$, and $g \in G$, set $gA := \{ga : a \in A\}$ and $Ag := \{ag : a \in A\}$.

(i) A *topological group* is G with Hausdorff topology such that $(g,h) \mapsto gh$ and $g \mapsto g^{-1}$ are continuous.
(ii) A *compact group* is a topological group whose underlying topological space is compact.
(iii) A Borel measure μ on $\mathcal{B}(G)$ is *left-invariant* if $\mu(gA) = \mu(A)$ for all $g \in G$ and $A \in \mathcal{B}(G)$.
(iv) A Borel measure μ on $\mathcal{B}(G)$ is *right-invariant* if $\mu(Ag) = \mu(A)$ for all $g \in G$ and $A \in \mathcal{B}(G)$.
(v) A *Haar measure* on a compact group G is a left-invariant Radon probability measure on $\mathcal{B}(G)$.

Definition 5.3.2. *Regular Actions on $C(G)$*
Let G be compact, $C(G)$ be continuous maps $G \to \mathbb{R}$ with $\|f\|_\infty := \max_{x\in G} |f(x)|$. For $g, x \in G$ and $f \in C(G)$:

(i) The *left regular action* is $\pi_{\mathrm{L}} : G \to \mathcal{L}(C(G))$, $[\pi_{\mathrm{L}}(g)f](x) := f(g^{-1}x)$.
(ii) The *right regular action* is $\pi_{\mathrm{R}} : G \to \mathcal{L}(C(G))$, $[\pi_{\mathrm{R}}(g)f](x) := f(xg^{-1})$.
(iii) A *probability functional* is $\psi \in C(G)^*$ with $\psi(\mathbf{1}_G) = 1$ and $f \ge 0 \Rightarrow \psi(f) \ge 0$ for all $f \in C(G)$.

Theorem 5.3.3. *Kakutani Fixed Probability Functional*
Let G be a compact group. Then there are probability functionals $\psi_{\mathrm{L}}, \psi_{\mathrm{R}} \in C(G)^*$ such that $\psi_{\mathrm{L}}(\pi_{\mathrm{L}}(g)f) = \psi_{\mathrm{L}}(f)$ and $\psi_{\mathrm{R}}(\pi_{\mathrm{R}}(g)f) = \psi_{\mathrm{R}}(f)$ for every $g \in G$ and $f \in C(G)$.

Proof. Let $\mathcal{P}(G) := \{\psi \in C(G)^* : \psi(\mathbf{1}_G) = 1,\ \forall f \in C(G)\ (f \ge 0 \Rightarrow \psi(f) \ge 0)\}$. Then $\delta_e \in \mathcal{P}(G)$, $\mathcal{P}(G)$ is convex and weak-* closed, and $\psi \in \mathcal{P}(G) \Rightarrow \|\psi\|_{\mathrm{op}} = 1$; hence $\mathcal{P}(G)$ is weak-* compact by Banach–Alaoglu.
For $g \in G$, set $T_{\mathrm{L}}(g)\psi := \psi \circ \pi_{\mathrm{L}}(g^{-1})$ and $T_{\mathrm{R}}(g)\psi := \psi \circ \pi_{\mathrm{R}}(g)$. Since $\pi_{\mathrm{L}}, \pi_{\mathrm{R}}$ preserve $\mathbf{1}_G$ and $C(G)_+$ and are norm-continuous in g, T_{L} and T_{R} are continuous affine G-actions on $\mathcal{P}(G)$.
Kakutani gives $\psi_{\mathrm{L}}, \psi_{\mathrm{R}} \in \mathcal{P}(G)$ with $T_{\mathrm{L}}(g)\psi_{\mathrm{L}} = \psi_{\mathrm{L}}$ and $T_{\mathrm{R}}(g)\psi_{\mathrm{R}} = \psi_{\mathrm{R}}$ for all $g \in G$.
Thus $\psi_{\mathrm{L}} \circ \pi_{\mathrm{L}}(g) = \psi_{\mathrm{L}}$ and $\psi_{\mathrm{R}} \circ \pi_{\mathrm{R}}(g) = \psi_{\mathrm{R}}$ for all $g \in G$.
Structural Commentary: Compactness lives in $\mathcal{P}(G)$; invariance comes from fixed points.
Key Insight: A fixed probability functional is invariant. □

Method 5.3.4. *Fixed Functional Pattern*
For a compact group G, set $\mathcal{P}(G) := \{\psi \in C(G)^* : \psi(\mathbf{1}_G) = 1,\ \forall f \in C(G)\ (f \ge 0 \Rightarrow \psi(f) \ge 0)\}$.

(i) Show $\mathcal{P}(G) \ne \varnothing$ via $\delta_x(f) := f(x)$, $x \in G$.
(ii) Show $\mathcal{P}(G)$ is weak-* compact convex: $\psi \in \mathcal{P}(G) \Rightarrow \|\psi\|_{\mathrm{op}} = 1$,
Banach–Alaoglu, and weak-* closed defining conditions.
(iii) Define $T_{\mathrm{L}}(g)\psi := \psi \circ \pi_{\mathrm{L}}(g^{-1})$ or $T_{\mathrm{R}}(g)\psi := \psi \circ \pi_{\mathrm{R}}(g)$.
For $T = T_{\mathrm{L}}, T_{\mathrm{R}}$, check $T(g)\mathcal{P}(G) \subseteq \mathcal{P}(G)$ and weak-* continuity.
(iv) Apply Kakutani: find $\psi \in \mathcal{P}(G)$ with $T(g)\psi = \psi$ for all $g \in G$.
(v) Apply Riesz–Markov, $\psi(f) = \int_G f \, d\mu$, and read $T(g)\psi = \psi$ as invariance of μ.

Structural Commentary: Compact state space plus adjoint action.
Key Insight: Haar measure is an invariant state represented by Riesz–Markov.

Proposition 5.3.5. *Translated Measures and Test Functions*
Let G be a compact group, let μ be a Borel measure on $\mathcal{B}(G)$, and fix $g \in G$.
Set $L_g(x) := gx$ and $R_g(x) := xg$ for $x \in G$.
For $A \in \mathcal{B}(G)$, define $\mu_g^{\mathrm{L}}(A) := \mu(L_g(A)) = \mu(gA)$ and $\mu_g^{\mathrm{R}}(A) := \mu(R_g(A)) = \mu(Ag)$.

(i) μ_g^{L} and μ_g^{R} are Borel measures on $\mathcal{B}(G)$.
(ii) If μ is Radon, then μ_g^{L} and μ_g^{R} are Radon.
(iii) For every $f \in C(G)$, $\int_G \pi_{\mathrm{L}}(g)f \, d\mu = \int_G f \, d\mu_g^{\mathrm{L}}$ and $\int_G \pi_{\mathrm{R}}(g)f \, d\mu = \int_G f \, d\mu_g^{\mathrm{R}}$.

Proof. For every $h \in G$, the maps L_h and R_h are homeomorphisms of G.

(i) Since L_g and R_g map Borel sets bijectively onto Borel sets, μ_g^{L} and μ_g^{R} are defined on $\mathcal{B}(G)$.
If $(A_n)_{n\geq 1}$ are pairwise disjoint Borel sets, then $L_g(\bigcup_n A_n) = \bigcup_n L_g(A_n)$ and $R_g(\bigcup_n A_n) = \bigcup_n R_g(A_n)$, with disjoint unions; countable additivity follows from countable additivity of μ.
(ii) Since $L_{g^{-1}}^{-1} = L_g$ and $R_{g^{-1}}^{-1} = R_g$, $\mu_g^{\mathrm{L}} = (L_{g^{-1}})_*\mu$ and $\mu_g^{\mathrm{R}} = (R_{g^{-1}})_*\mu$.
If μ is Radon, pushforward by a homeomorphism preserves compact sets, open sets, and Radon regularity.
Hence both translated measures are Radon.
(iii) For $f \in C(G)$, the pushforward identity gives $\int_G f \, d\mu_g^{\mathrm{L}} = \int_G f \circ L_{g^{-1}} \, d\mu = \int_G \pi_{\mathrm{L}}(g)f \, d\mu$
and $\int_G f \, d\mu_g^{\mathrm{R}} = \int_G f \circ R_{g^{-1}} \, d\mu = \int_G \pi_{\mathrm{R}}(g)f \, d\mu$.

Structural Commentary: Translated measures are inverse-translation pushforwards.
Key Insight: Measure translation is dual to test-function precomposition. □

Theorem 5.3.6. *Haar Measure on Compact Groups*
Let G be a compact group. Let $\mathcal{P}_{\mathrm{L}}(G)$ and $\mathcal{P}_{\mathrm{R}}(G)$ be the left- and right-invariant Radon probability measures on $\mathcal{B}(G)$, respectively.

(i) $\mathcal{P}_{\mathrm{L}}(G) \neq \emptyset$.
(ii) $\mathcal{P}_{\mathrm{R}}(G) \neq \emptyset$.
(iii) For every $(\lambda, \rho) \in \mathcal{P}_{\mathrm{L}}(G) \times \mathcal{P}_{\mathrm{R}}(G)$, $\lambda = \rho$.

Consequently, G admits a unique Haar measure, and this measure is right-invariant.

Proof. For a Radon probability measure σ on $\mathcal{B}(G)$, set $\Phi_\sigma(f) := \int_G f \, d\sigma$ for $f \in C(G)$.

(i) Choose ψ_{L} from Theorem 5.3.3. Riesz–Markov gives a Radon probability measure μ with $\psi_{\mathrm{L}} = \Phi_\mu$ on $C(G) = C_c(G)$. For $g \in G$, Proposition 5.3.5 gives $\Phi_{\mu_g^{\mathrm{L}}}(f) = \Phi_\mu(\pi_{\mathrm{L}}(g)f) = \Phi_\mu(f)$ for every $f \in C(G)$. Thus $\mu_g^{\mathrm{L}} = \mu$ by Theorem 5.2.13, so $\mu(gA) = \mu(A)$ for every $A \in \mathcal{B}(G)$ and $\mu \in \mathcal{P}_{\mathrm{L}}(G)$.
(ii) Choose ψ_{R} from Theorem 5.3.3. Riesz–Markov gives a Radon probability measure ν with $\psi_{\mathrm{R}} = \Phi_\nu$ on $C(G) = C_c(G)$. For $g \in G$, Proposition 5.3.5 gives $\Phi_{\nu_g^{\mathrm{R}}}(f) = \Phi_\nu(\pi_{\mathrm{R}}(g)f) = \Phi_\nu(f)$ for every $f \in C(G)$. Thus $\nu_g^{\mathrm{R}} = \nu$ by Theorem 5.2.13, so $\nu(Ag) = \nu(A)$ for every $A \in \mathcal{B}(G)$ and $\nu \in \mathcal{P}_{\mathrm{R}}(G)$.
(iii) Let $(\lambda, \rho) \in \mathcal{P}_{\mathrm{L}}(G) \times \mathcal{P}_{\mathrm{R}}(G)$, fix $f \in C(G)$, and set $h_f(x, y) := f(xy)$. Fubini applies to $h_f \in C(G \times G)$. Left-invariance gives $\int_G h_f(x, y) \, d\lambda(y) = \Phi_\lambda(f)$ for every x, and right-invariance gives $\int_G h_f(x, y) \, d\rho(x) = \Phi_\rho(f)$ for every y. Since $\lambda(G) = \rho(G) = 1$, Fubini gives $\Phi_\lambda(f) = \Phi_\rho(f)$. Hence $\lambda = \rho$ by Theorem 5.2.13.

Taking $\mu \in \mathcal{P}_{\mathrm{L}}(G)$ from (i) and $\nu \in \mathcal{P}_{\mathrm{R}}(G)$ from (ii), (iii) gives $\mu = \nu$, so μ is right-invariant. If η is Haar, then $\eta \in \mathcal{P}_{\mathrm{L}}(G)$ and (iii) gives $\eta = \nu$, so Haar measure is unique.
Structural Commentary: Fixed states give existence; Fubini gives uniqueness.
Key Insight: Invariance makes both iterated integrals constant. □

Proposition 5.3.7. *Point Masses of Haar Measure*
Let G be a compact group, and let μ be Haar measure on $\mathcal{B}(G)$.

(i) If G is finite, then every $A \subseteq G$ is Borel and $\mu(A) = \frac{|A|}{|G|}$.
(ii) If G is infinite, then $\mu(\{x\}) = 0$ for every $x \in G$.

Proof. Set $c := \mu(\{e\})$. For each $x \in G$, left-invariance gives $\mu(\{x\}) = \mu(x\{e\}) = c$.

(i) If G is finite, then G is discrete, so every $A \subseteq G$ is Borel. Since $1 = \mu(G) = \sum_{x \in G} \mu(\{x\}) = |G|\, c$, one has $c = \frac{1}{|G|}$, and finite additivity gives $\mu(A) = \sum_{a \in A} \mu(\{a\}) = \frac{|A|}{|G|}$.
(ii) If G is infinite and $c > 0$, choose $n \in \mathbb{N}$ with $n > \frac{1}{c}$ and distinct $x_1, \ldots, x_n \in G$.
Finite additivity gives $1 = \mu(G) \geq \sum_{j=1}^{n} \mu(\{x_j\}) = nc > 1$, a contradiction; hence $c = 0$.

Structural Commentary: Translation reduces point masses to one scalar.
Key Insight: Normalization fixes or kills that scalar. □

Proposition 5.3.8. *Nonempty Open Sets Have Positive Haar Measure*
Let G be a compact group, μ be Haar measure on $\mathcal{B}(G)$, $O \subseteq G$ be nonempty and open. Then $\mu(O) > 0$.

Proof. Choose $a \in O$. For each $x \in G$, $x = (xa^{-1})a \in (xa^{-1})O$, so $\{gO : g \in G\}$ is an open cover of G.
Compactness gives $g_1, \ldots, g_n \in G$ such that $G = \bigcup_{j=1}^n g_j O$. If $\mu(O) = 0$, then left-invariance and subadditivity give $1 = \mu(G) \leq \sum_{j=1}^n \mu(g_j O) = \sum_{j=1}^n \mu(O) = 0$, a contradiction. Hence $\mu(O) > 0$.
Structural Commentary: Compactness gives a finite translate cover.
Key Insight: Zero mass cannot cover total mass. □

6 Regular Approximation and Caratheodory Tests

6.1 Borel Approximation and Regular Measurable Sets

The Caratheodory definition is intrinsic to outer measure, but in $\mathbb{R}^p$ it has a topological shadow: measurable sets are exactly those that can be approximated by open, closed, compact, and cell-built sets up to arbitrarily small error. That bridge is also the route to the first nonmeasurable and non-Borel examples.

Lemma 6.1.1. *Outer Approximation by Open and G_δ Sets*
For $A \subseteq \mathbb{R}^p$:

(i) for every $\varepsilon > 0$, there exists open $G \subseteq \mathbb{R}^p$ with $A \subseteq G$ and $m(G) \le m_p^*(A) + \varepsilon$;
(ii) $m_p^*(A) = \inf\{m(G) : A \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\}$;
(iii) there exists G_δ set $H \supseteq A$ with $m(H) = m_p^*(A)$.

Proof.

(i) If $m_p^*(A) = +\infty$, take $G = \mathbb{R}^p$.
If $m_p^*(A) < +\infty$, choose an open-cell cover $A \subseteq \bigcup_k I_k$ with $\sum_k \mathrm{vol}_p(I_k) \le m_p^*(A) + \varepsilon$, and set $G := \bigcup_k I_k$.
Then G is open, $A \subseteq G$, and $m(G) \le \sum_k m(I_k) = \sum_k \mathrm{vol}_p(I_k) \le m_p^*(A) + \varepsilon$.
(ii) By (i), $\inf\{m(G) : A \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\} \le m_p^*(A) + \varepsilon$ for all $\varepsilon > 0$; let $\varepsilon \downarrow 0$.
Conversely, $A \subseteq G$ gives $m_p^*(A) \le m_p^*(G) = m(G)$ for open G; take the infimum over G.
(iii) If $m_p^*(A) = +\infty$, take $H = \mathbb{R}^p$.
If $m_p^*(A) < +\infty$, choose open $G_n \supseteq A$ with $m(G_n) \le m_p^*(A) + 1/n$ and set $H := \bigcap_n G_n$.
Then H is G_δ, $A \subseteq H$, and $m_p^*(A) \le m(H) \le m(G_n) \le m_p^*(A) + 1/n$ for all n, so $m(H) = m_p^*(A)$.

Structural Commentary: Outer measure can be realised by a Borel envelope from outside.
Key Insight: Infima over cell covers can be repackaged as open-set approximation. □

Corollary 6.1.2. *Null Sets Sit Inside Borel Null Sets*
Let $Z \subseteq \mathbb{R}^p$ with $m_p^*(Z) = 0$. Then there exists $B \subseteq \mathbb{R}^p$ such that:

(i) $Z \subseteq B$;
(ii) $B \in \mathcal{B}(\mathbb{R}^p)$;
(iii) $m(B) = 0$.

Proof. By Lemma 6.1.1, choose open sets $(G_n)_{n\ge1}$ in $\mathbb{R}^p$ and set $B := \bigcap_{n=1}^{\infty} G_n$ so that $Z \subseteq B$ and $m(B) = m_p^*(Z)$.

(i) The chosen envelope satisfies $Z \subseteq B$.
(ii) Since each G_n is open, $B = \bigcap_{n=1}^{\infty} G_n \in \mathcal{B}(\mathbb{R}^p)$.
(iii) Since $m_p^*(Z) = 0$, $m(B) = m_p^*(Z) = 0$.

Structural Commentary: A null set has a G_δ null envelope.
Key Insight: Outer approximation gives a Borel null superset. □

Theorem 6.1.3. *Open and G_δ Characterizations of Measurability*
For $E \subseteq \mathbb{R}^p$, the following are equivalent:

(i) $E \in \mathcal{L}^p$;
(ii) $\forall \varepsilon > 0$, $\exists$ open $G \subseteq \mathbb{R}^p$ such that $E \subseteq G$ and $m_p^*(G \setminus E) < \varepsilon$;
(iii) $\exists$ G_δ set $H \subseteq \mathbb{R}^p$ such that $E \subseteq H$ and $m_p^*(H \setminus E) = 0$;
(iv) $\exists$ G_δ set $H \subseteq \mathbb{R}^p$ and $Z \in \mathcal{L}^p$ such that $Z \subseteq H$, $m(Z) = 0$, and $E = H \setminus Z$.

Proof.

(i) $\Rightarrow$ (ii). If $m(E) < +\infty$, choose open $G \supseteq E$ with $m(G) \le m(E) + \varepsilon/2$.
Since $E \in \mathcal{L}^p$, $m_p^*(G \setminus E) = m(G) - m(E) < \varepsilon$. If $m(E) = +\infty$, set $E_1 := E \cap \{x : \|x\|_2 \le 1\}$ and $E_n := E \cap \{x : n-1 < \|x\|_2 \le n\}$ for $n \ge 2$. For each n, choose open $G_n \supseteq E_n$ with $m_p^*(G_n \setminus E_n) < \varepsilon 2^{-n}$; then $G := \bigcup_n G_n$ is open, $E \subseteq G$, and $G \setminus E \subseteq \bigcup_n (G_n \setminus E_n)$, so $m_p^*(G \setminus E) < \varepsilon$.
(ii) $\Rightarrow$ (iii). Choose open $G_n \supseteq E$ with $m_p^*(G_n \setminus E) < 1/n$, and set $H := \bigcap_n G_n$.
Then H is G_δ, $E \subseteq H$, and $H \setminus E \subseteq G_n \setminus E$ for all n; hence $m_p^*(H \setminus E) = 0$.
(iii) $\Rightarrow$ (i). Since H is G_δ, $H \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. Also $m_p^*(H \setminus E) = 0$ implies $H \setminus E \in \mathcal{L}^p$ by Proposition 3.3.7.
Thus $E = H \setminus (H \setminus E) \in \mathcal{L}^p$.
(iii) $\Leftrightarrow$ (iv). If (iii) holds, set $Z := H \setminus E$; then $Z \subseteq H$, $m(Z) = 0$, and $E = H \setminus Z$.
Conversely, if (iv) holds, then $E \subseteq H$ and $H \setminus E = Z$, so $m_p^*(H \setminus E) = 0$.

Structural Commentary: Lebesgue measurable sets are G_δ sets with null material removed.
Key Insight: Small open excess becomes a null G_δ excess after countably many approximations. □

Corollary 6.1.4. *Open Outer Regularity for Measurable Sets*
If $E \in \mathcal{L}^p$, then

(i) $\forall \varepsilon > 0$, $\exists$ open $G \supseteq E$ such that $m(G) \le m(E) + \varepsilon$;
(ii) $m(E) = \inf\{m(G) : E \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\}$.

Proof.

(i) By Theorem 6.1.3, choose open $G \supseteq E$ with $m_p^*(G \setminus E) < \varepsilon$.
If $m(E) < +\infty$, then $m(G) = m(E) + m(G \setminus E) \le m(E) + \varepsilon$; if $m(E) = +\infty$, the inequality is trivial.
(ii) From (i), $\inf_G m(G) \le m(E)$. Conversely, $E \subseteq G$ gives $m(E) \le m(G)$ by monotonicity; take $\inf_G$.

Structural Commentary: Measurable sets are measured from outside by open supersets.
Key Insight: Open excess controls the outer regular infimum. □

Theorem 6.1.5. *Closed and F_σ Characterizations of Measurability*
For $E \subseteq \mathbb{R}^p$, the following are equivalent:

(i) $E \in \mathcal{L}^p$;
(ii) $\forall \varepsilon > 0$, $\exists$ *closed* $F \subseteq E$ *such that* $m_p^*(E \setminus F) < \varepsilon$;
(iii) $\exists$ F_σ *set* $K \subseteq E$ *such that* $m_p^*(E \setminus K) = 0$;
(iv) $\exists$ F_σ *set* $K \subseteq E$ *and* $Z \in \mathcal{L}^p$ *such that* $Z \subseteq E$, $m(Z) = 0$, *and* $E = K \cup Z$.

Proof.

(*i*) $\Rightarrow$ (*ii*). Since $E^c \in \mathcal{L}^p$, Theorem 6.1.3 gives open $G \supseteq E^c$ with $m_p^*(G \setminus E^c) < \varepsilon$. Set $F := G^c$. Then F is closed, $F \subseteq E$, and $E \setminus F = G \setminus E^c$, so $m_p^*(E \setminus F) < \varepsilon$.
(*ii*) $\Rightarrow$ (*iii*). Choose closed $F_n \subseteq E$ with $m_p^*(E \setminus F_n) < 1/n$, and set $K := \bigcup_n F_n$. Then K is F_σ, $K \subseteq E$, and $E \setminus K \subseteq E \setminus F_n$ for all n; hence $m_p^*(E \setminus K) = 0$.
(*iii*) $\Rightarrow$ (*i*). Since K is F_σ, $K \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. Also $m_p^*(E \setminus K) = 0$ implies $E \setminus K \in \mathcal{L}^p$ by Proposition 3.3.7. Thus $E = K \cup (E \setminus K) \in \mathcal{L}^p$.
(*iii*) $\Leftrightarrow$ (*iv*). If (iii) holds, set $Z := E \setminus K$; then $Z \subseteq E$, $m(Z) = 0$, and $E = K \cup Z$. Conversely, if (iv) holds, then $E \setminus K \subseteq Z$, so $m_p^*(E \setminus K) = 0$.

Structural Commentary: Inside approximation is the complement of outside approximation.
Key Insight: Measurable sets are also F_σ sets with null material added. □

Corollary 6.1.6. *Closed Inner Regularity*
If $E \in \mathcal{L}^p$, then

(i) $\forall \varepsilon > 0$, $\exists$ *closed* $F \subseteq E$ *such that* $m(E) \leq m(F) + \varepsilon$;
(ii) $m(E) = \sup\{m(F) : F \subseteq E,\ F \subseteq \mathbb{R}^p \text{ closed}\}$.

Proof.

(i) By Theorem 6.1.5, choose closed $F \subseteq E$ with $m_p^*(E \setminus F) < \varepsilon$. Since $F \in \mathcal{L}^p$ and $E = F \sqcup (E \setminus F)$, $m(E) = m(F) + m(E \setminus F) \leq m(F) + \varepsilon$.
(ii) From (i), $m(E) \leq \sup_F m(F)$. Conversely, $F \subseteq E$ gives $m(F) \leq m(E)$; take $\sup_F$.

Structural Commentary: Measurable sets are measured from inside by closed subsets.
Key Insight: Small discarded measure controls the inner regular supremum. □

Corollary 6.1.7. *Lebesgue Regularity and Borel Completion on $\mathbb{R}$*
For $A \in \mathcal{L}^1$,

(i) $m(A) = \inf\{m(G) : A \subseteq G,\ G \text{ open in } \mathbb{R}\}$.
(ii) $m(A) = \sup\{m(K) : K \subseteq A,\ K \text{ compact in } \mathbb{R}\}$, *finite-measure case applied to* $A \cap (-n, n]$; $m(A) = +\infty$.
(iii) There exists $B \in \mathcal{B}(\mathbb{R})$ *with* $A \subseteq B$ *and* $m(B \setminus A) = 0$; *equivalently,* $\mathcal{L}^1$ *is the completion of* $\mathcal{B}(\mathbb{R})$.

Proof.

(i) For $\varepsilon > 0$, regular approximation gives open $G \supseteq A$ with $m(G \setminus A) < \varepsilon$, hence $m(G) \leq m(A) + \varepsilon$. Monotonicity gives the reverse inequality.
(ii) If $m(A) < +\infty$, choose closed $F \subseteq A$ with $m(A \setminus F) < \varepsilon$. Since $F \cap [-n, n] \uparrow F$, continuity from below gives compact $K = F \cap [-n, n]$ with $m(F) - m(K) < \varepsilon$, so $m(A) - m(K) < 2\varepsilon$. If $m(A) = +\infty$, apply the finite case to $A_n = A \cap (-n, n]$ and let $n \to \infty$.
(iii) Choose open $G_k \supseteq A$ with $m(G_k \setminus A) < 1/k$, and set $B = \bigcap_k G_k$. Then $B \in \mathcal{B}(\mathbb{R})$, $A \subseteq B$, and $m(B \setminus A) = 0$. Thus every Lebesgue set is Borel plus a null subset.
Conversely, Borel sets and null subsets are Lebesgue measurable.

Structural Commentary: Lebesgue sets are Borel sets plus null-set ambiguity.
Key Insight: Open outer approximation and compact inner approximation are the usable form of regularity. □

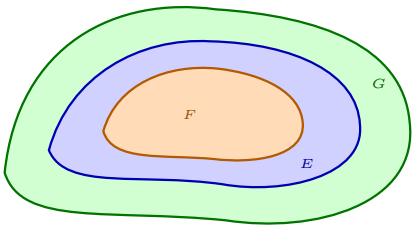

Figure 196: Regular approximation sandwiches E between F and G with small excess.

Theorem 6.1.8. *Lebesgue Measurable Non-Borel Sets Exist*

(i) There exists $E \subseteq \mathbb{R}$ *such that* $E \in \mathcal{L}^1$ *and* $E \notin \mathcal{B}(\mathbb{R})$.
(ii) $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}^1$.

Proof.

(i) Lemma 4.1.7 gives $|\mathcal{B}(\mathbb{R})| \leq \mathfrak{c}$. Let C be the Cantor set. Since $|C| = \mathfrak{c}$ and $m(C) = 0$, Proposition 3.3.7 gives $\mathcal{P}(C) \subseteq \mathcal{L}^1$, so $|\mathcal{L}^1| \geq 2^{\mathfrak{c}} > \mathfrak{c}$. Hence some $E \in \mathcal{L}^1$ is not Borel.
(ii) Proposition 4.1.11 gives $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}^1$, and (i) makes the containment proper.

Structural Commentary: Countable generation is smaller than completion by null subsets.
Key Insight: Completeness creates more measurable sets than topology can generate. □

6.2 Compact and Cell Approximation

Theorem 6.2.1. *Compact Approximation at Finite Outer Measure*
Let $E \subseteq \mathbb{R}^p$ with $m_p^*(E) < +\infty$. Then $E \in \mathcal{L}^p$ iff $\forall \varepsilon > 0$, $\exists$ compact $K \subseteq E$ with $m_p^*(E \setminus K) < \varepsilon$.

Proof.

($\Rightarrow$) Since $m(E) < +\infty$, choose closed $F \subseteq E$ with $m(E \setminus F) < \frac{\varepsilon}{2}$ by Theorem 6.1.5.
Let $Q_N := [-N, N]^p$. Since $F \cap Q_N \uparrow F$, continuity from below gives N with $m(F \setminus Q_N) < \frac{\varepsilon}{2}$.
Then $K := F \cap Q_N$ is compact, $K \subseteq E$, and $m_p^*(E \setminus K) = m(E \setminus K) \leq m(E \setminus F) + m(F \setminus Q_N) < \varepsilon$.

($\Leftarrow$) Choose compact $K_n \subseteq E$ with $m_p^*(E \setminus K_n) < \frac{1}{n}$ and set $K := \bigcup_{n \geq 1} K_n$.
Then K is F_σ, hence $K \in \mathcal{L}^p$, and $E \setminus K \subseteq E \setminus K_n$ for all n, so $m_p^*(E \setminus K) = 0$.
By Proposition 3.3.7, $E \setminus K \in \mathcal{L}^p$, hence $E = K \cup (E \setminus K) \in \mathcal{L}^p$.

Structural Commentary: Finite measure permits bounded truncation before inner closure.
Key Insight: Compact approximation is closed approximation plus a finite-measure tail estimate. □

Theorem 6.2.2. *Finite Cell Approximation in Symmetric Difference*
If $E \in \mathcal{L}^p$ and $m(E) < +\infty$, then $\forall \varepsilon > 0$, $\exists U = \bigcup_{j=1}^{N} I_j$ with I_j bounded open cells and $m(E \triangle U) < \varepsilon$.

Proof. Choose compact $K \subseteq E$ with $m(E \setminus K) < \frac{\varepsilon}{2}$ by Theorem 6.2.1, and open $G \supseteq E$ with $m_p^*(G \setminus E) < \frac{\varepsilon}{2}$ by Theorem 6.1.3. For each $x \in K$, choose a bounded open cell I_x with $x \in I_x$ and $\overline{I_x} \subseteq G$. Compactness gives $I_1, \ldots, I_N$ with $K \subseteq U := \bigcup_{j=1}^{N} I_j \subseteq G$. Then $E \triangle U \subseteq (E \setminus K) \cup (G \setminus E)$, so subadditivity gives $m(E \triangle U) < \varepsilon$.
Structural Commentary: Compactness turns a countable open cover into a finite geometric approximant.
Key Insight: Approximate from outside by cells and from inside by a compact core. □

Corollary 6.2.3. *Cell Approximation Variant Choices*
Let $E \in \mathcal{L}^p$ with $m(E) < +\infty$, and let $\varepsilon > 0$. There exists $A \subseteq \mathbb{R}^p$ with $m(E \triangle A) < \varepsilon$ in each format:

(i) A is a finite union of bounded open cells;
(ii) A is a finite union of bounded closed cells;
(iii) A is a finite union of bounded half-open cells of a fixed orientation.

Proof.

(i) By Theorem 6.2.2, choose $A = \bigcup_{j=1}^{N} I_j$ with I_j bounded open cells and $m(E \triangle A) < \varepsilon$.

(ii) Choose a finite open-cell union $U = \bigcup_{j=1}^{M} J_j$ with $m(E \triangle U) < \frac{\varepsilon}{2}$. If $M = 0$, take $A = \emptyset$; otherwise choose closed cells $C_j \subseteq J_j$ with $m(J_j \setminus C_j) < \frac{\varepsilon}{2M}$ and set $A := \bigcup_{j=1}^{M} C_j$. Then $m(U \setminus A) < \frac{\varepsilon}{2}$, so $m(E \triangle A) < \varepsilon$.

(iii) Choose a finite open-cell union U with $m(E \triangle U) < \varepsilon$. Refine by the finite coordinate grid from the endpoints of the open cells, and let A be the union of the half-open grid atoms contained in U.
Then $U \triangle A$ is contained in finitely many cell faces, each outer-null, so $m(E \triangle A) < \varepsilon$.

Structural Commentary: Cell format changes only boundary-null pieces.
Key Insight: A compact core inside an open envelope permits finite cell variants. □

Corollary 6.2.4. *Real-Line Finite Open-Interval Approximation*
If $E \in \mathcal{L}^1$ and $m(E) < +\infty$, then $\forall \varepsilon > 0$, $\exists O = \bigcup_{j=1}^{N} I_j$ with I_j pairwise disjoint bounded open intervals and $m(E \setminus O) + m(O \setminus E) = m(E \triangle O) < \varepsilon$.

Proof. By Theorem 6.2.2 with $p = 1$, choose a finite union U of bounded open intervals with $m(E \triangle U) < \varepsilon$. Write $U = O = \bigcup_{j=1}^{N} I_j$ as the disjoint union of its open interval components.
Since $E \setminus O$ and $O \setminus E$ are disjoint measurable sets, additivity gives $m(E \triangle O) = m(E \setminus O) + m(O \setminus E) < \varepsilon$.
Structural Commentary: Merge finite open intervals into disjoint components.
Key Insight: A compact core makes the countable cover finite; interval merging makes it disjoint. □

6.3 Caratheodory Tests and Inner Measure

Theorem 6.3.1. *Outer Measure Additivity with One Measurable Set*
Let $E \in \mathcal{L}^p$ and $F \subseteq \mathbb{R}^p$. Then $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.

Proof. Since $E \in \mathcal{L}^p$, Caratheodory splitting: $m_p^*(E \cup F) = m(E) + m_p^*(F \setminus E)$; $m_p^*(F) = m_p^*(E \cap F) + m_p^*(F \setminus E)$.
Hence $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.
Structural Commentary: One measurable cut splits arbitrary sets exactly.
Key Insight: Split both $E \cup F$ and F along the same measurable cut E. □

Corollary 6.3.2. *Disjoint Additivity and Difference*
Let $E \in \mathcal{L}^p$ and $F \subseteq \mathbb{R}^p$.

(i) $E \cap F = \emptyset \implies m_p^*(E \cup F) = m(E) + m_p^*(F)$.
(ii) $E \subseteq F$, $m(E) < +\infty \implies m_p^*(F \setminus E) = m_p^*(F) - m(E)$.

Proof.

(i) By Theorem 6.3.1, $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.
Since $E \cap F = \emptyset$ and $m_p^*(\emptyset) = 0$, $m_p^*(E \cup F) = m(E) + m_p^*(F)$.
(ii) Since $E \subseteq F$, Caratheodory splitting gives $m_p^*(F) = m(E) + m_p^*(F \setminus E)$; subtract $m(E) < +\infty$.

Structural Commentary: Exact splitting yields disjoint sums and finite differences.
Key Insight: Disjointness removes the overlap; containment isolates the remainder. □

Remark 6.3.3. *Excision Needs Measurability*
Let $E \subseteq \mathbb{R}^p$ with $m_p^*(E) < +\infty$. Open approximation may give, $\forall \varepsilon > 0$, an open $G \supseteq E$ with $m(G) - m_p^*(E) < \varepsilon$, but this does not imply $m_p^*(G \setminus E) < \varepsilon$. If $E \in \mathcal{L}^p$, Corollary 6.3.2 gives $m_p^*(G \setminus E) = m(G) - m(E)$.
Key Insight: Small open excess becomes small set difference only after E is a Caratheodory splitter.

Theorem 6.3.4. *Finite Outer Measure Inner Test*
If $m_p^*(E) < +\infty$, then $E \in \mathcal{L}^p \iff \exists B \in \mathcal{L}^p,\ B \subseteq E,\ m(B) = m_p^*(E)$.

Proof.

($\Rightarrow$) Set $B := E$. Then $B \in \mathcal{L}^p$, $B \subseteq E$, and $m(B) = m(E) = m_p^*(E)$.
($\Leftarrow$) Assume $B \in \mathcal{L}^p$, $B \subseteq E$, and $m(B) = m_p^*(E)$.
Since $m(B) < +\infty$, the difference formula gives $m_p^*(E \setminus B) = m_p^*(E) - m(B) = 0$.
Thus $E \setminus B \in \mathcal{L}^p$, and $E = B \cup (E \setminus B) \in \mathcal{L}^p$.

Structural Commentary: Exact inner mass leaves only a null remainder.
Key Insight: Equality of outer mass converts containment into measurability. □

Theorem 6.3.5. *Single Finite Container Test*
Let $A \in \mathcal{L}^p$, $m(A) < +\infty$, and $E \subseteq A$. Then $E \in \mathcal{L}^p \iff m(A) = m_p^*(E) + m_p^*(A \setminus E)$.

Proof.

($\Rightarrow$) Since $E \in \mathcal{L}^p$ and $E \subseteq A$,
Caratheodory splitting of A by E gives $m(A) = m(E) + m(A \setminus E) = m_p^*(E) + m_p^*(A \setminus E)$.
($\Leftarrow$) Choose G_δ set $H \supseteq A \setminus E$ with $m(H) = m_p^*(A \setminus E)$. Since $A \setminus E \subseteq A \cap H \subseteq H$, $m(A \cap H) = m_p^*(A \setminus E)$.
Set $B := A \setminus H$. $B \in \mathcal{L}^p$, $B \subseteq E$; $m(B) = m(A) - m(A \cap H) = m_p^*(E)$. By Theorem 6.3.4, $E \in \mathcal{L}^p$.

Structural Commentary: Inside a finite measurable container, Caratheodory only needs to be tested once.
Key Insight: Outer-approximate the complement; the remainder has full outer mass. □

Corollary 6.3.6. *Bounded Inner Measure Test*
For bounded cell $Q \subseteq \mathbb{R}^p$ and $E \subseteq Q$, define $m_{*,Q}(E) := m(Q) - m_p^*(Q \setminus E)$. Then $E \in \mathcal{L}^p \iff m_{*,Q}(E) = m_p^*(E)$.

Proof.

($\Rightarrow$) If $E \in \mathcal{L}^p$, then $m(Q) = m(E) + m_p^*(Q \setminus E)$, hence $m_{*,Q}(E) = m_p^*(E)$.
($\Leftarrow$) If $m_{*,Q}(E) = m_p^*(E)$, then $m(Q) = m_p^*(E) + m_p^*(Q \setminus E)$, so $E \in \mathcal{L}^p$ by the finite-container test.

Structural Commentary: Bounded inner measure tests one finite container.
Key Insight: Equality of inner and outer measure is a bounded version of Caratheodory. □

7 Vitali, Steinhaus, and Nonmeasurable Sets

7.1 Vitali, Steinhaus, and Nonmeasurability

Definition 7.1.1. *Difference Sets and Vitali Selectors*

(i) *Difference set*: for $A \subseteq \mathbb{R}^p$, $A - A := \{x - y : x, y \in A\}$; $A \subseteq B \Longrightarrow A - A \subseteq B - B$.
(ii) *Rational equivalence*: $x \sim_{\mathbb{Q}^p} y \iff x - y \in \mathbb{Q}^p$; equivalently, $[x]_{\mathbb{Q}^p} = x + \mathbb{Q}^p$.
(iii) *Vitali selector*: choose $\mathcal{V} \subseteq \mathbb{R}^p$ such that $|\mathcal{V} \cap (x + \mathbb{Q}^p)| = 1$ for every $x \in \mathbb{R}^p$.
(iv) *Rational translate*: for $q \in \mathbb{Q}^p$, $\mathcal{V}_q := q + \mathcal{V}$.

Lemma 7.1.2. *Steinhaus Lemma for Compact Sets*
If $K \subseteq \mathbb{R}^p$ is compact and $m(K) > 0$, then $\exists \delta > 0$ such that $B(0, \delta) \subseteq K - K$.

Proof. Choose open $G \supseteq K$ with $m(G) < 2m(K)$. Since K is compact and G^c is closed, set $\delta := \operatorname{dist}(K, G^c) > 0$. If $\|x\|_2 < \delta$, then $x + K \subseteq G$. If $(x + K) \cap K = \emptyset$, translation invariance and additivity give
$2m(K) = m(x + K) + m(K) = m((x + K) \cup K) \leq m(G) < 2m(K)$, a contradiction.
Hence $(x + K) \cap K \neq \emptyset$, so $x = k_1 - k_2 \in K - K$. Thus $B(0, \delta) \subseteq K - K$.
Structural Commentary: A positive compact set overlaps all sufficiently small translates of itself.
Key Insight: Disjoint small translates cannot fit inside a tight open envelope. □

Theorem 7.1.3. *Steinhaus Theorem*
If $E \in \mathcal{L}^p$ and $m(E) > 0$, then $\exists \delta > 0$ such that $B(0, \delta) \subseteq E - E$.

Proof. Set $E_n := E \cap B(0, n)$. Since $E_n \uparrow E$ and $m(E) > 0$, choose n with $0 < m(E_n) < +\infty$.
By compact inner approximation, choose compact $K \subseteq E_n$ with $m(K) > 0$.
Lemma 7.1.2 gives $\delta > 0$ with $B(0, \delta) \subseteq K - K \subseteq E - E$.
Structural Commentary: Positive measurable sets contain positive compact cores.
Key Insight: Difference-set thickness follows from compact inner regularity. □

Lemma 7.1.4. *Bounded Disjoint Translates Force Nullity*
Let $E \in \mathcal{L}^1$ and $\Lambda \subseteq \mathbb{R}$ be bounded, with E bounded and Λ countably infinite.
If $(\lambda + E)_{\lambda \in \Lambda}$ are pairwise disjoint, then $m(E) = 0$.

Proof. Since E and Λ are bounded, $U := \bigcup_{\lambda\in\Lambda} (\lambda + E)$ is bounded, so $m(U) < +\infty$.
By disjointness, countable additivity, and translation invariance, $m(U) = \sum_{\lambda\in\Lambda} m(\lambda + E) = \sum_{\lambda\in\Lambda} m(E)$.
Since Λ is countably infinite, finiteness forces $m(E) = 0$.
Structural Commentary: Infinitely many disjoint bounded translates cannot all carry positive equal mass.
Key Insight: Boundedness gives a finite container; translation invariance makes every summand the same. □

Proposition 7.1.5. *Real-Line Vitali Choice Sets*
Let $E \subseteq \mathbb{R}$ with $m_1^*(E) > 0$, and write $x \sim y$ iff $x - y \in \mathbb{Q}$.
(i) There exists bounded $E_0 \subseteq E$ with $m_1^*(E_0) > 0$.
(ii) If $C \subseteq E_0$ contains exactly one point from each $\sim$-class meeting E_0, then $C \notin \mathcal{L}^1$.
In particular, E contains a nonmeasurable subset.

Proof.
(i) Since $E = \bigcup_{n\geq 1} (E \cap [-n, n])$, countable subadditivity gives n with $E_0 := E \cap [-n, n]$ satisfying $m_1^*(E_0) > 0$.
(ii) Choose $b > 0$ with $E_0 \subseteq [-b, b]$ and set $\Lambda_0 := [-2b, 2b] \cap \mathbb{Q}$. Suppose $C \in \mathcal{L}^1$.
From $\lambda + c = \lambda' + c'$ with $\lambda, \lambda' \in \Lambda_0$ and $c, c' \in C$, we get $c - c' = \lambda' - \lambda \in \mathbb{Q}$; selector uniqueness gives $c = c'$ and $\lambda = \lambda'$. Thus $(\lambda + C)_{\lambda\in\Lambda_0}$ is pairwise disjoint, and Lemma 7.1.4 gives $m(C) = 0$.
For every $x \in E_0$, choose $c \in C$ with $x - c \in \mathbb{Q}$; since $x, c \in [-b, b]$, $x - c \in \Lambda_0$, so $E_0 \subseteq \bigcup_{\lambda\in\Lambda_0} (\lambda + C)$.
The right side is a countable union of null measurable sets, contradicting $m_1^*(E_0) > 0$.
Structural Commentary: Bounded rational shifts force the contradiction.
Key Insight: Selectors are either null under disjoint shifts or cover too much by null shifts. □

Lemma 7.1.6. *Rational Translates of a Vitali Selector Partition $\mathbb{R}^p$*
Let $(q_i)_{i\geq 1}$ enumerate $\mathbb{Q}^p$.
(i) $\mathbb{R}^p = \bigcup_{i=1}^{\infty} \mathcal{V}_{q_i}$.
(ii) Each $x \in \mathbb{R}^p$ has a unique representation $x = q_i + v$ with $i \in \mathbb{N}$ and $v \in \mathcal{V}$; hence $\mathbb{R}^p = \bigsqcup_{i=1}^{\infty} \mathcal{V}_{q_i}$.

Proof.
(i) For $x \in \mathbb{R}^p$, choose the unique $v \in \mathcal{V} \cap (x + \mathbb{Q}^p)$. Then $x - v = q_i$ for some i, so $x = q_i + v \in \mathcal{V}_{q_i}$.
(ii) If $q_i + v = q_j + w$ with $v, w \in \mathcal{V}$, then $v - w = q_j - q_i \in \mathbb{Q}^p$.
Selector uniqueness gives $v = w$, hence $q_i = q_j$ and $i = j$.
Structural Commentary: Rational translates enumerate the equivalence classes selected by $\mathcal{V}$.
Key Insight: One representative per class makes the rational translates disjoint and exhaustive. □

Theorem 7.1.7. *Vitali Selectors Are Nonmeasurable*
If $\mathcal{V} \subseteq \mathbb{R}^p$ is a Vitali selector, then $\mathcal{V} \notin \mathcal{L}^p$.

Proof. Assume $\mathcal{V} \in \mathcal{L}^p$.
(i) If $m(\mathcal{V}) > 0$, Steinhaus gives $\delta > 0$ with $B(0, \delta) \subseteq \mathcal{V} - \mathcal{V}$. Choose $0 \neq q \in \mathbb{Q}^p \cap B(0, \delta)$. Then $q = v_1 - v_2$ for some $v_1, v_2 \in \mathcal{V}$, so $v_1 \sim_{\mathbb{Q}^p} v_2$; selector uniqueness gives $v_1 = v_2$, hence $q = 0$, contradiction.
(ii) If $m(\mathcal{V}) = 0$, then each translate $\mathcal{V}_{q_i}$ is null. By Lemma 7.1.6, $\mathbb{R}^p = \bigsqcup_i \mathcal{V}_{q_i}$, so every bounded cell is contained in a countable union of null sets and has measure 0, contradicting positive cell volume.
Structural Commentary: Positive and zero measure are both impossible.
Key Insight: Steinhaus forbids positive measure; countable translated partition forbids zero measure. □

Theorem 7.1.8. *Nonmeasurable Subsets Are Everywhere*
If $E \subseteq \mathbb{R}^p$ and $m_p^*(E) > 0$, then $\exists N \subseteq E$ with $N \notin \mathcal{L}^p$.

Proof. Let (q_i) enumerate $\mathbb{Q}^p$ and set $E_i := E \cap \mathcal{V}_{q_i}$.
If $E_i \in \mathcal{L}^p$ and $m(E_i) > 0$, then $E_i - E_i$ contains a ball about 0, so $0 \neq q \in \mathbb{Q}^p \cap (E_i - E_i)$.
But $E_i \subseteq q_i + \mathcal{V}$, so $q = (q_i + v_1) - (q_i + v_2) = v_1 - v_2$ with $v_1, v_2 \in \mathcal{V}$, contradicting selector uniqueness.
Hence each measurable E_i is null. If every E_i were measurable, then $E = \bigcup_i E_i$ would satisfy $m_p^*(E) = 0$, contradiction. Thus some $E_i \subseteq E$ is nonmeasurable.
Structural Commentary: Any positive outer-measure set meets the Vitali partition in a nonmeasurable piece.
Key Insight: Vitali pieces cannot be measurable and positive. □

Corollary 7.1.9. *Nonadditive Decompositions*
(i) If $E \in \mathcal{L}^p$ and $0 < m(E) < +\infty$, then $\exists B, C \notin \mathcal{L}^p$ such that $E = B \sqcup C$ and $m(E) < m_p^*(B) + m_p^*(C)$.
(ii) If $B \notin \mathcal{L}^p$, $m_p^*(B) < +\infty$, and H is G_δ with $B \subseteq H$, $m(H) = m_p^*(B)$, then $C := H \setminus B \notin \mathcal{L}^p$ and $m(H) = m_p^*(B \cup C) < m_p^*(B) + m_p^*(C)$.

Proof.
(i) Pick $B \subseteq E$, $B \notin \mathcal{L}^p$, and put $C := E \setminus B$. Then $C \notin \mathcal{L}^p$, else $B = E \setminus C \in \mathcal{L}^p$.
Subadditivity gives $\leq$; equality would force $B \in \mathcal{L}^p$ by the finite-container test in E.
(ii) Since $H \in \mathcal{L}^p$, $C \in \mathcal{L}^p$ would give $B = H \setminus C \in \mathcal{L}^p$, contradiction. Also $m_p^*(C) > 0$, else $C \in \mathcal{L}^p$.
Thus $B \cup C = H$ and $m(H) = m_p^*(B)$ give $m_p^*(B \cup C) = m(H) < m_p^*(B) + m_p^*(C)$.
Structural Commentary: Nonmeasurability appears exactly as strict failure of additive splitting.
Key Insight: Equality in a finite measurable container is a measurability test. □

Theorem 7.1.10. *A Nonmeasurable Set with Only Null Measurable Pieces*
There exists $\mathcal{U} \subseteq \mathbb{R}^p$ such that:

(i) if $F \in \mathcal{L}^p$ and $F \subseteq \mathcal{U}$, then $m(F) = 0$;
(ii) if $F \in \mathcal{L}^p$ and $F \subseteq \mathcal{U}^c$, then $m(F) = 0$;
(iii) $\mathcal{U} \notin \mathcal{L}^p$.

Proof. Let $e := (1, \dots, 1)$, $G_0 := \mathbb{Q}^p + (2\sqrt{2})\mathbb{Z}e$, $G_1 := \sqrt{2}e + G_0$, $G := G_0 \sqcup G_1 = \mathbb{Q}^p + \sqrt{2}\mathbb{Z}e$; G_0, G_1 dense.
Let $\mathcal{E}$ select one representative from each class modulo G, and set $\mathcal{U} := G_0 + \mathcal{E}$.
Then $\mathbb{R}^p = (G_0 + \mathcal{E}) \sqcup (G_1 + \mathcal{E})$ and $\mathcal{U}^c = G_1 + \mathcal{E} = \sqrt{2}e + \mathcal{U}$.

(i) Suppose $F \in \mathcal{L}^p$, $F \subseteq \mathcal{U}$, and $m(F) > 0$. By Steinhaus, $B(0, r) \subseteq F - F$ for some $r > 0$; choose $g_1 \in G_1 \cap B(0, r)$. Since $F \subseteq G_0 + \mathcal{E}$, $g_1 = (g_0 + s_1) - (g_0' + s_2)$ for some $g_0, g_0' \in G_0$ and $s_1, s_2 \in \mathcal{E}$. Thus $s_1 - s_2 = g_1 - g_0 + g_0' \in G$, so $s_1 = s_2$ and $g_1 = g_0 - g_0' \in G_0$, contradicting $G_0 \cap G_1 = \emptyset$.
(ii) If $F \in \mathcal{L}^p$ and $F \subseteq \mathcal{U}^c = \sqrt{2}\,e + \mathcal{U}$, then $F - \sqrt{2}\,e \subseteq \mathcal{U}$ and $m(F) = m(F - \sqrt{2}\,e) = 0$ by (i).
(iii) If $\mathcal{U} \in \mathcal{L}^p$, then (i)–(ii) give $m(\mathcal{U}) = m(\mathcal{U}^c) = 0$, so $m([0, 1]^p) = 0$, contradiction.

Structural Commentary: This construction partitions representatives using two dense cosets.
Key Insight: Positive measurable pieces force forbidden coset differences. □

8 Cantor–Lebesgue Singular Function

8.1 Cantor-Lebesgue Singular Function Construction

Definition 8.1.1. *Cantor–Lebesgue Singular Function*
Let P be the Cantor set. For $k \geq 1$, let $I_{k,1}, \dots, I_{k,2^k-1}$ list, left to right, the components of the stage-k removed set U_k. Set $U := \bigcup_{k \geq 1} U_k = [0, 1] \setminus P$.

(i) For $x \in U$, choose $k \geq 1$ and $1 \leq j \leq 2^k - 1$ with $x \in I_{k,j}$, and set $\varphi(x) := \frac{j}{2^k}$.
(ii) This value is independent of the chosen stage: if $I_{k,j} = I_{\ell,m}$ with $\ell \geq k$, then $m = 2^{\ell-k} j$ and $\frac{m}{2^\ell} = \frac{j}{2^k}$.
(iii) On P, set $\varphi(0) := 0$ and $\varphi(x) := \sup\{\varphi(t) : t \in U \cap [0, x]\}$ for $x \in P \setminus \{0\}$.

The resulting $\varphi : [0, 1] \to [0, 1]$ is the *Cantor–Lebesgue singular function*.

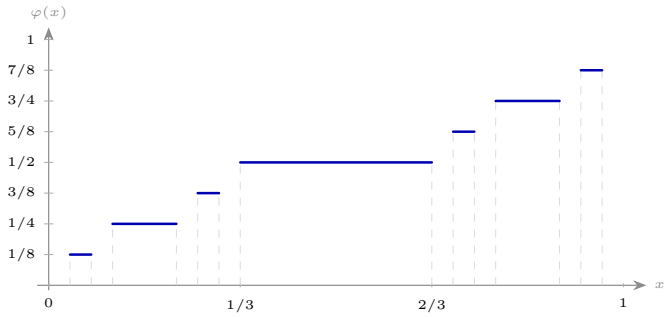

Figure 197: Horizontal segments show the Cantor–Lebesgue plateaus on the intervals removed through stage 3.

Proposition 8.1.2. *Properties of the Singular Function*
Let P be the Cantor set, $U := [0, 1] \setminus P$, and $\varphi : [0, 1] \to [0, 1]$ the Cantor–Lebesgue singular function. Then:

(i) φ is nondecreasing.
(ii) φ is continuous on $[0, 1]$.
(iii) $\varphi([0, 1]) = [0, 1]$.
(iv) $\varphi'(x) = 0$ for every $x \in U$.
(v) $m(U) = 1$.

Proof.

(i) At each stage k, the components $I_{k,1}, \dots, I_{k,2^k-1}$ occur left to right and carry heights $\frac{1}{2^k}, \dots, \frac{2^k-1}{2^k}$.
Thus the prescription on U is nondecreasing, and the supremum extension on P preserves order.
(ii) If $x_0 \in U$, then $x_0 \in J$ for some component J of U and $\varphi|_J$ is constant. If $x_0 \in P$, then for each k there are $a_k, b_k \in 2^{-k}\mathbb{Z} \cap [0, 1]$ with $a_k \leq \varphi(x_0) \leq b_k$, $b_k - a_k \leq 2^{-k}$, and $\varphi(y) \in [a_k, b_k]$ for all y near x_0.
Thus $\limsup_{y \to x_0} |\varphi(y) - \varphi(x_0)| \leq 2^{-k}$ for every k, hence φ is continuous at x_0.
(iii) The definition gives $\varphi(0) = 0$. For each k, the rightmost component of U_k has height $1 - 2^{-k}$, so $\varphi(1) = 1$.
By (ii) and the intermediate value theorem, $\varphi([0, 1]) = [0, 1]$.
(iv) For each component J of U, $\varphi|_J$ is constant; hence $\varphi'(x) = 0$ for every $x \in U$.
(v) Proposition 4.1.14 gives $m(P) = 0$; since $[0, 1] = P \sqcup U$, $m(U) = m([0, 1]) - m(P) = 1$.

Structural Commentary: Plateaus on U force variation onto P.
Key Insight: At stage k, neighboring plateau heights differ by 2^{-k}. □

Theorem 8.1.3. *Lebesgue Measurable Non-Borel Sets Exist*
There exist $A \subseteq [0, 1]$ and a homeomorphism $\psi : [0, 1] \to [0, 2]$ such that $A \in \mathcal{L}^1 \setminus \mathcal{B}(\mathbb{R})$ and $\psi(A) \notin \mathcal{L}^1$.

Proof. Let P be the Cantor set, $U := [0, 1] \setminus P$, φ the Cantor–Lebesgue singular function, and $\psi := \mathrm{id}_{[0,1]} + \varphi$.
Proposition 8.1.2 gives φ continuous and nondecreasing; hence ψ is continuous and strictly increasing.
Since $\psi(0) = 0$ and $\psi(1) = 2$, $\psi : [0, 1] \to [0, 2]$ is a homeomorphism.

Let $(J_k)_{k\geq 1}$ be the components of U. Since $\varphi|_{J_k}$ is constant, $\psi(J_k)$ is a translate of J_k and $m(\psi(J_k)) = m(J_k)$.
Thus $m(\psi(U)) = \sum_{k\geq 1} m(\psi(J_k)) = \sum_{k\geq 1} m(J_k) = m(U) = 1$.
Since $\psi(P)$ is compact and $[0,2] = \psi(P) \sqcup \psi(U)$, $m(\psi(P)) = 1$.
Choose $W \subseteq \psi(P)$ with $W \notin \mathcal{L}^1$ by Theorem 7.1.8, and set $A := \psi^{-1}(W)$. Then $A \subseteq P$ and $m(P) = 0$, so $A \in \mathcal{L}^1$. If $A \in \mathcal{B}(\mathbb{R})$, then $W = \psi(A) \in \mathcal{B}([0,2]) \subseteq \mathcal{L}^1$, contradiction. Hence $A \in \mathcal{L}^1 \setminus \mathcal{B}(\mathbb{R})$ and $\psi(A) = W \notin \mathcal{L}^1$.
Structural Commentary: Completion by null sets creates Lebesgue sets invisible to the Borel hierarchy.
Key Insight: A null Cantor set can be homeomorphically stretched to positive measure. □

Part XIV
Measure, Integration, and Functional Analysis

1 Measurable Functions and Lebesgue Integration

1.1 Measurable Functions

Definition 1.1.1. *Borel, Lebesgue, and Function Measurability*

(i) *Borel*: $\mathcal{B}(\mathbb{R}^p) := \sigma(\mathcal{O}_{\mathbb{R}^p})$; its elements are *Borel sets*.
(ii) *Lebesgue*: $\mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$, and $\mathcal{L}^p$ is obtained by adjoining subsets of Lebesgue null sets.
(iii) *Maps*: for measurable spaces $(X,\mathcal{M})$, $(Y,\mathcal{N})$, $f : X \to Y$ is *$(\mathcal{M},\mathcal{N})$-measurable* iff $\forall E \in \mathcal{N},\ f^{-1}(E) \in \mathcal{M}$.
(iv) *Borel maps*: $f : \mathbb{R}^p \to \mathbb{R}^q$ is *Borel measurable* iff $\forall B \in \mathcal{B}(\mathbb{R}^q),\ f^{-1}(B) \in \mathcal{B}(\mathbb{R}^p)$.
(v) *Thresholds*: write $\{f \bowtie a\} := \{x \in X : f(x) \bowtie a\}$ for $\bowtie \in \{>, \geq, <, \leq\}$.
Extended-real f is *$\mathcal{M}$-measurable* iff $\forall a \in \mathbb{R},\ \{f > a\} \in \mathcal{M}$.

Structural Commentary: Borel measurability tracks topology; Lebesgue measurability adds null-set completion.
Key Insight: Measurability is inverse-image stability.

Proposition 1.1.2. *Generator Test for Measurable Maps*
Let $f : X \to Y$, $\mathcal{A} \subseteq \mathcal{P}(Y)$, and $\mathcal{N} := \sigma(\mathcal{A})$.
(i) If $\forall A \in \mathcal{A},\ f^{-1}(A) \in \mathcal{M}$, then $f : X \to (Y,\mathcal{N})$ is measurable.
(ii) For real-valued f, $\forall a \in \mathbb{R},\ \{f > a\} \in \mathcal{M} \iff \forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) \in \mathcal{M}$.

Proof.
(i) Let $\mathcal{G} := \{E \subseteq Y : f^{-1}(E) \in \mathcal{M}\}$. Since $f^{-1}(E^c) = f^{-1}(E)^c$ and $f^{-1}(\bigcup_n E_n) = \bigcup_n f^{-1}(E_n)$, $\mathcal{G}$ is a σ-algebra. Thus $\mathcal{A} \subseteq \mathcal{G} \Longrightarrow \mathcal{N} = \sigma(\mathcal{A}) \subseteq \mathcal{G}$.
(ii) ($\Rightarrow$) Set $\mathcal{A} := \{(a,+\infty) : a \in \mathbb{R}\}$. Since $\forall a \in \mathbb{R},\ f^{-1}((a,+\infty)) = \{f > a\} \in \mathcal{M}$ and $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$, (i) gives $\forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) \in \mathcal{M}$.
(ii) ($\Leftarrow$) If $\forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) \in \mathcal{M}$, then $\forall a \in \mathbb{R},\ \{f > a\} = f^{-1}((a,+\infty)) \in \mathcal{M}$.

Structural Commentary: Measurability can be checked on any generating family of target sets.
Key Insight: Pullbacks turn generated σ-algebras into generated collections of preimages. □

Proposition 1.1.3. *Open-Set Test for Euclidean Targets*
Let $\Omega \subseteq \mathbb{R}^n$ be Lebesgue measurable and let $f : \Omega \to \mathbb{R}^m$. The following tests are equivalent:
(i) $\forall V \subseteq \mathbb{R}^m$ open, $f^{-1}(V)$ is measurable;
(ii) $\forall B \subseteq \mathbb{R}^m$ open boxes, $f^{-1}(B)$ is measurable;
(iii) $\forall j \in \{1,\dots,m\}$, $f_j : \Omega \to \mathbb{R}$ is threshold-measurable.
In particular, every continuous $f : \Omega \to \mathbb{R}^m$ is measurable.

Proof.
(i) $\Rightarrow$ *(ii)*. Open boxes are open.
(ii) $\Rightarrow$ *(i)*. By Lemma 4.1.7, $V = \bigcup_k B_k$ for rational boxes B_k, hence $f^{-1}(V) = \bigcup_k f^{-1}(B_k)$ is measurable.
(ii) $\Rightarrow$ *(iii)*. By *(ii)* $\Rightarrow$ *(i)*, $\{f_j > a\} = f^{-1}(\{y : y_j > a\})$ is measurable.
(iii) $\Rightarrow$ *(ii)*. If $B = \prod_{j=1}^m (a_j, b_j)$, then $f^{-1}(B) = \bigcap_{j=1}^m \big(\{f_j > a_j\} \cap \{f_j < b_j\}\big)$,
where $\{f_j < b_j\} = \Omega \setminus \bigcap_{\ell=1}^{\infty} \{f_j > b_j - \frac{1}{\ell}\}$. Thus each factor is measurable, hence so is $f^{-1}(B)$.

If f is continuous, $f^{-1}(V)$ is open relative to Ω by Theorem 5.1.18.
By Theorem 3.2.3, $f^{-1}(V) = W \cap \Omega$ for some open $W \subseteq \mathbb{R}^n$, hence it is measurable by Theorem 4.1.8.
Structural Commentary: Euclidean-valued measurability is just countable box algebra applied to inverse images.
Key Insight: Open sets reduce to rational boxes; vector functions reduce to coordinate threshold tests. □

Example 1.1.4. *Basic Measurable Functions*
Let $(X,\mathcal{M})$ be a measurable space.
(i) Constant real-valued functions are measurable, since each strict superlevel set is either X or $\emptyset$.
(ii) For $E \subseteq X$, the indicator $\mathbf{1}_E$ is measurable iff $E \in \mathcal{M}$.
(iii) Continuous maps between Euclidean measurable subsets are measurable by Proposition 1.1.3.
(iv) Monotone functions $\mathbb{R} \to \mathbb{R}$ are Borel measurable, since their strict superlevel sets are half-lines, $\mathbb{R}$, or $\emptyset$.

Key Insight: Constants, indicators of measurable sets, continuous functions, and monotone functions form the first reusable stock of measurable functions.

Theorem 1.1.5. *Closure Properties of Measurable Functions*
Let $(X,\mathcal{M})$ be a measurable space.
(i) The four tests $\forall a \in \mathbb{R},\ \{f > a\}, \{f \geq a\}, \{f < a\}, \{f \leq a\} \in \mathcal{M}$ are equivalent.
(ii) f measurable implies $|f|$ measurable.
(iii) If f_n are measurable, then $\sup_n f_n, \inf_n f_n, \limsup_n f_n, \liminf_n f_n$ are measurable.
(iv) If f, g real-valued measurable, $F : \mathbb{R}^2 \to \mathbb{R}$ continuous, then $F(f,g)$ measurable; so are $f+g$ and fg.

Proof.
(i) Use $\{f \geq a\} = \bigcap_{m \geq 1} \{f > a - \frac{1}{m}\}$; $\{f < a\} = X \setminus \{f \geq a\}$; $\{f \leq a\} = \bigcap_{m \geq 1} \{f < a + \frac{1}{m}\}$.
(ii) For $a > 0$, $\{|f| < a\} = \{f < a\} \cap \{f > -a\}$; for $a \leq 0$ the set is empty.

(iii) $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$. Also $\limsup_n f_n = \inf_m \sup_{n\geq m} f_n$; $\liminf_n f_n = \sup_m \inf_{n\geq m} f_n$; inf follow from (i).

(iv) For $G_a = \{(u,v) : F(u,v) > a\}$, write $G_a = \bigcup_k (\alpha_k, \beta_k) \times (\gamma_k, \delta_k)$.

Then $\{F(f,g) > a\} = \bigcup_k \big(\{\alpha_k < f < \beta_k\} \cap \{\gamma_k < g < \delta_k\}\big) \in \mathcal{M}$.

Structural Commentary: Measurability is countable set algebra applied to level sets.
Key Insight: Once threshold sets are measurable, ordinary limiting and algebraic operations stay measurable. □

Corollary 1.1.6. *Continuous Composition and Finite Lattice Operations*
Let $(X, \mathcal{M})$ be a measurable space and let $n \in \mathbb{N}_{\geq 1}$.
For $f_1, \dots, f_n : X \to \mathbb{R}$, set $\bigvee_{k=1}^n f_k := \max\{f_1, \dots, f_n\}$ and $\bigwedge_{k=1}^n f_k := \min\{f_1, \dots, f_n\}$.
(i) If $g : X \to \mathbb{R}$ is measurable and $\varphi : \mathbb{R} \to \mathbb{R}$ is continuous, then $\varphi \circ g$ is measurable.
(ii) If $f_1, \dots, f_n : X \to \mathbb{R}$ are measurable, then $\bigvee_{k=1}^n f_k$ and $\bigwedge_{k=1}^n f_k$ are measurable.

Proof.

(i) For $a \in \mathbb{R}$, $\{\varphi \circ g > a\} = g^{-1}(\varphi^{-1}((a, +\infty))) \in \mathcal{M}$, since $\varphi^{-1}((a, +\infty)) \in \mathcal{B}(\mathbb{R})$ and g is measurable.

(ii) Set $h_k = f_k$ for $1 \leq k \leq n$ and $h_k = f_n$ for $k > n$. Then (h_k) is measurable, $\bigvee_{k=1}^n f_k = \sup_{k\geq 1} h_k$, and $\bigwedge_{k=1}^n f_k = \inf_{k\geq 1} h_k$, hence both are measurable by Theorem 1.1.5(iii).

Structural Commentary: Pullbacks preserve Borel thresholds; finite lattices are finite suprema and infima.
Key Insight: Continuity and finite lattice operations preserve measurable threshold structure. □

Example 1.1.7. *Measurable Composition Can Fail*
Choose $A \in \mathcal{L}^1$ with $A \subseteq [0,1]$ and an increasing homeomorphism $\psi : [0,1] \to [0,2]$ such that $\psi(A) \notin \mathcal{L}^1$. Define

$$\widetilde{\psi}(x) := \begin{cases} x, & x \leq 0, \\ \psi(x), & 0 \leq x \leq 1, \\ x+1, & x \geq 1. \end{cases}$$

Then $\widetilde{\psi} : \mathbb{R} \to \mathbb{R}$ is a homeomorphism. Put $h := \mathbf{1}_A \circ \widetilde{\psi}^{-1}$.
The maps $\mathbf{1}_A$ and $\widetilde{\psi}^{-1}$ are measurable, but $h^{-1}((\frac{1}{2}, \frac{3}{2})) = \widetilde{\psi}(A) = \psi(A) \notin \mathcal{L}^1$.
Hence h is not Lebesgue measurable, so Lebesgue measurable functions are not composition-closed.

Definition 1.1.8. *Positive and Negative Parts*
For an extended-real-valued function f, define $f^+(x) := \sup\{f(x), 0\}$; $f^-(x) := \sup\{-f(x), 0\}$.
When f is finite-valued, $f = f^+ - f^-$ and $|f| = f^+ + f^-$; equivalently $f^+ = \frac{|f|+f}{2}$; $f^- = \frac{|f|-f}{2}$.

Corollary 1.1.9. *Measurability of Positive and Negative Parts*
If $f : X \to [-\infty, +\infty]$ is measurable, then $f^+, f^-, |f|$ are measurable.
If $f : X \to \mathbb{R}$, then f measurable $\iff$ f^+, f^- measurable.

Proof.

($\Rightarrow$) Since $\{-f > a\} = \{f < -a\}$, $-f$ is measurable.
Thus $f^+ = f \vee 0$, $f^- = (-f) \vee 0$, and $|f| = f \vee (-f)$ are measurable by finite sup-closure.

($\Leftarrow$) If $f : X \to \mathbb{R}$ and f^+, f^- are measurable, then $f = f^+ - f^-$ is measurable.

Structural Commentary: Sign decomposition is compatible with threshold-set measurability.
Key Insight: Reduce signs to suprema and finite algebra. □

Lemma 1.1.10. *Extended-Valued Measurability via the Finite Part*
Let $f : X \to \overline{\mathbb{R}}$ and set $A := f^{-1}(\{+\infty\})$, and $B := f^{-1}(\{-\infty\})$. Define $f_0(x) := \begin{cases} f(x), & x \notin A \cup B, \\ 0, & x \in A \cup B. \end{cases}$
Then f is measurable iff $A, B \in \mathcal{M}$ and $f_0 : X \to \mathbb{R}$ is measurable.

Proof.

($\Rightarrow$) If f is measurable, then $A = \bigcap_{n=1}^{\infty} \{f > n\}$; and $B = X \setminus \bigcup_{n=1}^{\infty} \{f > -n\}$, so $A, B \in \mathcal{M}$.

Also, $\{f_0 > \alpha\} = \begin{cases} \{f > \alpha\} \setminus A, & \alpha \geq 0, \\ \{f > \alpha\} \cup B, & \alpha < 0, \end{cases}$ so f_0 is measurable.

($\Leftarrow$) If $A, B \in \mathcal{M}$, f_0 is measurable, then $\{f > \alpha\} = \begin{cases} \{f_0 > \alpha\} \cup A, & \alpha \geq 0, \\ \{f_0 > \alpha\} \setminus B, & \alpha < 0, \end{cases}$ so $\{f > \alpha\} \in \mathcal{M}\ \forall \alpha \in \mathbb{R}$.

Structural Commentary: The infinite-level sets are measurable exactly when the finite core is measurable.
Key Insight: Assign harmless finite values at $\pm\infty$, then put the infinite fibers back by set algebra. □

Corollary 1.1.11. *Extended-Valued Algebra Conventions*
Let $f, g : X \to \overline{\mathbb{R}}$ be measurable.
(i) For $c \in \mathbb{R}$, cf, $|f|$, f^+, and f^- are measurable, with $0(\pm\infty) = 0$.
(ii) Let $E_1 := \{f = -\infty,\ g = +\infty\}$, $E_2 := \{f = +\infty,\ g = -\infty\}$, and $F := E_1 \cup E_2$.
If $h = f + g$ on $X \setminus F$ and $h = 0$ on F, then h is measurable.
(iii) With $0(\pm\infty) = 0$, fg is measurable.

Proof.

(i) By Lemma 1.1.10, $A_\pm(f) := \{f = \pm\infty\} \in \mathcal{M}$ and f_0 is measurable.
For $c \in \mathbb{R}$, the infinite fibers of cf are among $A_\pm(f)$ and $(cf)_0 = cf_0$, so cf is measurable.
The functions $|f|, f^+, f^-$ are measurable by Corollary 1.1.9.

(ii) $E_1, E_2 \in \mathcal{M}$. Set $f_n := \max\{-n, \min(f, n)\}$ and $g_n := \max\{-n, \min(g, n)\}$.
Then f_n, g_n are real-valued measurable and $(f_n + g_n)\mathbf{1}_{X\setminus F} \to h$ pointwise; hence h is measurable.
(iii) With f_n, g_n as in (ii), $f_n g_n$ measurable, $f_n g_n \to fg$ pointwise under $0(\pm\infty) = 0$; hence fg measurable.

Structural Commentary: Extended algebra is finite algebra plus measurable exceptional fibers.
Key Insight: Truncate first, use real-valued closure, then pass to pointwise limits. □

Definition 1.1.12. *Simple Functions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) *Indicator*: for $E \subseteq X$, $\mathbf{1}_E(x) := 1$ on E and $\mathbf{1}_E(x) := 0$ on $X \setminus E$.
(ii) *Simple*: a real-valued s is simple iff $s(X)$ is finite.
(iii) *Standard representation*: if $s(X) = \{c_1, \dots, c_n\}$ and $E_i := \{s = c_i\} \neq \emptyset$, then $X = \bigsqcup_{i=1}^{n} E_i$; $s = \sum_{i=1}^{n} c_i \mathbf{1}_{E_i}$.
The pairs (c_i, E_i) are unique up to reordering.
(iv) *Measurability*: s measurable iff $E_i \in \mathcal{M}$ $\forall i$; step functions are special case $X \subseteq \mathbb{R}$ with interval level sets.

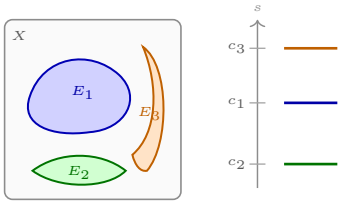

Figure 198: A simple function takes value c_i on each measurable piece E_i.

Proposition 1.1.13. *Simple Functions Form a Lattice Algebra*
Let $s, t : X \to \mathbb{R}$ be simple and $a \in \mathbb{R}$. Then as, $s + t$, st, $s \vee t$, and $s \wedge t$ are simple.
If s, t are measurable, so are these five functions.

Proof. Let $s(X) = \{c_i\}_{i=1}^m$ and $t(X) = \{d_j\}_{j=1}^n$. Each displayed range is contained in the corresponding finite set $\{ac_i\}$, $\{c_i + d_j\}$, $\{c_i d_j\}$, $\{c_i \vee d_j\}$, or $\{c_i \wedge d_j\}$, hence is finite.
If s, t are measurable, the atoms $P_{ij} := \{s = c_i\} \cap \{t = d_j\}$ are measurable and form a finite partition of X. Each displayed function is constant on every P_{ij}, so its level sets are finite unions of P_{ij}.
Structural Commentary: Algebra and lattice operations reduce to finite tables.
Key Insight: Refine two simple functions by the common partition of their level sets. □

Lemma 1.1.14. *Bounded Simple Approximation Brackets*
Let $(X, \mathcal{M})$ be a measurable space and let $f : X \to \mathbb{R}$ be bounded and $\mathcal{M}$-measurable. Then $\forall \varepsilon > 0$ $\exists$ simple $\mathcal{M}$-measurable functions $\varphi_\varepsilon, \psi_\varepsilon : X \to \mathbb{R}$ with $\varphi_\varepsilon \le f \le \psi_\varepsilon$ and $0 \le \psi_\varepsilon - \varphi_\varepsilon < \varepsilon$ on X.

Proof. Choose $c = y_0 < \cdots < y_N = d$ with $f(X) \subseteq (c, d)$ and $\max_k (y_k - y_{k-1}) < \varepsilon$.
Set $E_k := \{y_{k-1} \le f < y_k\} = \{f \ge y_{k-1}\} \cap \{f < y_k\} \in \mathcal{M}$; then $(E_k)_{k=1}^N$ partitions X.
Define $\varphi_\varepsilon := \sum_{k=1}^{N} y_{k-1} \mathbf{1}_{E_k}$ and $\psi_\varepsilon := \sum_{k=1}^{N} y_k \mathbf{1}_{E_k}$; these are $\mathcal{M}$-measurable simple functions.
For $x \in E_k$, $\varphi_\varepsilon(x) = y_{k-1} \le f(x) < y_k = \psi_\varepsilon(x)$ and $\psi_\varepsilon(x) - \varphi_\varepsilon(x) = y_k - y_{k-1} < \varepsilon$.
Structural Commentary: Pull back a finite range partition.
Key Insight: Mesh size controls the bracket gap. □

Theorem 1.1.15. *Approximation by Simple Functions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) $\forall f : X \to \mathbb{R}$ $\exists$ simple functions (s_n) with $s_n \to f$ pointwise.
(ii) For $f : X \to \overline{\mathbb{R}}$, f is $\mathcal{M}$-measurable iff f is the pointwise limit of $\mathcal{M}$-measurable simple functions.
(iii) If $f : X \to \overline{\mathbb{R}}$ is $\mathcal{M}$-measurable, the approximants in (ii) may be chosen with $|s_n| \le |f|$.
If $f \ge 0$, they may be chosen with $0 \le s_n \uparrow f$.

Proof. For $u : X \to [0, +\infty]$, set $Q_n u := 2^{-n} \lfloor 2^n (u \wedge n) \rfloor$.
Then $Q_n u(X) \subseteq 2^{-n}\{0, \dots, n2^n\}$, so $Q_n u$ is simple, and binary truncation gives $0 \le Q_n u \uparrow u$.
If u is measurable, the level sets of $Q_n u$ are threshold sets of $u \wedge n$, hence $Q_n u$ is measurable.

(i) For $f : X \to \mathbb{R}$, write $f = f^+ - f^-$ and set $s_n := Q_n f^+ - Q_n f^-$. Then s_n is simple and $s_n \to f$ pointwise.
(*ii*) ($\Rightarrow$) If $f : X \to \overline{\mathbb{R}}$ is measurable, then f^+, f^- are measurable and $s_n := Q_n f^+ - Q_n f^-$ is measurable simple. Since at each x at most one of $f^+(x), f^-(x)$ is infinite, $s_n(x) \to f(x)$ in $\overline{\mathbb{R}}$.
(*ii*) ($\Leftarrow$) If measurable simple $s_n \to f$ pointwise, then $f = \liminf_n s_n$ is measurable.
(*iii*) In (*ii*) $\Rightarrow$, $|s_n| \le Q_n f^+ + Q_n f^- \le |f|$. If $f \ge 0$, take $s_n := Q_n f$.

Structural Commentary: Simple functions quantize the range of f.
Key Insight: Dyadic range partitions give pointwise approximation while preserving measurability. □

Definition 1.1.16. *Complex-Valued Measurable Functions*
A map $f : X \to \mathbb{C}$ is *measurable* iff $\operatorname{Re} f, \operatorname{Im} f : X \to \mathbb{R}$ are measurable.
Equivalently, $f = \operatorname{Re} f + i \operatorname{Im} f$ with measurable real components.

Proposition 1.1.17. *Complex Measurable Closure*
If $f, g : X \to \mathbb{C}$ are measurable, $a \in \mathbb{C}$, and $h_n : X \to \mathbb{C}$ are measurable with $h_n \to h$ pointwise, then $f + g$, fg, af, and h are measurable.

Proof. Write $f = u + iv$, $g = r + is$, and $a = \alpha + i\beta$. Then $f + g = (u + r) + i(v + s)$, $af = (\alpha u - \beta v) + i(\alpha v + \beta u)$, and $fg = (ur - vs) + i(us + vr)$, so real closure gives measurability.
If $h_n \to h$, then $\operatorname{Re} h_n \to \operatorname{Re} h$ and $\operatorname{Im} h_n \to \operatorname{Im} h$, so pointwise-limit closure gives measurability.
Structural Commentary: Complex measurability is two real measurability statements.
Key Insight: Split into real and imaginary parts. □

Definition 1.1.18. *Almost Everywhere and In Measure*
Let E be measurable.

(i) *A.E. equality*: $f \sim_{\text{a.e.}} g$ on E iff $\mu(\{x \in E : f(x) \neq g(x)\}) = 0$.
(ii) *A.E. property*: a property holds a.e. on E iff its failure set in E is null.
(iii) *Null invariance*: $\sim_{\text{a.e.}}$ is an equivalence relation.
(iv) *A.E. convergence*: $f_n \to f$ a.e. on E iff $\exists N \subseteq E$ null such that $\forall x \in E \setminus N,\ f_n(x) \to f(x)$.
(v) *In measure*: $f_n \to f$ in measure on E iff $\forall \varepsilon > 0$, $\mu(\{x \in E : |f_n(x) - f(x)| > \varepsilon\}) \to 0$.

Remark 1.1.19. *Completion and Almost-Everywhere Limits*
If $E \subseteq N \in \mathcal{M}$, $\mu(N) = 0$, and $E \notin \mathcal{M}$, then $f_n \equiv 0$ is measurable and $f_n \to \mathbf{1}_E$ a.e.,
since $\{x : f_n(x) \not\to \mathbf{1}_E(x)\} \subseteq N$. Yet $\mathbf{1}_E$ is not $\mathcal{M}$-measurable.
In the completion, every $E \subseteq N$ is measurable, so null modifications preserve measurability.
Key Insight: Almost-everywhere statements are most stable on complete measure spaces.

Corollary 1.1.20. *Pointwise and Almost-Everywhere Limits*
Let (f_n) be $\overline{\mathbb{R}}$-valued measurable functions on $(X, \mathcal{M})$, and let $f : X \to \overline{\mathbb{R}}$.

(i) If $f_n \to f$ pointwise on X, then f is measurable.
(ii) If $(X, \mathcal{M}, \mu)$ is complete and $f_n \to f$ a.e. on X, then f is measurable.

Proof.

(i) Pointwise, $f = \liminf_n f_n = \limsup_n f_n$; closure under lim inf and lim sup gives measurability.

(ii) Choose $N \in \mathcal{M}$ with $\mu(N) = 0$ and $f_n \to f$ on $X \setminus N$.
Define $g = f$ and $g_n = f_n$ on $X \setminus N$, and $g = g_n = 0$ on N.
Then each g_n is measurable and $g_n \to g$ pointwise, so g is measurable by (i).
Since $\forall a \in \mathbb{R},\ \{f > a\} \triangle \{g > a\} \subseteq N$, completeness gives $\{f > a\} \in \mathcal{M}$; hence f is measurable.

Structural Commentary: Limits reduce to upper and lower limits; a.e. limits require completion.
Key Insight: Completion is exactly what lets null modifications preserve measurability. □

Proposition 1.1.21. *Completed Measurable Functions Have Original Representatives*
Let $(X, \mathcal{M}, \mu)$ be a measure space with completion $(X, \overline{\mathcal{M}}_\mu, \overline{\mu})$.
If $f : X \to \mathbb{R}$ is $\overline{\mathcal{M}}_\mu$-measurable, then $\exists g : X \to \mathbb{R}$ that is $\mathcal{M}$-measurable and satisfies $f = g$ μ-a.e.

Proof. For $q \in \mathbb{Q}$, let $A_q := \{f > q\} \in \overline{\mathcal{M}}_\mu$, let $A_q = E_q \cup Z_q$, with $E_q \in \mathcal{M}$ and $Z_q \subseteq N_q \in \mathcal{M}$, $\mu(N_q) = 0$.
Put $N := \bigcup_{q \in \mathbb{Q}} N_q$, so $N \in \mathcal{M}$ and $\mu(N) = 0$, and define $g(x) := \begin{cases} f(x), & x \notin N, \\ 0, & x \in N. \end{cases}$.
For $q \in \mathbb{Q}$, $\{g > q\} = (E_q \setminus N) \cup \begin{cases} N, & q < 0, \\ \emptyset, & q \geq 0, \end{cases}$ belongs to $\mathcal{M}$. Hence, for $a \in \mathbb{R}$, $\{g > a\} = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} \{g > q\} \in \mathcal{M}$.
Thus g is $\mathcal{M}$-measurable and $f = g$ on $X \setminus N$.
Structural Commentary: Completion only changes measurable functions on null sets.
Key Insight: Replace completed rational levels off one null set. □

Method 1.1.22. *Measure Proof Patterns*

(i) *Disjointize*. If $A_n \uparrow$, set $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$; then $A_n = \bigsqcup_{k \leq n} B_k$ and $\bigcup_n A_n = \bigsqcup_n B_n$.
(ii) *Approximate*. Prove on $\mathcal{E}$; use $d_\mu(C, D) := \mu^*(C \triangle D)$ and $\triangle$ estimates to pass limits through $\cup, \cap, \setminus$.
(iii) *Threshold*. For real f, show $\forall a \in \mathbb{R},\ \{f > a\} \in \mathcal{M}$; for $f : X \to \mathbb{R}^m$, use coordinate and open-set tests.
(iv) *Generators*. For $f : X \to (Y, \sigma(\mathcal{A}))$, check $\forall A \in \mathcal{A},\ f^{-1}(A) \in \mathcal{M}$.
(v) *Truncate*. For $f : X \to \overline{\mathbb{R}}$, prove results for $f_n := (-n) \vee (f \wedge n)$, then let $n \to \infty$.
(vi) *Topological bridge*. Use continuous open preimages.
Reduce Borel tests to countable interval or rectangle preimages.

Key Insight: Measure theory alternates between countable set algebra and approximation by elementary sets.

1.2 Lebesgue Integration

Lebesgue integration is calibrated on indicators, extended linearly to nonnegative simple functions, then extended by monotone approximation and positive–negative decomposition.

Perspective 1.2.1. *Simple Minorants Are the Right Primitive*
For Lebesgue measure on $\mathbb{R}$, bounded measurable functions supported on finite-measure sets form a useful intermediate class; arbitrary measure spaces need not admit that reduction.
Let $X \neq \emptyset$, $\mathcal{M} = \{\emptyset, X\}$, and $\mu(X) = +\infty$. If g is measurable and vanishes outside some $F \in \mathcal{M}$ with $\mu(F) < +\infty$, then $F = \emptyset$ and $g = 0$; hence finite-measure-support minorants of $\mathbf{1}_X$ have supremal integral 0, while the required normalization is $\int_X \mathbf{1}_X \, d\mu = \mu(X) = +\infty$.
Thus the defining minorants must include $\mathbf{1}_A$ $\forall A \in \mathcal{M}$, and more generally all nonnegative simple $s \leq f$.
Key Insight: The primitive calibration is $\int_X \mathbf{1}_A \, d\mu = \mu(A)$, with $\mu(A) = +\infty$ allowed.

Definition 1.2.2. *Lebesgue Integral*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $E \in \mathcal{M}$; let $\mathcal{S}^+$ be nonnegative measurable simple functions on X.
Write $s \in \mathcal{S}^+$ as $s = \sum_{i=1}^{n} c_i \mathbf{1}_{A_i}$ with $c_i \geq 0$ and pairwise disjoint $A_i \in \mathcal{M}$.
For measurable $f \geq 0$, put $\mathcal{S}^+(f) := \{s \in \mathcal{S}^+ : s \leq f\}$; for measurable $f : X \to \overline{\mathbb{R}}$, put $f^{\pm} := (\pm f) \vee 0$.

(i) *Simple*: $\mathcal{I}_E(s) := \sum_{i=1}^{n} c_i \mu(E \cap A_i)$, with $0 \cdot (+\infty) := 0$.
(ii) *Nonnegative*: $\int_E f \, d\mu := \sup_{s \in \mathcal{S}^+(f)} \mathcal{I}_E(s)$; f integrable on $E \iff \int_E f \, d\mu < +\infty$.
(iii) *Signed*: if $\min\{\int_E f^+ \, d\mu, \int_E f^- \, d\mu\} < +\infty$, set $\int_E f \, d\mu := \int_E f^+ \, d\mu - \int_E f^- \, d\mu$.
(iv) *L^1*: $f \in \mathscr{L}^1(\mu)$ on E iff f is measurable and $\int_E |f| \, d\mu < +\infty$.
When $E = X$, write $\int f \, d\mu := \int_X f \, d\mu$ and drop "on E".

Simple functions are finite linear combinations of indicators of measurable sets: coefficients carry size, while the level sets carry measurability.

Lemma 1.2.3. *Refining a Simple Integral*
Let $s \geq 0$ be simple measurable and suppose $s = \sum_{k=1}^{m} b_k \mathbf{1}_{F_k}$ with $b_k \geq 0$ and $F_k \in \mathcal{M}$.
Then $\forall E \in \mathcal{M}$, $\mathcal{I}_E(s) = \sum_{k=1}^{m} b_k \mu(E \cap F_k)$.

Proof. For $\eta \in \{0,1\}^m$, set $F_k^1 := F_k$, $F_k^0 := F_k^c$, discard empty atoms $G_\eta := \bigcap_{k=1}^{m} F_k^{\eta_k}$. On G_η, $s = c_\eta := \sum_{\eta_k = 1} b_k$.
Group equal c_η values to recover the standard representation; finite additivity gives
$\mathcal{I}_E(s) = \sum_{\eta} c_\eta \mu(E \cap G_\eta) = \sum_{k=1}^{m} b_k \sum_{\eta : \eta_k = 1} \mu(E \cap G_\eta) = \sum_{k=1}^{m} b_k \mu(E \cap F_k)$.
Structural Commentary: Refine nonstandard sums to one finite partition.
Key Insight: Refine first, then regroup by equal function values. □

Proposition 1.2.4. *Simple Integral Algebra*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, let $c \geq 0$, and let $s, t \in \mathcal{S}^+$. For fixed $s \in \mathcal{S}^+$, define $\nu_s : \mathcal{M} \to [0, +\infty]$ by $\nu_s(A) := \mathcal{I}_A(s)$. Then:
(i) $\mathcal{I}_E(cs) = c\mathcal{I}_E(s)$.
(ii) $\mathcal{I}_E(s + t) = \mathcal{I}_E(s) + \mathcal{I}_E(t)$.
(iii) ν_s is a measure on $\mathcal{M}$.

Proof. Write $s = \sum_i a_i \mathbf{1}_{A_i}$ and $t = \sum_j b_j \mathbf{1}_{B_j}$ on finite measurable partitions of X, with $a_i, b_j \geq 0$.

(i) $\mathcal{I}_E(cs) = \sum_i (ca_i)\mu(E \cap A_i) = c \sum_i a_i \mu(E \cap A_i) = c\mathcal{I}_E(s)$, with $0 \cdot (+\infty) = 0$.

(ii) On the common refinement $A_i \cap B_j$, $s + t = \sum_{i,j} (a_i + b_j) \mathbf{1}_{A_i \cap B_j}$.
$\mathcal{I}_E(s + t) = \sum_{i,j} (a_i + b_j) \mu(E \cap A_i \cap B_j) = \sum_i a_i \mu(E \cap A_i) + \sum_j b_j \mu(E \cap B_j)$ by Lemma 1.2.3.

(iii) For $A \in \mathcal{M}$, $\nu_s(A) = \sum_i a_i \mu(A \cap A_i)$.
Each $A \mapsto \mu(A \cap A_i)$ is a measure, and finite nonnegative sums of measures are measures.

Structural Commentary: Simple integration is weighted measure summation on finite partitions.
Key Insight: Refine to common atoms, then use finite additivity. □

Proposition 1.2.5. *Monotonicity for Nonnegative Integrals*
Let $f, g \geq 0$ be measurable and let $E, F \in \mathcal{M}$.
(i) If $f \leq g$ on E, then $\int_E f \, d\mu \leq \int_E g \, d\mu$.
(ii) If $E \subseteq F$, then $\int_E f \, d\mu \leq \int_F f \, d\mu$.

Proof.
(i) If $s \in \mathcal{S}^+(f)$, then $s\mathbf{1}_E \in \mathcal{S}^+(g)$, $\mathcal{I}_E(s) = \mathcal{I}_E(s\mathbf{1}_E) \leq \int_E g \, d\mu$. Taking $\sup_{s \in \mathcal{S}^+(f)}$ gives $\int_E f \, d\mu \leq \int_E g \, d\mu$.
(ii) If $E \subseteq F$ and $s \in \mathcal{S}^+(f)$, then $\mathcal{I}_E(s) \leq \mathcal{I}_F(s)$. Taking $\sup_{s \in \mathcal{S}^+(f)}$ gives $\int_E f \, d\mu \leq \int_F f \, d\mu$.

Structural Commentary: The nonnegative integral is a supremum over simple functions below the integrand.
Key Insight: More room below the integrand, or over a larger set, can only increase the supremum. □

Proposition 1.2.6. *Chebyshev Inequality for Nonnegative Integrals*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, and let $f : X \to [0, +\infty]$ be measurable.
For $\lambda > 0$, set $E_\lambda := E \cap \{f \geq \lambda\}$. Then $E_\lambda \in \mathcal{M}$ and $\mu(E_\lambda) \leq \lambda^{-1} \int_E f \, d\mu$.

Proof. Measurability gives $E_\lambda \in \mathcal{M}$.
Since $\lambda \mathbf{1}_{E_\lambda} \leq f$ on E, monotonicity gives $\lambda \mu(E_\lambda) = \int_E \lambda \mathbf{1}_{E_\lambda} \, d\mu \leq \int_E f \, d\mu$; divide by $\lambda > 0$.
Structural Commentary: Level-set minorants convert mass to measure.
Key Insight: Use $\lambda \mathbf{1}_{E_\lambda} \leq f$ on E. □

Method 1.2.7. *Splitting a Simple Minorant Under a Sum*
Let $E \in \mathcal{M}$, $f, g \geq 0$ measurable, and $\omega \in \mathcal{S}^+(f + g)$ simple.
(i) *Grid*: for $n \geq 1$, use $D_n := \{0, \frac{1}{n}, \ldots, 1\}$.

(ii) *Split*: define $\varphi_n(x) := \max\{\theta\omega(x) : \theta \in D_n,\ \theta\omega(x) \le f(x)\}$; $\psi_n(x) := \big((1-\frac{1}{n})\,\omega(x) - \varphi_n(x)\big)^+$.
(iii) *Bounds*: φ_n, ψ_n are simple, $\varphi_n \le f$, $\psi_n \le g$, and $(1-\frac{1}{n})\omega \le \varphi_n + \psi_n$.
(iv) *Use*: simple additivity gives $(1-\frac{1}{n})\mathcal{I}_E(\omega) \le \int_E f\,d\mu + \int_E g\,d\mu$; then let $n \to \infty$ and take $\sup_{\omega \in \mathcal{S}^+(f+g)}$.

Key Insight: Grid-split ω between f and g, losing only $\frac{1}{n}$.

Proposition 1.2.8. *Homogeneity and Additivity for Nonnegative Integrals*
Let $f, g \ge 0$ be measurable, let $c \ge 0$, and let $E \in \mathcal{M}$. All identities are in $[0, +\infty]$, with $0 \cdot (+\infty) = 0$.
(i) Homogeneity: $\int_E cf\,d\mu = c\int_E f\,d\mu$.
(ii) Additivity: $\int_E (f+g)\,d\mu = \int_E f\,d\mu + \int_E g\,d\mu$.

Proof.
(i) If $c = 0$, both sides are 0; if $c > 0$, $s \mapsto cs$ bijects $\mathcal{S}^+(f)$ with $\mathcal{S}^+(cf)$ and satisfies $\mathcal{I}_E(cs) = c\mathcal{I}_E(s)$. Hence taking suprema gives the claim.
(ii) If $s \in \mathcal{S}^+(f)$ and $t \in \mathcal{S}^+(g)$, then $s + t \in \mathcal{S}^+(f+g)$ and $\mathcal{I}_E(s) + \mathcal{I}_E(t) = \mathcal{I}_E(s+t) \le \int_E (f+g)\,d\mu$, so taking $\sup_{s,t}$ gives $\int_E f\,d\mu + \int_E g\,d\mu \le \int_E (f+g)\,d\mu$.
Conversely, for $\omega \in \mathcal{S}^+(f+g)$, choose simple $\varphi_n \le f$ and $\psi_n \le g$ with $(1-\frac{1}{n})\omega \le \varphi_n + \psi_n$.
Then $\big(1-\frac{1}{n}\big)\,\mathcal{I}_E(\omega) \le \mathcal{I}_E(\varphi_n) + \mathcal{I}_E(\psi_n) \le \int_E f\,d\mu + \int_E g\,d\mu$. Let $n \to \infty$, then take $\sup_{\omega \in \mathcal{S}^+(f+g)}$.

Structural Commentary: The nonnegative integral is linear before signed integrability is introduced.
Key Insight: Scale minorants; split sum-minorants on a finite grid. □

Proposition 1.2.9. *Signed Decompositions Give the Same Integral*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, and let $f, f_1, f_2 : X \to \overline{\mathbb{R}}$ be measurable. Assume $f_i \ge 0$, $\int_E f_i\,d\mu < +\infty$ for $i = 1, 2$, and $f = f_1 - f_2$ on E. Then:
(i) $f \in \mathscr{L}^1(\mu)$ on E.
(ii) $\int_E f\,d\mu = \int_E f_1\,d\mu - \int_E f_2\,d\mu$.

Proof.
(i) Since $-f_2 \le f \le f_1$ on E, $f^+ \le f_1$ and $f^- \le f_2$ on E; monotonicity gives $\int_E f^{\pm}\,d\mu < +\infty$.
(ii) Since $f^+ - f^- = f_1 - f_2$ on E, $f^+ + f_2 = f_1 + f^-$ on E.
Nonnegative additivity gives $\int_E f^+\,d\mu + \int_E f_2\,d\mu = \int_E f_1\,d\mu + \int_E f^-\,d\mu$; subtract the finite terms.

Structural Commentary: Finite positive decompositions determine one signed integral.
Key Insight: Move all nonnegative terms to one equality, integrate, then cancel finite terms. □

Perspective 1.2.10. *Step Functions versus Simple Functions*
Riemann: $\sup \sum_i m_i \ell(I_i)$ over interval partitions. Lebesgue: $\sup\{\mathcal{I}(s) : 0 \le s \le f,\ s = \sum_i c_i \mathbf{1}_{E_i},\ E_i \in \mathcal{M}\}$.

Structural Commentary: Riemann partitions intervals; Lebesgue partitions level sets.
Key Insight: Same finite constants, measurable pieces.

Perspective 1.2.11. *Undergraph Picture for Nonnegative Functions*
For measurable $f : \mathbb{R} \to [0, +\infty]$, set $\mathcal{U}_f := \{(x, y) \in \mathbb{R} \times [0, +\infty) : 0 \le y < f(x)\}$.
Geometrically, $\int f\,dm = m_2(\mathcal{U}_f)$; analytically, the integral is built as $\sup \mathcal{I}(s)$ over simple $0 \le s \le f$.

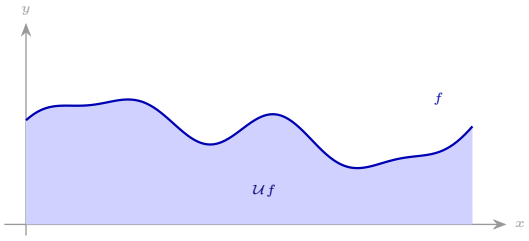

Figure 199: The integral of $f \ge 0$ is modeled by the measure of its undergraph.

Proposition 1.2.12. *Basic Integral Properties*
Let $E \in \mathcal{M}$.
(i) If f is measurable and bounded on E, and $\mu(E) < +\infty$, then $f \in \mathscr{L}^1(\mu)$ on E.
(ii) If $a \le f(x) \le b$ on E and $\mu(E) < +\infty$, then $a\mu(E) \le \int_E f\,d\mu \le b\mu(E)$.
(iii) If $f, g \in \mathscr{L}^1(\mu)$ on E and $f \le g$ on E, then $\int_E f\,d\mu \le \int_E g\,d\mu$.
(iv) If $f \in \mathscr{L}^1(\mu)$ on E, then $cf \in \mathscr{L}^1(\mu)$ on E for finite c, and $\int_E cf\,d\mu = c\int_E f\,d\mu$.
(v) If $\mu(E) = 0$ and f is measurable, then $\int_E f\,d\mu = 0$.
(vi) If $f \in \mathscr{L}^1(\mu)$ on E, $A \in \mathcal{M}$, and $A \subseteq E$, then $f \in \mathscr{L}^1(\mu)$ on A.

Proof.
(i) If $|f| \le M$ on E, then $0 \le f^{\pm} \le M\mathbf{1}_E$ on E, so $\int_E f^{\pm}\,d\mu \le M\mu(E) < +\infty$.
(ii) Since $a\mathbf{1}_E \le f \le b\mathbf{1}_E$ on E, order for the positive integral gives $a\mu(E) \le \int_E f\,d\mu \le b\mu(E)$.
(iii) Since $g - f \ge 0$ on E, $\int_E (g-f)\,d\mu \ge 0$, hence $\int_E f\,d\mu \le \int_E g\,d\mu$.
(iv) For $c \ge 0$, scale the simple approximants. For $c < 0$, use $(cf)^+ = (-c)f^-$ and $(cf)^- = (-c)f^+$.
(v) If $\mu(E) = 0$, then $\forall s \ge 0$ simple, $\mathcal{I}_E(s) = 0$, so $\int_E f^{\pm}\,d\mu = 0$.

(vi) Since $A \subseteq E$, $\mathcal{I}_A(s) \le \mathcal{I}_E(s)$ for $s \ge 0$ simple; hence $\int_A f^{\pm}\,d\mu \le \int_E f^{\pm}\,d\mu < +\infty$.

Structural Commentary: Estimate $f^{\pm}$ by simpler functions.
Key Insight: Monotonicity makes bounds, null sets, and restrictions immediate. □

Theorem 1.2.13. *The Integral as a Measure or Charge*
For $A \in \mathcal{M}$, define $\phi_f(A) := \int_A f\,d\mu$ when $f \ge 0$ is measurable, and define $\lambda_f(A) := \int_A f\,d\mu$ when $f \in \mathscr{L}^1(\mu)$.
(i) If $f \ge 0$ is measurable, then $\phi_f : \mathcal{M} \to [0,+\infty]$ is a measure.
(ii) If $f \in \mathscr{L}^1(\mu)$, then $\lambda_f : \mathcal{M} \to \mathbb{R}$ is finite-valued and countably additive.

Proof.
(i) For simple $s \ge 0$, $A \mapsto \mathcal{I}_A(s)$ is a measure, so $\phi_f(\emptyset) = 0$.
If $A = \bigsqcup_n A_n$, $s \in \mathcal{S}^+(f)$, then $\mathcal{I}_A(s) = \sum_n \mathcal{I}_{A_n}(s) \le \sum_n \phi_f(A_n)$; taking $\sup_s$ gives $\phi_f(A) \le \sum_n \phi_f(A_n)$. Conversely, $\forall N$, finite additivity gives $\sum_{n=1}^{N} \phi_f(A_n) = \phi_f(\bigcup_{n=1}^{N} A_n) \le \phi_f(A)$.
Let $N \to \infty$. Hence $\phi_f(A) = \sum_n \phi_f(A_n)$.
(ii) Since $f \in \mathscr{L}^1(\mu)$, $\phi_{f^+}(X), \phi_{f^-}(X) < +\infty$ and $\lambda_f = \phi_{f^+} - \phi_{f^-}$ is finite-valued.
If $A = \bigsqcup_n A_n$, then item (i) gives $\lambda_f(A) = \sum_n \phi_{f^+}(A_n) - \sum_n \phi_{f^-}(A_n) = \sum_n \lambda_f(A_n)$.

Structural Commentary: Setwise integration turns densities into measures.
Key Insight: Sup simple measures; subtract finite signs. □

Corollary 1.2.14. *Domain Additivity and Continuity for Integrable Functions*
Let $(X,\mathcal{M},\mu)$ be a measure space, let $E \in \mathcal{M}$, let $f \in \mathscr{L}^1(\mu)$ on E, and set $\mathcal{M}_E := \{A \in \mathcal{M} : A \subseteq E\}$.
(i) If $(A_n) \subseteq \mathcal{M}_E$ are pairwise disjoint, then $\int_{\bigsqcup_n A_n} f\,d\mu = \sum_n \int_{A_n} f\,d\mu$ and $\sum_n \left|\int_{A_n} f\,d\mu\right| < +\infty$.
(ii) If $A_n, A \in \mathcal{M}_E$ and $A_n \uparrow A$, then $\int_A f\,d\mu = \lim_{n\to\infty} \int_{A_n} f\,d\mu$.
(iii) If $A_n, A \in \mathcal{M}_E$ and $A_n \downarrow A$, then $\int_A f\,d\mu = \lim_{n\to\infty} \int_{A_n} f\,d\mu$.

Proof. Let $\mu_E := \mu|_{\mathcal{M}_E}$ and define $\lambda(A) := \int_A f\,d\mu$, $\nu(A) := \int_A |f|\,d\mu$ on $\mathcal{M}_E$.
By Theorem 1.2.13, λ is a finite signed measure, $\nu(E) < +\infty$, and $|\lambda(A)| \le \nu(A)$.
(i) Additivity gives $\lambda(\bigsqcup_n A_n) = \sum_n \lambda(A_n)$, and $\sum_n |\lambda(A_n)| \le \sum_n \nu(A_n) = \nu(\bigsqcup_n A_n) \le \nu(E) < +\infty$.
(ii) Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \ge 2$.
Since $A = \bigsqcup_n B_n$ and $A_N = \bigsqcup_{n\le N} B_n$, (i) gives $\lambda(A_N) = \sum_{n\le N} \lambda(B_n) \to \sum_n \lambda(B_n) = \lambda(A)$.
(iii) Set $C_n := A_1 \setminus A_n$ and $C := A_1 \setminus A$. Then $C_n \uparrow C$, so (ii) gives $\lambda(C_n) \to \lambda(C)$.
Hence $\lambda(A_n) = \lambda(A_1) - \lambda(C_n) \to \lambda(A_1) - \lambda(C) = \lambda(A)$.

Structural Commentary: Domain integrals form a finite signed measure.
Key Insight: Monotone domain limits are signed-measure continuity. □

Proposition 1.2.15. *Finite-Measure Support*
Let $(X,\mathcal{M},\mu)$ be a measure space, let $E \in \mathcal{M}$, and let $h : E \to \mathbb{R}$ be bounded and measurable.
Suppose $h = 0$ on $E \setminus E_0$ for some $E_0 \in \mathcal{M}$ with $E_0 \subseteq E$ and $\mu(E_0) < +\infty$. Then:
(i) $h \in \mathscr{L}^1(\mu)$ on E.
(ii) $\int_E h\,d\mu = \int_{E_0} h\,d\mu$.
(iii) If $E_1 \in \mathcal{M}$, $E_1 \subseteq E$, $\mu(E_1) < +\infty$, and $h = 0$ on $E \setminus E_1$, then $\int_{E_0} h\,d\mu = \int_{E_1} h\,d\mu$.

Proof.
(i) Choose $M < +\infty$ with $|h| \le M$ on E.
Since $h = 0$ on $E \setminus E_0$, $0 \le h^{\pm} \le M\mathbf{1}_{E_0}$ on E, so $\int_E h^{\pm}\,d\mu \le M\mu(E_0) < +\infty$.
(ii) Since $\int_{E\setminus E_0} h\,d\mu = 0$, $\int_E h\,d\mu = \int_{E_0} h\,d\mu$ by Corollary 1.2.14.
(iii) Apply item (ii) with E_0 and with E_1: $\int_{E_0} h\,d\mu = \int_E h\,d\mu = \int_{E_1} h\,d\mu$.

Structural Commentary: Finite support turns boundedness into integrability.
Key Insight: The integral only sees where h can be nonzero. □

Corollary 1.2.16. *Compact Continuous Functions on Finite Measure Spaces*
Let X be a compact topological space, let $\mathcal{M}$ be a σ-algebra containing the topology of X, and let $\mu(X) < +\infty$.
If $f : X \to \mathbb{R}$ is continuous, then:
(i) f is measurable.
(ii) $f \in \mathscr{L}^1(\mu)$.

Proof.
(i) $\forall O \subseteq \mathbb{R}$ open, $f^{-1}(O)$ is open in X by continuity, hence $f^{-1}(O) \in \mathcal{M}$.
(ii) The continuous image $f(X)$ is compact in $\mathbb{R}$, hence bounded.
Since $\mu(X) < +\infty$, $f \in \mathscr{L}^1(\mu)$ by Proposition 1.2.12(i).

Structural Commentary: Compactness supplies boundedness, while the measure hypothesis supplies finite mass.
Key Insight: Continuous on compact plus finite measure implies bounded measurable, hence integrable. □

Corollary 1.2.17. *Null and Finiteness Consequences for $f \ge 0$*
Let $f : X \to [0,+\infty]$ be measurable and $\phi_f(A) := \int_A f\,d\mu$.
(i) $\mu(A) = 0 \Longrightarrow \phi_f(A) = 0$ for $A \in \mathcal{M}$.
(ii) $\int_X f\,d\mu = 0 \Longleftrightarrow f = 0$ a.e.

(iii) $\int_X f\,d\mu < +\infty \Longrightarrow \mu\{f = +\infty\} = 0$ *and* $\{f > 0\}$ *is* σ*-finite.*

Proof.

(i) If $\mu(A) = 0$ and $0 \le s \le f$ is simple, then $\mathcal{I}_A(s) = 0$; hence $\phi_f(A) = \sup_s \mathcal{I}_A(s) = 0$.

(ii) ($\Rightarrow$) Set $E_n := \{f > \frac{1}{n}\}$. By Proposition 1.2.6, $\mu(E_n) \le n \int_X f\,d\mu = 0$.
Thus $\forall n,\ \mu(E_n) = 0$, and $\{f > 0\} = \bigcup_n E_n$ is null.

(ii) ($\Leftarrow$) If $N := \{f > 0\}$ null, every simple $0 \le s \le f$ vanishes on $X \setminus N$, so $\mathcal{I}_X(s) = \mathcal{I}_N(s) = 0$; take $\sup_s$.

(iii) Let $F_n := \{f \ge n\}$. By Proposition 1.2.6, $\mu(F_n) \le n^{-1} \int_X f\,d\mu$, so $\mu\{f = +\infty\} \le \mu(F_n) \to 0$.
Also $\{f > 0\} = \bigcup_n \{f > \frac{1}{n}\}$ and $\mu\{f > \frac{1}{n}\} \le n \int_X f\,d\mu < +\infty$.

Structural Commentary: Finite nonnegative integrals control threshold sets.
Key Insight: Threshold sets convert integral bounds into measure bounds. □

Theorem 1.2.18. *Absolute Integrability and Domination*
Let $(X, \mathcal{M}, \mu)$ *be a measure space and let* $E \in \mathcal{M}$. *Then:*

(i) If $f \in \mathscr{L}^1(\mu)$ *on* E, *then* $|f| \in \mathscr{L}^1(\mu)$ *on* E *and* $\left|\int_E f\,d\mu\right| \le \int_E |f|\,d\mu$.

(ii) If $f : X \to \overline{\mathbb{R}}$ *is measurable,* $g \in \mathscr{L}^1(\mu)$ *on* E, *and* $|f| \le g$ *on* E, *then* $f \in \mathscr{L}^1(\mu)$ *on* E.

Proof.

(i) Set $a := \int_E f^+\,d\mu$ and $b := \int_E f^-\,d\mu$. Since $f \in \mathscr{L}^1(\mu)$ on E, $a, b < +\infty$.
Hence $\int_E |f|\,d\mu = \int_E (f^+ + f^-)\,d\mu = a + b < +\infty$, and $\left|\int_E f\,d\mu\right| = |a - b| \le a + b = \int_E |f|\,d\mu$.

(ii) Since $|f| \le g$ on E, $g \ge 0$ on E and $\int_E g\,d\mu < +\infty$.
Also $0 \le f^\pm \le |f| \le g$ on E, so monotonicity gives $\int_E f^\pm\,d\mu < +\infty$.

Structural Commentary: Both claims reduce to finite positive and negative parts.
Key Insight: Use $|f| = f^+ + f^-$ and $0 \le f^\pm \le g$. □

Proposition 1.2.19. *Linearity of the Lebesgue Integral*
Let $(X, \mathcal{M}, \mu)$ *be a measure space, let* $E \in \mathcal{M}$, *let* $f, g \in \mathscr{L}^1(\mu)$ *on* E, *and let* $a, b \in \mathbb{R}$.
Then:

(i) For $c \in \mathbb{R}$ *and* $u \in \mathscr{L}^1(\mu)$ *on* E, $cu \in \mathscr{L}^1(\mu)$ *on* E *and* $\int_E cu\,d\mu = c\int_E u\,d\mu$.

(ii) $f + g \in \mathscr{L}^1(\mu)$ *on* E *and* $\int_E (f+g)\,d\mu = \int_E f\,d\mu + \int_E g\,d\mu$.

(iii) $af + bg \in \mathscr{L}^1(\mu)$ *on* E *and* $\int_E (af+bg)\,d\mu = a\int_E f\,d\mu + b\int_E g\,d\mu$.

Proof.

(i) Since $(cu)^\pm = c^+u^\pm + c^-u^\mp$, both $(cu)^+$ and $(cu)^-$ have finite integral. Nonnegative linearity gives $\int_E cu\,d\mu = (c^+a_u + c^-b_u) - (c^+b_u + c^-a_u) = c(a_u - b_u)$, where $a_u := \int_E u^+\,d\mu$ and $b_u := \int_E u^-\,d\mu$.

(ii) Since $|f+g| \le |f| + |g|$ and $|f| + |g| \in \mathscr{L}^1(\mu)$ on E, $f + g \in \mathscr{L}^1(\mu)$ on E by Theorem 1.2.18(ii).
Integrate $(f+g)^+ + f^- + g^- = (f+g)^- + f^+ + g^+$ over E and cancel finite terms.

(iii) By (i), $af, bg \in \mathscr{L}^1(\mu)$ on E, $\int_E af\,d\mu = a\int_E f\,d\mu$, and $\int_E bg\,d\mu = b\int_E g\,d\mu$.
Applying (ii) to af, bg gives $af + bg \in \mathscr{L}^1(\mu)$ on E and $\int_E (af+bg)\,d\mu = a\int_E f\,d\mu + b\int_E g\,d\mu$.

Structural Commentary: Signed linearity follows from finite sign pieces.
Key Insight: Integrate one nonnegative equality, then cancel finite terms. □

Proposition 1.2.20. *L^1 Level Sets, Null Changes, and Indefinite Integrals*
Let $f, g \in \mathscr{L}^1(\mu)$ *on* X, *and write* $\lambda_u(A) := \int_A u\,d\mu$.

(i) For $a > 0$, $\mu\{|f| \ge a\} \le a^{-1}\int_X |f|\,d\mu < +\infty$; *hence* $\{f \ne 0\}$ *is* σ*-finite.*

(ii) If measurable $h = 0$ *a.e., then* $h \in \mathscr{L}^1(\mu)$ *and* $\lambda_h \equiv 0$.

(iii) $\lambda_f \ge 0$ *on* $\mathcal{M} \Longrightarrow f \ge 0$ *a.e.;* $\lambda_f \equiv 0 \Longrightarrow f = 0$ *a.e.*

(iv) $\lambda_f = \lambda_g$ *on* $\mathcal{M} \Longleftrightarrow f = g$ *a.e.*

Proof.

(i) Since $a\mathbf{1}_{\{|f| \ge a\}} \le |f|$, integration gives $a\mu\{|f| \ge a\} \le \int_X |f|\,d\mu$. Also $\{f \ne 0\} = \bigcup_{k \ge 1} \{|f| \ge \frac{1}{k}\}$.

(ii) Let $N := \{h \ne 0\}$. For $A \in \mathcal{M}$, $h^\pm \mathbf{1}_A$ vanish off $A \cap N$, a null set.
Hence $\int_A h^+\,d\mu = \int_A h^-\,d\mu = 0$, so $\lambda_h(A) = 0$.

(iii) For $k \ge 1$, set $N_k := \{f \le -\frac{1}{k}\}$. By (i), $\mu(N_k) < +\infty$, and $\lambda_f(N_k) \le -k^{-1}\mu(N_k)$.
If $\lambda_f \ge 0$, then $\mu(N_k) = 0$, so $\{f < 0\} = \bigcup_k N_k$ is null. If $\lambda_f \equiv 0$, apply this to f and $-f$.

(iv) If $\lambda_f = \lambda_g$, then $\lambda_{f-g} \equiv 0$, so $f - g = 0$ a.e. by (iii). Conversely, if $f = g$ a.e., apply (ii) to $h = f - g$.

Structural Commentary: An L^1 function is detected by its integrals over all measurable sets, up to null sets.
Key Insight: Test sign on the level sets where the function is bounded away from 0. □

Proposition 1.2.21. *Finite-Measure Indicators by Intervals and Continuous Functions*
Let $A \subseteq \mathbb{R}$ *be Lebesgue measurable with* $m(A) < +\infty$. *Then* $\forall \varepsilon > 0$*:*

(i) $\exists G = \bigcup_{j=1}^{N} I_j$, I_j *open intervals, such that* $m(A \triangle G) < \varepsilon$.

(ii) $\exists f \in C(\mathbb{R})$ *such that* $0 \le f \le 1$ *and* $\int_{\mathbb{R}} |\mathbf{1}_A - f|\,dm < \varepsilon$.

Proof.

(i) Choose open $O \supseteq A$ with $m(O \setminus A) < \frac{\varepsilon}{2}$ and $m(O) < +\infty$.
Write $O = \bigsqcup_{j \geq 1} I_j$ with open intervals I_j, choose N with $m(O \setminus G) < \frac{\varepsilon}{2}$ for $G := \bigcup_{j=1}^{N} I_j$.
Then $A \setminus G \subseteq O \setminus G$ and $G \setminus A \subseteq O \setminus A$, so $m(A \triangle G) < \varepsilon$.

(ii) Apply (i) with $\frac{\varepsilon}{2}$ to get finite open G. Choose compact $K \subseteq G$ with $m(G \setminus K) < \frac{\varepsilon}{2}$, and choose piecewise-linear f with $0 \leq f \leq 1$, $f = 1$ on K, $f = 0$ on $\mathbb{R} \setminus G$.
Then $\int_{\mathbb{R}} |\mathbf{1}_G - f| \, dm \leq m(G \setminus K) < \frac{\varepsilon}{2}$, hence $\int_{\mathbb{R}} |\mathbf{1}_A - f| \, dm \leq m(A \triangle G) + \int_{\mathbb{R}} |\mathbf{1}_G - f| \, dm < \varepsilon$.

Structural Commentary: Finite-measure sets can be made geometrically simple up to small L^1 error.
Key Insight: Approximate by intervals, then smooth the indicator. □

Example 1.2.22. *Counting and Unit-Mass Integrals*

(i) *Counting*: for counting measure $\#$ on $(\mathbb{N}, 2^{\mathbb{N}})$ and $f : \mathbb{N} \to \mathbb{R}$,
$f \in \mathscr{L}^1(\#) \iff \sum_{n=1}^{\infty} |f(n)| < +\infty$, and then $\int_{\mathbb{N}} f \, d\# = \sum_{n=1}^{\infty} f(n)$.

(ii) *Unit mass*: for $(X, \mathcal{M})$, $p \in X$, and $\delta_p(E) := \mathbf{1}_E(p)$, $\int_X f \, d\delta_p = f(p)$ for measurable $f : X \to [0, +\infty]$.
If $f : X \to \mathbb{R}$ is measurable, then $f \in \mathscr{L}^1(\delta_p)$ and the same formula holds.

For δ_p, verify first on indicators, then extend to simple functions by finite linearity, to nonnegative functions by monotone simple approximation, and to real-valued functions by positive–negative splitting.

Example 1.2.23. *L^1 Is Not Closed Under Squaring*
On $X = \mathbb{N}$ with $\mathcal{M} = 2^{\mathbb{N}}$, define $\mu(E) := \sum_{n \in E} 4^{-n}$ and $f(n) := \frac{2^n}{\sqrt{n}}$.

Then $\int_X |f| \, d\mu = \sum_{n=1}^{\infty} \frac{1}{2^n \sqrt{n}} < +\infty$; $\int_X f^2 \, d\mu = \sum_{n=1}^{\infty} \frac{1}{n} = +\infty$.
The first series is dominated by the geometric series $\sum 2^{-n}$, while the second diverges by Corollary 6.1.12.
Thus $f \in \mathscr{L}^1(\mu)$ need not imply $f^2 \in \mathscr{L}^1(\mu)$.

1.3 Convergence Theorems

This subsection collects the main limit-interchange tools: monotone convergence, Fatou, and dominated convergence.

Theorem 1.3.1. *Monotone Convergence*
Let $E \in \mathcal{M}$, let f_n, f be measurable on E, and assume $0 \leq f_n \uparrow f$. Then $\int_E f_n \, d\mu \uparrow \int_E f \, d\mu$.

Proof. By monotonicity, $\int_E f_n \, d\mu \uparrow L \leq \int_E f \, d\mu$.
For $s \in \mathcal{S}^+(f)$ and $0 < c < 1$, set $E_n := \{x \in E : f_n(x) \geq c s(x)\}$.
Since $f_n \uparrow f$ and $s \leq f$, $E_n \uparrow E$, and $\int_E f_n \, d\mu \geq \int_{E_n} f_n \, d\mu \geq c \int_{E_n} s \, d\mu$.
Continuity from below for $\phi_s(A) := \int_A s \, d\mu$ gives $L \geq c \int_E s \, d\mu$. Let $c \uparrow 1$, then take $\sup_{s \in \mathcal{S}^+(f)}$.

Structural Commentary: Nonnegative integration commutes with monotone limits.
Key Insight: Simple minorants of the limit are eventually captured below the sequence. □

Theorem 1.3.2. *Series Form of Monotone Convergence*
Let $E \in \mathcal{M}$, let $f_n \geq 0$ be measurable, and set $f := \sum_{n=1}^{\infty} f_n$ on E. Then $\int_E f \, d\mu = \sum_{n=1}^{\infty} \int_E f_n \, d\mu$.

Proof. Set $S_N := \sum_{n=1}^{N} f_n$. Then $0 \leq S_N \uparrow f$.

By Theorem 1.3.1, $\int_E f \, d\mu = \lim_{N \to \infty} \int_E S_N \, d\mu = \lim_{N \to \infty} \sum_{n=1}^{N} \int_E f_n \, d\mu = \sum_{n=1}^{\infty} \int_E f_n \, d\mu$.

Structural Commentary: Nonnegative series are monotone partial sums.
Key Insight: Integrate partial sums, then pass upward. □

Corollary 1.3.3. *Beppo Levi Bounded-Integral Criterion*
Let $E \in \mathcal{M}$, let (f_n) be measurable on E, and assume $0 \leq f_n \uparrow f$ with $\sup_n \int_E f_n \, d\mu < +\infty$.
Then $f < +\infty$ a.e. on E and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu < +\infty$.

Proof. The monotone limit f is measurable, and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu \leq \sup_n \int_E f_n \, d\mu < +\infty$ by Theorem 1.3.1. Thus $f < +\infty$ a.e. on E by Corollary 1.2.17.

Structural Commentary: Bounded mass survives monotone limits.
Key Insight: Finite nonnegative integral forces a.e. finiteness. □

Proposition 1.3.4. *L^1 Approximation and Bounded Multipliers*
Let $f \in \mathscr{L}^1(\mu)$ on X.
(i) $\forall \varepsilon > 0$ $\exists$ a measurable simple function φ with $\int_X |f - \varphi| \, d\mu < \varepsilon$.
(ii) If g is bounded measurable, then $fg \in \mathscr{L}^1(\mu)$.

Proof.

(i) Choose simple $0 \leq s_n \uparrow f^+$ and $0 \leq t_n \uparrow f^-$. Then $\int_X (f^+ - s_n) \, d\mu \to 0$ and $\int_X (f^- - t_n) \, d\mu \to 0$.
For large n, $\varphi := s_n - t_n$ satisfies $\int_X |f - \varphi| \, d\mu \leq \int_X (f^+ - s_n) \, d\mu + \int_X (f^- - t_n) \, d\mu < \varepsilon$.

(ii) If $|g| \leq M$, then fg is measurable and $\int_X |fg| \, d\mu \leq M \int_X |f| \, d\mu < +\infty$.

Structural Commentary: L^1 is simple-closed and bounded-multiplier stable.
Key Insight: Approximate signs separately; dominate products by $M|f|$. □

Theorem 1.3.5. *Fatou's Lemma*
Let $E \in \mathcal{M}$ and let $f_n : E \to [0, +\infty]$ be measurable. Then $\int_E \liminf_n f_n \, d\mu \le \liminf_n \int_E f_n \, d\mu$.

Proof. Set $g_n := \inf_{i \ge n} f_i$. Then $0 \le g_n \uparrow \liminf_i f_i$ and $\forall i \ge n$, $g_n \le f_i$, so $\int_E g_n \, d\mu \le \inf_{i \ge n} \int_E f_i \, d\mu$.
By Theorem 1.3.1, $\int_E \liminf_i f_i \, d\mu = \lim_n \int_E g_n \, d\mu \le \lim_n \inf_{i \ge n} \int_E f_i \, d\mu = \liminf_n \int_E f_n \, d\mu$.
Structural Commentary: Replace lim inf by increasing tail infima.
Key Insight: Tail infima are monotone minorants. □

Example 1.3.6. *Strictness in Fatou's Lemma*
Fatou's inequality can be strict by concentration or escape of mass.

(i) *Concentration*: On $E = (0, 1)$ with Lebesgue measure m, set $f_n := n\mathbf{1}_{(0, \frac{1}{n})}$.
Then $f_n \to 0$ pointwise and $\forall n$, $\int_E f_n \, dm = 1$.
(ii) *Escape*: On $E = \mathbb{R}$ with Lebesgue measure m, set $g_n := \mathbf{1}_{(n, n+1)}$.
Then $g_n \to 0$ pointwise and $\forall n$, $\int_E g_n \, dm = 1$.

In either case, with $h_n = f_n$ or $h_n = g_n$, $\int_E \liminf_n h_n \, dm = 0 < 1 = \liminf_n \int_E h_n \, dm$.

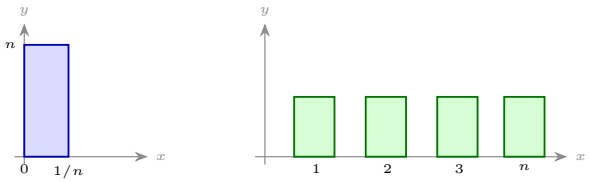

Figure 200: Strict Fatou behavior: $f_n = n\mathbf{1}_{(0, \frac{1}{n})}$ and $g_n = \mathbf{1}_{(n, n+1)}$ have integral 1 but vanish pointwise.

Theorem 1.3.7. *Dominated Convergence*
Let $E \in \mathcal{M}$, let f_n, f be measurable on E, and let $g \in \mathscr{L}^1(\mu)$ on E.
Suppose $f_n \to f$ a.e. on E and $\forall n$, $|f_n| \le g$ on E. Then:
(i) $f \in \mathscr{L}^1(\mu)$ on E.
(ii) $\lim_{n \to \infty} \int_E f_n \, d\mu = \int_E f \, d\mu$.

Proof. Choose null $N \subseteq E$ such that $f_n \to f$ on $E \setminus N$.
Changing values on N does not change integrals, so assume $f_n \to f$ everywhere on E.

(i) The pointwise limit f is measurable on E, and $|f| \le g$ on E; hence $f \in \mathscr{L}^1(\mu)$ on E by Theorem 1.2.18(ii).
(ii) Put $a_n := \int_E f_n \, d\mu$. Fatou applied to $g + f_n \ge 0$ gives $\int_E (g + f) \, d\mu \le \int_E g \, d\mu + \liminf_n a_n$.
Fatou applied to $g - f_n \ge 0$ gives $\int_E (g - f) \, d\mu \le \int_E g \, d\mu - \limsup_n a_n$.
Subtracting $\int_E g \, d\mu < +\infty$ gives $\int_E f \, d\mu \le \liminf_n a_n \le \limsup_n a_n \le \int_E f \, d\mu$.

Structural Commentary: Domination reduces signed convergence to two Fatou bounds.
Key Insight: Apply Fatou to $g + f_n$ and $g - f_n$. □

Theorem 1.3.8. *General Dominated Convergence*
Let $E \in \mathcal{M}$, let f_n, f, g be measurable on E, and let $g_n \ge 0$ be integrable on E.
Suppose $f_n \to f$ and $g_n \to g$ a.e. on E, with $\forall n$, $|f_n| \le g_n$ a.e. on E.
Assume $\lim_n \int_E g_n \, d\mu = \int_E g \, d\mu < +\infty$. Then:
(i) $\forall n$, $f_n \in \mathscr{L}^1(\mu)$ on E, and $f \in \mathscr{L}^1(\mu)$ on E.
(ii) $\lim_{n \to \infty} \int_E f_n \, d\mu = \int_E f \, d\mu$.

Proof. Choose null $N \subseteq E$ so that $f_n \to f$, $g_n \to g$, and $\forall n$, $|f_n| \le g_n$ on $E \setminus N$.
Changing values on N does not change integrals, so assume these relations hold on E.

(i) Since $|f_n| \le g_n \in \mathscr{L}^1(\mu)$, $f_n \in \mathscr{L}^1(\mu)$ on E by Theorem 1.2.18(ii).
Passing to the limit gives $|f| \le g$, and $\int_E g \, d\mu < +\infty$, hence $f \in \mathscr{L}^1(\mu)$ on E.
(ii) Put $a_n := \int_E f_n \, d\mu$, $b_n := \int_E g_n \, d\mu$, and $b := \int_E g \, d\mu$; then $b_n \to b$.
Fatou applied to $g_n + f_n \ge 0$ gives $b + \int_E f \, d\mu \le \liminf_n (b_n + a_n) = b + \liminf_n a_n$.
Fatou applied to $g_n - f_n \ge 0$ gives $b - \int_E f \, d\mu \le \liminf_n (b_n - a_n) = b - \limsup_n a_n$.
Subtracting $b < +\infty$ gives $\limsup_n a_n \le \int_E f \, d\mu \le \liminf_n a_n$, hence $a_n \to \int_E f \, d\mu$.

Structural Commentary: Varying domination works when masses converge.
Key Insight: Fatou on $g_n \pm f_n$ traps the limit. □

Corollary 1.3.9. *Monotone Convergence Almost Everywhere*
Let $E \in \mathcal{M}$ and $0 \le f_n \uparrow f$ a.e. on E. Then $\int_E f_n \, d\mu \uparrow \int_E f \, d\mu$.

Proof. Choose $N \subseteq E$ with $\mu(N) = 0$ and $f_n \uparrow f$ on $E \setminus N$. Set $g_n := f_n \mathbf{1}_{E \setminus N}$ and $g := f\mathbf{1}_{E \setminus N}$.
Then $0 \le g_n \uparrow g$, so $\int_E g_n \, d\mu \uparrow \int_E g \, d\mu$. Since $f_n = g_n$ and $f = g$ off N, the integrals over N vanish.
Structural Commentary: Monotone convergence ignores null exceptional sets.
Key Insight: Remove the exceptional set, prove the theorem there, then add back zero. □

Example 1.3.10. *Counting Measure Integrals Are Series*

Let μ be counting measure on $(\mathbb{N}, 2^{\mathbb{N}})$. For $f : \mathbb{N} \to [0, +\infty]$, set $s_N := \sum_{n=1}^{N} f(n)\mathbf{1}_{\{n\}}$; then $0 \le s_N \uparrow f$, with the case $f(k) = +\infty$ giving both sides $+\infty$. Hence $\int_{\mathbb{N}} f\, d\mu = \lim_N \int_{\mathbb{N}} s_N\, d\mu = \lim_N \sum_{n=1}^{N} f(n) = \sum_{n=1}^{\infty} f(n)$.

Structural Commentary: Counting measure turns integration into summation.
Key Insight: Finite-support truncations are the simple-function partial sums.

Example 1.3.11. *Boundary Tests for Monotone Convergence and Fatou*

On $([0, +\infty), \mathcal{B}, m)$:

(i) $f_n := \mathbf{1}_{[0,n]} \uparrow \mathbf{1}_{[0,+\infty)}$ and $\int f_n\, dm = n \uparrow +\infty$; monotone convergence allows $+\infty$.
(ii) $g_n := n^{-1}\mathbf{1}_{[n,+\infty)} \downarrow 0$ uniformly, but $\int g_n\, dm = +\infty\ \forall n$; decreasing convergence needs finite mass.
(iii) $w_n := -n^{-1}\mathbf{1}_{[0,n]} \rightrightarrows 0$, but $\int w_n\, dm = -1$; Fatou can fail without $f_n \ge 0$ or $f_n \ge -h \in L^1$.

Theorem 1.3.12. *Fatou with an Integrable Lower Bound*

Let $h \ge 0$, $h \in \mathscr{L}^1(\mu)$, and let f_n be measurable with $f_n \ge -h$.
Then, whenever the signed integrals are defined in $\overline{\mathbb{R}}$, $\int \liminf_{n\to\infty} f_n\, d\mu \le \liminf_{n\to\infty} \int f_n\, d\mu$.

Proof. Set $F_n := f_n + h \ge 0$. Fatou gives $\int \liminf_n F_n\, d\mu \le \liminf_n \int F_n\, d\mu$.
Since $\liminf_n F_n = h + \liminf_n f_n$ and $\int F_n\, d\mu = \int f_n\, d\mu + \int h\, d\mu$, subtract the finite term $\int h\, d\mu$.
Structural Commentary: Fatou survives controlled negative parts.
Key Insight: Add an integrable floor to make the sequence nonnegative. □

Remark 1.3.13. *Why the Negative Example Has No Floor*

For $w_n := -n^{-1}\mathbf{1}_{[0,n]}$, any $h \in L^1$ with $-h \le w_n\ \forall n$ must satisfy $h \ge n^{-1}$ on $[0, n]$.
Thus $h \ge k^{-1}$ on $[k-1, k]$, and $\int_0^N h\, dm \ge H_N := \sum_{k=1}^{N} \frac{1}{k} \to +\infty$, contradicting $h \in L^1$.

Perspective 1.3.14. *Domination Is the Safeguard*

$f_n \to f$ a.e. need not imply $\int f_n \to \int f$; domination $|f_n| \le g \in L^1$ restores the limit interchange.

Example 1.3.15. *A Nonuniform Limit with Vanishing Integrals*

For $x > 0$, set $f_n(x) := e^{-nx}x^{-\frac{1}{2}}$. Then $f_n \to 0$ pointwise, but $\forall \delta > 0$, $\sup_{0<x<\delta} f_n(x) = +\infty$.
For $n \ge 1$, $0 \le f_n \le g := e^{-x}x^{-\frac{1}{2}}$; $g \le x^{-\frac{1}{2}}\mathbf{1}_{(0,1]} + e^{-x}\mathbf{1}_{[1,\infty)} \in \mathscr{L}^1(0, \infty)$.
Thus dominated convergence gives $I_n := \int_0^\infty e^{-nx}x^{-\frac{1}{2}}\, dx \to 0$.
Structural Commentary: Uniform convergence is unnecessary once one integrable envelope controls the sequence.
Key Insight: Domination handles the singular behavior at 0 and the tail at infinity simultaneously.

Proposition 1.3.16. *Uniform Convergence on Finite Measure Spaces*

Assume $\mu(X) < +\infty$.

(i) If $0 \le f_n$ are measurable and $f_n \rightrightarrows f$, then $f \ge 0$ is measurable and $\int_X f\, d\mu = \lim_n \int_X f_n\, d\mu$ in $[0, +\infty]$.
(ii) If $f_n \in \mathscr{L}^1(\mu)$ and $f_n \rightrightarrows f$, then $f \in \mathscr{L}^1(\mu)$ and $\int_X f_n\, d\mu \to \int_X f\, d\mu$ in $\mathbb{R}$.

Proof.

(i) Measurability and $f \ge 0$ pass to pointwise limits. For $\varepsilon > 0$ and $n \gg 1$, $f \le f_n + \varepsilon\mathbf{1}_X$ and $f_n \le f + \varepsilon\mathbf{1}_X$.
Thus $\int_X f\, d\mu \le \int_X f_n\, d\mu + \varepsilon\mu(X)$ and $\int_X f_n\, d\mu \le \int_X f\, d\mu + \varepsilon\mu(X)$. Let $\varepsilon \downarrow 0$.
(ii) Choose N with $|f - f_N| \le 1$. Then $f = f_N + (f - f_N) \in \mathscr{L}^1(\mu)$, and
$\left|\int_X f_n\, d\mu - \int_X f\, d\mu\right| \le \int_X |f_n - f|\, d\mu \le \mu(X) \sup_X |f_n - f| \to 0$.

Structural Commentary: On finite measure spaces, uniform error controls integral error.
Key Insight: Finiteness of $\mu(X)$ turns a supremum bound into an integral bound. □

Theorem 1.3.17. *Bounded Convergence on Finite Measure Spaces*

Let $\mu(X) < +\infty$, and let $f_n, f : X \to \mathbb{R}$ be measurable.
Suppose $f_n \to f$ a.e. and $\exists M < +\infty$ such that $\forall n$, $|f_n| \le M$ a.e. Then:

(i) $f \in \mathscr{L}^1(\mu)$.
(ii) $\lim_{n\to\infty} \int_X f_n\, d\mu = \int_X f\, d\mu$.

Proof. Since $|f_n| \le M$ a.e. and $f_n \to f$ a.e., $|f| \le M$ a.e.
Set $G := M\mathbf{1}_X$. Then $G \in \mathscr{L}^1(\mu)$, since $\int_X G\, d\mu = M\mu(X) < +\infty$.

(i) Since $|f| \le G$ a.e., $f \in \mathscr{L}^1(\mu)$ by Theorem 1.2.18(ii).
(ii) Since $\forall n$, $|f_n| \le G$ a.e., $\int_X f_n\, d\mu \to \int_X f\, d\mu$ by Theorem 1.3.7(ii).

Structural Commentary: On a finite-measure space, a uniform pointwise bound is an integrable dominator.
Key Insight: Bounded convergence is the finite-measure special case of dominated convergence. □

Example 1.3.18. *Why the Finite-Measure Hypothesis Matters*

On $(\mathbb{R}, m)$, let $f_n := n^{-1}\mathbf{1}_{[0,n]}$. Then $\|f_n\|_\infty = \frac{1}{n} \to 0$, but $\int_{\mathbb{R}} f_n\, dm = 1\ \forall n$.
Thus uniform convergence need not commute with integrals when $m(\mathbb{R}) = +\infty$; Fatou only yields $0 \le 1$.

Example 1.3.19. *Why Domination Cannot Be Dropped*

On $(\mathbb{R}, m)$, let $f_n := n\mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}$. Then $f_n \to 0$ pointwise, but $\int_{\mathbb{R}} f_n\, dm = 1\ \forall n$.
Thus pointwise convergence without domination need not commute with integrals; Fatou only yields $0 \le 1$.

Theorem 1.3.20. *Absolutely Summable L^1 Series*
Let $f_n \in \mathscr{L}^1(\mu)$ and $\sum_{n\geq 1} \|f_n\|_1 < +\infty$.
Set $S_N := \sum_{n=1}^{N} f_n$. Then $\exists f : X \to \mathbb{R}$ measurable such that:
(i) $S_N \to f$ absolutely a.e.
(ii) $f \in \mathscr{L}^1(\mu)$.
(iii) $\int_X f\,d\mu = \sum_{n\geq 1} \int_X f_n\,d\mu$.

Proof. Set $G := \sum_{n\geq 1} |f_n|$. By monotone convergence, $\int_X G\,d\mu = \sum_{n\geq 1} \|f_n\|_1 < +\infty$, so $G < +\infty$ a.e. and $G \in \mathscr{L}^1(\mu)$. Define $f := \lim_N S_N$ on $\{G < +\infty\}$ and $f := 0$ elsewhere.

(i) Since $G < +\infty$ a.e., $\sum_{n\geq 1} |f_n| < +\infty$ a.e.; hence $S_N \to f$ absolutely a.e.
(ii) Since $|f| \leq G$ a.e. and $G \in \mathscr{L}^1(\mu)$, $f \in \mathscr{L}^1(\mu)$ by Theorem 1.2.18(ii).
(iii) Since $|S_N| \leq G$ and $S_N \to f$ a.e., dominated convergence gives

$$\int_X f\,d\mu = \lim_N \int_X S_N\,d\mu = \lim_N \sum_{n=1}^{N} \int_X f_n\,d\mu = \sum_{n\geq 1} \int_X f_n\,d\mu.$$

Structural Commentary: Absolute summability in L^1 gives a.e. absolute convergence and termwise integration.
Key Insight: First integrate the nonnegative majorant $\sum |f_n|$. □

Proposition 1.3.21. *L^1 Convergence Controls Absolute Integrals*
If $f_n, f \in \mathscr{L}^1(\mu)$ and $\|f_n - f\|_1 \to 0$, then:
(i) $\|f_n\|_1 \to \|f\|_1$.
(ii) $\int_X |f|\,d\mu = \lim_n \int_X |f_n|\,d\mu$.

Proof.

(i) The reverse triangle inequality gives $|\|f_n\|_1 - \|f\|_1| \leq \|f_n - f\|_1 \to 0$.
(ii) Since $\|h\|_1 = \int_X |h|\,d\mu$, item (i) gives $\int_X |f_n|\,d\mu \to \int_X |f|\,d\mu$.

Structural Commentary: L^1 convergence is strong enough to converge the L^1 sizes.
Key Insight: The reverse triangle inequality converts function convergence into norm convergence. □

Proposition 1.3.22. *Finite Integral Localizes to Finite Measure*
Let $f : X \to [0, +\infty]$ be measurable and $\int_X f\,d\mu < +\infty$. For $n \geq 1$, set $E_n := \{\frac{1}{n} < f \leq n\}$. Then:
(i) $\forall n,\ \mu(E_n) < +\infty$.
(ii) $\int_{X\setminus E_n} f\,d\mu \downarrow 0$.
Consequently, $\forall \varepsilon > 0\ \exists E \in \mathcal{M}$ with $\mu(E) < +\infty$ and $\int_{X\setminus E} f\,d\mu < \varepsilon$.

Proof.

(i) Since $f > \frac{1}{n}$ on E_n, $\mu(E_n) \leq n \int_{E_n} f\,d\mu \leq n \int_X f\,d\mu < +\infty$.
(ii) Since $f\mathbf{1}_{E_n} \uparrow f$ a.e., monotone convergence gives $\int_{E_n} f\,d\mu \uparrow \int_X f\,d\mu$.
Thus $\int_{X\setminus E_n} f\,d\mu = \int_X f\,d\mu - \int_{E_n} f\,d\mu \downarrow 0$.

Given $\varepsilon > 0$, choose n with $\int_{X\setminus E_n} f\,d\mu < \varepsilon$ and set $E = E_n$.
Structural Commentary: Finite integral mass can be captured on a finite-measure part of the space.
Key Insight: Keep the levels where f is bounded above and away from zero. □

Proposition 1.3.23. *Setwise Convergence from Total Mass Convergence*
Let $f_n, f : X \to [0, +\infty]$ be measurable, $f_n \to f$ pointwise, and $\int_X f\,d\mu = \lim_{n\to\infty} \int_X f_n\,d\mu < +\infty$.
Then $\forall E \in \mathcal{M}$, $\int_E f\,d\mu = \lim_{n\to\infty} \int_E f_n\,d\mu$.

Proof. Fix $E \in \mathcal{M}$ and set $E^c := X \setminus E$. By Fatou, $\int_E f\,d\mu \leq \liminf_n \int_E f_n\,d\mu$ and $\int_{E^c} f\,d\mu \leq \liminf_n \int_{E^c} f_n\,d\mu$.
Since $\int_X f_n\,d\mu = \int_E f_n\,d\mu + \int_{E^c} f_n\,d\mu \to \int_X f\,d\mu < +\infty$, $\limsup_n \int_E f_n\,d\mu \leq \int_X f\,d\mu - \int_{E^c} f\,d\mu = \int_E f\,d\mu$.
Thus $\liminf_n \int_E f_n\,d\mu = \limsup_n \int_E f_n\,d\mu = \int_E f\,d\mu$.
Structural Commentary: Total mass plus Fatou on complements fixes each part.
Key Insight: A lower bound on the complement is an upper bound on the chosen set. □

Example 1.3.24. *Why the Finite Total Mass Assumption Matters*
Let $X = \mathbb{N} \times \{0, 1\}$ with counting measure, set $E := \mathbb{N} \times \{1\}$.
Define $f := \mathbf{1}_{\mathbb{N}\times\{0\}}$; $f_n := \mathbf{1}_{\{1,\ldots,n\}\times\{0\}} + n\mathbf{1}_{\{n\}\times\{1\}}$.
Then $f_n \to f$ pointwise and $\int_X f\,d\mu = \lim_n \int_X f_n\,d\mu = +\infty$; $\int_E f\,d\mu = 0$, $\int_E f_n\,d\mu = n$.
Thus Proposition 1.3.23 can fail when the common total mass is infinite.

Definition 1.3.25. *Function Truncates*
For $f : X \to \mathbb{R}$ and $n \in \mathbb{N}$, define the *nth truncate*

$$T_n f(x) := (-n) \vee (f(x) \wedge n) = \begin{cases} -n, & f(x) < -n, \\ f(x), & -n \leq f(x) \leq n, \\ n, & f(x) > n. \end{cases}$$

Proposition 1.3.26. *Truncates Characterize Integrability*
Let $f : X \to \mathbb{R}$ be measurable and set $f_n := T_n f$ for $n \in \mathbb{N}$.
(i) $f \in \mathscr{L}^1(\mu) \iff \sup_n \|f_n\|_1 < +\infty$.
(ii) If $f \in \mathscr{L}^1(\mu)$, then $\int_X f_n \, d\mu \to \int_X f \, d\mu$.

Proof. $\forall n$, $T_n : \mathbb{R} \to [-n, n]$ is continuous, so f_n is measurable; moreover $f_n \to f$, $|f_n| \le |f|$, and $|f_n| \uparrow |f|$.
(i) ($\Rightarrow$) If $f \in \mathscr{L}^1(\mu)$, then $\|f_n\|_1 \le \|f\|_1$ for all n, so $\sup_n \|f_n\|_1 < +\infty$.
(i) ($\Leftarrow$) If $\sup_n \|f_n\|_1 < +\infty$, then MCT gives $\|f\|_1 = \lim_{n\to\infty} \|f_n\|_1 = \sup_n \|f_n\|_1 < +\infty$, so $f \in \mathscr{L}^1(\mu)$.
(ii) If $f \in \mathscr{L}^1(\mu)$, then DCT applied to $f_n \to f$ with $|f_n| \le |f|$ gives $\int_X f_n \, d\mu \to \int_X f \, d\mu$.
Structural Commentary: Truncation reduces L^1 questions to bounded approximants.
Key Insight: MCT reads $\|f\|_1$ from $|f_n| \uparrow |f|$; DCT passes signed integrals. □

1.4 Parameter Integrals and Differentiation

Corollary 1.4.1. *Continuity Under a Lebesgue Integral*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in [a,b]$, and suppose $\forall t \in [a,b]$, $|f_t| \le g \in \mathscr{L}^1(\mu)$. Then:
(i) If $t_0 \in [a,b]$ and $f_t \to f_{t_0}$ a.e. as $t \to t_0$, then $\lim_{t\to t_0} \int_X f_t \, d\mu = \int_X f_{t_0} \, d\mu$.
(ii) If $t \mapsto f(x,t)$ is continuous on $[a,b]$ for a.e. x, then $F(t) := \int_X f_t \, d\mu$ is continuous on $[a,b]$.

Proof.
(i) For any $t_n \to t_0$, set $u_n := f_{t_n}$. Then $u_n \to f_{t_0}$ a.e. and $|u_n| \le g$, so dominated convergence gives $\int_X u_n \, d\mu \to \int_X f_{t_0} \, d\mu$. The sequential criterion gives the claim.
(ii) If $t \mapsto f(x,t)$ is continuous a.e., then $\forall t_0 \in [a,b]$, $f_t \to f_{t_0}$ a.e. as $t \to t_0$.
By item (i), $\forall t_0 \in [a,b]$, $F(t) \to F(t_0)$, so F is continuous on $[a,b]$.
Structural Commentary: Parameter continuity reduces to dominated convergence along sequences.
Key Insight: The parameter disappears once t_n is frozen into a sequence of functions. □

Theorem 1.4.2. *Differentiation Under a Lebesgue Integral*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in [a,b]$, and suppose $f_{t_0} \in \mathscr{L}^1(\mu)$ for some $t_0 \in [a,b]$.
Assume $t \mapsto f(x,t)$ is continuous on $[a,b]$ and differentiable on (a,b) $\forall x \in X$, and $\exists g \in \mathscr{L}^1(\mu)$ such that $\forall (x,t) \in X \times (a,b)$, $|\partial_t f(x,t)| \le g(x)$. For $t \in (a,b)$, set $D_t(x) := \partial_t f(x,t)$; set $F(t) := \int_X f_t \, d\mu$. Then:
(i) $\forall t \in (a,b)$, D_t is measurable.
(ii) $\forall t \in [a,b]$, $f_t \in \mathscr{L}^1(\mu)$.
(iii) F is differentiable on (a,b), and $\forall t \in (a,b)$, $F'(t) = \int_X D_t \, d\mu$.

Proof.
(i) Fix $t \in (a,b)$ and choose $h_k \to 0$ with $t + h_k \in [a,b]$, $h_k \neq 0$.
The quotients $\frac{f_{t+h_k} - f_t}{h_k}$ are measurable and converge pointwise to D_t, so D_t is measurable.
(ii) By the mean value theorem, $|f_t - f_{t_0}| \le |t - t_0| \, g$, so $|f_t| \le |f_{t_0}| + |t - t_0| \, g \in \mathscr{L}^1(\mu)$.
(iii) If $t_n \to t \in (a,b)$ and $t_n \neq t$, then $\frac{F(t_n) - F(t)}{t_n - t} = \int_X \frac{f_{t_n} - f_t}{t_n - t} \, d\mu$. The integrand converges pointwise to D_t and has modulus at most g, so dominated convergence gives $\frac{F(t_n) - F(t)}{t_n - t} \to \int_X D_t \, d\mu$.
Structural Commentary: Bounded parameter derivatives dominate the difference quotients.
Key Insight: Differentiating the integral is just applying dominated convergence to secant slopes. □

Corollary 1.4.3. *A Pointwise Difference-Quotient Test*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in [a,b]$, let $f_{t_0} \in \mathscr{L}^1(\mu)$ for some $t_0 \in [a,b]$, and fix $t_1 \in [a,b]$.
Assume $\partial_t f(x,t_1)$ exists $\forall x$. For $t \neq t_1$, set $Q_t(x) := \frac{f_t(x) - f_{t_1}(x)}{t - t_1}$, $|Q_t| \le g$ for some $g \in \mathscr{L}^1(\mu)$.
Set $F(t) := \int_X f_t \, d\mu$. Then:
(i) $\forall t \in [a,b]$, $f_t \in \mathscr{L}^1(\mu)$.
(ii) F is differentiable at t_1, and $F'(t_1) = \int_X \partial_t f(x,t_1) \, d\mu$.

Proof.
(i) If $t_0 \neq t_1$, then $|f_{t_1}| \le |f_{t_0}| + |t_0 - t_1| \, g$; if $t_0 = t_1$, this is given.
Hence $f_{t_1} \in \mathscr{L}^1(\mu)$. For $t \neq t_1$, $|f_t| \le |f_{t_1}| + |t - t_1| \, g$, so $f_t \in \mathscr{L}^1(\mu)$.
(ii) For any $t_n \to t_1$ with $t_n \neq t_1$, $Q_{t_n} \to \partial_t f(\cdot, t_1)$ pointwise and $|Q_{t_n}| \le g$.
Dominated convergence gives $\frac{F(t_n) - F(t_1)}{t_n - t_1} \to \int_X \partial_t f(x,t_1) \, d\mu$.
Structural Commentary: One parameter point needs only domination of the secants into that point.
Key Insight: A global derivative bound is sufficient, but not necessary, for a one-point formula. □

Example 1.4.4. *Differentiating Under the Integral Sign*
For $t > 0$, set $F(t) := \int_0^\infty x^2 e^{-tx} \, dx$. For $t_0 > 0$ and $0 < |h| < \frac{t_0}{2}$, put $Q_h(x) := \frac{x^2 e^{-(t_0+h)x} - x^2 e^{-t_0 x}}{h}$.
Then $Q_h \to -x^3 e^{-t_0 x}$ pointwise and $|Q_h| \le x^3 e^{-\frac{t_0 x}{2}} \in \mathscr{L}^1(0,\infty)$. Hence $F'(t_0) = -\int_0^\infty x^3 e^{-t_0 x} \, dx$.
Structural Commentary: Parameter differentiation is a limit-interchange problem.
Key Insight: Prove a local-in-parameter integrable majorant for the difference quotients.

Example 1.4.5. *Laplace Transform Powers and Factorials*
For $t > 0$, set $I(t) := \int_0^\infty e^{-tx}\,dx$. Then $I(t) = \lim_{b\to+\infty} \frac{1-e^{-tb}}{t} = t^{-1}$.

For $[a,b] \subset (0,+\infty)$ and $k \geq 0$, $0 \leq x^k e^{-tx} \leq x^k e^{-ax} \in \mathscr{L}^1(0,\infty)$, so repeated differentiation under the integral gives $I^{(n)}(t) = (-1)^n \int_0^\infty x^n e^{-tx}\,dx$. Since $I^{(n)}(t) = (-1)^n n! t^{-n-1}$, $\forall n \in \mathbb{N}\cup\{0\}$, $\int_0^\infty x^n e^{-x}\,dx = n!$.
Structural Commentary: A parameter integral turns powers of x into derivatives in t.
Key Insight: The exponential envelope e^{-ax} controls all nearby parameter values $t \geq a$.

Corollary 1.4.6. *One Lebesgue and One Riemann Integral*
Let $f_t(x) := f(x,t)$, $t \in [a,b]$, with f_t measurable, $t \mapsto f(x,t)$ continuous a.e., and $\forall t \in [a,b]$, $|f_t| \leq g \in \mathscr{L}^1(\mu)$. Then, with Riemann integrals in t, $\int_a^b \left[\int_X f_t\,d\mu\right] dt = \int_X \left[\int_a^b f(x,t)\,dt\right] d\mu(x)$.

Proof. Set $h_t(x) := \int_a^t f(x,s)\,ds$. Then h_t is measurable, $|h_t| \leq (b-a)g$, and $\partial_t h_t = f_t$.
By differentiation under the integral, $H(t) := \int_X h_t\,d\mu$ satisfies $H'(t) = \int_X f_t\,d\mu$.
Hence $\int_a^b \left[\int_X f_t\,d\mu\right] dt = H(b) - H(a) = \int_X \left[\int_a^b f(x,t)\,dt\right] d\mu(x)$.
Structural Commentary: Finite parameter integrals move through by differentiating primitives.
Key Insight: Integrate first in the parameter up to t, then differentiate the resulting Lebesgue integral. □

Corollary 1.4.7. *Improper Parameter Interchange*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in \mathbb{R}$, and let $t \mapsto f(x,t)$ be continuous for a.e. x. Suppose $\exists g, h \in \mathscr{L}^1(\mu)$ such that $|f_t| \leq g$ $\forall t$ and $\int_{\mathbb{R}} |f(x,t)|\,dt \leq h(x)$ for a.e. x. Put $G(t) := \int_X f_t\,d\mu$, $H(x) := \int_{\mathbb{R}} f(x,t)\,dt$, and $H_R(x) := \int_{-R}^R f(x,t)\,dt$. Then:

(i) $H \in \mathscr{L}^1(\mu)$.
(ii) $\lim_{R\to+\infty} \int_{-R}^R G(t)\,dt = \int_X H\,d\mu$.

Thus $\int_{\mathbb{R}} G := \lim_{R\to+\infty} \int_{-R}^R G(t)\,dt$ exists and equals $\int_X H\,d\mu$.

Proof.
(i) Since $|H| \leq \int_{\mathbb{R}} |f(x,t)|\,dt \leq h$ a.e. and $h \in \mathscr{L}^1(\mu)$, $H \in \mathscr{L}^1(\mu)$ by Theorem 1.2.18(ii).
(ii) For $R > 0$, finite-interval interchange gives $\int_{-R}^R G(t)\,dt = \int_X H_R\,d\mu$.
Since $H_R \to H$ a.e. and $|H_R| \leq h$, dominated convergence gives $\int_X H_R\,d\mu \to \int_X H\,d\mu$.
Structural Commentary: Finite interchanges pass to improper limits under an L^1 bound.
Key Insight: Truncate in t, interchange on finite intervals, then pass $R \to +\infty$ by domination in x. □

1.5 Riemann-Lebesgue Comparison

Lemma 1.5.1. *Monotone Integral Squeeze*
Let $E \in \mathcal{M}$, $(\varphi_n), (\psi_n) \subseteq \mathscr{L}^1(\mu)$ on E, and $f : E \to \mathbb{R}$. Assume $\varphi_n \uparrow$, $\psi_n \downarrow$, $\varphi_n \leq f \leq \psi_n$ on E, and $\int_E (\psi_n - \varphi_n)\,d\mu \to 0$. Put $\varphi := \lim_n \varphi_n$ and $\psi := \lim_n \psi_n$. Then:

(i) $\varphi = \psi$ *a.e. on* E.
(ii) For $h := \varphi$, $f = h$ a.e., $h \in \mathscr{L}^1(\mu)$ on E, and $\int_E \varphi_n\,d\mu, \int_E \psi_n\,d\mu \to \int_E h\,d\mu$.

Proof.
(i) The limits φ, ψ are measurable and $\varphi \leq f \leq \psi$.
Since $0 \leq \psi - \varphi \leq \psi_n - \varphi_n$, $0 \leq \int_E (\psi - \varphi)\,d\mu \leq \int_E (\psi_n - \varphi_n)\,d\mu \to 0$; hence $\psi = \varphi$ a.e.
(ii) Let h be the common a.e. value from (i). Then $f = h$ a.e. and $|h| \leq |\varphi_1| + |\psi_1|$ a.e., so $h \in \mathscr{L}^1(\mu)$ on E.
Monotone convergence applied to $\varphi_n - \varphi_1 \uparrow h - \varphi_1$ and $\psi_1 - \psi_n \uparrow \psi_1 - h$ gives $\int_E \varphi_n\,d\mu, \int_E \psi_n\,d\mu \to \int_E h\,d\mu$.
Structural Commentary: Vanishing gaps collapse monotone envelopes.
Key Insight: Integral gaps force the two monotone limits to coincide a.e. □

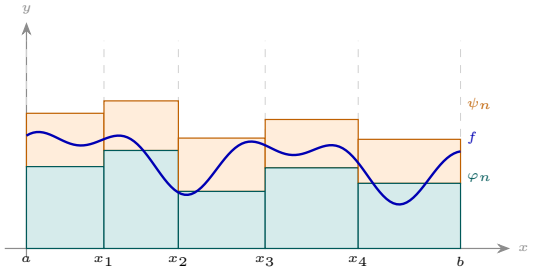

Figure 201: Step envelopes squeeze f to a common a.e. limit.

Theorem 1.5.2. *Comparison with the Riemann Integral*
Let m be Lebesgue measure on $[a,b]$ and let $f \in \mathcal{R}[a,b]$.

(i) $f \in \mathscr{L}^1(m)$.
(ii) $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.

Proof. Choose nested partitions P_n with $U(P_n,f) - L(P_n,f) \to 0$. Let φ_n, ψ_n be lower and upper step envelopes.
(i) Lemma 1.5.1 gives $f \in \mathscr{L}^1(m)$.

(ii) Since $\int \varphi_n\,dm = L(P_n,f)$ and $\int \psi_n\,dm = U(P_n,f)$, Lemma 1.5.1 gives $\int_{[a,b]} f\,dm = \lim_n L(P_n,f) = \lim_n U(P_n,f) = \int_a^b f(x)\,dx$.

Structural Commentary: Darboux envelopes squeeze the Lebesgue integral.
Key Insight: Vanishing Darboux gap identifies the two integral notions. □

Corollary 1.5.3. *Riemann Integrable Functions as Monotone Step Limits*
Let $f:[a,b]\to\mathbb{R}$ be Riemann integrable. Then $\exists(\varphi_n),(\psi_n)$ step-function sequences such that

(i) $\varphi_n\uparrow$, $\psi_n\downarrow$, and $\varphi_n\le f\le\psi_n$ on $[a,b]$;
(ii) $\lim_n \int_a^b \varphi_n\,dx = \lim_n \int_a^b \psi_n\,dx = \int_a^b f\,dx$;
(iii) $\varphi_n,\psi_n\to f$ a.e.; hence f is Lebesgue measurable and $\int_{[a,b]} f\,dm = \int_a^b f\,dx$.

Proof.

(i) Choose nested partitions P_n with $\max_{I\in P_n}\ell(I)\to 0$ and $U(P_n,f)-L(P_n,f)\to 0$.
Let φ_n,ψ_n be the lower and upper step envelopes on P_n; refinement gives $\varphi_n\uparrow$, $\psi_n\downarrow$, and $\varphi_n\le f\le\psi_n$.
(ii) Since $\int_a^b \varphi_n\,dx = L(P_n,f)$ and $\int_a^b \psi_n\,dx = U(P_n,f)$, by Darboux gap condition, both limits equal $\int_a^b f\,dx$.
(iii) By Lemma 1.5.1, $\exists h\in\mathscr{L}^1(m)$ with $h=f$ a.e., $\varphi_n,\psi_n\to h$ a.e., and $\int_{[a,b]} h\,dm = \lim_n \int_a^b \varphi_n\,dx = \int_a^b f\,dx$.

Structural Commentary: Riemann integrability is monotone step squeezing.
Key Insight: Refining partitions make Darboux sums into pointwise a.e. step approximations. □

Corollary 1.5.4. *Nonnegative Continuous Functions and Riemann Integrals*

(i) If $f\ge 0$ is continuous on $[a,b]$, then $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.
(ii) If $f\ge 0$ is continuous on $[0,+\infty)$, then $\int_{[0,+\infty)} f\,dm = \lim_{b\to+\infty}\int_0^b f(x)\,dx$ in $[0,+\infty]$.

Proof.

(i) Continuous f on $[a,b]$ is Riemann integrable; by Riemann–Lebesgue comparison, $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.
(ii) Put $f_b := f\mathbf{1}_{[0,b]}$. By (i), $\int_{[0,b]} f\,dm = \int_0^b f(x)\,dx$. As $b\uparrow+\infty$, $f_b\uparrow f$; monotone convergence gives the left limit, and the right limit is the nonnegative improper Riemann integral.

Structural Commentary: Compact comparison plus monotone truncation.
Key Insight: Truncate to compact intervals and pass upward. □

Theorem 1.5.5. *Lebesgue's Criterion for Riemann Integrability*
Let $f:[a,b]\to\mathbb{R}$ be bounded with discontinuity set D_f. Then $f\in\mathcal{R}[a,b]\iff m(D_f)=0$.

Proof. Set $\omega_f(x) := \lim_{r\downarrow 0}\operatorname{diam} f([a,b]\cap(x-r,x+r))$ and $D_k := \{x:\omega_f(x)\ge\frac{1}{k}\}$; then $D_f = \bigcup_{k\ge1} D_k$.

($\Rightarrow$) Fix k and $\varepsilon>0$. Choose P with $U(P,f)-L(P,f)<\frac{\varepsilon}{k}$. Set $\mathcal{B} := \{I\in P:\operatorname{int} I\cap D_k\neq\emptyset\}$.
Then $\frac{1}{k}\sum_{I\in\mathcal{B}}\ell(I)\le U(P,f)-L(P,f)$, so $\sum_{I\in\mathcal{B}}\ell(I)<\varepsilon$.
These intervals plus the partition points cover D_k; hence $m(D_k)=0$ and $m(D_f)=0$.
($\Leftarrow$) Suppose $m(D_f)=0$ and $|f|\le B$. If $B=0$, $f\in\mathcal{R}[a,b]$. For $B>0$ and $\varepsilon>0$, choose k with $\frac{b-a}{k}<\frac{\varepsilon}{2}$ and cover the compact null set D_k by finitely many intervals of total length $<\frac{\varepsilon}{4B}$.
Refining by this cover gives $U(P,f)-L(P,f)<\frac{b-a}{k}+\frac{\varepsilon}{2}<\varepsilon$, hence $f\in\mathcal{R}[a,b]$.

Structural Commentary: Darboux gaps measure oscillation.
Key Insight: Riemann failure is positive-measure discontinuity. □

Remark 1.5.6. *Small Discontinuity Sets*
The same idea underlies Theorem 1.5.5: discontinuity sets of measure zero, in the sense of Definition 4.1.13, do not obstruct Riemann integrability. Dirichlet's function fails because every point is discontinuous.

Example 1.5.7. *Dirichlet Function: Lebesgue but Not Riemann*
Let $D := \mathbf{1}_{\mathbb{Q}\cap[0,1]}$. Since $\mathbb{Q}\cap[0,1]$ is countable, $D=0$ a.e. on $[0,1]$, so $\int_{[0,1]} D\,dm = 0$.
Every nondegenerate subinterval $I\subseteq[0,1]$ meets both $\mathbb{Q}$ and $\mathbb{R}\setminus\mathbb{Q}$, hence $\inf_I D = 0$ and $\sup_I D = 1$.
Thus all lower Darboux sums are 0, all upper Darboux sums are 1, and D is not Riemann integrable.
Structural Commentary: Null support, dense oscillation.
Key Insight: A dense null set can be harmless for Lebesgue integration and fatal for Riemann integration.

Perspective 1.5.8. *Lebesgue's Criterion as a Roadmap*
For bounded $f:[a,b]\to\mathbb{R}$ with discontinuity set D_f, $f\in\mathcal{R}[a,b]\iff m(D_f)=0$.
Thus finite, countable, and null Cantor-type D_f are allowed, while $D_f=[a,b]$ is not.

1.6 Littlewood and Lusin Approximation

The measurable world is often controlled by ordinary topological objects after discarding a small exceptional set. In the real-line finite-measure setting, this principle has three recurring forms: measurable sets are nearly elementary, measurable functions are nearly continuous, and a.e. convergence is nearly uniform.

Principle 1.6.1. *Littlewood's Three Principles, Finite-Measure Form*
For subsets of the real line with finite measure:

(i) measurable sets are nearly finite unions of intervals; see Corollary 6.2.4;
(ii) measurable functions are nearly continuous; this is Lusin's theorem below;
(iii) a.e. pointwise convergence is nearly uniform; see Egoroff's theorem, Theorem 1.6.3.

Key Insight: The word "nearly" means after changing or discarding a set of arbitrarily small measure.

Definition 1.6.2. *Almost Uniform Convergence*
Let (f_n) be measurable functions on X.

(i) $f_n \to f$ *almost uniformly* if $\forall \delta > 0\ \exists E_\delta \in \mathcal{M}$ with $\mu(E_\delta) < \delta$ and $f_n \rightrightarrows f$ on $X \setminus E_\delta$.
(ii) (f_n) is *almost uniformly Cauchy* if $\forall \delta > 0\ \exists E_\delta \in \mathcal{M}$ with $\mu(E_\delta) < \delta$ s.t. (f_n) uniformly Cauchy on $X \setminus E_\delta$.

Theorem 1.6.3. *Egoroff's Theorem*
If $\mu(X) < +\infty$, $f_n \to f$ a.e., with all functions finite-valued off null set, then $f_n \to f$ almost uniformly.

Proof. Absorb the null exceptional set into every later E_δ, and assume $\forall x \in X,\ f_n(x) \to f(x)$.
For $m, n \in \mathbb{N}$, set $E_n(m) := \bigcup_{k=n}^{\infty} \{|f_k - f| \geq \frac{1}{m}\}$. Then $E_n(m) \downarrow \emptyset$, so $\mu(E_n(m)) \to 0$ by Corollary 5.1.1.
Given $\delta > 0$, choose k_m with $\mu(E_{k_m}(m)) < \delta 2^{-m}$ and set $E_\delta := \bigcup_{m \geq 1} E_{k_m}(m)$.
Then $\mu(E_\delta) < \delta$, and $x \notin E_\delta$, $k \geq k_m$ imply $|f_k(x) - f(x)| < \frac{1}{m}$; hence $f_n \rightrightarrows f$ on $X \setminus E_\delta$.
Structural Commentary: Finite measure turns decreasing bad sets with empty intersection into small bad sets.
Key Insight: A.e. convergence is almost uniform on finite measure spaces. □

Example 1.6.4. *Egoroff Needs a Finite-Valued Limit*
On $([0,1], \lambda)$, set $f_n := n\mathbf{1}_{[0,1]}$. Then $\forall x,\ f_n(x) \to +\infty$, so the pointwise limit is not finite-valued. If $f_n \to f$ almost uniformly for finite-valued measurable f, then for some E with $\lambda(E) < \frac{1}{2}$, $f_n \rightrightarrows f$ on $[0,1] \setminus E$; this nonempty set has $f_n(x) = n \to +\infty$, contradicting finite-valued pointwise convergence.

Lemma 1.6.5. *Continuous Extension from a Closed Subset of $\mathbb{R}$*
Let $F \subseteq \mathbb{R}$ be closed and let $h : F \to \mathbb{R}$ be continuous. Then $\exists g \in C(\mathbb{R})$ such that $g|_F = h$.

Proof. Cases $F = \emptyset$ and $F = \mathbb{R}$ are immediate. Otherwise $\mathbb{R} \setminus F = \bigsqcup_\alpha I_\alpha$, set $g = h$ on F; on each component:

(i) If $I_\alpha = (a, b)$, set $g(x) := \frac{b-x}{b-a}h(a) + \frac{x-a}{b-a}h(b)$.
(ii) If $I_\alpha = (-\infty, b)$, set $g(x) := h(b)$.
(iii) If $I_\alpha = (a, +\infty)$, set $g(x) := h(a)$.

The formulas give continuity off F; at $x \in F$, adjacent finite endpoints converge to x.
Hence continuity of h gives $g(y) \to h(x)$ as $y \to x$. Thus $g \in C(\mathbb{R})$ and $g|_F = h$.
Structural Commentary: Fill complementary intervals.
Key Insight: Interpolate between boundary values. □

Proposition 1.6.6. *Simple Functions Are Nearly Continuous*
Let $E \in \mathcal{L}^1$ with $m(E) < +\infty$, and $s : E \to \mathbb{R}$ measurable simple. $\forall \varepsilon > 0$, $\exists F \subseteq E$ and $g : \mathbb{R} \to \mathbb{R}$ such that:

(i) F is closed.
(ii) $g \in C(\mathbb{R})$ and $g = s$ on F.
(iii) $m(E \setminus F) < \varepsilon$.

Proof. If $E = \emptyset$, take $F = \emptyset$ and $g = 0$. Otherwise $s = \sum_{i=1}^{N} a_i \mathbf{1}_{E_i}$ with disjoint measurable E_i and $E = \bigsqcup_i E_i$. Choose closed $F_i \subseteq E_i$ with $m(E_i \setminus F_i) < \frac{\varepsilon}{N}$, set $F := \bigcup_i F_i$, and define $h : F \to \mathbb{R}$ by $h|_{F_i} := a_i$.

(i) Each F_i is closed and $N < +\infty$, so F is closed; also $F \subseteq E$.
(ii) For $x \in F_i$, the closed set $\bigcup_{j \neq i} F_j$ misses x, so h is locally constant at x in F; hence $h \in C(F)$.
$\exists g \in C(\mathbb{R})$ with $g|_F = h = s|_F$ by Lemma 1.6.5.
(iii) $m(E \setminus F) \leq \sum_i m(E_i \setminus F_i) < \varepsilon$.

Structural Commentary: Closed cores isolate simple levels.
Key Insight: Approximate each level set from inside, then extend. □

Theorem 1.6.7. *Lusin's Theorem*
Let $E \in \mathcal{L}^1$ with $m(E) < +\infty$, $f : E \to \mathbb{R}$ measurable. $\forall \varepsilon > 0$, $\exists F \subseteq E$ and $g : \mathbb{R} \to \mathbb{R}$ such that:

(i) F is closed.
(ii) $g \in C(\mathbb{R})$ and $g = f$ on F.
(iii) $m(E \setminus F) < \varepsilon$.

Proof. Take simple $s_n : E \to \mathbb{R}$ with $s_n \to f$ pointwise.
Pick $A \subseteq E$ with $s_n \to f$ uniformly on A and $m(E \setminus A) < \frac{\varepsilon}{4}$, and closed $F_0 \subseteq A$ with $m(A \setminus F_0) < \frac{\varepsilon}{4}$.
For $n \geq 1$, choose closed $F_n \subseteq E$ with $s_n|_{F_n} \in C(F_n)$ and $m(E \setminus F_n) < \varepsilon 2^{-(n+1)}$. Set $F := \bigcap_{n \geq 0} F_n$.

(i) F is an intersection of closed sets and $F \subseteq E$.
(ii) Since $F \subseteq F_0 \subseteq A$, the continuous functions $s_n|_F$ converge uniformly to $f|_F$, so $f|_F \in C(F)$.
Since F is closed and $f|_F \in C(F)$, Lemma 1.6.5 gives $\exists g \in C(\mathbb{R})$ with $g|_F = f|_F$.
(iii) $m(E \setminus F) \leq m(E \setminus F_0) + \sum_{n \geq 1} m(E \setminus F_n) < \varepsilon$.

Structural Commentary: Discard a small set to gain continuity.
Key Insight: Simple cores plus Egoroff give uniform continuous limits. □

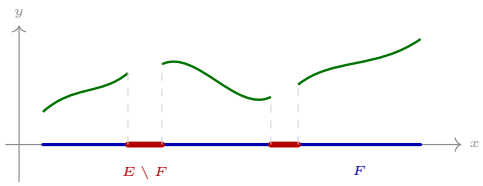

Figure 202: Lusin's theorem: keep F, discard $E \setminus F$, and extend with $g = f$ on F.

1.7 Convex Functions and Jensen's Inequality

Convexity is the geometric condition behind Jensen's inequality: averages move inside the argument, while supporting lines convert the nonlinear statement into a linear integral estimate.

Definition 1.7.1. *Convex Function on an Interval*
Let $I \subseteq \mathbb{R}$ be an interval and let $\varphi : I \to \mathbb{R}$.

(i) φ is *convex* iff $\forall x_1, x_2 \in I\ \forall \lambda \in [0,1]$, $\varphi(\lambda x_1 + (1-\lambda)x_2) \le \lambda\varphi(x_1) + (1-\lambda)\varphi(x_2)$.
(ii) Equivalently, $\forall x_1 < x < x_2$ in I, $\varphi(x) \le \frac{x_2 - x}{x_2 - x_1}\varphi(x_1) + \frac{x - x_1}{x_2 - x_1}\varphi(x_2)$.

Key Insight: Graphs lie below enclosing chords.

Lemma 1.7.2. *Chordal Slope Lemma*
Let $I \subseteq \mathbb{R}$ be an interval, let $\varphi : I \to \mathbb{R}$ be convex, and let $x_1, x, x_2 \in I$ satisfy $x_1 < x < x_2$.
Then $\frac{\varphi(x) - \varphi(x_1)}{x - x_1} \le \frac{\varphi(x_2) - \varphi(x_1)}{x_2 - x_1} \le \frac{\varphi(x_2) - \varphi(x)}{x_2 - x}$.

Proof. Since $x = \frac{x_2 - x}{x_2 - x_1}x_1 + \frac{x - x_1}{x_2 - x_1}x_2$, convexity gives $(x_2 - x_1)\varphi(x) \le (x_2 - x)\varphi(x_1) + (x - x_1)\varphi(x_2)$.
Rearrange, $(x_2 - x_1)(\varphi(x) - \varphi(x_1)) \le (x - x_1)(\varphi(x_2) - \varphi(x_1))$; $(x_2 - x)(\varphi(x_2) - \varphi(x_1)) \le (x_2 - x_1)(\varphi(x_2) - \varphi(x))$.
Divide by $(x - x_1)(x_2 - x_1) > 0$ and $(x_2 - x)(x_2 - x_1) > 0$, respectively.
Structural Commentary: One chord inequality, two rearrangements.
Key Insight: Convex secant slopes are nondecreasing. □

Proposition 1.7.3. *Increasing Derivative Gives Convexity*
Let $I \subseteq \mathbb{R}$ be open and let $\varphi : I \to \mathbb{R}$ be differentiable. If φ' is increasing on I, then φ is convex on I.
In particular, this holds if φ is twice differentiable and $\varphi'' \ge 0$ on I.

Proof. For $a < b$ in I, set $S(a,b) := \frac{\varphi(b) - \varphi(a)}{b - a}$. Let $u < v$ and $0 < \lambda < 1$, and put $x = \lambda u + (1-\lambda)v$.
By MVT, $S(u,x) = \varphi'(c_1)$, $S(x,v) = \varphi'(c_2)$ for some $c_1 \in (u,x)$, $c_2 \in (x,v)$. As $c_1 < c_2$, $S(u,x) \le S(x,v)$.
Clearing denominators, $\varphi(x) \le \frac{v - x}{v - u}\varphi(u) + \frac{x - u}{v - u}\varphi(v) = \lambda\varphi(u) + (1-\lambda)\varphi(v)$; the cases $\lambda = 0, 1$ are equality.
If $\varphi'' \ge 0$, for $a < b$, MVT gives $\varphi'(b) - \varphi'(a) = \varphi''(d)(b - a) \ge 0$ for some $d \in (a,b)$, so φ' is increasing.
Structural Commentary: MVT orders secant slopes.
Key Insight: Monotone slopes give convexity. □

Example 1.7.4. *Basic Convex Functions*
By Proposition 1.7.3, $x \mapsto x^p$ for $p \ge 1$, $x \mapsto e^{ax}$, and $x \mapsto \log(\frac{1}{x})$ are convex on $(0, +\infty)$, $\mathbb{R}$, and $(0, +\infty)$.

Proposition 1.7.5. *Finite Jensen Inequality*
Let $I \subseteq \mathbb{R}$ be an interval, $\varphi : I \to \mathbb{R}$ be convex, $x_1, \dots, x_n \in I$, $\lambda_1, \dots, \lambda_n \ge 0$ with $n \in \mathbb{N}$, $\sum_{k=1}^{n} \lambda_k = 1$.
Set $\bar{x} := \sum_{k=1}^{n} \lambda_k x_k$. Then $\bar{x} \in I$ and $\varphi(\bar{x}) \le \sum_{k=1}^{n} \lambda_k \varphi(x_k)$.

Proof. Induct on $m = \#\{k : \lambda_k > 0\}$. If $m = 1$, equality holds.
For $m \ge 2$, relabel so $\lambda_1, \dots, \lambda_m > 0$ and $\lambda_k = 0$ for $k > m$. Let $\Lambda = 1 - \lambda_m$ and $y = \Lambda^{-1} \sum_{k=1}^{m-1} \lambda_k x_k$.
By induction, $y \in I$ and $\varphi(y) \le \Lambda^{-1} \sum_{k=1}^{m-1} \lambda_k \varphi(x_k)$.
Since $\bar{x} = \Lambda y + \lambda_m x_m$, convexity gives $\bar{x} \in I$ and $\varphi(\bar{x}) \le \Lambda\varphi(y) + \lambda_m\varphi(x_m) \le \sum_{k=1}^{n} \lambda_k \varphi(x_k)$.
Structural Commentary: Induct by peeling off one weight.
Key Insight: Jensen iterates convexity. □

Lemma 1.7.6. *One-Sided Slopes and Supporting Lines*
Let $I \subseteq \mathbb{R}$ be open and let $\varphi : I \to \mathbb{R}$ be convex.
(i) $\forall x \in I$, $\varphi'(x-)$ and $\varphi'(x+)$ exist and are finite.
(ii) If $u < v$ in I, then $\varphi'(u-) \le \varphi'(u+) \le \frac{\varphi(v) - \varphi(u)}{v - u} \le \varphi'(v-) \le \varphi'(v+)$.
(iii) If $\alpha \in I$ and $\varphi'(\alpha-) \le m_0 \le \varphi'(\alpha+)$, then $\forall t \in I$, $\varphi(t) \ge \varphi(\alpha) + m_0(t - \alpha)$.

Proof. For $a < b$ in I, write $S(a,b) := \frac{\varphi(b) - \varphi(a)}{b - a}$.

(i) Fix $x \in I$ and choose $a < x < b$ in I.
If $a < r < s < x$, then $S(a,x) \le S(r,x) \le S(s,x) \le S(x,b)$ by Lemma 1.7.2, so $\lim_{r \uparrow x} S(r,x)$ exists in $\mathbb{R}$.
If $x < r < s < b$, the same lemma gives $S(a,x) \le S(x,r) \le S(x,s) \le S(x,b)$, so $\lim_{r \downarrow x} S(x,r)$ exists in $\mathbb{R}$.

(ii) Let $u < v$ in I. For $r < u < s < v$ and $u < q < v < t$, the chordal slope inequalities give $S(r,u) \le S(u,s) \le S(u,v) \le S(q,v) \le S(v,t)$. Letting $r \uparrow u$, $s \downarrow u$, $q \uparrow v$, and $t \downarrow v$ gives asserted chain.

(iii) If $t = \alpha$, equality holds. If $t > \alpha$, then (ii) gives $m_0 \le \varphi'(\alpha+) \le S(\alpha, t)$, hence $\varphi(t) \ge \varphi(\alpha) + m_0(t-\alpha)$.
If $t < \alpha$, then (ii) gives $S(t,\alpha) \le \varphi'(\alpha-) \le m_0$, which rearranges to the same estimate.

Structural Commentary: Monotone secant slopes have finite one-sided limits.
Key Insight: One-sided slope bounds produce supporting lines. □

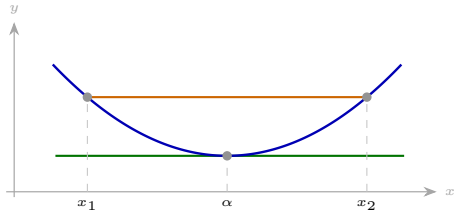

Figure 203: Color code: $y = \varphi(x)$ lies below the chord through x_1, x_2 and above the supporting line at α.

Theorem 1.7.7. *Jensen's Inequality on $[0,1]$*
Let m be Lebesgue measure on $[0,1]$, let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex, and let $f, \varphi \circ f \in L^1(m)$.
Then $\varphi\left(\int f\, dm\right) \le \int \varphi(f)\, dm$.

Proof. Set $\alpha := \int f\, dm$. By Lemma 1.7.6, choose $m_0 \in [\varphi'(\alpha-), \varphi'(\alpha+)]$; $\varphi(t) \ge \varphi(\alpha) + m_0(t-\alpha)\ \forall t \in \mathbb{R}$.
Substituting $t = f$ and integrating gives $\int \varphi(f)\, dm \ge \varphi(\alpha) + m_0(\int f\, dm - \alpha) = \varphi(\alpha)$.
Structural Commentary: Support at the mean.
Key Insight: The linear term vanishes. □

Remark 1.7.8. *Integrability in Jensen*
Linear growth, $|\varphi(t)| \le c_1 + c_2\,|t|$, makes $\varphi \circ f \in L^1([0,1])$ by Proposition 4.3.4.
If $\varphi \circ f \ge 0$, Jensen's right side may be extended-valued.

2 Product Measures and Fubini Theory

2.1 Product Measures

Product measure builds measurable rectangles and reads higher-dimensional integrals through slices. The abstract construction comes first; the Euclidean statements below are its geometric specialization.

Definition 2.1.1. *Product Measurable Spaces and Sections*
Let $(X, \mathcal{M})$, $(Y, \mathcal{N})$ be measurable spaces and $Z := X \times Y$.

(i) A *measurable rectangle* is $A \times B$ with $A \in \mathcal{M}$ and $B \in \mathcal{N}$.
Set $\mathcal{Z}_0 := \{ \bigcup_{j=1}^{n} (A_j \times B_j) : A_j \in \mathcal{M},\ B_j \in \mathcal{N}\}$ and $\mathcal{M} \otimes \mathcal{N} := \sigma\{A \times B : A \in \mathcal{M},\ B \in \mathcal{N}\}$.

(ii) For $E \subseteq Z$, the *sections* are $E_x := \{y \in Y : (x,y) \in E\}$ and $E^y := \{x \in X : (x,y) \in E\}$.

(iii) For $F : Z \to \overline{\mathbb{R}}$, the *function sections* are $F_x(y) := F(x,y)$ and $F^y(x) := F(x,y)$, where defined.

Lemma 2.1.2. *Finite Rectangle Algebra*
(i) If $R_j = A_j \times B_j$, then $R_1 \cap R_2 = (A_1 \cap A_2) \times (B_1 \cap B_2)$ and $R_1 \setminus R_2 = \big[(A_1 \cap A_2) \times (B_1 \setminus B_2)\big] \sqcup \big[(A_1 \setminus A_2) \times B_1\big]$.
(ii) $\mathcal{Z}_0$ is an algebra on Z.
(iii) $\forall E \in \mathcal{Z}_0\ \exists n \ge 1$, $A_j \in \mathcal{M}$, and $B_j \in \mathcal{N}$ such that $E = \bigsqcup_{j=1}^{n} (A_j \times B_j)$.

Proof.

(i) For $(x,y) \in Z$, $(x,y) \in R_1 \cap R_2 \iff x \in A_1 \cap A_2$ and $y \in B_1 \cap B_2$.
Also, $(x,y) \in R_1 \setminus R_2$ iff $x \in A_1$, $y \in B_1$, and either $x \notin A_2$ or $y \notin B_2$.
Splitting by $x \in A_2$ gives the disjoint union in (i).

(ii) If $E = \bigcup_j R_j$ and $F = \bigcup_k S_k$, then $E \cap F = \bigcup_{j,k} (R_j \cap S_k) \in \mathcal{Z}_0$ by (i).
Repeatedly applying (i) gives $R_j \setminus F \in \mathcal{Z}_0$, hence $E \setminus F = \bigcup_j (R_j \setminus F) \in \mathcal{Z}_0$.
Since $Z = X \times Y \in \mathcal{Z}_0$, complements are differences from Z.

(iii) If $E = \bigcup_{j=1}^{n} R_j$, set $S_j := R_j \setminus \bigcup_{i<j} R_i$. Then $S_j \in \mathcal{Z}_0$ by (ii), the S_j are disjoint, and $E = \bigsqcup_j S_j$.
Writing each S_j as a finite union of rectangles and relabeling gives required finite disjoint representation.

Structural Commentary: Product rectangles behave like elementary sets for the product space.
Key Insight: Before extending to a σ-algebra, make all finite overlaps into disjoint rectangle pieces. □

Theorem 2.1.3. *Product Measure Theorem*
For measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$, $\exists \pi : \mathcal{M} \otimes \mathcal{N} \to [0, +\infty]$ a measure such that $\forall (A,B) \in \mathcal{M} \times \mathcal{N}$, $\pi(A \times B) = \mu(A)\nu(B)$, where $0 \cdot (+\infty) = 0$. If μ, ν are σ-finite, π is unique; write $\pi = \mu \otimes \nu$.

Proof. For $E = \bigsqcup_{j=1}^{n} (A_j \times B_j) \in \mathcal{Z}_0$, set $\pi_0(E) := \sum_{j=1}^{n} \mu(A_j)\nu(B_j)$.
If $A \times B = \bigsqcup_{j \ge 1} (A_j \times B_j)$, then $\mathbf{1}_{A\times B} = \sum_{j \ge 1} \mathbf{1}_{A_j \times B_j}$, so monotone convergence in y, then x,
gives $\mu(A)\nu(B) = \sum_{j \ge 1} \mu(A_j)\nu(B_j)$. Refining two finite disjoint rectangle decompositions by intersections gives equal π_0-values; hence

π_0 is well defined. If $E = \bigsqcup_k E_k \in \mathcal{Z}_0$ with $E_k \in \mathcal{Z}_0$, write $E = \bigsqcup_i R_i$ and $E_k = \bigsqcup_\ell R_{k\ell}$.
Since $R_i = \bigsqcup_{k,\ell}(R_i \cap R_{k\ell})$, the rectangle comparison yields $\pi_0(E) = \sum_k \pi_0(E_k)$; hence π_0 is a premeasure.
Extend π_0 to $\sigma(\mathcal{Z}_0) = \mathcal{M} \otimes \mathcal{N}$. If μ, ν are σ-finite, then $\mathcal{Z}_0$ is π_0-σ-finite, so the extension is unique.
Structural Commentary: Rectangles fix the product measure.
Key Insight: Monotone convergence makes rectangle mass respect countable disjoint refinements. □

Lemma 2.1.4. *Measurable Sections*
(i) $E \in \mathcal{M} \otimes \mathcal{N}$ implies $\forall x \in X,\ E_x \in \mathcal{N}$ and $\forall y \in Y,\ E^y \in \mathcal{M}$.
(ii) If $F : Z \to \overline{\mathbb{R}}$ is $\mathcal{M} \otimes \mathcal{N}$-measurable, then F_x is $\mathcal{N}$-measurable $\forall x \in X$ and F^y is $\mathcal{M}$-measurable $\forall y \in Y$.

Proof.
(i) Fix $x \in X$ and set $\mathcal{C}_x := \{E \subseteq Z : E_x \in \mathcal{N}\}$. Since $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$, $\mathcal{C}_x$ is a σ-algebra. Also $(A \times B)_x = B$ if $x \in A$ and $\emptyset$ otherwise; hence $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{C}_x$. The E^y proof is identical.
(ii) For $a \in \mathbb{R}$, set $H_a := \{(u, v) \in Z : F(u, v) > a\} \in \mathcal{M} \otimes \mathcal{N}$. By (i), $(H_a)_x \in \mathcal{N}$ and $(H_a)^y \in \mathcal{M}$.
Since $(H_a)_x = \{y : F_x(y) > a\}$ and $(H_a)^y = \{x : F^y(x) > a\}$, all threshold sets are measurable.
Structural Commentary: Product measurability is stable under freezing one coordinate.
Key Insight: Section operations preserve the set algebra used to generate the product σ-algebra. □

Proposition 2.1.5. *Section Measures Integrate to Product Measure*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite, $\pi := \mu \otimes \nu$, and $E \in \mathcal{M} \otimes \mathcal{N}$. Set $f_E(x) := \nu(E_x)$ and $g_E(y) := \mu(E^y)$.
Then f_E, g_E are measurable and $\int_X f_E \, d\mu = \pi(E) = \int_Y g_E \, d\nu$.

Proof. Assume $\mu(X), \nu(Y) < +\infty$ and let $\mathcal{C}$ be the class of E satisfying the claim.
For $E = A \times B$, $f_E = \nu(B)\mathbf{1}_A$ and $g_E = \mu(A)\mathbf{1}_B$, so $A \times B \in \mathcal{C}$; finite additivity gives $\mathcal{Z}_0 \subseteq \mathcal{C}$.
If $E_n \uparrow E$ in $\mathcal{C}$, then $(E_n)_x \uparrow E_x$ and $(E_n)^y \uparrow E^y$; MCT gives $E \in \mathcal{C}$.
If $E_n \downarrow E$, finite total mass gives downward continuity and DCT, hence $E \in \mathcal{C}$.
Thus $\mathcal{C}$ is a monotone class containing $\mathcal{Z}_0$, so $\mathcal{C} = \mathcal{M} \otimes \mathcal{N}$. For σ-finite μ, ν, choose $X_n \uparrow X$, $Y_n \uparrow Y$ with $\mu(X_n), \nu(Y_n) < +\infty$, apply the finite case to $E_n := E \cap (X_n \times Y_n)$, then let $n \to \infty$ by MCT.
Structural Commentary: Cavalieri's principle is already true for abstract σ-finite products.
Key Insight: Prove rectangles, close under monotone limits, then exhaust by finite-measure rectangles. □

Perspective 2.1.6. *Rectangle Premeasure and the Completed Product*
Let $\mathcal{R} := \{A \times B : A \in \mathcal{M},\ B \in \mathcal{N}\}$ and define $\lambda(A \times B) := \mu(A)\nu(B)$ on $\mathcal{R}$.
(i) Lemma 2.1.2 and Theorem 2.1.3 imply that λ is a semiring premeasure.
(ii) Theorem 3.4.7 extends λ to a complete measure $\overline{\lambda}$ on the Caratheodory σ-algebra $\mathcal{C}_\lambda$.
(iii) The restriction $\overline{\lambda}|_{\mathcal{M}\otimes\mathcal{N}}$ is $\mu \otimes \nu$; the full space $(X \times Y, \mathcal{C}_\lambda, \overline{\lambda})$ is the completed product.
Thus $\mathcal{C}_\lambda$ may contain $N' \subseteq N$ with $N \in \mathcal{M} \otimes \mathcal{N}$, $(\mu \otimes \nu)(N) = 0$, and $N' \notin \mathcal{M} \otimes \mathcal{N}$.
This is why some section-measurability forms of Fubini require completeness.
Key Insight: Rectangles set mass; completion adds null subsets.

2.2 Fubini and Tonelli Theorems

Theorem 2.2.1. *Tonelli's Theorem for σ-Finite Products*
Let $(X, \mathcal{M}, \mu)$, $(Y, \mathcal{N}, \nu)$ be σ-finite, Let $Z := X \times Y$, $\pi := \mu \otimes \nu$.
Let $F : Z \to [0, +\infty]$ be $\mathcal{M} \otimes \mathcal{N}$-measurable. Put $I_F(x) := \int_Y F_x \, d\nu$ and $J_F(y) := \int_X F^y \, d\mu$.
(i) I_F is $\mathcal{M}$-measurable.
(ii) J_F is $\mathcal{N}$-measurable.
(iii) $\int_X I_F \, d\mu = \int_Z F \, d\pi = \int_Y J_F \, d\nu$.

Proof. For $F = \mathbf{1}_E$, the claims follow from Proposition 2.1.5.
Finite linearity gives them for nonnegative simple $\Phi = \sum_{j=1}^n a_j \mathbf{1}_{E_j}$. Choose simple $0 \le \Phi_n \uparrow F$.
(i) Since I_{Φ_n} is $\mathcal{M}$-measurable and $I_{\Phi_n}(x) = \int_Y (\Phi_n)_x \, d\nu \uparrow \int_Y F_x \, d\nu = I_F(x)$, I_F is $\mathcal{M}$-measurable.
(ii) Since J_{Φ_n} is $\mathcal{N}$-measurable and $J_{\Phi_n}(y) = \int_X (\Phi_n)^y \, d\mu \uparrow \int_X F^y \, d\mu = J_F(y)$, J_F is $\mathcal{N}$-measurable.
(iii) From $\int_X I_{\Phi_n} \, d\mu = \int_Z \Phi_n \, d\pi = \int_Y J_{\Phi_n} \, d\nu$, MCT in X, Z, and Y gives $\int_X I_F \, d\mu = \int_Z F \, d\pi = \int_Y J_F \, d\nu$.
Structural Commentary: Approximate by nonnegative simple functions.
Key Insight: MCT passes sections and integrals together. □

Theorem 2.2.2. *Fubini's Theorem for σ-Finite Products*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite, $\pi := \mu \otimes \nu$, and $F \in \mathscr{L}^1(\pi)$.
(i) $F_x \in \mathscr{L}^1(\nu)$ for μ-a.e. x and $F^y \in \mathscr{L}^1(\mu)$ for ν-a.e. y.
(ii) With $f(x) := \int_Y F_x \, d\nu$ and $g(y) := \int_X F^y \, d\mu$ on those full-measure sets, and 0 elsewhere, $f \in \mathscr{L}^1(\mu)$, $g \in \mathscr{L}^1(\nu)$, and $\int_X f \, d\mu = \int_Z F \, d\pi = \int_Y g \, d\nu$.

Proof.
(i) Tonelli on $|F|$ gives $\int_X (\int_Y |F_x| \, d\nu) d\mu = \int_Z |F| \, d\pi = \int_Y (\int_X |F^y| \, d\mu) d\nu < +\infty$, hence finite a.e.
(ii) Tonelli on F^+ and F^- gives finite identities; subtract them to obtain $\int_X f \, d\mu = \int_Z F \, d\pi = \int_Y g \, d\nu$.
Moreover $|f| \le \int_Y |F_x| \, d\nu$ and $|g| \le \int_X |F^y| \, d\mu$, so (i) gives $f, g \in \mathscr{L}^1$.
Structural Commentary: Fubini is Tonelli plus subtraction.
Key Insight: $\int |F| < +\infty$ makes the two nonnegative Tonelli identities finite enough to subtract. □

Proposition 2.2.3. *Product Borel and Measurable Building Blocks*
The following hold.

(i) $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$.
(ii) *If* $f : X \to \mathbb{R}$ *is* $\mathcal{M}$*-measurable and* $g : Y \to \mathbb{R}$ *is* $\mathcal{N}$*-measurable, then* $(x, y) \mapsto f(x)g(y)$ *is* $\mathcal{M} \otimes \mathcal{N}$*-measurable.*
(iii) *If* $u : \mathbb{R} \to \mathbb{R}$ *is Borel measurable, then* $(x, y) \mapsto u(x - y)$ *is* $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$*-measurable.*

Proof.

(i) Open rectangles give $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$.
Conversely, every open $O \subseteq \mathbb{R}^2$ is a countable union of rational rectangles, each in $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$.
(ii) For $\tilde{f}(x, y) := f(x)$ and $\tilde{g}(x, y) := g(y)$, $\{\tilde{f} > a\} = f^{-1}((a, +\infty)) \times Y$ and $\{\tilde{g} > a\} = X \times g^{-1}((a, +\infty))$.
Thus $\tilde{f}, \tilde{g}$ are measurable, so $\tilde{f}\tilde{g}$ is measurable.
(iii) The map $s(x, y) := x - y$ is continuous on $\mathbb{R}^2$, hence measurable by (i); therefore $u \circ s$ is measurable.

Structural Commentary: Product measurability supports two-variable formulas.
Key Insight: Check coordinate pullbacks on rectangles, then use ordinary measurable-function closure. □

Proposition 2.2.4. *Counting Factor Products*
Let $X = \mathbb{N}$, $\mathcal{M} = 2^{\mathbb{N}}$, μ *be counting measure, and* $(Y, \mathcal{N}, \nu)$ *be a measure space.*

(i) *For* $E \subseteq \mathbb{N} \times Y$, $E \in 2^{\mathbb{N}} \otimes \mathcal{N} \iff \forall n,\ E_n \in \mathcal{N}$, *and then* $E = \bigsqcup_{n \geq 1} (\{n\} \times E_n)$.
(ii) *The unique product measure is* $\pi = \mu \otimes \nu$ *with* $\pi(E) = \sum_{n \geq 1} \nu(E_n)$ *for* $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$.
(iii) *For* $F : \mathbb{N} \times Y \to \mathbb{R}$, F *is product-measurable* $\iff \forall n,\ F_n$ *is* $\mathcal{N}$*-measurable.*
(iv) *For product-measurable* F, $F \in \mathscr{L}^1(\pi) \iff \sum_{n \geq 1} \int_Y |F_n|\, d\nu < +\infty$;
then $\int_{\mathbb{N} \times Y} F\, d\pi = \sum_{n \geq 1} \int_Y F_n\, d\nu = \int_Y \sum_{n \geq 1} F_n\, d\nu$, *with* $\sum_n F_n$ *absolutely convergent* ν*-a.e.*

Proof.

(i) ($\Rightarrow$) If $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$, then $\forall n,\ E_n \in \mathcal{N}$ by Lemma 2.1.4.
(i) ($\Leftarrow$) If $\forall n,\ E_n \in \mathcal{N}$, then each $\{n\} \times E_n \in 2^{\mathbb{N}} \otimes \mathcal{N}$ and $E = \bigsqcup_{n \geq 1} (\{n\} \times E_n)$, hence $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$.
(ii) Define $\pi(E) := \sum_{n \geq 1} \nu(E_n)$.
If $(E^k)_k$ are disjoint, then $(\bigsqcup_k E^k)_n = \bigsqcup_k E^k_n$, so $\pi(\bigsqcup_k E^k) = \sum_n \sum_k \nu(E^k_n) = \sum_k \pi(E^k)$.
Also $\pi(\{n\} \times A) = \nu(A)$, and (i) forces uniqueness.
(iii) ($\Rightarrow$) If F is product-measurable, then $\forall n$, F_n is measurable by Lemma 2.1.4.
(iii) ($\Leftarrow$) If $\forall n$, F_n is measurable, then $\forall a \in \mathbb{R}$, $H_a := \{F > a\}$ satisfies $(H_a)_n = \{F_n > a\} \in \mathcal{N}$,
By (i), $H_a \in 2^{\mathbb{N}} \otimes \mathcal{N}$, so F is product-measurable.
(iv) For $G \geq 0$, (ii) gives the identity for $G = \mathbf{1}_E$; linearity and MCT give $\int_{\mathbb{N} \times Y} G\, d\pi = \sum_{n \geq 1} \int_Y G_n\, d\nu$.
(iv) Apply the preceding identity to $G = |F|$: $F \in \mathscr{L}^1(\pi) \iff \sum_{n \geq 1} \int_Y |F_n|\, d\nu < +\infty$.
If finite, then $\int_Y \sum_n |F_n|\, d\nu < +\infty$, so $\sum_n F_n$ converges absolutely a.e..
Applying the nonnegative identity to F^+ and F^- and subtracting gives the integral formulas.

Structural Commentary: A product with counting measure is a measurable sequence of fibers.
Key Insight: Counting atoms turn products into sums. □

Example 2.2.5. *A Diagonal Failure Without* σ*-Finiteness*
Let $X = Y = [0, 1]$, $\mathcal{M} = \mathcal{N} = \mathcal{B}([0, 1])$, $\mu = m$, $\nu = \#$, and $D := \{(x, y) \in [0, 1]^2 : x = y\}$.
Then $D \in \mathcal{M} \otimes \mathcal{N}$, $D_x = \{x\}$, and $D^y = \{y\}$, so $\nu(D_x) = 1$ and $\mu(D^y) = 0$.
Hence $\int_X \nu(D_x)\, d\mu = 1$ but $\int_Y \mu(D^y)\, d\nu = 0$, so Tonelli–Fubini can fail without σ-finite factors.

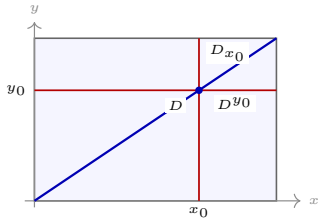

Figure 204: Diagonal sections: $\nu(D_x) = 1$ but $\mu(D^y) = 0$.

Example 2.2.6. *Nonunique Products Without* σ*-Finiteness*
Let $X = \mathbb{R}$, $\mathcal{M} = 2^{\mathbb{R}}$, and $\mu(A) = 0$ if A is countable, $\mu(A) = +\infty$ otherwise.
Define $\lambda_i : \mathcal{M} \otimes \mathcal{M} \to \{0, +\infty\}$ by $\lambda_i(E) = 0$ exactly on the following null ideals:
$\mathcal{I}_1 = \{E : E = G \cup H,\ \mathrm{pr}_x\, G$ countable, $\mathrm{pr}_y\, H$ countable$\}$
and $\mathcal{I}_2 = \{E : E = G \cup H \cup K,\ \mathrm{pr}_x\, G$ countable, $\mathrm{pr}_y\, H$ countable, $\mathrm{pr}_{y=x}\, K$ countable$\}$.
Both ideals are closed under countable unions, hence λ_i are measures.
For a rectangle $A \times B$, both measures equal $\mu(A)\mu(B)$: countable sides are null, and uncountable-by-uncountable rectangles are not in either ideal.
For $E = \{(x, y) : x + y = 0\}$, $\mathrm{pr}_{y=x}\, E$ is a point, so $\lambda_2(E) = 0$ but $\lambda_1(E) = +\infty$.

Structural Commentary: Without σ-finiteness, rectangle values need not determine a product measure.
Key Insight: Extra null lines preserve rectangles but change non-rectangles.

Example 2.2.7. *Double Series as Product Integrals*
With counting measure on $\mathbb{N}^2$, Tonelli gives, for $a_{mn} \geq 0$,
$\sum_m \sum_n a_{mn} = \sum_{(m,n)\in\mathbb{N}^2} a_{mn} = \sum_n \sum_m a_{mn} \in [0, +\infty]$.
Fubini needs integrability: set $a_{nn} = 1$, $a_{n,n+1} = -1$, and $a_{mn} = 0$ otherwise.
Then $\sum_m \sum_n a_{mn} = 0$, while $\sum_n \sum_m a_{mn} = 1$, and $\sum_{m,n} |a_{mn}| = +\infty$.

Proposition 2.2.8. *Separated Products*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite. Let $f \in \mathscr{L}^1(\mu)$, $g \in \mathscr{L}^1(\nu)$, and $H(x,y) := f(x)g(y)$.
(i) $H \in \mathscr{L}^1(\mu \otimes \nu)$.
(ii) $\int_{X\times Y} H\, d(\mu\otimes\nu) = (\int_X f\, d\mu)(\int_Y g\, d\nu)$.

Proof.
(i) Proposition 2.2.3 gives measurability. Tonelli gives $\int_{X\times Y} |H|\, d(\mu\otimes\nu) = (\int_X |f|\, d\mu)(\int_Y |g|\, d\nu) < +\infty$.
(ii) Since $H \in \mathscr{L}^1(\mu\otimes\nu)$, Fubini gives $\int_{X\times Y} H\, d(\mu\otimes\nu) = (\int_X f\, d\mu)(\int_Y g\, d\nu)$.
Structural Commentary: Tonelli handles size; Fubini handles sign.
Key Insight: Separated products integrate as products of integrals. □

Corollary 2.2.9. *Equal Section Measures*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite.
If $E, F \in \mathcal{M}\otimes\mathcal{N}$ and $\forall x \in X$, $\nu(E_x) = \nu(F_x)$, then $(\mu\otimes\nu)(E) = (\mu\otimes\nu)(F)$.

Proof. Set $f_E(x) := \nu(E_x)$ and $f_F(x) := \nu(F_x)$.
Since $f_E = f_F$, $(\mu\otimes\nu)(E) = \int_X f_E\, d\mu = \int_X f_F\, d\mu = (\mu\otimes\nu)(F)$ by Proposition 2.1.5.
Structural Commentary: Product mass is recovered from vertical section masses.
Key Insight: Equal fibers give equal product mass. □

Definition 2.2.10. *Convolution*
For Borel $f, g : \mathbb{R} \to \mathbb{R}$, if $y \mapsto f(x-y)g(y)$ is integrable for a.e. x,
define the *convolution* $(f * g)(x) := \int_{\mathbb{R}} f(x-y)g(y)\, dm(y)$ on that full-measure set and 0 elsewhere.

Proposition 2.2.11. *Convolution of L^1 Functions*
Let $f, g \in \mathscr{L}^1(m)$ be Borel.
(i) $f * g$ is defined for a.e. $x \in \mathbb{R}$.
(ii) $f * g \in \mathscr{L}^1(m)$.
(iii) $\int_{\mathbb{R}} |f * g|\, dm \leq (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm)$.

Proof. Set $\Phi(x,y) := f(x-y)g(y)$. By Proposition 2.2.3, Φ is product-measurable.
For fixed y, translation invariance gives $\int_{\mathbb{R}} |f(x-y)|\, dm(x) = \int_{\mathbb{R}} |f|\, dm$.
Hence Tonelli yields $\int_{\mathbb{R}^2} |\Phi|\, d(m\otimes m) = (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm) < +\infty$.
(i) Fubini gives $y \mapsto \Phi(x,y) \in \mathscr{L}^1(m)$ for a.e. x, so $f * g$ is a.e. defined.
(ii) Since $|f * g(x)| \leq \int_{\mathbb{R}} |\Phi(x,y)|\, dm(y)$ a.e., Fubini gives $\int_{\mathbb{R}} |f * g|\, dm < +\infty$.
(iii) The same estimate gives $\int_{\mathbb{R}} |f * g|\, dm \leq \int_{\mathbb{R}^2} |\Phi|\, d(m\otimes m) = (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm)$.
Structural Commentary: Apply Fubini to $\Phi(x,y) = f(x-y)g(y)$.
Key Insight: Translation invariance fixes the x-slice norm. □

Definition 2.2.12. *Products and Slices*
For $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^k$, set $A \times B \subseteq \mathbb{R}^{n+k}$.
For $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ define $E_x := \{y \in \mathbb{R}^k : (x,y) \in E\}$ as *x-slice*, and $E^y := \{x \in \mathbb{R}^n : (x,y) \in E\}$ as *y-slice*.
For $f : E \to \mathbb{R}$, write $f_x(y) = f^y(x) := f(x,y)$ where defined.

Theorem 2.2.13. *Measurable Products*
Let $A \in \mathcal{L}^n$ and $B \in \mathcal{L}^k$.
(i) $A \times B \in \mathcal{L}^{n+k}$.
(ii) $m_{n+k}(A\times B) = m_n(A)m_k(B)$, with $0 \cdot \infty = \infty \cdot 0 = 0$.

Proof. Boxes satisfy $m_{n+k}(I\times J) = m_n(I)m_k(J)$, hence so do open sets by disjoint box decompositions.
For bounded finite-measure A, B, choose $F_A \subseteq A \subseteq G_A$, $F_B \subseteq B \subseteq G_B$ with F_A, F_B closed, G_A, G_B bounded open, and $m_n(G_A \setminus F_A), m_k(G_B \setminus F_B) < \varepsilon$.
Then $F_A \times F_B \subseteq A\times B \subseteq G_A \times G_B$ and $m_n(G_A)m_k(G_B) - m_n(F_A)m_k(F_B) \to 0$.
(i) The squeeze proves $A\times B \in \mathcal{L}^{n+k}$ in the bounded finite-measure case.
For general A, B, with $A_N := A \cap [-N,N]^n$ and $B_N := B \cap [-N,N]^k$, $A\times B = \bigcup_N (A_N \times B_N) \in \mathcal{L}^{n+k}$.
(ii) The squeeze also gives $m_{n+k}(A\times B) = m_n(A)m_k(B)$ in the bounded finite-measure case.
Since $A_N \times B_N \uparrow A \times B$, $m_{n+k}(A\times B) = \lim_N m_n(A_N)m_k(B_N) = m_n(A)m_k(B)$, with $0\cdot\infty = \infty\cdot 0 = 0$.
Structural Commentary: Product measure comes from boxes and regularity.
Key Insight: Regularity turns rough measurable products into box products plus zero-set errors. □

Theorem 2.2.14. *Zero Slices*
If $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ is Lebesgue measurable, then E is null iff E_x is null for a.e. x.

Proof.
($\Rightarrow$) If $m(E) = 0$, cover E by an open W with small measure.
For $X_\alpha(W) = \{x : m(W_x) > \alpha\}$, $m(W) \geq \alpha m(X_\alpha(W))$; letting $m(W) \to 0$ gives $m(E_x) = 0$ for a.e. x.

($\Leftarrow$) Remove the exceptional null base so every slice is null. For compact $K \subseteq E$, cover each K_x by a small-measure slice neighborhood; compactness gives finitely many product neighborhoods covering K, whose total measure is arbitrarily small by Theorem 2.2.13. Thus $m(K) = 0$, hence $m(E) = 0$ by regularity.

Structural Commentary: Nullity can be checked fiberwise after removing an exceptional base set.
Key Insight: Product measure turns small slice covers into small total measure. □

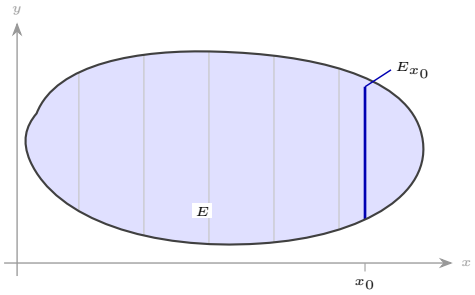

Figure 205: A planar set can be read by vertical slices; Fubini integrates the slice data over the base variable.

Theorem 2.2.15. *Cavalieri's Principle for Lebesgue Measure*
Let $n, k \in \mathbb{N}$ and $E \in \mathcal{L}^{n+k}$. For $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^k$, set $E_x := \{y : (x, y) \in E\}$ and $E^y := \{x : (x, y) \in E\}$.

(i) $E_x \in \mathcal{L}^k$ for a.e. x, and $f_E(x) := m_k(E_x)$ has a measurable representative with $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$.
(ii) $E^y \in \mathcal{L}^n$ for a.e. y, and $g_E(y) := m_n(E^y)$ has a measurable representative with $m_{n+k}(E) = \int_{\mathbb{R}^k} g_E \, dm_k$.

Proof. Let $\mathsf{C}_x(E)$ denote: $E_x \in \mathcal{L}^k$ for a.e. x, f_E has a measurable representative, and $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$.

(i) Rectangles satisfy $\mathsf{C}_x(A \times B)$ since $f_{A\times B} = m_k(B)\mathbf{1}_A$; finite additivity gives finite unions.
For bounded $Q = I \times J$, $\mathcal{C}_Q := \{E \subseteq Q : \mathsf{C}_x(E)\}$ is monotone:
$E_j \uparrow E$ gives $f_{E_j} \uparrow f_E$ and MCT, while $E_j \downarrow E$ gives $f_{E_j} \downarrow f_E \le m_k(J)\mathbf{1}_I$ and DCT.
Thus $\mathcal{C}_Q$ contains all Borel $E \subseteq Q$, and exhaustion by $Q_N = [-N, N]^{n+k}$ gives all Borel E.
For $E \in \mathcal{L}^{n+k}$, choose Borel B with $m_{n+k}(E \triangle B) = 0$; then $m_k((E \triangle B)_x) = 0$ a.e. by Theorem 2.2.14.
Hence $E_x \in \mathcal{L}^k$ a.e., $f_E = f_B$ a.e., and $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$.

(ii) Let $\tau : \mathbb{R}^k \times \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^k$ be $\tau(y, x) = (x, y)$ and put $\widetilde{E} := \tau^{-1}(E)$.
Apply (i) to $\widetilde{E} \subseteq \mathbb{R}^k \times \mathbb{R}^n$ gives $\widetilde{E}_y = E^y$ a.e., $g_E(y) = m_n(\widetilde{E}_y)$, $m_{n+k}(E) = m_{n+k}(\widetilde{E}) = \int_{\mathbb{R}^k} g_E \, dm_k$.

Structural Commentary: Slices reduce product measure to base integrals.
Key Insight: Extend from rectangles, then pass null differences. □

Perspective 2.2.16. *Fubini Proof Ladder*
Fubini ladder: $\mathbf{1}_E \to$ simple $\Phi \ge 0 \to f \ge 0$ by MCT, then $f = f^+ - f^-$ when $\int |f| < +\infty$.
Theorems 2.2.1 and 2.2.2 are the abstract form.
Key Insight: Fubini is a slice theorem, proof runs through indicators, simple functions, monotone convergence.

Proposition 2.2.17. *Undergraphs and Level-Set Measurability*
Let $f : \mathbb{R}^n \to [0, \infty]$, set $F_a := \{x : f(x) > a\}$, and let $\mathcal{U}f := \{(x, t) \in \mathbb{R}^n \times [0, \infty) : 0 < t < f(x)\}$.

(i) f is Lebesgue measurable iff $\forall q \in \mathbb{Q}_{>0}$, $F_q \in \mathcal{L}^n$.
(ii) If f is Lebesgue measurable, then $\mathcal{U}f \in \mathcal{L}^{n+1}$ and $\forall t > 0$, $(\mathcal{U}f)^t = F_t$.

Proof.

(i) ($\Rightarrow$) If f is measurable, each $F_q = f^{-1}((q, +\infty])$ is in $\mathcal{L}^n$.

(i) ($\Leftarrow$) If $\forall q \in \mathbb{Q}_{>0}$, $F_q \in \mathcal{L}^n$, then for $a \ge 0$, $F_a = \bigcup_{q\in\mathbb{Q},\ q>a} F_q \in \mathcal{L}^n$, while for $a < 0$, $F_a = \mathbb{R}^n$.
Thus all strict superlevel sets are measurable, so f is measurable.

(ii) By (i), every F_r with $r \in \mathbb{Q}_{>0}$ is measurable. Since $\mathcal{U}f = \bigcup_{q,r\in\mathbb{Q},\ 0<q<r} F_r \times (q, r)$, $\mathcal{U}f \in \mathcal{L}^{n+1}$.
For $t > 0$, $(\mathcal{U}f)^t = \{x : 0 < t < f(x)\} = F_t$.

Structural Commentary: Measurability is encoded by rational superlevels.
Key Insight: Undergraph slices are superlevel sets. □

Theorem 2.2.18. *Fubini–Tonelli for Nonnegative Functions*
Let $f : \mathbb{R}^n \times \mathbb{R}^k \to [0, \infty]$ be Lebesgue measurable.
Define $F(x) := \int_{\mathbb{R}^k} f_x \, dm_k$, f_x is measurable and arbitrarily elsewhere; define $G(y) := \int_{\mathbb{R}^n} f^y \, dm_n$ analogously.

(i) f_x is measurable for a.e. x, F is measurable, and $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^n} F \, dm_n$.
(ii) f^y is measurable for a.e. y, G is measurable, and $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^k} G \, dm_k$.

Proof. Set $U := \mathcal{U}f \subseteq \mathbb{R}^{n+k} \times [0, \infty)$. Then $U \in \mathcal{L}^{n+k+1}$ by Proposition 2.2.17.

(i) By Cavalieri, U_x is measurable for a.e. x, $H(x) := m_{k+1}(U_x)$ is measurable, and $m_{n+k+1}(U) = \int_{\mathbb{R}^n} H \, dm_n$.
For such x, $U_x = \mathcal{U}(f_x)$, so f_x is measurable and $H(x) = \int_{\mathbb{R}^k} f_x \, dm_k = F(x)$.
Since $m_{n+k+1}(U) = \int_{\mathbb{R}^{n+k}} f \, dm_{n+k}$, (i) follows.

(ii) Let $\tau(y, x) = (x, y)$ and $\tilde{f} := f \circ \tau$.
Applying (i) to $\tilde{f}$ gives $\tilde{f}_y = f^y$ a.e., $\tilde{F} = G$, and $\int_{\mathbb{R}^k} G \, dm_k = \int_{\mathbb{R}^{k+n}} \tilde{f} \, dm_{k+n} = \int_{\mathbb{R}^{n+k}} f \, dm_{n+k}$.

Structural Commentary: Cavalieri applied to the undergraph.
Key Insight: $(\mathcal{U}f)_x = \mathcal{U}(f_x)$. □

Theorem 2.2.19. *Fubini for L^1 Functions*
Let $f \in L^1(\mathbb{R}^2)$.

(i) $f(x,\cdot) \in L^1(\mathbb{R})$ for a.e. x; defining $F(x) := \int_{\mathbb{R}} f(x,y)\,dy$ there and $F = 0$ elsewhere gives $F \in L^1(\mathbb{R})$.
(ii) $f(\cdot,y) \in L^1(\mathbb{R})$ for a.e. y; defining $G(y) := \int_{\mathbb{R}} f(x,y)\,dx$ there and $G = 0$ elsewhere gives $G \in L^1(\mathbb{R})$.
(iii) $\int_{\mathbb{R}} F\,dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G\,dy$.

Proof. From Theorems 2.2.13–2.2.14, the identity holds for indicators.
By linearity and MCT, every measurable $h \geq 0$ satisfies $(*)$ $\int_{\mathbb{R}}(\int_{\mathbb{R}} h(x,y)\,dy)dx = \int_{\mathbb{R}^2} h = \int_{\mathbb{R}}(\int_{\mathbb{R}} h(x,y)\,dx)dy$.

(i) Applying $(*)$ to $|f|$ gives $\int_{\mathbb{R}}\int_{\mathbb{R}} |f(x,y)|\,dy\,dx = \int_{\mathbb{R}^2} |f| < +\infty$, so $f(x,\cdot) \in L^1(\mathbb{R})$ a.e.
Let $F^{\pm}(x) := \int_{\mathbb{R}} f^{\pm}(x,y)\,dy$; then $F = F^+ - F^-$ a.e. and $\int_{\mathbb{R}} |F|\,dx \leq \int_{\mathbb{R}}(F^+ + F^-)\,dx = \int_{\mathbb{R}^2} |f| < +\infty$.
(ii) Let $\tau(y,x) = (x,y)$ and $\tilde{f} := f \circ \tau$. Since τ preserves m_2, $\tilde{f} \in L^1(\mathbb{R}^2)$.
Applying (i) to $\tilde{f}$ gives $\tilde{f}(y,\cdot) = f(\cdot,y) \in L^1(\mathbb{R})$ a.e. and $\tilde{F} = G \in L^1(\mathbb{R})$.
(iii) Let $G^{\pm}(y) := \int_{\mathbb{R}} f^{\pm}(x,y)\,dx$. By $(*)$, $\int_{\mathbb{R}} F^{\pm}\,dx = \int_{\mathbb{R}^2} f^{\pm} = \int_{\mathbb{R}} G^{\pm}\,dy$. Since these quantities are finite and $F = F^+ - F^-$, $G = G^+ - G^-$ a.e., subtraction gives $\int_{\mathbb{R}} F\,dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G\,dy$.

Structural Commentary: Indicators, simple functions, MCT, then $f^{\pm}$.
Key Insight: Absolute integrability permits subtraction. □

2.3 Linear Changes of Variables and Euclidean Invariance

Perspective 2.3.1. *Coordinate Splitting as a Product Identification*
For $n = m + k$, let $\Phi : \mathbb{R}^n \to \mathbb{R}^m \times \mathbb{R}^k$ be $\Phi(x_1,\ldots,x_n) = ((x_1,\ldots,x_m),(x_{m+1},\ldots,x_n))$. Then Φ sends rectangles to product rectangles and preserves rectangular volume, so m_n is the completion of $\Phi_{\#}^{-1}(m_m \otimes m_k)$. Equivalently, integration over $\mathbb{R}^{m+k}$ matches iterated integration over $\mathbb{R}^m \times \mathbb{R}^k$ whenever Fubini applies.
Key Insight: Coordinate splitting preserves Lebesgue mass.

Proposition 2.3.2. *Lipschitz Images of Lebesgue Sets*
Let $\Psi : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz and let $E \in \mathcal{L}^n$. Then $\Psi(E) \in \mathcal{L}^n$.
Hence every linear map $\mathbb{R}^n \to \mathbb{R}^n$ preserves Lebesgue measurability.

Proof. Let L be a Lipschitz constant.

(i) $\forall I \subseteq \mathbb{R}^n$ bounded cells, $m_n^*(\Psi(I)) \leq C_{n,L}\,\mathrm{vol}_n(I)$. Indeed, partition I into grid cells of side at most δ.
Each image has diameter at most $L\sqrt{n}\,\delta$, hence lies in a cube of volume at most $C_{n,L}\delta^n$. Summing and letting $\delta \downarrow 0$ gives the estimate.
(ii) If $m_n^*(N) = 0$ and $\varepsilon > 0$, cover N by cells (I_k) with $\sum_k \mathrm{vol}_n(I_k) < \varepsilon$.
Then $\Psi(N) \subseteq \bigcup_k \Psi(I_k)$, so (i) gives $m_n^*(\Psi(N)) \leq C_{n,L}\varepsilon$; hence $m_n^*(\Psi(N)) = 0$.
(iii) If E is bounded, choose F_σ set $F \subseteq E$ with $m_n^*(E \setminus F) = 0$. Write $F = \bigcup_j F_j$ with each F_j compact.
Then $\Psi(F) = \bigcup_j \Psi(F_j)$ is Borel, $\Psi(E \setminus F)$ is null by (ii), and $\Psi(E) = \Psi(F) \cup \Psi(E \setminus F) \in \mathcal{L}^n$.
(iv) For general E, use $E = \bigcup_{r \geq 1} (E \cap [-r,r]^n)$ and apply (iii) to each bounded term.

Structural Commentary: Reduce to compact pieces plus null error.
Key Insight: Lipschitz maps preserve null sets. □

Theorem 2.3.3. *Linear Change of Variables for Nonnegative Functions*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ be invertible and let $f : \mathbb{R}^n \to [0,+\infty]$ be Lebesgue measurable.

(i) $f \circ T$ is Lebesgue measurable.
(ii) $\int f \circ T\,dm_n = |\det T|^{-1} \int f\,dm_n$.

Proof.

(i) Since $(f \circ T)^{-1}((a,+\infty]) = T^{-1}(f^{-1}((a,+\infty]))$ and T^{-1} is Lipschitz, Proposition 2.3.2 gives measurability.
(ii) The $n = 1$ identity follows from $m_1(\alpha E + \beta) = |\alpha|\,m_1(E)$.
Let $\mathcal{M}_+$ denote the measurable maps $\mathbb{R}^n \to [0,+\infty]$, set $J_S(h) := \int h(Sx)\,dm_n(x)$.
Set $\mathcal{A}_n := \{S \in GL(n) : J_S(h) = |\det S|^{-1} \int h\,dm_n\ \forall h \in \mathcal{M}_+\}$.
If $S_1, S_2 \in \mathcal{A}_n$, then $J_{S_1S_2}(h) = |\det S_2|^{-1} J_{S_1}(h) = |\det(S_1S_2)|^{-1} \int h\,dm_n$, so $S_1S_2 \in \mathcal{A}_n$.
The generators $D_j(c) : x_j \mapsto cx_j$, $P_{j,j+1} : x_j \leftrightarrow x_{j+1}$, and $H_{jk}(c) : x_j \mapsto x_j + cx_k$ lie in $\mathcal{A}_n$ for $c \neq 0$ by the $n = 1$ identity, Fubini, and slice translation invariance, with determinant factors $|c|$, 1, and 1.
Row reduction gives $T \in \mathcal{A}_n$; hence $J_T(f) = |\det T|^{-1} \int f\,dm_n$.

Structural Commentary: Verify elementary row operations, then compose.
Key Insight: Only scalings change mass. □

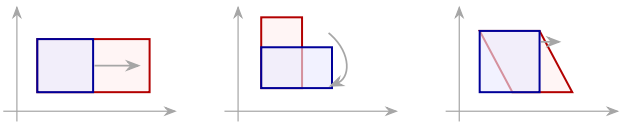

Figure 206: Original and image cells under scale, swap, and shear; Fubini checks each coordinatewise.

Corollary 2.3.4. *Invertible Linear Images of Lebesgue Sets*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ be invertible and let $E \subseteq \mathbb{R}^n$.
(i) $E \in \mathcal{L}^n \iff T(E) \in \mathcal{L}^n$.
(ii) If $E \in \mathcal{L}^n$, then $m_n(T(E)) = |\det T|\, m_n(E)$.

Proof.
(i) Since T and T^{-1} are Lipschitz, Proposition 2.3.2 gives $E \in \mathcal{L}^n \iff T(E) \in \mathcal{L}^n$.
(ii) For $E \in \mathcal{L}^n$, Theorem 2.3.3 applied to $\mathbf{1}_{T(E)}$ gives $m_n(E) = \int \mathbf{1}_{T(E)}(Tx)\, dm_n(x) = |\det T|^{-1}\, m_n(T(E))$.
Structural Commentary: Measurability is Lipschitz; mass follows from indicators.
Key Insight: Indicators turn integral change into set change. □

Corollary 2.3.5. *Linear Change of Variables for L^1 Functions*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ be invertible, let $f \in \mathscr{L}^1(m_n)$, and set $c_T := |\det T|^{-1}$.
(i) $f \circ T \in \mathscr{L}^1(m_n)$.
(ii) $\int f \circ T\, dm_n = c_T \int f\, dm_n$.

Proof.
(i) Apply Theorem 2.3.3 to $|f|$: $\int |f \circ T|\, dm_n = c_T \int |f|\, dm_n < +\infty$, hence $f \circ T \in \mathscr{L}^1(m_n)$.
(ii) Apply Theorem 2.3.3 to $f^{\pm}$: $\int f^{\pm} \circ T\, dm_n = c_T \int f^{\pm}\, dm_n < +\infty$.
Hence $\int f \circ T\, dm_n = \int f^+ \circ T\, dm_n - \int f^- \circ T\, dm_n = c_T \int f\, dm_n$.
Structural Commentary: Reduce signed integrals to finite positive parts.
Key Insight: Absolute integrability permits subtraction. □

Corollary 2.3.6. *Linear Images and Rigid Motions*
Let $E \in \mathcal{L}^n$.
(i) If $T \in \mathcal{L}(\mathbb{R}^n)$ is singular, then $T(E)$ is Lebesgue measurable and $m_n(T(E)) = 0$.
(ii) If $\Psi(x) = Ux + x_0$, where U is a linear isometry, then $\Psi(E) \in \mathcal{L}^n$ and $m_n(\Psi(E)) = m_n(E)$.

Proof.
(i) $T(E) \in \mathcal{L}^n$ by Proposition 2.3.2. Set $W := \operatorname{range} T$, $H := \mathbb{R}^{n-1} \times \{0\}$; choose invertible S with $S(W) \subseteq H$.
Since $H = \bigcup_N (H \cap [-N, N]^n)$ is null, $m_n(W) = 0$ by Corollary 2.3.4; hence $m_n(T(E)) = 0$.
(ii) Since U is an isometry, $|\det U| = 1$ by Proposition 1.1.19.
Thus $U(E) \in \mathcal{L}^n$ and $m_n(U(E)) = m_n(E)$ by Corollary 2.3.4.
By translation invariance, $\Psi(E) = U(E) + x_0 \in \mathcal{L}^n$ and $m_n(\Psi(E)) = m_n(U(E)) = m_n(E)$.
Structural Commentary: Singular maps land in null subspaces; rigid motions preserve determinant mass.
Key Insight: Lebesgue measure records determinant size, not the coordinate description of the set. □

3 Radon-Nikodym and Differentiation

3.1 Signed and Absolutely Continuous Measures

This subsection first decomposes finite signed set functions into measures, then compares measures by absolute continuity and singularity. Later subsections develop the local Euclidean counterpart of the Radon–Nikodym viewpoint.

Definition 3.1.1. *Positive, Negative, and Null Sets for a Charge*
Let λ be a charge on $(X, \mathcal{M})$.
(i) $\mathcal{P}_\lambda := \{P \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap P) \geq 0\}$, the *positive* sets.
(ii) $\mathcal{N}_\lambda := \{N \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap N) \leq 0\}$, the *negative* sets.
(iii) $\mathcal{Z}_\lambda := \{M \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap M) = 0\}$, the *null* sets.

Proposition 3.1.2. *Positive and Negative Set Closure*
For a charge λ:
(i) $A \in \mathcal{M}$ and $A \subseteq P \in \mathcal{P}_\lambda$ imply $A \in \mathcal{P}_\lambda$.
(ii) $A \in \mathcal{M}$ and $A \subseteq N \in \mathcal{N}_\lambda$ imply $A \in \mathcal{N}_\lambda$.
(iii) $\mathcal{P}_\lambda$ is closed under countable unions.
(iv) $\mathcal{N}_\lambda$ is closed under countable unions.

Proof.
(i) If $A \subseteq P \in \mathcal{P}_\lambda$, then $\lambda(E \cap A) = \lambda((E \cap A) \cap P) \geq 0\ \forall E \in \mathcal{M}$.
(ii) If $A \subseteq N \in \mathcal{N}_\lambda$, then $\lambda(E \cap A) = \lambda((E \cap A) \cap N) \leq 0\ \forall E \in \mathcal{M}$.
(iii) Let $(P_k) \subseteq \mathcal{P}_\lambda$ and set $P := \bigcup_{k \geq 1} P_k$. For $E \in \mathcal{M}$, set $F_1 := E \cap P_1$ and $F_k = E \cap P_k \setminus \bigcup_{j=1}^{k-1} P_j$ for $k \geq 2$.
Then (F_k) is pairwise disjoint, $\bigcup_k F_k = E \cap P$, and $F_k \subseteq P_k$, so $\lambda(F_k) \geq 0$.
Countable additivity gives $\lambda(E \cap P) = \sum_k \lambda(F_k) \geq 0$.
(iv) Apply the disjointization in (iii) to $(N_k) \subseteq \mathcal{N}_\lambda$; every disjoint piece has nonpositive charge, so $\bigcup_k N_k \in \mathcal{N}_\lambda$.
Structural Commentary: Sign sets are hereditary and countable-union stable.
Key Insight: Test sets can always be disjointized against a countable sign cover. □

Theorem 3.1.3. *Hahn Decomposition Theorem*
Let λ be a charge on $(X, \mathcal{M})$. Then $\exists P, N \in \mathcal{M}$ such that
(i) $P \cap N = \emptyset$ and $X = P \sqcup N$.

(ii) $P \in \mathcal{P}_\lambda$.
(iii) $N \in \mathcal{N}_\lambda$.
Such a pair (P, N) *is a Hahn decomposition of* X *for* λ.

Proof.

(i) Put $\alpha = \sup\{\lambda(A) : A \in \mathcal{P}_\lambda\}$. If $\alpha = +\infty$, choose $B_m \in \mathcal{P}_\lambda$, $\lambda(B_m) > m$, and set $C_m = \bigcup_{j=1}^{m} B_j$; then $C_m \in \mathcal{P}_\lambda$, $C_m \uparrow C$, and $\lambda(C_m) \geq \lambda(B_m) > m$, contradicting continuity from below and $\lambda(C) \in \mathbb{R}$.
Choose $A_n \in \mathcal{P}_\lambda$, $\lambda(A_n) \to \alpha$, replace A_n by $\bigcup_{j=1}^{n} A_j$, and set $P = \bigcup_{n\geq 1} A_n$, $N = X \setminus P$.
Hence $P, N \in \mathcal{M}$, $P \cap N = \emptyset$, and $X = P \sqcup N$.
(ii) For $E \in \mathcal{M}$, $E \cap A_n \uparrow E \cap P$ and $\lambda(E \cap A_n) \geq 0$, so continuity from below gives $\lambda(E \cap P) \geq 0$.
Thus $P \in \mathcal{P}_\lambda$, and $\lambda(P) = \alpha$.
(iii) Suppose $E \in \mathcal{M}$, $E \subseteq N$, and $\lambda(E) > 0$. Then $E \notin \mathcal{P}_\lambda$, otherwise $P \cup E \in \mathcal{P}_\lambda$ and $\lambda(P \cup E) > \alpha$.
Set $R_0 := E$. Inductively, if $R_{k-1} \notin \mathcal{P}_\lambda$, let $n_k := \min\{n \geq 1 : \exists H \in \mathcal{M},\ H \subseteq R_{k-1},\ \lambda(H) \leq -\frac{1}{n}\}$.
Choose such E_k, and set $R_k := R_{k-1} \setminus E_k$. If some $R_m \in \mathcal{P}_\lambda$, then $\lambda(R_m) = \lambda(E) - \sum_{k\leq m} \lambda(E_k) > 0$, so $P \cup R_m \in \mathcal{P}_\lambda$ and $\lambda(P \cup R_m) > \alpha$, impossible. Hence all $R_k \notin \mathcal{P}_\lambda$. Put $F_m := \bigcup_{k\leq m} E_k$ and $F := \bigcup_{k\geq 1} E_k$.
Since $\lambda(F_m) = \sum_{k\leq m} \lambda(E_k) \leq -\sum_{k\leq m} \frac{1}{n_k}$ and $F_m \uparrow F$, by finiteness of $\lambda(F)$, $\sum_k \frac{1}{n_k} < +\infty$, hence $n_k \to +\infty$.
If $G \in \mathcal{M}$ and $G \subseteq E \setminus F$, then $G \subseteq R_{k-1}$ $\forall k$, by minimality $\lambda(G) > -\frac{1}{n_k - 1}$ $\forall k \gg 1$; hence $\lambda(G) \geq 0$ and $E \setminus F \in \mathcal{P}_\lambda$. Since $\lambda(E \setminus F) = \lambda(E) - \lambda(F) > 0$, $P \cup (E \setminus F) \in \mathcal{P}_\lambda$ and $\lambda(P \cup (E \setminus F)) > \alpha$, contradiction.

Structural Commentary: A charge splits X into nonnegative-piece and nonpositive-piece regions.
Key Insight: Maximize positive mass; purify any leftover positive piece. □

Lemma 3.1.4. *Hahn Parts Are Unique up to Null Sets*
Let (P_i, N_i), $i = 1, 2$, *be Hahn decompositions for a charge* λ. $\forall E \in \mathcal{M}$*:*
(i) $\lambda(E \cap P_1) = \lambda(E \cap P_2)$.
(ii) $\lambda(E \cap N_1) = \lambda(E \cap N_2)$.

Proof. Fix $E \in \mathcal{M}$. For $\{i, j\} = \{1, 2\}$,
(i) $E \cap (P_i \setminus P_j) \subseteq P_i \cap N_j$, so $\lambda(E \cap (P_i \setminus P_j)) = 0$. Hence $\lambda(E \cap P_i) = \lambda(E \cap P_1 \cap P_2)$ for $i = 1, 2$.
(ii) $E \cap (N_i \setminus N_j) \subseteq N_i \cap P_j$, so $\lambda(E \cap (N_i \setminus N_j)) = 0$. Hence $\lambda(E \cap N_i) = \lambda(E \cap N_1 \cap N_2)$ for $i = 1, 2$.

Structural Commentary: Hahn decompositions are not unique as sets, but the measures they define are unique.
Key Insight: Positive-and-negative overlap is necessarily null. □

Definition 3.1.5. *Jordan Variations of a Charge*
Let λ be a charge, (P, N) a Hahn decomposition, and $E \in \mathcal{M}$.
(i) $\lambda^+(E) := \lambda(E \cap P)$, the *positive variation*.
(ii) $\lambda^-(E) := -\lambda(E \cap N)$, the *negative variation*.
(iii) $|\lambda|(E) := \lambda^+(E) + \lambda^-(E)$, the *total variation*.
These are independent of (P, N) by Lemma 3.1.4.

Theorem 3.1.6. *Jordan Decomposition Theorem*
Let λ *be a charge.*
(i) $\lambda = \lambda^+ - \lambda^-$.
(ii) If $\lambda = \mu - \nu$ *for finite measures* μ, ν*, then* $\lambda^+ \leq \mu$.
(iii) Under the same hypothesis, $\lambda^- \leq \nu$.

Proof. Let (P, N) be a Hahn decomposition and fix $E \in \mathcal{M}$.
(i) $\lambda(E) = \lambda(E \cap P) + \lambda(E \cap N) = \lambda^+(E) - \lambda^-(E)$.
(ii) $\lambda^+(E) = \mu(E \cap P) - \nu(E \cap P) \leq \mu(E)$.
(iii) $\lambda^-(E) = \nu(E \cap N) - \mu(E \cap N) \leq \nu(E)$.

Structural Commentary: Jordan decomposition is the minimal finite-measure difference.
Key Insight: Positive and negative Hahn pieces isolate the irreducible positive masses. □

Corollary 3.1.7. *Variation by Subsets and Null Sets*
For a charge λ *and* $E, M \in \mathcal{M}$*:*
(i) $\lambda^+(E) = \sup\{\lambda(F) : F \in \mathcal{M}, F \subseteq E\}$.
(ii) $\lambda^-(E) = -\inf\{\lambda(F) : F \in \mathcal{M}, F \subseteq E\}$.
(iii) $M \in \mathcal{Z}_\lambda \Leftrightarrow |\lambda|(M) = 0$.
(iv) $\pi_\lambda = \lambda^+$.
(v) $\nu_\lambda = |\lambda|$.

Proof.

(i) Let (P, N) be Hahn. If $F \in \mathcal{M}$ and $F \subseteq E$, then $\lambda(F) = \lambda^+(F) - \lambda^-(F) \leq \lambda^+(E)$, with equality at $F = E \cap P$.
(ii) Under the same hypotheses, $\lambda(F) \geq -\lambda^-(E)$, with equality at $F = E \cap N$.
(iii) ($\Rightarrow$) If $M \in \mathcal{Z}_\lambda$, then $\lambda^+(M) = \lambda^-(M) = 0$, so $|\lambda|(M) = 0$.

(iii) ($\Leftarrow$) If $|\lambda|(M) = 0$, then $\lambda^\pm(G \cap M) = 0$ $\forall G \in \mathcal{M}$, hence $M \in \mathcal{Z}_\lambda$.
(iv) This is (i) and the definition of π_λ.
(v) $\sum_j |\lambda(A_j)| \leq |\lambda|(E)$ $\forall$ finite measurable partitions (A_j) of E; the Hahn partition attains equality.

Structural Commentary: Variation is intrinsic to λ.
Key Insight: The positive part is the largest charge obtainable inside a set. □

Theorem 3.1.8. *Charges Form a Banach Space*
Let $\mathrm{ca}(X,\mathcal{M}) := \{\lambda : \mathcal{M} \to \mathbb{R} : \lambda \text{ is a charge}\}$ *and* $\|\lambda\|_{\mathrm{var}} := |\lambda|(X)$.
With pointwise operations, $(\mathrm{ca}(X,\mathcal{M}), \|\cdot\|_{\mathrm{var}})$ *is Banach.*

Proof.

(i) Charge linearity gives closure under pointwise operations. The variation axioms follow from $|\lambda|(X) = 0 \Rightarrow |\lambda|(E) = 0\ \forall E$, $|a\lambda| = |a|\,|\lambda|$, and $\sum_j |(\lambda+\eta)(A_j)| \le \sum_j |\lambda(A_j)| + \sum_j |\eta(A_j)|$ over finite measurable partitions (A_j).

(ii) Let (λ_n) be variation-Cauchy.
Since $|\lambda_n(E) - \lambda_m(E)| \le \|\lambda_n - \lambda_m\|_{\mathrm{var}}$, set $\lambda(E) := \lim_n \lambda_n(E)$; then λ is finitely additive.
For fixed n and finite disjoint (A_j), $\sum_j |(\lambda - \lambda_n)(A_j)| \le \limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}}$.

(iii) For disjoint (E_k), set $E := \bigcup_k E_k$, $R_K := \bigcup_{k>K} E_k$, and choose n with $\limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}} < \varepsilon$.
Since $\lambda_n(R_K) \to 0$, for large K, $|\lambda(R_K)| \le |\lambda_n(R_K)| + |(\lambda - \lambda_n)(R_K)| < 2\varepsilon$.
Hence $\lambda(R_K) \to 0$, so λ is countably additive.

(iv) Taking suprema in (ii) over finite partitions gives $\|\lambda - \lambda_n\|_{\mathrm{var}} \le \limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}} \to 0$.
Hence every variation-Cauchy sequence converges in variation to a charge.

Structural Commentary: Total variation is the signed-measure analogue of an L^1 norm.
Key Insight: Variation-Cauchy controls every finite partition. □

Theorem 3.1.9. *Variations of Integral Charges*
Let $f \in \mathscr{L}^1(\mu)$ *and* $\lambda(E) := \int_E f\,d\mu$. *For* $E \in \mathcal{M}$:
(i) $\lambda^+(E) = \int_E f^+\,d\mu$.
(ii) $\lambda^-(E) = \int_E f^-\,d\mu$.
(iii) $|\lambda|(E) = \int_E |f|\,d\mu$.

Proof. Set $P := \{f \ge 0\}$ and $N := \{f < 0\}$.
For $A \in \mathcal{M}$, $\lambda(A \cap P) = \int_{A\cap P} f\,d\mu \ge 0$ and $\lambda(A \cap N) = \int_{A\cap N} f\,d\mu \le 0$, so (P,N) is Hahn. Fix $E \in \mathcal{M}$.

(i) $\lambda^+(E) = \lambda(E \cap P) = \int_E f^+\,d\mu$.
(ii) $\lambda^-(E) = -\lambda(E \cap N) = \int_E f^-\,d\mu$.
(iii) $|\lambda|(E) = \lambda^+(E) + \lambda^-(E) = \int_E |f|\,d\mu$.

Structural Commentary: Measure variations match function parts.
Key Insight: Sign sets of the density are Hahn sets. □

Definition 3.1.10. *Absolute Continuity and Singularity*
Let λ, μ be measures and let η be a charge on $(X,\mathcal{M})$.

(i) *Absolute continuity for measures*: $\lambda \ll \mu \Leftrightarrow \forall E \in \mathcal{M}, \mu(E) = 0 \Rightarrow \lambda(E) = 0$.
(ii) *Absolute continuity for charges*: $\eta \ll \mu \Leftrightarrow |\eta| \ll \mu$.
(iii) *Mutual singularity*: $\lambda \perp \mu \Leftrightarrow \exists A, B \in \mathcal{M}$ with $A \cap B = \emptyset$, $X = A \cup B$, $\lambda(A) = 0$, and $\mu(B) = 0$.

Proposition 3.1.11. *Absolute Continuity Calculus*
Let λ, μ, ν *be measures and* η *a charge.*
(i) $\lambda \ll \lambda$.
(ii) $\lambda \ll \mu \ll \nu \Rightarrow \lambda \ll \nu$.
(iii) $\ll$ *is not symmetric.*
(iv) $(\forall E \in \mathcal{M}, \mu(E) = 0 \Rightarrow \eta(E) = 0) \Rightarrow \eta^+, \eta^-, |\eta| \ll \mu$.

Proof.

(i) Reflexivity follows directly from the null-set definition.
(ii) If $\lambda \ll \mu \ll \nu$ and $\nu(E) = 0$, then $\mu(E) = 0$, hence $\lambda(E) = 0$.
(iii) On $X = \{0,1\}$, $\mathcal{M} = 2^X$, $\delta_0 \ll \#$, but $\# \not\ll \delta_0$ since $\delta_0(\{1\}) = 0$ and $\#\{1\} = 1$.
(iv) If $\mu(E) = 0$ and $F \in \mathcal{M}$ satisfies $F \subseteq E$, then $\mu(F) = 0$ and $\eta(F) = 0$.
Hence $E \in \mathcal{Z}_\eta$ and $|\eta|(E) = 0$ by Corollary 3.1.7; therefore $\eta^{\pm}(E) \le |\eta|(E) = 0$.

Structural Commentary: Absolute continuity is null-set order; charges reduce to variation.
Key Insight: Null sets pass to measurable subsets, so all signed pieces vanish there. □

Corollary 3.1.12. *Absolutely Continuous and Singular Forces Zero*
Let λ, μ *be measures and let* η *be a charge.*
(i) $\lambda \ll \mu$ *and* $\lambda \perp \mu$ *imply* $\lambda = 0$.
(ii) $|\eta| \perp \mu$ *implies* $\eta^+ \perp \mu$.
(iii) $|\eta| \perp \mu$ *implies* $\eta^- \perp \mu$.

Proof.

(i) Choose $X = A \cup B$ with $\lambda(A) = 0$ and $\mu(B) = 0$. Since $\lambda \ll \mu$, $\lambda(B) = 0$, hence $\lambda(X) = 0$.
(ii) Choose $X = A \cup B$ with $|\eta|(A) = 0$ and $\mu(B) = 0$. Since $\eta^+ \le |\eta|$, $\eta^+(A) = 0$.
(iii) With the same $X = A \cup B$, $\eta^-(A) \le |\eta|(A) = 0$ and $\mu(B) = 0$; hence $\eta^- \perp \mu$.

Structural Commentary: Absolute continuity and singular support are incompatible except at zero.
Key Insight: A measure supported on a null set must vanish when it also respects that null ideal. □

Example 3.1.13. *Dominating a Countable Family of Measures*
Let (μ_n) be measures with $\mu_n(X) \le 1$, and set $\lambda(E) := \sum_{n\ge 1} 2^{-n}\mu_n(E)$.

Then λ is finite, since $\lambda(X) \le 1$, and countably additive by Tonelli for nonnegative series.
The bound $2^{-n}\mu_n(E) \le \lambda(E)$ implies $\lambda(E) = 0 \Rightarrow \mu_n(E) = 0$ and $\mu_n \ll \lambda\ \forall n$.
Key Insight: A summable positive mixture packages countably many null-set tests into one reference measure.

Lemma 3.1.14. *Epsilon–Delta Form of Absolute Continuity*
Let λ be finite and μ a measure on $(X, \mathcal{M})$. Then $\lambda \ll \mu \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0, \forall E \in \mathcal{M}, \mu(E) < \delta \Rightarrow \lambda(E) < \varepsilon$.

Proof.
($\Rightarrow$) Suppose the ε-δ condition fails. Then $\exists \varepsilon > 0$ and $E_n \in \mathcal{M}$ with $\mu(E_n) < 2^{-n}$ and $\lambda(E_n) \ge \varepsilon$.
Let $F_n = \bigcup_{k\ge n} E_k$ and $F = \bigcap_n F_n$. Then $\mu(F_n) \le 2^{-n+1}$, so $\mu(F) = 0$.
Since $F_n \downarrow F$, $\lambda(F_n) \ge \varepsilon$, and λ is finite, continuity from above gives $\lambda(F) \ge \varepsilon$, contradicting $\lambda \ll \mu$.
($\Leftarrow$) If $\mu(E) = 0$, then $\forall \delta > 0,\ \mu(E) < \delta$; hence $\forall \varepsilon > 0,\ \lambda(E) < \varepsilon$, so $\lambda(E) = 0$.

Structural Commentary: For finite target measure, absolute continuity is uniform smallness on small μ-sets.
Key Insight: A failing sequence is converted into a decreasing null tail with positive λ-mass. □

Example 3.1.15. *Finite Target Is Needed*
On $(\mathbb{N}, 2^{\mathbb{N}})$, let $\lambda = \#$ and $\mu(E) := \sum_{n\in E} 2^{-n}$. Since $\mu(E) = 0 \Leftrightarrow E = \emptyset$, $\lambda \ll \mu$.

But for $E_n = \{n\}$, $\mu(E_n) = 2^{-n} \to 0$ while $\lambda(E_n) = 1$, so the ε-δ condition fails for $\varepsilon = 1$.

Example 3.1.16. *Mutual Absolute Continuity Need Not Preserve Finiteness*
On $(\mathbb{N}, 2^{\mathbb{N}})$, set $\mu(E) := \sum_{n\in E} 2^{-n}$ and $\lambda(\emptyset) = 0$, $\lambda(E) = +\infty$ for $E \ne \emptyset$.

Then $\mu(\mathbb{N}) = 1$, while λ is not σ-finite. Since both measures have null ideal $\{\emptyset\}$, $\lambda \ll \mu$ and $\mu \ll \lambda$.
Key Insight: Absolute continuity compares null sets; it does not compare finiteness, σ-finiteness, or size.

3.2 Radon-Nikodym Theorem

Theorem 3.2.1. *Radon–Nikodym Theorem*
Let λ, μ be σ-finite measures on $(X, \mathcal{M})$ with $\lambda \ll \mu$.

(i) Representation: $\exists$ measurable $f : X \to [0, +\infty)$ such that $\lambda(E) = \int_E f\, d\mu$ for $E \in \mathcal{M}$.
(ii) Uniqueness: Any two representing densities agree μ-a.e.

The μ-a.e. class of f is denoted by $\frac{d\lambda}{d\mu}$.

Proof.
(i) Assume first $\lambda(X), \mu(X) < +\infty$. For $c > 0$, let (P_k, N_k) be Hahn for $\lambda - kc\mu$, set $A_k := N_k \setminus \bigcup_{j<k} A_j$, and set $B := X \setminus \bigcup_{k\ge 1} A_k$.
If $k \ge 2$, then $A_k \subseteq N_k \cap P_{k-1}$.
Hence $E \subseteq A_k$ implies $(k-1)c\mu(E) \le \lambda(E) \le kc\mu(E)$, with the lower bound $0 \le \lambda(E)$ when $k = 1$.
Since $B \subseteq P_k$ for $k \ge 1$, $\lambda(B) \ge kc\mu(B)\ \forall k \ge 1$; finiteness and $\lambda \ll \mu$ give $\mu(B) = \lambda(B) = 0$. Set $f_c := \sum_{k\ge 1} (k-1)c\, \mathbf{1}_{A_k}$.
Countable additivity over (A_k) gives $0 \le \lambda(E) - \int_E f_c\, d\mu \le c\mu(E)$ for $E \in \mathcal{M}$.
Let $c_n := 2^{-n}$ and $f_n := f_{c_n}$. For $E \in \mathcal{M}$, $\left|\int_E (f_n - f_m)\, d\mu\right| \le (c_n + c_m)\mu(E)$; by Proposition 1.2.20, $|f_n - f_m| \le c_n + c_m$ μ-a.e. Thus $\sum_n \|f_{n+1} - f_n\|_1 < +\infty$. By Tonelli, f_n converges μ-a.e. and in $L^1(\mu)$ to a measurable $f \ge 0$. Letting $n \to \infty$ in the approximation inequality gives $\lambda(E) = \int_E f\, d\mu$.
For the σ-finite case, choose a measurable partition $X = \bigsqcup_r Y_r$ with $\lambda(Y_r), \mu(Y_r) < +\infty$. Apply the finite case on each Y_r, obtaining f_r, and set $f := \sum_r f_r \mathbf{1}_{Y_r}$.
Then $\lambda(E) = \sum_r \lambda(E \cap Y_r) = \int_E f\, d\mu$; also $f < +\infty$ μ-a.e. because $\int_{Y_r} f\, d\mu = \lambda(Y_r) < +\infty\ \forall r$.

(ii) If f, g represent λ, then $f, g \in L^1(\mu|_{Y_r})$ and $f = g$ μ-a.e. on Y_r by Proposition 1.2.20.
Taking the countable union of the exceptional sets gives $f = g$ μ-a.e. on X.

Structural Commentary: Hahn decompositions of $\lambda - c\mu$ locate density bands.
Key Insight: Summably refined band approximations converge without invoking later L^1 completeness. □

Example 3.2.2. *A Non-σ-Finite Failure*
Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}([0, 1])$, $\mu = \#$, and $\lambda = m$. Since $\mu(E) = 0 \Leftrightarrow E = \emptyset$, $\lambda \ll \mu$.
If $\lambda(E) = \int_E f\, d\mu$ for some $f \ge 0$, then $\forall x$, $0 = \lambda(\{x\}) = \int_{\{x\}} f\, d\mu = f(x)$, so $\lambda([0, 1]) = 0$, contradiction.
Thus RN can fail when μ is not σ-finite.

Example 3.2.3. *Another Non-σ-Finite Failure*
Let X be uncountable, $\mathcal{M} := \{E \subseteq X : E \text{ or } X \setminus E \text{ countable}\}$, $\mu(E) := \#E$ if E is finite and $+\infty$ otherwise, and $\lambda(E) := 0$ if E is countable and $+\infty$ otherwise. Then $\mu(E) = 0 \Leftrightarrow E = \emptyset$, so $\lambda \ll \mu$.
If $\lambda(E) = \int_E f\, d\mu$ with $f \ge 0$, then $\forall x$, $0 = \lambda(\{x\}) = \int_{\{x\}} f\, d\mu = f(x)$, forcing $\lambda(X) = 0$, contradiction.
Structural Commentary: Singleton tests determine f, but not $\lambda(X)$ without σ-finiteness.
Key Insight: Non-σ-finite domination can lose global mass.

Remark 3.2.4. *When the Radon–Nikodym Derivative Is Integrable*
In Theorem 3.2.1, $\frac{d\lambda}{d\mu} \in L^1(\mu) \Leftrightarrow \lambda(X) < +\infty$, since $\int_X \frac{d\lambda}{d\mu}\, d\mu = \lambda(X)$.

Theorem 3.2.5. *Extended-Valued Radon–Nikodym Theorem*
Let μ be σ-finite, let λ be a measure on $(X, \mathcal{M})$, and assume $\lambda \ll \mu$.

(i) Representation: $\exists f : X \to [0, +\infty]$ measurable such that $\lambda(E) = \int_E f\, d\mu$ for $E \in \mathcal{M}$.
(ii) Uniqueness: Any two representing densities agree μ-a.e.

Proof.

(i) Assume first $\mu(X) < +\infty$ and set $\alpha := \sup\{\mu(A) : A \in \mathcal{M}, \lambda(A) < +\infty\}$. Choose $(A_n) \subseteq \mathcal{M}$ with $\lambda(A_n) < +\infty$ and $\mu(A_n) \to \alpha$, and set $C_n := \bigcup_{j \le n} A_j$, $N := \bigcup_n C_n$, and $P := X \setminus N$. Then $\lambda|_N$ is σ-finite.

For $E \subseteq P$, $\lambda(E) < +\infty \Rightarrow \mu(E) = 0$, since $\mu(E) > 0$ would give $\mu(C_n \cup E) = \mu(C_n) + \mu(E) > \alpha$ eventually. Hence $\mu(E) > 0 \Rightarrow \lambda(E) = +\infty$, while $\mu(E) = 0 \Rightarrow \lambda(E) = 0$ by $\lambda \ll \mu$.

By Theorem 3.2.1, $\exists g : N \to [0, +\infty)$ measurable such that $\lambda(E \cap N) = \int_{E \cap N} g \, d\mu$. Define $f = g$ on N and $f = +\infty$ on P; preceding dichotomy gives $\lambda(E) = \int_E f \, d\mu$. For σ-finite μ, $X = \bigsqcup_r Y_r$, $\mu(Y_r) < +\infty$.

Let f_r be the finite-μ density on Y_r and set $f := \sum_r f_r \mathbf{1}_{Y_r}$. Then $\lambda(E) = \sum_r \lambda(E \cap Y_r) = \int_E f \, d\mu$.

(ii) If f, h represent λ, fix $r, m, n \in \mathbb{N}$ and set $A^r_{m,n} := Y_r \cap \{f \le m\} \cap \{h \ge f + \frac{1}{n}\}$.
Since $\int_A f \, d\mu = \int_A h \, d\mu$ for measurable $A \subseteq Y_r$ and $\int_{A^r_{m,n}} f \, d\mu < +\infty$, $\mu(A^r_{m,n}) = 0$.

Taking the union over r, m, n gives $\mu\{f < h\} = 0$; symmetry gives $\mu\{h < f\} = 0$.

Structural Commentary: Maximal finite-λ regions separate ordinary density from infinite density.
Key Insight: Outside the maximal σ-finite part, every positive-μ set has infinite λ-mass. □

Theorem 3.2.6. *Lebesgue Decomposition Theorem*
Let λ, μ be σ-finite measures on $(X, \mathcal{M})$.

(i) $\exists$ measures λ_s, λ_a with $\lambda = \lambda_s + \lambda_a$, $\lambda_s \perp \mu$, and $\lambda_a \ll \mu$.
(ii) If $(\tilde\lambda_s, \tilde\lambda_a)$ is another such pair, then $\tilde\lambda_s = \lambda_s$ and $\tilde\lambda_a = \lambda_a$.

Proof.

(i) Put $\rho := \lambda + \mu$ and choose $f = \frac{d\lambda}{d\rho}$, $g = \frac{d\mu}{d\rho}$. With $A := \{g = 0\}$ and $B := \{g > 0\}$, set $\lambda_s(E) := \lambda(E \cap A)$ and $\lambda_a(E) := \lambda(E \cap B)$.
Then $\lambda = \lambda_s + \lambda_a$. Since $\mu(A) = \int_A g \, d\rho = 0$, $\lambda_s \perp \mu$.
If $\mu(E) = 0$, then $g = 0$ ρ-a.e. on E, so $\rho(E \cap B) = 0$ and $\lambda_a(E) = 0$; hence $\lambda_a \ll \mu$.

(ii) Let $(\tilde\lambda_s, \tilde\lambda_a)$ be another such pair. Since $\lambda_s \perp \mu$ and $\tilde\lambda_a \ll \mu$, $\lambda_s \perp \tilde\lambda_a$; choose $X = C \cup D$ with $\lambda_s(C) = 0$ and $\tilde\lambda_a(D) = 0$. Then $\lambda_s(E) = \lambda_s(E \cap D) \le \lambda(E \cap D) = \tilde\lambda_s(E \cap D) \le \tilde\lambda_s(E)$.
By symmetry, $\lambda_s = \tilde\lambda_s$, hence $\lambda_a = \tilde\lambda_a$.

Structural Commentary: Split λ into μ-visible and μ-singular parts.
Key Insight: Use $\rho = \lambda + \mu$ and split where $\frac{d\mu}{d\rho} = 0$. □

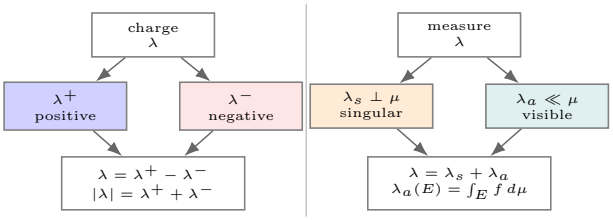

Figure 207: Charges split by sign; measures split into $\lambda_s \perp \mu$ and $\lambda_a \ll \mu$ parts.

Proposition 3.2.7. *Radon–Nikodym Calculus*
Let all measures below be σ-finite.

(i) $\lambda \ll \mu$, $g \ge 0$ measurable $\Rightarrow \int_X g \, d\lambda = \int_X g \frac{d\lambda}{d\mu} \, d\mu$.
(ii) $\nu \ll \lambda \ll \mu \Rightarrow \frac{d\nu}{d\mu} = \frac{d\nu}{d\lambda}\frac{d\lambda}{d\mu}$ μ-a.e.
(iii) $\lambda_1, \lambda_2 \ll \mu \Rightarrow \frac{d(\lambda_1+\lambda_2)}{d\mu} = \frac{d\lambda_1}{d\mu} + \frac{d\lambda_2}{d\mu}$ μ-a.e.
(iv) $\lambda \ll \mu \ll \lambda \Rightarrow \frac{d\lambda}{d\mu} = \left(\frac{d\mu}{d\lambda}\right)^{-1}$ μ-a.e.

Proof.

(i) For $g = \mathbf{1}_E$, $\int_X g \, d\lambda = \lambda(E) = \int_E \frac{d\lambda}{d\mu} \, d\mu = \int_X g \frac{d\lambda}{d\mu} \, d\mu$.
By linearity, identity holds for nonnegative simple g.
For measurable $g \ge 0$, take simple $g_n \uparrow g$ and apply monotone convergence to both sides.

(ii) Let $h := \frac{d\nu}{d\lambda}\frac{d\lambda}{d\mu}$. $\forall E \in \mathcal{M}$, (i) with $g = \mathbf{1}_E \frac{d\nu}{d\lambda}$ gives $\nu(E) = \int_E \frac{d\nu}{d\lambda} \, d\lambda = \int_E h \, d\mu$.
Hence h represents ν relative to μ, so $h = \frac{d\nu}{d\mu}$ μ-a.e.

(iii) Let $h := \frac{d\lambda_1}{d\mu} + \frac{d\lambda_2}{d\mu}$. $\forall E \in \mathcal{M}$, $\int_E h \, d\mu = \lambda_1(E) + \lambda_2(E) = (\lambda_1 + \lambda_2)(E)$.
Hence h represents $\lambda_1 + \lambda_2$ relative to μ, so uniqueness gives the identity.

(iv) Apply (ii) to $\mu \ll \lambda \ll \mu$. Since $\frac{d\mu}{d\mu} = 1$, $1 = \frac{d\mu}{d\lambda}\frac{d\lambda}{d\mu}$ μ-a.e.
Since $\mu \ll \lambda$, $\frac{d\mu}{d\lambda} > 0$ μ-a.e.; divide to obtain $\frac{d\lambda}{d\mu} = \left(\frac{d\mu}{d\lambda}\right)^{-1}$ μ-a.e.

Structural Commentary: Radon–Nikodym derivatives obey chain and linearity rules.
Key Insight: Prove identities by integrating over arbitrary measurable sets, then invoke uniqueness. □

Corollary 3.2.8. *Radon–Nikodym for Charges*
Let μ be σ-finite and let λ be a charge with $\lambda \ll \mu$. Then $\exists f \in \mathscr{L}^1(\mu)$ such that $\forall E \in \mathcal{M}$:

(i) $\lambda(E) = \int_E f \, d\mu$.
(ii) $\lambda^+(E) = \int_E f^+ \, d\mu$.
(iii) $\lambda^-(E) = \int_E f^- \, d\mu$.
(iv) $|\lambda|(E) = \int_E |f| \, d\mu$.

Proof. Since $|\lambda| \ll \mu$, $\lambda^{\pm} \ll \mu$. By Theorem 3.2.1, choose $f_{\pm} \geq 0$ with $\lambda^{\pm}(E) = \int_E f_{\pm}\, d\mu$.
Mutual singularity of λ^+ and λ^- gives $f_+ f_- = 0$ μ-a.e. Set $f := f_+ - f_-$.
Then $f^+ = f_+$ and $f^- = f_-$ μ-a.e., while $\int_X |f|\, d\mu = |\lambda|(X) < +\infty$.

(i) $\lambda(E) = \lambda^+(E) - \lambda^-(E) = \int_E f\, d\mu$.
(ii) $\lambda^+(E) = \int_E f_+\, d\mu = \int_E f^+\, d\mu$.
(iii) $\lambda^-(E) = \int_E f_-\, d\mu = \int_E f^-\, d\mu$.
(iv) $|\lambda|(E) = \lambda^+(E) + \lambda^-(E) = \int_E |f|\, d\mu$.

Structural Commentary: Signed absolute continuity is reduced to the two positive Jordan pieces.
Key Insight: Differentiate λ^+ and λ^- separately, then subtract. □

Example 3.2.9. *Radon–Nikodym Classification of Basic Stieltjes Generators*
Let m be Lebesgue measure on $\mathcal{B}(\mathbb{R})$, and let λ_i be the Borel–Stieltjes measure generated by g_i.

(i) $g_1(x) = 2x$: $\lambda_1 = 2m$, so $\lambda_1 \ll m$, $\frac{d\lambda_1}{dm} = 2$, $m \ll \lambda_1$, $\frac{dm}{d\lambda_1} = \frac{1}{2}$, and $\lambda_1(\mathbb{R}) = +\infty$.
(ii) $g_2(x) = \arctan x$: $\lambda_2(E) = \int_E (1+x^2)^{-1}\, dm$, so $\frac{d\lambda_2}{dm} = (1+x^2)^{-1}$, $\lambda_2(\mathbb{R}) = \pi$, $m \ll \lambda_2$, and $\frac{dm}{d\lambda_2} = 1 + x^2$.
(iii) $g_3 = \mathbf{1}_{[0,+\infty)}$: $\lambda_3 = \delta_0$, so $\lambda_3 \perp m$, $\lambda_3(\mathbb{R}) = 1$, $\lambda_3 \not\ll m$, and $m \not\ll \lambda_3$.
(iv) $g_4(x) = x\mathbf{1}_{[0,+\infty)}(x)$: $\lambda_4(E) = m(E \cap [0,+\infty))$, so $\lambda_4 \ll m$, $\frac{d\lambda_4}{dm} = \mathbf{1}_{[0,+\infty)}$, $m \not\ll \lambda_4$, and $\lambda_4(\mathbb{R}) = +\infty$.

Thus $\lambda_1, \lambda_2, \lambda_4 \ll m$, $\lambda_3 \perp m$, only λ_2, λ_3 are finite, and $m \ll \lambda_i$ exactly for $i = 1, 2$.
Structural Commentary: Stieltjes generators split naturally into density, atom, and restriction examples.
Key Insight: A positive density makes null sets equivalent; an atom lives on a Lebesgue-null set.

3.3 Differentiation of Measures

Definition 3.3.1. *Averages and Absolute Continuity*

(i) *Average*: for $f \in \mathscr{L}^1(E)$ and measurable $A \subseteq E$ with $m(A) > 0$, set $[f : A] := m(A)^{-1} \int_A f\, dm$.
(ii) *Density ratio*: for measurable B and cube or ball Q, $m(Q) > 0$, set $[B : Q] := \frac{m(B \cap Q)}{m(Q)}$.
(iii) *Indefinite integral*: for $f \in \mathscr{L}^1([a,b])$, $F(x) = F(a) + \int_a^x f(t)\, dt$.
(iv) *Absolute continuity*: $F : [a,b] \to \mathbb{R}$ is absolutely continuous iff $\forall \varepsilon > 0\ \exists \delta > 0$: if $(a_j, b_j)_{j=1}^N \subseteq [a,b]$ is a finite pairwise disjoint family, then $\sum_j (b_j - a_j) < \delta \Rightarrow \sum_j |F(b_j) - F(a_j)| < \varepsilon$.

Perspective 3.2.2. *Absolute Continuity as Interval Bookkeeping*
If $F(x) = \int_a^x f(t)\, dt$, absolute continuity controls total F-variation on disjoint intervals of small total length.

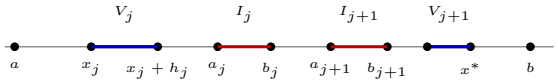

Figure 208: Interval bookkeeping for absolute continuity.

Perspective 3.3.3. *Vitali Covering Control*
Select balls disjointly by decreasing radius.
If B is rejected, then $B \cap B_j \neq \emptyset$ for some earlier B_j with $r(B) \leq r(B_j)$, hence $B \subseteq 5B_j$.
Key Insight: Disjoint balls add; fixed dilations cover misses.

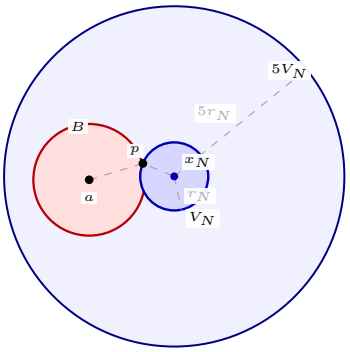

Figure 209: A comparable missed ball lies inside the fivefold dilation of the selected ball.

Definition 3.3.4. *Vitali Coverings and Density Points*
Let (M, d) be a metric space and let $A \subseteq M$.

(i) $\mathcal{V}$ is a *Vitali covering* of A iff $\forall p \in A\ \forall r > 0\ \exists V \in \mathcal{V}$ such that $p \in V$ and $\operatorname{diam}(V) < r$.
(ii) For measurable $E \subseteq \mathbb{R}^n$, its *density-point set* is $\operatorname{dp}(E) := \{p \in \mathbb{R}^n : \lim_{Q \downarrow p} [E : Q] = 1\}$, with Q cubes, equivalently balls, shrinking to p.

Theorem 3.3.5. *Vitali Covering Lemma for Balls and Cubes*
Let $E \subseteq \mathbb{R}^n$, $m^*(E) < +\infty$, and let $\mathcal{V}$ be a Vitali cover of E by closed balls or axis-parallel cubes.
Then $\exists (V_j)_{j \geq 1} \subseteq \mathcal{V}$ pairwise disjoint with $m^*(E \setminus \bigcup_{j \geq 1} V_j) = 0$.

Proof. Fix open $O \supseteq E$ with $m(O) < m^*(E) + \varepsilon$, and replace $\mathcal{V}$ by $\{V \in \mathcal{V} : V \subseteq O\}$.
Greedily choose a maximal disjoint (V_j) by dyadic diameter scales.
For a fixed dilation $V \mapsto V^*$, if $W \cap V \neq \emptyset$ and $\operatorname{diam} W \leq 2 \operatorname{diam} V$, then $W \subseteq V^*$ and $m(V^*) \leq C_n m(V)$.
Let $R = E \setminus \bigcup_j V_j$. Fix N and $p \in R$.
Since $V_1, \ldots, V_N$ are closed, choose $W \in \mathcal{V}$ with $p \in W$ and $W \cap \bigcup_{j \leq N} V_j = \emptyset$. Maximality gives some $j > N$ with $W \cap V_j \neq \emptyset$ and comparable diameter, hence $p \in W \subseteq V_j^*$. Thus $R \subseteq \bigcup_{j>N} V_j^*$.
Therefore $m^*(R) \leq C_n \sum_{j>N} m(V_j) \to 0$, because $\sum_j m(V_j) = m(\bigsqcup_j V_j) \leq m(O) < +\infty$. Hence $m^*(R) = 0$.
Structural Commentary: Vitali selection gives a disjoint cover modulo null sets.
Key Insight: $R \subseteq \bigcup_{j>N} V_j^*$ and disjoint tails vanish. □

Theorem 3.3.6. *Lebesgue Density Theorem*
For measurable $E \subseteq \mathbb{R}^n$, $\lim_{Q \downarrow p} [E : Q] = \mathbf{1}_E(p)$ for a.e. p, with $Q \downarrow p$ through cubes or balls.

Proof. Assume $m(E) < +\infty$. For $a \in (0,1)$ set $E_a := \{p \in E : \liminf_{Q \downarrow p} [E : Q] < a\}$.
Cubes $Q \ni p$ with $[E : Q] < a$ form a Vitali cover of E_a. For open $O \supseteq E_a$ with $m(O) < m(E_a) + \varepsilon$, choose disjoint $(Q_j) \subseteq O$ covering E_a modulo null sets by Theorem 3.3.5.
Hence $m(E_a) \leq \sum_j m(E_a \cap Q_j) \leq \sum_j m(E \cap Q_j) < a \sum_j m(Q_j) \leq a\, m(O) < a(m(E_a) + \varepsilon)$.
Let $\varepsilon \downarrow 0$; since $a < 1$, $m(E_a) = 0$. Thus $p \in E \setminus \bigcup_{a \in \mathbb{Q} \cap (0,1)} E_a$ has density 1. Apply same argument to bounded pieces of E^c to get density 0 a.e. on E^c; then exhaust arbitrary E by bounded pieces.
Structural Commentary: Measurable sets are locally 0 or 1 a.e.
Key Insight: Vitali turns bad density into $m(E_a) \leq a\, m(E_a)$. □

Corollary 3.3.7. *Density Points Represent a Measurable Set*
For measurable $E \subseteq \mathbb{R}^n$, $m(E \triangle \operatorname{dp}(E)) = 0$.

Proof. By Theorem 3.3.6, $[E : Q] \to 1$ at a.e. point of E, $[E : Q] \to 0$ at a.e. point of E^c; hence $E \triangle \operatorname{dp}(E)$ null.
Structural Commentary: The pointwise density dichotomy is converted into equality modulo null sets.
Key Insight: Density-one membership recovers the measurable set almost everywhere. □

Corollary 3.3.8. *Density Points Commute with Slicing Almost Everywhere*
Let $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ be measurable. For a.e. $x \in \mathbb{R}^n$, $\operatorname{dp}(E_x) = \{y : (x,y) \in \operatorname{dp}(E)\}$ up to a null subset of $\mathbb{R}^k$.

Proof. Set $D = \operatorname{dp}(E)$ and $N = E \triangle D$.
By Corollary 3.3.7, $m_{n+k}(N) = 0$; hence $m_k(N_x) = 0$ for a.e. x by Theorem 2.2.14.
Intersecting with the Fubini good set, E_x, D_x are measurable and $m_k(E_x \triangle D_x) = m_k((E \triangle D)_x) = 0$.
$m_k(E_x \triangle \operatorname{dp}(E_x)) = 0$ by Corollary 3.3.7 in $\mathbb{R}^k$. Therefore $m_k(\operatorname{dp}(E_x) \triangle D_x) = 0$.
Structural Commentary: Almost-everywhere local fullness is compatible with almost-everywhere slicing.
Key Insight: Null ambient errors have null slices for almost every base point. □

Lemma 3.3.9. *Large Averages Have Small Outer Measure*
Let $Q_0 \subseteq \mathbb{R}^n$ be a cube, let $0 \leq g \in \mathscr{L}^1(Q_0)$, and let $\lambda > 0$.
Set $A_\lambda := \{p \in Q_0 : \forall r > 0\ \exists$ a cube $Q \subseteq Q_0$, $p \in Q$, $\operatorname{diam} Q < r$, $[g : Q] > \lambda\}$. Then $m^*(A_\lambda) \leq \frac{1}{\lambda} \int_{Q_0} g\, dm$.

Proof. Let $\mathcal{V}_\lambda := \{Q \subseteq Q_0 : Q \text{ is a cube and } [g : Q] > \lambda\}$. This is a Vitali cover of A_λ; choose disjoint $(Q_j) \subseteq \mathcal{V}_\lambda$ covering A_λ modulo a null set. Then $\lambda m^*(A_\lambda) \leq \lambda \sum_j m(Q_j) < \sum_j \int_{Q_j} g\, dm \leq \int_{Q_0} g\, dm$.
Structural Commentary: Weak type for local averages.
Key Insight: Disjoint Vitali cubes turn pointwise large averages into one integral bound. □

Theorem 3.3.10. *Lebesgue Average Value Theorem*
For $f \in \mathscr{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, $\lim_{Q \downarrow p} [f : Q] = f(p)$ for a.e. $p \in \mathbb{R}^n$ with Q cubes, equivalently balls, shrinking to p.

Proof. Fix cube Q_0; to prove claim for a.e. $p \in Q_0^\circ$. For $\alpha > 0$, let $B_\alpha = \left\{ p \in Q_0^\circ : \limsup_{Q \downarrow p} |[f : Q] - f(p)| > \alpha \right\}$.
Choose $h \in C(Q_0)$ with $\|f - h\|_{L^1(Q_0)} < \eta$, and set $g = |f - h|$. Since $[h : Q] \to h(p)$, $B_\alpha \subseteq \{g > \frac{\alpha}{3}\} \cup A_{\frac{\alpha}{3}}(g)$.
By Markov and Lemma 3.3.9, $m^*(B_\alpha) \leq \frac{3}{\alpha} \int_{Q_0} g + \frac{3}{\alpha} \int_{Q_0} g < \frac{6\eta}{\alpha}$. Let $\eta \downarrow 0$; then $m(B_\alpha) = 0$.
Hence $B(Q_0) := \{p \in Q_0^\circ : \limsup_{Q \downarrow p} |[f : Q] - f(p)| > 0\} = \bigcup_{\alpha \in \mathbb{Q}_{>0}} B_\alpha$ is null.
With $Q_N := [-N, N]^n$, $\mathbb{R}^n = \bigcup_N Q_N^\circ$, so $\bigcup_N B(Q_N)$ is null.
Structural Commentary: Local averages recover the original integrable function almost everywhere.
Key Insight: Continuous approximation handles the good part; Vitali's estimate makes the bad part small. □

Theorem 3.3.11. *Lebesgue's Antiderivative Theorem*
For $f \in \mathscr{L}^1([a,b])$, define $F(x) := c + \int_a^x f(t)\, dt$.
(i) $F \in AC([a,b])$.
(ii) $F' = f$ a.e. on $[a,b]$.

Proof.

(i) Given $\varepsilon > 0$, choose $\delta > 0$ such that $m(A) < \delta$ implies $\int_A |f|\, dm < \varepsilon$.
If (α_j, β_j) are disjoint and $\sum_j (\beta_j - \alpha_j) < \delta$, then $\sum_j |F(\beta_j) - F(\alpha_j)| \le \int_{\bigcup_j (\alpha_j,\beta_j)} |f|\, dm < \varepsilon$.

(ii) For $x, x+h \in [a,b]$, let $I_h(x) := (\min\{x, x+h\}, \max\{x, x+h\})$. Then $\frac{F(x+h)-F(x)}{h} = [f : I_h(x)]$.
Extend f by 0 outside $[a,b]$; by Theorem 3.3.10 in one dimension, $[f : I_h(x)] \to f(x)$ for a.e. x.

Structural Commentary: The indefinite Lebesgue integral is the correct antiderivative for integrable functions.
Key Insight: Differentiating the integral is the same as shrinking the averaging interval. □

3.4 Monotone and Bounded-Variation Differentiation

Monotone functions are the one-dimensional model case for differentiation almost everywhere. Bounded-variation functions enter by Jordan decomposition: finite oscillation is exactly enough to write a function as the difference of two monotone functions.

Definition 3.4.1. *Bounded Variation*
Let $F : [a,b] \to \mathbb{R}$. For $a \le u < v \le b$, let $\mathcal{P}[u,v] := \{P = (x_0, \dots, x_N) : N \in \mathbb{N},\ u = x_0 < \cdots < x_N = v\}$.

(i) For $P = (x_0, \dots, x_N) \in \mathcal{P}[u,v]$, define $V(F,P) := \sum_{j=1}^{N} |F(x_j) - F(x_{j-1})|$.

(ii) Define $V_u^v(F) := \sup_{P \in \mathcal{P}[u,v]} V(F,P)$ for $u < v$, and $V_u^u(F) := 0$.

(iii) F has *bounded variation* on $[a,b]$ iff $F \in BV([a,b]) := \{G : [a,b] \to \mathbb{R} : V_a^b(G) < +\infty\}$.

Theorem 3.4.2. *Variation Function and Jordan Decomposition*
Let $F : [a,b] \to \mathbb{R}$. For $F \in BV([a,b])$, set $T_F : [a,b] \to \mathbb{R}$ by $T_F(a) := 0$ and $T_F(x) := V_a^x(F)$ for $a < x \le b$.

(i) Variation additivity: If $F \in BV([a,b])$, then $T_F(v) - T_F(u) = V_u^v(F)$ for $a \le u \le v \le b$.
(ii) Variation monotonicity: If $F \in BV([a,b])$, then T_F is nondecreasing.
(iii) Compensated monotonicity: If $F \in BV([a,b])$, then $F + T_F$ is nondecreasing.
(iv) Decomposition: If $F \in BV([a,b])$, then $F = (F + T_F) - T_F$ is a difference of nondecreasing functions.
(v) Converse: If $F = G - H$ for nondecreasing $G, H : [a,b] \to \mathbb{R}$,
then $F \in BV([a,b])$ and $V_a^b(F) \le G(b) - G(a) + H(b) - H(a)$.

Proof.

(i) Assume $F \in BV([a,b])$ and fix $a \le u \le v \le b$. The cases $u = a$ and $u = v$ are immediate; for $a < u < v$, concatenation gives $V_a^v(F) \ge V_a^u(F) + V_u^v(F)$, while inserting u into $P \in \mathcal{P}[a,v]$ and using the triangle inequality gives $V(F,P) \le V_a^u(F) + V_u^v(F)$. Taking suprema yields $V_a^v(F) = V_a^u(F) + V_u^v(F)$, QED.

(ii) By (i), $T_F(v) - T_F(u) = V_u^v(F) \ge 0$.

(iii) Since $|F(v) - F(u)| \le T_F(v) - T_F(u)$, $(F + T_F)(v) - (F + T_F)(u) \ge 0$.

(iv) The identity $F = (F + T_F) - T_F$ and (ii)–(iii) give the required monotone difference.

(v) Assume $F = G - H$ for nondecreasing $G, H : [a,b] \to \mathbb{R}$.
For $P = (x_0, \dots, x_N) \in \mathcal{P}[a,b]$, by monotonicity and triangle inequality,

$$V(F,P) \le \sum_{j=1}^{N} (G(x_j) - G(x_{j-1})) + \sum_{j=1}^{N} (H(x_j) - H(x_{j-1})) = G(b) - G(a) + H(b) - H(a).$$

Taking the supremum over P gives $V_a^b(F) \le G(b) - G(a) + H(b) - H(a) < +\infty$.

Structural Commentary: Bounded variation is exactly controlled monotone difference.
Key Insight: The variation function supplies the missing monotone part. □

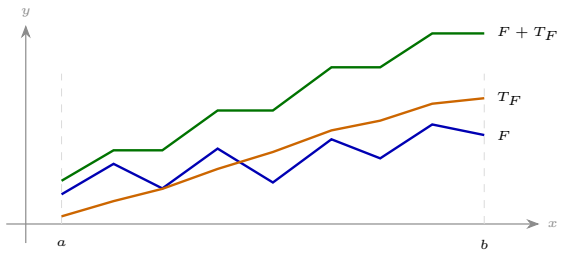

Figure 210: Variation T_F turns F into the increasing function $F + T_F$.

Example 3.4.3. *Basic Bounded-Variation Tests*

(i) If $F : [a,b] \to \mathbb{R}$ is nondecreasing, then $F \in BV([a,b])$ and $V_a^b(F) = F(b) - F(a)$.

(ii) If $F : [a,b] \to \mathbb{R}$ is L-Lipschitz, then $F \in BV([a,b])$ and $V_a^b(F) \le L(b-a)$.

(iii) Let $F : [0,1] \to \mathbb{R}$ satisfy $F(0) = 0$ and $F(x) = x\cos(\frac{\pi}{2x})$ for $0 < x \le 1$. Then F is continuous and $F \notin BV([0,1])$: for $P_n = (0, \frac{1}{2n}, \frac{1}{2n-1}, \dots, \frac{1}{2}, 1) \in \mathcal{P}[0,1]$, $V(F,P_n) = \sum_{k=1}^{n} \frac{1}{k} \to +\infty$.

Definition 3.4.4. *Upper and Lower Derivatives*
Let $I \subseteq \mathbb{R}$ be an interval, $F : I \to \mathbb{R}$, and $x \in I^\circ$.
For $h > 0$, set $Q_h(x) := \{t \in \mathbb{R} : 0 < |t| \le h,\ x + t \in I\}$ and $\Delta_t F(x) := \frac{F(x+t)-F(x)}{t}$.

(i) The *upper derivative* at x is $\overline{D}F(x) := \lim_{h \downarrow 0} \sup_{t \in Q_h(x)} \Delta_t F(x)$.

(ii) The *lower derivative* at x is $\underline{D}F(x) := \lim_{h \downarrow 0} \inf_{t \in Q_h(x)} \Delta_t F(x)$.

They exist in $\overline{\mathbb{R}}$, satisfy $\underline{D}F(x) \le \overline{D}F(x)$, and $F'(x)$ exists iff $\underline{D}F(x) = \overline{D}F(x) \in \mathbb{R}$.

Lemma 3.4.5. *Weak Bound for Upper Derivatives*
Let $F : [a,b] \to \mathbb{R}$ be nondecreasing. For $\alpha > 0$, set $E_\alpha := \{x \in (a,b) : \overline{D}F(x) \ge \alpha\}$.
Then:

(i) $m^*(E_\alpha) \le \frac{F(b)-F(a)}{\alpha}$ *for every* $\alpha > 0$.
(ii) $m^*\{x \in (a,b) : \overline{D}F(x) = +\infty\} = 0$.

Proof.

(i) Fix $\alpha > 0$ and $0 < \alpha' < \alpha$. For $x \in E_\alpha$ and $\delta > 0$, choose $[c,d] \subset (a,b)$ with $x \in [c,d]$, $d-c < \delta$, and $F(d) - F(c) > \alpha'(d-c)$. These intervals form a Vitali cover of E_α.
By Theorem 3.3.5, choose pairwise disjoint $I_k = [c_k, d_k]$ covering E_α modulo a null set.
Then $\alpha' m^*(E_\alpha) \le \alpha' \sum_k (d_k - c_k) < \sum_k (F(d_k) - F(c_k)) \le F(b) - F(a)$. Let $\alpha' \uparrow \alpha$.

(ii) Set $E_\infty := \{x \in (a,b) : \overline{D}F(x) = +\infty\}$. Since $E_\infty \subseteq E_n$ $\forall n \in \mathbb{N}$, (i) gives $m^*(E_\infty) \le \frac{F(b)-F(a)}{n} \to 0$.

Structural Commentary: Vitali selection converts local slope lower bounds into total rise.
Key Insight: Large local slopes are paid for by total monotone rise. □

Lemma 3.4.6. *Dini Gap Sets Are Null*
Let $F : [a,b] \to \mathbb{R}$ be nondecreasing and let $\alpha < \beta$.
Set $E_{\alpha,\beta} := \{x \in (a,b) : \underline{D}F(x) < \alpha < \beta < \overline{D}F(x)\}$. Then $m^*(E_{\alpha,\beta}) = 0$.

Proof. If $\alpha \le 0$, then $E_{\alpha,\beta} = \emptyset$ because every secant quotient of nondecreasing F is nonnegative.
Assume $0 < \alpha < \beta$, set $E = E_{\alpha,\beta}$, and fix $\varepsilon > 0$. Choose open O with $E \subseteq O \subseteq (a,b)$ and $m(O) < m^*(E) + \varepsilon$.
Since $\underline{D}F(x) < \alpha$ on E, intervals $[c,d] \subset O$ with $x \in [c,d]$, $F(d) - F(c) < \alpha(d-c)$ form Vitali cover of E.
Select pairwise disjoint $I_k = [c_k, d_k]$ covering E modulo a null set.
$\forall k$, modulo endpoints, $E \cap I_k \subseteq \{x \in (c_k, d_k) : \overline{D}(F|_{I_k})(x) \ge \beta\}$, so $m^*(E \cap I_k) \le \frac{F(d_k)-F(c_k)}{\beta}$ by Lemma 3.4.5.
Hence $m^*(E) \le \sum_k m^*(E \cap I_k) \le \frac{1}{\beta} \sum_k (F(d_k) - F(c_k)) < \frac{\alpha}{\beta} \sum_k (d_k - c_k) \le \frac{\alpha}{\beta} m(O)$.
Let $\varepsilon \downarrow 0$; then $m^*(E) \le \frac{\alpha}{\beta} m^*(E)$, and since $m^*(E) < +\infty$ and $\frac{\alpha}{\beta} < 1$, $m^*(E) = 0$.
Structural Commentary: Vitali intervals turn a slope gap into a contraction estimate.
Key Insight: Dini gaps force excess rise. □

Perspective 3.4.7. *Right Secants and Differentiation*
Right quotients $g_n(x) = n\big(F(x + \frac{1}{n}) - F(x)\big)$ encode local slope data.
Key Insight: Secants turn differentiation into slope estimates.

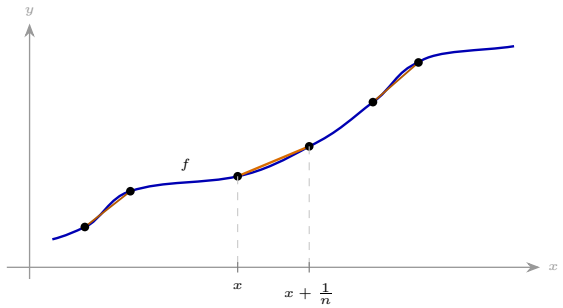

Figure 211: $g_n(x)$ is the slope of the short right secant from x to $x + \frac{1}{n}$.

Proposition 3.4.8. *Divided Differences and Moving Averages*
Let $f \in \mathscr{L}^1([a,b])$, extend f to $[a, b+1]$ by $f(x) = f(b)$ for $b < x \le b+1$, and fix $0 < h \le 1$.
For $x \in [a,b]$, set $\operatorname{Diff}_h f(x) := \frac{f(x+h)-f(x)}{h}$ and $\operatorname{Av}_h f(x) := h^{-1} \int_x^{x+h} f\, dm$.
Then $\forall a \le u < v \le b$, $\int_u^v \operatorname{Diff}_h f\, dm = \operatorname{Av}_h f(v) - \operatorname{Av}_h f(u)$.

Proof. For $a \le u < v \le b$, translation invariance gives $\int_u^v f(\cdot + h)\, dm = \int_{u+h}^{v+h} f\, dm$.
Hence $\int_u^v \operatorname{Diff}_h f\, dm = h^{-1}(\int_{u+h}^{v+h} f\, dm - \int_u^v f\, dm) = h^{-1}(\int_v^{v+h} f\, dm - \int_u^{u+h} f\, dm) = \operatorname{Av}_h f(v) - \operatorname{Av}_h f(u)$.
Structural Commentary: Difference quotients integrate to endpoint moving averages.
Key Insight: The interior contributions cancel after translation. □

Theorem 3.4.9. *Lebesgue Differentiability Theorem for Monotone Functions*
If $F : [a,b] \to \mathbb{R}$ is monotone, then F' exists a.e. on $[a,b]$.

Proof. Assume F is nondecreasing. Set $G_{\alpha,\beta} := \{\underline{D}F < \alpha < \beta < \overline{D}F\}$ for $\alpha, \beta \in \mathbb{Q}$, $\alpha < \beta$, and $E_\infty := \{\overline{D}F = +\infty\}$. Lemma 3.4.6 gives $m(G_{\alpha,\beta}) = 0$, and Lemma 3.4.5 gives $m(E_\infty) = 0$.
Outside $E_\infty \cup \bigcup_{\alpha<\beta} G_{\alpha,\beta}$, $\underline{D}F = \overline{D}F \in \mathbb{R}$, so F' exists. If F is nonincreasing, apply the result to $-F$.
Structural Commentary: Null Dini gaps plus finite upper derivative give differentiability.
Key Insight: Monotone rise controls exceptional slopes. □

Corollary 3.4.10. *Lipschitz and BV Differentiability*
Let $F : [a,b] \to \mathbb{R}$.

(i) If F is Lipschitz, then F' exists a.e. on $[a,b]$.

(ii) If $F \in BV([a,b])$, then F' exists a.e. on $[a,b]$.

Proof.

(i) If F is L-Lipschitz, then $G_+(x) := F(x) + Lx$ and $G_-(x) := Lx - F(x)$ are nondecreasing.
Their derivatives exist a.e. by Theorem 3.4.9, so $F' = \frac{G'_+ - G'_-}{2}$ a.e.

(ii) By Theorem 3.4.2, $F = G - H$ for nondecreasing G, H.
Their derivatives exist a.e. by Theorem 3.4.9, so $F' = G' - H'$ a.e.

Structural Commentary: Lipschitz and BV functions are reduced to finite combinations of monotone functions.
Key Insight: Almost-everywhere differentiability is stable under finite linear combinations. □

Corollary 3.4.11. *Absolute Continuity Implies BV Differentiability*
If $F \in AC([a,b])$, then:

(i) $F \in BV([a,b])$.
(ii) F' exists a.e. on $[a,b]$.

Proof.

(i) Choose $\delta > 0$ from absolute continuity with $\varepsilon = 1$, and choose $a = x_0 < \cdots < x_N = b$ with $x_j - x_{j-1} < \delta$. After inserting the x_j into a partition P, the increments within each $[x_{j-1}, x_j]$ have total length below δ and total F-variation below 1. Thus $V(F,P) \le N$ and $F \in BV([a,b])$.

(ii) By (i), $F \in BV([a,b])$; by Corollary 3.4.10 $\exists G \subseteq [a,b]$, $m([a,b] \setminus G) = 0$ and $F'(x) \in \mathbb{R}$ $\forall x \in G$.

Structural Commentary: Absolute continuity bounds variation; BV decomposition then gives differentiability.
Key Insight: One absolute-continuity scale controls every partition after a finite interval decomposition. □

Corollary 3.4.12. *Derivative Bounds for Monotone and BV Functions*
Let $F : [a,b] \to \mathbb{R}$.

(i) If F is nondecreasing, then $F' \ge 0$ a.e.
(ii) If F is nondecreasing, then $F' \in \mathscr{L}^1([a,b])$ and $\int_a^b F'\,dm \le F(b) - F(a)$.
(iii) If $F \in BV([a,b])$, then F' exists a.e.
(iv) If $F \in BV([a,b])$, then $F' \in \mathscr{L}^1([a,b])$.

Proof.

(i) Every secant quotient of a nondecreasing function is nonnegative; hence $F' \ge 0$ a.e. by Theorem 3.4.9.

(ii) Extend F by $F(b)$ on $(b, b+1]$, set $g_n(x) := n(F(x + \frac{1}{n}) - F(x))$. Then $g_n \ge 0$, $g_n \to F'$ a.e. on (a,b).
By Proposition 3.4.8, $\int_a^b g_n\,dm = F(b) - n\int_a^{a+\frac{1}{n}} F\,dm \le F(b) - F(a)$.
Fatou's lemma gives $\int_a^b F'\,dm \le F(b) - F(a) < +\infty$.

(iii) Corollary 3.4.10(ii) gives $\exists E \subseteq [a,b]$ with $m([a,b] \setminus E) = 0$ and $F'(x) \in \mathbb{R}$ $\forall x \in E$.

(iv) Write $F = G - H$ with G, H nondecreasing by Theorem 3.4.2. By (ii), $G', H' \in \mathscr{L}^1([a,b])$; on the full-measure set where G', H' exist, $F' = G' - H'$, so $F' \in \mathscr{L}^1([a,b])$.

Structural Commentary: BV inherits finite derivative mass from monotone parts.
Key Insight: Fatou turns right-secant bounds into an integral bound for the derivative. □

3.5 Absolute Continuity and BV Decomposition

Absolute continuity is the part of bounded variation detected by integration. The remaining bounded-variation behavior is singular: it may be monotone and continuous, but its derivative vanishes almost everywhere.

Proposition 3.5.1. *Absolutely Continuous Variation Function*
Let $F \in AC([a,b])$ and let $T_F(x) := V_a^x(F)$ be its total variation function.

(i) $T_F \in AC([a,b])$.
(ii) $F = (F + T_F) - T_F$ is a difference of two nondecreasing absolutely continuous functions.

Proof.

(i) Let $\varepsilon > 0$ and choose $\delta > 0$ from absolute continuity of F for the challenge $\frac{\varepsilon}{2}$.
Suppose $(c_k, d_k)_{k=1}^n$ are disjoint subintervals of (a,b) with $\sum_k (d_k - c_k) < \delta$.
Choose partitions P_k of $[c_k, d_k]$ such that $V_{c_k}^{d_k}(F) \le V(F, P_k) + \frac{\varepsilon}{2n}$.
Let $\mathcal{I}$ be all subintervals of all P_k; then $\mathcal{I}$ is disjoint up to endpoints and $\sum_{I \in \mathcal{I}} |I| = \sum_k (d_k - c_k)$.
Hence absolute continuity of F gives $\sum_k V(F, P_k) < \frac{\varepsilon}{2}$, thus $\sum_{k=1}^n |T_F(d_k) - T_F(c_k)| = \sum_{k=1}^n V_{c_k}^{d_k}(F) < \varepsilon$.

(ii) By (i), $T_F \in AC([a,b])$, hence $F + T_F \in AC([a,b])$.
For $x < y$, $T_F(y) - T_F(x) = V_x^y(F) \ge 0$ and $(F + T_F)(y) - (F + T_F)(x) = F(y) - F(x) + V_x^y(F) \ge 0$.
Thus T_F and $F + T_F$ are nondecreasing, and $F = (F + T_F) - T_F$.

Structural Commentary: Absolute continuity controls local variation, not just endpoint displacement.
Key Insight: Partition each local variation, then apply one absolute-continuity test to all pieces. □

Theorem 3.5.2. *Absolute Continuity via Divided Differences*
Let $F \in C([a,b])$, extend F constantly by $F(b)$ on $(b, b+1]$. Let $\operatorname{Diff}_h F(x) := \frac{F(x+h) - F(x)}{h}$, $0 < h \le 1$, $x \in [a,b]$.
Then $F \in AC([a,b])$ iff $\{\operatorname{Diff}_h F : 0 < h \le 1\}$ is uniformly integrable:
$\forall \varepsilon > 0\ \exists \delta > 0\ \forall E \in \mathcal{L}^1$, $E \subseteq [a,b]$ *and* $m(E) < \delta$ *imply* $\sup_{0<h\le 1} \int_E |\operatorname{Diff}_h F|\,dm < \varepsilon$.

Proof.

($\Leftarrow$) Assume the divided differences are uniformly integrable. Given $\varepsilon > 0$, choose $\delta > 0$ for $\frac{\varepsilon}{2}$.
For disjoint (c_k, d_k) with total length below δ, set $E := \bigcup_k (c_k, d_k)$.

By Proposition 3.4.8, $\sum_k |\mathrm{Av}_h F(d_k) - \mathrm{Av}_h F(c_k)| \le \int_E |\mathrm{Diff}_h F|\, dm < \frac{\varepsilon}{2}$.

Letting $h \downarrow 0$ gives $\sum_k |F(d_k) - F(c_k)| \le \frac{\varepsilon}{2} < \varepsilon$.

($\Rightarrow$) Suppose $F \in AC([a,b])$. By Proposition 3.5.1, write $F = U - V$ with U, V nondecreasing and absolutely continuous; since $|\mathrm{Diff}_h F| \le \mathrm{Diff}_h U + \mathrm{Diff}_h V$, assume F nondecreasing.
Let $\widetilde{F}$ be the constant extension to $[a, b+1]$.
Choose $\eta > 0$ for $\frac{\varepsilon}{2}$ in the absolute continuity of $\widetilde{F}$ and set $\delta := \frac{\eta}{2}$. If $m(E) < \delta$, take open $O \supseteq E \cap (a,b)$ with $O \cap (a,b) = \bigsqcup_j (\alpha_j, \beta_j)$ and $m(O \cap (a,b)) < \eta$; set $O_N := \bigcup_{j \le N} (\alpha_j, \beta_j)$.

For $0 < t \le h$, $\sum_{j \le N} ((\beta_j + t) - (\alpha_j + t)) < \eta$, so $\sum_{j \le N} (\widetilde{F}(\beta_j + t) - \widetilde{F}(\alpha_j + t)) < \frac{\varepsilon}{2}$.

Hence $\int_{O_N} \mathrm{Diff}_h F\, dm = \frac{1}{h} \int_0^h \sum_{j \le N} (\widetilde{F}(\beta_j + t) - \widetilde{F}(\alpha_j + t))\, dt < \frac{\varepsilon}{2}$.

Let $N \to \infty$ and use $E \subseteq O$ to get $\sup_{0<h\le 1} \int_E \mathrm{Diff}_h F\, dm < \varepsilon$.

Structural Commentary: Absolute continuity is uniform small-set control for translated secants.
Key Insight: Moving averages turn endpoint increments into integrals of divided differences. □

Theorem 3.5.3. *Fundamental Theorem for Absolutely Continuous Functions*
Let $F \in AC([a,b])$.

(i) F' exists a.e. on $[a,b]$.
(ii) $F' \in \mathscr{L}^1([a,b])$.
(iii) $F(y) - F(x) = \int_x^y F'\, dm$ for $a \le x \le y \le b$.

Proof.

(i) Corollary 3.4.11(ii) gives $\exists E \subseteq [a,b]$ with $m([a,b] \setminus E) = 0$ and $F'(x) \in \mathbb{R}$ $\forall x \in E$.
(ii) Corollary 3.4.11(i) gives $F \in BV([a,b])$, so Corollary 3.4.12(iv) gives $F' \in \mathscr{L}^1([a,b])$.
(iii) Set $f_n := \mathrm{Diff}_{\frac{1}{n}} F$. By Theorem 3.5.2, (f_n) is uniformly integrable; by (i), $f_n \to F'$ a.e.

Fix $a \le x < y \le b$ and $\varepsilon > 0$. Choose $M > 0$ with $\int_{\{|F'|>M\}} |F'|\, dm < \frac{\varepsilon}{6}$ and then $\delta > 0$ such that $M\delta < \frac{\varepsilon}{6}$ and $m(A) < \delta$ implies $\sup_n \int_A |f_n|\, dm < \frac{\varepsilon}{3}$. By Egorov's theorem, $f_n \to F'$ uniformly on $[x,y] \setminus A$ for some measurable A with $m(A) < \delta$. For large n, $\int_x^y |f_n - F'|\, dm < \varepsilon$.
Thus $\int_x^y f_n\, dm \to \int_x^y F'\, dm$. By Proposition 3.4.8, $\int_x^y f_n\, dm = \mathrm{Av}_{\frac{1}{n}} F(y) - \mathrm{Av}_{\frac{1}{n}} F(x) \to F(y) - F(x)$.

Structural Commentary: Uniform integrability is the condition that permits the discrete secant formula to pass to the derivative under the integral sign.
Key Insight: Uniform small-set control upgrades a.e. secant convergence to convergence of interval integrals. □

Corollary 3.5.4. *Indefinite Integral Characterization of Absolute Continuity*
A function $F : [a,b] \to \mathbb{R}$ satisfies:

(i) $F \in AC([a,b])$ iff $\exists g \in \mathscr{L}^1([a,b])$ such that $F(x) = F(a) + \int_a^x g\, dm$ for $x \in [a,b]$.
(ii) Any such g satisfies $g = F'$ a.e.

Proof.

(i) ($\Rightarrow$) If $F \in AC([a,b])$, Theorem 3.5.3 gives $F' \in \mathscr{L}^1([a,b])$ and $F(x) = F(a) + \int_a^x F'\, dm$; take $g = F'$.
(i) ($\Leftarrow$) If $F(x) = F(a) + \int_a^x g\, dm$ with $g \in \mathscr{L}^1([a,b])$, Theorem 3.3.11 gives $F \in AC([a,b])$.
(ii) If $F(x) = F(a) + \int_a^x g\, dm$, then Theorem 3.3.11 gives $F' = g$ a.e.

Structural Commentary: The two directions are the fundamental theorem and the antiderivative theorem.
Key Insight: Absolute continuity is exactly representability by an L^1 density. □

Proposition 3.5.5. *Radon–Nikodym on an Interval Recovers Absolute Continuity*
Let λ be a finite measure on $\mathcal{B}([a,b])$ with $\lambda \ll m$, let $f := \frac{d\lambda}{dm}$, and set $F(x) := \lambda([a,x])$.

(i) $F(y) - F(x) = \int_x^y f\, dm$ for $a \le x \le y \le b$.
(ii) $F \in AC([a,b])$.
(iii) $F' = f$ a.e. on $[a,b]$.

Proof. By Theorem 3.2.1, $f \in L^1(m)$, $\lambda(E) = \int_E f\, dm$ for $E \in \mathcal{B}([a,b])$. Since $\lambda \ll m$, $\lambda(\{x\}) = 0$ for $x \in [a,b]$.

(i) For $a \le x \le y \le b$, $\lambda(\{x\}) = 0$ gives $F(y) - F(x) = \lambda((x,y]) = \int_{(x,y]} f\, dm = \int_x^y f\, dm$.
(ii) Taking $x = a$ in (i), $F(y) = F(a) + \int_a^y f\, dm$ for $y \in [a,b]$; Corollary 3.5.4(i) gives $F \in AC([a,b])$.
(iii) Since $F = F(a) + \int_a^{(\cdot)} f\, dm$, Corollary 3.5.4(ii) gives $F' = f$ a.e. on $[a,b]$.

Structural Commentary: Interval RN representation differentiates cumulative mass.
Key Insight: No atoms, so interval increments equal density integrals. □

Remark 3.5.6. *Lebesgue–Stieltjes Integrals*
Let $g : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, and let μ_g be its Borel–Stieltjes measure.
For $E \in \mathcal{B}(\mathbb{R})$ and bounded Borel $f : E \to \mathbb{R}$, define $\int_E f\, dg := \int_E f\, d\mu_g$.
If $J \subseteq \mathbb{R}$ is an interval, $\mu_g|_J \ll m|_J$, and $h := \frac{d\mu_g}{dm}$ on J, then $\int_E f\, dg = \int_E fh\, dm$ for every Borel $E \subseteq J$.
For $a < b$ with $[a,b] \subseteq J$, $F_a : [a,b] \to \mathbb{R}$, $F_a(x) := \mu_g([a,x])$, is absolutely continuous and satisfies $F_a' = h$ a.e. by Proposition 3.5.5. If $g|_J$ is absolutely continuous, then $h = g'$ a.e. on J.

Example 3.5.7. *Cantor–Lebesgue Function Is Not Absolutely Continuous*
Let $\varphi : [0,1] \to [0,1]$ be the Cantor–Lebesgue function: φ is continuous, nondecreasing, $\varphi(0) = 0$, $\varphi(1) = 1$, and constant on every complementary interval of the Cantor set.
At stage n, write the remaining intervals as $I_{n,k} = [c_{n,k}, d_{n,k}]$, $1 \le k \le 2^n$, so $\sum_k |I_{n,k}| = (\frac{2}{3})^n$.

Since φ is constant on the deleted gaps, $\sum_{k=1}^{2^n} (\varphi(d_{n,k}) - \varphi(c_{n,k})) = 1$.

By continuity, choose $c_{n,k} < \alpha_{n,k} < \beta_{n,k} < d_{n,k}$ such that $\sum_k (\beta_{n,k} - \alpha_{n,k}) < (\frac{2}{3})^n$ and $\sum_k (\varphi(\beta_{n,k}) - \varphi(\alpha_{n,k})) > \frac{1}{2}$. Since $(\frac{2}{3})^n \to 0$, absolute continuity fails for $\varepsilon = \frac{1}{2}$.
Key Insight: The increase of φ lies on intervals with arbitrarily small total length.

Proposition 3.5.8. *Lipschitz, Absolutely Continuous, and BV Classes*
Let $[a,b]$ be nondegenerate, and let $\mathcal{F}_{\mathrm{Lip}}$ denote the Lipschitz functions on $[a,b]$.
(i) $\mathcal{F}_{\mathrm{Lip}} \subsetneq AC([a,b]) \subsetneq BV([a,b])$.
(ii) For $\mathcal{C} \in \{\mathcal{F}_{\mathrm{Lip}}, AC([a,b]), BV([a,b])\}$, $F, G \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$ imply $\alpha F + \beta G \in \mathcal{C}$.
(iii) If $F \in \mathcal{C}$ for some $\mathcal{C} \in \{\mathcal{F}_{\mathrm{Lip}}, AC([a,b]), BV([a,b])\}$, then $T_F(x) := V_a^x(F)$ lies in $\mathcal{C}$.

Proof.
(i) If F is L-Lipschitz, then $\sum_j |F(d_j) - F(c_j)| \le L \sum_j (d_j - c_j)$, so $F \in AC([a,b])$.
Corollary 3.4.11(i) gives $AC([a,b]) \subseteq BV([a,b])$.
Strictness: $x \mapsto \sqrt{\frac{x-a}{b-a}}$ lies in $AC([a,b]) \setminus \mathcal{F}_{\mathrm{Lip}}$, and $x \mapsto \varphi(\frac{x-a}{b-a})$, with φ Cantor–Lebesgue, lies in $BV([a,b]) \setminus AC([a,b])$ because φ is nondecreasing and its Cantor-stage intervals have total length $(\frac{2}{3})^n \to 0$ but total φ-increase 1.
(ii) If F, G are Lipschitz with constants L_F, L_G, then $\alpha F + \beta G$ is Lipschitz with constant $|\alpha| L_F + |\beta| L_G$.
If $F, G \in AC([a,b])$, the interval criterion gives $\alpha F + \beta G \in AC([a,b])$.
If $F, G \in BV([a,b])$, then $V(\alpha F + \beta G, P) \le |\alpha| V(F,P) + |\beta| V(G,P)$, so $V_a^b(\alpha F + \beta G) < +\infty$.
(iii) If F is L-Lipschitz, then $T_F(v) - T_F(u) = V_u^v(F) \le L(v-u)$.
If $F \in AC([a,b])$, Proposition 3.5.1(i) gives $T_F \in AC([a,b])$.
If $F \in BV([a,b])$, then T_F is nondecreasing and $V_a^b(T_F) = T_F(b) - T_F(a) = V_a^b(F) < +\infty$.

Structural Commentary: $\mathcal{F}_{\mathrm{Lip}} \subsetneq AC \subsetneq BV$ is a strict control hierarchy.
Key Insight: $\sqrt{\cdot}$ breaks Lipschitz; Cantor mass breaks absolute continuity. □

Proposition 3.5.9. *Algebra and Safe Composition*
Let $F, G \in AC([a,b])$.
(i) For $\alpha, \beta \in \mathbb{R}$, $\alpha F + \beta G \in AC([a,b])$.
(ii) $FG \in AC([a,b])$.
(iii) If $\Phi : \mathbb{R} \to \mathbb{R}$ is Lipschitz, then $\Phi \circ G \in AC([a,b])$.
(iv) If G is strictly monotone and $\Phi \in AC(J)$ on a compact interval $J \supseteq G([a,b])$, then $\Phi \circ G \in AC([a,b])$.

Proof.
(i) For disjoint (c_k, d_k), $\sum_k |(\alpha F + \beta G)(d_k) - (\alpha F + \beta G)(c_k)| \le |\alpha| \sum_k |F(d_k) - F(c_k)| + |\beta| \sum_k |G(d_k) - G(c_k)|$. Thus the right side is $< \varepsilon$ whenever $\sum_k (d_k - c_k) < \delta$ for a common AC scale of F, G, QED.
(ii) Let $|F| \le M_F$ and $|G| \le M_G$. For $\Delta F_k := F(d_k) - F(c_k)$ and $\Delta G_k := G(d_k) - G(c_k)$, $|F(d_k)G(d_k) - F(c_k)G(c_k)| \le M_F |\Delta G_k| + M_G |\Delta F_k|$. Choose a common AC scale with $\sum_k |\Delta F_k| < \frac{\varepsilon}{2(1+M_G)}$ and $\sum_k |\Delta G_k| < \frac{\varepsilon}{2(1+M_F)}$; then $\sum_k |F(d_k)G(d_k) - F(c_k)G(c_k)| < \varepsilon$.
(iii) If Φ is L-Lipschitz, then $\sum_k |\Phi(G(d_k)) - \Phi(G(c_k))| \le L \sum_k |G(d_k) - G(c_k)|$.
Choose the AC scale of G for $\frac{\varepsilon}{L}$ if $L > 0$; the case $L = 0$ is trivial.
(iv) Suppose G is strictly increasing. Choose $\eta > 0$ from $AC(\Phi)$ for ε and $\delta > 0$ from $AC(G)$ for η.
If $\sum_k (d_k - c_k) < \delta$, then $\sum_k (G(d_k) - G(c_k)) < \eta$.
Since $(G(c_k), G(d_k))$ are disjoint, $\sum_k |\Phi(G(d_k)) - \Phi(G(c_k))| < \varepsilon$.
If G is strictly decreasing, replace $(G(c_k), G(d_k))$ by $(G(d_k), G(c_k))$.

Structural Commentary: AC is stable under algebra and controlled composition.
Key Insight: Lipschitz bounds sizes; monotonicity preserves disjoint intervals. □

Example 3.5.10. *Absolutely Continuous Composition Can Fail*
Define $f, g : [-1,1] \to \mathbb{R}$ by $f(x) := \operatorname{sgn}(x) |x|^{\frac{1}{3}}$, $g(0) := 0$, and $g(x) := x^2 \cos \frac{\pi}{2x}$ for $x \ne 0$. Then:
(i) $f, g \in AC([-1,1])$.
(ii) $f \circ g \notin BV([-1,1])$, hence $f \circ g \notin AC([-1,1])$.

Proof.
(i) Let $h(t) := \frac{1}{3} |t|^{-\frac{2}{3}} \mathbf{1}_{t \ne 0}$. Then $h \in L^1([-1,1])$ and $f(x) = \int_0^x h \, dm$, so $f \in AC([-1,1])$.
For $x \ne 0$, $g'(x) = 2x \cos \frac{\pi}{2x} + \frac{\pi}{2} \sin \frac{\pi}{2x}$, hence $|g'| \le 2 + \frac{\pi}{2}$ off 0.
Together with $|g(x)| \le x^2$, this gives g Lipschitz, so $g \in AC([-1,1])$.
(ii) For $k \ge 1$, $g(\frac{1}{2k-1}) = 0$ and $g(\frac{1}{2k}) = \frac{(-1)^k}{4k^2}$. With $P_n := \{0\} \cup \{\frac{1}{j} : 1 \le j \le 2n\}$,
$V(f \circ g, P_n) \ge \sum_{k=1}^{n} \left| f(g(\frac{1}{2k})) - f(g(\frac{1}{2k-1})) \right| = 4^{-\frac{1}{3}} \sum_{k=1}^{n} k^{-\frac{2}{3}} \to +\infty$. Thus $f \circ g \notin BV([-1,1])$.

Key Insight: The cube root amplifies the inner oscillations until variation diverges. □

Definition 3.5.11. *Singular Bounded-Variation Function*
A function $S \in BV([a,b])$ is *singular* if $S' = 0$ a.e. on $[a,b]$.

Theorem 3.5.12. *Lebesgue Decomposition of a BV Function*
Let $F \in BV([a,b])$. Define $G(x) := F(a) + \int_a^x F'\,dm$, $H(x) := F(x) - G(x)$.

(i) $G \in AC([a,b])$ and $G' = F'$ a.e.
(ii) $H \in BV([a,b])$, $H' = 0$ a.e., $H(a) = 0$, and $F = G + H$.
(iii) If $F = G_1 + H_1$ with $G_1 \in AC([a,b])$, $H_1' = 0$ a.e., and $H_1(a) = 0$, then $G_1 = G$ and $H_1 = H$.

Proof.

(i) By Corollary 3.4.12, $F' \in \mathscr{L}^1([a,b])$. By Theorem 3.3.11, $G \in AC([a,b])$ and $G' = F'$ a.e.
(ii) Since $F, G \in BV([a,b])$, $H = F - G \in BV([a,b])$.
Moreover, $H' = F' - G' = 0$ a.e., $H(a) = 0$, and $F = G + H$.
(iii) If $F = G_1 + H_1$ is another normalized decomposition, then $G_1 - G = H - H_1$ is absolutely continuous, has derivative 0 a.e., and vanishes at a. By Theorem 3.5.3, $G_1 - G = 0$, hence $G_1 = G$ and $H_1 = H$.

Structural Commentary: A BV function splits into an L^1-density part and a derivative-invisible singular part.
Key Insight: Integrating the a.e. derivative separates the absolutely continuous and singular parts. □

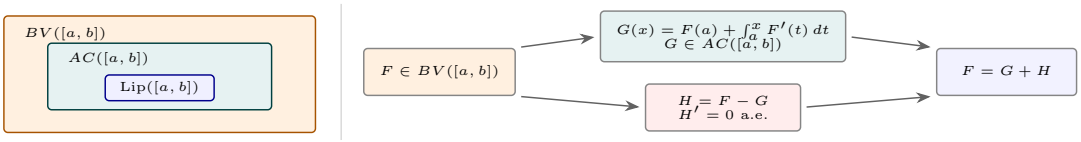

Figure 212: Lip $\subsetneq AC \subsetneq BV$, and each $F \in BV$ decomposes as $F = G + H$ with $G \in AC$ and $H' = 0$ a.e.

Corollary 3.5.13. *Absolutely Continuous Singular Functions Are Constant*

(i) If $F \in AC([a,b])$ and $F' = 0$ a.e., then F is constant.
(ii) The Cantor–Lebesgue function is nonconstant and singular.
(iii) The Cantor–Lebesgue function is not absolutely continuous.

Proof.

(i) By Theorem 3.5.3, $F(y) - F(x) = \int_x^y F'\,dm = 0$.
(ii) The function φ is nondecreasing, hence in $BV([0,1])$, and $\varphi' = 0$ on the full-measure set $[0,1] \setminus P$ by Proposition 8.1.2. Since $\varphi(0) = 0$ and $\varphi(1) = 1$, it is nonconstant.
(iii) If $\varphi \in AC([0,1])$, then $\varphi' = 0$ a.e. by (ii), so (i) gives φ constant, contradicting $\varphi(0) = 0 \neq 1 = \varphi(1)$.

Structural Commentary: The fundamental theorem makes absolute continuity exclude nonconstant singularity.
Key Insight: Derivative-invisible increase can occur in BV but not in absolute continuity. □

3.6 Nikodym Metric and Setwise Measure Limits

The Nikodym metric turns measurable sets modulo μ-null symmetric difference into a complete metric space. This lets Baire category control setwise limits of absolutely continuous measures.

Definition 3.6.1. *Setwise Convergence and Uniform Absolute Continuity*
Let $(X, \mathcal{M})$ be a measurable space.

(i) For $A, B \in \mathcal{M}$, $A \triangle B := (A \setminus B) \cup (B \setminus A)$.
(ii) Finite measures ν_n *converge setwise* to $\nu : \mathcal{M} \to [0, +\infty]$ iff $\nu_n(E) \to \nu(E)$ for every $E \in \mathcal{M}$.
(iii) For $\nu_n \ll \mu$ for all n, (ν_n) is *uniformly absolutely continuous* w.r.t. μ iff
$\forall \varepsilon > 0\ \exists \delta > 0$: $\mu(E) < \delta \Rightarrow \sup_n \nu_n(E) < \varepsilon$ for all $E \in \mathcal{M}$.

Lemma 3.6.2. *Nikodym Symmetric-Difference Calculus*
For $A, B, C \in \mathcal{M}$ and a finite measure ν:

(i) $A \triangle C \subseteq (A \triangle B) \cup (B \triangle C)$.
(ii) $|\nu(A) - \nu(B)| \leq \nu(A \triangle B)$.

Proof.

(i) If $x \in A \triangle C$, then $\mathbf{1}_A(x) \neq \mathbf{1}_C(x)$. Hence $\mathbf{1}_A(x) \neq \mathbf{1}_B(x)$ or $\mathbf{1}_B(x) \neq \mathbf{1}_C(x)$, so $x \in (A \triangle B) \cup (B \triangle C)$.
(ii) Since $\nu(A) - \nu(B) = \nu(A \setminus B) - \nu(B \setminus A)$, $|\nu(A) - \nu(B)| \leq \nu(A \setminus B) + \nu(B \setminus A) = \nu(A \triangle B)$.

Structural Commentary: Set inclusion gives the triangle inequality; disjoint differences give the mass bound.
Key Insight: Nikodym distance measures exactly the part where two measurable sets disagree. □

Definition 3.6.3. *Nikodym Metric Space*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space.

(i) *Null equivalence*: $A \sim_\mu B \Leftrightarrow \mu(A \triangle B) = 0$.
(ii) *Quotient space*: $\mathcal{M}_\mu := \mathcal{M}/\!\sim_\mu$, with class notation $[A] \in \mathcal{M}_\mu$.
(iii) *Metric*: $\rho_\mu([A],[B]) := \mu(A \triangle B)$.

By Lemma 3.6.2 and Proposition 3.1.12, ρ_μ is a well-defined metric on $\mathcal{M}_\mu$.

Proposition 3.6.4. *Nikodym Pseudometric*
Let $E \subseteq \mathbb{R}$ be Lebesgue measurable with $m(E) < \infty$, let X be the collection of Lebesgue measurable subsets of E, and for $A, B \in X$ set $A \triangle B := (A \setminus B) \cup (B \setminus A)$ and $\rho(A,B) := m(A \triangle B)$. Define $A \sim_\rho B$ iff $\rho(A,B) = 0$.

(i) ρ is a pseudometric on X.

(ii) $\widetilde{\rho}([A],[B]) := \rho(A,B)$ *is a metric on* $X/\!\sim_\rho$.
(iii) For all $A, B \in X$, $\rho(A,B) = \int_E |\mathbf{1}_A - \mathbf{1}_B|\, dm$.

Proof.

(i) For $A, B, C \in X$, $A\Delta B \subseteq E$, $A\Delta A = \emptyset$, $A\Delta B = B\Delta A$, and $A\Delta C \subseteq (A\Delta B) \cup (B\Delta C)$.
Hence $0 \le \rho(A,B) \le m(E) < \infty$, $\rho(A,A) = 0$, $\rho(A,B) = \rho(B,A)$, and $\rho(A,C) \le \rho(A,B) + \rho(B,C)$.
(ii) By (i), ρ is a pseudometric and $A \sim_\rho B \iff \rho(A,B) = 0$.
Proposition 3.1.12 gives that $\widetilde{\rho}([A],[B]) := \rho(A,B)$ is a metric on $X/\!\sim_\rho$.
(iii) For $t \in E$, $|\mathbf{1}_A(t) - \mathbf{1}_B(t)| = \mathbf{1}_{A\Delta B}(t)$; integrating gives $\int_E |\mathbf{1}_A - \mathbf{1}_B|\, dm = m(A\Delta B) = \rho(A,B)$.

Structural Commentary: Symmetric difference measures set disagreement.
Key Insight: Quotienting null disagreement gives definiteness. □

Theorem 3.6.5. *Completeness of the Nikodym Metric*
If $(X, \mathcal{M}, \mu)$ *is a finite measure space, then* $(\mathcal{M}_\mu, \rho_\mu)$ *is complete.*

Proof. Let (A_n) be ρ_μ-Cauchy. Choose $n_k \uparrow \infty$ with $\rho_\mu(A_{n_{k+1}}, A_{n_k}) < 2^{-k}$, and set $D_k := A_{n_{k+1}} \triangle A_{n_k}$.
Since $\mu(\bigcup_{k\ge N} D_k) \le \sum_{k\ge N} 2^{-k} \to 0$, $N_0 := \bigcap_N \bigcup_{k\ge N} D_k$ is null. Put $A := \bigcup_N \bigcap_{k\ge N} A_{n_k}$.
Then $A_{n_k} \triangle A \subseteq N_0 \cup \bigcup_{j\ge k} D_j$, so $\rho_\mu(A_{n_k}, A) \le \sum_{j\ge k} 2^{-j} \to 0$. Since (A_n) is Cauchy, $\rho_\mu(A_n, A) \to 0$.

Structural Commentary: Summable jumps give an a.e. eventual set limit.
Key Insight: Summable symmetric differences confine infinitely changing membership to a null set. □

Proposition 3.6.6. *Absolutely Continuous Measures Are Nikodym-Continuous*
Let $(X, \mathcal{M}, \mu)$ *be finite, let* ν *be a finite measure with* $\nu \ll \mu$, *and define* $\Phi_\nu([E]) := \nu(E)$.
(i) $\Phi_\nu : \mathcal{M}_\mu \to \mathbb{R}$ *is well-defined.*
(ii) Φ_ν *is uniformly continuous.*

Proof.

(i) If $A \sim_\mu B$, then $\mu(A\triangle B) = 0$, hence $\nu(A\triangle B) = 0$ and $\nu(A) = \nu(B)$.
(ii) By Lemma 3.6.2, $|\nu(A) - \nu(B)| \le \nu(A\triangle B)$. Given $\varepsilon > 0$, choose $\delta > 0$ by Lemma 3.1.14 such that $\mu(E) < \delta$ implies $\nu(E) < \varepsilon$.
Thus $\rho_\mu(A,B) < \delta$ implies $|\Phi_\nu(A) - \Phi_\nu(B)| < \varepsilon$.

Structural Commentary: Null-set identification defines the map; absolute continuity controls its modulus.
Key Insight: Absolute continuity converts small Nikodym distance into small change of mass. □

Proposition 3.6.7. *Uniform Absolute Continuity and Radon–Nikodym Derivatives*
Let $(X, \mathcal{M}, \mu)$ *be finite, let* (ν_n) *be finite measures with* $\nu_n \ll \mu$, *and set* $f_n = \frac{d\nu_n}{d\mu}$. *The following are equivalent.*

(i) (ν_n) *is uniformly absolutely continuous with respect to* μ.
(ii) The functions $\Phi_{\nu_n} : \mathcal{M}_\mu \to \mathbb{R}$ *are uniformly equicontinuous.*
(iii) The functions Φ_{ν_n} *are equicontinuous at* $\emptyset$.
(iv) $\{f_n : n \ge 1\}$ *is uniformly integrable:* $\forall \varepsilon > 0\ \exists \delta > 0\ \forall E \in \mathcal{M}$, $\mu(E) < \delta \Rightarrow \sup_n \int_E f_n\, d\mu < \varepsilon$.

Proof.

(i) $\Rightarrow$ (ii) Given $\varepsilon > 0$, choose $\delta > 0$ with $\mu(E) < \delta \Rightarrow \sup_n \nu_n(E) < \varepsilon$.
If $\rho_\mu(A,B) = \mu(A\triangle B) < \delta$, then $|\Phi_{\nu_n}(A) - \Phi_{\nu_n}(B)| = |\nu_n(A) - \nu_n(B)| \le \nu_n(A\triangle B) < \varepsilon$ uniformly in n.
(ii) $\Rightarrow$ (iii) Given $\varepsilon > 0$, uniform equicontinuity gives $\delta > 0$ such that
$\rho_\mu(A,B) < \delta \Rightarrow \sup_n |\Phi_{\nu_n}(A) - \Phi_{\nu_n}(B)| < \varepsilon$. With $B = \emptyset$, this is equicontinuity at $\emptyset$.
(iii) $\Rightarrow$ (i) Given $\varepsilon > 0$, choose $\delta > 0$ such that $\rho_\mu(E, \emptyset) < \delta \Rightarrow \sup_n |\Phi_{\nu_n}(E) - \Phi_{\nu_n}(\emptyset)| < \varepsilon$.
Since $\rho_\mu(E,\emptyset) = \mu(E)$ and $\Phi_{\nu_n}(\emptyset) = 0$, $\mu(E) < \delta \Rightarrow \sup_n \nu_n(E) < \varepsilon$.
(i) $\Leftrightarrow$ (iv) For $E \in \mathcal{M}$, $\nu_n(E) = \int_E f_n\, d\mu$. Thus $\sup_n \nu_n(E) < \varepsilon \Leftrightarrow \sup_n \int_E f_n\, d\mu < \varepsilon$ for every small-μ set E.

Structural Commentary: Uniform AC of ν_n equals uniform integrability of f_n.
Key Insight: Radon–Nikodym turns set bounds into integral bounds. □

Theorem 3.6.8. *Setwise Limits under Uniform Absolute Continuity*
Let $(X, \mathcal{M}, \mu)$ *be finite, let* ν_n *be finite measures with* $\nu_n \ll_{\mathrm{u}} \mu$, $\sup_n \nu_n(X) < +\infty$, *and* $\nu_n(E) \to \nu(E)\ \forall E \in \mathcal{M}$.

(i) ν *is a finite measure.*
(ii) $\nu \ll \mu$.

Proof.

(i) Since $\nu(X) = \lim_n \nu_n(X) \le \sup_n \nu_n(X)$, ν is finite-valued. Finite additivity passes to the pointwise limit.
For pairwise disjoint (E_k), set $E := \bigcup_k E_k$ and $R_N := \bigcup_{k>N} E_k$.
Given $\varepsilon > 0$, uniform absolute continuity and $\mu(R_N) \downarrow 0$ give $\nu_n(R_N) < \varepsilon\ \forall n$ and all sufficiently large N, hence $\nu(R_N) \le \varepsilon$.
Therefore $\nu(E) = \sum_{k=1}^{N} \nu(E_k) + \nu(R_N) \to \sum_k \nu(E_k)$.
(ii) If $\mu(A) = 0$, then for every $\varepsilon > 0$, uniform absolute continuity gives $\sup_n \nu_n(A) < \varepsilon$.
Passing to the setwise limit, $\nu(A) \le \varepsilon\ \forall \varepsilon > 0$, hence $\nu(A) = 0$.

Structural Commentary: Uniform tail control yields countable additivity and preserves the null ideal.
Key Insight: Uniform absolute continuity makes the tail of any disjoint union uniformly small. □

Lemma 3.6.9. *One-Point Equicontinuity Spreads*
Let $(X, \mathcal{M}, \mu)$ be finite, and let (ν_n) be finite measures with $\nu_n \ll \mu$.
If the maps Φ_{ν_n} are equicontinuous at one point $E_0 \in \mathcal{M}_\mu$, then they are uniformly equicontinuous on $\mathcal{M}_\mu$.

Proof. Fix $\varepsilon > 0$ and choose $\delta > 0$ so that $\rho_\mu(E, E_0) < \delta$ implies $\sup_n |\nu_n(E) - \nu_n(E_0)| < \frac{\varepsilon}{2}$.
If $\mu(C) < \delta$, then both $E_0 \setminus C$ and $E_0 \cup (C \setminus E_0)$ are within δ of E_0, so $\nu_n(C \cap E_0) < \frac{\varepsilon}{2}$, $\nu_n(C \setminus E_0) < \frac{\varepsilon}{2}$ $\forall n$.
Hence $\sup_n \nu_n(C) < \varepsilon$. Taking $C = A \triangle B$ gives $\sup_n |\nu_n(A) - \nu_n(B)| < \varepsilon$ whenever $\rho_\mu(A, B) < \delta$.
Structural Commentary: Split a small set into its portions inside and outside the equicontinuity base point.
Key Insight: Smallness near one set controls both the part inside it and the part outside it. □

Method 3.6.10. *Baire Mechanism for Vitali–Hahn–Saks*
Regard each absolutely continuous measure ν_n as $\Phi_{\nu_n} : \mathcal{M}_\mu \to \mathbb{R}$, $\Phi_{\nu_n}(E) := \nu_n(E)$.
(i) *Translate*: Setwise convergence becomes pointwise convergence of (Φ_{ν_n}).
(ii) *Apply Baire*: Completeness of $\mathcal{M}_\mu$ supplies a point of equicontinuity.
(iii) *Spread*: Lemma 3.6.9 upgrades one-point equicontinuity to uniform equicontinuity.
(iv) *Return to measures*: Proposition 3.6.7 converts uniform equicontinuity into uniform absolute continuity.

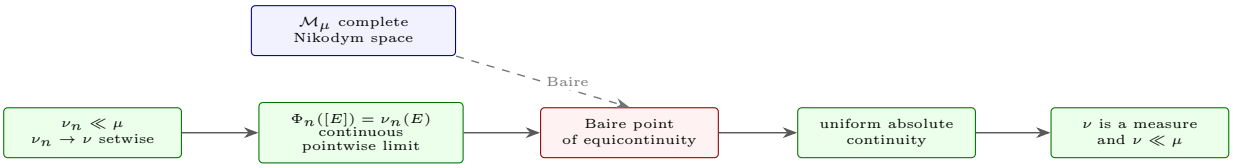

Figure 213: Vitali–Hahn–Saks mechanism: setwise convergence becomes pointwise convergence on $\mathcal{M}_\mu$; Baire gives uniform absolute continuity.

Theorem 3.6.11. *Vitali–Hahn–Saks Theorem*
Let $(X, \mathcal{M}, \mu)$ be finite, and let (ν_n) be finite measures with $\nu_n \ll \mu$, $\sup_n \nu_n(X) < +\infty$, and $\nu_n \to \nu$ setwise.

(i) (ν_n) is uniformly absolutely continuous with respect to μ.
(ii) ν is a finite measure.
(iii) $\nu \ll \mu$.

Proof.
(i) Set $\Phi_n([E]) := \nu_n(E)$. By Proposition 3.6.6, each Φ_n is continuous on complete $\mathcal{M}_\mu$.
If $A \sim_\mu B$, then $\nu_n(A) = \nu_n(B)$ $\forall n$, hence $\nu(A) = \nu(B)$.
Thus $\Phi([E]) := \nu(E)$ is well-defined and $\Phi_n \to \Phi$ pointwise. Since $|\Phi_n([E])| \le \sup_n \nu_n(X)$ and $\nu_n(E) \to \nu(E)$, Φ is real-valued.
Theorem 5.1.34 gives an equicontinuity point $E_0 \in \mathcal{M}_\mu$ for (Φ_n).
Lemma 3.6.9 gives uniform equicontinuity on $\mathcal{M}_\mu$, and Proposition 3.6.7 gives $\nu_n \ll_{\mathrm{u}} \mu$.
(ii) By (i), $\nu_n \ll_{\mathrm{u}} \mu$; Theorem 3.6.8(i) gives that ν is a finite measure.
(iii) By (i), $\nu_n \ll_{\mathrm{u}} \mu$; Theorem 3.6.8(ii) gives $\nu \ll \mu$.
Structural Commentary: Baire gives one equicontinuity point; Nikodym calculus spreads it.
Key Insight: Setwise convergence plus bounded total mass forces uniform null-set control. □

Corollary 3.6.12. *Nikodym Bounded Setwise Limit Theorem*
If finite measures (ν_n) on $(X, \mathcal{M})$ converge setwise to ν and $\sup_n \nu_n(X) < +\infty$, then ν is finite.

Proof. Define $\mu(E) := \sum_{n=1}^{\infty} 2^{-n} \nu_n(E)$. Then μ is a finite measure by countable additivity of nonnegative series, and $\nu_n \ll \mu$ $\forall n$ because $2^{-n}\nu_n \le \mu$. Thus ν is a finite measure by Theorem 3.6.11(ii).
Structural Commentary: A summable mixture places the sequence under one finite dominating measure.
Key Insight: A summable mixture builds one finite measure that dominates the whole sequence. □

4 L^p Spaces and Modes of Convergence

4.1 L^p Spaces

Lebesgue spaces turn integrability into normed linear geometry. The essential point is that an integral norm cannot distinguish functions that differ only on null sets, so the actual vectors are almost-everywhere equivalence classes. Their bounded linear functionals are themselves represented by integration against functions in the conjugate space.

Definition 4.1.1. *Norms and Seminorms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
(i) A *seminorm* is $N : \mathcal{V} \to [0, \infty)$ with $N(\alpha v) = |\alpha| N(v)$ and $N(u + v) \le N(u) + N(v)$ $\forall u, v \in \mathcal{V}$, $\alpha \in \mathbb{F}$.
(ii) A *norm* is a seminorm with $N(v) = 0 \iff v = 0$. A norm N induces $d_N(u, v) := N(u - v)$.

Example 4.1.2. *Norms and Seminorms Before Quotienting*
(i) On $\mathbb{F}^n$, $\|u\|_p := (\sum_{j=1}^{n} |u_j|^p)^{\frac{1}{p}}$ for $1 \le p < \infty$ and $\|u\|_\infty := \max_{1 \le j \le n} |u_j|$ are norms.
Minkowski gives $\|u + v\|_p \le \|u\|_p + \|v\|_p$.

(ii) On sequences, $\ell^p := \{a : \sum_n |a_n|^p < +\infty\}$ with $\|a\|_p := (\sum_n |a_n|^p)^{\frac{1}{p}}$.
$\ell^\infty := \{a : \sup_n |a_n| < +\infty\}$ with $\|a\|_\infty := \sup_n |a_n|$.

(iii) On $B(X)$, $\|f\|_\infty := \sup_{x\in X} |f(x)|$ is a norm. On $C[0,1]$, $N_0(f) := |f(0)|$ and $N_1(f) := \sup_{0\le x\le \frac{1}{2}} |f(x)|$ are seminorms. On $C^1[a,b]$, $N_D(f) := \sup_{a\le x\le b} |f'(x)|$ is a seminorm with kernel the constant functions.

Key Insight: Seminorms become norms after quotienting out their zero directions.

Remark 4.1.3. *Integral Norm on Continuous Functions*
On $C[0,1]$, $N(f) := \int_0^1 |f(x)|\,dx$ is a norm: if $f(x_0) \neq 0$, continuity gives $N(f) > 0$. It is not complete, since continuous ramps can be Cauchy in N and converge in L^1 to a discontinuous step function.

Definition 4.1.4. *L^p Spaces and Mean Convergence*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) For $1 \le p < \infty$, the *p-integrable class* is $\mathscr{L}^p(\mu) := \{f : X \to \mathbb{F} : f \text{ measurable}, \int_X |f|^p\,d\mu < +\infty\}$.

(ii) The *L^p space* and *L^p norm* are $L^p(\mu) := \mathscr{L}^p(\mu)/=_{\text{a.e.}}$ and $\|f\|_p := (\int_X |f|^p\,d\mu)^{\frac{1}{p}}$.

(iii) *L^p convergence*, or *mean convergence of order p*, means $f_n \to f \iff \|f_n - f\|_p \to 0$.
L^p-Cauchy means $\|f_m - f_n\|_p \to 0$ as $m, n \to \infty$.

(iv) *Essentially bounded* means $|f| \le M$ a.e. for some $M < +\infty$; $L^\infty(\mu)$ is the a.e.-quotient of such functions, with *essential sup norm* $\|f\|_\infty := \inf\{M \ge 0 : |f| \le M \text{ a.e.}\} = \inf_{\mu(N)=0} \sup_{x\in X\setminus N} |f(x)|$.

Proposition 4.1.5. *L^1 as a Quotient of a Seminorm*
On $\mathscr{L}^1(\mu)$ set $N_\mu(f) := \int_X |f|\,d\mu$.

(i) N_μ is a seminorm.
(ii) $N_\mu(f) = 0 \iff f = 0$ a.e.
(iii) $L^1(\mu) := \mathscr{L}^1(\mu)/=_{\text{a.e.}}$ is normed by $\|[f]\|_1 := \int_X |f|\,d\mu$; write f for $[f]$.

Proof.

(i) If $f, g \in \mathscr{L}^1(\mu)$ and $a, b \in \mathbb{R}$, then $|af + bg| \le |a|\,|f| + |b|\,|g| \in \mathscr{L}^1(\mu)$, so $\mathscr{L}^1(\mu)$ is linear. Also $N_\mu(af) = |a|\,N_\mu(f)$ and $N_\mu(f+g) \le N_\mu(f) + N_\mu(g)$.
(ii) ($\Rightarrow$) If $N_\mu(f) = 0$, set $E_n := \{|f| \ge \frac{1}{n}\}$. Then $\frac{1}{n}\mu(E_n) \le \int_{E_n} |f|\,d\mu \le N_\mu(f) = 0$, $\{f \neq 0\} = \bigcup_n E_n$ null.
(ii) ($\Leftarrow$) If $f = 0$ a.e., then $|f| = 0$ a.e., hence $N_\mu(f) = 0$.
(iii) If $f = g$ a.e., then $|f| = |g|$ a.e. and $\int_X |f|\,d\mu = \int_X |g|\,d\mu$, so $\|[f]\|_1$ is well defined. The seminorm axioms descend from (i), and $\|[f]\|_1 = 0 \Rightarrow f = 0$ a.e. by (ii), hence $[f] = [0]$.

Structural Commentary: The seminorm kernel is exactly the a.e.-zero functions.
Key Insight: Quotient by null differences before calling the integral a norm. □

Definition 4.1.6. *Complex Integrals and L^2*

(i) For $f = u + iv$, *complex measurability* means measurability of u, v.
(ii) On $E \in \mathcal{M}$, *complex integrability* means $u, v \in \mathscr{L}^1(\mu)$ on E; set $\int_E f\,d\mu := \int_E u\,d\mu + i\int_E v\,d\mu$.
(iii) *$L^2(\mu)$*: complex measurable f satisfying $\int_X |f|^2\,d\mu < +\infty$, modulo a.e. equality; $\|f\|_2^2 := \int_X |f|^2\,d\mu$.
(iv) The *L^2 inner product* is $\langle f, g\rangle_{L^2} := \int_X f\overline{g}\,d\mu$; it induces $\|f\|_2$.

Proposition 4.1.7. *Complex L^1 and Dominated Convergence*
The following statements hold.

(i) For complex measurable f on E, f is integrable $\iff |f| \in \mathscr{L}^1(\mu)$ on E; then $\left|\int_E f\,d\mu\right| \le \int_E |f|\,d\mu$.
(ii) If $f_n \to f$ a.e. on E and $|f_n| \le g \in \mathscr{L}^1(\mu)$, then f is integrable and $\int_E f\,d\mu = \lim_n \int_E f_n\,d\mu$.

Proof.

(i) ($\Rightarrow$) Write $f = u + iv$. If $u, v \in \mathscr{L}^1$, then $|f| \le |u| + |v|$, so $|f| \in \mathscr{L}^1$.
(i) ($\Leftarrow$) If $|f| \in \mathscr{L}^1$, then $|u|, |v| \le |f|$, so $u, v \in \mathscr{L}^1$ and f is integrable.
If $\int_E f\,d\mu = re^{i\theta}$ and $h = e^{-i\theta}f$, then $\left|\int_E f\,d\mu\right| = r = \operatorname{Re}\int_E h\,d\mu = \int_E \operatorname{Re} h\,d\mu \le \int_E |h|\,d\mu = \int_E |f|\,d\mu$.
(ii) Write $f_n = u_n + iv_n$ and $f = u + iv$. Since $|u_n|, |v_n| \le g$ and $u_n \to u$, $v_n \to v$ a.e., real DCT gives $\int_E u_n\,d\mu \to \int_E u\,d\mu$ and $\int_E v_n\,d\mu \to \int_E v\,d\mu$; hence the complex identity.

Structural Commentary: Complex integration is two real integrals.
Key Insight: Rotate the integral to make its value real and nonnegative before applying the real inequality. □

Theorem 4.1.8. *L^2 Geometry*
For $f, g \in L^2(\mu)$, the following hold.

(i) Schwarz: $fg \in L^1(\mu)$ and $\int_X |fg|\,d\mu \le \|f\|_2\,\|g\|_2$.
(ii) Minkowski: $f + g \in L^2(\mu)$ and $\|f+g\|_2 \le \|f\|_2 + \|g\|_2$.
(iii) Completeness: modulo equality a.e., $(L^2(\mu), \|\cdot\|_2)$ is complete.

Proof.

(i) Since $2|fg| \le |f|^2 + |g|^2$, $fg \in L^1(\mu)$.
For $\lambda \in \mathbb{R}$, $0 \le Q(\lambda) := \int_X (|f| + \lambda|g|)^2\,d\mu = \|f\|_2^2 + 2\lambda\int_X |fg|\,d\mu + \lambda^2\|g\|_2^2$; hence $\Delta(Q) \le 0$.
(ii) By (i), $\|f+g\|_2^2 \le \int_X (|f| + |g|)^2\,d\mu \le (\|f\|_2 + \|g\|_2)^2$.

(iii) Let (f_n) be Cauchy. Choose $n_k \uparrow \infty$ with $\|f_{n_{k+1}} - f_{n_k}\|_2 \le 2^{-k}$, and set $h_N := \sum_{k<N} |f_{n_{k+1}} - f_{n_k}|$.

Then $\|h_N\|_2 \le \sum_{k<N} 2^{-k}$, so Fatou gives $h := \lim_N h_N \in L^2$ and $f_{n_k} \to f$ a.e. for some f.

For fixed n, Fatou gives $\|f - f_n\|_2 \le \liminf_k \|f_{n_k} - f_n\|_2$; the Cauchy property sends the right side to 0.

Structural Commentary: Schwarz and Minkowski supply geometry; rapid subsequences prove completeness.
Key Insight: Positivity of the integral transfers Euclidean quadratic estimates to function classes. □

Example 4.1.9. *Mazur Homeomorphism Between L^1 and L^2*
Let $E \subseteq \mathbb{R}$ be Lebesgue measurable. Define $\Phi(f) := \operatorname{sgn}(f)\,|f|^{\frac{1}{2}}$ on $L^1(E)$ and $\Psi(g) := \operatorname{sgn}(g)\,|g|^2$ on $L^2(E)$.
Then $\Phi : L^1(E) \to L^2(E)$ is a homeomorphism with inverse $\Psi : L^2(E) \to L^1(E)$.

Proof. $|\Phi(f)|^2 = |f|$ and $|\Psi(g)| = |g|^2$, so the maps are well-defined; pointwise $\Psi\Phi = \mathrm{id}_{L^1}$ and $\Phi\Psi = \mathrm{id}_{L^2}$.
For $a, b \in \mathbb{R}$, $\left|\operatorname{sgn}(a)\,|a|^{\frac{1}{2}} - \operatorname{sgn}(b)\,|b|^{\frac{1}{2}}\right|^2 \le 2\,|a-b|$, hence $\|\Phi(f) - \Phi(h)\|_2^2 \le 2\,\|f-h\|_1$.
Also $\left|\operatorname{sgn}(a)\,|a|^2 - \operatorname{sgn}(b)\,|b|^2\right| \le 2\,|a-b|\,(|a|+|b|)$, so $\|\Psi(g) - \Psi(k)\|_1 \le 2\,\|g-k\|_2\,(\|g\|_2 + \|k\|_2)$ by Cauchy–Schwarz. Thus Φ and Ψ are continuous.
Key Insight: Nonlinear power maps can homeomorphically identify different L^p norm topologies. □

Definition 4.1.10. *Conjugate Indices*
For $p, q \in (1, \infty)$, p and q are *conjugate indices* if $p^{-1} + q^{-1} = 1$, equivalently $p + q = pq$.
The unique self-conjugate index is $p = q = 2$.

Lemma 4.1.11. *Young's Inequality*
If $A, B \ge 0$, $1 < p < \infty$, and $p^{-1} + q^{-1} = 1$, then:
(i) $AB \le \frac{A^p}{p} + \frac{B^q}{q}$;
(ii) equality holds iff $A^p = B^q$.

Proof.
(i) If $A, B > 0$, set $\alpha := \frac{1}{p}$ and $t := \frac{A^p}{B^q}$. Since $q^{-1} = 1 - \alpha$, weighted AM–GM gives $t^\alpha \le \alpha t + 1 - \alpha$.
Multiplying by B^q yields $AB \le \frac{A^p}{p} + \frac{B^q}{q}$. If $AB = 0$, the inequality is immediate.
(ii) For $A, B > 0$, equality in weighted AM–GM is $t = 1$, i.e. $A^p = B^q$.
If $AB = 0$, equality in (i) holds iff $A = B = 0$, again equivalent to $A^p = B^q$.
Structural Commentary: Young's inequality is weighted AM–GM in conjugate exponents.
Key Insight: Rescale to A^p and B^q, then use convexity. □

Theorem 4.1.12. *Hölder and Minkowski for Sequence Spaces*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
(i) If $p, q > 1$ and $p^{-1} + q^{-1} = 1$, then $\mathbf{x} \in \ell^p(\mathbb{F})$, $\mathbf{y} \in \ell^q(\mathbb{F})$ imply $\sum_{n\ge1} |x_n y_n| \le \|\mathbf{x}\|_p\,\|\mathbf{y}\|_q$.
(ii) If $1 \le p < \infty$ and $\mathbf{x}, \mathbf{y} \in \ell^p(\mathbb{F})$, then $\mathbf{x} + \mathbf{y} \in \ell^p(\mathbb{F})$ and $\|\mathbf{x}+\mathbf{y}\|_p \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$.

Proof.
(i) If $\|\mathbf{x}\|_p\,\|\mathbf{y}\|_q = 0$, the claim is immediate.
Otherwise set $X := \|\mathbf{x}\|_p$, $Y := \|\mathbf{y}\|_q$, $u_n := |x_n|/X$, and $v_n := |y_n|/Y$. Young's inequality gives $u_n v_n \le u_n^p/p + v_n^q/q$; summing over $1 \le n \le N$ and letting $N \to \infty$ gives $\sum_{n\ge1} |x_n y_n| \le XY$.
(ii) For $p = 1$, apply $|x_n + y_n| \le |x_n| + |y_n|$ and sum. For $p > 1$, put $s_N := (\sum_{n=1}^N |x_n + y_n|^p)^{1/p}$. If $s_N = 0$, the bound is trivial; if $s_N > 0$, then $\sum_{n=1}^N |x_n + y_n|^p \le \sum_{n=1}^N (|x_n| + |y_n|)\,|x_n + y_n|^{p-1} \le (\|\mathbf{x}\|_p + \|\mathbf{y}\|_p) s_N^{p-1}$ by (i), so $s_N \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$. Let $N \to \infty$.
Structural Commentary: Hölder controls products; Minkowski controls sums.
Key Insight: The metric property of ℓ^p is exactly Minkowski's inequality. □

Example 4.1.13. *Sequence Spaces as Vector Spaces*
Over $\mathbb{C}$, the sequence spaces $c_0 = \{(z_n) : z_n \to 0\}$, $\ell^\infty = \{(z_n) : \sup_n |z_n| < \infty\}$, and, for $1 \le p < \infty$, $\ell^p = \left\{(z_n) : \sum_{n=1}^{\infty} |z_n|^p < \infty\right\}$ are vector spaces under componentwise operations; for ℓ^p, closure under addition follows from Minkowski's inequality.

Example 4.1.14. *Function and Sequence Metric Spaces*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
(i) On $C_{\mathbb{F}}[a,b]$, $d_\infty(f,g) := \sup_{t\in[a,b]} |f(t) - g(t)|$ and $d_1(f,g) := \int_a^b |f(t) - g(t)|\,dt$ are metrics.
Definiteness of d_1 uses continuity.
(ii) On $\ell^\infty(\mathbb{F})$, $d_\infty(\mathbf{x}, \mathbf{y}) := \sup_{n\ge1} |x_n - y_n|$ is a metric.
(iii) For $1 \le p < \infty$, $d_p(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_p = (\sum_{n\ge1} |x_n - y_n|^p)^{1/p}$ is a metric on $\ell^p(\mathbb{F})$ by Theorem 4.1.12.
Structural Commentary: Norm estimates produce metrics on function and sequence spaces.
Key Insight: Infinite-dimensional examples require estimates, not just coordinate formulas.

Theorem 4.1.15. *Hölder's Inequality*
Let $1 < p < \infty$, $p^{-1} + q^{-1} = 1$, $f \in L^p(\mu)$, and $g \in L^q(\mu)$.
(i) $fg \in L^1(\mu)$.
(ii) $\|fg\|_1 \le \|f\|_p\,\|g\|_q$.

Proof. If $\|f\|_p\,\|g\|_q = 0$, then $f = 0$ a.e. or $g = 0$ a.e., so both claims hold.
Assume $\|f\|_p\,\|g\|_q > 0$ and set $A := \frac{|f|}{\|f\|_p}$, $B := \frac{|g|}{\|g\|_q}$. Young gives $\frac{|fg|}{\|f\|_p\|g\|_q} \le \frac{A^p}{p} + \frac{B^q}{q}$.
(i) Since $\int A^p\,d\mu = \int B^q\,d\mu = 1$, the majorant $\frac{A^p}{p} + \frac{B^q}{q}$ lies in $L^1(\mu)$; hence $fg \in L^1(\mu)$.

(ii) Integrating the displayed inequality gives $\frac{\|fg\|_1}{\|f\|_p\|g\|_q} \le \frac{1}{p}\int A^p\,d\mu + \frac{1}{q}\int B^q\,d\mu = p^{-1}+q^{-1} = 1$.

Structural Commentary: Normalize both factors to unit L^p and L^q size.
Key Insight: Products are controlled by conjugate powers. □

Corollary 4.1.16. *Cauchy–Bunyakovsky–Schwarz in L^2*
If $f, g \in L^2(\mu)$, then:
(i) $fg \in L^1(\mu)$;
(ii) $\left|\int_X fg\,d\mu\right| \le \int_X |fg|\,d\mu \le \|f\|_2\,\|g\|_2$.

Proof.
(i) Since $2^{-1}+2^{-1}=1$, Theorem 4.1.15(i) gives $fg \in L^1(\mu)$.
(ii) Theorem 4.1.15(ii) gives $\int_X |fg|\,d\mu = \|fg\|_1 \le \|f\|_2\,\|g\|_2$, and $\left|\int_X fg\,d\mu\right| \le \int_X |fg|\,d\mu$.

Structural Commentary: Hölder at $p=q=2$ is Cauchy–Schwarz.
Key Insight: Pair two L^2 factors through $2^{-1}+2^{-1}=1$. □

Theorem 4.1.17. *Minkowski's Inequality*
If $1 \le p < \infty$ and $f, h \in L^p(\mu)$, then:
(i) $f+h \in L^p(\mu)$;
(ii) $\|f+h\|_p \le \|f\|_p + \|h\|_p$.

Proof. For $p=1$, integrate $|f+h| \le |f|+|h|$. Let $p>1$ and $q := \frac{p}{p-1}$.
(i) Since $|f+h|^p \le 2^{p-1}(|f|^p+|h|^p)$, $f+h \in L^p(\mu)$. Also $|f+h|^{p-1} \in L^q(\mu)$ because $(p-1)q = p$.
(ii) Pointwise, $|f+h|^p \le (|f|+|h|)\,|f+h|^{p-1}$.
By Hölder, $\|f+h\|_p^p \le (\|f\|_p + \|h\|_p)\left\||f+h|^{p-1}\right\|_q = (\|f\|_p+\|h\|_p)\,\|f+h\|_p^{p-1}$.
If $\|f+h\|_p > 0$, divide by $\|f+h\|_p^{p-1}$; otherwise the claim is trivial.

Structural Commentary: Hölder turns the pointwise triangle inequality into a norm triangle inequality.
Key Insight: Minkowski is exactly the closure and triangle estimate for L^p. □

Theorem 4.1.18. *L^p Is Normed*
For $1 \le p < \infty$, $(L^p(\mu), \|\cdot\|_p)$ is a normed linear space.

Proof.
(i) If $f = f'$ and $g = g'$ a.e., then $\alpha f + \beta g = \alpha f' + \beta g'$ a.e..
Hence pointwise operations descend to $L^p(\mu)$. Minkowski gives $\alpha f + \beta g \in L^p(\mu)$.
(ii) $\|\alpha f\|_p^p = \int_X |\alpha f|^p\,d\mu = |\alpha|^p\,\|f\|_p^p$, so $\|\alpha f\|_p = |\alpha|\,\|f\|_p$.
(iii) $\|f\|_p = 0 \Rightarrow \int_X |f|^p\,d\mu = 0 \Rightarrow f = 0$ a.e.; hence $[f] = 0$ in $L^p(\mu)$.
(iv) Minkowski gives $\|f+g\|_p \le \|f\|_p + \|g\|_p$.

Structural Commentary: Quotienting by a.e. equality turns pointwise operations into vector-space operations.
Key Insight: Minkowski supplies the nontrivial norm axiom. □

Definition 4.1.19. *Completeness Language*
Let $(V, \|\cdot\|)$ be a normed space.
(i) *Complete*: V is complete iff every Cauchy sequence $(x_n) \subset V$ has $x \in V$ with $\|x_n - x\| \to 0$.
(ii) *Banach*: A Banach space is a complete normed space.
(iii) *L^p criterion*: $L^p(\mu)$ is complete iff every L^p-Cauchy sequence converges in $\|\cdot\|_p$.

Lemma 4.1.20. *Convergent Implies Cauchy*
If $f_n \to f$ in $L^p(\mu)$, then (f_n) is Cauchy in $L^p(\mu)$.

Proof. Given $\varepsilon > 0$, choose N so $n \ge N$ implies $\|f_n - f\|_p < \frac{\varepsilon}{2}$.
Then $m, n \ge N$ gives $\|f_m - f_n\|_p \le \|f_m - f\|_p + \|f - f_n\|_p < \varepsilon$ by Minkowski.
Structural Commentary: A convergent sequence has uniformly small tails around its limit.
Key Insight: Use the common limit as the bridge between two tails. □

Definition 4.1.21. *Rapidly Cauchy Sequence*
Let $(V, \|\cdot\|)$ be a normed space. A sequence $(x_k) \subset V$ is *rapidly Cauchy* if $\exists (\varepsilon_k) \subset (0, +\infty)$ such that:
(i) $\sum_{k=1}^{\infty} \varepsilon_k < +\infty$.
(ii) $\|x_{k+1} - x_k\| \le \varepsilon_k^2$ for every $k \ge 1$.

Proposition 4.1.22. *Rapidly Cauchy Subsequences*
In a normed space, the following hold.
(i) Every rapidly Cauchy sequence is Cauchy.
(ii) Every Cauchy sequence has a rapidly Cauchy subsequence.

Proof.
(i) Let (x_k) be rapidly Cauchy with witnesses (ε_k). For $m > n$, $\|x_m - x_n\| \le \sum_{j=n}^{m-1} \|x_{j+1} - x_j\| \le \sum_{j=n}^{\infty} \varepsilon_j^2$.
Since $\sum_j \varepsilon_j < +\infty$, eventually $\varepsilon_j \le 1$, so $\sum_{j=n}^{\infty} \varepsilon_j^2 \le \sum_{j=n}^{\infty} \varepsilon_j \to 0$.
(ii) Let (x_n) be Cauchy. Choose $n_1 < n_2 < \cdots$ such that $m, n \ge n_k \Rightarrow \|x_m - x_n\| \le 4^{-k}$.
Set $y_k := x_{n_k}$ and $\varepsilon_k := 2^{-k}$. Then $\sum_k \varepsilon_k < +\infty$ and $\|y_{k+1} - y_k\| \le 4^{-k} = \varepsilon_k^2$, so (y_k) is rapidly Cauchy.

Structural Commentary: A Cauchy tail contains a summably controlled thread.
Key Insight: Replace arbitrary tail control by a geometric subsequence. □

Theorem 4.1.23. *Riesz–Fischer Completeness Theorem*
For $1 \le p < \infty$, the following hold.

(i) Completeness: $(L^p(\mu), \|\cdot\|_p)$ is complete.
(ii) A.E. subsequence: if $f_n \to f$ in $L^p(\mu)$, then some subsequence converges to a representative of f a.e.

Proof.

(i) Let (f_n) be Cauchy. Choose $g_k = f_{n_k}$ with $\|g_{k+1} - g_k\|_p \le 4^{-k}$. Set $G_N := |g_1| + \sum_{k<N} |g_{k+1} - g_k|$.
By Minkowski, $\|G_N\|_p \le \|g_1\|_p + \sum_{k<N} 4^{-k} \le \|g_1\|_p + \frac{1}{3}$; Fatou gives $G := \lim_N G_N \in L^p$.
The series $h := g_1 + \sum_k (g_{k+1} - g_k)$ converges a.e. with $|h| \le G$ and $g_k \to h$ a.e.
Since $|g_k - h| \le 2G$, DCT gives $\|g_k - h\|_p \to 0$. Given $\varepsilon > 0$, choose k with $\|g_k - h\|_p < \frac{\varepsilon}{2}$ and then N with $n \ge N \Rightarrow \|f_n - g_k\|_p < \frac{\varepsilon}{2}$; hence $\|f_n - h\|_p \to 0$.
(ii) If $f_n \to f$ in L^p, then (f_n) is Cauchy. Applying (i) gives a subsequence $f_{n_k} \to h$ a.e. and in L^p.
Since also $f_{n_k} \to f$ in L^p, uniqueness in normed space gives $[h] = f$ in L^p, so h is a representative of f.

Structural Commentary: A rapid subsequence converges a.e.; DCT gives norm convergence.
Key Insight: Fatou finds the pointwise limit, dominated convergence brings it back to L^p. □

Proposition 4.1.24. *Essential Supremum Test*
If $f \in L^\infty(\mu)$, then:

(i) $|f| \le \|f\|_\infty$ a.e.;
(ii) $A < \|f\|_\infty \Rightarrow \mu(\{|f| > A\}) > 0$.

Proof.

(i) Let $M := \|f\|_\infty$. For each k, $N_k := \{|f| > M + \frac{1}{k}\}$ is null. On $X \setminus \bigcup_k N_k$, $|f| \le M + \frac{1}{k}$ $\forall k$, hence $|f| \le M$.
(ii) If $A < M$ and $\mu(\{|f| > A\}) = 0$, then $|f| \le A$ a.e., so A is an essential upper bound. Contradiction.

Structural Commentary: The essential supremum is the least a.e. upper bound.
Key Insight: The sharp bound cannot be lowered on a full-measure set. □

Theorem 4.1.25. *L^∞ Is Banach*
$(L^\infty(\mu), \|\cdot\|_\infty)$ is a Banach space.

Proof.

(i) Pointwise operations descend modulo a.e. equality, and $\|\alpha f\|_\infty = |\alpha| \|f\|_\infty$. Proposition 4.1.24(i) gives definiteness; outside the union of two null sets, $|f + g| \le \|f\|_\infty + \|g\|_\infty$, hence $\|f + g\|_\infty \le \|f\|_\infty + \|g\|_\infty$.
(ii) Let (f_n) be L^∞-Cauchy. For each m, n, choose null N_{mn} with $|f_m - f_n| \le \|f_m - f_n\|_\infty$ on $X \setminus N_{mn}$, and set $N := \bigcup_{m,n} N_{mn}$.
On $X \setminus N$, (f_n) is uniformly Cauchy, hence converges uniformly to a measurable f.
(iii) The uniform convergence off N gives $\|f_n - f\|_\infty \to 0$.
For fixed n_0, $|f| \le |f_{n_0}| + \sup_{n \ge n_0} \|f_n - f_{n_0}\|_\infty$ a.e.; hence $f \in L^\infty(\mu)$.

Structural Commentary: Completeness is uniform convergence off one null set.
Key Insight: Essential bounds become ordinary uniform bounds off a common null set. □

Proposition 4.1.26. *Simple Approximation, Support, and Tails*
Let $1 \le p < \infty$ and $f \in L^p(\mu)$.

(i) Simple measurable functions are dense in $L^p(\mu)$; also in $L^\infty(\mu)$ for $\|\cdot\|_\infty$.
(ii) $\{|f| > 0\}$ is σ-finite.
(iii) For $E_n := \{|f| \ge n\}$, $\mu(E_n) \to 0$.
(iv) $\forall \varepsilon > 0$ $\exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset$ implies $\|f \mathbf{1}_F\|_p < \varepsilon$.

Proof.

(i) Choose simple $s_n \to f$ a.e. with $|s_n| \le |f|$ in the real case; treat real and imaginary parts separately in the complex case. Then $|f - s_n|^p \le 2^p |f|^p$, so DCT gives $\|f - s_n\|_p \to 0$.
For $f \in L^\infty$, discard null N with $|f| \le \|f\|_\infty + \eta$ on $X \setminus N$, then quantize that bounded range.
(ii) Put $A_k := \{|f| \ge \frac{1}{k}\}$. Then $\mu(A_k) \le k^p \|f\|_p^p < +\infty$ and $\{|f| > 0\} = \bigcup_k A_k$.
(iii) $n^p \mu(E_n) \le \int_{E_n} |f|^p \, d\mu \le \|f\|_p^p$, hence $\mu(E_n) \le n^{-p} \|f\|_p^p \to 0$.
(iv) With A_k as in (ii), $A_k \uparrow \{|f| > 0\}$ and $|f|^p \mathbf{1}_{X \setminus A_k} \to 0$ a.e. DCT gives $\int_{X \setminus A_k} |f|^p \, d\mu \to 0$.
Choose $E_\varepsilon = A_k$ with this integral $< \varepsilon^p$; if $F \cap E_\varepsilon = \emptyset$, then $\|f \mathbf{1}_F\|_p^p \le \int_{X \setminus E_\varepsilon} |f|^p \, d\mu < \varepsilon^p$.

Structural Commentary: L^p mass is simple-approximable and finite-piece capturable.
Key Insight: Integrability turns level sets into finite-measure exhaustions. □

Proposition 4.1.27. *Shell Criterion on Finite Measure Spaces*
Let $\mu(X) < +\infty$, f be measurable, and $E_n := \{n - 1 \le |f| < n\}$ for $n \ge 1$.
For $1 \le p < \infty$, $f \in L^p(\mu) \iff \sum_{n \ge 1} n^p \mu(E_n) < +\infty$.

Proof.

($\Rightarrow$) If $f \in L^p(\mu)$, then $\mu(E_1) \le \mu(X) < +\infty$ and, for $n \ge 2$, $n^p \mathbf{1}_{E_n} \le 2^p |f|^p \mathbf{1}_{E_n}$.
Thus $\sum_n n^p \mu(E_n) \le \mu(X) + 2^p \int_X |f|^p \, d\mu < +\infty$.

($\Leftarrow$) If $\sum_n n^p\mu(E_n) < +\infty$, then $|f|^p = \sum_n |f|^p \mathbf{1}_{E_n} \le \sum_n n^p \mathbf{1}_{E_n}$, hence $\int_X |f|^p \, d\mu < +\infty$.

Structural Commentary: The finite base shell handles small values; weighted shells control tails.
Key Insight: Shell weights characterize L^p mass. □

Proposition 4.1.28. *Inclusions, Interpolation, and Multipliers*

(i) If $\mu(X) < +\infty$ and $1 \le r \le p < \infty$, then $L^p(\mu) \subseteq L^r(\mu)$ and $\|f\|_r \le \|f\|_p \, \mu(X)^{\frac{1}{r}-\frac{1}{p}}$.
(ii) On $(\mathbb{N}, 2^{\mathbb{N}}, \#)$, $1 \le p \le s < \infty$ implies $\ell^p \subseteq \ell^s$ and $\|a\|_s \le \|a\|_p$.
(iii) If $1 \le p_1 < p_2 < \infty$ and $f \in L^{p_1}(\mu) \cap L^{p_2}(\mu)$, then $f \in L^p(\mu)$ for $p_1 \le p \le p_2$.
(iv) If $1 \le p \le \infty$, then $L^p(\mu) \cdot L^\infty(\mu) \subseteq L^p(\mu)$ and $\|fg\|_p \le \|f\|_p \|g\|_\infty$.
(v) $L^\infty(\mu) \subseteq L^1(\mu) \iff \mu(X) < +\infty$. If $\mu(X) = 1$, then $\|f\|_\infty = \lim_{p\to\infty} \|f\|_p$ for $f \in L^\infty(\mu)$.

Proof.

(i) For $r = p$, $L^p \subseteq L^p$ and the estimate reads $\|f\|_p \le \|f\|_p$, i.e. equality.
For $r < p$, Hölder applied to $|f|^r \cdot 1$ gives $\int_X |f|^r \, d\mu \le \|f\|_p^r \, \mu(X)^{1-\frac{r}{p}}$; take rth roots.
(ii) For $a \in \ell^p$, $\forall n$, $|a_n| \le \|a\|_p$, hence $\sum_n |a_n|^s \le \|a\|_p^{s-p} \sum_n |a_n|^p \le \|a\|_p^s$.
(iii) For $p \in [p_1, p_2]$, $|f|^p \le |f|^{p_1} \mathbf{1}_{\{|f|\le 1\}} + |f|^{p_2} \mathbf{1}_{\{|f|>1\}}$.
(iv) For $p < \infty$, $|fg|^p \le \|g\|_\infty^p \, |f|^p$ a.e.; for $p = \infty$, $|fg| \le \|f\|_\infty \|g\|_\infty$ a.e.
(v) ($\Rightarrow$) If $L^\infty(\mu) \subseteq L^1(\mu)$, then $\mathbf{1}_X \in L^\infty(\mu)$ lies in $L^1(\mu)$, so $\mu(X) = \int_X \mathbf{1}_X \, d\mu < +\infty$.
(v) ($\Leftarrow$) If $\mu(X) < +\infty$ and $f \in L^\infty(\mu)$, then $\int_X |f| \, d\mu \le \|f\|_\infty \, \mu(X) < +\infty$. If $\mu(X) = 1$, then $\|f\|_p \le \|f\|_\infty$.
For $A < \|f\|_\infty$, $\mu(\{|f| > A\}) > 0$ and $\|f\|_p \ge A\mu(\{|f| > A\})^{\frac{1}{p}} \to A$. Let $A \uparrow \|f\|_\infty$.

Structural Commentary: Inclusions depend on domain size and pointwise bounds.
Key Insight: Check whether size comes from the function values or from the measure of the domain. □

Example 4.1.29. *Membership Models*

(i) On $(\mathbb{N}, 2^{\mathbb{N}}, \#)$, $a_n = \frac{1}{n}$ satisfies $\|a\|_p^p = \sum_{n\ge 1} n^{-p}$; hence $a \in \ell^p \iff p > 1$, while $a \in \ell^\infty \setminus \ell^1$.
(ii) On $\mathbb{N}$ with $\lambda(\{n\}) = n^{-2}$, $\lambda(\mathbb{N}) < +\infty$ and $f(n) = n^{\frac{1}{p_0}}$ satisfies $\int_{\mathbb{N}} |f|^p \, d\lambda = \sum_{n\ge 1} n^{\frac{p}{p_0}-2}$.
Hence $f \in L^p(\lambda) \iff 1 \le p < p_0$. In particular, $f(n) = \sqrt{n}$ gives $p_0 = 2$.
(iii) On $(0, \infty)$, $f(x) = x^{-\frac{1}{2}}(1 + |\log x|)^{-1}$ gives $\int_0^\infty |f|^p \, dx = \int_{-\infty}^\infty e^{(1-\frac{p}{2})t}(1 + |t|)^{-p} \, dt$ under $x = e^t$.

Structural Commentary: L^p membership reduces to series and endpoint tests.
Key Insight: Different ends impose different p-conditions.

Proposition 4.1.30. *Hölder Dual Norm Attainment*
Let $1 < p < \infty$, $p^{-1} + q^{-1} = 1$, and let $f \in L^p(\mu)$ be real-valued.

(i) $\sup_{\|g\|_q \le 1} \left|\int_X fg \, d\mu\right| = \|f\|_p$.
(ii) If $f \ne 0$, the supremum is attained by $g_0 = \|f\|_p^{-\frac{p}{q}} \operatorname{sign}(f) \, |f|^{p-1}$.

Proof.

(i) For $\|g\|_q \le 1$, Hölder gives $\left|\int_X fg \, d\mu\right| \le \|f\|_p \|g\|_q \le \|f\|_p$.
If $f = 0$ this is sharp; if $f \ne 0$, item (ii) gives an admissible g_0 with $\int_X fg_0 \, d\mu = \|f\|_p$.
(ii) Since $(p-1)q = p$, $\|g_0\|_q^q = \|f\|_p^{-p} \int_X |f|^p \, d\mu = 1$ and $\int_X fg_0 \, d\mu = \|f\|_p^{-\frac{p}{q}} \int_X |f|^p \, d\mu = \|f\|_p$.

Structural Commentary: Equality in Hölder is obtained by matching the conjugate powers.
Key Insight: The L^q test function points in the power direction of f. □

Definition 4.1.31. *Bounded Linear Functionals on L^p*

(i) A *linear functional* on real $L^p(\mu)$ is a linear map $G : L^p(\mu) \to \mathbb{R}$.
(ii) G is *bounded* if $\exists M < +\infty$ such that $\forall f \in L^p(\mu)$, $|G(f)| \le M \|f\|_p$.
(iii) The *operator norm* is $\|G\| := \sup\{|G(f)| : f \in L^p(\mu), \ \|f\|_p \le 1\}$.
(iv) G is *positive* if $f \ge 0$ a.e. implies $G(f) \ge 0$.

Lemma 4.1.32. *Positive Split of a Bounded Functional*
Let $1 \le p < +\infty$ and $G \in (L^p(\mu))^*_{\mathbb{R}}$. Then $G = G^+ - G^-$ for positive bounded linear functionals $G^\pm$.

Proof. For $f \ge 0$, set $G^+(f) := \sup\{G(g) : 0 \le g \le f\}$.
This is finite since $|G(g)| \le \|G\| \|g\|_p \le \|G\| \|f\|_p$, and positive homogeneity is immediate.
For $f_1, f_2 \ge 0$, $G^+(f_1) + G^+(f_2) \le G^+(f_1 + f_2)$; conversely, if $0 \le h \le f_1 + f_2$, set $h_1 = (h - f_2)^+$ and $h_2 = h - h_1$, so $h = h_1 + h_2$ with $0 \le h_j \le f_j$, hence $G(h) \le G^+(f_1) + G^+(f_2)$.
Thus G^+ is additive on the positive cone. Extend by $G^+(f) = G^+(f^+) - G^+(f^-)$; then G^+ is positive, linear, and bounded. Let $G^- = G^+ - G$. Since $G^+(f) \ge G(f)$ for $f \ge 0$, G^- is positive and bounded.
Structural Commentary: Function-lattice functionals split into positive parts.
Key Insight: Split any competitor $h \le f_1 + f_2$ into the portions assigned to f_1 and f_2. □

Theorem 4.1.33. *Riesz Representation for L^1*
Let $(X, \mathcal{M}, \mu)$ be σ-finite and $G \in (L^1(\mu))^*_{\mathbb{R}}$.

(i) $\exists g \in L^\infty(\mu)$ such that $\forall f \in L^1(\mu)$, $G(f) = \int_X fg \, d\mu$.
(ii) $\|G\| = \|g\|_\infty$.

(iii) If G is positive, then $g \geq 0$ a.e.

Proof.

(i) First let $\mu(X) < +\infty$ and $G \geq 0$. Set $\lambda(E) := G(\mathbf{1}_E)$. Finite additivity is linearity; if $E_n \uparrow E$, then $\mathbf{1}_{E_n} \to \mathbf{1}_E$ in L^1, hence $\lambda(E_n) \to \lambda(E)$. Thus λ is finite, and $\lambda \ll \mu$ since $\mu(E) = 0$ gives $\mathbf{1}_E = 0$ in L^1. Radon–Nikodym gives $g \geq 0$ with $\lambda(E) = \int_E g \, d\mu$, so simple functions are represented.
By (ii), $g \in L^\infty$, hence L^1-density of simple functions extends the identity to all $f \in L^1$. For σ-finite μ, apply the finite case to $f \mapsto G(f\mathbf{1}_{X_n})$ on $X_n \uparrow X$, $\mu(X_n) < +\infty$, and patch densities on overlaps. For general G, use the positive split $G = G^+ - G^-$ and subtract the two densities.

(ii) The pairing estimate gives $\|G\| \leq \|g\|_\infty$. Conversely, if $A > \|G\|$ and $E \subseteq \{|g| > A\}$ has $\mu(E) < +\infty$, then with $f = \operatorname{sign}(g)\mathbf{1}_E$, $A\mu(E) < \int_E |g| \, d\mu = G(f) \leq \|G\| \, \mu(E)$, so $\mu(E) = 0$.
A σ-finite exhaustion gives $\forall A > \|G\|$, $\mu(\{|g| > A\}) = 0$, hence $\|g\|_\infty \leq \|G\|$.

(iii) If G is positive and $B = \{g < 0\}$, then $\forall E \subseteq B$ of finite measure, $0 \leq G(\mathbf{1}_E) = \int_E g \, d\mu \leq 0$ only if $\int_E (-g) \, d\mu = 0$. Hence $\mu(E) = 0$ on $\{g < -\frac{1}{k}\}$ $\forall k$, and a σ-finite exhaustion gives $\mu(B) = 0$.

Structural Commentary: A bounded functional on L^1 is integration against an essentially bounded density.
Key Insight: Indicators give a measure; Radon–Nikodym gives the density. □

Theorem 4.1.34. *Riesz Representation for L^p, $1 < p < +\infty$*
Let $(X, \mathcal{M}, \mu)$ be any measure space, $1 < p < +\infty$, $q = \frac{p}{p-1}$, and $G \in (L^p(\mu))^*_{\mathbb{R}}$.

(i) Representation: $\exists g \in L^q(\mu)$ such that $\forall f \in L^p(\mu)$, $G(f) = \int_X fg \, d\mu$.
(ii) Norm: every representing g satisfies $\|G\| = \|g\|_q$.
(iii) Uniqueness: if $g, h \in L^q(\mu)$ represent G, then $g = h$ a.e.

Proof.

(i) Finite μ: for $G \geq 0$, set $\lambda(E) := G(\mathbf{1}_E)$. Since $E_n \uparrow E \Rightarrow \mathbf{1}_{E_n} \to \mathbf{1}_E$ in L^p, λ is finite.
Since $\mu(E) = 0 \Rightarrow \mathbf{1}_E = 0$ in L^p, $\lambda \ll \mu$. Let $g = \frac{d\lambda}{d\mu}$.
By simple approximation, $G(\phi) = \int_X \phi g \, d\mu$ for bounded ϕ.
For $g_m := g \wedge m$, $\int_X g_m^q \, d\mu \leq G(g_m^{q-1}) \leq \|G\| \, (\int_X g_m^q \, d\mu)^{1/p}$; hence $g \in L^q$, by L^p-density $G(f) = \int_X fg \, d\mu$.
General finite G follows from $G = G^+ - G^-$. For σ-finite μ, choose $X_n \uparrow X$, $\mu(X_n) < +\infty$, represent $G_n : u \mapsto G(\tilde{u})$ by g_n, and patch the compatible g_n to g.
Since $\|g_n\|_q = \|G_n\| \leq \|G\|$, by MCT $g \in L^q$; $f\mathbf{1}_{X_n} \to f$ in $L^p \Rightarrow G(f) = \int_X fg \, d\mu$.
For arbitrary μ, choose $\|f_n\|_p = 1$, $G(f_n) \to \|G\|$, and $X_0 := \bigcup_{n,k} \{|f_n| > 1/k\}$.
If $E \subseteq X \setminus X_0$, $\mu(E) < +\infty$, and $a = G(\mathbf{1}_E)$, then $G(f_n) \pm ta \leq \|G\| \, (1 + t^p \mu(E))^{1/p}$; after $n \to \infty$ and $t \downarrow 0$, $a = 0$. Thus G vanishes off X_0; represent on X_0 and extend by 0.

(ii) For representing g, by Hölder, $\|G\| \leq \|g\|_q$; by Proposition 4.1.30, $\|g\|_q = \sup_{\|f\|_p \leq 1} \left| \int_X fg \, d\mu \right| \leq \|G\|$.

(iii) If g, h represent G, then $k := g - h$ satisfies $\int_X fk \, d\mu = 0$ $\forall f \in L^p(\mu)$.
By Proposition 4.1.30 applied to $k \in L^q(\mu)$ with exponents q, p, $\|k\|_q = 0$; hence $g = h$ a.e.

Structural Commentary: For $1 < p < \infty$, every bounded functional comes from the conjugate space L^q.
Key Insight: Localize to σ-finite support, then patch. □

Definition 4.1.35. *Bounded Finitely Additive Measures*
Let $(X, \mathcal{M}, \mu)$ be a measure space.
For $E \in \mathcal{M}$, let $\Pi(E)$ be the finite families $(E_1, \ldots, E_n)$ with $E_k \in \mathcal{M}$, $E_k \subseteq E$, and $E_i \cap E_j = \emptyset$ for $i \neq j$.
For $\nu : \mathcal{M} \to \mathbb{R}$, set $|\nu|_{\text{fa}}(E) := \sup_{(E_k) \in \Pi(E)} \sum_k |\nu(E_k)|$ and $\|\nu\|_{\text{var}} := |\nu|_{\text{fa}}(X)$.

Define $\text{BFA}(X, \mathcal{M}, \mu)$ as the set of $\nu : \mathcal{M} \to \mathbb{R}$ satisfying:

(i) *Finite additivity*: $\nu(\emptyset) = 0$ and $\nu(\bigsqcup_{k=1}^n E_k) = \sum_{k=1}^n \nu(E_k)$.
(ii) *Null compatibility*: $\mu(E) = 0 \Rightarrow \nu(E) = 0$.
(iii) *Bounded variation*: $\|\nu\|_{\text{var}} < +\infty$.

Key Insight: BFA requires finite additivity, null compatibility, and finite variation, not countable additivity.

Theorem 4.1.36. *Kantorovitch Representation for $(L^\infty)^*$*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

For $\nu \in \text{BFA}(X, \mathcal{M}, \mu)$ and simple $\varphi = \sum_{k=1}^n c_k \mathbf{1}_{E_k}$ over a finite measurable partition, set $I_\nu(\varphi) := \sum_{k=1}^n c_k \nu(E_k)$.

(i) Extension: I_ν descends uniquely to $T_\nu \in (L^\infty(\mu))^*_{\mathbb{R}}$.
(ii) Isometry: $\|T_\nu\| = \|\nu\|_{\text{var}}$.
(iii) Surjectivity: if $S \in (L^\infty(\mu))^*_{\mathbb{R}}$ and $\nu_S(E) := S(\mathbf{1}_E)$, then $\nu_S \in \text{BFA}(X, \mathcal{M}, \mu)$ and $S = T_{\nu_S}$.

Thus $\nu \mapsto T_\nu$ is an isometric isomorphism $\text{BFA}(X, \mathcal{M}, \mu) \cong (L^\infty(\mu))^*_{\mathbb{R}}$.

Proof.

(i) Refinement and finite additivity make I_ν well defined.
For simple φ, discard $N = \{|\varphi| > \|\varphi\|_\infty\}$; since $|\nu|_{\text{fa}}(N) = 0$, $|I_\nu(\varphi)| \leq \|\varphi\|_\infty \|\nu\|_{\text{var}}$.
Hence uniform simple approximation extends I_ν to bounded measurable f.
If $\operatorname{supp} f \subseteq N$ and $\mu(N) = 0$, then $|\nu|_{\text{fa}}(N) = 0$, so $I_\nu(f) = 0$; therefore the extension depends only on the L^∞ class and defines T_ν with $\|T_\nu\| \leq \|\nu\|_{\text{var}}$.

(ii) For $(E_k) \in \Pi(X)$, set $c_k := \operatorname{sign}(\nu(E_k))$ and $\varphi := \sum_k c_k \mathbf{1}_{E_k}$.
Then $\|\varphi\|_\infty \leq 1$ and $\sum_k |\nu(E_k)| = T_\nu(\varphi) \leq \|T_\nu\|$.
Supremizing over $\Pi(X)$ gives $\|\nu\|_{\text{var}} \leq \|T_\nu\|$, and item (i) gives equality.

(iii) For $S \in (L^\infty(\mu))^*_{\mathbb{R}}$, $\nu_S(\emptyset) = 0$ and finite additivity follow from linearity.
If $\mu(E) = 0$, then $\mathbf{1}_E = 0$ in L^∞, so $\nu_S(E) = 0$.
For $(E_k) \in \Pi(X)$, with $c_k = \operatorname{sign}(\nu_S(E_k))$ and $\varphi = \sum_k c_k \mathbf{1}_{E_k}$, $\sum_k |\nu_S(E_k)| = S(\varphi) \le \|S\|$.
Hence $\nu_S \in \mathrm{BFA}(X, \mathcal{M}, \mu)$. Since S and T_{ν_S} agree on simple functions, uniform simple density gives $S = T_{\nu_S}$.

Structural Commentary: Simple extensions give T_ν; indicators recover ν.
Key Insight: Finite additivity supplies the full L^∞ dual. □

Proposition 4.1.37. *The Countably Additive Part*
For $g \in L^1(\mu)$, set $\nu_g(E) := \int_E g\, d\mu$.
(i) $\nu_g \in \mathrm{BFA}(X, \mathcal{M}, \mu)$.
(ii) $\|\nu_g\|_{\mathrm{var}} = \|g\|_1$.
(iii) $T_{\nu_g}(f) = \int_X fg\, d\mu$ for $f \in L^\infty(\mu)$; hence $L^1(\mu)$ embeds isometrically into $(L^\infty(\mu))^*$.

Proof.
(i) For disjoint $E_1, \dots, E_n$, $\nu_g(\bigsqcup_k E_k) = \sum_k \nu_g(E_k)$ by finite additivity of the integral, and
$\mu(E) = 0 \Rightarrow \nu_g(E) = 0$. Also, for $(E_k) \in \Pi(X)$, $\sum_k |\nu_g(E_k)| \le \sum_k \int_{E_k} |g|\, d\mu \le \|g\|_1$, so $\nu_g \in \mathrm{BFA}(X, \mathcal{M}, \mu)$.
(ii) Corollary 3.1.7 gives $\|\nu_g\|_{\mathrm{var}} = |\nu_g|(X)$, and Theorem 3.1.9 gives $|\nu_g|(X) = \int_X |g|\, d\mu = \|g\|_1$.
(iii) If $f = \sum_k c_k \mathbf{1}_{E_k}$ is simple, then $T_{\nu_g}(f) = \sum_k c_k \nu_g(E_k) = \int_X fg\, d\mu$.
For $f \in L^\infty(\mu)$, choose simple $f_n \to f$ uniformly; then $T_{\nu_g}(f_n) \to T_{\nu_g}(f)$ and $\int_X f_n g\, d\mu \to \int_X fg\, d\mu$.

Structural Commentary: Variation fixes the norm; simple approximation extends the pairing.
Key Insight: Countably additive representatives are exactly the familiar L^1 integral pairings. □

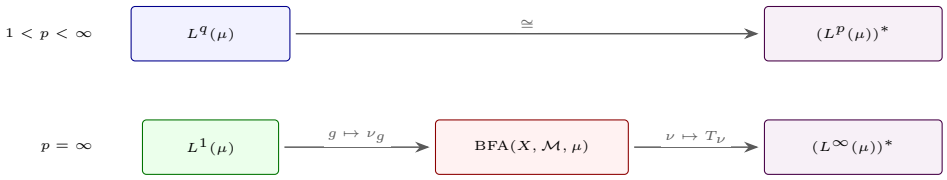

Figure 214: $L^q(\mu) \cong (L^p(\mu))^*$ for $1 < p < \infty$; $L^1(\mu)$ is the countably additive part of BFA $\cong (L^\infty(\mu))^*$.

Remark 4.1.38. *L^2 Riesz Gives Another Radon–Nikodym Route*
Let λ, μ be finite and $\lambda \ll \mu$. Put $\rho = \lambda + \mu$ and $G(h) := \int h\, d\lambda$ on $L^2(\rho)$.
If $G(h) = \int hg\, d\rho$, indicator tests give $0 \le g \le 1$ ρ-a.e. and $\mu(\{g = 1\}) = 0$.
Since $\int h(1-g)\, d\lambda = \int hg\, d\mu$, take $E_N = E \cap \{1 - g \ge \frac{1}{N}\}$ and $h = \frac{\mathbf{1}_{E_N}}{1-g} \in L^2(\rho)$ to get $\lambda(E_N) = \int_{E_N} \frac{g}{1-g}\, d\mu$. Let $N \to \infty$ by monotone convergence and $\lambda \ll \mu$: $\lambda(E) = \int_E \frac{g}{1-g}\, d\mu$.
Key Insight: Riesz representation on $L^2(\lambda + \mu)$ encodes the density in the ratio $\frac{g}{1-g}$.

Lemma 4.1.39. *Absolute Continuity of L^p Mass*
For $1 \le p < \infty$ and $f \in L^p(\mu)$, $\forall \varepsilon > 0\, \exists \delta > 0$ such that $\mu(E) < \delta$ implies $\|f\mathbf{1}_E\|_p < \varepsilon$.

Proof. Let $h = |f|^p \in L^1$. Choose $M > 0$ with $\int_{\{h > M\}} h\, d\mu < \frac{\varepsilon^p}{2}$, and set $\delta = \frac{\varepsilon^p}{2M}$.
If $\mu(E) < \delta$, then $\|f\mathbf{1}_E\|_p^p = \int_E h\, d\mu \le M\mu(E) + \int_{\{h > M\}} h\, d\mu < \varepsilon^p$.
Structural Commentary: L^p mass is absolutely continuous with respect to measure.
Key Insight: Truncate the tail; small measure controls the bounded part. □

Proposition 4.1.40. *Local Mass Limits for L^p Cauchy Sequences*
For $1 \le p < \infty$ and $f_n \in L^p(\mu)$, set $\beta_n(E) := \|f_n \mathbf{1}_E\|_p$ for $E \in \mathcal{M}$. The following hold.
(i) $|\beta_n(E) - \beta_m(E)| \le \|f_n - f_m\|_p$ for $E \in \mathcal{M}$.
(ii) If (f_n) is L^p-Cauchy, then $\forall E \in \mathcal{M}$, $\beta(E) := \lim_n \beta_n(E)$ exists.

Proof.
(i) Since $f_n \mathbf{1}_E = f_m \mathbf{1}_E + (f_n - f_m)\mathbf{1}_E$, $\beta_n(E) - \beta_m(E) \le \|f_n - f_m\|_p$.
Interchanging m, n gives $\beta_m(E) - \beta_n(E) \le \|f_n - f_m\|_p$; hence $|\beta_n(E) - \beta_m(E)| \le \|f_n - f_m\|_p$.
(ii) Fix $E \in \mathcal{M}$ and $\varepsilon > 0$. Choose N such that $m, n \ge N \Rightarrow \|f_n - f_m\|_p < \varepsilon$.
Then item (i) gives $|\beta_n(E) - \beta_m(E)| < \varepsilon$, so $(\beta_n(E))$ converges in $\mathbb{R}$.

Structural Commentary: Restricting to E is a 1-Lipschitz operation on L^p mass.
Key Insight: Local mass inherits global Cauchy control. □

Proposition 4.1.41. *Uniform Local Mass Control for L^p Cauchy Sequences*
Let $1 \le p < \infty$, f_n be L^p-Cauchy, and $\beta_n(E) := \|f_n \mathbf{1}_E\|_p$. Then $\forall \varepsilon > 0$:
(i) $\exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset$ implies $\sup_n \beta_n(F) < \varepsilon$;
(ii) $\exists \delta > 0$ such that $\mu(E) < \delta$ implies $\sup_n \beta_n(E) < \varepsilon$.

Proof. Choose N so $n \geq N$ implies $\|f_n - f_N\|_p < \frac{\varepsilon}{3}$.

(i) Choose finite-measure E_j for $1 \leq j \leq N$ with $F \cap E_j = \emptyset \Rightarrow \beta_j(F) < \frac{\varepsilon}{3}$, and set $E_\varepsilon = \bigcup_{j \leq N} E_j$.
For $n \leq N$ this works; for $n > N$, $\beta_n(F) \leq \|f_n - f_N\|_p + \beta_N(F) < \varepsilon$.

(ii) Choose $\delta_j > 0$ for $1 \leq j \leq N$ with $\mu(E) < \delta_j \Rightarrow \beta_j(E) < \frac{\varepsilon}{3}$, and set $\delta = \min_{j \leq N} \delta_j$.
For $n \leq N$ this works; for $n > N$, $\beta_n(E) \leq \|f_n - f_N\|_p + \beta_N(E) < \varepsilon$.

Structural Commentary: Cauchy families are uniformly tight and absolutely continuous.
Key Insight: Control finitely many early terms directly and the tail by Cauchy closeness. □

Method 4.1.42. *Lebesgue Proof Patterns*

(i) *Approximate*: for $f \geq 0$, use $s \in \mathcal{S}^+(f)$ and $\int_E f\, d\mu = \sup_{s \in \mathcal{S}^+(f)} \mathcal{I}_E(s)$.
(ii) *Promote*: prove for $\mathbf{1}_A$, extend to $s = \sum_i c_i \mathbf{1}_{A_i}$, then pass through $s_n \uparrow f$.
(iii) *Ignore null sets*: if $f \sim_{\text{a.e.}} g$ on E, replace f by g inside integrals over measurable $A \subseteq E$.
(iv) *Split signs*: use $f = f^+ - f^-$ and $h = u + iv$.
(v) *Exchange limits*: reduce Fatou and dominated convergence to $0 \leq f_n \uparrow f$.
(vi) *Slice*: prove boxes first; use regularity and null slices to pass to products and Fubini.

Key Insight: Reduce to simple functions, monotone convergence, null sets, and product boxes.

4.2 Modes of Convergence

Throughout this subsection $(X, \mathcal{M}, \mu)$ is a fixed measure space and, unless otherwise stated, functions are real-valued and measurable. The point is not to rank convergence modes linearly, but to know which hypotheses convert one mode into another.

Definition 4.2.1. *Pointwise, Uniform, and Almost-Everywhere Convergence*
Let (f_n) and f be functions on X.

(i) $f_n \to f$ *pointwise* if $\forall x \in X$, $\varepsilon > 0$, $\exists N(\varepsilon, x)$ s.t. $n \geq N(\varepsilon, x)$ implies $|f_n(x) - f(x)| < \varepsilon$.
(ii) $f_n \to f$ *uniformly* if $\forall \varepsilon > 0$ $\exists N(\varepsilon)$ s.t. $n \geq N(\varepsilon)$ implies $|f_n(x) - f(x)| < \varepsilon$ $\forall x \in X$.
(iii) $f_n \to f$ *almost everywhere* if $\exists A \in \mathcal{M}$ with $\mu(A) = 0$ s.t. $\forall x \in X \setminus A$, $\varepsilon > 0$, $\exists N(\varepsilon, x)$ s.t. $n \geq N(\varepsilon, x)$ implies $|f_n(x) - f(x)| < \varepsilon$.

Proposition 4.2.2. *First Implications and First Failures*
Let $f_n, f : X \to \mathbb{R}$.

(i) $f_n \rightrightarrows f \Rightarrow f_n \to f$ pointwise $\Rightarrow f_n \to f$ a.e.; converses fail.
(ii) $|X| < +\infty$ and $f_n \to f$ pointwise $\Rightarrow f_n \rightrightarrows f$.
(iii) $\mu(A) = 0 \Rightarrow A = \emptyset$ and $f_n \to f$ a.e. $\Rightarrow f_n \to f$ pointwise.

Proof.

(i) If $f_n \rightrightarrows f$, then $\forall \varepsilon > 0$ $\exists N_\varepsilon$ such that $\forall n \geq N_\varepsilon$ $\forall x \in X$, $|f_n(x) - f(x)| < \varepsilon$; hence pointwise convergence with $N(\varepsilon, x) = N_\varepsilon$.
Pointwise convergence is a.e. convergence with $A = \emptyset$.
Conversely, on $([0,1], \lambda)$, $x^n \to \mathbf{1}_{\{1\}}$ pointwise but not uniformly, and $\mathbf{1}_{\{0\}} \to 0$ a.e. but not pointwise.
(ii) If X is finite and $f_n \to f$ pointwise, then $\forall \varepsilon > 0$ $\forall x \in X$, choose $N(\varepsilon, x)$.
Set $N_\varepsilon = \max_{x \in X} N(\varepsilon, x)$; then $\forall n \geq N_\varepsilon$ $\forall x \in X$, $|f_n(x) - f(x)| < \varepsilon$, so $f_n \rightrightarrows f$.
(iii) If $f_n \to f$ a.e., then $\exists A \in \mathcal{M}$ with $\mu(A) = 0$ and $\forall x \in X \setminus A$, $f_n(x) \to f(x)$.
Since every null set is empty, $A = \emptyset$, so $f_n \to f$ pointwise.

Structural Commentary: Uniformity is bounded x-dependence of $N(\varepsilon, x)$.
Key Insight: Uniformity controls indices; a.e. convergence controls null exceptions. □

Proposition 4.2.3. *Uniform Convergence Gives L^p Convergence on Finite Measure Spaces*
Let $1 \leq p < \infty$, $\mu(X) < +\infty$, and $f_n \in L^p(\mu)$. If $f_n \rightrightarrows f$ on X, then $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.

Proof. For $\varepsilon > 0$, choose N with $n \geq N \Rightarrow |f_n - f| \leq \varepsilon$ on X.
Then $n \geq N \Rightarrow \|f_n - f\|_p \leq (\int_X \varepsilon^p\, d\mu)^{\frac{1}{p}} = \varepsilon \mu(X)^{\frac{1}{p}}$, so $\|f_n - f\|_p \to 0$.
Taking $\varepsilon = 1$, choose n_0 with $|f_{n_0} - f| \leq 1$ on X; hence $f - f_{n_0} \in L^p(\mu)$ and $f = f_{n_0} + (f - f_{n_0}) \in L^p(\mu)$.

Structural Commentary: Uniform error becomes L^p error after multiplication by $\mu(X)^{\frac{1}{p}}$.
Key Insight: Finite measure converts a pointwise sup-bound into an integral bound. □

Theorem 4.2.4. *Dominated Almost-Everywhere Convergence Gives L^p Convergence*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, and $f_n \to f$ a.e. Suppose $\exists g \in L^p(\mu)$ such that $\forall n$, $|f_n| \leq g$ a.e.

(i) $f \in L^p(\mu)$.
(ii) $\|f_n - f\|_p \to 0$.

Proof.

(i) Choose $N \in \mathcal{M}$ with $\mu(N) = 0$ such that on $X \setminus N$, $f_n \to f$ and $|f_n| \leq g$ $\forall n$.
Then $|f| \leq g$ a.e., so $|f|^p \leq g^p \in \mathscr{L}^1(\mu)$.
(ii) Since $|f_n - f|^p \to 0$ a.e. and $|f_n - f|^p \leq (2g)^p \in \mathscr{L}^1(\mu)$ a.e.,
Theorem 1.3.7 yields $\|f_n - f\|_p^p = \int_X |f_n - f|^p\, d\mu \to 0$.

Structural Commentary: Apply ordinary dominated convergence to the pth-power error.
Key Insight: A single L^p envelope controls both membership of the limit and convergence in mean. □

Corollary 4.2.5. *Uniform Boundedness on Finite Measure Spaces*
Let $\mu(X) < +\infty$, $0 \leq K < +\infty$, and $f_n \to f$ a.e. Assume $\forall n$, $|f_n| \leq K$ a.e. Then for every $1 \leq p < \infty$:

(i) $f \in L^p(\mu)$.

(ii) $\|f_n - f\|_p \to 0$.

Proof. Fix $1 \le p < \infty$ and set $g := K\mathbf{1}_X$. Then $\int_X g^p \, d\mu = K^p \mu(X) < +\infty$, so $g \in L^p(\mu)$ and $|f_n| \le g$ a.e.
(i) Since $f_n \to f$ a.e. and $|f_n| \le g \in L^p(\mu)$ a.e., Theorem 4.2.4(i) gives $f \in L^p(\mu)$.
(ii) The same hypotheses and Theorem 4.2.4(ii) give $\|f_n - f\|_p \to 0$.
Structural Commentary: Finite measure turns a uniform a.e. bound into an L^p envelope.
Key Insight: Bounded a.e. convergence on finite measure is dominated convergence. □

Example 4.2.6. *Uniform Need Not Imply L^p on Infinite Measure*
On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, fix $1 \le p < \infty$ and set $f_n := n^{-\frac{1}{p}} \mathbf{1}_{[0,n]}$. Then $\|f_n\|_\infty = n^{-\frac{1}{p}} \to 0$, so $f_n \rightrightarrows 0$.
However $\|f_n\|_p^p = \int_0^n n^{-1} \, d\lambda = 1$, hence $\|f_n\|_p \nrightarrow 0$.
Structural Commentary: The support grows exactly enough to preserve L^p mass.
Key Insight: Uniform smallness alone does not control integrals on infinite measure spaces.

Example 4.2.7. *Almost Everywhere and In-Measure Need Not Imply L^p*
On $[0,1]$ with Lebesgue measure, set $f_n := n\mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}$.
$\forall x \in [0,1]$, $x \in [\frac{1}{n}, \frac{2}{n}]$ for only finitely many n, hence $f_n(x) \to 0$.
If $\alpha > 0$ and $n \ge \alpha$, then $\lambda\{|f_n| \ge \alpha\} = \frac{1}{n} \to 0$, so $f_n \to 0$ in measure.
Yet for $1 \le p < \infty$, $\|f_n\|_p^p = n^p \lambda([\frac{1}{n}, \frac{2}{n}]) = n^{p-1} \nrightarrow 0$.
Given $\delta > 0$, choose $0 < \eta < \delta$; on $[0,1] \setminus [0, \eta]$, $\forall n > \frac{2}{\eta}$, $f_n = 0$, so $f_n \to 0$ almost uniformly.
Thus a.e., in-measure, and almost-uniform convergence do not imply L^p convergence.

Theorem 4.2.8. *Pointwise A.E. Convergence Plus Norms*
Let $1 \le p < \infty$, $f_n, f \in L^p(\mu)$, $f_n \to f$ a.e. Then $f_n \to f$ in $L^p(\mu)$ iff $\lim_{n\to\infty} \int_X |f_n|^p \, d\mu = \int_X |f|^p \, d\mu$.

Proof.
($\Rightarrow$) If $\|f_n - f\|_p \to 0$, then $\left| \|f_n\|_p - \|f\|_p \right| \le \|f_n - f\|_p$, hence $\int_X |f_n|^p \, d\mu = \|f_n\|_p^p \to \|f\|_p^p = \int_X |f|^p \, d\mu$.
($\Leftarrow$) Assume $\int_X |f_n|^p \, d\mu \to \int_X |f|^p \, d\mu$. Set $h_n := \frac{|f_n|^p + |f|^p}{2} - \left| \frac{f_n - f}{2} \right|^p \ge 0$ by convexity.
Since $h_n \to |f|^p$ a.e., Fatou gives $\int_X |f|^p \, d\mu \le \liminf_n \int_X h_n \, d\mu = \int_X |f|^p \, d\mu - 2^{-p} \limsup_n \int_X |f_n - f|^p \, d\mu$.
Thus $\limsup_n \|f_n - f\|_p^p \le 0$, so $f_n \to f$ in $L^p(\mu)$.
Structural Commentary: A.e. convergence identifies the limit; p-mass convergence prevents loss.
Key Insight: Fatou applied to the convexity defect forces the error norm to vanish. □

Definition 4.2.9. *Convergence and Cauchy Convergence in Measure*
Let (f_n) be measurable functions on X.
(i) $f_n \to f$ *in measure* if $\forall \alpha > 0$, $\mu\{x \in X : |f_n(x) - f(x)| \ge \alpha\} \to 0$.
(ii) (f_n) is *Cauchy in measure* if $\forall \alpha > 0$, $\mu\{x \in X : |f_m(x) - f_n(x)| \ge \alpha\} \to 0$ as $m, n \to \infty$.
The same definitions result with $>$ in place of $\ge$.

Proposition 4.2.10. *L^p Convergence Implies Convergence in Measure*
Let $1 \le p < \infty$. If $\|f_n - f\|_p \to 0$, then $f_n \to f$ in measure.

Proof. For $\alpha > 0$, set $E_n(\alpha) := \{|f_n - f| \ge \alpha\}$.
Since $\|f_n - f\|_p^p \ge \int_{E_n(\alpha)} |f_n - f|^p \, d\mu \ge \alpha^p \mu(E_n(\alpha))$, $\mu(E_n(\alpha)) \le \alpha^{-p} \|f_n - f\|_p^p \to 0$.
Structural Commentary: Large-error sets are controlled by the integral of the pth-power error.
Key Insight: This is Chebyshev's inequality in deterministic measure form. □

Example 4.2.11. *L^p Convergence Without Pointwise Convergence*
Let $X = [0,1]$ with Lebesgue measure and list $I_{m,k} := [\frac{k-1}{m}, \frac{k}{m}]$, $1 \le k \le m$, row by row in m.
Let $f_n := \mathbf{1}_{I_n}$, where I_n is the nth interval. If I_n lies in row $m(n)$, then $m(n) \to \infty$ and
$\|f_n\|_p^p = \lambda(I_n) = \frac{1}{m(n)} \to 0$; hence $f_n \to 0$ in L^p, therefore in measure by Proposition 4.2.10.
$\forall x \in [0,1]$, every row contains some $I_{m,k}$ with $x \in I_{m,k}$, giving a subsequence with value 1 at x; for infinitely many rows, some $I_{m,k}$ omits x, giving a subsequence with value 0 at x. Thus $(f_n(x))$ converges nowhere.
The subfamily $I_{m,2} = [\frac{1}{m}, \frac{2}{m}]$, $m \ge 2$, gives a subsequence converging pointwise to 0 on all of X.

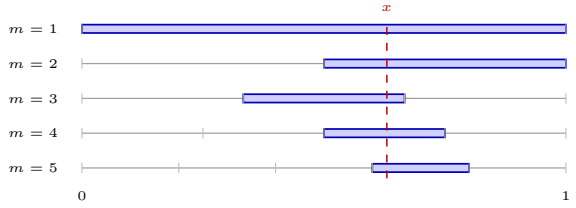

Figure 215: In each row, one shrinking typewriter interval contains the fixed point x.

Theorem 4.2.12. *Riesz Subsequence Principle for Convergence in Measure*
If (f_n) is Cauchy in measure, then there exist a subsequence (g_k) and a measurable f such that:
(i) $g_k \to f$ *a.e.*
(ii) $g_k \to f$ *in measure.*

(iii) $\forall \varepsilon > 0\ \exists E \in \mathcal{M}$ *with* $\mu(E) < \varepsilon$ *such that* $g_k \to f$ *uniformly on* $X \setminus E$.

Proof.

(i) Choose $n_1 < n_2 < \cdots$ so that $m, n \geq n_k \Rightarrow \mu\{|f_m - f_n| \geq \frac{1}{2^k}\} < \frac{1}{2^k}$, and set $g_k := f_{n_k}$.
For $E_k := \{|g_{k+1} - g_k| \geq \frac{1}{2^k}\}$, $\mu(E_k) < \frac{1}{2^k}$. Set $F_k := \bigcup_{j=k}^{\infty} E_j$. Then $F_k \downarrow$, $\mu(F_k) \leq \sum_{j=k}^{\infty} \frac{1}{2^j} = \frac{1}{2^{k-1}}$.
For $i > j \geq k$, $x \notin F_k$, $|g_i(x) - g_j(x)| \leq \sum_{r=j}^{i-1} |g_{r+1}(x) - g_r(x)| < \frac{1}{2^{j-1}}$.
Hence (g_j) is uniformly Cauchy on $X \setminus F_k$ $\forall k$.
Let $F := \bigcap_k F_k$, so $\mu(F) = 0$. Define $f = \lim_j g_j$ on $X \setminus F$, $f = 0$ on F. f is measurable, $g_j \to f$ a.e.

(ii) Letting $i \to \infty$ gives $j \geq k$, $x \notin F_k \Rightarrow |f(x) - g_j(x)| \leq \frac{1}{2^{j-1}} \leq \frac{1}{2^{k-1}}$. Given $\alpha, \varepsilon > 0$, choose k with $\frac{1}{2^{k-1}} < \alpha$ and $\mu(F_k) < \varepsilon$; then $j \geq k \Rightarrow \{|f - g_j| \geq \alpha\} \subseteq F_k$, so $g_j \to f$ in measure.

(iii) Given $\varepsilon > 0$, choose k with $\mu(F_k) < \varepsilon$.
The estimate in (ii) gives $\sup_{x \in X \setminus F_k} |f(x) - g_j(x)| \leq \frac{1}{2^{j-1}} \to 0$, so $g_j \to f$ uniformly on $X \setminus F_k$.

Structural Commentary: Cauchy in measure lets us select summably small exceptional sets.
Key Insight: Outside the tail union of exceptional sets, the subsequence is uniformly Cauchy. □

Corollary 4.2.13. *Completeness for Convergence in Measure*
(f_n) *is Cauchy in measure iff* $\exists f$ *measurable with* $f_n \to f$ *in measure.*

Proof.

(⇒) If (f_n) is Cauchy in measure, then $f_{n_k} \to f$ in measure for some measurable f by Theorem 4.2.12.
For $\alpha > 0$, $\{|f_n - f| \geq \alpha\} \subseteq \{|f_n - f_{n_k}| \geq \frac{\alpha}{2}\} \cup \{|f_{n_k} - f| \geq \frac{\alpha}{2}\}$.
Choose k large, then n large, to get $f_n \to f$ in measure.

(⇐) For $\alpha > 0$, $\{|f_m - f_n| \geq \alpha\} \subseteq \{|f_m - f| \geq \frac{\alpha}{2}\} \cup \{|f_n - f| \geq \frac{\alpha}{2}\}$.
Hence $\mu\{|f_m - f_n| \geq \alpha\} \to 0$ as $m, n \to \infty$.

Structural Commentary: The subsequential in-measure limit attracts the whole Cauchy sequence.
Key Insight: Convergence in measure is complete after measurable limits are admitted. □

Corollary 4.2.14. *Uniqueness for Convergence in Measure*
If $f_n \to f$ *and* $f_n \to g$ *in measure, then* $f = g$ *a.e.*

Proof. For $\alpha > 0$, $\{|f - g| \geq \alpha\} \subseteq \{|f - f_n| \geq \frac{\alpha}{2}\} \cup \{|f_n - g| \geq \frac{\alpha}{2}\}$, so $\mu\{|f - g| \geq \alpha\} = 0$.
Hence $\mu\{|f - g| > 0\} = 0$ and $f = g$ a.e.
Structural Commentary: Two in-measure limits cannot differ on a positive-measure set.
Key Insight: In-measure limits are unique modulo null sets. □

Corollary 4.2.15. *In-Measure Limits Have A.E. Subsequences*
If $f_n \to f$ *in measure, then some subsequence* f_{n_k} *converges to* f *a.e.*

Proof. Since $f_n \to f$ in measure, (f_n) is Cauchy in measure.
Choose a subsequence $f_{n_k} \to h$ a.e. and in measure by Theorem 4.2.12.
Then $h = f$ a.e. by Corollary 4.2.14; after enlarging the exceptional null set, $f_{n_k} \to f$ a.e.
Structural Commentary: Convergence in measure contains an almost-everywhere convergent subsequence.
Key Insight: Select a Cauchy-in-measure subsequence, then identify its limit by uniqueness. □

Proposition 4.2.16. *Subsequence and Indicator Consequences*

(i) $f_n \to f$ *in measure* ⇒ *every subsequence* $f_{n_k} \to f$ *in measure; if* (f_n) *is Cauchy in measure, then every* (f_{n_k}) *is Cauchy in measure.*
(ii) If $\|f_n - f\|_p \to 0$ *and* $\|f_{n_k} - g\|_p \to 0$, *then* $f = g$ *a.e.*
(iii) If $f_n = \mathbf{1}_{E_n}$ *and* $\|f_n - f\|_p \to 0$, *then* $f = \mathbf{1}_E$ *a.e. for some* $E \in \mathcal{M}$.

Proof.

(i) For $\alpha > 0$, $\mu\{|f_{n_k} - f| \geq \alpha\} \to 0$ since $n_k \to \infty$; the Cauchy claim is identical with $m_k, n_\ell \to \infty$.

(ii) $f_n \to f$ and $f_{n_k} \to g$ in measure by Proposition 4.2.10; also $f_{n_k} \to f$ in measure by (i).
Thus $f = g$ a.e. by Corollary 4.2.14.

(iii) By Proposition 4.2.10 and Theorem 4.2.12, some $f_{n_k} \to f$ a.e.
Since $f_{n_k} \in \{0, 1\}$, $f \in \{0, 1\}$ a.e.; with $E := \{f = 1\}$, $E \in \mathcal{M}$ and $f = \mathbf{1}_E$ a.e.

Structural Commentary: Subsequence permanence and a.e. uniqueness transfer mode limits to representatives.
Key Insight: In-measure subsequences preserve limits; indicator limits remain indicators a.e. □

Proposition 4.2.17. *Algebra of Convergence in Measure on Finite Measure Spaces*
Let $(X, \mathcal{M}, \mu)$ *satisfy* $\mu(X) < +\infty$. *All functions below are measurable and finite-valued a.e.*

(i) If $f_n \to f$ *and* $g_n \to g$ *in measure, then* $\forall a, b \in \mathbb{R}$, $af_n + bg_n \to af + bg$ *in measure.*
(ii) If $u_n \to 0$ *in measure and* h *is measurable and finite-valued a.e., then* $hu_n \to 0$ *in measure.*
(iii) If $f_n \to f$ *and* $g_n \to g$ *in measure, then* $f_n g_n \to fg$ *in measure.*
(iv) If $f_n \to f$ *in measure, then* $f_n^2 \to f^2$ *in measure.*

Proof.

(i) Fix $\alpha > 0$. Since $\{|a(f_n - f) + b(g_n - g)| \geq \alpha\} \subseteq \{|a(f_n - f)| \geq \frac{\alpha}{2}\} \cup \{|b(g_n - g)| \geq \frac{\alpha}{2}\}$, the measure of the right-hand side tends to 0 by convergence in measure of f_n and g_n.

(ii) Fix $\alpha, \eta > 0$. Since h is finite a.e. and $\mu(X) < +\infty$, choose $M > 0$ with $\mu\{|h| > M\} < \eta$.
Then $\{|hu_n| \geq \alpha\} \subseteq \{|h| > M\} \cup \{|u_n| \geq \frac{\alpha}{M}\}$, so $\limsup_n \mu\{|hu_n| \geq \alpha\} \leq \eta$. Let $\eta \downarrow 0$.

(iii) Set $u_n := f_n - f$ and $v_n := g_n - g$. Then $u_n, v_n \to 0$ in measure and $f_n g_n - fg = fv_n + gu_n + u_n v_n$. By (ii), $fv_n \to 0$ and $gu_n \to 0$ in measure. For $\alpha > 0$, $\{|u_n v_n| \geq \alpha\} \subseteq \{|u_n| \geq \sqrt{\alpha}\} \cup \{|v_n| \geq \sqrt{\alpha}\}$, so $u_n v_n \to 0$ in measure. Apply (i) to the three summands.
(iv) Apply (iii) with $g_n = f_n$ and $g = f$.

Structural Commentary: Finite measure makes finite measurable multipliers harmless after truncation.
Key Insight: Products split into fixed-multiplier errors and a product of two small-in-measure errors. □

Theorem 4.2.18. *Dominated Convergence in Measure Gives L^p Convergence*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, $f_n \to f$ in measure, and $|f_n| \leq g$ a.e. for some $g \in L^p(\mu)$. Then:
(i) $f \in L^p(\mu)$;
(ii) $\|f_n - f\|_p \to 0$.

Proof. Since $f_n \to f$ in measure, (f_n) is Cauchy in measure.

(i) By Theorem 4.2.12, some subsequence (u_j) satisfies $u_j \to h$ a.e. and in measure; $h = f$ a.e. by Corollary 4.2.14. Since $|u_j| \leq g$ a.e., $f \in L^p(\mu)$ by Theorem 4.2.4.
(ii) If norm convergence fails, then $\exists \varepsilon > 0$ and a subsequence (g_k) with $\|g_k - f\|_p \geq \varepsilon$. The argument in (i) produces a subsubsequence (h_j) with $h_j \to f$ a.e.; then $\|h_j - f\|_p \to 0$ by Theorem 4.2.4, a contradiction.

Structural Commentary: Reduce convergence in measure to almost-everywhere convergence along a subsequence.
Key Insight: Domination turns every in-measure subsequence into an L^p-convergent subsubsequence. □

Lemma 4.2.19. *Almost Uniform Cauchy Sequences Have Almost Uniform Limits*
If (f_n) is almost uniformly Cauchy, then $\exists f$ measurable such that $f_n \to f$ almost uniformly and a.e.

Proof. Choose $E_k \in \mathcal{M}$ with $\mu(E_k) < 2^{-k}$ and (f_n) uniformly Cauchy on $X \setminus E_k$. Set $F_k := \bigcup_{j=k}^{\infty} E_j$ and $F := \bigcap_k F_k$; then $F_k \downarrow$, $\mu(F_k) < 2^{-(k-1)}$, $\mu(F) = 0$, and (f_n) is uniformly Cauchy on $X \setminus F_k$. By Theorem 7.1.7, f_n has a uniform limit on each $X \setminus F_k$; these limits agree on overlaps, so define measurable f on $X \setminus F$, and set $f = 0$ on F. Hence $f_n \to f$ a.e. and uniformly on each $X \setminus F_k$; since $\mu(F_k) \to 0$, $f_n \to f$ almost uniformly.
Structural Commentary: Tail unions convert many small exceptional sets into nested uniform domains.
Key Insight: Almost uniform Cauchy means uniformly Cauchy after discarding a tail-small exceptional set. □

Theorem 4.2.20. *Almost Uniform Convergence and Convergence in Measure*
The following hold.
(i) If $f_n \to f$ almost uniformly, then $f_n \to f$ in measure.
(ii) If $f_n \to f$ in measure, then $\exists (f_{n_k})$ with $f_{n_k} \to f$ almost uniformly.

Proof.
(i) If $f_n \rightrightarrows f$ on $X \setminus E_\delta$ with $\mu(E_\delta) < \delta$, then $\forall \alpha > 0$ and sufficiently large n, $\{|f_n - f| \geq \alpha\} \subseteq E_\delta$. Thus $\limsup_n \mu\{|f_n - f| \geq \alpha\} \leq \delta$, and let $\delta \downarrow 0$.
(ii) Convergence in measure makes (f_n) Cauchy in measure. Choose $f_{n_k} \to h$ almost uniformly and in measure by Theorem 4.2.12. Uniqueness gives $h = f$ a.e.; absorb $\{h \neq f\}$ into the exceptional sets.

Structural Commentary: Almost uniform convergence is uniform convergence off arbitrarily small sets.
Key Insight: Convergence in measure always hides an almost-uniform subsequence. □

Corollary 4.2.21. *Finite-Measure Almost-Everywhere Convergence Implies In-Measure Convergence*
If $\mu(X) < +\infty$ and $f_n \to f$ a.e., with f_n, f finite-valued off a null set, then $f_n \to f$ in measure.

Proof. By Theorem 1.6.3, $f_n \to f$ almost uniformly; hence $f_n \to f$ in measure by Theorem 4.2.20.
Structural Commentary: Finite measure upgrades a.e. convergence through almost uniform convergence.
Key Insight: On finite measure spaces, a.e. convergence implies in-measure convergence. □

Corollary 4.2.22. *Finite-Measure Subsequence Test for Convergence in Measure*
Let $(X, \mathcal{M}, \mu)$ satisfy $\mu(X) < +\infty$, and let $f_n, f : X \to \overline{\mathbb{R}}$ be measurable and finite-valued a.e.
Then $f_n \to f$ in measure iff every subsequence of (f_n) has a further subsequence converging to f a.e.

Proof.
($\Rightarrow$) If $f_n \to f$ in measure, then every subsequence also converges to f in measure by Proposition 4.2.16. Choose a further subsequence converging to f a.e. by Corollary 4.2.15.
($\Leftarrow$) Suppose $f_n \not\to f$ in measure. Then $\exists \alpha, \varepsilon > 0$ and subsequence (f_{n_k}) with $\mu\{|f_{n_k} - f| \geq \alpha\} \geq \varepsilon$ $\forall k$. But some $f_{n_{k_j}} \to f$ a.e., hence in measure by Corollary 4.2.21, contradicting the uniform lower bound.

Structural Commentary: Subsequence tests characterize in-measure convergence.
Key Insight: Nonconvergence leaves a bad subsequence. □

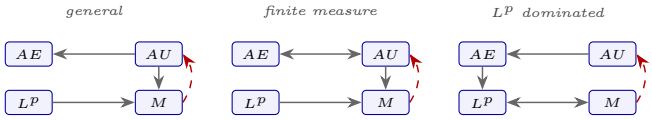

Figure 216: Modes of convergence in three settings: solid arrows are implications; dashed arrows yield convergent subsequences.

Method 4.2.23. *Counterexample Dictionary*

(i) On $(\mathbb{R},\lambda)$, $f_n = n^{-\frac{1}{p}}\mathbf{1}_{[0,n]}$ satisfies $f_n \rightrightarrows 0$ but $\|f_n\|_p = 1$.
(ii) On $([0,1],\lambda)$, $f_n = n\mathbf{1}_{[\frac{1}{n},\frac{2}{n}]}$ satisfies $f_n \to 0$ a.e., in measure, almost uniformly, but $\|f_n\|_p^p = n^{p-1} \nrightarrow 0$.
(iii) Typewriter intervals on $[0,1]$: if $f_n = \mathbf{1}_{I_n}$ and $\lambda(I_n) \to 0$ rowwise, then $\|f_n\|_p \to 0$ and $f_n \to 0$ in measure, but $(f_n(x))$ converges nowhere.
(iv) On $(\mathbb{R},\lambda)$, $f_n = \mathbf{1}_{[n,n+1]}$ satisfies $f_n \to 0$ pointwise but $\forall n,\ \lambda\{|f_n| \geq \frac{1}{2}\} = 1$, so $f_n \nrightarrow 0$ in measure.
(v) On $(\mathbb{R},\lambda)$, $f_n = \mathbf{1}_{[n,n+1]}$ is not almost uniformly convergent to 0: if $\lambda(E) < 1$ and $f_n \rightrightarrows 0$ on $\mathbb{R}\setminus E$, then $[n,n+1] \subseteq E$ for sufficiently large n, impossible.

Proposition 4.2.24. *An Integral Metric for Convergence in Measure on Finite Measure Spaces*
Let $\mu(X) < +\infty$ and $r(h) := \int_X |h|\,(1+|h|)^{-1}\,d\mu$. Then $f_n \to f$ in measure iff $r(f_n - f) \to 0$.

Proof.
($\Rightarrow$) Set $h_n := f_n - f$. For $\eta > 0$ and $A_n := \{|h_n| \geq \eta\}$, $r(h_n) \leq \mu(A_n) + \eta\mu(X)$. Let $n \to \infty$, then $\eta \downarrow 0$.
($\Leftarrow$) For $\alpha > 0$, $\alpha(1+\alpha)^{-1}\mathbf{1}_{\{|h_n|\geq\alpha\}} \leq |h_n|\,(1+|h_n|)^{-1}$, hence $\mu\{|h_n| \geq \alpha\} \leq (1+\alpha)\alpha^{-1} r(h_n) \to 0$.
Structural Commentary: The bounded error transform metrizes convergence in measure on finite measure spaces.
Key Insight: Small integral of the bounded error transform is equivalent to small large-error sets. □

Corollary 4.2.25. *Fatou and Dominated Convergence with In-Measure Hypotheses*
(i) $f_n \geq 0$ and $f_n \to f$ in measure $\Rightarrow \int_X f\,d\mu \leq \liminf_n \int_X f_n\,d\mu$.
(ii) $f_n \to f$ in measure and $|f_n| \leq g \in \mathscr{L}^1(\mu) \Rightarrow \int_X f_n\,d\mu \to \int_X f\,d\mu$.

Proof.
(i) Choose f_{n_k} with $\int f_{n_k}\,d\mu \to \liminf_n \int f_n\,d\mu$. By Theorem 4.2.12 and uniqueness, some $f_{n_{k_j}} \to f$ a.e.
Fatou gives $\int f\,d\mu \leq \liminf_j \int f_{n_{k_j}}\,d\mu = \liminf_n \int f_n\,d\mu$.
(ii) By Theorem 4.2.18 with $p=1$, $\|f_n - f\|_1 \to 0$, hence $\left|\int_X (f_n - f)\,d\mu\right| \leq \|f_n - f\|_1 \to 0$.
Structural Commentary: In-measure hypotheses reduce to a.e. subsequences or L^1 convergence.
Key Insight: Fatou and DCT survive convergence in measure through subsequences and domination. □

The next criterion packages the local mass controls needed to reverse convergence in measure into norm convergence.

Theorem 4.2.26. *Vitali Convergence Criterion for L^p*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, and let f be measurable. Consider the following conditions.
(i) $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.
(ii) $f_n \to f$ in measure.
(iii) $\forall \varepsilon > 0\ \exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F |f_n|^p\,d\mu < \varepsilon^p$.
(iv) $\forall \varepsilon > 0\ \exists \delta(\varepsilon) > 0$ such that $\mu(E) < \delta(\varepsilon) \Rightarrow \sup_n \int_E |f_n|^p\,d\mu < \varepsilon^p$.
Precisely, (i) is equivalent to the conjunction of (ii)–(iv).

Proof.
(i) Assume (i). Then (ii) holds by Proposition 4.2.10, and (f_n) is L^p-Cauchy. Conditions (iii)–(iv) follow from Proposition 4.1.41 after raising its norm bounds to the pth power.
(ii) Assume (ii)–(iv). Fix $\eta > 0$, choose E_η from (iii) and $\delta(\eta)$ from (iv), and put $F = X \setminus E_\eta$.
Then $\|(f_n - f_m)\mathbf{1}_F\|_p \leq \|f_n\mathbf{1}_F\|_p + \|f_m\mathbf{1}_F\|_p < 2\eta$.
If $\mu(E_\eta) > 0$, set $\alpha := \eta\mu(E_\eta)^{-\frac{1}{p}}$ and $H_{nm} := E_\eta \cap \{|f_n - f_m| \geq \alpha\}$.
By (ii), (f_n) is Cauchy in measure, so $\mu(H_{nm}) < \delta(\eta)$ for large m,n. On $E_\eta \setminus H_{nm}$, $|f_n - f_m| < \alpha$; on H_{nm}, use $|f_n - f_m| \leq |f_n| + |f_m|$. Hence $\|(f_n - f_m)\mathbf{1}_{E_\eta}\|_p \leq \alpha\mu(E_\eta)^{\frac{1}{p}} + \|f_n\mathbf{1}_{H_{nm}}\|_p + \|f_m\mathbf{1}_{H_{nm}}\|_p < 3\eta$ by (iv). If $\mu(E_\eta) = 0$, the E_η term is 0.
Thus $\|f_n - f_m\|_p < 5\eta$ eventually, so (f_n) is L^p-Cauchy.
By Riesz–Fischer, $f_n \to h$ in L^p; by Proposition 4.2.10 and Corollary 4.2.14, $h = f$ a.e., so (i) holds.
Structural Commentary: Vitali = in-measure convergence + tightness + uniform small-set control.
Key Insight: Tightness and small-set control convert in-measure convergence into an L^p Cauchy estimate. □

4.3 Composition and Convergence Modes

Proposition 4.3.1. *Composition and Pointwise Modes*
Let $\varphi : \mathbb{R} \to \mathbb{R}$.
(i) φ continuous, $f_n \to f$ a.e. $\Rightarrow \varphi \circ f_n \to \varphi \circ f$ a.e.
(ii) φ uniformly continuous preserves uniform, a.u., and in-measure convergence.
(iii) If φ is discontinuous, composition need not preserve a.e. convergence.
(iv) Non-uniform continuity can destroy uniform, even in-measure, convergence.

Proof.
(i) Outside a null set, $f_n(x) \to f(x)$; continuity gives $\varphi(f_n(x)) \to \varphi(f(x))$.
(ii) Given $\varepsilon > 0$, choose $\delta > 0$ with $|s-t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$. Uniform and almost-uniform convergence use $\sup|f_n - f| < \delta$ on X or $X \setminus E_\eta$; in measure, $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \delta\}$.
(iii) If φ is discontinuous at a, choose $t_n \to a$ and $\alpha > 0$ with $|\varphi(t_n) - \varphi(a)| \geq \alpha$.
On $X = \{*\}$, $f_n(*) = t_n \to a = f(*)$, but $\varphi(f_n(*)) \nrightarrow \varphi(f(*))$.
(iv) If φ is not uniformly continuous, choose a_n, b_n with $|a_n - b_n| \to 0$ and $|\varphi(a_n) - \varphi(b_n)| \geq \alpha > 0$.
On $(\mathbb{N},\#)$, set $f(k) = b_k$ and $f_n(k) = a_n$ for $k = n$, $f_n(k) = b_k$ for $k \neq n$.
Then $\|f_n - f\|_\infty \to 0$, but $\forall n,\ \#\{|\varphi(f_n) - \varphi(f)| \geq \alpha\} = 1$.

Structural Commentary: Continuity is pointwise; uniform continuity gives one global δ.
Key Insight: Composition needs the same uniformity as the convergence mode. □

Proposition 4.3.2. *Composition and Convergence in Measure*
Let $f_n, f : X \to \mathbb{R}$ be measurable with $f_n \to f$ in measure, and let $\varphi : \mathbb{R} \to \mathbb{R}$.

(i) If φ is uniformly continuous, then $\forall \varepsilon > 0$, $\mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \to 0$.
(ii) If $\mu(X) < +\infty$ and φ is continuous, then $\forall \varepsilon > 0$, $\mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \to 0$.

Proof.

(i) Fix $\varepsilon > 0$ and choose $\delta > 0$ with $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$.
Then $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \delta\}$, so the measures tend to 0.

(ii) Fix $\varepsilon, \eta > 0$. Since $\mu(X) < +\infty$ and f is finite-valued, choose M with $\mu\{|f| > M\} < \eta$. Uniform continuity of φ on $[-M-1, M+1]$ gives $0 < \delta < 1$ such that $s, t \in [-M-1, M+1]$ and $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$. Hence $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f| > M\} \cup \{|f_n - f| \geq \delta\}$, so $\limsup_n \mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \leq \eta \downarrow 0$.

Structural Commentary: Use global uniformity, or truncate to a bounded range.
Key Insight: On finite measure spaces, measurable finite-valued functions are almost bounded. □

Example 4.3.3. *Continuity Alone Is Not Enough on Infinite Measure*
On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, set $f(x) = x$, $f_n(x) = x + \frac{1}{n}$, and $\varphi(t) = t^2$.
Then $\|f_n - f\|_\infty = \frac{1}{n} \to 0$, hence $f_n \rightrightarrows f$ and $f_n \to f$ in measure.
However $\varphi(f_n) - \varphi(f) = \frac{2x}{n} + \frac{1}{n^2}$; for $E_n := \{|\varphi(f_n) - \varphi(f)| \geq 1\}$, $[\frac{n}{2}, +\infty) \subseteq E_n$, so $\lambda(E_n) = +\infty$.
Thus $\varphi \circ f_n \not\to \varphi \circ f$ in measure.

Proposition 4.3.4. *Growth Conditions for L^p Composition*
Let $1 \leq p < \infty$, let $\varphi : \mathbb{R} \to \mathbb{R}$ be measurable, and let $f_n, f : X \to \mathbb{R}$ be measurable.

(i) If $\exists A, B \geq 0$ such that $\forall t$, $|\varphi(t)| \leq A|t| + B$, $\mu(X) < +\infty$, and $f \in L^p(\mu)$, then $\varphi \circ f \in L^p(\mu)$.
(ii) If $\exists A \geq 0$ such that $\forall t$, $|\varphi(t)| \leq A|t|$, and $f \in L^p(\mu)$, then $\varphi \circ f \in L^p(\mu)$.
(iii) If $\mu(X) < +\infty$, φ is uniformly continuous, $f_n, f \in L^p(\mu)$, and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.
(iv) If φ is Lipschitz, $\varphi(0) = 0$, $f_n, f \in L^p(\mu)$, and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.

Proof.

(i) $|\varphi(f)|^p \leq 2^{p-1}(A^p |f|^p + B^p)$, so $\|\varphi \circ f\|_p^p \leq 2^{p-1}(A^p \|f\|_p^p + B^p \mu(X)) < +\infty$.

(ii) $|\varphi(f)|^p \leq A^p |f|^p$, so $\|\varphi \circ f\|_p \leq A \|f\|_p < +\infty$.

(iii) Uniform continuity gives constants $C, D \geq 0$ with $|\varphi(t)| \leq C|t| + D$ and $|\varphi(s) - \varphi(t)| \leq C|s - t| + D$.
Hence all compositions lie in L^p by (i). Let $h_n := |f_n - f|$.
Given $\eta > 0$, choose $\delta > 0$ with $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \eta$ and set $H_n := \{h_n \geq \delta\}$.
Then $\int_X |\varphi(f_n) - \varphi(f)|^p \, d\mu \leq \eta^p \mu(X) + 2^{p-1} \int_{H_n} (C^p h_n^p + D^p) \, d\mu$.
Since $\int h_n^p \, d\mu \to 0$ and $\mu(H_n) \leq \delta^{-p} \int h_n^p \, d\mu \to 0$, take $\limsup_n$ and then $\eta \downarrow 0$.

(iv) If L is Lipschitz, $|\varphi(f)| \leq L|f|$ and $\|\varphi \circ f_n - \varphi \circ f\|_p \leq L \|f_n - f\|_p \to 0$.

Structural Commentary: Growth gives membership; continuity estimates transfer L^p convergence.
Key Insight: Finite measure makes small uniform errors integrable; Lipschitz bounds give norm control. □

Example 4.3.5. *Uniform Continuity Does Not Suffice on Infinite Measure*
On $(\mathbb{R}, \lambda)$, fix $1 \leq p < \infty$, set $f = 0$, $f_n := n^{-1} \mathbf{1}_{[0, n^{\frac{p}{2}}]}$, and $\varphi(t) := \sqrt{|t|}$.
Then φ is uniformly continuous, $\varphi(0) = 0$, $\|f_n - f\|_p^p = n^{-\frac{p}{2}} \to 0$, but $\forall n$, $\|\varphi \circ f_n - \varphi \circ f\|_p^p = 1$.
Thus uniform continuity need not preserve L^p convergence on infinite measure spaces.

Lemma 4.3.6. *L^p Convergence Allows Continuous Composition in Measure*
Let $1 \leq p < \infty$, $f_n, f \in L^p(\mu)$, $\|f_n - f\|_p \to 0$, and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous.
Then $\varphi \circ f_n \to \varphi \circ f$ in measure.

Proof. Fix $\alpha, \eta > 0$. Choose M with $\mu\{|f| > M\} < \eta$. By uniform continuity of φ on $[-M-1, M+1]$, choose $0 < \delta < 1$ such that $s, t \in [-M-1, M+1]$ and $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \alpha$.
Then $\{|\varphi(f_n) - \varphi(f)| \geq \alpha\} \subseteq \{|f| > M\} \cup \{|f_n - f| \geq \delta\}$, so $\limsup_n \mu\{|\varphi(f_n) - \varphi(f)| \geq \alpha\} \leq \eta$. Let $\eta \downarrow 0$.

Structural Commentary: L^p convergence localizes to a large bounded range where continuity is uniform.
Key Insight: Norm convergence gives in-measure control; bounded-range continuity transfers it. □

Theorem 4.3.7. *Composition Growth Criteria in L^p*
Let $1 \leq p < \infty$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous.

(i) If $\mu(X) < +\infty$ and $\exists K > 0$ such that $|\varphi(t)| \leq K|t|$ for $|t| \geq K$, then $f \in L^p(\mu) \Rightarrow \varphi \circ f \in L^p(\mu)$.
(ii) Under (i), if $f_n, f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.
(iii) If $\exists K \geq 0$ such that $\forall t$, $|\varphi(t)| \leq K|t|$, then $f \in L^p(\mu) \Rightarrow \varphi \circ f \in L^p(\mu)$ on any measure space.
(iv) Under (iii), if $f_n, f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$ on any measure space.

Proof.

(i) Let $M := \sup_{|t| \leq K} |\varphi(t)|$, $C := K + M$. Then $\forall t$, $|\varphi(t)| \leq C(1 + |t|)$; hence $|\varphi(f)|^p \leq 2^{p-1} C^p (1 + |f|^p) \in L^1(\mu)$.

(ii) Let C be as in (i). Then $\varphi \circ f_n \to \varphi \circ f$ in measure by Lemma 4.3.6.
Since (f_n) is L^p-Cauchy, Proposition 4.1.41 and the growth bound from (i) supply the finite-tail and small-set estimates. Thus $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$ by Theorem 4.2.26.

(iii) $|\varphi(f)|^p \leq K^p |f|^p \in L^1(\mu)$.

(iv) $\varphi \circ f_n \to \varphi \circ f$ in measure by Lemma 4.3.6. Since (f_n) is L^p-Cauchy and $|\varphi(f_n)|^p \le K^p |f_n|^p$, the finite-tail and small-set estimates for $(\varphi \circ f_n)$ follow from Proposition 4.1.41.
Thus $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$ by Theorem 4.2.26.

Structural Commentary: Local growth needs finite measure; global growth does not.
Key Insight: Vitali converts in-measure composition plus inherited mass control into L^p convergence. □

Method 4.3.8. *Atomic Counterexamples for Composition Growth*

(i) If finite-measure tail growth fails, choose $|t_n| \ge n$ with $|\varphi(t_n)| > n^2 |t_n|$. On $\mathbb{N}$, set $\mu(\{n\}) = (n^2 |t_n|^p)^{-1}$ and $f(n) = t_n$. Then $\mu(\mathbb{N}) < +\infty$, $\int |f|^p \, d\mu = \sum_n n^{-2} < +\infty$, but $\int |\varphi(f)|^p \, d\mu \ge \sum_n n^{2p-2} = +\infty$.

(ii) If the same tail growth fails, choose $|t_n| \ge n$ with $|\varphi(t_n) - \varphi(0)| \ge n |t_n|$. With $\mu(\{n\}) = (n^p |t_n|^p)^{-1}$, $f = 0$, and $f_n = t_n \mathbf{1}_{\{n\}}$, one has $\mu(\mathbb{N}) < +\infty$, $\|f_n - f\|_p = \frac{1}{n} \to 0$, but $\|\varphi \circ f_n - \varphi \circ f\|_p \ge 1$.

(iii) If global linear growth fails and $\varphi(0) = 0$, the constructions in (i)–(ii), without requiring $\mu(\mathbb{N}) < +\infty$, give membership and convergence counterexamples on arbitrary measure spaces.

(iv) If $\varphi(0) \ne 0$, then on $(\mathbb{N}, \#)$, $0 \in L^p$ but $\varphi \circ 0$ is a nonzero constant and hence not in L^p; the constant sequence $f_n = 0$ gives the convergence failure.

4.4 Uniform Integrability and Vitali Convergence

Uniform integrability is the L^1 condition that rules out concentration of mass on tiny sets. On infinite-measure spaces it must be paired with tightness, which rules out mass escaping to remote finite-measure tails.

Lemma 4.4.1. *Finite-Measure Lebesgue Sets Have Small Pieces*
Let $E \in \mathcal{L}^1$, $m(E) < +\infty$, and $\delta > 0$. Then $\exists E_1, \dots, E_N \in \mathcal{L}^1$ such that:

(i) $E_i \cap E_j = \emptyset$ *for* $i \ne j$.

(ii) $E = \bigsqcup_{k=1}^{N} E_k$.

(iii) $m(E_k) < \delta$ *for* $1 \le k \le N$.

Proof. Choose n_0 with $T := E \setminus [-n_0, n_0]$ and $m(T) < \delta$. Partition $[-n_0, n_0]$ into finitely many disjoint intervals I_j with $m(I_j) < \delta$, and let (E_k) be the nonempty sets among T and $E \cap I_j$.

(i) Since $T \cap [-n_0, n_0] = \emptyset$ and the I_j are disjoint, the E_k are pairwise disjoint.

(ii) $E = T \bigsqcup (E \cap [-n_0, n_0]) = T \bigsqcup \bigsqcup_j (E \cap I_j) = \bigsqcup_{k=1}^{N} E_k$.

(iii) $m(T) < \delta$ and $m(E \cap I_j) \le m(I_j) < \delta$.

Structural Commentary: Finite measure gives a small tail T; a fine partition makes $E \setminus T$ small-piece.
Key Insight: One small tail plus a fine finite grid gives a finite small-piece decomposition. □

Proposition 4.4.2. *L^1 Absolute Continuity on Small Sets*
Let $E \in \mathcal{L}^1$, let $\mathcal{L}^1_E = \{A \in \mathcal{L}^1 : A \subseteq E\}$, let $m_E = m|_{\mathcal{L}^1_E}$, and let $f : E \to \mathbb{R}$ be measurable.
Consider the condition $(*)$: $\forall \varepsilon > 0 \ \exists \delta > 0$ such that $A \in \mathcal{L}^1_E$ and $m(A) < \delta$ imply $\int_A |f| \, dm < \varepsilon$.

(i) If $f \in L^1(m_E)$, then $(*)$ holds.
(ii) If $m(E) < +\infty$ and $(*)$ holds, then $f \in L^1(m_E)$.

Proof.

(i) Since $f \in L^1(m_E)$, Lemma 4.1.39 on $(E, \mathcal{L}^1_E, m_E)$ with $p = 1$ gives $\forall \varepsilon > 0 \ \exists \delta > 0$ such that $A \in \mathcal{L}^1_E$ and $m_E(A) < \delta$ imply $\int_A |f| \, dm_E < \varepsilon$. Since $m_E(A) = m(A)$ and $\int_A |f| \, dm_E = \int_A |f| \, dm$, $(*)$ holds.

(ii) Choose $\delta > 0$ from $(*)$ with $\varepsilon = 1$. By Lemma 4.4.1, write $E = \bigsqcup_{k=1}^{N} E_k$ with $E_k \in \mathcal{L}^1_E$ and $m(E_k) < \delta$. Then $(*)$ gives $\int_{E_k} |f| \, dm < 1$ for each k, hence $\int_E |f| \, dm = \sum_{k=1}^{N} \int_{E_k} |f| \, dm < N < +\infty$.

Structural Commentary: L^1 gives small-set control; finite partitions recover L^1.
Key Insight: A finite small-piece decomposition turns local integral bounds into one global finite integral. □

Definition 4.4.3. *Uniform Integrability and Tightness*
Let $\mathcal{F}$ be a family of measurable functions $h : X \to \mathbb{R}$ on $(X, \mathcal{M}, \mu)$.

(i) $\mathcal{F}$ is *uniformly integrable* on X if $\forall \varepsilon > 0 \, \exists \delta > 0$ such that $A \in \mathcal{M}$ and $\mu(A) < \delta$ imply $\sup_{h \in \mathcal{F}} \int_A |h| \, d\mu < \varepsilon$.

(ii) $\mathcal{F}$ is *tight* on X if $\forall \varepsilon > 0 \, \exists X_\varepsilon \in \mathcal{M}$ with $\mu(X_\varepsilon) < +\infty$ and $\sup_{h \in \mathcal{F}} \int_{X \setminus X_\varepsilon} |h| \, d\mu < \varepsilon$.

On finite measure spaces, tightness is automatic by taking $X_\varepsilon = X$.
Key Insight: UI controls mass on small A; tightness controls mass on $X \setminus X_\varepsilon$.

Example 4.4.4. *Concentration versus Escape*

(i) *Concentration*: on $(0, 1)$, $u_n := n \mathbf{1}_{(0, \frac{1}{n})}$. For every $\delta > 0$, choose $n > \delta^{-1}$ and $A_n = (0, \frac{1}{n})$.
Then $m(A_n) < \delta$ but $\int_{A_n} u_n \, dm = 1$, so (u_n) is not uniformly integrable.

(ii) *Escape*: on $\mathbb{R}$, let $I_n := (n, n+1)$ and $v_n := \mathbf{1}_{I_n}$. Since $\int_A v_n \, dm \le m(A)$, (v_n) is uniformly integrable.
If $m(E_0) < +\infty$ and $\sup_n \int_{\mathbb{R} \setminus E_0} v_n \, dm < \frac{1}{2}$, then $m(E_0 \cap I_n) > \frac{1}{2}$ for all n, contradiction.

Structural Commentary: Concentration occurs even on $m(X) < +\infty$; escape is an $m(X) = +\infty$ obstruction.
Key Insight: Uniform integrability controls small-set mass; tightness keeps mass in finite-measure cores.

Proposition 4.4.5. *Basic Sources of Uniform Integrability and Tightness*
Let $(X, \mathcal{M}, \mu)$ be a measure space.
(i) If $0 \le g \in L^1(\mu)$ and $\forall h \in \mathcal{F}$, $|h| \le g$ a.e., then $\mathcal{F}$ is uniformly integrable and tight.
(ii) If $f_1, \dots, f_N \in L^1(\mu)$, then $\{f_1, \dots, f_N\}$ is uniformly integrable and tight.

Proof. Let $\varepsilon > 0$.
(i) Choose $\delta > 0$ with $\mu(A) < \delta \Rightarrow \int_A g \, d\mu < \varepsilon$ by Lemma 4.1.39; hence $\sup_{h \in \mathcal{F}} \int_A |h| \, d\mu < \varepsilon$.
Choose finite-measure X_ε with $\int_{X \setminus X_\varepsilon} g \, d\mu < \varepsilon$ by Proposition 1.3.22; hence $\sup_{h \in \mathcal{F}} \int_{X \setminus X_\varepsilon} |h| \, d\mu < \varepsilon$.
(ii) For each j, apply (i) to $\{f_j\}$ with ε: choose $\delta_j > 0$ such that $\mu(A) < \delta_j \Rightarrow \int_A |f_j| \, d\mu < \varepsilon$, and choose $X_{\varepsilon,j}$ with $\mu(X_{\varepsilon,j}) < +\infty$ and $\int_{X \setminus X_{\varepsilon,j}} |f_j| \, d\mu < \varepsilon$. Set $\delta := \min_{1 \le j \le N} \delta_j$ and $X_\varepsilon := \bigcup_{j=1}^{N} X_{\varepsilon,j}$.
Then $\mu(A) < \delta \Rightarrow \sup_j \int_A |f_j| \, d\mu < \varepsilon$, while $\mu(X_\varepsilon) < +\infty$ and $\sup_j \int_{X \setminus X_\varepsilon} |f_j| \, d\mu < \varepsilon$.

Structural Commentary: An envelope gives common bounds; finite families use $\min_j \delta_j$ and $\bigcup_j X_{\varepsilon,j}$.
Key Insight: Domination gives uniform estimates; finite families allow minima and finite unions. □

Corollary 4.4.6. *L^p-Bounded Families Are Uniformly Integrable*
Let $1 < p < \infty$, q conjugate to p. If $\mathcal{F} \subseteq L^p(\mu)$ and $\forall h \in \mathcal{F}$, $\|h\|_p \le M$, then $\mathcal{F}$ is uniformly integrable.

Proof. If $M = 0$, every $h \in \mathcal{F}$ vanishes a.e. and the claim is immediate. Else, for measurable A with $\mu(A) < +\infty$,
by Hölder's inequality, $\int_A |h| \, d\mu = \int_X |h| \, \mathbf{1}_A \, d\mu \le \|h\|_p \, \|\mathbf{1}_A\|_q \le M \mu(A)^{\frac{1}{q}}$.
Given $\varepsilon > 0$, take $\delta = (\frac{\varepsilon}{M})^q$. Then $\mu(A) < \delta$ implies $\sup_{h \in \mathcal{F}} \int_A |h| \, d\mu < \varepsilon$.

Structural Commentary: Apply Hölder to $h\mathbf{1}_A$ and choose one measure threshold for the whole family.
Key Insight: Hölder converts a uniform L^p bound into uniform small-set L^1 control. □

Proposition 4.4.7. *Uniform Integrability by Large-Value Tails*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space and let (f_n) be bounded in $L^1(\mu)$. The following are equivalent.
(i) (f_n) is uniformly integrable.
(ii) $\forall \varepsilon > 0 \ \exists M > 0 : \ \sup_n \int_{\{|f_n| \ge M\}} |f_n| \, d\mu < \varepsilon$.

Proof. Let $C = \sup_n \|f_n\|_1 < +\infty$.
($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ from uniform integrability and $M > C/\delta$.
By Chebyshev, $\mu(\{|f_n| \ge M\}) \le M^{-1} \|f_n\|_1 \le C/M < \delta$, so $\sup_n \int_{\{|f_n| \ge M\}} |f_n| \, d\mu < \varepsilon$.
($\Leftarrow$) Given $\varepsilon > 0$, choose $M > 0$ with $\sup_n \int_{\{|f_n| \ge M\}} |f_n| \, d\mu < \frac{\varepsilon}{2}$ and set $\delta := \frac{\varepsilon}{2M}$.
If $\mu(A) < \delta$, then $\int_A |f_n| \, d\mu \le \int_{\{|f_n| \ge M\}} |f_n| \, d\mu + M\mu(A) < \varepsilon$.

Structural Commentary: Chebyshev shrinks tails; truncation recovers UI.
Key Insight: UI iff large tails vanish uniformly. □

Theorem 4.4.8. *Vitali Convergence for L^1 Families*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $f_n \in L^1(\mu)$, let $f : X \to \mathbb{R}$ be measurable, and assume $f_n \to f$ a.e.
(i) If $\mu(X) < +\infty$ and $\{f_n : n \ge 1\}$ is uniformly integrable, then $f \in L^1(\mu)$ and $\|f_n - f\|_1 \to 0$.
(ii) If $\{f_n : n \ge 1\}$ is uniformly integrable and tight, then $f \in L^1(\mu)$ and $\|f_n - f\|_1 \to 0$.
(iii) Under either hypothesis, $\int_X f_n \, d\mu \to \int_X f \, d\mu$.

Proof.
(i) Since $\mu(X) < +\infty$ and $f_n \to f$ a.e., $f_n \to f$ in measure by Corollary 4.2.21. Uniform integrability is the small-set hypothesis of Theorem 4.2.26 with $p = 1$; the finite-tail hypothesis holds with $E_\varepsilon = X$, QED.
(ii) Fix $\alpha, \eta > 0$ and set $B_n^\alpha := \{|f_n - f| \ge \alpha\}$.
By tightness, choose $X_0 \in \mathcal{M}$ with $\mu(X_0) < +\infty$, $\sup_n \int_{X \setminus X_0} |f_n| \, d\mu < \frac{\alpha\eta}{4}$. By Fatou, $\int_{X \setminus X_0} |f| \, d\mu \le \frac{\alpha\eta}{4}$.
Then $\alpha\mu(B_n^\alpha \cap (X \setminus X_0)) \le \int_{X \setminus X_0} (|f_n| + |f|) \, d\mu < \frac{\alpha\eta}{2}$. Since $f_n \to f$ a.e. on finite X_0, $\mu(B_n^\alpha \cap X_0) < \frac{\eta}{2}$ eventually by Corollary 4.2.21. Thus $\mu(B_n^\alpha) < \eta$ eventually, so $f_n \to f$ in measure.
By Theorem 4.2.26 with $p = 1$, using uniform integrability and tightness, $f \in L^1(\mu)$ and $\|f_n - f\|_1 \to 0$.
(iii) In either case, $\left|\int_X f_n \, d\mu - \int_X f \, d\mu\right| = \left|\int_X (f_n - f) \, d\mu\right| \le \|f_n - f\|_1 \to 0$.

Structural Commentary: Tightness gives finite cores; UI upgrades in-measure convergence to L^1.
Key Insight: Uniform integrability prevents concentration; tightness prevents escape. □

Corollary 4.4.9. *Nonnegative Integral Vanishing Criterion*
Let $f_n \in L^1(\mu)$ with $f_n \ge 0$ a.e. The following are equivalent.
(i) $\int_X f_n \, d\mu \to 0$.
(ii) $\|f_n\|_1 \to 0$.
(iii) $f_n \to 0$ in measure, and $\{f_n : n \ge 1\}$ is uniformly integrable and tight.

Proof.
(i) $\Rightarrow$ (ii) Since $f_n \ge 0$, $\|f_n\|_1 = \int_X f_n \, d\mu \to 0$.
(ii) $\Rightarrow$ (iii) By Theorem 4.2.26 with $p = 1$ and limit 0, $\|f_n\|_1 \to 0$ implies $f_n \to 0$ in measure,
$\forall \varepsilon > 0 \ \exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ and $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F f_n \, d\mu < \varepsilon$,
and $\forall \varepsilon > 0 \ \exists \delta > 0$ with $\mu(A) < \delta \Rightarrow \sup_n \int_A f_n \, d\mu < \varepsilon$; hence tightness and uniform integrability.

(*iii*) $\Rightarrow$ (*i*) Condition (iii) is hypotheses (ii)–(iv) of Theorem 4.2.26 with $p = 1$ and limit 0, so $\|f_n\|_1 \to 0$.
Since $f_n \geq 0$, $\int_X f_n \, d\mu = \|f_n\|_1 \to 0$.

Structural Commentary: For nonnegative functions, integral convergence to zero is L^1 convergence to zero.
Key Insight: Vitali's conditions characterize when nonnegative mass vanishes. □

Corollary 4.4.10. *Domination Gives Tail and Small-Set Control*
Let $1 \leq p < \infty$, $g \in L^p(\mu)$, and $\forall n$, $|f_n| \leq g$ a.e. Then $\forall \varepsilon > 0$:
(i) $\exists E_\varepsilon \in \mathcal{M}$ *with* $\mu(E_\varepsilon) < +\infty$ *such that* $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F |f_n|^p \, d\mu < \varepsilon^p$.
(ii) $\exists \delta(\varepsilon) > 0$ *such that* $\mu(E) < \delta(\varepsilon) \Rightarrow \sup_n \int_E |f_n|^p \, d\mu < \varepsilon^p$.

Proof.
(i) Choose E_ε with $\mu(E_\varepsilon) < +\infty$ and $F \cap E_\varepsilon = \emptyset \Rightarrow \int_F |g|^p \, d\mu < \varepsilon^p$ by Proposition 4.1.26.
Since $|f_n|^p \leq |g|^p$ a.e., $\sup_n \int_F |f_n|^p \, d\mu < \varepsilon^p$.
(ii) Choose $\delta(\varepsilon) > 0$ with $\mu(E) < \delta(\varepsilon) \Rightarrow \int_E |g|^p \, d\mu < \varepsilon^p$ by Lemma 4.1.39.
Since $|f_n|^p \leq |g|^p$ a.e., $\sup_n \int_E |f_n|^p \, d\mu < \varepsilon^p$.

Structural Commentary: One L^p envelope supplies both tail tightness and small-set control.
Key Insight: Domination transfers both estimates from g to every f_n. □

4.5 Approximation and Separability in L^p

Approximation results explain why L^p spaces are large enough to contain rough measurable functions but still small enough to be controlled by countable data when $p < \infty$. The guiding pattern is to localize the L^p mass, approximate measurable atoms by elementary sets, and then code the elementary sets with rational parameters.

Definition 4.5.1. *Dense Subsets and Separable Normed Spaces*
Let $A \subseteq B$ be subsets of a normed space $(V, \|\cdot\|)$.
(i) A is *dense in* B if $\forall b \in B \; \forall \varepsilon > 0 \; \exists a \in A$ with $\|a - b\| < \varepsilon$.
(ii) V is *separable* if it has a countable dense subset.
Equivalently, A is dense in B iff every $b \in B$ is the norm-limit of a sequence from A.

Example 4.5.2. *Sequence-Space Separability*
(i) $\ell^\infty(\mathbf{F})$ is not separable. The set $\{0,1\}^{\mathbb{N}}$ is uncountable, and $s \neq t$ gives $d_\infty(s,t) = 1$; the balls $B_{1/3}(s)$ are pairwise disjoint, so no countable dense set can meet all of them.
(ii) For $1 \leq p < \infty$, $\ell^p(\mathbf{F})$ is separable. Let D be the countable set of finite-support sequences with rational coordinates, using rational real and imaginary parts if $\mathbf{F} = \mathbb{C}$.
For $x \in \ell^p$ and $\varepsilon > 0$, choose N with $\sum_{n>N} |x_n|^p < (\varepsilon/2)^p$, then choose $y \in D$ with support in $\{1, \ldots, N\}$ and $\sum_{n=1}^N |x_n - y_n|^p < (\varepsilon/2)^p$. Thus $\|x - y\|_p < \varepsilon$.

Structural Commentary: Separability is the existence of a countable set meeting every nonempty open ball.
Key Insight: Finite rational data are dense in ℓ^p, while ℓ^∞ contains an uncountable 1-separated set.

Theorem 4.5.3. *Separability and C_c Density in L^p*
Let $E \subseteq \mathbb{R}$ be Lebesgue measurable and let $1 \leq p < \infty$.
If $C_c(E)$ denotes restrictions to E of continuous functions on $\mathbb{R}$ with bounded support, then:
(i) $L^p(E)$ is separable;
(ii) $C_c(E)$ is dense in $L^p(E)$.

Proof.
(i) Let $\mathbb{F}_0 := \mathbb{Q}$ if real, $\mathbb{F}_0 := \mathbb{Q} + i\mathbb{Q}$ if complex. Let $\mathcal{D} := \operatorname{span}_{\mathbb{F}_0}\{\mathbf{1}_{E \cap (r,s)} : r, s \in \mathbb{Q}, \; r < s\}$.
Then $\mathcal{D}$ is countable. Extend $f \in L^p(E)$ by 0 to $\tilde{f} \in L^p(\mathbb{R})$.
By Proposition 4.1.26, it suffices to approximate $s = \sum_{\ell=1}^M a_\ell \mathbf{1}_{A_\ell}$ with $m(A_\ell) < +\infty$.
For each ℓ, Corollary 6.2.4 and rational endpoint perturbation give a finite union V_ℓ of rational open intervals with $m(A_\ell \triangle V_\ell)$ small, hence $\|\mathbf{1}_{E \cap A_\ell} - \mathbf{1}_{E \cap V_\ell}\|_p \leq m(A_\ell \triangle V_\ell)^{1/p}$. Approximate a_ℓ by $q_\ell \in \mathbb{F}_0$; then $\sum_{\ell=1}^M q_\ell \mathbf{1}_{E \cap V_\ell} \in \mathcal{D}$ approximates f arbitrarily well. Hence $L^p(E)$ is separable.
(ii) For rational $r < s$ and $0 < \eta < \frac{s-r}{2}$, choose $\psi_\eta \in C_c(\mathbb{R})$ with $0 \leq \psi_\eta \leq 1$, $\psi_\eta = 1$ on $[r + \eta, s - \eta]$, and $\psi_\eta = 0$ on $\mathbb{R} \setminus (r, s)$. Then $\|\psi_\eta|_E - \mathbf{1}_{E \cap (r,s)}\|_p^p \leq m((r, r+\eta) \cup (s - \eta, s)) \leq 2\eta \to 0$.
Thus each generator of $\mathcal{D}$ lies in $\overline{C_c(E)}^{L^p(E)}$; by finite linearity and (i), $C_c(E)$ is dense in $L^p(E)$.

Structural Commentary: Finite-measure atoms reduce to rational intervals, then to continuous cutoffs.
Key Insight: Countable rational steps lie in $\overline{C_c(E)}^{L^p(E)}$. □

Remark 4.5.4. *L^1 Does Not Exhaust the Dual of L^∞*
Let $I = [a,b]$ with $a < b$, and let $\iota : L^1(I) \to (L^\infty(I))^*$ be $\iota(g)(f) := \int_I fg \, dm$. The map ι from Proposition 4.1.37 is not onto. If ι were onto, then $(L^\infty(I))^*$ would be separable by Theorem 4.5.3.
Let (Φ_n) be dense in its unit sphere and choose $u_n \in L^\infty(I)$ with $\|u_n\|_\infty = 1$ and $|\Phi_n(u_n)| > \frac{1}{2}$.
If $Y := \overline{\operatorname{span}}\{u_n : n \geq 1\} \neq L^\infty(I)$, Hahn–Banach gives $\Phi \in (L^\infty(I))^*$ with $\|\Phi\| = 1$ and $\Phi|_Y = 0$.
Choosing n with $\|\Phi - \Phi_n\| < \frac{1}{4}$ gives $|\Phi_n(u_n)| < \frac{1}{4}$, a contradiction. Hence $L^\infty(I)$ would be separable.
But $v_t := \mathbf{1}_{[a,t]}$ satisfies $\|v_t - v_s\|_\infty = 1$ for $s < t$, an uncountable separated family, so $L^\infty(I)$ is not separable.

Example 4.5.5. *L^∞ on an Interval Need Not Be Separable*
Let $I = [a, b]$ with $a < b$. For $t \in I$, set $u_t := \mathbf{1}_{[a,t]}$. If $s < t$, then $\|u_t - u_s\|_\infty = 1$, since $u_t - u_s = \mathbf{1}_{(s,t]}$ a.e.
Thus $\{B(u_t, \frac{1}{3}) : t \in I\}$ is pairwise disjoint. If countable $D \subseteq L^\infty(I)$ were dense, choose $d_t \in D \cap B(u_t, \frac{1}{3})$ for each $t \in I$; disjointness makes $t \mapsto d_t$ injective, impossible. Hence $L^\infty(I)$ is not separable.
Key Insight: Uncountably many distance-1 indicators preclude countable density.

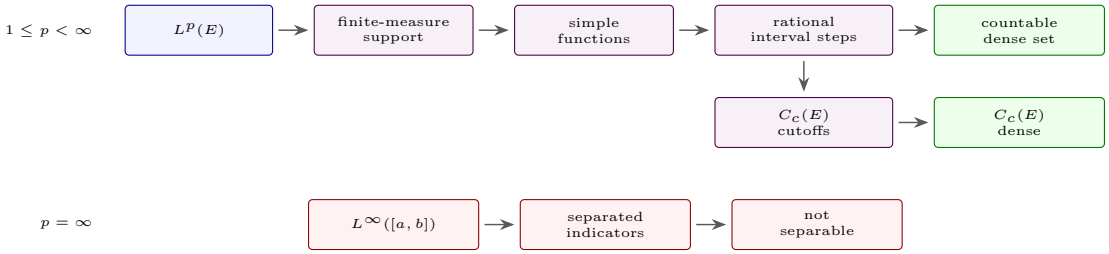

Figure 217: For $1 \le p < \infty$, rational interval steps and $C_c(E)$ are dense in $L^p(E)$; $L^\infty([a, b])$ is nonseparable.

4.6 Weak Convergence in L^p

Weak convergence in L^p keeps only the scalar information detected by conjugate functions. For $1 < p < +\infty$, reflexivity turns boundedness into weak subsequential compactness; at $p = 1$, bounded mass can still concentrate and destroy weak compactness unless uniform integrability is imposed.

Theorem 4.6.1. *L^p Weak Convergence by Dual Tests*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $q = \frac{p}{p-1}$, $(f_n) \subseteq L^p(\mu)$, and $f \in L^p(\mu)$, all real-valued. Then $f_n \rightharpoonup f$ in $L^p(\mu)$ iff $\int_X f_n g \, d\mu \to \int_X fg \, d\mu$ for all $g \in L^q(\mu)$.

Proof.
($\Rightarrow$) For $g \in L^q(\mu)$, $\Lambda_g(h) := \int_X hg \, d\mu$ is in $(L^p(\mu))^*$ by Holder.
Hence $f_n \rightharpoonup f$ implies $\int_X f_n g \, d\mu = \Lambda_g(f_n) \to \Lambda_g(f) = \int_X fg \, d\mu$.
($\Leftarrow$) Let $\Lambda \in (L^p(\mu))^*$. By Theorem 4.1.34, choose $g \in L^q(\mu)$ with $\Lambda(h) = \int_X hg \, d\mu \ \forall h \in L^p(\mu)$.
The assumed integral test yields $\Lambda(f_n) \to \Lambda(f)$, hence $f_n \rightharpoonup f$.

Structural Commentary: Dual tests are exactly L^q pairings.
Key Insight: Weak convergence in L^p is conjugate-function testing. □

Proposition 4.6.2. *Weak Limits and Norm Lower Semicontinuity*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $(f_n) \subseteq L^p(\mu)$, and $f_n \rightharpoonup f$ in real $L^p(\mu)$. Then:
(i) $\sup_n \|f_n\|_p < +\infty$;
(ii) if $f_n \rightharpoonup h$ in $L^p(\mu)$, then $h = f$ in $L^p(\mu)$;
(iii) $\|f\|_p \le \liminf_{n\to\infty} \|f_n\|_p$.

Proof.
(i) By Proposition 6.1.4 applied to $u_n = f_n$ in $L^p(\mu)$, $f_n \rightharpoonup f$ implies $\sup_n \|f_n\|_p < +\infty$.
(ii) Let $q = \frac{p}{p-1}$, $f_n \rightharpoonup h$. For $g \in L^q(\mu)$, by Theorem 4.6.1, $\int_X (f - h)g \, d\mu = \lim_n \int_X f_n g \, d\mu - \lim_n \int_X f_n g \, d\mu = 0$.
By Proposition 4.1.30, $\|f - h\|_p = 0$, so $h = f$ in $L^p(\mu)$.
(iii) By Proposition 6.1.4, $u \mapsto \|u\|_p$ is weakly lower semicontinuous on $L^p(\mu)$;
with $u_n = f_n$ and $u = f$, $\|f\|_p \le \liminf_{n\to\infty} \|f_n\|_p$.

Structural Commentary: General weak facts handle bounds; L^q tests force uniqueness.
Key Insight: Weak limits are norm-controlled and conjugate-tested. □

Theorem 4.6.3. *Reflexivity of L^p, $1 < p < +\infty$*
Let $(X, \mathcal{M}, \mu)$ be σ-finite, $1 < p < +\infty$, $q = \frac{p}{p-1}$, and $Jf(\Phi) := \Phi(f)$.
For $r \in \{p, q\}$ with conjugate r', set $T_r u \in (L^{r'}(\mu))^*$ by $(T_r u)(v) := \int_X uv \, d\mu$. Then:
(i) T_p and T_q are isometric isomorphisms;
(ii) $J : L^p(\mu) \to (L^p(\mu))^{**}$ is onto; hence real $L^p(\mu)$ is reflexive.

Proof.
(i) By Theorem 4.1.34, $T_p : L^p(\mu) \to (L^q(\mu))^*$ and $T_q : L^q(\mu) \to (L^p(\mu))^*$ are isometric isomorphisms.
(ii) Let $S \in (L^p(\mu))^{**}$. Since $S \circ T_q \in (L^q(\mu))^*$, by (i), $f \in L^p(\mu)$, $S(T_q g) = (T_p f)(g) = \int_X fg \, d\mu \ \forall g \in L^q(\mu)$.
For $\Phi \in (L^p(\mu))^*$, by (i) $\Phi = T_q g$; $S(\Phi) = S(T_q g) = \int_X fg \, d\mu = \Phi(f) = Jf(\Phi)$. Hence $S = Jf$, J is onto.

Structural Commentary: Represent both dual layers by conjugate pairings.
Key Insight: Every bidual functional is evaluation at an L^p function. □

Definition 4.6.4. *Weak Sequential Compactness*
Let $\mathcal{X}$ be a normed space and $K \subseteq \mathcal{X}$.
Then K is *weakly sequentially compact* if every $(x_n) \subseteq K$ admits $n_k \uparrow \infty$ and $x \in K$ with $x_{n_k} \rightharpoonup x$ in $\mathcal{X}$.

Theorem 4.6.5. *Riesz Weak Compactness Theorem*
Let $(X, \mathcal{M}, \mu)$ be σ-finite, $1 < p < +\infty$, $q = \frac{p}{p-1}$, and $(f_n) \subseteq L^p(\mu)$ be real-valued with $M := \sup_n \|f_n\|_p < +\infty$.
Then there exist $n_k \uparrow \infty$ and $f \in L^p(\mu)$ such that:
(i) $f_{n_k} \rightharpoonup f$ in $L^p(\mu)$;
(ii) $\int_X f_{n_k} g \, d\mu \to \int_X fg \, d\mu$ for all $g \in L^q(\mu)$.

Proof. Set $B_M := \{h \in L^p(\mu) : \|h\|_p \le M\}$. By Theorem 4.6.3, $L^p(\mu)$ is reflexive; hence B_M is weakly compact. By Eberlein–Smulian, B_M is weakly sequentially compact.
(i) Since $(f_n) \subseteq B_M$ and B_M is weakly sequentially compact, $\exists n_k \uparrow \infty$ and $f \in B_M$ with $f_{n_k} \rightharpoonup f$ in $L^p(\mu)$.
(ii) From (i) and Theorem 4.6.1, $\forall g \in L^q(\mu)$, $\int_X f_{n_k} g \, d\mu \to \int_X fg \, d\mu$.
Structural Commentary: Reflexive bounded balls are weakly sequentially compact.
Key Insight: Bounded L^p sequences have conjugate-test convergent subsequences. □

Corollary 4.6.6. *The L^p Unit Ball Is Weakly Sequentially Compact*
Let $(X, \mathcal{M}, \mu)$ be σ-finite, $1 < p < +\infty$, and $B_p := \{f \in L^p(\mu) : \|f\|_p \le 1\}$.
Then B_p is weakly sequentially compact in $L^p(\mu)$.

Proof. Let $(f_n) \subseteq B_p$. By Theorem 4.6.5, $\exists n_k \uparrow \infty$ and $f \in L^p(\mu)$ with $f_{n_k} \rightharpoonup f$.
By Proposition 4.6.2, $\|f\|_p \le \liminf_{k\to\infty} \|f_{n_k}\|_p \le 1$, so $f \in B_p$.
Structural Commentary: Weak limits stay in B_p by lower semicontinuity.
Key Insight: Closed norm balls are weakly sequentially closed. □

Theorem 4.6.7. *Radon–Riesz Principle*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $(f_n) \subseteq L^p(\mu)$, and $f \in L^p(\mu)$, all real-valued.
Then $f_n \rightharpoonup f$ and $\|f_n\|_p \to \|f\|_p$ iff $\|f_n - f\|_p \to 0$.

Proof.
($\Rightarrow$) If $\|f\|_p = 0$, then $\|f_n - f\|_p \le \|f_n\|_p + \|f\|_p \to 0$. Let $a := \|f\|_p > 0$.
Otherwise, after subsequences, $\|f_n - f\|_p \ge \varepsilon > 0$ and, with $a_n := \|f_n\|_p$, $u_n := \frac{f_n}{a_n}$, $u := \frac{f}{a}$, one has $u_n \rightharpoonup u$, $\|u_n\|_p = \|u\|_p = 1$, and $\|u_n - u\|_p \ge \eta > 0$. By uniform convexity, $\left\|\frac{u_n+u}{2}\right\|_p \le 1 - \delta$.
Since $\frac{u_n+u}{2} \rightharpoonup u$, weak lower semicontinuity implies $1 = \|u\|_p \le \liminf_n \left\|\frac{u_n+u}{2}\right\|_p \le 1 - \delta$, contradiction.
($\Leftarrow$) Assume $\|f_n - f\|_p \to 0$. Then $\left|\|f_n\|_p - \|f\|_p\right| \le \|f_n - f\|_p \to 0$.
For $\Lambda \in (L^p(\mu))^*$, $|\Lambda(f_n) - \Lambda(f)| \le \|\Lambda\| \, \|f_n - f\|_p \to 0$, hence $f_n \rightharpoonup f$.
Structural Commentary: Uniform convexity turns weak lower semicontinuity into strong convergence.
Key Insight: Weak direction plus norm size fixes the vector. □

Remark 4.6.8. *L^2 Proof of Radon–Riesz*
For $p = 2$, $f_n \rightharpoonup f$ in $L^2(\mu)$, $\|f_n\|_2 \to \|f\|_2$. Then $\int_X f_n f \, d\mu \to \|f\|_2^2$,
$\|f_n - f\|_2^2 = \|f_n\|_2^2 - 2\int_X f_n f \, d\mu + \|f\|_2^2 \to 0$. The converse follows from norm convergence in $L^2(\mu)$.
Key Insight: Weak convergence fixes the cross term.

Corollary 4.6.9. *Strong Subsequence Criterion*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $(f_n) \subseteq L^p(\mu)$, and $f_n \rightharpoonup f$ in real $L^p(\mu)$.
Then $\exists n_k \uparrow \infty$ with $\|f_{n_k} - f\|_p \to 0$ iff $\|f\|_p = \liminf_{n\to\infty} \|f_n\|_p$.

Proof. Set $L := \liminf_{n\to\infty} \|f_n\|_p$.
($\Rightarrow$) If $\|f_{n_k} - f\|_p \to 0$, then $\|f_{n_k}\|_p \to \|f\|_p$.
Since $L \le \lim_k \|f_{n_k}\|_p = \|f\|_p$ and, by Proposition 4.6.2, $\|f\|_p \le L$, one has $L = \|f\|_p$.
($\Leftarrow$) Assume $L = \|f\|_p$. Choose $n_k \uparrow \infty$ with $\|f_{n_k}\|_p \to L$.
Since $f_{n_k} \rightharpoonup f$ and $\|f_{n_k}\|_p \to \|f\|_p$, Theorem 4.6.7 implies $\|f_{n_k} - f\|_p \to 0$.
Structural Commentary: Match the liminf subsequence to Radon–Riesz.
Key Insight: Strong subsequences occur exactly when no norm is lost. □

Theorem 4.6.10. *Dunford–Pettis Criterion in L^1*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space and let (f_n) be bounded in $L^1(\mu)$. The following are equivalent.
(i) (f_n) is uniformly integrable.
(ii) $\forall (f_{n_k}) \subseteq (f_n)$ $\exists (f_{n_{k_j}}) \subseteq (f_{n_k})$ converging weakly in $L^1(\mu)$.

Proof.
($\Rightarrow$) Let (u_j) be a subsequence of (f_n), set $C := \sup_j \|u_j\|_1$, and put $T_m(t) := (-m) \vee (t \wedge m)$.
By Proposition 4.4.7, $\sup_j \|u_j - T_m(u_j)\|_1 \to 0$. For fixed m and sign $\pm$, $\left\|T_m(u_j^{\pm})\right\|_2^2 \le m \left\|u_j^{\pm}\right\|_1 \le mC$.
By bounded Hilbert weak subsequence extraction and diagonalization over $(m, \pm)$, pass to a subsequence, still (u_j), such that $\int_E T_m(u_j^{\pm}) \, d\mu$ converges for every $E \in \mathcal{M}$ and $m \in \mathbb{N}$. Set $\nu_j^{\pm}(E) := \int_E u_j^{\pm} \, d\mu$.
For r, s, m and $E \in \mathcal{M}$, $\left|\nu_r^{\pm}(E) - \nu_s^{\pm}(E)\right| \le \left|\int_E (T_m(u_r^{\pm}) - T_m(u_s^{\pm})) \, d\mu\right| + 2\sup_j \left\|u_j^{\pm} - T_m(u_j^{\pm})\right\|_1$.
Hence $(\nu_j^{\pm}(E))_j$ is Cauchy. Put $\nu^{\pm}(E) := \lim_j \nu_j^{\pm}(E)$.
By Theorem 3.6.11, $\nu^{\pm}$ are finite measures and $\nu^{\pm} \ll \mu$. Write $h^{\pm} = \frac{d\nu^{\pm}}{d\mu}$ and $h := h^+ - h^- \in L^1(\mu)$.
Then $\int_X \varphi u_j \, d\mu \to \int_X \varphi h \, d\mu$ for simple φ; by Lemma 1.1.14 and $\sup_j \|u_j\|_1 + \|h\|_1 < +\infty$, the convergence holds for every $\varphi \in L^\infty(\mu)$. Thus $u_j \rightharpoonup h$ in $L^1(\mu)$.

($\Leftarrow$) Suppose (f_n) is not uniformly integrable. Since finite subfamilies are uniformly integrable by Proposition 4.4.5(ii), choose a subsequence (u_j) and sets $A_j \in \mathcal{M}$ with $\mu(A_j) \to 0$ and $\int_{A_j} |u_j|\, d\mu \geq \varepsilon_0 > 0$.

By (ii), some (u_{j_ℓ}) converges weakly in $L^1(\mu)$ to u. Set $\eta_\ell(E) := \int_E u_{j_\ell}\, d\mu$ and $\eta(E) := \int_E u\, d\mu$.
Then $\eta_\ell(E) \to \eta(E)$ for every $E \in \mathcal{M}$, and $\sup_\ell |\eta_\ell|(X) = \sup_\ell \|u_{j_\ell}\|_1 < +\infty$ by Theorem 3.1.9.
The signed form of Theorem 3.6.11 implies $\eta_\ell \ll_u \mu$ in variation:
$\forall \varepsilon > 0\ \exists \delta > 0$ such that $\mu(A) < \delta \Rightarrow \sup_\ell |\eta_\ell|(A) < \varepsilon$. Since $|\eta_\ell|(A) = \int_A |u_{j_\ell}|\, d\mu$, the subsequence (u_{j_ℓ}) is uniformly integrable, contradicting $\int_{A_{j_\ell}} |u_{j_\ell}|\, d\mu \geq \varepsilon_0$ and $\mu(A_{j_\ell}) \to 0$.

Structural Commentary: Truncate, pass to setwise limits, apply VHS.
Key Insight: UI is weak sequential compactness in finite L^1. □

Corollary 4.6.11. *Dominated L^1 Families Have Weak Subsequences*
Let $(X, \mathcal{M}, \mu)$ be finite. If $g \in L^1(\mu)$, $\forall n\, |f_n| \leq g$ a.e., then (f_n) has subsequence that converges weakly in $L^1(\mu)$.

Proof. Since $|f_n| \leq g$ a.e., $\|f_n\|_1 \leq \|g\|_1\ \forall n$, (f_n) is bounded in $L^1(\mu)$. By Proposition 4.4.5(i), $\{f_n : n \geq 1\}$ is uniformly integrable. Theorem 4.6.10, $(i) \Rightarrow (ii)$, yields a weakly convergent subsequence.
Structural Commentary: Domination verifies Dunford–Pettis through a common integrable envelope.
Key Insight: One L^1 envelope prevents concentration throughout the sequence. □

Corollary 4.6.12. *Bounded L^p Families Are Weakly Compact in L^1*
Let $(X, \mathcal{M}, \mu)$ be finite, let $1 < p < +\infty$, and let (f_n) be bounded in $L^p(\mu)$.
Then (f_n) has a subsequence that converges weakly in $L^1(\mu)$.

Proof. Let q be conjugate to p and $M := \sup_n \|f_n\|_p < +\infty$. By Hölder, $\|f_n\|_1 \leq M\mu(X)^{\frac{1}{q}}\ \forall n$, so (f_n) is bounded in $L^1(\mu)$. By Corollary 4.4.6, $\{f_n : n \geq 1\}$ is uniformly integrable.
Hence Theorem 4.6.10, $(i) \Rightarrow (ii)$, yields a weakly convergent subsequence in $L^1(\mu)$.
Structural Commentary: Hölder gives L^1 boundedness, uniform integrability before Dunford–Pettis is applied.
Key Insight: On finite measure spaces, an L^p bound with $p > 1$ rules out L^1 concentration. □

Example 4.6.13. *A Bounded L^1 Sequence with No Weakly Convergent Subsequence*
In $L^1[0,1]$, $I_n = [0, \frac{1}{n}]$ $f_n = n\mathbf{1}_{I_n}$. Then $\|f_n\|_1 = 1\ \forall n$, and no subsequence of (f_n) converges weakly in $L^1[0,1]$.

Proof. Assume $f_{n_k} \rightharpoonup f$ in $L^1[0,1]$. For $0 < c < d \leq 1$, $\mathbf{1}_{[c,d]} \in L^\infty[0,1]$ and $[c,d] \cap I_{n_k} = \varnothing$ for k large.
Hence $\int_c^d f\, dm = \lim_{k\to\infty} \int_c^d f_{n_k}\, dm = 0$. Thus $f = 0$ a.e. on $(0,1]$.
Testing by $\mathbf{1}_{[0,1]}$ yields $0 = \int_0^1 f\, dm = \lim_{k\to\infty} \int_0^1 f_{n_k}\, dm = 1$, contradiction.
Key Insight: Unit mass concentrates on shrinking supports. □

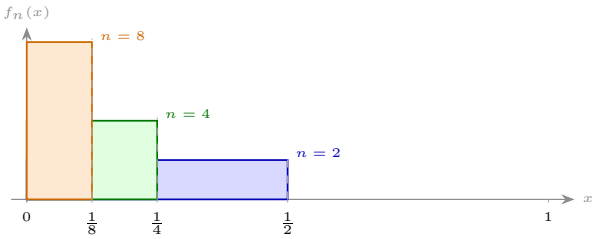

Figure 218: $f_n = n\mathbf{1}_{[0,\frac{1}{n}]}$: unit L^1 mass on shrinking support.

Remark 4.6.14. *Endpoint Failure and Uniform Integrability*
For $f_n = n\mathbf{1}_{[0,\frac{1}{n}]} \in L^1[0,1]$, $\|f_n\|_1 = 1$ but $\int_{\{f_n > R\}} f_n\, dm = 1$ whenever $n > R$.
Hence (f_n) is not uniformly integrable; the $p = 1$ failure is concentration, not lack of boundedness.

Theorem 4.6.15. *Banach–Saks Theorem*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $(f_n) \subseteq L^p(\mu)$, and $f_n \rightharpoonup f$ in real $L^p(\mu)$.
Then $\exists n_k \uparrow \infty$ such that, with $A_k := \frac{1}{k}\sum_{j=1}^k f_{n_j}$, $\|A_k - f\|_p \to 0$.

Proof. Set $h_n := f_n - f$. Then $h_n \rightharpoonup 0$ and $\sup_n \|h_n\|_p < +\infty$ by Proposition 4.6.2.
Since real $L^p(\mu)$ is uniformly convex, by the selection lemma there is $n_k \uparrow \infty$ with $\left\|\frac{1}{k}\sum_{j=1}^k h_{n_j}\right\|_p \to 0$.
Since $\frac{1}{k}\sum_{j=1}^k h_{n_j} = A_k - f$, $\|A_k - f\|_p \to 0$.
Structural Commentary: Reduce to weakly null; select Cesaro means by uniform convexity.
Key Insight: Convex averaging turns weak nullity into norm decay. □

Remark 4.6.16. *The L^2 Selection Estimate*
For $p = 2$, replace f_n by $f_n - f$ and assume $f_n \rightharpoonup 0$ in $L^2(\mu)$ with $\|f_n\|_2^2 \leq M$. Set $n_1 = 1$ and $S_k := \sum_{j=1}^k f_{n_j}$.

(i) Given S_k, choose $n_{k+1} > n_k$ so that $\left|\int_X S_k f_{n_{k+1}}\, d\mu\right| \leq 1$.
(ii) Then $\|S_{k+1}\|_2^2 = \|S_k\|_2^2 + 2\int_X S_k f_{n_{k+1}}\, d\mu + \|f_{n_{k+1}}\|_2^2 \leq \|S_k\|_2^2 + 2 + M$.
(iii) Hence $\|S_k/k\|_2^2 \leq \frac{2+M}{k}$ by induction.

Thus $S_k/k \to 0$ in $L^2(\mu)$.
Key Insight: Weak convergence makes new terms nearly orthogonal to old sums.

Definition 4.6.17. *Closed Convex Sets and Convex Functionals*
Let C be a subset of a normed space $\mathcal{X}$.

(i) C is *convex* if $\lambda f + (1-\lambda)g \in C$ whenever $f, g \in C$ and $0 \le \lambda \le 1$.
(ii) C is *closed* if every strongly convergent sequence in C has its limit in C.
(iii) A real-valued functional $T : C \to \mathbb{R}$ is *continuous* if $f_n \to f$ strongly in C implies $T(f_n) \to T(f)$.
(iv) If C is convex, then T is *convex* if $T(\lambda f + (1-\lambda)g) \le \lambda T(f) + (1-\lambda)T(g)$ $\forall f, g \in C$ and $0 \le \lambda \le 1$.

Example 4.6.18. *Basic Closed Convex Sets in L^p*
Let $E \subseteq \mathbb{R}$ be measurable and $1 \le p < +\infty$.

(i) If $g \in L^p(E)$ and $g \ge 0$ a.e., then $C_g := \{f \in L^p(E) : |f| \le g \text{ a.e.}\}$ is closed and convex.
(ii) $B := \{f \in L^p(E) : \|f\|_p \le 1\}$ is closed and convex.

Proof.

(i) If $f, h \in C_g$ and $0 \le \lambda \le 1$, then $|\lambda f + (1-\lambda)h| \le \lambda |f| + (1-\lambda)|h| \le g$ a.e., so C_g is convex.
If $f_n \in C_g$ and $f_n \to f$ in $L^p(E)$, choose $f_{n_k} \to f$ a.e. by Theorem 4.1.23; then $|f| \le g$ a.e., so $f \in C_g$.
(ii) If $f, h \in B$ and $0 \le \lambda \le 1$, then by Minkowski, $\|\lambda f + (1-\lambda)h\|_p \le \lambda \|f\|_p + (1-\lambda)\|h\|_p \le 1$.
If $f_n \in B$ and $f_n \to f$ in $L^p(E)$, then $\|f\|_p \le \|f - f_n\|_p + \|f_n\|_p \le \|f - f_n\|_p + 1$, hence $\|f\|_p \le 1$.

Key Insight: Norm limits preserve a.e. bounds and norm bounds. □

Lemma 4.6.19. *Weak Limits and Convex Lower Semicontinuity*
Let $E \subseteq \mathbb{R}$ be measurable, $1 < p < +\infty$, $C \subseteq L^p(E)$ closed, bounded, and convex.
Let $T : C \to \mathbb{R}$ be continuous and convex. If $(f_n) \subseteq C$ and $f_n \rightharpoonup f$ in $L^p(E)$, then, with $\alpha := \liminf_{n\to\infty} T(f_n)$:

(i) $f \in C$;
(ii) $T(f) \le \alpha$.

Proof.

(i) By Theorem 4.6.15, choose $n_k \uparrow \infty$ with $a_k := \frac{1}{k}\sum_{j=1}^{k} f_{n_j} \to f$ in $L^p(E)$.
Since $a_k \in \operatorname{co}\{f_{n_1}, \dots, f_{n_k}\} \subseteq C$ and C is closed, $f \in C$.
(ii) Choose $m_k \uparrow \infty$ with $T(f_{m_k}) \to \alpha$.

By Theorem 4.6.15, choose $g_j := f_{m_{k_j}}$ such that $b_k := \frac{1}{k}\sum_{j=1}^{k} g_j \to f$ in $L^p(E)$.

Then $b_k \in C$ and $T(g_j) \to \alpha$; by continuity and convexity, $T(f) = \lim_k T(b_k) \le \lim_k \frac{1}{k}\sum_{j=1}^{k} T(g_j) = \alpha$.

Structural Commentary: Banach–Saks supplies strong convex averages; convexity transfers the liminf.
Key Insight: Weak limits inherit closed convex constraints and convex lsc. □

Theorem 4.6.20. *Direct Method for Convex Functionals on L^p*
Let $E \subseteq \mathbb{R}$ be measurable, $1 < p < +\infty$, $C \subseteq L^p(E)$ nonempty, closed, bounded, and convex.
Let $T : C \to \mathbb{R}$ be continuous and convex. Then $\exists f_0 \in C$ such that $T(f_0) = \inf_{h\in C} T(h)$.

Proof.

(i) If $\inf_C T = -\infty$, choose $f_n \in C$ with $T(f_n) \to -\infty$.
Since C is bounded, by Theorem 4.6.5, $\exists n_k \uparrow \infty$ and $f \in L^p(E)$ with $f_{n_k} \rightharpoonup f$.
By Lemma 4.6.19, $f \in C$ and $T(f) \le \liminf_k T(f_{n_k}) = -\infty$, contradicting $T : C \to \mathbb{R}$. Thus $\inf_C T > -\infty$.
(ii) Let $c := \inf_C T$ and choose $f_n \in C$ with $T(f_n) \to c$.
Since C is bounded, by Theorem 4.6.5, $\exists n_k \uparrow \infty$ and $f_0 \in L^p(E)$ with $f_{n_k} \rightharpoonup f_0$.
By Lemma 4.6.19, $f_0 \in C$ and $T(f_0) \le \liminf_k T(f_{n_k}) = c$. Since $c \le T(f_0)$, $T(f_0) = c$.

Structural Commentary: Weak compactness supplies a limit; convex lsc keeps the infimum.
Key Insight: Direct method equals compactness plus lower semicontinuity. □

Corollary 4.6.21. *Integral Convex Functionals Have Unit-Ball Minimizers*
Let $\mu(E) < +\infty$, $1 < p < +\infty$, and $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous convex with $|\varphi(s)| \le c_1 + c_2|s|^p$ for some $c_1, c_2 \ge 0$. Set $B := \{f \in L^p(E) : \|f\|_p \le 1\}$ and $T(f) := \int_E \varphi(f)\,dm$. Then:

(i) $T : B \to \mathbb{R}$ is continuous and convex;
(ii) $\exists f_0 \in B$ such that $T(f_0) = \min_{f\in B} T(f)$.

Proof.

(i) For $f \in B$, $|\varphi(f)| \le c_1 + c_2|f|^p \in L^1(E)$, so $T(f) \in \mathbb{R}$.
Convexity of φ implies $T(\lambda f + (1-\lambda)h) \le \lambda T(f) + (1-\lambda)T(h)$ for $f, h \in B$, $0 \le \lambda \le 1$.
If $f_n \to f$ in $L^p(E)$, every subsequence admits $f_{n_k} \to f$ a.e. with $|f_{n_k}| \le g$ for some $g \in L^p(E)$.
Then $|\varphi(f_{n_k})| \le c_1 + c_2 g^p \in L^1(E)$, so by dominated convergence $T(f_{n_k}) \to T(f)$.
By the subsequence criterion, $T(f_n) \to T(f)$.
(ii) The set B is nonempty, bounded, convex by Minkowski, and closed since $f_n \in B$, $f_n \to f$ in $L^p(E)$ imply $\|f\|_p \le \liminf_n \|f_n\|_p \le 1$.
By Theorem 4.6.20, $\exists f_0 \in B$ with $T(f_0) = \min_{f\in B} T(f)$.

Structural Commentary: p-growth ⇒ continuity; direct method ⇒ attainment.
Key Insight: Growth converts strong L^p convergence to dominated convergence. □

5 Banach and Hilbert Spaces

5.1 Normed and Banach Spaces

Normed spaces make vector spaces metric; Banach spaces add completeness. This is the setting for infinite series, fixed points, bounded linear maps, and operator norms.

Definition 5.1.1. *Normed and Banach Spaces*
Let $\mathcal{X}$ be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

(i) *Norm*: $\|\cdot\| : \mathcal{X} \to [0,\infty)$; $\|x\| = 0 \iff x = 0$, $\|\lambda x\| = |\lambda| \, \|x\|$; $\|x + y\| \le \|x\| + \|y\|$ for $x, y \in \mathcal{X}$, $\lambda \in \mathbb{F}$.
(ii) *Normed space*: $(\mathcal{X}, \|\cdot\|)$ carries the metric $d(x, y) := \|x - y\|$.
(iii) *Banach space*: $(\mathcal{X}, \|\cdot\|)$ is complete in the norm metric.

Definition 5.1.2. *Basic Normed-Space Geometry*
Let $\mathcal{X}$ be a normed space.

(i) *Open unit ball*: $B_{\mathcal{X}} := \{x \in \mathcal{X} : \|x\| < 1\}$.
(ii) *Closed unit ball*: $\overline{B}_{\mathcal{X}} := \{x \in \mathcal{X} : \|x\| \le 1\}$.
(iii) *Unit vector*: $x \in \mathcal{X}$ satisfies $\|x\| = 1$.
(iv) *Minkowski sum*: $A + B := \{a + b : a \in A,\ b \in B\}$ for nonempty $A, B \subseteq \mathcal{X}$.
(v) *Scalar multiple*: $\lambda A := \{\lambda a : a \in A\}$ for nonempty $A \subseteq \mathcal{X}$ and $\lambda \in \mathbb{F}$.
(vi) *Closed linear span*: $\operatorname{cspan}(S) := \overline{\operatorname{span}(S)}$ for $S \subseteq \mathcal{X}$.
(vii) *Equivalent norms*: $\|\cdot\|_1 \sim \|\cdot\|_2$ iff $\exists c_1, c_2 > 0$ such that $\forall x \in \mathcal{X}$, $c_1 \|x\|_1 \le \|x\|_2 \le c_2 \|x\|_1$.

Proposition 5.1.3. *Linear Operations, Closed Spans, and Equivalent Norms*
Let $\mathcal{X}$ be a normed space.

(i) The norm map $x \mapsto \|x\|$, addition $\mathcal{X} \times \mathcal{X} \to \mathcal{X}$, and scalar multiplication $\mathbb{F} \times \mathcal{X} \to \mathcal{X}$ are continuous.
(ii) If $M \le \mathcal{X}$, then $\overline{M} \le \mathcal{X}$; for $S \subseteq \mathcal{X}$, $\operatorname{cspan}(S)$ is the least closed subspace containing S.
(iii) Equivalent norms have the same convergent and Cauchy sequences, topology, and compact sets.

Proof.

(i) $|\|x\| - \|y\|| \le \|x - y\|$, so $x \mapsto \|x\|$ is 1-Lipschitz.
Also $\|(x + h) + (y + k) - (x + y)\| \le \|h\| + \|k\|$, so addition is continuous.
If $\lambda_n \to \lambda$ and $x_n \to x$, then $\|\lambda_n x_n - \lambda x\| \le |\lambda_n| \, \|x_n - x\| + |\lambda_n - \lambda| \, \|x\| \to 0$, since (λ_n) is bounded.
(ii) Let $x, y \in \overline{M}$ and choose $x_n, y_n \in M$ with $x_n \to x$, $y_n \to y$.
For $\alpha, \beta \in \mathbb{F}$, $\alpha x_n + \beta y_n \in M$ and $\alpha x_n + \beta y_n \to \alpha x + \beta y$ by (i); hence $\overline{M} \le \mathcal{X}$.
If C is closed, $C \le \mathcal{X}$, and $S \subseteq C$, then $\operatorname{span}(S) \subseteq C$, so $\operatorname{cspan}(S) = \overline{\operatorname{span}(S)} \subseteq C$.
(iii) If $c \|z\|_1 \le \|z\|_2 \le C \|z\|_1$ for all $z \in \mathcal{X}$, then $B_1(x, \frac{\varepsilon}{C}) \subseteq B_2(x, \varepsilon) \subseteq B_1(x, \frac{\varepsilon}{c})$. Thus the topologies agree, the identity maps are bi-Lipschitz, convergent and Cauchy sequences agree, and compact sets agree.

Structural Commentary: Norms turn the algebraic operations into continuous operations.
Key Insight: Closing a subspace preserves linearity because linear combinations commute with limits. □

Definition 5.1.4. *Bounded Linear Maps and Operator Norm*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces and let $T : \mathcal{X} \to \mathcal{Y}$ be linear.

(i) *Bounded*: $\exists C \ge 0$ such that $\forall x \in \mathcal{X}$, $\|Tx\|_{\mathcal{Y}} \le C \|x\|_{\mathcal{X}}$.
(ii) *Operator norm*: $\|T\|_{\mathrm{op}} := \sup_{\|x\|_{\mathcal{X}} \le 1} \|Tx\|_{\mathcal{Y}} = \sup_{x \ne 0} \frac{\|Tx\|_{\mathcal{Y}}}{\|x\|_{\mathcal{X}}}$.
(iii) *Operator space*: $\mathcal{L}(\mathcal{X}, \mathcal{Y}) := \{T : \mathcal{X} \to \mathcal{Y} : T \text{ is bounded and linear}\}$ and $\mathcal{L}(\mathcal{X}) := \mathcal{L}(\mathcal{X}, \mathcal{X})$.

Theorem 5.1.5. *Bounded Equals Continuous for Linear Maps*
For a linear map $T : \mathcal{X} \to \mathcal{Y}$ between normed spaces, the following are equivalent:

(i) T is bounded.
(ii) T is continuous at 0.
(iii) T is continuous everywhere.

Proof.

(i) $\Rightarrow$ *(iii)* If T is bounded, then $\|Tx - Ty\| \le \|T\|_{\mathrm{op}} \|x - y\|$ for $x, y \in \mathcal{X}$; hence T is Lipschitz.
(iii) $\Rightarrow$ *(ii)* Since $0 \in \mathcal{X}$, continuity on $\mathcal{X}$ implies continuity at 0.
(ii) $\Rightarrow$ *(i)* Choose $\delta > 0$ with $\|z\| < \delta \Rightarrow \|Tz\| < 1$. For $x \ne 0$, set $z := \frac{\delta x}{2\|x\|}$.
Then $\|z\| < \delta$ and $\|Tx\| = \frac{2\|x\|}{\delta} \|Tz\| \le \frac{2}{\delta} \|x\|$. Thus $\|T\|_{\mathrm{op}} \le \frac{2}{\delta}$.

Structural Commentary: Continuity at 0 rescales to a global operator bound.
Key Insight: Linear continuity is boundedness. □

Theorem 5.1.6. *Bounded Operators Form a Banach Space*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces.

(i) $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is normed by $\|\cdot\|_{op}$ under pointwise operations.
(ii) If $\mathcal{Y}$ is Banach, then $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is Banach.

Proof.

(i) If $S, T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ and $\alpha, \beta \in \mathbb{F}$, then $\|(\alpha S + \beta T)x\| \le (|\alpha| \, \|S\|_{\mathrm{op}} + |\beta| \, \|T\|_{\mathrm{op}}) \|x\|$, so $\alpha S + \beta T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$.
Also $\|\alpha T\|_{\mathrm{op}} = |\alpha| \, \|T\|_{\mathrm{op}}$, $\|S + T\|_{\mathrm{op}} \le \|S\|_{\mathrm{op}} + \|T\|_{\mathrm{op}}$, and $\|T\|_{\mathrm{op}} = 0 \Rightarrow Tx = 0$ for all $x \in \mathcal{X}$.

(ii) Assume $\mathcal{Y}$ is Banach and let (T_n) be Cauchy in $\|\cdot\|_{\text{op}}$.
For each $x \in \mathcal{X}$, $\|T_n x - T_m x\| \le \|T_n - T_m\|_{\text{op}} \|x\|$, so define $Tx := \lim_n T_n x$.
Then $T(\alpha x + \beta y) = \alpha Tx + \beta Ty$. Given $\varepsilon > 0$, choose N with $\|T_n - T_m\|_{\text{op}} < \varepsilon$ for $m, n \ge N$.
Letting $m \to \infty$ gives $\|(T_n - T)x\| \le \varepsilon \|x\|$, so $\|T_n - T\|_{\text{op}} \le \varepsilon$.
Hence $T_N - T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, and $T_n \to T$.

Structural Commentary: Pointwise limits are linear; operator-norm Cauchy control is uniform in x.
Key Insight: Completeness of $\mathcal{Y}$ lifts to $\mathcal{L}(\mathcal{X}, \mathcal{Y})$. □

Corollary 5.1.7. *Neumann Series*
Let $\mathcal{X}$ be Banach and let $T \in \mathcal{L}(\mathcal{X})$ satisfy $\|T\|_{op} < 1$.

Then $I - T$ has bounded inverse, with Neumann series converging in $\mathcal{L}(\mathcal{X})$: $(I - T)^{-1} = \sum_{n=0}^{\infty} T^n$.

Proof. Let $S_N := \sum_{n=0}^{N} T^n$. For $M > N$, $\|S_M - S_N\|_{\text{op}} \le \sum_{n=N+1}^{M} \|T\|_{\text{op}}^n \le \frac{\|T\|_{\text{op}}^{N+1}}{1 - \|T\|_{\text{op}}} \to 0$, so $S_N \to S \in \mathcal{L}(\mathcal{X})$.
Since $(I - T)S_N = I - T^{N+1} = S_N(I - T)$ and $\|T^{N+1}\|_{\text{op}} \le \|T\|_{\text{op}}^{N+1} \to 0$.
Passing to the limit gives $(I - T)S = S(I - T) = I$.
Structural Commentary: Operator-norm convergence turns the geometric identity into an inverse.
Key Insight: If $\|T\|_{\text{op}} < 1$, then $\sum T^n = (I - T)^{-1}$. □

Proposition 5.1.8. *Quotient Norm and Quotient Map*
Let $M \le \mathcal{X}$ be closed, let $\mathcal{X}/M := \{x + M : x \in \mathcal{X}\}$, and let $q : \mathcal{X} \to \mathcal{X}/M$ be $q(x) := x + M$.
Set $\|x + M\|_{\mathcal{X}/M} := \inf_{m \in M} \|x - m\|$.

(i) $\|\cdot\|_{\mathcal{X}/M}$ is a norm on $\mathcal{X}/M$.
(ii) q is continuous with $\|q\|_{op} \le 1$.
(iii) $\forall x \in \mathcal{X}$ and $r > 0$, $q(B_r(x)) = B_r(qx)$.
(iv) q is open.

Proof.
(i) If $x' = x + m_0$ with $m_0 \in M$, then $\inf_{m \in M} \|x' - m\| = \inf_{m \in M} \|x - m\|$, so the formula is well defined.
Also $\|x + M\| = 0 \iff x \in \overline{M} = M$. For $\alpha \neq 0$,
$\|\alpha(x + M)\| = \inf_{m \in M} \|\alpha x - m\| = |\alpha| \inf_{m \in M} \|x - \alpha^{-1} m\| = |\alpha| \|x + M\|$; $\alpha = 0$ is trivial.
For $m, n \in M$, $\|(x + y) - (m + n)\| \le \|x - m\| + \|y - n\|$; take infima.
(ii) Since $\|q(x)\| \le \|x - 0\| = \|x\|$, $q \in \mathcal{L}(\mathcal{X}, \mathcal{X}/M)$ and $\|q\|_{\text{op}} \le 1$.
(iii) If $z \in B_r(x)$, then $\|qz - qx\| \le \|z - x\| < r$, so $q(B_r(x)) \subseteq B_r(qx)$. Conversely, if $\eta = qz$ and $\|\eta - qx\| < r$, choose $m \in M$ with $\|z - x - m\| < r$; then $z - m \in B_r(x)$ and $q(z - m) = \eta$.
(iv) If $U \subseteq \mathcal{X}$ is open and $y \in q(U)$, choose $x \in U$ with $qx = y$ and $r > 0$ with $B_r(x) \subseteq U$.
By (iii), $B_r(y) = q(B_r(x)) \subseteq q(U)$; hence $q(U)$ is open.

Structural Commentary: Quotient norm is distance to the closed subspace M.
Key Insight: Closedness makes zero distance equivalent to membership. □

Theorem 5.1.9. *Banach Quotients*
If M is a closed linear subspace of a Banach space $\mathcal{X}$, then $\mathcal{X}/M$ is Banach in the quotient norm.

Proof. Let (ξ_n) be Cauchy in $\mathcal{X}/M$. Choose $n_1 < n_2 < \cdots$ with $\|\xi_{n_{k+1}} - \xi_{n_k}\| < 2^{-k}$ and choose $u_k \in \mathcal{X}$ with $q(u_k) = \xi_{n_{k+1}} - \xi_{n_k}$, $\|u_k\| < 2^{-k}$. Since $\sum_k \|u_k\| < +\infty$ and $\mathcal{X}$ is Banach, $u := \sum_{k \ge 1} u_k$ exists.
For $K \ge 1$, $\xi_{n_K} = \xi_{n_1} + q(\sum_{k<K} u_k) \to \xi_{n_1} + q(u)$.
Since (ξ_n) is Cauchy and has a convergent subsequence, $\xi_n \to \xi_{n_1} + q(u)$.
Structural Commentary: Rapid quotient increments lift to an absolutely convergent series.
Key Insight: A Cauchy sequence with a convergent subsequence converges. □

Proposition 5.1.10. *Operator Factor through the Kernel Quotient*
Let $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, set $M := \ker T$, and let $\widetilde{T} : \mathcal{X}/M \to \mathcal{Y}$ be induced by Theorem 2.2.12.

(i) $\widetilde{T}$ is bounded.
(ii) $\|\widetilde{T}\|_{op} = \|T\|_{op}$.

Proof. Since $q(x) = q(y) \iff x - y \in M = \ker T$, the formula $\widetilde{T}(qx) := Tx$ is well defined and $T = \widetilde{T}q$.
For $m \in M$, $\|\widetilde{T}(qx)\| = \|Tx\| = \|T(x - m)\| \le \|T\|_{\text{op}} \|x - m\|$; taking $\inf_{m \in M}$ gives $\|\widetilde{T}(qx)\| \le \|T\|_{\text{op}} \|qx\|$.

(i) The estimate gives $\widetilde{T} \in \mathcal{L}(\mathcal{X}/M, \mathcal{Y})$.
(ii) The estimate gives $\|\widetilde{T}\|_{\text{op}} \le \|T\|_{\text{op}}$. Since $T = \widetilde{T}q$ and $\|q\|_{\text{op}} \le 1$, $\|T\|_{\text{op}} \le \|\widetilde{T}\|_{\text{op}} \|q\|_{\text{op}} \le \|\widetilde{T}\|_{\text{op}}$.

Structural Commentary: Collapsing $\ker T$ makes T descend without changing its norm.
Key Insight: The quotient infimum is taken over exactly the invisible directions of T. □

Example 5.1.11. *Standard Banach Space Examples*
Standard Banach spaces include the following canonical examples.

(i) If S is a set and $B(S) := \{f : S \to \mathbb{F} : \|f\|_\infty < +\infty\}$, then $(B(S), \|\cdot\|_\infty)$ is Banach.
(ii) $\ell^\infty := \{x = (x_n) : \sup_n |x_n| < +\infty\}$ is Banach under $\|x\|_\infty := \sup_n |x_n|$.

(iii) For $1 \le p < \infty$, $\ell^p := \{x = (x_n) : \sum_n |x_n|^p < +\infty\}$ is Banach under $\|x\|_p := (\sum_n |x_n|^p)^{1/p}$.
(iv) For a measure space $(X, \mathcal{M}, \mu)$ and $1 \le p \le \infty$, $L^p(\mu)$ is Banach.
(v) $L^2(\mu)$ is Hilbert with $\langle f, g\rangle := \int_X f\overline{g}\, d\mu$.

Example 5.1.12. *Completeness of ℓ^∞ and ℓ^p*
For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $(\ell^\infty(\mathbb{F}), \|\cdot\|_\infty)$ and $(\ell^p(\mathbb{F}), \|\cdot\|_p)$, $1 \le p < \infty$, are complete.

Proof. For ℓ^∞, let $\mathbf{x}^{(n)} = (x_j^{(n)})_j$ be Cauchy. Then $x_j^{(n)} \to y_j \in \mathbb{F}$ for each j.
If $\|\mathbf{x}^{(n)} - \mathbf{x}^{(m)}\|_\infty < \varepsilon$ for $m, n \ge N$, then $|x_j^{(n)} - y_j| \le \varepsilon$ for all j, so $\|\mathbf{x}^{(n)} - \mathbf{y}\|_\infty \le \varepsilon$.
Taking $n = N$ gives $\mathbf{y} \in \ell^\infty$, hence $\mathbf{x}^{(n)} \to \mathbf{y}$.
For ℓ^p, let $\mathbf{x}^{(n)}$ be Cauchy and define $\mathbf{y} = (y_j)$ by coordinatewise limits.
If $\|\mathbf{x}^{(n)} - \mathbf{x}^{(m)}\|_p < \varepsilon$ for $m, n \ge N$, then for each r, $\sum_{j=1}^r |x_j^{(n)} - y_j|^p = \lim_{m\to\infty} \sum_{j=1}^r |x_j^{(n)} - x_j^{(m)}|^p \le \varepsilon^p$.
Letting $r \to \infty$ gives $\|\mathbf{x}^{(n)} - \mathbf{y}\|_p \le \varepsilon$ for $n \ge N$. Taking $n = N$ gives $\mathbf{y} \in \ell^p$, hence $\mathbf{x}^{(n)} \to \mathbf{y}$.
Structural Commentary: Completeness of scalar coordinates combines with uniform tail control.
Key Insight: Norm-Cauchy control keeps coordinatewise limits inside the space. □

Example 5.1.13. *Two Metrics on $C[0,1]$*
For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $C_\mathbb{F} := C([0,1], \mathbb{F})$ is complete under $d_\infty(f,g) := \sup_{x\in[0,1]} |f(x) - g(x)|$ by Theorem 8.1.1, but not under $d_1(f,g) := \int_0^1 |f(x) - g(x)|\, dx$.

Proof. For $n \ge 2$, define $f_n \in C_\mathbb{F}$ by

$$f_n(x) := \begin{cases} 0, & 0 \le x \le \frac{1}{2}, \\ n(x - \frac{1}{2}), & \frac{1}{2} \le x \le \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \le x \le 1. \end{cases}$$

Let $s := \mathbf{1}_{(1/2,1]}$. Then $d_1(f_n, s) = 1/(2n)$, hence (f_n) is d_1-Cauchy. If $f_n \to g \in C_\mathbb{F}$ in d_1, then $d_1(g, s) = 0$, so $g = s$ a.e. Continuity forces $g = 0$ on $[0, 1/2]$ and $g = 1$ on $(1/2, 1]$, impossible at $1/2$.
Structural Commentary: The integral metric permits a continuous ramp to collapse to a jump.
Key Insight: Supremum control preserves continuity; integral control can miss jumps. □

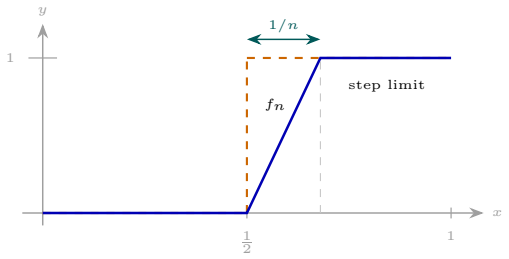

Figure 219: The continuous ramps f_n converge in the integral metric to a discontinuous step function.

Theorem 5.1.14. *Finite-Dimensional Normed Spaces*
Let $\mathcal{X}$ be a finite-dimensional normed space.
(i) $\mathcal{X}$ is Banach.
(ii) Every subspace of $\mathcal{X}$ is closed.
(iii) $\overline{B}_\mathcal{X}$ is compact.

Proof. Choose a basis $e_1, \dots, e_n$ and set $A : \mathbb{F}^n \to \mathcal{X}$, $Aa := \sum_{i=1}^n a_i e_i$.
Then A is linear bijective and $\|Aa\| \le \sum_i |a_i|\, \|e_i\| \le C\, \|a\|_2$.
Since $a \mapsto \|Aa\|$ is continuous, positive on $\{a : \|a\|_2 = 1\}$, $\exists c > 0$ with $c\, \|a\|_2 \le \|Aa\| \le C\, \|a\|_2$ $\forall a \in \mathbb{F}^n$.
(i) If (x_k) is Cauchy in $\mathcal{X}$, then $a_k := A^{-1}x_k$ is Cauchy in $\mathbb{F}^n$ by $c\, \|a_k - a_\ell\|_2 \le \|x_k - x_\ell\|$.
Since $\mathbb{F}^n$ is complete, $a_k \to a$, hence $x_k \to Aa$.
(ii) If $M \le \mathcal{X}$, then M is finite-dimensional, hence Banach by (i). If $x_k \in M$ and $x_k \to x$ in $\mathcal{X}$, then (x_k) converges in M to some $y \in M$; uniqueness of limits gives $x = y \in M$. Thus M is closed.
(iii) Since $\overline{B}_\mathcal{X} \subseteq A(\overline{B}_2(0, \frac{1}{c}))$ and $A^{-1}(\overline{B}_\mathcal{X})$ is closed in the compact Euclidean ball $\overline{B}_2(0, \frac{1}{c})$, $A^{-1}(\overline{B}_\mathcal{X})$ is compact; continuity of A gives compactness of $\overline{B}_\mathcal{X}$.
Structural Commentary: The coordinate map is a bi-Lipschitz linear isomorphism.
Key Insight: Finite-dimensional normed spaces inherit completeness and compactness from $\mathbb{F}^n$. □

Theorem 5.1.15. *Riesz Lemma*
Let M be a closed proper linear subspace of a normed space $\mathcal{X}$.
Then $\forall \varepsilon \in (0,1)$ $\exists u \in \mathcal{X}$ such that $\|u\| = 1$ and $\forall m \in M$, $\|u - m\| > 1 - \varepsilon$.

Proof. Choose $x \in \mathcal{X} \setminus M$ and set $d := \inf_{m\in M} \|x - m\|$. Since M is closed, $d > 0$.
Choose $m_0 \in M$ with $d \le \|x - m_0\| < \frac{d}{1-\varepsilon}$ and set $u := \frac{x - m_0}{\|x - m_0\|}$.
Then $\|u\| = 1$. For $m \in M$, $m_0 + \|x - m_0\|\, m \in M$, so $\|u - m\| = \frac{\|x - (m_0 + \|x - m_0\| m)\|}{\|x - m_0\|} \ge \frac{d}{\|x - m_0\|} > 1 - \varepsilon$.

Structural Commentary: A positive distance to M survives normalization.
Key Insight: Normalize an almost-minimizer of $\operatorname{dist}(x, M)$. □

Corollary 5.1.16. *Compact Unit Balls Characterize Finite Dimension*
For a normed space $\mathcal{X}$, $\overline{B}_{\mathcal{X}}$ is compact iff $\dim \mathcal{X} < \infty$.

Proof.
($\Leftarrow$) If $\dim \mathcal{X} < \infty$, then $\overline{B}_{\mathcal{X}}$ is compact by Theorem 5.1.14(iii).
($\Rightarrow$) Suppose $\dim \mathcal{X} = \infty$. Choose $x_1 \in \mathcal{X}$ with $\|x_1\| = 1$. Given $x_1, \dots, x_n$, set $M_n := \operatorname{span}\{x_1, \dots, x_n\}$.
Then M_n is finite-dimensional, closed, and proper, so Theorem 5.1.15 with $\varepsilon = \frac{1}{2}$ gives $x_{n+1} \in \mathcal{X}$ with $\|x_{n+1}\| = 1$ and $\operatorname{dist}(x_{n+1}, M_n) > \frac{1}{2}$. Since $x_i \in M_n$ for $i \le n$, $\|x_{n+1} - x_i\| > \frac{1}{2}$.
Thus $(x_n) \subseteq \overline{B}_{\mathcal{X}}$ has no Cauchy subsequence, so $\overline{B}_{\mathcal{X}}$ is not compact.
Structural Commentary: Riesz iteration builds a separated sequence in $\overline{B}_{\mathcal{X}}$.
Key Insight: Compact unit ball forces finite dimension. □

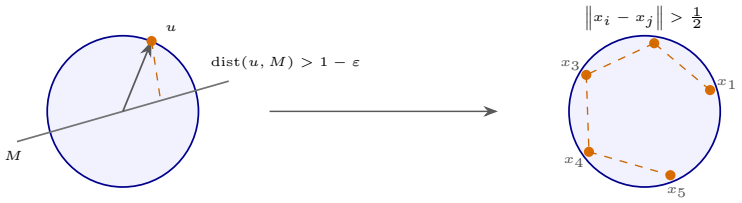

Figure 220: Riesz iteration: $M_n = \operatorname{span}\{x_1, \dots, x_n\}$, $\operatorname{dist}(x_{n+1}, M_n) > \frac{1}{2} \Rightarrow \|x_i - x_j\| > \frac{1}{2}$ $(i \neq j)$; hence no Cauchy subsequence in $\overline{B}_{\mathcal{X}}$.

5.2 Hilbert Spaces

Hilbert spaces are complete inner-product spaces. They extend the orthogonal geometry of Definition 1.1.3 to function and sequence spaces, supporting projection, Fourier expansion, and Riesz representation.

Definition 5.2.1. *Hilbert Space*
Let $\mathcal{H}$ be an inner-product space.
(i) *Hilbert space*: $\mathcal{H}$ is complete in the norm $\|x\| := \sqrt{\langle x, x \rangle}$.
(ii) *Orthonormal family*: $(e_i)_{i \in I} \subseteq \mathcal{H}$ satisfies $\forall i, j \in I$, $\langle e_i, e_j \rangle = \delta_{ij}$.

Proposition 5.2.2. *Continuity of the Inner Product*
Let $\mathcal{V}$ be an inner-product space, let $(x_n), (y_n) \subseteq \mathcal{V}$, and let $x, y \in \mathcal{V}$.
(i) $x_n \to x \Longrightarrow \|x_n\| \to \|x\|$.
(ii) $x_n \to x$ and $y_n \to y \Longrightarrow \langle x_n, y_n \rangle \to \langle x, y \rangle$.

Proof.
(i) $|\|x_n\| - \|x\|| \le \|x_n - x\| \to 0$, hence $\|x_n\| \to \|x\|$.
(ii) Since (y_n) is bounded, choose $M < \infty$ with $\|y_n\| \le M$.
Cauchy–Schwarz gives $\left|\langle x_n, y_n \rangle - \langle x, y \rangle\right| \le M \|x_n - x\| + \|x\| \|y_n - y\| \to 0$.
Structural Commentary: Cauchy–Schwarz turns norm convergence in either coordinate into scalar convergence.
Key Insight: Inner products are continuous because their errors are controlled by products of norms. □

Theorem 5.2.3. *Hilbert Completion*
Let $\mathcal{V}$ be an inner-product space.
(i) $\exists (\widehat{\mathcal{V}}, J)$ such that $\widehat{\mathcal{V}}$ is Hilbert, $J : \mathcal{V} \to \widehat{\mathcal{V}}$ is a linear isometry, and $\overline{J(\mathcal{V})} = \widehat{\mathcal{V}}$.
(ii) If $(\mathcal{K}, I)$ also satisfies (i), then $\exists! U : \widehat{\mathcal{V}} \to \mathcal{K}$ a surjective linear isometry with $U \circ J = I$.

Proof.
(i) Let $C(\mathcal{V})$ be the Cauchy sequences, set $(x_n) \sim (y_n) \iff \|x_n - y_n\| \to 0$, and define $\widehat{\mathcal{V}} := C(\mathcal{V})/\sim$.
Set $[x_n] + [y_n] := [x_n + y_n]$, $a[x_n] := [ax_n]$, $Jx := [(x)]$, and $\langle [x_n], [y_n] \rangle_{\widehat{\mathcal{V}}} := \lim_{n \to \infty} \langle x_n, y_n \rangle_{\mathcal{V}}$.
If $(x_n), (y_n)$ are bounded by M, then $\left|\langle x_n, y_n \rangle - \langle x_m, y_m \rangle\right| \le M \|y_n - y_m\| + M \|x_n - x_m\| \to 0$.
The same estimate for equivalent representatives makes the limit well defined.
Also $\|[x_n] - [y_n]\| = \lim_n \|x_n - y_n\|$, the metric-completion norm; hence $\widehat{\mathcal{V}}$ is Hilbert and $\overline{J(\mathcal{V})} = \widehat{\mathcal{V}}$.
(ii) Since $(\widehat{\mathcal{V}}, J)$ and $(\mathcal{K}, I)$ are metric completions of $\mathcal{V}$, there is a unique surjective isometry $U : \widehat{\mathcal{V}} \to \mathcal{K}$ with $U \circ J = I$. For $x, y \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $U(aJx + bJy) = U(J(ax + by)) = I(ax + by) = aUJx + bUJy$; by density and continuity, U is linear. Since $\|Uz - Uw\| = \|z - w\|$, polarization gives $\langle Uz, Uw \rangle_{\mathcal{K}} = \langle z, w \rangle_{\widehat{\mathcal{V}}}$.
Structural Commentary: Complete the metric first, then recover the inner product by taking limits.
Key Insight: Cauchy–Schwarz makes the lifted inner product well defined. □

Theorem 5.2.4. *Complete, Closed, and Finite-Dimensional Subspaces*
Let $\mathcal{S}$ be a subspace of an inner-product space $\mathcal{V}$.
(i) If $\mathcal{S}$ is complete in the inherited metric, then $\mathcal{S}$ is closed in $\mathcal{V}$.
(ii) If $\mathcal{V}$ is Hilbert, then $\mathcal{S}$ is Hilbert if and only if $\mathcal{S}$ is closed.
(iii) If $\mathcal{S}$ is finite-dimensional, then $\mathcal{S}$ is complete and closed in $\mathcal{V}$.

Proof.

(i) Let $v \in \overline{\mathcal{S}}$ and choose $s_n \in \mathcal{S}$ with $s_n \to v$ in $\mathcal{V}$. Then (s_n) is Cauchy in $\mathcal{S}$, so $s_n \to s \in \mathcal{S}$ by completeness. Uniqueness of limits gives $v = s$, hence $\overline{\mathcal{S}} = \mathcal{S}$.

(ii) Assume $\mathcal{V}$ Hilbert. If $\mathcal{S}$ is closed and $(s_n) \subseteq \mathcal{S}$ is Cauchy, then $s_n \to v \in \mathcal{V}$ and $v \in \mathcal{S}$, so $\mathcal{S}$ is complete. Conversely, if $\mathcal{S}$ is Hilbert, then $\mathcal{S}$ is complete, hence closed by (i).

(iii) Let $\dim \mathcal{S} = m$ and choose an orthonormal basis $e_1, \dots, e_m$.

If (s_n) is Cauchy in $\mathcal{S}$ and $s_n = \sum_{j=1}^{m} a_{n,j} e_j$, then $|a_{n,j} - a_{k,j}| = |\langle s_n - s_k, e_j \rangle| \le \|s_n - s_k\|$.

Thus $a_{n,j} \to a_j \in \mathbb{F}$ for each j, and $s_n \to s := \sum_{j=1}^{m} a_j e_j \in \mathcal{S}$. Hence $\mathcal{S}$ is complete, so closed by (i).

Structural Commentary: In a complete ambient space, closed subspaces are exactly complete subspaces.
Key Insight: Finite orthonormal coordinates reduce completeness to $\mathbb{F}^m$. □

Definition 5.2.5. *Series in an Inner-Product Space*
Let $(x_k) \subseteq \mathcal{V}$.

(i) The *partial sums* are $s_n := \sum_{k=1}^{n} x_k$.

(ii) The *series* $\sum_{k=1}^{\infty} x_k$ *converges* to $s \in \mathcal{V}$ iff $s_n \to s$.

(iii) The series is *absolutely convergent* iff $\sum_{k=1}^{\infty} \|x_k\| < \infty$.

Theorem 5.2.6. *Completeness via Absolutely Convergent Series*
$\mathcal{V}$ is complete $\iff$ every series $\sum_{k=1}^{\infty} x_k$ in $\mathcal{V}$ with $\sum_{k=1}^{\infty} \|x_k\| < \infty$ converges in $\mathcal{V}$.

Proof.

($\Rightarrow$) Let $s_n := \sum_{k=1}^{n} x_k$ and assume $\sum_k \|x_k\| < +\infty$. For $n > m$, $\|s_n - s_m\| \le \sum_{k=m+1}^{n} \|x_k\| \to 0$ as $m, n \to \infty$. Thus (s_n) is Cauchy, hence convergent by completeness.

($\Leftarrow$) Let (x_n) be Cauchy. Let $N_1 < N_2 < \cdots$ such that $i, j \ge N_k \Rightarrow \|x_i - x_j\| < 2^{-k}$, $y_k := x_{N_{k+1}} - x_{N_k}$. Then $\sum_k \|y_k\| \le \sum_k 2^{-k} < +\infty$, so $\sum_k y_k$ converges. Since $x_{N_m} = x_{N_1} + \sum_{k<m} y_k$, the subsequence (x_{N_m}) converges; because (x_n) is Cauchy, (x_n) converges.

Structural Commentary: Absolute tails make partial sums Cauchy; Cauchy tails contain summable increments.
Key Insight: Completeness is summability of norm-controlled series. □

Example 5.2.7. *ℓ^2 and the Parallelogram Test*
Let $\ell^2 := \{x = (x_n) : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}$. With $\langle x, y \rangle := \sum_{n=1}^{\infty} x_n \overline{y_n}$ and $\|x\|_2 = \langle x, x \rangle^{\frac{1}{2}}$, Cauchy coordinate limits and finite-set estimates give completeness, so ℓ^2 is Hilbert.
For $p \neq 2$, $\|\cdot\|_p$ is not inner-product-induced: with $x = (1, 1, 0, \dots)$ and $y = (1, -1, 0, \dots)$,

$\|x + y\|_p^2 + \|x - y\|_p^2 = 8$ while $2\|x\|_p^2 + 2\|y\|_p^2 = 4 \cdot 2^{\frac{2}{p}}$, so the parallelogram identity holds only for $p = 2$.
Structural Commentary: Completeness alone does not make a Banach space Hilbert.
Key Insight: The parallelogram identity detects whether a norm really came from an inner product.

Example 5.2.8. *The Supremum Norm Is Not Hilbertian*
For $a < b$, $C([a, b], \mathbb{F})$ with $\|h\|_\infty := \sup_{t \in [a,b]} |h(t)|$ is Banach but not Hilbertian: take $f(t) = 1$ and $g(t) = \frac{t-a}{b-a}$, so $\|f\|_\infty = \|g\|_\infty = 1$, $\|f + g\|_\infty = 2$, $\|f - g\|_\infty = 1$, and $\|f + g\|_\infty^2 + \|f - g\|_\infty^2 = 5 \neq 4 = 2\|f\|_\infty^2 + 2\|g\|_\infty^2$, violating the parallelogram identity.
Structural Commentary: Banach need not mean Hilbert.
Key Insight: One failed parallelogram identity rules out every possible inner product inducing the norm.

Theorem 5.2.9. *Hilbert Projection Theorem*
Let $C \subseteq \mathcal{H}$ be nonempty, closed, and convex. Then $\forall x \in \mathcal{H}$ $\exists! p \in C$ such that $\|x - p\| = \inf_{q \in C} \|x - q\|$.

Proof. Let $d := \inf_{q \in C} \|x - q\|$, $q_n \in C$, $\|x - q_n\| \to d$. Since $\frac{q_n + q_m}{2} \in C$, by parallelogram identity,

$\|q_n - q_m\|^2 = 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4\left\|x - \frac{q_n + q_m}{2}\right\|^2 \le 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4d^2 \to 0.$

Thus (q_n) is Cauchy. Hence $q_n \to p \in \mathcal{H}$; closedness gives $p \in C$, and $\|x - p\| = d$.
If $p_1, p_2 \in C$ both minimize, the same identity gives $\|p_1 - p_2\|^2 \le 2d^2 + 2d^2 - 4d^2 = 0$.
Structural Commentary: Convexity and parallelogram force minimizing sequences to converge in C.
Key Insight: Midpoint minimality turns near minimizers into a Cauchy sequence. □

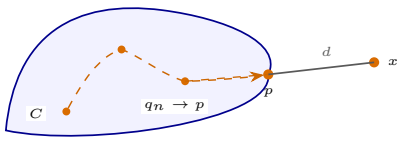

Figure 221: Hilbert projection: $q_n \in C$, $\|x - q_n\| \to d = \operatorname{dist}(x, C) \Rightarrow q_n \to p \in C$ and $\|x - p\| = d$.

Corollary 5.2.10. *Orthogonal Projection onto a Closed Subspace*
Let $M \leq \mathcal{H}$ be closed.

(i) $\forall x \in \mathcal{H}\ \exists! p \in M$ *such that* $x - p \in M^\perp$.
(ii) $\mathcal{H} = M \odot M^\perp$.

Proof.

(i) By Theorem 5.2.9, choose the unique $p \in M$ minimizing $\|x - p\|$, and set $r := x - p$.
For $y \in M$ and $t \in \mathbb{R}$, $p + ty \in M$, so $0 \leq \|r - ty\|^2 - \|r\|^2 = -2t \operatorname{Re}\langle r, y\rangle + t^2 \|y\|^2$.
Thus $\operatorname{Re}\langle r, y\rangle = 0$; over $\mathbb{C}$, replace y by iy to get $\operatorname{Im}\langle r, y\rangle = 0$. Hence $r \in M^\perp$.
If $x = p_1 + r_1 = p_2 + r_2$ with $p_i \in M$, $r_i \in M^\perp$, then $p_1 - p_2 = r_2 - r_1 \in M \cap M^\perp = \{0\}$.
(ii) By (i), each $x \in \mathcal{H}$ has $x = p + r$ with $p \in M$, $r \in M^\perp$, so $\mathcal{H} = M + M^\perp$.
Since $M \cap M^\perp = \{0\}$, the sum is direct: $\mathcal{H} = M \odot M^\perp$.

Structural Commentary: Nearest points are characterized by orthogonal residuals.
Key Insight: Every vector splits uniquely into M and $M^\perp$ parts. □

Definition 5.2.11. *Closed Linear Span and Total Sets*
For $A \subseteq \mathcal{H}$:

(i) *Closed linear span*: $\operatorname{cspan}(A) := \overline{\operatorname{span}(A)}$.
(ii) *Total*: A is total in $\mathcal{H}$ iff $\operatorname{cspan}(A) = \mathcal{H}$.

Theorem 5.2.12. *Double Orthogonal Complements in Hilbert Spaces*
Let $\mathcal{H}$ be a Hilbert space.

(i) $\forall A \subseteq \mathcal{H}$, $\operatorname{cspan}(A) = A^{\perp\perp}$.
(ii) $\forall \mathcal{S} \leq \mathcal{H}$, $\overline{\mathcal{S}} = \mathcal{S}^{\perp\perp}$.
(iii) $\forall \mathcal{K} \leq \mathcal{H}$, $\mathcal{K}$ *closed* $\iff \mathcal{K} = \mathcal{K}^{\perp\perp}$.
(iv) $\forall A \subseteq \mathcal{H}$, $\operatorname{span}(A)$ *dense in* $\mathcal{H} \iff A^\perp = \{\mathbf{0}\}$.

Proof.

(i) Let $B := \operatorname{cspan}(A)$. Since $A \subseteq B$, $B^\perp \subseteq A^\perp$. If $y \in A^\perp$, by $y \perp \operatorname{span}(A)$, continuity of $z \mapsto \langle z, y\rangle$, $y \perp B$, so $A^\perp = B^\perp$. By Corollary 5.2.10, $\mathcal{H} = B \odot B^\perp$; $B^{\perp\perp} = B$. Therefore $A^{\perp\perp} = B^{\perp\perp} = B = \operatorname{cspan}(A)$.
(ii) With $A = \mathcal{S}$ in (i), $\mathcal{S}^{\perp\perp} = \operatorname{cspan}(\mathcal{S}) = \overline{\operatorname{span}(\mathcal{S})} = \overline{\mathcal{S}}$.
(iii) By (ii), $\mathcal{K}^{\perp\perp} = \overline{\mathcal{K}}$; thus $\mathcal{K} = \mathcal{K}^{\perp\perp} \iff \mathcal{K} = \overline{\mathcal{K}} \iff \mathcal{K}$ is closed.
(iv) By (i), $\operatorname{span}(A)$ is dense $\iff A^{\perp\perp} = \mathcal{H}$. If $A^\perp = \{\mathbf{0}\}$, then $A^{\perp\perp} = \{\mathbf{0}\}^\perp = \mathcal{H}$.
Conversely, if $A^{\perp\perp} = \mathcal{H}$ and $y \in A^\perp$, then $y \in A^{\perp\perp}$, so $\|y\|^2 = \langle y, y\rangle = 0$; hence $A^\perp = \{\mathbf{0}\}$.

Structural Commentary: Double orthogonal complement equals closed linear span.
Key Insight: Missing closed-span directions are exactly nonzero orthogonal vectors. □

Example 5.2.13. *Completeness Is Needed for Double Complements*
Let $\mathcal{V} := c_{00}(\mathbb{N}; \mathbb{R}) \subseteq \ell^2$ with inherited inner product, $Lx := \sum_{n=1}^{\infty} \frac{x_n}{n}$, and $\mathcal{K} := \ker L$.

Then $|Lx| \leq \|x\|_2 \left(\sum_{n=1}^{\infty} n^{-2}\right)^{\frac{1}{2}}$, so L is continuous and $\mathcal{K}$ is a closed subspace of $\mathcal{V}$.

If $y \in \mathcal{K}^\perp$, then $e_1 - ne_n \in \mathcal{K}$ for $n \geq 2$, hence $0 = \langle y, e_1 - ne_n\rangle = y_1 - ny_n$.
Since $y \in c_{00}$, choose $n \geq 2$ with $y_n = 0$, so $y_1 = 0$ and then $\forall n,\ y_n = 0$.
Thus $\mathcal{K}^\perp = \{\mathbf{0}\}$ and $\mathcal{K}^{\perp\perp} = \mathcal{V}$. But $e_1 \notin \mathcal{K}$, hence $\mathcal{K} \neq \mathcal{V} = \mathcal{K}^{\perp\perp}$.
Structural Commentary: Closedness need not recover double complements without completeness.
Key Insight: The identity $\mathcal{K} = \mathcal{K}^{\perp\perp}$ is genuinely Hilbert-space geometry.

Perspective 5.2.14. *Hilbert and Hamel Basis Caution*
In Hilbert-space theory, an *orthonormal basis* usually means a complete orthonormal family: vectors are recovered by convergent Fourier expansion, not necessarily by finite linear combinations.
A *Hamel basis* is a vector-space basis and is much stronger in infinite dimension.
For example, the standard $(\mathbf{e}_j)$ in $\ell^2(\mathbb{F})$ is a complete orthonormal family, but $\operatorname{span}\{\mathbf{e}_j : j \geq 1\} = \mathbb{F}^{(\mathbb{N})} \subsetneq \ell^2(\mathbb{F})$.

Definition 5.2.15. *Hilbert Basis*

(i) A *Hilbert basis*, or complete orthonormal family, is a maximal orthonormal set under inclusion.
(ii) A *Hamel basis* is a vector-space basis.
(iii) Here *orthonormal basis* means Hilbert basis outside finite-dimensional Hamel context.

Proposition 5.2.16. *Finite Hilbert Bases Are Hamel Bases*
Let $\mathcal{O} = \{\mathbf{e}_1, \ldots, \mathbf{e}_m\}$ be orthonormal in $\mathcal{V}$ and set $W = \operatorname{span}(\mathcal{O})$.
Then $\mathcal{O}$ is a Hilbert basis $\iff W = \mathcal{V} \iff \mathcal{O}$ is an orthonormal Hamel basis.

Proof. Since $\mathcal{O}$ is orthonormal, it is linearly independent; hence $W = \mathcal{V} \iff \mathcal{O}$ is an orthonormal Hamel basis.
It remains to show maximality $\iff W = \mathcal{V}$.

($\Rightarrow$) Suppose $\mathcal{O}$ is maximal orthonormal and $W \neq \mathcal{V}$. Choose $\mathbf{v} \notin W$ and set $\mathbf{u} := \mathbf{v} - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j\rangle\, \mathbf{e}_j$.

Then $\langle \mathbf{u}, \mathbf{e}_i\rangle = 0$ for $1 \leq i \leq m$, and $\mathbf{u} \neq \mathbf{0}$ since $\mathbf{v} \notin W$.
Thus $\mathcal{O} \cup \{\mathbf{u}/\|\mathbf{u}\|\}$ is orthonormal, contradiction. Hence $W = \mathcal{V}$.

($\Leftarrow$) Suppose $W = \mathcal{V}$. If $\mathbf{w} \perp \mathcal{O}$, write $\mathbf{w} = \sum_{j=1}^{m} a_j \mathbf{e}_j$. Then $a_j = \langle \mathbf{w}, \mathbf{e}_j\rangle = 0$ for every j, so $\mathbf{w} = \mathbf{0}$.

Hence no nonzero vector can be adjoined orthogonally to $\mathcal{O}$; $\mathcal{O}$ is maximal orthonormal.

Structural Commentary: In finite orthonormal families, maximality equals spanning.
Key Insight: Any missing vector has a nonzero orthogonal residual. □

Example 5.2.17. *A Hilbert Basis Need Not Be a Hamel Basis*
In $\ell^2(\mathbb{F})$, let $\mathbf{e}_j = (0,\dots,0,1,0,\dots)$. Then $\{\mathbf{e}_j : j \geq 1\}$ is orthonormal and maximal, since $\forall j,\ \mathbf{x} \perp \mathbf{e}_j$ implies $\forall j,\ x_j = \langle \mathbf{x}, \mathbf{e}_j \rangle = 0$. Its finite span is $\mathbb{F}^{(\mathbb{N})} \subsetneq \ell^2(\mathbb{F})$, so it is a Hilbert basis but not a Hamel basis.

Theorem 5.2.18. *Hilbert Basis Characterizations*
Let $\mathcal{O} \subseteq \mathcal{H}$ be orthonormal. The following are equivalent:

(i) $\mathcal{O}$ *is maximal orthonormal.*
(ii) $\mathcal{O}^\perp = \{\mathbf{0}\}$.
(iii) $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) If $0 \neq v \in \mathcal{O}^\perp$, then $\mathcal{O} \cup \{v/\|v\|\}$ is orthonormal, contradicting maximality. Hence $\mathcal{O}^\perp = \{\mathbf{0}\}$.
(*ii*) $\Rightarrow$ (*iii*) By Theorem 5.2.12, $\operatorname{cspan}(\mathcal{O}) = \mathcal{O}^{\perp\perp} = (\mathcal{O}^\perp)^\perp = \{\mathbf{0}\}^\perp = \mathcal{H}$.
(*iii*) $\Rightarrow$ (*i*) If $\mathcal{O} \subsetneq \mathcal{O} \cup \{u\}$ is orthonormal with $\|u\| = 1$, then $u \in \mathcal{O}^\perp$.
By (iii) and Theorem 5.2.12(iv), $\mathcal{O}^\perp = \{\mathbf{0}\}$, contradiction. Thus $\mathcal{O}$ is maximal orthonormal.

Structural Commentary: Maximality, trivial orthogonal complement, and closed span are equivalent.
Key Insight: A Hilbert basis is total, not necessarily Hamel. □

Definition 5.2.19. *Arbitrary Sums in a Hilbert Space*
Let $(x_k)_{k\in K} \subseteq \mathcal{H}$.

(i) *Finite partial sums*: $\mathcal{F}(K) := \{J \subseteq K : |J| < \infty\}$, ordered by $\subseteq$, and $s_J := \sum_{k\in J} x_k$.
(ii) *Arbitrary convergence*: $\sum_{k\in K} x_k = x \iff \forall \varepsilon > 0\, \exists S \in \mathcal{F}(K)\, \forall T \in \mathcal{F}(K) : S \subseteq T \Rightarrow \|s_T - x\| \leq \varepsilon$.
(iii) *Nonnegative scalar sum*: for $(a_k)_{k\in K} \subseteq [0,\infty)$, $\sum_{k\in K} a_k < \infty \iff \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} a_k < \infty$.

Proposition 5.2.20. *Basic Properties of Arbitrary Sums*
Let $(x_k)_{k\in K}, (y_k)_{k\in K} \subseteq \mathcal{H}$ satisfy $\sum_{k\in K} x_k = x$ and $\sum_{k\in K} y_k = y$.

(i) $\forall a, b \in \mathbb{F},\ \sum_{k\in K} (ax_k + by_k) = ax + by$.
(ii) $\forall z \in \mathcal{H},\ \sum_{k\in K} \langle x_k, z\rangle = \langle x, z\rangle$.
(iii) $\forall z \in \mathcal{H},\ \sum_{k\in K} \langle z, x_k\rangle = \langle z, x\rangle$.

Proof. For $J \in \mathcal{F}(K)$, set $s_J := \sum_{k\in J} x_k$ and $t_J := \sum_{k\in J} y_k$.

(i) Since $s_J \to x$ and $t_J \to y$, $as_J + bt_J \to ax + by$.
(ii) For $z \in \mathcal{H}$ and $J \in \mathcal{F}(K)$, $\sum_{k\in J} \langle x_k, z\rangle = \langle s_J, z\rangle$; since $s_J \to x$, Proposition 5.2.2 gives $\langle s_J, z\rangle \to \langle x, z\rangle$.
(iii) For $z \in \mathcal{H}$ and $J \in \mathcal{F}(K)$, $\sum_{k\in J} \langle z, x_k\rangle = \langle z, s_J\rangle$; since $s_J \to x$, Proposition 5.2.2 gives $\langle z, s_J\rangle \to \langle z, x\rangle$.

Structural Commentary: Arbitrary sums are ordinary finite sums plus a convergence net.
Key Insight: Once the vector sum converges, continuous linear tests may pass through the sum. □

Theorem 5.2.21. *Arbitrary Orthonormal Sums*
Let $(u_k)_{k\in K} \subseteq \mathcal{H}$ be orthonormal.

(i) For $(r_k)_{k\in K} \subseteq \mathbb{F}$, $\sum_{k\in K} r_k u_k$ converges iff $\sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 < \infty$.
(ii) Under the conditions in (i), $\left\| \sum_{k\in K} r_k u_k \right\|^2 = \sum_{k\in K} |r_k|^2$.
(iii) $\forall (x_k)_{k\in K} \subseteq \mathcal{H}$, $\sum_{k\in K} x_k$ convergent implies $|\{k \in K : x_k \neq \mathbf{0}\}| \leq \aleph_0$.

Proof.

(*i*) ($\Rightarrow$) Let $s_J := \sum_{k\in J} r_k u_k$. If $s_J \to x$, choose $S \in \mathcal{F}(K)$ with $S \subseteq T \Rightarrow \|s_T - x\| \leq 1$.
For $J \in \mathcal{F}(K)$, $\sum_{k\in J} |r_k|^2 \leq \|s_{J\cup S}\|^2 \leq (\|x\| + 1)^2$.
(*i*) ($\Leftarrow$) Let $M := \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 < +\infty$. Given $\varepsilon > 0$, choose $S \in \mathcal{F}(K)$ with $M - \sum_{k\in S} |r_k|^2 < \varepsilon^2$.
If $S \subseteq J, L$, then $\|s_J - s_L\|^2 = \sum_{k\in J\triangle L} |r_k|^2 \leq \sum_{k\in (J\cup L)\setminus S} |r_k|^2 < \varepsilon^2$.
Thus (s_J) is Cauchy, hence converges in $\mathcal{H}$.
(*ii*) If $s_J \to s := \sum_{k\in K} r_k u_k$, then $\|s\|^2 = \lim_J \|s_J\|^2 = \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 = \sum_{k\in K} |r_k|^2$.
(*iii*) If $s_J := \sum_{k\in J} x_k$ converges, then (s_J) is Cauchy.
For each $n \geq 1$, choose $I_n \in \mathcal{F}(K)$ such that $I_n \subseteq J, L \Rightarrow \|s_J - s_L\| \leq \frac{1}{n}$.
Then $F \in \mathcal{F}(K \setminus I_n)$ implies $\left\| \sum_{k\in F} x_k \right\| = \|s_{I_n\cup F} - s_{I_n}\| \leq \frac{1}{n}$.
With $I := \bigcup_{n\geq 1} I_n$, I is countable; if $k \notin I$, then $\|x_k\| \leq \frac{1}{n}$ for all n, so $x_k = \mathbf{0}$.

Structural Commentary: Orthonormal finite sums turn net tails into square-sum tails.
Key Insight: Convergent arbitrary sums have countable support. □

Theorem 5.2.22. *Fourier Expansion in a Hilbert Space*
Let $\mathcal{O} = (u_k)_{k\in K} \subseteq \mathcal{H}$ be orthonormal. For $x \in \mathcal{H}$, set $\widehat{x}_{\mathcal{O}} := \sum_{k\in K} \langle x, u_k\rangle\, u_k$.

(i) $\widehat{x}_{\mathcal{O}}$ converges in $\mathcal{H}$.
(ii) $\|\widehat{x}_{\mathcal{O}}\|^2 = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2 \le \|x\|^2$.
(iii) $\widehat{x}_{\mathcal{O}}$ is the unique minimizer of $\|x - y\|$ over $y \in \operatorname{cspan}(\mathcal{O})$.
(iv) $\|\widehat{x}_{\mathcal{O}}\| = \|x\| \iff x \in \operatorname{cspan}(\mathcal{O})$.

Proof.
(i) For $J \in \mathcal{F}(K)$, finite Bessel gives $\sum_{k\in J} \left|\langle x, u_k\rangle\right|^2 \le \|x\|^2$.
Hence $\sup_J \sum_{k\in J} \left|\langle x, u_k\rangle\right|^2 < +\infty$, so $\sum_{k\in K} \langle x, u_k\rangle\, u_k$ converges by Theorem 5.2.21(i).
(ii) Theorem 5.2.21(ii) gives $\|\widehat{x}_{\mathcal{O}}\|^2 = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2$; finite Bessel gives the inequality $\sum_{k\in K} \left|\langle x, u_k\rangle\right|^2 \le \|x\|^2$.
(iii) For $i \in K$, Proposition 5.2.20 gives $\langle \widehat{x}_{\mathcal{O}}, u_i\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \langle u_k, u_i\rangle = \langle x, u_i\rangle$, so $r := x - \widehat{x}_{\mathcal{O}} \perp \mathcal{O}$.
By continuity, $r \perp M := \operatorname{cspan}(\mathcal{O})$. For $y \in M$, $x - y = r + (\widehat{x}_{\mathcal{O}} - y)$ orthogonally,
so $\|x - y\|^2 = \|r\|^2 + \|\widehat{x}_{\mathcal{O}} - y\|^2 \ge \|r\|^2 = \|x - \widehat{x}_{\mathcal{O}}\|^2$, with equality iff $y = \widehat{x}_{\mathcal{O}}$.
(iv) ($\Rightarrow$) As $x = \widehat{x}_{\mathcal{O}} + r$ orthogonally, $\|x\|^2 = \|\widehat{x}_{\mathcal{O}}\|^2 + \|r\|^2$. If $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$, then $r = 0$, $x = \widehat{x}_{\mathcal{O}} \in \operatorname{cspan}(\mathcal{O})$.
(iv) ($\Leftarrow$) If $x \in \operatorname{cspan}(\mathcal{O})$, the minimizer of $\|x - y\|$ over $y \in \operatorname{cspan}(\mathcal{O})$ is x; by (iii), $\widehat{x}_{\mathcal{O}} = x$, $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$.
Structural Commentary: Fourier expansion is orthogonal projection onto $\operatorname{cspan}(\mathcal{O})$.
Key Insight: Bessel bounds the coefficients; completeness sums them. □

Corollary 5.2.23. *Parseval Characterization of Hilbert Bases*
For an orthonormal family $\mathcal{O} = \{u_k : k \in K\}$ in $\mathcal{H}$, the following are equivalent:

(i) $\mathcal{O}$ is a Hilbert basis.
(ii) $\forall x \in \mathcal{H}$, $x = \sum_{k\in K} \langle x, u_k\rangle\, u_k$.
(iii) $\forall x, y \in \mathcal{H}$, Parseval's identity $\langle x, y\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \overline{\langle y, u_k\rangle}$ holds.
(iv) $\forall x \in \mathcal{H}$, $\|x\|^2 = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2$.

Proof.
(i) $\Rightarrow$ (ii) By Theorem 5.2.18, $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$. For $x \in \mathcal{H}$, both x and $\widehat{x}_{\mathcal{O}}$ minimize $\|x - y\|$ over $y \in \operatorname{cspan}(\mathcal{O})$.
By uniqueness in Theorem 5.2.22(iii), $x = \widehat{x}_{\mathcal{O}} = \sum_{k\in K} \langle x, u_k\rangle\, u_k$.
(ii) $\Rightarrow$ (iii) For $x, y \in \mathcal{H}$, (ii) and Proposition 5.2.20 give $\langle x, y\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \langle u_k, y\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \overline{\langle y, u_k\rangle}$.
(iii) $\Rightarrow$ (iv) Put $y = x$: $\|x\|^2 = \langle x, x\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \overline{\langle x, u_k\rangle} = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2$.
(iv) $\Rightarrow$ (i) By Theorem 5.2.22(ii), (iv) gives $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$ for every $x \in \mathcal{H}$.
By Theorem 5.2.22(iv), $x \in \operatorname{cspan}(\mathcal{O})$ for every x, so $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$. Theorem 5.2.18 gives (i).
Structural Commentary: Totality, Fourier expansion, Parseval, and norm equality are equivalent.
Key Insight: Parseval preserves all Hilbert geometry. □

Theorem 5.2.24. *Existence and Cardinality of Hilbert Bases*
Let $\mathcal{H}$ be a Hilbert space.

(i) $\mathcal{H}$ admits a Hilbert basis.
(ii) If $\mathcal{B}, \mathcal{C}$ are Hilbert bases of $\mathcal{H}$, then $|\mathcal{B}| = |\mathcal{C}|$.

Proof.
(i) Let $\mathfrak{O}$ be the orthonormal subsets of $\mathcal{H}$, ordered by $\subseteq$. Chains in $\mathfrak{O}$ have orthonormal unions, so Principle 6.1.3 gives maximal $\mathcal{O} \in \mathfrak{O}$; by Theorem 5.2.18, $\mathcal{O}$ is a Hilbert basis.
(ii) Let $\mathcal{B} = \{b_i : i \in I\}$ and $\mathcal{C} = \{c_j : j \in J\}$ be Hilbert bases.
If $|I| < \infty$, Proposition 5.2.16 gives $\dim \mathcal{H} = |I|$; orthonormality gives $|J| \le \dim \mathcal{H}$, hence $|J| < \infty$, and Proposition 5.2.16 gives $|J| = \dim \mathcal{H}$. The case $|J| < \infty$ is symmetric.
Assume I, J infinite. For $i \in I$, set $J_i := \{j \in J : \langle b_i, c_j\rangle \neq 0\}$. By Theorem 5.2.21(iii), $|J_i| \le \aleph_0$.
Since $\mathcal{B}^\perp = \{\mathbf{0}\}$, $j \notin \bigcup_{i\in I} J_i$ is impossible; hence $J = \bigcup_{i\in I} J_i$ and $|J| \le \sum_{i\in I} |J_i| \le \aleph_0\, |I| = |I|$.
Symmetry gives $|I| \le |J|$; Theorem 6.2.3 gives $|I| = |J|$.
Structural Commentary: Zorn gives existence; countable coordinate supports compare cardinalities.
Key Insight: Hilbert-basis size is intrinsic. □

Definition 5.2.25. *Hilbert Dimension*
For a Hilbert space $\mathcal{H}$ with Hilbert basis $\mathcal{B}$, define $\operatorname{hdim}(\mathcal{H}) := |\mathcal{B}|$; this is well defined by Theorem 5.2.24.

Proposition 5.2.26. *The Hilbert Space $\ell^2(K)$*
Let K be a set and set $\ell^2(K) := \{f : K \to \mathbb{F} : \sum_{k\in K} |f(k)|^2 < \infty\}$.

(i) $\ell^2(K)$ is Hilbert under $\langle f, g\rangle := \sum_{k\in K} f(k)\overline{g(k)}$.
(ii) The coordinate family $(\delta_k)_{k\in K}$ is a Hilbert basis of $\ell^2(K)$.

Proof.

(i) For $f, g \in \ell^2(K)$ and finite $F \subseteq K$, $\sum_{k\in F} \left|f(k)\overline{g(k)}\right| \le (\sum_{k\in F} |f(k)|^2)^{1/2}(\sum_{k\in F} |g(k)|^2)^{1/2} \le \|f\|_2 \|g\|_2$, so $\langle f, g\rangle$ is well defined. If (f_n) is Cauchy, then $|f_m(k) - f_n(k)| \le \|f_m - f_n\|_2$ for each k, so set $f(k) := \lim_n f_n(k)$.
Given $\varepsilon > 0$, choose N with $\|f_m - f_n\|_2 < \varepsilon$ for $m, n \ge N$.
Then for finite $F \subseteq K$ and $n \ge N$, $\sum_{k\in F} |f(k) - f_n(k)|^2 = \lim_m \sum_{k\in F} |f_m(k) - f_n(k)|^2 \le \varepsilon^2$.
Hence $f - f_n \in \ell^2(K)$, $\|f - f_n\|_2 \le \varepsilon$, $f \in \ell^2(K)$, and $f_n \to f$.

(ii) $\langle \delta_j, \delta_k\rangle = \mathbf{1}_{\{j=k\}}$. If $f \perp \delta_k$ for every $k \in K$, then $f(k) = \langle f, \delta_k\rangle = 0$ for every k, so $(\delta_k)_{k\in K}^{\perp} = \{\mathbf{0}\}$.
By Theorem 5.2.18, $(\delta_k)_{k\in K}$ is a Hilbert basis.

Structural Commentary: Coordinatewise limits and uniform finite-subset bounds prove completeness.
Key Insight: Square-summability is precisely the norm condition for arbitrary orthogonal coordinates. □

Corollary 5.2.27. *Classification by Hilbert Dimension*
For Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, $\mathcal{H}_1 \cong_{\text{iso}} \mathcal{H}_2$ iff $\operatorname{hdim}(\mathcal{H}_1) = \operatorname{hdim}(\mathcal{H}_2)$.

Proof.

(⇒) Let $U : \mathcal{H}_1 \to \mathcal{H}_2$ be a surjective linear isometry and let $\mathcal{B}$ be a Hilbert basis of $\mathcal{H}_1$.
Then $U\mathcal{B}$ is orthonormal and $\operatorname{cspan}(U\mathcal{B}) = U(\operatorname{cspan}\mathcal{B}) = \mathcal{H}_2$.
Thus $U\mathcal{B}$ is a Hilbert basis and $\operatorname{hdim}(\mathcal{H}_1) = |\mathcal{B}| = |U\mathcal{B}| = \operatorname{hdim}(\mathcal{H}_2)$.

(⇐) Let $\operatorname{hdim}(\mathcal{H}_1) = \operatorname{hdim}(\mathcal{H}_2) = |I|$, and choose Hilbert bases $(b_i)_{i\in I}$ and $(c_i)_{i\in I}$.
Define $\Phi_1 x := (i \mapsto \langle x, b_i\rangle)$ and $\Phi_2 y := (i \mapsto \langle y, c_i\rangle)$. By Corollary 5.2.23, $\|\Phi_1 x\|_2 = \|x\|$, $\|\Phi_2 y\|_2 = \|y\|$.
If $a = (a_i) \in \ell^2(I)$, then $\sum_{i\in I} a_i b_i$ and $\sum_{i\in I} a_i c_i$ converge by Theorem 5.2.21, with images a under Φ_1 and Φ_2.
Hence $U := \Phi_2^{-1}\Phi_1 : \mathcal{H}_1 \to \mathcal{H}_2$ is a surjective linear isometry.

Structural Commentary: Map both spaces isometrically onto the same coordinate model.
Key Insight: Hilbert spaces are classified by the number of orthogonal coordinate directions. □

Corollary 5.2.28. *Separable Hilbert Spaces*
$\mathcal{H}$ separable $\iff \operatorname{hdim}(\mathcal{H}) \le \aleph_0$.

Proof.

(⇒) Let $D \subseteq \mathcal{H}$ be countable dense and let $\mathcal{O}$ be a Hilbert basis. $\forall u \in \mathcal{O}$, choose $d_u \in D$ with $\|u - d_u\| < \frac{1}{3}$.
If $u \ne v$ and $d_u = d_v$, then $\sqrt{2} = \|u - v\| \le \|u - d_u\| + \|d_v - v\| < \frac{2}{3}$, impossible.
Hence $u \mapsto d_u$ injects $\mathcal{O}$ into D, so $\operatorname{hdim}(\mathcal{H}) = |\mathcal{O}| \le \aleph_0$.

(⇐) Let $\mathcal{O}$ be a Hilbert basis with $|\mathcal{O}| \le \aleph_0$. The finite $\mathbb{Q}$-span of $\mathcal{O}$ over $\mathbb{R}$, respectively finite $(\mathbb{Q} + i\mathbb{Q})$-span over $\mathbb{C}$, is countable and dense in $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$.

Structural Commentary: Countable bases and countable dense sets force each other.
Key Insight: Separability of a Hilbert space is exactly countability of its orthogonal coordinate directions. □

Corollary 5.2.29. *Infinite Hilbert Bases Are Not Hamel Bases*
If $\operatorname{hdim}(\mathcal{H})$ is infinite, no Hilbert basis of $\mathcal{H}$ is a Hamel basis.

Proof. Let $\mathcal{O}$ be Hilbert basis, choose distinct $u_n \in \mathcal{O}$.
Since $\sum_n 4^{-n} < \infty$, $x := \sum_n 2^{-n} u_n \in \mathcal{H}$ by Theorem 5.2.21.

For every n, continuity of $\langle \cdot, u_n\rangle$ gives $\langle x, u_n\rangle = 2^{-n} \ne 0$. If $\mathcal{O}$ were Hamel, then $x = \sum_{j=1}^{N} a_j v_j$ with $v_j \in \mathcal{O}$; choosing n with $u_n \notin \{v_1, \dots, v_N\}$ gives $\langle x, u_n\rangle = 0$, contradiction.
Structural Commentary: Infinite Hilbert coordinates allow square-summable infinite expansions.
Key Insight: A Hilbert basis spans by limits, while a Hamel basis spans by finite sums. □

Perspective 5.2.30. *Bounded Functionals*
Here $\mathcal{H}^* = \mathcal{L}(\mathcal{H}, \mathbb{F})$ by Definition 5.1.4 and Theorem 5.1.5.

Proposition 5.2.31. *Operator Norm Algebra*
Let $S, T \in \mathcal{L}(\mathcal{H})$ and $a \in \mathbb{F}$.

(i) $\|aT\|_{op} = |a| \|T\|_{op}$.
(ii) $\|S + T\|_{op} \le \|S\|_{op} + \|T\|_{op}$.
(iii) $\|ST\|_{op} \le \|S\|_{op} \|T\|_{op}$.
(iv) $\mathcal{L}(\mathcal{H})$ is a normed algebra under composition.

Proof.

(i) $\|aT\|_{\text{op}} = \sup_{\|x\|\le 1} \|aTx\| = |a| \sup_{\|x\|\le 1} \|Tx\| = |a| \|T\|_{\text{op}}$.

(ii) For $\|x\| \le 1$, $\|(S + T)x\| \le \|Sx\| + \|Tx\| \le \|S\|_{\text{op}} + \|T\|_{\text{op}}$; taking sup gives the claim.

(iii) For $\|x\| \le 1$, $\|STx\| \le \|S\|_{\text{op}} \|Tx\| \le \|S\|_{\text{op}} \|T\|_{\text{op}}$; taking sup gives the claim.

(iv) Items (i)–(ii), with $\|T\|_{\text{op}} = 0 \Rightarrow T = 0$, give a norm on $\mathcal{L}(\mathcal{H})$.
Item (iii) gives closure and submultiplicativity for associative composition.

Structural Commentary: Composition multiplies stretch bounds.
Key Insight: Bounded operators behave like matrices with a compatible norm. □

Theorem 5.2.32. *Riesz Representation for Hilbert Spaces*
For $\varphi \in \mathcal{H}^*$:

(i) $\exists! y \in \mathcal{H}$ *such that* $\forall x \in \mathcal{H}$, $\varphi(x) = \langle x, y\rangle$.

(ii) The representing vector satisfies $\|\varphi\|_{op} = \|y\|$.

Proof.

(i) If $\varphi = 0$, take $y = 0$. If $\varphi \neq 0$, set $M := \ker\varphi$, $a \notin M$, $a = p + u$, $p \in M$, $u \in M^{\perp}$ by Corollary 5.2.10.
Then $u \neq 0$ and $\varphi(u) = \varphi(a) \neq 0$. For $x \in \mathcal{H}$, $x - \frac{\varphi(x)}{\varphi(u)}u \in M$, hence $\langle x, u\rangle = \frac{\varphi(x)}{\varphi(u)}\|u\|^2$.
Thus $y := \overline{\varphi(u)}\,\|u\|^{-2}\,u$ satisfies $\varphi(x) = \langle x, y\rangle$ for every x.
If z also represents φ, then $\langle x, y - z\rangle = 0$ for every x; with $x = y - z$, $z = y$.

(ii) For every x, $|\varphi(x)| = \big|\langle x, y\rangle\big| \leq \|x\|\,\|y\|$, so $\|\varphi\|_{\mathrm{op}} \leq \|y\|$.
If $y \neq 0$, then $\|\varphi\|_{\mathrm{op}} \geq \frac{|\varphi(y)|}{\|y\|} = \|y\|$; if $y = 0$, both norms vanish.

Structural Commentary: The kernel is a hyperplane; its orthogonal direction represents the functional.
Key Insight: Bounded functionals are exactly inner products with fixed vectors. □

Corollary 5.2.33. *Riesz Identifies the Continuous Dual*
Define $R : \mathcal{H} \to \mathcal{H}^*$ *by* $Ry := (x \mapsto \langle x, y\rangle)$.

(i) R *is an isometric bijection.*
(ii) R *is conjugate-linear over* $\mathbb{C}$ *and linear over* $\mathbb{R}$.

Proof.

(i) For $y \in \mathcal{H}$, $|Ry(x)| = \big|\langle x, y\rangle\big| \leq \|x\|\,\|y\|$, so $Ry \in \mathcal{H}^*$. By Theorem 5.2.32, $\mathcal{H}^* = R(\mathcal{H})$ and $\|Ry\|_{\mathrm{op}} = \|y\|$.
If $Ry = Rz$, then $0 = \|R(y - z)\|_{\mathrm{op}} = \|y - z\|$; hence R is bijective and isometric.

(ii) For $x, y, z \in \mathcal{H}$ and $\alpha, \beta \in \mathbb{C}$, $R(\alpha y + \beta z)(x) = \langle x, \alpha y + \beta z\rangle = \overline{\alpha}Ry(x) + \overline{\beta}Rz(x)$.
Over $\mathbb{R}$, $R(\alpha y + \beta z) = \alpha Ry + \beta Rz$.

Structural Commentary: Continuous functionals are represented by vectors.
Key Insight: Riesz identifies $\mathcal{H}^*$ with $\mathcal{H}$, up to complex conjugation. □

Theorem 5.2.34. *Bounded Hilbert Adjoints*
Let $T \in \mathcal{L}(\mathcal{H})$.

(i) $\exists! T^* \in \mathcal{L}(\mathcal{H})$ *such that* $\forall u, v \in \mathcal{H}$, $\langle Tu, v\rangle = \langle u, T^*v\rangle$.
(ii) $\|T^*\|_{op} = \|T\|_{op}$.
(iii) $(T^*)^* = T$.

Proof.

(i) For $v \in \mathcal{H}$, set $\phi_v(u) := \langle Tu, v\rangle$. Since $|\phi_v(u)| \leq \|T\|_{\mathrm{op}}\,\|v\|\,\|u\|$, $\phi_v \in \mathcal{H}^*$.
By Theorem 5.2.32, $\exists! T^*v \in \mathcal{H}$ with $\phi_v(u) = \langle u, T^*v\rangle$ for every u.
For $\alpha, \beta \in \mathbb{F}$, $\phi_{\alpha v + \beta w} = \overline{\alpha}\phi_v + \overline{\beta}\phi_w$, so uniqueness gives $T^*(\alpha v + \beta w) = \alpha T^*v + \beta T^*w$.
Also $\|T^*v\| = \|\phi_v\|_{\mathrm{op}} \leq \|T\|_{\mathrm{op}}\,\|v\|$, hence $T^* \in \mathcal{L}(\mathcal{H})$.
If A also satisfies the adjoint identity, then $\langle u, (T^* - A)v\rangle = 0$ for every u, v, so $A = T^*$.

(ii) If $\mathcal{H} = \{\mathbf{0}\}$, immediate. Else $\|T\|_{\mathrm{op}} = \sup_{\|u\|=\|v\|=1}\big|\langle Tu, v\rangle\big| = \sup_{\|u\|=\|v\|=1}\big|\langle u, T^*v\rangle\big| = \|T^*\|_{\mathrm{op}}$.

(iii) For $u, v \in \mathcal{H}$, $\langle T^*u, v\rangle = \overline{\langle v, T^*u\rangle} = \overline{\langle Tv, u\rangle} = \langle u, Tv\rangle$.
Thus T satisfies the adjoint identity for T^*; by uniqueness, $(T^*)^* = T$.

Structural Commentary: Riesz turns the scalar tests induced by an operator back into vectors.
Key Insight: The adjoint is the dual action transported through the Hilbert-space Riesz map. □

Corollary 5.2.35. *Hilbert Adjoint Algebra Identities*
Let $S, T \in \mathcal{L}(\mathcal{H})$ *and* $a \in \mathbb{F}$.

(i) $(aS + T)^* = \overline{a}S^* + T^*$.
(ii) $(ST)^* = T^*S^*$.

Proof.

(i) For $u, v \in \mathcal{H}$, $\langle (aS + T)u, v\rangle = a\langle Su, v\rangle + \langle Tu, v\rangle = \langle u, \overline{a}S^*v + T^*v\rangle$.
Uniqueness of adjoints gives $(aS + T)^* = \overline{a}S^* + T^*$.

(ii) For $u, v \in \mathcal{H}$, $\langle STu, v\rangle = \langle Tu, S^*v\rangle = \langle u, T^*S^*v\rangle$; uniqueness gives $(ST)^* = T^*S^*$.

Structural Commentary: Move each operator across one inner product and invoke adjoint uniqueness.
Key Insight: Taking adjoints reverses composition and conjugates scalars. □

Definition 5.2.36. *Self-Adjoint, Nonnegative, and Coercive Operators*
Let $T \in \mathcal{L}(\mathcal{H})$.

(i) *Self-adjoint*: $T = T^*$.
(ii) *Nonnegative*: $T = T^*$ and $\forall h \in \mathcal{H}$, $\langle Th, h\rangle \geq 0$; write $T \geq 0$.
(iii) *Coercive*: if $\mathcal{H}$ is real, $\exists c > 0$ such that $\forall h \in \mathcal{H}$, $\langle Th, h\rangle \geq c\,\|h\|^2$.

Proposition 5.2.37. *Positive Order, Polarization, and Reducing Complements*
Let $\mathcal{H}$ *be a real Hilbert space.*

(i) If $T \geq 0$ *and* $S \in \mathcal{L}(\mathcal{H})$, *then* $S^*TS \geq 0$.
(ii) If $T \in \mathcal{L}(\mathcal{H})$ *is self-adjoint and* $Q_T(u) := \langle Tu, u\rangle$, *then* $\langle Tu, v\rangle = \frac{1}{4}\left[Q_T(u + v) - Q_T(u - v)\right]$.
(iii) If $T \in \mathcal{L}(\mathcal{H})$ *is self-adjoint and* $Q_T \equiv 0$, *then* $T = 0$.
(iv) If $T \in \mathcal{L}(\mathcal{H})$ *is self-adjoint and* $V \leq \mathcal{H}$ *satisfies* $T(V) \subseteq V$, *then* $T(V^{\perp}) \subseteq V^{\perp}$.

Proof.

(i) Since $T = T^*$, $(S^*TS)^* = S^*T^*S = S^*TS$. For $h \in \mathcal{H}$, $\langle S^*TSh, h\rangle = \langle TSh, Sh\rangle \geq 0$; hence $S^*TS \geq 0$.

(ii) For $u, v \in \mathcal{H}$, self-adjointness gives $Q_T(u \pm v) = Q_T(u) \pm 2\langle Tu, v\rangle + Q_T(v)$.
Subtracting gives $Q_T(u+v) - Q_T(u-v) = 4\langle Tu, v\rangle$.
(iii) By (ii), $\langle Tu, v\rangle = 0$ for all $u, v \in \mathcal{H}$; with $v = Tu$, $\|Tu\|^2 = 0$ for every u, so $T = 0$.
(iv) If $w \in V^\perp$ and $v \in V$, then $Tv \in V$ and $\langle Tw, v\rangle = \langle w, Tv\rangle = 0$; hence $Tw \in V^\perp$.

Structural Commentary: Adjoints transfer positivity and invariance; polarization recovers mixed terms.
Key Insight: For self-adjoint operators, the diagonal quadratic form controls the mixed inner products. □

Theorem 5.2.38. *Self-Adjoint Operator Norm from the Quadratic Form*
If $\mathcal{H}$ is real and $T \in \mathcal{L}(\mathcal{H})$ is self-adjoint, then $\|T\|_{op} = \sup_{\|u\|=1} \left|\langle Tu, u\rangle\right|$.

Proof. Let $\eta := \sup_{\|u\|=1} \left|\langle Tu, u\rangle\right|$. By Cauchy–Schwarz, $\eta \le \|T\|_{\mathrm{op}}$.
If $\eta = 0$, then $Q_T \equiv 0$ by scaling, so $T = 0$ by Proposition 5.2.37(iii).
Assume $\eta > 0$ and set $A_\pm := \eta I \pm T$. For $u \in \mathcal{H}$, $\left|\langle Tu, u\rangle\right| \le \eta \|u\|^2$, hence $\langle A_\pm u, u\rangle \ge 0$ and $A_\pm \ge 0$.
By Proposition 5.2.37(i), $A_+^* A_- A_+ \ge 0$ and $A_-^* A_+ A_- \ge 0$.
Since $A_\pm = A_\pm^*$ and $A_+A_- = A_-A_+ = \eta^2 I - T^2$, their sum is $2\eta(\eta^2 I - T^2) \ge 0$, so $\eta^2 I - T^2 \ge 0$.
Thus $\|Tu\|^2 = \langle T^2 u, u\rangle \le \eta^2 \|u\|^2$ for every u, hence $\|T\|_{\mathrm{op}} \le \eta$.
Structural Commentary: Self-adjointness lets order information about $\eta I \pm T$ control T^2.
Key Insight: A bounded self-adjoint operator is normed by its one-vector quadratic tests. □

Proposition 5.2.39. *Closed Range and the Adjoint Kernel*
If $T \in \mathcal{L}(\mathcal{H})$ has closed range, then $\mathcal{H} = \operatorname{range}(T) \odot \ker(T^*)$.

Proof. For $z \in \mathcal{H}$, $z \in \ker(T^*) \iff \forall u \in \mathcal{H},\ \langle u, T^* z\rangle = 0 \iff \forall u \in \mathcal{H},\ \langle Tu, z\rangle = 0 \iff z \in \operatorname{range}(T)^\perp$.
Thus $\ker(T^*) = \operatorname{range}(T)^\perp$, and Corollary 5.2.10 gives $\mathcal{H} = \operatorname{range}(T) \odot \ker(T^*)$ since $\operatorname{range}(T)$ is closed.
Structural Commentary: Identify the orthogonal complement of the range by moving T across the inner product.
Key Insight: The directions missed by T are exactly the vectors annihilated by T^*. □

Theorem 5.2.40. *Coercive Operators Are Invertible*
Let $\mathcal{H}$ be real and let $T \in \mathcal{L}(\mathcal{H})$ satisfy $\exists c > 0$ such that $\forall h \in \mathcal{H}$, $\langle Th, h\rangle \ge c\|h\|^2$.
(i) T is bijective.
(ii) $T^{-1} \in \mathcal{L}(\mathcal{H})$ and $\left\|T^{-1}\right\|_{op} \le c^{-1}$.

Proof. For $u \ne \mathbf{0}$, $c\|u\|^2 \le \langle Tu, u\rangle \le \|Tu\|\,\|u\|$, hence $\|Tu\| \ge c\|u\|$; the same holds for $u = \mathbf{0}$, so $\ker T = \{\mathbf{0}\}$.
If $Tu_n \to y$, then $\|u_n - u_m\| \le c^{-1}\|Tu_n - Tu_m\|$; hence $u_n \to u$ and $Tu = y$, so $\operatorname{range}(T)$ is closed.
Also $\langle T^* h, h\rangle = \langle Th, h\rangle \ge c\|h\|^2$, so $\ker(T^*) = \{\mathbf{0}\}$.
(i) By Proposition 5.2.39, $\mathcal{H} = \operatorname{range}(T) \odot \ker(T^*) = \operatorname{range}(T)$; with $\ker T = \{\mathbf{0}\}$, T is bijective.
(ii) For $y \in \mathcal{H}$, put $u = T^{-1}y$. Then $c\left\|T^{-1}y\right\| = c\|u\| \le \|Tu\| = \|y\|$, so $T^{-1} \in \mathcal{L}(\mathcal{H})$ and $\left\|T^{-1}\right\|_{\mathrm{op}} \le c^{-1}$.

Structural Commentary: Coercivity closes the range and kills both kernels; adjoint splitting gives surjectivity.
Key Insight: Coercivity supplies both a lower Lipschitz bound and, through the adjoint, surjectivity. □

Theorem 5.2.41. *Lax–Milgram Lemma*
Let $\mathcal{H}$ be a real Hilbert space and let $B : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ be bilinear.
Suppose there are constants $M, c > 0$ such that $|B(u,v)| \le M\|u\|\,\|v\|$ and $B(h,h) \ge c\|h\|^2$ $\forall u, v, h \in \mathcal{H}$.
Then $\forall \psi \in \mathcal{H}^*$ $\exists! h \in \mathcal{H}$ such that $\psi(u) = B(h, u)$ $(u \in \mathcal{H})$.

Proof. Let $R : \mathcal{H} \to \mathcal{H}^*$ be the real Riesz isometry, $Ry(u) = \langle u, y\rangle$, and define $S : \mathcal{H} \to \mathcal{H}^*$ by $Sh(u) = B(h, u)$. For $h, u \in \mathcal{H}$, $|Sh(u)| \le M\|h\|\,\|u\|$, hence $S \in \mathcal{L}(\mathcal{H}, \mathcal{H}^*)$.
Set $A := R^{-1}S \in \mathcal{L}(\mathcal{H})$. For $h \in \mathcal{H}$, $\langle Ah, h\rangle = \langle h, Ah\rangle = (RAh)(h) = Sh(h) = B(h,h) \ge c\|h\|^2$.
By Theorem 5.2.40, A is bijective, hence $S = RA$ is bijective.
For $\psi \in \mathcal{H}^*$, set $h := A^{-1}R^{-1}\psi$; then $\forall u \in \mathcal{H}$, $\psi(u) = (RAh)(u) = Sh(u) = B(h, u)$.
If h_1, h_2 solve this identity, then $S(h_1 - h_2) = 0$, so $h_1 = h_2$.
Structural Commentary: A coercive bilinear form is a Riesz-represented coercive operator in disguise.
Key Insight: Solving $B(h, \cdot) = \psi$ is solving an invertible operator equation after Riesz representation. □

Theorem 5.2.42. *Orthonormal Systems in L^2*
Let $(\phi_n)_{n\ge1} \subseteq L^2(\mu)$ be orthonormal, $\langle \phi_n, \phi_m\rangle = \delta_{nm}$, and $c_n(f) := \langle f, \phi_n\rangle = \int_X f\overline{\phi_n}\,d\mu$.

(i) $\forall f \in L^2(\mu)$, $\sum_{n=1}^{\infty} |c_n(f)|^2 \le \|f\|_2^2$.

(ii) If $(c_n) \in \ell^2$, then $s_N := \sum_{k=1}^{N} c_k \phi_k$ converges in $L^2(\mu)$.

(iii) If (ϕ_n) is complete, then $\forall f \in L^2(\mu)$, $\|f\|_2^2 = \sum_{n=1}^{\infty} |c_n(f)|^2$.

Proof.

(i) Set $s_N(f) := \sum_{k=1}^{N} c_k(f)\phi_k$. Finite orthonormality gives $\|f - s_N(f)\|_2^2 = \|f\|_2^2 - \sum_{k=1}^{N} |c_k(f)|^2 \ge 0$.
Monotone convergence of the partial sums yields $\sum_{n=1}^{\infty} |c_n(f)|^2 \le \|f\|_2^2$.

(ii) If $m < N$, then $\|s_N - s_m\|_2^2 = \sum_{k=m+1}^{N} |c_k|^2$. Since $(c_n) \in \ell^2$, (s_N) is Cauchy.
By completeness, $\exists g \in L^2(\mu)$ with $s_N \to g$.

(iii) Assume (ϕ_n) is complete. By (i), $(c_n(f)) \in \ell^2$; by (ii), $s_N(f) \to g$.
For fixed j, $\langle s_N(f), \phi_j\rangle = c_j(f)$ for $N \geq j$, hence $\langle f-g, \phi_j\rangle = c_j(f) - \lim_N \langle s_N(f), \phi_j\rangle = 0$.
Thus $f - g \perp \{\phi_j : j \geq 1\}$, so $f = g$; therefore $\|f\|_2^2 = \lim_N \|s_N(f)\|_2^2 = \sum_{n=1}^{\infty} |c_n(f)|^2$.

Structural Commentary: Complete orthonormal systems coordinatize L^2 by square-summable sequences.
Key Insight: Parseval is Pythagoras after passing from finite coordinates to an L^2 limit. □

Method 5.2.43. *Hilbert Space Proof Patterns*

(i) *Complete*: Cauchy tails, completions, and convergent subsequences force limits.
(ii) *Parallelogram*: test Hilbertian norms and make distance-minimizing sequences Cauchy.
(iii) *Project*: for closed M, write $x = p + r$, $p \in M$, $r \in M^\perp$.
(iv) *Separate*: density and equality via $M^\perp = \{\mathbf{0}\}$ and $M^{\perp\perp} = \overline{M}$.
(v) *Sum*: arbitrary convergence is net convergence over $\mathcal{F}(K)$; control finite tails.
(vi) *Expand*: orthonormal finite identities + Bessel bounds + completeness $\Rightarrow$ Fourier expansions.
(vii) *Classify*: Hilbert bases give hdim, separability, and $\mathcal{H} \cong \ell^2(K)$.
(viii) *Dualize*: Riesz identifies $\mathcal{H}^*$ with $\mathcal{H}$, producing adjoints and coercive operator equations.

Key Insight: Hilbert-space arguments are finite orthogonal geometry plus completeness.

5.3 Core Functional Analytic Theorems

The core theorems explain why complete normed spaces behave rigidly under linear maps. Hahn–Banach supplies norm-preserving extensions, while Baire category turns pointwise control into uniform boundedness.

Theorem 5.3.1. *Hahn–Banach Extension Theorem*
Let $\mathcal{X}$ be a real normed space, $M \leq \mathcal{X}$, $p : \mathcal{X} \to \mathbb{R}$ sublinear, and $\varphi : M \to \mathbb{R}$ linear with $\varphi \leq p$ on M.

(i) If $x_0 \in \mathcal{X} \setminus M$ and $M_0 := M + \mathbb{R}x_0$, then $\exists \varphi_0 : M_0 \to \mathbb{R}$ linear such that $\varphi_0|_M = \varphi$ and $\varphi_0 \leq p$ on M_0.
(ii) $\exists \Phi : \mathcal{X} \to \mathbb{R}$ linear such that $\Phi|_M = \varphi$ and $\Phi \leq p$ on $\mathcal{X}$.

Proof.

(i) For $y, z \in M$, $\varphi(y) + \varphi(z) = \varphi(y+z) \leq p(y+z) \leq p(y - x_0) + p(z + x_0)$; $\varphi(y) - p(y - x_0) \leq p(z + x_0) - \varphi(z)$.
Set $a := \sup_{y \in M} (\varphi(y) - p(y - x_0))$, $b := \inf_{z \in M} (p(z + x_0) - \varphi(z))$. Then $-p(-x_0) \leq a \leq b \leq p(x_0)$; $\alpha \in [a, b]$.
Since $x_0 \notin M$, $y + tx_0$ has unique coordinates, so define $\varphi_0(y + tx_0) := \varphi(y) + t\alpha$.
If $t > 0$, then $\varphi_0(y + tx_0) = t(\varphi(y/t) + \alpha) \leq tp(y/t + x_0) = p(y + tx_0)$.
If $t < 0$, then $\varphi_0(y + tx_0) = (-t)(\varphi(y/(-t)) - \alpha) \leq (-t)p(y/(-t) - x_0) = p(y + tx_0)$.
If $t = 0$, domination is $\varphi \leq p$ on M.
(ii) Let $\mathscr{E}$ be (N, ψ), $M \leq N \leq \mathcal{X}$, $\psi : N \to \mathbb{R}$ linear, $\psi|_M = \varphi$, $\psi \leq p$ on N, ordered by extension.
Chains have upper bound $(\bigcup N_i, \psi_*)$, where $\psi_*|_{N_i} = \psi_i$, so Principle 6.1.3 gives maximal $(N_*, \psi_*) \in \mathscr{E}$.
If $N_* \neq \mathcal{X}$, choose $x_0 \in \mathcal{X} \setminus N_*$; by (i), ψ_* extends to $N_* + \mathbb{R}x_0$, contradiction. Thus $N_* = \mathcal{X}$ and $\Phi := \psi_*$.

Structural Commentary: Extend across one new direction while preserving domination, then use maximality.
Key Insight: Sublinearity leaves a nonempty interval of admissible values on each added direction. □

Corollary 5.3.2. *Norm-Preserving Extensions and Norming Functionals*
Let $\mathcal{X}$ be a normed space and let $M \subseteq \mathcal{X}$ be a linear subspace.

(i) A bounded linear functional $\varphi : M \to \mathbb{F}$ admits an extension $\Phi \in \mathcal{X}^*$ with $\|\Phi\|_{op} = \|\varphi\|_{op}$.
(ii) $\forall 0 \neq x \in \mathcal{X}$ $\exists \Phi \in \mathcal{X}^*$ such that $\|\Phi\|_{op} = 1$ and $\Phi(x) = \|x\|$.

Proof.

(i) If $\mathbb{F} = \mathbb{R}$, apply Theorem 5.3.1 with $p(u) := \|\varphi\|_{\mathrm{op}} \|u\|$ to get $\Phi|_M = \varphi$ and $\Phi \leq p$.
Since $-\Phi(u) = \Phi(-u) \leq p(u)$, $|\Phi(u)| \leq \|\varphi\|_{\mathrm{op}} \|u\| \forall u \in \mathcal{X}$, so $\|\Phi\|_{\mathrm{op}} \leq \|\varphi\|_{\mathrm{op}}$; restriction gives equality.
If $\mathbb{F} = \mathbb{C}$, apply the real case to $\operatorname{Re}\varphi$ and set $\Phi(u) := F(u) - iF(iu)$.
Then $\Phi|_M = \varphi$, Φ is complex-linear, and $|\Phi(u)| = \sup_{|\alpha|=1} F(\alpha u) \leq \|\varphi\|_{\mathrm{op}} \|u\|$, so again $\|\Phi\|_{\mathrm{op}} = \|\varphi\|_{\mathrm{op}}$.
(ii) Fix $0 \neq x \in \mathcal{X}$, put $M_x := \mathbb{F}x$, and define $\varphi_x(\lambda x) := \lambda \|x\|$.
Then $\|\varphi_x\|_{\mathrm{op}} = 1$ and $\varphi_x(x) = \|x\|$; by (i), choose $\Phi \in \mathcal{X}^*$ with $\Phi|_{M_x} = \varphi_x$ and $\|\Phi\|_{\mathrm{op}} = 1$.

Structural Commentary: Preserve the norm, then extend a one-dimensional norming functional.
Key Insight: Hahn–Banach continues a bounded scalar test without weakening its norm bound. □

Corollary 5.3.3. *Natural Embedding Is an Isometry*
For a normed space $\mathcal{X}$, the canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$, $Jx(\Phi) = \Phi(x)$, satisfies $\|Jx\|_{op} = \|x\|$ $(x \in \mathcal{X})$.

Proof. $\forall \Phi \in \mathcal{X}^*$, $|Jx(\Phi)| = |\Phi(x)| \leq \|\Phi\|_{\mathrm{op}} \|x\|$, so $\|Jx\|_{\mathrm{op}} \leq \|x\|$. If $x \neq 0$, choose $\Phi \in \mathcal{X}^*$ with $\|\Phi\|_{\mathrm{op}} = 1$ and $\Phi(x) = \|x\|$ by Corollary 5.3.2; hence $\|Jx\|_{\mathrm{op}} \geq \|x\|$. The case $x = 0$ is immediate.
Structural Commentary: The upper bound is duality; a norming functional attains the matching lower bound.
Key Insight: The double dual records every vector without changing its norm. □

Example 5.3.4. *A Dominated Extension of the Limit Functional*
On real ℓ^∞, set $p((x_n)) := \limsup_{n\to\infty} x_n$. This functional is subadditive and positively homogeneous.
If $c \subseteq \ell^\infty$ is the subspace of convergent sequences and $L((x_n)) := \lim_{n\to\infty} x_n$, then $L \leq p$ on c.
By Theorem 5.3.1, L extends to a linear functional $\Lambda : \ell^\infty \to \mathbb{R}$ with $\Lambda(x) \leq \limsup_{n\to\infty} x_n$ $(x \in \ell^\infty)$.
This is the Hahn–Banach domination step behind limit-extension arguments on ℓ^∞.

Corollary 5.3.5. *Positive Riesz Representation on $C([a,b])$*
Let $J=[a,b]$ and let $G\in C(J)^*$ be positive, i.e. $f\ge 0$ on J implies $G(f)\ge 0$.

(i) $\exists \gamma:\mathcal{B}(\mathbb{R})\to[0,+\infty]$ *a finite Borel measure with* $\operatorname{supp}\gamma\subseteq J$ *and* $\forall f\in C(J)$, $G(f)=\int_J f\,d\gamma$.
(ii) $\|G\|_{op}=G(\mathbf{1}_J)=\gamma(J)$.

Proof.

(i) By Theorem 5.2.14, $\exists$ finite Radon μ on J such that $G(f)=\int_J f\,d\mu$ for $f\in C(J)$.
Define $\gamma(E):=\mu(E\cap J)$ for $E\in\mathcal{B}(\mathbb{R})$. Then $\gamma(\mathbb{R}\setminus J)=0$, so $\operatorname{supp}\gamma\subseteq J$, and $\int_J f\,d\gamma=\int_J f\,d\mu=G(f)$.
(ii) If $\|f\|_\infty\le 1$, then $\mathbf{1}_J\pm f\ge 0$, hence $-G(\mathbf{1}_J)\le G(f)\le G(\mathbf{1}_J)$ and $|G(f)|\le G(\mathbf{1}_J)$.
Thus $\|G\|_{\mathrm{op}}\le G(\mathbf{1}_J)$; taking $f=\mathbf{1}_J$ gives $\|G\|_{\mathrm{op}}\ge G(\mathbf{1}_J)$. Also $G(\mathbf{1}_J)=\int_J \mathbf{1}_J\,d\gamma=\gamma(J)$.

Structural Commentary: Specialize Riesz–Markov to the interval, then extend the measure by zero.
Key Insight: Positivity converts the order unit $\mathbf{1}_J$ into the exact dual norm. □

Corollary 5.3.6. *Signed Riesz Representation by a Charge*
Let $J=[a,b]$, and suppose $G\in C(J)^*$ admits $G=G^+-G^-$ with positive $G^\pm\in C(J)^*$.
Then $\exists\lambda:\mathcal{B}(\mathbb{R})\to\mathbb{R}$ a charge with $\operatorname{supp}\lambda\subseteq J$ and $\forall f\in C(J)$, $G(f)=\int_J f\,d\lambda$.

Proof. By Corollary 5.3.5, choose finite Borel measures $\gamma^\pm$ on $\mathbb{R}$ such that $\gamma^\pm(\mathbb{R}\setminus J)=0$ and $G^\pm(f)=\int_J f\,d\gamma^\pm$ for $f\in C(J)$. Define $\lambda(E):=\gamma^+(E)-\gamma^-(E)$ for $E\in\mathcal{B}(\mathbb{R})$.
Then λ is a charge, $\lambda(\mathbb{R}\setminus J)=0$, and for $f\in C(J)$, $G(f)=G^+(f)-G^-(f)=\int_J f\,d\gamma^+-\int_J f\,d\gamma^-=\int_J f\,d\lambda$.
Structural Commentary: Subtract the positive representing measures.
Key Insight: Signed representation is positive representation plus subtraction. □

Theorem 5.3.7. *Uniform Boundedness Principle*
Let $\mathcal{X}$ be Banach, $\mathcal{Y}$ normed. If $\mathcal{F}\subseteq\mathcal{L}(\mathcal{X},\mathcal{Y})$ pointwise bounded, $\forall x\in\mathcal{X}$, $\sup_{T\in\mathcal{F}}\|Tx\|<\infty$, then $\sup_{T\in\mathcal{F}}\|T\|_{op}<\infty$.

Proof. For $T\in\mathcal{F}$, $f_T(x):=\|Tx\|$ is continuous. Pointwise boundedness and Theorem 5.1.33 give $x_0\in\mathcal{X}$, $r>0$, and $C<\infty$ with $\sup_{T\in\mathcal{F}}\|Tz\|\le C$ for $z\in B(x_0,r)$. Set $M_0:=\sup_{T\in\mathcal{F}}\|Tx_0\|<\infty$.
If $\|y\|<r$, then $\sup_{T\in\mathcal{F}}\|Ty\|\le\sup_{T\in\mathcal{F}}\|T(x_0+y)\|+\sup_{T\in\mathcal{F}}\|Tx_0\|\le C+M_0$.
For $\|x\|=1$ and $0<t<r$, $t\sup_{T\in\mathcal{F}}\|Tx\|=\sup_{T\in\mathcal{F}}\|T(tx)\|\le C+M_0$. By $t\uparrow r$, $\sup_{T\in\mathcal{F}}\|T\|_{\mathrm{op}}\le\frac{C+M_0}{r}<\infty$.
Structural Commentary: Baire controls one ball; translation and scaling transfer control to the unit ball.
Key Insight: Baire and linearity turn pointwise bounds into one uniform operator bound. □

Corollary 5.3.8. *Banach–Steinhaus Pointwise Limit Theorem*
Let $\mathcal{X}$ be Banach, $\mathcal{Y}$ normed, $(T_n)\subseteq\mathcal{L}(\mathcal{X},\mathcal{Y})$. Suppose $\forall x\in\mathcal{X}$, T_nx converges in $\mathcal{Y}$, $Tx:=\lim_{n\to\infty}T_nx$.

(i) $\sup_n\|T_n\|_{op}<\infty$.
(ii) T *is linear.*
(iii) $T\in\mathcal{L}(\mathcal{X},\mathcal{Y})$ *and* $\|T\|_{op}\le\liminf_{n\to\infty}\|T_n\|_{op}$.

Proof.

(i) For each $x\in\mathcal{X}$, convergence of (T_nx) gives $\sup_n\|T_nx\|<\infty$; hence Theorem 5.3.7 gives $\sup_n\|T_n\|_{\mathrm{op}}<\infty$.
(ii) For $\alpha,\beta\in\mathbb{F}$ and $x,y\in\mathcal{X}$, $T(\alpha x+\beta y)=\lim_n T_n(\alpha x+\beta y)=\alpha Tx+\beta Ty$.
(iii) Let $L:=\liminf_n\|T_n\|_{\mathrm{op}}<\infty$. For $x\in\mathcal{X}$, $\|Tx\|=\lim_n\|T_nx\|\le L\|x\|$, so $T\in\mathcal{L}(\mathcal{X},\mathcal{Y})$ and $\|T\|_{\mathrm{op}}\le L$.

Structural Commentary: Uniform boundedness controls the family before taking its pointwise limit.
Key Insight: Completeness prevents the norms of pointwise-convergent operators from escaping. □

5.4 Open Mapping, Closed Graph, and Complemented Subspaces

The Baire-category principle has three operator-theoretic consequences that should be read together: closed range gives uniform preimage estimates, open maps are detected at the unit ball, and closed graphs force boundedness.

Theorem 5.4.1. *Closed Range Estimate*
Let $\mathcal{X},\mathcal{Y}$ be Banach spaces, let $T\in\mathcal{L}(\mathcal{X},\mathcal{Y})$, and set $\mathcal{R}:=T(\mathcal{X})$. The following are equivalent.

(i) $\mathcal{R}$ *is closed in* $\mathcal{Y}$.
(ii) $\exists M>0$ *such that* $\forall y\in\mathcal{R}$ $\exists x\in\mathcal{X}$ *with* $Tx=y$ *and* $\|x\|\le M\|y\|$.

Proof.

(*ii*) $\Rightarrow$ (*i*) Let $y\in\overline{\mathcal{R}}$. Choose $y_n\in\mathcal{R}$ with $y_n\to y$ and $\|y_n-y_{n-1}\|\le 2^{-n}$ for $n\ge 2$.
Pick $x_1,u_n\in\mathcal{X}$ with $Tx_1=y_1$, $Tu_n=y_n-y_{n-1}$, and $\|u_n\|\le M2^{-n}$.
Since $\sum_{n\ge2}\|u_n\|<\infty$, $x:=x_1+\sum_{n\ge2}u_n$ exists and $Tx=\lim_N y_N=y$, so $y\in\mathcal{R}$.

(*i*) $\Rightarrow$ (*ii*) If $\mathcal{R}=\{\mathbf{0}\}$, take $M=1$. Otherwise $\mathcal{R}$ is Banach and $\mathcal{R}=\bigcup_{n\ge1}n\overline{T(B_{\mathcal{X}})}^{\mathcal{R}}$.

By Baire and balanced convexity, $\exists c>0$ with $\overline{B}_{\mathcal{R}}\subseteq c\overline{T(B_{\mathcal{X}})}^{\mathcal{R}}$.
Thus each $0\ne r\in\mathcal{R}$ has $u\in\mathcal{X}$ with $\|u\|\le c\|r\|$ and $\|r-Tu\|\le\frac12\|r\|$.
For $y\ne0$, iterate from $r_0=y$ to get $r_k=r_{k-1}-Tu_k$, $\|r_k\|\le2^{-k}\|y\|$, and $\|u_k\|\le c2^{1-k}\|y\|$.
Then $x:=\sum_{k\ge1}u_k$ satisfies $Tx=y$ and $\|x\|\le2c\|y\|$. For $y=0$, take $x=0$.

Structural Commentary: Baire category turns surjectivity onto a closed range into uniform preimage control.
Key Insight: Closed range is equivalent to solving $Tx=y$ with a norm bound proportional to $\|y\|$. □

Proposition 5.4.2. *Relative Openness and Closed Range*
Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces, $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$.
Then $T : \mathcal{X} \to T(\mathcal{X})$ is open, for subspace topology on range, iff $T(\mathcal{X})$ closed in $\mathcal{Y}$.

Proof. Let $\mathcal{R} := T(\mathcal{X})$.

($\Rightarrow$) If $T : \mathcal{X} \to \mathcal{R}$ is open, then $\exists r > 0$ with $B_{\mathcal{R}}(0, r) \subseteq T(B_{\mathcal{X}})$.
For $0 \neq y \in \mathcal{R}$, choose $u \in B_{\mathcal{X}}$ with $Tu = \frac{r}{2\|y\|} y$, and set $x := \frac{2\|y\|}{r} u$.
Then $Tx = y$ and $\|x\| \leq \frac{2}{r} \|y\|$; for $y = 0$, take $x = 0$. By Theorem 5.4.1, $\mathcal{R}$ is closed.

($\Leftarrow$) If $\mathcal{R}$ is closed, then by Theorem 5.4.1, $\exists M > 0$ such that every $y \in \mathcal{R}$ has $x \in \mathcal{X}$ with $Tx = y$ and $\|x\| \leq M \|y\|$. Let $U \subseteq \mathcal{X}$ be open and $x_0 \in U$. Choose $\rho > 0$ with $x_0 + \rho B_{\mathcal{X}} \subseteq U$. If $z \in B_{\mathcal{R}}(Tx_0, \frac{\rho}{M})$, choose $h \in \mathcal{X}$ with $Th = z - Tx_0$ and $\|h\| < \rho$; then $z = T(x_0 + h) \in T(U)$. Thus $T(U)$ is open in $\mathcal{R}$.

Structural Commentary: For linear maps, openness is detected at the unit ball around the origin.
Key Insight: Relative openness is the topological form of uniform preimage control. □

Theorem 5.4.3. *Open Mapping Theorem*
Every surjective $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ between Banach spaces is open.

Proof. Since T is surjective, $T(\mathcal{X}) = \mathcal{Y}$, $\mathcal{Y}$ is closed in itself. By Proposition 5.4.2, $T : \mathcal{X} \to T(\mathcal{X}) = \mathcal{Y}$ open.
Structural Commentary: Surjectivity identifies the relative range with the codomain.
Key Insight: A Banach-space surjection has uniformly controlled preimages near the origin. □

Corollary 5.4.4. *Bounded Inverse Theorem*
If $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ is bijective between Banach spaces, then $T^{-1} \in \mathcal{L}(\mathcal{Y}, \mathcal{X})$.

Proof. Set $U := T^{-1} : \mathcal{Y} \to \mathcal{X}$. By Theorem 5.4.3, T is open: if $O \subseteq \mathcal{X}$ is open, then $U^{-1}(O) = T(O)$ is open in $\mathcal{Y}$. Thus U is continuous; since U is linear, Theorem 5.1.5 gives $U \in \mathcal{L}(\mathcal{Y}, \mathcal{X})$.
Structural Commentary: Openness of the forward bijection is continuity of the inverse.
Key Insight: Banach completeness prevents a bounded linear bijection from having an unbounded inverse. □

Corollary 5.4.5. *One-Sided Control of Banach Norms*
Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on $\mathcal{X}$ such that $(\mathcal{X}, \|\cdot\|_1)$ and $(\mathcal{X}, \|\cdot\|_2)$ are Banach.
Assume $\exists c > 0$ such that $\|x\|_2 \leq c \|x\|_1$ for all $x \in \mathcal{X}$.

(i) $\exists C > 0$ such that $\|x\|_1 \leq C \|x\|_2$ for all $x \in \mathcal{X}$.
(ii) $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.

Proof.

(i) Let $I : (\mathcal{X}, \|\cdot\|_1) \to (\mathcal{X}, \|\cdot\|_2)$ be the identity. Since $\|Ix\|_2 \leq c \|x\|_1$, $I \in \mathcal{L}$ and is bijective.
By Corollary 5.4.4, $I^{-1} \in \mathcal{L}$, hence $\exists C > 0$ such that $\|x\|_1 = \left\|I^{-1}x\right\|_1 \leq C \|x\|_2$.

(ii) For all $x \in \mathcal{X}$, $\|x\|_2 \leq c \|x\|_1$ by hypothesis and $\|x\|_1 \leq C \|x\|_2$ by (i).
Hence the identity maps between the two normed structures are Lipschitz.

Structural Commentary: Apply the bounded inverse theorem to the identity between the two normed structures.
Key Insight: Completeness upgrades one-sided norm control to two-sided equivalence. □

Definition 5.4.6. *Closed Operators and Closed Complements*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces.

(i) *Closed operator*: $T : \mathcal{X} \to \mathcal{Y}$ is linear and $\forall (x_n) \subseteq \mathcal{X}$, $x_n \to x$ and $Tx_n \to y$ imply $Tx = y$.
(ii) *Closed linear complement*: $W \leq \mathcal{X}$ satisfies $\mathcal{X} = V \oplus W$ and is closed.

Theorem 5.4.7. *Closed Graph Theorem*
Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces and let $T : \mathcal{X} \to \mathcal{Y}$ be linear. Set $\Gamma(T) := \{(x, Tx) : x \in \mathcal{X}\}$.

(i) T is bounded iff T is closed.
(ii) T is closed iff $\Gamma(T)$ is closed in $\mathcal{X} \times \mathcal{Y}$.

Proof. On $\mathcal{X} \times \mathcal{Y}$, use $\|(x, y)\| := \|x\| + \|y\|$.

(i) ($\Rightarrow$) If T is bounded, $x_n \to x$ and $Tx_n \to y$ imply $Tx_n \to Tx$; hence $y = Tx$, so T is closed.
(i) ($\Leftarrow$) Assume T is closed and set $\|x\|_* := \|x\| + \|Tx\|$. If (x_n) is $\|\cdot\|_*$-Cauchy, then $x_n \to x$ in $\mathcal{X}$ and $Tx_n \to y$ in $\mathcal{Y}$; by closedness, $y = Tx$, so $x_n \to x$ in $\|\cdot\|_*$. Thus $(\mathcal{X}, \|\cdot\|_*)$ is Banach.
Since $\|x\| \leq \|x\|_*$, by Corollary 5.4.5, $\exists C > 0$ such that $\|x\|_* \leq C \|x\|$; hence $\|Tx\| \leq C \|x\|$.
(ii) ($\Rightarrow$) If T is closed and $(x_n, Tx_n) \to (x, y)$, then $x_n \to x$ and $Tx_n \to y$, so $y = Tx$ and $(x, y) \in \Gamma(T)$.
(ii) ($\Leftarrow$) If $\Gamma(T)$ is closed, $x_n \to x$, and $Tx_n \to y$, then $(x_n, Tx_n) \to (x, y) \in \Gamma(T)$; hence $y = Tx$.

Structural Commentary: The graph norm turns closedness of the graph into completeness.
Key Insight: Graph-norm completeness and domination force equivalence with the original norm. □

Theorem 5.4.8. *Closed Complements and Continuous Projections*
Let V be a closed subspace of a Banach space $\mathcal{X}$. The following are equivalent.

(i) $\exists W \leq \mathcal{X}$ closed such that $\mathcal{X} = V \oplus W$.
(ii) $\exists P \in \mathcal{L}(\mathcal{X})$ such that $P^2 = P$ and $P(\mathcal{X}) = V$.

Proof.

(i) $\Rightarrow$ *(ii)* Let $\mathcal{X} = V \oplus W$ with W closed, and define $P(v + w) := v$. Then $P^2 = P$ and $P(\mathcal{X}) = V$.
If $x_n \to x$ and $Px_n \to v$, then $v \in V$ and $(I - P)x_n \to x - v \in W$; by uniqueness in $V \oplus W$, $Px = v$.
Thus P is closed, so $P \in \mathcal{L}(\mathcal{X})$ by Theorem 5.4.7.

(ii) $\Rightarrow$ *(i)* Let $P \in \mathcal{L}(\mathcal{X})$ with $P^2 = P$ and $P(\mathcal{X}) = V$. By Proposition 2.4.2, $\mathcal{X} = V \oplus \ker P$.
Since $\ker P = P^{-1}(\{\mathbf{0}\})$ and P is continuous, $\ker P$ is closed.

Structural Commentary: Closed complementary coordinates are exactly bounded projection coordinates.
Key Insight: The Closed Graph Theorem converts algebraic splitting into continuous splitting. □

Theorem 5.4.9. *Complemented Ranges Are Closed*
Let $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ act between Banach spaces. If $\operatorname{range}(T)$ has a closed linear complement in $\mathcal{Y}$, then it is closed.

Proof. Set $\mathcal{R} := T(\mathcal{X})$, $\mathcal{Y} = \mathcal{R} \oplus Y_0$, Y_0 closed. Put $Z := \mathcal{X} \times Y_0$, $\|(x, y_0)\|_Z := \|x\| + \|y_0\|$,
and $S : Z \to \mathcal{Y}$, $S(x, y_0) := Tx + y_0$. Then Z is Banach, $S \in \mathcal{L}(Z, \mathcal{Y})$, and $S(Z) = \mathcal{Y}$.
By Theorem 5.4.1, $\exists M > 0$ such that $\forall y \in \mathcal{Y}\ \exists (x, y_0) \in Z$: $y = Tx + y_0$ and $\|x\| + \|y_0\| \le M \|y\|$. If $y \in \mathcal{R}$,
by direct-sum uniqueness $y_0 = 0$; $\forall y \in \mathcal{R}\ \exists x \in \mathcal{X}$: $Tx = y$, $\|x\| \le M \|y\|$. By Theorem 5.4.1, $\mathcal{R}$ closed.
Structural Commentary: A closed complement transfers preimage control to the range summand.
Key Insight: Complementability supplies the missing uniform preimage estimate for closed range. □

Corollary 5.4.10. *Finite-Codimensional Operator Ranges Are Closed*
Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces and let $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$. If $\operatorname{codim}_{\mathcal{Y}} T(\mathcal{X}) < \infty$, then $T(\mathcal{X})$ is closed.

Proof. Set $\mathcal{R} := T(\mathcal{X})$. Since $\operatorname{codim}_{\mathcal{Y}} \mathcal{R} < \infty$, choose $Y_0 \le \mathcal{Y}$ with $\mathcal{Y} = \mathcal{R} \oplus Y_0$ and $\dim Y_0 < \infty$.
By Theorem 5.1.14, Y_0 is closed; by Theorem 5.4.9, $\mathcal{R}$ is closed.
Structural Commentary: Finite codimension produces a finite-dimensional closed complement.
Key Insight: Only finitely many missing codomain directions force the operator range to be closed. □

6 Weak Topologies, Compact Operators, and Spectral Theory

6.1 Weak Topologies and Duality

Weak topologies replace norm control by scalar tests. They are usually too weak to preserve distances, but strong enough to expose compactness phenomena that are invisible in norm topology.

Definition 6.1.1. *Dual Space, Weak Topology, and Weak-* Topology*
For a normed space $\mathcal{X}$ over $\mathbb{F}$, its continuous dual is $\mathcal{X}^* := \mathcal{L}(\mathcal{X}, \mathbb{F})$.

(i) The *weak topology* on $\mathcal{X}$ is the weakest topology making $\varphi : \mathcal{X} \to \mathbb{F}$ continuous $\forall \varphi \in \mathcal{X}^*$.
(ii) The *weak-* topology* on $\mathcal{X}^*$ is the weakest topology making the evaluation $\varphi \mapsto \varphi(x)$ continuous $\forall x \in \mathcal{X}$.
(iii) The *strong topology* is the norm topology.

For $m \in \mathbb{N}$, $\varepsilon > 0$, $\varphi_\bullet = (\varphi_1, \dots, \varphi_m) \in (\mathcal{X}^*)^m$, and $x_\bullet = (x_1, \dots, x_m) \in \mathcal{X}^m$:

(i) Weak: $N(x; \varphi_\bullet; \varepsilon) := \{y \in \mathcal{X} : |\varphi_j(y - x)| < \varepsilon\ \forall j\}$.
(ii) Weak-*: $N_*(\varphi; x_\bullet; \varepsilon) := \{\psi \in \mathcal{X}^* : |(\psi - \varphi)(x_j)| < \varepsilon\ \forall j\}$.

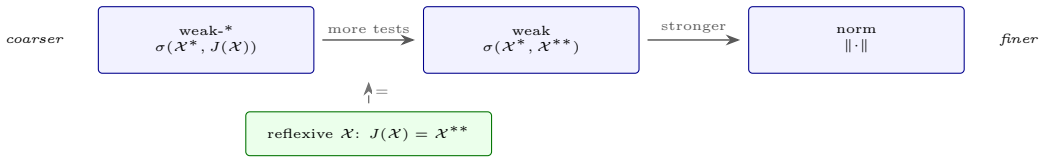

Figure 222: Topologies on $\mathcal{X}^*$: weak-* $\subseteq$ weak $\subseteq$ norm; reflexivity identifies the first two.

Definition 6.1.2. *Weak and Weak-* Convergence*

(i) $(x_n) \subseteq \mathcal{X}$ *converges weakly* to x, written $x_n \rightharpoonup x$, iff $\varphi(x_n) \to \varphi(x)\ \forall \varphi \in \mathcal{X}^*$.
(ii) $(\varphi_n) \subseteq \mathcal{X}^*$ *converges weak-** to $\varphi \in \mathcal{X}^*$ iff $\varphi_n(x) \to \varphi(x)\ \forall x \in \mathcal{X}$.

Proposition 6.1.3. *Strong Convergence Implies Weak Convergence*
If $x_n \to x$ in norm in a normed space $\mathcal{X}$, then $x_n \rightharpoonup x$.

Proof. $\forall \varphi \in \mathcal{X}^*$, $|\varphi(x_n) - \varphi(x)| = |\varphi(x_n - x)| \le \|\varphi\|_{\mathrm{op}} \|x_n - x\| \to 0$.
Structural Commentary: Apply an arbitrary bounded scalar test to the norm error.
Key Insight: Norm convergence controls every bounded linear test. □

Proposition 6.1.4. *Weakly Convergent Sequences Are Bounded*
If $x_n \rightharpoonup x$ in a normed space $\mathcal{X}$, then:

(i) $\sup_n \|x_n\| < +\infty$.
(ii) $\|x\| \le \liminf_{n \to \infty} \|x_n\|$.

Proof.

(i) $\forall \varphi \in \mathcal{X}^*$, the scalar sequence $(\varphi(x_n))$ is bounded. The space $\mathcal{X}^*$ is Banach by Theorem 5.1.6, so the Uniform Boundedness Principle, Theorem 5.3.7, applies to $Jx_n : \mathcal{X}^* \to \mathbb{F}$, $Jx_n(\varphi) := \varphi(x_n)$.
Since $\|Jx_n\|_{\mathrm{op}} = \|x_n\|$ by Corollary 5.3.3, the sequence is norm bounded.
(ii) If $x \neq 0$, choose $\varphi \in \mathcal{X}^*$ with $\|\varphi\|_{\mathrm{op}} = 1$ and $\varphi(x) = \|x\|$ by Corollary 5.3.2.
Then $\|x\| = \lim_{n \to \infty} |\varphi(x_n)| \le \liminf_{n \to \infty} \|x_n\|$. The case $x = 0$ is immediate.

Structural Commentary: Uniform boundedness proves (i); one norming functional proves (ii).
Key Insight: Scalar-test boundedness forces norm boundedness. □

Definition 6.1.5. *Reflexivity*
For a normed space $\mathcal{X}$, the canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$ is defined by $Jx(\varphi) := \varphi(x)$.
A Banach space $\mathcal{X}$ is *reflexive* if $J(\mathcal{X}) = \mathcal{X}^{**}$.

Theorem 6.1.6. *Banach–Alaoglu Theorem*
For a normed space $\mathcal{X}$, the closed unit ball $\overline{B}_{\mathcal{X}^*}$ is weak- compact.*

Proof. Let $B := \overline{B}_{\mathcal{X}}$, $B^* := \overline{B}_{\mathcal{X}^*}$, and $D := \{a \in \mathbb{F} : |a| \le 1\}$. By Theorem 6.3.13, D^B is compact.
The injection $R : B^* \hookrightarrow D^B$, $R\varphi = \varphi|_B$, identifies $\sigma(\mathcal{X}^*, \mathcal{X})|_{B^*}$ with pointwise convergence on B.
Let $f \in \overline{R(B^*)} \subseteq D^B$. If $x_j \in B$, $a_j \in \mathbb{F}$, and $\sum_j a_j x_j \in B$, then $f(\sum_j a_j x_j) = \sum_j a_j f(x_j)$ by passage from the identities for $R\varphi$.
Define $\Phi(0) := 0$ and $\Phi(x) := \|x\| f(x/\|x\|)$ for $x \neq 0$.
Then Φ is linear, $|\Phi(x)| \le \|x\|$, and $R\Phi = f$; hence $R(B^*)$ is closed in D^B. Therefore B^* is weak-* compact.
Structural Commentary: $B^* \hookrightarrow D^B$; closed linear constraints plus Tychonoff give compactness.
Key Insight: Weak-* topology records precisely the coordinates used in Tychonoff compactness. □

Remark 6.1.7. *Compactness and Sequences*
Weak-* compactness need not be weak-* sequential compactness. For $\mathcal{X} = \ell^\infty$, let $\varphi_n(x) := x_n$.
Then $\varphi_n \in \overline{B}_{\mathcal{X}^*}$, but for any subsequence (n_k), choosing $x \in \ell^\infty$ with $x_{n_k} = (-1)^k$ makes $\varphi_{n_k}(x) = (-1)^k$, so no subsequence is weak-* convergent. Helly's theorem gives a sequential substitute when $\mathcal{X}$ is separable.

Theorem 6.1.8. *Helly Selection for Bounded Dual Sequences*
Let $\mathcal{X}$ be a separable normed space. If $(T_n) \subseteq \mathcal{X}^*$ and $\sup_n \|T_n\|_{op} < +\infty$, then $\exists$ a subsequence (T_{n_k}) and $T \in \mathcal{X}^*$ such that $\forall x \in \mathcal{X}$, $T_{n_k}(x) \to T(x)$.

Proof. Let $D = \{x_j : j \ge 1\} \subseteq \mathcal{X}$ be dense and set $M := \sup_n \|T_n\|_{\mathrm{op}}$.
If $M = 0$, take $T = 0$; assume $M > 0$. By diagonal extraction, choose (n_k) such that $T_{n_k} x_j$ converges $\forall j$.
For $x \in \mathcal{X}$ and $\varepsilon > 0$, choose j with $\|x - x_j\| < \varepsilon/(4M)$ and then k, ℓ large with $|T_{n_k} x_j - T_{n_\ell} x_j| < \varepsilon/2$.
Then $|T_{n_k} x - T_{n_\ell} x| \le 2M \|x - x_j\| + |T_{n_k} x_j - T_{n_\ell} x_j| < \varepsilon$, so $(T_{n_k} x)$ is Cauchy. Define $Tx := \lim_k T_{n_k} x$.
Linearity passes to limits, and $|Tx| \le M \|x\|$; hence $T \in \mathcal{X}^*$ and $T_{n_k} x \to Tx$ $\forall x \in \mathcal{X}$.
Structural Commentary: Diagonalize on dense D; use $M = \sup_n \|T_n\|_{\mathrm{op}}$ to extend D-limits to $\mathcal{X}$.
Key Insight: A countable dense set reduces pointwise convergence to diagonal convergence. □

Corollary 6.1.9. *Embedding into a Continuous Function Space*
For every normed space $\mathcal{X}$, $\exists$ compact Hausdorff K and a linear isometry $\Phi : \mathcal{X} \to C(K)$.

Proof. Let $K := \overline{B}_{\mathcal{X}^*}$ with the weak-* topology. By Theorem 6.1.6, K is compact; since evaluations separate points of K, K is Hausdorff. Define $\Phi : \mathcal{X} \to C(K)$ by $\Phi(x)(\varphi) := \varphi(x)$. Each $\Phi(x)$ is weak-* continuous, Φ is linear, and by Corollary 5.3.3, $\|\Phi(x)\|_\infty = \|x\|$.
Structural Commentary: $K = \overline{B}^{w^*}_{\mathcal{X}^*}$ turns scalar tests into compact coordinates.
Key Insight: Evaluation turns each vector into a continuous function without changing its norm. □

Proposition 6.1.10. *Natural Embedding and Weak Topologies*
For a normed space $\mathcal{X}$, the canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$ is a homeomorphism $(\mathcal{X}, w) \to (J(\mathcal{X}), w^*|_{J(\mathcal{X})})$.

Proof. By Corollary 5.3.3, J is injective. Fix $x_0 \in \mathcal{X}$, $\varphi_\bullet = (\varphi_1, \dots, \varphi_m) \in (\mathcal{X}^*)^m$, and $\varepsilon > 0$.
Since $(Jx - Jx_0)(\varphi_j) = \varphi_j(x - x_0)$, $J^{-1}(N_*(Jx_0; \varphi_\bullet; \varepsilon) \cap J(\mathcal{X})) = N(x_0; \varphi_\bullet; \varepsilon)$.
Thus J and $J^{-1} : J(\mathcal{X}) \to \mathcal{X}$ are continuous.
Structural Commentary: Pullbacks of weak-* tests are weak tests.
Key Insight: J preserves precisely the scalar tests that define the weak topology. □

Theorem 6.1.11. *Kakutani's Weak Compactness Criterion*
For a Banach space $\mathcal{X}$, the following are equivalent.
(i) $\mathcal{X}$ is reflexive.
(ii) The closed unit ball $\overline{B}_{\mathcal{X}}$ is weakly compact.

Proof.
(*i*) $\Rightarrow$ (*ii*) If $J(\mathcal{X}) = \mathcal{X}^{**}$, then by Corollary 5.3.3, $J(\overline{B}_{\mathcal{X}}) = \overline{B}_{\mathcal{X}^{**}}$.
By Theorem 6.1.6 applied to $\mathcal{X}^*$, $\overline{B}_{\mathcal{X}^{**}}$ is weak-* compact. By Proposition 6.1.10, $\overline{B}_{\mathcal{X}}$ is weakly compact.
(*ii*) $\Rightarrow$ (*i*) By Proposition 6.1.10, $K := J(\overline{B}_{\mathcal{X}})$ is weak-* compact, hence weak-* closed, in $\mathcal{X}^{**}$.
If $F \in \overline{B}_{\mathcal{X}^{**}} \setminus K$, weak-* separation yields $\varphi \in \mathcal{X}^*$ with $\operatorname{Re} F(\varphi) > \sup_{\|x\| \le 1} \operatorname{Re} Jx(\varphi) = \|\varphi\|_{\mathrm{op}}$.
But $\operatorname{Re} F(\varphi) \le |F(\varphi)| \le \|F\|_{\mathrm{op}} \|\varphi\|_{\mathrm{op}} \le \|\varphi\|_{\mathrm{op}}$, a contradiction. Thus $\overline{B}_{\mathcal{X}^{**}} \subseteq K$.
For $F \in \mathcal{X}^{**}$ with $F \neq 0$, $F/\|F\|_{\mathrm{op}} \in K$, so $F \in J(\mathcal{X})$; hence $J(\mathcal{X}) = \mathcal{X}^{**}$.

Structural Commentary: Reflexivity identifies $J(\overline{B}_{\mathcal{X}})$ with $\overline{B}_{\mathcal{X}^{**}}$; weak-* separation forces the converse.
Key Insight: Reflexivity identifies the weak-* compact bidual ball with the embedded original ball. □

Corollary 6.1.12. *Closed Bounded Convex Sets in Reflexive Spaces*
Let $\mathcal{X}$ be reflexive Banach. If $C \subseteq \mathcal{X}$ is norm-closed, bounded, and convex, then C is weakly compact.

Proof. If $C = \emptyset$, the claim is trivial. Choose $r > 0$ with $C \subseteq r\overline{B}_{\mathcal{X}}$. By Theorem 6.1.11, $r\overline{B}_{\mathcal{X}}$ is weakly compact.
For $x_0 \notin C$, by Hahn–Banach separation, $\varphi \in \mathcal{X}^*$ and $\alpha \in \mathbb{R}$ with $\operatorname{Re}\varphi(x_0) > \alpha \ge \sup_{c \in C} \operatorname{Re}\varphi(c)$; hence $\{\operatorname{Re}\varphi > \alpha\}$ is weakly open, contains x_0, and misses C. Thus C is weakly closed, so C is weakly compact. *Structural Commentary*: $C \subseteq r\overline{B}_{\mathcal{X}}$ and C weakly closed inside a weakly compact ball.
Key Insight: Convex norm-closed sets remain closed when the topology is weakened to scalar tests. □

Theorem 6.1.13. *Goldstine's Theorem*
For a normed space $\mathcal{X}$, the weak- closure of $J(\overline{B}_{\mathcal{X}})$ in $\mathcal{X}^{**}$ is $\overline{B}_{\mathcal{X}^{**}}$.*

Proof. Set $C := \overline{J(\overline{B}_{\mathcal{X}})}^{\sigma(\mathcal{X}^{**},\mathcal{X}^{*})}$. J is isometric by Corollary 5.3.3, $\overline{B}_{\mathcal{X}^{**}} = \bigcap_{\varphi\in\overline{B}_{\mathcal{X}^{*}}} \{F \in \mathcal{X}^{**} : |F(\varphi)| \le 1\}$ is weak-* closed, hence $C \subseteq \overline{B}_{\mathcal{X}^{**}}$. Suppose $F \in \overline{B}_{\mathcal{X}^{**}} \setminus C$. By weak-* Hahn–Banach separation,
$\exists \varphi \in \mathcal{X}^{*}$ and $\alpha \in \mathbb{R}$ such that $\operatorname{Re} F(\varphi) > \alpha \ge \sup_{\|x\|\le 1} \operatorname{Re} Jx(\varphi) = \sup_{\|x\|\le 1} \operatorname{Re}\varphi(x) = \|\varphi\|_{\mathrm{op}}$.
But $\operatorname{Re} F(\varphi) \le |F(\varphi)| \le \|F\|_{\mathrm{op}}\|\varphi\|_{\mathrm{op}} \le \|\varphi\|_{\mathrm{op}}$, a contradiction. Hence $\overline{B}_{\mathcal{X}^{**}} \subseteq C$.
Structural Commentary: $J(\overline{B}_{\mathcal{X}}) \subseteq \overline{B}_{\mathcal{X}^{**}}$; weak-* separation excludes $\overline{B}_{\mathcal{X}^{**}} \setminus C$.
Key Insight: $J(\overline{B}_{\mathcal{X}})$ is weak-* dense in $\overline{B}_{\mathcal{X}^{**}}$ even when $J\mathcal{X} \neq \mathcal{X}^{**}$. □

Theorem 6.1.14. *Eberlein–Smulian Theorem for Unit Balls*
Let $\mathcal{X}$ be Banach and set $B := \overline{B}_{\mathcal{X}}$. The following are equivalent:
(i) B is compact in $(\mathcal{X}, w)$.
(ii) Every $(x_n) \subseteq B$ has a subsequence $x_{n_k} \rightharpoonup x$ for some $x \in B$.

Proof. By Eberlein–Smulian, for bounded $A \subseteq \mathcal{X}$, A is relatively weakly compact iff every sequence in A has a weakly convergent subsequence with limit in $\overline{A}^{w}$. By Hahn–Banach separation, B is weakly closed, so $\overline{B}^{w} = B$.
$(i) \Rightarrow (ii)$ Apply the criterion with $A = B$.
$(ii) \Rightarrow (i)$ The criterion makes B relatively weakly compact; since $\overline{B}^{w} = B$, B is weakly compact.
Structural Commentary: For $B = \overline{B}_{\mathcal{X}}$, weak compactness $\iff$ weak subsequential compactness.
Key Insight: Weak compactness of B is sequence-detectable. □

Corollary 6.1.15. *Reflexive Unit-Ball Characterization*
Let $\mathcal{X}$ be Banach, set $B := \overline{B}_{\mathcal{X}}$, and let $J : \mathcal{X} \to \mathcal{X}^{**}$ be canonical. The following are equivalent:
(i) $J(\mathcal{X}) = \mathcal{X}^{**}$.
(ii) B is compact in $(\mathcal{X}, w)$.
(iii) Every $(x_n) \subseteq B$ has a subsequence $x_{n_k} \rightharpoonup x$ for some $x \in B$.

Proof.
$(i) \Rightarrow (ii)$ $J(\mathcal{X}) = \mathcal{X}^{**} \Rightarrow B$ is compact in $(\mathcal{X}, w)$ by Theorem 6.1.11.
$(ii) \Rightarrow (iii)$ By Theorem 6.1.14, B weakly compact implies every $(x_n) \subseteq B$ has $x_{n_k} \rightharpoonup x \in B$.
$(iii) \Rightarrow (i)$ By Theorem 6.1.14, (iii) implies B weakly compact; by Theorem 6.1.11, $J(\mathcal{X}) = \mathcal{X}^{**}$.
Structural Commentary: $J(\mathcal{X}) = \mathcal{X}^{**} \iff B_w$ compact $\iff B_w$ subsequentially compact.
Key Insight: Reflexivity is unit-ball compactness. □

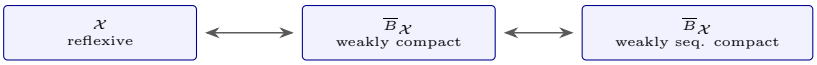

Figure 223: For Banach $\mathcal{X}$: $\mathcal{X}$ reflexive $\iff \overline{B}_{\mathcal{X}}$ weakly compact $\iff \overline{B}_{\mathcal{X}}$ weakly sequentially compact.

Proposition 6.1.16. *Hilbert Weak Convergence Tests*
Let $\mathcal{H}$ be a Hilbert space, let $(h_n), (u_n) \subseteq \mathcal{H}$, and let $h, u \in \mathcal{H}$.
(i) $h_n \rightharpoonup h$ if and only if $\forall u \in \mathcal{H}$, $\langle h_n, u\rangle \to \langle h, u\rangle$.
(ii) If $h_n \rightharpoonup h$ and $u_n \to u$ strongly, then $\langle h_n, u_n\rangle \to \langle h, u\rangle$.

Proof.
(i) $(\Rightarrow)$ For $u \in \mathcal{H}$, $\phi_u(x) := \langle x, u\rangle \in \mathcal{H}^{*}$, so $\langle h_n, u\rangle = \phi_u(h_n) \to \phi_u(h) = \langle h, u\rangle$.
(i) $(\Leftarrow)$ For $\phi \in \mathcal{H}^{*}$, $u_\phi \in \mathcal{H}$ with $\phi(x) = \langle x, u_\phi\rangle$ by Theorem 5.2.32; $\phi(h_n) = \langle h_n, u_\phi\rangle \to \langle h, u_\phi\rangle = \phi(h)$.
(ii) Let $C := \sup_n \|h_n\| < +\infty$ by Proposition 6.1.4(i).
Since $|\langle h_n, u_n\rangle - \langle h, u\rangle| \le C\|u_n - u\| + |\langle h_n - h, u\rangle|$, the limit is 0.
Structural Commentary: $\mathcal{H}^{*} = \{\langle \cdot, u\rangle : u \in \mathcal{H}\}$; boundedness controls $u_n - u$.
Key Insight: $h_n \rightharpoonup h \iff \langle h_n, u\rangle \to \langle h, u\rangle$ for all fixed $u \in \mathcal{H}$. □

Theorem 6.1.17. *Bounded Hilbert Sequences Have Weak Subsequences*
Let $\mathcal{H}$ be Hilbert and let $(h_n) \subseteq \mathcal{H}$ satisfy $C := \sup_n \|h_n\| < +\infty$. Then $\exists n_k \uparrow \infty$ and $h \in \mathcal{H}$ such that $h_{n_k} \rightharpoonup h$.

Proof. Set $M := \operatorname{cspan}\{h_n : n \ge 1\}$. Then M is a separable Hilbert space.
If $\dim M < +\infty$, then $C\overline{B}_M$ is compact by Theorem 5.1.14(iii); choose $h_{n_k} \to h \in M$ strongly, hence weakly.
Assume $\dim M = \infty$. Choose a Hilbert basis $(e_j)_{j\ge 1}$ of M by Corollary 5.2.28.
Diagonalize so $\langle h_{n_k}, e_j\rangle \to a_j$ $\forall j$. For $m \ge 1$, $\sum_{j=1}^{m} |a_j|^2 = \lim_k \sum_{j=1}^{m} |\langle h_{n_k}, e_j\rangle|^2 \le C^2$ by Theorem 5.2.22(ii).
Hence $a \in \ell^2$ and $h := \sum_j a_j e_j \in M$ by Theorem 5.2.21(i). If $v \in \operatorname{span}\{e_j : j \ge 1\}$, then $\langle h_{n_k} - h, v\rangle \to 0$.
For $u \in M$, $\varepsilon > 0$, $\|u - v\| < \varepsilon$ for v; then $\limsup_k |\langle h_{n_k} - h, u\rangle| \le (C + \|h\|)\varepsilon$, so $\langle h_{n_k} - h, u\rangle \to 0$.
For $u \in \mathcal{H}$, write $u = P_M u + u_\perp$, $u_\perp \in M^{\perp}$. Since $h_{n_k} - h \in M$, $\langle h_{n_k} - h, u\rangle = \langle h_{n_k} - h, P_M u\rangle \to 0$.
Thus $h_{n_k} \rightharpoonup h$ by Proposition 6.1.16.
Structural Commentary: Work in $M = \operatorname{cspan}\{h_n\}$; diagonal limits on (e_j) extend by $\overline{\operatorname{span}}\{e_j\} = M$ and $M^{\perp}$.
Key Insight: Bounded Hilbert sequences have countably coordinatized weak limits. □

Theorem 6.1.18. *Hilbert Spaces Are Reflexive*
For every Hilbert space $\mathcal{H}$, the canonical embedding $J : \mathcal{H} \to \mathcal{H}^{**}$ is onto.

Proof. Set $B := \overline{B}_{\mathcal{H}}$. If $(h_n) \subseteq B$, then by Theorem 6.1.17, $h_{n_k} \rightharpoonup h$ for some $h \in \mathcal{H}$.
By Proposition 6.1.4, $\|h\| \le \liminf_k \|h_{n_k}\| \le 1$, so $h \in B$.
Thus B is weakly subsequentially compact, hence $J(\mathcal{H}) = \mathcal{H}^{**}$ by Corollary 6.1.15.
Structural Commentary: B_w subsequentially compact $\Longrightarrow J(\mathcal{H}) = \mathcal{H}^{**}$.
Key Insight: Hilbert coordinates make B_w sequence-compact. □

Theorem 6.1.19. *Banach–Saks Theorem in Hilbert Spaces*

Let $\mathcal{H}$ be Hilbert and let $h_n \rightharpoonup h$. Then $\exists n_k \uparrow \infty$ such that $\frac{1}{k}\sum_{j=1}^{k} h_{n_j} \to h$ strongly.

Proof. Replace h_n by $h_n - h$; let $h = 0$. Put $C := \sup_n \|h_n\|^2 < +\infty$.

Let $n_1 = 1$, $n_1 < \cdots < n_k$, $s_k := \sum_{j=1}^{k} h_{n_j}$; $n_{k+1} > n_k$ with $\left|\langle s_k, h_{n_{k+1}}\rangle\right| \le 1$, since $\langle s_k, h_n\rangle \to 0$.

Then $\|s_{k+1}\|^2 = \|s_k\|^2 + 2\operatorname{Re}\langle s_k, h_{n_{k+1}}\rangle + \|h_{n_{k+1}}\|^2 \le \|s_k\|^2 + C + 2$, so $\|s_k\|^2 \le (C+2)k$.

Hence $\left\|\frac{1}{k}s_k\right\|^2 \le \frac{C+2}{k} \to 0$. For the original sequence, $\frac{1}{k}\sum_{j=1}^{k} h_{n_j} = h + \frac{1}{k}\sum_{j=1}^{k}(h_{n_j} - h) \to h$.

Structural Commentary: Choose n_{k+1} so $\left|\langle s_k, h_{n_{k+1}}\rangle\right| \le 1$; then $\|s_k\|^2 = O(k)$.
Key Insight: Weak-null subsequences can have $o(k)$ Cesàro norm. □

Theorem 6.1.20. *Radon–Riesz Theorem in Hilbert Spaces*
Let $\mathcal{H}$ be Hilbert and let $h_n \rightharpoonup h$. Then $h_n \to h$ in norm iff $\|h_n\| \to \|h\|$.

Proof.
($\Rightarrow$) If $\|h_n - h\| \to 0$, then $|\|h_n\| - \|h\|| \le \|h_n - h\| \to 0$.
($\Leftarrow$) Since $h_n \rightharpoonup h$, $\langle h_n, h\rangle \to \|h\|^2$. If also $\|h_n\| \to \|h\|$, then $\|h_n - h\|^2 = \|h_n\|^2 - 2\operatorname{Re}\langle h_n, h\rangle + \|h\|^2 \to 0$.

Structural Commentary: Weak convergence fixes $\langle h_n, h\rangle$; norm convergence fixes $\|h_n\|^2$.
Key Insight: $\|h_n - h\|^2$ has no remaining term. □

Theorem 6.1.21. *Metrizing Weak Topologies on the Unit Ball*
Let $\mathcal{X}$ be normed, $W \le \mathcal{X}^*$ separable and point-separating $B := \overline{B}_{\mathcal{X}}$. Choose $(\varphi_k)_{k\ge1}$ dense in $W \cap \overline{B}_{\mathcal{X}^*}$. Then $\rho(u,v) := \sum_{k\ge1} 2^{-k}|\varphi_k(u-v)|$ is a metric on B inducing $\sigma(\mathcal{X}, W)|_B$.

Proof. For $u, v \in B$, $|\varphi_k(u-v)| \le 2$, hence $\rho(u,v) < +\infty$; symmetry and the triangle inequality are termwise.
If $\rho(u,v) = 0$, then $\varphi_k(u-v) = 0$ for all k; for $\psi \in W \setminus \{0\}$, density gives $\varphi_{k_j} \to \frac{\psi}{\|\psi\|}$, hence $\psi(u-v) = 0$.
Since W separates points, $u = v$, so ρ is a metric. Fix $u \in B$. It remains to compare neighborhoods at u.
($\Rightarrow$) Given $r > 0$, choose N with $\sum_{k>N} 2^{1-k} < \frac{r}{2}$ and $\delta > 0$ with $\delta \sum_{k\le N} 2^{-k} < \frac{r}{2}$.
Then $V := \{v \in B : |\varphi_k(v-u)| < \delta, 1 \le k \le N\}$ is a $\sigma(\mathcal{X}, W)|_B$-neighborhood of u.
By $v \in V$, $\rho(u,v) \le \delta \sum_{k\le N} 2^{-k} + \sum_{k>N} 2^{1-k} < r$, hence $V \subseteq B_\rho(u, r)$.
($\Leftarrow$) Let $U = \{v \in B : |\psi_j(v-u)| < \varepsilon, 1 \le j \le m\}$ with $\psi_j \in W$ and $\varepsilon > 0$. Put $K := \{j : \psi_j \ne 0\}$.
For $j \in K$, choose k_j with $\left\|\frac{\psi_j}{\|\psi_j\|} - \varphi_{k_j}\right\| < \frac{\varepsilon}{4\|\psi_j\|}$, then choose $r > 0$ with $2^{k_j} r < \frac{\varepsilon}{2\|\psi_j\|}$ $\forall j \in K$.
If $\rho(u,v) < r$, then $\left|\varphi_{k_j}(v-u)\right| < 2^{k_j} r$, $\|v-u\| \le 2$,
so $|\psi_j(v-u)| \le \|\psi_j\|\left|\varphi_{k_j}(v-u)\right| + \|\psi_j\|\left\|\frac{\psi_j}{\|\psi_j\|} - \varphi_{k_j}\right\|\|v-u\| < \varepsilon$ for $j \in K$.
For $j \notin K$, $\psi_j = 0$. Thus $B_\rho(u,r) \subseteq U$.
Therefore ρ induces $\sigma(\mathcal{X}, W)|_B$.
Structural Commentary: $\rho = \sum_{k\ge1} 2^{-k}|\varphi_k(\cdot)|$ metrizes the bounded test topology generated by W.
Key Insight: On B, dense countable tests determine all W-tests. □

Corollary 6.1.22. *Metrizability of Weak Unit Balls*
Let $\mathcal{X}$ be normed, write $B_{\mathcal{Y}} := \overline{B}_{\mathcal{Y}}$ for normed $\mathcal{Y}$, and let $J : \mathcal{X} \to \mathcal{X}^{**}$ be canonical.
(i) If $\mathcal{X}^*$ is separable, then $(B_{\mathcal{X}}, \sigma(\mathcal{X}, \mathcal{X}^*))$ is metrizable.
(ii) If $\mathcal{X}$ is separable, then $(B_{\mathcal{X}^*}, \sigma(\mathcal{X}^*, J(\mathcal{X})))$ is metrizable.
(iii) If $\mathcal{X}$ is reflexive Banach, then $(B_{\mathcal{X}}, \sigma(\mathcal{X}, \mathcal{X}^*))$ is metrizable iff $\mathcal{X}$ is separable.

Proof.
(i) $\mathcal{X}^*$ is separable and point-separating on $\mathcal{X}$: if $x \ne y$, then $\exists \varphi \in \mathcal{X}^*$ with $\varphi(x-y) \ne 0$.
Theorem 6.1.21 with $(\mathcal{Y}, W) = (\mathcal{X}, \mathcal{X}^*)$ metrizes $(B_{\mathcal{X}}, \sigma(\mathcal{X}, \mathcal{X}^*))$.
(ii) Since J is an isometry, $\mathcal{X}$ separable implies $J(\mathcal{X})$ separable.
If $\varphi \ne \psi$ in $\mathcal{X}^*$, choose $x \in \mathcal{X}$ with $(\varphi - \psi)(x) \ne 0$, hence $Jx(\varphi - \psi) \ne 0$, $J(\mathcal{X})$ point-separating on $\mathcal{X}^*$.
Theorem 6.1.21 with $(\mathcal{Y}, W) = (\mathcal{X}^*, J(\mathcal{X}))$ metrizes $(B_{\mathcal{X}^*}, \sigma(\mathcal{X}^*, J(\mathcal{X})))$.
(iii) ($\Rightarrow$) Let $B := B_{\mathcal{X}}$, B_w metrizable. By reflexivity, Theorem 6.1.11, B_w is compact metrizable, separable.
Choose countable $D \subseteq B$ with $\overline{D}^w = B$. Set $K := \overline{\operatorname{co} D}^{\|\cdot\|}$. If $K \ne B$, choose $x \in B \setminus K$.
Hahn–Banach gives $\varphi \in \mathcal{X}^*$ with $\operatorname{Re}\varphi(x) > \sup_{y\in K}\operatorname{Re}\varphi(y)$.
Since $D \subseteq K$ and $x \in \overline{D}^w$, weak continuity gives $\operatorname{Re}\varphi(x) \le \sup_{y\in K}\operatorname{Re}\varphi(y)$, contradiction.
Thus $B = K = \overline{\operatorname{co}_{\mathbb{Q}} D}^{\|\cdot\|}$; $\operatorname{co}_{\mathbb{Q}} D$ is countable, so B is norm separable, and $\mathcal{X} = \bigcup_{m\ge1} mB$ is separable.
(iii) ($\Leftarrow$) Assume $\mathcal{X}$ separable. By (ii), Theorem 6.1.6, $J(\mathcal{X}) = \mathcal{X}^{**}$, $B_{\mathcal{X}^*}$ compact metrizable in $\sigma(\mathcal{X}^*, \mathcal{X}^{**})$.
Pick countable dense $D \subseteq B_{\mathcal{X}^*}$. The ($\Rightarrow$) argument in $\mathcal{X}^*$ gives $B_{\mathcal{X}^*} = \overline{\operatorname{co}_{\mathbb{Q}} D}^{\|\cdot\|}$.
Since $\operatorname{co}_{\mathbb{Q}} D$ is countable, $B_{\mathcal{X}^*}$ is norm separable and $\mathcal{X}^* = \bigcup_{m\ge1} mB_{\mathcal{X}^*}$ is separable, so (i) applies.

Structural Commentary: W separable metrizes $B_{\sigma(\mathcal{X},W)}$; compact metrizable B_w yields $B = \overline{\operatorname{co}_{\mathbb{Q}} D}^{\|\cdot\|}$.
Key Insight: Reflexive $B_{\mathcal{X}}$ is weakly metrizable iff $\mathcal{X}$ is separable. □

Theorem 6.1.23. *Whole Weak Topologies Are Not Metrizable in Infinite Dimension*
Let $\mathcal{X}$ be Banach with $\dim \mathcal{X} = +\infty$.

(i) $(\mathcal{X}, \sigma(\mathcal{X}, \mathcal{X}^*))$ *is not metrizable.*
(ii) $(\mathcal{X}^*, \sigma(\mathcal{X}^*, J(\mathcal{X})))$ *is not metrizable.*

Proof.

(i) Suppose ρ metrizes $\sigma(\mathcal{X}, \mathcal{X}^*)$. $\forall n$, choose $U_n \subseteq B_\rho(0, \frac{1}{n})$ with $U_n = \{x : |\varphi_{n,j}(x)| < \varepsilon_n,\ 1 \le j \le m_n\}$.
Since $\bigcap_{j=1}^{m_n} \ker \varphi_{n,j} \neq \{0\}$, choose x_n in this kernel with $\|x_n\| = n$.
Then $x_n \in U_n$, so $\rho(x_n, 0) < \frac{1}{n}$ and $x_n \rightharpoonup 0$, contradicting Proposition 6.1.4(i).

(ii) Suppose ρ_* metrizes $\sigma(\mathcal{X}^*, J(\mathcal{X}))$. $\forall n$, choose finite-dim $M_n \le \mathcal{X}$ with $M_n^\perp := \{\varphi : \varphi|_{M_n} = 0\} \subseteq B_{\rho_*}(0, \frac{1}{n})$.
Choose $y_n \notin M_n$ and define ψ_n on $M_n + \operatorname{span}\{y_n\}$ by $\psi_n|_{M_n} = 0$ and $\psi_n(y_n) = 1$.
Extend ψ_n to $\mathcal{X}$ by Corollary 5.3.2(i), and rescale to $\varphi_n \in M_n^\perp$ with $\|\varphi_n\|_{\text{op}} = n$.
Then $\rho_*(\varphi_n, 0) < \frac{1}{n}$, so $\varphi_n \to 0$ weak-*. By Theorem 5.3.7, $(\|\varphi_n\|_{\text{op}})$ is bounded, a contradiction.

Structural Commentary: Finite tests leave nonzero kernels in $\mathcal{X}$ and nonzero annihilators in $\mathcal{X}^*$.
Key Insight: Metric convergence would force bounded norms; finite scalar tests cannot. □

Method 6.1.24. *Weak Compactness Proof Patterns*

(i) *Compactify*: replace $\|x\|$ by finite coordinates $(\varphi_1(x), \dots, \varphi_m(x))$.
(ii) *Embed*: use $J : \mathcal{X} \to \mathcal{X}^{**}$ and compare $\sigma(\mathcal{X}, \mathcal{X}^*)$ with $\sigma(\mathcal{X}^{**}, \mathcal{X}^*)|_{J(\mathcal{X})}$.
(iii) *Separate*: if $F \notin J(B)$, choose $\varphi \in \mathcal{X}^*$ separating F from $J(B)$.
(iv) *Metrize on balls*: for separable $W \le \mathcal{X}^*$, use $\rho(u, v) = \sum_{k \ge 1} 2^{-k} |\varphi_k(u - v)|$ on bounded B.
(v) *Hilbertize*: replace φ by $\langle \cdot, h \rangle$ and use orthogonal coordinates/projections.

Key Insight: Weak compactness is compactness for scalar coordinates $\varphi(x)$ on bounded sets.

6.2 Compact Operators

Compact operators carry bounded sets to relatively compact sets. Their spectral and solvability theory therefore resembles matrix theory inside infinite-dimensional spaces.

Definition 6.2.1. *Compact and Finite-Rank Operators*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces and let $T : \mathcal{X} \to \mathcal{Y}$ be linear.

(i) *Compact*: $\overline{T(B)}$ is compact in $\mathcal{Y}$ $\forall B \subseteq \mathcal{X}$ bounded.
(ii) *Finite rank*: $\dim \operatorname{range}(T) < \infty$.

Proposition 6.2.2. *Compactness Tests on Hilbert Spaces*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$, and set $B := \overline{B}_{\mathcal{H}}$. The following are equivalent:

(i) K *is compact.*
(ii) $K(B)$ *is totally bounded.*
(iii) Every bounded $(h_n) \subseteq \mathcal{H}$ has a subsequence (h_{n_j}) such that Kh_{n_j} converges in norm.

Proof. For bounded $E \subseteq \mathcal{H}$, choose $r > 0$ with $E \subseteq rB$; then $K(E) \subseteq K(rB) = rK(B)$.

(i) $\Rightarrow$ *(ii)* K compact $\Rightarrow$ $\overline{K(B)}$ compact $\Rightarrow$ $\overline{K(B)}$ totally bounded $\Rightarrow$ $K(B)$ totally bounded.
(ii) $\Rightarrow$ *(iii)* (h_n) bounded $\Rightarrow \exists r > 0$, $\{h_n : n \ge 1\} \subseteq rB$. Then $(Kh_n) \subseteq K(rB) = rK(B)$ with $rK(B)$ totally bounded, so $\exists (n_j)$ such that (Kh_{n_j}) is Cauchy; $\mathcal{H}$ complete $\Rightarrow \exists y \in \mathcal{H}$, $\|Kh_{n_j} - y\| \to 0$.
(iii) $\Rightarrow$ *(i)* Let $E \subseteq \mathcal{H}$ be bounded. If $K(E)$ is not totally bounded, then $\exists \varepsilon > 0$ and $(x_n) \subseteq E$ such that $n \neq m \Rightarrow \|Kx_n - Kx_m\| \ge \varepsilon$. By (iii), $\exists (n_j)$ and $y \in \mathcal{H}$ with $\|Kx_{n_j} - y\| \to 0$, contradiction. Hence $K(E)$ is totally bounded, so $\overline{K(E)}$ is totally bounded and complete; Theorem 4.1.34 gives $\overline{K(E)}$ compact.

Structural Commentary: $E \subseteq rB \Rightarrow K(E) \subseteq rK(B)$; $K(B)$ totally bounded $\Rightarrow (h_n) \subseteq rB \Rightarrow \exists (n_j) : (Kh_{n_j})$ Cauchy.
Key Insight: K compact iff K sends bounded sequences to norm-precompact sequences. □

Proposition 6.2.3. *Finite Rank Operators Are Compact*
Let $\mathcal{X}, \mathcal{Y}$ be normed, $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, $R := \operatorname{range}(T)$. If $\dim R < +\infty$, then T is compact.

Proof. Let $B \subseteq \mathcal{X}$ be bounded. Then $T(B) \subseteq R$ is bounded, so $\overline{T(B)}^{\,R}$ is compact by Theorem 5.1.14.
Since $R \le \mathcal{Y}$ is finite-dimensional, R is closed in $\mathcal{Y}$; hence $\overline{T(B)}^{\,\mathcal{Y}} = \overline{T(B)}^{\,R}$ is compact. Thus T is compact.
Structural Commentary: $T(B) \subseteq R$, $\dim R < +\infty$, and bounded subsets of R have compact closure.
Key Insight: Finite-dimensional range makes bounded images relatively compact. □

Theorem 6.2.4. *Norm Limits of Compact Operators*
Let $\mathcal{X}$ be normed, $\mathcal{Y}$ Banach, $T_n, T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$. If each T_n compact and $\|T_n - T\|_{op} \to 0$, then T is compact.

Proof. Let $B \subseteq \mathcal{X}$ be bounded, $M > 0$, $\|x\| \le M$ for $x \in B$. For $\varepsilon > 0$, choose N with $\|T - T_N\|_{\text{op}} < \frac{\varepsilon}{2M}$.
Since $\overline{T_N(B)}$ is compact, choose $y_1, \dots, y_r \in \mathcal{Y}$ with $T_N(B) \subseteq \bigcup_{j=1}^r B_{\mathcal{Y}}(y_j, \frac{\varepsilon}{2})$.
Then $T(B) \subseteq \bigcup_{j=1}^r B_{\mathcal{Y}}(y_j, \varepsilon)$, so $T(B)$ is totally bounded. Since $\mathcal{Y}$ is complete, $\overline{T(B)}$ is compact; T is compact. *Structural Commentary*: $\|T - T_N\|_{\text{op}}$ transfers finite nets from $T_N(B)$ to $T(B)$ for bounded B.
Key Insight: Norm limits preserve relative compactness of bounded images. □

Proposition 6.2.5. *Compact Operators Form an Operator Ideal*
Let $\mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be normed spaces.

(i) If $K_1, \dots, K_m : \mathcal{X} \to \mathcal{Y}$ are compact, then $\sum_{j=1}^m K_j$ is compact.
(ii) If $K : \mathcal{X} \to \mathcal{Y}$ is compact, $A \in \mathcal{L}(\mathcal{Y}, \mathcal{Z})$, and $B \in \mathcal{L}(\mathcal{W}, \mathcal{X})$, then $AKB : \mathcal{W} \to \mathcal{Z}$ is compact.

Proof.

(i) Let $E \subseteq \mathcal{X}$ be bounded, $C_j := \overline{K_j(E)}$. Then each C_j is compact, and $\sum_{j=1}^{m} K_j(E) \subseteq \sigma(C_1 \times \cdots \times C_m)$ for the continuous map $\sigma(y_1, \ldots, y_m) := \sum_{j=1}^{m} y_j$. Thus $\sum_{j=1}^{m} K_j(E)$ is relatively compact.

(ii) Let $E \subseteq \mathcal{W}$ be bounded. Then $B(E)$ is bounded, $C := \overline{K(B(E))}$ is compact, and $AKB(E) \subseteq A(C)$. Since $A(C)$ is compact and closed in $\mathcal{Z}$, $\overline{AKB(E)} \subseteq A(C)$ is compact.

Structural Commentary: Products of compact image closures control sums; bounded precomposition and continuous postcomposition control AKB.
Key Insight: Compact operators form a two-sided ideal in $\mathcal{L}$. □

Example 6.2.6. *Diagonal Compact Operators*
Let $(e_k)_{k\geq 1}$ be a Hilbert basis of separable $\mathcal{H}$ and let $\lambda = (\lambda_k) \in \ell^\infty$ with $\lambda_k \to 0$.
For $h \in \mathcal{H}$, set $Th := \sum_{k=1}^{\infty} \lambda_k \langle h, e_k\rangle e_k$ and $T_N h := \sum_{k=1}^{N} \lambda_k \langle h, e_k\rangle e_k$.
By Parseval: $\sum_k |\lambda_k \langle h, e_k\rangle|^2 \leq \|\lambda\|_\infty^2 \|h\|^2 \Rightarrow T \in \mathcal{L}(\mathcal{H})$.
For $N \geq 1$, $\operatorname{range} T_N \subseteq \operatorname{span}\{e_1, \ldots, e_N\} \Rightarrow \operatorname{rank} T_N \leq N \Rightarrow T_N$ compact by Proposition 6.2.3.
Also $\|(T - T_N)h\|^2 = \sum_{k>N} |\lambda_k|^2 |\langle h, e_k\rangle|^2 \leq \sup_{k>N} |\lambda_k|^2 \|h\|^2 \Rightarrow \|T - T_N\|_{\text{op}} \leq \sup_{k>N} |\lambda_k| \to 0$.
Hence $T_N \to T$ in $\|\cdot\|_{\text{op}}$ and T is compact by Theorem 6.2.4.

Theorem 6.2.7. *Weak-to-Strong Characterization*
Let $\mathcal{H}$ be Hilbert, $K \in \mathcal{L}(\mathcal{H})$. Then K is compact iff $h_n \rightharpoonup h$ implies $\|Kh_n - Kh\| \to 0$ $\forall (h_n) \subseteq \mathcal{H}$ and $h \in \mathcal{H}$.

Proof.

($\Rightarrow$) Let $h_n \rightharpoonup h$. By Proposition 6.1.4(i), (h_n) bounded.
For every subsequence (n_j), By Proposition 6.2.2(iii), $\exists (j_\ell)$ and $y \in \mathcal{H}$ with $\left\|Kh_{n_{j_\ell}} - y\right\| \to 0$.
For $u \in \mathcal{H}$, $\langle y, u\rangle = \lim_\ell \left\langle Kh_{n_{j_\ell}}, u\right\rangle = \lim_\ell \left\langle h_{n_{j_\ell}}, K^* u\right\rangle = \langle h, K^* u\rangle = \langle Kh, u\rangle$, so $y = Kh$.
If $\|Kh_n - Kh\| \not\to 0$, then $\exists \varepsilon > 0$, (n_j) with $\left\|Kh_{n_j} - Kh\right\| \geq \varepsilon$, contradicting $\exists (j_\ell) : \left\|Kh_{n_{j_\ell}} - Kh\right\| \to 0$.

($\Leftarrow$) Let (h_n) be bounded. By Theorem 6.1.17, $\exists (n_j)$, $h \in \mathcal{H}$ with $h_{n_j} \rightharpoonup h$. By hypothesis, $\left\|Kh_{n_j} - Kh\right\| \to 0$.
Hence $\forall (h_n)$ bounded, $\exists (n_j)$ and $y \in \mathcal{H}$ with $\left\|Kh_{n_j} - y\right\| \to 0$; By Proposition 6.2.2(iii), K compact.

Structural Commentary: K compact, $h_n \rightharpoonup h \Rightarrow Kh_{n_{j_\ell}} \to y = Kh$; (h_n) bounded $\Rightarrow h_{n_j} \rightharpoonup h \Rightarrow Kh_{n_j} \to Kh$.
Key Insight: K compact iff $h_n \rightharpoonup h$ forces $Kh_n \to Kh$ in norm. □

Definition 6.2.8. *Orthogonal Projection Sequence*
Let $(e_k)_{k\geq 1}$ be a Hilbert basis for a separable Hilbert space $\mathcal{H}$.
The induced *orthogonal projection sequence* $(P_n)_{n\geq 1}$ is defined by:

(i) $P_n h := \sum_{k=1}^{n} \langle h, e_k\rangle e_k$ for $h \in \mathcal{H}$.
(ii) $\operatorname{range}(P_n) = \operatorname{span}\{e_1, \ldots, e_n\}$ and $\|P_n\|_{\text{op}} = 1$.
(iii) $P_n h \to h$ for every $h \in \mathcal{H}$.

Proposition 6.2.9. *Finite-Rank Approximation by Orthogonal Projections*
Let (P_n) be the orthogonal projection sequence induced by a Hilbert basis of a separable Hilbert space $\mathcal{H}$.
For $T \in \mathcal{L}(\mathcal{H})$, T is compact iff $\|P_n T - T\|_{op} \to 0$.

Proof.

($\Rightarrow$) $B := \overline{B}_{\mathcal{H}}$, $C := \overline{T(B)}$; T compact $\Rightarrow C$ compact.
$\varepsilon > 0 \Rightarrow \exists y_1, \ldots, y_r \in C, N \in \mathbb{N} : C \subseteq \bigcup_{j=1}^{r} B_{\mathcal{H}}(y_j, \frac{\varepsilon}{3})$ and $n \geq N \Rightarrow \max_{1\leq j\leq r} \|(P_n - I)y_j\| < \frac{\varepsilon}{3}$.
Thus $n \geq N$, $x \in B \Rightarrow \exists j : \|Tx - y_j\| < \frac{\varepsilon}{3}$ and $\|(P_n T - T)x\| \leq \|Tx - y_j\| + \|(P_n - I)y_j\| + \|y_j - Tx\| < \varepsilon$.
Hence $\|P_n T - T\|_{\text{op}} = \sup_{x\in B} \|(P_n T - T)x\| \to 0$.

($\Leftarrow$) $\operatorname{range}(P_n T) \subseteq \operatorname{range}(P_n)$, $\dim \operatorname{range}(P_n) < \infty \Rightarrow P_n T$ compact by Proposition 6.2.3.
Also $P_n T \to T$ in $\|\cdot\|_{\text{op}}$; Theorem 6.2.4 $\Rightarrow T$ compact.

Structural Commentary: Compact $C = \overline{T(B)}$ upgrades $P_n y \to y$ to uniform convergence on C.
Key Insight: T compact iff finite-rank projections approximate T in operator norm. □

Corollary 6.2.10. *Finite-Rank Operators Are Dense in Compact Operators*
Let $\mathcal{H}$ be separable, and let $\mathcal{K}(\mathcal{H})$ and $\mathcal{F}(\mathcal{H})$ denote the compact and finite-rank operators in $\mathcal{L}(\mathcal{H})$.

(i) $\mathcal{K}(\mathcal{H})$ is a norm-closed linear subspace of $\mathcal{L}(\mathcal{H})$.
(ii) $\overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{op}} = \mathcal{K}(\mathcal{H})$.

Proof.

(i) Proposition 6.2.5: $K_1, K_2 \in \mathcal{K}(\mathcal{H})$, $\alpha, \beta \in \mathbb{F} \Rightarrow \alpha K_1 + \beta K_2 \in \mathcal{K}(\mathcal{H})$.
Theorem 6.2.4: $K_n \in \mathcal{K}(\mathcal{H})$, $\|K_n - K\|_{\text{op}} \to 0 \Rightarrow K \in \mathcal{K}(\mathcal{H})$.

(ii) ($\Rightarrow$) $\mathcal{F}(\mathcal{H}) \subseteq \mathcal{K}(\mathcal{H})$, (i) $\Rightarrow \overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{\text{op}}} \subseteq \mathcal{K}(\mathcal{H})$.

(ii) ($\Leftarrow$) $\dim \mathcal{H} < +\infty \Rightarrow \mathcal{F}(\mathcal{H}) = \mathcal{L}(\mathcal{H}) \supseteq \mathcal{K}(\mathcal{H})$. If $\dim \mathcal{H} = +\infty$,
take (P_n); $T \in \mathcal{K}(\mathcal{H}) \Rightarrow P_n T \in \mathcal{F}(\mathcal{H})$, $\|P_n T - T\|_{\text{op}} \to 0 \Rightarrow T \in \overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{\text{op}}}$ by Proposition 6.2.9.

Structural Commentary: $\mathcal{K}(\mathcal{H})$ is norm closed, and $P_nT \in \mathcal{F}(\mathcal{H})$ with $P_nT \to T$ for compact T.
Key Insight: On separable $\mathcal{H}$, $\mathcal{K}(\mathcal{H}) = \overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{\mathrm{op}}}$. □

Theorem 6.2.11. *Schauder's Theorem*
Let $\mathcal{H}$ be Hilbert. If $K \in \mathcal{L}(\mathcal{H})$ is compact, then K^* is compact.

Proof. ByTheorem 5.2.34,$K^* \in \mathcal{L}(\mathcal{H})$. By Theorem 6.2.7, it suffices to prove $h_n \rightharpoonup h \Rightarrow \|K^*(h_n - h)\| \to 0$.
Put $u_n := h_n - h$. Then $u_n \rightharpoonup 0$, $M := \sup_n \|u_n\| < \infty$, and $\forall v \in \mathcal{H}$, $\langle K^*u_n, v\rangle = \langle u_n, Kv\rangle \to 0$, so $K^*u_n \rightharpoonup 0$.
Compactness of K gives $\|KK^*u_n\| \to 0$, hence $0 \le \|K^*u_n\|^2 = \langle KK^*u_n, u_n\rangle \le M\|KK^*u_n\| \to 0$.
Structural Commentary: $u_n \rightharpoonup 0 \Rightarrow K^*u_n \rightharpoonup 0$, K compact $\Rightarrow KK^*u_n \to 0$; $\|K^*u_n\|^2 \le M\|KK^*u_n\|$.
Key Insight: $\|K^*u_n\|^2 = \langle KK^*u_n, u_n\rangle$. □

Proposition 6.2.12. *Injective Compact Perturbations of the Identity Are Onto*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$ be compact, and set $A := I + K$. If $\ker A = \{0\}$, then $A(\mathcal{H}) = \mathcal{H}$.

Proof.
(i) If $\inf_{\|h\|=1}\|Ah\| = 0$, choose $\|h_n\| = 1$, $Ah_n \to 0$. By Proposition 6.2.2(iii), $Kh_{n_j} \to y$,
hence $h_{n_j} = Ah_{n_j} - Kh_{n_j} \to -y := h$, $\|h\| = 1$, and $Ah = 0$, contradiction.
Thus $\exists c > 0 : \|Ah\| \ge c\|h\|\ \forall h \in \mathcal{H}$.
(ii) Fix $g \in \mathcal{H}$ and put $M := \mathrm{cspan}\{K^m g : m \ge 0\}$. Then $g \in M$, $K(M) \subseteq M$, and M is separable.
If $\dim M < +\infty$, then $A|_M : M \to M$ injective $\Rightarrow$ $A|_M$ surjective $\Rightarrow g \in A(\mathcal{H})$.
(iii) Assume $\dim M = +\infty$. Let (P_n) be projection sequence of Hilbert basis of M,
$H_n := P_n(M)$, $\dim H_n < +\infty$. By Proposition 6.2.9, $\|P_nK - K\|_{\mathrm{op}} \to 0$. Let N,
$n \ge N \Rightarrow \|P_nK - K\|_{\mathrm{op}} < \frac{c}{2}$. Then $n \ge N$, $h \in H_n \Rightarrow \|(I + P_nK)h\| \ge \|Ah\| - \|(P_nK - K)h\| \ge \frac{c}{2}\|h\|$, so $I + P_nK : H_n \to H_n$
injective $\Rightarrow$ surjective.
(iv) Choose $u_n \in H_n$ with $(I + P_nK)u_n = P_ng$. Then $\|u_n\| \le \frac{2}{c}\|P_ng\| \le \frac{2}{c}\|g\|$.
By Theorem 6.1.17, $u_{n_j} \rightharpoonup u$ for some subsequence.
By Theorem 6.2.7, $Ku_{n_j} \to Ku$, $\|P_{n_j}Ku_{n_j} - Ku_{n_j}\| \le \|P_{n_j}K - K\|_{\mathrm{op}}\|u_{n_j}\| \to 0$.
Since $P_{n_j}g \to g$, $u_{n_j} = P_{n_j}g - P_{n_j}Ku_{n_j} \to g - Ku$; by weak uniqueness $u = g - Ku$, hence $Au = g$.
Since $g \in \mathcal{H}$ was arbitrary, $A(\mathcal{H}) = \mathcal{H}$.
Structural Commentary: $\|Ah\| \ge c\|h\|$ and $(I + P_nK)u_n = P_ng \Rightarrow Au = g$.
Key Insight: Compactness removes approximate kernels and passes projected solutions to a true solution. □

Definition 6.2.13. *Fredholm Operator and Index*
Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{L}(\mathcal{H})$.
(i) *Fredholm*: $\mathrm{range}(T)$ is closed, $\dim\ker T < \infty$, and $\mathrm{codim}\,\mathrm{range}(T) < \infty$.
(ii) *Index*: $\mathrm{ind}\,T := \dim\ker T - \mathrm{codim}\,\mathrm{range}(T)$ when T is Fredholm.

Theorem 6.2.14. *Riesz–Schauder for Compact Perturbations of the Identity*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$ be compact, and set $A := I + K$. Then:
(i) $\dim\ker A < \infty$.
(ii) $\mathrm{range}(A)$ *is closed.*
(iii) $\mathrm{codim}\,\mathrm{range}(A) = \dim\ker(A^*) = \dim\ker A < \infty$.
(iv) A is Fredholm with $\mathrm{ind}\,A = 0$.
(v) A is injective iff it is surjective.

Proof.
(i) $\dim\ker A = +\infty \Rightarrow \exists(u_n) \subseteq \ker A$ with $\langle u_n, u_m\rangle = \delta_{nm}$.
Then $Ku_n = -u_n$ and $n \ne m \Rightarrow \|Ku_n - Ku_m\| = \sqrt{2}$, contradicting Proposition 6.2.2(iii).
(ii) Let $H_0 := (\ker A)^\perp$. If $\inf\{\|Ah\| : h \in H_0, \|h\| = 1\} = 0$, choose $h_n \in H_0$, $\|h_n\| = 1$, $Ah_n \to 0$.
By Proposition 6.2.2(iii), $Kh_{n_j} \to y$, so $h_{n_j} = Ah_{n_j} - Kh_{n_j} \to -y := h \in H_0$, $Ah = 0$.
Thus $h \in \ker A \cap H_0 = \{0\}$, contradiction. Hence $\exists c > 0 : \|Ah\| \ge c\|h\|$ on H_0.
Also $A(H_0) = A(\mathcal{H})$. If $Ah_n \to y$ with $h_n \in H_0$, then $\|h_n - h_m\| \le c^{-1}\|Ah_n - Ah_m\|$.
Thus $h_n \to h \in H_0$ and $Ah = y$; hence $\mathrm{range}(A)$ is closed.
(iii) By Theorem 6.2.11, K^* compact, so (i)–(ii) apply to $A^* = I + K^*$.
By Proposition 5.2.39, $\mathcal{H} = \mathrm{range}(A) \odot \ker(A^*)$, $\mathrm{codim}\,\mathrm{range}(A) = \dim\ker(A^*)$.
If $\dim\ker A < \dim\ker(A^*)$, choose injective $L : \ker A \to \ker(A^*)$ with $L(\ker A) \ne \ker(A^*)$, $F := LP_{\ker A}$.
Then $\mathrm{rank}\,F < +\infty$, $K + F$ compact. For $x = z + n$, $z \in H_0$, $n \in \ker A$,
$(A + F)x = Az + Ln \in \mathrm{range}(A) \odot L(\ker A)$; injectivity of L gives $\ker(A + F) = \{0\}$ and
$\mathrm{range}(A + F) = \mathrm{range}(A) \odot L(\ker A) \ne \mathcal{H}$, contradicting Proposition 6.2.12. Thus $\dim\ker A \ge \dim\ker(A^*)$.
Apply the same argument to A^* and use $(K^*)^* = K$ to get $\dim\ker A = \dim\ker(A^*)$.
(iv) (i)–(iii) give $\mathrm{range}(A)$ closed, $\dim\ker A < +\infty$, $\mathrm{codim}\,\mathrm{range}(A) < +\infty$.
Also $\dim\ker A = \mathrm{codim}\,\mathrm{range}(A)$; hence A is Fredholm and $\mathrm{ind}\,A = 0$.
(v) (iii) gives $\ker A = \{0\} \iff \mathrm{codim}\,\mathrm{range}(A) = 0 \iff \mathrm{range}(A) = \mathcal{H}$.
Structural Commentary: Compactness $\Rightarrow \dim\ker A < \infty$, $\mathrm{range}(A)$ closed; $F = LP_{\ker A} \Rightarrow \dim\ker A = \dim\ker(A^*)$.
Key Insight: For $A = I + K$ compactly perturbed, injectivity and surjectivity have the same finite defect. □

Corollary 6.2.15. *Fredholm Alternative*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$ be compact, let $\lambda \in \mathbb{F} \setminus \{0\}$, and set $T_\lambda := \lambda I - K$.
(i) $\mathrm{range}(T_\lambda) = \mathcal{H}$ *iff* $\ker T_\lambda = \{0\}$.
(ii) If the conditions in (i) hold, then $T_\lambda^{-1} \in \mathcal{L}(\mathcal{H})$.

Proof. Let $\widetilde{K} := -\lambda^{-1}K$; $\widetilde{K}$ compact, $T_\lambda = \lambda(I + \widetilde{K})$, $\ker T_\lambda = \ker(I + \widetilde{K})$, $\mathrm{range}(T_\lambda) = \mathcal{H}$ iff $\mathrm{range}(I + \widetilde{K}) = \mathcal{H}$.
(i) ($\Rightarrow$) $\mathrm{range}(T_\lambda) = \mathcal{H} \Rightarrow \mathrm{range}(I + \widetilde{K}) = \mathcal{H} \Rightarrow \ker(I + \widetilde{K}) = \{0\}$ by Theorem 6.2.14(v), hence $\ker T_\lambda = \{0\}$.

(*i*) ($\Leftarrow$) $\ker T_\lambda = \{0\} \Rightarrow \ker(I + \widetilde{K}) = \{0\} \Rightarrow \operatorname{range}(I + \widetilde{K}) = \mathcal{H}$ by Theorem 6.2.14(v), hence $\operatorname{range}(T_\lambda) = \mathcal{H}$.
(*ii*) (i) $\Rightarrow T_\lambda$ bijective; Corollary 5.4.4 $\Rightarrow T_\lambda^{-1} \in \mathcal{L}(\mathcal{H})$.

Structural Commentary: $T_\lambda = \lambda(I + \widetilde{K})$ with $\widetilde{K}$ compact; bijective T_λ has bounded inverse.
Key Insight: $\forall f$, $T_\lambda u = f$ is solvable iff $\ker T_\lambda = \{0\}$. □

Theorem 6.2.16. *Fredholm Index Zero as an Invertible Plus Compact Operator*
Let $\mathcal{H}$ be Hilbert and let $T \in \mathcal{L}(\mathcal{H})$. The following are equivalent:

(i) T is Fredholm and $\operatorname{ind} T = 0$.
(ii) $\exists S, K \in \mathcal{L}(\mathcal{H})$ such that $T = S + K$, S is invertible, and K is compact.

Proof.

(*i*) $\Rightarrow$ (*ii*) $N := \ker T$, $R := \operatorname{range}(T)$, $Z := \ker(T^*)$.
By Proposition 5.2.39, $\mathcal{H} = R \odot Z$ and $\dim Z = \operatorname{codim} R$. $\operatorname{ind} T = 0 \Rightarrow \dim N = \dim Z < +\infty$.
$A : N \to Z$ isomorphism, P_N orthogonal projection, $K := AP_N$, $S := T - K$.
Then $\operatorname{rank} K < +\infty \Rightarrow K$ compact and $T = S + K$. For $h = v + u$, $v \in N^\perp$, $u \in N$, $Sh = Tv - Au \in R \odot Z$ and $Sh = 0 \Rightarrow Tv = 0$, $Au = 0 \Rightarrow v = 0$, $u = 0$; hence $\ker S = \{0\}$.
Also $T(N^\perp) = R$, $A(N) = Z$, and $y = r + z \in R \odot Z \Rightarrow \exists v \in N^\perp, u \in N : Tv = r, Au = -z, S(v + u) = y$; hence $\operatorname{range} S = \mathcal{H}$ and $S^{-1} \in \mathcal{L}(\mathcal{H})$ by Corollary 5.4.4.
(*ii*) $\Rightarrow$ (*i*) $C := S^{-1}K$, $A := I + C$. By Proposition 6.2.5(ii), C is compact and $T = SA$.
By Theorem 6.2.14(iv), A is Fredholm, $\operatorname{ind} A = 0$. S isomorphism $\Rightarrow \ker T = \ker A$, $\operatorname{range} T = S(\operatorname{range} A)$, $\operatorname{codim} \operatorname{range} T = \operatorname{codim} \operatorname{range} A$. Hence T is Fredholm and $\operatorname{ind} T = 0$.

Structural Commentary: $K = AP_{\ker T}$ makes $S = T - K$ invertible; $T = S(I + C)$ with C compact.
Key Insight: $\operatorname{ind} T = 0$ iff T is invertible modulo compact operators. □

Corollary 6.2.17. *Index-Zero Fredholm Operators Are Closed under Composition*
Let $\mathcal{H}$ be Hilbert, $S, T \in \mathcal{L}(\mathcal{H})$ be Fredholm with $\operatorname{ind} S = \operatorname{ind} T = 0$. Then ST is Fredholm and $\operatorname{ind}(ST) = 0$.

Proof. By Theorem 6.2.16, $S = S_0 + K_S$, $T = T_0 + K_T$, with S_0, T_0 invertible and K_S, K_T compact.
Let $U := S_0T_0$, $C := S_0K_T + K_ST_0 + K_SK_T$. Then $ST = U + C$, U invertible, by Proposition 6.2.5, C compact.
Theorem 6.2.16 $\Rightarrow ST$ Fredholm and $\operatorname{ind}(ST) = 0$.
Structural Commentary: $(S_0 + K_S)(T_0 + K_T) = S_0T_0 + C$, $C \in \mathcal{K}(\mathcal{H})$.
Key Insight: Invertible modulo compact operators are closed under products. □

Corollary 6.2.18. *Compact Perturbations Preserve Index-Zero Fredholm Operators*
Let $\mathcal{H}$ be Hilbert and let $T, K \in \mathcal{L}(\mathcal{H})$. If T is Fredholm, $\operatorname{ind} T = 0$, and K is compact, then:

(i) $T + K$ is Fredholm.
(ii) $\operatorname{ind}(T + K) = 0$.

Proof. By Theorem 6.2.16, $T = S + C$, S invertible, C compact. Set $\widetilde{C} := C + K$.
By Proposition 6.2.5, $\widetilde{C} \in \mathcal{K}(\mathcal{H})$ and $T + K = S + \widetilde{C}$.

(i) $T + K = S + \widetilde{C}$, S invertible, $\widetilde{C}$ compact $\Rightarrow T + K$ Fredholm by Theorem 6.2.16.
(ii) $T + K = S + \widetilde{C}$, S invertible, $\widetilde{C}$ compact $\Rightarrow \operatorname{ind}(T + K) = 0$ by Theorem 6.2.16.

Structural Commentary: $T = S + C$, $K \in \mathcal{K}(\mathcal{H}) \Rightarrow T + K = S + \widetilde{C}$, $\widetilde{C} = C + K \in \mathcal{K}(\mathcal{H})$.
Key Insight: ind = 0 is stable under compact perturbation. □

6.3 Spectral Theory of Operators

For bounded operators, spectrum replaces the finite-dimensional eigenvalue list. The resolvent tracks invertibility, while the spectrum records every obstruction to it.

Definition 6.3.1. *Spectrum and Resolvent*
Let $\mathcal{X}$ be a complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$.

(i) *Resolvent set*: $\rho(T) := \{\lambda \in \mathbb{C} : \lambda I - T$ is bijective and $(\lambda I - T)^{-1} \in \mathcal{L}(\mathcal{X})\}$.
(ii) *Spectrum*: $\sigma(T) := \mathbb{C} \setminus \rho(T)$.

Theorem 6.3.2. *Spectrum of a Complex Bounded Operator*
Let $\mathcal{X}$ be a nonzero complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$. Then:

(i) $\sigma(T) \neq \varnothing$;
(ii) $\sigma(T)$ is compact in $\mathbb{C}$;
(iii) $\sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq \|T\|_{op}\}$.

Proof. For $\lambda \in \rho(T)$, $R(\lambda) := (\lambda I - T)^{-1}$. If $\mu \in \rho(T)$, $|\lambda - \mu| \, \|R(\mu)\|_{\mathrm{op}} < 1$; $\lambda I - T = (I + (\lambda - \mu)R(\mu))(\mu I - T)$.
By Corollary 5.1.7, $\lambda \in \rho(T)$, $R(\lambda) = R(\mu)(I + (\lambda - \mu)R(\mu))^{-1}$. If $|\lambda| > \|T\|_{\mathrm{op}}$, then $\lambda I - T = \lambda(I - \lambda^{-1}T)$.
By Corollary 5.1.7, $\lambda \in \rho(T)$ and $\|R(\lambda)\|_{\mathrm{op}} \leq (|\lambda| - \|T\|_{\mathrm{op}})^{-1}$.

(i) Suppose $\sigma(T) = \varnothing$. By the local Neumann formula above, $R : \mathbb{C} \to \mathcal{L}(\mathcal{X})$ is entire.
For $x \in \mathcal{X}$ and $\Phi \in \mathcal{X}^*$, $f_{x,\Phi}(\lambda) := \Phi(R(\lambda)x)$ is an entire scalar function. By the estimate above, $f_{x,\Phi}(\lambda) \to 0$ as $|\lambda| \to +\infty$, and continuity bounds $f_{x,\Phi}$ on compact disks. By Liouville's theorem, $f_{x,\Phi} = 0$. By Corollary 5.3.3, bounded linear functionals separate points, so $R(0) = 0$, contradicting $(-T)R(0) = I$ on the nonzero space $\mathcal{X}$.
(ii) By the local Neumann formula, $\rho(T)$ is open, so $\sigma(T)$ is closed in $\mathbb{C}$.
By outside-disk formula, $\sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq \|T\|_{\mathrm{op}}\}$, hence $\sigma(T)$ is bounded. Thus $\sigma(T)$ is compact.
(iii) If $|\lambda| > \|T\|_{\mathrm{op}}$, then the outside-disk condition holds. By the outside-disk Neumann argument, $\lambda \in \rho(T)$.
Therefore every $\lambda \in \sigma(T)$ satisfies $|\lambda| \leq \|T\|_{\mathrm{op}}$.

Structural Commentary: By Neumann, $\rho(T)$ is open and $\sigma(T) \subseteq \{\lambda : |\lambda| \leq \|T\|_{\text{op}}\}$. By Liouville, $\sigma(T) \neq \varnothing$.
Key Insight: Complex Banach spectra are nonempty compact obstructions to invertibility. □

Definition 6.3.3. *Spectral Radius*
Let $\mathcal{X}$ be a nonzero complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$. The *spectral radius* of T is $r(T) := \sup_{\lambda \in \sigma(T)} |\lambda|$.

Theorem 6.3.4. *Spectral Radius Formula*
Let $\mathcal{X}$ be a nonzero complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$.
Then the sequence $(\|T^n\|_{op}^{1/n})_{n\geq 1}$ converges in $[0,\infty)$, and $\lim_{n\to\infty} \|T^n\|_{op}^{1/n} = r(T)$.

Proof. Set $a_n := \|T^n\|_{\text{op}}$ for $n \geq 0$, so $a_0 = 1$ and $a_{m+n} \leq a_m a_n$. If $a_m = 0$ for some $m \geq 1$, then T is nilpotent.
Also $\|T^n\|_{\text{op}}^{1/n} \to 0$, $\sigma(T) = \{0\}$ by finite Neumann series and Theorem 6.3.2.
Hence assume $a_n > 0$ for all $n \geq 1$. For fixed $m \geq 1$ and $n = qm + r$ with $0 \leq r < m$, $a_n^{1/n} \leq a_m^{q/n} a_r^{1/n}$.
By $\limsup$ in n, $\limsup_n a_n^{1/n} \leq a_m^{1/m}$, hence $\limsup_n a_n^{1/n} \leq \inf_{m\geq 1} a_m^{1/m} \leq \liminf_n a_n^{1/n}$.
Thus $L := \lim_n a_n^{1/n} = \inf_{m\geq 1} a_m^{1/m}$. If $|\lambda| > L$, choose n with $a_n < |\lambda|^n$.
Then $\|T^n/\lambda^n\|_{\text{op}} < 1$, so Corollary 5.1.7 makes $\lambda^n I - T^n$ invertible.
With $Q_n := \sum_{j=0}^{n-1} \lambda^{n-1-j} T^j$, $\lambda^n I - T^n = (\lambda I - T)Q_n = Q_n(\lambda I - T)$; as these are polynomials in T,
$(\lambda I - T)^{-1} = Q_n(\lambda^n I - T^n)^{-1}$. Thus $\lambda \in \rho(T)$, so $\sigma(T) \subseteq \{z \in \mathbb{C} : |z| \leq L\}$ and $r(T) \leq L$.
Let $R > r(T)$, $R_0 > \max\{R, \|T\|_{\text{op}}\}$. For $|z| = R_0$, by Neumann expansion, $(zI - T)^{-1} = \sum_{k\geq 0} z^{-k-1} T^k$.
By Cauchy's coefficient formula $T^n = \frac{1}{2\pi i} \int_{|z|=R_0} z^n (zI - T)^{-1}\, dz$.
Since $\{z : R \leq |z| \leq R_0\} \subseteq \rho(T)$, Cauchy's theorem moves the contour to $|z| = R$.
With $M_R := \sup_{|z|=R} \left\|(zI - T)^{-1}\right\|_{\text{op}} < +\infty$, $a_n \leq M_R R^{n+1}$, hence $L \leq R$. By letting $R \downarrow r(T)$, $L \leq r(T)$.
Structural Commentary: By submultiplicativity, L exists; Neumann, $r(T) \leq L$; resolvent bounds, $L \leq r(T)$.
Key Insight: The spectral radius is asymptotic operator-power growth. □

Definition 6.3.5. *Quadratic Form and Rayleigh Quotient*
Let $\mathcal{H}$ be a real Hilbert space and let $T \in \mathcal{L}(\mathcal{H})$ be self-adjoint.

(i) *Quadratic form*: $Q_T(h) := \langle Th, h\rangle$ for $h \in \mathcal{H}$.
(ii) *Rayleigh quotient*: $R_T(h) := \frac{\langle Th,h\rangle}{\langle h,h\rangle}$ for $h \in \mathcal{H} \setminus \{0\}$.

A unit-sphere maximizer of Q_T is equivalently a maximizer of R_T on $\mathcal{H} \setminus \{0\}$.

Example 6.3.6. *A Self-Adjoint Operator without Eigenvectors*
Let $a < b$, set $H := L^2([a,b];\mathbb{C})$, and define $(Tf)(x) := xf(x)$.
With $M := \max\{|a|, |b|\}$, $\|Tf\|_2 \leq M \|f\|_2$, so $T \in \mathcal{L}(H)$.
For $u, v \in H$, $\langle Tu, v\rangle = \int_a^b xu(x)\overline{v(x)}\, dx = \int_a^b u(x)\overline{xv(x)}\, dx = \langle u, Tv\rangle$, hence $T = T^*$.
If $Tf = \lambda f$ for $\lambda \in \mathbb{C}$, then $(x - \lambda)f(x) = 0$ a.e. Let $E_\lambda := \{x \in [a,b] : x = \lambda\}$.
Since $m(E_\lambda) = 0$, $f = 0$ a.e. on $[a,b] \setminus E_\lambda$ and on E_λ, hence $f = 0$ in H.
Thus no nonzero $f \in H$ satisfies $Tf = \lambda f$, so T has no eigenvectors.

Lemma 6.3.7. *Hilbert–Schmidt Lemma*
Let $0 \neq \mathcal{H}$ be real Hilbert space, $T = T^* \in \mathcal{L}(\mathcal{H})$ compact, $Q_T(h) := \langle Th, h\rangle$. Then $\exists \lambda \in \mathbb{R}$, $h_* \in \mathcal{H}$ such that:

(i) $\|h_*\| = 1$;
(ii) $Th_* = \lambda h_*$;
(iii) $|\lambda| = \|T\|_{op} = \sup_{\|h\|=1} |Q_T(h)|$.

Proof. If $T = 0$, choose $\|h_*\| = 1$ and set $\lambda = 0$; then (i)–(iii) hold. Assume $T \neq 0$ and put $\eta := \|T\|_{\text{op}}$.
By Theorem 5.2.38, $\eta = \sup_{\|h\|=1} |Q_T(h)| > 0$. Choose $\|x_n\| = 1$ and $s_n \in \{-1, 1\}$ with $Q_{s_n T}(x_n) \to \eta$.
Passing to a subsequence, $s_n = s$; write $h_n := x_n$ and $A := sT$.
By Theorems 6.1.17 and 6.2.7, after passing to a further subsequence, $h_n \rightharpoonup h_*$ and $Ah_n \to Ah_*$ in norm.
Since $Q_A(h_n) - Q_A(h_*) = \langle Ah_n - Ah_*, h_n\rangle + \langle Ah_*, h_n - h_*\rangle \to 0$, $Q_A(h_*) = \eta$. Set $\lambda := s\eta$.

(i) By weak lower semicontinuity, $\|h_*\| \leq 1$.
Since $|Q_A(g)| \leq \eta \|g\|^2$ $\forall g \in \mathcal{H}$, $\eta = Q_A(h_*) \leq \eta \|h_*\|^2$. As $\eta > 0$, $\|h_*\| = 1$.
(ii) By (i), $R_A(h_*) = \eta$. For $\|h\| = 1$, $R_A(h) = Q_A(h) \leq |Q_T(h)| \leq \eta$, so h_* maximizes R_A.
For $h_0 \in \mathcal{H}$, $\phi(t) := R_A(h_* + th_0)$ has $\phi'(0) = 0$, hence $0 = 2\langle Ah_* - \eta h_*, h_0\rangle$.
Since h_0 is arbitrary, $Ah_* = \eta h_*$. Thus $Th_* = s\eta h_* = \lambda h_*$.
(iii) By the definition of λ and by Theorem 5.2.38, $|\lambda| = \eta = \|T\|_{\text{op}} = \sup_{\|h\|=1} |Q_T(h)|$.

Structural Commentary: By compactness, signed Q_T attains $\|T\|_{\text{op}}$; by stationarity, $Th_* = \lambda h_*$.
Key Insight: A compact self-adjoint norm maximizer is an eigenvector. □

Corollary 6.3.8. *Extremal Rayleigh Values Are Eigenvalues*
Let $\mathcal{H}$ be a nonzero real Hilbert space, let $T \in \mathcal{L}(\mathcal{H})$ be compact and self-adjoint, and set $Q_T(h) := \langle Th, h\rangle$, $\alpha := \inf_{\|h\|=1} Q_T(h)$, $\beta := \sup_{\|h\|=1} Q_T(h)$.

(i) If $\beta > 0$, then $\exists h_\beta \in \mathcal{H}$ such that $\|h_\beta\| = 1$ and $Th_\beta = \beta h_\beta$.
(ii) If $\alpha < 0$, then $\exists h_\alpha \in \mathcal{H}$ such that $\|h_\alpha\| = 1$ and $Th_\alpha = \alpha h_\alpha$.
(iii) If $\beta > 0$ and $(h_n) \subset \mathcal{H}$ satisfies $\|h_n\| = 1$, $Q_T(h_n) \to \beta$, then $\exists n_j \uparrow \infty$ and $h_\beta \in \mathcal{H}$ with $h_{n_j} \to h_\beta$ in norm, $\|h_\beta\| = 1$, and $Th_\beta = \beta h_\beta$.

Proof.

(i) Assume $\beta > 0$. Choose $\|u_n\| = 1$ with $Q_T(u_n) \to \beta$.
By Theorems 6.1.17 and 6.2.7, after passing to a subsequence, $u_n \rightharpoonup u_*$ and $Tu_n \to Tu_*$ in norm.
Since $Q_T(u_n) - Q_T(u_*) = \langle Tu_n - Tu_*, u_n\rangle + \langle Tu_*, u_n - u_*\rangle \to 0$, $Q_T(u_*) = \beta$.
By weak lower semicontinuity, $\|u_*\| \le 1$. As $Q_T(g) \le \beta \|g\|^2\ \forall g \in \mathcal{H}$, $\beta = Q_T(u_*) \le \beta \|u_*\|^2$; $\|u_*\| = 1$.
Thus u_* maximizes R_T. For $h \in \mathcal{H}$, $\phi(t) := R_T(u_* + th)$ has $\phi'(0) = 0$, so $0 = 2\langle Tu_* - \beta u_*, h\rangle$.
Hence $Tu_* = \beta u_*$; take $h_\beta := u_*$.

(ii) Assume $\alpha < 0$. Apply (i) to $-T$. Its upper Rayleigh value is $\beta_{-T} = \sup_{\|h\|=1} \langle -Th, h\rangle = -\alpha > 0$, so $\exists h_\alpha$ with $\|h_\alpha\| = 1$ and $(-T)h_\alpha = (-\alpha)h_\alpha$. Hence $Th_\alpha = \alpha h_\alpha$.

(iii) Assume $\beta > 0$, $\|h_n\| = 1$, $Q_T(h_n) \to \beta$. By compactness step in (i), $h_{n_j} \rightharpoonup h_\beta$, $Th_{n_j} \to Th_\beta$ in norm.
As in (i), $Q_T(h_\beta) = \beta$, $\|h_\beta\| = 1$, and $Th_\beta = \beta h_\beta$.
By Theorem 6.1.20, since $\|h_{n_j}\| = \|h_\beta\| = 1$, $h_{n_j} \to h_\beta$ in norm.

Structural Commentary: By compactness, attain β; by $-T$, attain α; by Radon–Riesz, $\rightharpoonup + \|\cdot\| \to 1 \Rightarrow \to$.
Key Insight: Compact self-adjoint Rayleigh extrema are eigenvalues. □

Theorem 6.3.9. *Hilbert–Schmidt Theorem for Compact Self-Adjoint Operators*
Let $\mathcal{H}$ be a real Hilbert space and let $K \in \mathcal{L}(\mathcal{H})$ be compact and self-adjoint.
If $\operatorname{rank} K = +\infty$, then $\exists$ orthonormal sequence $(\psi_k)_{k\ge1} \subset \mathcal{H}$ and nonzero scalars $(\lambda_k)_{k\ge1} \subset \mathbb{R}$ such that:

(i) $K\psi_k = \lambda_k\psi_k$ *for all* $k \ge 1$, $|\lambda_{k+1}| \le |\lambda_k|$, *and* $\lambda_k \to 0$;
(ii) $\operatorname{cspan}\{\psi_k : k \ge 1\} = (\ker K)^\perp$;
(iii) $Kh = \sum_{k\ge1} \lambda_k \langle h, \psi_k\rangle \psi_k$ *for every* $h \in \mathcal{H}$, *with norm convergence.*

If $\operatorname{rank} K < +\infty$, then:

(iv) $\exists m \ge 0$, *nonzero* $\lambda_1, \dots, \lambda_m \in \mathbb{R}$, *and an orthonormal basis* $(\psi_k)_{k=1}^m$ *of* $\operatorname{range} K = (\ker K)^\perp$ *such that* $K\psi_k = \lambda_k\psi_k$ *and* $Kh = \sum_{k=1}^m \lambda_k \langle h, \psi_k\rangle \psi_k$ *for every* $h \in \mathcal{H}$.

Proof.

(i) Put $M_0 := \{0\}$ and $H_0 := \mathcal{H}$. If $K|_{H_j} \ne 0$, then H_j is K-invariant and $K|_{H_j}$ is compact self-adjoint.
By Lemma 6.3.7, $\psi_{j+1} \in H_j$, $\lambda_{j+1} \ne 0$ with $\|\psi_{j+1}\| = 1$, $K\psi_{j+1} = \lambda_{j+1}\psi_{j+1}$, and $|\lambda_{j+1}| = \left\|K|_{H_j}\right\|_{\mathrm{op}}$.
Set $M_{j+1} := M_j \oplus \operatorname{span}\{\psi_{j+1}\}$, $H_{j+1} := M_{j+1}^\perp$. Since $K(M_{j+1}) \subseteq M_{j+1}$, $K = K^*$, $K(H_{j+1}) \subseteq H_{j+1}$.
Thus the construction continues while $K|_{H_j} \ne 0$, and $|\lambda_{j+1}| \le |\lambda_j|$.
If $\operatorname{rank} K = +\infty$, it cannot stop, since $K|_{H_m} = 0$ implies $\operatorname{range} K \subseteq M_m$.
If $\lambda_k \not\to 0$, then for some $\varepsilon > 0$, $\|K\psi_j - K\psi_k\|^2 = \lambda_j^2 + \lambda_k^2 \ge 2\varepsilon^2$ for infinitely many $j \ne k$, contradiction.

(ii) Set $M := \operatorname{cspan}\{\psi_k : k \ge 1\}$. Since $K(M) \subseteq M$ and $K = K^*$, $K(M^\perp) \subseteq M^\perp$.
If $K|_{M^\perp} \ne 0$, by Lemma 6.3.7 choose $\phi \in M^\perp$, $\mu \ne 0$ with $\|\phi\| = 1$ and $K\phi = \mu\phi$.
Then $\phi \in H_{k-1}\ \forall k$, so $|\mu| \le \left\|K|_{H_{k-1}}\right\|_{\mathrm{op}} = |\lambda_k| \to 0$, contradiction. Hence $M^\perp \subseteq \ker K$.
Conversely, $x \in \ker K$ implies $\langle x, \psi_k\rangle = \lambda_k^{-1}\langle Kx, \psi_k\rangle = 0\ \forall k$, so $\ker K \subseteq M^\perp$.
Thus $M^\perp = \ker K$ and $M = (\ker K)^\perp$ by Theorem 5.2.12.

(iii) For $h \in \mathcal{H}$, write $h = m + z$ with $m \in M = (\ker K)^\perp$ and $z \in \ker K$. By Theorem 5.2.22,
$m = \sum_{k\ge1} \langle h, \psi_k\rangle \psi_k$ in norm. By continuity of K, $Kh = Km = \sum_{k\ge1} \lambda_k \langle h, \psi_k\rangle \psi_k$ in norm.

(iv) If $\operatorname{rank} K < +\infty$, the construction stops at some $m \ge 0$; otherwise $\{\psi_k\} \subset \operatorname{range} K$ is an infinite orthonormal set in finite-dimensional $\operatorname{range} K$. Let $M_m := \operatorname{span}\{\psi_1, \dots, \psi_m\}$.
Since $K|_{M_m^\perp} = 0$, $\operatorname{range} K \subseteq M_m$; since $\psi_k = \lambda_k^{-1}K\psi_k$, $M_m \subseteq \operatorname{range} K$. Hence $M_m = \operatorname{range} K$.
Also $M_m^\perp = \ker K$ by argument in (ii), so $\operatorname{range} K = (\ker K)^\perp$. For $h = m_0 + z$ with $m_0 \in M_m$, $z \in \ker K$,
by the finite orthonormal expansion of m_0, $Kh = Km_0 = \sum_{k=1}^m \lambda_k \langle h, \psi_k\rangle \psi_k$.

Structural Commentary: Iterated extremal eigenvectors exhaust $(\ker K)^\perp$; Fourier expansion diagonalizes K.
Key Insight: Nonzero compact self-adjoint eigendirections are finite or countable, accumulating only at 0. □

Perspective 6.3.10. *Hilbert Operator Spectral Theory*
On a complex Hilbert space $\mathcal{H}$, normal $T \in \mathcal{L}(\mathcal{H})$ is diagonalized by spectral functional calculus.
By Theorem 6.3.9, compact self-adjoint $K \in \mathcal{L}(\mathcal{H})$ has nonzero eigenvalues (λ_k) and an orthonormal eigenfamily (ψ_k) spanning $(\ker K)^\perp$, so $\mathcal{H} = \ker K \oplus \operatorname{cspan}\{\psi_k\}$ and $Kh = \sum_k \lambda_k \langle h, \psi_k\rangle \psi_k\ \forall h \in \mathcal{H}$.
The sum is finite iff $\operatorname{rank} K < +\infty$; otherwise $\lambda_k \to 0$.

6.4 Measure Duality and Riesz Representation

On a compact Hausdorff space, Riesz–Markov turns positive functionals on $C(K)$ into Radon measures. The remaining step is order-theoretic: every bounded functional is the difference of two positive functionals, exactly as a signed measure is the difference of its Jordan variations.

Proposition 6.4.1. *Positive Functionals on $C(K)$ Are Continuous*
Let K be compact Hausdorff, $C(K) = C(K;\mathbb{R})$, $\|\cdot\|_\infty$, $L : C(K) \to \mathbb{R}$ linear with $L(C(K)_+) \subseteq [0, +\infty)$. Then:

(i) $L \in C(K)^*$.
(ii) $\|L\|_{op} = L(\mathbf{1}_K)$.

Proof.

(i) For $f \in C(K)$, $-\|f\|_\infty \mathbf{1}_K \le f \le \|f\|_\infty \mathbf{1}_K$; $-L(\mathbf{1}_K)\|f\|_\infty \le L(f) \le L(\mathbf{1}_K)\|f\|_\infty$; $|L(f)| \le L(\mathbf{1}_K)\|f\|_\infty$.
(ii) By (i), $\|L\|_{\mathrm{op}} \le L(\mathbf{1}_K)$. If $K = \varnothing$, both sides are 0; if $K \ne \varnothing$, then $\|\mathbf{1}_K\|_\infty = 1$, so $\|L\|_{\mathrm{op}} \ge L(\mathbf{1}_K)$.

Structural Commentary: $-\|f\|_\infty \mathbf{1}_K \le f \le \|f\|_\infty \mathbf{1}_K$ yields $|L(f)| \le L(\mathbf{1}_K)\,\|f\|_\infty$.
Key Insight: Positivity plus $\mathbf{1}_K$ controls $\|L\|_{\mathrm{op}}$. □

Proposition 6.4.2. *Positive Decomposition of Functionals on $C(K)$*
Let K be compact Hausdorff and let $L \in C(K)^*$. Then $\exists L_+, L_- \in C(K)^*$ positive such that:
(i) $L = L_+ - L_-$.
(ii) $\|L\|_{op} = L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K)$.

Proof.
(i) For $f \in C(K)_+$, set $P(f) := \sup_{0\le g\le f} L(g)$; boundedness of L implies $P(f) < +\infty$.
If $0 \le h \le f_1 + f_2$, then $h = (h \wedge f_1) + (h - h \wedge f_1)$, terms in $[0, f_1]$ and $[0, f_2]$; $P(f_1 + f_2) \le P(f_1) + P(f_2)$.
The reverse inequality follows from $0 \le g_i \le f_i \Rightarrow 0 \le g_1 + g_2 \le f_1 + f_2$; homogeneity is immediate.
Define $L_+(u) := P(u^+) - P(u^-)$ and $L_- := L_+ - L$. Then L_+ is positive linear, $L_-(f) = P(f) - L(f) \ge 0$ for $f \in C(K)_+$, and by Proposition 6.4.1(i), $L_\pm \in C(K)^*$.
(ii) Set $A := L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K)$. By (i) and Proposition 6.4.1(ii), $\|L\|_{\mathrm{op}} \le A$.
For $\varepsilon > 0$, choose $0 \le \varphi \le \mathbf{1}_K$ with $L(\varphi) > P(\mathbf{1}_K) - \varepsilon = L_+(\mathbf{1}_K) - \varepsilon$.
Since $\|2\varphi - \mathbf{1}_K\|_\infty \le 1$ and $L_-(\mathbf{1}_K) = L_+(\mathbf{1}_K) - L(\mathbf{1}_K)$, $\|L\|_{\mathrm{op}} \ge L(2\varphi - \mathbf{1}_K) > A - 2\varepsilon$. Let $\varepsilon \downarrow 0$.
Structural Commentary: $P(f) = \sup_{0\le g\le f} L(g)$ is the positive part on $C(K)_+$; $2\varphi - \mathbf{1}_K$ tests $\|L\|_{\mathrm{op}}$.
Key Insight: $L = L_+ - L_-$ and $\|L\|_{\mathrm{op}} = L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K)$. □

Theorem 6.4.3. *Riesz Representation for $C(K)^*$*
Let K be compact Hausdorff. Let $\mathcal{M}_{\mathrm{Rad}}(K)$ be the finite signed measures ν on K with $\nu^\pm$ Radon.
Let $\|\nu\|_{\mathrm{var}} := |\nu|(K)$, and define $(T\nu)(f) := \int_K f\,d\nu$ for $f \in C(K)$. Then:
(i) T is linear and well defined.
(ii) $\|T(\nu)\|_{op} = \|\nu\|_{\mathrm{var}}$ for all $\nu \in \mathcal{M}_{\mathrm{Rad}}(K)$.
(iii) T is bijective.

Proof.
(i) For $\nu \in \mathcal{M}_{\mathrm{Rad}}(K)$ and $f \in C(K)$, $|T(\nu)(f)| \le \int_K |f|\,d\,|\nu| \le \|f\|_\infty \|\nu\|_{\mathrm{var}}$.
Hence $T(\nu) \in C(K)^*$ and $\|T(\nu)\|_{\mathrm{op}} \le \|\nu\|_{\mathrm{var}}$. Linearity is signed-integral linearity.
(ii) Set $L := T(\nu)$. By Proposition 6.4.2(i), Theorem 5.2.14, $L = L_+ - L_-$, $L_\pm = T(\mu_\pm)$ for finite Radon $\mu_\pm$.
Since $T(\nu^+ + \mu_-) = T(\nu^- + \mu_+)$, by uniqueness in Theorem 5.2.14, $\nu^+ + \mu_- = \nu^- + \mu_+$.
By Theorem 3.1.6(ii),(iii), $\nu^\pm \le \mu_\pm$. Therefore $\|\nu\|_{\mathrm{var}} \le \mu_+(K) + \mu_-(K) = L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K) = \|L\|_{\mathrm{op}}$ by Proposition 6.4.2(ii).
Combine with (i).
(iii) Let $L \in C(K)^*$. Proposition 6.4.2(i), Theorem 5.2.14: $L = L_+ - L_-$, $L_\pm = T(\mu_\pm)$ for finite Radon $\mu_\pm$.
Set $\nu := \mu_+ - \mu_-$. By Theorem 3.1.6(ii),(iii), $\nu^\pm \le \mu_\pm$.
If $0 \le \theta \le \mu$ and μ is finite Radon, then θ is Radon since the same open and compact approximants work with no larger error.
Hence $\nu \in \mathcal{M}_{\mathrm{Rad}}(K)$, $T(\nu) = L$. Thus T is surjective; injectivity follows from (ii).
Structural Commentary: Positive Riesz–Markov: $L_\pm = T(\mu_\pm)$; Jordan minimality: $\|T(\nu)\|_{\mathrm{op}} = \|\nu\|_{\mathrm{var}}$.
Key Insight: $C(K)^* \cong \mathcal{M}_{\mathrm{Rad}}(K)$ isometrically via $L(f) = \int_K f\,d\nu$. □

Example 6.4.4. *Point Evaluation Is a Dirac Measure*
Let $K \neq \emptyset$ be compact Hausdorff and fix $x_0 \in K$. Define $L_{x_0} \in C(K)^*$ by $L_{x_0}(f) := f(x_0)$.
Then $L_{x_0} = T(\delta_{x_0})$, since $\forall f \in C(K)$, $T(\delta_{x_0})(f) = \int_K f\,d\delta_{x_0} = f(x_0)$. Hence $\|L_{x_0}\|_{\mathrm{op}} = \|\delta_{x_0}\|_{\mathrm{var}} = 1$.

Remark 6.4.5. *Interval Form and Stieltjes Integrators*
Let $a < b$ and let ν be a finite signed Radon measure on $[a, b]$, extended by 0 to $\mathbb{R}$. Set $g(x) := \nu((-\infty, x])$.
By Proposition 4.1.5 applied to $\nu^\pm$, $g \in BV(\mathbb{R})$ is right-continuous and $g(-\infty) = 0$.
For $a \le c \le d \le b$, $\nu((c, d]) = g(d) - g(c)$ and $\nu(\{a\}) = g(a) - g(a-)$ with $g(a-) = 0$.
Thus, in the Lebesgue–Stieltjes convention, $\forall f \in C([a, b])$, $T(\nu)(f) = \int_{[a,b]} f\,d\nu = \int_{[a,b]} f\,dg$.

Corollary 6.4.6. *Weak-* Compactness for Radon Measures*
Let K be compact Hausdorff and set $\mathcal{M} := \mathcal{M}_{\mathrm{Rad}}(K)$, the finite signed measures ν on K with $\nu^\pm$ Radon.
Let $\|\nu\|_{\mathrm{var}} := |\nu|(K)$ and let $\tau := \sigma(\mathcal{M}, C(K))$ be induced by $\langle \nu, f\rangle := \int_K f\,d\nu$.
If $A \subseteq \mathcal{M}$ is τ-closed and $\sup_{\nu\in A} \|\nu\|_{\mathrm{var}} < +\infty$, then A is τ-compact.

Proof. Let $(T\nu)(f) := \int_K f\,d\nu$. By Theorem 6.4.3, $T : \mathcal{M} \to C(K)^*$ is an isometric bijection identifying τ with $\sigma(C(K)^*, C(K))$.
Choose $R < +\infty$ with $A \subseteq R\overline{B}_{\mathcal{M}}$. Then $T(A)$ is weak-* closed and $T(A) \subseteq R\overline{B}_{C(K)^*}$.
By Theorem 6.1.6, $R\overline{B}_{C(K)^*}$ is weak-* compact; hence $T(A)$ and $A = T^{-1}(T(A))$ are weak-* compact.
Structural Commentary: Riesz isometry: reduce to weak-* closed $T(A) \subseteq R\overline{B}_{C(K)^*}$; Banach–Alaoglu: compact.
Key Insight: $T : \mathcal{M}_{\mathrm{Rad}}(K) \cong C(K)^*$ transfers Banach–Alaoglu compactness to Radon measures. □

Corollary 6.4.7. *Bounded Measures on Compact Metric Spaces Have Weak-* Subsequences*
Let K be compact metric and let $\mu_n \ge 0$ be Borel measures on K with $M := \sup_n \mu_n(K) < +\infty$.
Then $\exists n_k \uparrow \infty$ and a Radon measure $\mu \ge 0$ on K such that $\int_K f\,d\mu_{n_k} \to \int_K f\,d\mu$ for all $f \in C(K)$.

Proof. Set $T_n(f) := \int_K f\,d\mu_n$. By Proposition 6.4.1(ii), $T_n \in C(K)^*$, $T_n \ge 0$, and $\|T_n\|_{\mathrm{op}} = \mu_n(K) \le M$.
Since $C(K)$ is separable by Theorem 9.2.10, by Theorem 6.1.8, $\exists n_k \uparrow \infty$ and $T \in C(K)^*$ with $T_{n_k}(f) \to T(f)$ for all $f \in C(K)$.
Positivity passes to T, so by Theorem 5.2.14, $T(f) = \int_K f\,d\mu$ for some Radon $\mu \ge 0$.
Structural Commentary: Bound (T_n) in $C(K)^*$; use separability and Helly; represent the positive limit.
Key Insight: Bounded positive measures on compact metric K are weak-* sequentially compact on $C(K)$. □

Part XV
Probability Foundations and Distributions

1 Probability Spaces and Random Variables

1.1 Probability Spaces and Event Algebra

Probability assigns numbers to experimental outcomes; measure theory specifies which subsets are events.

Definition 1.1.1. *Sample Spaces, Events, and Probability Spaces*

(i) A *sample space* is a set Ω of possible outcomes.
(ii) An *outcome* or *realization* is a point $\omega \in \Omega$.
(iii) An *event* is a subset $A \subseteq \Omega$; the admissible events form a *σ-algebra* $\mathcal{F} \subseteq \mathcal{P}(\Omega)$.
(iv) A *probability measure* on $(\Omega, \mathcal{F})$ is a measure $\mathbb{P} : \mathcal{F} \to [0,1]$ with $\mathbb{P}(\Omega) = 1$.
(v) The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Key Insight: Probability is countably additive size normalized so that the whole sample space has mass 1.

Method 1.1.2. *Sample Space Construction Method*
Choose Ω as the set of complete outcomes and encode the target condition as an event $A \subseteq \Omega$:

(i) *Finite product method*: for m discrete trials with state set S, use $\Omega = S^m$ and coordinate constraints.
For two coin tosses, $S = \{H, T\}$, $\Omega = S^2$, and first toss heads gives $A = \{\omega \in \Omega : \omega_1 = H\} = \{HH, HT\}$.
(ii) *Interval method*: for real-valued measurements, use $\Omega = \mathbb{R}$ and translate inequalities into intervals.
$10 < \omega \leq 23$ gives $A = (10, 23]$.
(iii) *Cylinder method*: for infinite sequences with state set S, use $\Omega = S^{\mathbb{N}}$ and constrain finitely many coordinates; for coin tosses, $S = \{H, T\}$ and first head on toss 3 gives $A = \{\omega \in \Omega : \omega_1 = T, \omega_2 = T, \omega_3 = H\}$.

Example 1.1.3. *Introductory Sample Spaces and Events*

(i) If a commute passes three traffic lights, each with outcome stop s or continue c, then $\Omega = \{c, s\}^3$. The event that the first light is a stop is $\{sss, ssc, scc, scs\}$.
(ii) If a print queue length is modeled as random, one possible sample space is $\Omega = \{0, 1, 2, \ldots\}$; with a known capacity N, one may instead use $\Omega = \{0, 1, \ldots, N\}$.
(iii) If the measured waiting time between events is nonnegative, a natural sample space is $\Omega = [0, \infty)$.

Key Insight: Choose Ω at the event resolution needed.

Definition 1.1.4. *Event Algebra, Indicators, and Monotone Events*

(i) For $A \in \mathcal{F}$, the *complement* is $A^c := \Omega \setminus A$.
(ii) For $A, B \in \mathcal{F}$, the *union* and *intersection* are $A \cup B$ and $A \cap B$.
(iii) For $A, B \in \mathcal{F}$, the *difference* is $A \setminus B := A \cap B^c$.
(iv) *Commutativity*: $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
(v) *Associativity*: $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.
(vi) *Distributivity*: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
(vii) *De Morgan laws*: $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.
(viii) Events $(A_i)_{i \in I}$ are *disjoint* or *mutually exclusive* if $A_i \cap A_j = \emptyset$ for $i \neq j$.
(ix) Events $(A_i)_{i \in I}$ form a *partition* of Ω if $A_i \cap A_j = \emptyset$ for $i \neq j$ and $\bigcup_{i \in I} A_i = \Omega$.
(x) The *indicator* of $A \in \mathcal{F}$ is $\mathbf{1}_A : \Omega \to \{0, 1\}$, where $\mathbf{1}_A(\omega) = 1$ for $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ for $\omega \notin A$.
(xi) A sequence (A_n) is *increasing* to A, written $A_n \uparrow A$, if $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_{n \geq 1} A_n$.
(xii) A sequence (A_n) is *decreasing* to A, written $A_n \downarrow A$, if $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_{n \geq 1} A_n$.

Proposition 1.1.5. *Indicator Algebra for Unions and Intersections*
For events $A, B \in \mathcal{F}$: $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B = \min\{\mathbf{1}_A, \mathbf{1}_B\}$, and $\mathbf{1}_{A \cup B} = \max\{\mathbf{1}_A, \mathbf{1}_B\}$.

Proof. Fix $\omega \in \Omega$ and set $a := \mathbf{1}_A(\omega)$, $b := \mathbf{1}_B(\omega)$. Then $a, b \in \{0, 1\}$, $\mathbf{1}_{A \cap B}(\omega) = ab = \min\{a, b\}$, and $\mathbf{1}_{A \cup B}(\omega) = \max\{a, b\}$. Since ω was arbitrary, the identities hold pointwise.
Structural Commentary: Boolean operations on events become arithmetic operations on 0-1 variables.
Key Insight: Indicators let event algebra be manipulated inside sums and expectations. □

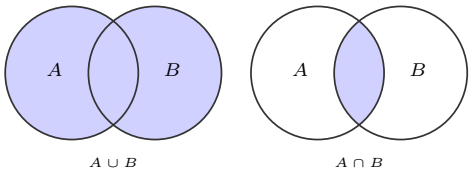

Figure 224: Set operations become event operations in a probability space.

Example 1.1.6. *Endpoint Sensitivity in Monotone Limits*
Let $\Omega = \mathbb{R}$.

(i) If $A_i = [0, \frac{1}{i}]$, then $\bigcup_{i=1}^{\infty} A_i = [0, 1]$ and $\bigcap_{i=1}^{\infty} A_i = \{0\}$.
(ii) If $A_i = (0, \frac{1}{i})$, then $\bigcup_{i=1}^{\infty} A_i = (0, 1)$ and $\bigcap_{i=1}^{\infty} A_i = \emptyset$.

Key Insight: Monotone set limits remember whether boundary points are included.

Proposition 1.1.7. *First Consequences of the Probability Axioms*
For events $A, B \in \mathcal{F}$:

(i) Null event: $\mathbb{P}(\emptyset) = 0$.
(ii) Monotonicity: $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$.
(iii) Bounds: $0 \leq \mathbb{P}(A) \leq 1$.
(iv) Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
(v) Disjoint additivity: $A \cap B = \emptyset \Rightarrow \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
(vi) Two-set inclusion-exclusion: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof.

(i) Countable additivity on $\Omega = \Omega \sqcup \emptyset \sqcup \emptyset \sqcup \cdots$ gives $1 = 1 + \sum_{n \geq 1} \mathbb{P}(\emptyset)$, hence $\mathbb{P}(\emptyset) = 0$.
(ii) If $A \subseteq B$, then $B = A \sqcup (B \setminus A)$, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.
(iii) Since $\emptyset \subseteq A \subseteq \Omega$, (i) and (ii) give $0 = \mathbb{P}(\emptyset) \leq \mathbb{P}(A) \leq \mathbb{P}(\Omega) = 1$.
(iv) Since $\Omega = A \sqcup A^c$, $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$, so $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
(v) If $A \cap B = \emptyset$, then $A \cup B = A \sqcup B$, so $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
(vi) Decompose $A \cup B = (A \setminus B) \sqcup (A \cap B) \sqcup (B \setminus A)$, $A = (A \setminus B) \sqcup (A \cap B)$, and $B = (A \cap B) \sqcup (B \setminus A)$. Hence $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Structural Commentary: The algebraic identities are set partitions in disguise.
Key Insight: Split overlapping sets into disjoint pieces before adding probabilities. □

Example 1.1.8. *Two Coin Tosses and Inclusion-Exclusion*
Let $\Omega = \{HH, HT, TH, TT\}$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$ for all $\omega \in \Omega$. Set $H_1 = \{HH, HT\}$ and $H_2 = \{HH, TH\}$. Since $H_1 \cap H_2 = \{HH\}$, $\mathbb{P}(H_1 \cup H_2) = \mathbb{P}(H_1) + \mathbb{P}(H_2) - \mathbb{P}(H_1 \cap H_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$.

Proposition 1.1.9. *Three-Set Inclusion-Exclusion and Bonferroni*
For events $A, B, C \in \mathcal{F}$:

(i) $\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$.
(ii) $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$.

Proof.

(i) Let $U = A \cup B$. Then $A \cup B \cup C = U \cup C$ and $U \cap C = (A \cap C) \cup (B \cap C)$.
By two-set inclusion-exclusion, $\mathbb{P}(U \cup C) = \mathbb{P}(U) + \mathbb{P}(C) - \mathbb{P}(U \cap C)$, $\mathbb{P}(U) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(U \cap C) = \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) - \mathbb{P}(A \cap B \cap C)$. Substitution gives (i).
(ii) Since $\mathbb{P}(A \cup B) \leq 1$ and $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, rearrange.

Structural Commentary: Inclusion-exclusion alternates between overcounting and correcting overlaps.
Key Insight: Bonferroni inequalities are one-sided forms of inclusion-exclusion. □

Theorem 1.1.10. *Continuity of Probabilities*

(i) Continuity from below: if $A_n \uparrow A$, then $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.
(ii) Continuity from above: if $A_n \downarrow A$, then $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.

Proof.

(i) Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then (B_n) is disjoint, $A_n = \bigcup_{j=1}^{n} B_j$, and $A = \bigcup_{j=1}^{\infty} B_j$, so countable additivity gives $\mathbb{P}(A_n) = \sum_{j=1}^{n} \mathbb{P}(B_j) \to \sum_{j=1}^{\infty} \mathbb{P}(B_j) = \mathbb{P}(A)$.
(ii) Set $C_n = A_1 \setminus A_n$. Then $C_n \uparrow A_1 \setminus A$, so (i) gives $\mathbb{P}(C_n) \to \mathbb{P}(A_1 \setminus A)$; since $A_1 = A_n \sqcup C_n$ and $A_1 = A \sqcup (A_1 \setminus A)$, finite additivity gives $\mathbb{P}(A_n) = \mathbb{P}(A_1) - \mathbb{P}(C_n) \to \mathbb{P}(A_1) - \mathbb{P}(A_1 \setminus A) = \mathbb{P}(A)$.

Structural Commentary: Monotone limits convert countable additivity into limit rules.
Key Insight: Increasing uses increments; decreasing uses complements. □

Definition 1.1.11. *Events Occurring Infinitely Often and Eventually Always*
For events $(A_n)_{n \geq 1}$:

(i) *Tail union*: $B_n := \bigcup_{i=n}^{\infty} A_i$.
(ii) *Tail intersection*: $C_n := \bigcap_{i=n}^{\infty} A_i$.
(iii) *Limsup*: $\limsup_{n \to \infty} A_n := \bigcap_{n=1}^{\infty} B_n = \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} A_i$.
(iv) *Liminf*: $\liminf_{n \to \infty} A_n := \bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} A_i$.
(v) *Infinitely often*: $\omega \in \limsup_n A_n$ iff $\omega \in A_n$ for infinitely many n.
(vi) *Eventually always*: $\omega \in \liminf_n A_n$ iff $\omega \in A_n$ for all but finitely many n.

Proposition 1.1.12. *Tail Monotonicity*
For events $(A_n)_{n \geq 1}$ with tail union B_n and tail intersection C_n:

(i) Tail unions: $B_{n+1} \subseteq B_n$ for all n, hence $B_n \downarrow \bigcap_{n \geq 1} B_n$.
(ii) Tail intersections: $C_n \subseteq C_{n+1}$ for all n, hence $C_n \uparrow \bigcup_{n \geq 1} C_n$.

Proof.

(i) Since $\{i : i \geq n+1\} \subseteq \{i : i \geq n\}$, $B_{n+1} = \bigcup_{i=n+1}^{\infty} A_i \subseteq \bigcup_{i=n}^{\infty} A_i = B_n$.

(ii) Since $\{i : i \geq n+1\} \subseteq \{i : i \geq n\}$, $C_n = \bigcap_{i=n}^{\infty} A_i \subseteq \bigcap_{i=n+1}^{\infty} A_i = C_{n+1}$.

Structural Commentary: Tail shifts turn recurrence and permanence into monotone limits.
Key Insight: Tail unions track recurrence; tail intersections track permanence. □

Proposition 1.1.13. *Union Bound and Countable Intersections of Almost-Sure Events*
For events $(A_n)_{n\geq 1}$:

(i) Union bound: $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

(ii) Countable almost-sure intersection: if $\mathbb{P}(A_n) = 1$ for all n, then $\mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = 1$.

Proof.

(i) Set $B_n := A_n \setminus \bigcup_{j<n} A_j$. Then (B_n) is disjoint, $\bigcup_n B_n = \bigcup_n A_n$, and $B_n \subseteq A_n$.
Hence $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(B_n) \leq \sum_n \mathbb{P}(A_n)$.

(ii) If $\mathbb{P}(A_n) = 1$ for all n, then $\mathbb{P}(A_n^c) = 0$ for all n.
By (i), $\mathbb{P}((\bigcap_n A_n)^c) = \mathbb{P}(\bigcup_n A_n^c) \leq \sum_n \mathbb{P}(A_n^c) = 0$, so $\mathbb{P}(\bigcap_n A_n) = 1$.

Structural Commentary: Countable subadditivity is countable additivity after disjointization.
Key Insight: Almost sure statements survive countable intersections. □

1.2 Finite Probability, Counting, and Independence

Definition 1.2.1. *Uniform Probability on a Countable Space*
Let Ω be countable.

(i) A probability measure $\mathbb{P}$ on $(\Omega, \mathcal{P}(\Omega))$ is *uniform* if $\exists c \in [0,1]$ such that $\mathbb{P}(\{\omega\}) = c$ for all $\omega \in \Omega$.
(ii) The constant c is the *common atom mass*.

Proposition 1.2.2. *No Uniform Probability on a Countably Infinite Set*
Let $\Omega = \{0, 1, 2, \ldots\}$. No uniform probability measure exists on $(\Omega, \mathcal{P}(\Omega))$.

Proof. Assume uniform $\mathbb{P}$ exists, and let c be its common atom mass.
Since $\Omega = \bigsqcup_{n=0}^{\infty} \{n\}$, countable additivity gives $1 = \mathbb{P}(\Omega) = \sum_{n=0}^{\infty} \mathbb{P}(\{n\}) = \sum_{n=0}^{\infty} c$.
If $c = 0$, the sum is 0; if $c > 0$, the sum diverges. Both contradict $1 = \mathbb{P}(\Omega)$.
Structural Commentary: Countable additivity forces total mass to equal the sum of the atom masses.
Key Insight: Equal mass on countably many atoms is incompatible with total mass 1. □

Method 1.2.3. *Finite Uniform Probability by Counting*
Let $0 < |\Omega| < \infty$ and assume the *uniform model* $\mathbb{P}(\{\omega\}) = |\Omega|^{-1}$ for all $\omega \in \Omega$.
For $A \subseteq \Omega$, $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) = \frac{|A|}{|\Omega|}$, so probability reduces to counting.

Method 1.2.4. *Counting Workflow for Uniform Finite Probability*
For a finite uniform sample space Ω and an event $A \subseteq \Omega$:

(i) choose complete outcomes so that each $\omega \in \Omega$ is equally likely;
(ii) count $|\Omega|$; and count $|A|$, usually by products, complements, or disjoint cases;
(iii) conclude $\mathbb{P}(A) = |A| \,/\, |\Omega|$.

Key Insight: The hard step is often choosing equally likely outcomes, not applying the ratio.

Example 1.2.5. *Simpson's Paradox in Two Urn Games*
Let $p(U)$ be the red probability in urn U, and let B_i, W_i be the black and white urns in game i.
Within games, $p(B_1) = 5/11 > 3/7 = p(W_1)$ and $p(B_2) = 6/9 > 9/14 = p(W_2)$.
After pooling by color into B and W, $p(B) = 11/20 < 12/21 = p(W)$, so the aggregate winner is white.
Key Insight: A within-group advantage can reverse after aggregation when the group sizes change.

Example 1.2.6. *Complement Counting in Uniform Models*

(i) For plates $\Omega = L^3 \times D^3$ with $|L| = 26$ and $|D| = 10$, $|\Omega| = 26^3 \cdot 10^3$. If E is the event of no repeated letters and no repeated digits, then $|E| = (26)_3(10)_3$, so $\mathbb{P}(E) = \frac{(26)_3(10)_3}{26^3 \cdot 10^3}$.
(ii) For $1 \leq n \leq 365$ iid uniform birthdays, take $\Omega = \{1, \ldots, 365\}^n$. If C is the event of a shared birthday, then $C^c = D_n(\{1, \ldots, 365\})$, so $\mathbb{P}(C) = 1 - \frac{(365)_n}{365^n}$; for $n = 23$, $\mathbb{P}(C) > 1/2$.

Key Insight: "At least one collision" is often easier to count by subtracting the no-collision event.

Definition 1.2.7. *Independent Events*

(i) Events $A, B \in \mathcal{F}$ are *independent*, written $A \perp\!\!\!\perp B$, if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
(ii) A family $(A_i)_{i \in I} \subseteq \mathcal{F}$ is *pairwise independent* if A_i and A_j are independent whenever $i \neq j$.
(iii) A family $(A_i)_{i \in I} \subseteq \mathcal{F}$ is *mutually independent* if $\mathbb{P}(\bigcap_{i \in J} A_i) = \prod_{i \in J} \mathbb{P}(A_i)$ for every finite $J \subseteq I$.

Key Insight: Mutual independence means multiplicativity for every finite subfamily, not just pairs.

Example 1.2.8. *Pairwise Independence Need Not Be Mutual*
Let $\Omega = \{HH, HT, TH, TT\}$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}\ \forall \omega \in \Omega$.
Set $A = \{HH, HT\}$, $B = \{HH, TH\}$, $C = \{HT, TH\}$.
Then $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$, $\mathbb{P}(A \cap B) = \mathbb{P}(A \cap C) = \mathbb{P}(B \cap C) = \frac{1}{4}$, so A, B, C pairwise independent.
But $A \cap B \cap C = \emptyset$, so $\mathbb{P}(A \cap B \cap C) = 0 \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$.
Key Insight: Pairwise checks do not control higher-order intersections.

Proposition 1.2.9. *Basic Independence Tests*
Let $A, B \in \mathcal{F}$.

(i) If $A \cap B = \emptyset$ and $\mathbb{P}(A)\mathbb{P}(B) > 0$, then $A \not\perp\!\!\!\perp B$.
(ii) If $A \perp\!\!\!\perp B$, then $A^c \perp\!\!\!\perp B$, $A \perp\!\!\!\perp B^c$, and $A^c \perp\!\!\!\perp B^c$.
(iii) If $\mathbb{P}(A) \in \{0, 1\}$, then $A \perp\!\!\!\perp C$ for every $C \in \mathcal{F}$.
(iv) If $A \perp\!\!\!\perp A$, then $\mathbb{P}(A) \in \{0, 1\}$.

Proof.

(i) Since $A \cap B = \emptyset$, $\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B)$.
(ii) If $A \perp\!\!\!\perp B$, then $\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = (1 - \mathbb{P}(A))\mathbb{P}(B) = \mathbb{P}(A^c)\mathbb{P}(B)$
and $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A)\mathbb{P}(B^c)$.
Applying the first identity to $A \perp\!\!\!\perp B^c$ gives $A^c \perp\!\!\!\perp B^c$.
(iii) Let $C \in \mathcal{F}$. If $\mathbb{P}(A) = 0$, then $0 \leq \mathbb{P}(A \cap C) \leq \mathbb{P}(A) = 0$, so $\mathbb{P}(A \cap C) = 0 = \mathbb{P}(A)\mathbb{P}(C)$.
If $\mathbb{P}(A) = 1$, then $\mathbb{P}(A^c) = 0$, hence $A^c \perp\!\!\!\perp C$ by the previous case and $A \perp\!\!\!\perp C$ by (ii).
(iv) Let $p = \mathbb{P}(A)$. From $A \perp\!\!\!\perp A$, $p = \mathbb{P}(A \cap A) = p^2$, hence $p \in \{0, 1\}$.

Structural Commentary: Independence is probabilistic, not visual; disjointness usually means strong dependence.
Key Insight: Complements preserve independence because B splits into $A \cap B$ and $A^c \cap B$. □

Method 1.2.10. *Independence, Complement, and Decomposition Methods*

(i) *Verify independence*: compute $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(A \cap B)$; conclude $A \perp\!\!\!\perp B$ iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
(ii) *Use complements*: if A^c is simpler, compute $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$.
If $A^c = \bigcap_{j=1}^{n} C_j$ with independent C_j, then $\mathbb{P}(A) = 1 - \prod_{j=1}^{n} \mathbb{P}(C_j)$.
(iii) *Disjoint decomposition*: write $E = \bigsqcup_j A_j$ with disjoint A_j, then $\mathbb{P}(E) = \sum_j \mathbb{P}(A_j)$.
(iv) *Geometric summation*: if $\mathbb{P}(A_j) = ar^{j-1}$ for $j \geq 1$ and $|r| < 1$, then $\mathbb{P}(E) = \sum_{j \geq 1} ar^{j-1} = \frac{a}{1-r}$.

Example 1.2.11. *Independence, Complements, and First Success*

(i) *Independence by counting*: for a fair die, $\Omega = \{1, \ldots, 6\}$, $A = \{2, 4, 6\}$, and $B = \{1, 2, 3, 4\}$.
Then $\mathbb{P}(A \cap B) = \frac{2}{6} = (\frac{3}{6})(\frac{4}{6}) = \mathbb{P}(A)\mathbb{P}(B)$, hence $A \perp\!\!\!\perp B$.
(ii) *Complement method*: for independent fair tosses $X_1, \ldots, X_{10}$, let $A = \{\exists j \leq 10 : X_j = H\}$
and $T_j = \{X_j = T\}$. Then $A^c = \bigcap_{j=1}^{10} T_j$ and $\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \prod_{j=1}^{10} \mathbb{P}(T_j) = 1 - (\frac{1}{2})^{10}$.
(iii) *First-success decomposition*: let $p_1 = \frac{1}{3}$, $p_2 = \frac{1}{4}$.
Let A_j be player 1's first success on turn j before player 2 succeeds.
Then $E = \bigsqcup_{j \geq 1} A_j$, $\mathbb{P}(A_j) = ((1 - p_1)(1 - p_2))^{j-1} p_1 = (\frac{1}{2})^{j-1}/3$, and $\mathbb{P}(E) = \sum_{j \geq 1} \mathbb{P}(A_j) = \frac{2}{3}$.

Example 1.2.12. *Reliability and Match Probabilities*
Let A_i be the event that component i works, and assume all relevant events are mutually independent.

(i) Series: if $\mathbb{P}(A_i^c) = p$ for $1 \leq i \leq n$ and $W_s = \bigcap_{i=1}^{n} A_i$, then $\mathbb{P}(W_s) = (1 - p)^n$ and $\mathbb{P}(W_s^c) = 1 - (1 - p)^n$.
(ii) Parallel: if $W_p = \bigcup_{i=1}^{n} A_i$, then $\mathbb{P}(W_p^c) = \mathbb{P}(\bigcap_{i=1}^{n} A_i^c) = p^n$.
(iii) DNA match: let $N_j \in \{A, C, G, T\}$ with $\mathbb{P}(N_j = x) = p_{xj}$ for $j = 1, 2$.
If $N_1 \perp\!\!\!\perp N_2$, then $\mathbb{P}(N_1 = N_2) = \sum_{x \in \{A,C,G,T\}} p_{x1} p_{x2}$.

Key Insight: Independence multiplies intersections; complements handle failures; matches sum disjoint cases.

1.3 Random Variables, Laws, and Distribution Functions

Definition 1.3.1. *Random Variables, Laws, and Distribution Functions*

(i) A *real-valued random variable* on $(\Omega, \mathcal{F}, \mathbb{P})$ is map $X : \Omega \to \mathbb{R}$ s.t. $\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}\ \forall B \in \mathcal{B}(\mathbb{R})$.
(ii) Its *law* or *distribution* is $\mathcal{L}(X) : \mathcal{B}(\mathbb{R}) \to [0, 1]$ given by $\mathcal{L}(X)(B) := \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B)$.
(iii) Its *cumulative distribution function* is $F_X : \mathbb{R} \to [0, 1]$ given by $F_X(x) := \mathbb{P}(X \leq x) = \mathcal{L}(X)((-\infty, x])$.

Definition 1.3.2. *Distribution Functions and Densities*

(i) A *distribution function* is $F : \mathbb{R} \to [0, 1]$ such that $x \leq y \Rightarrow F(x) \leq F(y)$, $F(x) = \lim_{t \downarrow x} F(t)$ for all $x \in \mathbb{R}$, $\lim_{x \to -\infty} F(x) = 0$, and $\lim_{x \to \infty} F(x) = 1$.
(ii) A *density* for F is an integrable $g : \mathbb{R} \to [0, \infty)$ with $\int_{-\infty}^{\infty} g(x)\,dx = 1$ and $F(t) = \int_{-\infty}^{t} g(x)\,dx\ \forall t \in \mathbb{R}$.
If g is continuous, then $F'(t) = g(t)$.

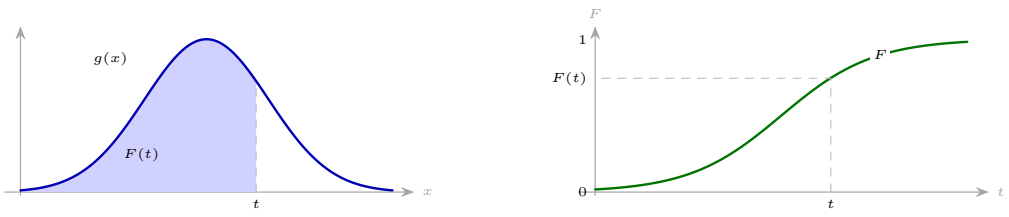

Figure 225: A normal density accumulates area into the distribution function $\Phi(t) = \int_{-\infty}^{t} g(x)\,dx$.

Definition 1.3.3. *Discrete and Continuous Random Variables*

(i) X is *discrete* if $X(\Omega) \subseteq S$ for some finite or countable $S = \{x_1, x_2, \ldots\} \subseteq \mathbb{R}$.
(ii) The *probability mass function* of discrete X is $f_X : S \to [0,1]$, $f_X(x) := \mathbb{P}(X = x)$, $\sum_{x \in S} f_X(x) = 1$.
(iii) The *discrete CDF* is $F_X(x) = \mathbb{P}(X \le x) = \sum_{x_i \le x} f_X(x_i)$.
(iv) X is *continuous* if its law has a density $f_X : \mathbb{R} \to [0, \infty)$.
(v) The *continuous probabilities* satisfy $\mathbb{P}(a < X < b) = \int_a^b f_X(t)\,dt$ and $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$.
(vi) The *point-mass distinction*: for continuous X, $\mathbb{P}(X = x) = 0$ for every fixed x, so $f_X(x) \neq \mathbb{P}(X = x)$.

A law may be mixed. It can have atoms and a continuous density on the remaining part.

Remark 1.3.4. *Mixed Laws Are Natural*
For a component lifetime X, one may have $\mathbb{P}(X = 0) = p > 0$ and, conditional on $X > 0$, a density g on $(0, \infty)$.
Then $\mathcal{L}(X)$ has an atom plus a continuous part, so it is mixed.

Example 1.3.5. *Counting Heads in Two Tosses*
Let $\Omega = \{H, T\}^2$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$, and set $X(\omega) = |\{j \in \{1,2\} : \omega_j = H\}|$.
Then $f_X(k) = \mathbb{P}(X = k) = \binom{2}{k} 2^{-2}$ for $k \in \{0,1,2\}$, so $(f_X(0), f_X(1), f_X(2)) = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$.
Hence $F_X(x) = 0$ for $x < 0$, $F_X(x) = \frac{1}{4}$ for $0 \le x < 1$, $F_X(x) = \frac{3}{4}$ for $1 \le x < 2$, $F_X(x) = 1$ for $x \ge 2$.
Key Insight: A CDF is defined for every real x, even when a discrete random variable takes only a few values.

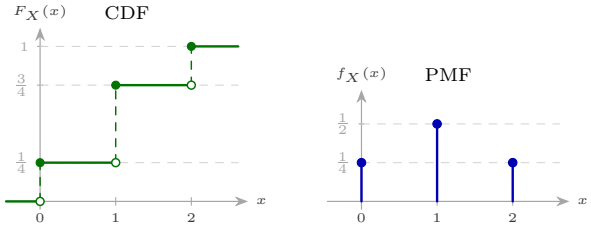

Figure 226: CDF and PMF for the number of heads in two independent fair coin tosses.

Proposition 1.3.6. *CDF Probability Calculus*
Let X have CDF F and define $F(x^-) := \lim_{y \uparrow x} F(y)$.

(i) Point mass: $\mathbb{P}(X = x) = F(x) - F(x^-)$.
(ii) Half-open interval: if $x < y$, then $\mathbb{P}(x < X \le y) = F(y) - F(x)$.
(iii) Right tail: $\mathbb{P}(X > x) = 1 - F(x)$.
(iv) Continuous endpoints: if X is continuous and $a < b$, then $\mathbb{P}(a < X < b) = \mathbb{P}(a \le X < b) = \mathbb{P}(a < X \le b) = \mathbb{P}(a \le X \le b)$.

Proof.

(i) Let $E_n = \{x - \frac{1}{n} < X \le x\}$. Then $E_n \downarrow \{X = x\}$ and $\mathbb{P}(E_n) = F(x) - F(x - \frac{1}{n})$.
Hence $\mathbb{P}(X = x) = \lim_n \mathbb{P}(E_n) = F(x) - F(x^-)$.
(ii) Since $\{X \le y\} = \{X \le x\} \sqcup \{x < X \le y\}$, $F(y) = F(x) + \mathbb{P}(x < X \le y)$.
(iii) Since $\{X > x\} = \{X \le x\}^c$, $\mathbb{P}(X > x) = 1 - \mathbb{P}(X \le x) = 1 - F(x)$.
(iv) If X is continuous, then $\mathbb{P}(X = t) = 0$ for all $t \in \mathbb{R}$. For $I_1 = (a,b)$, $I_2 = [a,b)$, $I_3 = (a,b]$, and $I_4 = [a,b]$, each $X^{-1}(I_j) \backslash X^{-1}(I_1) \subseteq \{X = a\} \cup \{X = b\}$, hence all four probabilities are equal.

Structural Commentary: CDF calculus subtracts nested lower-level events.
Key Insight: CDF jumps are point masses; continuous laws have none. □

Definition 1.3.7. *Survival and Hazard Functions*
Let $T \ge 0$ have CDF F and density f, and set $D := \{t \ge 0 : 1 - F(t) > 0\}$.

(i) *Survival Function*: $S : [0, \infty) \to [0,1]$, $S(t) := \mathbb{P}(T > t) = 1 - F(t)$.
(ii) *Hazard Function*: $h : D \to [0, \infty)$, $h(t) := \frac{f(t)}{1 - F(t)} = \frac{f(t)}{S(t)}$.

Structural Commentary: On intervals where S is differentiable and positive, $h(t) = -\frac{d}{dt} \log S(t)$.

Proposition 1.3.8. *Hazard Determines the Density*
Let $T \ge 0$ have survival $S(t) := \mathbb{P}(T > t)$ with $S \in C^1([0, \infty))$, $S(t) > 0$, $S(0) = 1$, density $f(t) := -S'(t)$, and hazard $h(t) := \frac{f(t)}{S(t)}$.
Put $H(t) := \int_0^t h(s)\,ds$. Then, for $t \ge 0$:

(i) Survival: $S(t) = e^{-H(t)}$.

(ii) Density: $f(t) = h(t)e^{-H(t)}$.

Proof. Fix $t \geq 0$.

(i) *Survival*: Since $(\log S)' = S'/S = -h$, $\log S(t) - \log S(0) = -H(t)$. Since $S(0) = 1$, $S(t) = e^{-H(t)}$.
(ii) *Density*: Since $h = f/S$, $f(t) = h(t)S(t) = h(t)e^{-H(t)}$.

Structural Commentary: $h = -S'/S = -(\log S)'$, so H is the cumulative log-decay of survival.
Key Insight: The hazard determines the law through $S = e^{-H}$ and $f = hS$. □

Example 1.3.9. *Constant and Power Hazards*
For $t \geq 0$:

(i) *Exponential*: If $F(t) = 1 - e^{-\lambda t}$ with $\lambda > 0$, then $S(t) = e^{-\lambda t}$, $f(t) = \lambda e^{-\lambda t}$, and $h(t) = \lambda$.
(ii) *Weibull-Type*: If $F(t) = 1 - e^{-\alpha t^\beta}$ with $\alpha, \beta > 0$, then $S(t) = e^{-\alpha t^\beta}$, $f(t) = \alpha\beta t^{\beta-1} e^{-\alpha t^\beta}$, $h(t) = \alpha\beta t^{\beta-1}$.

Key Insight: Power hazards increase for $\beta > 1$, decrease for $0 < \beta < 1$, and are constant for $\beta = 1$.

Definition 1.3.10. *Quantiles and Equality in Distribution*

(i) For a CDF F, the *quantile function* is $F^{-1}(q) := \inf\{x \in \mathbb{R} : F(x) \geq q\}$, $q \in (0, 1)$.
(ii) If F is continuous and strictly increasing, then F^{-1} is the ordinary inverse: $F(F^{-1}(q)) = q$ for $q \in (0, 1)$ and $F^{-1}(F(x)) = x$ for $x \in \mathbb{R}$.
(iii) The *pth quantile* is $x_p = F^{-1}(p)$; under (ii), $F(x_p) = p$ uniquely. Here $x_{\frac{1}{2}}$ is the median and $x_{\frac{1}{4}}, x_{\frac{3}{4}}$ are the lower and upper quartiles.
(iv) Random variables X, Y are *equal in distribution*, written $X \overset{\text{dist}}{=} Y$, iff $F_X = F_Y$ on $\mathbb{R}$.

Example 1.3.11. *Value at Risk as a Quantile*
Let $V_0 > 0$, $R = \frac{V_1 - V_0}{V_0}$, and $F_R(x) = \mathbb{P}(R \leq x)$.
The loss is $L = V_0 - V_1 = -V_0 R$, so $\{L \geq v\} = \{R \leq -\frac{v}{V_0}\}$ and $\mathbb{P}(L \geq v) = F_R(-\frac{v}{V_0})$.
If F_R is invertible and $\mathbb{P}(L \geq v_\alpha) = \alpha$, then $v_\alpha = -V_0 F_R^{-1}(\alpha)$.
Key Insight: VaR is a lower-tail return quantile converted back into a dollar loss.

1.4 Standard Distribution Families and Approximations

Definition 1.4.1. *Core Discrete Distributions*
Common discrete laws include:

(i) *Point mass*: $X \sim \delta_a$ for $a \in \mathbb{R}$ if $\mathbb{P}(X = a) = 1$.
(ii) *Discrete uniform*: $X \sim \text{Unif}\{1, \dots, k\}$ for $k \in \mathbb{N}$ if $\mathbb{P}(X = x) = \frac{1}{k}$ for $x = 1, \dots, k$.
(iii) *Bernoulli*: $X \sim \text{Bernoulli}(p)$ for $p \in [0, 1]$ if $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. Equivalently $\mathbb{P}(X = x) = p^x(1 - p)^{1-x}$ for $x \in \{0, 1\}$.
(iv) *Binomial*: $X \sim \text{Binomial}(n, p)$ for $n \in \mathbb{N}$ and $p \in [0, 1]$ if $\mathbb{P}(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$ for $x = 0, \dots, n$.
(v) *Geometric*: $X \sim \text{Geom}(p)$, $p \in (0, 1]$, if $\mathbb{P}(X = k) = p(1 - p)^{k-1}$ for $k \geq 1$; X counts trials to first success.
(vi) *Negative binomial*: $X \sim \text{NegBin}(r, p)$, $r \in \mathbb{N}$, $p \in (0, 1]$, if $\mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1 - p)^{k-r}$ for $k \geq r$; X counts trials to the rth success.
(vii) *Hypergeometric*: $X \sim \text{Hypergeom}(n, r, m)$ if $\mathbb{P}(X = k) = \dfrac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}$ for $\max\{0, m - n + r\} \leq k \leq \min\{r, m\}$.
(viii) *Poisson*: $X \sim \text{Poisson}(\lambda)$ for $\lambda \geq 0$ if $\mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ for $x = 0, 1, 2, \dots$.

Definition 1.4.2. *Core Continuous Distributions*
Common continuous laws include:

(i) *Uniform*: $X \sim \text{Uniform}(a, b)$ for $a < b$ if $f_X(x) = (b - a)^{-1} \mathbf{1}_{[a,b]}(x)$.
(ii) *Normal*: $X \sim N(\mu, \sigma^2)$ for $\mu \in \mathbb{R}$, $\sigma > 0$, if $f_X(x) = (\sigma\sqrt{2\pi})^{-1} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$ for $x \in \mathbb{R}$. $N(0, 1)$ has density ϕ and CDF Φ.
(iii) *Exponential*: $X \sim \text{Exp}(\lambda)$ for $\lambda > 0$ if $f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0,\infty)}(x)$.
(iv) *Gamma*: $X \sim \text{Gamma}(\alpha, \lambda)$ for $\alpha, \lambda > 0$ if $f_X(x) = \lambda^\alpha \Gamma(\alpha)^{-1} x^{\alpha-1} e^{-\lambda x} \mathbf{1}_{[0,\infty)}(x)$; $\text{Exp}(\lambda) = \text{Gamma}(1, \lambda)$.
(v) *Beta*: $X \sim \text{Beta}(\alpha, \beta)$ for $\alpha, \beta > 0$ if $f_X(x) = \Gamma(\alpha + \beta)(\Gamma(\alpha)\Gamma(\beta))^{-1} x^{\alpha-1}(1 - x)^{\beta-1} \mathbf{1}_{(0,1)}(x)$.
(vi) *Cauchy*: $X \sim \text{Cauchy}(0, 1)$ if $f_X(x) = (\pi(1 + x^2))^{-1}$ for $x \in \mathbb{R}$.
(vii) χ^2: $X \sim \chi^2_p$ for $p > 0$ if $f_X(x) = (\Gamma(\frac{p}{2}) 2^{\frac{p}{2}})^{-1} x^{\frac{p}{2}-1} e^{-\frac{x}{2}} \mathbf{1}_{(0,\infty)}(x)$.

Definition 1.4.3. *Normal Right-Tail Quantile*
For $0 < \alpha < 1$ and $Z \sim N(0, 1)$, $z(\alpha) := \Phi^{-1}(1 - \alpha)$, so $\mathbb{P}(Z > z(\alpha)) = \alpha$.

Method 1.4.4. *Discrete Distribution Model Recipes*

(i) *Indicators*: for $A \in \mathcal{F}$, $\mathbf{1}_A \sim \text{Bernoulli}(\mathbb{P}(A))$.
(ii) *Binomial counts*: if $B_1, \dots, B_n$ are independent $\text{Bernoulli}(p)$ variables, then $\sum_{i=1}^{n} B_i \sim \text{Binomial}(n, p)$.
(iii) *Geometric waiting*: the probability of first success on trial k is $(1 - p)^{k-1} p$.
(iv) *Negative-binomial waiting*: for iid $B_i \sim \text{Bernoulli}(p)$ and $T_r = \min\{k : \sum_{i=1}^{k} B_i = r\}$, $\{T_r = k\} = \{B_k = 1, \sum_{i=1}^{k-1} B_i = r - 1\}$, so $\mathbb{P}(T_r = k) = \binom{k-1}{r-1} p^r (1 - p)^{k-r}$.
(v) *Hypergeometric sampling*: if $M \subseteq [n]$, $|M| = r$, and S is a uniform m-subset, then $X = |S \cap M|$ and $\mathbb{P}(X = k) = \dfrac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}$ for $\max\{0, m - n + r\} \leq k \leq \min\{r, m\}$.

Key Insight: Name the event; PMFs come from independence or subset counts.

Proposition 1.4.5. *Poisson Limit of Rare Binomial Counts*
Let $\lambda > 0$, $k \in \mathbb{N}_0$, and $X_n \sim \text{Binomial}(n, \frac{\lambda}{n})$ for all sufficiently large n. Then $\lim_{n\to\infty} \mathbb{P}(X_n = k) = \frac{e^{-\lambda}\lambda^k}{k!}$.

Proof. For large n, $\mathbb{P}(X_n = k) = \binom{n}{k}(\frac{\lambda}{n})^k(1-\frac{\lambda}{n})^{n-k} = \frac{\lambda^k}{k!}\prod_{j=0}^{k-1}(1-\frac{j}{n})(1-\frac{\lambda}{n})^n(1-\frac{\lambda}{n})^{-k}$.
Since $\prod_{j=0}^{k-1}(1-\frac{j}{n}) \to 1$, $(1-\frac{\lambda}{n})^n \to e^{-\lambda}$, and $(1-\frac{\lambda}{n})^{-k} \to 1$, the limit is $\frac{e^{-\lambda}\lambda^k}{k!}$.
Structural Commentary: Rare binomial masses split into fixed-k choices and an exponential no-event factor.
Key Insight: Large n, small p, fixed np turns binomial masses into Poisson masses. □

Method 1.4.6. *Recursive Binomial and Poisson Mass Computation*
Let $q := 1 - p$.

(i) If $X \sim \text{Binomial}(n, p)$, $p_k := \mathbb{P}(X = k)$, and $q > 0$, then $p_0 = q^n$ and $\frac{p_k}{p_{k-1}} = \frac{(n-k+1)p}{kq}$ for $1 \le k \le n$.

(ii) If $X \sim \text{Poisson}(\lambda)$ and $p_k := \mathbb{P}(X = k)$, then $p_0 = e^{-\lambda}$ and $\frac{p_k}{p_{k-1}} = \frac{\lambda}{k}$ for $k \ge 1$.

Key Insight: Successive mass ratios avoid repeatedly evaluating large factorials.

Example 1.4.7. *Majority Decoding as a Binomial Tail*
Let $n = 2m + 1$ and let $C_i \sim \text{Bernoulli}(p)$ be independent corruption indicators.
Then $X = \sum_{i=1}^{n} C_i \sim \text{Binomial}(n, p)$ and the majority decoder is correct iff $X \le m$.
For $n = 5$, $p = 0.1$, $\mathbb{P}(X \le 2) = 0.9^5 + 5(0.1)(0.9)^4 + 10(0.1)^2(0.9)^3 \approx 0.9914$.
Key Insight: Repetition coding converts reliability into a lower binomial tail.

Example 1.4.8. *Poisson Approximation for Rare Repeated Events*
Let $I_j = \mathbf{1}_{\{\text{double six on roll } j\}}$ for $1 \le j \le 100$.
Then $I_j \sim \text{Bernoulli}(\frac{1}{36})$ iid, $X = \sum_{j=1}^{100} I_j \sim \text{Binomial}(100, \frac{1}{36})$, and with $\lambda = \frac{100}{36} = \frac{25}{9}$, $\mathbb{P}(X = k) \approx \frac{e^{-\frac{25}{9}}(\frac{25}{9})^k}{k!}$.
Key Insight: Rare repeated event counts are often well approximated by a Poisson law with $\lambda = np$.

Proposition 1.4.9. *Exponential Memorylessness*
If $T \sim \text{Exp}(\lambda)$ with $\lambda > 0$, then $\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t)$ for all $s, t \ge 0$.

Proof. For $u \ge 0$, $\mathbb{P}(T > u) = e^{-\lambda u}$. Hence $\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T>s+t)}{\mathbb{P}(T>s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t)$.
Structural Commentary: Conditioning on survival to time s leaves the residual tail unchanged.
Key Insight: Exponential tails convert elapsed time into a cancellable factor. □

Remark 1.4.10. *Poisson Counts and Exponential Gaps*
In homogeneous Poisson process, counts satisfy $N(t) \sim \text{Poisson}(\lambda t)$; interarrival gaps satisfy $S_i \sim \text{Exp}(\lambda)$ iid.

Proposition 1.4.11. *Standardizing Normal Variables*
For $\sigma, \sigma_i > 0$:

(i) Standardization: if $X \sim N(\mu, \sigma^2)$, then $\frac{X-\mu}{\sigma} \sim N(0, 1)$.

(ii) Affine reconstruction: if $Z \sim N(0, 1)$, then $\mu + \sigma Z \sim N(\mu, \sigma^2)$.

(iii) Independent sums: if $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then $\sum_{i=1}^{n} X_i \sim N(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} \sigma_i^2)$.

(iv) Interval probabilities: if $X \sim N(\mu, \sigma^2)$ and $a < b$, then $\mathbb{P}(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$.

Proof.

(i) Let $Y = \frac{X-\mu}{\sigma}$. By change of variables, $f_Y(z) = \sigma f_X(\mu + \sigma z) = (2\pi)^{-\frac{1}{2}} e^{-\frac{z^2}{2}}$, so $Y \sim N(0, 1)$.

(ii) Let $Y = \mu + \sigma Z$. Then $f_Y(y) = \sigma^{-1}\phi(\frac{y-\mu}{\sigma}) = (\sigma\sqrt{2\pi})^{-1}\exp(-\frac{(y-\mu)^2}{2\sigma^2})$, hence $Y \sim N(\mu, \sigma^2)$.

(iii) It suffices by induction to treat $n = 2$.
Let $X_i \sim N(\mu_i, v_i)$ be independent, $S = X_1 + X_2$, $v = v_1 + v_2$, $w = \frac{v_1 v_2}{v}$, and $a_s = \frac{v_2\mu_1 + v_1(s-\mu_2)}{v}$.
Then $f_S(s) = \int_{\mathbb{R}} f_1(u) f_2(s-u)\, du$, and completing the square gives
$f_1(u)f_2(s-u) = (2\pi v)^{-\frac{1}{2}} e^{-\frac{(s-\mu_1-\mu_2)^2}{2v}} (2\pi w)^{-\frac{1}{2}} e^{-\frac{(u-a_s)^2}{2w}}$.
The second factor integrates to 1, so $S \sim N(\mu_1 + \mu_2, v_1 + v_2)$; induction gives (iii).

(iv) Let $Z = \frac{X-\mu}{\sigma}$. Since $\sigma > 0$, $\{a < X < b\} = \{\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\}$.
Hence $\mathbb{P}(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$.

Structural Commentary: Normal laws are stable under affine maps and independent addition.
Key Insight: Standardize to $N(0, 1)$, compute with Φ, then transform back. □

Example 1.4.12. *Normal Tail and Quantile Standardization*
Let $X \sim N(3, 5)$ and $Z = \frac{X-3}{\sqrt{5}} \sim N(0, 1)$. Then $\mathbb{P}(X > 1) = \mathbb{P}(Z > -\frac{2}{\sqrt{5}}) = 1 - \Phi(-\frac{2}{\sqrt{5}}) \approx 0.81$.
If $\mathbb{P}(X < q) = 0.2$, then $\Phi(\frac{q-3}{\sqrt{5}}) = 0.2$, so $q = 3 + \sqrt{5}\,\Phi^{-1}(0.2) \approx 3 - 0.8416\sqrt{5} \approx 1.1181$.

2 Joint Distributions, Multivariate Laws, and Transformations

2.1 Joint Distributions and Independence

Definition 2.1.1. *Joint and Marginal Distributions*

(i) The *joint mass function* of discrete X, Y is $f_{X,Y}(x,y) := \mathbb{P}(X = x, Y = y)$.
(ii) The *discrete marginals* are $f_X(x) = \sum_y f_{X,Y}(x,y)$ and $f_Y(y) = \sum_x f_{X,Y}(x,y)$.
(iii) The *continuous marginals* are $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$, and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$.
(iv) The *joint CDF* is $F_{X,Y}(x,y) := \mathbb{P}(X \le x, Y \le y)$. (X,Y) has joint density $f_{X,Y}$.

Example 2.1.2. *A Three-Toss Joint Mass Table*

Let $\Omega = \{H,T\}^3$ with $\mathbb{P}(\{\omega\}) = 2^{-3}$, set $X = \mathbf{1}_{\{\omega_1 = H\}}$, $Y = \sum_{i=1}^{3} \mathbf{1}_{\{\omega_i = H\}}$, and write $p_{xy} := \mathbb{P}(X = x, Y = y)$.

	$Y=0$	$Y=1$	$Y=2$	$Y=3$
$X=0$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0
$X=1$	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$

Thus $p_{0,2} = \frac{1}{8}$, $\sum_y p_{0y} = \sum_y p_{1y} = \frac{1}{2}$, and $\sum_x p_{xj} = \binom{3}{j} 2^{-3}$ for $j = 0, \dots, 3$.

Proposition 2.1.3. *Rectangular Increments of a Joint CDF*
Let $F(x,y) := \mathbb{P}(X \le x, Y \le y)$.

(i) If $x_1 < x_2$ and $y_1 < y_2$, then $\mathbb{P}(x_1 < X \le x_2,\ y_1 < Y \le y_2) = F(x_2,y_2) - F(x_1,y_2) - F(x_2,y_1) + F(x_1,y_1)$.
(ii) If (X,Y) has density $f_{X,Y}$, $F \in C^2$, then $F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$; $f_{X,Y}(x,y) = \partial_x \partial_y F(x,y)$.

Proof.

(i) Set $A_{ij} := \{X \le x_i,\ Y \le y_j\}$ and $R := \{x_1 < X \le x_2,\ y_1 < Y \le y_2\}$.
Pointwise, $\mathbf{1}_R = \mathbf{1}_{A_{22}} - \mathbf{1}_{A_{12}} - \mathbf{1}_{A_{21}} + \mathbf{1}_{A_{11}}$; applying $\mathbb{E}$ gives the identity.
(ii) The density assumption gives $F(x,y) = \mathbb{P}(X \le x, Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$.
Since $F \in C^2$, the two-variable FTC gives $\partial_x \partial_y F = f_{X,Y}$.

Structural Commentary: Joint CDFs encode southwest rectangles; finite rectangles are four-corner increments.
Key Insight: Mixed derivatives are infinitesimal rectangular increments. □

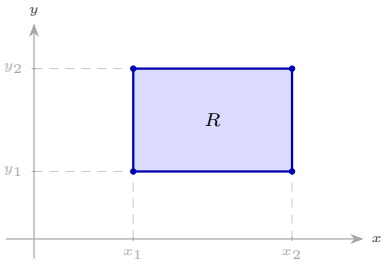

Figure 227: The rectangular probability is a four-term increment of the joint CDF.

Example 2.1.4. *A Non-Rectangular Bivariate Density*
Let $S = \{(x,y) : x^2 \le y \le 1\}$ and $f_{X,Y}(x,y) = cx^2 y \mathbf{1}_S(x,y)$.
Since $S = \{(x,y) : -1 \le x \le 1,\ x^2 \le y \le 1\}$, $1 = c\int_{-1}^{1}\int_{x^2}^{1} x^2 y\,dy\,dx = \frac{4c}{21}$, so $c = \frac{21}{4}$.
For $E = \{X \ge Y\}$, $S \cap E = \{(x,y) : 0 \le x \le 1,\ x^2 \le y \le x\}$, hence $\mathbb{P}(E) = (\frac{21}{4})\int_0^1\int_{x^2}^{x} x^2 y\,dy\,dx = \frac{3}{20}$.
Key Insight: In bivariate integration, pick one variable first and let the other range over the corresponding slice.

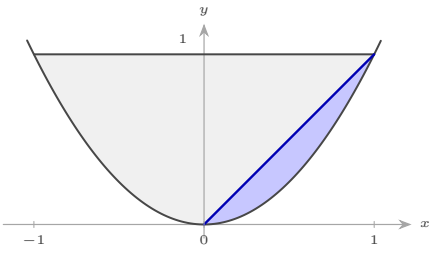

Figure 228: Gray: support $x^2 \le y \le 1$; blue: event $X \ge Y$ $(x^2 \le y \le x)$.

Definition 2.1.5. *Independent Random Variables and IID Samples*

(i) Random variables $X_1, \dots, X_n$ are *independent* if $\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i)\ \forall B_i \in \mathcal{B}(\mathbb{R})$.
(ii) They are *identically distributed* if $\mathcal{L}(X_1) = \cdots = \mathcal{L}(X_n)$.
(iii) They are *iid* if they are independent and identically distributed.
(iv) The *random vector* $X = (X_1, \dots, X_n)$ is a *random sample* when $X_1, \dots, X_n$ are iid from a common law.

Proposition 2.1.6. *Factorization Tests for Independence*

(i) Discrete factorization: discrete X, Y are independent iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all x, y.
(ii) Density factorization: if (X, Y) has joint density $f_{X,Y}$ and marginals f_X, f_Y, then X, Y are independent iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ a.e.
(iii) Rectangular support test: if $f_{X,Y}(x,y) = g(x)h(y)\mathbf{1}_{A\times B}(x,y)$ for measurable A, B, then X, Y are independent after normalization.

Proof.

(i) ($\Rightarrow$) If $X \perp\!\!\!\perp Y$, then $f_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) = f_X(x)f_Y(y)$.
(i) ($\Leftarrow$) If $f_{X,Y} = f_X f_Y$, then for A, B, $\mathbb{P}(X \in A, Y \in B) = \sum_{x\in A}\sum_{y\in B} f_X(x)f_Y(y) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$.
(ii) ($\Rightarrow$) If $X \perp\!\!\!\perp Y$, the product law has density $f_X f_Y$, so uniqueness of densities gives $f_{X,Y} = f_X f_Y$ a.e.
(ii) ($\Leftarrow$) If $f_{X,Y} = f_X f_Y$ a.e., then for Borel A, B, $\mathbb{P}(X \in A, Y \in B) = \int_A \int_B f_X(x)f_Y(y)\,dy\,dx = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$.
(iii) Let $G = \int_A g$ and $H = \int_B h$. Normalization gives $GH = 1$, while $f_X(x) = g(x)H\mathbf{1}_A(x)$ and $f_Y(y) = h(y)G\mathbf{1}_B(y)$, hence $f_X(x)f_Y(y) = f_{X,Y}(x,y)$ a.e.; apply (ii).

Structural Commentary: Independence is product structure of the joint law.
Key Insight: Independence means the joint law factors. □

2.2 Multivariate Models and Copulas

Definition 2.2.1. *Two Multivariate Families*

(i) *Multinomial*: for $n \in \mathbb{N}$ and $p \in [0,1]^k$ with $\sum_{j=1}^k p_j = 1$, $X \sim \text{Multinomial}(n,p)$ if $\mathbb{P}(X = x) = n!(x_1!\cdots x_k!)^{-1}\prod_{j=1}^k p_j^{x_j}$ for $x \in \mathbb{N}_0^k$ with $\sum_{j=1}^k x_j = n$; then $X_j \sim \text{Binomial}(n, p_j)$.
(ii) *Multivariate normal*: for $\mu \in \mathbb{R}^k$ and symmetric positive definite $\Sigma \in \mathbb{R}^{k\times k}$, $X \sim N(\mu,\Sigma)$ if $f_X(x) = ((2\pi)^k \det\Sigma)^{-\frac{1}{2}}\exp(-\frac{(x-\mu)^t\Sigma^{-1}(x-\mu)}{2})$ for $x \in \mathbb{R}^k$. $N(0, I_k)$ has independent $N(0,1)$ coordinates.

Proposition 2.2.2. *Multivariate Normal Affine Standardization*

Let $k \in \mathbb{N}$, $\mu \in \mathbb{R}^k$, $\Sigma \in \mathbb{R}^{k\times k}$ with $\Sigma \succ 0$, and $A = \Sigma^{\frac{1}{2}}$.

(i) For $X \in \mathbb{R}^k$, $X \sim N(\mu,\Sigma) \iff A^{-1}(X-\mu) \sim N(0, I_k)$.
(ii) For $Z \in \mathbb{R}^k$, $Z \sim N(0, I_k) \iff \mu + AZ \sim N(\mu,\Sigma)$.

Proof.

(i) ($\Rightarrow$) Put $Z = A^{-1}(X-\mu)$. With $x = \mu + Az$, $|\det A| = (\det\Sigma)^{\frac{1}{2}}$ and $(Az)^t\Sigma^{-1}(Az) = z^t z$. By change of variables, $f_Z(z) = f_X(\mu+Az)|\det A| = (2\pi)^{-\frac{k}{2}}\exp(-\frac{z^t z}{2})$.
(i) ($\Leftarrow$) Put $Z = A^{-1}(X-\mu) \sim N(0, I_k)$, so $X = \mu + AZ$. With $z = A^{-1}(x-\mu)$ and $|\det A^{-1}| = (\det\Sigma)^{-\frac{1}{2}}$, $f_X(x) = f_Z(A^{-1}(x-\mu))|\det A^{-1}| = ((2\pi)^k\det\Sigma)^{-\frac{1}{2}}\exp(-\frac{(x-\mu)^t\Sigma^{-1}(x-\mu)}{2})$.
(ii) ($\Rightarrow$) If $Z \sim N(0, I_k)$ and $X := \mu + AZ$, then $A^{-1}(X-\mu) = Z$; by (i) ($\Leftarrow$), $X \sim N(\mu,\Sigma)$.
(ii) ($\Leftarrow$) If $X := \mu + AZ \sim N(\mu,\Sigma)$, then $Z = A^{-1}(X-\mu)$; by (i) ($\Rightarrow$), $Z \sim N(0, I_k)$.

Structural Commentary: Positive definite covariance is an affine deformation of the standard normal.
Key Insight: Standardize by $\Sigma^{-\frac{1}{2}}$; reconstruct by $\Sigma^{\frac{1}{2}}$. □

Proposition 2.2.3. *Basic Multivariate Normal Calculus*

Let $X = (X_a, X_b) \sim N(\mu,\Sigma)$ with $X_a \in \mathbb{R}^r$, $X_b \in \mathbb{R}^s$, $\mu = (\mu_a,\mu_b)$, and $\Sigma = \begin{pmatrix}\Sigma_{aa} & \Sigma_{ab}\\ \Sigma_{ba} & \Sigma_{bb}\end{pmatrix} \succ 0$.

(i) Marginal: $X_a \sim N(\mu_a, \Sigma_{aa})$.
(ii) Conditional: $X_b \mid X_a = x_a \sim N(\mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a-\mu_a), \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab})$.
(iii) Linear form: for any vector a, $a^tX \sim N(a^t\mu, a^t\Sigma a)$.

Proof.

(i) Let $P_a X = X_a$. By Proposition 2.2.2, $X = \mu + \Sigma^{\frac{1}{2}}Z$ with $Z \sim N(0, I)$. Hence $X_a = P_a\mu + P_a\Sigma^{\frac{1}{2}}Z \sim N(\mu_a, P_a\Sigma P_a^t) = N(\mu_a,\Sigma_{aa})$.
(ii) Set $S := \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}$ and $m(x_a) := \mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a-\mu_a)$. The block determinant and inverse identities give $\det\Sigma = \det\Sigma_{aa}\det S$ and $(x-\mu)^t\Sigma^{-1}(x-\mu) = (x_a-\mu_a)^t\Sigma_{aa}^{-1}(x_a-\mu_a) + (x_b-m(x_a))^tS^{-1}(x_b-m(x_a))$. Thus $f_{b|a}(x_b \mid x_a) = \frac{f_{a,b}(x_a,x_b)}{f_a(x_a)}$ is the $N(m(x_a), S)$ density.
(iii) With $X = \mu + \Sigma^{\frac{1}{2}}Z$, $a^tX = a^t\mu + (\Sigma^{\frac{1}{2}}a)^tZ$. Since Z has independent $N(0,1)$ coordinates, $(\Sigma^{\frac{1}{2}}a)^tZ \sim N(0, a^t\Sigma a)$, hence $a^tX \sim N(a^t\mu, a^t\Sigma a)$.

Structural Commentary: Marginals, conditionals, and linear projections of Gaussian vectors remain Gaussian.
Key Insight: Gaussian calculus is affine calculus plus Schur complements. □

Definition 2.2.4. *Copulas*

(i) A *copula* is a joint CDF $C : [0,1]^2 \to [0,1]$.
(ii) *Uniform marginals*: $C(u,1) = u$ and $C(1,v) = v$ for $u, v \in [0,1]$.
(iii) If $C \in C^2$, its *copula density* is $c(u,v) := \partial_u\partial_v C(u,v)$.

Proposition 2.2.5. *Constructing a Joint Law from Marginals and a Copula*
Let F_X, F_Y be continuous CDFs with densities f_X, f_Y, and let C be a copula.
Define $F_{X,Y}(x,y) := C(F_X(x), F_Y(y))$. Then $F_{X,Y}$ has marginals F_X, F_Y.
If $C \in C^2$ with density $c = \partial_u \partial_v C$, then $f_{X,Y}(x,y) = c(F_X(x), F_Y(y)) f_X(x) f_Y(y)$.

Proof. $F_{X,Y}(x,\infty) = C(F_X(x), 1) = F_X(x)$ and $F_{X,Y}(\infty, y) = C(1, F_Y(y)) = F_Y(y)$.
If densities exist, $\partial_x \partial_y C(F_X(x), F_Y(y)) = (\partial_u \partial_v C)(F_X(x), F_Y(y)) f_X(x) f_Y(y)$ by the chain rule.
Structural Commentary: Marginals and dependence are separated into F_X, F_Y and C.
Key Insight: Marginals alone do not determine a joint law; the copula supplies the dependence. □

Example 2.2.6. *Farlie–Morgenstern Copulas*
For $|\alpha| \le 1$, $C_\alpha(u,v) = uv\{1 + \alpha(1-u)(1-v)\}$ is a copula with density $c_\alpha(u,v) = 1 + \alpha(1-2u)(1-2v)$.
Here $\alpha = 0$ gives the independence copula uv, while $\alpha = -1$ and $\alpha = 1$ give densities $2u + 2v - 4uv$ and $2 - 2u - 2v + 4uv$ on $[0,1]^2$, respectively; all have uniform marginals.

2.3 Transformations of Random Variables

Method 2.3.1. *One-Variable Transformations*
Let $Y = r(X)$.
(i) *Discrete pushforward*: if X is discrete, then $f_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}(X \in r^{-1}(\{y\})) = \sum_{x : r(x) = y} f_X(x)$.
(ii) *CDF method*: for $Y = r(X)$, compute $F_Y(y) = \int_{\{x : r(x) \le y\}} f_X(x)\,dx$; $f_Y(y) = F_Y'(y)$ where differentiable.
(iii) *Monotone change*: if r is strictly monotone on $\operatorname{supp} X$ with inverse s, then $f_Y(y) = f_X(s(y))\,|s'(y)|\,\mathbf{1}_{r(\operatorname{supp} X)}(y)$.
Key Insight: The CDF method works because it translates the transformed event back into an event about X.

Example 2.3.2. *Log Transform of an Exponential Variable*
Let $X \sim \mathrm{Exp}(1)$ and $Y = \log X$. Since $r(x) = \log x$ has inverse $s(y) = e^y$ with $s'(y) = e^y$, the change-of-variables formula gives $f_Y(y) = f_X(e^y) e^y = e^y e^{-e^y}$ for $y \in \mathbb{R}$.

Example 2.3.3. *A Nonmonotone Square Transform*
Let $X \sim \mathrm{Uniform}(-1, 3)$ and $Y = X^2$. Since $f_X = \frac{1}{4}$ on $[-1,3]$, $r^{-1}(\{y\}) \cap [-1,3] = \{-\sqrt{y}, \sqrt{y}\}$ for $0 < y < 1$ and $\{\sqrt{y}\}$ for $1 < y < 9$.
Hence $f_Y(y) = \frac{1}{4\sqrt{y}} \mathbf{1}_{(0,1)}(y) + \frac{1}{8\sqrt{y}} \mathbf{1}_{(1,9)}(y)$.
Key Insight: Non-one-to-one transformations require splitting the range into cases.

Example 2.3.4. *Squaring a Standard Normal*
Let $Z \sim N(0,1)$ and $X = Z^2$. For $x > 0$, $F_X(x) = \mathbb{P}(|Z| \le \sqrt{x}) = \Phi(\sqrt{x}) - \Phi(-\sqrt{x})$,
so $f_X(x) = x^{-\frac{1}{2}} \phi(\sqrt{x}) = (2\pi)^{-\frac{1}{2}} x^{-\frac{1}{2}} e^{-\frac{x}{2}}$. Hence $X \sim \chi^2_1 = \mathrm{Gamma}(\frac{1}{2}, \frac{1}{2})$ in the rate convention.

Example 2.3.5. *Reciprocal of a Uniform Variable*
Let $U \sim \mathrm{Uniform}(0,1)$ and $V = \frac{1}{U}$. Then $F_V(v) = 0$ for $v < 1$, while for $v \ge 1$, $F_V(v) = \mathbb{P}(U \ge \frac{1}{v}) = 1 - \frac{1}{v}$.
Hence $f_V(v) = v^{-2} \mathbf{1}_{[1,\infty)}(v)$.

Proposition 2.3.6. *Probability Integral Transform and Inverse Transform*
Let $F : I \to (0,1)$ be a continuous strictly increasing CDF with inverse $F^{-1} : (0,1) \to I$.
(i) If $X \sim F$, then $F(X) \sim \mathrm{Uniform}(0,1)$.
(ii) If $U \sim \mathrm{Uniform}(0,1)$ and $Y = F^{-1}(U)$, then $F_Y = F$.

Proof.
(i) For $0 < u < 1$, $\mathbb{P}(F(X) \le u) = \mathbb{P}(X \le F^{-1}(u)) = F(F^{-1}(u)) = u$, with endpoint values $0, 1$.
(ii) For $x \in I$, $\mathbb{P}(Y \le x) = \mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$.
Structural Commentary: F maps its law to uniform scale; F^{-1} maps uniform scale back.
Key Insight: Uniform random numbers become target laws by applying inverse CDFs. □

Method 2.3.7. *Several-Variable Transformations*
For $Z = r(X, Y)$ with joint density $f_{X,Y}$:
(i) *Sublevel set*: define $A_z := \{(x,y) : r(x,y) \le z\}$.
(ii) *CDF integral*: compute $F_Z(z) = \mathbb{P}(Z \le z) = \iint_{A_z} f_{X,Y}(x,y)\,dx\,dy$.
(iii) *Density*: differentiate where possible, $f_Z(z) = F_Z'(z)$.

Example 2.3.8. *Sum of Two Independent Uniforms*
Let $X_1, X_2 \sim \mathrm{Uniform}(0,1)$ be independent and $Y = X_1 + X_2$.
Since $f_{X_1,X_2} = \mathbf{1}_{[0,1]^2}$, $F_Y(y)$ is the area of $A_y = \{(x_1, x_2) \in [0,1]^2 : x_1 + x_2 \le y\}$,
so $F_Y(y) = \frac{y^2}{2} \mathbf{1}_{[0,1]}(y) + (1 - \frac{(2-y)^2}{2}) \mathbf{1}_{(1,2]}(y) + \mathbf{1}_{(2,\infty)}(y)$. Hence $f_Y(y) = y \mathbf{1}_{(0,1)}(y) + (2-y) \mathbf{1}_{(1,2)}(y)$.

Proposition 2.3.9. *Two-Dimensional Density Change of Variables*
Let (X,Y) have density $f_{X,Y}$ on $D \subseteq \mathbb{R}^2$, and let $T : D \xrightarrow{C^1} E$ be one-to-one with inverse h and $\det DT \ne 0$ on D. For $(U,V) = T(X,Y)$, $f_{U,V}(u,v) = f_{X,Y}(h(u,v))\,|\det Dh(u,v)|\,\mathbf{1}_E(u,v)$.

Proof. For Borel $B \subseteq E$, $\mathbb{P}((U,V) \in B) = \mathbb{P}((X,Y) \in h(B))$
$= \int_{h(B)} f_{X,Y}(x,y)\,dx\,dy = \int_B f_{X,Y}(h(u,v))\,|\det Dh(u,v)|\,du\,dv$ (by Theorem 5.1.12)
Structural Commentary: Transforming a density means preserving probability mass while rescaling area.
Key Insight: The absolute Jacobian accounts for local volume distortion. □

Example 2.3.10. *Sums and Quotients*
Let (X,Y) have joint density $f_{X,Y}$.
(i) Sum: for $S = X + Y$, $f_S(s) = \int_{\mathbb{R}} f_{X,Y}(x, s-x)\,dx$; if $X \perp\!\!\!\perp Y$, then $f_S(s) = \int_{\mathbb{R}} f_X(x) f_Y(s-x)\,dx$.

(ii) Quotient: for $Z = \frac{Y}{X}$, $f_Z(z) = \int_{\mathbb{R}} |x|\, f_{X,Y}(x, xz)\, dx$; if $X \perp\!\!\!\perp Y$, then $f_Z(z) = \int_{\mathbb{R}} |x|\, f_X(x) f_Y(xz)\, dx$.

Key Insight: Add one auxiliary coordinate, change variables, integrate it out.

Example 2.3.11. *Polar Transform of Two Standard Normals*
Let $X, Y \overset{\text{iid}}{\sim} N(0,1)$, $R = (X^2+Y^2)^{\frac{1}{2}}$, and $\Theta = \operatorname{atan2}(Y, X)$.

Polar change of variables gives $f_{R,\Theta}(r,\theta) = (2\pi)^{-1} r e^{-\frac{r^2}{2}} \mathbf{1}_{[0,\infty)}(r)\mathbf{1}_{[0,2\pi)}(\theta)$,
so $R \perp\!\!\!\perp \Theta$, $\Theta \sim \text{Uniform}(0, 2\pi)$, and $R^2 \sim \text{Exp}(\frac{1}{2})$.
Hence for iid $U_1, U_2 \sim \text{Uniform}(0,1)$, $(\sqrt{-2\log U_1}\cos(2\pi U_2), \sqrt{-2\log U_1}\sin(2\pi U_2)) \sim N(0, I_2)$.

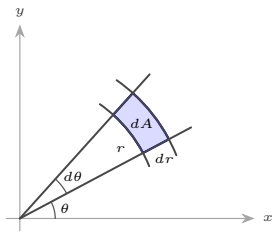

Figure 229: In polar coordinates, a small patch has area approximately $r\, dr\, d\theta$.

3 Expectation and Integration in Probability

3.1 Expectation, Variance, and LOTUS

Expectation is the averaging operation attached to a probability law. In the notation of Section 1.1.1, if X has law F_X, then integration against F_X is the common notation that covers both sums for discrete variables and ordinary integrals for variables with densities.

Definition 3.1.1. *Expectation Through the Law*
Let X be real-valued with distribution function F_X.

(i) *Expectation*: if $\int_{\mathbb{R}} |x|\, dF_X(x) < \infty$, then $\mathbb{E}(X) := \int_{\mathbb{R}} x\, dF_X(x)$.
(ii) *Discrete form*: if X has support S, mass function f_X, $\mathbb{E}(X) = \sum_{x\in S} x f_X(x)$ when $\sum_{x\in S} |x| f_X(x) < \infty$.
(iii) *Density form*: if X has density f_X, then $\mathbb{E}(X) = \int_{\mathbb{R}} x f_X(x)\, dx$ when $\int_{\mathbb{R}} |x| f_X(x)\, dx < \infty$.
(iv) *Function form*: for measurable $r : \mathbb{R} \to \mathbb{R}$, if $\int_{\mathbb{R}} |r(x)|\, dF_X(x) < \infty$, then $\mathbb{E}(r(X)) := \int_{\mathbb{R}} r(x)\, dF_X(x)$.

Proposition 3.1.2. *Indicators Turn Probabilities into Expectations*
For $A \in \mathcal{F}$, let $\mathbf{1}_A : \Omega \to \{0,1\}$ be $\mathbf{1}_A(\omega) = 1$ for $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ for $\omega \notin A$. Then $\mathbb{E}(\mathbf{1}_A) = \mathbb{P}(A)$.

Proof. Since $\Omega = A \sqcup A^c$ and $\mathbf{1}_A = 1$ on A, 0 on A^c, $\mathbb{E}(\mathbf{1}_A) = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A)$.
Structural Commentary: Indicators encode events as 0-1 random variables.
Key Insight: Probabilities are expectations of indicators. □

Example 3.1.3. *Expectation of the Two-Toss Head Count*
Let $\Omega = \{H,T\}^2$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$, and let $X(\omega)$ be the number of heads in ω.

Then $f_X(k) = \binom{2}{k} 2^{-2}$ for $k = 0,1,2$, so $\mathbb{E}(X) = \sum_{k=0}^{2} k f_X(k) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1$.

Example 3.1.4. *First Expectation Computations*

(i) If $X \sim \text{Bernoulli}(p)$, then $\mathbb{E}X = 0(1-p) + 1p = p$.
(ii) If $X \sim \text{Uniform}(-1, 3)$, then $\mathbb{E}X = (\frac{1}{4}) \int_{-1}^{3} x\, dx = 1$.
(iii) If $X \sim \text{Cauchy}(0,1)$, then $\mathbb{E}|X| = (\frac{2}{\pi}) \int_0^\infty x(1+x^2)^{-1}\, dx = \infty$, so $\mathbb{E}X$ does not exist.

Theorem 3.1.5. *Expectation of a Function*
Let r be measurable and assume the displayed expectations exist.

(i) One variable: for real-valued X, $\mathbb{E}(r(X)) = \int_{\mathbb{R}} r(x)\, dF_X(x)$.
(ii) Several variables: for (X,Y) with joint law $F_{X,Y}$, $\mathbb{E}(r(X,Y)) = \int_{\mathbb{R}^2} r(x,y)\, dF_{X,Y}(x,y)$.

Proof.

(i) Since $\mathcal{L}(X) = X_*\mathbb{P}$, $\mathbb{E}(r(X)) = \int_\Omega r(X(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} r(x)\, d\mathcal{L}(X)(x) = \int_{\mathbb{R}} r(x)\, dF_X(x)$.
(ii) Let $W = (X,Y)$. Since $\mathcal{L}(W) = W_*\mathbb{P} = F_{X,Y}$,
$\mathbb{E}(r(X,Y)) = \int_\Omega r(W(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}^2} r(x,y)\, d\mathcal{L}(W)(x,y) = \int_{\mathbb{R}^2} r(x,y)\, dF_{X,Y}(x,y)$.

Structural Commentary: Expectations of transformations integrate over the transformed law.
Key Insight: Pushforward laws let functions move from random variables to their values. □

Remark 3.1.6. *Law-Based Integration*
For measurable r, $\mathbb{E}(r(X))$ depends only on the law of X: $\mathbb{E}(r(X)) = \int_{\mathbb{R}} r(x)\, dF_X(x)$ whenever integrable.
Discrete laws give $\sum_x r(x) f_X(x)$; density laws give $\int_{\mathbb{R}} r(x) f_X(x)\, dx$.

Example 3.1.7. *Using the Function Rule*

(i) If $X \sim \text{Uniform}(0,1)$ and $Y = e^X$, then $\mathbb{E}Y = \mathbb{E}(e^X) = \int_0^1 e^x\, dx = e - 1$.
(ii) If $X \sim \text{Uniform}(0,1)$ and $Y = \max\{X, 1-X\}$, then $\mathbb{E}Y = \int_0^{\frac{1}{2}} (1-x)\, dx + \int_{\frac{1}{2}}^1 x\, dx = \frac{3}{4}$.
(iii) If $(X,Y) \sim \text{Uniform}([0,1]^2)$ and $Z = X^2 + Y^2$, then $\mathbb{E}Z = \int_0^1 \int_0^1 (x^2+y^2)\, dx\, dy = \frac{2}{3}$.

Definition 3.1.8. *Variance and Covariance*

(i) The *variance* of X is $\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}X)^2]$, assuming $\mathbb{E}(X^2) < \infty$.
(ii) The *standard deviation* of X is $\mathrm{sd}(X) := \sqrt{\mathrm{Var}(X)}$.
(iii) The *covariance* of X, Y is $\mathrm{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, assuming $\mathbb{E}(X^2), \mathbb{E}(Y^2) < \infty$.

3.2 Moments, Sample Statistics, and Random Vectors

Definition 3.2.1. *Moments, Correlation, and Sample Statistics*

(i) The *kth moment* of X is $\mathbb{E}(X^k)$, assuming $\mathbb{E}|X|^k < \infty$.
(ii) The *correlation* of X, Y is $\mathrm{Corr}(X, Y) := \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}$, assuming $\mathrm{Var}(X), \mathrm{Var}(Y) > 0$.
(iii) The *sample mean* of $X_1, \ldots, X_n$ is $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$.
(iv) The *sample variance* of $X_1, \ldots, X_n$ is $S_n^2 := (n-1)^{-1} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$.

Theorem 3.2.2. *Expectation, Variance, and Covariance Algebra*
Assume all displayed quantities exist and $a_i, a, b \in \mathbb{R}$.

(i) Linearity: $\mathbb{E}(\sum_i a_i X_i) = \sum_i a_i \mathbb{E}X_i$.
(ii) Independent product: if $X_1, \ldots, X_n$ are independent, then $\mathbb{E}(\prod_i X_i) = \prod_i \mathbb{E}X_i$.
(iii) Variance identities: $\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$ *and* $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$.
(iv) Covariance identity: $\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y$.
(v) Variance of a linear combination: $\mathrm{Var}(\sum_i a_i X_i) = \sum_i a_i^2 \mathrm{Var}(X_i) + 2 \sum_{i<j} a_i a_j \mathrm{Cov}(X_i, X_j)$.

Proof.

(i) $\mathbb{E}(\sum_i a_i X_i) = \int_\Omega \sum_i a_i X_i \, d\mathbb{P} = \sum_i a_i \int_\Omega X_i \, d\mathbb{P} = \sum_i a_i \mathbb{E}X_i$.
(ii) By independence, $\mathcal{L}(X_1, \ldots, X_n) = \prod_i F_{X_i}$; $\mathbb{E}(\prod_i X_i) = \int_{\mathbb{R}^n} \prod_i x_i \, d\prod_i F_{X_i}(x_i) = \prod_i \int_{\mathbb{R}} x_i \, dF_{X_i}(x_i) = \prod_i \mathbb{E}X_i$.
(iii) $\mathrm{Var}(X) = \mathbb{E}(X^2 - 2X\mathbb{E}X + (\mathbb{E}X)^2) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$.
Also $\mathrm{Var}(aX + b) = \mathbb{E}[(aX + b - a\mathbb{E}X - b)^2] = a^2 \mathrm{Var}(X)$.
(iv) $\mathrm{Cov}(X, Y) = \mathbb{E}(XY - X\mathbb{E}Y - Y\mathbb{E}X + \mathbb{E}X\mathbb{E}Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y$.
(v) With $S = \sum_i a_i X_i$, (iv) gives

$$\mathrm{Var}(S) = \mathrm{Cov}(S, S) = \sum_{i,j} a_i a_j \mathrm{Cov}(X_i, X_j) = \sum_i a_i^2 \mathrm{Var}(X_i) + 2 \sum_{i<j} a_i a_j \mathrm{Cov}(X_i, X_j).$$

Structural Commentary: Expectation is linear; variance and covariance are quadratic and bilinear consequences.
Key Insight: Expand products, then use linearity. □

Example 3.2.3. *Indicator Sums as a Method*

(i) If $X \sim \mathrm{Binomial}(n, p)$, write $X = \sum_{i=1}^{n} I_i$ with independent $I_i \sim \mathrm{Bernoulli}(p)$.
Then $\mathbb{E}X = np$ and $\mathrm{Var}(X) = np(1-p)$.
(ii) Let $\xi_i \in \{-1, 1\}$ with $\mathbb{P}(\xi_i = -1) = p$ and $\mathbb{P}(\xi_i = 1) = 1 - p$, independently, and $X_n = \sum_{i=1}^{n} \xi_i$.
Then $\mathbb{E}X_n = n(1-2p)$ and $\mathrm{Var}(X_n) = 4np(1-p)$.
(iii) Let $F_n = c \prod_{i=1}^{n} M_i$ with independent $M_i \in \{2, \frac{1}{2}\}$ and $\mathbb{P}(M_i = 2) = \mathbb{P}(M_i = \frac{1}{2}) = \frac{1}{2}$.
Then $\mathbb{E}F_n = c \prod_{i=1}^{n} \mathbb{E}M_i = c(\frac{5}{4})^n$.

Proposition 3.2.4. *Zero Variance Means Almost Surely Constant*
For $X \in L^2$ and $m := \mathbb{E}X$, $\mathrm{Var}(X) = 0$ iff $X = m$ a.s.

Proof.

($\Rightarrow$) If $\mathrm{Var}(X) = 0$, then $\mathbb{E}[(X - m)^2] = 0$. Since $(X - m)^2 \geq 0$, $\mathbb{P}((X - m)^2 > 0) = 0$, hence $X = m$ a.s.
($\Leftarrow$) If $X = m$ a.s., then $(X - m)^2 = 0$ a.s., so $\mathrm{Var}(X) = \mathbb{E}[(X - m)^2] = 0$.

Structural Commentary: Variance is the mean squared distance from the mean.
Key Insight: Zero nonnegative expectation forces zero almost surely. □

Proposition 3.2.5. *Tail Integral Formula*
If $X \geq 0$ has density f_X, distribution function F_X, and $\mathbb{E}X < \infty$,
then $\mathbb{E}X = \int_0^\infty \mathbb{P}(X > x)\, dx = \int_0^\infty (1 - F_X(x))\, dx$.

Proof. For $b > 0$, integration by parts gives $\int_0^b x f_X(x)\, dx = -b(1 - F_X(b)) + \int_0^b (1 - F_X(x))\, dx$.
Since $\mathbb{E}X < \infty$, $b(1 - F_X(b)) \leq \int_b^\infty x f_X(x)\, dx \to 0$. Letting $b \to \infty$ yields $\mathbb{E}X = \int_0^\infty (1 - F_X(x))\, dx = \int_0^\infty \mathbb{P}(X > x)\, dx$.
Structural Commentary: A nonnegative variable is integrated by summing its survival probabilities.
Key Insight: Expectation equals area under the tail. □

Example 3.2.6. *Maximum of Uniform Samples*
Let $X_1, \ldots, X_n$ be independent $\mathrm{Uniform}(0, 1)$ variables and $Y_n = \max_i X_i$.
Then $F_{Y_n}(y) = \mathbb{P}(X_1 \leq y, \ldots, X_n \leq y) = y^n$ for $0 \leq y \leq 1$. So $\mathbb{E}Y_n = \int_0^1 (1 - F_{Y_n}(y))\, dy = \int_0^1 (1 - y^n)\, dy = \frac{n}{n+1}$.

Theorem 3.2.7. *Mean and Variance of the Basic Sample Statistics*
Let $X_1, \dots, X_n$ be iid with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$.
(i) Sample-mean expectation: $\mathbb{E}\overline{X}_n = \mu$.
(ii) Sample-mean variance: $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$.
(iii) Unbiased sample variance: $\mathbb{E}S_n^2 = \sigma^2$.

Proof.
(i) $\mathbb{E}\overline{X}_n = n^{-1}\sum_{i=1}^{n}\mathbb{E}X_i = n^{-1}n\mu = \mu$.
(ii) Independence gives $\mathrm{Var}(\overline{X}_n) = n^{-2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = n^{-2}n\sigma^2 = \frac{\sigma^2}{n}$.
(iii) Since $\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\overline{X}_n - \mu)^2$. Thus $\mathbb{E}[(n-1)S_n^2] = n\sigma^2 - n(\frac{\sigma^2}{n}) = (n-1)\sigma^2$.
Structural Commentary: Averaging preserves mean and scales variance.
Key Insight: Center at μ, then subtract the fluctuation of $\overline{X}_n$. □

Lemma 3.2.8. *Expected Squared Deviation from a Sample Mean*
Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = \mu_i$ and $\mathrm{Var}(X_i) = \sigma^2$ for $1 \le i \le n$.
Set $\overline{X} := \frac{1}{n}\sum_{i=1}^{n}X_i$ and $\overline{\mu} := \frac{1}{n}\sum_{i=1}^{n}\mu_i$. Then $\mathbb{E}(X_i - \overline{X})^2 = (\mu_i - \overline{\mu})^2 + \frac{n-1}{n}\sigma^2$ for $1 \le i \le n$.

Proof. Set $U_i := X_i - \overline{X}$. Then $\mathbb{E}U_i = \mu_i - \overline{\mu}$, $\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$, and $\mathrm{Cov}(X_i, \overline{X}) = \frac{\sigma^2}{n}$ by independence.
Hence $\mathrm{Var}(U_i) = \mathrm{Var}(X_i) + \mathrm{Var}(\overline{X}) - 2\,\mathrm{Cov}(X_i, \overline{X}) = \sigma^2 + \frac{\sigma^2}{n} - 2\frac{\sigma^2}{n} = \frac{n-1}{n}\sigma^2$.
Thus $\mathbb{E}U_i^2 = (\mathbb{E}U_i)^2 + \mathrm{Var}(U_i) = (\mu_i - \overline{\mu})^2 + \frac{n-1}{n}\sigma^2$.
Structural Commentary: $X_i - \overline{X}$ has mean $\mu_i - \overline{\mu}$ and variance $\mathrm{Var}(X_i - \overline{X}) = \frac{n-1}{n}\sigma^2$.
Key Insight: $\mathrm{Cov}(X_i, \overline{X}) = \frac{\sigma^2}{n}$ removes exactly one $\frac{\sigma^2}{n}$ from the raw variance σ^2. □

Definition 3.2.9. *Mean Vector and Covariance Matrix*
For a random vector $X = (X_1, \dots, X_k)^t$:
(i) *Mean vector*: $\mathbb{E}X := (\mathbb{E}X_1, \dots, \mathbb{E}X_k)^t$.
(ii) *Covariance matrix*: $\mathrm{Var}(X) := [\mathrm{Cov}(X_i, X_j)]_{i,j=1}^{k}$.

Proposition 3.2.10. *Linear Transformations of Random Vectors*
Let $X \in \mathbb{R}^k$ satisfy $\mathbb{E}X = \mu$ and $\mathrm{Var}(X) = \Sigma$.
i. Linear form: for $a \in \mathbb{R}^k$, $\mathbb{E}(a^tX) = a^t\mu$ and $\mathrm{Var}(a^tX) = a^t\Sigma a$.
ii. Matrix transform: for $A \in \mathbb{R}^{m\times k}$, $\mathbb{E}(AX) = A\mu$ and $\mathrm{Var}(AX) = A\Sigma A^t$.

Proof.
i. $\mathbb{E}(a^tX) = a^t\mathbb{E}X = a^t\mu$, and $\mathrm{Var}(a^tX) = \mathbb{E}[(a^t(X-\mu))^2] = a^t\mathbb{E}[(X-\mu)(X-\mu)^t]a = a^t\Sigma a$.
ii. $\mathbb{E}(AX) = A\mathbb{E}X = A\mu$, and $\mathrm{Var}(AX) = \mathbb{E}[(AX - A\mu)(AX - A\mu)^t] = A\mathbb{E}[(X-\mu)(X-\mu)^t]A^t = A\Sigma A^t$.
Structural Commentary: Linear maps push means forward and conjugate covariance matrices.
Key Insight: Center first; then move constants outside expectations. □

Example 3.2.11. *Multinomial Covariance Structure*
If $X \sim \mathrm{Multinomial}(n, p)$ with $p = (p_1, \dots, p_k)$, then $\mathbb{E}X = np$ and $\mathrm{Var}(X_i) = np_i(1-p_i)$.
For $i \neq j$, $X_i + X_j \sim \mathrm{Binomial}(n, p_i + p_j)$, so $n(p_i+p_j)(1-p_i-p_j) = np_i(1-p_i) + np_j(1-p_j) + 2\,\mathrm{Cov}(X_i, X_j)$.
Hence $\mathrm{Cov}(X_i, X_j) = -np_ip_j$.

3.3 Probability Inequalities

Theorem 3.3.1. *Markov and Chebyshev Inequalities*
i. Markov: if $X \ge 0$ and $\mathbb{E}X < \infty$, then for $t > 0$, $\mathbb{P}(X \ge t) \le \frac{\mathbb{E}X}{t}$.
ii. Chebyshev: if $\mu := \mathbb{E}X$ and $\sigma^2 := \mathrm{Var}(X) < \infty$, then for $t > 0$, $\mathbb{P}(|X - \mu| \ge t) \le \frac{\sigma^2}{t^2}$.

Proof.
i. Since $X \ge t\mathbf{1}_{\{X \ge t\}}$, taking expectations gives $\mathbb{E}X \ge t\mathbb{P}(X \ge t)$.
ii. Apply (i) to $Y := (X-\mu)^2 \ge 0$ with threshold t^2, then $\mathbb{P}(|X-\mu| \ge t) = \mathbb{P}(Y \ge t^2) \le \frac{\mathbb{E}Y}{t^2} = \frac{\sigma^2}{t^2}$.
Structural Commentary: Tail bounds convert moment control into probability control.
Key Insight: Dominate the tail indicator by a scaled nonnegative random variable. □

Example 3.3.2. *A Distribution-Free Error-Rate Bound*
Let $X_1, \dots, X_n$ be independent Bernoulli(p) error indicators and $\overline{X}_n := n^{-1}\sum_{i=1}^{n}X_i$. Since $\mathbb{E}\overline{X}_n = p$ and $\mathrm{Var}(\overline{X}_n) = \frac{p(1-p)}{n}$, Chebyshev gives, for $\varepsilon > 0$, $\mathbb{P}(|\overline{X}_n - p| > \varepsilon) \le \frac{p(1-p)}{n\varepsilon^2} \le \frac{1}{4n\varepsilon^2}$.

Theorem 3.3.3. *Cauchy–Schwarz and Jensen*
If $X, Y \in L^2$, then $\mathbb{E}|XY| \le \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}$.
If $g : I \to \mathbb{R}$ convex, $X \in I$ a.s., $X, g(X) \in L^1$, then $\mathbb{E}g(X) \ge g(\mathbb{E}X)$; for concave g, inequality reverses.

Proof. For Cauchy–Schwarz, apply the L^2 inner-product inequality to $|X|$ and $|Y|$.
For Jensen, set $m := \mathbb{E}X$ and take an affine support $\ell(x) := g(m) + s(x-m)$ with $\ell \le g$ on I.
Then $\mathbb{E}g(X) \ge \mathbb{E}\ell(X) = g(m) + s(\mathbb{E}X - m) = g(\mathbb{E}X)$. For concave g, apply the convex case to $-g$.
Structural Commentary: Cauchy–Schwarz controls products; Jensen moves convexity through expectation.
Key Insight: Use Hilbert-space geometry for products and supporting lines for convex functions. □

Theorem 3.3.4. *Hoeffding's Bounded-Sum Bound*
Let $Y_1, \dots, Y_n$ be independent, $\mathbb{E}Y_i = 0$, $a_i \le Y_i \le b_i$ a.s., and $V := \sum_{i=1}^n (b_i - a_i)^2$. For $S_n := \sum_{i=1}^n Y_i$ and $t, \varepsilon > 0$, $\mathbb{P}(S_n \ge \varepsilon) \le \exp(-t\varepsilon + \frac{t^2 V}{8})$; if $V > 0$, then $\mathbb{P}(S_n \ge \varepsilon) \le \exp(-\frac{2\varepsilon^2}{V})$.

Proof. For $t > 0$, Markov gives $\mathbb{P}(S_n \ge \varepsilon) = \mathbb{P}(e^{tS_n} \ge e^{t\varepsilon}) \le e^{-t\varepsilon}\mathbb{E}e^{tS_n}$.
Independence gives $\mathbb{E}e^{tS_n} = \prod_{i=1}^n \mathbb{E}e^{tY_i}$, and Hoeffding's lemma gives $\mathbb{E}e^{tY_i} \le \exp(\frac{t^2(b_i-a_i)^2}{8})$.
Hence $\mathbb{P}(S_n \ge \varepsilon) \le \exp(-t\varepsilon + \frac{t^2 V}{8})$. If $V > 0$, minimize the exponent at $t = \frac{4\varepsilon}{V}$ to obtain $\exp(-\frac{2\varepsilon^2}{V})$.
Structural Commentary: Hoeffding turns bounded independent increments into exponential tail decay.
Key Insight: Exponentiate the sum, factor the MGF, then optimize t. □

Corollary 3.3.5. *Hoeffding for Bernoulli Averages*
If $X_1, \dots, X_n$ independent Bernoulli(p) *variables, $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$, then $\mathbb{P}(|\overline{X}_n - p| > \varepsilon) \le 2e^{-2n\varepsilon^2}$.*

Proof. Set $Y_i^+ := \frac{X_i - p}{n}$.
Then $\mathbb{E}Y_i^+ = 0$, $Y_i^+ \in [-\frac{p}{n}, \frac{1-p}{n}]$, and $V = \sum_i (\frac{1}{n})^2 = \frac{1}{n}$, by Theorem 3.3.4, $\mathbb{P}(\overline{X}_n - p \ge \varepsilon) \le e^{-2n\varepsilon^2}$.
Apply same argument to $Y_i^- := \frac{p - X_i}{n}$, $\mathbb{P}(p - \overline{X}_n \ge \varepsilon) \le e^{-2n\varepsilon^2}$; the union bound yields the claim.
Structural Commentary: Bernoulli averages are sums of bounded centered increments.
Key Insight: Control both tails separately, then add them. □

3.4 Moment Generating Functions

Definition 3.4.1. *Moment Generating Function*
The *moment generating function* of X is $\psi_X(t) := \mathbb{E}(e^{tX})$ on $D_X := \{t \in \mathbb{R} : \mathbb{E}e^{tX} < \infty\}$.
If $(-\delta, \delta) \subseteq D_X$ for some $\delta > 0$, then $\psi_X^{(k)}(0) = \mathbb{E}(X^k)$ for all $k \ge 1$.

Proposition 3.4.2. *Basic MGF Properties*
Assume the MGFs below are finite on a common open interval about 0.

(i) Affine transform: if $a, b \in \mathbb{R}$ and $Y = aX + b$, then $\psi_Y(t) = e^{bt}\psi_X(at)$.

(ii) Independent sum: if $X_1, \dots, X_n$ are independent and $Y := \sum_{i=1}^n X_i$, then $\psi_Y(t) = \prod_{i=1}^n \psi_{X_i}(t)$.

(iii) Uniqueness: if $\psi_X(t) = \psi_Y(t)$ for all t in an open interval containing 0, then $\mathcal{L}(X) = \mathcal{L}(Y)$.

Proof.
(i) $\psi_Y(t) = \mathbb{E}e^{t(aX+b)} = e^{bt}\mathbb{E}e^{(at)X} = e^{bt}\psi_X(at)$.
(ii) For admissible t, $e^{tX_1}, \dots, e^{tX_n}$ are independent and integrable.
Hence $\psi_Y(t) = \mathbb{E}e^{t\sum_{i=1}^n X_i} = \mathbb{E}\prod_{i=1}^n e^{tX_i} = \prod_{i=1}^n \mathbb{E}e^{tX_i} = \prod_{i=1}^n \psi_{X_i}(t)$.
(iii) Let $\mu := \mathcal{L}(X)$ and $\nu := \mathcal{L}(Y)$. If $\psi_X = \psi_Y$ on $(-\delta, \delta)$, then $\int_{\mathbb{R}} e^{tx}\,d\mu(x) = \int_{\mathbb{R}} e^{tx}\,d\nu(x)$ for all $|t| < \delta$.
Uniqueness of the two-sided Laplace transform gives $\mu = \nu$.
Structural Commentary: MGFs turn affine maps and independent sums into algebraic operations.
Key Insight: Exponentials convert sums into products, and the MGF near 0 determines the law. □

Example 3.4.3. *MGF Computations*
(i) If $X \sim \text{Exp}(1)$, then $\psi_X(t) = \mathbb{E}e^{tX} = \int_0^\infty e^{(t-1)x}\,dx = (1-t)^{-1}$ for $t < 1$.
Hence $\mathbb{E}X = \psi_X'(0) = 1$ and $\text{Var}(X) = \psi_X''(0) - (\psi_X'(0))^2 = 1$.
(ii) If $X \sim \text{Binomial}(n, p)$, then $X = \sum_{i=1}^n B_i$ with independent $B_i \sim \text{Bernoulli}(p)$.
Hence $\psi_X(t) = \prod_{i=1}^n \psi_{B_i}(t) = (pe^t + 1 - p)^n$.

Method 3.4.4. *MGF Calculus for Classical Laws*
If $\psi_X(t) = \mathbb{E}e^{tX}$ exists near 0, then $\mathbb{E}X = \psi_X'(0)$ and $\text{Var}(X) = \psi_X''(0) - \psi_X'(0)^2$.
(i) If $X \sim \text{Poisson}(\lambda)$, then $\psi_X(t) = \exp(\lambda(e^t - 1))$, $\mathbb{E}X = \lambda$, and $\text{Var}(X) = \lambda$.
(ii) If $X \sim \text{Gamma}(\alpha, \lambda)$ with rate λ, then $\psi_X(t) = (\frac{\lambda}{\lambda - t})^\alpha$ for $t < \lambda$, $\mathbb{E}X = \frac{\alpha}{\lambda}$, and $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.
(iii) If $Z \sim N(0,1)$, then $\psi_Z(t) = e^{\frac{t^2}{2}}$; for $Y = \mu + \sigma Z$, $\psi_Y(t) = \exp(\mu t + \frac{\sigma^2 t^2}{2})$.

Corollary 3.4.5. *Independent Sums from MGFs*
Let $X \perp\!\!\!\perp Y$, and interpret all MGFs on a common neighborhood of 0.

(i) If $\lambda, \mu > 0$, $X \sim \text{Poisson}(\lambda)$, and $Y \sim \text{Poisson}(\mu)$, then $X + Y \sim \text{Poisson}(\lambda + \mu)$.

(ii) If $X \sim \text{Gamma}(\alpha_1, \lambda)$ and $Y \sim \text{Gamma}(\alpha_2, \lambda)$ with $\alpha_1, \alpha_2, \lambda > 0$, then $X + Y \sim \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$.

(iii) If $\mu, \nu \in \mathbb{R}$, $\sigma, \tau > 0$, $X \sim N(\mu, \sigma^2)$, and $Y \sim N(\nu, \tau^2)$, then $X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2)$.

Proof.

(i) $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = \exp(\lambda(e^t-1))\exp(\mu(e^t-1)) = \exp((\lambda+\mu)(e^t-1))$, QED.

(ii) For $t < \lambda$, $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = (\frac{\lambda}{\lambda-t})^{\alpha_1+\alpha_2}$, QED.

(iii) $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = \exp(\mu t + \frac{\sigma^2t^2}{2})\exp(\nu t + \frac{\tau^2t^2}{2}) = \exp((\mu+\nu)t + \frac{(\sigma^2+\tau^2)t^2}{2})$, QED.

Structural Commentary: Independence turns convolution into multiplication of MGFs.
Key Insight: Closed distribution families are stable when their MGF parameters add. □

Method 3.4.6. *Common Means, Variances, and MGFs*
MGFs are stated on their natural neighborhood of 0; values at $t = 0$ are by continuity.
The geometric and negative-binomial rows use the total-trials convention.

Distribution	$\mathbb{E}X$	$\mathrm{Var}(X)$	$\psi_X(t)$
point mass at a	a	0	e^{at}
Bernoulli(p)	p	$p(1-p)$	$1-p+pe^t$
Binomial(n,p)	np	$np(1-p)$	$(1-p+pe^t)^n$
Geometric(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
NegBin(r,p)	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$
Poisson(λ)	λ	λ	$e^{\lambda(e^t-1)}$
Uniform(a,b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{(b-a)t}$
$N(\mu,\sigma^2)$	μ	σ^2	$\exp(\mu t + \frac{\sigma^2t^2}{2})$
Exp(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Gamma(α,λ)	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$
Beta(α,β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	not useful
χ^2_p	p	$2p$	$(1-2t)^{-\frac{p}{2}}$

4 Order Statistics, Extrema, and Moment Methods

4.1 Order Statistics, Extrema, and Moment Methods

Order statistics describe the sorted values of an independent sample. They turn joint density calculations into formulas for minima, maxima, medians, ranges, and distribution-free tolerance intervals.

Definition 4.1.1. *Order Statistics*
For real-valued $X_1, \dots, X_n$,

(i) The *order statistics* are the pointwise rearrangement $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ of $X_1, \dots, X_n$.

(ii) $X_{(1)} = \min_{1\le i\le n} X_i$, $X_{(n)} = \max_{1\le i\le n} X_i$, and if $n = 2m+1$, then $X_{(m+1)}$ is the sample median.

Proposition 4.1.2. *Minimum and Maximum of an IID Continuous Sample*
Let $n \ge 1$ and $X_1, \dots, X_n \overset{\text{iid}}{\sim} F$, where F is differentiable with density $f = F'$.
Set $V = X_{(1)} = \min_i X_i$ and $U = X_{(n)} = \max_i X_i$; then for $u, v \in \mathbb{R}$, $F_U(u) = F(u)^n$, $f_U(u) = nf(u)F(u)^{n-1}$, $F_V(v) = 1-(1-F(v))^n$, and $f_V(v) = nf(v)(1-F(v))^{n-1}$.

Proof. Since $\{U \le u\} = \bigcap_{i=1}^n \{X_i \le u\}$, independence gives $F_U(u) = \prod_{i=1}^n F(u) = F(u)^n$.
Differentiation gives $f_U(u) = nf(u)F(u)^{n-1}$.
Since $\{V > v\} = \bigcap_{i=1}^n \{X_i > v\}$, independence gives $1 - F_V(v) = \prod_{i=1}^n (1-F(v)) = (1-F(v))^n$.
Differentiation of $F_V(v) = 1-(1-F(v))^n$ gives $f_V(v) = nf(v)(1-F(v))^{n-1}$.
Structural Commentary: Extremes convert order events into products over independent coordinates.
Key Insight: Maxima use all observations to the left; minima use all observations to the right. □

Example 4.1.3. *Series and Parallel Exponential Lifetimes*
Let $T_1, \dots, T_n \overset{\text{ind}}{\sim} \text{Exp}(\lambda)$ be component lifetimes.

(i) For a series system, all components must survive, so system failure occurs at the first component failure: $V = \min_i T_i = T_{(1)}$. Thus $F_V(v) = 1 - e^{-n\lambda v}$ and $f_V(v) = n\lambda e^{-n\lambda v}$ for $v \ge 0$; hence $V \sim \text{Exp}(n\lambda)$.

(ii) For a parallel system, one surviving component suffices, so system failure occurs only after all components fail: $U = \max_i T_i = T_{(n)}$. Thus $F_U(u) = (1-e^{-\lambda u})^n$ and $f_U(u) = n\lambda e^{-\lambda u}(1-e^{-\lambda u})^{n-1}$ for $u \ge 0$.

Theorem 4.1.4. *Density of the kth Order Statistic*
Let $X_1, \dots, X_n$ be iid real-valued random variables with differentiable CDF F and density $f = F'$.
For $1 \le k \le n$ and $x \in \mathbb{R}$, $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} f(x)F(x)^{k-1}(1-F(x))^{n-k}$.

Proof. $N_x = \sum_{i=1}^n \mathbf{1}_{\{X_i \le x\}}$; $N_x \sim \text{Bin}(n, F(x))$, $\{X_{(k)} \le x\} = \{N_x \ge k\}$; $F_{X_{(k)}}(x) = \sum_{j=k}^n \binom{n}{j} F(x)^j (1-F(x))^{n-j}$.
With $p = F(x)$, $\frac{d}{dp}\sum_{j=k}^n \binom{n}{j} p^j(1-p)^{n-j} = \frac{n!}{(k-1)!(n-k)!}p^{k-1}(1-p)^{n-k}$, and the chain rule gives stated density.
Structural Commentary: The multinomial count of left, middle, and right observations supplies the coefficient.
Key Insight: An order statistic is located by counting how many sample points fall on each side. □

Corollary 4.1.5. *Uniform Order Statistics Are Beta*
If $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(0,1)$, *then for* $1 \le k \le n$, $X_{(k)} \sim \text{Beta}(k, n-k+1)$ *with density* $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k} \mathbf{1}_{(0,1)}(x)$.

Proof. On $(0,1)$, $F(x) = x$ and $f(x) = 1$, so Theorem 4.1.4 gives $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k}$.
Also $B(k, n-k+1) = \frac{(k-1)!(n-k)!}{n!}$, hence this density is the $\text{Beta}(k, n-k+1)$ density.
Structural Commentary: Uniform order statistics reduce the general density to the beta normalizing constant.
Key Insight: Counting positions in a uniform sample produces beta powers of left and right mass. □

Proposition 4.1.6. *Joint Density of Minimum and Maximum*
Let $n \ge 2$, *let* $X_1, \dots, X_n \overset{\text{iid}}{\sim} F$ *with differentiable CDF* F, *density* $f = F'$; $V = X_{(1)}$, $U = X_{(n)}$, $R = U - V$.

(i) For $v < u$, $f_{V,U}(v,u) = n(n-1)f(v)f(u)(F(u) - F(v))^{n-2}$, *and* $f_{V,U}(v,u) = 0$ *for* $v \ge u$.
(ii) If $F(t) = t$ *on* $[0,1]$, *then* $f_{V,U}(v,u) = n(n-1)(u-v)^{n-2}\mathbf{1}_{\{0<v<u<1\}}$.
(iii) In the uniform case, $f_R(r) = n(n-1)r^{n-2}(1-r)\mathbf{1}_{(0,1)}(r)$, *with* $F_R(r) = 0$ *for* $r < 0$, $F_R(r) = nr^{n-1} - (n-1)r^n$ *for* $0 \le r \le 1$, *and* $F_R(r) = 1$ *for* $r \ge 1$.

Proof.
(i) For $v < u$, $G(v,u) = \mathbb{P}(V \ge v, U \le u) = \mathbb{P}(v \le X_i \le u,\ 1 \le i \le n) = (F(u) - F(v))^n$, so $f_{V,U}(v,u) = -\partial_v \partial_u G(v,u) = n(n-1)f(v)f(u)(F(u) - F(v))^{n-2}$.
(ii) If $F(t) = t$ and $f(t) = 1$ on $(0,1)$, by (i), $f_{V,U}(v,u) = n(n-1)(u-v)^{n-2}$ on $0 < v < u < 1$, else 0.
(iii) For $0 < r < 1$, $f_R(r) = \int_0^{1-r} f_{V,U}(v, v+r)\, dv = n(n-1)r^{n-2}(1-r)$. Integration gives $F_R(r) = \int_0^r f_R(s)\, ds = nr^{n-1} - (n-1)r^n$; outside $[0,1]$, the support gives $F_R(r) \in \{0,1\}$.
Structural Commentary: Mixed derivatives select the two boundary observations.
Key Insight: Fixed extremes leave only interior mass. □

Example 4.1.7. *Distribution-Free Tolerance Interval*
Let $X_1, \dots, X_n$ be iid with continuous CDF F, and set $Q = F(X_{(n)}) - F(X_{(1)})$.
By the probability integral transform, $U_i = F(X_i) \overset{\text{iid}}{\sim} \text{Uniform}(0,1)$, so $Q \overset{d}{=} U_{(n)} - U_{(1)}$.
Therefore, for $0 \le \alpha \le 1$, $\mathbb{P}(Q > \alpha) = 1 - F_R(\alpha) = 1 - n\alpha^{n-1} + (n-1)\alpha^n$. For $n = 100$ and $\alpha = .95$, $\mathbb{P}(Q > .95) \approx .96$: $[X_{(1)}, X_{(n)}]$ covers at least .95 of the F-mass with probability about .96.

The expectation calculus developed above also supports a useful collection of moment methods: identifying whether the mean exists, replacing difficult distributions of functions by direct expectation, controlling spread by covariance, and using generating functions or Taylor expansion when direct integration is cumbersome.

Example 4.1.8. *Existence and First Distributional Means*
(i) If $X \sim \text{Geom}(p)$ on $\{1, 2, \dots\}$ and $q = 1 - p$, then $\mathbb{E}X = p \sum_{k \ge 1} k q^{k-1} = p \frac{d}{dq} \frac{q}{1-q} = \frac{1}{p}$.
(ii) If $X \sim \text{Poisson}(\lambda)$, then $\mathbb{E}X = e^{-\lambda} \sum_{k \ge 1} \frac{k\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{j \ge 0} \frac{\lambda^j}{j!} = \lambda$.
(iii) If $X \sim \text{Cauchy}(0,1)$, then $\mathbb{E}|X| = \frac{2}{\pi} \int_0^\infty x(1+x^2)^{-1}\, dx = \infty$, so X is not integrable and $\mathbb{E}X$ is undefined.
(iv) If $\mathbb{P}(Y = 2^k) = 2^{-(k+1)}$ for $k \ge 0$, then $\mathbb{E}Y = \sum_{k \ge 0} 2^{-1} = \infty$; truncation is essential.
Key Insight: Expectation is a finite absolute average, not merely a formal symmetry or a plausible center.

Method 4.1.9. *Indicator Sums Without Independence*
(i) For a finite index set I and events $(A_i)_{i \in I}$, if $W = \sum_{i \in I} \mathbf{1}_{A_i}$, then $\mathbb{E}W = \sum_{i \in I} \mathbb{P}(A_i)$.
(ii) The identity uses only linearity of expectation; the events A_i need not be independent.

Example 4.1.10. *Counting Word Occurrences*
Let $S_1, \dots, S_N \overset{\text{iid}}{\sim} \text{Uniform}\{A, C, G, T\}$ and fix a word $w \in \{A, C, G, T\}^q$.
For $1 \le m \le N - q + 1$, set $I_m = \mathbf{1}_{\{(S_m, \dots, S_{m+q-1}) = w\}}$ and $W = \sum_{m=1}^{N-q+1} I_m$.
Then $\mathbb{P}(I_m = 1) = 4^{-q}$ and $\mathbb{E}W = (N - q + 1)4^{-q}$.
Overlaps may create dependence; linearity does not require independence.

Example 4.1.11. *Coupon Collection by Waiting Times*
Draw uniformly from n coupon types, and let X_r be the waiting time for the rth new type. Then $X_r \sim \text{Geom}(\frac{n-r+1}{n})$ on $\{1, 2, \dots\}$, so $\mathbb{E}X_r = \frac{n}{n-r+1}$.
For $T = \sum_{r=1}^n X_r$, linearity gives $\mathbb{E}T = \sum_{r=1}^n \frac{n}{n-r+1} = nH_n$, where $H_n = \sum_{j=1}^n \frac{1}{j} \sim \log n + \gamma$.

Theorem 4.1.12. *Correlation Bound and Equality Case*
Let $X, Y \in L^2$ *have means* μ_X, μ_Y *and standard deviations* $0 < \sigma_X, \sigma_Y < \infty$, *set* $\rho = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$, *and define* $Z_X = \frac{X - \mu_X}{\sigma_X}$, $Z_Y = \frac{Y - \mu_Y}{\sigma_Y}$.

(i) $\mathbb{E}Z_X = \mathbb{E}Z_Y = 0$, $\text{Var}(Z_X) = \text{Var}(Z_Y) = 1$, *and* $\text{Cov}(Z_X, Z_Y) = \rho$.
(ii) $-1 \le \rho \le 1$, *equivalently* $|\rho| \le 1$.
(iii) $\rho = \pm 1$ *iff* $Z_Y = \pm Z_X$ *a.s., equivalently* $Y = \mu_Y \pm (\frac{\sigma_Y}{\sigma_X})(X - \mu_X)$ *a.s., with the same sign.*

Proof.
(i) $\mathbb{E}Z_X = 0$, $\text{Var}(Z_X) = \sigma_X^{-2} \text{Var}(X) = 1$, and similarly for Z_Y. Moreover, $\text{Cov}(Z_X, Z_Y) = \sigma_X^{-1}\sigma_Y^{-1} \text{Cov}(X, Y) = \rho$.

(ii) Since $0 \le \mathrm{Var}(Z_X \pm Z_Y) = 2 \pm 2\rho$, both $1+\rho \ge 0$ and $1-\rho \ge 0$, hence $|\rho| \le 1$.
(*iii*) ($\Rightarrow$) If $\rho = 1$, then $\mathbb{E}(Z_X - Z_Y)^2 = 0$, so $Z_X = Z_Y$ a.s..
If $\rho = -1$, then $\mathbb{E}(Z_X + Z_Y)^2 = 0$, so $Z_X = -Z_Y$ a.s.
(*iii*) ($\Leftarrow$) If $Z_Y = \pm Z_X$, then $\rho = \mathrm{Cov}(Z_X, \pm Z_X) = \pm 1$, and $\frac{Y-\mu_Y}{\sigma_Y} = \pm\frac{X-\mu_X}{\sigma_X}$
iff $Y = \mu_Y \pm (\frac{\sigma_Y}{\sigma_X})(X - \mu_X)$ a.s.

Structural Commentary: Correlation is centred covariance after unit-variance normalization.
Key Insight: Perfect correlation means no residual scatter around an affine line. □

Example 4.1.13. *Bivariate Uniform Association*
For (X, Y) with uniform marginals on $[0,1]$, $\mathbb{E}X = \mathbb{E}Y = \frac{1}{2}$ and $\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{1}{12}$.
Hence $\mathrm{Cov}(X,Y) = \mathbb{E}(XY) - \frac{1}{4}$ and $\rho = 12\,\mathrm{Cov}(X,Y)$.
If $\mathbb{E}(XY) = \frac{2}{9}$, then $\mathrm{Cov}(X,Y) = -\frac{1}{36}$, $\rho = -\frac{1}{3}$; if $\mathbb{E}(XY) = \frac{5}{18}$, then $\mathrm{Cov}(X,Y) = \frac{1}{36}$, $\rho = \frac{1}{3}$.
The sign of ρ records negative or positive association.

Method 4.1.14. *Moment Propagation by Taylor Expansion*
Let the relevant variables have finite second moments, and let g be C^2 near the relevant mean point.
(i) If $Y = g(X)$, $\mu = \mathbb{E}X$, and $\sigma^2 = \mathrm{Var}(X)$, then $\mathbb{E}Y \approx g(\mu)$ and $\mathrm{Var}(Y) \approx [g'(\mu)]^2\sigma^2$.
(ii) The second-order mean correction is $\mathbb{E}g(X) \approx g(\mu) + \frac{1}{2}g''(\mu)\sigma^2$.
(iii) For $Z = g(\mathbf{X})$, $\mu = \mathbb{E}\mathbf{X}$, and $\Sigma = \mathrm{Cov}(\mathbf{X})$, $\mathrm{Var}(Z) \approx \nabla g(\mu)^\top \Sigma \nabla g(\mu)$ and $\mathbb{E}Z \approx g(\mu) + \frac{1}{2}\,\mathrm{tr}(\Sigma H_g(\mu))$.

Structural Commentary: Taylor expansion propagates moments through local derivatives.
Key Insight: Smooth nonlinear functions are locally linear for variance and locally quadratic for mean correction.

Example 4.1.15. *Approximate Moments of a Ratio*
Let $Z = \frac{Y}{X}$, $\mu_X \neq 0$, $\mu = (\mu_X, \mu_Y)$, and write $\sigma_X^2 = \mathrm{Var}(X)$, $\sigma_Y^2 = \mathrm{Var}(Y)$, $\sigma_{XY} = \mathrm{Cov}(X,Y)$.
For $g(x,y) = \frac{y}{x}$, Taylor propagation gives $\mathbb{E}Z \approx \frac{\mu_Y}{\mu_X} + \frac{\sigma_X^2\mu_Y}{\mu_X^3} - \frac{\sigma_{XY}}{\mu_X^2}$
and $\mathrm{Var}(Z) \approx \frac{\sigma_X^2\mu_Y^2}{\mu_X^4} + \frac{\sigma_Y^2}{\mu_X^2} - \frac{2\sigma_{XY}\mu_Y}{\mu_X^3}$. The approximation is ill-conditioned as $|\mu_X| \downarrow 0$.

5 Normal-Derived Distributions

5.1 Normal-Derived Distribution Families

Independent $N(0,1)$ variables generate the normal-derived inference laws used below: χ^2 sums, Student t ratios, and F ratios.

Proposition 5.1.1. *Chi-Square Sums*
Let $n \ge 1$ and $Z_1, \dots, Z_n \overset{\text{ind}}{\sim} N(0,1)$.

(i) For $V = \sum_{i=1}^n Z_i^2$, $V \sim \chi_n^2 = \mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$ in the rate convention,
$f_V(v) = 2^{-\frac{n}{2}}\Gamma(\frac{n}{2})^{-1}v^{\frac{n}{2}-1}e^{-\frac{v}{2}}\mathbf{1}_{(0,\infty)}(v)$, $\psi_V(t) = (1-2t)^{-\frac{n}{2}}$ for $t < \frac{1}{2}$, $\mathbb{E}V = n$, and $\mathrm{Var}(V) = 2n$.
(ii) If $U \sim \chi_m^2$, $W \sim \chi_n^2$, and $U \perp\!\!\!\perp W$, then $U + W \sim \chi_{m+n}^2$.

Proof.

(i) For $Z \sim N(0,1)$ and $t < \frac{1}{2}$, $\mathbb{E}e^{tZ^2} = (2\pi)^{-\frac{1}{2}}\int_{\mathbb{R}} e^{-\frac{(1-2t)z^2}{2}}\,dz = (1-2t)^{-\frac{1}{2}}$.
By independence, $\psi_V(t) = \prod_{i=1}^n \mathbb{E}e^{tZ_i^2} = (1-2t)^{-\frac{n}{2}}$. By MGF uniqueness, $V \sim \mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$.
Differentiating ψ_V at 0 gives $\mathbb{E}V = n$ and $\mathrm{Var}(V) = 2n$.
(ii) If $U \perp\!\!\!\perp W$, then $\psi_{U+W}(t) = (1-2t)^{-\frac{m}{2}}(1-2t)^{-\frac{n}{2}} = (1-2t)^{-\frac{m+n}{2}}$, hence $U + W \sim \chi_{m+n}^2$.

Structural Commentary: Chi-square degrees of freedom add by MGF exponents.
Key Insight: Squared independent standard normals form gamma sums. □

Definition 5.1.2. *Normal Sample Statistics*
(i) *Sample*: $X_1, \dots, X_n$ are independent with $X_i \sim N(\mu, \sigma^2)$, $n \ge 2$, and $\sigma^2 > 0$.
(ii) *Sample mean*: $\overline{X} := n^{-1}\sum_{i=1}^n X_i$ estimates the location μ.
(iii) *Sample variance*: $S^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$ estimates the scale σ^2.

Proposition 5.1.3. *Normal Sample Mean*
Let $X_1, \dots, X_n$ be independent with $X_i \sim N(\mu, \sigma^2)$, and set $\overline{X} := n^{-1}\sum_{i=1}^n X_i$. Then $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.

Proof. Since independent normal sums are normal, $\overline{X}$ is normal.
Also $\mathbb{E}\overline{X} = n^{-1}\sum_i \mathbb{E}X_i = \mu$ and $\mathrm{Var}(\overline{X}) = n^{-2}\sum_i \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$. Thus $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.

Structural Commentary: Averaging preserves normality and divides variance by n.
Key Insight: The sample mean is an exact normal pivot center, not an asymptotic approximation. □

Theorem 5.1.4. *Mean-Residual Independence*
Let $X_1, \dots, X_n$ be independent with $X_i \sim N(\mu, \sigma^2)$, set $\overline{X} := n^{-1}\sum_i X_i$, $R := (X_1 - \overline{X}, \dots, X_n - \overline{X})$,
and $S^2 := (n-1)^{-1}\sum_i R_i^2$. Then $\overline{X} \perp\!\!\!\perp R$, hence $\overline{X} \perp\!\!\!\perp S^2$.

Proof. Let $Z_i := \frac{X_i-\mu}{\sigma}$, so $Z \sim N_n(0, I_n)$. Choose orthogonal Q with first row $n^{-\frac{1}{2}}\mathbf{1}^\top$, set $Y := QZ$.
Then $Y \sim N_n(0, I_n)$, $Y_1 \perp\!\!\!\perp (Y_2, \ldots, Y_n)$. Since $\overline{X} = \mu + \frac{\sigma Y_1}{\sqrt{n}}$ and $R = \sigma(I - n^{-1}\mathbf{1}\mathbf{1}^\top)Z = \sigma Q^\top(0, Y_2, \ldots, Y_n)^\top$, $\overline{X}$ is function of Y_1, R is function of $(Y_2, \ldots, Y_n)$. Therefore $\overline{X} \perp\!\!\!\perp R$, by $S^2 = (n-1)^{-1}\sum_i R_i^2$, $\overline{X} \perp\!\!\!\perp S^2$.

Structural Commentary: Mean and residuals are orthogonal projections.
Key Insight: For Gaussian vectors, orthogonal components are independent. □

Theorem 5.1.5. *Sample Variance Chi-Square Law*
Given independent $X_1, \ldots, X_n$; $X_i \sim N(\mu, \sigma^2)$; $S^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$. *Then* $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$.

Proof. Let $Z_i := \frac{X_i-\mu}{\sigma}$, so $Z = (Z_1, \ldots, Z_n) \sim N_n(0, I_n)$.
Choose orthogonal matrix Q with first row $n^{-\frac{1}{2}}(1, \ldots, 1)$, set $Y := QZ$.
Orthogonal invariance gives $Y \sim N_n(0, I_n)$ with independent coordinates.
Then $Y_1 = \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}$ and $\sum_{j=2}^n Y_j^2 = \sum_{i=1}^n (Z_i - \overline{Z})^2 = \frac{(n-1)S^2}{\sigma^2}$.
Thus $\frac{(n-1)S^2}{\sigma^2}$ is the sum of squares of $n-1$ independent $N(0,1)$ variables, hence χ^2_{n-1}.
Structural Commentary: Cantering leaves $n-1$ residual directions.
Key Insight: The sample variance is the squared Gaussian residual length scaled by σ^2. □

Definition 5.1.6. *Student t Distribution*
If $Z \sim N(0,1)$, $U \sim \chi^2_n$, $Z \perp\!\!\!\perp U$, then $T = Z/\sqrt{\frac{U}{n}}$ has *Student t distribution* with $df = n \geq 1$, $T \sim t_n$.

Proposition 5.1.7. *Student t Density*
Let $n \geq 1$ *and* $T = Z/\sqrt{\frac{U}{n}}$ *with* $Z \sim N(0,1)$, $U \sim \chi^2_n$, *and* $Z \perp\!\!\!\perp U$.

(i) For $t \in \mathbb{R}$, $f_T(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$.
(ii) For all $t \in \mathbb{R}$, $f_T(-t) = f_T(t)$.
(iii) For each fixed $t \in \mathbb{R}$, $\lim_{n\to\infty} f_T(t) = \phi(t) = (2\pi)^{-\frac{1}{2}}e^{-\frac{t^2}{2}}$.

Proof.

(i) For fixed $u > 0$, $z = t\sqrt{\frac{u}{n}}$ has Jacobian $\sqrt{\frac{u}{n}}$, so $f_T(t) = \int_0^\infty \phi(t\sqrt{\frac{u}{n}})\sqrt{\frac{u}{n}} f_U(u)\, du$.
Substituting $\phi(z) = (2\pi)^{-\frac{1}{2}}e^{-\frac{z^2}{2}}$ and $f_U(u) = 2^{-\frac{n}{2}}\Gamma(\frac{n}{2})^{-1}u^{\frac{n}{2}-1}e^{-\frac{u}{2}}\mathbf{1}_{(0,\infty)}(u)$ gives
$f_T(t) = \frac{1}{\sqrt{n\pi}2^{\frac{n+1}{2}}\Gamma(\frac{n}{2})}\int_0^\infty u^{\frac{n+1}{2}-1}e^{-\frac{(1+\frac{t^2}{n})u}{2}}\, du$. Then $F_T(t) = \Gamma(\frac{n+1}{2})2^{\frac{n+1}{2}}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$, yielding density.
(ii) By (i), $f_T(t) = c_n h_n(t^2)$ for $c_n > 0$, so $f_T(-t) = f_T(t)$ for all t.
(iii) For fixed t, $f_T(t) = (2\pi)^{-\frac{1}{2}}\frac{\Gamma(\frac{n+1}{2})}{\sqrt{\frac{n}{2}}\Gamma(\frac{n}{2})}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$, gamma ratio tends to 1, last factor tends to $e^{-\frac{t^2}{2}}$.

Structural Commentary: Student t is a normal scale mixture with random chi-square scale.
Key Insight: Random denominator uncertainty thickens the normal tails. □

Example 5.1.8. *t_1 is Cauchy*
If $T \sim t_1$, then $f_T(t) = \Gamma(1)(\sqrt{\pi}\Gamma(\frac{1}{2}))^{-1}(1+t^2)^{-1} = \frac{1}{\pi(1+t^2)}$, so $T \sim \text{Cauchy}(0,1)$.

Definition 5.1.9. *F Distribution*
If $U \sim \chi^2_m$, $V \sim \chi^2_n$, $U \perp\!\!\!\perp V$, then $W = (\frac{U}{m})/(\frac{V}{n})$ has *F distribution*, $df = (m,n)$, $W \sim F_{m,n}$.

Proposition 5.1.10. *F Density*
Let $m, n \geq 1$ *and* $W = (\frac{U}{m})/(\frac{V}{n})$, *where* $U \sim \chi^2_m$, $V \sim \chi^2_n$, *and* $U \perp\!\!\!\perp V$.

(i) $f_W(w) = \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})}(\frac{m}{n})^{\frac{m}{2}}w^{\frac{m}{2}-1}(1 + \frac{mw}{n})^{-\frac{m+n}{2}}\mathbf{1}_{(0,\infty)}(w)$.
(ii) If $n > 2$, *then* $\mathbb{E}W = \frac{n}{n-2}$.

Proof.

(i) For $w > 0$, set $u = \frac{mwv}{n}$; the Jacobian is $\frac{mv}{n}$, so $f_W(w) = \int_0^\infty f_U(\frac{mwv}{n})f_V(v)(\frac{mv}{n})\, dv$. By substitution,
$f_W(w) = Cw^{\frac{m}{2}-1}\int_0^\infty v^{\frac{m+n}{2}-1}e^{-\frac{(1+\frac{mw}{n})v}{2}}\, dv$; $C = \frac{(\frac{m}{n})^{\frac{m}{2}}}{2^{\frac{m+n}{2}}\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})}$, by gamma integral, QED.
(ii) Since $U \perp\!\!\!\perp V$, $\mathbb{E}W = (\frac{n}{m})\mathbb{E}U\,\mathbb{E}(V^{-1}) = n\,\mathbb{E}(V^{-1}) = \frac{n}{n-2}$ for $n > 2$.

Structural Commentary: An F variable is a ratio of independent chi-square averages.
Key Insight: Denominator variability controls the right tail and the mean condition $n > 2$. □

Corollary 5.1.11. *Reciprocal F and Squared t*

(i) For $m, n \geq 1$, *if* $W \sim F_{m,n}$, *then* $W^{-1} \sim F_{n,m}$.
(ii) If $T \sim t_n$, *then* $T^2 \sim F_{1,n}$.

Proof.

(i) Write $W = (\frac{U}{m})/(\frac{V}{n})$ with $U \sim \chi^2_m$, $V \sim \chi^2_n$, and $U \perp\!\!\!\perp V$; then $W^{-1} = (\frac{V}{n})/(\frac{U}{m}) \sim F_{n,m}$.
(ii) Write $T = Z/\sqrt{\frac{V}{n}}$ with $Z \sim N(0,1)$, $V \sim \chi^2_n$, and $Z \perp\!\!\!\perp V$; since $Z^2 \sim \chi^2_1$, $T^2 = (\frac{Z^2}{1})/(\frac{V}{n}) \sim F_{1,n}$.

Structural Commentary: Reciprocals swap F degrees of freedom, and squaring t removes the normal sign.
Key Insight: Both identities are immediate from the same chi-square-ratio representation. □

Part XVI
Probability and Stochastic Systems

1 Conditional Probability and Conditional Expectation

1.1 Conditional Probability

Definition 1.1.1. *Conditional Probability*
For $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the *conditional probability* of A given B is $\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, i.e. the probability of A under the normalised restriction of $\mathbb{P}$ to B.

Example 1.1.2. *Diagnostic Table*
Let T be the event of a positive test and D the event of disease, with observed counts

	D	D^c	total
T	25	14	39
T^c	18	78	96
total	43	92	135

Let $N(\cdot)$ be counts, $\widehat{\mathbb{P}}(A \mid B) = \frac{N(A \cap B)}{N(B)}$.
Thus $\widehat{\mathbb{P}}(D \mid T) = \frac{25}{39}$, $\widehat{\mathbb{P}}(D^c \mid T^c) = \frac{78}{96}$, $\widehat{\mathbb{P}}(D^c \mid T) = \frac{14}{39}$, and $\widehat{\mathbb{P}}(D \mid T^c) = \frac{18}{96}$.

Proposition 1.1.3. *Conditioning on a Fixed Event Gives a Probability Measure*
For $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, define $\mathbb{P}_B : \mathcal{F} \to [0,1]$ by $\mathbb{P}_B(A) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$.
Then $\mathbb{P}_B$ is a probability measure on $(\Omega, \mathcal{F})$.

Proof. For $A \in \mathcal{F}$, $\mathbb{P}_B(A) \geq 0$, and $\mathbb{P}_B(\Omega) = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1$.
If $(A_i)_{i \geq 1}$ are pairwise disjoint, then $(A_i \cap B)_{i \geq 1}$ are pairwise disjoint and $(\bigcup_i A_i) \cap B = \bigcup_i (A_i \cap B)$.
Thus $\mathbb{P}_B(\bigcup_i A_i) = \frac{\mathbb{P}(\bigcup_i (A_i \cap B))}{\mathbb{P}(B)} = \sum_i \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_i \mathbb{P}_B(A_i)$.
Structural Commentary: Conditioning normalizes the restriction of $\mathbb{P}$ to B.
Key Insight: Once B is fixed, conditional probability is ordinary probability on the restricted world. □

Remark 1.1.4. *Conditional Probability Is Not Symmetric*
In general, $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$; conditioning fixes the reference population. For disease event D and positive-test event T, large $\mathbb{P}(T \mid D)$ need not imply large $\mathbb{P}(D \mid T)$ when $\mathbb{P}(D)$ is small.

Proposition 1.1.5. *Multiplication and Chain Rules*
Let $A, B, C \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, $\mathbb{P}(B) > 0$, and $\mathbb{P}(B \cap C) > 0$.
(i) $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$.
(ii) $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)\mathbb{P}(C)$.

Proof.
(i) For $\mathbb{P}(F) > 0$, $\mathbb{P}(E \mid F)\mathbb{P}(F) = \mathbb{P}(E \cap F)$; apply this with $(E, F) = (A, B)$ and (B, A).
(ii) Since $\mathbb{P}(B \cap C) > 0$, $\mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)\mathbb{P}(C) = \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(B \cap C)} \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(C)} \mathbb{P}(C) = \mathbb{P}(A \cap B \cap C)$.
Structural Commentary: The chain rule orders information as C, then B within C, then A within $B \cap C$.
Key Insight: A joint probability factors into conditional increments along a chosen order. □

Example 1.1.6. *Cards Without Replacement*
Draw two cards from a standard deck without replacement. Let A be the event that the first card is the ace of clubs and B the event that the second card is the queen of diamonds. Since $\mathbb{P}(A) = \frac{1}{52}$ and $\mathbb{P}(B \mid A) = \frac{1}{51}$, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) = \frac{1}{52 \cdot 51}$. The factor $\mathbb{P}(B \mid A)$ uses the 51-card deck remaining after A occurs.

Proposition 1.1.7. *Independence Through Conditioning*
Let $A, B \in \mathcal{F}$ with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. The following are equivalent:
(i) A and B are independent.
(ii) $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.
(iii) $\mathbb{P}(B \mid A) = \mathbb{P}(B)$.

Proof.
(i) $\Rightarrow$ *(ii)* If A and B are independent, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, so $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A)$.
(ii) $\Rightarrow$ *(iii)* If $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, so $\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \mathbb{P}(B)$.
(iii) $\Rightarrow$ *(i)* If $\mathbb{P}(B \mid A) = \mathbb{P}(B)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A) = \mathbb{P}(A)\mathbb{P}(B)$, hence A and B are independent.
Structural Commentary: Each condition is $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ normalized by a positive marginal.
Key Insight: Independence means conditioning on either event preserves the probability of the other. □

Theorem 1.1.8. *Law of Total Probability*
(i) Let $A_1, \ldots, A_k \in \mathcal{F}$ form a finite partition of Ω with $\mathbb{P}(A_i) > 0$ for all i.
(ii) For every $B \in \mathcal{F}$, $\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$.

Proof.

(i) The partition assumptions give $B = \bigsqcup_{i=1}^{k} (B \cap A_i)$, hence $\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \cap A_i)$.

(ii) For each i, $\mathbb{P}(B \cap A_i) = \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$; substituting in (i) gives the formula.

Structural Commentary: The partition $\{A_i\}_{i=1}^{k}$ decomposes B into disjoint cases $B \cap A_i$.
Key Insight: Total probability is weighted averaging over cases. □

Example 1.1.9. *Second Draw by Total Probability*
An urn has 3 red balls, 1 blue ball; draw twice without replacement, let R_i be the event that draw i is red.
By total probability, $\mathbb{P}(R_2) = \mathbb{P}(R_2 \mid R_1)\mathbb{P}(R_1) + \mathbb{P}(R_2 \mid R_1^c)\mathbb{P}(R_1^c) = \frac{2}{3} \cdot \frac{3}{4} + 1 \cdot \frac{1}{4} = \frac{3}{4}$.
Key Insight: The second draw has the same marginal red probability as the first.

Definition 1.1.10. *Prior and Posterior Probabilities*
Let $A_1, \ldots, A_k \in \mathcal{F}$ be a partition with $\mathbb{P}(A_i) > 0$ for all i, and let $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$.

(i) $\mathbb{P}(A_i)$ is the *prior probability* of A_i.

(ii) $\mathbb{P}(A_i \mid B)$ is the *posterior probability* of A_i after conditioning on B.

Theorem 1.1.11. *Bayes' Theorem*

Let $A_1, \ldots, A_k \in \mathcal{F}$ satisfy $A_i \cap A_j = \varnothing$ for $i \neq j$, $\bigcup_{i=1}^{k} A_i = \Omega$, and $\mathbb{P}(A_i) > 0$ for all i.

If $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, then for each i, $\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\sum_{j=1}^{k} \mathbb{P}(B|A_j)\mathbb{P}(A_j)}$.

Proof. For fixed i, $\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\mathbb{P}(B)}$.
Since $B = \bigsqcup_{j=1}^{k} (B \cap A_j)$, $\mathbb{P}(B) = \sum_{j=1}^{k} \mathbb{P}(B \cap A_j) = \sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)$. Substitution gives the formula.
Structural Commentary: Bayes reweights priors by likelihoods, then normalises.
Key Insight: Posterior equals likelihood times prior divided by evidence. □

Corollary 1.1.12. *Posterior Mass Must Balance*
Let $A_1, \ldots, A_k \in \mathcal{F}$ be a partition of Ω with $\mathbb{P}(A_i) > 0$ for all i, and let $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$.
If $\mathbb{P}(A_1 \mid B) < \mathbb{P}(A_1)$, then there exists $i \in \{2, \ldots, k\}$ such that $\mathbb{P}(A_i \mid B) > \mathbb{P}(A_i)$.

Proof. Suppose $\mathbb{P}(A_i \mid B) \leq \mathbb{P}(A_i)$ for all $i \geq 2$
As $\sum_{i=1}^{k} \mathbb{P}(A_i \mid B) = \sum_{i=1}^{k} \mathbb{P}(A_i) = 1$, hence $1 = \sum_{i=1}^{k} \mathbb{P}(A_i \mid B) < \mathbb{P}(A_1) + \sum_{i=2}^{k} \mathbb{P}(A_i) = 1$, a contradiction.
Structural Commentary: Posterior mass is conserved across the partition.
Key Insight: A decrease in one cell must be offset by an increase in another. □

Example 1.1.13. *Transition Probabilities*
Let states U, M, L denote upper, middle, lower groups, and let $P = (p_{ij})$ with $p_{ij} = \mathbb{P}(j_2 \mid i_1)$:

	U_2	M_2	L_2
U_1	.45	.48	.07
M_1	.05	.70	.25
L_1	.01	.50	.49

For $\pi_1 = (.10, .40, .50)$, total probability gives $\pi_2 = \pi_1 P$ and $\mathbb{P}(U_2) = .45(.10) + .05(.40) + .01(.50) = .07$.
Bayes gives the inverse query $\mathbb{P}(U_1 \mid U_2) = \frac{\mathbb{P}(U_2|U_1)\mathbb{P}(U_1)}{\mathbb{P}(U_2)} = \frac{.45(.10)}{.07} \approx .64$.
Key Insight: Use πP forward and Bayes backward.

Example 1.1.14. *Base Rates in a Medical Test*
Let D be disease and T a positive test, with joint law

	D	D^c
T	0.0081	0.0900
T^c	0.0009	0.9010

Then $\mathbb{P}(T \mid D) = \frac{0.0081}{0.0081+0.0009} = 0.9$, $\mathbb{P}(T^c \mid D^c) = \frac{0.9010}{0.9010+0.0900} \approx 0.91$, $\mathbb{P}(D \mid T) = \frac{0.0081}{0.0081+0.0900} \approx 0.083$.
Key Insight: A rare condition can have small posterior probability after a positive test.

Example 1.1.15. *Rare Classes and Screening Errors*
Let L denote lying and $T = L^c$ truth, with $\mathbb{P}(+ \mid L) = .88$, $\mathbb{P}(+ \mid T) = .14$, $\mathbb{P}(L) = .01$, and $\mathbb{P}(T) = .99$.
Bayes gives $\mathbb{P}(T \mid +) = \frac{\mathbb{P}(+|T)\mathbb{P}(T)}{\mathbb{P}(+|T)\mathbb{P}(T)+\mathbb{P}(+|L)\mathbb{P}(L)} = \frac{(.14)(.99)}{(.14)(.99)+(.88)(.01)} \approx .94$.
Key Insight: Rare-class positives can mostly come from the common class.

Example 1.1.16. *Monty Hall*
The player chooses door 1. Let $\omega = (\omega_1, \omega_2)$, where ω_1 is the prize door and ω_2 is Monty's opened door.
Monty opens neither door 1 nor the prize door, and chooses uniformly from $\{2, 3\}$ when $\omega_1 = 1$.
The induced probabilities are $\mathbb{P}(1, 2) = \mathbb{P}(1, 3) = \frac{1}{6}$ and $\mathbb{P}(2, 3) = \mathbb{P}(3, 2) = \frac{1}{3}$.
Let $S = \{\omega_1 = 1\}$ be the event that staying wins.
Given $\omega_2 = 2$, the staying-win probability is $\mathbb{P}(S \mid \omega_2 = 2) = \frac{\frac{1}{6}}{\frac{1}{6}+\frac{1}{3}} = \frac{1}{3}$.
Thus switching wins with probability $\frac{2}{3}$, and the case $\omega_2 = 3$ is symmetric.
Key Insight: Once Monty's rule is part of the sample space, switching has conditional probability $\frac{2}{3}$.

Example 1.1.17. *Three Cards and Conditional Sampling*
Cards are GG, RR, and GR. Choose a card uniformly and then a side uniformly.
Let V be the event that the visible side is green and H the event that the hidden side is green.
Among the three equally likely green visible sides, two have green hidden side, so $\mathbb{P}(H \mid V) = \frac{2}{3}$.

Example 1.1.18. *Conditioning Changes the Reference Class*
Let $X \sim \text{Binomial}(5, \frac{1}{4})$ be number of blue-eyed children, Y be event that youngest child has blue eyes.

(i) Given $X \geq 1$, $\mathbb{P}(X \geq 3 \mid X \geq 1) = \frac{\binom{5}{3}(\frac{1}{4})^3(\frac{3}{4})^2 + \binom{5}{4}(\frac{1}{4})^4(\frac{3}{4}) + (\frac{1}{4})^5}{1-(\frac{3}{4})^5} = \frac{106}{781}$.

(ii) Given Y, $\mathbb{P}(X \geq 3 \mid Y) = 1 - \left(\frac{3}{4}\right)^4 - 4\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^3 = \frac{67}{256}$.

Key Insight: Similar information can determine different conditioning events.

Example 1.1.19. *Bayes with Several Categories*
Let $O \in \{M, W, L\}$ denote operating system with probabilities $(0.30, 0.50, 0.20)$.
Let I be infection, with $\mathbb{P}(I \mid O = M) = 0.65$, $\mathbb{P}(I \mid O = W) = 0.82$, and $\mathbb{P}(I \mid O = L) = 0.50$.
Bayes' theorem gives $\mathbb{P}(W \mid I) = \frac{0.82 \cdot 0.50}{0.65 \cdot 0.30 + 0.82 \cdot 0.50 + 0.50 \cdot 0.20} = \frac{82}{141}$.

Example 1.1.20. *Posterior and Predictive Probabilities*
Let C be uniform on $\{C_1, \ldots, C_5\}$, and suppose coin C_i has head probability p_i.
The head-probability vector is $(p_1, p_2, p_3, p_4, p_5) = \left(0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\right)$. Let H_n be the event that toss n is heads.
The posterior after one head is $\mathbb{P}(C_i \mid H_1) = \frac{p_i}{p_1+p_2+p_3+p_4+p_5} = \left(0, \frac{1}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{5}\right)_i$.
The predictive probability is $\mathbb{P}(H_2 \mid H_1) = \sum_{i=1}^{5} p_i \mathbb{P}(C_i \mid H_1) = \frac{3}{4}$.
If B_4 is the event that the first head occurs on toss 4,
the posterior is $\mathbb{P}(C_i \mid B_4) = \frac{(1-p_i)^3 p_i}{\sum_{j=1}^{5}(1-p_j)^3 p_j}$, and $\left(\mathbb{P}(C_i \mid B_4)\right)_{i=1}^{5} = \left(0, \frac{27}{46}, \frac{16}{46}, \frac{3}{46}, 0\right)$.

Key Insight: Bayes' theorem updates hidden parameters; prediction averages over the posterior.

Definition 1.1.21. *Conditional Mass and Density Functions*

(i) *Discrete*: if X, Y are discrete and $f_Y(y) > 0$, the conditional mass function is
$f_{X|Y}(x \mid y) := \mathbb{P}(X = x \mid Y = y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$.

(ii) *Continuous*: if X, Y have a joint density and $f_Y(y) > 0$, the conditional density is
$f_{X|Y}(x \mid y) := \frac{f_{X,Y}(x,y)}{f_Y(y)}$, so $\mathbb{P}(X \in A \mid Y = y) = \int_A f_{X|Y}(x \mid y)\, dx$ for measurable A.

Proposition 1.1.22. *Joint Factorizations and Total Probability for Conditional Laws*
Assume the displayed sums or integrals are well defined.

(i) If (X, Y) is discrete, $p_Y(y) > 0$, then $p_{X,Y}(x,y) = p_{X|Y}(x \mid y) p_Y(y)$, $p_X(x) = \sum_y p_{X|Y}(x \mid y) p_Y(y)$.

(ii) If (X, Y) has density, $f_X(x) > 0$, then $f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x)$, $f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y \mid x) f_X(x)\, dx$.

Proof.

(i) For $p_Y(y) > 0$, $p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$.
Hence $p_{X,Y}(x,y) = p_{X|Y}(x \mid y) p_Y(y)$ and $p_X(x) = \sum_y p_{X,Y}(x,y) = \sum_y p_{X|Y}(x \mid y) p_Y(y)$.

(ii) For $f_X(x) > 0$, $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$.
Hence $f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x)$ and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx = \int_{-\infty}^{\infty} f_{Y|X}(y \mid x) f_X(x)\, dx$.

Structural Commentary: Conditional laws disintegrate a joint law into a marginal law and a fiber law.
Key Insight: Total probability is marginalization after conditional factorization. □

Example 1.1.23. *Conditional Masses from the Three-Toss Table*
Let Y be the number of heads in three fair tosses, and let $X = 1$ iff the first toss is heads.
Since $p_Y(1) = \frac{3}{8}$, $p_{X,Y}(0,1) = \frac{2}{8}$, and $p_{X,Y}(1,1) = \frac{1}{8}$, one has $p_{X|Y}(0 \mid 1) = \frac{2}{3}$ and $p_{X|Y}(1 \mid 1) = \frac{1}{3}$.
Thus, given $Y = 1$, the first toss is more likely tails.

Example 1.1.24. *Uniform Square*
If $(X, Y) \sim \text{Uniform}([0,1]^2)$, then for $0 \leq y \leq 1$, $0 \leq x \leq 1$, $f_{X|Y}(x \mid y) = 1$. Hence $X \mid Y = y \sim \text{Uniform}(0, 1)$.

Example 1.1.25. *Conditioning in the Parabolic Region*
Let (X, Y) have density $f(x,y) = \frac{21}{4} x^2 y$, $x^2 \leq y \leq 1$. Then $f_X(x) = \frac{21}{8} x^2 (1 - x^4)$, $-1 \leq x \leq 1$.
For $x^2 \leq y \leq 1$, $f_{Y|X}(y \mid x) = \frac{\frac{21}{4} x^2 y}{\frac{21}{8} x^2 (1-x^4)} = \frac{2y}{1-x^4}$.
At $x = \frac{1}{2}$, $f_{Y|X}(y \mid \frac{1}{2}) = \frac{32y}{15}$, so $\mathbb{P}(Y \geq \frac{3}{4} \mid X = \frac{1}{2}) = \int_{\frac{3}{4}}^{1} \frac{32y}{15}\, dy = \frac{7}{15}$.

Example 1.1.26. *Poisson Thinning*
Let $\lambda > 0$, $0 \leq p \leq 1$, $N \sim \text{Poisson}(\lambda)$, and $X \mid N = n \sim \text{Binomial}(n, p)$.
For $k \in \mathbb{N}_0$, by total probability, $\mathbb{P}(X = k) = \sum_{n=k}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \binom{n}{k} p^k (1-p)^{n-k} = e^{-\lambda p} \frac{(\lambda p)^k}{k!}$, thus $X \sim \text{Poisson}(\lambda p)$.

Example 1.1.27. *A Triangular Exponential Joint Density*
Let $\lambda > 0$ and $f_{X,Y}(x,y) = \lambda^2 e^{-\lambda y} \mathbf{1}_{\{0 \leq x \leq y\}}$.
For $x \geq 0$, $f_X(x) = \int_x^{\infty} \lambda^2 e^{-\lambda y}\, dy = \lambda e^{-\lambda x}$; for $y \geq 0$, $f_Y(y) = \int_0^y \lambda^2 e^{-\lambda y}\, dx = \lambda^2 y e^{-\lambda y}$.
Hence $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \lambda e^{-\lambda(y-x)} \mathbf{1}_{[x,\infty)}(y)$ and $f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = y^{-1} \mathbf{1}_{[0,y]}(x)$.
Thus $Y \mid X = x \sim x + \text{Exp}(\lambda)$ and $X \mid Y = y \sim \text{Uniform}(0, y)$.

Example 1.1.28. *Bivariate Normal Conditional Density*
Let (X,Y) be bivariate normal with means μ_X,μ_Y, standard deviations $\sigma_X,\sigma_Y>0$, and correlation $\rho\in(-1,1)$.
For fixed x, $f_{Y\mid X}(y\mid x)=\frac{f_{X,Y}(x,y)}{f_X(x)}=\{\sigma_Y\sqrt{2\pi(1-\rho^2)}\}^{-1}\exp\left[-\frac{(y-\mu_Y-\rho\sigma_Y\sigma_X^{-1}(x-\mu_X))^2}{2\sigma_Y^2(1-\rho^2)}\right]$.
Thus $Y\mid X=x\sim N(\mu_Y+\rho\sigma_Y\sigma_X^{-1}(x-\mu_X),\sigma_Y^2(1-\rho^2))$; if $\rho=0$, this law is independent of x.

1.2 Conditional Expectation

Definition 1.2.1. *Conditional Expectation Given a Random Variable*
Let X,Y be random variables, and assume the relevant sums or integrals exist.
(i) *Discrete case*: If X,Y are discrete, then $\mathbb{E}(X\mid Y=y):=\sum_x x\,p_{X\mid Y}(x\mid y)$.
(ii) *Density case*: If X,Y have conditional density $f_{X\mid Y}$, then $\mathbb{E}(X\mid Y=y):=\int_{\mathbb{R}} x f_{X\mid Y}(x\mid y)\,dx$
(iii) *Measurable transforms*: For measurable r,

$$\mathbb{E}(r(X,Y)\mid Y=y)=\begin{cases}\sum_x r(x,y)p_{X\mid Y}(x\mid y), & \text{discrete case,}\\ \int_{\mathbb{R}} r(x,y)f_{X\mid Y}(x\mid y)\,dx, & \text{density case.}\end{cases}$$

(iv) *Random-variable form*: $\mathbb{E}(X\mid Y)$ is the function $g(Y)$ with $g(y)=\mathbb{E}(X\mid Y=y)$.

Example 1.2.2. *A Conditional Uniform Model*
Let $X\sim\text{Uniform}(0,1)$ and, conditionally on $X=x$, let $Y\sim\text{Uniform}(x,1)$.
For $0\le x<1$ and $x<y<1$, $f_{Y\mid X}(y\mid x)=\frac{1}{1-x}$, so $\mathbb{E}(Y\mid X=x)=\frac{1}{1-x}\int_x^1 y\,dy=\frac{1+x}{2}$; $\mathbb{E}(Y\mid X)=\frac{1+X}{2}$.

Theorem 1.2.3. *Tower Property*
Let X,Y be real-valued random variables whose joint law is either discrete or admits a density, and let $r:\mathbb{R}^2\to\mathbb{R}$ be measurable. If $\mathbb{E}|Y|<\infty$ and $\mathbb{E}|r(X,Y)|<\infty$, then $\mathbb{E}[\mathbb{E}(Y\mid X)]=\mathbb{E}Y$ and $\mathbb{E}[\mathbb{E}(r(X,Y)\mid X)]=\mathbb{E}(r(X,Y))$.

Proof. Let $h(x,y)$ denote either y or $r(x,y)$, and set $g_h(x):=\mathbb{E}(h(X,Y)\mid X=x)$.
In the density case, $g_h(x)=\int h(x,y)f_{Y\mid X}(y\mid x)\,dy$ and $f(x,y)=f_X(x)f_{Y\mid X}(y\mid x)$,
so Fubini gives $\mathbb{E}[g_h(X)]=\int\int h(x,y)f(x,y)\,dy\,dx=\mathbb{E}h(X,Y)$.
In the discrete case, $\mathbb{E}[g_h(X)]=\sum_x\sum_y h(x,y)p_{Y\mid X}(y\mid x)p_X(x)=\sum_x\sum_y h(x,y)p_{X,Y}(x,y)=\mathbb{E}h(X,Y)$.
Taking $h(x,y)=y$ and $h(x,y)=r(x,y)$ gives the two identities.
Structural Commentary: Conditional expectation first averages over fibers $\{X=x\}$, and the outer expectation averages those fiber means over the law of X.
Key Insight: Iterated expectation removes the conditioning variable without changing the total mean. □

Example 1.2.4. *Using the Tower Property*
Let $X\sim\text{Uniform}(0,1)$ and, conditionally on $X=x$, let $Y\sim\text{Uniform}(x,1)$.
Since $\mathbb{E}(Y\mid X)=\frac{1+X}{2}$ and $\mathbb{E}X=\frac12$, the tower property gives $\mathbb{E}Y=\mathbb{E}[\mathbb{E}(Y\mid X)]=\mathbb{E}\left[\frac{1+X}{2}\right]=\frac34$.

Proposition 1.2.5. *Pulling Out What Is Known*
Let X,Y be real-valued random variables whose joint law is either discrete or admits a density, and let $r,s:\mathbb{R}\to\mathbb{R}$ be measurable with $\mathbb{E}|s(Y)|<\infty$, $\mathbb{E}|r(X)s(Y)|<\infty$, and $\mathbb{E}|r(X)|<\infty$. Then $\mathbb{E}(r(X)s(Y)\mid X)=r(X)\mathbb{E}(s(Y)\mid X)$ and $\mathbb{E}(r(X)\mid X)=r(X)$.

Proof. Let K_x be the conditional law of Y given $X=x$, so $K_x(\{y\})=p_{Y\mid X}(y\mid x)$ in the discrete case and $K_x(dy)=f_{Y\mid X}(y\mid x)\,dy$ in the density case. Set $m_s(x):=\int s(y)K_x(dy)=\mathbb{E}(s(Y)\mid X=x)$, with the integral read as a sum in the discrete case. For admissible x, $\mathbb{E}(r(X)s(Y)\mid X=x)=\int r(x)s(y)K_x(dy)=r(x)m_s(x)$, and $\mathbb{E}(r(X)\mid X=x)=\int r(x)K_x(dy)=r(x)$. Evaluating at $x=X$ gives the identities.
Structural Commentary: Conditioning on X makes every function of X fixed inside remaining average over Y.
Key Insight: Known factors can be treated as constants under conditional expectation. □

Corollary 1.2.6. *Constant Conditional Mean Implies Uncorrelated*
Let $X,Y\in L^2$ and suppose $\mathbb{E}(X\mid Y)=c$ a.s. for some constant $c\in\mathbb{R}$. Then $c=\mathbb{E}X$ and $\text{Cov}(X,Y)=0$.

Proof. The tower property gives $\mathbb{E}X=\mathbb{E}[\mathbb{E}(X\mid Y)]=c$.
Since Y is known when conditioning on Y, $\mathbb{E}(XY\mid Y)=Y\mathbb{E}(X\mid Y)=cY$.
Taking expectations gives $\mathbb{E}(XY)=c\mathbb{E}Y=\mathbb{E}X\,\mathbb{E}Y$, hence $\text{Cov}(X,Y)=\mathbb{E}(XY)-\mathbb{E}X\,\mathbb{E}Y=0$.
Structural Commentary: A constant conditional mean kills covariance with all functions of Y, hence with Y.
Key Insight: Mean independence implies uncorrelatedness, but not conversely. □

Definition 1.2.7. *Conditional Variance*
Let X,Y be real-valued random variables with finite second conditional moment for each x under consideration.
(i) The conditional mean is $\mu(x):=\mathbb{E}(Y\mid X=x)$.
(ii) The conditional variance at x is $v(x):=\text{Var}(Y\mid X=x):=\mathbb{E}((Y-\mu(x))^2\mid X=x)$.
(iii) The conditional-variance random variable is $\text{Var}(Y\mid X):=v(X)$.

Theorem 1.2.8. *Law of Total Variance*
Let X,Y be real-valued random variables with $\mathbb{E}Y^2<\infty$. Set $b(x):=\mathbb{E}(Y\mid X=x)$ and $v(x):=\text{Var}(Y\mid X=x)$, and write $b(X)$ and $v(X)$ for the induced random variables. Then $\text{Var}(Y)=\mathbb{E}[v(X)]+\text{Var}(b(X))$.

Proof. Let $m:=\mathbb{E}Y$ and $e:=Y-b(X)$.
The tower property gives $\mathbb{E}b(X)=m$, and $\mathbb{E}(e\mid X)=\mathbb{E}(Y\mid X)-b(X)=0$.
Since $Y-m=e+(b(X)-m)$, expansion gives $\text{Var}(Y)=\mathbb{E}e^2+2\mathbb{E}[e(b(X)-m)]+\mathbb{E}(b(X)-m)^2$.
The cross term is zero because $\mathbb{E}[e(b(X)-m)]=\mathbb{E}[(b(X)-m)\mathbb{E}(e\mid X)]=0$.
Also $\mathbb{E}e^2=\mathbb{E}[\mathbb{E}(e^2\mid X)]=\mathbb{E}[v(X)]$, $\mathbb{E}(b(X)-m)^2=\text{Var}(b(X))$. Thus $\text{Var}(Y)=\mathbb{E}[v(X)]+\text{Var}(b(X))$.
Structural Commentary: Variance splits into within-fiber variance and between-fiber mean variation.
Key Insight: Total variance equals average conditional variance plus variance of conditional expectation. □

Example 1.2.9. *Random Binomial Proportion*
Let $Q \sim \text{Uniform}(0,1)$ and, conditionally on $Q = q$, let $X \sim \text{Binomial}(n, q)$.
Then $\mathbb{E}(X \mid Q) = nQ$, so $\mathbb{E}X = \mathbb{E}(nQ) = \frac{n}{2}$. Also $\text{Var}(X \mid Q) = nQ(1-Q)$.
Hence $\mathbb{E}[\text{Var}(X \mid Q)] = n\int_0^1 q(1-q)\,dq = \frac{n}{6}$, $\text{Var}(\mathbb{E}(X \mid Q)) = \text{Var}(nQ) = \frac{n^2}{12}$. Thus $\text{Var}(X) = \frac{n}{6} + \frac{n^2}{12}$.

1.3 Conditional Moment Methods and Prediction

Conditional expectation is not only a way to compute means. It is also the geometric projection rule behind random-sum formulas and least-squares prediction.

Proposition 1.3.1. *Random Sums*
Let $N \in \mathbb{N}_0$ a.s. with $\mathbb{E}N^2 < \infty$, and let $(X_i)_{i\geq 1}$ be iid, independent of N, with $\mathbb{E}X_1 = m$ and $\text{Var}(X_1) = v < \infty$.
Set $T = \sum_{i=1}^{N} X_i$, with $T = 0$ on $\{N = 0\}$.

(i) $\mathbb{E}T = m\mathbb{E}N$.
(ii) $\text{Var}(T) = v\mathbb{E}N + m^2\,\text{Var}(N)$.

Proof.

(i) On $\{N = n\}$, $T = \sum_{i=1}^{n} X_i$, so $\mathbb{E}(T \mid N = n) = nm$ and $\mathbb{E}(T \mid N) = mN$; hence $\mathbb{E}T = \mathbb{E}\mathbb{E}(T \mid N) = m\mathbb{E}N$.

(ii) On $\{N = n\}$, independence gives $\text{Var}(T \mid N = n) = nv$, so $\text{Var}(T \mid N) = vN$.
Total variance gives $\text{Var}(T) = \mathbb{E}(vN) + \text{Var}(mN) = v\mathbb{E}N + m^2\,\text{Var}(N)$.

Structural Commentary: The randomness of the number of summands contributes the second variance term.
Key Insight: First condition on how many summands exist; then average over that count. □

Proposition 1.3.2. *MGF of a Random Sum*
Let $T = \sum_{i=1}^{N} X_i$, where $N \in \mathbb{N}_0$ is independent of iid $(X_i)_{i\geq 1}$.
If ψ_N and ψ_X are finite at the relevant arguments, then $\psi_T(t) = \psi_N(\log \psi_X(t))$.

Proof. On $\{N = n\}$, independence gives $\mathbb{E}(e^{tT} \mid N = n) = \mathbb{E}e^{t\sum_{i=1}^{n} X_i} = \prod_{i=1}^{n} \psi_X(t) = [\psi_X(t)]^n$.
Hence $\psi_T(t) = \mathbb{E}[\psi_X(t)^N] = \mathbb{E}e^{N\log\psi_X(t)} = \psi_N(\log\psi_X(t))$.
Structural Commentary: Conditioning on N turns the random sum into a fixed product of MGFs.
Key Insight: Random counts compose MGFs by substituting $\log\psi_X(t)$ into ψ_N. □

Example 1.3.3. *Compound Poisson Count*
Let $N \sim \text{Poisson}(\mu)$ and, given N, let $T = \sum_{i=1}^{N} X_i$ with iid $X_i \sim \text{Poisson}(\lambda)$ independent of N.
Since $\psi_N(s) = \exp(\mu(e^s - 1))$ and $\psi_X(t) = \exp(\lambda(e^t - 1))$, one has $\psi_T(t) = \psi_N(\log\psi_X(t)) = \exp(\mu(e^{\lambda(e^t-1)} - 1))$.

Proposition 1.3.4. *Best Constant Predictor*
For $Y \in L^2$, $\arg\min_{c\in\mathbb{R}} \mathbb{E}(Y-c)^2 = \mathbb{E}Y$, and $\min_{c\in\mathbb{R}} \mathbb{E}(Y-c)^2 = \text{Var}(Y)$.

Proof. Let $\mu = \mathbb{E}Y$. Since $\mathbb{E}(Y-\mu) = 0$, $\mathbb{E}(Y-c)^2 = \mathbb{E}[(Y-\mu)+(\mu-c)]^2 = \text{Var}(Y) + (\mu-c)^2$.
Since $(\mu-c)^2 \geq 0$ with equality iff $c = \mu$, the unique minimizer is $c = \mu$.
Structural Commentary: The mean is the orthogonal projection of Y onto the constants.
Key Insight: Best constant prediction equals the average. □

Theorem 1.3.5. *Conditional Expectation as Best Predictor*
Let $X, Y \in L^2$ and $\mathcal{G} := \{h(X) : h(X) \in L^2\}$. Let $M := \mathbb{E}(Y \mid X)$, then $M \in \mathcal{G}$, $\mathbb{E}(Y-M)^2 \leq \mathbb{E}(Y-Z)^2 \forall Z \in \mathcal{G}$.

Proof. For $Z \in \mathcal{G}$, both M and Z are $\sigma(X)$-measurable.
Since $\mathbb{E}(Y - M \mid X) = 0$, $\mathbb{E}[(Y-M)(M-Z)] = \mathbb{E}[(M-Z)\mathbb{E}(Y-M \mid X)] = 0$.
Hence $\mathbb{E}(Y-Z)^2 = \mathbb{E}(Y-M)^2 + \mathbb{E}(M-Z)^2 \geq \mathbb{E}(Y-M)^2$.
Structural Commentary: Prediction from X is performed fiber by fiber over the conditional laws of Y.
Key Insight: The optimal nonlinear least-squares predictor is the conditional mean. □

Definition 1.3.6. *Best Linear Predictor*
For $X, Y \in L^2$, set $\mathcal{A}_X := \{\alpha + \beta X : \alpha, \beta \in \mathbb{R}\}$.
A *best linear predictor* of Y from X is any $\widehat{Y}^* \in \arg\min_{Z\in\mathcal{A}_X} \mathbb{E}(Y-Z)^2$.

Theorem 1.3.7. *Best Linear Predictor*
Let $X, Y \in L^2$, $\sigma_X^2 := \text{Var}(X) > 0$, $\mu_X := \mathbb{E}X$, and $\mu_Y := \mathbb{E}Y$.

(i) $\arg\min_{Z\in\mathcal{A}_X} \mathbb{E}(Y-Z)^2 = \{\alpha^* + \beta^* X\}$, *where* $\beta^* = \frac{\text{Cov}(X,Y)}{\sigma_X^2}$ *and* $\alpha^* = \mu_Y - \beta^*\mu_X$.

(ii) $\min_{Z\in\mathcal{A}_X} \mathbb{E}(Y-Z)^2 = \text{Var}(Y) - \frac{\text{Cov}(X,Y)^2}{\sigma_X^2}$.

(iii) If $\text{Var}(Y) > 0$ *and* $\rho = \text{Corr}(X,Y)$, *then the minimum error is* $\text{Var}(Y)(1-\rho^2)$.

Proof.

(i) Let $q(\alpha,\beta) := \mathbb{E}(Y-\alpha-\beta X)^2$. For fixed β, $q(\alpha,\beta) = \text{Var}(Y-\beta X) + (\mu_Y - \alpha - \beta\mu_X)^2$,
so $\alpha(\beta) = \mu_Y - \beta\mu_X$ uniquely minimizes over α.

(ii) Then $q(\alpha(\beta),\beta) = \mathrm{Var}(Y-\beta X) = \mathrm{Var}(Y) + \beta^2\sigma_X^2 - 2\beta\,\mathrm{Cov}(X,Y)$.
Since $\sigma_X^2 > 0$, this quadratic has unique minimizer $\beta^* = \frac{\mathrm{Cov}(X,Y)}{\sigma_X^2}$.

(iii) Substituting β^* gives $\mathrm{Var}(Y) - \frac{\mathrm{Cov}(X,Y)^2}{\sigma_X^2}$. If $\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$, then this equals $\mathrm{Var}(Y)(1-\rho^2)$.

Structural Commentary: Linear prediction projects Y onto $\mathcal{A}_X$.
Key Insight: The optimal slope is normalized covariance. □

Example 1.3.8. *Bivariate Normal Prediction*
If (X,Y) is bivariate normal with means μ_X,μ_Y, standard deviations $\sigma_X,\sigma_Y > 0$, and correlation $\rho \in (-1,1)$, then $Y \mid X \sim N(\mu_Y + \rho\sigma_Y\sigma_X^{-1}(X-\mu_X), \sigma_Y^2(1-\rho^2))$.
Thus $\mathbb{E}(Y \mid X) = \mu_Y + \rho\sigma_Y\sigma_X^{-1}(X-\mu_X)$ is linear, with conditional variance $\sigma_Y^2(1-\rho^2)$.

Example 1.3.9. *Regression to the Mean*
Let $\mathbb{E}X = \mathbb{E}Y = \mu$, $\mathrm{Var}(X) = \mathrm{Var}(Y) = \sigma^2 > 0$, and $\mathrm{Corr}(X,Y) = \rho$. Best linear predictor is $\widehat{Y} = \mu + \rho(X-\mu)$.
For $0 < \rho < 1$, $|\widehat{Y}-\mu| = \rho|X-\mu| < |X-\mu|$, so predicted scores regress toward μ.

2 Probabilistic Convergence

2.1 Modes of Probabilistic Convergence

Pointwise convergence is too strict for most random sequences. The useful replacement is to ask whether probabilities, laws, or moments of the errors become small.

Definition 2.1.1. *Modes of Probabilistic Convergence*
Let $(X_n)_{n\ge1}$ and X be real-valued random variables with distribution functions F_n and F.

(i) *Convergence in probability*: $X_n \xrightarrow{\mathbb{P}} X$ iff $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$.
(ii) *Convergence in distribution*: $X_n \rightsquigarrow X$ iff $F_n(t) \to F(t)$ at every continuity point t of F.
(iii) *Convergence in L^p*: for $p \ge 1$, $X_n \xrightarrow{L^p} X$ iff $\mathbb{E}\,|X_n - X|^p \to 0$.
(iv) *Convergence in quadratic mean*: $X_n \xrightarrow{qm} X$ iff $X_n \xrightarrow{L^2} X$.
(v) *Almost-sure convergence*: $X_n \xrightarrow{\text{a.s.}} X$ iff $\mathbb{P}(\{\omega \in \Omega : X_n(\omega) \to X(\omega)\}) = 1$.
(vi) *Constant limits*: if $X = c$ almost surely for a constant c, then $X_n \xrightarrow{\mathbb{P}} c$, $X_n \rightsquigarrow c$, $X_n \xrightarrow{L^p} c$, $X_n \xrightarrow{qm} c$, and $X_n \xrightarrow{\text{a.s.}} c$ denote convergence to that constant random variable in the corresponding modes.

Example 2.1.2. *A Shrinking Normal Law*
Let $X_n \sim N(0, \frac{1}{n})$, and let F be the distribution function of δ_0.
Since $\sqrt{n}X_n \sim N(0,1)$, $F_n(t) = \mathbb{P}(X_n \le t) = \Phi(\sqrt{n}t)$, so $F_n(t) \to 0$ for $t < 0$ and $F_n(t) \to 1$ for $t > 0$.
Hence $F_n(t) \to F(t)$ at every continuity point $t \ne 0$ of F, and therefore $X_n \rightsquigarrow 0$.

Theorem 2.1.3. *Basic Implications Between Convergence Modes*
Let X_n and X be real-valued random variables with distribution functions F_n and F, and let $c \in \mathbb{R}$.

(i) $X_n \xrightarrow{qm} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$.
(ii) $X_n \xrightarrow{\mathbb{P}} X \Rightarrow X_n \rightsquigarrow X$.
(iii) $X_n \rightsquigarrow c \Rightarrow X_n \xrightarrow{\mathbb{P}} c$.
(iv) $X_n \xrightarrow{\text{a.s.}} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$.
(v) $X_n \xrightarrow{L^1} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$, *and* $X_n \xrightarrow{qm} X \Rightarrow X_n \xrightarrow{L^1} X$.

Proof.

(i) If $X_n \xrightarrow{qm} X$, then for each $\varepsilon > 0$, Markov gives $\mathbb{P}(|X_n - X| > \varepsilon) \le \varepsilon^{-2}\mathbb{E}\,|X_n - X|^2 \to 0$.
(ii) Let x be a continuity point of F.
For each $\varepsilon > 0$, $F(x-\varepsilon) - \mathbb{P}(|X_n - X| > \varepsilon) \le F_n(x) \le F(x+\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon)$,
so $F(x-\varepsilon) \le \liminf_n F_n(x) \le \limsup_n F_n(x) \le F(x+\varepsilon)$; letting $\varepsilon \downarrow 0$ gives $F_n(x) \to F(x)$.
(iii) For $\varepsilon > 0$, the distribution function of c is continuous at $c-\varepsilon$ and $c+\varepsilon$.
Hence $\mathbb{P}(|X_n - c| > \varepsilon) \le F_n(c-\varepsilon) + 1 - F_n(c+\varepsilon) \to 0$.
(iv) Fix $\varepsilon > 0$ and set $Y_n = \mathbf{1}_{\{|X_n - X| > \varepsilon\}}$. If $X_n \to X$ a.s., then $Y_n \to 0$ a.s. and $0 \le Y_n \le 1$.
Hence dominated convergence gives $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{E}Y_n \to 0$.
(v) If $X_n \xrightarrow{L^1} X$, then $\mathbb{P}(|X_n - X| > \varepsilon) \le \varepsilon^{-1}\mathbb{E}\,|X_n - X| \to 0$ by Markov.
If $X_n \xrightarrow{qm} X$, then $\mathbb{E}\,|X_n - X| \le (\mathbb{E}\,|X_n - X|^2)^{\frac{1}{2}} \to 0$ by Cauchy–Schwarz.

Structural Commentary: Strong modes give tail bounds; distribution convergence gives c.d.f. control.
Key Insight: Convert to tail bounds; use c.d.f. squeezes when laws enter. □

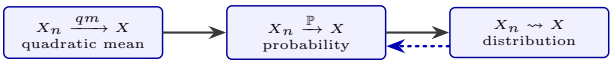

Figure 230: Solid: always; dotted: point-mass limit.

Example 2.1.4. *Why the Reverse Arrows Fail*

(i) Let $U \sim \mathrm{Uniform}(0,1)$ and $X_n = \sqrt{n}\mathbf{1}_{(0,\frac{1}{n})}(U)$. Then for each $\varepsilon > 0$, $\mathbb{P}(|X_n| > \varepsilon) \le \frac{1}{n} \to 0$,
so $X_n \xrightarrow{\mathbb{P}} 0$, but $\mathbb{E}X_n^2 = 1$ for all n; hence $X_n \not\to 0$ in quadratic mean.

(ii) Let $X \sim N(0,1)$ and set $X_n = -X$. Then $X_n \sim X$ for all n, so $X_n \rightsquigarrow X$,
but for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(2|X| > \varepsilon) > 0$ for all n; hence $X_n \not\to X$ in probability.

Remark 2.1.5. *A Warning About Expectations*
Convergence in probability does not imply convergence of expectations.
If $\mathbb{P}(X_n = n^2) = \frac{1}{n}$, $\mathbb{P}(X_n = 0) = 1 - \frac{1}{n}$, then $\forall \varepsilon > 0$, $\mathbb{P}(|X_n| > \varepsilon) = \frac{1}{n} \to 0$, so $X_n \xrightarrow{\mathbb{P}} 0$, $\mathbb{E}X_n = n \to +\infty$.

Definition 2.1.6. *Asymptotic Uniform Integrability*
A sequence (X_n) is *asymptotically uniformly integrable* if $\lim_{M\to\infty} \limsup_{n\to\infty} \mathbb{E}(|X_n| \mathbf{1}_{\{|X_n|>M\}}) = 0$.

Proposition 2.1.7. *Recovering Expectations from Probability Convergence*
Let $b \in \mathbb{R}$. If $X_n \xrightarrow{\mathbb{P}} b$ and (X_n) is asymptotically uniformly integrable, then $\mathbb{E}X_n \to b$.

Proof. For $M > |b|$, set $\tau_M(x) = (-M) \vee (x \wedge M)$. As τ_M bounded, continuous; $\tau_M(b) = b$, $\tau_M(X_n) \xrightarrow{\mathbb{P}} b$.
For $\eta > 0$, $\mathbb{E}|\tau_M(X_n) - b| \le \eta + 2M\mathbb{P}(|\tau_M(X_n) - b| > \eta)$, so $\mathbb{E}\tau_M(X_n) \to b$ after $n \to \infty$ and $\eta \downarrow 0$.
Moreover, $|X_n - \tau_M(X_n)| \le |X_n| \mathbf{1}_{\{|X_n|>M\}}$, hence $\limsup_n |\mathbb{E}X_n - b| \le \limsup_n \mathbb{E}(|X_n| \mathbf{1}_{\{|X_n|>M\}})$.
Letting $M \to \infty$ and using asymptotic uniform integrability gives $\mathbb{E}X_n \to b$.
Structural Commentary: Truncation makes convergence bounded; asymptotic uniform integrability kills tails.
Key Insight: Probability convergence handles the truncations; tail control restores expectations. □

Theorem 2.1.8. *Continuous Mapping and Slutsky Rules*
Let $g : \mathbb{R} \to \mathbb{R}$ be continuous, and let $c \in \mathbb{R}$.

(i) $X_n \xrightarrow{\mathbb{P}} X \Rightarrow g(X_n) \xrightarrow{\mathbb{P}} g(X)$.
(ii) $X_n \rightsquigarrow X \Rightarrow g(X_n) \rightsquigarrow g(X)$.
(iii) $X_n \rightsquigarrow X$ *and* $Y_n \xrightarrow{\mathbb{P}} c \Rightarrow X_n + Y_n \rightsquigarrow X + c$ *and* $X_n Y_n \rightsquigarrow cX$.
(iv) $X_n \xrightarrow{\mathbb{P}} X$ *and* $Y_n \xrightarrow{\mathbb{P}} Y \Rightarrow X_n + Y_n \xrightarrow{\mathbb{P}} X + Y$ *and* $X_n Y_n \xrightarrow{\mathbb{P}} XY$.

Proof.
(i) Fix $\varepsilon, \eta > 0$. Choose K, $\mathbb{P}(|X| > K) < \eta$, $\delta \le 1$ s.t. $|x - y| < \delta$, $|x|, |y| \le K + 1$; $|g(x) - g(y)| < \varepsilon$.
Then $\mathbb{P}(|g(X_n) - g(X)| > \varepsilon) \le \mathbb{P}(|X| > K) + \mathbb{P}(|X_n - X| > \delta)$, so $\limsup$ at most η.
(ii) Let t be a continuity point of the law of $g(X)$ and set $A_t = \{x : g(x) \le t\}$.
Since $\partial A_t \subseteq \{x : g(x) = t\}$ and $\mathbb{P}(g(X) = t) = 0$, the continuity-set form of weak convergence gives
$\mathbb{P}(g(X_n) \le t) = \mathbb{P}(X_n \in A_t) \to \mathbb{P}(X \in A_t) = \mathbb{P}(g(X) \le t)$.
(iii) If $Z_n \rightsquigarrow Z$ and $R_n \xrightarrow{\mathbb{P}} 0$, then $Z_n + R_n \rightsquigarrow Z$ by the c.d.f. squeeze
$F_{Z_n}(t - \eta) - \mathbb{P}(|R_n| > \eta) \le F_{Z_n + R_n}(t) \le F_{Z_n}(t + \eta) + \mathbb{P}(|R_n| > \eta)$,
followed by $n \to \infty$ and $\eta \downarrow 0$ through continuity points of F_Z.
Since $X_n + c \rightsquigarrow X + c$ by (ii), applying this with $Z_n = X_n + c$ and $R_n = Y_n - c$ gives $X_n + Y_n \rightsquigarrow X + c$.
Also $X_n \rightsquigarrow X$ implies (X_n) is tight, so $\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \le \mathbb{P}(|X_n| > K) + \mathbb{P}(|Y_n - c| > \frac{\varepsilon}{K})$.
Letting $n \to \infty$, then $K \uparrow \infty$ gives $X_n(Y_n - c) \xrightarrow{\mathbb{P}} 0$.
Since $cX_n \rightsquigarrow cX$ by (ii), the additive perturbation just proved gives $X_n Y_n = cX_n + X_n(Y_n - c) \rightsquigarrow cX$.
(iv) For sums, $\mathbb{P}(|(X_n + Y_n) - (X + Y)| > \varepsilon) \le \mathbb{P}(|X_n - X| > \frac{\varepsilon}{2}) + \mathbb{P}(|Y_n - Y| > \frac{\varepsilon}{2}) \to 0$.
For products, (X_n) and Y are tight, $\mathbb{P}(|X_n(Y_n - Y)| > \frac{\varepsilon}{2}) \le \mathbb{P}(|X_n| > K) + \mathbb{P}(|Y_n - Y| > \frac{\varepsilon}{2K})$,
while $\mathbb{P}(|Y(X_n - X)| > \frac{\varepsilon}{2}) \le \mathbb{P}(|Y| > K) + \mathbb{P}(|X_n - X| > \frac{\varepsilon}{2K})$; let $n \to \infty$ and then $K \uparrow \infty$.
Structural Commentary: Continuity transports convergence; Slutsky absorbs probability-small errors.
Key Insight: Control the bounded high-probability part; let the tails vanish. □

2.2 Probability on Metric Spaces

3 Limit Theorems

3.1 Laws of Large Numbers

The law of large numbers formalizes the idea that a large sample average is close to the mean of the distribution.

Perspective 3.1.1. *Law of the Sample Mean*
If $X_1, \ldots, X_n$ are IID, mean μ, variance σ^2, then $\overline{X}_n := n^{-1} \sum_{i=1}^n X_i$ is a random variable with law $\mathcal{L}(\overline{X}_n)$.
By Theorem 3.2.7, $\mathbb{E}\overline{X}_n = \mu$ and $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$, so $\mathcal{L}(\overline{X}_n)$ concentrates around μ as $n \to \infty$.

Theorem 3.1.2. *Weak Law of Large Numbers*
Let $(X_i)_{i\ge1}$ be IID real-valued random variables with $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 < +\infty$. Set $\overline{X}_n := n^{-1} \sum_{i=1}^n X_i$.
Then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$, equivalently $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \to 0$ for every $\varepsilon > 0$.

Proof. For $n \ge 1$, $\mathbb{E}\overline{X}_n = n^{-1} \sum_{i=1}^n \mathbb{E}X_i = \mu$ and, by independence, $\mathrm{Var}(\overline{X}_n) = n^{-2} \sum_{i=1}^n \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$.
For every $\varepsilon > 0$, Chebyshev gives $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \le \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0$, hence $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.
Structural Commentary: Averaging keeps the mean fixed and divides variance by n.
Key Insight: Chebyshev turns vanishing variance into convergence in probability. □

Proposition 3.1.3. *Weak LLN with Vanishing Average Variance*
Let $(X_i)_{i\ge1}$ be independent, $X_i \in L^2$, with $\mu_i := \mathbb{E}X_i$ and $\sigma_i^2 := \mathrm{Var}(X_i)$.
Set $\overline{X}_n := n^{-1} \sum_{i=1}^n X_i$ and $m_n := n^{-1} \sum_{i=1}^n \mu_i$. If $m_n \to \mu$ and $n^{-2} \sum_{i=1}^n \sigma_i^2 \to 0$, then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. $\mathbb{E}\overline{X}_n = m_n$. By independence, $\mathrm{Var}(\overline{X}_n) = n^{-2}\sum_{i=1}^{n}\sigma_i^2 \to 0$. Fix $\varepsilon > 0$. $\forall$ large n, $|m_n - \mu| < \frac{\varepsilon}{2}$.

Hence $\{|\overline{X}_n - \mu| > \varepsilon\} \subseteq \{|\overline{X}_n - m_n| > \frac{\varepsilon}{2}\}$. By Chebyshev, $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \le \frac{4\,\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} \to 0$.
Structural Commentary: Identical distribution is not essential for a weak law.
Key Insight: It is enough that the mean target stabilises and the average variance vanishes. □

Example 3.1.4. *Coin Proportions and Chebyshev Scale*

Let $X_i \sim \mathrm{Bernoulli}(p)$ be IID head indicators, $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$. Then $\mathbb{E}\overline{X}_n = p$, $\mathrm{Var}(\overline{X}_n) = \frac{p(1-p)}{n}$, $\overline{X}_n \xrightarrow{\mathbb{P}} p$.

For $p = \frac{1}{2}$, Chebyshev gives $\mathbb{P}(0.4 \le \overline{X}_n \le 0.6) \ge 1 - \frac{1}{4n(0.1)^2} = 1 - \frac{25}{n}$, at least 0.7 for $n \ge 84$.

If $n = 10000$ fair tosses produce 5067 heads, then $\overline{X}_n = 0.5067$, while $\mathrm{Var}(\overline{X}_n) = \frac{1}{4n} = 2.5 \cdot 10^{-5}$ and $\mathrm{sd}(\overline{X}_n) = 0.005$. The observed deviation is about 1.34 standard deviations from $\frac{1}{2}$.

Theorem 3.1.5. *Strong Law of Large Numbers*

Let $(X_i)_{i\ge1}$ be IID real-valued random variables with $X_1 \in L^1$, $\mu := \mathbb{E}X_1$, and $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$.

Then $\overline{X}_n \xrightarrow{\text{a.s.}} \mu$, equivalently $\mathbb{P}(\lim_{n\to\infty} \overline{X}_n = \mu) = 1$.

Proof. After replacing X_i by $X_i - \mu$, assume $\mu = 0$. Set $Y_i := X_i \mathbf{1}_{\{|X_i| \le i\}}$.

Since $\sum_{i=1}^{\infty}\mathbb{P}(X_i \ne Y_i) = \sum_{i=1}^{\infty}\mathbb{P}(|X_1| > i) < +\infty$, Borel–Cantelli gives $X_i = Y_i$ eventually a.s.

Hence $n^{-1}\sum_{i=1}^{n}(X_i - Y_i) \to 0$ a.s. For $Z_i := Y_i - \mathbb{E}Y_i$, independence gives

$\sum_{i=1}^{\infty}\frac{\mathrm{Var}(Z_i)}{i^2} \le \sum_{i=1}^{\infty}\frac{\mathbb{E}(X_1^2\mathbf{1}_{\{|X_1|\le i\}})}{i^2} < +\infty$, by Kolmogorov's variance criterion, $n^{-1}\sum_{i=1}^{n} Z_i \to 0$ a.s.

By dominated convergence, $\mathbb{E}Y_i \to 0$, hence Cesaro averaging gives $n^{-1}\sum_{i=1}^{n}\mathbb{E}Y_i \to 0$.

Combining $X_i - Y_i$, Z_i, and $\mathbb{E}Y_i$ gives $\overline{X}_n \to 0$ a.s.; adding back μ completes the proof.
Structural Commentary: Truncate tails; centered truncations have summable variance.
Key Insight: Replace by an eventually equal truncated sequence, then average the centered part. □

Corollary 3.1.6. *Strong Implies Weak*

Let $(X_i)_{i\ge1}$ be IID real-valued random variables, $X_1 \in L^1$, $\mu := \mathbb{E}X_1$, $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$. Then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. By Theorem 3.1.5, $\overline{X}_n \to \mu$ a.s. For $\varepsilon > 0$, set $Y_n := \mathbf{1}_{\{|\overline{X}_n - \mu| > \varepsilon\}}$.

Then $Y_n \to 0$ a.s. and $0 \le Y_n \le 1$, so dominated convergence gives $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) = \mathbb{E}Y_n \to 0$.
Structural Commentary: Almost-sure convergence gives pointwise tail indicators tending to zero.
Key Insight: Dominate the indicators by 1 and integrate. □

Proposition 3.1.7. *Consistency of Empirical Moments and Variance*
Let $(X_i)_{i\ge1}$ be iid, $X_1 \in L^2$, $\mu := \mathbb{E}X_1$, and $\sigma^2 := \mathrm{Var}(X_1)$.

For $n \ge 2$, set $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$ and $S_n^2 := (n-1)^{-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$. Then $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2$.

More generally, if $X_1 \in L^r$ for an integer $r \ge 1$, then $n^{-1}\sum_{i=1}^{n} X_i^r \xrightarrow{\mathbb{P}} \mathbb{E}X_1^r$.

Proof. Since $(X_i^r)_{i\ge1}$ is iid and integrable, the strong law gives $n^{-1}\sum_{i=1}^{n} X_i^r \to \mathbb{E}X_1^r$ a.s., hence in probability.

Let $m_2 := \mathbb{E}X_1^2 = \sigma^2 + \mu^2$. The cases $r = 1, 2$ give $\overline{X}_n \to \mu$ and $n^{-1}\sum_{i=1}^{n} X_i^2 \to m_2$ a.s.

Since $S_n^2 = \frac{n}{n-1}(n^{-1}\sum_{i=1}^{n} X_i^2 - \overline{X}_n^2)$, continuous mapping gives $S_n^2 \to m_2 - \mu^2 = \sigma^2$ a.s., hence in probability.
Structural Commentary: Sample moments are averages of transformed observations.
Key Insight: Apply the strong law to X_i^r, then pass through the continuous variance formula. □

Remark 3.1.8. *Why Integrability Matters*
If $X_1 \notin L^1$, the mean $\mathbb{E}X_1$ need not exist and sample averages need not converge to any finite deterministic limit; integrability is the condition that makes a law of large numbers meaningful.

3.2 Central Limit Theory

The law of large numbers says that $\overline{X}_n$ is close to μ for large n. The central limit theorem describes the scale and shape of the remaining fluctuation.

Theorem 3.2.1. *Central Limit Theorem*
Let $(X_i)_{i\ge1}$ be IID real-valued random variables, $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 \in (0, +\infty)$.

Let $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$, $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$.

Then $Z_n \rightsquigarrow N(0,1)$, equivalently $\mathbb{P}(Z_n \le z) \to \Phi(z) := \int_{-\infty}^{z}(2\pi)^{-\frac{1}{2}}e^{-\frac{x^2}{2}}\,dx$ $\forall z \in \mathbb{R}$.

Proof. Set $Y_i := \frac{X_i-\mu}{\sigma}$, so $\mathbb{E}Y_i = 0$, $\mathrm{Var}(Y_i) = 1$, and $Z_n = n^{-\frac{1}{2}} \sum_{i=1}^{n} Y_i$.

For $\varphi(t) := \mathbb{E}e^{\mathrm{i}tY_1}$, the finite second moment expansion gives $\varphi(u) = 1 - \frac{u^2}{2} + o(u^2)$ as $u \to 0$.

By independence, $\varphi_{Z_n}(t) = [\varphi(\frac{t}{\sqrt{n}})]^n = [1 - \frac{t^2}{2n} + o(\frac{1}{n})]^n \to e^{-\frac{t^2}{2}}$ for every $t \in \mathbb{R}$.

Since $e^{-\frac{t^2}{2}}$ is the characteristic function of $N(0,1)$, Levy's continuity theorem gives $Z_n \rightsquigarrow N(0,1)$; continuity of Φ gives the displayed c.d.f. convergence.

Structural Commentary: Centering and scaling turn the sum into unit-variance IID increments.

Key Insight: Characteristic functions turn normalized sums into powers, whose limit is Gaussian. □

Method 3.2.2. *Equivalent CLT Approximation Forms*

Let $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$ with mean μ and variance $\sigma^2 \in (0, +\infty)$, and set $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$.

(i) Standardized form: $Z_n \approx N(0,1)$.
(ii) Centered form: $\sqrt{n}(\overline{X}_n - \mu) \approx N(0, \sigma^2)$.
(iii) Sample-mean form: $\overline{X}_n \approx N(\mu, \frac{\sigma^2}{n})$.

Use any equivalent form to approximate probabilities involving $\overline{X}_n$.

Example 3.2.3. *Measurement Error and Sample Size*

Let $X_1, \ldots, X_n$ be independent measurements with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$.
For $c > 0$, the CLT gives $\mathbb{P}(|\overline{X}_n - \mu| < c) \approx \Phi(\frac{c\sqrt{n}}{\sigma}) - \Phi(\frac{-c\sqrt{n}}{\sigma})$.
For $(n, \sigma, c) = (16, 1, 0.5)$, this is $\Phi(2) - \Phi(-2) \approx 0.954$.

Lemma 3.2.4. *MGF Continuity Theorem*

Let Z_n, Z be real-valued with MGFs $\psi_n(t) := \mathbb{E}e^{tZ_n}$ and $\psi(t) := \mathbb{E}e^{tZ}$ finite on $(-a, a)$ for some $a > 0$.
If $\psi_n(t) \to \psi(t)$ for all $t \in (-a, a)$, then $Z_n \rightsquigarrow Z$.

Proof. For each $s \in \mathbb{R}$, the characteristic functions satisfy $\varphi_n(s) = \mathbb{E}e^{\mathrm{i}sZ_n} = \psi_n(\mathrm{i}s)$ by analytic continuation on the strip $\{z : |\mathrm{Re}\, z| < a\}$.
Pointwise MGF convergence near 0 gives $\varphi_n(s) \to \varphi(s)$, the characteristic function of Z; Levy's continuity theorem yields $Z_n \rightsquigarrow Z$.

Structural Commentary: Local MGF convergence determines local characteristic-function convergence.

Key Insight: MGFs near 0 identify the weak limit. □

Proposition 3.2.5. *MGF Proof of the IID CLT*

Let $(X_i)_{i\geq 1}$ be iid with $\mathbb{E}X_1 = 0$, $\mathrm{Var}(X_1) = \sigma^2 \in (0, +\infty)$, and MGF $M(t) := \mathbb{E}e^{tX_1}$ finite near 0.
Set $S_n := \sum_{i=1}^{n} X_i$ and $Z_n := \frac{S_n}{\sigma\sqrt{n}}$. Then $Z_n \rightsquigarrow N(0,1)$, equivalently $\mathbb{P}(Z_n \leq x) \to \Phi(x)$ for all $x \in \mathbb{R}$.

Proof. For fixed t near 0, independence gives $M_{Z_n}(t) = \mathbb{E}e^{\frac{tS_n}{\sigma\sqrt{n}}} = \prod_{i=1}^{n} \mathbb{E}e^{\frac{tX_i}{\sigma\sqrt{n}}} = [M(\frac{t}{\sigma\sqrt{n}})]^n$.

Since $M(0) = 1$, $M'(0) = 0$, and $M''(0) = \sigma^2$, Taylor expansion gives $M(s) = 1 + \frac{\sigma^2 s^2}{2} + o(s^2)$ as $s \to 0$.

Hence $M_{Z_n}(t) = (1 + \frac{t^2}{2n} + o(\frac{1}{n}))^n \to e^{\frac{t^2}{2}}$, the MGF of $N(0,1)$.
Lemma 3.2.4 gives $Z_n \rightsquigarrow N(0,1)$, and continuity of Φ gives the c.d.f. convergence.

Structural Commentary: The MGF proof needs the stronger near-zero MGF assumption.

Key Insight: Standardization gives $\frac{t^2}{2n}$; independence exponentiates. □

Example 3.2.6. *A Poisson Error-Count Approximation*

Let $X_1, \ldots, X_{125} \sim \mathrm{Poisson}(5)$ be IID and set $\overline{X}_{125} = 125^{-1} \sum_{i=1}^{125} X_i$.

Since $\mu = 5$ and $\sigma^2 = 5$, the CLT gives $\mathbb{P}(\overline{X}_{125} < 5.5) \approx \Phi(\frac{\sqrt{125}(5.5-5)}{\sqrt{5}}) = \Phi(2.5) \approx 0.9938$.

Example 3.2.7. *Normal Approximation to a Poisson Count*

Let $X_\lambda \sim \mathrm{Poisson}(\lambda)$, $Z_\lambda := \frac{X_\lambda - \lambda}{\sqrt{\lambda}}$. Then $M_{Z_\lambda}(t) = \exp(-t\sqrt{\lambda} + \lambda(e^{\frac{t}{\sqrt{\lambda}}} - 1))$ and $\log M_{Z_\lambda}(t) = \frac{t^2}{2} + o(1)$, so $Z_\lambda \rightsquigarrow N(0,1)$. If $X \sim \mathrm{Poisson}(900)$, then $\mathbb{P}(X > 950) = \mathbb{P}(\frac{X-900}{30} > \frac{5}{3}) \approx 1 - \Phi(\frac{5}{3}) \approx 0.04779$, close to 0.04712.

Example 3.2.8. *Normal Approximation to the Binomial*

If $X \sim \mathrm{Binomial}(n, p)$, $np > 5$, $n(1-p) > 5$, then $X \approx N(np, np(1-p))$; $\frac{X-np}{\sqrt{np(1-p)}} \approx N(0,1)$.

For $X \sim \mathrm{Binomial}(100, \frac{1}{2})$, $\mathbb{P}(X \geq 60) \approx 1 - \Phi(\frac{60-50}{5}) = 1 - \Phi(2) \approx 0.0228$.

Example 3.2.9. *Skewed Summands Need More Averaging*

Let $E_1, \ldots, E_n$ be iid $\mathrm{Exp}(1)$ and $G_n := \sum_{i=1}^{n} E_i \sim \mathrm{Gamma}(n, 1)$. Then $\frac{G_n - n}{\sqrt{n}} \rightsquigarrow N(0,1)$.

Skewness makes the normal approximation slower for small n.

Example 3.2.10. *Products and Lognormal Shape*

Let $Y_n = y_0 \prod_{i=1}^{n} X_i$ with iid $X_i > 0$, and suppose $\mu_\ell := \mathbb{E}\log X_1$ and $\sigma_\ell^2 := \mathrm{Var}(\log X_1) \in (0, +\infty)$.

Since $\log Y_n = \log y_0 + \sum_{i=1}^{n} \log X_i$, the CLT gives $\log Y_n \approx N(\log y_0 + n\mu_\ell, n\sigma_\ell^2)$, so Y_n is approximately lognormal. Likewise $V_n = v_0 \prod_{i=1}^{n} R_i$ when iid log-returns $\log R_i$ have finite positive variance.

Theorem 3.2.11. *Studentized Central Limit Theorem*

Let $(X_i)_{i\geq 1}$ be IID real-valued random variables with $\mathbb{E}X_1 = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 \in (0, +\infty)$.
Set $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$, $S_n^2 := (n-1)^{-1} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$, and $T_n := S_n^{-1}\sqrt{n}(\overline{X}_n - \mu)$. Then $T_n \rightsquigarrow N(0,1)$.

Proof. The CLT gives $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N(0,1)$, and Proposition 3.1.7 gives $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2$, hence $S_n \xrightarrow{\mathbb{P}} \sigma$.
Thus $\frac{\sigma}{S_n} \xrightarrow{\mathbb{P}} 1$, and Slutsky's rule from Theorem 2.1.8 gives $T_n = Z_n(\frac{\sigma}{S_n}) \rightsquigarrow N(0,1)$.
Structural Commentary: Studentization replaces the unknown σ by the consistent estimate S_n.
Key Insight: Slutsky permits multiplying the CLT statistic by $\frac{\sigma}{S_n} \to 1$. □

Theorem 3.2.12. *Berry–Esseen Rate*
Let $(X_i)_{i\geq 1}$ be IID real-valued random variables, $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 \in (0,+\infty)$, $\rho_3 := \mathbb{E}\,|X_1-\mu|^3 < +\infty$.
Let $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$, $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$, $\Phi(z) := \int_{-\infty}^z (2\pi)^{-\frac{1}{2}} e^{-\frac{x^2}{2}}\,dx$; $\sup_{z\in\mathbb{R}} |\mathbb{P}(Z_n \leq z) - \Phi(z)| \leq \frac{33}{4}\frac{\rho_3}{\sigma^3\sqrt{n}}$.

Proof. Set $Y_i := \frac{X_i-\mu}{\sigma}$, $\rho := \mathbb{E}\,|Y_1|^3 = \frac{\rho_3}{\sigma^3}$, and $W_n := n^{-\frac{1}{2}}\sum_{i=1}^n Y_i = Z_n$.
For $\varphi(t) := \mathbb{E}e^{itY_1}$, Taylor's formula with $\mathbb{E}Y_1 = 0$, $\mathbb{E}Y_1^2 = 1$, $\mathbb{E}\,|Y_1|^3 = \rho$ gives $\varphi(t) = 1 - \frac{t^2}{2} + O(\rho\,|t|^3)$ near 0.
Thus $\varphi_{W_n}(t) = [\varphi(\frac{t}{\sqrt{n}})]^n$ differs from $e^{-\frac{t^2}{2}}$ by $O(\rho n^{-\frac{1}{2}})$ in Esseen's smoothing integral.
Standard optimised smoothing bound gives $\sup_z |\mathbb{P}(W_n \leq z) - \Phi(z)| \leq \frac{33}{4}\frac{\rho}{\sqrt{n}}$. Substitute $\rho = \frac{\rho_3}{\sigma^3}$, QED.
Structural Commentary: Berry–Esseen bounds the CLT error by standardised third absolute moment.
Key Insight: Normalise, compare characteristic functions, then apply Esseen smoothing. □

Theorem 3.2.13. *Multivariate Central Limit Theorem*
Let $(X_i)_{i\geq 1}$ be IID random vectors in $\mathbb{R}^k$ with $\mathbb{E}X_1 = \mu \in \mathbb{R}^k$ and $\mathrm{Cov}(X_1) = \Sigma$, and set $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$.
Then $\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N_k(0,\Sigma)$.

Proof. For $a \in \mathbb{R}^k$, set $Y_{i,a} := a^t(X_i - \mu)$ and $s_a^2 := a^t\Sigma a$.
Then $(Y_{i,a})_{i\geq 1}$ are IID, $\mathbb{E}Y_{1,a} = 0$, $\mathrm{Var}(Y_{1,a}) = s_a^2$, and $a^t\sqrt{n}(\overline{X}_n - \mu) = n^{-\frac{1}{2}}\sum_{i=1}^n Y_{i,a}$.
If $s_a^2 > 0$, the scalar CLT gives $a^t\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N(0,s_a^2)$; if $s_a^2 = 0$, then $\mathbb{E}Y_{1,a}^2 = 0$, so the same convergence holds with degenerate limit $N(0,0)$. Since $a^tZ \sim N(0,s_a^2)$ for $Z \sim N_k(0,\Sigma)$, by Cramer–Wold, QED.
Structural Commentary: Multivariate convergence is checked through one-dimensional projections.
Key Insight: Project first, apply the scalar CLT, then use Cramer–Wold. □

Theorem 3.2.14. *Delta Method*
Let $(Y_n)_{n\geq 1}$ be real-valued random variables, let $\mu \in \mathbb{R}$ and $\sigma > 0$, and assume $\sigma^{-1}\sqrt{n}(Y_n - \mu) \rightsquigarrow N(0,1)$.
If $g : \mathbb{R} \to \mathbb{R}$ is differentiable at μ and $g'(\mu) \neq 0$, then $(|g'(\mu)|\,\sigma)^{-1}\sqrt{n}(g(Y_n) - g(\mu)) \rightsquigarrow N(0,1)$.
Equivalently $g(Y_n) \approx N(g(\mu), \frac{(g'(\mu))^2\sigma^2}{n})$.

Proof. Set $A_n := \sigma^{-1}\sqrt{n}(Y_n - \mu)$ and define $\rho(\mu) := 0$, $\rho(y) := \frac{g(y)-g(\mu)}{y-\mu} - g'(\mu)$ for $y \neq \mu$.
Then $A_n \rightsquigarrow Z \sim N(0,1)$, $A_n = O_{\mathbb{P}}(1)$, and $Y_n - \mu = \frac{\sigma A_n}{\sqrt{n}} \xrightarrow{\mathbb{P}} 0$.
Differentiability gives $\rho(y) \to 0$ as $y \to \mu$, hence $\rho(Y_n) \xrightarrow{\mathbb{P}} 0$.
For all n, $g(Y_n) - g(\mu) = (g'(\mu) + \rho(Y_n))(Y_n - \mu)$. Thus, with $T_n := (|g'(\mu)|\,\sigma)^{-1}\sqrt{n}(g(Y_n) - g(\mu))$,
$T_n = \mathrm{sgn}(g'(\mu))A_n(1 + \frac{\rho(Y_n)}{g'(\mu)}) \rightsquigarrow \mathrm{sgn}(g'(\mu))Z \sim N(0,1)$ by Theorem 2.1.8.
Structural Commentary: The delta method is first-order Taylor expansion at the limit point.
Key Insight: The derivative rescales the fluctuation; the remainder is negligible in probability. □

Theorem 3.2.15. *Multivariate Delta Method*
Let $(Y_n)_{n\geq 1}$ be random vectors in $\mathbb{R}^k$ with $\sqrt{n}(Y_n - \mu) \rightsquigarrow N_k(0,\Sigma)$, and let $g : \mathbb{R}^k \to \mathbb{R}$ be differentiable at μ with gradient $\nabla g(\mu)$.
Then $\sqrt{n}(g(Y_n) - g(\mu)) \rightsquigarrow N(0, \nabla g(\mu)^t\Sigma\nabla g(\mu))$.

Proof. Set $A_n := \sqrt{n}(Y_n - \mu)$, so $A_n \rightsquigarrow N_k(0,\Sigma)$ and $Y_n \xrightarrow{\mathbb{P}} \mu$.
Differentiability gives $g(Y_n) - g(\mu) = \nabla g(\mu)^t(Y_n - \mu) + r(Y_n)\,\|Y_n - \mu\|$ with $r(Y_n) \xrightarrow{\mathbb{P}} 0$.
Since $A_n = O_{\mathbb{P}}(1)$, $\sqrt{n}r(Y_n)\,\|Y_n - \mu\| = r(Y_n)\,\|A_n\| \xrightarrow{\mathbb{P}} 0$.
Thus $\sqrt{n}(g(Y_n) - g(\mu)) = \nabla g(\mu)^tA_n + o_{\mathbb{P}}(1) \rightsquigarrow N(0, \nabla g(\mu)^t\Sigma\nabla g(\mu))$ by Theorem 2.1.8.
Structural Commentary: Multivariate delta method linearizes g by its gradient.
Key Insight: The gradient projection carries the Gaussian limit; the differentiability remainder vanishes. □

Proposition 3.2.16. *Delta Method*
If $\sqrt{n}(T_n - \theta) \Rightarrow N(0,\Sigma)$ and $g : \mathbb{R}^d \to \mathbb{R}^k$ is differentiable at θ with derivative $Dg(\theta)$,
then $\sqrt{n}(g(T_n) - g(\theta)) \Rightarrow N(0, Dg(\theta)\Sigma Dg(\theta)^\top)$.

Proof. Differentiability gives $g(T_n) - g(\theta) = Dg(\theta)(T_n - \theta) + r_n$ with $\frac{\|r_n\|}{\|T_n-\theta\|} \to 0$ in probability.
Since $\sqrt{n}(T_n - \theta) = O_P(1)$, $\sqrt{n}r_n = o_P(1)$; $\sqrt{n}(g(T_n) - g(\theta)) = Dg(\theta)\sqrt{n}(T_n - \theta) + o_P(1)$, by Slutsky QED.
Structural Commentary: The derivative transports the limiting normal fluctuation through g.
Key Insight: First-order linearisation determines asymptotic variance. □

Example 3.2.17. *Exponentiating a Sample Mean*
Let $(X_i)_{i=1}^n$ be IID with mean μ and variance $\sigma^2 < +\infty$, and set $W_n := e^{\overline{X}_n}$.
For $g(s) = e^s$, $g'(\mu) = e^\mu$, so the delta method gives $W_n \approx N(e^\mu, \frac{e^{2\mu}\sigma^2}{n})$.

Example 3.2.18. *Product and Ratio of Sample Means*

Let $X_i = (X_{1i}, X_{2i})^t$ be IID with mean $\mu = (\mu_1, \mu_2)^t$ and covariance $\Sigma = (\sigma_{ij})$, and set $\overline{X}_{jn} := n^{-1} \sum_{i=1}^{n} X_{ji}$.

For $g(x_1, x_2) = x_1 x_2$, $\nabla g(\mu) = (\mu_2, \mu_1)^t$, so $\sqrt{n}(\overline{X}_{1n}\overline{X}_{2n} - \mu_1\mu_2) \rightsquigarrow N(0, \mu_2^2\sigma_{11} + 2\mu_1\mu_2\sigma_{12} + \mu_1^2\sigma_{22})$.

If $\mu_2 \neq 0$, then for $h(x_1, x_2) = \frac{x_1}{x_2}$ and $a := \nabla h(\mu) = (\frac{1}{\mu_2}, -\frac{\mu_1}{\mu_2^2})^t$, $\sqrt{n}(\frac{\overline{X}_{1n}}{\overline{X}_{2n}} - \frac{\mu_1}{\mu_2}) \rightsquigarrow N(0, a^t\Sigma a)$.

4 Stochastic Processes

4.1 Process-Level Probability

Definition 4.1.1. *Stochastic Process, State Space, and Index Set*

(i) A *stochastic process* is a family $(X_t)_{t\in T}$ of random variables.
(ii) The set T is the *index set*, often representing time.
(iii) Each X_t takes values in a *state space* $\mathcal{X}$.

Typical choices are discrete time $T = \{0, 1, 2, \ldots\}$ and continuous time $T = [0, \infty)$.

Example 4.1.2. *IID Sequences as Processes*
An IID sequence $(X_n)_{n\geq 1}$ is a stochastic process indexed by $T = \{1, 2, \ldots\}$.

Example 4.1.3. *Discrete States and Discrete Time*
If $\mathcal{X} = \{\text{sunny}, \text{cloudy}\}$ and X_n is the weather on day n, then $(X_n)_{n\geq 1}$ is a stochastic process with discrete state space and discrete index set.

Example 4.1.4. *The Empirical CDF as a Process*

Let $X_1, \ldots, X_n$ be IID with CDF F on $[0, 1]$, and define $\widehat{F}_n(t) = n^{-1} \sum_{i=1}^{n} \mathbf{1}\{X_i \leq t\}$.

Then $(\widehat{F}_n(t))_{t\in[0,1]}$ is a stochastic process with continuous index set $[0, 1]$ and state space $[0, 1]$.

Remark 4.1.5. *Sample Paths*
A *sample path* is one realized function $t \mapsto X_t(\omega)$ of a stochastic process.
The state space and index set may each be discrete or continuous.

Proposition 4.1.6. *Finite-Dimensional Chain Rule*
Let $X_{1:n} = (X_1, \ldots, X_n)$ have joint mass function or density $f_{1:n}$.

At points where the needed marginals are positive, $f_{1:n}(x_{1:n}) = f_1(x_1) \prod_{i=2}^{n} f_{i|1:i-1}(x_i \mid x_{1:i-1})$.

Proof. For $i \geq 2$, the definition of conditional density gives $f_{1:i}(x_{1:i}) = f_{i|1:i-1}(x_i \mid x_{1:i-1}) f_{1:i-1}(x_{1:i-1})$.
Iterating this identity from $i = n$ down to 2 yields the product formula.
Structural Commentary: The joint law factors by revealing one coordinate at a time.
Key Insight: Conditional densities convert successive joint marginals into a telescoping product. □

4.2 Filtrations and Adapted Processes

4.3 Martingales in Discrete and Continuous Time

4.4 Markov Chains and Markov Processes

Definition 4.4.1. *Probability Vectors and Markov Matrices*
A vector $\mathbf{x} \in \mathbb{R}^n$ is a *probability vector* if its entries are nonnegative and sum to 1. Let $\mathbf{e} := (1, \ldots, 1)^t$. A matrix $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{R})$ is a *Markov matrix* if $a_{ij} \geq 0$ for all i, j and $A\mathbf{e} = \mathbf{e}$; equivalently, every row sum is 1.

Lemma 4.4.2. *Basic Properties of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $1 \in \mathrm{spec}(A)$, and each A^p is Markov with entries in $[0, 1]$.

Proof. Since $A\mathbf{e} = \mathbf{e}$, 1 is an eigenvalue. Inductively, A^p has nonnegative entries and $A^p\mathbf{e} = \mathbf{e}$, so A^p is Markov. Each nonnegative row sums to 1, so all entries lie in $[0, 1]$.
Structural Commentary: Markov structure is closed under powers.
Key Insight: Nonnegative row sums of 1 bound every entry. □

Theorem 4.4.3. *Unit-Circle Spectrum of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $\rho(A) = 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Lemma 4.4.2 makes A power bounded. Theorem 3.3.7 gives $\rho(A) \leq 1$ and $\dim E_\lambda(A) = m_A(\lambda)$ on $|\lambda| = 1$. Since $A\mathbf{e} = \mathbf{e}$, $1 \in \mathrm{spec}(A)$, so $\rho(A) \geq 1$. Thus $\rho(A) = 1$.
Structural Commentary: Bounded Markov powers rule out unit-circle Jordan growth.
Key Insight: $A\mathbf{e} = \mathbf{e}$ forces the eigenvalue 1. □

Theorem 4.4.4. *Positive Markov Matrices Converge to a Stationary Distribution*
Let $n \geq 2$, $\mathbf{e} := (1, \ldots, 1)^t$, and $A = (a_{ij}) \in \mathrm{Mat}(n, n, \mathbb{R})$ satisfy $a_{ij} > 0$ and $A\mathbf{e} = \mathbf{e}$.
Let $\Delta_{n-1} := \{\mathbf{u} \in \mathbb{R}^n : \mathbf{u} \geq 0,\ \mathbf{e}^t\mathbf{u} = 1\}$. Then $\exists!\, \mathbf{x} \in \Delta_{n-1}$ s.t. $A^t\mathbf{x} = \mathbf{x}$, $\mathbf{x} > 0$, $\lim_{p\to\infty} A^p = \mathbf{e}\mathbf{x}^t$.

Proof. Let $A\mathbf{y} = \lambda\mathbf{y}$ with $\mathbf{y} \neq 0$ and $|\lambda| = 1$, and choose k with $|y_k| = \|\mathbf{y}\|_\infty$.

Since $\sum_j a_{kj} = 1$, then $\|\mathbf{y}\|_\infty = |\lambda y_k| = \left|\sum_{j=1}^{n} a_{kj} y_j\right| \leq \sum_{j=1}^{n} a_{kj} |y_j| \leq \|\mathbf{y}\|_\infty$. Thus all inequalities are equalities.
Since $a_{kj} > 0$, $|y_j| = \|\mathbf{y}\|_\infty$ $\forall j$; equality in the triangle inequality gives $\mathbf{y} = c\mathbf{e}$, $c \neq 0$.
Hence $\lambda = 1$, $E_1(A) = \mathrm{span}\{\mathbf{e}\}$, and $\mathrm{spec}(A) \cap \{z \in \mathbb{C} : |z| = 1\} = \{1\}$.
Theorem 4.4.3 makes 1 semisimple, so $A = S([1] \oplus J)S^{-1}$ with $S = [\mathbf{e}\ Y]$ and $\rho(J) < 1$.

First row of S^{-1} is $\mathbf{x}^t$. Then $\mathbf{x}^t A = \mathbf{x}^t$, $\mathbf{x}^t\mathbf{e} = 1$, and $A^p = S([1] \oplus J^p)S^{-1} \longrightarrow S([1] \oplus 0)S^{-1} = \mathbf{e}\mathbf{x}^t$.
Since $A^p \geq 0$, the limit gives $\mathbf{x} \geq 0$; since $A^t\mathbf{x} = \mathbf{x}$, $\mathbf{x}^t\mathbf{e} = 1$, and $A^t > 0$, we get $\mathbf{x} > 0$.
Finally, $\dim E_1(A^t) = n - \operatorname{rank}(A^t - I_n) = n - \operatorname{rank}(A - I_n) = \dim E_1(A) = 1$, so any $\mathbf{u} \in \Delta_{n-1}$ with $A^t\mathbf{u} = \mathbf{u}$ satisfies $\mathbf{u} = c\mathbf{x}$ and $\mathbf{e}^t\mathbf{u} = 1 = \mathbf{e}^t\mathbf{x}$, hence $\mathbf{u} = \mathbf{x}$.
Structural Commentary: Positivity leaves one simple unit-circle block; all other blocks decay.
Key Insight: The limit is the rank-one projection $\mathbf{e}\mathbf{x}^t$. □

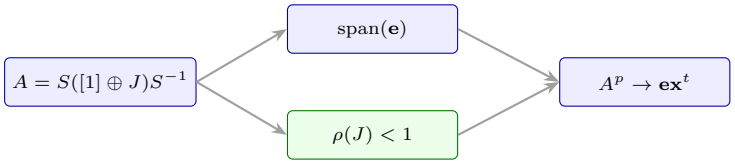

Figure 231: The line span($\mathbf{e}$) remains; all other Jordan directions decay, so $A^p \to \mathbf{e}\mathbf{x}^t$.

The preceding results are matrix facts. In probability language, a row-stochastic matrix becomes a transition matrix for a discrete-time chain.

Definition 4.4.5. *Markov Chain and Transition Matrix*
(i) Let $\mathcal{X}$ be countable and let $(X_n)_{n\geq 0}$ take values in $\mathcal{X}$.
(ii) Process is *Markov chain* if $\mathbb{P}(X_n = x \mid X_0, \ldots, X_{n-1}) = \mathbb{P}(X_n = x \mid X_{n-1})$ $\forall$ admissible n and $x \in \mathcal{X}$.
(iii) It is *homogeneous* if $p_{ij} := \mathbb{P}(X_{n+1} = j \mid X_n = i)$ is independent of n.
(iv) The matrix $P = [p_{ij}]$ is the *transition matrix*; $p_{ij} \geq 0$ and $\sum_{j\in\mathcal{X}} p_{ij} = 1$ for each i.

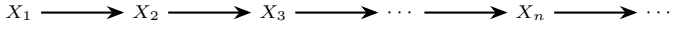

Figure 232: A Markov chain remembers the current state, not the full past.

Example 4.4.6. *Random Walk with Absorbing Barriers*
Let $\mathcal{X} = \{1, \ldots, N\}$ and $q = 1 - p$. The absorbing random walk has $p_{11} = p_{NN} = 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $2 \leq i \leq N - 1$, and all other transition probabilities equal to 0.

Definition 4.4.7. *n-Step Transition Probabilities*
For a homogeneous Markov chain, the *n-step transition probability* from i to j is
$p_{ij}(n) := \mathbb{P}(X_{m+n} = j \mid X_m = i)$; set $P_n = [p_{ij}(n)]$.

Theorem 4.4.8. *Chapman-Kolmogorov Equations*
Let $(X_r)_{r\geq 0}$ be homogeneous Markov chain on countable $\mathcal{X}$. Let $p_{ij}(r) := \mathbb{P}_i(X_r = j)$, $P_r = (p_{ij}(r))_{i,j\in\mathcal{X}}$.
$\forall i, j \in \mathcal{X}$ and $m, n \in \mathbb{N}_0$, $p_{ij}(m+n) = \sum_{k\in\mathcal{X}} p_{ik}(m)p_{kj}(n)$; equivalently, $P_{m+n} = P_m P_n$ and $P_n = P^n$ for $P = P_1$.

Proof. Fix $i, j \in \mathcal{X}$ and $m, n \in \mathbb{N}_0$. By total probability, the Markov property, and homogeneity,
$p_{ij}(m+n) = \mathbb{P}_i(X_{m+n} = j) = \sum_{k\in\mathcal{X}} \mathbb{P}_i(X_{m+n} = j \mid X_m = k)\mathbb{P}_i(X_m = k) = \sum_{k\in\mathcal{X}} p_{kj}(n)p_{ik}(m)$.
Thus $(P_m P_n)_{ij} = \sum_{k\in\mathcal{X}} p_{ik}(m)p_{kj}(n) = p_{ij}(m+n)$ for all i, j, so $P_{m+n} = P_m P_n$.
With $P_0 = I$ and $P = P_1$, induction gives $P_n = P^n$.
Structural Commentary: Transition over $m + n$ steps factors through the intermediate state after m steps.
Key Insight: Conditioning over intermediate states is matrix multiplication. □

Lemma 4.4.9. *Marginal Evolution*
Let $(X_n)_{n\geq 0}$ be a homogeneous Markov chain on countable $\mathcal{X}$ with $P = P_1 = (p_{ij})$, where $p_{ij}(n) := \mathbb{P}_i(X_n = j)$ and $P_n = (p_{ij}(n))$. Define row laws $\mu_n(j) := \mathbb{P}(X_n = j)$. Then $\mu_n = \mu_0 P^n$ for all $n \in \mathbb{N}_0$.

Proof. $\forall j \in \mathcal{X}$, by total probability $\mu_n(j) = \sum_{i\in\mathcal{X}} \mathbb{P}(X_n = j \mid X_0 = i)\mathbb{P}(X_0 = i) = \sum_{i\in\mathcal{X}} \mu_0(i)p_{ij}(n) = (\mu_0 P_n)_j$.
By Theorem 4.4.8, $P_n = P^n$, so $\mu_n(j) = (\mu_0 P^n)_j$ for every $j \in \mathcal{X}$; hence $\mu_n = \mu_0 P^n$.
Structural Commentary: Marginals evolve by multiplying the initial law by the transition semigroup.
Key Insight: Conditioning on X_0 turns distribution evolution into row-vector matrix multiplication. □

Definition 4.4.10. *Communication Classes*
(i) State i *reaches* j, written $i \to j$, if $p_{ij}(n) > 0$ for some $n \geq 0$.
(ii) States i, j *communicate*, written $i \leftrightarrow j$, if $i \to j$ and $j \to i$.
(iii) A *communication class* is an equivalence class under $\leftrightarrow$.
(iv) The chain is *irreducible* if all states lie in one communication class.
(v) A set C is *closed* if $p_{ij} = 0$ for all $i \in C$ and $j \notin C$; a singleton closed class is an *absorbing state*.

Proposition 4.4.11. *Communication Classes Partition the State Space*
Let $(X_n)_{n\geq 0}$ be a homogeneous Markov chain on countable $\mathcal{X}$ with n-step probabilities $p_{ij}(n)$.
Define $i \to j$ iff $\exists n \in \mathbb{N}_0$ such that $p_{ij}(n) > 0$, and $i \leftrightarrow j$ iff $i \to j$ and $j \to i$.
Then $\leftrightarrow$ is equivalence relation, and $\mathcal{X} = \bigsqcup_{C\in\mathcal{C}} C$ where $\mathcal{C} := \{[i]_\leftrightarrow : i \in \mathcal{X}\}$ is set of communication classes.

Proof. Since $p_{ii}(0) = 1$, $i \leftrightarrow i$ for all $i \in \mathcal{X}$; symmetry is immediate from the definition.
If $i \leftrightarrow j$ and $j \leftrightarrow k$, choose $m, n, r, s \in \mathbb{N}_0$ with $p_{ij}(m) > 0$, $p_{jk}(n) > 0$, $p_{kj}(r) > 0$, and $p_{ji}(s) > 0$.
By Theorem 4.4.8, $p_{ik}(m+n) \geq p_{ij}(m)p_{jk}(n) > 0$ and $p_{ki}(r+s) \geq p_{kj}(r)p_{ji}(s) > 0$, hence $i \leftrightarrow k$.
Thus $\leftrightarrow$ is reflexive, symmetric, and transitive.
Its equivalence classes are pairwise disjoint, cover $\mathcal{X}$, and are exactly the communication classes.
Structural Commentary: Communication classes are the equivalence classes generated by mutual reachability.
Key Insight: Positive transition paths concatenate by Chapman-Kolmogorov. □

Definition 4.4.12. *Recurrent and Transient States*

(i) Let $\tau_i^+ := \inf\{n \geq 1 : X_n = i\}$ be the first return time to i.
(ii) State i is *recurrent* or *persistent* if $\mathbb{P}_i(\tau_i^+ < \infty) = 1$.
(iii) State i is *transient* if $\mathbb{P}_i(\tau_i^+ < \infty) < 1$.

Theorem 4.4.13. *Recurrence Criterion*

(i) State i is recurrent iff $S_i := \sum_{n=0}^{\infty} p_{ii}(n) = \infty$.
(ii) State i is transient iff $S_i < \infty$.

Proof. Let $N_i := \sum_{n=0}^{\infty} \mathbf{1}\{X_n = i\}$ and $r_i := \mathbb{P}_i(\tau_i^+ < \infty)$. By Tonelli, $\mathbb{E}_i N_i = \sum_{n=0}^{\infty} p_{ii}(n) = S_i$.

By the Markov property after returns to i, $\mathbb{P}_i(N_i \geq k) = r_i^{k-1}$ for $k \geq 1$, hence $\mathbb{E}_i N_i = \sum_{k=1}^{\infty} r_i^{k-1}$.

(i) (⇒) If i is recurrent, then $r_i = 1$, so $S_i = \mathbb{E}_i N_i = \infty$.
(i) (⇐) If $S_i = \infty$, then $r_i < 1$ would give $\mathbb{E}_i N_i = (1 - r_i)^{-1} < \infty$, a contradiction; hence $r_i = 1$.
(ii) (⇒) If i is transient, then $r_i < 1$, so $S_i = \mathbb{E}_i N_i = (1 - r_i)^{-1} < \infty$.
(ii) (⇐) If $S_i < \infty$, then $r_i = 1$ would give $\mathbb{E}_i N_i = \infty$, a contradiction; hence $r_i < 1$.

Structural Commentary: Recurrence is detected by the expected total number of returns.
Key Insight: Visits to i repeat with geometric return probability r_i. □

Definition 4.4.14. *Period and Ergodicity*

(i) For recurrent i, it is *positive recurrent* if $\mathbb{E}_i \tau_i^+ < \infty$ and *null recurrent* otherwise.
(ii) The *period* of i is $d(i) := \gcd\{n \geq 1 : p_{ii}(n) > 0\}$.
(iii) State i is *aperiodic* if $d(i) = 1$.
(iv) State i is *ergodic* if it is recurrent, positive recurrent, and aperiodic.
(v) The chain is *ergodic* if every state is ergodic.

Definition 4.4.15. *Stationary and Limiting Distributions*

(i) A probability row vector π is *stationary* or *invariant* if $\pi = \pi P$.
(ii) A finite chain has *limiting distribution* π if $p_{ij}(n) \to \pi_j$ for all states i, j.

Equivalently, every row of P^n converges to π.

Remark 4.4.16. *Stationary Need Not Mean Convergent*
For the deterministic three-cycle on $\{1, 2, 3\}$, set $p_{12} = p_{23} = p_{31} = 1$ and all other $p_{ij} = 0$.
Then $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ satisfies $\pi P = \pi$, but $P^{n+3} = P^n$, so P^n does not converge.

Theorem 4.4.17. *Finite Ergodic Chain Convergence*
Let P be the transition matrix of a finite irreducible aperiodic Markov chain on $\mathcal{X}$.
There is a unique $\pi \in \Delta(\mathcal{X})$ with $\pi P = \pi$, and $P^n \to \mathbf{e}\pi$.

For every bounded $g : \mathcal{X} \to \mathbb{R}$, $N^{-1} \sum_{n=1}^{N} g(X_n) \to \pi(g) := \sum_{j \in \mathcal{X}} g(j)\pi_j$ almost surely.

Proof. Finite irreducibility and aperiodicity imply P is primitive: $P^m > 0$ for some $m \geq 1$.
Row-vector form of Theorem 4.4.4 on $A = P^m$, gives unique $\pi \in \Delta(\mathcal{X})$ with $\pi P^m = \pi$, $(P^m)^k \to \mathbf{e}\pi$.
Since $(\pi P)P^m = \pi P$ and $\pi P \in \Delta(\mathcal{X})$, uniqueness gives $\pi P = \pi$.
For $n = mk + r$ with $0 \leq r < m$, $P^n = (P^m)^k P^r \to \mathbf{e}\pi P^r = \mathbf{e}\pi$.
The finite-state Markov-chain strong law yields $N^{-1} \sum_{n=1}^{N} g(X_n) \to \pi(g)$ almost surely for bounded g.
Structural Commentary: Irreducibility gives one communicating class; aperiodicity removes cyclic obstruction.
Key Insight: Primitivity makes P^n converge to the rank-one projection $\mathbf{e}\pi$. □

Definition 4.4.18. *Detailed Balance*
Let $P = [p_{ij}]$ be a transition matrix on $\mathcal{X}$.
A probability row vector π satisfies *detailed balance* for P if $\pi_i p_{ij} = \pi_j p_{ji}$ for all $i, j \in \mathcal{X}$.

Theorem 4.4.19. *Detailed Balance Implies Stationarity*
Let $P = [p_{ij}]$ be row-stochastic on countable $\mathcal{X}$, and let $\pi \in \Delta(\mathcal{X})$. If $\pi_i p_{ij} = \pi_j p_{ji}$ $\forall i, j \in \mathcal{X}$, then $\pi P = \pi$.

Proof. $\forall j \in \mathcal{X}$, $(\pi P)_j = \sum_{i \in \mathcal{X}} \pi_i p_{ij} = \sum_{i \in \mathcal{X}} \pi_j p_{ji} = \pi_j \sum_{i \in \mathcal{X}} p_{ji} = \pi_j$, since row j of P sums to 1. QED.
Structural Commentary: Detailed balance gives pairwise cancellation of probability fluxes.
Key Insight: Reversibility is stronger than stationarity. □

Proposition 4.4.20. *Detailed Balance Implies Stationarity*
Let $\pi(dx) = f(x)\,dx$ and let P be a Markov kernel on state space X.
If $\pi(dx)P(x, dy) = \pi(dy)P(y, dx)$ as measures on $\mathsf{X} \times \mathsf{X}$, then $\pi P = \pi$.

Proof. For measurable $A \subseteq \mathsf{X}$, $\pi P(A) = \int P(x,A)\,\pi(dx) = \int \mathbf{1}_A(y)\,\pi(dx)P(x,dy)$.
By detailed balance, this equals $\int \mathbf{1}_A(y)\,\pi(dy)P(y,dx) = \int_A P(y,\mathsf{X})\,\pi(dy) = \pi(A)$, proving $\pi P = \pi$.
Structural Commentary: Detailed balance makes the joint law of consecutive states symmetric.
Key Insight: Marginalising the symmetric joint measure gives the same one-step density. □

Proposition 4.4.21. *Two-State Chain Limit*
Let $0 < a, b < 1$, $P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$. *Set* $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$. *Then* $\pi P = \pi$, π *unique,* $P^n \to \mathbf{e}\pi = \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$.

Proof. Solving $\pi P = \pi$ with $\pi\mathbf{e} = 1$ gives $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$.
The characteristic polynomial is $(\lambda - 1)(\lambda - (1-a-b))$, so $\operatorname{spec}(P) = \{1, 1-a-b\}$ and $|1-a-b| < 1$.
Thus stationary component is projected onto $\mathbf{e}\pi$, transverse component decays geometrically, giving $P^n \to \mathbf{e}\pi$.
Structural Commentary: The dynamics split into the stationary line and one transverse eigendirection.
Key Insight: The non-unit eigenvalue $1 - a - b$ controls convergence. □

4.5 Stationary Processes and Ergodicity

Definition 4.5.1. *Strict and Weak Stationarity*
Let $(X_t)_{t\in\mathbb{Z}}$ be a real-valued stochastic process.

(i) *Strict stationarity*: $\forall k \in \mathbb{N},\ t_1, \ldots, t_k, h \in \mathbb{Z}$, $(X_{t_1}, \ldots, X_{t_k}) \overset{\text{dist}}{=} (X_{t_1+h}, \ldots, X_{t_k+h})$.
(ii) *Weak stationarity*: $\exists \mu \in \mathbb{R}$ and $\gamma : \mathbb{Z} \to \mathbb{R}$ with $\forall t, h \in \mathbb{Z}$, $X_t \in L^2$, $\mathbb{E}X_t = \mu$, and $\operatorname{Cov}(X_t, X_{t-h}) = \gamma(h)$.

Key Insight: (i) plus L^2 implies (ii); (ii) fixes only μ and γ, not finite-dimensional laws.

4.6 Gaussian Processes

4.7 Point Processes and Renewal Structure

Counting processes model random event arrivals over time. The Poisson process is the basic model in which disjoint time intervals contribute independent counts and short intervals usually contain either zero or one event.

Definition 4.7.1. *Counting Process*
A *counting process* is a stochastic process $\{X(t) : t \geq 0\}$ with state space $\{0, 1, 2, \ldots\}$ such that $X(t)$ records the number of events observed up to and including time t.

Definition 4.7.2. *Poisson Process with Intensity Function*
Let $\lambda : [0,\infty) \to [0,\infty)$ be locally integrable.
A counting process $X = \{X(t) : t \geq 0\}$ is a *Poisson process* with *intensity function* λ if:

(i) $X(0) = 0$.
(ii) For $0 = t_0 < t_1 < \cdots < t_n$, the increments $\Delta_k X := X(t_k) - X(t_{k-1})$, $1 \leq k \leq n$, are independent.
(iii) For each $t \geq 0$, as $h \downarrow 0$, $\mathbb{P}(X(t+h) - X(t) = 1) = \lambda(t)h + o(h)$ and $\mathbb{P}(X(t+h) - X(t) \geq 2) = o(h)$.

Theorem 4.7.3. *Increment Distribution*
Let X *be a Poisson process with intensity* λ, *and set* $\Lambda(s,t) := \int_s^t \lambda(u)\,du$ *for* $0 \leq s \leq t$.
Then $X(t) - X(s) \sim \text{Poisson}(\Lambda(s,t))$. *In particular,* $X(t) \sim \text{Poisson}(\Lambda(0,t))$, *so* $\mathbb{E}X(t) = \operatorname{Var}(X(t)) = \Lambda(0,t)$.

Proof. Fix $s \geq 0$ and let $p_k(u) := \mathbb{P}(X(s+u) - X(s) = k)$ with $p_{-1} := 0$. By small-time conditions and independent increments, $p_k(u+h) = p_k(u)(1 - \lambda(s+u)h + o(h)) + p_{k-1}(u)\lambda(s+u)h + o(h)$.
Hence $p_k'(u) = -\lambda(s+u)p_k(u) + \lambda(s+u)p_{k-1}(u)$, with $p_0(0) = 1$ and $p_k(0) = 0$ for $k \geq 1$.
Writing $M(u) := \Lambda(s, s+u)$, the unique solution is $p_k(u) = e^{-M(u)}\frac{M(u)^k}{k!}$ by induction on k.
Taking $u = t - s$ yields $X(t) - X(s) \sim \text{Poisson}(\Lambda(s,t))$. Mean-variance identities follow from Poisson law.
Structural Commentary: The infinitesimal jump rules form a triangular system for the increment probabilities.
Key Insight: Integrating the local rate λ gives the Poisson mean over the interval. □

Definition 4.7.4. *Homogeneous Poisson Process*
If $\lambda(t) \equiv \lambda > 0$, then $X(t)$ is a *homogeneous Poisson process* with *rate* λ. In this case, $X(t) \sim \text{Poisson}(\lambda t)$.

Definition 4.7.5. *Waiting and Interarrival Times*
For a homogeneous Poisson process:

(i) Set $W_0 = 0$ and $W_n := \inf\{t \geq 0 : X(t) \geq n\}$ for $n \geq 1$; $W_1, W_2, \ldots$ are *waiting times*.
(ii) Set $S_n := W_{n+1} - W_n$ for $n \geq 0$; $S_0, S_1, \ldots$ are *sojourn* or *interarrival times*.

Theorem 4.7.6. *Poisson Interarrival Times*
Let X *be a homogeneous Poisson process with rate* $\lambda > 0$.
Then $S_0, S_1, \ldots \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, *with density* $f(s) = \lambda e^{-\lambda s}$ *for* $s \geq 0$.
For $n \geq 1$, $W_n = \sum_{r=0}^{n-1} S_r \sim \text{Gamma}(n, \lambda)$, *so* $\mathbb{E}W_n = \frac{n}{\lambda}$ *and* $\operatorname{Var}(W_n) = \frac{n}{\lambda^2}$.

Proof. For $t \geq 0$, $\mathbb{P}(S_0 > t) = \mathbb{P}(X(t) = 0) = e^{-\lambda t}$, so $S_0 \sim \text{Exp}(\lambda)$.
By independent stationary increments, $\mathbb{P}(S_n > t \mid \mathcal{F}_{W_n}) = \mathbb{P}(X(W_n + t) - X(W_n) = 0 \mid \mathcal{F}_{W_n}) = e^{-\lambda t}$.
Thus each $S_n \sim \text{Exp}(\lambda)$ and is independent of the past, so the S_n are IID.
Since $W_n = \sum_{r=0}^{n-1} S_r$, the Gamma law and its mean and variance follow.
Structural Commentary: Arrival times are cumulative sums of IID interarrival gaps.
Key Insight: No events in a length-t interval has probability $e^{-\lambda t}$. □

Proposition 4.7.7. *Conditional Binomial Count*
Let X be a homogeneous Poisson process with rate $\lambda > 0$.
For $s, t > 0$ and $n \in \mathbb{N}_0$, $X(t) \mid X(t+s) = n \sim \text{Binomial}\left(n, \frac{t}{t+s}\right)$.

Proof. Let $Y = X(t)$ and $Z = X(t+s) - X(t)$. Then $Y \sim \text{Poisson}(\lambda t)$, $Z \sim \text{Poisson}(\lambda s)$, and $Y \perp Z$.
For $0 \le k \le n$, $\mathbb{P}(Y = k \mid Y + Z = n) = \frac{\frac{(\lambda t)^k (\lambda s)^{n-k}}{k!(n-k)!}}{\frac{(\lambda(t+s))^n}{n!}} = \binom{n}{k} \left(\frac{t}{t+s}\right)^k \left(\frac{s}{t+s}\right)^{n-k}$; QED.
Structural Commentary: Conditioning on the total allocates independent Poisson counts.
Key Insight: Independent Poisson counts become multinomial after conditioning on their sum. □

Remark 4.7.8. *Uniform Event-Time Diagnostic*
For a homogeneous Poisson process on $[0, T]$, conditional on $X(T) = n$, the event times $(W_1, \ldots, W_n)$ have the law of the order statistics of n IID Uniform$(0, T)$ variables.
Marked nonuniformity in the observed event times is evidence against homogeneity.

Proposition 4.7.9. *Odd Poisson Count*
Let X be a homogeneous Poisson process with rate $\lambda > 0$. For $t \ge 0$, $\mathbb{P}(X(t) \in 2\mathbb{N}_0 + 1) = \frac{1 - e^{-2\lambda t}}{2}$.

Proof. Let $\mu = \lambda t$. Since $X(t) \sim \text{Poisson}(\mu)$, $\mathbb{P}(X(t) \in 2\mathbb{N}_0 + 1) = e^{-\mu} \sum_{k=0}^{\infty} \frac{\mu^{2k+1}}{(2k+1)!} = e^{-\mu} \sinh \mu = \frac{1 - e^{-2\mu}}{2}$.
Structural Commentary: Split the Poisson mass by parity.
Key Insight: The odd terms of e^{μ} sum to $\sinh \mu$. □

Proposition 4.7.10. *Nearest Neighbor in a Planar Poisson Process*
Let Φ be a planar Poisson point process with intensity $\lambda > 0$, fix $x_0 \in \mathbb{R}^2$, and set $R := \inf_{Y \in \Phi} \|Y - x_0\|$.
Then for $t \ge 0$, $\mathbb{P}(R > t) = e^{-\lambda \pi t^2}$, and $\mathbb{E}R = \frac{1}{2\sqrt{\lambda}}$.

Proof. Let $B_t := B(x_0, t)$, $N_t := \#(\Phi \cap B_t)$. Then $N_t \sim \text{Poisson}(\lambda \pi t^2)$, $\{R > t\} = \{N_t = 0\}$,
so $\mathbb{P}(R > t) = e^{-\lambda \pi t^2}$. Since $R \ge 0$, $\mathbb{E}R = \int_0^{\infty} \mathbb{P}(R > t)\, dt = \int_0^{\infty} e^{-\lambda \pi t^2}\, dt = \frac{1}{2\sqrt{\lambda}}$.
Structural Commentary: Nearest-neighbor distance is encoded by empty disks around x_0.
Key Insight: The tail of R is the zero-count probability of a Poisson disk count. □

5 Stochastic Calculus and SDEs

5.1 Brownian Motion

5.2 Stochastic Integration

5.3 Itô Calculus

5.4 Stochastic Differential Equations

5.5 Change of Measure and Girsanov Theory

5.6 Diffusions and PDE Connections

5.7 Stochastic Flows and Advanced Dynamics

Part XVII
Dynamical and PDE Systems

1 Dynamical Systems and ODEs

1.1 ODE Foundations and Flows

Ordinary differential equations prescribe the evolution of an unknown curve through relations among the curve and its derivatives. Their order, dependence on time, and linearity determine the natural state variables and solution methods.

Definition 1.1.1. *Implicit Scalar ODE and Regular Point*
Let $k \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R}^{k+2}$ be open, and $F \in C(U, \mathbb{R})$.
For $I \subseteq \mathbb{R}$, $x \in C^k(I, \mathbb{R})$, set $J_x^k(t) := (t, x(t), x^{(1)}(t), \dots, x^{(k)}(t))$.

(i) $F(t, y_0, \dots, y_k) = 0$ is an *implicit scalar ODE written through order k*; its k-jet form is $F(J_x^k(t)) = 0$.
(ii) x is a *solution* on I iff $J_x^k(t) \in U$ and $F(J_x^k(t)) = 0$ $\forall t \in I$.
(iii) $F \in C^1(U, \mathbb{R})$; $p = (t_*, y_0^*, \dots, y_k^*) \in F^{-1}(0)$ is *regular w.r.t. highest derivative* iff $\partial_{y_k} F(p) \neq 0$.

Key Insight: $\partial_{y_k} F(p) \neq 0$ locally solves the relation for $x^{(k)}$.

Proposition 1.1.2. *Local Explicit Form of a Regular Implicit ODE*
Let $k \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R}^{k+2}$ be open, $F \in C^1(U, \mathbb{R})$, $u_* = (t_*, y_0^*, \dots, y_{k-1}^*)$, and $p = (u_*, y_k^*) \in U$ with $F(p) = 0$ and $\partial_{y_k} F(p) \neq 0$. Then $\exists$ open $V \subseteq \mathbb{R}^{k+1}$, $W \subseteq \mathbb{R}$ with $u_* \in V$, $y_k^* \in W$, and a unique $f \in C^1(V, \mathbb{R})$ such that $f(V) \subseteq W$, $f(u_*) = y_k^*$, $V \times W \subseteq U$, and:

(i) $F(u, v) = 0$ iff $v = f(u)$ $\forall (u, v) \in V \times W$.
(ii) If $x \in C^k(I, \mathbb{R})$ and $J_x^k(I) \subseteq V \times W$, then $F(J_x^k(t)) = 0$ $\forall t \in I$ iff $x^{(k)}(t) = f(t, x(t), \dots, x^{(k-1)}(t))$ $\forall t \in I$.

Proof. Let $u = (t, y_0, \dots, y_{k-1})$, $v = y_k$, $\widehat{U} := \{(v, u) : (u, v) \in U\}$, and $\widehat{F}(v, u) := F(u, v)$.
Since $\partial_v \widehat{F}(y_k^*, u_*) = \partial_{y_k} F(p) \neq 0$, by Theorem 4.2.3, there are W, V and a unique $f \in C^1(V, \mathbb{R})$ with $W \times V \subseteq \widehat{U}$, $f(V) \subseteq W$, $f(u_*) = y_k^*$, and $\widehat{F}(v, u) = 0$ iff $v = f(u)$.

(i) For $(u, v) \in V \times W$, $F(u, v) = \widehat{F}(v, u)$, so $F(u, v) = 0$ iff $v = f(u)$.
(ii) Set $u_x(t) := (t, x(t), \dots, x^{(k-1)}(t))$ and $v_x(t) := x^{(k)}(t)$. Since $(u_x(t), v_x(t)) \in V \times W$ $\forall t \in I$, (i) gives $F(J_x^k(t)) = 0$ iff $v_x(t) = f(u_x(t))$ $\forall t \in I$.

Structural Commentary: IFT solves $F(u, y_k) = 0$ for y_k.
Key Insight: $\partial_{y_k} F(p) \neq 0$ turns $F = 0$ into $y_k = f(u)$ locally. □

Definition 1.1.3. *Explicit ODE Systems and Classes*
Let $k, n \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R} \times (\mathbb{R}^n)^k$ open, and $f : U \to \mathbb{R}^n$.
Use $X_x(t) := (x(t), x^{(1)}(t), \dots, x^{(k-1)}(t))$, $\xi = (\xi_0, \dots, \xi_{k-1})$, and $\pi(U) := \{\xi \in (\mathbb{R}^n)^k : \exists t \in \mathbb{R},\ (t, \xi) \in U\}$.

(i) $x^{(k)}(t) = f(t, X_x(t))$ is an *explicit ODE system of order k*.
(ii) For an interval $I \subseteq \mathbb{R}$, $x \in C^k(I, \mathbb{R}^n)$ is a *solution* iff $\forall t \in I$, $(t, X_x(t)) \in U$ and $x^{(k)}(t) = f(t, X_x(t))$.
(iii) The system is *autonomous* iff $\exists F : \pi(U) \to \mathbb{R}^n$ with $f(t, \xi) = F(\xi)$ $\forall (t, \xi) \in U$.
(iv) On an interval I, the system is *linear* iff $\exists g \in C(I, \mathbb{R}^n)$ and $A_j \in C(I, \mathbb{R}^{n \times n})$, $0 \leq j < k$,
with $f(t, \xi) = g(t) + \sum_{j=0}^{k-1} A_j(t) \xi_j$ $\forall (t, \xi) \in U \cap (I \times (\mathbb{R}^n)^k)$.
(v) A linear system on I is *homogeneous* iff (iv) holds with $g = 0$.
(vi) A solution $x : I \to \mathbb{R}^n$ is *global* iff $I = \mathbb{R}$.
(vii) A solution $x : I \to \mathbb{R}^n$ is *maximal* iff $\nexists$ solution $\bar{x} : \bar{I} \to \mathbb{R}^n$ with $I \subsetneq \bar{I}$ and $\bar{x}|_I = x$.

Key Insight: The state lift X_x has dimension kn.

Proposition 1.1.4. *Reduction to First Order*
Let $k, n \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R} \times (\mathbb{R}^n)^k$ open, and $f : U \to \mathbb{R}^n$.
For $y = (y_1, \dots, y_k)$, define $G(t, y) := (y_2, \dots, y_k, f(t, y))$.
For an interval $I \subseteq \mathbb{R}$ and $x \in C^k(I, \mathbb{R}^n)$, set $y := (x, x^{(1)}, \dots, x^{(k-1)})$ and assume $(t, y(t)) \in U$ $\forall t \in I$.
Then $y \in C^1(I, (\mathbb{R}^n)^k)$, and $x^{(k)}(t) = f(t, y(t))$ $\forall t \in I$ iff $y'(t) = G(t, y(t))$ $\forall t \in I$.

Proof.
(⇒) Since $x \in C^k(I, \mathbb{R}^n)$, then $y \in C^1(I, (\mathbb{R}^n)^k)$. The identities $y_j' = x^{(j)} = y_{j+1}$, $1 \leq j < k$, and $y_k' = x^{(k)} = f(t, y(t))$ imply $y' = G(\,\cdot\,, y)$ on I.
(⇐) Since $y_k = x^{(k-1)}$, the kth coordinate gives $x^{(k)}(t) = y_k'(t) = G_k(t, y(t)) = f(t, y(t))$ $\forall t \in I$.

Structural Commentary: Coordinates shift; the last coordinate carries f.
Key Insight: Adjoin derivatives to get first order. □

Proposition 1.1.5. *Autonomization by Adjoining Time*
Let $m \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^m$ be open, $f : U \to \mathbb{R}^m$, and $\widetilde{f}(s, \xi) := (1, f(s, \xi))$. For an interval $I \subseteq \mathbb{R}$:

(i) If $y \in C^1(I, \mathbb{R}^m)$, $(t, y(t)) \in U$, and $y' = f(t, y)$, then $z(t) := (t, y(t))$ solves $z' = \widetilde{f}(z)$.
(ii) If $z = (s, y) \in C^1(I, \mathbb{R}^{m+1})$, $z(I) \subseteq U$, $z' = \widetilde{f}(z)$, and $s(t_0) = t_0$ for some $t_0 \in I$, then $s = \mathrm{id}_I$ and $y'(t) = f(t, y(t))$ on I.

Proof.
(i) For $z = (t, y)$, $z' = (1, y') = (1, f(t, y)) = \widetilde{f}(t, y) = \widetilde{f}(z)$.
(ii) Since $(s', y') = z' = \widetilde{f}(s, y) = (1, f(s, y))$, $s' = 1$ and $y' = f(s, y)$.
With $s(t_0) = t_0$, $s = \mathrm{id}_I$; hence $y'(t) = f(t, y(t))$.

Structural Commentary: The clock coordinate has $s' = 1$.
Key Insight: $y' = f(t, y)$ becomes $z' = \widetilde{f}(z)$ with $z = (t, y)$. □

Example 1.1.6. *Newtonian Motion as a First-Order System*
Let $q : \mathbb{R} \to \mathbb{R}^3$ be the position of a particle of mass $m > 0$, let $v := q'$, and let $F : \mathbb{R}^3 \to \mathbb{R}^3$ be a position-dependent force. Newton's equation $mq''(t) = F(q(t))$ becomes $(q, v)' = (v, m^{-1}F(q))$ by Proposition 1.1.4. For uniform gravity $F(q) = -mge_3$, where $e_3 := (0, 0, 1)^{\mathsf{T}}$ and $g > 0$, the initial state $(q(0), v(0)) = (q_0, v_0)$ gives $q(t) = q_0 + v_0 t - \frac{g}{2}e_3 t^2$. Thus position and velocity together determine the mechanical state.

Proposition 1.1.7. *Scalar Autonomous Separation*
Let $D \subseteq \mathbb{R}$ be open, $f \in C(D, \mathbb{R})$, $D_f := \{x \in D : f(x) \neq 0\}$, and $x_0 \in D_f$.
Let C be the component of D_f containing x_0, and set $H(x) := \int_{x_0}^{x} \frac{du}{f(u)}$, $K := H(C)$. Then:

(i) K is an open interval and $H : C \to K$ is a differentiable bijection.
(ii) $\phi := H^{-1} : K \to C$ is the unique C-valued solution of $x' = f(x)$ with $x(0) = x_0$.
(iii) K is maximal among intervals of C-valued solutions through x_0, and ϕ is global iff $K = \mathbb{R}$.

Proof.

(i) Since D_f is open, C is an open interval. By Theorem 2.2.1, $H' = 1/f \neq 0$ on C.
Since f has constant sign on C by Theorem 5.1.43, H is strictly monotone.
Hence $H : C \to K = H(C)$ is bijective and K is an open interval.
(ii) By Corollary 1.1.20, $\phi \in C^1(K, C)$ and $\phi' = 1/(H' \circ \phi) = f \circ \phi$. Since $H(x_0) = 0$, $\phi(0) = x_0$.
If $u : L \to C$ solves $u' = f(u)$ with $0 \in L$ and $u(0) = x_0$, then $(H \circ u)' = H'(u)u' = 1$ and $(H \circ u)(0) = 0$.
Hence $H(u(t)) = t$, so $L \subseteq K$ and $u = \phi|_L$.
(iii) If $\bar{\phi} : \bar{K} \to C$ solves $x' = f(x)$, $K \subseteq \bar{K}$, and $\bar{\phi}|_K = \phi$, then (ii) gives $\bar{K} \subseteq K$; hence $\bar{K} = K$.
Thus K is maximal. Since $\operatorname{dom}\phi = K$, ϕ is global iff $K = \mathbb{R}$.

Structural Commentary: Monotone coordinate, inverse trajectory.
Key Insight: Time is accumulated reciprocal speed. □

Method 1.1.8. *Separation of Variables*
Let $D \subseteq \mathbb{R}$ be open, $J \subseteq \mathbb{R}$ open interval, $f \in C(D, \mathbb{R})$, $g \in C(J, \mathbb{R})$, $t_0 \in J$, $x_0 \in D_f := \{x \in D : f(x) \neq 0\}$.
Let $C \ni x_0$ be the component of D_f and $H(x) := \int_{x_0}^{x} \frac{du}{f(u)}$.
Since f has constant sign on C and $H' = 1/f$, $H : C \to H(C)$ is a differentiable bijection.

(i) Set $G(t) := \int_{t_0}^{t} g(s)\,ds$ and let L be the component of $\{t \in J : G(t) \in H(C)\}$ containing t_0.
(ii) Output $x := H^{-1} \circ G$ on L; then $x' = f(x)g$ on L and $x(t_0) = x_0$.

Example 1.1.9. *Global Growth, Finite-Time Blow-Up, and Nonuniqueness*
The separation coordinate distinguishes three basic behaviors.

(i) For $x' = x$ and $x(0) = x_0 > 0$, $H(x) = \log\left(\frac{x}{x_0}\right)$ and $x(t) = x_0 e^t$ on $\mathbb{R}$.
(ii) For $x' = x^2$ and $x(0) = x_0 > 0$, $H(x) = x_0^{-1} - x^{-1}$ and $x(t) = \frac{x_0}{1 - x_0 t}$ on $(-\infty, x_0^{-1})$; $x(t) \to +\infty$ as $t \uparrow x_0^{-1}$.
(iii) For $x' = \sqrt{|x|}$ and $x(0) = 0$, $a \in [0, +\infty)$ determines $x_a(t) := \begin{cases} 0, & t \leq a, \\ \frac{(t-a)^2}{4}, & t \geq a. \end{cases}$ Then $x_a \in C^1(\mathbb{R})$, $x_a' = \sqrt{|x_a|}$, $x_a(0) = 0$, and if $0 \leq a < b$, then $x_a = x_b$ on $(-\infty, a]$ while $x_a > x_b$ on $(a, b]$; hence the IVP is nonunique.

First-order equations can also be rewritten as fixed-point problems on complete function spaces.

Definition 1.1.10. *Scalar Initial-Value Problem*
Let $O \subseteq \mathbb{R}^2$ be open, $(x_0, y_0) \in O$, $g : O \to \mathbb{R}$, and $\mathrm{IVP}(O, g, x_0, y_0) := \{y' = g(x, y),\ y(x_0) = y_0\}$.
For $J \subseteq \mathbb{R}$ with $x_0 \in J$, $f : J \to \mathbb{R}$ solves $\mathrm{IVP}(O, g, x_0, y_0)$ iff

(i) J is open and f is differentiable.
(ii) $\Gamma_f := \{(x, f(x)) : x \in J\} \subseteq O$.
(iii) $f'(x) = g(x, f(x))\ \forall x \in J$, and $f(x_0) = y_0$.

Structural Commentary: $\Gamma_f \subseteq O$ makes $g(\,\cdot\,, f)$ defined; $f(x_0) = y_0$ selects the curve.

Proposition 1.1.11. *Differential and Integral Forms*
Let $O \subseteq \mathbb{R}^2$ be open, $(x_0, y_0) \in O$, $g \in C(O, \mathbb{R})$, $J \subseteq \mathbb{R}$ open with $x_0 \in J$, and $f \in C(J, \mathbb{R})$.
Let $\Gamma_f := \{(x, f(x)) : x \in J\} \subseteq O$. Then f solves $\mathrm{IVP}(O, g, x_0, y_0)$ iff $f(x) = y_0 + \int_{x_0}^{x} g(t, f(t))\,dt\ \forall x \in J$.

Proof.

($\Rightarrow$) If $f'(x) = g(x, f(x))$ on J, $f(x_0) = y_0$, by Theorem 2.2.2, $f(x) - y_0 = \int_{x_0}^{x} f'(t)\,dt = \int_{x_0}^{x} g(t, f(t))\,dt\ \forall x \in J$.
($\Leftarrow$) If $f(x) = y_0 + \int_{x_0}^{x} g(t, f(t))\,dt\ \forall x \in J$, then $g(\,\cdot\,, f(\,\cdot\,))$ is continuous on J.
By Theorem 2.2.1, $f'(x) = g(x, f(x))$ on J. Evaluating at $x = x_0$ gives $f(x_0) = y_0$.

Structural Commentary: Differentiation and integration are inverse along the continuous graph.
Key Insight: The IVP is equivalent to its Volterra integral equation. □

Theorem 1.1.12. *Picard Local Existence Theorem*
Let $O \subseteq \mathbb{R}^2$ be open, $(x_0, y_0) \in O$, and $g \in C(O, \mathbb{R})$.
Assume $|g(x, y_1) - g(x, y_2)| \leq M\,|y_1 - y_2|$ for some $M > 0$ whenever $(x, y_1), (x, y_2) \in O$.
Then $\exists a, b, \ell > 0$ with $Q := [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b] \subseteq O$ and $J_\ell := (x_0 - \ell, x_0 + \ell)$ such that:

(i) The IVP $y' = g(x, y)$, $y(x_0) = y_0$, has a differentiable solution $f : J_\ell \to \mathbb{R}$ with $\Gamma_f \subseteq Q$.
(ii) If $h : J_\ell \to \mathbb{R}$ is a differentiable solution with $\Gamma_h \subseteq Q$, then $h = f$.

Proof.

(i) Choose $a, b > 0$ with $Q := [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b] \subseteq O$.
Let $K := \sup_Q |g| < +\infty$, and choose $0 < \ell \le a$ with $\ell K \le b$ and $\ell M < 1$.
Put $I_\ell := [x_0 - \ell, x_0 + \ell]$, $\mathcal{X}_\ell := \{u \in C(I_\ell, \mathbb{R}) : \|u - y_0\|_\infty \le b\}$, and $(Tu)(x) := y_0 + \int_{x_0}^{x} g(t, u(t))\, dt$.
The space $\mathcal{X}_\ell$ is complete, and for $u, v \in \mathcal{X}_\ell$, $\|Tu - y_0\|_\infty \le \ell K \le b$ and $\|Tu - Tv\|_\infty \le \ell M \|u - v\|_\infty$.
Thus $T : \mathcal{X}_\ell \to \mathcal{X}_\ell$ is a contraction; by Theorem 4.5.3, $\exists! u_* \in \mathcal{X}_\ell$ with $Tu_* = u_*$.
For $f := u_*|_{J_\ell}$, $\Gamma_f \subseteq Q$; by Proposition 1.1.11, $f' = g(\cdot, f)$ and $f(x_0) = y_0$.

(ii) Let $h : J_\ell \to \mathbb{R}$ solve the IVP with $\Gamma_h \subseteq Q$. For $0 < r < \ell$, put $I_r := [x_0 - r, x_0 + r]$,
$\mathcal{X}_r := \{u \in C(I_r, \mathbb{R}) : \|u - y_0\|_\infty \le b\}$, and $(T_r u)(x) := y_0 + \int_{x_0}^{x} g(t, u(t))\, dt$.
Since $rM < 1$, T_r is a contraction. By Proposition 1.1.11, $T_r(h|_{I_r}) = h|_{I_r}$ and $T_r(u_*|_{I_r}) = u_*|_{I_r}$.
Uniqueness gives $h = f$ on I_r. Hence $h = f$ on $\bigcup_{0<r<\ell} I_r = J_\ell$.

Structural Commentary: Box control plus short-time contraction.
Key Insight: Picard iteration is Banach iteration. □

Method 1.1.13. *Picard Iteration*
Let $Q := [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b] \subseteq O$, assume $|g(x, y)| \le K$ for $(x, y) \in Q$,
$|g(x, y_1) - g(x, y_2)| \le M |y_1 - y_2|$ whenever $(x, y_1), (x, y_2) \in Q$. Let $\ell > 0$ with $\ell \le a$, $\ell K \le b$, and $\ell M < 1$.
Set $I_\ell := [x_0 - \ell, x_0 + \ell]$, $J_\ell := (x_0 - \ell, x_0 + \ell)$, $\mathcal{X}_\ell := \{u \in C(I_\ell, \mathbb{R}) : \|u - y_0\|_\infty \le b\}$,
$(Tu)(x) := y_0 + \int_{x_0}^{x} g(t, u(t))\, dt$. Then

(i) Initialize $f_0(x) \equiv y_0$ on I_ℓ.
(ii) Iterate $f_{n+1} := Tf_n$, i.e. $f_{n+1}(x) = y_0 + \int_{x_0}^{x} g(t, f_n(t))\, dt$ for $x \in I_\ell$.
(iii) Let $f_\infty := \lim_{n\to\infty} f_n$ uniformly on I_ℓ, and output $f := f_\infty|_{J_\ell}$.

By Theorem 4.5.3, the limit exists and is the unique fixed point of T; Proposition 1.1.11 identifies its restriction as the unique local solution with graph in Q.

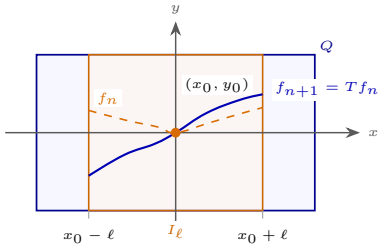

Figure 233: Picard setup: choose ℓ with $\ell K \le b$ so $T(\mathcal{X}_\ell) \subseteq \mathcal{X}_\ell$ and graphs stay in Q.

The same fixed-point formulation applies to vector fields and supports quantitative iteration estimates.

Definition 1.1.14. *Vector IVP and Local State Lipschitz Condition*
Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f : U \to \mathbb{R}^n$, and $(t_0, x_0) \in U$.

(i) The *initial-value problem* is $x' = f(t, x)$, $x(t_0) = x_0$; a solution on an interval $I \ni t_0$ is $x \in C^1(I, \mathbb{R}^n)$ with $(t, x(t)) \in U$ and $x'(t) = f(t, x(t))$ $\forall t \in I$.
(ii) f is *locally Lipschitz in the state, uniformly in time* iff $\forall$ compact $V \subseteq U$, $\exists L_V < +\infty$ such that $\|f(t, x) - f(t, y)\| \le L_V \|x - y\|$ whenever $(t, x), (t, y) \in V$.

Key Insight: State-Lipschitz control makes the integral operator asymptotically contractive under iteration.

Lemma 1.1.15. *C^1 Vector Fields Are Locally State-Lipschitz*
Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, and $f \in C^1(U, \mathbb{R}^n)$.
$\forall$ compact $K \subseteq U$, $\exists L_K < +\infty$ such that $\|f(t, x) - f(t, y)\| \le L_K \|x - y\|$ whenever $(t, x), (t, y) \in K$.

Proof. Fix compact $K \subseteq U$. If $K = \emptyset$, take $L_K = 0$. Otherwise choose $r > 0$, $K_r := \{z : \operatorname{dist}(z, K) \le r\} \subseteq U$.
Set $A := \max_{K_r} \|D_x f\|$, $M := \max_K \|f\|$, and $L_K := \max\{A, \frac{2M}{r}\}$. For $(t, x), (t, y) \in K$, put $\rho := \|x - y\|$.
If $\rho < r$, then $\sigma(s) := (t, x + s(y - x)) \in K_r$ for $s \in [0, 1]$; by Theorem 3.2.1, $\|f(t, x) - f(t, y)\| \le A\rho$.
If $\rho \ge r$, then $\|f(t, x) - f(t, y)\| \le 2M \le \frac{2M}{r}\rho$. Thus $\|f(t, x) - f(t, y)\| \le L_K \rho$.
Structural Commentary: Nearby pairs use $D_x f$; separated pairs use boundedness.
Key Insight: Compactness yields one state-Lipschitz constant. □

Theorem 1.1.16. *Quantitative Vector Picard–Lindelöf Theorem*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_0, x_0) \in U$. Choose $T, \delta > 0$ and set $V := [t_0, t_0 + T] \times \overline{B}_\delta(x_0) \subseteq U$.
Assume $L \ge 0$ and $\|f(t, x) - f(t, y)\| \le L \|x - y\|$ $\forall (t, x), (t, y) \in V$.
Set $M := \max_V \|f\|$, $T_0 := \min\{T, \delta/M\}$ with $\delta/0 := +\infty$, and $I := [t_0, t_0 + T_0]$.
Let $X := C(I, \mathbb{R}^n)$, $\|u\|_\infty := \max_{t\in I} \|u(t)\|$, and $\mathcal{C} := \{u \in X : \|u - x_0\|_\infty \le \delta\}$.
Define $Ku := x_0 + \int_{t_0}^{(\cdot)} f(s, u(s))\, ds$, $u_0 \equiv x_0$, and $u_{m+1} := Ku_m$.

(i) $(u_m) \subseteq \mathcal{C}$ and $u_m \to \phi$ uniformly for some $\phi \in \mathcal{C}$ satisfying $K\phi = \phi$; hence $\phi \in C^1(I, \mathbb{R}^n)$ solves the IVP.
(ii) ϕ is the unique solution on I whose graph lies in V.
(iii) If $M_0 := \max_{t\in I} \|f(t, x_0)\|$, then $\|\phi - u_m\|_\infty \le T_0 M_0 e^{LT_0} \frac{(LT_0)^m}{m!}$ $\forall m \ge 1$.

Proof.

(i) For $u \in \mathcal{C}$, $\|Ku - x_0\|_\infty \le MT_0 \le \delta$; hence $K(\mathcal{C}) \subseteq \mathcal{C}$.
For $u, v \in \mathcal{C}$, induction on $m \ge 1$ yields $\|K^m u(t) - K^m v(t)\| \le \frac{L^m (t - t_0)^m}{m!} \|u - v\|_\infty$ for $t \in I$.

Hence $\|u_{m+1}-u_m\|_\infty \le \frac{(LT_0)^m}{m!}\|Ku_0-u_0\|_\infty$, so (u_m) is uniformly Cauchy.
By Theorem 8.1.1, X is complete; since $\mathcal{C}$ is closed, $u_m \to \phi \in \mathcal{C}$.
The case $m=1$ implies K is continuous on $\mathcal{C}$, so $K\phi = \lim_{m\to\infty} Ku_m = \lim_{m\to\infty} u_{m+1} = \phi$.
Thus $\phi(t) = x_0 + \int_{t_0}^t f(s,\phi(s))\,ds$, so $\phi \in C^1(I,\mathbb{R}^n)$, $\phi(t_0)=x_0$, and $\phi' = f(\cdot,\phi)$.

(ii) If $Ku=u$ and $Kv=v$ in $\mathcal{C}$, then $\|u-v\|_\infty = \|K^m u - K^m v\|_\infty \le \frac{(LT_0)^m}{m!}\|u-v\|_\infty \ \forall m \ge 1$.
Choose m with $(LT_0)^m/m! < 1$; hence $u=v$.
If y solves the IVP on I with graph in V, then $y \in \mathcal{C}$ and integration implies $Ky = y$; hence $y=\phi$.

(iii) Since $\|Ku_0-u_0\|_\infty \le T_0 M_0$, (i) implies $\|\phi-u_m\|_\infty \le \sum_{j=m}^{\infty}\|u_{j+1}-u_j\|_\infty \le T_0M_0\sum_{j=m}^{\infty}\frac{(LT_0)^j}{j!}$.

For $a \ge 0$, $(m+k)! \ge m!k!$ implies $\sum_{j=m}^{\infty}\frac{a^j}{j!} \le \frac{a^m}{m!}e^a$; set $a = LT_0$.

Structural Commentary: Factorial iterate bounds replace the short-time contraction restriction.
Key Insight: Iterates contract after sufficiently many integrations. □

Theorem 1.1.17. *Picard–Lindelöf with an Integrable Lipschitz Bound*
Let $U \subseteq \mathbb{R}\times\mathbb{R}^n$ be open, $f \in C(U,\mathbb{R}^n)$, $(t_0,x_0)\in U$, and choose $T,\delta>0$ with $V := [t_0,t_0+T]\times\overline{B}_\delta(x_0) \subseteq U$.
Set $b(t) := \max_{x\in\overline{B}_\delta(x_0)}\|f(t,x)\|$. Let $L:[t_0,t_0+T]\to[0,+\infty)$ be integrable,
satisfy $\|f(t,x)-f(t,y)\| \le L(t)\|x-y\|$ for $(t,x),(t,y)\in V$. Choose $0<T_0\le T$ with $\int_{t_0}^{t_0+T_0} b(s)\,ds \le \delta$.
Put $I := [t_0,t_0+T_0]$, $A(t) := \int_{t_0}^t L(s)\,ds$, $A := A(t_0+T_0)$, and $B_0 := \int_{t_0}^{t_0+T_0}\|f(s,x_0)\|\,ds$.
Let $\mathcal{C} := \{u\in C(I,\mathbb{R}^n) : \|u-x_0\|_\infty \le \delta\}$, $(Ku)(t) := x_0 + \int_{t_0}^t f(s,u(s))\,ds$, $u_0 \equiv x_0$, and $u_{m+1} := Ku_m$. Then:
(i) $K(\mathcal{C}) \subseteq \mathcal{C}$ and $u_m \to \phi \in \mathcal{C}$ uniformly, where $K\phi=\phi$.
(ii) $\phi \in C^1(I,\mathbb{R}^n)$ is the unique solution of $x' = f(t,x)$, $x(t_0)=x_0$, whose graph lies in V.
(iii) $\|\phi-u_m\|_\infty \le B_0 e^A \frac{A^m}{m!}$ for $m \ge 0$.

Proof. For $u\in\mathcal{C}$ and $t\in I$, $\|Ku(t)-x_0\| \le \int_{t_0}^t b(s)\,ds \le \delta$; hence $K(\mathcal{C})\subseteq\mathcal{C}$.
Induction and $dA = L\,ds$ give $\|K^m u(t) - K^m v(t)\| \le \frac{A(t)^m}{m!}\|u-v\|_\infty$ for $u,v\in\mathcal{C}$ and $m\ge 1$.
With $u = Ku_0$ and $v=u_0$, $\|u_{j+1}-u_j\|_\infty \le B_0\frac{A^j}{j!}$ for $j\ge 0$. Thus (u_m) is uniformly Cauchy.

(i) By Theorem 8.1.1, $C(I,\mathbb{R}^n)$ is complete; closedness of $\mathcal{C}$ gives $u_m\to\phi\in\mathcal{C}$.
The case $m=1$ makes K continuous, so $K\phi = \lim_{m\to\infty}Ku_m = \lim_{m\to\infty}u_{m+1} = \phi$.

(ii) By Theorem 2.2.1, the fixed-point identity implies $\phi\in C^1(I,\mathbb{R}^n)$, $\phi(t_0)=x_0$, and $\phi' = f(\cdot,\phi)$.
If y is a solution with graph in V, then $y\in\mathcal{C}$ and $Ky=y$.
If $Ku=u$ and $Kv=v$, then $\|u-v\|_\infty \le \frac{A^m}{m!}\|u-v\|_\infty$ for $m\ge 1$; choose m with $\frac{A^m}{m!}<1$, hence $u=v$.

(iii) By telescoping, $\|\phi-u_m\|_\infty \le B_0\sum_{j=m}^{\infty}\frac{A^j}{j!}$. Since $(m+r)! \ge m!r!$, $\sum_{j=m}^{\infty}\frac{A^j}{j!} \le e^A\frac{A^m}{m!}$.

Structural Commentary: Integrated Lipschitz mass gives factorial iterate bounds.
Key Insight: Repeated integration contracts eventually. □

Remark 1.1.18. *Carathéodory Solutions*
Let $I\subseteq\mathbb{R}$, $t_0\in I$, $x_0\in\mathbb{R}^n$, $\delta>0$, $V := I\times\overline{B}_\delta(x_0)$, and $f: V\to\mathbb{R}^n$.
Suppose $f(\cdot,x)$ is measurable, $f(t,\cdot)$ is continuous for a.e. t, and $\exists b, L\in L^1(I,[0,+\infty))$ with $\|f(t,x)\|\le b(t)$ and $\|f(t,x)-f(t,y)\| \le L(t)\|x-y\|$ for a.e. t and $x,y\in\overline{B}_\delta(x_0)$.
Then a fixed point of $(Kx)(t) := x_0 + \int_{t_0}^t f(s,x(s))\,ds$ lies in $AC(I,\mathbb{R}^n)$ and satisfies $x' = f(\cdot,x)$ a.e.
Generally AC cannot be replaced by C^1. For $f(t,x)=\operatorname{sign}(t)x$, $x(t) := e^{|t|}$ satisfies $x(0)=1$,
$x(t) = 1+\int_0^t \operatorname{sign}(s)x(s)\,ds$, and $x' = \operatorname{sign}(t)x$ for $t\ne 0$, but $x\notin C^1((-1,1))$.

Lemma 1.1.19. *Time Regularity of ODE Solutions*
Let $I\subseteq\mathbb{R}$ be an interval, $k,n\in\mathbb{N}$, $k\ge 1$, $U\subseteq\mathbb{R}\times\mathbb{R}^n$ open, and $f\in C^k(U,\mathbb{R}^n)$.
Let $x\in C^1(I,\mathbb{R}^n)$ and $\gamma := (\mathrm{id}_I, x)$. If $\gamma(I)\subseteq U$ and $x' = f\circ\gamma$, then $x\in C^{k+1}(I,\mathbb{R}^n)$.

Proof. Let P_j mean $x\in C^j(I,\mathbb{R}^n)$. Since $x\in C^1$, P_1 holds.
If P_j holds and $1\le j\le k$, then $\gamma\in C^j(I,U)$, so $x' = f\circ\gamma\in C^j(I,\mathbb{R}^n)$; hence P_{j+1}. Thus P_{k+1}.
Structural Commentary: Bootstrap $x' = f\circ\gamma$.
Key Insight: Vector-field regularity lifts the trajectory by one derivative. □

Lemma 1.1.20. *Generalized Gronwall Inequality*
Let $\alpha,\beta,\psi\in C([0,T],\mathbb{R})$ with $\beta\ge 0$ and $\psi(t) \le \alpha(t) + \int_0^t \beta(s)\psi(s)\,ds \ \forall t\in[0,T]$.

(i) $\psi(t) \le \alpha(t) + \int_0^t \alpha(s)\beta(s)\exp\left(\int_s^t\beta(r)\,dr\right)ds \ \forall t\in[0,T]$.

(ii) If $\alpha(s)\le\alpha(t)$ whenever $0\le s\le t\le T$, then $\psi(t)\le\alpha(t)\exp\left(\int_0^t\beta(s)\,ds\right)\ \forall t\in[0,T]$.

Proof.

(i) Put $q(t) := \int_0^t\beta(s)\psi(s)\,ds$ and $w(t) := \exp\left(-\int_0^t\beta(s)\,ds\right)$. Then $(wq)' = w\beta(\psi-q) \le w\beta\alpha$.
Since $q(0)=0$, integration gives $q(t)\le\int_0^t\alpha(s)\beta(s)\exp\left(\int_s^t\beta(r)\,dr\right)ds$; combine this with $\psi\le\alpha+q$.

(ii) Under the monotonicity hypothesis, (i) gives $\psi(t)\le\alpha(t)\left(1+\int_0^t\beta(s)\exp\left(\int_s^t\beta(r)\,dr\right)ds\right)$.
For $q(s) := \exp\left(\int_s^t\beta(r)\,dr\right)$, $q' = -\beta q$, so the parenthesis equals $1+q(0)-q(t) = \exp\left(\int_0^t\beta(s)\,ds\right)$.

Structural Commentary: An integrating factor controls the accumulated integral term.
Key Insight: Exponential growth is the exact cost of repeated feedback. □

Theorem 1.1.21. *Continuous Dependence on Initial Data and Vector Fields*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f, g \in C(U, \mathbb{R}^n)$, $I := [t_0, T]$, and $x, y \in C^1(I, \mathbb{R}^n)$ satisfy $x' = f(\cdot, x)$, $y' = g(\cdot, y)$.
For $w : I \to \mathbb{R}^n$, set $\Gamma_w := \{(t, w(t)) : t \in I\}$. Assume $\Gamma_x \cup \Gamma_y \subseteq V \subseteq U$ with V compact.
Assume $\exists L \geq 0$ such that $\|f(t,p) - f(t,q)\| \leq L \|p - q\|$ whenever $(t,p), (t,q) \in V$.
Set $a := \|x(t_0) - y(t_0)\|$, $E := f - g$, $M := \max_V \|E\|$, $\rho_0(h) := h$, and $\rho_L(h) := L^{-1}(e^{Lh} - 1)$ for $L > 0$.
Then $\|x(t) - y(t)\| \leq e^{L(t-t_0)} a + M \rho_L(t - t_0)$ $\forall t \in I$.

Proof. Put $\Delta := x - y$, $\psi := \|\Delta\|$, and $R(s) := f(s, x(s)) - f(s, y(s)) + E(s, y(s))$.
Then $\Delta(t) = \Delta(t_0) + \int_{t_0}^t R(s)\, ds$ and $\|R(s)\| \leq L\psi(s) + M$. Hence $\psi(t) \leq a + \int_{t_0}^t (L\psi(s) + M)\, ds$.
If $L = 0$, then $\psi(t) \leq a + M(t - t_0) = a + M\rho_0(t - t_0)$.
If $L > 0$, set $\chi := \psi + M/L$. Then $\chi(t) \leq a + M/L + \int_{t_0}^t L\chi(s)\, ds$.
By Lemma 1.1.20(ii), $\chi(t) \leq (a + M/L)e^{L(t-t_0)}$, hence $\psi(t) \leq e^{L(t-t_0)} a + M\rho_L(t - t_0)$.
Structural Commentary: Split state error from field error; apply Gronwall.
Key Insight: Initial and model errors propagate exponentially. □

Theorem 1.1.22. *Local Lipschitz Dependence on the Initial Pair*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_*, x_*) \in U$. Assume f is locally state-Lipschitz:
$\forall$ compact $K \subseteq U$, $\exists L_K < +\infty$ such that $\|f(t,p) - f(t,q)\| \leq L_K \|p - q\|$ whenever $(t,p), (t,q) \in K$.
Then $\exists I, B, V, \Phi$ with $I \ni t_*$ compact, $B \ni x_*$ an open ball, $V \subseteq U$ compact, and $\Phi : I \times I \times B \to \mathbb{R}^n$.
Set $H := \sup I - \inf I$, $M := \max_V \|f\|$, choose a state-Lipschitz constant L for f on V.
Let $D((r,x),(q,y)) := \|x - y\| + M\,|r - q|$. Then:
(i) $\Phi(r, r, x) = x$, $(t, \Phi(t,r,x)) \in V$, and $\partial_t \Phi(t,r,x) = f(t, \Phi(t,r,x))$ for $(t,r,x) \in I \times I \times B$.
(ii) $\|\Phi(t,r,x) - \Phi(s,q,y)\| \leq e^{LH} D((r,x),(q,y)) + M\,|t - s|$ for $(t,r,x),(s,q,y) \in I \times I \times B$.

Proof. Choose $a, \delta > 0$ with $Q := [t_* - a, t_* + a] \times \overline{B}_\delta(x_*) \subseteq U$, and set $M_Q := \max_Q \|f\|$.
Choose $0 < h < a/2$ with $2hM_Q < \delta/2$, set $I := [t_* - h, t_* + h]$, $B := B_{\delta/2}(x_*)$, and $V := I \times \overline{B}_\delta(x_*)$.
Set $H := 2h$, $M := \max_V \|f\|$, and choose a state-Lipschitz constant L for f on V.

(i) Fix $(r,x) \in I \times B$, set $I_r^+ := [r, t_* + h]$ and $I_r^- := [t_* - h, r]$.
Since $I_r^\pm \times \overline{B}_{\delta/2}(x) \subseteq Q$ and $\left|I_r^\pm\right| \leq 2h$, Theorem 1.1.16, applied forward on I_r^+ and backward on I_r^-, gives a unique $u_{r,x} \in C^1(I, \mathbb{R}^n)$ with $u_{r,x}(r) = x$ and $u'_{r,x} = f(\cdot, u_{r,x})$.
For $t \in I$, $\|u_{r,x}(t) - x\| \leq M_Q\,|t - r| < \delta/2$, hence $\|u_{r,x}(t) - x_*\| < \delta$ and $(t, u_{r,x}(t)) \in V$.
Set $\Phi(t,r,x) := u_{r,x}(t)$; then $\Phi(r,r,x) = x$, $(t, \Phi(t,r,x)) \in V$, and $\partial_t \Phi(t,r,x) = f(t, \Phi(t,r,x))$.
(ii) Let $(t,r,x),(s,q,y) \in I \times I \times B$, $p := \max\{r,q\}$, and $D_0 := D((r,x),(q,y))$.
By (i) and $\|\partial_t \Phi\| \leq M$, $\|\Phi(p,r,x) - \Phi(p,q,y)\| \leq D_0$.
By Lemma 1.1.20(ii), after reversing parameter when $t < p$, $\|\Phi(t,r,x) - \Phi(t,q,y)\| \leq e^{L|t-p|} D_0 \leq e^{LH} D_0$.
The bound $\|\Phi(t,q,y) - \Phi(s,q,y)\| \leq M\,|t - s|$ and the triangle inequality give (ii).
Structural Commentary: Compare at the later start time, then apply Gronwall.
Key Insight: Initial-time error first becomes state error. □

Theorem 1.1.23. *Smooth Dependence and First Variational Equations*
Let $n, p \in \mathbb{N}$, $\Omega \subseteq \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^p$ be open, $f \in C^1(\Omega, \mathbb{R}^n)$, and $(t_0, x_0, \lambda_0) \in \Omega$.
Then $\exists I, B, \Lambda_0, \phi$ with $I \ni t_0$ compact, $B \ni x_0$ and $\Lambda_0 \ni \lambda_0$ open, and $\phi : I \times B \times \Lambda_0 \to \mathbb{R}^n$.
For $(x, \lambda) \in B \times \Lambda_0$, set $\gamma_{x,\lambda}(t) := (t, \phi(t,x,\lambda), \lambda)$.
Write $A_{x,\lambda} := D_x f \circ \gamma_{x,\lambda}$ and $G_{x,\lambda} := D_\lambda f \circ \gamma_{x,\lambda}$. The following hold:
(i) $\phi \in C^1(I \times B \times \Lambda_0, \mathbb{R}^n)$, $\phi(t_0, x, \lambda) = x$, and $\partial_t \phi(t,x,\lambda) = f(t, \phi(t,x,\lambda), \lambda)$ $\forall (t,x,\lambda) \in I \times B \times \Lambda_0$.
(ii) $D_x \phi(\cdot, x, \lambda)$ is the unique solution of $Y' = A_{x,\lambda} Y$, $Y(t_0) = I_n$, in $C^1(I, \mathbb{R}^{n \times n})$ $\forall (x,\lambda) \in B \times \Lambda_0$.
(iii) $D_\lambda \phi(\cdot, x, \lambda)$ is the unique solution of $Z' = A_{x,\lambda} Z + G_{x,\lambda}$, $Z(t_0) = 0_{n \times p}$, in $C^1(I, \mathbb{R}^{n \times p})$ $\forall (x,\lambda) \in B \times \Lambda_0$.

Proof. Put $z := (x, \lambda)$, $z_0 := (x_0, \lambda_0)$, $F : \Omega \to \mathbb{R}^{n+p}$ by $F(t,z) := (f(t,x,\lambda), 0)$; then $F \in C^1(\Omega, \mathbb{R}^{n+p})$.
Choose $a, r > 0$ with $W := [t_0 - a, t_0 + a] \times \overline{B}_{2r}(z_0) \subseteq \Omega$. Set $H := \max_W \|F\|$.
Let $\tau := a$ if $H = 0$, and $\tau := \min\{a, \frac{r}{H}\}$ if $H > 0$. For $z \in B_r(z_0)$, $[t_0 - \tau, t_0 + \tau] \times \overline{B}_r(z) \subseteq W$.
By Lemma 1.1.15, choose a state-Lipschitz constant L_W for F on W, and set $I := [t_0 - \tau, t_0 + \tau]$.
By Theorem 1.1.16, applied forward and backward, $\forall z \in B_r(z_0)$, $\exists! \Phi_z \in C^1(I, \mathbb{R}^{n+p})$ with $\Phi_z(t_0) = z$ and $\Phi'_z = F(\cdot, \Phi_z)$. Define $\Phi(t,z) := \Phi_z(t)$. Choose open B, Λ_0 with $x_0 \in B$, $\lambda_0 \in \Lambda_0$, and $B \times \Lambda_0 \subseteq B_r(z_0)$.

(i) Since the last p components of F vanish, $\Phi(t,x,\lambda) = (\phi(t,x,\lambda), \lambda)$.
Hence $\phi(t_0, x, \lambda) = x$ and $\partial_t \phi(t,x,\lambda) = f(t, \phi(t,x,\lambda), \lambda)$.
By Theorem 1.1.21, applied forward and backward, $\|\Phi_z - \Phi_w\|_\infty \leq e^{\tau L_W} \|z - w\|$ for $z, w \in B_r(z_0)$.
Hence Φ and $\partial_t \Phi = F(t, \Phi)$ are continuous.
Fix $z \in B_r(z_0)$ and set $\mathcal{L}_z(t) := D_z F(t, \Phi_z(t))$. Since $\mathcal{L}_z$ is bounded on I, apply Theorem 1.1.16 finitely many times forward and backward to obtain $P_z \in C^1(I, \mathbb{R}^{(n+p)\times(n+p)})$ with $P'_z = \mathcal{L}_z P_z$ and $P_z(t_0) = I_{n+p}$. Set $C_0 := \max_W \|D_z F\|$; By Lemma 1.1.20(ii) $\|P_z\|_\infty \leq e^{2\tau C_0}$.
For $h \to 0$ with $z + h \in B_r(z_0)$, put $\Delta_h := \Phi_{z+h} - \Phi_z$ and $e_h := \Delta_h - P_z h$.
Set $\eta_{h,s} := \Phi_z + s\Delta_h$, $\mathcal{R}_h(t,s) := D_z F(t, \eta_{h,s}(t)) - \mathcal{L}_z(t)$.
Then $R_h(t) := \int_0^1 \mathcal{R}_h(t,s)\, ds$, $\varepsilon_h := \|R_h\|_\infty$; $\varepsilon_h \to 0$; $e'_h = \mathcal{L}_z e_h + R_h \Delta_h$, $e_h(t_0) = 0$, $\|\Delta_h\|_\infty \leq e^{\tau L_W} \|h\|$.
By Lemma 1.1.20(ii), $\|e_h\|_\infty = O(\varepsilon_h \|h\|) = o(\|h\|)$; thus $D_z \Phi = P_z$.
For $z, w \in B_r(z_0)$, $P_z - P_w$ satisfies $(P_z - P_w)' = \mathcal{L}_z(P_z - P_w) + (\mathcal{L}_z - \mathcal{L}_w)P_w$ and $(P_z - P_w)(t_0) = 0$.
By Lemma 1.1.20(ii), $\|P_z - P_w\|_\infty \leq 2\tau e^{4\tau C_0} \|\mathcal{L}_z - \mathcal{L}_w\|_\infty \to 0$ as $z \to w$. Thus Φ, hence ϕ, is C^1.
(ii) Fix (x,λ) and set $z = (x,\lambda)$. Put $Y := D_x \phi(\cdot, x, \lambda)$, $A := A_{x,\lambda}$, and $G := G_{x,\lambda}$.
Here $\mathcal{L}_z(\xi, \mu) = (A\xi + G\mu, 0)$ and $D_x \Phi = (Y, 0)$. The x-columns of $P'_z = \mathcal{L}_z P_z$ give $Y' = AY$, $Y(t_0) = I_n$.
If $\widetilde{Y}$ also solves this IVP, $W := Y - \widetilde{Y}$ satisfies $W' = AW$, $W(t_0) = 0$; By Lemma 1.1.20(ii) $W = 0$.
(iii) Fix (x,λ) and set $z = (x,\lambda)$. Put $Z := D_\lambda \phi(\cdot, x, \lambda)$, $A := A_{x,\lambda}$, and $G := G_{x,\lambda}$.
Here $\mathcal{L}_z(\xi, \mu) = (A\xi + G\mu, 0)$. Since $D_\lambda \Phi = (Z, I_p)$, the λ-columns give $Z' = AZ + G$, $Z(t_0) = 0_{n \times p}$.
If $\widetilde{Z}$ also solves this IVP, $W := Z - \widetilde{Z}$ satisfies $W' = AW$, $W(t_0) = 0$; By Lemma 1.1.20(ii) $W = 0$.

Structural Commentary: Lift parameters; differentiate the extended flow.
Key Insight: Flow derivatives satisfy variational equations. □

Theorem 1.1.24. *Higher-Order Smooth Dependence*
Let $k, n, p \in \mathbb{N}$, $k \geq 1$, $\Omega \subseteq \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^p$ open, $f \in C^k(\Omega, \mathbb{R}^n)$, and $q_0 = (t_0, x_0, \lambda_0) \in \Omega$.
Then $\exists I \ni t_0$ compact, $B \ni x_0$ open in $\mathbb{R}^n$, $\Lambda_0 \ni \lambda_0$ open in $\mathbb{R}^p$, and $\phi : D \to \mathbb{R}^n$, with $D := I \times B \times \Lambda_0$.
For $\Gamma_\phi(t, x, \lambda) := (t, \phi(t, x, \lambda), \lambda)$:
(i) $\Gamma_\phi(D) \subseteq \Omega$, $\phi(t_0, x, \lambda) = x$, and $\partial_t \phi = f \circ \Gamma_\phi$ on D.
(ii) $\phi \in C^k(D, \mathbb{R}^n)$.

Proof. Let $\mathsf{H}_{r,d}$ mean: every $F \in C^r(\Omega_d, \mathbb{R}^d)$, $\Omega_d \subseteq \mathbb{R} \times \mathbb{R}^d$ open, has a local C^r flow near each $(\tau, z_*) \in \Omega_d$.
By Theorem 1.1.23(i) with a dummy parameter, $\mathsf{H}_{1,d}$ holds $\forall d$. Assume $\mathsf{H}_{j,d}$ for all d. Let $F \in C^{j+1}(\Omega_d, \mathbb{R}^d)$, $(\tau, z_*) \in \Omega_d$, and let Ψ be its local flow. By $\mathsf{H}_{j,d}$, $\Psi \in C^j$. Set $Q := D_z \Psi$ and $E_d := \mathrm{Mat}(d, d, \mathbb{R})$.
By Theorem 1.1.23(ii), $\partial_t Q = D_z F(t, \Psi) Q$ and $Q(\tau, z) = I_d$.
Set $\mathcal{H}(t, \zeta, P) := (F(t, \zeta), D_z F(t, \zeta) P)$ on $\Omega_d \times E_d$, and set $Y := (\Psi, Q)$. Then $\mathcal{H} \in C^j$ and $Y_t = \mathcal{H}(t, Y)$, $Y(\tau, z) = (z, I_d)$.
By $\mathsf{H}_{j,d+d^2}$, the flow Θ of $\mathcal{H}$ is C^j. By uniqueness, $Y(t, z) = \Theta(t, z, I_d)$, so $Q \in C^j$.
By $D_z \Psi = Q \in C^j$ and $\partial_t \Psi = F(t, \Psi) \in C^j$, $\Psi \in C^{j+1}$. By induction, $\mathsf{H}_{r,d}$ holds for all $r, d \in \mathbb{N}$.
Set $m := n + p$, $z := (x, \lambda)$, $z_0 := (x_0, \lambda_0)$, $\widetilde{\Omega} := \{(t, z) : (t, x, \lambda) \in \Omega\}$, and $\widetilde{F}(t, z) := (f(t, x, \lambda), 0_p)$.
By $\mathsf{H}_{k,m}$, after shrinking, choose compact $I \ni t_0$, open $U_0 \ni z_0$, and $\Phi \in C^k(I \times U_0, \mathbb{R}^m)$ with $\partial_t \Phi = \widetilde{F}(t, \Phi)$ and $\Phi(t_0, z) = z$.
Choose $B \ni x_0$ and $\Lambda_0 \ni \lambda_0$ with $B \times \Lambda_0 \subseteq U_0$. Set $D := I \times B \times \Lambda_0$, $\pi_x := \pi_{\mathbb{R}^n}$, $\pi_\lambda := \pi_{\mathbb{R}^p}$, and $\phi := \pi_x \circ \Phi$.
(i) By $\partial_t(\pi_\lambda \circ \Phi) = 0$ and $(\pi_\lambda \circ \Phi)(t_0, x, \lambda) = \lambda$, $\Phi = (\phi, \lambda)$ on D.
By this identity and $(t, \Phi) \in \widetilde{\Omega}$, $\Gamma_\phi(D) \subseteq \Omega$; by $\Phi(t_0, x, \lambda) = (x, \lambda)$, $\phi(t_0, x, \lambda) = x$.
By $\partial_t \Phi = \widetilde{F}(t, \Phi)$ and $\phi = \pi_x \circ \Phi$, $\partial_t \phi = f \circ \Gamma_\phi$ on D.
(ii) By $\Phi \in C^k(I \times U_0, \mathbb{R}^m)$ and linearity of π_x, $\phi \in C^k(D, \mathbb{R}^n)$.
Structural Commentary: Bootstrap through the variational flow.
Key Insight: Parameters are frozen coordinates. □

Proposition 1.1.25. *Regular Perturbation Expansion*
Let $k, n \in \mathbb{N}$, $k \geq 1$, $\Omega \subseteq \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}$ open, $f \in C^k(\Omega, \mathbb{R}^n)$, and $q_0 = (t_0, x_0, \varepsilon_0) \in \Omega$.
Let $D := I \times \Lambda_0$, where $I \ni t_0$ is compact and $\Lambda_0 \ni \varepsilon_0$ is open, and let $\phi \in C^k(D, \mathbb{R}^n)$.
Set $\Gamma_\phi(t, \varepsilon) := (t, \phi(t, \varepsilon), \varepsilon)$ and $h_\varepsilon := \varepsilon - \varepsilon_0$.
Assume $\Gamma_\phi(D) \subseteq \Omega$, $\phi(t_0, \varepsilon) = x_0$, and $\partial_t \phi = f \circ \Gamma_\phi$.
For $0 \leq j \leq k$, set $Z_j := (z_0, \ldots, z_j)$.
For $g = g(t, Z_j, \varepsilon)$ and $0 \leq j < k$, set $\mathcal{D}_j g := \partial_\varepsilon g + \sum_{r=0}^{j} D_{z_r} g \, z_{r+1}$.
Set $f_0(t, z_0, \varepsilon) := f(t, z_0, \varepsilon)$ and $f_{j+1} := \mathcal{D}_j f_j$ for $0 \leq j < k$.
For $0 \leq j \leq k$, set $\phi_j(t) := \partial_\varepsilon^j \phi(t, \varepsilon_0)$ and $\Phi_j(t) := (\phi_0(t), \ldots, \phi_j(t))$.
Set $T_k(t, \varepsilon) := \sum_{j=0}^{k} \frac{\phi_j(t)}{j!} h_\varepsilon^j$.
Set $r_k(t, \varepsilon) := \phi(t, \varepsilon) - T_k(t, \varepsilon)$ and $\rho_k(\varepsilon) := \sup_{t \in I} \|r_k(t, \varepsilon)\|$.
Then:
(i) For $0 \leq j \leq k$, $\phi_j' = f_j(t, \Phi_j, \varepsilon_0)$ and $\phi_j(t_0) = \delta_{j0} x_0$.
(ii) $\rho_k(\varepsilon) = o(|h_\varepsilon|^k)$.
(iii) If $f \in C^{k+1}(\Omega, \mathbb{R}^n)$, then, after shrinking I and Λ_0, $\rho_k(\varepsilon) = O(|h_\varepsilon|^{k+1})$.

Proof. For $0 \leq j \leq k$, set $z_j := \partial_\varepsilon^j \phi$, $\widehat{Z}_j := (z_0, \ldots, z_j)$, and $R_j := f_j(t, \widehat{Z}_j, \varepsilon)$.
For $g = g(t, \varepsilon)$, set $(\mathcal{I} g)(t, \varepsilon) := \int_{t_0}^{t} g(s, \varepsilon) \, ds$.
(i) Since $R_0 = f(t, \phi, \varepsilon)$, the IVP is $z_0 = x_0 + \mathcal{I} R_0$. By the chain rule, $\partial_\varepsilon R_j = R_{j+1}$ for $0 \leq j < k$.
Base $j = 0$ is the IVP. If $z_j = \delta_{j0} x_0 + \mathcal{I} R_j$, then by differentiation under $\mathcal{I}$, $z_{j+1} = \mathcal{I} R_{j+1}$.
By induction, $z_j = \delta_{j0} x_0 + \mathcal{I} R_j$ for $0 \leq j \leq k$.
At $\varepsilon = \varepsilon_0$, $z_j = \phi_j$ and $R_j = f_j(t, \Phi_j, \varepsilon_0)$.
By the preceding identity, $\phi_j = \delta_{j0} x_0 + \mathcal{I}(f_j(\,\cdot\,, \Phi_j, \varepsilon_0))$. Differentiating in t proves (i).
(ii) Put $h := h_\varepsilon$ and $G_t(\eta) := \partial_\varepsilon^k \phi(t, \eta) - \phi_k(t)$.
By integral Taylor, $r_k(t, \varepsilon) = \frac{h^k}{(k-1)!} \int_0^1 (1 - s)^{k-1} G_t(\varepsilon_0 + sh) \, ds$.
Set $\omega(h) := \sup_{(t,s) \in I \times [0,1]} \|G_t(\varepsilon_0 + sh)\|$. By uniform continuity, $\omega(h) \to 0$.
By the formula above, $\rho_k(\varepsilon) \leq \frac{|h|^k}{k!} \omega(h) = o(|h|^k)$.
(iii) If $f \in C^{k+1}$, by Theorem 1.1.24, choose $\widetilde{I} \ni t_0$ and $\widetilde{\Lambda} \ni \varepsilon_0$.
Let $\widetilde{\phi} \in C^{k+1}(\widetilde{I} \times \widetilde{\Lambda}, \mathbb{R}^n)$ satisfy the same IVP.
By local uniqueness from Theorem 1.1.16, after shrinking $I \times \Lambda_0$, $\widetilde{\phi} = \phi$, so $\phi \in C^{k+1}(I \times \Lambda_0, \mathbb{R}^n)$.
Choose compact $J \Subset \Lambda_0$ with $\varepsilon_0 \in J^\circ$.
Set $C_J := \max_{(t,\eta) \in I \times J} \left\| \partial_\varepsilon^{k+1} \phi(t, \eta) \right\|$.
For $\varepsilon \in J$, by the integral remainder, $\rho_k(\varepsilon) \leq \frac{C_J |h_\varepsilon|^{k+1}}{(k+1)!}$.
Structural Commentary: Differentiate, then Taylor expand.
Key Insight: Coefficients satisfy forced IVPs. □

Example 1.1.26. *Linear Drag as a Regular Perturbation*
Let $g > 0$, $\varepsilon \geq 0$, and $v' = -\varepsilon v - g$, $v(0) = 0$. Then $v(t, \varepsilon) = \frac{g(e^{-\varepsilon t} - 1)}{\varepsilon}$ if $\varepsilon > 0$, and $v_0(t) := v(t, 0) = -gt$.

At $\varepsilon_0 = 0$, $v_1' = -v_0 = gt$, $v_1(0) = 0$, hence $v_1(t) = \frac{gt^2}{2}$ and $v(t,\varepsilon) = v_0(t) + \varepsilon v_1(t) + o(\varepsilon)$ on compact t-intervals.
For fixed $\varepsilon > 0$, $t \to +\infty$ gives $v(t,\varepsilon) \to -\frac{g}{\varepsilon}$ but $v_0(t) + \varepsilon v_1(t) \to +\infty$; the expansion is local for bounded t.

Theorem 1.1.27. *Unique Maximal Solution*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_0, x_0) \in U$.
Assume $\forall$ compact $V \subseteq U$, $\exists L_V < +\infty$ such that $\|f(t,p) - f(t,q)\| \le L_V \|p - q\|$ whenever $(t,p),(t,q) \in V$.
For an open interval $J \ni t_0$, write $u \in \mathcal{A}(J)$ iff $u \in C^1(J, \mathbb{R}^n)$, $u(t_0) = x_0$, $(t, u(t)) \in U$, and $u' = f(\cdot, u)$.
Let $\mathcal{S}$ be all such pairs (J,u). For $(J,u),(K,v) \in \mathcal{S}$, write $(J,u) \preceq (K,v)$ iff $J \subseteq K$ and $v|_J = u$. Then:

(i) $\exists$ a maximal $(I_*, \phi_*) \in \mathcal{S}$ with $I_* = (T_-, T_+)$.
(ii) $(J,u) \preceq (I_*, \phi_*)$ $\forall (J,u) \in \mathcal{S}$; hence the maximal solution is unique.

Proof. By Theorem 1.1.16, applied forward and backward, $\mathcal{S} \neq \emptyset$. For $(J,u),(K,v) \in \mathcal{S}$, set $E := \{t \in J \cap K : u(t) = v(t)\}$.
Then $t_0 \in E$, E is closed, and local uniqueness at each $s \in E$ makes E open in $J \cap K$.
Since $J \cap K$ is an interval, $E = J \cap K$; thus $u = v$ on $J \cap K$ $\forall (J,u),(K,v) \in \mathcal{S}$.

(i) Set $I_* := \bigcup_{(J,u)\in\mathcal{S}} J$ and let $\phi_*(t) := u(t)$ for any $(J,u) \in \mathcal{S}$ with $t \in J$. Agreement on overlaps makes ϕ_* well defined.
Since all J are open intervals containing t_0, $I_* = (T_-, T_+)$. Locally $\phi_* = u$, so $(I_*, \phi_*) \in \mathcal{S}$.
If $(I_*, \phi_*) \preceq (\widehat{I}, \widehat{\phi}) \in \mathcal{S}$, then $I_* \subseteq \widehat{I} \subseteq I_*$; hence $\widehat{I} = I_*$.
(ii) If $(J,u) \in \mathcal{S}$, then $J \subseteq I_*$ and overlap agreement gives $u = \phi_*|_J$; hence $(J,u) \preceq (I_*, \phi_*)$.
If (K,ψ) is maximal, then $(K,\psi) \preceq (I_*, \phi_*)$; maximality gives $K = I_*$ and $\psi = \phi_*$.

Structural Commentary: Local uniqueness makes solution germs compatible.
Key Insight: The maximal solution is their union. □

Lemma 1.1.28. *Extension Criterion at a Finite Endpoint*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and assume f is locally Lipschitz in x on compact subsets of U.
Let $\phi \in C^1((T_-, T_+), \mathbb{R}^n)$ satisfy $\phi' = f(\cdot, \phi)$, $(t, \phi(t)) \in U$, and $T_+ < +\infty$.
Then $\exists \varepsilon > 0$ and a solution $\widehat{\phi} : (T_-, T_+ + \varepsilon) \to \mathbb{R}^n$ with $\widehat{\phi}|_{(T_-,T_+)} = \phi$ iff $\exists t_m \uparrow T_+$ and $y \in \mathbb{R}^n$ such that $\phi(t_m) \to y$ and $(T_+, y) \in U$.

Proof.
($\Rightarrow$) Let $\widehat{\phi}$ be such an extension, choose $t_m \uparrow T_+$, and put $y := \widehat{\phi}(T_+)$. Then $\phi(t_m) = \widehat{\phi}(t_m) \to y$ and $(T_+, y) \in U$.
($\Leftarrow$) Choose $r, \eta > 0$ with $W := [T_+ - \eta, T_+ + \eta] \times \overline{B}_r(y) \subseteq U$ and set $M := \max_W \|f\|$.
Choose m large enough that $t_m > T_+ - \eta$, $a := \|\phi(t_m) - y\| < \frac{r}{2}$, and $Mh < \frac{r}{2}$ for $h := T_+ - t_m$; put $J := [t_m, T_+)$ and $B := B_r(y)$.
If $\tau \in J$ is the first exit from B, then $r \le a + M(\tau - t_m) \le a + Mh < r$, impossible. Hence $\phi(J) \subseteq B$.
For $s,t \in J$, $\|\phi(t) - \phi(s)\| \le M|t - s|$, so $\phi(t)$ is Cauchy as $t \uparrow T_+$. Since $\phi(t_m) \to y$, $\lim_{t\uparrow T_+} \phi(t) = y$.
By Theorem 1.1.16, $\exists \varepsilon > 0$ and a solution $\widetilde{\phi}$ on $[T_+, T_+ + \varepsilon)$ with $\widetilde{\phi}(T_+) = y$. Define $\widehat{\phi} = \phi$ on (T_-, T_+) and $\widehat{\phi} = \widetilde{\phi}$ on $[T_+, T_+ + \varepsilon)$.
Continuity of f gives $\phi'(t) \to f(T_+, y) = \widetilde{\phi}'(T_+)$; hence $\widehat{\phi} \in C^1((T_-, T_+ + \varepsilon), \mathbb{R}^n)$ and is the required extension.

Structural Commentary: A convergent endpoint subsequence traps the tail for gluing.
Key Insight: Finite endpoint limits are exactly extendible states. □

Corollary 1.1.29. *Compact Escape at a Finite Maximal Endpoint*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and $f \in C(U, \mathbb{R}^n)$.
Assume $\forall C \Subset U$, $\exists L_C < +\infty$ such that $(t,p),(t,q) \in C \Rightarrow \|f(t,p) - f(t,q)\| \le L_C \|p - q\|$.
Let $J := (T_-, T_+)$ and let $\phi : J \to \mathbb{R}^n$ be maximal for $x' = f(t,x)$, with $\gamma(t) := (t, \phi(t)) \in U$. Then:

(i) $T_+ < +\infty \Rightarrow \forall K \Subset U$, $\exists \tau < T_+ : \gamma((\tau, T_+)) \cap K = \emptyset$.
(ii) $T_- > -\infty \Rightarrow \forall K \Subset U$, $\exists \tau > T_- : \gamma((T_-, \tau)) \cap K = \emptyset$.
(iii) If $U = \mathbb{R} \times \mathbb{R}^n$, then $T_+ < +\infty \Rightarrow \|\phi(t)\| \to +\infty$ as $t \uparrow T_+$ and $T_- > -\infty \Rightarrow \|\phi(t)\| \to +\infty$ as $t \downarrow T_-$.

Proof.
(i) If (i) fails for some $K \Subset U$, then $\exists t_m \uparrow T_+$ with $\gamma(t_m) \in K$. By compactness, after passing to a subsequence, $\gamma(t_m) \to (T_+, y) \in K \subseteq U$. By Lemma 1.1.28, ϕ admits a proper extension past T_+, contradiction.
(ii) Set $\widetilde{U} := \{(s,x) : (-s,x) \in U\}$, $\widetilde{f}(s,x) := -f(-s,x)$, $\widetilde{J} := (-T_+, -T_-)$, and $u(s) := \phi(-s)$.
Then $u : \widetilde{J} \to \mathbb{R}^n$ is maximal for $u' = \widetilde{f}(\cdot, u)$. For $K \Subset U$, put $\widetilde{K} := \{(-t,x) : (t,x) \in K\} \Subset \widetilde{U}$.
By (i) applied to u at $-T_-$, $\exists a < -T_-$ such that $(s, u(s)) \notin \widetilde{K}$ for $a < s < -T_-$.
With $\tau := -a$, $\gamma((T_-, \tau)) \cap K = \emptyset$.
(iii) Assume $U = \mathbb{R} \times \mathbb{R}^n$ and $T_+ < +\infty$. For $R > 0$, set $K_R := [T_+ - 1, T_+] \times \overline{B}_R(0)$.
By (i), $\exists \tau < T_+$ such that $\gamma((\tau, T_+)) \cap K_R = \emptyset$, hence $\|\phi(t)\| > R$ for $\max\{\tau, T_+ - 1\} < t < T_+$.
Since R is arbitrary, $\|\phi(t)\| \to +\infty$ as $t \uparrow T_+$. If $T_- > -\infty$, apply (ii) to $K_R^- := [T_-, T_- + 1] \times \overline{B}_R(0)$ to obtain $\|\phi(t)\| > R$ near T_-; hence $\|\phi(t)\| \to +\infty$ as $t \downarrow T_-$.

Structural Commentary: Compact recurrence creates an endpoint extension.
Key Insight: Finite maximal time forces compact escape. □

Corollary 1.1.30. *Global Existence under Linear Growth*
Let $f : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ be continuous.
Assume (LL): $\forall$ compact $V \subseteq \mathbb{R} \times \mathbb{R}^n$, $\exists L_V < +\infty$ such that $\|f(t,p) - f(t,q)\| \le L_V \|p - q\|$ $\forall (t,p),(t,q) \in V$.
Assume (LG): $\forall T > 0$, $\exists M_T, L_T \ge 0$ such that $\|f(t,x)\| \le M_T + L_T \|x\|$ $\forall (t,x) \in [-T, T] \times \mathbb{R}^n$.
If $\phi : (T_-, T_+) \to \mathbb{R}^n$ is maximal and $\phi' = f(\cdot, \phi)$, then $(T_-, T_+) = \mathbb{R}$.

Proof. Fix $t_0 \in (T_-, T_+)$, $x_0 := \phi(t_0)$. If $T_+ < +\infty$, choose $T > \max\{|t_0|, |T_+|\}$ and M_T, L_T from (LG), and set $C_T := (\|x_0\| + M_T(T_+ - t_0))e^{L_T(T_+ - t_0)}$.
With $q(t) := \|\phi(t)\|$, $q(t) \le \|x_0\| + \int_{t_0}^t (M_T + L_T q(s))\, ds$ on $[t_0, T_+)$. Lemma 1.1.20(ii) gives $q(t) \le (\|x_0\| + M_T(t - t_0))e^{L_T(t - t_0)} \le C_T < +\infty$.
Thus $\Gamma_+ := \{(t, \phi(t)) : t \in [t_0, T_+)\} \subseteq K_T := [-T, T] \times \overline{B}_{C_T}(0)$. For $t_m \uparrow T_+$, compactness gives $\phi(t_{m_j}) \to y$ and $(T_+, y) \in K_T$. By (LL) and Lemma 1.1.28, ϕ extends past T_+, contradiction. Hence $T_+ = +\infty$.
Set $g(s,x) := -f(t_0 - s, x)$ and $u(s) := \phi(t_0 - s)$ on $(t_0 - T_+, t_0 - T_-)$. Then $u' = g(\cdot, u)$, u is maximal, and g satisfies (LL),(LG) on

bounded s-intervals. Apply right-endpoint result to u, then $t_0 - T_- = +\infty$, $T_- = -\infty$.
Structural Commentary: Gronwall bounds; compactness forces extension.
Key Insight: Linear growth rules out finite escape. □

Theorem 1.1.31. *Peano Local Existence Theorem*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_0, x_0) \in U$. Choose $T, \delta > 0$ with $V := [t_0, t_0 + T] \times \overline{B}_\delta(x_0) \subseteq U$.
Set $M := \max_V \|f\|$, $T_0 := \min\{T, \delta/M\}$ with $\delta/0 := +\infty$, and $I := [t_0, t_0 + T_0]$.
For $y : I \to \mathbb{R}^n$, write $\Gamma_y := \{(t, y(t)) : t \in I\}$. Then:

(i) $\exists \phi \in C(I, \mathbb{R}^n)$ with $\Gamma_\phi \subseteq V$ and $\phi = x_0 + \int_{t_0}^{(\cdot)} f(s, \phi(s))\, ds$ on I.
(ii) This ϕ lies in $C^1(I, \mathbb{R}^n)$ and satisfies $\phi(t_0) = x_0$, $\phi' = f(\cdot\,, \phi)$.

Proof. For $N \geq 1$, set $h_N := T_0/N$, $t_j := t_0 + j h_N$, $\tau_N(t) := t_j$ on $[t_j, t_{j+1})$ for $0 \leq j < N$, and $\tau_N(t_N) := t_{N-1}$.
Construct x_N recursively by $x_N(t_0) := x_0$ and $x_N(t) := x_N(t_j) + (t - t_j) f(t_j, x_N(t_j))$ for $t \in [t_j, t_{j+1}]$, $0 \leq j < N$.
Write $(\mathcal{I}G)(t) := \int_{t_0}^t G(s)\, ds$, $Y_N := (\tau_N, x_N \circ \tau_N)$, and $F_N := f \circ Y_N$; then $x_N = x_0 + \mathcal{I} F_N$.
By induction, $\|x_N(t) - x_0\| \leq M(t - t_0) \leq \delta$ and $\|x_N(t) - x_N(s)\| \leq M\,|t - s|$.
Hence $\Gamma_{x_N} \subseteq V$ and (x_N) is uniformly bounded/equicontinuous.

(i) By Theorem 8.2.1, applied coordinatewise, $x_{N_k} \to \phi \in C(I, \mathbb{R}^n)$ uniformly.
Then $\Gamma_\phi \subseteq V$. Set $Y_\phi := (\mathrm{id}_I, \phi)$ and $f_\phi := f \circ Y_\phi$.
Since $\|\mathrm{id}_I - \tau_N\|_\infty \leq h_N$ and $\|x_N - x_N \circ \tau_N\|_\infty \leq M h_N$, $Y_{N_k}(I), Y_\phi(I) \subseteq V$ and $Y_{N_k} \to Y_\phi$ uniformly.
By uniform continuity of $f|_V$, $F_{N_k} \to f_\phi$ uniformly.
Passing to the limit in $x_{N_k} = x_0 + \mathcal{I} F_{N_k}$ gives $\phi = x_0 + \mathcal{I} f_\phi$.
(ii) Since $f_\phi \in C(I, \mathbb{R}^n)$, by Theorem 2.2.1 applied to (i), $\phi \in C^1(I, \mathbb{R}^n)$, $\phi(t_0) = x_0$, and $\phi' = f_\phi$.

Structural Commentary: Euler polygons plus Arzelà–Ascoli produce a convergent approximate trajectory.
Key Insight: Continuity yields existence through compactness, without uniqueness. □

Definition 1.1.32. *Osgood Modulus of Continuity*
Let $\rho \in C([0, +\infty), [0, +\infty))$.

(i) ρ is a *modulus of continuity* iff $\rho(0) = 0$ and ρ is nondecreasing.
(ii) A modulus ρ satisfies the *Osgood condition* iff $\int_0^1 \frac{dr}{\rho(r)} = +\infty$, interpreted as an improper extended integral; equivalently, $\int_0^a \frac{dr}{\rho(r)} = +\infty\ \forall a > 0$.

Key Insight: Osgood divergence prevents a zero separation from becoming positive in finite time.

Example 1.1.33. *Lipschitz, Hölder, and Almost-Lipschitz Moduli*
For $0 < \alpha < 1$, set $\rho_1(r) := r$, $\rho_\alpha(r) := r^\alpha$, $\rho_{\log}(0) := 0$, and $\rho_{\log}(r) := r(1 + |\log r|)$ for $r > 0$.
These functions are continuous, nondecreasing, and vanish at 0.
Since $\int_0^1 \frac{dr}{r} = +\infty$ and $\int_0^1 r^{-\alpha}\, dr = \frac{1}{1-\alpha} < +\infty$, ρ_1 is Osgood while the Hölder modulus ρ_α is not.
For the almost-Lipschitz modulus, $s = -\log r$ gives $\int_0^1 \frac{dr}{r(1+|\log r|)} = \int_0^{+\infty} \frac{ds}{1+s} = +\infty$, so $\rho_{\log}$ is Osgood.

Theorem 1.1.34. *Osgood Uniqueness Criterion*
Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, $(t_0, x_0) \in U$, $T > t_0$, and $\delta > 0$.
Put $J := [t_0, T]$, $V := J \times \overline{B}_\delta(x_0)$, and assume $V \subseteq U$.
Let $\rho \in C([0, +\infty), [0, +\infty))$ be nondecreasing, $\rho(0) = 0$, and $\int_0^a \frac{dr}{\rho(r)} = +\infty\ \forall a > 0$.
For $(t, p), (t, q) \in V$, set $A_t(p, q) := |\langle p - q, f(t, p) - f(t, q)\rangle|$.
Assume $\exists C \geq 0$ such that $A_t(p, q) \leq C\,\|p - q\|\,\rho(\|p - q\|)\ \forall (t, p), (t, q) \in V$.
For $w \in C(J, \mathbb{R}^n)$, set $\Gamma_w := \{(t, w(t)) : t \in J\}$.
If $u, v \in C^1(J, \mathbb{R}^n)$ satisfy $\Gamma_u \cup \Gamma_v \subseteq V$, $u(t_0) = v(t_0) = x_0$, $u' = f(\cdot\,, u)$, and $v' = f(\cdot\,, v)$, then $u = v$ on J.

Proof. Assume $S := \{t \in (t_0, T] : u(t) \neq v(t)\} \neq \emptyset$ and fix $t \in S$. Put $\Delta := u - v$ and $d := \|\Delta\|$.
For $s \in J$, set $E(s) := f(s, u(s)) - f(s, v(s))$. The set $Z_t := \{s \in [t_0, t] : d(s) = 0\}$ is nonempty and compact; set $a := \max Z_t$.
Then $a < t$, $d(a) = 0$, and $d > 0$ on $(a, t]$. On $(a, t]$, $d \in C^1$ and $d'(s) = \frac{\langle \Delta(s), E(s)\rangle}{d(s)}$. Hence $d'(s) \leq \frac{A_s(u(s), v(s))}{d(s)} \leq C\rho(d(s))$.
For $m \geq 1$, set $H_m(r) := \int_0^r \frac{dq}{\rho(q) + \frac{1}{m}}$. For $s \in (a, t]$, set $b_m(s) := \frac{\rho(d(s))}{\rho(d(s)) + \frac{1}{m}} \leq 1$.
If $a < r < t$, then on $[r, t]$, $(H_m \circ d)'(s) = \frac{d'(s)}{\rho(d(s)) + \frac{1}{m}} \leq C b_m(s) \leq C$.
Thus $H_m(d(t)) - H_m(d(r)) \leq C(t - r)$. Letting $r \downarrow a$ gives $H_m(d(t)) \leq C(t - a)$.
By Theorem 1.3.1, $H_m(d(t)) \uparrow \int_0^{d(t)} \frac{dq}{\rho(q)} = +\infty$, contradiction. Hence $S = \emptyset$ and $u = v$.
Structural Commentary: Regularize $1/\rho$; separate variables.
Key Insight: Divergence forbids first separation. □

Proposition 1.1.35. *Time Translation and the Local Flow Law*
Let $D \subseteq \mathbb{R}^n$ be open and $f : D \to \mathbb{R}^n$. For $(a, y) \in \mathbb{R} \times D$, let $\mathcal{S}(a, y)$ contain the pairs (I, u) such that $I \ni a$ is an open interval, $u \in C^1(I, D)$, $u(a) = y$, and $u' = f(u)$.
Assume $\mathcal{S}(a, y) \neq \emptyset\ \forall (a, y) \in \mathbb{R} \times D$, and $u = v$ on $I \cap J\ \forall (a, y) \in \mathbb{R} \times D$, $\forall (I, u), (J, v) \in \mathcal{S}(a, y)$.
For $x \in D$, let $\phi_x : I_x \to D$ be the maximal solution with $0 \in I_x$, $\phi_x(0) = x$, and $\phi_x' = f(\phi_x)$.
Define $\mathcal{D}_\Phi := \{(t, x) \in \mathbb{R} \times D : t \in I_x\}$ and $\Phi_t(x) := \phi_x(t)$ for $(t, x) \in \mathcal{D}_\Phi$.

(i) If $\phi : I \to D$, $0 \in I$, and $\phi' = f(\phi)$, then $\psi_\tau(t) := \phi(t - \tau)$ on $\tau + I$ satisfies $\psi_\tau' = f(\psi_\tau)\ \forall \tau \in \mathbb{R}$.
Its initial value is $\psi_\tau(\tau) = \phi(0)$.
(ii) $(0, x) \in \mathcal{D}_\Phi$ and $\Phi_0(x) = x\ \forall x \in D$; hence $\Phi_0 = \mathrm{id}_D$.
(iii) If $(s, x), (t, \Phi_s(x)), (t + s, x) \in \mathcal{D}_\Phi$, then $\Phi_{t+s}(x) = \Phi_t(\Phi_s(x))$.

Proof.

(i) For $t \in \tau + I$, $\psi_\tau'(t) = \phi'(t - \tau) = f(\phi(t - \tau)) = f(\psi_\tau(t))$ and $\psi_\tau(\tau) = \phi(0)$.
(ii) Since $0 \in I_x$, $(0, x) \in \mathcal{D}_\Phi$ and $\Phi_0(x) = \phi_x(0) = x\ \forall x \in D$.

(iii) Set $R := (I_x - s) \cap I_{\Phi_s(x)}$. Then $0, t \in R$ and $(r+s, x), (r, \Phi_s(x)) \in \mathcal{D}_\Phi$ for $r \in R$.
Define $u, v : R \to D$ by $u(r) := \Phi_{r+s}(x)$, $v(r) := \Phi_r(\Phi_s(x))$.
Then $u' = f(u)$ by (i), $v' = f(v)$, $u(0) = v(0) = \Phi_s(x)$.
Uniqueness gives $u = v$ on R, so $\Phi_{t+s}(x) = \Phi_t(\Phi_s(x))$.

Structural Commentary: Shift time; use uniqueness.
Key Insight: Autonomy makes elapsed time compose. □

1.2 Linear Systems and Stability

Scalar first-order linear equations admit an explicit evolution factor and a variation-of-constants representation.

Proposition 1.2.1. *Scalar Variation of Constants*
Let $I \subseteq \mathbb{R}$ be open, $a, g \in C(I, \mathbb{R})$, and $(t_0, x_0) \in I \times \mathbb{R}$.
Put $E(t,s) := \exp\left(\int_s^t a(r)\,dr\right)$ and $\phi(t) := E(t,t_0)x_0 + \int_{t_0}^t E(t,s)g(s)\,ds$. Then:

(i) $\phi \in C^1(I, \mathbb{R})$ and $(\phi' - a\phi, \phi(t_0)) = (g, x_0)$.
(ii) If $x \in C^1(I, \mathbb{R})$ and $(x' - ax, x(t_0)) = (g, x_0)$, then $x = \phi$.

Proof.
(i) Let $B(t) := x_0 + \int_{t_0}^t E(t_0, s)g(s)\,ds$. By additivity, $E(t,s) = E(t,t_0)E(t_0,s)$, so $\phi(t) = E(t,t_0)B(t)$.
By Theorem 2.2.1, $B \in C^1$, $\partial_t E(t,t_0) = a(t)E(t,t_0)$, and $B'(t) = E(t_0,t)g(t)$.
Hence $\phi \in C^1$, $\phi'(t) = a(t)\phi(t) + E(t,t_0)E(t_0,t)g(t) = a(t)\phi(t) + g(t)$, and $\phi(t_0) = x_0$.
(ii) If $(x' - ax, x(t_0)) = (g, x_0)$, then $\frac{d}{dt}(E(t_0,t)x(t)) = E(t_0,t)(x'(t) - a(t)x(t)) = E(t_0,t)g(t)$.
By Theorem 2.2.2, $E(t_0,t)x(t) = x_0 + \int_{t_0}^t E(t_0,s)g(s)\,ds$.
Multiplying by $E(t,t_0)$ and using $E(t,t_0)E(t_0,s) = E(t,s)$ gives $x(t) = \phi(t)$.

Structural Commentary: $E(t_0,t)$ removes $a(t)x$; integration accumulates g.
Key Insight: $(E(t_0,t)x)' = E(t_0,t)g$. □

Proposition 1.2.2. *Fiber-Preserving Change of Variables*
Let $\Omega, \widetilde{\Omega} \subseteq \mathbb{R}^2$ be open, $f : \Omega \to \mathbb{R}$, and $T : \Omega \to \widetilde{\Omega}$ be a C^1 diffeomorphism.
Assume $T(t,x) = (\sigma(t), \eta(t,x))$ and $T^{-1}(s,y) = (\tau(s), \xi(s,y))$. Put $G := \partial_t\eta + \partial_x\eta\, f$.
Define $\widetilde{f} : \widetilde{\Omega} \to \mathbb{R}$ by $\widetilde{f}(s,y) := \tau'(s)G(\tau(s), \xi(s,y))$.
Let $I \subseteq \mathbb{R}$ be an interval, $x \in C^1(I, \mathbb{R})$, $\Gamma_x := \{(t, x(t)) : t \in I\} \subseteq \Omega$, and $J := \sigma(I)$.
Set $y : J \to \mathbb{R}$ by $y(s) := \eta(\tau(s), x(\tau(s)))$. Then $y \in C^1(J, \mathbb{R})$, and $x' = f(\cdot, x)$ on I iff $y' = \widetilde{f}(\cdot, y)$ on J.

Proof. Since $\det DT(t,x) = \sigma'(t)\partial_x\eta(t,x) \neq 0$, $\sigma' \neq 0$, $\partial_x\eta \neq 0$, and $\tau'(\sigma(t))\sigma'(t) = 1$.

($\Rightarrow$) If $x' = f(\cdot, x)$ and $s \in J$, then $y'(s) = \tau'(s)G(\tau(s), x(\tau(s)))$. Since $\xi(s, y(s)) = x(\tau(s))$, $y'(s) = \widetilde{f}(s, y(s))$.
($\Leftarrow$) Let $t \in I$. Differentiating $y(\sigma(t)) = \eta(t, x(t))$ gives $\sigma'(t)y'(\sigma(t)) = \partial_t\eta(t,x(t)) + \partial_x\eta(t,x(t))x'(t)$.
Since $y' = \widetilde{f}(\cdot, y)$, $\sigma'(t)y'(\sigma(t)) = G(t, x(t))$. Hence $\partial_x\eta(t,x(t))[x'(t) - f(t,x(t))] = 0$, so $x' = f(\cdot, x)$.

Structural Commentary: Transform graphs; apply the chain rule.
Key Insight: $\widetilde{f}$ is the pushed-forward vector field. □

Method 1.2.3. *Homogeneous First-Order Reduction*
Let $D \subseteq \mathbb{R}$ be open, let $f \in C(D, \mathbb{R})$, and let $I \subseteq \mathbb{R} \setminus \{0\}$ be an interval. To solve $x' = f(x/t)$ on a branch for which $x(t)/t \in D$:

(i) Set $y := x/t$, so $x = ty$.
(ii) Differentiate to obtain $y' = (tx' - x)/t^2 = [f(y) - y]/t$.
(iii) Each $c \in D$ satisfying $f(c) = c$ gives the constant-ratio solution $x(t) = ct$.
(iv) On every subinterval where $f(y) \neq y$, separate $dy/(f(y) - y) = dt/t$, solve for y, and recover $x = ty$.

Method 1.2.4. *Affine-Fractional First-Order Reduction*
Let $D \subseteq \mathbb{R}$ be open, let $f \in C(D, \mathbb{R})$, put $A(t,x) := ax + bt + c$ and $B(t,x) := \alpha x + \beta t + \gamma$, and set $\Delta := a\beta - \alpha b$.
Work on a solution interval on which $B(t, x(t)) \neq 0$ and $A(t, x(t))/B(t, x(t)) \in D$.

(i) If $\Delta \neq 0$, there is a unique (t_*, x_*) with $A(t_*, x_*) = B(t_*, x_*) = 0$. Set $s := t - t_*$ and $y := x - x_*$.
Then $\frac{dy}{ds} = f(\frac{ay+bs}{\alpha y+\beta s})$; for $s \neq 0$, $z := y/s$ satisfies $s\frac{dz}{ds} = f(\frac{az+b}{\alpha z+\beta}) - z$.
(ii) If $\Delta = 0$ and $(a,b) \neq (0,0)$, choose the unique $\lambda \in \mathbb{R}$ with $(\alpha, \beta) = \lambda(a,b)$ and set $\widehat{f}(u) := f(\frac{u+c}{\lambda u+\gamma})$ wherever the quotient lies in D.
If $a \neq 0$, use $u := ax + bt$: $u' = a\widehat{f}(u) + b$, with $x = \frac{u-bt}{a}$. If $a = 0$, then $x' = \widehat{f}(bt)$.

Example 1.2.5. *Euler Equation Under Logarithmic Time*
For $t > 0$, set $s := \log t$ and $y(s) := x(e^s)$. Then $x'(t) = t^{-1}y'(s)$ and $x''(t) = t^{-2}(y''(s) - y'(s))$.
Hence Euler equation $t^2x'' + 3tx' + x = 2/t$ becomes constant-coefficient equation $y'' + 2y' + y = 2e^{-s}$.

Proposition 1.2.6. *Exact First-Order Equations*
Let $R := I \times J$ be an open rectangle, let $p, q \in C^1(R, \mathbb{R})$, and fix $(x_0, y_0) \in R$. Then:

(i) If $F \in C^2(R, \mathbb{R})$ satisfies $\partial_y F = p$ and $\partial_x F = q$, then $\partial_x p = \partial_y q$ on R.
(ii) If $\partial_x p = \partial_y q$ on R, then $F_0(x,y) := \int_{x_0}^x q(s, y_0)\,ds + \int_{y_0}^y p(x,r)\,dr$ belongs to $C^2(R, \mathbb{R})$ and satisfies $\partial_y F_0 = p$, $\partial_x F_0 = q$.
(iii) Under the hypothesis of (ii), every $F_1 \in C^2(R, \mathbb{R})$ satisfying $\partial_y F_1 = p$, $\partial_x F_1 = q$ has the form $F_1 = F_0 + c$ for a unique $c \in \mathbb{R}$.
(iv) Under the hypothesis of (ii), for every interval $L \subseteq I$ and every $y \in C^1(L, J)$, $p(\cdot, y)y' + q(\cdot, y) = 0$ iff $F_0(\cdot, y)$ is constant on L.

Proof.
(i) By Corollary 3.2.8, $\partial_x p = \partial_x\partial_y F = \partial_y\partial_x F = \partial_y q$.
(ii) The one-variable fundamental theorem gives $\partial_y F_0 = p$ and $\partial_x F_0 = q(x, y_0) + \int_{y_0}^y \partial_x p(x,r)\,dr$. By the hypothesis and the same theorem, $\partial_x F_0 = q(x,y_0) + \int_{y_0}^y \partial_y q(x,r)\,dr = q(x,y)$. Since these first partials are C^1, $F_0 \in C^2(R, \mathbb{R})$.

(iii) Set $H := F_1 - F_0$. Then $\partial_x H = \partial_y H = 0$. Applying the one-variable fundamental theorem first along the horizontal segment and then along the vertical segment gives $H(x,y) = H(x_0,y_0)$ for every $(x,y) \in R$. Thus $F_1 = F_0 + c$, and evaluation at (x_0,y_0) makes c unique.
(iv) The chain rule gives $(F_0(\,\cdot\,,y))' = q(\,\cdot\,,y) + p(\,\cdot\,,y)y'$. Therefore the differential equation holds exactly when this derivative vanishes, which on an interval is equivalent to constancy.

Structural Commentary: Build a potential; differentiate along graphs.
Key Insight: Mixed partials are the compatibility condition. □

Method 1.2.7. *Integrating Factor Depending Only on x*
Let $R := I \times J \subseteq \mathbb{R}^2$ be an open rectangle, let $p, q \in C^1(R,\mathbb{R})$, assume $p \neq 0$ on R, and fix $x_* \in I$. For $p(x,y)y' + q(x,y) = 0$:
(i) Set $r(x,y) := (\partial_y q - \partial_x p)/p$.
(ii) If $r(x,y) = r_0(x)$ on R, set $\mu(x) := \exp(\int_{x_*}^{x} r_0(s)\,ds)$.
(iii) Multiply by μ: $\partial_x(\mu p) = \mu(r_0 p + \partial_x p) = \mu \partial_y q = \partial_y(\mu q)$, so $\mu p\, y' + \mu q = 0$ is exact by Proposition 1.2.6.

For example, on either half-plane $x > 0$ or $x < 0$, $xy' + 3x - 2y = 0$ has $r_0(x) = -3/x$, so one may take the nonvanishing factor $\mu(x) = x^{-3}$.
A potential is $F(x,y) = y/x^2 - 3/x$, and the solutions are $y = 3x + Cx^2$.

Method 1.2.8. *Bernoulli Linearization*
Let $I \subseteq \mathbb{R}$ be open, let $a, b \in C(I,\mathbb{R})$, and let $n \in \mathbb{R} \setminus \{0,1\}$. On an interval $J \subseteq I$, consider a positive solution $x \in C^1(J,(0,+\infty))$ of $x' = a(t)x + b(t)x^n$.
(i) Set $y := x^{1-n}$.
(ii) Linearize: $y' = (1-n)x^{-n}x' = (1-n)a(t)y + (1-n)b(t)$ on J.
(iii) Solve $y' = \alpha y + \beta$ with $\alpha := (1-n)a$, $\beta := (1-n)b$ (Proposition 1.2.1); recover $x = y^{1/(1-n)}$ on $\{y > 0\}$.

Method 1.2.9. *Riccati Linearization from a Particular Solution*
Let $I \subseteq \mathbb{R}$ be open, $a, b, c \in C(I,\mathbb{R})$, and let $x_p \in C^1(I,\mathbb{R})$ solve $x_p' = ax_p + bx_p^2 + c$.
Let $x \in C^1(J,\mathbb{R})$ solve $x' = ax + bx^2 + c$ on an interval $J \subseteq I$ on which $x \neq x_p$.
(i) Set $u := x - x_p$ and $y := u^{-1}$ on J.
(ii) Subtract and invert: $u' = (a + 2bx_p)u + bu^2$, so $y' = -(a + 2bx_p)y - b$.
(iii) Solve $y' = \alpha y + \beta$ with $\alpha := -(a + 2bx_p)$, $\beta := -b$ (Proposition 1.2.1); recover $x = x_p + y^{-1}$ on $\{y \neq 0\}$.

Theorem 1.2.10. *Autonomous Matrix Evolution and Variation of Constants*
Let $A \in \mathbb{C}^{n\times n}$, $t_0 \in \mathbb{R}$, $x_0 \in \mathbb{C}^n$, and $g \in C(\mathbb{R},\mathbb{C}^n)$. For the matrix exponential from Definition 6.5.1, the following statements hold:
(i) $e^{0A} = I_n$, and $\forall s,t \in \mathbb{R}$, $e^{(t+s)A} = e^{tA}e^{sA}$, $(e^{tA})^{-1} = e^{-tA}$, and $\frac{d}{dt}e^{tA} = Ae^{tA} = e^{tA}A$.
(ii) $\exists!\, x \in C^1(\mathbb{R},\mathbb{C}^n)$ with $(x' - Ax, x(t_0)) = (0, x_0)$, namely $x(t) = e^{(t-t_0)A}x_0$.
(iii) $\exists!\, x \in C^1(\mathbb{R},\mathbb{C}^n)$ with $(x' - Ax, x(t_0)) = (g, x_0)$, namely $x(t) = e^{(t-t_0)A}x_0 + \int_{t_0}^{t} e^{(t-s)A}g(s)\,ds$.

Proof. Fix a submultiplicative matrix norm. Since $\|t^k A^k/k!\| \le |t|^k \|A\|^k/k!$, the defining series converges absolutely for every $t \in \mathbb{C}$. On bounded real t-intervals, it and its termwise derivative converge uniformly.
(i) The identities at $t = 0$ and for the derivative follow from the series. Absolute convergence permits the Cauchy product, and the binomial theorem gives $e^{tA}e^{sA} = \sum_{m=0}^{\infty} \frac{(t+s)^m A^m}{m!} = e^{(t+s)A}$. Taking $s = -t$ gives $e^{tA}e^{-tA} = I_n$.
(ii) Differentiation gives $x' = Ae^{(t-t_0)A}x_0 = Ax$ and $x(t_0) = x_0$.
If y is another solution, then $(e^{-(t-t_0)A}y(t))' = 0$, so $y(t) = e^{(t-t_0)A}x_0$.
(iii) Put $u(t) := e^{-(t-t_0)A}x(t)$. Then $u'(t) = e^{-(t-t_0)A}g(t)$, hence $u(t) = x_0 + \int_{t_0}^{t} e^{-(s-t_0)A}g(s)\,ds$. Multiplication by $e^{(t-t_0)A}$ and (i) give the formula. Conversely, differentiation verifies the equation and initial value; the difference of two solutions is zero by (ii).

Structural Commentary: The inverse exponential removes the homogeneous drift.
Key Insight: Matrix evolution transports forcing impulses from s to t. □

Proposition 1.2.11. *Pure Modes for Powers and Exponentials*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be diagonalizable with $A = S\Lambda S^{-1}$, where $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n)$ and the columns of S are $\mathbf{v}_1,\dots,\mathbf{v}_n$. If $\mathbf{x}_0 = \sum_{j=1}^{n} c_j\mathbf{v}_j$, then $A^k\mathbf{x}_0 = \sum_{j=1}^{n} c_j\lambda_j^k\mathbf{v}_j$ for $k \ge 0$; and $e^{tA}\mathbf{x}_0 = \sum_{j=1}^{n} c_j e^{\lambda_j t}\mathbf{v}_j$ for $t \in \mathbb{C}$.

Proof. Let $c = (c_1,\dots,c_n)^t$, so $\mathbf{x}_0 = Sc$.
Then $A^k = S\Lambda^k S^{-1}$, and Definition 6.5.1 gives $e^{tA} = S\left(\sum_{r=0}^{\infty} \frac{t^r\Lambda^r}{r!}\right)S^{-1} = S\,\mathrm{diag}(e^{\lambda_1 t},\dots,e^{\lambda_n t})S^{-1}$.
Applying these two formulas to Sc gives the stated identities.
Structural Commentary: Eigenbasis coordinates decouple powers and exponentials.
Key Insight: Diagonalization turns coupled evolution into independent scalar modes. □

Theorem 1.2.12. *Principal Matrix, Liouville Formula, and Variation of Constants*
Let $A \in C(\mathbb{R},\mathbb{C}^{n\times n})$. Then $\exists!\,\Pi : \mathbb{R}^2 \to \mathbb{C}^{n\times n}$ such that $\forall s,t \in \mathbb{R}$, $\partial_t\Pi(t,s) = A(t)\Pi(t,s)$ and $\Pi(s,s) = I_n$. The following statements hold:
(i) $\forall r,s,t \in \mathbb{R}$, $\Pi(t,r)\Pi(r,s) = \Pi(t,s)$ and $\Pi(t,s)^{-1} = \Pi(s,t)$.
(ii) $\forall s,t \in \mathbb{R}$, $\det\Pi(t,s) = \exp\left(\int_s^t \mathrm{tr}\,A(r)\,dr\right)$.
(iii) If $g \in C(\mathbb{R},\mathbb{C}^n)$ and $x_s \in \mathbb{C}^n$, then $\exists!\, x \in C^1(\mathbb{R},\mathbb{C}^n)$ with $(x' - Ax, x(s)) = (g, x_s)$, namely $x(t) = \Pi(t,s)x_s + \int_s^t \Pi(t,r)g(r)\,dr$ for $t \in \mathbb{R}$.

The matrix $\Pi(t,s)$ is the principal matrix solution from s; any matrix solution $X' = AX$ with $\det X(s) \neq 0$ is a fundamental matrix solution.

Proof. Identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$. For the canonical basis $e_1, \dots, e_n$, Theorem 1.1.27 and Corollary 1.1.30 give a unique global solution of $x' = A(t)x$, $x(s) = e_j$; use these solutions as the columns of $\Pi(t,s)$. Existence and uniqueness of Π hold columnwise.

(i) For fixed r, s, both $t \mapsto \Pi(t,r)\Pi(r,s)$ and $t \mapsto \Pi(t,s)$ solve $X' = A(t)X$ and agree at $t = r$; hence they coincide. Applying the cocycle law to the triples (s,t,s) and (t,s,t) gives $I_n = \Pi(s,t)\Pi(t,s) = \Pi(t,s)\Pi(s,t)$, so $\Pi(t,s)^{-1} = \Pi(s,t)$. If $X(t) := \Pi(t,0)$, then $\Pi(t,s) = X(t)X(s)^{-1}$; hence Π is jointly continuous.

(ii) Let $X(t) := \Pi(t,s)$ and write its columns as $x_1(t), \dots, x_n(t)$. The Leibniz formula and $x_j' = Ax_j$ give $\frac{d}{dt} \det X(t) = \sum_{j=1}^{n} \det(x_1, \dots, Ax_j, \dots, x_n)$ $\operatorname{tr}(A(t)) \det X(t)$. Multiplying by $\exp(-\int_s^t \operatorname{tr} A(r)\, dr)$ and using $\det X(s) = 1$ gives the claimed formula.

(iii) Differentiating the asserted formula gives $x' = A(t)x + g(t)$ and $x(s) = x_s$. If x, y have the same data, then $z := x - y$ solves $z' = A(t)z$, $z(s) = 0$, so the defining property of Π gives $z(t) = \Pi(t,s)z(s) = 0$.

Structural Commentary: Columnwise existence builds Π; uniqueness forces its cocycle law.
Key Insight: $\Pi(t,s)$ transports states and forcing between arbitrary times. □

Method 1.2.13. *Solving $\mathbf{x}'(t) = A\mathbf{x}(t)$ by Jordan Form*
Let $A \in \mathbb{C}^{n\times n}$ and $\mathbf{x}_0 \in \mathbb{C}^n$. To solve $\mathbf{x}'(t) = A\mathbf{x}(t)$ with $\mathbf{x}(0) = \mathbf{x}_0$:

(i) By Theorem 6.2.1, choose $S \in GL_n(\mathbb{C})$ with $A = SJS^{-1}$, where $J = \bigoplus_{i=1}^{r} J_{k_i}(\lambda_i)$.
(ii) Set $\mathbf{y}(t) = S^{-1}\mathbf{x}(t)$ and $\mathbf{y}_0 = S^{-1}\mathbf{x}_0$. Then $\mathbf{y}'(t) = J\mathbf{y}(t)$ and $\mathbf{x}(t) = S\mathbf{y}(t)$.
(iii) Split $\mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_r)$ according to the Jordan blocks. Then $\mathbf{y}_i'(t) = J_{k_i}(\lambda_i)\mathbf{y}_i(t)$.
(iv) By Proposition 6.5.3, $\mathbf{y}_i(t) = e^{tJ_{k_i}(\lambda_i)}\mathbf{y}_i(0) = e^{\lambda_i t} \sum_{q=0}^{k_i-1} \frac{t^q}{q!} J_{k_i}(0)^q \mathbf{y}_i(0)$ for $1 \le i \le r$.
Differentiating verifies the block equation.
(v) Reassemble $\mathbf{y}(t)$ and transform back: $\mathbf{x}(t) = S\mathbf{y}(t)$.

Thus a size-k block with eigenvalue λ contributes entries of the form $e^{\lambda t}p(t)$, where $\deg p < k$.

Perspective 1.2.14. *Jordan Functions and Polynomial Factors*

For a Jordan block $J_k(\lambda) = \lambda I_k + N$ with $N^k = 0$, any polynomial f satisfies $f(J_k(\lambda)) = \sum_{r=0}^{k-1} \frac{f^{(r)}(\lambda)}{r!} N^r$.

The nilpotent part contributes derivative terms; for the exponential these become polynomial factors in t by Proposition 6.5.3. As $t \to +\infty$, $\operatorname{Re}\lambda < 0$ gives blockwise decay and $\operatorname{Re}\lambda > 0$ gives growth.
If $\operatorname{Re}\lambda = 0$, the block evolution is bounded iff $N = 0$.

Example 1.2.15. *Nilpotent Jordan Block Evolution*

Let $N := J_3(0) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, so $N^3 = 0$. By Proposition 6.5.3, $e^{tN} = I_3 + tN + \frac{t^2}{2}N^2 = \left(\begin{smallmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{smallmatrix}\right)$. Nilpotent Jordan blocks create polynomial, rather than exponential, growth along generalized-eigenvector chains.

Theorem 1.2.16. *Spectral Criteria for Autonomous Linear Stability*
Let $A \in \mathbb{C}^{n\times n}$ and set $\alpha(A) := \max\{\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\}$. Then:

(i) $e^{tA} \to 0$ *as* $t \to +\infty$ *iff* $\alpha(A) < 0$.
(ii) $\sup_{t\ge 0} \|e^{tA}\| < +\infty$ *iff* $\alpha(A) \le 0$ *and* $\forall \lambda \in \operatorname{spec}(A)$, $\operatorname{Re}\lambda = 0 \implies \dim\ker(A - \lambda I_n) = m_A(\lambda)$.
(iii) If $\alpha(A) < 0$, *then* $\forall \gamma \in (0, -\alpha(A))$, $\exists C_\gamma > 0$ *such that* $\|e^{tA}\| \le C_\gamma e^{-\gamma t}$ $\forall t \ge 0$.

Proof. Choose $S \in GL_n(\mathbb{C})$ with $A = SJS^{-1}$ and use Proposition 6.5.3 on every block $J_k(\lambda)$. Fixed similarity changes all norms by at most the factor $\|S\|\,\|S^{-1}\|$.

(i) Each block exponential is $e^{\lambda t}\sum_{r=0}^{k-1} \frac{t^r N^r}{r!}$. It tends to zero exactly when $\operatorname{Re}\lambda < 0$; if $\operatorname{Re}\lambda \ge 0$, its diagonal entry $e^{\lambda t}$ does not tend to zero.
(ii) If $\operatorname{Re}\lambda < 0$, every polynomial factor is dominated by $e^{\operatorname{Re}\lambda\, t}$ and the block is bounded. If $\operatorname{Re}\lambda = 0$, the block is bounded exactly when $N = 0$, equivalently $k = 1$. If $\operatorname{Re}\lambda > 0$, its diagonal is unbounded.
(iii) Fix $0 < \gamma < -\alpha(A)$. For a block of size k, $e^{\gamma t}\|e^{tJ_k(\lambda)}\|$ is bounded because every $t^r e^{(\operatorname{Re}\lambda + \gamma)t}$ tends to zero. Take the maximum block bound and include the fixed similarity factor.

Structural Commentary: Jordan blocks separate exponential rates from nilpotent polynomial factors.
Key Insight: Boundary-spectrum nilpotence is the sole obstruction to boundedness. □

Definition 1.2.17. *Hurwitz Matrix*
A matrix $A \in \mathbb{C}^{n\times n}$ is *Hurwitz* iff $\max\{\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\} < 0$.

Perspective 1.2.18. *Planar Linear Phase Classification*
For $x' = Ax$ with $A \in \mathbb{R}^{2\times 2}$, two real eigenvalues of negative sign give a sink, two of positive sign give a source, and opposite signs give a saddle.
A nonreal pair $\alpha \pm i\beta$, $\beta \ne 0$, gives a spiral sink when $\alpha < 0$, a spiral source when $\alpha > 0$, and a center when $\alpha = 0$. A defective repeated real eigenvalue λ adds a polynomial shear along its generalized eigenvector; it still gives decay for $\lambda < 0$ and growth for $\lambda > 0$. For $\lambda = 0$, $e^{tA} = I_2 + tA$ in the defective case, so generalized-eigenvector directions grow linearly while the eigenspace consists of equilibria. These labels describe the origin; similarity changes the coordinates of the phase portrait without changing its spectral type.

Theorem 1.2.19. *Basis for Constant-Coefficient Linear ODEs*
Let $p(z) := z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0 = \prod_{j=1}^{m} (z - \lambda_j)^{a_j}$ over $\mathbb{C}$, where the λ_j are distinct and $\sum_{j=1}^{m} a_j = n$. Then:

(i) The functions $x_{j,k}(t) := t^k e^{\lambda_j t}$, $1 \le j \le m$, $0 \le k < a_j$, form a basis of the complex solutions of $p(D)x = 0$, where $D = d/dt$.
(ii) If $c_0, \dots, c_{n-1} \in \mathbb{R}$, write $\lambda_j = \alpha_j + i\beta_j$. The functions $t^k e^{\lambda_j t}$ for the real roots $\beta_j = 0$, together with $t^k e^{\alpha_j t}\cos(\beta_j t)$ and $t^k e^{\alpha_j t}\sin(\beta_j t)$ for one root from each conjugate pair with $\beta_j > 0$, form a real basis, where always $0 \le k < a_j$.

Proof. For $q \in \mathbb{C}[t]$, $(D-\lambda)(e^{\lambda t}q) = e^{\lambda t}q'$. Thus $(D-\lambda_j)^{a_j} x_{j,k} = 0$ for $k < a_j$, so every listed function solves $p(D)x = 0$.

(i) Suppose $\sum_{j=1}^{m} e^{\lambda_j t} q_j(t) = 0$ with $\deg q_j < a_j$. Fix j and apply $\prod_{i \neq j} (D-\lambda_i)^{a_i}$. All terms except the jth vanish, while the remaining term is $e^{\lambda_j t} R_j(D) q_j$, where $R_j(z) := \prod_{i \neq j} (z + \lambda_j - \lambda_i)^{a_i}$ and $R_j(0) \neq 0$.
On polynomials of degree below a_j, $R_j(D) - R_j(0)I$ is nilpotent because it is divisible by D. Thus $R_j(D)$ is invertible, so $q_j = 0$. The n listed functions are linearly independent.
Writing $y := (x, x', \dots, x^{(n-1)})$ converts $p(D)x = 0$ into a first-order companion system. Theorem 1.2.12(iii), with zero forcing, makes $y(0) \mapsto y$ a complex-linear bijection from $\mathbb{C}^n$ onto the global solution space. Thus that space has complex dimension n and the list is a basis.

(ii) Real coefficients pair nonreal roots by conjugation. Taking real and imaginary parts of the paired complex basis functions gives the stated real solutions; the change from each conjugate pair to its real and imaginary parts is invertible over $\mathbb{R}$.

Structural Commentary: Factor $p(D)$; isolate each exponential-polynomial root space.
Key Insight: Root multiplicity becomes polynomial degree. □

Method 1.2.20. *Exponential-Polynomial Forcing*
Let $p \in \mathbb{C}[z] \setminus \{0\}$, $q \in \mathbb{C}[t] \setminus \{0\}$, and $\beta \in \mathbb{C}$, and consider $p(D)x = q(t)e^{\beta t}$. Write $p(z) = (z-\beta)^a h(z)$ with $a \in \mathbb{N}_0$ and $h(\beta) \neq 0$. Set $d := \deg q$ and $\mathbb{C}_{\le d}[t] := \{u \in \mathbb{C}[t] : \deg u \le d\}$.

(i) Solve $h(D+\beta)u = q$ in $\mathbb{C}_{\le d}[t]$; the operator is triangular with nonzero diagonal $h(\beta)$ and hence invertible.
(ii) Choose the polynomial r with $r^{(a)} = u$ and vanishing coefficients in degrees $0, \dots, a-1$.
(iii) Set $x_p(t) := e^{\beta t} r(t)$. Then $p(D)x_p = e^{\beta t} h(D+\beta) D^a r = q(t)e^{\beta t}$ and $\deg r \le \deg q + a$.

The factor t^a in r is the resonance correction when β is a root of p.

Example 1.2.21. *Damping and Resonance*
For $x'' + 2\eta x' + \omega_0^2 x = 0$, with $\eta, \omega_0 > 0$, the roots are $-\eta \pm \sqrt{\eta^2 - \omega_0^2}$. Thus $\eta > \omega_0$, $\eta = \omega_0$, and $\eta < \omega_0$ give over-damping, critical damping, and under-damping, respectively; homogeneous solutions decay.
For the forced equation $x'' + 2\eta x' + \omega_0^2 x = \cos(\omega t)$, a complex particular solution is $z_p(t) = e^{i\omega t}/(\omega_0^2 - \omega^2 + 2i\eta\omega)$, and $x_p = \operatorname{Re} z_p$. If $\eta = 0$ and $\omega = \omega_0$, Method 1.2.20 instead gives $x_p(t) = t\sin(\omega_0 t)/(2\omega_0)$, whose amplitude grows linearly.
For a series RLC circuit, $\eta = R/(2L)$ and $\omega_0 = 1/\sqrt{LC}$; the same three regimes and resonance mechanism govern the current.

Method 1.2.22. *Reduction of Order for a Second Solution*
Let $I \subseteq \mathbb{R}$ be open, let $q_0, q_1 \in C(I, \mathbb{C})$, let $\phi_1 \in C^2(I, \mathbb{C})$ solve $x'' + q_1 x' + q_0 x = 0$, and restrict to an interval $J \subseteq I$ on which $\phi_1 \neq 0$. Fix $t_* \in J$.

(i) Set $x = c\phi_1$. Cancellation of the equation for ϕ_1 gives $\phi_1 c'' + (2\phi_1' + q_1\phi_1)c' = 0$.
(ii) With $d := c'$, solve $d' + (2\phi_1'/\phi_1 + q_1)d = 0$ to obtain $d(t) = C\phi_1(t)^{-2}\exp(-\int_{t_*}^{t} q_1(r)\,dr)$.
(iii) Taking $C = 1$ gives $\phi_2(t) := \phi_1(t)\int_{t_*}^{t} \phi_1(s)^{-2}\exp(-\int_{t_*}^{s} q_1(r)\,dr)\,ds$. Then $W(\phi_1, \phi_2)(t) = \phi_1\phi_2' - \phi_1'\phi_2 = \exp(-\int_{t_*}^{t} q_1(r)\,dr) \neq 0$. Thus ϕ_1, ϕ_2 are linearly independent; uniqueness for the associated two-dimensional first-order system makes them a basis of the solution space on J.

Theorem 1.2.23. *Floquet Normal Form*
Let $A \in C(\mathbb{R}, \mathbb{C}^{n\times n})$ satisfy $A(t+T) = A(t)$ for some $T > 0$, let Π be its principal matrix solution, and set $M(t_0) := \Pi(t_0 + T, t_0)$. Then:

(i) $\forall s, t \in \mathbb{R}$, $\Pi(t+T, s+T) = \Pi(t, s)$.
(ii) $M(t_0)$ *is invertible.*
(iii) $\exists Q(t_0) \in \mathbb{C}^{n\times n}$ *and a T-periodic* $P(\,\cdot\,, t_0) \in C^1(\mathbb{R}, GL_n(\mathbb{C}))$ *such that* $\Pi(t, t_0) = P(t, t_0)e^{(t-t_0)Q(t_0)}$ *and* $P(t_0, t_0) = I_n$.
(iv) $\forall t_0, t_1 \in \mathbb{R}$, $M(t_1) = \Pi(t_1, t_0)M(t_0)\Pi(t_1, t_0)^{-1}$*; consequently, the spectrum and Jordan block sizes of $M(t_0)$ are independent of t_0.*

Proof.

(i) Both $t \mapsto \Pi(t+T, s+T)$ and $t \mapsto \Pi(t, s)$ solve $X' = A(t)X$ and equal I_n at $t = s$, so uniqueness gives the identity.
(ii) Theorem 1.2.12(i) gives $M(t_0)^{-1} = \Pi(t_0, t_0 + T)$.
(iii) By (ii), every Jordan block of $M(t_0)$ has the form $J = \rho I + N$ with $\rho \neq 0$ and $N^k = 0$. Choose $\ell \in \mathbb{C}$ with $e^{\ell} = \rho$ and set $L := \ell I + \sum_{q=1}^{k-1} \frac{(-1)^{q+1}}{q}(\rho^{-1}N)^q$. In the quotient algebra $\mathbb{C}[z]/(z^k)$, the finite logarithm and exponential series satisfy $\exp(\log(1+z)) = 1+z$; substituting $z = \rho^{-1}N$ gives $e^L = J$. Reassembling and conjugating these block logarithms produces $R \in \mathbb{C}^{n\times n}$ with $e^R = M(t_0)$.
Set $Q(t_0) := T^{-1}R$ and $P(t, t_0) := \Pi(t, t_0)e^{-(t-t_0)Q(t_0)}$. Then $P(t_0, t_0) = I_n$. By (i) and the cocycle law, $\Pi(t+T, t_0) = \Pi(t, t_0)M(t_0)$; because $M(t_0) = e^{TQ(t_0)}$, this identity gives $P(t+T, t_0) = P(t, t_0)$. Both factors defining P are C^1 and invertible.
(iv) Periodicity from (i) and the cocycle law give $\Pi(t_1 + T, t_0) = \Pi(t_1, t_0)M(t_0)$ and also $\Pi(t_1 + T, t_0) = M(t_1)\Pi(t_1, t_0)$. Right multiplication by $\Pi(t_1, t_0)^{-1}$ proves the similarity identity, and similarity preserves the spectrum and Jordan block sizes.

Structural Commentary: Take a logarithm of one-period transport; factor the remainder.
Key Insight: Periodic evolution is constant exponential growth in periodic coordinates. □

Definition 1.2.24. *Floquet Multipliers and Exponents*
Let $A \in C(\mathbb{R}, \mathbb{C}^{n\times n})$ be T-periodic, let Π be its principal matrix solution, set $M(t_0) := \Pi(t_0 + T, t_0)$, and choose $Q(t_0)$ with $e^{TQ(t_0)} = M(t_0)$.

(i) The eigenvalues ρ of $M(t_0)$ are the *Floquet multipliers*.
(ii) The eigenvalues γ of a Floquet matrix $Q(t_0)$ are *Floquet exponents*. Spectral mapping gives $\operatorname{spec}(M(t_0)) = \{e^{T\gamma} : \gamma \in \operatorname{spec}(Q(t_0))\}$ and $m_{M(t_0)}(\rho) = \sum_{\substack{\gamma \in \operatorname{spec}(Q(t_0)) \\ e^{T\gamma} = \rho}} m_{Q(t_0)}(\gamma)$. Thus every exponent determines a multiplier by $\rho = e^{T\gamma}$, and the scalar logarithm determines γ only modulo $2\pi i/T$.

Corollary 1.2.25. *Floquet Multiplier Stability Criterion*
Let $A \in C(\mathbb{R}, \mathbb{C}^{n\times n})$ be T-periodic, let Π be its principal matrix solution, and set $M(t_0) := \Pi(t_0+T, t_0)$. Then:
(i) $\sup_{t\ge t_0} \|\Pi(t,t_0)\| < +\infty$ *iff* $|\rho| \le 1$ *for* $\rho \in \operatorname{spec}(M(t_0))$ *and* $\dim\ker(M(t_0) - \rho I_n) = m_{M(t_0)}(\rho)$ *whenever* $|\rho| = 1$.
(ii) $\Pi(t,t_0) \to 0$ *as* $t \to +\infty$ *iff* $\max\{|\rho| : \rho \in \operatorname{spec}(M(t_0))\} < 1$.

Proof. Write $t = t_0 + kT + \tau$, $k \in \mathbb{N}_0$, $0 \le \tau < T$. Periodicity and the cocycle law give by induction $\Pi(t,t_0) = \Pi(t_0+\tau, t_0)M(t_0)^k$. The first factor and its inverse are uniformly bounded for $0 \le \tau \le T$.
(i) Hence $\Pi(\,\cdot\,, t_0)$ is bounded iff $(M(t_0)^k)$ is bounded. By Theorem 3.3.7, this is equivalent to the stated multiplier and multiplicity conditions.
(ii) Likewise, $\Pi(t,t_0) \to 0$ iff $M(t_0)^k \to 0$. By Theorem 3.3.5, this is equivalent to $\rho(M(t_0)) < 1$.
Structural Commentary: Sample at integer periods; reduce continuous evolution to powers of M.
Key Insight: Floquet stability is matrix-power stability of one-period transport. □

Example 1.2.26. *Hill Equation and Its Floquet Discriminant*
Let $q \in C(\mathbb{R}, \mathbb{R})$ be T-periodic and consider $x'' + q(t)x = 0$. Its first-order matrix is $A(t) = \left(\begin{smallmatrix} 0 & 1 \\ -q(t) & 0 \end{smallmatrix}\right)$. Let Π be the principal matrix solution and $M(t_0) := \Pi(t_0+T, t_0)$. Since $\operatorname{tr} A(t) = 0$, by Theorem 1.2.12(ii), $\det M(t_0) = 1$.
With $\Delta := \frac{1}{2}\operatorname{tr} M(t_0)$, the multiplier equation is $\rho^2 - 2\Delta\rho + 1 = 0$. If $|\Delta| < 1$, the multipliers are distinct conjugates on the unit circle and all solutions are bounded. If $|\Delta| > 1$, one real multiplier has modulus above 1 and an unbounded solution exists. At $|\Delta| = 1$, boundedness holds exactly when $M(t_0) = \Delta I_2$; otherwise its nontrivial Jordan block produces linear growth in the period count.
The Mathieu equation $x'' + \omega^2(1 + \varepsilon\cos t)x = 0$ is the standard periodic-coefficient example; its stability regions are determined by $\Delta(\omega, \varepsilon)$.

Theorem 1.2.27. *Perturbation Bounds for Linear Evolution*
Let $A, B \in C(\mathbb{R}, \mathbb{C}^{n\times n})$, let Π_A, Π_{A+B} be the corresponding principal matrix solutions, and fix $t_0 \in \mathbb{R}$. Then:
(i) If $\|\Pi_A(t,s)\| \le Ce^{-\alpha(t-s)}$ *for all* $t \ge s \ge t_0$ *and* $\|B(t)\| \le b$ *for all* $t \ge t_0$, *where* $C, \alpha > 0$, $b \ge 0$, *and* $Cb < \alpha$, *then* $\|\Pi_{A+B}(t,s)\| \le Ce^{-(\alpha - Cb)(t-s)}$ *for all* $t \ge s \ge t_0$.
(ii) If $\|\Pi_A(t,s)\| \le C$ $\forall t \ge s \ge t_0$ *and* $B_1 := \int_{t_0}^{\infty} \|B(r)\|\,dr < +\infty$, *then* $\|\Pi_{A+B}(t,s)\| \le Ce^{CB_1}$ $\forall t \ge s \ge t_0$.

Proof. For fixed $s \ge t_0$, by Theorem 1.2.12(iii), applied columnwise with forcing $B(t)x(t)$, $\Pi_{A+B}(t,s) = \Pi_A(t,s) + \int_s^t \Pi_A(t,r)B(r)\Pi_{A+B}(r,s)\,dr$.
(i) Put $y(t) := e^{\alpha(t-s)}\|\Pi_{A+B}(t,s)\|$. The hypotheses give $y(t) \le C + Cb\int_s^t y(r)\,dr$. By Lemma 1.1.20(ii), after translating s to 0, $y(t) \le Ce^{Cb(t-s)}$, which is the asserted estimate.
(ii) Put $y(t) := \|\Pi_{A+B}(t,s)\|$. Then $y(t) \le C + C\int_s^t \|B(r)\|\,y(r)\,dr$. The same Gronwall estimate gives $y(t) \le C\exp(C\int_s^t \|B(r)\|\,dr) \le Ce^{CB_1}$.
Structural Commentary: Variation of constants converts perturbation into Gronwall feedback.
Key Insight: Perturbation mass subtracts from decay or multiplies boundedness. □

Corollary 1.2.28. *Vanishing Perturbations of a Hurwitz Matrix*
Let $A \in \mathbb{C}^{n\times n}$ be Hurwitz, let $B \in C(\mathbb{R}, \mathbb{C}^{n\times n})$, and suppose $\|B(t)\| \to 0$ as $t \to +\infty$. Set $\delta_A := \min\{-\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\}$. Then $\forall \gamma \in (0, \delta_A)$, $\exists K_\gamma > 0$ such that $\|\Pi_{A+B}(t,0)\| \le K_\gamma e^{-\gamma t}$ $\forall t \ge 0$.

Proof. Fix $0 < \gamma < \eta < \delta_A$. By Theorem 1.2.16(iii), $\exists C_\eta > 0$ with $\left\|e^{(t-s)A}\right\| \le C_\eta e^{-\eta(t-s)}$ for $t \ge s$.
Choose $t_0 \ge 0$ such that $\|B(t)\| \le (\eta - \gamma)/C_\eta$ for $t \ge t_0$. By Theorem 1.2.27(i), $\|\Pi_{A+B}(t,t_0)\| \le C_\eta e^{-\gamma(t-t_0)}$ for $t \ge t_0$.
The cocycle law yields $\Pi_{A+B}(t,0) = \Pi_{A+B}(t,t_0)\Pi_{A+B}(t_0,0)$.
Set $K_1 := C_\eta e^{\gamma t_0}\|\Pi_{A+B}(t_0,0)\|$; then $\|\Pi_{A+B}(t,0)\| \le K_1 e^{-\gamma t}$ for $t \ge t_0$.
Continuity gives $K_0 := \max_{0\le t\le t_0} e^{\gamma t}\|\Pi_{A+B}(t,0)\| < +\infty$. Taking $K_\gamma := \max\{K_0, K_1\}$ proves the claim.
Structural Commentary: Wait until B is smaller than the spectral decay margin.
Key Insight: A vanishing perturbation eventually preserves any smaller decay rate. □

1.3 Nonlinear Stability Theory

Exact formulas are often unnecessary for proving that scalar trajectories remain ordered or trapped. Differential inequalities provide comparison curves for this purpose.

Definition 1.3.1. *Upper and Lower Solutions*
Let $U \subseteq \mathbb{R}^2$ be open, $f : U \to \mathbb{R}$, and $J \subseteq \mathbb{R}$ an open interval. For $u \in C^1(J, \mathbb{R})$ with $\Gamma_u := \{(t, u(t)) : t \in J\} \subseteq U$:
(i) *Upper solution*: $u'(t) \ge f(t, u(t))$ $\forall t \in J$; it is *strict* iff $u'(t) > f(t, u(t))$ $\forall t \in J$.
(ii) *Lower solution*: $u'(t) \le f(t, u(t))$ $\forall t \in J$; it is *strict* iff $u'(t) < f(t, u(t))$ $\forall t \in J$.
Key Insight: Differential inequalities encode one-sided barriers without solving the ODE.

Definition 1.3.2. *Scalar Isocline*
Let $U \subseteq \mathbb{R}^2$ be open, $f : U \to \mathbb{R}$, and $c \in \mathbb{R}$.
For $x' = f(t,x)$, the set $\mathcal{I}_c := \{(t,x) \in U : f(t,x) = c\}$ is the *isocline of slope c*.
Let $J \subseteq \mathbb{R}$ be open and $w \in C^1(J, \mathbb{R})$ satisfy $\Gamma_w := \{(t, w(t)) : t \in J\} \subseteq \mathcal{I}_c$.
(i) If $w'(t) > c$ $\forall t \in J$, then w is a strict upper solution.
(ii) If $w'(t) < c$ $\forall t \in J$, then w is a strict lower solution.
Key Insight: An isocline fixes the ODE slope, so only the graph slope remains to be compared.

Lemma 1.3.3. *Strict Scalar Comparison*
Let $U \subseteq \mathbb{R}^2$ be open, $f \in C(U, \mathbb{R})$, $J \subseteq \mathbb{R}$ open, and $[t_0, T] \subseteq J$ with $t_0 < T$.
For $z \in C^1(J, \mathbb{R})$, set $\Gamma_z := \{(t, z(t)) : t \in J\}$, and let $I_+ := (t_0, T]$. Suppose $x \in C^1(J, \mathbb{R})$, $\Gamma_x \subseteq U$, and $x' = f(\,\cdot\,, x)$.
(i) If $u \in C^1(J, \mathbb{R})$, $\Gamma_u \subseteq U$, $u'(t) > f(t, u(t))$ $\forall t \in J$, and $x(t_0) \le u(t_0)$, then $x(t) < u(t)$ $\forall t \in I_+$.
(ii) If $v \in C^1(J, \mathbb{R})$, $\Gamma_v \subseteq U$, $v'(t) < f(t, v(t))$ $\forall t \in J$, and $v(t_0) \le x(t_0)$, then $v(t) < x(t)$ $\forall t \in I_+$.

Proof.

(i) Set $\Delta := u - x$. At every $t \in [t_0, T]$ with $\Delta(t) = 0$, one has $x(t) = u(t)$ and $\Delta'(t) = u'(t) - f(t, u(t)) > 0$. Thus $\Delta(t_0) \geq 0$ implies $\Delta > 0$ on $(t_0, t_0 + \varepsilon]$ for some $\varepsilon > 0$.
If $Z := \{t \in (t_0, T] : \Delta(t) = 0\} \neq \emptyset$, let $t_1 := \inf Z$. Continuity gives $\Delta(t_1) = 0$ and $\Delta > 0$ on (t_0, t_1), so the left derivative satisfies $\Delta'(t_1) \leq 0$. This contradicts $\Delta'(t_1) > 0$. Hence $Z = \emptyset$ and $\Delta > 0$ on I_+.

(ii) Set $\Delta := x - v$. At every zero of Δ, $\Delta' = f(\cdot\,, v) - v' > 0$. The argument from (i), with $x - v$ in place of $u - x$, gives $\Delta > 0$ on I_+.

Structural Commentary: A first contact forces $\Delta' \leq 0$, contradicting the strict residual sign.
Key Insight: Strict differential inequalities forbid barrier crossing. □

Definition 1.3.4. *Local State Lipschitz Condition*
Let $U \subseteq \mathbb{R}^2$ be open and $f : U \to \mathbb{R}$. Then f is *locally Lipschitz in the state, uniformly in time* iff, for every compact $K \subseteq U$, there exists $L_K < +\infty$ such that $|f(t, x) - f(t, y)| \leq L_K |x - y|$ whenever $(t, x), (t, y) \in K$.

Theorem 1.3.5. *Scalar Lipschitz Comparison*
Let $U \subseteq \mathbb{R}^2$ be open and $f \in C(U, \mathbb{R})$. Assume that for every compact $K \subseteq U$, there exists $L_K < +\infty$ such that $|f(t, p) - f(t, q)| \leq L_K |p - q|$ whenever $(t, p), (t, q) \in K$.
Let $I := [t_0, T] \subseteq J$, $J \subseteq \mathbb{R}$ open, $t_0 < T$. For $z : J \to \mathbb{R}$, set $\Gamma_z(A) := \{(t, z(t)) : t \in A\}$, $R_z := z' - f(\cdot\,, z)$.
Suppose $x, y \in C^1(J, \mathbb{R})$, $\Gamma_x(J) \cup \Gamma_y(J) \subseteq U$, and $R_x \leq R_y$ on I. Then:

(i) If $x(t_0) \leq y(t_0)$, then $x \leq y$ on I.
(ii) If $t_1 \in I$ and $x(t_1) < y(t_1)$, then $x < y$ on $[t_1, T]$.

Proof.

(i) Set $\Delta := x - y$ and suppose $S := \{t \in I : \Delta(t) > 0\} \neq \emptyset$. Since $\Delta(t_0) \leq 0$, every component C of S has a left endpoint $\alpha \in I$ with $\Delta(\alpha) = 0$. Choose $t_2 \in C$, so $\Delta > 0$ on $(\alpha, t_2]$.
With $K := \Gamma_x([\alpha, t_2]) \cup \Gamma_y([\alpha, t_2])$ and $L = L_K$, $R_x \leq R_y$ gives $\Delta' \leq f(\cdot\,, x) - f(\cdot\,, y) \leq L\Delta$ on $(\alpha, t_2]$.
Thus $(e^{-L(t-\alpha)}\Delta)' \leq 0$, so $\Delta(t_2) \leq e^{L(t_2-\alpha)}\Delta(\alpha) = 0$, contradicting $t_2 \in S$.

(ii) Set $\delta := y - x$. If $Z := \{t \in (t_1, T] : \delta(t) = 0\} \neq \emptyset$, let $t_2 := \inf Z$. Then $\delta > 0$ on $[t_1, t_2)$ and $\delta(t_2) = 0$.
With $K := \Gamma_x([t_1, t_2]) \cup \Gamma_y([t_1, t_2])$ and $L = L_K$, by $R_x \leq R_y$, $\delta' \geq f(\cdot\,, y) - f(\cdot\,, x) \geq -L\delta$ on $[t_1, t_2)$.
Hence $(e^{L(t-t_1)}\delta)' \geq 0$ and $0 = e^{L(t_2-t_1)}\delta(t_2) \geq \delta(t_1) > 0$, contradiction.

Structural Commentary: On a positive component, the integrating factor makes the order defect nonincreasing.
Key Insight: A nonpositive order defect cannot become positive. □

Lemma 1.3.6. *A Convergent Autonomous State Is an Equilibrium*
Let $f \in C(\mathbb{R}, \mathbb{R})$ and let $x \in C^1([t_0, +\infty), \mathbb{R})$ satisfy $x' = f(x)$. If $x(t) \to x_* \in \mathbb{R}$ as $t \to +\infty$, then:

(i) $x'(t) \to 0$.
(ii) $f(x_*) = 0$.

Proof.

(i) By continuity, $x'(t) = f(x(t)) \to f(x_*) := L$. If $L > 0$, then $x'(t) \geq L/2$ eventually, so integration forces $x(t) \to +\infty$. If $L < 0$, the analogous estimate forces $x(t) \to -\infty$. Both conclusions contradict $x(t) \to x_*$, so $L = 0$ and $x'(t) \to 0$.

(ii) The identity $L = f(x_*)$ from (i) now gives $f(x_*) = 0$.

Structural Commentary: The differential equation converts state convergence into derivative convergence.
Key Insight: A nonzero limiting velocity is incompatible with a finite limiting state. □

Example 1.3.7. *A Trapping Wedge for $x' = x^2 - t^2$*
For $f(t, x) := x^2 - t^2$ on $(0, +\infty) \times \mathbb{R}$, set $u(t) := t$ and $v(t) := -t$.
Then $u' = 1 > 0 = f(t, u)$ and $v' = -1 < 0 = f(t, v)$.
If $t_0 > 0$, $x_0 \in [-t_0, t_0]$, and $x \in C^1([t_0, T_+), \mathbb{R})$ satisfies $x' = x^2 - t^2$ and $x(t_0) = x_0$, then Lemma 1.3.3 gives $-t < x(t) < t$ $\forall t \in (t_0, T_+)$.

Proposition 1.3.8. *Central-Wedge Convergence for $x' = x^2 - t^2$*
Let $t_0 > 0$, $-t_0 < x_0 < t_0$, and let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = x^2 - t^2$, $x(t_0) = x_0$.

(i) $-t < x(t) < t$ for $t \in (t_0, T_+)$.
(ii) $T_+ = +\infty$.
(iii) $x(t) + t \to 0$ as $t \to +\infty$.

Proof.

(i) Let $u(t) := t$ and $v(t) := -t$. Since $u' - (u^2 - t^2) = 1 > 0$, $v' - (v^2 - t^2) = -1 < 0$, and $v(t_0) < x_0 < u(t_0)$, Lemma 1.3.3 implies $v(t) < x(t) < u(t)$ on every compact subinterval of $[t_0, T_+)$.

(ii) By (i), $x' = (x - t)(x + t) < 0$, so x is decreasing.
If $T_+ < +\infty$, then $-T_+ < x(t) \leq x_0$ for $t < T_+$, hence $\ell := \lim_{t \uparrow T_+} x(t) \in [-T_+, x_0]$.
Lemma 1.1.28 extends x through (T_+, ℓ), contradicting maximality. Hence $T_+ = +\infty$.

(iii) Fix $T \geq \max\{t_0, 2\}$, and set $r(t) := \sqrt{t^2 - 2}$, $w(t) := -r(t)$ for $t \geq T$. Since $r^2 - t^2 = w^2 - t^2 = -2$, $r' = t/r > 0 > -2$, and $w' = -t/r \geq -\sqrt{2} > -2$, both r, w are strict upper solutions.
Let $d(t) := t - x(t)$. By (i), $d > 0$ and $t + x > 0$, so $d' = 1 + t^2 - x^2 = 1 + d(t + x) > 1$.
Since $d(t) \to +\infty$ and $t - r(t) \to 0$, choose $T_1 \geq T$ with $d(T_1) > T_1 - r(T_1)$, i.e. $x(T_1) < r(T_1)$.
By strict comparison, $x < r$ for $t > T_1$. If $x(T_1) \leq w(T_1)$, then by strict comparison $x < w$ for $t > T_1$.
If $w(T_1) < x(T_1) < r(T_1)$, then while $w < x < r$, one has $x' < -2$ and $w' \geq -\sqrt{2}$, hence $(x - w)' < -(2 - \sqrt{2})$. Global existence forces a first contact $x(\tau) = w(\tau)$ in finite time, and strict comparison from τ gives $x < w$ thereafter.
Therefore eventually $0 < x(t) + t < t - r(t) = \frac{2}{t + \sqrt{t^2 - 2}} \to 0$.

Structural Commentary: Two nested isocline barriers squeeze the trajectory against $x = -t$.
Key Insight: Every trajectory entering the central wedge is asymptotic to its lower edge. □

Proposition 1.3.9. *Finite-Time Blow-Up Above the Upper Isocline*
Let $t_0 \geq 0$, $x_0 \geq \sqrt{t_0^2+2}$, and let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = x^2 - t^2$, $x(t_0) = x_0$.

(i) $x(t) > \sqrt{t^2+2}$ and $x'(t) > 2$ for $t \in (t_0, T_+)$.
(ii) $T_+ < +\infty$ and $x(t) \to +\infty$ as $t \uparrow T_+$.

Proof.

(i) Set $b(t) := \sqrt{t^2+2}$. Then $b^2 - t^2 = 2$ and $b' = \frac{t}{b} < 1 < 2$, so $b' < b^2 - t^2$.
Since $x_0 \geq b(t_0)$, Lemma 1.3.3 implies $b(t) < x(t)$ for $t > t_0$; hence $x' = x^2 - t^2 > b^2 - t^2 = 2$.

(ii) Suppose $T_+ = +\infty$. By (i), $x(t) > x_0 + 2(t - t_0)$; choose $t_1 \geq t_0$ with $x(t) > \sqrt{2}t > 0$ for $t \geq t_1$.
Then $x' = x^2 - t^2 > \frac{1}{2}x^2$ and $(\frac{1}{x})' = -\frac{x'}{x^2} < -\frac{1}{2}$ on $[t_1, +\infty)$, so $\frac{1}{x(t)} < \frac{1}{x(t_1)} - \frac{1}{2}(t - t_1)$.
This contradicts $x > 0$ for $t > t_1 + \frac{2}{x(t_1)}$. Thus $T_+ < +\infty$. Also $x' > 2$, so $\ell := \lim_{t \uparrow T_+} x(t) \in (x_0, +\infty]$.
If $\ell < +\infty$, Lemma 1.1.28 extends x through (T_+, ℓ), contradicting maximality. Hence $\ell = +\infty$.

Structural Commentary: Above $b(t) = \sqrt{t^2+2}$, linear escape enters the Riccati regime.
Key Insight: Once $x' \geq \frac{1}{2}x^2$, reciprocal decay forces finite-time escape. □

Theorem 1.3.10. *Separatrix Classification for $x' = x^2 - t^2$*
For $a \in \mathbb{R}$, let $x_a : [0, T_a) \to \mathbb{R}$ be the maximal forward solution of $x' = x^2 - t^2$, $x_a(0) = a$.

(i) $\exists!\, a_* \in [0, \sqrt{2}]$ such that $T_{a_*} = +\infty$, $t \leq x_{a_*}(t) \leq \sqrt{t^2+2}$ for $t \geq 0$, and $x_{a_*}(t) - t \to 0$ as $t \to +\infty$.
(ii) If $a > a_*$, then $T_a < +\infty$ and $x_a(t) \to +\infty$ as $t \uparrow T_a$.
(iii) If $a < a_*$, then $T_a = +\infty$ and $x_a(t) + t \to 0$ as $t \to +\infty$.

Proof. Set $\ell(t) := t$ and $b(t) := \sqrt{t^2+2}$. For $T > 0$, let p_T, q_T solve $x' = x^2 - t^2$ backward from $p_T(T) = \ell(T)$ and $q_T(T) = b(T)$. Under $s = T - t$, the reversed equation is $y' = (T-s)^2 - y^2$, while $\widehat{\ell}(s) := T - s$ and $\widehat{b}(s) := \sqrt{(T-s)^2+2}$ satisfy $\widehat{\ell}' < (T-s)^2 - \widehat{\ell}^2$ and $\widehat{b}' > (T-s)^2 - \widehat{b}^2$.
Lemma 1.3.3 and Theorem 1.3.5 trap and order the reversed solutions while they exist. Corollary 1.1.29 then gives existence on $[0, T]$, with $\ell \leq p_T < q_T \leq b$ on $[0, T]$ and strict outer inequalities on $[0, T)$.
Set $\alpha_T := p_T(0)$, $\beta_T := q_T(0)$, and $K_T := [\alpha_T, \beta_T]$. Order preservation gives $a \in K_T$ iff x_a exists on $[0, T]$ and $\ell \leq x_a \leq b$ there. Thus $S \geq T$ implies $K_S \subseteq K_T$, and $(K_n)_{n \geq 1}$ is a nested family of nonempty compact subintervals of $[0, \sqrt{2}]$.

(i) Choose $a_* \in \bigcap_{n \geq 1} K_n \subseteq [0, \sqrt{2}]$. Then x_{a_*} is global, $\ell \leq x_{a_*} \leq b$, and $0 \leq x_{a_*}(t) - t \leq b(t) - t = \frac{2}{b(t)+t} \to 0$.
If $a_1 < a_2$ lie in $\bigcap_{n \geq 1} K_n$, then $\delta := x_{a_2} - x_{a_1}$ satisfies $\delta(0) > 0$ and $\delta' = (x_{a_1} + x_{a_2})\delta \geq 0$.
Thus $0 < a_2 - a_1 \leq \delta(t) \leq b(t) - t \to 0$, impossible, so $\bigcap_{n \geq 1} K_n = \{a_*\}$.

(ii) Let $a > a_*$. Choose n with $a > \beta_n$. By comparison, $x_a > q_n > t$, so $x_a > 0$ and $x_a' > 0$ while both exist.
If $T_a \leq n$, then $\ell_a := \lim_{t \uparrow T_a} x_a(t) \in (a, +\infty]$. Lemma 1.1.28 excludes $\ell_a < +\infty$, so $x_a(t) \to +\infty$.
If $T_a > n$, then $x_a(n) > q_n(n) = \sqrt{n^2+2}$; Proposition 1.3.9 implies $T_a < +\infty$ and $x_a(t) \to +\infty$ as $t \uparrow T_a$.

(iii) Let $a < a_*$. Choose n with $a < \alpha_n$. By comparison, $x_a < p_n < t$.
On $[0, T_a \wedge n)$, whenever $x_a < -t$ one has $x_a' > 0$; after a contact with $-t$, strict comparison prevents a later downward crossing.
Hence $\min\{a, -n\} \leq x_a(t) < n$ on this interval. Corollary 1.1.29 therefore gives $T_a > n$.
Now $x_a(n) < n$. If $-n < x_a(n) < n$, Proposition 1.3.8 gives $T_a = +\infty$ and $x_a(t) + t \to 0$.
If $x_a(n) \leq -n$, set $h := x_a + t$. While $h < 0$, one has $h' = x_a^2 - t^2 + 1 > 1$ and $x_a(n) \leq x_a(t) < -t$. Compact escape gives existence through the finite time $n - h(n)$, so h has a first zero $\tau \in [n, n - h(n)]$. Strict comparison from τ gives $-t < x_a(t)$ for $t > \tau$, and continuity gives $x_a(t_2) < t_2$ for some $t_2 > \tau$.
Proposition 1.3.8 implies $T_a = +\infty$ and $x_a(t) + t \to 0$.

Structural Commentary: Nested corridors isolate the threshold separating blow-up from lower-edge attraction.
Key Insight: The orbit $x_{a_*} \sim t$ separates blow-up from attraction to $-t$. □

Theorem 1.3.11. *Asymptotic Stability by Linearization*
Let $U \subseteq \mathbb{R}^n$ be open with $0 \in U$, let $f \in C^1(U, \mathbb{R}^n)$ satisfy $f(0) = 0$, and set $A := Df(0)$. Assume $\operatorname{Re}\lambda < 0$ for every $\lambda \in \operatorname{spec}(A)$. Then $\exists r, K, \gamma > 0$ such that, for every $x_0 \in \overline{B}_r(0)$:

(i) The IVP $x' = f(x)$, $x(0) = x_0$, has a unique solution $x \in C^1([0, +\infty), U)$.
(ii) The solution satisfies $\|x(t)\| \leq Ke^{-\gamma t}\|x_0\|$ $\forall t \geq 0$.

Proof. Choose $\alpha \in (0, \min\{-\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\})$. By Theorem 1.2.16(iii), $\exists C \geq 1$ with $\|e^{tA}\| \leq Ce^{-\alpha t}$ for $t \geq 0$. Set $g(x) := f(x) - Ax$. Since $Dg(0) = 0$, choose $b \in (0, \alpha/C)$ and $R > 0$ such that $\overline{B}_R(0) \subseteq U$ and $\|g(x)\| \leq b\|x\|$ for $\|x\| \leq R$.
Set $r := R/(2C)$, $K := C$, and $\gamma := \alpha - Cb > 0$. Fix $x_0 \in \overline{B}_r(0)$. By Lemma 1.1.15 and Theorem 1.1.27, the autonomous field $(t, x) \mapsto f(x)$ has a unique maximal solution $x : [0, T_+) \to U$ with $x(0) = x_0$.

(i) Work before the first exit from $B_R(0)$. By Theorem 1.2.10(iii), $x(t) = e^{tA}x_0 + \int_0^t e^{(t-s)A}g(x(s))\,ds$. For $y(t) := e^{\alpha t}\|x(t)\|$, this gives $y(t) \leq C\|x_0\| + Cb\int_0^t y(s)\,ds$.
Lemma 1.1.20(ii) yields $\|x(t)\| \leq Ce^{-\gamma t}\|x_0\| \leq R/2$, so no exit occurs. If $T_+ < +\infty$, the graph remains in $[0, T_+] \times \overline{B}_{R/2}(0) \Subset \mathbb{R} \times U$, contradicting Corollary 1.1.29. Hence $T_+ = +\infty$; uniqueness is part of Theorem 1.1.27.

(ii) The estimate obtained in (i) is valid for every $t \geq 0$ and is exactly $\|x(t)\| \leq Ke^{-\gamma t}\|x_0\|$.

Structural Commentary: Split off Ax; absorb the differentiable remainder by Gronwall.
Key Insight: A Hurwitz linear part dominates its $o(\|x\|)$ remainder. □

1.4 Bifurcation Theory

A parameter can change both the position of equilibria and their number and stability. The harvested logistic equation is a complete scalar model of such a transition.

Definition 1.4.1. *Scalar Saddle-Node Pattern*
Let $\Lambda \subseteq \mathbb{R}$ be an interval, let $F : \mathbb{R} \times \Lambda \to \mathbb{R}$, and set $F_h(x) := F(x,h)$.
For $(x_c, h_c) \in \mathbb{R} \times \Lambda$, the family $(F_h)_{h\in\Lambda}$ has a *saddle-node pattern* at (x_c, h_c) iff $\exists \varepsilon > 0$, a neighborhood $X \subseteq \mathbb{R}$ of x_c, and $\sigma \in \{-1, 1\}$ such that the sets $H_\pm := \{h \in \Lambda : 0 < \pm\sigma(h - h_c) < \varepsilon\}$ are nonempty and:

(i) *Critical section*: $F_{h_c}^{-1}(0) \cap X = \{x_c\}$.
(ii) *Two-equilibrium side*: $\exists x_-, x_+ : H_+ \to X$ such that $F_h^{-1}(0) \cap X = \{x_-(h), x_+(h)\}$ and $x_-(h) < x_+(h)\ \forall h \in H_+$, with $x_\pm(h) \to x_c$ as $h \to h_c$ through H_+.
(iii) *Zero-equilibrium side*: $F_h^{-1}(0) \cap X = \emptyset\ \forall h \in H_-$.

Key Insight: A saddle-node is local creation or annihilation of two equilibria.

Proposition 1.4.2. *Equilibrium Branches of the Harvested Logistic Equation*
For $h \in [0, +\infty)$, let $F_h : \mathbb{R} \to \mathbb{R}$ be given by $F_h(x) := (1-x)x - h$. Set $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, and $x_\pm(h) := \frac{1}{2}(1 \pm \sqrt{1-4h})$ for $h \in [0, h_c]$.

(i) $F_h^{-1}(0) = \{x_-(h), x_+(h)\}$ *and* $x_-(h) < x_+(h)\ \forall h \in [0, h_c)$.
(ii) $F_{h_c}^{-1}(0) = \{x_c\}$ *and* $F_{h_c}(x) = -(x - x_c)^2\ \forall x \in \mathbb{R}$.
(iii) $F_h^{-1}(0) = \emptyset\ \forall h \in (h_c, +\infty)$.
(iv) $x_\pm(h) \to x_c$ *as* $h \uparrow h_c$.
(v) The family $(F_h)_{h\in[0,+\infty)}$ has a saddle-node pattern at (x_c, h_c).

Proof. The identity $F_h(x) = h_c - h - (x - x_c)^2$ holds $\forall (x,h) \in \mathbb{R} \times [0, +\infty)$; hence $F_h(x) = 0$ iff $(x - x_c)^2 = h_c - h$.

(i) For $h \in [0, h_c)$, $h_c - h > 0$ and $x_c \pm \sqrt{h_c - h} = x_\pm(h)$, which gives the two distinct zeros.
(ii) Substitution of $h = h_c$ gives $F_{h_c}(x) = -(x - x_c)^2$ and therefore $F_{h_c}^{-1}(0) = \{x_c\}$.
(iii) For $h > h_c$, the relation $(x - x_c)^2 = h_c - h < 0$ has no solution in $\mathbb{R}$.
(iv) The identity $x_\pm(h) = x_c \pm \sqrt{h_c - h}$ and the limit $h_c - h \downarrow 0$ give $x_\pm(h) \to x_c$.
(v) Take $X = \mathbb{R}$, $\sigma = -1$, and $\varepsilon = \frac{1}{8}$. Then $H_+ = (\frac{1}{8}, \frac{1}{4})$ and $H_- = (\frac{1}{4}, \frac{3}{8})$; items (i)–(iv) establish the critical zero, the two coalescing branches on H_+, and the absence of zeros on H_-.

Structural Commentary: Completing the square controls the zero count on both parameter sides.
Key Insight: At h_c, two simple roots merge into one double root. □

Lemma 1.4.3. *Scalar Phase-Line Barriers and Limits*
Let $F \in C^1(\mathbb{R}, \mathbb{R})$, and let $x : [t_0, T_+) \to \mathbb{R}$, $T_+ \in (t_0, +\infty]$, be a maximal forward solution of $x' = F(x)$.

(i) If $e \in F^{-1}(0)$ and $\exists \tau \in [t_0, T_+)$ such that $x(\tau) = e$, then $x(t) = e\ \forall t \in [t_0, T_+)$.
(ii) If x is monotone and $x([t_0, T_+))$ is bounded, then $T_+ = +\infty$ and $\exists \ell \in F^{-1}(0)$ such that $x(t) \to \ell$ as $t \to +\infty$.

Proof.

(i) The constant function $u \equiv e$ satisfies $u' = F(u)$ and $u(\tau) = x(\tau)$. Set $S := \{t \in [t_0, T_+) : x(t) = e\}$. Continuity makes S relatively closed, while local uniqueness makes S relatively open. Since $\tau \in S$ and $[t_0, T_+)$ is connected, $S = [t_0, T_+)$.
(ii) Monotone boundedness gives $\ell := \lim_{t\uparrow T_+} x(t) \in \mathbb{R}$. If $T_+ < +\infty$, by Lemma 1.1.28, x extends through (T_+, ℓ), contradicting maximality. Thus $T_+ = +\infty$.
Suppose $F(\ell) \neq 0$. By continuity and $x(t) \to \ell$, $\exists c > 0$ and $t_1 \geq t_0$ such that $F(x(t))$ has constant sign and $|F(x(t))| \geq c\ \forall t \geq t_1$. Hence $|x(t) - x(s)| = \left|\int_s^t F(x(r))\, dr\right| \geq c(t - s)\ \forall t > s \geq t_1$, contradicting $x(t) \to \ell$. Therefore $F(\ell) = 0$.

Structural Commentary: Uniqueness makes equilibria barriers; maximal extension identifies bounded monotone limits.
Key Insight: A finite phase-line limit must be a zero of F. □

Theorem 1.4.4. *Autonomous Scalar Phase-Line Classification*
Let $F \in C^1(\mathbb{R}, \mathbb{R})$ and let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F(x)$ with $x(t_0) = x_0$.
Set $E := F^{-1}(0)$. If $x_0 \notin E$, set $\sigma := \operatorname{sgn}(F(x_0))$ and $Z_\sigma := \{e \in E : \sigma(e - x_0) > 0\}$.

(i) If $x_0 \in E$, then $x \equiv x_0$.
(ii) If $x_0 \notin E$, then $(\sigma x)'(t) > 0\ \forall t \in [t_0, T_+)$.
(iii) If $x_0 \notin E$ and $Z_\sigma \neq \emptyset$, then $\exists! e_\sigma \in Z_\sigma$ such that $\sigma e_\sigma = \inf\{\sigma e : e \in Z_\sigma\}$. In this case, $T_+ = +\infty$ and $\sigma(x(t) - e_\sigma) \uparrow 0$ as $t \to +\infty$.
(iv) If $x_0 \notin E$ and $Z_\sigma = \emptyset$, then $\sigma x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Proof.

(i) Since $x_0 \in E$, by Lemma 1.4.3(i), $x(t) = x_0\ \forall t \in [t_0, T_+)$.
(ii) Let C be the component of $\mathbb{R} \setminus E$ containing x_0. Continuity gives $\sigma F > 0$ on C. If x left C, continuity would give $x(\tau) \in E$ for some τ, and Lemma 1.4.3(i) would force $x \equiv x(\tau)$, contradicting $x_0 \notin E$. Thus $x([t_0, T_+)) \subseteq C$ and $(\sigma x)' = \sigma F(x) > 0$.
(iii) Since $F(x_0) \neq 0$, continuity gives $\delta > 0$ such that $E \cap [x_0 - \delta, x_0 + \delta] = \emptyset$. Let $m := \inf\{\sigma e : e \in Z_\sigma\}$ and choose $(e_n) \subseteq Z_\sigma$ with $\sigma e_n \to m$. Then (e_n) converges to $e_\sigma := \sigma m$. Since E is closed, $e_\sigma \in E$; the zero-free neighborhood gives $\sigma(e_\sigma - x_0) > 0$, so $e_\sigma \in Z_\sigma$. Its defining equality makes it unique.
No zero lies strictly between x_0 and e_σ. By (ii) and Lemma 1.4.3(i), σx is increasing and bounded above by σe_σ. Lemma 1.4.3(ii) gives $T_+ = +\infty$ and $x(t) \to \ell \in E$. Since $\sigma x_0 < \sigma\ell \leq \sigma e_\sigma$, one has $\ell \in Z_\sigma$; minimality gives $\sigma e_\sigma \leq \sigma\ell$, hence $\ell = e_\sigma$.
(iv) By (ii), σx is strictly increasing. If it were bounded above, Lemma 1.4.3(ii) would give $x(t) \to \ell \in E$ with $\sigma(\ell - x_0) > 0$, so $\ell \in Z_\sigma$, a contradiction. Therefore $\sigma x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Structural Commentary: Components of $\mathbb{R} \setminus E$ fix the sign; the nearest zero supplies the barrier.
Key Insight: A trajectory approaches its next zero or escapes in the sign direction. □

Theorem 1.4.5. *Subcritical Harvest Phase Line*
Let $h_c := \frac{1}{4}$, $F_h(x) := (1-x)x - h$, $0 \le h < h_c$, and $x_\pm := \frac{1}{2}(1 \pm \sqrt{1-4h})$.
Let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F_h(x)$ with $x(t_0) = x_0$.

(i) If $x_0 \in \{x_-, x_+\}$, then $T_+ = +\infty$ and $x \equiv x_0$.
(ii) If $x_0 < x_-$, then $x'(t) < 0$ $\forall t \in [t_0, T_+)$ and $x(t) \to -\infty$ as $t \uparrow T_+$.
(iii) If $x_- < x_0 < x_+$, then $T_+ = +\infty$, $x'(t) > 0$ $\forall t \ge t_0$, and $x(t) \to x_+$ as $t \to +\infty$.
(iv) If $x_0 > x_+$, then $T_+ = +\infty$, $x'(t) < 0$ $\forall t \ge t_0$, and $x(t) \to x_+$ as $t \to +\infty$.

Proof. The factorization $F_h = -(\cdot - x_-)(\cdot - x_+)$ gives $F_h < 0$ on $(-\infty, x_-) \cup (x_+, +\infty)$ and $F_h > 0$ on (x_-, x_+).

(i) Since $x_0 \in F_h^{-1}(0)$, by Theorem 1.4.4(i), $x \equiv x_0$. The constant solution is defined on $[t_0, +\infty)$, so maximality gives $T_+ = +\infty$.
(ii) Here $\operatorname{sgn}(F_h(x_0)) = -1$, and no zero lies below x_0. By Theorem 1.4.4(ii),(iv), $x' < 0$ and $-x(t) \uparrow +\infty$ as $t \uparrow T_+$.
(iii) Here $\operatorname{sgn}(F_h(x_0)) = 1$, and x_+ is the nearest zero above x_0. By Theorem 1.4.4(ii),(iii), $T_+ = +\infty$, $x' > 0$, and $x(t) \to x_+$.
(iv) Here $\operatorname{sgn}(F_h(x_0)) = -1$, and x_+ is the nearest zero below x_0. By Theorem 1.4.4(ii),(iii), $T_+ = +\infty$, $x' < 0$, and $x(t) \to x_+$.

Structural Commentary: The factorization fixes three sign intervals; the phase-line theorem handles each.
Key Insight: The upper root attracts every initial state above the lower threshold. □

Theorem 1.4.6. *Critical Harvest Phase Line*
Let $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, $F_h(x) := (1-x)x - h$, and $h = h_c$.
Let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F_{h_c}(x)$ with $x(t_0) = x_0$.

(i) If $x_0 = x_c$, then $T_+ = +\infty$ and $x \equiv x_c$.
(ii) If $x_0 > x_c$, then $T_+ = +\infty$, $x'(t) < 0$ $\forall t \ge t_0$, and $x(t) \to x_c$ as $t \to +\infty$.
(iii) If $x_0 < x_c$, then $x'(t) < 0$ $\forall t \in [t_0, T_+)$ and $x(t) \to -\infty$ as $t \uparrow T_+$.

Proof. The identity $F_{h_c}(x) = -(x - x_c)^2$ gives $F_{h_c}^{-1}(0) = \{x_c\}$ and $F_{h_c} < 0$ on $\mathbb{R} \setminus \{x_c\}$.

(i) Since $x_0 \in F_{h_c}^{-1}(0)$, by Theorem 1.4.4(i), $x \equiv x_c$. The constant solution is defined on $[t_0, +\infty)$, so maximality gives $T_+ = +\infty$.
(ii) Here $\operatorname{sgn}(F_{h_c}(x_0)) = -1$, and x_c is the nearest zero below x_0. By Theorem 1.4.4(ii),(iii), $T_+ = +\infty$, $x' < 0$, and $x(t) \to x_c$.
(iii) Here $\operatorname{sgn}(F_{h_c}(x_0)) = -1$, and no zero lies below x_0. By Theorem 1.4.4(ii),(iv), $x' < 0$ and $-x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Structural Commentary: The double zero is the sole barrier; the sign is negative on both sides.
Key Insight: The critical equilibrium attracts from above and repels toward $-\infty$ below. □

Theorem 1.4.7. *Supercritical Harvest Phase Line*
Let $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, $F_h(x) := (1-x)x - h$, and $h > h_c$.
Let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F_h(x)$ with $x(t_0) = x_0$. Then:

(i) $x'(t) < 0$ $\forall t \in [t_0, T_+)$.
(ii) $x(t) \to -\infty$ as $t \uparrow T_+$.

Proof. For $h > h_c$, $F_h = -(\cdot - x_c)^2 - (h - h_c) < 0$ on $\mathbb{R}$ and $F_h^{-1}(0) = \emptyset$.

(i) By Theorem 1.4.4(ii), with $\sigma = -1$, $x' < 0$.
(ii) Since $F_h^{-1}(0) = \emptyset$, by Theorem 1.4.4(iv), $-x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Structural Commentary: The completed square removes every zero, so the phase-line escape case applies.
Key Insight: Above critical harvest, every trajectory escapes toward $-\infty$. □

Corollary 1.4.8. *Critical Dimensional Harvest Rate*
Let $\kappa, N_{\max} > 0$, let $H \ge 0$, and set $G_H(N) := \kappa(1 - \frac{N}{N_{\max}})N - H$ and $H_c := \frac{\kappa N_{\max}}{4}$.
Let $N : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $N' = G_H(N)$ with $N(t_0) = N_0$.

(i) $G_H^{-1}(0) \cap (0, +\infty) \ne \emptyset$ iff $H \in [0, H_c]$, and $H_c = \max\{\eta \ge 0 : G_\eta^{-1}(0) \cap (0, +\infty) \ne \emptyset\}$.
(ii) For $H \in [0, H_c)$, set $N_\pm(H) := \frac{N_{\max}}{2}(1 \pm \sqrt{1 - \frac{4H}{\kappa N_{\max}}})$. Then $G_H^{-1}(0) = \{N_-(H), N_+(H)\}$ and $G_{H_c}^{-1}(0) = \{\frac{N_{\max}}{2}\}$.
(iii) If $H \in [0, H_c)$ and $N_0 > N_-(H)$, then $T_+ = +\infty$ and $N(t) \to N_+(H)$ as $t \to +\infty$.

Proof. Set $h := \frac{H}{\kappa N_{\max}}$, $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, $x_\pm(h) := \frac{1}{2}(1 \pm \sqrt{1-4h})$ for $h \in [0, h_c]$, and $F_h(x) := (1-x)x - h$. Then $G_H(N) = \kappa N_{\max} F_h(\frac{N}{N_{\max}})$.
For the solution, set $x(\tau) := \frac{N(t_0 + \tau/\kappa)}{N_{\max}}$, $x_0 := \frac{N_0}{N_{\max}}$, and $\widehat{T}_+ := \kappa(T_+ - t_0)$. Thus $x : [0, \widehat{T}_+) \to \mathbb{R}$ is the maximal forward solution of $\frac{dx}{d\tau} = F_h(x)$ with $x(0) = x_0$.

(i) By Proposition 1.4.2(i)–(iii), $F_h^{-1}(0) \cap (0, +\infty) \ne \emptyset$ iff $h \in [0, h_c]$. Since $H = \kappa N_{\max} h$ and $H_c = \kappa N_{\max} h_c$, the asserted equivalence and maximum follow.
(ii) By Proposition 1.4.2(i),(ii), $F_h^{-1}(0) = \{x_-(h), x_+(h)\}$ for $h < h_c$ and $F_{h_c}^{-1}(0) = \{x_c\}$. Multiplication by $N_{\max}$ gives $N_\pm(H) = N_{\max} x_\pm(h)$ and $N_{\max} x_c = \frac{N_{\max}}{2}$.
(iii) The inequality $N_0 > N_-(H)$ is equivalent to $x_0 > x_-(h)$. By Theorem 1.4.5(i),(iii),(iv), $\widehat{T}_+ = +\infty$ and $x(\tau) \to x_+(h)$ as $\tau \to +\infty$. Hence $T_+ = +\infty$ and $N(t) \to N_{\max} x_+(h) = N_+(H)$.

Structural Commentary: State and time scaling reduce the dimensional model to the universal quadratic family.
Key Insight: The threshold is one quarter of the unharvested growth scale. □

1.5 Dynamical Systems Structure

For a time-periodic scalar equation, one period of continuous evolution can be compressed into a discrete map. Periodic solutions then become fixed points, and their long-time behavior can be studied through iteration.

Definition 1.5.1. *Scalar Poincaré Map*
Let $f \in C^1(\mathbb{R}^2, \mathbb{R})$ satisfy $f(t+1, x) = f(t, x)\ \forall (t, x) \in \mathbb{R}^2$.

(i) For $x_0 \in \mathbb{R}$, let $\phi_{x_0} : I_{x_0} \to \mathbb{R}$ be the unique maximal solution of $x' = f(t, x)$ with $0 \in I_{x_0}$ and $\phi_{x_0}(0) = x_0$; set $\phi(t, x_0) := \phi_{x_0}(t)$.
(ii) The *one-period domain* is $\mathcal{D} := \{x_0 \in \mathbb{R} : [0, 1] \subseteq I_{x_0}\}$.
(iii) The *Poincaré map* is $P : \mathcal{D} \to \mathbb{R}$, $P(x_0) := \phi(1, x_0)$.

Key Insight: P compresses one period of continuous evolution into one discrete step.

Proposition 1.5.2. *Periodic Orbits and Ordered Returns*
Let $f \in C^1(\mathbb{R}^2, \mathbb{R})$ satisfy $f(t+1, x) = f(t, x)$.
For $x_0 \in \mathbb{R}$, let $\phi_{x_0} : I_{x_0} \to \mathbb{R}$ be the unique maximal solution of $x' = f(t, x)$ with $0 \in I_{x_0}$ and $\phi_{x_0}(0) = x_0$.
Set $\mathcal{D} := \{x_0 \in \mathbb{R} : [0, 1] \subseteq I_{x_0}\}$ and $P : \mathcal{D} \to \mathbb{R}$, $P(x_0) := \phi_{x_0}(1)$.

(i) For $x_0 \in \mathcal{D}$, $P(x_0) = x_0 \iff I_{x_0} = \mathbb{R}$ and $\phi_{x_0}(t+1) = \phi_{x_0}(t)\ \forall t \in \mathbb{R}$.
(ii) For $x_0, y_0 \in \mathcal{D}$, $x_0 < y_0 \implies P(x_0) < P(y_0)$.

Proof.

(i) Suppose first that $P(x_0) = x_0$. For $n \in \mathbb{Z}$, define $u_n : [n, n+1] \to \mathbb{R}$ by $u_n(t) := \phi_{x_0}(t - n)$. Then $u_n' = f(t-n, u_n) = f(t, u_n)$, while $u_n(n+1) = \phi_{x_0}(1) = x_0 = u_{n+1}(n+1)$. The one-periodicity of f makes the one-sided derivatives agree at every integer, so the u_n glue to a one-periodic $u \in C^1(\mathbb{R}, \mathbb{R})$ satisfying $u' = f(t, u)$ and $u(0) = x_0$. Local uniqueness and maximality give $I_{x_0} = \mathbb{R}$ and $\phi_{x_0} = u$.
Conversely, if $I_{x_0} = \mathbb{R}$ and ϕ_{x_0} is one-periodic, then $P(x_0) = \phi_{x_0}(1) = \phi_{x_0}(0) = x_0$.
(ii) Choose an open interval J with $[0, 1] \subseteq J \subseteq I_{x_0} \cap I_{y_0}$. Since $f \in C^1$, it is state-Lipschitz on compact subsets of $\mathbb{R}^2$. By Theorem 1.3.5(ii), $\phi_{x_0}(t) < \phi_{y_0}(t)\ \forall t \in [0, 1]$; evaluation at $t = 1$ gives $P(x_0) < P(y_0)$.

Structural Commentary: Glue one-period segments; compare distinct initial states by scalar order preservation.
Key Insight: Periodic trajectories are fixed points of a strictly increasing return map. □

Proposition 1.5.3. *Periodic Scalar Linear Forcing*
Let $a \in \mathbb{R}$ and $g \in C(\mathbb{R}, \mathbb{R})$ satisfy $g(t+1) = g(t)$ and $g(t) \geq 0\ \forall t \in \mathbb{R}$. Set $I_a := \int_0^1 e^{-as} g(s)\, ds$.
For $x_0 \in \mathbb{R}$, let $x(\cdot\,; x_0)$ be the solution of $x' = ax + g(t)$ with $x(0; x_0) = x_0$, and set $P(x_0) := x(1; x_0)$.
If $a \neq 0$, put $x_{0,*} := -\frac{e^a}{e^a - 1} I_a$ and $x_*(t) := e^{at}\big(x_{0,*} + \int_0^t e^{-as} g(s)\, ds\big)$.

(i) $a \neq 0 \implies x_*$ is the unique one-periodic solution.
(ii) If $a = 0$ and $g \equiv 0$, the one-periodic solutions are $t \mapsto c$, $c \in \mathbb{R}$.
(iii) $a = 0$ and $g \not\equiv 0 \implies$ no one-periodic solution exists.

Proof. By Proposition 1.2.1, $x(t; x_0) = e^{at}\big(x_0 + \int_0^t e^{-as} g(s)\, ds\big)$ and $P(x_0) - x_0 = (e^a - 1)x_0 + e^a I_a$.
If $P(x_0) = x_0$, then $d(t) := x(t+1; x_0) - x(t; x_0)$ satisfies $d' = ad$ and $d(0) = 0$, so $d \equiv 0$. The converse is immediate; hence the one-periodic solutions correspond to $\mathrm{Fix}(P)$.

(i) If $a \neq 0$, then $e^a - 1 \neq 0$ and $\mathrm{Fix}(P) = \{x_{0,*}\}$. Substitution of $x_0 = x_{0,*}$ into the solution formula gives x_*.
(ii) If $a = 0$ and $g \equiv 0$, then $x(t; x_0) = x_0\ \forall (t, x_0) \in \mathbb{R}^2$.
(iii) If $a = 0$ and $g \not\equiv 0$, continuity and nonnegativity give $I_0 = \int_0^1 g(s)\, ds > 0$. Thus $P(x_0) - x_0 = I_0 \neq 0\ \forall x_0 \in \mathbb{R}$.

Structural Commentary: Variation of constants makes the return displacement affine in x_0.
Key Insight: $P(x_0) - x_0 = (e^a - 1)x_0 + e^a I_a$. □

Proposition 1.5.4. *Concave Periodic Return-Map Geometry*
Let $f, g \in C^2(\mathbb{R}, \mathbb{R})$ satisfy $f''(u) < 0\ \forall u \in \mathbb{R}$, $g(t+1) = g(t)$ and $g(t) \geq 0\ \forall t \in \mathbb{R}$, and $g \not\equiv 0$.
Let $W \subseteq \mathbb{R}^2$ be open and $(h, x) \in W$. For $(\eta, y) \in W$, let $\phi_\eta(\cdot\,, y)$ be the unique solution of $z' = f(z) + \eta g(t)$, $z(0) = y$, on $[0, 1]$.
Set $P_\eta(y) := \phi_\eta(1, y)$, $\phi(t) := \phi_h(t, x)$, $A(t) := f'(\phi(t))$, and $\theta(t) := \partial_x \phi_h(t, x)$.

(i) $P_h'(x) = \exp\Big(\int_0^1 A(s)\, ds\Big) > 0$.
(ii) $P_h''(x) = P_h'(x) \int_0^1 f''(\phi(s))\theta(s)\, ds < 0$.
(iii) $\partial_h P_h(x) = \int_0^1 \exp\Big(\int_s^1 A(r)\, dr\Big) g(s)\, ds > 0$.

Proof. Set $\mathcal{F}(t, z, \eta) := f(z) + \eta g(t)$. Since $\mathcal{F} \in C^2(\mathbb{R}^3, \mathbb{R})$, by Theorem 1.1.24, applied successively along the solution arcs over $[0, 1]$, the flow has the required two state derivatives and one parameter derivative.

(i) $\theta' = A\theta$ and $\theta(0) = 1$. Hence $\theta(t) = \exp\Big(\int_0^t A(s)\, ds\Big) > 0$, so $P_h'(x) = \theta(1)$.
(ii) Put $\zeta := \partial_x^2 \phi_h(\cdot\,, x)$. Then $\zeta' = A\zeta + f''(\phi)\theta^2$ and $\zeta(0) = 0$.
By Proposition 1.2.1, $\zeta(1) = \theta(1) \int_0^1 f''(\phi(s))\theta(s)\, ds$. Since $f'' < 0$ and $\theta > 0$, $P_h''(x) = \zeta(1) < 0$.
(iii) Put $\psi := \partial_h \phi_h(\cdot\,, x)$. Then $\psi' = A\psi + g$ and $\psi(0) = 0$.
By Proposition 1.2.1, $\psi(1) = \int_0^1 \exp\Big(\int_s^1 A(r)\, dr\Big) g(s)\, ds$. The integrand is nonnegative and not identically zero, so $\psi(1) > 0$ and $\partial_h P_h(x) = \psi(1)$.

Structural Commentary: Differentiate the flow; solve each scalar variational equation by an integrating factor.
Key Insight: $f'' < 0$ bends the return map; $g \geq 0$ moves it upward. □

Proposition 1.5.5. *Geometry of the Periodically Harvested Return Map*
Let $g(t) := 1 - \sin(2\pi t)$ and $F_\eta(t, z) := (1 - z)z - \eta g(t)$.
Let $W \subseteq \mathbb{R}^2$ be open and $(h, x) \in W$. For $(\eta, y) \in W$, let $\phi_\eta(\cdot\,, y)$ be the unique solution of $z' = F_\eta(t, z)$, $z(0) = y$, on $[0, 1]$.
Set $P_\eta(y) := \phi_\eta(1, y)$ and $\theta_h(t, x) := \partial_x \phi_h(t, x)$.

(i) $P_h'(x) = \exp\Big(\int_0^1 (1 - 2\phi_h(s, x))\, ds\Big) > 0$.

(ii) $P_h''(x) = -2P_h'(x)\int_0^1 \theta_h(s,x)\,ds < 0$.
(iii) $\partial_h P_h(x) = -\int_0^1 \exp\Big(\int_s^1 (1-2\phi_h(r,x))\,dr\Big)\,g(s)\,ds < 0$.

Proof. Put $f(z) := (1-z)z$, $\widetilde{W} := \{(-\eta, y) : (\eta, y) \in W\}$, and $Q_\lambda(y) := P_{-\lambda}(y)$ for $(\lambda, y) \in \widetilde{W}$. Then Q_λ is the return map for $z' = f(z) + \lambda g(t)$, and $f'' = -2$, while $g \geq 0$ and $g \not\equiv 0$. Set $E(s) := \exp\Big(\int_s^1 (1-2\phi_h(r,x))\,dr\Big)$.

(i) By Proposition 1.5.4(i), applied at $(-h, x)$, $P_h'(x) = Q'_{-h}(x) = \exp\Big(\int_0^1 (1-2\phi_h(s,x))\,ds\Big) > 0$.
(ii) By Proposition 1.5.4(ii), $P_h''(x) = Q''_{-h}(x) = -2P_h'(x)\int_0^1 \theta_h(s,x)\,ds < 0$.
(iii) By the chain rule and Proposition 1.5.4(iii), $\partial_h P_h(x) = -\partial_\lambda Q_{-h}(x) = -\int_0^1 E(s)g(s)\,ds < 0$.

Structural Commentary: Reflect the parameter; specialize the general concave return-map identities.
Key Insight: Harvesting shifts the increasing concave return map downward. □

Example 1.5.6. *The Unharvested Return Map*
For $x \in [0, +\infty)$, let $\phi_0(\,\cdot\,, x)$ solve $z' = (1-z)z$ with $z(0) = x$, and set $P_0(x) := \phi_0(1, x)$.
Direct differentiation verifies $\phi_0(t,x) = \frac{e^t x}{1+(e^t-1)x}$ for $t \geq 0$, so $P_0(x) = \frac{ex}{1+(e-1)x}$. Solving $P_0(x) = x$ gives $\mathrm{Fix}(P_0) = \{0, 1\}$. By Proposition 1.5.2(i), these fixed points correspond to the constant one-periodic solutions $z \equiv 0$ and $z \equiv 1$.

Theorem 1.5.7. *Iteration of an Increasing Concave Map*
Let $I \subseteq \mathbb{R}$ be an interval, $P : I \to I$ continuous, strictly increasing, strictly concave, and $\mathrm{Fix}(P) = \{x_1, x_2\}$ with $x_1 < x_2$. For $x \in I$, set $x_0 := x$ and $x_{n+1} := P(x_n)$.
(i) If $x > x_1$, then $x_n \to x_2$.
(ii) If $x = x_1$, then $x_n = x_1$ $\forall n \geq 0$.
(iii) If $x < x_1$, then $x_{n+1} < x_n$ $\forall n \geq 0$ and $x_n \downarrow \inf I$ in $\overline{\mathbb{R}}$; if $\inf I = -\infty$, then $x_n \to -\infty$.

Proof. For $a, b \in I$ with $a < b$, set $m(a,b) := \frac{P(b)-P(a)}{b-a}$. If $a < b < c$, strict concavity at $b = (1-\lambda)a + \lambda c$, $\lambda := \frac{b-a}{c-a} \in (0,1)$, gives $m(a,b) > m(a,c) > m(b,c)$. Since $P(x_j) = x_j$, $m(x_1, x_2) = 1$.

(i) If $x \in (x_1, x_2)$, then $m(x_1, x) > m(x_1, x_2) = 1$ and monotonicity gives $x < P(x) < x_2$. Hence $x_n \uparrow \ell \in (x_1, x_2]$, and continuity gives $P(\ell) = \ell$, so $\ell = x_2$. If $x = x_2$, the sequence is constant.
If $x > x_2$, then $m(x_2, x) < m(x_1, x_2) = 1$ and monotonicity gives $x_2 < P(x) < x$. Hence $x_n \downarrow \ell \in [x_2, x)$, and continuity again gives $\ell = x_2$.
(ii) Since $P(x_1) = x_1$, induction gives $x_n = x_1$ $\forall n \geq 0$.
(iii) If $x < x_1$, then $m(x, x_1) > m(x_1, x_2) = 1$, so $P(x) < x$. Induction gives $x_{n+1} < x_n < x_1$ $\forall n \geq 0$, and $x_n \downarrow \ell \in \overline{\mathbb{R}}$ with $\ell \geq \inf I$. If $\ell > \inf I$, then $\ell \in I$, continuity gives $P(\ell) = \ell < x_1$, and this contradicts $\mathrm{Fix}(P) = \{x_1, x_2\}$. Thus $\ell = \inf I$.

Structural Commentary: Nested secant slopes determine the sign of $P(x) - x$ on three intervals.
Key Insight: The lower fixed point is a threshold; the upper one attracts from above. □

Corollary 1.5.8. *Periodic Logistic Threshold Structure*
Let $g(t) := 1 - \sin(2\pi t)$. For $(\eta, y) \in \mathbb{R}^2$, let $\phi_{\eta,y} : I_{\eta,y} \to \mathbb{R}$ be the unique maximal solution of $z' = (1-z)z - \eta g(t)$ with $0 \in I_{\eta,y}$ and $\phi_{\eta,y}(0) = y$.
Set $\mathcal{D}_\eta := \{y \in \mathbb{R} : [0,1] \subseteq I_{\eta,y}\}$ and $P_\eta : \mathcal{D}_\eta \to \mathbb{R}$, $P_\eta(y) := \phi_{\eta,y}(1)$. Fix $h \geq 0$ and an interval $J \subseteq \mathcal{D}_h$.
(i) $\#\,\mathrm{Fix}(P_h|_J) \leq 2$; hence there are at most two one-periodic solutions with initial values in J.
(ii) If $P_h(J) \subseteq J$, $\mathrm{Fix}(P_h|_J) = \{x_1, x_2\}$ with $x_1 < x_2$, and $x \in J$ satisfies $x > x_1$, then $P_h^n(x) \to x_2$.
(iii) If $P_h(J) \subseteq J$, $\mathrm{Fix}(P_h|_J) = \{x_1, x_2\}$ with $x_1 < x_2$, and $x \in J$ satisfies $x < x_1$, then $P_h^n(x) \downarrow \inf J$ in $\overline{\mathbb{R}}$.
(iv) $\partial_h P_h(x) < 0$ $\forall x \in \mathcal{D}_h$.

Proof. Fix $x \in \mathcal{D}_h$. The solution arc $\{(t, \phi_{h,x}(t), h) : t \in [0,1]\}$ is compact. By Theorem 1.1.24, applied successively along a finite partition of this arc, $\exists$ an open neighborhood $W_x \subseteq \mathbb{R}^2$ of (h, x) on which the time-1 flow is defined and C^2. By Proposition 1.5.5, $P_h'(x) > 0$, $P_h''(x) < 0$, and $\partial_h P_h(x) < 0$. Thus P_h is strictly increasing and strictly concave on J.

(i) If $a < b < c$ were fixed points in J, strict concavity would give $P_h(b) > L_{a,c}(b) = b$, where $L_{a,c}$ is the chord through (a,a) and (c,c), contradicting $P_h(b) = b$. Hence $\#\,\mathrm{Fix}(P_h|_J) \leq 2$. By Proposition 1.5.2(i), these fixed points are exactly the initial values in J of one-periodic solutions.
(ii) Under the stated hypotheses, $P_h|_J : J \to J$ is continuous, strictly increasing, and strictly concave. By Theorem 1.5.7(i), $P_h^n(x) \to x_2$.
(iii) Under the stated hypotheses, by Theorem 1.5.7(iii), $P_h^n(x) \downarrow \inf J$ in $\overline{\mathbb{R}}$.
(iv) The local calculation preceding (i) gives $\partial_h P_h(x) < 0$ $\forall x \in \mathcal{D}_h$.

Structural Commentary: Smooth dependence supplies concavity; graph geometry and iteration classify fixed points.
Key Insight: Two periodic states form a threshold with an attracting upper branch. □

1.6 Semigroups and Evolution Viewpoint

1.7 Measure-Preserving Dynamics and Ergodicity

Measure-preserving dynamics study maps whose pullbacks keep mass fixed. The central question is whether time averages along one orbit recover space averages with respect to an invariant probability measure.

Definition 1.7.1. *Measure-Preserving and Ergodic Transformations*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $T : X \to X$.
For $E, A, B \in \mathcal{M}$, write $T^{-1}E := \{x \in X : T(x) \in E\}$ and $A \triangle B := (A \setminus B) \cup (B \setminus A)$.
(i) T is *measurable* iff $T^{-1}E \in \mathcal{M}$ $\forall E \in \mathcal{M}$.
(ii) If T is measurable, the *pushforward measure* is $(T_*\mu)(E) := \mu(T^{-1}E)$, $E \in \mathcal{M}$.
(iii) A measurable T is *measure preserving* iff $T_*\mu = \mu$.
(iv) For measurable T, $A \in \mathcal{M}$ is *invariant modulo μ* iff $\mu(A \triangle T^{-1}A) = 0$.
(v) If $0 < \mu(X) < +\infty$ and T is measure preserving, then T is *ergodic* iff $\mu(A \triangle T^{-1}A) = 0 \implies \mu(A) \in \{0, \mu(X)\}$ $\forall A \in \mathcal{M}$.

Key Insight: Pullbacks preserve mass; ergodic invariants are trivial.

Proposition 1.7.2. *Integral Test for Measure Preservation*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space and $T : X \to X$ measurable. Then $T_*\mu = \mu$ iff $\forall g \in L^1(X,\mu)$, $g \circ T \in L^1(X,\mu)$ and $\int_X g \circ T \, d\mu = \int_X g \, d\mu$.

Proof.
($\Rightarrow$) If $T_*\mu = \mu$, then $\int_X \mathbf{1}_E \circ T \, d\mu = \mu(T^{-1}E) = \mu(E) = \int_X \mathbf{1}_E \, d\mu \ \forall E \in \mathcal{M}$. Linearity gives the identity for nonnegative simple functions. For $0 \le g \in L^1(X,\mu)$, choose simple $0 \le s_n \uparrow g$; monotone convergence applied to s_n and $s_n \circ T$ gives the identity and $g \circ T \in L^1(X,\mu)$. Apply this conclusion to $g^{\pm}$ for general $g \in L^1(X,\mu)$.
($\Leftarrow$) Since $\mu(X) < +\infty$, $\mathbf{1}_E \in L^1(X,\mu) \ \forall E \in \mathcal{M}$. Taking $g = \mathbf{1}_E$ gives $(T_*\mu)(E) = \int_X \mathbf{1}_E \circ T \, d\mu = \int_X \mathbf{1}_E \, d\mu = \mu(E)$.
Structural Commentary: Indicators encode sets; approximation extends the identity to observables.
Key Insight: Measure preservation is equality of integrals after pullback. □

Remark 1.7.3. *Invariant Sets as Invariant Indicators*
If $T_*\mu = \mu$ and $A \in \mathcal{M}$, then $\mu(A \triangle T^{-1}A) = 0 \iff \mathbf{1}_A \circ T = \mathbf{1}_A$ μ-a.e.

Proposition 1.7.4. *Invariant Functions Detect Ergodicity*
Let $(X, \mathcal{M}, \mu)$ satisfy $0 < \mu(X) < +\infty$, and let $T : X \to X$ be measurable with $T_*\mu = \mu$. Then T is ergodic iff $\forall$ measurable $g : X \to \overline{\mathbb{R}}$, $g \circ T = g$ μ-a.e. implies $\exists c \in \overline{\mathbb{R}}$ such that $g = c$ μ-a.e.

Proof.
($\Rightarrow$) Assume T is ergodic and $g \circ T = g$ μ-a.e. For $a \in \mathbb{Q}$, set $A_a := \{g < a\}$. Then $\mu(A_a \triangle T^{-1}A_a) = 0$, so $\mu(A_a) \in \{0, \mu(X)\}$. With $\inf \varnothing := +\infty$, set $c := \inf\{a \in \mathbb{Q} : \mu(A_a) = \mu(X)\} \in \overline{\mathbb{R}}$. If $a < c$, then $\mu(A_a) = 0$; if $b > c$, monotonicity and the definition of c give $\mu(X \setminus A_b) = 0$. Countable unions give $\mu(g < c) = \mu(g > c) = 0$, including $c = \pm\infty$; hence $g = c$ μ-a.e.
($\Leftarrow$) Assume every measurable T-invariant $g : X \to \overline{\mathbb{R}}$ is constant μ-a.e. If $A \in \mathcal{M}$ and $\mu(A \triangle T^{-1}A) = 0$, then $\mathbf{1}_A \circ T = \mathbf{1}_A$ μ-a.e. Thus $\mathbf{1}_A = c$ μ-a.e. for some $c \in \{0, 1\}$, so $\mu(A) = c\mu(X) \in \{0, \mu(X)\}$.
Structural Commentary: Level sets convert invariant functions into invariant sets.
Key Insight: Ergodicity collapses invariant observables to constants. □

Method 1.7.5. *Krylov–Bogoliubov Averaging*
Let $X \neq \varnothing$ be compact metric, let $f \in C(X,X)$, and let $\mathcal{P}(X)$ denote the Borel probability measures on X.

(i) Choose $\eta \in \mathcal{P}(X)$ and set $\mu_n := \frac{1}{n} \sum_{k=0}^{n-1} (f^k)_*\eta \in \mathcal{P}(X)$.
(ii) For $g \in C(X)$, record $\int_X g \, d\mu_n = \frac{1}{n} \sum_{k=0}^{n-1} \int_X g \circ f^k \, d\eta$.
(iii) By Corollary 6.4.7, choose $n_j \uparrow \infty$ and a positive Radon measure μ with $\int_X g \, d\mu_{n_j} \to \int_X g \, d\mu \ \forall g \in C(X)$; testing $g = 1$ gives $\mu \in \mathcal{P}(X)$.
(iv) Use $\int_X g \circ f \, d\mu_n - \int_X g \, d\mu_n = \frac{1}{n} \int_X (g \circ f^n - g) \, d\eta$ and the bound $2 \|g\|_\infty / n$. Passing to n_j gives equality of the $C(X)$-integrals of the Radon measures $f_*\mu$ and μ. By Theorem 5.2.13, $f_*\mu = \mu$.

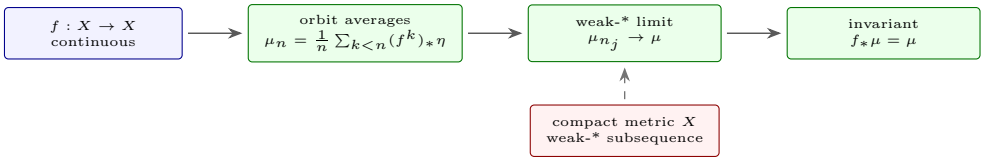

Figure 234: Krylov–Bogoliubov: $\mu_n = \frac{1}{n}\sum_{k<n}(f^k)_*\eta \xrightarrow{w^*} \mu$ and $\frac{1}{n}(g \circ f^n - g) \to 0$, hence $f_*\mu = \mu$.

Theorem 1.7.6. *Krylov–Bogoliubov Invariant Measure Theorem*
Let $X \neq \varnothing$ be compact metric, let $f \in C(X,X)$, and let $\mathcal{P}(X)$ denote the Borel probability measures on X. Then $\exists \mu \in \mathcal{P}(X)$ such that $f_*\mu = \mu$.

Proof. Choose $\eta \in \mathcal{P}(X)$ and set $\mu_n := \frac{1}{n} \sum_{k=0}^{n-1} (f^k)_*\eta \in \mathcal{P}(X)$. By Corollary 6.4.7, $\exists n_j \uparrow \infty$ and a positive Radon measure μ such that $\int_X g \, d\mu_{n_j} \to \int_X g \, d\mu \ \forall g \in C(X)$. Taking $g = 1$ gives $\mu(X) = 1$, so $\mu \in \mathcal{P}(X)$.
For $g \in C(X)$, $\int_X g \circ f \, d\mu_n - \int_X g \, d\mu_n = \frac{1}{n} \int_X (g \circ f^n - g) \, d\eta$, whose absolute value is at most $2 \|g\|_\infty / n$. Passing to n_j gives $\int_X g \, d(f_*\mu) = \int_X g \, d\mu \ \forall g \in C(X)$. Both measures are Radon, and $C_c(X) = C(X)$; hence $f_*\mu = \mu$ by Theorem 5.2.13. *Structural Commentary*: Average pushforwards; extract a weak-* limit; cancel endpoints.
Key Insight: Compactness turns orbit averages into an invariant probability. □

Proposition 1.7.7. *Ergodic Measures Are Extreme Invariant Measures*
Let X be compact metric, let $f \in C(X,X)$, let $\mathcal{P}(X)$ denote the Borel probability measures on X, and set $\mathcal{I}_f(X) := \{\lambda \in \mathcal{P}(X) : f_*\lambda = \lambda\}$. For $\mu \in \mathcal{I}_f(X)$, the following are equivalent:

(i) $\mu \in \mathrm{Ext}(\mathcal{I}_f(X))$;
(ii) f is μ-ergodic.

Proof.
(i) $\Rightarrow$ (ii) If f is not μ-ergodic, choose $A \in \mathcal{B}(X)$ with $\mu(A \triangle f^{-1}A) = 0$ and $0 < \mu(A) < 1$. Set $\nu(E) := \mu(E \cap A)/\mu(A)$ and $\rho(E) := \mu(E \cap A^c)/\mu(A^c)$. For $E \in \mathcal{B}(X)$, invariance modulo μ gives $\mu(f^{-1}E \cap A) = \mu(f^{-1}(E \cap A)) = \mu(E \cap A)$, and the same identity holds with A^c. Thus $\nu, \rho \in \mathcal{I}_f(X)$, while $\mu = \mu(A)\nu + \mu(A^c)\rho$ and $\nu(A) \neq \rho(A)$, contradicting (i).

(*ii*) $\Rightarrow$ (*i*) Let $\mu = \theta\nu + (1-\theta)\rho$ with $0 < \theta < 1$ and $\nu, \rho \in \mathcal{I}_f(X)$. Since $\theta\nu \le \mu$, by Theorem 3.2.1, $h := d\nu/d\mu$ satisfies $0 \le h \le \theta^{-1}$ μ-a.e. Invariance of ν and μ gives $\int_X (h\circ f)h\,d\mu = \int_X h\circ f\,d\nu = \int_X h\,d\nu = \int_X h^2\,d\mu$ and $\int_X (h\circ f)^2\,d\mu = \int_X h^2\,d\mu$. Hence $\int_X (h\circ f - h)^2\,d\mu = 0$. By Proposition 1.7.4, $h = c$ μ-a.e.; since $1 = \int_X h\,d\mu$, $c = 1$ and $\nu = \mu$. The convex identity then gives $\rho = \mu$.

Structural Commentary: Invariant sets split measures; invariant densities test splits.
Key Insight: Extreme invariant measures are exactly ergodic measures. □

Corollary 1.7.8. *Existence of an Ergodic Invariant Measure*
Let $X \neq \varnothing$ be compact metric, let $f \in C(X,X)$, let $\mathcal{P}(X)$ denote the Borel probability measures on X, and set $\mathcal{I}_f(X) := \{\mu \in \mathcal{P}(X) : f_*\mu = \mu\}$. Then $\exists \mu \in \mathcal{I}_f(X)$ such that f is μ-ergodic.

Proof. By Theorem 1.7.6, $\mathcal{I}_f(X) \neq \varnothing$, and it is convex. Let a net $(\mu_\alpha) \subseteq \mathcal{I}_f(X)$ converge weak-* among finite signed Radon measures to μ. Positivity on $C(X)_+$ and the identity $\int_X 1\,d\mu = 1$ pass to the limit, so $\mu \in \mathcal{P}(X)$. For $g \in C(X)$, $\int_X g\circ f\,d\mu = \lim_\alpha \int_X g\circ f\,d\mu_\alpha = \lim_\alpha \int_X g\,d\mu_\alpha = \int_X g\,d\mu$; hence $f_*\mu = \mu$ by Theorem 5.2.13. Thus $\mathcal{I}_f(X)$ is weak-* closed and $\|\lambda\|_{\mathrm{var}} = 1$ $\forall \lambda \in \mathcal{I}_f(X)$, so it is weak-* compact by Corollary 6.4.6.

By Theorem 9.2.10, choose a dense sequence $(g_j)_{j\ge1}$ in $C(X,\mathbb{R})$. Set $K_0 := \mathcal{I}_f(X)$ and, recursively, $a_j := \max\{\int_X g_j\,d\lambda : \lambda \in K_{j-1}\}$ and $K_j := \{\lambda \in K_{j-1} : \int_X g_j\,d\lambda = a_j\}$. Weak-* continuity of $\lambda \mapsto \int_X g_j\,d\lambda$ makes K_j nonempty and compact, and affinity makes it a face of K_{j-1}. Hence $K_\infty := \bigcap_{j\ge0} K_j \neq \varnothing$. If $\lambda, \nu \in K_\infty$, their integrals agree on the dense family (g_j), hence on $C(X,\mathbb{R})$. By Theorem 5.2.13, $\lambda = \nu$; write $K_\infty = \{\mu\}$.

If $\mu = \theta\lambda + (1-\theta)\nu$ with $0 < \theta < 1$ and $\lambda, \nu \in \mathcal{I}_f(X)$, the face property applied successively to $K_1, K_2, \dots$ gives $\lambda, \nu \in K_\infty$, so $\lambda = \nu = \mu$. Thus $\mu \in \mathrm{Ext}(\mathcal{I}_f(X))$, and f is μ-ergodic by Proposition 1.7.7. *Structural Commentary*: Compact nested faces isolate one extreme invariant measure.
Key Insight: Countably many observables isolate an ergodic invariant measure. □

Theorem 1.7.9. *Asymptotic Ergodic Averages*
Let $(X, \mathcal{M}, \mu)$ satisfy $0 < \mu(X) < +\infty$, and let $T : X \to X$ be measurable with $T_*\mu = \mu$. For $g \in L^1(X,\mu)$, set $U_T g := g \circ T$ and $A_n g := \frac{1}{n}\sum_{k=0}^{n-1} U_T^k g$. Then T is ergodic iff $A_n g \to \mu(X)^{-1}\int_X g\,d\mu$ μ-a.e. $\forall g \in L^1(X,\mu)$.

Proof.

(⇒) First establish the maximal inequality. For $v \in L^1(X,\mu)$, set $S_0 v := 0$ and $S_n v := \sum_{k=0}^{n-1} v\circ T^k$. For $N \ge 1$, set $M_N v := \max_{0\le n\le N} S_n v$ and $E_N := \{M_N v > 0\}$. On E_N, $v + M_N v \circ T \ge M_N v$. Since $M_N v = 0$ on E_N^c, $\int_{E_N} M_N v\,d\mu = \int_X M_N v\,d\mu$. Since $M_N v \circ T \ge 0$ and $T_*\mu = \mu$, $\int_{E_N} M_N v \circ T\,d\mu \le \int_X M_N v\,d\mu$. Integrating the pointwise inequality therefore gives $\int_{E_N} v\,d\mu \ge 0$. Since $E_N \uparrow E := \{\sup_{n\ge1} S_n v > 0\}$, dominated convergence gives $\int_E v\,d\mu \ge 0$.

Fix $g \in L^1(X,\mu)$. For $\varepsilon > 0$, $\sum_{n\ge1} \mu(|g\circ T^n| > \varepsilon n) = \sum_{n\ge1} \mu(|g| > \varepsilon n) \le \varepsilon^{-1}\int_X |g|\,d\mu$. By Theorem 5.1.6, applied with $\varepsilon = 1/m$, $m \ge 1$, $g\circ T^n/n \to 0$ μ-a.e. Hence $A_n g \circ T - A_n g = (g\circ T^n - g)/n \to 0$ μ-a.e., so $\overline{g} := \limsup_n A_n g$ and $\underline{g} := \liminf_n A_n g$ are T-invariant. By Proposition 1.7.4, $\exists a, b \in \overline{\mathbb{R}}$ such that $(\overline{g}, \underline{g}) = (a,b)$ μ-a.e. and $b \le a$.

Set $m := \mu(X)^{-1}\int_X g\,d\mu$. If $a > m$, choose $m < c < a$. Then $\limsup_n A_n(g-c) = a - c > 0$ μ-a.e., so the maximal inequality applied to $v = g - c$ gives $\int_X (g-c)\,d\mu \ge 0$, contradicting $c > m$. Thus $a \le m$. If $b < m$, choose $b < c < m$ and apply the maximal inequality to $v = c - g$; this gives $\int_X (c-g)\,d\mu \ge 0$, contradicting $c < m$. Therefore $b \ge m$, and $b \le a \le m \le b$ proves $A_n g \to m$ μ-a.e.

(⇐) If $A \in \mathcal{M}$ and $\mu(A \triangle T^{-1}A) = 0$, then $A_n \mathbf{1}_A = \mathbf{1}_A$ μ-a.e. $\forall n \ge 1$. The assumed limit gives $\mathbf{1}_A = \mu(A)/\mu(X)$ μ-a.e., so $\mu(A) \in \{0, \mu(X)\}$.

Structural Commentary: Hopf's maximal inequality squeezes invariant limsup and liminf.
Key Insight: Ergodicity collapses both asymptotic envelopes to the space mean. □

Proposition 1.7.10. *Koopman Isometry Characterization*
Let $(X, \mathcal{M}, \mu)$ satisfy $\mu(X) < +\infty$, and let $T : X \to X$ be measurable. Then $T_*\mu = \mu$ iff $\forall p \in [1, +\infty)$ and $\forall g \in L^p(X,\mu)$, the pullback $U_T g := g \circ T$ belongs to $L^p(X,\mu)$ and $\|U_T g\|_p = \|g\|_p$.

Proof.

(⇒) If $T_*\mu = \mu$, Proposition 1.7.2 applied to $|g|^p$ gives $\|U_T g\|_p^p = \int_X |g\circ T|^p\,d\mu = \int_X |g|^p\,d\mu = \|g\|_p^p$ $\forall p \in [1,+\infty)$ and $g \in L^p(X,\mu)$.
(⇐) Apply the $p = 1$ identity to $g = \mathbf{1}_E$, $E \in \mathcal{M}$: $(T_*\mu)(E) = \|\mathbf{1}_E \circ T\|_1 = \|\mathbf{1}_E\|_1 = \mu(E)$.

Structural Commentary: Test pullback norms on powers and indicators.
Key Insight: Koopman isometries encode measure preservation. □

Proposition 1.7.11. *Linear Preservation of Lebesgue Measure*
Let $n \ge 1$ and $T \in \mathcal{L}(\mathbb{R}^n)$. Then:

(i) $T_* m_n = m_n \iff T \in GL_n(\mathbb{R})$ and $|\det T| = 1$.
(ii) For $D := \mathrm{diag}(2, \frac{1}{2}) \in GL_2(\mathbb{R})$, $D_* m_2 = m_2$.

Proof.

(i) If $T_* m_n = m_n$ and T were singular, then $L := T(\mathbb{R}^n)$ would satisfy $m_n(L) = 0$ by Corollary 2.3.6, whereas $(T_* m_n)(L) = m_n(T^{-1}L) = m_n(\mathbb{R}^n) = +\infty$. Thus $T \in GL_n(\mathbb{R})$. For $E \in \mathcal{L}^n$ with $0 < m_n(E) < +\infty$, by Corollary 2.3.4, $m_n(E) = m_n(T^{-1}E) = |\det T|^{-1} m_n(E)$, so $|\det T| = 1$. Conversely, if $T \in GL_n(\mathbb{R})$ and $|\det T| = 1$, then $(T_* m_n)(E) = m_n(T^{-1}E) = m_n(E)$ $\forall E \in \mathcal{L}^n$ by the same corollary.
(ii) Since $\det D = 1$, claim (i) gives $D_* m_2 = m_2$.

Structural Commentary: Exclude singular range; apply determinant scaling to preimages.
Key Insight: Linear volume preservation is exactly unit determinant. □

Theorem 1.7.12. *Poincaré Recurrence*
Let $(X, \mathcal{M}, \mu)$ satisfy $\mu(X) < +\infty$, let $T : X \to X$ be measurable with $T_*\mu = \mu$, and let $A \in \mathcal{M}$. Set $R_A := \{x \in A : \#\{n \geq 1 : T^n x \in A\} = +\infty\}$. Then $\mu(A \setminus R_A) = 0$.

Proof. Set $E := A \cap \bigcap_{n\geq 1} T^{-n}(A^c)$, so $E \cap T^{-m}A = \varnothing$ $\forall m \geq 1$. If $x \in T^{-i}E \cap T^{-j}E$ with $0 \leq i < j$, then $T^i x \in E \cap T^{-(j-i)}E \subseteq E \cap T^{-(j-i)}A$, a contradiction. Thus $(T^{-k}E)_{k\geq 0}$ is pairwise disjoint. For $N \geq 0$, measure preservation gives $(N+1)\mu(E) = \mu(\bigcup_{k=0}^{N} T^{-k}E) \leq \mu(X)$; letting $N \to \infty$ yields $\mu(E) = 0$.

If $x \in A \setminus R_A$, let $r \geq 0$ be its last visit time to A. Then $T^r x \in E$, so $A \setminus R_A \subseteq \bigcup_{r\geq 0} T^{-r}E$. Countable subadditivity and $\mu(T^{-r}E) = \mu(E) = 0$ give $\mu(A \setminus R_A) = 0$. *Structural Commentary*: The nonreturn set is wandering; finite-return points eventually enter it.
Key Insight: Finite measure forbids positive-measure wandering sets. □

2 Partial Differential Equations

2.1 First-Order PDE and Characteristics

2.2 Elliptic Equations

2.3 Parabolic Equations

2.4 Hyperbolic Equations

3 Weak Solutions and Evolution Equations

3.1 Sobolev Spaces for PDE

Distributions test against smooth compactly supported functions; Sobolev spaces package weak derivatives in normed or Hilbert spaces as a bridge to PDE.

Definition 3.1.1. *Test Functions and Distributions*
Let $\Omega \subseteq \mathbb{R}^n$ be open. A *test function* is a smooth function $\varphi \in C_c^\infty(\Omega)$. A *distribution* on Ω is a continuous linear functional on the test-function space, written $u \in \mathcal{D}'(\Omega)$ and paired as $\langle u, \varphi \rangle$.

Definition 3.1.2. *Weak Derivative*
Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and let α be a multi-index. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is the *weak derivative* $D^\alpha u$ if

$$\int_\Omega u\, D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_\Omega v\, \varphi \, dx \qquad \text{for every } \varphi \in C_c^\infty(\Omega).$$

Definition 3.1.3. *Sobolev Spaces*
For $1 \leq p \leq \infty$ and an integer $k \geq 0$, the Sobolev space $W^{k,p}(\Omega)$ consists of functions $u \in L^p(\Omega)$ whose weak derivatives $D^\alpha u$ lie in $L^p(\Omega)$ for every $|\alpha| \leq k$. The norm is

$$\|u\|_{W^{k,p}} := \sum_{|\alpha|\leq k} \|D^\alpha u\|_{L^p}$$

with the usual supremum modification when $p = \infty$.

3.2 Weak Solutions and Variational Methods

3.3 Evolution Equations

Part XVIII
Numerical and Stochastic Computation

1 Numerical Analysis

1.1 Floating Point and Error Analysis

1.2 Nonlinear Equations and Optimization Basics

Method 1.2.1. *Newton and Secant Iterations*
To solve $f(x) = 0$:

(i) Newton uses tangent-line zero: $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$, $n \geq 1$ when $f'(x_{n-1}) \neq 0$.

(ii) Secant uses two-point slope: $x_n = x_{n-1} - \frac{f(x_{n-1})(x_{n-2}-x_{n-1})}{f(x_{n-2})-f(x_{n-1})}$, $n \geq 2$ when the denominator is nonzero.

These define iterates only; local convergence is proved with Taylor's theorem (Theorem 1.1.25).

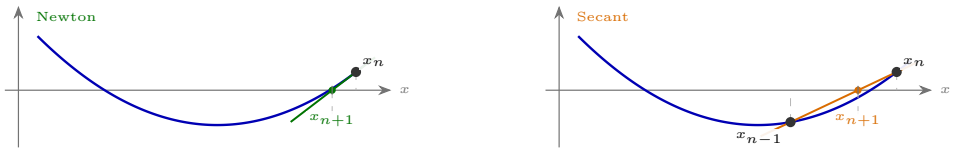

Figure 235: Newton: tangent-line zero. Secant: two-point line zero.

1.3 Interpolation and Approximation

1.4 Numerical Differentiation and Quadrature

1.5 Numerical Stability and Convergence

2 Numerical Differential Systems

2.1 Numerical ODEs

Numerical one-step methods replace the integral evolution of an initial-value problem by finitely many vector-field evaluations on a time grid.

Method 2.1.1. *Explicit Euler Method*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$, $f : U \to \mathbb{R}^n$, $(t_0, x_0) \in U$, and $T > 0$. For $x' = f(t, x)$, $x(t_0) = x_0$, choose $N \in \mathbb{N}$ with $N \geq 1$, set $h := T/N$ and $t_m := t_0 + mh$.

(i) Initialize $x_0^{(h)} := x_0$.

(ii) For $0 \leq m < N$ with $(t_m, x_m^{(h)}) \in U$, compute $x_{m+1}^{(h)} := x_m^{(h)} + hf(t_m, x_m^{(h)})$.

(iii) On each completed mesh interval, define the continuous Euler polygon by $x_h(t) := x_m^{(h)} + (t - t_m)f(t_m, x_m^{(h)})$ for $t_m \leq t \leq t_{m+1}$.

Thus $x_h' = f(t_m, x_m^{(h)})$ on each open mesh interval, while the nodes satisfy the explicit recurrence in (ii).

Theorem 2.1.2. *Uniform Error of the Euler Polygon*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, $(t_0, x_0) \in U$, and choose $T, \delta > 0$ with $V := [t_0, t_0 + T] \times \overline{B}_\delta(x_0) \subseteq U$.
Set $M := \max_V \|f\|$, assume $MT \leq \delta$, and assume $\exists L \geq 0$ such that $\|f(t, x) - f(t, y)\| \leq L \|x - y\|$ $\forall (t, x), (t, y) \in V$.
Let $\phi \in C^1([t_0, t_0 + T], \mathbb{R}^n)$ satisfy $\phi' = f(t, \phi)$, $\phi(t_0) = x_0$, and $(t, \phi(t)) \in V$ $\forall t \in [t_0, t_0 + T]$.
For $N \in \mathbb{N}$ with $N \geq 1$, set $h := T/N$, $t_m := t_0 + mh$, $x_0^{(h)} := x_0$, $x_{m+1}^{(h)} := x_m^{(h)} + hf(t_m, x_m^{(h)})$, and $x_h(t) := x_m^{(h)} + (t - t_m)f(t_m, x_m^{(h)})$ for $t_m \leq t \leq t_{m+1}$.
Define $\Delta(h) := \sup\{\|f(s, y) - f(t, x)\| : (s, y), (t, x) \in V,\ |s - t| \leq h,\ \|y - x\| \leq Mh\}$. Then:

(i) $(t, x_h(t)) \in V$ $\forall t \in [t_0, t_0 + T]$, and $\|x_h(t) - x_h(s)\| \leq M |t - s|$ $\forall s, t \in [t_m, t_{m+1}]$, $0 \leq m < N$.
(ii) With $h = T/N$, as $N \to \infty$, $\Delta(h) \to 0$ and $\|x_h - \phi\|_\infty \leq Te^{LT}\Delta(h)$; hence $x_h \to \phi$ uniformly.
(iii) If $\exists L_0 \geq 0$ such that $\|f(t, x) - f(s, x)\| \leq L_0 |t - s|$ $\forall (t, x), (s, x) \in V$, then $\Delta(h) \leq (L_0 + LM)h$ and $\|x_h - \phi\|_\infty \leq Te^{LT}(L_0 + LM)h$.

Proof.

(i) The initial graph point (t_0, x_0) lies in V. If $\left\|x_m^{(h)} - x_0\right\| \leq mhM$, then $(t_m, x_m^{(h)}) \in V$ and $\left\|x_{m+1}^{(h)} - x_0\right\| \leq \left\|x_m^{(h)} - x_0\right\| + h\left\|f(t_m, x_m^{(h)})\right\| \leq (m + 1)hM$.
Thus induction gives $\left\|x_m^{(h)} - x_0\right\| \leq mhM$. For $t \in [t_m, t_{m+1}]$, $\|x_h(t) - x_0\| \leq M(t - t_0) \leq MT \leq \delta$. Hence the graph lies in V, and the affine slope gives the meshwise Lipschitz bound.

(ii) Let $\tau_h(t) := t_m$ for $t \in [t_m, t_{m+1})$. The Euler polygon satisfies $x_h(t) = x_0 + \int_{t_0}^t f(\tau_h(s), x_h(\tau_h(s)))\, ds$.
Uniform continuity of $f|_V$ gives $\Delta(h) \to 0$. With $e(t) := \|x_h(t) - \phi(t)\|$, item (i) gives $\|x_h(s) - x_h(\tau_h(s))\| \leq Mh$, so $e(t) \leq \int_{t_0}^t (Le(s) + \Delta(h))\, ds$.
By Lemma 1.1.20(ii), $e(t) \leq (t - t_0)e^{L(t-t_0)}\Delta(h) \leq Te^{LT}\Delta(h)$.

(iii) For pairs in the definition of $\Delta(h)$, $\|f(s, y) - f(t, x)\| \leq L \|y - x\| + L_0 |s - t| \leq (LM + L_0)h$. Combine this with (ii).

Structural Commentary: Consistency supplies $\Delta(h)$; Gronwall propagates the local defect.
Key Insight: Uniform continuity is enough for convergence; time-Lipschitz control gives first order. □

Remark 2.1.3. *Euler Step-Doubling Control*
From a node (t_m, x_m), compare the full-step prediction $z_h := x_m + hf(t_m, x_m)$ with the two-half-step prediction $z_{1/2} := x_m + \frac{h}{2}f(t_m, x_m)$, $z_{2/2} := z_{1/2} + \frac{h}{2}f(t_m + h/2, z_{1/2})$. If $\|z_h - z_{2/2}\|$ exceeds a prescribed tolerance, reduce h and repeat the step. This is a practical error indicator rather than an a priori error bound.

Method 2.1.4. *Classical Fourth-Order Runge–Kutta Method*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$, $f : U \to \mathbb{R}^n$, $h > 0$, and $(t_m, x_m) \in U$, where x_m approximates a solution of $x' = f(t, x)$ at t_m. Set $t_{m+1} := t_m + h$ and assume every displayed argument of f below lies in U.

(i) Compute $k_{1,m} := f(t_m, x_m)$ and $k_{2,m} := f(t_m + h/2, x_m + hk_{1,m}/2)$.
(ii) Compute $k_{3,m} := f(t_m + h/2, x_m + hk_{2,m}/2)$ and $k_{4,m} := f(t_m + h, x_m + hk_{3,m})$.
(iii) Update $x_{m+1} := x_m + \frac{h}{6}(k_{1,m} + 2k_{2,m} + 2k_{3,m} + k_{4,m})$.

The four stages sample two midpoint predictions and one endpoint prediction before forming the weighted increment.

Method 2.1.5. *Heun's Improved Euler Method*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$, $f : U \to \mathbb{R}^n$, $h > 0$, and $(t_m, x_m) \in U$, where x_m approximates a solution of $x' = f(t, x)$ at t_m. Set $t_{m+1} := t_m + h$ and assume the predicted endpoint used below lies in U.

(i) Predict the endpoint by $y_m := x_m + hf(t_m, x_m)$.
(ii) Correct with the averaged slopes: $x_{m+1} := x_m + \frac{h}{2}[f(t_m, x_m) + f(t_{m+1}, y_m)]$.

Thus Heun replaces Euler's single left-endpoint slope by the average of the initial slope and its predicted endpoint value.

Proposition 2.1.6. *Heun-Type One-Step Error*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C^2(U, \mathbb{R}^n)$, $I \subseteq \mathbb{R}$ an open interval, $t_0 \in I$, and $\phi \in C^1(I, \mathbb{R}^n)$ satisfy $(t, \phi(t)) \in U$ and $\phi'(t) = f(t, \phi(t))$ $\forall t \in I$, with $\phi(t_0) = x_0$.
Set $f_0 := f(t_0, x_0)$. Then $\exists h_* > 0$ such that, $\forall h \in (0, h_*)$, setting $t_1 := t_0 + h$ and $y_0 := x_0 + hf_0$ gives $t_1 \in I$ and $(t_1, y_0), (t_1, x_0), (t_0, y_0) \in U$. As $h \downarrow 0$:

(i) For $H_h := x_0 + \frac{h}{2}[f(t_0, x_0) + f(t_1, y_0)]$, $\|\phi(t_1) - H_h\| = O(h^3)$.
(ii) For $\widetilde{H}_h := x_0 + \frac{h}{2}[f(t_1, x_0) + f(t_0, y_0)]$, $\left\|\phi(t_1) - \widetilde{H}_h\right\| = O(h^3)$.

Proof. Set $G_0 := \partial_t f(t_0, x_0) + D_x f(t_0, x_0) f_0$. By Lemma 1.1.19, $\phi \in C^3$ locally, so $\phi(t_1) = x_0 + hf_0 + \frac{h^2}{2}G_0 + O(h^3)$.

(i) Taylor expansion of f at (t_0, x_0) gives $f(t_1, y_0) = f_0 + h\partial_t f(t_0, x_0) + hD_x f(t_0, x_0) f_0 + O(h^2) = f_0 + hG_0 + O(h^2)$.
Hence $H_h = x_0 + hf_0 + h^2 G_0/2 + O(h^3)$, which matches the expansion of $\phi(t_1)$.
(ii) Separately, $f(t_1, x_0) = f_0 + h\partial_t f(t_0, x_0) + O(h^2)$ and $f(t_0, y_0) = f_0 + hD_x f(t_0, x_0) f_0 + O(h^2)$.
Thus $\widetilde{H}_h = x_0 + hf_0 + h^2 G_0/2 + O(h^3)$, proving the second estimate.

Structural Commentary: Expand the exact flow and both predicted slopes through second order.
Key Insight: Averaging reconstructs the quadratic term of the exact solution. □

Example 2.1.7. *Euler and RK4 for Exponential Growth*
For $x' = x$, $x(0) = 1$, and $h = 0.1$, Euler gives $x_m = (1 + h)^m$, hence $x_{10} = 1.1^{10} \approx 2.5937424601$.
Classical RK4 gives $x_{m+1} = R(h)x_m$ with $R(h) = 1 + h + h^2/2 + h^3/6 + h^4/24$, so $x_{10} = R(0.1)^{10} \approx 2.7182797441$.
Since $x(1) = e \approx 2.7182818285$, the respective absolute errors at $t = 1$ are approximately 1.2454×10^{-1} and 2.0843×10^{-6}.

2.2 Geometric Numerical Integration

2.3 Finite Difference PDE Methods

2.4 Finite Element Methods

2.5 Numerical SDEs

2.6 Computational Stability Across ODE/PDE/SDE

3 Monte Carlo and Stochastic Computation

3.1 Monte Carlo Foundations

Monte Carlo methods turn integrals into averages of simulated random variables. This is especially useful when the integral is high-dimensional or when the target density is known only up to a normalizing constant.

Method 3.1.1. *Basic Monte Carlo Integration*

(i) Let $I = \int h(x) f(x)\, dx$, where f is a probability density.
(ii) For IID draws $X_1, \ldots, X_N \sim f$, define $\widehat{I}_N = N^{-1} \sum_{i=1}^{N} h(X_i)$; then $\widehat{I}_N \xrightarrow{p} I$ by the law of large numbers.
(iii) With $Y_i = h(X_i)$, set $s^2 = (N-1)^{-1} \sum_{i=1}^{N} (Y_i - \widehat{I}_N)^2$ and $\widehat{\mathrm{se}} = \frac{s}{\sqrt{N}}$.
(iv) An approximate $1 - \alpha$ Monte Carlo interval is $\widehat{I}_N \pm z_{\frac{\alpha}{2}} \widehat{\mathrm{se}}$.

Example 3.1.2. *Monte Carlo Integration from the LLN*
Let $X_1, \ldots, X_n$ be iid Uniform$(0, 1)$ and $f \in L^1([0, 1])$.
The estimator $\widehat{I}_n(f) := n^{-1} \sum_{i=1}^{n} f(X_i)$ satisfies $\widehat{I}_n(f) \xrightarrow{\mathbb{P}} \mathbb{E} f(X_1) = \int_0^1 f(x)\, dx$.
For $f(x) = (2\pi)^{-\frac{1}{2}} e^{-\frac{x^2}{2}}$ on $[0, 1]$, a 1000-point estimate near 0.3417 is consistent with $\int_0^1 f(x)\, dx \approx 0.3413$.

Example 3.1.3. *Integrating on an Interval*
For $X \sim$ Uniform(a, b), $I = \int_a^b h(x)\, dx = (b - a)\mathbb{E}[h(X)]$.
Given IID draws $X_1, \ldots, X_N \sim$ Uniform(a, b), estimate I by $\widehat{I}_N = (b - a)N^{-1} \sum_{i=1}^{N} h(X_i)$.

Example 3.1.4. *Estimating a Normal CDF*

Let $Z \sim N(0,1)$ and $\Phi(x) = \mathbb{P}(Z \le x)$. For IID draws $Z_1, \dots, Z_N \sim N(0,1)$, use $h(s) = \mathbf{1}\{s \le x\}$ and $\widehat{\Phi}(x) = N^{-1} \sum_{i=1}^{N} \mathbf{1}\{Z_i \le x\}$.

Thus $\widehat{\Phi}(x)$ is the empirical fraction of draws not exceeding x.

Method 3.1.5. *Inverse-Transform Generation*

(i) *Exponential*: if $U \sim \text{Uniform}(0,1)$ and $\lambda > 0$, then $T = -\lambda^{-1} \log(1-U) \sim \text{Exp}(\lambda)$. (or $T = -\lambda^{-1} \log U$)

(ii) *Weibull*: if $F(x) = 1 - \exp(-(\frac{x}{\alpha})^{\beta})$ for $x \ge 0$ and $\alpha, \beta > 0$,
then $F^{-1}(u) = \alpha[-\log(1-u)]^{\frac{1}{\beta}}$ and $W = \alpha[-\log(1-U)]^{\frac{1}{\beta}} \sim F$.

(iii) *Integer-valued inverse CDF*: if F is integer-valued and $U \sim \text{Uniform}(0,1)$, set $Y = \inf\{k \in \mathbb{Z} : F(k) \ge U\}$.
Equivalently $F(k-1) < U \le F(k)$. Then $F_Y = F$.

Key Insight: The simulation problem is reduced to solving $u = F(x)$ for x.

Method 3.1.6. *Rejection Sampling*

Let f be a target density on an interval $[a,b]$, possibly with infinite endpoints.

Choose an envelope M such that $M(x) \ge f(x)$ on $[a,b]$ and $A := \int_a^b M(t)\,dt < \infty$. Let $m(x) := \frac{M(x)}{A}$.

(i) Generate T with density m.

(ii) Generate $U \sim \text{Uniform}(0,1)$ independently of T.

(iii) Accept $X = T$ if $U \le \frac{f(T)}{M(T)}$; otherwise reject T and repeat.

Key Insight: Sample under an easy envelope and keep only points that fall below the target curve.

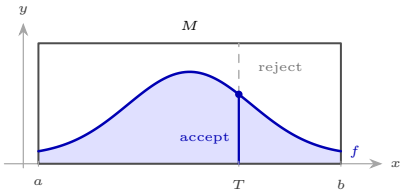

Figure 236: Rejection sample accepts when the vertical draw lies below $f(T)$.

Proposition 3.1.7. *Rejection Sampling Correctness*

Let f be a density, let $M \ge f$ with $A := \int M(x)\,dx < \infty$, and set $m := \frac{M}{A}$. If $T \sim m$, $U \sim \text{Uniform}(0,1)$ independently, and $\mathcal{A} := \{U \le \frac{f(T)}{M(T)}\}$, then $\mathbb{P}(\mathcal{A}) = \frac{1}{A}$ and $T \mid \mathcal{A}$ has density f.

Proof. For measurable B, $\mathbb{P}(T \in B, \mathcal{A}) = \int_B \frac{m(x)f(x)}{M(x)}\,dx = A^{-1} \int_B f(x)\,dx$.

Taking $B = \mathbb{R}$ gives $\mathbb{P}(\mathcal{A}) = A^{-1}$, and hence $\mathbb{P}(T \in B \mid \mathcal{A}) = \int_B f(x)\,dx$.

Structural Commentary: Acceptance reweights the envelope m back to f.

Key Insight: Conditional on acceptance, the normalized accepted density is exactly the target. □

3.2 Variance Reduction

Method 3.2.1. *Importance Sampling*

(i) For $I = \int h(x)f(x)\,dx$, choose a tractable proposal density g with $g(x) > 0$ wherever $h(x)f(x) \ne 0$.

(ii) Then $I = \int h(x)f(x)g(x)^{-1}g(x)\,dx = \mathbb{E}_g\left[\frac{h(X)f(X)}{g(X)}\right]$.

(iii) For IID draws $X_1, \dots, X_N \sim g$, use $\widehat{I}_N = N^{-1} \sum_{i=1}^{N} \frac{h(X_i)f(X_i)}{g(X_i)}$.

Remark 3.2.2. *Choosing the Proposal*

The importance weights have large, possibly infinite, variance when g is too small on regions where $|h|\,f$ is large.
Choose g to approximate the shape of $|h|\,f$ while giving it tails at least as thick as the target integrand.

Theorem 3.2.3. *Optimal Importance Density*

Let $I = \int h(x)f(x)\,dx$ and $A = \int |h(x)|\,f(x)\,dx \in (0,\infty)$.

For proposal densities g with $g(x) > 0$ wherever $|h(x)|\,f(x) > 0$, define $W_g = \frac{h(X)f(X)}{g(X)}$ for $X \sim g$.

The variance $\text{Var}_g(W_g)$ is minimised by $g^*(x) = \frac{|h(x)|f(x)}{A}$, and the minimum variance is $A^2 - I^2$.

Proof. For each admissible g, $\mathbb{E}_g W_g = I$, so $\text{Var}_g(W_g) = M(g) - I^2$, where $M(g) = \mathbb{E}_g(W_g^2) = \int \frac{h(x)^2 f(x)^2}{g(x)}\,dx$.

For any admissible g, Cauchy-Schwarz gives $A^2 = (\int |h(x)|\,f(x)g(x)^{-\frac{1}{2}} g(x)^{\frac{1}{2}}\,dx)^2 \le M(g) \int g(x)\,dx = M(g)$.

Equality holds precisely when $|h(x)|\,f(x)g(x)^{-\frac{1}{2}} = c\,g(x)^{\frac{1}{2}}$ a.e., equivalently $g(x) \propto |h(x)|\,f(x)$.

Normalization gives $g^*(x) = \frac{|h(x)|f(x)}{A}$, hence $M(g) \ge M(g^*) = A^2$ and $\text{Var}_g(W_g) \ge A^2 - I^2 = \text{Var}_{g^*}(W_{g^*})$.

Structural Commentary: Fixed mean reduces variance to second-moment minimization.

Key Insight: Cauchy-Schwarz is sharp exactly when the proposal is proportional to $|h|\,f$. □

3.3 Markov Chain Monte Carlo

Monte Carlo integration in Method 3.1.1 uses independent draws from the target density. Markov chain Monte Carlo instead constructs a Markov chain $X_0, X_1, \dots$ whose stationary density is the target f. When the chain mixes well, sample averages from the chain are used as approximations to expectations under f.

Method 3.3.1. *Metropolis-Hastings*
Let f be a target density available up to proportionality, and let $q(\cdot \mid x)$ be a tractable proposal density.
Choose X_0 arbitrarily; given $X_i = x$, define X_{i+1} as follows.

(i) Draw $Y \sim q(\cdot \mid x)$.
(ii) Set $r(x, Y) = \min\left\{\frac{f(Y)q(x|Y)}{f(x)q(Y|x)}, 1\right\}$.
(iii) Set $X_{i+1} = Y$ with probability $r(x, Y)$, and set $X_{i+1} = x$ otherwise.

Key Insight: Only ratios of f enter r, so unknown normalising constants cancel.

Proposition 3.3.2. *Detailed Balance for Metropolis-Hastings*
Let $\Phi(x, y) = f(x)q(y \mid x)$ and assume $\Phi(x, y) > 0$ iff $\Phi(y, x) > 0$.
For $x \neq y$, define $\alpha(x, y) = \min\left\{\frac{\Phi(y,x)}{\Phi(x,y)}, 1\right\}$ when $\Phi(x, y) > 0$ and $\alpha(x, y) = 0$ otherwise.
Set $p(x, y) = q(y \mid x)\alpha(x, y)$. Then $f(x)p(x, y) = f(y)p(y, x)$ for $x \neq y$.

Proof. If $\Phi(x, y) = 0$, the support condition gives $\Phi(y, x) = 0$, hence $p(x, y) = p(y, x) = 0$.
If $\Phi(x, y) > 0$, then $f(x)p(x, y) = \Phi(x, y) \min\left\{\frac{\Phi(y,x)}{\Phi(x,y)}, 1\right\} = \min\{\Phi(x, y), \Phi(y, x)\}$.
Interchanging x, y gives the same expression for $f(y)p(y, x)$, QED.
Structural Commentary: The acceptance rule symmetrizes the weighted proposal flow.
Key Insight: Both directions reduce to the same minimum of the two proposal fluxes. □

Method 3.3.3. *Random-Walk Metropolis-Hastings*

(i) Use the proposal $Y = X_i + \varepsilon_i$, where ε_i has a symmetric density.
(ii) Then $q(y \mid x) = q(x \mid y)$, so the acceptance probability is $r(x, y) = \min\left\{\frac{f(y)}{f(x)}, 1\right\}$.
(iii) For $\varepsilon_i \sim N(0, b^2)$, small b gives slow local moves, large b gives frequent rejection. Tune b for adequate mixing.

Example 3.3.4. *Cauchy Target*
For Cauchy target $f(x) = \frac{1}{\pi(1+x^2)}$ and symmetric random-walk proposal, $r(x, y) = \min\left\{\frac{1+x^2}{1+y^2}, 1\right\}$.
The proposal scale changes mixing efficiency, not the stationary target.

Method 3.3.5. *Independence Metropolis-Hastings*

(i) Use a state-independent proposal $Y \sim g$, so $q(y \mid x) = g(y)$.
(ii) The acceptance probability is $r(x, y) = \min\{\frac{f(y)g(x)}{f(x)g(y)}, 1\}$.
(iii) Choose g as a tractable approximation to the target with sufficient mass on important regions.

Method 3.3.6. *Gibbs Sampling*

(i) Let (X, Y) have joint density $f_{X,Y}$ with tractable conditionals $f_{X|Y}$ and $f_{Y|X}$.
(ii) Starting from (X_0, Y_0), iterate $X_{n+1} \sim f_{X|Y}(\cdot \mid Y_n)$ and $Y_{n+1} \sim f_{Y|X}(\cdot \mid X_{n+1})$.
(iii) In higher dimension, update one coordinate or block at a time from its full conditional.

Method 3.3.7. *Metropolis within Gibbs*

(i) If $f_{X_j|X_{-j}}$ is not directly simulable, update X_j by Metropolis-Hastings conditional on X_{-j}.
(ii) In the bivariate case, propose $Z \sim q(\cdot \mid X_n)$ and accept $X_{n+1} = Z$ with probability $\min\{\frac{f(Z,Y_n)q(X_n|Z)}{f(X_n,Y_n)q(Z|X_n)}, 1\}$.
(iii) Then update Y analogously, conditional on the newly updated X.

3.4 Sequential Monte Carlo

3.5 Rare-Event Simulation

Example 3.5.1. *Rare Tail Probability*
For $Z \sim N(0, 1)$, write $I = \mathbb{P}(Z > 3) = \int \mathbf{1}\{x > 3\} f(x)\, dx$.
If g is centered near the right tail and $X_1, \dots, X_N \sim g$, estimate I by $\widehat{I}_N = N^{-1} \sum_{i=1}^{N} \frac{\mathbf{1}\{X_i > 3\} f(X_i)}{g(X_i)}$.

3.6 Stochastic Optimization

Definition 3.6.1. *Online Estimator*
Let $(X_t)_{t \geq 1}$ be observations in $\mathcal{X}$, let $\widehat{\theta}_0 \in \Theta \subseteq \mathbb{R}^d$, let $(a_t)_{t \geq 0}$ be positive step sizes, and let $H : \Theta \times \mathcal{X} \to \mathbb{R}^d$.
$\widehat{\theta}_{t+1} := \widehat{\theta}_t + a_t H(\widehat{\theta}_t, X_{t+1})$ for $t \geq 0$.
Structural Commentary: $(\widehat{\theta}_t)$ is adapted to $\mathcal{F}_t := \sigma(\widehat{\theta}_0, X_1, \dots, X_t)$.
Key Insight: Under stationarity, analyze the recursion through $h(\theta) := \mathbb{E}H(\theta, X)$.

Method 3.6.2. *Minibatch Stochastic Gradient Descent*
Let $J_n(\theta) = n^{-1} \sum_{i=1}^{n} L(f_\theta(x_i), y_i)$, $|B_k| = b$, and $\eta_k > 0$.

(i) Sample B_k uniformly.
(ii) Compute $g_k = b^{-1} \sum_{i \in B_k} \nabla_\theta L(f_{\theta_k}(x_i), y_i)$.
(iii) Update $\theta_{k+1} = \theta_k - \eta_k g_k$.

Uniform sampling gives $\mathbb{E}[g_k \mid \theta_k] = \nabla J_n(\theta_k)$; larger b trades lower noise for higher cost.

Definition 3.6.3. *Momentum and Adaptive Methods*
Let g_k denote the stochastic gradient at θ_k; $\odot$, division, and square roots are coordinatewise.

(i) *Momentum*: $v_k = \mu v_{k-1} + g_k$ and $\theta_{k+1} = \theta_k - \eta_k v_k$, with $0 \le \mu < 1$.
(ii) *AdaGrad*: $r_k = r_{k-1} + g_k \odot g_k$ and $\theta_{k+1} = \theta_k - \frac{\eta_k g_k}{\sqrt{r_k}+\varepsilon}$.
(iii) *RMSProp*: $r_k = \rho r_{k-1} + (1-\rho) g_k \odot g_k$ and $\theta_{k+1} = \theta_k - \frac{\eta_k g_k}{\sqrt{r_k}+\varepsilon}$, with $0 < \rho < 1$.
(iv) *Adam*: update (m_k, r_k) as exponential averages of $(g_k, g_k \odot g_k)$,
set $(\widehat{m}_k, \widehat{r}_k) = (\frac{m_k}{1-\beta_1^k}, \frac{r_k}{1-\beta_2^k})$, and $\theta_{k+1} = \theta_k - \frac{\eta_k \widehat{m}_k}{\sqrt{\widehat{r}_k}+\varepsilon}$.

Part XIX
Statistical Inference and Time Series

Statistical inference turns probabilistic models into procedures for learning from data. The probability, measure-theoretic, stochastic-process, optimization, and Monte Carlo material developed in Parts 11–14 is treated here as prerequisite structure.

1 Statistical Models and Experiments

1.1 Statistical Experiments and Likelihoods

A statistical problem starts with observable data and a family of probability laws that could have generated it. The model is not the data itself; it is the mathematical language in which uncertainty about the data-generating mechanism is expressed.

Definition 1.1.1. *Statistical Experiment and Model*

(i) A *statistical experiment* is $(\mathcal{X}, \mathcal{A}, \mathcal{P})$, where $(\mathcal{X}, \mathcal{A})$ is measurable and $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is a family of probability measures on $(\mathcal{X}, \mathcal{A})$.
(ii) Θ is the *parameter space*, $X \in \mathcal{X}$ is the *sample*, and $\mathcal{P}$ is the *statistical model*.
(iii) The model is *well specified* if the true law $P_0 \in \mathcal{P}$.
(iv) The model is *misspecified* if $P_0 \notin \mathcal{P}$.

Definition 1.1.2. *Identifiability and Functionals*

(i) The parameterization is *identifiable* if $P_\theta = P_{\theta'}$ implies $\theta = \theta'$.
(ii) A *statistical functional* is a map $T : \mathcal{P} \to \mathcal{T}$, such as $T(P) = \int x\, dP(x)$ or $T(P) = \text{median}(P)$.
(iii) In a parametric model, a *target parameter* may be a function $\psi(\theta)$ rather than the full parameter θ.

Key Insight: Estimation should target a law or functional that the experiment can actually distinguish.

Definition 1.1.3. *Dominated Models, Likelihood, Score, and Information*

(i) A model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is *dominated* by μ if $P_\theta \ll \mu$ for all $\theta \in \Theta$; write $p_\theta := \frac{dP_\theta}{d\mu}$.
(ii) Given x, the *likelihood* and *log-likelihood* are $L_x(\theta) := p_\theta(x)$ and $\ell_x(\theta) := \log p_\theta(x)$ when $p_\theta(x) > 0$.
(iii) If $\Theta \subseteq \mathbb{R}^d$ and differentiation is valid, the *score* is $S_\theta(X) := \nabla_\theta \ell_X(\theta)$.
(iv) The *Fisher information* is $I(\theta) := \mathbb{E}_\theta[S_\theta(X) S_\theta(X)^\top]$.

Definition 1.1.4. *Maximum Likelihood Estimator*

For likelihood $L_X(\theta)$ and log-likelihood $\ell_X(\theta)$, a *maximum likelihood estimator* is any measurable $\widehat{\theta}_n(X) \in \arg\max_{\theta \in \Theta} L_X(\theta) = \arg\max_{\theta \in \Theta} \ell_X(\theta)$; equivalently, $\widehat{\theta}_n \in \arg\min_{\theta \in \Theta} [-\ell_X(\theta)]$.

Example 1.1.5. *Bernoulli and Normal Likelihoods*

Let $X_1, \dots, X_n$ be independent observations.

(i) If $X_i \sim \text{Bernoulli}(p)$, then $L_x(p) = p^s (1-p)^{n-s}$, where $s := \sum_{i=1}^n x_i$.

(ii) If $X_i \sim N(\mu, \sigma^2)$ and σ^2 is known, then $\ell_x(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2$, maximized at $\widehat{\mu} = \overline{x}$.

Example 1.1.6. *Multinomial Cell Probabilities*

Let $X = (X_1, \dots, X_m)$ have total count n.

(i) If $X \sim \text{Multinomial}(n; p_1, \dots, p_m)$ and $\sum_{j=1}^m p_j = 1$, then $L_x(p) \propto \prod_{j=1}^m p_j^{x_j}$ and $\widehat{p}_j = \frac{x_j}{n}$.

(ii) If $p_j = p_j(\theta)$, then $L_x(\theta) \propto \prod_{j=1}^m p_j(\theta)^{x_j}$ and $\widehat{\theta} \in \arg\max_\theta \sum_{j=1}^m x_j \log p_j(\theta)$.

Remark 1.1.7. *Likelihood as Relative Evidence*

For fixed data x, the likelihood is a function of θ, not a probability distribution on Θ.
Ratios $\frac{L_x(\theta_1)}{L_x(\theta_0)}$ compare how well two model laws explain the same observation.

1.2 Sufficiency, Exponential Families, and Local Asymptotics

The shape of a model determines which summaries of the data preserve information and which approximations become valid in large samples.

Definition 1.2.1. *Statistics, Sufficiency, and Ancillarity*

(i) A *statistic* is a measurable map $T : \mathcal{X} \to \mathcal{T}$.
(ii) T is *sufficient* if $\mathcal{L}_\theta(X \mid T)$ is independent of θ.
(iii) T is *ancillary* if $\mathcal{L}_\theta(T)$ is independent of θ.

Theorem 1.2.2. *Factorization Criterion*

Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be dominated by μ, with $p_\theta := \frac{dP_\theta}{d\mu}$. A statistic $T : \mathcal{X} \to \mathcal{T}$ is sufficient iff there exist measurable $g_\theta : \mathcal{T} \to [0, \infty)$ and $h : \mathcal{X} \to [0, \infty)$ such that $p_\theta(x) = g_\theta(T(x))h(x)$ μ-a.e. for all θ.

Proof.

(⇒) If T is sufficient, write $\nu_\theta := \mathcal{L}_\theta(T)$ and choose a θ-free kernel $K_t = \mathcal{L}_\theta(X \mid T = t)$, so $P_\theta(A) = \int K_t(A)\nu_\theta(dt)$. Choosing densities for ν_θ and for K_t on each fiber gives $p_\theta(x) = g_\theta(T(x))h(x)$.
(⇐) If $p_\theta = g_\theta \circ T\, h$, then on the fiber $T = t$ the factor $g_\theta(t)$ cancels: for $c(t) := \int_{T^{-1}\{t\}} h\, d\mu_t > 0$, $\mathbb{P}_\theta(X \in A \mid T = t) = c(t)^{-1} \int_{A \cap T^{-1}\{t\}} h\, d\mu_t$, independent of θ.

Structural Commentary: Factorisation separates parameter information $g_\theta(T)$ from residual shape h.
Key Insight: Sufficiency keeps exactly the likelihood's θ-dependent part. □

Example 1.2.3. *Bernoulli and Normal Sufficient Statistics*
Let $X_1, \dots, X_n$ be independent and let $x = (x_1, \dots, x_n)$.

(i) If $X_i \sim \text{Bernoulli}(\theta)$, then $p_\theta(x) = \theta^s(1-\theta)^{n-s}$ with $s := \sum_i x_i$.
Hence $T = \sum_i X_i$ is sufficient for θ, and $\mathcal{L}_\theta(X \mid T = t)$ is uniform on $\{x : \sum_i x_i = t\}$.

(ii) If $X_i \sim N(\mu, \sigma^2)$ with (μ, σ^2) unknown, then $p_{\mu,\sigma^2}(x) \propto (\sigma^2)^{-\frac{n}{2}} \exp\{-\frac{\sum_i x_i^2 - 2\mu \sum_i x_i + n\mu^2}{2\sigma^2}\}$.
Hence $T = (\sum_i X_i, \sum_i X_i^2)$ is sufficient for (μ, σ^2).

Corollary 1.2.4. *Likelihood Estimators Use Sufficient Statistics*
Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be dominated with densities p_θ, and let T be sufficient with $p_\theta(x) = g_\theta(T(x))h(x)$.
If $a(t) \in \arg\max_{\theta \in \Theta} g_\theta(t)$ exists, then $\widehat{\theta}(X) := a(T(X))$ is an MLE.

Proof. For fixed x, $L_x(\theta) = p_\theta(x) = h(x)g_\theta(T(x))$. If $h(x) > 0$, then $\arg\max_\theta L_x(\theta) = \arg\max_\theta g_\theta(T(x))$.
If $h(x) = 0$, then $L_x(\theta) = 0$ for all θ. Thus an MLE may be chosen as $a(T(x))$.
Structural Commentary: The θ-free factor $h(x)$ is irrelevant to maximization.
Key Insight: MLEs can depend on data only through a sufficient statistic. □

Theorem 1.2.5. *Rao–Blackwell Improvement*
Let T be sufficient and let $\widehat{\psi} \in L^2(P_\theta)$ estimate a real target $\psi(\theta)$. Choose the θ-free version $\widetilde{\psi}(T) := \mathbb{E}_\theta(\widehat{\psi} \mid T)$.
Then $\mathbb{E}_\theta \widetilde{\psi} = \mathbb{E}_\theta \widehat{\psi}$ and $\mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2 \le \mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2$ for all θ, with equality iff $\widehat{\psi} = \widetilde{\psi}$ P_θ-a.s.

Proof. Sufficiency makes $\mathcal{L}_\theta(X \mid T)$ independent of θ, so $\widetilde{\psi}$ is a T-statistic. Also $\mathbb{E}_\theta \widetilde{\psi} = \mathbb{E}_\theta \widehat{\psi}$.
Since $\mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi} \mid T) = 0$ and $\widetilde{\psi} - \psi(\theta)$ is T-measurable, $\mathbb{E}_\theta[(\widehat{\psi} - \widetilde{\psi})(\widetilde{\psi} - \psi(\theta))] = 0$.
Hence $\mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2 = \mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi})^2 + \mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2 \ge \mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2$, with equality iff $\mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi})^2 = 0$.
Structural Commentary: Rao–Blackwellization projects an estimator onto the $\sigma(T)$-measurable statistics.
Key Insight: Conditioning on a sufficient statistic preserves bias and removes residual noise. □

Definition 1.2.6. *Exponential Family*

(i) A dominated model $\{P_\eta : \eta \in \mathcal{N}\}$ is a *k-parameter exponential family* if $\mathcal{N} \subseteq \mathbb{R}^k$ and $p_\eta(x) = \exp\{\eta^\top T(x) - A(\eta)\}h(x)$.
(ii) η is the *natural parameter*, $T(X)$ is the *natural statistic*, and h is the carrier density.
(iii) $A(\eta)$ is the *log-partition function*: $A(\eta) = \log \int \exp\{\eta^\top T(x)\}h(x)\, d\mu(x)$ when finite.

Proposition 1.2.7. *Moment Identities in Full Exponential Families*
Let $p_\eta(x) = \exp\{\eta^\top T(x) - A(\eta)\}h(x)$ be a full exponential family with open natural parameter space, and assume differentiation under the integral is valid. Then $\nabla A(\eta) = \mathbb{E}_\eta T(X)$ and $\nabla^2 A(\eta) = \text{Cov}_\eta(T(X))$.

Proof. From $1 = \int p_\eta\, d\mu$, differentiation gives $0 = \int (T - \nabla A(\eta)) p_\eta\, d\mu$, hence $\nabla A(\eta) = \mathbb{E}_\eta T(X)$.
For coordinates i, j, differentiating $\mathbb{E}_\eta T_i = \int T_i p_\eta\, d\mu$ gives $\partial_j \mathbb{E}_\eta T_i = \mathbb{E}_\eta[T_i(T_j - \partial_j A(\eta))] = \text{Cov}_\eta(T_i, T_j)$, QED.
Structural Commentary: The log-partition function cumulatively encodes the natural statistic.
Key Insight: First and second derivatives of A give mean and covariance. □

Definition 1.2.8. *Local Asymptotic Normality*
For experiments $\{P_\theta^{(n)} : \theta \in \Theta \subseteq \mathbb{R}^d\}$, set $\theta_{n,h} := \theta_0 + \frac{h}{\sqrt{n}}$.

(i) The sequence is *locally asymptotically normal* at θ_0 with information $I(\theta_0)$ if,
for each fixed $h \in \mathbb{R}^d$, $\log(\frac{dP^{(n)}_{\theta_{n,h}}}{dP^{(n)}_{\theta_0}}) = h^\top \Delta_n - \frac{1}{2} h^\top I(\theta_0) h + o_{P^{(n)}_{\theta_0}}(1)$.

(ii) The *central sequence* satisfies $\Delta_n \Rightarrow N(0, I(\theta_0))$ under $P^{(n)}_{\theta_0}$.

Remark 1.2.9. *Why Local Asymptotics Matter*
LAN replaces local likelihood ratios by the Gaussian quadratic experiment $h^\top \Delta - \frac{1}{2} h^\top I(\theta_0) h$, giving normal approximations for estimators, tests, and confidence regions.

1.3 Design, Randomization, and Observational Comparisons

Comparative inference is only as meaningful as the experiment or study that produced the groups. The same two-sample statistic can estimate a causal treatment effect in a randomised experiment, describe an association in an observational study, or amplify bias if the groups were formed by a hidden mechanism.

Definition 1.3.1. *Controls, Randomisation, and Blinding*

(i) A *controlled experiment* compares $Y \mid T = 1$ and $Y \mid T = 0$ under common conditions.
(ii) *Randomisation* assigns $T \in \{0, 1\}$ so that T is independent of pre-treatment factors C by design.
(iii) *Blinding* conceals T when knowledge of treatment could affect Y or its measurement.

Definition 1.3.2. *Placebo Effect and Confounding*

(i) A *placebo effect* is variation in Y caused by belief in treatment rather than the active mechanism.
(ii) Placebos and blinding separate expectation effects from the causal effect of T.
(iii) A pre-treatment variable C is *confounded* with treatment T for response Y if $\mathcal{L}(C \mid T = 1) \neq \mathcal{L}(C \mid T = 0)$ and C affects Y.

(iv) An *observational study* compares groups without assigning T.
Group differences may reflect confounding rather than the factor of interest.

Example 1.3.3. *Mammary Artery Ligation*
A blinded trial randomised patients after preliminary incisions to ligation ($T = 1$) or sham surgery ($T = 0$). Similar outcomes show that controls remove background improvement; blinding limits placebo, observer effects.

Principle 1.3.4. *Confounding, Blocking, and Pairing*
Randomisation balances unmeasured covariates C in probability; blocking, matching, and pairing condition on known influential C before randomisation. Pairing is one-factor blocking; blocked and factorial designs extend this control to multiple factors.

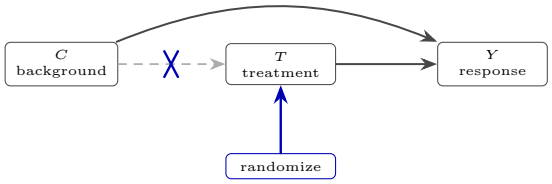

Figure 237: Randomization breaks the confounding path from C to T.

Example 1.3.5. *Uncontrolled Adjustments and Position Effects*
In the Lanarkshire milk experiment, post-randomisation movement between groups made T depend on perceived nourishment C, reintroducing selection bias; weighing children in clothes also confounded measurement with socioeconomic status. In the FD&C Red No. 40 study, cage location was systematic in sex and dose, so position effects confounded treatment comparisons; randomising or blocking cage location would protect inference.

Remark 1.3.6. *Comparative Study Checklist*
Before interpreting group comparisons, record sampling, treatment assignment T, controls, blinding, blocking or pairing factors C, dependence among repeated measurements, and the number of tested comparisons.
These determine whether the analysis supports causality, association, or hypothesis generation.

2 Estimation Theory

2.1 Estimators, Bias, Variance, and Risk

An estimator is a statistic interpreted as an answer to a target question. Its quality depends on the target, the loss function, and the model law under which performance is measured.

Definition 2.1.1. *Estimator, Loss, and Risk*
(i) A target parameter is a map $\psi : \Theta \to \Psi$.
(ii) An *estimator* of $\psi(\theta)$ is a statistic $\delta(X) \in \mathcal{D}$.
(iii) A *loss function* is $L : \Theta \times \mathcal{D} \to [0, \infty]$.
(iv) The *risk* of δ is $R(\theta, \delta) := \mathbb{E}_\theta L(\theta, \delta(X))$.
(v) For squared-error loss $L(\theta, d) = \|d - \psi(\theta)\|^2$, $R(\theta, \delta)$ is mean squared error.

Definition 2.1.2. *Sampling Distribution and Standard Error*
(i) For a statistic $T_n = T_n(X_1, \dots, X_n)$ under P_θ, its *sampling distribution* is $\mathcal{L}_\theta(T_n)$.
(ii) If $T_n \in L^2(P_\theta)$, its *standard error* is $\mathrm{SE}_\theta(T_n) := \sqrt{\mathrm{Var}_\theta(T_n)}$.
(iii) An *estimated standard error* estimates $\mathrm{SE}_\theta(T_n)$, typically by plug-in, asymptotics, or resampling.

Definition 2.1.3. *Bias, Variance, and Consistency*
For real-valued $\widehat{\psi}$, $\mathrm{Bias}_\theta(\widehat{\psi}) := \mathbb{E}_\theta\widehat{\psi} - \psi(\theta)$, and $\mathrm{Var}_\theta(\widehat{\psi}) := \mathbb{E}_\theta(\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi})^2$.
(i) $\widehat{\psi}$ is *unbiased* if $\mathrm{Bias}_\theta(\widehat{\psi}) = 0$ for all θ.
(ii) $\widehat{\psi}_n$ is *consistent* at θ if $\widehat{\psi}_n \to \psi(\theta)$ in P_θ-probability.

Proposition 2.1.4. *Bias-Variance Decomposition*
Let $\widehat{\psi} \in L^2(P_\theta)$ be real-valued. Under squared-error loss, $\mathrm{MSE}_\theta(\widehat{\psi}) := \mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2 = \mathrm{Var}_\theta(\widehat{\psi}) + \mathrm{Bias}_\theta(\widehat{\psi})^2$.

Proof. Write $\widehat{\psi} - \psi(\theta) = (\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi}) + (\mathbb{E}_\theta\widehat{\psi} - \psi(\theta))$. Square, take $\mathbb{E}_\theta$ gives the identity since $\mathbb{E}_\theta(\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi}) = 0$.
Structural Commentary: MSE splits into random fluctuation plus squared systematic error.
Key Insight: Accuracy requires controlling both variance and bias. □

Example 2.1.5. *Sample Mean*
If $X_1, \dots, X_n$ are independent with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$,
then $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$ satisfies $\mathbb{E}\overline{X}_n = \mu$, $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$, and $\overline{X}_n \xrightarrow{P} \mu$.

Definition 2.1.6. *Plug-In and M-Estimators*
(i) A *plug-in estimator* estimates $T(P)$ by $T(\widehat{P}_n)$, where $\widehat{P}_n$ is empirical or fitted.
(ii) An *M-estimator* satisfies $\widehat{\theta}_n \in \arg\min_{\theta \in \Theta} M_n(\theta)$ for an empirical criterion M_n.
(iii) Common choices are $M_n =$ negative log-likelihood, squared loss, or absolute loss.

Definition 2.1.7. *Location M-Estimates*
For data $x_1, \dots, x_n \in \mathbb{R}$, scale $\sigma > 0$, and loss $\Psi : \mathbb{R} \to \mathbb{R}$, $\widehat{\mu} \in \arg\min_{\mu \in \mathbb{R}} \sum_{i=1}^{n} \Psi(\frac{x_i - \mu}{\sigma})$.
Key Insight: Quadratic loss gives $\overline{x}$; absolute or subquadratic loss improves outlier resistance.

2.2 Information Bounds and Asymptotic Efficiency

Information inequalities describe the best possible local accuracy under regular models. Efficient procedures are those whose asymptotic error reaches these bounds.

Lemma 2.2.1. *Two Forms of Fisher Information*
For a regular one-parameter dominated model, let $\ell_\theta(x) := \log p_\theta(x)$ *and* $S_\theta(X) := \partial_\theta \ell_\theta(X)$.
Then $I(\theta) := \mathbb{E}_\theta S_\theta(X)^2 = -\mathbb{E}_\theta \partial_\theta^2 \ell_\theta(X)$.

Proof. Differentiating $\int p_\theta \, d\mu = 1$ gives $\mathbb{E}_\theta S_\theta = 0$.
Differentiating, $0 = \partial_\theta \mathbb{E}_\theta S_\theta = \mathbb{E}_\theta[\partial_\theta S_\theta + S_\theta^2] = \mathbb{E}_\theta[\partial_\theta^2 \ell_\theta + S_\theta^2]$, since $\partial_\theta p_\theta = S_\theta p_\theta$. Thus $I(\theta) = \mathbb{E}_\theta S_\theta^2 = -\mathbb{E}_\theta \partial_\theta^2 \ell_\theta$.
Structural Commentary: Fisher information is both score variance and expected log-likelihood curvature.
Key Insight: Normalisation forces the mean score to vanish, linking slope variance to curvature. □

Theorem 2.2.2. *Cramer–Rao Bound*
Let $X_1, \dots, X_n \sim P_\theta$ *be iid from a regular one-parameter dominated model with one-observation information* $I(\theta) > 0$. *If* $\widehat{\psi} \in L^2(P_\theta)$ *is unbiased for differentiable* $\psi(\theta)$, *then* $\operatorname{Var}_\theta(\widehat{\psi}) \geq \frac{\psi'(\theta)^2}{nI(\theta)}$.

Proof. Let $S_n(\theta) := \partial_\theta \log L_n(\theta) = \sum_{i=1}^n S_\theta(X_i)$. Regularity gives $\mathbb{E}_\theta S_n = 0$ and $\operatorname{Var}_\theta(S_n) = \mathbb{E}_\theta S_n^2 = nI(\theta)$.
Differentiating $\mathbb{E}_\theta \widehat{\psi} = \psi(\theta)$ gives $\psi'(\theta) = \mathbb{E}_\theta[\widehat{\psi} S_n] = \operatorname{Cov}_\theta(\widehat{\psi}, S_n)$.
Thus $\psi'(\theta)^2 \leq \operatorname{Var}_\theta(\widehat{\psi}) \operatorname{Var}_\theta(S_n) = \operatorname{Var}_\theta(\widehat{\psi}) nI(\theta)$ by Cauchy–Schwarz.
Structural Commentary: The score is the direction in which unbiased estimators must covary to track $\psi(\theta)$.
Key Insight: Information lower-bounds variance through Cauchy–Schwarz. □

Example 2.2.3. *Poisson Mean Attains the Bound*
For $X \sim \text{Poisson}(\lambda)$, $\ell_\lambda(X) = X \log \lambda - \lambda - \log(X!)$ and $\partial_\lambda^2 \ell_\lambda(X) = -\frac{X}{\lambda^2}$, so $I(\lambda) = -\mathbb{E}_\lambda \partial_\lambda^2 \ell_\lambda(X) = \frac{1}{\lambda}$.
For iid $X_1, \dots, X_n$, any unbiased estimator T of λ satisfies $\operatorname{Var}_\lambda(T) \geq \frac{\lambda}{n}$.
$\overline{X}$ attains equality since $\mathbb{E}\overline{X} = \lambda$ and $\operatorname{Var}(\overline{X}) = \frac{\lambda}{n}$.

Definition 2.2.4. *Asymptotic Efficiency*
For a scalar regular parameter with information $I(\theta_0) > 0$, an estimator sequence $\widehat{\theta}_n$ is *asymptotically efficient* at θ_0 if $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$, i.e. it attains the inverse Fisher-information variance.

Theorem 2.2.5. *Regular MLE Asymptotic Normality*
Let $X_1, \dots, X_n \sim P_{\theta_0}$ *iid in a regular one-parameter model, with* θ_0 *interior,* $I(\theta_0) > 0$, *and a consistent interior MLE* $\widehat{\theta}_n$. *Then* $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$.

Proof. Let $S_n(\theta) := \partial_\theta \ell_n(\theta)$ and $J_n(\theta) := -n^{-1} \partial_\theta S_n(\theta)$.
Since $S_n(\widehat{\theta}_n) = 0$, Taylor expansion gives $0 = S_n(\theta_0) - nJ_n(\widetilde{\theta}_n)(\widehat{\theta}_n - \theta_0)$ for some $\widetilde{\theta}_n$ between $\widehat{\theta}_n$ and θ_0.
Hence $\sqrt{n}(\widehat{\theta}_n - \theta_0) = J_n(\widetilde{\theta}_n)^{-1} n^{-\frac{1}{2}} S_n(\theta_0)$.
Regularity gives $J_n(\widetilde{\theta}_n) \xrightarrow{P} I(\theta_0)$ and $n^{-\frac{1}{2}} S_n(\theta_0) \Rightarrow N(0, I(\theta_0))$, so Slutsky gives the result.
Structural Commentary: Local likelihood curvature converts score noise into estimator error.
Key Insight: The MLE is asymptotically score divided by information. □

Remark 2.2.6. *Efficiency and Approximation*
Finite-sample efficiency means equality in the Cramer–Rao bound.
Asymptotic efficiency means $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$; finite-sample bias or boundary effects still matter.

2.3 Method of Moments and Parametric Likelihood Examples

Method 2.3.1. *Method of Moments*

(i) Let $X_1, \dots, X_n$ be independent from P_θ, and set $\mu_k(\theta) := \mathbb{E}_\theta X^k$ for $k = 1, \dots, r$.

(ii) If $\mu(\theta) = (\mu_1(\theta), \dots, \mu_r(\theta))$ identifies $\theta \in \Theta \subseteq \mathbb{R}^r$, define $\widehat{\mu}_k := n^{-1} \sum_{i=1}^n X_i^k$.

(iii) The *method of moments* estimator solves $\mu_k(\widehat{\theta}) = \widehat{\mu}_k$ for $k = 1, \dots, r$.

Key Insight: Match population moments to sample moments.

Proposition 2.3.2. *Consistency of Continuous Moment Estimates*
Let $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu_0$ *and let* g *be continuous at* μ_0 *with* $g(\mu_0) = \theta_0$. *Then* $\widehat{\theta}_n := g(\widehat{\mu}_n) \xrightarrow{P_{\theta_0}} \theta_0$. *Thus method-of-moments estimators are consistent whenever* $\widehat{\mu}_n \xrightarrow{P} \mu(\theta_0)$, *inversion map* $g = \mu^{-1}$ *is continuous at* $\mu(\theta_0)$.

Proof. Since $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu_0$ and g is continuous at μ_0, by continuous mapping theorem, $g(\widehat{\mu}_n) \xrightarrow{P_{\theta_0}} g(\mu_0) = \theta_0$.
For MoM, if the required moments are finite, the weak law gives $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu(\theta_0)$.
Structural Commentary: Moment consistency is convergence plus continuous inversion.
Key Insight: Estimate moments first, then pass through g. □

Example 2.3.3. *Moment Estimation for Small Open Probabilities*
Let $N \sim \text{Binomial}(m, p)$ be the number of open channels and let $S = cN$ be total current.
Then $\mathbb{E}S = cmp$, $\operatorname{Var}(S) = c^2 mp(1-p)$, and $\frac{\operatorname{Var}(S)}{\mathbb{E}S} = c(1-p) \approx c$ for small p.
Thus independent measurements estimate c by the empirical ratio $\frac{\widehat{\operatorname{Var}}(S)}{\widehat{\mathbb{E}}S}$, without knowing m.

Example 2.3.4. *Poisson Mean*
Let $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ be independent. Since $\mathbb{E}_\lambda X = \lambda$, $\widehat{\lambda}_{\text{mom}} = \overline{X}$.
Also $\ell(\lambda) = \sum_{i=1}^{n} \{X_i \log \lambda - \lambda - \log(X_i!)\}$ and $\ell'(\lambda) = \frac{S}{\lambda} - n$, where $S = \sum_i X_i$, so $\widehat{\lambda}_{\text{mle}} = \frac{S}{n} = \overline{X}$.
If true value is λ_0, then $S \sim \text{Poisson}(n\lambda_0)$. Hence $\mathbb{E}_{\lambda_0}\widehat{\lambda} = \lambda_0$, $\text{Var}_{\lambda_0}(\widehat{\lambda}) = \frac{\lambda_0}{n}$, $\text{SE}_{\lambda_0}(\widehat{\lambda}) = \sqrt{\frac{\lambda_0}{n}}$, $\widehat{\text{SE}}(\widehat{\lambda}) = \sqrt{\frac{\widehat{\lambda}}{n}}$.

Example 2.3.5. *Negative Binomial Overdispersion*
For overdispersed counts, negative-binomial mass $f(x \mid m, k) = (1 + \frac{m}{k})^{-k}\Gamma(k+x)\{x!\Gamma(k)\}^{-1}(\frac{m}{m+k})^x$, $x \in \mathbb{N}_0$,
has $\mathbb{E}X = m$ and $\text{Var}(X) = m + \frac{m^2}{k}$. By MoM $\widehat{m} = \overline{X}$, $\widehat{k} = \frac{\overline{X}^2}{\widehat{\sigma}^2 - \overline{X}}$ when $\widehat{\sigma}^2 > \overline{X}$.

Example 2.3.6. *Normal Mean and Variance*
For iid $X_i \sim N(\mu, \sigma^2)$, $\mu_1 = \mu$, $\mu_2 = \mu^2 + \sigma^2$. Thus by MoM $\widehat{\mu} = \overline{X}$; $\widehat{\sigma}^2 = n^{-1}\sum_{i=1}^{n} X_i^2 - \overline{X}^2 = n^{-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$.
The normal MLEs are the same; $\widehat{\sigma}^2$ uses denominator n, not the unbiased denominator $n-1$.

Example 2.3.7. *Gamma Shape and Rate*
Let X_i be iid gamma with shape α and rate λ, so $\mathbb{E}X = \frac{\alpha}{\lambda}$ and $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.
With $\widehat{\sigma}^2 := n^{-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$, MoM gives $\widehat{\alpha}_{\text{mom}} = \frac{\overline{X}^2}{\widehat{\sigma}^2}$ and $\widehat{\lambda}_{\text{mom}} = \frac{\overline{X}}{\widehat{\sigma}^2}$.
The log-likelihood is $\ell(\alpha, \lambda) = n\alpha \log \lambda + (\alpha - 1)\sum_i \log X_i - \lambda \sum_i X_i - n \log \Gamma(\alpha)$.
The MLE satisfies $\widehat{\lambda} = \frac{\widehat{\alpha}}{\overline{X}}$ and $\log \widehat{\alpha} - \log \overline{X} + n^{-1}\sum_i \log X_i - \psi(\widehat{\alpha}) = 0$, where $\psi = \frac{\Gamma'}{\Gamma}$.
The MoM estimate is a natural starting value.

Example 2.3.8. *Angular Density*
Let $f(x \mid \alpha) = \frac{1+\alpha x}{2}$ on $[-1, 1]$, with $\alpha \in [-1, 1]$. Since $\mathbb{E}_\alpha X = \int_{-1}^{1} \frac{x(1+\alpha x)}{2}\,dx = \frac{\alpha}{3}$, MoM gives $\widehat{\alpha} = 3\overline{X}$.
Also $\text{Var}_\alpha(X) = \frac{1}{3} - \frac{\alpha^2}{9}$, so $\text{Var}_\alpha(\widehat{\alpha}) = 9\,\text{Var}_\alpha(\overline{X}) = \frac{3-\alpha^2}{n}$.

3 Hypothesis Testing and Confidence

3.1 Tests, Likelihood Ratios, and Error Control

Testing formalises decisions between hypotheses. The central quantities are the probability of false rejection under the null and the probability of rejection under alternatives.

Definition 3.1.1. *Hypotheses, Tests, Size, and Power*
(i) A *null hypothesis* is a subset $\Theta_0 \subseteq \Theta$ of the parameter space.
(ii) An *alternative* is a subset $\Theta_1 \subseteq \Theta$ disjoint from Θ_0, so $\Theta_0 \cap \Theta_1 = \emptyset$.
(iii) Randomised *test* is a measurable map $\varphi : \mathcal{X} \to [0, 1]$, where $\varphi(x)$ is rejection probability at observation x.
(iv) The *size* of φ is $\alpha(\varphi) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\varphi(X)]$.
(v) The *power function* of φ is $\beta_\varphi(\theta) = \mathbb{E}_\theta[\varphi(X)]$.

Definition 3.1.2. *Simple and Composite Hypotheses; Type Errors*
(i) A *simple* hypothesis specifies one law; a *composite* hypothesis specifies several.
(ii) A *type I error* rejects a true H_0; its probability is bounded by $\alpha(\varphi)$.
(iii) For a simple alternative P_1, a *type II error* is failure to reject H_0 under P_1, with probability $\mathbb{E}_1[1 - \varphi(X)]$.
(iv) The *power* against P_1 is $1 - \mathbb{E}_1[1 - \varphi(X)]$.

Theorem 3.1.3. *Neyman–Pearson Lemma*
Let $P_1 \ll P_0$, $\Lambda := \frac{dP_1}{dP_0}$, $\alpha \in [0, 1]$. If $c \geq 0$ and $\gamma \in [0, 1]$ make $\varphi^* := \mathbf{1}_{\{\Lambda > c\}} + \gamma \mathbf{1}_{\{\Lambda = c\}}$ satisfy $\mathbb{E}_0\varphi^* = \alpha$, then φ^* is most powerful among level-α tests: for every $\psi : \mathcal{X} \to [0, 1]$ with $\mathbb{E}_0\psi \leq \alpha$, $\mathbb{E}_1\psi \leq \mathbb{E}_1\varphi^*$.

Proof. For any level-α competitor ψ, $(\varphi^* - \psi)(\Lambda - c) \geq 0$ pointwise.
Since $dP_1 = \Lambda\, dP_0$, $\mathbb{E}_1(\varphi^* - \psi) = \mathbb{E}_0[\Lambda(\varphi^* - \psi)] \geq c\mathbb{E}_0(\varphi^* - \psi) = c(\alpha - \mathbb{E}_0\psi) \geq 0$.
Structural Commentary: Null controls size constraint, while alternative supplies power objective.
Key Insight: Use the rejection budget where $\Lambda = \frac{dP_1}{dP_0}$ is largest. □

Remark 3.1.4. *Choosing the Null*
At level a, for simple P_0 versus P_1, the *Neyman–Pearson problem* is $\sup\{\mathbb{E}_1\varphi : 0 \leq \varphi \leq 1,\ \mathbb{E}_0\varphi \leq a\}$.
Thus H_0 is the protected claim: choose it so the false-rejection probability $\mathbb{E}_0\varphi$ is the error to be bounded by a.
The choice of H_0 is substantive, not determined by $\{P_0, P_1\}$.

Definition 3.1.5. *Uniformly Most Powerful Test*
For a composite alternative Θ_1, a level-α test φ is *uniformly most powerful* if, for every $\theta \in \Theta_1$, it has power at least as large as every other level-α test at that same θ.

Example 3.1.6. *Normal Mean, Known Variance*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ^2. For $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1 > \mu_0$, the likelihood ratio is increasing in $\overline{X}$, so the level-α test is $\varphi^* = \mathbf{1}_{\{\overline{X} > \mu_0 + \frac{z_{1-\alpha}\sigma}{\sqrt{n}}\}}$, where $z_{1-\alpha}$ is the $N(0,1)$ quantile. Since the cutoff is independent of μ_1, φ^* is uniformly most powerful for $H_1 : \mu > \mu_0$; no single likelihood-ratio direction is uniformly most powerful for $H_1 : \mu \neq \mu_0$.

Definition 3.1.7. *Generalized Likelihood-Ratio Test*
For $H_0 : \theta \in \Theta_0 \subseteq \Theta$, the *generalized likelihood ratio* is $\lambda(x) := \frac{\sup_{\theta \in \Theta_0} L_x(\theta)}{\sup_{\theta \in \Theta} L_x(\theta)} \in [0, 1]$.
A GLR test rejects H_0 for small $\lambda(x)$, typically $\lambda(x) \leq c_\alpha$ with c_α chosen to give level α.

Theorem 3.1.8. *Large-Sample Likelihood-Ratio Calibration*

(i) Let $H_0 : \theta \in \Theta_0 \subseteq \Theta$, $\dim \Theta = d$, $\dim \Theta_0 = d_0$, $r := d - d_0$, *and* $X^{(n)} \sim P_{\theta_0}$ *with* $\theta_0 \in \Theta_0$.

(ii) Let $L_n(\theta) := e^{\ell_n(\theta)}$ *and* $\lambda_n := \frac{\sup_{\theta \in \Theta_0} L_n(\theta)}{\sup_{\theta \in \Theta} L_n(\theta)}$.

(iii) Assume local coordinates with $\theta_0 = 0$, $\Theta_0 = \{(u, 0) : u \in \mathbb{R}^{d_0}\}$, $h = (u, v) \in \mathbb{R}^{d_0} \times \mathbb{R}^r$, *and* $\ell_n(\frac{h}{\sqrt{n}}) - \ell_n(0) = h^\top Z_n - \frac{\|h\|^2}{2} + o_{P_{\theta_0}}(1)$, *where* $Z_n = (U_n, V_n) \Rightarrow N_d(0, I_d)$, $U_n \in \mathbb{R}^{d_0}$, $V_n \in \mathbb{R}^r$.

(iv) Then under H_0, $-2 \log \lambda_n \Rightarrow \chi^2_r$.

Proof.

(i) Since $L_n = e^{\ell_n}$, $\log \lambda_n = \sup_{\theta \in \Theta_0} \ell_n(\theta) - \sup_{\theta \in \Theta} \ell_n(\theta)$, hence $-2 \log \lambda_n = 2\{\sup_{\theta \in \Theta} \ell_n(\theta) - \sup_{\theta \in \Theta_0} \ell_n(\theta)\}$.

(ii) In the local coordinates, write $\theta = \frac{h}{\sqrt{n}}$ for the full model and $\theta = \frac{(u,0)}{\sqrt{n}}$ for the null model.

With $Q_n(h) := h^\top Z_n - \frac{\|h\|^2}{2}$ and $Q_{0,n}(u) := Q_n(u, 0) = u^\top U_n - \frac{\|u\|^2}{2}$, (iii) and cancellation of $\ell_n(0)$ give $-2 \log \lambda_n = 2\{\sup_{h \in \mathbb{R}^d} Q_n(h) - \sup_{u \in \mathbb{R}^{d_0}} Q_{0,n}(u)\} + o_{P_{\theta_0}}(1)$.

(iii) For $a \in \mathbb{R}^m$, $x^\top a - \frac{\|x\|^2}{2} = \frac{\|a\|^2}{2} - \frac{\|x-a\|^2}{2}$, so supremum over $x \in \mathbb{R}^m$ is $\frac{\|a\|^2}{2}$ at $x = a$.

Applying this with $a = Z_n$ and $a = U_n$ gives $\sup_{h \in \mathbb{R}^d} Q_n(h) = \frac{\|Z_n\|^2}{2}$ and $\sup_{u \in \mathbb{R}^{d_0}} Q_{0,n}(u) = \frac{\|U_n\|^2}{2}$,

hence $-2 \log \lambda_n = \|Z_n\|^2 - \|U_n\|^2 + o_{P_{\theta_0}}(1) = \|V_n\|^2 + o_{P_{\theta_0}}(1)$.

(iv) Since $Z_n = (U_n, V_n) \Rightarrow N_d(0, I_d)$, marginal convergence gives $V_n \Rightarrow Y$ with $Y \sim N_r(0, I_r)$.
Therefore $\|V_n\|^2 \Rightarrow \|Y\|^2 \sim \chi^2_r$ by continuous mapping, and Slutsky gives $-2 \log \lambda_n \Rightarrow \chi^2_r$.

Structural Commentary: The likelihood ratio compares unconstrained fit with null-constrained fit.
Key Insight: Wilks' statistic is the squared Gaussian fluctuation in the $r = d - d_0$ constrained directions. □

Example 3.1.9. *Two-Sided Normal Mean GLRT*

Let $X_1, \ldots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ^2, and test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.

Since $\widehat{\mu} = \overline{X}$, $\lambda(X) = \frac{L_n(\mu_0)}{L_n(\overline{X})} = \exp\{-\frac{n(\overline{X}-\mu_0)^2}{2\sigma^2}\}$, so $-2 \log \lambda(X) = \frac{n(\overline{X}-\mu_0)^2}{\sigma^2}$.

Under H_0, $\frac{\sqrt{n}(\overline{X}-\mu_0)}{\sigma} \sim N(0, 1)$; hence $-2 \log \lambda(X) \sim \chi^2_1$,

and the level-α GLR rejects iff $\frac{n(\overline{X}-\mu_0)^2}{\sigma^2} > \chi^2_{1,1-\alpha}$, equivalently $|\overline{X} - \mu_0| > \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}$.

Example 3.1.10. *Sign Test as a Binomial GLRT*

Let $X \sim \text{Binomial}(n, p)$ and test $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$.

For observed x, $\widehat{p} = \frac{x}{n}$ and $\lambda(x) = \frac{L_x(\frac{1}{2})}{L_x(\widehat{p})} = \frac{(\frac{1}{2})^n}{[(\frac{x}{n})^x(1-\frac{x}{n})^{n-x}]}$, $0^0 = 1$ at $x = 0, n$.

Since $\lambda(x)$ decreases as $|x - \frac{n}{2}|$ increases, a level-α GLR rejects on the largest values of $|X - \frac{n}{2}|$, with cutoff chosen under $X \sim \text{Binomial}(n, \frac{1}{2})$; this is the two-sided sign test.

Definition 3.1.11. *p-Value*

A *p-value* for $H_0 : \theta \in \Theta_0$ is a statistic $p : \mathcal{X} \to [0, 1]$ such that $\sup_{\theta \in \Theta_0} \mathbb{P}_\theta\{p(X) \leq u\} \leq u$ for every $u \in [0, 1]$.

Then $\varphi_\alpha(X) := \mathbf{1}_{\{p(X) \leq \alpha\}}$ has size $\alpha(\varphi_\alpha) \leq \alpha$.

Proposition 3.1.12. *Continuous Null p-Values Are Uniform*

Let $H_0 : \theta \in \Theta_0$ *and let* $T : \mathcal{X} \to \mathbb{R}$ *satisfy* $\mathbb{P}_\theta(T \leq t) = F_0(t)$ *for all* $\theta \in \Theta_0$, *where* F_0 *is continuous.*
For the right-tail convention, set $p(X) := 1 - F_0(T(X))$.
Then $p(X) \sim \text{Uniform}(0, 1)$ *under every* $\theta \in \Theta_0$, *and* $\varphi_\alpha(X) := \mathbf{1}_{\{p(X) \leq \alpha\}}$ *has size* α *for* $\alpha \in [0, 1]$.

Proof. Fix $\theta \in \Theta_0$ and set $U := F_0(T(X))$.
By probability integral transform, Proposition 2.3.6, $U \sim \text{Uniform}(0, 1)$ under $\mathbb{P}_\theta$,
hence $p(X) = 1 - U \sim \text{Uniform}(0, 1)$. Thus $\mathbb{E}_\theta \varphi_\alpha = \mathbb{P}_\theta(p(X) \leq \alpha) = \alpha \ \forall \theta \in \Theta_0$, so $\alpha(\varphi_\alpha) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta \varphi_\alpha = \alpha$.

Structural Commentary: A continuous null CDF converts the statistic to a canonical uniform scale.
Key Insight: Exact calibration is the probability integral transform applied under H_0. □

Remark 3.1.13. *p-Values Are Tail Probabilities*

A p-value is a null tail probability, e.g. $p(x) = \mathbb{P}_{H_0}\{T(X) \geq T(x)\}$ for a right-tail statistic T.
It is not $\mathbb{P}(H_0 \mid x)$; posterior probabilities require priors and an alternative likelihood.

Definition 3.1.14. *Multiple-Testing Error Counts and FDR*

For a multiple-testing rule over M hypotheses.
Let M_0 be the number of true nulls and $M_1 = M - M_0$ the number of false nulls.
Let V be false rejections, S true rejections, U true nonrejections, and T false nonrejections.
The total number of discoveries is $R = V + S$.

(i) *Family-wise error rate*: $\text{FWER} := \mathbb{P}(V \geq 1)$.
(ii) *False discovery proportion*: $\text{FDP} := \frac{V}{\max(R,1)}$.
(iii) *False discovery rate*: $\text{FDR} := \mathbb{E}[\text{FDP}]$.
(iv) *Power*: One aggregate power measure is $\mathbb{E}[S]/M_1$ when $M_1 > 0$.

Thus FWER controls the chance of any false discovery.
FDR controls the expected false fraction among the reported discoveries.

Remark 3.1.15. *Error Rates*

For one test φ, size is $\alpha(\varphi) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta \varphi$. For a family of tests, let R be the number of rejections and V the number of false rejections; common targets are $\text{FWER} = \mathbb{P}\{V \geq 1\}$ and $\text{FDR} = \mathbb{E}[\frac{V}{R \vee 1}]$, while directional or selective rates replace V by the corresponding error count.

Method 3.1.16. *Fishing Expeditions and Bonferroni Control*

(i) For N null hypotheses, let R_i be the event that null i is rejected.
(ii) By the union bound, $\mathbb{P}(\bigcup_{i=1}^{N} R_i) \le \sum_{i=1}^{N} \mathbb{P}(R_i)$.
(iii) To control family-wise error at level α, test each null at level $\frac{\alpha}{N}$; then $\mathbb{P}(\text{at least one false rejection}) \le \alpha$.
(iv) Without such correction, many tests can produce small p-values even when all nulls are true.

Definition 3.1.17. *Positive FDR, Local FDR, and q-Values*

Let $Z_j = 0$ indicate a true null, $Z_j = 1$ a false null, t_j a test statistic, and Γ a rejection region.

(i) *pFDR*: For false discoveries V and discoveries R, $\mathrm{pFDR} := \mathbb{E}[V/R \mid R > 0]$.
(ii) *Mixture model*: If $\mathbb{P}(Z_j = 0) = \pi_0$, $\mathcal{L}(t_j \mid Z_j = 0) = F_0$, and $\mathcal{L}(t_j \mid Z_j = 1) = F_1$, then $\mathcal{L}(t_j) = \pi_0 F_0 + (1 - \pi_0) F_1$.
(iii) *Bayes form*: For a rejection region Γ, $\mathrm{pFDR}(\Gamma) = \mathbb{P}(Z_j = 0 \mid t_j \in \Gamma)$.
(iv) *Local FDR*: At value t_0, $\mathrm{lfdr}(t_0) := \mathbb{P}(Z_j = 0 \mid t_j = t_0)$.
(v) *q-value*: For feature j, $q_j := \inf\{\alpha : j \in \mathcal{R}_\alpha\}$, where $\mathcal{R}_\alpha$ is the discovery set at FDR level α.

Remark 3.1.18. *FDR Versus Family-Wise Error*

With false discoveries V and total discoveries R, $\mathrm{FWER} := \mathbb{P}(V \ge 1)$ controls any false rejection, while $\mathrm{FDR} := \mathbb{E}[V/\max(R, 1)]$ controls the expected false fraction among discoveries. Thus FDR is less conservative for large screening lists and measures list reliability directly.

Method 3.1.19. *Benjamini–Hochberg Step-Up Rule*

For M hypotheses with p-values $p_1, \dots, p_M$ and target level $\alpha \in (0, 1)$:

(i) *Order*: Let $p_{(1)} \le \cdots \le p_{(M)}$ be the order statistics.
(ii) *Step-up index*: Set $L := \max\{j : p_{(j)} \le \alpha j / M\}$, with $L := 0$ if the set is empty.
(iii) *Rejections*: Reject $\mathcal{R} := \{m : p_m \le p_{(L)}\}$ if $L > 0$, and reject none if $L = 0$.
(iv) *Guarantee*: If true-null p-values are independent and uniform, then $\mathrm{FDR} \le (M_0/M)\alpha \le \alpha$, where M_0 is the number of true nulls.

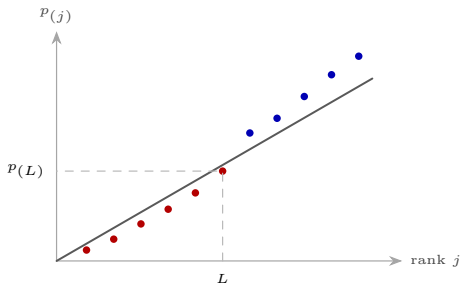

Figure 238: Benjamini–Hochberg threshold: red points reject, blue points do not; line is $\alpha j/M$.

3.2 Confidence Sets, Pivots, and Asymptotic Intervals

Confidence procedures attach uncertainty to estimators by controlling coverage probability under repeated sampling.

Definition 3.2.1. *Confidence Sets and Coverage*

A random set $C(X) \subseteq \Psi$ is a $1 - \alpha$ *confidence set* for $\psi : \Theta \to \Psi$ if $\inf_{\theta \in \Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1 - \alpha$.

The left side is the *coverage guarantee*.

Definition 3.2.2. *Pivots and Confidence Construction*

(i) A *pivot* is a statistic $Q(X, \theta)$ whose distribution under $\mathbb{P}_\theta$ is free of θ.
(ii) If constants a_α, b_α satisfy $\mathbb{P}_\theta\{a_\alpha \le Q(X, \theta) \le b_\alpha\} = 1 - \alpha$ for all $\theta \in \Theta$, then $C(X) := \{\theta \in \Theta : a_\alpha \le Q(X, \theta) \le b_\alpha\}$ is an exact $1 - \alpha$ confidence set for θ.

Corollary 3.2.3. *Hoeffding Confidence Interval for a Bernoulli Parameter*

Let $X_1, \dots, X_n$ be independent Bernoulli(p) *variables, $\widehat{p}_n := \overline{X}_n$, and $\alpha \in (0, 1)$.*

Set $\varepsilon_n := \sqrt{(2n)^{-1} \log(\frac{2}{\alpha})}$ and $C_n := [\widehat{p}_n - \varepsilon_n, \widehat{p}_n + \varepsilon_n]$.

(i) Coverage: $\mathbb{P}(p \in C_n) \ge 1 - \alpha$.
(ii) Length: $|C_n| = 2\varepsilon_n \le 0.05$ whenever $n \ge 800 \log(\frac{2}{\alpha})$.

Proof.

(i) Since $\{p \notin C_n\} = \{|\widehat{p}_n - p| > \varepsilon_n\}$, Corollary 3.3.5 gives $\mathbb{P}(p \notin C_n) \le 2e^{-2n\varepsilon_n^2} = \alpha$.
(ii) If $n \ge 800 \log(\frac{2}{\alpha})$, then $\varepsilon_n^2 \le \frac{1}{1600}$, hence $|C_n| = 2\varepsilon_n \le 0.05$.

Structural Commentary: Confidence intervals invert a concentration inequality.
Key Insight: Choose ε_n so the Hoeffding failure bound equals α. □

Proposition 3.2.4. *Test–Confidence Duality*

For each $\psi_0 \in \Psi$, let $\varphi_{\psi_0} : \mathcal{X} \to \{0, 1\}$ satisfy $\sup_{\theta : \psi(\theta) = \psi_0} \mathbb{E}_\theta \varphi_{\psi_0} \le \alpha$.

Then $C(X) := \{\psi_0 \in \Psi : \varphi_{\psi_0}(X) = 0\}$ is a $1 - \alpha$ confidence set for $\psi(\theta)$.

Proof. Fix $\theta \in \Theta$ and set $\psi_\theta := \psi(\theta)$.
By definition, $\{\psi(\theta) \notin C(X)\} = \{\varphi_{\psi_\theta}(X) = 1\}$, so $\mathbb{P}_\theta\{\psi(\theta) \notin C(X)\} = \mathbb{E}_\theta \varphi_{\psi_\theta} \le \sup_{\eta:\psi(\eta)=\psi_\theta} \mathbb{E}_\eta \varphi_{\psi_\theta} \le \alpha$.
Hence $\mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1-\alpha$ for every θ, and $\inf_{\theta\in\Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1-\alpha$.
Structural Commentary: Inverting all accepted point-null tests turns non-rejection regions into a confidence set.
Key Insight: Coverage failure equals rejection of the true null value. □

Proposition 3.2.5. *Confidence–Test Converse*
Let $C(X) \subseteq \Psi$ satisfy $\inf_{\theta\in\Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1-\alpha$.
For $\psi_0 \in \Psi$, let $\varphi_{\psi_0}(X) := \mathbf{1}_{\{\psi_0 \notin C(X)\}}$ for testing $H_0 : \psi(\theta) = \psi_0$. Then φ_{ψ_0} is level α: $\sup_{\theta:\psi(\theta)=\psi_0} \mathbb{E}_\theta \varphi_{\psi_0} \le \alpha$.

Proof. Fix θ with $\psi(\theta) = \psi_0$. Then $\mathbb{E}_\theta \varphi_{\psi_0} = \mathbb{P}_\theta\{\psi_0 \notin C(X)\} = \mathbb{P}_\theta\{\psi(\theta) \notin C(X)\} = 1 - \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \le \alpha$.
Taking $\sup_{\theta:\psi(\theta)=\psi_0}$ gives the level bound.
Structural Commentary: A confidence set tests a proposed value by checking membership.
Key Insight: Rejection of ψ_0 is exactly non-coverage when ψ_0 is true. □

Remark 3.2.6. *Acceptance Regions and Confidence Regions*
For level-α point-null tests with acceptance regions $A_{\psi_0} \subseteq \mathcal{X}$, the inversion $C(X) := \{\psi_0 \in \Psi : X \in A_{\psi_0}\}$ is the set of null values not rejected at X. Conversely, any confidence set $C(X)$ induces point-null tests $\varphi_{\psi_0}(X) = \mathbf{1}_{\{\psi_0 \notin C(X)\}}$.

Definition 3.2.7. *Wald, Score, and Likelihood Intervals*
For scalar regular θ with estimator $\widehat{\theta}$, standard error $\widehat{\mathrm{se}}(\widehat{\theta})$, score U_n, and likelihood ratio $\lambda_n(\theta_0)$:

(i) *Wald*: use $\widehat{\theta} \pm z_{1-\frac{\alpha}{2}} \widehat{\mathrm{se}}(\widehat{\theta})$.
(ii) *Score*: invert $\frac{|U_n(\theta_0)|}{\sqrt{I_n(\theta_0)}} \le z_{1-\frac{\alpha}{2}}$ over θ_0.
(iii) *Likelihood*: invert $-2\log\lambda_n(\theta_0) \le \chi^2_{1,1-\alpha}$ over θ_0.

Key Insight: Wald, score, and likelihood intervals are first-order equivalent in regular models.

Example 3.2.8. *Poisson Mean Wald Interval*
Let $X_1, \dots, X_n$ be independent Poisson(λ) variables.
Then $\widehat{\lambda} = \overline{X}$ and $\frac{\sqrt{n}(\overline{X}-\lambda)}{\sqrt{\lambda}} \Rightarrow N(0,1)$, so the plug-in standard error is $\widehat{\mathrm{se}}(\widehat{\lambda}) = \sqrt{\frac{\overline{X}}{n}}$.
The Wald interval is $C_\alpha(X) = [\overline{X} - z_{1-\frac{\alpha}{2}}\sqrt{\frac{\overline{X}}{n}}, \overline{X} + z_{1-\frac{\alpha}{2}}\sqrt{\frac{\overline{X}}{n}}]$, with asymptotic coverage $1-\alpha$.

3.3 Goodness-of-Fit and Diagnostic Plots

Goodness-of-fit methods compare an observed empirical pattern with the pattern predicted by a probability model. Formal tests give calibrated evidence against a specified model, while diagnostic plots show where the model fits or fails.

Definition 3.3.1. *Multinomial Goodness-of-Fit Model*

(i) *Observed counts*: $O = (O_1, \dots, O_m)$ records cell frequencies from n trials, with $O_i \in \mathbb{N}_0$ and $\sum_{i=1}^m O_i = n$.
(ii) *Null model*: $\mathcal{M}_0 = \{p(\theta) = (p_1(\theta), \dots, p_m(\theta)) : \theta \in \Theta_0\} \subseteq \Delta_{m-1}$ is parametric.
(iii) *Saturated model*: $\Delta_{m-1} = \{p \in [0,1]^m : \sum_i p_i = 1\}$ allows every multinomial probability vector.
(iv) *Fitted counts*: if $\widehat{\theta}$ is the null MLE, then $E_i := np_i(\widehat{\theta})$ for $i = 1, \dots, m$.
(v) *Degrees of freedom*: if $\dim\Theta_0 = k$, then $\mathrm{df} = (m-1) - k = m - k - 1$.

Definition 3.3.2. *Pearson and Likelihood-Ratio Goodness-of-Fit Statistics*

(i) *Inputs*: $O = (O_1, \dots, O_m)$ are observed counts, $E = (E_1, \dots, E_m)$ are fitted null counts with $E_i > 0$, and the null model has k free parameters.
(ii) *Pearson statistic*: $X^2 := \sum_{i=1}^m \frac{(O_i - E_i)^2}{E_i}$.
(iii) *Likelihood-ratio statistic*: $G^2 := 2\sum_{i=1}^m O_i \log(\frac{O_i}{E_i})$, with $0\log(\frac{0}{E_i}) := 0$.
(iv) *Null calibration*: under the regular null model, $X^2 \Rightarrow \chi^2_{m-k-1}$ and $G^2 = -2\log\lambda(X) \Rightarrow \chi^2_{m-k-1}$.
Moreover $G^2 - X^2 = o_P(1)$ under H_0.

Example 3.3.3. *Hardy–Weinberg Fit*
For genotype cells (M, MN, N), the Hardy–Weinberg model is $p(\theta) = ((1-\theta)^2, 2\theta(1-\theta), \theta^2)$.
With $n = 1029$ and $\widehat{\theta} = 0.4247$, the observed and fitted counts are

	M	MN	N
O_i	342	500	187
E_i	340.6	502.8	185.6

Since $m = 3$ and $k = 1$, $\mathrm{df} = m - k - 1 = 1$. Pearson statistic is $X^2 = 0.0319$, so $p = \mathbb{P}\{\chi^2_1 \ge 0.0319\}$ is large.
The data give no evidence against Hardy–Weinberg equilibrium.

Method 3.3.4. *Reading Cell Contributions*

(i) Compute signed residuals $r_i := O_i - E_i$ and Pearson contributions $c_i := \frac{(O_i - E_i)^2}{E_i}$.
(ii) Large c_i identify cells driving the rejection; the sign of r_i indicates overfit or underfit in that cell.
(iii) If some E_i are small, group adjacent or tail cells before testing so the chi-square approximation is reliable.

Remark 3.3.5. *Grouped Counts and Expected Frequencies*

Let $Y_1, \dots, Y_N$ be observations, let $A_1, \dots, A_m$ partition sample space, with observed counts $O_j := \sum_{i=1}^{N} \mathbf{1}_{\{Y_i \in A_j\}}$.

Given a fitted parametric law $P_{\widehat{\theta}}$, set $p_j(\widehat{\theta}) := P_{\widehat{\theta}}(Y \in A_j)$; the fitted expected count is $E_j := N p_j(\widehat{\theta})$.
For grouped Poisson tails, $p_j(\widehat{\theta})$ is a sum of Poisson masses, $(O_1, \dots, O_m)$ is modelled as multinomial under $P_{\widehat{\theta}}$.

Example 3.3.6. *Poisson Clumps*
For grouped bacterial clump frequencies from $n = 400$ grid squares, fit $H_0 : X \sim \text{Poisson}(\lambda)$ with $\widehat{\lambda} = 2.44$ and grouped tails so E_i are adequate. The Pearson statistic is $X^2 = 75.4$ with df $= 6$, so $p = \mathbb{P}\{\chi^2_6 \geq 75.4\} < 0.005$; the largest $c_i = \frac{(O_i - E_i)^2}{E_i}$ are in the first and last cells, indicating excess small and large counts.

Definition 3.3.7. *Overdispersed Counts*
Count data are *overdispersed* relative to a Poisson model if $\text{Var}(X) > \mathbb{E}X$; empirically, this is $\widehat{\sigma}^2 > \overline{X}$.

Definition 3.3.8. *Poisson Dispersion Test*

The data is $x_1, \dots, x_n$ are independent counts and $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$.

(i) *Models*: $H_0 : x_i \sim \text{Poisson}(\lambda)$ with common λ; the saturated alternative uses separate rates λ_i.

(ii) *Likelihood-ratio statistic*: $G^2_D := 2 \sum_{i=1}^{n} x_i \log(\frac{x_i}{\overline{x}})$, with $0 \log(\frac{0}{\overline{x}}) := 0$.

(iii) *Pearson approximation*: $T := \overline{x}^{-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$.

(iv) *Calibration*: under H_0, G^2_D and T are calibrated by χ^2_{n-1}
Large values indicate overdispersion relative to $\text{Var}(X) = \mathbb{E}X$.

Example 3.3.9. *Dispersion in Bacterial Clumps*
For bacterial clump counts with $n = 400$, $\overline{x} = 2.44$ and $s^2 = 4.59$, so $\frac{s^2}{\overline{x}} = 1.88 > 1$.
Dispersion statistic is $T \approx 752.7$ with df $= n - 1 = 399$, so $p = \mathbb{P}\{\chi^2_{399} \geq 752.7\} \approx 0$.
The data show strong overdispersion relative to a Poisson model.

Definition 3.3.10. *Hanging Rootograms and Chi-Grams*
Let n_j be observed bin counts and $\widehat{n}_j$ fitted counts in ordered bins.

(i) A *hanging histogram* plots raw residuals $r_j := n_j - \widehat{n}_j$, whose variance depends on $\widehat{n}_j$.

(ii) A *hanging rootogram* plots $q_j := \sqrt{n_j} - \sqrt{\widehat{n}_j}$, using $\text{Var}(\sqrt{n_j}) \approx \frac{1}{4}$ for count data.

(iii) A *hanging chi-gram* plots standardized residuals $z_j := \frac{n_j - \widehat{n}_j}{\sqrt{\widehat{n}_j}}$.

Runs of same-sign residuals, tail spikes, or systematic curvature indicate model departure.

Method 3.3.11. *Probability Plots*
Let $X_{(1)} \leq \cdots \leq X_{(n)}$ be order statistics and let F be the candidate continuous strictly increasing CDF.

(i) A *P-P plot* compares $F(X_{(k)})$ with $u_k := \frac{k}{n+1}$ for $k = 1, \dots, n$.

(ii) A *Q-Q plot* compares $X_{(k)}$ with $F^{-1}(u_k)$ for $k = 1, \dots, n$.

The model is plausible when the plotted points are approximately linear.
For $F(x) = G(\frac{x - \mu}{\sigma})$, Q-Q slope and intercept estimate σ and μ.

Method 3.3.12. *Probability Plots for Grouped Data*

Let bins boundaries $x_1 < \cdots < x_{m+1}$, counts $n_1, \dots, n_m$, total $n := \sum_{j=1}^{m} n_j$, cumulative counts $N_j := \sum_{i=1}^{j} n_i$.

Let G be the standardised CDF in the proposed location-scale family $F_{\mu,\sigma}(x) = G(\frac{x - \mu}{\sigma})$.

(i) Plot empirical boundary quantiles x_{j+1} against theoretical quantiles $G^{-1}(\frac{N_j}{n+1})$ for $j = 1, \dots, m$.

(ii) Use this grouped Q-Q plot when individual order statistics are unavailable.

Remark 3.3.13. *Grouped Normality Tests*
For binned normal-fit testing, use $X^2 = \sum_j \frac{(O_j - E_j)^2}{E_j}$ with $E_j = n\{F_{\widehat{\mu},\widehat{\sigma}}(b_{j+1}) - F_{\widehat{\mu},\widehat{\sigma}}(b_j)\}$.

The usual χ^2_{m-k-1} calibration requires $(\widehat{\mu}, \widehat{\sigma})$ from the grouped counts; ungrouped estimates change the limit.

Definition 3.3.14. *Normality Diagnostics*
Let $X_{(1)} \leq \cdots \leq X_{(n)}$, $\overline{X} := n^{-1} \sum_i X_i$, and $s^2 := n^{-1} \sum_i (X_i - \overline{X})^2$.

(i) *Normal Q-Q plot*: plot $X_{(k)}$ against $\Phi^{-1}(\frac{k}{n+1})$; normality predicts approximate linearity.

(ii) *Skewness*: $b_1 := n^{-1} \sum_{i=1}^{n} \frac{(X_i - \overline{X})^3}{s^3}$ measures asymmetry; large $|b_1|$ indicates nonnormality.

(iii) *Kurtosis*: $b_2 := n^{-1} \sum_{i=1}^{n} \frac{(X_i - \overline{X})^4}{s^4}$ measures tail weight and peakedness; normality gives $b_2 \approx 3$.

(iv) *Plot correlation*: Q-Q point correlation; small values indicate curvature.

3.4 Normal Sampling Pivots

For normal samples, the sample mean and sample variance separate exactly. This is the finite-sample source of the classical z, χ^2, and Student t pivots used in normal-theory confidence intervals and tests.

Corollary 3.4.1. *Student Pivot for a Normal Mean*
Let $X_1, \dots, X_n$ be independent; $X_i \sim N(\mu, \sigma^2)$, $\overline{X} := n^{-1} \sum_i X_i$, $S^2 := (n-1)^{-1} \sum_i (X_i - \overline{X})^2$; $\frac{\overline{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$.

Proof. Let $Z := \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}$ and $V := \frac{(n-1)S^2}{\sigma^2}$. By normal mean law, $Z \sim N(0,1)$. By sample-variance chi-square law, $V \sim \chi^2_{n-1}$. By mean-residual independence, $Z \perp\!\!\!\perp V$. Therefore $\frac{\overline{X}-\mu}{\frac{S}{\sqrt{n}}} = \frac{Z}{\sqrt{\frac{V}{n-1}}} \sim t_{n-1}$.

Structural Commentary: Replacing σ by S yields a Student pivot.
Key Insight: Student's t is a normal variable divided by an independent chi-square scale estimate. □

Method 3.4.2. *Exact Normal Mean Intervals*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables, with $\overline{X} := n^{-1}\sum_i X_i$ and $S^2 := (n-1)^{-1}\sum_i (X_i - \overline{X})^2$.

Below cases have exact coverage $1-\alpha$.

(i) *Known variance*: if σ is known, $C_\mu(X) = [\overline{X} - \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}, \overline{X} + \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}]$

(ii) *Unknown variance*: if σ is unknown, $C_\mu(X) = [\overline{X} - \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}, \overline{X} + \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}]$

(iii) *Variance*: $C_{\sigma^2}(X) = [\frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}]$

Example 3.4.3. *Normal Mean with Known Variance*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ.
The pivot $Z_\mu := \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim N(0,1)$ gives $\mathbb{P}_\mu\{|Z_\mu| \le z_{1-\frac{\alpha}{2}}\} = 1-\alpha$.
Solving for μ gives the exact interval $C_\alpha(X) = [\overline{X} - \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}, \overline{X} + \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}]$.
Test Connection: The two-sided level-α test of $H_0 : \mu = \mu_0$ rejects iff $\mu_0 \notin C_\alpha(X)$.

Example 3.4.4. *Normal Mean and Variance with Unknown Variance*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables, and set $S^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$.
The exact pivots are $T_\mu := \frac{\sqrt{n}(\overline{X}-\mu)}{S} \sim t_{n-1}$ and $W_{\sigma^2} := \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$. Writing $t_{\nu,p}$ and $\chi^2_{\nu,p}$ for p-quantiles, pivot inversion gives $C_\mu(X) = [\overline{X} - \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}, \overline{X} + \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}]$ and $C_{\sigma^2}(X) = [\frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}]$.
Both intervals have exact marginal coverage $1-\alpha$; $C_{\sigma^2}(X)$ is asymmetric because the χ^2 pivot is not symmetric.

3.5 Two-Sample Normal Inference

Two-sample inference compares the location of two response distributions. In a controlled experiment, one sample may receive a treatment and the other a control; in an observational comparison, the same formulas may be computed but their causal interpretation depends on the design.

Definition 3.5.1. *Two Independent Samples and the Location Contrast*
$X_1, \dots, X_n \sim F$ are independent, $Y_1, \dots, Y_m \sim G$ are independent, and the two samples are independent.

(i) *Means*: $\mu_X := \mathbb{E}X_i$ and $\mu_Y := \mathbb{E}Y_j$ summarize the two average levels, assuming both exist.
(ii) *Location contrast*: $\Delta := \mu_X - \mu_Y$ measures the shift from group Y to group X.
(iii) *Estimator*: $\widehat{\Delta} := \overline{X} - \overline{Y}$ estimates Δ, where $\overline{X} := n^{-1}\sum_i X_i$ and $\overline{Y} := m^{-1}\sum_j Y_j$.

Proposition 3.5.2. *Known Common Variance Pivot*
Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be independent samples, $X_i \sim N(\mu_X, \sigma^2)$, $Y_j \sim N(\mu_Y, \sigma^2)$, known $\sigma > 0$.
Set $\Delta := \mu_X - \mu_Y$, $\widehat{\Delta} := \overline{X} - \overline{Y}$, and $\mathrm{se}_0 := \sigma\sqrt{n^{-1} + m^{-1}}$.

(i) $\widehat{\Delta} \sim N(\Delta, \mathrm{se}_0^2)$ and $Z := \frac{\widehat{\Delta}-\Delta}{\mathrm{se}_0} \sim N(0,1)$.
(ii) $C_\alpha(X,Y) := [\widehat{\Delta} - z_{1-\frac{\alpha}{2}}\,\mathrm{se}_0, \widehat{\Delta} + z_{1-\frac{\alpha}{2}}\,\mathrm{se}_0]$ satisfies $\mathbb{P}_\Delta\{\Delta \in C_\alpha(X,Y)\} = 1-\alpha$.

Proof.

(i) Since $\overline{X} \sim N(\mu_X, \frac{\sigma^2}{n})$, $\overline{Y} \sim N(\mu_Y, \frac{\sigma^2}{m})$, and $\overline{X} \perp\!\!\!\perp \overline{Y}$, $\widehat{\Delta} = \overline{X} - \overline{Y} \sim N(\Delta, \frac{\sigma^2}{n} + \frac{\sigma^2}{m}) = N(\Delta, \mathrm{se}_0^2)$. Standardization gives $Z \sim N(0,1)$.
(ii) By (i), $\mathbb{P}_\Delta\{|Z| \le z_{1-\frac{\alpha}{2}}\} = 1-\alpha$, equivalently $\mathbb{P}_\Delta\{\Delta \in C_\alpha(X,Y)\} = 1-\alpha$.

Structural Commentary: The contrast is a linear combination of independent sample means.
Key Insight: Known common variance gives an exact normal pivot for Δ. □

Definition 3.5.3. *Pooled Sample Variance*
Let $X_1, \dots, X_n \sim N(\mu_X, \sigma^2)$ and $Y_1, \dots, Y_m \sim N(\mu_Y, \sigma^2)$ be independent samples with common unknown σ^2.

(i) *Sample variances*: $S_X^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$ and $S_Y^2 := (m-1)^{-1}\sum_{j=1}^m (Y_j - \overline{Y})^2$.
(ii) *Pooled variance*: $S_p^2 := \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$.
(iii) *Degrees-of-freedom weighting*: S_p^2 weights S_X^2 and S_Y^2 by $n-1$ and $m-1$, respectively.

Theorem 3.5.4. *Pooled Two-Sample Student Pivot*
Let $X_i \sim N(\mu_X, \sigma^2)$, $i = 1, \dots, n$, and $Y_j \sim N(\mu_Y, \sigma^2)$, $j = 1, \dots, m$, be mutually independent, with unknown common $\sigma^2 > 0$. Set $\Delta := \mu_X - \mu_Y$ and $\nu := n + m - 2$. Then $T := \frac{(\overline{X}-\overline{Y})-\Delta}{S_p\sqrt{n^{-1}+m^{-1}}} \sim t_\nu$.

Proof. Let $Z := \frac{(\overline{X}-\overline{Y})-\Delta}{\sigma\sqrt{n^{-1}+m^{-1}}}$ and $W := \frac{\nu S_p^2}{\sigma^2} = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{\sigma^2}$.
Since $\overline{X} - \overline{Y} \sim N(\Delta, \sigma^2(n^{-1}+m^{-1}))$, $Z \sim N(0,1)$. Let $U_X := \frac{(n-1)S_X^2}{\sigma^2}$ and $U_Y := \frac{(m-1)S_Y^2}{\sigma^2}$.
By normal mean-residual independence and sample independence, Z, U_X, U_Y mutually independent,

with $U_X \sim \chi^2_{n-1}$, $U_Y \sim \chi^2_{m-1}$. Hence $W = U_X + U_Y \sim \chi^2_\nu$ and $W \perp\!\!\!\perp Z$. Since $S_p^2 = \frac{\sigma^2 W}{\nu}$, $T = \frac{Z}{\sqrt{\frac{W}{\nu}}} \sim t_\nu$.

Structural Commentary: The pooled variance combines the two residual sums of squares.
Key Insight: A common variance permits replacing σ by an independent pooled estimate. □

Method 3.5.5. *Pooled Two-Sample t Procedures*
Under the equal-variance normal model, set $\Delta := \mu_X - \mu_Y$, $\nu := n + m - 2$, and $\mathrm{se}_p := S_p\sqrt{n^{-1} + m^{-1}}$.

(i) *Interval*: $C_\alpha := [(\overline{X} - \overline{Y}) - t_{\nu,1-\frac{\alpha}{2}}\,\mathrm{se}_p, (\overline{X} - \overline{Y}) + t_{\nu,1-\frac{\alpha}{2}}\,\mathrm{se}_p]$ is exact $1-\alpha$ for Δ.
(ii) *Statistic*: for $H_0 : \Delta = 0$, use $t_0 := \frac{\overline{X}-\overline{Y}}{\mathrm{se}_p}$, with $t_0 \sim t_\nu$ under H_0.
(iii) *Rejection*: reject H_0 for $|t_0| > t_{\nu,1-\frac{\alpha}{2}}$ under $H_1 : \Delta \neq 0$
reject H_0 for $t_0 > t_{\nu,1-\alpha}$ under $H_1 : \Delta > 0$
reject H_0 for $t_0 < -t_{\nu,1-\alpha}$ under $H_1 : \Delta < 0$.
(iv) *Duality*: the two-sided test rejects iff $0 \notin C_\alpha$.

Example 3.5.6. *Latent Heat of Fusion: Two Methods*
For methods A and B, let $n_A = 13$, $n_B = 8$, $\overline{x}_A = 80.02$, $\overline{x}_B = 79.98$, $s_A = 0.024$, $s_B = 0.031$, and $S_p = 0.027$.
Then $\widehat{\Delta} = \overline{x}_A - \overline{x}_B = 0.04$, $\widehat{\mathrm{se}} = S_p\sqrt{13^{-1} + 8^{-1}} \approx 0.012$, pooled 95% interval for $\Delta = \mu_A - \mu_B$ is
$\approx (0.015, 0.065)$. For $H_0 : \Delta = 0$, $t_0 = \frac{\widehat{\Delta}}{\widehat{\mathrm{se}}} \approx 3.33$ with $\nu = 19$, strong evidence that method means differ.

Proposition 3.5.7. *Likelihood-Ratio View of the Pooled Test*
Let $X_i \sim N(\mu_X, \sigma^2)$, $i = 1, \ldots, n$, and $Y_j \sim N(\mu_Y, \sigma^2)$, $j = 1, \ldots, m$, be mutually independent, with unknown common $\sigma^2 > 0$. Set $\Delta := \mu_X - \mu_Y$, $\nu := n + m - 2$, and $T := \frac{\overline{X}-\overline{Y}}{S_p\sqrt{n^{-1}+m^{-1}}}$.
For $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$, $\lambda(X, Y) = (1 + \frac{T^2}{\nu})^{-\frac{n+m}{2}}$.
Hence the GLR rejects for large $|T|$; at level α, reject iff $|T| > t_{\nu,1-\frac{\alpha}{2}}$.

Proof. Set $N := n+m$, $\nu := N-2$, $d := \overline{X}-\overline{Y}$, $R_1 := \sum_i (X_i - \overline{X})^2 + \sum_j (Y_j - \overline{Y})^2$, $\widehat{\mu}_0 := \frac{n\overline{X}+m\overline{Y}}{N}$, and $R_0 := \sum_i (X_i - \widehat{\mu}_0)^2 + \sum_j (Y_j - \widehat{\mu}_0)^2$.
Under H_1, $(\widehat{\mu}_X, \widehat{\mu}_Y, \widehat{\sigma}_1^2) = (\overline{X}, \overline{Y}, \frac{R_1}{N})$; under H_0, $(\widehat{\mu}, \widehat{\sigma}_0^2) = (\widehat{\mu}_0, \frac{R_0}{N})$.
The maximised Gaussian likelihood is proportional to $\widehat{\sigma}^{-N}$, hence $\lambda = (\frac{R_1}{R_0})^{\frac{N}{2}}$.
ANOVA decomposition gives $R_0 = R_1 + \{\frac{nm}{N}\}d^2$, while $T^2 = \frac{d^2}{(\frac{R_1}{\nu})(n^{-1}+m^{-1})} = \frac{\nu\{\frac{nm}{N}\}d^2}{R_1}$.
Therefore $\frac{R_0}{R_1} = 1 + \frac{T^2}{\nu}$ and $\lambda = (1 + \frac{T^2}{\nu})^{-\frac{N}{2}}$, decreasing in T^2; small λ iff large $|T|$.
Structural Commentary: The null fit adds only the between-sample mean discrepancy to the unrestricted fit.
Key Insight: The GLR depends on the data through the pooled t statistic. □

Method 3.5.8. *Unequal-Variance Two-Sample Procedure*
For independent samples without the equal-variance assumption, set $\widehat{v} := \frac{S_X^2}{n} + \frac{S_Y^2}{m}$.

(i) *Standard error*: estimate $\mathrm{Var}(\overline{X} - \overline{Y})$ by $\widehat{v}$.
(ii) *Welch statistic*: for $H_0 : \Delta = \Delta_0$, use $T_W := \frac{(\overline{X}-\overline{Y})-\Delta_0}{\sqrt{\widehat{v}}}$.
(iii) *Degrees of freedom*: calibrate T_W by t_ν, where $\nu := \frac{\widehat{v}^2}{\frac{(S_X^2/n)^2}{n-1} + \frac{(S_Y^2/m)^2}{m-1}}$.
(iv) *Use*: the procedure avoids pooling variances, at the cost of approximate calibration.

Corollary 3.5.9. *Variance Ratio Test Under Normality*
Let $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$ be independent samples, with $n, m \geq 2$.
Set $\rho := \frac{\sigma_X^2}{\sigma_Y^2}$ and $R := \frac{S_X^2}{S_Y^2}$. Then $\frac{R}{\rho} \sim F_{n-1,m-1}$.

Proof. Let $U := \frac{(n-1)S_X^2}{\sigma_X^2}$ and $V := \frac{(m-1)S_Y^2}{\sigma_Y^2}$.
Normal residual chi-square laws give $U \sim \chi^2_{n-1}$ and $V \sim \chi^2_{m-1}$, and sample independence gives $U \perp\!\!\!\perp V$.
Thus $\frac{\frac{U}{n-1}}{\frac{V}{m-1}} = \frac{R}{\rho} \sim F_{n-1,m-1}$. If $H_0 : \rho = 1$, then $R \sim F_{n-1,m-1}$, yielding exact normal-theory tests and confidence intervals for ρ; the calibration is normality-sensitive.
Structural Commentary: The comparison standardises two independent residual variance estimates.
Key Insight: Normality turns each sample variance into an independent chi-square pivot. □

Method 3.5.10. *Approximate Power for Balanced Two-Sample z Tests*
For planning, assume balanced samples of size n, known common σ, and target effect $\Delta := \mu_X - \mu_Y > 0$.

(i) *Statistic*: use $Z := \frac{\overline{X}-\overline{Y}}{\sigma\sqrt{\frac{2}{n}}}$ and reject two-sided at level α when $|Z| > z_{1-\frac{\alpha}{2}}$.
(ii) *Shift*: under $\Delta > 0$, $Z \sim N(\delta, 1)$ with $\delta := (\frac{\Delta}{\sigma})\sqrt{\frac{n}{2}}$.
(iii) *Power*: $\pi(\Delta) := 1 - \Phi(z_{1-\frac{\alpha}{2}} - \delta) + \Phi(-z_{1-\frac{\alpha}{2}} - \delta)$.
(iv) *Sample size*: if the lower-tail term is negligible, target power $1-\beta$ gives $n \approx \frac{2\sigma^2(z_{1-\frac{\alpha}{2}} + z_{1-\beta})^2}{\Delta^2}$.

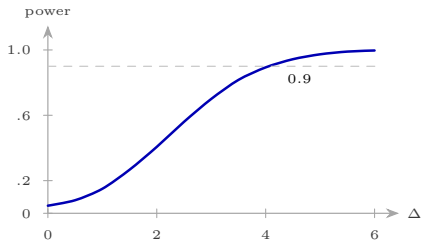

Figure 239: Power rises as the detectable mean difference Δ increases.

Remark 3.5.11. *Transformations and Normal Diagnostics*
Normal-theory two-sample procedures are often CLT-robust for large samples, but small skewed or outlier-prone samples require diagnostics. Transformations may stabilize spread and improve symmetry, but they change the estimand, e.g. a log-scale mean difference corresponds to a ratio-scale comparison.

3.6 Paired-Sample Inference

Paired comparisons arise when observations are naturally matched: before and after measurements on the same unit, littermates, matched subjects, or two measurement methods applied to the same specimen. Pairing changes the stochastic structure because the two samples are no longer independent.

Definition 3.6.1. *Paired Differences*

(i) *Pairs*: (X_i, Y_i), $i = 1, \dots, n$, are independent across i; dependence within each pair is allowed.

(ii) *Differences*: $D_i := X_i - Y_i$ and $\overline{D} := n^{-1} \sum_{i=1}^{n} D_i$.

(iii) *Mean*: if $\mathbb{E}X_i = \mu_X$ and $\mathbb{E}Y_i = \mu_Y$, then $\mathbb{E}D_i = \mu_X - \mu_Y$.

(iv) *Variance*: if $\mathrm{Var}(X_i) = \sigma_X^2$, $\mathrm{Var}(Y_i) = \sigma_Y^2$, and $\mathrm{Cov}(X_i, Y_i) = \sigma_{XY}$, then $\mathrm{Var}(D_i) = \sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}$.

Proposition 3.6.2. *Variance Effect of Pairing*
Let (X_i, Y_i), $i = 1, \dots, n$, *be independent pairs with* $\mathrm{Var}(X_i) = \sigma_X^2$, $\mathrm{Var}(Y_i) = \sigma_Y^2$, *and* $\mathrm{Corr}(X_i, Y_i) = \rho$.
Set $D_i := X_i - Y_i$ *and* $\overline{D} := n^{-1} \sum_i D_i$. *Then* $\mathrm{Var}(\overline{D}) = n^{-1}(\sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y)$.

Proof. Since $\mathrm{Cov}(X_i, Y_i) = \rho\sigma_X\sigma_Y$, $\mathrm{Var}(D_i) = \mathrm{Var}(X_i - Y_i) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$.
Independence across pairs gives $\mathrm{Var}(\overline{D}) = n^{-2} \sum_i \mathrm{Var}(D_i) = \frac{\mathrm{Var}(D_i)}{n}$.

If $\sigma_X = \sigma_Y = \sigma$, substitution gives $\mathrm{Var}(\overline{D}) = \frac{2\sigma^2(1-\rho)}{n}$; independent unpaired samples of size n per group have $\mathrm{Var}(\overline{X} - \overline{Y}) = \frac{2\sigma^2}{n}$, so the paired-to-unpaired variance ratio is $1 - \rho$.
Structural Commentary: Pairing replaces two marginal measurements by one within-pair contrast.
Key Insight: Positive within-pair correlation cancels shared variation and lowers variance. □

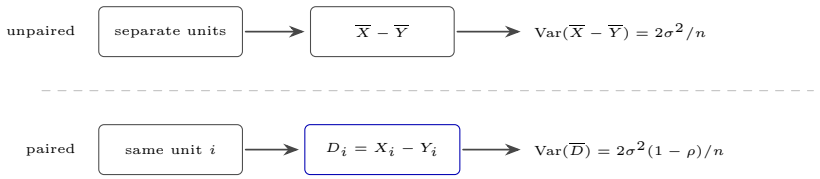

Figure 240: Pairing cancels shared variation; positive within-pair correlation lowers variance.

Method 3.6.3. *Paired t Procedure*
Let $D_1, \dots, D_n \sim N(\mu_D, \sigma_D^2)$, where $\mu_D := \mu_X - \mu_Y$.

(i) *Variance*: $S_D^2 := (n-1)^{-1} \sum_{i=1}^{n} (D_i - \overline{D})^2$.

(ii) *Pivot*: $T := \frac{\overline{D} - \mu_D}{S_D/\sqrt{n}} \sim t_{n-1}$.

(iii) *Interval*: $C_\alpha := [\overline{D} - \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}, \overline{D} + \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}]$ is exact $1 - \alpha$ for μ_D.

(iv) *Test*: for $H_0 : \mu_D = 0$, reject two-sided at level α iff $|\overline{D}| > \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}$.

For large n, use same rule by CLT; for small nonnormal differences, prefer rank or sign methods.

Example 3.6.4. *Platelet Aggregation Before and After Smoking*
For $n = 11$ before-after platelet measurements, let $D_i = \text{after}_i - \text{before}_i$ with $\overline{D} = 10.27$ and $\frac{S_D}{\sqrt{11}} = 2.40$.
Then $t_0 = \frac{\overline{D}}{\frac{S_D}{\sqrt{11}}} = 4.28$ with $\nu = 10$, so $p < 0.01$ two-sided; a 90% confidence interval is approximately $(5.9, 14.6)$, indicating a positive mean increase after smoking.

Example 3.6.5. *Mercury Measurement Methods*
For $n = 25$ paired mercury measurements, let $D_i = M_{1i} - M_{2i}$ with $\overline{D} = 0.04$ and $S_D = 0.116$.
Then $t_0 = \frac{\overline{D}}{\frac{S_D}{\sqrt{25}}} = 1.724$ with $\nu = 24$, giving $p \approx 0.094$ two-sided.
Plot of D_i against measurement level should be checked: level-dependent discrepancies not captured by $\overline{D}$ alone.

Remark 3.6.6. *Pairing Is Not Extra Independent Sampling*
If unit i contributes repeated measurements $Z_{i1}, \dots, Z_{ik}$, independence is across units, not within unit.
Treating all nk observations as iid ignores within-unit dependence and overstates the effective sample size.

4 Decision Theory

4.1 Loss, Risk, Bayes, and Minimax Rules

Decision theory compares statistical procedures by their consequences, not only by their formulas. The same data model can lead to different optimal rules under different loss functions.

Definition 4.1.1. *Statistical Decision Problem and Risk*

(i) *Problem*: a statistical decision problem is $(\{P_\theta : \theta \in \Theta\}, \mathcal{A}, L)$ with $L : \Theta \times \mathcal{A} \to [0, \infty]$.
(ii) *Rule*: a decision rule is a measurable $\delta : \mathcal{X} \to \mathcal{A}$.
(iii) *Risk*: $R(\theta, \delta) := \mathbb{E}_\theta[L(\theta, \delta(X))]$.

Definition 4.1.2. *Bayes Risk, Minimaxity, and Admissibility*

(i) *Bayes risk*: for prior Π on Θ, $r(\Pi, \delta) := \int R(\theta, \delta)\, d\Pi(\theta)$.
(ii) *Bayes rule*: δ_Π is Bayes if $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.
(iii) *Maximum risk*: $M(\delta) := \sup_{\theta \in \Theta} R(\theta, \delta)$; $\delta^\star$ is minimax if $M(\delta^\star) = \inf_\delta M(\delta)$.
(iv) *Admissibility*: δ inadmissible if $\exists \delta'$ with $R(\theta, \delta') \le R(\theta, \delta)\ \forall \theta$ and $<$ for some θ; otherwise δ admissible.

Proposition 4.1.3. *Posterior Risk Characterization*
Let Π be a prior, let $\Pi(\cdot \mid x)$ be a posterior, and set $\rho_x(a) := \int L(\theta, a)\, d\Pi(\theta \mid x)$.
If a measurable selector $\delta_\Pi(x) \in \arg\min_{a \in \mathcal{A}} \rho_x(a)$ exists, then δ_Π is Bayes: $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.

Proof. By iterated expectation, $r(\Pi, \delta) = \mathbb{E}_\Pi[\rho_X(\delta(X))]$, where X has the prior predictive law.
Since $\rho_x(\delta_\Pi(x)) \le \rho_x(\delta(x))\ \forall$ rule δ and almost every x, $r(\Pi, \delta_\Pi) \le r(\Pi, \delta)\ \forall \delta$. Thus $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.
Structural Commentary: Bayes risk is the prior-predictive average of posterior risks.
Key Insight: Global Bayes optimization reduces to pointwise posterior-risk minimization. □

Example 4.1.4. *Bayes Rules Under Common Losses*
For a real parameter and posterior distribution $\Pi(\cdot \mid x)$:

(i) Squared-error loss is minimized by the posterior mean.
(ii) Absolute-error loss is minimized by a posterior median.
(iii) Zero-one loss on a finite action set is minimized by a posterior mode.

Remark 4.1.5. *Least Favorable Priors*
If a prior $\Pi^\star$ has Bayes risk equal to the minimax value, then any Bayes rule for $\Pi^\star$ with constant risk on the support of $\Pi^\star$ is minimax. Such priors are called least favorable.

4.2 Shrinkage, Empirical Bayes, and Regularization

Risk can often be reduced by intentionally pulling noisy estimates toward a structured target. The price is bias; the benefit is variance reduction.

Definition 4.2.1. *Shrinkage Estimator*

(i) *Input*: let $\widehat{\theta} = \widehat{\theta}(X)$ be a data-based estimator.
(ii) *Target*: choose θ_{tar} as a reference value or lower-complexity model.
(iii) *Weight*: let $a(X) \in [0, 1]$ control the amount of shrinkage.
(iv) *Estimator*: $\delta(X) := a(X)\widehat{\theta} + \{1 - a(X)\}\theta_{\text{tar}}$.

Example 4.2.2. *Ridge Regression as Shrinkage*
For design matrix X, response y, and $\lambda \ge 0$, ridge regression sets $\widehat{\beta}_\lambda \in \arg\min_\beta \{\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2\}$.

If $X^\top X$ is invertible, $\widehat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$, shrinking $\widehat{\beta}_{\text{LS}}$ toward 0 most in weak eigendirections.

Theorem 4.2.3. *James–Stein Phenomenon*
Let $X \sim \mathcal{N}_d(\theta, \sigma^2 I_d)$ with $d \ge 3$, known σ^2, and squared-error risk $R(\theta, \delta) := \mathbb{E}_\theta \|\delta(X) - \theta\|^2$.
Define $\delta_{\text{JS}}(X) := (1 - \frac{(d-2)\sigma^2}{\|X\|^2})X$ for $X \ne 0$, with arbitrary value at $X = 0$.
Then $R(\theta, \delta_{\text{JS}}) < R(\theta, X) = d\sigma^2$ for every $\theta \in \mathbb{R}^d$.

Proof. Let $a := (d-2)\sigma^2$ and $g(x) := -\frac{ax}{\|x\|^2}$, so $\delta_{\text{JS}}(X) = X + g(X)$.
Stein's identity gives $R(\theta, X + g) = d\sigma^2 + \mathbb{E}_\theta[\|g(X)\|^2 + 2\sigma^2 \nabla \cdot g(X)]$.
For $x \ne 0$, $\|g(x)\|^2 = \frac{a^2}{\|x\|^2}$ and $\nabla \cdot g(x) = -\frac{a(d-2)}{\|x\|^2}$.
Thus $R(\theta, \delta_{\text{JS}}) = d\sigma^2 - (d-2)^2\sigma^4 \mathbb{E}_\theta[\|X\|^{-2}] < d\sigma^2$, with $\mathbb{E}_\theta[\|X\|^{-2}] < \infty$ since $d \ge 3$.
Structural Commentary: Shrinkage trades bias toward 0 for enough variance reduction to lower total risk.
Key Insight: In dimension $d \ge 3$, the usual estimator X is inadmissible under squared-error loss. □

Definition 4.2.4. *Empirical Bayes*

(i) *Hierarchy*: $\theta_i \mid \eta \sim \Pi_\eta$ and $X_i \mid \theta_i \sim P_{\theta_i}$ for $i = 1, \dots, m$.
(ii) *Hyperparameter*: estimate η by $\widehat{\eta} = \widehat{\eta}(X_1, \dots, X_m)$ from the marginal law of the data.
(iii) *Plug-in Bayes rule*: use the Bayes rule computed under $\Pi_{\widehat{\eta}}$.

Remark 4.2.5. *Regularization as Decision-Theoretic Structure*
Penalization, shrinkage, and empirical Bayes impose structure to reduce risk, typically by trading increased bias for reduced variance. The criterion is $R(\theta, \delta)$, not unbiasedness.

5 Bayesian Inference

5.1 Priors, Posteriors, and Conjugacy

Bayesian inference treats unknown parameters as uncertain quantities and updates a prior distribution by the likelihood supplied by the data.

Definition 5.1.1. *Prior, Posterior, and Predictive Distribution*
Let Π be a prior on Θ and $p_\theta(x)$ a dominated likelihood.

(i) *Prior*: Π encodes uncertainty about $\theta \in \Theta$ before observing x.
(ii) *Evidence*: $m(x) := \int_\Theta p_\vartheta(x)\, d\Pi(\vartheta)$, assumed finite and positive.
(iii) *Posterior*: $d\Pi(\theta \mid x) := \frac{p_\theta(x)\, d\Pi(\theta)}{m(x)}$.
(iv) *Predictive*: for future X_{new}, $\mathbb{P}(X_{\text{new}} \in A \mid x) = \int_\Theta P_\theta(A)\, d\Pi(\theta \mid x)$.

Definition 5.1.2. *Conjugate Prior Family*

(i) *Conjugacy*: $\{\Pi_\eta : \eta \in H\}$ is conjugate if $\Pi_\eta(\cdot \mid x) = \Pi_{\eta(x)}$ for some $\eta(x) \in H$.

Proposition 5.1.3. *Posterior Odds and Likelihood Ratios*
Let H_0, H_1 be simple hypotheses with $\mathbb{P}(H_i) = \pi_i > 0$ and likelihood $p_i(x) = p(x \mid H_i)$ for $i = 0, 1$.
For $m(x) := \pi_0 p_0(x) + \pi_1 p_1(x) > 0$, $\mathbb{P}(H_i \mid x) = \frac{\pi_i p_i(x)}{m(x)}$ and $\frac{\mathbb{P}(H_0 \mid x)}{\mathbb{P}(H_1 \mid x)} = (\frac{\pi_0}{\pi_1})(\frac{p_0(x)}{p_1(x)})$.

Proof. Bayes' rule gives $\mathbb{P}(H_i \mid x) = \frac{\pi_i p_i(x)}{m(x)}$ for $i = 0, 1$. Therefore $\frac{\mathbb{P}(H_0 \mid x)}{\mathbb{P}(H_1 \mid x)} = \frac{\frac{\pi_0 p_0(x)}{m(x)}}{\frac{\pi_1 p_1(x)}{m(x)}} = (\frac{\pi_0}{\pi_1})(\frac{p_0(x)}{p_1(x)})$.
Structural Commentary: Posterior odds factor as prior odds times the data likelihood ratio.
Key Insight: Bayes updating multiplies odds by $\frac{p_0(x)}{p_1(x)}$. □

Example 5.1.4. *Beta–Binomial Updating*
If $X \mid p \sim \text{Binomial}(n, p)$ and $p \sim \text{Beta}(a, b)$, then $p \mid X = x \sim \text{Beta}(a + x, b + n - x)$.
Thus $\mathbb{E}[p \mid x] = \frac{a+x}{a+b+n}$, a weighted average of $\frac{a}{a+b}$ and $\frac{x}{n}$.

Example 5.1.5. *Gamma–Poisson Updating*
Let $X_1, \dots, X_n \mid \lambda$ be iid Poisson(λ) and $\lambda \sim \text{Gamma}(\alpha, \nu)$ in rate form.
Writing $s := \sum_{i=1}^{n} x_i$, the posterior kernel is $\pi(\lambda \mid x) \propto \lambda^{\alpha+s-1} e^{-(\nu+n)\lambda}$, hence $\lambda \mid x \sim \text{Gamma}(\alpha + s, \nu + n)$.
Thus $\mathbb{E}[\lambda \mid x] = \frac{\alpha+s}{\nu+n}$, $\text{Var}(\lambda \mid x) = \frac{\alpha+s}{(\nu+n)^2}$, $\text{mode}(\lambda \mid x) = \frac{\alpha+s-1}{\nu+n}$ if $\alpha + s > 1$.

Example 5.1.6. *Normal Mean with Normal Prior*
Let $X_1, \dots, X_n \mid \mu$ be iid $N(\mu, \sigma^2)$ with known σ^2, and let $\mu \sim N(m_0, \tau_0^2)$.
Then $\mu \mid X \sim N(m_n, \tau_n^2)$, where $\tau_n^{-2} = \tau_0^{-2} + n\sigma^{-2}$ and $m_n = \tau_n^2(\tau_0^{-2} m_0 + n\sigma^{-2}\overline{X})$.

Example 5.1.7. *Known Mean and Unknown Precision*
Let $X_1, \dots, X_n \mid \xi$ be iid $N(\theta_0, \xi^{-1})$ with known θ_0, and let $\xi \sim \text{Gamma}(\alpha, \lambda)$ in rate form.
Writing $q := \sum_{i=1}^{n} (x_i - \theta_0)^2$, the posterior is $\xi \mid x \sim \text{Gamma}(\alpha + \frac{n}{2}, \lambda + \frac{q}{2})$.
With a flat gamma prior, $\sigma^2 = \xi^{-1}$ centers near the MLE $\frac{q}{n}$.

Definition 5.1.8. *Improper Prior*
An *improper prior* is a σ-finite prior measure Π on Θ with $\Pi(\Theta) = \infty$. For likelihood $p_\theta(x)$, it is usable only when $0 < m_\Pi(x) := \int_\Theta p_\theta(x)\, d\Pi(\theta) < \infty$, in which case $d\Pi(\theta \mid x) := \frac{p_\theta(x)\, d\Pi(\theta)}{m_\Pi(x)}$ is a proper posterior.

Remark 5.1.9. *Conditioning Viewpoint*
Bayesian updating is conditioning under the joint law $dP(\theta, x) = p_\theta(x)\, d\Pi(\theta)\, dx$; the posterior $\Pi(\cdot \mid x)$ is a regular conditional law of θ given $X = x$.

5.2 Computation, Prediction, and Asymptotic Bayes

Except in conjugate models, posterior summaries are integrals that usually require approximation. Monte Carlo and optimization methods from numerical computation become statistical tools here.

Remark 5.2.1. *Bayesian Integrals as a Motivation*
For prior density $\pi(\theta)$ and likelihood $\mathcal{L}(\theta)$, write $Z = \int \mathcal{L}(u)\pi(u)\, du$.
Then $\pi(\theta \mid X^n) = Z^{-1}\mathcal{L}(\theta)\pi(\theta)$ and $\mathbb{E}[\theta \mid X^n] = Z^{-1} \int \theta\, \mathcal{L}(\theta)\pi(\theta)\, d\theta$.
For $\theta = (\theta_1, \dots, \theta_k)$, high-dimensional marginalisation is often analytically impractical.

Definition 5.2.2. *Bayesian Credible Set*
Let $\Pi(\cdot \mid x)$ be the posterior on Θ.

(i) *Credible set*: $C(x) \subseteq \Theta$ has credibility $1 - \alpha$ if $\Pi(C(x) \mid x) \geq 1 - \alpha$.
(ii) *Conditioning*: credibility is a posterior probability conditional on the observed data x.

Definition 5.2.3. *Posterior Monte Carlo Integration*
Let $\Pi(\cdot \mid x)$ be the posterior on Θ.

(i) *Monte Carlo sample*: let $\theta^{(1)}, \dots, \theta^{(M)}$ be draws, or approximate draws, from $\Pi(\cdot \mid x)$.
(ii) *Posterior mean*: for integrable g, $\int g(\theta)\, d\Pi(\theta \mid x) \approx M^{-1} \sum_{m=1}^{M} g(\theta^{(m)})$.
(iii) *Dependent draws*: MCMC uses dependent samples but targets the same posterior integrals.

Method 5.2.4. *Simulating a Posterior Function*

(i) Let $\delta = g(p_1, \dots, p_k)$ be the posterior target.

(ii) Draw posterior samples $(P_1^{(i)}, \dots, P_k^{(i)})$ for $i = 1, \dots, N$.

(iii) Set $\delta^{(i)} = g(P_1^{(i)}, \dots, P_k^{(i)})$, use empirical distribution of $\delta^{(i)}$ to estimate posterior distribution of δ.

Example 5.2.5. *Two Binomial Proportions*

Suppose $X \sim \text{Binomial}(n, p_1)$ and $Y \sim \text{Binomial}(m, p_2)$ are independent, with independent flat priors on p_1, p_2.
Then $p_1 \mid X \sim \text{Beta}(X+1, n-X+1)$ and $p_2 \mid Y \sim \text{Beta}(Y+1, m-Y+1)$.
For $\delta = p_2 - p_1$, draw $P_1^{(i)} \sim \text{Beta}(X+1, n-X+1)$ and $P_2^{(i)} \sim \text{Beta}(Y+1, m-Y+1)$, set $\delta^{(i)} = P_2^{(i)} - P_1^{(i)}$, and summarize the empirical distribution of $\delta^{(i)}$.

Method 5.2.6. *Gibbs Sampling from Full Conditionals*

(i) *Kernel*: unknown mean θ, precision ξ, flat-prior posterior satisfies $\pi(\theta, \xi \mid x) \propto \xi^{\frac{n}{2}-1} \exp\{-\frac{\xi}{2} \sum_i (x_i - \theta)^2\}$.

(ii) *Decomposition*: $\sum_i (x_i - \theta)^2 = \sum_i (x_i - \overline{x})^2 + n(\theta - \overline{x})^2$.

(iii) *Mean conditional*: $\theta \mid \xi, x \sim N(\overline{x}, (n\xi)^{-1})$.

(iv) *Precision conditional*: $\xi \mid \theta, x \sim \text{Gamma}(\frac{n}{2}, \frac{1}{2} \sum_i (x_i - \theta)^2)$ in rate form.

(v) *Gibbs sampler*: alternate conditionals; post-burn-in averages estimate posterior summaries.

Method 5.2.7. *Laplace Approximation*

(i) *Setup*: let $\pi_n(\theta \mid x) \propto \exp\{-n q_n(\theta)\}$, with interior minimiser $\widehat{\theta}$.

(ii) *Curvature*: set $H_n(\widehat{\theta}) := \nabla^2 q_n(\widehat{\theta})$, assumed positive definite.

(iii) *Quadratic approximation*: $q_n(\theta) \approx q_n(\widehat{\theta}) + \frac{1}{2}(\theta - \widehat{\theta})^\top H_n(\widehat{\theta})(\theta - \widehat{\theta})$.

(iv) *Normal approximation*: $\Pi(\cdot \mid x) \approx N(\widehat{\theta}, \{n H_n(\widehat{\theta})\}^{-1})$.

Theorem 5.2.8. *Bernstein–von Mises Principle*

Let $X_1, \dots, X_n \sim P_{\theta_0}$ in a regular d-dimensional model with nonsingular information $I(\theta_0)$.
Assume the prior density π is positive and continuous near θ_0, and let $\widehat{\theta}_n$ be an efficient center.
Then $\Pi(\sqrt{n}(\theta - \widehat{\theta}_n) \in \cdot \mid X_1, \dots, X_n) \Rightarrow N_d(0, I(\theta_0)^{-1})$ in P_{θ_0}-probability.

Proof. Set $h := \sqrt{n}(\theta - \widehat{\theta}_n)$. Local asymptotic normality gives $\ell_n(\widehat{\theta}_n + \frac{h}{\sqrt{n}}) - \ell_n(\widehat{\theta}_n) = -\frac{1}{2} h^\top I(\theta_0) h + o_{P_{\theta_0}}(1)$ uniformly on bounded h. Prior continuity gives $\frac{\pi(\widehat{\theta}_n + \frac{h}{\sqrt{n}})}{\pi(\theta_0)} \to 1$ in P_{θ_0}-probability on bounded h.
After normalization, the posterior density of h converges to the $N_d(0, I(\theta_0)^{-1})$ density, with tail mass outside bounded sets negligible. Thus the displayed posterior weak convergence holds.
Structural Commentary: Locally, the likelihood is quadratic and the prior is flat to first order.
Key Insight: In regular finite-dimensional models, posterior uncertainty is asymptotically frequentist normal. □

Remark 5.2.9. *Alignment and Limits*

In regular fixed-dimensional models, Bayesian credible sets based on the Bernstein–von Mises normal approximation have asymptotic frequentist coverage. This alignment can fail in high-dimensional, nonparametric, weakly identified, or boundary problems, where the prior may remain first-order relevant.

5.3 Bayesian Normal Comparisons

The normal two-sample model also has a Bayesian interpretation. With diffuse priors, the resulting posterior formulas resemble frequentist t intervals, but the conditioning and interpretation are different.

Definition 5.3.1. *Common-Precision Two-Sample Normal Model*

(i) *Common-Precision Model*: $\theta = (\mu_X, \mu_Y, \xi) \in \mathbb{R}^2 \times (0, \infty)$ with shared precision $\xi = \sigma^{-2}$ and variance ξ^{-1}.

(ii) *Conditional model*: Given (μ_X, μ_Y, ξ), the variables $X_1, \dots, X_n, Y_1, \dots, Y_m$ are independent, with $X_i \mid (\mu_X, \xi) \sim N(\mu_X, \xi^{-1})$ and $Y_j \mid (\mu_Y, \xi) \sim N(\mu_Y, \xi^{-1})$.

(iii) *Improper prior*: $\pi(\mu_X, \mu_Y, \xi) \propto \xi^{-1}$ on $\mu_X, \mu_Y \in \mathbb{R}$ and $\xi > 0$.

(iv) *Observed data*: Write $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_m)$.

(v) *Sample means*: $\overline{x} := n^{-1} \sum_{i=1}^n x_i$ and $\overline{y} := m^{-1} \sum_{j=1}^m y_j$.

(vi) *Sample variances*: $s_X^2 := (n-1)^{-1} \sum_{i=1}^n (x_i - \overline{x})^2$ and $s_Y^2 := (m-1)^{-1} \sum_{j=1}^m (y_j - \overline{y})^2$.

(vii) *Pooled variance*: $s_p^2 := \frac{(n-1)s_X^2 + (m-1)s_Y^2}{m+n-2}$.

(viii) *Posterior kernel*: $\pi(\mu_X, \mu_Y, \xi \mid x, y) \propto \xi^{\frac{m+n}{2}-1} \exp\left\{-\frac{\xi}{2}\left[\sum_{i=1}^n (x_i - \mu_X)^2 + \sum_{j=1}^m (y_j - \mu_Y)^2\right]\right\}$.

Proposition 5.3.2. *Conditional Posterior Factors*

For $n, m \geq 2$, put $\nu := m + n - 2$ and take Gamma *shape-rate.*
In the common-precision model, $\pi(\mu_X, \mu_Y, \xi \mid x, y) = \pi(\mu_X \mid \xi, x, y)\pi(\mu_Y \mid \xi, x, y)\pi(\xi \mid x, y)$,
where $(\mu_X, \mu_Y) \mid \xi, x, y \sim N(\overline{x}, (n\xi)^{-1}) \otimes N(\overline{y}, (m\xi)^{-1})$ and $\xi \mid x, y \sim \text{Gamma}(\frac{\nu}{2}, \frac{\nu s_p^2}{2})$. $\mu_X \perp\!\!\!\perp \mu_Y \mid (\xi, x, y)$.

Proof. Let $S_X(\mu_X) := \sum_{i=1}^n (x_i - \mu_X)^2$, $S_Y(\mu_Y) := \sum_{j=1}^m (y_j - \mu_Y)^2$, and $A := (n-1)s_X^2 + (m-1)s_Y^2 = \nu s_p^2$.
Completing squares gives $S_X(\mu_X) = (n-1)s_X^2 + n(\mu_X - \overline{x})^2$ and $S_Y(\mu_Y) = (m-1)s_Y^2 + m(\mu_Y - \overline{y})^2$.
Thus $\pi(\mu_X, \mu_Y, \xi \mid x, y) \propto \xi^{\frac{m+n}{2}-1} \exp\{-\frac{\xi A}{2}\} \exp\{-\frac{n\xi(\mu_X - \overline{x})^2}{2}\} \exp\{-\frac{m\xi(\mu_Y - \overline{y})^2}{2}\}$.

For fixed ξ, last two factors are independent normal kernels for μ_X, μ_Y with variances $(n\xi)^{-1}$, $(m\xi)^{-1}$.
Also $\int \exp\{-\frac{n\xi(\mu_X-\overline{x})^2}{2}\}\,d\mu_X \propto (n\xi)^{-\frac{1}{2}}$ and $\int \exp\{-\frac{m\xi(\mu_Y-\overline{y})^2}{2}\}\,d\mu_Y \propto (m\xi)^{-\frac{1}{2}}$.
Hence marginalising μ_X, μ_Y leaves $\pi(\xi \mid x, y) \propto \xi^{\frac{\nu}{2}-1}\exp\{-\frac{\xi\nu s_p^2}{2}\}$.
Structural Commentary: Completing squares isolates location kernels and pooled residual $A = \nu s_p^2$.
Key Insight: Marginalising μ_X, μ_Y leaves the pooled gamma law for ξ. □

Proposition 5.3.3. *Posterior t Law for a Mean Difference*
In the common-precision model with $n, m \geq 2$, set $\Delta := \mu_X - \mu_Y$, $d := \overline{x} - \overline{y}$, $c := n^{-1} + m^{-1}$, $\nu := m + n - 2$, and $T := \frac{\Delta - d}{s_p\sqrt{c}}$. Then $T \mid x, y \sim t_\nu$.

Proof. By Proposition 5.3.2, $\Delta \mid \xi, x, y \sim N(d, \frac{c}{\xi})$ and $\xi \mid x, y \sim \text{Gamma}(\frac{\nu}{2}, \frac{\nu s_p^2}{2})$.
Equivalently, take independent $E \sim N(0,1)$, $V \sim \chi^2_\nu$, $\xi = \frac{V}{\nu s_p^2}$, $\Delta = d + \sqrt{\frac{c}{\xi}}\,E$. Thus $T = \frac{\Delta - d}{s_p\sqrt{c}} = \frac{E}{\sqrt{\frac{V}{\nu}}}$, QED.
Structural Commentary: Normal given ξ; Student after mixing over ξ.
Key Insight: The t law is exactly a standard normal divided by an independent estimated-scale factor $\sqrt{\frac{V}{\nu}}$. □

Remark 5.3.4. *Same Algebra, Different Conditioning*
The statistic has the same algebra as the pooled two-sample t pivot in Theorem 3.5.4: frequentist inference treats Δ as fixed and X, Y as random, whereas the posterior treats x, y as fixed and $\Delta \mid x, y$ as random.

Example 5.3.5. *Latent Heat Posterior Comparison*
For method contrast $\Delta = \mu_X - \mu_Y$, the common-precision posterior gives $\mathbb{P}(\Delta > 0 \mid X, Y) \approx 0.998$ and posterior interval $\mathbb{P}(0.015 \leq \Delta \leq 0.065 \mid X, Y) \approx 0.95$, numerically matching the pooled normal-theory interval.

6 Nonparametric and High-Dimensional Statistics

6.1 Empirical Processes, Density, and Regression Smoothing

Nonparametric statistics replaces finite-dimensional parameter assumptions by smoothness, shape, sparsity, or complexity restrictions.

Definition 6.1.1. *Empirical Measure and Empirical Process*
Let $X_1, \ldots, X_n$ be observations in $(\mathcal{X}, \mathcal{A})$ with reference law P.

(i) *Empirical measure*: $\widehat{P}_n := n^{-1}\sum_{i=1}^n \delta_{X_i}$, so $\widehat{P}_n f = n^{-1}\sum_{i=1}^n f(X_i)$.

(ii) *Empirical process*: For measurable $\mathcal{F}$, $\mathbb{G}_n(f) := \sqrt{n}(\widehat{P}_n f - Pf)$ for $f \in \mathcal{F}$.

Proposition 6.1.2. *Fixed-Point Behavior of the Empirical CDF*
Let $X_1, \ldots, X_n$ be independent with common CDF F.
For $x \in \mathbb{R}$, set $I_i(x) := \mathbf{1}_{(-\infty, x]}(X_i)$ and $F_n(x) := n^{-1}\sum_{i=1}^n I_i(x)$. Then, for each fixed $x \in \mathbb{R}$:

(i) Law: $nF_n(x) \sim \text{Binomial}(n, F(x))$.
(ii) Mean: $\mathbb{E}F_n(x) = F(x)$.
(iii) Variance: $\text{Var}(F_n(x)) = n^{-1}F(x)(1 - F(x))$.

Proof. Fix $x \in \mathbb{R}$ and put $p := F(x)$ and $I_i := I_i(x)$.

(i) Since $I_i \sim \text{Bernoulli}(p)$ independently, $nF_n(x) = \sum_{i=1}^n I_i \sim \text{Binomial}(n, p)$.

(ii) Linearity gives $\mathbb{E}F_n(x) = n^{-1}\sum_{i=1}^n \mathbb{E}I_i = n^{-1}np = p = F(x)$.

(iii) Independence gives $\text{Var}(F_n(x)) = n^{-2}\sum_{i=1}^n \text{Var}(I_i) = n^{-2}np(1-p) = n^{-1}F(x)(1 - F(x))$.

Structural Commentary: Pointwise, $F_n(x) = n^{-1}\sum_{i=1}^n I_i(x)$ with $I_i(x) \sim \text{Bernoulli}(F(x))$ independently.
Key Insight: The empirical CDF at one point is a normalised binomial count. □

Remark 6.1.3. *Pointwise and Simultaneous CDF Uncertainty*
Proposition 6.1.2 is pointwise: x is fixed and $F_n(x)$ is a normalised binomial count.
Simultaneous uncertainty controls $\|F_n - F\|_\infty := \sup_x |F_n(x) - F(x)|$, yielding bands and diagnostics.

Proposition 6.1.4. *Two-Point Covariance of the Empirical CDF*
Let $X_1, \ldots, X_n$ be independent with common CDF F.
For $w \in \mathbb{R}$, set $I_i(w) := \mathbf{1}_{(-\infty, w]}(X_i)$, $F_n(w) := n^{-1}\sum_{i=1}^n I_i(w)$. For $u, v \in \mathbb{R}$, put $m := \min\{u, v\}$. Then:

(i) Covariance: $\text{Cov}\big(F_n(u), F_n(v)\big) = n^{-1}\big(F(m) - F(u)F(v)\big)$.
(ii) Sign: $\text{Cov}\big(F_n(u), F_n(v)\big) \geq 0$.

Proof. Fix $u, v \in \mathbb{R}$ and set $m := \min\{u, v\}$.

(i) *Covariance*: $\text{Cov}(F_n(u), F_n(v)) = n^{-2}\sum_{i,j}\text{Cov}(I_i(u), I_j(v))$.

For $i \neq j$ the term is 0 by independence, while for $i = j$,
$\text{Cov}(I_i(u), I_i(v)) = \mathbb{E}[I_i(u)I_i(v)] - \mathbb{E}I_i(u)\mathbb{E}I_i(v) = F(m) - F(u)F(v)$.
Hence $\text{Cov}(F_n(u), F_n(v)) = n^{-1}(F(m) - F(u)F(v))$.

(ii) *Sign*: If $u \le v$, then $F(m) - F(u)F(v) = F(u)(1 - F(v)) \ge 0$.
If $v \le u$, then $F(m) - F(u)F(v) = F(v)(1 - F(u)) \ge 0$.

Structural Commentary: $F_n(u)$ and $F_n(v)$ average iid Bernoulli pairs, so covariance scales by $1/n$.
Key Insight: Cross-index independence leaves only the same-observation overlap $\{X_i \le m\}$. □

Theorem 6.1.5. *Glivenko–Cantelli Theorem*
Let $X_1, X_2, \dots$ be i.i.d. real variables with CDF F, and set $\widehat{F}_n(t) := n^{-1} \sum_{i=1}^{n} \mathbf{1}_{\{X_i \le t\}}$.

Then $\sup_{t \in \mathbb{R}} \left|\widehat{F}_n(t) - F(t)\right| \to 0$ almost surely.

Proof. Add independent $R_i \sim \mathrm{Unif}[0,1]$ and set $U_i := F(X_i-) + R_i\{F(X_i) - F(X_i-)\}$.
Then U_i are i.i.d. Unif$[0,1]$, and with $H_n(u) := n^{-1} \sum_i \mathbf{1}_{\{U_i \le u\}}$ one has $\widehat{F}_n(t) = H_n(F(t))$ a.s.

Hence $\sup_t \left|\widehat{F}_n(t) - F(t)\right| \le \sup_{0 \le u \le 1} |H_n(u) - u|$.
Fix $k \ge 1$, $u_j := \frac{j}{k}$, let $M_{n,k} := \max_{0 \le j \le k} |H_n(u_j) - u_j|$.
By the SLLN, $M_{n,k} \to 0$ a.s. If $u_{j-1} \le u \le u_j$, monotonicity gives $H_n(u_{j-1}) - u_j \le H_n(u) - u \le H_n(u_j) - u_{j-1}$,
Hence $\sup_{0 \le u \le 1} |H_n(u) - u| \le M_{n,k} + k^{-1}$, and $\limsup_{n \to \infty} \sup_{0 \le u \le 1} |H_n(u) - u| \le k^{-1}$ a.s. $\forall k$. Let $k \to \infty$.

Structural Commentary: Finite-grid SLLN plus monotonicity gives uniform CDF control.
Key Insight: Monotonicity bounds between-grid drift by the mesh. □

Proposition 6.1.6. *Approximate Variance of a Sample Quantile*
Let $Y_1, \dots, Y_n$ be iid with continuous CDF F and density f. Fix $p \in (0,1)$, set $x_p := F^{-1}(p)$ with $f(x_p) > 0$, choose k with $\frac{k}{n+1} \approx p$, and put $\widehat{x}_p := Y_{(k)}$. Then:

(i) Probability Scale: $U_{(k)} := F(Y_{(k)})$ has kth uniform order-statistic law. $\mathrm{Var}(U_{(k)}) = \frac{k(n+1-k)}{(n+1)^2(n+2)} \approx \frac{p(1-p)}{n}$.

(ii) Quantile Scale: $\mathrm{Var}(\widehat{x}_p) \approx \frac{p(1-p)}{n f(x_p)^2}$.

Proof.
(i) By probability integral transform, $U_i := F(Y_i)$ are iid Uniform$(0,1)$, $U_{(k)} = F(Y_{(k)})$.
Theorem 4.1.4 gives the stated exact variance, hence $\mathrm{Var}(U_{(k)}) \approx \frac{p(1-p)}{n}$ when $\frac{k}{n+1} \approx p$.
(ii) Let $g := F^{-1}$. Since $g'(p) = \frac{1}{f(x_p)}$ and $\widehat{x}_p = g(U_{(k)})$, delta approximation $g(U_{(k)}) \approx g(p) + g'(p)(U_{(k)} - p)$
gives $\mathrm{Var}(\widehat{x}_p) \approx g'(p)^2 \mathrm{Var}(U_{(k)}) \approx \frac{p(1-p)}{n f(x_p)^2}$.

Structural Commentary: Quantile variance is probability-scale variance multiplied by $(F^{-1})'(p)^2$.
Key Insight: Low density near x_p inflates sample-quantile variability by $1/f(x_p)^2$. □

Proposition 6.1.7. *Distribution-Free Median Interval*
Let $X_1, \dots, X_n$ be independent with continuous CDF F and median η, so $F(\eta) = \frac{1}{2}$.
Let $X_{(1)} \le \cdots \le X_{(n)}$ and $1 \le k \le \lfloor \frac{n}{2} \rfloor$. Then:

(i) Interval: $C_k := [X_{(k)}, X_{(n-k+1)}]$.

(ii) Coverage: $\mathbb{P}(\eta \in C_k) = 1 - \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$.

Proof.
(i) Let $B_- := \sum_{i=1}^{n} \mathbf{1}_{\{X_i < \eta\}}$, $B_+ := \sum_{i=1}^{n} \mathbf{1}_{\{X_i > \eta\}}$. By continuity, $B_- + B_+ = n$ a.s. and $B_- \sim \mathrm{Binomial}(n, \frac{1}{2})$.
(ii) $\eta \notin C_k$ iff $B_- < k$ or $B_+ < k$. Since $k \le \lfloor \frac{n}{2} \rfloor$, these two events are disjoint.
(iii) By symmetry, $\mathbb{P}(\eta \notin C_k) = 2\mathbb{P}(B_- < k) = 2 \sum_{j=0}^{k-1} \binom{n}{j} 2^{-n} = \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$, so $\mathbb{P}(\eta \in C_k) = 1 - \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$.

Structural Commentary: The coverage depends only on the binomial count of observations below η, not on F.
Key Insight: Continuity turns median coverage into a symmetric binomial tail. □

Definition 6.1.8. *Kernel Density Estimator*
(i) *Kernel and bandwidth*: Let $\int K(u)\,du = 1$ and $h > 0$.
(ii) *Estimator*: For data $X_1, \dots, X_n$, $\widehat{f}_h(x) := (nh)^{-1} \sum_{i=1}^{n} K(\frac{x - X_i}{h})$.

Larger h lowers variance and raises bias; smaller h does the reverse.

Definition 6.1.9. *Nonparametric Regression*
(i) *Model*: $Y = m(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X] = 0$, so $m(x) = \mathbb{E}[Y \mid X = x]$.
(ii) *Weights*: For kernel K and bandwidth $h > 0$, set $w_i(x) := K(\frac{x - X_i}{h})$.
(iii) *Estimator*: If $\sum_i w_i(x) > 0$, define $\widehat{m}_h(x) := \frac{\sum_{i=1}^{n} w_i(x) Y_i}{\sum_{i=1}^{n} w_i(x)}$.

Definition 6.1.10. *Kernel-Weighted Local Average*
Let $(X_i, Y_i)_{i=1}^n$ be regression data and fix a query point x_0.
(i) *Local kernel*: For a nonnegative profile D and width $h_\lambda(x_0) > 0$, set $K_\lambda(x_0, x) := D(\|x - x_0\| / h_\lambda(x_0))$.

(ii) *Estimator*: If $\sum_{i=1}^{n} K_\lambda(x_0, X_i) > 0$, local-constant (Nadaraya–Watson) estimate is $\widehat{m}(x_0) := \frac{\sum_{i=1}^{n} K_\lambda(x_0, X_i) Y_i}{\sum_{i=1}^{n} K_\lambda(x_0, X_i)}$.

Metric windows set $h_\lambda(x_0) = \lambda$; nearest-neighbor windows set $h_\lambda(x_0) = \operatorname{dist}(x_0, X_{(k)}(x_0))$.

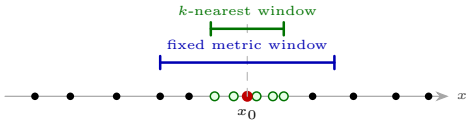

Figure 241: Fixed metric and k-nearest windows around x_0.

Example 6.1.11. *Common Local-Smoothing Kernels*
In one dimension, common compact profiles are $D_E(t) := \frac{3}{4}(1-t^2)\mathbf{1}_{\{|t|\le 1\}}$ for Epanechnikov smoothing and $D_T(t) := (1-|t|^3)^3\mathbf{1}_{\{|t|\le 1\}}$ for tri-cube smoothing. Both have support $[-1,1]$; Gaussian profiles have full support, and normalization constants cancel in regression-weight averages but matter for density estimation.

Method 6.1.12. *Local Linear Regression Smoothing*
Fix x_0 and bandwidth $h > 0$.

(i) *Weighted fit*: Set $(\widehat{\beta}_0(x_0), \widehat{\beta}_1(x_0)) \in \operatorname*{argmin}_{\beta_0,\beta_1} \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1(X_i - x_0))^2 K(\frac{X_i - x_0}{h})$.

(ii) *Estimate*: Define $\widehat{m}_h(x_0) := \widehat{\beta}_0(x_0)$.

Smaller h gives a more variable fit; larger h gives more bias.

Method 6.1.13. *Local Polynomial Regression*
Fix a query point $x_0 \in \mathbb{R}^p$ and degree $d \in \mathbb{N}_0$.

(i) *Basis*: Let $\mathcal{A}_d := \{\alpha \in \mathbb{N}_0^p : |\alpha| \le d\}$ and $b_d(u) := (u^\alpha)_{\alpha\in\mathcal{A}_d}$, where $u^\alpha := \prod_{j=1}^{p} u_j^{\alpha_j}$.

(ii) *Weights*: Set $K_i(x_0) := K_\lambda(x_0, X_i)$.

(iii) *Weighted fit*: For $\beta = (\beta_\alpha)_{\alpha\in\mathcal{A}_d}$, choose $\widehat{\beta}(x_0) \in \operatorname*{argmin}_{\beta\in\mathbb{R}^{|\mathcal{A}_d|}} \sum_{i=1}^{n} K_i(x_0)(Y_i - b_d(X_i - x_0)^\top\beta)^2$.

(iv) *Estimator*: With $\mathbf{0} \in \mathcal{A}_d$ the constant multi-index, define $\widehat{m}_d(x_0) := \widehat{\beta}_{\mathbf{0}}(x_0)$.

$d = 0$ is local constant; $d = 1$ is local linear with boundary correction.
Larger d trades lower curvature bias for higher variance.

Proposition 6.1.14. *Equivalent Kernel Moments*
Fix $x_0 \in \mathbb{R}$ and set $u_i := X_i - x_0$, $K_i := K_\lambda(x_0, X_i)$, $z_i := (1, u_i)^\top$, $Z := (z_1^\top, \dots, z_n^\top)^\top$, $W := \operatorname{diag}(K_1, \dots, K_n)$, $Y := (Y_1, \dots, Y_n)^\top$, and $G := Z^\top W Z$.
Assume G is nonsingular, let $e_1 := (1,0)^\top$, $\widehat{\beta}(x_0) := G^{-1}Z^\top WY$, and $\widehat{m}(x_0) := e_1^\top\widehat{\beta}(x_0)$.

(i) Equivalent weights: $\widehat{m}(x_0) = \sum_{i=1}^{n} \ell_i(x_0) Y_i$ with $\ell_i(x_0) := K_i z_i^\top G^{-1} e_1$.

(ii) Moment constraints: $\sum_{i=1}^{n} \ell_i(x_0) = 1$ and $\sum_{i=1}^{n} u_i \ell_i(x_0) = 0$.

(iii) Affine reproduction: If $Y_i = a + bX_i$, then $\widehat{m}(x_0) = a + bx_0$.

Proof.

(i) *Equivalent weights*: The weighted criterion $(Y - Z\beta)^\top W(Y - Z\beta)$ has normal equations $G\widehat{\beta} = Z^\top WY$, so $\widehat{\beta} = G^{-1}Z^\top WY$. Since G^{-1} is symmetric, $\widehat{m}(x_0) = e_1^\top G^{-1}Z^\top WY = \sum_{i=1}^{n} K_i e_1^\top G^{-1} z_i Y_i = \sum_{i=1}^{n} \ell_i(x_0) Y_i$.

(ii) *Moment constraints*: With $\ell(x_0) := (\ell_1(x_0), \dots, \ell_n(x_0))^\top = WZG^{-1}e_1$, $Z^\top\ell(x_0) = Z^\top WZG^{-1}e_1 = e_1$, hence $\sum_i \ell_i(x_0) = 1$ and $\sum_i u_i \ell_i(x_0) = 0$.

(iii) *Affine reproduction*: If $Y_i = a + bX_i = (a + bx_0) + bu_i$, then $\widehat{m}(x_0) = (a + bx_0)\sum_i \ell_i(x_0) + b\sum_i u_i\ell_i(x_0) = a + bx_0$.

Structural Commentary: Weighted projection onto $\operatorname{span}\{1, x - x_0\}$ imposes two weight moments.
Key Insight: $Z^\top\ell = e_1$ encodes constant reproduction and zero first moment. □

Definition 6.1.15. *Structured Kernel Metric*
Fix $x_0, x \in \mathbb{R}^p$, bandwidth $\lambda > 0$, profile $D \ge 0$, and a positive semidefinite matrix $A \succeq 0$.

(i) *Quadratic scale*: Define $\rho_A(x, x_0) := (x - x_0)^\top A(x - x_0)$.

(ii) *Kernel*: Set $K_{\lambda,A}(x_0, x) := D(\rho_A(x, x_0)/\lambda)$.

A sets coordinate weights, correlations, and active directions.
Standardise predictors first since ρ_A is unit-dependent.

Method 6.1.16. *Local Likelihood and Varying Coefficients*
Let $\ell_i(\theta)$ be the log-likelihood contribution of observation i under parameter θ.

(i) *Local likelihood*: At target x_0, set $K_i(x_0) := K_\lambda(x_0, X_i)$ and choose $\widehat{\theta}(x_0) \in \operatorname*{argmax}_\theta \sum_{i=1}^{n} K_i(x_0)\ell_i(\theta)$.

(ii) *Local parameterisation*: In generalised models, replace a constant natural parameter by $\eta_i(\gamma; x_0) := b_d(X_i - x_0)^\top\gamma$ and maximize $\sum_i K_i(x_0)\ell_i(\eta_i(\gamma; x_0))$ over γ.

(iii) *Varying coefficients*: For predictors (U, Z) with $U \in \mathbb{R}^q$, model $m(U, Z) := \alpha(Z) + \sum_{r=1}^{q} \beta_r(Z) U_r$.

(iv) *Local fit*: At z_0, choose $(\widehat{\alpha}, \widehat{\beta}) \in \operatorname{argmin}_{a,b} \sum_i K_\lambda(z_0, Z_i)(Y_i - a - b^\top U_i)^2$ and set $(\widehat{\alpha}(z_0), \widehat{\beta}(z_0)) := (\widehat{\alpha}, \widehat{\beta})$.

The coefficients are ordinary regression coefficients locally in U, but functions of the index variable Z.

Remark 6.1.17. *Bandwidth, Boundaries, and Dimension*
Let $N_\lambda(x_0) := \{i : K_\lambda(x_0, X_i) > 0\}$ and $n_\lambda(x_0) := |N_\lambda(x_0)|$.

(i) *Bias–variance*: Larger λ usually increases $n_\lambda(x_0)$, lowering variance, raising bias; smaller λ does reverse.
(ii) *Windows*: Metric windows fix radius; k-nearest-neighbor windows choose $\lambda(x_0)$ so that $n_{\lambda(x_0)}(x_0) \approx k$.
(iii) *Dimension*: In $\mathbb{R}^p$, neighborhoods must expand with p to keep $n_\lambda(x_0)$ stable, reducing local resolution.
(iv) *Cost*: Without indexing, one query evaluates $K_\lambda(x_0, X_i)$ for all i, costing $O(n)$.

Method 6.1.18. *Bandwidth Choice by Leave-One-Out Cross-Validation*

(i) *Candidates*: Fix bandwidths $h_1, \dots, h_M > 0$.

(ii) *Leave-one-out score*: Let $\widehat{m}_{h_k,-i}$ be local linear fit excluding (X_i, Y_i). Set $\mathrm{CV}(h_k) := \sum_{i=1}^{n} (Y_i - \widehat{m}_{h_k,-i}(X_i))^2$.

(iii) *Selection*: Choose $\widehat{h} \in \operatorname{argmin}_{1 \le k \le M} \mathrm{CV}(h_k)$, with a visual smoothness check.

Remark 6.1.19. *Complexity Control*
For a function class $\mathcal{F}$, estimation is stable when $\sup_{f \in \mathcal{F}} \left| \widehat{P}_n f - P f \right|$ is small with high probability or a.s.
Smoothness and entropy control this supremum in place of finite dimension.

6.2 Sparse High-Dimensional Inference

High-dimensional statistics studies regimes where the number of candidate parameters is comparable to, or much larger than, the sample size. Estimation is possible only after adding structure.

Definition 6.2.1. *High-Dimensional Regime and Sparsity*

(i) *High dimension*: A sequence has dimension $p = p_n$, often with $p_n \gg n$.
(ii) *Sparsity*: For $\theta \in \mathbb{R}^p$, $\|\theta\|_0 := |\{j : \theta_j \neq 0\}|$; θ is s-sparse if $\|\theta\|_0 \le s$.

Definition 6.2.2. *Lasso Estimator*

(i) *Lasso*: For $y = X\beta_0 + \varepsilon$ and $\lambda \ge 0$, $\widehat{\beta}_\lambda \in \arg\min_{\beta \in \mathbb{R}^p} \{(2n)^{-1} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1\}$.

The convex ℓ^1 penalty promotes sparsity.

Definition 6.2.3. *Restricted Eigenvalue Condition*

(i) *Restricted eigenvalue*: For $S \subseteq [p] := \{1, \dots, p\}$, $C(S) := \{v : \|v_{S^c}\|_1 \le 3 \|v_S\|_1\}$, and $\kappa > 0$,
X satisfies $\mathrm{RE}(S, \kappa)$ if $n^{-\frac{1}{2}} \|Xv\|_2 \ge \kappa \|v_S\|_2$ for all $v \in C(S)$.

Theorem 6.2.4. *Basic Lasso Rate Form*
Let $y = X\beta_0 + \varepsilon$, $S := \mathrm{supp}(\beta_0)$, $|S| \le s$, and let $\widehat{\beta}_\lambda$ be any lasso solution.
If X satisfies $\mathrm{RE}(S, \kappa)$ and $\lambda > 0$ with $\lambda \ge 2 \left\| \frac{X^\top \varepsilon}{n} \right\|_\infty$, then $n^{-1} \left\| X(\widehat{\beta}_\lambda - \beta_0) \right\|_2^2 \le \frac{9 s \lambda^2}{\kappa^2}$ and $\left\| \widehat{\beta}_\lambda - \beta_0 \right\|_1 \le \frac{12 s \lambda}{\kappa^2}$.

Proof. Set $\delta := \widehat{\beta}_\lambda - \beta_0$. The lasso optimality inequality gives $(2n)^{-1} \|X\delta\|_2^2 + \lambda \|\beta_0 + \delta\|_1 \le n^{-1} \varepsilon^\top X\delta + \lambda \|\beta_0\|_1$.
Since $\lambda \ge 2 \left\| \frac{X^\top \varepsilon}{n} \right\|_\infty$, $n^{-1} \varepsilon^\top X\delta \le \frac{\lambda}{2} \|\delta\|_1$, and since $\beta_{0,S^c} = 0$, $\|\beta_0\|_1 - \|\beta_0 + \delta\|_1 \le \|\delta_S\|_1 - \|\delta_{S^c}\|_1$.
Thus $(2n)^{-1} \|X\delta\|_2^2 + \frac{\lambda}{2} \|\delta_{S^c}\|_1 \le \frac{3\lambda}{2} \|\delta_S\|_1$, so $\delta \in C(S)$. By $\mathrm{RE}(S, \kappa)$, $\|\delta_S\|_1 \le \sqrt{s} \|\delta_S\|_2 \le \frac{\sqrt{s} \|X\delta\|_2}{\kappa \sqrt{n}}$.
Hence $n^{-1} \|X\delta\|_2^2 \le \frac{3\lambda \sqrt{s} \|X\delta\|_2}{\kappa \sqrt{n}}$, giving $n^{-1} \|X\delta\|_2^2 \le \frac{9 s \lambda^2}{\kappa^2}$; also $\|\delta\|_1 \le 4 \|\delta_S\|_1 \le \frac{12 s \lambda}{\kappa^2}$.
Structural Commentary: Basic inequality plus RE converts score control into prediction and ℓ^1 error bounds.
Key Insight: The ℓ^1 penalty forces the error into the RE cone. □

Method 6.2.5. *Featurewise Two-Sample Screening*
Let x_{ij} be feature $j \in [M]$ for observation i, with groups C_1, C_2 and $n_\ell := |C_\ell|$.

(i) *Group means*: For $\ell \in \{1, 2\}$, set $\bar{x}_{\ell j} := n_\ell^{-1} \sum_{i \in C_\ell} x_{ij}$.

(ii) *Pooled scale*: Set $\widehat{\sigma}_j^2 := (n_1 + n_2 - 2)^{-1} \{ \sum_{i \in C_1} (x_{ij} - \bar{x}_{1j})^2 + \sum_{i \in C_2} (x_{ij} - \bar{x}_{2j})^2 \}$ and $\mathrm{se}_j := \widehat{\sigma}_j \sqrt{n_1^{-1} + n_2^{-1}}$.

(iii) *Statistic*: Define $t_j := (\bar{x}_{2j} - \bar{x}_{1j}) / \mathrm{se}_j$; large $|t_j|$ flags a featurewise mean shift.
(iv) *Family*: Screening acts on $\{t_j : 1 \le j \le M\}$, so error control targets the family, not a single statistic.

Remark 6.2.6. *Statistical and Computational Structure*
High-dimensional inference couples concentration bounds, convex optimization, and selection error control; computation and statistics are part of the same guarantee.

6.3 Rank-Based Comparative Tests

Rank methods replace numerical values by their relative order. They are invariant under strictly increasing transformations and reduce the influence of extreme observations, which makes them natural companions to the two-sample and paired t procedures.

Definition 6.3.1. *Mann–Whitney and Rank-Sum Statistics*
Let $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$ be independent continuous samples.
Rank the pooled observations increasingly.

(i) *Pooled size*: $N := m + n$.
(ii) *Y-ranks*: $R_j^Y := \operatorname{rank}(Y_j)$ in the pooled sample.
(iii) *Rank sum*: $T_Y := \sum_{j=1}^{m} R_j^Y$.
(iv) *Mann–Whitney statistic*: $U_Y := T_Y - \frac{m(m+1)}{2}$.

Under continuity, $U_Y = \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{1}_{\{X_i < Y_j\}}$ a.s., so U_Y counts cross-sample pairs with $Y_j > X_i$.

Theorem 6.3.2. *Null Mean and Variance of the Rank Sum*
Let $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$ be independent continuous samples.
Let rank-sum T_Y and Mann–Whitney statistic U_Y. Assume $F = G$ and set $N := m + n$.

(i) Rank law: $T_Y = \sum_{r \in A} r$, where A is uniform over all m-subsets of $\{1, \dots, N\}$.
(ii) Rank-sum moments: $\mathbb{E}T_Y = \frac{m(N+1)}{2}$ and $\operatorname{Var}(T_Y) = \frac{mn(N+1)}{12}$.
(iii) Mann–Whitney moments: $\mathbb{E}U_Y = \frac{mn}{2}$ and $\operatorname{Var}(U_Y) = \frac{mn(N+1)}{12}$.

Proof.

(i) Continuity gives no ties and $F = G$ makes labels exchangeable, so the Y-rank set $A \subseteq \{1, \dots, N\}$ is uniform over all m-subsets and $T_Y = \sum_{r \in A} r$.
(ii) For ranks $1, \dots, N$, $\bar{r} = \frac{N+1}{2}$ and $S_r^2 = \frac{N^2-1}{12}$; sampling m ranks without replacement gives $\mathbb{E}T_Y = m\bar{r} = \frac{m(N+1)}{2}$ and $\operatorname{Var}(T_Y) = \frac{mS_r^2(N-m)}{N-1} = \frac{mn(N+1)}{12}$.
(iii) Since $U_Y = T_Y - \frac{m(m+1)}{2}$, shifting by a constant gives $\mathbb{E}U_Y = \frac{m(N+1)}{2} - \frac{m(m+1)}{2} = \frac{mn}{2}$ and $\operatorname{Var}(U_Y) = \operatorname{Var}(T_Y) = \frac{mn(N+1)}{12}$.

Structural Commentary: Under $F = G$, ranks reduce to simple random sampling without replacement.
Key Insight: U_Y is just a centered rank sum, so only the mean shifts. □

Method 6.3.3. *Mann–Whitney Test*
Given independent samples $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$, test $H_0 : F = G$:

(i) *Compute*: Pool and rank $m + n$ observations, use midranks for ties. Compute T_Y, $U_Y = T_Y - \frac{m(m+1)}{2}$.
(ii) *Calibrate*: For small m, n, enumerate m-subsets of $\{1, \dots, m+n\}$ to get the exact null law.
For large m, n, use $Z := \frac{U_Y - \frac{mn}{2}}{\sqrt{\frac{mn(m+n+1)}{12}}} \approx N(0,1)$, with tie correction when needed.
(iii) *Decide*: Compute the relevant one- or two-sided tail probability under the chosen null law.
Reject H_0 at level α when $p \le \alpha$.

Example 6.3.4. *Latent Heat Data by Ranks*
For the latent-heat data, take $m = 8$, $n = 13$, $N = 21$, and midranks for ties.
The smaller-sample rank sum is $R = 51$, its complement is $R' = m(N+1) - R = 125$, and $R^* = \min(R, R') = 51$.
Under equality of the two method distributions, $\mu_T = \frac{m(N+1)}{2} = 88$ and $\sigma_T = \sqrt{\frac{mn(N+1)}{12}} \approx 13.8$,
hence $z = \frac{R - \mu_T}{\sigma_T} \approx -2.68$ and the two-sided normal p-value is ≈ 0.007.
The exact rank-sum test therefore rejects equality at level 0.01.

Definition 6.3.5. *Mann–Whitney Effect Probability*
Let $X \sim F$ and $Y \sim G$ be independent, with samples $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$.

(i) *Effect probability*: $\pi := \mathbb{P}(X < Y)$.
(ii) *Estimator*: $\widehat{\pi} := (mn)^{-1} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{1}_{\{X_i < Y_j\}} = \frac{U_Y}{mn}$.

Thus π is the probability that a random Y exceeds a random X.

Method 6.3.6. *Shift Intervals from Pairwise Differences*
Assume the shift model $G(x) = F(x - \Delta)$.

(i) *Compute*: Set $D_{ij} := Y_j - X_i$ and order $D_{(1)} \le \cdots \le D_{(mn)}$.
(ii) *Calibrate*: Choose k so the two-sided rank test accepts a candidate shift when $k \le U_Y(\Delta) \le mn - k$.
(iii) *Invert*: Report $[D_{(k)}, D_{(mn-k+1)}]$ as the inverted shift interval.

The Hodges–Lehmann point estimate is $\widehat{\Delta}_{HL} := \operatorname{median}\{D_{ij} : 1 \le i \le n, 1 \le j \le m\}$.

Definition 6.3.7. *Signed-Rank Statistic*
Let $D_1, \dots, D_n$ be paired differences.

(i) *Nonzero index set*: $I := \{i : D_i \neq 0\}$.
(ii) *Absolute ranks*: $R_i := \operatorname{rank}(|D_i|)$ among $\{|D_\ell| : \ell \in I\}$, using midranks for ties.
(iii) *Signed-rank statistic*: $W_+ := \sum_{i \in I} R_i \mathbf{1}_{\{D_i > 0\}}$.

Theorem 6.3.8. *Signed-Rank Mean and Variance*
Under the signed-rank null, assume no zero differences and no ties in $|D_i|$.

(i) Mean: $\mathbb{E}W_+ = \frac{n(n+1)}{4}$.

(ii) Variance: $\mathrm{Var}(W_+) = \frac{n(n+1)(2n+1)}{24}$.

Proof. Let $\mathcal{R}$ be the absolute-rank information and let I_k indicate positivity at absolute-rank k.
Then $W_+ = \sum_{k=1}^{n} kI_k$ and, under symmetry, $I_k \mid \mathcal{R}$ are independent Bernoulli($\frac{1}{2}$).

(i) *Mean*: $\mathbb{E}(W_+ \mid \mathcal{R}) = \frac{1}{2}\sum_{k=1}^{n} k = \frac{n(n+1)}{4}$; the right side is deterministic, hence $\mathbb{E}W_+ = \frac{n(n+1)}{4}$.

(ii) *Variance*: $\mathrm{Var}(W_+ \mid \mathcal{R}) = \frac{1}{4}\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{24}$, while $\mathbb{E}(W_+ \mid \mathcal{R})$ is deterministic.

Total variance gives $\mathrm{Var}(W_+) = \frac{n(n+1)(2n+1)}{24}$.

Structural Commentary: Condition on absolute ranks; only independent fair signs remain.
Key Insight: W_+ is a weighted sum of Bernoulli signs with weights $1, \ldots, n$. □

Method 6.3.9. *Signed-Rank Test*
For paired differences D_i, test $H_0 : D_i \overset{d}{=} -D_i$.

(i) *Compute*: Discard zero differences, rank nonzero $|D_i|$ using midranks for ties, and compute W_+.
(ii) *Calibrate*: Conditional on the nonzero absolute ranks, enumerate the 2^r sign assignments, where $r := |I|$, to get the null law of W_+.
(iii) *Decide*: Compute the relevant tail probability and reject H_0 at level α when $p \le \alpha$.

The sign test is the coarser paired alternative using only signs, not ranks.

Example 6.3.10. *Paired Rank Analyses*
Two paired-rank calculations are:

(i) *Platelet data*: With one negative signed rank, use W_-; exact signed-rank test rejects symmetry at 0.01.
(ii) *Mercury data*: Dropping one zero difference and using midranks gives $n = 24$ and $W_+ = 194.5$. Under H_0, $\mathbb{E}_0 W_+ = \frac{24\cdot 25}{4} = 150$ and $\mathrm{Var}_0(W_+) = \frac{24\cdot 25\cdot 49}{24} = 1225$, so $Z = \frac{194.5-150}{35} \approx 1.27$ and two-sided $p \approx 0.20$; there is no strong evidence of a systematic shift.

Remark 6.3.11. *Rank Tests and Normal Theory*
Rank tests use order information, require no normal model, and are robust to outliers; under normal models their efficiency relative to t tests is often close to 1.

7 Time Series Models

7.1 Stationarity, Autocovariance, and Linear Processes

Time-series statistics studies data ordered by time, where dependence is part of the signal rather than a nuisance to be removed. The process-level notions of strict and weak stationarity are fixed in Definition 4.5.1.

Definition 7.1.1. *Autocorrelation, Partial Autocorrelation, and Spectral Density*
Let $(X_t)_{t\in\mathbb{Z}}$ be weakly stationary with autocovariance $\gamma(h) := \mathrm{Cov}(X_t, X_{t-h})$ and $\gamma(0) > 0$.
For $h \ge 2$ and $t \in \mathbb{Z}$, set $\mathcal{H}_{t,h} := \mathrm{span}\{1, X_{t-1}, \ldots, X_{t-h+1}\}$ let $P_{t,h}$ be L^2-orthogonal projection onto $\mathcal{H}_{t,h}$.
Assume $\mathrm{Var}(X_t - P_{t,h}X_t) > 0$ and $\mathrm{Var}(X_{t-h} - P_{t,h}X_{t-h}) > 0$ $\forall h \ge 2,\ t \in \mathbb{Z}$.

(i) *Autocorrelation function*: $\rho : \mathbb{Z} \to [-1, 1]$, $\rho(h) := \frac{\gamma(h)}{\gamma(0)}$.

(ii) *Partial autocorrelation function*: $\alpha : \mathbb{N}_0 \to [-1, 1]$, $\alpha(0) := 1$, $\alpha(1) := \frac{\gamma(1)}{\gamma(0)}$, and $\alpha(h) := \mathrm{Corr}(X_t - P_{t,h}X_t, X_{t-h} - P_{t,h}X_{t-h})$ $\forall h \ge 2,\ t \in \mathbb{Z}$.

(iii) *Spectral density*: If $\sum_{h\in\mathbb{Z}} |\gamma(h)| < \infty$, $f_X : [-\pi, \pi] \to \mathbb{C}$, $f_X(\lambda) := (2\pi)^{-1} \sum_{h\in\mathbb{Z}} \gamma(h)e^{-ih\lambda}$ $\forall \lambda \in [-\pi, \pi]$.

Definition 7.1.2. *White Noise and Linear Process*
Let $\sigma^2 \ge 0$, $\mu \in \mathbb{R}$, and $(\psi_j)_{j\ge 0} \subset \mathbb{R}$ satisfy $\sum_{j=0}^{\infty} |\psi_j| < \infty$; all processes below are real-valued, indexed by $\mathbb{Z}$.

(i) *White noise*: $(\varepsilon_t)_{t\in\mathbb{Z}} \subset L^2$, $\mathbb{E}\varepsilon_t = 0$ and $\mathrm{Var}(\varepsilon_t) = \sigma^2$ $\forall t \in \mathbb{Z}$, and $\mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0$ $\forall s, t \in \mathbb{Z}$ with $s \ne t$.

(ii) *Linear process*: $(X_t)_{t\in\mathbb{Z}}$ satisfies $X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ $\forall t \in \mathbb{Z}$ for some $(\varepsilon_t)_{t\in\mathbb{Z}}$ satisfying (i).

Structural Commentary: $\sum_j |\psi_j| < \infty$ gives an L^2-stable one-sided innovation filter.

Example 7.1.3. *Autoregression of Order One*
Let $(\varepsilon_t)_{t\in\mathbb{Z}}$ be white noise, variance σ^2, $|\phi| < 1$. For $X_t = \phi X_{t-1} + \varepsilon_t$, stationary solution is $X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$.
Thus $\gamma(h) = \mathrm{Cov}(X_t, X_{t-h}) = \frac{\sigma^2\phi^{|h|}}{1-\phi^2}$, $\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^{|h|}$. Autocorrelation decays geometrically at rate $|\phi|$.

Remark 7.1.4. *Process-Level Prerequisites*
Time-series inference uses filtrations $(\mathcal{F}_t)$, adapted X_t, predictors $\mathbb{E}(X_{t+h} \mid \mathcal{F}_t)$, martingale differences ε_t, and stationarity or ergodicity assumptions.

7.2 ARIMA, State-Space, and Volatility Models

Classical time-series models combine deterministic filters, stochastic innovations, and likelihood-based estimation.

Definition 7.2.1. *ARMA Model*
Let $p, q \in \mathbb{N}_0$, $\phi_i \in \mathbb{R}$ $(1 \le i \le p)$, $\theta_j \in \mathbb{R}$ $(1 \le j \le q)$, and let B denote the backshift, $(BX)_t = X_{t-1}$.
Set $\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$ and $\theta(z) := 1 + \sum_{j=1}^{q} \theta_j z^j$; empty sums are zero.
$(X_t)_{t\in\mathbb{Z}}$ is ARMA(p, q) with innovations $(\varepsilon_t)_{t\in\mathbb{Z}}$ iff (ε_t) is white noise and $\phi(B)X_t = \theta(B)\varepsilon_t$ $\forall t \in \mathbb{Z}$.
Structural Commentary: Causal iff $\phi(z) \neq 0$ for $|z| \le 1$; invertible iff $\theta(z) \neq 0$ for $|z| \le 1$.
Key Insight: AR roots control recursion; MA roots control innovation recovery.

Definition 7.2.2. *ARIMA Model*
Let $p, q, d \in \mathbb{N}_0$, $\phi_i \in \mathbb{R}$ $(1 \le i \le p)$, $\theta_j \in \mathbb{R}$ $(1 \le j \le q)$, and let B denote the backshift.
Set $\Delta^d := (1 - B)^d$, $\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$, and $\theta(z) := 1 + \sum_{j=1}^{q} \theta_j z^j$.
$(X_t)_{t\in\mathbb{Z}}$ is ARIMA(p, d, q) with innovations $(\varepsilon_t)_{t\in\mathbb{Z}}$ iff (ε_t) is white noise and $\phi(B)\Delta^d X_t = \theta(B)\varepsilon_t$ $\forall t \in \mathbb{Z}$.
Structural Commentary: $Y_t := \Delta^d X_t$ satisfies ARMA(p, q) with the same innovations.
Key Insight: $d > 0$ shifts stationarity from X_t to $\Delta^d X_t$.

Definition 7.2.3. *Filtering Problem*
Let $(Z_t, Y_t)_{t\ge1}$ be a hidden-state model with state space $(E, \mathcal{E})$, and set $\mathcal{F}_t^Y := \sigma(Y_1, \dots, Y_t)$.
For $A \in \mathcal{E}$, $\pi_t(A) := \mathbb{P}(Z_t \in A \mid \mathcal{F}_t^Y)$; the task is recursive propagation or approximation of $(\pi_t)_{t\ge1}$.
Structural Commentary: (π_t) is an $\mathcal{F}_t^Y$-adapted probability-kernel-valued process.
Key Insight: Conditioning converts hidden dynamics into an observable belief-state recursion.

Definition 7.2.4. *Linear Gaussian State-Space Model*
Let $m, r, n \in \mathbb{N}$, $F \in \mathbb{R}^{m\times m}$, $G \in \mathbb{R}^{m\times r}$, $H \in \mathbb{R}^{n\times m}$, and $Q \in \mathbb{R}^{r\times r}$, $R \in \mathbb{R}^{n\times n}$ with $Q, R \succeq 0$.
Let (η_t) and (ε_t) be mutually independent Gaussian noises with $\eta_t \sim N(0, Q)$ and $\varepsilon_t \sim N(0, R)$.
Let α_0 be independent Gaussian. $(\alpha_t, Y_t)_{t\in\mathbb{N}_0}$ is *linear Gaussian state-space* iff
$\alpha_t \in \mathbb{R}^m$, $Y_t \in \mathbb{R}^n$, $\alpha_{t+1} = F\alpha_t + G\eta_t$, $Y_t = H\alpha_t + \varepsilon_t$ $\forall t \in \mathbb{N}_0$.
Key Insight: With $\mathcal{F}_t^Y := \sigma(Y_0, \dots, Y_t)$, Kalman filtering recurses $(a_t, P_t) := (\mathbb{E}(\alpha_t \mid \mathcal{F}_t^Y), \mathrm{Var}(\alpha_t \mid \mathcal{F}_t^Y))$.

Definition 7.2.5. *ARCH and GARCH Volatility*
Let $p, q \in \mathbb{N}_0$, $\omega > 0$, $\alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_p \ge 0$, and let $(z_t)_{t\in\mathbb{Z}}$ be standardized white noise.
All processes below are real-valued and indexed by $\mathbb{Z}$; set $A := \sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j$.

(i) *ARCH(q)*: $(X_t, \sigma_t)_{t\in\mathbb{Z}}$, $\sigma_t \ge 0$, satisfies $X_t = \sigma_t z_t$ and $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i X_{t-i}^2$ $\forall t \in \mathbb{Z}$.

(ii) *GARCH(p, q)*: $(X_t, \sigma_t)_{t\in\mathbb{Z}}$, $\sigma_t \ge 0$, satisfies $X_t = \sigma_t z_t$ and $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i X_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2$ $\forall t \in \mathbb{Z}$.

Structural Commentary: $\omega > 0$, $\alpha_i, \beta_j \ge 0$ force $\sigma_t^2 \ge 0$; by stationary GARCH, $A < 1$ has $\mathbb{E}X_t^2 = \frac{\omega}{1-A}$.

Remark 7.2.6. *Likelihood by Filtering*
If one-step densities exist, likelihood filtering has $S_t = \mathcal{T}_\vartheta(S_{t-1}, W_t)$ and $\ell_T(\vartheta) = \sum_{t=1}^{T} \log p_\vartheta(W_t \mid S_{t-1})$.
Inference depends on stability of S_t.

Method 7.2.7. *Estimating a Finite Transition Matrix*

(i) Observe finite homogeneous chain $X_0, \dots, X_n$, let $n_{ij} := \sum_{t=0}^{n-1} \mathbf{1}\{X_t = i, X_{t+1} = j\}$, $n_i := \sum_j n_{ij}$.

(ii) The row likelihood is proportional to $\prod_j p_{ij}^{n_{ij}}$ subject to $\sum_j p_{ij} = 1$.

(iii) If $n_i > 0$, the rowwise MLE is $\widehat{p}_{ij} = \frac{n_{ij}}{n_i}$.

(iv) If $n_i = 0$, row i is not identifiable without a convention or prior.

8 Sequential Inference and Resampling

8.1 Resampling, Bootstrap, and Permutation Methods

Resampling methods approximate sampling distributions by recomputing statistics on data-generated pseudo-samples.

Definition 8.1.1. *Nonparametric Bootstrap*
For $X_{1:n} \sim P^{\otimes n}$, set $\widehat{P}_n := n^{-1} \sum_{i=1}^{n} \delta_{X_i}$. Conditional on $X_{1:n}$, let $X_{1:n}^* \sim \widehat{P}_n^{\otimes n}$ and $\widehat{P}_n^* := n^{-1} \sum_{i=1}^{n} \delta_{X_i^*}$.
For any law functional T, $\mathcal{L}(T(\widehat{P}_n^*) - T(\widehat{P}_n) \mid X_{1:n})$ approximates $\mathcal{L}(T(\widehat{P}_n) - T(P))$.
Key Insight: P is replaced by $\widehat{P}_n$ before pseudo-sampling.

Method 8.1.2. *Bootstrap Standard Error*
Let $\widehat{\psi} = s(X_{1:n})$ be observed statistic, $\mathcal{R}_*(X_{1:n})$ be the chosen bootstrap law on pseudo-samples, $B \ge 2$.

(i) *Resample*: $X_{1:n}^{*b} \sim \mathcal{R}_*(X_{1:n})$ for $b = 1, \dots, B$.

(ii) *Recompute*: $\widehat{\psi}^{*b} := s(X_{1:n}^{*b})$ for $b = 1, \dots, B$.

(iii) *Estimate*: $\widehat{\mathrm{se}}_*(\widehat{\psi}) := \{(B-1)^{-1} \sum_{b=1}^{B} (\widehat{\psi}^{*b} - \overline{\psi}^*)^2\}^{\frac{1}{2}}$, where $\overline{\psi}^* := B^{-1} \sum_{b=1}^{B} \widehat{\psi}^{*b}$.

Method 8.1.3. *Bootstrap Variability for Location Summaries*
Let $T = T(x_1, \dots, x_n)$ be a location summary and let $B \geq 1$.

(i) *Resample*: For $1 \leq b \leq B$, draw $x_1^{*(b)}, \dots, x_n^{*(b)}$ iid from $\widehat{F}_n := n^{-1} \sum_{i=1}^{n} \delta_{x_i}$.

(ii) *Recompute*: Set $T_b^* := T(x_1^{*(b)}, \dots, x_n^{*(b)})$ and $\overline{T}^* := B^{-1} \sum_{b=1}^{B} T_b^*$.

(iii) *Spread*: Estimate variability by $\widehat{\mathrm{se}}_*(T) := \left((B-1)^{-1} \sum_{b=1}^{B} (T_b^* - \overline{T}^*)^2 \right)^{1/2}$.

Key Insight: This is useful for medians and trimmed means, where analytic standard errors can be tail-sensitive.

Definition 8.1.4. *Parametric Bootstrap*
Let $\{P_\theta : \theta \in \Theta\}$ be parametric model for $X_{1:n} := (X_1, \dots, X_n)$, let $\widehat{\theta} := \widehat{\theta}(X_{1:n})$, and let S be a statistic.
$X_1^*, \dots, X_n^* \mid X_{1:n} \sim P_{\widehat{\theta}}^{\otimes n}$, and $\mathcal{L}(S(X_{1:n}^*) \mid X_{1:n})$ is the fitted-model resampling law for $S(X_{1:n})$.

Key Insight: The model class $\{P_\theta\}$ is retained; only θ is replaced by $\widehat{\theta}$.

Method 8.1.5. *Parametric Bootstrap Standard Errors and Intervals*
Let $\{P_\theta : \theta \in \Theta\}$ be parametric model, $\widehat{\psi} := s(X_{1:n})$ estimate $\psi(\theta)$, fix $B \geq 2$, fix confidence level $1 - \alpha$.

(i) *Fit*: $\widehat{\theta} := \widehat{\theta}(X_{1:n})$.

(ii) *Simulate*: $X_{1:n}^{*b} \sim P_{\widehat{\theta}}^{\otimes n}$ and $\widehat{\psi}^{*b} := s(X_{1:n}^{*b})$ for $b = 1, \dots, B$.

(iii) *Standard error*: $\widehat{\mathrm{se}}_*(\widehat{\psi}) := \{(B-1)^{-1} \sum_{b=1}^{B} (\widehat{\psi}^{*b} - \overline{\psi}^*)^2\}^{\frac{1}{2}}$, where $\overline{\psi}^* := B^{-1} \sum_{b=1}^{B} \widehat{\psi}^{*b}$.

(iv) *Pivotal interval*: for empirical quantiles $q_{\frac{\alpha}{2}}^*$ and $q_{1-\frac{\alpha}{2}}^*$ of $\{\widehat{\psi}^{*b} - \widehat{\psi}\}_{b=1}^B$, use $[\widehat{\psi} - q_{1-\frac{\alpha}{2}}^*, \widehat{\psi} - q_{\frac{\alpha}{2}}^*]$.

The percentile interval uses empirical quantiles of $\{\widehat{\psi}^{*b}\}_{b=1}^B$ directly.
The pivotal and percentile intervals coincide when the bootstrap error law is symmetric about 0.

Remark 8.1.6. *Plug-In Monte Carlo*
The parametric bootstrap estimates model-dependent quantities by replacing the unknown parameter with its estimate before simulation. If $h(X, \theta)$ is a loss, error indicator, or other accuracy summary, then Monte Carlo from $P_{\widehat{\theta}}$ approximates $\mathbb{E}_\theta h(X, \theta)$ by the empirical average of $h(X^{*b}, \widehat{\theta})$.

Definition 8.1.7. *Permutation Test*
Let a finite group G act on the sample space $\mathcal{X}$, let P_0 satisfy $gX \overset{\text{dist}}{=} X$ under P_0 $\forall g \in G$, and let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic. With U_G uniform on G and independent of X, $\mathcal{L}(T(U_G X) \mid X)$, the empirical law on $\{T(gX) : g \in G\}$, is the reference law for tail probabilities.

Key Insight: Calibration follows from the invariance $gX \overset{\text{dist}}{=} X$ under P_0.

Method 8.1.8. *Permutation Plug-In FDR Estimate*
Let t_j be observed statistics for $j \in [M]$, and let $t_j^{(k)}$ be the statistic after label permutation $k \in [K]$.

(i) *Observed discoveries*: For cutpoint $C > 0$, set $R_{\text{obs}}(C) := \sum_{j=1}^{M} \mathbf{1}_{\{|t_j| > C\}}$.

(ii) *Null discoveries*: Estimate the expected false discoveries by $\widehat{\mathbb{E}}(V; C) := K^{-1} \sum_{k=1}^{K} \sum_{j=1}^{M} \mathbf{1}_{\{|t_j^{(k)}| > C\}}$.

(iii) *Plug-in FDR*: If $R_{\text{obs}}(C) > 0$, set $\widehat{\mathrm{FDR}}(C) := \widehat{\mathbb{E}}(V; C) / R_{\text{obs}}(C)$.

(iv) *Permutation p-values*: Use $\{t_j^{(k)}\}$ to compare $|t_j|$ with featurewise or pooled permutation null statistics.

Remark 8.1.9. *Asymmetric Cutpoints and SAM*
Let $t_{(1)} \leq \cdots \leq t_{(M)}$ and $\bar{t}_{(j)}^0 := K^{-1} \sum_{k=1}^{K} t_{(j)}^{(k)}$, where $t_{(j)}^{(k)}$ is the jth ordered permutation statistic.
For $d_j := t_{(j)} - \bar{t}_{(j)}^0$, $a > 0$, set $c_+(a) := \min\{t_{(j)} : d_j > a\}$, $c_-(a) := \max\{t_{(j)} : d_j < -a\}$ when nonempty.
Reject $t_j \in A(c_-, c_+) := (-\infty, c_-] \cup [c_+, \infty)$, estimate FDR by Method 8.1.8, $A(c_-, c_+)$ replace $\{|t| > C\}$.

Remark 8.1.10. *Validity Depends on the Resampling Law*
Bootstrap validity is asymptotic; permutation validity is finite-sample exact under invariance; both require the resampling law to match the target sampling variation.

8.2 Sequential Experiments, Filtering, and Online Inference

Sequential methods update decisions as observations arrive. They must account for optional stopping, adaptive sampling, and computational constraints.

Definition 8.2.1. *Sequential Test and Stopping Time*
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $I \subseteq \mathbb{N}_0$ be a time index, let $(\mathcal{F}_t)_{t \in I}$ be a filtration, and let $(D, \mathcal{D})$ be a decision space. For any $\tau : \Omega \to I \cup \{\infty\}$, set $\mathcal{F}_\tau := \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \ \forall t \in I\}$.

(i) *Stopping time*: $\tau : \Omega \to I \cup \{\infty\}$ satisfies $\{\tau \leq t\} \in \mathcal{F}_t$ $\forall t \in I$.

(ii) *Sequential test*: (τ, d) with τ a stopping time and $d : (\Omega, \mathcal{F}_\tau) \to (D, \mathcal{D})$ measurable.

Key Insight: Stop and decide using only pre-stopping information.

Definition 8.2.2. *Sequential Probability Ratio Test*
Let P_0 and P_1 be simple iid models with one-observation densities p_0 and p_1 for $(X_t)_{t \geq 1}$,
let $\Lambda_0 := 1$, $\Lambda_t := \prod_{i=1}^{t} \frac{p_1(X_i)}{p_0(X_i)}$, and choose $0 < A < B < \infty$.
$\tau := \inf\{t \geq 1 : \Lambda_t \in (0, A] \cup [B, \infty)\}$, with terminal decision $d_\tau = P_0$ on $\{\Lambda_\tau \leq A\}$ and $d_\tau = P_1$ on $\{\Lambda_\tau \geq B\}$.
Key Insight: First exit of $\log \Lambda_t$ from $(\log A, \log B)$ converts cumulative likelihood evidence into a stopping rule.

Remark 8.2.3. *Sequential Cross-References*
Sequential inference uses filtrations and martingales from stochastic processes, stability of recursions from dynamical systems, and simulation methods from Monte Carlo computation.

8.3 Two-Sample Resampling Methods

Two-sample resampling procedures must preserve the source of randomness relevant to the question. Bootstrap procedures usually resample within groups to estimate sampling variability under the fitted empirical distributions; permutation procedures pool labels when the null asserts exchangeability.

Method 8.3.1. *Two-Sample Nonparametric Bootstrap*

Let $x_{1:n}$ and $y_{1:m}$ be independent samples from F and G, let $\widehat{F}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i}$ and $\widehat{G}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j}$.

Let $T(F,G)$ be a two-sample functional, and fix $B \geq 2$.

(i) *Resample*: $x^{*b}_{1:n} \sim \widehat{F}_n^{\otimes n}$ and $y^{*b}_{1:m} \sim \widehat{G}_m^{\otimes m}$ independently for $b = 1,\dots,B$.

(ii) *Recompute*: with $\widehat{F}^{*b}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i^{*b}}$ and $\widehat{G}^{*b}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j^{*b}}$, set $\widehat{T}^{*b} := T(\widehat{F}^{*b}_n, \widehat{G}^{*b}_m)$.

(iii) *Summarize*: $s^*_T := \mathrm{sd}\{\widehat{T}^{*b} : 1 \leq b \leq B\}$; $I^*_{1-\alpha} := [q^*_{\frac{\alpha}{2}}, q^*_{1-\frac{\alpha}{2}}]$, where q^*_γ is the empirical γ-quantile.

This is the two-sample version of Method 8.1.2.

Example 8.3.2. *Bootstrap for $\pi = \mathbb{P}(X < Y)$*
Let $x_{1:n} \sim F^{\otimes n}$ and $y_{1:m} \sim G^{\otimes m}$ independently.
Set $\widehat{F}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i}$, $\widehat{G}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j}$, and $\widehat{\pi} := (mn)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbf{1}_{\{x_i<y_j\}}$.
For $b = 1,\dots,B$, draw $x^{*b}_{1:n} \sim \widehat{F}_n^{\otimes n}$ and $y^{*b}_{1:m} \sim \widehat{G}_m^{\otimes m}$ independently.
Set $\widehat{\pi}^{*b} := (mn)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbf{1}_{\{x_i^{*b}<y_j^{*b}\}}$, and use $s^*_\pi := \mathrm{sd}\{\widehat{\pi}^{*b} : 1 \leq b \leq B\}$.
Key Insight: Separate resampling targets $F \otimes G$ variation; rank-sum calibration imposes $F = G$.

Method 8.3.3. *Bootstrap for a Hodges–Lehmann Shift*
For independent samples $x_{1:n}$ and $y_{1:m}$ and $B \geq 2$, set $\widehat{\Delta}_{\mathrm{HL}} := \mathrm{median}\{y_j - x_i : 1 \leq i \leq n,\ 1 \leq j \leq m\}$, resample the two groups separately for $b = 1,\dots,B$, and use the empirical spread or quantiles of the recomputed medians $\widehat{\Delta}^{*b}_{\mathrm{HL}}$ for a standard error or interval.

Method 8.3.4. *Permutation Test for a Difference of Means*

Let $x_{1:n}$ be control responses, let $y_{1:m}$ be treatment responses, set $\overline{x} := n^{-1}\sum_{i=1}^{n}x_i$ and $\overline{y} := m^{-1}\sum_{j=1}^{m}y_j$.

Let $z_{1:N}$ with $N = n + m$ be the pooled responses, and set $T_{\mathrm{obs}} := \overline{y} - \overline{x}$.
Let H_0 denote no treatment effect, set $\mathcal{S}_{m,N} := \{S \subseteq \{1,\dots,N\} : |S| = m\}$, and use $S \mid H_0 \sim \mathrm{Unif}(\mathcal{S}_{m,N})$.

(i) *Relabel*: $S \in \mathcal{S}_{m,N}$, by enumeration or Monte Carlo draws from $\mathrm{Unif}(\mathcal{S}_{m,N})$.

(ii) *Recompute*: $T^*_S := m^{-1}\sum_{i\in S}z_i - n^{-1}\sum_{i\notin S}z_i$.

(iii) *Calibrate*: $p_{\mathrm{exact}} := \binom{N}{m}^{-1}\#\{S \in \mathcal{S}_{m,N} : |T^*_S| \geq |T_{\mathrm{obs}}|\}$, Monte Carlo approximation if $\binom{N}{m}$ large.

Remark 8.3.5. *Permutation and Rank Tests*
The permutation test for means and the Mann–Whitney test both use the randomization principle that, under the null, labels are exchangeable. They differ in the statistic: the permutation mean test uses the numerical values and is sensitive to outliers, while the Mann–Whitney test first replaces values by ranks.

Part XX
Statistical Data Analysis and Applied Models

Applied statistical analysis organizes data summaries, sampling designs, contingency tables, designed experiments, and regression models into reusable workflows. The inferential tools of Part 15 supply the testing, estimation, and uncertainty language used here.

1 Survey Sampling

1.1 Finite Population Parameters and Simple Random Sampling

Survey sampling treats a finite population as fixed and the sample-selection mechanism as random. The inferential target is often a population mean, total, or proportion, while the random variables come from the sampling design.

Definition 1.1.1. *Finite Population Parameters*
Let $U := \{1, \dots, N\}$ and fix $x_i \in \mathbb{R}$, $i \in U$.

(i) *Population mean*: $\mu := \frac{1}{N} \sum_{i=1}^{N} x_i$.

(ii) *Population total*: $\tau := \sum_{i=1}^{N} x_i = N\mu$.

(iii) *Population variance*: $\sigma^2 := \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^{N} x_i^2 - \mu^2$.

(iv) *Population proportion*: $\{x_i : i \in U\} \subseteq \{0, 1\} \implies p := \mu = \frac{1}{N} \sum_{i=1}^{N} x_i$ and $\sigma^2 = p(1-p)$.

Definition 1.1.2. *Simple Random Sampling*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, and fix $x_i \in \mathbb{R}$, $i \in U$.
Let S be a random n-element subset of U; for a random ordering $J_1, \dots, J_n$ of S, set $X_k := x_{J_k}$.

(i) *Simple random sample*: $\mathbb{P}(S = s) = \frac{1}{\binom{N}{n}}$, $\forall s \subseteq U$ with $|s| = n$.

(ii) *Sample mean*: $\overline{X} := \frac{1}{n} \sum_{i=1}^{n} X_i$.

(iii) *Total estimator*: $T := N\overline{X}$.

(iv) *Sample proportion*: $\{x_i : i \in U\} \subseteq \{0, 1\} \implies \widehat{p} := \overline{X}$.

Method 1.1.3. *Sequential Simple Random Sampling*
Fix target size $n \ge 1$, assume terminal stream length is $N \ge n$, N unknown while streaming.
Population units arrive sequentially as $1, \dots, N$. Let $S_t \subseteq \{1, \dots, t\}$ denote current reservoir after t arrivals.

(i) *Initialize*: for $1 \le t \le n$, set $S_t = \{1, \dots, t\}$.
(ii) *Admit*: for $t > n$, draw $B_t \sim \text{Bernoulli}(\frac{n}{t})$.
(iii) *Update*: if $B_t = 0$, set $S_t = S_{t-1}$; if $B_t = 1$, draw $J_t \mid S_{t-1} \sim \text{Unif}(S_{t-1})$ and set $S_t = (S_{t-1} \setminus \{J_t\}) \cup \{t\}$.
(iv) *Output*: after the stream ends, return S_N.

Then for every $t \ge n$, $S_t \sim \text{Unif}\{s \subseteq \{1, \dots, t\} : |s| = n\}$.
Hence S_N is a simple random sample without replacement.

Example 1.1.4. *Capture–Recapture Likelihood*
Suppose t population units are tagged and a later sample of size m contains $r > 0$ tagged units.
For population size n, likelihood is $L_n = \frac{\binom{t}{r}\binom{n-t}{m-r}}{\binom{n}{m}}$. Since $\frac{L_n}{L_{n-1}} = \frac{(n-t)(n-m)}{n(n-t-m+r)}$, L_n increases for $n < \frac{mt}{r}$ and decreases for $n > \frac{mt}{r}$.
Hence $\widehat{n} = \lfloor \frac{mt}{r} \rfloor$; for $(t, m, r) = (10, 20, 4)$, $\widehat{n} = 50$.

Lemma 1.1.5. *Law of One Sample Position*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Let $\mu := \frac{1}{N} \sum_{a=1}^{N} x_a$, $\sigma^2 := \frac{1}{N} \sum_{a=1}^{N} (x_a - \mu)^2$, $S \sim \text{Unif}\{s \subseteq U : |s| = n\}$.
If $(J_1, \dots, J_n)$ is a random ordering of S, set $X_k := x_{J_k}$.
Let $\zeta_1, \dots, \zeta_m$ be distinct values in $\{x_a : a \in U\}$, let $n_j := \#\{a \in U : x_a = \zeta_j\}$. For each $k \in \{1, \dots, n\}$:

(i) Distribution: the selected value X_k has the finite-population law $\mathbb{P}(X_k = \zeta_j) = \frac{n_j}{N}$ for $j = 1, \dots, m$.
(ii) Moments: the finite-population law has mean $\mathbb{E}X_k = \mu$ and variance $\text{Var}(X_k) = \sigma^2$.

Proof.

(i) For every $a \in U$, $\mathbb{P}(J_k = a) = \frac{1}{N}$, so $\mathbb{P}(X_k = \zeta_j) = \sum_{a : x_a = \zeta_j} \mathbb{P}(J_k = a) = \frac{n_j}{N}$.

(ii) From (i), $\mathbb{E}X_k = \frac{1}{N} \sum_{a=1}^{N} x_a = \mu$. By finite-population law, $\text{Var}(X_k) = \frac{1}{N} \sum_{a=1}^{N} (x_a - \mu)^2 = \sigma^2$.

Structural Commentary: Each sample position is marginally uniform on U.
Key Insight: Without replacement changes joint dependence, not the one-position marginal distribution. □

Theorem 1.1.6. *Unbiased SRS Estimators*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N} \sum_{a=1}^{N} x_a$, $\tau := \sum_{a=1}^{N} x_a = N\mu$. Let $S \sim \text{Unif}\{s \subseteq U : |s| = n\}$.
Given random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n} \sum_{k=1}^{n} X_k$, and $T := N\overline{X}$. Then:

(i) Mean estimator: the sample mean is design-unbiased for the finite-population mean, $\mathbb{E}\overline{X} = \mu$.

(ii) Total estimator: the expansion estimator is design-unbiased for the finite-population total, $\mathbb{E}T = \tau$.

Proof.

(i) For each k, $\mathbb{P}(J_k = a) = \frac{1}{N}\ \forall a \in U$, so $\mathbb{E}X_k = \frac{1}{N}\sum_{a=1}^{N} x_a = \mu$. By linearity $\mathbb{E}\overline{X} = \frac{1}{n}\sum_{k=1}^{n}\mathbb{E}X_k = \mu$.

(ii) Since $T = N\overline{X}$, $\mathbb{E}T = N\mathbb{E}\overline{X} = N\mu = \tau$.

Structural Commentary: $T = N\overline{X}$ rescales the mean estimate.
Key Insight: Equal position marginals give unbiasedness; independence is unnecessary. □

Lemma 1.1.7. *Negative Dependence Without Replacement*
Let $U := \{1, \dots, N\}$, $2 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\tau := \sum_{a=1}^{N} x_a$, $\sigma^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu)^2$. Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
For random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$. If $i, j \in \{1, \dots, n\}$ and $i \ne j$, then $\mathrm{Cov}(X_i, X_j) = -\frac{\sigma^2}{N-1}$.

Proof. Let $(Y_1, \dots, Y_N)$ be a uniformly random ordering of $(x_1, \dots, x_N)$.
Each Y_a has variance σ^2, and symmetry gives $\mathrm{Cov}(Y_a, Y_b) = \gamma$ for $a \ne b$.
Since $\sum_{a=1}^{N} Y_a = \tau$ is fixed, $0 = \mathrm{Var}(\sum_{a=1}^{N} Y_a) = N\sigma^2 + N(N-1)\gamma$.
Thus $\gamma = -\frac{\sigma^2}{N-1}$, and (X_i, X_j) has same law as any two distinct positions in full random ordering.
Structural Commentary: A fixed total forces negative pairwise covariance.
Key Insight: Without replacement, one large draw leaves less for the rest. □

Theorem 1.1.8. *Variance and Finite Population Correction*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N}\sum_{a=1}^{N} x_a$ and $\sigma^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu)^2$. Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, and $T := N\overline{X}$. Then:

(i) Mean variance: the sample mean has design variance $\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right)$.
(ii) Total variance: the total estimator has design variance $\mathrm{Var}(T) = N^2\,\mathrm{Var}(\overline{X})$.

Proof.

(i) Each X_k has variance σ^2. Let $(Y_1, \dots, Y_N)$ be a uniformly random ordering of $(x_1, \dots, x_N)$, set $\tau := \sum_{a=1}^{N} x_a$, and let γ be the common covariance of Y_a, Y_b for $a \ne b$.
Since $\sum_{a=1}^{N} Y_a = \tau$, $0 = \mathrm{Var}(\sum_{a=1}^{N} Y_a) = N\sigma^2 + N(N-1)\gamma$, so $\gamma = -\frac{\sigma^2}{N-1}$.
Therefore $\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\left(n\sigma^2 + n(n-1)\left[-\frac{\sigma^2}{N-1}\right]\right) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right)$.

(ii) The total estimator satisfies $T = N\overline{X}$, so $\mathrm{Var}(T) = \mathrm{Var}(N\overline{X}) = N^2\,\mathrm{Var}(\overline{X})$.

Structural Commentary: The factor $\frac{N-n}{N-1}$ is the *finite population correction*.
Key Insight: Averaging gives $\frac{1}{n}$ scaling; without replacement subtracts negative covariance. □

Corollary 1.1.9. *Estimated Standard Errors*
Let $U := \{1, \dots, N\}$, $1 < n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$, and let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$. For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $T := N\overline{X}$, and $s^2 := \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \overline{X})^2$.
Then practical plug-in estimated variances are:

(i) Mean: the plug-in estimated variance of the sample mean is $s^2_{\overline{X}} = \frac{s^2}{n}\left(1 - \frac{n}{N}\right)$.
(ii) Total: the plug-in estimated variance of the total estimator is $s^2_T = N^2 s^2_{\overline{X}}$.
(iii) Proportion: if $x_a \in \{0, 1\}\ \forall a \in U$ and $\widehat{p} := \overline{X}$, then plug-in estimated variance of $\widehat{p}$ is $s^2_{\widehat{p}} = \frac{\widehat{p}(1-\widehat{p})}{n-1}\left(1 - \frac{n}{N}\right)$.

Proof. Set $c_N := 1 - \frac{n}{N}$.

(i) Replace unknown finite-population variance by s^2, correction by c_N gives $s^2_{\overline{X}} = (\frac{s^2}{n})c_N = \frac{s^2}{n}(1 - \frac{n}{N})$.

(ii) Since $T = N\overline{X}$, variance scales quadratically, so $s^2_T = N^2 s^2_{\overline{X}}$.

(iii) If $X_k \in \{0, 1\}$ and $\widehat{p} = \overline{X}$, then $s^2 = \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \widehat{p})^2 = \frac{1}{n-1}(n\widehat{p} - n\widehat{p}^2) = \frac{n\widehat{p}(1-\widehat{p})}{n-1}$.
Substituting this into (i) gives $s^2_{\widehat{p}} = \frac{\widehat{p}(1-\widehat{p})c_N}{n-1} = \frac{\widehat{p}(1-\widehat{p})}{n-1}(1 - \frac{n}{N})$.

Structural Commentary: Plug-in SEs use s^2 and the $1 - \frac{n}{N}$ correction.
Key Insight: Binary-data variance depends only on $\widehat{p}$. □

Example 1.1.10. *Finite Population Correction in a Hospital Survey*
For a finite population of $N = 393$ hospitals, let x_i denote discharges and suppose the finite-population standard deviation is $\sigma \approx 589.7$. Under simple random sampling without replacement with $n = 32$, the sample-mean standard deviation is $\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}\sqrt{1 - \frac{n-1}{N-1}} \approx \frac{589.7}{\sqrt{32}}\sqrt{1 - \frac{31}{392}} \approx 100.0$. Since $\frac{n}{N} = \frac{32}{393}$ is small, the finite-population correction is near 1, and the standard error is mainly governed by the $\frac{1}{\sqrt{n}}$ averaging scale.

1.2 Normal Approximation and Confidence Intervals

The exact design formulas in Theorem 1.1.8 give the center and scale of the sampling distribution. For large samples, a normal approximation turns those formulas into error probabilities and confidence intervals.

Method 1.2.1. *Normal Approximation for SRS*

(i) Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, $x_i \in \mathbb{R}$ for $i \in U$, and $S \sim \text{Unif}\{s \subseteq U : |s| = n\}$.

(ii) For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := n^{-1} \sum_{k=1}^{n} X_k$, $T := N\overline{X}$, $\mu := N^{-1} \sum_{i=1}^{N} x_i$, $\sigma^2 := N^{-1} \sum_{i=1}^{N} (x_i - \mu)^2$, $\tau := N\mu$, and $\sigma^2_{\overline{X}} := \frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)$.

(iii) Let Φ be the standard normal cdf. For large n, use $\overline{X} \approx N(\mu, \sigma^2_{\overline{X}})$.

(iv) If $\sigma_{\overline{X}} > 0$, then $\mathbb{P}(|\overline{X} - \mu| \le \delta) \approx 2\Phi(\frac{\delta}{\sigma_{\overline{X}}}) - 1$ for $\delta > 0$, and $T - \tau = N(\overline{X} - \mu) \approx N(0, N^2\sigma^2_{\overline{X}})$.

Method 1.2.2. *Approximate SRS Confidence Intervals*

(i) Let S be an SRS without replacement, with targets μ, τ and estimators $\overline{X}, T$.

(ii) Fix $0 < \alpha < 1$, let Φ be the standard normal cdf, set $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$, and use plug-in SEs $s_{\overline{X}}, s_T$, and, when binary, $s_{\widehat{p}}$.

(iii) $\text{CI}_{1-\alpha}(\mu) = \overline{X} \pm z(\frac{\alpha}{2}) s_{\overline{X}}$ and $\text{CI}_{1-\alpha}(\tau) = T \pm z(\frac{\alpha}{2}) s_T$.

(iv) In the dichotomous case, with population proportion p and sample proportion $\widehat{p}$, use $\text{CI}_{1-\alpha}(p) = \widehat{p} \pm z(\frac{\alpha}{2}) s_{\widehat{p}}$.

(v) Intervals are random before sampling; after observation, parameters are fixed and coverage is procedural.

Example 1.2.3. *Generic SRS Intervals*

Suppose finite population has $N = 8000$ units. Simple random sample of $n = 100$ gives $\overline{X} = 1.6$ for variable with sample standard deviation $s = 0.8$. Use $z(0.025) \approx 1.96$, suppose sample gives $\widehat{p} = 0.12$ for binary attribute.

(i) *Mean*: $s_{\overline{X}} = \frac{s}{\sqrt{n}} \sqrt{1 - \frac{n}{N}} = \frac{0.8}{10} \sqrt{1 - \frac{100}{8000}} \approx 0.0795$, then 95% CI for μ is $1.6 \pm 1.96(0.0795) = (1.44, 1.76)$.

(ii) *Total*: $T = N\overline{X} = 8000(1.6) = 12800$ and $s_T = N s_{\overline{X}} \approx 636$, then 95% CI for τ is $(11553, 14047)$.

(iii) *Proportion*: $s_{\widehat{p}} = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n-1} \left(1 - \frac{n}{N}\right)} = \sqrt{\frac{0.12(0.88)}{99} \left(1 - \frac{100}{8000}\right)} \approx 0.0325$, then 95% CI for p is $(0.056, 0.184)$.
The estimated attribute total is $N\widehat{p} = 960$ with $N s_{\widehat{p}} \approx 260$ and interval $(451, 1469)$.

These intervals quantify SRS uncertainty for fixed finite-population quantities.

Method 1.2.4. *Sample Size from Desired Half-Width*

(i) Fix $0 < \alpha < 1$, half-width $h > 0$, planning SD $\sigma > 0$, and planned size n.
Let Φ be the standard normal cdf and set $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$.

(ii) Ignoring the finite population correction, $h \approx z(\frac{\alpha}{2}) \frac{\sigma}{\sqrt{n}}$ gives $n \approx \left(\frac{z(\frac{\alpha}{2})\sigma}{h}\right)^2$.

(iii) For total based on population proportion over N units, use planning value $p_0 \in [0, 1]$, SE $N\sqrt{\frac{p_0(1-p_0)}{n}}$.

(iv) With negligible finite population correction, halving interval width requires roughly $4n$ observations.

1.3 Ratio Estimation

Ratio estimation uses an auxiliary variable whose population mean or total is known. It is especially useful when the auxiliary variable is strongly correlated with the target variable.

Definition 1.3.1. *Population and Sample Ratios*

Let $U := \{1, \dots, N\}$ and fixed pairs $(x_i, y_i) \in \mathbb{R}^2$, $i \in U$.

Set $\mu_x := \frac{1}{N} \sum_{i=1}^{N} x_i$, $\mu_y := \frac{1}{N} \sum_{i=1}^{N} y_i$, $\tau_x := \sum_{i=1}^{N} x_i$, and $\tau_y := \sum_{i=1}^{N} y_i$, with $\mu_x \neq 0$.

For SRS pairs $(X_i, Y_i)_{i=1}^n$, set $\overline{X} := \frac{1}{n} \sum_{i=1}^{n} X_i$ and $\overline{Y} := \frac{1}{n} \sum_{i=1}^{n} Y_i$, with $\overline{X} \neq 0$.

(i) *Population ratio*: $r := \sum_{i=1}^{N} y_i / \sum_{i=1}^{N} x_i = \frac{\mu_y}{\mu_x}$.

(ii) *Sample ratio*: $R := \frac{\overline{Y}}{\overline{X}}$.

(iii) *Ratio mean estimator*: $\overline{Y}_R := \mu_x R$ when μ_x is known.

(iv) *Ratio total estimator*: $T_R := \tau_x R$ when τ_x is known.

Key Insight: $\{i \in U : x_i = 0\} = \emptyset$ does not imply $r = \frac{1}{N} \sum_{i=1}^{N} \frac{y_i}{x_i}$.

Definition 1.3.2. *Population Covariance and Correlation*

Let $U := \{1, \dots, N\}$ and fixed pairs $(x_i, y_i) \in \mathbb{R}^2$, $i \in U$.

Set $\mu_x := \frac{1}{N} \sum_{i=1}^{N} x_i$, $\mu_y := \frac{1}{N} \sum_{i=1}^{N} y_i$, $\sigma_x^2 := \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu_x)^2$, and $\sigma_y^2 := \frac{1}{N} \sum_{i=1}^{N} (y_i - \mu_y)^2$.

For sample pairs $(X_i, Y_i)_{i=1}^n$, set $\overline{X} := \frac{1}{n} \sum_{i=1}^{n} X_i$ and $\overline{Y} := \frac{1}{n} \sum_{i=1}^{n} Y_i$.

(i) *Population covariance*: $\sigma_{xy} := \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu_x)(y_i - \mu_y)$.

(ii) *Population correlation*: $\rho := \frac{\sigma_{xy}}{\sigma_x \sigma_y}$ when $\sigma_x, \sigma_y > 0$.

(iii) *Sample covariance*: $s_{xy} := \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})$.

Theorem 1.3.3. *Approximate Moments of the Ratio Estimator*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, *and let* $(x_a, y_a) \in \mathbb{R}^2$ *be fixed for* $a \in U$.
Set $\lambda_n := \frac{1}{n}(1 - \frac{n-1}{N-1})$, $\mu_x := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\mu_y := \frac{1}{N}\sum_{a=1}^{N} y_a$, $r := \frac{\mu_y}{\mu_x}$
with $\mu_x \neq 0$, $\sigma_x^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu_x)^2$, $\sigma_y^2 := \frac{1}{N}\sum_{a=1}^{N}(y_a - \mu_y)^2$, *and* $\sigma_{xy} := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu_x)(y_a - \mu_y)$.
Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$, *let* $(J_1, \dots, J_n)$ *be a random ordering of* S, *set* $(X_k, Y_k) := (x_{J_k}, y_{J_k})$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $\overline{Y} := \frac{1}{n}\sum_{k=1}^{n} Y_k$,
and $R := \frac{\overline{Y}}{\overline{X}}$ *on* $\{\overline{X} \neq 0\}$. *The subscript* d *denotes design moments.*

(i) Variance: first-order Taylor gives $\mathrm{Var}_d(R) \approx \frac{\lambda_n}{\mu_x^2}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$.
(ii) Mean: second-order Taylor gives $\mathbb{E}_d R \approx r + \frac{\lambda_n}{\mu_x^2}(r\sigma_x^2 - \sigma_{xy})$.

Proof. Let $(A_q, B_q)_{q=1}^N$ be a uniformly random ordering of the population pairs; the ordered sample has the law of the first n entries. For $q \neq r$, write $\gamma_x := \mathrm{Cov}_d(A_q, A_r)$, $\gamma_y := \mathrm{Cov}_d(B_q, B_r)$, and $\gamma_{xy} := \mathrm{Cov}_d(A_q, B_r)$.
The fixed sums give $0 = \mathrm{Var}_d(\sum_{q=1}^{N} A_q) = N\sigma_x^2 + N(N-1)\gamma_x$, $0 = \mathrm{Var}_d(\sum_{q=1}^{N} B_q) = N\sigma_y^2 + N(N-1)\gamma_y$,
and $0 = \mathrm{Cov}_d(\sum_{q=1}^{N} A_q, \sum_{q=1}^{N} B_q) = N\sigma_{xy} + N(N-1)\gamma_{xy}$.
Hence $\gamma_x = -\frac{\sigma_x^2}{N-1}$, $\gamma_y = -\frac{\sigma_y^2}{N-1}$, $\gamma_{xy} = -\frac{\sigma_{xy}}{N-1}$, $V_x := \mathrm{Var}_d(\overline{X}) = \lambda_n\sigma_x^2$, $V_y := \mathrm{Var}_d(\overline{Y}) = \lambda_n\sigma_y^2$,
and $C_{xy} := \mathrm{Cov}_d(\overline{X}, \overline{Y}) = \lambda_n\sigma_{xy}$. Let $\Sigma := \begin{pmatrix} V_x & C_{xy} \\ C_{xy} & V_y \end{pmatrix}$.
For $g(x, y) = \frac{y}{x}$ and $\mu := (\mu_x, \mu_y)$, $g(\mu) = r$, $\nabla g(\mu) = (-\frac{r}{\mu_x}, \frac{1}{\mu_x})^\top$, and $H_g(\mu) = \begin{pmatrix} 2r/\mu_x^2 & -1/\mu_x^2 \\ -1/\mu_x^2 & 0 \end{pmatrix}$.

(i) *Variance*: $\mathrm{Var}_d(R) \approx \nabla g(\mu)^\top \Sigma \nabla g(\mu) = \frac{\lambda_n}{\mu_x^2}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$.
(ii) *Mean*: $\mathbb{E}_d R \approx g(\mu) + \frac{1}{2}\operatorname{tr}(H_g(\mu)\Sigma) = r + \frac{\lambda_n}{\mu_x^2}(r\sigma_x^2 - \sigma_{xy})$.

Structural Commentary: The ratio estimator is locally linear for variance and locally quadratic for bias.
Key Insight: Ratio variance is scaled residual variation in $Y - rX$. □

Corollary 1.3.4. *Estimated Standard Error of a Ratio*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 < n \le N$, *and let* $(x_a, y_a) \in \mathbb{R}^2$ *be fixed for* $a \in U$.
Set $\lambda_n := \frac{1}{n}(1 - \frac{n-1}{N-1})$, $\mu_x := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\tau_x := \sum_{a=1}^{N} x_a$, $\tau_y := \sum_{a=1}^{N} y_a$, *and* $r := \frac{\tau_y}{\tau_x}$ *with* $\tau_x \neq 0$.
Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$. *For a random ordering* $(J_1, \dots, J_n)$ *of* S,
set $(X_i, Y_i) := (x_{J_i}, y_{J_i})$, $\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$, *and* $\overline{Y} := \frac{1}{n}\sum_{i=1}^{n} Y_i$.
Assume $\overline{X} \neq 0$ *and set* $R := \frac{\overline{Y}}{\overline{X}}$, $s_x^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$, $s_y^2 := \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2$, $s_{xy} := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$, *and* $q_R := R^2 s_x^2 + s_y^2 - 2Rs_{xy}$. *If* $0 < \alpha < 1$ *and* $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$ *for the standard normal cdf* Φ, *then:*

(i) Ratio variance: $s_R^2 := \frac{\lambda_n}{\overline{X}^2} q_R$ *estimates* $\mathrm{Var}_d(R)$.
(ii) Ratio interval: $\mathrm{CI}_{1-\alpha}(r) := R \pm z(\frac{\alpha}{2}) s_R$.
(iii) Mean and total: for $\overline{Y}_R := \mu_x R$ *and* $T_R := \tau_x R$, $\mathrm{SE}(\overline{Y}_R) = |\mu_x|\, s_R$ *and* $\mathrm{SE}(T_R) = |\tau_x|\, s_R$ *when the corresponding auxiliary constant is known.*

Proof.

(i) The plug-in map $(\mu_x, r, \sigma_x^2, \sigma_y^2, \sigma_{xy}) \mapsto (\overline{X}, R, s_x^2, s_y^2, s_{xy})$ in Theorem 1.3.3 yields $\widehat{\mathrm{Var}}_d(R) = \frac{\lambda_n}{\overline{X}^2}(R^2 s_x^2 + s_y^2 - 2Rs_{xy}) = s_R^2$.
(ii) If $Z_R := (R - r)/s_R \approx \mathcal{N}(0, 1)$, then $\mathbb{P}_d(|Z_R| \le z(\frac{\alpha}{2})) \approx 1 - \alpha$, equivalently $r \in R \pm z(\frac{\alpha}{2}) s_R$.
(iii) For known c, $\widehat{\mathrm{Var}}_d(cR) = c^2 s_R^2$, so $\mathrm{SE}(cR) = |c|\, s_R$; set $c = \mu_x$ and $c = \tau_x$.

Structural Commentary: Plug in sample moments to estimate the ratio variance.
Key Insight: Ratio uncertainty is residual uncertainty in $Y - RX$. □

Method 1.3.5. *When the Ratio Estimate Helps*
Let $U = \{1, \dots, N\}$ contain fixed pairs (x_i, y_i) with $\mu_x, \mu_y > 0$, $\sigma_x, \sigma_y > 0$, σ_{xy}, $\rho := \frac{\sigma_{xy}}{\sigma_x \sigma_y}$, and $r := \frac{\mu_y}{\mu_x}$.
For an SRS $(X_k, Y_k)_{k=1}^n$, set $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $\overline{Y} := \frac{1}{n}\sum_{k=1}^{n} Y_k$, $R := \frac{\overline{Y}}{\overline{X}}$, and $\overline{Y}_R := \mu_x R$.
Ignoring finite-population corrections:

(i) *Variances*: $V_0 := \mathrm{Var}(\overline{Y}) = \frac{\sigma_y^2}{n}$ and $V_R^* := \frac{1}{n}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$ approximate $\mathrm{Var}(\overline{Y}_R)$.
(ii) *Improvement*: $V_R^* < V_0 \iff 2\sigma_{xy} > r\sigma_x^2 \iff 2\rho\sigma_y > r\sigma_x$.
(iii) *Coefficient form*: with $C_x := \frac{\sigma_x}{\mu_x}$ and $C_y := \frac{\sigma_y}{\mu_y}$, $V_R^* < V_0 \iff \rho > \frac{1}{2}\frac{C_x}{C_y}$.
(iv) *Favorable case*: large positive ρ and small C_x/C_y.

Example 1.3.6. *Mortgage-Payment Ratio*
For an SRS of $n = 100$ recent house buyers, let Y_i be monthly mortgage payment and X_i gross income.
Suppose $\overline{X} = 3100$, $\overline{Y} = 868$, $s_x = 1200$, $s_y = 250$, and $\widehat{\rho} := s_{xy}/(s_x s_y) = 0.85$.

(i) *Ratio*: $R = \frac{\overline{Y}}{\overline{X}} = \frac{868}{3100} = 0.28$.
(ii) *Variance input*: $q := R^2 s_x^2 + s_y^2 - 2R\widehat{\rho} s_x s_y \approx 32596$.

(iii) *Standard error*: ignoring the finite-population correction, $s_R \approx \frac{\sqrt{q}}{\sqrt{100}\,\overline{X}} \approx 0.00582$.
(iv) *Interval*: $\mathrm{CI}_{0.95}(r) = R \pm 1.96 s_R \approx 0.28 \pm 1.96(0.00582) = (0.269, 0.291)$.

Thus the population mortgage-payment-to-income ratio is about 28%.

Remark 1.3.7. *Bias and Resampling Corrections*
For SRS from fixed pairs (x_i, y_i), set $R := \overline{Y}/\overline{X}$, $\overline{Y}_R := \mu_x R$, and $T_R := \tau_x R$.
(i) Since $g(x,y) = y/x$ is nonlinear, generally $\mathbb{E}_d R \neq r$, hence $\mathbb{E}_d \overline{Y}_R \neq \mu_y$ and $\mathbb{E}_d T_R \neq \tau_y$.
(ii) Bias corrections target $B_d(R) := \mathbb{E}_d R - r$; use resampling when Taylor bias matters.

1.4 Stratified Random Sampling

Stratified sampling partitions the population into subpopulations and samples within each one. It can guarantee representation of important groups and can reduce variance when strata are internally homogeneous or have different means.

Definition 1.4.1. *Strata and the Stratified Mean*

Let U be a finite population, $L \geq 1$, nonempty subsets $U_1, \ldots, U_L \subseteq U$, $N_l := |U_l|$, $N := \sum_{l=1}^{L} N_l$,
and fixed values $x_{il} \in \mathbb{R}$ for $i = 1, \ldots, N_l$, $l = 1, \ldots, L$.
For stratum samples of sizes n_l, write sampled values as X_{il}, $i = 1, \ldots, n_l$.

(i) *Strata*: $U = \bigsqcup_{l=1}^{L} U_l$.
(ii) *Stratum weight*: $W_l := \frac{N_l}{N}$.
(iii) *Stratum mean*: $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$.
(iv) *Stratum variance*: $\sigma_l^2 := \frac{1}{N_l} \sum_{i=1}^{N_l} (x_{il} - \mu_l)^2$.
(v) *Stratum sample mean*: $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$.
(vi) *Stratified mean*: $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$.

Structural Commentary: With $\mu := \frac{1}{N} \sum_{l=1}^{L} \sum_{i=1}^{N_l} x_{il}$, $\mu = \sum_{l=1}^{L} W_l \mu_l$.

Theorem 1.4.2. *Mean and Variance of the Stratified Estimate*

Let $L \geq 1$, $N_l > 1$, $1 \leq n_l \leq N_l$, $N := \sum_{l=1}^{L} N_l$, $W_l := \frac{N_l}{N}$, and $\lambda_l := \frac{1}{n_l}(1 - \frac{n_l - 1}{N_l - 1})$.

For fixed $x_{il} \in \mathbb{R}$, $i = 1, \ldots, N_l$, $l = 1, \ldots, L$, set $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$, $\sigma_l^2 := \frac{1}{N_l} \sum_{i=1}^{N_l} (x_{il} - \mu_l)^2$, and $\mu := \sum_{l=1}^{L} W_l \mu_l$.

Independently sample $S_l \sim \mathrm{Unif}\{s \subseteq \{1, \ldots, N_l\} : |s| = n_l\}$; for random orderings $(J_{il})_{i=1}^{n_l}$ of S_l, set $X_{il} := x_{J_{il},l}$, $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$,

and $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$. Let $\mathbb{E}_d, \mathrm{Var}_d$ denote design moments.

(i) Mean: $\mathbb{E}_d \overline{X}_s = \mu$.
(ii) Variance: $\mathrm{Var}_d(\overline{X}_s) = \sum_{l=1}^{L} W_l^2 \lambda_l \sigma_l^2$.
(iii) Small fractions: if $n_l/N_l \ll 1$ for all l, then $\mathrm{Var}_d(\overline{X}_s) \approx \sum_{l=1}^{L} W_l^2 \frac{\sigma_l^2}{n_l}$.

Proof.

(i) For $a \in \{1, \ldots, N_l\}$, $\mathbb{P}_d(J_{il} = a) = 1/N_l$, hence $\mathbb{E}_d X_{il} = \mu_l$, $\mathbb{E}_d \overline{X}_l = \mu_l$, and $\mathbb{E}_d \overline{X}_s = \sum_{l=1}^{L} W_l \mu_l = \mu$.

(ii) Independence across strata gives $\mathrm{Var}_d(\overline{X}_s) = \sum_{l=1}^{L} W_l^2 \mathrm{Var}_d(\overline{X}_l)$.

For stratum l, let $(Y_{1,l}, \ldots, Y_{N_l,l})$ uniformly order $\{x_{il} : 1 \leq i \leq N_l\}$, put $\gamma_l := \mathrm{Cov}_d(Y_{a,l}, Y_{b,l})$ for $a \neq b$.

As $\sum_{a=1}^{N_l} Y_{a,l}$ is fixed, $0 = N_l \sigma_l^2 + N_l(N_l - 1)\gamma_l$, so $\gamma_l = -\frac{\sigma_l^2}{N_l - 1}$, $\mathrm{Var}_d(\overline{X}_l) = \frac{1}{n_l^2}(n_l \sigma_l^2 + n_l(n_l - 1)\gamma_l) = \lambda_l \sigma_l^2$.

(iii) If $n_l/N_l \ll 1$, then $1 - \frac{n_l - 1}{N_l - 1} \approx 1$, so $\lambda_l \sigma_l^2 \approx \sigma_l^2/n_l$.

Structural Commentary: Stratified estimation is a weighted sum of independent within-stratum estimators.
Key Insight: Independent strata make variance add as $\sum_l W_l^2 \mathrm{Var}_d(\overline{X}_l)$. □

Corollary 1.4.3. *Totals and Estimated Variances*

Let $L \geq 1$, $N_l > 1$, $1 < n_l \leq N_l$, $N := \sum_{l=1}^{L} N_l$, $W_l := \frac{N_l}{N}$, and $f_l := n_l/N_l$.

Fix $x_{il} \in \mathbb{R}$ for $i = 1, \ldots, N_l$, $l = 1, \ldots, L$, set $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$, $\mu := \sum_{l=1}^{L} W_l \mu_l$, and $\tau := N\mu$.

Assume independently $S_l \sim \mathrm{Unif}\{s \subseteq \{1, \ldots, N_l\} : |s| = n_l\}$; for random orderings $(J_{il})_{i=1}^{n_l}$ of S_l.

Set $X_{il} := x_{J_{il},l}$, $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$, $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$, $T_s := N\overline{X}_s$, and $s_l^2 := \frac{1}{n_l - 1} \sum_{i=1}^{n_l} (X_{il} - \overline{X}_l)^2$.

For $0 < \alpha < 1$, let $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$.

(i) Total moments: $\mathbb{E}_d T_s = \tau$ and $\mathrm{Var}_d(T_s) = N^2 \mathrm{Var}_d(\overline{X}_s)$.

(ii) Estimated mean variance: $s^2_{\overline{X}_s} := \sum_{l=1}^{L} W_l^2(1-f_l)\frac{s_l^2}{n_l}$.

(iii) Intervals: $\mathrm{CI}_{1-\alpha}(\mu) = \overline{X}_s \pm z(\frac{\alpha}{2})s_{\overline{X}_s}$ *and* $\mathrm{CI}_{1-\alpha}(\tau) = T_s \pm z(\frac{\alpha}{2})Ns_{\overline{X}_s}$.

Proof.

(i) Since $T_s = N\overline{X}_s$ and $\tau = N\mu$, Theorem 1.4.2 gives $\mathbb{E}_d T_s = N\mu = \tau$ and $\mathrm{Var}_d(T_s) = N^2\,\mathrm{Var}_d(\overline{X}_s)$.

(ii) Let $S_l^2 := \frac{1}{N_l-1}\sum_{i=1}^{N_l}(x_{il}-\mu_l)^2$. Then $\mathrm{Var}_d(\overline{X}_l) = \frac{(1-f_l)S_l^2}{n_l}$, $\mathbb{E}_d s_l^2 = S_l^2$, so $S_l^2 \mapsto s_l^2$; $s^2_{\overline{X}_s} = \sum_{l=1}^{L} W_l^2(1-f_l)\frac{s_l^2}{n_l}$.

(iii) With $\frac{\overline{X}_s-\mu}{s_{\overline{X}_s}} \approx \mathcal{N}(0,1)$, inversion gives $\mathrm{CI}_{1-\alpha}(\mu)$; multiplying by N gives $\mathrm{CI}_{1-\alpha}(\tau)$.

Structural Commentary: Totals and intervals scale the stratified mean.
Key Insight: Estimated variance adds weighted stratum variances. □

Theorem 1.4.4. *Neyman Allocation*

Let $L \geq 1$, $W_l > 0$, $\sum_{l=1}^{L} W_l = 1$, $\sigma_l > 0$, $S_\sigma := \sum_{k=1}^{L} W_k\sigma_k$, *and* $n > 0$.

On $\mathcal{A} := \{(n_1,\dots,n_L) \in (0,\infty)^L : \sum_{l=1}^{L} n_l = n\}$, *set* $V(n_1,\dots,n_L) := \sum_{l=1}^{L}\frac{W_l^2\sigma_l^2}{n_l}$.

Then V *has the unique minimiser* $n_l^* := n\frac{W_l\sigma_l}{S_\sigma}$, $l = 1,\dots,L$.

Proof. V is strictly convex on $(0,\infty)^L$ since $\partial_{ll}V = \frac{2W_l^2\sigma_l^2}{n_l^3} > 0$ and $\partial_{lm}V = 0$ for $l \neq m$.

For $\mathcal{L}(n,\lambda) := V(n) + \lambda(\sum_{l=1}^{L} n_l - n)$, the first-order equations are $-\frac{W_l^2\sigma_l^2}{n_l^2} + \lambda = 0$, hence $n_l = \frac{W_l\sigma_l}{\sqrt{\lambda}}$.
The constraint gives $\frac{1}{\sqrt{\lambda}} = \frac{n}{S_\sigma}$, so $n_l = n_l^*$; strict convexity gives uniqueness.
Structural Commentary: The constrained optimum balances marginal variance reduction across strata.
Key Insight: A stratum receives more sample not only when it is large, but also when it is internally variable. □

Corollary 1.4.5. *Allocation Comparisons*

Let $L \geq 1$, $W_l > 0$, $\sum_{l=1}^{L} W_l = 1$, $\mu_l \in \mathbb{R}$, $\sigma_l > 0$, $\mu := \sum_{l=1}^{L} W_l\mu_l$, $\overline{\sigma} := \sum_{l=1}^{L} W_l\sigma_l$, *and* $n > 0$.

Ignoring finite-population corrections, let $\overline{X}_{so}$ *be the optimal-allocation stratified mean,* $\overline{X}_{sp}$ *the proportional-allocation stratified mean with* $n_l = nW_l$, *and* $\overline{X}$ *the ordinary SRS mean.*

(i) Allocation variances: $\mathrm{Var}(\overline{X}_{so}) \approx \frac{\overline{\sigma}^2}{n}$ *and* $\mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l\sigma_l^2$.

(ii) Optimal-allocation gain: $\mathrm{Var}(\overline{X}_{sp}) - \mathrm{Var}(\overline{X}_{so}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l(\sigma_l-\overline{\sigma})^2$.

(iii) Proportional-allocation gain: $\mathrm{Var}(\overline{X}) - \mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l(\mu_l-\mu)^2$.

Proof.

(i) With finite-population corrections suppressed, $\mathrm{Var}(\overline{X}_s) \approx \sum_{l=1}^{L}\frac{W_l^2\sigma_l^2}{n_l}$. Neyman allocation gives $n_l = \frac{nW_l\sigma_l}{\overline{\sigma}}$, hence $\mathrm{Var}(\overline{X}_{so}) \approx \frac{\overline{\sigma}^2}{n}$; proportional allocation $n_l = nW_l$ gives $\mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l\sigma_l^2$.

(ii) Since $\sum_{l=1}^{L} W_l = 1$, $\sum_{l=1}^{L} W_l\sigma_l^2 - \overline{\sigma}^2 = \sum_{l=1}^{L} W_l(\sigma_l-\overline{\sigma})^2$.

(iii) For ordinary SRS, $\mathrm{Var}(\overline{X}) \approx \frac{1}{n}\{\sum_{l=1}^{L} W_l\sigma_l^2 + \sum_{l=1}^{L} W_l(\mu_l-\mu)^2\}$; subtract the proportional-allocation variance.

Structural Commentary: Proportional handles means; optimal also handles spreads.
Key Insight: Mean gaps help stratification; spread gaps help Neyman allocation. □

Example 1.4.6. *Four-Stratum Stratification Summary*
Let $N = 393$ and strata $l \in \{A,B,C,D\}$ have rounded summaries $(N_l) = (98,98,98,99)$, $(W_l) = (.249,.249,.249,.251)$, $(\mu_l) = (182.9, 526.5, 956.3, 1591.2)$, and $(\sigma_l) = (103.4, 204.8, 243.5, 419.2)$. Let n be the total sample size, ignore finite-population corrections, and write $\overline{X}_s$ for the stratified mean.

(i) *Equal allocation*: if $n_l = n/4$, then $\mathrm{Var}(\overline{X}_s) \approx \sum_{l=1}^{4} W_l^2\sigma_l^2/(n/4) \approx 72042.6/n$, so $\mathrm{SE}(\overline{X}_s) \approx 268.4/\sqrt{n}$.

(ii) *SRS benchmark*: with overall standard deviation $\sigma \approx 589.7$, $\mathrm{SE}(\overline{X}) \approx 589.7/\sqrt{n}$.

(iii) *Neyman allocation*: $a_l := \frac{W_l\sigma_l}{\sum_{k=1}^{4} W_k\sigma_k} \approx (.106,.210,.250,.434)$ for (A,B,C,D).

Stratification reduces variance, and Neyman allocation assigns the largest share to the stratum with largest σ_l.

Remark 1.4.7. *Model Limits and Other Designs*
For a finite-population survey:

(i) *Systematic*: fixed-interval sampling after a random start is SRS-like only when list order is noninformative; periodic order can bias selection.

(ii) *Cluster*: sampling groups lowers cost but can inflate variance when within-cluster correlation is positive.

(iii) *Scope*: design-based SEs measure random selection variation, not nonresponse, coverage error, selection bias, interviewer effects, or wording effects.

2 Exploratory Data Analysis and Data Summaries

2.1 Empirical Distribution and Survival Summaries

Exploratory summaries first treat the observed values as a batch of numbers. A stochastic model may later explain their variability, but the displays themselves are useful before a model has been chosen.

Definition 2.1.1. *Empirical CDF and Empirical Survival Function*
Let $x_1, \dots, x_n \in \mathbb{R}$ be data; for survival summaries assume $x_i \geq 0$ for $1 \leq i \leq n$.

(i) *Empirical CDF*: $F_n : \mathbb{R} \to [0,1]$, $F_n(x) := n^{-1}\#\{i \in \{1,\dots,n\} : x_i \leq x\}$.
(ii) *Empirical Survival Function*: $S_n : [0,\infty) \to [0,1]$, $S_n(t) := 1 - F_n(t) = n^{-1}\#\{i \in \{1,\dots,n\} : x_i > t\}$.

Key Insight: F_n and S_n encode complementary empirical tails, with $S_n(t) = 1 - F_n(t)$ for $t \geq 0$.

Remark 2.1.2. *Step-Function Convention*
Let $v_1 < \cdots < v_m$ be the distinct observed values, with multiplicities $r_1, \dots, r_m$. Then F_n is right-continuous, constant on each component of $\mathbb{R} \setminus \{v_1, \dots, v_m\}$, and has jump $\Delta F_n(v_j) = \frac{r_j}{n}$.

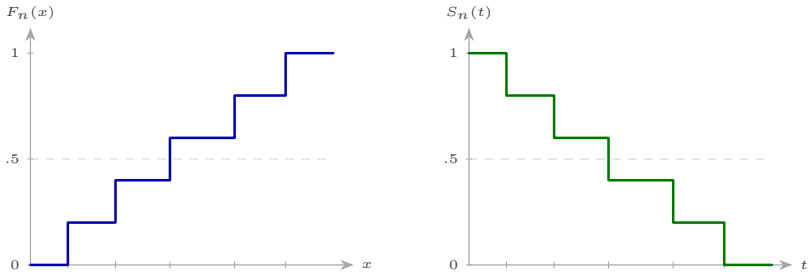

Figure 242: Empirical CDF (blue) rises as empirical survival (green) falls.

Method 2.1.3. *Empirical Log-Survival Plot*
Suppose $0 \leq T_{(1)} < \cdots < T_{(n)}$ and there are no ties. Set $p_i := \frac{i}{n+1}$ and $y_i := \log(1 - p_i)$.

(i) *Plot*: Draw $P^{(n)}_{\log S} := \{(T_{(i)}, y_i) : 1 \leq i \leq n\}$; using $n+1$ keeps y_n finite.
(ii) *Hazard Reading*: Since $(\log S)' = -h$, more negative local slopes indicate larger hazards.
(iii) *Tail Variability*: $\mathrm{Var}(\log[1 - F_n(t)]) \approx n^{-1}\frac{F(t)}{1-F(t)}$, which grows as $F(t) \uparrow 1$.

2.2 Quantile and Shape Displays

Distributional shape can be read from cumulative displays, density displays, or quantile comparisons. These views emphasize different questions: cumulative proportion, local concentration, and relative change between two distributions.

Definition 2.2.1. *Quantile–Quantile Plot*
Let F and G be continuous strictly increasing CDFs with quantile inverses $F^{-1}, G^{-1} : (0,1) \to \mathbb{R}$. For equal-sized batches $x_1, \dots, x_n$ and $y_1, \dots, y_n$, let $x_{(1)} \leq \cdots \leq x_{(n)}$ and $y_{(1)} \leq \cdots \leq y_{(n)}$.

(i) *Q–Q Plot*: $Q_{F,G} := \{(F^{-1}(p), G^{-1}(p)) : 0 < p < 1\} \subseteq \mathbb{R}^2$.
(ii) *Empirical Q–Q Plot*: $Q^{(n)}_{x,y} := \{(x_{(i)}, y_{(i)}) : 1 \leq i \leq n\} \subseteq \mathbb{R}^2$.

Key Insight: $(u, v) \in Q_{F,G}$ implies $F(u) = G(v)$, so plotted pairs match common cumulative probability levels.

Proposition 2.2.2. *Location and Scale Effects in Q–Q Plots*
Let F, G be continuous strictly increasing CDFs.
Set $x_p := F^{-1}(p)$, $y_p := G^{-1}(p)$ for $p \in (0,1)$, and $Q_{F,G} := \{(x_p, y_p) : 0 < p < 1\}$. Then:
(i) Location: If $y_p = x_p + h$ $\forall p \in (0,1)$ with $h \in \mathbb{R}$, then $G(y) = F(y - h)$ $\forall y \in \mathbb{R}$, $Q_{F,G} \subseteq \{(x,y) : y = x + h\}$.
(ii) Scale: If $y_p = cx_p$ $\forall p \in (0,1)$ with $c > 0$, then $G(y) = F(\frac{y}{c})$ $\forall y \in \mathbb{R}$, $Q_{F,G} \subseteq \{(x,y) : y = cx\}$.

Proof.
(i) *Location*: Fix $y \in \mathbb{R}$ and set $p := G(y)$. Then $p \in (0,1)$, $y = y_p = x_p + h$, and $x_p = y - h$, so $G(y) = p = F(x_p) = F(y - h)$. Also $(x_p, y_p) \in Q_{F,G}$ implies $y_p = x_p + h$.
(ii) *Scale*: Fix $y \in \mathbb{R}$ and set $p := G(y)$. Then $p \in (0,1)$, $y = y_p = cx_p$, and $x_p = \frac{y}{c}$, so $G(y) = p = F(x_p) = F(\frac{y}{c})$. Also $(x_p, y_p) \in Q_{F,G}$ implies $y_p = cx_p$.

Structural Commentary: A Q–Q plot is the graph of the quantile transport $F^{-1}(p) \mapsto G^{-1}(p)$.
Key Insight: Straight Q–Q lines encode location shifts and scale changes. □

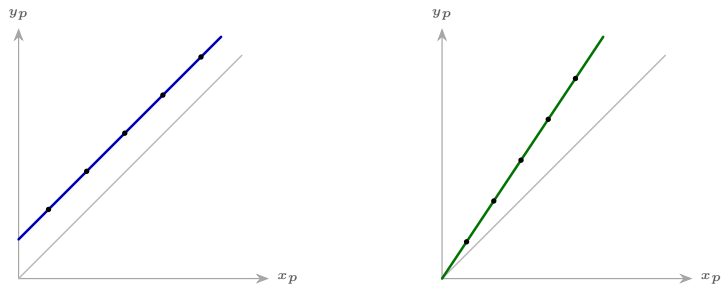

Figure 243: Q–Q lines: blue $y = x + h$ for location shift, green $y = cx$ for scale change.

Example 2.2.3. *Normal and Exponential Q–Q Lines*
For $p \in (0,1)$, set $x_p := F^{-1}(p)$ and $y_p := G^{-1}(p)$.

(i) *Normal Location*: If $F = N(0,1)$ and $G = N(1,1)$, then $y_p = x_p + 1$, so the Q–Q line is $y = x + 1$.
(ii) *Normal Location–Scale*: If $F = N(0,1)$ and $G = N(1,4)$, then $y_p = 1 + 2x_p$, so the Q–Q line is $y = 1 + 2x$.
(iii) *Exponential Rates*: If $F = \text{Exp}(\lambda_1)$ and $G = \text{Exp}(\lambda_2)$, then $x_p = -\frac{\log(1-p)}{\lambda_1}$ and $y_p = -\frac{\log(1-p)}{\lambda_2} = \frac{\lambda_1}{\lambda_2} x_p$, so the Q–Q line is $y = \frac{\lambda_1}{\lambda_2} x$.

Method 2.2.4. *Interpolated Empirical Quantiles*
Let $X_{(1)} \le \cdots \le X_{(n)}$ and set $p_k := \frac{k}{n+1}$ for $1 \le k \le n$.

(i) *Grid*: Assign $X_{(k)}$ to probability p_k.
(ii) *Interpolation*: If $p_k \le p \le p_{k+1}$ for $1 \le k < n$, set $\lambda := \frac{p - p_k}{p_{k+1} - p_k} = (n+1)p - k$, $\widehat{x}_p := (1-\lambda)X_{(k)} + \lambda X_{(k+1)}$.
(iii) *Q–Q Use*: For unequal samples, evaluate both interpolated quantile functions on a common grid $p \in (0,1)$.

Definition 2.2.5. *Histogram, Density Display, and Stem-and-Leaf Plot*
Let $x_1, \ldots, x_n \in \mathbb{R}$ be data. For disjoint intervals $B_1, \ldots, B_m$ with widths $w_j > 0$.
Set $N_j := \#\{i \in \{1, \ldots, n\} : x_i \in B_j\}$. Let K be a kernel, $h > 0$, and $b > 0$ be a chosen scale.

(i) *Histogram*: $H_j^c := N_j$, $H_j^p := \frac{N_j}{n}$, or $H_j^d := \frac{N_j}{n w_j}$ on B_j.
(ii) *Density Display*: $\widehat{f}_h : \mathbb{R} \to [0, \infty)$, $\widehat{f}_h(x) := (nh)^{-1} \sum_{i=1}^{n} K(\frac{x - x_i}{h})$.
(iii) *Stem-and-Leaf Plot*: (s_i, ℓ_i) satisfies $x_i = b s_i + \ell_i$, $s_i \in \mathbb{Z}$, $\ell_i \in [0, b)$.
Entries with common s_i are printed together in increasing s_i.

Structural Commentary: For density-scaled heights $H_j^d = \frac{N_j}{n w_j}$, $\sum_j H_j^d w_j = 1$ when the bins cover the sample. *Key Insight*: Smaller w_j or h retains more local variation; larger w_j or h increases smoothing.

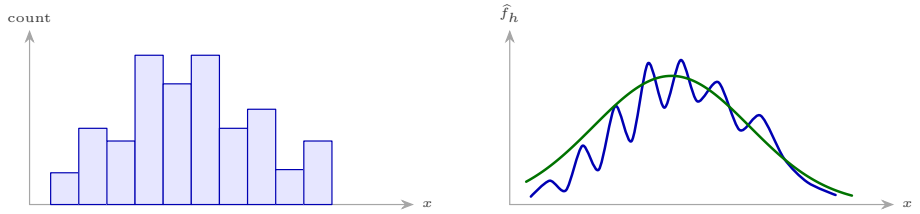

Figure 244: Shape displays: narrow bins at left; small h (blue) is rougher than large h (green).

Remark 2.2.6. *What Shape Displays Preserve*

(i) *ECDF*: Preserves ranks and supports direct quantile reading.
(ii) *Histogram*: Preserves bin counts (N_j), not exact values.
(iii) *Kernel Estimate*: Preserves smoothed local mass; bandwidth h controls resolution.
(iv) *Stem–Leaf*: Preserves coarse shape and recoverable numerical values.

2.3 Location, Scale, and Robust Summaries

Numerical summaries compress a batch more aggressively than a plot. They should therefore be chosen with attention to the feature being summarized and to the influence of extreme observations.

Definition 2.3.1. *Location Summaries for a Batch*
Let $x_1, \ldots, x_n \in \mathbb{R}$ have order statistics $x_{(1)} \le \cdots \le x_{(n)}$.
When a trim fraction is used, let $0 \le \alpha < \frac{1}{2}$ and $m_\alpha := \lfloor n\alpha \rfloor$.

(i) *Arithmetic Mean*: $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$.
(ii) *Median*: $\widetilde{x} := x_{\left(\frac{n+1}{2}\right)}$ if n is odd, and $\widetilde{x} := \frac{x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}}{2}$ if n is even.
(iii) $100\alpha\%$ *Trimmed Mean*: $\overline{x}_\alpha := (n - 2m_\alpha)^{-1} \sum_{i=m_\alpha+1}^{n-m_\alpha} x_{(i)}$.

Structural Commentary: Under $x_i \mapsto x_i + \Delta$, $\overline{x} \mapsto \overline{x} + \frac{\Delta}{n}$; $\widetilde{x}$ and $\overline{x}_\alpha$ depend only on central order statistics.
Key Insight: Extreme magnitudes affect $\overline{x}$ directly, but affect $\widetilde{x}$ and $\overline{x}_\alpha$ mainly through rank changes.

Remark 2.3.2. *Outliers and Center*
Replacing one x_i by M sends $\overline{x}$ to $\overline{x} + \frac{M - x_i}{n}$, hence unbounded as $M \to \pm\infty$.
Once an observation is extreme, moving it farther outward leaves $\widetilde{x}$ unchanged and leaves $\overline{x}_\alpha$ unchanged if it remains trimmed. This is descriptive robustness, not an independence or model.

Remark 2.3.3. *Compare More Than One Center*
Let $\mathcal{C} := \{\overline{x}, \widetilde{x}, \overline{x}_\alpha, \widehat{\mu}_\Psi\}$. For symmetric light-tailed data, range($\mathcal{C}$) is typically small; a large range signals skewness, heavy tails, outliers, or mixture structure.

Definition 2.3.4. *Dispersion Summaries*
Let $x_1, \ldots, x_n \in \mathbb{R}$ have mean $\overline{x}$, median $\widetilde{x}$, and selected sample quartiles Q_1, Q_3.

(i) *Sample Variance*: $s^2 := (n-1)^{-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$.
(ii) *Sample Standard Deviation*: $s := \sqrt{s^2}$.
(iii) *Interquartile Range*: $\mathrm{IQR} := Q_3 - Q_1$.
(iv) *Median Absolute Deviation*: $\mathrm{MAD} := \mathrm{median}\{|x_i - \widetilde{x}| : 1 \le i \le n\}$.

Structural Commentary: s^2 is residual-based; IQR and MAD are order-based.
Key Insight: A single extreme value can dominate s, but not IQR or MAD unless it changes central ranks.

Remark 2.3.5. *Normal Scale Calibrations*
For approximately normal data, $\frac{\mathrm{IQR}}{1.35}$ and $\frac{\mathrm{MAD}}{0.675}$ are common robust scale estimates for the normal standard deviation. The constants calibrate the population normal IQR and median absolute deviation to σ.

2.4 Boxplots, Scatterplots, and Relationship Summaries

Boxplots and scatterplots are compact exploratory displays. A boxplot summarizes one batch or compares several batches; a scatterplot displays relationships between paired variables.

Method 2.4.1. *Boxplot Construction*
A standard boxplot is built from the median, lower quartile Q_1, upper quartile Q_3, and $\mathrm{IQR} = Q_3 - Q_1$:

(i) Draw a box from Q_1 to Q_3 and mark the median inside the box.
(ii) Draw whiskers to the most extreme observations lying within 1.5 IQR of the nearer quartile.
(iii) Mark observations beyond the whiskers individually as possible outliers.

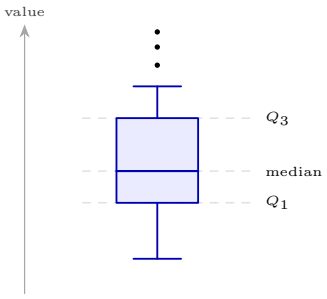

Figure 245: Boxplot: box Q_1 to Q_3, line median, dots possible outliers.

Remark 2.4.2. *Comparing Batches with Boxplots*

(i) *Location*: Differences in medians M_g indicate group shifts.
(ii) *Spread*: Differences in $\mathrm{IQR}_g = Q_{3,g} - Q_{1,g}$ indicate central-spread changes.
(iii) *Tails*: Unequal whiskers or one-sided outliers indicate skewness or tail imbalance.

Definition 2.4.3. *Scatterplot, Pearson Correlation, and Rank Correlation*
Let $(x_i, y_i) \in \mathbb{R}^2$ for $1 \le i \le n$, set $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$ and $\overline{y} := n^{-1} \sum_{i=1}^{n} y_i$.
Let R_i, S_i be chosen ranks of x_i, y_i, using average ranks for ties.
Set $\overline{R} := n^{-1} \sum_{i=1}^{n} R_i$ and $\overline{S} := n^{-1} \sum_{i=1}^{n} S_i$. Set $a_i := x_i - \overline{x}$, $b_i := y_i - \overline{y}$, $u_i := R_i - \overline{R}$, and $v_i := S_i - \overline{S}$.
Let $S_{xx} := \sum_i a_i^2$, $S_{yy} := \sum_i b_i^2$, $S_{xy} := \sum_i a_i b_i$, $S_{RR} := \sum_i u_i^2$, $S_{SS} := \sum_i v_i^2$, and $S_{RS} := \sum_i u_i v_i$.

(i) *Scatterplot*: $P := \{(x_i, y_i) : 1 \le i \le n\} \subset \mathbb{R}^2$, displayed in the xy-plane.
(ii) *Pearson Correlation Coefficient*: if $S_{xx} S_{yy} > 0$, $r := \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$.
(iii) *Rank Correlation Coefficient*: if $S_{RR} S_{SS} > 0$, $r_{\mathrm{rank}} := \frac{S_{RS}}{\sqrt{S_{RR} S_{SS}}}$.

Structural Commentary: r is affine-scale invariant; r_{rank} is monotone-transform invariant.
Key Insight: r measures linear association; r_{rank} measures rank association.

Method 2.4.4. *Scatterplot Diagnostics*
For paired data $P = \{(x_i, y_i) : 1 \le i \le n\}$:

(i) *Raw Plot*: Inspect P for outliers, curvature, clusters, and nonconstant spread.
(ii) *Transform*: If scale obscures structure, compare $(g(x_i), h(y_i))$ for domain-meaningful g, h such as log.

(iii) *Correlation*: Use r only when the displayed relation is approximately linear.
(iv) *Multiple Variables*: For $z_i \in \mathbb{R}^d$ with $d > 2$, inspect all pairwise plots before relying on pairwise summaries.

Example 2.4.5. *Changing Scale Can Reveal Structure*
If $x_i > 0$ and $\max_i x_i / \min_i x_i \gg 1$, the raw plot $P = \{(x_i, y_i)\}$ may compress variation. The transformed plot $P_{\log} := \{(\log x_i, y_i)\}$ can reveal monotone or linear structure, and generally $r(P_{\log}) \neq r(P)$ because Pearson correlation depends on the chosen coordinates.

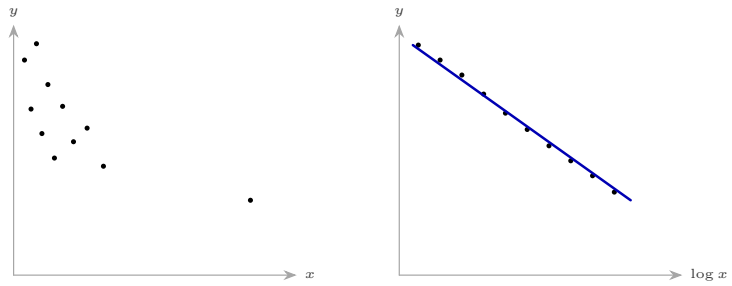

Figure 246: Scatterplots: raw x compresses points; $\log x$ reveals a linear trend.

Remark 2.4.6. *Exploration Before Formal Inference*
Exploratory displays choose the inferential target: transformations $g(X)$, influential subsets, comparable spreads IQR_g, and data-quality checks should be settled before intervals, tests, or models are trusted.

3 Categorical Data Analysis

3.1 Fisher's Exact Test and Fixed Margins

Categorical data record counts in classes rather than measurements on a continuous scale. In a two-way table, the rows and columns encode two classifications, and the cells record how many units fall in each cross-classification. The probability model is therefore built from multinomial and hypergeometric count laws rather than from normal sampling pivots.

Definition 3.1.1. *Two-Way Contingency Table*
Let $I, J \in \mathbb{N}$ and $n_{ij} \in \mathbb{N}_0$ for $1 \le i \le I$, $1 \le j \le J$. Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, and $n_{\cdot\cdot} := \sum_{i=1}^{I}\sum_{j=1}^{J} n_{ij}$.
For corresponding random counts, write N_{ij} and use capitalized totals.
(i) *Two-way contingency table*: $(n_{ij})_{1\le i\le I,\ 1\le j\le J}$, indexed by row and column classifications.
(ii) *Margins*: $\{n_{i\cdot} : 1 \le i \le I\}$, $\{n_{\cdot j} : 1 \le j \le J\}$, and $n_{\cdot\cdot}$.
Structural Commentary: Fixed-margin exact tests condition on the margins of random table $N = (N_{ij})$.

Method 3.1.2. *Fisher's Exact Test for a 2×2 Table*
Let a 2×2 table have fixed margins

	1	2	Total
1	N_{11}	N_{12}	$n_{1\cdot}$
2	N_{21}	N_{22}	$n_{2\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 2}$	$n_{\cdot\cdot}$

Let R_1, R_2 be fixed disjoint row sets with $|R_i| = n_{i\cdot}$.
Let C be the random column-1 set with $|C| = n_{\cdot 1}$, and set $\Omega_0 := \{A \subseteq R_1 \cup R_2 : |A| = n_{\cdot 1}\}$.
(i) $H_0 : \mathbb{P}_0(C = A) = \binom{n_{\cdot\cdot}}{n_{\cdot 1}}^{-1}$ $\forall A \in \Omega_0$.
(ii) $N_{11} = |C \cap R_1|$, $N_{12} = n_{1\cdot} - N_{11}$, $N_{21} = n_{\cdot 1} - N_{11}$, and $N_{22} = n_{2\cdot} - n_{\cdot 1} + N_{11}$.
(iii) $\mathcal{S} := \{x \in \mathbb{Z} : 0 \le x \le n_{1\cdot},\ 0 \le n_{\cdot 1} - x \le n_{2\cdot}\}$ and $p_0(x) := \mathbb{P}_0(N_{11} = x) = \dfrac{\binom{n_{1\cdot}}{x}\binom{n_{2\cdot}}{n_{\cdot 1}-x}}{\binom{n_{\cdot\cdot}}{n_{\cdot 1}}}$, $x \in \mathcal{S}$.
(iv) $p_+(x_{\mathrm{obs}}) := \sum_{x\in\mathcal{S}:x\ge x_{\mathrm{obs}}} p_0(x)$, $p_-(x_{\mathrm{obs}}) := \sum_{x\in\mathcal{S}:x\le x_{\mathrm{obs}}} p_0(x)$, and $p_2(x_{\mathrm{obs}}) := \sum_{x\in\mathcal{S}:p_0(x)\le p_0(x_{\mathrm{obs}})} p_0(x)$.
Reject when the selected value is $\le \alpha$.

Example 3.1.3. *Promotion Decisions and Fixed Margins*
In randomised label experiment with 24 male-labeled,24 female-labeled files, the observed table is

	Male	Female	Total
Promote	21	14	35
Hold file	3	10	13
Total	24	24	48

Under H_0 and fixed margins $(35, 13)$, $(24, 24)$, N_{11} satisfies $\mathbb{P}_0(N_{11} = x) = \dfrac{\binom{35}{x}\binom{13}{24-x}}{\binom{48}{24}}$, $x \in \mathcal{S} := \{11, \dots, 24\}$.
At level 0.05, $R^{(2)}_{0.05} := \{11, 12, 13, 14, 21, 22, 23, 24\}$; since $x_{\mathrm{obs}} = 21 \in R^{(2)}_{0.05}$, reject H_0.

Remark 3.1.4. *What Is Exact Here*
For $N = (N_{ij})$ and $\mathcal{M} := (\{N_{i\cdot}\}_i, \{N_{\cdot j}\}_j)$, exactness means evaluating $\mathcal{L}_0(N \mid \mathcal{M})$ without asymptotics. Conditioning is design-justified when $\mathcal{M}$ is ancillary to H_0; rejection interpretation is design-dependent.

3.2 Chi-Square Tests for Contingency Tables

For an $I \times J$ table with observed counts n_{ij} and fitted null counts E_{ij}, use $X^2 := \sum_{i=1}^{I} \sum_{j=1}^{J} (n_{ij} - E_{ij})^2 / E_{ij}$; in contingency tables, E_{ij} is determined by the row-column null model.

Definition 3.2.1. *Homogeneity of Multinomial Distributions*

Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and $\Delta_{I-1} := \{\pi \in [0,1]^I : \sum_{i=1}^{I} \pi_i = 1\}$.

For independent columns $j \in [J]$, let $(N_{1j}, \dots, N_{Ij}) \sim \text{Multinomial}(n_{\cdot j}; \pi_{1j}, \dots, \pi_{Ij})$ with observed counts n_{ij}.
The column multinomial laws are *homogeneous* iff $H_0 : \exists \pi \in \Delta_{I-1}$ with $\pi_{ij} = \pi_i \ \forall (i,j) \in [I] \times [J]$; under H_0, π_i is the common category probability.

Proposition 3.2.2. *Common Probability MLE Under Homogeneity*

Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, $\Delta_{I-1} := \{\pi \in [0,1]^I : \sum_{i=1}^{I} \pi_i = 1\}$, and $N_{\cdot j} := (N_{1j}, \dots, N_{Ij})$, independently for $j \in [J]$. Under H_0, $N_{\cdot j} \sim \text{Multinomial}(n_{\cdot j}; \pi)$ for a common $\pi \in \Delta_{I-1}$.

With observed counts n_{ij}, set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$ and $n_{\cdot\cdot} := \sum_{i=1}^{I} n_{i\cdot} > 0$.

Then $\widehat{\pi}_i = \frac{n_{i\cdot}}{n_{\cdot\cdot}}$ and $E_{ij} = n_{\cdot j} \widehat{\pi}_i = \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$ for $(i,j) \in [I] \times [J]$.

Proof. Up to factors independent of π, $L(\pi) \propto \prod_{j=1}^{J} \prod_{i=1}^{I} \pi_i^{n_{ij}} = \prod_{i=1}^{I} \pi_i^{n_{i\cdot}}$, hence $\ell(\pi) = \sum_{i=1}^{I} n_{i\cdot} \log \pi_i$ on Δ_{I-1}.

Stationarity of $\ell(\pi) + \lambda(1 - \sum_{i=1}^{I} \pi_i)$ gives $\frac{n_{i\cdot}}{\pi_i} - \lambda = 0$ when $n_{i\cdot} > 0$ and $\widehat{\pi}_i = 0$ when $n_{i\cdot} = 0$,
Normalisation gives $\lambda = n_{\cdot\cdot}$ and $\widehat{\pi}_i = \frac{n_{i\cdot}}{n_{\cdot\cdot}}$. Therefore $E_{ij} = \mathbb{E}_{\widehat{\pi}} N_{ij} = n_{\cdot j} \widehat{\pi}_i$.
Structural Commentary: H_0 reduces IJ cell probabilities to one shared $\pi \in \Delta_{I-1}$.
Key Insight: $\widehat{\pi}$ is the pooled row proportion, and E_{ij} redistributes each column total by $\widehat{\pi}_i$. □

Definition 3.2.3. *Homogeneity Fitted Counts*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, $n_{\cdot\cdot} := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$, and $\widehat{\pi}_i := \frac{n_{i\cdot}}{n_{\cdot\cdot}}$.
The homogeneity *fitted counts* are $E_{ij} := n_{\cdot j} \widehat{\pi}_i = \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$ for $(i,j) \in [I] \times [J]$.

Method 3.2.4. *Pearson Test of Homogeneity*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, $n_{\cdot\cdot} := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$, and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$. Assume $E_{ij} > 0$ for cells used.

(i) *Statistic*: $X^2 := \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$.

(ii) *Calibration*: If $n_{\cdot\cdot}$ large, $\min_{i,j} E_{ij}$ not small, then $X^2 \overset{H_0}{\approx} \chi^2_\nu$; $\nu := J(I-1) - (I-1) = (I-1)(J-1)$.
(iii) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_{(I-1)(J-1)} \geq X^2_{\text{obs}}\} \leq \alpha$.

Example 3.2.5. *Word Counts by Text Group*
Let $W = \{a, an, this, that, with, without\}$, let $G_1, \dots, G_4$ be text groups. Observed word-count table n_{wj} is

Word	G_1	G_2	G_3	G_4
a	147	186	101	83
an	25	26	11	29
this	32	39	15	15
that	94	105	37	22
with	59	74	28	43
without	18	10	10	4
Total	375	440	202	196

For $C_A := \{1, 2, 3\}$, by homogeneity test: $X^2_A = 12.27$, $\nu_A = (6-1)(3-1) = 10$, $p_A = \mathbb{P}\{\chi^2_{10} \geq 12.27\} \approx 0.27$.
Pooling C_A and comparing it with $C_B := \{4\}$ gives $X^2_{A,B} = 32.81$, $\nu_{A,B} = 6 - 1 = 5$, and $p_{A,B} < 0.001$.
$w \in \{an, that\}$ gives the largest Pearson contributions, localising nonhomogeneity by row.

Definition 3.2.6. *Independence in a Contingency Table*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and $N = (N_{ij}) \sim \text{Multinomial}(n; (\pi_{ij})_{[I] \times [J]})$ with $n > 0$, observed counts n_{ij}.
Set $\pi_{i\cdot} := \sum_{j=1}^{J} \pi_{ij}$, $\pi_{\cdot j} := \sum_{i=1}^{I} \pi_{ij}$, $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, and $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$.
Rows and columns are *independent* iff $H_0 : \pi_{ij} = \pi_{i\cdot} \pi_{\cdot j} \ \forall (i,j) \in [I] \times [J]$; then $\widehat{\pi}_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n^2}$ and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n}$.

Method 3.2.7. *Pearson Test of Independence*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$, $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n}$. Assume $E_{ij} > 0$ for cells used.

(i) *Statistic*: $X^2 := \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$.

(ii) *Calibration*: If n is large and $\min_{i,j} E_{ij}$ is not small, then $X^2 \overset{H_0}{\approx} \chi^2_\nu$ with $\nu := (I-1)(J-1)$.
(iii) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_\nu \geq X^2_{\text{obs}}\} \leq \alpha$.

(iv) *Degrees of Freedom*: $\nu = (IJ-1) - ((I-1)+(J-1)) = (I-1)(J-1)$.

Example 3.2.8. *Binary Classifications*
Let $n = 1436$, $[n] := \{1, \dots, n\}$, and $(A_k, B_k) \in \{a_1, a_2\} \times \{b_1, b_2\}$ for $k \in [n]$.
Set $n_{ij} := \#\{k \in [n] : A_k = a_i,\ B_k = b_j\}$ and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n}$. The observed and independence-fitted tables are

n_{ij}

$a_i \backslash b_j$	b_1	b_2	Total
a_1	550	61	611
a_2	681	144	825
Total	1231	205	1436

E_{ij}

$a_i \backslash b_j$	b_1	b_2
a_1	523.8	87.2
a_2	707.2	117.8

Under $H_0 : A \perp B$, $X^2 = \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{(n_{ij} - E_{ij})^2}{E_{ij}} = 16.01$, $\nu = 1$, and $p < 0.001$; reject H_0.
Thus $A \not\perp B$; absent randomised A, inference is associative, not causal.

Remark 3.2.9. *Homogeneity Versus Independence*
Both tests use $E_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$, $X^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$, and $\nu = (I-1)(J-1)$.

(i) *Homogeneity*: $N_{\cdot j} \sim \text{Multinomial}(n_{\cdot j}; \pi_{\cdot j})$ independently for $j \in [J]$, with fixed $n_{\cdot j}$, $H_0 : \pi_{\cdot 1} = \cdots = \pi_{\cdot J}$.
(ii) *Independence*: $N \sim \text{Multinomial}(n_{\cdot\cdot}; (\pi_{ij})_{[I]\times[J]})$ with $H_0 : \pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}\ \forall (i,j) \in [I] \times [J]$.

If $\pi_{\cdot j} > 0$, then independence iff $\pi_{i|j} := \frac{\pi_{ij}}{\pi_{\cdot j}} = \pi_{i\cdot}\ \forall (i,j) \in [I] \times [J]$.

Method 3.2.10. *Likelihood-Ratio Version*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ with null fitted counts E_{ij} for $(i,j) \in [I] \times [J]$.
Set $n_{\cdot\cdot} := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$ and $\nu := (I-1)(J-1)$.
Let H_0 be the homogeneity or independence null producing E_{ij}, and assume $E_{ij} > 0$ whenever $n_{ij} > 0$.

(i) *Statistic*: $G^2 := 2 \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} \log\left(\frac{n_{ij}}{E_{ij}}\right)$.
(ii) *Zero Counts*: $n_{ij} = 0$ contributes 0 to G^2.
(iii) *Calibration*: Under the large-sample likelihood-ratio conditions of Theorem 3.1.8, $G^2 \overset{H_0}{\approx} \chi^2_\nu$.
(iv) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_\nu \geq G^2_{\text{obs}}\} \leq \alpha$.

3.3 Matched Pairs and McNemar's Test

Matched categorical data require a different table from unmatched samples. The pair, not the individual measurement, is the independent sampling unit.

Definition 3.3.1. *Matched Binary Pair Table*
Let $(Y_k^{(1)}, Y_k^{(2)}) \in \{0,1\}^2$ for $k \in [n]$, and set $R_k := Y_k^{(1)} + 1$, $S_k := Y_k^{(2)} + 1$.
For $(r,s) \in \{1,2\}^2$, set $n_{rs} := \#\{k \in [n] : (R_k, S_k) = (r,s)\}$, so $n = \sum_{r=1}^{2} \sum_{s=1}^{2} n_{rs}$.

(i) *Matched Binary Pair Table*: $N := (n_{rs})_{(r,s)\in\{1,2\}^2}$.
(ii) *Concordant Cells*: $C := \{(r,s) \in \{1,2\}^2 : r = s\} = \{(1,1),(2,2)\}$; counts $(n_{rs})_{(r,s)\in C} = (n_{11}, n_{22})$.
(iii) *Discordant Cells*: $D := \{(r,s) \in \{1,2\}^2 : r \neq s\} = \{(1,2),(2,1)\}$; counts $(n_{rs})_{(r,s)\in D} = (n_{12}, n_{21})$.
Key Insight: (n_{12}, n_{21}) determines marginal change; (n_{11}, n_{22}) records agreement.

Definition 3.3.2. *Marginal Homogeneity for Matched Pairs*
Let $\pi = (\pi_{rs})_{(r,s)\in\{1,2\}^2} \in [0,1]^4$ satisfy $\sum_{r=1}^{2} \sum_{s=1}^{2} \pi_{rs} = 1$. Set $\pi_{r\cdot} := \sum_{s=1}^{2} \pi_{rs}$ and $\pi_{\cdot s} := \sum_{r=1}^{2} \pi_{rs}$.
The matched-pair law is *marginally homogeneous* iff $H_0 : (\pi_{1\cdot}, \pi_{2\cdot}) = (\pi_{\cdot 1}, \pi_{\cdot 2})$, equivalently $\pi_{12} = \pi_{21}$.
Key Insight: H_0 constrains only the discordant probabilities π_{12}, π_{21}.

Method 3.3.3. *McNemar's Test*
Let $n_{rs} \in \mathbb{N}_0$ and cell probabilities π_{rs} for $(r,s) \in \{1,2\}^2$, with $n := \sum_{r=1}^{2} \sum_{s=1}^{2} n_{rs} > 0$. Set $d := n_{12} + n_{21}$.

(i) *Null*: $H_0 : \pi_{12} = \pi_{21}$.
(ii) *MLE Under H_0*: $\widehat{\pi}_{11} = \frac{n_{11}}{n}$, $\widehat{\pi}_{22} = \frac{n_{22}}{n}$, and $\widehat{\pi}_{12} = \widehat{\pi}_{21} = \frac{d}{2n}$.
(iii) *Statistic*: If $d > 0$, $X^2 := \frac{(n_{12} - n_{21})^2}{d}$.
(iv) *Calibration*: If d is large, then $X^2 \overset{H_0}{\approx} \chi^2_1$.
(v) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_1 \geq X^2_{\text{obs}}\} \leq \alpha$.

Example 3.3.4. *Matched Case-Control Pairs*
In a matched case-control study, 85 cases were each paired with one control, and a binary exposure was recorded for both members of each pair.

	Control	
Case	Unexposed	Exposed
Unexposed	37	7
Exposed	15	26

An unmatched chi-square test treats cases and controls as independent samples.
McNemar's test uses only discordant pairs; here $X^2 = (7-15)^2/(7+15) = 2.91$ and $p \approx 0.09$, so accounting for the matching weakens the evidence of an exposure difference.

Example 3.3.5. *Matched Case-Crossover Exposure*
In a matched case-crossover study, each subject is observed during an event window and a comparable control window, with a binary exposure recorded in both windows.

Event window	Control window		
	Exposed	Unexposed	Total
Exposed	13	157	170
Unexposed	24	505	529
Total	37	662	699

The discordant counts are 157 and 24, so $X^2 = (157-24)^2/(157+24) = 97.7$, giving strong evidence that exposure prevalence differs between the two windows. This does not by itself establish causation, since time-varying covariates may affect both exposure and event risk.

Remark 3.3.6. *Matched Does Not Mean Independent*
McNemar's null hypothesis is marginal homogeneity, not independence of paired responses. Thus the analysis must preserve the matching structure rather than treat the two margins as independent samples.

3.4 Odds Ratios and Case-Control Sampling

Odds ratios summarize association in a binary exposure–binary outcome table. They are especially useful because the same cross-product ratio can be estimated under several sampling designs, including retrospective case-control designs where risks themselves cannot be estimated.

Definition 3.4.1. *Odds and Odds Ratio*
Let A satisfy $0 < \mathbb{P}(A) < 1$.
Let X and D be binary exposure and disease events with $0 < \mathbb{P}(X) < 1$ and $0 < \mathbb{P}(D \mid X), \mathbb{P}(D \mid \overline{X}) < 1$.

(i) *Odds*: $\operatorname{odds}(A) := \frac{\mathbb{P}(A)}{1-\mathbb{P}(A)}$.
(ii) *Odds ratio*: $\Delta := \frac{\operatorname{odds}(D|X)}{\operatorname{odds}(D|\overline{X})}$.

Structural Commentary: If $o = \operatorname{odds}(A)$, then $\mathbb{P}(A) = \frac{o}{1+o}$; $p \mapsto p/(1-p)$ is a bijection $(0,1) \to (0,\infty)$.

Proposition 3.4.2. *Cross-Product Formula*
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $X, D \in \mathcal{F}$, and set $X_0 := \overline{X}$, $X_1 := X$, $D_0 := \overline{D}$, $D_1 := D$.
Let $\pi_{ij} := \mathbb{P}(X_i \cap D_j) > 0$ for $i, j \in \{0,1\}$, with margins $\pi_{i\cdot} := \pi_{i0} + \pi_{i1}$ and $\pi_{\cdot j} := \pi_{0j} + \pi_{1j}$.

	$\overline{D}$	D	
$\overline{X}$	π_{00}	π_{01}	$\pi_{0\cdot}$
X	π_{10}	π_{11}	$\pi_{1\cdot}$
	$\pi_{\cdot 0}$	$\pi_{\cdot 1}$	1

With $\Delta := \frac{\operatorname{odds}(D|X)}{\operatorname{odds}(D|\overline{X})}$, the cross-product identity is $\Delta = \frac{\pi_{11}\pi_{00}}{\pi_{01}\pi_{10}}$.

Proof. Since $\pi_{1\cdot} = \pi_{10} + \pi_{11}$ and $\pi_{0\cdot} = \pi_{00} + \pi_{01}$, $\mathbb{P}(D \mid X) = \frac{\pi_{11}}{\pi_{1\cdot}}$ and $1 - \mathbb{P}(D \mid X) = \frac{\pi_{10}}{\pi_{1\cdot}}$, so $\operatorname{odds}(D \mid X) = \frac{\pi_{11}}{\pi_{10}}$.
Similarly, $\operatorname{odds}(D \mid \overline{X}) = \frac{\pi_{01}}{\pi_{00}}$. Hence $\Delta = \frac{\frac{\pi_{11}}{\pi_{10}}}{\frac{\pi_{01}}{\pi_{00}}} = \frac{\pi_{11}\pi_{00}}{\pi_{01}\pi_{10}}$.
Structural Commentary: $\pi_{1\cdot}, \pi_{0\cdot}$ enter conditional probabilities but cancel after passage to odds.
Key Insight: Δ equals diagonal product $\pi_{11}\pi_{00}$ divided by off-diagonal product $\pi_{01}\pi_{10}$. □

Method 3.4.3. *Estimating an Odds Ratio by Design*
Let $N = (n_{ij})_{i,j\in\{0,1\}}$ be a 2×2 table for (X_i, D_j), where $X_1 := X$, $X_0 := \overline{X}$, $D_1 := D$, and $D_0 := \overline{D}$.
Assume $n_{ij} > 0$, set $n_{i\cdot} := n_{i0} + n_{i1}$, $n_{\cdot j} := n_{0j} + n_{1j}$, $n := \sum_{i=0}^{1}\sum_{j=0}^{1} n_{ij}$.

(i) *Simple Random Sampling*: $\widehat{\pi}_{ij} := \frac{n_{ij}}{n}$ estimates π_{ij}; hence $\widehat{\Delta} := \frac{n_{00}n_{11}}{n_{01}n_{10}}$ estimates Δ.
(ii) *Prospective Sampling*: $n_{0\cdot}, n_{1\cdot}$ fixed; $\frac{n_{01}}{n_{0\cdot}}$, $\frac{n_{11}}{n_{1\cdot}}$ estimate $\mathbb{P}(D \mid \overline{X})$, $\mathbb{P}(D \mid X)$; $\mathbb{P}(X)$ not identified.
(iii) *Retrospective Case-Control Sampling*: $n_{\cdot 0}, n_{\cdot 1}$ are fixed; disease risks are not identified, but $\widehat{\Delta} = \frac{n_{11}}{n_{01}} / \frac{n_{10}}{n_{00}}$ is estimable from exposure odds.

Key Insight: Design-fixed margins cancel in $\widehat{\Delta}$, so odds ratio is estimable even when risks or margins are not.

Example 3.4.4. *Unmatched Case-Control Odds Ratio*
Consider an unmatched case-control table with binary exposure:

	Exposed	Unexposed
Case	67	34
Control	43	64

The estimated odds ratio is $\widehat{\Delta} = \frac{67\cdot 64}{43\cdot 34)} = 2.93$, indicating a positive exposure–disease association.

Method 3.4.5. *Parametric Bootstrap for an Odds-Ratio Standard Error*
Let $a, b, c, d \in \mathbb{N}$ denote the observed counts in rows $D, \overline{D}$ and columns $X, \overline{X}$: $a := n_{DX}$, $b := n_{D\overline{X}}$, $c := n_{\overline{D}X}$, and $d := n_{\overline{D}\overline{X}}$. Set $m_D := a + b$, $m_{\overline{D}} := c + d$, $\widehat{p}_D := \frac{a}{m_D}$, $\widehat{p}_{\overline{D}} := \frac{c}{m_{\overline{D}}}$, and $\widehat{\Delta} := \frac{ad}{bc}$.

(i) *Bootstrap Law*: For $r \in [R]$, draw independently $A_r^* \sim \operatorname{Binomial}(m_D, \widehat{p}_D)$ and $C_r^* \sim \operatorname{Binomial}(m_{\overline{D}}, \widehat{p}_{\overline{D}})$.
(ii) *Replicate*: If $0 < A_r^* < m_D$ and $0 < C_r^* < m_{\overline{D}}$, set $\widehat{\Delta}_r^* := \frac{A_r^*(m_{\overline{D}} - C_r^*)}{(m_D - A_r^*)C_r^*}$.
(iii) *Standard Error*: With $\overline{\Delta}^* := \frac{1}{R}\sum_{r=1}^{R}\widehat{\Delta}_r^*$, set $\widehat{\operatorname{se}}_*(\widehat{\Delta}) := \left(\frac{1}{R-1}\sum_{r=1}^{R}(\widehat{\Delta}_r^* - \overline{\Delta}^*)^2\right)^{1/2}$.

Key Insight: Resample exposure counts conditional on fixed row totals; $\operatorname{sd}\{(\widehat{\Delta}_r^*)_{r\in[R]}\}$ estimates $\operatorname{se}(\widehat{\Delta})$.

Remark 3.4.6. *Odds Ratios Are Not Automatically Risks*
The odds ratio $\Delta = \text{odds}(D \mid X)/\text{odds}(D \mid \overline{X})$ differs from risk contrasts such as $\mathbb{P}(D \mid X) - \mathbb{P}(D \mid \overline{X})$.
In retrospective sampling, $n_{\cdot 0}, n_{\cdot 1}$ are design-fixed, absolute risks are not identified without population margins; Δ remains estimable because those margins cancel in the cross-product ratio.

3.5 Higher-Order Tables and Sampling Design

Two-way tables are the simplest categorical-data structure. Higher-order tables introduce several possible notions of dependence: full independence, conditional independence given a third factor, marginal homogeneity, and interactions among factors.

Definition 3.5.1. *Two-Way Log-Linear Form*
Let $I, J \in \mathbb{N}$ and let $\pi_{ij} > 0$ be two-way cell probabilities for $1 \le i \le I$, $1 \le j \le J$,
with $\pi_{i\cdot} := \sum_{j=1}^{J} \pi_{ij}$, $\pi_{\cdot j} := \sum_{i=1}^{I} \pi_{ij}$, $\alpha_i := \log \pi_{i\cdot}$, and $\beta_j := \log \pi_{\cdot j}$.

(i) *Independence form*: $\pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$ $\forall 1 \le i \le I$, $1 \le j \le J \iff \log \pi_{ij} = \alpha_i + \beta_j$ $\forall 1 \le i \le I$, $1 \le j \le J$.
(ii) *Interaction terms*: $\gamma_{ij} := \log \pi_{ij} - \alpha_i - \beta_j$, giving $\log \pi_{ij} = \alpha_i + \beta_j + \gamma_{ij}$.

Key Insight: $\gamma_{ij} = 0$ $\forall 1 \le i \le I$, $1 \le j \le J \iff \pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$ $\forall 1 \le i \le I$, $1 \le j \le J$.

Remark 3.5.2. *Three-Way Interactions*
Let $\pi_{ijk} > 0$ for $(i, j, k) \in [I] \times [J] \times [K]$. A log-linear model with all two-factor interactions and no three-factor interaction is $\log \pi_{ijk} = \mu + \alpha_i + \beta_j + \eta_k + \delta_{ij} + \varepsilon_{ik} + \gamma_{jk}$.
The absent term λ_{ijk} encodes the constraint of no third-order interaction; nested models are compared by deleting interaction terms and using the corresponding loss of degrees of freedom.

Example 3.5.3. *Graduate Admissions and Aggregation*
In the Berkeley admissions example, the aggregate admission rate for men was higher than for women.
Within the six listed majors, however, women were not consistently admitted at lower rates.
The apparent aggregate disadvantage was driven by application patterns: women applied more often to majors with lower admission rates. Major was therefore confounded with sex in the observational comparison.

Principle 3.5.4. *Choose the Test from the Sampling Mechanism*
For a contingency table $N = (n_{ij})$, let $n_{i\cdot}$ and $n_{\cdot j}$ denote row and column margins.
Select the test from the design-fixed quantities:

(i) *Fixed Margins*: If $(n_{i\cdot})$ and $(n_{\cdot j})$ are fixed in a 2×2 table, use Fisher's exact test.
(ii) *Independent Samples*: If rows are independent multinomial samples with fixed $(n_{i\cdot})$, test homogeneity of column probabilities.
(iii) *Single Multinomial Sample*: If $N \sim \text{Multinomial}(n, (\pi_{ij}))$, test independence, i.e. $\pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$.
(iv) *Matched Pairs*: If observations are paired counts n_{rs}, test marginal homogeneity using discordant cells.
(v) *Case-Control Sampling*: If $(n_{\cdot 0}, n_{\cdot 1})$ are fixed by disease status, estimate odds ratios. Absolute risks require population margins.

Key Insight: The same numeric table can encode different null hypotheses under different sampling laws.

Remark 3.5.5. *Use Counts, Not Percentages*
Let $Q(O, E) := \sum_i \frac{(O_i - E_i)^2}{E_i}$. Pearson's chi-square calibration applies to counts: $Q(O/n, E/n) = \frac{1}{n} Q(O, E)$ and $Q(100O/n, 100E/n) = \frac{100}{n} Q(O, E)$. Thus relative frequencies or percentages give a rescaled statistic unless the original count scale is restored.

4 Analysis of Variance and Designed Experiments

4.1 One-Way Layouts and the F Test

Analysis of variance compares several mean levels by splitting total variation into variation within treatment groups and variation between treatment groups. The method is a multisample continuation of the equal-variance two-sample procedure, with design and randomisation assumptions kept explicit.

Definition 4.1.1. *Balanced One-Way Layout*
For $I, J \in \mathbb{N}$, one factor has levels $i = 1, \dots, I$, each with exactly J independent replicates Y_{ij}, $j = 1, \dots, J$.
The normal fixed-effects model satisfies $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $\sum_{i=1}^{I} \alpha_i = 0$, and independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Structural Commentary: $\mathbb{E}Y_{ij} = \mu + \alpha_i$, and $\sum_{i=1}^{I} \alpha_i = 0$ identifies μ as the average treatment mean.

Key Insight: $H_0 : \alpha_i = 0$ $\forall i$ iff all treatment means are equal.

Definition 4.1.2. *One-Way Sums of Squares*
For a balanced one-way layout, set $\overline{Y}_{i\cdot} := \frac{1}{J} \sum_{j=1}^{J} Y_{ij}$ and $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ} \sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij}$.

(i) *Total*: $SS_{\text{tot}} := \sum_{i=1}^{I} \sum_{j=1}^{J} (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$; grand-mean dispersion.
(ii) *Within-treatment*: $SS_W := \sum_{i=1}^{I} \sum_{j=1}^{J} (Y_{ij} - \overline{Y}_{i\cdot})^2$; residual dispersion within treatments.
(iii) *Between-treatment*: $SS_B := J \sum_{i=1}^{I} (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$; treatment-mean dispersion scaled by J.

Key Insight: $SS_{\text{tot}} = SS_W + SS_B$ separates within-treatment noise from between-treatment mean differences.

Proposition 4.1.3. *One-Way Sum-of-Squares Decomposition*
For $I, J \in \mathbb{N}$ and observations Y_{ij}, set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$,

$SS_{\text{tot}} := \sum_i\sum_j (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$, $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, and $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$. Then $SS_{\text{tot}} = SS_W + SS_B$.

Proof. Set $r_{ij} := Y_{ij} - \overline{Y}_{i\cdot}$ and $b_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$. Then $Y_{ij} - \overline{Y}_{\cdot\cdot} = r_{ij} + b_i$ and $\sum_j r_{ij} = 0\ \forall i$.

Thus $SS_{\text{tot}} = \sum_i\sum_j (r_{ij} + b_i)^2 = \sum_i\sum_j r_{ij}^2 + 2\sum_i b_i \sum_j r_{ij} + J\sum_i b_i^2 = SS_W + SS_B$.

Structural Commentary: $Y_{ij} - \overline{Y}_{\cdot\cdot} = r_{ij} + b_i$, where $r_{ij} := Y_{ij} - \overline{Y}_{i\cdot}$ and $b_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$.
Key Insight: $\sum_j r_{ij} = 0\ \forall i$ annihilates the cross term $2\sum_i b_i \sum_j r_{ij}$. □

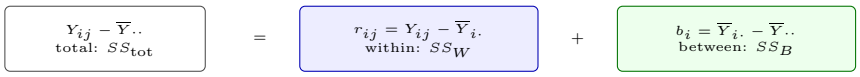

Figure 247: Squaring and summing gives $SS_{\text{tot}} = SS_W + SS_B$; large MS_B/MS_W suggests treatment effects.

Theorem 4.1.4. *Expected One-Way Sums of Squares*
For $I, J \in \mathbb{N}$, let $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = 0$, independent errors satisfying

$\mathbb{E}\varepsilon_{ij} = 0$, $\mathrm{Var}(\varepsilon_{ij}) = \sigma^2$. Set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$.

Then $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$; $\mathbb{E}(SS_W) = I(J-1)\sigma^2$

and $\mathbb{E}(SS_B) = (I-1)\sigma^2 + J\sum_i \alpha_i^2$. Hence $S_p^2 := \frac{SS_W}{I(J-1)}$ satisfies $\mathbb{E}S_p^2 = \sigma^2$.

Proof. For fixed i, Lemma 3.2.8 with $\mu_{ij} = \mu + \alpha_i$ gives $\mathbb{E}\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2 = (J-1)\sigma^2$, hence $\mathbb{E}SS_W = I(J-1)\sigma^2$.

For $\overline{Y}_{i\cdot}$, $\mathbb{E}\overline{Y}_{i\cdot} = \mu + \alpha_i$, $\mathrm{Var}(\overline{Y}_{i\cdot}) = \frac{\sigma^2}{J}$, and $\frac{1}{I}\sum_i (\mu + \alpha_i) = \mu$. By Lemma 3.2.8, $(\overline{Y}_{i\cdot})_{i=1}^I$ gives

$\mathbb{E}\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2 = \sum_i \alpha_i^2 + \frac{I-1}{J}\sigma^2$, so $\mathbb{E}SS_B = J\sum_i \alpha_i^2 + (I-1)\sigma^2$. Thus $\mathbb{E}S_p^2 = \frac{\mathbb{E}SS_W}{I(J-1)} = \sigma^2$.
Structural Commentary: $\mathbb{E}SS_W$ depends only on σ^2; $\mathbb{E}SS_B$ adds $J\sum_i \alpha_i^2$.
Key Insight: Under $\alpha_i = 0\ \forall i$, $\mathbb{E}\{\frac{SS_B}{I-1}\} = \mathbb{E}\{\frac{SS_W}{I(J-1)}\} = \sigma^2$. □

Theorem 4.1.5. *Normal-Theory One-Way F Test*
For $I, J \in \mathbb{N}$, let $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = 0$ and independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$, $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$.
$MS_W := \frac{SS_W}{I(J-1)}$, $MS_B := \frac{SS_B}{I-1}$; $\frac{SS_W}{\sigma^2} \sim \chi^2_{I(J-1)}$.
If $H_0 : \alpha_i = 0\ \forall i$, then $\frac{SS_B}{\sigma^2} \sim \chi^2_{I-1}$, $SS_B \perp\!\!\!\perp SS_W$, $F := \frac{MS_B}{MS_W} \sim F_{I-1, I(J-1)}$.

Proof. Let $C_m := \mathrm{Id}_m - m^{-1}\mathbf{1}_m\mathbf{1}_m^\top$, so $C_m^2 = C_m = C_m^\top$ and $\mathrm{rank}\, C_m = m - 1$.
For each i, $Z_i := \frac{1}{\sigma}(Y_{i1} - \mu - \alpha_i, \ldots, Y_{iJ} - \mu - \alpha_i)^\top \sim N_J(0, \mathrm{Id}_J)$ independently,
$\frac{1}{\sigma^2}\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2 = Z_i^\top C_J Z_i \sim \chi^2_{J-1}$. Summing over independent treatments gives $\frac{SS_W}{\sigma^2} \sim \chi^2_{I(J-1)}$.
Under H_0, $\widetilde{Z} := \sqrt{J}\sigma^{-1}(\overline{Y}_{1\cdot} - \mu, \ldots, \overline{Y}_{I\cdot} - \mu)^\top \sim N_I(0, \mathrm{Id}_I)$, and $\frac{SS_B}{\sigma^2} = \widetilde{Z}^\top C_I \widetilde{Z} \sim \chi^2_{I-1}$.
For each i, $C_J Z_i \perp\!\!\!\perp \mathbf{1}_J^\top Z_i$ by orthogonal normal components.
Hence $SS_W \perp\!\!\!\perp (\overline{Y}_{1\cdot}, \ldots, \overline{Y}_{I\cdot})$ and $SS_W \perp\!\!\!\perp SS_B$. Thus $F = \frac{SS_B}{I-1} / \frac{SS_W}{I(J-1)} = \frac{MS_B}{MS_W} \sim F_{I-1, I(J-1)}$.
Structural Commentary: $IJ = 1 + I(J-1) + (I-1)$ partitions df into mean, residual, and contrast subspaces.
Key Insight: Under H_0, MS_B and MS_W are independent σ^2 estimators, so $\frac{MS_B}{MS_W}$ has an F law. □

Example 4.1.6. *Generic Balanced One-Way ANOVA*
Let $I = 7$, $J = 10$, and $Y_{ij} = \mu_i + \varepsilon_{ij}$ with independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$. Suppose the ANOVA table is

Component	*df*	*SS*	*MS*	*F*
Treatment	6	.125	.021	5.66
Error	63	.231	.0037	
Total	69	.356		

Then $F = 5.66$ has null law $F_{6,63}$, $p = \mathbb{P}(F_{6,63} \ge 5.66) < .01$, and $H_0 : \mu_1 = \cdots = \mu_7$ is rejected at level .01.
If diagnostics are adequate, between-treatment variation exceeds within-treatment noise.

Method 4.1.7. *Unequal Sample Sizes*

For $I \in \mathbb{N}$, treatment i has observations Y_{ij}, $1 \le j \le J_i$, with $J_i \in \mathbb{N}$ and $N := \sum_{i=1}^I J_i$.

(i) $\overline{Y}_{i\cdot} := \frac{1}{J_i}\sum_{j=1}^{J_i} Y_{ij}$ and $\overline{Y}_{\cdot\cdot} := \frac{1}{N}\sum_{i=1}^I\sum_{j=1}^{J_i} Y_{ij}$.

(ii) $SS_W := \sum_{i=1}^I\sum_{j=1}^{J_i} (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := \sum_{i=1}^I J_i(\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$, $SS_{\text{tot}} := \sum_{i=1}^I\sum_{j=1}^{J_i} (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$
satisfy $SS_{\text{tot}} = SS_W + SS_B$.

(iii) $\mathrm{df}_W = N - I$ and $\mathrm{df}_B = I - 1$.

(iv) If $Y_{ij} = \mu_i + \varepsilon_{ij}$ with independent $\varepsilon_{ij} \sim N(0, \sigma^2)$ and $H_0 : \mu_1 = \cdots = \mu_I$, then $F := \frac{\frac{SS_B}{I-1}}{\frac{SS_W}{N-I}} \sim F_{I-1,N-I}$.

Corollary 4.1.8. *Two Treatments Give $F = t^2$*
Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ be independent normal samples with common variance, means $\overline{X}, \overline{Y}$, pooled variance $S_p^2 := \frac{SS_W}{n+m-2}$, and $t := \frac{\overline{X}-\overline{Y}}{S_p\sqrt{n^{-1}+m^{-1}}}$. The one-way ANOVA statistic for the two groups satisfies $F = t^2$.

Proof. Let $N := n + m$ and $\overline{Z} := \frac{n\overline{X}+m\overline{Y}}{N}$. Then $SS_B = n(\overline{X} - \overline{Z})^2 + m(\overline{Y} - \overline{Z})^2 = \frac{nm}{N}(\overline{X} - \overline{Y})^2$.
Since $df_B = 1$ and $MS_W = \frac{SS_W}{N-2} = S_p^2$, $F = \frac{SS_B}{MS_W} = \frac{(\overline{X}-\overline{Y})^2}{S_p^2(n^{-1}+m^{-1})} = t^2$.
Structural Commentary: Two groups leave one between-group contrast.
Key Insight: ANOVA squares the pooled t contrast. □

The same one-way decomposition also supports a random-effects interpretation, where groups are sampled from a population of possible group levels rather than fixed in advance.

Definition 4.1.9. *Balanced One-Way Random Effects Model*
A *balanced one-way random-effects model* has I groups and common replication J, with $Y_{ij} = \mu + A_i + \varepsilon_{ij}$ for $1 \le i \le I$, $1 \le j \le J$, independent A_i satisfying $\mathbb{E}A_i = 0$ and $\mathrm{Var}(A_i) = \sigma_A^2$, and independent errors ε_{ij} independent of (A_i) with $\mathbb{E}\varepsilon_{ij} = 0$ and $\mathrm{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$.
Structural Commentary: $\mathrm{Var}(Y_{ij}) = \sigma_A^2 + \sigma_\varepsilon^2$, and $\mathrm{Cov}(Y_{ij}, Y_{ij'}) = \sigma_A^2$ for $j \ne j'$ within the same group.

Proposition 4.1.10. *Random-Effects Expected Mean Squares*
Assume $Y_{ij} = \mu + A_i + \varepsilon_{ij}$ for $1 \le i \le I$, $1 \le j \le J$, with independent A_i and ε_{ij}, $\mathbb{E}A_i = \mathbb{E}\varepsilon_{ij} = 0$, $\mathrm{Var}(A_i) = \sigma_A^2$, and $\mathrm{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$. Then $\mathbb{E}(MS_W) = \sigma_\varepsilon^2$ and $\mathbb{E}(MS_B) = \sigma_\varepsilon^2 + J\sigma_A^2$.
The method-of-moments estimators are $\widehat{\sigma}_\varepsilon^2 = MS_W$ and $\widehat{\sigma}_A^2 = \frac{MS_B - MS_W}{J}$, truncated at 0 if required.

Proof. Condition on (A_i). Since $Y_{ij} - \overline{Y}_{i\cdot} = \varepsilon_{ij} - \overline{\varepsilon}_{i\cdot}$, $\mathbb{E}(SS_W \mid A) = I(J-1)\sigma_\varepsilon^2$, hence $\mathbb{E}(MS_W) = \sigma_\varepsilon^2$.
Also $\overline{Y}_{i\cdot} = \mu + A_i + \overline{\varepsilon}_{i\cdot}$ has variance $\sigma_A^2 + \frac{\sigma_\varepsilon^2}{J}$.
Hence the one-way between mean square satisfies $\mathbb{E}(MS_B) = J\,\mathrm{Var}(\overline{Y}_{i\cdot}) = \sigma_\varepsilon^2 + J\sigma_A^2$.
Solving these two moment equations gives the estimators.
Structural Commentary: Within estimates error; between adds group heterogeneity.
Key Insight: The excess $MS_B - MS_W$ estimates $J\sigma_A^2$. □

Example 4.1.11. *Dye Strength Batch Variation*
Suppose $I = 6$ batches each have $J = 6$ subsamples, with $\overline{y}_{\cdot\cdot} = 93.61$
and batch means $(92.00, 90.50, 91.25, 98.25, 91.17, 98.50)$.
Given $SS_B = 415.47$ and $SS_W = 134.08$, the mean squares are $MS_B = 83.09$ and $MS_W = 4.47$.
The moment estimates are $\widehat{\sigma}_\varepsilon^2 = MS_W = 4.47$ and $\widehat{\sigma}_A^2 = \frac{MS_B - MS_W}{6} = 13.10$.
Batch variation dominates measurement-error variation.

Remark 4.1.12. *Assumptions and Robustness*
For $Y_{ij} = \mu_i + \varepsilon_{ij}$, the one-way F calibration assumes independent $\varepsilon_{ij} \sim N(0, \sigma^2)$. Balance softens variance heterogeneity; clustered, paired, blocked, or repeated data require dependence modeling.

4.2 Multiple Comparisons After ANOVA

The one-way F test assesses $H_0 : \mu_1 = \cdots = \mu_I$; after rejection, inference targets contrasts $\mu_a - \mu_b$ with error control over the chosen comparison family.

Definition 4.2.1. *Studentized Range Quantile*
Let $Q_{I,\nu}$ denote a random variable with the studentized range distribution for I treatment means and ν error degrees of freedom. The *studentized range quantile* is $q_{I,\nu}(\alpha) := \inf\{q \in \mathbb{R} : \mathbb{P}(Q_{I,\nu} > q) \le \alpha\}$.
Structural Commentary: Balanced Tukey comparisons use pooled scale S_p.

Method 4.2.2. *Tukey Pairwise Comparisons*
In a balanced normal one-way layout, let $\mathcal{P} := \{(a,b) : 1 \le a < b \le I\}$, $\nu := I(J-1)$, and $h_\alpha := q_{I,\nu}(\alpha)S_p/\sqrt{J}$.

(i) For $(a,b) \in \mathcal{P}$, the simultaneous $100(1-\alpha)\%$ confidence interval for $\mu_a - \mu_b$ is $(\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}) \pm h_\alpha$.
(ii) Reject $H_{0,ab} : \mu_a = \mu_b$ iff $\left|\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}\right| > h_\alpha$.

Simultaneity over $\mathcal{P}$ uses the studentized range cutoff.

Example 4.2.3. *Generic Tukey Pairwise Calculation*
Consider a balanced one-way layout with $I = 7$, $J = 10$, $S_p = .06$, and ordered sample means
$(\overline{Y}_{1\cdot}, \overline{Y}_{3\cdot}, \overline{Y}_{7\cdot}, \overline{Y}_{2\cdot}, \overline{Y}_{5\cdot}, \overline{Y}_{6\cdot}, \overline{Y}_{4\cdot}) = (4.062, 4.003, 3.998, 3.997, 3.957, 3.955, 3.920)$.
With $q_{7,60}(.05) = 4.31$, the Tukey half-width is $h_{.05} = 4.31(.06)/\sqrt{10} \approx .082$.
Rejected pairs at simultaneous level .05 are $(1,4), (1,5), (1,6), (3,4)$; remaining pairwise nulls not rejected.

Method 4.2.4. *Bonferroni Pairwise Comparisons*
Let $\mathcal{C} \subseteq \{(a,b) : 1 \le a < b \le I\}$ be the planned pairwise family with $|\mathcal{C}| = k$.
Set $\nu := N - I$ and $h_{ab} := t_{\nu,1-\alpha/(2k)}S_p\sqrt{1/J_a + 1/J_b}$.

(i) For $(a,b) \in \mathcal{C}$, the Bonferroni confidence interval for $\mu_a - \mu_b$ is $(\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}) \pm h_{ab}$.
(ii) Reject $H_{0,ab} : \mu_a = \mu_b$ iff $\left|\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}\right| > h_{ab}$.

The family satisfies FWER $\le \alpha$ and allows unequal treatment sizes through J_a, J_b.

Example 4.2.5. *Generic Bonferroni Pairwise Calculation*
For $I = 7$ treatments and $k = \binom{7}{2} = 21$ planned pairwise comparisons at $\alpha = .05$, take $t_{60,1-.05/(2\cdot 21)} \approx 3.16$.
The resulting simultaneous 95% Bonferroni half-width is $h \approx .085$.
Rejected pairwise nulls are $(1,4), (1,5), (1,6)$; all other planned pairwise nulls are not rejected.

Remark 4.2.6. *The Global Test Does Not Name the Difference*
Rejecting $H_0 : \mu_1 = \cdots = \mu_I$ establishes only $\exists\, a < b$ with $\mu_a \neq \mu_b$.
It does not identify such pairs or provide simultaneous sign guarantees.
Select Tukey, Bonferroni, or planned-contrast procedures according to the preassigned comparison family.

4.3 Rank-Based One-Way ANOVA

When normality or outlier sensitivity is a concern, a one-way comparison can be made after replacing observations by pooled ranks.

Definition 4.3.1. *Kruskal–Wallis Ranks*
Let $\mathcal{Y} := \{Y_{ij} : 1 \le i \le I, 1 \le j \le J_i\}$ be an unequal-size one-way sample with $N := \sum_{i=1}^{I} J_i$.
Assign increasing pooled ranks to $\mathcal{Y}$, averaging ranks over ties.

(i) *Pooled rank*: $R_{ij} := \mathrm{rank}_{\mathrm{pool}}(Y_{ij})$.

(ii) *Treatment mean rank*: $\overline{R}_{i\cdot} := \frac{1}{J_i} \sum_{j=1}^{J_i} R_{ij}$.

(iii) *Grand mean rank*: $\overline{R}_{\cdot\cdot} := \frac{N+1}{2}$.

Method 4.3.2. *Kruskal–Wallis Test*
Let $SS_B^{(R)} := \sum_{i=1}^{I} J_i(\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2$ and $K := \frac{12}{N(N+1)} SS_B^{(R)} = \frac{12}{N(N+1)} \sum_{i=1}^{I} J_i \overline{R}_{i\cdot}^{\,2} - 3(N+1)$.

(i) The exact null law enumerates all allocations of pooled ranks to group sizes $J_1, \ldots, J_I$.

(ii) For moderate samples, $K \overset{H_0}{\approx} \chi^2_{I-1}$.

(iii) Reject $H_0 : F_1 = \cdots = F_I$ at level α when $\mathbb{P}\{\chi^2_{I-1} \ge K_{\mathrm{obs}}\} \le \alpha$.

Example 4.3.3. *Generic Kruskal–Wallis Rank Calculation*
For $I = 7$ groups, suppose the pooled ranks give $K_{\mathrm{obs}} = 29.51$.
Using $K \overset{H_0}{\approx} \chi^2_6$, the tail probability satisfies $\mathbb{P}\{\chi^2_6 \ge 29.51\} < .005$.
Reject $H_0 : F_1 = \cdots = F_7$; the group response distributions are not all equal.

Corollary 4.3.4. *Two Groups Recover Mann–Whitney*
For $I = 2$, let $T_i := \sum_{j=1}^{J_i} R_{ij}$ and $N = J_1 + J_2$. Then $K = \frac{12}{J_1 J_2 (N+1)} \left(T_1 - \frac{J_1(N+1)}{2}\right)^2$, so the Kruskal–Wallis rejection order is the two-sided Mann–Whitney rank-sum order.

Proof. Let $G := \frac{N+1}{2}$. Since $T_1 + T_2 = NG$, $\overline{R}_{1\cdot} - G = \frac{T_1 - J_1 G}{J_1}$ and $\overline{R}_{2\cdot} - G = -\frac{T_1 - J_1 G}{J_2}$.
Hence $SS_B^{(R)} = (T_1 - J_1 G)^2 (\frac{1}{J_1} + \frac{1}{J_2}) = \frac{N(T_1 - J_1 G)^2}{J_1 J_2}$, and the formula for K follows.
Since $U_1 = T_1 - \frac{J_1(J_1+1)}{2}$ is affine in T_1, both tests order evidence by $|T_1 - J_1 G|$.
Structural Commentary: Two groups leave one rank contrast.
Key Insight: K is a squared centered rank sum. □

Method 4.3.5. *Rank-Based Follow-Up Comparisons*
After rejecting global rank null, let $\mathcal{C} \subseteq \{(a,b) : 1 \le a < b \le I\}$ be planned follow-up family with $|\mathcal{C}| = k$.

(i) For $(a,b) \in \mathcal{C}$, test $H_{0,ab} : F_a = F_b$ by a two-sample rank-sum statistic and let p_{ab} be its two-sided p-value.

(ii) Reject $H_{0,ab}$ iff $p_{ab} \le \frac{\alpha}{k}$, equivalently $k p_{ab} \le \alpha$.

(iii) The selected family satisfies FWER $\le \alpha$ and is most efficient when $\mathcal{C}$ is fixed before inspection.

Remark 4.3.6. *What Rank ANOVA Tests*
The null is $H_0 : F_1 = \cdots = F_I$, not merely $\mu_1 = \cdots = \mu_I$. Under comparable distributional shapes, rejection is read as a location difference; the rank form avoids normality and reduces sensitivity to extremes.

4.4 Two-Way Layouts and Interaction

A two-way layout studies two factors at once. It estimates main effects for each factor and checks whether the effect of one factor changes across levels of the other.

Definition 4.4.1. *Two-Way Additive Parametrization*
For one observation Y_{ij} at factor levels $i = 1, \ldots, I$ and $j = 1, \ldots, J$,
set $\overline{Y}_{i\cdot} := J^{-1} \sum_{j=1}^{J} Y_{ij}$, $\overline{Y}_{\cdot j} := I^{-1} \sum_{i=1}^{I} Y_{ij}$, and $\overline{Y}_{\cdot\cdot} := (IJ)^{-1} \sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij}$.

(i) *Intercept*: $\widehat{\mu} := \overline{Y}_{\cdot\cdot}$.

(ii) *Factor A, B coefficient*: $\widehat{\alpha}_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$, $1 \le i \le I$ and $\widehat{\beta}_j := \overline{Y}_{\cdot j} - \overline{Y}_{\cdot\cdot}$, $1 \le j \le J$.

(iii) *Additive fitted value*: $\widehat{Y}_{ij} := \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j$.

(iv) *Interaction residual*: $\widehat{\delta}_{ij} := Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}_{\cdot\cdot}$, so $Y_{ij} = \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j + \widehat{\delta}_{ij}$.

Structural Commentary: Exact fits satisfy zero-sum effects and margins: $\sum_i \widehat{\alpha}_i = \sum_j \widehat{\beta}_j = \sum_i \widehat{\delta}_{ij} = \sum_j \widehat{\delta}_{ij} = 0$.

Remark 4.4.2. *Interaction and Parallel Profiles*
Under additivity, $\widehat{Y}_{ij} - \widehat{Y}_{i'j} = \widehat{\alpha}_i - \widehat{\alpha}_{i'}$ is independent of j, so factor-A profiles are parallel across B. Nonparallel profiles indicate $\delta_{ij} \neq 0$ structure: the contrast between factor levels depends on the conditioning level.

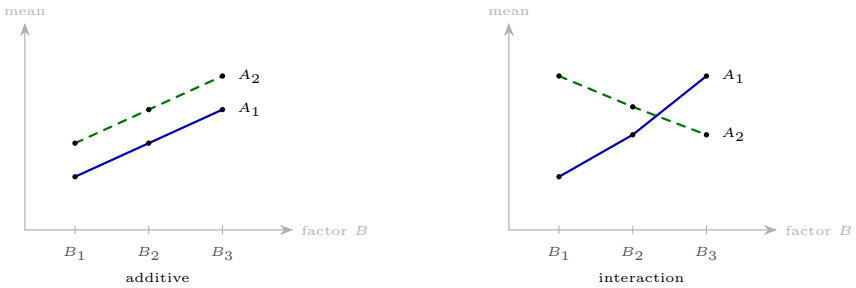

Figure 248: Parallel profiles indicate additivity; changing gaps indicate interaction.

Definition 4.4.3. *Balanced Two-Way Normal Model*
A *balanced two-way normal fixed-effects model* observes Y_{ijk} for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$, with common cell size $K > 1$, and assumes $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$ for mutually independent $\varepsilon_{ijk} \sim N(0, \sigma^2)$,
under $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$, and $\sum_j \delta_{ij} = 0\ \forall i$.
Key Insight: $\delta_{ij} = 0\ \forall (i,j)$ iff the cell means are additive in the two factors.

Definition 4.4.4. *Two-Way Sums of Squares*
Let Y_{ijk} be balanced observations for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$.

(i) *Means*: $\overline{Y}_{ij\cdot} := \frac{1}{K}\sum_k Y_{ijk}$, $\overline{Y}_{i\cdot\cdot} := \frac{1}{JK}\sum_j\sum_k Y_{ijk}$, $\overline{Y}_{\cdot j\cdot} := \frac{1}{IK}\sum_i\sum_k Y_{ijk}$, $\overline{Y}_{\cdots} := \frac{1}{IJK}\sum_i\sum_j\sum_k Y_{ijk}$.

(ii) *Between-cell components*: $SS_A := JK\sum_i(\overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdots})^2$, $SS_B := IK\sum_j(\overline{Y}_{\cdot j\cdot} - \overline{Y}_{\cdots})^2$,
and $SS_{AB} := K\sum_i\sum_j(\overline{Y}_{ij\cdot} - \overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdot j\cdot} + \overline{Y}_{\cdots})^2$.

(iii) *Within-cell and total*: $SS_E := \sum_i\sum_j\sum_k(Y_{ijk} - \overline{Y}_{ij\cdot})^2$, $SS_{\text{tot}} := \sum_i\sum_j\sum_k(Y_{ijk} - \overline{Y}_{\cdots})^2$,
and $SS_{\text{tot}} = SS_A + SS_B + SS_{AB} + SS_E$.

Structural Commentary: Orthogonal projections split total variation.

Theorem 4.4.5. *Expected Two-Way Sums of Squares*
Assume $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$ with independent $\varepsilon_{ijk} \sim N(0,\sigma^2)$,
and $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$; $\sum_j \delta_{ij} = 0\ \forall i$.

(i) $\mathbb{E}(SS_A) = (I-1)\sigma^2 + JK\sum_i \alpha_i^2$.

(ii) $\mathbb{E}(SS_B) = (J-1)\sigma^2 + IK\sum_j \beta_j^2$.

(iii) $\mathbb{E}(SS_{AB}) = (I-1)(J-1)\sigma^2 + K\sum_i\sum_j \delta_{ij}^2$.

(iv) $\mathbb{E}(SS_E) = IJ(K-1)\sigma^2$.

Proof. Let $Y = m + \varepsilon \in \mathbb{R}^{IJK}$ with $m_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij}$ and $\varepsilon \sim N(0, \sigma^2 I_{IJK})$.
For an orthogonal projection P, $\mathbb{E}\,\|PY\|^2 = \|Pm\|^2 + \sigma^2 \operatorname{rank} P$.

(i) $P_A m$ has value α_i repeated over j,k, so $\|P_A m\|^2 = JK\sum_i \alpha_i^2$ and $\operatorname{rank} P_A = I - 1$.

(ii) $P_B m$ has value β_j repeated over i,k, so $\|P_B m\|^2 = IK\sum_j \beta_j^2$ and $\operatorname{rank} P_B = J - 1$.

(iii) $P_{AB} m$ has value δ_{ij} repeated over k, so $\|P_{AB} m\|^2 = K\sum_i\sum_j \delta_{ij}^2$ and $\operatorname{rank} P_{AB} = (I-1)(J-1)$.

(iv) P_E centers within cells; since m_{ijk} is constant in k for fixed (i,j), $P_E m = 0$ and $\operatorname{rank} P_E = IJ(K-1)$.

Applying the projection identity to P_A, P_B, P_{AB}, P_E gives the four expectations.
Structural Commentary: Balanced orthogonality separates noise degrees of freedom from effect energy.
Key Insight: Each expectation is σ^2 times rank plus squared signal norm. □

Theorem 4.4.6. *Two-Way Normal-Theory F Tests*
Assume observations Y_{ijk} for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$ satisfy $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$
with independent $\varepsilon_{ijk} \sim N(0,\sigma^2)$, $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$, and $\sum_j \delta_{ij} = 0\ \forall i$.
Let $d_E := IJ(K-1)$ and $MS_E := \frac{SS_E}{d_E}$.

(i) $SS_A, SS_B, SS_{AB}, SS_E$ are independent, and $\frac{SS_E}{\sigma^2} \sim \chi^2_{d_E}$.

(ii) Under $H_A : \alpha_i = 0\ \forall i$, $\frac{SS_A}{\sigma^2} \sim \chi^2_{I-1}$; under $H_B : \beta_j = 0\ \forall j$, $\frac{SS_B}{\sigma^2} \sim \chi^2_{J-1}$;
under $H_{AB} : \delta_{ij} = 0\ \forall (i,j)$, $\frac{SS_{AB}}{\sigma^2} \sim \chi^2_{(I-1)(J-1)}$.

(iii) $F_A := \frac{\frac{SS_A}{I-1}}{MS_E} \sim F_{I-1,d_E}$ under H_A, $F_B := \frac{\frac{SS_B}{J-1}}{MS_E} \sim F_{J-1,d_E}$ under H_B,
and $F_{AB} := \frac{\frac{SS_{AB}}{(I-1)(J-1)}}{MS_E} \sim F_{(I-1)(J-1),d_E}$ under H_{AB}.

Proof.

(i) Vectorise $Y = m + \varepsilon \in \mathbb{R}^{IJK}$ with $\varepsilon \sim N(0, \sigma^2 I_{IJK})$.
Let P_A, P_B, P_{AB}, P_E be the mutually orthogonal projections onto the factor-A, factor-B, interaction, and within-cell subspaces, so $SS_R = \|P_R Y\|^2$ for $R \in \{A, B, AB, E\}$.

Orthogonal projections of a spherical normal vector are independent, and $\frac{\|P_R Y\|^2}{\sigma^2} \sim \chi^2_{d_R}(\lambda_R)$ with $d_R = \operatorname{rank} P_R$ and $\lambda_R = \frac{\|P_R m\|^2}{\sigma^2}$. Since $d_E = IJ(K-1)$ and $\lambda_E = 0$, claim (i) follows.

(ii) Here $d_A = I-1$, $d_B = J-1$, and $d_{AB} = (I-1)(J-1)$. Under H_A, H_B, H_{AB} respectively, $\lambda_A, \lambda_B, \lambda_{AB}$ vanish, so the central chi-square laws in claim (ii) follow.

(iii) In each null case the numerator chi-square in claim (ii) is independent of SS_E by claim (i); dividing by its degrees of freedom and by $\frac{SS_E}{d_E} = MS_E$ gives the stated F ratios in claim (iii).

Structural Commentary: Balanced normal ANOVA is an orthogonal Gaussian projection decomposition.
Key Insight: Null effects remove noncentrality, leaving independent chi-square ratios. □

Example 4.4.7. *Generic Transformed Two-Way ANOVA*
Consider balanced two-way study with factor A having 2 levels, factor B having 3 levels, $K = 18$ observations per cell. After log transformation stabilises cell spread, ANOVA table on transformed scale is

Source	*df*	*SS*	*MS*	*F*
Factor A	1	2.074	2.074	5.99
Factor B	2	15.588	7.794	22.53
Interaction	2	.810	.405	1.17
Error	102	35.296	.346	
Total	107	53.768		

For factor A, $F_A = 5.99$ gives $p < .025$, and an estimated log contrast .28 corresponds to the original-scale multiplier $e^{.28} \approx 1.32$ with approximate interval $(1.06, 1.65)$. For interaction, $F_{AB} = 1.17$ gives $p = .31$, so there is no strong evidence that the A contrast depends on the level of B.

Remark 4.4.8. *Transformations and Diagnostics*
Two-way ANOVA inherits the normal-theory assumptions of the one-way model. Residual plots, normal probability plots, profile plots, and plots of cell spread against cell mean help decide whether a transformation is needed. A transformation can stabilize variance and improve symmetry, but it also changes additive effects on the transformed scale into multiplicative or otherwise nonlinear effects on the original scale.

Remark 4.4.9. *Factorial Economy and Incomplete Designs*
Factorial designs use each observation to estimate multiple factor contrasts and interactions.
As the factor set grows, full factorials may be replaced by fractional factorials that omit cells while preserving chosen estimable effects; when blocks cannot contain all treatments, incomplete block designs retain partial blocking without all block-treatment pairs.

4.5 Randomised Blocks and Friedman's Test

Randomised block designs compare treatments within relatively homogeneous blocks. Blocking removes major background variation from the treatment comparison, much as pairing removes shared variation.

Definition 4.5.1. *Randomised Block Design*
A randomised block design has J blocks, each containing one experimental unit for each of the I treatments; within every block, treatments are assigned by an independent random permutation.
The additive block model is $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$ for $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = \sum_j \beta_j = 0$.

Key Insight: Blocks absorb heterogeneity; $\alpha_i - \alpha_{i'}$ is a within-block treatment contrast.

Method 4.5.2. *Randomized Block F Test*
Treat $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$ as a two-way layout with one observation per treatment-block cell and no interaction.

(i) Define $MS_A := \frac{SS_A}{I-1}$, $MS_B := \frac{SS_B}{J-1}$, and $MS_{AB} := \frac{SS_{AB}}{(I-1)(J-1)}$.

(ii) Under additivity, $\mathbb{E}(MS_A) = \sigma^2 + \frac{J}{I-1}\sum_i \alpha_i^2$, $\mathbb{E}(MS_B) = \sigma^2 + \frac{I}{J-1}\sum_j \beta_j^2$, and $\mathbb{E}(MS_{AB}) = \sigma^2$.

(iii) Test $H_0 : \alpha_1 = \cdots = \alpha_I = 0$ by $F := \frac{MS_A}{MS_{AB}}$.

(iv) Under the normal additive null, $F \sim F_{I-1,(I-1)(J-1)}$.

Remark 4.5.3. *Interaction in a Block Design*
If $\delta_{ij} \neq 0$, then MS_{AB} estimates σ^2 plus treatment-block interaction energy.
The denominator of $F = \frac{MS_A}{MS_{AB}}$ is inflated, so the additive-model treatment test is conservative.

Example 4.5.4. *Generic Randomized Block ANOVA*
Suppose $I = 7$ treatments are compared across $J = 10$ blocks, with each block receiving all treatments in randomized order. The randomized-block ANOVA table is

Source	*df*	*SS*	*MS*	*F*
Treatments	6	53013	8835	2.85
Blocks	9	103280	11476	3.71
Interaction	54	167130	3095	
Total	69	323422		

$F = 2.85$ on $(6, 54)$ degrees of freedom gives $p < .025$, rejecting equal treatment effects.
Blocking isolates nuisance block heterogeneity.

Method 4.5.5. *Friedman's Rank Test for Blocks*
For observations Y_{ij} from treatment i in block j, rank $Y_{1j}, \ldots, Y_{Ij}$ within each block.

(i) Let R_{ij} be within-block rank, use average ranks for ties, set $\overline{R}_{i\cdot} := \frac{1}{J}\sum_j R_{ij}$, $\overline{R}_{\cdot\cdot} := \frac{I+1}{2}$.

(ii) Define $Q := \frac{12J}{I(I+1)}\sum_i (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2$.

(iii) Under $H_0 : F_1 = \cdots = F_I$, use $Q \overset{H_0}{\approx} \chi^2_{I-1}$ for moderate J.

(iv) Reject at level α when $\mathbb{P}\{\chi^2_{I-1} \ge Q_{\text{obs}}\} \le \alpha$.

Example 4.5.6. *Generic Friedman Rank Calculation*
For $I = 7$ treatments, suppose the average within-block ranks are

Treatment	T_1	T_2	T_3	T_4	T_5	T_6	T_7
Average rank	5.10	4.90	2.30	3.50	3.05	4.90	4.25

With $\overline{R}_{\cdot\cdot} = 4$, $\sum_i (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2 = 6.935$ and $Q = 14.86$.
Using $Q \overset{H_0}{\approx} \chi^2_6$, the tail probability is below .025, so the treatment response distributions are not all equal.

Corollary 4.5.7. *Two Treatments Recover the Sign Test*
For $I = 2$, set $D_j := Y_{1j} - Y_{2j}$ and $B := \#\{j : D_j > 0\}$. Friedman's statistic is a function of B alone, so the two-treatment Friedman test is equivalent to the paired sign test.

Proof. Ignoring ties, $R_{1j} = 2\mathbf{1}_{\{D_j>0\}} + \mathbf{1}_{\{D_j<0\}} = 1 + \mathbf{1}_{\{D_j>0\}}$.
Hence $\sum_j R_{1j} = J + B$, so $\overline{R}_{1\cdot}$, $\overline{R}_{2\cdot}$, and Q are determined by B.
Under H_0, signs are exchangeable and $B \sim \mathrm{Bin}(J, 1/2)$, which is the paired sign-test calibration.
Structural Commentary: Two within-block ranks encode only the sign of D_j.
Key Insight: The rank-sum statistic collapses to a binomial sign count. □

5 Linear Models and Regression

5.1 Least Squares Curve Fitting and Linearization

Least squares begins as a geometric fitting rule and becomes a statistical model only after assumptions are made about the errors. The optimization step is asymmetric in the two plotted variables: ordinary regression measures vertical prediction errors.

Definition 5.1.1. *Least-Squares Fit to a Parametric Family*
Let $[n] := \{1, \dots, n\}$, let $(x_i, y_i)_{i\in[n]} \subseteq \mathcal{X} \times \mathbb{R}$, let $\Theta \subseteq \mathbb{R}^p$, and let $m : \mathcal{X} \times \Theta \to \mathbb{R}$.

(i) *Criterion*: $S : \Theta \to [0, \infty)$, $S(\theta) := \sum_{i=1}^{n} \big(y_i - m(x_i; \theta)\big)^2$.

(ii) *Estimate*: $\widehat{\theta} \in \arg\min_{\theta\in\Theta} S(\theta)$.

(iii) *Residual*: $\widehat{e}_i := y_i - m(x_i; \widehat{\theta})$, $i \in [n]$.

Key Insight: $y_i = m(x_i; \widehat{\theta}) + \widehat{e}_i$, so $\widehat{e}_i > 0$ iff $y_i > m(x_i; \widehat{\theta})$.

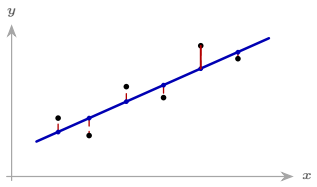

Figure 249: Least squares minimizes squared vertical residuals.

Definition 5.1.2. *Linear in the Parameters*
Let $\mathcal{X}$ be an input set, let $\Theta \subseteq \mathbb{R}^p$, and let $m : \mathcal{X} \times \Theta \to \mathbb{R}$. m is linear in β if and only if
$\exists$ fixed maps $\phi_0, \dots, \phi_{p-1} : \mathcal{X} \to \mathbb{R}$ with $m(x; \beta) = \sum_{j=0}^{p-1} \beta_j \phi_j(x)$ $\forall x \in \mathcal{X}$ and $\beta \in \Theta$.

Structural Commentary: Polynomial regression uses $\phi_j(x) = x^j$ and estimates only the coefficients β_j.

Proposition 5.1.3. *Least-Squares Line*
Let $[n] := \{1, \dots, n\}$, $(x_i, y_i)_{i\in[n]} \subseteq \mathbb{R}^2$, $\overline{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$, $\overline{y} := \frac{1}{n}\sum_{i=1}^{n} y_i$, assume $D := \sum_{i=1}^{n} (x_i - \overline{x})^2 > 0$.

For $b = (b_0, b_1) \in \mathbb{R}^2$, set $S(b) := \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$.

Then $\exists! \widehat{\beta} = (\widehat{\beta}_0, \widehat{\beta}_1) \in \arg\min_{b\in\mathbb{R}^2} S(b)$; $\widehat{y}(x) := \widehat{\beta}_0 + \widehat{\beta}_1 x$ satisfies $\widehat{\beta}_1 = \frac{1}{D}\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})$, $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$.

Proof. Set $u_i := x_i - \overline{x}$, $v_i := y_i - \overline{y}$, and $c(b) := b_0 + b_1\overline{x} - \overline{y}$.
Since $\sum_{i=1}^{n} u_i = \sum_{i=1}^{n} v_i = 0$, $S(b) = \sum_{i=1}^{n} (v_i - b_1 u_i)^2 + n c(b)^2$.
Thus $c(\widehat{\beta}) = 0$, so $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1\overline{x}$, and $\widehat{\beta}_1$ uniquely minimizes $Q(t) := \sum_{i=1}^{n} (v_i - t u_i)^2$.
Since $Q'(t) = -2\sum_{i=1}^{n} u_i(v_i - t u_i)$ and $\sum_{i=1}^{n} u_i^2 = D > 0$, the unique critical point is $t = \frac{1}{D}\sum_{i=1}^{n} u_i v_i$.
Structural Commentary: The fitted line passes through $(\overline{x}, \overline{y})$.
Key Insight: Centering turns the slope into covariance divided by variance. □

Example 5.1.4. *Linear and Nonlinear Parameters*
For fixed $\omega = (\omega_1, \omega_2)$, $f_{\theta,\omega}(t) := A\cos(\omega_1 t) + B\sin(\omega_1 t) + C\cos(\omega_2 t) + D\sin(\omega_2 t) = \theta^t \phi_\omega(t)$
with $\theta = (A, B, C, D)^t$ and $\phi_\omega(t) = (\cos(\omega_1 t), \sin(\omega_1 t), \cos(\omega_2 t), \sin(\omega_2 t))^t$.
Estimating ω makes $f_{\theta,\omega}$ nonlinear in (θ, ω).
For fixed $\lambda = (\alpha, \beta)$, $g_{\eta,\lambda}(t) := Ae^{-\alpha t} + Be^{-\beta t} = \eta^t \psi_\lambda(t)$ with $\eta = (A, B)^t$ and $\psi_\lambda(t) = (e^{-\alpha t}, e^{-\beta t})^t$.
Estimating λ makes $g_{\eta,\lambda}$ nonlinear in (η, λ).

Method 5.1.5. *Linearizing Transformations*
Let $[n] := \{1, \dots, n\}$, $K > 0$, $C > 0$, $e_A \in \mathbb{R}$, $T > 0$, and $\alpha(T) := C \exp\left(-\frac{e_A}{KT}\right)$.
For observations $(T_i, \alpha_i)_{i \in [n]} \subset (0, \infty)^2$, set $x := \frac{1}{T}$, $z := \log \alpha(T)$, $x_i := \frac{1}{T_i}$, and $z_i := \log \alpha_i$.

(i) $z = a + bx$ with $a := \log C$ and $b := -\frac{e_A}{K}$.

(ii) $(\widehat{a}, \widehat{b}) \in \underset{(a,b) \in \mathbb{R}^2}{\arg\min} \sum_{i=1}^{n} (z_i - a - bx_i)^2$, $\widehat{C} := e^{\widehat{a}}$, and $\widehat{e}_A := -K\widehat{b}$.

(iii) $\widehat{\varepsilon}_i := z_i - \widehat{a} - \widehat{b}x_i$ is diagnosed against x_i and $\widehat{z}_i := \widehat{a} + \widehat{b}x_i$.
Systematic structure indicates misspecification on the transformed scale.

5.2 Simple Linear Regression Inference and Diagnostics

Simple linear regression adds a noise model to the least-squares line. The x_i values are treated as fixed design points, and randomness enters through the observed responses.

Definition 5.2.1. *Standard Simple Linear Regression Model*
Let $[n] := \{1, \dots, n\}$, let $x_i \in \mathbb{R}$ be fixed for $i \in [n]$, let $\beta_0, \beta_1 \in \mathbb{R}$, and let $\sigma^2 > 0$.

(i) *Standard model*: $Y_i = \beta_0 + \beta_1 x_i + e_i$, $\mathbb{E}e_i = 0$, $\mathrm{Var}(e_i) = \sigma^2$, and $\mathrm{Cov}(e_i, e_j) = 0$ for $i \neq j$, $i, j \in [n]$.

(ii) *Normal model*: $Y_i = \beta_0 + \beta_1 x_i + e_i$ with $e_1, \dots, e_n$ independent and $e_i \sim N(0, \sigma^2)$, $i \in [n]$.

Structural Commentary: $(x_i)_{i \in [n]}$ is fixed; randomness enters through $(e_i)_{i \in [n]}$.

Theorem 5.2.2. *Mean and Variance of the Simple Regression Coefficients*
Let $[n] := \{1, \dots, n\}$, let $x_i \in \mathbb{R}$, and set $\overline{x} := \frac{1}{n} \sum_{i=1}^{n} x_i$ and $D := \sum_{i=1}^{n} (x_i - \overline{x})^2 > 0$.
Let $\beta := (\beta_0, \beta_1)$ and assume $Y_i = \beta_0 + \beta_1 x_i + e_i$, $\mathbb{E}e_i = 0$, $\mathrm{Var}(e_i) = \sigma^2$, and $\mathrm{Cov}(e_i, e_j) = 0$ for $i \neq j$.
If $\widehat{\beta} = (\widehat{\beta}_0, \widehat{\beta}_1)$ minimizes $\sum_{i=1}^{n} (Y_i - b_0 - b_1 x_i)^2$ over $b = (b_0, b_1) \in \mathbb{R}^2$, then $\mathbb{E}\widehat{\beta} = \beta$; $\mathrm{Var}(\widehat{\beta}) = \sigma^2 \begin{pmatrix} \frac{1}{n} + \frac{\overline{x}^2}{D} & -\frac{\overline{x}}{D} \\ -\frac{\overline{x}}{D} & \frac{1}{D} \end{pmatrix}$.

Proof. Set $u_i := x_i - \overline{x}$ and $\overline{e} := \frac{1}{n} \sum_{i=1}^{n} e_i$.

Proposition 5.1.3 gives $\widehat{\beta}_1 - \beta_1 = D^{-1} \sum_{i=1}^{n} u_i e_i$ and $\widehat{\beta}_0 - \beta_0 = \overline{e} - \overline{x}(\widehat{\beta}_1 - \beta_1)$.

Because $\sum_{i=1}^{n} u_i = 0$, $\mathbb{E}(\widehat{\beta} - \beta) = 0$, $\mathrm{Var}(\widehat{\beta}_1) = D^{-2} \sum_{i=1}^{n} u_i^2 \sigma^2 = \frac{\sigma^2}{D}$, $\mathrm{Cov}(\overline{e}, \widehat{\beta}_1) = \frac{1}{n} D^{-1} \sum_{i=1}^{n} u_i \sigma^2 = 0$, and $\mathrm{Var}(\overline{e}) = \frac{\sigma^2}{n}$. Therefore $\mathrm{Var}(\widehat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\overline{x}^2 \sigma^2}{D}$ and $\mathrm{Cov}(\widehat{\beta}_0, \widehat{\beta}_1) = -\frac{\overline{x}\sigma^2}{D}$.
Structural Commentary: The slope is a fixed linear combination of the responses.
Key Insight: Centering the design isolates the uncertainty in the slope. □

Definition 5.2.3. *Residual Sum of Squares and Error Scale*
Let $[n] := \{1, \dots, n\}$ with $n > 2$, and let $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ for $i \in [n]$.

(i) *Residual Sum of Squares*: $\mathrm{RSS} := \sum_{i=1}^{n} (Y_i - \widehat{y}_i)^2$.

(ii) *Error Scale*: $s^2 := \frac{\mathrm{RSS}}{n-2}$ and $s := \sqrt{s^2}$.

Key Insight: $n - 2$ subtracts the fitted coefficients $\widehat{\beta}_0, \widehat{\beta}_1$.

Corollary 5.2.4. *Simple Regression t Intervals and Tests*
Let $n > 2$, $[n] := \{1, \dots, n\}$, $\alpha \in (0, 1)$, $\beta \in \mathbb{R}^2$, $\sigma^2 > 0$, $x = (x_1, \dots, x_n)^t \in \mathbb{R}^n$, $X := (\mathbf{1}, x) \in \mathbb{R}^{n \times 2}$ with $\mathrm{rank}(X) = 2$, $Y = X\beta + e$, and $e \sim N(0, \sigma^2 I_n)$.
Set $\widehat{\beta} := (X^t X)^{-1} X^t Y$, $s^2 := \frac{\|Y - X\widehat{\beta}\|_2^2}{n-2}$, $\overline{x} := \frac{1}{n} \sum_{i=1}^{n} x_i$, $D := \sum_{i=1}^{n} (x_i - \overline{x})^2$, $c_0 := \frac{1}{n} + \frac{\overline{x}^2}{D}$, and $c_1 := D^{-1}$.
For $j \in \{0, 1\}$ and $\beta_{j0} \in \mathbb{R}$, set $s_{\widehat{\beta}_j} := s\sqrt{c_j}$.
Then $T_j := \frac{\widehat{\beta}_j - \beta_j}{s_{\widehat{\beta}_j}} \sim t_{n-2}$, $C_j(Y) := [\widehat{\beta}_j - t_{n-2,1-\frac{\alpha}{2}} s_{\widehat{\beta}_j}, \widehat{\beta}_j + t_{n-2,1-\frac{\alpha}{2}} s_{\widehat{\beta}_j}]$ satisfies $\mathbb{P}_\beta(\beta_j \in C_j(Y)) = 1 - \alpha$,
and the two-sided level-α test of $H_0 : \beta_j = \beta_{j0}$ rejects iff $\left| \frac{\widehat{\beta}_j - \beta_{j0}}{s_{\widehat{\beta}_j}} \right| > t_{n-2,1-\frac{\alpha}{2}}$.

Proof. Let $P := X(X^t X)^{-1} X^t$, $M := I - P$, and $G := (X^t X)^{-1} = (g_{jk})_{j,k=0}^{1}$.
Then $\widehat{\beta} - \beta = GX^t e$, $g_{00} = c_0$, $g_{11} = c_1$, $\mathrm{Var}(\widehat{\beta}_j) = \sigma^2 c_j$, and $Z_j := \frac{\widehat{\beta}_j - \beta_j}{\sigma\sqrt{c_j}} \sim N(0, 1)$.
Also $\frac{(n-2)s^2}{\sigma^2} = \frac{e^t M e}{\sigma^2}$, $M = M^t = M^2$, $\mathrm{rank}(M) = n - 2$, and $MX = 0$, hence $U := \frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{n-2}$ and $\mathrm{Cov}(GX^t e, Me) = GX^t \sigma^2 M = 0$. Joint normality gives $Z_j \perp U$, so $T_j = \frac{Z_j}{\sqrt{\frac{U}{n-2}}} \sim t_{n-2}$.
The interval and rejection rule follow from $\mathbb{P}(|T_j| \leq t_{n-2,1-\frac{\alpha}{2}}) = 1 - \alpha$.
Structural Commentary: $T_j = \frac{Z_j}{\sqrt{\frac{U}{n-2}}}$ with $Z_j \sim N(0, 1)$, $U \sim \chi^2_{n-2}$, and $Z_j \perp U$.
Key Insight: Normality and orthogonal residual projection make replacing σ by s exact. □

Example 5.2.5. *Calibration by Peak Area*
Let $[n] := \{1, \dots, n\}$, with known standard concentrations x_i and measured peak areas A_i, $i \in [n]$.
Fit $A_i = \beta_0 + \beta_1 x_i + e_i$ by least squares, giving $\widehat{A}(x) = \widehat{\beta}_0 + \widehat{\beta}_1 x$.
For a new area A_*, the inverse calibration estimate is $\widehat{x}_* := \frac{A_* - \widehat{\beta}_0}{\widehat{\beta}_1}$ when $\widehat{\beta}_1 \neq 0$.
In the reported fit, $\widehat{\beta}_0 = .0729$, $\widehat{\beta}_1 = 10.77$, $s = .068$, $\mathrm{se}(\widehat{\beta}_0) = .0297$, and $\mathrm{se}(\widehat{\beta}_1) = .27$, hence $t_0 = 2.45$, $t_1 = 40.20$, and the reported

95% intervals are $C_0 = (.011, .135)$ and $C_1 = (10.21, 11.33)$.
Thus $\widehat{\beta}_1 > 0$ makes peak area a monotone calibration signal.

Definition 5.2.6. *Homoscedastic and Heteroscedastic Errors*
Let $[n] := \{1, \dots, n\}$, let $e = (e_1, \dots, e_n)^t$ be errors at design points x_i, and set $v_i := \mathrm{Var}(e_i)$.

(i) *Homoscedastic*: $\exists \sigma^2 \geq 0$ with $v_i = \sigma^2$, $i \in [n]$.
(ii) *Heteroscedastic*: $\exists i, j \in [n]$ with $v_i \neq v_j$.

Key Insight: $\mathrm{Var}(e) \neq \sigma^2 I_n$ invalidates formulas requiring $\mathrm{Var}(e) = \sigma^2 I_n$.

Method 5.2.7. *Residual Diagnostics*
After fitting a regression, examine residuals against the fitted values and against important predictors.

(i) A random horizontal band supports the chosen mean form.
(ii) Curvature suggests a transformation or added nonlinear term.
(iii) Changing spread suggests heteroscedasticity.
(iv) A normal probability plot checks large departures from normality and possible outliers.
(v) Points with extreme x values may have high leverage and can strongly affect the fitted line.

These checks refine the modeling assumptions; they do not replace them.

Example 5.2.8. *Diagnostic Patterns*
Let $[n] := \{1, \dots, n\}$, $r_i := Y_i - \widehat{Y}_i$, $f_i := \widehat{Y}_i$, $\ell_i \in \mathcal{G}$,
$m_r(x) := \mathbb{E}(r_i \mid x_i = x)$, $v_r(f) := \mathrm{Var}(r_i \mid f_i = f)$, and $m_a := \mathbb{E}(r_i \mid \ell_i = a)$.

(i) $m_r \not\equiv 0$ or m_r non-affine; transform variables or add nonlinear terms.
(ii) v_r nonconstant; transform Y_i or use heteroscedastic errors.
(iii) $\exists a, b \in \mathcal{G}$ with $m_a \neq m_b$ or $\exists i \neq j$ with $\mathrm{Cov}(r_i, r_j \mid \ell_i = \ell_j) \neq 0$; add group effects or correlated errors.

Proposition 5.2.9. *Correlation and the Regression Line*
Let $[n] := \{1, \dots, n\}$, *let* $(x_i, y_i) \in \mathbb{R}^2$ *for* $i \in [n]$.
Set $\overline{x} := \frac{1}{n}\sum_{i=1}^n x_i$, $\overline{y} := \frac{1}{n}\sum_{i=1}^n y_i$, $s_{xx} := \frac{1}{n}\sum_{i=1}^n (x_i - \overline{x})^2$, $s_{yy} := \frac{1}{n}\sum_{i=1}^n (y_i - \overline{y})^2$, $s_{xy} := \frac{1}{n}\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})$.
If $s_{xx}, s_{yy} > 0$, $r := \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}}$, $(\widehat{\beta}_0, \widehat{\beta}_1)$ *minimizes* $\sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$ *over* $(b_0, b_1) \in \mathbb{R}^2$,
and $\widehat{y}(x) := \widehat{\beta}_0 + \widehat{\beta}_1 x$, *then* $\widehat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$, $r = \widehat{\beta}_1 \sqrt{\frac{s_{xx}}{s_{yy}}}$, *and* $\frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}} = r \frac{x - \overline{x}}{\sqrt{s_{xx}}}$.

Proof. Set $D := \sum_{i=1}^n (x_i - \overline{x})^2 = n s_{xx}$ and $N := \sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y}) = n s_{xy}$.
Proposition 5.1.3 gives $\widehat{\beta}_1 = \frac{N}{D} = \frac{s_{xy}}{s_{xx}}$ and $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$.
Hence $r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}} = \widehat{\beta}_1 \sqrt{\frac{s_{xx}}{s_{yy}}}$ and $\widehat{y}(x) - \overline{y} = \widehat{\beta}_1 (x - \overline{x})$, so $\frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}} = r \frac{x - \overline{x}}{\sqrt{s_{xx}}}$.
Structural Commentary: $z_x := \frac{x - \overline{x}}{\sqrt{s_{xx}}}$ and $z_{\widehat{y}} := \frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}}$ give $z_{\widehat{y}} = r z_x$.
Key Insight: $\widehat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$ becomes $r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}}$ after standardising both coordinates. □

Remark 5.2.10. *Regression to the Mean*
With $s_{xx}, s_{yy} > 0$ and $0 < r < 1$, set $z_x := \frac{x - \overline{x}}{\sqrt{s_{xx}}}$ and $z_{\widehat{y}} := \frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}}$.
Proposition 5.2.9 gives $z_{\widehat{y}} = r z_x$, hence $|z_{\widehat{y}}| < |z_x|$ for $z_x \neq 0$.

5.3 Matrix Linear Models and Normal Equations

Matrix notation separates the geometry of least squares from the statistical assumptions on the errors. It also makes multiple regression a direct extension of simple regression.

Definition 5.3.1. *Fixed-Design Linear Model*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta, b \in \mathbb{R}^p$, and $e \in \mathbb{R}^n$.

(i) *Model*: $Y = X\beta + e$.
(ii) *Least-Squares Estimator*: $\widehat{\beta} \in \arg\min_{b \in \mathbb{R}^p} \|Y - Xb\|_2^2$.

Structural Commentary: An intercept is encoded by $X_{\cdot 1} = \mathbf{1}$.

Theorem 5.3.2. *Normal Equations for the Linear Model*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, *and* $Q(b) := \|Y - Xb\|_2^2$ *for* $b \in \mathbb{R}^p$.

(i) $\widehat{\beta} \in \arg\min_{b \in \mathbb{R}^p} Q(b)$ *iff* $X^t X \widehat{\beta} = X^t Y$.
(ii) If $\mathrm{rank}(X) = p$, *then* $X^t X$ *is invertible and* $\arg\min_{b \in \mathbb{R}^p} Q(b) = \{(X^t X)^{-1} X^t Y\}$.

Proof.
(i) ($\Rightarrow$) If $\widehat{\beta}$ minimizes Q, then $\forall (t, h) \in \mathbb{R} \times \mathbb{R}^p$, $0 \leq Q(\widehat{\beta} + th) - Q(\widehat{\beta}) = 2th^t X^t (X\widehat{\beta} - Y) + t^2 \|Xh\|_2^2$.
Hence $\forall h \in \mathbb{R}^p$, $h^t X^t (X\widehat{\beta} - Y) = 0$, so $X^t X \widehat{\beta} = X^t Y$.
(i) ($\Leftarrow$) If $X^t X \widehat{\beta} = X^t Y$, then $\forall h \in \mathbb{R}^p$, $Q(\widehat{\beta} + h) - Q(\widehat{\beta}) = \|Xh\|_2^2 \geq 0$, so $\widehat{\beta}$ minimizes Q.
(ii) If $\mathrm{rank}(X) = p$, then $\forall u \in \mathbb{R}^p \setminus \{0\}$, $u^t X^t X u = \|Xu\|_2^2 > 0$, so $X^t X$ is invertible.
Applying (i) gives $\arg\min_{b \in \mathbb{R}^p} Q(b) = \{(X^t X)^{-1} X^t Y\}$.

Structural Commentary: $\widehat{Y} = X\widehat{\beta}$ and $\widehat{e} = Y - \widehat{Y}$ satisfy $X^t \widehat{e} = 0$, hence $\widehat{e} \perp \mathrm{col}(X)$.
Key Insight: The projection $\widehat{Y} \in \mathrm{col}(X)$ is unique; $\mathrm{rank}(X) = p$ makes $b \mapsto Xb$ injective, so $\widehat{\beta}$ is unique. □

Method 5.3.3. *Projection View of Least Squares*

(i) *Full rank*: If $\operatorname{rank}(X) = d$, define $H := X(X^\top X)^{-1}X^\top$ and $\widehat{y} := Hy$.
(ii) *Projection*: H is the orthogonal projection onto $\operatorname{col}(X)$; hence $y - \widehat{y} \perp \operatorname{col}(X)$.
(iii) *Rank deficiency*: If $\operatorname{rank}(X) < d$, $\widehat{\beta}$ may be nonunique, but $\widehat{y} = \operatorname{proj}_{\operatorname{col}(X)} y$ is unique.

Rank deficiency affects coefficient representation, not least-squares fitted values.

Definition 5.3.4. *Hat Matrix, Fitted Values, and Residuals*
Assume $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\operatorname{rank}(X) = p$, $\widehat{\beta} = (X^tX)^{-1}X^tY$, and $I = I_n$.

(i) *Hat Matrix*: $P := X(X^tX)^{-1}X^t$.
(ii) *Fitted Values*: $\widehat{Y} := X\widehat{\beta} = PY$.
(iii) *Residuals*: $\widehat{e} := Y - \widehat{Y} = (I - P)Y$.

Key Insight: $\operatorname{range}(P) = \operatorname{col}(X)$ and $PY = \widehat{Y}$.

Proposition 5.3.5. *Projection Properties of the Hat Matrix*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, $I = I_n$, $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, and $\widehat{e} := (I - P)Y$.
Then $P = P^t = P^2$, $I - P = (I - P)^t = (I - P)^2$, $\operatorname{range}(P) = \operatorname{col}(X)$, $\widehat{Y} = P_{\operatorname{col}(X)}Y$, and $X^t\widehat{e} = 0$.

Proof. Since $(X^tX)^t = X^tX$, $P^t = X((X^tX)^{-1})^tX^t = P$, and $P^2 = X(X^tX)^{-1}X^tX(X^tX)^{-1}X^t = P$.
Hence $(I - P)^t = I - P$ and $(I - P)^2 = I - 2P + P^2 = I - P$.
If $z = Xb \in \operatorname{col}(X)$, then $Pz = X(X^tX)^{-1}X^tXb = z$; also $\operatorname{range}(P) \subseteq \operatorname{col}(X)$, so $\operatorname{range}(P) = \operatorname{col}(X)$.
Finally, $X^t\widehat{e} = X^t(I - P)Y = X^tY - X^tX(X^tX)^{-1}X^tY = 0$, so $\widehat{e} \perp \operatorname{col}(X)$ and $\widehat{Y} = P_{\operatorname{col}(X)}Y$.
Structural Commentary: $P = P_{\operatorname{col}(X)}$ and $I - P = P_{\operatorname{col}(X)^\perp}$.
Key Insight: Fitting decomposes $Y = PY + (I - P)Y = \widehat{Y} + \widehat{e}$ with $\widehat{Y} \perp \widehat{e}$. □

Theorem 5.3.6. *Unbiased Estimate of the Error Variance*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, $\mathbb{E}e = 0$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, $\widehat{e} := Y - \widehat{Y}$, and $s^2 := \frac{\|\widehat{e}\|_2^2}{n-p}$. Then $\mathbb{E}\|\widehat{e}\|_2^2 = (n - p)\sigma^2$ and $\mathbb{E}s^2 = \sigma^2$.

Proof. Let $M := I_n - P$. Since $PX = X$, $\widehat{e} = MY = M(X\beta + e) = Me$.
Since $M = M^t = M^2$, $\|\widehat{e}\|_2^2 = e^tM^tMe = e^tMe$. For fixed $A \in \mathbb{R}^{n\times n}$ and centered e with $\operatorname{Var}(e) = \sigma^2 I_n$,
$\mathbb{E}(e^tAe) = \operatorname{tr}(A\sigma^2 I_n) = \sigma^2 \operatorname{tr}(A)$. Thus $\mathbb{E}\|\widehat{e}\|_2^2 = \sigma^2 \operatorname{tr}(M)$,
and $\operatorname{tr}(M) = n - \operatorname{tr}(P) = n - \operatorname{tr}((X^tX)^{-1}X^tX) = n - p$. Therefore $\mathbb{E}s^2 = \frac{1}{n-p}\mathbb{E}\|\widehat{e}\|_2^2 = \sigma^2$.
Structural Commentary: $M = I_n - P = P_{\operatorname{col}(X)^\perp}$ and $\operatorname{tr}(M) = n - p$.
Key Insight: $\widehat{e} = Me$, so residual variation is noise projected onto the $n - p$ directions orthogonal to $\operatorname{col}(X)$. □

Definition 5.3.7. *Standardized Residual*
Let $[n] := \{1, \dots, n\}$, $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, $P = (p_{ij}) := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, $\widehat{e} := Y - \widehat{Y}$, $s^2 := \frac{\|\widehat{e}\|_2^2}{n-p}$, and $s > 0$. The *standardized residual* is $r_i := \frac{\widehat{e}_i}{s\sqrt{1-p_{ii}}}$ for $i \in [n]$.
Key Insight: Under $\operatorname{Var}(e) = \sigma^2 I_n$, $\operatorname{Var}(\widehat{e}_i) = \sigma^2(1 - p_{ii})$.

Proposition 5.3.8. *Residuals Are Uncorrelated with Fitted Values*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, and $\widehat{e} := (I_n - P)Y$. Then $\operatorname{Cov}(\widehat{e}, \widehat{Y}) = 0$.

Proof. Since $\operatorname{Cov}(Y) = \operatorname{Cov}(e) = \sigma^2 I_n$ and $P = P^t = P^2$,
$\operatorname{Cov}(\widehat{e}, \widehat{Y}) = (I_n - P)\operatorname{Cov}(Y)P^t = \sigma^2(I_n - P)P = \sigma^2(P - P^2) = 0$.
Structural Commentary: $\widehat{Y} = PY$ and $\widehat{e} = (I_n - P)Y$ are images of Y under complementary projections.
Key Insight: Spherical error covariance makes orthogonal projection components uncorrelated. □

5.4 Inference, Prediction, and Multiple Regression

Once the linear model is written in matrix form, standard errors for coefficients and predictions are consequences of linear transformations of random vectors.

Theorem 5.4.1. *Mean and Covariance of Least-Squares Estimates*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, $\mathbb{E}e = 0$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$. Then $\mathbb{E}\widehat{\beta} = \beta$ and $\operatorname{Var}(\widehat{\beta}) = \sigma^2(X^tX)^{-1}$.

Proof. Let $A := (X^tX)^{-1}X^t$. Since $AX = I_p$, $\widehat{\beta} = A(X\beta + e) = \beta + Ae$.
Hence $\mathbb{E}\widehat{\beta} = \beta + A\mathbb{E}e = \beta$, and $\operatorname{Var}(\widehat{\beta}) = A\operatorname{Var}(e)A^t = \sigma^2(X^tX)^{-1}X^tX(X^tX)^{-1} = \sigma^2(X^tX)^{-1}$.
Structural Commentary: $\widehat{\beta} - \beta = (X^tX)^{-1}X^te$ is a fixed linear transformation of e.
Key Insight: X^tX controls precision: $\operatorname{Var}(\widehat{\beta}) = \sigma^2(X^tX)^{-1}$. □

Method 5.4.2. *Coefficient Inference in a Normal Linear Model*
Let $Y = X\beta + e$, $e \sim N_n(0, \sigma^2 I_n)$, $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, $\widehat{\beta} := (X^tX)^{-1}X^tY$,
$s^2 := \frac{1}{n-p}\left\|Y - X\widehat{\beta}\right\|_2^2$, $C = (c_{jk}) := (X^tX)^{-1}$, $j \in \{1, \dots, p\}$, and $\alpha \in (0, 1)$.

(i) *Standard Error*: $s_{\widehat{\beta}_j} := s\sqrt{c_{jj}}$.
(ii) *Pivot*: $T_j := \frac{\widehat{\beta}_j - \beta_j}{s_{\widehat{\beta}_j}} \sim t_{n-p}$.
(iii) *Confidence Interval*: $C_{j,1-\alpha} := \left[\widehat{\beta}_j - t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}, \widehat{\beta}_j + t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}\right]$.
(iv) *Test Statistic*: For $H_0 : \beta_j = \beta_{j0}$, $t_j := \frac{\widehat{\beta}_j - \beta_{j0}}{s_{\widehat{\beta}_j}}$.

If a coefficient CLT holds, $T_j \approx N(0,1)$.

Remark 5.4.3. *Linear-Model Inference After Selection*

Under fixed-design $y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I_n)$, and $\operatorname{rank}(X) = d$, $\widehat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1})$ and $\widehat{\sigma}^2 := \frac{\|y - X\widehat{\beta}\|_2^2}{n-d}$.
These give standard errors, t or asymptotic z statistics, and nested-model F tests when X is fixed in advance.
After data-driven selection of X or the model, these standard errors ignore selection.

Proposition 5.4.4. *Conditional Coverage Implies Unconditional Coverage*
Let $C(X,Y)$ be a confidence set for β_j, $\alpha \in (0,1)$, and $F := \{\operatorname{rank}(X) = p\}$.
If $\mathbb{P}(F) = 1$ and $\mathbb{P}(\beta_j \in C(X,Y) \mid X) = 1 - \alpha$ on F, then $\mathbb{P}(\beta_j \in C(X,Y)) = 1 - \alpha$.

Proof. Set $A := \{\beta_j \in C(X,Y)\}$. Since $\mathbb{P}(F) = 1$, $\mathbb{P}(A) = \mathbb{E}[\mathbb{P}(A \mid X)] = \mathbb{E}[(1-\alpha)\mathbf{1}_F + \mathbb{P}(A \mid X)\mathbf{1}_{F^c}] = 1 - \alpha$. *Structural Commentary*: $\mathbb{P}(F) = 1$ and $\mathbb{P}(A \mid X) = 1 - \alpha$ on F.
Key Insight: $\mathbb{P}(A \mid X) = 1 - \alpha$ a.s. implies $\mathbb{P}(A) = 1 - \alpha$. □

Proposition 5.4.5. *Mean Response at a New Design Point*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $e \sim N_n(0, \sigma^2 I_n)$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$, $s^2 := \frac{1}{n-p}\left\|Y - X\widehat{\beta}\right\|_2^2$, $x_0 \in \mathbb{R}^p$, $c_0 := x_0^t(X^tX)^{-1}x_0 > 0$,
$\mu_0 := x_0^t\beta$, $\widehat{\mu}_0 := x_0^t\widehat{\beta}$, $s_{\widehat{\mu}_0} := s\sqrt{c_0}$, and $\alpha \in (0,1)$. Then $\mathbb{E}\widehat{\mu}_0 = \mu_0$, $\operatorname{Var}(\widehat{\mu}_0) = \sigma^2 c_0$, $T_0 := \frac{\widehat{\mu}_0 - \mu_0}{s_{\widehat{\mu}_0}} \sim t_{n-p}$,
and $C_{0,1-\alpha} := \left[\widehat{\mu}_0 - t_{n-p,1-\frac{\alpha}{2}} s_{\widehat{\mu}_0}, \widehat{\mu}_0 + t_{n-p,1-\frac{\alpha}{2}} s_{\widehat{\mu}_0}\right]$ has coverage $1 - \alpha$ for μ_0.

Proof. Let $A := (X^tX)^{-1}X^t$, $P := XA$, and $M := I_n - P$.
Since $AX = I_p$, $\widehat{\mu}_0 - \mu_0 = x_0^tA(X\beta + e) - x_0^t\beta = x_0^tAe$, so $\mathbb{E}\widehat{\mu}_0 = \mu_0$ and $\operatorname{Var}(\widehat{\mu}_0) = x_0^tA(\sigma^2 I_n)A^tx_0 = \sigma^2 c_0$.
Normality gives $Z := \frac{\widehat{\mu}_0 - \mu_0}{\sigma\sqrt{c_0}} \sim N(0,1)$.

Since $M = M^t = M^2$, $\operatorname{rank}(M) = n - p$, and $MA^tx_0 = 0$, $Z \perp U := \frac{e^tMe}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p}$.
Thus $T_0 = \frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}$, and $\mathbb{P}(\mu_0 \in C_{0,1-\alpha}) = \mathbb{P}(|T_0| \le t_{n-p,1-\frac{\alpha}{2}}) = 1 - \alpha$.

Structural Commentary: $\widehat{\mu}_0 - \mu_0 = x_0^t(X^tX)^{-1}X^te$ and $s^2 = \frac{e^t(I_n - P)e}{n-p}$ use orthogonal components of e.
Key Insight: Estimating a mean response adds uncertainty only through $c_0 = x_0^t(X^tX)^{-1}x_0$. □

Corollary 5.4.6. *Prediction Interval for a New Observation*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $e \sim N_n(0, \sigma^2 I_n)$.
Let $x_0 \in \mathbb{R}^p$, $e_0 \sim N(0, \sigma^2)$ with $e_0 \perp e$, and $Y_0 := x_0^t\beta + e_0$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$, $\widehat{\mu}_0 := x_0^t\widehat{\beta}$, $s^2 := \frac{\|Y - X\widehat{\beta}\|_2^2}{n-p}$, $c_0 := x_0^t(X^tX)^{-1}x_0$, $s_{\text{pred}} := s\sqrt{1 + c_0}$, and $\alpha \in (0,1)$.
Then $\operatorname{Var}(Y_0 - \widehat{\mu}_0) = \sigma^2(1 + c_0)$, $T_{\text{pred}} := \frac{Y_0 - \widehat{\mu}_0}{s_{\text{pred}}} \sim t_{n-p}$,
and $I_{0,1-\alpha} := \left[\widehat{\mu}_0 - t_{n-p,1-\frac{\alpha}{2}} s_{\text{pred}}, \widehat{\mu}_0 + t_{n-p,1-\frac{\alpha}{2}} s_{\text{pred}}\right]$ has coverage $1 - \alpha$ for Y_0.

Proof. Let $A := (X^tX)^{-1}X^t$, $P := XA$, and $M := I_n - P$.
Since $AX = I_p$, $Y_0 - \widehat{\mu}_0 = e_0 - x_0^tAe$, so $\operatorname{Var}(Y_0 - \widehat{\mu}_0) = \sigma^2 + \sigma^2 x_0^tAA^tx_0 = \sigma^2(1 + c_0)$.
Normality gives $Z := \frac{Y_0 - \widehat{\mu}_0}{\sigma\sqrt{1+c_0}} \sim N(0,1)$. Since $AM = 0$, $Z \perp U := \frac{e^tMe}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p}$.
Thus $T_{\text{pred}} = \frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}$, and $\mathbb{P}(Y_0 \in I_{0,1-\alpha}) = \mathbb{P}(|T_{\text{pred}}| \le t_{n-p,1-\frac{\alpha}{2}}) = 1 - \alpha$.
Structural Commentary: $Y_0 - \widehat{\mu}_0 = e_0 - x_0^t(X^tX)^{-1}X^te$ combines future noise and estimation error.
Key Insight: Prediction variance is mean-response variance plus irreducible noise, $\sigma^2 c_0 + \sigma^2$. □

Definition 5.4.7. *Multiple Regression Coefficients*

Let $[n] := \{1, \dots, n\}$, $z_i = (x_{i1}, \dots, x_{i,p-1}) \in \mathbb{R}^{p-1}$ for $i \in [n]$, $u_1, \dots, u_{p-1}$ be the standard basis of $\mathbb{R}^{p-1}$, and $m(z) := \beta_0 + \sum_{k=1}^{p-1} \beta_k z_k$
In $Y_i = m(z_i) + e_i$, the *multiple regression coefficient* β_k satisfies $\forall z \in \mathbb{R}^{p-1}$ and $\forall \delta \in \mathbb{R}$, $m(z + \delta u_k) - m(z) = \delta\beta_k$ for $k \in \{1, \dots, p-1\}$.
Structural Commentary: Polynomial regression replaces z_i by transformed design columns such as z_{ik}^2.

Definition 5.4.8. *Coefficient of Determination*

Let $[n] := \{1, \dots, n\}$, $Y, \widehat{Y} \in \mathbb{R}^n$, $\overline{Y} := n^{-1}\sum_{i=1}^n Y_i$, $\mathbf{1} \in \operatorname{col}(X)$, and $\sum_{i=1}^n (Y_i - \overline{Y})^2 > 0$.

(i) *Residual Sum of Squares*: $\text{RSS} := \sum_{i=1}^n (Y_i - \widehat{Y}_i)^2$.

(ii) *Total Sum of Squares*: $\text{TSS} := \sum_{i=1}^n (Y_i - \overline{Y})^2$.

(iii) *Coefficient of Determination*: $R^2 := 1 - \frac{\text{RSS}}{\text{TSS}}$.

Structural Commentary: $R^2 = \operatorname{corr}(Y, \widehat{Y})^2$ for an intercept model.
Key Insight: $R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}$ is the observed response variation removed by the fitted model.

Remark 5.4.9. *Adjusted Coefficients*
Let X_{-j} be X without column x_j, and set $z_j := (I - \operatorname{proj}_{\operatorname{col}(X_{-j})})x_j$.

Then $\widehat{\beta}_j = \frac{z_j^\top y}{z_j^\top z_j}$, the simple-regression coefficient of y on the adjusted predictor z_j.

Thus $\widehat{\beta}_j$ uses only the part of x_j left after adjusting for other predictors.
If $x_j \approx \operatorname{col}(X_{-j})$, then $\|z_j\|_2$ is small and $\widehat{\beta}_j$ is unstable; this is collinearity.

Remark 5.4.10. *Collinearity and Interpretation*
Let X_{-j} be X without column $X_{\cdot j}$, set $d_j := \operatorname{dist}(X_{\cdot j}, \operatorname{col}(X_{-j}))$, and let $C = (c_{kl}) := (X^tX)^{-1}$.
If $0 < d_j \ll \|X_{\cdot j}\|_2$, then X^tX is nearly singular and $s_{\widehat{\beta}_j} = s\sqrt{c_{jj}}$ can be large even when $\widehat{Y} = X\widehat{\beta}$ is stable.
If $d_j = 0$, then $\operatorname{rank}(X) < p$ and $\widehat{\beta}$ is not unique.

Example 5.4.11. *Quadratic Terms as a Lack-of-Fit Check*
Compare $M_1 : Y_i = \beta_0 + \beta_1 x_i + e_i$ with $M_2 : Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + e_i$.
If $t_2 = \widehat{\beta}_2 / s_{\widehat{\beta}_2}$ is small and residual curvature is absent, retain M_1.
If $|t_2|$ is large and residual curvature decreases under M_2, use the quadratic term.
Add x^2 only when residual structure or scale supports $\beta_2 \neq 0$.

5.5 Random Design, Bootstrap, and Extensions

The fixed-design model conditions on the observed predictor values. Many observational studies are more naturally viewed as sampling random predictor-response pairs, with the fixed-design analysis recovered after conditioning on the realised design.

Definition 5.5.1. *Random-Design Linear Regression*
Let $[n] := \{1, \dots, n\}$, $(Y_i, \xi_i)_{i \in [n]}$ be independent copies of (Y, ξ), $\xi \in \mathbb{R}^p$, $\beta \in \mathbb{R}^p$, and $\sigma^2 > 0$.
A *Random-Design Linear Regression* is an iid sample $(Y_i, \xi_i)_{i \in [n]}$ satisfying $\mathbb{E}(Y \mid \xi = x) = x^t\beta$
and $\operatorname{Var}(Y \mid \xi = x) = \sigma^2$ for $x \in \operatorname{supp}(\xi)$.
Structural Commentary: Conditioning on $X_{i\cdot} = \xi_i^t$ gives $Y_{1:n} = X\beta + e$ with $e_i = Y_i - \xi_i^t\beta$.

Remark 5.5.2. *Unconditional Variability Can Differ*
Under random design, if $\mathbb{E}(\widehat{\beta} \mid X) = \beta$, then $\mathbb{E}\widehat{\beta} = \beta$.
Moreover, $\operatorname{Var}(\widehat{\beta}) = \mathbb{E}[\operatorname{Var}(\widehat{\beta} \mid X)] + \operatorname{Var}(\mathbb{E}[\widehat{\beta} \mid X]) = \sigma^2\mathbb{E}[(X^tX)^{-1}]$, not generally $\sigma^2(X^tX)^{-1}$ for frozen design.

Method 5.5.3. *Pairs Bootstrap for Random-Design Regression*
Let $D := \{(Y_i, x_i) : i \in [n]\}$, $x_i \in \mathbb{R}^p$, and let $T(D)$ be a fitted coefficient, prediction, or smooth statistic.

(i) Draw $D_b^* := \{(Y_{i_m}, x_{i_m}) : m \in [n]\}$ with $i_1, \dots, i_n$ sampled iid from $[n]$.
(ii) Fit least squares on D_b^* and compute $T_b^* := T(D_b^*)$.
(iii) Repeat for $b \in [B]$ and estimate $\operatorname{se}(T(D))$ by $\widehat{\operatorname{se}}_*(T) := \left(\frac{1}{B-1} \sum_{b=1}^{B} (T_b^* - \overline{T}^*)^2 \right)^{1/2}$, where $\overline{T}^* := B^{-1} \sum_{b=1}^{B} T_b^*$.

Perspective 5.5.4. *Extensions Beyond Ordinary Least Squares*

(i) *Robust regression*: Replace $\sum_i \widehat{e}_i^2$ by $\sum_i \Psi(\widehat{e}_i)$; $\Psi(u) = u^2$ gives least squares.
(ii) *Variable selection*: For selected predictors $S \subseteq \{1, \dots, p\}$, $\widehat{\beta}_S$ and $\operatorname{se}(\widehat{\beta}_S)$ are conditional on S.
(iii) *Influence*: Large leverage $h_{ii} = x_i^t(X^tX)^{-1}x_i$ can make observation i dominate $\widehat{\beta}$.
(iv) *Inverse calibration*: Given a fitted response curve $\widehat{m}$ and measured response y_0,
estimating x_0 from $\widehat{m}(x_0) = y_0$ is an inverse problem with uncertainty beyond $\operatorname{se}(\widehat{\beta})$.

Part XXI
Statistical Learning and Deep Learning

Statistical learning studies prediction rules, representation maps, and trained systems as mathematical objects. The statistical foundations of Parts 15 and 16 remain the language for experiments, inference, and model assessment; here the emphasis is on risk, generalization, optimization, architectures, and evaluation of learned predictors.

1 Learning Problems and Risk

1.1 Prediction, Loss Functions, and Risk

A learning problem turns observations into an action rule. Unlike inferential problems, the primary object is often the future loss of a predictor rather than a confidence statement about a parameter.

Definition 1.1.1. *Prediction Problem and Risk*
Inputs lie in $(\mathcal{X}, \mathcal{A})$, outputs in $(\mathcal{Y}, \mathcal{B})$, and $(X, Y) \sim P$ on $\mathcal{X} \times \mathcal{Y}$. Let $(\mathcal{A}_0, \mathcal{C})$ be an action space.

(i) *Predictor*: A predictor is a measurable map $f : \mathcal{X} \to \mathcal{A}_0$.
(ii) *Loss and risk*: For loss $\ell : \mathcal{A}_0 \times \mathcal{Y} \to [0, \infty]$, define $R(f) := \mathbb{E}_P \ell(f(X), Y)$.
(iii) *Bayes risk*: $R^* := \inf_f R(f)$, over measurable $f : \mathcal{X} \to \mathcal{A}_0$.

Definition 1.1.2. *Supervised Learning Terminology*
Let $S := \{(x_i, y_i) : 1 \le i \le n\}$ with $x_i = (x_{i1}, \dots, x_{ip})^\top \in \mathcal{X}$ and $y_i \in \mathcal{Y}$.

(i) *Features*: Coordinates x_{ij} are variables, inputs, predictors, independent variables, or features.
(ii) *Response*: Values y_i are outputs, responses, targets, or dependent variables.
(iii) *Regression*: If $\mathcal{Y} \subseteq \mathbb{R}^q$, the task is quantitative prediction.
(iv) *Classification*: If $\mathcal{Y} = \mathcal{G}$ is a finite unordered label set, the task is classification.
(v) *Ordered labels*: If $\mathcal{Y}$ is ordered categorical, order matters but arithmetic on labels usually does not.
(vi) *One-hot coding*: For $|\mathcal{G}| = K$, encode class g by $t_g \in \{0, 1\}^K$ with $(t_g)_g = 1$.
Classify vector scores $s(x) \in \mathbb{R}^K$ by $\operatorname{argmax}_k s_k(x)$.

If y_i are unobserved, the problem is unsupervised and targets structure in $\{x_i\}_{i=1}^n$.

Example 1.1.3. *Representative Learning Problems*

(i) *Email filtering*: x contains word and punctuation frequencies, $y \in \{\text{spam}, \text{ham}\}$, and asymmetric loss can penalize false deletion more than missed spam.
(ii) *Clinical regression*: $y = \log(\text{PSA})$ and x contains log cancer volume, prostate weight, age, and related measurements; correlated predictors are common.
(iii) *Digit recognition*: $x \in [0, 1]^{16 \times 16}$ or $[0, 1]^{256}$ and $y \in \{0, \dots, 9\}$; action set may add rejection option $\perp$.
(iv) *Microarrays*: An expression matrix $X \in \mathbb{R}^{G \times n}$ has genes as rows and samples as columns.
Typical first tasks are unsupervised clustering of genes or samples.

Example 1.1.4. *Common Prediction Losses*

(i) Regression commonly uses squared loss $\ell(a, y) = \|a - y\|_2^2$ or absolute loss $\ell(a, y) = |a - y|$.
(ii) Binary classification with labels in $\{-1, 1\}$ uses zero-one loss $\mathbf{1}_{\{ay \le 0\}}$ for a signed score a.
(iii) Probabilistic classification uses log loss $-\log q_y$ when q is a predicted distribution on labels.

Definition 1.1.5. *Bayes Predictors*

(i) *Conditional risk*: For action a, set $r_x(a) := \mathbb{E}[\ell(a, Y) \mid X = x]$; any $f^*(x) \in \operatorname{argmin}_a r_x(a)$ is Bayes.
(ii) *Squared loss*: If $\ell(a, y) = (a - y)^2$ and $\mathbb{E}[|Y| \mid X = x] < \infty$, then $f^*(x) = \mathbb{E}[Y \mid X = x]$.
(iii) *Absolute loss*: If $\ell(a, y) = |a - y|$, any conditional median $m(x)$
with $\mathbb{P}(Y \le m(x) \mid X = x) \ge \frac{1}{2}$ and $\mathbb{P}(Y \ge m(x) \mid X = x) \ge \frac{1}{2}$ is Bayes.
(iv) *Zero-one loss*: For labels $\mathcal{G}$ and $\ell(g, y) = \mathbf{1}_{\{g \ne y\}}$, a Bayes classifier is $g^*(x) \in \operatorname{argmax}_{g \in \mathcal{G}} \mathbb{P}(Y = g \mid X = x)$.

Remark 1.1.6. *Risk as a Prediction Target*
Here risk means predictive loss $R(f) = \mathbb{E}_P \ell(f(X), Y)$ under a data law P.
Decision-theoretic risk instead studies $R(\theta, \delta) = \mathbb{E}_\theta L(\theta, \delta(X))$ over a model family.

1.2 Empirical Risk Minimization

The population risk is the target, but only a sample is observed. Empirical risk minimisation replaces the unknown expectation by an average over the training data.

Definition 1.2.1. *Empirical Risk Minimization*
Let $S := ((x_1, y_1), \dots, (x_n, y_n))$.

(i) *Empirical risk*: For loss ℓ, define $\widehat{R}_S(f) := n^{-1} \sum_{i=1}^n \ell(f(x_i), y_i)$.
(ii) *Hypothesis class*: Let $\mathcal{F}$ be the admissible predictor class.
(iii) *ERM*: Any $\widehat{f} \in \operatorname{argmin}_{f \in \mathcal{F}} \widehat{R}_S(f)$ is an empirical risk minimiser.

Definition 1.2.2. *Regularised Empirical Risk*
Let $\Omega : \mathcal{F} \to [0, \infty]$ be a complexity penalty and $\lambda \ge 0$.
Define $Q_\lambda(f) := \widehat{R}_S(f) + \lambda \Omega(f)$ and choose $\widehat{f}_\lambda \in \operatorname{argmin}_{f \in \mathcal{F}} Q_\lambda(f)$.
The penalty Ω may encode smoothness, sparsity, norm control, margin size, or architecture.

Theorem 1.2.3. *Finite-Class ERM Bound*
Let $S = ((X_1, Y_1), \dots, (X_n, Y_n))$ be i.i.d. from P, let $\emptyset \neq \mathcal{F}$ be finite.
Assume $0 \le \ell(f(X), Y) \le 1$ for all $f \in \mathcal{F}$, and fix $\delta \in (0,1)$. Set $\varepsilon_{n,\delta} := \sqrt{\log(2|\mathcal{F}|/\delta)/(2n)}$.

(i) Uniform deviation: With probability at least $1-\delta$, $\sup_{f\in\mathcal{F}} \left|R(f) - \widehat{R}_S(f)\right| \le \varepsilon_{n,\delta}$.

(ii) ERM excess risk: If $\widehat{f} \in \operatorname{argmin}_{f\in\mathcal{F}} \widehat{R}_S(f)$, then $R(\widehat{f}) \le \inf_{f\in\mathcal{F}} R(f) + 2\varepsilon_{n,\delta}$.

Proof.

(i) For fixed f, set $Z_i(f) := \ell(f(X_i), Y_i) \in [0,1]$, so $R(f) = \mathbb{E}Z_i(f)$ and $\widehat{R}_S(f) = n^{-1}\sum_{i=1}^n Z_i(f)$.
Hoeffding and a union bound give $\mathbb{P}(\sup_{f\in\mathcal{F}} \left|\widehat{R}_S(f) - R(f)\right| > \varepsilon) \le 2|\mathcal{F}| e^{-2n\varepsilon^2}$; set $\varepsilon = \varepsilon_{n,\delta}$.

(ii) On the uniform-deviation event, choose $f^* \in \operatorname{argmin}_{f\in\mathcal{F}} R(f)$.
Since $\widehat{R}_S(\widehat{f}) \le \widehat{R}_S(f^*)$, $R(\widehat{f}) \le \widehat{R}_S(\widehat{f}) + \varepsilon_{n,\delta} \le \widehat{R}_S(f^*) + \varepsilon_{n,\delta} \le R(f^*) + 2\varepsilon_{n,\delta}$.

Structural Commentary: Union-bounded concentration gives uniform control; ERM turns it into risk control.
Key Insight: Control every gap $\left|R - \widehat{R}_S\right|$ by $\varepsilon_{n,\delta}$; the ERM comparison pays this gap twice. □

Remark 1.2.4. *Statistical and Computational Error*
In practice the selected predictor may be only an approximate minimizer of the empirical criterion.
Its excess risk is then decomposed into approximation error, estimation error, and optimisation error.

1.3 Surrogate Losses and Calibration

Many prediction losses are discontinuous or combinatorial. Learning algorithms often optimise a smoother surrogate and rely on calibration to connect that surrogate to the target loss.

Definition 1.3.1. *Surrogate Loss*
For binary labels $y \in \{-1, 1\}$, let $g : \mathcal{X} \to \mathbb{R}$ be score function. Define $m_g(x, y) := yg(x)$ and $\widehat{y}_g(x) := \operatorname{sign}(g(x))$.
(i) *Zero-one loss*: $\ell_{01}(g(x), y) := \mathbf{1}_{\{m_g(x,y)\le 0\}}$.
(ii) *Surrogate*: $\ell_\phi(g(x), y) := \phi(m_g(x, y))$ for tractable $\phi : \mathbb{R} \to [0, \infty)$.

Example 1.3.2. *Margin Surrogates*
(i) Hinge loss: $\phi(t) = \max\{0, 1 - t\}$.
(ii) Logistic loss: $\phi(t) = \log(1 + \exp(-t))$.
(iii) Exponential loss: $\phi(t) = \exp(-t)$.
(iv) Squared margin loss: $\phi(t) = (1 - t)^2$.

Definition 1.3.3. *Classification Calibration*
Let $R_\phi(g) := \mathbb{E}\phi(Yg(X))$, $R_{01}(g) := \mathbb{P}(Yg(X) \le 0)$, $R^*_\phi := \inf_g R_\phi(g)$, and $R^*_{01} := \inf_g R_{01}(g)$.
Margin surrogate ϕ is classification calibrated if $R_\phi(g_m) \to R^*_\phi$ implies $R_{01}(g_m) \to R^*_{01}$ $\forall$ score sequence (g_m).
Equivalently, $R_\phi(g) - R^*_\phi \to 0$ forces $R_{01}(g) - R^*_{01} \to 0$.

Remark 1.3.4. *Proper Scoring Rules*
A scoring rule $S(q, y)$ is proper if $\operatorname{argmin}_q \mathbb{E}[S(q, Y) \mid X = x] = \mathcal{L}(Y \mid X = x)$.
Thus log and Brier losses learn calibrated conditional probabilities.

1.4 Bias-Variance and Approximation-Estimation Tradeoffs

Learning quality depends both on the richness of the model class and on the variability introduced by fitting that class from a finite sample.

Proposition 1.4.1. *Bias-Variance Decomposition*
Let $Y = f_0(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X = x] = 0$ and $\operatorname{Var}(\varepsilon \mid X = x) = \sigma^2(x)$.
Let random S produce $\widehat{f}_S$, set $\bar{f}(x) := \mathbb{E}_S\widehat{f}_S(x)$, and let $Y_x \perp S$ be a fresh response at x.
Then $\mathbb{E}_{S,Y_x}[(Y_x - \widehat{f}_S(x))^2] = \sigma^2(x) + (f_0(x) - \bar{f}(x))^2 + \operatorname{Var}_S(\widehat{f}_S(x))$.

Proof. Write $Y_x - \widehat{f}_S(x) = \varepsilon_x + f_0(x) - \widehat{f}_S(x)$, where $\varepsilon_x := Y_x - f_0(x)$.
Since $\mathbb{E}[\varepsilon_x] = 0$ and Y_x is conditionally independent of S, $\mathbb{E}_{S,Y_x}[\varepsilon_x\{f_0(x) - \widehat{f}_S(x)\}] = 0$.
Thus $\mathbb{E}_{S,Y_x}[(Y_x - \widehat{f}_S(x))^2] = \sigma^2(x) + \mathbb{E}_S[(f_0(x) - \widehat{f}_S(x))^2]$.
Finally, $\mathbb{E}_S[(f_0(x) - \widehat{f}_S(x))^2] = (f_0(x) - \bar{f}(x))^2 + \mathbb{E}_S[(\widehat{f}_S(x) - \bar{f}(x))^2]$.
Structural Commentary: Squared prediction error equals noise plus fitted-function error.
Key Insight: Center $\widehat{f}_S(x)$ at $\bar{f}(x)$; the remaining cross term has mean zero. □

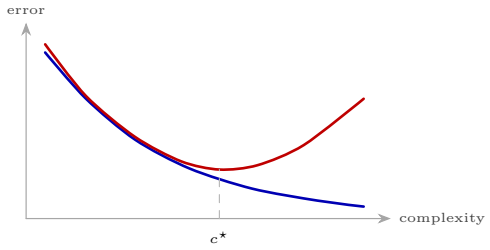

Figure 250: Bias–variance tradeoff: blue: training error, red: test error; dashed line marks test minimum.

Method 1.4.2. *Nearest-Neighbor Local Averaging*
Let $\mathcal{T} := \{(x_i, y_i) : 1 \le i \le n\}$ be training data in a metric space $(\mathcal{X}, d)$.
Let $N_k(x) \subseteq \{1, \dots, n\}$ be the indices of the k smallest distances $d(x_i, x)$.
For numerical responses, define the local average $\widehat{f}_k(x) := \frac{1}{k} \sum_{i \in N_k(x)} y_i$.
For class labels $g_i \in \mathcal{G}$, define $\widehat{p}_g(x) := \frac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}_{\{g_i = g\}}$ and classify by $\widehat{g}_k(x) \in \operatorname{argmax}_{g \in \mathcal{G}} \widehat{p}_g(x)$.
Thus k-nearest neighbours replace conditioning at $X = x$ by averaging over $N_k(x)$.

Remark 1.4.3. *Locality in High Dimension*
The parameter k trades local bias against variance: small k reacts strongly to S, while large k averages more but may wash out local structure. In $[0,1]^p$, a subcube with volume fraction r has side length $e_p(r) = r^{1/p}$, so $e_{10}(0.01) \approx 0.63$ and $e_{10}(0.10) \approx 0.80$. For N uniform points in the unit ball, the median nearest-neighbour radius is $d(p, N) = (1 - 2^{-1/N})^{1/p}$, about 0.52 for $(p, N) = (10, 500)$. Thus high-dimensional neighbourhoods large enough to contain data may not be local; useful local methods need structure or an adapted metric.

Definition 1.4.4. *Approximation and Estimation Errors*
Let $R^* := \inf_f R(f)$ and $R^*_{\mathcal{F}} := \inf_{f \in \mathcal{F}} R(f)$.

(i) *Approximation error*: $\mathcal{A}(\mathcal{F}) := R^*_{\mathcal{F}} - R^*$.
(ii) *Estimation error*: For a data-dependent $\widehat{f} \in \mathcal{F}$, $\mathcal{E}(\widehat{f}; \mathcal{F}) := R(\widehat{f}) - R^*_{\mathcal{F}}$.
(iii) *Excess risk*: $R(\widehat{f}) - R^* = \mathcal{A}(\mathcal{F}) + \mathcal{E}(\widehat{f}; \mathcal{F})$.

Larger $\mathcal{F}$ usually lowers $\mathcal{A}(\mathcal{F})$ and raises estimation difficulty.

Remark 1.4.5. *Model Complexity*
As $c(\mathcal{F})$ grows, $\widehat{R}_S(\widehat{f}_{\mathcal{F}})$ usually decreases, while $R(\widehat{f}_{\mathcal{F}}) - R^* = \mathcal{A}(\mathcal{F}) + \mathcal{E}(\widehat{f}_{\mathcal{F}}; \mathcal{F})$ trades smaller approximation error for larger estimation error. For zero-one loss, the relevant failure is crossing the wrong side of the decision boundary, not an additive squared-error bias–variance split.

2 Generalization Theory

2.1 Uniform Convergence and VC Theory

Generalisation asks when small training loss implies small population loss. Uniform convergence controls the gap for all predictors in a class at once.

Definition 2.1.1. *Uniform Convergence*
For samples S_n of size n and bounded loss, $\mathcal{F}$ has uniform convergence
if $\sup_{f \in \mathcal{F}} \left| R(f) - \widehat{R}_{S_n}(f) \right| \to 0$ in probability, or almost surely, as $n \to \infty$.

Definition 2.1.2. *Shattering and VC Dimension*
For $\mathcal{C} \subseteq 2^{\mathcal{X}}$, a finite $A \subseteq \mathcal{X}$ is shattered if $\{A \cap C : C \in \mathcal{C}\} = 2^A$,
and $\operatorname{VCdim}(\mathcal{C})$ is the supremum of $|A|$ over finite shattered $A \subseteq \mathcal{X}$.

Example 2.1.3. *Shattering Examples*
For $\mathcal{H}_p := \{\{x \in \mathbb{R}^p : a^\top x + b \ge 0\} : a \in \mathbb{R}^p, b \in \mathbb{R}\}$, $\operatorname{VCdim}(\mathcal{H}_p) = p + 1$; in particular, $\mathcal{H}_2$ shatters any three noncollinear points and shatters no four-point set.
Parameter count not decisive: one-parameter class $\mathcal{C} := \{\{x \in \mathbb{R} : \sin(\alpha x) > 0\} : \alpha \in \mathbb{R}\}$ has infinite VCdim.

Theorem 2.1.4. *VC Generalization Bound Form*
Let $\mathcal{F} \subseteq \{0,1\}^{\mathcal{X}}$ have $1 \le d := \operatorname{VCdim}(\mathcal{F}) < \infty$, let $S_n = ((X_i, Y_i))_{i=1}^n$ be i.i.d., and use $\ell_f(x, y) := \mathbf{1}_{\{f(x) \ne y\}}$, $R(f) := \mathbb{E}\ell_f(X, Y)$, and $\widehat{R}_{S_n}(f) := \frac{1}{n} \sum_{i=1}^n \ell_f(X_i, Y_i)$.
Define $\Pi_{\mathcal{F}}(m) := \max |\{(f(x_1), \dots, f(x_m)) : f \in \mathcal{F}\}|$, where the maximum is over $x_1, \dots, x_m \in \mathcal{X}$.

(i) Growth bound: If $m \ge d \ge 1$, then $\Pi_{\mathcal{F}}(m) \le \sum_{j=0}^{d} \binom{m}{j} \le \left(\frac{em}{d}\right)^d$.

(ii) Uniform deviation: For every $\delta \in (0,1)$ and $n \ge d$, with probability at least $1 - \delta$,

$$\sup_{f \in \mathcal{F}} \left| R(f) - \widehat{R}_{S_n}(f) \right| \le C \sqrt{\frac{d \log\left(\frac{2en}{d}\right) + \log\left(\frac{1}{\delta}\right)}{n}}$$

for a universal constant C.

Proof.

(i) *Growth bound*: Since no set of size $d+1$ is shattered, Sauer's lemma gives $\Pi_{\mathcal{F}}(m) \le \sum_{j=0}^{d} \binom{m}{j}$,
and the binomial-sum bound gives $\sum_{j=0}^{d} \binom{m}{j} \le \left(\frac{em}{d}\right)^d$ for $m \ge d$.

(ii) *Uniform deviation*: Symmetrization with an independent ghost sample and Hoeffding's inequality on the at most $\Pi_{\mathcal{F}}(2n)$ realized labelings give $\mathbb{P}(\sup_{f\in\mathcal{F}}\left|R(f)-\widehat{R}_{S_n}(f)\right|>\varepsilon)\le 8\Pi_{\mathcal{F}}(2n)\exp(-cn\varepsilon^2)$ for a universal $c>0$; applying (i) with $m=2n$ and solving the right side at level δ yields (ii).

Structural Commentary: VC dimension polynomially bounds sample labelings.
Key Insight: In finite-class concentration, use $|\mathcal{F}|\rightsquigarrow\Pi_{\mathcal{F}}(2n)\le\left(\frac{2en}{d}\right)^d$. □

Remark 2.1.5. *Growth Functions*
For $d=\mathrm{VCdim}(\mathcal{F})$, $\Pi_{\mathcal{F}}(m)$ is the effective number of labelings on m sample points. Sauer's lemma gives $\log\Pi_{\mathcal{F}}(m)\le d\log\left(\frac{em}{d}\right)$, so VC bounds replace $\log|\mathcal{F}|$ by the sample-level complexity $\log\Pi_{\mathcal{F}}(m)$.

Method 2.1.6. *Structural Risk Minimization*

(i) *Model ladder*: Choose $\mathcal{F}_1\subseteq\mathcal{F}_2\subseteq\cdots$ with complexities c_m, such as $c_m=\mathrm{VCdim}(\mathcal{F}_m)$.
(ii) *Uniform penalties*: Choose $\delta_m>0$ with $\sum_m\delta_m\le\delta$ and penalties $\psi_m(n,\delta_m)$ such that $\mathbb{P}(\sup_{f\in\mathcal{F}_m}\left|R(f)-\widehat{R}_{S_n}(f)\right|\le\psi_m(n,\delta_m))\ge 1-\delta_m$.
(iii) *SRM rule*: Choose $(\widehat{m},\widehat{f})\in\operatorname{argmin}_{m\in\mathbb{N},f\in\mathcal{F}_m}\{\widehat{R}_{S_n}(f)+\psi_m(n,\delta_m)\}$.

Larger m improves fit but raises ψ_m; SRM balances flexibility against search cost.

2.2 Rademacher Complexity and Covering Numbers

For real-valued predictors, complexity is often measured by how well the class can fit random noise.

Definition 2.2.1. *Rademacher Complexity*
Let $\mathcal{F}\subseteq\mathbb{R}^{\mathcal{X}}$, fix $S_n=(x_1,\dots,x_n)\in\mathcal{X}^n$, and let $\sigma_1,\dots,\sigma_n$ be i.i.d. Rademacher signs, $\mathbb{P}(\sigma_i=1)=\mathbb{P}(\sigma_i=-1)=\frac{1}{2}$. The empirical Rademacher complexity is $\widehat{\mathfrak{R}}_{S_n}(\mathcal{F}):=\mathbb{E}_\sigma\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\sigma_i f(x_i)$.

If $X_1,\dots,X_n\sim P_X$ i.i.d., the population complexity is $\mathfrak{R}_n(\mathcal{F}):=\mathbb{E}_{S_n}\widehat{\mathfrak{R}}_{S_n}(\mathcal{F})$.

Definition 2.2.2. *Covering Number*
For a metric space (T,d) and $\varepsilon>0$, $N(\varepsilon,T,d):=\inf\{m:\exists t_1,\dots,t_m\in T,\ T\subseteq\bigcup_{j=1}^m\overline{B}_d(t_j,\varepsilon)\}$.

The entropy $\log N(\varepsilon,T,d)$ measures metric size at scale ε, not parameter count.

Remark 2.2.3. *Entropy Methods*
Bounds on $\log N(\varepsilon,\mathcal{F},d)$ can be integrated over scales to control empirical processes.
This is the route from functional-analytic compactness to statistical-learning rates.

2.3 Concentration and Stability

Generalization can also follow from concentration of the training criterion or from stability of the learning algorithm itself.

Definition 2.3.1. *Subgaussian Random Variable*
Centered random variable Z is *subgaussian* with variance proxy σ^2 if $\mathbb{E}\exp(\lambda Z)\le\exp(\lambda^2\sigma^2/2)\ \forall\lambda\in\mathbb{R}$.
Subgaussian variables have Gaussian-type tail bounds.

Theorem 2.3.2. *Bounded Differences Inequality*
Let $X_1,\dots,X_n$ be independent, $X:=(X_1,\dots,X_n)$, and $F:\prod_{i=1}^n\mathcal{X}_i\to\mathbb{R}$.
For $x=(x_1,\dots,x_n)$, write $x^{(i\leftarrow x_i')}:=(x_1,\dots,x_{i-1},x_i',x_{i+1},\dots,x_n)$.
Assume $\left|F(x)-F(x^{(i\leftarrow x_i')})\right|\le c_i$ for all i,x,x_i', and set $v:=\sum_{i=1}^n c_i^2$.
Then, for every $t>0$, $\mathbb{P}\{F(X)-\mathbb{E}F(X)\ge t\}\le\exp\left(-\frac{2t^2}{v}\right)$ and $\mathbb{P}\{|F(X)-\mathbb{E}F(X)|\ge t\}\le 2\exp\left(-\frac{2t^2}{v}\right)$.

Proof. Let $\mathcal{G}_i:=\sigma(X_1,\dots,X_i)$, $M_i:=\mathbb{E}(F(X)\mid\mathcal{G}_i)$, and $\Delta_i:=M_i-M_{i-1}$.
Then $M_0=\mathbb{E}F(X)$, $M_n=F(X)$, and the bounded-differences hypothesis implies that, conditionally on $\mathcal{G}_{i-1}$, Δ_i has mean 0 and lies in an interval of length at most c_i. By Hoeffding's lemma, $\mathbb{E}(\exp(\lambda\Delta_i)\mid\mathcal{G}_{i-1})\le\exp\left(\frac{\lambda^2c_i^2}{8}\right)$ for $\lambda>0$. Iterating, $\mathbb{E}\exp(\lambda(F(X)-\mathbb{E}F(X)))\le\exp\left(\frac{\lambda^2v}{8}\right)$.
By Chernoff's bound, $\mathbb{P}\{F(X)-\mathbb{E}F(X)\ge t\}\le\exp\left(-\lambda t+\frac{\lambda^2v}{8}\right)$, optimised at $\lambda=\frac{4t}{v}$.
Applying the same bound to $-F$ and taking a union bound gives the two-sided inequality.
Structural Commentary: Coordinate sensitivity plus independence makes $F(X)$ subgaussian.
Key Insight: Doob increments reduce global concentration to bounds on one-coordinate changes c_i. □

Theorem 2.3.3. *Rademacher Generalization Bound*
Let $Z_i=(X_i,Y_i)\in\mathcal{Z}:=\mathcal{X}\times\mathcal{Y}$ be i.i.d. from P.
Let $S_n=(Z_1,\dots,Z_n)$, $\mathcal{G}:=\{g_f:\mathcal{Z}\to[0,1]:g_f(x,y)=\ell(f(x),y),f\in\mathcal{F}\}$.
For $g\in\mathcal{G}$, write $Pg:=\mathbb{E}g(Z)$ and $P_ng:=\frac{1}{n}\sum_{i=1}^n g(Z_i)$.

(i) Uniform deviation: $\forall\delta\in(0,1)$, with probability at least $1-\delta$, $\sup_{g\in\mathcal{G}}(Pg-P_ng)\le 2\widehat{\mathfrak{R}}_{S_n}(\mathcal{G})+3\sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2n}}$.

(ii) Risk bound: Hence, simultaneously for all $f\in\mathcal{F}$, $R(f)\le\widehat{R}_{S_n}(f)+2\widehat{\mathfrak{R}}_{S_n}(\ell\circ\mathcal{F})+3\sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2n}}$.

Proof.

(i) Let $\Phi(S_n) := \sup_{g\in\mathcal{G}}(Pg - P_n g)$. Changing one Z_i changes Φ and $\widehat{\mathfrak{R}}_{S_n}(\mathcal{G})$ by at most $1/n$, so McDiarmid gives $\Phi \le \mathbb{E}\Phi + \varepsilon$ and $\mathfrak{R}_n(\mathcal{G}) \le \widehat{\mathfrak{R}}_{S_n}(\mathcal{G}) + \varepsilon$ with probability at least $1-\delta$, for $\varepsilon := \sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2n}}$.
Symmetrisation gives $\mathbb{E}\Phi \le 2\mathfrak{R}_n(\mathcal{G})$, hence $\Phi \le 2\widehat{\mathfrak{R}}_{S_n}(\mathcal{G}) + 3\varepsilon$.

(ii) For $z = (x, y)$ and $g_f(z) = \ell(f(x), y)$, $Pg_f = R(f)$ and $P_n g_f = \widehat{R}_{S_n}(f)$.
Applying (i) to every $g_f \in \mathcal{G}$ gives the stated simultaneous bound.

Structural Commentary: Rademacher complexity measures sample-wise noise fitting.

Key Insight: Symmetrization turns $P - P_n$ into $\mathbb{E}_\sigma \sup_{g\in\mathcal{G}} \frac{1}{n}\sum_{i=1}^{n} \sigma_i g(Z_i)$. □

Definition 2.3.4. *Uniform Stability*
A learning algorithm A has *uniform stability* β_n for loss ℓ if, for any samples $S, S' \in \mathcal{Z}^n$ differing in one coordinate, $\sup_{z\in\mathcal{Z}} |\ell(A(S), z) - \ell(A(S'), z)| \le \beta_n$.

Remark 2.3.5. *Algorithm-Dependent Generalisation*
Stability controls generalisation through the map $S \mapsto A(S)$, not only through $\mathcal{F}$.
Thus regularisation or optimisation can generalise well even when the ambient class is large.

2.4 PAC-Bayes and Information-Theoretic Bounds

PAC-Bayes bounds analyze randomized predictors by comparing a data-dependent posterior distribution on hypotheses to a fixed prior.

Definition 2.4.1. *Randomised Predictor*
Let $\mathcal{H}$ be a hypothesis space and $Q \in \mathcal{P}(\mathcal{H})$. The *randomised predictor* Q samples $h \sim Q$ before prediction, with risk $R(Q) := \mathbb{E}_{h\sim Q} R(h)$ and empirical risk $\widehat{R}_S(Q) := \mathbb{E}_{h\sim Q}\widehat{R}_S(h)$.

Theorem 2.4.2. *PAC-Bayes Bound Form*
Let $S_n = (Z_1, \dots, Z_n)$ be i.i.d., let $\ell(h, Z) \in [0, 1]$, and let $\Pi \in \mathcal{P}(\mathcal{H})$ be independent of S_n.
For $Q \ll \Pi$, set $\mathrm{KL}(Q\|\Pi) := \mathbb{E}_Q \log \frac{dQ}{d\Pi}$ and $\mathrm{kl}(a, b) := a \log \frac{a}{b} + (1-a)\log\frac{1-a}{1-b}$.

(i) With probability at least $1-\delta$, every posterior $Q \ll \Pi$ satisfies $\mathrm{kl}(\widehat{R}_{S_n}(Q), R(Q)) \le \frac{\mathrm{KL}(Q\|\Pi) + \log\left(\frac{n+1}{\delta}\right)}{n}$.

(ii) On the same event, $R(Q) \le \widehat{R}_{S_n}(Q) + \sqrt{\frac{\mathrm{KL}(Q\|\Pi) + \log\left(\frac{n+1}{\delta}\right)}{2n}}$ for every $Q \ll \Pi$.

Proof.

(i) For fixed h, the bounded-loss kl moment bound gives $\mathbb{E}_{S_n} \exp(n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))) \le n+1$.
Thus $\mathbb{E}_{S_n}\mathbb{E}_{h\sim\Pi} \exp(n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))) \le n+1$, so Markov gives,
with probability at least $1-\delta$, $\mathbb{E}_\Pi \exp(n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))) \le \frac{n+1}{\delta}$.
On this event, the Donsker-Varadhan inequality with $\varphi(h) := n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))$ gives
$\mathbb{E}_Q\varphi \le \mathrm{KL}(Q\|\Pi) + \log \mathbb{E}_\Pi \exp\varphi \;\forall Q \ll \Pi$. By convexity of kl, $n\,\mathrm{kl}(\widehat{R}_{S_n}(Q), R(Q)) \le \mathbb{E}_Q\varphi$, proving (i).

(ii) If $R(Q) \ge \widehat{R}_{S_n}(Q)$, Pinsker gives $\mathrm{kl}(\widehat{R}_{S_n}(Q), R(Q)) \ge 2(R(Q) - \widehat{R}_{S_n}(Q))^2$.
if $R(Q) < \widehat{R}_{S_n}(Q)$, the claim is immediate. Combining with (i) gives (ii).

Structural Commentary: PAC-Bayes replaces class size by posterior-prior divergence.

Key Insight: A prior-average exponential tail plus $Q \ll \Pi$ yields all posteriors at cost $\mathrm{KL}(Q\|\Pi)$. □

Remark 2.4.3. *Prior and Posterior*
In PAC-Bayes, Π need not be subjective; it anchors the complexity term $\mathrm{KL}(Q\|\Pi)$.
Smaller $\mathrm{KL}(Q\|\Pi)$ means the data-dependent posterior Q stays closer to the fixed prior.

Definition 2.4.4. *Information-Theoretic Generalization*
Let $S_n = (Z_1, \dots, Z_n)$, $W = A(S_n)$ be learned output. *Information-theoretic generalization* bounds control $\mathrm{gen}(A) := \mathbb{E}[R(W) - \widehat{R}_{S_n}(W)]$ by dependence measures such as $I(W; S_n)$.

3 Unsupervised and Latent-Variable Learning

3.1 Clustering and Mixture Models

Unsupervised learning searches for structure in inputs without direct labels or a direct prediction loss. Clustering summarizes the input distribution by grouping observations, while mixture models replace hard groups by latent component probabilities.

Definition 3.1.1. *k-Means Clustering*
Let $x_1, \dots, x_n \in \mathbb{R}^d$ and $k \in \{1, \dots, n\}$. Put $\mathcal{C} = \{C : \{1, \dots, n\} \to \{1, \dots, k\}\}$, $\mathcal{M} = (\mathbb{R}^d)^k$,
$I_r(C) = \{i : C(i) = r\}$, and, for $m = (m_1, \dots, m_k)$, $\Phi(C, m) = \sum_{r=1}^{k} \sum_{i\in I_r(C)} \|x_i - m_r\|_2^2$.

(i) A *k-means solution* is any $(C^\star, m^\star) \in \operatorname{arg\,min}_{C\in\mathcal{C},\, m\in\mathcal{M}} \Phi(C, m)$.

(ii) Its *clusters* are $I_r^\star = I_r(C^\star)$, $r = 1, \dots, k$.

(iii) For fixed C, a *minimizing center* is $m_r = |I_r(C)|^{-1} \sum_{i\in I_r(C)} x_i$ if $I_r(C) \ne \varnothing$, and arbitrary if $I_r(C) = \varnothing$.

(iv) For fixed m, a *minimizing assignment* satisfies $C(i) \in \operatorname{arg\,min}_{1\le r\le k} \|x_i - m_r\|_2^2$ for each i.

Definition 3.1.2. *Object Dissimilarities*
Let $x_1, \dots, x_n$ be objects and let $d_{ii'} = d(x_i, x_{i'})$; when attributes are used, write $x_i = (x_{i1}, \dots, x_{ip})$.

(i) A *dissimilarity matrix* is $D = (d_{ii'}) \in [0,\infty)^{n\times n}$ with $d_{ii} = 0$ and $d_{ii'} = d_{i'i}$; directional inputs $a_{ii'}$ are symmetrized by $d_{ii'} = s(a_{ii'}, a_{i'i})$, where $s : [0,\infty)^2 \to [0,\infty)$ and $s(u,v) = s(v,u)$.

(ii) Given d_j and $w_j \geq 0$ with $\sum_{j=1}^{p} w_j = 1$, the *additive dissimilarity* is $d(x_i, x_{i'}) = \sum_{j=1}^{p} w_j d_j(x_{ij}, x_{i'j})$.

(iii) *Coordinate rules*: quantitative $d_j(a,b) = |a-b|$ or $(a-b)^2$; ordinal $r \mapsto (r-1/2)/M$ for $r = 1, \dots, M$; categorical $d_j(a,b) = L_j(a,b)$ with $L_j(a,a) = 0$ and $L_j(a,b) \geq 0$.

Remark 3.1.3. *Weights and Missing Values*
For $d(x_i, x_{i'}) = \sum_{j=1}^{p} w_j d_j(x_{ij}, x_{i'j})$, the empirical scale of coordinate j is $\bar{d}_j = \binom{n}{2}^{-1} \sum_{i<i'} d_j(x_{ij}, x_{i'j})$, so its effective contribution is $w_j \bar{d}_j$. If $O_{ii'} \subseteq \{1, \dots, p\}$ is the nonempty set of coordinates observed for both x_i and $x_{i'}$, pairwise deletion uses $d(x_i, x_{i'}) = \sum_{j \in O_{ii'}} \tilde{w}_{j,ii'} d_j(x_{ij}, x_{i'j})$ with $\tilde{w}_{j,ii'} = w_j / \sum_{\ell \in O_{ii'}} w_\ell$.
Alternatives are imputed values $\tilde{x}_{ij}$ before computing d or an added categorical level `missing`.

Definition 3.1.4. *Within- and Between-Cluster Scatter*
Fix $k \in \{1, \dots, n\}$ and a symmetric dissimilarity matrix $D = (d_{ii'})$.
Let $\mathcal{C}_k = \{C : \{1, \dots, n\} \to \{1, \dots, k\}\}$ and $I_r(C) = \{i : C(i) = r\}$.

(i) The *within-cluster scatter* is $W(C) = \frac{1}{2} \sum_{r=1}^{k} \sum_{i \in I_r(C)} \sum_{i' \in I_r(C)} d_{ii'}$ for $C \in \mathcal{C}_k$.

(ii) The *total scatter* is $T = \frac{1}{2} \sum_{i=1}^{n} \sum_{i'=1}^{n} d_{ii'}$.

(iii) The *between-cluster scatter* is $B(C) = T - W(C)$ for $C \in \mathcal{C}_k$.

As T is fixed, $\arg\min_{C \in \mathcal{C}_k} W(C) = \arg\max_{C \in \mathcal{C}_k} B(C)$.

Algorithm 2: k-Means Alternating Minimization

Require: Data $x_1, \dots, x_n \in \mathbb{R}^d$, number of clusters k, initial centers or assignments
Ensure: Cluster assignments $C(i)$ and centers $m_1, \dots, m_k$
repeat
Given assignments, set each m_r to the mean of the observations with $C(i) = r$
Given centers, assign each observation by $C(i) \leftarrow \arg\min_{1 \leq r \leq k} \|x_i - m_r\|_2^2$
until the assignments no longer change

Proposition 3.1.5. *k-Means Descent and Local Minima*
Let $\mathcal{C} = \{C : \{1, \dots, n\} \to \{1, \dots, k\}\}$, $\mathcal{M} = (\mathbb{R}^d)^k$, *and* $\Phi(C, m) = \sum_{r=1}^{k} \sum_{i : C(i) = r} \|x_i - m_r\|_2^2$.
Suppose $(C^t, m^t) \in \mathcal{C} \times \mathcal{M}$ *satisfies* $m^{t+1} \in \arg\min_{m \in \mathcal{M}} \Phi(C^t, m)$ *and* $C^{t+1} \in \arg\min_{C \in \mathcal{C}} \Phi(C, m^{t+1})$ *for each* t.

(i) The objective is nonincreasing: $\Phi(C^{t+1}, m^{t+1}) \leq \Phi(C^t, m^t)$.
(ii) If $C^{t+1} = C^t = C^\star$ *and* $m^{t+1} = m^t = m^\star$, *then* $(C^\star, m^\star)$ *is coordinatewise minimal:* $m^\star \in \arg\min_{m \in \mathcal{M}} \Phi(C^\star, m)$ *and* $C^\star \in \arg\min_{C \in \mathcal{C}} \Phi(C, m^\star)$.
Coordinatewise minimal need not be global; compare initialisations by terminal Φ.

Proof.
(i) By the center update, $\Phi(C^t, m^{t+1}) \leq \Phi(C^t, m^t)$. By the assignment update, $\Phi(C^{t+1}, m^{t+1}) \leq \Phi(C^t, m^{t+1})$. Combining gives the claim.
(ii) At a fixed point, $C^{t+1} = C^t = C^\star$ and $m^{t+1} = m^t = m^\star$ in the two update inclusions, giving the two coordinatewise optimality conditions.

Structural Commentary: Exact block minimization forces descent.
Key Insight: A stopped run certifies blockwise, not global, optimality. □

Algorithm 3: k-Medoids Alternating Representatives

Require: Pairwise dissimilarities $D(x_i, x_{i'})$, number of clusters k, initial assignments
Ensure: Cluster assignments $C(i)$ and representative indices $i_1^\star, \dots, i_k^\star$
repeat
for $r = 1, \dots, k$ **do**
Set $i_r^\star \leftarrow \arg\min_{i : C(i) = r} \sum_{C(i') = r} D(x_i, x_{i'})$
end for
Assign each observation by $C(i) \leftarrow \arg\min_{1 \leq r \leq k} D(x_i, x_{i_r^\star})$
until the assignments no longer change

Remark 3.1.6. *k-Medoids Versus k-Means*
Fix an assignment C and write $I_r = \{i : C(i) = r\}$.

(i) *k-means*: $m_r \in \arg\min_{m \in \mathbb{R}^d} \sum_{i \in I_r} \|x_i - m\|_2^2$, hence $m_r = |I_r|^{-1} \sum_{i \in I_r} x_i$ when $I_r \neq \varnothing$.

(ii) *k-medoids*: $i_r^\star \in \arg\min_{i \in I_r} \sum_{i' \in I_r} D(x_i, x_{i'})$, with representative $x_{i_r^\star}$.

k-medoids only requires pairwise dissimilarities and keeps representatives inside the data set.
Its representative update is usually costlier than the k-means empirical mean.

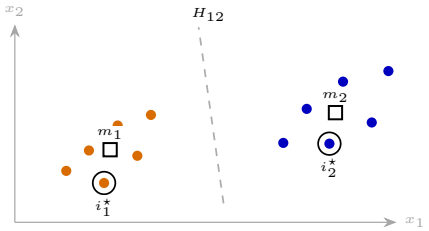

Figure 251: Open squares are centroids, circled points are medoids, the dashed line is nearest-center boundary.

Remark 3.1.7. *Choosing the Number of Clusters*
For k-means or k-medoids, choose k and an initialisation. Let W_k be terminal within-cluster scatter.
The elbow diagnostic inspects $k \mapsto W_k$; the gap statistic uses $G(k) = \mathbb{E}_0[\log W_k] - \log W_k$, where $\mathbb{E}_0$ is under a reference distribution on the data range.
With simulation standard error s'_{k+1}, one rule is $\widehat{k} = \min\{k : G(k) \geq G(k+1) - s'_{k+1}\}$.
This formalises the bend comparison but remains heuristic.

Method 3.1.8. *Hierarchical Clustering and Linkage*
Given dissimilarities $d_{ii'}$ and a linkage $L(G, H)$ between nonempty groups, agglomerative hierarchical clustering constructs nested partitions.

(i) Initialize $\mathcal{P}_0 = \{\{1\}, \ldots, \{n\}\}$.
(ii) At step t, choose distinct $G, H \in \mathcal{P}_t$ minimizing $L(G, H)$.
(iii) Set $\mathcal{P}_{t+1} = (\mathcal{P}_t \setminus \{G, H\}) \cup \{G \cup H\}$ and record merge height $L(G, H)$.
(iv) Stop at $\mathcal{P}_{n-1} = \{\{1, \ldots, n\}\}$.

For groups G, H with sizes N_G, N_H, common linkages are

(i) *Single*: $L_{\mathrm{SL}}(G, H) = \min_{i \in G,\, i' \in H} d_{ii'}$.
(ii) *Complete*: $L_{\mathrm{CL}}(G, H) = \max_{i \in G,\, i' \in H} d_{ii'}$.
(iii) *Group average*: $L_{\mathrm{GA}}(G, H) = (N_G N_H)^{-1} \sum_{i \in G} \sum_{i' \in H} d_{ii'}$.

Recorded merges form a dendrogram; divisive clustering reverses the process by recursive splits.

Remark 3.1.9. *Dendrograms and Cophenetic Distances*
For dendrogram merge height $h_T(i, i')$, the cut at height a partitions observations by $i \sim_a i'$ iff $h_T(i, i') \leq a$.
The cophenetic dissimilarity is $C_{ii'} = h_T(i, i')$, the first height at which i and i' share a cluster.
It is ultrametric: $C_{ii'} \leq \max\{C_{ih}, C_{i'h}\}$. Thus a dendrogram is an algorithmic hierarchical summary, not an exact representation of arbitrary input dissimilarities.

Method 3.1.10. *Self-Organizing Map*
A self-organizing map uses prototypes $m_1^t, \ldots, m_q^t \in \mathbb{R}^p$ indexed by grid locations $\ell_1, \ldots, \ell_q \in \mathcal{Q}_1 \times \mathcal{Q}_2$.
Let $\alpha_t > 0$ and $h_t : [0, \infty) \to [0, 1]$ be the learning rate and neighborhood kernel.

(i) Select an observation x_i.
(ii) Find a winner $j_t \in \arg\min_{1 \leq j \leq q} \left\| x_i - m_j^t \right\|_2^2$.
(iii) Update every prototype by $m_k^{t+1} = m_k^t + \alpha_t h_t(\|\ell_{j_t} - \ell_k\|_2)(x_i - m_k^t)$, $k = 1, \ldots, q$.
(iv) Increment t and repeat until the stopping rule or schedule terminates.

A common kernel is $h_t(s) = \mathbf{1}_{s \leq r_t}$ with decreasing radius r_t.
Early large neighborhoods enforce grid smoothness; if $h_t(s) = \mathbf{1}_{s=0}$, the method reduces to online k-means.

Remark 3.1.11. *Checking the Topological Constraint*
For q prototypes, compare $E_{\mathrm{SOM}} = \sum_i \left\| x_i - m_{C(i)} \right\|_2^2$ with the q-means optimum $E_q^\star = \min_{C, m} \sum_i \left\| x_i - m_{C(i)} \right\|_2^2$.
Since $E_q^\star \leq E_{\mathrm{SOM}}$, a large gap $E_{\mathrm{SOM}} - E_q^\star$ indicates an overly restrictive lattice constraint.

Method 3.1.12. *Spectral Clustering*
Given weights $w_{ii'} \geq 0$, set $W = (w_{ii'})$, $g_i = \sum_{i'=1}^{n} w_{ii'}$, $G = \mathrm{diag}(g_i)$, and $L = G - W$.
Often $w_{ii'} = s_{ii'} = \exp(-\frac{\|x_i - x_{i'}\|_2^2}{c})$ (restricted to mutual nearest-neighbor edges). For target k clusters:

(i) Construct W, G, and the Laplacian L.
(ii) Compute eigenvectors $u_1, \ldots, u_k$ for the k smallest nontrivial eigenvalues of L.
(iii) Form $U = (u_1, \ldots, u_k) \in \mathbb{R}^{n \times k}$ and represent observation i by row $U_{i\cdot}$.
(iv) Cluster the rows $U_{i\cdot}$ into k groups, typically by k-means.

$f^T L f = \frac{1}{2} \sum_{i=1}^{n} \sum_{i'=1}^{n} w_{ii'}(f_i - f_{i'})^2$, so small-eigenvalue directions are smooth across high-weight edges.

Remark 3.1.13. *Connected Components and Random Walks*
Let $L = G - W$ and, when $g_i > 0$, $P = G^{-1}W$.

(i) If the graph has components $A_1, \ldots, A_m$, then $\ker L = \mathrm{span}\{\mathbf{1}_{A_1}, \ldots, \mathbf{1}_{A_m}\}$.
(ii) The matrix P is a random-walk transition matrix.
A good spectral cluster A has large within-set transition mass $\sum_{i \in A} \sum_{i' \in A} P_{ii'}$ and small cross-set mass.

Nearly separated clusters replace exact zero eigenvalues by small eigenvalues.

Definition 3.1.14. *Finite Mixture Model*

Let $Z \in \{1, \dots, k\}$ be latent with $\mathbb{P}(Z = z) = \pi_z$, $\pi_z \geq 0$, and $\sum_{z=1}^{k} \pi_z = 1$.

Given component densities $p_z(\cdot; \theta_z)$, the marginal density is $p(x) = \sum_{z=1}^{k} \pi_z p_z(x; \theta_z)$.

Definition 3.1.15. *Mixture Responsibilities*

For fitted $(\widehat{\pi}_m, \widehat{\theta}_m)_{m=1}^{k}$, define $\widehat{r}_{im} = \widehat{\mathbb{P}}(Z_i = m \mid x_i) = \widehat{\pi}_m p_m(x_i; \widehat{\theta}_m) \left(\sum_{\ell=1}^{k} \widehat{\pi}_\ell p_\ell(x_i; \widehat{\theta}_\ell) \right)^{-1}$.

Then $\widehat{r}_i$ is soft membership and $\widehat{C}(i) \in \arg\max_{1 \leq m \leq k} \widehat{r}_{im}$ is a hard assignment.

Remark 3.1.16. *Mixtures as Density Estimates*

Gaussian mixtures give flexible density estimates $p(x) = \sum_{m=1}^{k} \pi_m \phi(x; \mu_m, \Sigma_m)$. If class-conditional mixtures $\widehat{p}_g(x)$ and class priors $\widehat{\pi}_g$ are fit separately, the Bayes classifier is $\widehat{g}(x) \in \arg\max_g \widehat{\pi}_g \widehat{p}_g(x)$.

Without labels, component-label agreement is discovered structure, not supervised training.

Remark 3.1.17. *Gaussian Mixtures as Soft k-Means*

For $p_m(x) = \phi(x; \mu_m, \sigma^2 I)$, the responsibility is $r_{im} = \pi_m \exp(-\frac{\|x_i - \mu_m\|_2^2}{2\sigma^2}) \left(\sum_{\ell=1}^{k} \pi_\ell \exp(-\frac{\|x_i - \mu_\ell\|_2^2}{2\sigma^2}) \right)^{-1}$.

Thus r_{im} is distance-weighted; as $\sigma^2 \downarrow 0$, it concentrates on nearest centers, yielding hard k-means assignments.

Method 3.1.18. *Expectation-Maximization*

Given $x_{1:n}$ and $p(x) = \sum_{m=1}^{k} \pi_m p_m(x; \theta_m)$, set $\ell(\pi, \theta) = \sum_{i=1}^{n} \log(\sum_{m=1}^{k} \pi_m p_m(x_i; \theta_m))$. EM alternates:

(i) *E-step*: compute $r_{im} = \frac{\pi_m p_m(x_i; \theta_m)}{\sum_{\ell=1}^{k} \pi_\ell p_\ell(x_i; \theta_\ell)}$.

(ii) *M-step*: update $\pi_m^{\text{new}} = n^{-1} \sum_{i=1}^{n} r_{im}$ and $\theta_m^{\text{new}} \in \arg\max_{\theta_m} \sum_{i=1}^{n} r_{im} \log p_m(x_i; \theta_m)$.

For Gaussian components, the M-step gives weighted means and variances or covariances.
With an exact M-step, ℓ is nondecreasing, but convergence may be to a local maximum.

Remark 3.1.19. *Identifiability*

For any permutation τ of $\{1, \dots, k\}$, $\sum_{m=1}^{k} \pi_m p_m(x; \theta_m) = \sum_{m=1}^{k} \pi_{\tau(m)} p_{\tau(m)}(x; \theta_{\tau(m)})$.

Thus labels are identifiable only up to permutation: density is invariant, component interpretation is not.

3.2 Dimension Reduction and Manifold Learning

Dimension reduction replaces high-dimensional observations, similarities, or dissimilarities by lower-dimensional coordinates that preserve selected structure. Linear methods preserve variance or inner products; manifold methods preserve neighborhoods, graph distances, or local affine geometry.

Definition 3.2.1. *Principal Component Analysis*

Let centered observations $x_1, \dots, x_n \in \mathbb{R}^p$ be the rows of $X \in \mathbb{R}^{n \times p}$, and let $1 \leq r \leq p$.

(i) Rank-r principal component analysis chooses $V = [v_1 \ \cdots \ v_r] \in \mathbb{R}^{p \times r}$, with $V^T V = I_r$ solving $\min_{V^T V = I_r} \|X - XVV^T\|_F^2$.

(ii) The columns $v_1, \dots, v_r$ are the principal directions, $\text{im}(V)$ is the rank-r principal subspace, and $x_i \mapsto VV^T x_i$ is the fitted projection.

(iii) Equivalently, V solves $\max_{V^T V = I_r} \text{tr}(V^T X^T X V)$,

with sequential directions $v_k \in \text{argmax}\{\sum_{i=1}^{n} (v^T x_i)^2 : \|v\|_2 = 1, \ v \perp v_1, \dots, v_{k-1}\}$.

Method 3.2.2. *PCA by SVD*

Let the centered observations be the rows of $X \in \mathbb{R}^{n \times p}$, and fix $1 \leq r \leq \min\{n, p\}$.

(i) Compute an SVD $X = UDV^T$ with singular values $d_1 \geq \cdots \geq 0$.
(ii) Set $V_r = [v_1 \ \cdots \ v_r]$, the first r right singular vectors.
(iii) Set the score matrix $Z_r = XV_r = U_r D_r$.
(iv) Set the fitted matrix $\widehat{X}_r = Z_r V_r^T = XV_r V_r^T = U_r D_r V_r^T$.

The columns of V_r are the rank-r PCA directions, Z_r contains the principal-component scores, and $\widehat{X}_r$ is the orthogonal rank-r reconstruction minimizing $\left\|X - \widehat{X}\right\|_F^2$.

Method 3.2.3. *Kernel PCA*

Let k be a positive definite kernel on observations $x_1, \dots, x_n$, fix r, and set $M = n^{-1}\mathbf{1}\mathbf{1}^T$.

(i) Form the Gram matrix $K = (k(x_i, x_{i'}))_{i,i'}$.
(ii) Center it by $\widetilde{K} = (I_n - M)K(I_n - M)$.
(iii) Compute $\widetilde{K} u_j = \lambda_j u_j$ with $\lambda_1 \geq \cdots \geq \lambda_r > 0$, and set $U_r = [u_1 \ \cdots \ u_r]$, $\Lambda_r = \text{diag}(\lambda_1, \dots, \lambda_r)$.
(iv) Return coordinates $Z = U_r \Lambda_r^{1/2}$.

Kernel PCA is spectral PCA on $\widetilde{K}$; radial kernels encode local similarity.

Method 3.2.4. *Sparse Principal Components*
Let centered data have rows x_i^T in $X \in \mathbb{R}^{n\times p}$, and let $v \in \mathbb{R}^p$ be a loading vector.

(i) *Variance form*: set $\mathcal{V}_t = \{a \in \mathbb{R}^p : a^T a = 1,\ \sum_{j=1}^{p} |a_j| \le t\}$ and choose $v \in \operatorname{argmax}_{a\in\mathcal{V}_t} a^T X^T X a$.

(ii) *Reconstruction form*: for one component, choose $(\theta, v) \in \operatorname{argmin}_{\|\theta\|_2=1} \sum_{i=1}^{n} \left\|x_i - \theta v^T x_i\right\|_2^2 + \lambda \|v\|_2^2 + \lambda_1 \|v\|_1$.

L_1 regularization sparsifies loadings; multiple components alternate sparse-loading and orthogonal-score updates.

Definition 3.2.5. *Multidimensional Scaling*
Let $D = (d_{ii'})$ be a symmetric dissimilarity matrix with $d_{ii} = 0$, fix $r < n$, and write $Z = (z_1^T, \dots, z_n^T)^T \in \mathbb{R}^{n\times r}$ with fitted distances $\delta_{ii'}(Z) = \|z_i - z_{i'}\|_2$. Multidimensional scaling chooses Z so that $\delta_{ii'}(Z)$ approximates $d_{ii'}$.

(i) *Least-squares MDS*, or Kruskal–Shepard scaling, minimizes $S_{\mathrm{MDS}}(Z) = \sum_{i\neq i'} (d_{ii'} - \delta_{ii'}(Z))^2$ over $Z \in \mathbb{R}^{n\times r}$.

(ii) *Sammon MDS*, defined when $d_{ii'} > 0$ for $i \neq i'$, minimizes $S_{\mathrm{Sammon}}(Z) = \sum_{i\neq i'} \frac{(d_{ii'} - \delta_{ii'}(Z))^2}{d_{ii'}}$ over $Z \in \mathbb{R}^{n\times r}$.

Sammon weighting penalizes relative errors more strongly for small dissimilarities.

Method 3.2.6. *Classical Scaling*
Let $H = I_n - n^{-1}\mathbf{1}\mathbf{1}^T$ and fix $r < n$.

(i) If centered similarities $S = (s_{ii'})$ are supplied, set $B = S$.
(ii) If Euclidean distances $\Delta = (\Delta_{ii'})$ are supplied, set $\Delta^{(2)} = (\Delta_{ii'}^2)$ and $B = -\frac{1}{2} H \Delta^{(2)} H$.
(iii) Compute $Be_j = \lambda_j e_j$ with $\lambda_1 \ge \cdots \ge \lambda_r > 0$, and set $E_r = [e_1\ \cdots\ e_r]$, $\Lambda_r = \operatorname{diag}(\lambda_1, \dots, \lambda_r)$.
(iv) Return $Z_r = E_r \Lambda_r^{1/2} \in \mathbb{R}^{n\times r}$, with row z_i^T as the coordinate of observation i.

The construction gives centered coordinates and fitted Gram matrix $Z_r Z_r^T = E_r \Lambda_r E_r^T$, the rank-$r$ positive-semidefinite approximation to B from its leading positive spectrum.

Method 3.2.7. *Nonmetric Scaling*
Given dissimilarities $D = (d_{ii'})$, coordinates $Z = (z_1^T, \dots, z_n^T)^T \in \mathbb{R}^{n\times r}$, and an increasing map θ, nonmetric scaling minimizes $S_{\mathrm{NM}}(Z, \theta) = \sum_{i\neq i'} (\|z_i - z_{i'}\|_2 - \theta(d_{ii'}))^2 / \sum_{i\neq i'} \|z_i - z_{i'}\|_2^2$.

(i) With θ fixed, update Z by minimizing $S_{\mathrm{NM}}(Z, \theta)$ over $Z \in \mathbb{R}^{n\times r}$.
(ii) With Z fixed, update θ by isotonic regression of the fitted distances $\|z_i - z_{i'}\|_2$ on the dissimilarities $d_{ii'}$.

Only the ordering of $d_{ii'}$ constrains $\theta(d_{ii'})$; their original scale is discarded.

Definition 3.2.8. *Encoder and Decoder*
Let $\mathcal{X}$ be a data space and $\mathcal{H}$ a representation space.

(i) An *encoder* is a map $e : \mathcal{X} \to \mathcal{H}$, with representation $h = e(x)$.
(ii) A *decoder* is a map $d : \mathcal{H} \to \mathcal{X}$, with reconstruction $\widehat{x} = d(h) = d(e(x))$.

Linear PCA is the case e, d linear with $\operatorname{im}(d)$ an orthogonal representation subspace.

Method 3.2.9. *Procrustes Shape Alignment*
Let $X_1, X_2 \in \mathbb{R}^{n\times p}$ have corresponding landmark rows.
Orthogonal Procrustes alignment solves $\min_{R^T R = I, \mu\in\mathbb{R}^p} \left\|X_2 - (X_1 R + \mathbf{1}\mu^T)\right\|_F$.

(i) Compute column means $\bar{x}_1, \bar{x}_2$ and centered matrices $\widetilde{X}_a = X_a - \mathbf{1}\bar{x}_a^T$, $a = 1, 2$.
(ii) Compute an SVD $\widetilde{X}_1^T \widetilde{X}_2 = U D V^T$.
(iii) Set $\widehat{R} = UV^T$ and $\widehat{\mu} = \bar{x}_2 - \widehat{R}^T \bar{x}_1$.
(iv) For scaled Procrustes, set $\widehat{\beta} = \operatorname{tr}(D) / \left\|\widetilde{X}_1\right\|_F^2$, $\widehat{\mu}_\beta = \bar{x}_2 - \widehat{\beta}\widehat{R}^T \bar{x}_1$, and use fitted configuration $\widehat{\beta} X_1 \widehat{R} + \mathbf{1}\widehat{\mu}_\beta^T$.

The unscaled solution uses $\widehat{\beta} = 1$ and preserves size; the scaled solution also matches overall scale.

Method 3.2.10. *Procrustes Shape Average*

For centered matched shapes $X_\ell \in \mathbb{R}^{n\times p}$, set $\Phi(M, \{R_\ell\}) = \sum_{\ell=1}^{L} \|X_\ell R_\ell - M\|_F^2$, $R_\ell^T R_\ell = I_p$.

A Procrustes shape average solves $\min_{M, \{R_\ell\}} \Phi(M, \{R_\ell\})$.

(i) Initialize $M^{(0)}$ as a centered observed shape or an initially aligned average.
(ii) Given $M^{(t)}$, compute $R_\ell^{(t)} \in \operatorname{argmin}_{R^T R = I_p} \left\|X_\ell R - M^{(t)}\right\|_F^2$ for each ℓ.
(iii) Update $M^{(t+1)} = L^{-1} \sum_{\ell=1}^{L} X_\ell R_\ell^{(t)}$.
(iv) Recenter $M^{(t+1)}$ and, when size is fixed, rescale it to the chosen norm.
(v) Repeat (ii)–(iv) until Φ changes by less than the chosen tolerance.

The template is determined only up to a common right rotation unless a final orientation convention is imposed.

Remark 3.2.11. *Manifold Hypothesis*
The manifold hypothesis asserts that observations $x_1, \dots, x_n \in \mathbb{R}^p$ satisfy $\operatorname{dist}(x_i, \mathcal{M}) \le \varepsilon$ for some r-dimensional set $\mathcal{M} \subset \mathbb{R}^p$, with $r \ll p$ and small $\varepsilon \ge 0$. For curved $\mathcal{M}$, ambient distances $\|x_i - x_j\|_2$ need not approximate intrinsic distances $d_{\mathcal{M}}(x_i, x_j)$. Manifold learning therefore estimates low-dimensional coordinates from neighborhoods, graph geodesics, or tangent spaces, rather than from global Euclidean variance alone.

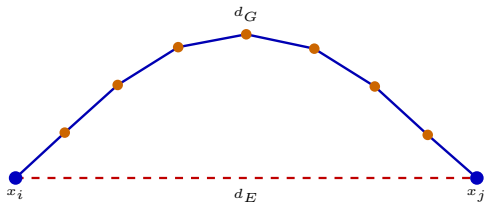

Figure 252: d_G follows nearby samples along the curve; d_E is the ambient shortcut.

Example 3.2.12. *Manifold Methods*
Isomap, locally linear embedding, Laplacian eigenmaps, and diffusion maps construct low-dimensional coordinates from a neighborhood graph or kernel on the sample.

Method 3.2.13. *Isomap*
Let $x_i \in \mathbb{R}^p$, fix r, set $H = I_n - n^{-1}\mathbf{1}\mathbf{1}^T$, and choose a connected neighborhood graph G on $[n]$.

(i) Assign edge weights $w_{ij} = \|x_i - x_j\|_2$ to neighboring pairs.
(ii) Compute shortest-path distances $d_G(i,j)$ and set $D_G^{(2)} = (d_G(i,j)^2)$.
(iii) Form the centered Gram estimate $B = -\frac{1}{2} H D_G^{(2)} H$.
(iv) Compute $Be_k = \lambda_k e_k$ with $\lambda_1 \geq \cdots \geq \lambda_r > 0$, set $E_r = [e_1 \ \cdots \ e_r]$, $\Lambda_r = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$.
(v) Return $Z = E_r \Lambda_r^{1/2}$.

Isomap replaces ambient distances by graph geodesic distances; the coordinate extraction is spectral.

Method 3.2.14. *Locally Linear Embedding*
Let $x_i \in \mathbb{R}^p$, fix K, r, and let $\mathcal{N}(i)$ be the K nearest neighbors of x_i.

(i) Set $\mathcal{W}_i = \{w : \sum_k w_k = 1,\ w_k = 0\ (k \notin \mathcal{N}(i))\}$ and $w_i \in \operatorname*{argmin}_{w \in \mathcal{W}_i} \left\| x_i - \sum_k w_k x_k \right\|_2^2$.
(ii) Form $W = (w_{ik})$ and $M = (I_n - W)^T (I_n - W)$.
(iii) Compute eigenvectors $u_1, \ldots, u_{r+1}$ of M for the smallest eigenvalues, with $u_1 \propto \mathbf{1}$.
(iv) Return $Y = [u_2 \ \cdots \ u_{r+1}] \in \mathbb{R}^{n \times r}$, with row y_i^T as the coordinate of x_i.

The affine constraint makes local weights translation-invariant; discarding u_1 gives mean-centered coordinates.

Method 3.2.15. *Local MDS*
Let $\mathcal{N} \subset \{(i,i') : i < i'\}$ be symmetric nearby pairs, let $z_i \in \mathbb{R}^r$, and fix $\tau > 0$.
Local MDS minimizes $S_{\text{local}}(Z) = \sum_{(i,i') \in \mathcal{N}} (d_{ii'} - \|z_i - z_{i'}\|_2)^2 - \tau \sum_{(i,i') \notin \mathcal{N}} \|z_i - z_{i'}\|_2$.

(i) The local stress term preserves distances on nearby pairs $(i,i') \in \mathcal{N}$.
(ii) The separation term penalizes collapse of non-neighboring pairs $(i,i') \notin \mathcal{N}$.

Thus the objective preserves local metric structure while repelling unrelated manifold regions.

Definition 3.2.16. *Principal Curves and Surfaces*
Let $X \in \mathbb{R}^p$ be a random vector and let $f : \Lambda \to \mathbb{R}^p$ be a smooth parametrized set.

(i) The *projection index* is $\lambda_f(x) \in \operatorname*{argmin}_{\lambda \in \Lambda} \|x - f(\lambda)\|_2$.
(ii) A *principal curve* has $\Lambda \subset \mathbb{R}$ and satisfies $f(\lambda) = \mathbb{E}\{X \mid \lambda_f(X) = \lambda\}$.
(iii) A *principal surface* is the same definition with $\Lambda \subset \mathbb{R}^q$, $q \geq 2$, and $f = (f_1, \ldots, f_p)$.

For finite data, alternate coordinate-wise smoothing against $\lambda_f(x_i)$ and reprojection onto $f(\Lambda)$.

Remark 3.2.17. *SOMs Versus Principal Surfaces*
Both self-organizing maps and principal surfaces approximate data by low-dimensional representatives.

(i) A principal surface is a smooth map $f : \Lambda \subset \mathbb{R}^q \to \mathbb{R}^p$ with fitted coordinates $\lambda_f(x_i)$.
(ii) A self-organizing map is a finite prototype family $\{m_a\}_{a \in A} \subset \mathbb{R}^p$ indexed by a lattice or graph A.

With many prototypes and kernel neighborhood weights, batch SOM updates resemble coordinate-wise surface smoothing, but one fit is smooth-parametric and the other is finite-prototype.

Remark 3.2.18. *Local versus Global Scaling*
Let $d_E(i,j) = \|x_i - x_j\|_2$ and let $\mathcal{N}$ be a neighborhood relation.

(i) *Global scaling* uses all pairs through $d_E(i,j)$ or centered inner products, and is appropriate when ambient distances encode the target geometry.
(ii) *Local scaling* uses $\mathcal{N}$, graph distances, or local reconstruction weights to preserve neighborhood order or local shape on a curved set.

Neighborhood scale is critical: too small disconnects, too large shortcuts.

3.3 Matrix Factorization and Topic Models

Many unsupervised models express data as combinations of latent factors.

Definition 3.3.1. *Low-Rank Matrix Factorization*
Let $X \in \mathbb{R}^{m \times n}$ and $1 \leq r \leq \min\{m,n\}$.
A *rank-r matrix factorization* is pair $(U,V) \in \mathbb{R}^{m \times r} \times \mathbb{R}^{n \times r}$ with fitted matrix $\widehat{X} = UV^\top$ and $\operatorname{rank}(\widehat{X}) \leq r$.
For a loss $\mathcal{L}$, choose (U,V) to minimize $\mathcal{L}(X, UV^\top)$; the columns encode latent row and column factors.

Definition 3.3.2. *Nonnegative Matrix Factorization*
Let $X \in \mathbb{R}_+^{n\times p}$ and $1 \le r \le \min\{n,p\}$.
A *nonnegative matrix factorization* is a pair $(W,H) \in \mathbb{R}_+^{n\times r} \times \mathbb{R}_+^{r\times p}$ with fitted matrix $\widehat{X} = WH$.
For loss $\mathcal{L}$, minimize $\mathcal{L}(X, WH)$ over $W, H \ge 0$, yielding additive parts-based factors.

Method 3.3.3. *Poisson NMF Multiplicative Updates*
For count data $X = (x_{ij})$, set $\mu = WH$ and $L(W,H) = \sum_{i,j}\{x_{ij}\log\mu_{ij} - \mu_{ij}\}$, up to constants.

(i) Initialize $W, H > 0$.
(ii) Set $\rho_{ik}^W = (\sum_j \frac{h_{kj}x_{ij}}{\mu_{ij}})/(\sum_j h_{kj})$ and update $w_{ik} \leftarrow w_{ik}\rho_{ik}^W$.
(iii) Recompute $\mu = WH$, set $\rho_{kj}^H = (\sum_i \frac{w_{ik}x_{ij}}{\mu_{ij}})/(\sum_i w_{ik})$, and update $h_{kj} \leftarrow h_{kj}\rho_{kj}^H$.
(iv) Repeat (ii)–(iii) until $L(W,H)$ stabilises.

The updates are local; different initialisations may give different NMF decompositions.

Remark 3.3.4. *NMF Non-Uniqueness*
An NMF $X = WH$ is not identifiable from nonnegativity alone: if A is invertible and $WA \ge 0$, $A^{-1}H \ge 0$, then $X = (WA)(A^{-1}H)$.
Interpretability therefore depends on additional geometry, constraints, or initialization.

Definition 3.3.5. *Archetypal Analysis*
Let $X \in \mathbb{R}^{n\times p}$ have observed rows x_i^T, and fix r archetypes.

(i) Observation weights $W = (w_{ik}) \in \mathbb{R}_+^{n\times r}$ satisfy $\sum_{k=1}^{r} w_{ik} = 1$.
(ii) Archetype weights $B = (b_{ki}) \in \mathbb{R}_+^{r\times n}$ satisfy $\sum_{i=1}^{n} b_{ki} = 1$.
(iii) Archetypes are the rows of $H = BX$, hence lie in $\mathrm{conv}\{x_1,\dots,x_n\}$.
(iv) Archetypal analysis chooses W, B minimising $J(W,B) = \|X - WBX\|_F^2$.

Archetypes are convex-hull extremes rather than central prototypes.

Remark 3.3.6. *Archetypes, k-Means, and NMF*
For prototype matrix H and assignment/mixture weights W:

(i) k-means uses one-hot rows of W.
(ii) NMF uses nonnegative W, H without requiring prototype rows of H to be observed-data mixtures.
(iii) Archetypal analysis uses simplex rows of W and constrains $H = BX$ with simplex rows of B.

Archetypes are convex-hull prototypes; k-means is hard-assigned and NMF is hull-free.

Definition 3.3.7. *Topic Model*
Fix K topics and a vocabulary of size p.

(i) Topic k is a word distribution $\beta_k \in \Delta_{p-1}$.
(ii) Document d has topic mixture $\theta_d \in \Delta_{K-1}$.
(iii) Token m in document d is generated by $z_{dm} \sim \mathrm{Cat}(\theta_d)$ and $w_{dm} \sim \mathrm{Cat}(\beta_{z_{dm}})$.
(iv) Latent Dirichlet allocation adds priors $\theta_d \sim \mathrm{Dir}(\alpha)$ and $\beta_k \sim \mathrm{Dir}(\eta)$.

Remark 3.3.8. *Regularized Factors*
For a factorization $X \approx UV^\top$, constraints select an identifiable and interpretable solution class.

(i) Low rank bounds latent dimension.
(ii) Sparsity restricts active variables or factors.
(iii) Nonnegativity enforces additive parts.
(iv) Probabilistic priors encode uncertainty and shrinkage.

These restrictions convert an underdetermined algebraic fit into a statistical learning problem.

3.4 Representation Learning Objectives

Representation learning trains maps that make downstream prediction, reconstruction, comparison, or generation easier.

Definition 3.4.1. *Autoencoder*
Let $x_1,\dots,x_n \in \mathcal{X}$ and let $\mathcal{H}$ be a representation space.

(i) An *autoencoder* is an encoder-decoder pair $e : \mathcal{X} \to \mathcal{H}$ and $d : \mathcal{H} \to \mathcal{X}$.
(ii) It is trained by minimizing a reconstruction loss such as $n^{-1}\sum_{i=1}^{n} \|x_i - d(e(x_i))\|_2^2$.

Bottlenecks, noise, sparsity, or latent-variable constraints rule out the unrestricted identity solution.

Definition 3.4.2. *Sparse Representation Objective*
Let $x_i \in \mathbb{R}^p$, dictionary $W \in \mathbb{R}^{p\times r}$, codes $h_i \in \mathbb{R}^r$, and $\lambda > 0$.
A *sparse representation objective* chooses $W, h_1,\dots,h_n$ to minimize $\sum_{i=1}^{n}\{\|x_i - Wh_i\|_2^2 + \lambda\|h_i\|_1\}$.
The reconstruction term fits x_i, while the L_1 term sparsifies h_i.

Definition 3.4.3. *Contrastive Objective*
Let $g : \mathcal{X} \to \mathbb{R}^r$ be a representation map, $s(u,v)$ a similarity score, and $\tau > 0$ a temperature.

(i) For an anchor x, sample a positive x^+ and negatives $x_1^-,\dots,x_K^-$.
(ii) With $a^+ = s(g(x), g(x^+))/\tau$ and $a_k^- = s(g(x), g(x_k^-))/\tau$, min loss i.e. $\log\left\{\exp(a^+) + \sum_{k=1}^{K}\exp(a_k^-)\right\} - a^+$.

The objective raises similarity to positives and lowers relative similarity to negatives.

3.5 Association Rules and Density-Ratio Views

Association-rule analysis searches for simple descriptions of regions where the input distribution has relatively large probability. The classical market-basket setting is the binary special case, but the same idea extends to conjunctions of coordinate conditions.

Definition 3.5.1. *Conjunctive Region*
Let $X = (X_1, \dots, X_p)$ take values in $\mathcal{S} = \prod_{j=1}^{p} \mathcal{S}_j$. For $\mathcal{J} \subseteq \{1, \dots, p\}$ and $s = (s_j)_{j \in \mathcal{J}}$ with $s_j \subseteq \mathcal{S}_j$, the *conjunctive region* is $R(\mathcal{J}, s) = \{x \in \mathcal{S} : x_j \in s_j \ \forall j \in \mathcal{J}\}$, with event $\{X \in R(\mathcal{J}, s)\} = \bigcap_{j \in \mathcal{J}} \{X_j \in s_j\}$.
Coordinates in $\{1, \dots, p\} \setminus \mathcal{J}$ are unrestricted; under the extension $s_j = \mathcal{S}_j$, they add no constraint.
For $\mathcal{S}_j \subseteq \mathbb{R}$, s_j is typically an interval; for categorical $\mathcal{S}_j$, s_j is typically a subset of levels.

Definition 3.5.2. *Market-Basket Item Sets*
Let $\mathcal{I} = \{1, \dots, K\}$ and $z_i = (z_{ik})_{k \in \mathcal{I}} \in \{0,1\}^{\mathcal{I}}$ for $i = 1, \dots, n$, where $z_{ik} = 1$ iff observation i contains item k.

(i) An *item set* is any $\mathcal{K} \subseteq \mathcal{I}$.
(ii) The containment indicator is $Y_i(\mathcal{K}) = \prod_{k \in \mathcal{K}} z_{ik}$; observation i contains $\mathcal{K}$ iff $Y_i(\mathcal{K}) = 1$.
(iii) The empirical support is $T(\mathcal{K}) = n^{-1} \sum_{i=1}^{n} Y_i(\mathcal{K})$.
(iv) For finite categorical supports, take $\mathcal{I} = \{(j, a) : 1 \le j \le p,\ a \in \mathcal{S}_j\}$, hence $K = \sum_{j=1}^{p} |\mathcal{S}_j|$.

Definition 3.5.3. *Association Rule, Confidence, and Lift*
Let $A, B \subseteq \mathcal{I}$ satisfy $A \cap B = \emptyset$, $A \neq \emptyset$, $B \neq \emptyset$, $T(A) > 0$, and $T(B) > 0$.
The *association rule* with antecedent A and consequent B is $A \Rightarrow B$.

(i) The rule support is $S(A \Rightarrow B) = T(A \cup B)$.
(ii) The confidence is $C(A \Rightarrow B) = \frac{T(A \cup B)}{T(A)}$.
(iii) The lift is $L(A \Rightarrow B) = \frac{C(A \Rightarrow B)}{T(B)} = \frac{T(A \cup B)}{T(A)T(B)}$.

Interpretation: $C(A \Rightarrow B)$ estimates prevalence of B given A; $L(A \Rightarrow B)$ compares it with $T(B)$.

Example 3.5.4. *A Market-Basket Rule*
Let $A = \{\text{peanut butter}, \text{jelly}\}$ and $B = \{\text{bread}\}$.

(i) Support $T(A \cup B) = 0.03$ means all three items appear in about 3% of baskets.
(ii) Confidence $C(A \Rightarrow B) = 0.82$ means about 82% of baskets containing A also contain bread.
(iii) If $T(B) = 0.43$, then $L(A \Rightarrow B) = 0.82/0.43 \approx 1.91$.

The rule predicts bread more often than its marginal prevalence.

Algorithm 4: Apriori Association-Rule Search

Require: Binary item data $z_1, \dots, z_n$, support threshold t, confidence threshold c
Ensure: Rules with support above t and confidence above c
Compute supports for all one-item sets and retain $\mathcal{F}_1 = \{\mathcal{K} : |\mathcal{K}| = 1,\ T(\mathcal{K}) > t\}$
Set $m \leftarrow 2$
while $\mathcal{F}_{m-1} \neq \emptyset$ **do**
  Form candidate item sets of size m whose every size-$(m-1)$ subset lies in $\mathcal{F}_{m-1}$
  Scan the data to compute $T(\mathcal{K})$ for each candidate
  Retain $\mathcal{F}_m = \{\mathcal{K} : |\mathcal{K}| = m,\ T(\mathcal{K}) > t\}$
  Set $m \leftarrow m + 1$
end while
For each retained item set $\mathcal{K}$, test nonempty proper antecedents $A \subset \mathcal{K}$
Report $A \Rightarrow \mathcal{K} \setminus A$ when $C(A \Rightarrow \mathcal{K} \setminus A) > c$

Remark 3.5.5. *Apriori Monotonicity and Limits*
Support is antitone: $\mathcal{L} \subseteq \mathcal{K} \Longrightarrow T(\mathcal{L}) \ge T(\mathcal{K})$.

(i) If $T(\mathcal{L}) < t$, then every $\mathcal{K} \supseteq \mathcal{L}$ also has $T(\mathcal{K}) < t$ and may be pruned.
(ii) A support threshold t can discard rules $A \Rightarrow B$ with large $C(A \Rightarrow B)$ or $L(A \Rightarrow B)$ but small $T(A \cup B)$.

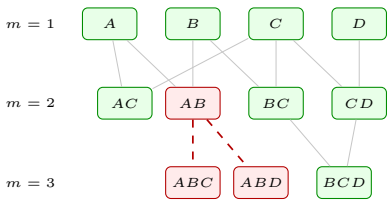

Figure 253: Apriori keeps frequent item sets; if $T(AB) \le t$, all supersets of AB are pruned.

Method 3.5.6. *Generalized Association Rules*
For $\mathcal{J} \subseteq \{1, \dots, p\}$ and $s_j \subseteq \mathcal{S}_j$, set $R(\mathcal{J}, s) = \bigcap_{j \in \mathcal{J}} \{X_j \in s_j\}$ and $\widehat{P}\{R(\mathcal{J}, s)\} = n^{-1} \sum_{i=1}^{n} \mathbf{1}\{x_{ij} \in s_j \ \forall j \in \mathcal{J}\}$.

(i) Allow s_j to be intervals for quantitative variables and subsets of levels for categorical variables.
(ii) Search over candidate regions with a heuristic rule learner.
(iii) Retain rules whose fitted score and observed support meet the chosen thresholds.

Method 3.5.7. *Unsupervised Learning as Supervised Density-Ratio Estimation*
Let g be the data density, g_0 a reference density, and $Y = 1$ label data while $Y = 0$ labels reference observations.

(i) Draw a reference sample from g_0 and pool it with the data, weighting the two classes equally.
(ii) Fit a probabilistic classifier $\widehat{\mu}(x) \approx \Pr(Y = 1 \mid X = x)$.
(iii) Use $\mu(x) = g(x)/(g(x) + g_0(x))$ to estimate the density ratio by $\widehat{g}(x)/g_0(x) = \widehat{\mu}(x)/(1 - \widehat{\mu}(x))$.

Uniform, Gaussian, and product-marginal g_0 target high density, nonnormality, and dependence, respectively.

Remark 3.5.8. *Rule Learners for Density Ratios*
After estimating $\widehat{g}/g_0$, any learner that returns regions $R = \bigcap_{j \in \mathcal{J}} \{X_j \in s_j\}$ can search for generalized item sets. Tree leaves and box-peeling regions have this conjunctive form.
The unsupervised target is large $\widehat{g}(x)/g_0(x)$ on R together with adequate empirical support, not predictive risk.

3.6 Graph Ranking and PageRank

Graph ranking turns link structure into a numerical importance score. PageRank is a spectral example: a page receives weight from incoming links, and a link from an already important page contributes more than a link from an unimportant one.

Definition 3.6.1. *PageRank Equations*

For pages indexed by $\{1, \dots, N\}$, define $L \in \{0,1\}^{N \times N}$ by $L_{ij} = 1$ iff page j links to page i, and set $c_j = \sum_{i=1}^{N} L_{ij}$.

Assume $c_j > 0$ for all j and fix $d \in (0,1)$.

A *PageRank score vector* is $p = (p_1, \dots, p_N) \in \mathbb{R}^N$ satisfying $p_i = (1-d) + d \sum_{j=1}^{N} \frac{L_{ij}}{c_j} p_j$ for $i = 1, \dots, N$.

Proposition 3.6.2. *PageRank Normalization*

Let $L \in \{0,1\}^{N \times N}$, $c_j = \sum_{i=1}^{N} L_{ij} > 0$, *and* $d \in (0,1)$.

If $p \in \mathbb{R}^N$ *satisfies* $p_i = (1-d) + d \sum_{j=1}^{N} \frac{L_{ij}}{c_j} p_j$ *for* $i = 1, \dots, N$, *then* $\sum_{i=1}^{N} p_i = N$.

Proof. Let $P = \sum_{i=1}^{N} p_i$. Summing over i gives $P = N(1-d) + dP$, since $\sum_{i=1}^{N} L_{ij} = c_j$.
Hence $(1-d)P = N(1-d)$, so $P = N$.
Structural Commentary: Column normalisation preserves total score up to the teleportation offset.
Key Insight: Summing cancels the link structure because each column has mass c_j. □

Method 3.6.3. *Eigenvector Form*
Let $\mathbf{e} = (1, \dots, 1)^T$, $D_c = \operatorname{diag}(c_1, \dots, c_N)$, and $A = (1-d)\frac{\mathbf{e}\mathbf{e}^T}{N} + dLD_c^{-1}$. The PageRank equations are $\mathbf{p} = A\mathbf{p}$ with $\mathbf{e}^T\mathbf{p} = N$. Since $\mathbf{e}^T A = \mathbf{e}^T$ and $A_{ij} \geq (1-d)/N > 0$, A is positive column-stochastic. Thus $\mathbf{p}$ is the positive eigenvector for eigenvalue 1 normalized by $\mathbf{e}^T\mathbf{p} = N$, and $\mathbf{p}/N$ is the stationary distribution in column-vector convention.

Algorithm 5: PageRank by Power Iteration

Require: Link matrix L, positive outlink counts c_j, damping parameter $0 < d < 1$, initial vector $\mathbf{p}_0 > 0$
Ensure: PageRank vector normalized by $\mathbf{e}^T\mathbf{p} = N$
  Form $A = (1-d)\mathbf{e}\mathbf{e}^T/N + dLD_c^{-1}$
  repeat
    $\mathbf{p}_k \leftarrow A\mathbf{p}_{k-1}$
    $\mathbf{p}_k \leftarrow N\mathbf{p}_k/(\mathbf{e}^T\mathbf{p}_k)$
  until the iterates stabilize

Remark 3.6.4. *Random-Surfer Normalization*

The probability-scale normalization uses $\pi_i = (1-d)/N + d \sum_{j=1}^{N} (L_{ij}/c_j)\pi_j$ and $\mathbf{e}^T\pi = 1$.

Equivalently, $p = N\pi$ satisfies the average-score form $\mathbf{e}^T\mathbf{p} = N$.
Random-surfer form: follow an outgoing link with probability d; jump uniformly with probability $1-d$.

Example 3.6.5. *Four-Page Network*
Let $d = 0.85$, $L = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $\mathbf{c} = L^T\mathbf{e} = (2,1,1,1)^T$, and $D_c = \operatorname{diag}(\mathbf{c})$.
Solving $\mathbf{p} = (1-d)\mathbf{e} + dLD_c^{-1}\mathbf{p}$ gives $\widehat{\mathbf{p}} = (1.49, 0.78, 1.58, 0.15)^T$. Since $L_{4j} = 0$ for all j, $\widehat{p}_4 = 1 - d = 0.15$.

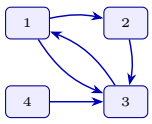

Figure 254: Four-page PageRank graph: $\widehat{\mathbf{p}} \approx (1.49, 0.78, 1.58, 0.15)^T$.

3.7 Latent Factor and Component Analysis

Latent factor methods model observed coordinates as mixtures of unobserved sources. The main question is which structure makes the sources interpretable: covariance structure, statistical independence, or non-Gaussian projections.

Definition 3.7.1. *Factor Analysis*
Let $X \in \mathbb{R}^p$ and $q < p$. A *factor-analysis model* is a representation $X = AS + \varepsilon$ with $A \in \mathbb{R}^{p\times q}$, $S \in \mathbb{R}^q$, $\varepsilon \in \mathbb{R}^p$, where: $\mathrm{Cov}(S) = I_q$, $\mathrm{Cov}(\varepsilon) = D_\varepsilon$ for diagonal $D_\varepsilon \in \mathbb{R}^{p\times p}$, and $\mathrm{Cov}(S, \varepsilon) = 0$.
The columns $a_1, \dots, a_q$ of A are the *factor loadings*, and $\Sigma := \mathrm{Cov}(X) = AA^T + D_\varepsilon$.
For any orthogonal $R \in \mathbb{R}^{q\times q}$, (A, S) and (AR^T, RS) yield the same Σ.
Hence (A, S) is identifiable only up to orthogonal rotation.

Definition 3.7.2. *Independent Component Analysis*
Let $X \in \mathbb{R}^p$. An *independent component model* is a representation $X = AS$ such that:

(i) $A \in \mathbb{R}^{p\times p}$ is nonsingular.
(ii) $S = (S_1, \dots, S_p)^T \in \mathbb{R}^p$ has mutually independent components.

If X is whitened, then $\mathrm{Cov}(X) = I_p$ and A may be taken orthogonal.
ICA estimates orthogonal A with $Y = A^T X$ approximately componentwise independent, typically via non-Gaussianity contrasts. Gaussian components are identifiable only up to orthogonal rotations.

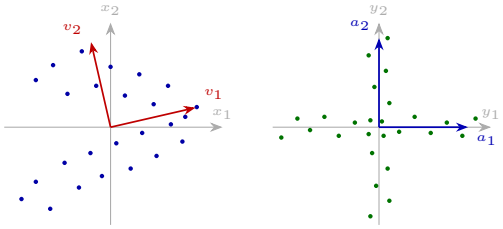

Figure 255: v_1, v_2: PCA variance axes; a_1, a_2: ICA rotation exposing independent non-Gaussian coordinates.

Method 3.7.3. *Entropy View of ICA*
For a density g, define $H(g) = -\int g(y) \log g(y)\, dy$.
Let $Y = (Y_1, \dots, Y_p)$ have joint density g_Y and marginal densities g_j.

(i) Set $H(Y) := H(g_Y)$, $H(Y_j) := H(g_j)$, and $I(Y) := \sum_{j=1}^p H(Y_j) - H(Y)$.
Then $I(Y) \geq 0$, with $I(Y) = 0$ iff $Y_1, \dots, Y_p$ are independent.
(ii) If X is whitened, A is orthogonal, and $Y = A^T X$, then $H(Y) = H(X)$.
Hence minimising $I(Y)$ over orthogonal A is equivalent to minimising $\sum_{j=1}^p H(Y_j)$.
(iii) Replace marginal-entropy minimisation by marginal non-Gaussianity maximisation,
e.g. through negentropy contrasts, since Gaussian marginals maximise entropy at fixed variance.

Remark 3.7.4. *Projection Pursuit and ICA*
Projection pursuit optimizes a non-Gaussianity index over $Y = a^T X$ or $Y = (a_1^T X, a_2^T X)$.
ICA uses the same type of index on $Y_j = a_j^T X$, but imposes an orthogonal system $(a_1, \dots, a_p)$ after whitening.

Method 3.7.5. *Product-Density ICA*
For whitened $X \in \mathbb{R}^p$ and orthogonal $A \in \mathbb{R}^{p\times p}$, set $Y = A^T X$.

(i) Assume the product density $f_Y(y) = \prod_{j=1}^p f_j(y_j)$.
(ii) Parameterize each marginal density by $f_j(s) = \phi(s)\exp(g_j(s))$ with $\int \phi(s)\exp(g_j(s))\, ds = 1$.
(iii) Estimate $A, g_1, \dots, g_p$ by penalized likelihood subject to $A^T A = I$.
For fixed A, the objective decomposes into p one-dimensional smooth density fits.

The roughness penalty shrinks g_j toward 0, the Gaussian baseline ϕ.

Algorithm 6: Product-Density ICA Alternation

Require: Whitened observations $x_1, \dots, x_n$, initial orthogonal matrix A
Ensure: Orthogonal unmixing matrix A and components $s_{ij} = a_j^T x_i$
repeat
  for each component j **do**
    Fit a smooth tilted-Gaussian density $f_j(s) = \phi(s)\exp(g_j(s))$ to the current scores $a_j^T x_i$
    Update $a_j \leftarrow \mathbb{E}\{X g_j'(a_j^T X)\} - \mathbb{E}\{g_j''(a_j^T X)\} a_j$
  end for
  Orthogonalize A: if $A = UDV^T$ is its SVD, set $A \leftarrow UV^T$
until the directions stabilize

4 Probabilistic Graphical and Causal Learning

4.1 Directed and Undirected Graphical Models

Graphical models use graph structure to encode conditional independence and to organize high-dimensional probability laws.

Definition 4.1.1. *Directed Graphical Model*
Let $G = (V, E)$ be a directed acyclic graph with $V = \{1, \dots, d\}$ and $\mathrm{pa}(j) = \{i \in V : i \to j \in E\}$.
For $x = (x_1, \dots, x_d)$, write $x_{\mathrm{pa}(j)} = (x_i)_{i\in\mathrm{pa}(j)}$. A law for $X = (X_1, \dots, X_d)$ with joint density or mass function p *factorizes according to* G if $p(x) = \prod_{j=1}^d p(x_j \mid x_{\mathrm{pa}(j)})$, with $p(x_j \mid x_\varnothing) := p(x_j)$.

Definition 4.1.2. *Markov Graph Vocabulary*
Let $G = (V, E)$ be an undirected graph, and let $A, B, C \subseteq V$ be disjoint.

(i) Vertices $j, k \in V$ are *adjacent* iff $\{j, k\} \in E$.
(ii) A *path* from u to v is $(v_0, \dots, v_m)$ with $v_0 = u$, $v_m = v$, and $\{v_{\ell-1}, v_\ell\} \in E$ for $\ell = 1, \dots, m$.
(iii) C *separates* A and B iff every path from $a \in A$ to $b \in B$ contains a vertex in C.
(iv) A *clique* is $K \subseteq V$ such that $\{j, k\} \in E$ for all distinct $j, k \in K$.
It is *maximal* iff no strictly larger clique contains K.

Definition 4.1.3. *Undirected Graphical Model*
Let $G = (V, E)$ be an undirected graph, let $\mathcal{C}$ be its maximal cliques, and let p be a positive joint density or mass function for $X = (X_j)_{j \in V}$. For $C \subseteq V$, write $x_C = (x_j)_{j \in C}$. The following are equivalent:

(i) p is *Markov with respect to* G: if $j \neq k$ and $\{j, k\} \notin E$, then $X_j \perp\!\!\!\perp X_k \mid X_{V \setminus \{j,k\}}$.
(ii) p factorizes as $p(x) = Z^{-1} \prod_{C \in \mathcal{C}} \psi_C(x_C)$ for clique potentials $\psi_C \geq 0$, with $Z = \sum_x \prod_{C \in \mathcal{C}} \psi_C(x_C)$ in the discrete case and $Z = \int \prod_{C \in \mathcal{C}} \psi_C(x_C)\, dx$ in the continuous case.

Proposition 4.1.4. *Gaussian Graphical Zeros*
Let $X \sim N_d(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Theta = \Sigma^{-1} = (\theta_{jk})$. For j, write $\theta_{-j,j} = (\theta_{\ell j})_{\ell \neq j}$ and $\mu_{-j} = (\mu_\ell)_{\ell \neq j}$. Then:

(i) For $z \in \mathbb{R}^{d-1}$, $\mathbb{E}[X_j \mid X_{-j} = z] = \mu_j - \frac{\theta_{-j,j}^T (z - \mu_{-j})}{\theta_{jj}}$.
(ii) For $j \neq k$, $X_j \perp\!\!\!\perp X_k \mid X_{-(j,k)}$ iff $\theta_{jk} = 0$.

Proof.

(i) Let $Y = X_j$ and $Z = X_{-j}$. Order coordinates as (Z, Y), with covariance block Σ_{ZZ}, cross-covariance σ_{Zj}.
Gaussian conditioning gives $\mathbb{E}[Y \mid Z = z] = \mu_j + (z - \mu_{-j})^T \Sigma_{ZZ}^{-1} \sigma_{Zj}$.
From $\Sigma\Theta = I$, the (Z, j) block gives $\Sigma_{ZZ}\theta_{-j,j} + \sigma_{Zj}\theta_{jj} = 0$, hence $\Sigma_{ZZ}^{-1}\sigma_{Zj} = -\frac{\theta_{-j,j}}{\theta_{jj}}$.
(ii) Let $A = \{j, k\}$ and $B = A^c$. The conditional Gaussian law of $X_A \mid X_B$ has precision $\Theta_{AA} = \begin{pmatrix} \theta_{jj} & \theta_{jk} \\ \theta_{jk} & \theta_{kk} \end{pmatrix}$. A bivariate Gaussian has independent coordinates iff its precision is diagonal, so $X_j \perp\!\!\!\perp X_k \mid X_B$ iff $\theta_{jk} = 0$.

Structural Commentary: Conditioning turns precision entries into regression interactions.
Key Insight: Gaussian conditional independence is zero conditional precision. □

Definition 4.1.5. *Ising Model*
Let $E \subseteq \{\{j, k\} : 1 \leq j < k \leq p\}$, set $x_0 = 1$, $\widetilde{E} = E \cup \{\{0, j\} : 1 \leq j \leq p\}$, and $\mathcal{N}(j) = \{k : \{j, k\} \in E\}$.
A *pairwise Ising model* for $X \in \{0, 1\}^p$ is specified by:

(i) Joint law $p_\Theta(x) = \exp\left\{ \sum_{\{j,k\} \in \widetilde{E}} \theta_{jk} x_j x_k - \Phi(\Theta) \right\}$, where $\Phi(\Theta) = \log \sum_{x \in \{0,1\}^p} \exp\left\{ \sum_{\{j,k\} \in \widetilde{E}} \theta_{jk} x_j x_k \right\}$.
(ii) Node conditionals $\mathbb{P}_\Theta(X_j = 1 \mid X_{-j} = x_{-j}) = \left(1 + \exp\left\{ -\theta_{0j} - \sum_{k \in \mathcal{N}(j)} \theta_{jk} x_k \right\}\right)^{-1}$ for $j = 1, \dots, p$.

For $\{j, k\} \in E$, θ_{jk} is the conditional log-odds interaction between nodes j and k.

Definition 4.1.6. *Restricted Boltzmann Machine*
Let $v \in \{0, 1\}^p$ be visible units, let $h \in \{0, 1\}^q$ be hidden units, and let $\sigma(t) = (1 + \exp(-t))^{-1}$.
A *restricted Boltzmann machine* is specified by:

(i) Graph restriction: no visible-visible or hidden-hidden edges.
(ii) Joint law $p(v, h) = Z^{-1} \exp(a^T v + b^T h + v^T W h)$ with $a \in \mathbb{R}^p$, $b \in \mathbb{R}^q$, and $W \in \mathbb{R}^{p \times q}$.
(iii) Conditional factorization: $p(v \mid h) = \prod_{j=1}^{p} \mathrm{Bern}(v_j; \sigma(a_j + W_{j\cdot} h))$ and $p(h \mid v) = \prod_{k=1}^{q} \mathrm{Bern}(h_k; \sigma(b_k + W_{\cdot k}^T v))$.

In supervised variants, v may concatenate feature and label units.

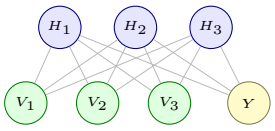

Figure 256: RBM: H_j connect only to visible features V_i and label Y; no within-layer edges.

Remark 4.1.7. *Separation*
For disjoint vertex sets A, B, C, separation is a graphical certificate for $X_A \perp\!\!\!\perp X_B \mid X_C$: DAGs use d-separation, while undirected graphs use path separation by C.
In a Markov graph, C separates A and B implies $X_A \perp\!\!\!\perp X_B \mid X_C$.
Missing edges give pairwise certificates; separation gives setwise certificates.

4.2 Inference and Learning in Graphical Models

Inference in a graphical model computes marginal, conditional, or most likely latent quantities from the joint law encoded by the graph.

Definition 4.2.1. *Inference Tasks*
Let $X = (X_j)_{j \in V}$ have joint law p, and let $A, B \subseteq V$ be disjoint. Standard *inference tasks* are:

(i) *Marginal inference*: compute $p(x_A)$.

(ii) *Conditional inference*: compute $p(x_A \mid x_B)$.
(iii) *MAP inference*: compute $x_A^\star \in \arg\max_{x_A} p(x_A \mid x_B)$, with $B = \varnothing$ for unconditional MAP.
(iv) *Latent posterior inference*: for latent Z and observed $X = x$, compute $p(z \mid x)$.

Method 4.2.2. *Message Passing*
Let $G = (V, E)$ be a graphical model with local potentials.

(i) *Tree belief propagation*: on trees, two-way edge messages yield exact node and edge marginals.
(ii) *Loopy belief propagation*: on cyclic graphs, iterated message updates yield approximate marginals.

Definition 4.2.3. *Variational Inference*
For latent-variable model $p(x, z)$, intractable $p(z \mid x)$, tractable family $\mathcal{Q}$, *variational inference* choose $q^\star \in \mathcal{Q}$:

(i) *KL projection*: $q^\star \in \arg\min_{q \in \mathcal{Q}} \mathrm{KL}(q(z) \| p(z \mid x))$.
(ii) *ELBO maximization*: equivalently, maximize $\mathcal{L}(q) := \mathbb{E}_q[\log p(x, z)] - \mathbb{E}_q[\log q(z)]$, since $\log p(x) = \mathcal{L}(q) + \mathrm{KL}(q(z) \| p(z \mid x))$.

Remark 4.2.4. *Learning Parameters and Structure*
Let G denote the graph and θ its model parameters.

(i) *Parameter learning*: for fixed G, estimate θ by likelihood or regularized likelihood.
(ii) *Structure learning*: estimate unknown G by optimizing a statistical score over graph candidates, often with structural penalties or combinatorial search.

Method 4.2.5. *Gaussian Graphical Model Learning*
For centered Gaussian data with empirical covariance S, let $\Theta = \Sigma^{-1} \succ 0$ and $\ell(\Theta) := \log\det\Theta - \mathrm{trace}(S\Theta)$.

(i) *Known graph*: for edge set E, estimate $\widehat{\Theta} \in \arg\max_{\Theta \succ 0} \ell(\Theta)$ subject to $\theta_{jk} = 0$ whenever $\{j, k\} \notin E$.
(ii) *Unknown graph*: graphical lasso solves $\widehat{\Theta} \in \arg\max_{\Theta \succ 0} \left\{ \ell(\Theta) - \lambda \sum_{j \neq k} |\theta_{jk}| \right\}$, penalise off-diagonal entries.

Graph selection is therefore shrinkage of inverse-covariance entries; zeros of $\widehat{\Theta}$ remove edges.

Figure 257: $\mathcal{D}$ is the data and S its empirical covariance. Y gives $X \perp \{Z, W\} \mid Y$; graphical lasso solves $\widehat{\Theta}_\lambda \in \arg\max_{\Theta \succ 0} \left\{ \log\det\Theta - \mathrm{tr}(S\Theta) - \lambda \sum_{j \neq k} |\theta_{jk}| \right\}$, and zeros of $\widehat{\Theta}_\lambda$ define missing edges in $\widehat{G}$.

Algorithm 7: Graphical Lasso by Block Coordinate Descent

Require: Empirical covariance $S \in \mathbb{R}^{p \times p}$, penalty $\lambda \geq 0$
Ensure: Covariance estimate W and sparse precision estimate $\widehat{\Theta} = W^{-1}$
Initialize $W \leftarrow S + \lambda I$ and keep the diagonal fixed
repeat
  for $j = 1, \ldots, p$ **do**
    Partition W into W_{11}, w_{12}, and w_{22} by isolating row and column j; partition S similarly to get s_{12}
    Define $\mathcal{S}_\lambda(z) = \mathrm{sign}(z)(|z| - \lambda)_+$
    repeat
      for $k = 1, \ldots, p-1$ **do**

$$\beta_k \leftarrow \frac{\mathcal{S}_\lambda\left(s_{12,k} - \sum_{\ell \neq k} (W_{11})_{k\ell}\beta_\ell\right)}{(W_{11})_{kk}}$$

      end for
    until β has converged for this block
    Update $w_{12} \leftarrow W_{11}\beta$ and symmetrically update row and column j of W
  end for
until W has converged
In the final cycle recover $\widehat{\theta}_{12} = -\widehat{\beta}\widehat{\theta}_{22}$ with $1/\widehat{\theta}_{22} = w_{22} - w_{12}^T\widehat{\beta}$

Method 4.2.6. *Discrete Markov Network Learning*
For a binary Markov network with edge statistic $T_{jk}(X) = X_j X_k$, likelihood equations are $\widehat{\mathbb{E}}T_{jk} = \mathbb{E}_\Theta T_{jk}$ for $\{j, k\} \in E$; exact $\mathbb{E}_\Theta$ sums over $x \in \{0, 1\}^p$.

(i) *Small graphs*: solve $\widehat{\mathbb{E}}T_{jk} = \mathbb{E}_\Theta T_{jk}$ by log-linear Poisson MLE, gradient/Newton updates, or iterative proportional fitting.
(ii) *Large graphs*: approximate $\mathbb{E}_\Theta T_{jk}$ by mean-field or Gibbs-sampling estimates.
(iii) *Hidden-node models*: use observed-data conditionals for $\widehat{\mathbb{E}}T_{jk}$; for RBMs, CD-$r$ starts at data and runs r alternating Gibbs steps $h \sim p(h \mid v)$, $v \sim p(v \mid h)$.

4.3 Causal Graphs, Interventions, and Identifiability

Causal learning asks which predictions remain valid under interventions, not merely under passive observation.

Definition 4.3.1. *Structural Causal Model*
Let J index endogenous variables $X = (X_j)_{j \in J}$ and exogenous disturbances $U = (U_j)_{j \in J}$.
A structural causal model is $M = (G, \{f_j\}_{j \in J}, P_U)$ such that:

(i) $G = (J, E)$ is a directed graph with $\mathrm{pa}(j) = \{k \in J : k \to j \in E\}$.

(ii) P_U is the joint law of U.
(iii) $X_j = f_j(X_{\mathrm{pa}(j)}, U_j)$ for every $j \in J$.
The edges $k \to j$ are the direct causal inputs to X_j.

Definition 4.3.2. *Intervention*
For $M = (G, \{f_k\}_{k\in J}, P_U)$ and $x \in \mathcal{X}_j$, set $\tilde{f}_j \equiv x$ and $\tilde{f}_k = f_k$ for $k \neq j$.
The *intervention* $\mathrm{do}(X_j = x)$ is the modified model $M_{j:=x} = (G_{j:=x}, \{\tilde{f}_k\}_{k\in J}, P_U)$, where $G_{j:=x}$ deletes all incoming edges to j; the induced law $P_{M_{j:=x}}$ is the interventional distribution.

Definition 4.3.3. *Identifiability*
Let $\mathcal{M}$ be the assumed model class and let P_M be the observational law induced by $M \in \mathcal{M}$.
A causal functional $\psi : \mathcal{M} \to \Theta$ is *identifiable* on $\mathcal{M}$ if $P_M = P_{M'}$ implies $\psi(M) = \psi(M')$ for all $M, M' \in \mathcal{M}$.

Remark 4.3.4. *Causal and Predictive Targets*
Let $\hat{y} = g(z)$ be trained under the observational law P_M. Small risk $R_{P_M}(g) = \mathbb{E}_{P_M}[\ell(Y, g(Z))]$ need not imply small interventional risk $R_{P_{M_{j:=x}}}(g)$ after $\mathrm{do}(X_j = x)$. Transport across regimes requires causal stability of the conditional mechanism used by g, not only predictive fit under P_M.

4.4 Counterfactuals and Policy Evaluation

Policy learning and evaluation combine prediction with causal questions about actions not actually taken.

Definition 4.4.1. *Potential Outcomes*
Let $A \in \mathcal{A}$ be a treatment.
(i) *Potential outcome*: $Y(a)$ is the outcome under $\mathrm{do}(A = a)$.
(ii) *Consistency*: $Y = Y(A)$.
(iii) *Individual effect*: for binary $A \in \{0, 1\}$, $\tau = Y(1) - Y(0)$.
(iv) *Average treatment effect*: $\mathrm{ATE} = \mathbb{E}[\tau]$.

Definition 4.4.2. *Ignorability and Propensity*
For binary treatment A and covariates X:
(i) *Ignorability*: $(Y(0), Y(1)) \perp\!\!\!\perp A \mid X$.
(ii) *Propensity score*: $e(x) = P(A = 1 \mid X = x)$.
(iii) *Overlap*: $0 < e(X) < 1$ almost surely.

Definition 4.4.3. *Off-Policy Evaluation*
Let $(X_i, A_i, Y_i)_{i=1}^n$ be logged under behavior policy $\mu(a \mid x)$, and let $\pi(a \mid x)$ be a target policy.
(i) *Target value*: $V(\pi) = \mathbb{E}_\pi[Y]$.
(ii) *Off-policy evaluation*: estimate $V(\pi)$ from data sampled under μ.
(iii) *Inverse-propensity weighting*: $\hat{V}_{\mathrm{IPW}}(\pi) = n^{-1} \sum_{i=1}^n \{\pi(A_i \mid X_i)/\mu(A_i \mid X_i)\} Y_i$.
(iv) *Overlap*: $\mu(a \mid x) > 0$ whenever $\pi(a \mid x) > 0$.

Remark 4.4.4. *Feedback*
If a predictor g induces actions through a policy $\pi_g(a \mid x)$, then future data are drawn from the post-deployment law P^{π_g} rather than the pre-deployment law P^0. Feedback is the dependence $g \mapsto \pi_g \mapsto P^{\pi_g}$; causal learning, reinforcement learning, and deployment monitoring all study or control this induced law.

5 Neural Network Foundations

5.1 Network Architectures and Expressivity

The migrated deep-feedforward-network notes describe neural networks as learned feature maps followed by prediction heads.

Definition 5.1.1. *Deep Feedforward Network*
Let $x \in \mathcal{X}$, set $h^{(0)} = x$, and fix depth L.
(i) *Layer recursion*: $h^{(k)} = g^{(k)}(W^{(k)} h^{(k-1)} + b^{(k)})$ for $k = 1, \ldots, L$.
(ii) *Network map*: $f_\theta : \mathcal{X} \to \mathcal{Y}$ is $f_\theta(x) = h^{(L)} = (f_L \circ \cdots \circ f_1)(x)$.
(iii) *Parameters*: $\theta = \{(W^{(k)}, b^{(k)})\}_{k=1}^L$.
(iv) *Training objective*: choose θ to minimize empirical risk $\hat{R}_n(f_\theta)$ or population risk $R(f_\theta)$.

Remark 5.1.2. *Learned Feature Maps*
Let ℓ be the task loss and let $R(f) = \mathbb{E}[\ell(Y, f(X))]$. Two representation parametrizations are:
(i) *fixed features*: $f_w(x) = a_w(\phi(x))$, with ϕ fixed and only the head parameter w optimized.
(ii) *learned features*: $f_{\theta,w}(x) = a_w(\phi_\theta(x))$, with θ and w optimized jointly.
Learning ϕ_θ adds adaptivity but usually destroys convexity.

Definition 5.1.3. *Projection Pursuit and Ridge Functions*
Let $x \in \mathbb{R}^p$.
(i) *Projection-pursuit form*: $f(x) = \sum_{m=1}^M g_m(\omega_m^T x)$ with $\|\omega_m\|_2 = 1$.
(ii) *Ridge coordinate*: $u_m = \omega_m^T x$.
(iii) *Ridge function*: $g_m(u_m)$ depends on x only through the scalar coordinate u_m.
(iv) *Learned components*: the directions ω_m and univariate profiles g_m are estimated from data.

Definition 5.1.4. *Output and Hidden Units*
Let $T \in \mathbb{R}^K$ denote output scores. Common output parametrizations are:
(i) *linear*: $f_k(x) = T_k$, for conditional means or real-valued responses.

(ii) *sigmoid*: $p(x) = \sigma(T) = \frac{1}{1+\exp(-T)}$, for Bernoulli probabilities.

(iii) *softmax*: $p_k(x) = \exp(T_k)/\sum_{\ell=1}^{K} \exp(T_\ell)$, for K-class probabilities.

Hidden units define activations $Z_m = \phi_m(a_m)$ from preactivations a_m, commonly using ReLU, leaky ReLU, maxout, tanh, or another nonlinear map.

Definition 5.1.5. *Single Hidden Layer Network*
Let $x \in \mathbb{R}^p$, $Z = (Z_1, \dots, Z_M)^T$, $T = (T_1, \dots, T_K)^T$. A single-hidden-layer feedforward network is defined by:

(i) $Z_m = \sigma(\alpha_{0m} + \alpha_m^T x)$ for $m = 1, \dots, M$.
(ii) $T_k = \beta_{0k} + \beta_k^T Z$ for $k = 1, \dots, K$.
(iii) $f_k(x) = g_k(T)$, with $g_k(T) = T_k$ for regression and $g_k(T) = \exp(T_k)/\sum_{\ell=1}^{K} \exp(T_\ell)$ for K-class classification.

In the classification case, $f(x) \in \Delta^{K-1}$.

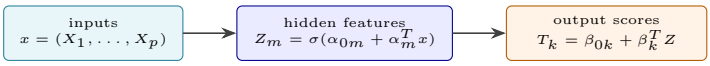

Figure 258: Single-hidden-layer feedforward network.

Remark 5.1.6. *Neural Networks as Ridge Expansions*
For scalar output, a one-hidden-layer network has $f(x) = \beta_0 + \sum_{m=1}^{M} \beta_m \sigma(\alpha_{0m} + \alpha_m^T x)$.
If $\alpha_m \neq 0$ and $\omega_m = \frac{\alpha_m}{\|\alpha_m\|_2}$, the mth term is $\beta_m \sigma(\alpha_{0m} + \|\alpha_m\|_2 \omega_m^T x)$, a ridge term in coordinate $\omega_m^T x$ with learned *direction* ω_m, *scale* $\|\alpha_m\|_2$, *threshold* α_{0m}, and *output coefficient* β_m.
Thus a shallow scalar network is a projection-pursuit expansion with parametrized ridge profiles.

Theorem 5.1.7. *Universal Approximation Theorem Form*
For compact $K \subset \mathbb{R}^d$, $h \in C(K)$, non-polynomial activation $\sigma : \mathbb{R} \to \mathbb{R}$, $\forall \varepsilon > 0$, $\exists\, M \in \mathbb{N}$, $a_m, b_m \in \mathbb{R}$, $w_m \in \mathbb{R}^d$ s.t. one-hidden-layer network $f_M(x) = \sum_{m=1}^{M} a_m \sigma(w_m^\top x + b_m)$ satisfies $\|f_M - h\|_\infty < \varepsilon$ on K.

Proof. Let $\mathcal{A}_\sigma := \operatorname{span}\{\sigma(w^\top x + b) : w \in \mathbb{R}^d,\ b \in \mathbb{R}\} \subset C(K)$.
The classical universal approximation theorem states that $\overline{\mathcal{A}_\sigma}^{\|\cdot\|_\infty} = C(K)$ for suitable non-polynomial σ.
Since $h \in C(K)$ and $\varepsilon > 0$, there is $f_M \in \mathcal{A}_\sigma$ with $\|f_M - h\|_\infty < \varepsilon$.
Thus $f_M(x) = \sum_{m=1}^{M} a_m \sigma(w_m^\top x + b_m)$ has one hidden layer and linear output.
Structural Commentary: Non-polynomial activations make ridge spans dense in $C(K)$.
Key Insight: Width supplies finite linear combinations of nonlinear ridge functions. □

Remark 5.1.8. *Depth and Expressivity*
Let $N_L(\varepsilon, h)$ be the minimum width of a depth-L network with $\|f - h\|_\infty \leq \varepsilon$ on a compact domain. Universal approximation gives $N_L(\varepsilon, h) < \infty$, not trainability; depth expressivity is $N_L(\varepsilon, h) \ll N_{L'}(\varepsilon, h)$ for some $L > L'$.

5.2 Backpropagation and Automatic Differentiation

Backpropagation is reverse-mode differentiation applied to the computational graph of the network and loss.

Definition 5.2.1. *Computational Graph*
A computational graph is a finite directed acyclic graph $G = (V, E)$ together with variables $(x_v)_{v \in V}$ and local maps $(f_v)_{v \in V \setminus S}$, where $\operatorname{pa}(v) = \{u \in V : (u, v) \in E\}$ and $S = \{v \in V : \operatorname{pa}(v) = \varnothing\}$.

(i) For $v \in S$, x_v is an input, parameter, or constant.
(ii) For $v \in V \setminus S$, $x_v = f_v((x_u)_{u \in \operatorname{pa}(v)})$.
(iii) An edge $u \to v$ records the direct dependence of x_v on x_u in f_v.

For scalar output $L = x_\ell$ and parameter sources $P \subseteq S$, reverse mode computes $(\partial L/\partial x_v)_{v \in V}$, hence $\nabla_{(x_p)_{p \in P}} L$.

Method 5.2.2. *Backpropagation*
Let $h^{(0)} = x$, $z^{(k)} = W^{(k)} h^{(k-1)} + b^{(k)}$, $h^{(k)} = \phi_k(z^{(k)})$ for $1 \leq k \leq K$, and $L = \ell(h^{(K)}, y)$.
Write $\bar{u} = \nabla_u L$ for the adjoint of u.

(i) Compute $z^{(k)}$ and $h^{(k)}$ for $k = 1, \dots, K$, then compute L.
(ii) Initialize $\bar{h}^{(K)} = \nabla_{h^{(K)}} \ell(h^{(K)}, y)$.
(iii) For $k = K, \dots, 1$, set $\bar{z}^{(k)} = J_{\phi_k}(z^{(k)})^T \bar{h}^{(k)}$ and $(\nabla_{W^{(k)}} L, \nabla_{b^{(k)}} L, \bar{h}^{(k-1)}) = (\bar{z}^{(k)} (h^{(k-1)})^T, \bar{z}^{(k)}, (W^{(k)})^T \bar{z}^{(k)})$.

Thus backpropagation is reverse-mode differentiation of L through the layerwise computational graph.

Method 5.2.3. *Single-Hidden-Layer Backpropagation Equations*
For one training case (x_i, y_i) in a single-hidden-layer network, let $m = 1, \dots, M$ index hidden units, $k = 1, \dots, K$ index output components, and $z_i = (z_{1i}, \dots, z_{Mi})^T$.

(i) Compute $a_{mi} = \alpha_{0m} + \alpha_m^T x_i$, $z_{mi} = \sigma(a_{mi})$, $t_{ki} = \beta_{0k} + \beta_k^T z_i$, and $f_k(x_i) = g_k(t_{ki})$.
(ii) Compute $R_i = \sum_{k=1}^{K} \{y_{ik} - f_k(x_i)\}^2$ and $\delta_{ki} = \frac{\partial R_i}{\partial t_{ki}} = -2\{y_{ik} - f_k(x_i)\} g_k'(t_{ki})$.
(iii) Compute $\nabla_{\beta_k} R_i = \delta_{ki} z_i$ and $\frac{\partial R_i}{\partial \beta_{0k}} = \delta_{ki}$.
(iv) Compute $s_{mi} = \frac{\partial R_i}{\partial a_{mi}} = \sigma'(a_{mi}) \sum_{k=1}^{K} \beta_{km} \delta_{ki}$.

(v) Compute $\nabla_{\alpha_m} R_i = s_{mi} x_i$ and $\frac{\partial R_i}{\partial \alpha_{0m}} = s_{mi}$.

Batch gradients are obtained by summing these per-case derivatives over i.

Remark 5.2.4. *Automatic Differentiation*
For an implemented graph map $F : \mathbb{R}^p \to \mathbb{R}^q$, automatic differentiation returns $DF(x)$ by chain-rule evaluation on the executed graph. It differentiates the floating-point computation up to roundoff, without finite differences or symbolic simplification.

5.3 Optimization for Deep Learning

The old optimization notes are migrated here only where the algorithms are used as training methods for neural networks. General numerical optimization belongs to Part 18.

Definition 5.3.1. *Neural Empirical Objective*

For data $(x_i, y_i)_{i=1}^n$, network $f_\theta : x \mapsto f(x;\theta)$, loss L, neural empirical objective is $J_n(\theta) = n^{-1} \sum_{i=1}^n L(f_\theta(x_i), y_i)$.

The population objective is $J(\theta) = \mathbb{E}[L(f_\theta(X), Y)]$ under the data-generating law.

Algorithm 8: Minibatch Backpropagation Step

Require: Minibatch B, parameters θ, learning rate η
Compute predictions $f(x;\theta)$ for all $(x, y) \in B$
Compute minibatch loss $\widehat{L}_B(\theta)$
Run reverse-mode automatic differentiation to compute $g = \nabla_\theta \widehat{L}_B(\theta)$
Update parameters $\theta \leftarrow \theta - \eta g$

Remark 5.3.2. *Deep Optimization Challenges*
For empirical loss $J_n(\theta)$, deep optimization is shaped by:

(i) *geometry*: large $\kappa(\nabla^2 J_n)$, saddles with $\nabla J_n \approx 0$ and indefinite $\nabla^2 J_n$, and plateaus with $\|\nabla J_n\| \approx 0$.
(ii) *gradient propagation*: large or small $\left\| \frac{\partial h^{(L)}}{\partial h^{(k)}} \right\|$ causes exploding or vanishing gradients.
(iii) *stochasticity*: $g_k = \nabla J_n(\theta_k) + \xi_k$, with $\mathbb{E}[\xi_k \mid \theta_k] = 0$ and $\mathrm{Var}(\xi_k \mid \theta_k) \neq 0$.
(iv) *stabilization*: initialization, normalization, gradient clipping, skip connections, and learning-rate schedules control scale, conditioning, and update variance.

These choices are part of the training specification, not post hoc implementation details.

Remark 5.3.3. *Second-Order Methods*
Let $J : \mathbb{R}^d \to \mathbb{R}$, $g_k = \nabla J(\theta_k)$, and $H_k = \nabla^2 J(\theta_k)$.

(i) *Newton*: choose p_k from $H_k p_k = -g_k$ and set $\theta_{k+1} = \theta_k + \eta_k p_k$.
(ii) *Conjugate gradient*: solve $H_k p = -g_k$ approximately using only products $H_k v$.
(iii) *BFGS/L-BFGS*: use $p_k = -B_k g_k$, where $B_k \approx H_k^{-1}$; L-BFGS stores few curvature pairs.

Large networks make explicit H_k storage or factorization prohibitive, so these methods usually act as preconditioners, curvature approximations, or analysis tools.

5.4 Regularization, Normalization, and Generalization

Deep-learning regularization combines penalties, stochastic training effects, architecture, and validation-based stopping.

Definition 5.4.1. *Parameter-Norm Regularization*
Minimize $\widetilde{J}_\lambda(\theta) = J(\theta) + \lambda \Omega(\theta)$, with $\lambda \geq 0$, $\Omega \geq 0$, and penalized non-intercept set $\mathcal{P}$.

(i) *ℓ^2 weight decay*: $\Omega_2(\theta) = \sum_{j \in \mathcal{P}} \theta_j^2$.
(ii) *ℓ^1 penalty*: $\Omega_1(\theta) = \sum_{j \in \mathcal{P}} |\theta_j|$.
(iii) *Single-hidden-layer weight decay*: for weights $\alpha_{m\ell}$ and β_{km}, $\Omega_2(\theta) = \sum_{m,\ell} \alpha_{m\ell}^2 + \sum_{k,m} \beta_{km}^2$.

Intercepts such as α_{0m} and β_{0k} are excluded unless explicitly included in $\mathcal{P}$.

Definition 5.4.2. *Dropout*
Let $h \in \mathbb{R}^d$, $q \in (0, 1]$, and $m_j \overset{\text{ind}}{\sim} \text{Bernoulli}(q)$.

(i) *Training mask*: replace h by $\widetilde{h} = \frac{m \odot h}{q}$.
(ii) *Subnetwork*: each mask $m \in \{0, 1\}^d$ selects the active coordinates $\{j : m_j = 1\}$ with shared parameters.
(iii) *Evaluation*: use the unmasked activation h when inverted dropout scaling is used in training.

Dropout optimizes an expected masked loss over tied-parameter subnetworks.

Method 5.4.3. *Early Stopping*
Let θ_t be the parameter after training step t and let $V(\theta)$ be the validation loss.

(i) Compute $V_t = V(\theta_t)$ after selected training steps t.
(ii) Select $t^\star \in \arg\min_{0 \leq t \leq T} V_t$.
(iii) Return $\widehat{\theta}_{\text{ES}} = \theta_{t^\star}$.

The stopping time $t^\star$ restricts the reachable parameter set and acts as an implicit regularization parameter.

Remark 5.4.4. *Starting Values, Scaling, and Multiple Fits*
For a nonconvex neural objective $J(\theta)$, the fitted parameter can depend on initialization and scaling.

(i) *Initialization*: avoid $\theta_0 = 0$ when symmetry identifies hidden units, and avoid large $\|\theta_0\|$ when $\sigma'(a) \approx 0$.
(ii) *Scaling*: standardize inputs, e.g. $x_j \leftarrow \frac{x_j - \hat{\mu}_j}{\hat{\sigma}_j}$, so coordinates have comparable optimization scale.
(iii) *Multiple fits*: run starts $r = 1, \ldots, R$, obtain $\widehat{\theta}^{(r)}$, and choose $\widehat{\theta}^{(r^\star)}$ by validation or penalized criterion.

Thus initialization law, preprocessing, and restart selection are part of the training specification.

Remark 5.4.5. *Hidden Units and Weight Decay*
Let M be hidden width, λ the weight-decay strength, and $t^\star$ the early-stopping time.

(i) *Small M*: the model class may have excessive approximation error.
(ii) *Large M*: overfitting can occur unless $\lambda > 0$, finite $t^\star$, or another constraint limits effective complexity.
(iii) *Selection*: choose $(M, \lambda, t^\star)$ by validation or another held-out criterion.

Weight decay and early stopping bias the fit toward smaller effective weights and simpler functions.

Definition 5.4.6. *Batch Normalization*
Let $H = (h_{ij}) \in \mathbb{R}^{b\times d}$ be minibatch activations, indexed by example i and coordinate j.

(i) *Batch statistics*: $\mu_j = \frac{1}{b}\sum_i h_{ij}$ and $s_j^2 = \frac{1}{b}\sum_i (h_{ij}-\mu_j)^2$.
(ii) *Normalization*: $\widehat{h}_{ij} = \frac{h_{ij}-\mu_j}{\sqrt{s_j^2+\varepsilon}}$.
(iii) *Affine output*: $y_{ij} = \gamma_j \widehat{h}_{ij} + \beta_j$, with learned γ_j, β_j.
(iv) *Evaluation*: replace μ_j and s_j^2 by running training estimates.

Remark 5.4.7. *Augmentation and Noise*
Let $\ell(y, f_\theta(x))$ be the task loss.

(i) *Data augmentation*: replace (X, Y) by $(T(X), Y)$ for random label-preserving transforms T.
(ii) *Label smoothing*: replace one-hot y by $y^{(\varepsilon)} = (1-\varepsilon)y + \varepsilon u$, where u is a reference class distribution.
(iii) *Noise injection*: train on $x + \xi$ or hidden activations $h + \xi$, with $\mathbb{E}\xi = 0$.
(iv) *Multitask learning*: minimize $\sum_{r=1}^{R} \omega_r \ell_r(y_r, f_{\theta,r}(x))$ with shared representation parameters.

These methods regularize through altered data, targets, perturbations, or shared representations.

6 Modern Deep Models

6.1 Convolutional and Geometric Architectures

The convolutional-network source is migrated here because convolution is an architectural prior for data with grid, group, or geometric structure.

Definition 6.1.1. *Convolution*
For $f, g : \mathbb{R} \to \mathbb{R}$ and $a, b : \mathbb{Z} \to \mathbb{R}$:

(i) *Continuous*: $f * g : D_c \ni t \mapsto \int_{-\infty}^{\infty} f(\tau) g(t-\tau)\, d\tau \in \mathbb{R}$; $D_c := \{t \in \mathbb{R} : \int_{\mathbb{R}} |f(\tau)g(t-\tau)|\, d\tau < \infty\}$.
(ii) *Discrete*: $a * b : D_d \ni n \mapsto \sum_{m\in\mathbb{Z}} a[m]b[n-m] \in \mathbb{R}$; $D_d := \{n \in \mathbb{Z} : \sum_{m\in\mathbb{Z}} |a[m]b[n-m]| < \infty\}$.

Definition 6.1.2. *Convolutional Neural Network*
Let $\Phi = L_k \circ \cdots \circ L_1$ be a neural network on array-valued activations.

(i) *Convolutional layer*: on $z : \mathbb{Z}^d \to \mathbb{R}^{C_{\text{in}}}$, $L_j z = \sigma(K * z + b)$ or $\sigma(K \star z + b)$, with finite-support K, $b \in \mathbb{R}^{C_{\text{out}}}$, pointwise σ, and cross-correlation $\star$.
(ii) *Convolutional neural network*: Φ is a CNN if at least one layer L_j is convolutional.
(iii) *Library convention*: many neural-network libraries implement $\star$ while naming the operation convolution.

Definition 6.1.3. *Sparse Interactions, Sharing, and Equivariance*
For $x : \mathbb{Z}^d \to \mathbb{R}^{C_{\text{in}}}$, finite $W \subset \mathbb{Z}^d$, kernels $K_w \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$, $b \in \mathbb{R}^{C_{\text{out}}}$, and pointwise σ,
set $Lx(u) := \sigma(\sum_{w\in W} K_w x(u-w) + b)$ and $(\tau_a x)(u) := x(u-a)$.

(i) *Sparse interaction*: $Lx(u)$ depends only on $\{x(u-w) : w \in W\}$.
(ii) *Parameter sharing*: the same $(K_w)_{w\in W}$ and b define $Lx(u)$ for every u.
(iii) *Translation equivariance*: $L(\tau_a x) = \tau_a(Lx)$ for every $a \in \mathbb{Z}^d$ where defined.

Pooling with finite U applies ρ to $\{Lx(u+w) : w \in U\}$; shift-stable ρ gives local approximate invariance.

Definition 6.1.4. *Pooling, Padding, and Stride*
Let $x : \Omega \to \mathbb{R}^C$ on finite $\Omega \subset \mathbb{Z}^d$, with $W \subset \mathbb{Z}^d$, $p \in \mathbb{N}_0^d$, $s \in \mathbb{N}^d$, offset r, statistic $\rho : (\mathbb{R}^C)^W \to \mathbb{R}^{C'}$,
and $\Omega^{(p)} := \Omega + \prod_i \{-p_i, \dots, p_i\}$.

(i) *Padding*: an extension $\tilde{x} : \Omega^{(p)} \to \mathbb{R}^C$ with $\tilde{x}|_\Omega = x$ and a boundary rule on $\Omega^{(p)} \setminus \Omega$.
(ii) *Stride*: evaluation sites $\Omega_{s,r} := \{u \in \Omega^{(p)} : \forall i,\ u_i \equiv r_i \pmod{s_i},\ u + W \subset \Omega^{(p)}\}$.
(iii) *Pooling*: $P_{\rho,W,s,r} x : \Omega_{s,r} \ni u \mapsto \rho((\tilde{x}(u+w))_{w\in W}) \in \mathbb{R}^{C'}$.

Remark 6.1.5. *Geometric Extensions*
Let X be a domain with local neighborhoods $\mathcal{N}(x) \subset X$ and symmetry group G.

(i) *Graph neural networks*: on $\mathcal{G} = (V, E)$, $h_v^{(\ell+1)} = \phi(h_v^{(\ell)}, \text{Agg}_{u\in\mathcal{N}(v)}\, m_{uv}^{(\ell)})$ with invariant Agg.
(ii) *Equivariant networks*: Φ is G-equivariant if $\Phi(\rho_{\text{in}}(g)x) = \rho_{\text{out}}(g)\Phi(x)$ for every $g \in G$.
(iii) *Manifold architectures*: for a manifold M, layers use chart, tangent, or geodesic neighborhoods $\mathcal{N}(p) \subset M$.

Each variant builds locality or symmetry into the model class.

6.2 Sequence Models and Attention

Sequence models process ordered data where context and position matter.

Definition 6.2.1. *Recurrent Neural Network*
For dim d_x, d_h, d_y, parameters $\Theta = (W_x, W_h, b, \phi, \psi, h_0)$, $T \geq 1$, an RNN is a map $\Phi_T : (\mathbb{R}^{d_x})^T \to (\mathbb{R}^{d_y})^T$:

(i) *State update*: $h_t = \phi(W_x x_t + W_h h_{t-1} + b)$ for $t = 1, \ldots, T$, with shared (W_x, W_h, b).
(ii) *Readout*: $y_t = \psi(h_t)$ and $\Phi_T(x_{1:T}) = (y_1, \ldots, y_T)$.

Remark 6.2.2. *Long-Term Dependencies*
For $h_t = F_t(h_{t-1}, x_t)$, set $J_t := \partial h_t / \partial h_{t-1}$ and $P_{s:T} := \prod_{t=s+1}^{T} J_t$.
Then $\nabla_{h_s} \mathcal{L} = P_{s:T}^{\top} \nabla_{h_T} \mathcal{L}$; $\|P_{s:T}\| \ll 1$ vanishes gradients and $\|P_{s:T}\| \gg 1$ explodes them.

(i) *Gating*: gates $g_t \in [0,1]^{d_h}$ interpolate between copy and update.
(ii) *Gradient clipping*: replace a gradient g by $\min\{1, c/\|g\|\}g$, so the clipped norm is at most c.
(iii) *Skip paths*: use additive routes such as $h_t = h_{t-1} + F_t(h_{t-1}, x_t)$, giving transition Jacobian $I + \partial F_t / \partial h_{t-1}$.

These mechanisms keep the long-horizon Jacobian product numerically controlled.

Definition 6.2.3. *Attention*
For $Q \in \mathbb{R}^{n_q \times d}$, $K \in \mathbb{R}^{n_k \times d}$, $V \in \mathbb{R}^{n_k \times d_v}$, define $\mathrm{Attn}(Q, K, V) \in \mathbb{R}^{n_q \times d_v}$ row-wise:

(i) *Scores*: $S := d^{-1/2} Q K^{\top} \in \mathbb{R}^{n_q \times n_k}$.
(ii) *Weights*: $A := \mathrm{softmax}_{\mathrm{row}}(S)$, $A_{ij} := \exp(S_{ij}) / \sum_{\ell=1}^{n_k} \exp(S_{i\ell})$.
(iii) *Output*: $\mathrm{Attn}(Q, K, V) := AV$, equivalently $(\mathrm{Attn}(Q, K, V))_{i\cdot} = \sum_{j=1}^{n_k} A_{ij} V_{j\cdot}$.

Remark 6.2.4. *Attention as Routing*
For token matrix X, set $Q = XW_q$, $K = XW_k$, $V = XW_v$, $S = d^{-1/2} Q K^{\top}$, and $A = \mathrm{softmax}_{\mathrm{row}}(S)$.

(i) *Content routing*: $y_i = \sum_j A_{ij} v_j$ routes value vectors by query-key compatibility.
(ii) *Masking*: a mask $M_{ij} \in \{0, -\infty\}$ replaces S by $S + M$, forcing $A_{ij} = 0$ for forbidden interactions.
(iii) *Positional encoding*: absolute codes use $x_i + e_i$; relative codes add a position term to S_{ij}.

Thus attention replaces a fixed recurrence neighborhood by data-dependent communication.

6.3 Transformers and Large-Scale Training

Transformers combine attention, feedforward blocks, normalization, and residual connections into architectures that scale well on sequence and multimodal data.

Definition 6.3.1. *Self-Attention*
For $X \in \mathbb{R}^{T \times d}$, head dimension d_h, and projection matrices W_Q, W_K, W_V, W_O:

(i) *Self-attention*: $Q = XW_Q$, $K = XW_K$, $V = XW_V$, and $\mathrm{SA}(X) := \mathrm{Attn}(Q, K, V)$.
(ii) *Multi-head attention*: with heads $H_r := \mathrm{Attn}(XW_Q^{(r)}, XW_K^{(r)}, XW_V^{(r)})$, set $\mathrm{MHA}(X) := [H_1 | \cdots | H_R] W_O$.

Definition 6.3.2. *Transformer Block*
For $X \in \mathbb{R}^{T \times d}$, let MHA be multi-head attention, FFN positionwise feedforward, and $\mathrm{LN}_1, \mathrm{LN}_2$ layer normalizations. A pre-norm block $B : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times d}$ is:

(i) *Attention sublayer*: $Z := X + \mathrm{MHA}(\mathrm{LN}_1 X)$.
(ii) *Feedforward sublayer*: $B(X) := Z + \mathrm{FFN}(\mathrm{LN}_2 Z)$.

Stacking blocks gives $B_L \circ \cdots \circ B_1(X)$.

Remark 6.3.3. *Large-Scale Training*
For data $\mathcal{D} = \{z_i\}_{i=1}^N$, loss $\ell_i(\theta)$, mini-batch B_t, and gradient estimate $\hat{g}_t := |B_t|^{-1} \sum_{i \in B_t} \nabla_\theta \ell_i(\theta_t)$:

(i) *Optimization*: choose update $\theta_{t+1} = U_t(\theta_t, \hat{g}_t)$, batch size $|B_t|$, and schedule η_t.
(ii) *Data*: sampling, filtering, weighting, and augmentation define the empirical training law $\widehat{P}_{\mathcal{D}}$.
(iii) *Numerics*: mixed precision uses low-precision arithmetic with scaling or high-precision accumulators.
(iv) *Parallelism*: workers compute local gradients $g_t^{(p)}$ and synchronize $\bar{g}_t = P^{-1} \sum_{p=1}^{P} g_t^{(p)}$.
(v) *Memory*: checkpointing stores selected activations and recomputes the rest during backpropagation.
(vi) *Evaluation*: validation loss and task metrics estimate out-of-sample performance.

Remark 6.3.4. *Inductive Bias*
Let $I = \{1, \ldots, T\}$ index tokens, $W_i \subset I$ be a local window, and $A_{ij}(x)$ be attention weights.

(i) *Locality*: convolution uses $y_i = F((x_j)_{j \in W_i})$.
(ii) *Global interaction*: attention uses $y_i = \sum_{j \in I} A_{ij}(x) V x_j$.
(iii) *Domain-specific bias*: impose masks, recurrence, sparsity, or $\Phi(\rho_{\mathrm{in}}(g)x) = \rho_{\mathrm{out}}(g)\Phi(x)$.

Useful bias matches the task's locality, symmetry, and dependency scale.

6.4 Generative Models

Generative modeling learns distributions, simulators, or conditional samplers rather than only point predictions.

Definition 6.4.1. *Latent-Variable Generative Model*
Let $Z \in \mathcal{Z}$, $X \in \mathcal{X}$, prior $p(z)$, decoder $p_\theta(x \mid z)$, and data law P_{data}.

(i) *Joint law*: $p_\theta(x, z) := p_\theta(x \mid z) p(z)$.
(ii) *Marginal law*: $p_\theta(x) := \int_{\mathcal{Z}} p_\theta(x \mid z) p(z)\, dz$.
(iii) *Learning*: choose θ so p_θ approximates P_{data}, e.g. by maximizing $\sum_{i=1}^n \log p_\theta(x_i)$.

Definition 6.4.2. *Variational Autoencoder*
For prior $p(z)$, decoder $p_\theta(x \mid z)$, encoder $q_\phi(z \mid x)$, and data $\{x_i\}_{i=1}^n$:

(i) *Generative model*: $p_\theta(x, z) := p_\theta(x \mid z)p(z)$.
(ii) *Variational posterior*: $q_\phi(z \mid x)$ approximates $p_\theta(z \mid x)$.
(iii) *ELBO*: $\mathcal{L}(\theta, \phi; x) := \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi(z \mid x) \| p(z))$.
(iv) *Training*: maximize $\sum_{i=1}^n \mathcal{L}(\theta, \phi; x_i)$ over (θ, ϕ).

Definition 6.4.3. *Generative Adversarial Network*
Let $Z \sim p_Z$, data law P_{data}, generator $G_\theta : \mathcal{Z} \to \mathcal{X}$, and discriminator $D_\psi : \mathcal{X} \to (0, 1)$.

(i) *Generator*: G_θ induces $P_\theta := (G_\theta)_{\#} p_Z$ on $\mathcal{X}$.
(ii) *Discriminator*: $D_\psi(x)$ estimates the probability that $x \sim P_{\text{data}}$ rather than $x \sim P_\theta$.
(iii) *Adversarial game*: $\min_\theta \max_\psi \mathbb{E}_{x \sim P_{\text{data}}} \log D_\psi(x) + \mathbb{E}_{z \sim p_Z} \log(1 - D_\psi(G_\theta(z)))$.

Definition 6.4.4. *Diffusion Model*
Let $X_0 \sim P_{\text{data}}$ and $X_0, \ldots, X_T$ be a Markov chain on $\mathcal{X}$.

(i) *Forward noising*: fixed kernels $q_t(x_t \mid x_{t-1})$ define $q(x_{1:T} \mid x_0) = \prod_{t=1}^T q_t(x_t \mid x_{t-1})$.
(ii) *Reverse model*: learned kernels $p_\theta(x_{t-1} \mid x_t)$ define $p_\theta(x_{0:T}) = p_T(x_T) \prod_{t=1}^T p_\theta(x_{t-1} \mid x_t)$.
(iii) *Training*: fit θ so $p_\theta(x_{t-1} \mid x_t)$ approximates the reverse conditionals of q.
(iv) *Sampling*: draw $X_T \sim p_T$ and iterate $X_{t-1} \sim p_\theta(\cdot \mid X_t)$ for $t = T, \ldots, 1$.

6.5 Reinforcement Learning and Decision Systems

Reinforcement learning studies agents whose actions affect future observations and rewards.

Definition 6.5.1. *Markov Decision Process*
An MDP is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with:

(i) *State-action spaces*: states $s \in \mathcal{S}$ and actions $a \in \mathcal{A}$.
(ii) *Transition law*: $P(\cdot \mid s, a) \in \mathcal{P}(\mathcal{S})$ and $S_{t+1} \sim P(\cdot \mid S_t, A_t)$.
(iii) *Reward and discount*: $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ and $\gamma \in [0, 1)$.
(iv) *Policy*: $\pi(\cdot \mid s) \in \mathcal{P}(\mathcal{A})$ and $A_t \sim \pi(\cdot \mid S_t)$.

Definition 6.5.2. *Value Functions*
For policy π, define the discounted return $G_t := \sum_{k=0}^{\infty} \gamma^k r(S_{t+k}, A_{t+k})$.

(i) *State value*: $V^\pi(s) := \mathbb{E}_\pi[G_0 \mid S_0 = s]$.
(ii) *Action value*: $Q^\pi(s, a) := \mathbb{E}_\pi[G_0 \mid S_0 = s, A_0 = a]$.

Definition 6.5.3. *Bellman Equation*
For an MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ and value function $V : \mathcal{S} \to \mathbb{R}$:

(i) *Policy Bellman equation*: $V^\pi(s) = \sum_a \pi(a \mid s)(r(s, a) + \gamma \sum_{s'} P(s' \mid s, a)V^\pi(s'))$.
(ii) *Optimal Bellman equation*: $V^\star(s) = \max_a (r(s, a) + \gamma \sum_{s'} P(s' \mid s, a)V^\star(s'))$.

Remark 6.5.4. *RL as a Learning System*
Let $\tau = (S_0, A_0, R_0, \ldots)$ denote a trajectory and P^π its law under policy π.

(i) *Prediction*: estimate V^π, Q^π, or transition/reward models from trajectories.
(ii) *Feedback*: policy changes alter the data law, $\pi \mapsto P^\pi$.
(iii) *Optimization*: choose π to maximize $J(\pi) := \mathbb{E}_{P^\pi} \sum_{t \geq 0} \gamma^t R_t$.
(iv) *Exploration*: sample actions to reduce uncertainty about rewards, transitions, or values.
(v) *Evaluation*: validate under sequential dependence and policy-induced distribution shift.

7 Learning Systems and Evaluation

7.1 Data Splitting, Validation, and Benchmarking

Evaluation estimates how the trained system behaves on data not used to fit it.

Definition 7.1.1. *Training, Validation, and Test Splits*
Let $\mathcal{D} = \mathcal{D}_{\text{tr}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{te}}$ be a disjoint partition of $\{(x_i, y_i)\}_{i=1}^n$, and fix a loss ℓ.

(i) Training: for each $\lambda \in \Lambda$, fit $\widehat{f}_\lambda$ using only $\mathcal{D}_{\text{tr}}$.
(ii) Validation: set $\widehat{\lambda} \in \arg\min_{\lambda \in \Lambda} \widehat{R}_{\text{val}}(\lambda)$, where $\widehat{R}_{\text{val}}(\lambda) = |\mathcal{D}_{\text{val}}|^{-1} \sum_{(x,y) \in \mathcal{D}_{\text{val}}} \ell(\widehat{f}_\lambda(x), y)$.
(iii) Test: after $\widehat{\lambda}$ is fixed, report $\widehat{R}_{\text{te}} = |\mathcal{D}_{\text{te}}|^{-1} \sum_{(x,y) \in \mathcal{D}_{\text{te}}} \ell(\widehat{f}_{\widehat{\lambda}}(x), y)$ as the final performance estimate.

If $\mathcal{D}_{\text{val}}$ is unavailable, cross-validation or bootstrap resampling approximates validation.
Any pre-reporting use of $\mathcal{D}_{\text{te}}$ makes $\widehat{R}_{\text{te}}$ adaptive and typically optimistic.

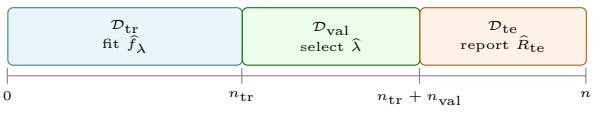

Figure 259: Train/validation/test split: $\mathcal{D} = \mathcal{D}_{\text{tr}} \sqcup \mathcal{D}_{\text{val}} \sqcup \mathcal{D}_{\text{te}}$, with $\mathcal{D}_{\text{tr}}$ fitting $\widehat{f}_\lambda$, $\mathcal{D}_{\text{val}}$ selecting $\widehat{\lambda}$, and $\mathcal{D}_{\text{te}}$ estimating final risk $\widehat{R}_{\text{te}}$.

Definition 7.1.2. *Model Selection and Model Assessment*
Let $\{\widehat{f}_\lambda : \lambda \in \Lambda\}$ be candidates, let $\widehat{R}_{\mathrm{val}}(\lambda)$ be validation error, and let $R(f) = \mathbb{E}\ell(f(X), Y)$.

(i) *Model selection*: choose $\widehat{\lambda} \in \Lambda$, typically $\widehat{\lambda} \in \arg\min_{\lambda\in\Lambda} \widehat{R}_{\mathrm{val}}(\lambda)$, and output $\widehat{f}_{\widehat{\lambda}}$.
(ii) *Model assessment*: after $\widehat{\lambda}$ is fixed, estimate $R(\widehat{f}_{\widehat{\lambda}})$ using data independent of training and selection.

Reusing selection data for assessment yields adaptive, typically optimistic error.

Definition 7.1.3. *Conditional and Expected Test Error*
Let $\widehat{f}_{\mathcal{T}}$ be trained on random $\mathcal{T}$, and let (X^0, Y^0) be an independent test point.

(i) *Conditional test error*: $\mathrm{Err}_{\mathcal{T}} = \mathbb{E}_{X^0,Y^0}[\ell(\widehat{f}_{\mathcal{T}}(X^0), Y^0) \mid \mathcal{T}]$.
(ii) *Expected test error*: $\mathrm{Err} = \mathbb{E}_{\mathcal{T}} \mathrm{Err}_{\mathcal{T}}$.

Reusable error estimators usually target Err, while $\mathrm{Err}_{\mathcal{T}}$ is the error conditional on the realized training set.

Method 7.1.4. *K-Fold Cross-Validation*
Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, loss ℓ, candidates Λ, and fold map $\kappa : \{1, \dots, n\} \to \{1, \dots, K\}$ be fixed.

(i) For each $\lambda \in \Lambda$ and $k = 1, \dots, K$, fit $\widehat{f}_\lambda^{-k}$ using only observations with $\kappa(i) \neq k$.
(ii) For each $\lambda \in \Lambda$, compute $\mathrm{CV}(\lambda) = n^{-1} \sum_{i=1}^n \ell(\widehat{f}_\lambda^{-\kappa(i)}(x_i), y_i)$.
(iii) Choose $\widehat{\lambda} \in \arg\min_{\lambda\in\Lambda} \mathrm{CV}(\lambda)$, or the one-standard-error candidate when that rule is specified.
(iv) Refit $\widehat{f}_{\widehat{\lambda}}$ on all of $\mathcal{D}$.

Leave-one-out has $K = n$; $K = 5$ or 10 often lowers variance and computation but fits on smaller training folds.

Remark 7.1.5. *What Cross-Validation Estimates*
For learning rule $\mathcal{A}$, let $\mathrm{Err}_{\mathcal{A}}(m) = \mathbb{E}_{\mathcal{T}_m,X^0,Y^0}\ell(\mathcal{A}(\mathcal{T}_m)(X^0), Y^0)$. K-fold cross-validation targets roughly $\mathrm{Err}_{\mathcal{A}}(n(1 - 1/K))$, not the conditional error of $\mathcal{A}(\mathcal{D})$. Leave-one-out has $m = n-1$ but dependent losses; fixed $K = 5$ or 10 lowers computation and often variance, with possible upward bias.

Definition 7.1.6. *Benchmark*
A benchmark is a fixed evaluation protocol $\mathcal{B} = (\mathcal{D}, \Pi, \mathcal{S}, m, \mathcal{R})$.

(i) *Data*: $\mathcal{D}$ specifies the population, sample, or data source.
(ii) *Preprocessing*: Π specifies transformations, filtering, feature construction, and allowed fitting scope.
(iii) *Splitting*: $\mathcal{S}$ specifies train-validation-test splits or resampling schemes.
(iv) *Metric*: m specifies the reported score.
(v) *Resources*: $\mathcal{R}$ specifies allowed external data, models, compute, and manual intervention.

Two reported scores are directly comparable only when computed under the same $\mathcal{B}$.

Remark 7.1.7. *Metric Choice*
Choose $m(f; \mathcal{D}_{\mathrm{te}})$ so its ordering of candidate rules matches the deployment objective $U(f)$.
Accuracy, log loss, calibration, ranking quality, cost, and latency need not agree.

Remark 7.1.8. *Data Leakage*
For protected evaluation data E, leakage is dependence of any pre-evaluation choice Z_{pre} on labels or covariates in E. Then error measured on E is adaptive, not honest out-of-sample error.

Method 7.1.9. *Cross-Validation with Screening or Preprocessing*
For a fold map $\kappa : \{1, \dots, n\} \to \{1, \dots, K\}$, set $\mathcal{D}_k = \{(x_i, y_i) : \kappa(i) = k\}$ and $\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k$.
Data-dependent pipeline choices must be fold-local:

(i) For each k, learn preprocessing, screening, feature selection, and tuning from $\mathcal{D}_{-k}$ only.
(ii) Fit $\widehat{f}^{-k}$ on $\mathcal{D}_{-k}$ after applying exactly those fold-local choices.
(iii) Predict and score $\mathcal{D}_k$ using $\widehat{f}^{-k}$ and the same fold-local choices.
(iv) Average the held-out losses over $k = 1, \dots, K$.

Full-data supervised choices leak held-out labels into $\widehat{f}^{-k}$ and bias cross-validation downward.
The protocol should state whether unsupervised preprocessing is fold-local or full-covariate.

Method 7.1.10. *Bootstrap Estimates of Prediction Error*
Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$; draw bootstrap samples S^{*b} and fit $\widehat{f}^{*b}$; set $C^{-i} = \{b : (x_i, y_i) \notin S^{*b}\}$, $I_{\mathrm{oob}} = \{i : C^{-i} \neq \varnothing\}$, and $\overline{\mathrm{err}} = n^{-1} \sum_i \ell(\widehat{f}(x_i), y_i)$.

(i) *Naive bootstrap*: $\widehat{\mathrm{Err}}_{\mathrm{boot}} = B^{-1} \sum_{b=1}^B n^{-1} \sum_{i=1}^n \ell(\widehat{f}^{*b}(x_i), y_i)$.
(ii) *Out-of-bag bootstrap*: $\widehat{\mathrm{Err}}^{(1)} = |I_{\mathrm{oob}}|^{-1} \sum_{i\in I_{\mathrm{oob}}} |C^{-i}|^{-1} \sum_{b\in C^{-i}} \ell(\widehat{f}^{*b}(x_i), y_i)$.
(iii) *.632 bootstrap*: $\widehat{\mathrm{Err}}^{(.632)} = .368\,\overline{\mathrm{err}} + .632\,\widehat{\mathrm{Err}}^{(1)}$.
(iv) *.632+ bootstrap*: $\widehat{\mathrm{Err}}^{(.632+)} = (1 - w_+)\overline{\mathrm{err}} + w_+\widehat{\mathrm{Err}}^{(1)}$, with $w_+ \in [.632, 1]$ increase with overfitting.

Naive reuse is most optimistic; out-of-bag and .632 variants reduce reuse bias.

Remark 7.1.11. *Limits of Reuse-Based Assessment*
Write $\widehat{R}_{\mathrm{reuse}}(\lambda) = R(\widehat{f}_\lambda) + b_\lambda + \varepsilon_\lambda$. Nearly constant b_λ can preserve candidate comparisons despite biased levels.
After choosing $\widehat{\lambda}$, estimating $R(\widehat{f}_{\widehat{\lambda}})$ requires data independent of all fitting, preprocessing, tuning, and iteration.

7.2 Robustness, Shift, and Distributional Uncertainty

Learning systems are deployed under distributions that may differ from the training law.

Definition 7.2.1. *Distribution Shift*
Let P be training law, Q deployment law on (X, Y). Distribution shift means $Q \neq P$; common cases are:

(i) *Covariate shift*: $Q_X \neq P_X$ and $Q_{Y|X} = P_{Y|X}$.
(ii) *Label shift*: $Q_Y \neq P_Y$ and $Q_{X|Y} = P_{X|Y}$.
(iii) *Concept shift*: $Q_{Y|X} \neq P_{Y|X}$.

Definition 7.2.2. *Adversarial Example*
Let $\mathcal{F}$ be a predictor class, ℓ a loss, $d_f : \mathcal{X} \to \mathcal{D}$ the decision rule induced by $f \in \mathcal{F}$, $\Delta(x)$ the admissible perturbation set at x, and $\mathcal{H}_y \subseteq \mathcal{D} \times \mathcal{D}$ the harmful decision-change relation for label y.

(i) *Adversarial example*: for fixed $f \in \mathcal{F}$ and (x, y), any $x' = x + \delta$ with $\delta \in \Delta(x)$ and $\big(\ell(f(x'), y) > \ell(f(x), y)\big) \vee \big((d_f(x), d_f(x')) \in \mathcal{H}_y\big)$.

(ii) *Adversarial training*: for $S = \{(x_i, y_i)\}_{i=1}^n$, choose $\widehat{f}_{\mathrm{adv}} \in \arg\min_{f \in \mathcal{F}} n^{-1} \sum_{i=1}^n \sup_{\delta \in \Delta(x_i)} \ell(f(x_i + \delta), y_i)$.

(iii) *Neural perturbation form*: minimize $\ell(y, f_\theta(x + \delta^\star))$, where $\delta^\star \in \arg\max_{\|\delta\| \le \rho} \ell(y, f_\theta(x + \delta))$.

Definition 7.2.3. *Distributionally Robust Optimization*
Let $\mathcal{F}$ be a hypothesis class, ℓ a loss, and $\mathcal{U} \subseteq \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ an uncertainty set of deployment laws.

(i) *Robust risk*: for $Q \in \mathcal{U}$ and $f \in \mathcal{F}$, $R_Q(f) = \mathbb{E}_{(X,Y)\sim Q} \ell(f(X), Y)$ and $R_{\mathcal{U}}(f) = \sup_{Q \in \mathcal{U}} R_Q(f)$.

(ii) *DRO solution*: any $\widehat{f}_{\mathrm{DRO}} \in \arg\min_{f \in \mathcal{F}} R_{\mathcal{U}}(f)$.

Remark 7.2.4. *Uncertainty*
Let $\mathcal{T}$ be the training data, P_{tr} the training law, and P_{dep} the deployment law.

(i) *Aleatoric uncertainty*: irreducible variability in $Y \mid X = x$ under the data-generating law.
(ii) *Epistemic uncertainty*: uncertainty in f from finite $\mathcal{T}$, misspecification, or nonidentifiability.
(iii) *Out-of-distribution uncertainty*: uncertainty from $P_{\mathrm{dep}} \neq P_{\mathrm{tr}}$ or inputs outside the training support.

Robust systems calibrate probabilities and abstain when uncertainty exceeds a policy threshold.

7.3 Interpretability, Fairness, and Safety

Evaluation also asks whether a model can be understood, audited, and constrained in the setting where it is used.

Definition 7.3.1. *Interpretability*
Let $\mathcal{F}, \mathcal{C}, \mathcal{E}, \mathcal{A}, \mathcal{B}$ denote predictor, query-context, explanation, audience, and audit-task spaces.

(i) *Interpretability specification*: $\mathcal{I} = (\mathcal{E}, a, b, \Phi)$ with $(a, b) \in \mathcal{A} \times \mathcal{B}$ and $\Phi : \mathcal{F} \times \mathcal{C} \to \mathcal{E}$.
(ii) *Explanation*: for $(f, c) \in \mathcal{F} \times \mathcal{C}$, $e = \Phi(f, c)$.

Typical codomains $\mathcal{E}$ encode attributions, counterfactuals, surrogate rules, or representation summaries.

Definition 7.3.2. *Fairness Criteria*
Let A be a protected attribute, $\widehat{Y}$ a decision, S a score, and Y the outcome.
A group fairness criterion imposes equalities or tolerances across values of A; common examples are:

(i) *Demographic parity*: $\mathbb{P}(\widehat{Y} = 1 \mid A = a)$ is constant in a.
(ii) *Equalized odds*: $\mathbb{P}(\widehat{Y} = 1 \mid Y = y, A = a)$ is constant in a for each y.
(iii) *Group calibration*: $\mathbb{P}(Y = 1 \mid S = s, A = a) = s$ for each relevant s and a.

These criteria need not be mutually compatible; the chosen condition should match the policy goal.

Definition 7.3.3. *Safety Constraint*
An additional requirement $f \in \mathcal{C}$ or $R_{\mathrm{safe}}(f) \le \tau$ beyond average-case loss is a *safety constraint*.
$\mathcal{C}$ encodes abstention, monotonicity, action bounds, review triggers, or worst-case risk limits.

Remark 7.3.4. *Beyond Accuracy*
For a deployed predictor f, evaluation beyond accuracy is a specification $\mathcal{E} = (\mathcal{I}, \mathcal{F}, \mathcal{C})$.

(i) *Interpretability*: $\mathcal{I}$ specifies the explanation object, audience, and audit protocol.
(ii) *Fairness*: $\mathcal{F}$ specifies protected attributes, group metrics, and required parity or error constraints.
(iii) *Safety*: $\mathcal{C}$ specifies forbidden actions, abstention rules, human review triggers, or worst-case risk bounds.

A scalar score is incomplete unless the relevant components of $\mathcal{E}$ are stated.

7.4 Deployment, Monitoring, and Feedback Loops

Deployment turns a fitted model into part of a larger stochastic system.

Definition 7.4.1. *Monitoring*
Given deployed f and stream $Z_t = (X_t, \widehat{Y}_t, Y_t^*)$, monitoring outputs (M_t, a_t) with $a_t \in \{0, 1\}$.

(i) *Distribution*: compare $P_t(X)$ and $P_t(\widehat{Y})$ with validation baselines.
(ii) *Performance*: estimate loss, calibration, and subgroup error when labels Y_t^* arrive.
(iii) *Operations*: track latency, missingness, failures, and data-pipeline changes.

An alarm $a_t = 1$ indicates that validation assumptions or service constraints may no longer hold.

Definition 7.4.2. *Feedback Loop*
A deployed policy $A_t = \pi(f(X_t), H_t)$ is a *feedback loop* if $P(X_{t+1}, Y_{t+1} \mid A_t) \neq P(X_{t+1}, Y_{t+1})$.
Future data are then endogenous to f.

Method 7.4.3. *Model Update Policy*
Given deployed f_t, data $\mathcal{D}_{\le t}$, and diagnostics M_t, a model update policy maps f_t to f_{t+1}:

(i) *Trigger*: retrain only when a scheduled time, alarm M_t, or performance threshold calls for it.
(ii) *Eligibility*: choose the data subset $\mathcal{D}_{\mathrm{upd}} \subseteq \mathcal{D}_{\le t}$ allowed for training.
(iii) *Validation*: accept candidate $\widetilde{f}$ only if it passes the benchmark and safety criteria against f_t.
(iv) *Deployment*: replace f_t by $\widetilde{f}$ under a rollout plan and rollback rule.

The policy determines future training data and evaluation conditions.

Remark 7.4.4. *Human-System Boundary*
For human-supported decisions, the deployed policy is $\pi = (f, I, A, O, M, G)$, not just predictor f.

(i) *Interface*: I determines what predictions, explanations, and uncertainty are shown.

(ii) *Action rule*: A maps model output and human input to decisions.
(iii) *Override*: O specifies when humans may ignore or modify f.
(iv) *Monitoring*: M tracks errors, drift, latency, and misuse.
(v) *Governance*: G assigns review, audit, and accountability.

System evaluation should target π rather than f alone.

Part XXII
Classical Supervised Learning

1 Linear Prediction and Generalized Linear Models

1.1 Linear Predictors and Least Squares

Linear prediction reduces inputs to scores of the form "intercept plus weighted features." Regression reads scores as predictions; classification compares or transforms them. Its appeal is geometric: projections, hyperplanes, and feature maps.

Definition 1.1.1. *Linear Predictor*
For $x \in \mathbb{R}^p$, $w \in \mathbb{R}^p$, and $b \in \mathbb{R}$, the *linear predictor* is $f_{w,b}(x) := w^\top x + b$.
Regression uses $f_{w,b}$ directly; classification transforms it by g. Score level sets are parallel hyperplanes.

Definition 1.1.2. *Least-Squares Linear Regression*
For $X \in \mathbb{R}^{n\times d}$ and $y \in \mathbb{R}^n$, *least-squares linear regression* chooses $\widehat{\beta} \in \operatorname*{argmin}_{\beta\in\mathbb{R}^d} \|y - X\beta\|_2^2$ and $\widehat{y} := X\widehat{\beta}$.

If $X^\top X$ is nonsingular, then $\widehat{\beta} = (X^\top X)^{-1}X^\top y$. Columns of X may be raw variables, transformations, indicators, interactions, or basis functions; linearity is in β.

Geometrically, least squares moves y to the closest point in $\operatorname{col}(X)$.

Method 1.1.3. *Multiple-Output Least Squares*
For $Y \in \mathbb{R}^{n\times K}$ and $B \in \mathbb{R}^{d\times K}$, set $Y = XB + E$.

(i) *Least squares*: If $\operatorname{rank}(X) = d$, then $\widehat{B} = (X^\top X)^{-1}X^\top Y$.
(ii) *Decoupling*: Writing $Y = (y_1, \dots, y_K)$ gives $\widehat{B} = (\widehat{\beta}_1, \dots, \widehat{\beta}_K)$, where each $\widehat{\beta}_k$ is the OLS fit of y_k on X.

Shared responses can be pooled by low-rank B or common-direction shrinkage.

Method 1.1.4. *Canonical Correlation and Reduced-Rank Regression*

(i) *CCA*: Choose (v_j, u_j) with large $\operatorname{corr}(Xv_j, Yu_j)$, orthogonal across j; at most $\min(d, K)$ pairs.
(ii) *Rank constraint*: Fit $Y \approx XB$ with $\operatorname{rank}(B) \le m$, restricting fitted responses to m dimensions.
(iii) *Objective*: $\widehat{B}_m^{\mathrm{rr}} \in \operatorname*{argmin}_{\operatorname{rank}(B)\le m} \sum_i (y_i - B^\top x_i)^\top \widehat{\Sigma}^{-1}(y_i - B^\top x_i)$.
(iv) *Role of m*: $m = K$ gives separate OLS; smaller m pools shared response structure.

Reduced rank assumes the predictable response variation lies in a low-dimensional subspace.

1.2 Linear Classification and Generalized Linear Models

Definition 1.2.1. *Discriminant Functions and Linear Boundaries*
For $\mathcal{G} = \{1, \dots, K\}$ and scores $\delta_k : \mathcal{X} \to \mathbb{R}$, set $\widehat{G}(x) \in \operatorname{argmax}_{k\in\mathcal{G}} \delta_k(x)$. Only score differences matter.

(i) *Pairwise boundary*: The k versus ℓ boundary lies in $\{x : \delta_k(x) = \delta_\ell(x)\}$.
(ii) *Linear boundary*: If $\delta_k - \delta_\ell$ is affine in x for all k, ℓ, then all pairwise decision boundaries are hyperplanes.
(iii) *Binary log-odds*: If $\log \frac{\mathbb{P}(G=1|X=x)}{\mathbb{P}(G=2|X=x)} = \beta_0 + \beta^\top x$, then the boundary is $\{x : \beta_0 + \beta^\top x = 0\}$.

Method 1.2.2. *Indicator-Matrix Regression Classifier*
For classes $g_i \in \{1, \dots, K\}$, set $t_k = e_k \in \mathbb{R}^K$, $Y_i = t_{g_i}$, and $\widetilde{x}_i^\top = (1, x_i^\top)$, coding labels as Euclidean targets.

(i) *Least-squares fit*: Choose $\widehat{B} \in \operatorname{argmin}_B \sum_{i=1}^n \|Y_i - B^\top \widetilde{x}_i\|_2^2$; if $\widetilde{X}^\top\widetilde{X}$ is nonsingular, $\widehat{B} = (\widetilde{X}^\top\widetilde{X})^{-1}\widetilde{X}^\top Y$.
(ii) *Scores*: For a new x, set $\widetilde{x} = (1, x^\top)^\top$ and $\widehat{f}(x) := \widehat{B}^\top\widetilde{x}$.
(iii) *Classification*: Predict $\widehat{G}(x) \in \operatorname{argmax}_k \widehat{f}_k(x)$, equivalently nearest t_k.

With an intercept, $\sum_k \widehat{f}_k(x) = 1$, but $\widehat{f}_k(x)$ may leave $[0, 1]$ and, for $K \ge 3$, *mask* middle classes.

Definition 1.2.3. *Generalized Linear Model*
A *generalized linear model* gives $Y \mid X = x$ an exponential-family law and makes a linked mean linear: $g(\mu(x)) = w^\top x + b$. Logistic regression uses the Bernoulli law and logit link $g(\mu) = \log \frac{\mu}{1-\mu}$.

Thus a GLM separates score, mean scale, and likelihood.

Definition 1.2.4. *Multiclass Logistic Regression*
With class K as baseline, set $\eta_k(x) := \beta_{k0} + \beta_k^\top x = \log \frac{\mathbb{P}(G=k|X=x)}{\mathbb{P}(G=K|X=x)}$ for $k < K$.

(i) *Softmax probabilities*: For $k < K$, $p_k(x;\theta) := \frac{\exp(\eta_k(x))}{1+\sum_{\ell=1}^{K-1}\exp(\eta_\ell(x))}$, and $p_K(x;\theta) := \frac{1}{1+\sum_{\ell=1}^{K-1}\exp(\eta_\ell(x))}$.
(ii) *Likelihood*: Fit θ by maximising $\ell(\theta) := \sum_{i=1}^n \log p_{g_i}(x_i;\theta)$.

The softmax gives $p(x;\theta) \in \Delta^{K-1}$; the reference class changes only the parametrisation.

Method 1.2.5. *Binary Logistic Regression by IRLS*
For $y_i \in \{0, 1\}$ and intercept-augmented rows $x_i^\top$, set $p_i(\beta) := (1 + \exp(-x_i^\top\beta))^{-1}$.

(i) *Derivatives*: With $p = (p_i)$ and $W := \operatorname{diag}(p_i(1-p_i))$, $\nabla\ell(\beta) = X^\top(y - p)$ and $\nabla^2\ell(\beta) = -X^\top WX$.
(ii) *Adjusted response*: At β^{old}, set $z := X\beta^{\mathrm{old}} + W^{-1}(y - p)$.
(iii) *IRLS update*: Set $\beta^{\mathrm{new}} := (X^\top WX)^{-1}X^\top Wz$.

IRLS is Newton's method as weighted least squares.
The log-likelihood is concave when the MLE is finite; step-halving controls Newton overshoot.

Method 1.2.6. *L_1-Regularized Logistic Regression*
With unpenalized intercept, maximize $\ell_\lambda(\beta_0,\beta) := \sum_{i=1}^{n}\{y_i\eta_i - \log(1+\exp(\eta_i))\} - \lambda\sum_{j=1}^{p}|\beta_j|$, where $\eta_i := \beta_0 + x_i^\top\beta$.

(i) *Probabilities*: Set $p_i := (1+\exp(-\eta_i))^{-1}$ and residual $r := y - p$.
(ii) *KKT conditions*: If $\widehat{\beta}_j \neq 0$, then $x_j^\top r = \lambda\,\mathrm{sign}(\widehat{\beta}_j)$; if $\widehat{\beta}_j = 0$, then $\left|x_j^\top r\right| \le \lambda$.

The KKT conditions are thresholded feature–residual correlations.

1.3 Discriminant Analysis and Naive Bayes

Method 1.3.1. *Gaussian Discriminant Analysis*
If $X \mid G = k$ has density f_k and prior π_k, then $\mathbb{P}(G=k \mid X=x) = \frac{f_k(x)\pi_k}{\sum_{\ell=1}^{K} f_\ell(x)\pi_\ell}$.

(i) *Score*: If $X \mid G = k \sim N(\mu_k,\Sigma_k)$, compare $\delta_k(x) := -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^\top\Sigma_k^{-1}(x-\mu_k) + \log\pi_k$.
(ii) *LDA*: If $\Sigma_k = \Sigma$ for all k, equivalent linear scores are $\delta_k^{\mathrm{LDA}}(x) := x^\top\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top\Sigma^{-1}\mu_k + \log\pi_k$.
(iii) *QDA*: If Σ_k varies with k, the quadratic term remains and the rule is quadratic discriminant analysis.

Estimate π_k, μ_k, Σ_k from training data; pooled covariance gives LDA and classwise covariances give QDA.
Shared covariance makes the boundaries linear.

Method 1.3.2. *Regularized Discriminant Analysis*
For $0 \le \alpha \le 1$, set $\widehat{\Sigma}_k(\alpha) := \alpha\widehat{\Sigma}_k + (1-\alpha)\widehat{\Sigma}$.

(i) *Endpoints*: $\alpha = 0$ gives LDA and $\alpha = 1$ gives QDA.
(ii) *Scalar shrinkage*: For $0 \le \gamma \le 1$, replace $\widehat{\Sigma}$ by $\widehat{\Sigma}(\gamma) := \gamma\widehat{\Sigma} + (1-\gamma)\widehat{\sigma}^2 I$.

Shrinkage reduces variance when covariance estimation is unstable.

Definition 1.3.3. *Naive Bayes Classifier*
For $x = (x_1,\dots,x_p)$, assume $f_j(x) = \prod_{k=1}^{p} f_{jk}(x_k)$ for each class j.

(i) *Posterior*: With priors π_j, $p_j(x) := \mathbb{P}(G=j \mid X=x) \propto \pi_j \prod_{k=1}^{p} f_{jk}(x_k)$.
(ii) *Additive log odds*: Relative to class J, $\log\frac{p_\ell(x)}{p_J(x)} = \log\frac{\pi_\ell}{\pi_J} + \sum_{k=1}^{p}\log\frac{f_{\ell k}(x_k)}{f_{Jk}(x_k)}$.

On the log scale, features contribute additively.
Even misspecified independence can reduce variance enough to classify well near boundaries.

Remark 1.3.4. *Least Squares Versus LDA*
For $K = 2$, least-squares regression on any distinct label coding gives $\widehat{\beta} \propto \widehat{\Sigma}^{-1}(\widehat{\mu}_2 - \widehat{\mu}_1)$, the LDA direction.
The cutpoint may differ unless classes are balanced or tuned separately.
For $K > 2$, LDA uses pooled-Mahalanobis centroid distances and avoids masking.

Remark 1.3.5. *Logistic Regression Versus LDA*
Both can yield linear log-odds. Logistic regression models class probabilities directly and leaves P_X free; LDA fits a joint Gaussian mixture for (X,G) and uses Bayes' rule.

Remark 1.3.6. *Whitening and Reduced-Rank LDA*
With common covariance W, LDA compares $W^{-1/2}x$ to whitened centroids, adjusted by $\log\pi_k$.
The whitened centroids span dimension at most $K-1$, so LDA reduces to this centroid subspace.
Fisher directions maximize $\frac{a^\top Ba}{a^\top Wa}$, equivalently solve $Ba = \lambda Wa$, where B is between-class scatter.

Remark 1.3.7. *Feature Maps*
A feature map $\phi : \mathcal{X} \to \mathcal{H}$ gives linear prediction in feature space, $f(x) = w^\top\phi(x) + b$, which is generally nonlinear in x. Kernel methods choose ϕ implicitly; neural networks learn ϕ from data.

2 Regularisation and Model Selection

2.1 Penalized Linear Estimation

Regularisation controls model complexity by penalising unstable parameter directions or by constraining the feasible set.

Definition 2.1.1. *Penalised Linear Estimation*
For empirical loss $L_n(\beta)$, a penalised estimator solves $\widehat{\beta}_\lambda \in \arg\min_\beta\{L_n(\beta) + \lambda\Omega(\beta)\}$.
The choices $\Omega(\beta) = \|\beta\|_2^2$ and $\Omega(\beta) = \|\beta\|_1$ give ridge and lasso penalties.

Definition 2.1.2. *Subset Selection, Ridge, and Lasso*
Let $y \in \mathbb{R}^n$ be centered and let $X \in \mathbb{R}^{n\times p}$ have centered, standardised columns.

(i) *Best subset*: for a size m, solve $\widehat{\beta}_m^{\mathrm{sub}} \in \underset{|\,\mathrm{supp}(\beta)|\le m}{\arg\min} \|y - X\beta\|_2^2$
(ii) *Ridge*: for $\lambda \ge 0$, solve $\widehat{\beta}_\lambda^{\mathrm{ridge}} \in \arg\min_\beta\{\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2\}$
(iii) *Lasso*: for $\lambda \ge 0$, solve $\widehat{\beta}_\lambda^{\mathrm{lasso}} \in \arg\min_\beta\left\{\frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1\right\}$

Centering separates the unpenalised intercept, estimated by the mean on the raw scale.

Remark 2.1.3. *Penalty Geometry*
Assume $X^TX = I$ and set $z_j = x_j^T y$. For subset size m, best subset selection keeps the m largest $|z_j|$.
Ridge and lasso have coordinates $\widehat{\beta}_{\lambda,j}^{\mathrm{ridge}} = \frac{z_j}{1+\lambda}$ and $\widehat{\beta}_{\lambda,j}^{\mathrm{lasso}} = \mathrm{sign}(z_j)(|z_j| - \lambda)_+$.
Thus ℓ^2 shrinkage is smooth, while ℓ^1 shrinkage can set coefficients exactly to zero.

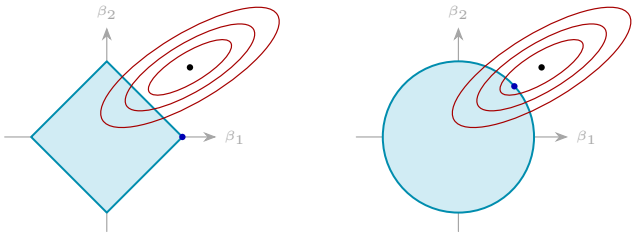

Figure 260: Left: lasso ℓ^1 diamond; right: ridge ℓ^2 disk.

Proposition 2.1.4. *Ridge Shrinkage and Effective Degrees*
Let $X \in \mathbb{R}^{n\times p}$ be centered with rank r and thin SVD $X = UDV^T$, where $D = \operatorname{diag}(d_1, \dots, d_r)$ and $d_j > 0$.
For $\lambda > 0$, set $S_\lambda := X(X^TX + \lambda I_p)^{-1}X^T$ and $s_j(\lambda) := \frac{d_j^2}{d_j^2+\lambda}$. Then $S_\lambda = U \operatorname{diag}(s_1(\lambda), \dots, s_r(\lambda))U^T$.
So $\widehat{y}_\lambda^{\text{ridge}} := X\widehat{\beta}_\lambda^{\text{ridge}} = S_\lambda y = \sum_{j=1}^{r} s_j(\lambda)u_j u_j^T y$ and $\operatorname{df}(\lambda) := \operatorname{tr}(S_\lambda) = \sum_{j=1}^{r} s_j(\lambda)$.

Proof. Since $X^TX = VD^2V^T$ and $X^T = VDU^T$, we have $V^T(X^TX + \lambda I_p)^{-1}V = (D^2 + \lambda I_r)^{-1}$.
Hence $S_\lambda = UDV^T(X^TX + \lambda I_p)^{-1}VDU^T = UD(D^2 + \lambda I_r)^{-1}DU^T = U \operatorname{diag}(s_1(\lambda), \dots, s_r(\lambda))U^T$.
Apply to y and take traces, using $U^TU = I_r$.
Structural Commentary: Ridge is diagonal in the left singular-vector basis of X.
Key Insight: Ridge keeps direction u_j by $s_j(\lambda)$; weak directions count less in the fit and $\operatorname{df}(\lambda)$. □

2.2 Validation and Selection Bias

Definition 2.2.1. *Model Selection by Validation*
Let $\mathcal{M}$ be a candidate set of models or hyperparameters.

(i) A *validation criterion* is $m \mapsto \widehat{R}_{\text{val}}(m)$; choose $\widehat{m} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_{\text{val}}(m)$.

(ii) In *K-fold cross-validation*, take folds $I_1, \dots, I_K$ and fits $\widehat{f}_m^{(-k)}$.
Define $\ell_{ik}(m) := \ell(y_i, \widehat{f}_m^{(-k)}(x_i))$, $\widehat{R}_{\text{cv}}(m) := \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in I_k} \ell_{ik}(m)$, and choose $\widehat{m}_{\text{cv}} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_{\text{cv}}(m)$.

Method 2.2.2. *One-Standard-Error Rule*
Let $(\mathcal{M}, \preceq)$ order candidates by complexity, with $\widehat{R}_{\text{cv}}, \widehat{\text{se}} : \mathcal{M} \to \mathbb{R}$.

(i) Choose $\widehat{m}_{\min} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_{\text{cv}}(m)$ and set $\widehat{R}_{\min} := \widehat{R}_{\text{cv}}(\widehat{m}_{\min})$.
(ii) Define $\mathcal{M}_{1\text{se}} := \{m \in \mathcal{M} : \widehat{R}_{\text{cv}}(m) \le \widehat{R}_{\min} + \widehat{\text{se}}(\widehat{m}_{\min})\}$.
(iii) Select $\widehat{m}_{1\text{se}}$ as the $\preceq$-minimal element of $\mathcal{M}_{1\text{se}}$; use any final test set only after this choice.

Example 2.2.3. *A Selection and Shrinkage Workflow*
Let $D = \{(x_i, y_i)\}_{i=1}^n$, where $y_i = \log(\text{PSA}_i)$ and $x_i = (\text{lcavol}_i, \text{lweight}_i, \text{age}_i, \text{lbph}_i, \text{svi}_i, \text{lcp}_i, \text{gleason}_i, \text{pgg45}_i)$.
Split indices into training and test sets T, E, standardize X_T, and choose each complexity parameter by cross-validation on T: subset size m, ridge/lasso penalty λ, or PCR/PLS dimension M.
The selected fit $\widehat{f}$ is then evaluated once on E by $\widehat{R}_{\text{test}} := |E|^{-1}\sum_{i\in E} \ell(y_i, \widehat{f}(x_i))$.

Remark 2.2.4. *Selection Bias*
If $\widehat{m} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_V(m)$ is chosen on validation set V, then $\widehat{R}_V(\widehat{m})$ is optimistically biased for $R(\widehat{m})$.
Estimate the selected model with an independent set E, using $\widehat{R}_E(\widehat{m})$, or use nested validation.

2.3 Derived Input Directions and Regularization Paths

Method 2.3.1. *Derived Input Directions*
Let X, y be centered. A derived-direction method chooses $Z_M = (z_1, \dots, z_M)$ with $z_m = Xv_m$ or data-adaptive analogues, then fits $\widehat{y}_M = P_{Z_M} y$, where $P_{Z_M} := Z_M(Z_M^TZ_M)^{-1}Z_M^T$; M is selected by validation.

(i) *Principal components regression*: for $X = UDV^T$, take $v_1, \dots, v_M$ as the leading right singular vectors,
set $z_m = Xv_m = d_m u_m$, and fit $\widehat{y}_M^{\text{pcr}} = \sum_{m=1}^{M} z_m \frac{z_m^T y}{z_m^T z_m} = \sum_{m=1}^{M} u_m u_m^T y$.

(ii) *Partial least squares*: set $X^{(0)} = X$ and, for $m = 1, \dots, M$, define $z_m = X^{(m-1)}X^{(m-1)T}y$ and
$X^{(m)} = (I - P_{z_m})X^{(m-1)}$, where $P_{z_m} := \frac{z_m z_m^T}{z_m^T z_m}$. With $Z_M = (z_1, \dots, z_M)$, fit $\widehat{y}_M^{\text{pls}} = P_{Z_M} y$.

Algorithm 9: Partial Least Squares Directions

Require: Centered response y, standardized columns $x_1, \dots, x_p$, number of directions $M \le p$
Ensure: Fitted vectors $\widehat{y}^{(1)}, \dots, \widehat{y}^{(M)}$
Set $\widehat{y}^{(0)} \leftarrow 0$ and $x_j^{(0)} \leftarrow x_j$ for $j = 1, \dots, p$
for $m = 1, \dots, M$ **do**
$z_m \leftarrow \sum_{j=1}^{p} (x_j^{(m-1)})^T y\, x_j^{(m-1)}$
$\widehat{\theta}_m \leftarrow \frac{z_m^T y}{z_m^T z_m}$
$\widehat{y}^{(m)} \leftarrow \widehat{y}^{(m-1)} + \widehat{\theta}_m z_m$

for $j = 1, \dots, p$ **do**
$x_j^{(m)} \leftarrow x_j^{(m-1)} - \frac{z_m^T x_j^{(m-1)}}{z_m^T z_m} z_m$
end for
end for

Remark 2.3.2. *PLS Versus PCR*
PCR chooses v_m from $X^T X v_m = d_m^2 v_m$, so directions are ordered only by predictor variance d_m^2.
PLS chooses $z_m = X^{(m-1)} X^{(m-1)T} y$ and deflates $X^{(m)} = (I - P_{z_m}) X^{(m-1)}$, so each direction depends on both variance in $X^{(m-1)}$ and covariance with y.

Method 2.3.3. *Piecewise-Linear Path Algorithms*
Let $\widehat{\beta}(\lambda) \in \arg\min_\beta \{R(\beta) + \lambda J(\beta)\}$.

(i) If R is quadratic or piecewise quadratic and J is piecewise linear, fixed active-set and sign regions give affine KKT equations in λ, so $\widehat{\beta}(\lambda)$ is piecewise linear.
(ii) *Least angle regression*: with residual $r = y - X\beta$ and active set $\mathcal{A}$, move $\beta_\mathcal{A}$ in a direction keeping $|x_j^T r|$ tied for $j \in \mathcal{A}$ until a new variable ties or the least-squares fit is reached.
(iii) *Lasso path*: follow the same tied-correlation path and remove $j \in \mathcal{A}$ when β_j reaches 0.
(iv) *Coordinate descent*: for fixed λ, update $\beta_j \leftarrow \arg\min_t \{\frac{1}{2} \|y - X_{-j}\beta_{-j} - x_j t\|_2^2 + \lambda(|t| + \|\beta_{-j}\|_1)\}$ and warm-start along $\lambda_1 > \cdots > \lambda_L$.

Method 2.3.4. *LAR, Lasso, and Correlation Ties*
Let $\widehat{\beta}_\lambda \in \arg\min_\beta \{\frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1\}$, $r_\lambda := y - X\widehat{\beta}_\lambda$, and $\mathcal{A}_\lambda := \{j : \widehat{\beta}_{\lambda,j} \neq 0\}$.

(i) *KKT*: $x_j^T r_\lambda = \lambda \operatorname{sign}(\widehat{\beta}_{\lambda,j})$ for $j \in \mathcal{A}_\lambda$, and $|x_j^T r_\lambda| \le \lambda$ for $j \notin \mathcal{A}_\lambda$.
(ii) *Correlation ties*: active predictors satisfy $|x_j^T r_\lambda| = \lambda$; inactive predictors satisfy $|x_j^T r_\lambda| \le \lambda$.
(iii) *LAR*: with $C = \|X^T r\|_\infty$, move in the equiangular active direction preserving $|x_j^T r| = C$ for $j \in \mathcal{A}$ until some $k \notin \mathcal{A}$ reaches $|x_k^T r| = C$.
(iv) *Lasso path*: on a fixed active set and sign pattern, the KKT equations are affine in λ. When some $\widehat{\beta}_{\lambda,j}$ reaches 0, remove j and recompute the active direction.

Algorithm 10: Least Angle Regression Path

Require: Centered response y and standardized columns $x_1, \dots, x_p$ with $\|x_j\|_2 = 1$
Ensure: Piecewise-linear coefficient path $\{\beta(t)\}$
Initialize $\beta \leftarrow 0$, $r \leftarrow y$, $c \leftarrow X^T r$, $C \leftarrow \|c\|_\infty$, and $\mathcal{A} \leftarrow \{j : |c_j| = C\}$
repeat
$s_\mathcal{A} \leftarrow \operatorname{sign}(c_\mathcal{A})$, $\widetilde{X}_\mathcal{A} \leftarrow X_\mathcal{A} \operatorname{diag}(s_\mathcal{A})$, and $G_\mathcal{A} \leftarrow \widetilde{X}_\mathcal{A}^T \widetilde{X}_\mathcal{A}$
$\alpha_\mathcal{A} \leftarrow (\mathbf{1}^T G_\mathcal{A}^{-1} \mathbf{1})^{-1/2}$, $w_\mathcal{A} \leftarrow \alpha_\mathcal{A} G_\mathcal{A}^{-1} \mathbf{1}$, $u_\mathcal{A} \leftarrow \widetilde{X}_\mathcal{A} w_\mathcal{A}$, and $a \leftarrow X^T u_\mathcal{A}$
$\gamma \leftarrow \min\{C/\alpha_\mathcal{A}, \min_{j \notin \mathcal{A}}^+ \{(C - c_j)/(\alpha_\mathcal{A} - a_j), (C + c_j)/(\alpha_\mathcal{A} + a_j)\}\}$
Record $\beta_\mathcal{A}(t) = \beta_\mathcal{A} + t \operatorname{diag}(s_\mathcal{A}) w_\mathcal{A}$ for $0 \le t \le \gamma$
$\beta_\mathcal{A} \leftarrow \beta_\mathcal{A} + \gamma \operatorname{diag}(s_\mathcal{A}) w_\mathcal{A}$ and $r \leftarrow r - \gamma u_\mathcal{A}$
$c \leftarrow X^T r$, $C \leftarrow \|c\|_\infty$, and $\mathcal{A} \leftarrow \mathcal{A} \cup \{j \notin \mathcal{A} : |c_j| = C\}$
until $C = 0$ or $\mathcal{A} = \{1, \dots, p\}$

Algorithm 11: Incremental Forward Stagewise Regression

Require: Centered response y, standardized columns $x_1, \dots, x_p$, small step size $\varepsilon > 0$
Ensure: A stagewise coefficient path
Initialize $\beta \leftarrow 0$ and $r \leftarrow y$
repeat
Choose j maximizing $|x_j^T r|$
$\delta \leftarrow \varepsilon \operatorname{sign}(x_j^T r)$
$\beta_j \leftarrow \beta_j + \delta$
$r \leftarrow r - \delta x_j$
until the residual has negligible correlation with every predictor

Remark 2.3.5. *Stagewise and Lasso Paths*
Let $c = X^T r$ and $j \in \arg\max_k |c_k|$. Forward stagewise uses $\beta_j \leftarrow \beta_j + \varepsilon \operatorname{sign}(c_j)$ and $r \leftarrow r - \varepsilon \operatorname{sign}(c_j) x_j$; as $\varepsilon \downarrow 0$, it follows maximal-correlation directions, tracking LAR/lasso until a sign reversal is required.
Boosting applies the same greedy infinitesimal update to additive fits.

Algorithm 12: Pathwise Coordinate Descent for the Lasso

Require: Centered response y, standardized columns x_j with $\|x_j\|_2 = 1$, decreasing penalties $\lambda_1 > \cdots > \lambda_L \ge 0$
Ensure: Lasso coefficient vectors $\widehat{\beta}(\lambda_\ell)$ on the penalty grid
Define $S(z, \lambda) = \operatorname{sign}(z)(|z| - \lambda)_+$
Initialize $\beta \leftarrow 0$ and $r \leftarrow y - X\beta$
for $\ell = 1, \dots, L$ **do**
repeat
for $j = 1, \dots, p$ **do**
$r \leftarrow r + x_j \beta_j$
$z_j \leftarrow x_j^T r$
$\beta_j \leftarrow S(z_j, \lambda_\ell)$
$r \leftarrow r - x_j \beta_j$
end for
until the coefficient vector has converged
$\widehat{\beta}(\lambda_\ell) \leftarrow \beta$
end for

2.4 Effective Degrees and Information Criteria

Definition 2.4.1. *Effective Degrees of Freedom*
Let $\widehat{y} = S_\lambda y$ be a linear smoother with $S_\lambda = S_\lambda(X)$ independent of y.
Its *effective degrees of freedom* are $\mathrm{df}(\lambda) := \mathrm{trace}(S_\lambda)$. If $S_\lambda = P_V$ with $\dim V = m$, then $\mathrm{df}(\lambda) = m$.
For shrinkage smoothers, $\mathrm{df}(\lambda)$ need not be integral.

Remark 2.4.2. *Optimism of Training Error*
Assume $y = \mu + \varepsilon$, $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2 I$, and $\widehat{y} = S_\lambda y$ with $S_\lambda = S_\lambda(X)$.
Let $y^0 = \mu + \varepsilon^0$ be independent, $\mathrm{err}_{\mathrm{tr}}(\lambda) := n^{-1}\|y - \widehat{y}\|_2^2$, and $\mathrm{Err}_{\mathrm{in}}(\lambda) := n^{-1}\mathbb{E}_{y^0}\|y^0 - \widehat{y}\|_2^2$.
Then $\mathrm{Opt}(\lambda) := \mathbb{E}\{\mathrm{Err}_{\mathrm{in}}(\lambda) - \mathrm{err}_{\mathrm{tr}}(\lambda)\} = \frac{2}{n}\sum_{i=1}^n \mathrm{Cov}(\widehat{y}_i, y_i) = \frac{2\sigma^2}{n}\,\mathrm{trace}(S_\lambda)$.
Thus $\mathrm{df}(\lambda) = \mathrm{trace}(S_\lambda)$ is the covariance penalty.

Method 2.4.3. *AIC, C_p, and BIC Selection*
Let Λ be candidate tuning values. For a criterion $C : \Lambda \to \mathbb{R}$, select $\widehat{\lambda} \in \arg\min_{\lambda\in\Lambda} C(\lambda)$.

(i) For squared error, set $\overline{\mathrm{err}}(\lambda) := \frac{1}{n}\sum_{i=1}^n (y_i - \widehat{y}_{\lambda,i})^2$ and $d_\lambda := \mathrm{df}(S_\lambda)$; use $C_p(\lambda) := \overline{\mathrm{err}}(\lambda) + \frac{2\widehat{\sigma}^2}{n} d_\lambda$.
(ii) For likelihood fits with maximized log-likelihood ℓ_λ, use $\mathrm{AIC}(\lambda) := -\frac{2}{n}\ell_\lambda + \frac{2}{n} d_\lambda$.
(iii) Use $\mathrm{BIC}(\lambda) := -\frac{2}{n}\ell_\lambda + \frac{\log n}{n} d_\lambda$; the larger penalty favors smaller d_λ as n grows.

Remark 2.4.4. *Bayesian and MDL Readings of BIC*
For models $\mathcal{M}_m$ with dimension d_m, maximised likelihood L_m, and nearly uniform prior π_m,
set $\mathrm{BIC}_m := -2\log L_m + d_m \log n$. By laplace approximation, posterior weight is $\widehat{\mathbb{P}}(\mathcal{M}_m \mid Z) = \frac{\pi_m \exp\{-\mathrm{BIC}_m/2\}}{\sum_{\ell=1}^M \pi_\ell \exp\{-\mathrm{BIC}_\ell/2\}}$, so $\arg\min_m \mathrm{BIC}_m$ is an approximate MAP model.
In MDL form, $\frac{1}{2}\mathrm{BIC}_m$ is model code length plus fitted-discrepancy code length up to constants.

Method 2.4.5. *Generalized Cross-Validation*
Let Λ be candidate tuning values and $\widehat{y}_\lambda = S_\lambda y$.

(i) Define $e_{\lambda i} := y_i - \widehat{y}_{\lambda i}$, $h_{\lambda i} := S_{\lambda,ii}$, and $d_\lambda := \mathrm{trace}(S_\lambda)$.
(ii) The leave-one-out residual is $e_{\lambda i}^{(-i)} = y_i - \widehat{f}_\lambda^{(-i)}(x_i) = \frac{e_{\lambda i}}{1 - h_{\lambda i}}$.
(iii) *Generalized cross-validation* replaces $h_{\lambda i}$ by $\bar{h}_\lambda := d_\lambda/n$ and sets $\mathrm{GCV}(\lambda) := \frac{1}{n}\sum_{i=1}^n \left(\frac{e_{\lambda i}}{1 - d_\lambda/n}\right)^2$.
(iv) Select $\widehat{\lambda}_{\mathrm{gcv}} \in \arg\min_{\lambda\in\Lambda} \mathrm{GCV}(\lambda)$; this uses d_λ instead of the full leverage diagonal.

3 Basis Expansions, Splines, and Wavelets

3.1 Basis Expansions, Splines, and Wavelet Features

Basis expansions keep the fitting step linear while moving nonlinearity into the representation of the inputs.

Definition 3.1.1. *Linear Basis Expansion*

A *linear basis expansion* chooses $h = (h_1, \ldots, h_M) : \mathcal{X} \to \mathbb{R}^M$ and fits $f_\beta(x) := \beta^T h(x) = \sum_{m=1}^M \beta_m h_m(x)$.

Typical h_m include coordinates, powers, interactions, one-dimensional transforms, and region indicators.
The model is linear in $h(x)$, but generally nonlinear in x.

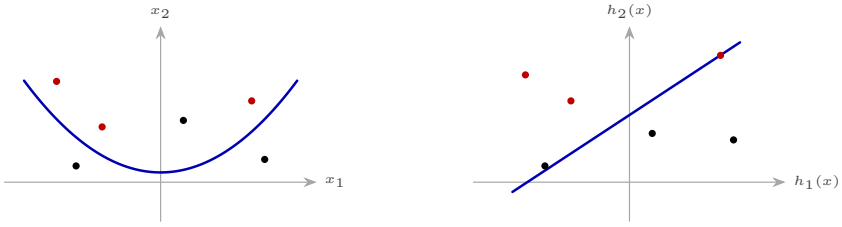

Figure 261: Basis expansion: boundary curved in x, linear in $h(x)$.

Remark 3.1.2. *Dictionaries Need Complexity Control*

For a dictionary $\mathcal{H} = \{h_m\}_{m=1}^M$, fitting $f_\beta(x) = \sum_{m=1}^M \beta_m h_m(x)$ is flexible but unstable when M is large or the h_m are nearly collinear. Complexity control restricts admissible fits: additive models impose $f(x) = \sum_j f_j(x_j)$, subset selection imposes $\|\beta\|_0 \le s$, and regularization minimizes empirical loss plus $\lambda\Omega(\beta)$.

Definition 3.1.3. *Regression Splines*
Let $\xi_1 < \cdots < \xi_K$ be knots.

(i) *Order-M regression spline* is piecewise polynomial $s : \mathbb{R} \to \mathbb{R}$ of degree $\le M-1$, $s \in C^{M-2}$ at knots.
(ii) *Cubic truncated-power basis* is basis $1, x, x^2, x^3, (x-\xi_1)_+^3, \ldots, (x-\xi_K)_+^3$ for cubic spline space.
(iii) *Natural cubic spline* is a cubic regression spline satisfying $s''(x) = 0$ for $x \notin [\xi_1, \xi_K]$.
(iv) *B-spline basis* is a locally supported basis for the same spline space.

Method 3.1.4. *Smoothing Splines as Penalized Fits*
Let $(x_i, y_i)_{i=1}^n \subset \mathbb{R}^2$, $y = (y_1, \ldots, y_n)^T$, $\lambda \ge 0$, and $\mathcal{F} := \{f : f'' \in L^2(\mathbb{R})\}$.

(i) The smoothing spline is $\widehat{f}_\lambda \in \arg\min_{f\in\mathcal{F}} \left\{\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_{\mathbb{R}} (f''(t))^2\, dt\right\}$.

(ii) $\widehat{f}_\lambda$ is a natural cubic spline with knots at the distinct observed inputs.
(iii) With $\widehat{y}_\lambda = (\widehat{f}_\lambda(x_1), \dots, \widehat{f}_\lambda(x_n))^T$, there is a smoother matrix S_λ such that $\widehat{y}_\lambda = S_\lambda y$ and $d_\lambda := \operatorname{trace}(S_\lambda)$.
(iv) For $e_{\lambda i} := y_i - \widehat{f}_\lambda(x_i)$ and $h_{\lambda i} := S_{\lambda,ii}$, leave-one-out CV is $\mathrm{CV}(\lambda) := \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_{\lambda i}}{1-h_{\lambda i}}\right)^2$.

Definition 3.1.5. *Tensor-Product and Thin-Plate Splines*
Let $\{h_{rj}\}_{j=1}^{J_r}$ be a univariate basis for coordinate x_r.
(i) A *two-way tensor-product basis* is $\{g_{jk}\}$ with $g_{jk}(x_1,x_2) := h_{1j}(x_1)h_{2k}(x_2)$ and dimension J_1J_2.
(ii) A *d-way tensor-product basis* is $\{g_{j_1,\dots,j_d}\}$ with $g_{j_1,\dots,j_d}(x) := \prod_{r=1}^{d} h_{rj_r}(x_r)$ and dimension $\prod_{r=1}^{d} J_r$.
(iii) A *thin-plate spline fit* is any $\widehat{f}_\lambda \in \arg\min_f Q_\lambda(f)$, where $Q_\lambda(f) := \sum_{i=1}^{n}(y_i - f(x_i))^2 + \lambda J(f)$, $D_{ab}f := \partial_a\partial_b f$, and $J(f) := \int_{\mathbb{R}^2}\{(D_{11}f)^2 + 2(D_{12}f)^2 + (D_{22}f)^2\}\,dx$.
(iv) A *thin-plate representer* is $\widehat{f}(x) = a_0 + a^\top x + \sum_{\ell=1}^{L} c_\ell\phi(\|x-\kappa_\ell\|)$; $\phi(r) := r^2\log r$, $\phi(0) := 0$, $\sum_{\ell=1}^{L} c_\ell(1,\kappa_\ell^\top) = 0$.

Method 3.1.6. *Spline-Filtered Logistic Features*
Let $(x_i, y_i)_{i=1}^n$ with $x_i \in \mathbb{R}^p$, $y_i \in \{0,1\}$, and let $H \in \mathbb{R}^{p\times M}$ be a smooth basis matrix with $M \ll p$.
(i) Restrict $\beta \in \mathbb{R}^p$ to $\mathrm{col}(H)$ by writing $\beta = H\theta$, $\theta \in \mathbb{R}^M$.
(ii) Define filtered features $z_i := H^T x_i$; then $x_i^T\beta = z_i^T\theta$.
(iii) Fit $\widehat{\theta}, \widehat{b} \in \arg\min_{\theta,b} \sum_{i=1}^{n}\{\log(1+\exp(b+z_i^T\theta)) - y_i(b+z_i^T\theta)\}$.
(iv) Set $\widehat{\beta} = H\widehat{\theta}$; smoothness is imposed by $\widehat{\beta} \in \mathrm{col}(H)$ and dimension drops from p to M.

Method 3.1.7. *Wavelet Shrinkage*
Let $y \in \mathbb{R}^n$ be observed on a regular lattice and let $W \in \mathbb{R}^{n\times n}$ be an orthonormal wavelet matrix.
(i) Transform y to wavelet coefficients $y^* := W^\top y$.
(ii) Estimate sparse coefficients by $\widehat{\theta}_\lambda \in \arg\min_{\theta\in\mathbb{R}^n} Q_\lambda(\theta)$, where $Q_\lambda(\theta) := \|y - W\theta\|_2^2 + 2\lambda\|\theta\|_1$.
(iii) Since $W^\top W = I$, $\widehat{\theta}_{\lambda,j} = \operatorname{sign}(y_j^*)(|y_j^*| - \lambda)_+$ for $j = 1,\dots,n$.
(iv) Reconstruct $\widehat{f}_\lambda := W\widehat{\theta}_\lambda = \sum_{j:|y_j^*|>\lambda}\operatorname{sign}(y_j^*)(|y_j^*|-\lambda)w_j$, where w_j is the jth wavelet atom.

4 Kernel Methods and RKHS Theory

4.1 Kernel Methods and RKHS Representations

Kernel methods keep the algorithm linear in an implicit feature space while computing only inner products.

Definition 4.1.1. *Positive Definite Kernel*
A map $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$ is a *positive definite kernel* if:
(i) $k(x,y) = k(y,x)$ for all $x, y \in \mathcal{X}$;
(ii) for every $n \in \mathbb{N}$ and $x_1,\dots,x_n \in \mathcal{X}$, $K = (k(x_i,x_j))_{i,j=1}^n$ satisfies $a^\top Ka = \sum_{i,j=1}^{n} a_ia_jk(x_i,x_j) \ge 0\ \forall a \in \mathbb{R}^n$.

Example 4.1.2. *Common Kernels*
For $x, y \in \mathbb{R}^p$, common positive definite kernels include:
(i) *Polynomial*: for $d \in \mathbb{N}_{>0}$, $k_d(x,y) := (\langle x,y\rangle + 1)^d = \langle h_d(x), h_d(y)\rangle$, encoding monomials of degree $\le d$. If $p = 2$ and $d = 2$, one feature map is $h_2(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2) \in \mathbb{R}^6$.
(ii) *Gaussian RBF*: for $\nu > 0$, $k_\nu(x,y) := \exp(-\nu\|x-y\|_2^2)$; it is radial and translation invariant, with $k_\nu(x,y)$ decreasing as $\|x-y\|_2$ increases.

Definition 4.1.3. *Reproducing Kernel Hilbert Space*
A Hilbert space $\mathcal{H} \subseteq \mathbb{R}^{\mathcal{X}}$ is a *reproducing kernel Hilbert space* if point evaluation is continuous; explicitly:
(i) for every $x \in \mathcal{X}$, the evaluation map $E_x : \mathcal{H} \to \mathbb{R}$, $E_x(f) := f(x)$, is continuous;
(ii) by Riesz, $k_x \in \mathcal{H}$ denotes the unique representer of E_x, so $f(x) = \langle f, k_x\rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}$;
(iii) the *reproducing kernel* is $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$, $k(x,y) := k_x(y) = \langle k_x, k_y\rangle_{\mathcal{H}}$, with $k_x = k(x,\cdot)$.

Theorem 4.1.4. *Representer Theorem Form*
Let $\mathcal{H}$ be an RKHS on $\mathcal{X}$ with kernel k.
Let $(x_i,y_i)_{i=1}^n \subseteq \mathcal{X}\times\mathcal{Y}$, $\lambda > 0$, $\ell : \mathbb{R}\times\mathcal{Y} \to (-\infty,+\infty]$, and $\Phi : [0,\infty) \to \mathbb{R}$ nondecreasing.
For $J(f) := \sum_{i=1}^{n}\ell(f(x_i),y_i) + \lambda\Phi(\|f\|_{\mathcal{H}})$ and $S := \operatorname{span}\{k(x_i,\cdot) : 1 \le i \le n\}$, if $f^\star \in \operatorname{argmin}_{f\in\mathcal{H}} J(f)$, then $Pf^\star \in S$ is also a minimiser, where $P : \mathcal{H} \to S$ is orthogonal projection.
Hence some minimiser has $f^\star(\cdot) = \sum_{i=1}^{n}\alpha_ik(x_i,\cdot)$; if Φ is strictly increasing, every minimiser has this form.

Proof. Since S is finite-dimensional, it is closed, so P exists.
Write $f = f_S + f_\perp$ with $f_S := Pf \in S$ and $f_\perp \in S^\perp$.
For each i, $k(x_i,\cdot) \in S$, hence $f_\perp(x_i) = \langle f_\perp, k(x_i,\cdot)\rangle_{\mathcal{H}} = 0$ and $f_S(x_i) = f(x_i)$.
Thus the empirical loss in J is unchanged by $f \mapsto f_S$.
By orthogonality, $\|f\|_{\mathcal{H}}^2 = \|f_S\|_{\mathcal{H}}^2 + \|f_\perp\|_{\mathcal{H}}^2$, so $\|f_S\|_{\mathcal{H}} \le \|f\|_{\mathcal{H}}$, hence $J(f_S) \le J(f)$ by monotonicity of Φ.
Applying this to a minimiser $f^\star$ gives $J(Pf^\star) = J(f^\star)$, so $Pf^\star$ is a minimiser in S.
Since $S = \{\sum_{i=1}^{n}\alpha_ik(x_i,\cdot) : \alpha \in \mathbb{R}^n\}$, finite expansion follows. If Φ strictly increasing, $f_\perp \ne 0$, then $\|f_S\|_{\mathcal{H}} < \|f\|_{\mathcal{H}}$ while loss is

unchanged, so $J(f_S) < J(f)$, contradicting minimality.
Structural Commentary: $E : \mathcal{H} \to \mathbb{R}^n$, $Ef = (f(x_i))_{i=1}^n$, has $E|_{S^\perp} = 0$ and $Ef = E(Pf)$; only S is data-visible.
Key Insight: Regularisation kills $S^\perp$, so optimise over $\alpha \in \mathbb{R}^n$ via $f_\alpha = \sum_{i=1}^n \alpha_i k(x_i, \cdot)$ and $Ef_\alpha = K\alpha$. □

Method 4.1.5. *Kernel Ridge Regression*
Let $(x_i, y_i)_{i=1}^n \subseteq \mathcal{X} \times \mathbb{R}$, $y := (y_1, \dots, y_n)^\top$, $\lambda > 0$, and $K_{ij} := k(x_i, x_j)$.

(i) *RKHS problem*. Solve $\widehat{f} \in \operatorname*{argmin}_{f \in \mathcal{H}} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}}^2$.

(ii) *Representer reduction*. Write $\widehat{f}(\cdot) = \sum_{i=1}^n \widehat{\alpha}_i k(x_i, \cdot)$; then $\widehat{f}(x_j) = (K\widehat{\alpha})_j$ and $\left\|\widehat{f}\right\|_{\mathcal{H}}^2 = \widehat{\alpha}^\top K \widehat{\alpha}$.

(iii) *Finite problem*. Minimise $\widehat{\alpha} \in \operatorname*{argmin}_{\alpha \in \mathbb{R}^n} \|y - K\alpha\|_2^2 + \lambda \alpha^\top K \alpha$.

(iv) *Solution*. Since $K \succeq 0$ and $\lambda > 0$, $K + \lambda I \succ 0$ and $\widehat{\alpha} = (K + \lambda I)^{-1} y$.
Hence $\widehat{y} = K\widehat{\alpha}$ and $\widehat{f}(x) = k_x^\top \widehat{\alpha}$, where $k_x := (k(x, x_1), \dots, k(x, x_n))^\top$.

Remark 4.1.6. *Localisation Kernels and RKHS Kernels*
Localisation kernel has weights $w_i(x_0) \propto K_\lambda(x_0, x_i)$. $w_i(x_0)$ is large when x_i close to x_0; small otherwise.
RKHS kernel is PSD: $K_{ij} = k(x_i, x_j)$ is Gram matrix, i.e., $k(x, y) = \langle h(x), h(y) \rangle_{\mathcal{F}}$ for some feature map h.
Thus: localisation kernels → local averaging; RKHS kernels → inner products for global regularisation.

Example 4.1.7. *Radial Basis Expansions*
Let $\xi_1, \dots, \xi_M \in \mathbb{R}^p$, $\lambda_j > 0$, $\beta \in \mathbb{R}^M$, and let $D : [0, \infty) \to [0, \infty)$ be a radial profile.

(i) *Unnormalized basis*: $\varphi_j(x) := D(\frac{\|x - \xi_j\|_2}{\lambda_j})$ and $f(x) := \sum_{j=1}^M \beta_j \varphi_j(x)$.

(ii) *Normalized basis*: for $\lambda_j = \lambda$, set $h_j(x) := D(\frac{\|x - \xi_j\|_2}{\lambda})/s(x)$, where $s(x) := \sum_{k=1}^M D(\frac{\|x - \xi_k\|_2}{\lambda}) > 0$.

Then $h_j(x) \geq 0$ and $\sum_{j=1}^M h_j(x) = 1$.

(iii) *Training-point centers*: $M = n$, $\xi_i = x_i$, $\beta_i = Y_i$ give $\widehat{f}(x) := \sum_{i=1}^n Y_i h_i(x)$ (Nadaraya–Watson local avg).

Remark 4.1.8. *Spline and Kernel Viewpoints*
Common form: $\min_f \sum_{i=1}^n \ell(f(x_i), y_i) + \lambda \|Pf\|_{\mathcal{H}}^2$, where P removes an unpenalised $\mathcal{U}$, gives $\widehat{f} = u + \sum_{i=1}^n \alpha_i k(x_i, \cdot)$ with $u \in \mathcal{U}$. Splines choose P, kernel methods choose k, and SVMs choose a margin loss ℓ.

5 Support Vector Machines and Margin Methods

5.1 Margins and SVM Primal Problems

Margin methods prefer classifiers that separate classes with a buffer, not merely with zero training error. The support-vector construction makes this preference explicit by trading margin width against margin violations.

Definition 5.1.1. *Affine Hyperplane and Signed Distance*
Let $w \in \mathbb{R}^p \setminus \{0\}$, $b \in \mathbb{R}$, and $q_{w,b}(x) := w^\top x + b$.

(i) *Affine hyperplane*: $H_{w,b} := \{x \in \mathbb{R}^p : q_{w,b}(x) = 0\}$, with normal direction w.

(ii) *Signed distance*: $d_{w,b}(x) := \frac{q_{w,b}(x)}{\|w\|_2}$.

(iii) *Scale invariance*: for $c > 0$, $H_{cw,cb} = H_{w,b}$ and $q_{cw,cb} = c q_{w,b}$.

Definition 5.1.2. *Margin*
Let $y_i \in \{-1, 1\}$ and $q(x) := w^\top x + b$.

(i) *Functional margin*: $\gamma_i(q) := y_i q(x_i)$.

(ii) *Geometric margin*: for $w \neq 0$, $\rho_i(w, b) := \frac{y_i(w^\top x_i + b)}{\|w\|_2}$.

(iii) *Correct classification*: $\gamma_i(q) > 0$; larger $\gamma_i(q)$ means larger signed confidence at fixed score scale.

Definition 5.1.3. *Hard-Margin Support Vector Machine*
For separable $(x_i, y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1, 1\}$:

(i) *Unit-margin scale*: $y_i(w^\top x_i + b) \geq 1$ for all i.

(ii) *Hard-margin primal*: $(\widehat{w}, \widehat{b}) \in \operatorname*{argmin}_{w,b} \frac{1}{2}\|w\|_2^2$ subject to $y_i(w^\top x_i + b) \geq 1$, $i = 1, \dots, n$.

(iii) *Margin width*: the slab $-1 \leq w^\top x + b \leq 1$ has width $\frac{2}{\|w\|_2}$, so the primal maximises geometric margin.

Definition 5.1.4. *Soft-Margin Support Vector Machine*
For possibly nonseparable $(x_i, y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1, 1\}$ and $C > 0$:

(i) *Slack variables*: $\xi_i \geq 0$ and $y_i(w^\top x_i + b) \geq 1 - \xi_i$, $i = 1, \dots, n$.

(ii) *Soft-margin primal*: $(\widehat{w}, \widehat{b}, \widehat{\xi}) \in \operatorname{argmin}_{w,b,\xi}\{\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n \xi_i\}$ subject to the slack constraints.

(iii) *Penalty parameter*: C prices $\sum_{i=1}^n \xi_i$; larger C penalises violations more strongly.

Remark 5.1.5. *Why the Standard Slack Scaling*
Before score normalisation, additive slack uses $y_i(w^\top x_i + b) \geq M - \xi_i$, relative slack uses $y_i(w^\top x_i + b) \geq M(1 - \xi_i)$.
Setting $M = 1$ gives convex constraint $y_i(w^\top x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$.
Thus ξ_i measures margin failure: 0 on/outside the margin, $(0, 1]$ inside but correct, and > 1 misclassified.

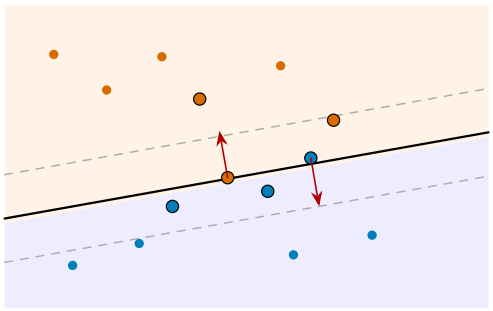

Figure 262: Soft-margin SVM: orange $y=+1$, blue $y=-1$; solid $f(x)=0$, dashed $f(x)=\pm 1$; circled support vectors, red slack arrows ξ_i.

Proposition 5.1.6. *Hinge-Loss Penalty Form*
Let $(x_i,y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1,1\}$, $C>0$, $f_{w,b}(x) := w^\top x + b$, and $\lambda := 1/C$. For soft-margin primal $\min_{w,b,\xi}\{\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n \xi_i\}$ *subject to $\xi_i \geq 0$ and $y_i f_{w,b}(x_i) \geq 1-\xi_i$, eliminating ξ and scaling by $\frac{1}{C}$ gives* $\min_{w,b}\sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+ + \frac{\lambda}{2}\|w\|_2^2$. *For fixed (w,b), minimal feasible slack is $\xi_i^\star = [1-y_i f_{w,b}(x_i)]_+$.*

Proof. Fix (w,b) and put $r_i := 1-y_i f_{w,b}(x_i)$. Feasibility gives $\xi_i \geq 0$ and $\xi_i \geq r_i$, hence $\xi_i \geq \max\{0,r_i\} = [r_i]_+$.
Since $C>0$, the objective is strictly increasing in each ξ_i, so $\xi_i^\star = [r_i]_+$.
Substitution yields $\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+$.
Multiply by $1/C>0$ preserves minimisers, $\sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+ + \frac{1}{2C}\|w\|_2^2 = \sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+ + \frac{\lambda}{2}\|w\|_2^2$.
Structural Commentary: The ξ_i decouple; eliminate them pointwise.
Key Insight: Soft-margin SVM is hinge empirical risk plus an ℓ_2 penalty on w; the intercept b is unpenalized. □

5.2 Dual Problems and Kernelization

Proposition 5.2.1. *Hard-Margin Dual and Support Vectors*
Let $(x_i,y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1,1\}$ be linearly separable. The primal $P : \min_{w,b}\frac{1}{2}\|w\|_2^2$ subject to $y_i(w^\top x_i + b) \geq 1$ has dual $D : \max_{\alpha\in\mathbb{R}_+^n} \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y_i y_j x_i^\top x_j$ subject to $\sum_{i=1}^n \alpha_i y_i = 0$.
At any optimum, $\widehat{w} = \sum_{i=1}^n \widehat{\alpha}_i y_i x_i$ and $\widehat{\alpha}_i > 0 \Rightarrow y_i(\widehat{w}^\top x_i + \widehat{b}) = 1$; these active points are the support vectors.

Proof. Let $g_i(w,b) := y_i(w^\top x_i + b) - 1$ and $\alpha_i \geq 0$. The Lagrangian is $L(w,b,\alpha) = \frac{1}{2}\|w\|_2^2 - \sum_{i=1}^n \alpha_i g_i(w,b)$.
Stationarity gives $0 = \nabla_w L = w - \sum_{i=1}^n \alpha_i y_i x_i$ and $0 = \partial_b L = -\sum_{i=1}^n \alpha_i y_i$, hence $w = \sum_{i=1}^n \alpha_i y_i x_i$ and $\sum_{i=1}^n \alpha_i y_i = 0$.
Substituting into L gives $D(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y_i y_j x_i^\top x_j$, with $\alpha \geq 0$ and $\sum_i \alpha_i y_i = 0$.
Strong duality holds by convexity and strict feasibility.
Complementary slackness gives $\widehat{\alpha}_i[y_i(\widehat{w}^\top x_i + \widehat{b}) - 1] = 0$, so $\widehat{\alpha}_i > 0$ forces $y_i(\widehat{w}^\top x_i + \widehat{b}) = 1$.
Structural Commentary: Stationarity leaves α; data enter via $y_i y_j x_i^\top x_j$ and $\sum_i \alpha_i y_i = 0$.
Key Insight: $\alpha_i > 0$ marks margin support; these points form $\widehat{w}$ and fix the separator. □

Proposition 5.2.2. *Soft-Margin Dual and Support Types*
Let $(x_i,y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1,1\}$ and $C>0$. The primal $P : \min_{w,b,\xi}\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n \xi_i$ subject to $y_i(w^\top x_i + b) \geq 1-\xi_i$ and $\xi_i \geq 0$ has dual $D : \max_{\alpha\in[0,C]^n} \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y_i y_j x_i^\top x_j$ subject to $\sum_{i=1}^n \alpha_i y_i = 0$.
At any optimum, $\widehat{w} = \sum_{i=1}^n \widehat{\alpha}_i y_i x_i$. With $\widehat{f}(x) := \widehat{w}^\top x + \widehat{b}$, $\widehat{\alpha}_i = 0 \Rightarrow y_i\widehat{f}(x_i) \geq 1$, $0 < \widehat{\alpha}_i < C \Rightarrow y_i\widehat{f}(x_i) = 1$, and $\widehat{\alpha}_i = C \Rightarrow y_i\widehat{f}(x_i) = 1 - \widehat{\xi}_i \leq 1$. The support vectors are exactly $\{i : \widehat{\alpha}_i > 0\}$.

Proof. Let $\alpha_i \geq 0$ multiply $y_i(w^\top x_i + b) - 1 + \xi_i \geq 0$ and $\mu_i \geq 0$ multiply $\xi_i \geq 0$.
The Lagrangian is $L = \frac{1}{2}\|w\|_2^2 + C\sum_i \xi_i - \sum_i \alpha_i[y_i(w^\top x_i + b) - 1 + \xi_i] - \sum_i \mu_i\xi_i$.
Stationarity gives $w = \sum_i \alpha_i y_i x_i$, $\sum_i \alpha_i y_i = 0$, and $C - \alpha_i - \mu_i = 0$, hence $0 \leq \alpha_i \leq C$.
Substitution gives dual; Slater gives strong duality.
Complementary slackness gives $\alpha_i[y_i f(x_i) - 1 + \xi_i] = 0$ and $(C-\alpha_i)\xi_i = 0$.
Thus $\alpha_i = 0$ implies $\xi_i = 0$ and $y_i f(x_i) \geq 1$; $0 < \alpha_i < C$ implies $\xi_i = 0$ and $y_i f(x_i) = 1$.
Hence $\alpha_i = C$ implies $y_i f(x_i) = 1 - \xi_i \leq 1$.
Structural Commentary: Slacks become box constraints $0 \leq \alpha_i \leq C$.
Key Insight: α_i labels type: 0 outside, $(0,C)$ on-margin, C inside/wrong-side. □

Method 5.2.3. *Kernelized Support Vector Classifier*
Let $(x_i, y_i)_{i=1}^n \subseteq \mathcal{X} \times \{-1, 1\}$, $C > 0$, $h : \mathcal{X} \to \mathcal{F}$, and $k(x, x') := \langle h(x), h(x') \rangle_{\mathcal{F}}$.
The kernelized dual is $\widehat{\alpha} \in \underset{\alpha \in [0,C]^n}{\operatorname{argmax}} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(x_i, x_j)$ subject to $\sum_{i=1}^n \alpha_i y_i = 0$.
The fitted score and classifier are $\widehat{f}(x) = \sum_{i=1}^n \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$ and $\widehat{G}(x) = \operatorname{sign}(\widehat{f}(x))$.
If $0 < \widehat{\alpha}_s < C$, then $\widehat{b} = y_s - \sum_{i=1}^n \widehat{\alpha}_i y_i k(x_s, x_i)$; average this over all such s for stability.

Proof. In feature space, the soft-margin primal uses $v \in \mathcal{F}$ and score $f_{v,b}(x) = \langle v, h(x) \rangle_{\mathcal{F}} + b$.
Its Lagrangian stationarity gives $v = \sum_{i=1}^n \alpha_i y_i h(x_i)$ and $\sum_{i=1}^n \alpha_i y_i = 0$, while slack stationarity gives $0 \le \alpha_i \le C$.
Substitution yields $\sum_i \alpha_i - \frac{1}{2} \left\| \sum_i \alpha_i y_i h(x_i) \right\|_{\mathcal{F}}^2 = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j k(x_i, x_j)$.
Hence $\widehat{f}(x) = \langle \widehat{v}, h(x) \rangle_{\mathcal{F}} + \widehat{b} = \sum_i \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$.
If $0 < \widehat{\alpha}_s < C$, KKT gives zero slack and $y_s \widehat{f}(x_s) = 1$, so $\widehat{f}(x_s) = y_s$ and the stated $\widehat{b}$ follows.
Structural Commentary: Coordinates vanish; optimisation uses only $K_{ij} = k(x_i, x_j)$.
Key Insight: The classifier is finite and support-vector based: only indices with $\widehat{\alpha}_i > 0$ contribute to $\widehat{f}$. □

Remark 5.2.4. *Hinge Loss and Probability-Oriented Losses*
Let $\eta(x) := \mathbb{P}(Y = 1 \mid X = x)$ and $m := yf(x)$.
(i) *Hinge*: $[1 - m]_+$ targets $\operatorname{sign}(2\eta(x) - 1)$ and is zero for $m \ge 1$.
(ii) *Logistic*: $\log(1 + e^{-m})$ targets $\log(\eta(x)/(1 - \eta(x)))$ and keeps a smooth positive tail.
(iii) *Squared*: squared loss targets $\mathbb{E}[Y \mid X = x] = 2\eta(x) - 1$ and keeps residual-magnitude leverage.

Method 5.2.5. *SVM Regularization Path View*
set $\lambda := 1/C$; as $\lambda \downarrow 0$ ($C \uparrow \infty$), hinge loss dominates norm regularisation.
(i) *Fit*: for each λ, solve for $f_\lambda(x) := w_\lambda^\top x + b_\lambda$ and multipliers $\alpha_i(\lambda)$.
(ii) *Classify states*: with $m_i(\lambda) := y_i f_\lambda(x_i)$, $m_i > 1 \Rightarrow \alpha_i = 0$, $m_i = 1 \Rightarrow 0 \le \alpha_i \le C$, and $m_i < 1 \Rightarrow \alpha_i = C$.
(iii) *Track*: fixed active set $\Rightarrow \alpha_i(\lambda)$ and f_λ are linear.
(iv) *Kernelize*: replace $w_\lambda^\top x$ by $\sum_{i=1}^n \alpha_i(\lambda) y_i k(x, x_i)$, so $f_\lambda(x) = \sum_{i=1}^n \alpha_i(\lambda) y_i k(x, x_i) + b_\lambda$.

5.3 Support Vector Regression and Perceptrons

Method 5.3.1. *Epsilon-Insensitive Support Vector Regression*
Let $(x_i, y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \mathbb{R}$, $\epsilon > 0$, $\lambda > 0$, and $f_{w,b}(x) := w^\top x + b$.
(i) *Define loss*: for residual r, set $V_\epsilon(r) := [|r| - \epsilon]_+$.
(ii) *Fit*: solve $\min_{w,b} \sum_{i=1}^n V_\epsilon(y_i - f_{w,b}(x_i)) + \frac{\lambda}{2} \|w\|_2^2$.
(iii) *Read tube*: $|r_i| \le \epsilon \Rightarrow V_\epsilon(r_i) = 0$ and $|r_i| > \epsilon \Rightarrow V_\epsilon(r_i) = |r_i| - \epsilon$.
(iv) *Kernelize*: $\widehat{f}(x) = \sum_{i=1}^n (\widehat{\alpha}_i^* - \widehat{\alpha}_i) k(x, x_i) + \widehat{b}$; tube sides have opposite signs.

Method 5.3.2. *Perceptron Update as a Contrast*
Choose $(w_0, b_0) \in \mathbb{R}^p \times \mathbb{R}$ and learning rate $\rho > 0$.
(i) *Scan*: at step t, choose (x_i, y_i) and compute $m_i(t) := y_i(w_t^\top x_i + b_t)$.
(ii) *Update*: if $m_i(t) \le 0$, set $(w_{t+1}, b_{t+1}) := (w_t, b_t) + \rho y_i (x_i, 1)$.
(iii) *Keep*: if $m_i(t) > 0$, set $(w_{t+1}, b_{t+1}) := (w_t, b_t)$.
(iv) *Contrast*: perceptron may separate but is order-dependent; SVM solves a convex max-margin problem.

Remark 5.3.3. *Support Vectors*
Let $\widehat{f}(x) = \sum_{i=1}^n \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$ and $S := \{i : \widehat{\alpha}_i > 0\}$.
Then $\widehat{f}(x) = \sum_{i \in S} \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$, and KKT gives $i \in S \Rightarrow y_i \widehat{f}(x_i) \le 1$. Thus only support vectors enter $\widehat{f}$.

Remark 5.3.4. *Margins and Generalization*
Let $m_i(f) := y_i f(x_i)$ and $\widehat{R}_\gamma(f) := n^{-1} \sum_{i=1}^n \mathbf{1}\{m_i(f) \le \gamma\}$.
Bounds control $R(f) := \mathbb{P}(Yf(X) \le 0)$ by $\widehat{R}_\gamma(f)$ plus a complexity proxy $\mathcal{C}_\gamma(f) \asymp \|f\|^2 / \gamma^2$.
For linear $f_{w,b}$ with $\min_i m_i(f) \ge 1$, the geometric margin is $\rho = 1/\|w\|_2$.

6 Trees and Ensemble Averaging

6.1 Tree Models and Bagging

Tree-based methods partition the input space into simple regions and fit a local rule in each region. Ensemble averaging methods keep this local modelling idea but stabilise it by resampling, voting, and decorrelating the individual trees.

Definition 6.1.1. *Decision Tree*
Let $\mathcal{X} \subseteq \mathbb{R}^p$ and let $\mathcal{A}$ be the prediction space.
A decision tree T is a finite binary recursive partition of $\mathcal{X}$ with a leaf value in $\mathcal{A}$ on each terminal region.
(i) *Split*: a split at node $R \subseteq \mathcal{X}$ is a feature-threshold pair (j, t) with $j \in \{1, \dots, p\}$ and $t \in \mathbb{R}$.
It produces children $R_L := R \cap \{x : x_j \le t\}$ and $R_R := R \cap \{x : x_j > t\}$.

(ii) *Terminal partition*: after splitting stops, the leaves of T are nonempty regions $R_1(T), \dots, R_M(T)$ with $\mathcal{X} = \bigcup_{m=1}^{M} R_m(T)$ and $R_m(T) \cap R_{m'}(T) = \varnothing$ for $m \neq m'$.

(iii) *Leaf rule*: a leaf assignment $a_m \in \mathcal{A}$ for $m = 1, \dots, M$ defines $\widehat{f}_T : \mathcal{X} \to \mathcal{A}$ by $\widehat{f}_T(x) := a_m$ for the unique m such that $x \in R_m(T)$.

6.2 CART Splits and Pruning

Definition 6.2.1. *CART Regression-Tree Notation and Scores*
Let $D_n = \{(x_i, y_i)\}_{i=1}^n \subseteq \mathcal{X} \times \mathbb{R}$. For $R \subseteq \mathcal{X}$, set $I_R := \{i : x_i \in R\}$ and $n_R := |I_R|$.

(i) *Cell loss*: if $n_R > 0$, define $\bar{y}_R := n_R^{-1} \sum_{i \in I_R} y_i$ and $\mathrm{SSE}(R) := \sum_{i \in I_R} (y_i - \bar{y}_R)^2$.

(ii) *Tree predictor*: if T has leaves $R_1(T), \dots, R_{|T|}(T)$, let $m_T(x)$ be the unique $m \in \{1, \dots, |T|\}$ with $x \in R_m(T)$ and set $\widehat{f}_T(x) := \bar{y}_{R_{m_T(x)}(T)}$.

(iii) *Split children*: for $a = (j, s) \in \{1, \dots, p\} \times \mathbb{R}$, set $R_L(a) := R \cap \{x : x_j \le s\}$ and $R_R(a) := R \cap \{x : x_j > s\}$.

(iv) *Admissible splits*: set $\mathcal{A}_R := \{a \in \{1, \dots, p\} \times \mathbb{R} : n_{R_L(a)} > 0, n_{R_R(a)} > 0\}$.

(v) *Split score*: for $a \in \mathcal{A}_R$, set $S_R(a) := \mathrm{SSE}(R_L(a)) + \mathrm{SSE}(R_R(a))$.

(vi) *Pruning score*: for $\alpha \ge 0$, set $C_\alpha(T) := \sum_{m=1}^{|T|} \mathrm{SSE}(R_m(T)) + \alpha |T|$.

Method 6.2.2. *CART-Style Regression Trees*

(i) *Grow*: split eligible leaves by $a_R \in \operatorname{argmin}_{a \in \mathcal{A}_R} S_R(a)$ until stopping, yielding T_0.

(ii) *Prune*: for each $\alpha \ge 0$, choose $T_\alpha \in \operatorname{argmin}_{T \preceq T_0} C_\alpha(T)$.

(iii) *Select*: choose $\widehat{\alpha} \ge 0$ by validation or cross-validation and return $T_{\widehat{\alpha}}$.

Deep growth lowers bias; the cost-complexity penalty $\alpha|T|$ controls variance by subtree selection.

Definition 6.2.3. *Classification Tree Impurity*
For node m, let $N_{mk} := |\{i : x_i \in R_m, y_i = k\}|$, $N_m := \sum_{k=1}^K N_{mk}$, $\widehat{p}_{mk} := \frac{N_{mk}}{N_m}$. Common impurity scores are:

(i) *Misclassification*: $I_{\mathrm{mis}}(m) := 1 - \max_{1 \le k \le K} \widehat{p}_{mk}$.

(ii) *Gini*: $I_{\mathrm{Gini}}(m) := \sum_{k=1}^K \widehat{p}_{mk}(1 - \widehat{p}_{mk}) = 1 - \sum_{k=1}^K \widehat{p}_{mk}^2$.

(iii) *Entropy*: $I_{\mathrm{ent}}(m) := -\sum_{k : \widehat{p}_{mk} > 0} \widehat{p}_{mk} \log \widehat{p}_{mk}$.

The usual leaf prediction is the majority class, $\widehat{k}(m) \in \operatorname{argmax}_{1 \le k \le K} \widehat{p}_{mk}$.

With unequal losses $L_{kk} = 0$, choose the class minimizing expected loss, $\widehat{\ell}(m) \in \operatorname{argmin}_{1 \le \ell \le K} \sum_{k=1}^K L_{k\ell} \widehat{p}_{mk}$.

Remark 6.2.4. *Tree Practicalities*
For categorical levels $\mathcal{A}_j$, split $B \subset \mathcal{A}_j$ versus B^c.
For binary responses, set $I_a := \{i : X_{ij} = a\}$ and order levels by $\widehat{p}_a := |I_a|^{-1} \sum_{i \in I_a} Y_i$.
Missing values can be an NA level or use a surrogate split g' maximising $\sum_i \mathbf{1}\{g'(x_i) = g(x_i)\}$.
Many candidate splits inflate gain; control growth with $N_m \ge n_{\min}$, depth limits, validation, or pruning.

6.3 Ensembles, Random Forests, and Stacking

Definition 6.3.1. *Bagging and Random Forests*
Let $D_n = \{(x_i, y_i)\}_{i=1}^n$, $B \ge 1$, and $m \le p$.

(i) *Bagging*: draw $D_1^*, \dots, D_B^*$ from D_n, fit $T_1, \dots, T_B$, and aggregate $\widehat{f}_{\mathrm{bag}}(x) := B^{-1} \sum_{b=1}^B T_b(x)$ for regression, with voting or averaged probabilities for classification.

(ii) *Random forest*: in addition to bagging, at each node R of T_b choose $M_{b,R} \subseteq \{1, \dots, p\}$ with $|M_{b,R}| = m$ and optimise the split over features in $M_{b,R}$ only.

Bagging averages bootstrap perturbations; random split-feature subsets reduce $\mathrm{corr}(T_b, T_{b'})$ before averaging.

Method 6.3.2. *Bagged Prediction and Votes*
Let $D_n := \{(x_i, y_i)\}_{i=1}^n$ and $B \ge 1$; base outputs are $\widehat{f}$ for regression and $(\widehat{g}, \widehat{p}_1, \dots, \widehat{p}_K)$ for classification.

(i) For $b = 1, \dots, B$, draw a bootstrap sample D_b^* from D_n and fit the base learner, giving $\widehat{f}^{*b}$ or $(\widehat{g}^{*b}, \widehat{p}_{b1}, \dots, \widehat{p}_{bK})$.

(ii) Return the task-specific aggregate: for regression, $\widehat{f}_{\mathrm{bag}}(x) := B^{-1} \sum_{b=1}^B \widehat{f}^{*b}(x)$; for classification, either $\widehat{g}_{\mathrm{vote}}(x) \in \operatorname{argmax}_{1 \le k \le K} \sum_{b=1}^B \mathbf{1}\{\widehat{g}^{*b}(x) = k\}$ or $\bar{p}_k(x) := B^{-1} \sum_{b=1}^B \widehat{p}_{bk}(x)$ with $\widehat{g}_{\mathrm{prob}}(x) \in \operatorname{argmax}_{1 \le k \le K} \bar{p}_k(x)$.

Averaging reduces perturbation variance for unstable base learners.
Random split-feature subsets additionally reduce $\mathrm{corr}(T_b, T_{b'})$.

Theorem 6.3.3. *Squared-Error Benefit of Averaging*
Fix x, let $Z := \widehat{f}^*(x)$ be perturbation-random and independent of $Y \mid X = x$, and set $f_{\mathrm{ag}}(x) := \mathbb{E}_* Z$.
If $\mathbb{E}_* Z^2 < \infty$ and $\mathbb{E}_Y Y^2 < \infty$, then $\mathbb{E}_{*,Y}[(Y - Z)^2] = \mathbb{E}_Y[(Y - f_{\mathrm{ag}}(x))^2] + \mathbb{E}_*[(Z - f_{\mathrm{ag}}(x))^2]$.

Proof. Let $\mu := f_{\mathrm{ag}}(x) = \mathbb{E}_* Z$.
Since $Y - \mu$ and $Z - \mu$ are independent and $\mathbb{E}_*(Z - \mu) = 0$, $\mathbb{E}_{*,Y}[(Y - \mu)(Z - \mu)] = \mathbb{E}_Y(Y - \mu)\mathbb{E}_*(Z - \mu) = 0$.
Expanding $Y - Z = (Y - \mu) - (Z - \mu)$ gives $\mathbb{E}_{*,Y}[(Y - Z)^2] = \mathbb{E}_Y[(Y - \mu)^2] + \mathbb{E}_*[(Z - \mu)^2]$.

For independent fits Z_b, the average $\widehat{f}_B(x) := B^{-1}\sum_{b=1}^{B} Z_b$ has mean μ and perturbation variance $\mathrm{Var}_*(Z)/B$.
Structural Commentary: squared loss decomposes into target risk plus perturbation variance.
Key Insight: bagging estimates $\mathbb{E}_*\widehat{f}^*(x)$; unstable base learners gain most because $\mathrm{Var}_*(Z)$ is large. □

Theorem 6.3.4. *Correlated Ensemble Variance*
Let $Z_1,\dots,Z_B$ *have* $\mathrm{Var}(Z_b)=\sigma^2>0$ *and* $\mathrm{Corr}(Z_b,Z_c)=\rho$ *for* $b\neq c$.
For $\bar{Z}_B := \frac{1}{B}\sum_{b=1}^{B} Z_b$, $\mathrm{Var}(\bar{Z}_B)=\rho\sigma^2+\frac{1-\rho}{B}\sigma^2$; *hence, if* $\rho\geq 0$, $\mathrm{Var}(\bar{Z}_B)\to\rho\sigma^2$ *as* $B\to\infty$.

Proof. Since $\mathrm{Cov}(Z_b,Z_c)=\rho\sigma^2$ for $b\neq c$, $\mathrm{Var}(\bar{Z}_B)=B^{-2}\left\{\sum_{b=1}^{B}\mathrm{Var}(Z_b)+2\sum_{1\leq b<c\leq B}\mathrm{Cov}(Z_b,Z_c)\right\}$.
Substitution gives $\mathrm{Var}(\bar{Z}_B)=B^{-2}\left\{B\sigma^2+B(B-1)\rho\sigma^2\right\}=\rho\sigma^2+\frac{1-\rho}{B}\sigma^2$.
Structural Commentary: $\mathrm{Var}(\bar{Z}_B)=\rho\sigma^2+(1-\rho)\sigma^2/B$ separates shared and idiosyncratic variance.
Key Insight: increasing B shrinks only $(1-\rho)\sigma^2/B$; Random Forest lowers ρ by split randomisation. □

Algorithm 13: Random Forest

Require: Training data $(x_i,y_i)_{i=1}^n$, number of trees B, candidate split count $m\leq p$, tree stopping rule
Ensure: Trees $T_1,\dots,T_B$ and an aggregated predictor
  for $b=1,\dots,B$ **do**
    Draw a bootstrap resample S^{*b} of size n
    Grow a tree T_b on S^{*b}
    for each nonterminal node while growing T_b **do**
      Select m of the p predictors at random
      Choose the best split using only those m predictors
      Split the node and continue until the stopping rule is met
    end for
  end for
  For regression, return $\widehat{f}_{\mathrm{rf}}(x)=B^{-1}\sum_{b=1}^{B}T_b(x)$
  For classification, return a majority vote or averaged class-probability rule from $T_1(x),\dots,T_B(x)$

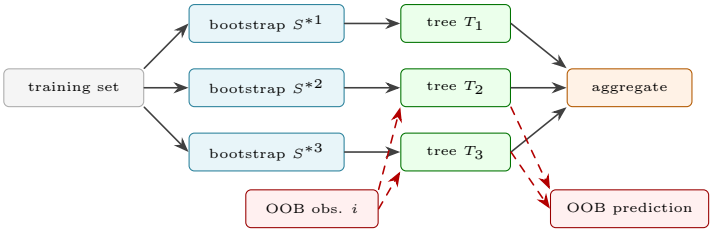

Figure 263: Blue bootstraps train green trees with m split candidates; black arrows aggregate, red dashed arrows give OOB prediction for i.

Method 6.3.5. *Out-of-Bag Error and Forest Importance*
Let I_b^* index bootstrap sample for T_b, $O_i := \{b : i\notin I_b^*\}$, $\mathcal{I}_{\mathrm{oob}} := \{i : |O_i|>0\}$, and L be prediction loss.

(i) For each $i\in\mathcal{I}_{\mathrm{oob}}$, construct $\widehat{h}_{\mathrm{oob}}(x_i)$: for regression, set $\widehat{h}_{\mathrm{oob}}(x_i) := |O_i|^{-1}\sum_{b\in O_i} T_b(x_i)$; for classification, use an OOB vote or averaged class-probability vector over $b\in O_i$.
(ii) Set $\widehat{R}_{\mathrm{oob}} := |\mathcal{I}_{\mathrm{oob}}|^{-1}\sum_{i\in\mathcal{I}_{\mathrm{oob}}} L(y_i,\widehat{h}_{\mathrm{oob}}(x_i))$ and update the estimate as B changes.

For feature j, let $\mathcal{S}_{bj}$ be j-splits of T_b, with gains ΔQ_{bs} and OOB losses $R_b^{\mathrm{oob}}, R_{bj}^{\pi}$ before/after permuting X_j.
Define $I_j^{\mathrm{split}} := \sum_{b=1}^{B}\sum_{s\in\mathcal{S}_{bj}}\Delta Q_{bs}$ and $I_j^{\pi} := B^{-1}\sum_{b=1}^{B}(R_{bj}^{\pi}-R_b^{\mathrm{oob}})$.
Both are fitted-forest summaries; I_j^{π} is not a refit-without-X_j loss contrast.

Remark 6.3.6. *Proximities, Tuning, and Overfitting*
Let $L_b(i)$ denote the terminal leaf of x_i in T_b, and set $O_{ij} := \{b : i,j\notin I_b^*\}$.
Define $P_{ij} := B^{-1}\sum_{b=1}^{B}\mathbf{1}\{L_b(i)=L_b(j)\}$ and, when $|O_{ij}|>0$, $P_{ij}^{\mathrm{oob}} := |O_{ij}|^{-1}\sum_{b\in O_{ij}}\mathbf{1}\{L_b(i)=L_b(j)\}$.
Thus P is a fitted-forest similarity; it is neither data-intrinsic nor generally a metric.
Tuning effects are parameter-specific: $m\downarrow$ lowers $\mathrm{corr}(T_b,T_{b'})$ and can raise bias; $n_{\min}\uparrow$ smooths terminal leaves; $B\uparrow$ lowers finite-ensemble Monte Carlo error but does not regularise the limiting forest.
Defaults $m_{\mathrm{cls}}\approx\sqrt{p}$ and $m_{\mathrm{reg}}\approx p/3$ are starting values only.

Method 6.3.7. *Error-Correcting Output Codes*
For training data $(x_i,y_i)_{i=1}^n$ with $y_i\in\{1,\dots,K\}$, let d_H denote Hamming distance.
A code matrix $C\in\{0,1\}^{K\times L}$ assigns class k the row $C_{k\cdot}$, and column ℓ induces binary labels $z_{i\ell} := C_{y_i\ell}$.

(i) Choose C with distinct rows, equivalently $d_H(C_{k\cdot},C_{k'\cdot})>0$ for $k\neq k'$.
(ii) For each $\ell=1,\dots,L$, fit $\widehat{p}_\ell(x)\approx\mathbb{P}(Z_\ell=1\mid X=x)$ to $(x_i,z_{i\ell})_{i=1}^n$.
(iii) For a test point x, compute $\delta_k(x) := \sum_{\ell=1}^{L}|C_{k\ell}-\widehat{p}_\ell(x)|$ for $k=1,\dots,K$ and return $\widehat{k}(x)\in\operatorname{argmin}_{1\leq k\leq K}\delta_k(x)$.

Row separation gives error correction; redundant columns reduce variance and absorb column errors.

Method 6.3.8. *Stacking and Bumping*
Let $\widehat{f}_1, \dots, \widehat{f}_M$ be fitted candidates, $\widehat{f}_m^{-i}(x_i)$ the out-of-sample prediction for observation i from candidate m, $\mathcal{W} \subseteq \mathbb{R}^M$ the admissible weight set, and L a loss. Commonly $\mathcal{W} := \{w \in \mathbb{R}^M : w_m \geq 0, \sum_{m=1}^{M} w_m = 1\}$.

(i) *Stacking*: set $R_{\mathrm{st}}(w) := \sum_{i=1}^{n} (y_i - \sum_{m=1}^{M} w_m \widehat{f}_m^{-i}(x_i))^2$.

Choose $\widehat{w}^{\mathrm{st}} \in \operatorname{argmin}_{w \in \mathcal{W}} R_{\mathrm{st}}(w)$, and return $\widehat{f}_{\mathrm{st}}(x) := \sum_{m=1}^{M} \widehat{w}_m^{\mathrm{st}} \widehat{f}_m(x)$.

(ii) *Bumping*: fit bootstrap candidates $\widehat{f}^{*1}, \dots, \widehat{f}^{*B}$, set $R_b^* := \sum_{i=1}^{n} L(y_i, \widehat{f}^{*b}(x_i))$.

Choose $\widehat{b} \in \operatorname{argmin}_{1 \leq b \leq B} R_b^*$, and return $\widehat{f}^{*\widehat{b}}$.

Coordinate weights $w = e_m$ give cross-validated model selection; general $w \in \mathcal{W}$ gives model averaging.
Stacking averages fixed candidates; bumping selects among resampled fits.

7 Additive Models and Adaptive Basis Methods

Additive and adaptive basis methods sit between rigid linear models and fully local learners. They keep prediction interpretable by building the fitted function from one-dimensional effects, boxes, hinges, or soft routing weights.

7.1 Generalized Additive Models

Definition 7.1.1. *Generalized Additive Model*
Let $X = (X_1, \dots, X_p)$, $x = (x_1, \dots, x_p)$, $\mu(x) := \mathbb{E}[Y \mid X = x]$, $\alpha \in \mathbb{R}$, and $f_j : \mathcal{X}_j \to \mathbb{R}$.

(i) *Additive mean*: $\mu(x) = \alpha + \sum_{j=1}^{p} f_j(x_j)$.

(ii) *Generalized additive model*: for a link g on the range of μ, $g(\mu(x)) = \alpha + \sum_{j=1}^{p} f_j(x_j)$.

(iii) *Identifiability*: with data $x_i = (x_{i1}, \dots, x_{ip})$, require $\sum_{i=1}^{n} f_j(x_{ij}) = 0$ for all j; this separates α from the f_j.

(iv) *Standard links*: Gaussian $g(t) = t$; binomial $g(t) = \log\left(\frac{t}{1-t}\right)$ or $\Phi^{-1}(t)$; counts $g(t) = \log t$.

Algorithm 14: Backfitting for an Additive Regression Model

Require: Training data $(x_i, y_i)_{i=1}^n$, univariate smoothers S_j for predictors $j = 1, \dots, p$
Ensure: Centered component fits $\widehat{f}_1, \dots, \widehat{f}_p$ and intercept $\widehat{\alpha}$
Set $\widehat{\alpha} \leftarrow n^{-1} \sum_{i=1}^n y_i$ and initialize $\widehat{f}_j(x_{ij}) \leftarrow 0$
repeat
  for $j = 1, \dots, p$ **do**
    $r_{ij} \leftarrow y_i - \widehat{\alpha} - \sum_{k \neq j} \widehat{f}_k(x_{ik})$ for $i = 1, \dots, n$
    $\widehat{f}_j \leftarrow S_j(r_{1j}, \dots, r_{nj})$
    Center $\widehat{f}_j$ by subtracting $n^{-1} \sum_i \widehat{f}_j(x_{ij})$
  end for
until the component functions change by less than the chosen tolerance

Method 7.1.2. *Local Scoring for Additive Logistic Regression*
For binary data $(x_i, y_i)_{i=1}^n$, $y_i \in \{0, 1\}$, use the additive predictor $\eta_i = \alpha + \sum_{j=1}^{p} f_j(x_{ij})$.

(i) *Mean*: set $p_i = \frac{1}{1+\exp(-\eta_i)}$ for $i = 1, \dots, n$.

(ii) *Working pair*: set $z_i = \eta_i + \frac{y_i - p_i}{p_i(1-p_i)}$ and $w_i = p_i(1 - p_i)$.

(iii) *Weighted fit*: update $(\alpha, f_1, \dots, f_p)$ by weighted backfitting, minimizing $\sum_{i=1}^{n} w_i \{z_i - \alpha - \sum_{j=1}^{p} f_j(x_{ij})\}^2$.

(iv) *Iteration*: recompute η_i and repeat (i)–(iii) until the additive predictor converges.

7.2 Adaptive Basis and Rule Methods

Method 7.2.1. *MARS: Adaptive Piecewise-Linear Basis Search*
For data $(x_i, y_i)_{i=1}^n$ and $x \in \mathbb{R}^p$, use reflected hinges $(x_j - t)_+$ and $(t - x_j)_+$ with knots $t \in \{x_{ij} : i = 1, \dots, n\}$.
A MARS fit is $\widehat{f}(x) = \widehat{\beta}_0 + \sum_{m=1}^{M} \widehat{\beta}_m h_m(x)$, with each h_m a selected hinge product of order $\leq d_{\max}$.

(i) *Forward pass*: add the admissible hinge pair or product giving the largest decrease in $\sum_{i=1}^{n} (y_i - \widehat{f}(x_i))^2$.

(ii) *Backward pass*: prune the forward sequence.

Score model λ by $\mathrm{GCV}(\lambda) = \frac{\sum_{i=1}^{n} (y_i - \widehat{f}_\lambda(x_i))^2}{\{1 - M(\lambda)/n\}^2}$, with effective size $M(\lambda)$.

(iii) *Selection*: choose the pruned model with smallest GCV.

The cap $d_{\max}$ controls interaction order; $d_{\max} = 1$ gives an additive piecewise-linear model.

Method 7.2.2. *PRIM Box Peeling*
For data $(x_i, y_i)_{i=1}^n$, a box $B = \prod_{j=1}^p [a_j, b_j]$, set $I(B) = \{i : x_i \in B\}$ and $\bar{y}(B) = |I(B)|^{-1} \sum_{i \in I(B)} y_i$.

Fix peeling fraction $\rho \in (0,1)$ and minimum size $m_{\min}$.

(i) *Initialise*: take B to be the full training box.
(ii) *Peel*: while $|I(B)| > m_{\min}$, replace B by the one-face deletion $B' \subset B$ removing a ρ fraction of $I(B)$ with maximal $\bar{y}(B')$.
(iii) *Paste*: expand a face when the resulting box B' increases $\bar{y}(B')$.
(iv) *Select*: choose B^* from the peeling-pasting path by validation or constraints; remove $I(B^*)$ and repeat for additional boxes.

PRIM returns high-mean rectangular regions rather than a recursive partition.

Algorithm 15: Patient Rule Induction Method

Require: Data $(x_i, y_i)_{i=1}^n$, peel fraction ρ, minimum size $m_{\min}$, box count q
Ensure: Selected boxes $\mathcal{S} = (B_1^*, \dots, B_q^*)$
  $D \leftarrow \{1, \dots, n\}$, $\mathcal{S} \leftarrow ()$
  for $r = 1, \dots, q$ **do**
    $I_D(B) := \{i \in D : x_i \in B\}$, $\bar{y}_D(B) := |I_D(B)|^{-1} \sum_{i \in I_D(B)} y_i$
    $B \leftarrow \prod_{j=1}^p [\min_{i \in D} x_{ij}, \max_{i \in D} x_{ij}]$, $\mathcal{P} \leftarrow \{B\}$
    repeat
      $\mathcal{C}_\rho(B) \leftarrow \{B' \subset B : |I_D(B')| \geq (1-\rho)|I_D(B)|\}$ among one-face deletions
      $B \leftarrow \operatorname{argmax}_{B' \in \mathcal{C}_\rho(B)} \bar{y}_D(B')$, $\mathcal{P} \leftarrow \mathcal{P} \cup \{B\}$
    until $|I_D(B)| \leq m_{\min}$
    $\mathcal{E}(B) \leftarrow \{B' \supset B : \bar{y}_D(B') > \bar{y}_D(B)\}$ among one-face expansions
    if $\mathcal{E}(B) \neq \varnothing$ **then**
      $B \leftarrow \operatorname{argmax}_{B' \in \mathcal{E}(B)} \bar{y}_D(B')$, $\mathcal{P} \leftarrow \mathcal{P} \cup \{B\}$
    end if
    $B_r^* \leftarrow \text{Select}(\mathcal{P})$, $\mathcal{S} \leftarrow (\mathcal{S}, B_r^*)$, $D \leftarrow D \setminus I_D(B_r^*)$
  end for

Remark 7.2.3. *Adaptive Structure Versus Tree Structure*

(i) *CART*: represents $\mathcal{X}$ by a binary-tree partition; leaves carry constant local fits.
(ii) *PRIM*: represents high-response regions by boxes $B \subset \mathcal{X}$ without requiring a tree partition.
(iii) *MARS*: represents $\widehat{f}$ by hinge products h_m and coefficients $\widehat{\beta}_m$, linear in $\widehat{\beta}$.

CART emphasizes tree interpretability; PRIM keeps box rules; MARS trades boxes for smoother hinge bases.

Remark 7.2.4. *Soft Trees and Mixtures of Experts*
Let node v have children $1, \dots, K_v$, gates γ_{vk}, leaves $\mathcal{L}$, leaf path $P(\ell)$, and expert $m_\ell(x; \theta_\ell)$.

(i) *Gate*: $g_{vk}(x) = \frac{\exp(\gamma_{vk}^\top x)}{\sum_{r=1}^{K_v} \exp(\gamma_{vr}^\top x)}$, so $\sum_{k=1}^{K_v} g_{vk}(x) = 1$.
(ii) *Leaf weight*: $\pi_\ell(x) = \prod_{(v,k) \in P(\ell)} g_{vk}(x)$.
(iii) *Prediction*: $\widehat{m}(x) = \sum_{\ell \in \mathcal{L}} \pi_\ell(x) m_\ell(x; \theta_\ell)$.
(iv) *Fitting*: estimate (γ, θ) by maximizing the mixture likelihood, often via EM-type alternating updates.

Soft trees replace hard branch indicators by input-dependent probabilities while retaining tree-local experts.

Remark 7.2.5. *Missing Predictors*
For observation i, let $O_i = \{j : x_{ij} \text{ observed}\}$ and $M_i = \{1, \dots, p\} \setminus O_i$; specify prediction from x_{i,O_i}.

(i) *Trees*: use surrogate splits or a categorical missing level.
(ii) *Single imputation*: replace x_{ij}, $j \in M_i$, by $\widehat{x}_{ij}$ before fitting additive or basis models.
(iii) *Model-based imputation*: estimate $\widehat{x}_{ij}$ from a feature model $\widehat{m}_j(x_{i,O_i})$.
(iv) *Multiple imputation*: fit on completed data sets $X^{(1)}, \dots, X^{(L)}$ to propagate imputation uncertainty.

Single imputation treats $\widehat{x}_{ij}$ as observed; multiple imputation propagates missing-data uncertainty.

8 Boosting and Additive Trees

8.1 Stagewise Boosting

Boosting is a stagewise way to build a strong predictor from simple base learners. The key object is an additive score: later terms are trained against the current fit rather than fit independently.

Definition 8.1.1. *Boosting*
Let $\mathcal{H}$ be a weak-learner class and set $F_0 \equiv 0$.

(i) *Additive score*: $F_M(x) = \sum_{m=1}^M \alpha_m h_m(x)$, with $h_m \in \mathcal{H}$ and $\alpha_m \in \mathbb{R}$.
(ii) *Stagewise update*: choose (α_m, h_m) using residuals, gradients, or weights from F_{m-1}; $F_m = F_{m-1} + \alpha_m h_m$.
(iii) *Prediction*: regression uses $F_M(x)$; binary classification uses $\operatorname{sign} F_M(x)$; multiclass scores use $\operatorname{argmax}_k F_{Mk}(x)$.

Boosting is the stagewise construction of F_M from weak learners.

Method 8.1.2. *Forward Stagewise Additive Modeling*
Let $b(x; \gamma)$ range over a basis class and let $L(y, f)$ be a pointwise loss.

(i) *Initialize*: choose f_0.

(ii) *Fit step*: at stage m, take $(\beta_m, \gamma_m) \in \operatorname{argmin}_{\beta,\gamma} \sum_{i=1}^{n} L\{y_i, f_{m-1}(x_i) + \beta b(x_i;\gamma)\}$.
(iii) *Update*: set $f_m(x) = f_{m-1}(x) + \beta_m b(x;\gamma_m)$.
(iv) *Stop*: after M stages, return f_M.

Previously added terms are fixed; for squared-error loss, step (ii) fits the current residuals.

Remark 8.1.3. *Stagewise Paths and Lasso Geometry*

Let $\mathcal{T} = \{T_1, \dots, T_K\}$ be a tree dictionary and $f_\alpha(x) = \sum_{k=1}^{K} \alpha_k T_k(x)$.

For large K, direct least squares is ill posed; a penalized target is $\min_\alpha \{\sum_{i=1}^{n} (y_i - f_\alpha(x_i))^2 + \lambda J(\alpha)\}$.

With $J(\alpha) = \sum_{k=1}^{K} |\alpha_k|$, active trees are sparse.

Small-step boosting traces a forward-stagewise path through this dictionary; shrinkage shortens each move.

8.2 AdaBoost and Exponential Loss

Algorithm 16: AdaBoost.M1

Require: Binary training data (x_i, y_i) with $y_i \in \{-1, 1\}$, weighted classification learner, iterations M
Ensure: Classifier $G(x)$

```
Initialize w_i ← 1/n for i = 1, ..., n
for m = 1, ..., M do
    Fit a classifier G_m using observation weights w_i
    err_m ← (Σ_{i=1}^n w_i 1_{y_i ≠ G_m(x_i)}) / (Σ_{i=1}^n w_i)
    α_m ← log((1 − err_m)/err_m)
    for i = 1, ..., n do
        w_i ← w_i exp{α_m 1_{y_i ≠ G_m(x_i)}}
    end for
end for
return G(x) = sign(Σ_{m=1}^M α_m G_m(x))
```

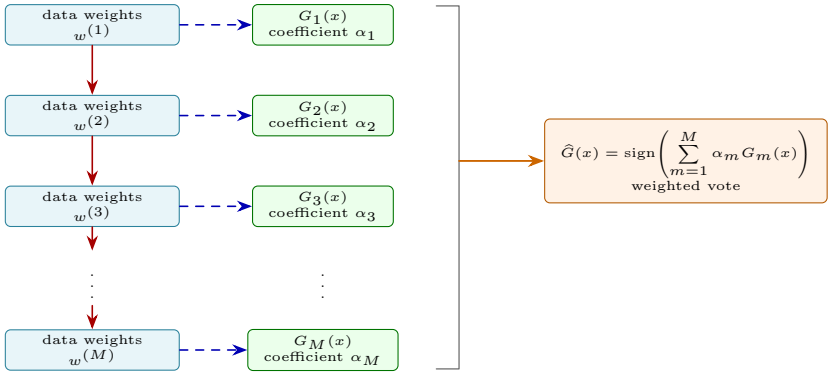

Figure 264: AdaBoost alternates fitting weak learners with updating observation weights, then aggregates them by a weighted vote.

Proposition 8.2.1. *AdaBoost as Exponential-Loss Stagewise Fitting*

For $y_i \in \{-1, 1\}$, use $L(y, f) = \exp(-yf)$. Fix stage m, current score f_{m-1}, weights $w_i^{(m)} := \exp\{-y_i f_{m-1}(x_i)\}$, and classifier $G : \mathcal{X} \to \{-1, 1\}$. Set $Q_m(\beta, G) := \sum_{i=1}^{n} L\{y_i, f_{m-1}(x_i) + \beta G(x_i)\}$ and $E_m(G) := \sum_{i=1}^{n} w_i^{(m)} \mathbf{1}_{\{y_i \neq G(x_i)\}}$.

(i) For fixed $\beta > 0$, $\operatorname{argmin}_G Q_m(\beta, G) = \operatorname{argmin}_G E_m(G)$.

(ii) If $G_m \in \operatorname{argmin}_G E_m(G)$ and $\mathrm{err}_m := E_m(G_m) / \sum_{i=1}^{n} w_i^{(m)}$, then $\beta_m = \frac{1}{2} \log\left(\frac{1-\mathrm{err}_m}{\mathrm{err}_m}\right)$ minimizes $Q_m(\beta, G_m)$.

(iii) The induced weights satisfy $w_i^{(m+1)} = w_i^{(m)} \exp\{-\beta_m y_i G_m(x_i)\}$. With $\alpha_m = 2\beta_m$, a misclassified point has relative factor $\exp(\alpha_m)$.

Remark 8.2.2. *Exponential Loss and Class Probabilities*

For $Y \in \{-1, 1\}$ and $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$, the population minimizer of $\mathbb{E}\{\exp(-Yf(X)) \mid X = x\}$ is $f^*(x) = \frac{1}{2} \log \frac{\eta(x)}{1-\eta(x)}$. Hence $\operatorname{sign} f^*(x)$ is the Bayes rule. The binomial deviance $\log\{1 + \exp(-2Yf(X))\}$ has the same minimizer but penalizes large negative margins less aggressively on finite samples.

8.3 Gradient Boosted Trees and Rule Ensembles

Definition 8.3.1. *Additive Tree Expansion*

Let $R_1, \dots, R_J$ be terminal regions, $\gamma_j \in \mathbb{R}$, and $\Theta = \{(R_j, \gamma_j) : 1 \le j \le J\}$.

(i) *Tree*: $T(x; \Theta) = \sum_{j=1}^{J} \gamma_j \mathbf{1}_{\{x \in R_j\}}$.

(ii) *Additive expansion*: $f_M(x) = \sum_{m=1}^{M} T(x;\Theta_m)$.

(iii) *Terminal fit*: with regions R_{jm} fixed, $\widehat{\gamma}_{jm} \in \operatorname{argmin}_{\gamma} \sum_{x_i \in R_{jm}} L\{y_i, f_{m-1}(x_i) + \gamma\}$.

Algorithm 17: Gradient Tree Boosting

Require: Training data (x_i, y_i), differentiable loss $L(y, f)$, tree sizes J_m, learning rate $0 < \nu \leq 1$, iterations M
Ensure: Additive predictor f_M

$f_0(x) \leftarrow \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma)$
for $m = 1, \ldots, M$ **do**
  for $i = 1, \ldots, n$ **do**
    $r_{im} \leftarrow -\left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f=f_{m-1}}$
  end for
  Fit a regression tree to targets r_{im}, giving terminal regions R_{jm}, $j = 1, \ldots, J_m$
  for $j = 1, \ldots, J_m$ **do**
    $\gamma_{jm} \leftarrow \arg\min_{\gamma} \sum_{x_i \in R_{jm}} L\{y_i, f_{m-1}(x_i) + \gamma\}$
  end for
  $f_m(x) \leftarrow f_{m-1}(x) + \nu \sum_{j=1}^{J_m} \gamma_{jm} \mathbf{1}_{\{x \in R_{jm}\}}$
end for
return f_M

Remark 8.3.2. *Pseudo-Residuals*
At stage m, pseudo-residuals are negative pointwise gradients $r_{im} := -\partial_f L(y_i, f)|_{f=f_{m-1}(x_i)}$.
Gradient boosting fits the next tree to (x_i, r_{im}).

(i) *Squared error*: $r_{im} = y_i - f_{m-1}(x_i)$.
(ii) *Absolute error*: $r_{im} = \operatorname{sign}\{y_i - f_{m-1}(x_i)\}$, away from zero residuals.
(iii) *Multinomial deviance*: $r_{ikm} = \mathbf{1}_{\{y_i = \mathcal{G}_k\}} - p_k(x_i)$, where p_k is the current softmax probability.

Algorithm 18: Gradient Boosting for K-Class Classification

Require: Encoded labels $y_{ik} \in \{0, 1\}$, classes $k = 1, \ldots, K$, tree sizes J_m, iterations M
Ensure: Class-score functions $\widehat{f}_1, \ldots, \widehat{f}_K$
Initialize $f_{k0}(x) \leftarrow 0$ for $k = 1, \ldots, K$
for $m = 1, \ldots, M$ **do**
  $p_k(x) \leftarrow \frac{\exp(f_{k,m-1}(x))}{\sum_{\ell=1}^{K} \exp(f_{\ell,m-1}(x))}$ for $k = 1, \ldots, K$
  for $k = 1, \ldots, K$ **do**
    Compute $r_{ikm} \leftarrow y_{ik} - p_k(x_i)$ for $i = 1, \ldots, n$
    Fit a regression tree to targets r_{ikm}, giving terminal regions R_{jkm}, $j = 1, \ldots, J_m$
    for $j = 1, \ldots, J_m$ **do**
      $\gamma_{jkm} \leftarrow \frac{K-1}{K} \frac{\sum_{x_i \in R_{jkm}} r_{ikm}}{\sum_{x_i \in R_{jkm}} |r_{ikm}|(1 - |r_{ikm}|)}$
    end for
    $f_{km}(x) \leftarrow f_{k,m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jkm} \mathbf{1}_{\{x \in R_{jkm}\}}$
  end for
end for
return $\widehat{f}_k(x) = f_{kM}(x)$ for $k = 1, \ldots, K$

Remark 8.3.3. *Tree Size, Shrinkage, and Subsampling*
Boosted-tree complexity is controlled by iterations M, terminal counts J_m, shrinkage ν, and sampling.

(i) *Iterations*: larger M adds more trees and increases model capacity.
(ii) *Tree size*: a common J caps interactions; stumps ($J = 2$) give main effects, and larger J allows order up to $J - 1$.
(iii) *Shrinkage*: use $f_m = f_{m-1} + \nu T_m$ with $0 < \nu \leq 1$; smaller ν usually requires larger M.
(iv) *Subsampling*: fit T_m on random observations to reduce cost and add averaging-like randomness.

These tuning choices trade optimization speed, interaction depth, and test risk.

Method 8.3.4. *Relative Importance and Partial Dependence*
Let f be a fitted tree ensemble. For $S \subset \{1, \ldots, p\}$, write $C = S^c$ and $x_i = (x_{iS}, x_{iC})$.

(i) *Relative importance*: for predictor j, sum the split-improvement scores over tree nodes splitting on j.
For an ensemble, average over trees and rescale if desired.

(ii) *Partial dependence*: define $f_S(x_S) = \mathbb{E}_{X_C}\{f(x_S, X_C)\}$ and estimate $\widehat{f}_S(x_S) = n^{-1} \sum_{i=1}^{n} f(x_S, x_{iC})$.

Partial dependence averages over the marginal distribution of X_C.
It equals $\mathbb{E}\{f(X) \mid X_S = x_S\}$ only under compatible dependence structure.

Principle 8.3.5. *Bet on Sparsity*

Let $\mathcal{T} = \{T_1, \ldots, T_K\}$ and $f_\alpha = \sum_{k=1}^{K} \alpha_k T_k$, with support $S(\alpha) = \{k : \alpha_k \neq 0\}$.

An ℓ^1 penalty is useful when a low-risk predictor has $|S(\alpha)| \ll K$ and $|S(\alpha)|$ is small relative to n and noise.
Dense signals with many small α_k remain hard to estimate.
Sparsity is meaningful only relative to n, noise level, and $\mathcal{T}$.

Method 8.3.6. *Lasso Post-Processing of Ensembles*
Let a fitted ensemble generate the fixed dictionary $\mathcal{T}_L = \{T_1, \dots, T_M\}$ and let L be the chosen prediction loss.

(i) *Fit path*: compute $\widehat{\alpha}(\lambda) \in \operatorname{argmin}_{\alpha}\{\sum_{i=1}^{n} L(y_i, \alpha_0 + \sum_{m=1}^{M} \alpha_m T_m(x_i)) + \lambda \sum_{m=1}^{M} |\alpha_m|\}$.
(ii) *Select*: choose λ by validation or an information criterion.
(iii) *Return*: use $\widehat{f}_\lambda(x) = \widehat{\alpha}_0 + \sum_{m:\widehat{\alpha}_m(\lambda)\neq 0} \widehat{\alpha}_m(\lambda) T_m(x)$.

The post-processed model can use fewer trees.
A useful generated dictionary is relevant and not nearly duplicated, with subsampling one way to diversify it.

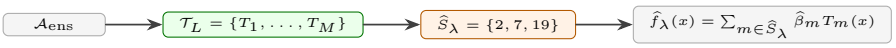

Figure 265: An ensemble generator $\mathcal{A}_{\text{ens}}$ builds the dictionary $\mathcal{T}_L$; lasso selects the active set $\widehat{S}_\lambda$ for the sparse predictor $\widehat{f}_\lambda$.

Method 8.3.7. *Rule Ensembles*
Let $T_1, \dots, T_M$ be fitted trees. A node path p defines the rule $r_p(x) = \mathbf{1}_{\{x \text{ satisfies all splits on } p\}}$.

(i) *Extract*: form $\mathcal{T}^m_{\text{rule}} = \{r_p : p \text{ is a node path in } T_m\}$.
(ii) *Aggregate*: set $\mathcal{T}_{\text{rule}} = \bigcup_{m=1}^{M} \mathcal{T}^m_{\text{rule}}$.
(iii) *Fit*: regress y on $\mathcal{T}_{\text{rule}}$ by lasso or another regularized linear method.
(iv) *Augment*: optionally add original variables x_j as linear terms.

Rules encode split interactions; added linear terms encode main linear effects.

9 High-Dimensional Supervised Learning

High-dimensional supervised learning studies prediction problems in which the number of candidate features is comparable to, or much larger than, the number of observations. The central difficulty is not merely computational size: many rules can fit the training data too well, while the scientific question often asks which features or feature groups are driving the prediction.

9.1 High-Dimensional Linear Rules

Principle 9.1.1. *Less Fitting in the $p \gg n$ Regime*
Let $X \in \mathbb{R}^{n\times p}$, $r = \operatorname{rank}(X)$, and $p \gg n$.

(i) If $r < p$, least-squares coefficients are nonunique since $\ker(X) \neq \{\mathbf{0}\}$.
(ii) Full-covariance discriminant rules are singular when covariance rank is $< p$.
Pooled rank is $\leq n - K$, and class-k rank is $\leq n_k - 1$.
(iii) Impose effective dimension $< p$ via shrinkage, diagonal covariance, row-space fitting, sparsity, or screening.

Measure complexity by $\operatorname{df}(\lambda)$, as in Proposition 2.1.4: ridge keeps coordinates; sparse rules select features.

Method 9.1.2. *Diagonal LDA and Nearest Shrunken Centroids*
For classes $k = 1, \dots, K$ and features $j = 1, \dots, p$, let $\bar{x}_{kj}$ be the class mean, $\bar{x}_{\cdot j}$ the overall mean, s_j the pooled within-class standard deviation, π_k the class prior, n_k the class size, $m_k := (\frac{1}{n_k} - \frac{1}{n})^{1/2}$, and $s_0 > 0$.

(i) *Diagonal LDA*: score x by $\delta_k(x) := -\sum_{j=1}^{p} \frac{(x_j - \bar{x}_{kj})^2}{s_j^2} + 2\log \pi_k$ and set $\widehat{G}(x) := \arg\max_k \delta_k(x)$.
(ii) *Nearest shrunken centroids*: for $\Delta \geq 0$, set $d_{kj} := \frac{\bar{x}_{kj} - \bar{x}_{\cdot j}}{m_k(s_j + s_0)}$, $d'_{kj} := \operatorname{sign}(d_{kj})(|d_{kj}| - \Delta)_+$,
and $\bar{x}'_{kj} := \bar{x}_{\cdot j} + m_k(s_j + s_0)d'_{kj}$; evaluate the diagonal LDA score with $\bar{x}'_{kj}$ in place of $\bar{x}_{kj}$.

Diagonal LDA is nearest-centroid classification after coordinatewise standardization. In nearest shrunken centroids, feature j is removed iff $d'_{kj} = 0$ for all k; increasing Δ increases shrinkage and sparsity.

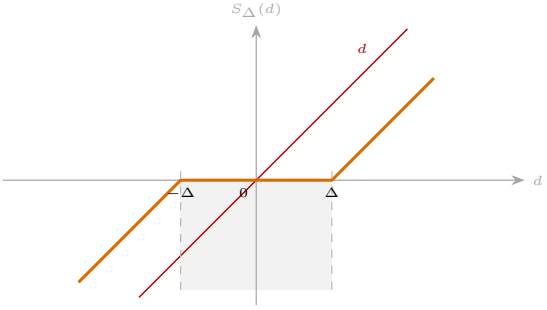

Figure 266: Soft thresholding $S_\Delta(d) = \operatorname{sign}(d)(|d| - \Delta)_+$. The shaded band $|d| \leq \Delta$ maps to zero; outside it, contrasts shrink toward zero by Δ.

Method 9.1.3. *Quadratic Regularization for Linear Rules*
Let $\eta_k(x) := \beta_{k0} + x^T\beta_k$, $\lambda > 0$, and $\gamma \in [0,1]$.

(i) *L_2 multinomial logits*: set $p_{\beta,k}(x) := \frac{\exp(\eta_k(x))}{\sum_{\ell=1}^{K}\exp(\eta_\ell(x))}$ and choose $\widehat{\beta} \in \arg\max_{\beta}\{\sum_{i=1}^{n}\log p_{\beta,g_i}(x_i) - \frac{\lambda}{2}\sum_{k=1}^{K}\|\beta_k\|_2^2\}$; leave β_{k0} unpenalized.
(ii) *Covariance shrinkage*: set $\widehat{\Sigma}_\gamma := \gamma\widehat{\Sigma} + (1-\gamma)\operatorname{diag}(\widehat{\Sigma})$ and use $\widehat{\Sigma}_\gamma$ in the discriminant score.

Both variants impose quadratic stabilization without sparsity. The endpoint $\gamma = 0$ gives diagonal LDA; increasing γ reintroduces off-diagonal covariance information.

Proposition 9.1.4. *Row-Space Reduction for Quadratic Penalties*
Let $X \in \mathbb{R}^{n\times p}$ have rank r and thin SVD $X = UDV^T = RV^T$, where $R := UD \in \mathbb{R}^{n\times r}$ and $V^TV = I_r$.
Fix $y \in \mathbb{R}^n$ and $\lambda > 0$.

(i) Ridge regression satisfies $\widehat{\beta}_\lambda^{\text{ridge}} = V(R^TR + \lambda I_r)^{-1}R^Ty$.
(ii) If $F(\beta_0,\beta) := \sum_{i=1}^{n} L(y_i, \beta_0 + x_i^T\beta) + \lambda\|\beta\|_2^2$ has a minimiser, then one satisfies $\widehat{\beta} \in \operatorname{row}(X) = \operatorname{col}(V)$.

Thus quadratic-penalty fitting may be performed in the r row-space coordinates r_i^T and mapped back by $\widehat{\beta} = V\widehat{\theta}$.

Proof. Every $\beta \in \mathbb{R}^p$ decomposes uniquely as $\beta = V\theta + z$, where $\theta = V^T\beta$ and $z \in \operatorname{col}(V)^\perp = \ker(X)$.
Then $X\beta = R\theta$ and $\|\beta\|_2^2 = \|\theta\|_2^2 + \|z\|_2^2$.

(i) For ridge, dropping z preserves fitted values and weakly decreases the penalty, so a minimiser has $z = 0$.
Thus $\widehat{\theta} \in \arg\min_{\theta}\{\|y - R\theta\|_2^2 + \lambda\|\theta\|_2^2\}$, giving $\widehat{\theta} = (R^TR + \lambda I_r)^{-1}R^Ty$ and $\widehat{\beta} = V\widehat{\theta}$.
(ii) If $(\widehat{\beta}_0, \widehat{\beta})$ minimises F and $\widehat{\beta} = V\widehat{\theta} + \widehat{z}$, then $F(\widehat{\beta}_0, V\widehat{\theta}) \le F(\widehat{\beta}_0, \widehat{\beta})$, with the same linear predictors and no larger penalty. Hence $(\widehat{\beta}_0, V\widehat{\theta})$ is also a minimiser and $V\widehat{\theta} \in \operatorname{col}(V)$.

Structural Commentary: Kernel components affect only the quadratic penalty, not the fitted values.
Key Insight: Quadratic penalties reduce fitting to $\operatorname{row}(X)$. □

9.2 Sparse and Structured Penalties

Method 9.2.1. *Sparse and Structured Penalties*
Let $\lambda, \lambda_1, \lambda_2 \ge 0$ and $\alpha \in [0,1]$. Sparse penalties replace quadratic shrinkage by nonsmooth coordinate penalties.

(i) *Lasso multinomial logits*: maximize $\sum_{i=1}^{n}\log\mathbb{P}(g_i \mid x_i) - \lambda\sum_{k=1}^{K}\sum_{j=1}^{p}|\beta_{kj}|$.
(ii) *Elastic net*: replace $|\beta_{kj}|$ by $\alpha|\beta_{kj}| + (1-\alpha)\beta_{kj}^2$.
(iii) *Fused lasso*: for ordered features, use $\lambda_1\sum_{j=1}^{p}|\beta_j| + \lambda_2\sum_{j=1}^{p-1}|\beta_{j+1} - \beta_j|$.

The L_1 term selects coordinates; the quadratic term shrinks correlated features.
Adjacent differences favor piecewise-constant coefficients.

Method 9.2.2. *Dantzig Selector*
Assume centered y and standardised columns x_j of X.
For residual $r(\beta) := y - X\beta$, the Dantzig selector has two equivalent constrained forms.

(i) *Correlation budget*: choose $\widehat{\beta}_s^{\text{DS}} \in \arg\min_{\beta}\|\beta\|_1$ subject to $\|X^Tr(\beta)\|_\infty \le s$.
(ii) *Coefficient budget*: choose $\widehat{\beta}_t^{\text{DS}} \in \arg\min_{\|\beta\|_1\le t}\|X^Tr(\beta)\|_\infty$.

Both Dantzig and lasso control $\|\beta\|_1$ and monitor residual correlations $x_j^Tr(\beta)$.
Lasso minimises residual sum of squares, while Dantzig directly bounds $\|X^Tr(\beta)\|_\infty$.

Method 9.2.3. *Grouped Lasso*
Let $\{1,\dots,p\} = G_1 \sqcup \cdots \sqcup G_L$, with $|G_\ell| = p_\ell$, design block $X_\ell \in \mathbb{R}^{n\times p_\ell}$, coefficient block $\beta_\ell \in \mathbb{R}^{p_\ell}$, and $\lambda \ge 0$.
The grouped lasso solves $(\widehat{\beta}_0, \widehat{\beta}_1, \dots, \widehat{\beta}_L) \in \arg\min_{\beta_0,\beta_1,\dots,\beta_L}\left\{\left\|y - \beta_0\mathbf{1} - \sum_{\ell=1}^{L}X_\ell\beta_\ell\right\|_2^2 + \lambda\sum_{\ell=1}^{L}\sqrt{p_\ell}\,\|\beta_\ell\|_2\right\}$.
Since $\|\beta_\ell\|_2 = 0$ iff $\beta_\ell = \mathbf{0}$, the penalty selects or removes entire groups; $\sqrt{p_\ell}$ normalizes group size.

Method 9.2.4. *Relaxed, Adaptive, and Nonconvex Lasso Variants*
Let $\widehat{\beta}_\lambda^{\text{lasso}}$ be the lasso estimate, $A_\lambda := \{j : \widehat{\beta}_{\lambda,j}^{\text{lasso}} \ne 0\}$, $\widetilde{\beta}$ an initial estimate, $\nu > 0$, and $a \ge 2$.

(i) *Relaxed lasso*: fit the lasso to get A_λ, then refit on X_{A_λ} by least squares or with penalty $\phi\lambda$, $0 \le \phi \le 1$.
(ii) *Adaptive lasso*: set $w_j := \frac{1}{|\widetilde{\beta}_j|^\nu}$ and minimize $\frac{1}{2}\|y - X\beta\|_2^2 + \lambda\sum_{j=1}^{p}w_j|\beta_j|$.
(iii) *SCAD*: use the nonconvex penalty $J_{a,\lambda}$ with $J'_{a,\lambda}(t) := \lambda\operatorname{sign}(t)\left[\mathbf{1}_{\{|t|\le\lambda\}} + \frac{(a\lambda - |t|)_+}{(a-1)\lambda}\mathbf{1}_{\{|t|>\lambda\}}\right]$.

All three reduce lasso shrinkage bias for large signals.
Relaxed and adaptive lasso keep convex refits or weighted convex fits, while SCAD is nonconvex.

Remark 9.2.5. *Three Roles for Feature Selection*
In high-dimensional learning, feature selection has three distinct roles.

(i) *Preprediction screening*: reduce variance before fitting, as in nearest shrunken centroids.
(ii) *Sparse prediction*: estimate a sparse predictive rule, as in lasso or elastic-net logistic regression.
(iii) *Scientific discovery*: report selected features for separate assessment.
Multiplicity control is then inferential, not predictive.

9.3 Latent Factors and Distance-Based Classifiers

Algorithm 19: Supervised Principal Components

Require: Standardized feature matrix X, outcome y, association scores s_j, thresholds $0 \le \theta_1 < \cdots < \theta_K$, candidate component counts $\mathcal{M}$
Ensure: A selected principal-component predictor
  for $k = 1, \dots, K$ **do**
    $A_k \leftarrow \{j : |s_j| > \theta_k\}$
    for $m \in \mathcal{M}$ **do**
      Compute the first m principal components of the submatrix X_{A_k}
      Fit the outcome model using those components as predictors
      Estimate held-out prediction error by cross-validation
    end for
  end for
  Choose (θ_k, m) by the cross-validation criterion and refit using the selected columns

Remark 9.3.1. *Screening Before Components*
Let s_j measure the marginal association between feature x_j and outcome y, and set $A_\theta := \{j : |s_j| > \theta\}$.
Supervised principal components compute PCs of X_{A_θ} rather than all of X.

(i) *Regression score*: take s_j as a standardized univariate regression coefficient for y on x_j.
(ii) *Survival score*: take s_j as the one-feature Cox score for x_j.

Ordinary PCs follow variance in X alone; screening makes the component outcome-guided.
The resulting component is a denoised latent factor from weak but aligned features.

Method 9.3.2. *Preconditioned Lasso*
Given X, y, and $\lambda > 0$, preconditioning uses a denoised response before sparse refitting.

(i) Fit supervised principal components and compute fitted values $\widehat{y} = (\widehat{y}_1, \dots, \widehat{y}_n)^T$.
(ii) Fit $\widehat{\beta}^{\mathrm{pre}} \in \arg\min_\beta \{\frac{1}{2} \|\widehat{y} - X\beta\|_2^2 + \lambda \|\beta\|_1\}$.

Stage (i) denoises the target; stage (ii) selects original features that reproduce $\widehat{y}$.

Method 9.3.3. *Inner-Product and Distance-Only Classifiers*
Let $K_{ii'} := \langle x_i, x_{i'} \rangle$ be a positive semidefinite Gram matrix, g_i class labels, $n_k = \#\{i : g_i = k\}$, and $M := \frac{1}{n}\mathbf{1}\mathbf{1}^T$.

(i) *Nearest neighbor*: compute $\|x_i - x_{i'}\|^2 = K_{ii} + K_{i'i'} - 2K_{ii'}$.
(ii) *Nearest centroid*: for test point x_0, compute $\|x_0 - \bar{x}_k\|^2 = \langle x_0, x_0 \rangle - \frac{2}{n_k} \sum_{g_i = k} \langle x_0, x_i \rangle + \frac{1}{n_k^2} \sum_{g_i = k} \sum_{g_{i'} = k} \langle x_i, x_{i'} \rangle$.
(iii) *Distance-only input*: for squared distances $\Delta_{ii'}^2$, set $B_{ii'} := -\frac{1}{2}\Delta_{ii'}^2$ and $\widetilde{K} := (I - M)B(I - M)$.

Thus nearest-neighbor, nearest-centroid, kernel PCA, and related classifiers require only inner products or centered distances, not explicit feature vectors.

Example 9.3.4. *String Kernels*
For strings over a finite alphabet, let $\mathcal{A}_m$ be the set of length-m substrings, and let $\phi_a(x)$ count occurrences of a in string x. Define $\Phi_m(x) := (\phi_a(x))_{a \in \mathcal{A}_m}$ and $K_m(x, x') := \langle \Phi_m(x), \Phi_m(x') \rangle$.
Although $|\mathcal{A}_m|$ grows exponentially in m, $K_m(x, x')$ can often be computed without forming $\Phi_m(x)$.
Thus support-vector or centroid classifiers can use sequence similarities directly.

10 Prototype and Nearest-Neighbor Methods

Prototype and nearest-neighbor classifiers are model-free rules built from distances in feature space. They can fit irregular decision boundaries and are often strong predictors, but their behavior is controlled by geometry: scaling, metric choice, neighborhood size, and storage of representative points. The local-averaging view appears in Method 1.4.2; this section records the classification and metric-design side.

10.1 Prototype Classifiers

Definition 10.1.1. *Labeled Prototype Rule*
Let $\mathcal{D}_n = \{(x_i, g_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^p$ and $g_i \in \{1, \dots, K\}$. Fix prototype counts $R_1, \dots, R_K$, distance d on $\mathbb{R}^p$.

(i) A labeled prototype set is $\mathcal{M} = \{(m_j(k), k) : m_j(k) \in \mathbb{R}^p,\ 1 \le j \le R_k,\ 1 \le k \le K\}$.
(ii) For $x \in \mathbb{R}^p$, choose $(\widehat{j}, \widehat{k}) \in \underset{1 \le k \le K,\ 1 \le j \le R_k}{\arg\min} d(x, m_j(k))$ and define $\widehat{G}(x) = \widehat{k}$.
(iii) If $\mathcal{M} = \{(x_i, g_i) : 1 \le i \le n\}$, then $\widehat{G}$ is the one-nearest-neighbor classifier.

For quantitative features, Euclidean distance requires comparable coordinate scales.

Method 10.1.2. *k-Means Prototypes*
Let $I_k = \{i : g_i = k\}$, fix R prototypes per class, and use $d(x, z) = \|x - z\|_2$.

(i) For each $k = 1, \dots, K$, choose $m_1(k), \dots, m_R(k) \in \mathbb{R}^p$ to minimize $\sum_{i \in I_k} \min_{1 \le j \le R} \|x_i - m_j(k)\|_2^2$.
(ii) For $x \in \mathbb{R}^p$, choose $(\widehat{j}, \widehat{k}) \in \underset{1 \le k \le K,\ 1 \le j \le R}{\arg\min} \|x - m_j(k)\|_2$ and set $\widehat{G}(x) = \widehat{k}$.

The stored labeled prototype set is $\mathcal{M} = \{(m_j(k), k) : 1 \le j \le R,\ 1 \le k \le K\}$.
The rule has piecewise-linear Euclidean decision boundaries; since classes are compressed independently, prototypes need not move away from competing classes near a boundary.

Algorithm 20: Learning Vector Quantization

Require: Labeled data $(x_i, g_i)_{i=1}^n$, prototypes per class R, learning rates $\epsilon_t \downarrow 0$

Ensure: Labeled prototypes $m_j(k)$
  Initialize $m_1(k), \dots, m_R(k)$ from training observations in class k, for each k
  for $t = 1, 2, \dots$ **do**
    Sample a training index i
    Choose the closest prototype (j, k) to x_i over all classes and prototype indices
    if $g_i = k$ **then**
      $m_j(k) \leftarrow m_j(k) + \epsilon_t\big(x_i - m_j(k)\big)$
    else
      $m_j(k) \leftarrow m_j(k) - \epsilon_t\big(x_i - m_j(k)\big)$
    end if
  end for

Remark 10.1.3. *Boundary-Aware Prototype Motion*
For a sampled pair (x_i, g_i), let $(j^\star, k^\star) \in \underset{1\le k\le K,\ 1\le j\le R_k}{\arg\min} d(x_i, m_j(k))$ and let $\epsilon_t > 0$.

(i) If $g_i = k^\star$, set $m_{j^\star}(k^\star) \leftarrow m_{j^\star}(k^\star) + \epsilon_t(x_i - m_{j^\star}(k^\star))$.
(ii) If $g_i \neq k^\star$, set $m_{j^\star}(k^\star) \leftarrow m_{j^\star}(k^\star) - \epsilon_t(x_i - m_{j^\star}(k^\star))$.

Thus every sampled class can move the nearest prototype of any class.
Unlike classwise k-means, LVQ is an online algorithmic rule rather than exact minimization of a fixed criterion.

Method 10.1.4. *Gaussian Mixtures as Soft Prototypes*
Let $\varphi_p(\cdot; \mu, \Sigma)$ be the p-variate Gaussian density. For class k, fix a prior ρ_k, component weights α_{kj}, centers $\mu_{kj} \in \mathbb{R}^p$, covariances $\Sigma_{kj} \succ 0$, and $j = 1, \dots, R_k$, with $\rho_k, \alpha_{kj} \ge 0$, $\sum_{k=1}^{K} \rho_k = 1$, and $\sum_{j=1}^{R_k} \alpha_{kj} = 1$.

(i) Define the class density $f_k(x) = \sum_{j=1}^{R_k} \alpha_{kj} \varphi_p(x; \mu_{kj}, \Sigma_{kj})$.

(ii) For $x \in \mathbb{R}^p$, set $Z(x) = \sum_{\ell=1}^{K} \rho_\ell f_\ell(x) = \sum_{\ell=1}^{K} \sum_{r=1}^{R_\ell} \rho_\ell \alpha_{\ell r} \varphi_p(x; \mu_{\ell r}, \Sigma_{\ell r})$.

(iii) Compute $\tau_k(x) = \frac{\rho_k f_k(x)}{Z(x)}$ and $\omega_{kj}(x) = \frac{\rho_k \alpha_{kj} \varphi_p(x; \mu_{kj}, \Sigma_{kj})}{Z(x)}$. Set $\widehat{G}(x) \in \underset{1\le k\le K}{\arg\max}\, \tau_k(x)$.

The pairs (μ_{kj}, Σ_{kj}) are soft prototypes and $\omega_{kj}(x)$ gives fractional membership.
If $\Sigma_{kj} = \sigma^2 I$ and all $\rho_k \alpha_{kj} > 0$, then as $\sigma^2 \downarrow 0$ the dominant component is nearest center μ_{kj} away from ties.

10.2 Nearest Neighbors

Method 10.2.1. *k-Nearest-Neighbor Classification*
Fix a distance d on the feature space and an integer $k \in \{1, \dots, n\}$.

(i) For a query x_0, compute $d_i = d(x_i, x_0)$ for $i = 1, \dots, n$.
(ii) Let $N_k(x_0) \subset \{1, \dots, n\}$ be indices of k smallest values among $d_1, \dots, d_n$. Set $\widehat{G}(x_0) \in \underset{g\in\{1,\dots,K\}}{\arg\max} \sum_{i\in N_k(x_0)} \mathbf{1}_{\{g_i = g\}}$.

Small k gives lower bias and higher variance; larger k smooths the decision boundary by averaging over more labels. The tuning parameter k is usually chosen by validation or cross-validation.

Proposition 10.2.2. *Asymptotic One-Nearest-Neighbor Error*
Assume $K \ge 2$ and fix x. Let $p_k = \mathbb{P}(G = k \mid X = x)$, choose $k^\star \in \underset{1\le k\le K}{\arg\max}\, p_k$, and set $L^\star(x) = 1 - p_{k^\star}$.

In the asymptotic one-nearest-neighbor idealisation, the query label G and nearest-neighbor label G' are independent draws from $(p_1, \dots, p_K)$, and the classifier predicts G'.

(i) $L_{1\mathrm{NN}}(x) = \mathbb{P}(G' \neq G) = \sum_{k=1}^{K} p_k(1 - p_k) = 1 - \sum_{k=1}^{K} p_k^2$.

(ii) $L^\star(x) \le L_{1\mathrm{NN}}(x) \le 2L^\star(x) - \frac{K}{K-1}\big(L^\star(x)\big)^2 \le 2L^\star(x)$.

Proof.

(i) Independence gives $\mathbb{P}(G' = G) = \sum_{k=1}^{K} p_k^2$, hence $L_{1\mathrm{NN}}(x) = 1 - \sum_{k=1}^{K} p_k^2$.

(ii) Let $p_\star = p_{k^\star}$ and $q = 1 - p_\star = L^\star(x)$.

Since $p_k \le p_\star$ for every k, $\sum_{k=1}^{K} p_k^2 \le p_\star \sum_{k=1}^{K} p_k = p_\star$, so $L_{1\mathrm{NN}}(x) \ge 1 - p_\star = L^\star(x)$.

Also $\sum_{k\neq k^\star} p_k = q$, so Cauchy-Schwarz gives $\sum_{k\neq k^\star} p_k^2 \ge \frac{q^2}{K-1}$.

Therefore $\sum_{k=1}^{K} p_k^2 \ge (1-q)^2 + \frac{q^2}{K-1}$ and $L_{1\mathrm{NN}}(x) \le 2q - \frac{Kq^2}{K-1} \le 2q = 2L^\star(x)$.

Structural Commentary: Dominant mass gives the lower bound; uniform residual mass gives the upper bound. *Key Insight*: One-nearest-neighbor error is label-disagreement probability at fixed x. □

Example 10.2.3. *Tuning Depends on the Problem*
Let $X = (X_1, \dots, X_{10})$ have independent $\mathrm{Unif}(0, 1)$ coordinates and let $Y \in \{0, 1\}$.

(i) *Halfspace*: $Y = \mathbf{1}_{\{X_1 > 1/2\}}$; the decision boundary is the single hyperplane $X_1 = 1/2$.
(ii) *Checkerboard*: $Y = \mathbf{1}_{\{\mathrm{sign}(\prod_{j=1}^{3}(X_j - 1/2)) > 0\}}$.

The planes $X_j = 1/2$ for $j = 1, 2, 3$ form eight cells with alternating labels.

Across repeated simulations, the preferred neighborhood size k or prototype count R depended on the case. For memory-based and prototype rules, tuning is geometry-dependent, not a universal complexity constant.

Example 10.2.4. *Image Scene Classification by Local Pixel Features*
For each pixel s, let $b_s \in \mathbb{R}^4$ be its four spectral-band vector and let $g_s \in \{1, \dots, 7\}$ be its land-use label.
Replace b_s by the local feature vector $z_s = (b_t : t \in \mathcal{N}_3(s)) \in \mathbb{R}^{36}$, where $\mathcal{N}_3(s)$ is the 3×3 pixel patch centered at s. A five-nearest-neighbor rule on the features z_s had reported test error about 9.5%, showing that local spatial features can make nearest-neighbor classification competitive for irregular class boundaries.

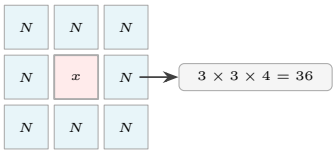

Figure 267: A 3×3 patch over four bands gives 36 features per pixel.

10.3 Invariant Distances and Adaptive Neighborhoods

Method 10.3.1. *Invariant Metrics and Tangent Distance*
Let $x, x' \in \mathbb{R}^p$, let $\mathcal{A}$ be a transformation family, and write $T_\alpha x$ for the transformed input.
For local transformations, let $V_x \in \mathbb{R}^{p \times r_x}$ span the tangent directions at x.

(i) *Orbit metric*: $d_{\text{inv}}(x, x') = \inf_{\alpha, \beta \in \mathcal{A}} \left\| T_\alpha x - T_\beta x' \right\|_2$ compares the transformation orbits of x and x'.

(ii) *Tangent metric*: $d_{\tan}(x, x') = \inf_{a,b} \|x + V_x a - x' - V_{x'} b\|_2$ compares tangent affine spaces.

The tangent metric discounts small local transformations without making large transformations cost-free.
Data augmentation encodes similar invariance by storing transformed copies, which becomes costly when many transformation directions are needed.

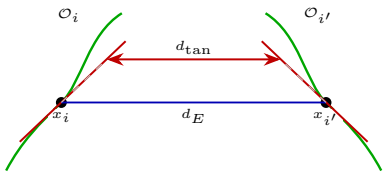

Figure 268: Tangent distance uses local tangents instead of orbits; d_E is ordinary distance.

Method 10.3.2. *Discriminant Adaptive Nearest Neighbors*
DANN replaces the fixed Euclidean metric by a query-dependent quadratic metric.
For a query x_0, choose a pilot index set $P(x_0)$, final neighbor count h, and $\epsilon > 0$. From $\{(x_i, g_i) : i \in P(x_0)\}$ estimate local class proportions π_k, class means $\bar{x}_k$, local mean $\bar{x}$, and pooled within-class covariance $\mathbf{W} \succ 0$.

(i) Compute the local between-class covariance $\mathbf{B} = \sum_{k=1}^{K} \pi_k (\bar{x}_k - \bar{x})(\bar{x}_k - \bar{x})^T$.

(ii) Form $\mathbf{B}^\star = \mathbf{W}^{-1/2} \mathbf{B} \mathbf{W}^{-1/2}$ and $\mathbf{\Sigma}_{x_0} = \mathbf{W}^{-1/2} (\mathbf{B}^\star + \epsilon I) \mathbf{W}^{-1/2}$.

(iii) With $D_{x_0}(x_i, x_0) = (x_i - x_0)^T \mathbf{\Sigma}_{x_0} (x_i - x_0)$, let $N_h(x_0)$ index h smallest distances.

(iv) Set $\widehat{G}(x_0) \in \arg\max_{g \in \{1, \dots, K\}} \sum_{i \in N_h(x_0)} \mathbf{1}_{\{g_i = g\}}$.

Whitening by $\mathbf{W}$ removes local within-class scale.
$\mathbf{B}^\star + \epsilon I$ stretches neighborhoods along directions with weak local class separation, and ϵ prevents degeneracy.

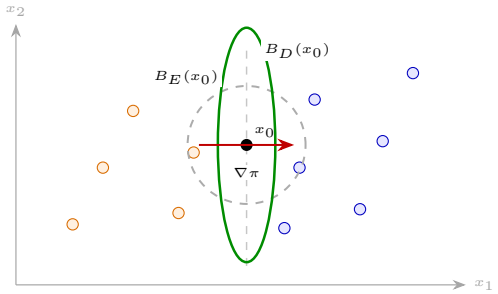

Figure 269: DANN replaces the Euclidean ball by a local ellipsoid, narrow along the class-change direction and elongated where class probabilities vary slowly.

Method 10.3.3. *Global Dimension Reduction for Nearest Neighbors*
Fix a target dimension $L \le p$. Let $\mathbf{B}_i$ be the local between-class matrix estimated at training point x_i.

(i) Compute the global class-separation matrix $\overline{\mathbf{B}} = N^{-1} \sum_{i=1}^{N} \mathbf{B}_i$.

(ii) Let $(\theta_\ell, e_\ell)_{\ell=1}^{p}$ be eigenpairs of $\overline{\mathbf{B}}$ with $\theta_1 \ge \cdots \ge \theta_p$.

(iii) Project by $P_L x = (e_1^T x, \dots, e_L^T x) \in \mathbb{R}^L$ and run nearest-neighbor classification on $P_L x_i$.

The rank-L summary $\overline{\mathbf{B}}_{[L]} = \sum_{\ell=1}^{L} \theta_\ell e_\ell e_\ell^T$ solves $\min_{\mathrm{rank}(M)=L} \sum_{i=1}^{N} \mathrm{trace}[(\mathbf{B}_i - M)^2]$.
The reduced space is selected from local class-separation information, not unsupervised variance alone.

Remark 10.3.4. *Computation and Storage*
With N stored observations in $\mathbb{R}^p$, direct nearest-neighbor search costs $O(Np)$ per query and stores $O(Np)$ coordinates. Storage can be reduced by replacing the retained set $S \subset \{1, \dots, N\}$.

(i) *Prototype compression*: store learned representatives rather than all training observations.
(ii) *Editing*: delete observations whose local neighborhoods have inconsistent labels.
(iii) *Condensing*: grow a small retained set by adding i only if it misclassifies (x_i, g_i).

Fast search reduces query cost in favorable geometries; compression, editing, and condensing reduce storage by keeping representatives or boundary-relevant observations.

11 Flexible and Mixture Discriminant Analysis

11.1 Flexible and Mixture Discriminant Rules

Linear discriminant analysis is useful partly because it is simple: each class is represented by a centroid, distances are measured in a common covariance metric, and the decision boundaries are linear. Its limitations are equally structural. A single centroid may not represent an inhomogeneous class, linear boundaries may be too rigid, and high-dimensional predictors may require smoothness or shrinkage beyond ordinary covariance pooling. Flexible, penalized, and mixture discriminant methods keep the LDA comparison geometry while changing the regression fit, the metric, or the number of class prototypes.

Method 11.1.1. *Flexible and Penalized Discriminant Analysis*
Optimal scoring represents LDA by regressions from x_i to transformed labels.
Let $g_i \in \{1, \dots, K\}$, $1 \le L \le K-1$, label scores $\theta_\ell : \{1, \dots, K\} \to \mathbb{R}$, and fitted score maps $\eta_\ell : \mathcal{X} \to \mathbb{R}$, with centering and orthogonality constraints on the θ_ℓ removing constant score directions.

(i) *Flexible discriminant analysis*: minimize $\mathcal{L}_{\mathrm{FDA}} = n^{-1} \sum_{\ell=1}^{L} \sum_{i=1}^{n} \{\theta_\ell(g_i) - \eta_\ell(x_i)\}^2 + \lambda \sum_{\ell=1}^{L} J(\eta_\ell)$ over θ_ℓ and η_ℓ from a chosen smoother class: linear, spline, additive, MARS, kernel, or similar.
(ii) *Penalized discriminant analysis*: impose $\eta_\ell(x) = h(x)^\top \beta_\ell$ and roughness penalty $\beta_\ell^\top \Omega \beta_\ell$.
Equivalently classify by $D_\lambda(x,k) = \big(h(x) - \bar{h}_k\big)^\top (\widehat{\Sigma}_{W,h} + \lambda\Omega)^{-1} \big(h(x) - \bar{h}_k\big)$, with $\bar{h}_k$ and $\widehat{\Sigma}_{W,h}$ the class-k centroid and pooled within-class covariance in h-coordinates.

For $\eta_\ell(x) = x^\top \beta_\ell$ and no roughness penalty, the fitted directions recover LDA up to scale; FDA changes the regression class to obtain nonlinear boundaries, while PDA shrinks rough or unstable h-directions.

Algorithm 21: FDA Coordinates by Optimal Scoring

Require: Class labels $g_i \in \{1, \dots, K\}$, indicator response matrix $Y \in \mathbb{R}^{n \times K}$, linear smoother S_λ for the chosen regression method
Ensure: Nonconstant discriminant coordinate functions $\eta_1, \dots, \eta_{K-1}$
Fit the multivariate regression of Y on the features: $\widehat{Y} \leftarrow S_\lambda Y$
Let $\eta^*(x)$ denote the K fitted regression functions evaluated at a new input x
Set $D_\pi \leftarrow Y^T Y / n$
Solve $Y^T \widehat{Y} \theta = \gamma D_\pi \theta$ and normalize eigenvectors so that $\Theta^T D_\pi \Theta = I$
Discard the constant scoring direction
for each remaining eigenvector θ_ℓ **do**
  Set $\eta_\ell(x) \leftarrow \theta_\ell^T \eta^*(x)$
end for

Remark 11.1.2. *FDA as LDA After a Regression Map*
Let $\eta = (\eta_1, \dots, \eta_L) : \mathcal{X} \to \mathbb{R}^L$ and $z = \eta(x)$. FDA applies LDA in z-space, so linear discriminants in z pull back to decision boundaries in x determined by the regression class for η.
Polynomial η gives algebraic boundaries, quadratic η gives quadratic boundaries, and additive, spline, or kernel η gives the corresponding structured nonlinear boundaries.

Method 11.1.3. *Mixture Discriminant Analysis*
Mixture discriminant analysis replaces the class-k LDA prototype by R_k Gaussian subclasses while sharing one covariance Σ. Let $\pi_{kr} \ge 0$, $\sum_{r=1}^{R_k} \pi_{kr} = 1$, centers μ_{kr}, class prior ρ_k, $f_k(x) = \sum_{r=1}^{R_k} \pi_{kr} \phi(x; \mu_{kr}, \Sigma)$, and $p_k(x) = \Pr(G = k \mid X = x) = \rho_k f_k(x) / \sum_{j=1}^{K} \rho_j f_j(x)$.
If $R_k = 1$ for all k, this is ordinary common-covariance Gaussian discrimination.

(i) *E-step*: for $g_i = k$, set $r_{ikr} = \pi_{kr} \phi(x_i; \mu_{kr}, \Sigma) / \sum_{s=1}^{R_k} \pi_{ks} \phi(x_i; \mu_{ks}, \Sigma)$ for $r = 1, \dots, R_k$.
(ii) *M-step*: update π_{kr}, μ_{kr}, and Σ by weighted Gaussian maximum likelihood with weights r_{ikr}.
(iii) *Blurred-response variant*: use subclass columns c_{kr} and row i equal to $(r_{ik1}, \dots, r_{ikR_k})$ on class-k columns, zero elsewhere; then fit weighted DA with optional rank restrictions, FDA, or PDA.

Remark 11.1.4. *MDA and Prototype Rules*
MDA is a probabilistic prototype classifier with prototypes μ_{kr}, weights π_{kr}, and shared metric Σ^{-1}. It assigns x by posterior scores $p_k(x) \propto \rho_k \sum_{r=1}^{R_k} \pi_{kr} \phi(x; \mu_{kr}, \Sigma)$, not by a hard nearest-center rule.
Relative to classwise k-means, it estimates π_{kr} and Σ; relative to an unconstrained Gaussian mixture, each component (k, r) is nested inside the observed class k.

Definition 11.1.5. *Kernel Density Classification*
Let $G \in \{1, \dots, J\}$ be the class label.
Estimate $\pi_j := \mathbb{P}(G = j)$ by $\widehat{\pi}_j$ and $f_j(x) := f_{X \mid G=j}(x)$ by a kernel density estimate $\widehat{f}_j(x)$.

(i) *Posterior*: Define $\widehat{p}_j(x) := \widehat{\mathbb{P}}(G = j \mid X = x) := \dfrac{\widehat{\pi}_j \widehat{f}_j(x)}{\sum_{k=1}^{J} \widehat{\pi}_k \widehat{f}_k(x)}$ whenever the denominator is positive.
(ii) *Classifier*: The plug-in Bayes rule is $\widehat{g}(x) \in \operatorname{argmax}_{1 \le j \le J} \widehat{p}_j(x)$.

Full density estimation is unnecessary if only the boundary is needed.

Remark 11.1.6. *Computational Scale*
Let N be the sample size, p the predictor dimension, and T_{fit} the cost of the chosen smoother.

(i) *LDA/PDA*: covariance-scale linear algebra costs about $Np^2 + p^3$ before low-rank reduction.
(ii) *FDA*: cost is T_{fit}; additive or adaptive basis fits can be nearly linear in N for fixed model size, while dense kernel or spline smoothers can cost at least an $N \times N$ solve.
(iii) *MDA*: cost multiplies the common-covariance discriminant fit by EM iterations and depends on initialization because mixture likelihoods are nonconvex.

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